FANUC Series 30*i*/300*i*/300*i*s-MODEL A
FANUC Series 31*i*/310*i*/310*i*s-MODEL A
FANUC Series 32*i*/320*i*/320*i*s-MODEL A

Common to Lathe System/Machining Center System

USER'S MANUAL

B-63944EN/03

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In this manual we have tried as much as possible to describe all the various matters.
However, we cannot describe all the matters which must not be done, or which cannot be done, because there are so many possibilities.
Therefore, matters which are not especially described as possible in this manual should be regarded as "impossible".

SAFETY PRECAUTIONS

This section describes the safety precautions related to the use of CNC units.
It is essential that these precautions be observed by users to ensure the safe operation of machines equipped with a CNC unit (all descriptions in this section assume this configuration). Note that some precautions are related only to specific functions, and thus may not be applicable to certain CNC units.
Users must also observe the safety precautions related to the machine, as described in the relevant manual supplied by the machine tool builder. Before attempting to operate the machine or create a program to control the operation of the machine, the operator must become fully familiar with the contents of this manual and relevant manual supplied by the machine tool builder.

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DEFINITION OF WARNING, CAUTION, AND NOTE

This manual includes safety precautions for protecting the user and preventing damage to the machine. Precautions are classified into **Warning** and **Caution** according to their bearing on safety. Also, supplementary information is described as a **Note**. Read the **Warning**, **Caution**, and **Note** thoroughly before attempting to use the machine.

> ⚠ **WARNING**
> Applied when there is a danger of the user being injured or when there is a danger of both the user being injured and the equipment being damaged if the approved procedure is not observed.

> ⚠ **CAUTION**
> Applied when there is a danger of the equipment being damaged, if the approved procedure is not observed.

> **NOTE**
> The Note is used to indicate supplementary information other than Warning and Caution.

- Read this manual carefully, and store it in a safe place.

GENERAL WARNINGS AND CAUTIONS

⚠ WARNING

1 Never attempt to machine a workpiece without first checking the operation of the machine. Before starting a production run, ensure that the machine is operating correctly by performing a trial run using, for example, the single block, feedrate override, or machine lock function or by operating the machine with neither a tool nor workpiece mounted. Failure to confirm the correct operation of the machine may result in the machine behaving unexpectedly, possibly causing damage to the workpiece and/or machine itself, or injury to the user.

2 Before operating the machine, thoroughly check the entered data.
Operating the machine with incorrectly specified data may result in the machine behaving unexpectedly, possibly causing damage to the workpiece and/or machine itself, or injury to the user.

3 Ensure that the specified feedrate is appropriate for the intended operation. Generally, for each machine, there is a maximum allowable feedrate. The appropriate feedrate varies with the intended operation. Refer to the manual provided with the machine to determine the maximum allowable feedrate.
If a machine is run at other than the correct speed, it may behave unexpectedly, possibly causing damage to the workpiece and/or machine itself, or injury to the user.

4 When using a tool compensation function, thoroughly check the direction and amount of compensation.
Operating the machine with incorrectly specified data may result in the machine behaving unexpectedly, possibly causing damage to the workpiece and/or machine itself, or injury to the user.

⚠ WARNING

5 The parameters for the CNC and PMC are factory-set. Usually, there is not need to change them. When, however, there is not alternative other than to change a parameter, ensure that you fully understand the function of the parameter before making any change.
Failure to set a parameter correctly may result in the machine behaving unexpectedly, possibly causing damage to the workpiece and/or machine itself, or injury to the user.

6 Immediately after switching on the power, do not touch any of the keys on the MDI panel until the position display or alarm screen appears on the CNC unit.
Some of the keys on the MDI panel are dedicated to maintenance or other special operations. Pressing any of these keys may place the CNC unit in other than its normal state. Starting the machine in this state may cause it to behave unexpectedly.

7 The User's Manual and programming manual supplied with a CNC unit provide an overall description of the machine's functions, including any optional functions. Note that the optional functions will vary from one machine model to another. Therefore, some functions described in the manuals may not actually be available for a particular model. Check the specification of the machine if in doubt.

8 Some functions may have been implemented at the request of the machine-tool builder. When using such functions, refer to the manual supplied by the machine-tool builder for details of their use and any related cautions.

⚠ CAUTION

The liquid-crystal display is manufactured with very precise fabrication technology. Some pixels may not be turned on or may remain on. This phenomenon is a common attribute of LCDs and is not a defect.

NOTE

Programs, parameters, and macro variables are stored in nonvolatile memory in the CNC unit. Usually, they are retained even if the power is turned off.

Such data may be deleted inadvertently, however, or it may prove necessary to delete all data from nonvolatile memory as part of error recovery.

To guard against the occurrence of the above, and assure quick restoration of deleted data, backup all vital data, and keep the backup copy in a safe place.

WARNINGS AND CAUTIONS RELATED TO PROGRAMMING

This section covers the major safety precautions related to programming. Before attempting to perform programming, read the supplied User's Manual carefully such that you are fully familiar with their contents.

⚠ WARNING

1 **Coordinate system setting**
If a coordinate system is established incorrectly, the machine may behave unexpectedly as a result of the program issuing an otherwise valid move command. Such an unexpected operation may damage the tool, the machine itself, the workpiece, or cause injury to the user.

2 **Positioning by nonlinear interpolation**
When performing positioning by nonlinear interpolation (positioning by nonlinear movement between the start and end points), the tool path must be carefully confirmed before performing programming. Positioning involves rapid traverse. If the tool collides with the workpiece, it may damage the tool, the machine itself, the workpiece, or cause injury to the user.

3 **Function involving a rotation axis**
When programming polar coordinate interpolation or normal-direction (perpendicular) control, pay careful attention to the speed of the rotation axis. Incorrect programming may result in the rotation axis speed becoming excessively high, such that centrifugal force causes the chuck to lose its grip on the workpiece if the latter is not mounted securely. Such mishap is likely to damage the tool, the machine itself, the workpiece, or cause injury to the user.

4 **Inch/metric conversion**
Switching between inch and metric inputs does not convert the measurement units of data such as the workpiece origin offset, parameter, and current position. Before starting the machine, therefore, determine which measurement units are being used. Attempting to perform an operation with invalid data specified may damage the tool, the machine itself, the workpiece, or cause injury to the user.

⚠ **WARNING**

5 **Constant surface speed control**
When an axis subject to constant surface speed control approaches the origin of the workpiece coordinate system, the spindle speed may become excessively high. Therefore, it is necessary to specify a maximum allowable speed. Specifying the maximum allowable speed incorrectly may damage the tool, the machine itself, the workpiece, or cause injury to the user.

6 **Stroke check**
After switching on the power, perform a manual reference position return as required. Stroke check is not possible before manual reference position return is performed. Note that when stroke check is disabled, an alarm is not issued even if a stroke limit is exceeded, possibly damaging the tool, the machine itself, the workpiece, or causing injury to the user.

7 **Tool post interference check**
A tool post interference check is performed based on the tool data specified during automatic operation. If the tool specification does not match the tool actually being used, the interference check cannot be made correctly, possibly damaging the tool or the machine itself, or causing injury to the user. After switching on the power, or after selecting a tool post manually, always start automatic operation and specify the tool number of the tool to be used.

8 **Absolute/incremental mode**
If a program created with absolute values is run in incremental mode, or vice versa, the machine may behave unexpectedly.

9 **Plane selection**
If an incorrect plane is specified for circular interpolation, helical interpolation, or a canned cycle, the machine may behave unexpectedly. Refer to the descriptions of the respective functions for details.

10 **Torque limit skip**
Before attempting a torque limit skip, apply the torque limit. If a torque limit skip is specified without the torque limit actually being applied, a move command will be executed without performing a skip.

⚠ **WARNING**

11 **Programmable mirror image**
Note that programmed operations vary considerably when a programmable mirror image is enabled.

12 **Compensation function**
If a command based on the machine coordinate system or a reference position return command is issued in compensation function mode, compensation is temporarily canceled, resulting in the unexpected behavior of the machine.
Before issuing any of the above commands, therefore, always cancel compensation function mode.

WARNINGS AND CAUTIONS RELATED TO HANDLING

This section presents safety precautions related to the handling of machine tools. Before attempting to operate your machine, read the supplied User's Manual carefully, such that you are fully familiar with their contents.

⚠ WARNING

1 **Manual operation**
When operating the machine manually, determine the current position of the tool and workpiece, and ensure that the movement axis, direction, and feedrate have been specified correctly. Incorrect operation of the machine may damage the tool, the machine itself, the workpiece, or cause injury to the operator.

2 **Manual reference position return**
After switching on the power, perform manual reference position return as required.
If the machine is operated without first performing manual reference position return, it may behave unexpectedly. Stroke check is not possible before manual reference position return is performed.
An unexpected operation of the machine may damage the tool, the machine itself, the workpiece, or cause injury to the user.

3 **Manual numeric command**
When issuing a manual numeric command, determine the current position of the tool and workpiece, and ensure that the movement axis, direction, and command have been specified correctly, and that the entered values are valid.
Attempting to operate the machine with an invalid command specified may damage the tool, the machine itself, the workpiece, or cause injury to the operator.

4 **Manual handle feed**
In manual handle feed, rotating the handle with a large scale factor, such as 100, applied causes the tool and table to move rapidly. Careless handling may damage the tool and/or machine, or cause injury to the user.

⚠ **WARNING**

5 **Disabled override**
If override is disabled (according to the specification in a macro variable) during threading, rigid tapping, or other tapping, the speed cannot be predicted, possibly damaging the tool, the machine itself, the workpiece, or causing injury to the operator.

6 **Origin/preset operation**
Basically, never attempt an origin/preset operation when the machine is operating under the control of a program. Otherwise, the machine may behave unexpectedly, possibly damaging the tool, the machine itself, the tool, or causing injury to the user.

7 **Workpiece coordinate system shift**
Manual intervention, machine lock, or mirror imaging may shift the workpiece coordinate system. Before attempting to operate the machine under the control of a program, confirm the coordinate system carefully.
If the machine is operated under the control of a program without making allowances for any shift in the workpiece coordinate system, the machine may behave unexpectedly, possibly damaging the tool, the machine itself, the workpiece, or causing injury to the operator.

8 **Software operator's panel and menu switches**
Using the software operator's panel and menu switches, in combination with the MDI panel, it is possible to specify operations not supported by the machine operator's panel, such as mode change, override value change, and jog feed commands. Note, however, that if the MDI panel keys are operated inadvertently, the machine may behave unexpectedly, possibly damaging the tool, the machine itself, the workpiece, or causing injury to the user.

9 **RESET key**
Pressing the RESET key stops the currently running program. As a result, the servo axes are stopped. However, the RESET key may fail to function for reasons such as an MDI panel problem. So, when the motors must be stopped, use the emergency stop button instead of the RESET key to ensure security.

⚠ WARNING

10 **Manual intervention**
If manual intervention is performed during programmed operation of the machine, the tool path may vary when the machine is restarted. Before restarting the machine after manual intervention, therefore, confirm the settings of the manual absolute switches, parameters, and absolute/incremental command mode.

11 **Feed hold, override, and single block**
The feed hold, feedrate override, and single block functions can be disabled using custom macro system variable #3004. Be careful when operating the machine in this case.

12 **Dry run**
Usually, a dry run is used to confirm the operation of the machine. During a dry run, the machine operates at dry run speed, which differs from the corresponding programmed feedrate. Note that the dry run speed may sometimes be higher than the programmed feed rate.

13 **Cutter and tool nose radius compensation in MDI mode**
Pay careful attention to a tool path specified by a command in MDI mode, because cutter or tool nose radius compensation is not applied. When a command is entered from the MDI to interrupt in automatic operation in cutter or tool nose radius compensation mode, pay particular attention to the tool path when automatic operation is subsequently resumed. Refer to the descriptions of the corresponding functions for details.

14 **Program editing**
If the machine is stopped, after which the machining program is edited (modification, insertion, or deletion), the machine may behave unexpectedly if machining is resumed under the control of that program. Basically, do not modify, insert, or delete commands from a machining program while it is in use.

WARNINGS RELATED TO DAILY MAINTENANCE

⚠ WARNING

1 **Memory backup battery replacement**

When replacing the memory backup batteries, keep the power to the machine (CNC) turned on, and apply an emergency stop to the machine. Because this work is performed with the power on and the cabinet open, only those personnel who have received approved safety and maintenance training may perform this work.

When replacing the batteries, be careful not to touch the high-voltage circuits (marked ⚠ and fitted with an insulating cover).

Touching the uncovered high-voltage circuits presents an extremely dangerous electric shock hazard.

NOTE

The CNC uses batteries to preserve the contents of its memory, because it must retain data such as programs, offsets, and parameters even while external power is not applied.

If the battery voltage drops, a low battery voltage alarm is displayed on the machine operator's panel or screen.

When a low battery voltage alarm is displayed, replace the batteries within a week. Otherwise, the contents of the CNC's memory will be lost.

Refer to the Section “Method of replacing battery” in the User’s Manual (Common to T/M series) for details of the battery replacement procedure.

⚠ WARNING

2 **Absolute pulse coder battery replacement**

When replacing the memory backup batteries, keep the power to the machine (CNC) turned on, and apply an emergency stop to the machine. Because this work is performed with the power on and the cabinet open, only those personnel who have received approved safety and maintenance training may perform this work.

When replacing the batteries, be careful not to touch the high-voltage circuits (marked ⚠ and fitted with an insulating cover).

Touching the uncovered high-voltage circuits presents an extremely dangerous electric shock hazard.

NOTE

The absolute pulse coder uses batteries to preserve its absolute position.

If the battery voltage drops, a low battery voltage alarm is displayed on the machine operator's panel or screen.

When a low battery voltage alarm is displayed, replace the batteries within a week. Otherwise, the absolute position data held by the pulse coder will be lost.

Refer to the FANUC SERVO MOTOR αi series Maintenance Manual for details of the battery replacement procedure.

⚠ WARNING

3 Fuse replacement

Before replacing a blown fuse, however, it is necessary to locate and remove the cause of the blown fuse.

For this reason, only those personnel who have received approved safety and maintenance training may perform this work.

When replacing a fuse with the cabinet open, be careful not to touch the high-voltage circuits (marked ⚠ and fitted with an insulating cover). Touching an uncovered high-voltage circuit presents an extremely dangerous electric shock hazard.

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I. GENERAL

1 GENERAL

About this manual

This manual consists of the following parts:

I. GENERAL
Describes chapter organization, applicable models, related manuals, and notes for reading this manual.

II. PROGRAMMING
Describes each function: Format used to program functions in the NC language, explanations, and limitations.

III. OPERATION
Describes the manual operation and automatic operation of a machine, procedures for inputting and outputting data, and procedures for editing a program.

IV. MAINTENANCE
Describes procedures for daily maintenance and replacing batteries.

APPENDIX
Lists parameters, valid data ranges, and alarms.

NOTE

1 This manual describes the functions common to the lathe system and machining center system. For the functions specific to the lathe system or machining center system, refer to the User's Manual (Lathe System) (B-63944EN-1) or the User's Manual (Machining Center System) (B-63944EN-2).

2 Some functions described in this manual may not be applied to some products. For detail, refer to the Descriptions manual (B-63942EN).

3 This manual does not detail the parameters not mentioned in the text. For details of those parameters, refer to the Parameter Manual (B-63950EN).
Parameters are used to set functions and operating conditions of a CNC machine tool, and frequently-used values in advance. Usually, the machine tool builder factory-sets parameters so that the user can use the machine tool easily.

4 This manual describes not only basic functions but also optional functions. Look up the options incorporated into your system in the manual written by the machine tool builder.

Applicable models

This manual describes the models indicated in the table below.
In the text, the abbreviations indicated below may be used.

Model name	Abbreviation	
FANUC Series 30*i*-MODEL A	30*i* –A	Series 30*i*
FANUC Series 300*i*-MODEL A	300*i*–A	Series 300*i*
FANUC Series 300*i*s-MODEL A	300*i*s–A	Series 300*i*s
FANUC Series 31*i*-MODEL A	31*i* –A	Series 31*i*
FANUC Series 31*i*-MODEL A5	31*i* –A5	
FANUC Series 310*i*-MODEL A	310*i*–A	Series 310*i*
FANUC Series 310*i*-MODEL A5	310*i*–A5	
FANUC Series 310*i*s-MODEL A	310*i*s–A	Series 310*i*s
FANUC Series 310*i*s-MODEL A5	310*i*s–A5	
FANUC Series 32*i*-MODEL A	32*i* –A	Series 32*i*
FANUC Series 320*i*-MODEL A	320*i*–A	Series 320*i*
FANUC Series 320*i*s-MODEL A	320*i*s–A	Series 320*i*s

NOTE
1 For an explanatory purpose, the following descriptions may be used according to the types of path control used:
- T series: For the lathe system
- M series: For the machining center system

2 Unless otherwise noted, the model names 31*i*/310*i*/310*i*s-A, 31*i*/310*i*/310*i*s-A5, and 32*i*/320*i*/320*i*s-A are collectively referred to as 30*i*/300*i*/300*i*s. However, this convention is not necessarily observed when item 3 below is applicable.

3 Some functions described in this manual may not be applied to some products.
For details, refer to the Descriptions (B-63942EN).

Special symbols

This manual uses the following symbols:

- **M**

Indicates a description that is valid only for the machine center system set as system control type (in parameter No. 0983).
In a general description of the method of machining, a machining center system operation is identified by a phase such as "for milling machining".

- **T**

Indicates a description that is valid only for the lathe system set as system control type (in parameter No. 0983).
In a general description of the method of machining, a lathe system operation is identified by a phrase such as "for lathe cutting".

-

Indicates the end of a description of a system control type.
When a system control type mark mentioned above is not followed by this mark, the description of the system control type is assumed to continue until the next item or paragraph begins. In this case, the next item or paragraph provides a description common to the control types.

- **IP**

Indicates a combination of axes such as X_ Y_ Z_
In the underlined position following each address, a numeric value such as a coordinate value is placed (used in PROGRAMMING.).

- **;**

Indicates the end of a block. It actually corresponds to the ISO code LF or EIA code CR.

Related manuals of Series 30*i*/300*i*/300*i*s- MODEL A Series 31*i*/310*i*/310*i*s- MODEL A Series 32*i*/320*i*/320*i*s- MODEL A

The following table lists the manuals related to Series 30*i*/300*i* /300*i*s-A, Series 31*i*/310*i* /310*i*s-A, Series 32*i*/320*i* /320*i*s-A. This manual is indicated by an asterisk(*).

Table 1 Related manuals

Manual name	Specification number	
DESCRIPTIONS	B-63942EN	
CONNECTION MANUAL (HARDWARE)	B-63943EN	
CONNECTION MANUAL (FUNCTION)	B-63943EN-1	
USER'S MANUAL (Common to Lathe System/Machining Center System)	B-63944EN	*
USER'S MANUAL (For Lathe System)	B-63944EN-1	
USER'S MANUAL (For Machining Center System)	B-63944EN-2	
MAINTENANCE MANUAL	B-63945EN	
PARAMETER MANUAL	B-65950EN	
Programming		
Macro Compiler / Macro Executor PROGRAMMING MANUAL	B-63943EN-2	
Macro Compiler OPERATOR'S MANUAL	B-66264EN	
C Language Executor OPERATOR'S MANUAL	B-63944EN-3	
PMC		
PMC PROGRAMMING MANUAL	B-63983EN	
Network		
PROFIBUS-DP Board OPERATOR'S MANUAL	B-63994EN	
Fast Ethernet / Fast Data Server OPERATOR'S MANUAL	B-64014EN	
DeviceNet Board OPERATOR'S MANUAL	B-64044EN	
FL-net Board OPERATOR'S MANUAL	B-64164EN	
Operation guidance function		
MANUAL GUIDE *i* (Common to Lathe System/Machining Center System) OPERATOR'S MANUAL	B-63874EN	
MANUAL GUIDE *i* (For Machining Center System) OPERATOR'S MANUAL	B-63874EN-2	
MANUAL GUIDE *i* (Set-up Guidance Functions) OPERATOR'S MANUAL	B-63874EN−1	

Related manuals of SERVO MOTOR αi/βi series

The following table lists the manuals related to SERVO MOTOR αi/βi series

Table 2 Related manuals

Manual name	Specification number
FANUC AC SERVO MOTOR αi series DESCRIPTIONS	B-65262EN
FANUC AC SPINDLE MOTOR αi series DESCRIPTIONS	B-65272EN
FANUC AC SERVO MOTOR βi series DESCRIPTIONS	B-65302EN
FANUC AC SPINDLE MOTOR βi series DESCRIPTIONS	B-65312EN
FANUC SERVO AMPLIFIER αi series DESCRIPTIONS	B-65282EN
FANUC SERVO AMPLIFIER βi series DESCRIPTIONS	B-65322EN
FANUC SERVO MOTOR αis series FANUC SERVO MOTOR αi series FANUC AC SPINDLE MOTOR αi series FANUC SERVO AMPLIFIER αi series MAINTENANCE MANUAL	B-65285EN
FANUC SERVO MOTOR βis series FANUC AC SPINDLE MOTOR βi series FANUC SERVO AMPLIFIER βi series MAINTENANCE MANUAL	B-65325EN
FANUC AC SERVO MOTOR αi series FANUC AC SERVO MOTOR βi series FANUC LINEAR MOTOR LiS series FANUC SYNCHRONOUS BUILT-IN SERVO MOTOR DiS series PARAMETER MANUAL	B-65270EN
FANUC AC SPINDLE MOTOR αi/βi series, BUILT-IN SPINDLE MOTOR Bi series PARAMETER MANUAL	B-65280EN

The above servo motors and the corresponding spindles can be connected to the CNC covered in this manual. In the αi SV series, they can be connected only to upgrade versions. In the βi SVSP series, they cannot be connected.

This manual mainly assumes that the FANUC SERVO MOTOR αi series of servo motor is used. For servo motor and spindle information, refer to the manuals for the servo motor and spindle that are actually connected.

1.1 NOTES ON READING THIS MANUAL

⚠ CAUTION

1 The function of an CNC machine tool system depends not only on the CNC, but on the combination of the machine tool, its magnetic cabinet, the servo system, the CNC, the operator's panels, etc. It is too difficult to describe the function, programming, and operation relating to all combinations. This manual generally describes these from the stand-point of the CNC. So, for details on a particular CNC machine tool, refer to the manual issued by the machine tool builder, which should take precedence over this manual.

2 In the header field of each page of this manual, a chapter title is indicated so that the reader can reference necessary information easily.
By finding a desired title first, the reader can reference necessary parts only.

3 This manual describes as many reasonable variations in equipment usage as possible. It cannot address every combination of features, options and commands that should not be attempted.
If a particular combination of operations is not described, it should not be attempted.

1.2 NOTES ON VARIOUS KINDS OF DATA

⚠ CAUTION

Machining programs, parameters, offset data, etc. are stored in the CNC unit internal non-volatile memory. In general, these contents are not lost by the switching ON/OFF of the power. However, it is possible that a state can occur where precious data stored in the non-volatile memory has to be deleted, because of deletions from a maloperation, or by a failure restoration. In order to restore rapidly when this kind of mishap occurs, it is recommended that you create a copy of the various kinds of data beforehand.

II. PROGRAMMING

1 GENERAL

Chapter 1, "GENERAL", consists of the following sections:

1.1 TOOL MOVEMENT ALONG WORKPIECE PARTS FIGURE-INTERPOLATION 14
1.2 FEED-FEED FUNCTION 16
1.3 PART DRAWING AND TOOL MOVEMENT 17
1.4 CUTTING SPEED - SPINDLE FUNCTION 27
1.5 SELECTION OF TOOL USED FOR VARIOUS MACHINING - TOOL FUNCTION 28
1.6 COMMAND FOR MACHINE OPERATIONS - AUXILIARY FUNCTION 29
1.7 PROGRAM CONFIGURATION 30
1.8 TOOL MOVEMENT RANGE - STROKE 33

1.1 TOOL MOVEMENT ALONG WORKPIECE PARTS FIGURE-INTERPOLATION

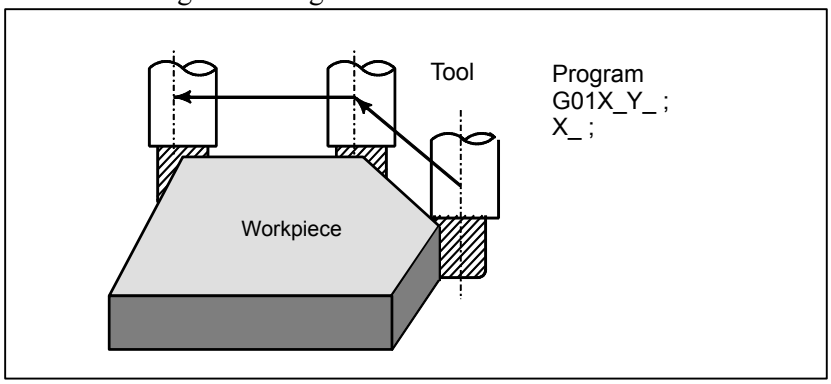
The tool moves along straight lines and arcs constituting the workpiece parts figure (See II-4).

Explanation

The function of moving the tool along straight lines and arcs is called the interpolation.

- Tool movement along a straight line

- For milling machining



- For lathe cutting

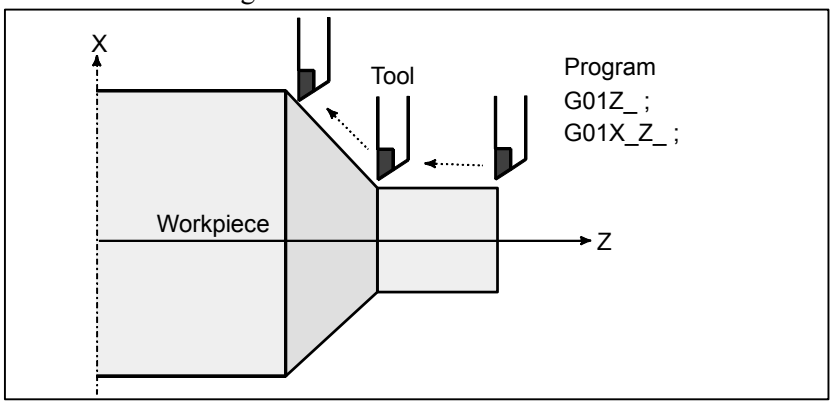
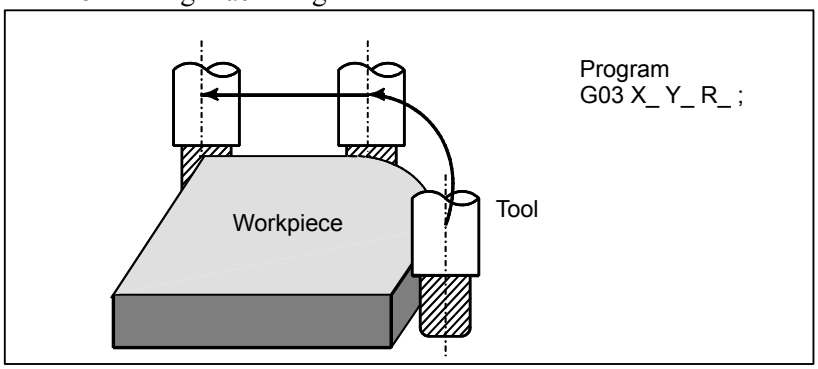


Fig. 1.1 (a) Tool movement along a straight line

- Tool movement along an arc

- For milling machining



- For lathe cutting

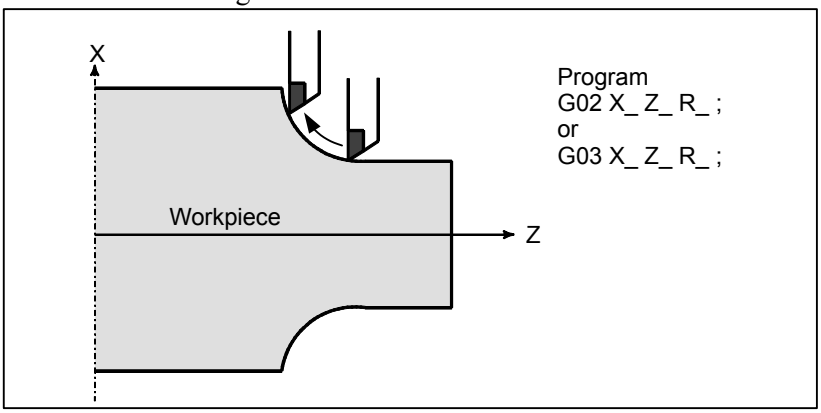


Fig. 1.1 (b) Tool movement along an arc

The term interpolation refers to an operation in which the tool moves along a straight line or arc in the way described above.
Symbols of the programmed commands G01, G02, ... are called the preparatory function and specify the type of interpolation conducted in the control unit.

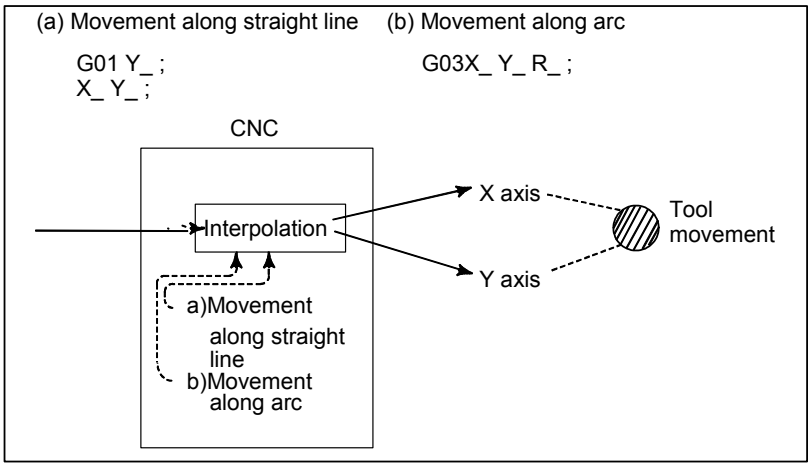


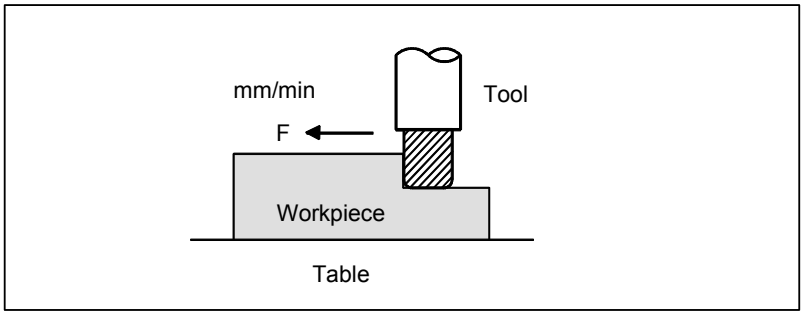
Fig. 1.1 (c) Interpolation function

NOTE
Some machines move tables instead of tools but this manual assumes that tools are moved against workpieces.

1.2 FEED-FEED FUNCTION

Movement of the tool at a specified speed for cutting a workpiece is called the feed.

- For milling machining



- For lathe cutting

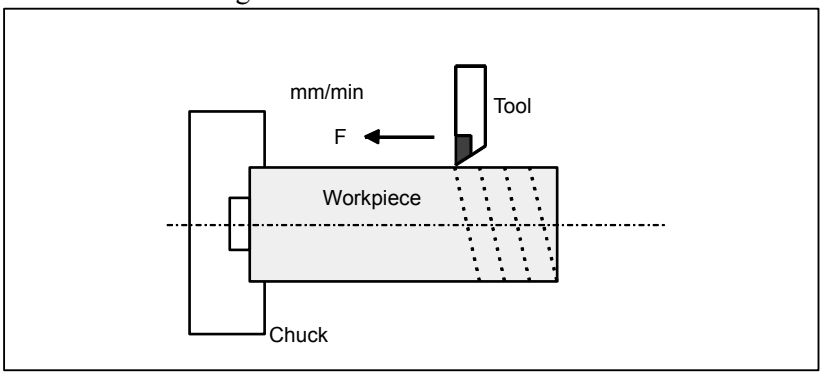


Fig. 1.2 (a) Feed function

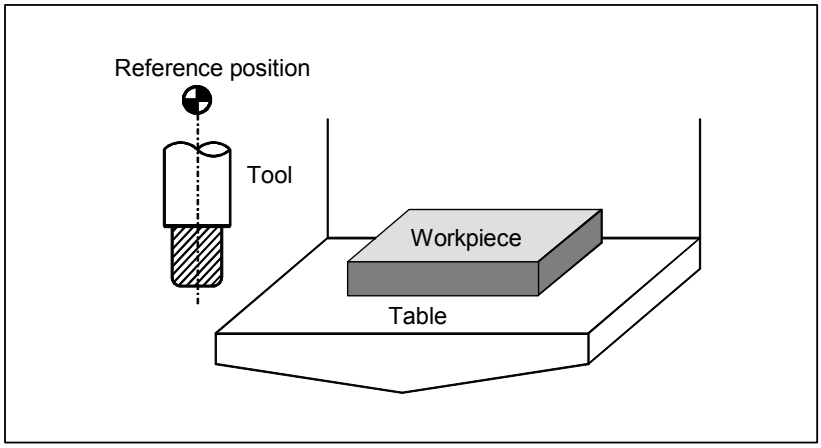
Feedrates can be specified by using actual numerics. For example, to feed the tool at a rate of 150 mm/min, specify the following in the program:
F150.0
The function of deciding the feed rate is called the feed function (See II-5).

1.3 PART DRAWING AND TOOL MOVEMENT

1.3.1 Reference Position (Machine-specific Position)

A CNC machine tool is provided with a fixed position. Normally, tool change and programming of absolute zero point as described later are performed at this position. This position is called the reference position.

- For milling machining



- For lathe cutting

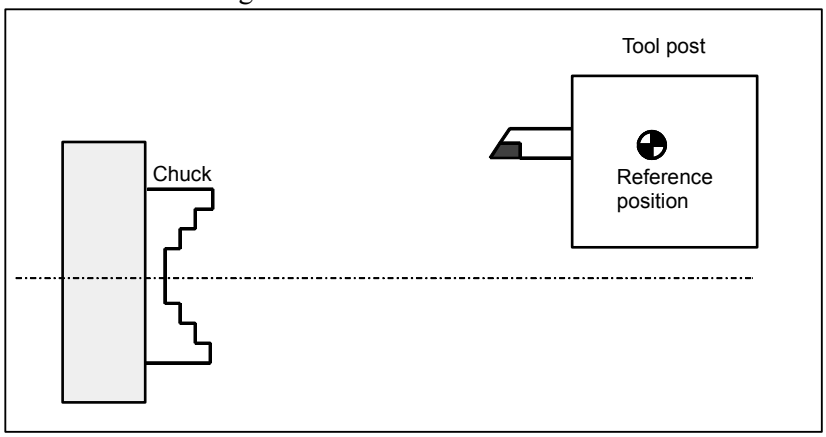


Fig. 1.3.1 (a) Reference position

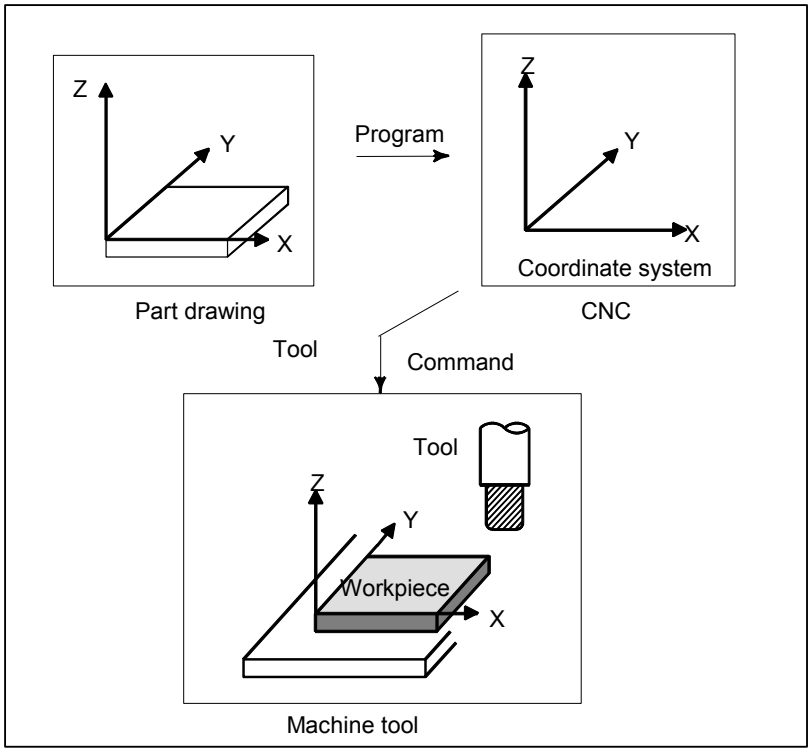
Explanation

The tool can be moved to the reference position in two ways:

1. Manual reference position return (See III-3.1)
   Reference position return is performed by manual button operation.
2. Automatic reference position return (See II-6)
   In general, manual reference position return is performed first after the power is turned on. In order to move the tool to the reference position for tool change thereafter, the function of automatic reference position return is used.

1.3.2 Coordinate System on Part Drawing and Coordinate System Specified by CNC - Coordinate System

- For milling machining



- For lathe cutting

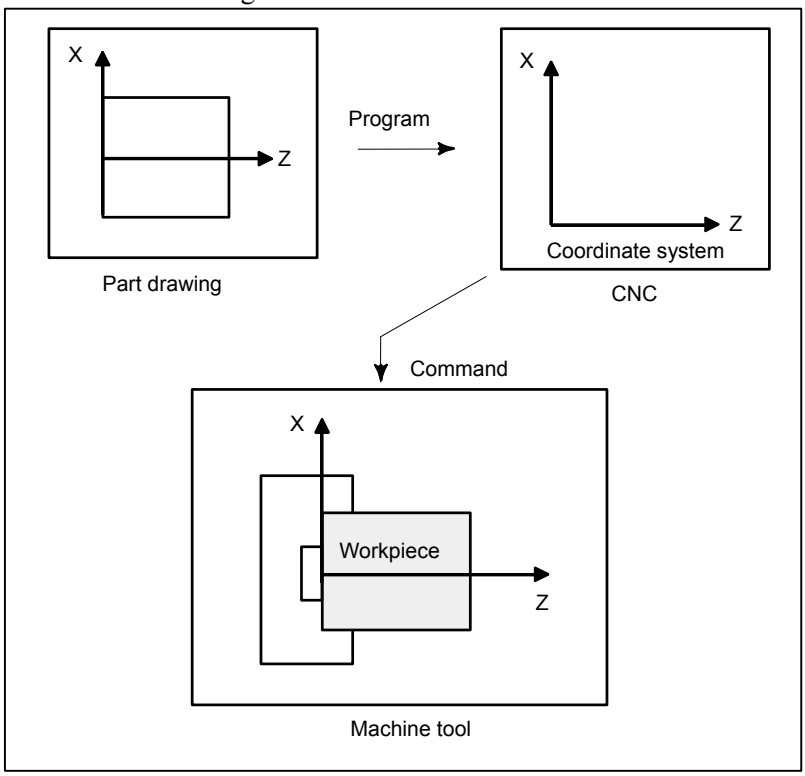


Fig. 1.3.2 (a) Coordinate system

Explanation

- Coordinate system

The following two coordinate systems are specified at different locations: (See II-7)

1 Coordinate system on part drawing
The coordinate system is written on the part drawing. As the program data, the coordinate values on this coordinate system are used.

2. Coordinate system specified by the CNC
The coordinate system is prepared on the actual machine tool table. This can be achieved by programming the distance from the current position of the tool to the zero point of the coordinate system to be set.

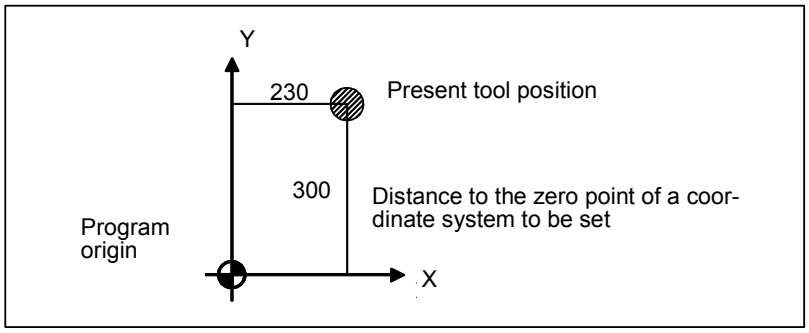
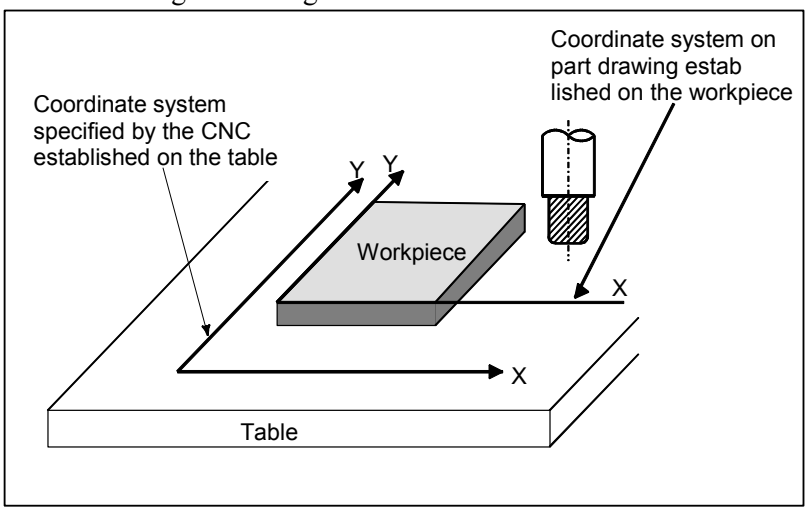


Fig. 1.3.2 (b) Coordinate system specified by the CNC

Concrete programming methods for setting coordinate systems specified by the CNC are explained in II-7, "COORDINATE SYSTEM".

The positional relation between these two coordinate systems is determined when a workpiece is set on the table.

- For milling machining



- For lathe cutting

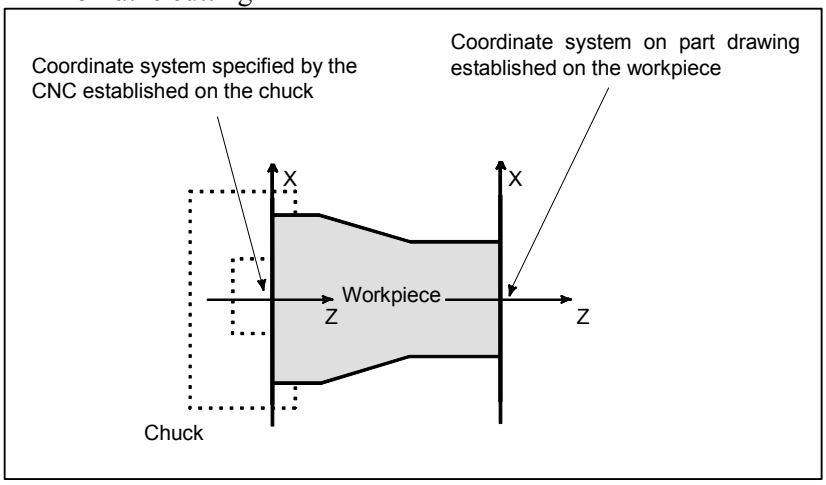


Fig. 1.3.2 (c) Coordinate system specified by CNC and coordinate system on part drawing

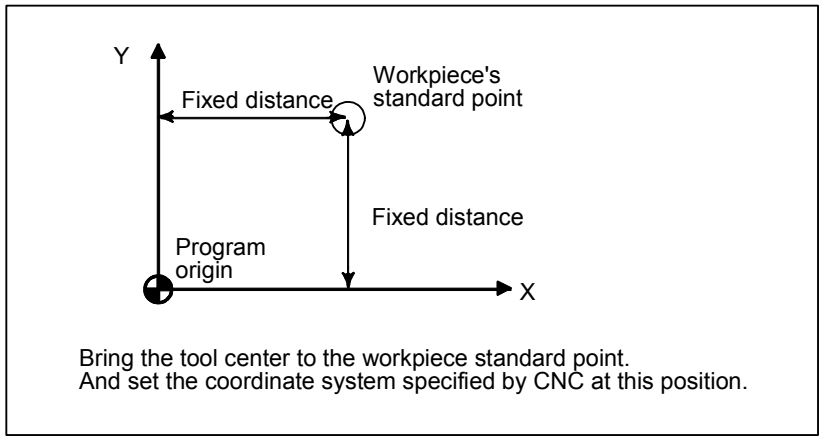
The tool moves on the coordinate system specified by the CNC in accordance with the command program generated with respect to the coordinate system on the part drawing, and cuts a workpiece into a shape on the drawing.
Therefore, in order to correctly cut the workpiece as specified on the drawing, the two coordinate systems must be set at the same position.

- Methods of setting the two coordinate systems in the same position

M

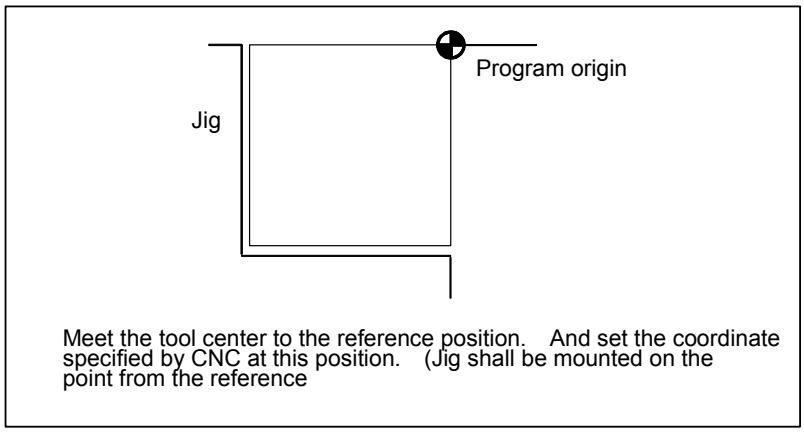
To set the two coordinate systems at the same position, simple methods shall be used according to workpiece shape, the number of machinings.

1. Using a standard plane and point of the workpiece.



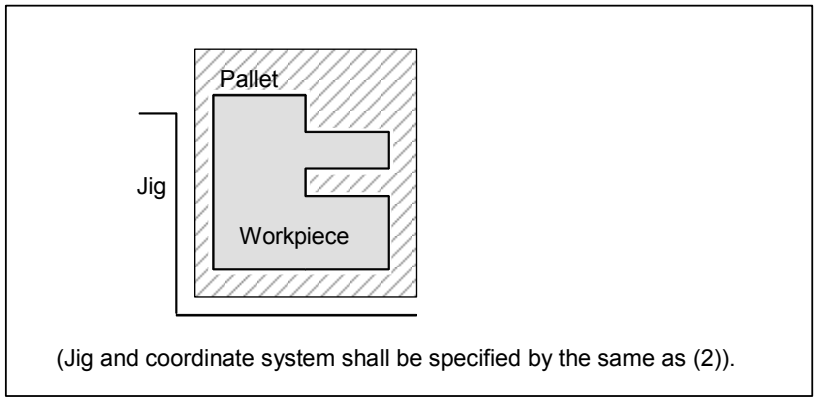
Bring the tool center to the workpiece standard point.
And set the coordinate system specified by CNC at this position.

2. Mounting a workpiece directly against the jig



Meet the tool center to the reference position. And set the coordinate specified by CNC at this position. (Jig shall be mounted on the point from the reference

3. Mounting a workpiece on a pallet, then mounting the workpiece and pallet on the jig

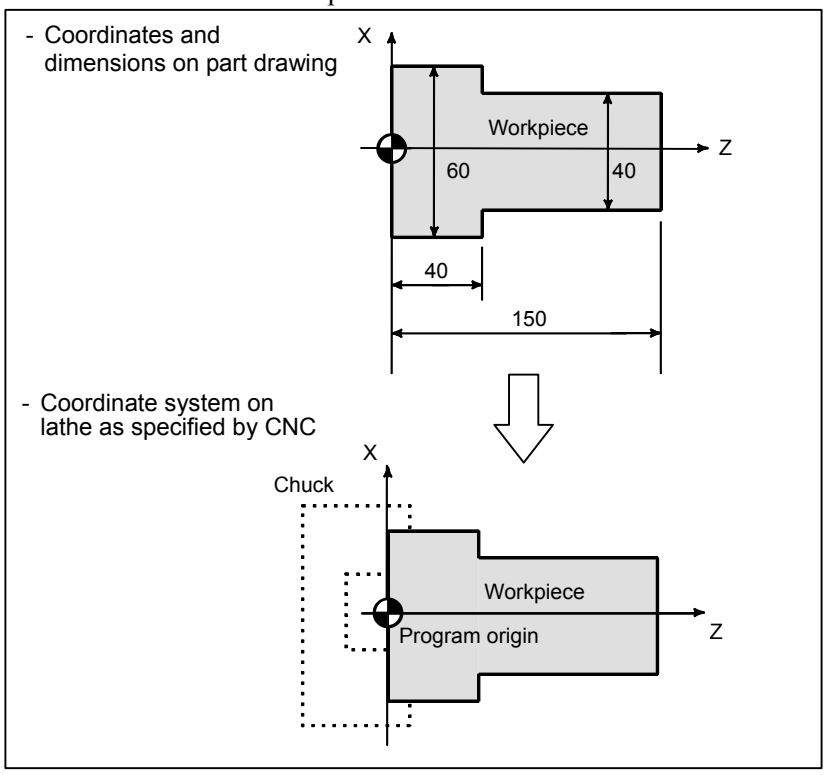


(Jig and coordinate system shall be specified by the same as (2)).

T

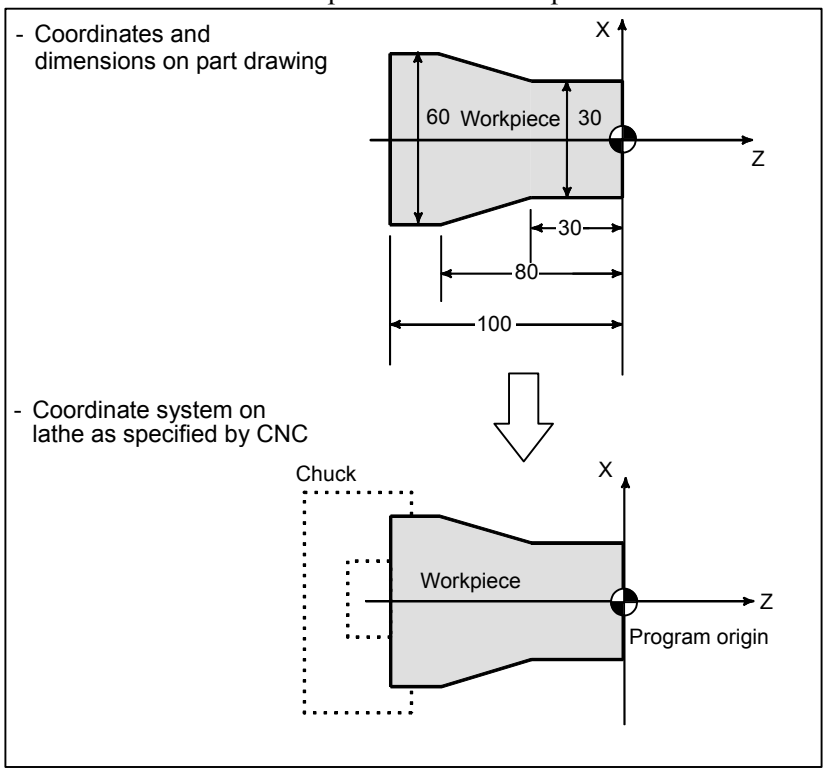
The following method is usually used to define two coordinate systems at the same location.

1 When coordinate zero point is set at chuck face



When the coordinate system on the part drawing and the coordinate system specified by the CNC are set at the same position, the program origin can be set on the chuck face.

2. When coordinate zero point is set at workpiece end face.



When the coordinate system on the part drawing and the coordinate system specified by the CNC are set at the same position, the program origin can be set on the end face of the workpiece.

1.3.3 How to Indicate Command Dimensions for Moving the Tool (Absolute, Incremental Commands)

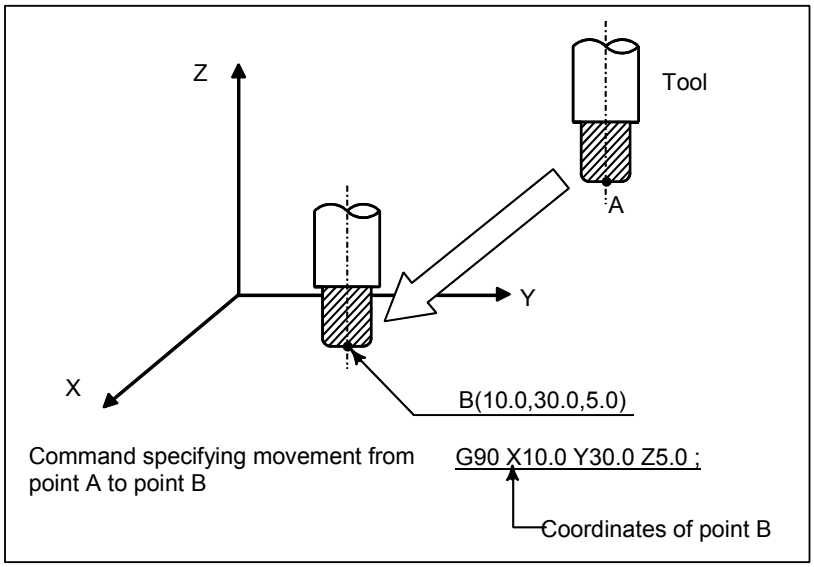
Explanation

Command for moving the tool can be indicated by absolute command or incremental command (See II-8.1).

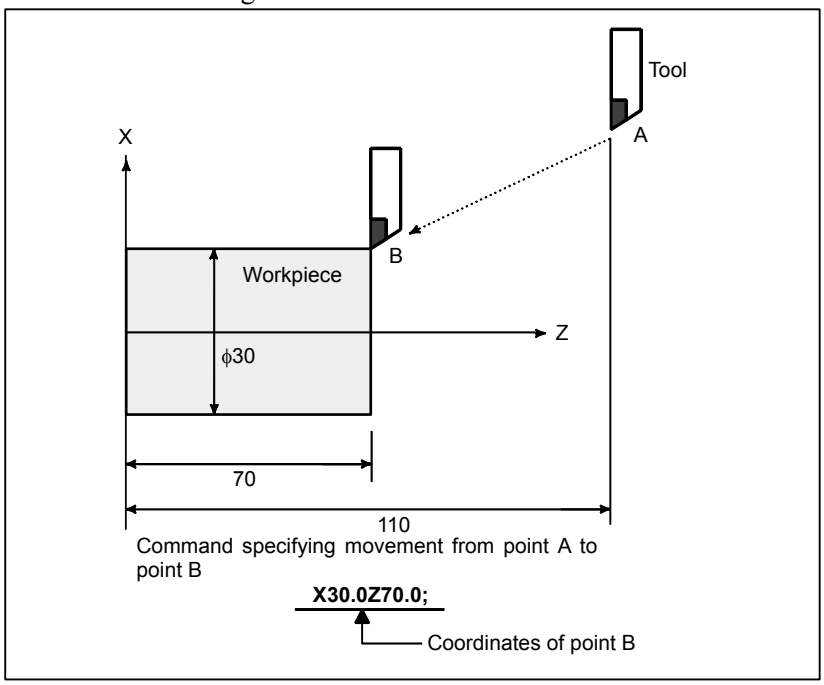
- Absolute command

The tool moves to a point at "the distance from zero point of the coordinate system" that is to the position of the coordinate values.

- For milling machining



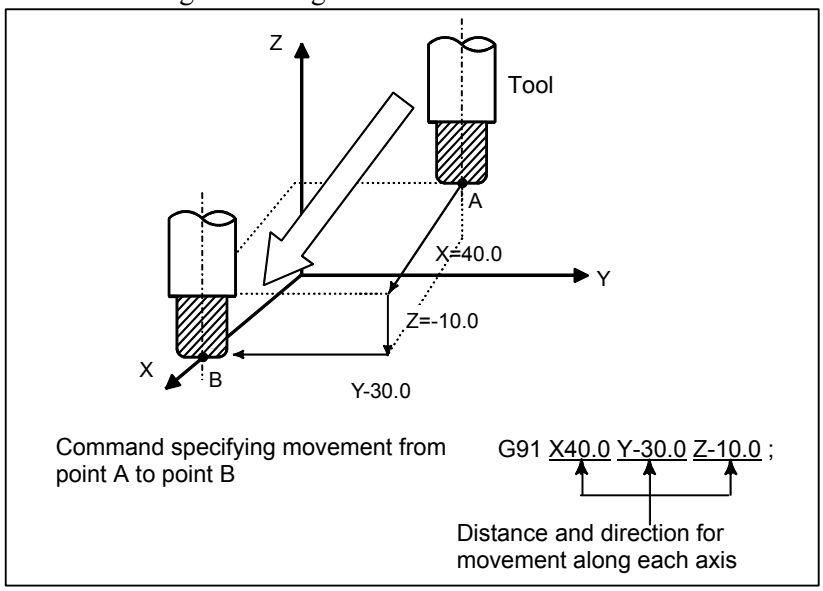
- For lathe cutting



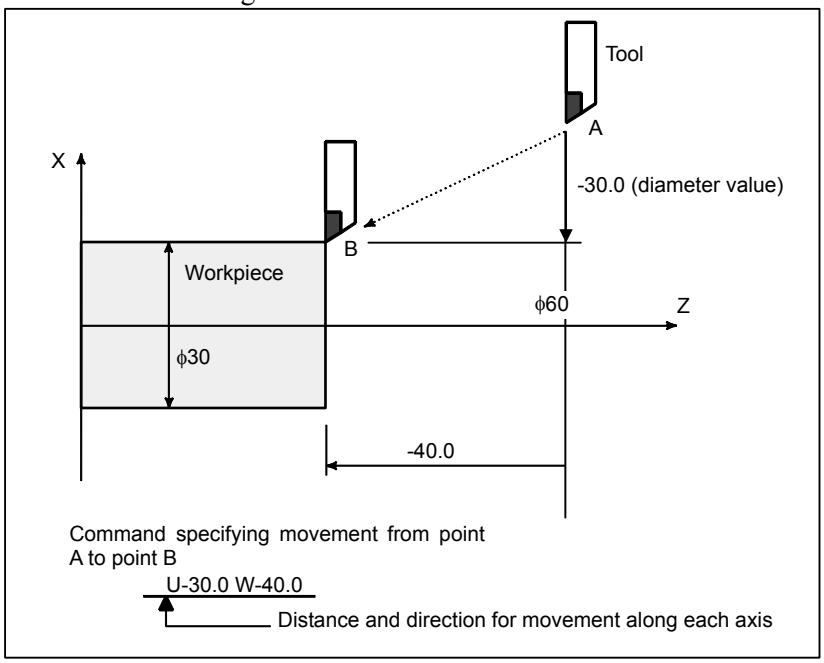
- Incremental command

Specify the distance from the previous tool position to the next tool position.

- For milling machining



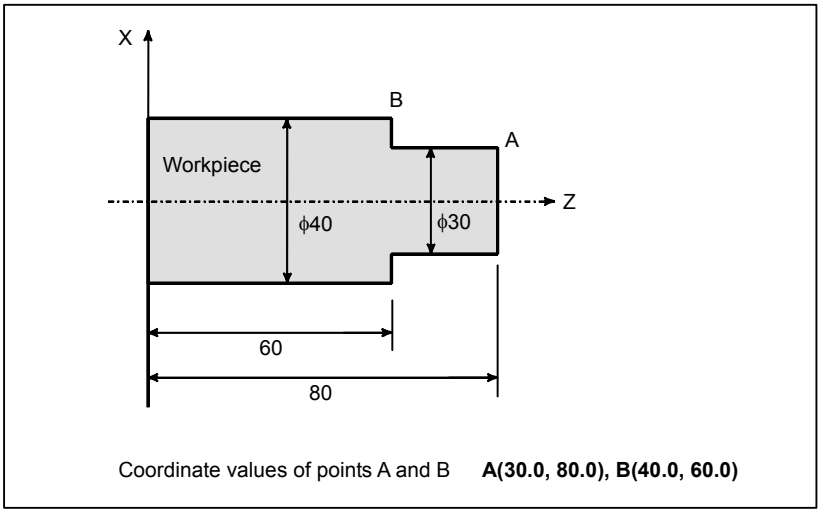
- For lathe cutting



- Diameter programming / radius programming

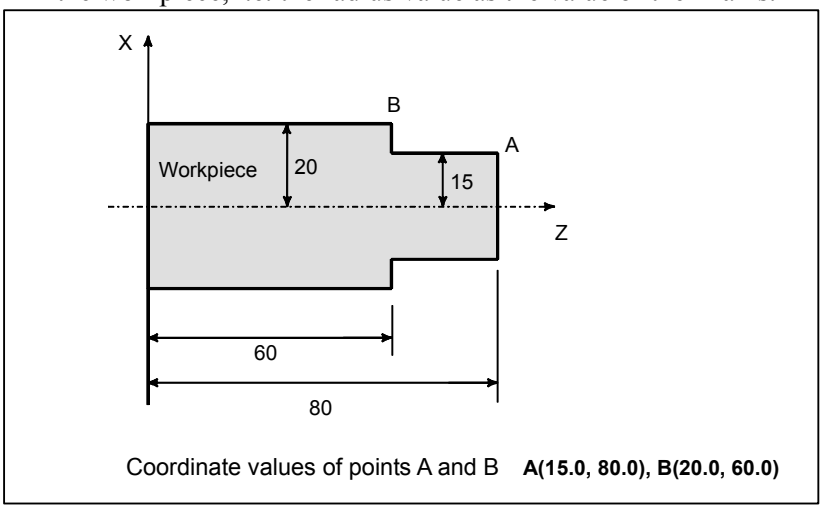
Dimensions of the X axis can be set in diameter or in radius. Diameter programming or radius programming is employed independently in each machine.

1. Diameter programming
   In diameter programming, specify the diameter value indicated on the drawing as the value of the X axis.



Coordinate values of points A and B **A(30.0, 80.0), B(40.0, 60.0)**

2. Radius programming
   In radius programming, specify the distance from the center of the workpiece, i.e. the radius value as the value of the X axis.

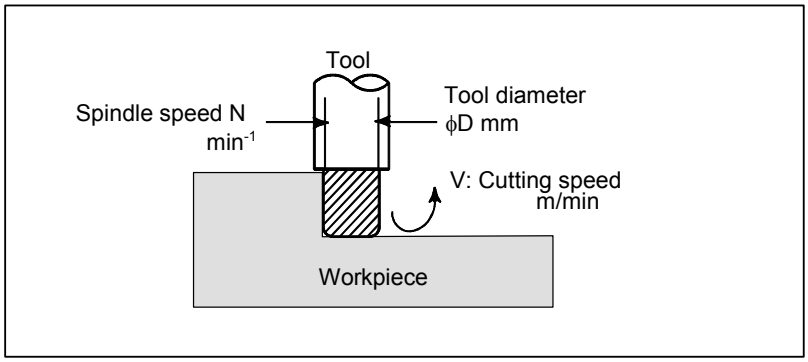


Coordinate values of points A and B **A(15.0, 80.0), B(20.0, 60.0)**

1.4 CUTTING SPEED - SPINDLE FUNCTION

The speed of the tool with respect to the workpiece when the workpiece is cut is called the cutting speed.
As for the CNC, the cutting speed can be specified by the spindle speed in min^{-1} unit.

- For milling machining

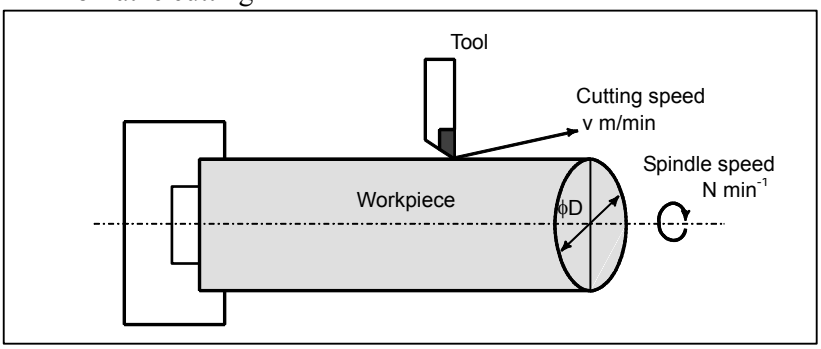


<When a workpiece should be machined with a tool 100 mm in diameter at a cutting speed of 80 m/min.>
The spindle speed is approximately 250 min^{-1}, which is obtained from $N=1000v/\pi D$. Hence the following command is required:

S250;

Commands related to the spindle speed are called the spindle speed function (See II-9).

- For lathe cutting



<When a workpiece 200 mm in diameter should be machined at a cutting speed of 300 m/min.>
The spindle speed is approximately 478 min^{-1}, which is obtained from $N=1000v/\pi D$. Hence the following command is required:

S478;

Commands related to the spindle speed are called the spindle speed function (See II-9).
The cutting speed v (m/min) can also be specified directly by the speed value. Even when the workpiece diameter is changed, the CNC changes the spindle speed so that the cutting speed remains constant.
This function is called the constant surface speed control function (See II-9.3).

1.5 SELECTION OF TOOL USED FOR VARIOUS MACHINING - TOOL FUNCTION

Overview

For each of various types of machining (such as drilling, tapping, boring, and milling for milling machining, or rough machining, semifinish machining, finish machining, threading, and grooving for lathe cutting), a necessary tool is to be selected. When a number is assigned to each tool and the number is specified in the program, the corresponding tool is selected.

Examples

M

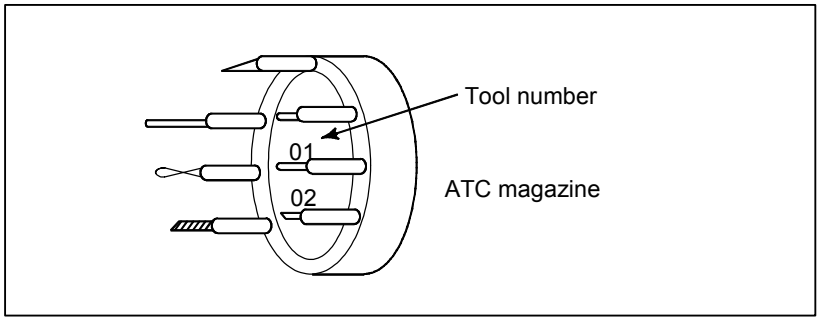


Fig. 1.5 (a) Tool used for various machining

<When No.01 is assigned to a drilling tool>
When the tool is stored at location 01 in the ATC magazine, the tool can be selected by specifying T01. This is called the tool function (See II-10).

T

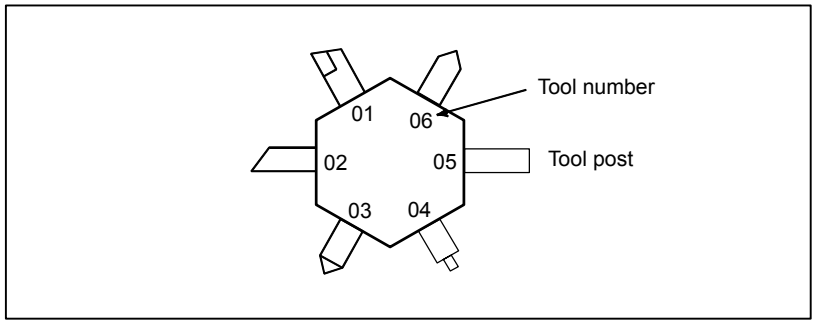


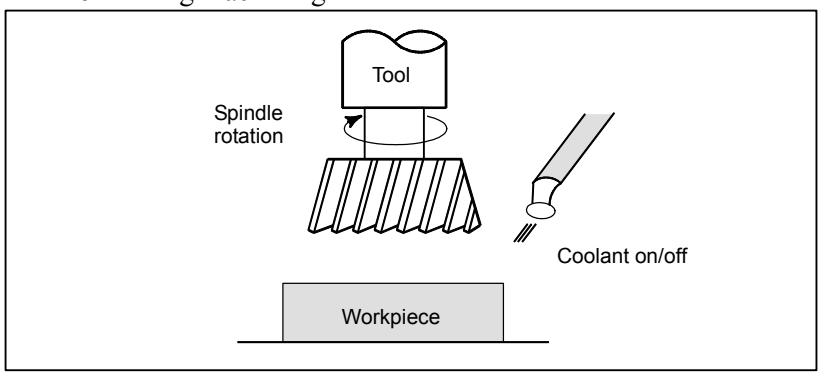
Fig. 1.5 (b) Tool used for various machining

<When No.01 is assigned to a roughing tool>
When the tool is stored at location 01 of the tool post, the tool can be selected by specifying T0101. This is called the tool function (See II-10).

1.6 COMMAND FOR MACHINE OPERATIONS - AUXILIARY FUNCTION

When a workpiece is actually machined with a tool, the spindle is rotated, coolant is supplied, and the chuck is opened/closed. So, control needs to be exercised on the spindle motor of the machine, coolant valve on/off operation, and chuck open/close operation.

- For milling machining



- For lathe cutting

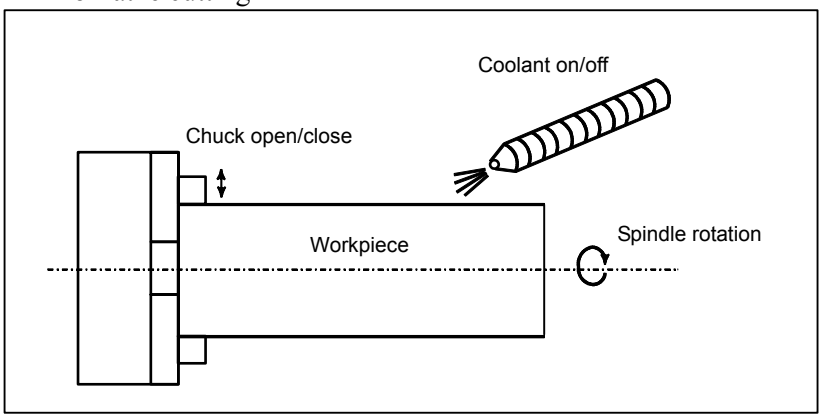


Fig. 1.6 (a) Auxiliary function

The function of specifying the on-off operations of the components of the machine is called the auxiliary function. In general, the function is specified by an M code (See II-11).
For example, when M03 is specified, the spindle is rotated clockwise at the specified spindle speed.

1.7 PROGRAM CONFIGURATION

A group of commands given to the CNC for operating the machine is called the program. By specifying the commands, the tool is moved along a straight line or an arc, or the spindle motor is turned on and off. In the program, specify the commands in the sequence of actual tool movements.

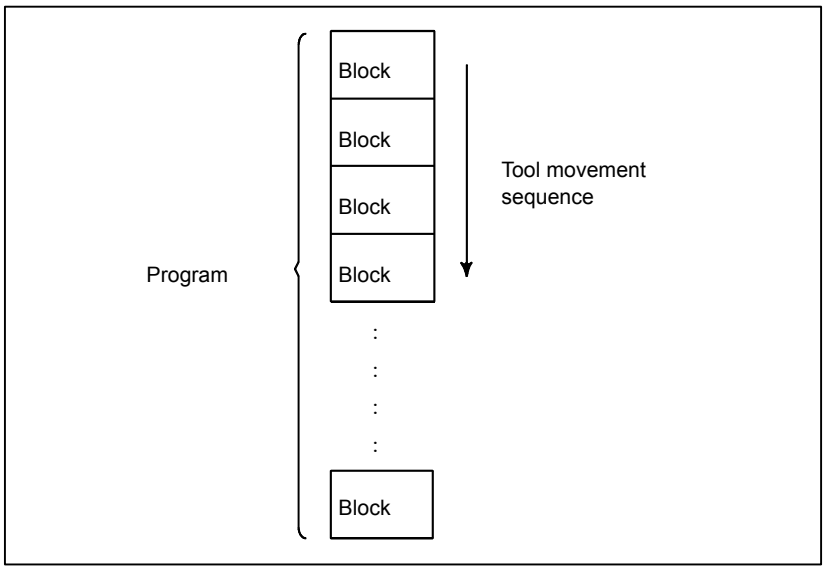


Fig. 1.7 (a) Program configuration

A group of commands at each step of the sequence is called the block. The program consists of a group of blocks for a series of machining. The number for discriminating each block is called the sequence number, and the number for discriminating each program is called the program number (See II-13).

Explanation

The block and the program have the following configurations.

- Block

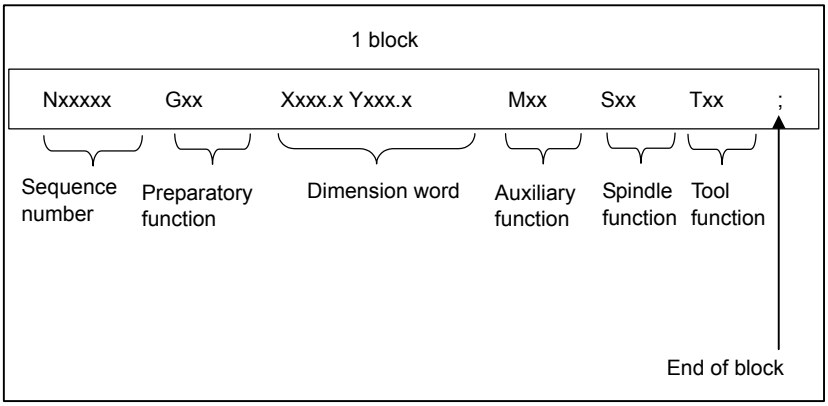


Fig. 1.7 (b) Block configuration

A block starts with a sequence number to identify the block and ends with an end-of-block code.
This manual indicates the end-of-block code by; (LF in the ISO code and CR in the EIA code).
The contents of the dimension word depend on the preparatory function. In this manual, the portion of the dimension word may be represent as IP_.

- Program

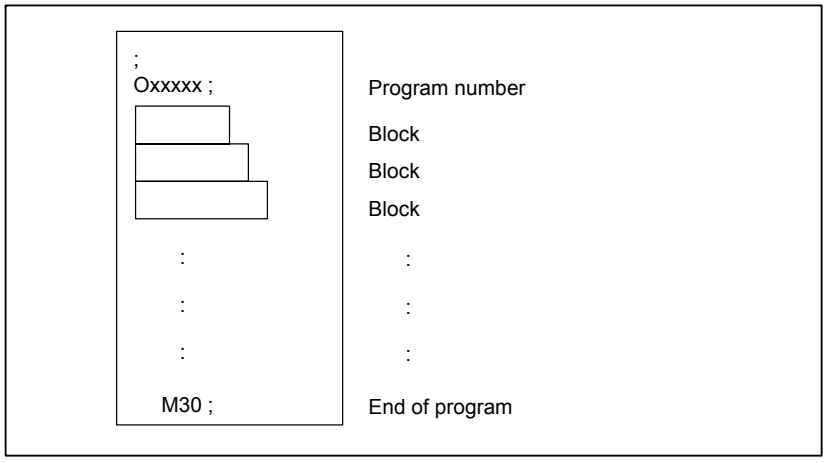


Fig. 1.7 (c) Program configuration

Normally, a program number is specified after the end-of-block (;) code at the beginning of the program, and a program end code (M02 or M30) is specified at the end of the program.

- Main program and subprogram

When machining of the same pattern appears at many portions of a program, a program for the pattern is created. This is called the subprogram. On the other hand, the original program is called the main program. When a subprogram execution command appears during execution of the main program, commands of the subprogram are executed. When execution of the subprogram is finished, the sequence returns to the main program.

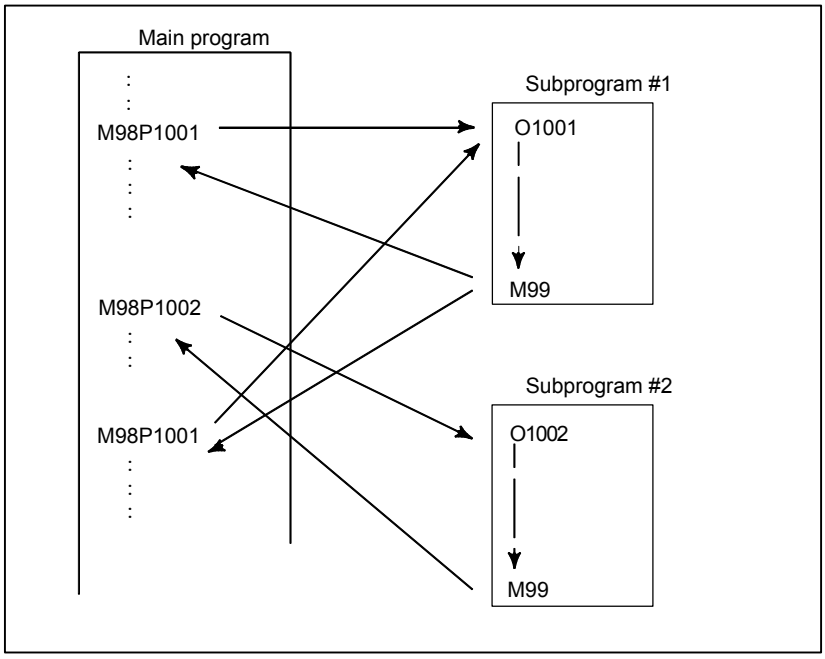
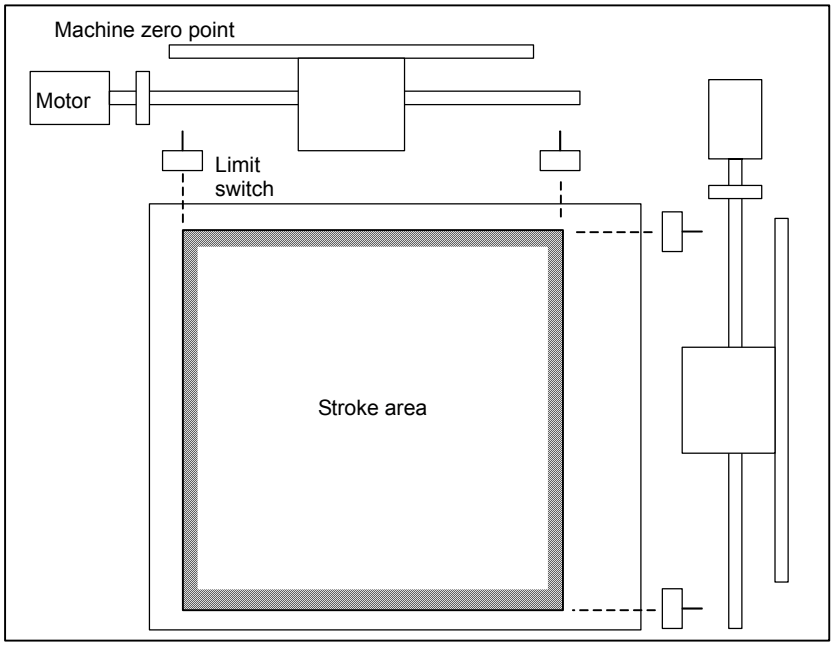


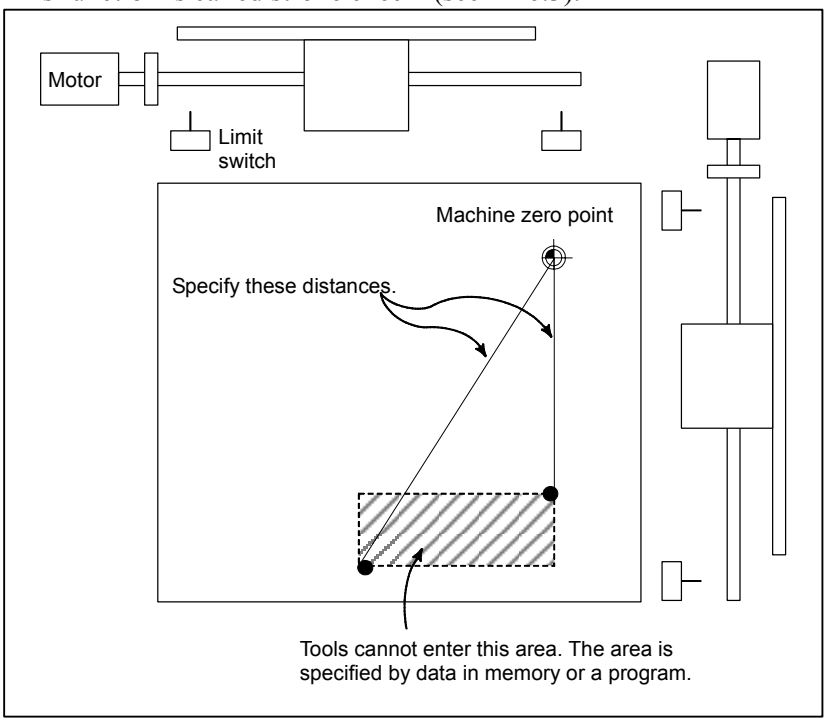
Fig. 1.7 (d) Subprogram execution

1.8 TOOL MOVEMENT RANGE - STROKE

Limit switches are installed at the ends of each axis on the machine to prevent tools from moving beyond the ends. The range in which tools can move is called the stroke.



Besides strokes defined with limit switches, the operator can define an area which the tool cannot enter using a program or data in memory. This function is called stroke check (see III-6.3).



2 CONTROLLED AXES

Chapter 2, "CONTROLLED AXES", consists of the following sections:

2.1 NUMBER OF CONTROLLED AXES.....35
2.2 NAMES OF AXES 36
2.3 INCREMENT SYSTEM.....37
2.4 MAXIMUM STROKE.....38

2.1 NUMBER OF CONTROLLED AXES

Explanation

The number of controlled axes used with this NC system depends on the model and system control type as indicated below.

		Series 30*i*-A Series 300*i*-A Series 300*i*s-A	Series 31*i*-A5 Series 310*i*-A5 Series 310*i*s-A5	Series 31*i*-A Series 310*i*-A Series 310*i*s-A	Series 32*i*-A Series 320*i*-A Series 320*i*s-A
Number of basic controlled axes	Lathe system	2 axes	2 axes	2 axes	2 axes
	Machining center system	3 axes	3 axes	3 axes	3 axes
Controlled axes expansion (total) (including Cs axes and PMC axes)		Max. 32 axes	Max. 20 axes	Max. 20 axes	Max. 9 axes
Basic simultaneously controlled axes (each path)		2 axes	2 axes	2 axes	2 axes
Simultaneously controlled axes expansion (total / each path)		Max. 24 axes	Max. 12 axes	Max. 12 axes	Max. 5 axes

NOTE

1 The maximum number of controlled axes that can be used is limited depending on the option configuration. Refer to the manual provided by the machine tool builder for details.

2 The number of simultaneously controllable axes for manual operation (jog feed, manual reference position return, or manual rapid traverse) is 1 or 3 (1 when bit 0 (JAX) of parameter No. 1002 is set to 0 and 3 when it is set to 1).

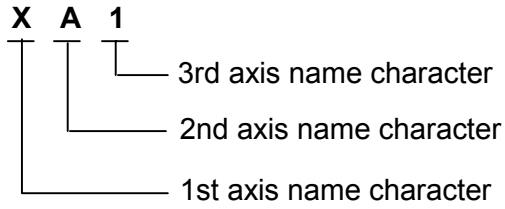
2.2 NAMES OF AXES

Explanation

The move axes of machine tools are assigned names. These names are referred to as addresses or axis names. Axis names are determined according to the machine tool. The naming rules comply with standards such as the ISO standards.

With complex machines, one character would become insufficient for representing axis names. So, up to three characters can be used for axis names. A move axis may be named "X", "X1", or "XA1". The first character of the three characters is called the first axis name character, the second character is called the second axis name character, and third character is called the third axis name character.

Example)



NOTE

1 Axis names are predetermined according to the machine used. Refer to the manual supplied by the machine tool builder.

2 Since many ordinary machines use one character to represent each address, one-character addresses are used in the description in this manual.

2.3 INCREMENT SYSTEM

Explanation

The increment system consists of the least input increment (for input) and least command increment (for output). The least input increment is the least increment for programming the travel distance. The least command increment is the least increment for moving the tool on the machine. Both increments are represented in mm, inches, or deg.
Five types of increment systems are available as indicated in Table 2.3 (a). For each axis, an increment system can be set using a bit from bit 0 to bit 3 (ISA, ISC, ISD, or ISE) of parameter No. 1013.
IS-C, IS-D, and IS-E are optional functions.

Table 2.3 (a) Increment system

Name of increment system	Least input increment		Least command increment	
IS-A	0.01	mm	0.01	mm
	0.001	inch	0.001	inch
	0.01	deg	0.01	deg
IS-B	0.001	mm	0.001	mm
	0.0001	inch	0.0001	inch
	0.001	deg	0.001	deg
IS-C	0.0001	mm	0.0001	mm
	0.00001	inch	0.00001	inch
	0.0001	deg	0.0001	deg
IS-D	0.00001	mm	0.00001	mm
	0.000001	inch	0.000001	inch
	0.00001	deg	0.00001	deg
IS-E	0.000001	mm	0.000001	mm
	0.0000001	inch	0.0000001	inch
	0.000001	deg	0.000001	deg

The least command increment is either metric or inch depending on the machine tool. Set metric or inch to the parameter INM (No.0100#0).
For selection between metric and inch for the least input increment, G code (G20 or G21) or a setting parameter selects it.
Combined use of the inch system and the metric system is not allowed. There are functions that cannot be used between axes with different unit systems (circular interpolation, cutter compensation, etc.). For the increment system, see the machine tool builder's manual.

NOTE
1 The unit (mm or inch) in the table is used for indicating a diameter value for diameter programming (when bit 3 (DIA) of parameter No. 1006 is set to 1) or a radius value for radius programming.
2 Some increment systems are unavailable depending on the model. For details, refer to “Descriptions” (B-63942EN).

2.4 MAXIMUM STROKE

Explanation

The maximum stroke controlled by this CNC is shown in the table below:
Maximum stroke = Least command increment × 999999999
(99999999 for IS-A)
Commands that exceed the maximum stroke are not permitted.

Table 2.4 (a) Maximum strokes

Name of increment system	Least input increment		Maximum stroke	
IS-A	0.01	mm	±999999.99	mm
	0.001	inch	±99999.999	inch
	0.01	deg	±999999.99	deg
IS-B	0.001	mm	±999999.999	mm
	0.0001	inch	±99999.9999	inch
	0.001	deg	±999999.999	deg
IS-C	0.0001	mm	±99999.9999	mm
	0.00001	inch	±9999.99999	inch
	0.0001	deg	±99999.9999	deg
IS-D	0.00001	mm	±9999.99999	mm
	0.000001	inch	±999.999999	inch
	0.00001	deg	±9999.99999	deg
IS-E	0.000001	mm	±999.999999	mm
	0.0000001	inch	±99.9999999	inch
	0.000001	deg	±999.999999	deg

NOTE
1 The actual stroke depends on the machine tool.
2 The unit (mm or inch) in the table is used for indicating a diameter value for diameter programming (when bit 3 (DIA) of parameter No. 1006 is set to 1) or a radius value for radius programming.
3 Some increment systems are unavailable depending on the model. For details, refer to "Descriptions" (B-63942EN).

3 PREPARATORY FUNCTION (G FUNCTION)

A number following address G determines the meaning of the command for the concerned block.
G codes are divided into the following two types.

Type	Meaning
One-shot G code	The G code is effective only in the block in which it is specified.
Modal G code	The G code is effective until another G code of the same group is specified.

(Example)

G01 and G00 are modal G codes in group 01.

```
G01 X_ ;
    Z_ ;   } G01 is effective in this range.
    X_ ;
G00 Z_ ;   } G00 is effective in this range.
    X_ ;
G01 X_ ;
  :
```

T

There are three G code systems in the lathe system: A, B, and C (Table 3.1(a)). Select a G code system using bits 6 (GSB) and 7 (GSC) of parameter No. 3401. To use G code system B or C, the corresponding option is needed. Generally, User's Manual describes the use of G code system A, except when the described item can use only G code system B or C. In such cases, the use of G code system B or C is described.

Explanation

1. When the clear state (bit 6 (CLR) of parameter No. 3402) is set at power-up or reset, the modal G codes are placed in the states described below.
   (1) The modal G codes are placed in the states marked with ◤ as indicated in Table.
   (2) G20 and G21 remain unchanged when the clear state is set at power-up or reset.
   (3) Which status G22 or G23 at power on is set by bit 7 (G23) of parameter No. 3402. However, G22 and G23 remain unchanged when the clear state is set at reset.
   (4) The user can select G00 or G01 by setting parameter G01 (No. 3402#0).
   (5) The user can select G90 or G91 by setting parameter G91 (No. 3402#3).
   When G code system B or C is used in the lathe system, setting bit 3 (G91) of parameter No. 3402 determines which code, either G90 or G91, is effective.
   (6) In the machining center system, the user can select G17, G18, or G19 by setting bits 1 (G18) and 2 (G19) of parameter No. 3401.
2. G codes other than G10 and G11 are one-shot G codes.
3. When a G code not listed in the G code list is specified, or a G code that has no corresponding option is specified, alarm PS0010 occurs.
4. Multiple G codes can be specified in the same block if each G code belongs to a different group. If multiple G codes that belong to the same group are specified in the same block, only the last G code specified is valid.
5. If a G code belonging to group 01 is specified in a canned cycle for drilling, the canned cycle for drilling is cancelled. This means that the same state set by specifying G80 is set. Note that the G codes in group 01 are not affected by a G code specifying a canned cycle for drilling.
6. G codes are indicated by group.
7. The group of G60 is switched according to the setting of bit 0 (MDL) of parameter No. 5431. (When the MDL bit is set to 0, the 00 group is selected. When the MDL bit is set to 1, the 01 group is selected.)

T

8. When G code system A is used, absolute or incremental programming is specified not by a G code (G90/G91) but by an address word (X/U, Z/W, C/H, Y/V). Only the initial level is provided at the return point of the canned cycle for drilling..

3.1 G CODE LIST IN THE MACHINING CENTER SYSTEM

M

Table 3.1 (a) G code list

G code	Group	Function	
G00	01	Positioning (rapid traverse)	
G01		Linear interpolation (cutting feed)	
G02		Circular interpolation CW or helical interpolation CW	
G03		Circular interpolation CCW or helical interpolation CCW	
G02.2, G03.2		Involute interpolation CW/CCW	
G02.3, G03.3		Exponential interpolation CW/CCW	
G02.4, G03.4		Three-dimensional coordinate system conversion CW/CCW	
G04	00	Dwell	
G05		AI contour control (high-precision contour control compatible command), High-speed cycle machining	
G05.1		AI contour control / Nano smoothing / Smooth interpolation	
G05.4		HRV3, 4 on/off	
G06.2	01	NURBS interpolation	
G07	00	Hypothetical axis interpolation	
G07.1 (G107)		Cylindrical interpolation	
G08		AI contour control (advanced preview control compatible command)	
G09		Exact stop	
G10		Programmable data input	
G10.6		Tool retract and recover	
G10.9		Programmable switching of diameter/radius specification	
G11		Programmable data input mode cancel	
G12.1	21	Polar coordinate interpolation mode	
G13.1		Polar coordinate interpolation cancel mode	
G15	17	Polar coordinates command cancel	
G16		Polar coordinates command	
G17	02	XpYp plane selection	Xp: X axis or its parallel axis Yp: Y axis or its parallel axis Zp: Z axis or its parallel axis
G18		ZpXp plane selection	
G19		YpZp plane selection	
G20 (G70)	06	Input in inch	
G21 (G71)		Input in mm	
G22	04	Stored stroke check function on	
G23		Stored stroke check function off	
G25	19	Spindle speed fluctuation detection off	
G26		Spindle speed fluctuation detection on	
G27	00	Reference position return check	
G28		Automatic return to reference position	
G29		Movement from reference position	
G30		2nd, 3rd and 4th reference position return	
G30.1		Floating reference position return	
G31		Skip function	
G31.8		EGB-axis skip	
G33	01	Threading	
G34		Variable lead threading	
G35		Circular threading CW	
G36		Circular threading CCW	

Table 3.1 (a) G code list

G code	Group	Function
G37	00	Automatic tool length measurement
G38		Tool radius/tool nose radius compensation : preserve vector
G39		Tool radius/tool nose radius compensation : corner circular interpolation
G40	07	Tool radius/tool nose radius compensation : cancel 3-dimensional cutter compensation : cancel
G41		Tool radius/tool nose radius compensation : left 3-dimensional cutter compensation : left
G42		Tool radius/tool nose radius compensation : right 3-dimensional cutter compensation : right
G41.2		3-dimensional cutter compensation : left (type 1)
G41.3		3-dimensional cutter compensation : leading edge offset
G41.4		3-dimensional cutter compensation : left (type 1) (FS16i-compatible command)
G41.5		3-dimensional cutter compensation : left (type 1) (FS16i-compatible command)
G41.6		3-dimensional cutter compensation : left (type 2)
G42.2		3-dimensional cutter compensation : right (type 1)
G42.4		3-dimensional cutter compensation : right (type 1) (FS16i-compatible command)
G42.5		3-dimensional cutter compensation : right (type 1) (FS16i-compatible command)
G42.6		3-dimensional cutter compensation : right (type 2)
G40.1	19	Normal direction control cancel mode
G41.1		Normal direction control on : left
G42.1		Normal direction control on : right
G43	08	Tool length compensation +
G44		Tool length compensation -
G43.1		Tool length compensation in tool axis direction
G43.4		Tool center point control (type 1)
G43.5		Tool center point control (type 2)
G45	00	Tool offset : increase
G46		Tool offset : decrease
G47		Tool offset : double increase
G48		Tool offset : double decrease
G49 (G49.1)	08	Tool length compensation cancel
G50	11	Scaling cancel
G51		Scaling
G50.1	22	Programmable mirror image cancel
G51.1		Programmable mirror image
G50.2	31	Polygon turning cancel
G51.2		Polygon turning
G50.4	00	Cancel synchronous control
G50.5		Cancel composite control
G50.6		Cancel superimposed control
G51.4		Start synchronous control
G51.5		Start composite control
G51.6		Start superimposed control
G52	00	Local coordinate system setting
G53		Machine coordinate system setting
G53.1		Tool axis direction control

Table 3.1 (a) G code list

G code	Group	Function
G54 (G54.1)	14	Workpiece coordinate system 1 selection
G55		Workpiece coordinate system 2 selection
G56		Workpiece coordinate system 3 selection
G57		Workpiece coordinate system 4 selection
G58		Workpiece coordinate system 5 selection
G59		Workpiece coordinate system 6 selection
G54.2	23	Rotary table dynamic fixture offset
G54.4	33	Workpiece setting error compensation
G60	00	Single direction positioning
G61	15	Exact stop mode
G62		Automatic corner override
G63		Tapping mode
G64		Cutting mode
G65	00	Macro call
G66	12	Macro modal call A
G66.1		Macro modal call B
G67		Macro modal call A/B cancel
G68	16	Coordinate system rotation start or three-dimensional coordinate conversion mode on
G69		Coordinate system rotation cancel or three-dimensional coordinate conversion mode off
G68.2		Tilted working plane command
G68.3		Tilted working plane command by tool axis direction
G68.4		Tilted working plane command (incremental multi-command)
G70.7	00	Finishing cycle
G71.7		Outer surface rough machining cycle
G72.7		End rough machining cycle
G73.7		Closed loop cutting cycle
G74.7		End cutting off cycle
G75.7		Outer or inner cutting off cycle
G76.7		Multiple threading cycle
G72.1		Figure copying (rotary copy)
G72.2		Figure copying (linear copy)
G73	09	Peck drilling cycle
G74		Left-handed tapping cycle
G75	01	Plunge grinding cycle
G76	09	Fine boring cycle
G77	01	Plunge direct sizing/grinding cycle
G78		Continuous-feed surface grinding cycle
G79		Intermittent-feed surface grinding cycle
G80	09	Canned cycle cancel Electronic gear box : synchronization cancellation
G80.4	34	Electronic gear box: synchronization cancellation
G81.4		Electronic gear box: synchronization start
G80.5	24	Electronic gear box 2 pair: synchronization cancellation
G81.5		Electronic gear box 2 pair: synchronization start
G81	09	Drilling cycle or spot boring cycle Electronic gear box : synchronization start
G81.1	00	Chopping

Table 3.1 (a) G code list

G code	Group	Function
G82	09	Drilling cycle or counter boring cycle
G83		Peck drilling cycle
G84		Tapping cycle
G84.2		Rigid tapping cycle (FS15 format)
G84.3		Left-handed rigid tapping cycle (FS15 format)
G85		Boring cycle
G86		Boring cycle
G87		Back boring cycle
G88		Boring cycle
G89		Boring cycle
G90	03	Absolute programming
G91		Incremental programming
G91.1	00	Checking the maximum incremental amount specified
G92		Setting for workpiece coordinate system or clamp at maximum spindle speed
G92.1		Workpiece coordinate system preset
G93	05	Inverse time feed
G94		Feed per minute
G95		Feed per revolution
G96	13	Constant surface speed control
G97		Constant surface speed control cancel
G96.1	00	Spindle indexing execution (waiting for completion)
G96.2		Spindle indexing execution (not waiting for completion)
G96.3		Spindle indexing completion check
G96.4		SV speed control mode ON
G98	10	Canned cycle : return to initial level
G99		Canned cycle : return to R point level
G107	00	Cylindrical interpolation
G112	21	Polar coordinate interpolation mode
G113		Polar coordinate interpolation mode cancel
G160	20	Infeed control cancel
G161		Infeed control

3.2 G CODE LIST IN THE LATHE SYSTEM

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Table 3.2 (a) G code list

G code system			Group	Function
A	B	C		
G00	G00	G00	01	Positioning (Rapid traverse)
G01	G01	G01		Linear interpolation (Cutting feed)
G02	G02	G02		Circular interpolation CW or helical interpolation CW
G03	G03	G03		Circular interpolation CCW or helical interpolation CCW
G02.2	G02.2	G02.2		Involute interpolation CW
G02.3	G02.3	G02.3		Exponential interpolation CW
G02.4	G02.4	G02.4		Three-dimensional coordinate system conversion CW
G03.2	G03.2	G03.2		Involute interpolation CCW
G03.3	G03.3	G03.3		Exponential interpolation CCW
G03.4	G03.4	G03.4		Three-dimensional coordinate system conversion CCW
G04	G04	G04	00	Dwell
G05	G05	G05		AI contour control (command compatible with high precision contour control), High-speed cycle machining
G05.1	G05.1	G05.1		AI contour control / Nano smoothing / Smooth interpolation
G05.4	G05.4	G05.4		HRV3, 4 on/off
G06.2	G06.2	G06.2	01	NURBS interpolation
G07	G07	G07	00	Hypothetical axis interpolation
G07.1 (G107)	G07.1 (G107)	G07.1 (G107)		Cylindrical interpolation
G08	G08	G08		Advanced preview control
G09	G09	G09		Exact stop
G10	G10	G10		Programmable data input
G10.6	G10.6	G10.6		Tool retract and recover
G10.9	G10.9	G10.9		Programmable switching of diameter/radius specification
G11	G11	G11		Programmable data input mode cancel
G12.1 (G112)	G12.1 (G112)	G12.1 (G112)	21	Polar coordinate interpolation mode
G13.1 (G113)	G13.1 (G113)	G13.1 (G113)		Polar coordinate interpolation cancel mode
G17	G17	G17	16	XpYp plane selection
G18	G18	G18		ZpXp plane selection
G19	G19	G19		YpZp plane selection
G20	G20	G70	06	Input in inch
G21	G21	G71		Input in mm
G22	G22	G22	09	Stored stroke check function on
G23	G23	G23		Stored stroke check function off
G25	G25	G25	08	Spindle speed fluctuation detection off
G26	G26	G26		Spindle speed fluctuation detection on

Table 3.2 (a) G code list

G code system			Group	Function
A	B	C		
G27	G27	G27	00	Reference position return check
G28	G28	G28		Return to reference position
G29	G29	G29		Movement from reference position
G30	G30	G30		2nd, 3rd and 4th reference position return
G30.1	G30.1	G30.1		Floating reference point return
G31	G31	G31		Skip function
G31.8	G31.8	G31.8		EGB-axis skip
G32	G33	G33	01	Threading
G34	G34	G34		Variable lead threading
G35	G35	G35		Circular threading CW
G36	G36	G36		Circular threading CCW (When bit 3 (G36) of parameter No. 3405 is set to 1) or Automatic tool offset (X axis) (When bit 3 (G36) of parameter No. 3405 is set to 0)
G37	G37	G37		Automatic tool offset (Z axis) (When bit 3 (G36) of parameter No. 3405 is set to 0)
G37.1	G37.1	G37.1		Automatic tool offset (X axis) (When bit 3 (G36) of parameter No. 3405 is set to 1)
G37.2	G37.2	G37.2		Automatic tool offset (Z axis) (When bit 3 (G36) of parameter No. 3405 is set to 1)
G38	G38	G38		Tool radius/tool nose radius compensation: with vector held
G39	G39	G39		Tool radius/tool nose radius compensation: corner rounding interpolation
G40	G40	G40	07	Tool radius/tool nose radius compensation : cancel
G41	G41	G41		Tool radius/tool nose radius compensation : left
G42	G42	G42		Tool radius/tool nose radius compensation : right
G41.2	G41.2	G41.2		3-dimensional cutter compensation : left (type 1)
G41.3	G41.3	G41.3		3-dimensional cutter compensation : (leading edge offset)
G41.4	G41.4	G41.4		3-dimensional cutter compensation : left (type 1) (FS16i-compatible command)
G41.5	G41.5	G41.5		3-dimensional cutter compensation : left (type 1) (FS16i-compatible command)
G41.6	G41.6	G41.6		3-dimensional cutter compensation : left (type 2)
G42.2	G42.2	G42.2		3-dimensional cutter compensation : right (type 1)
G42.4	G42.4	G42.4		3-dimensional cutter compensation : right (type 1) (FS16i-compatible command)
G42.5	G42.5	G42.5		3-dimensional cutter compensation : right (type 1) (FS16i-compatible command)
G42.6	G42.6	G42.6		3-dimensional cutter compensation : right (type 2)
G40.1	G40.1	G40.1	19	Normal direction control cancel mode
G41.1	G41.1	G41.1		Normal direction control left on
G42 .1	G42 .1	G42 .1		Normal direction control right on

Table 3.2 (a) G code list

G code system			Group	Function
A	B	C		
G43	G43	G43	23	Tool length compensation + (Parameter TCT (No.5040#3) must be "1".)
G44	G44	G44		Tool length compensation - (Parameter TCT (No.5040#3) must be "1".)
G43.1	G43.1	G43.1		Tool length compensation in tool axis direction (Parameter TCT (No.5040#3) must be "1".)
G43.4	G43.4	G43.4		Tool center point control (type 1) (Parameter TCT (No.5040#3) must be "1".)
G43.5	G43.5	G43.5		Tool center point control (type 2) (Parameter TCT (No.5040#3) must be "1".)
G43.7 (G44.7)	G43.7 (G44.7)	G43.7 (G44.7)		Tool offset (Parameter TCT (No.5040#3) must be "1".)
G49 (G49.1)	G49 (G49.1)	G49 (G49.1)		Tool length compensation cancel (Parameter TCT (No.5040#3) must be "1".)
G50	G92	G92	00	Coordinate system setting or max spindle speed clamp
G50.3	G92.1	G92.1		Workpiece coordinate system preset
-	G50	G50	18	Scaling cancel
-	G51	G51		Scaling
G50.1	G50.1	G50.1	22	Programmable mirror image cancel
G51.1	G51.1	G51.1		Programmable mirror image
G50.2 (G250)	G50.2 (G250)	G50.2 (G250)	20	Polygon turning cancel
G51.2 (G251)	G51.2 (G251)	G51.2 (G251)		Polygon turning
G50.4	G50.4	G50.4	00	Cancel synchronous control
G50.5	G50.5	G50.5		Cancel composite control
G50.6	G50.6	G50.6		Cancel superimposed control
G51.4	G51.4	G51.4		Start synchronous control
G51.5	G51.5	G51.5		Start composite control
G51.6	G51.6	G51.6		Start superimposed control
G52	G52	G52		Local coordinate system setting
G53	G53	G53		Machine coordinate system setting
G53.1	G53.1	G53.1		Tool axis direction control
G54 (G54.1)	G54 (G54.1)	G54 (G54.1)	14	Workpiece coordinate system 1 selection
G55	G55	G55		Workpiece coordinate system 2 selection
G56	G56	G56		Workpiece coordinate system 3 selection
G57	G57	G57		Workpiece coordinate system 4 selection
G58	G58	G58		Workpiece coordinate system 5 selection
G59	G59	G59		Workpiece coordinate system 6 selection
G54.4	G54.4	G54.4	26	Workpiece setting error compensation
G60	G60	G60	00	Single direction positioning
G61	G61	G61	15	Exact stop mode
G62	G62	G62		Automatic corner override mode
G63	G63	G63		Tapping mode
G64	G64	G64		Cutting mode
G65	G65	G65	00	Macro call
G66	G66	G66	12	Macro modal call A
G66.1	G66.1	G66.1		Macro modal call B
G67	G67	G67		Macro modal call A/B cancel

Table 3.2 (a) G code list

G code system			Group	Function
A	B	C		
G68	G68	G68	04	Mirror image on for double turret or balance cutting mode
G68.1	G68.1	G68.1	17	Coordinate system rotation start or three-dimensional coordinate system conversion mode on
G68.2	G68.2	G68.2		Tilted working plane command
G68.3	G68.3	G68.3		Tilted working plane command by tool axis direction
G68.4	G68.4	G68.4		Tilted working plane command (incremental multi-command)
G69	G69	G69	04	Mirror image off for double turret or balance cutting mode cancel
G69.1	G69.1	G69.1	17	Coordinate system rotation cancel or three-dimensional coordinate system conversion mode off
G70	G70	G72	00	Finishing cycle
G71	G71	G73		Stock removal in turning
G72	G72	G74		Stock removal in facing
G73	G73	G75		Pattern repeating cycle
G74	G74	G76		End face peck drilling cycle
G75	G75	G77		Outer diameter/internal diameter drilling cycle
G76	G76	G78		Multiple-thread cutting cycle
G71	G71	G72	01	Traverse grinding cycle
G72	G72	G73		Traverse direct sizing/grinding cycle
G73	G73	G74		Oscillation grinding cycle
G74	G74	G75		Oscillation direct sizing/grinding cycle
G80	G80	G80	10	Canned cycle cancel for drilling Electronic gear box : synchronization cancellation
G80.4	G80.4	G80.4	28	Electronic gear box: synchronization cancellation
G81.4	G81.4	G81.4		Electronic gear box: synchronization start
G80.5	G80.5	G80.5	27	Electronic gear box 2 pair: synchronization cancellation
G81.5	G81.5	G81.5		Electronic gear box 2 pair: synchronization start
G81	G81	G81	10	Spot drilling (FS15-T format) Electronic gear box : synchronization start
G82	G82	G82		Counter boring (FS15-T format)
G83	G83	G83		Cycle for face drilling
G83.1	G83.1	G83.1		High-speed peck drilling cycle (FS15-T format)
G83.5	G83.5	G83.5		High-speed peck drilling cycle
G83.6	G83.6	G83.6		Peck drilling cycle
G84	G84	G84		Cycle for face tapping
G84.2	G84.2	G84.2		Rigid tapping cycle (FS15-T format)
G85	G85	G85		Cycle for face boring
G87	G87	G87		Cycle for side drilling
G87.5	G87.5	G87.5		High-speed peck drilling cycle
G87.6	G87.6	G87.6		Peck drilling cycle
G88	G88	G88		Cycle for side tapping
G89	G89	G89		Cycle for side boring
G90	G77	G20	01	Outer diameter/internal diameter cutting cycle
G92	G78	G21		Threading cycle
G94	G79	G24		End face turning cycle
G91.1	G91.1	G91.1	00	Maximum specified incremental amount check
G96	G96	G96	02	Constant surface speed control
G97	G97	G97		Constant surface speed control cancel

Table 3.2 (a) G code list

G code system			Group	Function
A	B	C		
G96.1	G96.1	G96.1	00	Spindle indexing execution (waiting for completion)
G96.2	G96.2	G96.2		Spindle indexing execution (not waiting for completion)
G96.3	G96.3	G96.3		Spindle indexing completion check
G96.4	G96.4	G96.4		SV speed control mode ON
G93	G93	G93	05	Inverse time feed
G98	G94	G94		Feed per minute
G99	G95	G95		Feed per revolution
-	G90	G90	03	Absolute programming
-	G91	G91		Incremental programming
-	G98	G98	11	Canned cycle : return to initial level
-	G99	G99		Canned cycle : return to R point level

4 INTERPOLATION FUNCTIONS

Interpolation functions specify the way to make an axis movement (in other words, a movement of the tool with respect to the workpiece or table).

Chapter 4, "INTERPOLATION FUNCTIONS", consists of the following sections:

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4.22 TORQUE LIMIT SKIP 145
4.23 THREE-DIMENSIONAL CIRCULAR INTERPOLATION.149

4.1 POSITIONING (G00)

The G00 command moves a tool to the position in the workpiece system specified with an absolute or an incremental programming at a rapid traverse rate.
In the absolute programming, coordinate value of the end point is programmed.
In the incremental programming the distance the tool moves is programmed.

Format

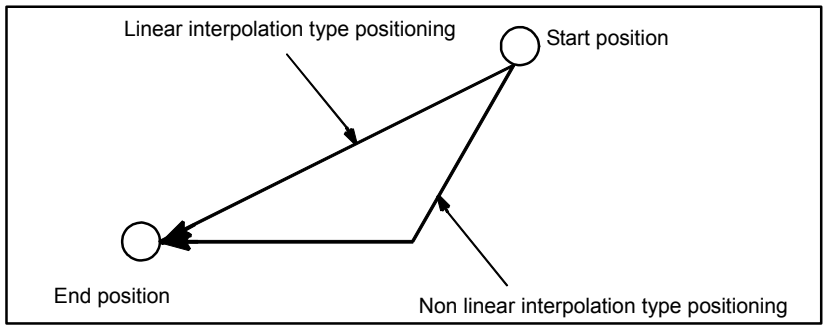
G00 IP_ ;

IP_ : For an absolute programming, the coordinates of an end point, and for an incremental programming, the distance the tool moves.

Explanation

Either of the following tool paths can be selected according to bit 1 (LRP) of parameter No. 1401.

- Nonlinear interpolation type positioning
  The tool is positioned with the rapid traverse rate for each axis separately. The tool path is normally straight.
- Linear interpolation type positioning
  The tool is positioned within the shortest possible time at a speed that is not more than the rapid traverse rate for each axis.



The rapid traverse rate in G00 command is set to the parameter No. 1420 for each axis independently by the machine tool builder. In the positioning mode actuated by G00, the tool is accelerated to a predetermined speed at the start of a block and is decelerated at the end of a block. Execution proceeds to the next block after confirming the in-position.
"In-position " means that the feed motor is within the specified range. This range is determined by the machine tool builder by setting to parameter (No. 1826).

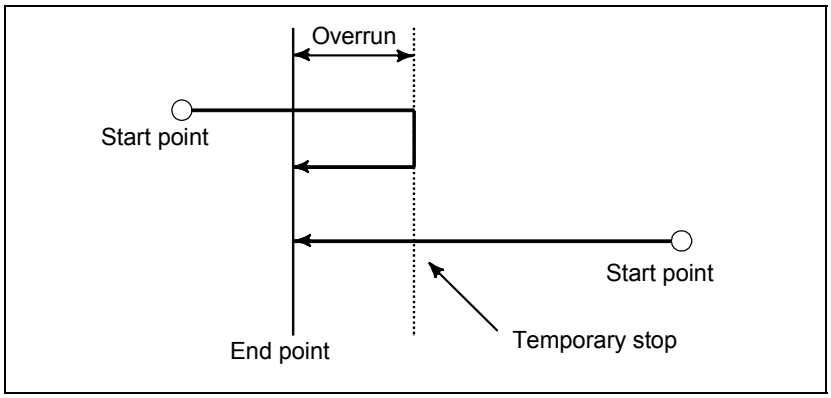
Limitation

The rapid traverse rate cannot be specified in the address F.
Even if linear interpolation type positioning is specified, nonlinear type interpolation positioning is used in the following cases. Therefore, be careful to ensure that the tool does not foul the workpiece.

- G28 specifying positioning between the reference and intermediate positions.
- G53

4.2 SINGLE DIRECTION POSITIONING (G60)

For accurate positioning without play of the machine (backlash), final positioning from one direction is available.



Format

G60 IP_ ;

IP_ : For an absolute programming, the coordinates of an end point, and for an incremental programming, the distance the tool moves.

Explanation

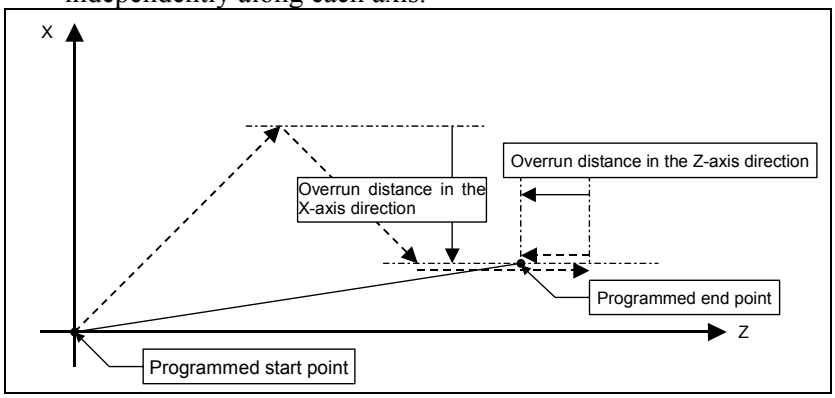
An overrun and a positioning direction are set by the parameter No. 5440. Even when a commanded positioning direction coincides with that set by the parameter, the tool stops once before the end point.
G60, which is a one-shot G-code, can be used as a modal G-code in group 01 by setting 1 to the bit 0 (MDL) of parameter No. 5431.
This setting can eliminate specifying a G60 command for every block. Other specifications are the same as those for a one-shot G60 command. When a one-shot G code is specified in the single direction positioning mode, the one-shot G command is effective like G codes in group 01.

(Example)

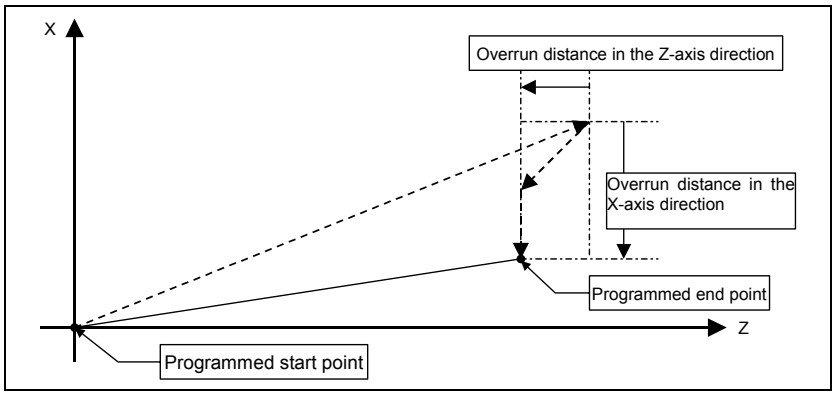
When one-shot G60 commands are used.	
G90; G60 X0Y0; G60 X100; G60 Y100; G04 X10; G00 X0Y0;	 } Single direction positioning
When modal G60 command is used.	
G90G60; X0Y0; X100; Y100; G04X10; G00X0 Y0;	Single direction positioning mode start } Single direction positioning Single direction positioning mode cancel

- Overview of operation

- **In the case of positioning of non-linear interpolation type (bit 1 (LRP) of parameter No. 1401 = 0)**
  As shown below, single direction positioning is performed independently along each axis.



- **In the case of positioning of linear interpolation type (bit 1 (LRP) of parameter No. 1401 = 1)**
  Positioning of interpolation type is performed until the tool once stops before or after a specified end point. Then, the tool is positioned independently along each axis until the end point is reached.



Limitation

- Single direction positioning is not performed along an axis for which no overrun distance is set in parameter No. 5440.
- Single direction positioning is not performed along an axis for which travel distance 0 is specified.
- The mirror image function is not applied in a parameter-set direction. Even in the mirror image mode, the direction of single direction positioning remains unchanged. If positioning of linear interpolation type is used, and the state of mirror image when a single direction positioning block is looked ahead differs from the state of mirror image when the execution of the block is started, an alarm is issued. When switching mirror image in the middle of a program, disable looking ahead by specifying a non-buffering M code. Then, switch mirror image when there is no look-ahead block.

- In the cylindrical interpolation mode (G07.1), single direction positioning cannot be used.
- In the polar coordinate interpolation mode (G12.1), single direction positioning cannot be used.
- When specifying single direction positioning on a machine that uses arbitrary angular axis control, first position the angular axis then specify the positioning of the Cartesian axis. If the reverse specification order is used, or the angular axis and Cartesian axis are specified in the same block, an incorrect positioning direction can result.
- In positioning at a restart position by program restart function, single direction positioning is not performed.

M

- During canned cycle for drilling, no single direction positioning is effected in drilling axis.
- The single direction positioning does not apply to the shift motion in the canned cycles of G76 and G87.

T

- The G-code for single direction positioning is always G60, if G-code system is A or B or C in all case.
- The single direction positioning can not be commanded during the multiple repetitive cycle (G70-G76).
- No single direction positioning is effected in the drilling or patting axis, during canned cycle for drilling (G83-G89) and the rigid tapping (G84, G88). However, it can be commanded for positioning.
- The single direction positioning can not be commanded during the canned cycle (G90, G92, G94).
- During the single direction positioning mode (G60), the following G-code can not be commanded.
  G07.1, G12.1, G70-G76, G90-G94.

4.3 LINEAR INTERPOLATION (G01)

Tools can move along a line.

Format

G01 IP_ **F_ ;**

IP_ : For an absolute programming, the coordinates of an end point, and for an incremental programming, the distance the tool moves.

F_ : Speed of tool feed (Feedrate)

Explanation

A tools move along a line to the specified position at the feedrate specified in F.
The feedrate specified in F is effective until a new value is specified. It need not be specified for each block.
The feedrate commanded by the F code is measured along the tool path. If the F code is not commanded, the feedrate is regarded as zero.
The feedrate of each axis direction is as follows.

G01 αα ββ γγ ζζ Ff ;

Feed rate of α axis direction : $F\alpha = \frac{\alpha}{L} \times f$

Feed rate of β axis direction : $F\beta = \frac{\beta}{L} \times f$

Feed rate of γ axis direction : $F\gamma = \frac{\gamma}{L} \times f$

Feed rate of ζ axis direction : $F\zeta = \frac{\zeta}{L} \times f$

$L = \sqrt{\alpha^2 + \beta^2 + \gamma^2 + \zeta^2}$

The feedrate of the rotary axis is commanded in the unit of deg/min (the unit is decimal point position).
When the straight line axis α (such as X, Y, or Z) and the rotating axis β (such as A, B, or C) are linearly interpolated, the feedrate is that in which the tangential feedrate in the α and β cartesian coordinate system is commanded by F (mm/min).
β-axis feedrate is obtained ; at first, the time required for distribution is calculated by using the above formula, then the β-axis feedrate unit is changed to deg/min.

A calculation example is as follows.
G91 G01 X20.0B40.0 F300.0 ;
This changes the unit of the C axis from 40.0 deg to 40mm with metric input. The time required for distribution is calculated as follows:

$$\frac{\sqrt{20^2+40^2}}{300} \fallingdotseq 0.14907(mm)$$

The feedrate for the C axis is

$$\frac{40}{0.14907} \fallingdotseq 268.3\,\mathrm{deg/\,min}$$

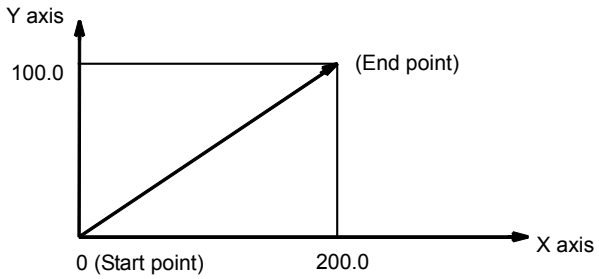
In simultaneous 3 axes control, the feedrate is calculated the same way as in 2 axes control.

Example

- Linear interpolation

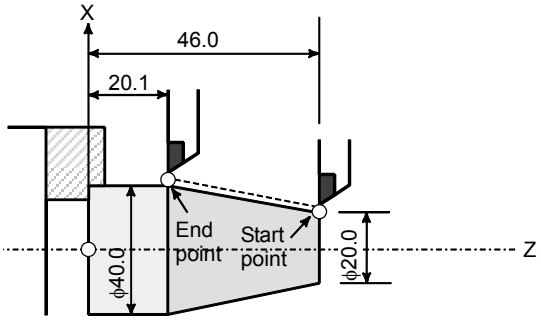
- For milling machining

(G91) G01X200.0Y100.0F200.0;

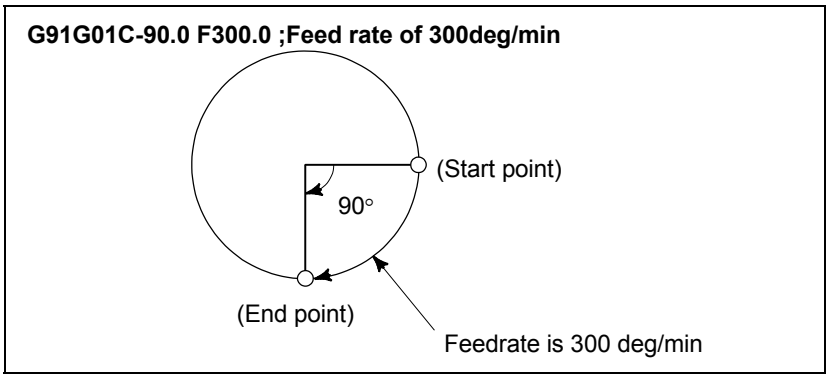


- For lathe cutting

(Diameter programming)
G01X40.0Z20.1F20; (Absolute programming)
or
G01U20.0W-25.9F20; (Incremental programming)



- Feedrate for the rotary axis



4.4 CIRCULAR INTERPOLATION (G02, G03)

The command below will move a tool along a circular arc.

Format

Arc in the XpYp plane

$$G17 \begin{Bmatrix} G02 \\ G03 \end{Bmatrix} Xp_\ Yp_ \begin{Bmatrix} I_\ J_ \\ R_ \end{Bmatrix} F_\ ;$$

Arc in the ZpXp plane

$$G18 \begin{Bmatrix} G02 \\ G03 \end{Bmatrix} Zp_\ Xp_ \begin{Bmatrix} I_\ K_ \\ R_ \end{Bmatrix} F_\ ;$$

Arc in the YpZp plane

$$G19 \begin{Bmatrix} G02 \\ G03 \end{Bmatrix} Yp_\ Zp_ \begin{Bmatrix} J_\ K_ \\ R_ \end{Bmatrix} F_\ ;$$

Command	Description
G17	Specification of arc on XpYp plane
G18	Specification of arc on ZpXp plane
G19	Specification of arc on YpZp plane
G02	Circular Interpolation : Clockwise direction (CW)
G03	Circular Interpolation : Counterclockwise direction (CCW)
Xp_	Command values of X axis or its parallel axis (set by parameter No. 1022)
Yp_	Command values of Y axis or its parallel axis (set by parameter No. 1022)
Zp_	Command values of Z axis or its parallel axis (set by parameter No. 1022)
I_	Xp axis distance from the start point to the center of an arc with sign
J_	Yp axis distance from the start point to the center of an arc with sign
K_	Zp axis distance from the start point to the center of an arc with sign
R_	Arc radius (with sign, radius value for lathe cutting)
F_	Feedrate along the arc

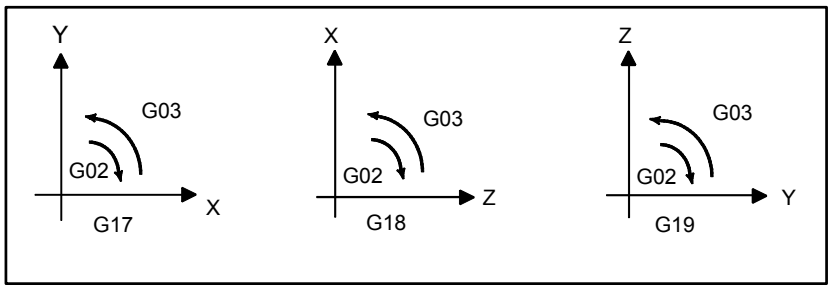
T

NOTE
The U-, V-, and W-axes can be used with G-codes B and C.

Explanation

- Direction of the circular interpolation

"Clockwise"(G02) and "counterclockwise"(G03) on the XpYp plane (ZpXp plane or YpZp plane) are defined when the XpYp plane is viewed in the positive-to-negative direction of the Zp axis (Yp axis or Xp axis, respectively) in the Cartesian coordinate system. See the figure below.



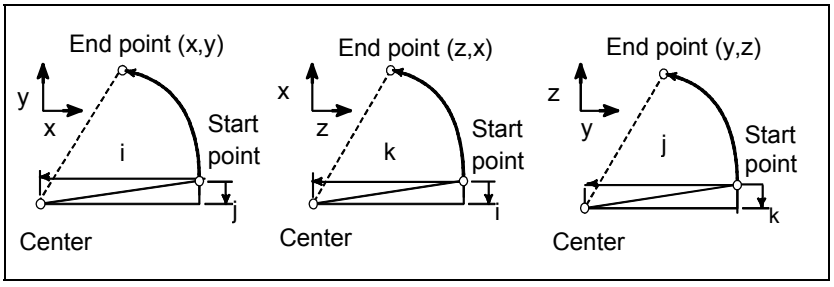
- Distance moved on an arc

The end point of an arc is specified by address Xp, Yp or Zp, and is expressed as an absolute or incremental value according to G90 or G91. For the incremental value, the distance of the end point which is viewed from the start point of the arc is specified with sign.

- Distance from the start point to the center of arc

The arc center is specified by addresses I, J, and K for the Xp, Yp, and Zp axes, respectively. The numerical value following I, J, or K, however, is a vector component in which the arc center is seen from the start point, and is always specified as an incremental value irrespective of G90 and G91, as shown below.

I, J, and K must be signed according to the direction.



I0, J0, and K0 can be omitted.

If the difference between the radius at the start point and that at the end point exceeds the permitted value in a parameter (No.3410), an alarm PS0020 occurs.

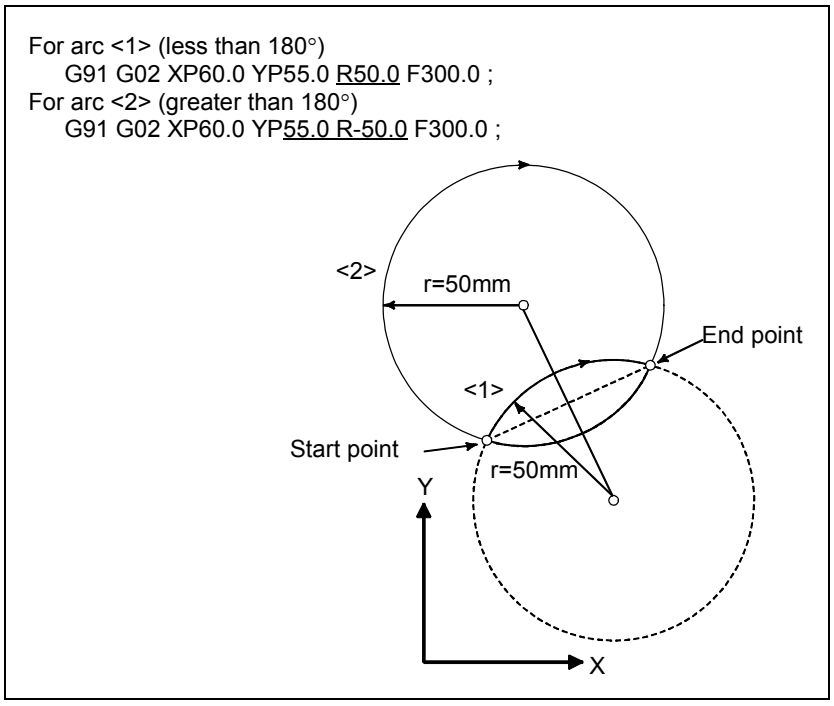
- Command for a circle

When Xp, Yp, and Zp are omitted (the end point is the same as the start point) and the center is specified with I, J, and K, a 360° arc (circle) is specified.

G02 I_ ; Command for a circle

- Arc radius

The distance between an arc and the center of a circle that contains the arc can be specified using the radius, R, of the circle instead of I, J, and K.
In this case, one arc is less than 180°, and the other is more than 180° are considered. When an arc exceeding 180° is commanded, the radius must be specified with a negative value. If Xp, Yp, and Zp are all omitted, if the end point is located at the same position as the start point and when R is used, an arc of 0° is programmed
G02R_ ; (The cutter does not move.)



- Feedrate

The feedrate in circular interpolation is equal to the feedrate specified by the F code, and the feedrate along the arc (the tangential feedrate of the arc) is controlled to be the specified feedrate.
The error between the specified feedrate and the actual tool feedrate is ±2% or less. However, this feedrate is measured along the arc after the tool radius compensation is applied

Limitation

- Simultaneously specifying R with I, J, and K

If I, J, K, and R addresses are specified simultaneously, the arc specified by address R takes precedence and the other are ignored.

- Specifying an axis that is not contained in the specified plane

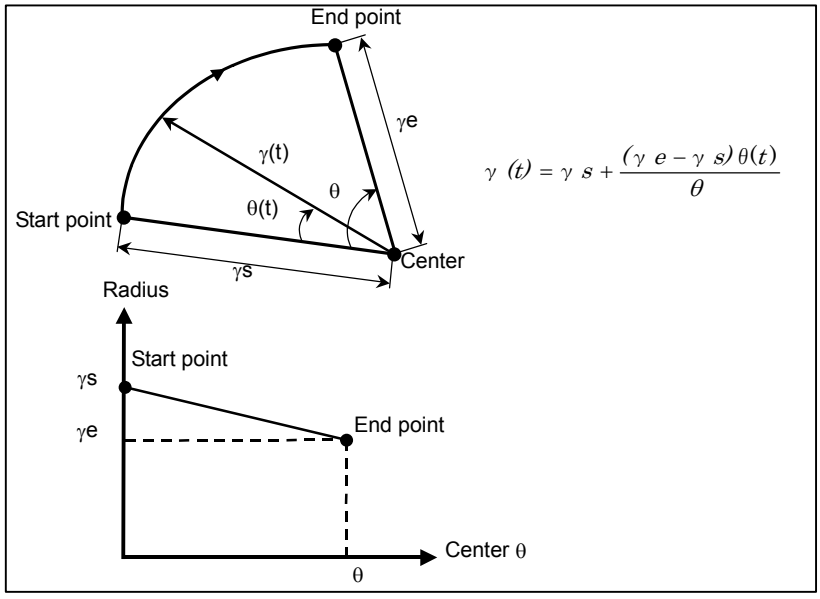
If an axis not comprising the specified plane is commanded, an alarm PS0028 occurs.
For example,
For milling machining:
If the X-axis and a U-axis parallel to the X-axis are specified when the XY plane is specified
For lathe cutting:
If the X-axis and a U-axis parallel to the X-axis are specified when the ZX plane is specified with G code system B or C

- Specifying a semicircle with R

When an arc having a center angle approaching 180° is specified, the calculated center coordinates may contain an error. In such a case, specify the center of the arc with I, J, and K.

- Difference in the radius between the start and end points

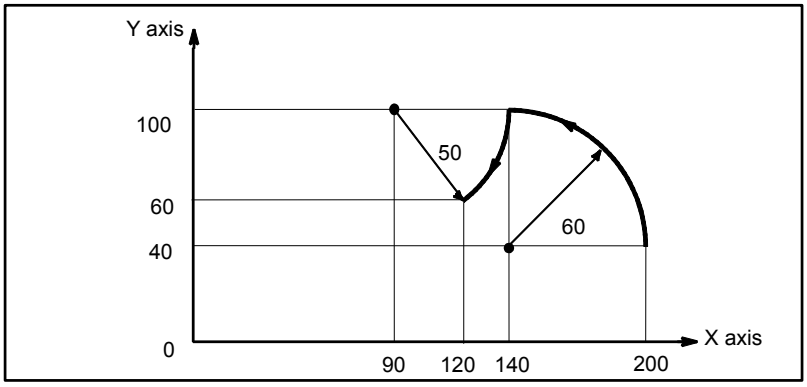
If the difference in the radius between the start and end points of the arc exceeds the value specified in parameter No. 3410, alarm PS0020 is generated.
When an end point does not lie on the arc, a spiral results, as shown below.



The arc radius changes linearly with the center angle θ(t). Spiral interpolation is performed using a circular command that specifies one arc radius for the start point and another arc radius for the end point. To use spiral interpolation, set a large value in parameter No. 3410, used to specify the limit on the arc radius error.

Example

M



The above tool path can be programmed as follows;

(1) In absolute programming

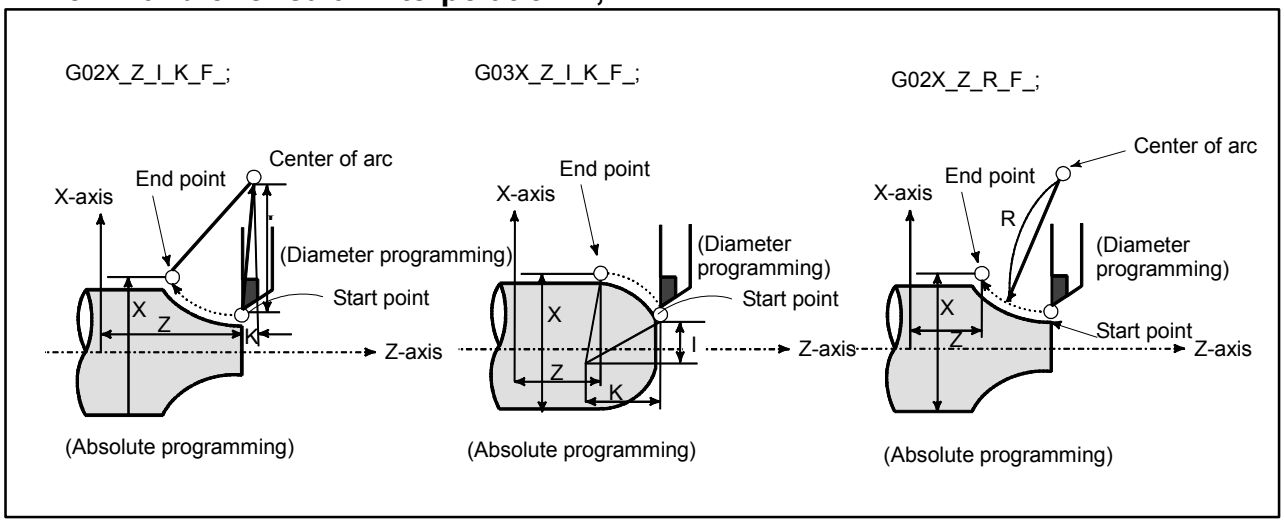
```
G92X200.0 Y40.0 Z0 ;
G90 G03 X140.0 Y100.0 R60.0 F300. ;
G02 X120.0 Y60.0 R50.0 ;
or
G92X200.0 Y40.0Z0 ;
G90 G03 X140.0 Y100.0I-60.0 F300. ;
G02 X120.0 Y60.0I-50.0 ;
```

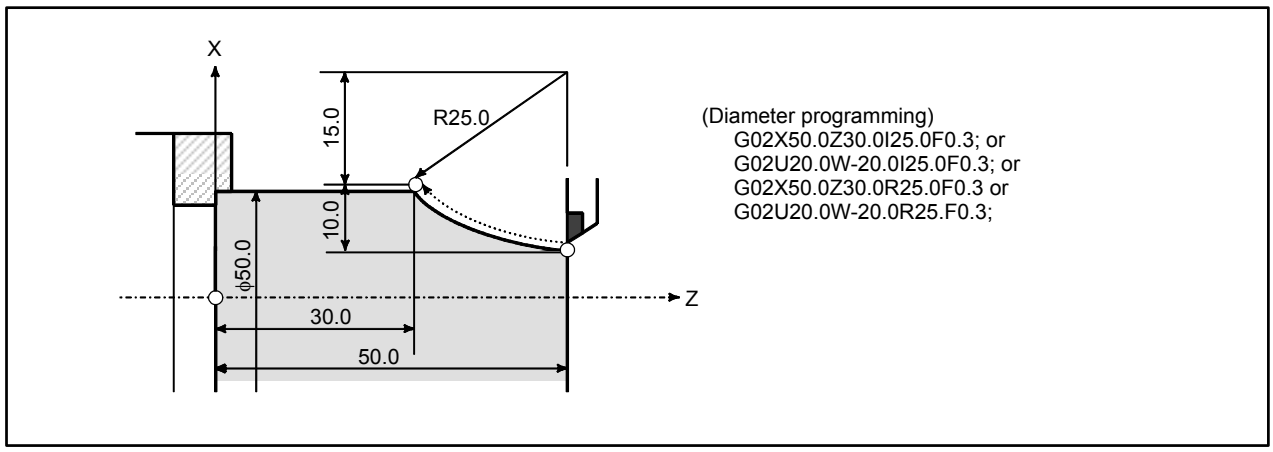
(2) In incremental programming

```
G91 G03 X-60.0 Y60.0 R60.0 F300. ;
G02 X-20.0 Y-40.0 R50.0 ;
or
G91 G03 X-60.0 Y60.0 I-60.0 F300. ;
G02 X-20.0 Y-40.0 I-50.0 ;
```

T

- Command of circular interpolation X, Z





4.5 HELICAL INTERPOLATION (G02, G03)

Helical interpolation which moved helically is enabled by specifying up to two other axes which move synchronously with the circular interpolation by circular commands.

Format

Arc in the XpYp plane

$$\mathbf{G17}\ \left\{\begin{matrix}\mathbf{G02}\\ \mathbf{G03}\end{matrix}\right\}\ \mathbf{Xp_\ Yp_}\ \left\{\begin{matrix}\mathbf{I_\ J_}\\ \mathbf{R_}\end{matrix}\right\}\ \mathbf{\alpha_\ (\beta_)\ F_\ ;}$$

Arc in the ZpXp plane

$$\mathbf{G18}\ \left\{\begin{matrix}\mathbf{G02}\\ \mathbf{G03}\end{matrix}\right\}\ \mathbf{Zp_\ Xp_}\ \left\{\begin{matrix}\mathbf{K_\ I_}\\ \mathbf{R_}\end{matrix}\right\}\ \mathbf{\alpha_\ (\beta_)\ F_\ ;}$$

Arc in the YpZp plane

$$\mathbf{G19}\ \left\{\begin{matrix}\mathbf{G02}\\ \mathbf{G03}\end{matrix}\right\}\ \mathbf{Yp_\ Zp_}\ \left\{\begin{matrix}\mathbf{J_\ K_}\\ \mathbf{R_}\end{matrix}\right\}\ \mathbf{\alpha_\ (\beta_)\ F_\ ;}$$

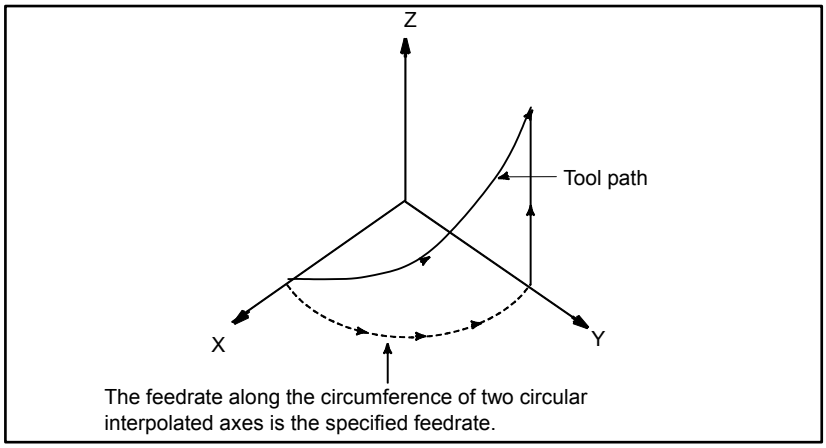
α, β : Any one axis where circular interpolation is not applied.
Up to two other axes can be specified.

Explanation

A tangential velocity of an arc in a specified plane or a tangential velocity about the linear axis can be specified as the feedrate, depending on the setting of bit 5 (HTG) of parameter No.1403.
An F command specifies a feedrate along a circular arc, when HTG is specified to 0. Therefore, the feedrate of the linear axis is as follows:

$$F \times \frac{\text{Length of linear axis}}{\text{Length of circular arc}}$$

Determine the feedrate so the linear axis feedrate does not exceed any of the various limit values.

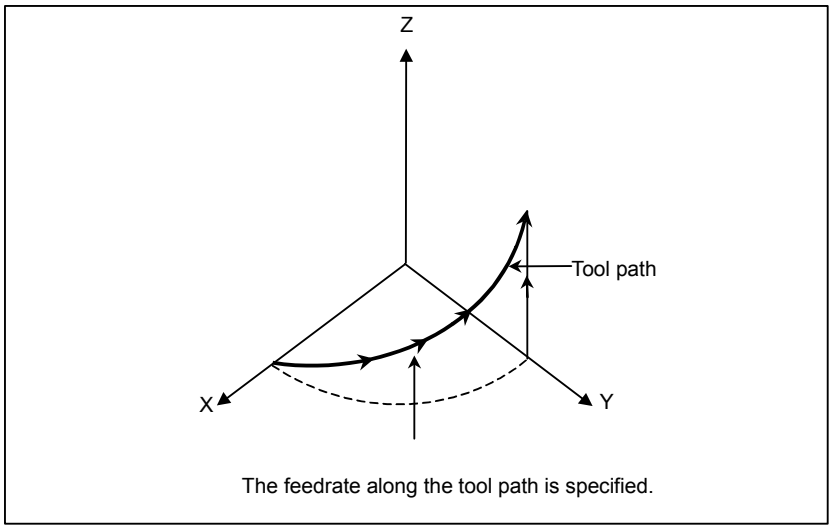


If HTG is set to 1, specify a feedrate along the tool path about the linear axis. Therefore, the tangential velocity of the arc is expressed as follows:

$$F \times \frac{\text{Length of arc}}{\sqrt{(\text{Length of arc})^2 + (\text{Length of linear axis})^2}}$$

The velocity along the linear axis is expressed as follows:

$$F \times \frac{\text{Length of linear axis}}{\sqrt{(\text{Length of arc})^2 + (\text{Length of linear axis})^2}}$$



Limitation

- Tool radius/tool nose radius compensation is applied only for a circular arc.
- Tool offset and tool length compensation cannot be used in a block in which a helical interpolation is commanded.

4.6 HELICAL INTERPOLATION B (G02, G03)

The helical interpolation B function differs from the helical interpolation function just in that circular interpolation and a movement on four axes outside the specified plane can be simultaneously performed.
For the restrictions and parameters, see the description of the helical interpolation function.

Format

Arc in the XpYp plane

$$\mathbf{G17}\ \left\{\begin{matrix}\mathbf{G02}\\ \mathbf{G03}\end{matrix}\right\}\ \mathbf{Xp_\ Yp_}\ \left\{\begin{matrix}\mathbf{I_\ J_}\\ \mathbf{R_}\end{matrix}\right\}\ \alpha_\ \beta_\ \gamma_\ \delta_\ \mathbf{F_\ ;}$$

Arc in the ZpXp plane

$$\mathbf{G18}\ \left\{\begin{matrix}\mathbf{G02}\\ \mathbf{G03}\end{matrix}\right\}\ \mathbf{Zp_\ Xp_}\ \left\{\begin{matrix}\mathbf{K_\ I_}\\ \mathbf{R_}\end{matrix}\right\}\ \alpha_\ \beta_\ \gamma_\ \delta_\ \mathbf{F_\ ;}$$

Arc in the YpZp plane

$$\mathbf{G19}\ \left\{\begin{matrix}\mathbf{G02}\\ \mathbf{G03}\end{matrix}\right\}\ \mathbf{Yp_\ Zp_}\ \left\{\begin{matrix}\mathbf{J_\ K_}\\ \mathbf{R_}\end{matrix}\right\}\ \alpha_\ \beta_\ \gamma_\ \delta_\ \mathbf{F_\ ;}$$

α, β, γ, δ : Any axis to which circular interpolation is not applied.
Up to four axes can be specified.

4.7 SPIRAL INTERPOLATION, CONICAL INTERPOLATION (G02, G03)

Spiral interpolation is enabled by specifying the circular interpolation command together with a desired number of revolutions or a desired increment (decrement) for the radius per revolution.
Conical interpolation is enabled by specifying the spiral interpolation command together with an additional axis of movement, as well as a desired increment (decrement) for the position along the additional axes per spiral revolution.

Format

- Spiral interpolation

XpYp plane

G17 { G02 / G03 } X_Y_I_J_Q_L_F_;

ZpXp plane

G18 { G02 / G03 } Z_X_K_I_Q_L_F_;

YpZp plane

G19 { G02 / G03 } Y_Z_J_K_Q_L_F_;

X, Y, Z : Coordinates of the end point
L : Number of revolutions (positive value without a decimal point) (*1)
Q : Radius increment or decrement per spiral revolution (*1, *2)
I, J, K : Signed distance from the start point to the center (same as the distance specified for circular interpolation)
F : Feedrate

(*1) Either the number of revolutions (L) or the radius increment or decrement (Q) can be omitted. When L is omitted, the number of revolutions is automatically calculated from the distance between the current position and the center, the position of the end point, and the radius increment or decrement. When Q is omitted, the radius increment or decrement is automatically calculated from the distance between the current position and the center, the position of the end point, and the number of revolutions. If both L and Q are specified but their values contradict, Q takes precedence. Generally, either L or Q should be specified. The L value must be a positive value without a decimal point. To specify four revolutions plus 90°, for example, round the number of revolutions up to five and specify L5.
(*2) The increment system for Q depends on the reference axis.

- Conical interpolation

XpYp plane

G17 {**G02** / **G03**} **X_ Y_ I_ J_ Z_ Q_ L_ F_ ;**

ZpXp plane

G18 {**G02** / **G03**} **Z_ X_ K_ I_ Y_ Q_ L_ F_ ;**

YpZp plane

G19 {**G02** / **G03**} **Y_ Z_ J_ K_ X_ Q_ L_ F_ ;**

X, Y, Z : Coordinates of the end point

L : Number of revolutions (positive value without a decimal point) (*1)

Q : Radius increment or decrement per spiral revolution (*1, *2)

I, J, K : Two of the three values represent a signed vector from the start point to the center. The remaining value is a height increment or decrement per spiral revolution in conical interpolation. (*1)

When the XpYp plane is selected:

The I and J values represent a signed vector from the start point to the center. The K value represents a height increment or decrement per spiral revolution.

When the ZpXp plane is selected:

The K and I values represent a signed vector from the start point to the center. The J value represents a height increment or decrement per spiral revolution.

When the YpZp plane is selected:

The J and K values represent a signed vector from the start point to the center. The I value represents a height increment or decrement per spiral revolution.

F : Feedrate (The tangential velocity about the linear axis is specified.)

(*1) One of the height increment/decrement (I, J, K), radius increment/decrement (Q), and the number of revolutions (L) must be specified. The other two items can be omitted.

- Sample command for the XpYp plane

G18 {**G02** / **G03**} **X_ Y_ I_ J_ Z_** {**K_** / **Q_** / **L_**} **F_ ;**

If both L and Q are specified, but their values contradict, Q takes precedence. If both L and a height increment or decrement are specified, but their values contradict, the height increment or decrement takes precedence. If both Q and a height increment or decrement are specified, but their values contradict, Q takes precedence. The L value must be a positive value without a decimal point. To specify four revolutions plus 90°, for example, round the number of revolutions up to five and specify L5.

(*2) The increment system for Q depends on the reference axis.

Explanation

- Function of spiral interpolation

Spiral interpolation in the XY plane is defined as follows:

$$(X - X_0)^2 + (Y - Y_0)^2 = (R + Q')^2$$

X_0: X coordinate of the center
Y_0: Y coordinate of the center
R : Radius at the beginning of spiral interpolation
Q' : Variation in radius

When the programmed command is assigned to this function, the following expression is obtained:

$$(X - X_S - I)^2 + (Y - Y_S - J)^2 = (R + (L' + \frac{\theta}{360})Q)^2$$

where

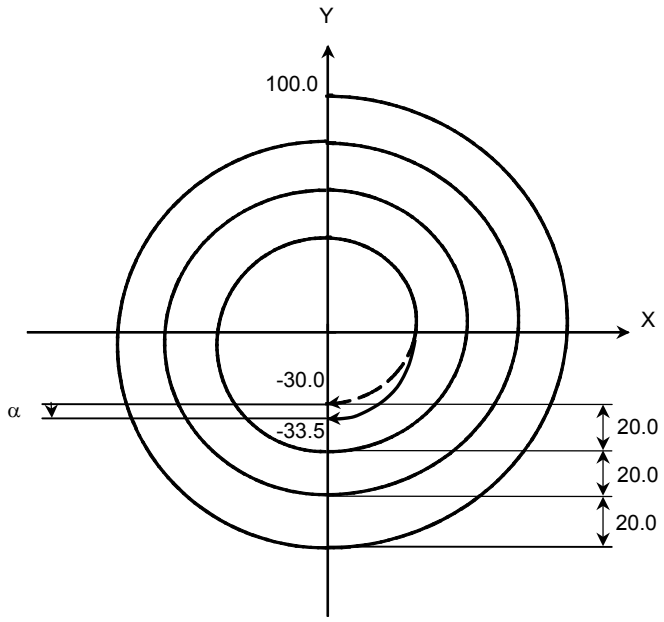
X_S: X coordinate of the start point
Y_S: Y coordinate of the start point
I : X coordinate of the vector from the start point to the center
J : Y coordinate of the vector from the start point to the center
R : Radius at the beginning of spiral interpolation
Q : Radius increment or decrement per spiral revolution
L' : (Current number of revolutions) - 1
θ : Angle between the start point and the current position (degrees)

- Controlled axes

For conical interpolation, two axes of a plane and two additional axes, that is, four axes in total, can be specified. A rotary axis can be specified as the additional axis.

- Difference between end points

If the difference between the programmed end point and the calculated end point of a spiral exceeds a value specified in parameter No. 3471 about any axis of a selected plane, an alarm PS5123 will be issued.
If the difference between the programmed height and calculated height of the end point a cone exceeds a value specified in parameter No. 3471, an alarm PS5123 will be issued. The figure below illustrates details.



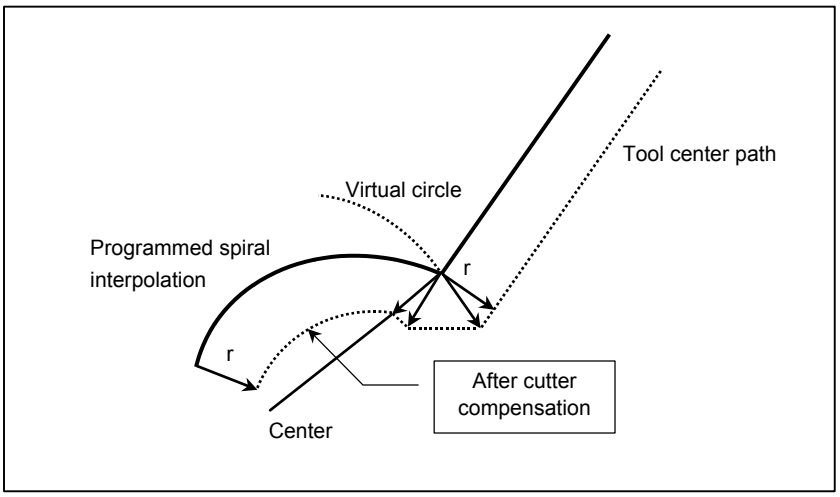
G90 G02 X0 Y-33.5 I0 J-100. { Q-20. / L4 } F300;

The coordinates of the programmed end point are (0, -33.5) while the coordinates of the calculated end point are (0, -30.0). A value greater than the difference (α: Tolerance) is specified in parameter No. 3471. If the end point is exceeded, an alarm PS5123 will be issued. The same is specified for the height of conical interpolation.

- Tool radius compensation

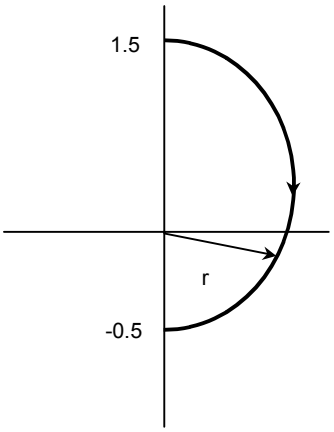
M

The spiral or conical interpolation command can be programmed in tool radius compensation mode. This compensation is performed in the same way as described in "When it is exceptional" in "Tool Movement in Offset Mode" section. A virtual circle centered on the center of spiral interpolation is thought at the end of a block. The tool path is obtained by performing tool radius compensation along the virtual circle and the blocks before and after the spiral interpolation. When the end point of the block is at the center of the spiral interpolation, no virtual circle can be drawn. If drawing is attempted, an alarm PS5124 is issued.



- Actual cutting feedrate

A constant speed is maintained in spiral interpolation or conical interpolation. The angular velocity near the center, however, may increase because of the small radius of the spiral. This can be avoided by maintaining the angular velocity after the radius of the spiral reaches a value specified in parameter No. 3472. Consequently, the actual cutting feedrate decreases.
An example is illustrated below.



Specified feedrate: F100.
Parameter 3472 (r) = 1.000
IS-B

While the radius of the spiral shown at the left is greater than the value specified in parameter No. 3472, the actual cutting feedrate is F100. As the radius decreases, the actual feedrate also decreases, the actual feedrate near the end point being about F65.

- Deceleration by acceleration

During spiral interpolation, the function of deceleration by acceleration is enabled. The feedrate may decrease as the tool approaches the center of the spiral.

- Dry run

When the dry run signal is inverted from 0 to 1 or from 1 to 0 during movement along an axis, the movement is accelerated or decelerated to the desired speed without first reducing the speed to zero.

Limitation

- Radius

During spiral interpolation and conical interplation, the addresses "C", "R", ",C", or ",R" cannot be specified.

- Feed functions

The functions of feed per rotation, inverse time feed, F command with one digit, and automatic corner override cannot be used.

- Retrace

A program including spiral or conical interpolation cannot be retraced.

- Polar coordinate interpolation, scaling, and normal direction control

Spiral interpolation and conical interpolation cannot be specified in these modes.

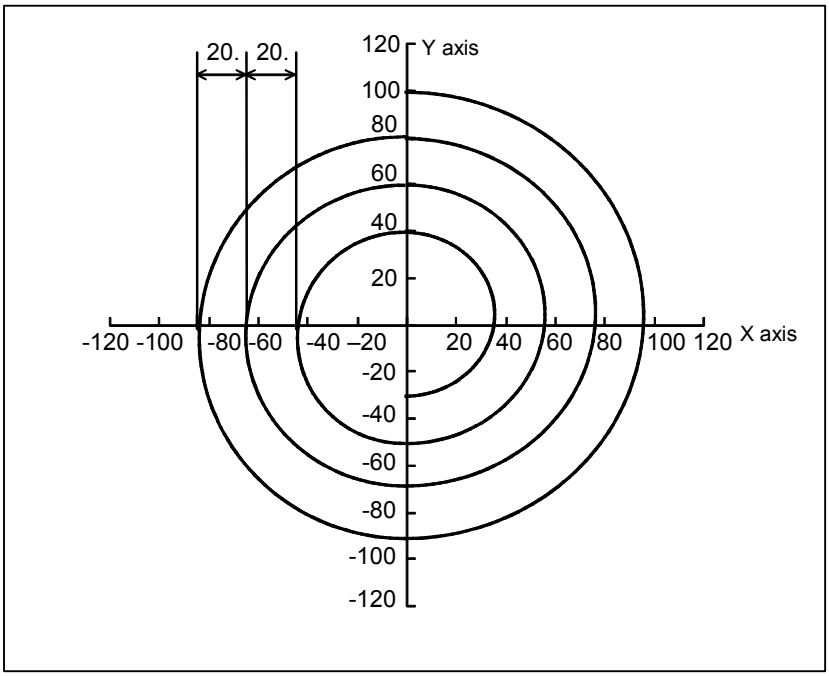
- Optional function

To use the conical interpolation function, the optional function for helical interpolation is also necessary.

Example

- Spiral interpolation

The path indicated below is programmed with absolute and incremental values as follows:



This sample path has the following values:

- Start point : (0, 100.0)
- End point (X, Y) : (0, -30.0)
- Distance to the center (I, J) : (0, -100.0)
- Radius increment or decrement (Q) : -20.0
- Number of revolutions (L) : 4

(1) With absolute values, the path is programmed as follows:

```
G90 G02 X0 Y-30.0 I0 J-100.0 { Q-20.0 / L4 } F300.0 ;
```

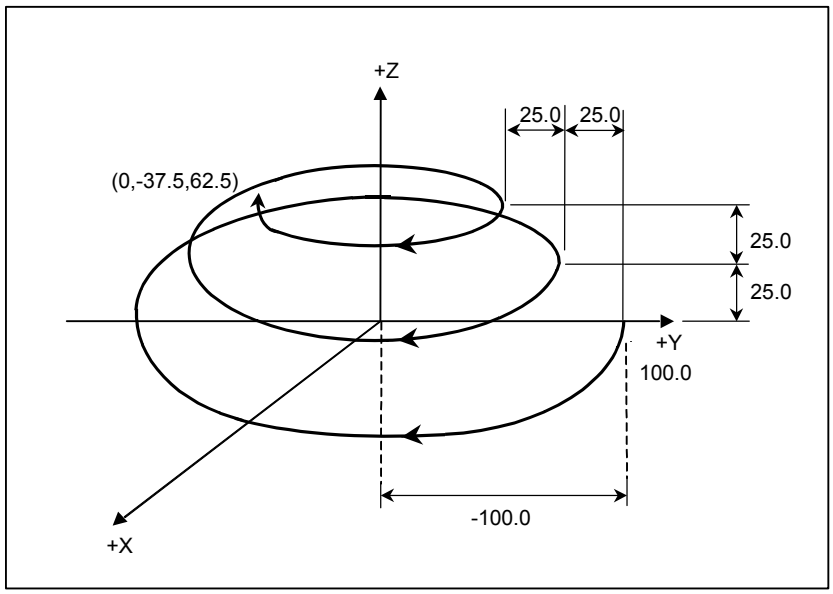
(2) With incremental values, the path is programmed as follows:

```
G91 G02 X0 Y-130.0 I0 J-100.0 { Q-20.0 / L4 } F300.0 ;
```

(Either the Q or L setting can be omitted.)

- Conical interpolation

The sample path shown below is programmed with absolute and incremental values as follows:



This sample path has the following values:

- Start point : (0, 100.0, 0)
- End point (X, Y, Z) : (0, -37.5, 62.5)
- Distance to the center (I, J) : (0, -100.0)
- Radius increment or decrement (Q) : -25.0
- Height increment or decrement (K) : 25.0
- Number of revolutions (L) : 3

(1) With absolute values, the path is programmed as follows:

```
G90 G02 X0 Y-37.5 Z62.5 I0 J-100.0 { K25.0 Q-25.0 L3 } F300.0 ;
```

(2) With incremental values, the path is programmed as follows:

```
G91 G02 X0 Y-137.5 Z62.5 I0 J-100.0 { K25.0 Q-25.0 L3 } F300.0;
```

(Either the Q or L setting can be omitted.)

4.8 POLAR COORDINATE INTERPOLATION (G12.1, G13.1)

Overview

Polar coordinate interpolation is a function that exercises contour control in converting a command programmed in a Cartesian coordinate system to the movement of a linear axis (movement of a tool) and the movement of a rotary axis (rotation of a workpiece). This function is useful for grinding a cam shaft.

Format

G12.1; Starts polar coordinate interpolation mode (enables polar coordinate interpolation).
Specify linear or circular interpolation using coordinates in a Cartesian coordinate system consisting of a linear axis and rotary axis (hypothetical axis).
G13.1; Polar coordinate interpolation mode is cancelled (for not performing polar coordinate interpolation).

Specify G12.1 and G13.1 in Separate Blocks.

Explanation

- Polar coordinate interpolation mode (G12.1)

The axes of polar coordinate interpolation (linear axis and rotary axis) should be specified in advance, with corresponding parameters. Specifying G12.1 places the system in the polar coordinate interpolation mode, and selects a plane (called the polar coordinate interpolation plane) formed by one linear axis and a hypothetical axis intersecting the linear axis at right angles. The linear axis is called the first axis of the plane, and the hypothetical axis is called the second axis of the plane. Polar coordinate interpolation is performed in this plane.

In the polar coordinate interpolation mode, both linear interpolation and circular interpolation can be specified by absolute or incremental programming.

Tool radius compensation can also be performed. The polar coordinate interpolation is performed for a path obtained after tool radius compensation.

The tangential velocity in the polar coordinate interpolation plane (Cartesian coordinate system) is specified as the feedrate, using F.

- Polar coordinate interpolation cancel mode (G13.1)

Specifying G13.1 cancels the polar coordinate interpolation mode.

- Polar coordinate interpolation plane

G12.1 starts the polar coordinate interpolation mode and selects a polar coordinate interpolation plane (Fig. 4.8 (a)). Polar coordinate interpolation is performed on this plane.

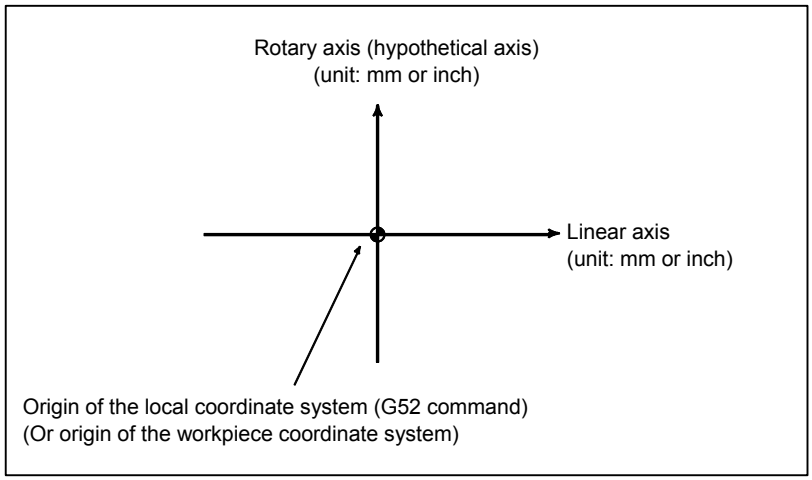


Fig. 4.8 (a) Polar coordinate interpolation plane

When the power is turned on or the system is reset, polar coordinate interpolation is canceled (G13.1).
The linear and rotation axes for polar coordinate interpolation must be set in parameters Nos. 5460 and 5461 beforehand.

> ⚠ **CAUTION**
> The plane used before G12.1 is specified (plane selected by G17, G18, or G19) is canceled. It is restored when G13.1 (canceling polar coordinate interpolation) is specified.
> When the system is reset, polar coordinate interpolation is canceled and the plane specified by G17, G18, or G19 is used.

- Distance moved and feedrate for polar coordinate interpolation

- The unit for coordinates on the hypothetical axis is the same as the unit for the linear axis (mm/inch).
  In the polar coordinate interpolation mode, program commands are specified with Cartesian coordinates on the polar coordinate interpolation plane. The axis address for the rotary axis is used as the axis address for the second axis (hypothetical axis) in the plane. Whether a diameter or radius is specified for the first axis in the plane is the same as for the rotary axis regardless of the specification for the first axis in the plane.
  The hypothetical axis is at coordinate 0 immediately after G12.1 is specified. Polar interpolation is started assuming the rotation angle of 0 for the position of the tool when G12.1 is specified.

Example)

When a value on the X-axis (linear axis) is input in millimeters

```
G12.1;
G01 X10. F1000. ;.... A 10-mm movement is made on the
                      Cartesian coordinate system.
C20. ; ..... A 20-mm movement is made on the
                      Cartesian coordinate system.
G13.1;
```

When a value on the X-axis (linear axis) is input in inches

```
G12.1;
G01 X10. F1000. ; .. A 10-inch movement is made on the
                      Cartesian coordinate system.
C20. ; ..... A 20-inch movement is made on the
                      Cartesian coordinate system.
G13.1;
```

- The unit for the feedrate is mm/min or inch/min.
  Specify the feedrate as a speed (relative speed between the workpiece and tool) tangential to the polar coordinate interpolation plane (Cartesian coordinate system) using F.

- G codes which can be specified in the polar coordinate interpolation mode

G01 .....Linear interpolation
G02, G03.....Circular interpolation
G02.2, G03.2.....Involute interpolation
G04 .....Dwell, Exact stop
G40, G41, G42.....Tool radius compensation
(Polar coordinate interpolation is applied to the path after tool radius compensation.)
G65, G66, G67.....Custom macro command
G90, G91.....Absolute programming, incremental programming
G94, G95.....Feed per minute, feed per revolution

- Circular interpolation in the polar coordinate plane

The addresses for specifying the radius of an arc for circular interpolation (G02 or G03) in the polar coordinate interpolation plane depend on the first axis in the plane (linear axis).

- I and J in the Xp-Yp plane when the linear axis is the X-axis or an axis parallel to the X-axis.
- J and K in the Yp-Zp plane when the linear axis is the Y-axis or an axis parallel to the Y-axis.
- K and I in the Zp-Xp plane when the linear axis is the Z-axis or an axis parallel to the Z-axis.

The radius of an arc can be specified also with an R command.

NOTE
In a lathe system, the parallel axes U, V, and W can be used in the G code system B or C.

- Movement along axes not in the polar coordinate interpolation plane in the polar coordinate interpolation mode

The tool moves along such axes normally, independent of polar coordinate interpolation.

- Current position display in the polar coordinate interpolation mode

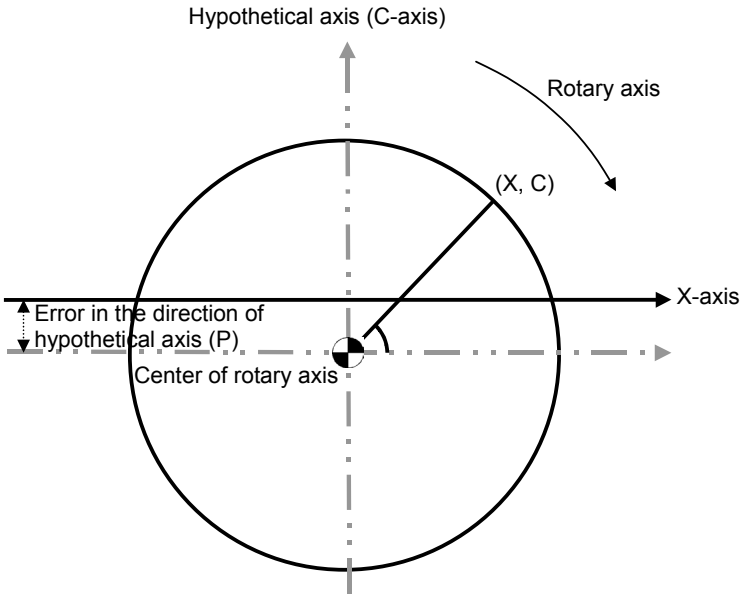
Actual coordinates are displayed. However, the remaining distance to move in a block is displayed based on the coordinates in the polar coordinate interpolation plane (Cartesian coordinates).

- Coordinate system for the polar coordinate interpolation

Basically, before G12.1 is specified, a local coordinate system (or workpiece coordinate system) where the center of the rotary axis is the origin of the coordinate system must be set.

- Compensation in the direction of the hypothetical axis in polar coordinate interpolation

If the first axis of the plane has an error from the center of the rotary axis in the hypothetical axis direction, in other words, if the rotary axis center is not on the X-axis, the hypothetical axis direction compensation function in the polar coordinate interpolation mode is used. With the function, the error is considered in polar coordinate interpolation. The amount of error is specified in parameter No. 5464.

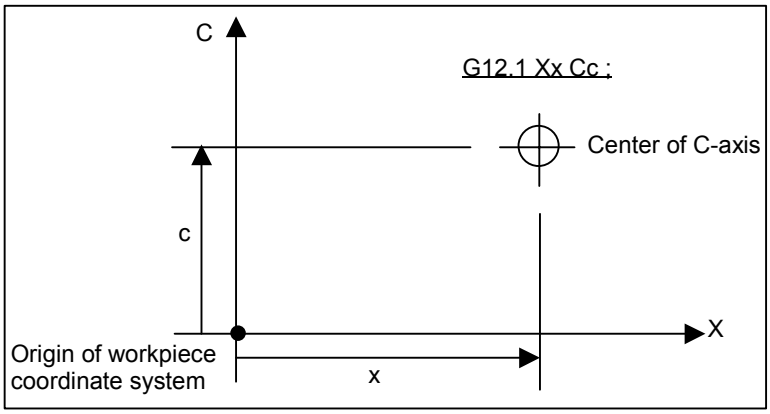


(X, C)	Point in the X-C plane (The center of the rotary axis is considered to be the origin of the X-C plane.)
X	X coordinate in the X-C plane
C	Hypothetical axis coordinate in the X-C plane
P	Error in the direction of the hypothetical axis (specified in parameter No. 5464)

- Shifting the coordinate system in polar coordinate interpolation

In the polar coordinate interpolation mode, the workpiece coordinate system can be shifted. The current position display function shows the position viewed from the workpiece coordinate system before the shift. The function to shift the coordinate system is enabled when bit 2 (PLS) of parameter No. 5450 is specified accordingly.
The shift can be specified in the polar coordinate interpolation mode, by specifying the position of the center of the rotary axis C (A, B) in the X-C (Y-A, Z-B) interpolation plane with reference to the origin of the workpiece coordinate system, in the following format.

```
G12.1 X_ C_ ;   (Polar coordinate interpolation for the X-axis and C-axis)
G12.1 Y_ A_ ;   (Polar coordinate interpolation for the Y-axis and A-axis)
G12.1 Z_ B_ ;   (Polar coordinate interpolation for the Z-axis and B-axis)
```



Limitation

- Changing the coordinate system during polar coordinate interpolation

In the G12.1 mode, the coordinate system must not be changed (G92, G52, G53, relative coordinate reset, G54 through G59, etc.).

- Tool radius/tool nose radius compensation

The polar coordinate interpolation mode (G12.1 or G13.1) cannot be started or terminated in the tool radius/tool nose radius compensation mode (G41 or G42). G12.1 or G13.1 must be specified in the tool radius/tool nose radius compensation canceled mode (G40).

- Tool length offset command

Tool length offset must be specified in the polar coordinate interpolation cancel mode before G12.1 is specified. It cannot be specified in the polar coordinate interpolation mode. Furthermore, no offset values can be changed in the polar coordinate interpolation mode.

- Tool offset command

A tool offset must be specified before the G12.1 mode is set. No offset can be changed in the G12.1 mode.

- Program restart

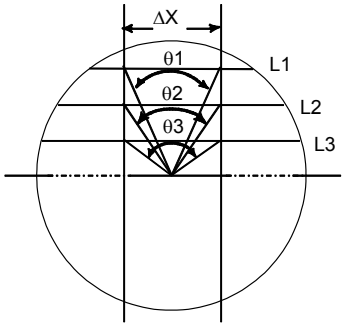
For a block in the G12.1 mode, the program and the block cannot be restarted.

- Cutting feedrate for the rotary axis

Polar coordinate interpolation converts the tool movement for a figure programmed in a Cartesian coordinate system to the tool movement in the rotary axis (C-axis) and the linear axis (X-axis). When the tool comes close to the center of the workpiece, the C-axis velocity component increases. If the maximum cutting feedrate for the C-axis (parameter No. 1430) is exceeded, the automatic feedrate override function and automatic speed clamp function are enabled.
If the maximum cutting feedrate for the X-axis is exceeded, the automatic feedrate override function and automatic speed clamp function are enabled.

⚠ WARNING

1 Consider lines L1, L2, and L3. ΔX is the distance the tool moves per time unit at the feedrate specified with address F in the Cartesian coordinate system. As the tool moves from L1 to L2 to L3, the angle at which the tool moves per time unit corresponding to ΔX in the Cartesian coordinate system increases from $\theta 1$ to $\theta 2$ to $\theta 3$. In other words, the C-axis component of the feedrate becomes larger as the tool moves closer to the center of the workpiece. The C component of the feedrate may exceed the maximum cutting feedrate for the C-axis because the tool movement in the Cartesian coordinate system has been converted to the tool movement for the C-axis and the X-axis.



L: Distance (in mm) between the tool center and workpiece center when the tool center is the nearest to the workpiece center
R: Maximum cutting feedrate (deg/min) of the C axis
Then, a speed specifiable with address F in polar coordinate interpolation can be given by the formula below. If the maximum cutting feedrate for the C-axis is exceeded, the automatic speed control function for polar coordinate interpolation automatically controls the feedrate.

$$F < L \times R \times \frac{\pi}{180} \quad \text{(mm/min)}$$

2 The following function cannot be used for the rotary axis of polar coordinate interpolation.
- Index table indexing function

- Automatic speed control for polar coordinate interpolation

If the velocity component of the rotary axis exceeds the maximum cutting feedrate in the polar coordinate interpolation mode, the speed is automatically controlled.

- Automatic override

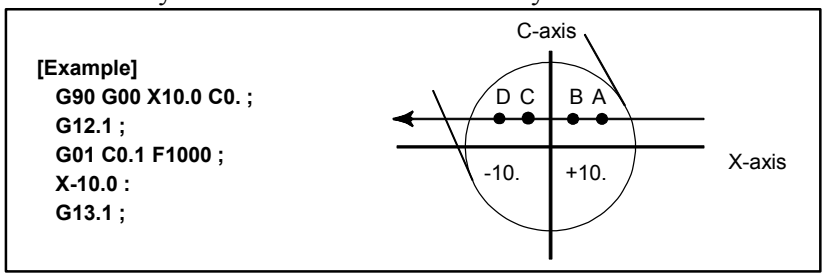
If the velocity component of the rotary axis exceeds the permissible velocity (maximum cutting feedrate multiplied by the permission factor specified in parameter No. 5463), the feedrate is automatically overridden as indicated below.

Override = (Permissible velocity) ÷ (Velocity component of rotary axis) × 100(%)

- Automatic speed clamp

If the velocity component of the rotary axis after automatic override still exceeds the maximum cutting feedrate, the speed of the rotary axis is automatically clamped. As a result, the velocity component of the rotary axis will not exceed the maximum cutting feedrate.

The automatic speed clamp function works only when the center of the tool is very close to the center of the rotary axis.



Automatic speed control for polar coordinate interpolation

Suppose that the maximum cutting feedrate of the rotary axis is 360 (3600 deg/min) and that the permission factor of automatic override for polar coordinate interpolation (parameter No. 5463) is 0 (90%). If the program indicated above is executed, the automatic override function starts working when the X coordinate becomes 2.273 (point A). The automatic speed clamp function starts working when the X coordinate becomes 0.524 (point B).

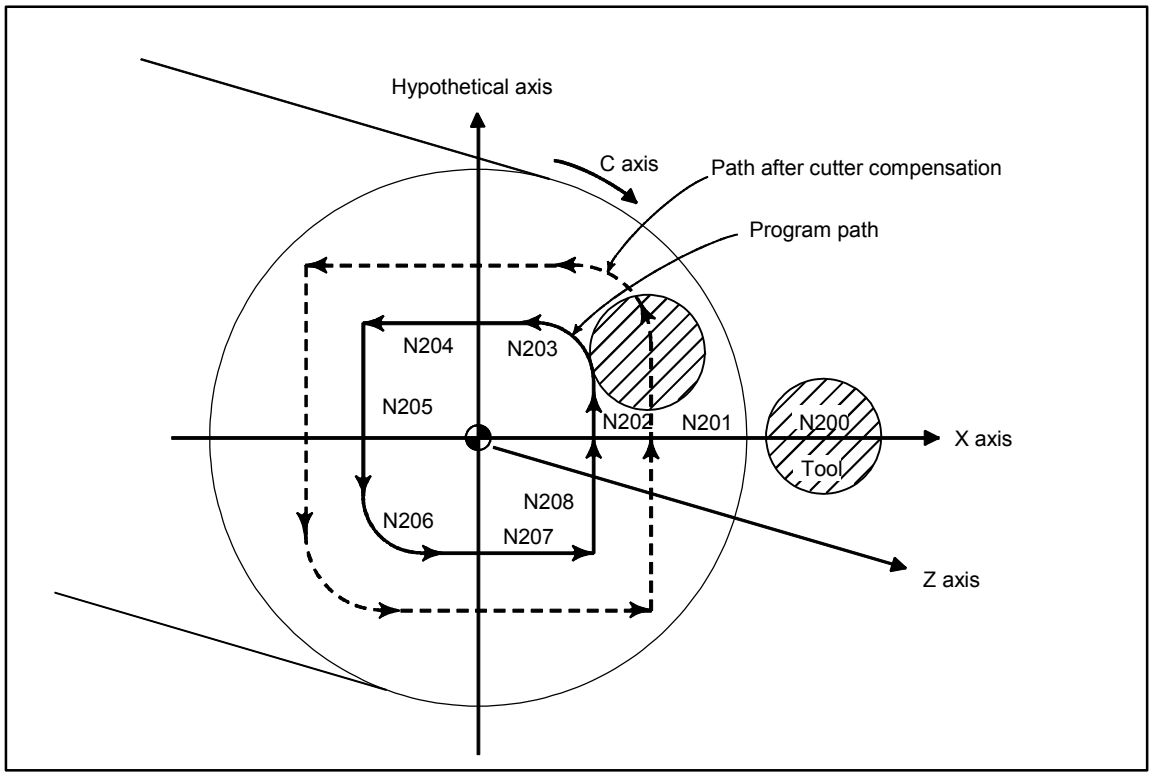
The minimum value of automatic override for this example is 3%. The automatic speed clamp function continues working until the X coordinate becomes -0.524 (point C). Then, the automatic override function works until the X coordinate becomes -2.273 (point D).

(The coordinates indicated above are the values in the Cartesian coordinate system.)

NOTE
1 While the automatic speed clamp function is working, the machine lock or interlock function may not be enabled immediately.
2 If a feed hold stop is made while the automatic speed clamp function is working, the automatic operation halt signal is output. However, the operation may not stop immediately.
3 The clamped speed may exceed the clamp value by a few percent.

Example

Sample program for polar coordinate interpolation in a Cartesian coordinate system consisting of the X-axis (a linear axis) and a hypothetical axis




```
O001;
   .
No10 T0101
   .
N0100 G90 G00 X60.0 C0 Z__;        Positioning to start point
N0200 G12.1;                       Start of polar coordinate interpolation
N0201 G42 G01 X20.0F__;
N0202 C10.0;
N0203 G03 X10.0 C20.0 R10.0;
N0204 G01 X-20.0;                  Geometry program
N0205 C-10.0;                      (program based on cartesian
N0206 G03 X-10.0 C-20.0 I10.0 J0;  coordinates on X
N0207 G01 X20.0;                   axis-hypothetical axis plane)
N0208 C0;
N0209 G40 X60.0;
N0210 G13.1;                       Cancellation of polar coordinate
N0300 Z__;                         interpolation
N0400 X__C__;
   .
N0900M30;
```

4.9 CYLINDRICAL INTERPOLATION (G07.1)

In cylindrical interpolation function, the amount of movement of a rotary axis specified by angle is converted to the amount of movement on the circumference to allow linear interpolation and circular interpolation with another axis.
Since programming is enabled with the cylinder side face expanded, programs such as a program for grooving cylindrical cams can be created very easily.

Format

G07.1 IP r; Starts the cylindrical interpolation mode (enables cylindrical interpolation).
:
:
:
G07.1 IP 0; The cylindrical interpolation mode is cancelled.

IP : An address for the rotary axis
r : The radius of the workpiece
Specify G07.1 IPr; and G07.1 IP0; in separate blocks.
G107 can be used instead of G07.1.

Explanation

- Plane selection (G17, G18, G19)

To specify a G code for plane selection, set the rotary axis in parameter No. 1022 as a linear axis that is one of the basic three axes of the basic coordinate system or an axis parallel to one of the basic axes. For example, when rotary axis C-axis is assumed to be parallel to the X-axis, specifying G17, axis address C, and Y at the same time can select a plane formed by the C-axis and Y-axis (the Xp-Yp plane).

T

NOTE
The U-, V-, and W-axes can be used with G-codes B and C.

- Feedrate

A feedrate specified in the cylindrical interpolation mode is the feedrate on the circumference.

- Circular interpolation (G02, G03)

Circular interpolation can be performed between the rotary axis set for cylindrical interpolation and another linear axis. Radius R is used in commands in the same way as described.
The unit for a radius is not degrees but millimeters (for metric input) or inches (for inch input).

<Example Circular interpolation between the Z axis and C axis>

For the C axis of parameter (No.1022), 5 (axis parallel with the X axis) is to be set. In this case, the command for circular interpolation is

G18 Z_C_;
G02 (G03) Z_C_R_;

For the C axis of parameter (No.1022), 6 (axis parallel with the Y axis) may be specified instead. In this case, however, the command for circular interpolation is

G19 C_Z_;
G02 (G03) Z_C_R_;

- Tool radius/tool nose radius compensation

To perform tool radius/tool nose radius compensation in the cylindrical interpolation mode, cancel any ongoing tool radius/tool nose radius compensation mode before entering the cylindrical interpolation mode. Then, start and terminate tool radius/tool nose radius compensation within the cylindrical interpolation mode.

- Cylindrical interpolation accuracy

In the cylindrical interpolation mode, the amount of travel of a rotary axis specified by an angle is once internally converted to a distance of a linear axis on the outer surface so that linear interpolation or circular interpolation can be performed with another axis. After interpolation, such a distance is converted back to an angle. For this conversion, the amount of travel is rounded to a least input increment.
So when the radius of a cylinder is small, the actual amount of travel can differ from a specified amount of travel. Note, however, that such an error is not accumulative.
If manual operation is performed in the cylindrical interpolation mode with manual absolute on, an error can occur for the reason described above.

$$\text{The actual amount of travel} = \left[\frac{\text{MOTION REV}}{2\times 2\pi\text{R}}\times\left[\text{Specified value}\times\frac{2\times 2\pi\text{R}}{\text{MOTION REV}}\right]\right]$$

MOTION REV : The amount of travel per rotation of the rotary axis (360°)
R : Workpiece radius
[ ] : Rounded to the least input increment

Limitation

- Arc radius specification in the circular interpolation

In the cylindrical interpolation mode, an arc radius cannot be specified with word address I, J, or K.

- Positioning

In the cylindrical interpolation mode, positioning operations (including those that produce rapid traverse cycles such as G28, G53, G73, G74, G76, G80 to G89) cannot be specified. Before positioning can be specified, the cylindrical interpolation mode must be cancelled. Cylindrical interpolation (G07.1) cannot be performed in the positioning mode (G00).

- Cylindrical interpolation mode setting

In the cylindrical interpolation mode, the cylindrical interpolation mode cannot be reset. The cylindrical interpolation mode must be cancelled before the cylindrical interpolation mode can be reset.

- Rotary axis

Only one rotary axis can be set for cylindrical interpolation. Therefore, it is impossible to specify more than one rotary axis in the G07.1 command.

- Rotary axis roll-over

If a rotary axis using the roll-over function is specified at the start of the cylindrical interpolation mode, the roll-over function is automatically disabled in the cylindrical interpolation mode. After the cylindrical interpolation mode is canceled, the roll-over function is enabled automatically.

- Rotary axis control function

If a rotary axis using the multiple rotary axis control function is specified at the start of the cylindrical interpolation mode, the rotary axis control function is automatically disabled in the cylindrical interpolation mode. After the cylindrical interpolation mode is canceled, the rotary axis control function is enabled automatically.

- Tool radius/tool nose radius compensation

If the cylindrical interpolation mode is specified when tool radius/tool nose radius compensation is already being applied, correct compensation is not performed. Specify compensation in the cylindrical interpolation mode.

- Canned cycle for drilling

Canned cycles (G73, G74, and G81 to G89 for M series / G80 to G89 for T series) for drilling, cannot be specified during cylindrical interpolation mode.

M

- Coordinate system setting

In the cylindrical interpolation mode, a workpiece coordinate system (G92, G54 to G59) or local coordinate system (G52) cannot be specified.

- Tool offset

A tool offset must be specified before the cylindrical interpolation mode is set. No offset can be changed in the cylindrical interpolation mode.

- Index table indexing function

Cylindrical interpolation cannot be specified when the index table indexing function is being used.

- Parallel axis

The rotary axis specified for cylindrical interpolation must not be a parallel axis.

T

- Coordinate system setting

In the cylindrical interpolation mode, a workpiece coordinate system G50 cannot be specified.

- Mirror image for double turret

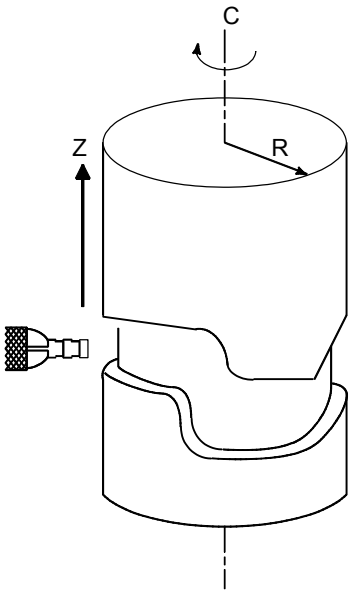
Mirror image for double turret, G68 and G69, cannot be specified during cylindrical interpolation mode.

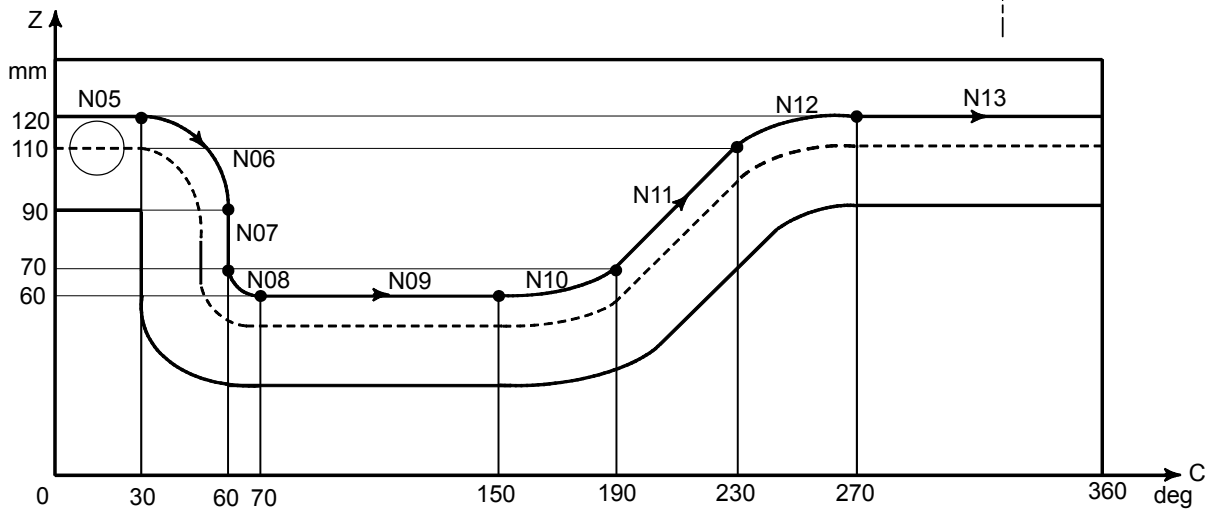
Example

Example of a Cylindrical Interpolation

```
O0001 (CYLINDRICAL INTERPOLATION );
N01 G00 G90 Z100.0 C0 ;
N02 G01 G91 G18 Z0 C0 ;
N03 G07.1 C57299 ;*
N04 G90 G01 G42 Z120.0 D01 F250. ;
N05 C30.0 ;
N06 G03 Z90.0 C60.0 R30.0 ;
N07 G01 Z70.0 ;
N08 G02 Z60.0 C70.0 R10.0 ;
N09 G01 C150.0 ;
N10 G02 Z70.0 C190.0 R75.0 ;
N11 G01 Z110.0 C230.0 ;
N12 G03 Z120.0 C270.0 R75.0 ;
N13 G01 C360.0 ;
N14 G40 Z100.0 ;
N15 G07.1 C0 ;
N16 M30 ;
```

(* A command with a decimal point can also be used.)





4.10 CUTTING POINT INTERPOLATION FOR CYLINDRICAL INTERPOLATION (G07.1)

The conventional cylindrical interpolation function controls the tool center so that the tool axis always moves along a specified path on the cylindrical surface, towards the rotary axis (cylindrical axis) of the workpiece. The cutting point interpolation for cylindrical interpolation function controls the tool so that the tangent of the tool and the cutting surface of a contour always passes the rotation center of the workpiece. This means that the cutting surface of the contour is always perpendicular to the cylinder. With this function, the figure on the cutting surface can always be kept constant regardless of the tool radius compensation value of the tool used.

Format

This command is specified in the same way as for the conventional cylindrical interpolation function.

G07.1 IPr ; Circular interpolation mode on start (enabling cylindrical interpolation)

:

G07.1 IP0 ; Circular interpolation mode cancel

IP : One rotary axis address

r : Cylinder radius of rotary axis

Specify each of G07.1 IPr; and G07.1 IP0; singly in a block.

G107 can also be specified.

Explanation

- Comparison with conventional circular interpolation

As shown in Fig. 4.10 (a), the tool is controlled in the offset axis (Y-axis) direction that is perpendicular to the tool center and workpiece rotation center.

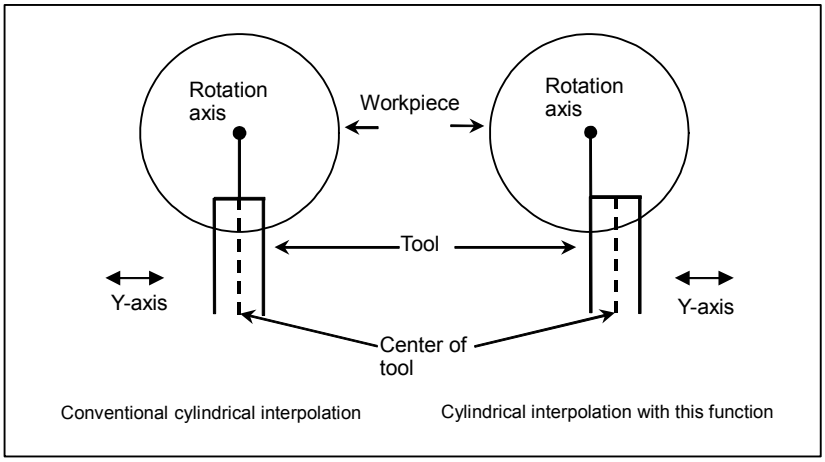


Fig. 4.10 (a) Comparison with conventional circular interpolation

- Cutting point compensation

(1) Cutting point compensation between blocks
As shown in Fig. 4.10 (b), cutting point compensation is achieved by moving between blocks N1 and N2.
(a) Let C1 and C2 be the heads of the vectors normal to N1 and N2 from S1, which is the intersection of the tool center paths of blocks N1 and N2
(b) After the tool moves to S1 according to the command of N1, the tool moves through V on the C-axis as a result of cutting point compensation, then through $-V \times \frac{\pi}{180} \times r$ along the Y-axis.

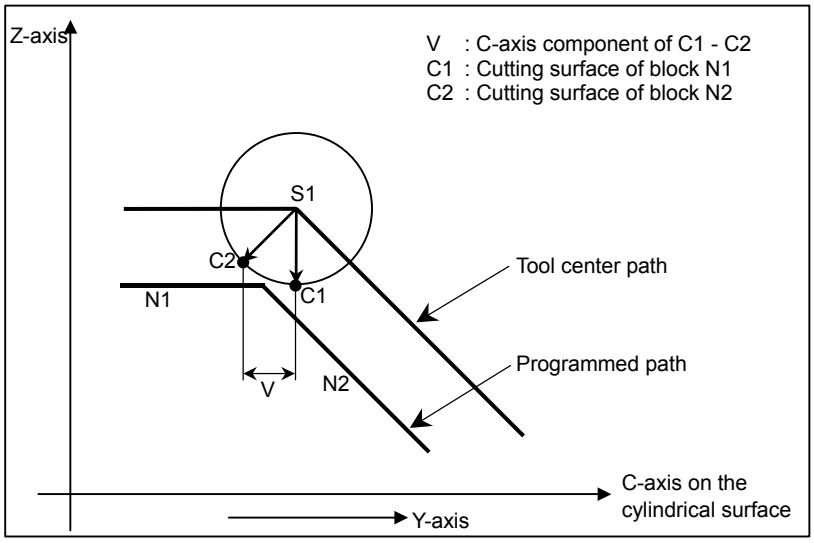


Fig.4.10 (b) Cutting point compensation between blocks

(2) Cutting point compensation in a circular command block
As shown in Fig. 4.10 (c), the movement required for cutting point compensation is made simultaneously with circular interpolation in block N1.
(a) Let C0 be the head of the vector normal to N1 from S0, which is the tool center position at the start point of circular block N1. Let C1 be the head of the similar vector at the end point.
(b) As the tool moves from S0 to S1, a superimposed movement is made by the C-axis component of (C1 - C2) (V in the figure) on the C-axis, and a superimposed movement is made by $-V \times \frac{\pi}{180} \times r$ along the Y-axis. along the Y-axis.
That is, the following expressions are valid. As movement is made through L as shown in Fig. 4.10 (c), the superimposed movements are made on the C-axis and Y-axis as follows:

$$\Delta C = \Delta V$$

$$\Delta Y = -\frac{\pi}{180}(\Delta V)r$$

ΔV : Cutting point compensation value (ΔV2 - ΔV1) for movement of ΔL
ΔV1 : C-axis component of the vector normal to N1 from the tool center of the start point of ΔL
ΔV2 : C-axis component of the vector normal to N1 from the tool center of the end point of ΔL
R : Arc radius

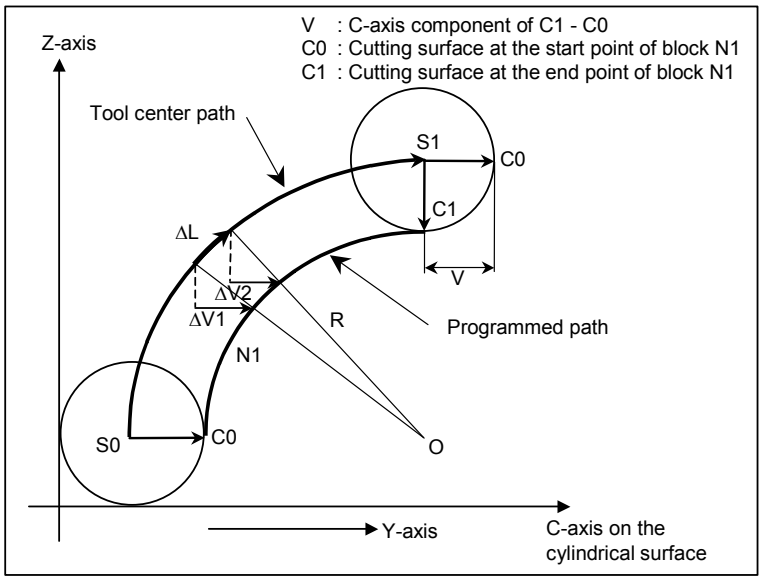


Fig.4.10 (c) Cutting point compensation in a circular command block

(3) When cutting point compensation is not applied between blocks
When, as shown in Fig. 4.10 (d) and Fig. 4.10 (e), the cutting point compensation value (V in the figures) is less than the value set in parameter No. 19534, one of the operations below is performed. (The operation that is performed depends on the setting of bit 6 (CYS) of parameter No.19530.

(a) When bit 6 (CYS) of parameter No.19530 is set to 1
Cutting point compensation is not applied between blocks N1 and N2, but is applied when block N2 is executed.

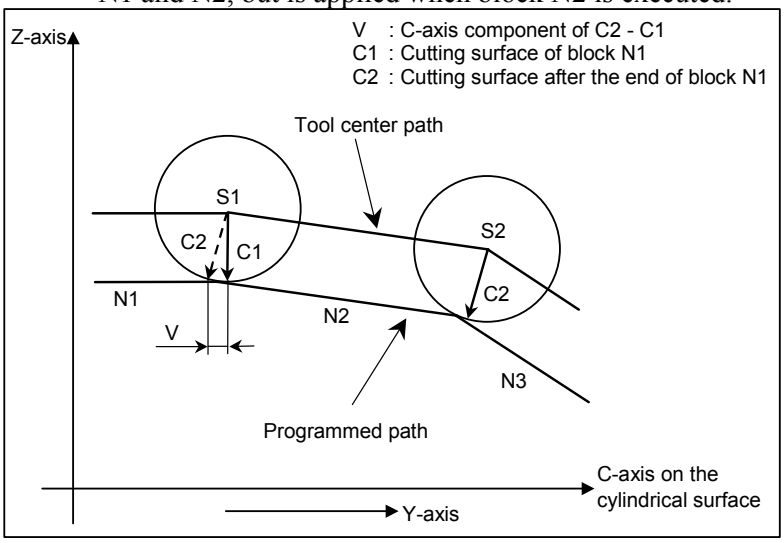


Fig.4.10 (d) When bit 6 (CYS) of parameter No.19530 is set to 1

(b) When bit 6 (CYS) of parameter No.19530 is set to 0
Cutting point compensation is not performed between blocks N1 and N2. Whether to apply cutting point compensation between block N2 and N3 is determined by taking the cutting point compensation value between blocks N2 and N3 (V in the figure) into consideration.

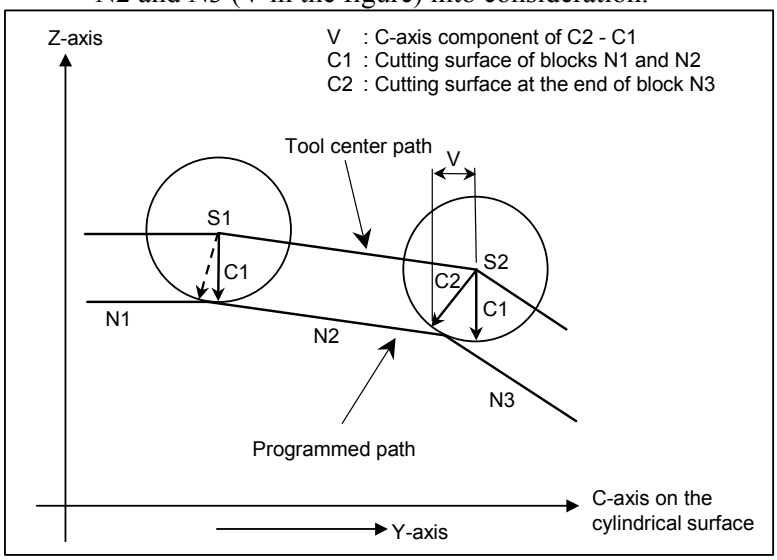


Fig.4.10 (e) When bit 6 (CYS) of parameter No.19530 is 0

(b) When the amount of travel (L1) of block N2 is less than the value set in parameter No. 19535, as shown in Fig. 4.10 (f), cutting point compensation is not applied between blocks N1 and N2. Instead, block N2 is executed with the cutting point compensation of the previous block. When the amount of travel (L2) of block N3 is greater than the value set in parameter No. 19535, cutting point compensation is applied between blocks N2 and N3.

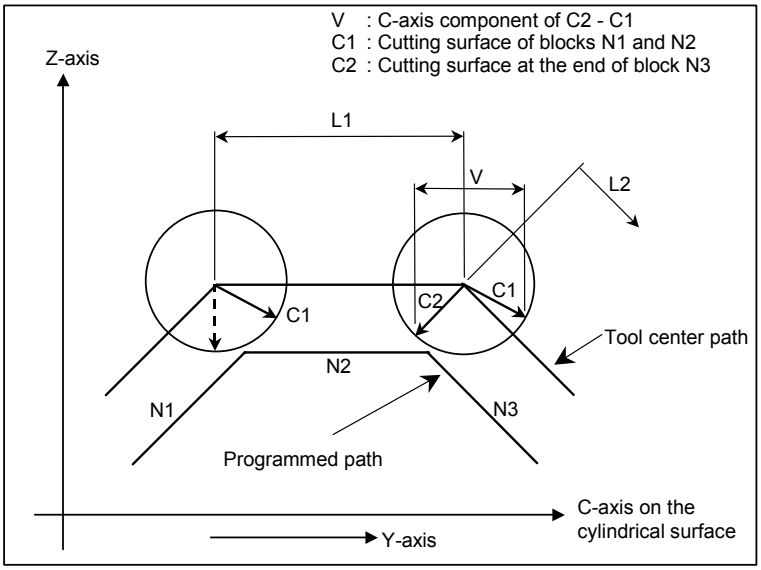


Fig.4.10 (f) When the amount of travel (L1) of block N2 is less than the parameter value

(d) When, as shown in Fig. 4.10 (g), the diameter of an arc (R in the figure) is less than the value set in parameter No. 19535, cutting point compensation is not applied simultaneously with circular interpolation

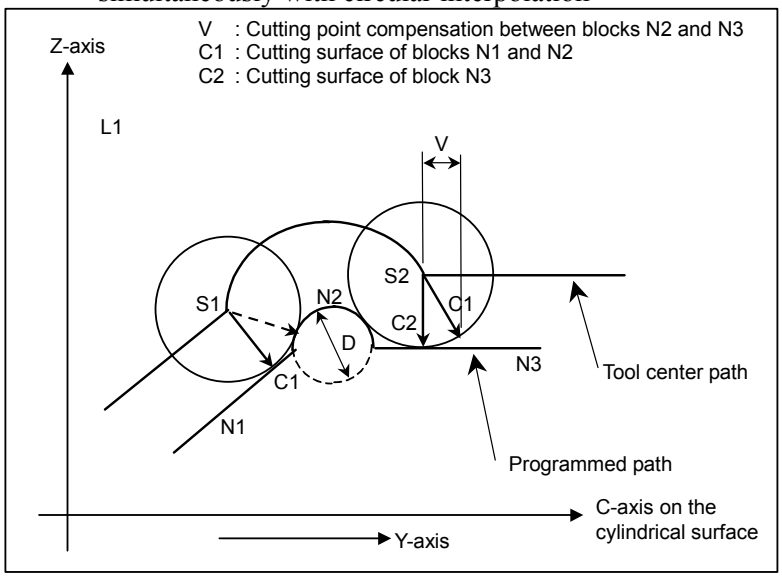


Fig.4.10 (g) When the diameter of an arc is less than the parameter value

- When cutting point compensation is used with normal direction control

When cutting point compensation is used with normal direction control, cutting point compensation between specified blocks is performed regardless of the method described in "**Cutting point compensation**" above, but is performed simultaneously with the movement of the normal direction control axis (C-axis).

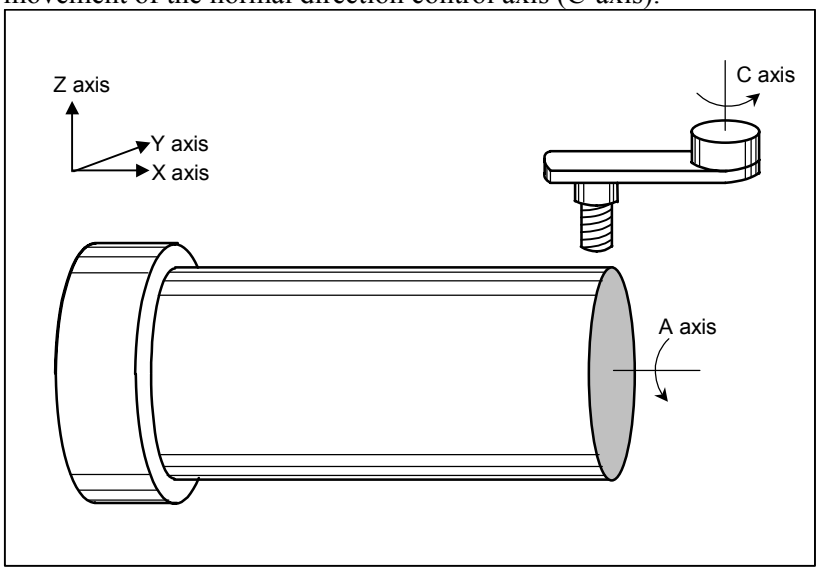


Fig. 4.10 (h) When used with normal direction control

(1) When the normal direction changes between blocks N1 and N2, cutting point compensation is also performed between blocks N1 and N2.
As shown in Fig. 4.10 (i), cutting point compensation described in (1) in "**Cutting point compensation**" is performed simultaneously with the movement made by normal direction control between blocks N1 and N2.

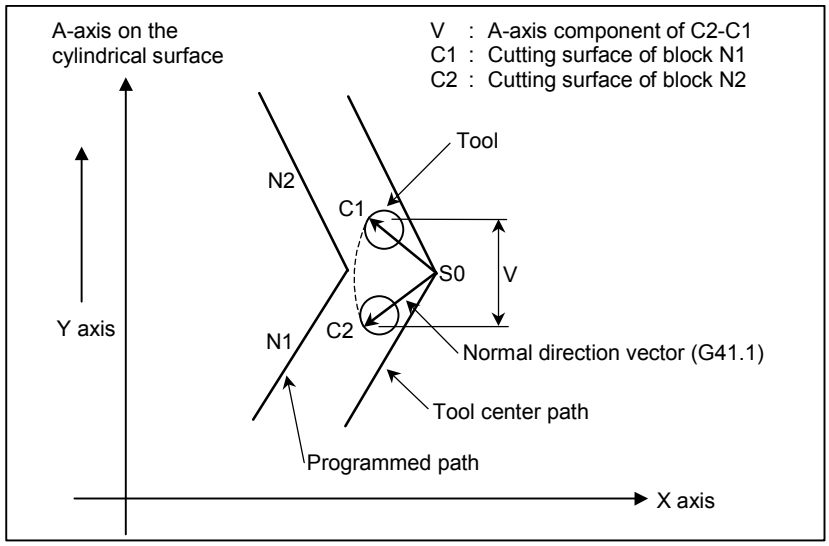


Fig. 4.10 (i) When the normal direction changes between blocks N1 and N2

(2) When gentle normal direction control changes the normal direction as a specified block is executed, cutting point compensation is performed simultaneously with the movement made by the block. When the normal direction control axis rotates by θ1 simultaneously with the movement made by blocks N1 to N2 as shown in Fig. 4.10 (j), cutting point compensation by the movement of vector V1 is also performed simultaneously with the movement in N2.

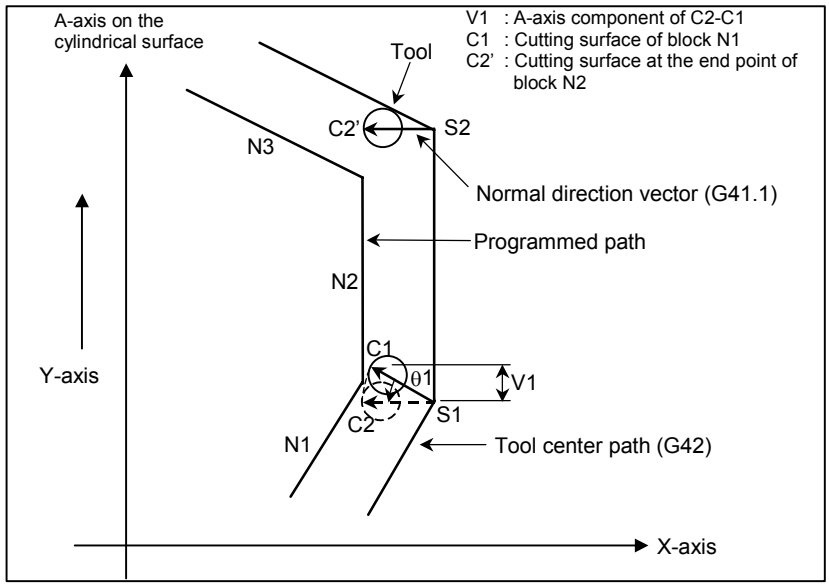


Fig. 4.10 (j) Gentle normal direction control

(3) When a specified block is executed while the normal direction control axis is held in the normal direction set at the end point of the previous block, cutting point compensation is not performed, and cutting point compensation applied in the previous block is maintained. As shown in Fig. 4.10 (k), when the movement amount in N2 (L1 in the figure) is smaller than the setting of parameter No. 5483, so the normal direction control axis does not rotate at S1 while the movement amount in N3 (L2 in the figure) is greater than the setting of parameter No. 5483, so the normal direction axis rotates at S2, cutting point compensation is not performed at S1 but is performed by the movement for vector V2 at S2.

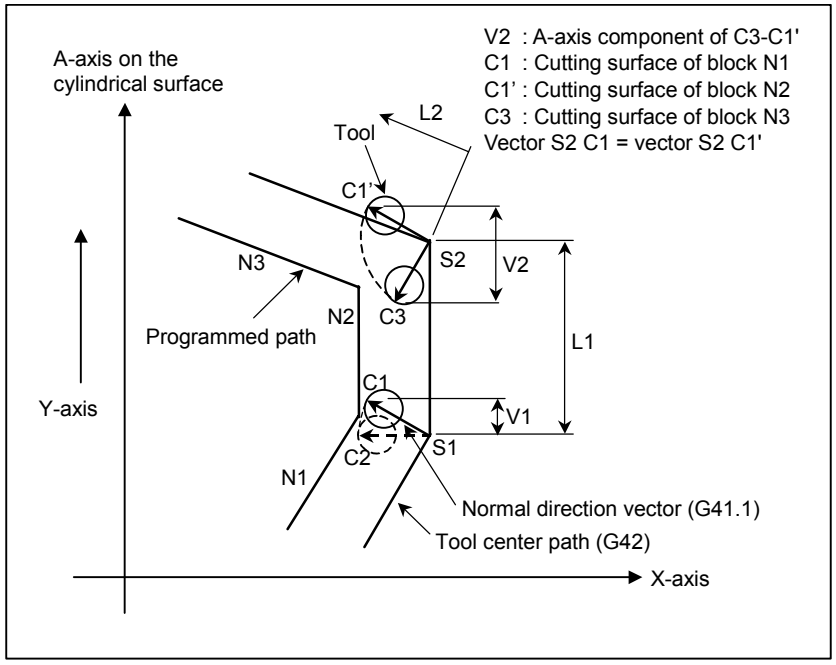


Fig. 4.10 (k) When the normal direction is the same as in the previous block

- Feedrate during cutting point compensation

(1) The tool moves at a specified feedrate while cutting point compensation is being applied between blocks.

(2) The actual speed indication and feedrate during circular interpolation are as described below.

Actual speed indication

The speed component of each axis after cutting point compensation at a point in time during circular interpolation is as follows:

$Fz' = Fz$: Speed component of linear axis

$Fc' = Fc + (Vce - Vcs)$: Speed component of rotary axis

$Fy' = -(Vce - Vcs)\frac{\pi r}{180}$: Speed component of offset axis

Fz : Speed component of a cylindrical interpolation linear axis before cutting point compensation

Fc : Speed component of cylindrical interpolation rotation axis before cutting point compensation
Vcs: Rotation axis component of a tool contact point vector (Vs in the figure) at the start point at a point in time
Vce: Rotation axis component of tool contact point vector (Ve in the figure) at the end point at a point in time
r : Radius of the cylinder of a rotary axis
Accordingly, the actual speed indication during circular interpolation is greater than the specified value when |Fc'| > |Fc|(inner offset of the arc). Conversely, the actual speed indication during circular interpolation is less than the specified value when |Fc'| < |Fc|(outer offset of the arc).

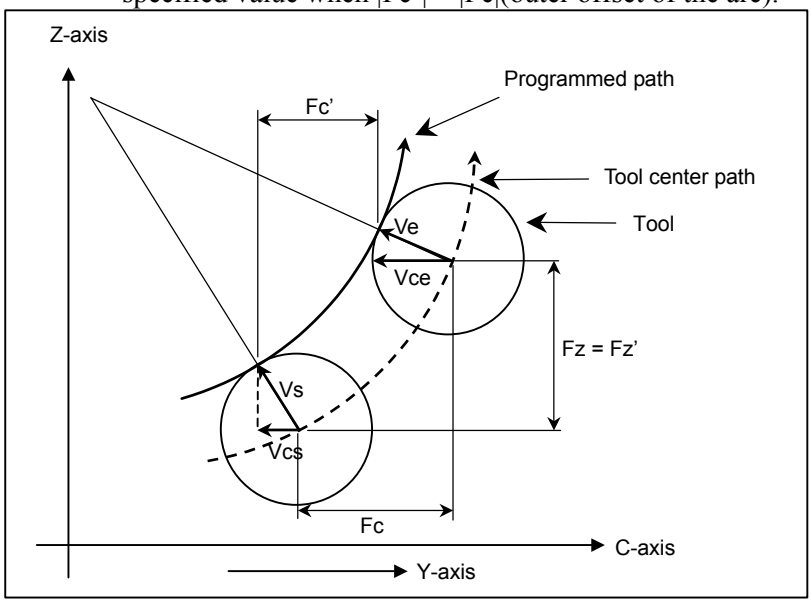


Fig.4.10 (l) Actual speed indication during circular interpolation

- Usable G codes

(1) In any of the following G code modes, cutting point interpolation for cylindrical interpolation can be specified:
G01, G02, G03: Linear interpolation, circular interpolation
G17, G18, G19: Plane selection
G22 : Stored stroke check function on
G64 : Cutting mode
G90, G91 : Absolute programming, incremental programming
G94 : Feed per minute

(2) Any of the following G codes can be specified in cutting point interpolation for cylindrical interpolation mode:
G01, G02, G03 : Linear interpolation, circular interpolation
G04 : Dwell
G40, G41, G42 : Tool radius compensation
G40.1 to G42.1 : Normal direction control
G64 : Cutting mode
G65 to G67 : Macro call
G90, G91 : Absolute programming, incremental programming

- Parameter

To enable this function, set bit 5 (CYA) of parameter No. 19530 to 1.

Limitation

- Overcutting during inner corner cutting

Theoretically, when the inner area of a corner is cut using linear interpolation as shown in Fig. 4.10 (m), this function slightly overcuts the inner walls of the corner. This overcutting can be avoided by specifying a value of R that is slightly greater than the radius of the tool at the corner.

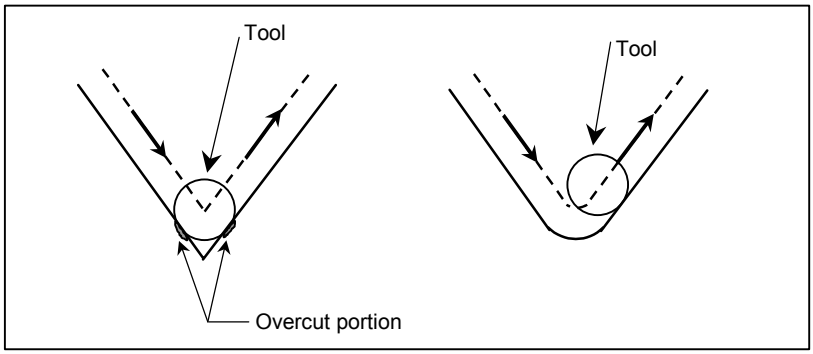


Fig.4.10 (m) Overcutting

- Setting the minimum input increment for an offset axis (Y-axis)

Set the same minimum input increment for an offset axis and linear axis when cylindrical interpolation is performed.

- Workpiece radius specification

When specifying the radius of a workpiece, use the minimum input increment (with no decimal point) for the linear axis used in cylindrical interpolation.

- Reference axis setting (parameter No. 1031)

When different increment systems are used for a linear axis and a rotary axis in cylindrical interpolation, specify the number of the linear axis in cylindrical interpolation to set the axis as the reference axis.

- Rotary axis roll-over function

When a rotary axis for which the rollover function is used is specified as a rotary axis to be used for cylindrical interpolation, the roll-over function is disabled in the cylindrical interpolation mode.
When the cylindrical interpolation is canceled, the rollover function is automatically enabled.

- Program restart

In a program restart operation, G07.1 for the cylindrical interpolation mode must not be specified.
Otherwise, PS0175 is issued.

Example

- Example of cutting point interpolation for cylindrical interpolation

The sample program below indicates the positional relationships between a workpiece and tool.

```
O0001 (CYLINDRICAL INTERPOLATION1) ;
N01 G00 G90 Z100.0 C0 ;
N02 G01 G91 G19 Z0 C0 ;
N03 G07.1 C57299 ;
N04 G01 G42 G90 Z120.0 D01 F250. ; ..... (1)
N05 C20.0 ; ..... (2)
N06 G02 Z110.0 C60.0 R10.0 ; ..... (3)
N07 G01 Z100.0 ; ..... (4)
N08 G03 Z60.0 C70.0 R40.0 ; ..... (5)
N09 G01 C100.0 ;
   :
M30 ;
```

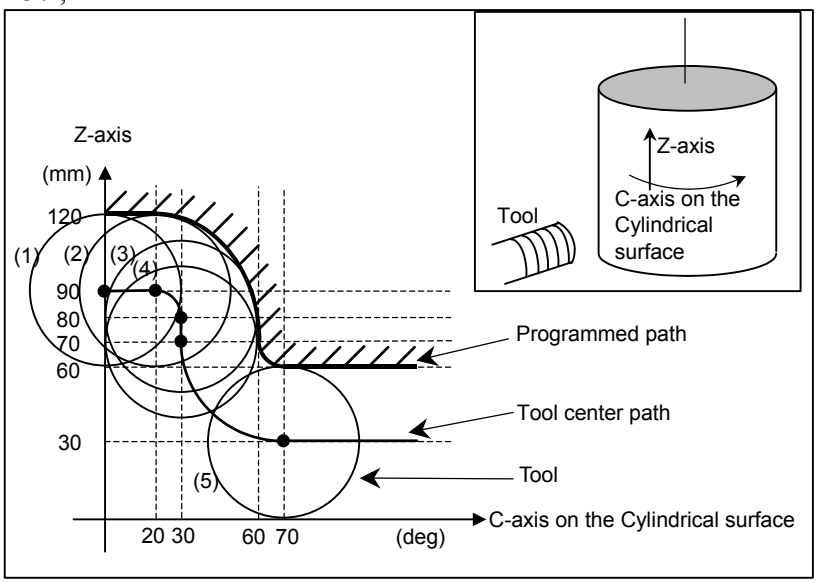


Fig. 4.10 (n) Path of sample program for cutting point interpolation for cylindrical interpolation

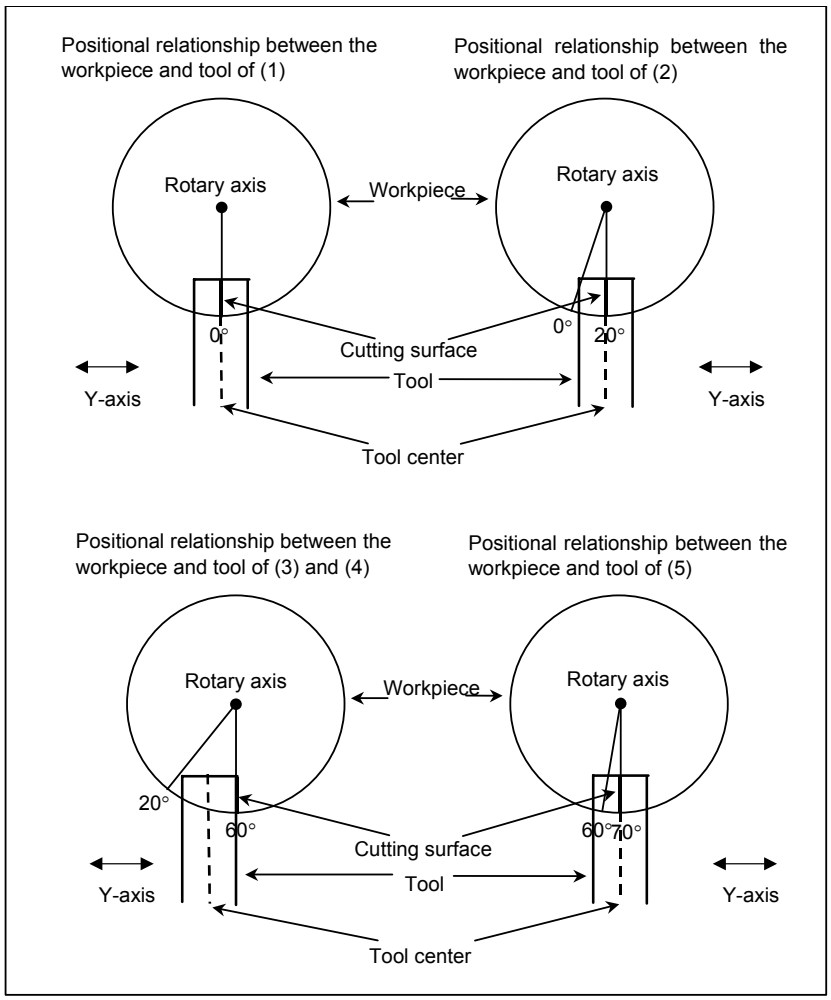


Fig. 4.10 (o) Positional relationships between workpiece and tool of sample program

The cutting surface in the rotary axis direction in (3) and (4) are uniform even if the tool radius compensation amount is modified.

- Example of specifying cutting point interpolation for cylindrical interpolation and normal direction control at the same time

Tool radius compensation No.01 is 30 mm.

```
O0002 (CYLINDRICAL INTERPOLATION2) ;
N01 G00 G90 X100.0 A0 ;
N02 G01 G91 G17 X0 A0 ;
N03 G07.1 C57299 ;
N04 G01 G41 G42.1 G90 X120.0 D01 F250. ;
N05 A20.0 ;
N06 G03 X80.0 A60.0 R40.0 ;
N07 G01 X70.0 ;
N08 G02 X70.0 A70.0 R10.0 ;
N09 G01 A150.0 ;
N10 G02 X70.0 A190.0 R85.0 ;
N11 G01 X110.0 A265.0 ;
N12 G03 X120.0 A305.0 R85.0 ;
N13 G01 A360.0 ;
N14 G40 G40.1 X100.0 ;
N15 G07.1 A0 ;
N16 M30 ;
```

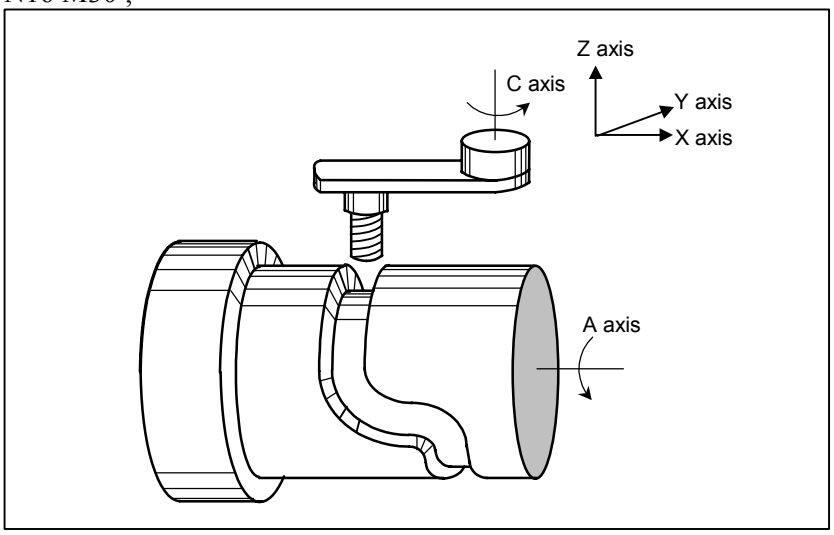


Fig. 4.10 (p) Example of specifying normal direction control at the same time

4.11 EXPONENTIAL INTERPOLATION (G02.3, G03.3)

Exponential interpolation exponentially changes the rotation of a workpiece with respect to movement on the rotary axis. Furthermore, exponential interpolation performs linear interpolation with respect to another axis. This enables tapered groove machining with a constant helix angle (constant helix taper machining). This function is best suited for grooving and grinding tools such as taper end mills.

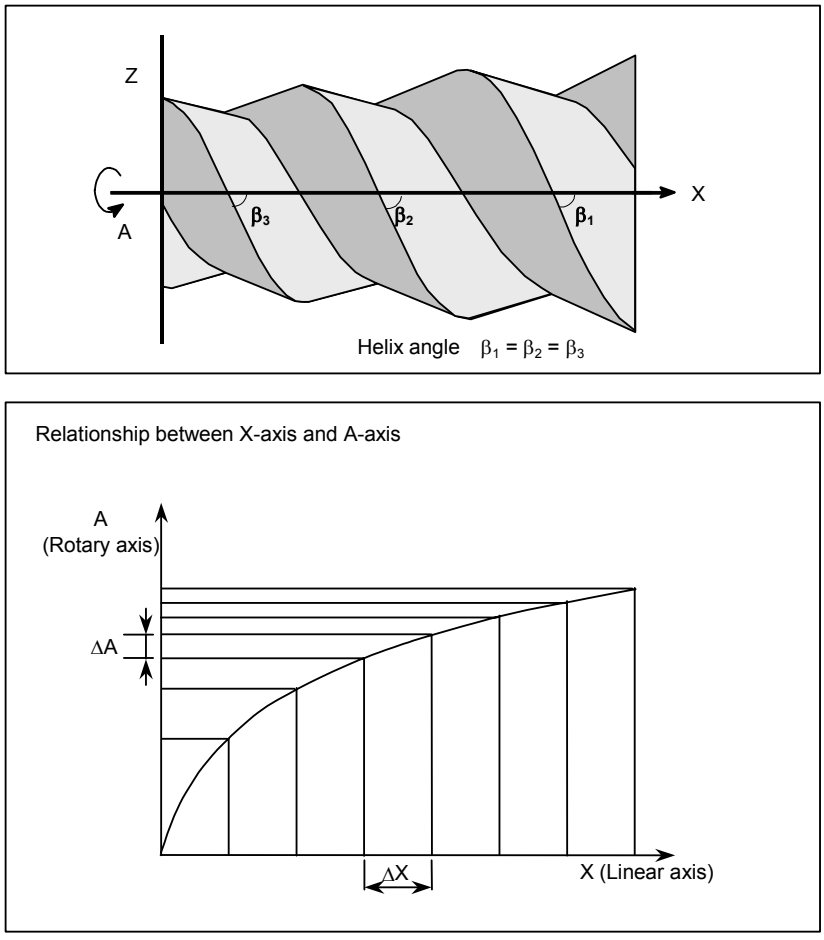


Fig. 4.11 (a) Exponential interpolation

Format

Positive rotation (ω = 0)

G02. 3 X_ Y_ Z_ I_ J_ K_ R_ F_ Q_ ;

Negative rotation (ω = 1)

G03. 3 X_ Y_ Z_ I_ J_ K_ R_ F_ Q_ ;

X_ : Specifies an end point with an absolute or incremental value.
Y_ : Specifies an end point with an absolute or incremental value.
Z_ : Specifies an end point with an absolute or incremental value.
I_ : Specifies angle I (from ±1 to ±89 deg in units of 0.001 deg).
J_ : Specifies angle J (from ±1 to ±89 deg in units of 0.001 deg).
K_ : Specifies the amount to divide the linear axis for exponential interpolation (span value). The specification unit depends on the reference axis. Specify a positive value.
The span value is specified in the manner specified in bit 0 (SPN) of parameter No. 5630. If SPN is set to 0, the division amount is specified in parameter No. 5643. If SPN is set to 1, the value specified with K becomes valid.
R_ : Specifies constant R for exponential interpolation. (See Explanations below.)
F_ : Specifies the initial feedrate.
Specified in the same way as an ordinary F code. Specify a composite feedrate including a feedrate on the rotary axis.
Q_ : Specifies the feedrate at the end point.
The same unit used for F is used. The CNC internally performs interpolation between the initial feedrate (F) and final feedrate (Q), depending on the travel distance on the linear axis.

Explanation

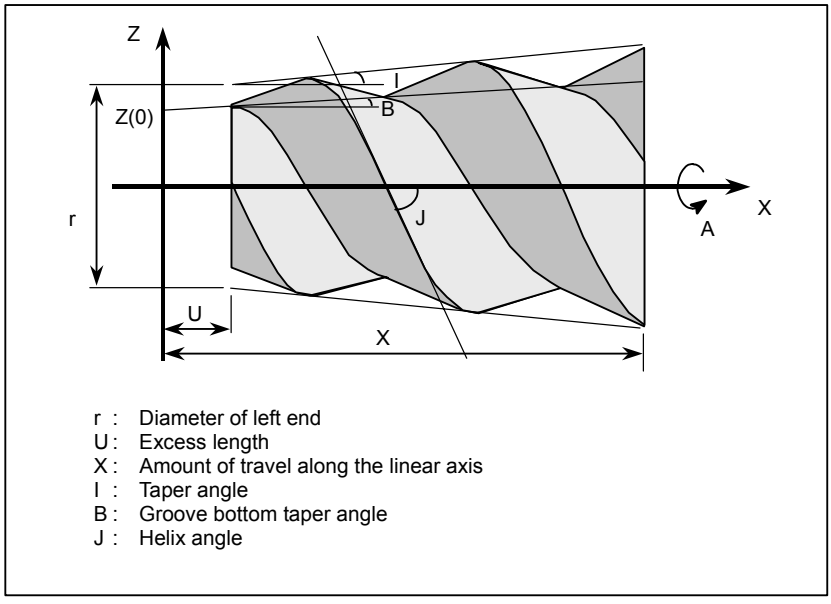


Fig. 4.11 (b) Constant helix machining for producing a tapered figure

In Fig. 4.11 (b) an absolute value on the X-axis, Z-axis, or A-axis is expressed as a function of workpiece rotation angle θ, such as $X(\theta)$, $Z(\theta)$, and $A(\theta)$.
Linear interpolation with the X-axis is performed for an axis other than the X-axis or A-axis. When $X(\theta) = 0$, $A(\theta) = 0$.

The relationship is expressed as follows.

$$Z(\theta) = \left\{ \frac{r}{2} - U * \tan(I) \right\} * (e^{\frac{\theta}{K}} - 1) * \frac{\tan(B)}{\tan(I)} + Z(0) \quad \cdots\cdots (1)$$

$$X(\theta) = \left\{ \frac{r}{2} - U * \tan(I) \right\} * (e^{\frac{\theta}{K}} - 1) * \frac{1}{\tan(I)} \quad \cdots\cdots (2)$$

$$A(\theta) = (-1)^{\omega} * \frac{360}{2\pi} * \theta$$

Where

$$K = \frac{\tan J}{\tan I}$$

ω: Helix direction (0: Positive, 1: Negative)

From Expressions (1) and (2), the following is obtained ;

$$Z(\theta) = \tan(B) * X(\theta) + Z(0) \quad \cdots\cdots (3)$$

From Expression (3), the Z-axis position is determined from a groove bottom taper angle (B) and X-axis position.
From Expression (1) and exponential definition expression (described later), the following is determined:

$$R = r/2 - U * \tan(I) \quad \cdots\cdots (4)$$

Constant R is determined from the left end diameter (r), excess length (U), and taper angle (I) according to Expression (4). Specify a taper angle (I) in address I, and specify a helix angle (J) in address J. Select a helix direction with G02.3 or G03.3.

- Exponential definition expressions

Exponential relational expressions for a linear axis and rotary axis are defined as follows:

$$X(\theta) = R * (e^{\frac{\theta}{K}} - 1) * \frac{1}{\tan(I)} \quad \cdots\cdots (5)$$

$$A(\theta) = (-1)^{\omega} * 360 * \frac{\theta}{2\pi} \quad \cdots\cdots (6)$$

Where,

$$K = \frac{\tan(J)}{\tan(I)}$$

$$\omega = 0/1$$

R, I, and J are constants, and θ represents an angle (radian).

- Span value K

A movement on an axis is carried out as linear interpolation in units of values obtained by dividing the movement on the X-axis by the span value (address K).
The following is obtained from Expression (5)

$$\theta(X) = K * \ln(\frac{X * \tan(I)}{R} + 1) \quad \text{..... (7)}$$

When there is movement from X_1 to X_2 on the linear axis, the amount of movement on the rotary axis is determined by:

$$\Delta\theta = K * \{\ln(\frac{X_2 * \tan(I)}{R} + 1) - \ln(\frac{X_1 * \tan(I)}{R} + 1)\}$$

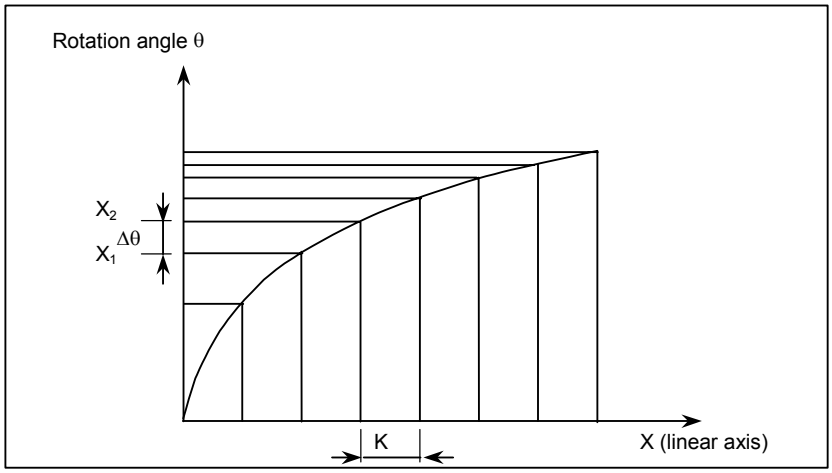


Fig. 4.11 (c) Span value K

- Rotation axis θ

In exponential interpolation, Expression (7) indicates the relationship between the X coordinate and the rotation angle θ about the A-axis. The expression in the parentheses of the natural logarithm ln in Expression (7) must satisfy Expression (8) indicated below, because of the condition of ln (the value in the parentheses is positive).

$$\frac{X * \tan(I)}{R} > -1 \quad \text{..... (8)}$$

If the value of X*tan(I)/R becomes -1 or less, the position comes to the right of point (A) in Fig. 4.11 (d). Because the resultant figure is unfeasible, an alarm PS5062 will be issued.

In exponential interpolation, the X, Y, Z, and U values are handled as coordinates in the workpiece coordinate system. If a positive value specified in incremental programming corresponds to a negative value in the workpiece coordinate system, the negative value is used in the calculation.

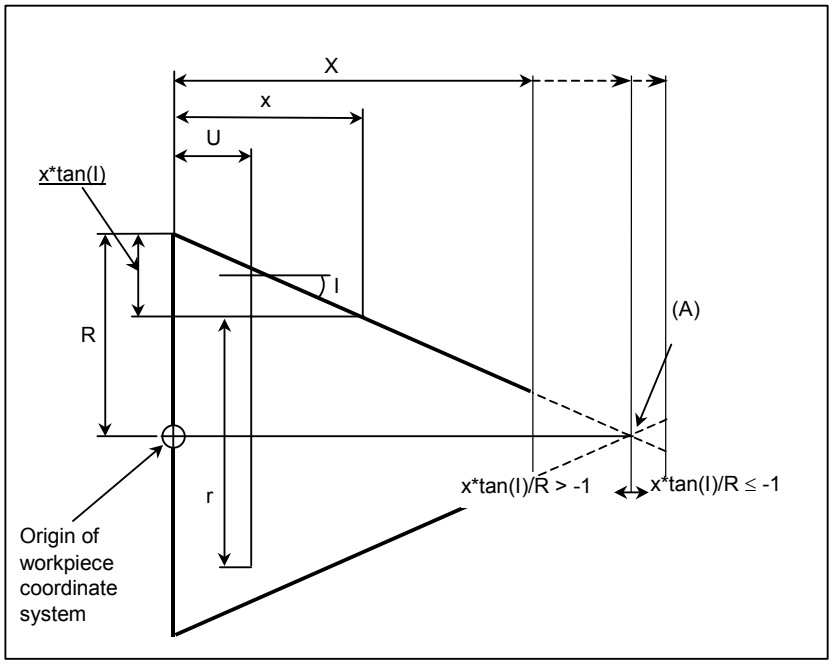


Fig. 4.11 (d) Rotation angle θ

- Taper angle I

The machining profile and the sign of taper angle I have the following relationships:

- If the profile tapers up toward the right, the I value is positive.
- If the profile tapers down toward the right, the I value is negative.

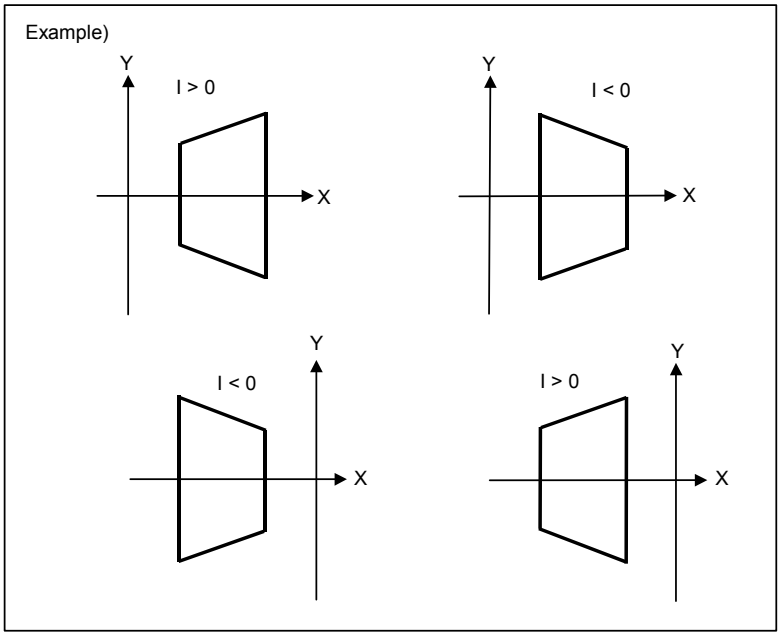


Fig. 4.11 (e) Taper angle I

- Helix angle J

The sign of the helix angle J is assigned as illustrated below.

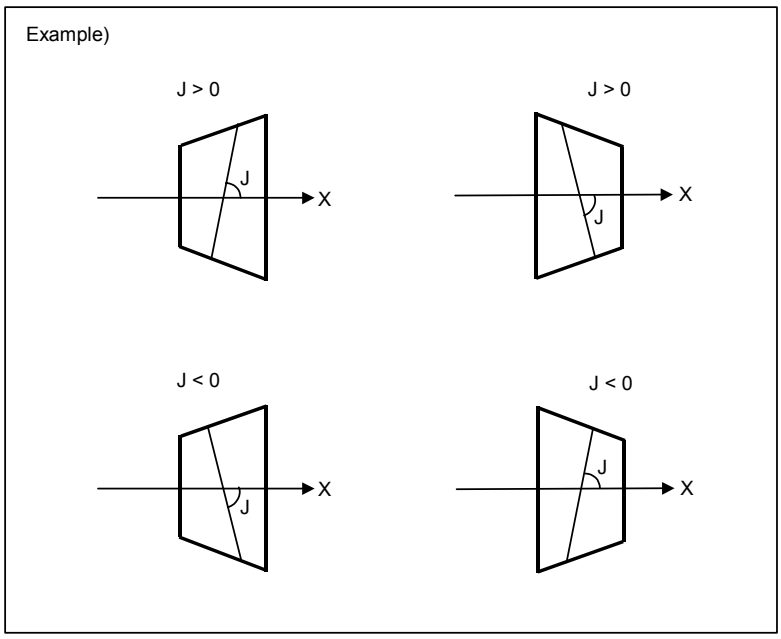
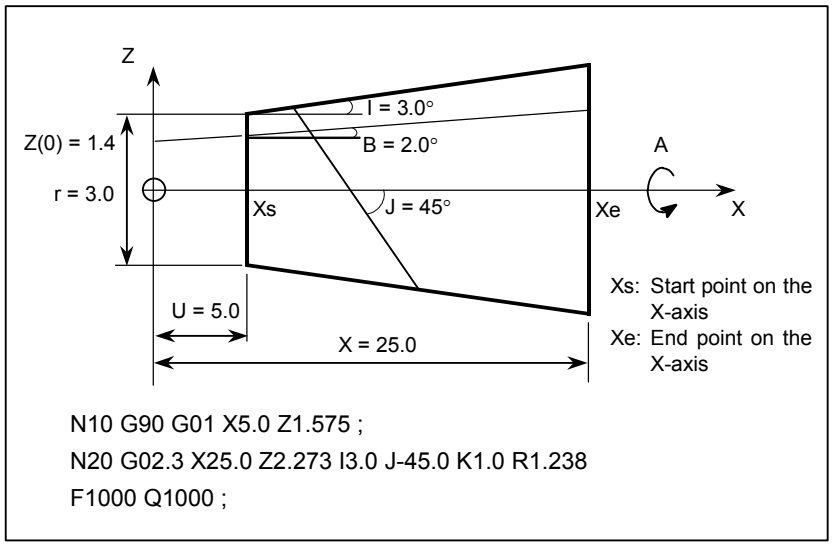


Fig. 4.11 (f) Helix angle J

Example




```
N10 G90 G01 X5.0 Z1.575 ;
N20 G02.3 X25.0 Z2.273 I3.0 J-45.0 K1.0 R1.238
F1000 Q1000 ;
```

The start point and end point of the Z-axis and the constant R are obtained from the following expressions:

Start point on the Z-axis $= \tan(B) * Xs + Z(0)$

End point on the Z-axis $= \tan(B) * Xe + Z(0)$

$R = r/2 - U * \tan(I)$

Limitation

- Cases where linear interpolation is performed

Even when the G02.3 or G03.3 mode is set, linear interpolation is performed in the following cases:

- When the linear axis specified in parameter No. 5641 is not specified, or the amount of movement on the linear axis is 0
- When the rotary axis specified in parameter No. 5642 is specified
- When the amount for dividing the linear axis (span value) is 0

- Tool compensation

The tool compensation functions (tool length compensation, tool radius/tool nose radius compensation, and 3-dimensional cutter compensation) cannot be used in the G02.3 or G03.3 mode.

> ⚠ **CAUTION**
> The amount for dividing the linear axis for exponential interpolation (span value) affects figure precision. However, if an excessively small value is set, the machine may stop during interpolation. Try to specify an optimal span value depending on the machine being used.

4.12 SMOOTH INTERPOLATION (G05.1)

Either of two types of machining can be selected, depending on the program command.

- For those portions where the accuracy of the figure is critical, such as at corners, machining is performed exactly as specified by the program command.
- For those portions having a large radius of curvature where a smooth figure must be created, points along the machining path are interpolated with a smooth curve, calculated from the polygonal lines specified with the program command (smooth interpolation).

In this manner, high-speed, high-precision machining can be performed.

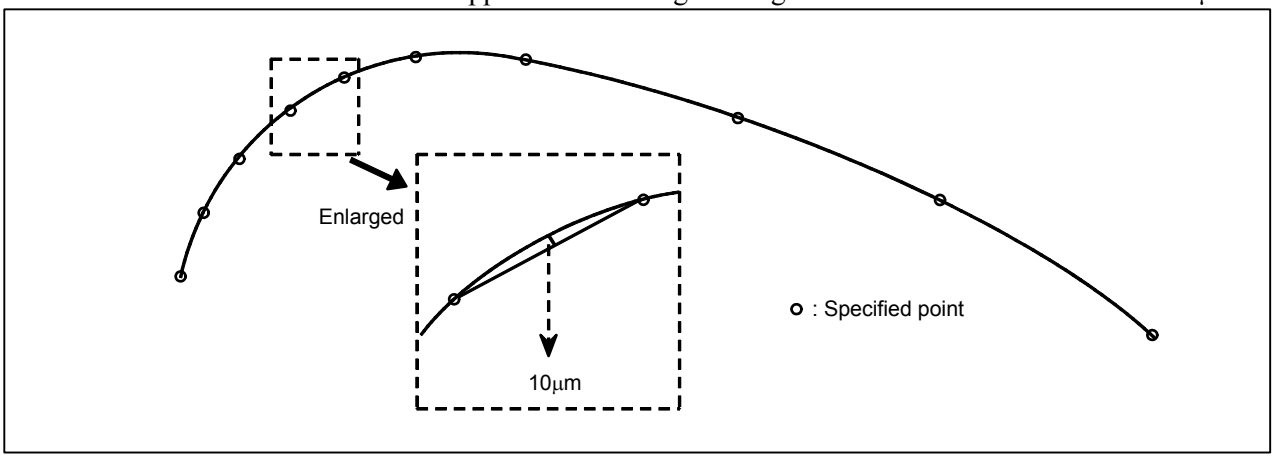
Format

```
G05.1Q2X0Y0Z0 ; Starting of smooth interpolation mode
    :
G05.1Q0 ; Cancellation of smooth interpolation mode
```

Explanation

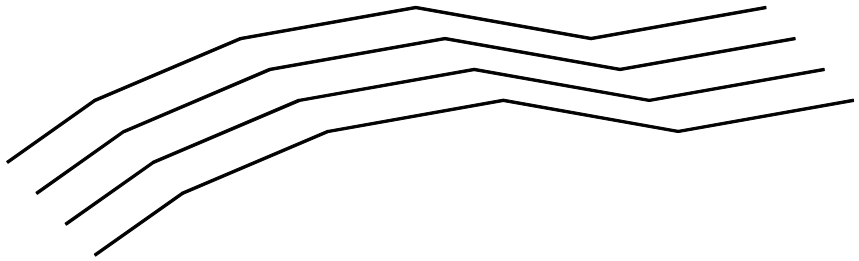
- Characteristics of smooth interpolation

To machine a part having sculptured surfaces, such as metal moldings used in automobiles and airplanes, a part program usually approximates the sculptured surfaces with minute line segments. As shown in the following figure, a sculptured curve is normally approximated using line segments with a tolerance of about 10 μm.



When a program approximates a sculptured curve with line segments, the length of each segment differs between those portions that have mainly a small radius of curvature and those that have mainly a large radius of curvature. The length of the line segments is short in those portions having a small radius of curvature, while it is long in those portions having a large radius of curvature. Because linear interpolation controls the tool movement exactly as programmed so as not to let the tool depart from the path specified by a part program, machining is performed exactly along line segments with which a sculptured curve is approximated. Consequently, when a curve with a large radius of curvature and a moderate curvature change is machined, the corners of line segments may become apparent. Such concave and convex portions, produced as a result of machining performed exactly as specified, are found troublesome when a smooth surface must be obtained by finishing.

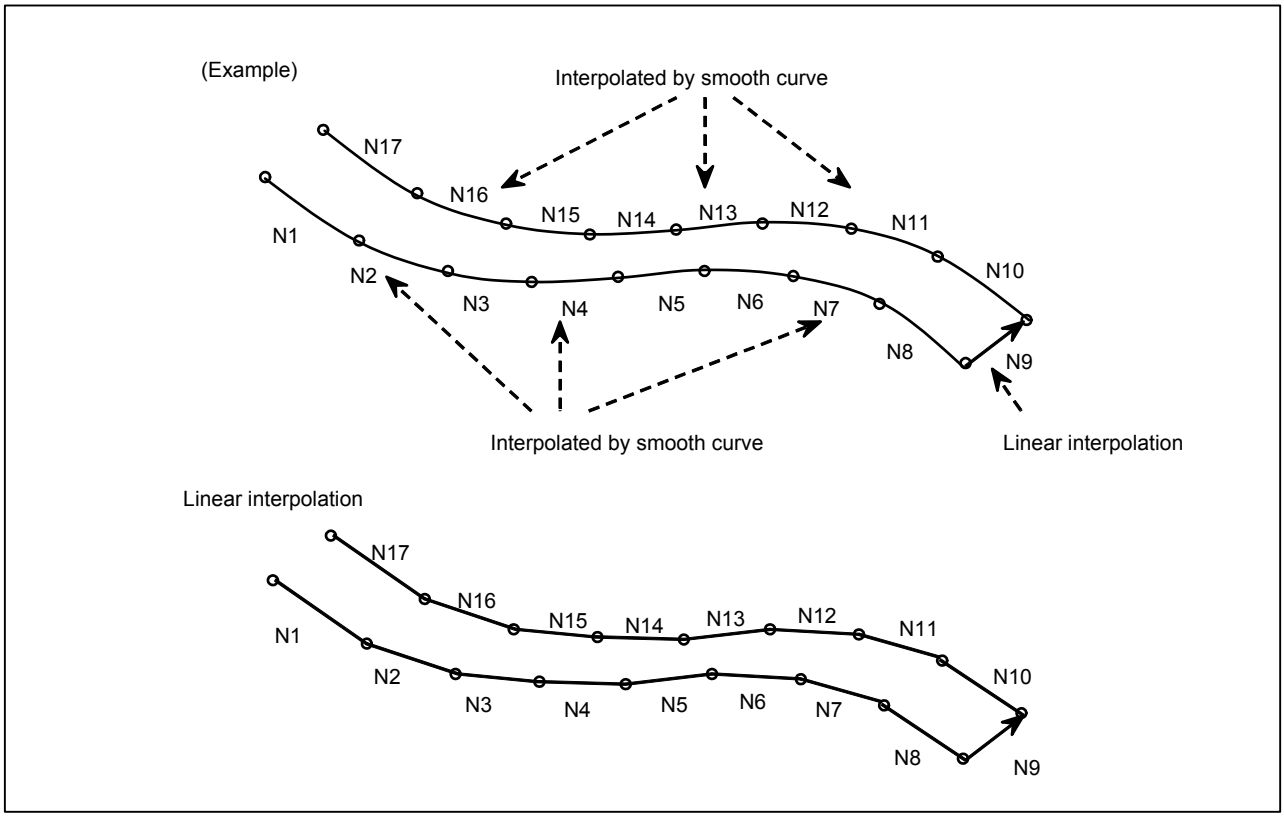
Profile	Portions having mainly a small radius of curvature	Portions having mainly a large radius of curvature
Example of machined parts	Automobile parts	Decorative parts, such as body side moldings
Length of line segment	Short	Long
Resulting surfaces produced using high-precision contour control	Smooth surface even when machining is performed exactly as specified by a program	Uneven surfaces may result when machining is performed exactly as specified by a program



Example of uneven surfaces (polygon) resulting from machining that precisely follows the line segments.

In smooth interpolation mode, the CNC automatically determines, according to the program command, whether an accurate figure is required, such as at corners, or a smooth figure is required where the radius of curvature is large. If a block specifies a travel distance or direction which differs greatly from that in the preceding block, smooth interpolation is not performed for that block. Linear interpolation is performed exactly as specified by the program command. Programming is thus very simple.

Specifying G5.1 Q2 also enables AI contour control to be turned on at the same time. The automatic velocity control by AI contour control reduces impacts on the machanical system. Specifying G5.1 Q0 cancels the smooth interpolation mode. The AI contour control mode is also canceled at the same time.



- Conditions for performing smooth interpolation

Smooth interpolation is performed when all the following conditions are satisfied. If any of the following conditions is not satisfied for a block, that block is executed without smooth interpolation then the conditions are checked for the next block.

(1) The machining length specified in the block is shorter than the length specified with parameter No. 8486.
(2) The machining length is other than 0.
(3) The modes are:
  G01 : Linear interpolation
  G13.1 : Polar coordinate interpolation cancel
  G15 : Polar coordinate command cancel
  G40 : Tool radius/tool nose radius compensation cancel (except for 3-dimensional cutter compensation)
  G64 : Cutting mode
  G80 : Canned cycle cancel
  G94 : Feed per minute
(4) Machining is specified only along the axes specified with G05.1Q2.
(5) The block is judged to be unsuitable for smooth interpolation, as performed with the internal algorithm of the CNC.

- Commands which cancel smooth interpolation

(1) Auxiliary and second auxiliary functions
(2) M98, M99: Subprogram call
  M198 : Calling a subprogram in external memory

- Checking the smooth interpolation mode

Diagnostic data (No. 5000#0) indicates whether the smooth interpolation mode is enabled in the current block.
If the smooth interpolation mode is enabled, "smooth interpolation on" bit is set to 1.

Limitation

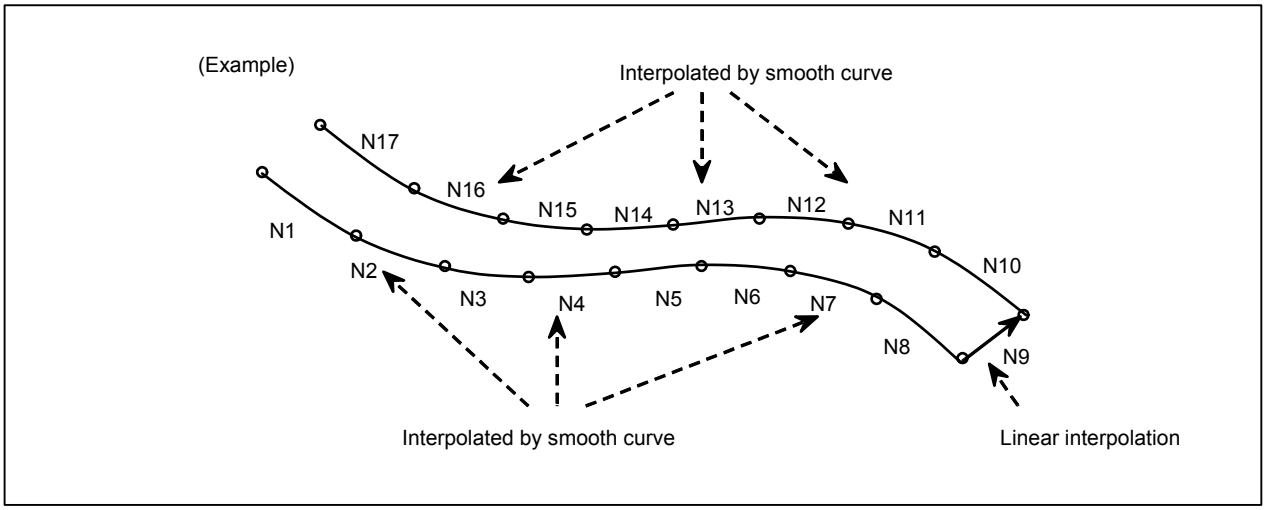
- Controlled axes

Smooth interpolation can be specified only for the X-, Y-, and Z-axes and any axes parallel to these axes (up to three axes at one time).

Example

<Example program for smooth interpolation>

```
                                N10  X-1000  Z350  ;
   .                            N11  X-1000  Z175  ;
   .                            N12  X-1000  Z25   ;
G91  ;                          N13  X-1000  Z- 50 ;
G05. 1  Q2  X0  Y0  Z0  ;       N14  X-1000  Z- 50 ;
N01  G01  X1000  Z-300  ;       N15  X-1000  Z50   ;
N02  X1000  Z-200  ;            N16  X-1000  Z200  ;
N03  X1000  Z-50  ;             N17  X-1000  Z300  ;
N04  X1000  Z50  ;              G05. 1  Q0  ;
N05  X1000  Z50  ;                 .
N06  X1000  Z-25  ;                .
N07  X1000  Z-175  ;
N08  X1000  Z-350  ;
N09  Y1000  ;
```



4.13 NANO SMOOTHING

Overview

When a desired sculptured surface is approximated by minute segments, the nano smoothing function generates a smooth curve inferred from the programmed segments and performs necessary interpolation.

The nano smoothing function infers a curve from a programmed figure approximated with segments within tolerance. If the spacing between adjacent inflection points or programmed points is not constant, this function can generate a smoother curve than the conventional smooth interpolation function.

The interpolation of the curve reduces the segment approximation error, and the nano-interpolation makes the cutting surface smoother.

The interpolation of the curve reduces the segment approximation error, and the nano interpolation makes the cutting surface smoother.

Nano smoothing 2 allows the basic three axes (X, Y, and Z) or their parallel axes as well as two rotation axes to be specified. This enables five-axis machining (including rotation axes) to produce very smooth finished surfaces and so finds applications such as machining of smooth workpiece side faces.

In addition, this function can be used with tool center point control (type 1 (G43.4)). This eliminates the necessity of modifying a program when the tool length is changed.

For this function, the AI contour control option is required.

For nano smoothing 2, the nano smoothing option is required.

Format

G5.1 Q3 Xp0 Yp0 Zp0 [α0] [β0] ; : Nano smoothing mode on
G5.1 Q0 ; : Nano smoothing mode off

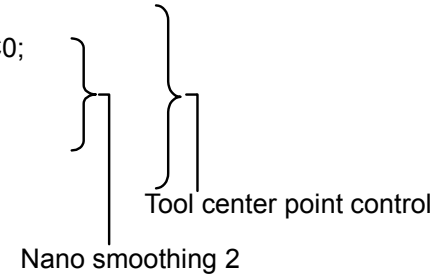
Xp : X-axis or an axis parallel to the X-axis
Yp : Y-axis or an axis parallel to the Y-axis
Zp : Z-axis or an axis parallel to the Z-axis
α, β : Rotary axis

NOTE
1 Specify G5.1 alone in a block. (Avoid specifying any other G code in the same block.)
2 Specify position 0 for the axis programmed in the nano smoothing mode on block. The specified axis is subjected to nano smoothing, but no movement is made even in the absolute programming mode.

Nano smoothing 2 enables specification of the basic three axes (X-, Y-, and Z-axes) or their parallel axes as axes on which to perform nano smoothing, as well as two rotary axes.
If executing nano smoothing 2 simultaneously with tool center point control (type 1 (G43.4)), enable tool center point control (G43.4) first and then enable nano smoothing 2 (G5.1 Q3). To cancel them, cancel nano smoothing 2 (G5.1 Q0) first and then cancel tool center point control (G49).

Example :

```
O0010
…
G43.4 H1 P0;
G5.1 Q3 X0 Y0 Z0 B0 C0;
…
G5.1 Q0;
G49;
…
M30;
```



- Nano smoothing mode

Specifying G5.1 Q3 selects the nano smoothing mode. An axis of nano smoothing is specified in the same block. The basic three axes (X, Y, and Z) and their parallel axes can be specified as the axis of nano smoothing. In the nano smoothing mode, nano smoothing is performed in a block which satisfies the conditions described below. Nano smoothing is cancelled in a block which does not satisfy the conditions.
Specifying G5.1 Q3 also enables AI contour control. The automatic velocity control by AI contour control reduces impacts on the mechanical system.
Specifying G5.1 Q0 cancels the nano smoothing mode. The AI contour control mode is also cancelled at the same time.
Making a reset also cancels the nano smoothing mode.

- Conditions to enable nano smoothing

Nano smoothing is enabled when the following conditions are satisfied. Nano smoothing is cancelled in a block which does not satisfy the conditions. A decision is made to perform nano smoothing from the next block.
In the programmed block length and the difference in angle between adjacent programmed blocks in the conditions listed below, only the basic three axes (or their parallel axes) are considered, and the rotation axes are excluded. When the nano smoothing mode is canceled in a block when these conditions are satisfied, nano smoothing for the rotation axes is not performed, either.

<1> The programmed block length is shorter than the length specified in parameter No. 8486.
<2> The programmed block length is longer than the length specified in parameter No. 8490.
<3> The difference in angle between adjacent programmed blocks is smaller than the value specified in parameter No. 8487.
<4> The current mode is one of the following:
- Linear interpolation
- Feed per minute
- Tool radius compensation cancel
- Canned cycle cancel
- Scaling cancel
- Macro modal call cancel
- Constant surface speed control cancel
- Cutting mode
- Coordinate system rotation / 3-dimensional coordinate system conversion cancel
- Polar coordinate cancel
- Normal-direction control cancel
- Polar coordinate interpolation cancel
- Programmable mirror image cancel
- Polygon turning cancel

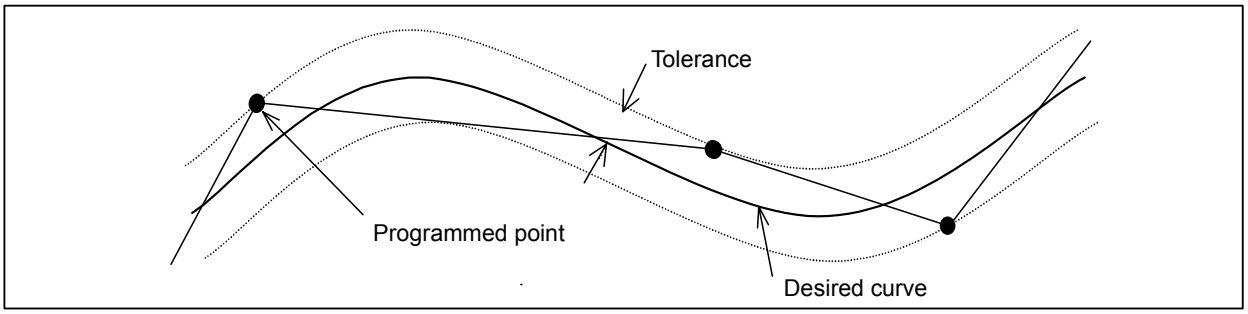
<5> No one-shot G code is specified in the block.
<6> Buffering is not suppressed in the block.
<7> A movement just on the axis of nano smoothing is specified in the block.

- Checking the nano smoothing mode

Diagnostic data (No. 5000) indicates whether the nano smoothing mode is enabled in the current block.
If the nano smoothing mode is enabled, the "smoothing on" bit is set to 1.

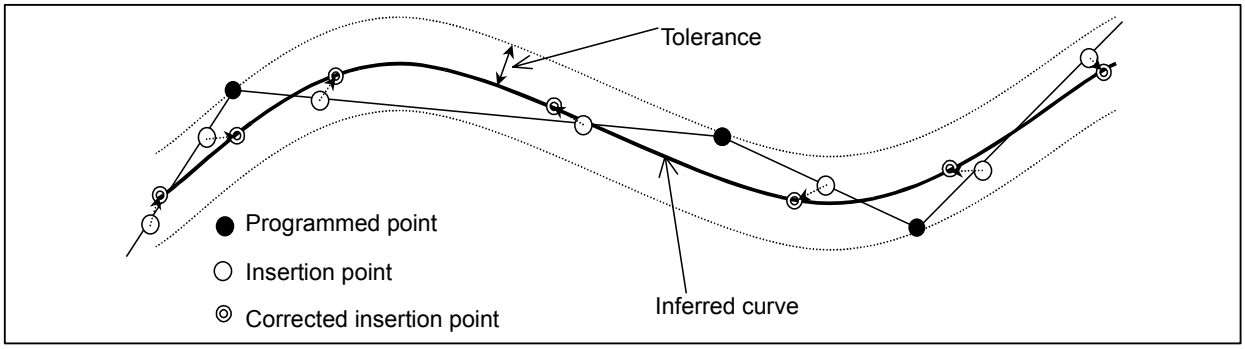
Explanation

Generally, a program approximates a sculptured surface with minute segments with a tolerance of about 10 μm.



Many programmed points are placed on the boundary of tolerance. The programmed points also have a rounding error owing to the least input increment of the CNC. The nano smoothing function creates multiple insertion points between adjacent programmed points so that a smooth curve can be created from the approximation segments. The desired curve is inferred from the insertion points of multiple blocks including buffered blocks.

Many insertion points are closer to the desired curve than the programmed points. A stable curve can be inferred with the insertion points created from multiple blocks including buffered blocks. Because the position of each insertion point is corrected in a unit smaller than the least input increment of the CNC within tolerance, the impact of rounding error is reduced.
Nano-interpolation is performed for the curve inferred from the corrected insertion points, so the resultant cutting surface becomes smooth.

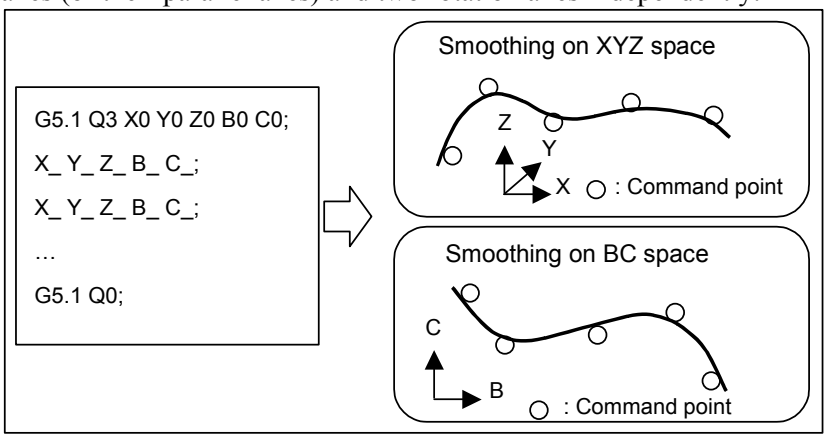


- Specifying the tolerance

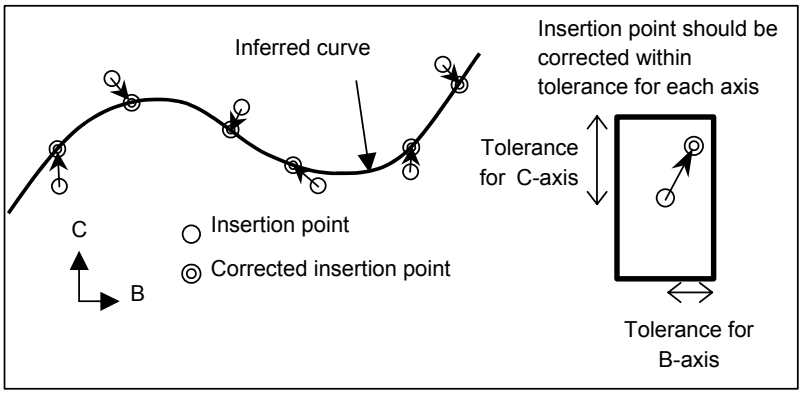
The tolerance of the program of nano smoothing is specified in parameter No. 19581.
The insertion points are corrected within tolerance, and a curve is inferred accordingly.
If 0 is specified in parameter No. 19581, the minimum travel distance in the increment system is considered to be the tolerance.

- Nano smoothing 2

Nano smoothing 2 performs smooth interpolation for the basic three axes (or their parallel axes) and two rotation axes independently.



Insertion points for the rotation axes are corrected so that each axis element of the distance between an insertion point and the corrected insertion point for both axes does not exceed the tolerance set for each axis in parameter No. 19587.



- Making a decision on the basis of the spacing between adjacent programmed points

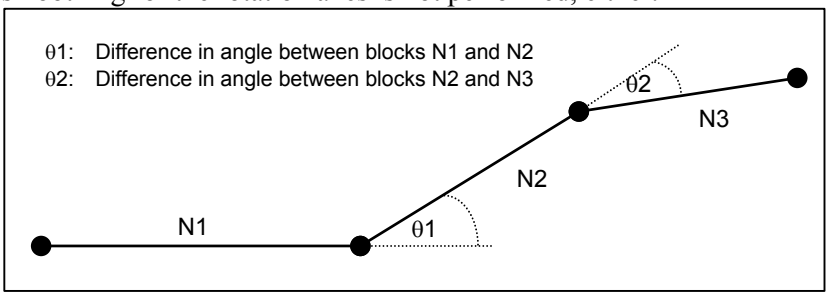
If the spacing between adjacent programmed points (block length) exceeds the value specified in parameter No. 8486 or falls below the value specified in parameter No. 8490 in the nano smoothing mode, the nano smoothing mode is cancelled at the start point of the block. Linear interpolation can be performed in the block.

When a decision is made on the basis of the spacing between adjacent programmed points, only the basic three axes (or their parallel axes) are considered, and the rotation axes are excluded. When the nano smoothing mode is canceled in a block, nano smoothing for the rotation axes is not performed, either.

If the values specified in the parameters are 0, no decision is made on the basis of the spacing between adjacent programmed points.

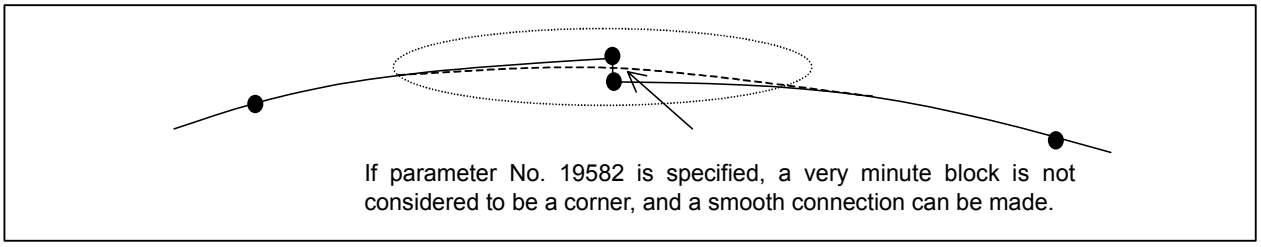
- Making a decision at a corner

If the difference in angle (see the following figure) between adjacent programmed blocks exceeds the value specified in parameter No. 8487 in the nano smoothing mode, the nano smoothing mode is cancelled at the corner.
The decision at the corner is made by considering the basic three axes (or their parallel axes) only; the rotation axes are not considered. When the nano smoothing mode is canceled in a block, nano smoothing for the rotation axes is not performed, either.



If the value specified in the parameter is 0, no decision is made at the corner on the basis of the difference in angle.
Very minute blocks created for some reasons such as a calculation error of CAM can be ignored, and a smooth connection can be made at a corner. To do this, specify parameter No. 19582 to the minimum travel distance with which a decision is made on the basis of difference in angle. Then, the decision at a corner is disabled for a block of which distance is less than the specified minimum travel distance.

However, a decision based on the spacing between adjacent programmed points specified in parameter No. 8490 has higher priority than the decision at a corner. Therefore, the value specified in parameter No. 19582 must be greater than the value specified in parameter No. 8490.



Limitation

- Single-block operation

When single-block operation is carried out in the nano smoothing mode, the operation stops at a corrected insertion point not at a programmed point.
Even in the nano smoothing mode, normal single-block operation is carried out for a block that does not satisfy the conditions of nano smoothing mode.

- Tool length compensation

To carry out tool length compensation, specify the command before specifying nano smoothing. Avoid changing the amount of compensation in the nano smoothing mode.
If G43, G44, or G49 is specified in a block between the block in which the command of nano smoothing mode on (G5.1 Q3) is specified and the block in which the command of nano smoothing mode off (G5.1 Q0) is specified, an alarm PS0343 will be issued.

- Tool radius/tool nose radius compensation

If tool radius/tool nose radius compensation is specified in the nano smoothing mode, the nano smoothing mode is cancelled. Then, when the command of tool radius/tool nose radius compensation cancel (G40) is specified, a decision is made whether to start nano smoothing from the next block. The startup and cancel operations of type C are always carried out for the tool radius/tool nose radius compensation specified in the nano smoothing mode, irrespective of the parameter setting.
A command related to tool radius/tool nose radius compensation should not be specified in the nano smoothing mode unless it is absolutely necessary.

- Interruption type custom macro

No interruption type custom macro can be used in the nano smoothing mode.
If the nano smoothing mode is specified while an interruption type custom macro is enabled or if an interruption type custom macro is enabled in the nano smoothing mode, an alarm PS0342 will be issued.

- Manual intervention

Manual intervention by specifying the manual absolute on command cannot be performed in the nano smoothing mode. If this is attempted, an alarm PS0340 will be issued at the cycle start after manual intervention.

- Rotary table dynamic fixture offset

The command of rotary table dynamic fixture offset (G54.2) must be cancelled before specifying the nano smoothing mode. These commands cannot be used in the nano smoothing mode. If an attempt is made to use one of these commands, an alarm PS0343 will be issued.

- Number of blocks that can be specified successively

Up to about 300,000,000 blocks can be specified successively in the nano smoothing mode. If more blocks are specified, an alarm PS0341 will be issued.
However, when a block which does not satisfy the conditions of the nano smoothing mode is encountered, the mode is canceled and the counted number of successive blocks is reset to 0.

- Continuity of a program

Curve interpolation is carried out for multiple programmed blocks including buffered blocks in the nano smoothing mode.
Therefore, the programmed commands must be executed continuously in the nano smoothing mode.
The continuity of a program may be lost, and continuous execution may not be performed, in some cases such as the following: A single-block stop is made in the nano smoothing mode; and another program is executed in the MDI mode. If this occurs, an alarm PS0344 will be issued.

- Restrictions on resumption of automatic operation

(1) Program restart
Curve interpolation is performed for corrected insertion points not for programmed points in the nano smoothing mode. Accordingly, when a sequence number is specified to restart the program, the operation cannot be restart from a programmed point in a block.
To restart a program, specify a block number, using the block counter displayed in the program screen.

(2) Block restart
A block cannot be restarted in the nano smoothing mode. When an attempt is made to restart a block, an alarm PS0344 will be issued.

(3) Tool retract and recover
The tool cannot be retracted or recovered in the nano smoothing mode.

(4) Retracing
Retracing cannot be performed in the nano smoothing mode.

(5) Active block cancel
The active block cancel function is temporarily disabled in the nano smoothing mode.

- Functions that cannot be used simultaneously

The nano smoothing function cannot be used simultaneously with the following functions.

- Parallel axis control
- Twin table control

- Tool center point control

Nano smoothing 2 can be used with tool center point control (type 1 (G43.4)). However, when it is used with tool center point control (type 2 (G43.5)) or tool posture control, alarm PS5421 (ILLEGAL COMMAND IN G43.4/G43.5) is issued.

4.14 NURBS INTERPOLATION (G06.2)

Many computer-aided design (CAD) systems used to design metal dies for automobiles and airplanes utilize non-uniform rational B-spline (NURBS) to express a sculptured surface or curve for the metal dies.

This function enables NURBS curve expression to be directly specified to the CNC. This eliminates the need for approximating the NURBS curve with minute line segments. This offers the following advantages:

1. No error due to approximation of a NURBS curve by small line segments
2. Short part program
3. No break between blocks when small blocks are executed at high speed
4. No need for high-speed transfer from the host computer to the CNC

When this function is used, a computer-aided machining (CAM) system creates a NURBS curve according to the NURBS expression output from the CAD system, after compensating for the length of the tool holder, tool diameter, and other tool elements. The NURBS curve is programmed in the NC format by using these three defining parameters: control point, weight, and knot.

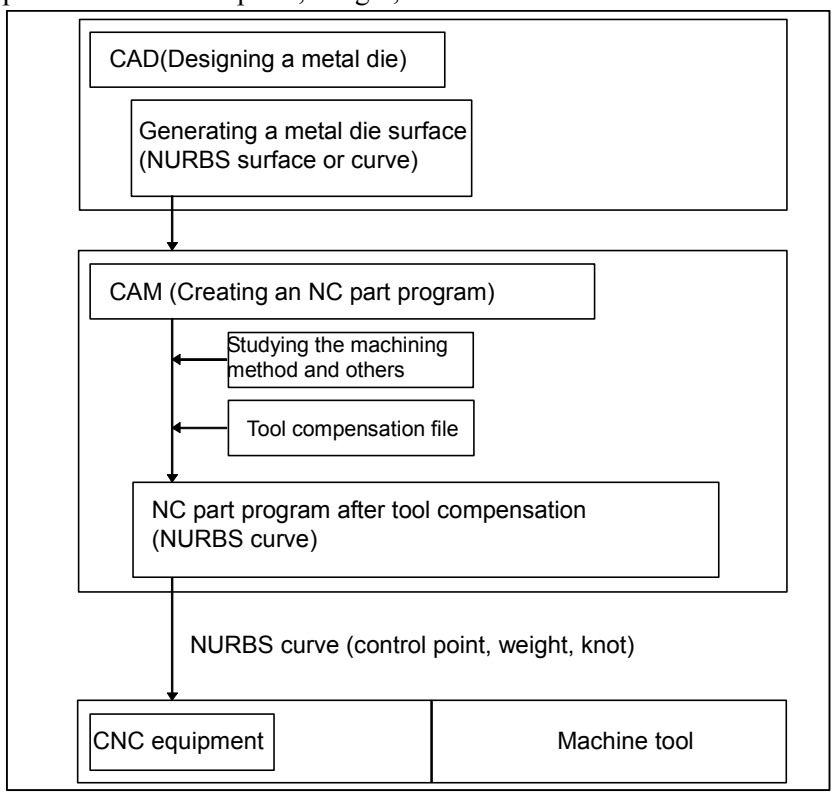


Fig. 4.14 (a) NC part program for machining a metal die according to a NURBS curve

NURBS interpolation can be performed for up to five axes (including two rotation axes). Therefore, NURBS interpolation can be performed for the basic three axes (X, Y, and Z) and two rotation axes at the same time. This enables five-axis machining to produce very smooth finished surfaces and so finds applications such as machining of smooth workpiece side faces.
In addition, NURBS interpolation can be used with tool center point control (type 1 (G43.4)). This eliminates the necessity of modifying a program when the tool length is changed.

Format

```
G06.2[P_] K_X_Y_Z_[α_] [β_][R_] [F_];
          K_X_Y_Z_[α_] [β_][R_];
          K_X_Y_Z_[α_] [β_][R_];
          K_X_Y_Z_[α_] [β_][R_];
            :
          K_X_Y_Z_[α_] [β_][R_];
          K_;
          :
          K_;
G01 . . .

G06.2 : Start NURBS interpolation mode
P_    : Rank of NURBS curve
X_Y_Z_: Control point
α_β_  : Control point (rotary axes)
R_    : Weight
K_    : Knot
F_    : Feedrate
```

If executing NURBS interpolation simultaneously with tool center point control (type 1 (G43.4)), enable tool center point control (G43.4) first and then enable NURBS interpolation (G06.2). To cancel them, end NURBS interpolation (G code command in group 01 other than G06.2) first and then cancel tool center point control (G49).

Example :

```
O0010
…
G43.4 H1 P0;
G06.2 K_ X_ Y_ Z_ B_ C_;
…
K_
G01 X_ Y_ Z_ B_ C_;
G49;
…
M30;
```

Tool center point control

NURBS interpolation

Explanation

- NURBS interpolation mode

NURBS interpolation mode is selected when G06.2 is programmed. G06.2 is a modal G code of group 01. NURBS interpolation mode ends when a G code of group 01 other than G06.2 (G00, G01, G02, G03, etc.) is specified.

- Rank of NURBS

A rank of NURBS can be specified with address P. The rank setting, if any, must be specified in the first block. If the rank setting is omitted, a rank of four (degree of three) is assumed for NURBS. The valid data range for P is 2 to 4. The P values have the following meanings:

P2: NURBS having a rank of two (degree of one)
P3: NURBS having a rank of three (degree of two)
P4: NURBS having a rank of four (degree of three) (default)

This rank is represented by k in the defining expression indicated in the description of NURBS curve below. For example, a NURBS curve having a rank of four has a degree of three. The NURBS curve can be expressed by the constants t^3, t^2, and t^1.

- Weight

The weight of a control point programmed in a single block can be defined. When the weight setting is omitted, a weight of 1.0 is assumed.

- Knot

The number of specified knots must equal the number of control points plus the rank value. In the blocks specifying the first to last control points, each control point and a knot are specified in an identical block. After these blocks, as many blocks (including only a knot) as the rank value are specified. The NURBS curve programmed for NURBS interpolation must start from the first control point and end at the last control point. The first k knots (where k is the rank) must have the same values as the last k knots (multiple knots). If the absolute coordinates of the start point of NURBS interpolation do not match the position of the first control point, alarm PS5117 is issued. (To specify incremental values, G06.2 X0 Y0 Z0 K_ must be programmed.)

- NURBS curve

Using these variables:
k : Rank
P*i* : Control point
Wi : Weight
Xi : Knot ($X_i < X_{i+1}$)
Knot vector $[X_0, X_1, \ldots, X_m]$ ($m = n + k$)
t : Spline parameter,
the spline basis function N can be expressed with the de Boor-Cox recursive formula, as indicated below:

$$N_{i,1}(t) = \begin{cases} 1(x_i < t < 1x_{i+1}) \\ 0(t < x_i, x_{i+1} < t) \end{cases}$$

$$N_{i,k}(t) = \frac{(t - x_i)N_{i,k-1}(t)}{x_{i+k-1} - x_i} + \frac{(x_{i+k} - t)N_{i+1,k-1}(t)}{x_{i+k} - x_{i+1}}$$

The NURBS curve P(t) of interpolation can be expressed as follows:

$$P(t) = \frac{\sum_{i=0}^{n} N_{i,k}(t) w_i p_i}{\sum_{i=0}^{n} N_{i,k}(t) w_i}$$

($X0 \leq t \leq Xm$)

- Reset

A reset during NURBS interpolation results in the clear state. The modal code of group 1 enters the state specified in bit 0 (G01) of parameter No. 3402.

Limitation

- Controlled axes

NURBS interpolation can be performed on up to five axes (including two rotary axes). The axes of NURBS interpolation must be specified in the first block. A new axis cannot be specified before the beginning of the next NURBS curve or before NURBS interpolation mode ends.

- Command in NURBS interpolation mode

In NURBS interpolation mode, any command other than the NURBS interpolation command (miscellaneous function and others) cannot be specified.

- Manual intervention

If manual intervention is attempted while manual absolute mode is set, alarm PS5118 is issued.

- Tool radius compensation

Tool radius compensation cannot be simultaneously executed. NURBS interpolation can only be specified after tool radius compensation has been canceled.

- 3-dimensional cutter compensation

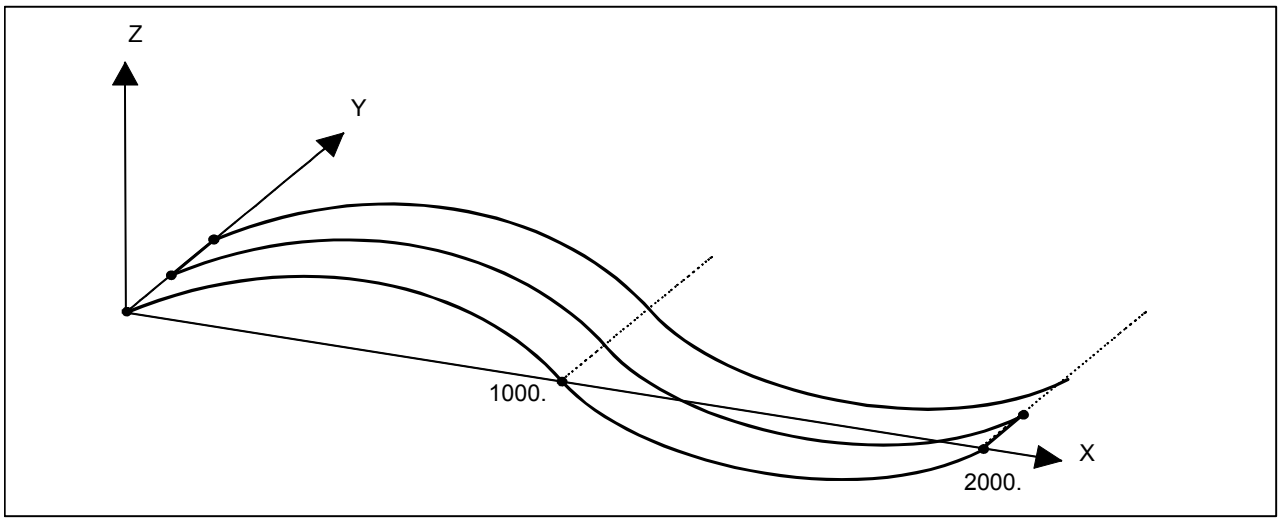
3-dimensional cutter compensation cannot be simultaneously executed. NURBS interpolation can only be specified after 3-dimensional cutter compensation has been canceled.

- Tool centerpoint control

Tool center point control (type 1 (G43.4)) can be simultaneously executed, but tool center point control (type 2 (G43.5)) and tool posture control cannot be simultaneously executed. If an attempt is made to execute tool center point control (type 2 (G43.5) or tool posture control simultaneously with NURBS interpolation, alarm PS5421 is issued.

Example

```
<Sample NURBS interpolation program>
G90;
...
G06.2   K0.     X0.     Z0.;
        K0.     X300.   Z100.;
        K0.     X700.   Z100.;
        K0.     X1300.  Z-100.;
        K0.5    X1700.  Z-100.;
        K0.5    X2000.  Z0.;
        K1.0;
        K1.0;
        K1.0;
        K1.0;
G01     Y0.5;
G06.2   K0.     X2000.  Z0.;
        K0.     X1700.  Z-100.;
        K0.     X1300.  Z-100.;
        K0.     X700.   Z100.;
        K0.5    X300.   Z100.;
        K0.5    X0.     Z0.;
        K1.0;
        K1.0;
        K1.0;
        K1.0;
G01     Y0.5;
G06.2 ...
...
G01 ...
```



4.14.1 NURBS Interpolation Additional Functions

In the FANUC Series 30*i*/31*i*, NURBS interpolation provides the following additional functions:

- Parametric feedrate control

The maximum feedrate of each segment is determined according to a specified feedrate and acceleration. For a series of segments, the feedrates at the start point and end point of the segments are determined as described below. During a movement from the start point to end point, the feedrate changes continuously. This function can be used only for NURBS interpolation when bit 5 (FDI) of parameter No. 8412 is set to 1.

1. Start point speed
   (1) At the start of NURBS interpolation
   If the amount of travel in the previous block is 0, the maximum speed in the first segment is regarded as the start point speed. If the amount of travel in the previous block is not 0, whichever is smaller of the specified speed in the previous block and the maximum speed in the first segment is regarded as the start point speed.
   Previous block: Block immediately preceding the block in which NURBS interpolation is specified.
   (2) Second and subsequent segments
   Whichever is smaller of the maximum speed in the previous segment and the maximum speed in this segment is regarded as the start point speed.

2. End point speed
   (1) In a segment other than the last
   Whichever is smaller of the maximum speed in this segment and the maximum speed in the next segment is regarded as the end point speed.
   (2) At the end of NURBS interpolation
   If the amount of travel in the next block is 0, the maximum speed in this segment is regarded as the end point speed. If the amount of travel in the next block is not 0, whichever is smaller of the maximum speed in this segment and the specified speed in the next block is regarded as the end point speed.

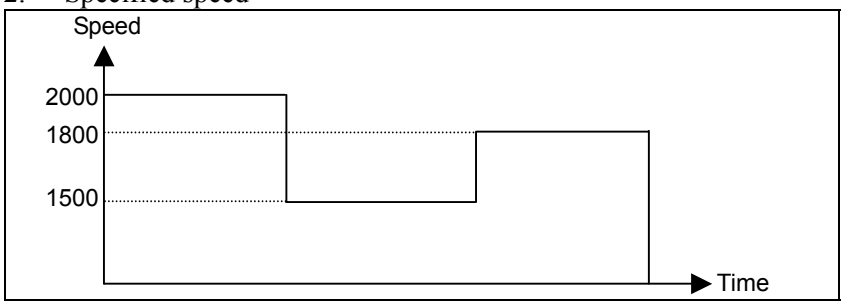
After the stop of a single block, the start point speed at a restart will be the maximum speed in this segment.

After the stop of a feed hold, the speed at a restart will be the same as the speed at the stop of the feed hold.

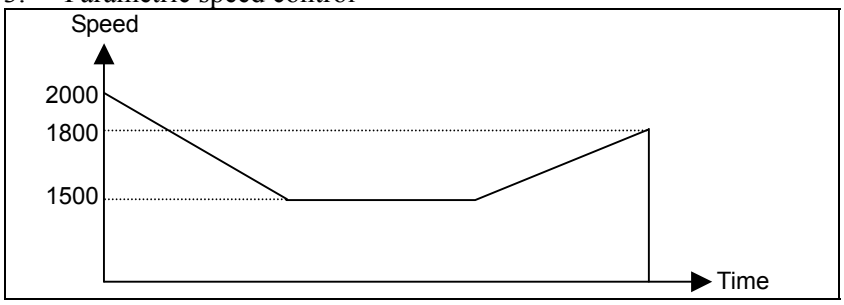
Example

1. Specified program
   G90 G06.2 X0. Y0. K0. F2000 ;
   X10. Y10. K0. F1500 ;
   X20. Y20. K0. F1800 ;
   X30. Y30. K0. ;
   X40. Y40. K1.
   X50. Y50. K2.
   K3.
   K3.
   K3.
   K3.

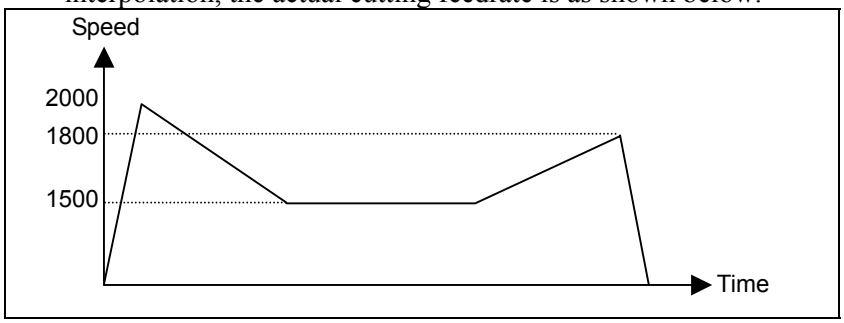
2. Specified speed



3. Parametric speed control



4. Because of the addition of acceleration/deceleration before interpolation, the actual cutting feedrate is as shown below.



NOTE
The actual acceleration/deceleration method is bell-shaped acceleration/deceleration.

- High-precision knot command

If bit 1 (HIK) of parameter No. 8412 is set to 1, knot commands with a whole number of up to 12 digits and a decimal fraction of up to 12 digits can be specified. This function can be used only for knot commands (address K) with a decimal point which are specified during NURBS interpolation.

If a high precision knot command is used, the knot command format is as shown below.

K (Number of A digits) . (Number of B digits)

where A + B must be 12 or less. The address K and the numbers including the decimal point must be 14 characters or less.

(Example)

Valid
K.999999999999
K1234.56789012
K999999999999.

Invalid (causing alarm PS0012 to be issued)
K0.999999999999
K1234.567891230
K1234.567891234
K999999999999.0

- Simple start command

If bit 0 (EST) of parameter No. 8412 is set to 1, a control command may be omitted at the first control point. Because the same value is set for the knot in the first block and the knot in the second block, the knot command may also be omitted in the first bock.

The format of a NURBS interpolation command using this function is as shown below.

```
G06.2 [P__ ] [K__ ] [IP__ ] [R__ ] [F__ ] ;
                K__ IP__ [R__ ] ;
                K__ IP__ [R__ ] ;
                K__ IP__ [R__ ] ;
                …
                K__ IP__ [R__ ] ;
                K__ ;
                …
                K__ ;
G01…
…
```

G06.2 : NURBS interpolation mode ON
P : Rank of the NURBS curve
IP : Control point
R : Weight
K : Knot
F : Speed

Words enclosed in [ ] are optional.

If a control point command is omitted in the first block, the current position as determined when NURBS interpolation is started is regarded as the first control point. Also, only the axis specified for the control point in the second block is handled as a NURBS interpolation axis and, therefore, in the second block, specify all axes subject to NURBS interpolation.

If a knot command is omitted in the first block, the knot is regarded as the same as the knot specified in the second block.

4.15 HYPOTHETICAL AXIS INTERPOLATION (G07)

In helical interpolation, when pulses are distributed with one of the circular interpolation axes set to a hypothetical axis, sine interpolation is enabled.
When one of the circular interpolation axes is set to a hypothetical axis, pulse distribution causes the speed of movement along the remaining axis to change sinusoidally. If the major axis for threading (the axis along which the machine travels the longest distance) is set to a hypothetical axis, threading with a fractional lead is enabled. The axis to be set as the hypothetical axis is specified with G07.

Format

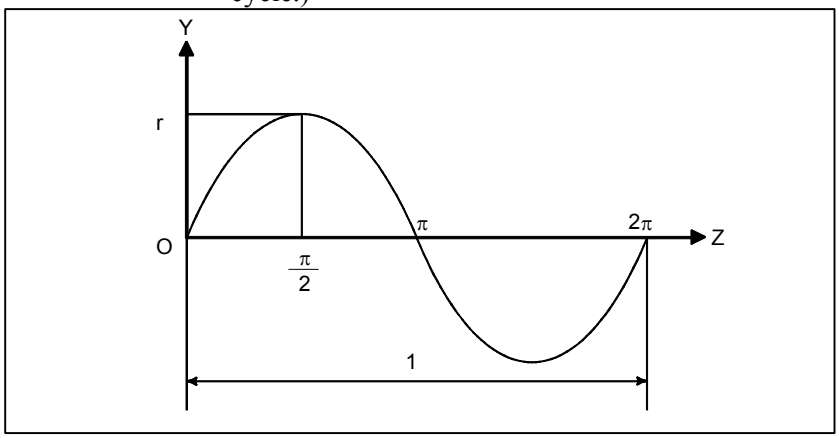
G07 α0 ; : Hypothetical axis setting
:
G07 α1 ; : Hypothetical axis cancel
Where, α is any one of the addresses of the controlled axes.

Explanation

- Sine interpolation

The a axis is regarded as a hypothetical axis for the period of time from the G07 α0 command until the G07 α1 command appears.
Suppose sine interpolation is performed for one cycle in the YZ plane. The hypothetical axis is then the X axis.
$X^2+Y^2 = r^2$ (r is the radius of an arc.)

$Y = rSIN(\frac{2\pi}{1}Z)$ (1 is the distance traveled along the Z-axis in one cycle.)



- Interlock, stroke limit, and external deceleration

Interlock, stroke limit, and external deceleration can also apply to the hypothetical axis.

- Handle interruption

An interrupt caused by the handle also applies to the hypothetical axis. This means that movement for a handle interrupt is performed.

Limitation

- Manual operation

The hypothetical axis can be used only in automatic operation. In manual operation, it is not used, and movement takes place.

- Move command

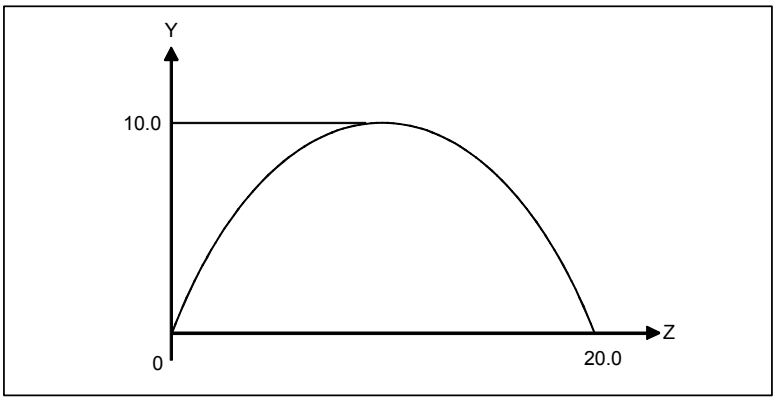
Specify hypothetical axis interpolation only in the incremental mode.

- Coordinate rotation

Hypothetical axis interpolation does not support coordinate rotation.

Example

- Sine interpolation



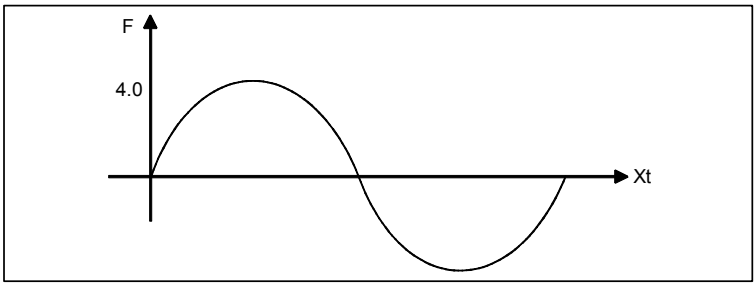

```
N001  G07  X0  ;
N002  G91  G17  G03  X-20.0  Y0.0  I-10.0  Z20.0  F100 ;
N003  G01  X10.0 ;
N004  G07  X1 ;
```

From the N002 to N003 blocks, the X-axis is set to a hypothetical axis. The N002 block specifies helical cutting in which the Z-axis is the linear axis. Since no movement takes place along the X axis, movement along the Y-axis is performed while performing sine interpolation along the Z-axis.
In the N003 block, there is no movement along the X-axis, and so the machine dwells until interpolation terminates.

- Changing the feedrate to form a sine curve

(Sample program)

```
G07Z0 ;              The Z-axis is set to a hypothetical axis.
G02X0Z0I10.0F4. ;    The feedrate on the X-axis changes sinusoidally.
G07Z1 ;              The use of the Z-axis as a hypothetical axis is
                     canceled.
```



4.16 VARIABLE LEAD THREADING (G34)

Specifying an increment or a decrement value for a lead per screw revolution enables variable lead threading to be performed.

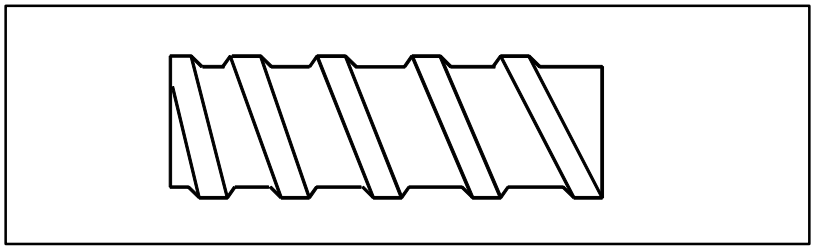


Fig. 4.16 (a) Variable lead screw

Format

G34 IP_ F_ K_ Q_ ;

IP_ : End point
F_ : Lead in longitudinal axis direction at the start point
K_ : Increment and decrement of lead per spindle revolution
Q_ : Shift amount of starting angle of thread cutting

Explanation

Address other than K are the same as in straight/taper thread cutting with G32.
The K value depends on the increment system of the reference axis, as indicated in Table 4.16 (a).
If the specified K value exceeds the range indicated in Table 4.16 (a), if the maximum lead is exceeded after a change due to the K value, or if the lead value is negative, an alarm PS0313 will be issued.

Table 4.16 (a) Range of valid K values

Increment system of reference axis	Metric input (mm/rev)	Inch input (inch/rev)
IS-A	±0.001 to ±500.000	±0.00001 to±50.00000
IS-B	±0.0001 to ±500.0000	±0.000001 to±50.000000
IS-C	±0.00001 to ±50.00000	±0.0000001 to ±5.0000000
IS-D	±0.000001 to ±5.000000	±0.00000001 to ±0.50000000
IS-E	±0.0000001 to ±0.5000000	±0.000000001to ±0.050000000

⚠ CAUTION
The "thread cutting cycle retract" is not effective for G34.

Example

Lead at the start point: 8.0 mm
Lead increment: 0.3 mm/rev

```
G34 Z-72.0 F8.0 K0.3 ;
```

4.17 CIRCULAR THREADING (G35, G36)

Using the G35 and G36 commands, a circular thread, having the specified lead in the direction of the major axis, can be machined.

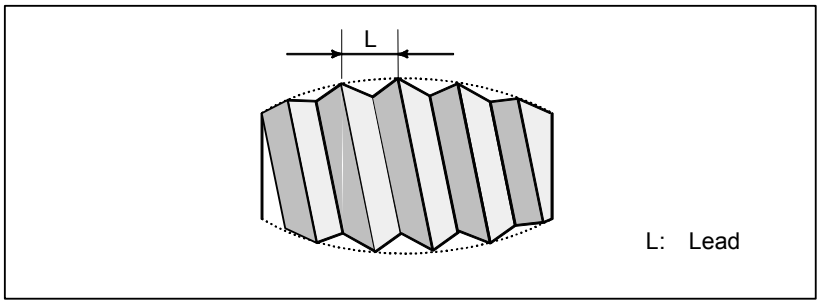


Fig. 4.17 (a) Circular threading

Format

A sample format for the G18 plane (Z-X plane) is indicated below. When using the format for the G17 plane (X-Y plane), change the addresses Z, X, K, and I to X, Y, I, and J respectively. When using the format for the G19 plane (Y-Z plane), change the addresses Z, X, K, and I to Y, Z, J, and K respectively.

M

$$\begin{Bmatrix} \mathbf{G35} \\ \mathbf{G36} \end{Bmatrix} \mathbf{X_\ Z_} \quad \begin{Bmatrix} \mathbf{I_\ K_} \\ \mathbf{R_} \end{Bmatrix} \mathbf{F_\ Q_\ ;}$$

G35 : Clockwise circular threading command
G36 : Counterclockwise circular threading command

X, Z : Specify the arc end point (in the same way as for G02, G03).
I, K : Specify the arc center relative to the start point, using relative coordinates (in the same way as for G02, G03).
R : Specify the arc radius.
F : Specify the lead in the direction of the major axis.
Q : Specify the shift of the threading start angle (0° to 360°, with least input increment of 0.001) (The value can be programmed with a decimal point.)

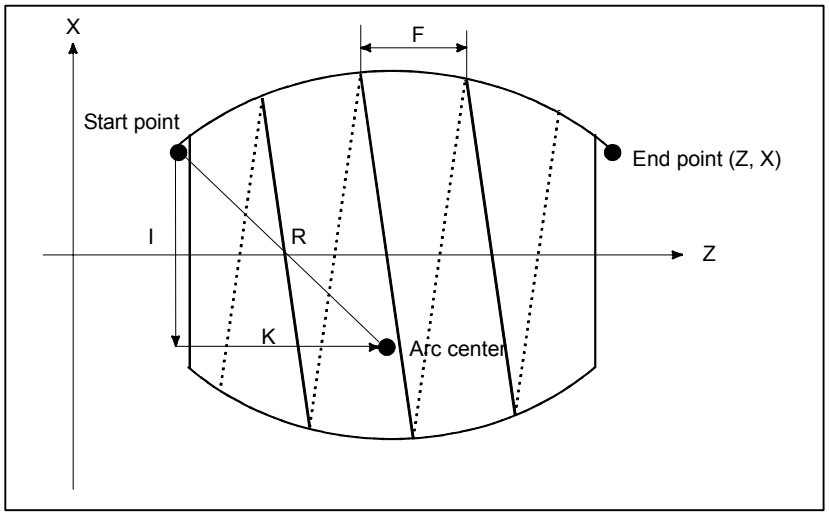
T

$$\left\{\begin{matrix}\mathbf{G35}\\\mathbf{G36}\end{matrix}\right\}\ \mathbf{X(U)_\ Z(W)_}\ \left\{\begin{matrix}\mathbf{I_\ K_}\\\mathbf{R_}\end{matrix}\right\}\ \mathbf{F_\ Q_\ ;}$$

G35 : Clockwise circular threading command
G36 : Counterclockwise circular threading command

X(U), Z(W) : Specify the arc end point (in the same way as for G02, G03).

I, K : Specify the arc center relative to the start point, using relative coordinates (in the same way as for G02, G03).
R : Specify the arc radius.
F : Specify the lead in the direction of the major axis.
Q : Specify the shift of the threading start angle (0° to 360°, with least input increment of 0.001) (The value cannot be programmed with a decimal point.)



Explanation

- Specifying the arc radius

If R is specified with I and K, only R is effective.

- Shift angle

If an angle greater than 360° is programmed, it is set to 360°.

M

- Specifying shift angle Q

To specify the shift angle Q, set bit 0 (GQS) of parameter No. 3451 to 1.

T

- Automatic tool compensation

The G36 command is used to specify the following two functions: Automatic tool compensation X and counterclockwise circular threading. The function for which G36 is to be used depends on bit 3 (G36) of parameter No. 3405.

- When parameter G36 is set to 0, the G36 command is used for automatic tool compensation X.
- When parameter G36 is set to 1, the G36 command is used for counterclockwise circular threading.

G37.1 can be used to specify automatic tool compensation X and G37.2 can be used to specify automatic tool compensation Z.

(Specification method)
G37.1 X_
G37.2 Z_

- G code when parameter G36 (bit 3 of No. 3405) is set to 1

G code	G code group	Function
G35	01	Clockwise circular threading
G36		Counterclockwise circular threading
G37	00	Automatic tool compensation Z
G37.1		Automatic tool compensation X
G37.2		Automatic tool compensation Z

Limitation

- Range of specifiable arc

An arc must be specified such that it falls within a range in which the major axis of the arc is always the Z-axis or always the X-axis, as shown in Fig. 4.17 (b), and (c). If the arc includes a point at which the major axis changes from the X-axis to Z-axis, or vice versa, as shown in Fig. 4.17 (d), an alarm PS5058 is issued.

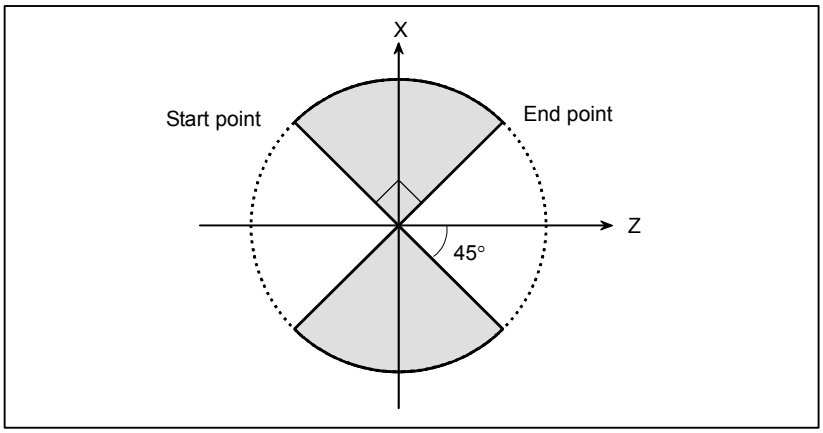


Fig. 4.17 (b) Range in which the Z-axis is the major axis

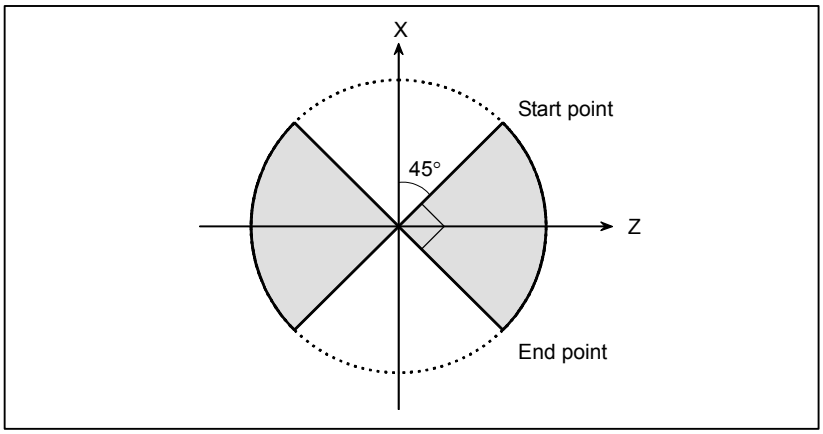


Fig. 4.17 (c) Range in which the X-axis is the major axis

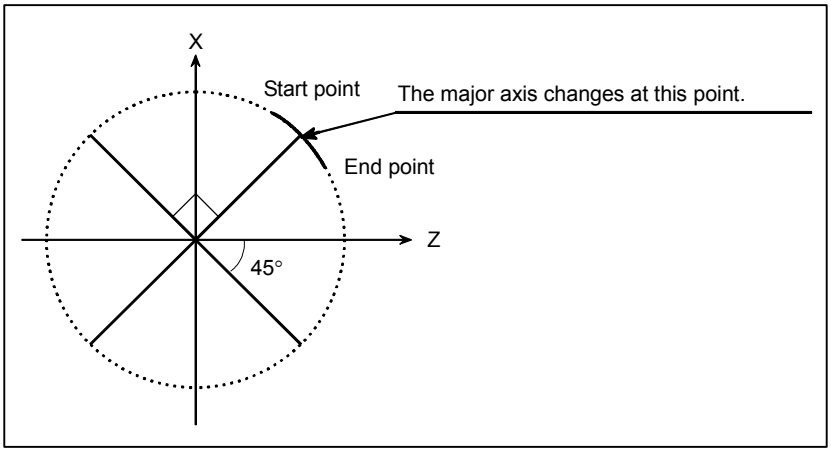


Fig. 4.17 (d) Example of arc specification which causes an alarm

- End point not on an arc

If the end point is not on an arc, a movement on an axis is made to a position of which coordinate matches the corresponding coordinate of the end point. Then, a movement is made on another axis to reach the end point.

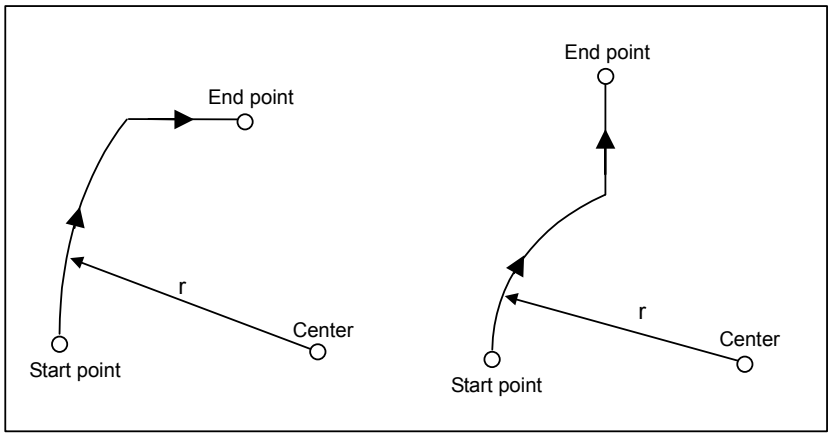


Fig. 4.17 (e) Movement when the end point is not on an arc

4.18 SKIP FUNCTION (G31)

Linear interpolation can be commanded by specifying axial move following the G31 command, like G01. If an external skip signal is input during the execution of this command, execution of the command is interrupted and the next block is executed.
The skip function is used when the end of machining is not programmed but specified with a signal from the machine, for example, in grinding. It is used also for measuring the dimensions of a workpiece.

Format

G31 IP ;
G31 : One-shot G code (If is effective only in the block in which it is specified)

Explanation

The coordinate values when the skip signal is turned on can be used in a custom macro because they are stored in the custom macro system variable #100151 to #100182, as follows. For systems with up to 20 axes, FS16-compatible system variables (#5061 to #5080) may also be used.

#100151 (#5061) : First axis coordinate value
#100151 (#5062) : Second axis coordinate value
:
#100170 (#5080) : 20th axis coordinate value

#100171 : 21st axis coordinate value
#100172 : 22nd axis coordinate value
:
#100182 : 32nd axis coordinate value

⚠ CAUTION
Disable feedrate override, dry run, and automatic acceleration/deceleration (however, these become available by setting bit 7 (SKF) of parameter No.6200 to 1.) when the feedrate per minute is specified, allowing for an error in the position of the tool when a skip signal is input. These functions are enabled when the feedrate per rotation is specified.

NOTE
If G31 command is issued while tool radius/tool nose radius compensation is applied, an alarm PS0035 is displayed. Cancel the tool radius compensation with the G40 command before the G31 command is specified.

Example

- The next block to G31 is an incremental programming

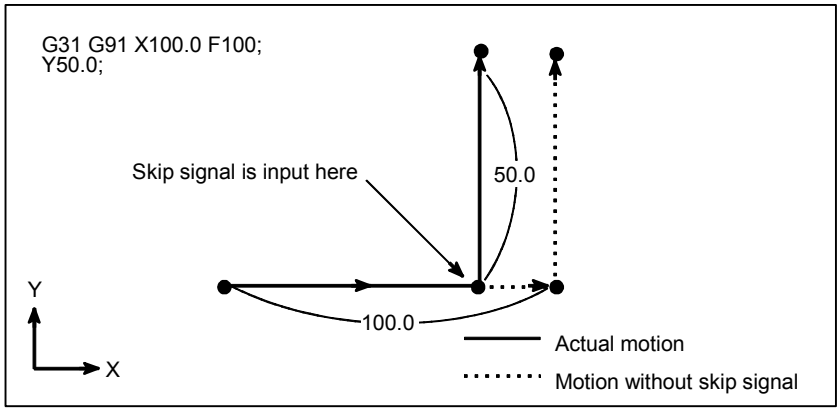


Fig. 4.18 (a) The next block is an incremental programming

- The next block to G31 is an absolute programming for 1 axis

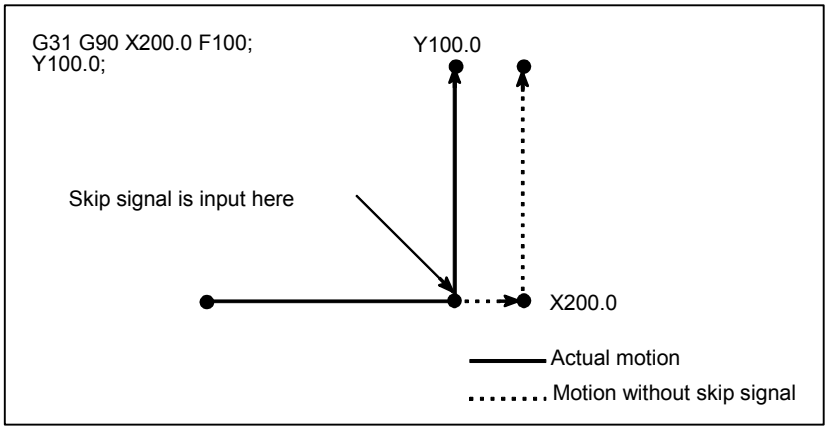


Fig. 4.18 (b) The next block is an absolute programming for 1 axis

- The next block to G31 is an absolute programming for 2 axes

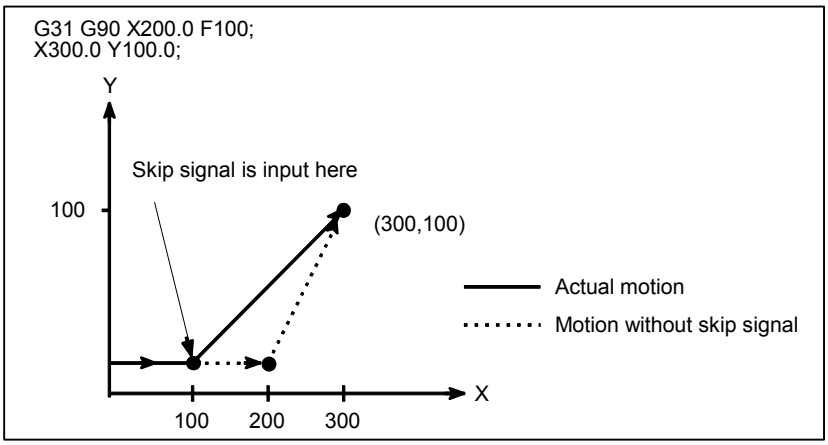


Fig. 4.18 (c) The next block is an absolute programming for 2 axes

4.19 MULTI-STEP SKIP (G31)

In a block specifying P1 to P4 after G31, the multi-step skip function stores coordinates in a custom macro variable when a skip signal (4-point or 8-point ; 8-point when a high-speed skip signal is used) is turned on. In the block where Q1 to Q4 are specified after G04, dwell can be skipped when skip signals (four or eight signals, or eight signals when high-speed skip signals are used) are input.
A skip signal from equipment such as a fixed-dimension size measuring instrument can be used to skip programs being executed.
In plunge grinding, for example, a series of operations from rough machining to spark-out can be performed automatically by applying a skip signal each time rough machining, semi-fine machining, fine-machining, or spark-out operation is completed.

Format

Move command
G31 IP_ F_ P_ ;
IP_ : End point
F_ : Feedrate
P_ : P1 to P4
Dwell
G04X(U,P)_ (Q_);
X(U,P)_ : Dwell time
Q_ : Q1 to Q4

Explanation

Multi-step skip is caused by specifying P1, P2, P3, or P4 in a G31 block. For an explanation of selecting (P1, P2, P3, or P4), refer to the manual supplied by the machine tool builder.
Specifying Q1, Q2, Q3, or Q4 in G04 (dwell command) enables dwell skip in a similar way to specifying G31. A skip may occur even if Q is not specified. For an explanation of selecting (Q1, Q2, Q3, or Q4), refer to the manual supplied by the machine tool builder.

- Correspondence to skip signals

Parameter Nos. 6202 to 6205 can be used to specify whether the 4-point or 8-point skip signal is used (when a high-speed skip signal is used). Specification is not limited to one-to-one correspondence. It is possible to specify that one skip signal correspond to two or more Pn's or Qn's (n=1, 2, 3, 4). Also, bits 0 (DS1) and 7 (DS8) parameter No.6206 can be used to specify dwell.

⚠ **CAUTION**
Dwell is not skipped when Qn is not specified and bits 0 (DS1) and 7 (DS8) parameter No.6206 are not set.

4.20 HIGH-SPEED SKIP SIGNAL (G31)

The skip function operates based on a high-speed skip signal (connected directly to the NC; not via the PMC) instead of an ordinary skip signal. In this case, up to eight signals can be input.
Delay and error of skip signal input is 0 - 2 msec at the NC side (not considering those at the PMC side).
This high-speed skip signal input function keeps this value to 0.1 msec or less, thus allowing high precision measurement.
For details, refer to the appropriate manual supplied from the machine tool builder.

Format

G31 IP ;

G31; One-shot G code (If is effective only in the block in which it is specified)

4.21 CONTINUOUS HIGH-SPEED SKIP FUNCTION

Overview

The continuous high-speed skip function is used to read absolute coordinates using high-speed skip signals HDI0 to HDI7. Inputting a high-speed skip signal in a G31P90 block causes absolute coordinates to be stored in custom macro variables #5061 to #5080. For a system with more than 20 axes, they are stored in variables #100151 to #100182. An axis movement does not stop even if a signal is input, so that the coordinates of multiple points can be read.
Using bit 5 (CSE) of parameter No. 6201, it is possible to enable both the rising and falling edges of a high-speed skip signal.
For details, refer to the manual supplied by the machine tool builder.

Format

G31 P90 IP ;

G31; One-shot G code (If is effective only in the block in which it is specified)

Explanation

- Custom macro variables

If a high-speed skip signal is input when G31P90 is issued, absolute coordinates are stored in custom macro variables #5061 to #5080. For a system exceeding 20 axes, they are stored in variables #100151 to #100182.
These variables are updated as soon as the axis reaches a new skip position. It is, therefore, necessary to specify a speed so that the axis reaches the next skip position in a period of time longer than it takes for the application to read the variable values.

System with 20 or less axes		System with more than 20 axes	
#5061	Coordinates on the first axis	#100151	Coordinates on the first axis
#5062	Coordinates on the second axis	#100152	Coordinates on the second axis
#5063	Coordinates on the third axis	#100153	Coordinates on the third axis
	:		:
#5079	Coordinates on the 19th axis	#100181	Coordinates on the 31st axis
#5080	Coordinates on the 20th axis	#100182	Coordinates on the 32nd axis

- High-speed skip signal

This function works with a high-speed skip signal only.
Select a high-speed skip signal with bits 0 to 7 (9S1 to 9S8) of parameter No. 6208.

- Ending a block

A G31P90 block is ended when the axis reaches the end point.

Limitation

The continuous high-speed skip function (G31P90) block must be a command for a single axis only. If an attempt is made to specify two or more axes, P/S alarm No. 5068 is issued.

4.22 TORQUE LIMIT SKIP

Overview

Executing the move command following G31P99 (or G31P98) while overriding the torque limit[*1] on the servo motor enables cutting feed in to be performed in the same way as in linear interpolation (G01). If, during the movement with this command, the torque of the servo motor reaches the torque limit value (torque limit on the servo motor multiplied by the override) due to pressing or other causes or a skip signal (including a high-speed skip signal) is input, any remaining move commands will be canceled, and the next block is executed. (The operation of canceling any remaining move commands and executing the next block is called a skip operation in the remainder of this document.)

It is possible to override the torque limit on the servo motor with the following command methods:

(1) Execute the torque limit override command in the PMC window.
(2) Execute an address Q command in the same block as that containing a G31P99 (or G31P98) command.

*1 : The torque limit on the servo motor is automatically set to a value conforming to the motor type setting.

Format

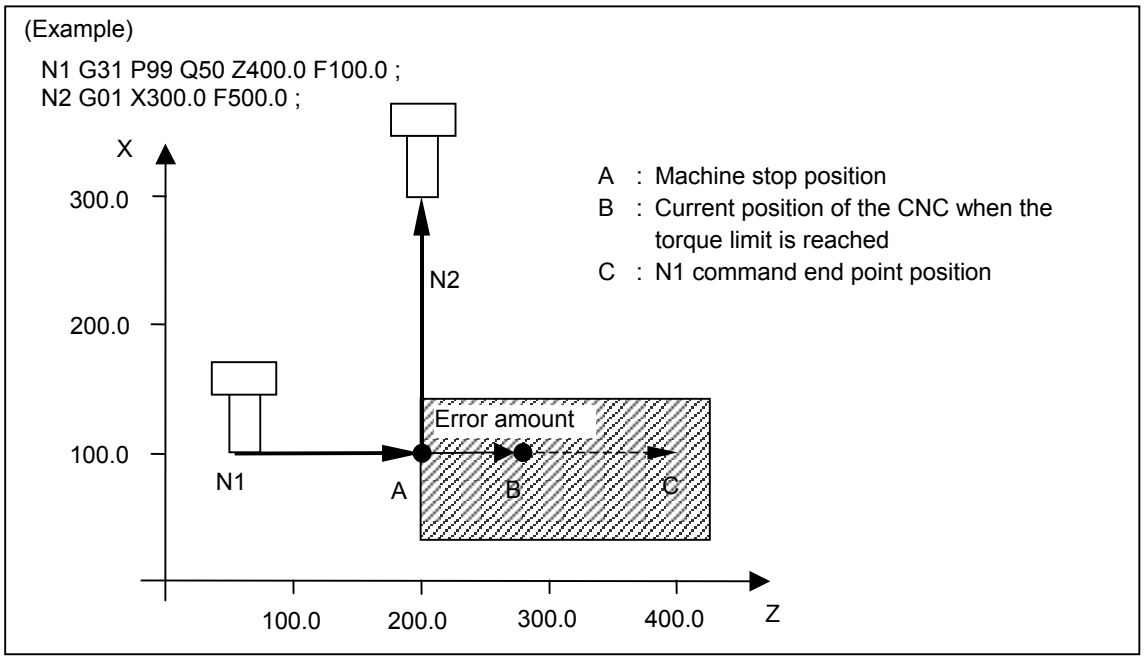
G31 P98 Q_ α_ F_
G31 P99 Q_ α_ F_

G31 : Skip command (one-shot G code)
P98 : Performs a skip operation if the torque of the servo motor reaches the limit value.
P99 : Performs a skip operation if the torque of the servo motor reaches the limit value or if a skip signal is input.
Q : Override value for the torque limit
Range of valid settings: 1 to 254 (%)
0 to 255 correspond to 0% to 100%.
The Q command is optional. If omitting it, specify the torque limit command in the PMC window beforehand. If it is omitted, and no torque limit override is set beforehand, alarm PS0035 is issued.
If an attempt is made to specify a value out of range, alarm PS0366 is issued.
The specified override value is effective only to the block in which it is specified.
At the end of the skip operation, the override value returns to the one assumed immediately before the G31 command.
α : Axis address on any one axis
F : Feedrate

- Conditions for performing a skip operation

Condition	Command	
	G31P98	G31P99
The torque limit value is reached.	A skip operation is performed.	A skip operation is performed.
A skip signal is input.	No skip operation is performed.	A skip operation is performed.

- Operation during a torque limit skip



A torque limit skip presses a specified axis against a previously prepared part or other item while the torque limit command is being executed on the servo motor, and then performs a skip operation when the servo motor reaches the torque limit value. A skip operation is performed with the fact that the torque limit value as detected in the servo motor is reached. It is, therefore, not necessary to input a skip signal using a separate sensor or other device unlike with normal skip functions.

(1) At point A, the machine comes in contact with the object under measurement and stops. At this time, because the torque limit value is not reached, no skip operation is performed, move commands are continuously output, and the current position of the CNC is updated.

(2) Because move commands are output but the machine remains stopped, there occurs a difference (error amount) between the current position of the CNC and the machine position, and torque is applied to the servo motor.

(3) When the torque limit value is reached, a skip operation is performed at machine stop position, point A, and the N2 command is executed. Assuming that the current position of the CNC when the torque limit is reached is point B, the error amount during the torque limit skip is (A - B).

- Torque limit command

If, with the torque limit skip command, no torque limit override value is specified with address Q and no torque limit command is issued from the PMC window, etc., alarm PS0035 is issued.

When no toque limit command is issued, the torque limit override value is either 0% or 100%.

If, in a torque limit skip command block, no torque limit override value is specified with address Q, a torque limit is to be specified as in the program example below.

(Program example)
O0012
:
Mxx (Specify a torque limit from the PMC via the window)
:
G31 P99 X200. F100.（Torque limit skip command）
:
G01 X100. F500.（Move command with the torque limit being still effective）
:
Myy (Cancel the torque limit from the PMC)
:
M30

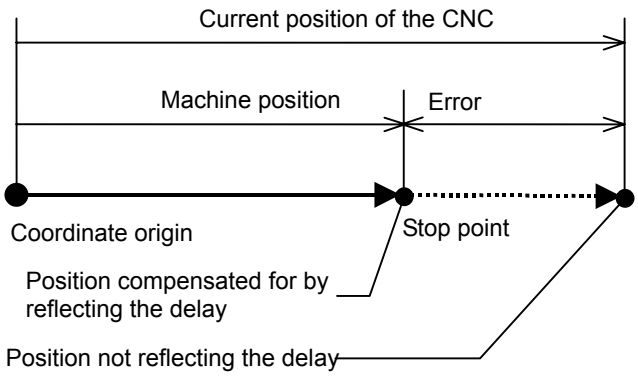
- Positional deviation limit during the torque limit command

While the torque limit skip command is being executed, the positional deviation limit check with the settings of parameters Nos. 1828 and 1829 is not performed. Instead, the positional deviation limit check with the settings of parameter No. 6287 is performed. If the positional deviation exceeds the limit, alarm SV0004 is issued and an instantaneous stop occurs.

- Custom macro variables

When the torque limit skip command is executed, the custom macro system variables #5061 to #5080 (skip signal position; for a system with more than 20 axes, variables #100151 to #100182 are used) store the coordinate position assumed at the end of the skip. In reality, there is a deviation due to the delay of the servo system between the machine position and the current position of the CNC when a skip operation is executed. This deviation can be determined from the positional deviation of the servo. By setting bit 2 (TSE) of parameter No. 6201, it is possible to select whether or not the skip signal position to be stored in system variables should be compensated for with the error (positional deviation) of the servo system.

Position during a skip operation



NOTE

1 Specify only a single axis with the torque limit skip command. If no axis is specified or an attempt is made to specify more than one, alarm PS0369 is issued.
2 Do not issue the torque limit skip command in 3-dimensional coordinate conversion mode or to a parallel axis. Otherwise, alarm PS0035 is issued.
3 Do not issue the torque limit skip command in G41 or G42 mode. Otherwise, alarm PS0035 is issued.
4 A torque limit arrival signal is output regardless of the torque limit skip command.
5 Do not issue the torque limit skip command for an axis being synchronized with synchronization control (such as synchronization control, twin table control, or electronic gear box).
6 Do not specify the torque limit skip command in a continuous block.
7 The higher the movement speed, the larger the error between the position at which the machine stops and the position at which a skip is actually detected. Also, the error increases as the speed is varied during movement. Do not vary the speed with override and so on.

4.23 THREE-DIMENSIONAL CIRCULAR INTERPOLATION

Overview

Specifying an intermediate and end point on an arc enables circular interpolation in a 3-dimensional space.

Format

The command format is as follows:

G02.4 X_{X1} Y_{Y1} Z_{Z1} $\alpha_{\alpha1}$ $\beta_{\beta1}$; First block (mid-point of the arc)

X_{X2} Y_{Y2} Z_{Z2} $\alpha_{\alpha2}$ $\beta_{\beta2}$; Second block (end point of the arc)

α,β : Arbitrary axes other than the 3-dimensional circular interpolation axis (up to two axes)
To be specified if necessary.

G03.4 can also be specified instead of G02.4.
G03.4 performs the same operation as G02.4.

Explanation

- G code group

G02.4 and G03.4 are modal G codes of group 01. They therefore remain effective until another G code in group 01 is specified.

- Start point, mid-point, and end point

An arc in a 3-dimensional space is uniquely defined with its start point (current position) and a specified intermediate point and end point, as shown below. Two command blocks are used to define this arc. The first command block specifies the tool path between the start point and intermediate point. The second command block specifies the tool path between the intermediate point and end point.

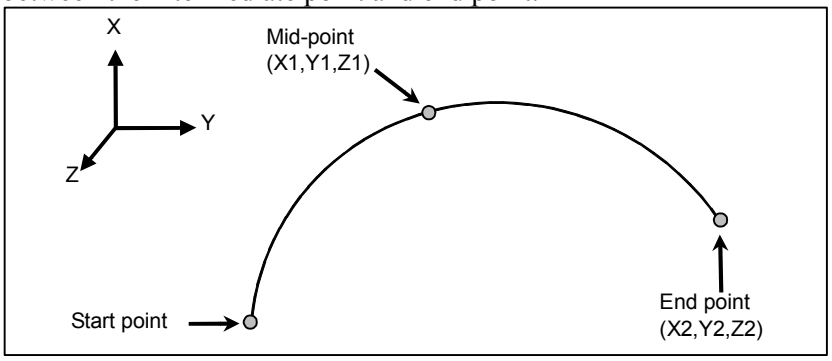


Fig. 4.23 (a) Start, Mid, and End Points

If the modal code is changed by specifying a code such as G01 with the end point not specified, the arc cannot be obtained, and alarm PS5432 is issued. During MDI operation, alarm PS5432 is also issued if a cycle start is applied with only the mid-point specified.

- Movement along axes other than the three-dimensional circular interpolation axis

In addition to the three-dimensional circular interpolation axis (X/Y/Z), up to two arbitrary axes (α/β) can be specified at a time. If / are omitted from the first block (mid-point specification) and are specified only in the second block (end point specification), the tool moves to the specified point along the α/β axes during movement from the mid-point of the arc to the end point. If α/β are omitted from the second block (end point specification) and are specified only in the first block (mid-point specification), the tool moves to the specified point along the α/β axes during movement from the start point of the arc to the mid-point.

- Incremental programming

With an incremental programming, the position of the mid-point relative to the start point must be specified in the first block, and the position of the end point relative to the mid-point must be specified in the second block.

- Direction of rotation

The direction of rotation cannot be specified. The movement is the same regardless of whether G02.4 or G03.4 is specified.

- Single block

When operation is performed using a single block, one cycle start causes movement from the start point to the end point. A single-block stop is not performed between the first block (mid-point specification) and the second block (end point specification).

- Start point assumed if three-dimensional circular interpolations are specified consecutively

If three-dimensional circular interpolations are specified consecutively, the end point in one interpolation is assumed to be the start point in the next interpolation.

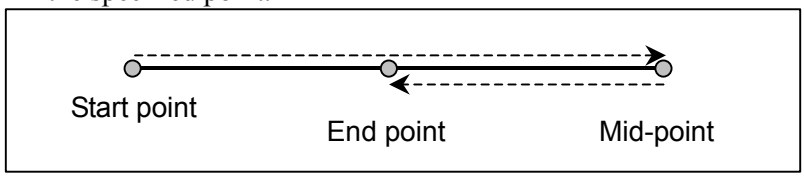
- Velocity commands

As the velocity command, specify the tangential velocity along the arc in the three-dimensional space.

Limitation

- Cases in which linear interpolation is performed

- f the start point, mid-point, and end-point are on the same line, linear interpolation is performed.
- If the start point coincides with the mid-point, the mid-point coincides with the end point, or the end point coincides with the start point, linear interpolation is performed up to the end point.
- If the start point, mid-point, and end-point are on the same line and the end point lies between the start point and the mid-point, the tool first moves with linear interpolation from the start point to the mid-point, then returns from the mid-point to the end point with linear interpolation. Thus, the tool always passes through the specified point.



- Whole circles

A whole circle (360° arc) cannot be specified. (This corresponds to the case in which linear interpolation is performed, as described earlier.)

- Compensation functions

Before using this function, cancel the compensation functions of group 07, such as cutter radius compensation.

- Manual absolute

While this function is in use, manual intervention is not possible with the manual absolute switch set to the ON position. If intervention is performed, alarm PS5433 is issued when operation restarts.

- Restrictions on commands

In the three-dimensional circular interpolation mode, the functions listed below can be used, but the states of these functions must not be modified.

- Inch input/metric input (An alarm is issued if a state modification is made with G20 or G21.)
- Mirror image (The state of a signal must not be changed.)
- One-digit F code feed (No feedrate change must be made using the manual handle.)

- Unusable commands

In the three-dimensional circular interpolation mode, the functions listed below must not be specified. Otherwise, an alarm is issued.

- Exponential interpolation..... G02.3,G03.3
- Dwell G04
- High-speed machining ..... G05 (Excluding G05P10000 and G05P0)
- AI contour control G05.1Q1,G5.1Q0

- Hypothetical axis interpolation G07
- Cylindrical interpolation G07.1
- Advanced preview control G08
  (Use AI high-precision contour control.)
- Polar coordinate interpolation G12.1,G13.1
- Polar coordinate command G15,G16
- Reference position return check G27
- Reference position return G28
- 2nd reference position return G30
- 3rd, 4th reference position return G30
- Skip G31
- Threading G33
- Automatic tool length measurement G37
- 3-dimensional cutter compensation G41
- Tool offset G45,G46,G47,G48
- Programmable mirror image G50.1,G51.1
- Local coordinate system G52
- Machine coordinate system G53
- Single direction positioning G60
- Tapping mode G63
- Macro call (Subprogram call is possible.) G65,G66,G67
- Figure copying G72.1,G72.2
- Canned cycle G73-G79,G80,G81-G89,G98,G99
- Workpiece coordinate system setting G92
- Workpiece coordinate system preset G92.1
- Feed per revolution G95
- Constant surface speed control G96,G97
- NURBS interpolation G06.2
- Workpiece coordinate system setting G54-G59,G54.1
- 3-dimensional coordinate system conversion G68
- Coordinate system rotation G68
- Scaling G50,G51
- Programmable mirror image G50.1,G51.1
- Exact stop mode G61
- Tool radius/tool nose radius compensation G38,G39,G40,G41,G42
- 3-dimensional cutter compensation ..... G40, G41.2, G41.3, G41.4, ... G41.5, G41.6, G42.2, G42.4, G42.5, G42.6
- Data setting G10
- Exact stop G09
- Auxiliary function
- 2nd auxiliary function
- Spindle function
- Tool function

- Unusable functions

If the following function is specified in the three-dimensional circular interpolation mode, a warning is output:

- MDI intervention

If any of the following functions is specified in the three-dimensional circular interpolation mode, the alarm (PS5196) is issued:

- Manual operation interrupt
- Tool retract and recover

In the three-dimensional circular interpolation mode, the following functions cannot be used:

- Sequence number comparison and stop (Stop operation cannot be performed with a sequence number in the three-dimensional circular interpolation mode.)
- Index table indexing
- Rotary axis control
- Macro executor (Execution macro)
- Manual handle interruption
- Optional chamfering/corner R

- Other limitations

When the following function is used, three-dimensional circular interpolation cannot be used:

- Arbitrary angular axis control

A limitation may be imposed on other NC command combinations. See the description of each function.

5 FEED FUNCTIONS

Chapter 5, "FEED FUNCTIONS", consists of the following sections:

5.1 OVERVIEW 155
5.2 RAPID TRAVERSE 157
5.3 CUTTING FEED 158
5.4 CUTTING FEEDRATE CONTROL 164
5.5 FEEDRATE INSTRUCTION ON IMAGINARY CIRCLE FOR A ROTARY AXIS 169
5.6 DWELL 174

5.1 OVERVIEW

The feed functions control the feedrate of the tool. The following two feed functions are available:

- Feed functions

1. Rapid traverse
   When the positioning command (G00) is specified, the tool moves at a rapid traverse feedrate set in the CNC (parameter No. 1420).
2. Cutting feed
   The tool moves at a programmed cutting feedrate.

- Override

Override can be applied to a rapid traverse rate or cutting feedrate using the switch on the machine operator's panel.

- Automatic acceleration/deceleration

To prevent a mechanical shock, acceleration/deceleration is automatically applied when the tool starts and ends its movement (Fig. 5.1(a)).

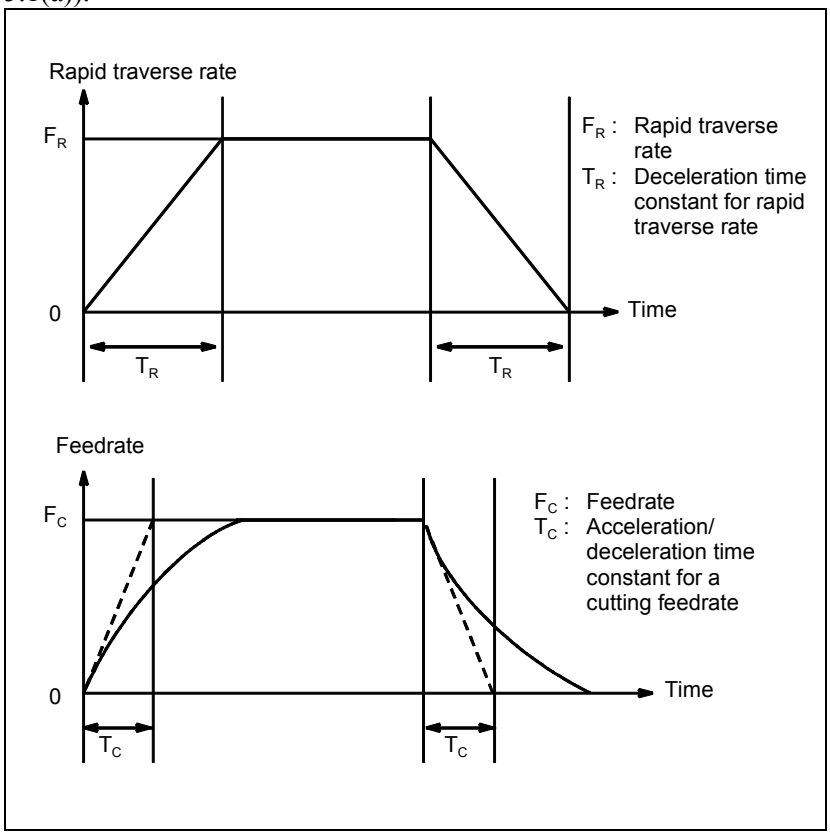


Fig. 5.1 (a) Automatic acceleration/deceleration (example)

- Tool path in a cutting feed

When the movement direction changes between a specified block and the next block during cutting feed, the tool path may be rounded because of the relationship between the time constant and feedrate (Fig. 5.1(b)).

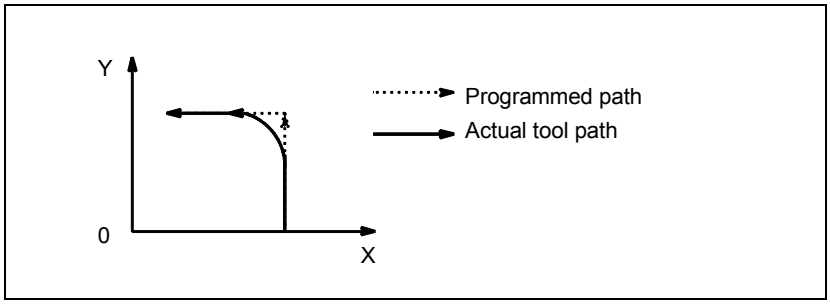


Fig. 5.1 (b) Example of tool path between two blocks

In circular interpolation, a radial error occurs (Fig. 5.1(c)).

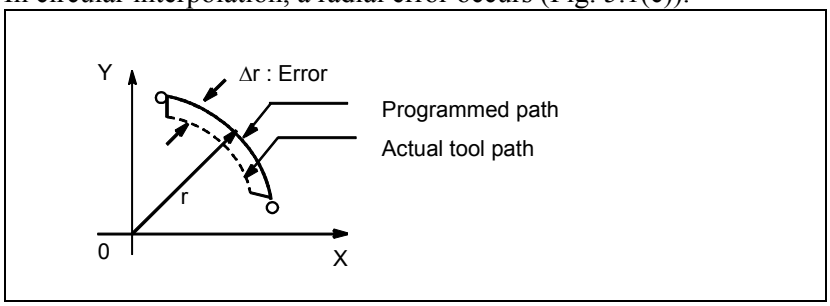


Fig. 5.1 (c) Example of radial error in circular interpolation

The rounded-corner path shown in Fig. 5.1(b) and the error shown in Fig. 5.1 (c) depend on the feedrate. So, the feedrate needs to be controlled for the tool to move as programmed.

5.2 RAPID TRAVERSE

Format

G00 IP_ ;
G00 : G code (group 01) for positioning (rapid traverse)
IP_ : Dimension word for the end point

Explanation

The positioning command (G00) positions the tool by rapid traverse. In rapid traverse, the next block is executed after the specified feedrate becomes 0 and the servo motor reaches a certain range set by the machine tool builder (in-position check).
A rapid traverse rate is set for each axis by parameter No. 1420, so no rapid traverse feedrate need be programmed.
The following overrides can be applied to a rapid traverse rate with the switch on the machine operator's panel:F0, 25%, 50%, 100%
F0: Allows a fixed feedrate to be set for each axis by parameter No. 1421.
For detailed information, refer to the appropriate manual of the machine tool builder.

5.3 CUTTING FEED

Overview

Feedrate of linear interpolation (G01), circular interpolation (G02, G03), etc. are commanded with numbers after the F code.
In cutting feed, the next block is executed so that the feedrate change from the previous block is minimized.

M

Four modes of specification are available:

1. Feed per minute (G94)
   After F, specify the amount of feed of the tool per minute.
2. Feed per revolution (G95)
   After F, specify the amount of feed of the tool per spindle revolution.
3. Inverse time feed (G93)
   Specify the inverse time (FRN) after F.
4. One-digit F code feed
   Specify a desired one-digit number after F. Then, the feedrate set with the CNC for that number is set.

T

Two modes of specification are available:

1. Feed per minute (G98)
   After F, specify the amount of feed of the tool per minute.
2. Feed per revolution (G99)
   After F, specify the amount of feed of the tool per spindle revolution.
3. Inverse time feed (G93)
   Specify the inverse time (FRN) after F.

Format

M

Feed per minute
G94 ; G code (group 05) for feed per minute
F_ ; Feedrate command (mm/min or inch/min)
Feed per revolution
G95 ; G code (group 05) for feed per revolution
F_ ; Feedrate command (mm/rev or inch/rev)
Inverse time feed (G93)
G93 ; Inverse time feed command G code (05 group)
F_ ; Feedrate command (1/min)
One-digit F code feed
Fn ;
n : Number from 1 to 9

T

Feed per minute
G98 ; G code (group 05) for feed per minute
F_ ; Feedrate command (mm/min or inch/min)
Feed per revolution
G99 ; G code (group 05) for feed per revolution
F_ ; Feedrate command (mm/rev or inch/rev)
Inverse time feed (G93)
G93 ; Inverse time feed command G code (05 group)
F_ ; Feedrate command (1/min)

Explanation

- Direction of the cutting feedrate

Cutting feed is controlled so that the tangential feedrate is always set at a specified feedrate.

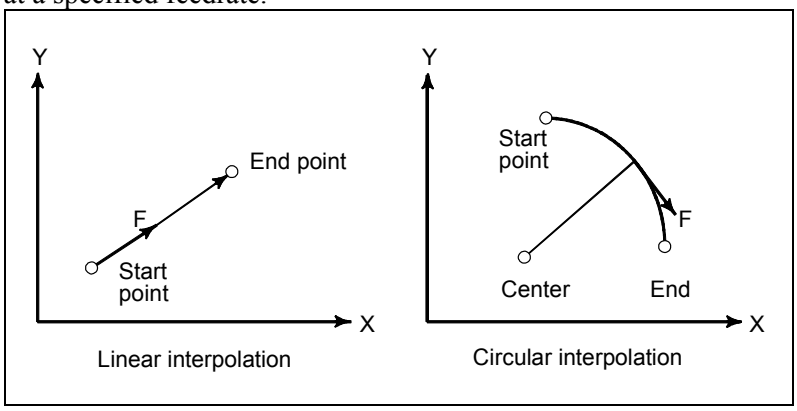


Fig. 5.3 (a) Tangential feedrate (F)

- Feed per minute

After specifying G code for feed per minute (in the feed per minute mode), the amount of feed of the tool per minute is to be directly specified by setting a number after F. G code for feed per minute is a modal code. Once a G code for feed per minute is specified, it is valid until G code for feed per revolution (feed per revolution) is specified.

M

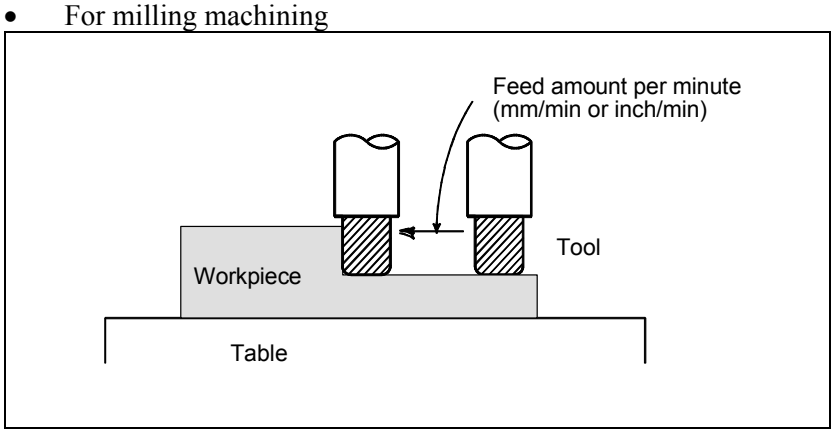
At power-on, the feed per minute mode is set.

T

At power-on, the feed per revolution mode is set.

An override from 0% to 254% (in 1% steps) can be applied to feed per minute with the switch on the machine operator's panel. For detailed information, see the appropriate manual of the machine tool builder.

- For milling machining



- For lathe cutting

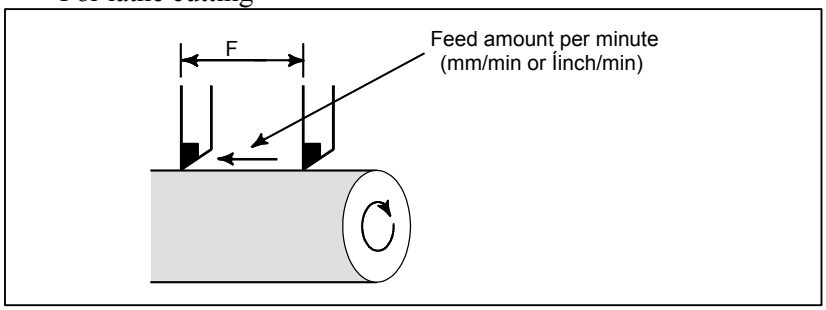


Fig. 5.3 (b) Feed per minute

> ⚠ **CAUTION**
> No override can be used for some commands such as for threading.

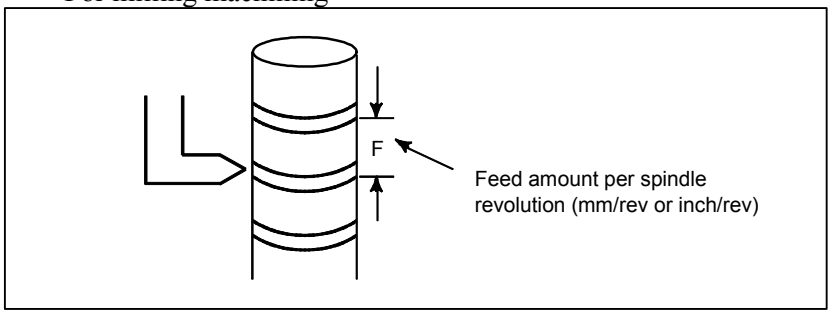
- Feed per revolution

After specifying G code for feed per revolution (in the feed per revolution mode), the amount of feed of the tool per spindle revolution is to be directly specified by setting a number after F. G code for feed per revolution is a modal code. Once a G code for feed per revolution is specified, it is valid until G code for feed per minute (feed per minute) is specified.
An override from 0% to 254% (in 1% steps) can be applied to feed per revolution with the switch on the machine operator's panel. For detailed information, see the appropriate manual of the machine tool builder.

T

If bit 0 (NPC) of parameter No. 1402 has been set to 1, feed per revolution commands can be specified even when a position coder is not being used. (The CNC converts feed per revolution commands to feed per minute commands.)

- For milling machining



- For lathe cutting

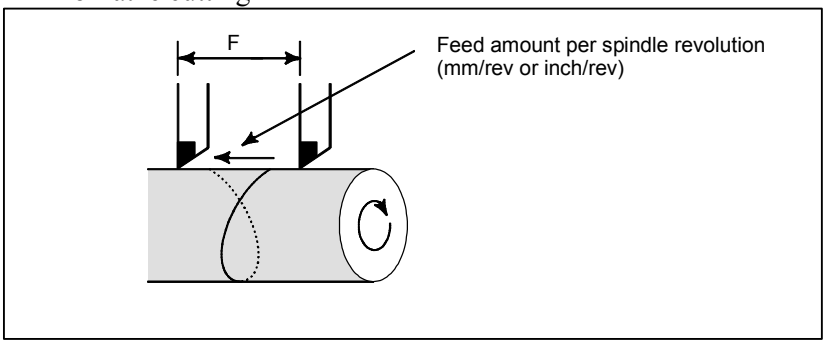


Fig. 5.3 (c) Feed per revolution

> ⚠ **CAUTION**
> When the speed of the spindle is low, feedrate fluctuation may occur.
> The slower the spindle rotates, the more frequently feedrate fluctuation occurs.

- Inverse time feed

When G code for inverse time feed is specified, the inverse time specification mode (G93 mode) is set. Specify the inverse time (FRN) with an F code.
A value from 0.001 to 9999.999 can be specified as FRN, regardless of whether the input mode is inches or metric, or the increment system is IS-B or IS-C.

F code specification value	FRN
F1	0.001
F1 (*1)	1.000
F1.0	1.000
F9999999	9999.999
F9999 (*1)	9999.000
F9999.999	9999.999

> **NOTE**
> *1 Value specified in fixed-point format with bit 0 (DPI) of parameter No. 3401 set to 1

G code for inverse time feed is a modal G code and belongs to group 05 (includes G code for feed per revolution and G code for feed per minute).
When an F value is specified in inverse time specification mode and the feedrate exceeds the maximum cutting feedrate, the feedrate is clamped to the maximum cutting feedrate.
In the case of circular interpolation, the feedrate is calculated not from the actual amount of movement in the block but from the arc radius. This means that actual machining time is longer when the arc radius is longer than the arc distance and shorter when the arc radius is shorter than the arc distance. Inverse time feed can also be used for cutting feed in a canned cycle.

NOTE
1 In the inverse time specification mode, an F code is not handled as a modal code and therefore needs to be specified in each block. If an F code is not specified, alarm PS0011 (FEED ZERO (COMMAND)) is issued.
2 When F0 is specified in inverse time specification mode, alarm PS0011 (FEED ZERO (COMMAND)) is issued.
3 Inverse time feed cannot be used when PMC axis control is in effect.
4 If the calculated cutting feedrate is smaller than the allowable range, alarm PS0011 (FEED ZERO (COMMAND)) is issued.

Example

- **For linear interpolation (G01)**

$$FRN = \frac{1}{time(min)} = \frac{feedrate}{distance}$$

Feedrate: mm/min (for metric input)
inch/min (for inch input)
Distance: mm (for metric input)
inch (for inch input)

- To end a block in 1 (min)

$$FRN = \frac{1}{time(\min)} = \frac{1}{1(\min)} = 1$$

Specify F1.0.

- To end a block in 10 (sec)

$$FRN = \frac{1}{time(\sec)/60} = \frac{1}{10/60(\sec)} = 6$$

Specify F6.0.

- To find the movement time required when F0.5 is specified

$$TIME(\min) = \frac{1}{FRN} = \frac{1}{0.5} = 2$$

2 (min) is required.

- To find the movement time required when F10.0 is specified

$$TIME(\min) = \frac{1 \times 60}{FRN} = \frac{60}{10} = 6$$

6 (sec) is required.

- **For circular interpolation (G02, G03)**

$$FRN = \frac{1}{time(\min)} = \frac{feedrate}{arcradius}$$

Feedrate: mm/min (for metric input)
inch/min (for inch input)
Arc radius: mm (for metric input)
inch (for inch input)

NOTE
In the case of circular interpolation, the feedrate is calculated not from the actual amount of movement in the block but from the arcadias.

- One-digit F code feed

M

When a one-digit number from 1 to 9 is specified after F, the feedrate set for that number in a parameter Nos. 1451 to 1459 is used. When F0 is specified, the rapid traverse rate is applied.
The feedrate corresponding to the number currently selected can be increased or decreased by turning on the switch for changing one-digit F feedrate on the machine operator's panel, then by rotating the manual pulse generator.
The increment/decrement, ΔF, in feedrate per scale of the manual pulse generator is as follows:

$$\Delta F = \frac{F\max}{100X}$$

Fmax : Feedrate upper limit for F1-F4 set by parameter (No.1460), or feedrate upper limit for F5-F9 set by parameter (No.1461)
X : Any value of 1-127 set by parameter No.1450

The feedrate set or altered is kept even while the power is off. The current feed rate is displayed on the LCD screen.

- Cutting feedrate clamp

Parameter No. 1430 can be used to specify the maximum cutting feedrate for each axis. When the cutting feedrate along an axis exceeds the maximum feedrate for the axis as a result of interpolation, the cutting feedrate is clamped to the maximum feedrate.

Reference

See Appendix D for range of feedrate command value.

5.4 CUTTING FEEDRATE CONTROL

Cutting feedrate can be controlled, as indicated in Table 5.4 (a).

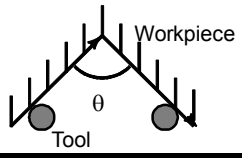
Table 5.4 (a) Cutting Feedrate Control

Function name		G code	Validity of G code	Description
Exact stop		G09	This function is valid for specified blocks only.	The tool is decelerated at the end point of a block, then an in-position check is made. Then the next block is executed.
Exact stop mode		G61	Once specified, this function is valid until G62, G63, or G64 is specified.	The tool is decelerated at the end point of a block, then an in-position check is made. Then the next block is executed.
Cutting mode		G64	Once specified, this function is valid until G61, G62, or G63 is specified.	The tool is not decelerated at the end point of a block, but the next block is executed.
Tapping mode		G63	Once specified, this function is valid until G61, G62, or G64 is specified.	The tool is not decelerated at the end point of a block, but the next block is executed. When G63 is specified, feedrate override and feed hold are invalid.
Automatic corner override	Automatic override for inner corners	G62	Once specified, this function is valid until G61, G63, or G64 is specified.	When the tool moves along an inner corner during tool radius compensation, override is applied to the cutting feedrate to suppress the amount of cutting per unit of time so that a good surface finish can be produced.
	Internal circular cutting feedrate change	-	This function is valid in the tool radius compensation mode, regardless of the G code.	The internal circular cutting feedrate is changed.

NOTE

1 The purpose of in-position check is to check that the servo motor has reached within a specified range (specified with a parameter by the machine tool builder).
In-position check is not performed when bit 5 (NCI) of parameter No. 1601 is set to 1.

2 Inner corner angle θ: $2° < \theta \leq \alpha \leq 178°$
(α is a set value)



Format

Exact stop	**G09 IP_ ;**
Exact stop mode	**G61 ;**
Cutting mode	**G64 ;**
Tapping mode	**G63 ;**
Automatic corner override	**G62 ;**

5.4.1 Exact Stop (G09, G61), Cutting Mode (G64), Tapping Mode (G63)

Explanation

The inter-block paths followed by the tool in the exact stop mode, cutting mode, and tapping mode are different (Fig. 5.4.1 (a)).

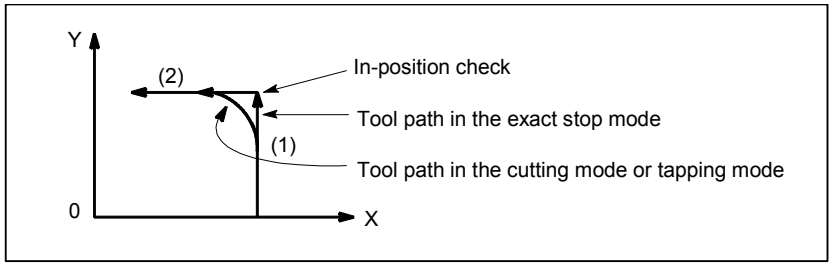


Fig. 5.4.1 (a) Example of tool paths from block (1) to block (2)

> ⚠ **CAUTION**
> The cutting mode (G64 mode) is set at power-on or system clear.

5.4.2 Automatic Corner Override

When tool radius compensation is performed, the movement of the tool is automatically decelerated at an inner corner and internal circular area. This reduces the load on the tool and produces a smoothly machined surface.

5.4.2.1 Automatic override for inner corners (G62)

Explanation

- Override condition

When G62 is specified, and the tool path with tool radius compensation applied forms an inner corner, the feedrate is automatically overridden at both ends of the corner.
There are four types of inner corners (Fig. 5.4.2(a)).
$2° \leq \theta \leq \theta p \leq 178°$ in Fig. 5.4.2(a)qp is a value set with parameter No. 1711. When θ is approximately equal to θp, the inner corner is determined with an error of 0.001° or less.

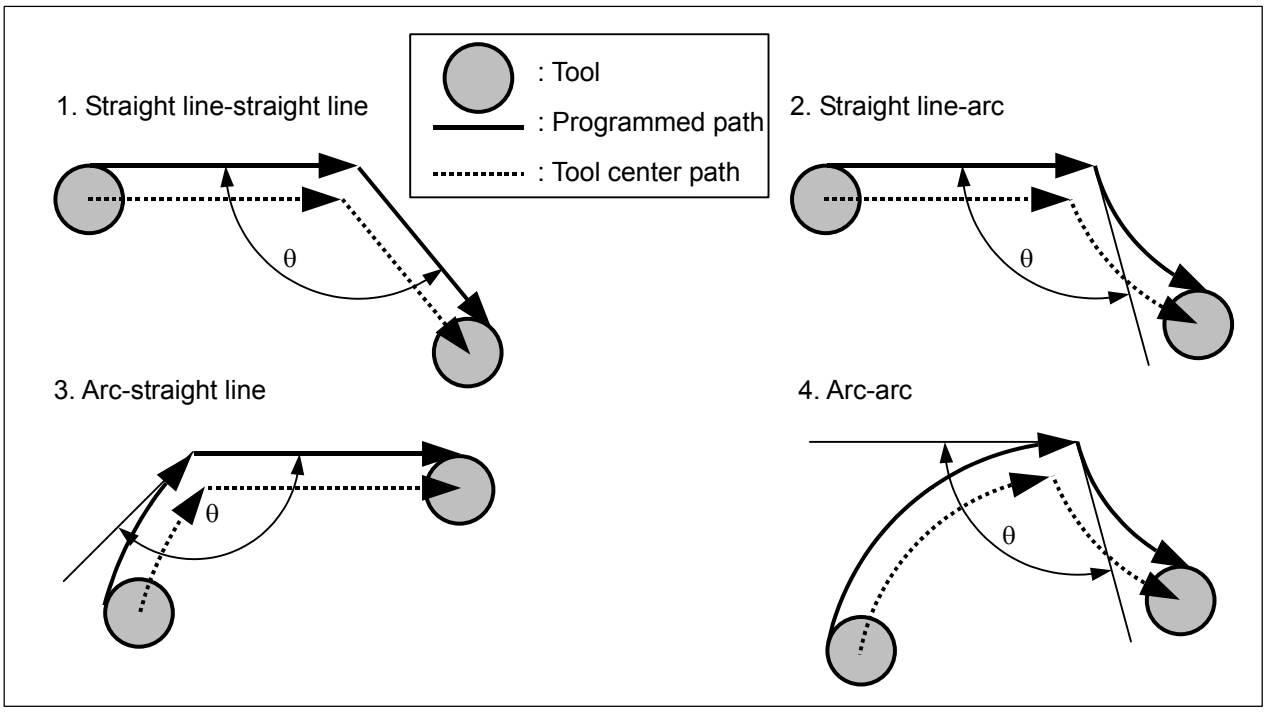


Fig. 5.4.2(a) Inner corner

- Override range

When a corner is determined to be an inner corner, the feedrate is overridden before and after the inner corner. The distances Ls and Le, where the feedrate is overridden, are distances from points on the tool center path to the corner (Fig. 5.4.2(b), Fig. 5.4.2(c), Fig. 5.4.2(d)). Ls and Le are set with parameter Nos. 1713 and 1714.

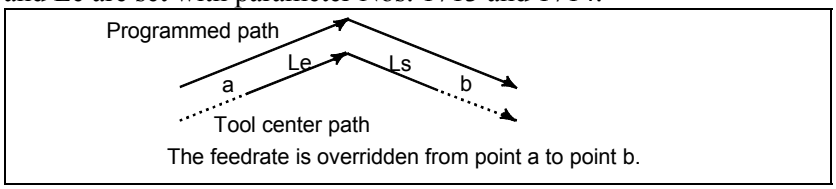


FIg. 5.4.2.1 (b) Override Range (Straight Line to Straight Line)

When a programmed path consists of two arcs, the feedrate is overridden if the start and end points are in the same quadrant or in adjacent quadrants (Fig. 5.4.2(c)).

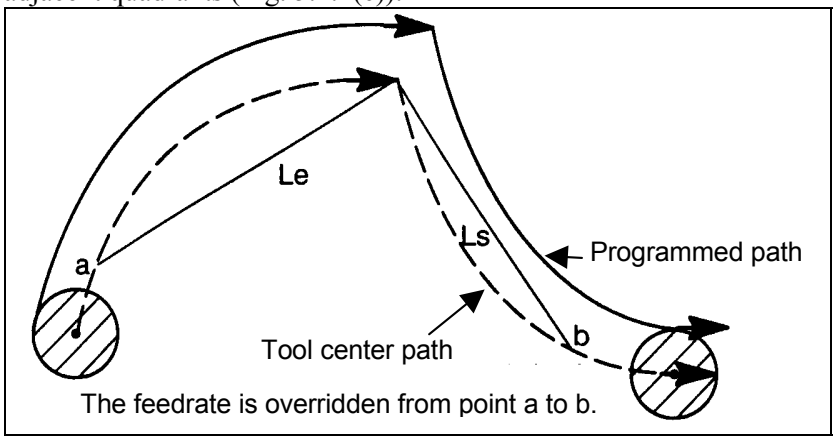


Fig. 5.4.2(c) Override Range (Arc to Arc)

Regarding program (2) of an arc, the feedrate is overridden from point a to point b and from point c to point d (Fig. 5.4.2(d)).

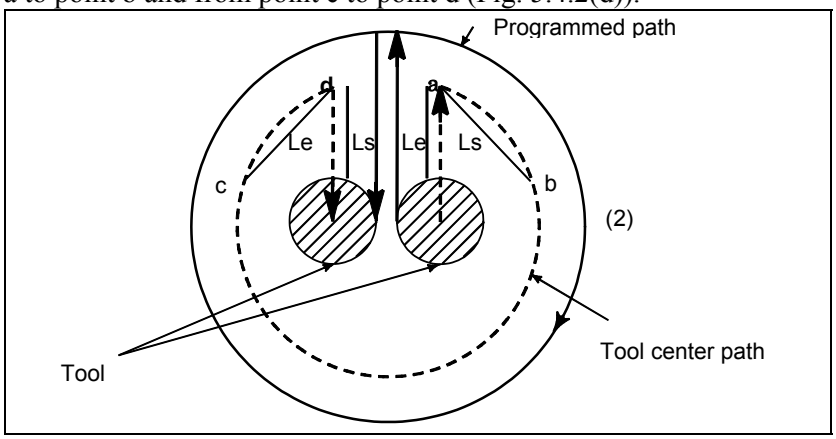


Fig. 5.4.2(d) Override Range (Straight Line to Arc, Arc to Straight Line)

- Override value

An override value is set with parameter No. 1712. An override value is valid even for dry run and one-digit F code feed specification.
In the feed per minute mode, the actual feedrate is as follows:

$$F = (\text{automatic override for inner corners}) \times (\text{feedrate override})$$

Limitation

- Acceleration/deceleration before interpolation

Override for inner corners is disabled during acceleration/deceleration before interpolation.

- Start-up/G41, G42

Override for inner corners is disabled if the corner is preceded by a start-up block or followed by a block including G41 or G42.

- Offset

Override for inner corners is not performed if the offset is zero.

5.4.2.2 Internal circular cutting feedrate change

For internally offset circular cutting, the feedrate on a programmed path is set to a specified feedrate (F) by specifying the circular cutting feedrate with respect to F, as indicated below (Fig. 5.4.2(e)). This function is valid in the tool radius compensation mode, regardless of the G62 code.

$$F = \frac{Rc}{Rp}$$

Rc : Tool center path radius
Rp : Programmed radius
It is also valid for the dry run and the one-digit F code feed command.

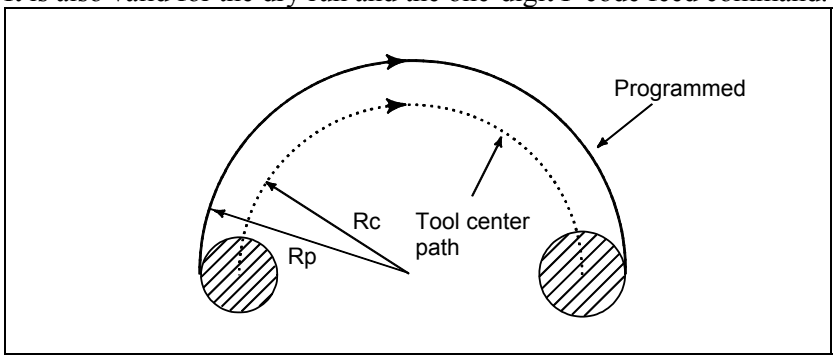


Fig. 5.4.2(e) Internal circular cutting feedrate change

If Rc is much smaller than Rp, Rc/Rp ≒ 0; the tool stops. A minimum deceleration ratio (MDR) is to be specified with parameter No. 1710. When Rc/Rp≤MDR, the feedrate of the tool is (F×MDR).

> ⚠ **CAUTION**
> When internal circular cutting must be performed together with override for inner corners, the feedrate of the tool is as follows:
> $$F \times \frac{Rc}{Rp} \times (\text{override for the inner corners}) \times (\text{feedrate override})$$

5.5 FEEDRATE INSTRUCTION ON IMAGINARY CIRCLE FOR A ROTARY AXIS

Overview

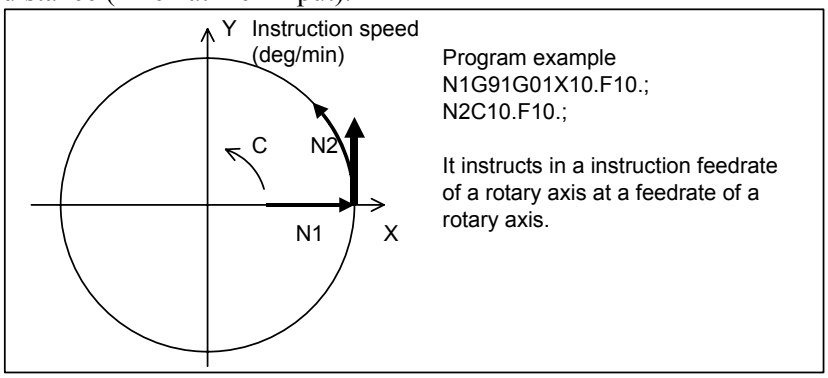
This function acquires movement feedrate on imaginary circle by synthetic movement distance is calculated from movement distance of a rotary axis by using instruction angle and the parameter of an imaginary radius (No.1465).
Then, movement feedrate on imaginary circle is feedrate of a rotary axis.

Explanation

Cutting feedrate

- Conventional method

In linear interpolation of a linear axis and a rotary axis, 1deg in movement angle of a rotary axis is interpolated as 1mm in movement distance (1inch at inch input).



Feedrate of liner axis(X axis) $F_X = F \times \frac{\Delta X}{L}\ (mm/min)$

Feedrate of rotary axis(C axis) $F_C = F \times \frac{\Delta C}{L}\ (deg/min)$

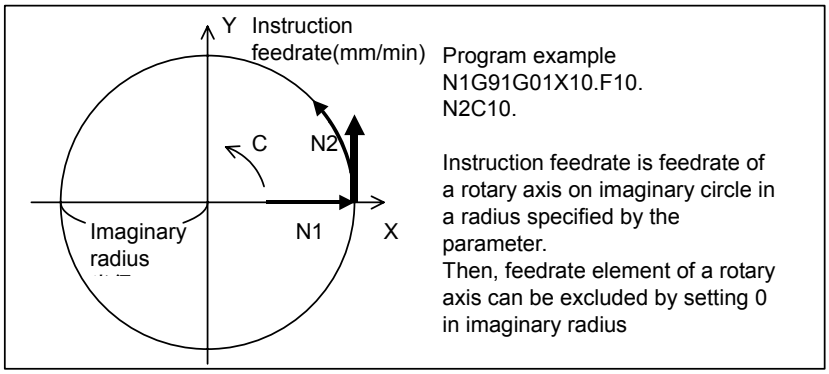
Synthetic movement distance

$$L = \sqrt{\Delta X^2 + \Delta Y^2 + \Delta Z^2 + \Delta B^2 + \Delta C^2}\ (mm)$$

Movement time $T = \frac{L}{F}\ (min)$

- Feedrate instruction on imaginary circle of a rotary axis

In this function, synthetic movement distance is obtained based on movement distance of a rotary axis requested from instruction angle and the parameter of an imaginary radius(parameter No.1465).



Feedrate of liner axis(X axis) $F_X = F \times \dfrac{\Delta X}{L'}\ (mm/min)$

Feedrate of rotary axis(C axis) $F_C = F \times \dfrac{\Delta C}{L'}\ (deg/min)$

Synthetic movement distance

$$L' = \sqrt{\Delta X^2 + \Delta Y^2 + \Delta Z^2 + \left(\frac{\pi \times l_B \times \Delta B}{180}\right)^2 + \left(\frac{\pi \times l_C \times \Delta C}{180}\right)^2}\ (mm)$$

Movement time $T' = \dfrac{L'}{F}\ (min)$

l_B , l_C : imaginary radius(parameter No.1465)

It becomes feedrate from which movement feedrate on imaginary circle is instructed by this. <Example 1 reference>

In this function, feedrate of a axis becomes L/L' times for the feedrate displayed on the NC screen from difference of the method of obtaining movement distance. Especially, a movement of a axis quickens when small value is set to an imaginary radius. Note input of the parameter enough.
Cutting feedrate is clamped based on the maximum cutting feedrate parameter (No.1430) and feedrate of an actual axis (data before this function is converted). Therefore, it is possible to instruct at feedrate more than setting the maximum cutting feedrate by setting big value in an imaginary radius (parameter No.1465). When small value is set in an imaginary radius, it is clamped at the feedrate following setting the maximum cutting feedrate.
Moreover, dry run becomes effective as for this function, too.

AI Contour Control

AI contour control is done to movement feedrate on imaginary circle.
Therefore, It is likely not to become feedrate in the calculation in this function by feedrate control of AI contour control. Then, feedrate of AI contour Control is clamped at parameter(No.1432). Moreover, it is clamped at parameter(No.8465), When the parameter(No.8465) is not 0.

0mm in imaginary radius

When an imaginary radius is assumed 0mm, synthesized distance is as follows because the movement distance of a rotary axis becomes 0mm.

$$L' = \sqrt{\Delta X^2 + \Delta Y^2 + \Delta Z^2}$$

A movement feedrate of a linear axis can be instruction feedrate F by excluding feedrate element of a rotary axis.<Example 2 reference>
Moreover, it moves at the maximum cutting feedrate in case of this setting and instruction only in a rotary axis.

Examples

- Example 1

When the follwing block is instructed on IS-B,
G91 G01 C10. F10. ;

The calculation is as follows, when 10.000(10mm) is set in an imaginary radius(parameter No.1465).

$$L' = \sqrt{\left(\frac{\pi \times l_C \times \Delta B}{180}\right)^2} = \sqrt{\left(\frac{\pi \times 10_{(mm)} \times 10_{(deg)}}{180}\right)^2} = 1.7453292\cdots_{(mm)}$$

$$F_C = 10_{(mm/\min)} \times \frac{10_{(deg)}}{1.7453292\cdots_{(mm)}} = 57.2957795\cdots_{(deg/min)}$$

$$T' = \frac{L'}{F} = \frac{1.7453292\cdots_{(mm)}}{10_{(mm/min)}} = 0.17453292\cdots_{(min)} = 10.4719755\cdots_{(sec)}$$

Therefor, the movement time becomes about 10.472(sec), and the rotation feedrate becomes about 57.296(deg/min). The feedrate on 10.000mm in an imaginary radius becomes 10.000mm/min at instruction feedrate in Fig.5.5(a).

The calclation is as follows, when 36.000(36mm) is set in an imaginary radius(parameter No.1465).

$$L' = \sqrt{\left(\frac{\pi \times l_C \times \Delta B}{180}\right)^2} = \sqrt{\left(\frac{\pi \times 36_{(mm)} \times 10_{(deg)}}{180}\right)^2} = 6.28318530\cdots_{(mm)}$$

$$F_C = 10_{(mm/min)} \times \frac{10_{(deg)}}{6.28318530\cdots_{(mm)}} = 15.9154943\cdots_{(deg/min)}$$

$$T' = \frac{L'}{F} = \frac{6.28318530\cdots_{(mm)}}{10_{(mm/min)}} = 0.628318530\cdots_{(min)} = 37.6991118\cdots_{(sec)}$$

Therefor, the movement time becomes about 37.700(sec), and the rotation feedrate becomes about 15.915(deg/min). The feedrate on 36.000mm in an imaginary radius becomes 10.000mm/min at instruction feedrate in Fig.5.5(a).

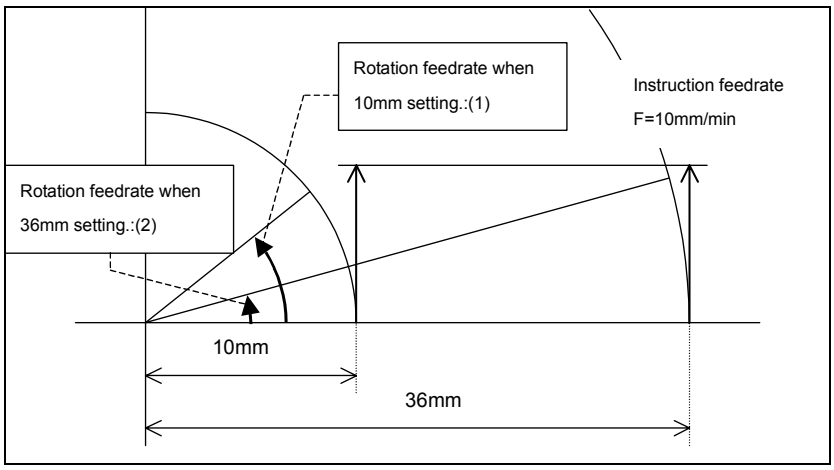


Fig. 5.5 (a)

- Example 2

When a machine into which direction of a tool changes by a rotary axis like Fig2, movement feedrate of a standard point can be assumed to be instruction feedrate F by making this function effective and setting 0 in imaginary radius(parameter No.1465) .

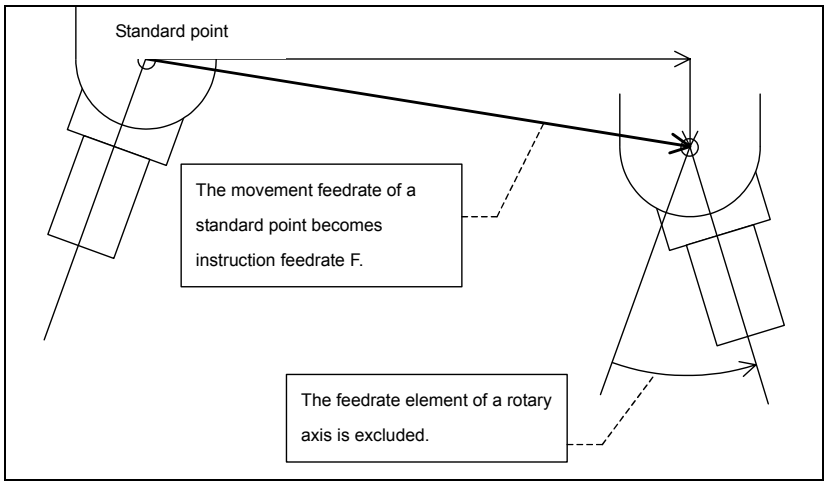


Fig. 5.5 (b)

Limitation

This function corresponds only the linear interpolation(G01).
However, it doesn't correspond to the following functions.

- Inverse time feed
- Feed per revolution
- Normal direction control
- High-speed cycle cutting
- Cylindrical interpolation
- Polar coordinate interpolation
- Tool center point control for 5-axis machining
- Axis control by PMC
- Manual numerical command

In this function, in case of using the parallel axis control and the twin table control, an imaginary radius parameter(No.1465) specifies the same value with a master axis and a slave axis.

NOTE

1 When the parameter ROTx (No.1006#0) and the parameter RFDx (No.1408#0) are 1, this function becomes effective.
2 The parameter RFDx (No.1408#0) and an imaginary radius (parameter No.1465) of this function can be rewriting by the programable parameter input(G10).
3 It moves at the maximum cutting feedrate when this function effectively and sets 0 in an imaginary radius (parameter No.1465) and it instructs only a rotary axis.
4 Note setting RFDx (No.1408#0) and an imaginary radius (parameter No.1465) enough. Especially, a movement of axis quickens compared with this function unused when small value is set to an imaginary radius.
5 In this function, the same value as the parameter value (No.1408, No.1465) of a master axis is used with a slave axis, when it uses axis synchronous control.

5.6 DWELL

Format

M

G04 X_; or G04 P_;
X_ : Specify a time or spindle speed (decimal point permitted)
P_ : Specify a time or spindle speed (decimal point not permitted)

T

G04 X_ ; or G04 U_ ; or G04 P_ ;
X_ : Specify a time or spindle speed (decimal point permitted)
U_ : Specify a time or spindle speed (decimal point permitted)
P_ : Specify a time or spindle speed (decimal point not permitted)

Explanation

By specifying a dwell, the execution of the next block is delayed by the specified time. (Dwell per second)
By setting bit 1 (DWL) of parameter No. 3405 in the feed per revolution mode, the execution of the next block is delayed until the rotation count of the spindle reaches the specified number. (Dwell per revolution)

Table 5.6 (a) Command value range of the dwell time (Command by X or U)

Increment system	Command value range	Dwell time unit
IS-A	0.01 to 999999.99	s or rev
IS-B	0.001 to 99999.999	
IS-C	0.0001 to 9999.9999	
IS-D	0.00001 to 999.99999	
IS-E	0.000001 to 99.999999	

Table 5.6 (b) Command value range of the dwell time (Command by P)

Increment system	Command value range	Dwell time unit
IS-A	1 to 99999999	0.01 s or rev
IS-B	1 to 99999999	0.001 s or rev
IS-C	1 to 99999999	0.0001 s or rev
IS-D	1 to 99999999	0.00001 s or rev
IS-E	1 to 99999999	0.000001 s or rev

In the case of dwell per second, the specification unit for dwell time specified with P can be fixed at 0.001 second by setting bit 7 (DWT) of parameter No. 1015 to 1.

NOTE

1 When X, U, or P is specified without a decimal point, the specification unit does not depend on inch/metric input. Depending on whether the X-axis is present, the following increment system is used:
   - When the X-axis is present
     The increment system of the X-axis is used.
   - When the X-axis is not present
     The increment system of the reference axis is used.

2 When P is specified, bit 7 (IPR) of parameter No. 1004 exercises no influence.

M

Specify dwell also to make an exact check in the cutting mode (G64 mode).
If the specification of P and X is omitted, an exact stop occurs.

6 REFERENCE POSITION

A CNC machine tool has a special position where, generally, the tool is exchanged or the coordinate system is set, as described later. This position is referred to as a reference position.

Chapter 6, "REFERENCE POSITION", consists of the following sections:

6.1 REFERENCE POSITION RETURN.....177
6.2 FLOATING REFERENCE POSITION RETURN (G30.1) 184

6.1 REFERENCE POSITION RETURN

Overview

- Reference position

The reference position is a fixed position on a machine tool to which the tool can easily be moved by the reference position return function. For example, the reference position is used as a position at which tools are automatically changed. Up to four reference positions can be specified by setting coordinates in the machine coordinate system in parameters (No. 1240 to 1243).

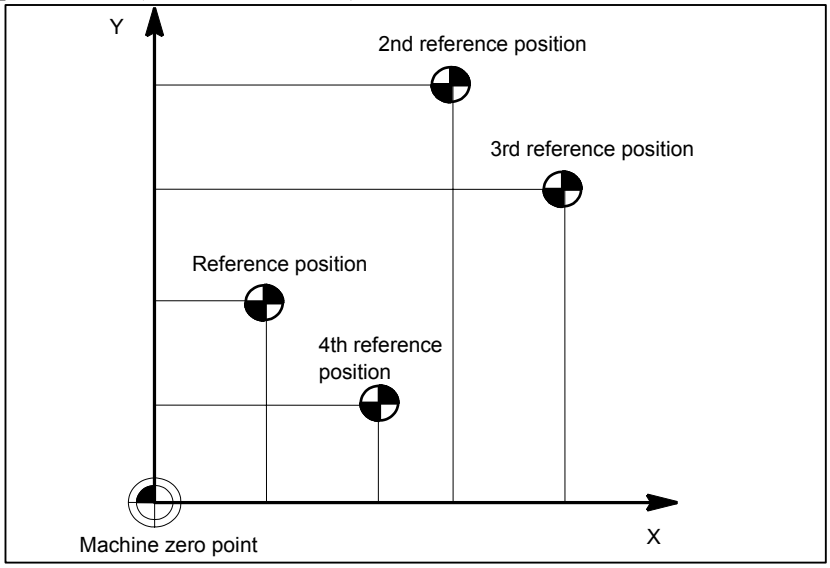


Fig. 6.1 (a) Machine zero point and reference positions

- Automatic reference position return (G28) and movement from the reference position (G29)

The automatic reference position return (G28) function returns tools automatically to the reference position via an intermediate position along a specified axis. When reference position return is completed, the lamp for indicating the completion of reference position return goes on. The return from reference position (G29) function moves tools from the reference position to a specified position via an intermediate position along a specified axis.

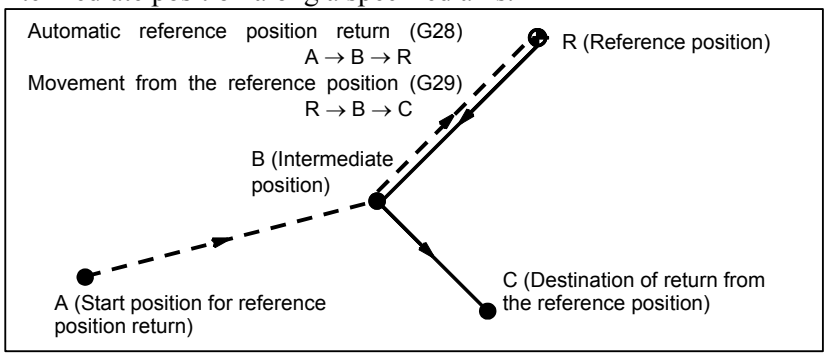


Fig. 6.1 (b) Reference position return and return form the reference position

- Reference position return check (G27)

The reference position return check (G27) is the function which checks whether the tool has correctly returned to the reference position as specified in the program. If the tool has correctly returned to the reference position along a specified axis, the lamp for the axis for indicating the completion of reference position return goes on.
If the tool has not reached the reference position, an alarm (PS0092) "ZERO RETURN CHECK (G27) ERROR" is issued.
When no movement was made along the axis, whether the current position is the reference position is checked.

Format

- Automatic reference position return and 2nd/3rd/4th reference position return

G28 IP_; Reference position return
G30 P2 IP_; 2nd reference position return (P2 can be omitted.)
G30 P3 IP_; 3rd reference position return
G30 P4 IP_; 4th reference position return

IP : Specify the intermediate position in the absolute coordinate system.
(absolute/incremental programming)

There is no need to calculate an actual travel distance between the intermediate position and the reference position.

- Movement from reference position

G29 IP_;

IP : Specify the destination of return from the reference position in the absolute coordinate system.
(absolute/incremental programming)
The intermediate position is determined by G28, G30, or G30.1 specified immediately before this command.

- Reference position return check

G27 IP_;

IP : Specify positioning to the reference position in the absolute coordinate system so as to return to the reference position. (absolute/incremental programming)

Explanation

- Automatic reference position return (G28)

Positioning to the intermediate or reference positions are performed at the rapid traverse rate of each axis.
Therefore, for safety, the compensation functions, such as the tool radius compensation and tool length compensation, should be cancelled before executing this command.
The coordinates for the intermediate position are stored in the CNC for the axes for which a value is specified in a G28 block. For the other axes, the previously specified coordinates are used.

(Example)

N1 G28 X40.0 ;
(The tool moves to the reference position along the X-axis and the intermediate position (X40.0) is stored.)
N2 G28 Y60.0 ;
(The tool moves to the reference position along the Y-axis and the intermediate position (Y60.0) is stored.)
N3 G29 X10.0 Y20.0 ;
(The tool moves to the position specified with G29 via the intermediate position (X40.0 Y60.0) previously specified with G28 along the X-axis and Y-axis.)

- 2nd, 3rd, and 4th reference position return (G30)

The 2nd, 3rd, and 4th reference position return (G30) function can be used after the reference positions are established.
The G30 command is generally used when the automatic tool changer (ATC) position differs from the reference position.

- Movement from the reference position (G29)

This function is executed after the tool is returned to the reference position by G28 or G30.
For incremental programming, the command value specifies the incremental value from the intermediate point.
The tool moves to the intermediate and specified positions at the feedrate specified with a parameter.
When the workpiece coordinate system is changed after the tool reaches the reference position through the intermediate point by the G28 command, the intermediate point also shifts to a new coordinate system. If G29 is then commanded, the tool moves to the commanded position through the intermediate point which has been shifted to the new coordinate system.
The same operations are performed also for G30 and G30.1 commands.
After the power is turned on, an alarm (PS0305) is issued if an attempt is made to execute G29 (movement from the reference position) before G28 (automatic reference position return), G30 (2nd, 3rd, and 4th reference position return), or G30.1 (floating reference position return) is executed.

- Reference position return check (G27)

G27 command positions the tool at rapid traverse rate. If the tool reaches the reference position, the lamp for indicating the completion of reference position return lights up.
When the tool returns to the reference position along only one axis, the lamp for the axis for indicating the completion of reference position return lights up.
After positioning, if the tool has not reached the reference position along a specified axis, an alarm (PS0092) "ZERO RETURN CHECK (G27) ERROR" is issued.
When no movement was made along the axis, whether the current position is the reference position is checked.

- Setting of the reference position return feedrate

Before a coordinate system is established with the first reference position return after power-on, the manual and automatic reference position return feedrates and automatic rapid traverse rate conform to the setting of parameter No. 1428 for each axis.
After a reference position is established upon the completion of reference position return, the manual reference position return feedrate conforms to the setting of the parameter No. 1428 for each axis.

NOTE

1 To this feedrate, a rapid traverse override (F0,25,50,100%) is applied, for which the setting is 100%.
2 After a reference position has been established upon the completion of reference position return, the automatic reference position return feedrate will conform to the ordinary rapid traverse rate.
3 When a value is set for parameter No. 1428, the feedrates conform to the parameter settings shown below.

	Before a coordinate system is established	After a coordinate system is established
Automatic reference position return (G28)	No. 1428	No.1420
Automatic rapid traverse (G00)	No.1428	No.1420
Manual reference position return (*1)	No.1428	No.1428 (*3)
Manual rapid traverse rate	No.1423 (*2)	No.1424

1420: Rapid traverse rate
1423: Jog feedrate
1424: Manual rapid traverse rate
1428: Reference position return feedrate

When parameter No. 1428 is set to 0, the feedrates conform to the parameter settings shown below.

	Before a coordinate system is established	After a coordinate system is established
Automatic reference position return (G28)	No. 1420	No.1420
Automatic rapid traverse (G00)	No.1420	No.1420
Manual reference position return (*1)	No.1424	No.1424 (*3)
Manual rapid traverse rate	No.1423 (*2)	No.1424

*1 By using JZR (bit 2 of parameter No. 1401), the manual reference position return feedrate can always be set as a jog feedrate.
*2 When RPD (bit 0 of parameter No. 1401) is 1, the setting of parameter No. 1424 (manual rapid traverse rate) is used.
When the setting of parameter No. 1424 (manual rapid traverse rate) is 0, parameter No. 1420 (rapid traverse rate) is used.
*3 When reference position return without dogs is performed in rapid traverse mode, or when manual reference position return is performed in rapid traverse mode regardless of deceleration dogs after a reference position is established, the reference position return feedrate for each of these functions (setting of DLF (bit 1 of parameter No. 1404)) is used.

Limitation

- Status the machine lock being turned on

The lamp for indicating the completion of reference position return does not go on when the machine lock is turned on, even when the tool has automatically returned to the reference position. In this case, it is not checked whether the tool has returned to the reference position even when a reference position return check command is specified.

- When automatic reference position return (G28) is executed if no reference position is established

When automatic reference position return (G28) is executed if no reference position is established, movement from the intermediate position in a reference position direction is the same as that in manual reference position return.
(This movement is referred to as a low-speed type of automatic reference position return (G28).)
In this case, the tool moves in the direction for reference position return specified in parameter ZMIx (bit 5 of No. 1006). Therefore the specified intermediate position must be a position to which reference position return is possible.

NOTE
When automatic reference position return (G28) is executed after a reference position is established, positioning is performed from the intermediate position to the reference position. This movement is referred to as a high-speed type of automatic reference position return (G28).

- Reference position return check in an offset mode

In an offset mode, the position to be reached by the reference position return check is the position obtained by adding the offset value.
Therefore, if the position with the offset value added is not the reference position, the lamp for indicating the completion of reference position return does not light up, but an alarm is displayed instead. Usually, cancel offsets before G27 is commanded.

- Lighting the lamp when the programmed position does not coincide with the reference position

When the machine tool system is an inch system with metric input, the lamp for indicating the completion of reference position return may also light up even if the programmed position is shifted from the reference position by the least setting increment. This is because the least setting increment of the machine tool system is smaller than its least command increment.

Example

G28G90X1000.0Y500.0 ;
(Programs movement from A to B. The tool moves to reference position R via intermediate position B.)
T1111 ;
(Changing the tool at the reference position)
G29X1300.0Y200.0 ;
(Programs movement from B to C. The tool moves from reference position R to C specified with G29 via intermediate position B.)

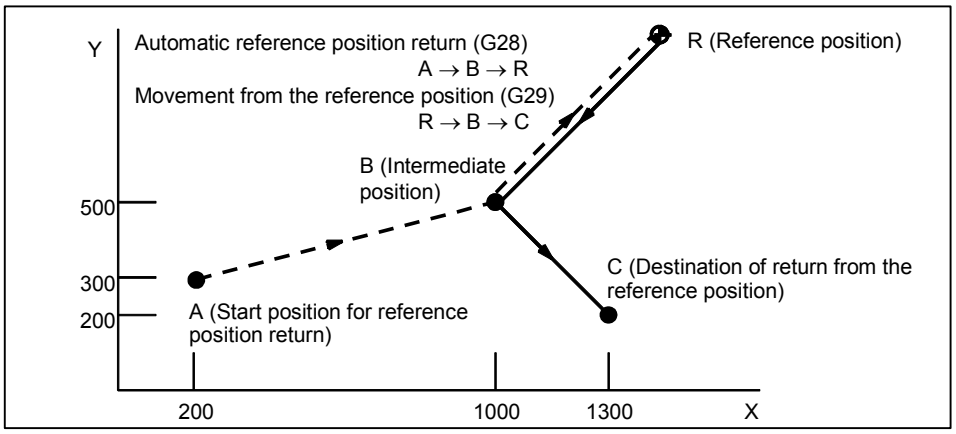


Fig. 6.1 (c) Reference position return and movement from the reference position

6.2 FLOATING REFERENCE POSITION RETURN (G30.1)

Tools ca be returned to the floating reference position.
A floating reference point is a position on a machine tool, and serves as a reference point for machine tool operation.
A floating reference point need not always be fixed, but can be moved as required.

Format

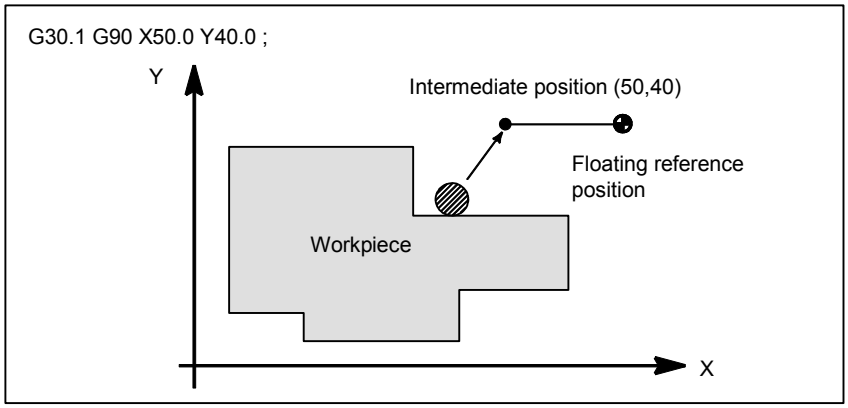
G30.1 IP_;

IP_ : Specify the intermediate position to the floating reference position in the absolute coordinate system. (absolute/incremental programming)

Explanation

Generally speaking, on a machining center or milling machine, cutting tools can be replaced only at specific positions. A position where tools can be replaced is defined as the second or third reference point. Using G30 can easily move the cutting tools back to these points.
On some machine tools, the cutting tools can be replaced at any position unless they interfere with the workpiece. With these machines, the cutting tools should be replaced at a position as close to the workpiece as possible so as to minimize the machine cycle time. For this purpose, the tool change position is to be changed, depending on the figure of the workpiece. This operation can easily be performed using this function.
That is, a tool change position suitable for the workpiece is memorized as a floating reference point. Then command G30.1 can easily cause return to the tool change position.
A floating reference point becomes a machine coordinate position memorized by pressing the soft key [SETFRP] on the current positions display screen. The coordinates of the floating reference position are stored with parameter No. 1244.
The G30.1 block first positions the tool at the intermediate point along the specified axes at rapid traverse rate, then further moves the tool from the intermediate point to the floating reference point at rapid traverse rate. Before using G30.1, cancel the compensation functions, such as tool radius compensation and tool length compensation.
A floating reference point is not lost even if power is turned off.
The function for movement from the reference position (G29) can be specified for moving the tool from the floating reference position.

Example


G30.1 G90 X50.0 Y40.0 ;
Y
Intermediate position (50,40)
Floating reference position
Workpiece
X

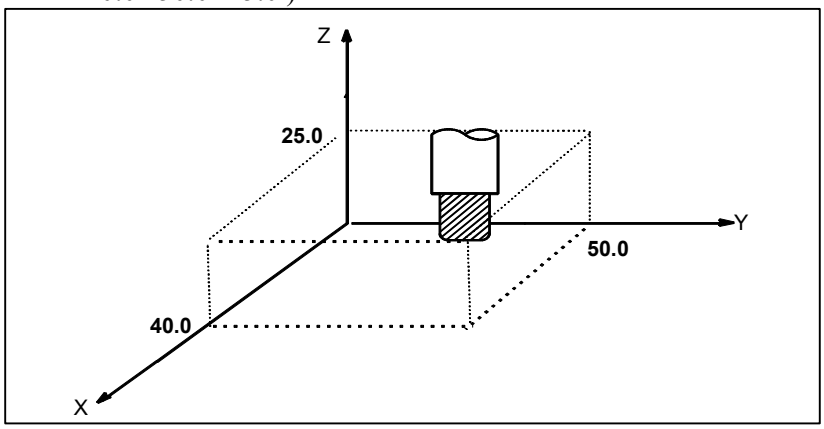
7 COORDINATE SYSTEM

By teaching the CNC a desired tool position, the tool can be moved to the position. Such a tool position is represented by coordinates in a coordinate system. Coordinates are specified using program axes.
When three program axes, the X-axis, Y-axis, and Z-axis, are used, coordinates are specified as follows:

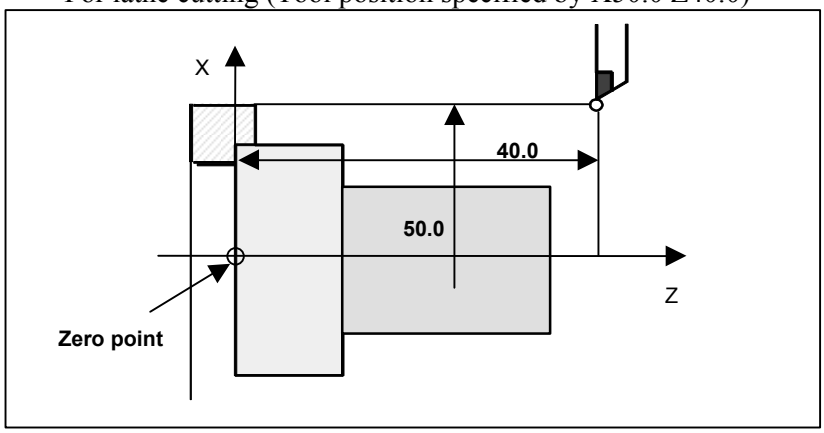
X_Y_Z_

This command is referred to as a dimension word.

- For milling machining (Tool position specified by X40.0Y50.0Z25.0)



- For lathe cutting (Tool position specified by X50.0 Z40.0)



Coordinates are specified in one of following three coordinate systems:

(1) Machine coordinate system
(2) Workpiece coordinate system
(3) Local coordinate system

The number of the axes of a coordinate system varies from one machine to another. So, in this manual, a dimension word is represented as IP_.

7.1 MACHINE COORDINATE SYSTEM

The point that is specific to a machine and serves as the reference of the machine is referred to as the machine zero point. A machine tool builder sets a machine zero point for each machine.
A coordinate system with a machine zero point set as its origin is referred to as a machine coordinate system.
A machine coordinate system is set by performing manual reference position return after power-on (see III-3.1). A machine coordinate system, once set, remains unchanged until the power is turned off.
The reference position is not always the origin of the machine coordinate system.
(See "Setting a machine coordinate system" described later.)

Format

M

(G90) G53 IP_ P1;
IP_: Absolute dimension word
P1: Enables the high-speed G53 function.

T

G53 IP_ P1;
IP_: Absolute dimension word
P1: Enables the high-speed G53 function.

Explanation

- Selecting a machine coordinate system (G53)

When a command is specified the position on a machine coordinate system, the tool moves to the position by rapid traverse. G53, which is used to select a machine coordinate system, is a one-shot G code; that is, it is valid only in the block in which it is specified on a machine coordinate system. Specify an absolute command for G53. When an incremental command is specified, the G53 command is ignored. When the tool is to be moved to a machine-specific position such as a tool change position, program the movement in a machine coordinate system based on G53.

- High-speed G53 function

This function enables the inter-rapid traverse block overlap function between machine coordinate selection command (G53) and positioning (rapid traverse) command (G00) blocks, thus making it possible to execute the next rapid traverse command (G00) without decelerating to a stop at the end of the machine coordinate selection command (G53). Therefore, high-speed positioning is available even when the machine coordinate selection command (G53) is used.
Specifying P1 in a G53 block enables the high-speed G53 function.

Limitation

- Cancel of the compensation function

When the G53 command is specified, cancel the compensation functions such as the cutter compensation, tool length compensation, tool nose radius compensation, and tool offset.

- G53 specification immediately after power-on

Since the machine coordinate system must be set before the G53 command is specified, at least one manual reference position return or automatic reference position return by the G28 command must be performed after the power is turned on. This is not necessary when an absolute-position detector is attached.

- Blocks in which the high-speed G53 function is usable

The high-speed G53 function is usable in the following combinations of commands:

- G53 → G00
- G53 → G53

The high-speed G53 function is unusable in the following combination of commands:

- G00 → G53

- Specification in the same block

M

Commands G50/G51, G50.1/G51.1, and G68/G69 cannot be specified in the same block where the G53 command is specified.

T

Commands G50/G51 (except for G code system A), G50.1/G51.1, and G68.1/G69.1 cannot be specified in the same block where the G53 command is specified.

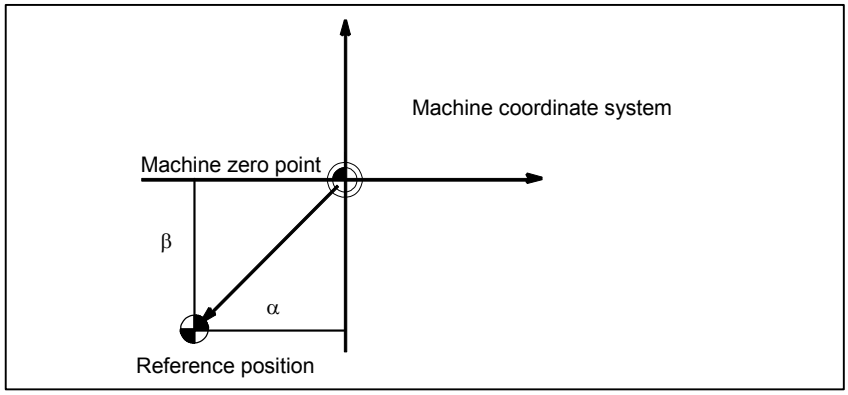
Note

NOTE
G53 is a G code for disabling buffering.

Reference

- Setting a machine coordinate system

When manual reference position return is performed after power-on, a machine coordinate system is set so that the reference position is at the coordinate values of (α, β) set using parameter No.1240.


Machine coordinate system
Machine zero point
β
α
Reference position

7.2 WORKPIECE COORDINATE SYSTEM

Overview

A coordinate system used for machining a workpiece is referred to as a workpiece coordinate system. A workpiece coordinate system is to be set with the CNC beforehand (setting a workpiece coordinate system).
A machining program sets a workpiece coordinate system (selecting a workpiece coordinate system).
A set workpiece coordinate system can be changed by shifting its origin (changing a workpiece coordinate system).

7.2.1 Setting a Workpiece Coordinate System

A workpiece coordinate system can be set using one of three methods:

(1) Method using a workpiece coordinate system setting G code
A workpiece coordinate system is set by specifying a value in the program after a workpiece coordinate system setting G code.
(2) Automatic setting
If bit 0 of parameter ZPR No. 1201 is set to 1, a workpiece coordinate system is automatically set when manual reference position return is performed (see III-3.1.).
This function is, however, disabled when the workpiece coordinate system option is being used.
(3) Method using a workpiece coordinate system selection G code
Six workpiece coordinate systems can be set beforehand using the MDI panel. Program commands G54 to G59 can be used to select the workpiece axis to be used. (see III-12.3.4.)

When using an absolute command, establish the workpiece coordinate system in any of the above ways.

Format

- Setting a workpiece coordinate system

M

```
(G90) G92 IP_
```

T

```
G50 IP_
```

Explanation

A workpiece coordinate system is set so that a point on the tool, such as the tool tip, is at specified coordinates.

M

If a coordinate system is set using G92 during tool length offset, a coordinate system in which the position before offset matches the position specified in G92 is set. Cutter compensation is cancelled temporarily with G92.

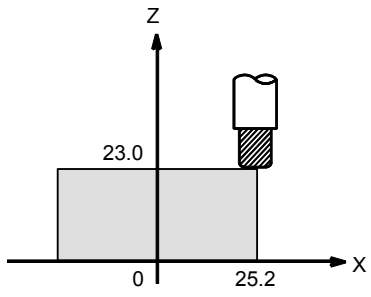
T

If IP_ is an incremental command value, the workpiece coordinate system is defined so that the current tool position coincides with the result of adding the specified incremental value to the coordinates of the previous tool position. If a coordinate system is set using G50 during offset, a coordinate system in which the position before offset matches the position specified in G50 is set.

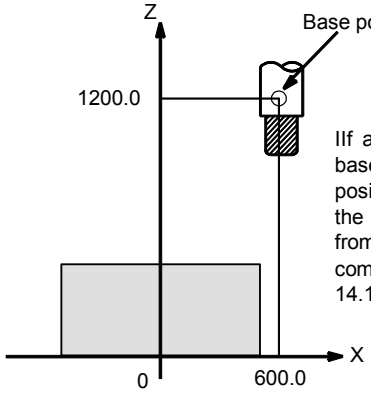
Example

M

(Example 1)
Setting the coordinate system by the G92X25.2Z23.0; command (The tool tip is the start point for the program.)

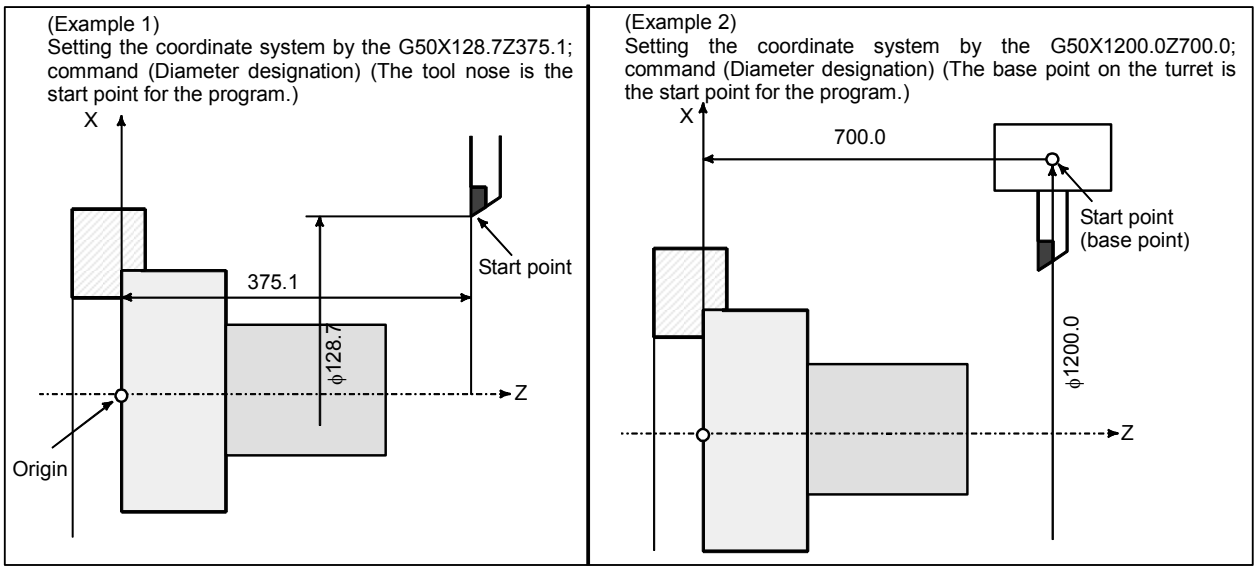


(Example 2)
Setting the coordinate system by the G92X600.0Z1200.0; command (The base point on the tool holder is the start point for the program.)



IIf an absolute command is issued, the base point moves to the commanded position. In order to move the tool tip to the commanded position, the difference from the tool tip to the base point is compensated by tool length offset (see II-14.1).

T



> ⚠ **CAUTION**
> The set workpiece coordinate system depends on diameter programming or radius programming.

Notes

- Command for setting a workpiece coordinate system in the tool length compensation mode

Executing a workpiece coordinate system setting G code command (G92 or, for G code system A in the lathe system, G50) presets a coordinate system in such a way that the specified position will be a pre-compensation position.
However, this G code cannot be used together with a block where tool length compensation vectors vary. For example, it cannot be used together with the following blocks.

> **EXAMPLE**
> 1 Block in which G43/G44 is issued
> 2 Block which is in the G43 or G44 mode and in which an H code is issued
> 3 Block which is in the G43 or G44 mode and in which G49 is issued
> 4 Block in which, in the G43 or G44 mode, compensation vectors are canceled using a G code such as G28 or G53 and then resumed again

When presetting a workpiece coordinate system, using the workpiece coordinate system setting G code, do not stop in the previous block to change a tool length compensation offset selected, for example, with the MDI.

7.2.2 Selecting a Workpiece Coordinate System

The user can choose from set workpiece coordinate systems as described below. (For information about the methods of setting, see II-7.2.1.)

(1) Once a workpiece coordinate system is set by a workpiece coordinate system setting G code or by automatic workpiece coordinate system setting, absolute commands indicate positions in the workpiece coordinate system.

(2) Choosing from six workpiece coordinate systems set using the MDI panel
By specifying a G code from G54 to G59, one of the workpiece coordinate systems 1 to 6 can be selected.
G54 : Workpiece coordinate system 1
G55 : Workpiece coordinate system 2
G56 : Workpiece coordinate system 3
G57 : Workpiece coordinate system 4
G58 : Workpiece coordinate system 5
G59 : Workpiece coordinate system 6
Workpiece coordinate system 1 to 6 are established after reference position return after the power is turned on. When the power is turned on, G54 coordinate system is selected.

When bit 2 (G92) of parameter No. 1202 is set to 1, executing the workpiece coordinate system setting G92 code command results in the issue of an alarm PS0010. This is designed to prevent the user from confusing coordinate systems.

> ⚠ **CAUTION**
> The set workpiece origin offset value depends on diameter programming or radius programming.

Example

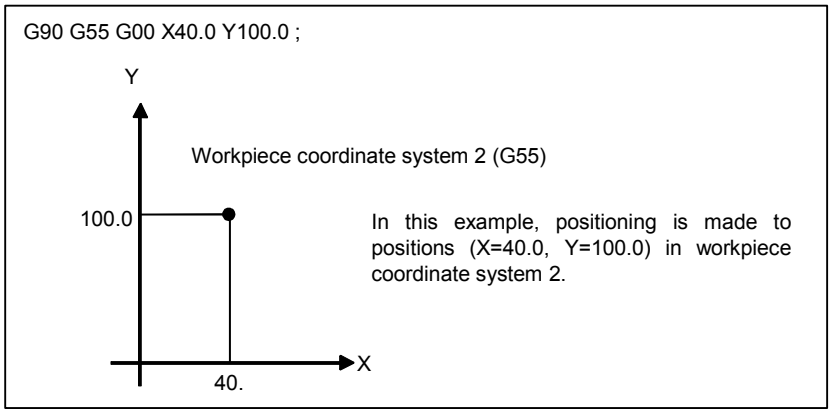


Fig. 7.2.2 (a)

7.2.3 Changing Workpiece Coordinate System

The six workpiece coordinate systems specified with G54 to G59 can be changed by changing an external workpiece origin offset value or workpiece origin offset value.
Three methods are available to change an external workpiece origin offset value or workpiece origin offset value.

(1) Inputting from the MDI panel (see III-12.3.4)
(2) Programming (using a programmable data input G code or a workpiece coordinate system setting G code)
(3) Using the external data input function
An external workpiece origin offset value can be changed by input signal to CNC. Refer to machine tool builder's manual for details.

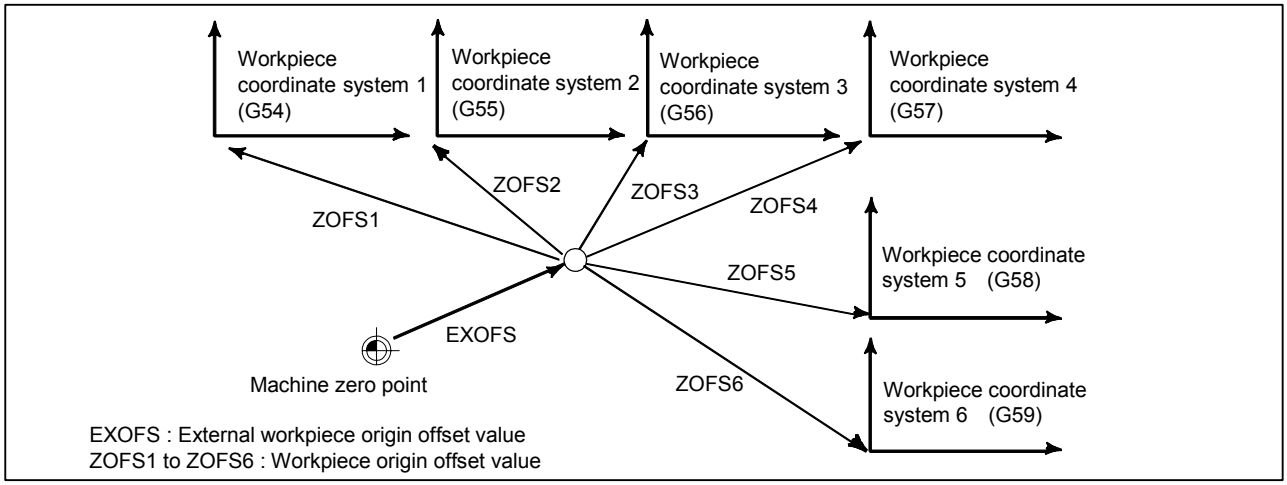


Fig. 7.2.3 (a) Changing an external workpiece origin offset value or workpiece origin offset value

Format

- Changing by inputting programmable data

G10 L2 Pp IP_;

p=0 : External workpiece origin offset value
p=1 to 6 : Workpiece origin offset value correspond to workpiece coordinate system 1 to 6
IP_ : For an absolute command, workpiece origin offset for each axis.
For an incremental command, value to be added to the set workpiece origin offset for each axis (the result of addition becomes the new workpiece origin offset).

- Changing by setting a workpiece coordinate system

M

```
G92 IP_ ;
```

T

```
G50 IP_ ;
```

Explanation

- Changing by inputting programmable data

By specifying a programmable data input G code, the workpiece origin offset value can be changed for each workpiece coordinate system.

- Changing by setting a workpiece coordinate system

By specifying a workpiece coordinate system setting G code, the workpiece coordinate system (selected with a code from G54 to G59) is shifted to set a new workpiece coordinate system so that the current tool position matches the specified coordinates (IP_).
Then, the amount of coordinate system shift is added to all the workpiece origin offset values. This means that all the workpiece coordinate systems are shifted by the same amount.

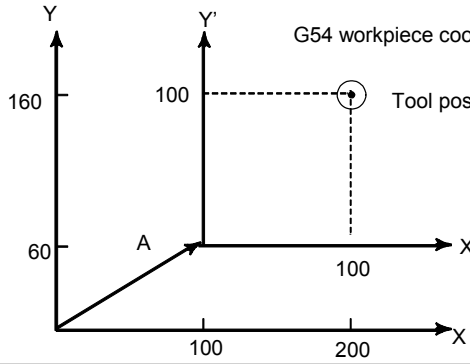
> ⚠ **CAUTION**
> When a coordinate system is set with workpiece coordinate system setting G92 code command after an external workpiece origin offset value is set, the coordinate system is not affected by the external workpiece origin offset value. When G92X100.0Z80.0; is specified, for example, the coordinate system having its current tool reference position at X = 100.0 and Z = 80.0 is set.

T

If IP is an incremental command value, the workpiece coordinate system is defined so that the current tool position coincides with the result of adding the specified incremental value to the coordinates of the previous tool position. (Coordinate system shift)

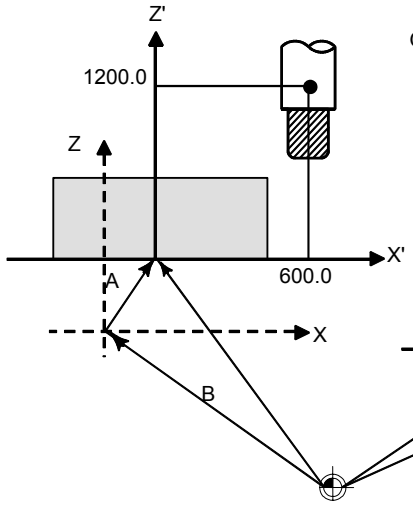
Example

M

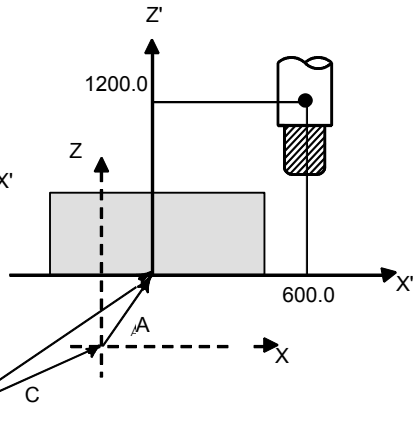


If G92X100Y100; is commanded when the tool is positioned at (200, 160) in G54 mode, workpiece coordinate system 1 (X' - Y') shifted by vector A is created.

G54 workpiece coordinate system



G55 workpiece coordinate system



Suppose that a G54 workpiece coordinate system is specified. Then, a G55 workpiece coordinate system where the black circle on the tool (figure at the left) is at (600.0,1200.0) can be set with the following command if the relative relationship between the G54 workpiece coordinate system and G55 workpiece coordinate system is set correctly:

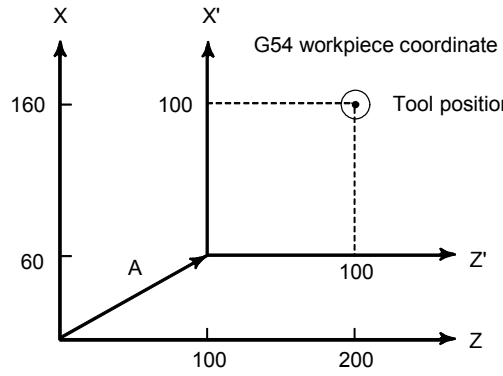
G92X600.0Z1200.0;

Also, suppose that pallets are loaded at two different positions. If the relative relationship of the coordinate systems of the pallets at the two positions is correctly set by handling the coordinate systems as the G54 workpiece coordinate system and G55 workpiece coordinate system, a coordinate system shift with G92 in one pallet causes the same coordinate system shift in the other pallet. This means that workpieces on two pallets can be machined with the same program just by specifying G54 or G55.

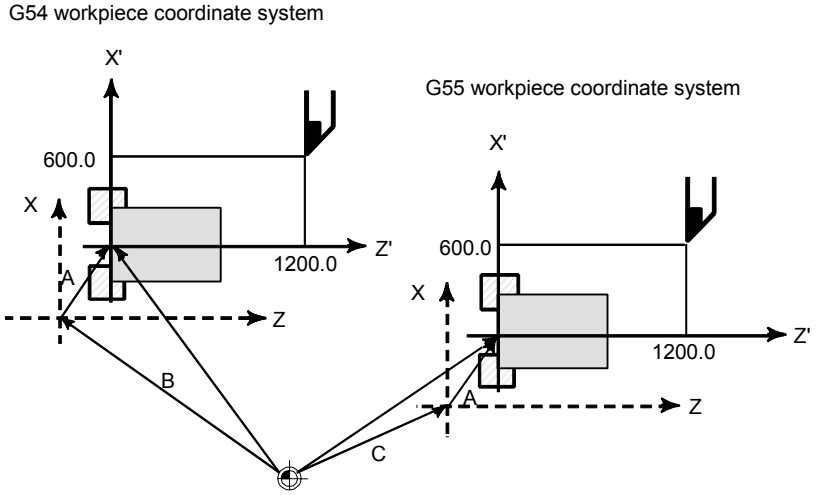
X' - Z' .....New workpiece coordinate system
X - Z .....Original workpiece coordinate system
A : Offset value created by G92
B : Workpiece origin offset value in the G54
C : Workpiece origin offset value in the G55

Example

T



If G50X100Z100; is commanded when the tool is positioned at (200, 160) in G54 mode, workpiece coordinate system 1 (X' - Z') shifted by vector A is created.



X' - Z' ..... New workpiece coordinate system
X - Z ..... Original workpiece coordinate system
A : Offset value created by G50
B : Workpiece origin offset value in G54
C : Workpiece origin offset value in G55

Suppose that a G54 workpiece coordinate system is specified. Then, a G55 workpiece coordinate system where the black tool nose point on the tool (figure at the left) is at (600.0,1200.0) can be set with the following command if the relative relationship between the G54 workpiece coordinate system and G55 workpiece coordinate system is set correctly:

G50X600.0Z1200.0;

Also, suppose that loading is performed at two different places and that the G54 and G55 workpiece coordinate systems are set for these two places. If the relative relationship between the coordinate systems in the two places is correctly set, a coordinate system shift with G50 in one loading place causes the same coordinate system shift in the other loading place. This means that workpieces on the two loading positions can be machined with the same program just by specifying G54 or G55.

7.2.4 Workpiece Coordinate System Preset (G92.1)

The workpiece coordinate system preset function presets a workpiece coordinate system shifted by manual intervention to the pre-shift workpiece coordinate system. The latter system is displaced from the machine zero point by a workpiece origin offset value.
There are two methods for using the workpiece coordinate system preset function. One method uses a programmed command. The other uses MDI operations on the absolute position display screen, relative position display screen, and overall position display screen (see III-12.1.4).

Format

M

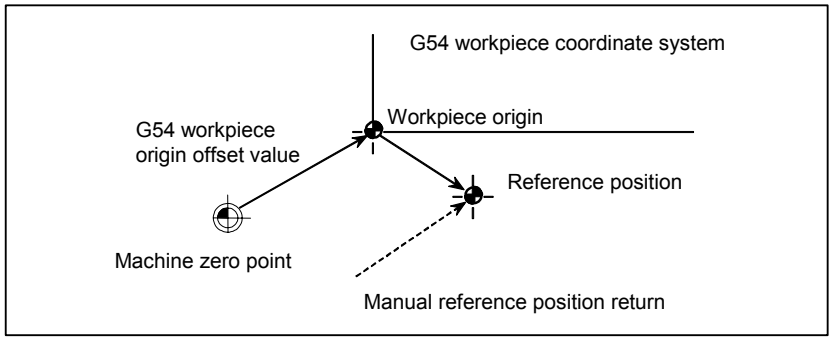
G92.1 IP 0 ;
IP 0 : Specifies axis addresses subject to the workpiece coordinate system preset operation. Axes that are not specified are not subject to the preset operation.

T

G50.3 IP0 ; (G92.1 IP 0; for G code system B or C)
IP 0 : Specifies axis addresses subject to the workpiece coordinate system preset operation. Axes that are not specified are not subject to the preset operation.

Explanation

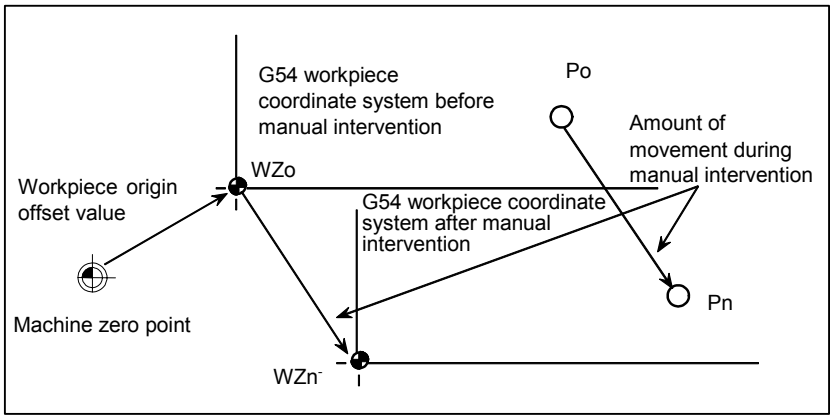
When manual reference position return operation is performed in the reset state, a workpiece coordinate system is shifted by the workpiece origin offset value from the machine coordinate system zero point. Suppose that the manual reference position return operation is performed when a workpiece coordinate system is selected with G54. In this case, a workpiece coordinate system is automatically set which has its origin displaced from the machine zero point by the G54 workpiece origin offset value; the distance from the origin of the workpiece coordinate system to the reference position represents the current position in the workpiece coordinate system.



If an absolute position detector is provided, the workpiece coordinate system automatically set at power-up has its origin displaced from the machine zero point by the G54 workpiece origin offset value. The machine position at the time of power-up is read from the absolute position detector and the current position in the workpiece coordinate system is set by subtracting the G54 workpiece origin offset value from this machine position. The workpiece coordinate system set by these operations is shifted from the machine coordinate system using the commands and operations listed below.

(a) Manual intervention performed when the manual absolute signal is off
(b) Move command executed in the machine lock state
(c) Movement by handle interruption
(d) Operation using the mirror image function
(e) Shifting the workpiece coordinate system by setting the local coordinate system or workpiece coordinate system

In the case of (a) above, the workpiece coordinate system is shifted by the amount of movement during manual intervention.



In the operation above, a workpiece coordinate system once shifted can be preset using G code (G92.1) specification or MDI operation to a workpiece coordinate system displaced by a workpiece origin offset value from the machine zero point.
Bit 3 (PPD) of parameter No. 3104 specifies whether to preset relative coordinates (RELATIVE) as well as absolute coordinates.
When no workpiece coordinate system option (G54 to G59) is selected, the workpiece coordinate system is preset to the coordinate system with its origin placed at the reference position.

Limitation

- Tool radius · tool nose radius compensation, tool length compensation, tool offset

When using the workpiece coordinate system preset function, cancel compensation modes: Tool radius · tool nose radius compensation, tool length compensation, and tool offset. If the function is executed without canceling these modes, compensation vectors are cancelled.

M

- Tool length compensation

When using the workpiece coordinate system preset function, cancel tool length compensation. If the function is executed without canceling these modes, compensation vectors are cancelled.

- Program restart

The workpiece coordinate system preset function is not executed during program restart.

- Prohibited modes

Do not use the workpiece coordinate system preset function when the scaling, coordinate system rotation, programmable image, or figure copy mode is set.

7.2.5 Addition of Workpiece Coordinate System Pair (G54.1 or G54)

Besides the six workpiece coordinate systems (standard workpiece coordinate systems) selectable with G54 to G59, 48 or 300 additional workpiece coordinate systems (additional workpiece coordinate systems) can be used.

Format

- Selecting the additional workpiece coordinate systems

G54.1Pn ; or **G54Pn ;**

Pn : Codes specifying the additional workpiece coordinate systems

n : 1 to 48 or 1 to 300

- Setting the workpiece origin offset value in the additional workpiece coordinate systems (G10)

G10L20Pn IP_;

Pn : Codes specifying the workpiece coordinate system for setting the workpiece origin offset value

n : 1 to 48 or 1 to 300

IP_ : Axis addresses and a value set as the workpiece origin offset

Explanation

- Selecting the additional workpiece coordinate systems

When a P code is specified together with G54.1 (G54), the corresponding coordinate system is selected from the additional workpiece coordinate systems (1 to 48 or 1 to 300).

A workpiece coordinate system, once selected, is valid until another workpiece coordinate system is selected. Standard workpiece coordinate system 1 (selectable with G54) is selected at power-on.

G54.1 P1 ..... Additional workpiece coordinate system 1
G54.1 P2 ..... Additional workpiece coordinate system 2
:
G54.1 P48 ..... Additional workpiece coordinate system 48
:
G54.1 P300 ..... Additional workpiece coordinate system 300

As with the standard workpiece coordinate systems, the following operations can be performed for a workpiece origin offset in an additional workpiece coordinate system:

(1) The workpiece origin offset value setting screen can be used to display and set a workpiece origin offset value.
(2) The G10 function enables a workpiece origin offset value to be set by programming (refer to II-7.2.3).
(3) A custom macro allows a workpiece origin offset value to be handled as a system variable.
(4) Workpiece origin offset data can be entered or output as external data.
(5) The PMC window function enables workpiece origin offset data to be read as program command modal data.

- Setting the workpiece origin offset value in the additional coordinate systems (G10)

When a workpiece origin offset value is specified using an absolute value, the specified value is the new offset value. When it is specified using an incremental value, the specified value is added to the current offset value to obtain a new offset value.

Limitation

- Specifying P codes

A P code must be specified after G54.1 (G54). If G54.1 is not followed by a P code in the same block, additional workpiece coordinate system 1 (G54.1P1) is assumed.
If a value not within the specifiable range is specified in a P code, an alarm PS0030 is issued.
P codes other than workpiece offset numbers cannot be specified in a G54.1 (G54) block.

Example 1) G54.1G04P1000;
Example 2) G54.1M98P48;

7.2.6 Automatic Coordinate System Setting

When bit 0 (ZPR) of parameter No. 1201 for automatic coordinate system setting is 1, a coordinate system is automatically determined when manual reference position return is performed.
Once α, β, and γ are set with parameter No. 1250, a workpiece coordinate system is set upon reference position return so that the base point on the tool holder or the tip of the basic tool is positioned at X = α, Y = β, and Z = γ.
This processing occurs as if the following are specified at the reference position:

M

G92Xα Yβ Zγ;

T

G50Xα Zγ;

This function cannot be used, however, when the workpiece coordinate system option is selected.

T

When the setting of a workpiece coordinate system shift amount is other than 0, a workpiece coordinate system shifted by the amount is set.

7.2.7 Workpiece Coordinate System Shift

T

Explanation

When the coordinate system actually set by the G50 command or the automatic system setting deviates from the programmed workpiece system, the set coordinate system can be shifted (see III-3.1).
Set the desired shift amount in the workpiece coordinate system shift memory.

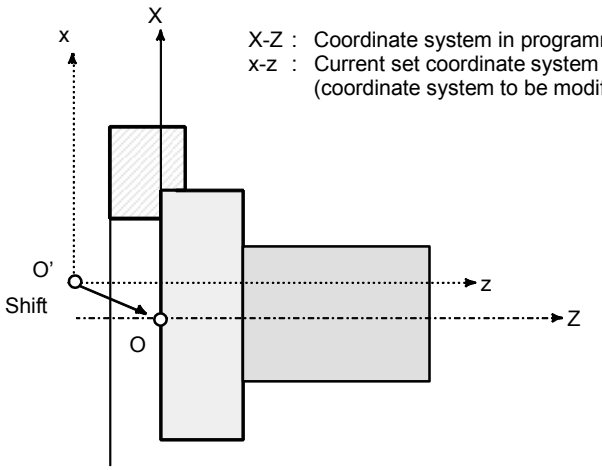


Fig. 7.2.7 (a) Workpiece coordinate system shift

Format

- Changing the workpiece coordinate system shift amount

G10 P0 IP_;

IP : Settings of an axis address and a workpiece coordinate system shift amount

⚠ CAUTION
A single block can contain a combination of X, Y, Z, C, U, V, W, and H (in G code system A). In this case, if commands are specified for the same axis, whichever appears later becomes valid.

Limitation

- Shift amount and coordinate system setting command

Specifying a coordinate system setting command (G50 or G92) invalidates the shift amount that has already been set.

Example)

When G50X100.0Z80.0; is specified, a coordinate system is set so that the current base position of the tool is at X =100.0 and Z = 80.0, regardless of which value has been set for the workpiece coordinate system shift amount.

- Shift amount and coordinate system setting

After a shift amount is set, when automatic coordinate system setting is performed upon manual reference position return, the set coordinate system is immediately shifted by the set amount.

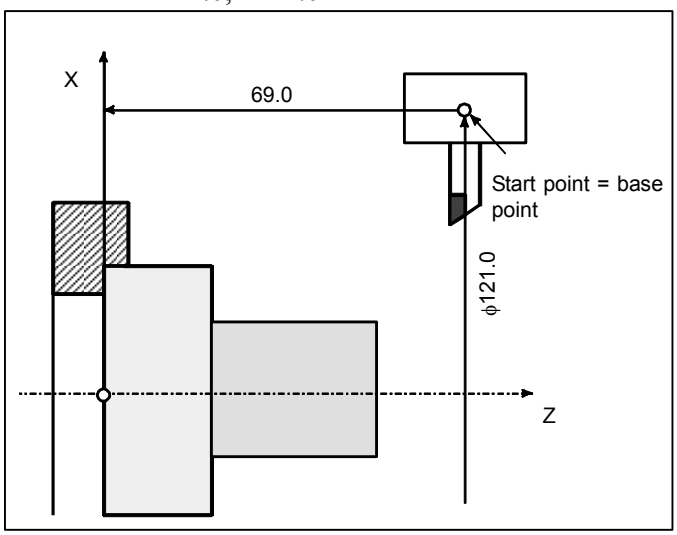
- Diameter and radius values

The workpiece coordinate system shift amount depends on diameter programming or radius programming.

Example)

Although the base point should be positioned at X = ϕ120.0 (diameter value) and Z = 70.0 from the workpiece origin, the actual position is at X = ϕ121.0 and Z = 69.0 from the origin. Set a shift amount as shown below:

X=1.0, Z=-1.0



7.3 LOCAL COORDINATE SYSTEM

When a program is created in a workpiece coordinate system, a child workpiece coordinate system can be set for easier programming. Such a child coordinate system is referred to as a local coordinate system.

Format

G52 IP_; Setting the local coordinate system
:
G52 IP 0 ; Canceling of the local coordinate system
IP_ : Origin of the local coordinate system

Explanation

By specifying G52 IP_;, a local coordinate system can be set in all the workpiece coordinate systems (G54 to G59). The origin of each local coordinate system is set at the position specified by IP_ in the workpiece coordinate system.
Once a local coordinate system is established, the coordinates in the local coordinate system are used in an axis shift command. The local coordinate system can be changed by specifying the G52 command with the origin of a new local coordinate system in the workpiece coordinate system.
To cancel the local coordinate system or specify the coordinate value in the workpiece coordinate system, match the origin of the local coordinate system with that of the workpiece coordinate system.

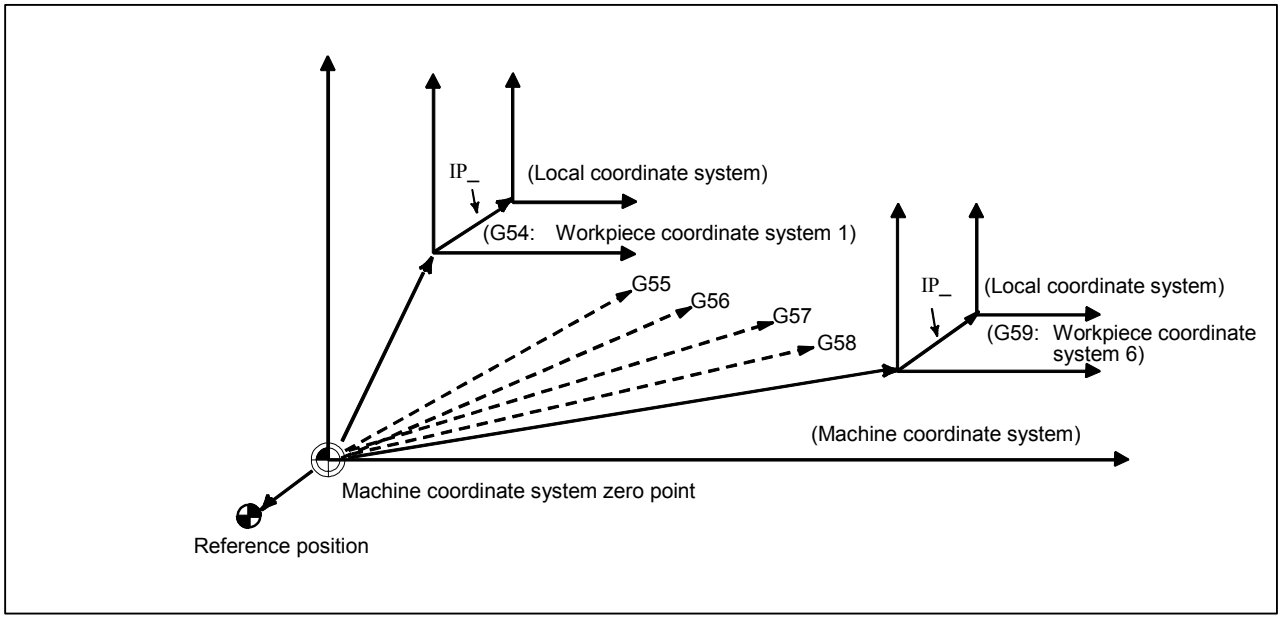


Fig. 7.3 (a) Setting the local coordinate system

⚠ CAUTION

1 When ZCL (bit 2 of parameter No.1201) is set to 1 and an axis returns to the reference position by the manual reference position return function, the origin of the local coordinate system of the axis matches that of the workpiece coordinate system. The same is true when the following command is issued:

G52α0;

α: Axis which returns to the reference position

2 The local coordinate system setting does not change the workpiece and machine coordinate systems.

3 Whether the local coordinate system is canceled at reset depends on the parameter setting. The local coordinate system is canceled when either bit 6 (CLR) of parameter No.3402 or bit 3 (RLC) of parameter No.1202 is set to 1. In three-dimensional coordinate conversion mode, however, the local coordinate system is not canceled when bit 2 (D3R) of parameter No. 5400 is set to 1.

4 When a workpiece coordinate system is set with the G92 (G50 for G code system A in the T series) command, the local coordinate system is canceled. However, the local coordinate system of an axis for which no coordinate system is specified in a G92 (G50 for G code system A in the T series) block remains unchanged.

5 G52 cancels the offset temporarily in tool radius · tool nose radius compensation.

6 Command a move command immediately after the G52 block in the absolute mode.

7.4 PLANE SELECTION

Select the planes for circular interpolation, cutter compensation, and drilling by G-code.
The following table lists G-codes and the planes selected by them.

Explanation

Table 7.4 (a) Plane selected by G code

G code	Selected plane	Xp	Yp	Zp
G17	Xp Yp plane	X-axis or an axis parallel to it	Y-axis or an axis parallel to it	Z-axis or an axis parallel to it
G18	Zp Xp plane			
G19	Yp Zp plane			

Xp, Yp, Zp are determined by the axis address appeared in the block in which G17, G18 or G19 is commanded.
When an axis address is omitted in G17, G18 or G19 block, it is assumed that the addresses of basic three axes are omitted.
Parameter No. 1022 is used to specify that an optional axis be parallel to the each axis of the X-, Y-, and Z-axes as the basic three axes.
The plane is unchanged in the block in which G17, G18 or G19 is not commanded.
The movement instruction is irrelevant to the plane selection.

M

When the power is turned on or the CNC is reset, G17 (XY plane), G18 (ZX plane), or G19 (YZ plane) is selected by bits 1 (G18) and 2 (G19) of parameter No. 3402).

T

When the power is turned on, G18 (ZX plane) is selected.

NOTE
U-, V-, and W-axes can be used with G-codes B and C.

Example

Plane selection when the X-axis is parallel with the U-axis.

```
G17X_Y_    XY plane,
G17U_Y_    UY plane
G18X_Z_    ZX plane
   X_Y_    Plane is unchanged (ZX plane)
G17        XY plane
G18        ZX plane
G17 U_     UY plane
G18Y_ ;    ZX plane, Y axis moves regardless without any
           relation to the plane.
```

8 COORDINATE VALUE AND DIMENSION

Chapter 8, "COORDINATE VALUE AND DIMENSION", consists of the following sections:

8.1 ABSOLUTE AND INCREMENTAL PROGRAMMING 210
8.2 INCH/METRIC CONVERSION (G20, G21).....212
8.3 DECIMAL POINT PROGRAMMING 216
8.4 DIAMETER AND RADIUS PROGRAMMING 218
8.5 DYNAMIC SWITCHING OF DIAMETER/RADIUS SPECIFICATION 219

8.1 ABSOLUTE AND INCREMENTAL PROGRAMMING

There are two ways to command travels of the tool; the absolute programming, and the incremental programming. In the absolute programming, coordinate value of the end position is programmed. The incremental programming is used to program the amount of a tool movement.

M

G90 and G91 are used to programming absolute or incremental programming, respectively.

T

Absolute programming or incremental programming is used depending on the programming used. See following tables.

G code system	A	B or C
Command method	Address word	G90, G91

Format

M

Absolute programming **G90 IP_ ;** Incremental programming **G91 IP_ ;**

T

- G code system A

	Absolute programming	Incremental programming
X axis move command	X	U
Z axis move command	Z	W
Y axis move command	Y	V
C axis move command	C	H

- G code system B or C

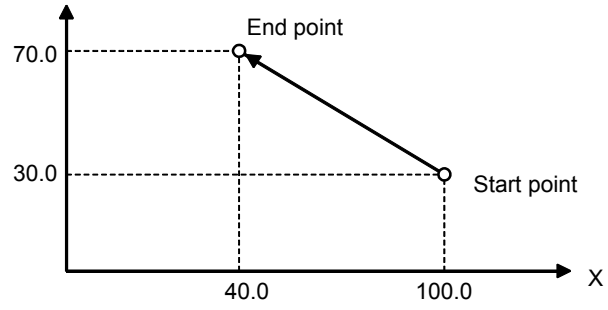
Absolute programming **G90 IP_ ;** Incremental programming **G91 IP_ ;**

Example

M

G90 X40.0 Y70.0 ; Absolute programming

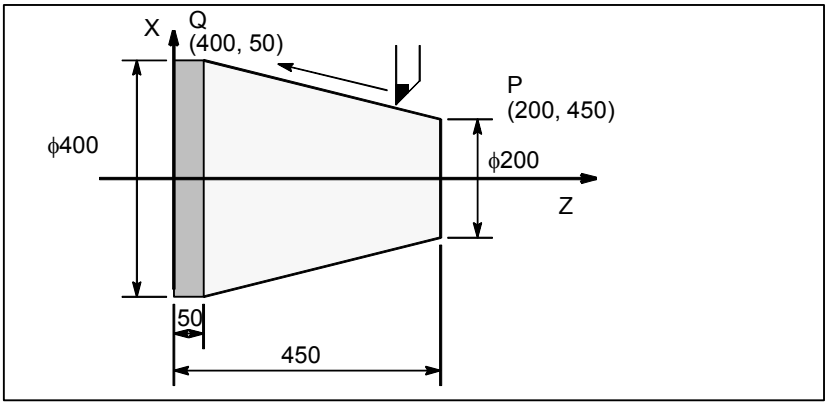
G91 X-60.0 Y40.0 ; Incremental programming



T

Tool movement from point P to point Q (diameter programming is used for the X-axis)

	G code system A	G code system B or C
Absolute programming	X400.0 Z50.0 ;	G90 X400.0 Z50.0 ;
Incremental programming	U200.0 W-400.0 ;	G91 X200.0 Z-400.0 ;



NOTE

1 Absolute programming and incremental programming can be used together in a block.
In the above example, the following command can be specified : X400.0 W-400.0 ; (in the G code system A)
2 When both X and U or Z and W are used together in a block, the one specified later is effective.
3 Incremental programming cannot be used when names of the axes are A and B during G code system A is selected.

8.2 INCH/METRIC CONVERSION (G20, G21)

Either inch or metric input (least input increment) can be selected by G code.

Format

G20 ;	Inch input
G21 ;	Metric input

This G code must be specified in an independent block before setting the coordinate system at the beginning of the program. After the G code for inch/metric conversion is specified, the unit of input data is switched to the least inch or metric input increment of increment system (II-2.3). The unit of data input for degrees remains unchanged. The unit systems for the following values are changed after inch/metric conversion:

- Feedrate commanded by F code
- Positional command
- Workpiece origin offset value
- Tool compensation value
- Unit of scale for manual pulse generator
- Movement distance in incremental feed
- Some parameters

When the power is turned on, the G code is the same as that held before the power was turned off.

⚠ WARNING
G20 and G21 must not be switched during a program.

NOTE
1 When the least input increment and the least command increment systems are different, the maximum error is half of the least command increment. This error is not accumulated.
2 The inch and metric input can also be switched using settings (see III-12.3.1).
3 If a function selected using bit 2 (IRF) of parameter No. 14000 or bit 0 (NIM) of parameter No. 11222 is not used, be sure to perform inch/metric conversion at the reference position (machine coordinate system origin).

Performing inch/metric conversion in the reference position (parameter No. 1240 is not 0)

Conventionally, inch/metric conversion must be performed at the machine coordinate system origin. However, setting bit 2 (IRF) of parameter No. 14000 to 1 enables inch/metric conversion to be performed in the reference position (parameter No. 1240).

If an attempt is made to perform inch/metric conversion when an axis with this function enabled is not in the reference position, alarm PS5362 is issued to cancel the attempt.
Before trying to perform inch/metric conversion, be sure to set the axis of interest to the reference position, using the G28 command, for example.

In addition, if the workpiece coordinate system has been shifted from the machine coordinate system, using the following commands or operations, bit 1 (CIM) of parameter No. 11222 can be used to select whether to issue alarm PS1298 or to clear the offset.

- Manual intervention performed with the manual absolute signal being off
- Move command issued with the machine locked
- Move command issued using a handle interrupt
- Mirror image-based operation
- Workpiece coordinate system shift caused by local coordinate system setting (G52) or workpiece coordinate system setting (G92)

Switching conditions

Performing inch/metric conversion in any position other than the reference position requires satisfying all of the following conditions. Failing to satisfy any of the conditions results in alarm PS1298 being issued. For electronic gear box 2 pair synchronization or electronic gear box synchronization, alarm PS1595 is issued.

- Positioning or linear interpolation
- Polar coordinate interpolation cancel mode
- Polar coordinate command cancel
- Spindle Speed fluctuation detection off
- Tool radius · tool nose radius compensation cancel / 3-dimensional tool compensation cancel
- Normal direction control cancel
- Tool length offset cancel
- Scaling cancel
- Programmable mirror image cancel
- Polygon turning cancel
- Macro modal call A/B cancel
- Coordinate system rotation or 3-dimensional coordinate system conversion mode off
- Mirror image for double turret or balanced cutting mode cancel (T series only)
- Canned cycle cancel

- Electronic gear box 2 pair synchronization cancel
- Electronic gear box synchronization cancel
- Constant surface speed control cancel

In addition, performing inch/metric conversion in any position other than the reference position requires the following options.

- Inch/metric conversion
- Workpiece coordinate system

Restrictions

When performing the following operations, make sure that the axis of interest is in the reference position (machine coordinate system origin).

- Inch/metric conversion based on bit 2 (INI) of setting parameter No. 0
- Inch/metric conversion based on programmable parameter input (G10)
- Inch/metric conversion based on custom macro variable No. 3005

Performing inch/metric conversion in positions other than the reference position

Setting bit 0 (NIM) of parameter No. 11222 enables inch/metric conversion to be performed even in positions other than the reference position.

In addition, if the workpiece coordinate system has been shifted from the machine coordinate system, using the following commands or operations, bit 1 (CIM) of parameter No. 11222 can be used to select whether to issue alarm PS1298 or to clear the offset.

- Manual intervention performed with the manual absolute signal being off
- Move command issued with the machine locked
- Move command issued using a handle interrupt
- Mirror image-based operation
- Workpiece coordinate system shift caused by local coordinate system setting (G52) or workpiece coordinate system setting (G92)

If an axis is under any of the following controls, however, no automatic coordinate system conversion based on this function can be carried out for the axis.

- Axis control by PMC
- Axis synchronous control (for slave axes when the master axis is a PMC axis)
- Flexible synchronous control (for slave axes when the master axis is a PMC axis)
- Chopping
- Spindle control with servo motor
- Spindle control with Cs contour control
- Servo/spindle synchronous control

Switching conditions

Performing inch/metric conversion in any position other than the reference position requires satisfying all of the following conditions. Failing to satisfy any of the conditions results in alarm PS1298 being issued. For electronic gear box 2 pair synchronization or electronic gear box synchronization, alarm PS1595 is issued.

- Positioning or linear interpolation
- Polar coordinate interpolation cancel mode
- Polar coordinate command cancel
- Spindle Speed fluctuation detection off
- Tool radius · tool nose radius compensation cancel / 3-dimensional tool compensation cancel
- Normal direction control cancel
- Tool length offset cancel
- Scaling cancel
- Programmable mirror image cancel
- Polygon turning cancel
- Macro modal call A/B cancel
- Coordinate system rotation or 3-dimensional coordinate system conversion mode off
- Mirror image for double turret or balanced cutting mode cancel (T series only)
- Canned cycle cancel
- Electronic gear box 2 pair synchronization cancel
- Electronic gear box synchronization cancel
- Constant surface speed control cancel

In addition, performing inch/metric conversion in any position other than the reference position requires the following options.

- Inch/metric conversion
- Workpiece coordinate system
- Workpiece coordinate system preset

Restrictions

When performing the following operations, make sure that the axis of interest is in the reference position (machine coordinate system origin).

- Inch/metric conversion based on bit 2 (INI) of setting parameter No. 0
- Inch/metric conversion based on programmable parameter input (G10)
- Inch/metric conversion based on custom macro variable No. 3005

8.3 DECIMAL POINT PROGRAMMING

Numerical values can be entered with a decimal point. A decimal point can be used when entering a distance, time, or speed. Decimal points can be specified with the following addresses:

M

X, Y, Z, U, V, W, A, B, C, I, J, K, Q, R, F

T

X, Y, Z, U, V, W, A, B, C, I, J, K, R, F

Explanation

There are two types of decimal point notation: calculator-type notation and standard notation.
When calculator-type decimal notation is used, a value without decimal point is considered to be specified in millimeters inch, or deg. When standard decimal notation is used, such a value is considered to be specified in least input increments. Select either calculator-type or standard decimal notation by using the bit 0 (DPI) of parameter No.3401.Values can be specified both with and without decimal point in a single program.

Example

Program command	Pocket calculator type decimal point programming	Standard type decimal point programming
X1000 Command value without decimal point	1000mm Unit :mm	1mm Unit : Least input increment (0.001mm)
X1000.0 Command value with decimal point	1000mm Unit :mm	1000mm Unit :mm

> ⚠ **CAUTION**
> When specifying a dimension word for a command G code in a block, be sure to place the dimension word after the command G code.

NOTE

1 A specified value less than the minimum unit is treated as described below.

Example 1)

When a value is specified directly at an address (in the case of IS-B)

X1.2345 ; Treated as X1.235

X-1.2345 ; Treated as X-1.234

Example 2)

When a value is assigned to a macro variable (in the case of IS-B)

#100=1.2345;

X#100 ; Treated as X1.235

#100=-1.2345;

X#100 ; Treated as X-1.234

2 When more than eight digits are specified, an alarm occurs. If a value is entered with a decimal point, the number of digits is also checked after the value is converted to an integer according to the least input increment.

Examples:

X1.23456789;

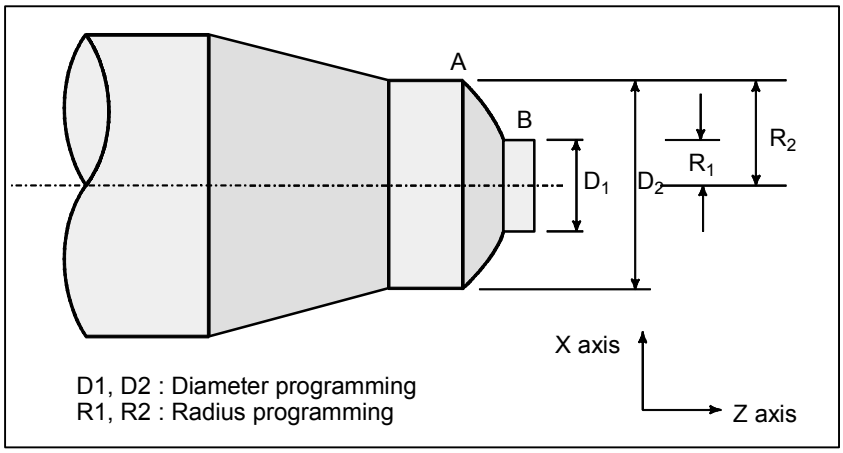
Alarm PS0003 occurs because more than eight digits are specified.

X123456.7;

If the least input increment is 0.001 mm, the value is converted to integer 123456700. Because the integer has more than eight digits, an alarm occurs.

8.4 DIAMETER AND RADIUS PROGRAMMING

Since the workpiece cross section is usually circular in CNC lathe control programming, its dimensions can be specified in two ways : Diameter and Radius



When the diameter is specified, it is called diameter programming and when the radius is specified, it is called radius programming.

Explanation

- Notes on diameter programming/radius programming for each command

Radius programming or diameter programming can be specified by bit 3 (DIA) of parameter No.1006. When using diameter programming, note the conditions listed in the Table 8.4 (a).

Table 8.4 (a) Notes on specifying diameter value

Item	Notes
X axis command	Specified with a diameter value
Incremental command	Specified with a diameter value In the above figure, specifies D2 minus D1 for tool path B to A.
Coordinate system setting (G50)	Specifies a coordinate value with a diameter value
Component of tool offset value	Bit 1 of parameter No.5004 determines either diameter or radius value
Parameters in canned cycle, such as cutting depth along X axis. (R)	Specifies a radius value
Radius designation in circular interpolation (R, I, K, and etc.)	Specifies a radius value
Feedrate along axis	Specifies change of radius/rev. or change of radius/min.
Display of axis position	Displayed as diameter value

8.5 DIAMETER AND RADIUS SETTING SWITCHING FUNCTION

Overview

Usually, whether to use diameter specification or radius specification to specify a travel distance on each axis is uniquely determined by the setting of bit 3 (DIAx) of parameter No. 1006. However, this function enables switching between diameter specification and radius specification by using a signal or G code.
Thus, a coordinate, program, and so forth can be specified by switching between diameter specification and radius specification for each controlled axis.

Explanation

- Selection of a diameter/radius specification switching method

Two methods are available for switching between diameter specification and radius specification:
1) Signal
2) G code
Use bit 5 (PGD) of parameter No. 3400 to determine which method to use.

- Switching method using a signal

For switching between diameter specification and radius specification, set the diameter/radius specification switch signal from DI1 to DI8 (input signals) corresponding to a desired axis, from 0 to 1.
If an input signal is set from 0 to 1, and radius specification is selected (with bit 3 (DIAx) of parameter No. 1006 = 0) for the axis corresponding to the input signal, the specification method switches to diameter specification; the specification method switches to radius specification if diameter specification is selected (with bit 3 (DIAx) of parameter No. 1006 = 1).
During switching, the diameter/radius specification switching in-progress signal from DM1 to DM8 (output signals) corresponding to a switched axis is output.
To return the diameter/radius specification of an axis to the original state, set the setting of the corresponding diameter/radius specification switch signal from DI1 to DI8, from 1 to 0.

NOTE

1 When operating an input signal by using an M code, for example, during automatic operation, perform a switching operation according to the method below to reflect the state of diameter/radius specification switching in the execution block correctly.
As an auxiliary function for switching, use an unbuffered M code (parameter No. 3411 and up).
Use the following sequences for a specified M code:
- When switching is performed
  M code → Input signal ON → Confirmation of output signal ON → FIN
- When switching is cancelled
  M code → Input signal OFF → Confirmation of output signal OFF → FIN

If a diameter/radius specification switch signal is operated during automatic operation without following the sequences above, alarm PS5320 is issued.

2 If a diameter/radius specification switch signal is operated while a movement is made on an axis subject to switching, alarm PS5320 is issued.

- Switching method using a G code (programmable diameter/radius specification switching)

The format of a G code for diameter/radius specification switching is as follows:

Format

G10.9 IP_ ;

IP_ : Address and command value of an specified axis subject to diameter/radius specification switching
Specify 0 or 1 as the command value.
0: Radius specification
1: Diameter specification

NOTE

1 Specify G10.9 in a single block specifying no other codes.

2 After an axis address, specify a command value without using the decimal point.

- Switching operation

According to the switching methods above, diameter/radius specification is internally switched as described below.

1) Switching using a signal
   - When parameter DIAx = 0 (radius specification) → Operation is performed with diameter specification.
   - When parameter DIAx = 1 (diameter specification) → Operation is performed with radius specification.

2) Switching using a G code
   - When the specified address value = 0 (radius specification) → Operation is performed with radius specification.
   - When the specified address value = 1 (diameter specification) → Operation is performed with diameter specification.

NOTE
1 When the diameter/radius specification switching state is to be cancelled using a reset or mode switching at the time of signal-based switching, the input signal needs to be operated.
2 Switching using a G code is cancelled by a reset.

⚠ **CAUTION**
When switching is performed from diameter specification to radius specification, the travel distance based on the same move command is doubled when compared with diameter specification. So, when switching from diameter specification to radius specification, ensure safety in machine operation.

- Selecting the machine coordinate system

When diameter and radius setting switching is taking place as directed with the diameter and radius setting switching function, the coordinates specified by a machine coordinate system select command (G53) vary depending on the value of bit 7 (PDM) of parameter No. 11222. If the parameter PDM is 0, diameter and radius switching occurs. If the parameter PDM is 1, the coordinates vary according to the value of bit 3 (DIAx) of parameter No. 1006.

Limitation

- Feedrate

A radius-based feedrate is specified in both of diameter specification and radius specification at all times.

- Data not switchable

The following data follows the setting of parameter DIAx, so that diameter and radius switching is not performed:

- Parameter
- Offset
- Workpiece coordinate system
- Scale display on the graphic screen

NOTE
For offset data, the settings of bit 1 (ORC) of parameter No. 5004 and bit 2 (ODI) of parameter No. 5004 have priority.

- Switchable data and commands

For the following data and commands, diameter and radius switching is performed according to the specified specification method:

- Programmed move command
- Current position display
- Workpiece coordinate system preset
- Movement based on the manual numeric command G00 or G01

- Use with other functions

Diameter/radius specification switching cannot be performed for an axis on which a movement is being made with any of the functions indicated below.
Moreover, none of the functions indicated below can be performed during diameter and radius switching.

- Synchronous/composite control
- Superimposed Control
- Axis synchronous control
- PMC axis control

9 SPINDLE SPEED FUNCTION (S FUNCTION)

The spindle speed can be controlled by specifying a value following address S.
Chapter 9, "SPINDLE SPEED FUNCTION (S FUNCTION)", consists of the following sections:

9.1 SPECIFYING THE SPINDLE SPEED WITH A CODE 224
9.2 SPECIFYING THE SPINDLE SPEED VALUE DIRECTLY (S5-DIGIT COMMAND) 224
9.3 CONSTANT SURFACE SPEED CONTROL (G96, G97) 225
9.4 SPINDLE POSITIONING FUNCTION 230
9.5 SPINDLE SPEED FLUCTUATION DETECTION 236
9.6 SPINDLE CONTROL WITH SERVO MOTOR.....241

9.1 SPECIFYING THE SPINDLE SPEED WITH A CODE

When a value is specified after address S, the code signal and strobe signal are sent to the machine to control the spindle rotation speed.
A block can contain only one S code. Refer to the appropriate manual provided by the machine tool builder for details such as the number of digits in an S code or the execution order when a move command and an S code command are in the same block.

9.2 SPECIFYING THE SPINDLE SPEED VALUE DIRECTLY (S5-DIGIT COMMAND)

The spindle speed can be specified directly by address S followed by a max. five-digit value (min^{-1}). The unit for specifying the spindle speed may vary depending on the machine tool builder. Refer to the appropriate manual provided by the machine tool builder for details.

9.3 CONSTANT SURFACE SPEED CONTROL (G96, G97)

Specify the surface speed (relative speed between the tool and workpiece) following S. The spindle is rotated so that the surface speed is constant regardless of the position of the tool.

Format

- Constant surface speed control command

G96S<u>xxxxx</u> ;

↑ Surface speed (m/min or feet/min)

This surface speed unit may change according to machine tool builder's specification.

- Constant surface speed control cancel command

G97S<u>xxxxx</u> ;

↑ Spindle speed (min^{-1})

This surface speed unit may change according to machine tool builder's specification.

- Constant surface speed controlled axis command

G96Pα ;

P0 : Axis set in the parameter (No. 3770)

P1 : X axis, P2 : Y axis, P3 : Z axis, P4 : 4th axis

P5 : 5th axis, P6 : 6th axis, P7 : 7th axis, P8 : 8th axis

NOTE

If multi-spindle control (spindle selecting based on address P or spindle name expansion) is enabled, axis specification based on address P is disabled. Use parameter No. 3770 for axis specification.

- Clamp of maximum spindle speed

G92 S_ ;

The maximum spindle speed (min^{-1}) follows S.

T

G50 S_ ;

The maximum spindle speed (min^{-1}) follows S.

NOTE

G50 can be used with G code system A.

Explanation

- Constant surface speed control command (G96)

G96 (constant surface speed control command) is a modal G code. After a G96 command is specified, the program enters the constant surface speed control mode (G96 mode) and specified S values are assumed as a surface speed. A G96 command must specify the axis along which constant surface speed control is applied. A G97 command cancels the G96 mode. When constant surface speed control is applied, a spindle speed higher than the value specified in G92S_; or G50S_; (maximum spindle speed) is clamped at the maximum spindle speed. When the power is turned on, the maximum spindle speed is not yet set and the speed is not clamped. S (surface speed) commands in the G96 mode are assumed as S = 0 (the surface speed is 0) until M03 (rotating the spindle in the positive direction) or M04 (rotating the spindle in the negative direction) appears in the program.

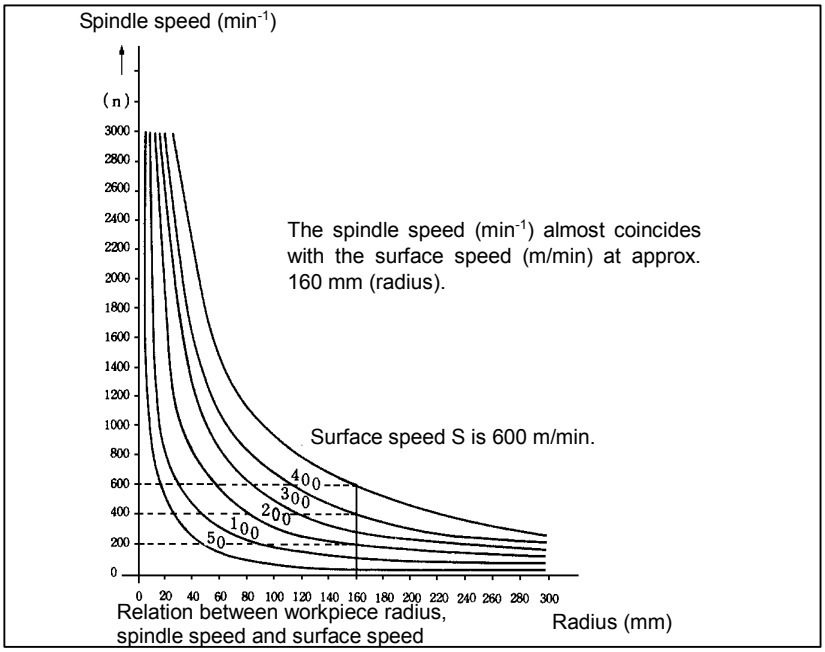


Fig. 9.3 (a) Relation between workpiece radius, spindle speed and surface speed

- Setting the workpiece coordinate system for constant surface speed control

To execute the constant surface speed control, it is necessary to set the workpiece coordinate system , and so the coordinate value at the center of the rotary axis, for example, Z axis, (axis to which the constant surface speed control applies) becomes zero.

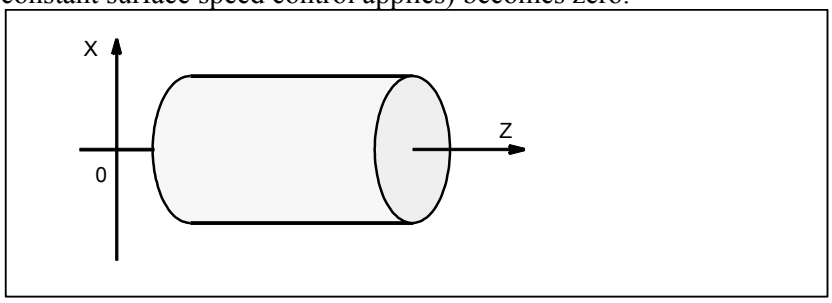
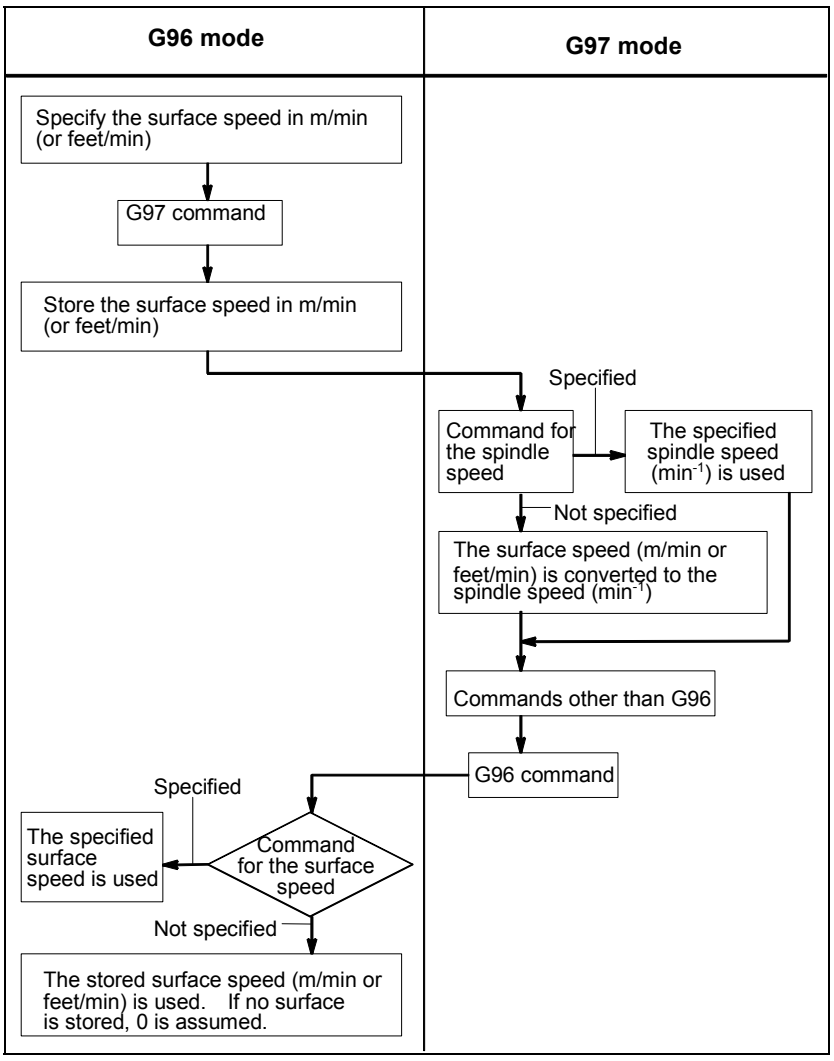


Fig. 9.3 (b) Example of the workpiece coordinate system for constant surface speed control

- Surface speed specified in the G96 mode



Limitation

- Constant surface speed control for threading

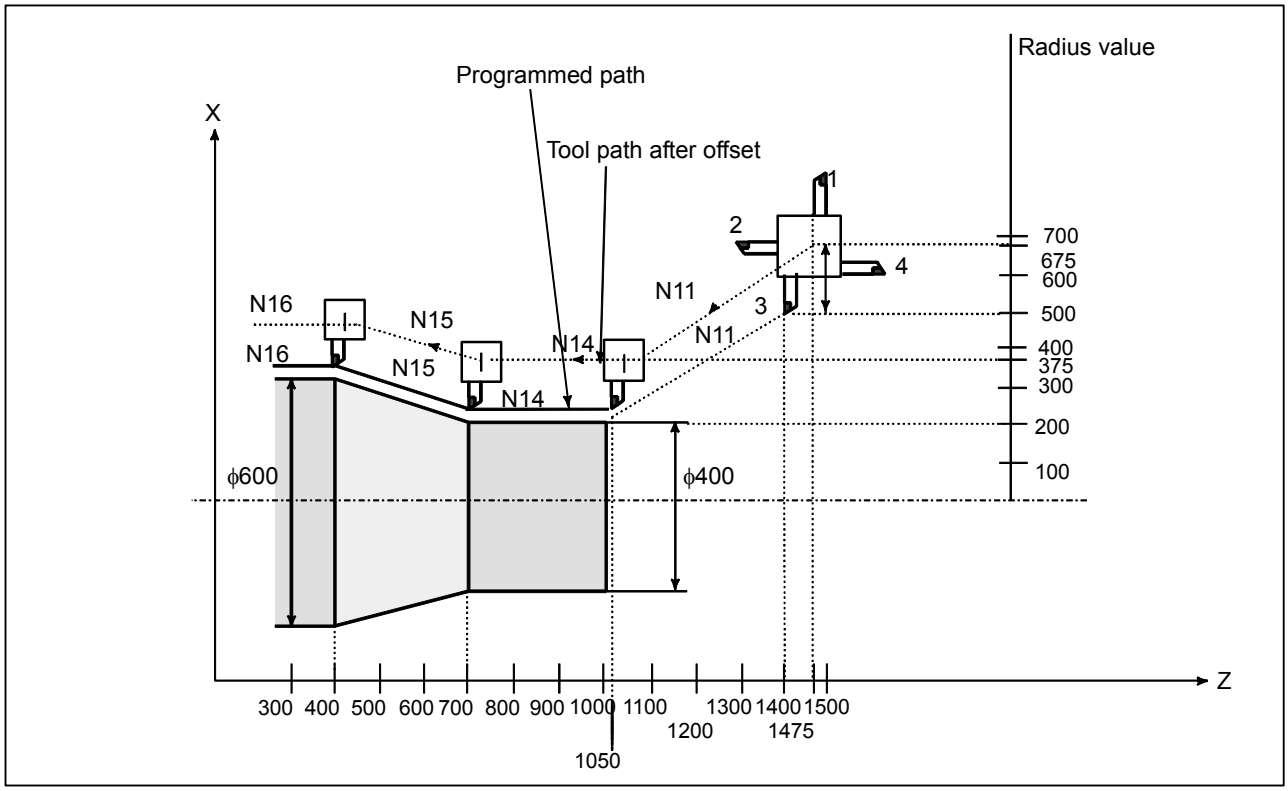
The constant surface speed control is also effective during threading. Accordingly, it is recommended that the constant surface speed control be invalidated with G97 command before starting the scroll threading and taper threading, because the response problem in the servo system may not be considered when the spindle speed changes.

- Constant surface speed control for rapid traverse (G00)

In a rapid traverse block specified by G00, the constant surface speed control is not made by calculating the surface speed to a transient change of the tool position, but is made by calculating the surface speed based on the position at the end point of the rapid traverse block, on the condition that cutting is not executed at rapid traverse.

Example

T



N8 G00 X1000.0Z1400.0 ;
N9 T33;
N11 X400.0Z1050.0;
N12 G50S3000 ; (Designation of max. spindle speed)
N13 G96S200 ; (Surface speed 200 m/min)
N14 G01 Z 700.0F1000 ;
N15 X600.0Z 400.0;
N16 Z_ ;

The CNC calculates the spindle speed which is proportional to the specified surface speed at the position of the programmed coordinate value on the X axis. This is not the value calculated according to the X axis coordinate after offset when offset is valid. At the end point N15 in the example above, the speed at 600 dia. (Which is not the turret center but the tool nose) is 200 m/min. If X axis coordinate value is negative, the CNC uses the absolute value.

9.4 SPINDLE POSITIONING FUNCTION

Overview

In turning, the spindle connected to the spindle motor is rotated at a certain speed to rotate the workpiece mounted on the spindle. This spindle control status is referred to as spindle rotation mode.
The spindle positioning function turns the spindle connected to the spindle motor by a certain angle to position the workpiece mounted on the spindle at a certain angle. This spindle control status is referred to as spindle positioning mode.

The spindle positioning function involves the following three operations:

1. Canceling the spindle rotation mode and entering the spindle positioning mode
   Place the spindle in the spindle positioning mode and establish a reference position by specifying a given M code (set with a parameter). (Spindle orientation)
2. Positioning the spindle in the spindle positioning mode
   The spindle is positioned in either of the two methods:
   1) Positioning with an arbitrary angle by an axis address
   2) Positioning with a semi-fixed angle by a given M code (set with a parameter)
3. Canceling the spindle positioning mode, and entering the spindle rotation mode
   Place the spindle in the spindle rotation mode by specifying a given M code (set with a parameter).

The least command increment, least input increment, and maximum value for the spindle positioning axis are as follows:

- Least command increment

  $\frac{360}{4096} \fallingdotseq 0.088$ deg (when the gear ratio of the spindle to the position coder
- Least input increment
  0.001 deg (IS-B)
- Maximum value
  ±999999.999 deg

9.4.1 Spindle Orientation

When spindle positioning is first performed after the spindle motor is used for normal spindle operation, or when spindle positioning is interrupted, the spindle orientation is required.
Orientation permits the spindle to stop at a predetermined position.
Orientation is directed by the M code set in parameter No. 4960. The direction of orientation can be set with a parameter.
For the serial spindle, it is set in bit 4 (RETSV) of parameter No. 4000.
With the grid shift function, the orientation position can be shifted in a range of 0 to 360 deg with parameter No. 4073 for a serial spindle.

- Feedrate during spindle orientation

An orientation feedrate for a serial spindle is determined by a spindle parameter setting.
In orientation, the serial spindle stops at the orientation position after several rotations of the spindle motor.

- Omission of orientation

By using bit 2 (ISZ) of parameter No. 4950, orientation upon switching to spindle positioning mode can be omitted if it is unnecessary (for example, when no start position is specified and incremental positioning from the current position is only required). More specifically, when an M code for switching to spindle positioning mode is specified, the spindle control mode is simply switched to spindle positioning mode and then the processing is completed without orientation.

- Program reference position

The position at which orientation is completed is assumed to be a program reference position. However, the program reference position can be changed through coordinate system setting (G92 or G50) or automatic coordinate system setting (bit 0 (ZPR) of parameter No. 1201).
When a setting is made to omit orientation, a program reference position is not established, and operation by an absolute command is unpredictable during spindle positioning with an axis address.

9.4.2 Spindle Positioning

The spindle can be positioned with a semi-fixed angle or arbitrary angle.

- Positioning with a semi-fixed angle

Use an M code to specify a positioning angle. The specifiable M code value may be one of the six values from Mα to M (α+5). Value α must be set in parameter No. 4962 beforehand. The positioning angles corresponding to Mα to M (α+5) are listed below. Value β must be set in parameter No. 4963 beforehand.

M-code (Ex.) β = α + 5	Positioning angle	(Ex.) β = 30°
Mα	β	30°
M (α + 1)	2β	60°
M (α + 2)	3β	90°
M (α + 3)	4β	120°
M (α + 4)	5β	150°
M (α + 5)	6β	180°

When the number of M codes to be used, value γ, is specified in parameter No. 4964, a specifiable M code value may be in a range of values from Mα to M (α + (γ - 1)), up to 255 values from Mα to M (α + (255 - 1)).

M code (Ex.) γ = 11	Positioning angle	(Ex.) β = 30°
Mα	β	30°
M (α + 1)	2β	60°
M (α + 2)	3β	90°
M (α + 3)	4β	120°
...	...	...
M (α + 11 - 1)	11β	330°

The direction of rotation can be specified in IDM (bit 1 of parameter 4950).

- Positioning with an arbitrary angle

Specify the position with an arbitrary angle using the axis address followed by a signed numeric value or numeric values. The axis address must be specified in the G00 mode.
(The explanation below is given assuming that a C axis address is set.)

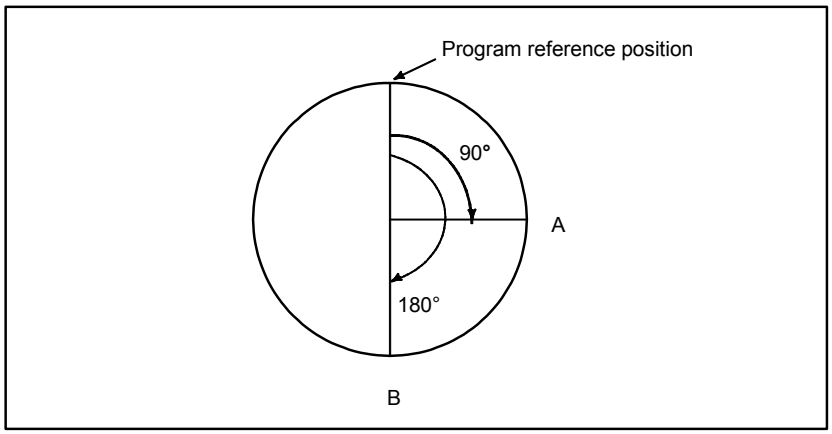
(Example) C-45000
C180.000

A numeric with the decimal point can be entered. The value must be specified in degrees.

(Example) C36.0=C36 degrees

- Absolute commands and incremental commands

Incremental commands are always used for positioning with a semi-fixed angle (using M codes).
The direction of rotation can be specified with bit 1 (IDM) of parameter No. 4950.
Absolute and incremental commands can be used for positioning with an arbitrary angle.
With absolute commands for positioning with an arbitrary angle, when the rotation axis rollover function is used (bit 0 (ROA) of parameter No. 1008 is 1), shortcut control is also enabled (bit 1 (RAB) of parameter No. 1008 is 0).



Command format		G-code system A in lathe system		G-code system B or C in lathe system, and machining center system	
		Address used	Command A-B in the above figure	Address used and G code	Command A-B in the above figure
Absolute command	Specify the end point with a distance from the program reference position.	C	C180.0 ;	G90, C	G90 C180 .;
Incremental command	Specify a distance from the start point to the end point.	H	H90.0 ;	G91, C	G91 C90 .;

- Feedrate during positioning

The feedrate during positioning equals the rapid traverse speed specified in parameter No. 1420.
For the specified speed, an override of 100%, 50%, 25%, and F0 (parameter No. 1421) can be applied.

9.4.3 Canceling Spindle Positioning

When modes are to be switched from spindle positioning to normal spindle rotation, the M code set in parameter No. 4961 must be specified.

Also, the spindle positioning mode is canceled and the spindle rotation mode is set in the following cases:

<1> A reset operation (including an emergency stop) occurs when a servo alarm is issued.

<2> A reset operation (including an emergency stop) occurs when a spindle alarm is issued.

<3> An orientation operation in progress is stopped due to a reset or alarm, or for some other reason.

<4> A reset operation (including an emergency stop) occurs when bit (IOR) 0 of parameter No. 4950 is 1.

⚠ CAUTION

1 During execution of spindle positioning sequences (canceling the spindle rotation mode and entering the spindle positioning mode, positioning the spindle in the spindle positioning mode, and canceling the spindle positioning mode and entering the spindle rotation mode), the automatic operation stop signal *SP is invalid. This means that automatic operation does not stop until all the sequences are completed, even when the *SP signal becomes 0.
2 Dry run and machine lock cannot be performed during spindle positioning.
3 Auxiliary function lock is disabled for M codes for the spindle positioning function.
4 The serial spindle Cs-axis contour control function and the spindle positioning function cannot be used at a time. If both options are specified, the spindle positioning function has priority.
5 The spindle positioning axis is handled as a controlled axis. Therefore, controlled axis-related signals (such as the overtravel signal) must be set.
6 When using the rigid tapping function and the spindle positioning function concurrently, do not specify rigid tapping in spindle positioning mode or spindle positioning in rigid tapping mode.

NOTE

1 M code commands for positioning of a spindle must be specified in a single block. Other commands must not be contained in the same block. (Also, M code commands for positioning of another spindle must not be contained in the same block.)
Even when the single-block, multiple-M code command function is also used, related M codes must be specified in a single block.
2 Even when the single-block, multiple-M code command function is also used, related M codes must be specified in a single block.
3 Axis address commands for positioning of a spindle must be specified in a single block. Other commands must not be contained in the same block. However, the following commands can be contained in the same block where axis address commands are specified:
G00, G90, G91, G92 (G-code systems B and C in the T series, M series)
G00, G50 (G-code system A in the T series)
4 M code commands for spindle positioning specify M codes that are not buffered.
5 Spindle positioning cannot be performed by manual operation (in jog feed, manual handle feed, manual numeric command, or other mode).
6 Spindle positioning cannot be performed by PMC axis control.
7 For spindle positioning, program restart and block restart operations cannot be performed. Use the MDI for these operations.
8 The stored stroke limit check is disabled for the spindle positioning axis.
9 The axis removal function is disabled for the spindle positioning axis.
10 The pitch error compensation function is disabled for the spindle positioning axis.
11 When a setting is made to omit spindle orientation, the reference position return completion signal does not become 1.
12 In spindle orientation, all-axis interlock and axis-specific interlock are checked only when a block is started. A signal is ignored if input during the execution of the block.
13 A difference between a specified travel distance and an actual travel distance is maintained until spindle positioning mode is canceled.

9.5 SPINDLE SPEED FLUCTUATION DETECTION

Overview

With this function, an overheat alarm (OH0704) is raised and the spindle speed fluctuation detection alarm signal SPAL is issued when the spindle speed deviates from the specified speed due to machine conditions.
This function is useful, for example, for preventing the seizure of the guide bushing.
G26 enables spindle speed fluctuation detection.
G25 disables spindle speed fluctuation detection.

Format

- Spindle fluctuation detection on

G26 Pp Qq Rr Ii ;

P: Time (in ms) from the issue of a new spindle rotation command (S command) to the start of checking whether the actual spindle speed is so fast that an overheat can occur.
When a specified speed is reached within the time period of P, a check is started at that time.

Q: Tolerance (%) of a specified spindle speed

$$q = \left| \frac{1 - actual\ spindle\ speed}{specified\ spndle\ speed} \right| \times 100$$

If a specified spindle speed lies within this range, it is regarded as having reached the specified value. Then, the checking of an actual spindle speed is started.

R: Spindle speed fluctuation (%) at which the actual spindle speed is so fast that an overheat can occur

$$r = \left| \frac{1 - speed\ that\ can\ cause\ overheat}{speciofied\ spndle\ speed} \right| \times 100$$

If the fluctuation of the actual spindle speed to the specified spindle speed exceeds the spindle speed fluctuation of R, the actual spindle speed is regarded as being so fast that an overheat can occur.

I: Spindle speed fluctuation width at which the actual spindle speed (min^{-1}) is so fast that an overheat can occur
If the fluctuation (width) between the specified and actual spindle speeds exceeds the spindle speed fluctuation width of I, the actual spindle speed is regarded as being so fast that an overheat can occur.

G26 enables the spindle speed fluctuation detection function. The values specified for P, Q, R, and **I** are set in the following parameters: No. 4914, No. 4911, No. 4912, and No. 4913, respectively. Each command address corresponds to a parameter number as listed below.

Command address	Parameter number
Q	No.4911
R	No.4912
I	No.4913
P	No.4914

If the P, Q, R, or I command address is omitted, the function detects the fluctuation of the actual spindle speed according to the value set in the corresponding parameter (No. 4914, No. 4911, No. 4912, or No. 4913).
The parameters (No. 4914, No. 4911, No. 4912, and No. 4913) for the spindle on which the currently selected position coder is mounted are used for the setting and spindle speed fluctuation detection check.

- Spindle fluctuation detection off

```
G25;
```

G25 disables the spindle speed fluctuation detection function.
When G25 is specified, the parameters (No. 4914, No. 4911, No. 4912, and No. 4913) are unchanged. When the power is turned on or after a reset (clear state (bit 6 (CLR) of parameter No. 3402 = 1)) is executed, the spindle speed fluctuation detection function is disabled (G25). For the clear state, also check the setting of bit 3 (C19) of parameter No. 3408 for the M series or bit 0 (C08) of parameter No. 3407 for the T series.

Explanation

The function for detecting spindle speed fluctuation checks whether the actual speed varies for the specified speed or not. Si or Sr, whichever is greater, is taken as the allowable fluctuation speed (Sm). An alarm (OH0704) is activated when the actual spindle speed varies for the commanded speed (Sc) under the condition that the variation width exceeds the allowable variation width (Sm).

$|Sc - Sa| > Sm$

Sc : Specified spindle speed
Sa : Actual spindle speed
Si : The allowable constant variation width which is independent of the specified spindle speed (parameter (No.4913))
Sr : The allowable variation width which is obtained by multiplying Sc (commanded spindle speed) by r (constant ratio). (r = parameter (No.4912))

Parameter FLR(No.4900#0)= 0	Parameter FLR(No.4900#0)= 1
$Sr = Sc \times \frac{r}{100}$	$Sr = Sc \times \frac{r}{1000}$

Sm: Si or Sr, whichever is greater

- Conditions to start spindle speed fluctuation detection

If the specified spindle speed Sc changes, spindle speed fluctuation detection starts when one of the conditions below is met:

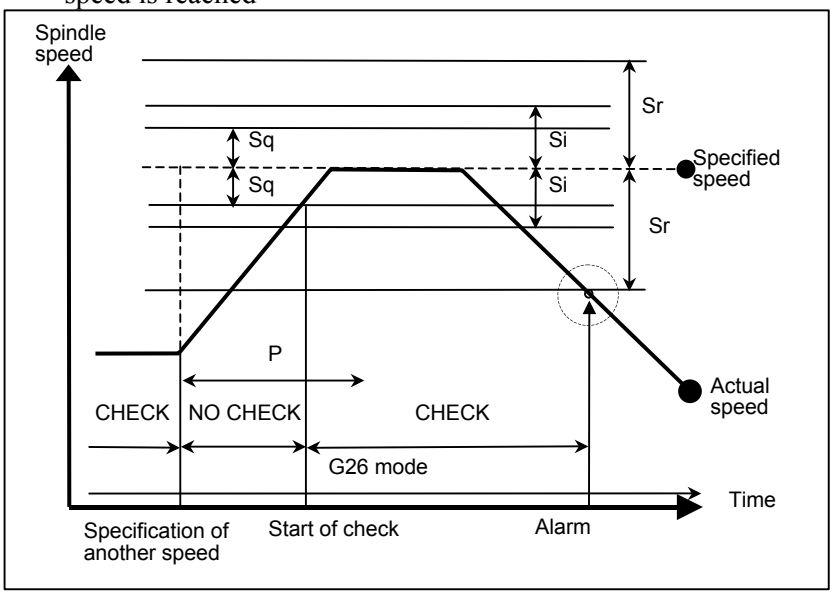
<1> The actual spindle speed falls in a range of (Sc - Sq) to (Sc + Sq)
Sc : Specified spindle speed
Sq : Tolerance within which the spindle is assumed to attain the programmed speed
(parameter (No.4911))

Parameter FLR= 0	Parameter FLR= 1
$Sq = Sc \times \frac{q}{100}$	$Sq = Sc \times \frac{q}{1000}$

<2> When time p specified in parameter No. 4914 elapses after the specified speed Sc changes.

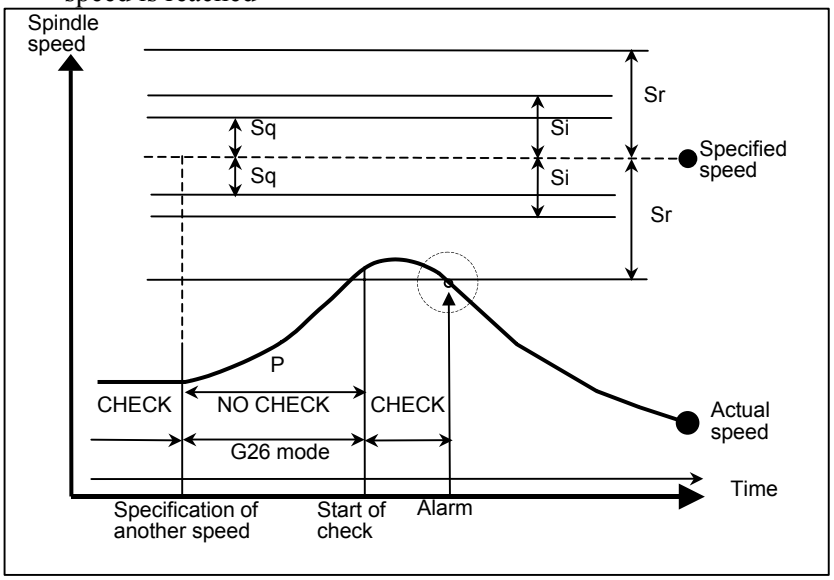
- Examples of spindle speed fluctuation detection

(Example 1)
When an alarm (OH0704) is issued after a specified spindle speed is reached



(Example 2)

When an alarm OH0704 is issued before a specified spindle speed is reached



Specified speed :

(Speed specified by address S and five-digit value) × (spindle override)

Actual speed : Speed detected with a position coder

p : Period after a change occurs in the actual spindle speed until detection starts

Parameter No.4914, address P

Sq : (Specified spindle speed) × (Detection start tolerance (q))

Parameter No.4911, address Q

Parameter FLR = 0	Parameter FLR = 1
$\frac{q}{100}$	$\frac{q}{1000}$

Sr : (Specified spindle speed) × (Allowable variation (r))

Parameter No.4912, address R

Parameter FLR = 0	Parameter FLR = 1
$\frac{r}{100}$	$\frac{r}{1000}$

Si : Allowable variation width

Parameter No.4913, address I

If the difference between the specified speed and actual speed exceeds both Sr and Si, an alarm OH0704 is raised.

- Relationship between spindle speed control and each spindle

Function \ Spindle	Serial spindle			
	1st spindle	2nd spindle	3rd spindle	4th spindle
Spindle speed fluctuation detection	Possible	Possible(*1)	Possible(*1)	Possible(*1)

NOTE
1 An optional function of multi spindle control is necessary.
2 The spindle speed fluctuation detection function is effective for a single spindle. The function cannot be executed for two or more spindles.
The spindle speed fluctuation detection function is effective for a spindle on which the currently selected position coder is mounted. Just a single position coder can be selected. Multiple position coders cannot be selected. For the selection of a position coder, see the section of "Multi spindle."
* Position coder selection signals (PC2SLC<Gn028.7>, PC3SLC<Gn026.0>, PC4SLC<Gn026.1>)
3 The parameters that become valid are the parameters of the spindle speed fluctuation detection function (No.4911, No.4912, No.4913, No.4914) for the spindle on which the currently selected position coder is mounted.

- Spindle for which to detect the spindle speed fluctuation

For the spindle for which to detect the spindle speed fluctuation, refer to the appropriate manual provided by the machine tool builder.

9.6 SPINDLE CONTROL WITH SERVO MOTOR

9.6.1 Spindle Control with Servo Motor

- Command with a program

This function provides SV speed control mode in which spindle rotation commands, S commands, are effective to a rotation axis with a servo motor and position control mode in which normal positioning can be performed. To place a servo motor under speed control, you must turn SV speed control mode ON (G96.4). Once SV speed control mode is turned ON,
S commands are effective to the servo motor until it is canceled.
You cannot specify positioning commands while SV speed control mode is ON. To perform positioning, you must cancel SV speed control mode.
To cancel SV speed control mode (turn position control mode ON), you must specify a spindle indexing command, G96.1/G96.2.
In position control mode, S commands are invalid as in the normal servo axis state. Note, however, S command information is stored; when SV speed control mode is ON, rotation starts at the already specified speed.

- Command with a signal

You can also turn SV speed control mode ON/OFF with the SV speed control mode signal (Gn521). When you turn SV speed control mode ON with the signal, and if you turn SV speed control mode OFF with a program command, you must specify the SV speed control mode signal or specify G96.4 to turn SV speed control mode ON again. You can check the state of SV speed control mode with the SV speed control mode in-progress signal (Fn521). When you turn SV speed control mode signal OFF while rotating, spindle indexing is executed. Then SV speed control mode is turned off. Spindle indexing is executed with R0(absolute position 0).

Format

G96.4 P_ ;	SV speed control mode ON
M03 (M04) S_ P_ ;	Rotation command
S: Spindle speed [min^{-1}] (numeric value of up to five digits)	
P: Spindle selection with multi-spindle control	

Using G96.4 and a spindle selection command P can turn the SV speed control mode ON for each axis separately. Likewise, using a spindle indexing command (G96.1/G96.2) and a spindle selection command P can cancel the SV speed control mode, that is, turn on the position control mode ON, for each axis separately.
To specify “G96.4 P_ ;”, use an independent block.
However, it is possible to specify "G96.4 P_ S_ ;".

9.6.2 Spindle Indexing Function

Format

G96.1 P_ R_ ;	After spindle indexing is completed, the operation of the next block is started.
G96.2 P_ R_ ;	Before spindle indexing is completed, the operation of the next block is started.
G96.3 P_ ;	After it is confirmed that spindle indexing is completed, the operation of the next block is started.

P: Spindle selection with multi-spindle control
R: Stoppage angle [deg] (0 to (parameter No. 1260))

To specify address P, use parameter No. 3781 (P code for selecting a spindle in multi-spindle control).
To turn the position control mode ON without performing spindle indexing, do so after canceling the SV speed control mode by issuing a G96.1 command with no R specified when the motor is at a halt.
When the motor is rotating, issuing a G96.1 (or G96.2) command with no R specified results in the motor coming to a halt by behaving in the same manner as for R0.

NOTE
1 When using G96.2, issue G96.3 before another spindle move command, to make sure that the spindle is at a complete halt.
2 To issue G96.1, G96.2, or G96.3, use an independent block.

Spindle indexing command

- Move command

(1) Command waiting for spindle indexing to be completed
If G96.1 is issued, the next block is executed after spindle indexing is completed.
(2) Command not waiting for spindle indexing to be completed
If G96.2 is issued, the next block can be executed before spindle indexing is completed.

- Movement completion check command

G96.3 is used to check to see if spindle indexing is completed. If it has not been completed, the next block waits for spindle indexing to be completed. If it is completed, the next block is executed.

- SV speed control mode cancellation

If G96.1 is used to perform spindle indexing, the SV speed control mode is canceled when spindle indexing is completed.
If G96.2 is used to perform spindle indexing, G96.3 can be used to check to see if spindle indexing is completed and, if completed, cancel the SV speed control mode. Issuing G96.2 not followed by G96.3 cannot cancel the SV speed control mode even if spindle indexing is completed.
Start SV speed control mode cancellation at a path to which the axis of interest belongs.

- Spindle indexing command during spindle rotation

Issuing G96.1 or G96.2 with a position specified during spindle rotation causes the spindle to stop at the specified position.
Example)

```
M03 S1000 ; ..... Rotation at S1000
G96.1 P1 R180. ; ..... Stoppage of rotation at the 180deg position
```

- Spindle indexing speed

Issuing G96.1 or G96.2 causes a move speed to be dedicated to spindle indexing. Specify the move speed for spindle indexing, using parameter No. 11012.

Spindle indexing command (absolute coordinate/machine coordinate)

Using bit 0 (SIC) of parameter No. 11005 can select which coordinate system, absolute or machine, is to be used in spindle indexing.

Example:

If the difference between the machine and absolute coordinates (machine coordinate – absolute coordinate) is 100.000:

- Spindle indexing performed by turning the SV speed control mode signal (Gn521) OFF (1 to 0) or issuing a spindle indexing G code (with no R specified)
  1. If bit 0 (SIC) of parameter No. 11005 = 0:
     Spindle indexing is performed with a machine coordinate of 100.000 and an absolute coordinate of 0.000.
  2. If bit 0 (SIC) of parameter No. 11005 = 1:
     Spindle indexing is performed with a machine coordinate of 0.000 and an absolute coordinate of 260.000.
- G code-based spindle indexing (with R specified)
  Assuming that spindle indexing is specified with R100.000:
  1. If bit 0 (SIC) of parameter No. 11005 = 0:
     Spindle indexing is performed with a machine coordinate of 200.000 and an absolute coordinate of 100.000.
  2. If bit 0 (SIC) of parameter No. 11005 = 1:
     Spindle indexing is performed with a machine coordinate of 100.000 and an absolute coordinate of 0.000.

10 TOOL FUNCTION (T FUNCTION)

Chapter 10, "TOOL FUNCTION (T FUNCTION)", consists of the following sections:

10.1 TOOL SELECTION FUNCTION.....245
10.2 TOOL MANAGEMENT FUNCTION.....247
10.3 TOOL MANAGEMENT EXTENSION FUNCTION 265
10.4 TOOL MANAGEMENT FUNCTION FOR OVERSIZE TOOLS 277
10.5 TOOL LIFE MANAGEMENT 279

10.1 TOOL SELECTION FUNCTION

By specifying an up to 8-digit numerical value following address T, a code signal and a strobe signal are transmitted to the machine tool. This is used to select tools on the machine.
One T code can be commanded in a block. Refer to the machine tool builder's manual for the number of digits commendable with address T and the correspondence between the T codes and machine operations.
When a move command and a T code are specified in the same block, the commands are executed in one of the following two ways:

(i) Simultaneous execution of the move command and T function commands.

(ii) Executing T function commands upon completion of move command execution.

The selection of either (i) or (ii) depends on the machine tool builder's specifications. Refer to the manual issued by the machine tool builder for details.

Explanations

T

The value after the T code indicates the desired tool. Part of the value is also used as a tool offset number that specifies the amount of tool offset or the like. The tool can be selected as follows according to the specification method and parameter settings.

Description of a T code (Note 1)		How to specify the offset number for each parameter setting (Note 2)
LGN (No.5002#1) = 0	LGN (No.5002#1) = 1	
T○○○○○○○ ○ ↑ ↑ Tool selection Tool geometry tool wear offset	T○○○○○○○ ○ ↑ ↑ Tool selection tool geometry offset Tool wear offset	The tool wear offset number is specified using the low-order one digit of a T code.
		When parameter (No.5028) is set to 1
T○○○○○○ ○○ ↑ ↑ Tool selection Tool geometry tool wear offset	T○○○○○○ ○○ ↑ ↑ Tool selection tool geometry offset Tool wear offset	The tool wear offset number is specified using the low-order two digits of a T code.
		When parameter (No.5028) is set to 2
T○○○○○ ○○○ ↑ ↑ Tool selection Tool geometry tool wear offset	T○○○○○ ○○○ ↑ ↑ Tool selection tool geometry offset Tool wear offset	The tool wear offset number is specified using the low-order three digits of a T code.
		When parameter (No.5028) is set to 3

NOTE
1 The maximum number of digits of a T code can be specified by parameter (No.3032) as 1 to 8.
2 When parameter (No.5028) is set to 0, the number of digits used to specify the offset number in a T code depends on the number of tool offsets.
Example)
When the number of tool offsets is 1 to 9:
Low-order one digit
When the number of tool offsets is 10 to 99:
Low-order two digits
When the number of tool offsets is 100 to 999:
Low-order three digits

Refer to the machine tool builder's manual for correspondence between the T-code and the tool and the number of digit to specify tool selection.

Example (T2+2)
N1G00X1000Z1400;
N2T0313; (Select tool No. 3 and tool offset value No. 13)
N3X400Z1050;

10.2 TOOL MANAGEMENT FUNCTION

Overview

The tool management function totally manages tool information including information about tool offset and tool life.

Explanation

A tool type number is specified with a T code. The tool type number is any number the user can define freely. With tool type numbers, tools can be grouped by various conditions such as life, compensation value, and cutting conditions. When each type is assumed to have a single tool, tool type numbers are equivalent to unique tool numbers.
For each tool, an information storage area is prepared in the CNC (SRAM). This area contains information such as a tool type number, tool life, tool status (such as a breakage condition), tool compensation number (H, D, G, or W), spindle speed (S), cutting feedrate (F), and freely-definable customize data. Such data is called tool management data.
A cartridge management table that links cartridge information and tool management data is provided so that the CNC can manage the cartridges of the machine and tool change operations. In addition, areas for managing the tools in the spindle and tool standby positions are prepared.
When a tool type number is specified using a T code command, a tool that has the tool type number and the shortest life is searched for, and the cartridge number and pot number of the location where the tool is stored are output to the PMC. Then, a tool change operation using the cartridge number and pot number is enabled by the PMC ladder sequence.
Tool information in the CNC is managed by the tool management data and cartridge management table (including the spindle management table and standby position table).

- Tool management data

As tool management data, information about each tool is registered with a tool management data number.
The following number of sets of tool management data can be used:

Tool management function 64 sets	64 sets in total
Tool management function 240 sets	240 sets in total
Tool management function 1000 sets	1000 sets in total

NOTE
For the number of tool management data sets, refer to the relevant manual issued by the machine tool builder.

- Details of data

The following details the tool management data registered for each data number:

- Tool type number (T code)

Item	Description
Data length	4byte
Valid data range	0,1 to 99,999,999

- Tool life counter

Item	Description
Data length	4byte
Unit of data	When the number of use times is specified: Times When time is specified: Seconds
Valid data range	When the number of use times is specified: 0 to 99,999,999 times When time is specified: 0 to 3,599,999 s (999 hours 59 minutes 59 seconds)

The value of an increment counter, that is, the number of use times (time) is indicated. The remaining life value is [the maximum tool life value minus tool life counter value].

- Maximum tool life value

Item	Description
Data length	4byte
Unit of data	When the number of use times is specified: Times When time is specified: Seconds
Valid data range	When the number of use times is specified: 0 to 99,999,999 times When time is specified: 0 to 3,599,999 s (999 hours 59 minutes 59 seconds)

- Notice life value

Item	Description
Data length	4byte
Unit of data	When the number of use times is specified: Times When time is specified: Seconds
Valid data range	When the number of use times is specified: 0 to 99,999,999 times When time is specified: 0 to 3,599,999 s (999 hours 59 minutes 59 seconds)

If a value other than 0 is set, the tool life expiration notice signal TLCHB <F064.3> or TLCHBx <F329.4 to 7> is output when the remaining life value of the tool (= maximum tool life value minus tool life counter value) has reached the set value.

- Tool life status

Item	Description
Data length	1byte
Detail data	0: Life management is not performed. 1: Tool not yet used 2: Life remains. 3: Life expired. 4: Tool breakage (skip)

The machine (PMC) determines tool breakage and stores corresponding information through the window. In tool management of the CNC, a broken tool is regarded as being equivalent to tools whose lives have expired.

- Tool information

Item	Description
Data length	1 byte (flag data)
#0 RGS	0: Tool management data is invalid. (-) 1: Tool management data is valid. (R)
#1 TIM	0: Type for counting the number of use times (C) 1: Type for counting time (T)
#2 BDT	0: Normal tool (N) 1: Oversize tool (B)
#3 LOC	0: Data accessible (U) 1: Data not accessible (L)
#4 SEN	When the tool life status indicates that tool management is not performed: 0: This tool is not searched. (-). 1: This tool is searched. (S)
#5 to #7	Reserved

When RGS is set to 0 in tool management data, the tool management data is regarded as not being registered even when values are set for other items.

NOTE
Make sure you specify the same life count type for all tools of the same type. You can check tools of the same type for any difference in life count type, using the "check function."

M

- Tool length compensation number (H)

Item	Description
Data length	2byte
Valid data range	0 to 999

- Cutter compensation number (D)

Item	Description
Data length	2byte
Valid data range	0 to 999

T

- Tool geometry compensation number (G)

Item	Description
Data length	2byte
Valid data range	0 to 999

- Tool wear compensation number (W)

Item	Description
Data length	2byte
Valid data range	0 to 999

NOTE
When the machine control type is the combined system type, tool length compensation and cutter compensation numbers are used for paths for the machining center system, and for paths for the lathe system, tool geometry compensation and tool wear compensation numbers are used.

- Spindle speed (S)

Item	Description
Data length	4byte
Unit of data	min^{-1}
Valid data range	1 to 99,999

- Feedrate (F)

Item	Description
Data length	4byte
Unit of data	mm/min, inch/min, deg/min, mm/rev, inch/rev
Valid data range	0 to 99,999,999

As additional tool management data, areas for setting customize data (5 data items including customize data 0 and customize data 1 to 4) are provided without defining specific usage. The user can use these customize data areas freely to set a warning life value, cutting resistance, override value, spindle current value, maximum and minimum S/F, and other items according to the targeted application.

- Customize data 0

Item	Description
Data length	1 byte (bit type)
Valid data range	0 or 1 on a bit-by-bit basis

- Customize data 1 to 4 (to 20) (to 40)

Item	Description
Data length	4byte
Valid data range	-99,999,999 to 99,999,999

NOTE
For the maximum number of tool management function customization data items, refer to the relevant manual issued by the machine tool builder.

- Cartridge management table

The storage status of tools in cartridges is managed with the cartridge management table.

- Multiple cartridge numbers can be defined. (Numbers from 1 to 4, up to four cartridges)
- The maximum number of pots for all cartridges is 64, 240, or 1000, which depends on whether a tool management data option is selected.
- Cartridge numbers and pot numbers can be assigned freely within cartridge management data by parameter setting (described later).
- The tool management data number adjacent to a pot number is linked with tool data defined in the tool management table. Therefore, the tool attached to the pot is indicated.
- Zero set as a data number indicates that no tool is attached.
- The cartridge management table can be read from and written to through the PMC window and FOCAS2.
- The spindle management table and standby position table are provided to indicate special cartridge positions.
- Spindle positions and standby positions, regarded as special cartridge positions, have fixed cartridge numbers 11 to 14 (the positions of the first to fourth spindles) and 21 to 24 (the first to fourth standby positions).
- With the PMC window, the spindle position table and standby position table can be read from and written to.
- Tool life counting is performed only for the tools at the spindle positions.

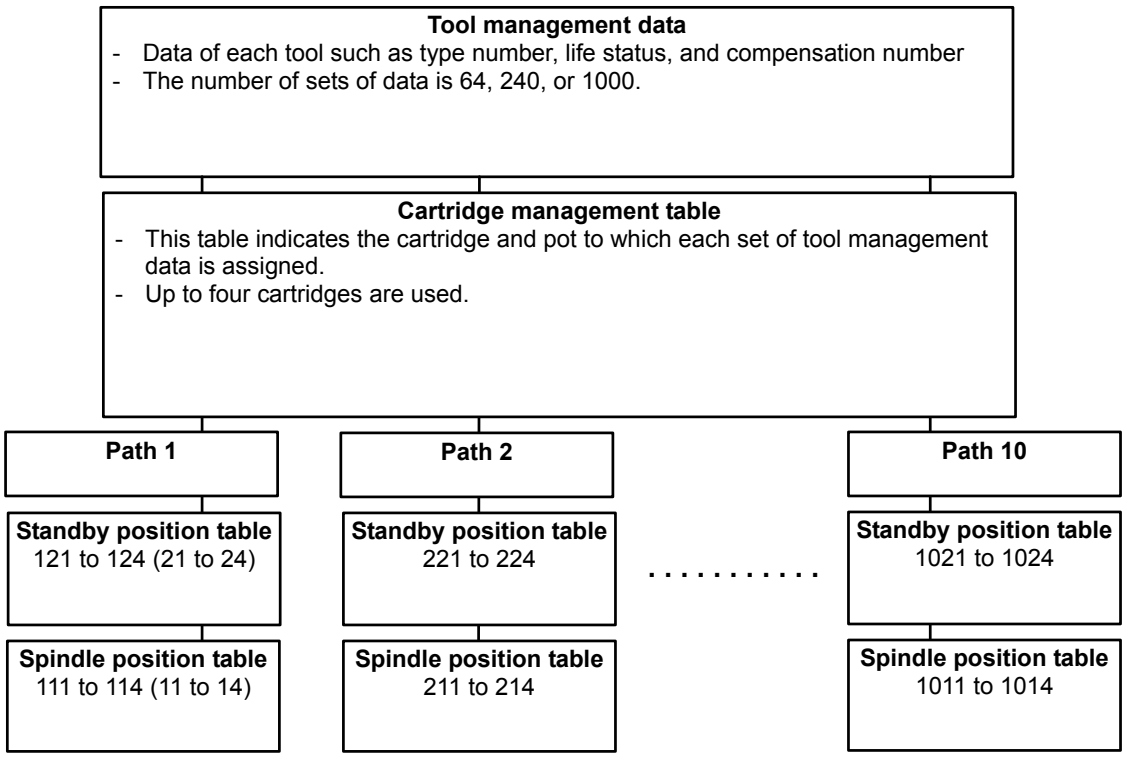
- Multi-path system

The tool management data and cartridge management table are common data among the paths. The spindle management table and standby position table, however, are treated as independent data for each path.
When the spindle table or standby position table is specified as a cartridge from the PMC window, specify the following, in which the path number is set in the hundred's place:

	Spindle position			
	First	**Second**	**Third**	**Fourth**
First path	111(11)	112(12)	113(13)	114(14)
Second path	211	212	213	214
Third path	311	312	313	314
Fourth path	411	412	413	414
Fifth path	511	512	513	514
Sixth path	611	612	613	614
Seventh path	711	712	713	714
Eighth path	811	812	813	814
Ninth path	911	912	913	914
Tenth path	1011	1012	1013	1014

	Standby position			
	First	**Second**	**Third**	**Fourth**
First path	121(21)	122(22)	123(23)	124(24)
Second path	221	222	223	224
Third path	321	322	323	324
Fourth path	421	422	423	424
Fifth path	521	522	523	524
Sixth path	621	622	623	624
Seventh path	721	722	723	724
Eighth path	821	822	823	824
Ninth path	921	922	923	924
Tenth path	1021	1022	1023	1024

NOTE
When specifying 111, 121, and so on to specify the first path, you may specify just 11, 21, and so on.



- M/T series

The tool management data and cartridge management table are data common to the M and T series.

- Tool life management and tool change

The CNC performs tool life management, regarding tools having the same tool type number as a group. When a tool type number (T code) is specified by an NC program, the tool management data registered in the CNC is searched to find a tool having the shortest life among the tools with the same tool type number.
The cartridge number and pot number corresponding to the searched tool are output as a T code signal to the PMC. Based on the output cartridge number and pot number, the PMC performs preparation for a tool change (to the next tool). A different tool can also be selected on the PMC side.
The CNC performs tool life counting for each tool that is at a spindle position in the spindle management table.

When the lives of all tools having the tool type number specified by the T code have expired, alarm PS5317 is issued. If there is a tool in the spindle position or standby position, that tool is selected to continue machining.
As the T code signal, a specified tool type number instead of the cartridge number and pot number can also be output directly by parameter setting.

There are two types of tool life management counting methods: counting the number of use times and counting cutting time. One of the counting methods is set in tool information of tool management data.

Other major specifications related to tool life management are as follows:
Tool type number (T code): Up to 8 digits (1 to 99,999,999)
Maximum tool life value:
99,999,999 times when the number of use times is specified
999 hours 59 minutes 59 seconds when time is specified
Life count interval when time is specified: 1 second
Tool life management count restart M code: Enabled
Tool life count override: Enabled

To specify a certain tool directly without performing tool life management by tool type number, use the following format:

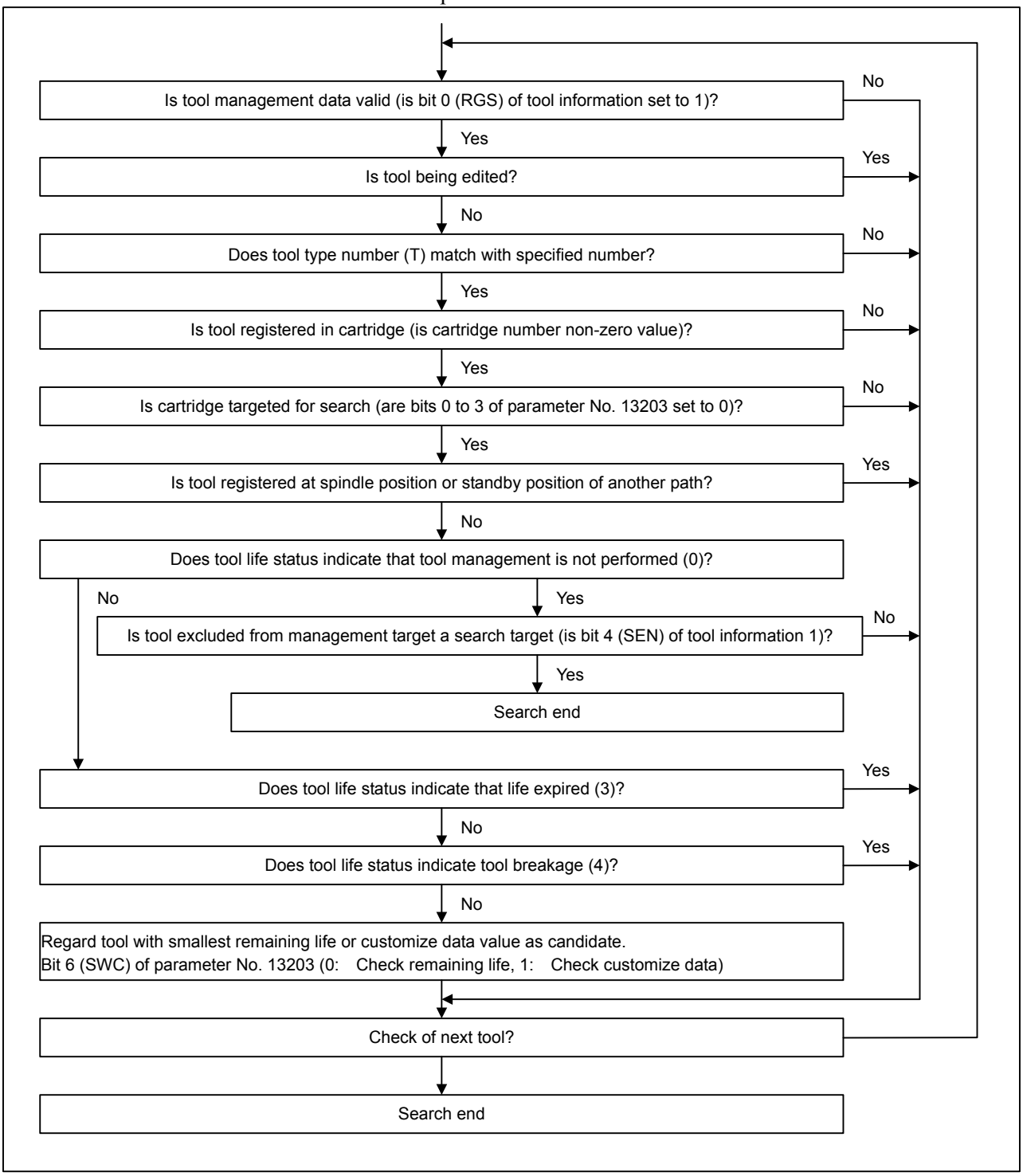
M (value in parameter No. 13252) T (cartridge number) (pot number) ;

When parameter No. 13252 is set to 333, the cartridge number is 2, and the pot number is 27, for example, the following command is specified:

M333 T20027;

- Tool search order

Tools having a tool type number (T) specified by a program are searched sequentially from tool management data number 1 while registered data contents are checked. The following shows how a search operation is made within the NC:



- System variables

The following tool management data of the tool being used as a spindle after a tool change by M06 and the tool to be used next which is specified by a T code can be read through custom macro variables:

Being used	Item	
#8401	Tool management data number	
#8402	Tool type number (T)	
#8403	Tool life counter	
#8404	Maximum tool life value	
#8405	Tool notice life value	
#8406	Tool life status	
#8407	Customize data 0 (bit)	
#8408	Tool information	
#8409	Tool length compensation number (H)	For machining center systems
#8410	Cutter compensation number (D)	
#8411	Spindle speed (S)	
#8412	Cutting feedrate (F)	
#8413	Tool geometry compensation number (G)	For lathe systems
#8414	Tool wear compensation number (W)	
#8431	Customize data 1	
#8432	Customize data 2	
#8433	Customize data 3	
#8434	Customize data 4	
#8435	Customize data 5	
#8436	Customize data 6	
#8437	Customize data 7	
#8438	Customize data 8	
#8439	Customize data 9	
#8440	Customize data 10	
#8441	Customize data 11	
#8442	Customize data 12	
#8443	Customize data 13	
#8444	Customize data 14	
#8445	Customize data 15	
#8446	Customize data 16	
#8447	Customize data 17	
#8448	Customize data 18	
#8449	Customize data 19	
#8450	Customize data 20	
#8451	Customize data 21	
#8452	Customize data 22	
#8453	Customize data 23	
#8454	Customize data 24	
#8455	Customize data 25	
#8456	Customize data 26	
#8457	Customize data 27	
#8458	Customize data 28	
#8459	Customize data 29	
#8460	Customize data 30	
#8461	Customize data 31	
#8462	Customize data 32	
#8463	Customize data 33	

Being used	Item
#8464	Customize data 34
#8465	Customize data 35
#8466	Customize data 36
#8467	Customize data 37
#8468	Customize data 38
#8469	Customize data 39
#8470	Customize data 40

When a cartridge number of a spindle position (11 to 14) or standby position (21 to 24) is specified in #8400, information about the corresponding position can be read.
If the spindle position table or standby position table has an empty pot, <empty> is read from #8402 to #8470.
Value 0 is read from #8401 (data number).

Therefore, machining conditions registered in tool management data can be specified directly by coding, for example, D#8410, H#8409, S#8411, and F#8412 with a tool change macro (such as M06). Similarly, customize data can be referenced by a custom macro, and machining programs can be customized according to the tool used.

NOTE
For the maximum number of tool management function customization data items, refer to the relevant manual issued by the machine tool builder.

- Specifying a tool compensation number

M

When parameter No. 13265 is 0, a compensation number registered as tool management data of a tool attached at a spindle position can be selected by specifying H99 or D99.
(99 is treated as a special number, so 99 cannot be specified directly as a compensation number.)
When other than 0 is set in parameter No. 13265, the number set in the parameter can be specified instead of 99. For example, if parameter No. 13265 is 3, specifying H3 specifies the tool length compensation number registered for the tool at the spindle position.

T

When the tool management function is not used, a tool compensation number is specified also with a T code; when the tool management function is used, the T code is used only to specify a tool type number, so a tool compensation number must be specified with address D.
Except the specifications for addresses, the specifications for the number of permissible digits (parameter No. 3032) and the number of digits consisting of a compensation number (parameter No. 5028), separation of geometry compensation numbers and wear compensation numbers (bit 1 (LGN) of parameter No. 5002), and so on are the same as for T.

If parameter No. 13265 is set to 0, when a compensation number registered for a tool attached at a spindle position is specified, the specification format varies according to the number of digits as follows, which is the same as for conventional T:
When the compensation number is 1 digit long: D9
When the compensation number is 2 digits long: D99
When the compensation number is 3 digits long: D999
Because 9, 99, or 999 is treated as a special number, it cannot be specified directly as a compensation number.
Unless parameter No. 13265 is set to 0, the number set in the parameter can be specified instead of 9, 99, or 999. When parameter No. 13265 is set to 3, specifying D3 specifies the tool geometry compensation number and tool wear compensation number registered for a tool attached at the spindle position.

Multi-path system

Depending on whether the local path is a machining center system or a lathe system, tool compensation numbers are specified by using one of the above methods.

Spindle selection

When specifying compensation numbers of a tool attached to a spindle other than the first spindle, specify the spindle number with address P within the same block that contains H/D. When specifying the first spindle, you can omit P.
D99 P3; Specifying compensation numbers registered for the tool attached at the third spindle
D99 ; Specifying compensation numbers registered for the tool attached at the first spindle

- Read/write operations for tool management data and cartridge management table

The tool management data and cartridge management table can be read from and written to by using the CNC/PMC data window library (FOCAS2). Therefore, a specific tool management system including all available tool data not registered in the CNC can be built easily by using the OPEN CNC.

Similarly, the tool management data and cartridge management table can also be read from and written to using the PMC window.

Tool management data of the tool being used as a spindle after a tool change operation performed by M06 and the tool being selected by specifying a T code can be read using a custom macro.

The user can modify tool management data by MDI from the tool management function screen of the CNC. Addition, modification, and deletion of the above tool management data can be made from a part program (G10). Such data can also be input to and output from external I/O equipment by using the ALL I/O screen and tool management function screen.

- G10 format

Addition, modification, and deletion are performed for the tool management data and cartridge management table from programs. For use of this function, the programmable data input option function is required.
If a format error is found in the commands from G10 L75/L76/L77 to G11, or if a value beyond the valid data range is specified, alarm PS5312 is issued. In such a case, correct the program. Within the range from G10 to G11, no decimal point can be specified with any address. If a decimal point is specified, alarm PS0007 results.

- Registering new tool management data

Tool management data can be registered. When data is punched out to an external device from the tool management data screen, this format is used. The specification of those items that are not registered may be omitted.

G10 L75 P1;
N_ ; Tool management data number specification
T_ C_ L_ I_ B_ Q_ H_ D_ S_ F_ J_ K_ ;
P0 R_ ; Customization data 0
P1 R_ ; Customization data 1
P2 R_ ; Customization data 2
P3 R_ ; Customization data 3
P4 R_ ; Customization data 4
N_ ; Tool management data number
:
G11;

N_	Tool management data No.	1 to 64 (1 to 240, 1 to 1000)
T_	Tool type No. (T)	0 to 99,999,999
C_	Tool life counter	0 to 99,999,999
L_	Maximum tool life	0 to 99,999,999
I_	Noticed life	0 to 99,999,999
B_	Tool life state	0 to 4
Q_	Tool information	Bit format (8 bits)
H_	Tool length compensation No. (H)	0 to 999 (M series)
D_	Cutter compensation No. (D)	0 to 999 (M series)
S_	Spindle speed (S)	0 to 99,999
F_	Cutting feedrate (F)	0 to 99,999,999
J_	Tool geometry compensation No. (G)	0 to 999 (T series)
K_	Tool geometry compensation No. (W)	0 to 999 (T series)
P_	Customization data No.	0 to 4 (0 to 20, 0 to 40)
R_	Customization data value	-99,999,999 to 99,999,999

Specify customization data in the following format:

P (customization-number) R (value)

Use the bit format only when specifying the customization data 0 (P0). Specify other data in the binary format. The specification of customization data that need not be set may be omitted.

Example)
G10 L75 P1;
N1 ; Tool management data No.1
T10000001 C0 L1000 B1 Q00000001 H1 D1 S4000 F10000 ;
P0 R11101101 ; Customization data 0
P4 R99999999 ; Customization data 4
N2 ; Tool management data No.2
:
G11 ;

Modifying tool management data

Tool management data can be modified.
The specification of those items that are not modified may be omitted.

```
G10 L75 P2 ;
N_ ;
T_ C_ L_ I_ B_ Q_ H_ D_ S_ F_ J_ K_ ;
P_ R_ ;
N_ ;
 :
G11 ;
```

Deleting tool management data

The data of a specified data number can be deleted from tool management data.
The cartridge management table data corresponding to a deleted tool management data number is also deleted. (The tool management data number in the cartridge management table is cleared to 0.)

```
G10 L75 P3 ;
N_ ;
N_ ;
 :
N_ ;
G11 ;
```

Registering new cartridge management table data

A tool management data number can be registered with a free pot in the cartridge management table.

```
G10 L76 P1 ;
N cartridge-number P pot-number R tool-management-data-number ;
N cartridge-number P pot-number R tool-management-data-number ;
N cartridge-number P pot-number R tool-management-data-number ;
N cartridge-number P pot-number R tool-management-data-number ;
G11 ;
```

For a spindle position table and standby position table, only cartridge number data is specified.
Example)
G10 L76 P2 ;
N11 R1; Changes the tool management data number of the spindle position to No. 1.
N21 R29; Changes the tool management data number of the standby position to No. 29.
G11 ;

Modifying the cartridge management table

Tool management data numbers in the cartridge management table can be modified.

```
G10 L76 P2 ;
N cartridge-number P pot-number R tool-management-data-number ;
N cartridge-number P pot-number R tool-management-data-number ;
N cartridge-number P pot-number R tool-management-data-number ;
N cartridge-number P pot-number R tool-management-data-number ;
G11 ;
```

For a spindle position table and standby position table, only cartridge number data is specified.
Example)
G10 L76 P2 ;
N11 R1; Changes the tool management data number of the spindle position to No. 1.
N21 R29; Changes the tool management data number of the standby position to No. 29.
G11 ;

Deleting cartridge management table data

Tool management data numbers can be deleted from the cartridge management table.

```
G10 L76 P3 ;
N cartridge-number P pot-number R tool-management-data-number ;
N cartridge-number P pot-number R tool-management-data-number ;
N cartridge-number P pot-number R tool-management-data-number ;
N cartridge-number P pot-number R tool-management-data-number ;
G11 ;
```

For a spindle position table and standby position table, only cartridge number data is specified.
Example)
G10 L76 P3 ;
N11 ; Deletes the tool management data number of the spindle position (clears the number to 0).
N21 ; Deletes the tool management data number of the standby position (clears the number to 0).
G11 ;

Naming customization data

The display name of customization data (0 to 40) can be set.

```
G10 L77 P1 ;
N_ ;
P_ R_ ;
P_ R_ ;
;
N_ ;
P_ R_ ;
P_ R_ ;
G11 ;
```

N_: Customization data No. (0 to 40)
P_: Character No. (1 to 16)
R_: Character code (ANK or shifted JIS)
- When a shifted JIS code is used, an area for two characters is used.
- Specify 0 to clear data.

- A set display name can be checked on the tool management data screen only.
- When no name is registered, a name such as the default "Customize 3" is displayed.
- A name consisting of up to 16 characters is displayed. For a name shorter than 16 characters, register 0 in the empty area. Those characters that are registered immediately before 0 are displayed.
- To clear data, set 0 as a character code.
- No character code data check is made.

When the name of customization data 3 is set as "MEASURE5", for example, specify the following:

Example)

```
G10 L77 P1 ;
N3 ;          Specifies customization data 3.
P1 R32 ;      character ‘space’ ASCII code 20h
P2 R77 ;      character “M” ASCII code 4Dh
P3 R69 ;      character “E” ASCII code 45h
P4 R65 ;      character “A” ASCII code 41h
P5 R83 ;      character “S” ASCII code 53h
P6 R85 ;      character “U” ASCII code 55h
P7 R82 ;      character “R” ASCII code 54h
P8 R69 ;      character “E” ASCII code 45h
P9 R53 ;      character “5” ASCII code 35h
P10 R0 ;      Clears data.  (Not displayed.  End)
G11 ;
```

Naming tool life states

The display name of a tool life state (0 to 4) can be set.

```
G10 L77 P2 ;
N_ ;
P_ R_ ;
P_ R_ ;
N_ ;
P_ R_ ;
P_ R_ ;
G11 ;
```

N_: Tool life state (0 to 4)
P_: Character No. (1 to 12)
R_: Character code (ANK or shifted JIS)
- When a shifted JIS code is used, an area for two characters is used.
- Specify 0 to clear data.

- A set display name can be checked on the tool management data screen only.
- If no name is registered, a default such as "Remaining" and "Unremaining" is displayed.
- A name consisting of up to 12 characters is displayed. For a name shorter than 12 characters, register 0 in the empty area. Those characters that are registered immediately before 0 are displayed.
- To clear data, set 0 as a character code.
- No character code data check is made.
- The defaults are as follows:
  - 0: Invalid
  - 1: Remaining
  - 2: Remaining
  - 3: Unremaining
  - 4: Broken

To set "NORMAL" as the name of tool life state 2 (Remaining), specify the following:

Example)

```
G10 L77 P2 ;
N2 ;          Specifies tool life state 2.
P1 R78 ;      character “N” ASCII code 4Eh
P2 R79 ;      character “O” ASCII code 4Fh
P3 R82 ;      character “R” ASCII code 52h
P4 R77 ;      character “M” ASCII code 4Dh
P5 R65 ;      character “A” ASCII code 41h
P6 R76 ;      character “L” ASCII code 4Ch
P7 R0 ;       Clears data.  (Not displayed.  End)
G11 ;
```

10.3 TOOL MANAGEMENT EXTENSION FUNCTION

Overview

The following functions have been added to the tool management function:

1. Customization of tool management data display
2. Setting of spindle position/standby position display
3. Input of customize data with the decimal point
4. Protection of various tool information items with the KEY signal
5. Selection of a tool life count period
6. Each tool data screen
7. Total life time display for tools of the same type

They will be explained in detail below.

10.3.1 Customization of Tool Management Data Display

With the tool management data screen display customization function, the display positions of screen elements (type number, tool information, life counter, and so forth) on the tool management screen can be changed and whether to display or hide such screen elements can be chosen using the G10 format. This function enables a customized tool management screen to be configured.

Format

```
G10L77P3;
N_ R_;
G11;
  N_: Tool management data screen display position number
  R_: Tool management data display item
```

Explanation

- number (N_)

- A display position number represents an ordinal number from the leftmost position on the tool management data screen.
- As N, a number from 1 to 200 can be specified.

- Tool management data display item (R_)

- For a value to be set (in R_), see the table below.
- For actual setting, consider a display width plus 1.

Items common to machining center systems and lathe systems

R	Item	Display width	Remarks
-1	End of setting	-	
0	Blank column	10	
1	No.	4	For spindle/standby positions for 10 paths
2	TYPE-NO.	8	
3	MG	4	
4	POT	5	

R	Item	Display width	Remarks
5	T-INFORMATION	10	
6	L-COUNT	10	
7	MAX-LIFE	10	
8	NOTICE-L	10	
9	L-STATE	6 or 12	The display width is switched by bit 1 of parameter No. 13201.
10	S (Spindle speed)	10	
11	F (Feedrate)	10	
12	Tool figure number (A)	3	

Offset-related items for machining center systems

R	Item	Display width	Remarks
20	H (Tool length compensation number)	4	
21	D (Cutter compensation number)	4	
22	OFFSET－M	10	Tool offset memory A
23	GEOMETRY－M	10	Tool offset memory B
24	WEAR－M	10	
25	GEOMETRY（H）	10	Tool offset memory C
26	WEAR（H）	10	
27	GEOMETRY（D）	10	
28	WEAR（D）	10	

Offset-related items for lathe systems

R	Item	Display width	Remarks
40	TG (Tool geometry compensation number)	4	
41	TW (Tool wear compensation number)	4	
42	OFFSET－X	10	
43	OFFSET－Z	10	
44	OFFSET－R	10	Tool nose radius compensation
45	TIP	10	
46	OFFSET－Y	10	Y-axis offset
47	OFFSET－B	10	B-axis control
48	GEOMETRY (X)	10	Tool geometry/wear compensation
49	WEAR (X)	10	
50	GEOMETRY (Z)	10	
51	WEAR (Z)	10	
52	GEOMETRY (R)	10	Tool nose radius compensation, Tool geometry/wear compensation
53	GEOMETRY TIP	10	
54	WEAR (R)	10	
55	WEAR TIP	10	
56	GEOMETRY (Y)	10	Y-axis offset, Tool geometry/ wear compensation
57	WEAR (Y)	10	
58	GEOMETRY (B)	10	B-axis control, Tool geometry/ wear compensation
59	WEAR (B)	10	
60	GEOMETRY (X2)	10	2nd geometry tool offset
61	GEOMETRY (Z2)	10	
62	GEOMETRY (Y2)	10	

Items related to customize data

R	Item	Display width	Remarks
80	CUSTOM 0	10	
81	CUSTOM 1	10	
82	CUSTOM 2	10	
83	CUSTOM 3	10	
84	CUSTOM 4	10	
85	CUSTOM 5	10	Tool management function customize data extension (5 to 20) or tool management function customize data extension (5 to 40)
86	CUSTOM 6	10	
87	CUSTOM 7	10	
88	CUSTOM 8	10	
89	CUSTOM 9	10	
90	CUSTOM 10	10	
91	CUSTOM 11	10	
92	CUSTOM 12	10	
93	CUSTOM 13	10	
94	CUSTOM 14	10	
95	CUSTOM 15	10	
96	CUSTOM 16	10	
97	CUSTOM 17	10	
98	CUSTOM 18	10	
99	CUSTOM 19	10	
100	CUSTOM 20	10	
101	CUSTOM 21	10	Tool management function customize data extension (5 to 40)
102	CUSTOM 22	10	
103	CUSTOM 23	10	
104	CUSTOM 24	10	
105	CUSTOM 25	10	
106	CUSTOM 26	10	
107	CUSTOM 27	10	
108	CUSTOM 28	10	
109	CUSTOM 29	10	
110	CUSTOM 30	10	
111	CUSTOM 31	10	
112	CUSTOM 32	10	
113	CUSTOM 33	10	
114	CUSTOM 34	10	
115	CUSTOM 35	10	
116	CUSTOM 36	10	
117	CUSTOM 37	10	
118	CUSTOM 38	10	
119	CUSTOM 39	10	
120	CUSTOM 40	10	

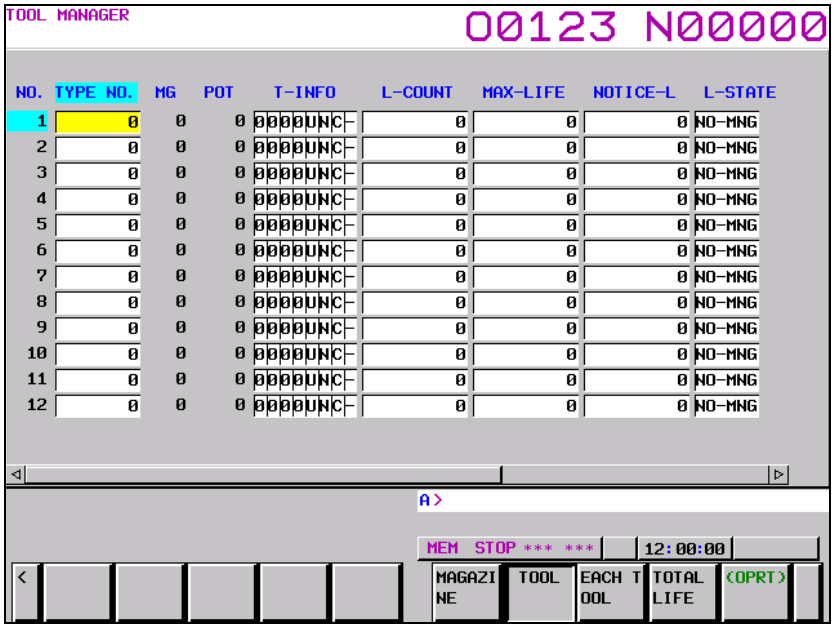
NOTE
1 If G10 L77 P3 is terminated normally, the power must be turned off before operation is continued.
2 The setting becomes effective after the power is turned off then back on.

Example

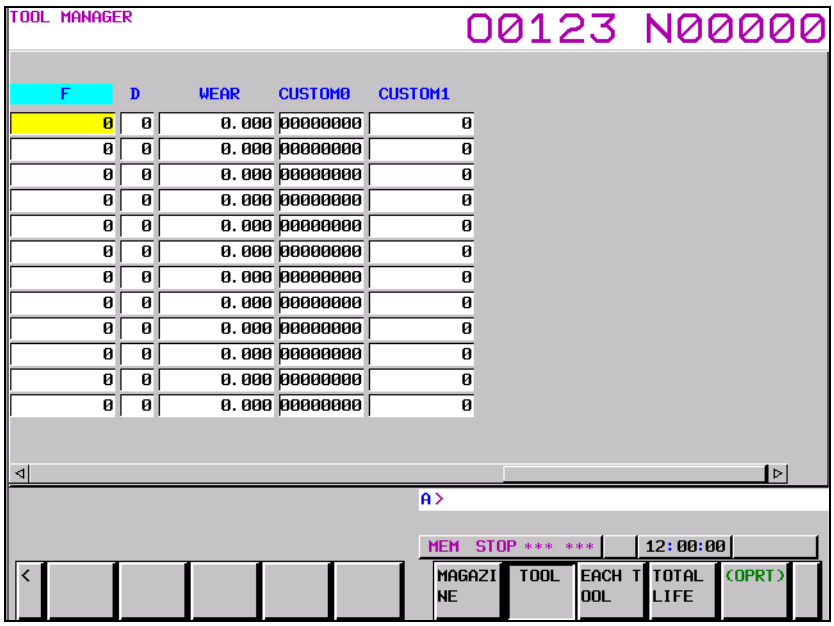
Example of setting tool offset memory A

G10L77P3;	Set tool management data screen display customization
N1 R1;	Set No. as number 1
N2 R2;	Set TYPE-NO. as number 2
N3 R3;	Set MG as number 3
N4 R4;	Set POT as number 4
N5 R5;	Set T-INFORMATION as number 5
N6 R6;	Set L-COUNT as number 6
N7 R7;	Set MAX-LIFE as number 7
N8 R8;	Set NOTICE-L as number 8
N9 R9;	Set L-STATE as number 9
N10 R11;	Set F (Feedrate) as number 10
N11 R21;	Set D as number 11
N12 R22;	Set OFFSET-M as number 12
N13 R80;	Set CUSTOM 0 as number 13
N14 R81;	Set CUSTOM 1 as number 14
N15 R-1;	End
G11;	Cancel the setting mode

On the tool management data screen, the first page displays "No., Type Number, MG, Pot, Tool Information, Life Counter, Maximum Life, Noticed Life, and Life State".
The second page displays "F (Feedrate), D (Cutter Compensation Number), Tool Offset-M, Customize Data 0, and Customize Data 1".



Example 1: Page 1



Example 1: Page 2

NOTE

1 This setting is enabled when bit 0 (TDC) of parameter No. 13201 is set to 1.
2 Up to 20 pages can be set.
3 Be sure to specify an end.
4 If an item that requires the corresponding option is specified without specifying the option, the item is displayed as a blank field.
5 No items after an end are displayed.
(Example) In the case of N1 → R1 (No.), N2 → R2 (Type Number), N 3 → R3 (MG), N4 → R-1 (End), N5 → R4 (Pot), only the items "No., Type Number, and MG" are displayed on the screen.
6 If G10 L77 P3 is specified, the setting immediately before is initialized.
7 If an invalid value is set in R (tool management data display item), a blank field is displayed.

10.3.2 Setting of Spindle Position / Standby Position Display

In MG on the tool management data screen, a spindle position or standby position is displayed as a number such as 11, 12, and 13. With the spindle position/standby position display setting function, three arbitrary characters can be displayed using the G10 format.

Format

```
G10L77P4;
N_;
P_R_;
G11;
```

N_: Spindle position/standby position setting
P_: Character number
R_: Character code

Explanation

- Spindle position/standby position setting (N_)

Specify a spindle position or standby position to be renamed. The table below indicates the values to be specified.

	Spindle position			
	First	Second	Third	Fourth
1st path	111	112	113	114
2nd path	211	212	213	214
3rd path	311	312	313	314
4th path	411	412	413	414
5th path	511	512	513	514
6th path	611	612	613	614
7th path	711	712	713	714
8th path	811	812	813	814
9th path	911	912	913	914
10th path	1011	1012	1013	1014

	Standby position			
	First	Second	Third	Fourth
1st path	121	122	123	124
2nd path	221	222	223	224
3rd path	321	322	323	324
4th path	421	422	423	424
5th path	521	522	523	524
6th path	621	622	623	624
7th path	721	722	723	724
8th path	821	822	823	824
9th path	921	922	923	924
10th path	1021	1022	1023	1024

- Character number (P_)

Specify a character number (1 to 3). Up to three characters are displayed. If a character string to be specified is shorter than three characters, specify 0 in the leading blank character position(s). A character string immediately before 0 is displayed.

- Character code (R_)

Set the name of a spindle position/standby position by using a character code (ASCII code or Shift JIS code).

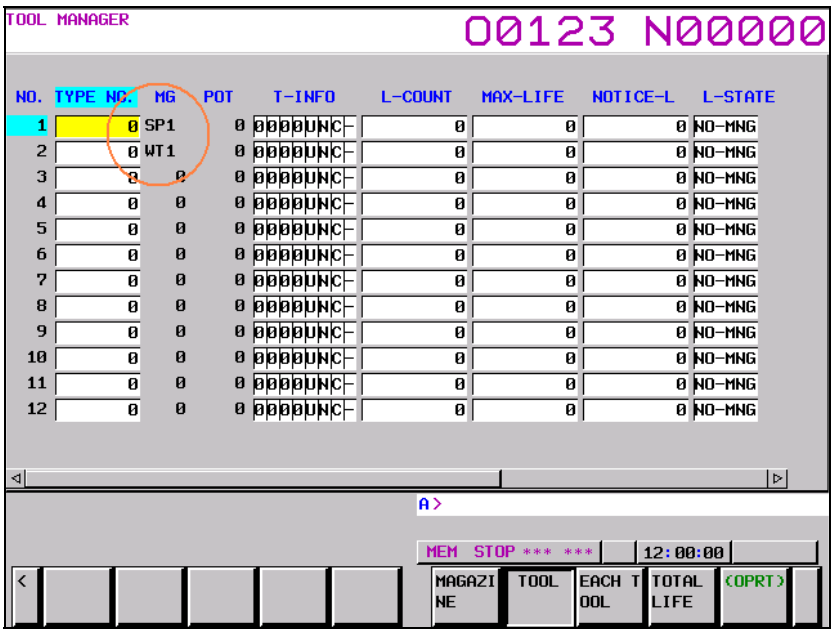
- For a character code, use ASCII code or Shift JIS code.
- No character code data check is made.

Example

When spindle 1 is named "SP1", and standby 1 is named "WT1"

G10L77P4;	Set spindle position/standby position
N111;	Specify spindle 1
P1 R83;	53h as ASCII code for "S"
P2 R80;	50h as ASCII code for "P"
P3 R49;	31h as ASCII code for "1"
N121;	Specify standby 1
P1 R87;	57h as ASCII code for "W"
P2 R84;	54h as ASCII code for "T"
P3 R49;	31h as ASCII code for "1"
G11;	Cancel the setting mode

In the MG item on the tool management data screen, spindle 1 is displayed as "SP1", and standby 1 is displayed as "WT1".



NOTE
Data registered becomes effective after the screen display is switched to the tool management screen.

10.3.3 Input of Customize Data with the Decimal Point

With the function for input of customize data with the decimal point, the number of decimal places can be set using the G10 format for each customize data item (customize data 1, ..., 40) to enable data input with the decimal point.

Format

```
G10L77P5;
N_R_;
G11;
  N_: Customize data number
  R_: Decimal point position
```

Explanation

- Customize data number (N_)

- A value of 1 to 4 can be specified for N.
- When tool management function customize data extension (5 to 20) is available, a value of 1 to 20 can be specified.
- When tool management function customize data extension (5 to 40) is available, a value of 1 to 40 can be specified.

- Decimal point position (R_)

Specify a decimal point position (0 to 7). When 0 is set, the number of decimal places of customize data cannot be input.

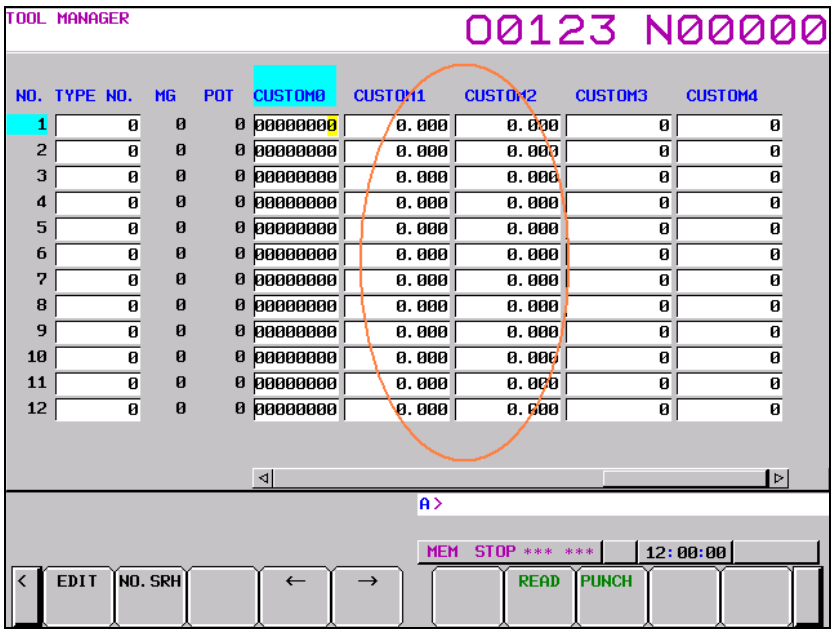
NOTE

1 If G10 L77 P5 is terminated normally, the power must be turned off before operation is continued.
2 The setting becomes effective after the power is turned off then back on.
3 When the number of decimal places is set for customize data, the number of decimal places can be input only through the MDI keys.
4 To set the number of decimal places for customize data 5 to 20, the customize data extension (5 to 20) option for the tool management function or the customize data extension (5 to 40) option for the tool management function is required.
5 To set the number of decimal places for customize data 21 to 40, the customize data extension (5 to 40) option for the tool management function is required.

Example 1

When customize data 1 and customize data 2 are input with three decimal places

G10L77P5;	Set the number of decimal places for customize data
N1 R3;	Set the number of decimal places to 3 for customize data 1
N2 R3;	Set the number of decimal places to 3 for customize data 2
G11;	Cancel the setting mode



- Use in the G10 format

Between G10 and G11, the decimal point cannot be specified at any address. So, if decimal point input for customize data is enabled with this function, a G10 L75/L76/L77 command using a custom macro variable (#8431 to #8450) reads and sets the value of the integer part.

Example 2

(Example 1)

Condition:

"3" is set as the decimal point position of customize data 1.
"1" is set as the decimal point position of customize data 2.

Operation:

Data is transferred from customize data 1 to customize data 2 by using a custom macro variable.

```
G10L77P5;          <1> Set customize data decimal point position
N1 R3;             <2> Set 3 as decimal point position of customize data 1
N2 R1;             <3> Set 1 as decimal point position of customize data 2
G11;               <4> Cancel the setting mode
;
G10 L75 P1;        <5> Register tool management data
N01;               <6> Register with No. 1
P1 R12345;         <7> Set ”12.345” for customize data 1
P2 R#8431;         <8> Set ”1.2” for customize data 2
G11;               <9> Cancel the setting mode
```

In Example 1, customize data 1 is directly set in customize data 2 by using a custom macro variable.
Customize data 1 holds "12.345". In step <8>, only the integer part is read and processed as "P2 R12".
So, "1.2" is set in customize data 2.

(Example 2)

Condition:

"3" is set as the decimal point position of customize data 1.
"1" is set as the decimal point position of customize data 2.

Operation:

By using the custom macro variable of customize data 1, the data is transferred to customize data 2 after being multiplied by 1000.

```
G10L77P5;          <1> Set customize data decimal point position
N1 R3;             <2> Set 3 as decimal point position of customize data 1
N2 R1;             <3> Set 1 as decimal point position of customize data 2
G11;               <4> Cancel the setting mode
;
G10 L75 P1;        <5> Register tool management data
N01;               <6> Register with No. 1
P1 R12345;         <7> Set ”12.345” for customize data 1
P2 R[#8431*1000];  <8> Set ”1234.5” for customize data 2
G11;               <9> Cancel the setting mode
```

In Example 2, customize data 1 is directly set in customize data 2 by using a custom macro variable.
Customize data 1 holds "12.345". In step <8>, the data is multiplied by 1000 to eliminate the fractional part.
So, this command is equivalent to "P2 R12345", so that "1234.5" is set in customize data 2.

10.3.4 Protection of Various Tool Information Items with the KEY Signal

When tool management data is in the edit state, various information items can be modified. By setting bit 0 of parameter No. 13204 to 1, tool management data can be protected with the KEY signal so that various information items are not registered, modified, and deleted.

10.3.5 Selection of a Tool Life Count Period

A tool life count period can be chosen between 1 sec and 8 msec on a tool-by-tool basis.

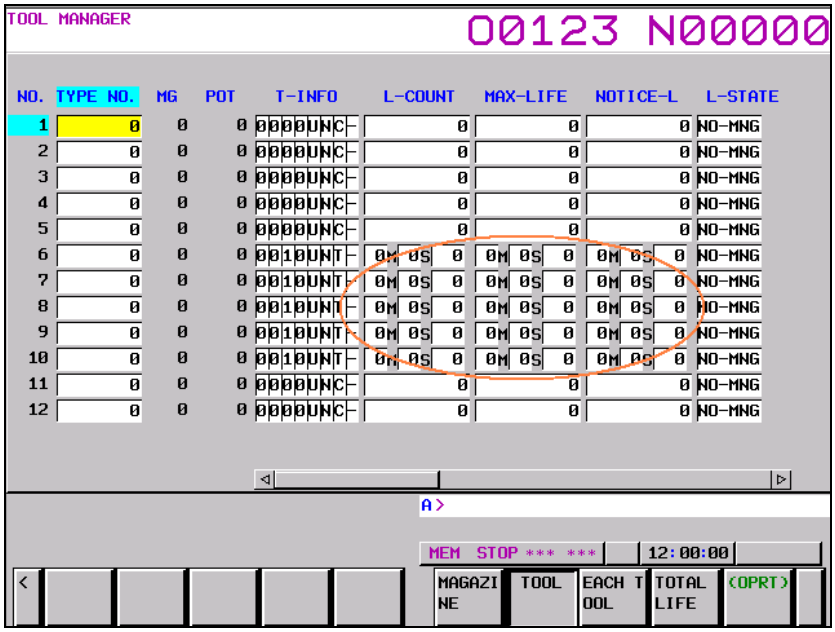
- Tool life count period selection

Bit 5 of tool information is used to make a life count period selection.

Item		Description
Data length		1 byte (flag data)
#5	REV	0: A life count period of 1 sec is used. 1: A life count period of 8 msec is used.

Range of count is as follows.
1sec: 0 to 3,599,999 seconds (999 hours 59 minutes 59 seconds)
8msec: 0 to 3,599,992 ms (59 minutes 59 seconds 992 milliseconds)

NOTE
This function is valid when the tool information TIM (#1) is set to 1.



10.3.6 Each tool Data Screen

All data for a specified tool can be extracted and displayed.

10.3.7 Total Life Time Display for Tools of The Same Type

The remaining lives of tools with the same type numbers are totaled, and totals are displayed in order by tool type number or by remaining life. Also, data of tools with the same tool type numbers are displayed in a list.

10.4 TOOL MANAGEMENT FUNCTION FOR OVERSIZE TOOLS

Overview

Tool management function for oversize tools is added to the tool management function.
The figure of an oversize tool can be defined freely, and the figure of each oversize tool is registered. When an oversize tool is stored in a cartridge, interference with tools stored in other pots is considered. This function is usable with cartridges of chain type and matrix type.

Format

By G10 data input, an oversize tool number can be set in the tool management data. Moreover, tool figure data can be set.

<Registration of new tool management data>

```
G10 L75 P1 ;
N_;
A_;
G11 ;
```

N_: Tool management data number
A_: Specify tool figure number (0 to 20).

<Modification of tool management data>

```
G10 L75 P2;
N_;
A_;
G11 ;
```

N_: Tool management data number
A_: Specify tool figure number (0 to 20).

<Registration of tool figure data>

```
G10 L77 P6 ;
N_P_Q_R_S_T_;
G11 ;
```

N_: Tool figure number
P_: Number of pots to be occupied in left direction (in steps of 0.5 pot)
Q_: Number of pots to be occupied in right direction (in steps of 0.5 pot)
R_: Number of pots to be occupied in upper direction (in steps of 0.5 pot)
S_: Number of pots to be occupied in lower direction (in steps of 0.5 pot)
T_: 0 for figure A, or 1 for figure B

NOTE

1 If a target tool is registered in a cartridge and interferes with other tools in registration or modification of tool figure data of the tool management data, alarm PS5360 is issued. (The data is not input.)
2 If a tool interferes with other tools in registration or modification to the cartridge management table, alarm PS5360 is issued. (The data is not input.)
3 If an attempt is made to modify tool figure data, and the tool for which the tool figure data number subject to modification is set is registered in the cartridge, alarm PS5360 is issued. (The data is not input.)

10.5 TOOL LIFE MANAGEMENT

Tools are classified into several groups, and a tool life (use count or use duration) is specified for each group in advance. Each time a tool is used, its life is counted, and when the tool life expires, a new tool that is sequenced next within the same group is selected automatically. With this function, the tool life can be managed while machining is being performed continuously. Data for tool life management consists of tool group numbers, tool life values, tool numbers, and codes for specifying a tool offset value. These data items are registered in the CNC.

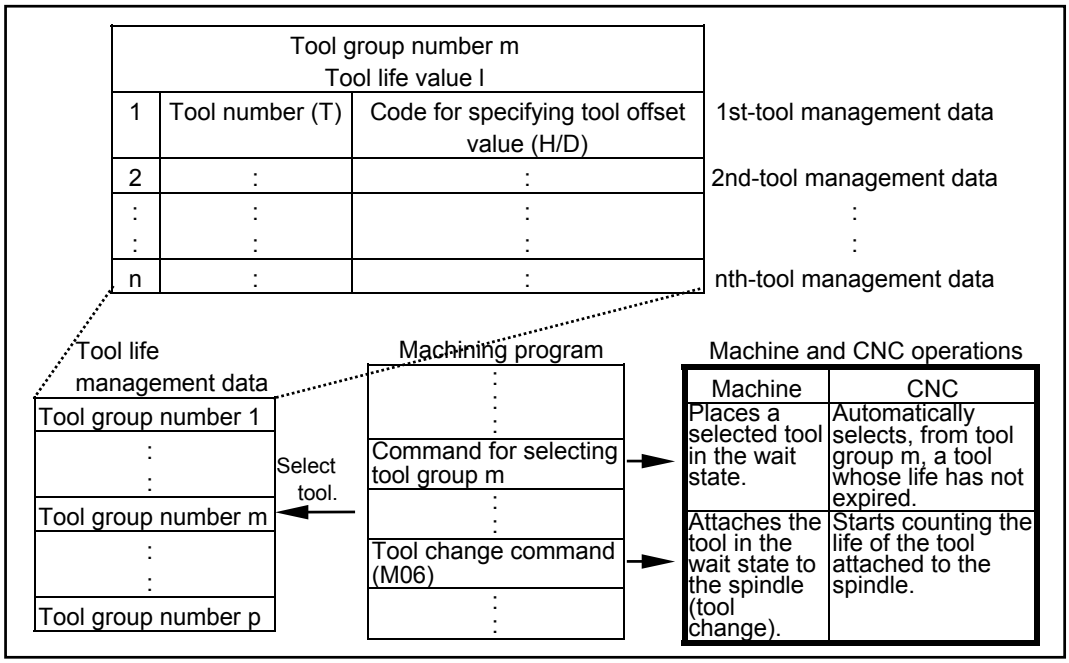


Fig. 10.5 (a) Tool selection from machining program

M

A group is selected by a T code, and tool life counting is started by the M06 command.

T

The T series has two tool change types (the turret type and the ATC type). The turret type uses only a T code to select a group, specify a tool offset value, and start tool life counting, whereas the ATC type, like the M series, uses a T code to select a group and the M06 command to start tool life counting. With the ATC type, only a D code is used for specifying a tool offset value. The tool change type is selected by bit 3 (TCT) of parameter No. 5040.

> ⚠ **CAUTION**
> This function cannot be used if bit 1 (LGN) of parameter No. 5002 is set to 1 to use the same number as the tool selection number to specify a geometric offset number.

- Life management B function

If the tool life management B function is enabled, the maximum tool life value can be extended, and the tool life expiration prior notice signal can be output to post tool life expiration in advance when the remaining life (the life value minus the life counter value) has reached the remaining life setting. The remaining life setting is registered as tool life management data in the CNC in advance.
The tool life management B function is enabled by setting bit 4 (LFB) of parameter No. 6805 to 1.

M

If the tool life management B function is enabled, the function for selecting a tool group by an arbitrary group number can be used.

T

The tool life management B function can be used. However, the function for selecting a tool group by an arbitrary group number can be used only if the tool change type is the ATC type (bit 3 (TCT) of parameter No. 5040 = 1).

- Maximum number of tool life management groups and multipath system

Up to 256 tool life management groups can be used in the entire CNC system.
For each path, set a maximum number of groups to be used in parameter No. 6813.
The maximum number of groups must be a multiple of the minimum number of groups (eight groups). If the maximum number of groups is 0, the tool life management function is disabled.
The total number of groups in the entire CNC system can be expanded to 1024 groups by using an option for additional tool life management groups.

Multipath system (when tool life management is performed for three paths)
Suppose that, of the total groups in the system (256 groups), 128 groups are allocated to path 1, 64 groups to path 2, and 64 groups to path 3. Then, set parameter No. 6813 as follows:

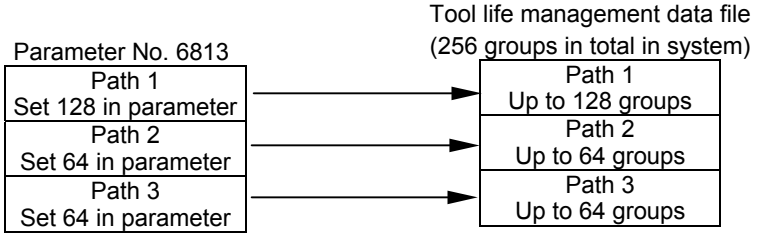


Fig. 10.5 (b) Group allocation in entire system

> ⚠ **CAUTION**
> Re-setting this parameter results in the tool life management data file being cleared at power-on. Once the parameter is re-set, register tool life management data for each path again.

10.5.1 Tool Life Management Data

Tool life management data consists of tool group numbers, tool numbers, codes for specifying a tool offset value, tool life values, arbitrary group numbers, and remaining life settings.
Whether to use arbitrary group numbers and remaining life settings can be specified by setting bit 5 (TGN) and bit 3 (GRP) of parameter No. 6802.

Explanation

- Tool group number

Up to 256 tool life management groups can be used in the entire CNC system.
In parameter No. 6813, set a maximum number of groups to be used. With the set maximum number of groups, up to four tools can be registered per group. The combination of the maximum number of registrable groups and the maximum number of tools can be changed by setting bit 0 (GS1) and bit 1 (GS2) of parameter No. 6800 as follows.

Table 10.5.1 Maximum numbers of registrable groups and tools

GS2 (No. 6800#1)	GS1 (No. 6800#0)	Number of groups	Number of tools
0	0	1/8 of maximum number of groups (parameter No. 6813)	32
0	1	1/4 of maximum number of groups (parameter No. 6813)	16
1	0	1/2 of maximum number of groups (parameter No. 6813)	8
1	1	Maximum number of groups (parameter No. 6813)	4

⚠ CAUTION
After changing the setting of bit 0 (GS1) and bit 1 (GS2) of parameter No. 6800, re-register tool life management data by issuing the G10L3 (registration after deletion of data for all groups). Otherwise, the newly set combination does not become valid.

NOTE
The total number of groups in the entire CNC system can be expanded to up to 1024 groups, using an option for additional tool life management groups.

- Tool number

A tool number is specified with a T code. A number consisting of up to eight digits (99999999) can be specified.

NOTE
The maximum number of digits usable in the T code is specified in parameter No. 3032.

- Codes for specifying a tool offset value

M

Codes for specifying a tool offset value include an H code (for tool length offset) and a D code (for cutter compensation). Numbers up to 999 (up to three digits long) can be registered as codes for specifying tool offset values. However, the number of registered codes must not exceed the number of tool offsets the CNC can use.

NOTE
If codes for specifying tool offset values are not used, the registration of these codes can be omitted.

T

If the tool change type is the turret type (bit 3 (TCT) of parameter No. 5040 = 0), neither the H code nor D code is used to specify a tool offset value. The T code includes a compensation code. If the tool change type is the ATC type (bit 3 (TCT) of parameter No. 5040 = 1), only the D code is used.

- Tool life value

A tool life value can be registered in terms of use duration or use count. The maximum value is as follows:
Up to 4300 minutes can be registered if duration specification is selected, or up to 65535 times can be registered if use count specification is selected.
When the tool life management B function is enabled, up to 100000 minutes (or 60000 minutes if the life is counted every 0.1 second) can be registered if duration specification is selected, or up to 99999999 times can be registered if use count specification is selected.

- Arbitrary group number

M

If a function for allowing specification of arbitrary group numbers is used (bit 5 (TGN) of parameter No. 6802 = 1), an arbitrary group number can be specified with a T code to select a tool life management group.
An arbitrary group number ranging from 1 to 99999999 can be specified.

T

Arbitrary group numbers can be used only if the tool change type is the ATC type (bit 3 (TCT) of parameter No. 5040 = 1).

- Remaining life setting

Bit 3 (GRP) of parameter No. 6802 is used to specify which value, a value set for each group or a parameter-set value (parameters No. 6844 and 6845), is to be used as the remaining life setting until a new tool is selected.
If a value set for each group is used (bit 3 (GRP) of parameter No. 6802 = 1), the value is registered as tool life management data.
A value ranging from 1 to 100000 minutes (or 60000 minutes if the life is counted every 0.1 second) can be specified if duration specification is selected, or a value up to 99999999 times can be specified if use count specification is selected.

10.5.2 Registering, Changing, and Deleting Tool Life Management Data

By programming, tool life management data can be registered in the CNC, and registered tool life management data can be changed or deleted.

Explanation

The program format varies depending on the following four types of operation:

- Registration after deletion of all groups

After all registered tool life management data is deleted, programmed tool life management data is registered.

- Change of tool life management data

Tool life management data can be set for a group for which no tool life management data is registered, and already registered tool life management data can be changed.

- Deletion of tool life management data

Tool life management data can be deleted.

- Setting of tool life count type

A count type (use duration or use count) can be set for each group separately.

Format

- Registration after deletion of all groups

M

Format	Meaning
G10L3; P-L-; T-H-D-; T-H-D-; : P-L-; T-H-D-; T-H-D-; : G11; M02(M30);	G10L3: Register data after deleting data of all groups. P-: Group number L-: Tool life value T-: Tool number H-: Code for specifying tool offset value (H code) D-: Code for specifying tool offset value (D code) G11: End of registration

T

Format	Meaning
G10L3; P-L-; T-(D-); T-(D-); : P-L-; T-(D-); T-(D-); : G11; M02(M30);	G10L3: Register data after deleting data of all groups. P-: Group number L-: Tool life value T-: For turret type (bit 3 (TCT) of parameter No. 5040 = 0), tool number and tool offset number For ATC type (bit 3 (TCT) of parameter No. 5040 = 1), tool number G11: End of registration (D-): For ATC type (bit 3 (TCT) of parameter No. 5040 = 1), code for specifying tool offset value

NOTE
No H code is used even if the tool change type is the ATC type (bit 3 (TCT) of parameter No. 5040 = 1).

If more than one offset value is to be used for the same tool within one process, specify the command as shown below (on the assumption that tool offset numbers are two digits long).

Format	Meaning
G10L3; P-L-; T0101; T0102; T0103; : G11; M02(M30);	 Tool number 01, tool offset number 01 Tool number 01, tool offset number 02 Tool number 01, tool offset number 03

- Change of tool life management data

M

Format	Meaning
G10L3P1; P-L-; T-H-D-; T-H-D-; : P-L-; T-H-D-; T-H-D-; : G11; M02(M30);	G10L3P1: Start changing group data. P-: Group number L-: Tool life value T-: Tool number H-: Code for specifying tool offset value (H code) D-: Code for specifying tool offset value (D code) G11: End of change of group

T

Format	Meaning
G10L3P1; P-L-; T-(D-); T-(D-); : P-L-; T-(D-); T-(D-); : G11; M02(M30);	G10L3P1: Start changing group data. P-: Group number L-: Tool life value T-: For turret type (bit 3 (TCT) of parameter No. 5040 = 0), tool number and tool offset number For ATC type (bit 3 (TCT) of parameter No. 5040 = 1), tool number G11: End of registration (D-): For ATC type (bit 3 (TCT) of parameter No. 5040 = 1), code for specifying tool offset value

NOTE
No H code is used even if the tool change type is the ATC type (bit 3 (TCT) of parameter No. 5040 = 1).

- Deletion of tool life management data

Format	Meaning
G10L3P2; P-; P-; P-; P-; : G11; M02(M30);	G10L3P2: Start deleting group data. P-: Group number G11: End of deletion

- Setting of tool life count type

Format	Meaning
G10L3 (or G10L3P1); P-L-Q-; T-H-D-; T-H-D-; : G11; M02(M30);	Q: Life count type (1: Use count. 2: Duration.)

> ⚠ **CAUTION**
> If the Q command is omitted, the life count type is set according to the setting of bit 2 (LTM) of parameter No. 6800.

- Arbitrary group number

M

If the tool life management B function is enabled (bit 4 (LFB) of parameter No. 6805 = 1), and use of the function for specifying an arbitrary group number is set (bit 5 (TGN) of parameter No. 6802 = 1), an arbitrary group number can be registered by programming a command.

Format	Meaning
G10L3; (or G10L3P1); P-L-T-; T-H-D-; T-H-D-; : G11; M02(M30);	T-: Arbitrary group number (1 to 99999999)

> **NOTE**
> 1 A group number that is the same as the group number of another group cannot be registered. An attempt to register such a group number results in alarm PS0431.
> 2 If an arbitrary group number (T) is 0, alarm PS0431 is issued.
> 3 If ever an arbitrary group number is specified, the T command must be specified. If the T command is omitted, alarm PS0149 is issued.

T

The function for specifying an arbitrary group number is available only if the tool change type is the ATC type (bit 3 (TCT) of parameter No. 5040 = 1).
If the tool change type is the ATC type, the format used for setting an arbitrary group number is the same as for the M series.

Remaining life setting

If the tool life management B function is enabled (bit 4 (LFB) of parameter No. 6805 = 1), and a remaining life value is to be set for each group separately (bit 3 (GRP) of parameter No. 6802 = 1), a remaining life value can be set as tool life management data by programming a command.

Format	Meaning
G10L3; (or G10L3P1); P-L-R-; T-H-D-; T-H-D-; : G11; M02(M30);	R-: Remaining life setting (remaining life until a new tool is selected)

NOTE
1 If the remaining life setting (R) is 0 or omitted, the remaining life is assumed to be 0. In this case, the tool life expiration prior notice function is disabled.
2 The remaining life setting (R) cannot exceed the life value (L). If an attempt is made to specify a value greater than the life value, alarm PS0431 is issued.
3 The remaining life setting (R) must be an unsigned value having no decimal point.
4 The unit of the remaining life setting (R) depends on the setting of bit 1 (FGL) of parameter No. 6805, which selects the unit of the tool life value.

Tool life value

- If the life management B function is disabled

If the tool life management B function is disabled (bit 4 (LFB) of parameter No. 6805 = 0), a tool life value is registered as a duration or a use count according to the setting of bit 2 (LTM) of parameter No. 6800 or the setting of a count type (Q command). The maximum values are listed below.

Table 10.5.2 (a) Life count types and maximum life values

LFB (No.6805#4)	LTM (No.6800#2)	Life count type	Maximum life value
0	0	Use count specification	65535 times
	1	Duration specification	4300 minutes

If the count type is duration specification, the unit of a life value specified with address L in a program may be one minute or 0.1 second, which is determined by the setting of bit 1 (FGL) of parameter No. 6805.

Table 10.5.2 (b) Life value unit and maximum value in L command

LFB (No.6805#4)	FGL (No.6805#1)	Life value unit	Maximum value in L command	Example
0	0	1 minute	4300	L100: Life value is 100 minutes.
	1	0.1 second	2580000	L1000: Life value is 100 seconds.

- If the life management B function is enabled

If the tool life management B function is enabled (bit 4 (LFB) of parameter No. 6805 = 1)

Table 10.5.2 (c) Life count types and maximum life values

LFB (No.6805#4)	LTM (No.6800#2)	Life count type	Maximum life value
1	0	Use count specification	99999999 times
	1	Duration specification	100000 minutes
			60000 minutes(Note)

Table 10.5.2 (d) Life value unit and maximum value in L command

LFB (No.6805#4)	FGL (No.6805#1)	Life value unit	Maximum value in L command	Example
1	0	1 minute	100000	L100: Life value is 100 minutes.
			60000(Note)	
	1	0.1 second	60000000	L1000: Life value is 100 seconds.
			36000000(Note)	

NOTE
If the life is counted every 0.1 second (bit 0 (FCO) of parameter No. 6805 = 1), the maximum value specified by duration is 60000 minutes.

10.5.3 Tool Life Management Commands in Machining Program

Explanation

M

- Commands

The following commands are used for tool life management:

TOOOOOOOO;

Specifies a tool group number.
The tool life management function selects, from a specified group, a tool whose life has not expired, and outputs its T code signal.
In OOOOOOOO, specify the sum of the tool life management ignore number specified in parameter No. 6810 and a desired group number.
Example: To specify tool group number 1 when the tool life management ignore number is 100, specify "T101;".

NOTE
If OOOOOOOO is not greater than a tool life management ignore number, the T code is treated as an ordinary T code.

If an arbitrary group number is used (bit 4 (LFB) of parameter No. 6805 = 1, and bit 5 (TGN) of parameter No. 6802 = 1), it must be a preset arbitrary group number rather than a tool group number specified in OOOOOOOO.
Example: To specify tool group number 1 when arbitrary group number 1234 is set for tool group number 1, specify T1234.

NOTE
1 If an arbitrary group number is used, the tool life management ignore number specified in parameter No. 6810 is not used.
2 If the group specified by OOOOOOOO is not present, the CNC assumes that a tool not controlled under the tool life management function is selected.

M06;

Terminates life management for the previously used tools, and begins counting the life of a new tool selected with the T code.

NOTE
1 M06 is treated as an M code not involving buffering.
2 If more than one M code is to be specified within the same block, specify M06 first among these M codes.

H99;

Selects the H code registered in tool life management data for the currently used tool to enable tool length offset. Parameter No. 13265 can be used to enable compensation according to an H code other than H99.

H00;

Cancels tool length offset.

D99;

Selects the D code registered in tool life management data for the currently used tool to perform cutter compensation.
Parameter No. 13266 can be used to enable compensation according to a D code other than D99.

D00;

Cancels cutter compensation.

NOTE
H99 and D99 must be specified after the M06 command. If a code other than H99 or D99 or a code other than the H or D code set, respectively, in parameter No. 13265 or 13266 is specified following M06, no H code or D code is selected for tool life management data.

- Types

For tool life management, the four tool change types (types A to D) listed below are used. Which type to use varies from one machine to another. For details, refer to the relevant manual of each machine tool builder.

Table 10.5.3 Differences among tool change types

Tool change type	A		B		C		D	
Parameters M6T and M6E	M6T	M6E	M6T	M6E	M6T	M6E	M6T	M6E
M6T (No.6800#7) M6E (No.6801#7)	0	0	1	0	1	0		1
Tool group number specified in the same block as the tool change command (M06)	Tool group already used		Tool group to be used next					
Time when tool life is counted	Life counting is performed for a tool in the specified tool group if M06 is specified next.						Life counting is performed if a tool in the tool group specified in the same block as M06 is specified.	
Remarks	If the T command (retract tool group) following the M06 command is not the currently used tool group, alarm PS0155 is issued (if bit 6 (IGI) of parameter No. 6800 = 0).		Normally, if a tool group number command is specified alone, type B is used. However, even if a tool group number command is specified alone with type C, no alarm is issued. (This means that there is no difference in operation between types B and C.)				If M06 is specified alone, alarm PS0153 is raised.	

NOTE
If a tool group number is specified and a new tool is selected, the new tool selection signal is output.

- Tool length compensation in tool axis direction
- Tool center point control

For these functions, a compensation value selected in tool life management is used for compensation.

T

If the tool change type is the ATC type (bit 3 (TCT) of parameter No. 5040 = 1), commands are specified in the same manner as for the M series except that neither H99 nor H00 is used for the T series. See the description for the M series. For details on the tool compensation types applied when the tool change type is the ATC type, refer to Section 5.7, "Extended Tool Selection," in II "Programming," of the lathe system user's manual (B-63944EN-1).
The command specification method for the turret type (bit 3 (TCT) of parameter No. 5040 = 0) is explained below.

⚠ **CAUTION**
This function cannot be used if the same number as the tool selection number is to be used to specify geometry offset number (bit 1 (LGN) of parameter No. 5002 = 1).

- Commands

T○○○○○○99;

The tool life management function ends counting the life of the tool used so far, selects, from the group specified by ○○○○○○, a tool whose life has not expired, outputs the T code signal for the tool, and starts counting the life of the tool.

Example:
Suppose that a T199 command (with a tool offset specified with the lower two digits) is issued to cause the life management function to select T10001 of tool group 1. Then, T code 100 is output, and the tool offset number 1 is selected.

If the condition that a new tool be selected is not met, and the second or subsequent selection of the same group is made since the entry of the control unit into the automatic operation start state from the reset state, the next to the currently selected T code is selected if more than one offset is registered.
If the third selection is made, for example, the third offset is selected from among multiple offsets registered for the same tool.

Example:

Suppose that the tool offset is specified with the lower two digits, and that two offset numbers are set for the same tool number in group 1 with the following two T codes:

T10001 T10002

The first T199 command issued since the entry of the control unit into the automatic operation start state from the reset state selects the first T code, T10001. Then, if T199 is issued again before the control unit is reset, the second T code, T10002, is selected. Furthermore, if T199 is issued again before the control unit is reset, the second T code, T10002, is selected because no third offset is present.

Setting bit 1 (TSM) of parameter No. 6801 set to 1 enables life counting to be performed for each T code separately even if T codes specifying multiple offsets for the same number are registered.

T○○○○○○88;

The offset of the tool whose life is currently counted by tool life management is canceled. The tool offset code is set to 00, and the tool number is output as a T code signal.

Example:

Suppose that the tool number of the tool currently used by the life management function is 100. Then, issuing a T188 command (with a tool offset specified with the lower two digits) outputs T code 100 and selects offset number 0, canceling the offset.

NOTE

If life counting is not performed, or if the specified tool does not belong to the group for which life counting is being performed, alarm PS0155 is issued.

The numbers of digits in ○○○○○○ and 99/88 vary as follows:

No.5028	99	88
1	T○○○○○○○ 9 ↑ ↑ Select group Start life counting	T○○○○○○○ 8 ↑ ↑ Select group Cancel tool offset
2	T○○○○○○ 99 ↑ ↑ Select group Start life counting	T○○○○○○ 88 ↑ ↑ Select group Cancel tool offset
3	T○○○○○ 999 ↑ ↑ Select group Start life counting	T○○○○○ 888 ↑ ↑ Select group Cancel tool offset

The maximum number of digits in T codes is set in parameter No. 3032.

The number of digits used for specifying an offset number is selected by parameter No. 5028. If 0 is selected, the number of digits depends on the number of tool offsets.

Example:

If there are one to nine tool offsets: Lowest digit
If there are 10 to 99 tool offsets: Lower two digits
If there are 100 to 999 tool offsets: Lower three digits

NOTE
Offset start and cancel operations involve compensation by moving a tool or by shifting the coordinate system. Using bit 6 (LWM) of parameter No. 5002 can select whether to perform a compensation operation if a T code is specified or if an axis move command is specified.
For details, refer to Subsection 5.1.5, “Offset Operation,” of the lathe system User's Manual (B-63944EN-1).

TOOOOOOΔΔ;

If the tool offset number in ΔΔ is neither 99 nor 88, the T code is treated as an ordinary T code. If life counting is being currently performed, it is ended.

Examples

M

- Tool change type A

If a block specifying a tool change command (M06) also contains a tool group command (T code), the T code is used as a command for returning the tool to its cartridge. By specifying a tool group number with a T code, the number of the tool used so far is output as a T code signal. If the specified tool number is not a tool number of the tool group of the tool being used, alarm PS0155 is issued. The alarm, however, can be suppressed by setting bit 6 (IGI) of parameter No. 6800 to 1.

Example:	**Suppose that the tool life management ignore number is 100.**
T101 ;	A tool whose life has not expired is selected from group 1.
:	(Suppose that tool number 010 is selected.)
:	
M06 ;	Tool life counting is performed for the tool in group 1.
:	(The life of tool number 010 is counted.)
:	
T102 ;	A tool whose life has not expired is selected from group 2.
:	(Suppose that tool number 100 is selected.)
:	
M06 T101 ;	Tool life counting is performed for the tool in group 2.
:	(The life of tool number 100 is counted.)
:	The number of the tool currently used (in group 1) is output
:	with a T code signal. (Tool number 010 is output.)
T103 ;	A tool whose life has not expired is selected from group 3.
:	(Suppose that tool number 200 is selected.)
:	
M06 T102 ;	Tool life counting is performed for the tool in group 3.
:	(The life of tool number 200 is counted.)
:	
G43 H99 ;	Tool length offset for the tool selected from group 3 is used.
:	
G41 D99 ;	Cutter compensation for the tool selected from group 3 is
:	used.
D00 ;	Cutter compensation is canceled.
:	
H00 ;	Tool length offset is canceled.

- Tool change types B and C

If a block specifying a tool change command (M06) also contains a tool group command (T code), the T code is used to specify a tool group number for which life counting is to be performed by the next tool change command.

Example: Suppose that the tool life management ignore number is 100.	
T101 ;	A tool whose life has not expired is selected from group 1.
:	(Suppose that tool number 010 is selected.)
:	
M06 T102 ;	Tool life counting is performed for the tool in group 1.
:	(The life of tool number 010 is counted.)
:	A tool whose life has not expired is selected from group 2.
:	(Suppose that tool number 100 is selected.)
M06 T103 ;	Tool life counting is performed for the tool in group 2.
:	(The life of tool number 100 is counted.)
:	A tool whose life has not expired is selected from group 3.
:	(Suppose that tool number 200 is selected.)
G43 H99 ;	Tool length offset for the tool selected from group 2 is used.
:	
:	
G41 D99 ;	Cutter compensation for the tool selected from group 2 is
:	used.
:	
D00 ;	Cutter compensation is canceled.
:	
:	
H00 ;	Tool length offset is canceled.
:	
:	
M06 T104 ;	Tool life counting is performed for the tool in group 3.
:	(The life of tool number 200 is counted.)
	A tool whose life has not expired is selected from group 4.

- Tool change type D

For a tool selected by a tool group command (T code), life counting is performed by a tool change command (M06) specified in the same block as the tool group command. Specifying a T code alone does not results in an alarm; however, specifying an M06 command alone results in alarm PS0153. However, the alarm can be suppressed by setting bit 7 (TAD) of parameter No. 6805 to 1.

Example: Suppose that the tool life management ignore number is 100.	
T101 M06 ; : : : :	A tool whose life has not expired is selected from group 1. (Suppose that tool number 010 is selected.) Tool life counting is performed for the tool in group 1. (The life of tool number 010 is counted.)
T102 M06 ; : : : G43 H99 ; : :	A tool whose life has not expired is selected from group 2. (Suppose that tool number 100 is selected.) Tool life counting is performed for the tool in group 2. (The life of tool number 100 is counted.) Tool length offset for the tool selected from group 2 is used.
G41 D99 ; : :	Cutter compensation for the tool selected from group 2 is used.
D00 ; : :	Cutter compensation is canceled.
H00 ; : :	Tool length offset is canceled.
T103 M06 ; : :	A tool whose life has not expired is selected from group 3. Tool life counting is performed for the tool in group 3.

T

For the ATC type (bit 3 (TCT) of parameter No. 5040 = 1), commands are specified in the same manner as for the M series except that D99 rather than H99 is used. For the ATC type, see the description for the M series.
The following shows how commands are specified if the turret type is used (bit 3 (TCT) of parameter No. 5040 = 0):

Example: Suppose that offset numbers are two digits long.	
T0199 ; : : : : : :	A tool whose life has not expired is selected from group 1. (Suppose that T1001 is selected. The tool number is 10, and the offset number is 01.) Tool life counting is performed for the tool in group 1. (The life of tool number 10 is counted.)
T0188 ; : : : :	The offset of the tool being used in group 1 is canceled. (Since the tool being used is T1001, the tool number is 10, and the offset number is 00.)
T0299 ; : : : : :	A tool whose life has not expired is selected from group 2. (Suppose that T2002 is selected. The tool number is 20, and the offset number is 02.) Tool life counting is performed for the tool in group 2. (The life of tool number 20 is counted.)
T0299 ; : : : : : :	If more than one offset number is specified for the currently used tool in group 2, the next offset number is selected. (Suppose that T2002 and T2003 are registered with tool number 20. In this case, T2003 is selected. The tool number is 20, and the offset number is 03.)
T0301 ; : :	Life counting for the tool in group 2 ends, and this command is treated as an ordinary T code. (The tool number is 03, and the offset number is 01.)

10.5.4 Tool Life Counting and Tool Selection

Either use count specification or duration specification is selected as the tool life count type according to the state of bit 2 (LTM) of parameter No. 6800. Life counting is performed for each group separately, and the life counter contents are preserved even after the power is turned off.

Table 10.5.4 Tool life management count types and intervals

Tool life count type	Use count specification	Time specification
Bit 2 (LTM) of parameter No. 6800	0	1
Life count interval	Incremented by one for tools used in one program Counting can be resumed by using tool life count restart M code (parameter No. 6811).	Bit 0 (FCO) of parameter No. 6805 0: Every second 1: Every 0.1 second Can be changed by override.

Explanation

M

- Use count specification (LTM=0)

If a tool group (T code) is specified, a tool whose life has not expired is selected from the specified tool group. The life counter for the selected tool is then incremented by one by a tool change command (M06). Unless a tool life count restart M code is specified, selection of a new tool and the increment operation can be performed only if a tool group number command and a tool change command are issued for the first time since the entry of the control unit into the automatic operation start state from the reset state.

> ⚠ **CAUTION**
> No matter how many times the same tool group number is specified in a program, the use count is not incremented, and no new tool is selected.

- Duration specification (LTM=1)

After all registered tool life management data is deleted, programmed tool life management data is registered.
If a tool group command (T code) is specified, a tool whose life has not expired is selected from the specified tool group. Then, life management for the selected tool is started by a tool change command (M06). Life management (counting) is performed by measuring the time during which the tool is actually used in the cutting mode at regular intervals (every second or 0.1 second). The life count interval is specified by bit 0 (FCO) of parameter No. 6805. The time required for single block stop, feed hold, rapid traverse, dwell, machine lock, and interlock operations is not counted.
Setting bit 2 (LFV) of parameter No. 6801 enables the life count to be overridden as directed by tool life count override signals. An override from 0 times to 99.9 times can be applied. If 0 times is specified, counting is not performed.

T

If the tool change type is the ATC type (bit 3 (TCT) of parameter No. 5040 = 1), the same specification method as for the M series applies. See the description for the M series.
The following explains the specification method for the turret type (bit 3 (TCT) of parameter No. 5040 = 0):

- Use count specification (LTM=0)

If a tool group command (T○○99 code) is issued, a tool whose life has not expired is selected from the specified tool group, and the life counter for the selected tool is incremented by one. Unless a tool life count restart M code is specified, selection of a new tool and the increment operation can be performed only if a tool group number command and a tool change command are issued for the first time since the entry of the control unit into the automatic operation start state from the reset state.

> ⚠ **CAUTION**
> No matter how many times the same tool group number is specified in a program, the use count is not incremented, and no new tool is selected.

- Duration specification (LTM=1)

If a tool group command (T○○99 code) is specified, a tool whose life has not expired is selected from the specified tool group, and tool management for the selected tool starts.
Life management (counting) is performed by measuring the time during which the tool is actually used in the cutting mode at regular intervals (every second or 0.1 second). The life count interval is specified by bit 0 (FCO) of parameter No. 6805. The time required for single block stop, feed hold, rapid traverse, dwell, machine lock, and interlock operations is not counted.

Setting bit 2 (LFV) of parameter No. 6801 enables the life count to be overridden according tool life count override signals. An override from 0 times to 99.9 times can be applied. If 0 times is specified, counting is not performed.

NOTE

1 When a tool is selected, tools are searched starting from the current tool toward the last tool to find a tool whose life has not expired. When the last tool is reached during this search, the search restarts from the first tool. If the search does find any tool whose life has not expired, the last tool is selected. When the tool currently used is changed by the tool skip signal, the next new tool is selected using the method described here.

2 If tool life counting indicates that the life of the last tool in a group has expired, the tool change signal is output. If the life count type is duration specification, the signal is output as soon as the life of the last tool in the group has expired. If the life count type is use count specification, the signal is output when the CNC is reset by a command such as M02 or M30 or when the tool life count restart M code is specified after the life of the last tool in the group has expired.

3 If a T command is specified, a group and a tool in the group are selected while the T command is buffered. This means that if a block to be buffered contains a T command specifying a group while machining is being performed with that group selected, the next T command is already buffered even if the tool life expires during machining, so the next tool is not selected. To prevent this, if the life count type is duration specification and a T command is to be specified to select the same group successively, insert an M code for suppressing buffering into a place immediately before the T command.

4 When the tool life is counted, the remaining life of a group (the life value minus the life counter value) is compared with the remaining life setting, and the status of the tool life expiration prior notice signal is changed according to the result of the comparison.

- M99

If the life count is specified by use count and bit 0 (T99) of parameter No. 6802 is 1, the tool change signal TLCH<Fn064.0> is output and the automatic operation is stopped if the life of at least one tool group has expired when the M99 command is executed. If the life count type is duration specification, the tool change signal is output immediately when the life of at least one tool group has expired; when the M99 command is specified, the automatic operation is stopped, but no more tool change signal is output.

M

If the life count is specified by use count, a tool group command (T code) issued after the M99 command selects, from a specified group, a tool whose life has not expired, and the next tool change command (M06) increments the tool life counter by one.

T

If the life count is specified by use count, when a tool group command (T code) is specified after the M99 command is specified, a tool whose life has not expired is selected from a specified group, and the tool life counter is incremented by one.
If the tool change type is the ATC type (bit 3 (TCT) of parameter No. 5040 = 1), the same specifications as for the M series apply.

10.5.5 Tool Life Count Restart M Code

Explanation

M

If the life count is specified by use count, the tool change signal is output if the life of at least one tool group has expired when a tool life count restart M code is issued. A tool group command (T code) issued after the tool life count restart M code selects, from a specified group, a tool whose life has not expired, and the next tool change command (M06) increments the tool life counter by one. This enables the tool life to be counted by a tool change command (M06) even if the command is not the first tool change command (M06) issued since the entry of the CNC into the automatic operation start state from the reset state. The tool life count restart M code is specified in parameter No. 6811.

Example:

Suppose that M16 is a tool life count restart M code and that the tool life management ignore number is 100.
Also suppose that the life count is specified by use count.

```
T101 ;   A tool whose life has not expired is selected from group 1.
  :
  :
M06 ;    Tool life management is performed for group 1.
  :      (The tool life counter is incremented by one.)
  :
T102 ;   A tool whose life has not expired is selected from group 2.
  :
  :
M06 ;    Tool life management is performed for group 2.
  :      (The tool life counter is incremented by one.)
  :
M16 ;    Tool life counting is restarted.
T101 ;   A tool whose life has not expired is selected from group 1.
  :
  :
M06 ;    Tool life management is performed for group 1.
  :      (The tool life counter is incremented by one.)
```

T

If the tool change type is the ATC type (bit 3 (TCT) of parameter No. 5040 = 1), the same specifications as for the M series apply. See the description for the M series.
The following explanation assumes that the tool change type is the turret type (bit 3 (TCT) of parameter No. 5040 = 0).
If the life count is specified by use count, the tool change signal is output if the life of at least one tool group has expired when a tool life count restart M code is specified. A tool group command (T code) issued after the tool life count restart M code selects a tool whose life has not expired from a specified group, and the tool life counter is incremented by one. This enables the tool life to be counted by a tool group command (T code) even if the command is not the first tool group command issued since the entry of the CNC into the automatic operation start state from the reset state. The tool life count restart M code is specified in parameter No. 6811.

Example:
Suppose that M16 is a tool life count restart M code.
Also suppose that the life count is specified by use count.

```
T199 ;  A tool whose life has not expired is selected from group 1.
  :     Tool life management is performed for group 1.
  :     (The tool life counter is incremented by one.)
  :
T299 ;  A tool whose life has not expired is selected from group 2.
  :     Tool life management is performed for group 2.
  :     (The tool life counter is incremented by one.)
  :
M16 ;   Tool life counting is restarted.
T199 ;  A tool whose life has not expired is selected from group 1.
  :     Tool life management is performed for group 1.
  :     (The tool life counter is incremented by one.)
```

M
T

NOTE
1 The tool life count restart M code is treated as an M code not involved in buffering.
2 If the life count type is use count specification, the tool change signal is output if the life of at least one tool group has expired when the tool life count restart M code is specified. If the life count type is duration specification, specifying the tool life count restart M code causes nothing.

10.5.6 Disabling Life Count

Explanation

M
T

If bit 6 (LFI) of parameter No. 6804 is “1”, the tool life count disable signal LFCIV can be used to select whether to cancel the tool life count.

If the tool life count disable signal LFCIV is “1”, the tool life count disabled signal LFCIF becomes “1”, and the tool life count is disabled.
If the tool life count disable signal LFCIV is “0”, the tool life count disabled signal LFCIF becomes “0”, and the tool life count is enabled.

NOTE
No buffering should occur when the state of the tool life count disable signal LFCIV is changed. So, use, for example, M codes not involved in buffering to change the signal state. If M06 (for the M series) or a tool change T code (for the T series) is issued in a block that follows directly a block in which an auxiliary function code with buffering enabled is used to turn on or off the tool life count disable signal LFCIV, it is likely that a command for specifying whether to count may become incorrect.

11 AUXILIARY FUNCTION

Overview

There are two types of auxiliary functions ; auxiliary function (M code) for specifying spindle start, spindle stop, program end, and so on, and secondary auxiliary function (B code) for specifying index table positioning.
When a move command and auxiliary function are specified in the same block, the commands are executed in one of the following two ways:

(1) Simultaneous execution of the move command and auxiliary function commands.
(2) Executing auxiliary function commands upon completion of move command execution.

The selection of either sequence depends on the machine tool builder's specification. Refer to the manual issued by the machine tool builder for details.

Chapter 11, "AUXILIARY FUNCTION", consists of the following sections:

11.1 AUXILIARY FUNCTION (M FUNCTION).....307
11.2 MULTIPLE M COMMANDS IN A SINGLE BLOCK.....308
11.3 M CODE GROUPING FUNCTION 309
11.4 SECOND AUXILIARY FUNCTIONS (B CODES) 314

11.1 AUXILIARY FUNCTION (M FUNCTION)

When a numeral is specified following address M, code signal and a strobe signal are sent to the machine. The machine uses these signals to turn on or off its functions.
Usually, only one M code can be specified in one block. Depending on the setting of bit 7 (M3B) of parameter No. 3404, up to three M codes can be specified.
Which M code corresponds to which machine function is determined by the machine tool builder. The machine processes all operations specified by M codes except those specified by M98, M99,M198 or called subprogram(Parameter No.6071 to 6079), or called custom macro (Parameter No.6080 to 6089). Refer to the machine tool builder's instruction manual for details.

Explanation

The following M codes have special meanings.

- M02,M30 (End of program)

This indicates the end of the main program.
Automatic operation is stopped and the CNC unit is reset. (This differs with the machine tool builder.)
After a block specifying the end of the program is executed, control returns to the start of the program.
Bits 5 (M02) and 4 (M30) of parameter No. 3404 can be used to disable M02, M30 from returning control to the start of the program.

- M00 (Program stop)

Automatic operation is stopped after a block containing M00 is executed. When the program is stopped, all existing modal information remains unchanged. The automatic operation can be restarted by actuating the cycle operation. (This differs with the machine tool builder.)

- M01 (Optional stop)

Similarly to M00, automatic operation is stopped after a block containing M01 is executed. This code is only effective when the Optional Stop switch on the machine operator's panel has been pressed.

- M98 (Calling of subprogram)

This code is used to call a subprogram. The code and strobe signals are not sent. See the subprogram II-13.3 for details.

- M99 (End of subprogram)

This code indicates the end of a subprogram.
M99 execution returns control to the main program. The code and strobe signals are not sent. See the subprogram section II-13.3 for details.

11.2 MULTIPLE M COMMANDS IN A SINGLE BLOCK

Usually, only one M code can be specified in one block. By setting bit 7 (M3B) of parameter No. 3404 to 1, however, up to three M codes can be specified simultaneously in one block.
Up to three M codes specified in one block are output to the machine at the same time. So, when compared with a case where a single M code is specified in one block, a reduced machining cycle time can be achieved.

Explanation

CNC allows up to three M codes to be specified in one block. However, some M codes cannot be specified at the same time due to mechanical operation restrictions. For detailed information about the mechanical operation restrictions on simultaneous specification of multiple M codes in one block, refer to the manual of each machine tool builder.
M00, M01, M02, M30, M98, M99, or M198 must not be specified together with another M code.
Some M codes other than M00, M01, M02, M30, M98, M99, and M198 cannot be specified together with other M codes; each of those M codes must be specified in a single block.
Such M codes include these which direct the CNC to perform internal operations in addition to sending the M codes themselves to the machine. To be specified, such M codes are M codes for calling program numbers 9001 to 9009 and M codes for disabling advance reading (buffering) of subsequent blocks. Meanwhile, multiple of M codes that direct the CNC only to send the M codes themselves (without performing internal operations) can be specified in a single block.
However, it is possible to specify multiple M codes that are sent to the machine in the same block unless they direct the CNC to perform internal operations. (Since the processing method depends on the machine, refer to the manual of the machine tool builder.)

Example

One M command in a single block	Multiple M commands in a single block
M40; M50; M60; G28G91X0Y0Z0; : : :	M40M50M60; G28G91X0Y0Z0; : : : : :

11.3 M CODE GROUPING FUNCTION

Overview

Classifying a maximum of 500 M codes into a maximum of 127 groups allows the user:

- To receive an alarm if an M code that must be specified alone is included when multiple M codes are specified in a block.
- To receive an alarm if M codes belonging to the same group are specified in the same block when multiple M codes are specified in a block.

11.3.1 Setting an M Code Group Number Using the Setting Screen

- Procedure for displaying the M code group setting screen

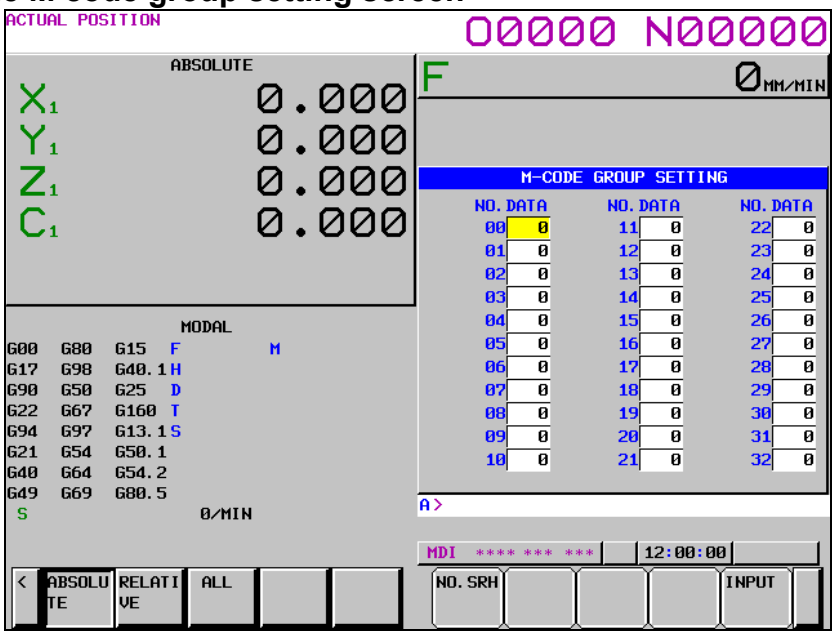


Fig. 11.3 (a) M code group setting screen

You can use the "M code group setting screen (Fig. 11.3 (a))" to set a group number for each M code.

Display the "M code group setting screen" using the following procedure:

(1) Press function key

and the continuous menu key several times. Soft key [M CODE GROUP] appears.
(2) Press soft key [M CODE GROUP].

In the "NO." field, M codes for which an M code group can be set are displayed.
An M code group can be set for the following M codes: M00 to M99, and any 400 M codes selected from M100 and subsequent M codes. For details of how to add the 100th and subsequent M codes, see the explanation of parameters Nos. 3441 to 3444.

In the "DATA" field, the M code group number corresponding to each M code is displayed.

- Setting a group number

To set an M code group number on the "M code group setting screen (Fig. 11.3 (a))," use he following procedure:

1 Select the MDI mode.
2 Set "PARAMETER WRITE" on the setting screen to 1.
3 Display the "M code group setting screen".
4 Move the cursor to the M code to be set using page keys and cursor keys. You can also enter the number of the M code to be set and press soft key [NO.SRH] to move the cursor to the M code.
5 Enter a group number and press soft key [INPUT] or the

key.

The valid range of M code group numbers is from 1 to 127 (127 groups). If a value of 0 is input, it is not registered as an M code group.

- Examples of setting parameters Nos. 3441 to 3444

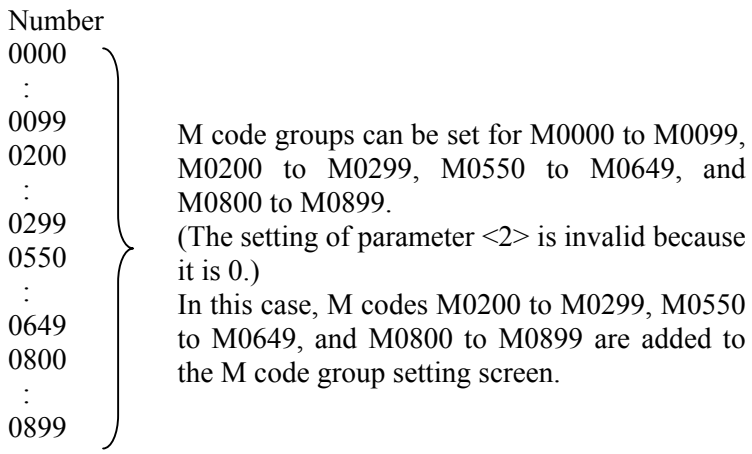
In the following examples, the number of digits of an M code is 4.
<1> to <4> indicate parameters Nos. 3441 to 3444.

(1) When <1> = 300, <2> = 400, <3> = 500, and <4> = 900 are set

Number
0000 to 0099 } 100 codes
0300 to 0399 } 100 codes
0400 to 0499 } 100 codes
0500 to 0599 } 100 codes
0900 to 0999 } 100 codes

M code groups can be set for M0000 to M0099, M0300 to M0599, and M0900 to M999.
M codes M0300 to M0599 and M0900 to M999 are added to the M code group setting screen.

(2) When <1> = 200, <2> = 0, <3> = 550, and <4> = 800 are set



11.3.2 Setting an M Code Group Number Using a Program

You can execute a program to set an M code group number and M code name. The command format is shown below.

Format

G10 L40 Pn Rg ;

Pn: “n” specifies an M code.

Rg: “g” specifies an M code group number.

NOTE

1 If the format is invalid, an alarm PS1144 is issued.

2 If an M code group cannot be set for the M code specified for the P command or if the group number specified for the R command is not within the range between 0 and 127, an alarm PS1305 is issued.

(Example)

Executing the following program sets M code group “1” to M03:

G10 L40 P03 R1 ;

11.3.3 M Code Group Check Function

When multiple M commands in a single block (enabled when bit 7 (M3B) of parameter No. 3404 is set to 1) are used, you can check the following items. You can also select whether to check the items using bit 1 (MGC) of parameter No. 3400.

(1) M code to be specified in a single block containing no other M codes
If an M code which must be specified in a single block containing no other M codes is specified together with another M code, an alarm PS5016 is issued.
(2) M codes in the same group
If multiple M codes in the same group are specified together, an alarm PS5016 is issued.

The valid range of group numbers is from 0 to 127 (128 groups).
Group numbers 0 and 1 have special meaning. Note the following points:

- Each M code with group number 1 is assumed to be an M code to be specified in a single block containing no other M codes.
- For each M code with group number 0, the "same group M code check" is ignored. That is, when multiple M codes with group number 0 are specified in a single block, the alarm is not issued.
- For each M code with group number 0, the "check for an M code to be specified in a single block containing no other M codes" is not ignored. That is, if an M code with group number 1 and an M code with group number 0 are specified in a single block, the alarm is issued.
- For M codes that are not to be output to the machine such as M98, M99, M198, and M codes for subprogram and macro calls (set in parameters Nos. 6071 to 6079 and Nos. 6080 to 6089 and with the macro executor), be sure to set 0 as the group number.
- For M00, M01, M02, M30, and M codes for which buffering is suppressed (set in parameters Nos. 3411 to 3432), be sure to set 1 as the group number.

11.4 SECOND AUXILIARY FUNCTIONS (B CODES)

Overview

If a value with a maximum of eight digits is specified after address B, the code signal and strobe signal are transferred for calculation of the rotation axis. The code signal is retained until the next B code is specified.
Only one B code can be specified for each block. When the maximum number of digits are specified by parameter No.3033, an alarm is issued if the number of digits of a command exceeds the specified number.
In addition, the address used for specifying the second auxiliary function can be changed to an address other than address B (address A, C, U, V, or W) by setting parameter No.3460.
However, the address used for the second auxiliary function cannot also be used as the address of the controlled axis. For details, refer to the manual available from the machine tool builder.

Explanation

- Range of specification

-99999999 to 99999999

- Output value

The value specified after the address of the second auxiliary function is output on the code signals B00 to B31. Note the following about a output value.

1. When a command with a decimal point or a negative command is disabled
   (When bit 0 (AUP) of parameter No.3450 is set to 0)

   When the second auxiliary function with no decimal point is specified, the specified value is output on the code signals as is, regardless of the desktop calculator decimal point setting (bit 0 (DPI) of parameter No.3401).
   Example:

Specified value	Output value
B10	10

   When the second auxiliary function with a decimal point is specified, alarm PS0007 is issued.
   When the second auxiliary function is specified with a negative value, alarm PS0006 is issued.

2. When a command with a decimal point or a negative command is enabled
(When bit 0 (AUP) of parameter No.3450 is set to 1)

When the desktop calculator decimal point setting is not specified (when bit 0 (DPI) of parameter No.3401 is set to 0), if the second auxiliary function with no decimal point is specified, the specified value is output on the code signals as is.
Example:

Specified value	Output value
B10	10

When desktop calculator decimal point input is specified (when bit 0 (DPI) of parameter No.3401 is set to 1), if the second auxiliary function with no decimal point is specified, the specified value multiplied by a magnification is output on the code signals. (Magnifications are shown in Table 11.4 (a).)
Example:

Specified value	Output value	
B10	10000	(When metric input is used and the reference axis is IS-B. The magnification is 1000.)

When the second auxiliary function with a decimal point is specified, the specified value multiplied by a magnification is output to the code signals. (Magnifications are shown in Table 11.4 (a).)
Example:

Specified value	Output value	
B10.	10000	(When metric input is used and the reference axis is IS-B. The magnification is 1000.)
B0.123	1230	(When inch input is used, the reference axis is IS-B, and parameter AUX is set to 1. The magnification is 10000.)

The magnification is determined as shown below according to the setting unit of the reference axis (specified by parameter No.1031) and bit 0 (AUX) of parameter No.3405.

Table 11.4 (a) Magnifications for an output value when the second auxiliary function with a decimal point is specified for desktop calculator decimal point input

Setting unit		Parameter AUX = 0	Parameter AUX = 1
Metric input system	Reference axis: IS-A	100×	100×
	Reference axis: IS-B	1000×	1000×
	Reference axis: IS-C	10000×	10000×
	Reference axis: IS-D	100000×	100000×
	Reference axis: IS-E	1000000×	1000000×
Inch input system	Reference axis: IS-A	100×	1000×
	Reference axis: IS-B	1000×	10000×
	Reference axis: IS-C	10000×	100000×
	Reference axis: IS-D	100000×	1000000×
	Reference axis: IS-E	1000000×	10000000×

⚠ CAUTION

If a decimal fraction remains after multiplying the specified value with a decimal point by a magnitude in Table 11.4 (a), the fraction is truncated.

Example:

Specified value	Output value
B0.12345	1234 (When inch input is used, the reference axis is IS-B, and parameter AUX is set to 1. The magnification is 10000.)

NOTE

If the number of digits of the specified value exceeds the allowable number of digits (set by parameter No.3033), alarm PS0003 is issued. When the specified value is multiplied by a magnitude in Table 11.4 (a), the allowable number of digits must be set for the resultant value.

Limitation

Addresses used for the second auxiliary functions (addresses specified with B or parameter No. 3460) cannot be used as the addresses used for controlled axis names.

12 PROGRAM MANAGEMENT

Chapter 12, "PROGRAM MANAGEMENT", consists of the following sections:

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12.2 FILES 323
12.3 RELATION WITH CONVENTIONAL FUNCTIONS 326

12.1 FOLDERS

Overview

Folders can be created in program memory.

12.1.1 Folder Configuration

The following folders can be created:

- Folder names are up to 32 characters long.
- The following characters can be used in folder names:
  Alphabetical characters (uppercase and lowercase letters), numeric characters, and the symbols below:
  – + _ .
  Because "." and ".." are reserved folder names, they cannot be used.

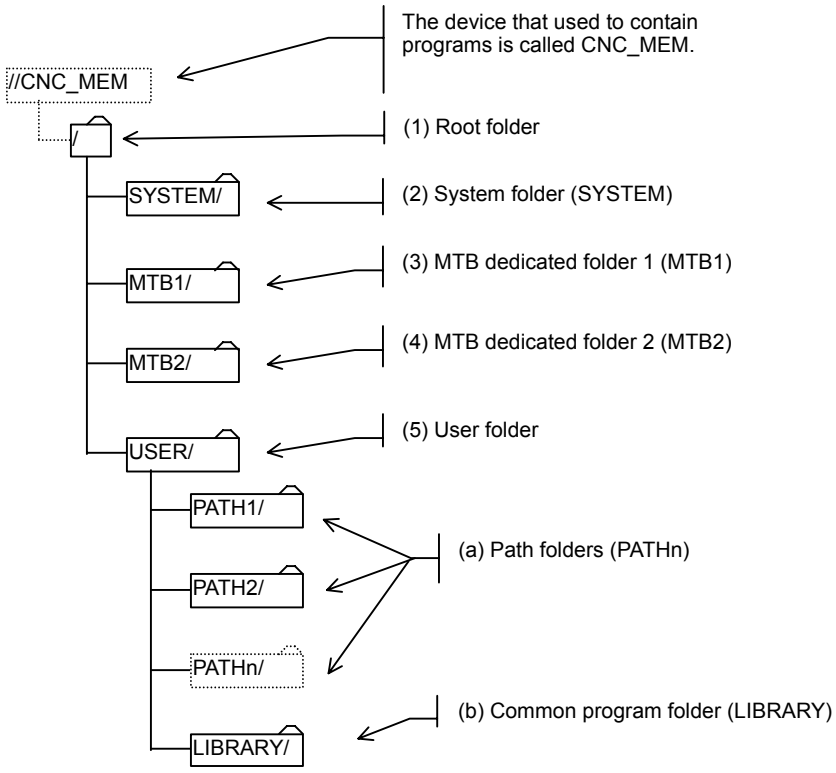
- Initial folders

When program memory is initialized, folders having predefined structures and names are created. These folders are called initial folders.

(1) Root folder
The parent folder of all folders
(2) System folder（SYSTEM）
Contains subprograms and macro programs of the system.
(3) MTB dedicated folder 1 (MTB1)
Contains subprograms and macro programs created by the machine tool builder.
(4) MTB dedicated folder 2 (MTB2)
Contains subprograms and macro programs created by the machine tool builder.
(5) User folder
Contains programs created by the user.
The following folders are further created in this folder:
  (a) Path folders (PATHn: As many folders as the number of paths are created.)
  Each contains main programs, subprograms, and macro programs used for the corresponding path.
  (b) Common program folder (LIBRARY)
  Contains subprograms and macro programs used commonly.

NOTE
The initial folders can neither be deleted nor renamed.

[Initial folder configuration]



- User created folders

Folders other than the initial folders are called user created folders.
User created folders can be created in the following initial folders:

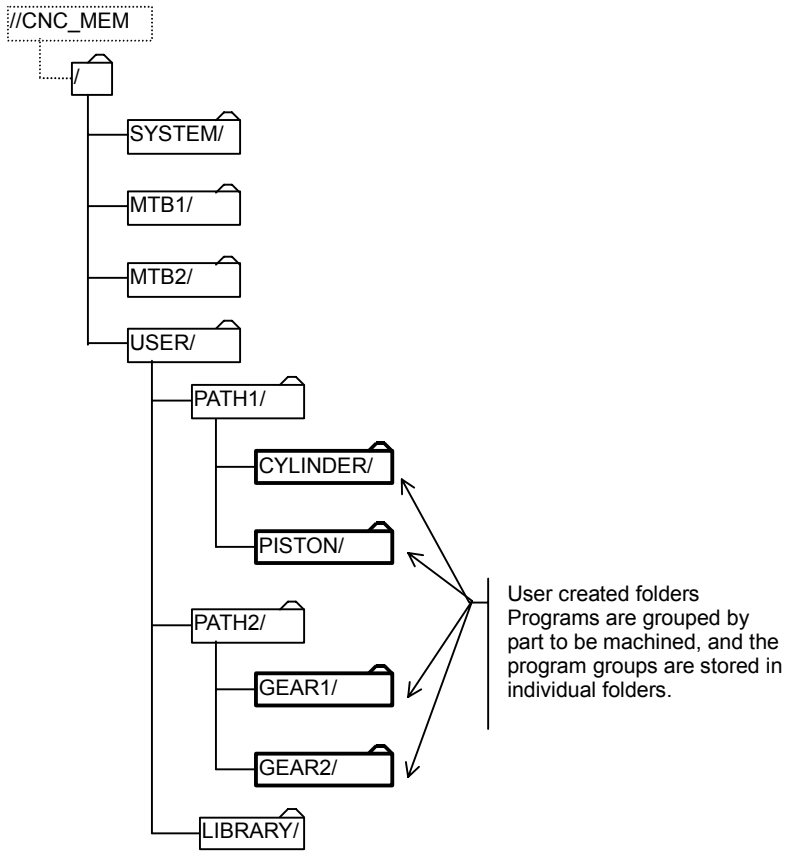
- User folder
- Path folders

User created folders can contain user created main programs, subprograms, and macro programs.

NOTE

1 Folder names must each be unique in the same folder.
2 Each time a user created folder is created, the number of programs that can be registered decreases by one.
3 Folder hierarchical levels of user created folders are limited.
Up to three hierarchical levels starting from the user folder (USER/) are permitted.

[Sample folder configuration]



12.1.2 Folder Attributes

The following attributes can be set for folders except the root folder:

- Edit disable
- Edit/display disable

- Edit disable

Editing of the programs and folders in a folder can be disabled.
A program in the folder can be output to an external device.
A program cannot be input (registered) from an external device to the folder.

- Edit/display disable

Editing and display of the programs and folders in a folder can be disabled.
When this attribute is set for a folder, the programs and folders in the folder become invisible. (The folder appears to contain nothing.)
A program in the folder cannot be output to an external device and a program cannot be input (registered) from an external device to a folder.

12.1.3 Default Folders

Default folders are folders on which operations are performed when no folder is specified. There are two types of default folders as follows:

- Foreground default folder
- Background default folder

- Foreground default folder

A folder used for foreground operations except automatic operations and program editing is set.
The target operations include:

- Program input/output
- External data input
- External workpiece number search

- Background default folder

A folder used for background operations is set.
The target operations include:

- Program input/output
- External I/O device control

NOTE

1 When the foreground or background default folder is not set, a path folder, which is an initial folder, is assumed.
2 Foreground and background default folder settings are stored in default folder setting files.
3 When a program file, program folder, or program folder management file is cleared, the default folder setting file is cleared at the same time.

12.2 FILES

Overview

Desired file names can be given to part programs in program memory.

12.2.1 File Name

File names can be set as follows:

- File names are up to 32 characters long.
- The following characters can be used in file names:
  Alphabetical characters (uppercase and lowercase letters), numeric characters, and the symbols below:
  – + _ .
  Because "." and ".." are reserved file names, they cannot be used.

- File names and program numbers

File names are associated with program numbers as explained below.
When the file name of a program consists of "O" plus a numeric with the following limitation, the program can be handled also by program number.

- The numeric must be a leading zero suppressed value from 1 to 9999.

When the file name of a program does not have the above format, the program cannot be handled by program number.
When a file name consisting of "O" plus a numeric does not satisfy the above limitation, the file cannot be created.

Example)

File names that can be treated as program numbers

O123	Program number 123
O1	Program number 1
O3000	Program number 3000
O9999	Program number 9999

File names that cannot be treated as program numbers

ABC
o123
O123.4

NOTE

1 File names must each be unique in the same folder.
2 When the file name of a program is not treated as a program number, the program is restricted as follows:
  - The program cannot be specified by program number.
  - Information output by program number is impossible..

- Displaying File Names and Program Numbers

The file name of the program selected or being executed as the main program is displayed as shown in Figs. 12.2.1 (a) to 12.2.2 (c).

- For file names that can be handled as program numbers, the program number is displayed.

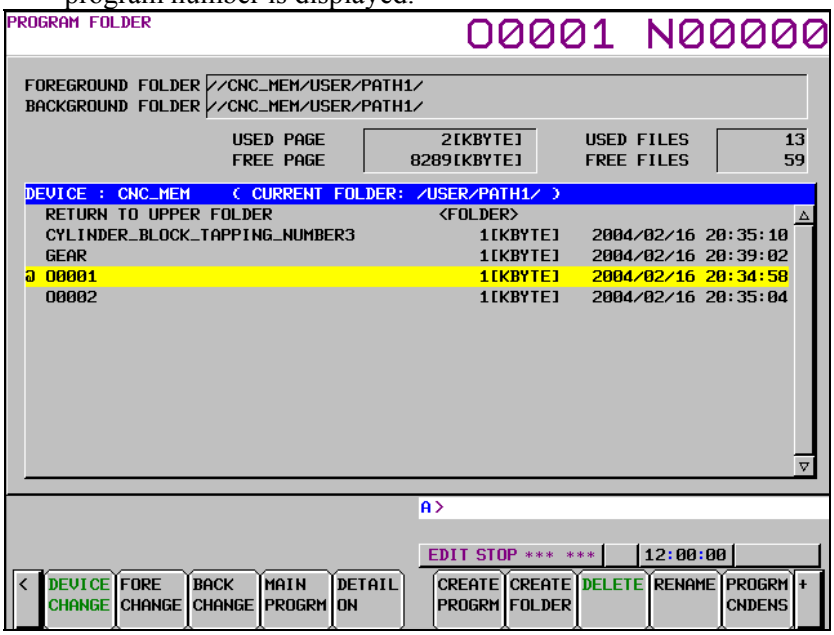


Fig. 12.2.1 (a) File name display 1

- For file names that cannot be handled as program numbers, the file name is displayed.
  The display layout varies as shown below, depending on the length of a file name.

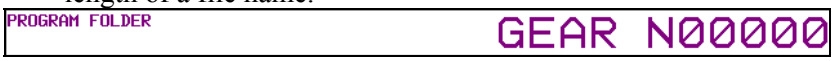


Fig. 12.2.1 (b) File name display 2

Fig. 12.2.1 (c) File name display 3

12.2.2 File Attributes

The following attributes can be set for files:

- Edit disable
- Edit/display disable
- Encoding
- Change protection level/output protection level

- Edit disable

Editing of a specified program can be disabled.
A program cannot be input (registered) from an external device to the folder.

- Edit/display disable

Editing and display of a specified program can be disabled.
When this attribute is set, the program becomes invisible.
(The folder appears not to contain this program.)
An output to an external device and an input from an external device (program registration) cannot be made.

- Encoding

A specified program can be encoded.
For details of encoding, see the description of the “Program encryption” function.
An output to an external device and an input from an external device (program registration) cannot be made.

- Change protection level/output protection level

With the 8-level data protection function, change and output protection can be provided for a specified program.
For details of the 8-level data protection function, see the description of the “Protection of data at eight levels” function.

12.3 RELATION WITH CONVENTIONAL FUNCTIONS

This section explains relation with conventional functions when folder names and file names are used.

12.3.1 Relation with Folders

This subsection explains how folders are used for operations and editing.

- Automatic operation
Main program

As the main program running for an automatic operation, a program in a desired folder can be selected.

Subprogram (called by M98/G72.1/G72.2)
Macro program (called by G65/G66/G66.1/M96)

- Subprogram call (M98)
- Macro call (simple call G65/modal call G66, G66.1)
- Macro interrupt (M96)
- Figure copying (G72.1, G72.2)

When a call listed above is made, folders are searched in the following order, and the program found first is called:
<1> Folders containing the main program
<2> Common program folder, which is an initial folder (LIBRARY)

Bit 7 (SCF) of parameter No. 3457 can be used to add the following search folders. (The folders are searched in the order listed below.) The folder to be actually enabled is separately set by bits 1 to 3 of parameters No. 3457.
<3> MTB dedicated folder 2, which is an initial folder (MTB2)
<4> MTB dedicated folder 1, which is an initial folder (MTB1)
<5> System folder, which is an initial folder (SYSTEM)

Subprogram (called by M code/specific address/2nd auxiliary function)
Macro program (called by G code/M code/T code/one touch macro)

- Subprogram call by M code/specific address/2nd auxiliary function code
- Macro call by G code/M code/T code
- One touch macro call

For programs called as listed above, the folders to be searched are set in bits 0 to 3 of parameter No.3457 in advance. (The search order is described below.)
The folders set as the search targets are searched, and the program found first is called.
<1> Common program folder, which is an initial folder (LIBRARY)
<2> MTB dedicated folder 2, which is an initial folder (MTB2)
<3> MTB dedicated folder 1, which is an initial folder (MTB1)
<4> System folder, which is an initial folder (SYSTEM)

- Program editing

A program in any folder can be edited.

- Program I/O

The following functions are performed for default folders:

- Program input from external devices
- Program output to external devices
  (Except the format with folder names)

The following functions are performed for a set background default folder:

- External I/O device control

- External data input

A set foreground default folder is searched for an external program.

- External workpiece number search

A set foreground default folder is searched for an external workpiece number.

- Macro executor

Programs called by execution macros, conversational macros, and auxiliary macros are programs in P-Code files regardless of the default folder.
For programs on the macro executor, the folder function and file name function are unavailable.

12.3.2 Relation with File Names

File names can be used with the following functions:

- Subprogram call (M98)
- Macro call (simple call G65/modal call G66, G66.1)
- Interruption type macro call (M96)
- Subprogram call in figure copying (G72.1, G72.2)
- Program I/O with external devices

- Subprogram call by file name
- Macro call by file name

- Subprogram call (M98)
- Macro call (G65/G66/G66.1)
- Interruption type macro call (M96)
- Figure copying (G72.1, G72.2)

When a program is called in the above functions, a subprogram call by file name and a macro call by file name can be used.

- Subprogram call by file name
  M98 <file-name> Lxx ;
- Macro call by file name
  G65 <file-name> Lxx argument-specification ;
  G66 <file-name> Lxx argument-specification ;
  G66.1 <file-name> Lxx argument-specification ;
- Interruption type macro call
  M96 <file-name> ;
- Subprogram call by file name in figure copying (G72.1, G72.2)
  G72.1 <file-name> Lxx Xxx Yxx Rxx ;
  G72.2 <file-name> Lxx Ixx Jxx ;

In the above subprogram and macro calls, the program having the file name indicated by <file-name> is called.

Format examples:

- Subprogram call
  M98 <R50> L1 ;
- Macro call
  G65 <R50> L1 A0 ;
  G66 <R50> L1 A1 ;
  G66.1 <R50> L1 A2 ;
- Interruption type macro call
  M96 <R50> ;
- Subprogram call in figure copy
  G72.1 <R50> L1 X0 Y0 R0 ;
  G72.2 <R50> L1 I0 J0 ;

NOTE
1 When characters in <> are read, they are treated in the same way as for characters in comments. So, note that these characters are treated differently from other significant information portions.
Refer to Appendix B “PROGRAM CODE LIST” for details.
2 The <file-name> word must be placed immediately after each word for calling (M98, G65, and so forth).

12.3.3 Related Parameters

This subsection lists the meanings of parameters related to program numbers and the folders and programs to be manipulated or executed.

Parameter No.	Bit No.	Description	Manipulation/execution target
3202	0 (NE8)	Disables or enables editing of programs O8000 to O8999.	Corresponding programs in all folders
	4 (NE9)	Disables or enables editing of programs O9000 to O9999.	Same as above
3204	3 (P8E)	Disables or enables editing of programs O80000000 to O8999999.	Same as above
	4 (P9E)	Disables or enables editing of programs O90000000 to O99999999.	Same as above
	5 (SPR)	Assumes or does not assume a particular program number in the nine thousands to be a number obtained by adding 90000000.	Same as above
3210/3211	-	Password/keyword for protecting programs in the nine thousands	Same as above
3222/3223	-	Program protection range (minimum value)/(maximum value)	Same as above
3404	2 (SBP)	In the subprogram call function, address P in the M198 block specifies a file/program number.	(Not dependent on the folder function and file name function)
6001	5 (TCS)	Calls or does not call a custom macro by T code.	The following initial folders can be selected by parameter setting: - Common program folder (LIBRARY) - MTB dedicated folder 2 (MTB2) - MTB dedicated folder 1 (MTB1) - System folder (SYSTEM)
6050 to 6059	-	G code for calling a custom macro with program No. 9010 to 9019	Same as above
6071 to 6079	-	M code for calling a subprogram with program No. 9001 to 9009	Same as above
6080 to 6089	-	M code for calling a custom macro with program No. 9020 to 9029	Same as above
6090/6091	-	ASCII code for calling a subprogram with program No. 9004/9005	Same as above
8341/8343	-	Target program number (for 4- and 8-digit O number) for sequence number comparison and stop	Foreground or background default folder depending on the mode

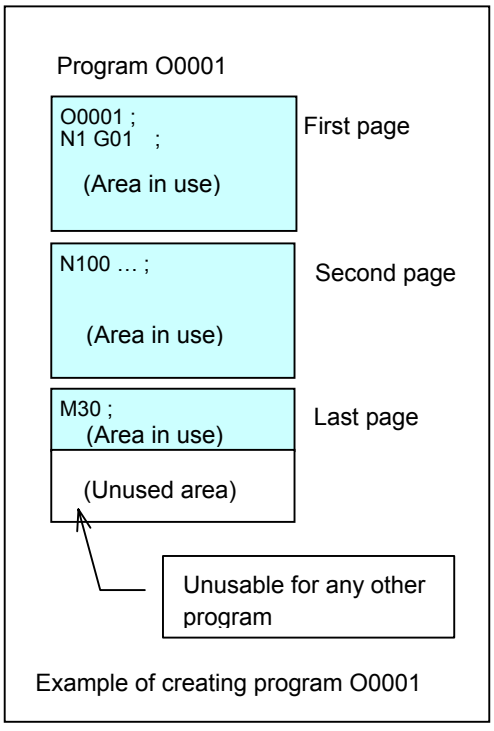
12.3.4 Part Program Storage Size / Number of Registerable Programs

The following table lists the combinations of program storage sizes and the total number of registrable programs.

Part program storage size	Number of registerable programs	Number of registerable programs expansion 1	Number of registerable programs expansion 2 *2
32Kbyte *1	63	-	-
64Kbyte	63	125	-
128Kbyte	63	250	-
256Kbyte	63	500	-
512Kbyte	63	1000	-
1Mbyte	63	1000	2000
2Mbyte	63	1000	4000
4Mbyte *2	63	1000	4000
8Mbyte *2	63	1000	4000

*1) For Series 32*i*-A only

*2) For Series 30*i*/31*i*-A, For Series 31*i*-A5 only



Example of creating program O0001

NOTE

1 Creating one folder results in the number of programs yet to be registerable decreasing one.

2 The program storage size means the maximum size of a program if the program is the one and only program registered.

3 If more than one program is registered, the total size of registerable programs reduces for the following reason.
The Series 30*i*/31*i*/32*i*-A and Series 31*i*-A5 manage programs in page units. The unit of program storage is managed also in page units. When a program is created, as many pages as necessary to store the program are secured, and the program is stored on these pages. Generally, the last program storage page has an unused area (left figure). This unused area cannot be used to store any other program. For the sake of program management, it is regarded as an area in use.
The Series 16*i*/18*i*/21*i* uses a similar way of management, but the unit of pages in it differs from that in the Series 30*i*/31*i*/32*i*-A and FS31*i*-A5. So, if more than one program is registered in the Series 30*i*/31*i*/32*i*-A or FS31*i*-A5, the total program size of registerable programs in the Series 30*i*/31*i*/32*i*-A or Series 31*i*-A5 differs from that in the Series 16*i*/18*i*/21*i*.

13 PROGRAM CONFIGURATION

Overview

- Main program and subprogram

There are two program types, main program and subprogram. Normally, the CNC operates according to the main program. However, when a command calling a subprogram is encountered in the main program, control is passed to the subprogram. When a command specifying a return to the main program is encountered in a subprogram, control is returned to the main program.

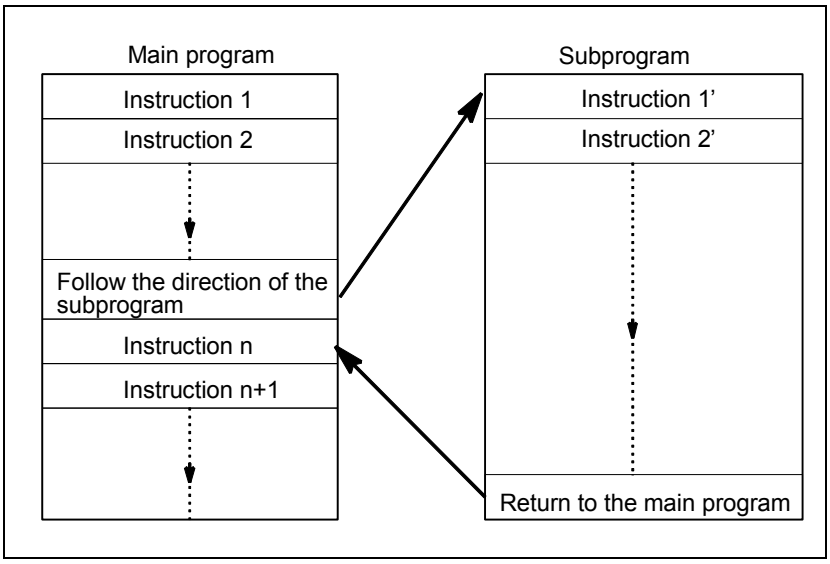


Fig. 13 (a) Main program and Subprogram

The CNC memory can hold up to 4000 main programs and subprograms (63 as standard). A main program can be selected from the stored main programs to operate the machine. See III-9 or III-10.4 for the methods of registering and selecting programs.

- Program components

A program consists of the following components:

Table 13 (a) Program components

Components	Descriptions
Program code start	Symbol indicating the start of a program file
Leader section	Used for the title of a program file, etc.
Program start	Symbol indicating the start of a program
Program section	Commands for machining
Comment section	Comments or directions for the operator
Program code end	Symbol indicating the end of a program file

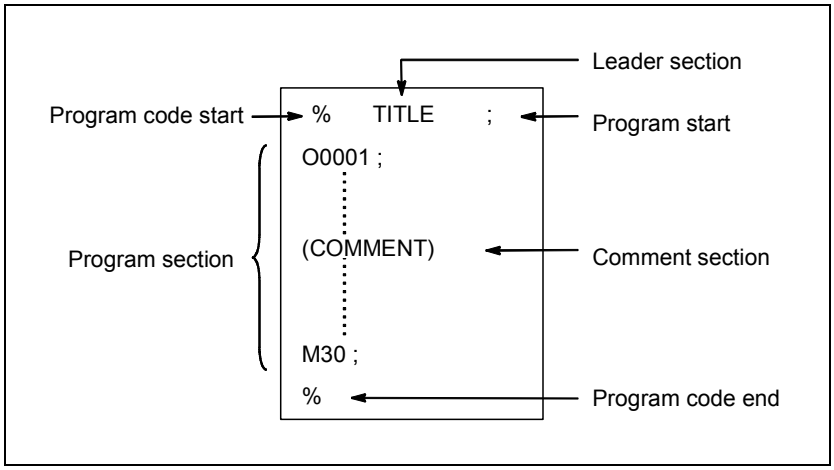


Fig. 13 (b) Program configuration

- Program section configuration

A program section consists of several blocks. A program section starts with a program number or file name and ends with a program end code.

Program section configuration	**Program section**
Program number	O0001 ;
Block 1	N1 G91 G00 X120.0 Y80.0 ;
Block 2	N2 G43 Z-32.0 H01 ;
: :	
Block n	Nn Z0 ;
Program end	M30 ;

A block contains information necessary for machining, such as a move command or coolant on/off command. Specifying a slash (/) at the start of a block disables the execution of some blocks (see "optional block skip" in II-13.2).

13.1 PROGRAM COMPONENTS OTHER THAN PROGRAM SECTIONS

This section describes program components other than program sections. See II-13.2 for a program section.

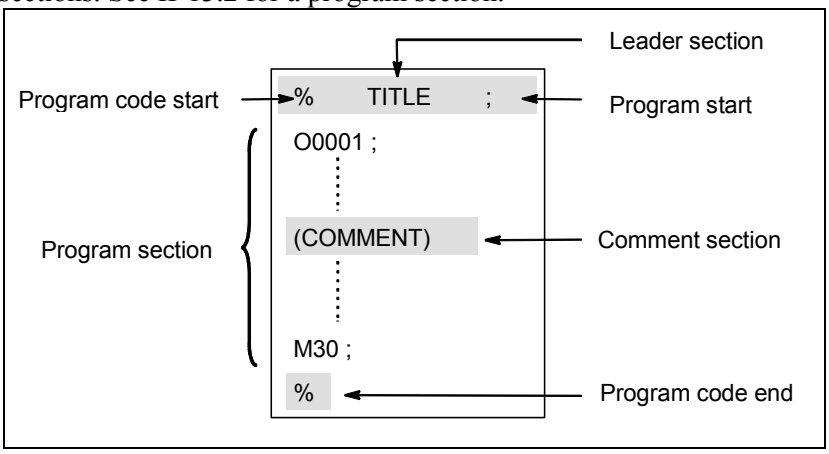


Fig. 13.1 (a) Program configuration

Explanation

- Program code start

The program code start indicates the start of a file that contains NC programs.
The mark is not required when programs are entered using SYSTEM P or ordinary personal computers. The mark is not displayed on the screen. However, if the file is output, the mark is automatically output at the start of the file.

Table 13.1 (a) Code of a program code start

Name	ISO code	EIA code	Notation in this manual
Program code start	%	ER	%

- Leader section

Data entered before the programs in a file constitutes a leader section.
When machining is started, the label skip state is usually set by turning on the power or resetting the system. In the label skip state, all information is ignored until the first end-of-block code is read. When a file is read into the CNC unit from an I/O device, leader sections are skipped by the label skip function.
A leader section generally contains information such as a file header. When a leader section is skipped, even a TV parity check is not made. So a leader section can contain any codes except the EOB code.

- Program start

The program start code is to be entered immediately after a leader section, that is, immediately before a program section.
This code indicates the start of a program, and is always required to disable the label skip function.
With SYSTEM P or ordinary personal computers, this code can be entered by pressing the return key.

Table 13.1 (b) Code of a program start

Name	ISO code	EIA code	Notation in this manual
Program start	LF	CR	;

NOTE
If one file contains multiple programs, the EOB code for label skip operation must not appear before a second or subsequent program number.

- Comment section

Any information enclosed by the control-out and control-in codes is regarded as a comment.
The user can enter a header, comments, directions to the operator, etc. in a comment section.

Table 13.1 (c) Codes of a control-in and a control-out

Name	ISO code	EIA code	Notation in this manual	Meaning
Control-out	(	2-4-5	(	Start of comment section
Control-in	)	2-4-7	)	End of comment section

When a program is read into memory for memory operation, comment sections, if any, are not ignored but are also read into memory. Note, however, that codes other than those listed in the code table in Appendix A are ignored, and thus are not read into memory.
When data in memory is output on external I/O device (See III-8), the comment sections are also output.
When a program is displayed on the screen, its comment sections are also displayed. However, those codes that were ignored when read into memory are not output or displayed.
During memory operation or DNC operation, all comment sections are ignored.
The TV check function can be used for a comment section by setting bit 1 (CTV) of parameter No. 0100.

> ⚠ **CAUTION**
> If a long comment section appears in the middle of a program section, a move along an axis may be suspended for a long time because of such a comment section. So a comment section should be placed where movement suspension may occur or no movement is involved.

> **NOTE**
> 1 If only a control-in code is read with no matching control-out code, the read control-in code is ignored.
> 2 The following codes cannot be used in the comment section:
> - EOB
> - % (ER for EIA)

- Program code end

A program code end is to be placed at the end of a file containing NC programs.
If programs are entered using the automatic programming system, the mark need not be entered.
The mark is not displayed on the screen. However, when a file is output, the mark is automatically output at the end of the file.
If an attempt is made to execute % when M02 or M30 is not placed at the end of the program, the alarm PS5010 is occurred.

Table 13.1 (d) Code of a program code end

Name	ISO code	EIA code	Notation in this manual
Program code end	%	ER	%

13.2 PROGRAM SECTION CONFIGURATION

This section describes elements of a program section. See II-13.1 for program components other than program sections.

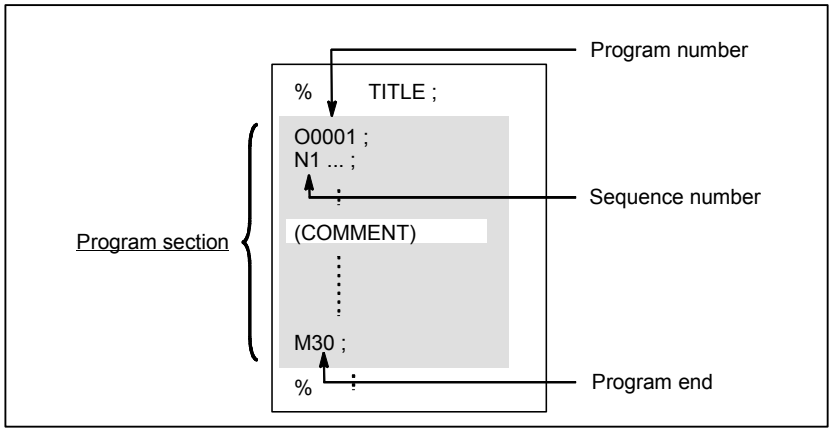


Fig. 13.2 (a) Program configuration

- Program number

A program number consisting of address O followed by a four-digit number is assigned to each program at the beginning registered in memory to identify the program. When the 8-digit number function is selected, the program number consists of eight digits.
In ISO code, the colon (:) can be used instead of O.
When no program number is specified at the start of a program, the sequence number (N....) at the start of the program is regarded as its program number. If a five-digit sequence number is used, the lower four digits are registered as a program number. If the lower four digits are all 0, the program number registered immediately before added to 1 is registered as a program number. Note, however, that N0 cannot be used for a program number.
If there is no program number or sequence number at the start of a program, a program number must be specified using the MDI panel when the program is stored in memory (See III-8.2 or III-9.1)

NOTE
Program numbers 8000 to 9999 may be used by machine tool builders, and the user may not be able to use these numbers.

- File name

A file name can be assigned instead of a program number.
When coding a file name, be sure to place the file name enclosed in "<" and ">" at the beginning of a program.
Example) % ;
<PARTS_1> ;
N1 ...
:
M30 ;
%

NOTE
A file name can be coded:
- At the beginning of a program
- Immediately after M98, G65, G66, G66.1, M96, G72.1, or G72.2

Do not code a file name in other than the above.

- Sequence number and block

A program consists of several commands. One command unit is called a block. One block is separated from another with an EOB of end of block code.

Table 13.2 (a) EOB code

Name	ISO code	EIA code	Notation in this manual
End of block (EOB)	LF	CR	;

At the head of a block, a sequence number consisting of address N followed by a number not longer than eight digits (1 to 99999999) can be placed. Sequence numbers can be specified in a random order, and any numbers can be skipped. Sequence numbers may be specified for all blocks or only for desired blocks of the program. In general, however, it is convenient to assign sequence numbers in ascending order in phase with the machining steps (for example, when a new tool is used by tool replacement, and machining proceeds to a new surface with table indexing.)

N300X200.0Z300.0 ; A sequence number is underlined.

Fig. 13.2 (b) Sequence number and block (example)

NOTE
N0 must not be used for the reason of file compatibility with other CNC systems.
Program number 0 cannot be used. So 0 must not be used for a sequence number regarded as a program number.

- TV check (Vertical parity check)

A parity check is made for each block of input data. If the number of characters in one block (starting with the code immediately after an EOB and ending with the next EOB) is odd, a P/S alarm (No.002) is output.
No TV check is made only for those parts that are skipped by the label skip function. Bit 1 (CTV) of parameter No. 0100 is used to specify whether comments enclosed in parentheses are counted as characters during TV check. The TV check function can be enabled or disabled by setting on the MDI unit (See III-12.3.1.).

- Block configuration (word and address)

A block consists of one or more words. A word consists of an address followed by a number some digits long. (The plus sign (+) or minus sign (-) may be prefixed to a number.)
For an address, one of the letters (A to Z) is used ; an address defines the meaning of a number that follows the address.

Word = Address + number (Example : X-1000)

Table 13.2 (b) indicates the usable addresses and their meanings.
The same address may have different meanings, depending on the preparatory function specification.

Table 13.2 (b) Major functions and addresses

Function	Address	Meaning
Program number	O(*)	Program number
Sequence number	N	Sequence number
Preparatory function	G	Specifies a motion mode (linear, arc, etc.)
Dimension word	X, Y, Z, U, V, W, A, B, C	Coordinate axis move command
	I, J, K	Coordinate of the arc center
	R	Arc radius
Feed function	F	Rate of feed per minute, Rate of feed per revolution
Spindle speed function	S	Spindle speed
Tool function	T	Tool number
Auxiliary function	M	On/off control on the machine tool
	B	Table indexing, etc.
Program number designation	P	Subprogram number
Number of repetitions	P, L	Number of subprogram repetitions
Parameter	P, Q	Canned cycle parameter

M

Offset number	D, H	Offset number
Dwell	P, X	Dwell time

T

Dwell	P, X, U	Dwell time

NOTE
(*) In ISO code, the colon (:) can also be used as the address of a program number.

N_	G_	X_ Y_	F_	S_	T_	M_	;
Sequence number	Preparatory function	Dimension word	Feed-function	Spindle speed function	Tool function	Auxiliary function	

Fig. 13.2 (c) 1 block (example)

- Major addresses and ranges of command values

Major addresses and the ranges of values specified for the addresses are shown below. Note that these figures represent limits on the CNC side, which are totally different from limits on the machine tool side. For example, the CNC allows a tool to traverse up to about 100 m (in millimeter input) along the X axis.
However, an actual stroke along the X axis may be limited to 2 m for a specific machine tool.
Similarly, the CNC may be able to control a cutting feedrate of up to 240 m/min, but the machine tool may not allow more than 3 m/min. When developing a program, the user should carefully read the manuals of the machine tool as well as this manual to be familiar with the restrictions on programming.

Table 13.2 (c) Major addresses and ranges of command values

Function		Address	Input in mm	Input in inch
Program number		O (*1)	1 to 99999999	1 to 99999999
Sequence number		N	1 to 99999999	1 to 99999999
Preparatory function		G	0 to 99.9	0 to 99.9
Dimension word	Increment system IS-A	X,Y,Z,U,V, W,A,B,C,I,J ,K,R (*2)	±999999.99 mm ±999999.99 deg.	±99999.999 inch *3 ±999999.99 deg.
	Increment system IS-B		±999999.999 mm ±999999.999 deg.	±99999.9999 inch *3 ±999999.999 deg.
	Increment system IS-C		±99999.9999 mm ±99999.9999 deg.	±9999.99999 inch *3 ±99999.9999 deg.
	Increment system IS-D		±9999.99999 mm ±9999.99999 deg.	±999.999999 inch *3 ±9999.99999 deg.
	Increment system IS-E		±999.999999 mm ±999.999999 deg.	±99.9999999 inch *3 ±999.999999 deg.
Feed per minute	Increment system IS-A	F	0.01 to 999000.00mm/min	0.001 to 96000.000inch/min
	Increment system IS-B		0.001 to 999000.000mm/min	0.0001 to 9600.0000inch/min
	Increment system IS-C		0.0001 to 99999.9999mm/min	0.00001 to 4000.00000inch/min
	Increment system IS-D		0.00001 to 9999.99999mm/min	0.000001 to 400.000000inch/min
	Increment system IS-E		0.000001 to 999.999999mm/min	0.0000001 to 40.0000000inch/min
Feed per revolution		F	0.001 to 50000mm/rev	0.0001 to 50.0000inch/rev
Spindle speed function		S (*4)	0 to 99999999	0 to 99999999
Tool function		T (*4)	0 to 99999999	0 to 99999999
Auxiliary function		M (*4)	0 to 99999999	0 to 99999999
		B (*4)	0 to 99999999	0 to 99999999
Offset number (M series only)		H, D	0 to 999	0 to 999

Function		Address	Input in mm	Input in inch
Dwell	Increment system IS-A	X, U (T series only)	0 to 999999.99 sec	0 to 999999.99 sec
	Increment system IS-B		0 to 99999.999 sec	0 to 99999.999 sec
	Increment system IS-C		0 to 9999.9999 sec	0 to 9999.9999 sec
	Increment system IS-D		0 to 999.99999 sec	0 to 999.99999 sec
	Increment system IS-E		0 to 99.999999 sec	0 to 99.999999 sec
Dwell		P	1 to 99999999	1 to 99999999
Designation of a program number		P	1 to 99999999	1 to 99999999
Number of subprogram repetitions		L	1 to 99999999	1 to 99999999
		P	0 to 9999	0 to 9999

*1 In ISO code, the colon (:) can also be used as the address of a program number.

*2 When address I, J, K, or R is used to specify the radius for circular interpolation, the specifiable range is as follows:

Increment system	Input in mm	Input in inch
IS-A	±999999999.99 mm	±99999999.999 inch
IS-B	±999999999.999 mm	±99999999.9999 inch
IS-C	±99999999.9999 mm	±9999999.99999 inch
IS-D	±9999999.99999 mm	±999999.999999 inch
IS-E	±999999.999999 mm	±99999.9999999 inch

*3 For inch input/millimeter machines, the maximum specifiable range of dimension words is as follows:

Increment system	Maximum specifiable range
IS-A	±39370.078 inch
IS-B	±39370.0787 inch
IS-C	±3937.00787 inch
IS-D	±393.700787 inch
IS-E	±39.3700787 inch

*4 The maximum value of addresses M, S, T, and B is 99999999. Note that, however, values longer than the permissible number of digits set in parameter No. 3030 to 3033 cannot be specified. The values and uses for some codes are limited by parameter setting. (For example, some M codes are not buffered.) For details, refer to the parameter manual.

- Optional block skip

When a slash followed by a number (/n (n=1 to 9)) is specified at the head of a block, and optional block skip signals BDT1 to BDT9 are set to 1 during automatic operation, the information (/n to the end of the block (EOB)) contained in the block for which /n corresponding to signal BDTn is specified is ignored.

Example 1)

```
/2  N123  X100.0  Y200.0 ;
```

Example 2)

```
//3  N123  X100.0  Y200.0  ;        → Incorrect
/1  /3  N123  X100.0  Y200.0  ;     → Correct
```

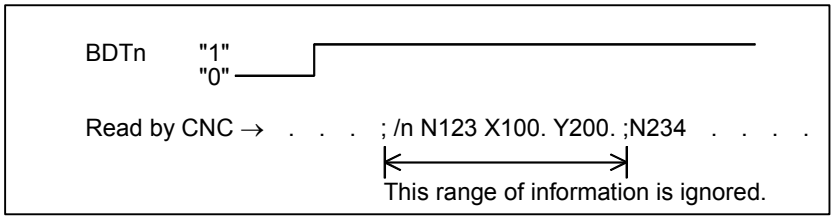
Input signal and program code

Input signal	Start code to be ignored
BDT1	/ or /1(NOTE)
BDT2	/2
BDT3	/3
BDT4	/4
BDT5	/5
BDT6	/6
BDT7	/7
BDT8	/8
BDT9	/9

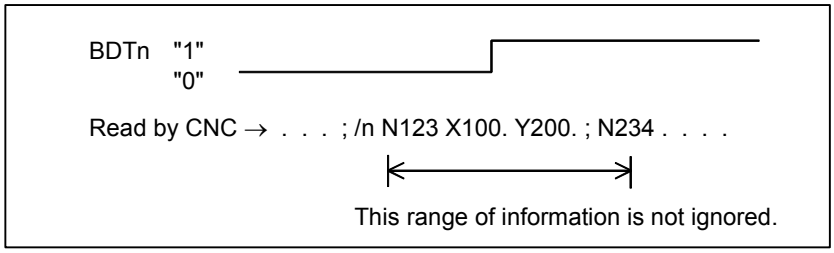
NOTE
1 Number 1 for /1 can be omitted. However, when two or more optional block skips are specified for one block, number 1 for /1 cannot be omitted.
2 Depending on the machine tool, all optional block skip signals (1 to 9) may not be usable. Refer to manuals of the machine tool builder to find which switches are usable.

The following shows the relationship between the timing at which optional block skip signals BDT1 to BDT9 are set to 1 and the range of information to be ignored.

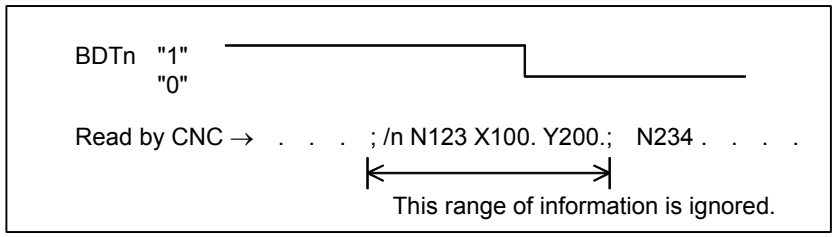
1. When the signal BDTn is set to 1 before the CNC starts reading a block that contains /n, the block is ignored.



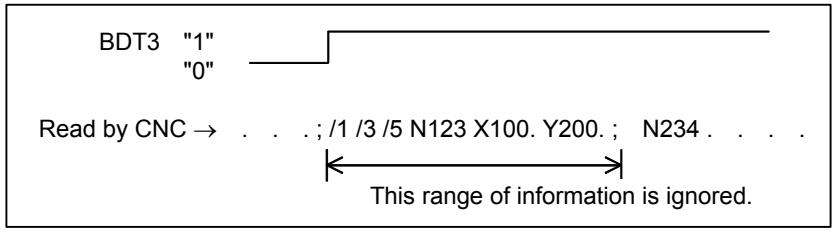
2. When the signal BDTn is set to 1 while the CNC is reading a block that contains /n, the block is not ignored.



3. When the signal BDTn is set to 0 while the CNC is reading a block that contains /n, the block is ignored.



4. Two or more optional block skips can be specified in one block. When the signal corresponding to any of the specified skips is set to 1, the block is ignored.



NOTE

1 This function is not used when a program is registered in memory. A block containing / is registered in memory regardless of the statuses of optional block skip signals. When a program in memory is also output regardless of the statuses of optional block skip signals.
In addition, the optional block skip function is enabled during a search for a sequence number.

2 Position of a slash
A slash (/) must be specified at the head of a block. If a slash is placed elsewhere, the information from the slash to immediately before the EOB code is ignored.

3 TV and TH check
When the optional block skip signal is set to 1, TH and TV checks are made for the skipped portions in the same way as when the optional block skip signal is set to 0.

- Program end

The end of a program is indicated by programming one of the following codes at the end of the program:

Table 13.2 (d) Code of a program end

Code	Meaning usage
M02	For main program
M03	
M99	For subprogram

If one of the program end codes is executed in program execution, the CNC terminates the execution of the program, and the reset state is set. When the subprogram end code is executed, control returns to the program that called the subprogram.

> ⚠ **CAUTION**
> A block containing an optional block skip code such as /M02 ; , /M30 ; , or /M99 ; is not regarded as the end of a program. (see "Optional block skip".)

13.3 SUBPROGRAM (M98, M99)

If a program contains a fixed sequence or frequently repeated pattern, such a sequence or pattern can be stored as a subprogram in memory to simplify the program.
A subprogram can be called from the main program.
A called subprogram can also call another subprogram.

Format

- Subprogram configuration

One subprogram

```
Oxxxx ;
  :
M99;
```

Subprogram number or subprogram file name (or the colon (:) optionally in the case of ISO)

Program end

M99 need not constitute a separate block as indicated below.

Example) X100.0Y100.0M99;

- Subprogram call

- When a subprogram with a 4-digit or shorter program number is called

M98 Pxxxx xxxx ;

Subprogram number

Number of times the subprogram is called repeatedly

or

M98 Pxxxx Lxxxxxxxx ;

Number of times the subprogram is called repeatedly

Subprogram number

- When a subprogram with a 5-digit or longer program number is called

M98 Pxxxxxxxx Lxxxxxxxx ;

Subprogram number

Number of times the subprogram is called repeatedly

- When a subprogram is called by file name

M98 <xxxx> Lxxxxxxxx ;

Subprogram file name

Number of times the subprogram is called repeatedly

- Called program and folders to be searched

The order in which folders are searched depends on the method of calling a subprogram. Folders are searched in sequence and the program found first is called. For details, see the "Managing Programs" chapter.

NOTE

1 When calling a subprogram of a 4-digit or shorter subprogram number repeatedly (P8 digit), make the subprogram number 4 digits by prefixing it with “0” if the subprogram number is shorter than 4 digits.
Example)
P100100: Call subprogram No. 100 ten times.
P50001: Call subprogram No. 1 five times.

2 If the number of times the subprogram is called repeatedly is omitted when a subprogram with a 4-digit or shorter program number is called, the subprogram is called just once.
In this case, it is not necessary to adjust the subprogram number length to 4 digits as described in Item 1 above.

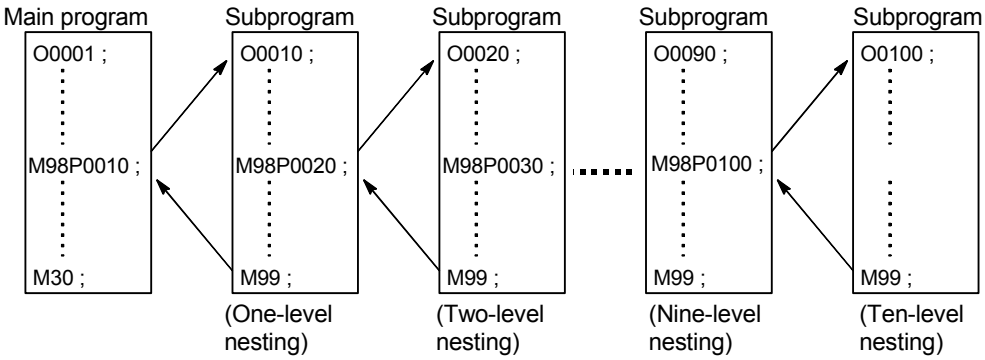
3 When calling a subprogram of a 4-digit or shorter program number repeatedly (P8 digit), do not specify address L in the same block.

4 When calling a subprogram with a 5-digit or longer program number, do not omit the specification of the number of repeats.

5 When calling a subprogram by file name, be sure to specify the file name immediately after M98.

Explanation

When the main program calls a subprogram, it is regarded as a one-level subprogram call. Thus, subprogram calls can be nested up to ten levels as shown below.



A single call command can repeatedly call a subprogram up to 99999999 times. For compatibility with automatic programming systems, in the first block, Nxxxxxxxx can be used instead of a subprogram number that follows O (or :). A sequence number after N is registered as a subprogram number.

NOTE
1. The M98 and M99 code signal and strobe signal are not output to the machine tool.
2. If the subprogram number specified by address P cannot be found, an alarm PS0078 is output.
3. When an attempt is made to call a subprogram by file name, but the specified file cannot be found, alarm PS0310 is issued.

Example

- M98 P51002 ;

  This command specifies "Call the subprogram (number 1002) five times in succession." A subprogram call command (M98P_) can be specified in the same block as a move command.

- X1000.0 M98 P1200 ;

  This example calls the subprogram (number 1200) after an X movement.

- Execution sequence of subprograms called from a main program

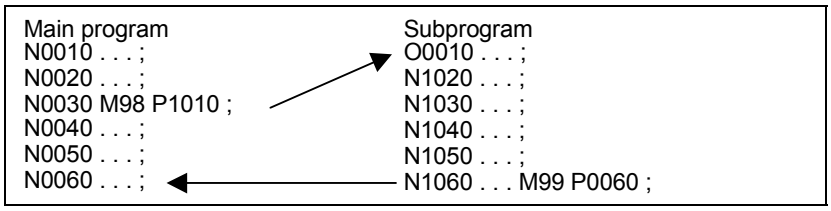
Main program		Subprogram
N0010 . . . ;	1	O1010 . . . ;
N0020 . . . ;	2	N1020 . . . ;
N0030 M98 P21010 ;	3	N1030 . . . ;
N0040 . . . ;		N1040 . . . ;
N0050 M98 P1010 ;		N1050 . . . ;
N0060 . . . ;		N1060 . . . M99 ;

A subprogram can call another subprogram in the same way as a main program calls a subprogram.

Special usage

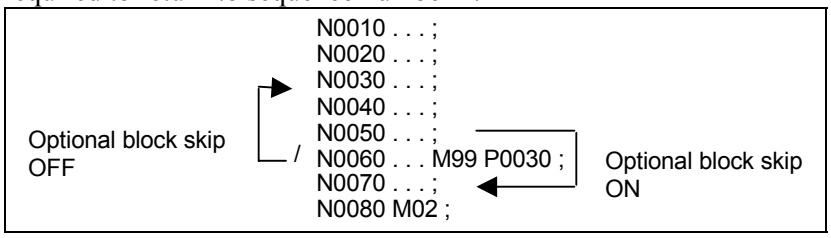
- Specifying the sequence number for the return destination in the main program

If P is used to specify a sequence number when a subprogram is terminated, control does not return to the block after the calling block, but returns to the block with the sequence number specified by P. Note, however, that P is ignored if the main program is operating in a mode other than memory operation mode.
This method consumes a much longer time than the normal return method to return to the main program.



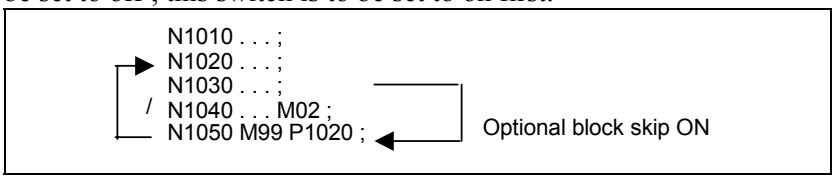
- Using M99 in the main program

If M99 is executed in a main program, control returns to the start of the main program. For example, M99 can be executed by placing /M99 ; at an appropriate location of the main program and setting the optional block skip function to off when executing the main program. When M99 is executed, control returns to the start of the main program, then execution is repeated starting at the head of the main program.
Execution is repeated while the optional block skip function is set to off.
If the optional block skip function is set to on, the /M99 ; block is skipped ; control is passed to the next block for continued execution.
If/M99Pn ; is specified, control returns not to the start of the main program, but to sequence number n. In this case, a longer time is required to return to sequence number n.



- Using a subprogram only

A subprogram can be executed just like a main program by searching for the start of the subprogram with the MDI.
(See III-10.4 for information about search operation.)
In this case, if a block containing M99 is executed, control returns to the start of the subprogram for repeated execution. If a block containing M99Pn is executed, control returns to the block with sequence number n in the subprogram for repeated execution. To terminate this program, a block containing /M02 ; or /M30 ; must be placed at an appropriate location, and the optional block switch must be set to off ; this switch is to be set to on first.

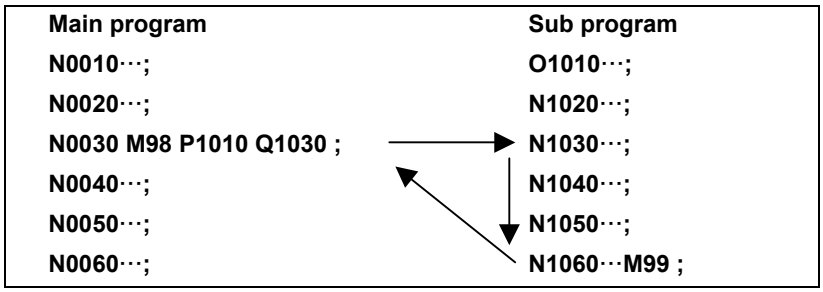


- Subprogram call with sequence number

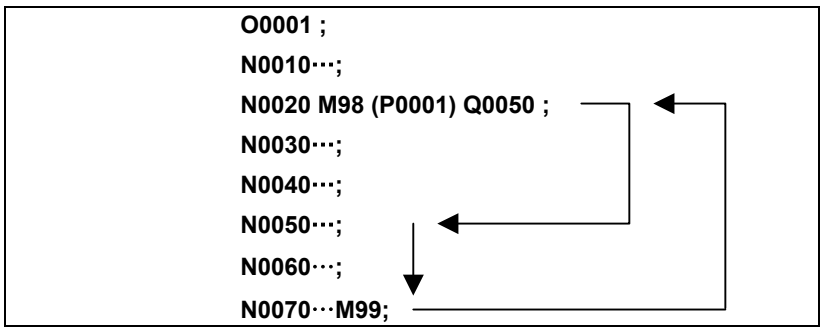
Setting bit 0 (SQC) of parameter No. 6005 to 1 can call a specified sequence number in the subprogram for execution.
In a subprogram call command, specify the letter Q followed by a sequence number to be called after the letter P for specifying a program number.

M98 Pxxxx Qxxxxx ;
Sequence number
Program number

This command causes program execution to start at the called sequence number in the subprogram. If a repetition count is specified, program execution is repeated from the specified sequence number.



This function enables a sequence number in the same program to be called for execution as shown below.
This method, however, requests the programmer to be aware of an allowable call nesting level. If an attempt is made to exceed the allowable nesting level, alarm PS0077 meaning "TOO MANY SUB,MACRO NESTING" is issued.



For a call within the same program, specification of Pxxxx in a block can be omitted when the block includes M98.

This function is usable only for subprogram calls by M98; it is unusable for non-M98 calls, such as macro calls or external subprogram calls based on M198.

14 FUNCTIONS TO SIMPLIFY PROGRAMMING

Chapter 14, "FUNCTIONS TO SIMPLIFY PROGRAMMING", consists of the following sections:

14.1 FIGURE COPYING (G72.1, G72.2).....351
14.2 THREE-DIMENSIONAL COORDINATE SYSTEM CONVERSION 359

14.1 FIGURE COPYING (G72.1, G72.2)

Machining can be repeated after moving or rotating the figure using a subprogram.

Format

- Rotational copying

Xp-Yp plane (specified by G17) :
G72.1 P_ L_ Xp_Yp_R_ ;
Zp-Xp plane (specified by G18) :
G72.1 P_ L_ Zp_Xp_R_ ;
Yp-Zp plane (specified by G19) :
G72.1 P_ L_ Yp_Zp_R _;

P :Subprogram number
L :Number of times the operation is repeated
Xp :Center of rotation on the Xp axis
(Xp : X-axis or an axis parallel to the X-axis)
Yp :Center of rotation on the Yp axis
(Yp: Y-axis or an axis parallel to the Y-axis)
Zp :Center of rotation on the Zp axis
(Zp: Z-axis or an axis parallel to the Z-axis)
R :Angular displacement
(A positive value indicates a counterclockwise angular displacement. Specify an incremental value.)
Specify a plane selection command (G17, G18, or G19) to select the plane on which the rotational copying is made.

- Linear copying

Xp-Yp plane (specified by G17) :
G72.2 P_ L_ I_ J_ ;
Zp-Xp plane (specified by G18) :
G72.2 P_ L_ K_ I_ ;
Yp-Zp plane (specified by G19) :
G72.2 P_ L_ J_ K_;
P :Subprogram number
L :Number of times the operation is repeated
I :Shift along the Xp axis
J :Shift along the Yp axis
K :Shift along the Zp axis
Specify a plane selection command (G17, G18, or G19) to select the plane on which the linear copying is made.

Explanation

- First block of the subprogram

Always specify a move command in the first block of a subprogram that performs a rotational or linear copying. If the first block contains only the program number such as O1234; and does not have a move command, movement may stop at the start point of the figure made by the n-th (n = 1, 2, 3, ...) copying.
Specify the first move command in the absolute mode.
(Example of an incorrect program)

```
O1234 ;
G00 G90 X100.0 Y200.0 ;
. . . . . ;
. . . . . ;
M99 ;
```

(Example of a correct program)

```
O1000 G00 G90 X100.0 Y200.0 ;
. . . . . ;
. . . . . ;
M99 ;
```

- Combination of rotational and linear copying

The linear copying command can be specified in a subprogram for a rotational copying. Also, the rotational copying command can be specified in a subprogram for a linear copying.

- Subprogram call

In a subprogram for rotational or linear copying, M98 for calling another subprogram or G65 for calling a macro can be specified.

- Specifying the center of rotation

The center of rotation specified with G72.1 is processed as an absolute position even in the incremental mode.

- Specifying address

In a block with G72.1, addresses other than P, L, Xp, Yp, Zp, or R are ignored. The subprogram number (P), coordinates of the center of rotation (Xp, Yp, Zp), and angular displacement (R) must be specified.
In a block with G72.2, addresses other than P, L, I, J, or K are ignored. The subprogram number (P) and shift (I, J, K) must be specified.

- Address P

If the subprogram number specified with P is not found, alarm PS0078 occurs. If P is not specified, alarm PS0076 occurs.

- Address L

If L is omitted, the repetition count is assumed to be 1 and the subprogram is called only once.

- Increment in angular displacement or shift

In a block with G72.1, an increment in angular displacement is specified with address R. The angular displacement of the figure made by the n-th rotation is calculated as follows : R × (n - 1).
In a block with G72.2, an increment in shift is specified with addresses I, J, and K. The shift of the figure made by the n-th movement is calculated as follows : (Programmed shift) × (n - 1).

- Nesting level of a subprogram

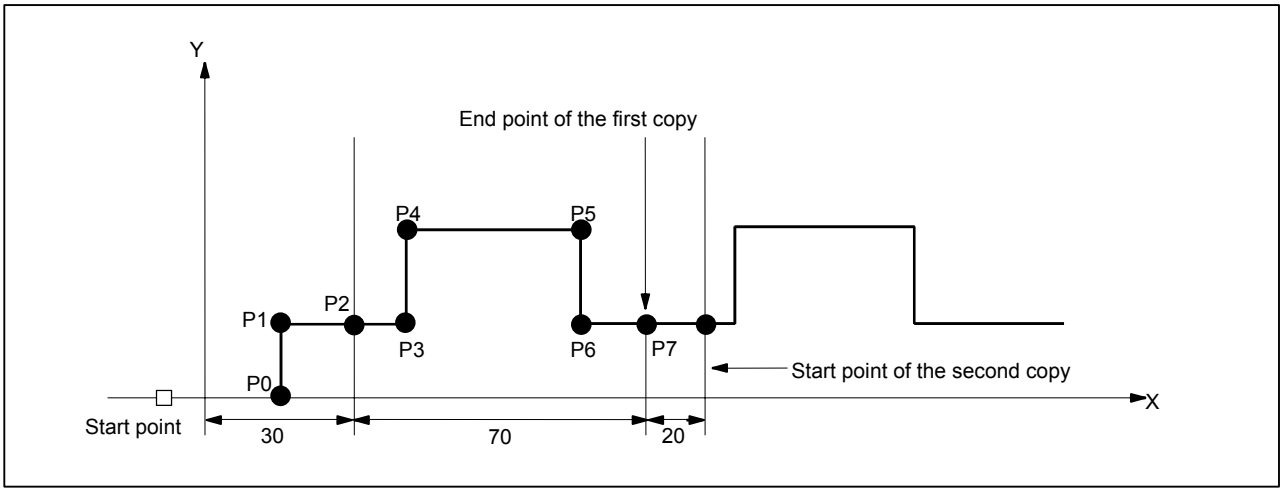
If a subprogram is called by G72.1 or G72.2, the nesting level is increased by one in the same manner as when M98 is specified.

- Block end position

The coordinates of a figure moved rotationally or linearly (block end position) can be read from #5001 and subsequent system variables of the custom macro of rotational or linear copying.

- Disagreement between end point and start point

If the end point of the figure made by the n-th copy does not agree with the start point of the figure to be made by the next (n + 1) copy, the figure is moved from the end point to the start point, then copying is started. (Generally, this disagreement occurs if an incorrect angular displacement or shift is specified.)




```
Main program
  O1000 ;
  N10 G92 X-20.0 Y0 ;
  N20 G00 G90 X0 Y0 ;
  N30 G01 G17 G41 X20. Y0 D01 F10 ;    (P0)
  N40 Y20. ;                           (P1)
  N50 X30. ;                           (P2)
  N60 G72.2 P2000 L3 I90. J0 ;
                    └─Although a shift of 70 mm was
                      required, I90.0 was specified instead
                      of I70.0. Since an incorrect shift was
                      specified, the end point of the figure
                      made by the n-th copy disagrees
                      with the start point of the figure to be
                      made by the next (n + 1) copy.

Subprogram
  O2000 G90 G01 X40. ;                 (P3)
  N100 Y40. ;                          (P4)
  N200 G01 X80. ;                      (P5)
  N300 G01 Y20. ;                      (P6)
  N400 X100. ;                         (P7)
  N500 M99;
```

Limitation

- Specifying two or more commands to copy a figure

G72.1 cannot be specified more than once in a subprogram for making a rotational copying (If this is attempted, alarm PS0160 will occur). G72.2 cannot be specified more than once in a subprogram for making a linear copying (If this is attempted, alarm PS0161 will occur).

- Commands that must not be specified

Within a program that performs a rotational or linear copying, the following must not be specified:

- Command for changing the selected plane (G17 to G19)
- Command for specifying polar coordinates
- Reference position return command
- Coordinate system rotation, scaling, programmable mirror image

The command for rotational or linear copying can be specified after a command for coordinate system rotation, scaling, or programmable mirror image is executed.

- Modes that must not be selected

The figure cannot be copied during chamfering, corner rounding, or tool offset.

- Unit system

The two axes of the plane for copying a figure must have an identical unit system.

- Single block

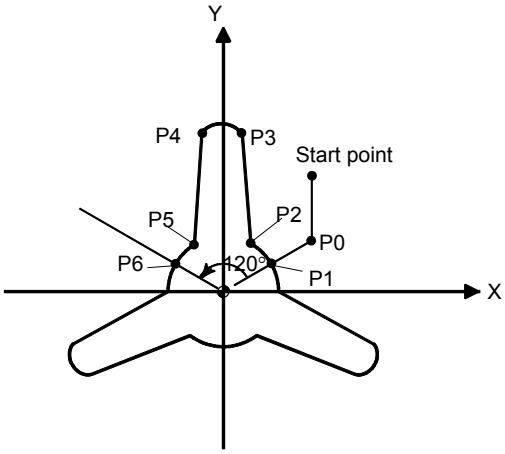
Single-block stops are not performed in a block with G72.1 or G72.2.

- Specifying tool radius compensation and the workpiece coordinate system

In a subprogram for copying a figure, the G code for tool radius / tool nose radius compensation or compensation amount (H or D code) cannot be changed. G92 and G54 to G59 cannot be changed either. Those codes must be specified before figure copying is started.

Example

- Rotational copying



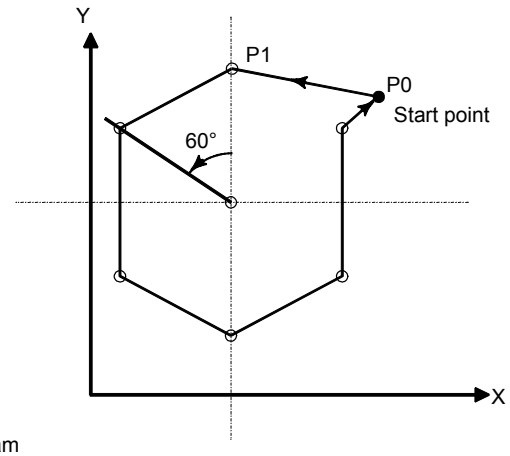
Main program

```
O1000 ;
N10 G92 X40.0 Y50.0 ;
N20 G00 G90 X_ Y_ ;                    (P0)
N30 G01 G17 G41 X_ Y_ D01 F10 ;        (P1)
N40 G72.1 P2000 L3 X0 Y0 R120.0 ;
N50 G40 G01 X_ Y_ I_ J_ ;              (P0)
N60 G00 X40.0 Y50.0 ;
N70 M30 ;
```

Sub program

```
O2000 G03 X_ Y_ R30.0 ;                (P2)
N100 G01 X_ Y_ ;                       (P3)
N200 G03 X_ Y_ R10.0 ;                 (P4)
N300 G01 X_ Y_ ;                       (P5)
N400 G03 X_ Y_ R30.0 ;                 (P6)
N500 M99;
```

- Rotational copying (spot boring)



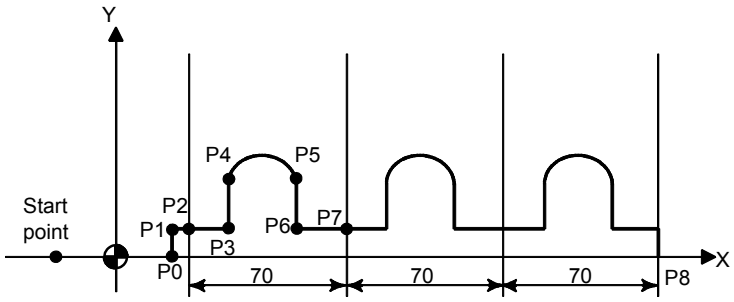
Main program

```
O3000 ;
N10 G92 G17 X80.0 Y50.0 ;                (P0)
N20 G72.1 P4000 L6 X0 Y0 R60.0 ;
N30 G80 G00 X80.0 Y50.0 ;                (P0)
N40 M30 ;
```

Subprogram

```
O4000 N100 G90 G81 X_ Y_ R_ Z_ F_ ;      (P1)
N200 M99 ;
```

- Linear copying



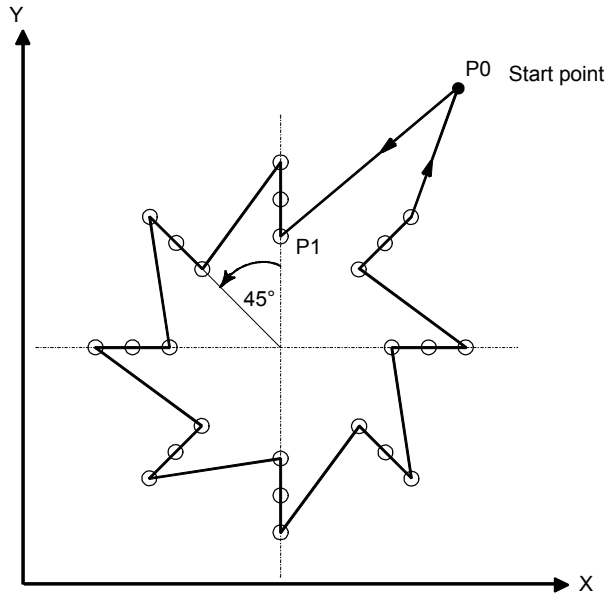
Main program

```
O1000 ;
N10 G92 X-20.0 Y0 ;
N20 G00 G90 X0 Y0 ;
N30 G01 G17 G41 X20. Y0 D01 F10 ;   (P0)
N40 Y20. ;                          (P1)
N50 X30. ;                          (P2)
N60 G72.2 P2000 L3 I70.0 J0 ;
N70 X_ Y_ ;                         (P8)
N80 X0 ;
N90 G00 G40 X-20.0 Y0 ;
N100 M30 ;
```

Subprogram

```
O2000 G90 G01 X_ ;                  (P3)
N100 Y_ ;                           (P4)
N200 G02 X_ I_ ;                    (P5)
N300 G01 Y_ ;                       (P6)
N400 X_ ;                           (P7)
N500 M99 ;
```

- Combination of rotational copying and linear copying (bolt hole circle)



Main program

```
O1000 ;
N10 G92 G17 X100.0 Y80.0 ;                (P0)
N20 G72.1 P2000 X0 Y0 L8 R45.0 ;
N30 G80 G00 X100.0 Y80.0 ;                (P0)
N40 M30 ;
```

Subprogram (rotational copy)

```
O2000 N100 G72.2 P3000 I0 J_ L3 ;
N200 M99 ;
```

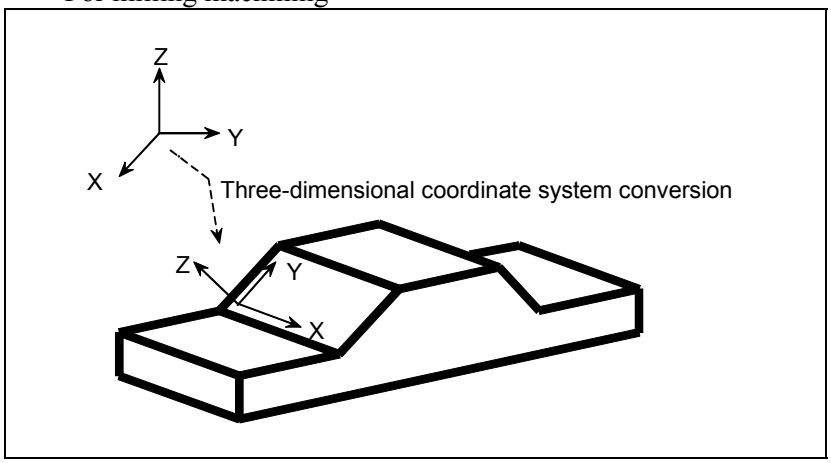
Subprogram (linear copy)

```
O3000 N110 G90 G81 X_ Y_ R_ Z_ F_ ;     (P1)
N210 M99 ;
```

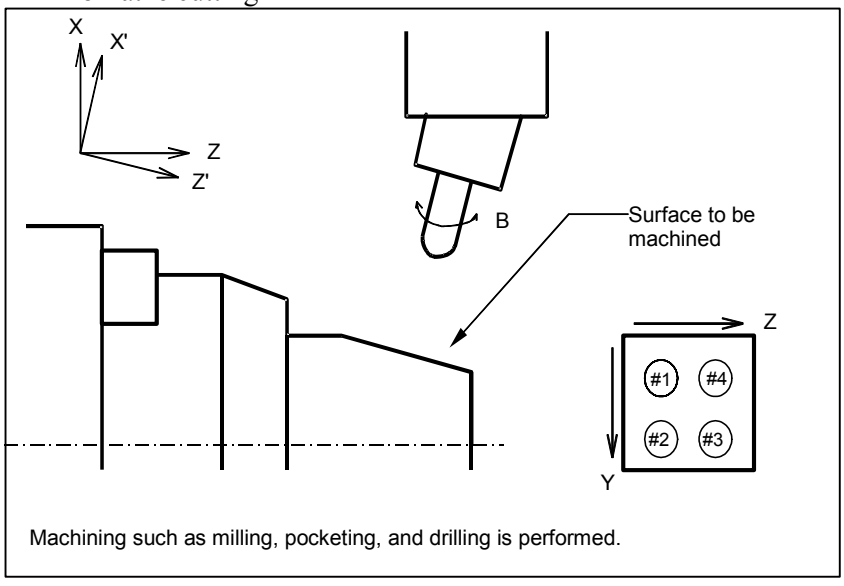
14.2 THREE-DIMENSIONAL COORDINATE SYSTEM CONVERSION

Coordinate system conversion about an axis can be carried out if the center of rotation, direction of the axis of rotation, and angular displacement are specified. This function is very useful in three-dimensional machining by a die-sinking machine or similar machine. For example, if a program specifying machining on the XY plane is converted by the three-dimensional coordinate system conversion function, the identical machining can be executed on a desired plane in three-dimensional space.

- For milling machining



- For lathe cutting



Format

M

G68 Xp_{X1} Yp_{y1} Zp_{z1} I_{i1} J_{j1} K_{k1} Rα ; Starting three-dimensional coordinate system conversion

:

: Three-dimensional coordinate system conversion mode

G69 ; Canceling three-dimensional coordinate system conversion

Xp, Yp, Zp : Center of rotation (absolute coordinates) on the X, Y, and Z axis or parallel axes

I, J, K : Direction of the axis of rotation

R : Angular displacement

T

G68.1 Xp_{X1} Yp_{y1} Zp_{z1} I_{i1} J_{j1} K_{k1} Rα ; Starting three-dimensional coordinate system conversion

:

: Three-dimensional coordinate system conversion mode

G69.1 ; Canceling three-dimensional coordinate system conversion

Xp, Yp, Zp : Center of rotation (absolute coordinates) on the X, Y, and Z axis or parallel axes

I, J, K : Direction of the axis of rotation

R : Angular displacement

NOTE
The G code of this function is hereinafter described using the format (G68/G69) for the machining center system in this section.

Explanation

- Command for three-dimensional coordinate system conversion (program coordinate system)

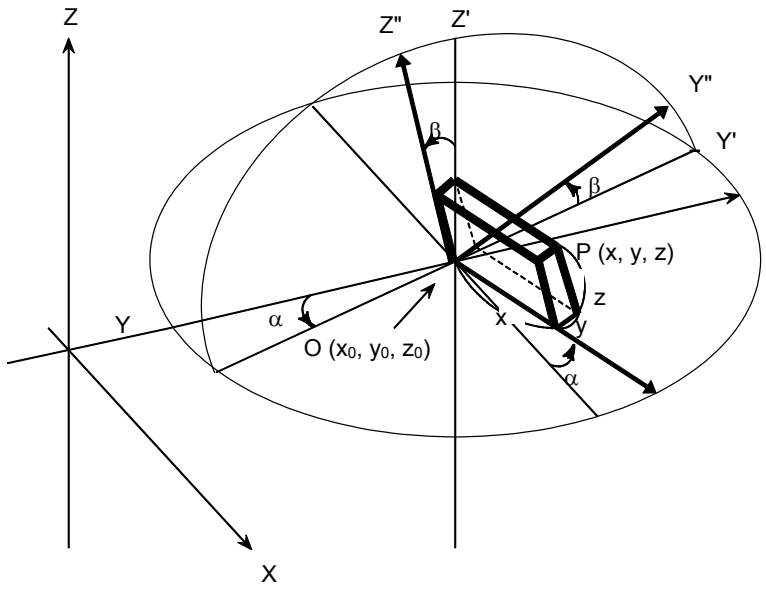
N1 G68 Xp x_1 Yp y_1 Zp z_1 I i_1 J j_1 K k_1 R α ;
N2 G68 Xp x_2 Yp y_2 Zp z_2 I i_2 J j_2 K k_2 R β ;
N3
:
Nn G69 ;

Three-dimensional coordinate system conversion can be executed twice.

In the N1 block, specify the center, direction of the axis of rotation, and angular displacement of the first rotation.

When this block is executed, the center of the original coordinate system is shifted to (x_1, y_1, z_1), then rotated around the vector (i_1, j_1, k_1) by angular displacement α. The new coordinate system is called X'Y'Z'. In the N2 block, specify the center, direction of the axis of rotation, and angular displacement of the second rotation. In the N2 block, specify coordinates and the angle with the coordinate system formed after the N1 block in Xp, Yp, Zp, I, J, K, and R. When the N2 block is executed, the X'Y'Z' coordinate system is shifted to (x_2, y_2, z_2), then rotated around the vector (i_2, j_2, k_2) by angular displacement β. The newest coordinate system is called X"Y"Z". In the subsequent N3 block, coordinates in the X"Y"Z" coordinate system are specified with Xp, Yp, and Zp. The X"Y"Z" coordinate system is called the program coordinate system.
If (Xp, Yp, Zp) is not specified in the N2 block, (Xp, Yp, Zp) in the N1 block is assumed to be the center of the second rotation (the N1 and N2 blocks have a common center of rotation). If the coordinate system is to be rotated only once, the N2 block need not be specified.

Example) G68 Xx_0 Yy_0 Zz_0 I0 J0 K1 Rα ;
G68 I1 J0 K0 Rβ ;



X, Y, Z	:	Workpiece coordinate system
X', Y', Z'	:	Coordinate system formed after the first conversion
X", Y", Z"	:	Coordinate system formed after the second conversion
α	:	Angular displacement of the first rotation
β	:	Angular displacement of the second rotation
O (x_0, y_0, z_0)	:	Center of rotation
P (x, y, z)	:	Coordinates in the X"Y"Z" coordinate system (program coordinate system)

- Format error

If one of the following format errors is detected, alarm PS5044 occurs:

1. When I, J, or K is not specified in a block with G68 (a parameter of coordinate system rotation is not specified)
2. When I, J, and K are all set to 0 in a block with G68
3. When R is not specified in a block with G68

- Center of rotation

Specify absolute coordinates with Xp, Yp, and Zp in the G68 block.

- Equation for three-dimensional coordinate system conversion

The following equation shows the general relationship between (x, y, z) in the program coordinate system and (X, Y, Z) in the original coordinate system (workpiece coordinate system).

$$\begin{pmatrix} X \\ Y \\ Z \end{pmatrix} = (M_1)\begin{pmatrix} x \\ y \\ z \end{pmatrix} + \begin{pmatrix} x_1 \\ y_1 \\ z_1 \end{pmatrix}$$

When conversion is carried out twice, the relationship is expressed as follows:

$$\begin{pmatrix} X \\ Y \\ Z \end{pmatrix} = (M_1)(M_2)\begin{pmatrix} x \\ y \\ z \end{pmatrix} + (M_1)\begin{pmatrix} x_2 \\ y_2 \\ z_2 \end{pmatrix} + \begin{pmatrix} x_1 \\ y_1 \\ z_1 \end{pmatrix}$$

X, Y, Z : Coordinates in the original coordinate system (workpiece coordinate system)
x, y, z : Programmed value (coordinates in the program coordinate system)
x_1, y_1, z_1 : Center of rotation of the first conversion
x_2, y_2, z_2 : Center of rotation of the second conversion (coordinates in the coordinate system formed after the first conversion)
M_1 : First conversion matrix
M_2 : Second conversion matrix

M_1 and M_2 are conversion matrices determined by an angular displacement and rotation axis. Generally, the matrices are expressed as shown below.

$$\begin{pmatrix} n_1^2 + (1-n_1^2)\cos\theta & n_1n_2(1-\cos\theta) - n_3\sin\theta & n_1n_3(1-\cos\theta) + n_2\sin\theta \\ n_1n_2(1-\cos\theta) + n_3\sin\theta & n_2^2 + (1-n_2^2)\cos\theta & n_2n_3(1-\cos\theta) - n_1\sin\theta \\ n_1n_3(1-\cos\theta) - n_2\sin\theta & n_2n_3(1-\cos\theta) + n_1\sin\theta & n_3^2 + (1-n_3^2)\cos\theta \end{pmatrix}$$

n_1 : Cosine of the angle made by the rotation axis and X-axis i/p
n_2 : Cosine of the angle made by the rotation axis and Y-axis j/p
n_3 : Cosine of the angle made by the rotation axis and Z-axis k/p
θ : Angular displacement
Value p is obtained by the following:

$$p = \sqrt{i^2 + j^2 + k^2}$$

Conversion matrices for rotation on two-dimensional planes are shown below:

(1) Coordinate system conversion on the XY plane

$$M = \begin{pmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

(2) Coordinate system conversion on the YZ plane

$$M = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta \\ 0 & \sin\theta & \cos\theta \end{pmatrix}$$

(3) Coordinate system conversion on the ZX plane

$$M = \begin{pmatrix} \cos\theta & 0 & \sin\theta \\ 0 & 1 & 0 \\ -\sin\theta & 0 & \cos\theta \end{pmatrix}$$

- Three basic axes and their parallel axes

Three-dimensional coordinate system conversion can be applied to a desired combination of three axes selected out of the basic three axes (X, Y, Z) and their parallel axes. The three-dimensional coordinate system subjected to three-dimensional coordinate system conversion is determined by axis addresses specified in the G68 block. If Xp, Yp, or Zp is not specified, X, Y, or Z of the basic three axes is assumed. However, if the basic three axes are not specified in parameter 1022, alarm PS0048 occurs.

In a single G68 block, both a basic axis and a parallel axis cannot be specified.

If this is attempted, alarm PS0047 occurs.

(Example)

When U-axis, V-axis, and W-axis are parallel to the X-axis, Y-axis, and Z-axis respectively

```
G68  X_ I_ J_ K_ R_ ;          XYZ coordinate system
G68  U_V_ Z_ I_ J_ K_ R_ ;     UVZ coordinate system
G68  W_ I_ J_ K_ R_ ;          XYW coordinate system
```

- Specifying the second conversion

Three-dimensional coordinate system conversion can be executed twice. The center of rotation of the second conversion must be specified with the axis addresses specified for the first conversion. If the axis addresses of the second conversion are different from the axis addresses of the first conversion, the different axis addresses are ignored. An attempt to execute three-dimensional coordinate system conversion three or more times causes alarm PS5043.

- Angular displacement R

A positive angular displacement R indicates a clockwise rotation along the axis of rotation. Specify angular displacement R in 0.001 degrees within the range of -360000 to 360000.
To specify angular displacement R in 0.00001 degrees (one hundred-thousandth), set bit 0 (FRD) of parameter No. 11630 to 1. In this case, angular displacement R is specified within the range of -36000000 to 36000000.

- G codes that can be specified

The following G codes can be specified in the three-dimensional coordinate system conversion mode:

G00 Positioning
G01 Linear interpolation
G02 Circular interpolation (clockwise)
G03 Circular interpolation (counterclockwise)
G04 Dwell
G10 Data setting
G17 Plane selection (XY)
G18 Plane selection (ZX)
G19 Plane selection (YZ)
G28 Reference position return
G29 Movement from the reference position
G30 Second, third, or fourth reference position return
G53 Selecting the machine coordinate system
G65 Custom macro call
G66 Custom macro modal call
G67 Custom macro modal call cancel
G40 Canceling tool radius / tool nose radius compensation
G41 Tool radius / tool nose radius compensation to the left
G42 Tool radius / tool nose radius compensation to the right
G73, G74, G76, G80 to G89 Canned cycle for drilling

M

G43 Increasing tool length compensation
G44 Decreasing tool length compensation
G45 Increasing the tool offset
G46 Decreasing the tool offset
G47 Doubling the tool offset
G48 Halving the tool offset
G49 Canceling tool length compensation
G50.1 Canceling programmable mirror image
G51.1 Programmable mirror image
G90 Absolute programming
G91 Incremental programming
G94 Feed per minute
G95 Feed per revolution
G98 Canned cycle (return to the initial level)
G99 Canned cycle (return to the level of point R)

T

G90 Absolute programming (when G code system B or C is used.)
G91 Incremental programming (when G code system B or C is used.)
G94 Feed per minute (when G code system B or C is used.)
G95 Feed per revolution(when G code system B or C is used.)
G98 Canned cycle (return to the initial level) (when G code system B or C is used.)
G99 Canned cycle (return to the level of point R) (when G code system B or C is used.)

- Rapid traverse rate in drilling of a canned cycle for drilling

In three-dimensional coordinate system conversion mode, rapid traverse rate in drilling by a canned cycle for drilling equals the cutting feedrate specified in parameter No. 5412. If the parameter is set to 0, the rapid traverse rate equals the maximum cutting feedrate.

- Compensation functions

M

If tool length compensation, tool radius / tool nose radius compensation, or tool offset is specified with three-dimensional coordinate system conversion, compensation is performed first, followed by three-dimensional coordinate system conversion.

T

If tool radius / tool nose radius compensation is specified with three-dimensional coordinate system conversion, compensation is performed first, followed by three-dimensional coordinate system conversion.

- Relationship between three-dimensional and two-dimensional coordinate system conversion

Three-dimensional and two-dimensional coordinate system conversion use identical G codes (G68 and G69). A G code specified with I, J, and K is processed as the command for three-dimensional coordinate system conversion. A G code not specified with I, J, and K is processed as the command for two-dimensional coordinate system conversion.

- Custom macro system variables

Coordinates on the workpiece coordinate system are assigned to system variables #5041 to #5048 (current position on each axis).

- Reset

If a reset occurs during three-dimensional coordinate system conversion mode, the mode is canceled and the continuous-state G code is changed to G69.
Bit 2 (D3R) of parameter No. 5400 determines whether just the G69.1 code is used to cancel the three-dimensional coordinate system conversion mode (G68.1). When this setting is selected, a CNC reset by a reset operation or by an input signal from the PMC will not cancel the three-dimensional coordinate system conversion mode.

- Absolute position display

The absolute coordinates based on the program or workpiece coordinate system can be displayed in the three-dimensional coordinate system conversion mode. Specify a desired coordinate system in bit 6 (DAK) of parameter No. 3106.

- Three-dimensional rigid tapping

By specifying the rigid tapping command in three-dimensional coordinate system conversion mode, tapping can be executed in the direction of the angle programmed by the three-dimensional coordinate system conversion command.
In three-dimensional coordinate system conversion mode, "Position Error Z", displayed on the spindle adjustment screen, is taken from the longitudinal tapping axis after three-dimensional conversion.
Positioning in three-dimensional coordinate system conversion mode must be linear interpolation positioning (bit 1 (LRP) of parameter No.1401 is set to 1).

Limitation

- Manual intervention

Three-dimensional coordinate system conversion does not affect the degree of manual intervention or manual handle interrupt.

- Positioning in the machine coordinate system

Three-dimensional coordinate system conversion does not affect positioning in the machine coordinate system (e.g. specified with G28, G30, or G53).

- Specifying rapid traverse

Specify linear rapid traverse when three-dimensional coordinate system conversion is executed. (Set bit 1 (LRP) of parameter No.1401 to 1.)

- Block with G68 or G69

In a block with G68 or G69, other G codes must not be specified. G68 must be specified with I, J, and K.

- Position display and compensation

To display the absolute position when three-dimensional coordinate system conversion is executed, set bits 4 to 7 (DRL, DRC, DAL, and DAC) of parameters No. 3104 to 0.

- Mirror image

M

Programmable mirror image can be specified, but external mirror image (mirror image by the mirror image signal or setting) cannot be specified. Three-dimensional coordinate system conversion is carried out after the programmable mirror image function is executed.

T

External mirror image (mirror image by the mirror image signal or setting) cannot be specified.

- Three-dimensional coordinate system conversion and other continuous-state commands

M

Canned cycles, G41, G42, or G51.1 must be nested between G68 and G69.
(Example)

```
G68 X100. Y100. Z100. I0. J0. K1. R45. ;
  :
G41 D01 ;
  :
G40 ;
  :
G69 ;
```

T

Canned cycles, G41 or G42 must be nested between G68.1 and G69.1.
(Example)

```
G68.1 X100. Y100. Z100. I0. J0. K1. R45. ;
  :
G41 X_ Z_ I_ K_ ;
  :
G40 ;
  :
G69.1 ;
```

T

- Relationship between three-dimensional coordinate system conversion and tool offset

When using a tool offset command, nest the tool offset command within the three-dimensional coordinate system conversion mode.
(Example)

```
G68.1 X100. Y100. Z100. I0. J0. K1. R45. ;
  :
T0101;
  :
T0100;
  :
G69.1 ;
```

- PMC axis control

In the three-dimensional coordinate system conversion mode, PMC axis control cannot be performed for the three axes related to the conversion (alarm).

- Manual operation

When manual feeding is performed during a three-dimensional coordinate system conversion, the tangent velocity in the coordinate system after conversion (program coordinate system) equals the lowest feedrate of those on the selected axes.

- Workpiece coordinate system

Avoid changing the workpiece coordinate system in the three-dimensional coordinate system conversion mode.

- Manual reference position return

Avoid making a manual reference position return in the three-dimensional coordinate system conversion mode.

- Cs contour axis

When specifying the Cs contour axis and rapid traverse at the same time in the three-dimensional coordinate system conversion mode, make a reference position return on the Cs contour axis in advance. If a reference position return is made in the first rapid traverse after the Cs contour axis is selected (bit 1 (NRF) of parameter No.3700 is set to 0), avoid specifying the reference position return command in the three-dimensional coordinate system conversion mode.

Example

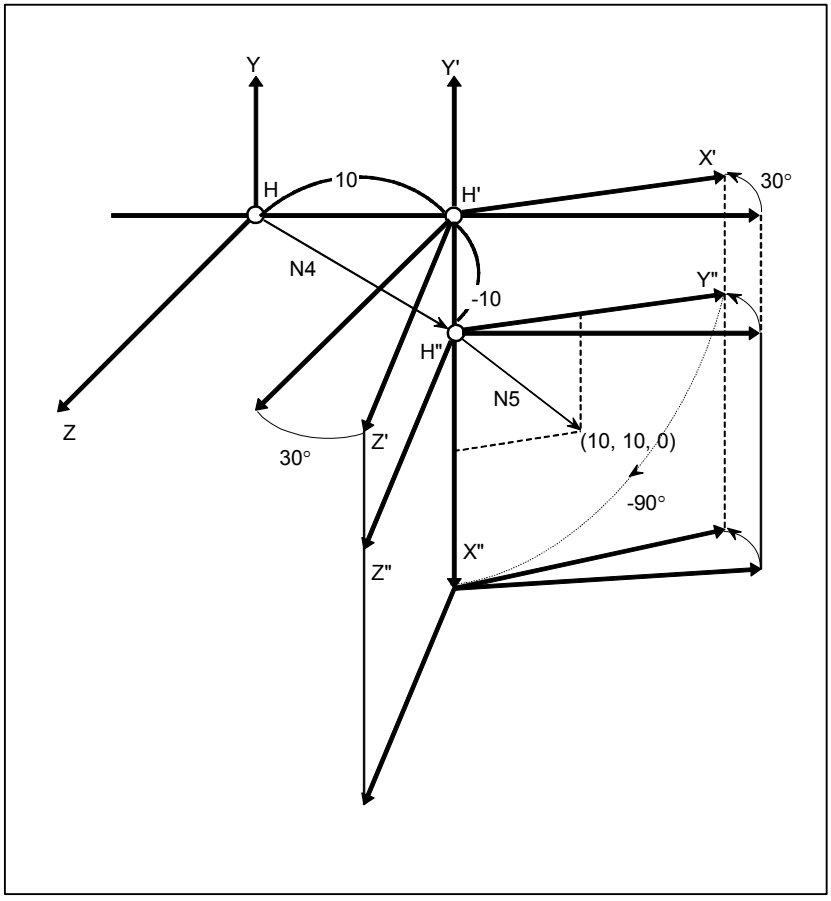
N1 G90 X0 Y0 Z0 ; Carries out positioning to zero point H.

N2 G68 X10. Y0 Z0 I0 J1 K0 R30. ; Forms new coordinate system X'Y'Z'.

N3 G68 X0 Y-10. Z0 I0 J0 K1 R-90. ; Forms other coordinate system X"Y"Z".
The origin agrees with (0, -10, 0) in coordinate system X'Y'Z.

N4 G90 X0 Y0 Z0 ; Carries out positioning to zero point H" on coordinate system X"Y"Z".

N5 X10. Y10. Z0 ; Carries out positioning to (10, 10, 0) on coordinate system X"Y"Z".



15 COMPENSATION FUNCTION

Chapter 15, "COMPENSATION FUNCTION", consists of the following sections:

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15.2 SCALING (G50, G51) 379
15.3 PROGRAMMABLE MIRROR IMAGE (G50.1, G51.1) 389
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15.5 WORKPIECE SETTING ERROR COMPENSATION.....397

15.1 TOOL LENGTH COMPENSATION (G43, G44, G49)

This function can be used by setting the difference between the tool length assumed during programming and the actual tool length of the tool used into the offset memory. It is possible to compensate the difference without changing the program.
Specify the direction of offset with G43 or G44. Select a tool length compensation value from the offset memory by entering the corresponding address and number (H code).

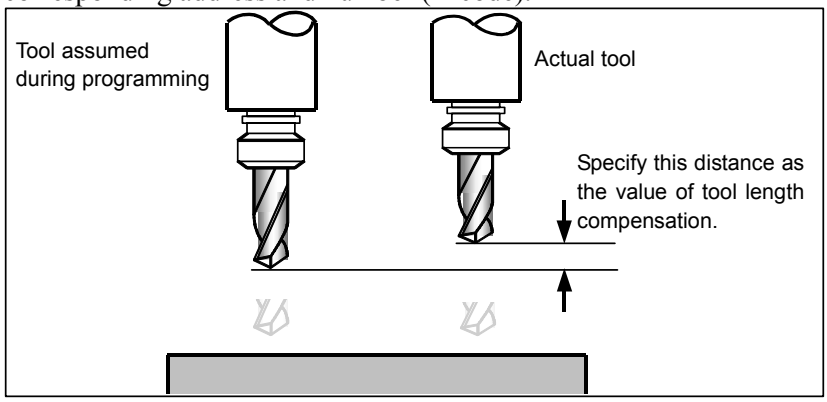


Fig. 15.1 (a) Tool length compensation

15.1.1 Overview

The following three methods of tool length compensation can be used, depending on the axis along which tool length compensation can be made.

- Tool length compensation A
  Compensates for the difference in tool length along the basic Z-axis.
- Tool length compensation B
  Compensates for the difference in tool length in the direction normal to a selected plane.
- Tool length compensation C
  Compensates for the difference in tool length along a specified axis.

Format

Type	Format	Description
Tool length compensation A	**G43 Z_ H_ ;** **G44 Z_ H_ ;**	G43 : Positive offset G44 : Negative offset G17 : XY plane selection G18 : ZX plane selection G19 : YZ plane selection α : Address of a specified axis H : Address for specifying the tool length compensation value X, Y, Z : Offset move command
Tool length compensation B	**G17 G43 Z_ H_ ;** **G17 G44 Z_ H_ ;** **G18 G43 Y_ H_ ;** **G18 G44 Y_ H_ ;** **G19 G43 X_ H_ ;** **G19 G44 X_ H_ ;**	
Tool length compensation C	**G43 α_H_ ;** **G44 α_H_ ;**	
Tool length compensation cancel	**G49 ; or H0 ;**	

Explanation

- Selection of tool length compensation

Select tool length compensation A, B, or C, by setting bits 0 (TLC) and 1 (TLB) of parameter No.5001 .

Parameter No.5001		Type
Bit 1 (TLB)	Bit 0 (TLC)	
0	0	Tool length compensation A
1	0	Tool length compensation B
0/1	1	Tool length compensation C

- Direction of the offset

When G43 is specified, the tool length compensation value (stored in offset memory) specified with the H code is added to the coordinates of the end position specified by a command in the program. When G44 is specified, the same value is subtracted from the coordinates of the end position. The resulting coordinates indicate the end position after compensation, regardless of whether the absolute or incremental mode is selected.
When the specification of an axis is omitted, a movement is made by the tool length compensation value.
G43 and G44 are modal G codes. They are valid until another G code belonging to the same group is used.

- Specification of the tool length compensation value

The tool length compensation value assigned to the number (offset number) specified in the H code is selected from offset memory and added to or subtracted from the moving command in the program.

Example

```
 :
H1 ;        The offset value of offset number 1 is
            selected.
 :
G43 Z_ ;    Offset is applied according to the offset
            value of offset number 1.
 :
H2 ;        Offset is applied according to the offset
            value of offset number 2.
 :
H0 ;        Offset is applied according to the offset
            value 0.
 :
H3 ;        Offset is applied according to the offset
            value of offset number 3.
 :
G49 ;       Offset is canceled.
 :
H4 ;        The offset value of offset number 4 is
            selected.
 :
```

A tool length compensation value is to be set in the offset memory corresponding to an offset number.

⚠ WARNING

When another offset number is specified, the tool length compensation value just changes to a new value. The new tool length compensation value is not added to the old tool length compensation value.

```
H1 : Tool length compensation value 20.0
H2 : Tool length compensation value 30.0
G90 G43 Z100.0 H1 ; Z will move to 120.0
G90 G43 Z100.0 H2 ; Z will move to 130.0
```

NOTE

The tool length compensation value corresponding to offset No. 0, that is, H0 always means 0. It is impossible to set any other tool length compensation value to H0.

- Performing tool length compensation along two or more axes

Tool length compensation B can be executed along two or more axes when the axes are specified in two or more blocks.
By setting bit 3 (TAL) of parameter No. 5001 to 1, tool length compensation C can also can be executed along two or more axes when the axes are specified in two or more blocks. If no axis is specified in the same block, the alarm PS0027 is issued. If two or more axes are specified in the same block, the alarm PS0336 is issued.

Example 1
When tool length compensation B is executed along the X-axis and Y-axis
G19 G43 H_ ; Offset in X axis
G18 G43 H_ ; Offset in Y axis

Example 2
When tool length compensation C is executed along the X-axis and Y-axis
G43 X_ H_ ; Offset in X axis
G43 Y_ H_ ; Offset in Y axis

Example 3
When an alarm is issued with tool length compensation C
G43 X_ Y_ H_ ; An alarm (PS0336) occurs

- Tool length compensation cancel

To cancel tool length compensation, specify G49 or H0. After G49 or H0 is specified, the system immediately cancels the offset mode.

NOTE

1 If offset is executed along two or more axes, offset along all axes is canceled by specifying G49. If H0 is used to specify cancellation, offset along only the axis normal to a selected plane is canceled in the case of tool length compensation B, or offset along only the last axis specified by G43 or G44 is canceled in the case of tool length compensation C.

2 If offset is executed along three or more axes, and offset along all axes is canceled using G49, the alarm PS0015 (TOO MANY SIMULTANEOUS AXES) may be issued. By using H0 together, for example, cancel offset so that the number of simultaneously controlled axes (the number of axes along which movements are made simultaneously) does not exceed the allowable range of the system.

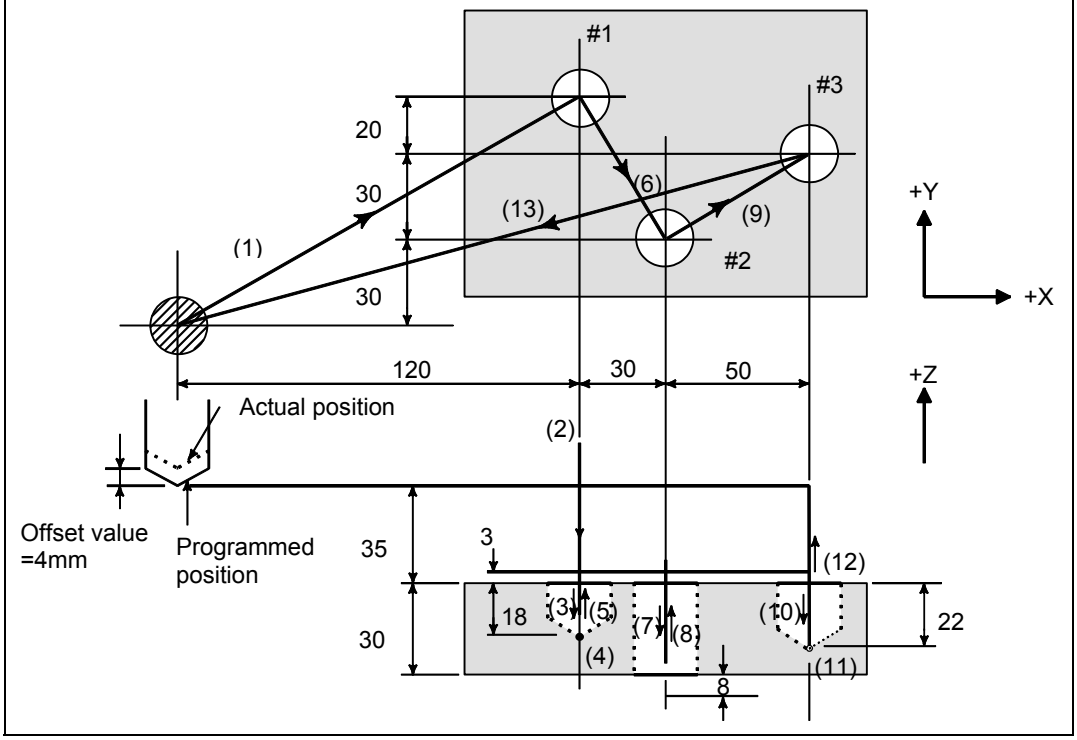
3 If tool length compensation and three-dimensional coordinate system conversion are canceled by a reset with three-dimensional coordinate system conversion performed during tool length compensation, the direction of tool length compensation vector cancellation becomes incorrect. By setting bit 6 (LVK) of parameter No. 5003 to 1 and setting bit 2 (D3R) of parameter No. 5400 to 1, ensure that the tool length compensation vector and three-dimensional coordinate system conversion are not canceled by a reset.

Example)

```
G43 H1 ;
G68 X_ Y_ Z_ I_ J_ K_ R_ ;
  :     }
  :     } Ensure that tool length compensation and three-dimensional
  :     } coordinate conversion are not canceled by a reset in this range.
G69 ;
G49 ;
```

Example

Tool length compensation (in boring holes #1, #2, and #3)



Program

H1=-4.0 (Tool length compensation value)

N1 G91 G00 X120.0 Y80.0 ;.....(1)
N2 G43 Z-32.0 H1 ; .....(2)
N3 G01 Z-21.0 F1000 ; .....(3)
N4 G04 P2000 ;.....(4)
N5 G00 Z21.0 ;.....(5)
N6 X30.0 Y-50.0 ;.....(6)
N7 G01 Z-41.0 ; .....(7)
N8 G00 Z41.0 ;.....(8)
N9 X50.0 Y30.0 ; .....(9)
N10 G01 Z-25.0 ; .....(10)
N11 G04 P2000 ; .....(11)
N12 G00 Z57.0 H0 ; .....(12)
N13 X-200.0 Y-60.0 ;.....(13)
N14 M2 ;

Notes

- Command for setting a workpiece coordinate system in the tool length compensation mode

Executing a workpiece coordinate system setting G code command (G92 or, for G code system A in the lathe system, G50) presets a coordinate system in such a way that the specified position will be a pre-compensation position.

However, this G code cannot be used together with a block where tool length compensation vectors vary. For details, see “Notes” in Subsection 7.2.1, “Setting a Workpiece Coordinate System.”

15.1.2 G53, G28, G30, and G30.1 Commands in Tool Length Compensation Mode

This section describes the tool length compensation cancellation and restoration performed when G53, G28, G30, or G31 is specified in tool length compensation mode. Also described is the timing of tool length compensation.

Explanation

- Tool length compensation vector cancellation

When G53, G28, G30, or G30.1 is specified in tool length compensation mode, tool length compensation vectors are canceled as described below. However, the previously specified modal G code remains displayed; modal code display is not switched to G49.

(1) When G53 is specified

Command	Specified axis	Operation
G53 IP_	Tool length compensation axis	Canceled upon movement being performed
	Other than tool length compensation axis	Not canceled
G49 G53 IP_	Tool length compensation axis	Canceled upon movement being performed
	Other than tool length compensation axis	Canceled

(IP_ : Dimension word)

> ⚠ **CAUTION**
> If tool length compensation is applied along multiple axes, the offset vector along the axis specified by G53 is canceled.

(2) When G28, G30, or G30.1 is specified

Command	Specified axis	Operation
G28 IP_	Tool length compensation axis	Not canceled at an intermediate point. Canceled at the reference position.
	Other than tool length compensation axis	Not canceled at an intermediate point. Canceled at the reference position.
G49 G28 IP_	Tool length compensation axis	Canceled when a movement is made to an intermediate point.
	Other than tool length compensation axis	Canceled when a movement is made to an intermediate point.

(IP_ : Dimension word)

> **⚠ CAUTION**
> If tool length compensation is applied along multiple axes, the offset vector along the axis on which a reference position return operation has been performed is canceled.

- Tool length compensation vector restoration

Tool length compensation vectors, canceled by specifying G53, G28, G30, or G30.1 in tool length compensation mode, are restored as described below.

Type	Parameter EVO (No.5001#6)	Restoration condition
A/B	0	The H command or G43 (G44) is specified.
	1	Restored by the next buffered block.
C		The H command or G43 (G44)IP_ is specified.

(IP_ : Dimension word)

> **⚠ CAUTION**
> If a tool length compensation vector is restored only with H_, G43, or G44 when tool length compensation is applied along multiple axes, the tool length compensation vector along only the axis normal to a selected plane is restored in the case of tool length compensation B, or the tool length compensation vector along only the last axis for which tool length compensation is specified is restored in the case of tool length compensation C. The tool length compensation vector along any other axes is not restored.

15.2 SCALING (G50, G51)

Overview

A programmed figure can be magnified or reduced (scaling).
Two types of scaling are available, one in which the same magnification rate is applied to each axis and the other in which different magnification rates are applied to different axes.
The magnification rate can be specified in the program.
Unless specified in the program, the magnification rate specified in the parameter is applied.

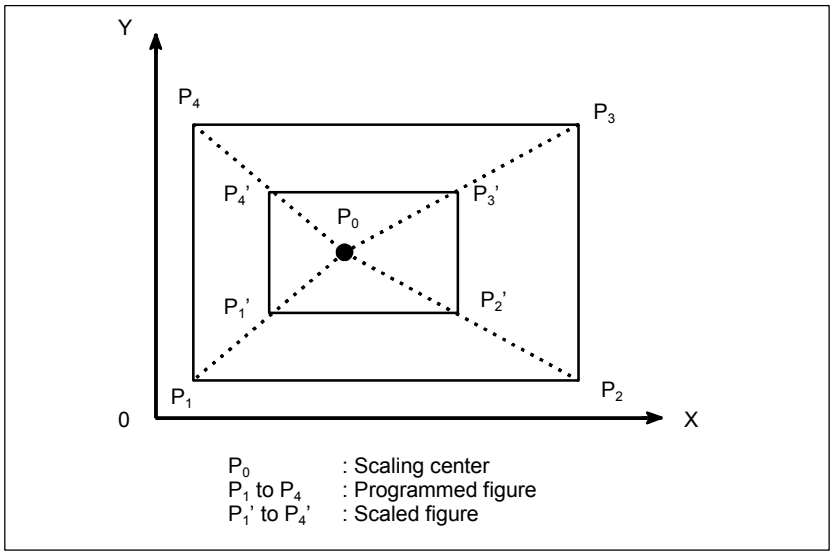


Fig. 15.2 (a) Scaling

Format

Scaling up or down along all axes at the same rate of magnification (When parameter XSC (No. 5400#6) = 0)

Format	Meaning of command
G51 IP_P_ ; Scaling start } Scaling is effective. (Scaling mode) **G50 ;** Scaling cancel	**IP_** : Absolute command for center coordinate value of scaling **P_** : Scaling magnification

Scaling up or down along each axes at a different rate of magnification (mirror image) (When parameter XSC (No. 5400#6) = 1)

Format	Meaning of command
G51 IP_I_J_K_; Scaling start } Scaling is effective. (Scaling mode) **G50 ;** Scaling cancel	**IP_** : Absolute command for center coordinate value of scaling **I_J_K_** : Scaling magnification for basic 3 axes (X, Y, and Z axes) respectively

T

NOTE
This function is available when the G-code system B or C is set.

⚠ CAUTION
1 Specify G51 in a separate block.
2 After the figure is enlarged or reduced, specify G50 to cancel the scaling mode.

NOTE
1 Entering electronic calculator decimal point input mode (bit 0 (DPI) of parameter No. 3401 = 1) does not cause the units of the magnification rates P, I, J, and K to change.
2 Setting the least input increment equal to 10 times the least command increment (bit 7 (IPR) of parameter No. 1004 = 1) does not cause the units of the magnification rates P, I, J, and K to change.
3 An attempt to specify 0 as a magnification rate causes alarm PS0142 to occur in a G51 block.

Explanation

- Axis for which scaling is to be enabled

For the axis for which scaling is to be enabled, set bit 0 (SCL) of parameter No. 5401 to 1.

- Minimum unit of scaling magnification

Least input increment of scaling magnification is: 0.001 or 0.00001.
It is 0.00001 (one hundred thousandth) if bit 7 (SCR) of parameter No. 5400 is 0 and 0.001 if it is 1.

- Scaling center

Even in incremental command (G91) mode, the scaling center coordinates IP_ specified in the G51 block are assumed those of an absolute position.
If the scaling center coordinates are omitted, the position assumed when G51 is specified is assumed the scaling center.

> ⚠ **CAUTION**
> With the move command subsequent to the G51 block, execute an absolute (G90 mode) position command.
> If no absolute position command is executed after the G51 block, the position assumed when G51 is specified is assumed the scaling center; once an absolute position command is executed, the scaling center assumes the coordinates specified in the G51 block, after that block.

- Scaling along each axis at the same rate of magnification

Set bit 6 (XSC) of parameter No. 5400 to 0.
If the scaling magnification P is not specified, the magnification set in parameter No. 5411 is used.
Decimal point input is not accepted as the magnification P. If decimal point input is made, alarm PS0007 will occur.
A negative value cannot be specified as the magnification P. If a negative value is specified, alarm PS0006 will occur.
The allowable magnification range is from 0.00001 to 9999.99999.

- Scaling of each axis, programmable mirror image (negative magnification)

Each axis can be scaled by different magnifications. Also when a negative magnification is specified, a mirror image is applied. The axis subject to the mirror image is the one that contains the scaling center.
Set bit 6 (XSC) of parameter No. 5400to 1 to validate each axis scaling (mirror image).
Using I, J, and K, specify the scaling magnifications for the basic 3 axes (X to Z axes). Use parameter No. 1022 to specify which axes to use as the basic 3 axes. For those of the X to Z axes for which I, J, and K are not specified and for axes other than the basic 3 axes, the magnification set with parameter No. 5421 is used.
A value other than 0 must be set to parameter No. 5421.
Decimal point programming can not be used to specify the rate of magnification (I, J, K).
Magnification can be set within the range of ±0.00001 to ±9999.99999.

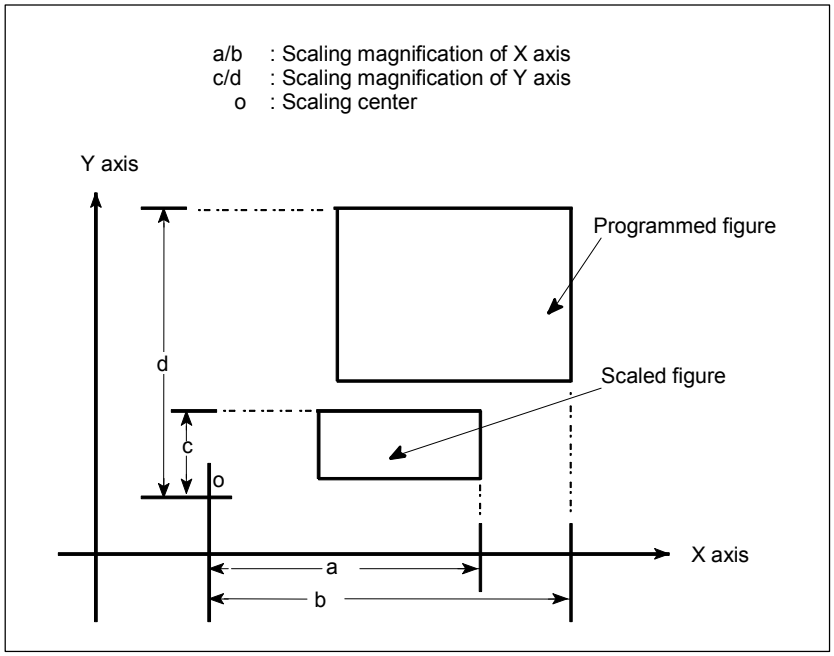


Fig. 15.2 (b) Scaling of each axis

⚠ CAUTION

Specifying the following commands at the same time causes them to be executed in the order indicated below:

<1> Programmable mirror image (G51.1)
<2> Scaling (G51) (including a mirror image with a negative magnification)
<3> Mirror image due to the external switch of the CNC or the settings of the CNC

In this case, the programmable mirror image is effective to the scaling center and magnification as well.

To specify G51.1 and G51 at the same time, specify them in this order; to cancel them, specify them in the reverse order.

- Scaling of circular interpolation

Even if different magnifications are applied to each axis in circular interpolation, the tool will not trace an ellipse.

```
G90 G00 X0.0 Y100.0 Z0.0;
G51 X0.0 Y0.0 Z0.0 I2000 J1000;
(A magnification of 2 is applied to the X-component and a magnification of 1 is applied to the Y-component.)
G02 X100.0 Y0.0  I0 J–100.0 F500;
```

Above commands are equivalent to the following command:

```
G90 G00 X0.0 Y100.0 Z0.0;
G02 X200.0 Y0.0 I0 J–100.0 F500;
(Because the end point is not on an arc, spiral interpolation is assumed.)
```

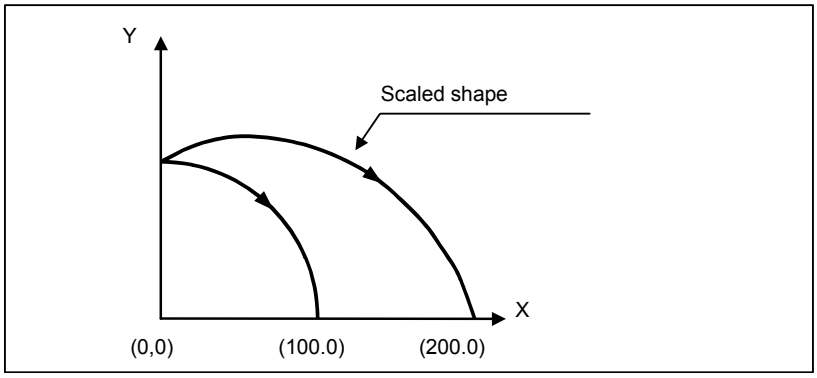


Fig. 15.2 (c) Scaling for circular interpolation1

Even for an R-specified arc, scaling is applied to each of I, J, and K after the radius value (R) is converted into a vector in the center direction of each axis.

If, therefore, the above G02 block contains the following R-specified arc, the operation will be same as that in which I and J are specified.

```
G02 X100.0 Y0.0 R100.0 F500 ;
```

- Scaling and coordinate system rotation

If both scaling and coordinate system rotation are specified at the same time, scaling is performed first, followed by coordinate system rotation. In this case, scaling is effective to the rotation center as well.
To specify both of them, specify scaling first and then coordinate system rotation. To cancel them, specify them in the reverse order.

Example
Main program

```
O1
G90 G00 X20.0 Y10.0 ;
M98 P1000 ;
G51 X20.0 Y10.0 I3000 J2000 ; (x 3 in the X direction and x 2
                               in the Y direction)
M98 P1000 ;
G17 G68 X35.0 Y20.0 R30. ;
M98 P1000 ;
G69 ;
G50 ;
M30 ;
```

Subprogram

```
O1000 ;
G01 X20.0 Y10.0 F500 ;
G01 X50.0 ;
G01 Y30.0 ;
G01 X20.0 ;
G01 Y10.0 ;
M99 ;
```

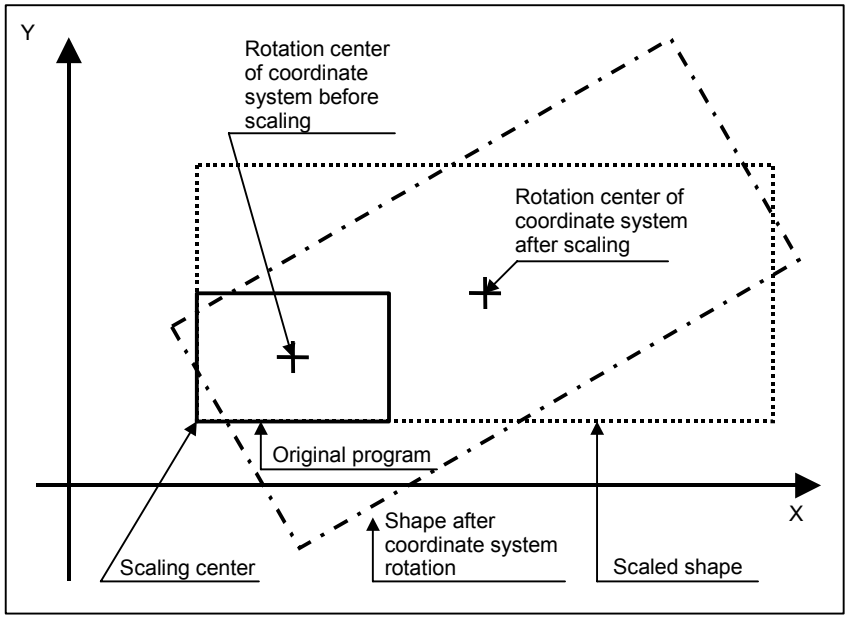


Fig. 15.2 (d) Scaling and coordinate system rotation

- Scaling and optional chamfering/corner R

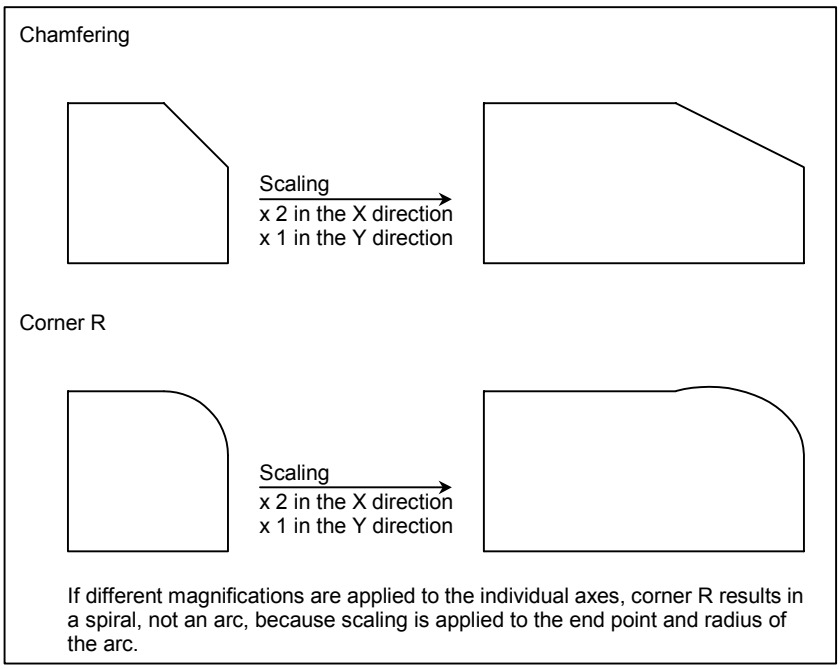


Fig. 15.2 (e) Scaling and optional chamfering/corner R

Limitation

- Tool compensation

This scaling is not applicable to tool radius · tool nose radius compensation values, tool length compensation values, and tool offset values (Fig. 15.2 (f)).

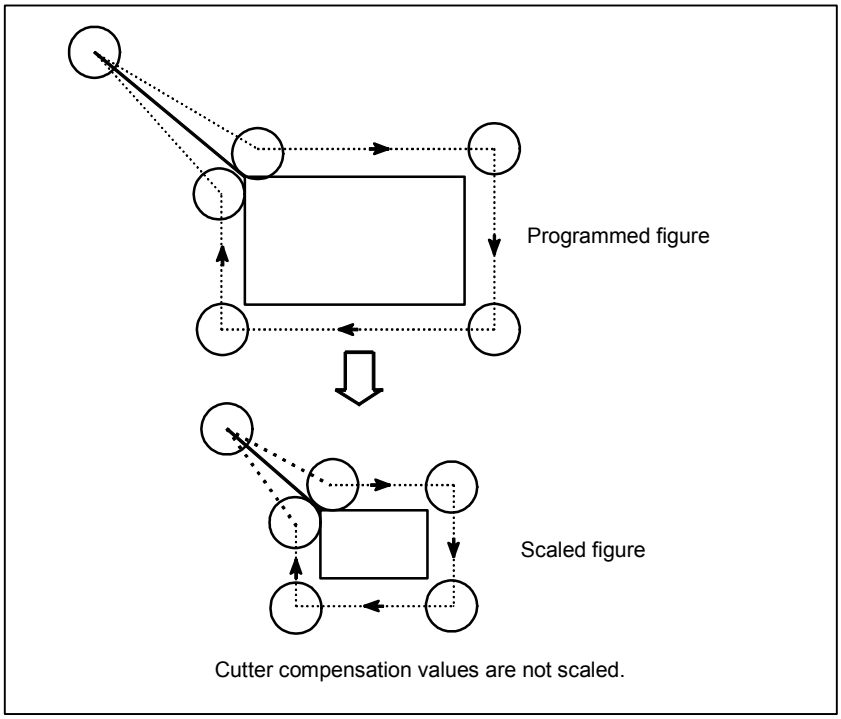


Fig. 15.2 (f) Scaling during cutter compensation

- Invalid scaling

M

Scaling is not applied to the travel distance during canned cycle shown below.

- Cut-in value Q and retraction value d of peck drilling cycle (G83, G73).
- Fine boring cycle (G76)
- Shift value Q of X and Y axes in back boring cycle (G87).

In manual operation, the travel distance cannot be increased or decreased using the scaling.

T

This function is available to G code systems B and C only; it is not available to G code system A.
During scaling, the following functions cannot be used. If any of them is specified, alarm PS0300 will occur.

- Finishing cycle (G70, G72)
- Outer surface rough cutting cycle (G71, G73)
- End rough cutting cycle (G72, G74)
- Closed loop cutting cycle (G73, G75)
- End cutting off cycle (G74, G76)
- Outer/inner surface cutting off cycle (G75, G77)
- Multiple threading cycle (G76, G78)
- Traverse grinding cycle (for grinding machines) (G71, G72)
- Traverse direct constant-size grinding cycle (for grinding machines) (G72, G73)
- Oscillation grinding cycle (for grinding machines) (G73, G74)
- Oscillation direct constant-size grinding cycle (for grinding machines) (G74, G75)
- Face drilling cycle (G83, G83)
- Face tapping cycle (G84, G84)
- Face boring cycle (G85, G85)
- Side drilling cycle (G87, G87)
- Side tapping cycle (G88, G88)
- Side boring cycle (G89, G89)
- Outer/inner surface turning cycle (G77, G20)
- Threading cycle (G78, G21)
- End turning cycle (G79, G24)

(The G code in system B is given first, followed by that of system C.)

In manual operation, the travel distance cannot be increased or decreased using the scaling.

⚠ CAUTION

1 If a parameter setting value is employed as a scaling magnification without specifying P, the setting value at G51 command time is employed as the scaling magnification, and a change of this value, if any, is not effective.
2 Before specifying the G code for reference position return (G27, G28, G29, G30, etc.) or coordinate system setting (G52 to G59, G92, etc.), cancel the scaling mode.
3 If scaling results are rounded by counting fractions of 5 and over as a unit and disregarding the rest, the move amount may become zero. In this case, the block is regarded as a no movement block, and therefore, it may affect the tool movement by cutter compensation. See the description of cutter compensation.
4 Refrain from scaling on a rotation axis for which the rollover function is enabled. Otherwise, the tool may rotate in a short-cut manner, possibly resulting in unexpected movement.

NOTE

1 The position display represents the coordinate value after scaling.
2 When a mirror image was applied to one axis of the specified plane, the following results:
(1) Circular command
..... Direction of rotation is reversed.
(2) Tool radius · tool nose radius compensation
..... Offset direction is reversed.
(3) Coordinate system rotation
..... Rotation angle is reversed.

Example

Sample program of a scaling in each axis

```
O1;
G51 X20.0 Y10.0 I750 J250; (× 0.75 in the X direction, × 0.25 in
                            the Y direction)
G00 G90 X60.0 Y50.0;
G01 X120.0 F100;
G01 Y90;
G01 X60;
G01 Y50;
G50;
M30;
```

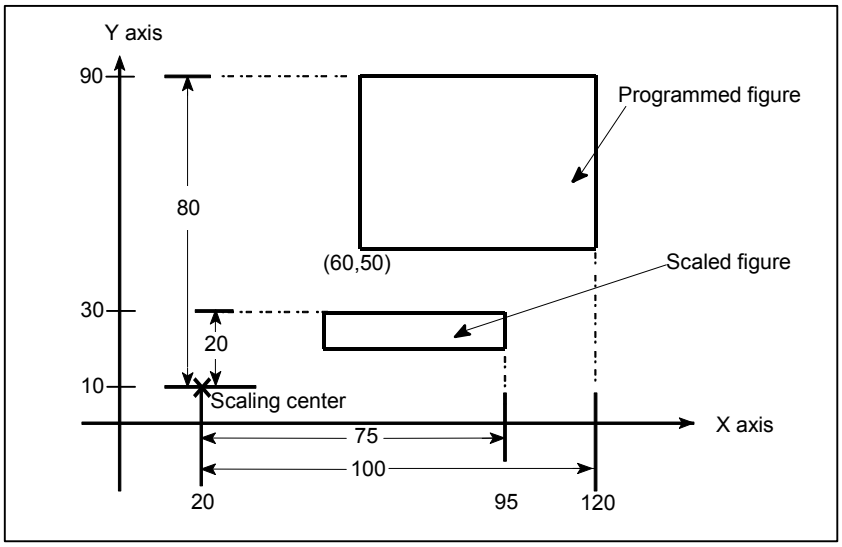
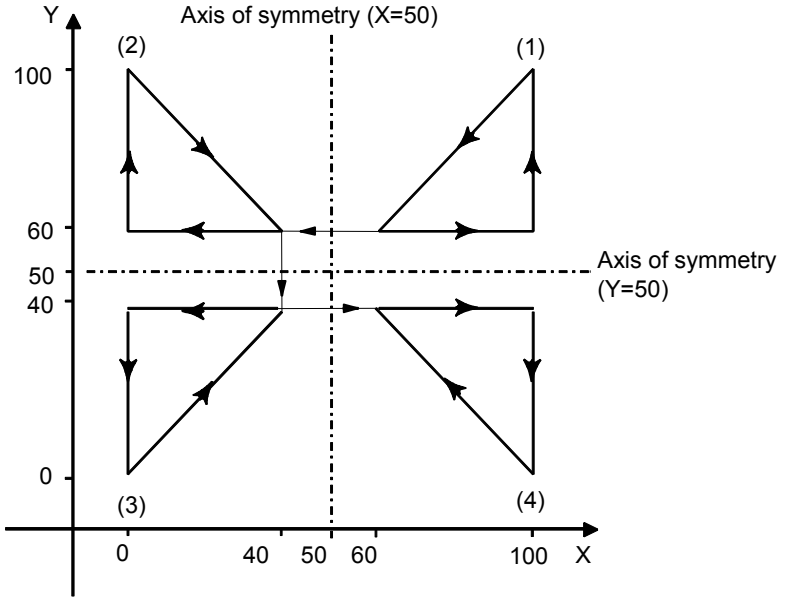


Fig. 15.2 (g) Program example of scaling in each axis

15.3 PROGRAMMABLE MIRROR IMAGE (G50.1, G51.1)

A mirror image of a programmed command can be produced with respect to a programmed axis of symmetry (Fig. 15.3 (a)).



(1) Original image of a programmed command
(2) Image symmetrical about a line parallel to the Y-axis and crossing the X-axis at 50
(3) Image symmetrical about point (50, 50)
(4) Image symmetrical about a line parallel to the X-axis and crossing the Y-axis at 50

Fig. 15.3 (a) Programmable mirror image

Format

G51.1 IP_ ; Setting a programmable image
: A mirror image of a command specified in
: these blocks is produced with respect to
: the axis of symmetry specified by G51.1
: IP_;.
G50.1 IP_ ; Canceling a programmable mirror image

IP_ : Point (position) and axis of symmetry for producing a mirror image when specified with G51.1.
Axis of symmetry for producing a mirror image when specified with G50.1. Point of symmetry is not specified.

Explanation

- Mirror image by setting

If the programmable mirror image function is specified when the command for producing a mirror image is also selected by a CNC external switch or CNC setting (see III-4.5), the programmable mirror image function is executed first.

- Mirror image on a single axis in a specified plane

Applying a mirror image to one of the axes on a specified plane changes the following commands as follows :

Command	Explanation
Circular command	G02 and G03 are interchanged.
Tool radius · tool nose radius compensation	G41 and G42 are interchanged.
Coordinate system rotation	CW and CCW (directions of rotation) are interchanged.

Limitation

- Scaling and coordinate system rotation

Processing proceeds from program mirror image to scaling and coordinate system rotation in the stated order. The commands should be specified in this order, and, for cancellation, in the reverse order. Do not specify G50.1 or G51.1 during scaling or coordinate system rotation mode.

- Commands related to reference position return and coordinate system

In programmable mirror image mode, G codes related to reference position return (G27, G28, G29, G30, etc.) and those for changing the coordinate system (G52 to G59, G92, etc.) must not be specified. If any of these G codes is necessary, specify it only after canceling the programmable mirror image mode.

15.4 NORMAL DIRECTION CONTROL (G40.1,G41.1,G42.1)

Overview

When a tool with a rotation axis (C-axis) is moved in the XY plane during cutting, the normal direction control function can control the tool so that the C-axis is always perpendicular to the tool path (Fig. 15.4 (a)).

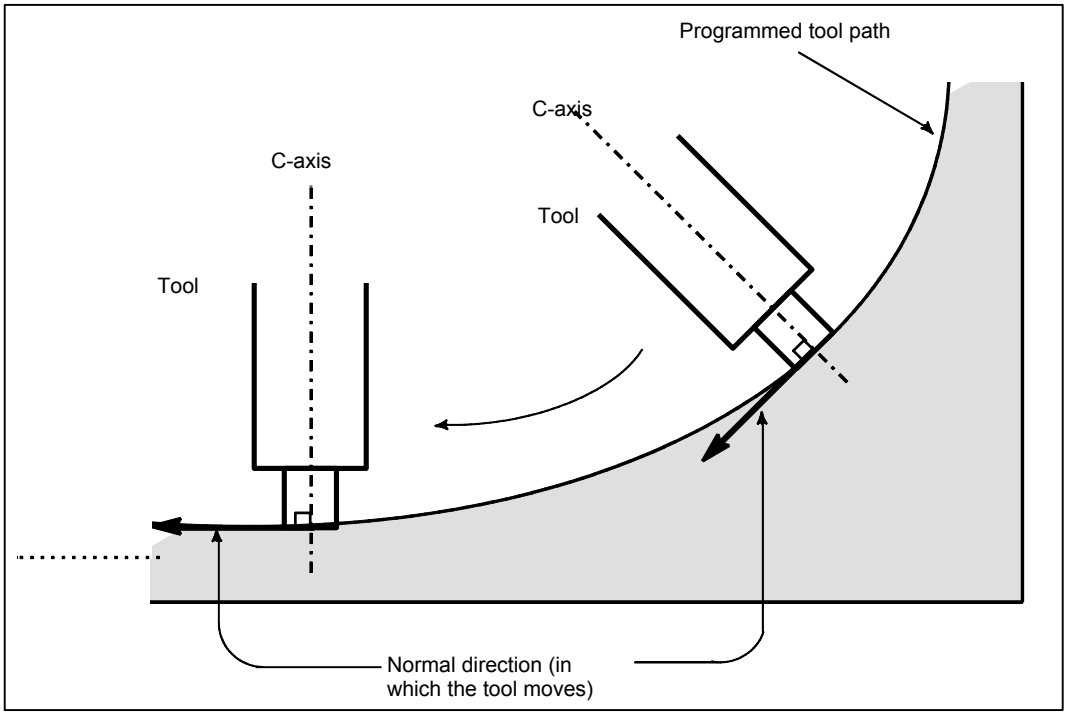
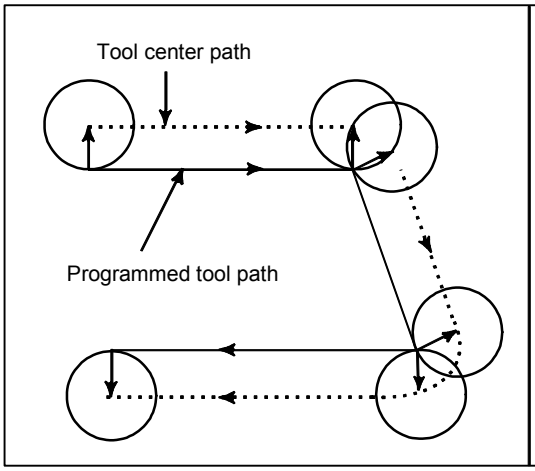


Fig. 15.4 (a) Sample Movement of the tool

Format

G41.1 ; Normal direction control, left
G42.1 ; Normal direction control, right
G40.1 ; Cancel normal direction control

The normal direction control, left (G41.1) command is used when the workpiece is on the right side of the tool as viewed while you are looking into the tool's way.
Once either G41.1 or G42.1 is issued, normal direction control is enabled (normal direction control mode).
Issuing G40.1 cancels the normal direction control mode.



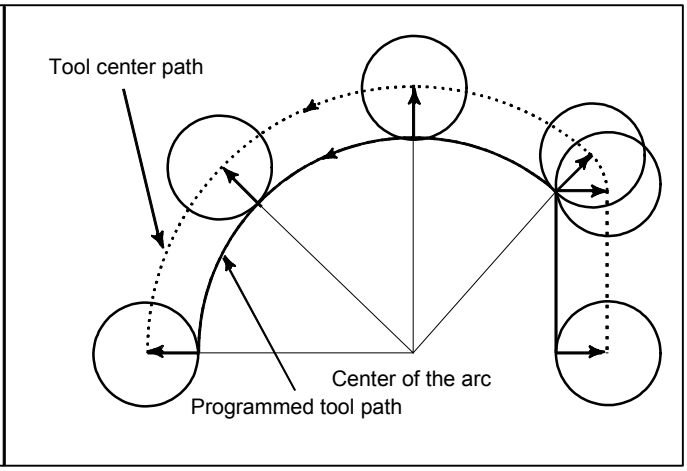


Fig. 15.4 (b) Normal direction control, left (G41.1) Fig. 15.4 (c) Normal direction control, right (G42.1)

Explanation

- Angle of the C axis

When viewed from the center of rotation around the C-axis, the angular displacement about the C-axis is determined as shown in Fig. 15.4 (d). The positive side of the X-axis is assumed to be 0, the positive side of the Y-axis is 90°, the negative side of the X-axis is 180°, and the negative side of the Y-axis is 270°.

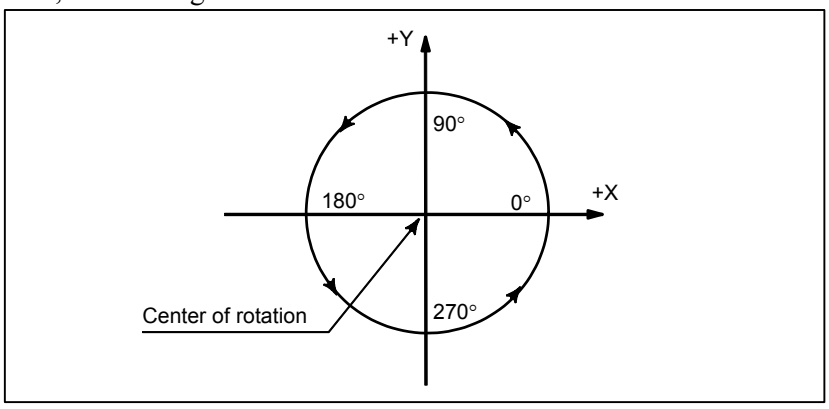


Fig. 15.4 (d) Angle of the C axis

- Normal direction control of the C axis

When the cancel mode is switched to the normal direction control mode, the C-axis becomes perpendicular to the tool path at the beginning of the block containing G41.1 or G42.1.

In the interface between blocks in the normal direction control mode, a command to move the tool is automatically inserted so that the C-axis becomes perpendicular to the tool path at the beginning of each block. The tool is first oriented so that the C-axis becomes perpendicular to the tool path specified by the move command, then it is moved along the X- and Y axes.

In the cutter compensation mode, the tool is oriented so that the C-axis becomes perpendicular to the tool path created after compensation.

In single-block operation, the tool is not stopped between a command for rotation of the tool and a command for movement along the X- and Y-axes. A single-block stop always occurs after the tool is moved along the X- and Y-axes.

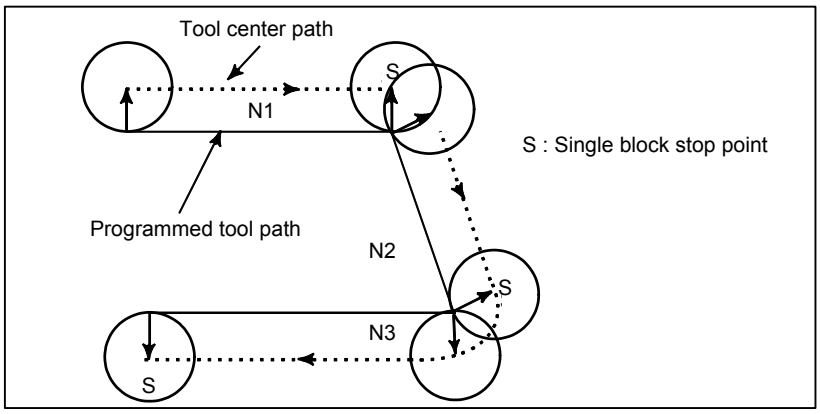


Fig. 15.4 (e) Point at which a single-block stop occurs in the normal direction control mode

Before circular interpolation is started, the C-axis is rotated so that the C-axis becomes normal to the arc at the start point. During circular interpolation, the tool is controlled so that the C-axis is always perpendicular to the tool path determined by circular interpolation.

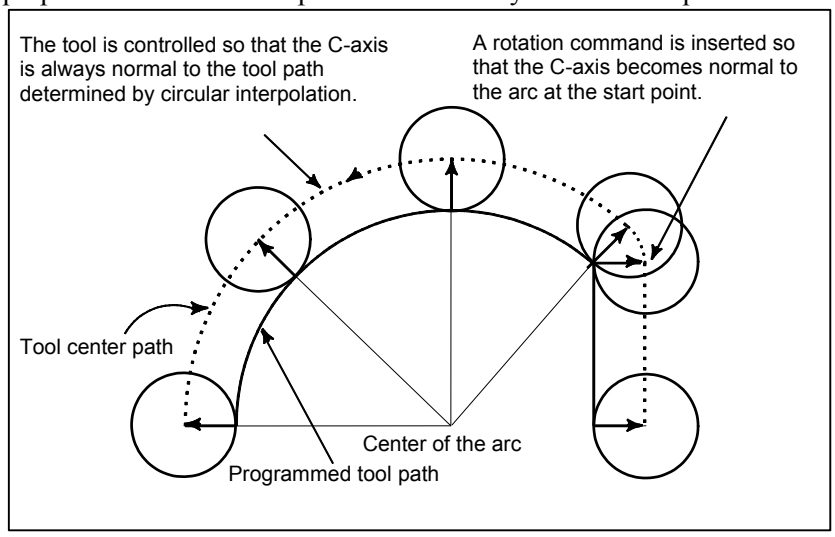


Fig. 15.4 (f) Normal direction control of the circular interpolation

NOTE

During normal direction control, the C axis always rotates through an angle less than 180 deg. I.e., it rotates in whichever direction provides the shorter route.

- C axis feedrate

Movement of the tool inserted at the beginning of each block is executed at the feedrate set in parameter 5481. If dry run mode is on at that time, the dry run feedrate is applied. If the tool is to be moved along the X-and Y-axes in rapid traverse (G00) mode, the rapid traverse feedrate is applied.
The feedrate of the C axis during circular interpolation is defined by the following formula.

$$F \times \frac{\text{Amount of movement of the C axis (deg)}}{\text{Length of arc (mm or inch)}} \text{ (deg/min)}$$

F : Feedrate (mm/min or inch/min) specified by the corresponding block of the arc

Amount of movement of the C axis :

The difference in angles at the beginning and the end of the block.

NOTE

If the feedrate of the C axis exceeds the maximum cutting speed of the C axis specified to parameter No. 1430, the feedrate of each of the other axes is clamped to keep the feedrate of the C axis below the maximum cutting speed of the C axis.

- Normal direction control axis

A C-axis to which normal-direction control is applied can be assigned to any axis with parameter No. 5480.

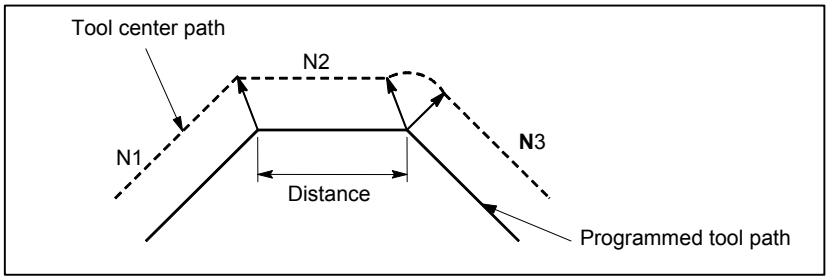
- Angle for which figure insertion is ignored

When the rotation angle to be inserted, calculated by normal-direction control, is smaller than the value set with parameter No. 5482, the corresponding rotation block is not inserted for the axis to which normal-direction control is applied. This ignored rotation angle is added to the next rotation angle to be inserted, the total angle being subject to the same check at the next block.
If an angle of 360 degrees or more is specified, the corresponding rotation block is not inserted.
If an angle of 180 degrees or more is specified in a block other than that for circular interpolation with a C-axis rotation angle of 180 degrees or more, the corresponding rotation block is not inserted.

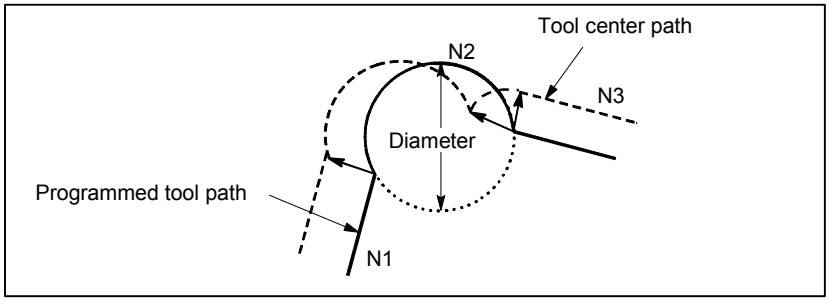
- Movement for which arc insertion is ignored

Specify the maximum distance for which machining is performed with the same normal direction as that of the preceding block.

- Linear movement
  When distance N2, shown below, is smaller than the set value, machining for block N2 is performed using the same direction as that for block N1.



- Circular movement
  When the diameter of block N2, shown below, is smaller than the set value, machining for block N2 is performed using the same normal direction as that for block N1. And control as compensation along the circular arc is not performed.



NOTE
1 Do not specify any command to the C axis during normal direction control. Any command specified at this time is ignored.
2 Before processing starts, it is necessary to correlate the workpiece coordinate of the C axis with the actual position of the C axis on the machine using the coordinate system setting (G92) or the like.
3 The helical cutting option is required to use this function. Helical cutting cannot be specified in the normal direction control mode.
4 Normal direction control cannot be performed by the G53 move command.
5 The C-axis must be a rotation axis.

T

NOTE
1 In the normal direction control mode, the following commands cannot be issued. An attempt to issue any of them results in alarm PS1471 being raised.
- Plane selection command (G17, G18, G19)
- Automatic reference position return(G28)
- 2nd/3rd/4th reference position return(G30)
- Move from the reference position (G29)

15.5 WORKPIECE SETTING ERROR COMPENSATION

When a workpiece is placed on the machine, the workpiece is not always placed at an ideal position.
With this function, a displaced workpiece can be machined according to the program.
This function can compensate for a workpiece setting error on a rotation axis used with a function involving rotation axis operation such as a function for tool center point control, function for 3-dimensional cutter compensation, and a tilted working plane command. So, a workpiece can be machined according to the program even during tool center point control, 3-dimensional cutter compensation, and tilted working plane command.

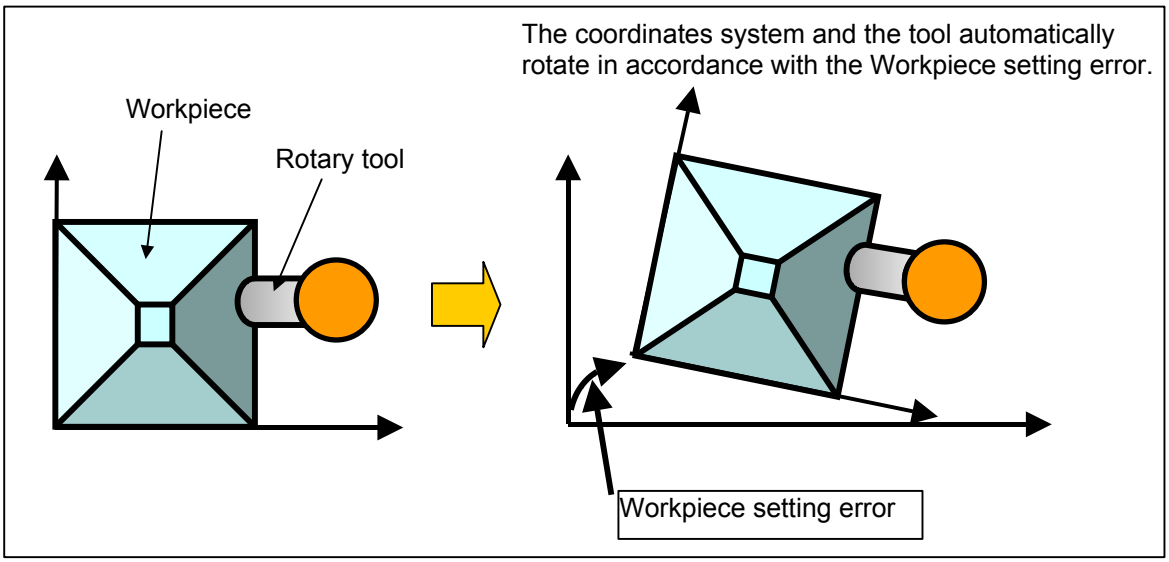


Fig. 15.5 (a) Outline of workpiece setting error

Format

```
G54.4 Pn ;      Start workpiece setting error compensation
  :
  :             } Workpiece setting error compensation mode
G54.4 P0 ;      Cancel workpiece setting error compensation

Pn  : Workpiece setting error specification code
n   : 1 to 7
```

A start block and cancellation block for workpiece setting error compensation suppress buffering.

Explanation

- Workpiece setting error

A workpiece setting error is defined by the following eight variables:

- X direction error Δx
- Y direction error Δy
- Z direction error Δz
- Rotation direction error Δa (rotation error on the X-axis in degrees)
- Rotation direction error Δb (rotation error on the Y-axis in degrees)
- Rotation direction error Δc (rotation error on the Z-axis in degrees)
- Table rotation axis position 1
- Table rotation axis position 2

[About Δx, Δy, and Δz]
Δx, Δy, and Δz represent the coordinate values of the origin of the workpiece coordinate system (X'Y'Z' in the figure below, which is hereinafter referred to as the "workpiece setting coordinate system") based on a displaced workpiece, as viewed from the original workpiece coordinate system (XYZ in the figure below).

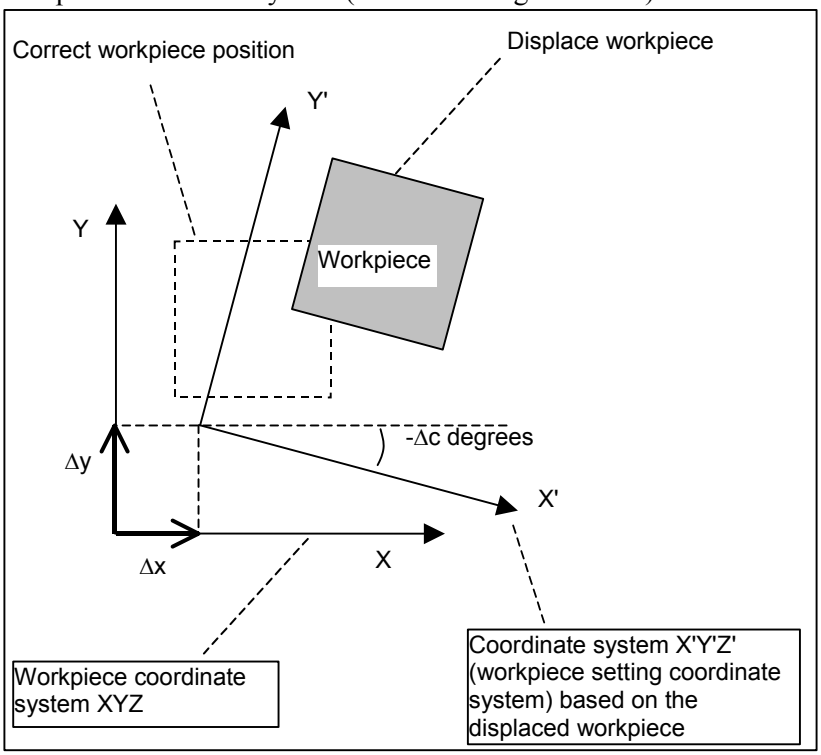


Fig. 15.5 (b) Errors in X, Y, and Z directions

[About Δa, Δb, and Δc]
Δa, Δb, and Δc are defined as described below.
The workpiece coordinate system obtained by rotating a workpiece coordinate system about the X-axis by angle Δa, about the Y-axis by angle Δb, and about the Z-axis by angle Δc then shifting that workpiece coordinate system by (Δx, Δy, Δz) from the origin of that workpiece coordinate system is supposed to match the "workpiece setting coordinate system".
(Note: In determination of positive/negative rotation, clockwise rotation is defined to be positive rotation.)

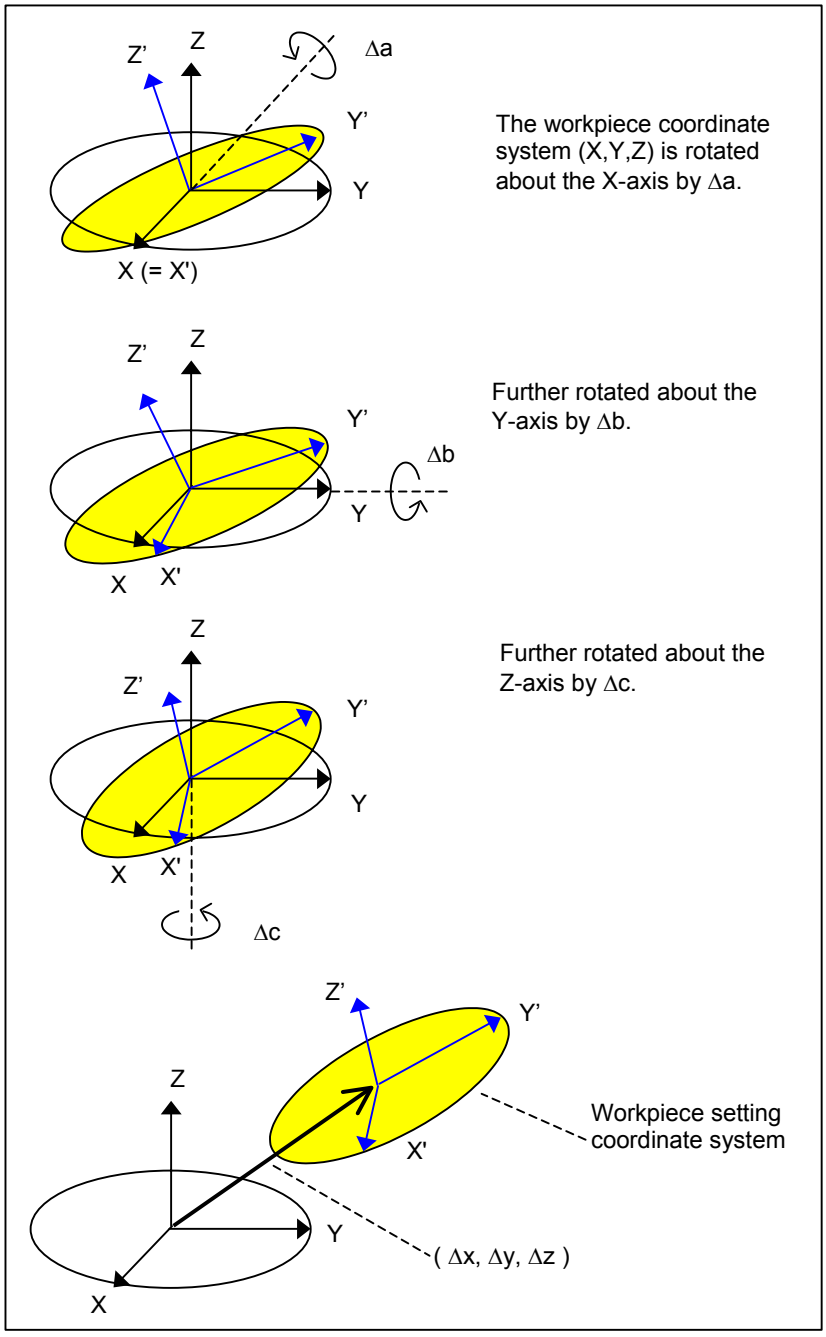


Fig. 15.5 (c) Errors in rotation directions

The example of Fig. 15.5 (b) assumes that Δa = Δb = 0 and Δc represents a nonzero value.

[About table rotation axis position 1 and table rotation axis position 2]

The table rotation axis position means the machine coordinate on the table rotation axis of a 5-axis machine with a table rotation axis (machine of table rotation type or composite type) when a workpiece setting error (displacement from the correct workpiece position to the actual workpiece position) is measured.
When two table rotation axes are used, set table rotation axis position 1 for the master rotation axis, and set table rotation axis position 2 for the slave rotation axis. (As for master rotation axis and slave rotation axis, refer to the Table 1 described later.)
When only one table rotation axis is used, set table rotation axis position 1 for the axis.
When no table rotation axis is used, or the machine used is not a 5-axis machine, table rotation axis position 1 and table rotation axis position 2 need not be set.
No setting can be made for a hypothetical axis.

In the descriptions above, X, Y, and Z represent the three basic axes, X, Y, and Z, specified by parameter No. 1022.
If the specification of any of the three basic axes, X, Y, and Z, is missing, alarm PS0436 is issued.

- Setting of a workpiece setting error on the workpiece setting error setting screen

A workpiece setting error is set on the workpiece setting error setting screen.
For explanations about the setting method, see Chapter 12, "SETTING AND DISPLAYING DATA," of Part III.

- Workpiece setting error of No. 00 (COMMON)

Each of the values set in No. 00 (COMMON) is added to the corresponding value of each of No. 01 through No. 07.
Note that rotation direction errors for No.00(COMMON) are not available.
Example:

Suppose that workpiece setting errors are set as follows:

No. 00 (COMMON)	No. 01	No. 02
X 10.000	x 0.500	x 0.800
Y 0.000	y 0.000	y 0.000
Z 0.000	z 0.000	z 0.000
	a 1.500	a 1.800
	b 0.000	b 0.000
	c 0.000	c 0.000

When the workpiece setting error of No. 01 is selected, workpiece setting error compensation is performed based on the following:
Δx = 10.000+0.500 = 10.500
Δa = 1.500

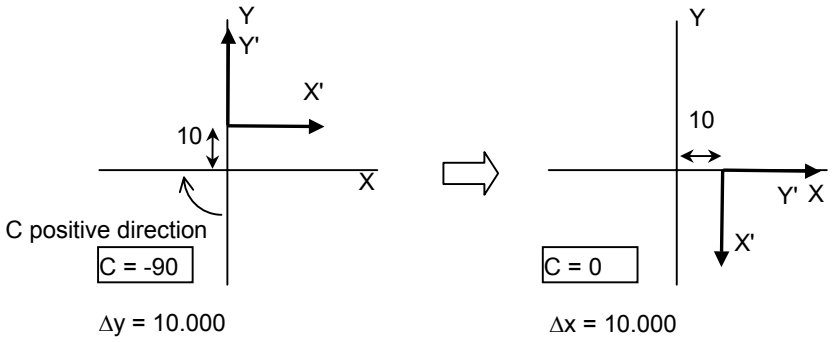
When the workpiece setting error of No. 02 is selected, workpiece setting error compensation is performed based on the following:
Δx = 10.000+0.800 = 10.800
Δa = 1.800

If the setting of table rotation axis position 1/table rotation axis position 2 differs between the workpiece setting error of No. 00 and each workpiece setting error when a 5-axis machine (machine of table rotation type or composite type) with table rotation axes is used, no simple additions are made. Instead, before additions are made, conversion is made to values based on the machine coordinate on a table rotation axis being 0.
Example:

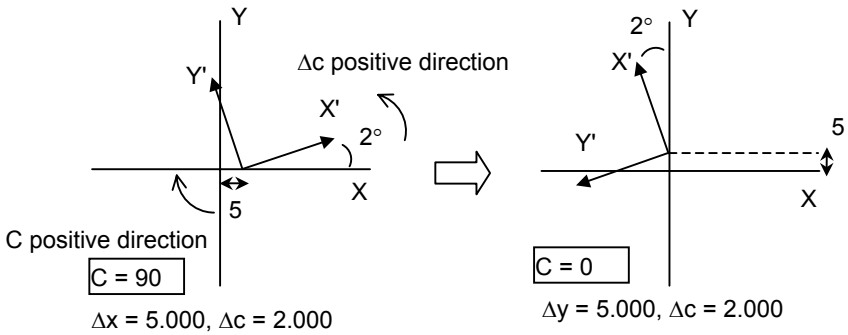
Suppose that workpiece setting errors are set as follows on a 5-axis machine with table rotation axis C about the Z-axis.

No. 00 (COMMON)	No. 01
X 0.000	x 5.000
y 10.000	y 0.000
z 0.000	z 0.000
	a 0.000
	b 0.000
	c 2.000
C -90.000	C 90.000

First, the error values of No. 00 are converted to those based on C = 0.000.



Next, the error values of No. 01 are converted to those based on C = 0.000.

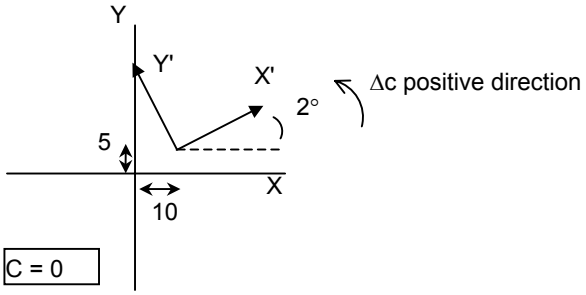


When the workpiece setting error of No. 01 is selected, workpiece setting error compensation is performed as follows based on the converted error values based on C = 0.000:

$\Delta x = 10.000+0.000 = 10.000$

$\Delta y = 0.000+5.000 = 5.000$

$\Delta c = 2.000$



Δx = 10.000, Δy = 5.000, Δc = 2.000

- Least input increment of data and valid data range

[About errors Δx, Δy, and Δz]

The unit of data is the unit of input, and follows the least input increment for the reference axis specified by parameter No. 1031.

- For metric input

Unit system of reference axis	IS-A	IS-B	IS-C	IS-D	IS-E
Least input increment (mm)	0.01	0.001	0.0001	0.00001	0.000001
Maximum settable value (mm)	±999,999.99	±999,999.999	±99,999.9999	±9,999.99999	±999.999999

- For inch input

Unit system of reference axis	IS-A	IS-B	IS-C	IS-D	IS-E
Least input increment (inch)	0.001	0.0001	0.00001	0.000001	0.0000001
Maximum settable value (inch)	±99,999.999	±99,999.9999	±9,999.99999	±999.999999	±99.9999999

[About errors Δa, Δb, and Δc]

With parameter No. 11201, the number of decimal places of the least input increment can be specified.

Parameter No. 11201	1	2	3	4
Least input increment (deg)	0.1	0.01	0.001	0.0001
Maximum settable value (deg)	±99,999,999.9	±9999,999.99	±999,999.999	±99,999.9999

Parameter No. 11201	5	6	7	8
Least input increment (deg)	0.00001	0.000001	0.0000001	0.00000001
Maximum settable value (deg)	±9,999.99999	±999.999999	±99.9999999	±9.99999999

Note, however, that a value from 1 to 8 can be specified in parameter No. 11201.

If a value not within the specifiable range is specified in parameter No. 11201, the least input increment of the reference axis is followed.

Unit system of reference axis	IS-A	IS-B	IS-C	IS-D	IS-E
Least input increment (deg)	0.01	0.001	0.0001	0.00001	0.000001
Maximum settable value (deg)	±999,999.99	±999,999.999	±99,999.9999	±9,999.99999	±999.999999

[About table rotation axis position 1 and table rotation axis position 2]
The unit of data follows the least input increment of each corresponding rotation axis.

Unit system of rotation axis	IS-A	IS-B	IS-C	IS-D	IS-E
Least input increment (deg)	0.01	0.001	0.0001	0.00001	0.000001
Maximum settable value (deg)	±999,999.99	±999,999.999	±99,999.9999	±9,999.99999	±999.999999

- "Active workpiece setting error"

In the workpiece setting error compensation mode, "Active workpiece setting error" on the workpiece setting error setting screen displays the currently selected workpiece setting error number and workpiece setting error (sum of the data of the currently selected workpiece setting error number and the data of No. 00).

If the workpiece setting error compensation mode is not set, "MODE OFF" is indicated, and 0 is displayed in each data field.

If a 5-axis machine (machine of table rotation type or composite type) with table rotation axes is used, and a nonzero value is set for table rotation axis position 1/table rotation axis position 2, error values obtained by conversion based on the machine coordinates on the table rotation axes being 0 are displayed.

- Setting of workpiece setting errors with custom macro variables

By using custom macro variables #26000 to #26077, workpiece setting errors can be read and written.
(The custom macro option is required.)

The numbers of macro variables correspond to Errors as follows :

	Error No.00 (COMMON)	Error No.01	Error No.02	Error No.03	Error No.04	Error No.05	Error No.06	Error No.07
X direction error Δx	#26000	#26010	#26020	#26030	#26040	#26050	#26060	#26070
Y direction error Δy	#26001	#26011	#26021	#26031	#26041	#26051	#26061	#26071
Z direction error Δz	#26002	#26012	#26022	#26032	#26042	#26052	#26062	#26072
Rotation direction error Δa	-	#26013	#26023	#26033	#26043	#26053	#26063	#26073
Rotation direction error Δb	-	#26014	#26024	#26034	#26044	#26054	#26064	#26074
Rotation direction error Δc	-	#26015	#26025	#26035	#26045	#26055	#26065	#26075
Table rotation axis position 1	#26006	#26016	#26026	#26036	#26046	#26056	#26066	#26076
Table rotation axis position 2	#26007	#26017	#26027	#26037	#26047	#26057	#26067	#26077

- Workpiece setting error compensation mode

By specifying G54.4 Pn (n: 1 to 7), the workpiece setting error compensation mode is set.
With Pn, select a workpiece setting error from No. 01 to No. 07.
In the workpiece setting error compensation mode, the program is executed in a "workpiece setting coordinate system" defined by shifting the workpiece coordinate system.
G54.4 is a modal G code that belongs to group 33 with the M series or group 26 with the T series.
By specifying G54.4 P0, the workpiece setting error compensation mode is canceled.

In a block for starting workpiece setting error compensation, the machine is not moved, but the absolute coordinates are changed by the error. So, the next absolute command makes a movement to a specified position in the workpiece setting coordinate system. This means that an absolute command is needed after a block for starting workpiece setting error compensation.

NOTE

1 Be sure to specify Pn in a block specifying G54.4. If P is not specified, or a number (n) not within the specifiable range is specified after P, alarm PS0437 is issued.
2 Specify G54.4 singly. If another G code or axis command is specified together, alarm PS0437 is issued.
3 Workpiece setting error compensation is valid for movement in automatic operation.
4 Any workpiece setting error between starting blocks remains valid until the workpiece setting error compensation mode is canceled.
5 Workpiece setting error compensation may not be specified doubly. If G54.4 Pn (n 0) is specified in the workpiece setting error compensation mode, alarm PS0437 is issued.
6 No absolute coordinate is changed in a block for starting workpiece setting error compensation unless the two angles stated below exceed the value specified in parameter No. 11204.
   (1) Angle made by the current absolute coordinate point and a singular point posture
   (2) Angle made by a point sifted by a workpiece setting error from the absolute coordinate point and a singular point posture
7 If all the errors are 0, issuing the G54.4 command cannot put the workpiece setting error compensation mode in effect.

- Tool direction compensation on a 5-axis machine

With a 5-axis machine, tool direction compensation must be performed by setting bit 0 (RCM) of parameter No. 11200 to 1. This means that rotation axis position compensation is performed to direct the tool as programmed relative to the workpiece.
For tool direction compensation, the following parameters must be set:

Parameter No.	Description
19680	Type of mechanical section
19681	Controlled axis number of the first rotation axis
19682	Axis direction of the first rotation axis
19683	Inclination angle when the first rotation axis is inclined
19684	Rotation direction of the first rotation axis
19685	Rotation angle when the first rotation axis is a hypothetical axis
19686	Controlled axis number of the second rotation axis
19687	Axis direction of the second rotation axis
19688	Inclination angle when the second rotation axis is inclined
19689	Rotation direction of the second rotation axis
19690	Rotation angle when the second rotation axis is a hypothetical axis
19696#0	Whether the first rotation axis is an ordinary rotation axis/hypothetical axis
19696#1	Whether the second rotation axis is an ordinary rotation axis/hypothetical axis
19697	Tool axis direction
19698	Reference angle RA
19699	Reference angle RB
19700	Rotary table position (X-axis of the basic three axes)
19701	Rotary table position (Y-axis of the basic three axes)
19702	Rotary table position (Z-axis of the basic three axes)
19703	Intersection offset vector between the first and second rotation axes of the table (X-axis of the basic three axes)
19704	Intersection offset vector between the first and second rotation axes of the table (Y-axis of the basic three axes)
19705	Intersection offset vector between the first and second rotation axes of the table (Z-axis of the basic three axes)

With a 5-axis machine, it is recommended that Workpiece setting error compensation is commanded after AICC(AI contouring control) is already active.(G05.1 Q1 is commanded or bit 0 (SHP) of parameter No.1604 is 1.) And, the following parameters for Acc/Deceleration are needed.

Parameter No.	Description
1432	Maximum cutting feedrate in the acceleration/deceleration before interpolation mode
1660	Maximum allowable acceleration rate in acceleration/deceleration before interpolation
1671	Maximum allowable acceleration rate for rapid traverse in acceleration/deceleration before interpolation
1672	Acceleration/deceleration change time for rapid traverse in bell-shaped acceleration/deceleration before interpolation
1737	Allowable acceleration rate for each axis when the deceleration function based on acceleration/deceleration under AI contour control.

Parameter No.	Description
1769	Time constant for acceleration/deceleration after cutting feed interpolation in the acceleration/deceleration before interpolation mode
1772	Acceleration/deceleration change time in bell-shaped acceleration/deceleration before interpolation
1783	Allowable speed difference in speed determination based on a corner speed difference

NOTE

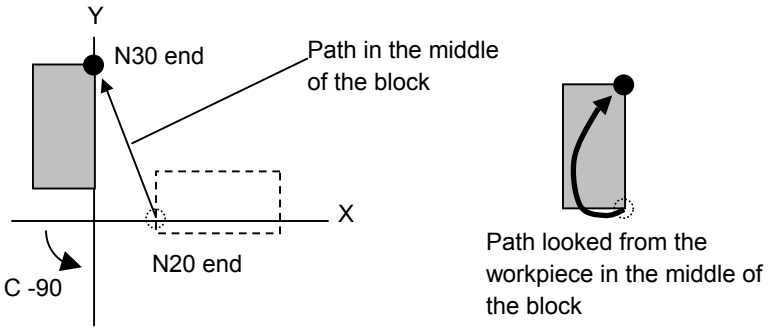
1 If any of the parameters above is set incorrectly, alarm PS0438 is issued.
2 Depending on the machine configuration, it may be physically impossible to orient the tool in the compensation direction. In such a case, alarm DS0030 is issued.
3 In a block for starting workpiece setting error compensation, the absolute coordinate on a rotation axis is changed considering the workpiece setting error. At this time, depending on the machine configuration, a rotation axis for orienting the tool in the tool direction as viewed from the workpiece setting coordinate system may be absent. In such a case, alarm PS0438 is issued.
4 In the workpiece setting error compensation mode, look-ahead acceleration/deceleration before interpolation is automatically valid. Moreover, be sure to set the following parameters:
   (1) Bit 1 (LRP) of parameter No. 1401=1:
   Linear rapid traverse
   (2) Bit 5 (FRP) of parameter No. 19501=1:
   Use acceleration/deceleration before interpolation for rapid traverse.
   (3) Parameter No. 1671:
   Maximum allowable acceleration rate for rapid traverse in acceleration/deceleration before interpolation
   (4) Parameter No. 1660:
   Maximum allowable acceleration rate in acceleration/deceleration before interpolationIf the parameters above are not set, alarm PS0438 is issued.
5 When TCP(Tool center point control) or G53.1 of TWP(Tilted working plane command) is used during Workpiece setting error compensation, Tool direction compensation is needed (Bit 0 (RCM) of parameter No.11200 is 1.) Otherwise, in the case of TCP, the alarm PS5421, in the case of G53.1 of TWP, the alarm PS5458 occurs.

If linear interpolation or circular interpolation is specified on a machine of table rotation type or composite type, linear interpolation or circular interpolation is performed as viewed from the workpiece on the table.

<Example>

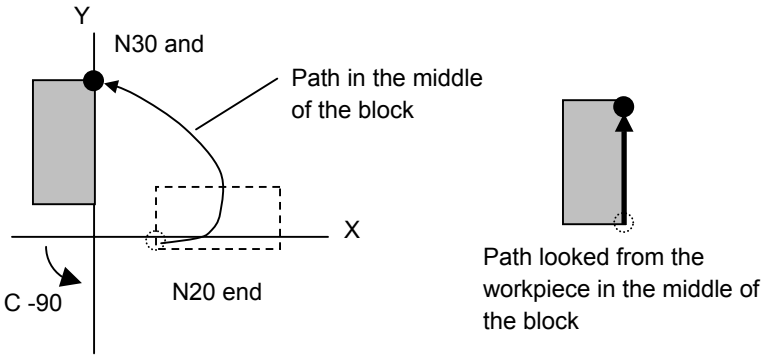
The machine is a table rotation type with A axis(master axis) around X axis and C axis (slave axis) around Z axis (when A=0). Firstly, suppose that Workpiece setting error is 0 and the following program is commanded.

```
O1
N10 G55                          (G55 is X50.0 Y0.0 Z0.0.)
N20 X0 Y0 Z0 A0 C0
N30 G01 X－50.0 Y150.0 C－90.0 F100.0
```



Next, suppose that Workpiece setting error has Δc=5.0, which is the error around Z axis, and N15,N16 are added to O2 as follows : When Tool direction compensation is available (Bit 0 (RCM) of parameter No.11200 is 1), the path will be as follows :

```
O2
N10 G55
N15 G05.1 Q1
N16 G54.4 P1
N20 X0 Y0 Z0 A0 C0
N30 G01 X－50.0 Y150.0 C－90.0 F100.0
```



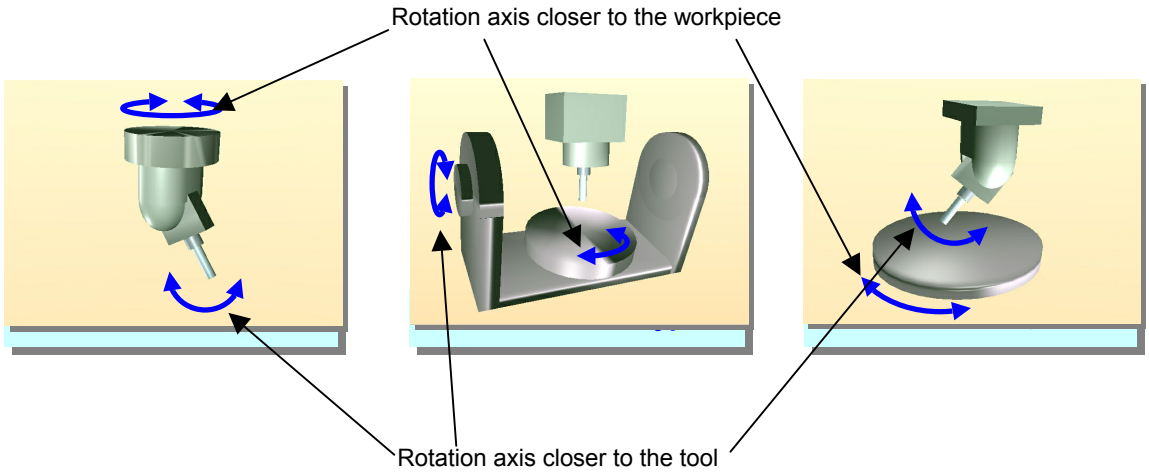
The path looked from the workpiece in the middle of the block is linear.

- "Rotation axis closer to the tool" and "rotation axis closer to the workpiece" on a 5-axis machine

When tool direction compensation is performed on a 5-axis machine, a singular point and singular point posture need to be considered. Here, a "rotation axis closer to the tool" and "rotation axis closer to the workpiece", which are used in a description of a singular point and singular point posture provided later, are explained.
On a 5-axis machine with two rotation axes, one rotation axis functions to tilt the tool toward the workpiece. This rotation axis is referred to as a "rotation axis closer to the tool".
The other rotation axis is referred to as a "rotation axis closer to the workpiece".
Depending on the type of mechanical section, the rotation axis closer to the tool and rotation axis closer to the workpiece are determined as indicated in Table 1.

Table 15.5 "Rotation axis closer to the tool" and "rotation axis closer to the workpiece"

Type of mechanical section (No. 19680)	Rotation axis closer to the tool	Rotation axis closer to the workpiece
Tool rotation type (2)	Slave axis	Master axis
Table rotation type (12)	Master axis	Slave axis
Composite type (21)	Tool rotation axis	Table rotation axis



- Singular point and singular point posture on a 5-axis machine

A tool posture is uniquely determined when the angles of the two rotation axes are determined. Usually, however, a combination of the angles of the two rotation axes for achieving a certain tool posture is not determined uniquely.

In particular, such a tool posture that the angle of the rotation axis closer to the tool is arbitrary is referred to as a "singular point posture". Moreover, such an angle of the rotation axis closer the tool that the tool posture becomes a singular point posture is referred to as a "singular point (or singular point angle)". When the angle of the rotation axis closer to the tool is a singular point, the center of the rotation axis closer to the workpiece and the tool posture (tool direction) are parallel with each other.

Example:

When the C-axis (about the Z-axis) is the master axis, the B-axis is the slave axis (about the Y-axis), and the reference tool axis direction is along the Z-axis on a 5-axis machine of tool rotation type, the singular point is B = …, 0 deg, 180 deg, ... At this time, an arbitrary C-axis angle represents a singular point posture (in the same direction as the Z-axis or in the direction opposite to the Z-axis).

If the angle of the B-axis is B = 0 (singular point) as shown in Fig. 4, for example, an arbitrary C-axis angle represents the same tool posture (singular point posture).

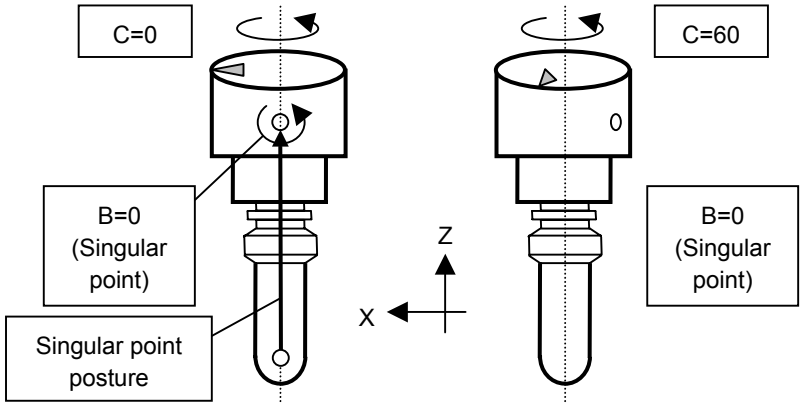


Fig. 15.5 (d) Singular point and singular point attitude

When the reference tool axis direction is inclined (parameter No. 19698, No. 19699), or the rotation axis is an angular axis (parameter No. 19682, No. 19683, No. 19687, No. 19688), for example, no singular point and singular point posture exist on some machines.

- Conditions to decide that Tool is in singular posture

When the angle between the tool and the singular posture is less than the parameter No.11204, it is decided that the tool is in singular posture.
In the descriptions below, the description 'when the tool is in singular…' means that 'when it is decided that the tool is in singular posture….'.
In addition, whether the tool is in singular point posture is decided at each end point of tool paths calculated in the NC. If the parameter No. 11204 value is very small and the feedrate and the override value are large, it is likely that the tool may pass by a singular point without deciding that the tool is in a singular point posture at each end point even when the tool path crosses the singular point. In this case, the tool will not be able to go beyond the singular point. To avoid this problem, set parameter No. 11204 with a value slightly larger than the quotient of the division of the highest rotation axis move speed (the highest cutting feedrate (parameter No. 1432) among the rotation axes or the rapid traverse rate (parameter No. 1420) whichever is higher) by 15000.

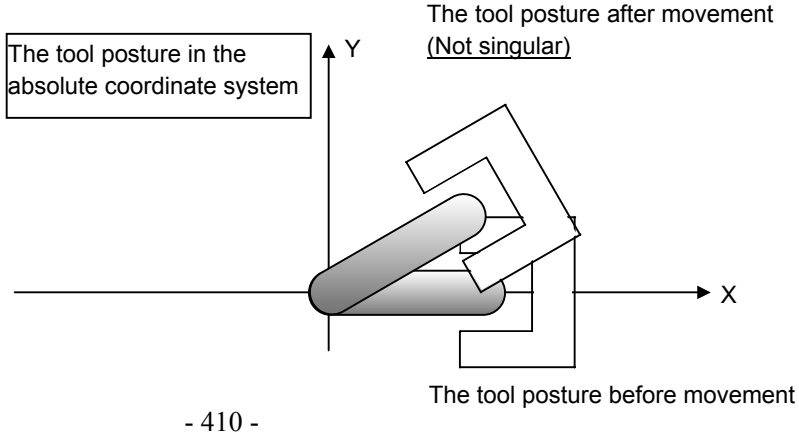
- Movement in the case that Tool direction is compensated

In the case that the tool direction is compensated (bit 0 (RCM) of parameter No.11200=1) on a 5 axis machine, the compensation is performed in every interpolation, and the rotation axes may move to positions which are different from the commanded positions.
In general, there are two pairs of rotation axes positions in the region between 0 degree to 360 degree to turn the tool in a certain direction.
And in the case that the tool is in singular point posture, the position of the rotation axis closer to the workpiece is not determined uniquely.
How the positions are determined is described as follows :

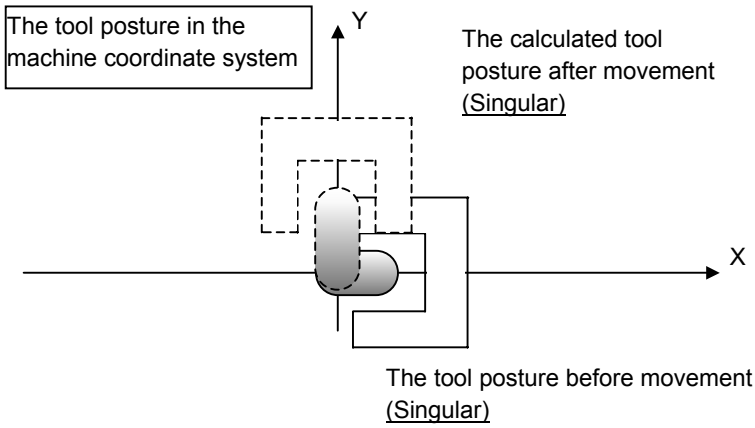
(1) In the case that the current machine position is singular and the position after movement in real time is also singular.
   (a) When the absolute position after movement in real time is not singular, the rotation axis closer to the workpiece does not move.

Example:

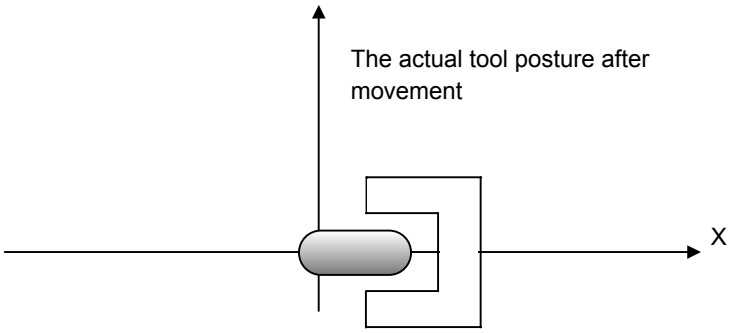
The rotation axis about the Z-axis is the master axis, the rotation axis about the Y-axis is the slave axis, and the reference tool axis direction is along the Z-axis on a 5-axis machine of tool rotation type.



Assume that the workpiece setting error around the Y-axis exists, and the tool posture after the compensation of tool direction becomes like following figure. (The tools before and after movement are in singular point posture.)



In this case, the rotation axis about the Z-axis (the rotation axis closer to the workpiece) does not move. (Only the rotation axis closer to the tool moves.)

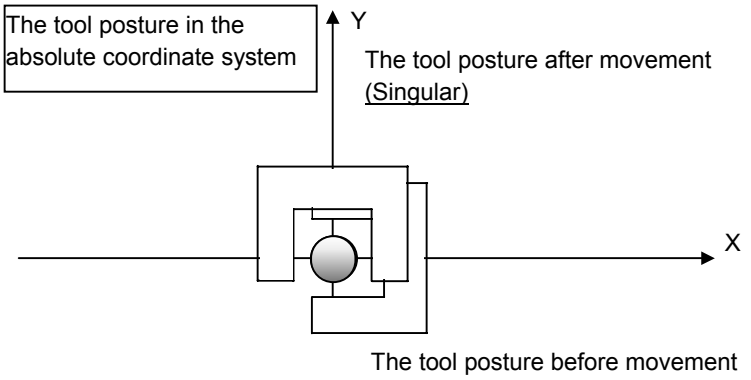


If the parameter No.11204 is set to the value except 0, the tools both before and after movement may be in singular point posture like the above example.
In this case, the rotation axis closer to the workpiece does not move. The rotation axis closer to the tool moves in the move direction in which the tool in the machine system (machine coordinate system) is decided to be in a singular point posture, but not move in the opposite direction.

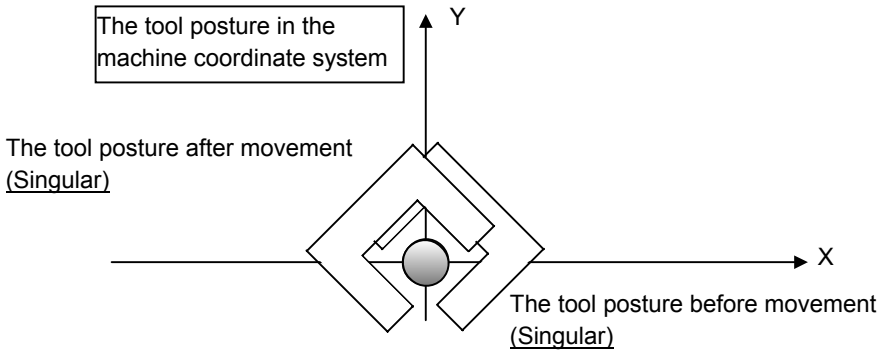
(b) When the absolute position after movement in real time is singular, the rotation axis closer to the workpiece moves as commanded.

Example:

The rotation axis about the Z-axis is the master axis, the rotation axis about the Y-axis is the slave axis, and the reference tool axis direction is along the Z-axis on a 5-axis machine of tool rotation type.



Assume that the workpiece setting error around the Z-axis exists, and the tool posture after the compensation of tool direction becomes like following figure.



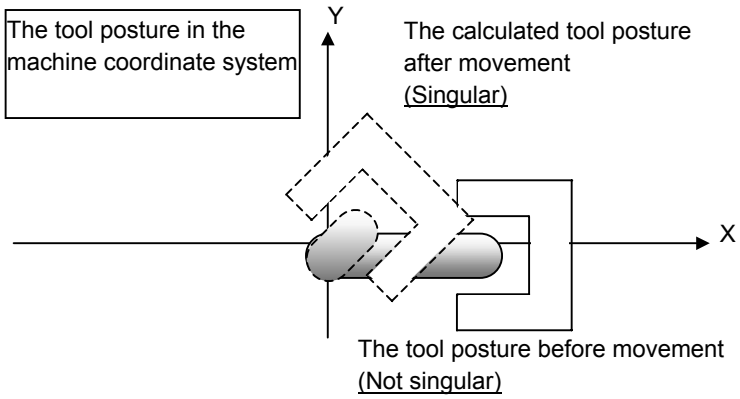
In this case, the rotation axis about Z-axis (the rotaion axis closer to the workpiece) moves as commanded.

(2) In the case that the current machine position is not singular and the position after movement in real time is singular.
: The rotation axis closer to workpiece does not move.

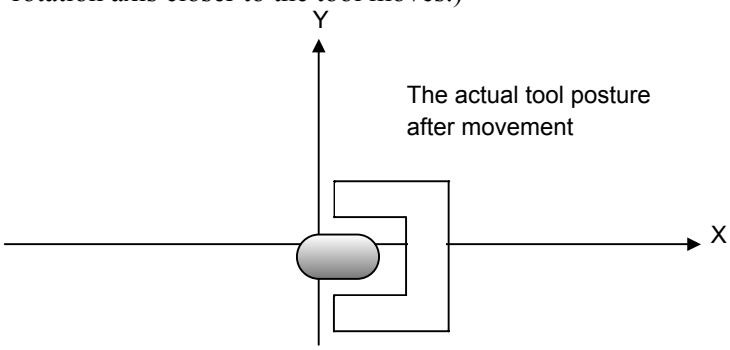
Example:

The rotation axis about the Z-axis is the master axis, the rotation axis about the Y-axis is the slave axis, and the reference tool axis direction is along the Z-axis on a 5-axis machine of tool rotation type.

Assume that the tool posture after the compensation of tool direction becomes like following figure.



In this case, the rotation axis about the Z-axis (the rotation axis closer to the workpiece) does not move. (Only the rotation axis closer to the tool moves.)

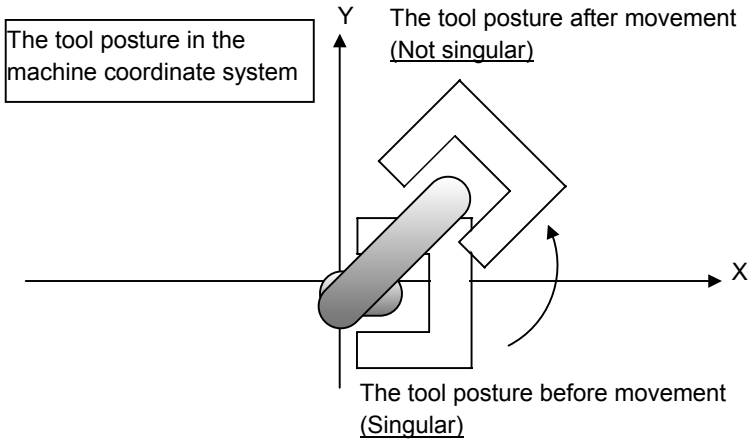


(3) In the case that the current machine position is singular and the position after movement in real time is not singular.
: In order to position the tool to the correct direction, there are two pairs of solutions of rotation axes angles within 0 – 360deg. A solution with which the rotation axis closer to workpiece moves shorter is selected.

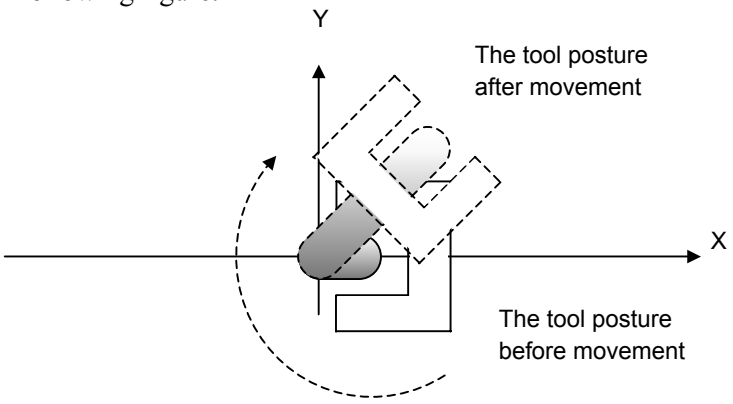
Example:

The rotation axis about the Z-axis is the master axis, the rotation axis about the Y-axis is the slave axis, and the reference tool axis direction is along the Z-axis on a 5-axis machine of tool rotation type.

Assume that the tool posture after the compensation of tool direction becomes like following figure.



In this case, the rotation axes move to the positions with which the rotation axis about the Z-axis (the rotation axis closer to workpiece) moves shorter. (The tool moves in the direction of the arrow in the above figure.)
The tool does not move in the direction of the arrow in the following figure.

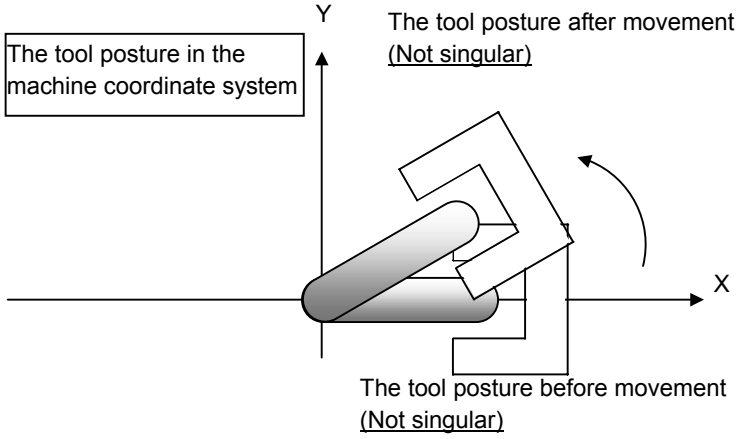


(4) In the case that the current machine position is not singular and the position after movement in real time is not singular.
: In order to position the tool to the correct direction, there are two pairs of solutions of rotation axes angles within 0 – 360deg. One solution with which the rotation axes don't pass the singular point is selected. And, the rotation axes move to the positions of the selected solution.

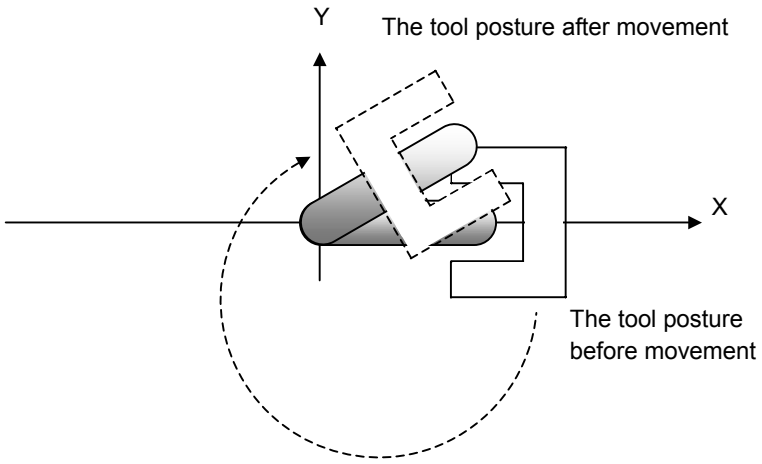
Example:

The rotation axis about the Z-axis is the master axis, the rotation axis about the Y-axis is the slave axis, and the reference tool axis direction is along the Z-axis on a 5-axis machine of tool rotation type.

Assume that the tool posture after the compensation of tool direction becomes like following figure.



In this case, the rotation axes move to the positions with which the rotation axes don't pass the singular point. (The tool moves in the direction of the arrow in the above figure.)
The tool does not move in the direction of the arrow in the following figure.



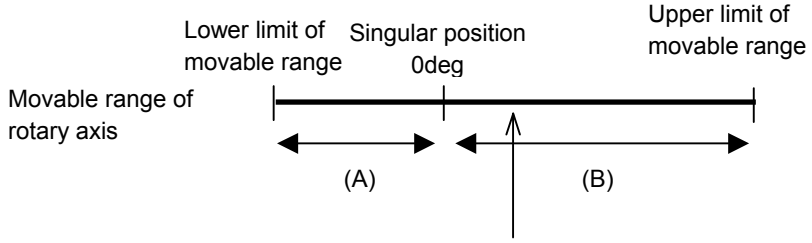
- When the tool posture is closer to a singular point posture on a 5-axis machine

If tool direction compensation is performed on a 5-axis machine, and the tool posture gets closer to a singular point posture during execution of a block, the rotation axis closer to the workpiece may make a large motion.

In addition, if parameter No. 11204 (angle for deciding on a singular point posture) is set with a value, it is likely that the tool direction and tool tip position may be incorrect while the tool is decided to be in a singular point posture. Furthermore, it is likely that the rotation axis closer to the tool may abruptly move by the parameter-set angle. So, do not set parameter No. 11204 with a value larger than necessary.

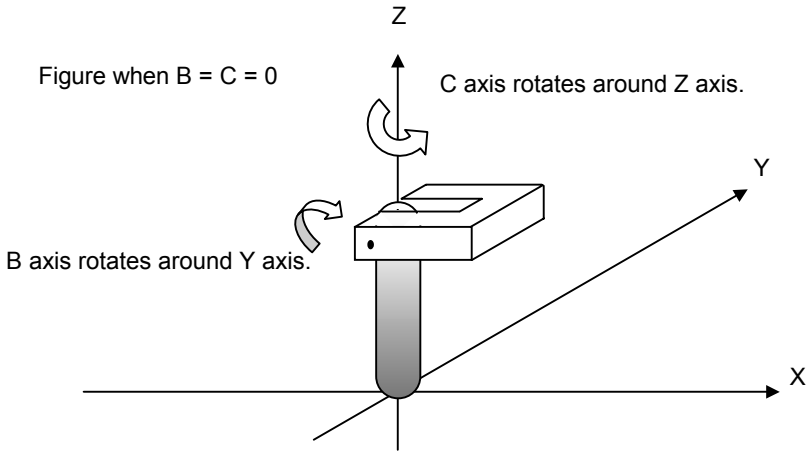
- Notes in the case that rotary axes have movable range

When the tool direction is compensated on a 5-axis machine, there can be the case that the machine position does not pass the singular posture during a progress of work at all.
In this case, in accordance with 4) described above, the solution with which the rotary axes don't pass the singular point is always selected. Therefore, in this case, the movement direction(area) depends on where the tool exists in comparison with the singular point, because the tool position is always selected so that the rotary axes don't pass the singular point.
In the case that rotary axes have movable range and a singular point exists in that range, Workpiece setting error compensation must be activated after the rotary axes have been moved to the range where the rotary axes should move, that is, the range (A) between the lower limit and the singular point or the range(B) between the upper limit and the singular point.
Generally speaking, if (A) and (B) are different, it is recommended that the larger one is selected.

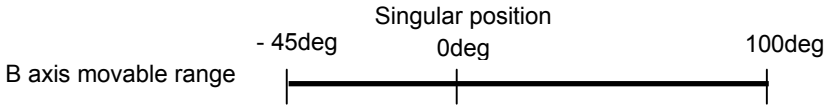


Workpiece setting error compensation must be activated after the rotary axes have been moved to the range where the rotary axes should move

The following is an example that the movement direction(area) depends on position of the rotary axes before starting workpiece setting error compensation.
The Master axis is C axis around Z axis, and the Slave axis is B axis around Y axis(when C=0). Tool direction is +Z when B=C=0.
Here, suppose B axis' movable range is –45 deg to +100 deg.

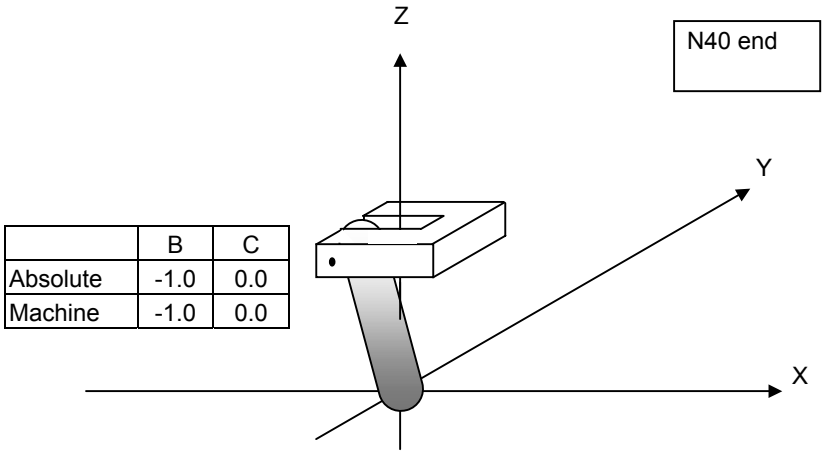


When B=0, the tool posture does not change even if C axis moves. That means the position B=0 is singular point.

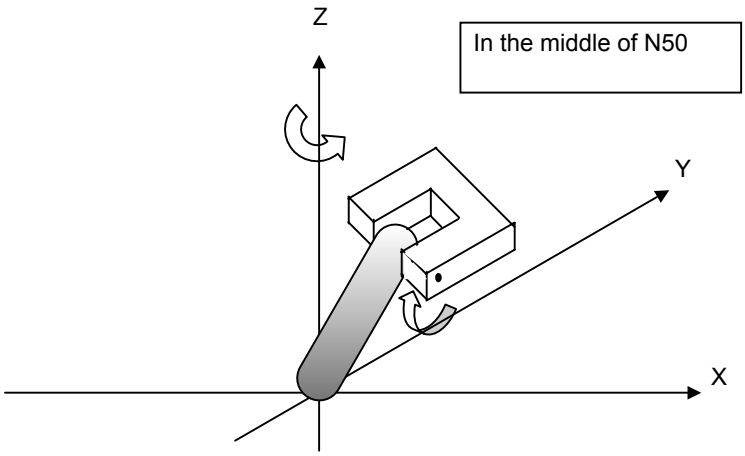


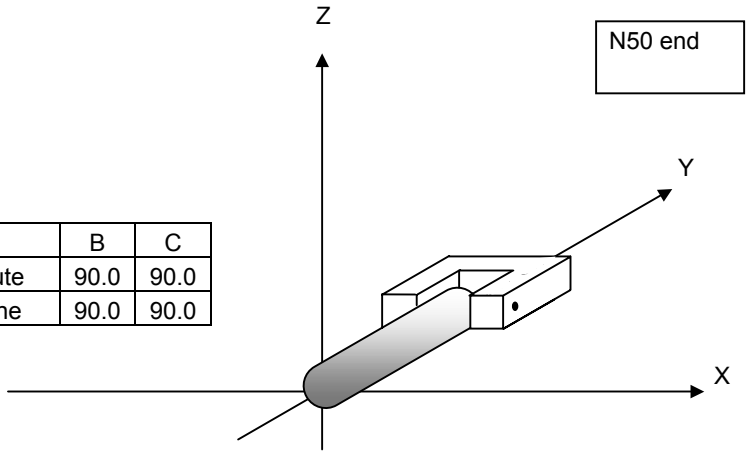
Firstly, suppose the following program without Workpiece setting error.

Program without Workpiece setting error
O1
N10 G5.1 Q1
N20 G90 G01 B-1.0 C0 F1000
N30 G43.4 H1
N40 X0 Y0 Z0
N50 B90.0 C90.0



	B	C
Absolute	-1.0	0.0
Machine	-1.0	0.0



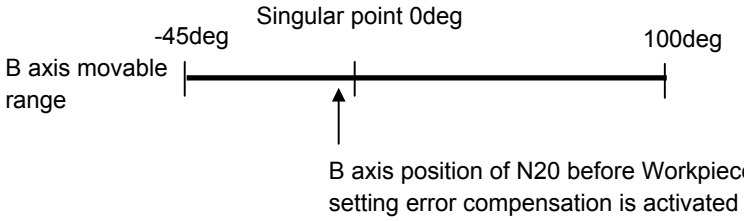


	B	C
Absolute	90.0	90.0
Machine	90.0	90.0

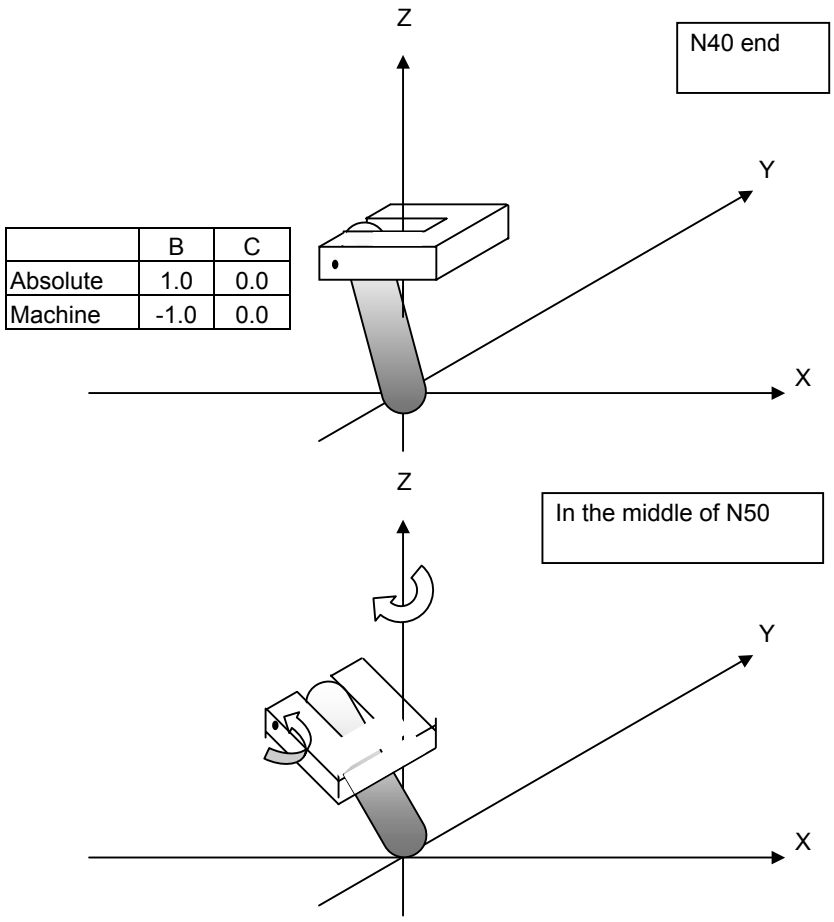
At N50, machine position moves to B90.0 and C90.0, as commanded.

Next, suppose there is the error -2.0deg around Y axis and Workpiece setting error Δb=-2.000 is set, and the block N25 is added as follows :

On O2, B axis position is –1.0 before Workpiece setting error compensation is activated, which is between the lower limit of movable range and the singular point of B axis.



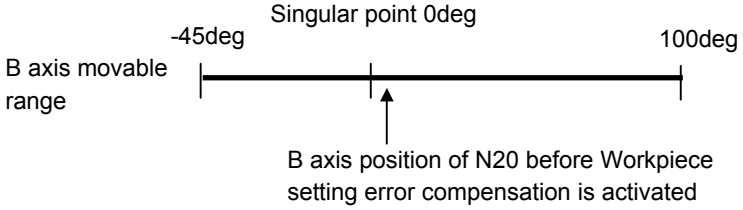
```
O2
N10 G5.1 Q1
N20 G90 G01 B-1.0 C0 F1000 ; B axis machine position is between
                             the lower limit and the singular
                             point
N25 G54.4 P1
N30 G43.4 H1
N40 X0 Y0 Z0
N50 B90.0 C90.0
```



	B	C
Absolute	1.0	0.0
Machine	-1.0	0.0

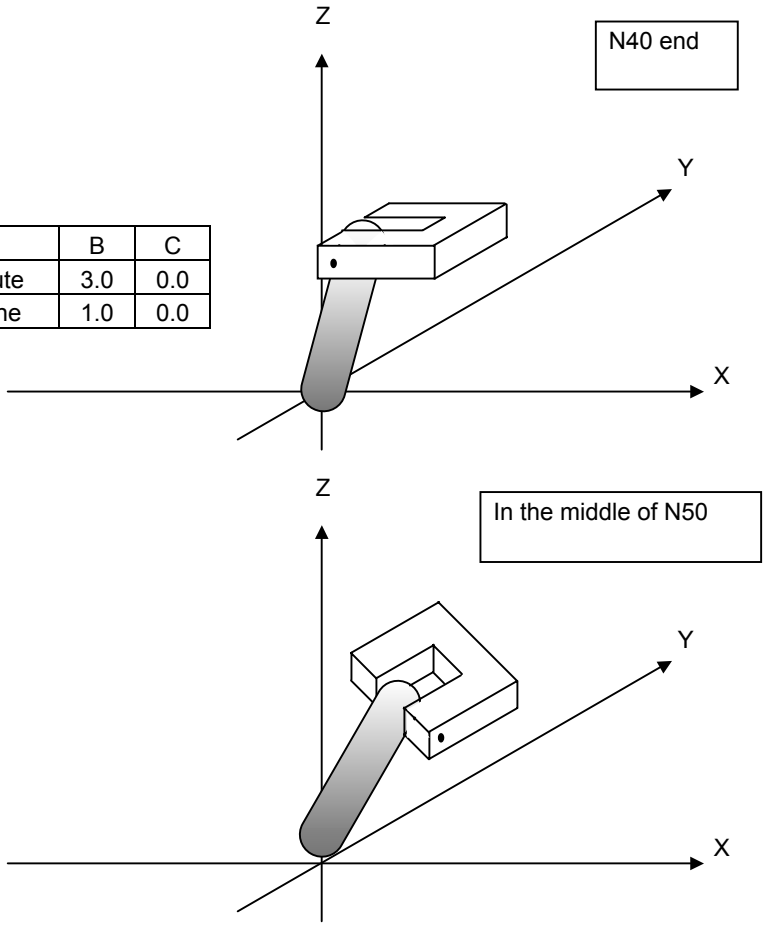
During N50, Machine position does not pass the singular point at all. So, B axis moves from the starting position(–1.0) to the position(B–90.0,C–90.0). As the result, B axis moves over the lower limit of B axis movable range.

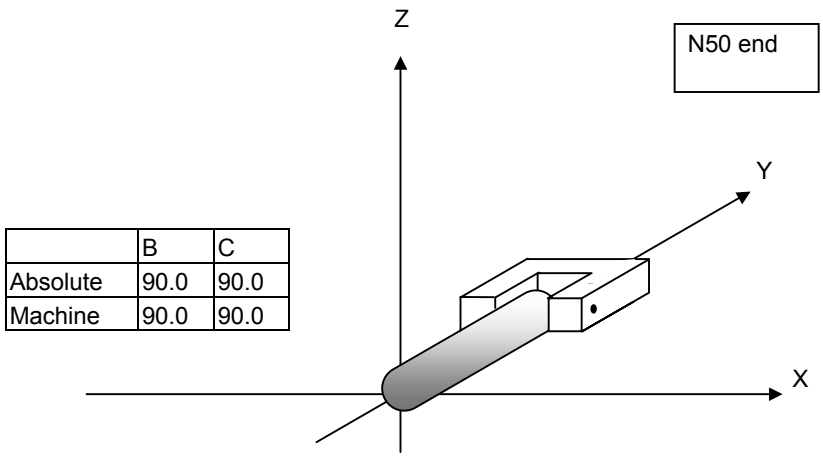
Therefore, suppose the program O3 in which B axis position is changed to 1.0, which is between the singular point and the upper limit, before Workpiece setting error compensation is activated.
Then, the range between the singular point and the upper limit of B axis movable range is the range where B axis should move.



```
O3
N10 G5.1 Q1
N20 G90 G01 B1.0 C0 F1000 ;   B axis machine position is between
                              the upper limit and the singular
                              point
N25 G54.4 P1
N30 G43.4 H1
N40 X0 Y0 Z0
N50 B90.0 C90.0
```

	B	C
Absolute	3.0	0.0
Machine	1.0	0.0





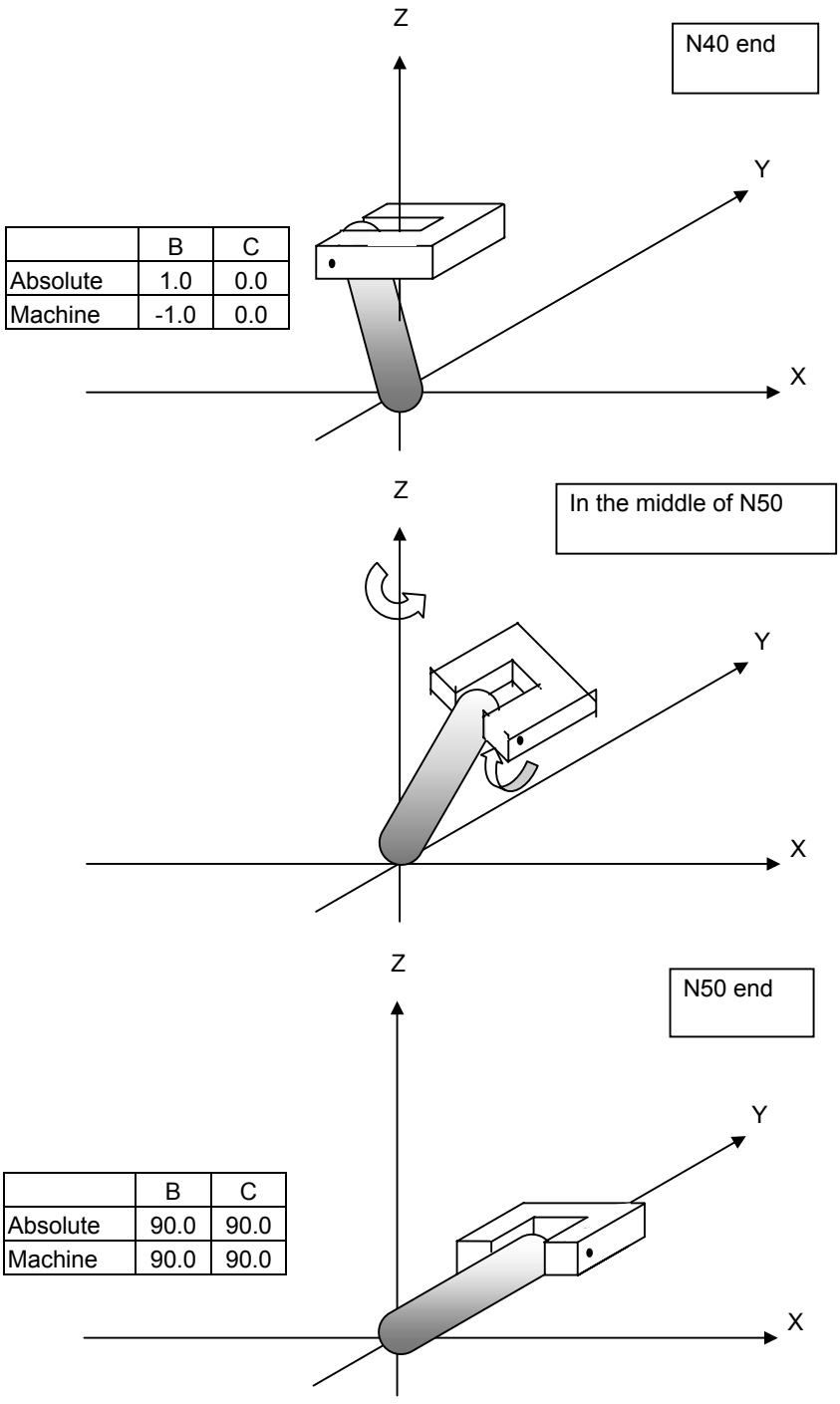
	B	C
Absolute	90.0	90.0
Machine	90.0	90.0

This time, machine position moves to B90.0,C90.0 during N50. As the result, B axis does not move over the lower limit of B axis movable range.

In O2, the case that B axis moves over the limit of movable range is the case that machine position does not make the singular point posture during N50.
When parameter No.11204 has 0, whether the tool posture is a singular point posture or not is judged strictly. So when the tool posture is almost a singular point posture, the tool posture is not regarded as a singular point posture.
When parameter No.11204 has a proper value and the tool posture in the machine coordinate system is regarded as a singular point posture, there is the possibility that the position passes the singular point.
In the above example, when parameter No.11204 has 0.1, the machine position come to make the singular point posture during N50 in O2, and B axis passes the singular point and moves to B90.0,C90.0 during N50.

```
O2
N10 G5.1 Q1
N20 G90 G01 B-1.0 C0 F1000 ;  B axis is between the lower limit
                              and the singular point.
N25 G54.4 P1
N30 G43.4 H1
N40 X0 Y0 Z0
N50 B90.0 C90.0
```

O2 is executed with the parameter No.11204=0.1.



	B	C
Absolute	1.0	0.0
Machine	-1.0	0.0

	B	C
Absolute	90.0	90.0
Machine	90.0	90.0

At N50, the machine position moves to B90.0,C90.0. As the result, B axis does not move over the lower limit of B axis movable range.

- Error for which the singular point processing is unnecessary

Even if tool direction compensation must be performed in a 5-axis machine, the above mentioned singular point processing becomes unnecessary if all workpiece setting errors in rotation directions are 0 (only X, Y, and Z directions) or there is only an error about the rotation axis closer to the workpiece.
In this case, rotation axis compensation is performed simply by shifting the rotation axis closer to the workpiece by a distance corresponding to the error; the above mentioned singular point processing is not carried out.

[Tool rotation type example]

In a tool rotation type, specify the C-axis (rotating about the Z-axis) and B-axis (rotating about the Y-axis), respectively, as the master and slave rotation axes.
The C-axis is a rotation axis closer to the workpiece.
No singular point posture processing is carried out if the workpiece setting errors in rotation directions are:
Rotation direction error Δa (rotation error about the X-axis): 0°
Rotation direction error Δb (rotation error about the Y-axis): 0°
Rotation direction error Δc (rotation error about the Z-axis): α°
The B-axis will move as directed. The C-axis will move to a position compensated by α° with respect to the specified position.

[Table rotation type example]

In a table rotation type, specify the A-axis (rotating about the X-axis) and C-axis (rotating about the Z-axis), respectively, as the master and slave rotation axes.
The C-axis is a rotation axis closer to the workpiece.
No singular point posture processing is carried out if the workpiece setting errors in rotation directions are:
Rotation direction error Δa (rotation error about the X-axis): 0°
Rotation direction error Δb (rotation error about the Y-axis): 0°
Rotation direction error Δc (rotation error about the Z-axis): α°
The A-axis will move as directed. The C-axis will move to a position compensated by α° with respect to the specified position.

[Composite type example]

In a composite type, specify the B-axis (rotating about the Y-axis) and C-axis (rotating about the Z-axis), respectively, as the tool and table rotation axes.
The C-axis is a rotation axis closer to the workpiece.
No singular point posture processing is carried out if the workpiece setting errors in rotation directions are:
Rotation direction error Δa (rotation error about the X-axis): 0°
Rotation direction error Δb (rotation error about the Y-axis): 0°
Rotation direction error Δc (rotation error about the Z-axis): α°
The B-axis will move as directed. The C-axis will move to a position compensated by α° with respect to the specified position.

- Absolute position display

Whether absolute coordinates in the workpiece setting error mode are to be displayed in the workpiece coordinate system or workpiece setting coordinate system can be chosen by using bit 6 (DAK) of parameter No. 3106.

- Custom macro variables

To system variables #5041 through #5048 (current position on each axis), the coordinates in the workpiece coordinate system are assigned.

- Range of rotary axes movements

In the case that Tilted working plane command or Type II of Tool center point control or Type II of Tool radius compensation for 5-axis machining is used, there are the parameters No.19741 to No.19744 to limit the range of rotary axes movements. When tool direction is compensated (bit 0 (RCM) of parameter No.11200=1), the rotary axes are compensated after the range of rotary axes movements are limited. Therefore, when the rotary axes are compensated, the rotary axes positions may be out of the range of rotary axes movements.
Therefore, in the case that the rotary axes have the range of movements, set the stored stroke limit check to prevent a movement over the range.

Examples

- Example 1

O1 represents a program that cuts each side of a square.

```
O1 ;
N10 G55 ;                              Set coordinate system
N20 G90 G00 X0 Y0 Z300.0 B0 C0 ;       Move to initial position
N30 G01 G43 H01 Z40.0 F500. ;          Start tool length compensation
                                           H01 is tool length compensation number.
N40 X50.0 Y50.0 Z20.0 ;                Z-axis height of machining plane is 20.0.
N50 X150.0 ;
N60 Y150.0 ;
N70 X50.0 ;
N80 Y50.0 ;
N90 X0 Y0 Z40.0 ;
N100 G49 Z300.0 ;                      Cancel tool length compensation
                                           Move to initial position on Z-axis
N110 M30;
```

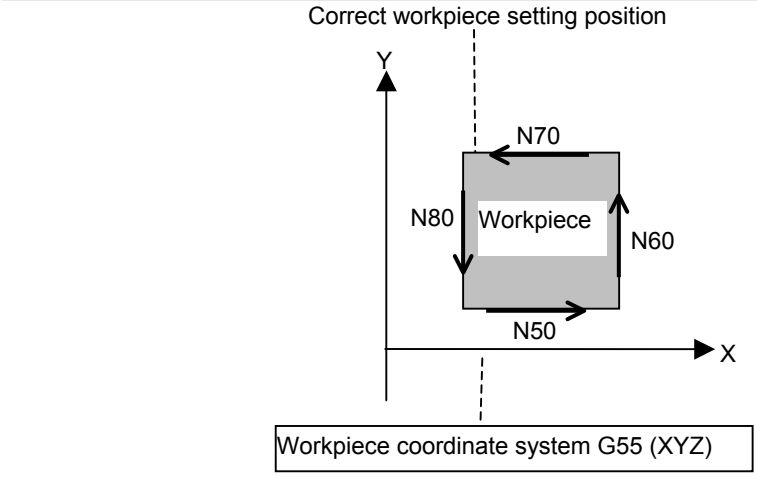


Fig. 15.5 (e) Operation when there is no workpiece setting error

Suppose that the workpiece is displaced from the "correct workpiece setting position" as shown in Fig. 15.5 (f).

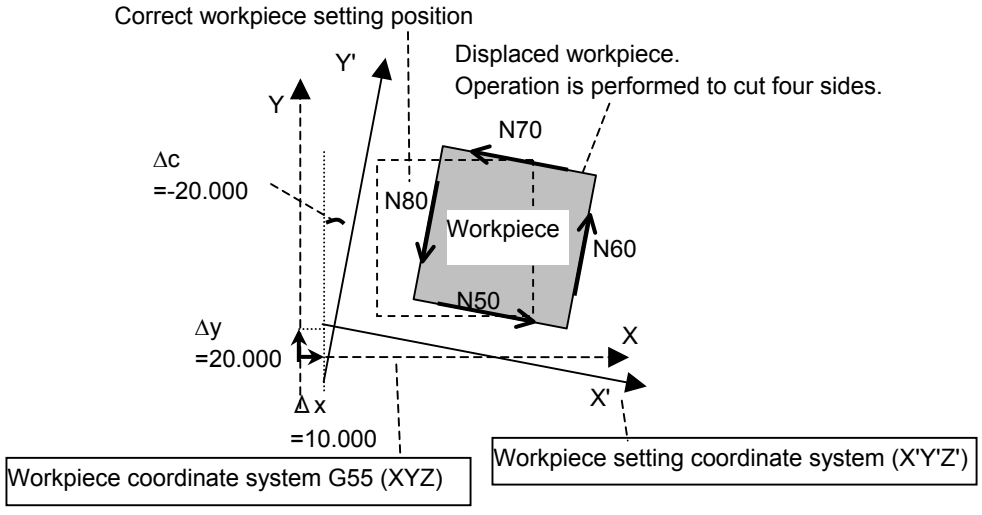


Fig. 15.5 (f) Operation when there is a workpiece setting error

The workpiece coordinate system, when rotated by -20.000 deg about the Z-axis, shifted by 10.000 in the X direction, and shifted by 20.000 in the Y direction, is to match the workpiece setting coordinate system.
At this time, set the following workpiece setting error values:

(Workpiece setting error No. 01)
x 10.000
y 20.000
z 0.000
a 0.000
b 0.000
c -20.000

To validate the workpiece setting error, add N15 and N115 to O1 as indicated below.

```
O1 ;
N10 G55 ;                          Set coordinate system
N15 G54.4 P1                       Workpiece setting error  compensation mode ON
N20 G90 G00 X0 Y0 Z300.0 B0 C0 ;   Move to initial position
N30 G01 G43 H01 Z40.0 F500. ;      Start tool length compensation
                                       H01 is tool length compensation number.
N40 X50.0 Y50.0 Z20.0 ;            Z-axis height of machining plane is 20.0.
N50 X150.0 ;
N60 Y150.0 ;
N70 X50.0 ;
N80 Y50.0 ;
N90 X0 Y0 Z40.0 ;
N100 G49 Z300.0 ;                  Cancel tool length compensation
                                       Move to initial position on Z-axis
N115 G54.4 P0 ;                    Workpiece setting error  compensation mode OFF
N110 M30;
```

When O1 is executed, the tool moves to cut each side of the displaced workpiece as shown by the solid lines in Fig. 15.5 (f) .

- Example 2

O2 is a program for cutting each side of a square by using tool center point control.
The machine is of tool rotation type, the C-axis is the master rotation axis and rotates about the Z-axis, and the B-axis is the slave axis and rotates about the Y-axis. For cutting on the plane normal to the movement direction, the tool is tilted 45 deg relative to the +Z direction.

```
O2 ;
N10 G55 ;                                  Set coordinate system
N20 G90 G00 X0 Y0 Z300.0 B0 C0 ;           Move to initial position
N30 G01 G43.4 H01 Z40.0 F500. ;            Start tool center point control
                                               H01 is tool length compensation number.
N40 X50.0 Y50.0 Z20.0 B45.0 C-90.0 ;       Z-axis height of machining plane is 20.0.
N50 X150.0 ;
N60 C0.0 ;
N70 Y150.0 ;
N80 C90.0 ;
N90 X50.0 ;
N100 C180.0 ;
N110 Y50.0 ;
N120 X0 Y0 Z40.0 B0.0 C0.0 ;
N130 G49 Z300.0 ;                          Cancel tool length compensation
                                               Move to initial position on Z- axis
N140 M30;
```

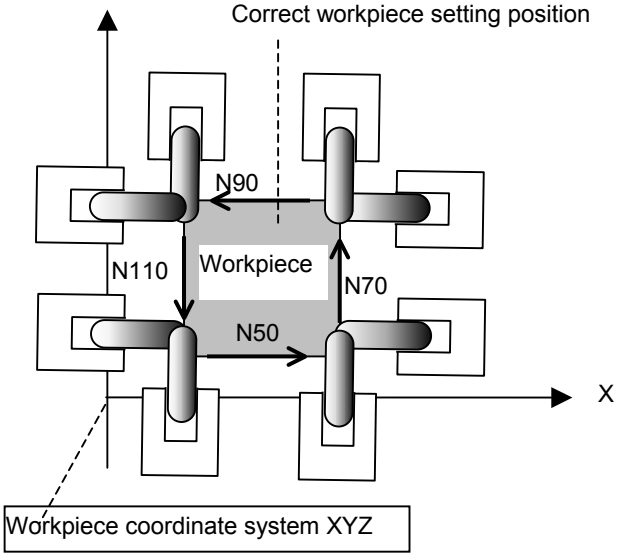


Fig. 15.5 (g) Operation of tool center point control when there is no workpiece setting error

Suppose that the workpiece is displaced from the correct workpiece setting position as with example 1, and set a workpiece setting error in the same was as in Example 1.
Moreover, add N15,N16,N135 and N136 to O2 as indicated below to validate the workpiece setting error as in the case of Example 1.

```
O2 ;
N10 G55 ;                              Set coordinate system
N15 G05.1 Q1                           AI contour control mode ON
N16 G54.4 P1                           Workpiece setting error compensation mode ON
N20 G90 G00 X0 Y0 Z300.0 B0 C0 ;       Move to initial position
N30 G01 G43.4 H01 Z40.0 F500. ;        Start tool center point control
                                           H01 is tool length compensation number.
N40 X50.0 Y50.0 Z20.0 B45.0 C-90.0 ;   Z-axis height of machining plane is 20.0.
N50 X150.0 ;
N60 C0.0 ;
N70 Y150.0 ;
N80 C90.0 ;
N90 X50.0 ;
N100 C180.0 ;
N110 Y50.0 ;
N120 X0 Y0 Z40.0 B0.0 C0.0 ;
N130 G49 Z300.0 ;                      Cancel tool length compensation
                                           Move to initial position on Z- axis
N135 G54.4 P0 ;                        Workpiece setting error compensation mode OFF
N136 G05.1 Q0 ;                        AI contour control mode OFF
N140 M30;
```

Set bit 0 (RCM) of parameter No. 11200 to 1 to perform tool direction compensation, then execute O2. When the tool moves to cut each side of the workpiece with the tool tip, the tool is tilted 45 deg relative to the +Z direction on the plane normal to the movement direction as shown in Fig. 15.5 (h).

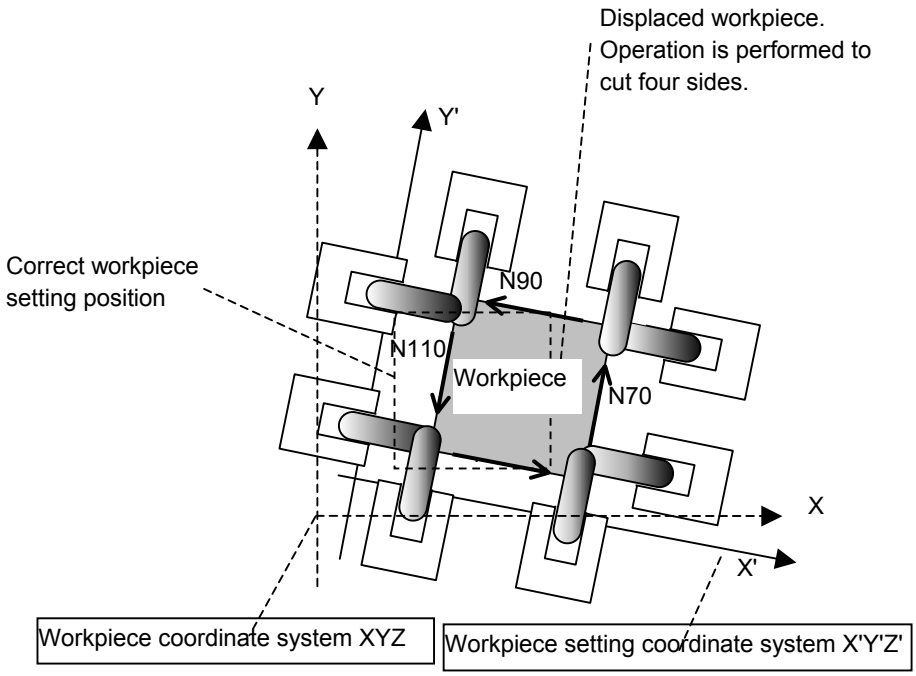


Fig. 15.5 (h) Operation of tool center point control when there is a workpiece setting error

Restrictions (general)

Described below is the restriction related to workpiece setting compensation errors.
Note that, for 5-axis machines of tool rotation or composite type, the restriction, “Restrictions (only with a 5-axis machine of table rotation or composite type),” described later, in addition to the above, is applied.

- Specifiable G codes

When workpiece setting error compensation is enabled, the G codes listed below can be specified.

G00 Positioning
G01 Linear interpolation
G04 Dwell
G05.1 Q0/Q1 AI contour control mode OFF/ON
G10 Programmable data input
G11 Programmable data input mode cancel
G17 Plane selection (XY)
G18 Plane selection (ZX)
G19 Plane selection (YZ)
G40 Tool radius · tool nose radius compensation / 3-dimensional cutter compensation cancel
G41.2/G41.4/G41.5 3-dimensional cutter compensation left (type 1) (only if also tool center point control is used)

G41.6 3-dimensional cutter compensation left (type 2)
(only if also tool center point control is used)
G42.2/G42.4/G42.5 3-dimensional cutter compensation right (type 1)
(only if also tool center point control is used)
G42.6 3-dimensional cutter compensation right (type 2)
(only if also tool center point control is used)
G43 Tool length compensation +
G43.4 Tool center point control (type 1)
G43.5 Tool center point control (type 2)
G49 (G49.1) Tool length compensation cancel
G53.1 Tool axis direction control
G65 Custom macro call
G66 Custom macro modal call
G67 Custom macro modal call cancel
G68.2 Tilted working plane command

M

G69 Coordinate system rotation/three-dimensional coordinate system conversion/tilted working plane command cancel
G90 Absolute programming
G91 Incremental programming
G94 Feed per minute
G95 Feed per revolution

T

G69.1 Coordinate system rotation/three-dimensional coordinate system conversion/tilted working plane command cancel
G90 Absolute programming (for G code system B and C)
G91 Incremental programming (for G code system B and C)
G94 Feed per minute (for G code system B and C)
G95 Feed per revolution (for G code system B and C)

- Modal G codes usable when workpiece setting error compensation is specified

In a modal G code state listed below, workpiece setting error compensation can be specified.

G00 Positioning
G01 Linear interpolation
G13.1 Polar coordinate interpolation mode cancel
G15 Polar coordinate command cancel
G17 Plane selection (XY)
G18 Plane selection (ZX)
G19 Plane selection (YZ)
G21 Metric input
G23 Stored stroke check function OFF
G25 Spindle speed fluctuation detection OFF

G40 Tool radius · tool nose radius compensation / 3-dimensional cutter compensation cancel
G49 (G49.1) Tool length compensation cancel
G50 Scaling cancel
G50.1 Programmable mirror image cancel
G50.2 Polygon turning cancel
G54 to G59, G54.1 Workpiece coordinate system selection
G64 Cutting mode
G67 Custom macro modal call cancel
G80 Canned cycle cancel
G80.5 (G80.8) Electronic gear box synchronization cancel
G97 Constant surface speed control cancel

M

G40.1 Normal direction control cancel
G54.2 P0 Rotary table dynamic fixture offset cancel
G69 Coordinate system rotation/three-dimensional coordinate system conversion/tilted working plane command cancel
G90 Absolute programming
G91 Incremental programming
G94 Feed per minute
G95 Feed per revolution
G98 Canned cycle initial level return
G99 Canned cycle R point level return

T

G69 Mirror image for double turret OFF/balanced cutting mode cancel
G69.1 Coordinate system rotation/three-dimensional coordinate system conversion/tilted working plane command cancel
G90 Absolute programming (for G code system B and C)
G91 Incremental programming (for G code system B and C)
G94 Feed per minute (for G code system B and C)
G95 Feed per revolution (for G code system B and C)
G98 Canned cycle initial level return (for G code system B and C)
G99 Canned cycle R point level return (for G code system B and C)

- Manual intervention and handle interruption

In the workpiece setting error compensation mode, do not perform manual intervention and handle interruption.

- Positioning in the machine coordinate system

In the workpiece setting error compensation mode, do not perform positioning in the machine coordinate system (such as G28, G30, or G53 command).

- Rapid traverse command

When using workpiece setting error compensation, specify linear rapid traverse (by setting set bit 1 (LRP) of parameter No. 1401 to 1).

- Relationships with other modal commands

The commands for the functions listed below must be nested with a workpiece setting error compensation command and must be placed between G54.4 P_ and G54.4 P0.
Namely, specify G54.4 P_ when the mode of a desired function below is off. Next, turn on then off the mode of the function, then specify G54.4 P0.

- Cutter compensation (G40, G41.2, G41.4, G41.5, G41.6, G42.2, G42.4, G42.5, G42.6)
- Tool length compensation (G43, G43.4, G43.5, G49)
- Tilted working plane command (G68.2, G69)

- PMC axis control

In the workpiece setting error compensation mode, do not perform PMC axis control to the axes which are for workpiece setting error compensation.

- Movement of start-up and cancel of Tool length offset

To perform tool length compensation (including tool center point control) in the workpiece setting error compensation mode, set bit 6 (TOS) of parameter No. 5006 to 1 (perform tool length compensation by shifting a coordinate system). If parameter TOS = 0, alarm PS0438 is issued.

- Mirror image

Do not use programmable mirror imaging or external mirror imaging (based on a mirror image signal or setting) during workpiece setting error compensation.

- Stroke limit check before movement

In the mode of Workpiece setting error compensation, Stroke limit check before movement is not available.

- Feedrate override

When tool direction is compensated (bit 0 (RCM) of parameter No.11200=1), the rotation axis compensation is performed to the movement which is modified by the override signals. If a large movement of rotary axis occurs for the compensation, the feedrate of the rotary axis is clamped by the speed which is generated by calculation that the parameters No.11202,No.11203 override the maximum cutting feedrate/rapid traverse.
While rotary axes move with this override, even when the override signals change, the actual feedrate may not change, and, even when feed hold or the like is done, the actual rotary axes movements may not stop immediately because automatic operation stops after the movement caused by rotation axis compensation has been completed.

> ⚠ **WARNING**
> Rotation axis compensation near a singular point causes a rotation axis to move largely, thus resulting in a symptom that it is impossible to change override or to reset feed hold during the axis movement being likely to occur. If it is necessary to stop the axis immediately during this symptom, use the emergency stop function.

- In the case that Tool posture control is used with Type-2 of Tool center point control

In the case that Tool posture control is used with Type-2 of Tool center point control, when the angle between a tool posture and the singular posture is less than the parameter No.19738, the tool posture at the end of the block is changed so that the tool posture passes the singular posture.
But, when tool direction is compensated (bit 0 (RCM) of parameter No.11200 = 1), the tool posture may not pass the singular posture because of the compensation after tool posture control.

- Distance-to-go display

The distance-to-go display is one before workpiece setting error compensation is performed.
It is not the actual machine's distance to go.

- Program restart

Do not restart a program if workpiece setting error compensation is being performed for it.

- Reset

To cancel the workpiece setting error compensation mode, be sure to use a reset by resetting bit 2 (D3R) of parameter No. 5400 to 0.
Performing tool length compensation (including tool center point control) in the workpiece setting error compensation mode causes the tool length compensation mode and tool length compensation vector to be canceled by a reset no matter what the setting of the following parameters:

- Bit 6 (CLR) of parameter No.3402
- Bit 0 (C08) of parameter No.3407
- Bit 6 (LVK) of parameter No.5003

- Setting not involving tool direction compensation

For 5-axis machines, be sure to select tool direction compensation (set bit 0 (RCM) of parameter No. 11200 to 1).
For machines other than 5-axis machines, workpiece setting error compensation is carried out with no tool direction compensation selected (bit 0 (RCM) of parameter No. 11200 is 0). However, do not specify rotation errors other than those about the tool direction. Otherwise, machining will not be performed properly. (Example: If the tool is in the Z direction, do not specify rotation errors about the X- or Y-axis.

- Synchronous slave axis absolute coordinates

If feed axis synchronization is in effect, no slave axis absolute coordinate is displayed properly.

- Workpiece coordinate system shift

Do not perform workpiece setting error compensation with G52 (local coordinate system setting) or G92/G50 (T series) (workpiece coordinate system shift) executed.
Once the workpiece coordinate system has been shifted, preset it using G92.1 or axis-specific workpiece coordinate system preset functions before starting workpiece setting error compensation.

- Canned cycle

In the workpiece setting error compensation mode, do not carry out canned cycles.

Restrictions (only with a 5-axis machine of table rotation or composite type)

For 5-axis machines of table rotation or composite type in particular, this restriction, in addition to "Restrictions (general)" above, is applied.

- Hypothetical axis used for a table rotation axis

If a hypothetical axis is used as a table rotation axis, compensation is performed with the angle of the table rotation axis set to 0 degree.

- Acceleration/deceleration at a corner

When a command for linear interpolation is specified, linear interpolation is performed as viewed from the workpiece on the table. So, even when the command specifies a linear interpolation, the control point may make a curved motion. This means that a corner operation may be performed with some commands.
So, if a small value is set as an allowable feedrate difference (parameter No. 1783) or as an allowable acceleration/deceleration rate (parameter No. 1660 and No. 1737) for a corner, the tool may be decelerated.

- G codes that must not be specified

Among the "Specifiable G codes" in "Restrictions (general)" above, the G codes listed below must not be specified when tool direction compensation is performed with a 5-axis machine of table rotation type or composite type.

M

G95 Feed per revolution

T

G95 Feed per revolution (for G code system B and C)

- Modal G codes not usable when workpiece setting error compensation is specified

Among the "Modal G codes usable when workpiece setting error compensation is specified" in "Restrictions (general)" above, the G codes listed below must not be specified with a 5-axis machine of table rotation type or composite type in the modal G code state indicated below.

M

G95 Feed per revolution

T

G95 Feed per revolution (for G code system B and C)

- Command for an axis not related to 5-axis machining

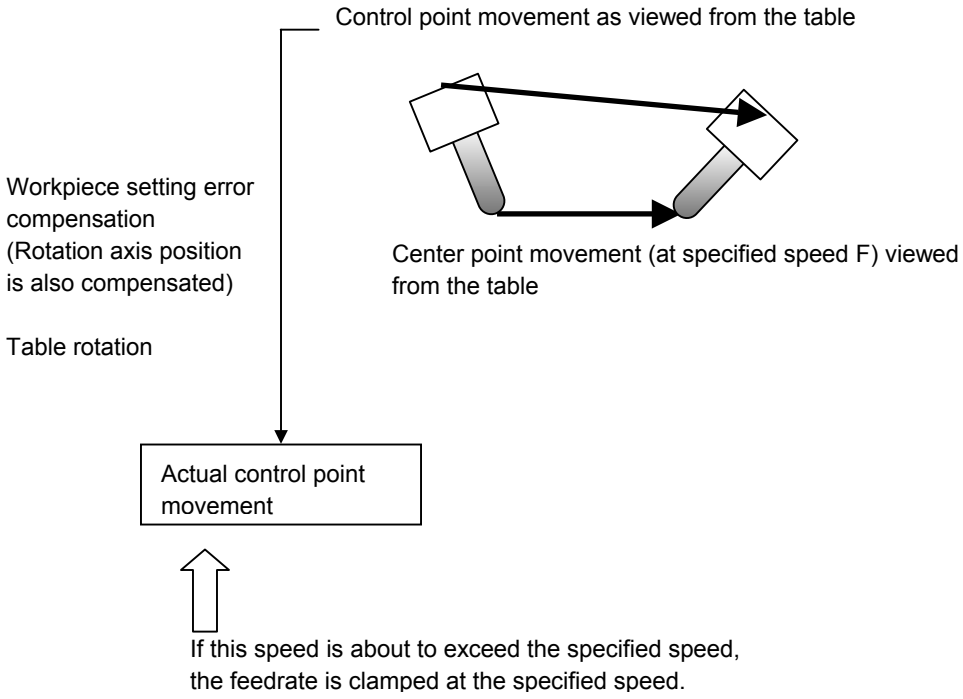
Those axes that are not related to 5-axis machining must not be specified. Otherwise, alarm PS0439 is issued.

- Combination with tool center point control

If bit 5 (WKP) of parameter No. 19696 = 0 (table coordinate system command), starting tool center point control with the table rotation axis position not set at 0 usually causes the workpiece coordinate system to be fixed at the table with it in the startup state, thus making it the table coordinate system.
If tool center point control is carried out in the workpiece setting error compensation mode, this specification becomes ineffective, and the table coordinate system is set up on the assumption that the workpiece coordinate system is fixed at the table with the table rotation axis position being at 0.

- Speed in the workpiece setting error compensation mode

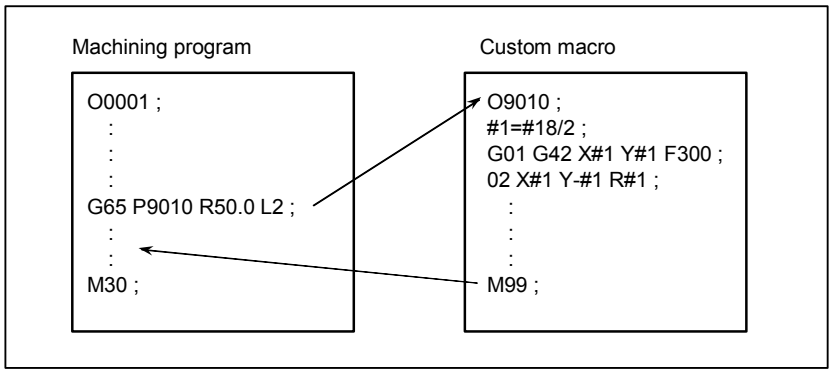
When workpiece setting error compensation is performed in a machine of table rotation or composite type, processing equivalent to tool center point control is carried out so that tool movement as viewed from the table can be controlled.
In this case, control point movement is calculated by assuming that the tool tip moves at the specified speed. If the resulting control point speed exceeds the specified speed, the feedrate is clamped at the specified speed.



Conversely, if the control point speed is lower than the specified speed, it is likely that the tool may move at a speed lower than the specified speed.

16 CUSTOM MACRO

Although subprograms are useful for repeating the same operation, the custom macro function also allows use of variables, arithmetic and logic operations, and conditional branches for easy development of general programs such as pocketing and user-defined canned cycles.
A machining program can call a custom macro with a simple command, just like a subprogram.



16.1 VARIABLES

An ordinary machining program specifies a G code and the travel distance directly with a numeric value; examples are G100 and X100.0.
With a custom macro, numeric values can be specified directly or using a variable number. When a variable number is used, the variable value can be changed by a program or using operations on the MDI panel.

```
#1=#2+100 ;
G01 X#1 F300 ;
```

Explanation

- Variable representation

When specifying a variable, specify a number sign (#) followed by a variable number.

#i (i = 1, 2, 3, 4, ...)

[Example] #5
#109
#1005

A variable can also be represented as follows using <expression> described in the section about arithmetic and logic operation commands.

#[<expression>]

[Example] #[#100]
#[#1001-1]
#[#6/2]

Variable #i shown in the following can be replaced with a variable of #[<expression>].

- Types of variables

Variables can be classified as local variables, common variables, and system variables according to the variable number. Each of those variables has its own usage and characteristics. Read-only system constants are also provided.

- Range of variable values

Local and common variables can have a value in the following ranges. If the result of calculation exceeds the range, an alarm PS0111 is issued.

When bit 0 (F16) of parameter No.6008 = 0

Maximum value: approx. $\pm 10^{308}$
Minimum value: approx. $\pm 10^{-308}$
Numeric data handled by a custom macro conforms to the IEEE standard and is handled as a double-precision real number. An error resulting from operation depends on the precision.

When bit 0 (F16) of parameter No.6008 = 1

Maximum value: approx. $\pm 10^{47}$
Minimum value: approx. $\pm 10^{-29}$

- Local variable (#1-#33)

A local variable is a variable that is used locally in a macro. That is, local variable #i used by a macro called at a certain time is different from that used by a macro called at another time, regardless of whether the two macros are the same. Therefore, for example, when macro A calls macro B during multiple calls or the like, it is impossible for macro B to corrupt a local variable used by macro A by erroneously using the variable.
A local variable is used to pass arguments. For information on correspondence between arguments and addresses, see the section about macro calling commands. The initial state of a local variable to which no arguments are passed is <null> and the user can freely use the variable. The attribute of a local variable is READ/WRITE enabled.

- Common variable (#100-#199, #500-#999)

A common variable is shared among the main program, subprograms called by the main program, and macros while a local variable is used locally in a macro. That is, #i used by a macro is the same as that used by another macro. Therefore, a resultant common variable obtained by using a macro can be used by another macro. The attribute of a common variable is basically READ/WRITE enabled. However, the common variable can be protected (its attribute is set to READ only) by specifying its variable number using parameters No.6031 and No.6032. A common variable can be freely used by the user even when its usage is not defined by the system. The number of common variables can be specified by selecting one of the following options.

(a) 100 common variables (specified only with the custom macro option)
The common variables #100 to #149 and #500 to #549 can be used. The variables #100 to #149 are cleared during power-down, but the variables #500 to #549 are retained during power-down.

(b) 600 common variables (specified with the custom macro option or custom macro common variable addition option)
The common variables #100 to #199 and #500 to #999 can be used. The variables #100 to #199 are cleared during power-down, but the variables #500 to #999 are retained during power-down.

- Write protection of a common variable

Multiple common variables (#500 to #999) can be protected (their attributes are set to READ only) by setting variable numbers in parameters No.6031 and No.6032. This protection is enabled for both Input/All Clear by MDI on the macro screen and write operation by a macro program. If the NC program specifies WRITE operation (used in the left side) for a common program in the set range, an alarm PS0116 is issued.

- System variable

A variable whose usage does not vary in the system. The attribute of a system variable is READ only, WRITE only, or READ/WRITE enabled depending on the nature of a system variable.

- System constant

A system constant can be referenced as with a variable even though its value is fixed. The attribute of a system constant is READ only.

- Omission of the decimal point

When a variable value is defined in a program, the decimal point can be omitted.

[Example]

When #1 = 123; is defined, the actual value of variable #1 is 123.000.

- Referencing variables

The value following an address can be replaced with a variable. When programming as <address>#i or <address>-#i, the variable value or the complement of it is used as the specified value of the address.

[Example] F#33 is the same as F1.5 when #33 = 1.5.
Z-#18 is the same as Z-20.0 when #18 = 20.0.
G#130 is the same as G3 when #130 = 3.0.

A variable cannot be referenced using address/, :, or O and N.

[Example] Programming such as O#27 or N#1 is not allowed.
n (n = 1 to 9) in the optional block skip /n cannot be a variable.

A variable number cannot be specified by a direct variable.

[Example] When replacing 5 in #5 with #30, specify #[#30] instead of ##30.
No values exceeding the maximum allowable value for each address can be specified.

[Example] When #140 = 120, G#140 exceeds the maximum allowable value.
When a variable is used as address data, the variable is automatically rounded off to the number of significant figures of each address or less.

[Example] For a machine with an increment system of 1/1000 mm (IS-B), when #1 = 12.3456, G00 X#1; becomes G00 X12.346;.

If <expression>, described later, is used, the value following an address can be replaced with <expression>.

<address>[<expression>] or <address>-[<expression>]

The program code shown above indicates the value of <expression> or the complement of the value is used as an address value. Note that a constant with no decimal point, enclosed in brackets ([ ]), is assumed to have a decimal point at the end.

[Example] X[#24+#18*COS[#1]]
Z-[#18+#26]

- Undefined variable

When the value of a variable is not defined, such a variable is referred to as a "null" variable. Variables #0 and #3100 are always null variables. They cannot be written to, but they can be read.

(a) Quotation
When an undefined variable is quotated, the address itself is also ignored.

Original command	G90 X100 Y#1
Equivalent command when #1 = <null>	G90 X100
Equivalent command when #1 = 0	G90 X100 Y0

(b) Definition/replacement, addition, multiplication
When a local variable or common variable is directly replaced with <null>, the result is <null>. When a system variable is directly replaced with <null> or the result of calculation including <null> is replaced, a variable value of 0 is assumed.

Original expression (local variable)	#2=#1	#2=#1*5	#2=#1+#1
Replacement result (when #1 = <null>)	<null>	0	0
Replacement result (when #1 = 0)	0	0	0

Original expression (common variable)	#100=#1	#100=#1*5	#100=#1+#1
Replacement result (when #1 = <null>)	<null>	0	0
Replacement result (when #1 = 0)	0	0	0

Original expression (system variable)	#2001=#1	#2001=#1*5	#2001=#1+#1
Replacement result (when #1 = <null>)	0	0	0
Replacement result (when #1 = 0)	0	0	0

(c) Comparison
<null> differs from 0 only for EQ and NE.
<null> is equal to 0 for GE, GT, LE, and LT.

- When <null> is assigned to #1

Conditional expression	#1 EQ #0	#1 NE 0	#1 GE #0	#1 GT 0	#1 LE #0	#1 LT 0
Evaluation result	Established (true)	Established (true)	Established (true)	Not established (false)	Established (true)	Not established (false)

- When 0 is assigned to #1

Conditional expression	#1 EQ #0	#1 NE 0	#1 GE #0	#1 GT 0	#1 LE #0	#1 LT 0
Evaluation result	Not established (false)	Not established (false)	Established (true)	Not established (false)	Established (true)	Not established (false)

- Specifying a system variable (constant) by its name

A system variable (constant) is specified by its variable number, but it can also be specified by its predetermined system variable (constant) name. A system variable (constant) name begins with an underscore (_), followed by up to seven uppercase letters, numerics, or underscores. For axis-dependent variables (such as coordinates) or variables having a lot of data of similar types (such as tool compensation), subscript [n] (n: integer) can be used to specify values. In this case, n can be specified in <expression> format (calculation format).

The command format must be specified in [#system-variable-name] format, as shown below.

[#_DATE]

[Example]

```
[#_DATE]=20040117 ;          : 2004.01.17 is assigned to #3011 (year month date).
[#_TIME]=161705 ;            : 16:17:05 is assigned to #3012 (time minute second).
#101=[#_ABSMT[1]] ;          : #5021 (machine coordinate value of the 1st axis) is read off and assigned to #101.
#102=[#_ABSKP[#500*2]] ; : #506x (skip position of [#500*2]th axis) is read off and assigned to #102.
```

If a value other than an integer is specified for subscript n, a variable value is referenced, assuming that the fractional portion is rounded off.

[Example]

```
[#_ABSIO[1.4999999]] : This value is assumed to be [#_ABSIO[1]], that is, #5001.
[#_ABSIO[1.5000000]] : This value is assumed to be [#_ABSIO[2]], that is, #5002.
```

NOTE

1 When the specified variable name is not registered, an alarm PS1098 is issued.
2 When a negative or other invalid subscript is specified, an alarm PS1099 is issued.

- System constant #0, #3100-#3102 (Attribute: R)

Constants used as fixed values in the system can be used as system variables. Such constants are called system constants. The system constants provided are shown below.

Constant number	Constant name	Description
#0, #3100	[#_EMPTY]	Null
#3101	[#_PI]	Circular constant π = 3.14159265358979323846
#3102	[#_E]	Base of natural logarithm e = 2.71828182845904523536

- Specifying a common variable by its name

Specifying a variable name set by the SETVN command described later allows reading from or writing to a common variable.
The command must be specified in the form [#common-variable-name] such as [#VAR500].

[Example]

```
X[#POS1] Y[#POS2] ;          : Specifying a position by the variable name
[#POS1] = #100+#101 ;        : Executing a assignment statement by the variable name
#[100+[#ABS]] = 500 ;        : Same as above (by a variable number)
#500 = [1000+[#POS2]*10] ;   : Reading a variable by a variable name
```

- Setting and specifying the name of a common variable (SETVN)

For the 50 common variables, #500 to #549, a name of up to eight characters can be specified by using a command as shown below.

SETVN n [VAR500, VAR501, VAR502,.....] ;

n represents the starting number of a common variable for which the name is specified.
VAR500 is the variable name of variable n, VAR501 is the variable name of variable n+1, and VAR502 is the variable name of variable number n+2, and so on. Each string is delimited by a comma (,). All codes that can be used as meaningful information in a program except control in, control out, [,], EOB, EOR, and : (colon in a program number) can be used. However, each name must begin with an alphabetical character. Variable names are not cleared on switch-off.
Specifying a set variable name allows reading from or writing to the common variable. The command must be specified in the form [#common-variable-name] such as [#VAR500].

[Example] SETVN 510[TOOL_NO, WORK_NO, COUNTER1, COUNTER2];

The command above names the variables as follows.

Variable	Name
#510	TOOL_NO
#511	WORK_NO
#512	COUNTER1
#513	COUNTER2

The names specified by the command can be used in a program. For example, when 10 is assigned to #510, the expression [#TOOL_NO]=10; can be used instead #510=10;.

NOTE

If the same name was specified for different common variables, only the variable which has the smaller variable number can be referenced with the specified name.

16.2 SYSTEM VARIABLES

System variables can be used to read and write internal CNC data such as tool compensation values and current position data. System variables are essential for automation and general-purpose program development.

List of system variables and constants

n represents a subscript.
R, W, and R/W are attributes of a variable and indicate read-only, write-only, and read/write enabled, respectively.

- Interface signals

System variable number	System variable name	Attribute	Description
#1000-#1031	[#_UI[n]]	R	Interface input signals (BIT), UI000-UI031 NOTE) Subscript n represents a BIT position (0-31).
#1032-#1035	[#_UIL[n]]	R	Interface input signals (LONG), UI000-UI031/ UI100-UI131/ UI200-UI231/UI300-UI331 NOTE) Subscript n (0-3): 0 = UI000-UI031, 1 = UI100-UI131, 2 = UI200-231, 3 = UI300-UI331
#1100-#1131	[#_UO[n]]	R/W	Interface output signals (BIT), UO000-UO031 NOTE) Subscript n represents a BIT position (0-31).
#1132-#1135	[#_UOL[n]]	R/W	Interface output signals (LONG), UO000-UO031/ UO100-UO131/UO200-UO231/UO300-UO331 NOTE) Subscript n (0-3): 0 = UO000-UO031, 1 = UO100-UO131, 2 = UO200-231, 3 = UO300-UO331

- Tool compensation value

M

Tool compensation memory A

System variable number	System variable name	Attribute	Description
#2001-#2200 #10001-#10999	[#_OFS[n]]	R/W	Tool compensation value Note) Subscript n represents a compensation number (1 to 200). When the number of sets is larger than 200, the numbers to the left can also be used. Note) Subscript n represents a compensation number (1 to 999).

Tool compensation memory B when bit 3 (V15) of parameter No.6000 = 0

System variable number	System variable name	Attribute	Description
#2001-#2200	[#_OFSW[n]]	R/W	Tool compensation value (wear) Note) Subscript n represents a compensation number (1 to 200).
#10001-#10999			The numbers to the left can also be used. Note) Subscript n represents a compensation number (1 to 999).
#2201-#2400	[#_OFSG[n]]	R/W	Tool compensation value (H code, geometry) Note) Subscript n represents a compensation number (1 to 200).
#11001-#11999			The numbers to the left can also be used. Note) Subscript n represents a compensation number (1 to 999).

Tool compensation memory B when bit 3 (V15) of parameter No.6000 = 1

System variable number	System variable name	Attribute	Description
#2001-#2200	[#_OFSG[n]]	R/W	Tool compensation value (geometry) Note) Subscript n represents a compensation number (1 to 200).
#10001-#10999			The numbers to the left can also be used. Note) Subscript n represents a compensation number (1 to 999).
#2201-#2400	[#_OFSW[n]]	R/W	Tool compensation value (H code, wear) Note) Subscript n represents a compensation number (1 to 200).
#11001-#11999			The numbers to the left can also be used. Note) Subscript n represents a compensation number (1 to 999).

Tool compensation memory C when bit 3 (V15) of parameter No.6000 = 0

System variable number	System variable name	Attribute	Description
#2001-#2200	[#_OFSHW[n]]	R/W	Tool compensation value (H code, wear) Note) Subscript n represents a compensation number (1 to 200).
#10001-#10999			The numbers to the left can also be used. Note) Subscript n represents a compensation number (1 to 999).
#2201-#2400	[#_OFSHG[n]]	R/W	Tool compensation value (H code, geometry) Note) Subscript n represents a compensation number (1 to 200).
#11001-#11999			The numbers to the left can also be used. Note) Subscript n represents a compensation number (1 to 999).
#12001-#12999	[#_OFSDW[n]]	R/W	Tool compensation value (D code, wear) Note) Subscript n represents a compensation number (1 to 999).
# 13001-#13999	[#_OFSDG[n]]	R/W	Tool compensation (D code, geometry) Note) Subscript n represents a compensation number (1 to 999).

Tool compensation memory C when bit 3 (V15) of parameter No.6000 = 1

System variable number	System variable name	Attribute	Description
#2001-#2200	[#_OFSHG[n]]	R/W	Tool compensation value (H code, geometry) Note) Subscript n represents a compensation number (1 to 200).
#10001-#10999			The numbers to the left can also be used. Note) Subscript n represents a compensation number (1 to 999).
#2201-#2400	[#_OFSHW[n]]	R/W	Tool compensation value (H code, wear) Note) Subscript n represents a compensation number (1 to 200).
#11001-#11999			The numbers to the left can also be used. Note) Subscript n represents a compensation number (1 to 999).
#2401-#2600	[#_OFSDG[n]]	R/W	Tool compensation value (D code, geometry) (Note 2) Note 1) Subscript n represents a compensation number (1 to 200). Note 2) Enabled when bit 5 (D15) of parameter No.6004 = 1.
#12001-#12999			The numbers to the left can also be used. Note) Subscript n represents a compensation number (1 to 999).
#2601-#2800	[#_OFSDW[n]]	R/W	Tool compensation value (D code, wear) (Note 2) Note 1) Subscript n represents a compensation number (1 to 200). Note 2) Enabled when bit 5 (D15) of parameter No.6004 = 1.
#13001-#13999			The numbers to the left can also be used. Note) Subscript n represents a compensation number (1 to 999).

- Tool compensation value

T

Without tool geometry/wear compensation memory

System variable number	System variable name	Attribute	Description
#2001-#2064 #10001-#10999	[#_OFSX[n]]	R/W	X-axis compensation value (*1) Note) Subscript n represents a compensation number (1 to 64). When the number of sets is larger than 64, the numbers to the left can be used. Note) Subscript n represents a compensation number (1 to 999).
#2101-#2164 #11001-#11999	[#_OFSZ[n]]	R/W	Z-axis compensation value (*1) Note) Subscript n represents a compensation number (1 to 64). When the number of sets is larger than 64, the numbers to the left can be used. Note) Subscript n represents a compensation number (1 to 999).
#2201-#2264 #12001-#12999	[#_OFSR[n]]	R/W	Tool nose radius compensation value Note) Subscript n represents a compensation number (1 to 64). When the number of sets is larger than 64, the numbers to the left can be used. Note) Subscript n represents a compensation number (1 to 999).
#2301-#2364 #13001-#13999	[#_OFST[n]]	R/W	Virtual tool tip T position Note) Subscript n represents a compensation number (1 to 64). When the number of sets is larger than 64, the numbers to the left can be used. Note) Subscript n represents a compensation number (1 to 999).
#2401-#2449 #14001-#14999	[#_OFSY[n]]	R/W	Y-axis compensation value (*1) Note) Subscript n represents a compensation number (1 to 49). When the number of sets is larger than 49, the numbers to the left can be used. Note) Subscript n represents a compensation number (1 to 999).

(*1) X-axis: X-axis of basic three axes, Z-axis: Z-axis of basic three axes, Y-axis: Y-axis of basic three axes

With tool geometry/wear compensation memory

System variable number	System variable name	Attribute	Description
#2001-#2064 #10001-#10999	[#_OFSXW[n]]	R/W	X-axis compensation value (wear) (*1) Note) Subscript n represents a compensation number (1 to 64). When the number of sets is larger than 64, the numbers to the left can be used. Note) Subscript n represents a compensation number (1 to 999).
#2101-#2164 #11001-#11999	[#_OFSZW[n]]	R/W	Z-axis compensation value (wear) (*1) Note) Subscript n represents a compensation number (1 to 64). When the number of sets is larger than 64, the numbers to the left can be used. Note) Subscript n represents a compensation number (1 to 999).
#2201-#2264 #12001-#12999	[#_OFSRW[n]]	R/W	Tool nose radius compensation value (wear) Note) Subscript n represents a compensation number (1 to 64). When the number of sets is larger than 64, the numbers to the left can be used. Note) Subscript n represents a compensation number (1 to 999).
#2301-#2364 #13001-#13999	[#_OFST[n]]	R/W	Virtual tool tip T position Note) Subscript n represents a compensation number (1 to 64). When the number of sets is larger than 64, the numbers to the left can be used. Note) Subscript n represents a compensation number (1 to 999).
#2401-#2449 #14001-#14999	[#_OFSYW[n]]	R/W	Y-axis compensation value (wear) (*1) Note) Subscript n represents a compensation number (1 to 49). When the number of sets is larger than 49, the numbers to the left can be used. Note) Subscript n represents a compensation number (1 to 999).
#2451-#2499 #19001-#19999	[#_OFSYG[n]]	R/W	Y-axis compensation value (geometry) (*1) Note) Subscript n represents a compensation number (1 to 49). When the number of sets is larger than 49, the numbers to the left can be used. Note) Subscript n represents a compensation number (1 to 999).
#2701-#2749 #15001-#15999	[#_OFSXG[n]]	R/W	X-axis compensation value (geometry) (*1) Note) Subscript n represents a compensation number (1 to 49). When the number of sets is larger than 49, the numbers to the left can be used. Note) Subscript n represents a compensation number (1 to 49).

System variable number	System variable name	Attribute	Description
#2801-#2849 #16001-#16999	[#_OFSZG[n]]	R/W	Z-axis compensation value (geometry) (*1) Note) Subscript n represents a compensation number (1 to 49). When the number of sets is larger than 49, the numbers to the left can be used. Note) Subscript n represents a compensation number (1 to 999).
#2901-#2964 #17001-#17999	[#_OFSRG[n]]	R/W	Tool nose radius compensation value (geometry) Note) Subscript n represents a compensation number (1 to 64). When the number of sets is larger than 64, the numbers to the left can be used. Note) Subscript n represents a compensation number (1 to 999).

(*1) X-axis: X-axis of basic three axes, Z-axis: Z-axis of basic three axes, Y-axis: Y-axis of basic three axes

- Workpiece coordinate system shift amount

T

System variable number	System variable name	Attribute	Description
#2501	[#_WKSFTX]	R/W	X-axis workpiece shift amount
#2601	[#_WKSFTZ]	R/W	Z-axis workpiece shift amount
#100751 to #100800	[#_WZ_SFT [n]]	R/W	nth axis workpiece shift amount Note) Subscript n represents an axis number (1 to 50).

X-axis: X-axis of basic three axes, Z-axis: Z-axis of basic three axes

- Automatic operation or the like

System variable number	System variable name	Attribute	Description
#3000	[#_ALM]	W	Macro alarm
#3001	[#_CLOCK1]	R/W	Clock 1 (ms)
#3002	[#_CLOCK2]	R/W	Clock 2 (hr)
#3003	[#_CNTL1]	R/W	Enable or disable the suppression of single block stop. Enable or disable the waiting of the auxiliary function completion signal.
#3003 bit 0	[#_M_SBK]	R/W	Enable or disable the suppression of single block stop.
#3003 bit 1	[#_M_FIN]	R/W	Enable or disable waiting for the auxiliary function completion signal.
#3004	[#_CNTL2]	R/W	Enable or disable feed hold. Enable or disable feedrate override. Enable or disable exact stop check.
#3004 bit 0	[#_M_FHD]	R/W	Enable or disable feed hold.
#3004 bit 1	[#_M_OV]	R/W	Enable or disable feedrate override.
#3004 bit 2	[#_M_EST]	R/W	Enable or disable exact stop check.
#3005	[#_SETDT]	R/W	Read/write setting data.

System variable number	System variable name	Attribute	Description
#3006	[#_MSGSTP]	W	Stop with a message.
#3007	[#_MRIMG]	R	Status of a mirror image (DI and setting)
#3008	[#_PRSTR]	R	Restarting/not restarting a program

- Time

System variable number	System variable name	Attribute	Description
#3011	[#_DATE]	R	Year/Month/Date
#3012	[#_TIME]	R	Hour/Minute/Second

- Number of parts

System variable number	System variable name	Attribute	Description
#3901	[#_PRTSA]	R/W	Total number of parts
#3902	[#_PRTSN]	R/W	Number of required parts

- Tool compensation memory

M

System variable number	System variable name	Attribute	Description
#3980	[#_OFSMEM]	R	Tool compensation memory information

- Main program number

System variable number	System variable name	Attribute	Description
#4000	[#_MAINO]	R	Main program number

- Modal information

M

System variable number	System variable name	Attribute	Description
#4001-#4030	[#_BUFG[n]]	R	Modal information on blocks that have been specified by last minute (G code) Note) Subscript n represents a G code group number.
#4102	[#_BUFB]	R	Modal information on blocks that have been specified by last minute (B code)
#4107	[#_BUFD]	R	Modal information on blocks that have been specified by last minute (D code)
#4108	[#_BUFE]	R	Modal information on blocks that have been specified by last minute (E code)
#4109	[#_BUFF]	R	Modal information on blocks that have been specified by last minute (F code)
#4111	[#_BUFH]	R	Modal information on blocks that have been specified by last minute (H code)
#4113	[#_BUFM]	R	Modal information on blocks that have been specified by last minute (M code)
#4114	[#_BUFN]	R	Modal information on blocks that have been specified by last minute (sequence number)
#4115	[#_BUFO]	R	Modal information on blocks that have been specified by last minute (program number)
#4119	[#_BUFS]	R	Modal information on blocks that have been specified by last minute (S code)
#4120	[#_BUFT]	R	Modal information on blocks that have been specified by last minute (T code)
#4130	[#_BUFWZP]	R	Modal information on blocks that have been specified by last minute (additional workpiece coordinate system number)
#4201-#4230	[#_ACTG[n]]	R	Modal information on the block currently being executed (G code) Note) Subscript n represents a G code group number.
#4302	[#_ACTB]	R	Modal information on the block currently being executed (B code)
#4307	[#_ACTD]	R	Modal information on the block currently being executed (D code)
#4308	[#_ACTE]	R	Modal information on the block currently being executed (E code)
#4309	[#_ACTF]	R	Modal information on the block currently being executed (F code)
#4311	[#_ACTH]	R	Modal information on the block currently being executed (H code)
#4313	[#_ACTM]	R	Modal information on the block currently being executed (M code)
#4314	[#_ACTN]	R	Modal information on the block currently being executed (sequence number)
#4315	[#_ACTO]	R	Modal information on the block currently being executed (program number)
#4319	[#_ACTS]	R	Modal information on the block currently being executed (S code)

System variable number	System variable name	Attribute	Description
#4320	[#_ACTT]	R	Modal information on the block currently being executed (T code)
#4330	[#_ACTWZP]	R	Modal information on the block currently being executed (additional workpiece coordinate system number)
#4401-#4430	[#_INTG[n]]	R	Modal information on interrupted blocks (G code) Note) Subscript n represents a G code group number.
#4502	[#_INTB]	R	Modal information on interrupted blocks (B code)
#4507	[#_INTD]	R	Modal information on interrupted blocks (D code)
#4508	[#_INTE]	R	Modal information on interrupted blocks (E code)
#4509	[#_INTF]	R	Modal information on interrupted blocks (F code)
#4511	[#_INTH]	R	Modal information on interrupted blocks (H code)
#4513	[#_INTM]	R	Modal information on interrupted blocks (M code)
#4514	[#_INTN]	R	Modal information on interrupted blocks (sequence number)
#4515	[#_INTO]	R	Modal information on interrupted blocks (program number)
#4519	[#_INTS]	R	Modal information on interrupted blocks (S code)
#4520	[#_INTT]	R	Modal information on interrupted blocks (T code)
#4530	[#_INTWZP]	R	Modal information on interrupted blocks (additional workpiece coordinate system number)

T

System variable number	System variable name	Attribute	Description
#4001-#4030	[#_BUFG[n]]	R	Modal information on blocks that have been specified by last minute (G code) Note) Subscript n represents a G code group number.
#4108	[#_BUFE]	R	Modal information on blocks that have been specified by last minute (E code)
#4109	[#_BUFF]	R	Modal information on blocks that have been specified by last minute (F code)
#4113	[#_BUFM]	R	Modal information on blocks that have been specified by last minute (M code)
#4114	[#_BUFN]	R	Modal information on blocks that have been specified by last minute (sequence number)
#4115	[#_BUFO]	R	Modal information on blocks that have been specified by last minute (program number)
#4119	[#_BUFS]	R	Modal information on blocks that have been specified by last minute (S code)

System variable number	System variable name	Attribute	Description
#4120	[#_BUFT]	R	Modal information on blocks that have been specified by last minute (T code)
#4130	[#_BUFWZP]	R	Modal information on blocks that have been specified by last minute (additional workpiece coordinate system number)
#4201-#4230	[#_ACTG[n]]	R	Modal information on the block currently being executed (G code) Note) Subscript n represents a G code group number.
#4308	[#_ACTE]	R	Modal information on the block currently being executed (E code)
#4309	[#_ACTF]	R	Modal information on the block currently being executed (F code)
#4313	[#_ACTM]	R	Modal information on the block currently being executed (M code)
#4314	[#_ACTN]	R	Modal information on the block currently being executed (sequence number)
#4315	[#_ACTO]	R	Modal information on the block currently being executed (program number)
#4319	[#_ACTS]	R	Modal information on the block currently being executed (S code)
#4320	[#_ACTT]	R	Modal information on the block currently being executed (T code)
#4330	[#_ACTWZP]	R	Modal information on the block currently being executed (additional workpiece coordinate system number)
#4401-#4430	[#_INTG[n]]	R	Modal information on interrupted blocks (G code) Note) Subscript n represents a G code group number.
#4508	[#_INTE]	R	Modal information on interrupted blocks (E code)
#4509	[#_INTF]	R	Modal information on interrupted blocks (F code)
#4513	[#_INTM]	R	Modal information on interrupted blocks (M code)
#4514	[#_INTN]	R	Modal information on interrupted blocks (sequence number)
#4515	[#_INTO]	R	Modal information on interrupted blocks (program number)
#4519	[#_INTS]	R	Modal information on interrupted blocks (S code)
#4520	[#_INTT]	R	Modal information on interrupted blocks (T code)
#4530	[#_INTWZP]	R	Modal information on interrupted blocks (additional workpiece coordinate system number)

- Position information

System variable number	System variable name	Attribute	Description
#5001-#5020	[#_ABSIO[n]]	R	End point position of the previous block (workpiece coordinate system) Note) Subscript n represents an axis number (1 to 20)
#100001-#100050			The numbers to the left can also be used. Note) Subscript n represents an axis number (1 to 50).
#5021-#5040	[#_ABSMT[n]]	R	Specified current position (machine coordinate system) Note) Subscript n represents an axis number (1 to 20).
#100051-#100100			The numbers to the left can also be used. Note) Subscript n represents an axis number (1 to 50).
#5041-#5060	[#_ABSOT[n]]	R	Specified current position (workpiece coordinate system) Note) Subscript n represents an axis number (1 to 20).
#100101-#100150			The numbers to the left can also be used. Note) Subscript n represents an axis number (1 to 50).
#5061-#5080	[#_ABSKP[n]]	R	Skip position (workpiece coordinate system) Note) Subscript n represents an axis number (1 to 20).
#100151-#100200			The numbers to the left can also be used. Note) Subscript n represents an axis number (1 to 50).

- Tool length compensation value

M

System variable number	System variable name	Attribute	Description
#5081-#5100	[#_TOFS[n]]	R	Tool length compensation value Note) Subscript n represents an axis number (1 to 20).
#100201-#100250			The numbers to the left can also be used. Note) Subscript n represents an axis number (1 to 50).

- Tool offset value

T

System variable number	System variable name	Attribute	Description
#5081, #100201	[#_TOFSWX]	R	X-axis tool offset (wear)
#5082, #100202	[#_TOFSWZ]		Z-axis tool offset (wear)
#5083, #100203	[#_TOFSWY]		Y-axis tool offset (wear)
#5084 - #5100	[#_TOFS[n]]		Tool offset (wear) for an arbitrary axis Note) Subscript n represents an axis number (4 to 20).
#100204 - #100250			Tool offset (wear) for an arbitrary axis Note) Subscript n represents an axis number (4 to 50).
#5121, #100901	[#_TOFSGX]	R	X-axis tool offset (geometry)
#5122, #100902	[#_TOFSGZ]		Y-axis tool offset (geometry)
#5123, #100903	[#_TOFSGY]		Z-axis tool offset (geometry)
#5124 - #5140	[#_TOFSG[n]]		Tool offset (geometry) for an arbitrary axis Note) Subscript n represents an axis number (4 to 20).
#100904 - #100950			Tool offset (geometry) for an arbitrary axis Note) Subscript n represents an axis number (4 to 50).

X-axis: X-axis of basic three axes, Z-axis: Z-axis of basic three axes, Y-axis: Y-axis of basic three axes

- Servo position deviation

System variable number	System variable name	Attribute	Description
#5101-#5120	[#_SVERR[n]]	R	Servo positional deviation Note) Subscript n represents an axis number (1 to 20).
#100251-#100300			The numbers to the left can also be used. Note) Subscript n represents an axis number (1 to 50).

- Manual handle interruption

System variable number	System variable name	Attribute	Description
#5121-#5140	[#_MIRTP[n]]	R	Manual handle interruption Note) Subscript n represents an axis number (1 to 20).
#100651-#100700			The numbers to the left can also be used. Note) Subscript n represents an axis number (1 to 50).

- Distance to go

System variable number	System variable name	Attribute	Description
#5181-#5200	[#_DIST[n]]	R	Distance to go Note) Subscript n represents an axis number (1 to 20).
#100801-#100850			The numbers to the left can also be used. Note) Subscript n represents an axis number (1 to 50).

- Workpiece origin offset value, extended workpiece origin offset value

M

System variable number	System variable name	Attribute	Description
#5201-#5220	[#_WZCMN[n]]	R/W	External workpiece origin offset value Note) Subscript n represents an axis number (1 to 20).
#5221-#5240	[#_WZG54[n]]	R/W	G54 workpiece origin offset value Note) Subscript n represents an axis number (1 to 20).
#5241-#5260	[#_WZG55[n]]	R/W	G55 workpiece origin offset value Note) Subscript n represents an axis number (1 to 20).
#5261-#5280	[#_WZG56[n]]	R/W	G56 workpiece origin offset value Note) Subscript n represents an axis number (1 to 20).
#5281-#5300	[#_WZG57[n]]	R/W	G57 workpiece origin offset value Note) Subscript n represents an axis number (1 to 20).
#5301-#5320	[#_WZG58[n]]	R/W	G58 workpiece origin offset value Note) Subscript n represents an axis number (1 to 20).
#5321-#5340	[#_WZG59[n]]	R/W	G59 workpiece origin offset value Note) Subscript n represents an axis number (1 to 20).
#100301-#100350	[#_WZCMN[n]]	R/W	External workpiece origin offset value Note) Subscript n represents an axis number (1 to 50).
#100351-#100400	[#_WZG54[n]]	R/W	G54 workpiece origin offset value Note) Subscript n represents an axis number (1 to 50).
#100401-#100450	[#_WZG55[n]]	R/W	G55 workpiece origin offset value Note) Subscript n represents an axis number (1 to 50).
#100451-#100500	[#_WZG56[n]]	R/W	G56 workpiece origin offset value Note) Subscript n represents an axis number (1 to 50).
#100501-#100550	[#_WZG57[n]]	R/W	G57 workpiece origin offset value Note) Subscript n represents an axis number (1 to 50).

System variable number	System variable name	Attribute	Description
#100551-#100600	[#_WZG58[n]]	R/W	G58 workpiece origin offset value Note) Subscript n represents an axis number (1 to 50).
#100601-#100650	[#_WZG59[n]]	R/W	G59 workpiece origin offset value Note) Subscript n represents an axis number (1 to 50).
Extended workpiece origin offset value			
#7001-#7020	[#_WZP1[n]]	R/W	G54.1P1 workpiece origin offset value Note) Subscript n represents an axis number (1 to 20).
#7021-#7040	[#_WZP2[n]]	R/W	G54.1P2 workpiece origin offset value Note) Subscript n represents an axis number (1 to 20).
: :	: :	: :	: :
#7941-#7960	[#_WZP48[n]]	R/W	G54.1P48 workpiece origin offset value Note) Subscript n represents an axis number (1 to 20).
#14001-#14020	[#_WZP1[n]]	R/W	G54.1P1 workpiece origin offset value Note) Subscript n represents an axis number (1 to 20).
#14021-#14040	[#_WZP2[n]]	R/W	G54.1P2 workpiece origin offset value Note) Subscript n represents an axis number (1 to 20).
: :	: :	: :	: :
#19981-#20000	[#_WZP300[n]]	R/W	G54.1P300 workpiece origin offset value Note) Subscript n represents an axis number (1 to 20).
#101001-#101050	[#_WZP1[n]]	R/W	G54.1P1 workpiece origin offset value Note) Subscript n represents an axis number (1 to 50).
#101051-#101100	[#_WZP2[n]]	R/W	G54.1P2 workpiece origin offset value Note) Subscript n represents an axis number (1 to 50).
: :	: :	: :	: :
#115901-#115950	[#_WZP299[n]]	R/W	G54.1P299 workpiece origin offset value Note) Subscript n represents an axis number (1 to 50).
#115951-#116000	[#_WZP300[n]]	R/W	G54.1P300 workpiece origin offset value Note) Subscript n represents an axis number (1 to 50).

T

System variable number	System variable name	Attribute	Description
#5201-#5220	[#_WZCMN[n]]	R/W	External workpiece origin offset value Note) Subscript n represents an axis number (1 to 20).
#5221-#5240	[#_WZG54[n]]	R/W	G54 workpiece origin offset value Note) Subscript n represents an axis number (1 to 20).
#5241-#5260	[#_WZG55[n]]	R/W	G55 workpiece origin offset value Note) Subscript n represents an axis number (1 to 20).
#5261-#5280	[#_WZG56[n]]	R/W	G56 workpiece origin offset value Note) Subscript n represents an axis number (1 to 20).
#5281-#5300	[#_WZG57[n]]	R/W	G57 workpiece origin offset value Note) Subscript n represents an axis number (1 to 20).
#5301-#5320	[#_WZG58[n]]	R/W	G58 workpiece origin offset value Note) Subscript n represents an axis number (1 to 20).
#5321-#5340	[#_WZG59[n]]	R/W	G59 workpiece origin offset value Note) Subscript n represents an axis number (1 to 20).
#100301-#100350	[#_WZCMN[n]]	R/W	External workpiece origin offset value Note) Subscript n represents an axis number (1 to 50).
#100351-#100400	[#_WZG54[n]]	R/W	G54 workpiece origin offset value Note) Subscript n represents an axis number (1 to 50).
#100401-#100450	[#_WZG55[n]]	R/W	G55 workpiece origin offset value Note) Subscript n represents an axis number (1 to 50).
#100451-#100500	[#_WZG56[n]]	R/W	G56 workpiece origin offset value Note) Subscript n represents an axis number (1 to 50).
#100501-#100550	[#_WZG57[n]]	R/W	G57 workpiece origin offset value Note) (1 to 50).
#100551-#100600	[#_WZG58[n]]	R/W	G58 workpiece origin offset value Note) Subscript n represents an axis number (1 to 50).
#100601-#100650	[#_WZG59[n]]	R/W	G59 workpiece origin offset value Note) Subscript n represents an axis number (1 to 50).
Extended workpiece origin offset value			
#7001-#7020	[#_WZP1[n]]	R/W	G54.1P1 workpiece origin offset value Note) Subscript n represents an axis number (1 to 20).
#7021-#7040	[#_WZP2[n]]	R/W	G54.1P2 workpiece origin offset value Note) Subscript n represents an axis number (1 to 20).
: :	: :	: :	: :

System variable number	System variable name	Attribute	Description
#7941-#7960	[#_WZP48[n]]	R/W	G54.1P48 workpiece origin offset value Note) Subscript n represents an axis number (1 to 20).
#101001-#101050	[#_WZP1[n]]	R/W	G54.1P1 workpiece origin offset value Note) Subscript n represents an axis number (1 to 50).
#101051-#101100	[#_WZP2[n]]	R/W	G54.1P2 workpiece origin offset value Note) Subscript n represents an axis number (1 to 50).
: :	: :	: :	: :
#115901-#115950	[#_WZP299[n]]	R/W	G54.1P299 workpiece origin offset value Note) (1 to 50).
#115951-#116000	[#_WZP300[n]]	R/W	G54.1P300 workpiece origin offset value Note) Subscript n represents an axis number (1 to 50).

- Skip position (detection unit)

System variable number	System variable name	Attribute	Description
#5421-#5440	[#_SKPDTC[n]]	R	Skip position (detection unit) Note) Subscript n represents an axis number (1 to 20).
#100701-#100750			The numbers to the left can also be used. Note) Subscript n represents an axis number (1 to 50).

- Rotary table dynamic fixture offset value

M

System variable number	System variable name	Attribute	Description
#5500	[#_FOFSP]	R	Number of the standard fixture offset being selected (P)
#5501-#5520	[#_FOFSVAL[n]]	R	Offset value of the standard fixture offset being selected Note) Subscript n represents an axis number (1 to 20).
#117001-#117050			The numbers to the left can also be used. Note) Subscript n represents an axis number (1 to 50).
#5521-#5540	[#_FOFS1[n]]	R/W	Standard fixture offset value (first set) Note) Subscript n represents an axis number (1 to 20).
#117051-#117100			The numbers to the left can also be used. Note) Subscript n represents an axis number (1 to 50).

System variable number	System variable name	Attribute	Description
#5541-#5560	[#_FOFS2[n]]	R/W	Standard fixture offset value (second set) Note) Subscript n represents an axis number (1 to 20).
#117101-#117150			The numbers to the left can also be used. Note) Subscript n represents an axis number (1 to 50).
#5561-#5580	[#_FOFS3[n]]	R/W	Standard fixture offset value (third set) Note) Subscript n represents an axis number (1 to 20).
#117151-#117200			The numbers to the left can also be used. Note) Subscript n represents an axis number (1 to 50).
#5581-#5600	[#_FOFS4[n]]	R/W	Standard fixture offset value (fourth set) Note) Subscript n represents an axis number (1 to 20).
#117201-#117250			The numbers to the left can also be used. Note) Subscript n represents an axis number (1 to 50).
#5601-#5620	[#_FOFS5[n]]	R/W	Standard fixture offset value (fifth set) Note) Subscript n represents an axis number (1 to 20).
#117251-#117300			The numbers to the left can also be used. Note) Subscript n represents an axis number (1 to 50).
#5621-#5640	[#_FOFS6[n]]	R/W	Standard fixture offset value (sixth set) Note) Subscript n represents an axis number (1 to 20).
#117301-#117350			The numbers to the left can also be used. Note) Subscript n represents an axis number (1 to 50).
#5641-#5660	[#_FOFS7[n]]	R/W	Standard fixture offset value (seventh set) Note) Subscript n represents an axis number (1 to 20).
#117351-#117400			The numbers to the left can also be used. Note) Subscript n represents an axis number (1 to 50).
#5661-#5680	[#_FOFS8[n]]	R/W	Standard fixture offset value (eighth set) Note) Subscript n represents an axis number (1 to 20).
#117401-#117450			The numbers to the left can also be used. Note) Subscript n represents an axis number (1 to 50).

- Second geometry tool offset value

T

System variable number	System variable name	Attribute	Description
#5801-#5832 #27001-#27999	[#_OFSX2G[n]]	R/W	Second geometry tool offset X-axis compensation value Note) Subscript n represents a compensation number (1 to 32). If the number of pairs is greater than 32, the numbers on the left are also permitted. Note) Subscript n represents a compensation number (1 to 999).
#5833-#5864 #28001-#28999	[#_OFSZ2G[n]]	R/W	Second geometry tool offset Z-axis compensation value Note) Subscript n represents a compensation number (1 to 32). If the number of pairs is greater than 32, the numbers on the left are also permitted. Note) Subscript n represents a compensation number (1 to 999).
#5865-#5896 #29001-#29999	[#_OFSY2G[n]]	R/W	Second geometry tool offset Y-axis compensation value Note) Subscript n represents a compensation number (1 to 32). If the number of pairs is greater than 32, the numbers on the left are also permitted. Note) Subscript n represents a compensation number (1 to 999).

- Dynamic standard tool compensation value

M

System variable number	System variable name	Attribute	Description
#118051-#118100	[#_DOFS1[n]]	R/W	Dynamic standard tool compensation value (first set) Note) Subscript n represents an axis number (1 to 50).
#118101-#118150	[#_DOFS2[n]]	R/W	Dynamic standard tool compensation value (second set) Note) Subscript n represents an axis number (1 to 50).
#118151-#118200	[#_DOFS3[n]]	R/W	Dynamic standard tool compensation value (third set) Note) Subscript n represents an axis number (1 to 50).
#118201-#118250	[#_DOFS4[n]]	R/W	Dynamic standard tool compensation value (fourth set) Note) Subscript n represents an axis number (1 to 50).

System variable number	System variable name	Attribute	Description
#118251-#118300	[#_DOFS5[n]]	R/W	Dynamic standard tool compensation value (fifth set) Note) Subscript n represents an axis number (1 to 50).
#118301-#118350	[#_DOFS6[n]]	R/W	Dynamic standard tool compensation value (sixth set) Note) Subscript n represents an axis number (1 to 50).
#118351-#118400	[#_DOFS7[n]]	R/W	Dynamic standard tool compensation value (seventh set) Note) Subscript n represents an axis number (1 to 50).
#118401-#118450	[#_DOFS8[n]]	R/W	Dynamic standard tool compensation value (eighth set) Note) Subscript n represents an axis number (1 to 50).

- Feedrate reduction ratio for rapid traverse overlap

System variable number	System variable name	Attribute	Description
#100851-#100900	[#_ROVLP [n]]	R/W	Feedrate reduction ratio for rapid traverse overlap Note) Subscript n represents an axis number (1 to 50).

- System constant

System constant number	System constant name	Attribute	Description
#0, #3100	[#_EMPTY]	R	Null
#3101	[#_PI]	R	Circular constant π = 3.14159265358979323846
#3102	[#_E]	R	Base of natural logarithm e = 2.71828182845904523536

Explanation

R, W, and R/W are attributes of a variable and represents read-only, write-only, and read/write enabled, respectively.

- Interface signal #1000-#1031, #1032, #1033-#1035 (Attribute: R)
#1100-#1115, #1132, #1133-#1135 (Attribute: R/W)

[Input signal]

The status of interface input signals can be obtained by reading the value of system variables #1000 to #1032.

Variable number	Variable name	Point	Interface input signal
#1000	[#_UI[0]]	1	UI000 (2^0)
#1001	[#_UI[1]]	1	UI001 (2^1)
#1002	[#_UI[2]]	1	UI002 (2^2)
#1003	[#_UI[3]]	1	UI003 (2^3)
#1004	[#_UI[4]]	1	UI004 (2^4)
#1005	[#_UI[5]]	1	UI005 (2^5)
#1006	[#_UI[6]]	1	UI006 (2^6)
#1007	[#_UI[7]]	1	UI007 (2^7)
#1008	[#_UI[8]]	1	UI008 (2^8)
#1009	[#_UI[9]]	1	UI009 (2^9)
#1010	[#_UI[10]]	1	UI010 (2^{10})
#1011	[#_UI[11]]	1	UI011 (2^{11})
#1012	[#_UI[12]]	1	UI012 (2^{12})
#1013	[#_UI[13]]	1	UI013 (2^{13})
#1014	[#_UI[14]]	1	UI014 (2^{14})
#1015	[#_UI[15]]	1	UI015 (2^{15})
#1016	[#_UI[16]]	1	UI016 (2^{16})
#1017	[#_UI[17]]	1	UI017 (2^{17})
#1018	[#_UI[18]]	1	UI018 (2^{18})
#1019	[#_UI[19]]	1	UI019 (2^{19})
#1020	[#_UI[20]]	1	UI020 (2^{20})
#1021	[#_UI[21]]	1	UI021 (2^{21})
#1022	[#_UI[22]]	1	UI022 (2^{22})
#1023	[#_UI[23]]	1	UI023 (2^{23})
#1024	[#_UI[24]]	1	UI024 (2^{24})
#1025	[#_UI[25]]	1	UI025 (2^{25})
#1026	[#_UI[26]]	1	UI026 (2^{26})
#1027	[#_UI[27]]	1	UI027 (2^{27})
#1028	[#_UI[28]]	1	UI028 (2^{28})
#1029	[#_UI[29]]	1	UI029 (2^{29})
#1030	[#_UI[30]]	1	UI030 (2^{30})
#1031	[#_UI[31]]	1	UI031 (2^{31})
#1032	[#_UIL[0]]	32	UI000-UI031
#1033	[#_UIL[1]]	32	UI100-UI131
#1034	[#_UIL[2]]	32	UI200-UI231
#1035	[#_UIL[3]]	32	UI300-UI331

Variable value	Input signal
1.0	Contact closed
0.0	Contact opened

Since the read value is 1.0 or 0.0 regardless of the unit system, the unit system must be considered when a macro is created.
The input signals at 32 points can be read at a time by reading from system variables #1032 to #1035.

$$\#1032 = \sum_{i=0}^{30} \#[1000+i] \times 2^i - \#1031 \times 2^{31}$$

$$\#[1032+n] = \sum_{i=0}^{30} \left\{2^i \times V_i\right\} - 2^{31} \times V_{31}$$

When $UIn_i = 0$, $V_i = 0$.
When $UIn_i = 1$, $V_i = 1$.
n = 0-3

[Output signal]
Interface output signals can be sent by assigning values to system variables #1100 to #1132 for sending interface signals.

Variable number	Variable name	Point	Interface input signal
#1100	[#_UO[0]]	1	UO000 (2^0)
#1101	[#_UO[1]]	1	UO001 (2^1)
#1102	[#_UO[2]]	1	UO002 (2^2)
#1103	[#_UO[3]]	1	UO003 (2^3)
#1104	[#_UO[4]]	1	UO004 (2^4)
#1105	[#_UO[5]]	1	UO005 (2^5)
#1106	[#_UO[6]]	1	UO006 (2^6)
#1107	[#_UO[7]]	1	UO007 (2^7)
#1108	[#_UO[8]]	1	UO008 (2^8)
#1109	[#_UO[9]]	1	UO009 (2^9)
#1110	[#_UO[10]]	1	UO010 (2^{10})
#1111	[#_UO[11]]	1	UO011 (2^{11})
#1112	[#_UO[12]]	1	UO012 (2^{12})
#1113	[#_UO[13]]	1	UO013 (2^{13})
#1114	[#_UO[14]]	1	UO014 (2^{14})
#1115	[#_UO[15]]	1	UO015 (2^{15})
#1116	[#_UO[16]]	1	UO016 (2^{16})
#1117	[#_UO[17]]	1	UO017 (2^{17})
#1118	[#_UO[18]]	1	UO018 (2^{18})
#1119	[#_UO[19]]	1	UO019 (2^{19})
#1120	[#_UO[20]]	1	UO020 (2^{20})
#1121	[#_UO[21]]	1	UO021 (2^{21})
#1122	[#_UO[22]]	1	UO022 (2^{22})
#1123	[#_UO[23]]	1	UO023 (2^{23})
#1124	[#_UO[24]]	1	UO024 (2^{24})

Variable number	Variable name	Point	Interface input signal
#1125	[#_UO[25]]	1	UO025 (2^{25})
#1126	[#_UO[26]]	1	UO026 (2^{26})
#1127	[#_UO[27]]	1	UO027 (2^{27})
#1128	[#_UO[28]]	1	UO028 (2^{28})
#1129	[#_UO[29]]	1	UO029 (2^{29})
#1130	[#_UO[30]]	1	UO030 (2^{30})
#1131	[#_UO[31]]	1	UO031 (2^{31})
#1132	[#_UOL[0]]	32	UO000-UO031
#1133	[#_UOL[1]]	32	UO100-UO131
#1134	[#_UOL[2]]	32	UO200-UO231
#1135	[#_UOL[3]]	32	UO300-UO331

Variable value	Input signal
1.0	Contact closed
0.0	Contact opened

The output signals at 32 points can be written at a time by writing to system variables #1132 to #1135. The signals can also be read.

$$\#1132 = \sum_{i=0}^{30} \#[1000+i] \times 2^i - \#1131 \times 2^{31}$$

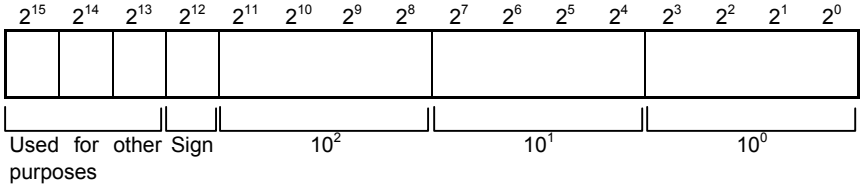
$$\#[1132+n] = \sum_{i=0}^{30} \left\{ 2^i \times V_i \right\} - 2^{31} \times V_{31}$$

When $UIn_i = 0$, $V_i = 0$.
When $UIn_i = 1$, $V_i = 1$.
n = 0-3

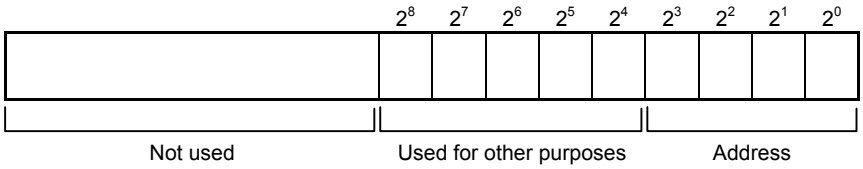
NOTE
1 When a value other than 1.0 or 0.0 is assigned to variables #1100 to #1131, it is assumed as follows.
<null> is assumed to be 0.
A value other than <null> or 0 is assumed to be 1.
Where, a value less than 0.00000001 is undefined.
2 When any of UI016 to UI031, UI100 to UI131, UI200 to UI231, UI300 to UI331, UO016 to UO031, UO200 to UO231, and UO300 to UO331 are used, parameter MIF (No.6001#0) must be set to 1.

Example

Structure of DI



Structure of DO



<1> Address switching signed BCD 3 digits are read.

Macro calling instruction

```
G65 P9100 D (address);
```

A custom macro body is created as follows.

```
O9100 ;
#1132 = #1132 AND 496 OR#7 ;   :  Address sending
G65 P9101 T60 ;                :  Timer macro
#100 = BIN[#1032 AND 4095] ;   :  BCD 3 digits are read.
IF [#1012 EQ  0] GOTO  9100 ;  :  A sign is attached.
#100 = -#100
N9100 M99
```

<2> Eight types of address switching signed BCD 6 digits (3-digit integer part + 3-digit fractional part) are read into #101.

Structure on the machine side

```
When DO 2^0 = 0:          Data with 3 decimal places
When DO 2^0 = 1:          Data with 3-digit integer part
When DO 2^3 to 2^1 = 000: No1 data when #1 = 0
When DO 2^3 to 2^1 = 001: No2 data when #2 = 0
      :
When DO 2^3 to 2^1 = 111:       No8 data when #8 = 0
```

Macro calling instruction

```
G65   P9101   D (data number);
```

A custom macro body is created as follows.

```
O9101 ;
G65 P9101 D[#1*2+1] ;
#101 = #100  ;
G65 P9100 D[#1*2] ;
#101 = #101 + #100/1000 ;
M99 ;
```

- Tool compensation value #2001-#2800, #10001-#13999 (Attribute: R/W)

M

The compensation values can be obtained by reading system variables #2001 to #2800 or #10001 to #13999 for tool compensation. The compensation values can also be changed by assigning values to the system variables.

<1> Tool compensation memory A

- When the number of compensations is 200 or less

Compensation number	Variable number	Variable name
1	#2001	[#_OFS[1]]
2	#2002	[#_OFS[2]]
3	#2003	[#_OFS[3]]
:	:	:
199	#2199	[#_OFS[199]]
200	#2200	[#_OFS[200]]

- When the number of compensations is more than 200 (For compensation with a compensation number of 200 or less, #2001 to #2200 can also be used.)

Compensation number	Variable number	Variable name
1	#10001	[#_OFS[1]]
2	#10002	[#_OFS[2]]
3	#10003	[#_OFS[3]]
:	:	:
998	#10998	[#_OFS[998]]
999	#10999	[#_OFS[999]]

<2> Tool compensation memory B

- When the number of compensations is 200 or less
  When bit 3 (V15) of parameter No.6000 = 0

Compensation number	Wear		Geometry	
	Variable number	Variable name	Variable number	Variable name
1	#2001	[#_OFSW[1]]	#2201	[#_OFSG[1]]
2	#2002	[#_OFSW[2]]	#2202	[#_OFSG[2]]
3	#2003	[#_OFSW[3]]	#2203	[#_OFSG[3]]
:	:	:	:	:
199	#2199	[#_OFSW[199]]	#2399	[#_OFSG[199]]
200	#2200	[#_OFSW[200]]	#2400	[#_OFSG[200]]

When bit 3 (V15) of parameter No.6000 = 1

Compensation number	Wear		Geometry	
	Variable number	Variable name	Variable number	Variable name
1	#2201	[#_OFSW[1]]	#2001	[#_OFSG[1]]
2	#2202	[#_OFSW[2]]	#2002	[#_OFSG[2]]
3	#2203	[#_OFSW[3]]	#2003	[#_OFSG[3]]
:	:	:	:	:
199	#2399	[#_OFSW[199]]	#2199	[#_OFSG[199]]
200	#2400	[#_OFSW[200]]	#2200	[#_OFSG[200]]

- When the number of compensations is more than 200 (For compensation with a compensation number of 200 or less, #2001 to #2400 can also be used.)
  When bit 3 (V15) of parameter No.6000 = 0

Compensation number	Geometry		Wear	
	Variable number	Variable name	Variable number	Variable name
1	#11001	[#_OFSG[1]]	#10001	[#_OFSW[1]]
2	#11002	[#_OFSG[2]]	#10002	[#_OFSW[2]]
3	#11003	[#_OFSG[3]]	#10003	[#_OFSW[3]]
:	:	:	:	:
998	#11998	[#_OFSG[998]]	#10998	[#_OFSW[998]]
999	#11999	[#_OFSG[999]]	#10999	[#_OFSW[999]]

When bit 3 (V15) of parameter No.6000 = 1

Compensation number	Geometry		Wear	
	Variable number	Variable name	Variable number	Variable name
1	#10001	[#_OFSG[1]]	#11001	[#_OFSW[1]]
2	#10002	[#_OFSG[2]]	#11002	[#_OFSW[2]]
3	#10003	[#_OFSG[3]]	#11003	[#_OFSW[3]]
:	:	:	:	:
998	#10998	[#_OFSG[998]]	#11998	[#_OFSW[998]]
999	#10999	[#_OFSG[999]]	#11999	[#_OFSW[999]]

<3> Tool offset memory C

- When the number of compensations is 200 or less
  When bit 3 (V15) of parameter No.6000 = 0

H code				
Compensation number	Geometry		Wear	
	Variable number	Variable name	Variable number	Variable name
1	#2201	[#_OFSHG[1]]	#2001	[#_OFSHW[1]]
2	#2202	[#_OFSHG[2]]	#2002	[#_OFSHW[2]]
3	#2203	[#_OFSHG[3]]	#2003	[#_OFSHW[3]]
:	:	:	:	:
199	#2399	[#_OFSHG[199]]	#2199	[#_OFSHW[199]]
200	#2400	[#_OFSHG[200]]	#2200	[#_OFSHW[200]]

When bit 3 (V15) of parameter No.6000 = 1

H code				
Compensation number	Geometry		Wear	
	Variable number	Variable name	Variable number	Variable name
1	#2001	[#_OFSHG[1]]	#2201	[#_OFSHW[1]]
2	#2002	[#_OFSHG[2]]	#2202	[#_OFSHW[2]]
3	#2003	[#_OFSHG[3]]	#2203	[#_OFSHW[3]]
:	:	:	:	:
199	#2199	[#_OFSHG[199]]	#2399	[#_OFSHW[199]]
200	#2200	[#_OFSHG[200]]	#2400	[#_OFSHW[200]]

D code				
Compensation number	Geometry		Wear	
	Variable number	Variable name	Variable number	Variable name
1	#2401	[#_OFSDG[1]]	#2601	[#_OFSDW[1]]
2	#2402	[#_OFSDG[2]]	#2602	[#_OFSDW[2]]
3	#2403	[#_OFSDG[3]]	#2603	[#_OFSDW[3]]
:	:	:	:	:
199	#2599	[#_OFSDG[199]]	#2799	[#_OFSDW[199]]
200	#2600	[#_OFSDG[200]]	#2800	[#_OFSDW[200]]

NOTE
1 When #2401 to #2800 are used for reading or writing of D codes, bit 5 (D15) of parameter No.6004 must be set to 1.
2 When bit 5 (D15) of parameter No.6004 is set to 1, system variables #2500 to #2806 for workpiece origin offset cannot be used. Use system variables #5201 to #5324.

- When the number of compensations is more than 200 (For compensation with a compensation number of 200 or less, #2001 to #2800 can also be used.)
  When bit 3 (V15) of parameter No.6000 = 0

H code				
Compensation number	Geometry		Wear	
	Variable number	Variable name	Variable number	Variable name
1	#11001	[#_OFSHG[1]]	#10001	[#_OFSHW[1]]
2	#11002	[#_OFSHG[2]]	#10002	[#_OFSHW[2]]
3	#11003	[#_OFSHG[3]]	#10003	[#_OFSHW[3]]
:	:	:	:	:
998	#11998	[#_OFSHG[998]]	#10998	[#_OFSHW[998]]
999	#11999	[#_OFSHG[999]]	#10999	[#_OFSHW[999]]

D code				
Compensation number	Geometry		Wear	
	Variable number	Variable name	Variable number	Variable name
1	#13001	[#_OFSDG[1]]	#12001	[#_OFSDW[1]]
2	#13002	[#_OFSDG[2]]	#12002	[#_OFSDW[2]]
3	#13003	[#_OFSDG[3]]	#12003	[#_OFSDW[3]]
:	:	:	:	:
998	#13998	[#_OFSDG[998]]	#12998	[#_OFSDW[998]]
999	#13999	[#_OFSDG[999]]	#12999	[#_OFSDW[999]]

When bit 3 (V15) of parameter No.6000 = 1

H code				
Compensation number	Geometry		Wear	
	Variable number	Variable name	Variable number	Variable name
1	#10001	[#_OFSHG[1]]	#11001	[#_OFSHW[1]]
2	#10002	[#_OFSHG[2]]	#11002	[#_OFSHW[2]]
3	#10003	[#_OFSHG[3]]	#11003	[#_OFSHW[3]]
:	:	:	:	:
998	#10998	[#_OFSHG[998]]	#11998	[#_OFSHW[998]]
999	#10999	[#_OFSHG[999]]	#11999	[#_OFSHW[999]]

D code				
Compensation number	Geometry		Wear	
	Variable number	Variable name	Variable number	Variable name
1	#12001	[#_OFSDG[1]]	#13001	[#_OFSDW[1]]
2	#12002	[#_OFSDG[2]]	#13002	[#_OFSDW[2]]
3	#12003	[#_OFSDG[3]]	#13003	[#_OFSDW[3]]
:	:	:	:	
998	#12998	[#_OFSDG[998]]	#13998	[#_OFSDW[998]]
999	#12999	[#_OFSDG[999]]	#13999	[#_OFSDW[999]]

- Tool compensation value #2001-#2964, #10001-#19999 (Attribute: R/W)

T

The compensation values can be obtained by reading system variables #2001 to #2964 or #10001 to #19999 for tool compensation. The compensation values can also be changed by assigning values to the system variables.

<1> Without tool geometry/wear compensation memory

- When the number of compensations is 64 or less

Compensation number	Variable number	Variable name	Description
1	#2001	[#_OFSX[1]]	X-axis compensation value (*1)
2	#2002	[#_OFSX[2]]	
3	#2003	[#_OFSX[3]]	
:	:	:	
63	#2063	[#_OFSX[63]]	
64	#2064	[#_OFSX[64]]	
1	#2101	[#_OFSZ[1]]	Z-axis compensation value (*1)
2	#2102	[#_OFSZ[2]]	
3	#2103	[#_OFSZ[3]]	
:	:	:	
63	#2163	[#_OFSZ[63]]	
64	#2164	[#_OFSZ[64]]	
1	#2201	[#_OFSR[1]]	Tool nose radius compensation value
2	#2202	[#_OFSR[2]]	
3	#2203	[#_OFSR[3]]	
:	:	:	
63	#2263	[#_OFSR[63]]	
64	#2264	[#_OFSR[64]]	
1	#2301	[#_OFST[1]]	Virtual tool tipT position
2	#2302	[#_OFST[2]]	
3	#2303	[#_OFST[3]]	
:	:	:	
63	#2363	[#_OFST[63]]	
64	#2364	[#_OFST[64]]	
1	#2401	[#_OFSY[1]]	Y-axis compensation value (*1)
2	#2402	[#_OFSY[2]]	
3	#2403	[#_OFSY[3]]	
:	:	:	
48	#2448	[#_OFSY[48]]	
49	#2449	[#_OFSY[49]]	

(*1) X-axis: X-axis of basic three axes, Z-axis: Z-axis of basic three axes, Y-axis: Y-axis of basic three axes

- When the number of compensations is more than 64 (For compensation with a compensation number of 64 or less, #2001 to #2449 can also be used.)

Compensation number	Variable number	Variable name	Description
1	#10001	[#_OFSX[1]]	X-axis compensation value (*1)
2	#10002	[#_OFSX[2]]	
3	#10003	[#_OFSX[3]]	
:	:	:	
998	#10998	[#_OFSX[998]]	
999	#10999	[#_OFSX[999]]	
1	#11001	[#_OFSZ[1]]	Z-axis compensation value (*1)
2	#11002	[#_OFSZ[2]]	
3	#11003	[#_OFSZ[3]]	
:	:	:	
998	#11998	[#_OFSZ[998]]	
999	#11999	[#_OFSZ[999]]	
1	#12001	[#_OFSR[1]]	Tool nose radius compensation value
2	#12002	[#_OFSR[2]]	
3	#12003	[#_OFSR[3]]	
:	:	:	
998	#12998	[#_OFSR[998]]	
999	#12999	[#_OFSR[999]]	
1	#13001	[#_OFST[1]]	Virtual tool tip T position
2	#13002	[#_OFST[2]]	
3	#13003	[#_OFST[3]]	
:	:	:	
998	#13998	[#_OFST[998]]	
999	#13999	[#_OFST[999]]	
1	#14001	[#_OFSY[1]]	Y-axis compensation value (*1)
2	#14002	[#_OFSY[2]]	
3	#14003	[#_OFSY[3]]	
:	:	:	
998	#14998	[#_OFSY[998]]	
999	#14999	[#_OFSY[999]]	

(*1) X-axis: X-axis of basic three axes, Z-axis: Z-axis of basic three axes, Y-axis: Y-axis of basic three axes

<2> With tool geometry/wear compensation memory

- When the number of compensations is 64 or less

Compensation number	Variable number	Variable name	Description
1	#2001	[#_OFSXW[1]]	X-axis compensation value (wear) (*1)
2	#2002	[#_OFSXW[2]]	
3	#2003	[#_OFSXW[3]]	
:	:	:	
63	#2063	[#_OFSXW[63]]	
64	#2064	[#_OFSXW[64]]	
1	#2101	[#_OFSZW[1]]	Z-axis compensation value (wear) (*1)
2	#2102	[#_OFSZW[2]]	
3	#2103	[#_OFSZW[3]]	
:	:	:	
63	#2163	[#_OFSZW[63]]	
64	#2164	[#_OFSZW[64]]	
1	#2201	[#_OFSRW[1]]	Tool nose radius compensation value (wear)
2	#2202	[#_OFSRW [2]]	
3	#2203	[#_OFSRW [3]]	
:	:	:	
63	#2263	[#_OFSRW [63]]	
64	#2264	[#_OFSRW [64]]	
1	#2301	[#_OFST[1]]	Virtual tool tip T position
2	#2302	[#_OFST[2]]	
3	#2303	[#_OFST[3]]	
:	:	:	
63	#2363	[#_OFST[63]]	
64	#2364	[#_OFST[64]]	
1	#2401	[#_OFSYW[1]]	Y-axis compensation value (wear) (*1)
2	#2402	[#_OFSYW [2]]	
3	#2403	[#_OFSYW [3]]	
:	:	:	
48	#2448	[#_OFSYW [48]]	
49	#2449	[#_OFSYW [49]]	
1	#2451	[#_OFSYG[1]]	Y-axis compensation value (geometry) (*1)
2	#2452	[#_OFSYG [2]]	
3	#2453	[#_OFSYG [3]]	
:	:	:	
48	#2498	[#_OFSYG [48]]	
49	#2499	[#_OFSYG [49]]	
1	#2701	[#_OFSXG[1]]	X-axis compensation value (geometry) (*1)
2	#2702	[#_OFSXG[2]]	
3	#2703	[#_OFSXG [3]]	
:	:	:	
48	#2748	[#_OFSXG [48]]	
49	#2749	[#_OFSXG [49]]	
1	#2801	[#_OFSZG[1]]	Z-axis compensation value (geometry) (*1)
2	#2802	[#_OFSZG[2]]	
3	#2803	[#_OFSZG[3]]	
:	:	:	
48	#2848	[#_OFSZG[48]]	
49	#2849	[#_OFSZG[49]]	

Compensation number	Variable number	Variable name	Description
1	#2901	[#_OFSRG[1]]	Tool nose radius compensation value (geometry)
2	#2902	[#_OFSRG[2]]	
3	#2903	[#_OFSRG[3]]	
:	:	:	
63	#2963	[#_OFSRG[63]]	
64	#2964	[#_OFSRG[64]]	
1	#19001	[#_OFSYG[1]]	Y-axis compensation value (geometry) (*1)
2	#19002	[#_OFSYG[2]]	
3	#19003	[#_OFSYG[3]]	
:	:	:	
998	#19998	[#_OFSYG[998]]	
999	#19999	[#_OFSYG[999]]	

(*1) X-axis: X-axis of basic three axes, Z-axis: Z-axis of basic three axes, Y-axis: Y-axis of basic three axes

- When the number of compensations is more than 64 (For compensation with a compensation number of 64 or less, #2001 to #2964 or #10001 to #19999 can also be used.)

Compensation number	Variable number	Variable name	Description
1	#10001	[#_OFSXW[1]]	X-axis compensation value (wear) (*1)
2	#10002	[#_OFSXW[2]]	
3	#10003	[#_OFSXW[3]]	
:	:	:	
998	#10998	[#_OFSXW[998]]	
999	#10999	[#_OFSXW[999]]	
1	#11001	[#_OFSZW[1]]	Z-axis compensation value (wear) (*1)
2	#11002	[#_OFSZW[2]]	
3	#11003	[#_OFSZW[3]]	
:	:	:	
998	#11998	[#_OFSZW[998]]	
999	#11999	[#_OFSZW[999]]	
1	#12001	[#_OFSRW[1]]	Tool nose radius compensation value (wear)
2	#12002	[#_OFSRW [2]]	
3	#12003	[#_OFSRW [3]]	
:	:	:	
998	#12998	[#_OFSRW [998]]	
999	#12999	[#_OFSRW [999]]	
1	#13001	[#_OFST[1]]	Virtual tool tip T position
2	#13002	[#_OFST[2]]	
3	#13003	[#_OFST[3]]	
:	:	:	
998	#13998	[#_OFST[998]]	
999	#13999	[#_OFST[999]]	
1	#14001	[#_OFSYW[1]]	Y-axis compensation value (wear) (*1)
2	#14002	[#_OFSYW [2]]	
3	#14003	[#_OFSYW [3]]	
:	:	:	
998	#14998	[#_OFSYW [998]]	
999	#14999	[#_OFSYW [999]]	

Compensation number	Variable number	Variable name	Description
1	#15001	[#_OFSXG[1]]	X-axis compensation value (geometry) (*1)
2	#15002	[#_OFSXG[2]]	
3	#15003	[#_OFSXG [3]]	
:	:	:	
998	#15998	[#_OFSXG [998]]	
999	#15999	[#_OFSXG [999]]	
1	#16001	[#_OFSZG[1]]	Z-axis compensation value (geometry) (*1)
2	#16002	[#_OFSZG[2]]	
3	#16003	[#_OFSZG[3]]	
:	:	:	
998	#16998	[#_OFSZG[998]]	
999	#16999	[#_OFSZG[999]]	
1	#17001	[#_OFSRG[1]]	Tool nose radius compensation value (geometry)
2	#17002	[#_OFSRG[2]]	
3	#17003	[#_OFSRG[3]]	
:	:	:	
998	#17998	[#_OFSRG[998]]	
999	#17999	[#_OFSRG[999]]	
1	#19001	[#_OFSYG[1]]	Y-axis compensation value (geometry) (*1)
2	#19002	[#_OFSYG[2]]	
3	#19003	[#_OFSYG[3]]	
:	:	:	
998	#19998	[#_OFSYG[998]]	
999	#19999	[#_OFSYG[999]]	

(*1) X-axis: X-axis of basic three axes, Z-axis: Z-axis of basic three axes, Y-axis: Y-axis of basic three axes

- Alarm #3000 (Attribute: W)

When an error is detected in a macro, an unit can enter the alarm state. In addition, an alarm message of up to 60 characters with alphabet and numerals can be specified between a control-out and a control-in after the expression. When an alarm message is not specified, a macro alarm is used instead.

Variable number	Variable name	Description
#3000	[#_ALM]	Macro alarm

When bit 1 (MCA) of parameter No.6008 = 0

#3000 = n (ALARM MESSAGE); (n: 0-200)

On the screen, the alarm number obtained by adding the value of #3000 to 3000 and alarm message appear.

(Example) #3000 = 1 (ALARM MESSAGE);

→ "3001 ALARM MESSAGE" appears on the alarm screen.

When bit 1 (MCA) of parameter No.6008 = 1

#3000 = n (ALARM MESSAGE); (n: 0-4095)

On the screen, the alarm number of #3000 and alarm message appear after MC.

(Example) #3000=1 (ALARM MESSAGE);

→ "MC0001 ALARM MESSAGE" appears on the alarm screen.

- Clock #3001, #3002 (Attribute: R/W)

The clock time can be obtained by reading system variables #3001 and #3002 for clocks. The time can be preset by entering a value in the system variables.

Type	Variable number	Variable name	Unit	At power-on	Count condition
Clock 1	#3001	[#_CLOCK1]	1 ms	Reset to 0	Anytime
Clock 2	#3002	[#_CLOCK2]	1 hour	Same as at power-down	When the STL signal is on

The clock accuracy is 16 ms. Clock 1 returns to 0 after a lapse of 2147483648 ms. Clock 2 returns to 0 after a lapse of 9544.37176 hours.

[Example]

```
Timer
Macro calling command
G65 P9101 T (wait time) ms ;
A macro is created as follows.
O9101;
#3001 = 0;   Initial setting
WHILE [#3001 LE #20] DO1:   Wait for a specified time
END1 ;
M99 ;
```

- Controlling of single block stop and waiting for the auxiliary function completion signal #3003 (Attribute: R/W)

Assigning the following values in system variable #3003 allows the specification of whether single block stop is disabled in the following blocks or whether a wait for the completion signal (FIN) of the auxiliary function (M, S, T, or B) before going to the next block is enabled. When a wait for completion signal is disabled, the distribution end signal (DEN) is not sent. Be careful not to specify the next auxiliary function without waiting for the completion signal.

Variable number and variable name	Value	Single block stop	Auxiliary function completion signal
#3003 [#_CNTL1]	0	Enabled	Waiting
	1	Disabled	Waiting
	2	Enabled	Not waiting
	3	Disabled	Not waiting

In addition, the following variable names can be used to enable or disable single block stop and a wait for the auxiliary function completion signal, individually.

Variable name	Value	Single block stop	Auxiliary function completion
[#_M_SBK]	0	Enabled	-
	1	Disabled	-
[#_M_FIN]	0	-	Waiting
	1	-	Not waiting

[Example]

Drill cycle (for incremental programming)
(G81 equivalent)
Macro calling command
G65 P9081 L Iterations R R point Z Z point;
A custom macro body is created as follows.

```
O9081 ;
#3003 = 1 ;
G00 Z#18 ;
G01 Z#26 ;
G00 Z-[ ROUND[#18] + ROUND[#26] ] ;
#3003 = 0 ;
M99 ;
```

Disable single block stop.
#18 corresponds to R and #26 to Z.

NOTE
#3003 is cleared by a reset.

- Enabling of feed hold, feedrate override, and exact stop check #3004 (Attribute: R/W)

Assigning the following values in system variable #3004 allows the specification of whether feed hold and feedrate override are enabled in the following blocks or whether exact stop in G61 mode or by G09 command is disabled.

Variable number and variable name	Value	Field hold	Feedrate override	Exact stop
#3004 [#_CNTL2]	0	Enabled	Enabled	Enabled
	1	Disabled	Enabled	Enabled
	2	Enabled	Disabled	Enabled
	3	Disabled	Disabled	Enabled
	4	Enabled	Enabled	Disabled
	5	Disabled	Enabled	Disabled
	6	Enabled	Disabled	Disabled
	7	Disabled	Disabled	Disabled

In addition, the following variable names can be used to enable or disable feed hold, feedrate override, and exact stop in G61 mode or by the G09 command, individually.

Variable number and variable name	Value	Feed hold	Feedrate override	Exact stop
[#_M_FHD]	0	Enabled	-	-
	1	Disabled	-	-
[#_M_OV]	0	-	Enabled	-
	1	-	Disabled	-
[#_M_EST]	0	-	-	Enabled
	1	-	-	Disabled

NOTE

1 These system variables are provided to maintain compatibility with conventional NC programs. It is recommended that functions provided by G63, G09, G61, and other G codes be used to enable or disable feed hold, feedrate override, and exact stop.

2 When the feed hold button is pressed during execution of a block for which feed hold is disabled:
   <1> If the feed hold button is kept pressed, operation stops after execution of the block. When single block stop is disabled, however, operation does not stop.
   <2> If the pressed feed hold button is released, the feed hold lamp lights, but operation does not stop until the end of the first block that was enabled.

3 #3004 is cleared by a reset.

4 If exact stop is disabled by #3004, the original exact stop position between cutting feed and positioning block is not affected. #3004 can temporarily disable exact stop in G61 mode or by the G09 command between cutting feed and cutting feed.

- Settings #3005 (Attribute: R/W)

Settings can be read and written.
Binary values are converted to decimals.

#3005

	#15	#14	#13	#12	#11	#10	#9	#8
Setting							FCV	

	#7	#6	#5	#4	#3	#2	#1	#0
Setting			SEQ			INI	ISO	TVC

#9 (FCV) : Whether to use the FANUC Series 15 program format conversion capability
#5 (SEQ) : Whether to automatically insert sequence numbers
#2 (INI) : Millimeter input or inch input
#1 (ISO) : Whether to use EIA or ISO as the output code
#0 (TVC) : Whether to make a TV check

- Stop with a message #3006 (Attribute: W)

When "#3006=1 (MESSAGE);" is commanded in the macro, the program executes blocks up to the immediately previous one and then stops. When a message of up to 60 characters with alphabet and numerals, which is enclosed by a control-in character and control-out character, is programmed in the same block, the message is displayed on the external operator message screen.

Variable number	Variable name	Description
#3006	[#_MSGSTP]	Stop with a message

- Status of a mirror image #3007 (Attribute: R)

The status of an mirror image (setting or DI) at that point in time can be obtained for each axis by reading #3007.

Variable number	Variable name	Description
#3007	[#_MRIMG]	Status of a mirror image

When the status is indicated in binary, each bit corresponds with an axis as follows.

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
nth axis	32	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
nth axis	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1

For the 32 bits, 0 indicates that a mirror image is disabled and 1 indicates that a mirror image is enabled.

[Example] When #3007 is 3, a mirror image is enabled for the 1st and 2nd axes.

NOTE
1 The status of a programmable mirror image is not reflected on this variable.
2 When the mirror image function is set for the same axis by the mirror image signal and setting, the signal value and setting value are ORed and then output.
3 When mirror image signals for axes other than the controlled axes are turned on, they are not read into system variable #3007.

- Status during restart of a program #3008 (Attribute: R)

Whether a program is restarting can be determined by reading #3008.

Variable number	Variable name	Description
#3008	[#_PRSTR]	0: Program is not restarting. 1: Program is restarting.

- Time #3011, #3012 (Attribute: R)

Year/month/date and hour/minute/second can be obtained by reading system variables #3011 and #3012. This variable is read-only. To change year/month/date and hour/minute/second, use the timer screen.

[Example] May 20, 2004, PM 04:17:05
#3011 = 20040520
#3012 = 161705

- Total number of parts and the number of required parts #3901 and #3902 (Attribute: R/W)

The number of required parts and the number of machined parts can be displayed on the screen by using the operation time and part number displaying function. When the (total) number of machined parts reaches the number of required parts, a signal indicating the fact is sent to the machine (PMC side).
The system variables can be used to read or write the total number of parts and the number of required parts.

Variable number	Variable name	Description
#3901	[#_PRTSA]	Total number of parts
#3902	[#_PRTSN]	Number of required parts

- Type of tool compensation memory #3980 (Attribute: R)

M

System variable #3980 can be used to read the type of compensation memory.

Variable number	Variable name	Description
#3980	[#_OFSMEM]	Types of tool compensation memory 0: Tool compensation memory A 1: Tool compensation memory B 2: Tool compensation memory C

- Main program number #4000 (Attribute: R)

System variable #4000 can be used to read the main program number regardless of the level of a subprogram.

Variable number	Variable name	Description
#4000	[#_MAINO]	Main program number

NOTE
1 The main program number indicates the number of the program that is first started.
2 When an O number is specified by MDI during execution of the main program or when the second O number is specified in DNC mode, the value of #4000 changes to the specified O number. In addition, when no programs are registered or when no O numbers are specified in DNC mode, the value of #4000 changes to 0.

- Modal information #4001-#4130, #4201-#4330, #4401-#4530 (Attribute: R)

The modal information specified before the previous block of the macro statement that reads system variables #4001 to #4130 can be obtained in the block currently being looked ahead, by reading system variables #4001 to #4130.
The modal information of the block currently being executed can be obtained by reading system variables #4201 to #4330.
The modal information specified before the block interrupted by an interruption type custom macro by reading system variables #4401 to #4530.
The unit used when it was specified is applied.

M

(Category: <1> Previous block, <2> Running block, <3> Interrupted block)

Category	Variable number	Variable name	Description
<1>	#4001	[#_BUFG[1]]	Modal information (G code: group 1)
<2>	#4201	[#_ACTG[1]]	
<3>	#4401	[#_INTG[1]]	
<1>	#4002	[#_BUFG[2]]	Modal information (G code: group 2)
<2>	#4202	[#_ACTG[2]]	
<3>	#4402	[#_INTG[2]]	
:	:	:	:
:	:	:	:
<1>	#4030	[#_BUFG[30]]	Modal information (G code: group 30)
<2>	#4230	[#_ACTG[30]]	
<3>	#4430	[#_INTG[30]]	
<1>	#4102	[#_BUFB]	Modal information (B code)
<2>	#4302	[#_ACTB]	
<3>	#4502	[#_INTB]	
<1>	#4107	[#_BUFD]	Modal information (D code)
<2>	#4307	[#_ACTD]	
<3>	#4507	[#_INTD]	
<1>	#4108	[#_BUFE]	Modal information (E code)
<2>	#4308	[#_ACTE]	
<3>	#4508	[#_INTE]	
<1>	#4109	[#_BUFF]	Modal information (F code)
<2>	#4309	[#_ACTF]	
<3>	#4509	[#_INTF]	
<1>	#4111	[#_BUFH]	Modal information (H code)
<2>	#4311	[#_ACTH]	
<3>	#4511	[#_INTH]	
<1>	#4113	[#_BUFM]	Modal information (M code)
<2>	#4313	[#_ACTM]	
<3>	#4513	[#_INTM]	
<1>	#4114	[#_BUFN]	Modal information (sequence number N)
<2>	#4314	[#_ACTN]	
<3>	#4514	[#_INTN]	
<1>	#4115	[#_BUFO]	Modal information (program number O)
<2>	#4315	[#_ACTO]	
<3>	#4515	[#_INTO]	

Category	Variable number	Variable name	Description
<1> <2> <3>	#4119 #4319 #4519	[#_BUFS] [#_ACTS] [#_INTS]	Modal information (S code)
<1> <2> <3>	#4120 #4320 #4520	[#_BUFT] [#_ACTT] [#_INTT]	Modal information (T code)
<1> <2> <3>	#4130 #4330 #4530	[#_BUFWZP] [#_ACTWZP] [#_INTWZP]	Modal information (additional workpiece coordinate system number P)

T

(Category: <1> Previous block, <2> Running block, <3> Interrupted block)

Category	Variable number	Variable name	Description
<1> <2> <3>	#4001 #4201 #4401	[#_BUFG[1]] [#_ACTG[1]] [#_INTG[1]]	Modal information (G code: group 1)
<1> <2> <3>	#4002 #4202 #4402	[#_BUFG[2]] [#_ACTG[2]] [#_INTG[2]]	Modal information (G code: group 2)
: :	: :	: :	: :
<1> <2> <3>	#4030 #4230 #4430	[#_BUFG[30]] [#_ACTG[30]] [#_INTG[30]]	Modal information (G code: group 30)
<1> <2> <3>	#4108 #4308 #4508	[#_BUFE] [#_ACTE] [#_INTE]	Modal information (E code)
<1> <2> <3>	#4109 #4309 #4509	[#_BUFF] [#_ACTF] [#_INTF]	Modal information (F code)
<1> <2> <3>	#4113 #4313 #4513	[#_BUFM] [#_ACTM] [#_INTM]	Modal information (M code)
<1> <2> <3>	#4114 #4314 #4514	[#_BUFN] [#_ACTN] [#_INTN]	Modal information (sequence number N)
<1> <2> <3>	#4115 #4315 #4515	[#_BUFO] [#_ACTO] [#_INTO]	Modal information (program number O)
<1> <2> <3>	#4119 #4319 #4519	[#_BUFS] [#_ACTS] [#_INTS]	Modal information (S code)
<1> <2> <3>	#4120 #4320 #4520	[#_BUFT] [#_ACTT] [#_INTT]	Modal information (T code)
<1> <2> <3>	#4130 #4330 #4530	[#_BUFWZP] [#_ACTWZP] [#_INTWZP]	Modal information (additional workpiece coordinate system number P)

NOTE

1 Previous block and running block
Since the CNC reads the block that is ahead of the block currently being executed by the machining program, the block being retrieved by the CNC is normally different from that currently being executed. The previous block indicates the block that is ahead of the block being retrieved by the CNC, that is, the block that is ahead of the program block in which #4001 to #4130 are specified.

2 When variables exceeding the number of control axes are specified, the alarm PS0115, “VARIABLE NO. OUT OF RANGE” occurs.

```
[Example]  O1234 ;
           N10 G00 X200. Y200. ;
           N20 G01 X1000. Y1000. F10. ;
                 :
                 :
           N50 G00 X500. Y500. ;
           N60 #1 = #4001 ;
```

Assume that the CNC is currently executing N20. If the CNC retrieved and processed the blocks up to N60 as shown above, the running block is N20 and the previous block is N50. Therefore, group 1 modal information in the running block is G01 and group 1 modal information in the previous block is G00.
When N60 #1 = #4201, #1 = 1.
When N60 #1 = #4001, #1 = 0.

- Position information #5001-#5080, #100001-#100200 (Attribute: R)

The end position of the previous block, the specified current position (for the machine coordinate system and workpiece coordinate system), and the skip signal position can be obtained by reading the values of system variables #5001 to #5080 or #100001 to -#100200.

Variable number	Variable name	Position information	Coordinate system	Tool position/tool length/cutter compensation	Reading operation during movement
#5001	[#_ABSIO[1]]	1st axis block end point position	Workpiece coordinate system	Not included	Enabled
#5002	[#_ABSIO[2]]	2nd axis block end point position			
:	:	:			
#5020	[#_ABSIO[20]]	20th axis block end point position			
#100001	[#_ABSIO[1]]	1st axis block end point position			
#100002	[#_ABSIO[2]]	2nd axis block end point position			
:	:	:			
#100050	[#_ABSIO[50]]	50th axis block end point position			
#5021	[#_ABSMT[1]]	1st axis current position	Machine coordinate system	Included	Disabled
#5022	[#_ABSMT[2]]	2nd axis current position			
:	:	:			
#5040	[#_ABSMT[20]]	20th axis current position			
#100051	[#_ABSMT[1]]	1st axis current position			
#100052	[#_ABSMT[2]]	2nd axis current position			
:	:	:			
#100100	[#_ABSMT[50]]	50th axis current position			
#5041	[#_ABSOT[1]]	1st axis current position	Workpiece coordinate system	Included	Disabled
#5042	[#_ABSOT[2]]	2nd axis current position			
:	:	:			
#5060	[#_ABSOT[20]]	20th axis current position			
#100101	[#_ABSOT[1]]	1st axis current position			
#100102	[#_ABSOT[2]]	2nd axis current position			
:	:	:			
#100150	[#_ABSOT[50]]	50th axis current position			
#5061	[#_ABSKP[1]]	1st axis skip position	Workpiece coordinate system	Included	Enabled
#5062	[#_ABSKP[2]]	2nd axis skip position			
:	:	:			
#5080	[#_ABSKP[20]]	20th axis skip position			
#100151	[#_ABSKP[1]]	1st axis skip position			
#100152	[#_ABSKP[2]]	2nd axis skip position			
:	:	:			
#100200	[#_ABSKP[50]]	50th axis skip position			

NOTE

1 When variables exceeding the number of control axes are specified, the alarm PS0115, “VARIABLE NO. OUT OF RANGE” occurs.
2 The position information for 20th or earlier axis can be used with #5001 to #5080.
3 The block end point position (ABSIO) of the skip (G31) is the position where the skip signal is turned on. If the skip signal is not turned on, the position is the end position of the block.
4 "Read operation during movement is disabled" means that the accurate reading of values during movement is not guaranteed.

- Tool length compensation value #5081-#5100, #100201-#100250 (Attribute: R)

M

Tool length compensation in the block currently being executed can be obtained for each axis by reading system variables #5081 to #5100 or #100201 to #100250.

Variable number	Variable name	Position information	Read operation during movement
#5081	[#_TOFS[1]]	1st axis tool length compensation value	Disabled
#5082	[#_TOFS[2]]	2nd axis tool length compensation value	
:	:	:	
#5100	[#_TOFS[20]]	20th axis tool length compensation value	
#100201	[#_TOFS[1]]	1st axis tool length compensation value	
#100202	[#_TOFS[2]]	2nd axis tool length compensation value	
:	:	:	
#100250	[#_TOFS[50]]	50th axis tool length compensation value	

NOTE
1 When variables exceeding the number of control axes are specified, the alarm PS0115, “VARIABLE NO. OUT OF RANGE” occurs.
2 The tool length compensation for 20th or earlier axis can also be used with #5081 to #5100.

- Tool offset #5081-#5083, #5121-#5123 (Attribute: R) #100201-#100250, #100901-#100950

T

Tool offset in the block currently being executed can be obtained for each axis by reading system variables #5081 to #5083 or #5121 to #5123. (X-axis: X-axis of basic three axes, Z-axis: Z-axis of basic three axes, Y-axis: Y-axis of basic three axes)
<1> Without tool geometry/wear compensation memory

Variable number	Variable name	Position information	Read operation during movement
#5081	[#_TOFSWX]	X-axis tool offset value	Disabled
#5082	[#_TOFSWZ]	Z-axis tool offset value	
#5083	[#_TOFSWY]	Y-axis tool offset value	
#5084	[#_TOFS[4]]	4th axis tool offset value	
:	:	:	
#5100	[#_TOFS[20]]	20th axis tool offset value	
#100201	[#_TOFSWX]	X-axis tool offset value	
#100202	[#_TOFSWZ]	Z-axis tool offset value	
#100203	[#_TOFSWY]	Y-axis tool offset value	
#100204	[#_TOFS[4]]	4th axis tool offset value	
:	:	:	
#100250	[#_TOFS[50]]	50th axis tool offset value	

<2> With tool geometry/wear compensation memory

Variable number	Variable name	Position information	Read operation during movement
#5081	[#_TOFSWX]	X-axis tool offset value (geometry)	Disabled
#5082	[#_TOFSWZ]	Z-axis tool offset value (geometry)	
#5083	[#_TOFSWY]	Y-axis tool offset value (geometry)	
#5084	[#_TOFS[4]]	4th axis tool offset value (geometry)	
:	:	:	
#5100	[#_TOFS[20]]	20th axis tool offset value (geometry)	
#100201	[#_TOFSWX]	X-axis tool offset value (geometry)	
#100202	[#_TOFSWZ]	Z-axis tool offset value (geometry)	
#100203	[#_TOFSWY]	Y-axis tool offset value (geometry)	
#100204	[#_TOFS[4]]	4th axis tool offset value (geometry)	
:	:	:	
#100250	[#_TOFS[50]]	50th axis tool offset value (geometry)	
#5121	[#_TOFSGX]	X-axis tool offset value (geometry)	
#5122	[#_TOFSGZ]	Z-axis tool offset value (geometry)	
#5123	[#_TOFSGY]	Y-axis tool offset value (geometry)	
#5124	[#_TOFSG[4]]	4th axis tool offset value (geometry)	
:	:	:	
#5140	[#_TOFSG[20]]	20th axis tool offset value (geometry)	
#100901	[#_TOFSGX]	X-axis tool offset value (geometry)	
#100902	[#_TOFSGZ]	Z-axis tool offset value (geometry)	
#100903	[#_TOFSGY]	Y-axis tool offset value (geometry)	
#100904	[#_TOFSG[4]]	4th axis tool offset value (geometry)	
:	:	:	
#100950	[#_TOFSG[50]]	50th axis tool offset value (geometry)	

When tool geometry/wear compensation memory exists, the system variable values vary as follows depending on parameter LWT (No.5002#2) and parameter LGT (No.5002#4).

Variable number	LWT=0 LGT=0	LWT=1 LGT=0	LWT=0 LGT=1	LWT=1 LGT=1
#5081 #5082 #5083	Wear compensation	0	Wear compensation	Wear compensation
#5121 #5122 #5123	Geometry compensation	Wear compensation + Geometry compensation	Geometry compensation	Geometry compensation

NOTE
1 The set value is read as the tool offset regardless of bit 1 (ORC) of parameter No.5004 and bit 0 (OWD) of parameter No.5040.
2 To read the tool offset (geometry) using #5121 to #5123, set bit 2 (VHD) of parameter No.6004 to 0.

- Servo position deviation #5101-#5120, #100251-#100300 (Attribute: R)

The servo position deviation for each axis can be obtained by reading system variables #5101 to #5120 or #100251 to #100300.

Variable number	Variable name	Position information	Read operation during movement
#5101	[#_SVERR[1]]	1st axis servo position deviation	Disabled
#5102	[#_SVERR[2]]	2nd axis servo position deviation	
:	:	:	
#5120	[#_SVERR[20]]	20th axis servo position deviation	
#100251	[#_SVERR[1]]	1st axis servo position deviation	
#100252	[#_SVERR[2]]	2nd axis servo position deviation	
:	:	:	
#100300	[#_SVERR[50]]	50th axis servo position deviation	

NOTE
1 When variables exceeding the number of control axes are specified, the alarm PS0115, “VARIABLE NO. OUT OF RANGE” occurs.
2 The servo position deviation for 20th or earlier axis can be used with #5101 to #5120.

- Manual handle interruption #5121-#5140, #100651-#100700 (Attribute: R)

The manual handle interruption for each axis can be obtained by reading system variables #5121 to #5140 or #100651 to #100700.

Variable number	Variable name	Position information	Read operation during movement
#5121	[#_MIRTP[1]]	1st axis manual handle interruption	Disabled
#5122	[#_MIRTP[2]]	2nd axis manual handle interruption	
:	:	:	
#5140	[#_MIRTP[20]]	20th axis manual handle interruption	
#100651	[#_MIRTP[1]]	1st axis manual handle interruption	
#100652	[#_MIRTP[2]]	2nd axis manual handle interruption	
:	:	:	
#100700	[#_MIRTP[50]]	50th axis manual handle interruption	

NOTE
1 When variables exceeding the number of control axes are specified, the alarm PS0115, “VARIABLE NO. OUT OF RANGE” occurs.
2 The manual handle interruption for 20th or earlier axis can be used with #5121 to #5140.

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NOTE
#5121 to #5140 are enabled only when bit 2 (VHD) of parameter No.6004 is set to 1.

- Distance to go #5181-#5200, #100801-#100850 (Attribute: R)

The distance to go value for each axis can be obtained by reading system variables #5181 to #5200 or #100801 to #100850.

Variable number	Variable name	Position information	Read operation during movement
#5181	[#_DIST[1]]	1st axis distance to go value	Disabled
#5182	[#_DIST[2]]	2nd axis distance to go value	
:	:	:	
#5200	[#_DIST[20]]	20th axis distance to go value	
#100801	[#_DIST[1]]	1st axis distance to go value	
#100802	[#_DIST[2]]	2nd axis distance to go value	
:	:	:	
#100850	[#_DIST[50]]	50th axis distance to go value	

NOTE
1 When variables exceeding the number of control axes are specified, the alarm PS0115, “VARIABLE NO. OUT OF RANGE” occurs.
2 The distance to go values for 20th or earlier axis can be used with #5181 to #5200.

- Workpiece coordinate system shift value #2501, #2601 (Attribute: R/W)

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The X-axis workpiece coordinate system shift value can be obtained by system variable #2501 while Z-axis workpiece coordinate system shift value by system variable #2601. Each workpiece coordinate system shift value can be obtained by system variable #100751-#100800. The X-axis or Z-axis workpiece coordinate system shift value can be changed by assigning values to the system variables. (X-axis: X-axis of basic three axes, Z-axis: Z-axis of basic three axes, Y-axis: Y-axis of basic three axes)

Variable number	Variable name	Description
#2501	[#_WZ_SFTX]	X-axis workpiece shift value
#2601	[#_WZ_SFTZ]	Z-axis workpiece shift value
#100751	[#_WZ_SFT[1]]	1st axis workpiece shift value
#100752	[#_WZ_SFT [2]]	2nd axis workpiece shift value
:	:	:
#100800	[#_WZ_SFT [50]]	50th axis workpiece shift value

NOTE
1 The value of a variable with a number greater than the number of controlled axes is undefined.
2 The X-axis of three basic axes can also be used in system variable #2501, and the Z-axis of three basic axes can also be used in system variable #2601.

- Workpiece origin offset value #5201-#5340, #100301-#100650 (Attribute: R/W)

The workpiece origin offset value can be obtained by reading system variables #5201 to #5340 or #100301 to #100650. The offset value can also be changed by assigning values to the system variables.

Variable number	Variable name	Controlled axis	Workpiece coordinate system
#5201	[#_WZCMN[1]]	1st axis external workpiece origin offset value	External workpiece origin offset value (applied to all coordinate systems)
#5202	[#_WZCMN[2]]	2nd axis external workpiece origin offset value	
:	:	:	
#5220	[#_WZCMN[20]]	20th axis external workpiece origin offset value	
#5221	[#_WZG54[1]]	1st axis workpiece origin offset value	G54
#5222	[#_WZG54[2]]	2nd axis workpiece origin offset value	
:	:	:	
#5240	[#_WZG54[20]]	20th axis workpiece origin offset value	
#5241	[#_WZG55[1]]	1st axis workpiece origin offset value	G55
#5242	[#_WZG55[2]]	2nd axis workpiece origin offset value	
:	:	:	
#5260	[#_WZG55[20]]	20th axis workpiece origin offset value	
#5261	[#_WZG56[1]]	1st axis workpiece origin offset value	G56
#5262	[#_WZG56[2]]	2nd axis workpiece origin offset value	
:	:	:	
#5280	[#_WZG56[20]]	20th axis workpiece origin offset value	
#5281	[#_WZG57[1]]	1st axis workpiece origin offset value	G57
#5282	[#_WZG57[2]]	2nd axis workpiece origin offset value	
:	:	:	
#5300	[#_WZG57[20]]	20th axis workpiece origin offset value	
#5301	[#_WZG58[1]]	1st axis workpiece origin offset value	G58
#5302	[#_WZG58[2]]	2nd axis workpiece origin offset value	
:	:	:	
#5320	[#_WZG58[20]]	20th axis workpiece origin offset value	
#5321	[#_WZG59[1]]	1st axis workpiece origin offset value	G59
#5322	[#_WZG59[2]]	2nd axis workpiece origin offset value	
:	:	:	
#5340	[#_WZG59[20]]	20th axis workpiece origin offset value	
#100301	[#_WZCMN[1]]	1st axis external workpiece origin offset value	External workpiece origin offset value (applied to all coordinate systems)
#100302	[#_WZCMN[2]]	2nd axis external workpiece origin offset value	
:	:	:	
#100350	[#_WZCMN[50]]	50th axis external workpiece origin offset value	
#100351	[#_WZG54[1]]	1st axis workpiece origin offset value	G54
#100352	[#_WZG54[2]]	2nd axis workpiece origin offset value	
:	:	:	
#100400	[#_WZG54[50]]	50th axis workpiece origin offset value	
#100401	[#_WZG55[1]]	1st axis workpiece origin offset value	G55
#100402	[#_WZG55[2]]	2nd axis workpiece origin offset value	
:	:	:	
#100450	[#_WZG55[50]]	50th axis workpiece origin offset value	
#100451	[#_WZG56[1]]	1st axis workpiece origin offset	G56
#100452	[#_WZG56[2]]	2nd axis workpiece origin offset	
:	:	:	
#100500	[#_WZG56[50]]	50th axis workpiece origin offset	

Variable number	Variable name	Controlled axis	Workpiece coordinate system
#100501	[#_WZG57[1]]	1st axis workpiece origin offset	G57
#100502	[#_WZG57[2]]	2nd axis workpiece origin offset	
:	:	:	
#100550	[#_WZG57[50]]	50th axis workpiece origin offset	
#100551	[#_WZG58[1]]	1st axis workpiece origin offset	G58
#100552	[#_WZG58[2]]	2nd axis workpiece origin offset	
:	:	:	
#100600	[#_WZG58[50]]	50th axis workpiece origin offset	
#100601	[#_WZG59[1]]	1st axis workpiece origin offset	G59
#100602	[#_WZG59[2]]	2nd axis workpiece origin offset	
:	:	:	
#100650	[#_WZG59[50]]	50th axis workpiece origin offset	

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The following variables can also be used when bit 5 (D15) of parameter No. 6004 is set to 0:

Axis	Function	Variable number
1st axis	External workpiece origin offset value	#2500
	G54 workpiece origin offset value	#2501
	G55 workpiece origin offset value	#2502
	G56 workpiece origin offset value	#2503
	G57 workpiece origin offset value	#2504
	G58 workpiece origin offset value	#2505
	G59 workpiece origin offset value	#2506
2nd axis	External workpiece origin offset value	#2600
	G54 workpiece origin offset value	#2601
	G55 workpiece origin offset value	#2602
	G56 workpiece origin offset value	#2603
	G57 workpiece origin offset value	#2604
	G58 workpiece origin offset value	#2605
	G59 workpiece origin offset value	#2606
3rd axis	External workpiece origin offset value	#2700
	G54 workpiece origin offset value	#2701
	G55 workpiece origin offset value	#2702
	G56 workpiece origin offset value	#2703
	G57 workpiece origin offset value	#2704
	G58 workpiece origin offset value	#2705
	G59 workpiece origin offset value	#2706
4th axis	External workpiece origin offset value	#2800
	G54 workpiece origin offset value	#2801
	G55 workpiece origin offset value	#2802
	G56 workpiece origin offset value	#2803
	G57 workpiece origin offset value	#2804
	G58 workpiece origin offset value	#2805
	G59 workpiece origin offset value	#2806

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The following variables can be used to maintain compatibility with conventional models.

Axis	Function	Variable number
1st axis	External workpiece origin offset value	#2550
	G54 workpiece origin offset value	#2551
	G55 workpiece origin offset value	#2552
	G56 workpiece origin offset value	#2553
	G57 workpiece origin offset value	#2554
	G58 workpiece origin offset value	#2555
	G59 workpiece origin offset value	#2556
2nd axis	External workpiece origin offset value	#2650
	G54 workpiece origin offset value	#2651
	G55 workpiece origin offset value	#2652
	G56 workpiece origin offset value	#2653
	G57 workpiece origin offset value	#2654
	G58 workpiece origin offset value	#2655
	G59 workpiece origin offset value	#2656
3rd axis	External workpiece origin offset value	#2750
	G54 workpiece origin offset value	#2751
	G55 workpiece origin offset value	#2752
	G56 workpiece origin offset value	#2753
	G57 workpiece origin offset value	#2754
	G58 workpiece origin offset value	#2755
	G59 workpiece origin offset value	#2756
4th axis	External workpiece origin offset value	#2850
	G54 workpiece origin offset value	#2851
	G55 workpiece origin offset value	#2852
	G56 workpiece origin offset value	#2853
	G57 workpiece origin offset value	#2854
	G58 workpiece origin offset value	#2855
	G59 workpiece origin offset value	#2856

NOTE
1 When variables exceeding the number of control axes are specified, the alarm PS0115, “VARIABLE NO. OUT OF RANGE” occurs.
2 The workpiece origin offset value for 20th or earlier axis can be used with #5201 to #5340.

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NOTE
To use variables #2500 to #2806, #5201 to #5340, and #100301 to #100650, optional variables for the workpiece coordinate systems are necessary.

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NOTE
To use variables #2550 to #2856, #5201 to #5340, and #100301 to #100650, optional variables for the workpiece coordinate systems are necessary.

- Workpiece origin offset value of the additional workpiece coordinate system

#7001-#7960, #101001-#116000 (Attribute: R/W)

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#14001-#20000 (Attribute: R/W)

The workpiece origin offset value of the additional workpiece coordinate system can be obtained by reading system variables #7001 to #7960, #14001 to #20000, #101001 to #116000. The offset value can also be changed by assigning values to the system variables.

Variable number	Variable name	Controlled axis	Additional workpiece system number
#7001	[#_WZP1[1]]	1st axis workpiece origin offset value	
#7002	[#_WZP1[2]]	2nd axis workpiece origin offset value	1
:	:	:	(G54.1 P1)
#7020	[#_WZP1[20]]	20th axis workpiece origin offset value	
#7021	[#_WZP2[1]]	1st axis workpiece origin offset value	
#7022	[#_WZP2[2]]	2nd axis workpiece origin offset value	2
:	:	:	(G54.1 P2)
#7040	[#_WZP2[20]]	20th axis workpiece origin offset value	
#7041	[#_WZP3[1]]	1st axis workpiece origin offset value	
#7042	[#_WZP3[2]]	2nd axis workpiece origin offset value	3
:	:	:	(G54.1 P3)
#7060	[#_WZP3[20]]	20th axis workpiece origin offset value	
:	:	:	:
#7941	[#_WZP48[1]]	1st axis workpiece origin offset value	
#7942	[#_WZP48[2]]	2nd axis workpiece origin offset value	48
:	:	:	(G54.1 P48)
#7960	[#_WZP48[20]]	20th axis workpiece origin offset value	

System variable number = 7000 + (Coordinate system number -1) × 20 + Axis number
Coordinate number: 1 to 48
Axis number: 1 to 20

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Variable number	Variable name	Controlled axis	Additional workpiece system number
#14001	[#_WZP1[1]]	1st axis workpiece origin offset value	
#14002	[#_WZP1[2]]	2nd axis workpiece origin offset value	1
:	:	:	(G54.1 P1)
#14020	[#_WZP1[20]]	20th axis workpiece origin offset value	
#14021	[#_WZP2[1]]	1st axis workpiece origin offset value	
#14022	[#_WZP2[2]]	2nd axis workpiece origin offset value	2
:	:	:	(G54.1 P2)
#14040	[#_WZP2[20]]	20th axis workpiece origin offset value	
#14041	[#_WZP3[1]]	1st axis workpiece origin offset value	
#14042	[#_WZP3[2]]	2nd axis workpiece origin offset value	3
:	:	:	(G54.1 P3)
#14060	[#_WZP3[20]]	20th axis workpiece origin offset value	
:	:	:	:
#19981	[#_WZP300[1]]	1st axis workpiece origin offset value	
#19982	[#_WZP300[2]]	2nd axis workpiece origin offset value	300
:	:	:	(G54.1 P300)
#20000	[#_WZP300[20]]	20th axis workpiece origin offset value	

System variable number = 14000 + (Coordinate system number -1) × 20 + Axis number
Coordinate number: 1 to 300
Axis number: 1 to 20

Variable number	Variable name	Controlled axis	Additional workpiece system number
#101001	[#_WZP1[1]]	1st axis workpiece origin offset value	
#101002	[#_WZP1[2]]	2nd axis workpiece origin offset value	1
:	:	:	(G54.1 P1)
#101050	[#_WZP1[50]]	50th axis workpiece origin offset value	
#101051	[#_WZP2[1]]	1st axis workpiece origin offset value	
#101052	[#_WZP2[2]]	2nd axis workpiece origin offset value	2
:	:	:	(G54.1 P2)
#101100	[#_WZP2[50]]	50th axis workpiece origin offset value	
#101101	[#_WZP3[1]]	1st axis workpiece origin offset value	
#101102	[#_WZP3[2]]	2nd axis workpiece origin offset value	3
:	:	:	(G54.1 P3)
#101150	[#_WZP3[50]]	50th axis workpiece origin offset value	
:	:	:	:
#115951	[#_WZP300[1]]	1st axis workpiece origin offset value	
#115952	[#_WZP300[2]]	2nd axis workpiece origin offset value	300
:	:	:	(G54.1 P300)
#116000	[#_WZP300[50]]	50th axis workpiece origin offset value	

System variable number = 101000 + (Coordinate system number -1) × 50 + Axis number
Coordinate number: 1 to 300
Axis number: 1 to 50

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NOTE
1 When variables exceeding the number of control axes are specified, the alarm PS0115, “VARIABLE NO. OUT OF RANGE” occurs.
2 The workpiece origin offset of additional workpiece coordinate system for 20th or earlier axis can be used with #7001 to #7960 or #14001 to #20000.
3 Optional variables for 48 additional workpiece coordinate systems are #7001 to #7960 (G54.1 P1 to G54.1 P48). Optional variables for 300 additional workpiece coordinate systems are #14001 to #20000, and #101001 to #116000 (G54.1 P1 to G54.1 P300). With these variables, #7001 to #7960 can also be used.

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NOTE
1 When variables exceeding the number of control axes are specified, the alarm PS0115, “VARIABLE NO. OUT OF RANGE” occurs.
2 Optional variables for 48 additional workpiece coordinate systems are #7001 to #7960 (G54.1 P1 to G54.1 P48).
3 Optional variables for 300 additional workpiece coordinate systems are #101001 to #116000 (G54.1 P1 to G54.1 P300). With these variables, #7001 to #7960 can also be used.

- Skip position (detection unit) #5421-#5440, #100701-#100750 (Attribute: R)

The skip position with the detection unit can be obtained by reading system variables #5421 to #5440 or #100701 to #100750. The values of these variables are integers, and values in the workpiece coordinate system.

Variable number	Variable name	Position information	Read operation during movement
#5421	[#_SKPDTC[1]]	1st axis skip position (detection unit)	Disabled
#5422	[#_SKPDTC[2]]	2nd axis skip position (detection unit)	
:	:	:	
#5440	[#_SKPDTC[20]]	20th axis skip position (detection unit)	
#100151	[#_SKPDTC[1]]	1st axis skip position (detection unit)	
#100152	[#_SKPDTC[2]]	2nd axis skip position (detection unit)	
:	:	:	
#100200	[#_SKPDTC[50]]	50th axis skip position (detection unit)	

NOTE
1 When variables exceeding the number of control axes are specified, the alarm PS0115, “VARIABLE NO. OUT OF RANGE” occurs.
2 The skip position (detection unit) for 20th or earlier axis can be used with #5421 to #5440.
3 To specify these variables, set bit 2 (DSK) of parameter No. 6210 to 1. (If an attempt is made to specify them when DSK is 0, alarm PS0115 is issued.)

- Reference fixture offset number being selected #5500 (Attribute: R)

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The reference fixture offset number being selected can be read by reading system variables #5500.

Variable number	Variable name	Description
#5500	[#_FOFSP]	Reference fixture offset number being selected

- Reference fixture offset value being selected #5501-#5520, #117001-#117050 (Attribute: R)

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The reference fixture offset being selected can be read by reading system variables #5501 to #5520 or #117001# to 117050.

Variable number	Variable name	Description
#5501	[#_FOFSVAL[1]]	1st axis reference fixture offset value being selected
#5502	[#_FOFSVAL[2]]	2nd axis reference fixture offset value being selected
:	:	:
#5520	[#_FOFSVAL[20]]	20th axis reference fixture offset value being selected
#117001	[#_FOFSVAL[1]]	1st axis reference fixture offset value being selected
#117002	[#_FOFSVAL[2]]	2nd axis reference fixture offset value being selected
:	:	:
#117050	[#_FOFSVAL[50]]	50th axis reference fixture offset value being selected

NOTE
1 When variables exceeding the number of control axes are specified, the alarm PS0115, “VARIABLE NO. OUT OF RANGE” occurs.
2 The reference fixture offset for 20th or earlier axis can be used with #5501 to #5520.

- Reference fixture offset value #5521-#5680, #117051-#117450 (Attribute: R/W)

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The reference fixture offset values in the rotary table dynamic fixture offset function can be obtained by reading system variables #5521 to #5680 or #117051 to #117450. The reference fixture offset values can also be changed by assigning values to the system values.

Variable number	Variable name	Controlled axis	Fixture offset number
#5521	[#_FOFS1[1]]	1st axis reference fixture offset value being selected	
#5522	[#_FOFS1[2]]	2nd axis reference fixture offset value being selected	1
:	:	:	(G54.2 P1)
#5540	[#_FOFS1[20]]	20th axis reference fixture offset value being selected	
#5541	[#_FOFS2[1]]	1st axis reference fixture offset value being selected	
#5542	[#_FOFS2[2]]	2nd axis reference fixture offset value being selected	2
:	:	:	(G54.2 P2)
#5560	[#_FOFS2[20]]	20th axis reference fixture offset value being selected	
:	:	:	:
#5661	[#_FOFS8[1]]	1st axis reference fixture offset value being selected	
#5662	[#_FOFS8[2]]	2nd axis reference fixture offset value being selected	8
:	:	:	(G54.2 P8)
#5680	[#_FOFS8[20]]	20th axis reference fixture offset value being selected	
#117051	[#_FOFS1[1]]	1st axis reference fixture offset value being selected	
#117052	[#_FOFS1[2]]	2nd axis reference fixture offset value being selected	1
:	:	:	(G54.2 P1)
#117100	[#_FOFS1[50]]	50th axis reference fixture offset value being selected	
#117101	[#_FOFS2[1]]	1st axis reference fixture offset value being selected	
#117102	[#_FOFS2[2]]	2nd axis reference fixture offset value being selected	2
:	:	:	(G54.2 P2)
#117150	[#_FOFS2[50]]	50th axis reference fixture offset value being selected	
:	:	:	:
#117401	[#_FOFS8[1]]	1st axis reference fixture offset value being selected	
#117402	[#_FOFS8[2]]	2nd axis reference fixture offset value being selected	8
:	:	:	(G54.2 P8)
#117450	[#_FOFS8[50]]	50th axis reference fixture offset value being selected	

NOTE
1 When variables exceeding the number of control axes are specified, the alarm PS0115, “VARIABLE NO. OUT OF RANGE” occurs.
2 The reference fixture offset values for 20th or earlier axis can be used with #5521 to #5580.

- Second geometry tool offset value #5801-#5896, #27001-#29999 (Attribute: R/W)

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By reading the values of system variables #5801 to #5896 and #27001 to #29999, it is possible to determine the second geometry tool offset value, and by assigning values to the system variables, it is possible to change the second geometry tool offset value.

- For compensation numbers up to 32

Compensation number	Variable number	Variable name	Attribute	Description
1	#5801	[#_OFSX2G[1]]	R/W	Second geometry tool offset X-axis compensation value
:	:	:	:	
32	#5832	[#_OFSX2G[32]]	R/W	
1	#5833	[#_OFSZ2G[1]]	R/W	Second geometry tool offset Z-axis compensation value
:	:	:	:	
32	#5864	[#_OFSZ2G[32]]	R/W	
1	#5865	[#_OFSY2G[1]]	R/W	Second geometry tool offset Y-axis compensation value
:	:	:	:	
32	#5896	[#_OFSY2G[32]]	R/W	

- For compensation numbers greater than 32 (Compensation values with compensation numbers of up to 32 can also be used in variables #5801 to #5896.)

Compensation number	Variable number	Variable name	Attribute	Description
1	#27001	[#_OFSX2G[1]]	R/W	Second geometry tool offset X-axis compensation value
:	:	:	:	
999	#27999	[#_OFSX2G[999]]	R/W	
1	#28001	[#_OFSZ2G[1]]	R/W	Second geometry tool offset Z-axis compensation value
:	:	:	:	
999	#28999	[#_OFSZ2G[999]]	R/W	
1	#29001	[#_OFSY2G[1]]	R/W	Second geometry tool offset Y-axis compensation value
:	:	:	:	
999	#29999	[#_OFSY2G[999]]	R/W	

- Dynamic reference tool compensation value #118051-#118450 (Attribute: R/W)

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The dynamic reference tool compensation value in the rotary head dynamic tool compensation function can be obtained by reading system variables #118051 to #118450. The dynamic reference tool compensation value can also be obtained by assigning values to the system variables.

Variable number	Variable name	Controlled axis	Dynamic tool offset number
#118051	[#_DOFS1[1]]	1st axis dynamic reference tool compensation value	
#118052	[#_DOFS1[2]]	2nd axis dynamic reference tool compensation value	1
:	:	:	(G43.2H1)
#118100	[#_DOFS1[50]]	50th axis dynamic reference tool compensation value	
#118101	[#_DOFS2[1]]	1st axis dynamic reference tool compensation value	
#118102	[#_DOFS2[2]]	2nd axis dynamic reference tool compensation value	2
:	:	:	(G43.2H2)
#118150	[#_DOFS2[50]]	50th axis dynamic reference tool compensation value	
#118151	[#_DOFS3[1]]	1st axis dynamic reference tool compensation value	
#118152	[#_DOFS3[2]]	2nd axis dynamic reference tool compensation value	3
:	:	:	(G43.2H3)
#118200	[#_DOFS3[50]]	50th axis dynamic reference tool compensation value	
#118201	[#_DOFS4[1]]	1st axis dynamic reference tool compensation value	
#118202	[#_DOFS4[2]]	2nd axis dynamic reference tool compensation value	4
:	:	:	(G43.2H4)
#118250	[#_DOFS4[50]]	50th axis dynamic reference tool compensation value	
#118251	[#_DOFS5[1]]	1st axis dynamic reference tool compensation value	
#118252	[#_DOFS5[2]]	2nd axis dynamic reference tool compensation value	5
:	:	:	(G43.2H5)
#118300	[#_DOFS5[50]]	50th axis dynamic reference tool compensation value	
:	:	:	:
#118401	[#_DOFS8[1]]	1st axis dynamic reference tool compensation value	
#118402	[#_DOFS8[2]]	2nd axis dynamic reference tool compensation value	8
:	:	:	(G43.2H8)
#118450	[#_DOFS8[50]]	50th axis dynamic reference tool compensation value	

NOTE
When variables exceeding the number of control axes are specified, the alarm PS0115, “VARIABLE NO. OUT OF RANGE” occurs.

- Feedrate reduction ratio for rapid traverse overlap #100851-#100900 (Attribute: R/W)

The feedrate reduction ratio for rapid traverse overlap can also be changed by setting values to the system variables #100851 to #100900.

Variable number	Variable name	Controlled axis
#100851	[#_ROVLP[1]]	1st axis feedrate reduction ratio for rapid traverse overlap
#100852	[#_ROVLP[2]]	2nd axis feedrate reduction ratio for rapid traverse overlap
:	:	:
#100900	[#_ROVLP[50]]	50th axis feedrate reduction ratio for rapid traverse overlap

- Switching between P-CODE variables and system variables (#10000 or later) #8570 (Attribute: R/W)

This system variable allows read/write operations of P-CODE variables (#10000 to #89999) for the macro executor function. For details on P-CODE variables, refer to the Macro Compiler / Macro Executor Programming Manual (B-63943EN-2).
System variable #8570 can be used to make variables #10000 or later correspond to either P-CODE variables or system variables.

#8570 setting	Specified variable	Corresponding variable
#8570 = 1	#10000	P-CODE variables (#10000)
	:	:
	#89999	P-CODE variables (#89999)
#8570 = 0	#10000	System variables (#10000)
	:	:
	#89999	System variables (#89999)

Example

```
#8570 = 0 ;
#10001 = 123 ; → Writing to system variable
                 #10001 (tool compensation)
#8570 = 1 ;
#10001 = 456 ; → Writing to P-CODE variable
                 #10001 (tool compensation)
```

NOTE

1 Variable #8570 can be used only when the macro executor function is enabled.
2 System variables (#10000 or later) always correspond to system variables specified by their variable names even when #8570 is 1.
3 When an attempt is made to access a variable that cannot be used with P-CODE variables (#10000 or later), an alarm PS0115 occurs.

16.3 ARITHMETIC AND LOGIC OPERATION

Various operations can be performed on variables. Program an arithmetic and logic operation in the same way as for a general arithmetic expression.

#i=<expression>

<Expression>

The expression to the right of the arithmetic and logic operation contains constants and/or variables combined by a function or operator. Variables #j and #k below can be replaced with a constant. If a constant used in an expression has no decimal point, it is assumed to end with a decimal point.

Table 16.3 (a) Arithmetic and logic operation

Type of operation	Operation	Description
<1> Definition or replacement	#i=#j	Definition or replacement of a variable
<2> Addition-type operations	#i=#j+#k	Addition
	#i=#j-#k	Subtraction
	#i=#j OR #k	Logical OR (bit by bit of 32 bits)
	#i=#j XOR #k	Exclusive OR (bit by bit of 32 bits)
<3> Multiplication-type operations	#i=#j*#k	Multiplication
	#i=#j/#k	Division
	#i=#j AND #k	Logical AND (bit by bit of 32 bits)
	#i=#j MOD #k	Remainder (A remainder is obtained after #j and #k are rounded to their nearest whole numbers. When #j is a negative value, #i is assumed to be a negative value.)
<4> Functions	#i=SIN[#j]	Sine (in degrees)
	#i=COS[#j]	Cosine (in degrees)
	#i=TAN[#j]	Tangent (in degrees)
	#i=ASIN[#j]	Arc sine
	#i=ACOS[#j]	Arc cosine
	#i=ATAN[#j]	Arc tangent (one argument), ATN can also be used.
	#i=ATAN[#j]/[#k]	Arc tangent (two arguments), ATN can also be used.
	#i=ATAN[#j,#k]	Arc tangent (two arguments), ATN can also be used.
	#i=SQRT[#j]	Square root, SQR can also be used.
	#i=ABS[#j]	Absolute value
	#i=BIN[#j]	Conversion from BCD to binary
	#i=BCD[#j]	Conversion from binary to BCD
	#i=ROUND[#j]	Rounding off, RND can also be used.
	#i=FIX[#j]	Rounding down to an integer
	#i=FUP[#j]	Rounding up to an integer
	#i=LN[#j]	Natural logarithm
	#i=EXP[#j]	Exponent using base e (2.718...)
	#i=POW[#j,#k]	Power (#j to the #kth power)
	#i=ADP[#j]	Addition of a decimal point

Explanation

- Angle units

The units of angles used with the SIN, COS, ASIN, ACOS, TAN, and ATAN functions are degrees. For example, 90 degrees and 30 minutes is represented as 90.5 degrees.

- ARCSIN #i = ASIN[#j];

- The solution ranges are as indicated below:
  When the NAT bit (bit 0 of parameter 6004) is set to 0: 270° to 90°
  When the NAT bit (bit 0 of parameter 6004) is set to 1: -90° to 90°
- When #j is beyond the range of -1 to 1, an alarm PS0119 is issued.
- A constant can be used instead of the #j variable.

- ARCCOS #i = ACOS[#j];

- The solution ranges from 180° to 0°.
- When #j is beyond the range of -1 to 1, an alarm PS0119 is issued.
- A constant can be used instead of the #j variable.

- ARCTAN #i = ATAN[#j]/[#k]; (two arguments)

- ATAN[#j,#k] is equivalent to ATAN[#j]/[#k].
- When point (#k,#j) on plane X-Y is given, this function returns the value of the arc tangent for the angle made by the point.
- A constant can be used instead of the #j variable.
- The solution ranges are as follows:
  When the NAT bit (bit 0 of parameter 6004) is set to 0: 0° to 360°

Example:

When #1 = ATAN[-1]/[-1]; is specified, #1 is 225.0.

When the NAT bit (bit 0 of parameter 6004) is set to 1: -180° to 180°

Example:

When #1 = ATAN[-1]/[-1]; is specified, #1 is -135.0.

- ARCTAN #i = ATAN[#j]; (one argument)

- When ATAN is specified with one argument, this function returns the main value of arc tangent ($-90° \leq ATAN[\#j] \leq 90°$). In other word, this function returns the same value as ATAN in calculator specifications.
- To use this function as the dividend of a division, be sure to enclose it with brackets ([]). If this function is not enclosed, ATAN[#j]/[#k] is assumed.

Example:

```
#100 = [ATAN[1]]/10 ; : Divides ATAN with one argument by 10.
#100 = ATAN[1]/[10] ; : Executes ATAN with two arguments.
#100 = ATAN[1]/10 ;   : Assumes ATAN with two arguments, but issues
                        an alarm PS1131 because the X coordinate
                        specification is not enclosed with brackets ([]).
```

- Natural logarithm #i = LN[#j];

- When the antilogarithm (#j) is zero or smaller, an alarm PS0119 is issued.
- A constant can be used instead of the #j variable.

- Exponential function #i = EXP[#j];

- When the result of the operation overflows, an alarm PS0119 is issued.
- A constant can be used instead of the #j variable.

- ROUND function

- When the ROUND function is included in an arithmetic or logic operation command, IF statement, or WHILE statement, the ROUND function rounds off at the first decimal place.

Example:

When #1=ROUND[#2]; is executed where #2 holds 1.2345, the value of variable #1 is 1.0.

- When the ROUND function is used in NC statement addresses, the ROUND function rounds off the specified value according to the least input increment of the address.

Example:

Creation of a drilling program that cuts according to the values of variables #1 and #2, then returns to the original position

Suppose that the increment system is 1/1000 mm, variable #1 holds 1.2345, and variable #2 holds 2.3456. Then,

G00 G91 X-#1; Moves 1.235 mm in negative direction.

G01 X-#2 F300; Moves 2.346 mm in negative direction.

G00 X[#1+#2]; Since 1.2345 + 2.3456 = 3.5801 in positive direction, the travel distance is 3.580, which does not return the tool to the original position.

This difference comes from whether addition is performed before or after rounding off. G00X-[ROUND[#1]+ROUND[#2]]; must be specified to return the tool to the original position.

- Add decimal point (ADP) function

- ADP[#n] (n = 1 to 33) can be executed to add a decimal point to an argument passed with no decimal point, in the subprogram.

Example:

In the subprogram called with G65 P_X10;, the value of ADP[#24] is a value to which a decimal point is added at its end (that is, 10.). Use this function when you do not want to consider the increment system in the subprogram. When bit 4 (CVA) of parameter No. 6007 is set to 1, however, the ADP function cannot be used because any argument is converted to 0.01 the moment it is passed.

NOTE

For compatibility among programs, it is recommended that the ADP function be not used, and decimal points be added in the argument specification for a macro call.

- Rounding up and down to an integer (FUP and FIX)

With CNC, when the absolute value of the integer produced by an operation on a number is greater than the absolute value of the original number, such an operation is referred to as rounding up to an integer. Conversely, when the absolute value of the integer produced by an operation on a number is less than the absolute value of the original number, such an operation is referred to as rounding down to an integer. Be particularly careful when handling negative numbers.

Example:

Suppose that #1=1.2 and #2=-1.2.
When #3=FUP[#1] is executed, 2.0 is assigned to #3.
When #3=FIX[#1] is executed, 1.0 is assigned to #3.
When #3=FUP[#2] is executed, -2.0 is assigned to #3.
When #3=FIX[#2] is executed, -1.0 is assigned to #3.

- Abbreviations of arithmetic and logic operation commands

When a function is specified in a program, the first two characters of the function name can be used to specify the function.

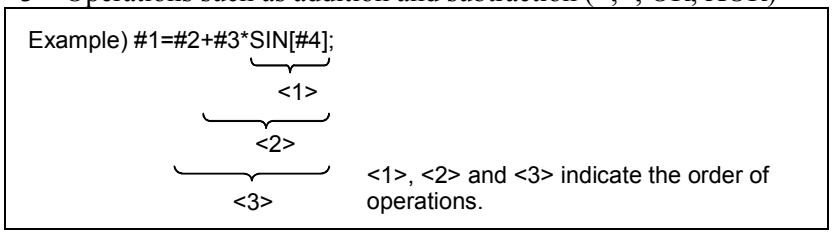
Example:

ROUND → RO
FIX → FI

NOTE
POW cannot be abbreviated.

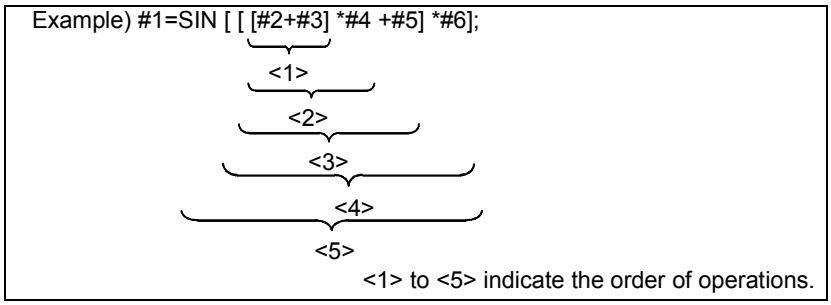
- Priority of operations

<1> Functions
<2> Operations such as multiplication and division (*, /, AND)
<3> Operations such as addition and subtraction (+, -, OR, XOR)



- Bracket nesting

Brackets are used to change the order of operations. Brackets can be used to a depth of five levels including the brackets used to enclose a function. When a depth of five levels is exceeded, an alarm PS0118 occurs.



Limitation

- **Caution concerning decreased precision**
  When bit 0 (F16) of parameter No. 6008 is set to 0

  - Addition and subtraction
    Note that when an absolute value is subtracted from another absolute value in addition or subtraction, the relative error may become 10^{-15} or greater.
    For example, assume that #1 and #2 have the following true values in the process of operation.
    (The following values are examples in the process of operation and cannot actually be specified from any program.)
    ```
    #1=9876543210.987654321
    #2=9876543210.987657777
    ```
    You cannot obtain the following result with operation #2-#1:
    ```
    #2-#1=0.000003456
    ```
    This is because the precision of custom macro variables is 15 decimal digits. With this precision, the values of #1 and #2 become:
    ```
    #1=9876543210.987650000
    #2=9876543210.987660000
    ```
    (Precisely, the actual values are slightly different from the above values because they are internally processed in binary.) Therefore, the result is:
    ```
    #2-1=0.000010000
    ```
    A large error occurs.
  - Logical expressions
    Be aware of errors that can result from conditional expressions using EQ, NE, GT, LT, GE, and LE because they are processed basically in the same way as addition and subtraction. For example, if the following statement is used to decide whether #1 is equal to #2 in the above example, a correct decision may not be resulted because errors may occur:
    ```
    IF [#1 EQ #2]
    ```
    Evaluate the difference between #1 and #2 with:
    ```
    IF [ABS [#1-#2]LT 0.1]
    ```
    Then, assume that the values are equal when the difference does not exceed the allowable error range.
  - Trigonometric functions
    The absolute error is guaranteed for trigonometric functions. However, the relative error is 10^{-15} or greater. Carefully perform multiplication or division after executing a trigonometric function.
  - FIX function
    When using the FIX function for the result of an operation, be careful with the precision. For example, when the following operations are performed, the value of #3 may not always be 2.
    ```
    N10 #1=0.002;
    N20 #2=#1*1000;
    N30 #3=FIX[#2];
    ```

This is because an error may occur in operation N20 and the result may not be

#2=2.0000000000000000

but a value a little smaller than 2 such as the following:

#2=1.9999999999999997

To prevent this, specify N30 as follows:

N30 #3=FIX[#2+0.001];

Generally, specify the FIX function as follows:

FIX[expression] → FIX[expression ±ε]

(Specify +ε when the value of the expression is positive or -ε when it is negative, and 0.1, 0.01, 0.001, ... for ε as required.)

NOTE
The operation result of exponential function #i=EXP[#j]; overflows when #j exceeds about 790.

When bit 0 (F16) of parameter No. 6008 is set to 1

Errors may occur when operations are performed.

Table 16.3 (b) Errors involved in operations

Operation	Average error	Maximum error	Type of error
a = b*c	1.55×10^{-10}	4.66×10^{-10}	Relative error(*1) $\left\lvert \frac{\varepsilon}{a} \right\rvert$
a = b / c	4.66×10^{-10}	1.88×10^{-9}	
$a = \sqrt{b}$	1.24×10^{-9}	3.73×10^{-9}	
a = b + c a = b – c	2.33×10^{-10}	5.32×10^{-10}	(*2) $\mathrm{MIN}\left\lvert \frac{\varepsilon}{b} \right\rvert \left\lvert \frac{\varepsilon}{c} \right\rvert$
a = SIN [b] a = COS [b]	5.0×10^{-9}	1.0×10^{-8}	Absolute error(*3) $\left\lvert \varepsilon \right\rvert$ degrees
a = ATAN [b] / [c]	1.8×10^{-6}	3.6×10^{-6}	

NOTE
1 The relative error depends on the result of the operation.
2 Smaller of the two types of errors is used.
3 The absolute error is constant, regardless of the result of the operation.
4 Function TAN performs SIN/COS.
5 Note that, in the case of natural logarithm #i=LN[#j]; and exponential function #i=EXP[#j];, the relative error may become 10^{-8} or greater.
6 The operation result of exponential function #i=EXP[#j]; overflows when #j exceeds about 110.

- The precision of variable values is about 8 decimal digits. When very large numbers are handled in an addition or subtraction, the expected results may not be obtained.

Example:

When an attempt is made to assign the following values to variables #1 and #2:

#1=9876543210123.456
#2=9876543277777.777

the values of the variables become:

#1=9876543200000.000
#2=9876543300000.000

In this case, when #3=#2-#1; is calculated, #3=100000.000 results. (The actual result of this calculation is slightly different because it is performed in binary.)

- Also be aware of errors that can result from conditional expressions using EQ, NE, GE, GT, LE, and LT.

Example:

IF[#1 EQ #2] is effected by errors in both #1 and #2, possibly resulting in an incorrect decision.

Therefore, instead find the difference between the two variables with IF[ABS[#1-#2]LT0.001].

Then, assume that the values of the two variables are equal when the difference does not exceed an allowable limit (0.001 in this case).

- Also, be careful when rounding down a value.

Example:

When #2=#1*1000; is calculated where #1=0.002;, the resulting value of variable #2 is not exactly 2 but 1.99999997.

Here, when #3=FIX[#2]; is specified, the resulting value of variable #3 is not 2.0 but 1.0.

In this case, round down the value after correcting the error so that the result is greater than the expected number, or round it off as follows:

#3=FIX[#2+0.001]
#3=ROUND[#2]

- Brackets

Brackets ([ ]) are used to enclose an expression.
Note that parentheses ( ) are used for comments.

- Divisor

When a divisor of zero is specified in a division, an alarm PS0112 occurs.

16.4 INDIRECT AXIS ADDRESS SPECIFICATION

Overview

When the custom macro function is enabled, you can use AX[(axis-number)] in an axis address specification to indirectly specify an axis with its axis number and not to directly specify it with its axis name.
You can also use AXNUM[(axis-name)] to obtain the axis number for an axis name.

Explanation

- Indirect axis address

You can use indirect axis address AX[ ] to specify an axis with its axis number. (An equal sign (=) is always required following AX[ ].)

AX[(axis-number)] = (numeric-value) ;

(axis-number): 1 to the number of controlled axes (number of controlled axes for each path in a multipath system)
(numeric-value): Value specified for the axis specified with its axis number

If an invalid axis number is specified, an alarm PS0331 occurs. If a value is put in a decimal place, the number is rounded off to an integer and the result is treated as the axis number.
You can also specify a variable (local, common, or system variable) for (axis-number). To specify an operation using a variable name in (axis-number), however, enclose the variable name with brackets ([ ]).

1. **AX[1]=100.0;**
   Specifies a value of 100.000 for the first axis.
2. **AX[#500]=200.0;**
   Specifies a value of 200.000 for the axis having the axis number indicated by the value stored in #500.
3. **AX[#500+1]=300.0;**
   Specifies a value of 300.000 for the axis having the axis number obtained by adding 1 to the value stored in #500.
4. **SETVN 500 [ABC];**
   AX[#ABC]=400.0;
   Specifies a value of 400.000 for the axis having the axis number indicated by the value stored in #ABC (#500).
5. **SETVN 500 [ABC];**
   AX[[#ABC]+1]=500.0;
   Specifies a value of 400.000 for the axis having the axis number obtained by adding 1 to the value stored in #ABC (#500).
6. **SETVN 500 [ABC];**
   AX[#ABC+1]=500.0;
   Alarm PS0331 occurs.

- AXNUM function

You can use AXNUM[ ] to obtain an axis number.

AXNUM[(axis-name)];

If an invalid axis name is specified, an alarm PS0332 occurs.

When the number of controlled axes is 3, the name of the first axis is X, that of the second axis is Y, and that of the third axis is Z

1. **#500=AXNUM[X];**
   A value of 1 is stored in #500.
2. **#501=AXNUM[Y];**
   A value of 2 is stored in #501.
3. **#502=AXNUM[Z];**
   A value of 3 is stored in #502.
4. **#503=AXNUM[A];**
   Alarm PS0332 occurs.

Example

Examples when the name of the first axis is X, that of the second axis is Y, and that of the third axis is Z1

```
N10 SETVN 500[AXIS1,AXIS2,AXIS3] ;
N20 [#AXIS1]=AXNUM[X] ;
N30 [#AXIS2]=AXNUM[Y] ;
N40 [#AXIS3]=AXNUM[Z1] ;
N50 G92 AX[#AXIS1]=0 AX[#AXIS2]=0 AX[#AXIS3]=0 ;
N60 G90G01F1000. ;
N70 AX[#AXIS1]=100.0 AX[#AXIS2]=100.0 AX[#AXIS3]=100.0 ;
N80 G02 AX[#AXIS1]=100. 0 AX[#AXIS2]= 0.0 R50.0 ;
N90 M02;
```

Limitation

When the custom macro function is enabled, AX and AXN cannot be used as an extended axis name. AX is assumed to be AX[ ] and AXN is assumed to be AXNUM[ ].

16.5 MACRO STATEMENTS AND NC STATEMENTS

The following blocks are referred to as macro statements:

- Blocks containing an arithmetic or logic operation (=)
- Blocks containing a control statement (such as GOTO, DO, END)
- Blocks containing a macro call command (such as macro calls by G65, G66, G66.1, G67, or other G codes, or by M codes)

Any block other than a macro statement is referred to as an NC statement.

Explanation

- Differences from NC statements

- Even when single block mode is on, the machine does not stop. Note, however, that the machine stops in the single block mode when bit 5 of parameter SBM No. 6000 is 1.
- Macro blocks are not regarded as blocks that involve no movement in the cutter compensation mode.

- NC statements that have the same property as macro statements

- NC statements that include a subprogram call command (such as subprogram calls by M98 or other M codes, or by T codes) and not include other command addresses except an O, file name, N, P, or L address have the same property as macro statements.
- NC statements that include M99 and those do not include other command addresses except an O, file name, N, P, or L address have the same property as macro statements.

16.6 BRANCH AND REPETITION

In a program, the flow of control can be changed using the GOTO statement and IF statement. Three types of branch and repetition operations are used:

Branch and repetition	┬	GOTO	(unconditional branch)
	├	IF	(conditional branch: if ..., then...)
	└	WHILE	(repetition while ...)

16.6.1 Unconditional Branch (GOTO Statement)

A branch to sequence number n occurs. When a sequence number outside of the range 1 to 99999999 is specified, an alarm PS1128 occurs. A sequence number can also be specified using an expression.

GOTOn ;	n: Sequence number (1 to 99999999)

Example:

```
GOTO 1;
GOTO #10;
```

⚠ WARNING

Do not specify multiple blocks with the same sequence number in a single program. It is very dangerous to specify such blocks because the destination of a branch from the GOTO statement is undefined.

NOTE

1 A backward branch takes more time as compared with a forward branch.
2 In the destination of GOTOn, that is, the block with sequence number n, the sequence number must appear at the beginning of the block. If the sequence number is not at the beginning of the block, a branch cannot be made.

16.6.2 GOTO Statement Using Stored Sequence Numbers

When the GOTO statement is executed in a custom macro control command, a sequence number search is made for sequence numbers stored at previous execution of the corresponding blocks at a high speed.
As a "sequence number stored at previous execution of the corresponding block", a unique sequence number within the same program or the sequence number of a subprogram call with which the block was executed is stored.
The storage type differs depending on the values of the following parameters.

(1) When bit 1 (MGO) of parameter No. 6000 is set to 1
- Fixed type: Up to 20 sequence numbers stored at execution of the corresponding blocks from the start of operation

(2) When bit 4 (HGO) of parameter No. 6000 is set to 1
- Variable type: Up to 30 sequence numbers stored at execution of the corresponding blocks before execution of the GOTO statement
- History type: Up to 10 sequence numbers stored by a sequence number search previously made using the GOTO statement

The stored sequence numbers are canceled in the following cases:
- Immediately after power-on
- After a reset
- Operation after program registration or editing (including background editing and MDI program editing)

⚠ WARNING

Do not specify multiple blocks with the same sequence number in a single program.

It is very dangerous to specify the sequence number of the branch destination before and after the GOTO statement and execute the GOTO statement because the branch destination changes according to the values of the parameters as shown below:

When bit 1 (MGO) or 4 (HGO) of parameter No. 6000 is set to 1	When both bits 1 (MGO) and 4 (HGO) of parameter No. 6000 are set to 0
: N10; : GOTO10; : N10; A branch to N10 before the GOTO statement occurs.	: N10; : GOTO10; : N10; A branch to N10 after the GOTO statement occurs.

When bit 1 (MGO) or 4 (HGO) of parameter No. 6000 is set to 1 and the GOTO statement is executed, the sequence number of the branch destination may not be contained in the sequence numbers stored at previous execution of the corresponding blocks. In this case, a branch to the sequence number in a block following the GOTO statement occurs (the destination is the same as when both bits are set to 0).

NOTE

When an external program is read and executed by DNC operation, the executed sequence numbers are not stored.

When a program registered in memory is executed by a subprogram call, the sequence numbers are stored.

⚠ CAUTION

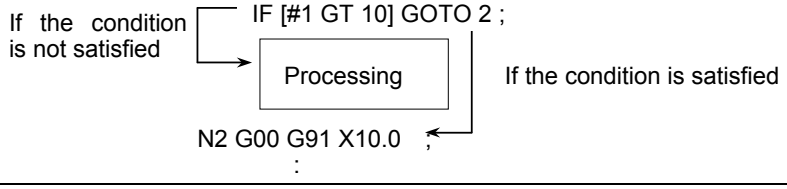
According to the restrictions on the GOTO statement, no branch to a sequence number within a DO-END loop cannot be made. If a program in which a branch to a sequence number within a loop occurs is executed, operation may differ depending on whether the GOTO statement using stored sequence numbers is used.

16.6.3 Conditional Branch (IF Statement)

Specify a <conditional expression> after IF.

IF[<conditional expression>]GOTOn

If the specified <conditional expression> is satisfied (true), a branch to sequence number n occurs. If the specified condition is not satisfied, the next block is executed.

If the value of variable #1 is greater than 10, a branch to sequence number N2 occurs.


IF[<conditional expression>]THEN

If the specified <conditional expression> is satisfied (true), a macro statement specified after THEN is executed.
Only a single macro statement is executed.

If the values of #1 and #2 are the same, 0 is assigned to #3.
IF[#1 EQ #2] THEN#3=0 ;

If the values of #1 and #2 are the same and those of #3 and #4 are also the same, 0 is assigned to #5.
IF[[#1 EQ #2] AND [#3 EQ #4]] THEN#5=0 ;

If the values of #1 and #2 are the same or those of #3 and #4 are the same, 0 is assigned to #5.
IF[[#1 EQ #2] OR [#3 EQ #4]] THEN#5=0 ;

Explanation

- <Conditional expression>

<Conditional expressions> are divided into <simple conditional expressions> and <complex conditional expressions>. In a <simple conditional expression>, a relational operator described in Table 16.6 (a) is specified between two variables or between a variable and constant to be compared. An <expression> can be used instead of a variable. With a <complex conditional expression>, an AND (logical AND), OR (logical OR), or XOR (exclusive OR) operation is performed for the results (true or false) of multiple <simple conditional expressions>.

- Relational operators

Relational operators each consist of two letters and are used to compare two values to determine whether they are equal or one value is smaller or greater than the other value. Note that the equal sign (=) and inequality sign (>, <) cannot be used as a relational operator.

Table 16.6 (a) Relational operators

Operator	Meaning
EQ	Equal to(=)
NE	Not equal to(≠)
GT	Greater than(>)
GE	Greater than or equal to(≥)
LT	Less than(<)
LE	Less than or equal to(≤)

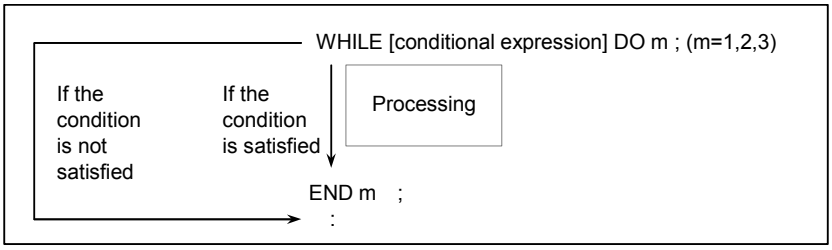
Sample program

The sample program below finds the total of numbers 1 to 10.

```
O9500;
  #1=0;..... Initial value of the variable to hold the sum
  #2=1;..... Initial value of the variable as an addend
N1 IF[#2 GT 10] GOTO 2;... Branch to N2 when the addend is greater than
     10
  #1=#1+#2;..... Calculation to find the sum
  #2=#2+1;..... Next addend
  GOTO 1; ..... Branch to N1
N2 M30;..... End of program
```

16.6.4 Repetition (WHILE Statement)

Specify a conditional expression after WHILE. While the specified condition is satisfied, the program from DO to END is executed. If the specified condition is not satisfied, program execution proceeds to the block after END.

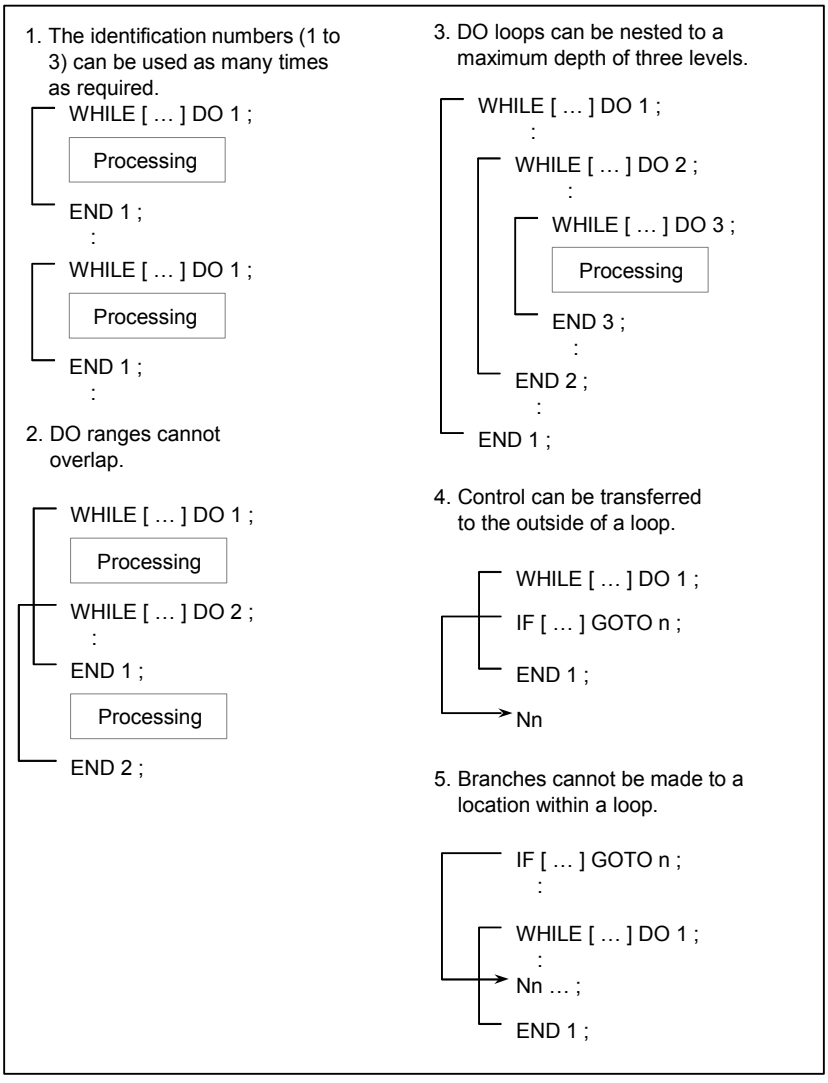


Explanation

While the specified condition is satisfied, the program from DO to END after WHILE is executed. If the specified condition is not satisfied, program execution proceeds to the block after END. The same format as for the IF statement applies. A number after DO and a number after END are identification numbers for specifying the range of execution. The numbers 1, 2, and 3 can be used. When a number other than 1, 2, and 3 is used, an alarm PS0126 occurs.

- Nesting

The identification numbers (1 to 3) in a DO-END loop can be used as many times as desired. Note, however, when a program includes crossing repetition loops (overlapped DO ranges), an alarm PS0124 occurs.



Limitation

- Infinite loops

When DO m is specified without specifying the WHILE statement, an infinite loop ranging from DO to END is produced.

- Processing time

When a branch to the sequence number specified in a GOTO statement occurs, the sequence number is searched for. For this reason, processing in the reverse direction takes a longer time than processing in the forward direction. Therefore, in the case of processing in the reverse direction, use the WHILE statement for repetition to reduce processing time.

- Undefined variable

In a conditional expression that uses EQ or NE, a <null> and zero have different effects. In other types of conditional expressions, a <null> is regarded as zero.

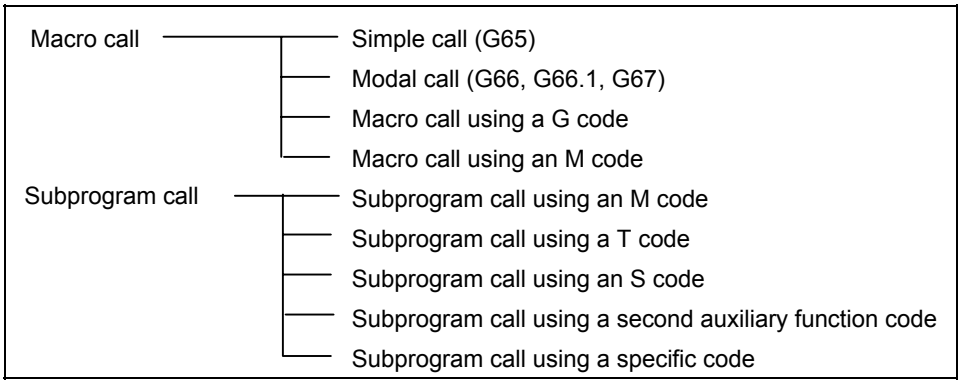
Sample program

The sample program below finds the total of numbers 1 to 10.

```
O0001;
#1=0;
#2=1;
WHILE[#2 LE 10]DO 1;
#1=#1+#2;
#2=#2+1;
END 1;
M30;
```

16.7 MACRO CALL

A macro program can be called using the following methods. The calling methods can roughly be divided into two types: macro calls and subprogram calls.
A macro program can also be called during MDI operation in the same way.



Limitation

- Call nesting

Macro calls can be nested to a depth of up to five levels and subprogram calls can be nested to a depth of up to ten levels; calls can be nested to a depth of up to 15 levels in total.

- Differences between macro calls and subprogram calls

Macro call (G66, G66.1, Ggg, or Mmm) differs from subprogram call (for example M98, Mmm, or Ttt) as described below.

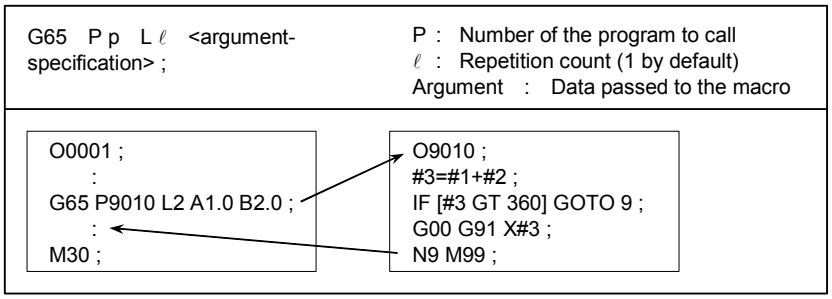
- With a macro call, an argument (data passed to a macro) can be specified. A subprogram call does not have this capability.
- If a macro call block contains another NC command (such as G01 X100.0 G65 Pp), an alarm PS0127 occurs.
- If a subprogram call block contains another NC command (such as G01 X100.0 M98 Pp), the subprogram is called after the command is executed.
- In any macro call block, the machine does not stop in the single block mode.
  If a subprogram call block contains another NC command (such as G01 X100.0 M98 Pp), the machine stops in the single block mode.
- With a macro call, the level of local variables changes. With a subprogram call, the level of local variables does not change. (See "Local variable levels" in Limitation of Subsection 16.7.1.)

- Called program and folders to be searched

The order in which folders are called depends on the method of calling a macro or subprogram.
Folders are searched in sequence and the program found first is called.
For details, see the "Managing Programs" chapter.

16.7.1 Simple Call (G65)

When G65 is specified, the custom macro specified at address P is called. Data (argument) can be passed to the custom macro program.



Explanation

- Call

- After G65, specify at address P the program number of the custom macro to call.
- When a number of repetitions is required, specify a number from 1 to 999999999 after address L. When L is omitted, 1 is assumed.
- By using argument specification, values are assigned to corresponding local variables.

- Argument specification

Two types of argument specification are available. Argument specification I uses letters other than G, L, O, N, and P once each. Argument specification II uses A, B, and C once each and also uses I, J, and K up to ten times. The type of argument specification is determined automatically according to the letters used.

- Argument specification I

Address	Variable number
A	#1
B	#2
C	#3
D	#7
E	#8
F	#9
H	#11

Address	Variable number
I	#4
J	#5
K	#6
M	#13
Q	#17
R	#18
S	#19

Address	Variable number
T	#20
U	#21
V	#22
W	#23
X	#24
Y	#25
Z	#26

- Addresses G, L, N, O, and P cannot be used in arguments.
- Addresses that need not be specified can be omitted. Local variables corresponding to an omitted address are set to null.
- Addresses do not need to be specified alphabetically. They conform to word address format.
  I, J, and K need to be specified alphabetically, however.
  Argument specification I is always used for I, J, and K by setting bit 7 (IJK) of parameter No. 6008 to 1

Example
- When bit 7 (IJK) of parameter No. 6008 is 0, I_J_K_ means that I = #4, J = #5, and K = #6 while K_J_I_ means K = #6, J = #8, and I= #10 because argument specification II is used.
- When bit 7 (IJK) of parameter No. 6008 is 1, K_J_I_ means that I = #4, J = #5, and K = #6, which is the same as with I_J_K_, because argument specification I is used.

- Argument specification II
  Argument specification II uses A, B, and C once each and uses I, J, and K up to ten times. Argument specification II is used to pass values such as three-dimensional coordinates as arguments.

Address	Variable number
A	#1
B	#2
C	#3
I_1	#4
J_1	#5
K_1	#6
I_2	#7
J_2	#8
K_2	#9
I_3	#10
J_3	#11

Address	Variable number
IK_3	#12
I_4	#13
J_4	#14
K_4	#15
I_5	#16
J_5	#17
K_5	#18
I_6	#19
J_6	#20
K_6	#21
I_7	#22

Address	Variable number
J_7	#23
K_7	#24
I_8	#25
J_8	#26
K_8	#27
I_9	#28
J_9	#29
K_9	#30
I_{10}	#31
J_{10}	#32
K_{10}	#33

- Subscripts of I, J, and K for indicating the order of argument specification are not written in the actual program.

NOTE
When bit 7 (IJK) of parameter No. 6008 is 1, argument II cannot be used.

Limitation

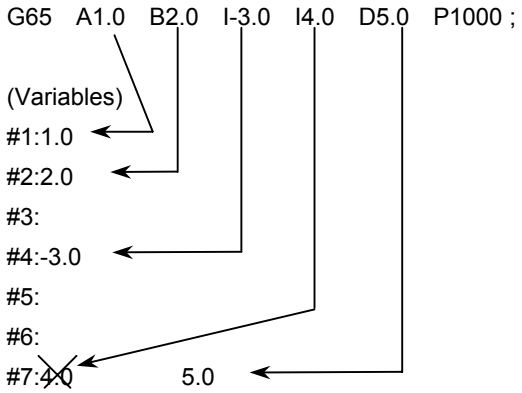
- Format

G65 must be specified before any argument.

- Mixture of argument specifications I and II

The CNC internally identifies argument specification I and argument specification II. If a mixture of argument specification I and argument specification II is specified, the type of argument specification specified later takes precedence.

[Example]



When both the I4.0 and D5.0 arguments are commanded for variable #7 in this example, the latter, D5.0, is valid.

- Position of the decimal point

The units used for argument data passed without a decimal point correspond to the least input increment of each address.

> ⚠ **CAUTION**
> The value of an argument passed without a decimal point may vary according to the system configuration of the machine. It is good practice to use decimal points in macro call arguments to maintain program compatibility.

- Extended axis name

The axis address of an extended axis name cannot be specified as an argument. If an attempt is made to specify it, alarm PS0129 is issued.

M

When a value is specified with no decimal point, the number of decimal places is determined as follows.

Address	For a non-axis address	For an axis address
D, E, H, M, S, or T	0	
Q or R	α (NOTE 2)	
A, C, I, J, K, X, Y, or Z	α (NOTE 2)	β (NOTE 3)
B, U, V(NOTE 1), or W	0	β (NOTE 3)
Second auxiliary function	γ (NOTE 4)	

Address	Metric input	Inch input
F (G93 mode)	3	
F (G94 mode)	0	2
F (G95 mode)	2 (NOTE 5)	4 (NOTE 5)

NOTE

1 When V is used in a call using a specific code, the number of decimal places is determined according to the setting for the reference axis.

2 α is determined according to the increment system for the reference axis (axis specified with parameter No. 1031) as listed in the table in NOTE 3.

3 β is determined according to the increment system for the corresponding axis address as listed in the following table.

Increment system	Linear axis (metric input)	Linear axis (inch input)	Rotation axis
IS-A	2	3	2
IS-B	3	4	3
IS-C	4	5	4
IS-D	5	6	5
IS-E	6	7	6

When bit 7 (IPR) of parameter No. 1004 is set to 1, the above values from which 1 is subtracted are used. When the increment system for an axis is IS-A, however, the setting of bit 7 (IPR) of parameter No. 1004 is not effective.

When calculator-type decimal notation for each axis is used (bit 0 (ADX) of parameter No. 3455 is set to 1), the number of decimal places is 0. When bit 7 (EAP) of parameter No. 3452 is set to 1, however, calculator-type decimal notation is not effective and the number of decimal places is determined as listed in the above table.

4 γ is determined according to the increment system for the reference axis (axis specified with parameter No. 1031) as listed in the following table. (When bit 7 (BDX) of parameter No. 3450 is set to 1, γ is also determined in the same way.)

Increment system for the reference axis	AUP (No.3450#0) = 0	AUP(3450#0) = 1			
		AUX (No.3405#0) = 0		AUX (No.3405#0) = 1	
		Metric	Inch	Metric	Inch
IS-A	0	2		2	3
IS-B		3		3	4
IS-C		4		4	5
IS-D		5		5	6
IS-E		6		6	7

5 When bit 1 (FR3) of parameter No. 1405 is 1, the values in the table need to be incremented by 1.

6 When calculator-type decimal notation is used (bit 0 (DPI) of parameter No. 3401 is set to 1), the number of decimal places is 0.

T

When a value is specified with no decimal point, the number of decimal places is determined as follows.

Address	For a non-axis address	For an axis address
H, M, Q, S, or T	0	
D or R	α (NOTE 1)	
A, B, C, I, J, K, U, V, W, X, Y, or Z	α (NOTE 1)	β (NOTE 2)
Second auxiliary function	γ (NOTE 3)	

Address	Metric input	Inch input
E, F (G98 mode)	0 (NOTE 4)	2 (NOTE 4)
E, F (G99 mode)	4	6

NOTE

1 α is determined according to the increment system for the reference axis (axis specified with parameter No. 1031) as listed in the table in NOTE 2.

2 β is determined according to the increment system for the corresponding axis address as listed in the following table.

Increment system	Linear axis (metric input)	Linear axis (inch input)	Rotation axis
IS-A	2	3	2
IS-B	3	4	3
IS-C	4	5	4
IS-D	5	6	5
IS-E	6	7	6

When bit 7 (IPR) of parameter No. 1004 is set to 1, the above values from which 1 is subtracted are used. When the increment system for an axis is IS-A, however, the setting of bit 7 (IPR) of parameter No. 1004 is not effective.

When calculator-type decimal notation for each axis is used (bit 0 (ADX) of parameter No. 3455 is set to 1), the number of decimal places is 0. When bit 7 (EAP) of parameter No. 3452 is set to 1, however, calculator-type decimal notation is not effective and the number of decimal places is determined as listed in the above table.

NOTE

3 γ is determined according to the increment system for the reference axis (axis specified with parameter No. 1031) as listed in the following table. (When bit 7 (BDX) of parameter No. 3450 is set to 1, γ is also determined in the same way.)

Increment system for the reference axis	AUP (No.3450#0) = 0	AUP(3450#0) = 1			
		AUX (No.3405#0) = 0		AUX (No.3405#0) = 1	
		Metric	Inch	Metric	Inch
IS-A	0	2		2	3
IS-B		3		3	4
IS-C		4		4	5
IS-D		5		5	6
IS-E		6		6	7

4 When bit 2 (FM3) of parameter No. 1404 is 1, the values in the table need to be incremented by 3.

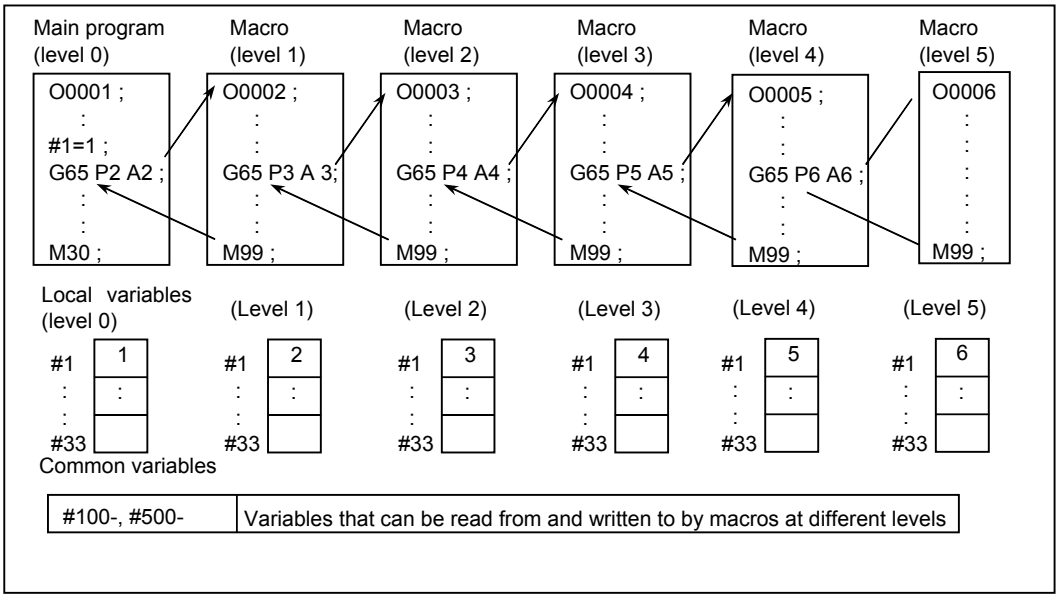
5 When calculator-type decimal notation is used (bit 0 (DPI) of parameter No. 3401 is set to 1), the number of decimal places is 0.

- Call nesting

Macro calls can be nested to a depth of up to five levels including simple calls (G65) and modal calls (G66/G66.1). Subprogram calls can be nested to a depth of up to 15 levels including macro calls.
A macro program can also be called during MDI operation in the same way.

- Local variable levels

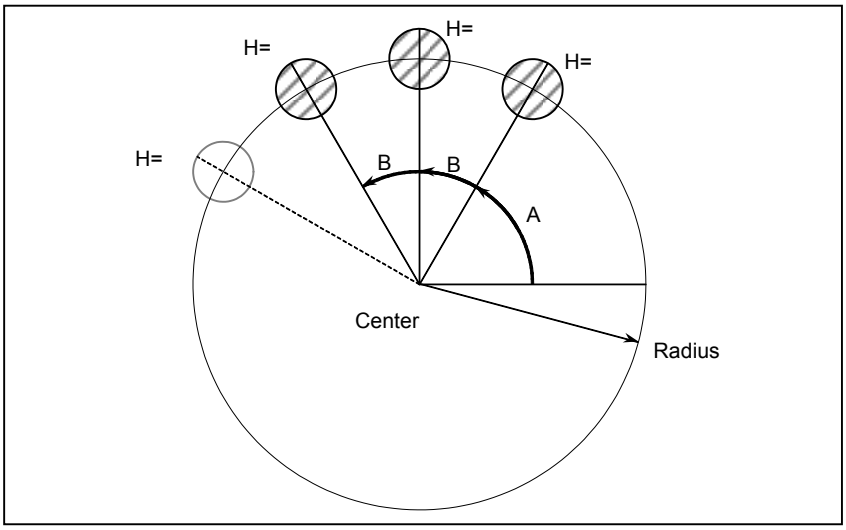
- Local variables from level 0 to 5 are provided for nesting.
- The level of the main program is 0.
- Each time a macro is called (with G66, G66.1, Ggg, or Mmm), the local variable level is incremented by one. The values of the local variables at the previous level are saved in the CNC.
- When M99 is executed in a macro program, control returns to the calling program. At that time, the local variable level is decremented by one; the values of the local variables saved when the macro was called are restored.



Sample program (bolt hole circle)

M

A macro is created which drills H holes at intervals of B degrees after a start angle of A degrees along the periphery of a circle with radius I. The center of the circle is (X,Y). Commands can be specified in either the absolute or incremental programming.
To drill in the clockwise direction, specify a negative value for B.



- Calling format

```
G65  P9100  Xx  Yy  Zz  Rr  Ff  Ii  Aa  Bb  Hh ;
```

X : X coordinate of the center of the circle
(absolute or incremental programming) ..... (#24)
Y : Y coordinate of the center of the circle
(absolute or incremental programming) ..... (#25)
Z : Hole depth..... (#26)
R : Coordinates of an approach point..... (#18)
F : Cutting feedrate ..... (#9)
I : Radius of the circle..... (#4)
A : Drilling start angle..... (#1)
B : Incremental angle
(clockwise when a negative value is specified)..... (#2)
H : Number of holes ..... (#11)

- Program calling a macro program

```
O0002;
G90 G92 X0 Y0 Z100.0;
G65 P9100 X100.0 Y50.0 R30.0 Z-50.0 F500 I100.0 A0 B45.0 H5;
M30;
```

- Macro program (called program)

```
O9100;
#3=#4003; ..... Stores G code of group 3.
G81 Z#26 R#18 F#9 K0; (Note) ... Drilling cycle.
..... Note:  L0 can also be used.
IF[#3 EQ 90]GOTO 1; ..... Branches to N1 in the G90 mode.
#24=#5001+#24; ..... Calculates the X coordinate of the center.
#25=#5002+#25; ..... Calculates the Y coordinate of the center.
N1 WHILE[#11 GT 0]DO 1;..... Until the number of remaining holes reaches 0
#5=#24+#4*COS[#1]; ..... Calculates a drilling position on the X-axis.
#6=#25+#4*SIN[#1]; ..... Calculates a drilling position on the Y-axis.
G90 X#5 Y#6; ..... Performs drilling after moving to the target
                   position.
#1=#1+#2; ..... Updates the angle.
#11=#11-1; ..... Decrements the number of holes.
END 1;
G#3 G80; ..... Returns the G code to the original state.
M99;
```

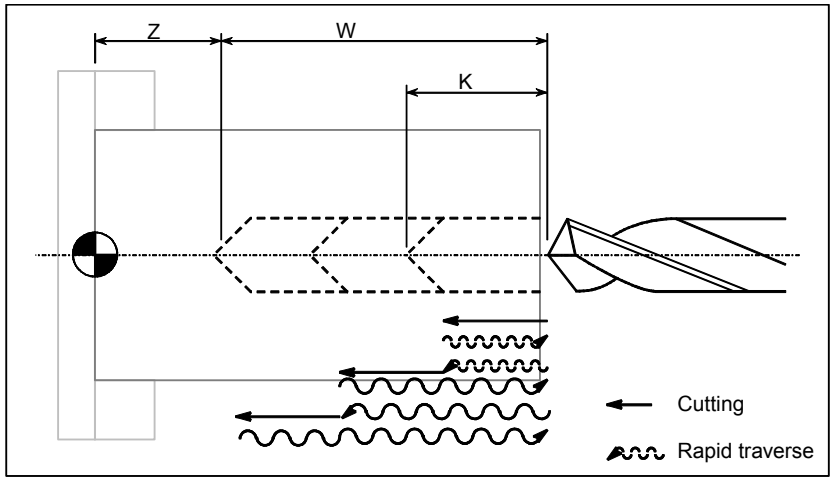
Meaning of variables:

- #3: Stores the G code of group 3.
- #5: X coordinate of the next hole to drill
- #6: Y coordinate of the next hole to drill

Sample program (Drill cycle)

T

Move the tool beforehand along the X- and Z-axes to the position where a drilling cycle starts. Specify Z or W for the depth of a hole, K for the depth of a cut, and F for the cutting feedrate to drill the hole.



- Calling format

G65 P9100 { Zz / Ww } Kk Ff ;

Z : Hole depth (absolute programming)
W : Hole depth (incremental programming)
K : Cutting amount per cycle
F : Cutting feedrate

- Program calling a macro program

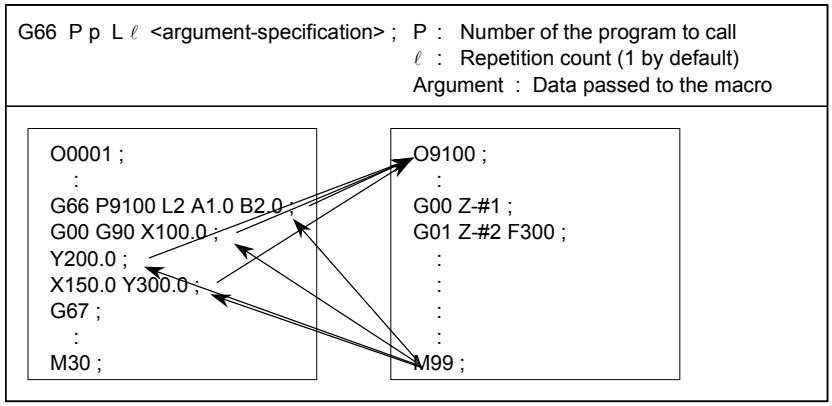
```
O0002;
G50 X100.0 Z200.0 ;
G00 X0 Z102.0 S1000 M03 ;
G65 P9100 Z50.0 K20.0 F0.3 ;
G00 X100.0 Z200.0 M05 ;
M30
```

- Macro program (called program)

```
O9100;
#1=0 ;..... Clear the data for the depth of the current hole.
#2=0 ;..... Clear the data for the depth of the preceding hole.
IF [#23 NE #0] GOTO 1 ;. If incremental programming, specifies the jump to N1.
IF [#26 EQ #0] GOTO 8 ;. If neither Z nor W is specified, an error occurs.
#23=#5002-#26 ;..... Calculates the depth of a hole.
N1 #1=#1+#6 ;..... Calculates the depth of the current hole.
IF [#1 LE #23] GOTO 2 ;. Determines whether the hole to be cut is too deep?
#1=#23 ;..... Clamps at the depth of the current hole.
N2 G00 W-#2 ; ..... Moves the tool to the depth of the preceding hole at the cutting feedrate.
G01 W- [#1-#2] F#9 ;..... Drills the hole.
G00 W#1 ;..... Moves the tool to the drilling start point.
IF [#1 GE #23] GOTO 9 ;. Checks whether drilling is completed.
#2=#1 ;..... Stores the depth of the current hole.
GOTO 1 ;
N9 M99
N8 #3000=1 (NOT Z OR W COMMAND)  Issues an alarm.
```

16.7.2 Modal Call: Call After the Move Command (G66)

Once G66 is issued to specify a modal call a macro is called after a block specifying movement along axes is executed. This continues until G67 is issued to cancel a modal call.



Explanation

- Call

- After G66, specify at address P a program number subject to a modal call.
- When a number of repetitions is required, a number from 1 to 999999999 can be specified at address L.
- As with a simple call (G65), data passed to a macro program is specified in arguments.
- In the G66 mode, a macro or subprogram can be called.

- Cancellation

When a G67 code is specified, modal macro calls are no longer performed in subsequent blocks.

- Call nesting

Macro calls can be nested to a depth of up to five levels including simple calls (G65) and modal calls (G66/G66.1). Subprogram calls can be nested to a depth of up to 15 levels including macro calls.

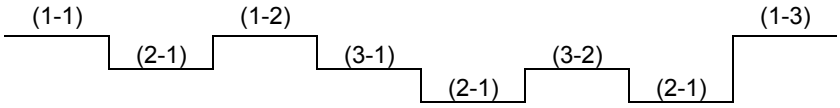
- Modal call nesting

For a single modal call (when G66 is specified only once), each time the move command is executed, the specified macro is called. When nested modal macro calls are specified, the macro at the next higher level is called each time the move command for a macro call is executed.
Macros are called in reverse order in which they are specified. Each time G67 is issued, the macros are canceled one by one in reverse order in which they are specified.

[Example]

```
G66 P9100 ;                O9100 ;                O9200 ;
X10.0 ;        (1-1)       Z50.0 ;     (2-1)      X60.0 ;        (3-1)
G66 P9200 ;                M99 ;                  Y70.0 ;        (3-2)
X15.0 ;        (1-2)                              M99;
G67 ;          Cancels P9200.
G67 ;          Cancels P9100.
X-25.0 ;       (1-3)
```

Execution order of the above program (blocks not containing the move command omitted)



* No modal call is performed after (1-3) because the mode is not the macro call mode.

Limitation

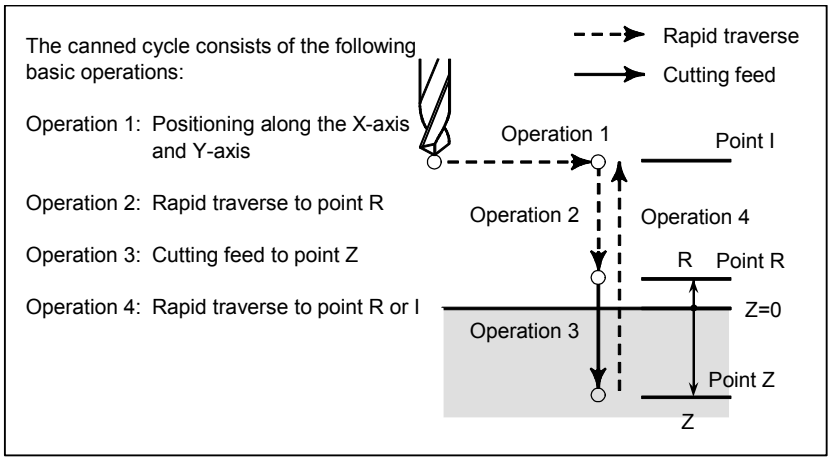
- G66 and G67 blocks are specified in pairs in the same program. If a G67 code is specified not in the G66 mode, an alarm PS1100 occurs. Bit 0 (G67) of parameter No. 6000 can be set to 1 to specify that the alarm does not occur in this case.
- In a G66 block, no macros can be called. Local variables (arguments) are set, however.
- G66 needs to be specified before any arguments.
- No macros can be called in a block which contains a code such as a auxiliary function that does not involve movement along an axis.
- Local variables (arguments) can only be set in G66 blocks. Note that local variables are not set each time a modal call is performed.

NOTE
If M99 is specified in a block in which a call is performed, it is executed after the call is performed.

Sample program

M

The same operation as the drilling canned cycle G81 is created using a custom macro and the machining program makes a modal macro call. For program simplicity, all drilling data is specified using absolute values.



- Calling format

```
G66 P9110 Zz Rr Ff Ll ;
```

Z : Coordinates of position Z (absolute programming only) .....(#26)
R : Coordinates of position R (absolute programming only).....(#18)
F : Cutting feedrate.....(#9)
L : Repetition count

- Program that calls a macro program

```
O0001;
G28 G91 X0 Y0 Z0;
G92 X0 Y0 Z50.0;
G00 G90 X100.0 Y50.0;
G66 P9110 Z-20.0 R5.0 F500;
G90 X20.0 Y20.0;
X50.0;
Y50.0;
X70.0 Y80.0;
G67;
M30;
```

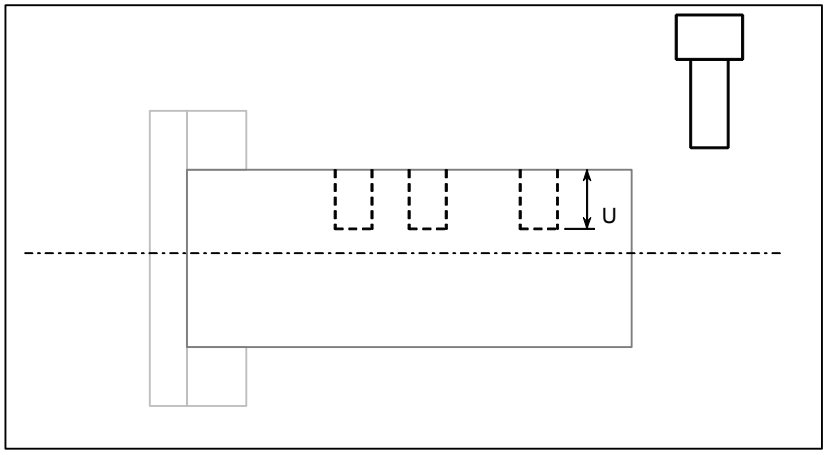
- Macro program (program called)

```
O9110;
  #1=#4001;  .....Stores G00/G01.
  #3=#4003;.....Stores G90/G91.
  #4=#4109;.....Stores the cutting feedrate.
  #5=#5003;.....Stores the Z coordinate at the start of drilling.
  G00 G90 Z#18; .....Positioning at position R
  G01 Z#26 F#9;.....Cutting feed to position Z
  IF[#4010 EQ 98]GOTO 1; Return to position I
  G00 Z#18;.....Positioning at position R
  GOTO 2;
N1 G00 Z#5; .....Positioning at position I
N2 G#1 G#3 F#4; .....Restores modal information.
  M99;
```

Sample program

T

This program makes a groove at a specified position.



- Calling format

```
G66 P9110 Uu Ff
```

U : Groove depth (incremental programming)
F : Cutting feed of grooving

- Program that calls a macro program

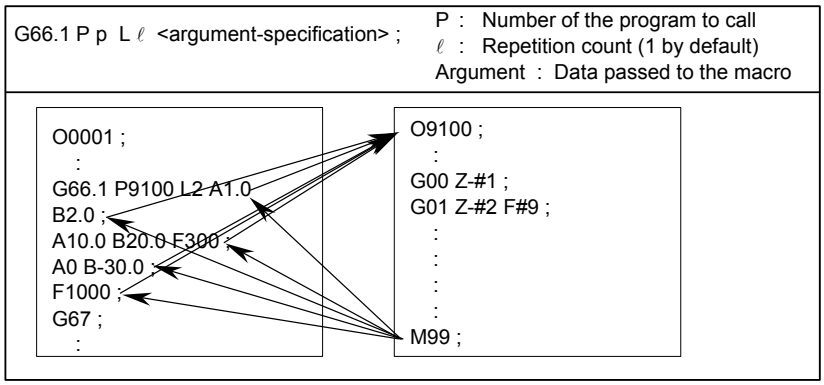
```
O0003 ;
G50 X100.0 Z200.0 ;
S1000 M03 ;
G66 P9110 U5.0 F0.5 ;
G00 X60.0 Z80.0 ;
Z50.0 ;
Z30.0 ;
G67 ;
G00 X00.0 Z200.0 M05 ;
M30;
```

- Macro program (program called)

```
O9110 ;
G01 U - #21 F#9 ;..... Cuts the workpiece.
G00 U#21 ; ..... Retracts the tool.
M99 ;
```

16.7.3 Modal Call: Each Block Call (G66.1)

In this macro call mode, the specified macro is unconditionally called for each NC command block. All data other than O, file name, N, and G codes that is specified in each block is not executed and is used as arguments. (The G code in the block in which G66.1 is specified is not used as an argument. Only the last G code specified in subsequent blocks is used as an argument.)
The NC command blocks having O or file name N have the same effect as when G65P is specified in the next position, the other NC command blocks have the same effect as when G65P is specified at the beginning.



[Example]

In the G66.1 P100 ; mode,
N001 G01 G91 X100 Y200 D1 R1000 ;
is the same as
N001 G65 P100 G01 G91 X100 Y200 D1 R1000 ;

Explanation

- Call

- After G66.1, specify at address P a program number subject to a modal call.
- When a number of repetitions is required, a number from 1 to 999999999 can be specified at address L.
- As with a simple call (G65), data passed to a macro program is specified in arguments.
- In the G66.1 mode, a macro or subprogram can be called.

- Cancellation

When a G67 code is specified, modal macro calls are no longer performed in subsequent blocks.

- Call nesting

Macro calls can be nested to a depth of up to five levels including simple calls (G65) and modal calls (G66/G66.1). Subprogram calls can be nested to a depth of up to 15 levels including macro calls.

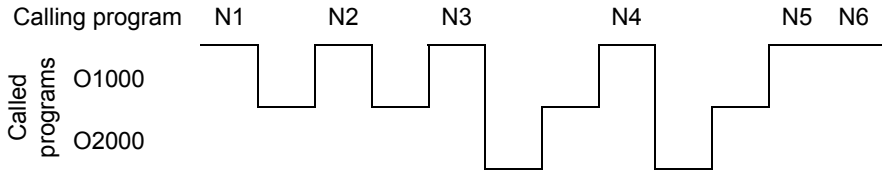
- Modal call nesting

For a single modal call (when G66.1 is specified only once), the specified macro is called for each NC command block. When nested modal macro calls are specified, the macro at the next higher level is also called in a block within a called macro in which an NC command is specified.
Macros are called in reverse order in which they are specified. Each time G67 is issued, the macros are canceled one by one in reverse order in which they are specified.

[Example] Axis specification switching

```
N1 G66.1 P1000 X10.0 ;          → Calls O1000 and executes Y10.0.
N2 X20.0 ;                      → Calls O1000 and executes Y20.0.
N3 G66.1 P2000 Y10.0 Z20.0 ;    → Calls O2000 and executes Y20.0
                                  Z10.0. Then, calls O1000 and
                                  executes X20.0 Z10.0.
N4 X10.0 Y20.0 Z30.0 ;          → Calls O2000 and executes X10.0
                                  Y30.0 Z20.0. Then, calls O1000
                                  and executes X30.0 Y10.0 Z20.0.
N5 G67 ;                        → Cancels P2000.
N6 G67 ;                        → Cancels P1000.

O1000 X#25 Y#24 Z#26 ;          (X-Y switching)
M99 ;

O2000 X#24 Y#26 Z#25 ;          (Y-Z switching)
M99 ;
```

Execution order of the above program (blocks not containing the move command omitted)



In N1 and N2 blocks, O1000 is called and the X and Y specifications are made to change places.
In N3 and N4 blocks, O2000 is called first and the Y and Z specifications are made to change places. For the switched specification, O1000 is called and the X and Y specifications are made to change places. Therefore, the X, Y, and Z specifications are switched to the Z, X, and Y specifications.

Limitation

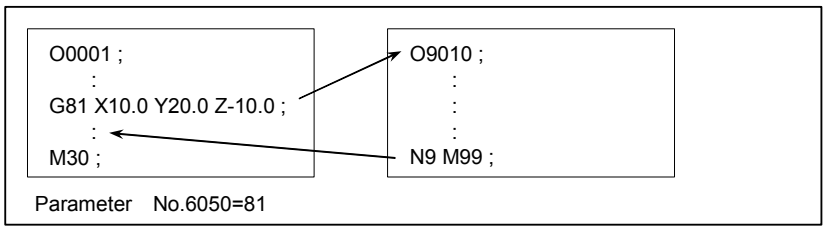
- G66.1 and G67 blocks are specified in pairs in the same program. If a G67 code is specified not in the G66.1 mode, an alarm PS1100 occurs. Bit 0 (G67) of parameter No. 6000 can be set to 1 to specify that the alarm does not occur in this case.
- G66.1 block
  (a) In a G66.1 block, a macro is called.
  (b) The correspondence between addresses specified as arguments and variables is the same as for simple calls.

- Block following a G66.1 block in which a call is performed (not including G66.1 blocks)
  (a) Addresses G, P, and L can also be used as arguments. Address G corresponds to #10; address L to #12; address P to #16. However, the restrictions on the input format of normal NC commands are put on the data. For example, ;G1000. P0.12 L-4 cannot be specified.
  (b) When multiple G codes are specified, only the last G code is used as an argument. O, file name, and N codes and G codes not in group 00 are passed to the next and subsequent blocks.

NOTE
1 In a block in which only an O number, file name, sequence number, EOB, macro statement, or M99 command is specified, a macro is not called for each block.
2 In each block, when an address other than O, file name, or N is specified, it is assumed to be an NC command and a macro is called for each block. When N is specified following an address other than O, file name, or N, it is used as an argument. In this case, N corresponds to variable #14 and the number of decimal places is 0.
3 If M99 is specified in a block in which a macro is called, it is executed after the call is performed.

16.7.4 Macro Call Using a G Code

By setting a G code number used to call a macro program in a parameter, the macro program can be called in the same way as for a simple call (G65).



Explanation

By setting a G code number from -9999 to 9999 used to call a custom macro program (O9010 to O9019) in the corresponding parameter (No.6050 to No.6059), the macro program can be called in the same way as with G65. To call custom macro program O9040 to O9049 using a G code with a decimal point, set bit 0 (DPG) of parameter No. 6007 to 1 and set the G code number in the corresponding parameter (Nos. 6060 to 6069).
The number of decimal places of a G code is 1. Set the value obtained by multiplying a desired G code number by 10 in the corresponding parameter.

[Example] When parameter No. 6060 is set to 234, custom macro program O9040 is called using G23.4.

When a negative G code is set, a modal call is performed. In this case, bit 3 (MGE) of parameter No. 6007 can be set to select the G66 or G66.1 mode.
For example, when a parameter is set so that macro program O9010 can be called with G81, a user-specific cycle created using a custom macro can be called without modifying the machining program.

- Correspondence between parameter numbers and program numbers

G code with no decimal point		G code with a decimal point	
Parameter number	**Program number**	**Parameter number**	**Program number**
6050	O9010	6060	O9040
6051	O9011	6061	O9041
6052	O9012	6062	O9042
6053	O9013	6063	O9043
6054	O9014	6064	O9044
6055	O9015	6065	O9045
6056	O9016	6066	O9046
6057	O9017	6067	O9047
6058	O9018	6068	O9048
6059	O9019	6069	O9049

- Repetition

As with a simple call, a number of repetitions from 1 to 999999999 can be specified at address L.

- Argument specification

As with a simple call, two types of argument specification are available: Argument specification I and argument specification II. The type of argument specification is determined automatically according to the addresses used.

Limitation

- Nesting of calls using G codes

- To call another program in a program called using a G code, only G65, M98, G66, or G66.1 can be used normally.
- When bit 6 (GMP) of parameter No. 6008 is set to 1, a call using an M, T, or S code, second auxiliary function, or specific code can be performed in a program called using a G code.

16.7.5 Macro Call Using a G Code (Specification of Multiple Definitions)

By setting the starting G code number used to call a macro program, the number of the starting program to be called, and the number of definitions, macro calls using multiple G codes can be defined.

Explanation

As many custom macros as the number specified in parameter No. 6040 can be called using as many G codes as the number specified in parameter No. 6040. The numeric value set in parameter No. 6038 indicates the starting G code number and the program number set in parameter No. 6039 indicates the starting program number. To disable this type of call, set 0 in parameter No. 6040.
When a negative G code is set in parameter No. 6038, modal calls are performed. In this case, bit 3 (MGE) of parameter No. 6007 can be set to select the G66 or G66.1 mode.
The number of repetitions and argument specification are set in the same way as with a macro call using a G code.
[Example]

Set parameter No. 6038 to 900, parameter No. 6039 to 10000000, and parameter No. 6040 to 100.

G900 → O1000
G901 → O1001
G902 → O1002
:
G999 → O1099

Custom macro calls (simple calls) for 100 combinations are defined as shown above. When parameter No. 6038 is changed to -900, custom macro calls (modal calls) for the same combinations are defined.

NOTE
1 The calls defined by this setting become all invalid in the following cases:
<1> A value outside the valid data range is set in one of the above parameters.
<2> (No. 6039 + No. 6040 - 1) > 99999999
2 Simple and modal calls cannot be mixed in the specification.
3 If the G code set in parameter Nos. 6050 to 6059 to call the corresponding macro program is within the G code range for calling programs using multiple G codes, the macro program corresponding to the G code set in parameter Nos. 6050 to 6059 is called.

16.7.6 Macro Call Using a G Code with a Decimal Point (Specification of Multiple Definitions)

When bit 0 (DPG) of parameter No. 6007, by setting the starting G code number with a decimal point used to call a macro program, the number of the starting program to be called, and the number of definitions, multiple macro calls using multiple G codes with a decimal point can be defined.

Explanation

As many custom macros as the number specified in parameter No. 6043 can be called using as many G codes with a decimal point as the number specified in parameter No. 6043. The numeric value set in parameter No. 6041 indicates the starting G code number with a decimal point and the program number set in parameter No. 6042 indicates the starting program number. To disable this type of call, set 0 in parameter No. 6043.

When a negative G code is set in parameter No. 6041, modal calls are performed. In this case, bit 3 (MGE) of parameter No. 6007 can be set to select the G66 or G66.1 mode.

The number of repetitions and argument specification are set in the same way as with a macro call using a G code.

[Example]

Set parameter No. 6041 to 900, parameter No. 6042 to 2000, and parameter No. 6043 to 100.

G90.0 → O2000
G90.1 → O2001
G90.2 → O2002
:
G99.9 → O2099

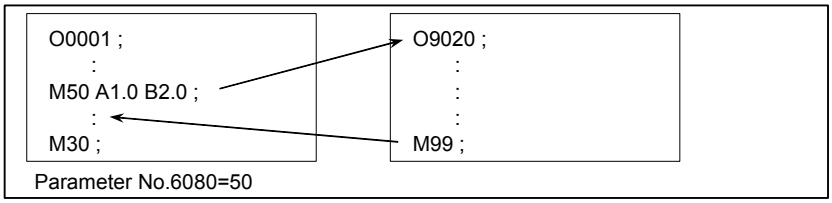
Custom macro calls (simple calls) for 100 combinations are defined as shown above. When parameter No. 6041 is changed to -900, custom macro calls (modal calls) for the same combinations are defined.

NOTE

1 The calls defined by this setting become all invalid in the following cases:
   <1> A value outside the valid data range is set in one of the above parameters.
   <2> (No. 6042 + No. 6043 - 1) > 99999999
   <3> Bit 0 (DPG) of parameter No. 6007 is set to 0 (this setting invalidates a macro call using a G code with a decimal point).
2 Simple and modal calls cannot be mixed in the specification.
3 If the G code set in parameter Nos. 6060 to 6069 to call the corresponding macro program is within the G code range for calling programs using multiple G codes, the macro program corresponding to the G code set in parameter Nos. 6060 to 6069 is called.

16.7.7 Macro Call Using an M Code

By setting an M code number used to call a macro program in a parameter, the macro program can be called in the same way as with a simple call (G65).



Explanation

By setting an M code number from 3 to 99999999 used to call custom macro program O9020 to O9029 in the corresponding parameter (Nos. 6080 to 6089), the macro program can be called in the same way as with G65.

- Correspondence between parameter numbers and program numbers

Parameter number	Corresponding program number
6080	O9020
6081	O9021
6082	O9022
6083	O9023
6084	O9024
6085	O9025
6086	O9026
6087	O9027
6088	O9028
6089	O9029

Example)

When parameter No. 6080 is set to 990, O9020 is called using M990.

- Repetition

As with a simple call, a number of repetitions from 1 to 99999999 can be specified at address L.

- Argument specification

As with a simple call, two types of argument specification are available: Argument specification I and argument specification II. The type of argument specification is determined automatically according to the addresses used.

Limitation

- An M code used to call a macro program must be specified at the start of a block.
- To call another program in a program called using an M code, only G65, M98, G66, or G66.1 can be used normally.
- When bit 6 (GMP) of parameter No. 6008 is set to 1, a call using a G code can be performed in a program called using an M code.

16.7.8 Macro Call Using an M Code (Specification of Multiple Definitions)

By setting the starting M code number used to call a macro program, the number of the starting program to be called, and the number of definitions, macro calls using multiple M codes can be defined.

Explanation

As many custom macros as the number specified in parameter No. 6049 can be called using as many M codes as the number specified in parameter No. 6049. The numeric value set in parameter No. 6047 indicates the starting M code number and the program number set in parameter No. 6048 indicates the starting program number. To disable this type of call, set 0 in parameter No. 6049.
The number of repetitions and argument specification are set in the same way as with a macro call using a M code.
[Example]

Set parameter No. 6047 to 9000, parameter No. 6048 to 4000, and parameter No. 6049 to 100.

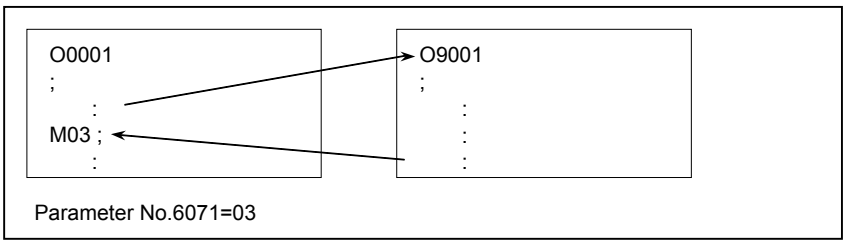
M90000000 → O4000
M90000001 → O4001
M90000002 → O4002
:
M90000099 → O4099

Custom macro calls (simple calls) for 100 combinations are defined as shown above.

NOTE
1 The calls defined by this setting become all invalid in the following cases:
<1> A value outside the valid data range is set in one of the above parameters.
<2> (No. 6048 + No. 6049 - 1) > 99999999
2 If the M code set in parameter Nos. 6080 to 6089 to call the corresponding macro program is within the M code range for calling programs using multiple M codes, the macro program corresponding to the M code set in parameter Nos. 6080 to 6089 is called.

16.7.9 Subprogram Call Using an M Code

By setting an M code number used to call a subprogram (macro program) in a parameter, the macro program can be called in the same way as with a subprogram call (M98).



Explanation

By setting an M code number from 3 to 99999999 used to call subprogram O9001 to O9009 in the corresponding parameters Nos. 6071 to 6079, the subprogram can be called in the same way as with M98.

- Correspondence between parameter numbers and program numbers

Parameter number	Program number
6071	O9001
6072	O9002
6073	O9003
6074	O9004
6075	O9005
6076	O9006
6077	O9007
6078	O9008
6079	O9009

- Repetition

As with a simple call, a number of repetitions from 1 to 99999999 can be specified at address L.

- Argument specification

Argument specification is not allowed.

- M code

An M code in a macro program that has been called is treated as an ordinary M code.

Limitation

- To call another program in a program called using an M code, only G65, M98, G66, or G66.1 can be used normally.
- When bit 6 (GMP) of parameter No. 6008 is set to 1, a call using a G code can be performed in a program called using an M code.

16.7.10 Subprogram Call Using an M Code (Specification of Multiple Definitions)

By setting the starting M code number used to call a subprogram, the number of the starting subprogram to be called, and the number of definitions, subprogram calls using multiple M codes can be defined.

Explanation

As many subprograms as the number specified in parameter No. 6046 can be called using as many M codes as the number specified in parameter No. 6046. The numeric value set in parameter No. 6044 indicates the starting M code number and the numeric value set in parameter No. 6045 indicates the starting subprogram number. To disable this type of call, set 0 in parameter No. 6046.

[Example]

Set parameter No. 6044 to 80000000, parameter No. 6045 to 30000000, and parameter No. 6046 to 100.

M80000000 → O3000
M80000001 → O3001
M80000002 → O3002
:
M80000099 → O3099

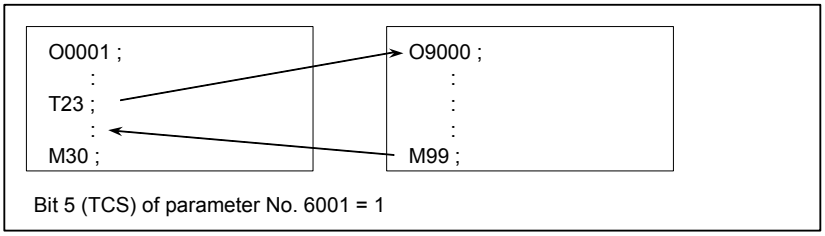
Subprogram calls for 100 combinations are defined as shown above.

NOTE

1 The calls defined by this setting become all invalid in the following cases:
   <1> A value outside the valid data range is set in one of the above parameters.
   <2> (No. 6045 + No. 6046 - 1) > 99999999
2 If the M code set in parameter Nos. 6071 to 6079 to call the corresponding subprogram is within the M code range for calling subprograms using multiple M codes, the subprogram corresponding to the M code set in parameter Nos. 6071 to 6079 is called.

16.7.11 Subprogram Calls Using a T Code

By enabling subprograms to be called with a T code in a parameter, a subprogram can be called each time the T code is specified in the machining program.



Explanation

- Call

By setting bit 5 (TCS) of parameter No. 6001 to 1, subprogram O9000 can be called each time a T code is specified in a machining program. A T code specified in a machining program is assigned to common variable #149.

- Repetition

As with a simple call, a number of repetitions from 1 to 99999999 can be specified at address L.

- Argument specification

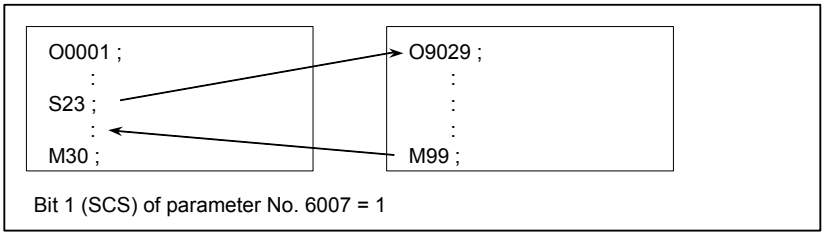
Argument specification is not allowed.

Limitation

- To call another program in a program called using a T code, only G65, M98, G66, or G66.1 can be used normally.
- When bit 6 (GMP) of parameter No. 6008 is set to 1, a call using a G code can be performed in a program called using a T code.

16.7.12 Subprogram Calls Using an S Code

By enabling subprograms to be called with an S code in a parameter, a subprogram can be called each time the S code is specified in the machining program.



Explanation

- Call

By setting bit 1 (SCS) of parameter No. 6007 to 1, subprogram O9029 can be called each time a S code is specified in a machining program. An S code specified in a machining program is assigned to common variable #147.

- Repetition

As with a simple call, a number of repetitions from 1 to 99999999 can be specified at address L.

- Argument specification

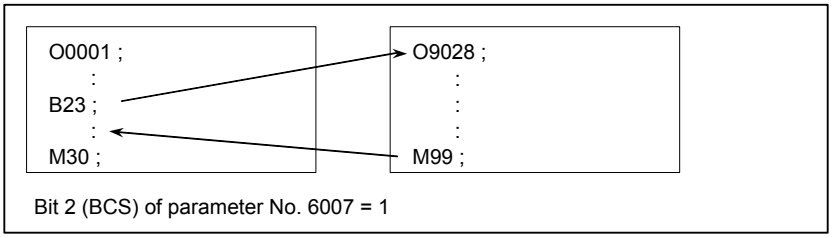
Argument specification is not allowed.

Limitation

- To call another program in a program called using an S code, only G65, M98, G66, or G66.1 can be used normally.
- When bit 6 (GMP) of parameter No. 6008 is set to 1, a call using a G code can be performed in a program called using an S code.

16.7.13 Subprogram Calls Using a Secondary Auxiliary Function

By enabling subprograms to be called with a secondary auxiliary function in a parameter, a subprogram can be called each time the secondary auxiliary function is specified in the machining program.



Explanation

- Call

By setting bit 2 (BCS) of parameter No. 6007 to 1, subprogram O9028 can be called each time a secondary auxiliary function code is specified in a machining program. A secondary auxiliary function specified in a machining program is assigned to common variable #146.

- Repetition

As with a simple call, a number of repetitions from 1 to 99999999 can be specified at address L.

- Argument specification

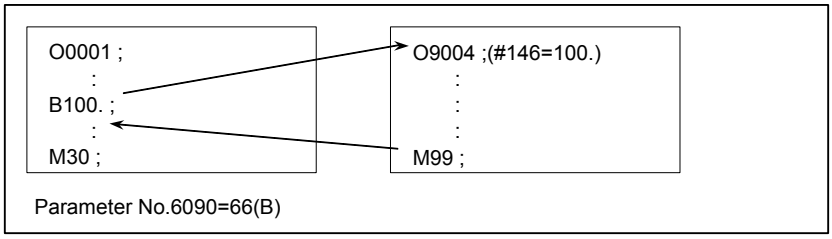
Argument specification is not allowed.

Limitation

- To call another program in a program called using a second auxiliary function, only G65, M98, G66, or G66.1 can be used normally.
- When bit 6 (GMP) of parameter No. 6008 is set to 1, a call using a G code can be performed in a program called using a second auxiliary function.

16.7.14 Subprogram Call Using a Specific Address

By enabling subprograms to be called with a specific address in a parameter, a subprogram can be called each time the specific address is specified in the machining program.



Explanation

- Call

By setting the code (ASCII code converted to decimal) corresponding to a specific address in parameter No. 6090 or No. 6091, the custom macro program, O9004 or O9005, corresponding to each parameter can be called when the specific address is specified in a machining program. The code value corresponding to a specific address specified in a machining program is assigned to the common variables (#146, #147). The table below indicates the addresses that can be set.

M

Address	Parameter setting
A	65
B	66
D	68
F	70
H	72
I	73
J	74
K	75
L	76
M	77
P	80
Q	81
R	82
S	83
T	84
V	86
X	88
Y	89
Z	90

NOTE
When address L is set, the number of repetitions cannot be set.

T

Address	Parameter setting
A	65
B	66
F	70
H	72
I	73
J	74
K	75
L	76
M	77
P	80
Q	81
R	82
S	83
T	84

NOTE
When address L is set, the number of repetitions cannot be set.

- Correspondence between parameter numbers and program numbers and between the parameter numbers and common variables

Parameter number	Program number	Common variable
6090	O9004	#146
6091	O9005	#147

- Repetition

As with a simple call, a number of repetitions from 1 to 99999999 can be specified at address L.

- Argument specification

Argument specification is not allowed.

Limitation

- To call another program in a program called using a specific code, only G65, M98, G66, or G66.1 can be used normally.
- When bit 6 (GMP) of parameter No. 6008 is set to 1, a call using a G code can be performed in a program called using a specific code.

Sample program

By using the subprogram call function that uses M codes, the cumulative usage time of each tool is measured.

Conditions

- The cumulative usage time of each of tools T01 to T05 is measured.
  No measurement is made for tools with numbers greater than T05.
- The following variables are used to store the tool numbers and measured times:

#501	Cumulative usage time of tool number 1
#502	Cumulative usage time of tool number 2
#503	Cumulative usage time of tool number 3
#504	Cumulative usage time of tool number 4
#505	Cumulative usage time of tool number 5

- Usage time starts being counted when the M03 command is specified and stops when M05 is specified. System variable #3002 is used to measure the time during which the cycle start lamp is on. The time during which the machine is stopped by feed hold and single block stop operation is not counted, but the time used to change tools and pallets is included.

Operation check

- Parameter setting

Set 3 in parameter No.6071, and set 5 in parameter No.6072.

- Variable value setting

Set 0 in variables #501 to #505.

- Program that calls a macro program

```
O0001;
T01 M06;
M03;
    :
M05; ..... Changes #501.
T02 M06;
M03;
    :
M05; ..... Changes #502.
T03 M06;
M03;
    :
M05; ..... Changes #503.
T04 M06;
M03;
    :
M05; ..... Changes #504.
T05 M06;
M03;
    :
M05; ..... Changes #505.
M30;
```

- Macro program (program called)

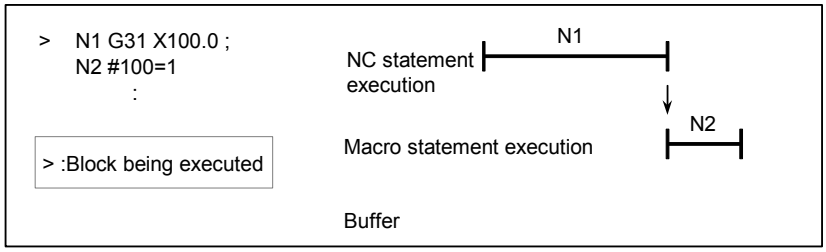
```
O9001(M03); .....Macro to start counting
  M01;
  IF[#4120 EQ 0]GOTO 9; .....No tool specified
  IF[#4120 GT 5]GOTO 9;.....Out-of-range tool number
  #3002=0;.....Clears the timer.
N9 M03;.....Rotates the spindle in the
  M99;                                  forward direction.

O9002(M05); .....Macro to end counting
  M01;
  IF[#4120 EQ 0]GOTO 9; .....No tool specified
  IF[#4120 GT 5]GOTO 9;.....Out-of-range tool number
  #[500+#4120]=#3002+#[500+#4120]; .....Calculates cumulative time.

N9 M05;.....Stops the spindle.
  M99;
```

16.8 PROCESSING MACRO STATEMENTS

For smooth machining, the CNC prereads the NC statement to be performed next. This operation is referred to as buffering. For example, many NC statements are buffered during look-ahead by AI contour control and so forth.
In the cutter compensation mode (G41 or G42), the CNC prereads the NC statements at least three blocks ahead to find intersections even if look-ahead by AI contour control and so forth is not applied.
Macro statements for arithmetic expressions and conditional branches are processed as soon as they are read into the buffer. Therefore, the timing of the macro statement execution is not always the specified order.
At the blocks containing M00, M01, M02 or M30, blocks containing M-codes for which buffering is suppressed by setting parameters Nos. 3411 to 3420 and No.3421 to 3432, and blocks containing prevention buffering G codes such as G31 or G53, the CNC stops to preread the NC statement after that. Then, the stop of the macro statement execution is guaranteed until such M codes or G codes complete its execution.

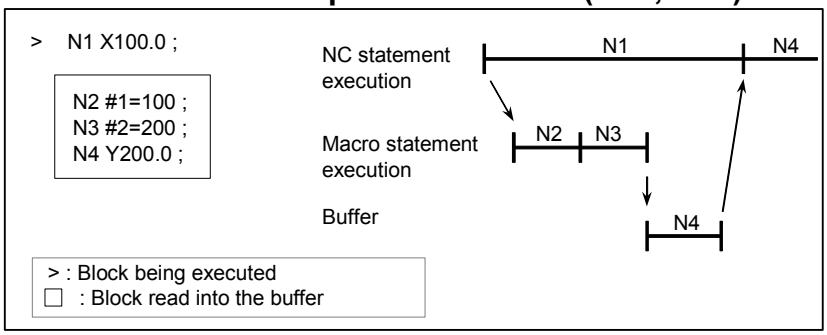
Explanation

- When the next block is not buffered (M codes that are not buffered, G31, etc.)



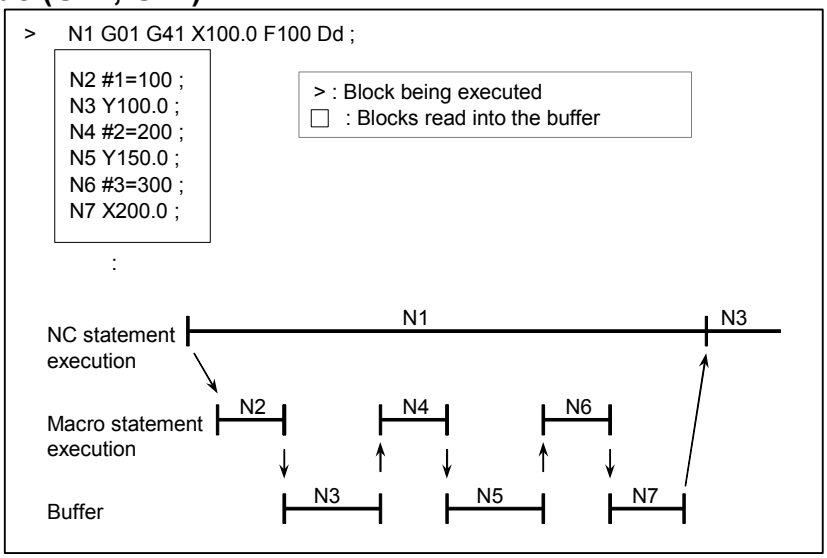
⚠CAUTION
In case that you need to execute the macro statement after completing the block just before the macro statement, specify M code or G code that is not buffered just before the macro statement. Specially, in case of reading/writing the system variables to control signals, coordinates, offset value, etc., it may different system variable data by the timing of the NC statement execution. To avoid this phenomenon, specify such M codes or G codes before the macro statement, if necessary.

- Buffering the next block in other than cutter compensation mode (G41, G42)



When N1 is being executed, the next NC statement (N4) is read into the buffer. The macro statements (N2, N3) between N1 and N4 are processed during execution of N1.

- In cutter compensation mode (G41, G42)



When N1 is being executed, the NC statements in the next three blocks (up to N7) are read into the buffer. The macro statements (N1, N4, N6) between N1 and N7 are processed during execution of N1.

16.9 REGISTERING CUSTOM MACRO PROGRAMS

Custom macro programs are similar to subprograms. They can be registered and edited in the same way as subprograms. The storage capacity is determined by the total length of tape used to store both custom macros and subprograms.

16.10 CODES AND RESERVED WORDS USED IN CUSTOM MACROS

In addition to the codes used in ordinary programs, the following codes are used in custom macro programs.

Explanation

- Codes

(1) When the ISO code is used or when bit 4 (ISO) of parameter No. 6008 is set to 0
(The codes are represented in hexadecimal.)

Meaning	Code
*	0AAh
=	0BDh
#	0A3h
[	0DBh
]	0DDh
?	03Fh
@	0C0h
&	0A6h
_	05Fh
O	0CFh

(2) When the EIA code is used or when the ISO code is used with bit 4 (ISO) of parameter No. 6008 set to 1

Meaning	Code
*	Code set in parameter No. 6010
=	Code set in parameter No. 6011
#	Code set in parameter No. 6012
[	Code set in parameter No. 6013
]	Code set in parameter No. 6014
?	Code set in parameter No. 6015
@	Code set in parameter No. 6016
&	Code set in parameter No. 6017
_	Code set in parameter No. 6018

For O, the same code as for O indicating a program number is used. Set a hole pattern for each of *, =, #, [,], ?, @, &, and _ in the ISO or EIA code in the corresponding parameter (Nos. 6010 to 6018).
The code 00h cannot be used. The code indicating an alphabetic character can be used for the code indicating a character listed above, but the code can be no longer used to indicate the original character.

- Reserved words

The following reserved words are used in custom macros:

AND, OR, XOR, MOD, EQ, NE, GT, LT, GE, LE,
SIN, COS, TAN, ASIN, ACOS, ATAN, ATN, SQRT, SQR, ABS, BIN, BCD, ROUND, RND, FIX, FUP, LN, EXP, POW, ADP, IF, GOTO, WHILE, DO, END, BPRNT, DPRNT, POPEN, PCLOS, SETVN, AX, AXNUM

System variable (constant) names and registered common variable names are also used as reserved words.

16.11 EXTERNAL OUTPUT COMMANDS

In addition to the standard custom macro commands, the following macro commands are available. They are referred to as external output commands.

- BPRNT
- DPRNT
- POPEN
- PCLOS

These commands are provided to output variable values and characters through the reader/puncher interface.

Explanation

Specify these commands in the following order:

Open command: POPEN
Before specifying a sequence of data output commands, specify this command to establish a connection to an external input/output device.

Data output command: BPRNT or DPRNT
Specify necessary data output.

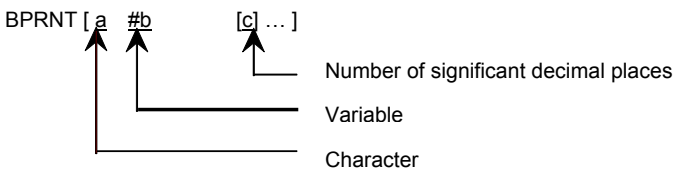
Close command: PCLOS
When all data output commands have completed, specify PCLOS to release a connection to an external input/output device.

- Open command POPEN

The POPEN command establishes a connection to an external input/output device. It must be specified before a sequence of data output commands. The CNC outputs a DC2 control code.

- Data output command BPRNT

The BPRNT command outputs characters and variable values in binary.



(i) Specified characters are converted to the codes according to the setting data (ISO) that is output at that time.
Specifiable characters are as follows:
- Letters (A to Z)
- Numbers
- Special characters (*, /, +, -, ?, @, &, _)

NOTE
1 An asterisk (*) is output by a space code.
2 When using ?, @, &, and/or _, use the ISO code as the punch code (setting data (ISO) = 1).

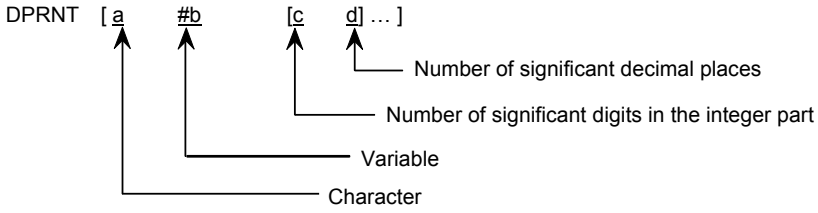
(ii) All variables are stored with a decimal point. Specify a variable followed by the number of significant decimal places enclosed in brackets. A variable value is treated as 2-word (32-bit) data, including the decimal digits. It is output as binary data starting from the highest byte.
(iii) When specified data has been output, an EOB code is output according to the setting code (ISO).
(iv) <Null> variables are regarded as 0.

Example
BPRNT [C** X#100 [3] Y#101 [3] M#10 [0]]
Variable value
#100=0.40956
#101=-1638.4
#10=12.34
are output as follows:

C3 A0 A0 D8 00 00 01 9A 59 FF E7 00 00 4D 00 00 00 0C 0A

C	sp	sp	X0000019A	YFFE70000	M0000000C	LF
	(* *)		(410)	(-1638400)	(12)	(;)

- Data output command DPRNT



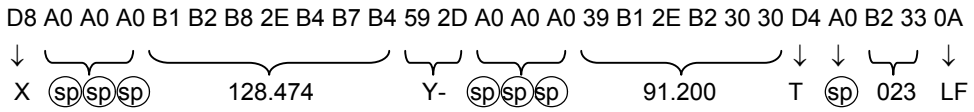
The DPRNT command outputs characters and each digit in the value of a variable according to the code set in the settings (ISO).

(i) For an explanation of the DPRNT command, see Items (i), (iii), and (iv) for the BPRNT command.

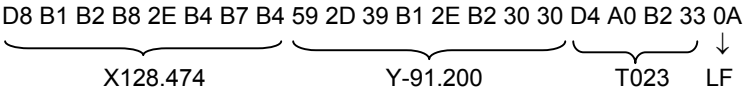
(ii) When outputting a variable, specify # followed by the variable number, then specify the number of digits in the integer part and the number of decimal places enclosed in brackets.
For the value of a variable, as many codes as the specified number of digits are output according to the settings one by one, starting with the highest digit. The decimal point is also output using the set code.
Each variable must be a numeric value consisting of up to nine digits. When high-order digits are zeros, these zeros are not output if bit 1 (PRT) of parameter No. 6001 is 1. If parameter PRT is 0, a space code is output each time a zero is encountered.
When the number of decimal places is not zero, digits in the decimal part are always output. If the number of decimal places is zero, no decimal point is output.
When bit 1 (PRT) of parameter No. 6001 is 0, a space code is output to indicate a positive number instead of +; if parameter PRT is 1, no code is output.

Example
DPRNT [X#2 [53] Y#5 [53] T#30 [20]]
Variable value
#2=128.47398
#5=-91.2
#30=123.456
are output as follows:

(1) Parameter PRT (No.6001#1) = 0



(2) Parameter PRT (No.6001#1) = 1



- Close command PCLOS

The PCLOS command releases a connection to an external input/output device. Specify this command when all data output commands have terminated. DC4 control code is output from the CNC.

- Required setting

Specify the specification number of an input/output device to be used for the I/O device specification number.
According to the above settings, set data items (such as the baud rate) for the reader/puncher interface.
Never specify the output device FANUC Cassette or Floppy for punching.
When specifying a DPRNT command to output data, specify whether leading zeros are output as spaces (by setting bit 1 (PRT) of parameter No. 6001 to 1 or 0).
To indicate the end of a line of data in ISO code, specify whether to use only an LF (bit 4 (CRO) of parameter 6001 is 0) or an LF and CR bit 4 (CRO) of parameter 6001 is 1).

NOTE
1 It is not necessary to always specify the open command (POPEN), data output command (BPRNT, DPRNT), and close command (PCLOS) together. Once an open command is specified at the beginning of a program, it does not need to be specified again except after a close command was specified.
2 Be sure to specify open commands and close commands in pairs. Specify the close command at the end of the program. However, do not specify a close command if no open command has been specified.
3 When a reset operation is performed while commands are being output by a data output command, output is stopped and subsequent data is erased. Therefore, when a reset operation is performed by a code such as M30 at the end of a program that performs data output, specify a close command at the end of the program so that processing such as M30 is not performed until all data is output.

16.12 RESTRICTIONS

- Sequence number search

A custom macro program cannot be searched for a sequence number.

- Single block

Even while a macro program is being executed, blocks can be stopped in the single block mode.
A block containing a macro call command (G66, G66.1, Ggg, Mmm, or G67) does not stop even when the single block mode is on.
Whether blocks containing arithmetic and logic operation commands and control commands are stopped depends on the settings of bits 5 (SBM) and 7 (SBV) of parameter No. 6000 as shown in the following table.

		Bit 5 (SBM) of parameter No. 6000	
		0	1
Bit 7 (SBV) of parameter No. 6000	0	Not stopped when the single block mode is on.	Can be stopped in the single block mode. (Variable #3003 cannot be used to disable single block stop. Single block stop is always enabled.)
	1	Can be stopped in the single bock mode. (Variable #3003 can be used to enable or disable single block stop.)	

Note that when a single block stop occurs at a macro statement in cutter compensation mode, the statement is assumed to be a block that does not involve movement, and proper compensation cannot be performed in some cases. (Strictly speaking, the block is regarded as specifying a movement with a travel distance 0.)

- Optional block skip

A / appearing in the middle of an <expression> (enclosed in brackets [ ] on the right-hand side of an arithmetic expression) is regarded as a division operator; it is not regarded as the specifier for an optional block skip code.

- Operation in EDIT mode

By setting bit 0 (NE8) of parameter No.3202 and bit 4 (NE9) of parameter No.3202 to 1, deletion and editing are disabled for custom macro programs and subprograms with program numbers 8000 to 8999 and 9000 to 9999. This prevents registered custom macro programs and subprograms from being destroyed by accident. When the entire memory is cleared, the contents of memory such as custom macro programs are deleted.

- Reset

With a reset operation, local variables and common variables #100 to #199 are cleared to null values. They can be prevented from clearing by setting bit 6 (CCV) of parameter No.6001. System variables #100 to #199 are not cleared.
A reset operation clears any called states of custom macro programs and subprograms, and any DO states, and returns control to the main program.

- Display of the PROGRAM RESTART

As with M98, the M and T codes used for subprogram calls are not displayed.

- Feed hold

When a feed hold is enabled during execution of a macro statement, the machine stops after execution of the macro statement. The machine also stops when a reset or alarm occurs.

- DNC operation

The control commands (such as GOTO and WHILE-DO) cannot be executed during DNC operation.
However, this restriction is removed when a program registered in program memory is called during DNC operation.

- Constant values that can be used in <expression>

+0.00000000001 to +999999999999
-999999999999 to -0.00000000001
The number of significant digits is 12 (decimal).
If this range is exceeded, an alarm PS0012 occurs.

16.13 INTERRUPTION TYPE CUSTOM MACRO

When a program is being executed, another program can be called by inputting an interrupt signal (UINT) from the machine.
This function is referred to as an interruption type custom macro function. Program an interrupt command in the following format:

Format

M96Pxxxxxxxx ;	Enables custom macro interrupt
M97 ;	Disables custom macro interrupt

Explanation

Use of the interruption type custom macro function allows the user to call a program during execution of an arbitrary block of another program. This allows programs to be operated to match situations which vary from time to time.

(1) When a tool abnormality is detected, processing to handle the abnormality is started by an external signal.
(2) A sequence of machining operations is interrupted by another machining operation without the cancellation of the current operation.
(3) At regular intervals, information on current machining is read.
Listed above are examples like adaptive control applications of the interruption type custom macro function.

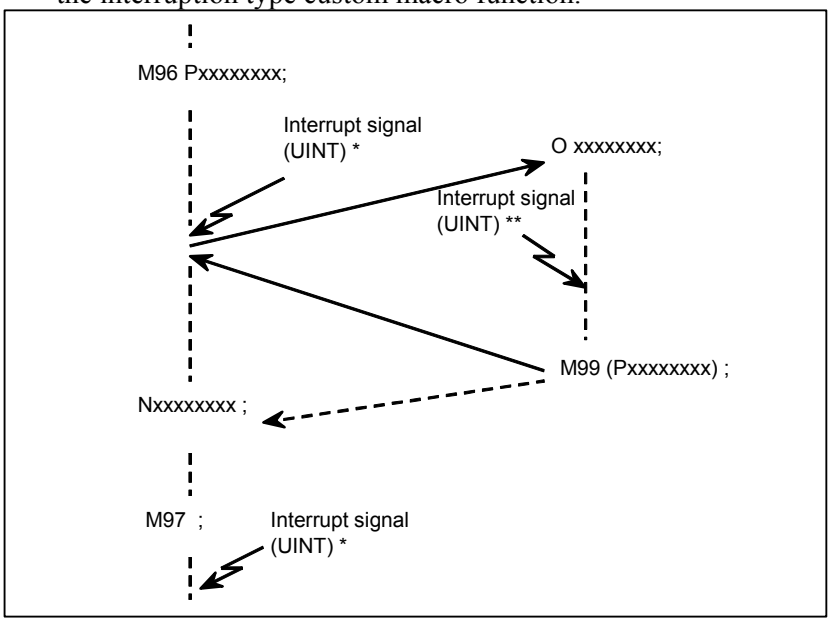


Fig 16.13 (a) Interruption type custom macro function

When M96Pxxxx is specified in a program, subsequent program operation can be interrupted by an interrupt signal (UINT) input to execute the program specified by Pxxxx. When the interrupt signal (UINT, marked with an asterisk (*) in Fig 16.13 (a)) is input during execution of the interrupt program or after M97, it is ignored.

16.13.1 Specification Method

Explanation

- Interrupt conditions

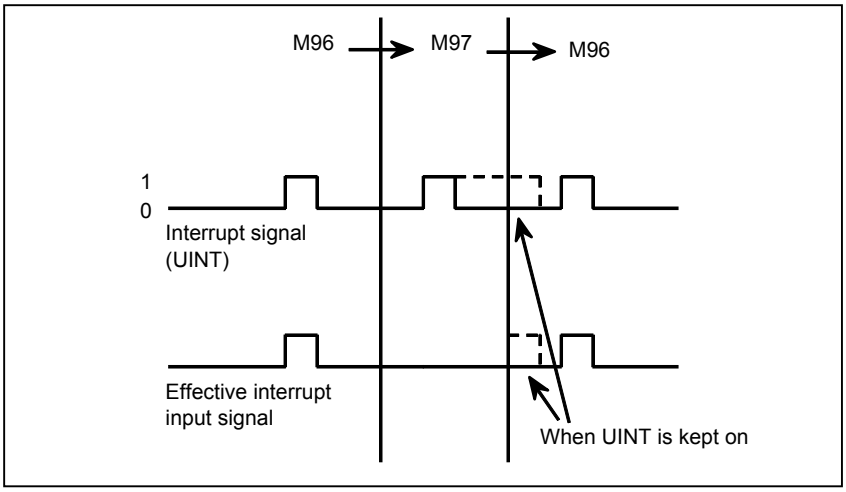
A custom macro interrupt is available only during program execution. It is enabled under the following conditions

- When memory operation, DNC operation, or MDI operation is selected
- When STL (start lamp) is on
- When a custom macro interrupt is not currently being processed

A macro interrupt cannot be performed during manual operation.

- Specification

Generally, the custom macro interrupt function is used by specifying M96 to enable the interrupt signal (UINT) and M97 to disable the signal.
Once M96 is specified, a custom macro interrupt can be initiated by the input of the interrupt signal (UINT) until M97 is specified or the CNC is reset. After M97 is specified or the CNC is reset, no custom macro interrupts are initiated even when the interrupt signal (UINT) is input. The interrupt signal (UINT) is ignored until another M96 command is specified.



The interrupt signal (UINT) becomes valid after M96 is specified. Even when the signal is input in M97 mode, it is ignored. When the signal input in M97 mode is kept on until M96 is specified, a custom macro interrupt is initiated as soon as M96 is specified (only when the status-triggered scheme is employed); when the edge-triggered scheme is employed, the custom macro interrupt is not initiated even when M96 is specified.

NOTE
For the status-triggered and edge-triggered schemes, see Item "Custom macro interrupt signal (UINT)" of II-16.13.2.

16.13.2 Details of Functions

Explanation

- Subprogram-type interrupt and macro-type interrupt

There are two types of custom macro interrupts: Subprogram-type interrupts and macro-type interrupts. The interrupt type used is selected by bit 5 (MSB) of parameter No.6003.

(a) Subprogram-type interrupt: When bit 5 (MSB) of parameter No.6003 is set to 1
An interrupt program is called as a subprogram.
This means that the levels of local variables remain unchanged before and after the interrupt.
This interrupt is not included in the nesting level of subprogram calls.

(b) Macro-type interrupt: When bit 5 (MSB) of parameter No.6003 is set to 0
An interrupt program is called as a custom macro.
This means that the levels of local variables change before and after the interrupt.
The interrupt is not included in the nesting level of custom macro calls.
When a subprogram call or a custom macro call is performed within the interrupt program, this call is included in the nesting level of subprogram calls or custom macro calls.
Arguments cannot be passed from the current program even when the custom macro interrupt is a macro-type interrupt. The local variables immediately after interruption are all cleared to null.

- M codes for custom macro interrupt control

In general, custom macro interrupts are controlled by M96 and M97. However, these M codes, may already being used for other purposes (such as an M function or macro M code call) by some machine tool builders.
For this reason, bit 4 (MPR) of parameter No.6003 is provided to set M codes for custom macro interrupt control.
When specifying this parameter to use the custom macro interrupt control M codes set by parameters, set parameters Nos.6033 and 6034 as follows:
Set the M code to enable custom macro interrupts in parameters Nos.6033, and set the M code to disable custom macro interrupts in parameter 6034.
When specifying that parameter-set M codes are not used, M96 and M97 are used as the custom macro control M codes regardless of the settings of parameters Nos.6033 and 6034.
The M codes used for custom macro interrupt control are processed internally (they are not output to external units). However, in terms of program compatibility, it is undesirable to use M codes other than M96 and M97 to control custom macro interrupts.

- Custom macro interrupts and NC statements

When performing a custom macro interrupt, the user may want to interrupt the NC statement being executed, or the user may not want to perform the interrupt until the execution of the current block is completed. Bit 2 (MIN) of parameter No.6003)is used to select whether to perform interrupts even in the middle of a block or to wait until the end of the block. The type of interrupt performed even in the middle of a block is called type I and the type of interrupt performed at the end of the block is called type II.

⚠ CAUTION

For interrupt type I, operation after control is returned differs depending on whether the interrupt program contains an NC statement.
When the program number block contains EOB (;), it is assumed to contain an NC statement.

(Program containing an NC statement)

```
O0013;
#101=#5041;
#102=#5042;
#103=#5043;
M99;
```

(Program containing no NC statement)

```
O0013#101=#5041;
#102=#5042;
#103=#5043;
M99;
```

Type I (when an interrupt is performed even in the middle of a block)

(i) When the interrupt signal (UINT) is input, any movement or dwell being performed is stopped immediately and the interrupt program is executed.

(ii) If there are NC statements in the interrupt program, the command in the interrupted block is lost and the NC statement in the interrupt program is executed. When control is returned to the interrupted program, the program is restarted from the next block after the interrupted block.

(iii) If there are no NC statements in the interrupt program, control is returned to the interrupted program by M99, then the program is restarted from the command in the interrupted block.

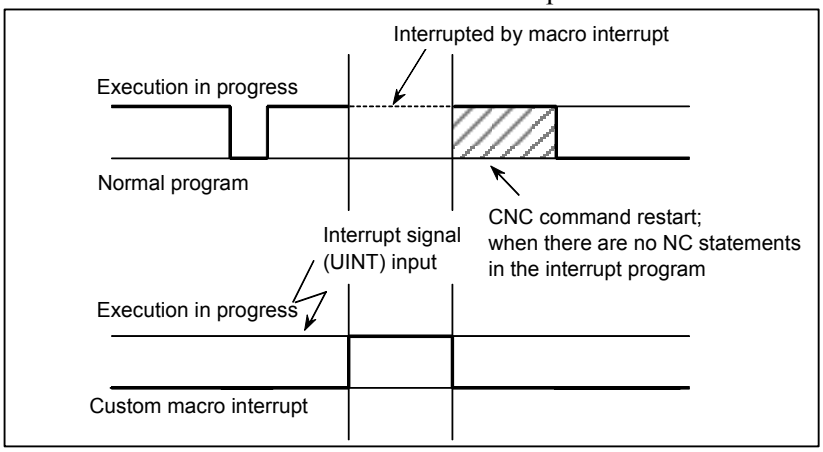


Fig. 16.13 (b) Custom macro interrupt and NC command (type I)

Type II (when an interrupt is performed at the end of the block)

(i) If the block being executed is not a block that consists of several cycle operations such as a drilling canned cycle and automatic reference position return (G28), an interrupt is performed as follows:
When an interrupt signal (UINT) is input, macro statements in the interrupt program are executed immediately unless an NC statement is encountered in the interrupt program. NC statements are not executed until the current block is completed.

(ii) If the block being executed consists of several cycle operations, an interrupt is performed as follows:
When the last movement in the cycle operations is started, macro statements in the interrupt program are executed unless an NC statement is encountered. NC statements are executed after all cycle operations are completed.

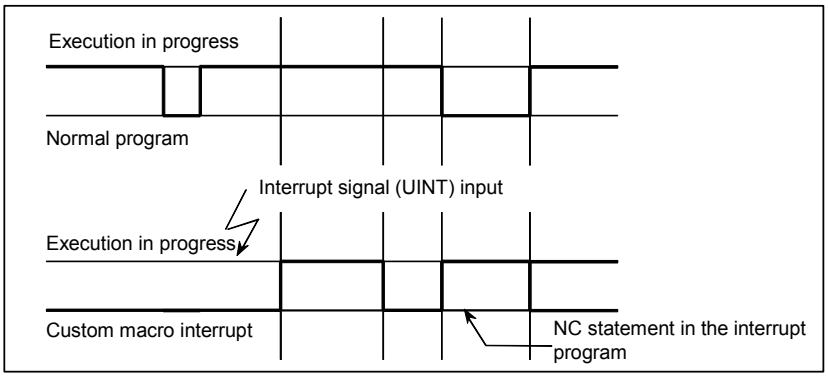


Fig. 16.13 (c) Custom macro interrupt and NC command (type II)

M

NOTE
During execution of a program for cycle operations, interrupt type II is performed regardless of whether bit 2 (MIN) of parameter No. 6003 is set to 0 or 1. Cycle operations are available for the following functions:
<1> Automatic reference position return
<2> Tool radius · tool nose radius compensation (generating multiple blocks using the specified block such as when the tool moves around the outside of an acute angle)
<3> Canned cycle
<4> Automatic tool length measurement
<5> Optional chamfering/corner R
<6> Exponential interpolation
<7> Normal direction control
<8> Cutting point interpolation for cylindrical interpolation

T

NOTE

During execution of a program for cycle operations, interrupt type II is performed regardless of whether bit 2 (MIN) of parameter No. 6003 is set to 0 or 1. Cycle operations are available for the following functions:

<1> Automatic reference position return
<2> Tool radius · tool nose radius compensation (generating multiple blocks using the specified block such as when the tool moves around the outside of an acute angle)
<3> Canned cycle (No interruption type custom macro can be used during execution of a multiple repetitive canned turning cycle, however.)
<4> Automatic tool length measurement
<5> Chamfering/corner R
<6> Exponential interpolation
<7> Cutting point interpolation for cylindrical interpolation

- Conditions for enabling and disabling the custom macro interrupt signal

The interrupt signal becomes valid after execution starts of a block that contains M96 for enabling custom macro interrupts. The signal becomes invalid when execution starts of a block that contains M97.
While an interrupt program is being executed, the interrupt signal becomes invalid. The signal become valid when the execution of the block that immediately follows the interrupted block in the main program is started after control returns from the interrupt program. In type I, if the interrupt program consists of only macro statements, the interrupt signal becomes valid when execution of the interrupted block is started after control returns from the interrupt program.

- Custom macro interrupt signal (UINT)

There are two schemes for custom macro interrupt signal (UINT) input: The status-triggered scheme and edge-triggered scheme. When the status-triggered scheme is used, the signal is valid when it is on. When the edge triggered scheme is used, the signal becomes valid on the rising edge when it switches from off to on status.
One of the two schemes is selected with bit 3 (TSE) of parameter No.6003.

When the status-triggered scheme is selected by this parameter, a custom macro interrupt is generated if the interrupt signal (UINT) is on at the time the signal becomes valid. By keeping the interrupt signal (UINT) on, the interrupt program can be executed repeatedly.

When the edge-triggered scheme is selected, the interrupt signal (UINT) becomes valid only on its rising edge. Therefore, the interrupt program is executed only momentarily (in cases when the program consists of only macro statements). When the status-triggered scheme is inappropriate, or when a custom macro interrupt is to be performed just once for the entire program (in this case, the interrupt signal may be kept on), the edge-triggered scheme is useful.

Except for the specific applications mentioned above, use of either scheme results in the same effects. The time from signal input until a custom macro interrupt is executed does not vary between the two schemes.

In the example shown in Fig 16.13 (d), an interrupt is executed four times when the status-triggered scheme is used; when the edge-triggered scheme is used, the interrupt is executed just once.

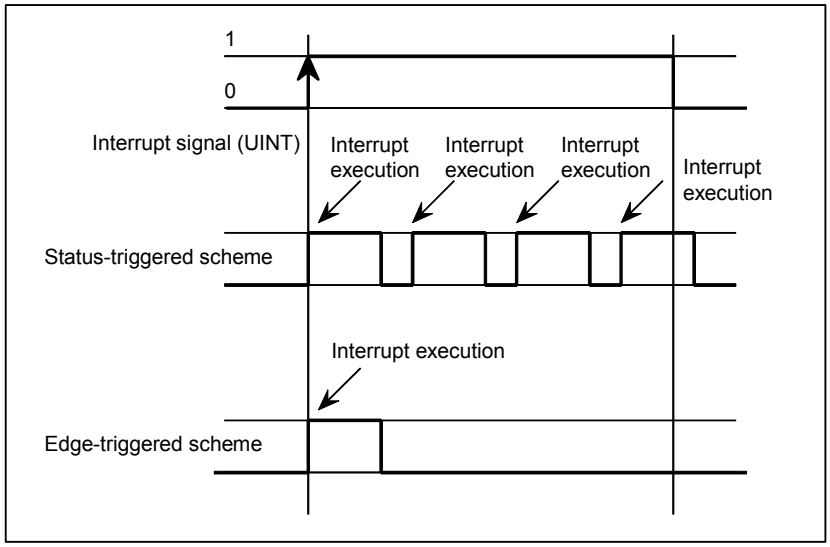


Fig. 16.13 (d) Custom macro interrupt signal

- Return from a custom macro interrupt

To return control from a custom macro interrupt to the interrupted program, specify M99. A sequence number in the interrupted program can also be specified using address P. If this is specified, the program is searched from the beginning for the specified sequence number. Control is returned to the first sequence number found.

NOTE

When an M99 block consists only of address O, N, P, L, or M, this block is regarded as belonging to the previous block in the program. Therefore, a single-block stop does not occur for this block. In terms of programming, the following <1> and <2> are basically the same. (The difference is whether Gxx is executed before M99 is recognized.)

<1> Gxx Xxxx ;
M99 ;
<2> Gxx Xxxx M99 ;

- Custom macro interrupt and modal information

A custom macro interrupt is different from a normal program call. It is initiated by an interrupt signal (UINT) during program execution. In general, any modifications of modal information made by the interrupt program should not affect the interrupted program.
For this reason, even when modal information is modified by the interrupt program, the modal information before the interrupt is restored when control is returned to the interrupted program by M99.

When control is returned from the interrupt program to the interrupted program by M99 Pxxxxxxxx, however, modal information can again be controlled by the program. In this case, the new continuous information modified by the interrupt program is passed to the interrupted program.

In this case, take the following action as required:
<1> The interrupt program provides modal information to be used after control is returned to the interrupted program.
<2> After control is returned to the interrupted program, modal information is specified again as necessary.

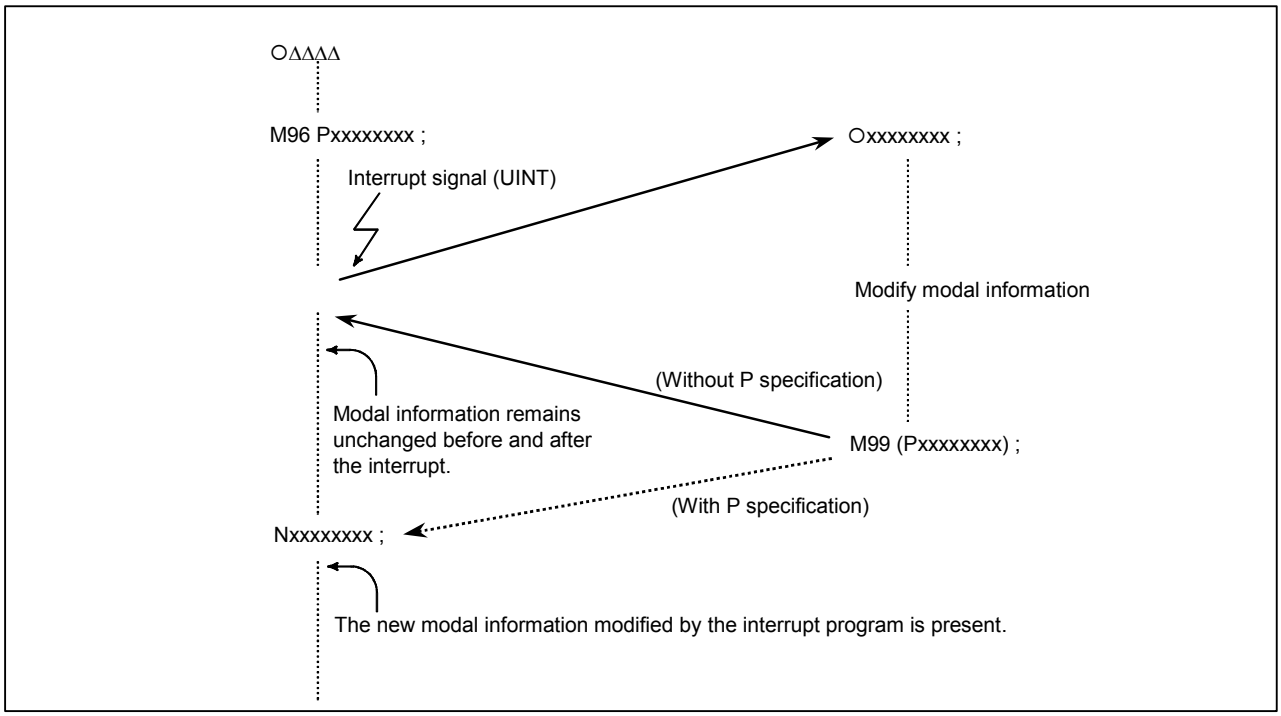


Fig. 16.13 (e) Custom macro interrupt and modal information

Modal information when control is returned by M99

The modal information present before the interrupt becomes valid. The new modal information modified by the interrupt program is made invalid.

Modal information when control is returned by M99 Pxxxxxxxx

The new modal information modified by the interrupt program remains valid even after control is returned.

Modal information which was valid in the interrupted block

The old modal information which was valid in the interrupted block can be read using custom macro system variables #4401 to #4530.

M

System variable	Modal information which was valid when a custom macro interrupt was generated
#4401	G code (group 01)
:	:
#4421	G code (group 21)
#4502	B code
#4507	D code
#4508	E code
#4509	F code
#4511	H code
#4513	M code
#4514	Sequence number
#4515	Program number
#4519	S code
#4520	T code
#4530	Additional workpiece coordinate system number

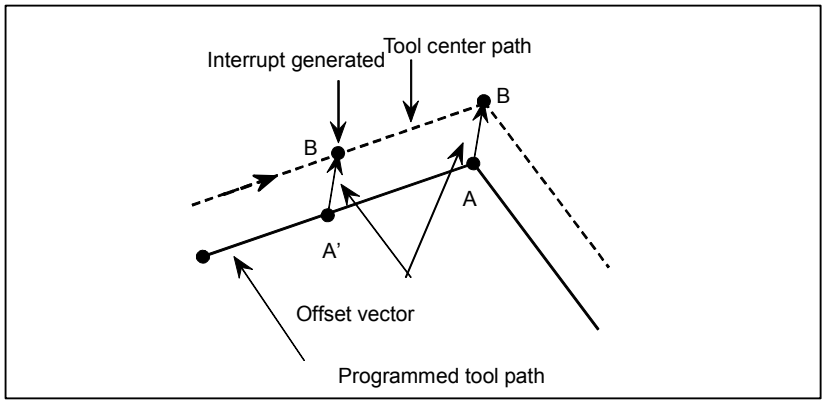
T

System variable	Modal information which was valid when a custom macro interrupt was generated
#4401	G code (group 01)
:	:
#4421	G code (group 21)
#4508	E code
#4509	F code
#4513	M code
#4514	Sequence number
#4515	Program number
#4519	S code
#4520	T code
#4530	Additional workpiece coordinate system number

- System variables (position information values) for the interrupt program

Position information can be read as follows.

Macro variable	Condition	Position information value
#5001 or above	Until the first NC statement appears	Coordinates of point A
	After an NC statement with no move command appears	Coordinates of point A'
	After an NC statement with a move command appears	Coordinates of the end point of the move command
#5021 or above		Machine coordinates of point B'
#5041 or above		Workpiece coordinates of point B'



- Custom macro interrupt and custom macro modal call

When the interrupt signal (UINT) is input and an interrupt program is called, the custom macro modal call is canceled (G67). However, when G66 is specified in the interrupt program, the custom macro modal call becomes valid. When control is returned from the interrupt program by M99, the modal call is restored to the state it was in before the interrupt was generated. When control is returned by M99 Pxxxxxxxx;, the modal call in the interrupt program remains valid.

- Custom macro interrupt and program restart

In program restart, when the interrupt signal (UINT) is input during dry run recovery after a search, the interrupt program is called after restart of all axes is completed.
That is, interrupt type II is assumed regardless of the parameter setting.

M

NOTE
1 Alarm PS1101 occurs in the following cases:
<1>An interrupt is generated in the programmable mirror image (G51.1) mode and another G51.1 is specified in the interrupt program.
<2>An interrupt is generated in the coordinate system rotation (G68) mode and another G68 is specified in the interrupt program.
<3>An interrupt is generated in the scaling (G51) mode and another G51 is specified in the interrupt program.
2 In program restart, do not input the interrupt signal (UINT) during dry run recovery after a search.

T

NOTE
1 Alarm PS1101 occurs in the following cases:
<1>An interrupt is generated in the programmable mirror image (G51.1) mode and another G51.1 is specified in the interrupt program.
<2>An interrupt is generated in the coordinate system rotation (G68.1) mode and another G68.1 is specified in the interrupt program.
<3>An interrupt is generated in the scaling (G51) mode and another G51 is specified in the interrupt program.
2 No interruption type custom macro can be used during execution of a multiple repetitive canned turning cycle.
3 In program restart, do not input the interrupt signal (UINT) during dry run recovery after a search.

17 REAL-TIME CUSTOM MACRO

Chapter 17, "REAL-TIME CUSTOM MACRO", consists of the following sections:

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17.7 AXIS CONTROL COMMAND.....608
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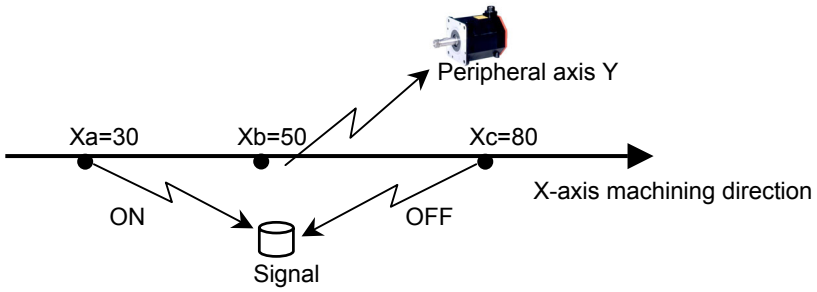
Overview

Used with an NC program, the real time custom macro function controls peripheral axes and signals.

If a macro statement is used together with an NC statement, a program using the conventional custom macro function executes the macro statement immediately when the macro statement is read. So, the macro statement cannot be executed independently of the NC statement.
On the other hand, the real time custom macro function enables the following control operations when a real time macro command (RTM command) is coded in an NC program.

- A real time macro command starts operation in synchronism with the NC statement and is executed independently. During NC program execution, a real time macro command can be executed at the same time.
- PMC interface signals can be read and written (with a restriction). In an NC program, a motion using a signal as a trigger can be coded.
- Variables dedicated to a real time macro command can be read and written.
- A real time macro command can exercise axis control. (The option is required.)
- Multiple real time macro commands can be executed at the same time. Multiple real time custom macro statements can be coded in an NC program and can be controlled independently of each other.

By using a real time custom macro, control on signals and peripheral axes can be programmed.
The example below turns on and off a signal and performs operation on a peripheral axis when machining points are passed.

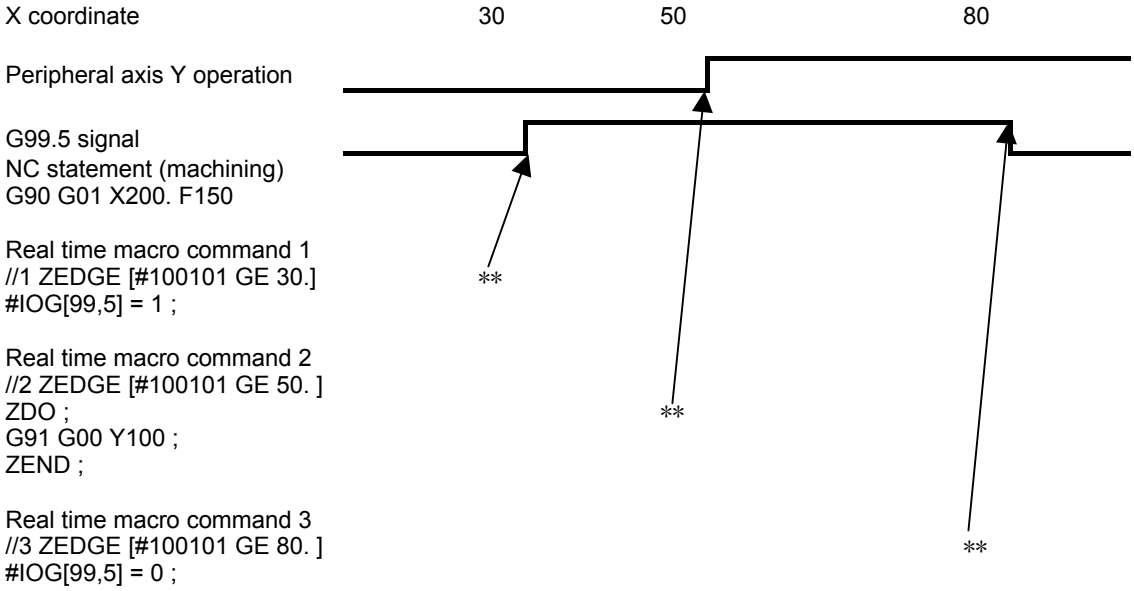


During movement for machining along the X-axis in the figure above:
<1> When point Xa is passed, the G99.5 signal is set to 1.
→ Real time macro command 1
<2> When point Xb is passed, positioning is started on peripheral axis Y. → Real time macro command 2
<3> When point Xc is passed, the G99.5 signal is set to 0.
→ Real time macro command 3

The operation above is programmed using real time macro commands.

```
Program
O0001 ;
G92 X0 ;
//1 ZEDGE [#100101 GE 30. ] #IOG[99,5] = 1 ;
//2 ZEDGE [#100101 GE 50.] ZDO ;
G91 G00 Y100 ;
ZEND ;
//3 ZEDGE [#100101GE 80. ] #IOG[99,5] = 0 ;
G90 G01 X200. F150 ;
M30 ;
```

The timing is as follows. (‘**’ represents the satisfaction of the condition.)



Explanation

To use the real time custom macro function, a real time macro command (RTM command) is coded in an NC program.

- Real time macro command (RTM command)

The real time macro command (RTM command) is a macro command that starts execution in phase with an NC statement in the program. After the execution of a real time macro command starts, the RTM command operates independently of the NC statement.
An RTM command is a statement dedicated to the real time custom macro function.
An RTM command consists of a set of one or more real time macro statements (RTM statements).

- Real time macro statement (RTM statement)

The real time macro statement (RTM statement) is a statement included in an RTM command.
One or more RTM statements make up an RTM command.
An RTM statement consists of a macro command and axis control command dedicated to the real time custom macro function.

The axis control command of an RTM statement is an RTM statement including an address. This command is used to exercise axis control.
Example

```
// ZDO ;
G90 G00 X100 ;
ZEND ;
```

(ZDO and ZEND are reserved words required for the axis control command of an RTM statement, and are detailed later.)

The macro command of an RTM statement is a macro statement used with an RTM statement for arithmetic/logical operation and signal control. (In the example below, #RV[1] and #RV[2] are variables dedicated to the real time macro command.)
Example

```
// #IOG[124, 5] = 1 ;
// #RV[1] = #RV[2] * 10 ;
```

Format

The real time macro command format is shown below.
The RTM command is a command with two slashes (//) prefixed at the start of a block.

```
//n <real-time-macro-statement>
  or
//n ZDO ;
<real-time-macro-statement>
:
ZEND ;
```

N: Modal ID (1 to 10) (Omissible)
When a proper number is coded in n, a modal real time macro command is specified. When n is omitted, a one-shot real time macro command is specified.

ZDO to ZEND are detailed later.

17.1 TYPES OF REAL TIME MACRO COMMANDS

17.1.1 Modal Real Time Macro Command / One-shot Real Time Macro Command

Explanation

A command with '//' followed by an RTM statement is referred to as a one-shot real time macro command (one-shot RTM command).
Example:

```
// #RV[1] = 30 ;
```

On the other hand, a command with '//' followed by number n (1 to 10) then an RTM statement is referred to as a modal real time macro command (modal RTM command).
Example:

```
//3 #RV[1] = 30 ;
```

A one-shot RTM command starts when the execution of the first following NC command starts. A started one-shot RTM command is valid until the end of the NC command.
A modal RTM command starts when the execution of the first following NC command starts, as in the case of a one-shot RTM command.
Unlike a one-shot RTM command, however, a started modal RTM command is valid until the automatic operation is completed.

- Start of a real time macro command

An RTM command starts when the execution of the first following NC command starts.

Example:
When NC command (1) starts execution in the program below, macro commands (2) and (4) are executed in succession without waiting for the end of (1).
On the other hand, RTM command (3)starts execution when NC command (5) starts execution after the end of NC command (1).

```
O0001 ;
G90 G00 X30. ;      (1) NC command
#100=0 ;            (2) Macro command
// #RV[0]=1 ;       (3) RTM command
#102=2;             (4) Macro command
G90 G00 X100. ;     (5) NC command
M30 ;
```

- End of a real time macro command

When one of the following conditions is satisfied, the RTM command is terminated.

Termination conditions common to one-shot RTM and modal RTM commands

- When RTM command processing is completed
- When a reset occurs

Termination condition specific to a one-shot RTM command

- When the execution of the NC command that started at the same time has ended

However, if the RTM statement being executed is an axis control command, the command is terminated when the execution of the block is completed.
If the execution of the Y10. block is completed before the X100. block in the command below, and the execution of the Y20. block starts, for example, the X100. command of the RTM statement is executed until the end.

```
// ZDO ;
X100 ;
ZEND ;
Y10. ;
Y20. ;
```

NOTE

1 No one-shot RTM command can be specified using any of the commands indicated below as a trigger. When using any of these commands as a trigger, use a modal RTM command.
- Command related to reference position return
- Command related to AI contour control
- Command related to canned cycles (rigid tapping, drilling cycle, etc.)
- Command related to cutter compensation
- Command related to tool length compensation
- Command related to automatic tool length measurement
- Command related to coordinate system rotation
  Command related to scaling
- Command related to programmable mirror image

2 If an RTM command is specified using, as a trigger, a block such as a block specifying NURBS interpolation or a T series multiple repetitive canned cycle that does not necessarily pass the start point or end point of the command, operation can start or end at a point other than the start point and end point. Do not use such a block as a trigger.

NOTE

3 Do not restart a program that includes an RTM command.

4 When an NC statement used as a trigger for an RTM command represents an auxiliary function, execution continues even if the FIN signal is awaited.
If the following program is executed, for example, the count-up operation of #RV[0] continues until the FIN signal of M55 is returned:

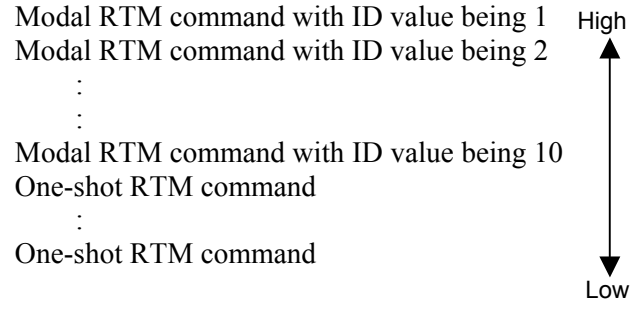
```
O0001 ;
// ZWHILE [1] #RV[0] = #RV[0]+1 ;
M55 ;
G91 X200. ;
   :
```

5 If M02 follows an RTM command, execution continues until a reset occurs, even when the program itself is terminated.
If the following program is executed, for example, the count-up operation of #RV[0] continues until a reset occurs:

```
O0001 ;
// ZWHILE [1] #RV[0] = #RV[0]+1 ;
M02 ;
```

- Priority of commands

If a modal RTM command and one-shot RTM command are specified at the same time, the modal RTM command is executed first.
If multiple modal RTM commands are specified at the same time, the commands are executed in ascending order of ID values.
No priority is applicable to one-shot RTM commands.
The order of execution is as follows:



Example 1)
Priority of modal RTM commands

```
O0001 ;
//1 #RV[0]=1 ;
//3 #RV[0]=3 ;
//2 #RV[0]=2 ;
M02 ;
```

When the program above is executed, the RTM commands are executed in the following order:
#RV[0]=1
#RV[0]=2
#RV[0]=3
So, the value of #RV[0] is 3.

Example 2)

Priority of modal RTM commands and a one-shot RTM command

```
O0001 ;
//3 #RV[0]=3 ;
//1 #RV[0]=1 ;
// #RV[0]=10 ;
//5 #RV[0]=5 ;
M02 ;
```

When the program above is executed, the RTM commands are executed in the following order:
//1 #RV[0]=1
//3 #RV[0]=3
//5 #RV[0]=5
// #RV[0]=10
So, the value of #RV[0] is 10.

Example 3)

Priority of one-shot RTM commands

When the following program is executed, the value of #RV[0] is undefined, that is, 1, 2, or 3.

```
O0001 ;
//#RV[0]=1 ;
//#RV[0]=2 ;
//#RV[0]=3 ;
G04P10 ;
M30 ;
```

Even if a priority is applicable, the order of execution or the order of termination can change when an RTM statement includes a control code, ZWHILE or ZEDGE, or axis control command.

Example 4)

The RTM command priority of #RV[0]=1 in a modal command with its ID value being 1 is higher than the priority of #RV[1]=1 in a modal command with its ID value being 2. However, #RV[0]=1 is executed after the end of the execution of the block specifying the axis control command G91 G00 X10., so that #RV[1]=1 is actually executed earlier than #RV[0]=1.

```
O0001 ;
//1 ZDO ;
G91 G00 X10. ;
#RV[0]=1 ;
```

```
ZEND ;
//2 #RV[1]=1 ;
G04 P10 ;
M30 ;
```

Example 5)

In the RTM command priority, ZEDGE in a modal command with its ID value being 1 is always a false control code (detailed later). The RTM command priority of #RV[0]=1 in a modal command with its ID value being 1 is higher than the priority of #RV[1]=1 and #RV[2]=1 in a modal command with their ID values being 2. However, #RV[0]=1 is executed after the condition of ZEDGE becomes true (that is, at the second time or later), so that #RV[1]=1 and #RV[2]=1 are executed earlier than #RV[0]=1.

```
O0001 ;
//1 ZEDGE [ #IOG[234.0] EQ 1 ] #RV[0]=1 ;
//2 ZDO ;
#RV[1]=1 ;
#RV[2]=1 ;
ZEND ;
G04 P10 ;
M30 ;
```

- Number of real time macro commands

A program can have multiple RTM commands coded.
Up to six one-shot RTM commands can be specified.
If one-shot RTM commands more than the maximum allowable number are specified, an alarm is issued.

Up to ten modal RTM commands can be specified. When specifying modal RTM commands, ensure that there is no duplicate ID.

If there is a duplicate ID, or an incorrect ID is specified, an alarm is issued.

In all paths, up to 16 RTM commands can be executed simultaneously.
When an axis control command is included, up to four commands can be executed simultaneously.

NOTE
1 In a block specifying an RTM statement, no NC command can be coded.
2 If the maximum specifiable number of commands or the maximum number of simultaneously executable commands is exceeded, a PS alarm is issued.

NOTE

3 If an NC statement to be used to trigger an RTM command is specified in a block (e.g., small block) that ends in a very short time, an RTM statement programmed to start at a different timing may be executed simultaneously. If the following is specified, for example, #RV[0]=1 and #RV[1]=2 may be executed simultaneously:

```
// #RV[0]=1 ;
G91 G01 X0.002 ; F5000
// #RV[1]=2 ;
X0.001 ;
```

4 When a function for reading multiple blocks in advance is used, the same ID must not be coded in the scope of blocks read in advance. In the following program, for example, while //1 #RV[0]=#100101 is being executed during AI contour control, //1 #RV[1]=#100101 and //1 #RV[2]=#100101 are also read in advance. So, an alarm may be issued.

```
//1 #RV[0]=#100101 ;
X#100 ;
//1 #RV[1]=#100101 ;
X#100 ;
//1 #RV[2]=#100101 ;
X#100 ;
```

5 When a function for reading multiple blocks in advance is used, up to three blocks among the blocks read in advance can trigger an RTM command.
For example, if the blocks up to the block of (2) are read in advance during execution of (1) in the program below, up to three blocks can trigger an RTM command. In the program below, the number of NC blocks that trigger an RTM command exceeds 3, so that the RTM command of (a) must not be coded.

```
X30. Y50. ; .....(1)
// Z-30. ;
// #RV[0]=#RV[0]+1 ;
X3. Y16. ; → NC block 1 triggering an RTM command
X-23. Y4. ;
// #RV[1]=#RV[1]+1 ;
//2 Z30. ;
X-2. Y9. ; → NC block 2 triggering an RTM command
X17. Y5. ;
// #RV[2]=#RV[2]+1 ;
X-2. Y9. ; → NC block 3 triggering an RTM command
// #RV[3]=#RV[3]+1 ; .....(a)
X-12. Y-3. ;
X-100. Y200. ; .....(2)
```

- Reserved words

The following reserved words are used with real time custom macros:

- Reserved words dedicated to real time custom macros
  ZDO, ZEND, ZONCE, ZWHILE, ZEDGE

- Reserved words shared with custom macros
  AND, OR, XOR, MOD, EQ, NE, GT, LT, GE, LE, SIN, COS, TAN, ASIN, ACOS, ATAN, ATN, SQRT, SQR, ABS, BIN, BCD, ROUND, RND, FIX, FUP, LN, EXP, POW

Be sure to fully spell out the reserved words for real time custom macros. For example, 'ZONCE' must not be coded as 'ZON' or 'ZONC'.

17.2 VARIABLES

Overview

With real time custom macros, the following variables can be handled:

- System variables dedicated to real time custom macros
- Variables (RTM variables) dedicated to real time custom macros
- System variables for some custom macros

The RTM variables mean the variables dedicated to real time custom macros.

List of usable variables

		Real time custom macro	Custom macro
Real time custom macro variables	System variables	Usable	Unusable
	RTM variables	Usable	Unusable
Custom macro variables	System variables	Partially usable	Usable
	Common variables	Unusable	Usable
	Local variables	Unusable	Usable

The variables (system variables and RTM variables) dedicated to real time custom macros are the variables specific to the real time custom macro function. Those variables cannot be used with the custom macro function.

17.2.1 Variables Dedicated To Real Time Custom Macros

These variables are dedicated to real time custom macros. The variables are classified as system variables and RTM variables.

17.2.1.1 System variables

System variables dedicated to real time custom macros

Format

#IOp [m, n] Bit-by-bit read/write
#IOpB [m] Byte-by-byte read/write
#IOpW [m] Word-by-word read/write
#IOpD [m] Double word-by-double word read/write

p : Signal type (X, G, F, Y, D, R)
m : Signal byte address
n : (Used for bit-by-bit read/write only) Signal address bit number (0 to 7)

PMC interface signals are read and written. Bit-by-bit and byte-by-byte read/write operations are possible.
The following signals can be used:

Variable name	Signal type	Read	Write
#IOX #IOXB #IOXW #IOXD	X	Possible	Impossible
#IOG #IOGB #IOGW #IOGD	G	Impossible	Possible
#IOF #IOFB #IOFW #IOFD	F	Possible	Impossible
#IOY #IOYB #IOYW #IOYD	Y	Impossible	Possible
#IOD #IODB #IODW #IODD	D	Possible	Possible
#IOR #IORB #IORW #IORD	R	Possible	Possible

For the valid signal address range, see the specifications of the PMC as well.

When writing to a signal, make the variable unprotected on the PMC signal protection screen (described later) beforehand.

Specify an address by using m and n.
Example:

#IOF[1, 3]	F1.3 bit type
#IOG[1, 5]	G1.5 bit type
#IOFB[32]	F32 byte type
#IOGB[12]	G12 byte type

Read/write operations are performed in the same as for an ordinary macro statement.
Example:

#RV[0]=#IOFB[32]	Assigns F32 to #RV[0].
#IOG[99.3] = 1	Sets G99.3 to 1.

If a signal at a nonexistent address is specified, a PS alarm is issued.

⚠ CAUTION

1 Controls handling other signals, such as a ladder or macro executor, must not write to a signal address being written to by an RTM statement. Ensure that a single control writes to the same byte signal address.
For example, when the G000.0 signal is written to by an RTM statement, do not write to the G000.7 signal from a ladder.
2 Ensure that the same F signal is not read from by an RTM statement and written to by the NC simultaneously.
3 Ensure that the same G signal is not written to by an RTM statement and read from by the NC simultaneously.
4 For word-by-word read/write, specify an even-numbered address and for double word-by-double word read/write, specify an address that is a multiple of four.

NOTE

The system variables described above are not supported for a multi-path PMC.

- PMC signal protection

Whether to enable a write to a signal handled by an RTM statement can be set. This function protects against a malfunction due to incorrect coding.

On the PMC signal protection screen, set whether to enable a write to a signal.
If an RTM statement makes an attempt to write to a signal write-protected on the PMC signal protection screen, a PS alarm is issued at the time of execution.

Set whether to enable a write to each address of Y and G on a byte-by-byte basis.
For each of addresses D and R, set a write-enabled range.

For the unwritable signals (X, F), the screen is not displayed.

Before changing an address on the PMC signal protection range specification screen, perform a clear operation first and then input a new address.

- Input/output

A value set for PMC signal protection can be input/output.

- Input/output format

After punching PMC signal protection, one file(DIDOENBL.TXT) is created.
Please execute input/output operation in EDIT mode.
The output format is as follows:

L Specification method
  0: Byte specification
  1: Range specification
Q Alphabetic signal address
  0: G, 2: Y, 5: R, 9: D
K (for use in range specification only) Index number (0 to 9)
R Address number for byte specification
  Start address number for range specification
P Protection value for byte specification
  0: Not writable
  1: Writable
  End address number
  for range specification

Example of output

```
%
L0Q2R0000P0
L0Q2R0001P1
:                    } Y0 to Y127 protect information
L0Q2R0127P1
:
L0Q0R0000P1
L0Q0R0001P1
:                    } G0 to G767 protect information
L0Q0R0767P0
L0Q0R1000P1
:                    } G1000 to G1767 protect information
L0Q0R1767P0
:                                                        } Byte specification
L1Q5K0R0002P0014 ---- R2 to R14 are writable (Index 0)
:                                                        } Range specification
L1Q9K9R0032P0127 ---- D32 to D127 are writable (Index 9)
M02
%
```

17.2.1.2 Real time macro variables (RTM variables)

The real time macro variables (RTM variables) are variables dedicated to real time custom macros.
The RTM variables are classified as volatile real time macro variables (volatile RTM variables) and nonvolatile real time macro variables (nonvolatile RTM variables).

The data of a nonvolatile RTM variable is preserved even when the power is turned off.
The data of a volatile RTM variable is cleared to 0 when the power is turned off.

Format

#RV [m] Volatile RTM variable
m: Volatile RTM variable number (0 to 99)
#RVS [n] Nonvolatile RTM variable
n: Nonvolatile RTM variable number (0 to 31)

NOTE
1 RTM variables can be used with an RTM statement only. RTM variables cannot be used with an NC statement and macro statement.
2 No RTM variable assumes a "null" value.
3 Volatile RTM variables are cleared to 0 by a reset. On the other hand, nonvolatile RTM variables are not cleared to 0 by a reset.

Explanation

- Input/output

RTM variables can be input/output in a specified format.
Both nonvolatile RTM variables and volatile RTM variables can be input/output.

- Input/output format

After punching RTM variables, one file(RTMMACRO.TXT) is created.
The output format is described below.
The value of an RTM variable is output in hexadecimal as a double-precision floating-point data bit image.
When a nonvolatile RTM variable is output, G10L87 is followed by an RTM variable number and a variable value.
When a volatile RTM variable is output, G10L88 is followed by an RTM variable number and a variable value.

Example of output

```
%
G10L87P0(3FE0000000000000)
G10L87P1(4000000000000000)
:                                   Nonvolatile RTM variable
G10L87P30(4010000000000000)
G10L87P31(4014000000000000)

G10L88P0(4008000000000000)
G10L88P1(3FD9999999800000)
:                                   Volatile RTM variable
G10L88P98(3FF0000000000000)
G10L88P99(4010000000000000)
M02
%
```

Execute input/output operation in EDIT mode.

17.2.2 Custom Macro Variables

With real time custom macros, a part of the custom macro variables (part of the system variables) can be handled.

17.2.2.1 System variables

With real time custom macros, position-related information among the system variables of the custom macros can be handled.

- Position information #100001 to #100182 (Attribute: Read only)

Block end position #100001 to #100032
Current position (machine coordinate system) #100051 to #100082
Current position (workpiece coordinate system) #100101 to #100132
Skip position #100151 to #100182

By readying the values of system variables #100001 to #100182, the end position of the previous block, the current positions (machine coordinate system and workpiece coordinate system), and skip signal position can be found.

Variable No.	Position information	Coordinate system
#100001 #100002 : #100032	Block end position on 1st axis Block end position on 2nd axis : Block end position on 32nd axis	Workpiece coordinate system
#100051 #100052 : #100082	Current position on 1st axis Current position on 2nd axis : Current position on 32nd axis	Machine coordinate system
#100101 #100102 : #100132	Current position on 1st axis Current position on 2nd axis : Current position on 32nd axis	Workpiece coordinate system
#100151 #100152 : #100182	Skip position on 1st axis Skip position on 2nd axis : Skip position on 32nd axis	Workpiece coordinate system

NOTE
1 The value of a variable with a number greater than the number of controlled axes is undefined.
2 The end position (ABSIO) of a skip (G31) block is the skip signal ON position if the skip signal is turned on. If the skip signal is not turned on, the block end point is undefined.
3 As block end position information #100001 to #100032, the end positions read in advance are obtained instead of the end positions of the block currently being executed.

- Servo positional deviation #100251 to #100282 (Attribute: Read only)

By reading the values of system variables #100251 to #100282, the servo positional deviation on each axis can be found.

Variable No.	Position information
#100251	Servo positional deviation on 1st axis
#100252	Servo positional deviation on 2nd axis
:	:
#100282	Servo positional deviation on 32nd axis

NOTE
The value of a variable with a number greater than the number of controlled axes is undefined.

- Remaining travel distance #100801 to #100832 (Attribute: Read only)

By reading the values of system variables #100801 to #100832, the remaining travel distance on each axis can be read.

Variable No.	Position information
#100801	Travel distance on 1st axis
#100802	Travel distance on 2nd axis
:	:
#100832	Travel distance on 32nd axis

NOTE
1 The value of a variable with a number greater than the number of controlled axes is undefined.
2 System variables not described here are unusable.
3 The name of a system variable cannot be specified.
Do not specify a command as indicated below.
//1 #RV[0]=[#_ABSOT[1]] ;

17.2.2.2 Local variables

The local variables(#1 to #33) cannot be used.

17.3 ARITHMETIC AND LOGICAL OPERATION

With the real time custom macros, the following arithmetic and logical operations can be specified:

Table 1.3 Arithmetic and logical operation

Type of operation	Operation	Description
(1) Definition, substitution	#i=#j	Definition or substitution of a variable
(2) Additive operation	#i=#j+#k	Addition
	#i=#j-#k	Subtraction
	#i=#j OR #k	Logical OR (Bit by bit for 32 bits)
	#i=#j XOR #k	Exclusive OR (Bit by bit for 32 bits)
(3) Multiplicative operation	#i=#j*#k	Multiplication
	#i=#j/#k	Division
	#i=#j AND #k	Logical AND (Bit by bit for 32 bits)
	#i=#j MOD #k	Remainder (#j and #k are rounded to an integer to find a remainder. When #j is negative, #i is also negative.)
(4) Function	#i=SIN[#j]	Sine (in deg)
	#i=COS[#j]	Cosine (in deg)
	#i=TAN[#j]	Tangent (in deg)
	#i=ASIN[#j]	Arcsine
	#i=ACOS[#j]	Arccosine
	#i=ATAN[#j]	Arctangent (1 argument). ATN is acceptable.
	#i=ATAN[#j]/[#k]	Arctangent (2 arguments). ATN is acceptable.
	#i=ATAN[#j,#k]	Ditto
	#i=SQRT[#j]	Square root. SQR is acceptable.
	#i=ABS[#j]	Absolute value
	#i=BIN[#j]	Binary conversion from BCD
	#i=BCD[#j]	BCD conversion from binary
	#i=ROUND[#j]	Rounding. RND is acceptable.
	#i=FIX[#j]	Discarding fractional digits
	#i=FUP[#j]	Rounding fractional digits upward to an integer
	#i=LN[#j]	Natural logarithm
	#i=EXP[#j]	Exponent with e (2.718……) used as the base
	#i=POW[#j,#k]	Power (#j to #k power)

- Constant specifiable in <expression>

+0.00000000001 to +999999999999
-999999999999 to -0.00000000001
Up to 12 decimal digits can be specified.
If the maximum allowable number of digits is exceeded, P/S0012 alarm is issued.

NOTE
1 The ADP function is not available.
2 With an RTM statement, the external output commands (BPRNT, DPRNT, POPEN, and PCLOS) are unavailable.
3 The FS16i compatibility specifications are not applicable. Bit 0 (F16) of parameter No. 6008 = 1 (with the operation result precision based on the FS16i compatibility specifications) is invalid.
4 For information on decreased precision, see "When bit 0 (F16) of parameter No. 6008 is set to 0" in "Caution concerning decreased precision" in the CUSTOM MACRO chapter.
5 Bit 0 (NAT) of parameter No. 6004 is valid in an RTM command.
6 The setting of bit 5 (SBM) of parameter No. 6000 and bit 7 (SBV) of parameter No. 6000 is invalid. For the single block specifications, see the separate item.

17.4 CONTROL ON REAL TIME MACRO COMMANDS

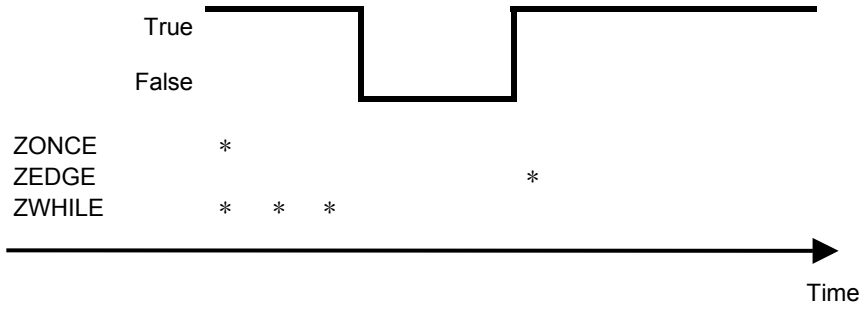
Explanation

By using a reserved word for controlling statements in an RTM command, the flow of the RTM command can be changed or multiple statements can be controlled as a set of statements. Four reserved words are used to control an RTM command as indicated below.

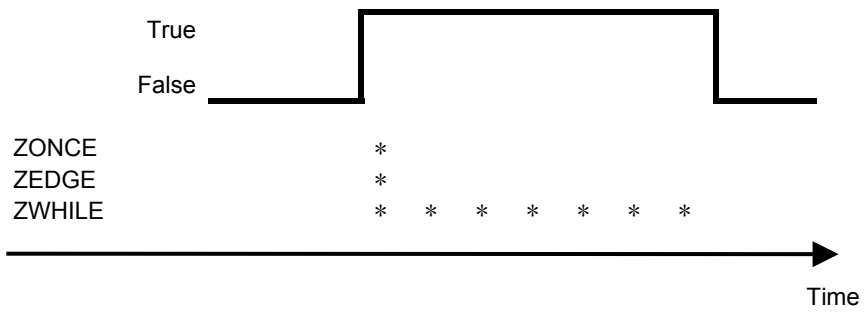
Reserved word name	Syntax	Meaning
ZONCE	// ZONCE A B	If A is true, B is executed.
ZEDGE	// ZEDGE A B	If A becomes true, B is executed.
ZWHILE	// ZWHILE A B	While A is true, B is executed repeatedly.
ZDO...ZEND	// ZDO B1 B2 B3 ZEND	(Multiple statements) B1, B2, and B3 are sequentially executed.

The timing chart of an RTM command using these reserved words is indicated below. (Multi-statement control ZDO...ZEND is excluded.)
When the condition of a each reserved word is True, it shows '*'.

When condition A makes transitions from True to False to True



When condition A makes transitions from False to True to False



17.4.1 Conditional Branch (ZONCE Statement)

After ZONCE, <conditional-expression> and <real-time-macro-statement> are coded.

- //(n) ZONCE [<conditional-expression>] <real-time-macro-statement>

If <conditional-expression> is true, <real-time-macro-statement> is executed. If <conditional-expression> is false, <real-time-macro-statement> is not executed, but control exits from the ZONCE command to terminate the execution. <conditional-expression> conforms to the custom macro specifications.

// ZONCE A B (If A is true, B is executed.)
If A is true, the command is terminated after executing B.
If A is false, the command is terminated without executing B.

If the workpiece coordinate on the first axis is greater than 30, the Y1.0 signal is output.
// ZONCE [#100101 GT 30.] #IOY[1,0] = 1 ;

If the workpiece coordinate on the first axis is greater than 30 and #RV[0] matches #RV[3], the Y1.0 signal is output.
// ZONCE[[#100101 GT 30.] AND [#RV[0] EQ #RV[3]]] #IOY[1,0]=1 ;

In <real-time-macro-statement>, multiple RTM statements can be coded.
In this case, code the following by using ZDO...ZEND of multi-statement structure:
// ZONCE [<conditional-expression>] ZDO ;
<real-time-macro-statement-1> ;
<real-time-macro-statement-2> ;
:
ZEND ;

If the workpiece coordinate on the second axis is equal to or less than 10, the rapid traverse override value is changed.
// ZONCE [#100102 LE 10.] ZDO ; **#IOG[14,0]=0 ;** **#IOG[14,1]=1 ;** **ZEND ;**

However, if <conditional-expression-1> specifies an axis control command, be sure to use ZDO...ZEND even when a single statement is used.

If the workpiece coordinate on the first axis is greater than 30 and #RV[0] matches #RV[3], a movement on the A-axis starts.
//1 ZONCE [[#100101 GT 30.] AND [#RV[0] EQ #RV[3]]] ZDO ; **G91 G00 A20. ;** **ZEND ;**

Similarly, use ZDO...ZEND for a multi-statement including an axis control command.

If the workpiece coordinate on the second axis is equal to or less than 10, a movement on the V-axis starts and the Y1.0 signal is set to 1.
//1 ZONCE [#100102 LE 10.] ZDO ; **G91 G00 V10. ;** **#IOY[1,0] = 1 ;** **ZEND ;**

17.4.2 Condition Transition (ZEDGE Statement)

After ZEDGE, <conditional-expression> and
<real-time-macro-statement> are coded.
// ZEDGE A B (If A becomes true, B is executed.)

- //(n) ZEDGE [<conditional-expression>] <real-time-macro-statement>

When <conditional-expression> makes a transition from false to true, an RTM statement specified after <conditional-expression> is executed. <conditional-expression> conforms to the custom macro specifications.
With the ZEDGE command, the result of the first evaluation of <conditional-expression> is always false.

Explanation

The difference between the ZONCE command and ZEDGE command is that if <conditional-expression> is false, the RTM statement itself is terminated in the case of ZONCE, while the evaluation of <conditional-expression> continues until <conditional-expression> becomes true in the case of ZEDGE. (With a one-shot real time macro command, the evaluation of <conditional-expression> continues until the execution of the NC statement that started operation simultaneously with the command ends.)

On the rising edge of the G address signal 4.3, the workpiece coordinate on the third axis is read.
// ZEDGE [#IOG[4,3] EQ 1] #RV[0]=#100103 ;

In the example above, even if the [#IOG[4,3] EQ 1] is true from the beginning, #RV[0]=#100103 of the RTM statement is not executed. #RV[0]=#100103 is executed when the result of evaluation of [#IOG[4,3] EQ 1] changes from false to true.

As in the case of ZONCE, multiple RTM statements can be coded in <real-time-macro-statement>.
If <conditional-expression-1> specifies an axis control command, be sure to use ZDO...ZEND even when a single statement is used.
Code the following by using ZDO...ZEND of multi-statement structure:
// ZEDGE [<conditional-expression>] ZDO ;
<real-time-macro-statement-1> ;

<real-time-macro-statement-2> ;
:
ZEND ;

On the falling edge of the X address signal, a movement on the B-axis is started and the Y1.0 signal is set to 1.
// ZEDGE [#IOX[1,3] EQ 0] ZDO ; **G91 G00 B10. ;** **#IOY[1,0] = 1 ;** **ZEND ;**

On the rising edge of the G address signal, a movement on the U-axis is started.
// ZEDGE [#IOG[4,3] EQ 1] ZDO ; **G91 G00 U25. ;** **ZEND ;**

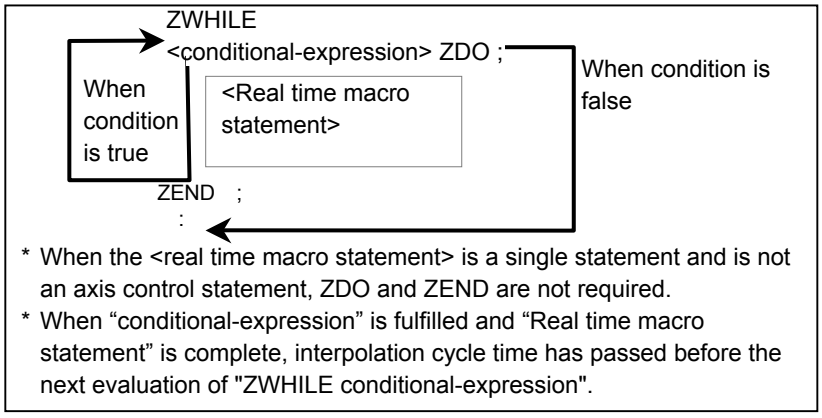
If the remaining travel distance on the second axis is equal to or less than 10, the rapid traverse override value is changed.
// ZEDGE [#100802 LE 10.] ZDO ; **#IOG[14,0]=0 ;** **#IOG[14,1]=1 ;** **ZEND ;**

17.4.3 Repetition (ZWHILE Statement)

After ZWHILE, a <conditional statement> is coded.
// ZWHILE A B (While A is true, B is executed repeatedly.)

- //(n) ZWHILE [<conditional-expression>] <real-time-macro-statement>

While <conditional-expression> is true, the RTM statement specified after <conditional-expression> is executed.
If <conditional-expression> is not satisfied, the ZWHILE statement is terminated, and the next block is processed.



* When the <real time macro statement> is a single statement and is not an axis control statement, ZDO and ZEND are not required.
* When “conditional-expression” is fulfilled and “Real time macro statement” is complete, interpolation cycle time has passed before the next evaluation of "ZWHILE conditional-expression".

Explanation

While <conditional-expression> is true, the command or commands between ZDO and ZEND after ZWHILE are executed.
If <conditional-expression> is not satisfied, the command after ZEND is processed.
The same <conditional-expression> and operators as for the ZONCE statement are used.

While the F address 234.1 signal is 1, an incremental movement on the U-axis is repeatedly performed, and #RV[0] is incremented by 1 each time.
//1 ZWHILE [#IOF[234,1] EQ 1] ZDO ; **G91 G00 U10. ;** **#RV[0] = #RV[0]+1 ;** **ZEND ;**

17.4.4 Multi-statement (ZDO...ZEND Statement)

- //(n) ZDO ;
<real-time-macro-statement-1> <real-time-macro-statement-2>
<real-time-macro-statement-3> …
ZEND ;

One or multiple statements enclosed in ZDO...ZEND are regarded as a one RTM statement (multiple-statement structure).

Explanation

The maximum number of RTM statements in one multi-statement (between ZDO...ZEND) is roughly five, but it is subject to change (Because it depends on the contents of RTM statements).

ZDO...ZEND is used in the following cases:

- When multiple <real time macro statements> are handled as a series of commands

If the workpiece coordinate on the first axis exceeds 30, the rapid traverse override value is changed.
// ZEDGE [#100101 GT 30.] ZDO ; **#IOG[14,0]=1 ;** **#IOG[14,1]=0 ;** **ZEND ;**

- When an axis control command is specified

On the rising edge of the X signal 7.0, a movement on the V-axis is started.
// ZEDGE [#IOX[7,0] EQ 1] ZDO ; **G91 G00 V35. ;** **ZEND ;**

- Nesting

ZONCE, ZEDGE, ZWHILE, and ZDO...ZEND cannot be nested and overlapped.
For details, see the following:

1. ZONCE, ZEDGE, ZWHILE, and ZDO...ZEND may be used any number times.

```
// ZWHILE [···] ZDO ;
    Processing
ZEND ;
  :
// ZONCE[···] ZDO ;
    Processing
ZEND ;
  :
```

2. One ZDO...ZEND range must not overlap another ZDO...ZEND range.

```
// ZONCE [···] ZDO ;
    Processing
// ZWHILE [···] ZDO ;
  :
ZEND ;
    Processing
ZEND ;
```

3. ZONCE, ZEDGE, ZWHILE, and ZDO...ZEND must not be nested.

```
// ZWHILE [···] ZDO ;
  :
  ZONCE [···] ZDO ;
    :
    ZEDGE [···] ZDO ;
      Processin
    ZEND ;
    :
  ZEND ;
  :
ZEND ;
```

- Endless loop

An endless loop is formed if the conditional expression enclosed in brackets after the ZWHILE statement is always satisfied.

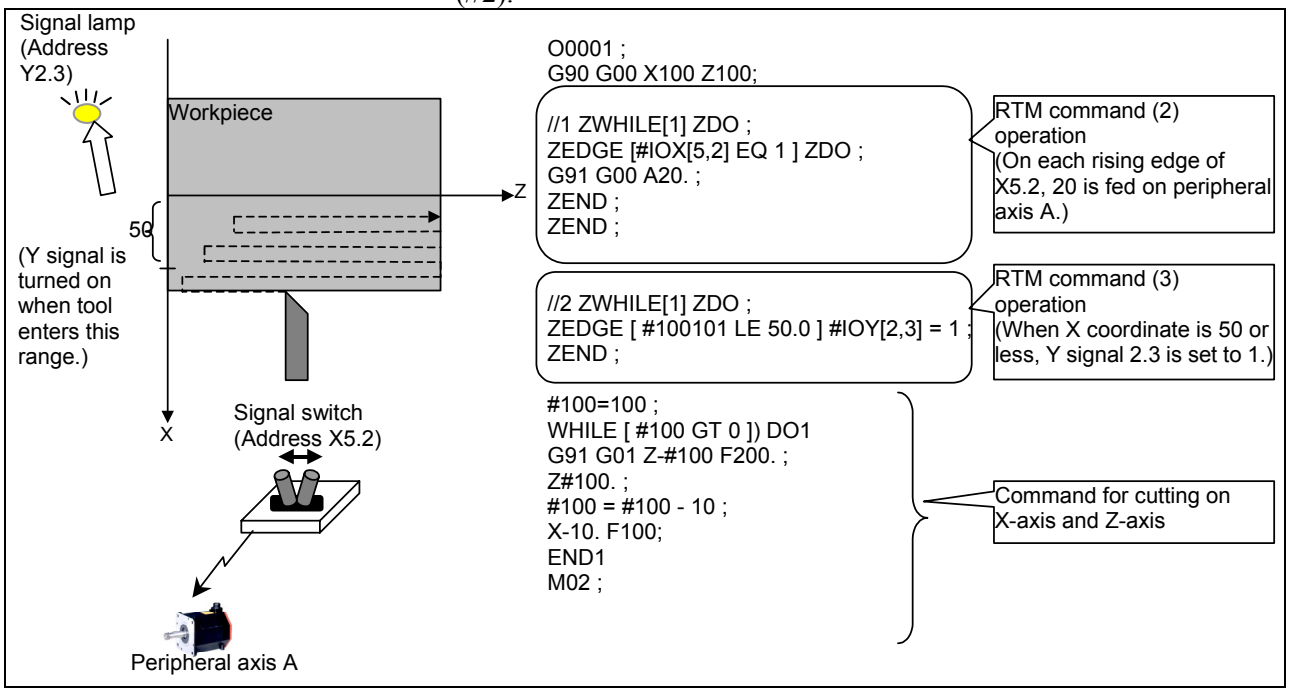
Example)
In the example below, #RV[0] is counted up unconditionally.
// ZWHILE [1] #RV[1]=#RV[1]+1 ;

Sample program

The sample program below exercises the following three control operations at the same time.

(1) A cutting operation is performed on the X-axis and Z-axis.
(2) On each rising edge of the X signal 5.2, 20 is fed on the peripheral axis A.
(3) When the workpiece coordinate on the X-axis (#100101) is equal to or less than 50, the Y signal 2.3 is set to 1.

The operation of (1) is coded in the main NC statement.
The operation of (2) is coded in the first modal RTM command (//1).
The operation of (3) is coded in the second modal RTM command (//2).

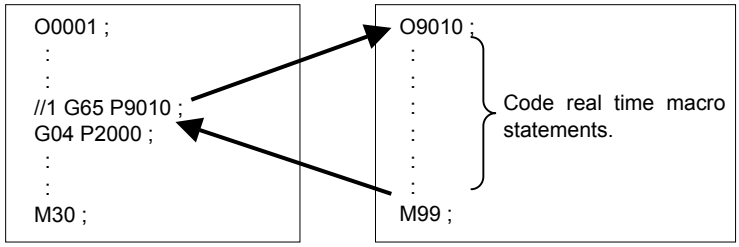


17.5 MACRO CALL

A series of RTM statements can be formed into a subprogram, which can be called from the main program.

When G65 is specified in an RTM command, the real time macro specified in address P is called.

G65 P p ; P : Number of real time macro program to be called



The execution of real time macro //1 including O9010 starts simultaneously with the G04 block of main program O0001.

Explanation

- Calling

In address P after G65, specify the program number of a real time custom macro to be called.

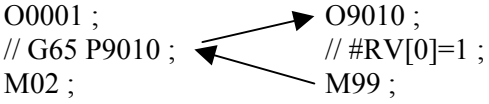
NOTE

1 In argument P of G65, only a constant value can be used. No value can be specified through a variable.
Example:
// G65 P9010 ; Correct
// G65 P#RV[0] ; Incorrect
2 The inclusion of another NC command (such as G01 X100.0 G65 Pp) is not allowed. If another NC command is included, PS0127 alarm is issued.
3 The G65 block for calling a real time macro does not make a single block stop.
4 On the other hand, a real time macro program called by real time macro calling makes a single block stop.

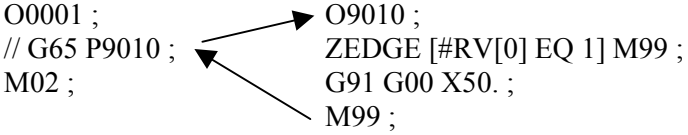
- Call destination real time program

In a called real time macro program, only an RTM statement can be coded.

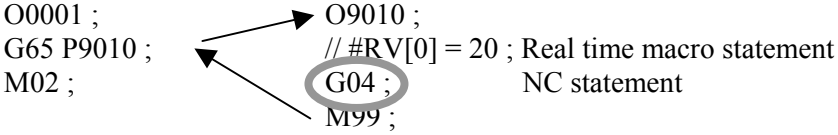
In a called real time macro program, no additional RTM command may be executed. (The RTM command symbol '//' may not be coded.) For example, do not execute the following program:



Code M99 in the last block only.
For example, do not execute the following program:



When executing an RTM statement in a subprogram, code an NC statement before the M99 block for returning to the calling program.



- Format

Specify G65 at the start of a block.

- Call nesting level

Macro call nesting is not allowed.

- Differences from macro call using a custom macro

Macro call using a real time custom macro and macro call using a custom macro differs as described below.

- In macro call using a custom macro, an argument (data to be passed to a macro) and the number of repetitions can be specified. In macro call using a real time custom macro, such information cannot be specified.
- With a real time custom macro, other types of macro call (macro call using G66, G66.1, G, and M code) and subprogram call are not allowed.

17.6 OTHERS

If an axis control command is followed by a macro command in an RTM command, the execution of the macro command starts when the axis control command is completed or deceleration starts.
If deceleration on the X-axis starts upon completion of distribution according to the axis control command of (1), for example, the macro command of (2) is executed. If acceleration/deceleration is not applied on the X-axis, (2) is executed upon completion of distribution on the X-axis.

```
// ZDO ;
G91 G00 X30 ;       (1) Axis control command of the RTM statement
#RV[0] = 1 ;        (2) Macro command of the RTM statement
ZEND ;
```

If an RTM command is executed in a custom macro WHILE DO to END loop, and the same RTM is looked ahead while the RTM command is being executed, a PS alarm may be issued.
For example, if the command below is executed, a PS alarm is issued.

```
WHILE [1] DO1 ;
// #RV[0] = #RV[0] + 1 ;
#100=#100+1 ;
END1 ;
```

In this example, insert G04 ; block between a #100=#100+1; and END1;.

17.7 AXIS CONTROL COMMAND

In an RTM statement, a G code for specifying a movement can be specified. For axis control, the PMC axis control interface is used. The specifications differ from the specifications for the G code used with an NC statement.

Format

```
// ZDO ;
G_ IP _ F_ ;
  :
ZEND ;
```

For an axis control command, ZDO...ZEND of multi-statement structure is used for coding even if a single statement (single RTM statement) is used.

⚠ CAUTION
Do not specify an NC statement for an axis to be controlled with an RTM statement. Moreover, do not exercise axis control with an RTM statement on an axis to be controlled with an NC statement.

NOTE
1 In one block, only one axis may be specified. Do not specify two or more axes in the same block.
2 In the three-dimensional coordinate conversion mode, the polygon mode, etc., an RTM statement concerning that axis cannot be specified.

Explanation

- Setting

For an axis control command in an RTM statement, the interface of a PMC axis control group set by a parameter beforehand is used.
For an axis to be controlled with an RTM statement, set a group to be used with parameter No. 8010. Then, set bit 0 (XRT) of parameter No. 8011 for the axis to 1.

NOTE
An axis for which bit 0 (XRT) of parameter No. 8011 is set to 1 is dedicated to real time custom macros, so that such an axis cannot be used with PMC axis control.

- Relationship with PMC axis control

Axis control based on an RTM statement uses the PMC axis control interface. So, the specifications for a move command in each block within an RTM statement are generally equivalent to the specifications for the PMC axis control command.
For the detailed specifications and restrictions related to axis control, refer to the specifications of PMC axis control as well.

NOTE
1 An axis used with an RTM statement must not be specified from PMC axis control.
2 A PMC axis control group used with an RTM statement must not be specified from PMC axis control.

- Operation command code

The table below indicates the G codes that can be specified in an RTM statement.
The ◤ mark in the table indicates modal G codes when an RTM command starts.

Code	Group	Meaning
◤ G00	01	Positioning
G01		Feed at specified feedrate
G28	00	Reference position return
G53		Machine coordinate system selection
G90	03	Absolute command NOTE 1
◤ G91		Incremental command NOTE 2
◤ G94	05	Feed per minute
G95		Feed per revolution

NOTE
1 The absolute command (G90) is valid for machine coordinate system selection (G53) only.
2 The incremental command (G91) is valid for positioning (G00), linear interpolation (G01), and reference position return (G28) only.
3 In T series G code system A as well, not G98/G99 but G90/G91 are valid with an RTM command.

- Modal information

Basically, modal information such as F and modal G codes in an RTM statement is independent in an NC statement and each RTM statement. Modal information in an RTM statement is initialized when the execution of the RTM statement is started (when a block specified with // is executed).
The initial value of modal information in an RTM statement is as follows:

State of the G codes marked with ◤ in the operation command code list
F code : F0

⚠ CAUTION

With the G codes (inch input/metric input) of group 06, the same information as the modal information of an NC statement is used in an RTM statement. Do not change the modal information of group 06 with an NC statement in a block after the first RTM statement coded in a program.

NOTE

With an RTM command, bit 0 (G01), bit 3 (G91), and bit 4 (FPM) of parameter No. 3402 are invalid when the power is turned on or in the cleared state.

Example 1)

Modal information can be controlled independently in an NC statement and each RTM statement.

```
O0001 ;
G90 G01 X100. Y100. F500. ;      (1)
//1 ZDO ;
Z50. ;                           (2)
G01 Z100. F100. ;                (3)
ZEND ;
//2 ZDO ;
G01 A30. F200. ;                 (4)
A50. ;                           (5)
ZEND ;
X200. ;                          (6)
:
```

The modal information of command (2) is G00, G91, and G94, regardless of the modal information (G90 G01 set in (1)) of the NC statement.

The modal information of command (5) is G01, G91, G94, and F200.. This modal information is not affected by the modal information (command (1)) of the NC statement and the modal information (commands (2) and (3)) in the RTM statement with modal ID=1.

The modal information of command (6) is not affected by the modal information (commands (2) to (5)) in the RTM statement, so that the modal information is G01, G90, and F500..

Example 2)

Modal information is initialized when the execution of each RTM command is started. Even if the same program includes RTM commands with the same ID, the modal information of the RTM command executed first is not inherited by the RTM command executed next.

```
O0001 ;
G90 G01 X100. Y100. F500. ;
//1 ZDO ;
G01 Z100. F3000. ;                (1)
ZEND ;
X200. ;
//1 ZDO ;
Z200. ;                           (2)
ZEND ;
X300. ;
:
```

The modal information of command (2) is G91 and G00, regardless of command (1).

- Single block stop

If an NC statement is placed in the single block stop state, for example, by the single block stop signal SBK, the axis control command of an RTM statement also comes to a single block stop.
However, the automatic operation start signal STL is turned off when all conditions on the NC statement side are satisfied, even if the axis control command of an RTM statement is not terminated.
If an RTM command consists of multiple statements and an axis control command is coded in multiple blocks, only the block of the RTM statement that is currently executing an axis command can be brought to a single block stop by setting the group-by-group single block stop signal (ESBKg) for PMC axis control corresponding to the axis to 1.

- Feed hold

Even when an NC statement enters the automatic operation halt state, the axis control command of an RTM statement does not stop immediately but stops at the time of termination of the block currently being executed. To stop also the axis control command of an RTM statement immediately when the NC statement enters the automatic operation halt state, control the temporary stop signal (ESTPg) for PMC axis control of the related group by monitoring the automatic operation halt in-progress signal (SPL).

- Reset

Even when the CNC is reset by an MDI reset, the external reset signal (ERS), or the reset and rewind signal (RRW), the axis control command of an RTM statement does not stop immediately but stops at the time of termination of the block currently being executed. When the emergency stop state is set, however, the RTM statement also stops immediately.
During execution of the axis control command of an RTM statement, the real time macro being executed can be stopped independently of the NC statement by setting the group-by-group reset signal (ECLRg) for PMC axis control corresponding to the axis to 1. To stop also the axis control command of an RTM statement when the CNC is reset, control the reset signal (ECLRg) for PMC axis control of the related group by monitoring the reset signal (RST).

- Alarm stop

Even when a PS alarm is issued with an NC statement, the axis control command of the RTM statement being executed does not stop immediately but stops at the time of termination of the block currently being executed. Moreover, even when an overtravel alarm is issued for an axis other than the axis controlled by the RTM statement being executed, the RTM statement being executed does not stop immediately but stops at the time of termination of the block currently being executed.

- Alarm reset

When the group-by-group alarm signal (EIALg) for PMC axis control corresponding to the axis specified by an RTM statement is set to 1, correct the cause of the alarm, then reset the CNC, and reset PMC axis control with the reset signal (ECLRg) for PMC axis control of the related group.
(Tip)

In the following cases, the alarm signal (EIALg) is set to 1:

(1) When a servo alarm is issued
(2) When an overtravel alarm is issued
(3) When P/S0130 alarm is issued
(4) When P/S0139 alarm is issued

- Interlock

In an RTM statement, the interlock signal (*IT) for an NC statement, interlock signal for each axis (*ITn), interlock signal for each axis direction (+ITn/-ITn) are invalid. Instead, the axis control temporary stop signal (ESTPg) for PMC axis control is valid in an RTM statement. When temporarily stopping the axis controlled by an RTM statement, control the corresponding axis control temporary stop signal (ESTPg) for PMC axis control.

The cutting block start interlock signal *CSL (G8.1) and the block start interlock signal *BSL (G8.3) are invalid.

- Machine lock

The same machine lock signals (all axes/each axis) as used with an NC statement are used. However, by disabling machine lock for PMC axis control with the following parameters, machine lock can be disabled for the axis being controlled by an RTM statement:

Bit 0 (MLE) of parameter No. 8001
Bit 1 (MLS) of parameter No. 8006

- Dry run

With bit 2 (OVE) of parameter No. 8001, whether to use the dry run signal (DRN) for an NC statement or the dry run signal (EDRN) for a PMC axis can be chosen. When enabling a dry run feedrate for rapid traverse in an RTM statement, set bit 3 (RDE) of parameter No. 8001 to 1. If a feed command specifying a feedrate is specified when the manual rapid traverse selection signal (RT or ERT) is set to 1, the cutting feedrate is used as the feedrate during dry run. ("Dry run feedrate × maximum manual feedrate override value" is not applicable.)

- In-position check

In the in-position state, the in-position signal (EINPg) is set to 1.
When bit 6 (NCI) of parameter No. 8004 is set to 1, no in-position check is made during axis control based on an RTM statement. The setting of bit 5 (NCI) of parameter No. 1601 for an NC statement is invalid. (However, the in-position signal (INPx) for an NC statement is affected by the value of NCI.)

- Manual absolute

The manual absolute signal (*ABSM) is invalid.

- External deceleration

By setting bit 0 (EDC) of parameter No. 8005 to 1, the external deceleration function can be used also for the axis being controlled by an RTM statement. Note, however, that the external deceleration function is unusable for feed with a specified feedrate (feed per revolution). For an external deceleration rate, the external deceleration rate setting for rapid traverse (parameter No. 1427, No. 1441, and No. 1444) is valid, regardless of the type of feed.
The external deceleration function for the axis being controlled by an RTM statement is also valid for feed with a specified feedrate (feed per minute), regardless of the settings of bit 4 (EDP) of parameter No. 1005 and bit 5 (EDM) of parameter No. 1005. (The enabling of rapid traverse only is impossible.)

- Mirror image

At enabling mirror image for the axis being controlled by an RTM statement, set bit 0 (EMR) of parameter No. 8008 to 1, and set either bit 0 (MIRx) of parameter No.0012 or mirror image signal MIx to '1'. Programmable mirror image is not available.

NOTE

1 The same group as the PMC axis control group used for axis control cannot be executed with an RTM statement.
2 No NC statement can specify an axis being controlled by an RTM statement.
3 An alarm is issued if, during execution of an RTM statement, an attempt is made to execute another RTM statement with the same ID. In the program below, for example, the RTM statement of (1) operates using the NC statement of (2) as a trigger, and the RTM statement of (3) operates using the NC statement of (4) as a trigger. An alarm is issued if (1) is still operating when (4) is started after completion of the interpolation of (2).

```
//1 ZDO ;
G91 G00 A100. ;   (1)  RTM statement
ZEND ;
G01 X30. Y20. ;   (2)  NC statement
//1 ZDO ;
G91 G00 B100. ;   (3)  RTM statement
ZEND ;
X40. Y50. ;       (4)  NC statement
```

4 When T series G code system A is used, address words (such as U, W, H, and V) for incremental commands must not be coded. Otherwise, alarm PS0413 is issued.
5 An RTM statement for performing axis control proceeds to the next command at the end of distribution. For example, in the program below, #RV[0] counts up at the time X50. distribution ends and deceleration starts.

```
// ZDO ;
G00 X50. ;
#RV[0]=#RV[0]+1 ;
ZEND ;
```

Command details

- Rapid traverse

A movement is made at a rapid traverse rate on an axis from the current position to the point separated by a specified value.

Format

```
// ZDO ;
G91 G00 IP _ ;
ZEND ;
```
G91 : G code for incremental command
IP _ : Travel distance

NOTE
1 Only one axis can be specified in one block.
2 The absolute command (G90) cannot be specified.
3 The rapid traverse overlap function cannot be used.
4 When using this function, be sure to set bit 0 (RPD) of parameter No. 8002 to 0.

- Rapid traverse override
  With bit 2 (OVE) of parameter No. 8001, whether to use the rapid traverse override signal (ROV) for an NC statement or the rapid traverse override signal (EROV) dedicated to PMC axis control can be chosen.

NOTE
Even if bit 4 (RF0) of parameter No. 1401 is set to 1, rapid traverse does not stop with a cutting feed override of 0%.

- Feed with a specified feedrate (feed per minute)

A movement is made at a feedrate specified in F on an axis from the current position to the point separated by a specified value.

Format

```
// ZDO ;
G94 G91 G01 IP _ F_ ;
ZEND ;
```
G94 : G code for feed per minute
G91 : G code for incremental command
IP __: Travel distance
F__ : Feedrate command (mm/min or inch/min)

NOTE
1 Only one axis can be specified in one block.
2 The absolute command (G90) cannot be specified.
3 The block overlap function cannot be used.
4 When IS-A is used, a feedrate below 10 mm/min is discarded.
5 The feedrate cannot be clamped to a parameter-set maximum cutting feedrate.
6 No time is added to the cumulative cutting time.
7 Even if this command is executed, the cutting in-progress signal is not output.
8 The operation between blocks for feed with a specified feedrate is performed in the cutting mode.
9 For this command, the feedrate arrival signal SAR (G029.4) cannot be used.

- Feedrate specification range
  The table below indicates the feedrate specification range.

	Linear axis		Rotation axis (deg/min)
	Metric input (mm/min)	Inch input (inch/min)	
IS-A	10. to 240000.	0.1 to 6553.5	10. to 240000.
IS-B	1. to 65535.	0.01 to 655.35	1. to 65535.
IS-C	0.1 to 6553.5	0.001 to 65.535	0.1 to 6553.5
IS-D	0.01 to 655.35	0.0001 to 6.5535	0.01 to 655.35
IS-E	0.001 to 65.535	0.00001 to 0.65535	0.001 to 65.535

NOTE
1 Be sure to set the following parameters to 0:
  Bit 3 (F10) of parameter No. 8002
  Bit 4 (EFD) of parameter No. 8006
  Bit 4 (PF1) of parameter No. 8002
  Bit 5 (PF2) of parameter No. 8002
  When a value other than 0 is set, the feedrate specification range changes, depending on the set parameter.
2 A value less than the lowest feedrates indicated above is discarded.

- Feedrate override
  With bit 2 (OVE) of parameter No. 8001, whether to use the feedrate override signal (*FV) for an NC statement or the feedrate override signal (*EFOV) dedicated to PMC axis control can be chosen.

NOTE
1 The second feedrate override function cannot be used.
2 The feedrate override function cannot be disabled using #3004.

- Override cancel
  With bit 2 (OVE) of parameter No. 8001, whether to use the feedrate override cancel signal (OVC) for an NC statement or the feedrate override cancel signal (EOVC) dedicated to PMC axis control can be chosen.

- Operation between blocks
  If feed with a specified feedrate is performed in succession in RTM statements, no stop occurs between blocks, but the next block is executed. With the accumulated pulse zero check signal (ELCKZg) for PMC axis control, an accumulated pulse zero check can be made between blocks specifying feed with a specified feedrate.

NOTE
1 The error detect signal (SMZ) cannot be used.
2 With the accumulated pulse zero check signal (ELCKZg) for PMC axis control, an accumulated pulse zero check can be made between blocks specifying feed with a specified feedrate for feed on the same axis only.

- Acceleration/deceleration time constant
  For an acceleration/deceleration time constant to be used for feed with a specified feedrate in an RTM statement when exponential acceleration/deceleration is used, whether to use the time constant for an NC statement or the time constant dedicated to PMC axis control can be chosen using parameter No. 8030.

NOTE
Look-ahead acceleration/deceleration before interpolation cannot be used.

- Feed with a specified feedrate (feed per revolution)

A movement is made at a feedrate specified in F on an axis from the current position to the point separated by a specified value.

Format

```
// ZDO ;
G95 G91 G01 IP _ F_ ;
ZEND ;
```

G95 : G code for feed per revolution
G91 : G code for incremental command
IP _ : Travel distance
F_ : Feedrate command (mm/rev or inch/rev)

NOTE

1 Only one axis can be specified in one block.
2 The absolute command (G90) cannot be specified.
3 The block overlap function cannot be used.
4 Be sure to set the parameters below to 0. If a value other than 0 is set, the feedrate specification range changes, depending on the set parameter.
FR1 (bit 6 of parameter No. 8002)
FR2 (bit 7 of parameter No. 8002)
5 The feedrate is clamped to the maximum cutting feedrate set in parameter No. 8022. (Parameter No. 1430 for an NC statement is not valid for this command.)
6 To use feed per revolution, a position coder is required at all times. (Bit 0 (NPC) of parameter No. 1402 is not applicable to this command.)
7 No time is added to the cumulative cutting time.
8 Even if this command is executed, the cutting in-progress signal is not output.
9 The operation between blocks for feed with a specified feedrate is performed in the cutting mode at all times.
10 For this command, the feedrate arrival signal SAR (G029.4) cannot be used.

- Feedrate specification range
  The table below indicates the feedrate specification range.

	Linear axis		Rotation axis (deg/rev)
	Metric input (mm/rev)	Inch input (inch/rev)	
T series	0.001 to 65.535	0.000001 to 0.65535	0.001 to 65.535
M series	0.01 to 500.00	0.0001 to 6.5535	0.01 to 500.00

- Feedrate override
  With bit 2 (OVE) of parameter No. 8001, whether to use the feedrate override signal (*FV) for an NC statement or the feedrate override signal (*EFV) dedicated to PMC axis control can be chosen.

NOTE
1 The second feedrate override function cannot be used.
2 The feedrate override function cannot be disabled using #3004.

- Override cancel
  With bit 2 (OVE) of parameter No. 8001, whether to use the feedrate override cancel signal (OVC) for an NC statement or the feedrate override cancel signal (EOVC) dedicated to PMC axis control can be chosen.

- Operation between blocks
  If feed with a specified feedrate is performed in succession in RTM statements, no stop occurs between blocks, but the next block is executed. With the accumulated pulse zero check signal (ELCKZg) for PMC axis control, an accumulated pulse zero check can be made between blocks specifying feed with a specified feedrate.

NOTE
1 The error detect signal (SMZ) cannot be used.
2 With the accumulated pulse zero check signal (ELCKZg) for PMC axis control, an accumulated pulse zero check can be made between blocks specifying feed with a specified feedrate for feed on the same axis only.

- Acceleration/deceleration time constant
  For an acceleration/deceleration time constant to be used for feed with a specified feedrate in an RTM statement when exponential acceleration/deceleration is used, whether to use the time constant for an NC statement or the time constant dedicated to PMC axis control can be chosen using parameter No. 8030.

NOTE
Look-ahead acceleration/deceleration before interpolation cannot be used.

- Reference position return

A movement is made at the rapid traverse rate to the first reference position on a specified axis. Upon completion of reference position return, the return completion lamp is turned on.

Format

```
// ZDO ;
G91 G28 IP 0 ;
ZEND ;
```

G91 : G code for incremental command

IP 0 : Specify an axis on which reference position return is to be performed.
Be sure to specify 0 as the travel distance.

NOTE

1 Only one axis can be specified in one block.
2 No intermediate point may be specified. (Be sure to specify 0 as the travel distance. Otherwise, alarm PS0410 is issued.) If a movement needs to be once made to an intermediate point, specify positioning with rapid traverse (G00)/machine coordinate system selection (G53) then specify this command.
3 The absolute command (G90) cannot be specified.
4 When using this function, be sure to set bit 0 (RPD) of parameter No. 8002 to 0.
5 Before this command can be specified, a reference position must be established. (With this command, do not specify an axis on which a reference position is not established.)
6 Before specifying this command with the T series, cancel tool offset and tool-nose radius compensation. Before specifying this command with the M series, cancel cutter compensation, tool length compensation, and tool offset.

- Rapid traverse override
  With bit 2 (OVE) of parameter No. 8001, whether to use the rapid traverse override signal for an NC statement or the rapid traverse override signal dedicated to PMC axis control can be chosen.

NOTE

Even if bit 4 (RF0) of parameter No. 1401 is set to 1, rapid traverse does not stop with a cutting feed override of 0%.

- Machine coordinate system selection

When a position in the machine coordinate system is specified, a movement is made to the position on the axis by rapid traverse. The G53 code for machine coordinate system selection is a one-shot G code, so that a command in the machine coordinate system is valid in the block that specifies G53. When moving the tool to a machine-specific position such as a tool change position, perform programming in the machine coordinate system based on G53.

Format

```
// ZDO ;
G90 G53 IP _ ;
ZEND ;
```

G90 : G code for absolute command

IP _ : Position in machine coordinate system

NOTE

1 Only one axis can be specified in one block.
2 The incremental command (G91) cannot be specified.
3 When using this function, be sure to set bit 0 (RPD) of parameter No. 8002 to 0.
4 Before this command can be specified, a reference position must be established. (With this command, do not specify an axis on which a reference position is not established.)
5 Before specifying this command with the T series, cancel tool offset and tool-nose radius compensation. Before specifying this command with the M series, cancel cutter compensation, tool length compensation, and tool offset.

- Rapid traverse override
  With bit 2 (OVE) of parameter No. 8001, whether to use the rapid traverse override signal for an NC statement or the rapid traverse override signal dedicated to PMC axis control can be chosen.

NOTE

Even if bit 4 (RF0) of parameter No. 1401 is set to 1, rapid traverse does not stop with a cutting feed override of 0%.

17.8 NOTES

- Address without the decimal point

In general, an NC address without the decimal point is subject to calculator-type decimal point input when bit 0 (DPI) of parameter No. 3401 or bit 0 (AXDx) of parameter No. 3455 is set to 1. In other cases, an NC address without the decimal point is regarded as a least input increment.

- Indirect axis address specification

When indirect axis address specification is used, use a direct reference argument. No indirect reference may be used.
Example)

AX[1]	Correct
AX[#RV[0]]	Incorrect

- Diameter specification/radius specification

For an axis to be controlled with an RTM statement, select radius specification. (Set bit 3 (DIA) of parameter No. 1006 to 0.)

- Alarm PS0224

If bit 0 (ZRN) of parameter No. 1005 is set to 0, and an axis control command is specified in an NC statement before a reference position return operation is not performed even once after power-up, alarm PS0224, "ZERO RETURN NOT FINISHED." is issued. In axis control using an RTM statement, the alarm is not issued even when bit 0 (ZRN) of parameter No. 1005 is set to 0, unless bit 6 (EZR) of parameter No. 8006 is set to 1.

- Axis removal

Even when axes are removed, interlock is not applied to the axis being controlled by an RTM statement.

- Stroke limit check before movement

No stroke limit check before movement is made with a block in an RTM statement.

- Others

For the axis being controlled by an RTM statement, the functions disabled for PMC axis control cannot be used.

- Other disabled functions

For the axis being controlled by an RTM statement, the functions listed below as examples are disabled.
Do not specify these functions for the axis being controlled by an RTM statement.
- Feed-forward
- Advanced preview feed-forward
- Fine acceleration/deceleration
- Automatic corner deceleration

- One-digit F code feed
- Scaling
- Coordinate system rotation
- Polar coordinate interpolation
- Balance cutting
- Feed stop
- Constant surface speed control
- Positioning function based on optimal acceleration, etc.

⚠ CAUTION

In an RTM statement, do not specify an axis used with any of the following functions:

- Axis synchronous control
  Be sure to set the axis synchronous control selection signal (SYNCn) and axis synchronous control manual feed selection signal (SYNCJn) of the axis to 0.
- Synchronous/composite control, superimposed control
- Index table indexing
- Three-dimensional coordinate system conversion
- Rigid tapping
- Compound machining
- Polygon turning
- Cs contour control
- AI contour control I/II

17.9 LIMITATION

Major general notes on RTM commands are provided below.

- Background drawing

The RTM command has no effect in background drawing. Do not specify an RTM command during background drawing.

- Interrupt-type custom macro

In an interrupt-type custom macro, no RTM command can be coded.

- Macro executor

In a macro executor, no RTM command can be coded.
Moreover, no macro executor can be coded from an RTM command.
In a series of programs, do not code a macro executor (execution macro) and RTM command at the same time.
In addition, observe the following:

- Do not code an RTM command in a program that uses an execution macro.
- Do not call an execution macro from a program with an RTM command coded.
- Do not code an RTM command in an execution macro.
- Do not use an execution macro as an RTM command trigger.

- Sequence number

No sequence number can be coded in an RTM command. No RTM command can be searched for a sequence number.

- Optional block skip

Optional block skip is unusable.
A slash (/) that appears in the middle of <expression> (enclosed in [ ] on the right-hand side of an arithmetic/logical expression) is regarded as a division operator; it is not regarded as an optional block skip code.

- Single block

When an RTM command is used, a single block stop generally occurs according to an NC statement, regardless of the values of bit 5 (SBM) of parameter No. 6000 and bit 7 (SBV) of parameter No. 6000. However, no single block stop occurs according to G65 for real time macro call.

- Processing

If there are many RTM statements between NC statements, the command flow may break.

- DNC operation

DNC operation is disabled.

- Operation in each event

If an event such as an emergency stop or alarm occurs during execution of an RTM command, the NC command and RTM command generally operate as indicated below.

Event	NC command	RTM command consisting of a macro command	RTM command including an axis control command
PS alarm issued in NC command execution	Operation stops when the block involving the PS alarm cause starts.	The RTM command stops when the NC command stops.	If the RTM statement being executed when the NC command stops is an axis control command, operation on the axis stops when the block stops.
Alarm issued in RTM command execution	Operation stops with the alarm.	The RTM statement stops at that point. Other RTM commands stop when the NC command stops.	The RTM statement stops at that point. Other RTM commands stop operation on the axis when axis control by the RTM statement being executed when the NC statement stops ends.
Single block (SBK signal)	Operation stops when the command being executed ends.	The RTM command is suspended when the NC command stops. When the NC command restarts, the RTM command restarts.	If the RTM statement being executed when the NC command stops is an axis control statement, operation on the axis stops when the block ends.
Feed hold (*SP signal)	A gradual stop occurs.	The RTM command stops when the NC command stops. When the NC command restarts, the RTM command restarts.	If the RTM statement being executed when the NC command stops is an axis control statement, operation on the axis stops when the block ends.
NC command stopped by a reset (ERS signal)	A gradual stop occurs, and the command also ends.	The RTM command stops when the NC command stops.	If the RTM statement being executed when the NC command stops is an axis control command, operation on the axis stops when the block ends.
Emergency stop	An immediate stop occurs, and the command also ends.	The command immediately ends.	Operation on the axis immediately stops, and the command also immediately ends.
End of the NC command	-	If the RTM command being executed is a one-shot RTM command, the command ends. If the RTM command being executed is a modal RTM command, the command continues execution.	If the one-shot RTM command being executed when the NC command ends is an axis control command, operation on the axis stops when the block ends.
Machine lock (MLK signal)	The NC command is valid.	The RTM command operates.	The RTM command can be enabled. For details, see the description of the axis control command.
Dry run (DRN signal)	The NC command is valid.	The RTM command operates.	The RTM command can be enabled. For details, see the description of the axis control command.
Interlock (*IT signal)	The NC command is valid.	The RTM command operates.	The RTM command can be enabled. For details, see the description of the axis control command.

18 PROGRAMMABLE PARAMETER INPUT (G10)

Overview

The values of parameters and pitch error compensation data can be entered in a program. This function is used for setting pitch error compensation data when attachments are changed or the maximum cutting feedrate or cutting time constants are changed to meet changing machining conditions.

Format

- Prameter entry mode

G10L52 ; Parameter entry mode setting
N_R_ ; For parameters other than the axis type or spindle type
N_P_R_ ; For axis type or spindle type parameters
:
G11 ; Parameter entry mode cancel
N_ : Parameter number
R_ : Parameter setting value (Leading zeros can be omitted.)
P_ : Axis number 1 to maximum controlled axis number (to be specified when an axis type parameter or spindle type parameter is specified)

NOTE
G10L52 cannot be used to enter pitch error compensation data.

- Pitch error compensation data entry mode

G10L50 ; Pitch error compensation data entry mode setting
N_R_ ; Pitch error compensation data entry
N_P_R_ ; For axis type parameters
:
G11 ; Pitch error compensation data entry mode cancel
N_ : Compensation position number for pitch errors compensation +10,000
R_ : Pitch error compensation data

NOTE
G10L50 cannot be used to enter parameter.

Explanation

- Setting value (R_)

Do not use a decimal point in the setting (R_) of a parameter or pitch error compensation data. As the value of R, a custom macro variable can be used.
When a parameter of real type is used, set an integer value in (R_) according to the increment system of the parameter.

- Axis number (P_)

As the axis number (P_), specify the order of a controlled axis to be displayed on the CNC display screen, by using an axis type parameter. For example, specify P2 for the control axis which is displayed second.
For a spindle type also, specify the order of an axis to be display on the CNC display screen.

⚠ WARNING

1 Do not fail to perform reference position return manually after changing the pitch error compensation data or backlash compensation data. Without this, the machine position can deviate from the correct position.
2 The canned cycle mode must be cancelled before entering of parameters. When not cancelled, the drilling motion may be activated.

⚠ CAUTION

Compatibility with the Series 16*i*/18*i*/21*i*:
This model has parameters that are not compatible with the Series 16*i*/18*i*/21*i*. So, before using this function, make a check according to the Parameter Manual (B-63950EN) of this model.

NOTE

Other NC statements cannot be specified while in parameter input mode.

Example

1. Set bit 2 (SBP) of bit type parameter No. 3404

G10L52 ;	Parameter entry mode
N3404 R 00000100 ;	SBP setting
G11 ;	Cancel parameter entry mode

2. Change the values for the Z-axis (3rd axis) and A-axis (4th axis) in axis type parameter No. 1322 (the coordinates of stored stroke limit 2 in the positive direction for each axis).
(When the increment systems for the 3rd and 4th axes are IS-B and millimeter machine, respectively)

G10L52 ;	Parameter entry mode
N1322P3R4500 ;	Change the value for the Z-axis to 4.500
N1322PP4R12000 ;	Change the value for the A-axis to 12.000
G11 ;	Cancel parameter entry mode

3. Change compensation point numbers 10 and 20 of pitch error compensation.

G10L50 ;	Pitch error compensation data entry mode
N10010R1 ;	Change the compensation point number from 10 to 1
N10020R5 ;	Change the compensation point number from 20 to 5
G11 ;	Pitch error compensation data entry mode

19 PATTERN DATA INPUT

Chapter 19, "PATTERN DATA INPUT", consists of the following sections:

19.1 OVERVIEW 630
19.2 EXPLANATION 631
19.3 EXPLANATION OF OPERATION 634
19.4 DEFINITION OF THE SCREEN 638

19.1 OVERVIEW

In the program of the fixed form processing with the custom macro, the operator select the processing pattern on the menu screen and specified the size, number and so on to the variable on the custom macro screen. As above mentioned, this function enables users to perform programming simply without programming using an existing NC language.

With the aid of this function, a machine tool builder can prepare the program of a hole machining cycle (such as a boring cycle or tapping cycle) using the custom macro function, and can store it into the program memory.

This cycle is assigned pattern names, such as BOR1, TAP3, and DRL2.

An operator can select a pattern from the menu of pattern names displayed on the screen.

Data (pattern data) which is to be specified by the operator should be created in advance with variables in a drilling cycle.

The operator can identify these variables using names such as DEPTH, RETURN RELIEF, FEED, MATERIAL or other pattern data names. The operator assigns values (pattern data) to these names.

The operator selects the pattern on the menu screen, and the selected pattern number is assigned to the system variable. The custom macro of the selected pattern can be started by starting a program then referring to the system variable in the program.

19.2 EXPLANATION

This function is consist of Pattern menu screen and Custom macro screen.
The process pattern is selected on the pattern menu screen.
Then the process pattern is selected, the custom macro screen is displayed.
On this custom macro screen, the variable with the name and coment is displayed according to the selected process pattern.
The process data can be input by referring to the variable name with the numerical value on the drawing.

The following is the example for the pattern menu and the custom macro.

(1) Pattern menu screen

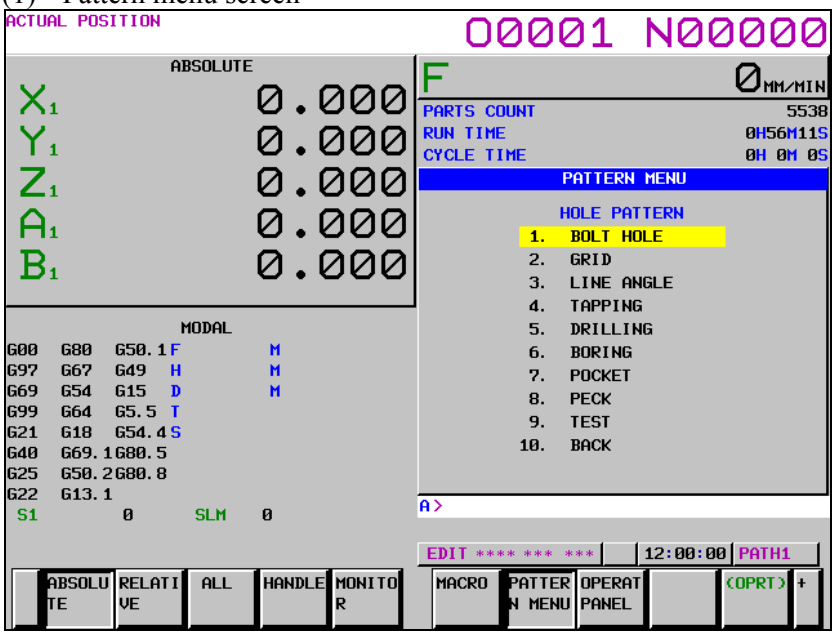


Fig. 19.2 (a) Pattern data menu screen (10.4-inch)

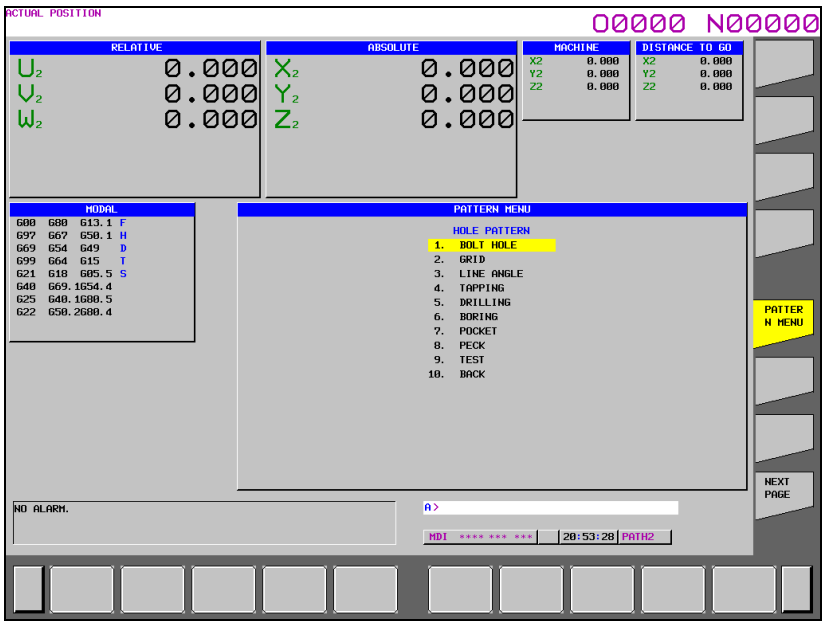


Fig. 19.2 (b) Pattern data menu screen (15-inch)

(2) Custom macro screen

The name of varialbe and comment can be displayed on the usual custom macro screen.

The menu title and pattern name on the pattern menu screen and the variable name on the custom macro screen can be defined

The position of the comment displayed on the custom macro screen can be selected by setting bit 0 (POC) of parameter No. 11318.

On 15-inch display unit, the comment is displayed at only bottom of the custom macro screen.

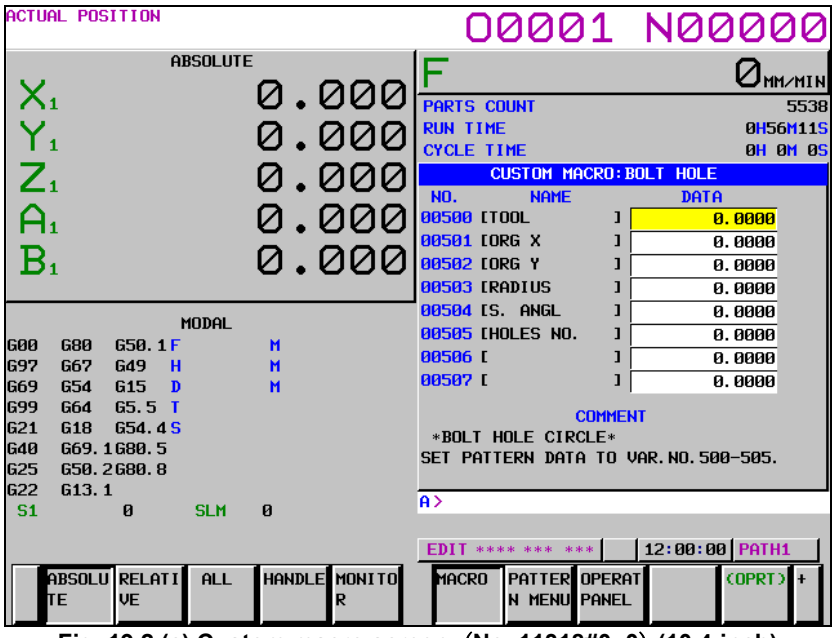


Fig. 19.2 (c) Custom macro screen（No. 11318#0=0）(10.4-inch)

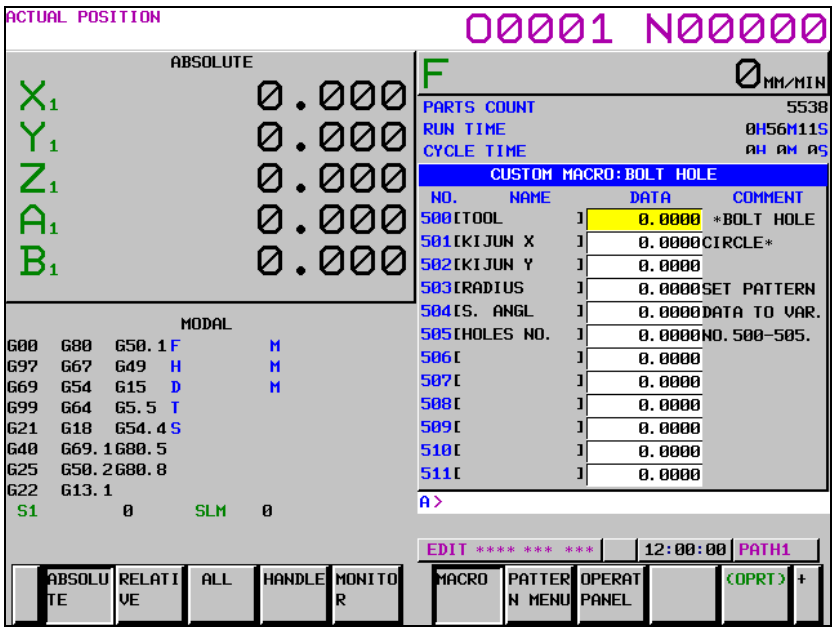


Fig. 19.2 (d) Custom macro screen (parameter No. 11318#0=1) (10.4-inch)

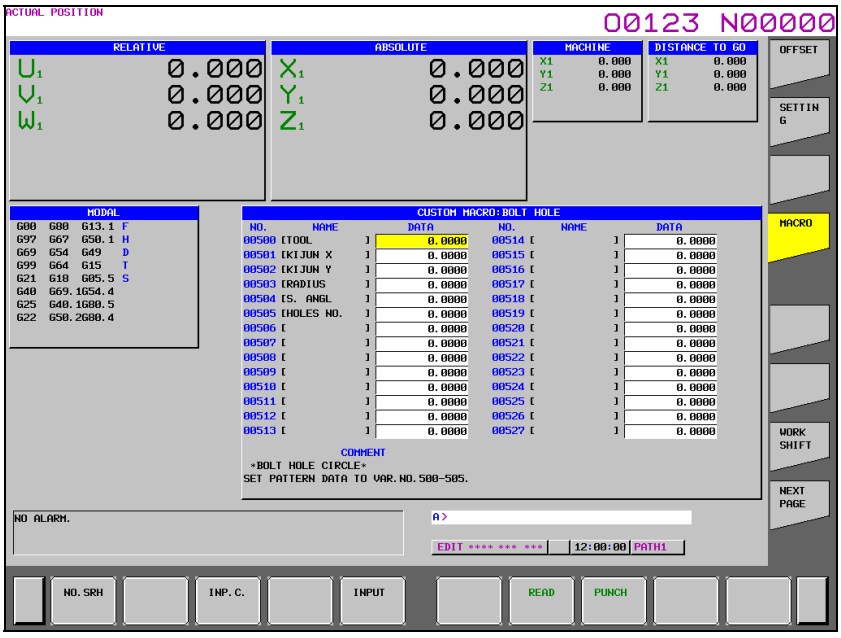


Fig. 19.2 (e) Custom macro screen (15-inch)

19.3 EXPLANATION OF OPERATION

The following explains how to display the pattern menu screen.

For 7.2-, 8.4-, and 10.4-inch LCDs

1 Press function key .
2 Press continous menu key .
3 Press soft key [PATTERN MENU].

For a 15-inch LCD

1 Press function key .
2 Press vertical soft key [NEXT PAGE] several times.
3 Press vertical soft key [PATTERN MENU].

Pattern menu screen

The following pattern menu is displayed.

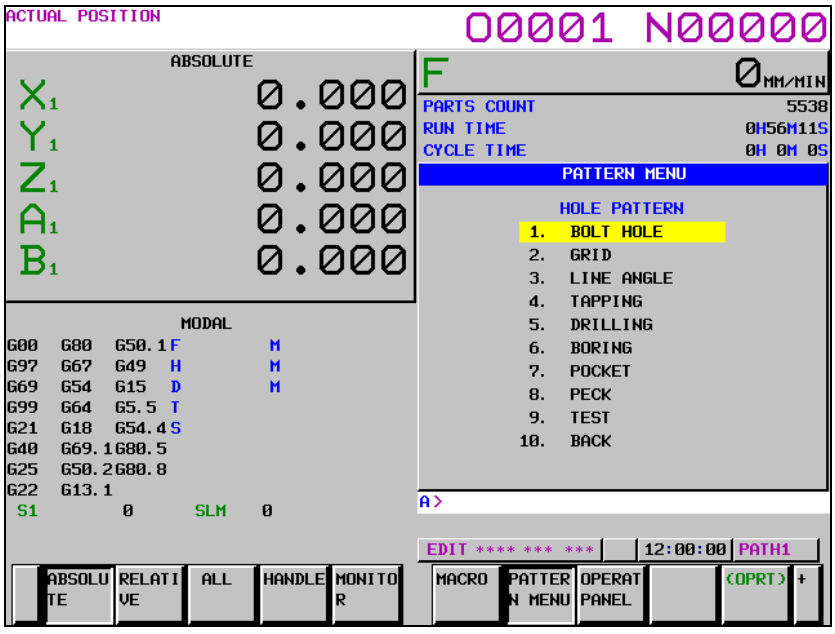


Fig. 19.3 (f) Pattern menu screen (10.4-inch)

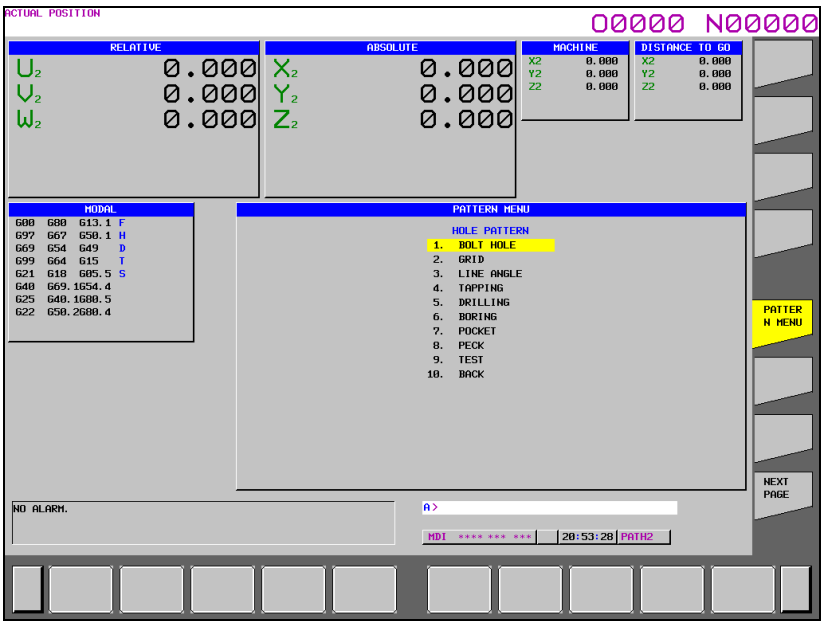


Fig. 19.3 (g) Pattern menu screen (15-inch)

Select the pattern on this screen
The following two methods are effective.

- Selection by cursor
  Move the cursor to the pattern name with the cursor move keys , and press the softkey [SELECT] or .
- Selection by setting of pattern number
  The number that is displayed in the left side of the pattern name is input and press the softkey [select] or .

The selected pattern number is registered to system variable #5900. The custom macro of the selected pattern can be started by starting a fixed program (external program No. search) with an external signal. This program refers to the system variable #5900 in the program. This system variable #5900 is kept after power-off.

Custom macro variable screen

The following custom macro screen is displayed.

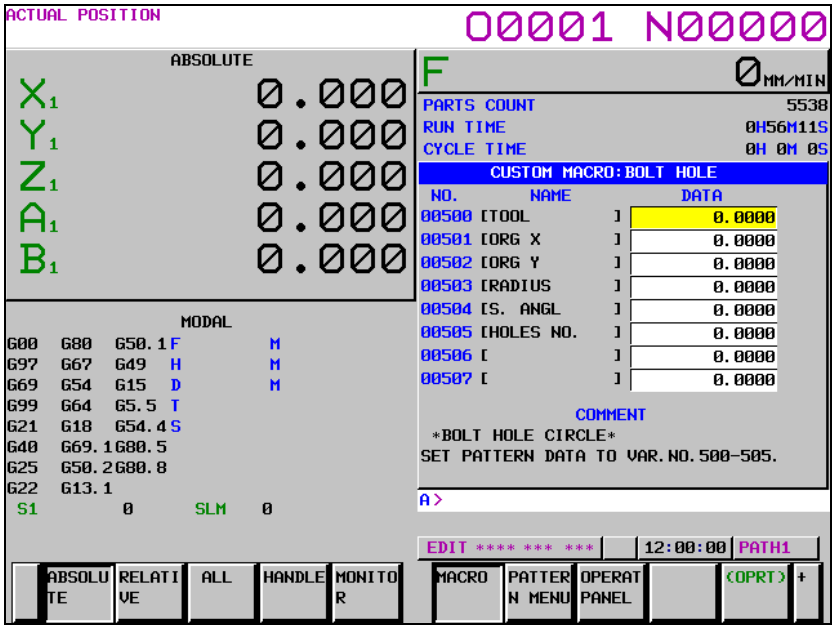


Fig. 19.3 (h) Custom macro screen when the pattern data is input (10.4-inch)

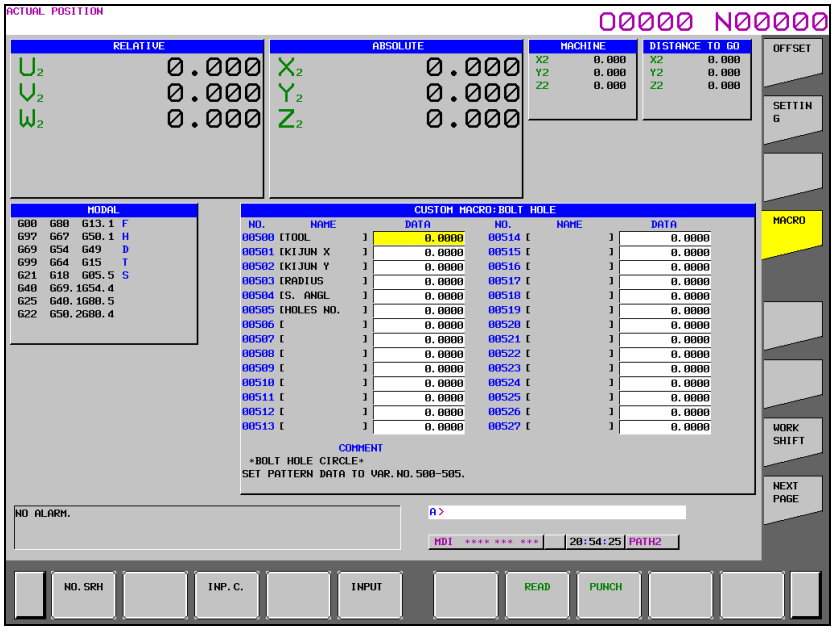


Fig. 19.3 (i) Custom macro screen when the pattern data is input (15-inch)

When the screen is changed to the custom macro screen, the macro variable number that is selected first is specifiied with the parameters Nos.6101 to 6110. The macro variables that variable name is not defined can be input, too.

NOTE
1 The variable name that is displayed cannot be used as the common variable name of the NC program.
2 When the common varialbe name is defined by SETVN command, the varialbe name defined by pattern data input function is given priority.
3 When bit 0 (POC) of parameter No.11318 is set to “1”, The variable number is three digit display. And the value of 12 digits or more is input, 11 digits from head of value are displayed.
Example)
Input: -123456789.123 → Display: -123456789.1

19.4 DEFINITION OF THE SCREEN

The definition of the screen is performed by NC program.

Program configuration

This function is consist of one program for the definition of pattern menu screen and maximum ten programs for the definition of custom macro screen.
Register the macro program for the pattern data input function to the following folder.
//CNC_MEM/USER/LIBRARY/PATH1 to PATH[n]

* [n] is the maximum path number of the system
* The PATH1 to PATH[n] should be made by the machine tool builder.

The program number is as follows

Table 19.4 (a) Numbers of subprograms employed in the pattern data input function

Sub program No.	Screen
O9500	Specifies character strings displayed on the pattern data menu.
O9501	Specifies a character string of the pattern data corresponding to pattern No.1
O9502	Specifies a character string of the pattern data corresponding to pattern No.2
O9503	Specifies a character string of the pattern data corresponding to pattern No.3
O9504	Specifies a character string of the pattern data corresponding to pattern No.4
O9505	Specifies a character string of the pattern data corresponding to pattern No.5
O9506	Specifies a character string of the pattern data corresponding to pattern No.6
O9507	Specifies a character string of the pattern data corresponding to pattern No.7
O9508	Specifies a character string of the pattern data corresponding to pattern No.8
O9509	Specifies a character string of the pattern data corresponding to pattern No.9
O9510	Specifies a character string of the pattern data corresponding to pattern No.10

Table 19.4 (b) Macro commands used in the pattern data input function

G code	H code	Function
G65	H90	Specifies the menu title.
G65	H91	Specifies the pattern name.
G65	H92	Specifies the pattern data title.
G65	H93	Specifies the variable name.
G65	H94	Specifies the comment.

Table 19.4 (c) System variables employed in the pattern data input function

System variable	Function
#5900	Pattern No. selected by user.

19.4.1 Definition of the Pattern Menu Screen

Menu title and pattern name are defined as follows.

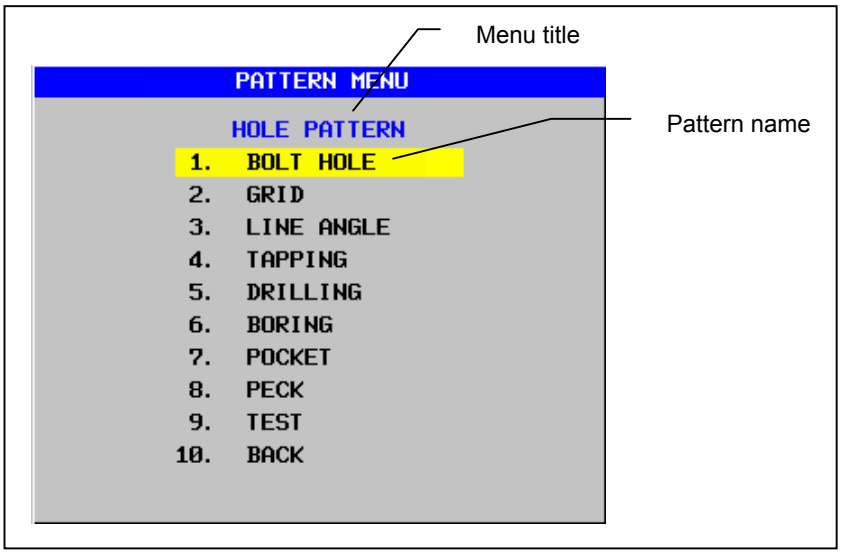


Fig. 19.4.1 (a) Pattern menu screen

Definition of menu title

The character string displayed in the menu title of the pattern menu screen is defined.
The menu title is specified up to 12 characters in a half size letter and up to 6 characters in a full size letter such as kanji character.

- Format

G65 H90 P_ Q_ R_ I_ J_ K_ ;

H90 : Specifies the menu title
P_ : The code of 1st and 2nd characters of title
Q_ : The code of 3rd and 4th characters of title
R_ : The code of 5th and 6th characters of title
I_ : The code of 7th and 8th characters of title
J_ : The code of 9th and 10th characters of title
K_ : The code of 11th and 12th characters of title

As for the way of setting the character-code, refer to the Subsection 19.4.3, "Setting the Character-codes".

Definition of pattern name

The character string displayed in the pattern name which becomes a menu item is defined.
The pattern name is specified up to 10 characters in a half size letter and up to 5 characters in a full size letter.

- Format

G65 H91 P_ Q_ R_ I_ J_ K_ ;

H91 : Specifies the pattern name
P_ : Specifies the menu number of the pattern name
The menu number = 1 to 10
Q_ : The code of 1st and 2nd characters of pattern name
R_ : The code of 3rd and 4th characters of pattern name
I_ : The code of 5th and 6th characters of pattern name
J_ : The code of 7th and 8th characters of pattern name
K_ : The code of 9thd and 10th characters of pattern name
As for the way of setting the character-code, refer to the Subsection 19.4.3, "Setting the Character-codes".

Example

The following is example for pattern menu screen.

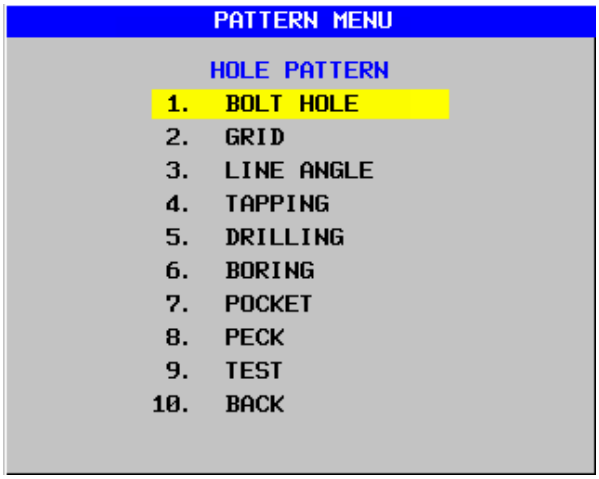


Fig. 19.4.1 (b) Pattern menu screen

```
O9500
N1 G65 H90 P072079 Q076069 R032080 I065084 J084069 K082078; .. "HOLE PATTERN"
N2 G65 H91 P1 Q066079 R076084 I032072 J079076 K069032; ..... "BOLT HOLE"
N3 G65 H91 P2 Q071082 R073068; ..... "GRID"
N4 G65 H91 P3 Q076073 R078069 I032065 J078071 K076069; ..... "LINE ANGLE"
N5 G65 H91 P4 Q084065 R080080 I073078 J071032; ..... "TAPPING"
N6 G65 H91 P5 Q068082 R073076 I076073 J078071;..... "DORILLING"
N7 G65 H91 P6 Q066079 R082073 I078071; ..... "BORING"
N8 G65 H91 P7 Q080079 R067075 I069084; ..... "POCKET"
N9 G65 H91 P8 Q080069 R067075; ..... "PECK"
N10 G65 H91 P9 Q084069 R083084; ..... "TEST"
N11 G65 H91 P10 Q066065 R067075; ..... "BACK"
N12 M99;
```

19.4.2 Definition of the Custom Macro Screen

The title, variable name and comment are defined as follows.

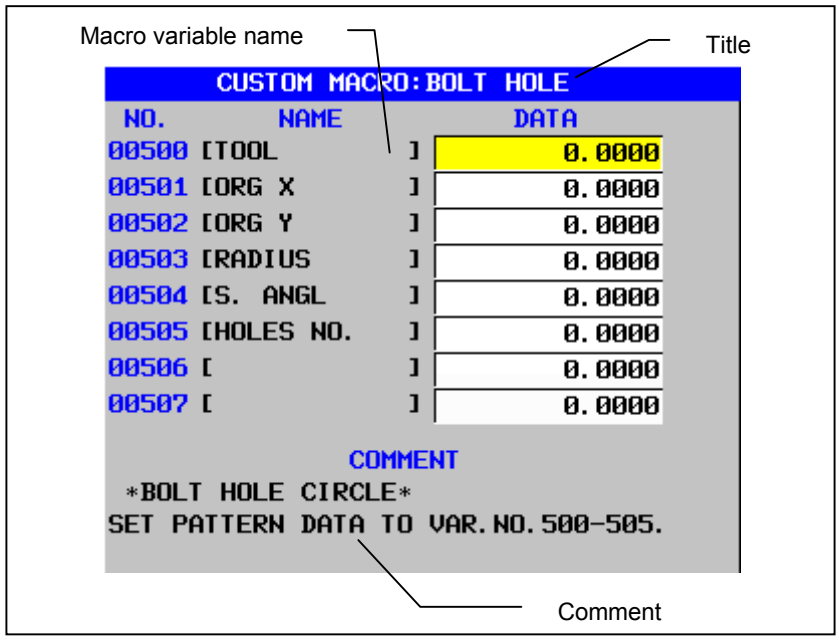


Fig. 19.4.2 (a) Custom macro screen

Definition of title

The character string displayed in the title of the custom macro screen is defined.
The title is specified up to 12 characters in a half size letter and up to 6 characters in a full size letter.

- Format

G65 H92 P_ Q_ R_ I_ J_ K_ ;

H92: Specifies the menu title
P_ : The code of 1st and 2nd characters of the menu title
Q_ : The code of 3rd and 4th characters of the menu title
R_ : The code of 5th and 6th characters of the menu title
I_ : The code of 7th and 8th characters of the menu title
J_ : The code of 9th and 10th characters of the menu title
K_ : The code of 11th and 12th characters of the menu title

As for the way of setting the character-code, refer to the Subsection 19.4.3, "Setting the Character-codes".

Definition of macro variable

The character string displayed in the macro variable name is defined. The macro variable is specified up to 10 characters in a half size letter and up to 5 characters in a full size letter.

The variable which can be used is as follows
#100 to 199 (100 variables)
#500 to 999 (500 variables), 600 variables in total

- Format

G65 H93 P_ Q_ R_ I_ J_ K_ ;

H93 : Specifies the variable name
P_ : Specifies the variable number
Specifies 100 to 199 or 500 to 999
Q_ : The code of 1st and 2nd characters of the variable name
R_ : The code of 3rd and 4th characters of the variable name
I_ : The code of 5th and 6th characters of the variable name
J_ : The code of 7th and 8th characters of the variable name
K_ : The code of 9th and 10th characters of the variable name
As for the way of setting the character-code, refer to the Subsection 19.4.3, "Setting the Character-codes".

Definition of a comment

The character string of the comment displayed on the custom macro screen is defined.
The comment is specified by up to 12 characters in a half size letter and up to 6 characters in a full size letter per one block.

- When the comment is displayed at the bottom（bit 0 (POC) of parameter No.11318=0）
  The number of blocks is defined up to maximum 9 blocks.
  One line is composition of 3 blocks and maximum 3 lines can be displayed.

  The comment is displayed in the order specified by program as the following figure.
  The block is arranged in defined order.

Block 1	Block 2	Block 3
Block 4	Block 5	Block 6
Block 7	Block 8	Block 9

Fig.19.4.2 (b) Block reyout of comment

- When the comment is displayed at the right（bit 0 (POC) of parameter No.11318=1）
  1 line is composed by 1 blocks, the maximum number of lines is 8 on the 7.2-inch and 8.4-inch display unit or 12 on the 10.4-inch display unit.

- Format

G65 H94 P_ Q_ R_ I_ J_ K_ ;

H94: Specifies the comment
P_ : The code of 1st and 2nd characters of comment
Q_ : The code of 3rd and 4th characters of comment
R_ : The code of 5th and 6th characters of comment
I_ : The code of 7th and 8th characters of comment
J_ : The code of 9th and 10th characters of comment
K_ : The code of 11th and 12th characters of comment
As for the way of setting the character-code, refer to the Subsection 19.4.3, "Setting the Character-codes".

Example

The following is example of the custom macro screen.

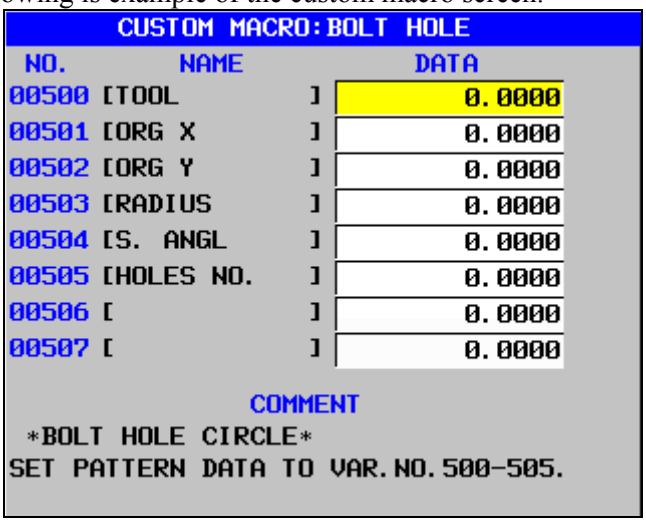


Fig. 19.4.2 (c) Custom macro screen（No. 11318#0=0）

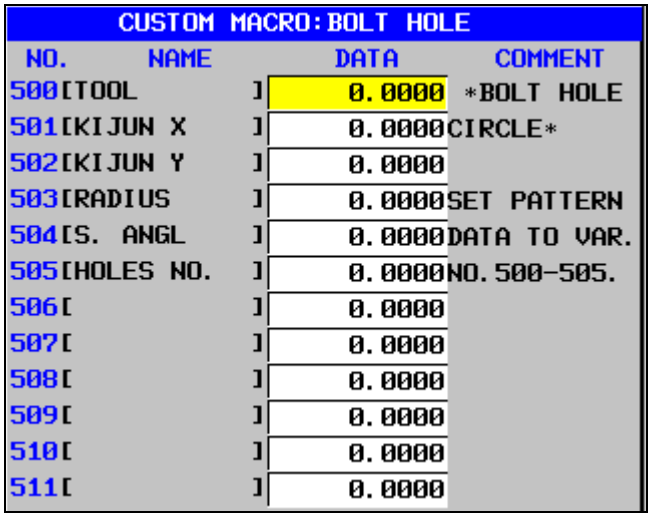


Fig. 19.4.2 (d) Custom macro screen（No. 11318#0=1）

```
O9501;
N1 G65 H92 P066079 Q076084 R032072 I079076 J069032; ....."BOLT HOLE"
N2 G65 H93 P500 Q084079 R079076; ....."TOOL"
N3 G65 H93 P501 Q079082 R071032 I08832;....."ORG X"
N4 G65 H93 P502 Q079082 R071032 I08932;....."ORG Y"
N5 G65 H93 P503 Q082065 R068073 I085803;....."RADIUS"
N6 G65 H93 P504 Q083046 R032065 I078071 J076032....."S. ANGL"
N7 G65 H93 P505 Q072079 R076079 I083032 J078079 K046032....."HOLES NO."
N8 G65 H94 P032042 Q066079 R076084 I032072 J079076 K069032;....." *BOLT HOLE"
N9 G65 H94 P067073 Q082067 R076069 I042032;....."CIRCLE*"
N10 G65 H94 P083069 Q084032 R080065 I084084 J069082 K078032;..."SET PATTERN"
N11 G65 H94 P068065 Q084065 R032084 I079032 J086065 K082046;..."DATA NO VAR."
N12 G65 H94 P078079 Q046053 R048048 I045053 J048053 K046032;..."NO500-505"
N13 M99;
```

19.4.3 Setting the Character-codes

The character cannot be used to specify the NC program.
Therefor, the code corresponding to the character is specified.
One character is consist of three figures in a half size letter and six figures in a full size letter.
The character code is specified for each address of the G65 instruction by six digits.

Refer to the table for the character code.
Example)

When "ABCDEFGH" is specified, the description of the code is as follows.
Encoded character string : 065 066 067 068 069 070 071 072

```
P065066 Q067068 R069070 I071072;
  AB      CD      EF      GH
```

NOTE

1 Space (032) is added ahead of the character-code, when the character-code of three digits or less is specified.
Example)
P065066 Q067; → " AB C "
032(space) is put at the end, when "ABC" is displayed.
P065066 Q067032; → " ABC "

2 It is assumed in that the space of two characters was defined in the address when there is an address not defined.
Example)
P065066 I067068; → "AB CD"

Characters and codes to be used for the patterndata input function

Character	Code	Comment	Character	Code	Comment
A	065		6	054	
B	066		7	055	
C	067		8	056	
D	068		9	057	
E	069			032	Space
F	070		!	033	Exclamation mark
G	071		”	034	Quotation mark
H	072		#	035	Hash sign
I	073		$	036	Dollar sign
J	074		%	037	Percent
K	075		&	038	Ampersand
L	076		’	039	Apostrophe
M	077		*	042	Asterisk
N	078		+	043	Plus sign
O	079		,	044	Comma
P	080		-	045	Minus sign
Q	081		.	046	Period
R	082		/	047	Slash
S	083		:	058	Colon
T	084		;	059	Semicolon
U	085		<	060	Left angle bracket
V	086		=	061	Equal sign
W	087		>	062	Right angle bracket
X	088		?	063	Question mark
Y	089		@	064	HAt”mark
Z	090		[	091	Left square bracket
0	048		¥	092	Yen sign
1	049		]	093	Right squar bracket
2	050		^	094	
3	051		_	095	Underscore
4	052				
5	053				

The characters and the codes of the katakana is as follows.

Character	Code	Comment	Character	Code	Comment
ア	177		ム	209	
イ	178		メ	210	
ウ	179		モ	211	
エ	180		ヤ	212	
オ	181		ユ	213	
カ	182		ヨ	214	
キ	183		ラ	215	
ク	184		リ	216	
ケ	185		ル	217	
コ	186		レ	218	
サ	187		ロ	219	
シ	188		ワ	220	
ス	189		ヲ	166	
セ	190		ン	221	
ソ	191		ァ	167	
タ	192		ィ	168	
チ	193		ゥ	169	
ツ	194		ェ	170	
テ	195		ォ	171	
ト	196		ャ	172	
ナ	197		ュ	173	
ニ	198		ョ	174	
ヌ	199		ッ	175	
ネ	200		″	222	Diacritical mark
ノ	201		°	223	Diacritical mark
ハ	202		。	161	Punctuation
ヒ	203		「	162	Left quotation mark
フ	204		」	163	Right quotation mark
ヘ	205		、	164	Comma
ホ	206		・	165	Point
マ	207			000	Space
ミ	208				

NOTE
Diacritical mark is one character.

The characters and the codes of the hiragana and the kanji are as follows. The following hiraganas and kanjis use two characters of the alphanumeric character.

ぁ	あ	ぃ	い	う	う	ぇ	え	ぉ	お
002 000	002 002	002 004	002 006	002 008	002 010	002 012	002 014	002 016	002 018
か	が	き	ぎ	く	ぐ	け	げ	こ	ご
002 020	002 022	002 024	002 026	002 028	002 030	002 032	002 034	002 036	002 038
さ	ざ	し	じ	す	ず	せ	ぜ	そ	ぞ
002 040	002 042	002 044	002 046	002 048	002 050	002 052	002 054	002 056	002 058
た	だ	ち	ぢ	っ	つ	づ	て	で	と
002 060	002 062	002 064	002 066	002 068	002 070	002 072	002 074	002 076	002 078
ど	な	に	ぬ	ね	の	は	ば	ぱ	ひ
002 080	002 082	002 084	002 086	002 088	002 090	002 092	002 094	002 096	002 098
び	ぴ	ふ	ぶ	ぷ	へ	べ	ぺ	ほ	ぼ
002 100	002 102	002 104	002 106	002 108	002 110	002 112	002 114	002 116	002 118
ぽ	ま	み	む	め	も	ゃ	や	ゅ	ゆ
002 120	002 122	002 124	002 126	002 128	002 130	002 132	002 134	002 136	002 138
ょ	よ	ら	り	る	れ	ろ	わ	わ	素
002 140	002 142	002 144	002 146	002 148	002 150	002 152	002 154	002 156	002 158
材	を	ん	種	類	棒	穴	成	形	質
002 160	002 162	002 164	002 166	002 168	002 170	002 172	002 174	002 176	002 178
寸	法	外	径	長	端	面	最	小	内
002 180	002 182	002 184	002 186	002 188	002 190	002 192	002 194	002 196	002 198
大	加	工	切	削	倣	正	途	中	荒
002 200	002 202	002 204	002 206	002 208	002 210	002 212	002 214	002 216	002 218
具	番	号	仕	上	込	点	方	向	速
002 220	002 222	002 224	002 226	002 228	002 230	002 232	002 234	002 236	002 238
度	送	量	開	始	深	主	軸		
002 240	002 242	002 244	002 246	002 248	002 250	002 252	002 254		
回	転	数	位	置	決	直	線	時	円
003 000	003 002	003 004	003 006	003 008	003 010	003 012	003 014	003 016	003 018
反	現	在	指	令	値	領	域	診	断
003 020	003 022	003 024	003 026	003 028	003 030	003 032	003 034	003 036	003 038
操	作	手	引	機	械	残	移	動	次
003 040	003 042	003 044	003 046	003 048	003 050	003 052	003 054	003 056	003 058
早	電	源	投	入	間	分	秒	自	運
003 060	003 062	003 064	003 066	003 068	003 070	003 072	003 074	003 076	003 078
負	荷	実	使	用	寿	命	新	規	除
003 080	003 082	003 084	003 086	003 088	003 090	003 092	003 094	003 096	003 098
隅	取	単	補	能	独	終	了	記	角
003 100	003 102	003 104	003 106	003 108	003 110	003 112	003 114	003 116	003 118
溝	刃	幅	広	設	定	ー	覧	表	部
003 120	003 122	003 124	003 126	003 128	003 130	003 132	003 134	003 136	003 138
炭	合	金	鋼	超	硬	先	付	摩	耗
003 140	003 142	003 144	003 146	003 148	003 150	003 152	003 154	003 156	003 158
仮	想	副	行	挿	消	去	山	高	準
003 160	003 162	003 164	003 166	003 168	003 170	003 172	003 174	003 176	003 178
備	完	後	弧	助	択	無	視	器	原
003 180	003 182	003 184	003 186	003 188	003 190	003 192	003 194	003 196	003 198

登	録	再	処	理	描	画	過	容	編
003 200	003 202	003 204	003 206	003 208	003 210	003 212	003 214	003 216	003 218
集	未	対	相	座	標	示	名	歯	変
003 220	003 222	003 224	003 226	003 228	003 230	003 232	003 234	003 236	003 238
呼	推	馬	力	系	選	達	閉		
003 240	003 242	003 244	003 246	003 248	003 250	003 252	003 254		
禁	復	帰	書	個	桁	稼	由	両	半
004 000	004 002	004 004	004 006	004 008	004 010	004 012	004 014	004 016	004 018
逃	底	逆	下	空	四	触	平	代	辺
004 020	004 022	004 024	004 026	004 028	004 030	004 032	004 034	004 036	004 038
格	子	周	心	本	群	停	止	巾	微
004 040	004 042	004 044	004 046	004 048	004 050	004 052	004 054	004 056	004 058
状	路	範	囲	倍	率	注	側	特	殊
004 060	004 062	004 064	004 066	004 068	004 070	004 072	004 074	004 076	004 078
距	離	連	続	増	隔	件	初	期	条
004 080	004 082	004 084	004 086	004 088	004 090	004 092	004 094	004 096	004 098
経	握	圧	扱	陰	隠	右	押	横	黄
004 100	004 102	004 104	004 106	004 108	004 110	004 112	004 114	004 116	004 118
億	屋	化	何	絵	階	概	該	巻	換
004 120	004 122	004 124	004 126	004 128	004 130	004 132	004 134	004 136	004 138
気	起	軌	技	疑	供	共	境	強	教
004 140	004 142	004 144	004 146	004 148	004 150	004 152	004 154	004 156	004 158
掘	繰	係	傾	型	検	権	研	肩	見
004 160	004 162	004 164	004 166	004 168	004 170	004 172	004 174	004 176	004 178
験	元	弦	減	孔	巧	控	更	校	構
004 180	004 182	004 184	004 186	004 188	004 190	004 192	004 194	004 196	004 198
根	左	差	雑	参	散	産	算	治	耳
004 200	004 202	004 204	004 206	004 208	004 210	004 212	004 214	004 216	004 218
式	失	修	十	従	勝	商	少	尚	昇
004 220	004 222	004 224	004 226	004 228	004 230	004 232	004 234	004 236	004 238
植	色	食	伸	信	侵	振	浸		
004 240	004 242	004 244	004 246	004 248	004 250	004 252	004 254		
真	暗	以	意	異	影	鋭	越	価	可
005 000	005 002	005 004	005 006	005 008	005 010	005 012	005 014	005 016	005 018
科	果	箇	課	各	拡	核	学	掛	漢
005 020	005 022	005 024	005 026	005 028	005 030	005 032	005 034	005 036	005 038
簡	観	関	含	却	客	休	急	業	曲
005 040	005 042	005 044	005 046	005 048	005 050	005 052	005 054	005 056	005 058
均	筋	継	計	軽	言	限	互	降	採
005 060	005 062	005 064	005 066	005 068	005 070	005 072	005 074	005 076	005 078
済	細	姿	思	写	射	斜	者	車	借
005 080	005 082	005 084	005 086	005 088	005 090	005 092	005 094	005 096	005 098
縦	重	出	述	術	渉	照	省	章	証
005 100	005 102	005 104	005 106	005 108	005 110	005 112	005 114	005 116	005 118
象	身	進	人	図	違	印	沿	遠	央
005 120	005 122	005 124	005 126	005 128	005 130	005 132	005 134	005 136	005 138
奥	往	応	会	解	改	割	活	願	基
005 140	005 142	005 144	005 146	005 148	005 150	005 152	005 154	005 156	005 158
奇	寄	岐	既	近	区	矩	駆	偶	旧
005 160	005 162	005 164	005 166	005 168	005 170	005 172	005 174	005 176	005 178

求	球	究	級	欠	結	口	語	誤	交
005 180	005 182	005 184	005 186	005 188	005 190	005 192	005 194	005 196	005 198
厚	項	刻	告	黒	財	策	糸	試	資
005 200	005 202	005 204	005 206	005 208	005 210	005 212	005 214	005 216	005 218
事	持	似	釈	弱	受	収	純	順	所
005 220	005 222	005 224	005 226	005 228	005 230	005 232	005 234	005 236	005 238
序	剰	場	常	飾	水	錐	据		
005 240	005 242	005 244	005 246	005 248	005 250	005 252	005 254		
制	整	製	前	全	然	則	属	即	他
006 000	006 002	006 004	006 006	006 008	006 010	006 012	006 014	006 016	006 018
多	存	谷	探	短	徴	鎮	調	頂	鉄
006 020	006 022	006 024	006 026	006 028	006 030	006 032	006 034	006 036	006 038
添	頭	同	導	道	熱	年	濃	箱	発
006 040	006 042	006 044	006 046	006 048	006 050	006 052	006 054	006 056	006 058
抜	伴	必	百	複	物	文	聞	併	忘
006 060	006 062	006 064	006 066	006 068	006 070	006 072	006 074	006 076	006 078
末	密	有	余	与	裏	立	略	青	席
006 080	006 082	006 084	006 086	006 088	006 090	006 092	006 094	006 096	006 098
石	積	赤	接	折	粗	創	双	捜	太
006 100	006 102	006 104	006 106	006 108	006 110	006 112	006 114	006 116	006 118
打	体	待	態	替	段	知	地	致	遅
006 120	006 122	006 124	006 126	006 128	006 130	006 132	006 134	006 136	006 138
追	通	伝	得	読	凸	凹	突	鈍	敗
006 140	006 142	006 144	006 146	006 148	006 150	006 152	006 154	006 156	006 158
杯	背	配	品	不	布	並	頁	別	片
006 160	006 162	006 164	006 166	006 168	006 170	006 172	006 174	006 176	006 178
返	勉	弁	保	明	滅	木	目	歪	揺
006 180	006 182	006 184	006 186	006 188	006 190	006 192	006 194	006 196	006 198
様	溶	要	抑	良	輪	和	話	枠	節
006 200	006 202	006 204	006 206	006 208	006 210	006 212	006 214	006 216	006 218
説	絶	千	専	浅	旋	総	走	退	台
006 220	006 222	006 224	006 226	006 228	006 230	006 232	006 234	006 236	006 238
第	題	卓	室	着	柱	鋳	丁		
006 240	006 242	006 244	006 246	006 248	006 250	006 252	006 254		
低	訂	肉	日	白	薄	比	皮	被	非
007 000	007 002	007 004	007 006	007 008	007 010	007 012	007 014	007 016	007 018
美	普	伏	歩	包	門	問	絡	列	万
007 020	007 022	007 024	007 026	007 028	007 030	007 032	007 034	007 036	007 038
利	訳	礼	乱	放	枚	約	練	油	劣
007 040	007 042	007 044	007 046	007 048	007 050	007 052	007 054	007 056	007 058
例	郭	戻	冷	垂	緑	紫	許	測	精
007 060	007 062	007 064	007 066	007 068	007 070	007 072	007 074	007 076	007 078
効	→	↗	↑	↖	←	↙	↓	↘	
007 080	007 082	007 084	007 086	007 088	007 090	007 092	007 094	007 096	007 098
				板	予	〃	家	装	管
007 100	007 102	007 104	007 106	007 108	007 110	007 112	007 114	007 116	007 118
粉	等					貫	安	α	β
007 120	007 122	007 124	007 126	007 128	007 130	007 132	007 134	007 136	007 138
程	抗	張	任	破	損	御	足	守	般
007 140	007 142	007 144	007 146	007 148	007 150	007 152	007 154	007 156	007 158

納	義	丸	汎	固	毎	当	的	詳	鳥
007 160	007 162	007 164	007 166	007 168	007 170	007 172	007 174	007 176	007 178
適	論	額	縁	温	給	界	混	監	締
007 180	007 182	007 184	007 186	007 188	007 190	007 192	007 194	007 196	007 198
護	己	称	樹	脂	料	落	確	認	報
007 200	007 202	007 204	007 206	007 208	007 210	007 212	007 214	007 216	007 218
排	性	生	績	判	搬	砥	θ	島	壁
007 220	007 222	007 224	007 226	007 228	007 230	007 232	007 234	007 236	007 238
				]	[	▗	■		
007 240	007 242	007 244	007 246	007 248	007 250	007 252	007 254		

20 HIGH-SPEED CUTTING FUNCTIONS

Chapter 20, "HIGH-SPEED CUTTING FUNCTIONS", consists of the following sections:

20.1 AI CONTOUR CONTROL FUNCTION I AND AI CONTOUR CONTROL FUNCTION II (G05.1) 653
20.2 MACHINING CONDITION SELECTING FUNCTION 670
20.3 JERK CONTROL 671
20.4 OPTIMUM TORQUE ACCELERATION/ DECELERATION 676
20.5 HIGH-SPEED CYCLE MACHINING 689

20.1 AI CONTOUR CONTROL FUNCTION I AND AI CONTOUR CONTROL FUNCTION II (G05.1)

Overview

The AI contour control I and AI contour control II functions are provided for high-speed, high-precision machining. This function enables suppression of acceleration/deceleration delays and servo delays that become larger with increases in the feedrate and reduction of machining profile errors.
There are two types of AI contour control; these two types are referred to temporarily as AI contour control I and AI contour control II. AI contour control I is designed mainly for part machining, and AI contour control II is for machining of successive minute straight lines such as mold machining and for machining by curve commands in NURBS and so on.
When a high-speed processing option is used with AI contour control II, high-speed operation processing is enabled, which reduces the interpolation cycle, therefore, faster and more precise machining is enabled. In addition, an option to expand read-ahead blocks to up to 1000 blocks is available.
In the descriptions below that are common to AI contour control I and AI contour control II, the term "AI contour control" is used.

Format

G05.1 Q_ ;
Q1 : AI contour control mode on
Q0 : AI contour control mode off

NOTE
1 Always specify G05.1 in an independent block.
2 The AI contour control mode is also canceled by a reset.
3 The AI contour control mode can be turned on at the start of automatic operation by setting bit 0 (SHP) of parameter No. 1604.

The AI contour control mode can be controlled also with the formats that have been used for the conventional advanced preview control, high-precision contour control, and AI high-precision contour control functions.

G08 P_ ;
P1 : AI contour control mode on
P0 : AI contour control mode off
G05 P_ ;
P10000 : AI contour control mode on
P0 : AI contour control mode off

NOTE
1 Always specify G08 and G05 in an independent block.
2 G05 can be specified only for AI contour control II.
3 The AI contour control mode is also canceled by a reset.
4 Valid functions are limited depending on the command format. For details, see the description of "Valid functions".

Valid functions

The functions listed below are valid in the AI contour control mode.

Valid functions are limited depending on the command format and whether AI contour control I or II is used.

Function	AI contour control I	AI contour control II	AI contour control II with high-speed processing
Number of blocks read ahead	30 (When G8 is specified: 1)	200 (When G8 is specified: 1)	600 (*1) (When G8 is specified: 1)
Look-ahead acceleration/deceleration before interpolation	Linear or bell-shaped acceleration/deceleration	Linear or bell-shaped acceleration/deceleration	Linear or bell-shaped acceleration/deceleration
Speed control with feedrate difference on each axis	Enabled	Enabled	Enabled
Speed control with acceleration in circular interpolation	Enabled	Enabled	Enabled
Speed control with acceleration on each axis	Enabled (When G8 is specified: Not enabled)	Enabled (When G8 is specified: Not enabled)	Enabled (When G8 is specified: Not enabled)
Smooth speed control	Not enabled	Enabled (When G8 is specified: Not enabled)	Enabled (When G8 is specified: Not enabled)
Speed control with cutting load	Not enabled	Enabled	Enabled
Disregard of feedrate command	Not enabled	Enabled	Enabled

*1 The number of blocks can optionally be expanded to 1000 blocks.

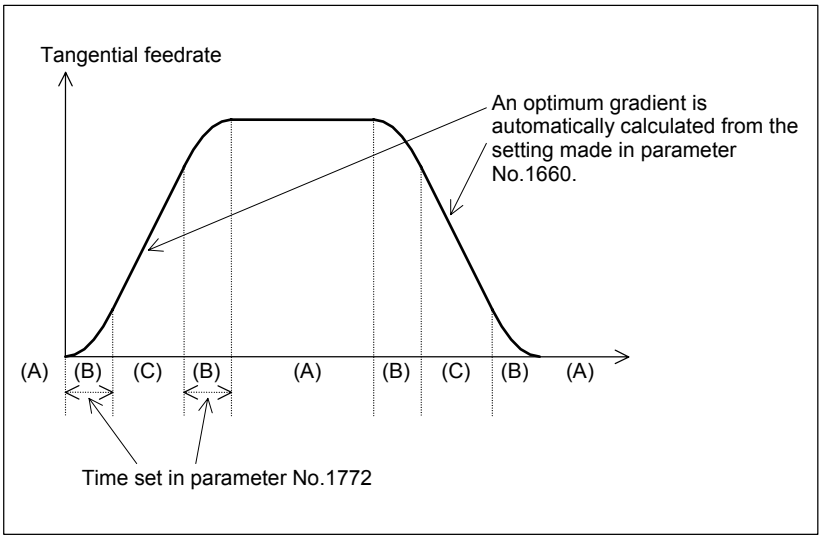
Explanation

- Look-ahead acceleration/deceleration before interpolation

There are two types of look-ahead acceleration/deceleration before interpolation, the linear acceleration/deceleration type and the bell-shaped acceleration/deceleration type.
Look-ahead bell-shaped acceleration/deceleration before interpolation produces smoother acceleration/deceleration.

- Setting an acceleration

A permissible acceleration for the linear acceleration/deceleration of each axis is set in parameter No.1660. For bell-shaped acceleration/deceleration, acceleration change time (B) (period of transition from constant speed state (A) to constant acceleration/ deceleration state (C)) is set in parameter No.1772. In the constant acceleration/deceleration state (C), acceleration/deceleration is performed with the maximum tangential acceleration not exceeding the permissible acceleration of each axis specified in parameter No.1660.
The acceleration change time specified in parameter No.1772 is held constant, regardless of the tangential acceleration.

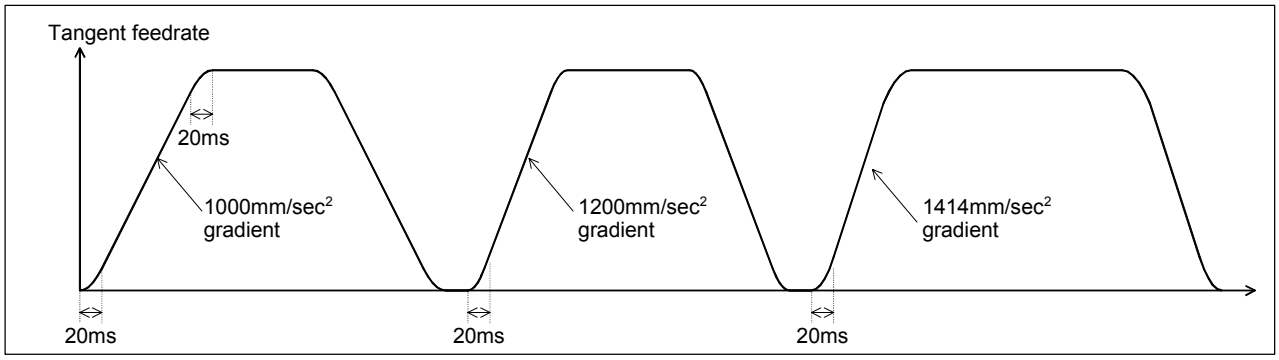


- Method of determining the tangent acceleration

Acceleration/deceleration is performed with the largest tangent acceleration/deceleration that does not exceed the acceleration set for each axis.
(Example)
X-axis permissible acceleration: 1000 mm/sec^2
Y-axis permissible acceleration: 1200 mm/sec^2
Acceleration change time: 20 msec
Program:

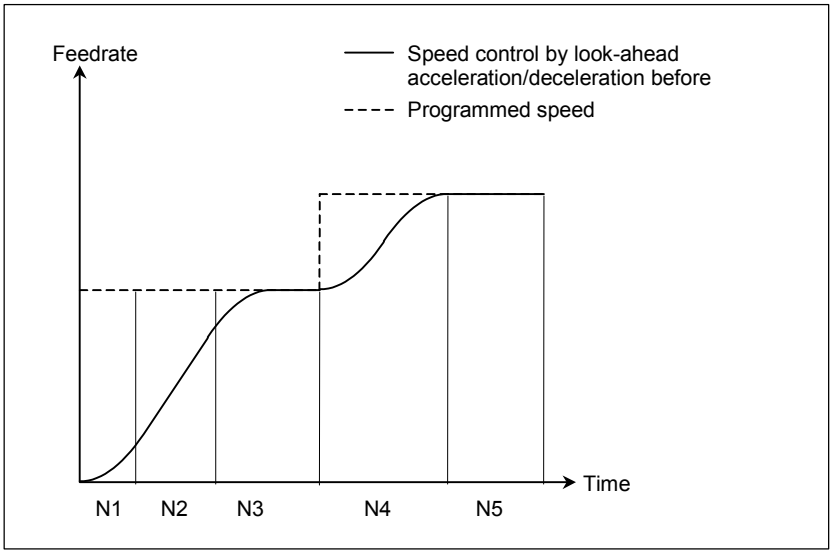
```
N1 G01 G91 X20. F6000 ..Move on the X-axis.
G04 X0.01
N2 Y20.....Move on the Y-axis.
G04 X0.01
N3 X20. Y20. .....Move in the XY direction (at 45 degrees).
```

The acceleration in N3 is 1414 mm/sec^2. At this point, the acceleration on the X-axis is equal to the set value (1000 mm/sec^2).



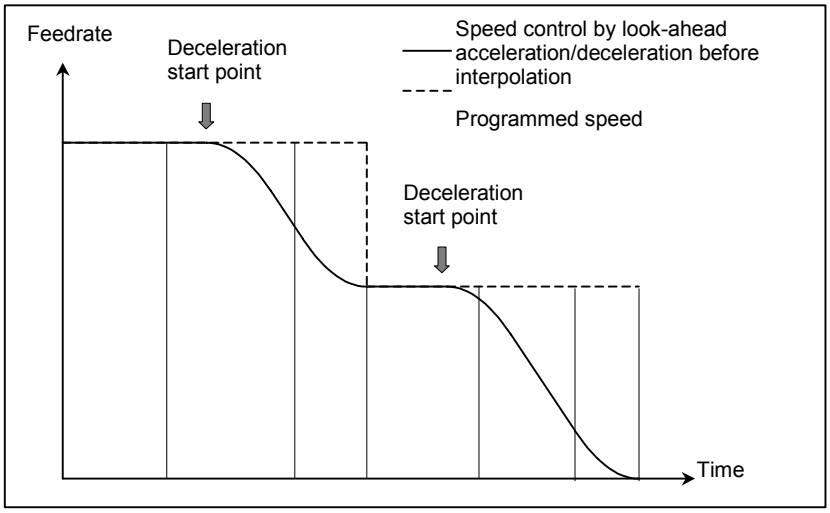
- Acceleration

Acceleration is performed so that the feedrate programmed for a block is attained at the beginning of the block.
Acceleration can be performed over several blocks.



- Deceleration

Deceleration starts in advance so that the feedrate programmed for a block is attained at the beginning of the block.
Deceleration can be performed over several blocks.

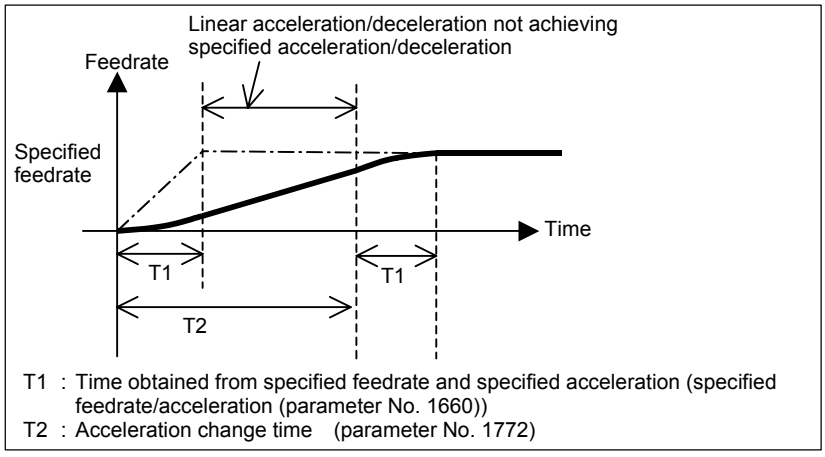


- Deceleration based on a distance

If the total distance of the blocks read ahead becomes shorter than or equal to the deceleration distance obtained from the current feedrate, deceleration starts.
If the total distance of the blocks read ahead during deceleration increases, acceleration is performed.
If the blocks of a small amount of travel are successively specified, deceleration and acceleration may be performed alternately, making the feedrate inconsistent.
To avoid this, decrease the programmed feedrate.

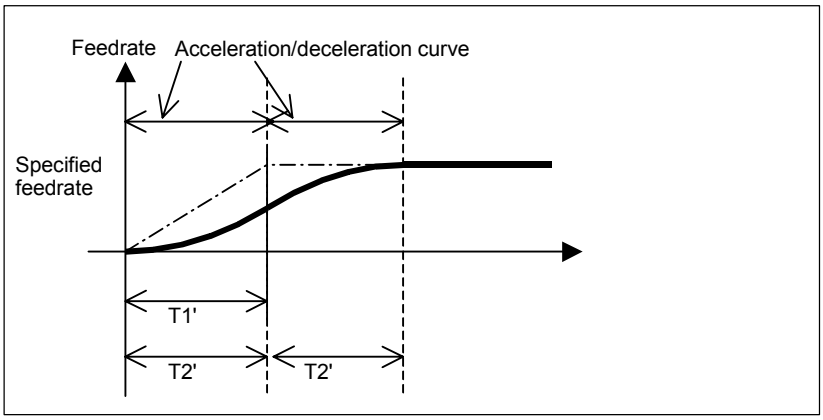
- Function for changing time constant of bell-shaped acceleration/deceleration

Bell-shaped acceleration/deceleration before interpolation is performed using the parameter-set acceleration and acceleration change time. If a low feedrate is specified, a linear acceleration/ deceleration may not achieve the specified acceleration as shown:



T1 : Time obtained from specified feedrate and specified acceleration (specified feedrate/acceleration (parameter No. 1660))
T2 : Acceleration change time (parameter No. 1772)

In such a case, set bit 3 (BCG) of parameter No. 7055 to 1. Then, the internal acceleration and vector time constant of acceleration/deceleration before interpolation are changed to make the acceleration/deceleration pattern as close as possible to the optimum bell-shaped acceleration/deceleration before interpolation based on a specified acceleration/deceleration reference speed, and so acceleration/deceleration time is reduced.



There are three methods for specifying the acceleration/deceleration reference speed.

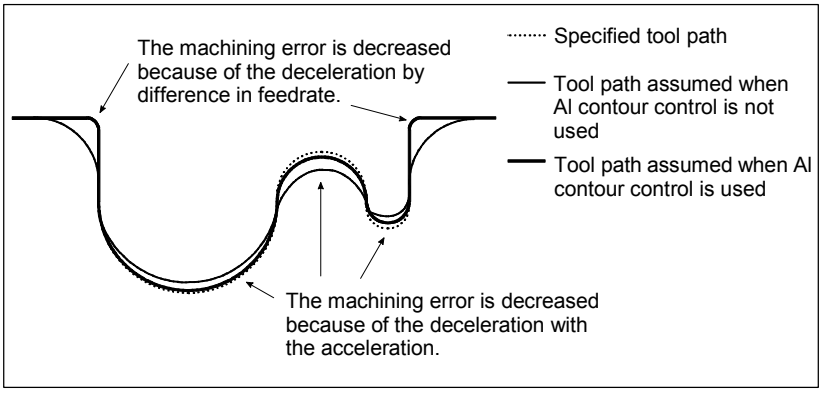
(1) Specifying the speed using an F in a G05.1 Q1 block
(2) Setting the speed on parameter No.7066
(3) Setting the speed specified with the F command issued at the start of cutting as the reference speed

When F is specified in a G05.1Q1 block, the specified feedrate is assumed to be the acceleration/deceleration reference speed. This command can be used only in the feed per minute mode.
If no F command is specified in a G05.1Q1 block, the feedrate specified in parameter No. 7066 is assumed to be the acceleration/deceleration reference speed. If 0 is set in parameter No. 7066, the F command specified in the cutting start block is assumed to be the acceleration/deceleration reference speed.

- Automatic feedrate control function

In AI contour control mode, the feedrate is automatically controlled by the reading-ahead of blocks.
The feedrate is determined using the following conditions. If the specified feedrate exceeds the determined feedrate, acceleration/ deceleration before interpolation is performed to achieve the determined feedrate.

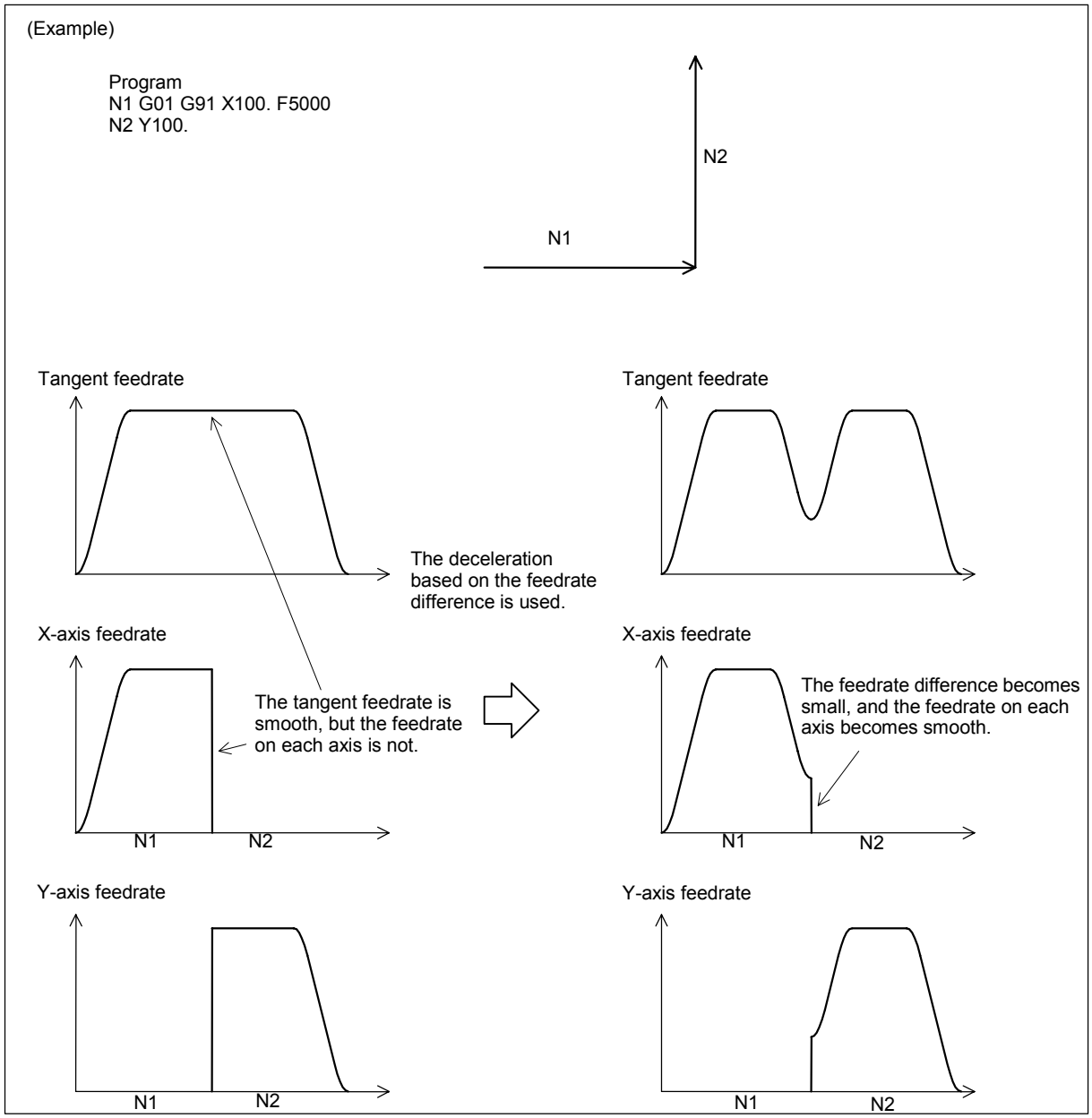
<1> Feedrate changes on each axis at a corner and the permissible feedrate change that has been set
<2> Expected acceleration on each axis and the permissible acceleration that has been set
<3> Cutting load that is expected from the travel direction on the Z-axis



For details, see the explanation of each function.

- Speed control based on the feedrate difference on each axis at a corner

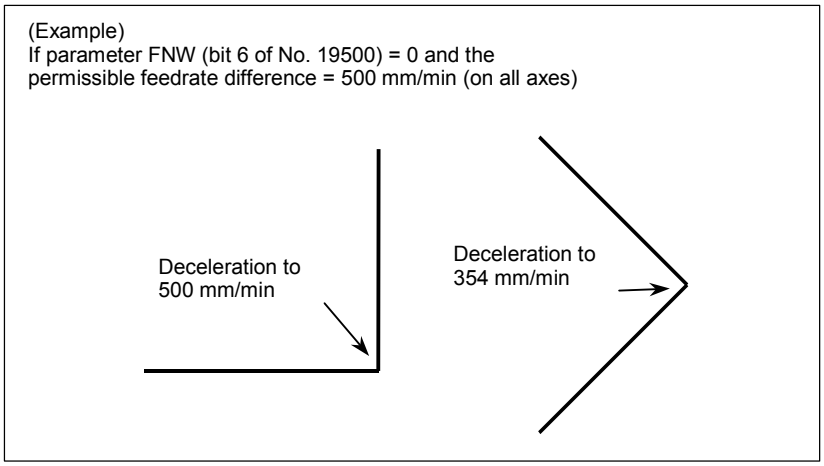
By using the speed control based on the feedrate difference on each axis at a corner, if a feedrate change occurs on an axis on each axis at a corner, the feedrate is determined so that any feedrate difference exceeding the permissible feedrate difference on that axis that has been set for parameter No. 1783 does not occur, and deceleration is automatically performed.



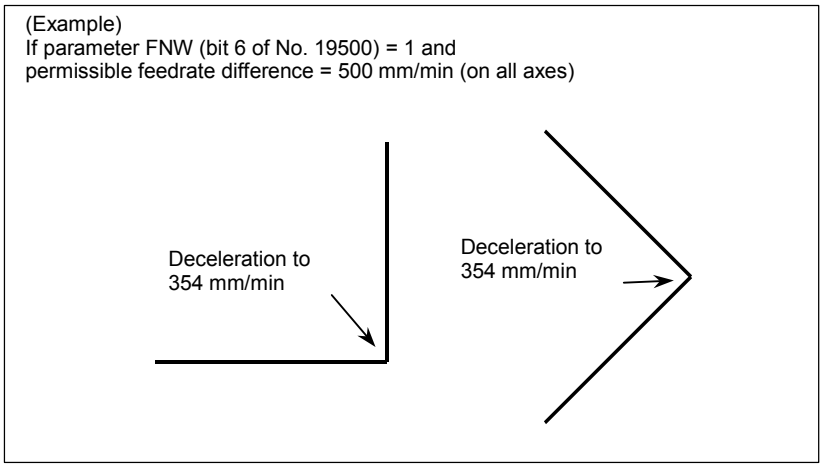
The method of deceleration based on the feedrate difference differs depending on the setting made for bit 6 (FNW) of parameter No. 19500.

If "0" is set, the largest feedrate that does not exceed the permissible feedrate difference set for parameter No. 1783 is assumed to be the deceleration feedrate.

In this case, the deceleration feedrate differs if the travel direction differs, even if the shape is the same.

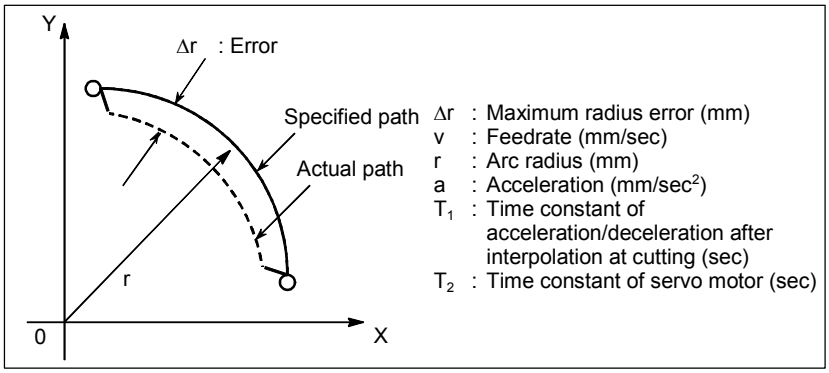


If "1" is set, the feedrate is determined not only with the condition that the permissible feedrate difference and permissible acceleration on each axis are not exceeded, but also that the deceleration feedrate is constant regardless of the travel direction if the shape is the same.
If 1 is set for this parameter, the deceleration feedrate determined with the feedrate difference may be up to 30% lower than that determined if 0 is set.



- Speed control with acceleration in circular interpolation

When high-speed cutting is performed in circular interpolation, helical interpolation, or spiral interpolation, the actual tool path has an error with respect to the programmed path. In circular interpolation, this error can be approximated from the equation given below.



$$\Delta r = \frac{1}{2}(T_1^2 + T_2^2)\frac{v^2}{r} = \frac{1}{2}(T_1^2 + T_2^2)\cdot a \quad \text{.....(Equation 1)}$$

In actual machining, permissible error Δr is given, so the maximum permissible acceleration a (mm/sec^2) in equation 1 is determined.
When a specified feedrate causes the radial error from an arc having a programmed radius to exceed the permissible error, speed control with acceleration in circular interpolation automatically clamps the arc-cutting feedrate by using parameter settings.

Let the permissible acceleration calculated from the permissible acceleration set for each axis be A. Then, maximum permissible feedrate v with programmed radius r is expressed as follows:

$$v = \sqrt{A \cdot r} \quad \text{.....(Equation 2)}$$

If a specified feedrate exceeds feedrate v obtained from equation 2, the feedrate is clamped at feedrate v automatically.
The permissible acceleration is specified in parameter No. 1735. If there is a difference in permissible acceleration between two axes for circular interpolation, the lower acceleration is regarded as the permissible acceleration.
If the radius of an arc is small, too small value can be calculated as deceleration v. In such a case, the lower feedrate limit can be set in parameter No. 1732 to prevent the feedrate from being decreased too much.

- Speed control with the acceleration on each axis

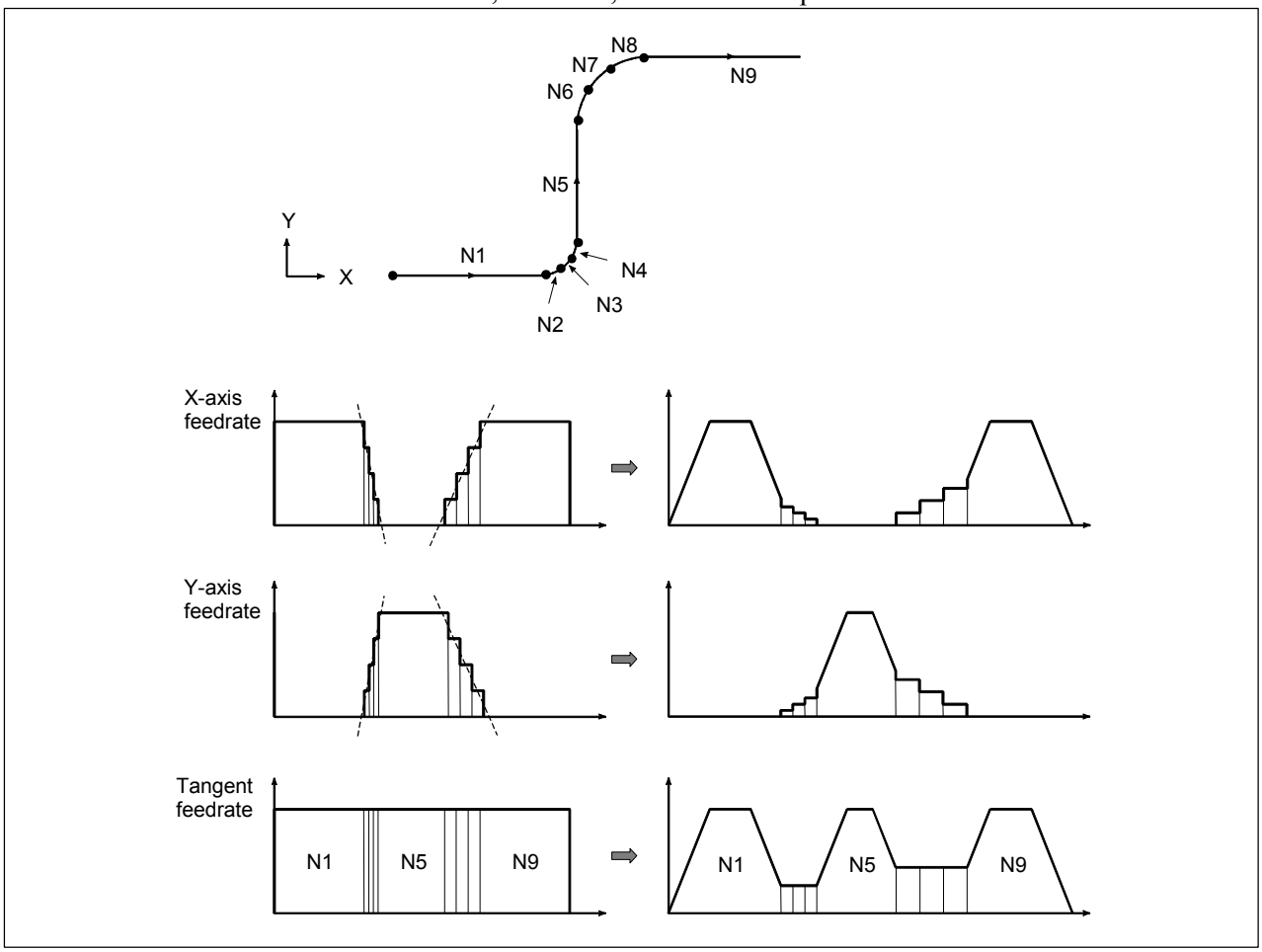
When consecutive small lines are used to form a curve, as in the example shown in the figure below, the feedrate differences on each axis at the individual corners are not very large. Thus, deceleration with the feedrate differences is not effective. Consecutive small feedrate differences, however, cause a large acceleration on each axis, as a whole.

In such a case, deceleration can be performed to reduce the impact on the machine and the machining error caused by too large an acceleration. The deceleration feedrate is determined to be the feedrate that does not cause the acceleration on each axis to exceed the permissible acceleration set for parameter No. 1737.

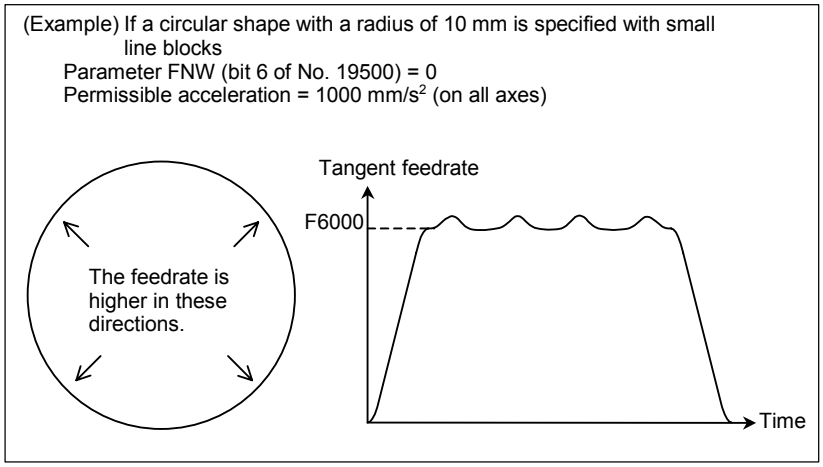
The deceleration feedrate is determined for each corner. The actual feedrate is the smaller of the deceleration feedrate determined at the start point of the block and that determined at the end point.

Depending on the specified figure, a very low deceleration feedrate may be calculated. In such a case, the lower feedrate limit can be set in parameter No. 1738 to prevent the feedrate from being decreased too much.

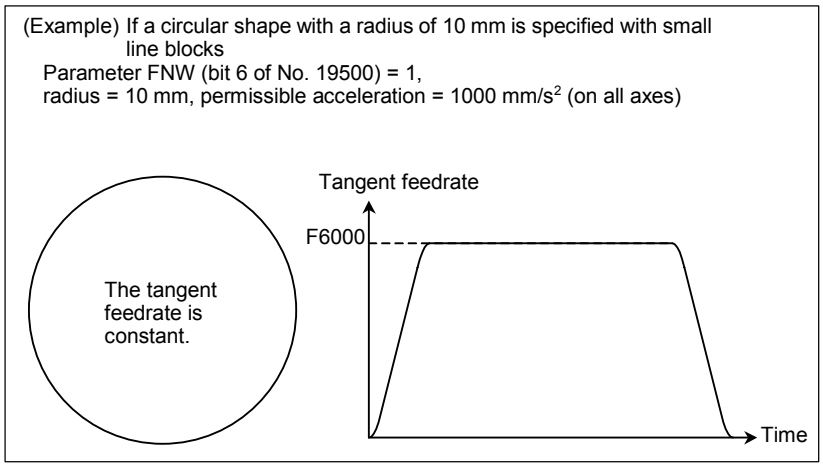
In the following example, the acceleration (gradient of the broken line in the feedrate graph) at too large at corners N2 to N4 and N6 to N8 and, therefore, deceleration is performed.



The method of determining the feedrate with the acceleration differs depending on the setting of bit 6 (FNW) of parameter No. 19500.
If "0" is set, the highest feedrate that does not cause the permissible acceleration set for parameter No. 1737 to be exceeded is assumed to be the deceleration feedrate. In this case, the deceleration feedrate differs depending on the travel direction even if the shape is the same, as shown in the figure below.



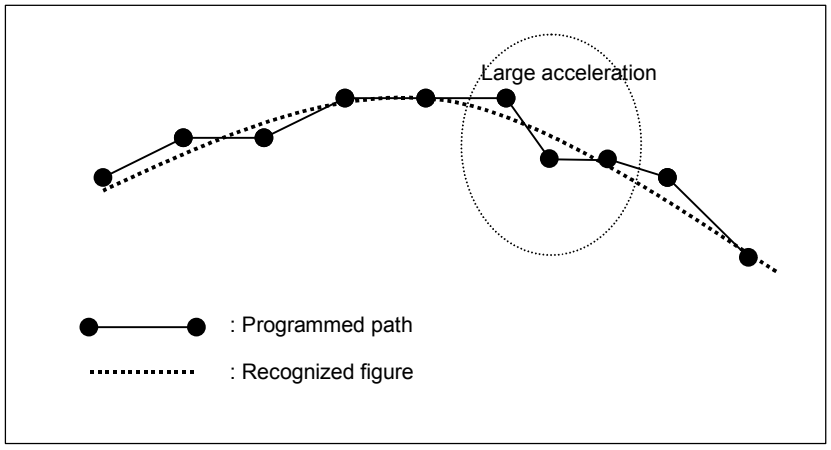
If "1" is set, the feedrate is determined with not only the condition that the permissible acceleration on each axis is not exceeded but also the condition that the deceleration feedrate is constant regardless of the travel direction if the shape is the same.
If 1 is set for this parameter, the deceleration feedrate determined with the feedrate difference or acceleration may be up to 30% lower than that determined if 0 is set.



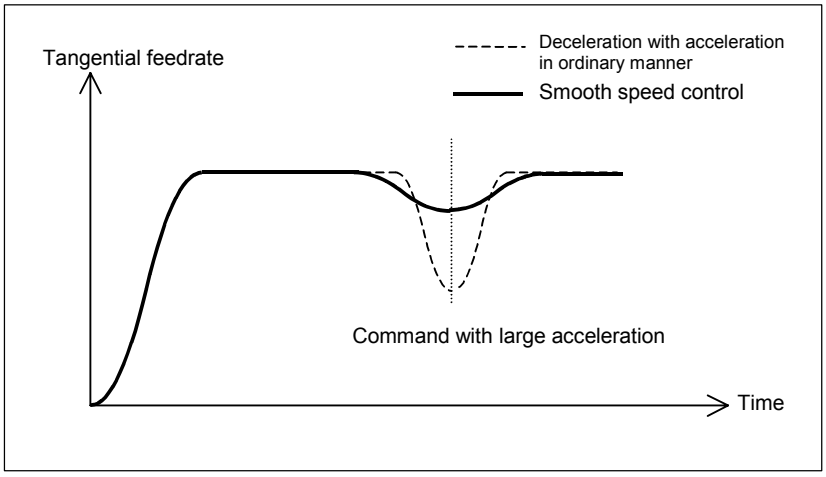
NOTE
In circular interpolation, the tangent feedrate is constant regardless of the setting of the parameter.

- Smooth speed control

In speed control with acceleration, the smooth speed control function recognizes the entire figure from preceding and following blocks including blocks read ahead to make a smooth feedrate determination.
When a curve is specified with successive minute straight lines, programmed values are rounded to the least input increment before issued, so the machining profile is approximated with a broken line.
When the feedrate is determined with acceleration in an ordinary manner, an optimum feedrate is automatically calculated exactly for a programmed figure, so a large acceleration may result depending on the command, which can lead to deceleration.
In such a case, the use of smooth speed control enables speed control by recognizing the entire figure, which provides smooth speed control while suppressing local deceleration, therefore increasing the feedrate.



Also for a part of a programmed figure in which a large acceleration would be required, the acceleration is obtained based on the figure recognized from multiple blocks, and the feedrate is determined so that the acceleration is within the permissible acceleration set in parameter No. 1737.

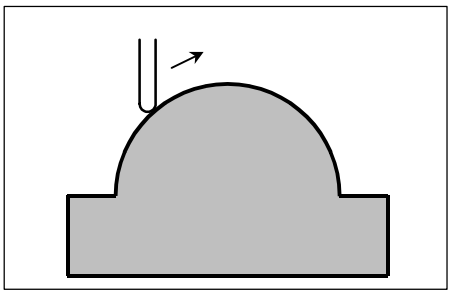


Smooth speed control obtains the acceleration by using the figure recognized from the preceding and following blocks including blocks read ahead, so smooth speed control is enabled even in parts in which the acceleration increases.
Smooth speed control is enabled under the following conditions:
<1> Speed control with acceleration is enabled in the AI contour control mode.
<2> Successive linear interpolation commands are specified.
<3> Bit 0 (HPF) of parameter No. 19503 is set to 1.

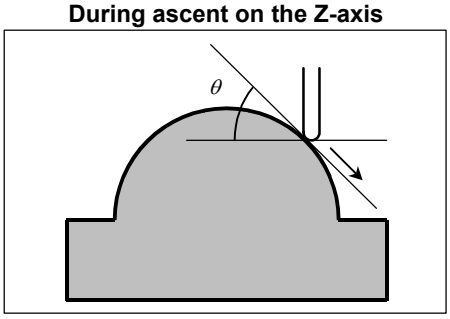
> ⚠ **CAUTION**
> When smooth speed control is used, the feedrate in a certain figure such as a corner may become larger than the feedrate obtained by ordinary speed control with acceleration. For corners, set parameter No. 1783, which is the permissible feedrate difference parameter for speed control with the feedrate difference at corners, to perform appropriate deceleration by speed control with the corner feedrate difference.

- Speed control with the cutting load

Usually, the cutting resistance produced when machining is performed with the bottom of the cutter as the tool lowers along the Z-axis is greater than the cutting resistance produced when machining is performed with the side of the cutter as the tool rises along the Z-axis. Therefore, deceleration is required.
In AI contour control, the tool travel direction on the Z-axis is used as a condition for calculating the machining feedrate.
This function is enabled when bit 4 (ZAG) of parameter No. 8451 is set to 1.



During ascent on the Z-axis



During descent on the Z-axis

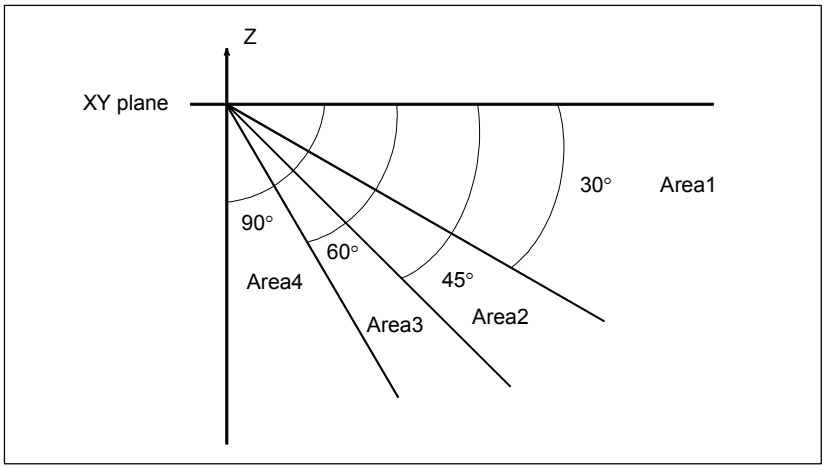
The descent angle θ during descent on the Z-axis (angle formed by the XY plane and the tool center path) is as shown in the figure. The descent angle is divided into four areas, and the override values for the individual areas are set for the following parameters:

Parameter No. 8456 for area 2
Parameter No. 8457 for area 3
Parameter No. 8458 for area 4

For area 1, however, no parameter is available, and an override of 100% is used at all times. The feedrate obtained according to other feedrate control is multiplied by the override value of the area to which descent angle θ belongs.

Area1 $0^\circ \leq \theta < 30^\circ$
Area2 $30^\circ \leq \theta < 45^\circ$
Area3 $45^\circ \leq \theta < 60^\circ$
Area4 $60^\circ \leq \theta < 90^\circ$

The feedrate can be overridden with an inclination by setting bit 1 (ZG2) of parameter No. 19515 to 1. In this case, specify the override value for area 1 in parameter No. 19516.



> ⚠ **CAUTION**
> 1 The speed control with the cutting feed is effective only when the tool is parallel with the Z-axis. Thus, it may not be possible to apply this function, depending on the structure of the machine used.
> 2 In the speed control with the cutting feed, the travel direction on the Z-axis is determined with the appropriate NC command. If, therefore, manual intervention is performed on the Z-axis with manual absolute on, or if a mirror image is applied on the Z-axis, the direction on the Z-axis cannot be determined. When using the speed control with the cutting load, do not use these functions.
> 3 When performing three-dimensional coordinate conversion, determine the descent angle on the Z-axis using the converted coordinate system.
> 4 Speed control with the cutting load is enabled for all interpolations in the AI contour control mode. This function, however, can be made valid only for linear interpolations by setting bit 4 (ZOL) of parameter No. 19503 to 1.

- Ignoring feedrate commands

In a block in which AI contour control is enabled, all feedrate commands (F commands) can be ignored by setting bit 7 (NOF) of parameter No. 8451.
The term feedrate commands, as used here, refer to the following commands:

<1> Modal F commands before the block in which AI contour control is enabled
<2> F commands and modal F commands in the block in which AI contour control is enabled

When the feedrate commands are ignored, it is assumed that the upper feedrate limit specified for parameter No. 8465 is specified.
Note, however, that any issued F commands and modal F commands are stored within the CNC.
Thus, in a block in which AI contour control changes from the enabled state to the disabled state, the modal values of the F commands described in <1> and <2> described above are used as modal F commands, instead of the modal values of the F commands calculated by AI contour control.

- Another example of determining the feedrate

If a specified feedrate exceeds the upper feedrate limit of AI contour control (in parameter No. 8465), the feedrate is clamped at the upper feedrate. The upper feedrate limit is clamped at the maximum cutting feedrate (parameter No. 1432).

Limitation

- Conditions for temporarily canceling the AI contour control mode

If one of the commands listed below is issued in the AI contour control mode, the AI contour control mode is canceled temporarily.
If the system becomes ready for AI contour control after it is canceled, the AI contour control mode is restored automatically.

- Positioning (rapid traverse)
- Single direction positioning
- Spindle positioning
- Rigid tapping
- Hypothetical axis interpolation
- Threading (single type, combined type)
- Electronic gear box
- Superimposed control
- When no move command is specified
- One-shot G code other than the following:
  Tool offset
  Cutter compensation vector retention
  Cutter compensation corner rounding
  Exact stop

- Functions that cannot be specified in the AI contour control mode

In the AI contour control mode, the functions listed below cannot be specified. Before specifying these functions, turn off the AI contour control mode; after the command ends, turn on the mode again.

- Threading
- Circular threading
- Variable lead threading

Threading, circular threading, and variable lead threading can be specified in the AI contour control mode by setting bit 1 (THA) of parameter No. 1611. However, the AI contour control mode is automatically canceled.

Notes

- About processing macro statements

In AI contour control mode, the NC statements of multiple blocks are looked ahead. Macro statements such as arithmetic expressions and conditional branches are processed as soon as they are read into the buffer. Therefore, the timing of the macro statement execution is not always the specified order.
In case that you need to execute the macro statement after completing the NC block just before the macro statement, specify M code or G code that is not buffered just before the macro statement. Specially, in case of reading/writing the system variables to control signals, coordinates, offset value, etc., it may be different system variable data by the timing of the NC statement execution. To avoid this phenomenon, specify such M codes or G codes before the macro statement, if necessary.

20.2 MACHINING CONDITION SELECTING FUNCTION

Overview

By setting a speed- or precision-focused parameter set in an AI contour control function and specifying a precision level in accordance with the machining conditions during machining, parameters suitable to the conditions can be automatically calculated so that machining can be performed.
This function is an optional one.

Format

- Changing the precision level using a program

In addition to being switched on the precision level selection screen, the precision level can be changed using a program in the format below.

```
G05.1 Q1 Rx ;
  x.....Level (1 to 10)
```

The precision level can also be changed in the format used with conventional "advanced preview control", "high precision contour control", and "AI high precision contour control".

```
G05 P10000 Rx ;
  x.....Level (1 to 10)
G08 P1 Rx ;
  x.....Level (1 to 10)
```

⚠ CAUTION
Once specified, a level remains effective even if the AI contour control mode is canceled.

20.3 JERK CONTROL

20.3.1 Speed Control with Change of Acceleration on Each Axis

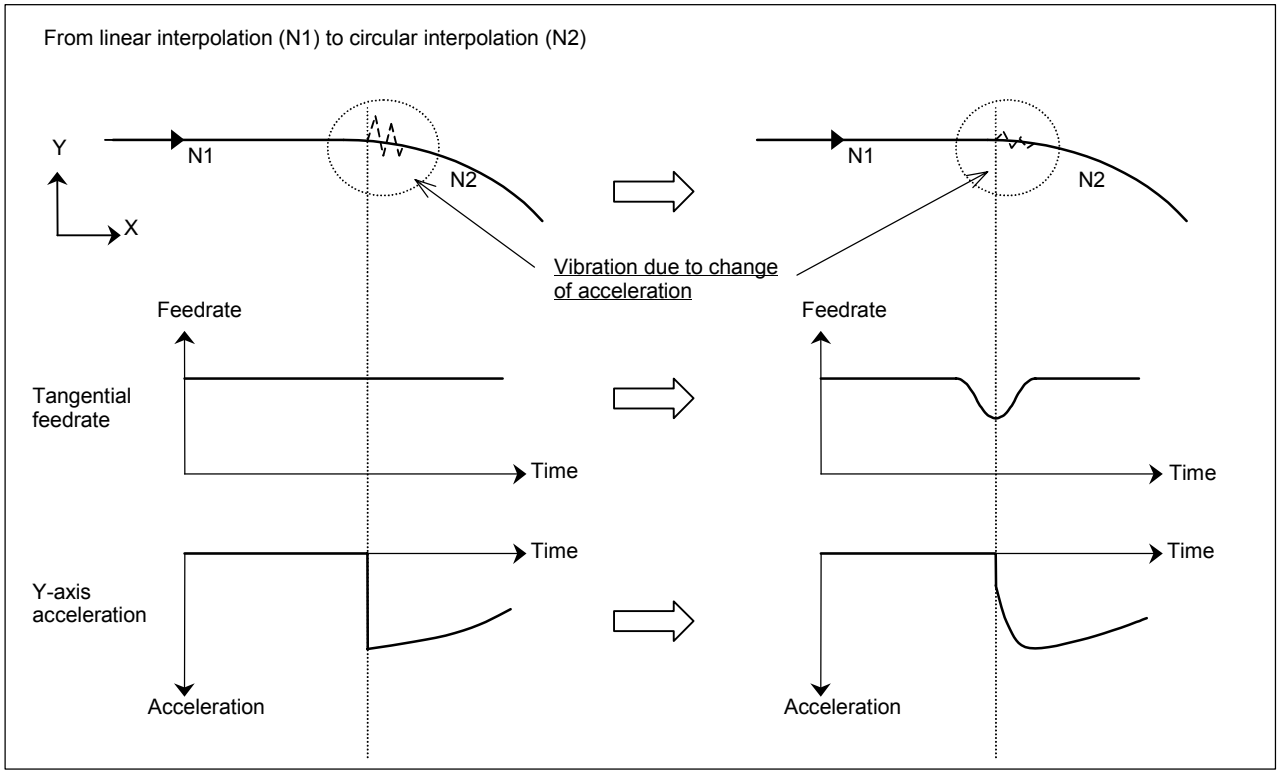
Overview

In portions in which acceleration changes largely, such as a portion where a programmed figure changes from a straight line to curve, vibration or shock on the machine may occur. Speed control with change of acceleration on each axis is a function to suppress machining errors due to vibration and machine shock generated by change of acceleration. This function obtains a feedrate so that change of acceleration is within the parameter-set permissible acceleration change amount for each axis, and performs deceleration by using acceleration/deceleration before interpolation.

> ⚠ **CAUTION**
> Before speed control with change of acceleration on each axis can be used, the options for jerk control and AI contour control are required.

Explanation

In the following example, the Y-axis acceleration changes largely at the contact point between a linear interpolation and circular interpolation, so deceleration is performed.



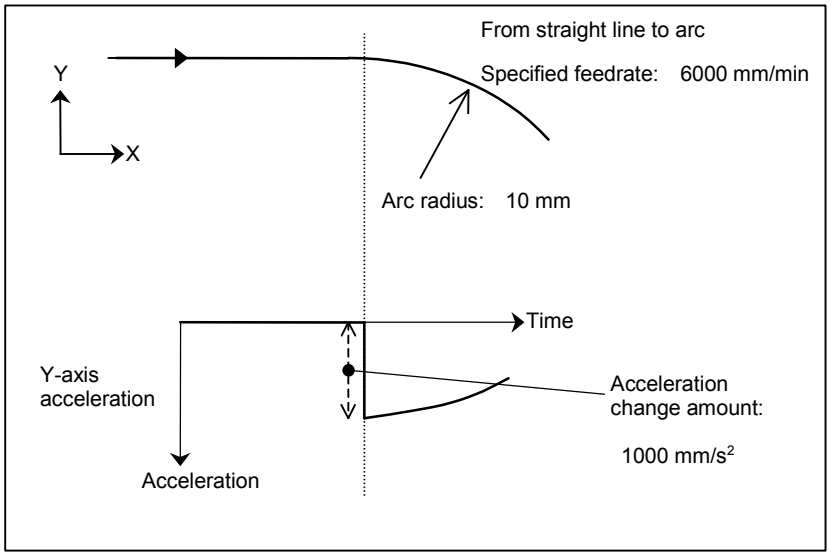
- Setting the permissible acceleration change amount

The permissible acceleration change amount for each axis is set in parameter No. 1788. When 0 is set in this parameter for a certain axis, speed control with change of acceleration is not performed for that axis.

- Parameter setting example

Suppose a figure shown below in which a straight line is followed by an arc. Let the specified feedrate and the arc radius be 6000 mm/min and 10 mm, respectively. Then, the Y-axis acceleration change amount at the contact point of the linear and arc portions is obtained as follows:

$$\frac{v^2}{r} = 1000 mm / s^2$$



To suppress the change of acceleration to 300 mm/s^2, set 300 mm/s^2 for the Y-axis in parameter No. 1788.

Note that the change of acceleration is determined from the interpolation data of the CNC, so it may differ from the theoretical value.

The actual machine is affected by acceleration/deceleration and other factors, so the value to be set in the parameter should be determined after adjustments are made.

- For successive linear interpolations

When there are successive linear interpolations, speed control with change of acceleration obtains the deceleration feedrate from the change in acceleration between the start point and end point of a specified block.

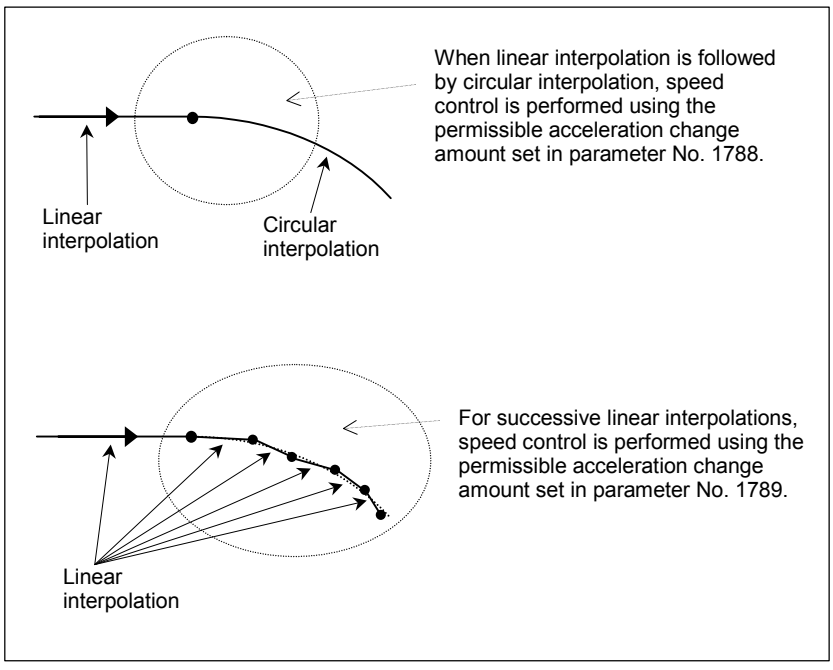
When a curve is specified using successive minute straight lines, programmed values are rounded to the least input increment before issued, so the machining profile is approximated with a broken line. The error due to rounding may increase change of acceleration, and especially when the line segments specified by blocks are short, deceleration is performed frequently. As a result, the machining speed cannot increase enough. In such a case, a relatively large value should be set in parameter No. 1789 as the permissible acceleration change amount for each axis in successive linear interpolations to improve the machining speed.

When a value other than 0 is set in parameter No. 1789 for an axis for which deceleration with change of acceleration is enabled, this setting is regarded as the permissible acceleration change amount at corners in which linear interpolations meet. (For portions where a linear interpolation and circular interpolation meet and where circular interpolations meet, the setting in parameter No. 1788 is used.)

When 0 is set in parameter No. 1789 for an axis, the setting in parameter No. 1788 specifying the ordinary permissible acceleration change amount is used even at a corner in which linear interpolations meet.

When smooth speed control is used in speed control with permissible acceleration in AI contour control (named temporarily), the deceleration feedrate is obtained from the change of acceleration calculated by smooth speed control.

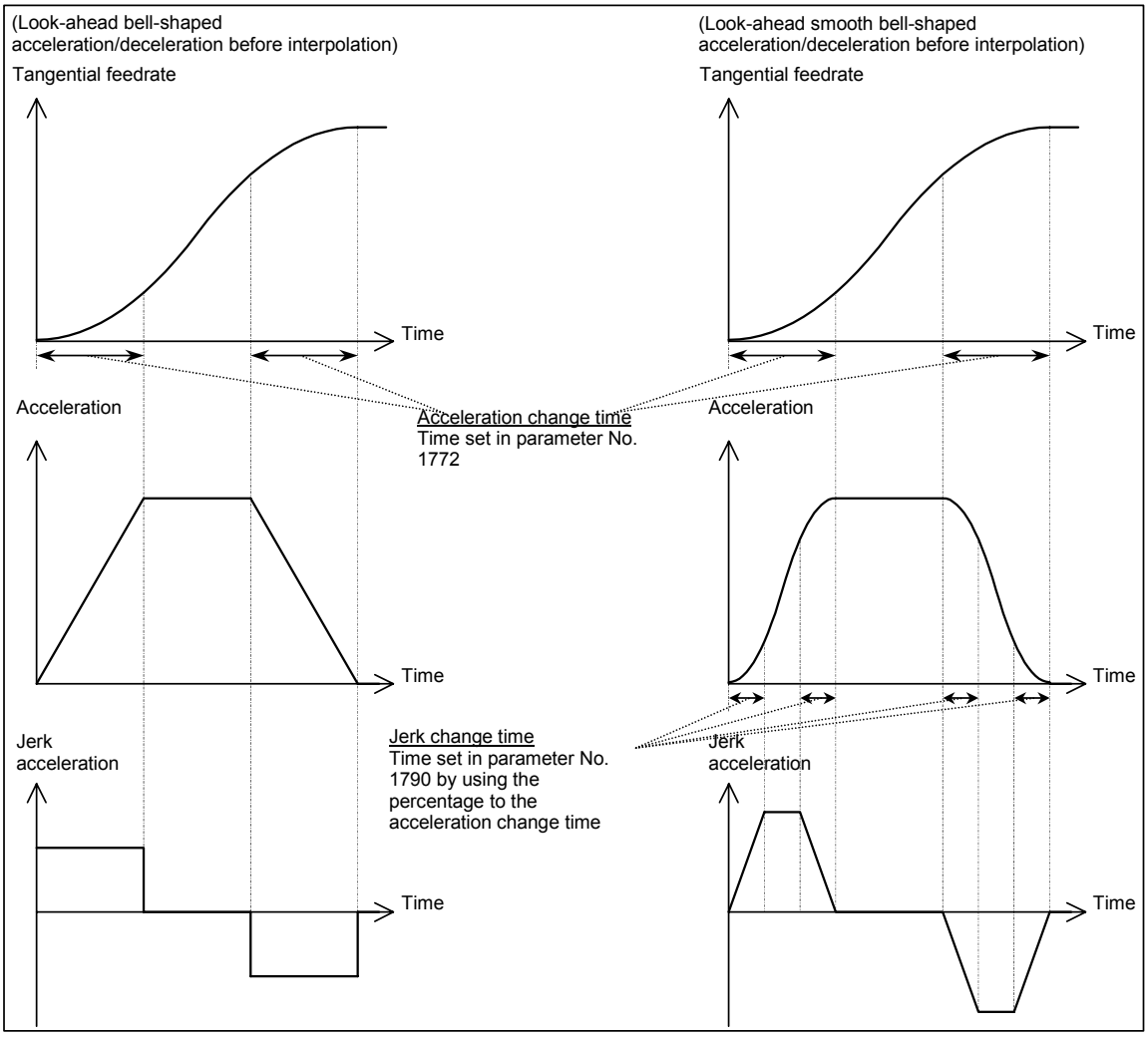
Therefore, the deceleration feedrate may be higher than the ordinary deceleration feedrate.



20.3.2 Look-Ahead Smooth Bell-Shaped Acceleration/Deceleration before Interpolation

Overview

In look-ahead bell-shaped acceleration/deceleration before interpolation performs smooth acceleration/deceleration by changing the acceleration at a constant rate in specified acceleration change time.
In look-ahead smooth bell-shaped acceleration/deceleration before interpolation, the jerk change time is specified in parameter No. 1790 by using the percentage to the acceleration change time for look-ahead bell-shaped acceleration/deceleration before interpolation, and change of acceleration is also controlled so that the change is bell-shaped. This enables smoother acceleration/deceleration, therefore reducing machine vibration and shock due to acceleration/ deceleration.



⚠ CAUTION
Before look-ahead smooth bell-shaped acceleration/deceleration before interpolation can be used, the option for jerk control and AI contour control II is required.

Explanation

- Setting the jerk change time

The jerk change time is set in parameter No. 1790 by using the percentage to the acceleration change time.
The actual jerk change time is represented by the percentage to the acceleration change time set in parameter No. 1772.
The jerk change time must be within a half of the acceleration change time, so the value to be set in the parameter ranges 0 to 50 (percent).
If 0 or a value beyond the specifiable range is specified in parameter No. 1790, look-ahead smooth bell-shaped acceleration/deceleration before interpolation is not enabled.

- Acceleration/deceleration before interpolation for linear type rapid traverse

When bell-shaped acceleration/deceleration is used in acceleration/ deceleration before interpolation for linear type rapid traverse, enabling look-ahead smooth bell-shaped acceleration/deceleration before interpolation applies smooth bell-shaped acceleration/deceleration to acceleration/deceleration before interpolation for linear type rapid traverse.
In this case, the jerk change time is represented by the percentage set in parameter No. 1790 to the acceleration change time set in parameter No. 1672.

- Optimum torque acceleration/deceleration

When bell-shaped acceleration/deceleration is used in optimum torque acceleration/deceleration, enabling look-ahead smooth bell-shaped acceleration/deceleration before interpolation applies smooth bell-shaped acceleration/deceleration to optimum torque acceleration/ deceleration.
In this case, the jerk change time is represented by the percentage set in parameter No. 1790 to the acceleration change time set in parameter No. 1672.

20.4 OPTIMUM TORQUE ACCELERATION/DECELERATION

Overview

This function enables execution of acceleration/deceleration adapted to the torque characteristics of the motor and the characteristics of the machine due to the friction and gravity of the machine.

Usually, because of the friction of the machine, gravity, the torque characteristics of the motor, and other factors, the acceleration/ deceleration performance (torque for acceleration/deceleration) is different with direction of movement, acceleration or deceleration. In this function, acceleration pattern of rapid traverse for the following situations, plus movement and acceleration, plus movement and deceleration, minus movement and acceleration, minus movement and deceleration can be set into parameters according to the torque for acceleration/deceleration of each situation.
Acceleration/deceleration can be performed according to these parameter setting, so that the most of the capability of the motor can be used and positioning time can be reduced.

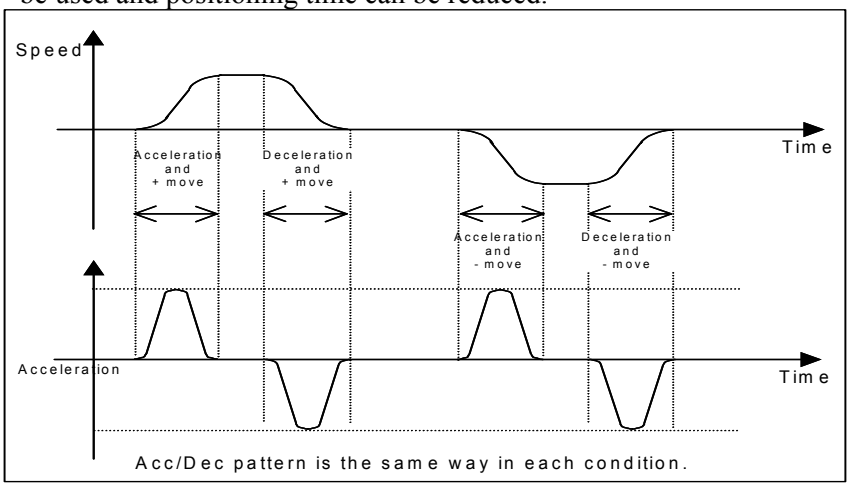


Fig. 20.4 (a) Conventional acceleration/declaration

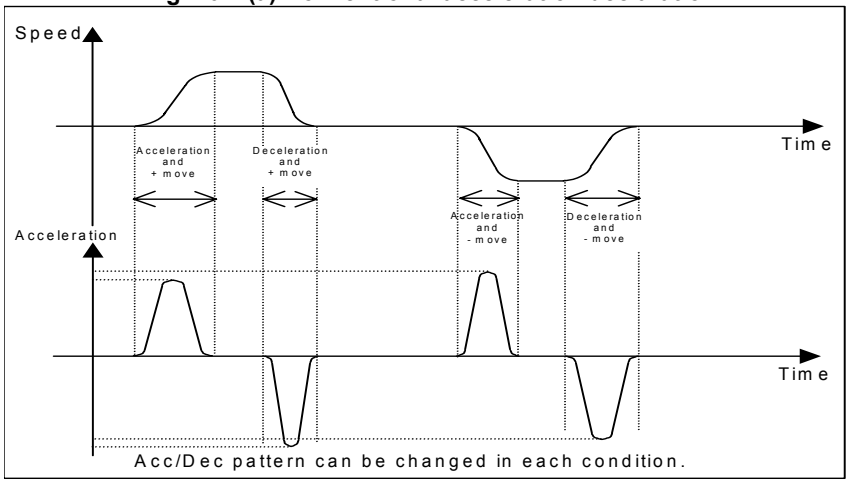


Fig. 20.4 (b) Acceleration/deceleration with this function

Explanation

Optimum torque acceleration/deceleration selects the acceleration pattern set with parameters on the basis of the axial movement direction and the acceleration/deceleration state, determines the acceleration for each axis from the current speed, and controls the tangential acceleration/deceleration for rapid traverse.

- Setting optimum torque acceleration/deceleration

When bit 0 (FAP) of parameter No. 19540 and bit 1 (LRP) of parameter No. 1401 are set to 1, and a value other than 0 is set for any one of the axes as the reference acceleration in parameter No. 1671 as shown below, rapid traverse is accelerated/decelerated by optimum torque acceleration/deceleration.

Table 20.4 (a) Optimum torque acceleration/deceleration

Parameter FAP (Optimum torque acc/dec)	Parameter LRP (Linear type positioning)	Reference acceleration	Bell-shaped acceleration change time	Acceleration pattern
1	1	Parameter No.1671	Parameter No.1672	See "Setting an acceleration pattern".

To enable bell-shaped acceleration/deceleration in addition to optimum torque acceleration/declaration, set the bell-shaped acceleration change time with parameter No.1672.

- Required conditions in addition to parameter setting

If the conditions for performing AI contour control and for acceleration/deceleration before interpolation are satisfied, optimum torque acceleration/deceleration is performed for accelerating and decelerating rapid traverse. Note, however, that if 1 is set to bit 0 (FAE) of parameter No. 11240, optimum torque acceleration/deceleration is enabled even if AI contour control is disabled. If rapid traverse is subject to optimum torque acceleration/deceleration, after-interpolation acceleration/deceleration does not apply to rapid traverse.

- Cases in which optimum torque acceleration/ deceleration is disabled

If optimum torque acceleration/deceleration is disabled by parameter settings, or the required conditions other than parameter settings are not satisfied, acceleration/deceleration after interpolation is used to accelerate and decelerate rapid traverse.

- Setting acceleration pattern data

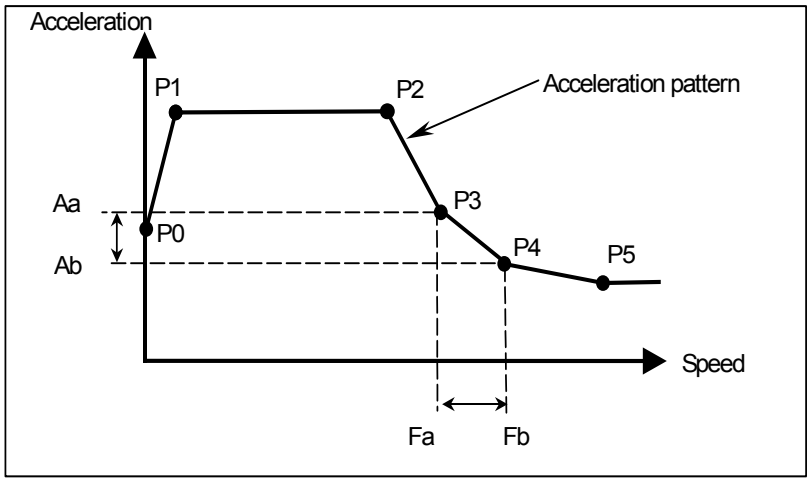


Fig. 20.4 (c) Setting acceleration pattern

Set the speed and the acceleration at each of the acceleration setting points P0 to P5 for each condition, plus movement and acceleration, plus movement and deceleration, minus movement and acceleration, minus movement and deceleration, and for each axis.
The line joining the acceleration setting points is regarded a acceleration pattern.
For example, while the speed is between Fa and Fb in the previous figure, the acceleration is calculated with Aa and Ab. Tangential acceleration is controlled not to exceed the calculated acceleration for each axis.

> ⚠ **CAUTION**
> It is not desirable to set an acceleration pattern in which a large acceleration is set right at a speed of 0, because this will shock the machine. For this reason, **be sure to set a low acceleration at a speed of 0**, as in the figure above.

Table 20.4 (b) Parameters for acceleration pattern

Acceleration setting point	Speed parameter	Acceleration parameter			
		During acceleration		During deceleration	
		During movement in plus direction	During movement in minus direction	During movement in plus direction	During movement in minus direction
P0	(Speed 0)	No.19545	No.19551	No.19557	No.19563
P1	No.19541	No.19546	No.19552	No.19558	No.19564
P2	No.19542	No.19547	No.19553	No.19559	No.19565
P3	No.19543	No.19548	No.19554	No.19560	No.19566
P4	No.19544	No.19549	No.19555	No.19561	No.19567
P5	No.1420	No.19550	No.19556	No.19562	No.19568

The speed at P0 is 0, and the speed at P5 is the rapid traverse rate specified with parameter No. 1420. The speeds at P1 to P4 are to be set into speed parameters Nos. 19541 to 19544 as ratio to the rapid traverse speed (parameter No. 1420).
Any acceleration setting point for which the speed parameters Nos. 19541 to 19544 is set to 0 will be skipped, and the next point whose speed parameter is set to a non-zero value will be joined together as acceleration pattern.
The accelerations at P0 to P5 are to be set into acceleration parameters Nos. 19545 to 19568 as ratio to the reference acceleration. If any of the acceleration parameters Nos. 19545 to 19568 is set to 0, the acceleration is assumed 100% (Reference acceleration). Acceleration parameters should be set to 0 at the acceleration setting point whose speed parameter is set to 0.
If this function is enabled and parameter No.1671 for an axis are set to 0, the following values are assumed as the reference acceleration for that axis:

1000.0mm/sec^2, 100.0inch/sec^2, 100.0 deg/sec^2

- Example of setting acceleration pattern data

In this example, the machine is equipped with the α*i*S 30/4000.
Motor speed at rapid traverse is 3000 (min^{-1}).

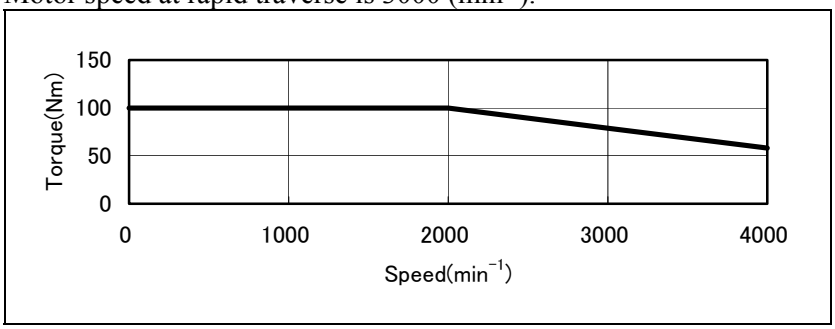


Fig. 20.4 (d) Speed-torque characteristics of model α*i*S 30/4000

Specifications of the motor model α*i*S 30/4000

Rotor inertia	:0.0099(Kgm^2)	
Maximum torque	:100(Nm)	Speed 0 to 2000(min^{-1})
Torque at rapid traverse	:79(Nm)	Speed 3000(min^{-1})
Minimum torque	:58(Nm)	Speed 4000(min^{-1})

It assumes that 10 (Nm) is needed for the friction torque, so that the torque for acceleration/deceleration is shown as the following figure.
Because the friction torque is different on each machine, it is necessary to observe the actual torque output on the machine.

Maximum torque	:90(=100-10)(Nm)	Speed 0 to 2000(min^{-1})
Torque at rapid traverse	:69(=79-10)(Nm)	Speed 3000(min^{-1})
Minimum torque	:48(=58-10)(Nm)	Speed 4000(min^{-1})

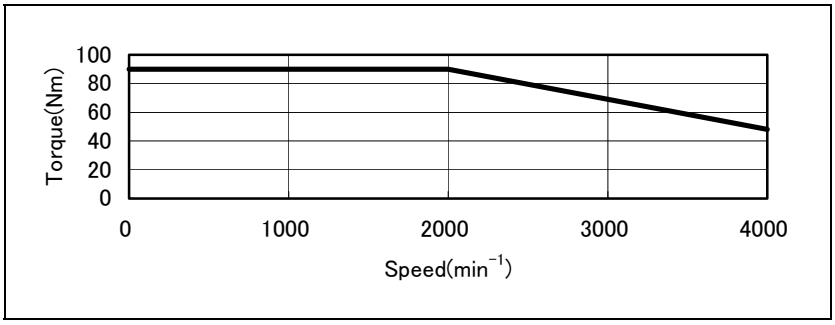


Fig. 20.4 (e) Torque for Acc/Dec with consideration of friction

Let the torque be x (Nm), the inertia be y (Kgm^2), and the ball screw pitch p (mm), then the acceleration A is calculated as follows:

$$A = \frac{x[N \cdot m]}{y[kg \cdot m^2]} \times \frac{p}{2\pi}[mm] = \frac{x([kg \cdot m/\sec^2][m])}{y[kg \cdot m^2]} \times \frac{p}{2\pi}[mm]$$

$$= \frac{x \times p}{2\pi \times y}[mm/\sec^2]$$

Machine specification is assumed as follows,
Ball screw pitch : 16 (mm)
Inertia : The machine inertia is to be 2.0 times higher than that of the rotor.
(Rotor inertia: 0.0099 (Kgm^2))
The acceleration at maximum torque 90 (Nm) is,

$$\frac{90 \times 16}{2\pi \times 3.0 \times 0.0099} = 7717[mm/\sec^2]$$

The acceleration at torque 69 (Nm) in rapid traverse 3000 (min^{-1}) is,

$$\frac{69 \times 16}{2\pi \times 3.0 \times 0.0099} = 5916[mm/\sec^2]$$

From the above data, the parameters related to acceleration pattern is given in the table below.
The example assumes that the acceleration is the same regardless of whether acceleration or deceleration is in progress or whether the movement is in the plus or minus direction.

- When setting the speed at P1, use the following value as a rough guide:
  If a bell-shaped acceleration change time (parameter No. 1672) is set, use as a rough guide the ratio of the following speed to the rapid traverse rate:
  Reference acceleration × T2/4
  where T2 is the bell-shaped acceleration change time (msec).
  For example, assume T2 to be 40 msec, then the speed at P1 should be approximately
  4124 [mm/sec^2]×40 [msec]/4
  =4124 [mm/sec^2]×60^2×40 [msec]/60000/4
  =2474 [mm/min]
- If no bell-shaped acceleration change time (parameter No. 1672) is set, use about 5% of the rapid traverse rate as a rough guide.

Table 20.4 (c) Example of setting parameters related to acceleration pattern

	Parameter No.	Setting	Unit	Remarks
Rapid traverse rate	1420	48000.	mm/ min	The ball screw pitch is assumed 16 mm, so that the rapid traverse rate is 48000 mm/min at the maximum speed 3000 (min^{-1}).
Reference acceleration	1671	4124.	msec	Reference acceleration is 4124mm/sec^2.
Speed at P1	19541	515	0.01%	- Assuming that parameter No. 1672 (bell-shaped acceleration change time) is set to 40 (msec), the speed at P1 will be 2474 mm/min from the calculation described on the previous page. Set its ratio to the rapid traverse rate, or 5.15%. 0.0515=2474/48000 - If parameter No. 1672 (bell-shaped acceleration change time) is not set, set about 5.00%.
Speed at P2	19542	6666	0.01%	Since the torque characteristic is constant (90 Nm) in speeds reaching 2000 min^{-1}, set P2 to the ratio of the speed (32000 mm/min) at 2000 min^{-1} to the rapid traverse rate (48000 mm/min). 0.6666=32000 / 48000
Speeds at P3 to P4	19543 to 19544	0	0.01%	P3 to P4 are skipped because the torque drops almost linearly from the speed 2000 (min^{-1}) to 3000 (min^{-1}).
Acceleration at P0	19545,19551, 19557,19563	9356	0.01%	Set half the acceleration at P1, or 9356.
Acceleration at P1	19546,19552, 19558,19564	18712	0.01%	At P1, 90(Nm) can be used for the acceleration/deceleration, so set the ratio 7717 (mm/sec^2) to 4124 (mm/sec^2). (1.8712 = 7717/4124)
Acceleration at P2	19547,19553 19559,19565	18712	0.01%	At P2, set the same speed as that at P1.
Acceleration at P3 to P4	19548 to 19549, 19554 to 19555, 19560 to 19561 19566 to 19567	0	0.01%	0 is set because P3 to P4 are skipped.
Acceleration at P5	19550,19556 19562,19568	14345	0.01%	At P5, 69(Nm) can be used for the acceleration/deceleration, so set the ratio 5916 (mm/sec^2) to 4124 (mm/sec^2). (1.4345 = 5916 / 4124)

With the above parameter settings, the acceleration pattern will be shown as the following figure. From speeds from 0 mm/min to 2474 mm/min, the acceleration as calculated in accordance with the acceleration pattern is applied; from speeds from 2474 mm/min to 32000 mm/min, an acceleration of 7716 mm/sec^2; and from speeds from 32000 mm/min to 48000 mm/min, the acceleration as calculated in accordance with the acceleration pattern.

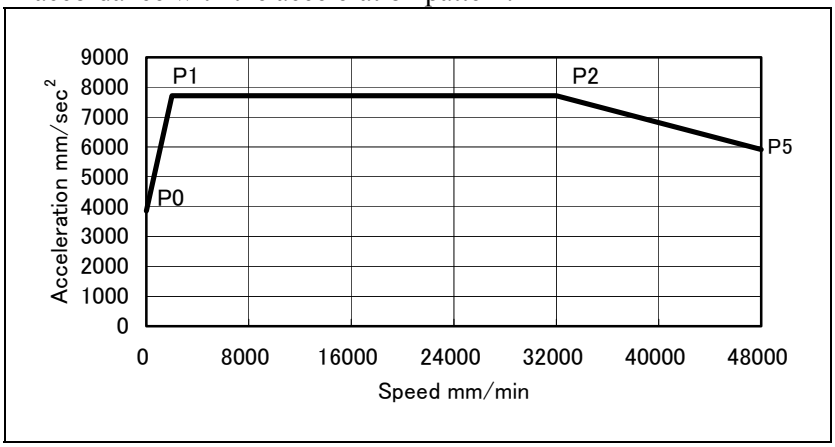


Fig. 20.4 (f) Acceleration pattern with consideration of friction

NOTE
The values in the model αiS 30/4000 speed-torque characteristic diagram are just typical ones. The values will change depending on the digital servo software, parameters, input voltage, and other factors.
The optimum acceleration will, therefore, change due to the characteristics of the machine.

- Examples of setting if the acceleration pattern differs depending on whether acceleration or deceleration is in progress and whether the movement is in the minus or plus direction

From the effect of gravity and friction, torque for acceleration/deceleration is different on each condition, such as acceleration, deceleration or plus move (up), minus move (down).
The following example is for the vertical axis and gravity and friction torque are assumed as follows.
Torque of gravity : 20 (Nm)
Torque of friction : 10 (Nm)
Because these value are different on each machine, it is necessary to observe the output torque on the actual machine for deciding acceleration pattern. The conditions are the same as the previous example.
(Condition)

Motor speed at rapid traverse : 3000(min^{-1})
Ball screw pitch : 16(mm)
Inertia : The machine inertia is to be 2.0 times higher than that of the rotor.
Rotor inertia : 0.0099(Kgm^2)

Maximum torque : 100(Nm) Speed 0 to 2000(min^{-1})
Torque at rapid traverse : 79(Nm) Speed 3000(min^{-1})
Minimum torque : 58(Nm) Speed 4000(min^{-1})

(1) In case of plus move (up) and acceleration
Because torque of Gravity and friction work against the output torque of motor, the torque for acceleration/deceleration is as follows.
Maximum torque : 70(=100-20-10) (Nm)
Speed 0 to 2000(min^{-1})
Torque at rapid traverse : 49(=79-20-10) (Nm)
Speed 3000(min^{-1})
Minimum torque : 28(=58-20-10) (Nm)
Speed 4000(min^{-1})

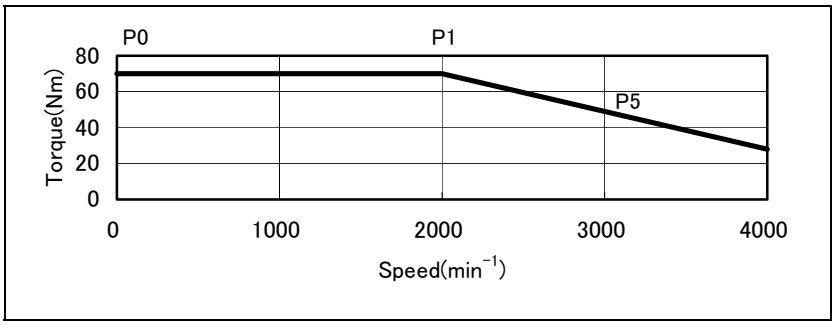


Fig. 20.4 (g) Torque for Acc/Dec in case of + move and acceleration

Parameter setting is as follows,

	Parameter No.	Setting	Unit	Remarks
Acceleration at P0	19545	7277	0.01%	Set half the acceleration at P1, or 7277.
Acceleration at P1-P2	19546,19547	14554	0.01%	At P1 and P2, 70(Nm) can be used for the acceleration/ deceleration, so set the ratio 6002 (mm/sec^2) to 4124 (mm/sec^2). (1.4554 = 6002 / 4124)
Acceleration at P3-P4	19548 to 19549	0	0.01%	0 is set because P3 to P4 are skipped.
Acceleration at P5	19550	10187	0.01%	At P5, 49(Nm) can be used for the acceleration/deceleration, so set the ratio 4201 (mm/sec^2) to 4124 (mm/sec^2). (1.0187 = 4201 / 4124)

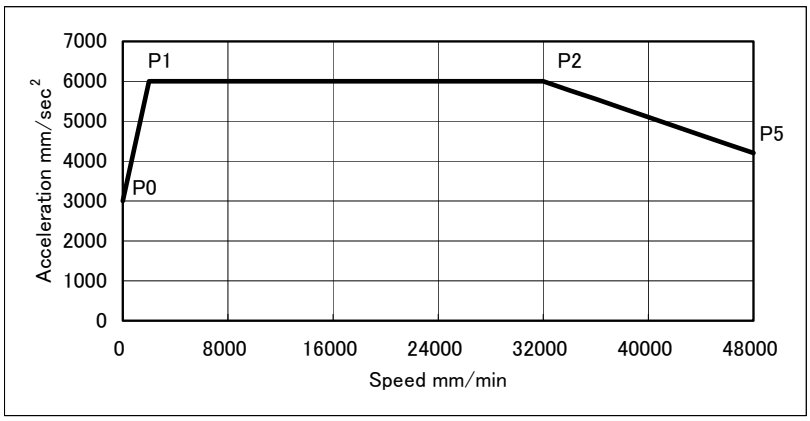


Fig. 20.4 (h) Acceleration pattern in case of + move and acceleration

(2) In case of plus move (up) and deceleration
Because torque of Gravity and friction work forward to the output torque of motor, the torque for acceleration/deceleration is as follows.

Maximum torque : 130(=100+20+10) (Nm)
Speed 0 to 2000(min^{-1})
Torque at rapid traverse : 109(=79+20+10) (Nm)
Speed 3000(min^{-1})
Minimum torque : 88(=58+20+10) (Nm)
Speed 4000(min^{-1})

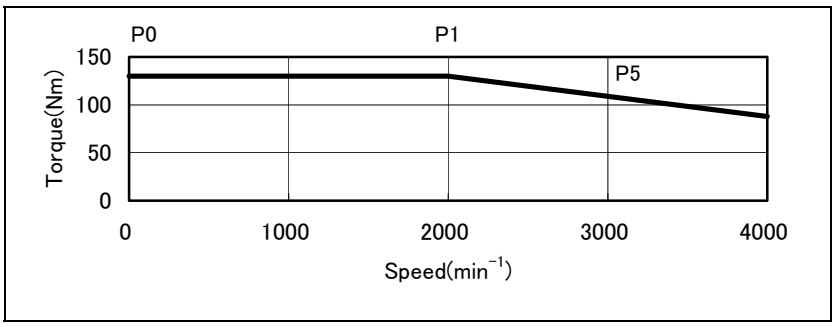


Fig. 20.4 (i) Torque for Acc/Dec in case of + move and deceleration

Parameter setting is as follows,

	Parameter No.	Setting	Unit	Remarks
Acceleration at P0	19557	13514	0.01%	Set half the acceleration at P1, or 13514.
Acceleration at P1-P2	19558, 19559	27027	0.01%	At P1 and P2, 130(Nm) can be used for the acceleration/ deceleration, so set the ratio 11146 (mm/sec^2) to 4124 (mm/sec^2). (2.7027 = 11146 / 4124)
Acceleration at P3-P4	19560 to 19561	0	0.01%	0 is set because P3 to P4 are skipped.
Acceleration at P5	19562	22662	0.01%	At P5, 109(Nm) can be used for the acceleration/deceleration, so set the ratio 9346 (mm/sec2) to 4124 (mm/sec2). (2.2662 = 9346 / 4124)

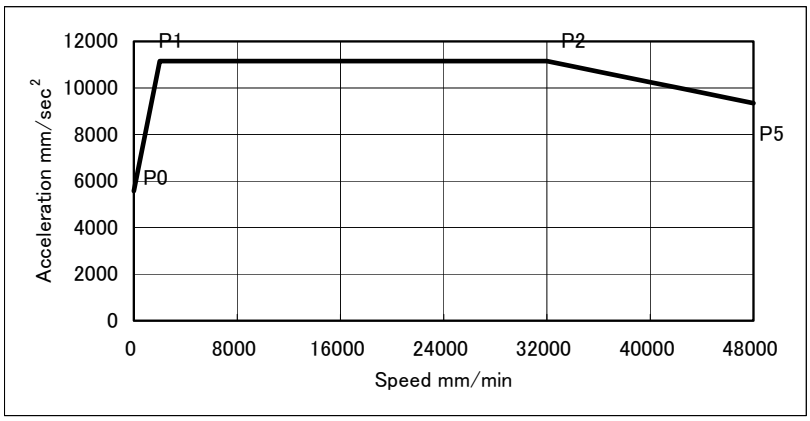


Fig. 20.4 (j) Acceleration pattern in case of + move and deceleration

(3) In case of minus move (down) and acceleration
Because torque of Gravity works forward to the output torque of motor and torque of friction works against the output torque of motor, torque for acceleration/deceleration is as follows.

Maximum torque : 110(=100+20-10) (Nm)
Speed 0 to 2000(min^{-1})
Torque at rapid traverse : 89(=79+20-10) (Nm)
Speed 3000(min^{-1})
Minimum torque : 68(=58+20-10) (Nm)
Speed 4000(min^{-1})

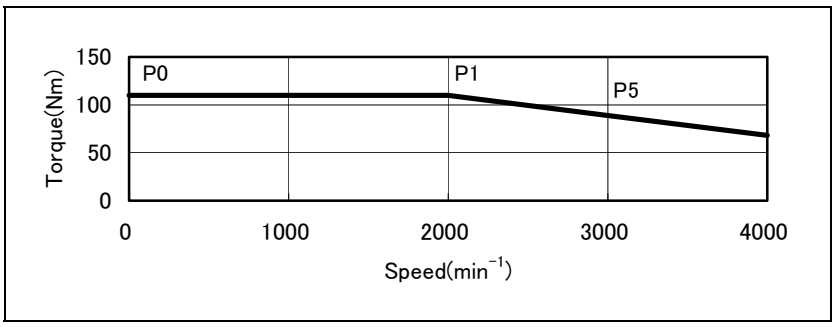


Fig. 20.4 (k) Torque for Acc/Dec in case of - move and acceleration

Parameter setting is as follows,

	Parameter No.	Setting	Unit	Remarks
Acceleration at P0	19551	11435	0.01%	Set half the acceleration at P1, or 11435.
Acceleration at P1-P2	19552,19553	22869	0.01%	At P1 and P2, 110(Nm) can be used for the acceleration/ deceleration, so set the ratio 9431 (mm/sec^2) to 4124 (mm/sec^2). (2.2869 = 9431 / 4124)
Acceleration at P3-P4	19554 to 19555	0	0.01%	0 is set because P3 to P4 are skipped.
Acceleration at P5	19556	18504	0.01%	At P5, 89(Nm) can be used for the acceleration/deceleration, so set the ratio 7631 (mm/sec^2) to 4124 (mm/sec^2). (1.8504 = 7631 / 4124)

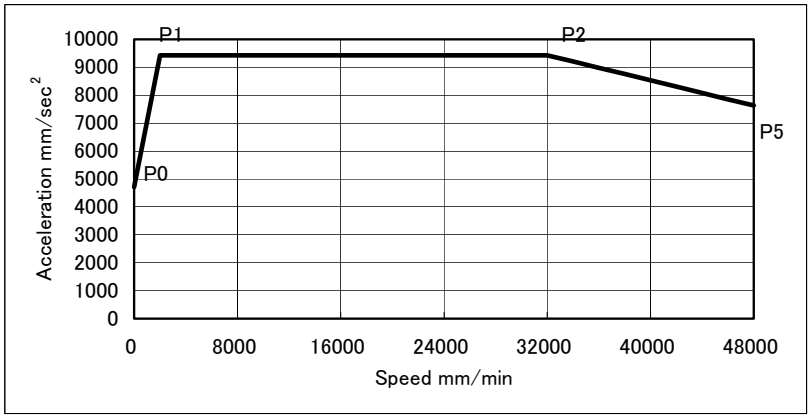


Fig. 20.4 (l) Acceleration pattern in case of - move and acceleration

(4) In case of minus move (down) and deceleration
Because torque of Gravity works against the output torque of motor and torque of friction works forward to the output torque of motor, torque for acceleration/deceleration is as follows.

Maximum torque : 90(=100-20+10) (Nm)
Speed 0 to 2000(min^{-1})

Torque at rapid traverse : 69(=79-20+10) (Nm)
Speed 3000(min^{-1})

Minimum torque : 48(=58-20+10) (Nm)
Speed 4000(min^{-1})

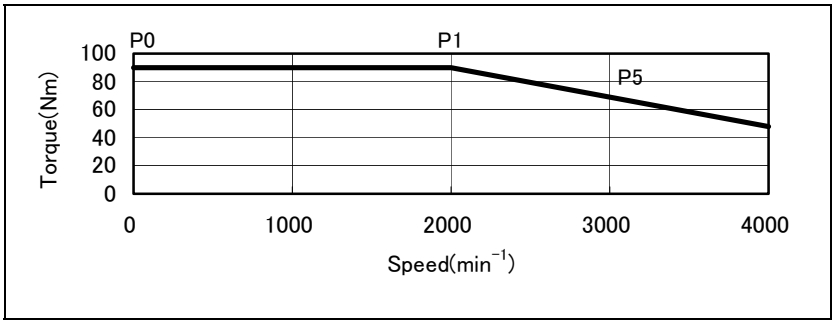


Fig. 20.4 (m) Torque for Acc/Dec in case of - move and deceleration

Parameter setting is as follows,

	Parameter No.	Setting	Unit	Remarks
Acceleration at P0	19563	9356	0.01%	Set half the acceleration at P1, or 9356.
Acceleration at P1-P2	19564,19565	18712	0.01%	At P1 and P2, 90(Nm) can be used for the acceleration/ deceleration, so set the ratio 7717 (mm/sec^2) to 4124 (mm/sec^2). (1.8712 = 7717 / 4124)
Acceleration at P3-P4	19566 to 19567	0	0.01%	0 is set because P3 to P4 are skipped.
Acceleration at P5	19568	14345	0.01%	At P5, 69(Nm) can be used for the acceleration/deceleration, so set the ratio 5916 (mm/sec^2) to 4124 (mm/sec^2). (1.4345 = 5916 / 4124)

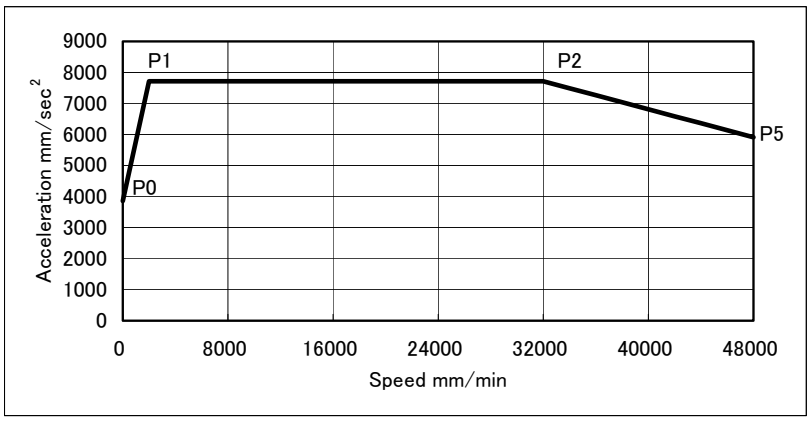


Fig. 20.4 (n) Acceleration pattern in case of - move and deceleration

Limitation

- Linear type positioning

Optimum torque acceleration/deceleration is not enabled unless linear-type positioning is set (bit 1 (LRP) of parameter No. 1401 is set to 1).

- Modes and conditions

Optimum torque acceleration/deceleration is enabled when AI contour control mode is active and the conditions under which look-ahead acceleration/deceleration before interpolation is performed are satisfied. Note, however, that if 1 is set to bit 0 (FAE) of parameter No. 11240, optimum torque acceleration/deceleration is enabled even if AI contour control is disabled.

- Target axes

Optimum torque acceleration/deceleration cannot be performed for a specific axis only. All axes operated by programmed commands are targeted for optimum torque acceleration/deceleration. This means that the PMC axes are excluded.

- Acceleration pattern

In the same direction move, it is necessary that the acceleration during deceleration is set 1/3 of one during acceleration at least.

Also, the acceleration pattern data must be set so that the time required for the deceleration from the rapid traverse rate to a speed of 0 does not exceed 4000 (msec). This does not include the acceleration change time of bell-shaped acceleration/deceleration.

If the deceleration acceleration ratio or the time required for the deceleration to a speed of 0 exceeds the above range, alarm (DS1710) will be issued at the time rapid traverse is executed.

A slight error will occur between the specified acceleration and the actual acceleration.

- Relationship with the customer board

Optimum torque acceleration/deceleration cannot be used from the customer board.

- Tool center point control

Optimum torque acceleration/deceleration is disabled in the tool center point control mode (except startup and cancellation). In this case, positioning is accelerated/decelerated with the reference acceleration.

20.5 HIGH-SPEED CYCLE MACHINING

This function converts the shape to be machined into a data group that can be subject to high-speed pulse distribution, using the macro executor, calls the data group with a CNC command (G05 command), and executes it.

Format

G05 P10xxx Lxxx ;

P10001 to P10999: Start number of the machining cycle to call

L1 to L999 : Repeat count of the machining cycle (assumed to be L1 if omitted).

Cycle data for up to 999 cycles can be created. Use address "P" to specify which cycle to use for machining. It is possible to call multiple cycles in succession using connection information (in a header) and execute them.
Use address "L" to specify the repeat count of the machining cycle. It is possible to specify the repeat count (in a header) for each cycle.
Connection of cycles and the repeat count are explained with the following example.

Example)

If
Cycle 1 Connection information 2 Repeat count 1
Cycle 2 Connection information 3 Repeat count 3
Cycle 3 Connection information 0 Repeat count 1
G05 P10001 L2;
the following cycles are executed in succession:
1,2,2,2,3,1,2,2,2,3

NOTE

1 If an attempt is made to issue the function in the modes below, an alarm is issued.
- Hypothetical axis interpolation (G07)
- Cylindrical interpolation (G07.1)
- Polar coordinate interpolation (G12.1)
- Polar coordinate command (G16)
- Spindle speed fluctuation detection (G26)
- Tool radius·Tool nose radius compensation (G41/G42)
- Normal direction control (G41.1/G42.1)
- Scaling (G51)
- Programmable mirror image (G51.1)
- Coordinate system rotation (G68)
- Canned cycle (G81 to G89)
- Constant surface speed control (G96)
- Macro interrupt (M96)

2 During high-speed cycle machining, single block stop is disabled.

3 During high-speed cycle machining, handle interruption is disabled.

21 AXIS CONTROL FUNCTIONS

Chapter 21, "AXIS CONTROL FUNCTIONS", consists of the following sections:

21.1 AXIS SYNCHRONOUS CONTROL 692
21.2 POLYGON TURNING (G50.2, G51.2).....709
21.3 SYNCHRONOUS, COMPOSITE AND SUPERIMPOSED CONTROL BY PROGRAM COMMAND (G50.4, G51.4, G50.5, G51.5, G50.6, AND G51.6).....715
21.4 ROTARY AXIS ROLL-OVER.....719
21.5 ARBITRARY ANGULAR AXIS CONTROL.....721
21.6 TOOL RETRACT AND RECOVER 732
21.7 ELECTRONIC GEAR BOX 736
21.8 TANDEM CONTROL 771

21.1 AXIS SYNCHRONOUS CONTROL

Overview

When a movement is made along one axis by using two servo motors as in the case of a large gantry machine, a command for one axis can drive the two motors by synchronizing one motor with the other. Moreover, by using a feedback signal from each motor, a positional difference (synchronous error) between the two motors is detected to compensate for the synchronous error. When a synchronous error exceeding a set value occurs, a synchronous error check can be made to issue an alarm and stop a movement along the axis.

An axis used as the reference for axis synchronous control is referred to as a master axis (M-axis), and an axis along which a movement is made in synchronism with the master axis is referred to as a slave axis (S-axis).

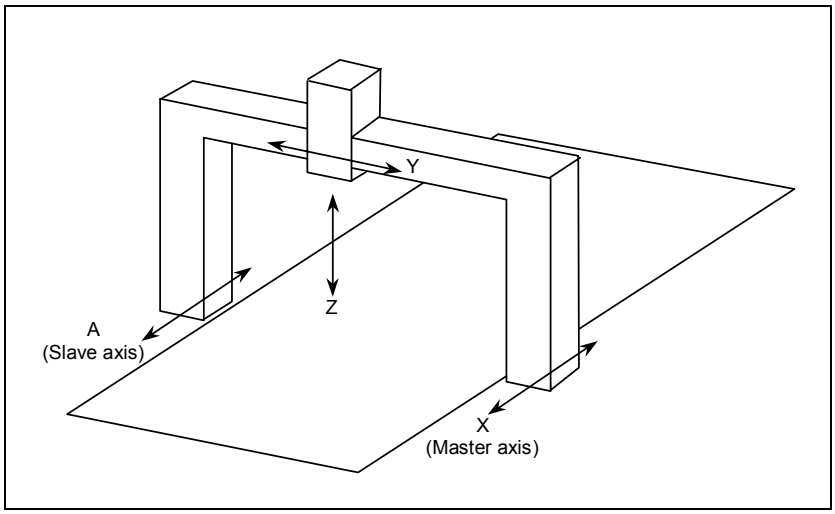


Fig. 21.1 (a) Example of machine with X and A being synchronous axes

Even when synchronous error compensation is not used, the synchronous establishment function can be used for automatic compensation to eliminate a machine coordinate error in cases such as emergency stop cancellation.

An external signal can be used to turn synchronization on and off. When synchronization is turned on and off using an external signal, synchronous error compensation cannot be used.

21.1.1 Axis Configuration for Axis Synchronous Control

Explanation

- Master axis and slave axis for axis synchronous control

An axis used as the reference for axis synchronous control is referred to as a master axis (M-axis), and an axis along which a movement is made in synchronism with the master axis is referred to as a slave axis (S-axis).
By setting the axis number of a master axis in the parameter No. 8311 of the slave axis, the axis configuration for axis synchronous control is determined.

- Synchronous operation and normal operation

Operation where axis synchronous control is turned on (enabled) to make a movement along the slave axis in synchronism with the master axis is referred to as synchronous operation. Operation where axis synchronous control is turned off (disabled) to make movements along the master axis and slave axis independently of each other is referred to as normal operation.

(Example)
Automatic operation when the master axis is the X-axis and the slave axis is the A-axis

In synchronous operation, movements are made along the X-axis and A-axis according to the programmed command Xxxxx for the master axis.
In normal operation, movements are made along the master axis and slave axis independently of each other as in the case of normal CNC control. The programmed command Xxxxx makes a movement along the X-axis. The programmed command Aaaaa makes a movement along the A-axis. The programmed command Xxxxx Aaaaa makes movements along the X-axis and A-axis at the same time.

The mode of operation can be switched between synchronous operation and normal operation by an input signal, or synchronous operation can be performed at all times. Which mode to use can be set using bit 5 (SCA) of parameter No. 8304.

- Switching between synchronous operation and normal operation by using an input signal

When bit 5 (SCA) of parameter No. 8304 is set to 0 for the slave axis, the signal SYNCx/SYNCJx (with x representing a slave axis number) is used to switch between synchronous operation and normal operation. When SYNCx/SYNCJx = 1, synchronous operation is selected. When SYNCx/SYNCJx = 0, normal operation is selected.
Synchronous error compensation cannot be used when the mode of operation is switched between synchronous operation and normal operation.
During feed axis synchronization control, the output signal SYNOx is set to "1".

- Setting for using synchronous operation at all times

When bit 5 (SCA) of parameter No. 8304 for the slave axis is set to 1, synchronous operation is performed at all times, regardless of the setting of the signal SYNCx/SYNCJx.

- Synchronous control axis name

The name of a master axis and the name of a slave axis may be the same or may be different from each other.

- Restrictions on using the same name for the master axis and slave axis

If the same axis name is assigned to the master axis and slave axis, manual operation only is allowed in normal operation. Automatic operation and manual numeric command cannot be performed.

- Setting of an axis name subscript

A subscript can be attached to an axis name like X1, X2, XM, and XS. If the same axis name is used for multiple axes, and a unique subscript is assigned to each of those axes, the axes can be distinguished from each other on the screen display, or which of those axes issued an alarm can be identified.
Set a subscript in parameter No. 3131.

- Setting of multiple slave axes

One master axis can have multiple slave axes.

(Example)

In the example below, movements along the X1-axis and X2-axis are made in synchronism with the XM-axis.

Axis name indication	Controlled axis number	Axis name Parameter (No. 1020)	Subscript Parameter (No.3131)	Master axis number Parameter (No.8311)	Operation
XM	1	88	77	0	
Y	2	89	0	0	
X1	3	88	49	1	A movement is made in synchronism with the XM-axis.
X2	4	88	50	1	A movement is made in synchronism with the XM-axis.

When one master axis has multiple slave axes, synchronous error compensation, synchronous establishment, and synchronous error check are performed for each slave axis independently.

- Combination with tandem control

Tandem control can be used with each of the master and slave axes. The same restriction on axis arrangement as imposed in the case of normal tandem control is imposed. No particular restriction is imposed on axis synchronous control.

- Axis selection on the screen display

On a screen such as the current position display screen, a slave axis is also displayed. The display of a slave axis can be disabled by setting bit 0 (NDP) of parameter No. 3115 to 1 and setting bit 1 (NDA) of parameter No. 3115 to 1.

- Axis selection in actual cutting feedrate display

By setting bit 2 (SAF) of parameter No. 8303 to 1 for a slave axis, the slave axis can be included in an actual cutting feedrate display calculation during synchronous operation.

- Axis synchronous control with an absolute-position detector

When bit 7 (SMA) of parameter No. 8302 is set to 1 to attach an absolute-position detector, and bit 4 (APZ) of parameter No. 1815 for an axis placed in synchronous operation is turned off, APZ for the axis (axes) placed together in synchronous operation is also turned off.

- Slave axis mirror image

By setting parameter No. 8312, a mirror image can be applied to a slave axis placed in synchronous operation. When the mirror image function is enabled, the direction in which the absolute and relative coordinates change is the same as for the machine coordinates.

At this time, synchronization error compensation, synchronization establishment, synchronization error check, and correction mode cannot be used.

The mirror image set by bit 0 (MIR) of parameter No. 0012 cannot be applied to the slave axis. Because this mirror image differs from the mirror image set by parameter MIR, it does not affect input signal MIx <G106> or output signal MMIx <F108>.

- External machine coordinate system shift

Bit 7 (SYE) of parameter No. 8304 can be set to 1 for the slave axis to shift the slave axis by the same amount as specified for the master axis when external machine coordinate system shift is specified by external data input/output for the master axis in synchronous control.

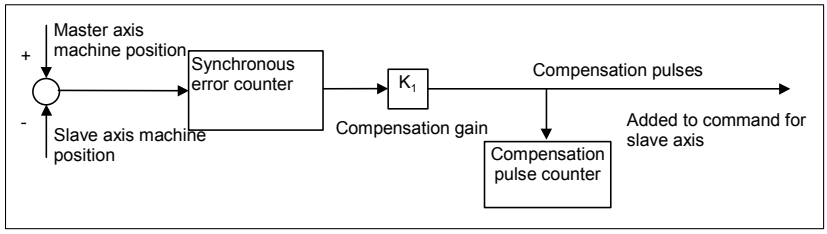
21.1.2 Synchronous Error Compensation

Explanation

When a synchronous error value exceeding the zero width set in parameter No. 8333 is detected, compensation pulses for synchronous error reduction are calculated and added onto the command pulses output for the slave axis. This compensation is not performed in servo-off state, servo alarm state, follow-up operation, and modification mode.
Compensation pulses are calculated by multiplying the synchronous error value between master and slave axes by a compensation gain.

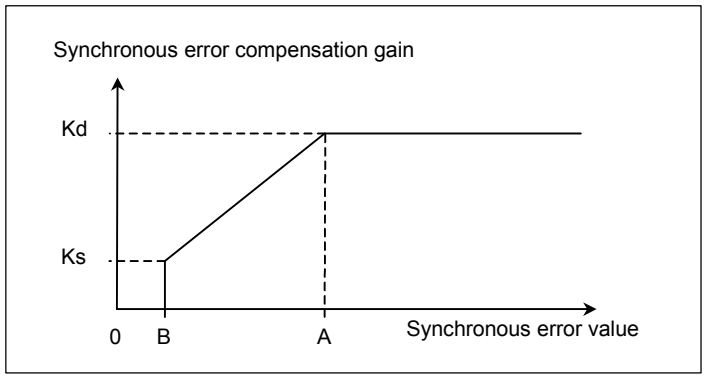
Compensation pulses = synchronous error × (Ci/1024)
Ci: Compensation gain (parameter No. 8334)

Before synchronous error compensation can be performed, bit 5 (SCA) of parameter No. 8304 must be set to 1 to perform synchronous operation at all times. Furthermore, the first synchronous establishment after power-up must already be performed by a manual reference position return or another operation.



- Synchronous error compensation smooth suppress function

When bit 6 (SMS) of parameter No. 8304 is set to 1, the synchronous error compensation smooth suppress function is enabled. With this function, another set of parameters for a synchronous error zero width and synchronous error compensation gain (B and Ks in the figure below) can be set. So, even a small synchronous error can be reduced smoothly as shown below.



A: Synchronous error zero width (parameter No. 8333)
B: Synchronous error zero width 2 (parameter No. 8335)
(0 < B < A)
Kd: Synchronous error compensation gain (parameter No. 8334)

Ks: Synchronous error compensation gain 2 (parameter No. 8336) (0 < Ks < Kd)
Er: Synchronous error value between the current master axis and slave axis
K: Current synchronous error compensation gain for Er

1. When Er < B, compensation is not performed. (K = 0)
2. When B < Er < A

$$K = Ks + \frac{(Er - B)(Kd - Ks)}{A - B}$$

Compensation is performed with the above gain:
3. When Er > A, compensation is performed with a gain of K = Kd.

21.1.3 Synchronous Establishment

Explanation

Upon power-up or after emergency stop cancellation, the machine positions on the master axis and slave axis under axis synchronous control are not always the same. In such a case, the synchronous establishment function matches the machine position on the master with that on the slave axis.

- Synchronous establishment method

The method of synchronous establishment used when synchronous error compensation is performed differs from that used when synchronous error is not performed.

- Synchronous establishment based on synchronous error compensation

When synchronous error compensation is performed (when bit 3 (CLP) of parameter No. 8304 for the slave axis is set to 1), synchronous establishment is performed in the same way as synchronous error compensation. This means that the positional difference between the master axis and slave axis is regarded as a synchronous error, and pulses produced by multiplying the positional difference by a synchronous error compensation gain are output for the slave axis. So, the parameter No. 8334 for specifying a synchronous error compensation gain must be set before synchronous establishment can be performed.

If the parameter No. 8333 for specifying a synchronous error compensation zero width is set, no further synchronous establishment is performed after the positional difference between the master axis and slave axis becomes the zero width or below.

- Synchronous establishment based on machine coordinates

To perform synchronous establishment when synchronous error compensation is disabled (when bit 3 (CLP) of parameter No. 8304 for the slave axis is set to 0), enable synchronous establishment based on machine coordinates by setting bit 7 (SOF) of parameter No. 8303 to 1. This method of synchronous establishment outputs the machine coordinate difference between the master axis and slave axis as command pulses for the slave axis to establish synchronization. A machine coordinate difference is output at a time as command pulses. So, if the compensation value is large, the machine abruptly makes a large movement. Taking this into consideration, set a maximum allowable compensation value to be used for synchronous establishment in parameter No. 8325. As a maximum allowable compensation value, set a maximum allowable value by which the machine may move abruptly. If a compensation value is larger than the value set in this parameter, an alarm SV0001 is issued, and synchronous establishment is not performed. Moreover, when parameter No. 8325 is set to 0, synchronous establishment is not performed.

The result of comparing the positional difference between the master axis and slave axis with a maximum allowable compensation value for synchronous establishment can be checked using the synchronous establishment enable state output signal SYNOF <F0211>.

- First synchronous establishment after power-up

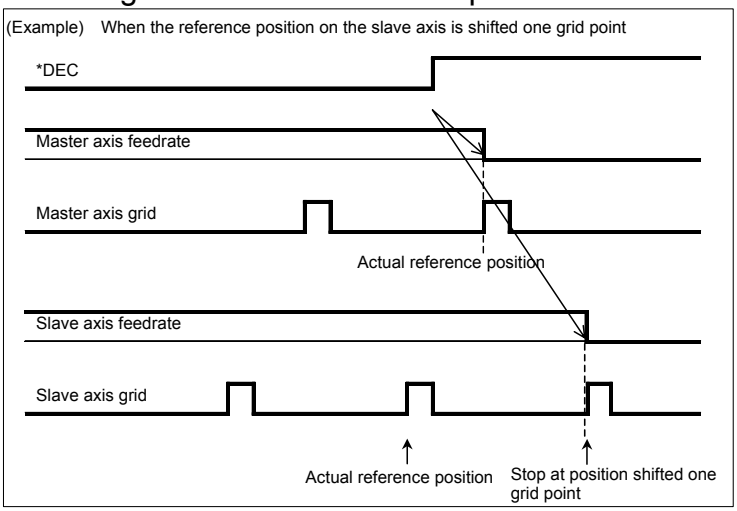
Two methods of performing the first synchronous establishment after power-up are available. One method is based on manual reference position return operation, and the other is based on absolute position detection.
Until this synchronous establishment is completed, synchronous error compensation is disabled. However, a synchronous error check is made.

- Synchronous establishment based on manual reference position return operation

When manual reference position return operation is performed along axes under axis synchronous control, the machine is placed at the reference position on the master axis and slave axis according to the same sequence as for normal reference position return operation.
When synchronous error compensation is used, the synchronous error counter is reset and synchronous error compensation is started upon completion of reference position return along both axes.
The sequence is the same as the grid method for one axis only. However, only the deceleration signal for the master axis is used. When the deceleration signal is set to 0, the machine gradually stops along the master axis and slave axis, then an FL feedrate is set. When the deceleration signal is set to 1, the machine moves to a grid point along each of the master axis and slave axis, then stops.

NOTE

When the grid position difference between the master axis and slave axis is large, a reference position shift can occur, depending on the timing of the *DEC signal set to 1. In the example below, the shift along the slave axis is so large that the position shifted one grid point from the actual reference position is regarded as the reference position.



In such a case, match the grid position according to Subsection 21.1.4, "Automatic Setting for Grid Position Matching."

- Synchronous establishment based on absolute position detection

When an absolute-position detector is used as the position detector, the machine positions on the master axis and slave axis are found at power-up time for automatic establish synchronization.

- Synchronous establishment after emergency stop cancellation, etc.

Synchronous establishment is also performed when servo position control is turned on, for example, at emergency stop cancellation, servo alarm cancellation, or servo-off cancellation time.
However, synchronous establishment is not performed at the time of axis removal cancellation. So, synchronous establishment based on manual reference position return operation is required as in the case of power-up time.

- One-direction synchronous establishment

When synchronous error compensation is disabled, synchronous establishment can be performed by setting bit 0 (SSO) of parameter No. 8305 to 1 to move the machine in one direction along the master axis and slave axis. The move direction depends on the reference position setting based on bit 0 (SSA) of parameter No. 8304. When SSA = 0, for example, the machine coordinate on the master axis or slave axis, whichever larger, is used as the reference point. So, the machine moves in the + direction along the axes.
When bit 1 (SSE) of parameter No. 8305 is set to 1, normal synchronous establishment is performed instead of one-direction synchronous establishment after an emergency stop.

21.1.4 Automatic Setting for Grid Position Matching

Explanation

Before axis synchronous control can be performed, the reference position on the master axis must be matched with the reference position on the slave axis. With this function, the CNC automatically matches the reference positions (grid positions) on the master axis and slave axis under axis synchronous control.

[Operation procedure]
The procedure below is usable when bit 0 (ATE) of parameter No. 8303 is set to 1.

1. Set bit 1 (ATS) of parameter No. 8303 to 1.
2. Turn off the power then turn on the power.
3. Set the REF mode (or JOG mode in the case of reference position setting without dogs) when synchronous operation is ready, and make movements in the reference position return direction along the master axis and slave axis.
4. The movements along the master axis and slave axis automatically stop, and a grid difference value is set in parameter No. 8326. At this time, bit 1 (ATS) of parameter No. 8303 is set to 0, and the power-off request alarm PW0000 is issued.
5. Turn off the power then turn on the power again.
6. Perform normal reference position return operation.

NOTE
1 Parameter setting
When bit 1 (ATS) of parameter No. 8303 is set, bit 4 (APZ) of parameter No. 1815 and parameter No. 8326 for the master axis and slave axis are set to 0.
When the operator sets parameter No. 8326 (MDI, G10L50), bit 0 (ATE) of parameter No. 8303 is set to 0.
2 This function cannot be used together with the reference position shift function.

21.1.5 Synchronous Error Check

Explanation

A synchronous error value is monitored at all times. If an error exceeding a certain limit is detected, an alarm is issued and the movement along the axis is stopped.
When synchronous error compensation is performed, a check considering a positional deviation as well is made.
When synchronous error compensation is not performed, a synchronous error check based on machine coordinates is made.

- Check made when synchronous error compensation is performed

When synchronous error compensation is performed, a check considering a positional deviation is made. The actual machine position shift considering a servo positional deviation as well is checked. Depending on the value of a synchronous error, one of two alarms is issued: an alarm (DS alarm) for deceleration stop, and an alarm (SV alarm) for turning off the servo system immediately. This check is enabled when a value other than 0 is set in the parameters Nos. 8331 and 8332 for specifying a maximum allowable synchronous error. When using this method of checking a synchronous error value, disable "synchronous error check based on machine coordinates" described later by setting parameter No. 8314 to 0.

- Synchronous error excessive alarm 1 (DS0002)

When a synchronous error exceeding the value set in parameter No. 8331 is detected, synchronous error excessive alarm 1 is issued. When synchronous error excessive alarm 1 is issued, the motor gradually stops. At this time, synchronous error compensation remains enabled, so that the synchronous error is reduced by compensation. Accordingly, as the synchronous error value becomes smaller than the maximum allowable value, the alarm can be reset. If the alarm cannot be reset, the synchronous error needs to be manually corrected by selecting the correction mode described later.

- Synchronous error excessive alarm 2 (SV0002)

When a synchronous error exceeding the value set in parameter No. 8332 is detected, synchronous error excessive alarm 2 is issued. Before synchronous establishment is performed at power-up time, a value obtained by multiplying the value set in parameter No. 8332 by the coefficient set in parameter No. 8330 is used for judgement. When synchronous error excessive alarm 2 is issued, the motor immediately stops as in the case of other servo alarms. Accordingly, the positional difference between the master axis and slave axis remains uncorrected, so that the alarm cannot be reset usually. In this case, the synchronous error needs to be manually corrected by selecting the correction mode described later.

- Synchronous error check based on machine coordinates

When synchronous error compensation is not performed, a synchronous error check based on machine coordinates is made.

The machine coordinate on the master axis is compared with that on the slave axis. When the error between the machine coordinates exceeds the value set in parameter No. 8314, the SV0005 alarm is issued, and the motor is stopped immediately.

A check can be made even in the emergency stop, servo off, and servo alarm states.

If a synchronous error check is made when the mode of operation is switched between synchronous operation and normal operation with an input signal, an error check is made even in normal operation. So, even if the axis synchronous control selection signal (SYNCx) or the axis synchronous control manual feed selection signal (SYNCJx) is set to 0 by mistake during synchronous operation, damage to the machine can be prevented.

The machine coordinates on the master axis and slave axis can be checked using the machine coordinate match state output signal SYNMT <F0210>.

- Synchronous error check based on a positional deviation value

The servo positional deviation value of the master axis and slave axis is monitored during axis synchronous control. When the positional deviation value exceeds the limit value set in parameter No. 8323, the DS0001 alarm is issued, and the axis synchronous control positional deviation error alarm signal <F403.0> is output.

The DS0001 alarm is issued to the master axis and slave axis.

When bit 4 (SYA) of parameter No. 8301 is set to 1, the positional deviation limit value of the master axis and slave axis is checked even if a servo-off occurs during axis synchronous control.

21.1.6 Methods of Alarm Recovery by Synchronous Error Check

Explanation

To recover from an alarm issued as a result of synchronous error check, two methods are available. One method uses the correction mode, and the other uses normal operation.
If the mode of operation is switched between synchronous operation and normal operation by using an input signal, only the method using normal operation can be used.
If synchronous operation is used at all times, only the method using the correction mode can be used.

- Procedure for correcting a synchronous error by using the correction mode

Use this method if synchronous operation is used at all times without using an input signal (when bit 5 (SCA) of parameter No. 8304 is set to 1).
When the correction mode is used, synchronous error check can be temporarily disabled, and a movement can be made along the master axis or slave axis to correct a synchronous error.
In the correction mode, synchronous error compensation and error check are not performed, so that an alarm DS0003 is issued as a warning.

1. Select the correction mode, and select an axis along which a movement is to be made by manual master axis feed. Set bit 2 (ADJ) of parameter No. 8304 of the master axis or slave axis to 1 to set the correction mode. Thus, by manual master axis feed, a movement can be made along the axis with this parameter set to 1.
   When this parameter is set to 1, the DS0003 (axis synchronous control correction mode) alarm is issued.

2. Reset the synchronous error excessive alarm.
   In this state, synchronous error compensation and error check are not performed. Be careful.

3. Select the manual mode (jog, incremental feed, or handle).

4. While checking the synchronous error value, make a movement along the master axis or slave axis in the direction that reduces the error.
   If one master axis has multiple slave axes, an attempt to reduce the synchronous error of one slave axis by master axis movement may increase the synchronous error of another slave axis, thus disabling a movement in any direction. In such a case, by setting bit 4 (MVB) of parameter No. 8304 to 1, a movement can be made in a direction that increases the synchronous error.

5. When the synchronous error is reduced to within the allowable value for suppressing the alarm, reset the value of bit 2 (ADJ) of parameter No. 8304 to the original value to switch from the correction mode to the normal synchronization mode.
   Synchronous error compensation and synchronous error check are restarted.

6. Reset the correction mode alarm.

- Method of recovery using normal operation

Use this method when switching between synchronous operation and normal operation by using an input signal.
Use the procedure below for recovery from alarm SV0005.

1. Set SYNCx/SYNCJx (with x representing a slave axis number) to 0 to select normal operation.

2. Set a value greater than the current value in the parameter No. 8314 for specifying a maximum allowable synchronous error, then reset the alarm.

3. Make a movement along the master axis or slave axis by using the manual handle so that the machine coordinates of the master axis and slave axis match to a maximum possible extent.

4. Return the value of parameter No. 8314 for specifying a maximum allowable synchronous error to the original value.

21.1.7 Axis Synchronous Control Torque Difference Alarm

Explanation

If a movement made along the master axis differs from a movement made along the slave axis during axis synchronous control, the machine can be damaged. To prevent such damage, the torque command difference between the two axes is observed. If the difference is abnormal, a servo alarm SV0420 can be issued.

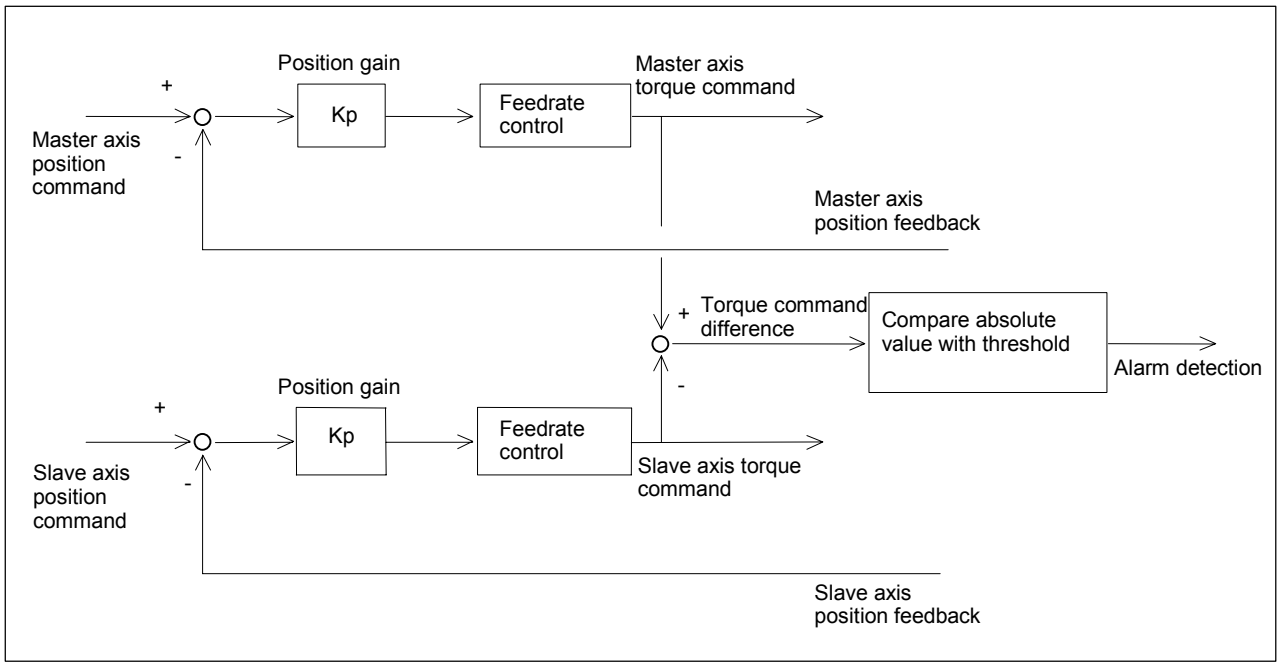


Fig. 21.1.7 (a) System configuration

[Method of use]

Specify the threshold parameter No. 2031 according to the procedure below.

1. Set 0 in parameter No. 2031, and disable the torque difference alarm detection function.
2. To check the absolute value of the torque difference between the synchronous axes, set the parameters below. Set the same value for the two axes placed under axis synchronous control.
   Parameter No. 2115 = 0
   Parameter No. 2151 is as described below.
   - For the servo software 90E0 series, set it to 434 if the setting of parameter No. 1023 is 1, 2, 5, 6, 9, 10… and to 6578 if it is 3, 4, 7, 8, 11, 12…
   - For the servo software 90D0 series, set it to 434.
3. Display the diagnostic screen by pressing the function key

   then the [DGNOS] soft key.
   Diagnose No. 0353 indicates the absolute value of the torque difference between the two axes.

4. Read the absolute torque difference value presented when normal operation is being performed. In the threshold parameter No. 2031, set a value obtained by adding some margin to the read absolute value.
   The absolute torque difference value can be observed with the Servo Guide Mate.

- Enabling/disabling of alarm detection

Alarm detection is enabled when the time set in parameter No. 8327 has elapsed after the servo ready signal SA <F000.6> is set to 1. When the input signal NSYNCA <G059.7> is set to 1, alarm detection is disabled.

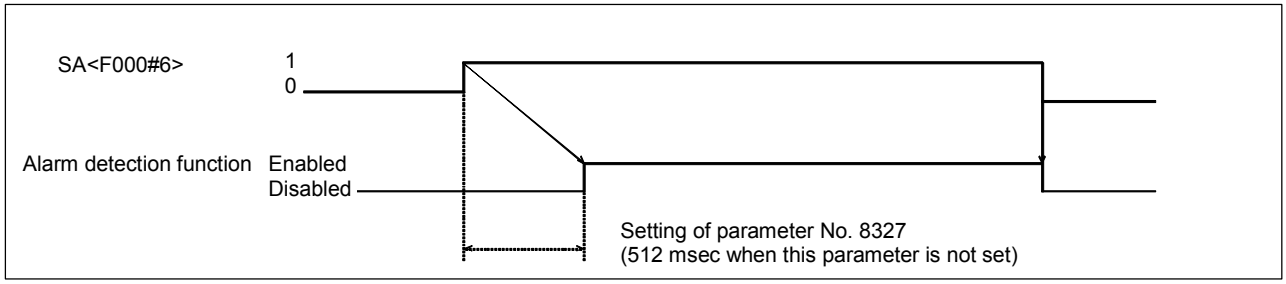


Fig. 21.1.7 (b) Timing chart

When the servo ready signal SA <F000.6> is set to 0, torque difference alarm detection is disabled.

NOTE
The servo axis number combination of the master axis and slave axis synchronized with each other must be such that an odd servo axis number is assigned to the master axis and the next servo axis number is assigned to the slave axis like (1,2) and (3,4).

⚠ CAUTION
1 When making a synchronous error check, ensure that the reference position on the master axis and the reference position on the slave axis must be at the same position.
2 In manual reference position return operation, the same operation is performed along the master axis and slave axis until a deceleration operation starts. After a deceleration operation starts, grid detection is performed for the master axis and slave axis independently of each other.
3 Pitch error compensation and backlash compensation are performed for the master axis and slave axis independently of each other.

NOTE

1 During axis synchronous control, a movement based on the reference position return check (G27), automatic reference position return (G28), 2nd/3rd/4th reference position return (G30), or machine coordinate system selection (G53) command is made as described below according to the setting of bit 7 (SRF) of parameter No. 8304.
   <1> When SRF = 0, the same movement as made along the master axis is made along the slave axis.
   <2> When SRF = 1, a movement is made along the slave axis to the specified position independently of a movement made along the master axis to the specified position.
2 A command not involving a movement along an axis such as the workpiece coordinate system setting command (G92) and local coordinate system setting command (G52) is set with the master axis according to the master axis programming.
3 During synchronous operation, the axis-by-axis signals such as for external deceleration, interlock, and machine lock are enabled for the master axis only. During synchronous operation, those signals for the slave axis are ignored.
4 When switching the synchronization state in a program, be sure to specify M codes (parameter No. 8337 and No. 8338) for turning synchronization on and off. By switching between the input signals SYNCx <G138> and SYNCJx <G140> from the PMC with the M codes, the synchronization state can be switched in the program.
5 When controlled axis removal is performed, the synchronization state is cancelled. When performing controlled axis removal, perform removal for the master axis and slave axis at the same time.
6 If a programmed command is specified for the slave axis during synchronous operation, an alarm PS0213 is issued.
   A programmed command can be specified for the slave axis when switching between synchronous operation and normal operation is set to 0 (with bit 5 (SCA) of parameter No. 8304 set to 0) to select normal operation.
7 Axis synchronous control and PMC axis control cannot be used at the same time.

21.2 POLYGON TURNING (G50.2, G51.2)

Polygon turning means machining a workpiece to a polygonal figure by rotating the workpiece and tool at a certain ratio.

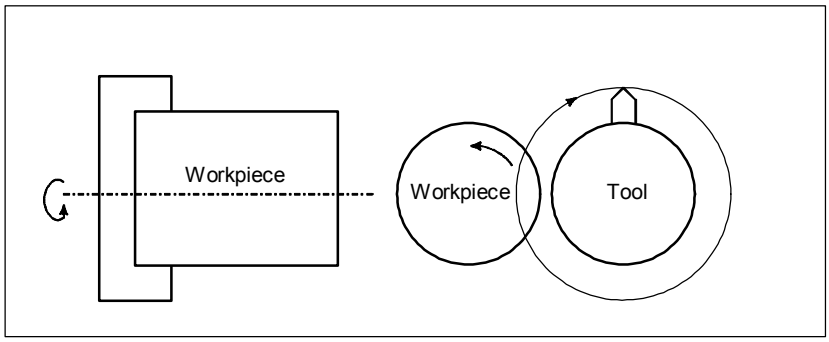


Fig. 21.2 (a) Polygon turning

By changing conditions which are rotation ratio of workpiece and tool and number of cutters, the workpiece can be machined to a square or hexagon. The machining time can be reduced as compared with polygonal figure machining using the polar coordinate interpolation. The machined figure, however, is not exactly polygonal. Generally, polygon turning is used for the heads of square and/or hexagon bolts or hexagon nuts.

As the tool rotary axis, one of the following can be used:

- CNC controlled axis (servo axis)
- Second spindle (Two serial spindles are connected.)

Polygonal machining performed using a servo axis as the tool rotary axis is referred to as polygon turning. Polygonal machining performed using the second spindle as the tool rotary axis is referred to as polygon turning with two spindles.

Function name	Workpiece axis	Tool rotary axis
Polygon turning	Spindle (Either an analog spindle or serial spindle is usable. However, a detector equivalent to a position coder is required.)	Servo axis
Polygon turning with two spindles	Spindle (Serial spindle)	Spindle (Serial spindle)

Explanation

A CNC controlled axis (servo axis) is assigned to the tool rotary axis. This rotary axis of tool is called Y-axis in the following description. As the workpiece axis (spindle), either a serial spindle or analog spindle can be used.

The Y-axis is controlled by the G51.2 command, so that the ratio of the rotation speeds of the spindle (previously specified by S-command) and the tool becomes the specified ratio.

When simultaneous start is specified by G51.2, the one-rotation signal sent from the position codes set on the spindle is detected. After one-rotation signal detection, the Y-axis is controlled using the rotation ratio of the spindle and Y-axis specified by P and Q. So, a position coder needs to be attached to the spindle. This control will be maintained until the polygon turning cancel command is executed (G50.2). Polygon turning is cancelled by any of the following in addition to the G50.2 command:

(1) Power off
(2) Emergency stop
(3) Servo alarm
(4) Reset (external reset signal ERS, reset/rewind signal RRW, and RESET key on the MDI panel)
(5) Occurrence of alarms PS0217 to PS0221, PS0314, and PS05018

NOTE

1 Before polygon turning, reference position return operation on the Y-axis needs to be specified to determine the rotation start position of the tool. This reference position return operation is performed by detecting a deceleration limit as in the case of manual reference position return operation. (By setting bit 7 (PLZ) of parameter No. 7600, reference position return operation can be performed without detecting a deceleration limit.)
2 The rotation direction on the Y-axis is determined by the sign of Q, and is not affected by the rotation direction of the position coder.
3 Among Y-axis position indications, the indication of a machine coordinate (MACHINE) changes within the amount of movement for 0 to 1 revolution as a movement is made on the Y-axis. The absolute coordinate and relative coordinate are not updated. So, when specifying an absolute-position command for the Y-axis after polygon turning mode cancellation, set a workpiece coordinate system after reference position return operation.

NOTE

4 For the Y-axis engaged in polygon turning, jog feed and handle feed are disabled.
5 For the Y-axis not engaged in polygon turning, a move command can be specified as in the case of other controlled axes.
6 The Y-axis engaged in polygon turning is not counted in the number of simultaneously controlled axes.
7 One workpiece must be machined using a fixed spindle speed until the workpiece is finished.
8 Polygon turning with two spindles cannot be used at the same time.
9 G50.2 is the G code for suppressing buffering.

⚠ **CAUTION**

1 During polygon turning, threading cannot be performed.
2 For the Y-axis engaged in synchronous operation, the signals below are valid or invalid:
Signals valid for the Y-axis
- Machine lock
- Servo-off

Signals invalid for the Y-axis
- Feed hold
- Interlock
- Override
- Dry run
(At dry run time, however, the one-rotation signal is not awaited in a G51.2 block.)

Format

G50.2 Polygon turning cancel

G51.2 P_ Q_ ;
P,Q: Rotation ratio of spindle and Y-axis
Specify range:
P: Integer from 1 to 999
Q: Integer from -999 to -1 or from 1 to 999
When Q is a positive value, Y-axis makes positive rotation.
When Q is a negative value, Y-axis makes negative rotation.

NOTE
Specify G50.2 and G51.2 in a single block.

Example

G00 X100. 0 Z20.0 S1000.0 M03 ;	
	(Workpiece rotation speed 1000 (min^{-1}))
G51.2 P1 Q2 ;	(Tool rotation start (tool rotation speed 2000 (min^{-1}))
G01 X80.0 F10.0 ;	(X-axis infeed)
G04 X2.0 ;	(Waiting 2 seconds)
G00 X100.0 ;	(X-axis escape)
G50.2 ;	(Tool rotation stop)
M05 S0 ;	(Spindle stop)

- Principle of polygon turning

The principle of polygon turning is explained below. In the figure below the radius of tool and workpiece are A and B, and the angular speeds of tool and workpiece are α and β. The origin of XY Cartesian coordinates is assumed to be the center of the workpiece.
Simplifying the explanation, consider that the tool center exists at the position
P_o (A, 0) on the workpiece periphery, and the tool nose starts from position P_{to} (A-B, 0).

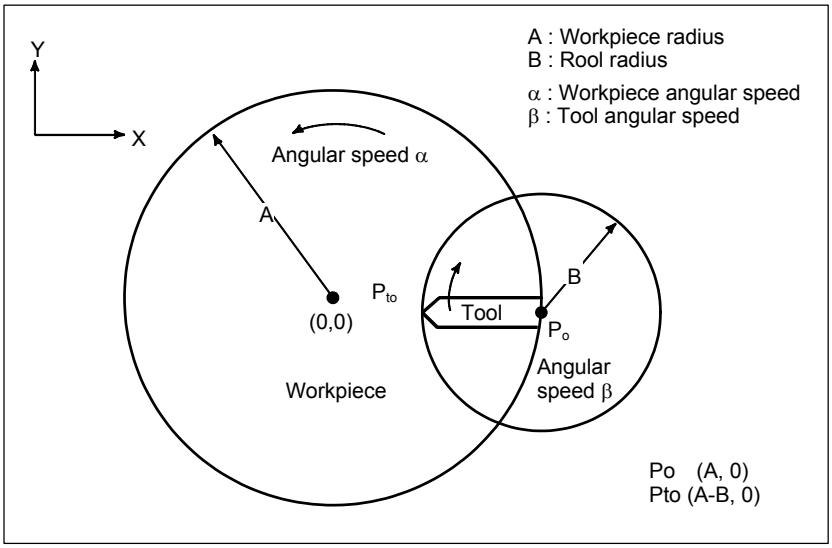


Fig. 21.2 (b) Principle of polygon turning

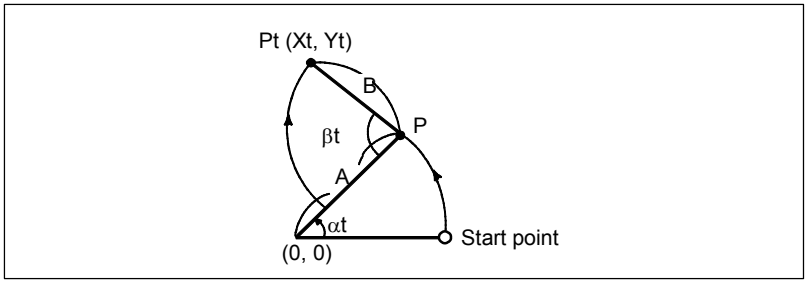


Fig. 21.2 (c) Tool nose position

In this case, the tool nose position Pt (Xt, Yt) after time t is expressed by equations 1 and 2:

$Xt=A\cos\alpha t-B\cos(\beta-\alpha)t$ (Equation 1)

$Yt=A\sin\alpha t+B\sin(\beta-\alpha)t$ (Equation 2)

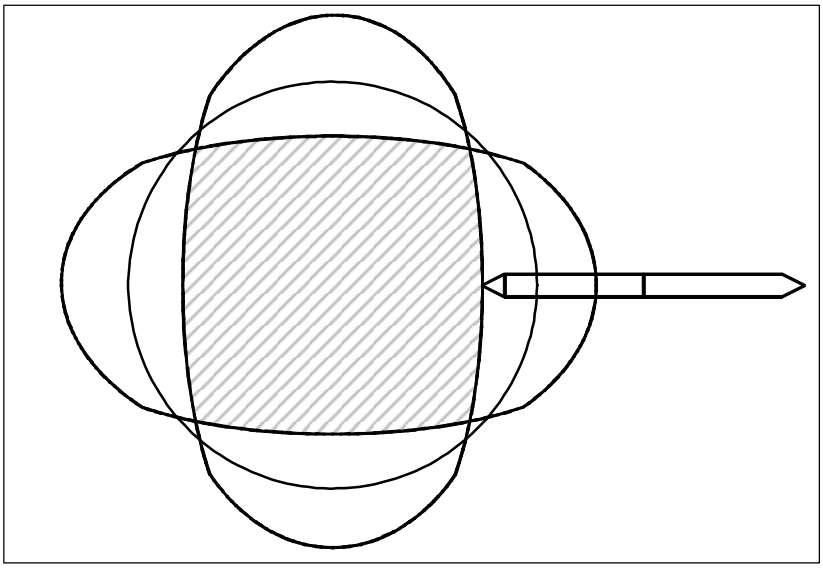
Assuming that the rotation ration of workpiece to tool is 1:2, namely, $\beta=2\alpha$, equations 1 and 2 are modified as follows:

$Xt=A\cos\alpha t-B\cos\alpha t=(A-B)\cos\alpha t$ (Equation 1)'

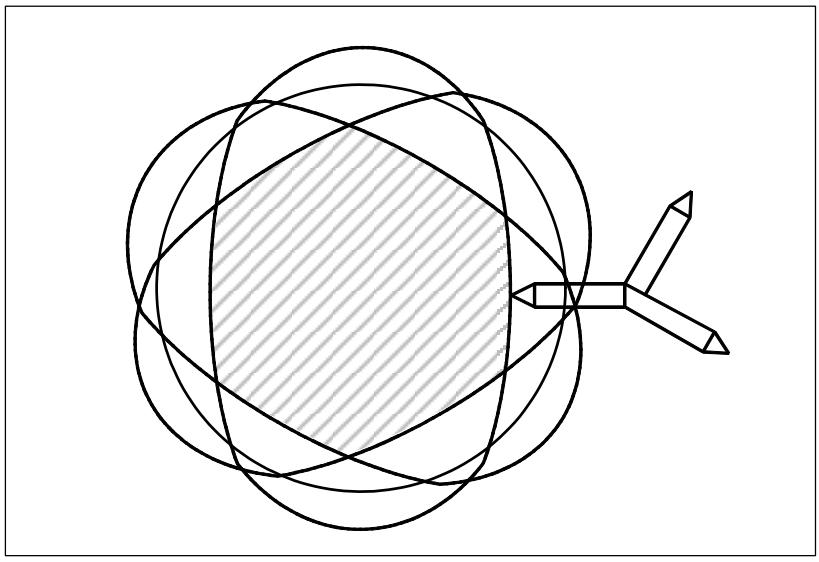
$Xt=A\sin\alpha t+B\sin\alpha t=(A+B)\sin\alpha t$ (Equation 2)'

These equations indicate that the tool nose path draws an ellipse with longer diameter A+B and shorter diameter A-B.

Then consider the case when one tool is set at 180° symmetrical positions, for a total of two. A square can be machined with these tools as shown below.



If three tools are set at every 120°, the machining figure will be a hexagon as shown below.



⚠ WARNING
For the maximum rotation speed of the tool, see the instruction manual supplied with the machine. Do not specify a spindle speed higher than the maximum tool speed or a ratio to the spindle speed that results in a speed higher than the maximum tool speed.

21.3 SYNCHRONOUS, COMPOSITE AND SUPERIMPOSED CONTROL BY PROGRAM COMMAND (G50.4, G51.4, G50.5, G51.5, G50.6, AND G51.6)

Synchronous control, composite control, and superimposed control can be started or canceled using a program command instead of a DI signal. Synchronous control, composite control, and superimposed control based on a DI signal is also possible.

For the basic operations of synchronous control, composite control, and superimposed control, see Section 8.5, "SYNCHRONOUS CONTROL AND COMPOSITE CONTROL", and Section 8.6, "SUPERIMPOSED CONTROL" in the CONNECTION MANUAL (FUNCTION) (B-63943EN-1).

Format

G51.4 P_ Q_ (L_) ; Synchronous control start (L is omissible.)
G50.4 Q_ ; Synchronous control cancel

P: Synchronous master axis ID number
Q: Synchronous slave axis ID number

L: Parking command
1: Master parking (slave parking cancel)
2: Slave parking (master parking cancel)
0: No parking (parking cancel)
(When L is omitted, the specification of L0 is assumed.)

G51.5 P_ Q_ ; Composite control start
G50.5 P_ Q_ ; Composite control cancel

P: Composite axis 1 ID number
Q: Composite axis 2 ID number

G51.6 P_ Q_ ; Superimposed control start
G50.6 Q_ ; Superimposed control cancel

P: Superimposed master axis ID number
Q: Superimposed slave axis ID number

As an ID number, set a unique value for identifying each axis in parameter No. 12600 for both of P and Q.

G51.4/G50.4, G51.5/G50.5, and G51.6/G50.6 are one-shot G codes of group 00.

Explanation

Synchronous control

Synchronous control is performed with the G51.4/G50.4 commands, instead of simultaneously controlled axis selection signals.

Parameter setting examples for a 3-path system

- Parameter No.12600

	Path 1	Path 2	Path 3
X	101	201	301
Z	102	202	302

- Parameter No.8180

	Path 1	Path 2	Path 3
X	0	0	0
Z	0	102	102

- Program example (M100 to M103 are synchronization M codes.)

Path 1	Path 2	Path 3	Operation
N10 M100 P123 ;	M100 P123 ;	M100 P123 ;	Synchronization between paths 1, 2, and 3
N20 G51.4 P102 Q202 ; N30 G51.4 P102 Q302 ;			Start of Z1-Z2 synchronous control Start of Z1-Z3 synchronous control
N40 M101 P123 ;	M101 P123 ;	M101 P123 ;	Synchronization between paths 1, 2, and 3
N50 G00 Z100.;	Synchronous slave movement	Synchronous slave movement	Z1-Z2 and Z1-Z3 synchronous control
N60 M102 P123 ;	M102 P123 ;	M102 P123 ;	Synchronization between paths 1, 2, and 3
N70 G50.4 Q202 ; N80 G50.4 Q302 ;			Cancellation of Z1-Z2 synchronous control Cancellation of Z1-Z3 synchronous control
N90 M103 P123 ;	M103 P123 ;	M103 P123 ;	Synchronization between paths 1, 2, and 3

- Start of synchronous control

N20 G51.4 P102 Q202 : Start of synchronous control with the Z1-axis being a master axis and the Z2-axis being a slave axis
N30 G51.4 P102 Q302 : Start of synchronous control with the Z1-axis being a master axis and the Z3-axis being a slave axis

- Cancellation of synchronous control

N70 G50.4 Q202 : Cancellation of synchronous control with the Z1-axis being a master axis and the Z2-axis being a slave axis
N80 G50.4 Q302 : Cancellation of synchronous control with the Z1-axis being a master axis and the Z3-axis being a slave axis

- Parking

Use the L command to specify the start and cancellation of the parking of synchronous axes.
If the L command is omitted or if the L0 command is issued, the parking of both synchronous master axis and synchronous slave axis is canceled.

- Parameter check

If the axis number corresponding to the P specified with G51.4 is not set in slave axis parameter No. 8180, alarm PS5339 is issued.

Composite control

Composite control is performed with the G51.5/G50.5 commands, instead of composite control axis selection signals.

Parameter setting examples for a 3-path system

- Parameter No.12600

	Path 1	Path 2	Path 3
X	101	201	301
Z	102	202	302

- Parameter No.8183

	Path 1	Path 2	Path 3
X	0	101	0
Z	0	102	0

- Program example (M100 to M103 are synchronization M codes.)

Path 1	Path 2	Path 3	Operation
N10 M100 P12 ;	M100 P12 ;	Independent operation	Synchronization between paths 1 and 2
N20 G51.5 P101 Q201 ; N30 G51.5 P102 Q202 ;			Start of X1-X2 composite control Start of Z1-Z2 composite control
N40 M101 P12 ;	M101 P12 ;		Synchronization between paths 1 and 2
N50 G00 X 100. Z100.;	Composite movement		X1-X2 and Z1-Z2 composite control
N60 M102 P12 ;	M102 P12 ;		Synchronization between paths 1 and 2
N70 G50.5 P101 Q201 ; N80 G50.5 P102 Q202 ;			Cancellation of X1-X2 composite control Cancellation of Z1-Z2 composite control
N90 M103 P12 ;	M103 P12 ;		Synchronization between paths 1 and 2

- Start of composite control

N20 G51.5 P101 Q201 : Start of composite control on the X1- and X2-axes
N30 G51.5 P102 Q202 : Start of composite control on the Z1- and Z2-axes

- Cancellation of composite control

N70 G50.5 P101 Q201 : Cancellation of composite control on the X1- and X2-axes
N80 G50.5 P102 Q202 : Cancellation of composite control on the Z1- and Z2-axes

- Parameter check

If the composite control axis corresponding to the P or Q specified with G51.5/G50.5 is not set in parameter No. 8183, alarm PS5339 is issued.

Superimposed control

Superimposed control is performed with the G51.6/G50.6 commands, instead of superimposed control axis selection signals.

Parameter setting examples for a 3-path system

- Parameter No.12600

	Path 1	Path 2	Path 3
X	101	201	301
Z	102	202	302

- Parameter No.8186

	Path 1	Path 2	Path 3
X	0	0	0
Z	0	0	102

- Program example (M100 to M103 are synchronization M codes.)

Path 1	Path 2	Path 3	Operation
N10 M100 P13 ;	Independent operation	M100 P13 ;	Synchronization between paths 1 and 3
N20 G51.6 P102 Q302 ;			Start of Z1-Z3 superimposed control
N30 M101 P13 ;		M101 P13 ;	Synchronization between paths 1 and 3
N40 G00 Z100.;		G00 Z-200.;	Z1-Z3 superimposed control (Z+100 superimposed on Z3)
N50 M102 P13 ;		M102 P13 ;	Synchronization between paths 1 and 3
N60 G50.6 Q302 ;			Cancellation of Z1-Z3 superimposed control
N70 M103 P13 ;		M103 P13 ;	Synchronization between paths 1 and 3

- Start of superimposed control

N20 G51.6 P102 Q302 : Start of superimposed control with the Z1-axis being a master axis and the Z3-axis being a slave axis

- Cancellation of superimposed control

N60 G50.6 Q302 : Cancellation of superimposed control with the Z1-axis being a master axis and the Z3-axis being a slave axis

- Parameter check

If the axis number corresponding to the P specified with G51.6 is not set in superimposed slave axis parameter No. 8186, alarm PS5339 is issued.

NOTE

1 If G codes (G50.4/G50.5/G50.6) for canceling synchronous, composite, and superimposed control with program commands are issued for axes under synchronous, composite, and superimposed control with DI signals, synchronous, composite, and superimposed control is canceled.

2 If the synchronous control axis selection signal, composite control axis selection signal, and superimposed control axis selection signal are changed from '1' to '0' for axes under synchronous, composite, and superimposed control with program commands, synchronous, composite, and superimposed control is canceled.

21.4 ROTARY AXIS ROLL-OVER

The roll-over function prevents coordinates for the rotary axis from overflowing. The roll-over function is enabled by setting bit 0 (ROAx) of parameter No. 1008 to 1.

21.4.1 Rotary Axis Roll-over

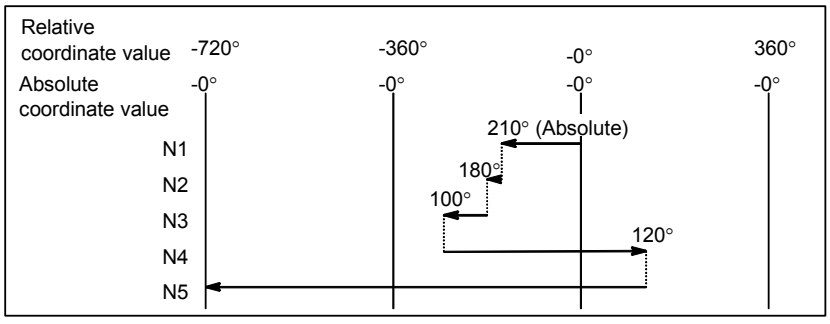
Explanation

For an incremental programming, the tool moves the angle specified in the command. For an absolute programming, the coordinates after the tool has moved are values set in parameter No. 1260, and rounded by the angle corresponding to one rotation. The tool moves in the direction in which the final coordinates are closest when bit 1 (RABx) of parameter No. 1008 is set to 0. Displayed values for relative coordinates are also rounded by the angle corresponding to one rotation when bit 2 (RRLx) of parameter No. 1008 is set to 1.

Example

Assume that axis A is the rotary axis and that the amount of movement per rotation is 360.000 (parameter No. 1260 = 360000). When the following program is executed using the roll-over function of the rotary axis, the axis moves as shown below.

G90 A0 ;	Sequence number	Actual movement value	Absolute coordinate value after movement end
N1 G90 A-150.0 ;	N1	-150	210
N2 G90 A540.0 ;	N2	-30	180
N3 G90 A-620.0 ;	N3	-80	100
N4 G91 A380.0 ;	N4	+380	120
N5 G91 A-840.0 ;	N5	-840	0



M

NOTE
This function cannot be used together with the index table indexing function.

21.4.2 Rotary Axis Control

This function controls a rotary axis as specified by an absolute command.
With this function, the sign of the value specified in the command is interpreted as the direction of rotation, and the absolute value of the specified value is interpreted as the coordinates of the target end position.

Explanation

This function is valid when rotary axis roll-over is enabled (bit 0 (ROAx) of parameter No. 1008 is set to 1).
If bit 3 (RAAx) of parameter No.1007 is set to 1, an absolute command specified for the roll-over rotary axis is interpreted as follows: The sign and absolute value of the value specified in the command represent the direction of rotation and the end position of movement respectively.
If bit 3 (RAAx) of parameter No.1007 is set to 0, the setting by bit 1 (RABx) of parameter No.1008 becomes significant.

NOTE
1 This function can be used only when the corresponding option is provided. (When a machining center system is used, this function cannot be used together with the index table indexing function.)
2 This function is valid for a roll-over rotary axis.
3 If bit 3 (RAAx) of parameter No.1007 is set to 1, bit 1 (RABx) of parameter No.1008 is ignored. To select a rotary motion of a shorter travel distance, set both RAAx and RABx to 0.
4 This function is not supported when the machine coordinate system of the PMC axis control function is selected.

21.5 ARBITRARY ANGULAR AXIS CONTROL

Overview

When the angular axis installed makes an angle other than 90° with the perpendicular axis, the Arbitrary angular axis control function controls the distance traveled along each axis according to the inclination angle as in the case where the angular axis makes 90° with the perpendicular axis.
Arbitrary axes can be specified as a set of an angular axis and perpendicular axis by parameter setting.
The actual distance traveled is controlled according to an inclination angle. However, a program, when created, assumes that the angular axis and perpendicular axis intersect at right angles. The coordinate system used at this time is referred to as the program coordinate system. (The program coordinate system may be referred to as the Cartesian coordinate system, and the actual move coordinate system may be referred to as the angular coordinate system or machine coordinate system.)

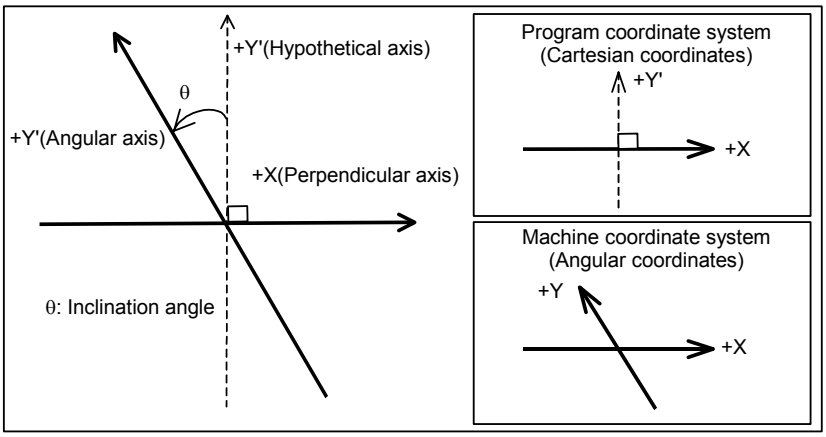


Fig. 21.5 (a)

Explanation

When the amounts of travel along the angular axis and the perpendicular axis are Ya and Xa, respectively, the amounts are controlled according to the formulas shown below.

$$Ya = \frac{Yp}{\cos\theta}$$

Xa,*Ya*: Actual distance
Xp,*Yp*: Programmed distance

The amount of travel along the perpendicular axis is corrected by the influence of travel along the angular axis, and is determined by the following formula:

$$Xa = Xp - C \times Yp \times \tan\theta$$

NOTE
The coefficient C is 2 in the case of diameter specification for the perpendicular axis (X) or 1 in the case of radius specification.

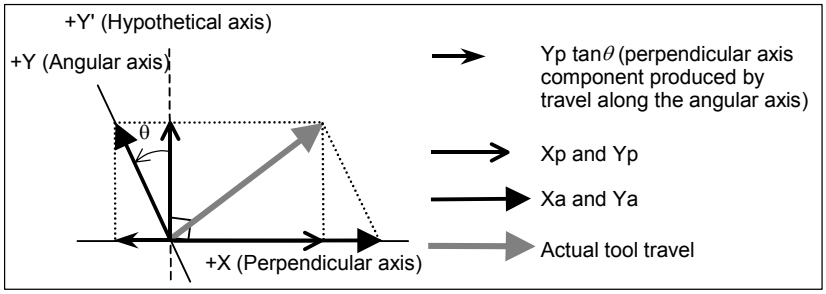


Fig. 21.5 (b)

- Feedrate

When the Y-axis is an angular axis, and the X-axis is a perpendicular axis, the feedrate along each axis is controlled as described below so that the feedrate in the tangent direction becomes Fp.
The feedrate component along the Y-axis is determined by the following expressions:

$$Fay = \frac{Fp}{\cos\theta}$$

Fa represents the actual feedrate.
Fp represents a programmed feedrate.

$$Fax = Fp - Fp \times \tan\theta$$

- Absolute and relative position display

An absolute and a relative position are indicated in the programmed Cartesian coordinate system.

- Machine position display

A machine position indication is provided in the machine coordinate system where an actual movement is taking place according to an inclination angle.

Method of use

The angular and perpendicular axes for which arbitrary angular axis control is to be applied must be specified beforehand, using parameters Nos. 8211 and 8212. When 0 is set in one of the parameters, the same number is specified in the parameters, or a number other than the controlled axis numbers is specified in a parameter, however, an angular axis and perpendicular axis are selected according to the table below.

	Angular axis	Perpendicular axis
M series	Y-axis of the basic three axes (axis with 2 set in parameter No. 1022)	Z-axis of the basic three axes (axis with 3 set in parameter No. 1022)
T series	X-axis of the basic three axes (axis with 1 set in parameter No. 1022)	Z-axis of the basic three axes (axis with 3 set in parameter No. 1022)

- Bit 0 (AAC) of parameter No. 8200 enables or disables the arbitrary angular axis control. If the function is enabled, the distance traveled along each axis is controlled according to an angular angle parameter No. 8210.

- By using bit 2 (AZR) of parameter No. 8200, whether to make a movement along the perpendicular axis by a movement made along the angular axis when a manual reference position return operation is performed along the angular axis can be chosen. When a movement along the perpendicular axis is enabled (AZR = 1), a reference position return operation along the perpendicular axis can be performed by a movement made along the angular axis.
- By setting the normal axis/angular axis control invalid signal NOZAGC to 1, angular axis control only for the angular axis can be available. In this time the angular axis are converted to those along the angular coordinate system without affecting commands to normal axis.
  Use this signal when operating each axis independently.

- Manual reference position return operation

A movement is made to the reference position (machine position) set in parameter No. 1240. By using bit 2 (AZR) of parameter No. 8200, whether to make a movement along the perpendicular axis when a reference position return operation is performed along the angular axis can be chosen.

- Automatic reference position return operation and floating reference position return operation (G28, G30, G30.1)

A movement to the middle point along the angular axis affects a movement along the perpendicular axis. As a movement from the middle point to the reference position along the angular axis, a selection can be made with bit 0 (ARF) of parameter No. 8209 between a Cartesian coordinate system operation (FS16*i* compatibility) and an angular coordinate system operation. If manual reference position return operation is not performed even once after the power is turned on, operation is performed in the same sequence as for manual reference position return operation. So, specify commands first for the angular axis then for the perpendicular axis.

Example 1)

When the Y-axis is an angular axis and the X-axis is a perpendicular axis

(1) If the angular axis is first specified then the perpendicular axis is specified, reference position return operation is performed normally.

```
G28Y_;
G28X_;
```

(2) If the perpendicular axis is first specified then the angular axis is specified, or if the perpendicular axis and the angular axis are specified at the same time, alarm PS0372 is issued when a movement is made along the perpendicular axis.

```
{G28X_;      or   {G28X_ Y_;
 G28Y_;
```

Example 2)

Automatic reference position return examples

(If the Y-axis is an angular axis, the X-axis is a perpendicular axis, and the inclination angle is -30°)

<1> Command for automatic reference position return along the Y-axis from point P2

>G91G28X200.

<2> Command for automatic reference position return along the X-axis from point P1

>G91G28Y100.

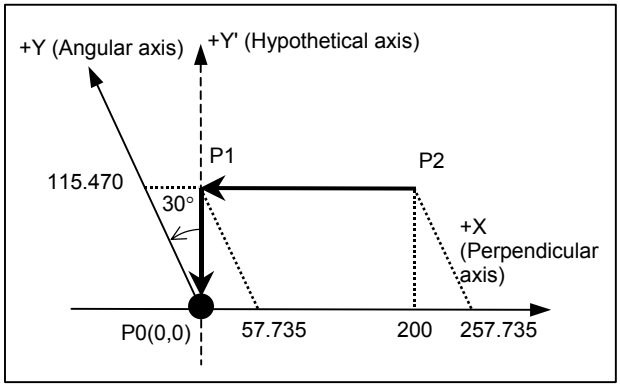
(1) If bit 0 (ARF) of parameter No. 8209 is 1 (FS16*i* compatibility)

<1> Coordinates of P1

(Absolute coordinates)		(Machine coordinates)	
X	0.000	X	57.735
Y	100.000	Y	115.470

<2> Coordinates of P0

(Absolute coordinates)		(Machine coordinates)	
X	0.000	X	0.000
Y	0.000	Y	0.000



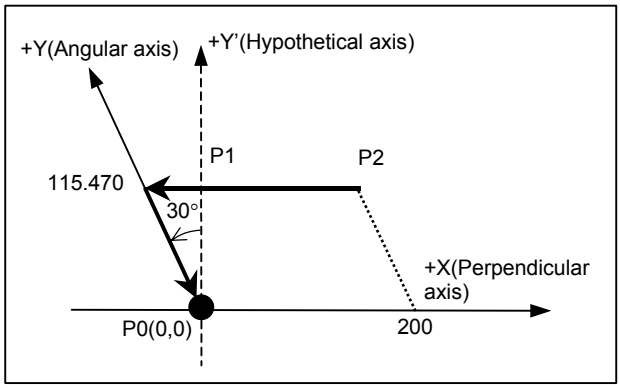
(2) If bit 0 (ARF) of parameter No. 8209 is 0

<1> Coordinates of P1

(Absolute coordinates)		(Machine coordinates)	
X	0.000	X	0.000
Y	100.000	Y	115.470

<2> Coordinates of P0

(Absolute coordinates)		(Machine coordinates)	
X	0.000	X	0.000
Y	0.000	Y	0.000



- Reference position return operation of high-speed type

When a reference position is already established and a reference position return operation of high-speed type is to be performed, the reference position return operation need not be performed in the order from the angular axis to the perpendicular axis.

- Machine coordinate selection (G53)

By specifying (G90)G53X_Y_: (when the Y-axis is an angular axis, the X-axis is a perpendicular axis, and the inclination angle is -30°), a movement is made by rapid traverse.
However, a movement along the angular axis (G53 command) does not affect a movement along the perpendicular axis, regardless of whether the perpendicular axis/angular axis control disable signal (NOZAGC) is turned on or off.
Example)

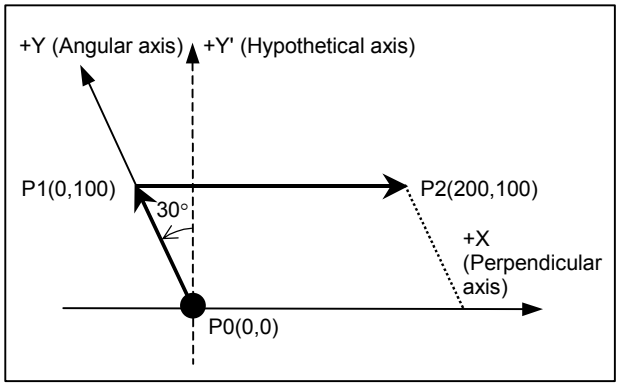
1 Move command for movement from point P0 to point P1
>G90G53Y100.
2 Move command for movement from point P1 to point P2
>G90G53X200.

<1> Coordinates of P1

(Absolute coordinates)		(Machine coordinates)	
X	-50.000	X	0.000
Y	86.603	Y	100.000

<2> Coordinates of P2

(Absolute coordinates)		(Machine coordinates)	
X	150.000	X	200.000
Y	86.603	Y	100.000



- Commands for linear interpolation and linear interpolation type positioning (G01, G00)

The tool moves to a specified position in the Cartesian coordinate system when the following is specified:
(G90)G00X_Y_; (when the Y-axis is an angular axis, the X-axis is a perpendicular axis, and the inclination angle is -30°)
or
(G90)G01X_Y_F_; (when the Y-axis is an angular axis, the X-axis is a perpendicular axis, and the inclination angle is -30°)

Example) Examples of positioning

1 Move command for movement from point P0 to point P1
>G90G00Y100.

2 Move command for movement from P1 to P2
G90G00X200.

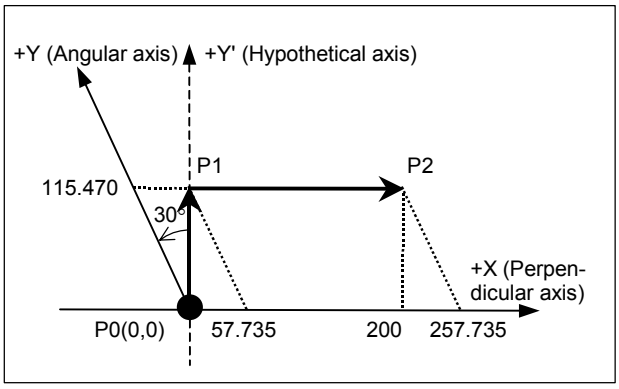
(1) When the perpendicular axis/angular axis control disable signal (NOZAGC) is set to 0

<1> Coordinates of P1

(Absolute coordinates)	(Machine coordinates)
X 0.000	X 57.735
Y 100.000	Y 115.470

<2> Coordinates of P2

(Absolute coordinates)	(Machine coordinates)
X 200.000	X 257.735
Y 100.000	Y 115.470



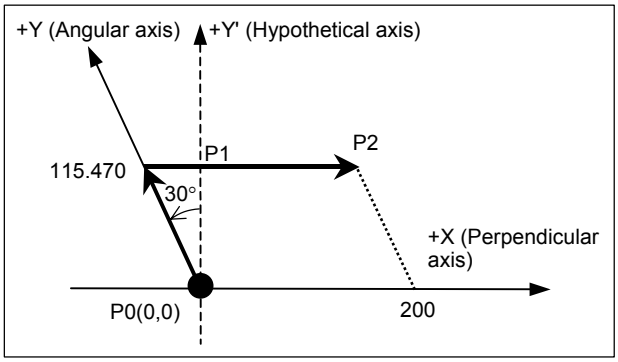
(2) When the perpendicular axis/angular axis control disable signal (NOZAGC) is set to 1

<1> Coordinates of P1

(Absolute coordinates)	(Machine coordinates)
X 0.000	X 0.000
Y 100.000	Y 115.470

<2> Coordinates of P2

(Absolute coordinates)	(Machine coordinates)
X 200.000	X 200.000
Y 100.000	Y 115.470



- Three-dimensional coordinate conversion

In the three-dimensional coordinate conversion mode, angular coordinate system conversion is applied to the workpiece coordinate system that has undergone three-dimensional coordinate conversion.

- Stored stroke limit

Stored stroke limits under arbitrary angular axis control can be set not in a angular coordinate system but in the Cartesian coordinate system by setting bits 0, 1, and 2 (AOT, AO2, and AO3) of parameter No. 8201.

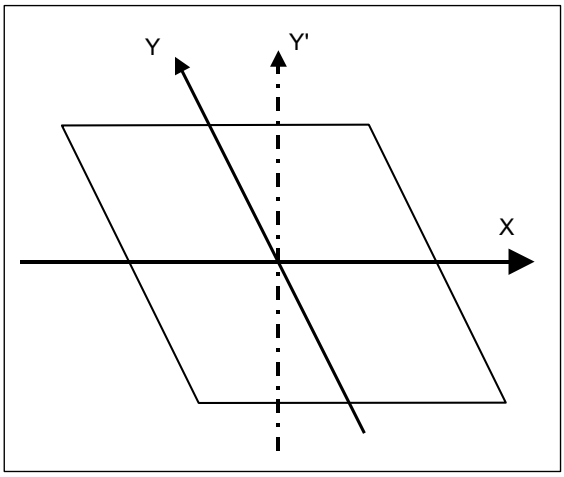


Fig. 21.5 (c) OT area in a angular coordinate system

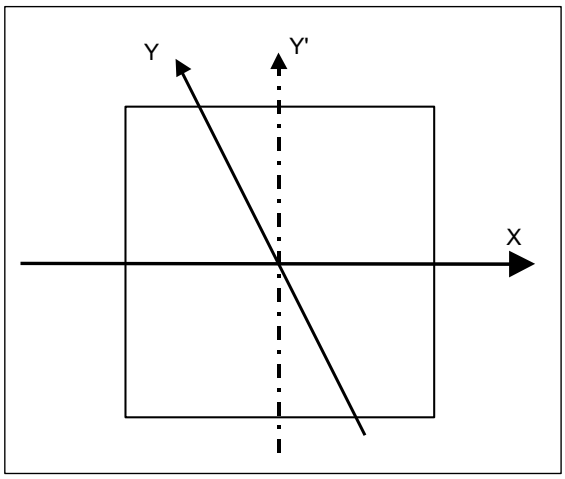


Fig. 21.5 (d) OT area in a Cartesian coordinate system

Machine coordinates include a value converted for the angular axis and a compensation value for the perpendicular axis, so that a angular machine coordinate system as shown in Fig. 21.5 (c) results.
A stored stroke limit is checked in the machine coordinate system, so that the limit area is slanted to form a rhombus as shown in Fig. 21.5 (c). In this case, the area cannot be identified intuitively. So, stroke limits are checked not in an actual angular machine coordinate system but in a virtual Cartesian machine coordinate system as shown in Fig. 21.5 (d).

The functions that operate in the Cartesian coordinate system are:

- Stored stroke check 1 (Both of I and II)
- Stored stroke check 2 (G22/G23)
- Stored stroke check 3

- Stored stroke check before move

The stored stroke check function before move does not work in a angular coordinate system. Unless this function is enabled, and the coordinate system is converted to the Cartesian coordinate system, no stroke check is made.

- Stroke limit external setting (function specific to the M series only and valid only for OT1)
- Bit 7 (BFA) of parameter No. 1300 for specifying whether to issue an alarm before or after a stroke limit is exceeded (valid for OT1 and OT3)

The stored stroke limit functions other than the above work in a angular coordinate system.

- Relationships between this function and axis-by-axis input/output signals

The table below indicates the relationships between this function and the meaning of each controlled axis signal.

The input/output signals are classified as signals valid for the program coordinate system (Cartesian coordinate system) and signals valid for the machine coordinate system (angular coordinate system). In the "Classification" column, "Cartesian" is indicated for a signal that is valid for the Cartesian coordinate system, and "Angular" is indicated for a signal that is valid for the angular coordinate system.

A signal valid for the Cartesian coordinate system means a signal valid for a specified axis, and a signal valid for the angular coordinate system is a signal valid for actual machine movement.

This means that when a movement is made along the perpendicular axis by a movement along the angular axis alone:

A signal valid for the Cartesian coordinate system is affected by a movement along the angular axis.

A signal valid for the angular coordinate system is not affected by a movement along the angular axis.

Input signal				
Signal name		**Address**	**Classification**	**Remarks**
Interlock for each axis	*ITx	G130	Cartesian	When a movement is made along the angular axis only, interlocking the perpendicular axis does not interlock a movement along the perpendicular axis made by a movement along the angular axis. Caution) When using the interlock signal for each axis, make both of the angular axis and perpendicular axis high.
Overtravel	*+Lx *-Lx	G114 G116	Angular	This signal is applied to each axis independently. (If the perpendicular axis is made high, no alarm is issued for the perpendicular axis even when an OT alarm is issued for the angular axis.)
Deceleration signal for reference position return	*DECx	X009	Angular	This signal is applied to each axis independently.
Servo-off signal	SVFx	G126	Angular	This signal is applied to each axis independently.
Control axis detach signal	DTCHx	G124	Angular	This signal is applied to each axis independently.
Feed axis direction selection signal	+Jx -Jx	G100 G102	Cartesian	A movement is made in the Cartesian coordinate system. (When the +J/-J signal for the angular axis is made high, a movement is made also along the perpendicular axis.)

Input signal				
Signal name		**Address**	**Classification**	**Remarks**
Mirror image	MIx	G106	Angular	Mirror image is applied to the angular coordinate system for each axis independently. Caution) Be sure to turn off the mirror image signal for the angular axis and perpendicular axis engaged in manual operation.
Manual feed interlock signal for each axis direction, tool compensation value write signal	+MIT1, +MIT2	X004.2, 4	Cartesian	Set the tool compensation parameter in the Cartesian coordinate system.
Machine lock for each axis	MLKx	G108	Angular	This signal is applied to each axis independently.

Output signal				
Signal name		**Address**	**Classification**	**Remarks**
In-position signal	INPx	F104	Angular	Applied to each axis independently.
Mirror image check signal	MMIx	F108	Angular	Applied to each axis independently.
Controlled axis removal in-progress signal	MDTCHx	F110	Angular	Applied to each axis independently.
Travel in-progress signal	MVx	F102	Angular	Applied to each axis independently.
Reference position return completion signal	ZPx	F094	Cartesian	Applied to each axis independently. (A manual reference position return operation and the first automatic reference position return operation after power-up need to be performed first for the angular axis.)
2nd reference position return completion signal	ZP2x	F096	Cartesian	Applied to each axis independently.
3rd reference position return completion signal	ZP3x	F098	Cartesian	Applied to each axis independently.
4th reference position return completion signal	ZP4x	F100	Cartesian	Applied to each axis independently.

Limitation

- Three-dimensional coordinate conversion

If the basic three axes in the three-dimensional coordinate conversion mode do no include a perpendicular axis and angular axis for arbitrary angular axis control, operation cannot be performed normally in a correct angular coordinate system.

- Linear scale with absolute address reference mark

- For both of the angular axis and perpendicular axis, a linear scale with an absolute address reference mark must be used.
- Reference position return operation must be first completed along the angular axis.
- Return operation cannot be performed along the perpendicular axis while return operation is being performed along the angular axis.

- Synchronous control

For synchronous control on axes related to arbitrary angular axis control, the angular axis and Cartesian axis on the master axis side and the angular axis and Cartesian axis on the slave axis side must be placed under synchronous control at the same time. Moreover, synchronous control can be exercised between angular axes only or between Cartesian axes only.
If an attempt is made to perform operation under a condition other than the above, the alarm PS0375 is issued.
Example)

Path 1		Path 2
X1 (Cartesian axis)	←Synchronous→	X2 (Cartesian axis)
Y1 (angular axis)	←Synchronous→	Y2 (angular axis)

- Composite control

For composite control on axes related to arbitrary angular axis control, the angular axis and Cartesian axis on the master axis side and the angular axis and Cartesian axis on the slave axis side must be placed under composite control at the same time. Moreover, composite control can be exercised between angular axes only or between Cartesian axes only.
If an attempt is made to perform operation under a condition other than the above, the alarm PS0375 is issued.
Example)

Path 1		Path 2
X1 (Cartesian axis)	←composite→	X2 (Cartesian axis)
Y1 (angular axis)	←composite→	Y2 (angular axis)

- Rigid tapping

As a rigid tapping axis, no angular axis can be used.

- Functions that cannot be used simultaneously

- Axis synchronous control, twin table control, parallel axis control, polygon turning, rigid tapping, hypothetical axis control, EGB function, PMC axis control, superimposed control

⚠ CAUTION

1 After arbitrary angular axis control parameter setting, be sure to perform manual reference position return operation.
2 Before manual reference position return operation is performed along the perpendicular axis, reference position return operation along the angular axis must be completed (with the reference position return completion signal for the angular axis (ZPx) set to 1). If reference position return operation is performed along the perpendicular axis first, an alarm PS0372 is issued.
3 When the setting is made so that the tool moves along the perpendicular axis during manual reference position return along the angular axis (bit 2 (AZK) of parameter No. 8200 is set to 0), if once manual reference position return has been performed along the angular axis, also perform manual reference position return along the perpendicular axis immediately after the operation.
4 Before attempting to manually move the tool along the angular and perpendicular axes simultaneously, set perpendicular/angular axis control disable signal NOZAGC to 1.
5 Once the tool has been moved along the angular axis when perpendicular/angular axis control disable signal NOZAGC has been set to 1, manual reference position return must be performed.
6 The same increment system must be used with the angular axis and perpendicular axis.
7 Before a perpendicular axis reference position return check can be made, angular axis reference position return operation must be completed.
8 No rotary axis must be set for the angular axis and perpendicular axis. A rotary axis may be specified only for a linear axis.
9 Set a position switch operation range (parameter Nos. 6930 to 6965) in a angular coordinate system.

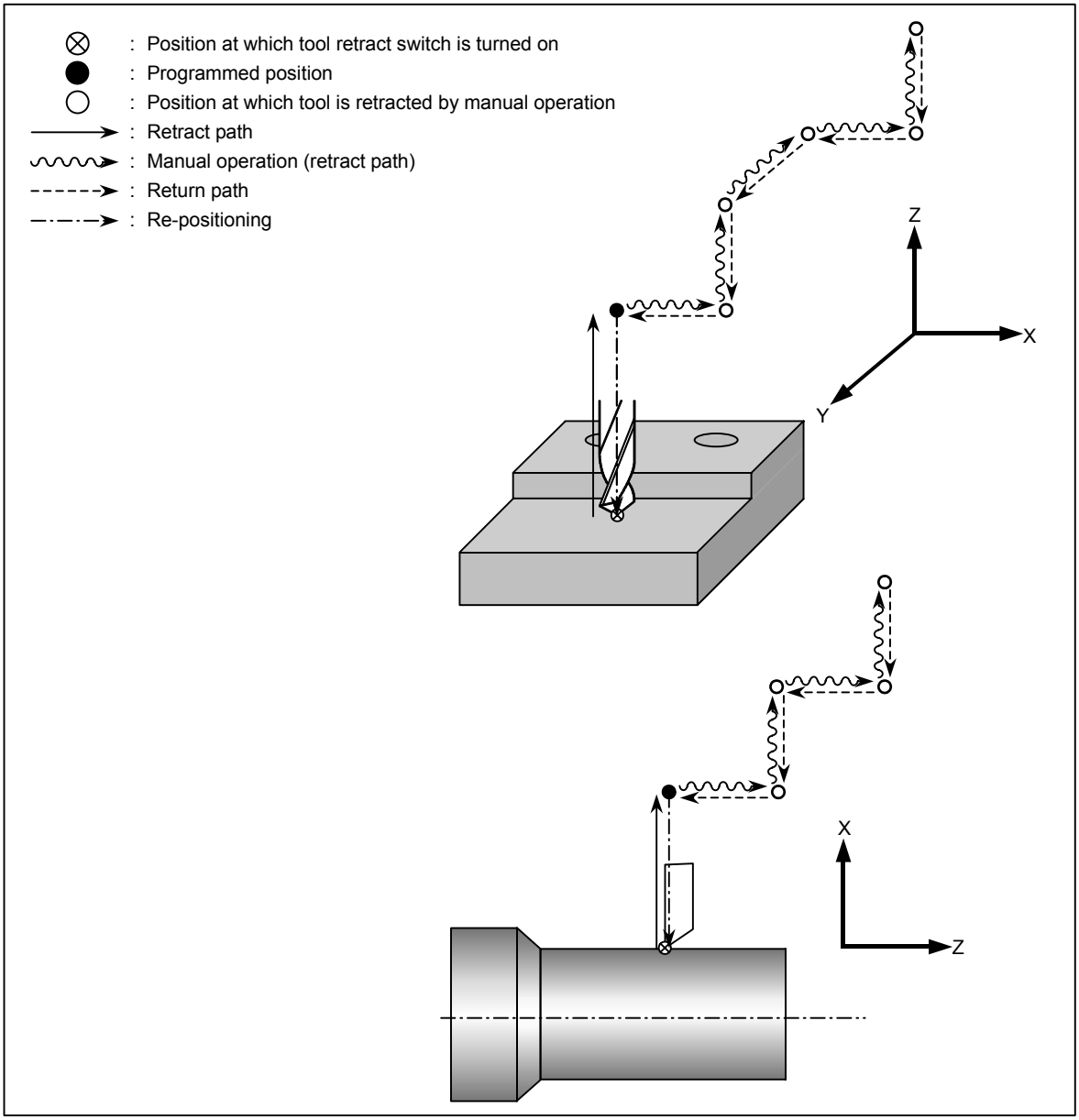
21.6 TOOL RETRACT AND RECOVER

Overview

To replace the tool damaged during machining or to check the status of machining, the tool can be withdrawn from a workpiece. The tool can then be advanced again to restart machining efficiently.
The tool retract and recover operation consists of the following four steps:

- Retract
  The tool is retracted to a predefined position using the TOOL WITHDRAW switch.
- Withdrawal
  The tool is moved to the tool-change position manually.
- Return
  The tool returns to the retract position.
- Repositioning
  The tool returns to the interrupted position.

For the tool retract and recover operations, see “Tool retract and recover ” in the Part III.



Format

Specify a retraction axis and distance in the following format:

Specify the amount of retraction, using G10.6.

G10.6 IP_ ;

IP: In incremental mode, retraction distance from the position where the retract signal is turned on
In the absolute mode, retraction distance to an absolute position
The specified amount of retraction is effective until G10.6 is next executed. To cancel the retraction, specify the following:
G10.6 ; (as a single block containing no other commands)

Explanation

- Retraction

When the TOOL WITHDRAW switch on the machine operator's panel is turned on during automatic operation or in the automatic operation stop or hold state, the tool is retracted the length of the programmed retraction distance. This operation is called retraction. The position at which retraction is completed is called the retraction position. The tool is retracted to the retraction position in linear interpolation at the dry run feedrate. When retraction is completed, the RETRACTIONS POSITION LED on the machine operator's panel goes on.

When the TOOL WITHDRAW switch is turned on during execution of a block in automatic operation, execution of the block is interrupted immediately and the tool is retracted. After retraction is completed, the system enters the automatic operation hold state.

If the retraction distance and direction are not programmed, retraction is not performed. In this state, the tool can be withdrawn and returned.

When the TOOL WITHDRAW switch is turned on in the automatic operation stop or hold state, the tool is retracted, then the automatic operation stop or hold state is entered again.

When the TOOL WITHDRAW switch is turned on, the tool withdraw mode is set. When the tool withdraw mode is set, the TOOL BEING WITHDRAWN LED on the machine operator's panel goes on.

- Withdrawal

When the manual mode is set, the tool can be moved manually (jog feed, incremental feed, handle feed, or manual numeral command) to replace the tool or measure a machined workpiece. This operation is called a withdrawal. The tool withdrawal path is automatically memorized by the CNC.

- Return

When the mode is returned to automatic operation mode and the TOOL RETURN switch on the machine operator's panel is turned off, the CNC automatically moves the tool to the retraction position by tracing the manually-moved tool path backwards. This operation is called a return. Upon completion of a return to the retraction position, the RETRACTIONS POSITION LED comes on.

- Repositioning

When the cycle start button is pressed while the tool is in the retraction position, the tool moves to the position where the TOOL WITHDRAW switch was turned on. This operation is called repositioning. Upon completion of repositioning, the TOOL BEING WITHDRAWN LED is turned off, indicating that the tool withdrawal mode has terminated. Operation after completion of repositioning depends on the automatic operation state when the tool withdrawal mode is set.

(1) When the tool withdrawal mode is set during automatic operation, operation is resumed after completion of repositioning.
(2) When the tool withdrawal mode is set when automatic operation is held or stopped, the original automatic operation hold or stop state is set after completion of repositioning. When the cycle start button is pressed again, automatic operation is resumed.

Limitation

- Offset

If the origin, presetting, or workpiece origin offset value (or External workpiece origin offset value) is changed after retraction is specified with G10.6 in absolute mode, the change is not reflected in the retraction position. After such changes are made, the retraction position must be respecified with G10.6.
When the tool is damaged, automatic operation can be interrupted with a tool withdrawal and return operation in order to replace the tool. Note that if the tool offset value is changed after tool replacement, the change is ignored when automatic operation is resumed from the start point or other point in the interrupted block.

- Machine lock, mirror image, and scaling

When withdrawing the tool manually in the tool withdrawal mode, never use the machine lock, mirror-image, or scaling function.

- Reset

Upon reset, the retraction data specified in G10.6 is cleared. Retraction data needs to be specified again.

- Retraction command

The tool withdrawal and return function is enabled even when the retraction command is not specified. In this case, retraction and repositioning are not performed.

> ⚠ **WARNING**
> The retraction axis and retraction distance specified in G10.6 need to be changed in an appropriate block according to the figure being machined. Be very careful when specifying the retraction distance; an incorrect retraction distance may damage the workpiece, machine, or tool.

21.7 ELECTRONIC GEAR BOX

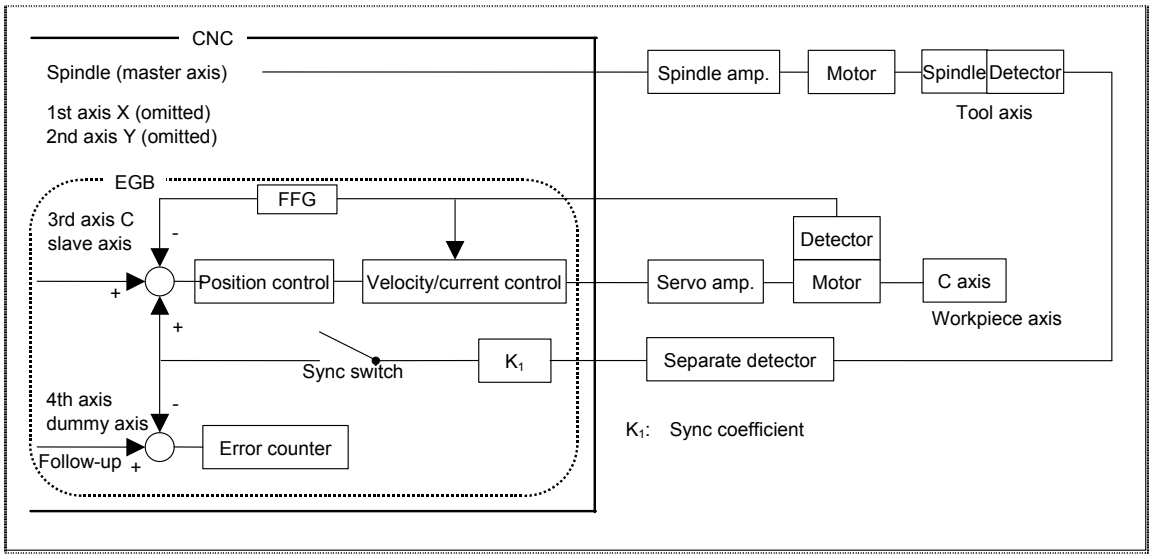
21.7.1 Electronic Gear Box

Overview

This function enables fabrication of high-precision gears, screws, and other components by rotating the workpiece in synchronization with a rotating tool or by moving the tool in synchronization with a rotating workpiece. The rate of synchronization can be specified with a program. The synchronization of tool and workpiece axes with this function adopts a system in which the synchronization is directly controlled by digital servo, so that the workpiece axis can follow up the speed fluctuations on the tool axis with no error, thereby allowing fabrication of high-precision cogwheels. In the subsequent explanation, the Electronic Gear Box is called the EGB.
There are several conditions in the setting and so on of the workpiece axis and the tool axis, and refer to the appropriate manual provided by the machine tool builder.

- Example of controlled axis configuration

Spindle : EGB master axis : Tool axis
1st axis : X axis
2nd axis : Y axis
3rd axis : C axis (EGB slave axis : Workpiece axis)
4th axis : C axis (EGB dummy axis : Cannot be used as a normal controlled axis.)



Format

	Parameter EFX (No.7731#0)=0	Parameter EFX(No.7731#0)=1	
		Parameter HBR (No.7731#5)=1	Parameter HBR (No.7731#5)=0
Start of synchronization	**G81 T__ (L__) (Q__ P__) ;**	**G81.4 R__ (L__) (Q__ P__) ;**	**G81.4 T__ (L__) (Q__ P__) ;**
Cancellation of synchronization	**G80 ;**	**G80.4 ;**	**G80.4 ;**
	(*1)	**(*2)**	**(*3)**

T(or R) : Number of teeth (Specifiable range: 1 to 1000)

L : Number of hob threads (Specifiable range: -200 to +200)
The sign of L determines the direction of rotation for the workpiece axis.
When L is positive, the direction of rotation for the workpiece axis is positive (+ direction).
When L is negative, the direction of rotation for the workpiece axis is negative (- direction).
When L is 0, it follows the setting of bit 3 (LZR) of parameter No.7701.
If L is not specified, the number of hob threads is assumed 1.

Q : Module or diametral pitch
Specify a module in the case of metric input.
(Unit: 0.00001mm, Specifiable range: 0.01 to 25.0mm)
Specify a diametral pitch in the case of inch input.
(Unit: 0.00001inch^{-1}, Specifiable range: 0.01 to 254.0 inch^{-1})

P : Gear helix angle
(Unit: 0.0001deg, Specifiable range: -90.0 to 90.0deg)

*1 Use it for machining centers.

*2 Use it for lathes.

*3 Use it for machining centers.
This format enables specification of the same G codes as those for lathes.

*4 When specifying Q and P, the user can use a decimal point.

NOTE
Specify G81, G80, G81.4, and G80.4 in a single block.

Explanation

- Master axis, slave axis, and dummy axis

The synchronization reference axis is called the master axis, while the axis along which movement is performed in synchronization with the master axis is called the slave axis. For example, if the workpiece moves in synchronization with the rotating tool as in a hobbing machine, the tool axis is the master axis and the workpiece axis is the slave axis.

Which axes to become the master and slave axes depends on the configuration of the machine. For details, refer to the manual issued by the machine tool builder.

A single servo axis is used exclusively so that digital servo can directly read the rotation position of the master axis. (This axis is called the EGB dummy axis.)

- Synchronization control

(1) Start of synchronization

If G81 is issued so that the machine enters synchronization mode, the synch switch of the EGB function is closed, and the synchronization of the tool and workpiece axes is started. During synchronization, the rotation about the tool and workpiece axes is controlled so that the relationship between T (number of teeth) and L (number of hob threads) is maintained. During synchronization, the synchronization relationship is maintained regardless of whether the operation is automatic or manual.

Specify P and Q to use helical gear compensation.

If only either P or Q is issued, alarm PS1594 is generated.

If, during synchronization, G81 is issued again without synchronization cancelation, alarm PS1595 is generated if ECN, bit 3 of parameter No. 7731, is 0. If ECN, bit 3 of parameter No. 7731, is 1, helical gear compensation is conducted with the synchronization coefficient being changed to the one newly specified with T and L commands if T and L commands are issued, and if T and L commands are not issued and only P and Q commands are issued, helical gear compensation is conducted with the synchronization coefficient kept intact. This allows consecutive fabrication of helical gears and super gears.

(2) Start of tool axis rotation

When the rotation of the tool axis starts, the rotation of the workpiece axis starts so that the synchronous relationship specified in the G81 block can be maintained.

The rotation direction of the workpiece axis depends on the rotation direction of the tool axis. That is, when the rotation direction of the tool axis is positive, the rotation direction of the workpiece axis is also positive; when the rotation direction of the tool axis is negative, the rotation direction of the workpiece axis is also negative. However, by specifying a negative value for L, the rotation direction of the workpiece axis can be made opposite to the rotation direction of the tool axis.

During synchronization, the machine coordinates of the workpiece axis and EGB axis are updated as synchronous motion proceeds. On the other hand, a synchronous move command has no effect on the absolute and relative coordinates.

(3) Termination of tool axis rotation

In synchronism with gradual stop of the tool axis, the workpiece axis is decelerated and stopped. By specifying the command below after the spindle stops, synchronization is canceled, and the EGB synchronization switch is opened.

(4) Cancellation of synchronization

When cancellation of synchronization is issued, the absolute coordinate on the workpiece axis is updated in accordance with the amount of travel during synchronization. Subsequently, absolute commands for the workpiece axis will be enabled.

For a rotation axis, the amount of travel during synchronization, as rounded to 360-degree units is added to the absolute coordinate.

In the G80 block, only O and N addresses can be specified.

By setting HBR, bit 0 of parameter No. 7700, to 0, it is possible to cancel synchronization with a reset. If the manual absolute signal is ON, the absolute coordinate is updated.

Synchronization is automatically canceled under the following conditions:

<1> An emergency stop is applied.
<2> A servo alarm is generated.
<3> Alarm PW0000 (POWER MUST BE OFF) is generated.
<4> An IO alarm is generated.

⚠ CAUTION

1 Feed hold, interlock, and machine lock are invalid to a slave axis in EGB synchronization.
2 Even if an OT alarm is issued for a slave axis in EGB synchronization, synchronization will not be canceled.
3 During synchronization, it is possible to execute a move command for a slave axis and other axes, using a program. The move command for a slave command must be an incremental one.

NOTE

1 If bit 0 (HBR) of parameter No. 7700 is set to 1, EGB synchronization will not be canceled due to a reset. Usually, set this parameter bit to 1.
2 In synchronous mode, it is not possible to specify G27, G28, G29, G30, G30.1, and G53 for a slave axis.
3 It is not possible to use controled axis detach for a slave axis.
4 During synchronization, manual handle interruption can be performed on the slave and other axes.
5 In synchronization mode, no inch/metric conversion commands (G20 and G21) cannot be issued.
6 In synchronous mode, only the machine coordinates on a slave axis are updated.
7 If bit 0 (EFX) of parameter No. 7731 is 0, no drilling canned cycle can be used. To use a drilling canned cycle, set bit 0 (EFX) of parameter No. 7731 to 1 and use G81.4 instead of G81 and G80.4 instead of G80.
8 If TDP, bit 0 of parameter No. 7702, is 1, the permissible range of T is 0.1 to 100 (1/10 of the specified value).
9 If, at the start of EGB synchronization (G81), L is specified as 0, synchronization starts with L assumed to be 1 if bit 3 (LZR) of parameter No.7701, is 0; if bit 3 (LZR) of parameter No.7701, is 1, synchronization is not started with L assumed to be 0. At this time, helical gear compensation is performed.
10 Feed per revolution is performed on the feedback pulses on the spindle. By setting ERV, bit 0 of parameter No. 7703, to 1, feed per revolution can be performed based on the speed on the synchronous slave axis.
11 Actual cutting feedrate display does not take synchronization pulses into consideration.
12 For an EGB slave axis, synchronous and cpmposite control cannot be executed.
13 In EGB synchronization mode, AI contour control mode is temporarily canceled.

- Helical gear compensation

For a helical gear, the workpiece axis is compensated for the movement along the Z-axis (axial feed axis) based on the torsion angle of the gear.
Helical gear compensation is performed with the following formulas:

$$\text{Compensation angle} = \frac{Z \times \sin(P)}{\pi \times T \times Q} \times 360 \text{ (for metric input)}$$

$$\text{Compensation angle} = \frac{Z \times Q \times \sin(P)}{\pi \times T} \times 360 \text{ (for inch input)}$$

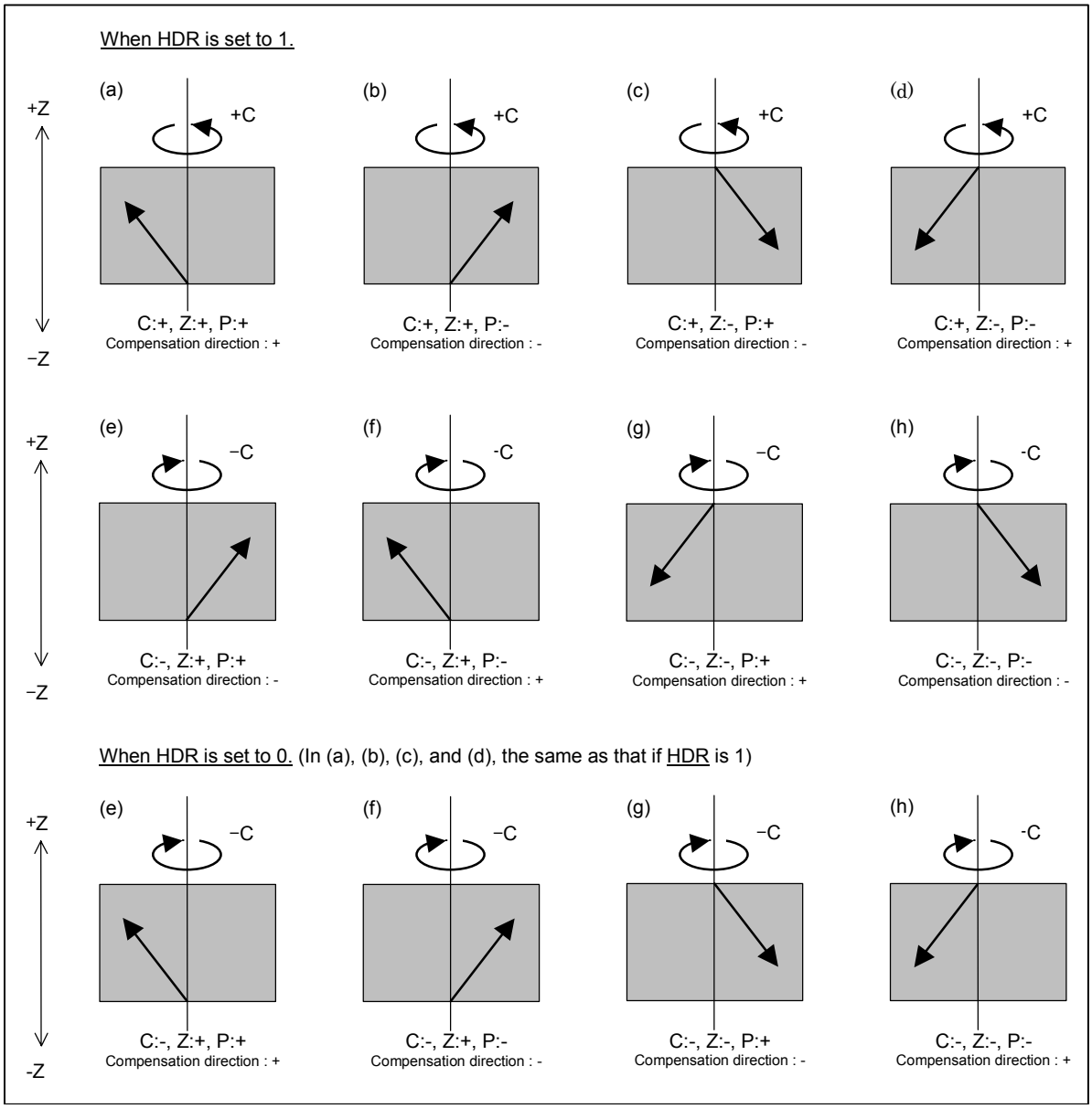
where

Compensation angle: Signed absolute value (deg)
Z :Amount of travel on the Z-axis after the specification of G81
P : Signed gear helix angle (deg)
π : Circular constant
T : Number of teeth
Q : Module (mm) or diametral pitch ($inch^{-1}$)
Use P, T, and Q specified in the G81 block.

In helical compensation, the machine coordinates on the workpiece axis and the absolute coordinates are updated with helical compensation.

- Direction of helical gear compensation

The direction depends on HDR, bit 2 of parameter No. 7700.



- Synchronization coefficient

A synchronization coefficient is internally represented using a fraction (Kn/Kd) to eliminate an error. The formula below is used for calculation.

$$\text{Synchronization coefficient} = \frac{K_n}{K_d} = \frac{L}{T} \times \frac{\beta}{\alpha}$$

where

L : Number of hob threads

T : Number of teeth

α : Number of pulses of the position detector per rotation about the master axis (parameter No. 7772)

β : Number of pulses of the position detector per rotation about the slave axis (parameter No. 7773)

Kn / Kd is a value resulting from reducing the right side of the above formula, but the result of reduction is subject to the following restrictions:

$-2147483648 \leq K_n \leq 2147483647$

$1 \leq K_d \leq 65535$

When this restriction is not satisfied, the alarm PS1596 is issued when G81 is specified.

Example

```
O1000 ;
N0010 M19 ;                 Tool axis orientation
N0020 G28 G91 C0 ;          Reference position return on the workpiece axis
N0030 G81 T20 L1 ;          Synchronous start on tool and workpiece axes
                            (Rotation about the workpiece axis by 18° per rotation about the tool axis)
N0040 S300 M03 ;            Rotation about the tool axis at 300min-1
N0050 G01 X__F__;           Movement along the X-axis (cut)
N0060 G01 Z__F__;           Movement along the Z-axis (machining)
----------;                 If necessary, axis commands such as C, X, and Z commands are allowed.
N0100 G01 X__F__;           Movement along the X-axis (escape)
N0110 M05 ;                 Stop on the tool axis
N0120 G80 ;                 Synchronous cancellation on tool and workpiece axes
N0130 M30 ;
```

- Retract function

(1) Retract function with an external signal
When the retract switch on the machine operator's panel is turned on, retraction is performed with the retract amount set in parameter No. 7741 and the feedrate set in parameter No. 7740. No movement is performed along an axis for which 0 is set as the retract amount.
For the retract switch, refer to the relevant manual provided by the machine tool builder.

(2) Retract function with an alarm
If, during EGB synchronization or automatic operation, a CNC alarm is issued, retraction is performed with the retract amount set in parameter No. 7741 and the speed set in parameter No. 7740.
This can prevent the tool and the object being machined from damage if a servo alarm is generated.
No movement is performed along an axis for which 0 is set as the retract amount.
For the retract switch, refer to the relevant manual provided by the machine tool builder.

Conditions under the retract function with an alarm can be executed
The conditions under which the retract function with an alarm can be executed can be changed using the settings of ARE, bit 1 of parameter No. 7703, and bit 2 (ARO) of parameter No. 7703.
The table below lists parameter settings and corresponding conditions.

ARE	ARO	Condition
1	0	EGB synchronization is in progress.
1	1	Both EGB synchronization and automatic operation are in progress.
0	0	Either EGB synchronization or automatic operation is in progress.
0	1	

⚠ CAUTION
1 Retraction is performed at the speed specified in parameter No. 7740.
2 Feed hold is not effective to movement during retraction.
3 Feedrate override is not effective to movement during retraction.

NOTE

1 During a retract operation, an interlock is effective to the retract axis.
2 During a retract operation, a machine lock is effective to the retract axis.
3 The retraction direction depends on the movement direction of the machine, regardless of whether an mirror image (signal and setting) is enabled or disabled. (No mirror image can be applied to the updating of absolute coordinates.)
4 If retraction is performed during automatic operation, automatic operation is halted simultaneously with a retract operation, but it is at the end of the retract operation that the operation state switches to the automatic operation halt state.
5 It is not possible to perform automatic operation during retraction.
6 The acceleration/deceleration of a retract operation is in the acceleration/deceleration state at the start of retraction.
7 Retract movement is performed with non-linear type positioning.
8 If, during a retract operation, a reset or an emergency stop is made, the operation is interrupted.
9 If, during a retract operation on multiple axes, an OT alarm or a malfunction prevention alarm is issued for a retract axis, the operation stops only on the axis for which the alarm is issued if bit 4 (RTS) of parameter No. 7731 is 0. If bit 4 (RTS) of parameter No. 7731 is 1, the retract operation is interrupted on all axes. If a servo alarm or non-axis malfunction prevention alarm is issued, the retract operation is interrupted on all axes regardless of the setting of bit 4 (RTS) of parameter No. 7731.
10 To enable the retract function with an alarm, bit 3 (ART) of parameter No.7702, must be set.
11 The retract function with an alarm does not perform a retract operation on the retract axis if an overtravel alarm or a servo alarm is geneated on the retract axis.
12 If a new alarm is issued during retraction with the retract function with an alarm, a retract operation is not performed.

21.7.2 Electronic Gear Box Automatic Phase Synchronization

Overview

In the electronic gear box (EGB), when synchronization start or cancellation is specified, synchronization is not started or canceled immediately. Instead, acceleration/deceleration is executed first. Therefore, synchronization can be started or canceled while the spindle is rotating. Also, synchronization ratio can be changed while the spindle is rotating.
At synchronization start, automatic phase synchronization is performed such that the workpiece axis position matches the position corresponding to the spindle one-rotation signal. With this synchronization, the same operation is performed as synchronization start caused by a one-rotation signal in hobbing synchronization when using the functions of a hobbing machine.
The spindle corresponds to the EGB master axis and the workpiece axis corresponds to an EGB slave axis.

Format

- Acceleration/deceleration type

```
G81  T _  L _  R1  ;   Synchronization start
G80  R1  ;             Synchronization cancellation
```

T : Number of teeth (range of valid settings: 1-1000)
L : Number of hob threads (range of valid settings: -200 to +200, excluding 0)
When L is positive, the direction of rotation about the workpiece axis is positive (+ direction).
When L is negative, the direction of rotation about the workpiece axis is negative (- direction).

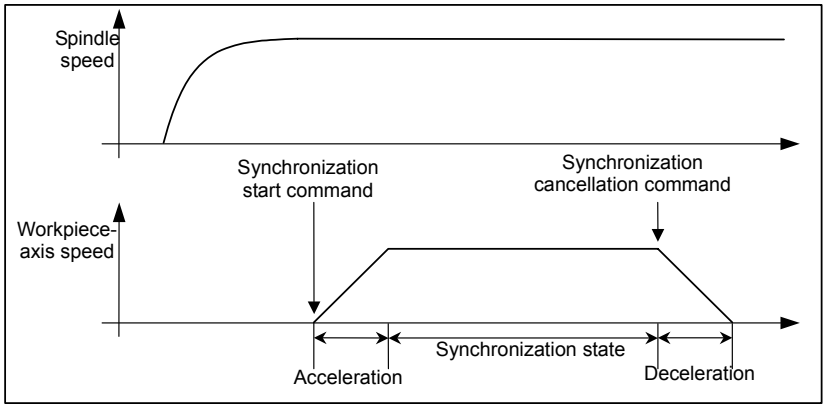
- Acceleration/deceleration plus automatic phase synchronization type

```
G81  T _  L _  R2  ;   Synchronization start
G80  R1  ;             Synchronization cancellation
```

T : Number of teeth (range of valid settings: 1-1000)
L : Number of hob threads (range of valid settings: -200 to +200, excluding 0)
When L is positive, the direction of rotation about the workpiece axis is positive (+ direction).
When L is negative, the direction of rotation about the workpiece axis is negative (- direction).

Explanation

- Acceleration/deceleration type



1. Specify G81R1 to start synchronization.
   When G81R1 is specified, the workpiece axis (slave axis) is subject to acceleration at the acceleration rate set in parameter No. 7778. When the synchronization speed is reached, the G81R1 block is terminated.
2. For cancellation, specify G80R1 while the tool is moved away from the workpiece.
3. When G80R1 is specified, deceleration is started immediately at the acceleration rate set in parameter No. 7778.
   When the speed is reduced to 0, the G80R1 block is terminated.

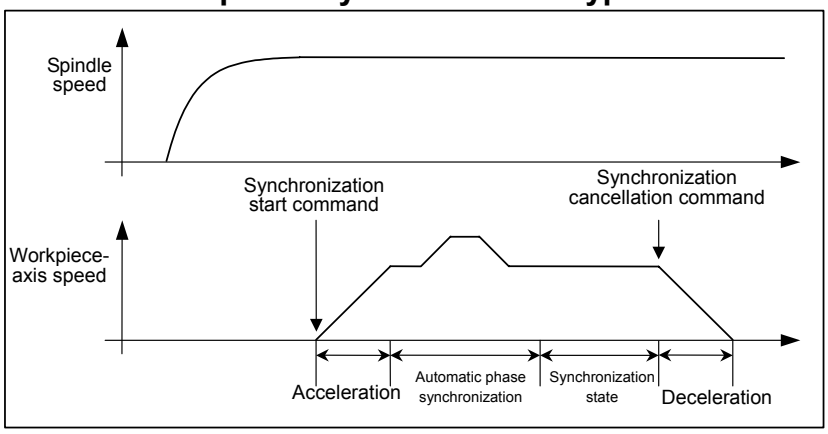
NOTE

1 During synchronization start/cancellation, acceleration/deceleration is linear.

2 In the automatic cancellation of synchronization due to one of the following causes, deceleration is performed and synchronization is canceled:
   <1> Reset
   <2> PW0000 (POWER MUST BE OFF)
   <3> IO alarm

3 If EFX, bit 0 of parameter No. 7731, is 0, the canned cycle for drilling cannot be used. To use the canned cycle for drilling, set EFX, bit 0 of parameter No. 7731, to 1 and use G81.4 instead of G81 and G80.4 instead of G80.

- Acceleration/deceleration plus automatic phase synchronization type



1. Move the workpiece axis to the position that corresponds to that of the one-rotation signal of the spindle.
2. Specify G81R2 to start synchronization.
   When G81R2 is specified, the workpiece axis is accelerated with the acceleration according to the acceleration rate set in the parameter No.7778. Upon completion of phase synchronization, the G81R2 block terminates.
3. For cancellation, specify G80R2 while the tool is moved away from the workpiece.
4. When G80R2 is issued, deceleration is started immediately according to the acceleration rate set in parameter No. 7778. When the speed reaches 0, the G80R2 block terminates.

⚠ CAUTION

1 Set the automatic phase synchronization speed in parameter No. 7776 and the movement direction in PHD, bit 7 of parameter No. 7702.

2 Phase synchronization acceleration/deceleration is performed with the rapid traverse linear acceleration/deceleration rate (time constant specified in parameter No. 1620).

3 The workpiece axis speed is the speed synchronized with spindle rotation with automatic phase synchronization speed being superposed. When setting the position deviation limit parameter No. 1828, take the superposition into consideration.

NOTE

1 The one-rotation signal used for automatic phase synchronization is issued not by the spindle position coder but by the separate pulse coder attached to the spindle and used to collect EGB feedback information. This means that the orientation position based on the one-rotation signal issued by the spindle position coder does not match the position used as the reference for the workpiece axis when establishing phase synchronization for automatic phase synchronization based on G81R2.
Moreover, the one-rotation signal of the separate pulse coder must be turned on for each rotation of the spindle.

2 With the use of parameter No.7777, the position at which the phase of the workpiece axis is matched can be shifted from the position corresponding to the one-rotation signal in automatic phase matching.

3 In automatic phase synchronization, when a synchronization command is issued again in a synchronization state, movement about the workpiece axis is performed such that the position corresponding to the one-rotation signal of the spindle matches the position about the workpiece axis specified in the G81R2 synchronization start command executed first.

4 In automatic phase synchronization, movement is performed about the workpiece axis from the current position to the nearest phase position in the phase synchronization movement direction specified by the parameter.

5 Linear acceleration/deceleration applies to synchronization start/cancellation.

6 The acceleration/deceleration plus automatic phase synchronization type can be executed by the PHS parameter, bit 6 of No.7702, without specifying an R2 command in a G81 or G80 block.

7 In the automatic cancellation of synchronization due to one of the following causes, deceleration is performed and synchronization canceled:
<1> Reset
<2> PW0000 (POWER MUST BE OFF)
<3> IO alarm

8 The acceleration rate parameter No.7778 must not be changed in the synchronization mode.

9 If the acceleration/deceleration parameter No. 7778 is 0, alarm PS1598 is issued when G81 is issued.

10 For the FS16*i* series, the acceleration/deceleration for automatic phase synchronization is divided into acceleration/deceleration and time constant, which are set in parameters Nos. 2135 and 2136. For the FS30i series, acceleration/deceleration is directly set in parameter No. 7778.

11 If EFX, bit 0 of parameter No. 7731, is 0, the canned cycle for drilling cannot be used. To use the canned cycle for drilling, set EFX, bit 0 of parameter No. 7731, to 1 and use G81.4 instead of G81 and G80.4 instead of G80.

Program example

- Acceleration/deceleration type

```
M03 ;                  Clockwise spindle rotation command
G81  T_  L_  R1 ;      Synchronization start command
G00  X_ ;              Positions the workpiece at the machining
                       position.

Machining in the synchronous state

G00  X_ ;              Retract the workpiece from the tool.
G81  T_  L_  R1 ;      Synchronization ratio change.
G00  X_ ;              Positions the workpiece at the machining
                       position.

Machining in the synchronous state

G00  X_ ;              Retract the workpiece from the tool.
G80  R1  ;             Synchronization cancel command
```

- Acceleration/deceleration plus automatic phase synchronization type

```
M03  ;                 Clockwise spindle rotation command
G00  G90  C_ ;         C-axis positioning
G81  T_  L_  R2 ;      Synchronization start command
G00  X_ ;              Positions the workpiece at the machining
                       position.

Machining in the synchronous state

G00  X_ ;              Retract the workpiece from the tool.
G81  T_  L_  R2 ;      Synchronization ratio change.
G00  X_ ;              Positions the workpiece at the machining
                       position.

Machining in the synchronous state

G00  X_ ;              Retract the workpiece from the tool.
G80  R2  ;             Synchronization cancel command
```

21.7.3 Skip Function for EGB Axis

Overview

This function enables the skip or high-speed skip signal (these signals are collectively called skip signals in the remainder of this manual) for the EBG slave axis in synchronization mode with the EGB (electronic gear box).
This function has features such as the following:

1 If a skip signal is input while an EGB axis skip command block is being executed, this block does not terminate until the specified number of skip signals have been input.
2 If a skip signal is input while an EGB axis skip command block is being executed, the tool remains in synchronous mode and moves, not stopping on the EGB slave axis.
3 The machine coordinates assumed when skip signals are input and the number of input skip signals are stored in specified custom macro variables.

Format

G81 T_ L_ ; EGB mode ON
G31.8 G91 α0 P_ Q_ (R_) ; EGB skip command

α: Specify an EGB slave axis. The specified value must always be "0".
P: Number of the first one of the custom macro variables used to store the machine coordinates assumed when skip signals are input.
Q: Number of skip signals that can be input during the execution of G31.8
(permissible range: 1 to 200).
R: Number of the custom macro variable used to store the number of input skip signals.
Specify it to check the number of input signals.

Explanation

G31.8 is a one-shot G code.
When G31.8 is executed, the machine coordinates assumed when skip signals are input are written in as many custom macro variables as the number specified in Q, starting with the one having the number specified in P, when the skip command block terminates.
Also, the number of input skip signals is written to the custom macro variable specified in R each time a skip signal is input.

Example

```
G81  T200  L2 ;                          EGB mode ON
X_ ;
Z_ ;
G31.8  G91  C0  P500  Q200  R1 ;   EGB skip command
```

After 200 skip signals have been input, the 200 skip positions on the C-axis that correspond to the respective skip signals are stored in custom macro variables #500 to #699.
Also, the number of input skip signals is stored in custom macro variable #1.

NOTE

1 When specifying this function, specify only a single EGB slave axis. If no axis is specified for two or more axes are specified, alarm PS1152 is generated.
2 If P is not specified, alarm PS1152 is generated.
3 If R is not specified, the number of input skip signals is not written to a custom macro variable.
4 The custom macro variable numbers specified in P and R must be existing ones. If a non-existent variable number is specified, alarm PS0115 is generated.
If a variable shortage occurs, alarm PS0115 is generated.
5 Whether to use conventional skip signals or high-speed skip signals with this function can be specified with HSS, bit 4 of parameter No. 6200.
If deciding to use high-speed skip signals, specify which high-speed signals to enable with 9S1 to 9S8, bits 0 to 7 of parameter No. 6208).
6 Skip positions are calculated from feedback pulses from the machine. Thus, they are free from errors due to delay in acceleration/deceleration and the servo system

21.7.4 Electronic Gear Box 2 Pair

Overview

The Electronic Gear Box is a function for rotating a workpiece in sync with a rotating tool, or to move a tool in sync with a rotating workpiece. With this function, the high-precision machining of gears, threads, and the like can be implemented. A desired synchronization ratio can be programmed.
Up to two sets of axes can be synchronized. A gear grinding machine can be controlled, for instance, by using one axis for rotating the workpiece in sync with the tool and another axis for performing dressing in sync with the tool.
The specification method differs depending on the configuration of the machine. For details, refer to the relevant manual provided by the machine tool builder.
The electronic gear box is hereinafter called an EGB function.

21.7.4.1 Specification method (G80.5, G81.5)

Format

G81.5 $\left\{\begin{matrix}\mathbf{Tt}\\ \mathbf{Pp}\end{matrix}\right\}$ $\left\{\begin{matrix}\beta\mathbf{j}\\ \beta\mathbf{0}\ \mathbf{L}l\end{matrix}\right\}$ **;** Synchronization start

↑ Amount of travel along the master axis

↑ Amount of travel along the slave axis

G80.5 β0 ; Synchronization end

Explanation

- Master axis, slave axis, and dummy axis

The synchronization reference axis is called the master axis, while the axis along which movement is performed in synchronization with the master axis is called the slave axis. For example, if the workpiece moves in synchronization with the rotating tool as in a hobbing machine, the tool axis is the master axis and the workpiece axis is the slave axis.
Which axes to become the master and slave axes depends on the configuration of the machine. For details, refer to the manual issued by the machine tool builder.
A single servo axis is used exclusively so that digital servo can directly read the rotation position of the master axis. (This axis is called the EGB dummy axis.)

- Synchronization start

When the ratio of the master-axis travel to the slave-axis travel is specified, synchronization starts.

Specify the master-axis travel in either of the following ways.

1 Master-axis speed
T t: Master-axis speed (1≤ t ≤1000)

2 Master-axis pulse count
P p : Master-axis pulse count (1≤ p ≤999999999)
Specify a pulse count on the condition that four pulses correspond to one period in the A and B phases.

Specify the slave-axis travel in either of the following ways.

1 Slave-axis travel
βj : Slave-axis address
j : Slave-axis travel indicated in units of the minimum travel increments(the range of valid settings for usual axis movement applies)
When j = 0, the specified command is regarded as being a command for the slave-axis speed, described below. In this case, if L is not specified, an alarm is output.

2 Slave-axis speed
β0 L±*l*
β : Slave-axis address
l : Slave axis speed(1≤ *l* ≤200)

- Synchronization end

1 Canceling synchronization for each axis by issuing a command
With a G80.5 β0 command, synchronization is canceled.
β is the address of the slave axis. Synchronization of the slave axis specified by β is canceled.
A cancellation command can be issued only for one axis in one block.
When β0 is not specified, the synchronization of all currently synchronized axes is canceled.
When a synchronization cancellation command is issued, the absolute coordinates for the slave axis are updated according to the amount of travel during synchronization. For a rotation axis, the value obtained by rounding off the amount of travel during synchronization to the nearest 360 degrees is added to the absolute coordinates.

2 Canceling synchronization by a reset
By setting HBR, bit 0 of parameter No. 7700, to 0, synchronization is canceled with a reset. If the manual absolute signal is ON, the absolute coordinates are updated.

3 Others
Synchronization is automatically canceled under the following conditions.

<1> Emergency stop
<2> Servo alarm
<3> Alarm PW0000 (indicating that the power should be turned off)
<4> An IO alarm is generated

⚠ CAUTION

1 Feed hold, interlock, and machine lock are invalid to a slave axis in EGB synchronization.
2 Even if an OT alarm is issued for a slave axis in EGB synchronization, synchronization will not be canceled.
3 During synchronization, it is possible to execute a move command for a slave axis and other axes, using a program. The move command for a slave command must be an incremental one.

NOTE

1 If bit 0 (HBR) of parameter No. 7700 is set to 1, EGB synchronization will not be canceled due to a reset. Usually, set this parameter bit to 1.
2 In synchronous mode, it is not possible to specify G27, G28, G29, G30, G30.1, and G53 for a slave axis.
3 It is not possible to use controled axis detach for a slave axis.
4 During synchronization, manual handle interruption can be performed on the slave and other axes.
5 In synchronization mode, no inch/metric conversion commands (G20 and G21) cannot be issued.
6 In synchronous mode, only the machine coordinates on a slave axis are updated.
7 If bit 0 (EFX) of parameter No. 7731 is 0, no drilling canned cycle can be used. To use a drilling canned cycle, set bit 0 (EFX) of parameter No. 7731 to 1.
8 If, during synchronization, G81.5 is issued again, alarm PS1594 is issued if bit 3 (ECN) of parameter No. 7731 is 0. If bit 3 (ECN) of parameter No. 7731 is 1, the synchronization coefficient can be changed to a newly specified one.
9 Actual cutting feedrate display does not take synchronization pulses into consideration.
10 For an EGB slave axis, synchronous and cpmposite control cannot be executed.
11 In EGB synchronization mode, AI contour control mode is temporarily canceled.

21.7.4.2 Description of commands compatible with those for a hobbing machine (G80, G81)

A command compatible with that for a hobbing machine can be used as a synchronization command.
Usually, a hobbing machine performs machining by synchronizing the workpiece axis (usually, the C-axis) to the hobbing axis (spindle).
If there are two synchronization sets with the EGB, which synchronization set to start with this specification method can be specified with parameter No. 7710.

Format

	Parameter EFX (No.7731#0)=0	Parameter EFX(No.7731#0)=1	
		Parameter HBR (No.7731#5)=1	Parameter HBR (No.7731#5)=0
Start of synchronization	**G81 T__ (L__) (Q__ P__) ;**	**G81.4 R__ (L__) (Q__ P__) ;**	**G81.4 T__ (L__) (Q__ P__) ;**
Cancellation of synchronization	**G80 ;**	**G80.4 ;**	**G80.4 ;**
	(*1)	**(*2)**	**(*3)**

T(or R) : Number of teeth (Specifiable range: 1 to 1000)
L : Number of hob threads (Specifiable range: -200 to +200)
The sign of L determines the direction of rotation for the workpiece axis.
When L is positive, the direction of rotation for the workpiece axis is positive (+ direction).
When L is negative, the direction of rotation for the workpiece axis is negative (- direction).
When L is 0, it follows the setting of bit 3 (LZR) of parameter No.7701.
If L is not specified, the number of hob threads is assumed 1.
Q : Module or diametral pitch
Specify a module in the case of metric input.
(Unit: 0.00001mm, Specifiable range: 0.01 to 25.0mm)
Specify a diametral pitch in the case of inch input.
(Unit: 0.00001inch^{-1}, Specifiable range: 0.01 to 254.0 inch^{-1})
P : Gear helix angle
(Unit: 0.0001deg, Specifiable range: -90.0 to 90.0deg)
*1 Use it for machining centers.
*2 Use it for lathes.
*3 Use it for machining centers.
This format enables specification of the same G codes as those for lathes.
*4 When specifying Q and P, the user can use a decimal point.

NOTE
Specify G81, G80, G81.4, and G80.4 in a single block.

Explanation

- Synchronization start

Specify P and Q to use helical gear compensation. In this case, if only one of P and Q is specified, alarm PS1594 is generated.
When G81 is issued so that the machine enters synchronization mode, the synchronization of the workpiece axis to the spindle is started.
During synchronization, control is performed such that the ratio of the spindle speed to the workpiece-axis speed is the same as that of T (number of teeth) to L (number of hob threads).
If, during synchronization, G81 is issued again without synchronization cancellation, alarm PS1595 is generated if ECN, bit 3 of parameter No. 7731, is 0. If ECN, bit 3 of parameter No. 7731, is 1, helical gear compensation is conducted with the synchronization coefficient being changed to the one newly specified with T and L commands if T and L commands are issued, and if T and L commands are not issued and only P and Q commands are issued, helical gear compensation is conducted with the synchronization coefficient kept intact. This allows consecutive fabrication of helical gears and super gears.

- Synchronization end

Synchronization of all synchronized axes is canceled.
When a synchronization cancellation command is issued, the absolute coordinates for the slave axis are updated according to the amount of travel during synchronization.
For a rotation axis, the value obtained by rounding off the amount of travel during synchronization to the nearest 360 degrees is added to the absolute coordinates.
In a G80 block, do not specify addresses other than O or N.

⚠ CAUTION
1 Feed hold, interlock, and machine lock are invalid to a slave axis in EGB synchronization.
2 Even if an OT alarm is issued for a slave axis in EGB synchronization, synchronization will not be canceled.
3 During synchronization, it is possible to execute a move command for a slave axis and other axes, using a program. The move command for a slave command must be an incremental one.

NOTE
1 If bit 0 (HBR) of parameter No. 7700 is set to 1, EGB synchronization will not be canceled due to a reset. Usually, set this parameter bit to 1.
2 In synchronous mode, it is not possible to specify G27, G28, G29, G30, G30.1, and G53 for a slave axis.

NOTE

3 It is not possible to use controled axis detach for a slave axis
4 During synchronization, manual handle interruption can be performed on the slave and other axes.
5 In synchronization mode, no inch/metric conversion commands (G20 and G21) cannot be issued.
6 In synchronous mode, only the machine coordinates on a slave axis are updated.
7 If bit 0 (EFX) of parameter No. 7731 is 0, no drilling canned cycle can be used. To use a drilling canned cycle, set bit 0 (EFX) of parameter No. 7731 to 1 and use G81.4 instead of G81 and G80.4 instead of G80.
8 If bit 0 (TDP) of parameter No. 7702, is 1, the permissible range of T is 0.1 to 100 (1/10 of the specified value).
9 If, at the start of EGB synchronization (G81), L is specified as 0, synchronization starts with L assumed to be 1 if bit 3 (LZR) of parameter No.7701, is 0; if bit 3 (LZR) of parameter No.7701, is 1, synchronization is not started with L assumed to be 0. At this time, helical gear compensation is performed.
10 Feed per revolution is performed on the feedback pulses on the spindle. By setting bit 0 (ERV) of parameter No. 7703, to 1, feed per revolution can be performed based on the speed on the synchronous slave axis.
11 Actual cutting feedrate display does not take synchronization pulses into consideration.
12 For an EGB slave axis, synchronous and cpmposite control cannot be executed.
13 In EGB synchronization mode, AI contour control mode is temporarily canceled.

- Helical gear compensation

For a helical gear, the workpiece axis is subjected to compensation for movement along the Z axis (axial feed axis) according to the twisted angle of the gear.
Helical gear compensation is performed with the following data.

$$\text{Compensation angle} = \frac{Z \times \sin(P)}{\pi \times T \times Q} \times 360 \text{ (for metric input)}$$

$$\text{Compensation angle} = \frac{Z \times Q \times \sin(P)}{\pi \times T} \times 360 \text{ (for inch input)}$$

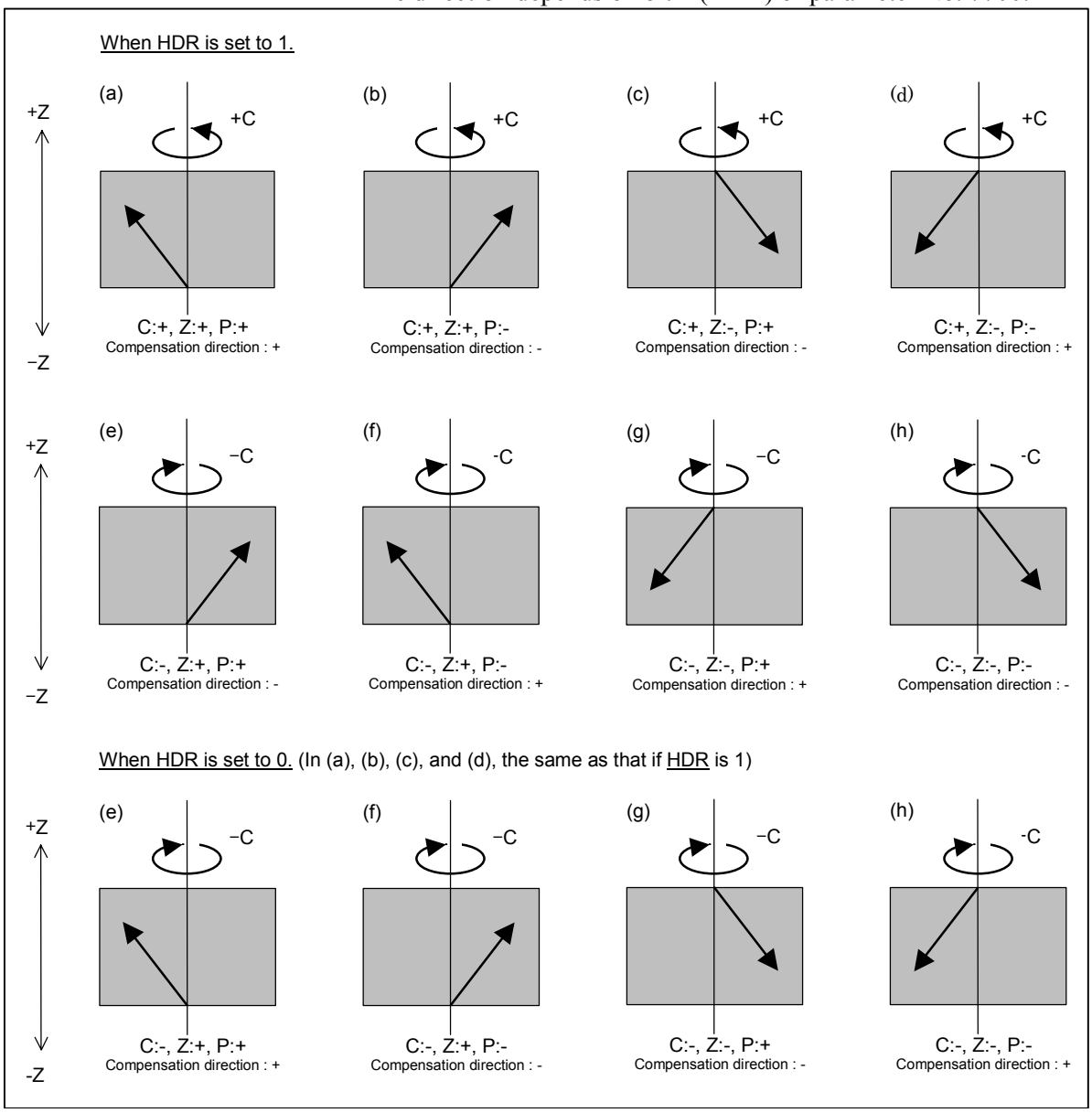
where

Compensation angle: Absolute value with sign (degrees)
Z :Amount of travel along the Z axis after a G81 command is issued (mm or inch)
P : Twisted angle of the gear with sign (degrees)
π : Circular constant
T : Number of teeth
Q : Module (mm) or diametral pitch ($inch^{-1}$)
Use P, T, and Q specified in the G81 block.

In helical compensation, the machine coordinates on the workpiece axis and the absolute coordinates are updated with helical compensation.

- Direction of helical gear compensation

The direction depends on bit 2 (HDR) of parameter No. 7700.



21.7.4.3 Controlled axis configuration example

- For gear grinders

Spindle : EGB master axis : Tool axis
1st axis : X axis
2nd axis : Y axis
3rd axis : C axis (EGB slave axis : Workpiece axis)
4th axis : C axis (EGB dummy axis : Cannot be used as a normal controlled axis)
5th axis : V axis (EGB slave axis : Dressing axis)
6th axis : V axis (EGB dummy axis : Cannot be used as a normal controlled axis)

Cannot be used as a normal controlled axis)

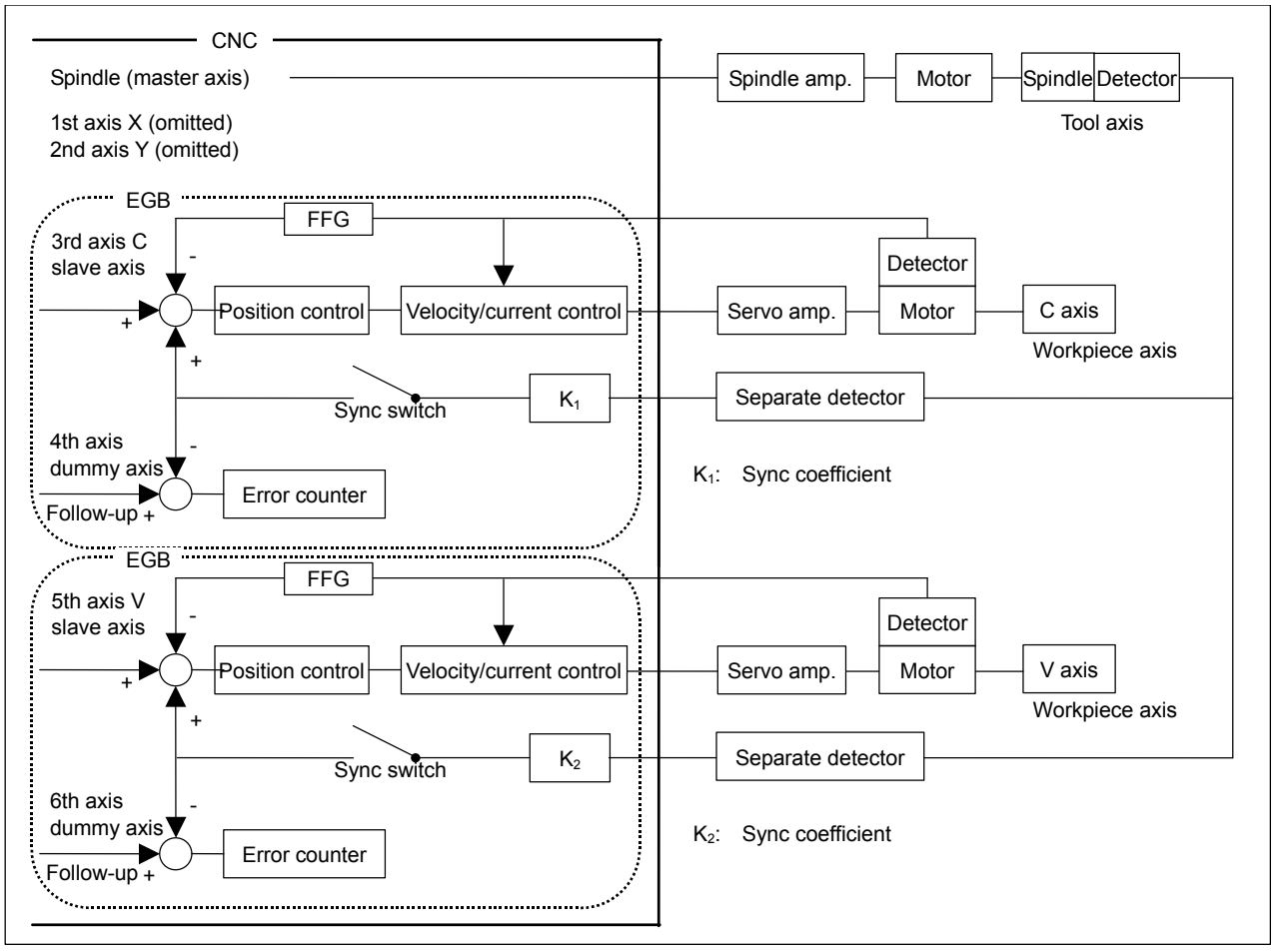


Fig. 21.7.4.3 (a)

21.7.4.4 Sample programs

- When the master axis is the spindle, and the slave axis is the C-axis

(1) G81.5 T10 C0 L1 ;
Synchronization between the master axis and C-axis is started at the ratio of one rotation about the C-axis to ten rotations about the master axis.

(2) G81.5 T10 C0 L-1 ;
Synchronization between the master axis and C-axis is started at the ratio of one rotation about the C-axis to ten rotations about the master axis.
In this case, however, the direction of rotation is opposite to that of (1).

(3) G81.5 T1 C3.26 ;
Synchronization between the master axis and C-axis is started at the ratio of a 3.26-degree rotation about the C-axis per one rotation about the master axis.

(4) G81.5 P10000 C-0.214 ;
Synchronization between the master axis and C-axis is started at the ratio of a -0.214 degree rotation about the C-axis to 10,000 feedback pulses from the pulse coder of the master axis.

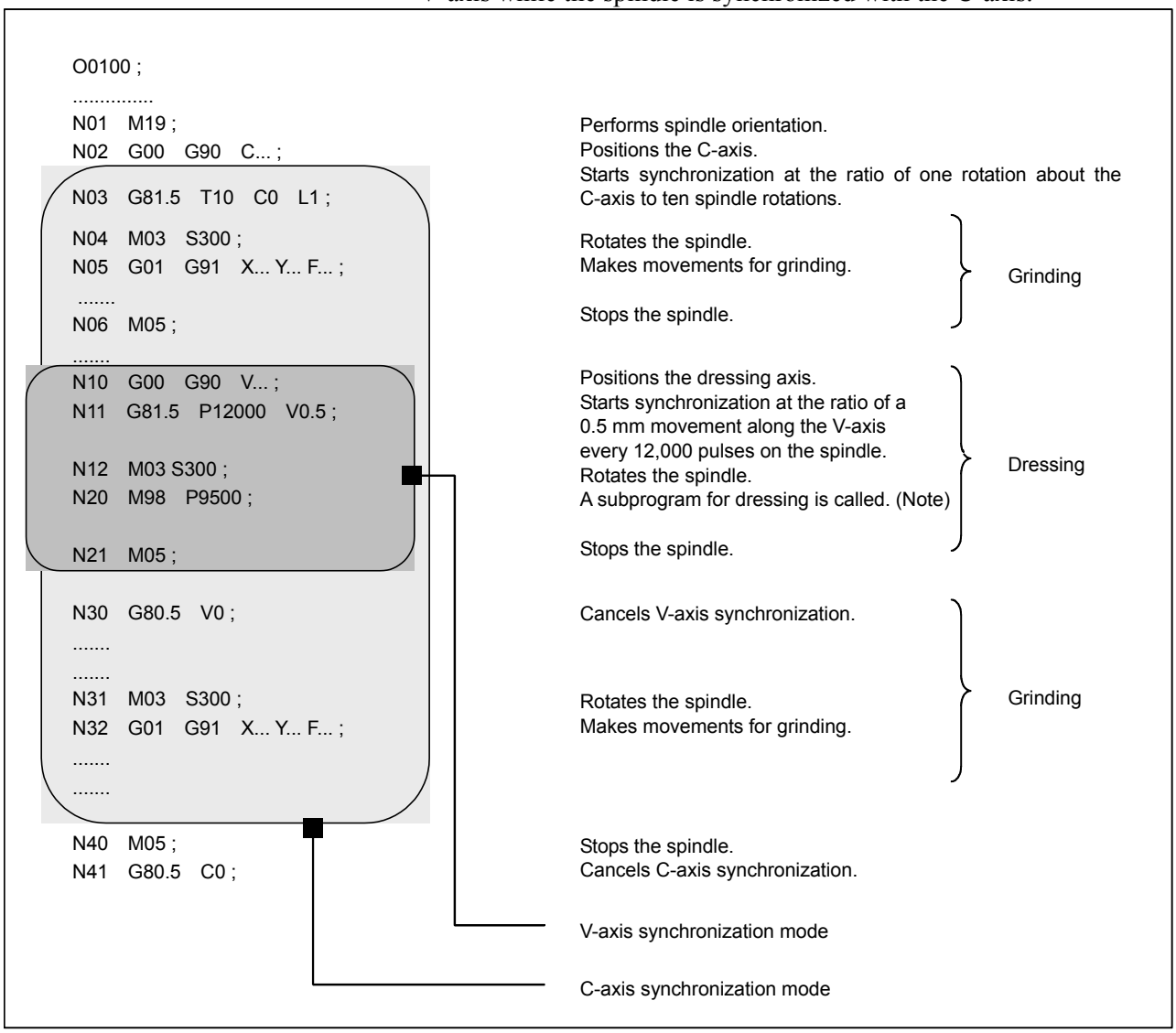
- When the master axis is the spindle, the slave axis is the V-axis (linear axis), and inch/metric conversion is performed

(1) For a millimeter machine and metric input
G81.5 T1 V1.0 ;
Synchronization between the master axis and V-axis is started at the ratio of a 1.00 mm movement along the V-axis per rotation about the master axis.

(2) For a millimeter machine and inch input
G81.5 T1 V1.0 ;
Synchronization between the master axis and V-axis is started at the ratio of a 1.0 inch movement (25.4 mm) along the V-axis per rotation about the master axis.

- When two groups of axes are synchronized simultaneously

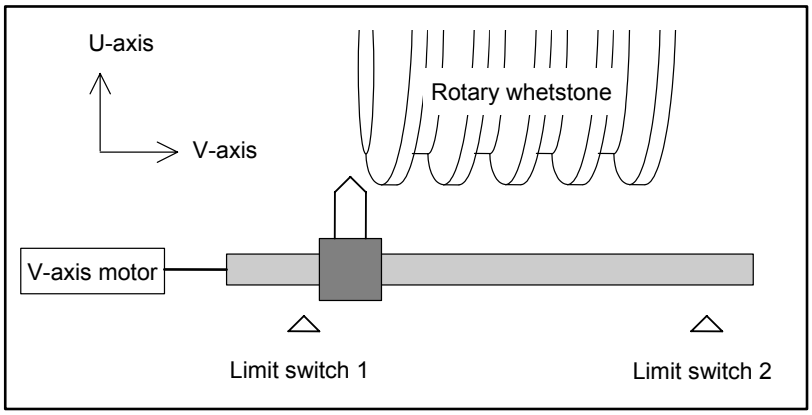
Based on the controlled axis configuration described in Fig. 21.7.4.3 (a), the sample program below synchronizes the spindle with the V-axis while the spindle is synchronized with the C-axis.



Thus, the synchronizations of two groups can be started and canceled independently of each other.

- Example of use of dressing

Gear grinder in the following machine configuration




```
O9500 ;
N01  G01  G91  U__  F100 ;   Dressing axis approach
N02  M03  S100 ;             The M03 command causes the PMC to rotate the
                             whetstone in the positive direction.
                             In accordance with this, the tool moves along the V-axis
                             in the + direction.  When the tool reaches the position of
                             limit switch 2 on the V-axis, the PMC stops the
                             whetstone and returns FIN.
N02  U__  V__ ;              Movement to the next dressing position
N03  M04  S100 ;             The M04 command causes the PMC to rotate the
                             whetstone in the negative direction.
                             In accordance with this, the tool moves along the V-axis
                             in the - direction.  When the tool reaches the position of
                             limit switch 1 on the V-axis, the PMC stops the
                             whetstone and returns FIN.
N04  U__  V__ ;              Movement to the next dressing position
                             If required, N02 to N04 are repeated to conduct dressing.
.....
.....

M99 ;
```

NOTE

If the V-axis (linear axis) is synchronized with the spindle as in dressing, the V-axis travel range is determined by the rotation of the spindle. To perform dressing with the tool moving back and forth along the V-axis in a certain range, therefore, the PMC must perform an operation in which the tool is stopped temporarily and is reversed when it reaches a certain position on the V-axis.
In the above example, limit switches are provided to determine the range of travel along the V-axis and the PMC performs control so that the whetstone rotates until the tool reaches the position of each limit switch on the V-axis.
By using the position switch function instead of limit switches, dressing can be performed as in the following example, without the need to mount limit switches to the machine. By rewriting the operating ranges of the position switches (parameters Nos. 6930 to 6945 and Nos. 6950 to 6965) using the G10 programming Parameter input, the range of travel along the V-axis can be specified using a program.

- Command specification for hobbing machines

Based on the controlled axis configuration described in Fig. 21.7.4.3 (a), the sample program below sets the C-axis (in parameter 7710) for starting synchronization with the spindle according to the command specification method for hobbing machines.

```
O1234 ;
N0010  M19  ;                  Performs tool axis orientation.
N0020  G28  G91  C0  ;         Performs reference position return along the workpiece axis.
N0030  G81  T20  L1 ;          Starts synchronization with the spindle and C-axis at the ratio of
                               a 18° rotation about the C-axis to one spindle rotation.

N0040  S300  M03 ;             Rotates the spindle at 300 min-1.
N0050  G01  X...  F... ;       Makes a movement along the X-axis (for cutting).
N0060  G01  Y...  F... ;       Makes a movement along the Y-axis (for grinding).
                               Axes such as the C-axis, X-axis, and Y-axis can be specified as
                               required.
N0100  G01  X...  F... ;       Makes a movement along the X-axis (for retraction).
N0110  M05 ;                   Stops the spindle.
N0120  G80 ;                   Cancels the synchronization between the spindle and C-axis.
N0130  M30 ;
```

21.7.4.5 Synchronization ratio specification range

The programmed ratio (synchronization ratio) of a movement along the slave axis to a movement along the master axis is converted to a detection unit ratio inside the NC. If such converted data (detection unit ratio) exceeds a certain allowable data range in the NC, synchronization cannot be established correctly, and an alarm PS1596 is issued.
Even when a programmed master axis movement and a programmed slave axis movement are within specifiable ranges, a detection unit ratio obtained by conversion can exceed the allowable range, thus resulting in an alarm.
Let K be a synchronization ratio. The internal data corresponding K is the amount of slave axis movement (Kn) represented in the detection unit divided by the amount of master axis movement (Kd) represented in the detection unit;
this fraction is represented as Kn/Kd (reduced to its lowest terms) as indicated below.

$$K = \frac{Kn}{Kd} = \frac{\text{Amount of slave axis movement represented in the detection unit}}{\text{Amount of master axis movement represented in the detection unit}}$$

Kn and Kd must lie within the following ranges:

$$-2147483648 \leq Kn \leq 2147483647$$
$$1 \leq Kd \leq 65535$$

When Kn or Kd exceeds its allowable range above, an alarm PS1596 is issued.
In conversion to the detection unit, when the CMR (command multiplication: parameter 1820) is a fraction or when inch/millimeter conversion is used, the fraction is directly converted without modification so that no error can occur in the conversion of specified amounts of movement.
During conversion, the amount of movement is multiplied by 254/100 for inch input on a millimeter machine, and 100/254 for metric input on an inch machine. Thus, Kn and Kd can become large numbers. If a synchronization ratio cannot be reduced to its lowest terms, an alarm condition is likely to occur.

- Example 1)

Based on the controlled axis configuration described in Fig. 21.7.4.3 (a), suppose that the spindle and V-axis are as follows:

Spindle pulse coder	: 72000pulse/rev (4 pulses for one A/B phase cycle)
C-axis least command increment	: 0.001deg
C-axis CMR	: 5
V-axis least command increment	: 0.001mm
V-axis CMR	: 5

Then, the C-axis detection unit is 0.0002 degree. The V-axis detection unit is 0.0002 mm. In this case, the synchronization ratio (Kn, Kd) is related with a command as indicated below. Here, let Pm and Ps be the amounts of movements represented in the detection unit on the master axis and slave axis specified in a synchronization start command, respectively.

(1) When the master axis is the spindle, and the slave axis is the C-axis

(a) Command: G81.5 T10 C0 L1 ;
Operation : Synchronization between the spindle and C-axis is started at the ratio of one rotation about the C-axis to ten spindle rotations.
Pm : (Number of pulses per spindle rotation) × (10 rotations) → 72000 × 10
Ps : (Amount of movement per rotation about the C-axis) × CMR × (one rotation) → 360000 × 5 × 1

$$\frac{Kn}{Kd} = \frac{360000 \times 5 \times 1}{72000 \times 10} = \frac{5}{2}$$

Both Kn and Kd are within the allowable range. No alarm is output.

(b) Command: G81.5 T10 C0 L-1 ;
Operation : Synchronization between the spindle and C-axis is started at the ratio of one rotation about the C-axis to ten spindle rotations.
In this case, however, the direction of rotation is opposite to that of (a) above.
Pm : (Number of pulses per spindle rotation) × (10 rotations) → 72000 × 10
Ps : (Amount of movement per rotation about the C-axis) × CMR × (one rotation) → -360000 × 5 × 1

$$\frac{Kn}{Kd} = \frac{-360000 \times 5 \times 1}{72000 \times 10} = \frac{-5}{2}$$

Both Kn and Kd are within the allowable range. No alarm is output.

(c) Command: G81.5 T1 C3.263 ;
Operation : Synchronization between the spindle and C-axis is started at the ratio of a 3.263-degree rotation about the C-axis to one spindle rotation.
Pm : (Number of pulses per spindle rotation) × (one rotation) → 72000 × 1
Ps : (Amount of C-axis movement) × CMR → 3263 × 5

$$\frac{Kn}{Kd} = \frac{3263 \times 5}{72000 \times 1} = \frac{3263}{14400}$$

Both Kn and Kd are within the allowable range. No alarm is output.

In this sample program, when T1 is specified for the master axis, the synchronization ratio (fraction) of the CMR of the C-axis to the denominator Kd can always be reduced to lowest terms, thus Kd falls in the allowable range. So, the specifiable range of C is as follows:

$-99999999 \leq C \leq 99999999$

(d) Command: G81.5 T10 C3.263 ;
Operation : Synchronization between the spindle and C-axis is started at the ratio of a 3.263-degree rotation about the C-axis to ten spindle rotations.
Pm : (Number of pulses per spindle rotation) × (10 rotations) → 72000 × 10
Ps : (Amount of C-axis movement) × CMR → 3263 × 5

$$\frac{Kn}{Kd} = \frac{3263 \times 5}{72000 \times 10} = \frac{3263}{144000}$$

In this case, an alarm is issued because Kd exceeds the specifiable range.

(e) Command: G81.5 P10000 C-0.214 ;
Operation : Synchronization between the spindle and C-axis is started at the ratio of a -0.214 degree rotation of the C-axis to 10,000 feedback pulses from the pulse coder of the spindle.
Pm : (Specified number of feedback pulses from the pulse coder of the spindle) → 10000
Ps : (Amount of C-axis movement) × CMR → -214 × 5

$$\frac{Kn}{Kd} = \frac{-214 \times 5}{10000} = \frac{-107}{1000}$$

Both Kn and Kd are within the allowable range. No alarm is output.

(2) When the master axis is the spindle, the slave axis is the V-axis (linear axis), and inch/metric conversion is performed
(a) For a millimeter machine and metric input
Command: G81.5 T1 V1.0 ;
Operation : Synchronization between the spindle and V-axis is started at the ratio of a 1.0 inch movement (25.4 mm) along the V-axis per spindle rotation.
Pm : (Number of pulses per spindle rotation) × (one rotation) → 72000 × 1

Ps : (Amount of V-axis movement) × CMR → 1000 × 5

$$\frac{Kn}{Kd} = \frac{1000 \times 5}{72000} = \frac{5}{72}$$

Both Kn and Kd are within the allowable range. No alarm is output.

(b) For a millimeter machine and inch input
Command: G81.5 T1 V1.0 ;
Operation : Synchronization between the spindle and V-axis is started at the ratio of a 1.0inch (25.4mm) movement along the V-axis per spindle rotation.
Pm : (Number of pulses per spindle rotation) × (one rotation) → 72000 × 1
Ps : (Amount of V-axis movement) × CMR × 254 ÷ 100 → 10000 × 5 × 254 ÷ 100

$$\frac{Kn}{Kd} = \frac{10000 \times 5 \times 254}{72000 \times 100} = \frac{127}{72}$$

Both Kn and Kd are within the allowable range. No alarm is output.

(c) For a millimeter machine and inch input
Command: G81.5 T1 V0.0013 ;
Operation : Synchronization between the spindle and V-axis is started at the ratio of a 0.0013 inch (0.03302 mm) movement along the V-axis per spindle rotation.
Pm : (Number of pulses per spindle rotation) × (one rotation) → 72000 × 1
Ps : (Amount of V-axis movement) × CMR × 254 ÷ 100 → 13 × 5 × 254 ÷ 100

$$\frac{Kn}{Kd} = \frac{13 \times 5 \times 254}{72000 \times 100} = \frac{1651}{720000}$$

In this case, an alarm is issued because Kd exceeds the specifiable range.

- Example 2)

Based on the controlled axis configuration described in Fig. 21.7.4.3 (a), suppose that the spindle and V-axis are as follows:

Spindle pulse coder : 72000pulse/rev (4 pulses for one A/B phase cycle)
C-axis least command increment : 0.001deg
C-axis CMR : 1/2
V-axis least command increment : 0.001mm
V-axis CMR : 1/2

Then, the C-axis detection unit is 0.002 degree. The V-axis detection unit is 0.002 mm.

In this case, the synchronization ratio (Kn, Kd) is related with a command as indicated below. Here, let Pm and Ps be the amounts of movements represented in the detection unit for the master axis and slave axis specified in a synchronization start command, respectively.

(1) When the master axis is the spindle, and the slave axis is the C-axis

(a) Command: G81.5 T1 C3.263 ;
Operation : Synchronization between the spindle and C-axis is started at the ratio of a 3.263-degree rotation about the C-axis per spindle rotation.
Pm : (Number of pulses per spindle rotation) × (one rotation) → 72000 × 1
Ps : (Amount of C-axis movement) × CMR → 3263 × 1 ÷ 2

$$\frac{Kn}{Kd} = \frac{3263 \times 1}{72000 \times 2} = \frac{3263}{144000}$$

In this case, an alarm is issued because Kd exceeds the specifiable range.

(b) Command: G81.5 T1 C3.26 ;
The value specified for C slightly differs from that specified for C in (a).
Operation : Synchronization between the spindle and C-axis is started at the ratio of a 3.26-degree rotation about the C-axis per spindle rotation.
Pm : (Number of pulses per spindle rotation) × (one rotation) → 72000 × 1
Ps : (Amount of C-axis movement) × CMR → 3260 × 1 ÷ 2

$$\frac{Kn}{Kd} = \frac{3260 \times 1}{72000 \times 2} = \frac{163}{7200}$$

(a) causes an alarm to be output because the values cannot be reduced. (b) causes no alarm because the ratio of the travel distances can be reduced to a simple ratio.

21.7.4.6 Retract function

See "Retract function" in the Subsection 21.7.4, "Electronic Gear Box".

21.8 TANDEM CONTROL

When enough torque for driving a large table cannot be produced by only one motor, two motors can be used for movement along a single axis.Positioning is performed by the main motor only. The submotor is used only to produce torque. With this tandem control function, the torque produced can be doubled.

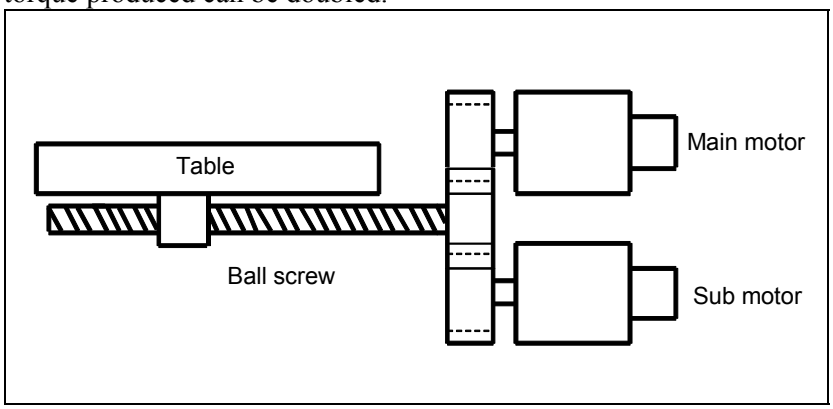


Fig. 21.8 (a) Example of operation

In general, the NC regards tandem control as being performed for one axis. However, for servo parameter management and servo alarm monitoring, tandem control is regarded as being performed for two axes.
For details, refer to the relevant manual published by the machine tool builder.

22 5-AXIS MACHINING FUNCTION

Chapter 22, "5-AXIS MACHINING FUNCTION", consists of the following sections:

22.1 TOOL CENTER POINT CONTROL 773
22.2 TOOL POSTURE CONTROL 828
22.3 TILTED WORKING PLANE COMMAND 838
22.4 INCLINED ROTARY AXIS CONTROL 894
22.5 3-DIMENSIONAL CUTTER COMPENSATION 897

22.1 TOOL CENTER POINT CONTROL

Overview

On a 5-axis machine having two rotary axes that turn a tool or table, this function performs tool length compensation constantly, even in the middle of a block, and exerts control so that the tool center point moves along the specified path. (See Fig. 22.1 (a).)
There are three different types of 5-axis machines - <1> those that rotate the tool only, <2> those that rotate the table only, and <3> those that rotate both the tool and table. (See Fig. 22.1 (d).)
This function is intended to perform machining on such 5-axis machines having rotary axes that turn a tool or table as well as three orthogonal axes (X-, Y-, and Z-axes) by accomplishing tool length compensation while changing the attitude of the tool. It enables the tool center point to move along the specified path even if the tool's direction changes with respect to the workpiece.
A coordinate system used for programming the tool center point control is called the programming coordinate system.
A coordinate system fixed on the table can be used as the programming coordinate system, which makes CAM programming easy. This coordinate system is particularly called the table coordinate system.
A workpiece coordinate system fixed on a machine coordinate system can be employed as the programming coordinate system as well.
In any case, the cutting speed can be controlled easily because the tool center point moves at a specified speed with respect to the table (workpiece).
The commands that can be issued during tool center point control are positioning (G00), linear interpolation (G01), circular interpolation (G02, G03), and helical interpolation (G02, G03).

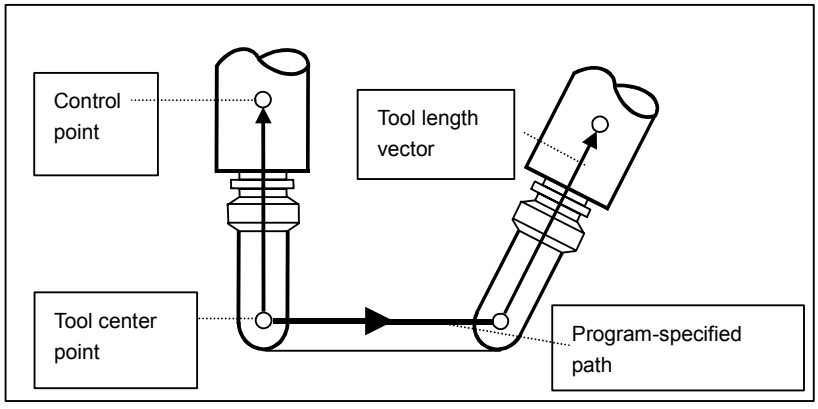


Fig. 22.1 (a) Path of the tool center point

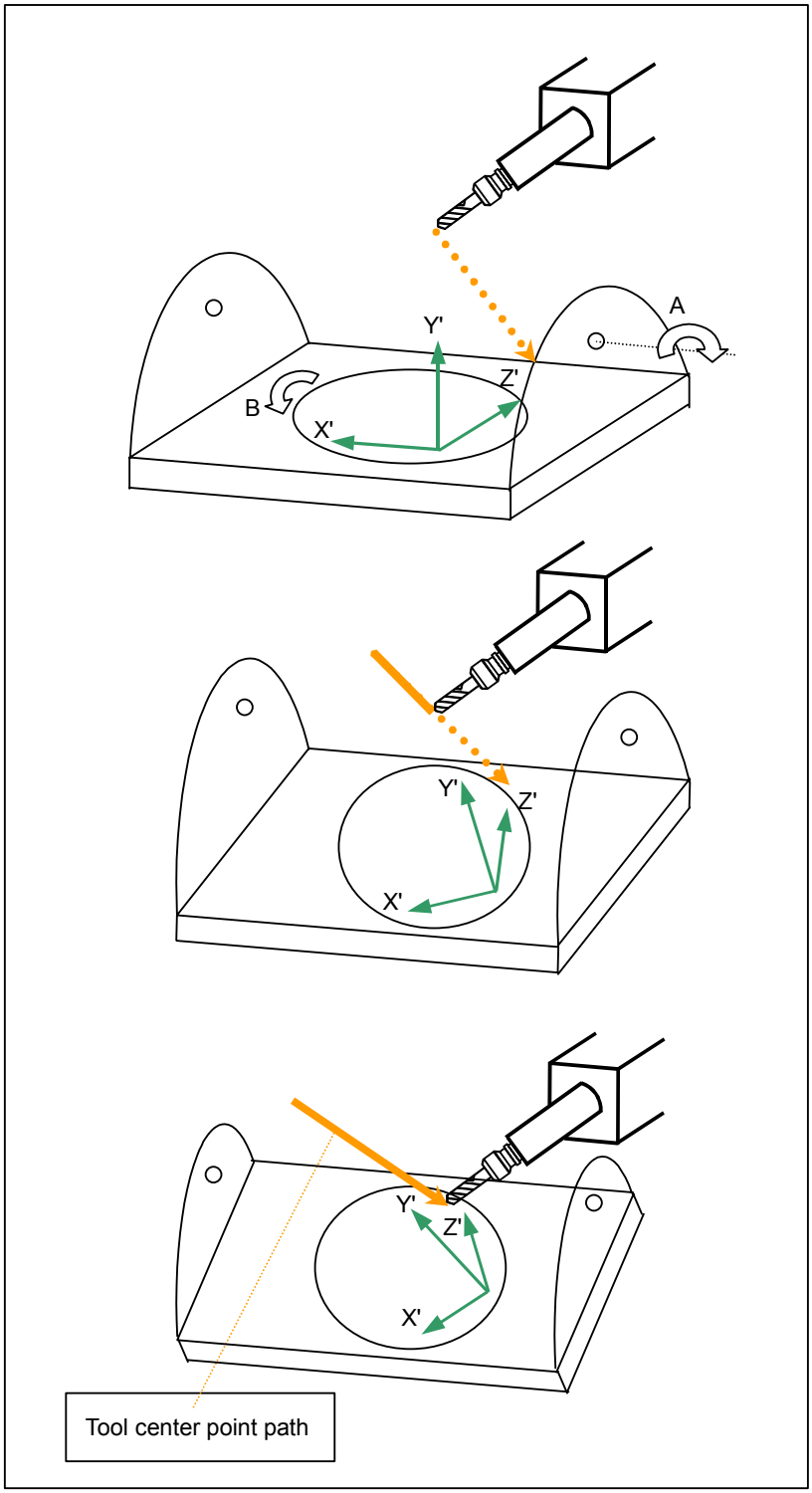


Fig. 22.1 (b) Path of the tool center point

When a coordinate system fixed on the table is used as the programming coordinate system, programming can be performed without worrying about the rotation of the table because the programming coordinate system does not move with respect to the table, although the position and direction of the workpiece fixed on the table change due to its rotation. When a straight line is specified, the tool center point moves along a straight path with respect to the workpiece as instructed. (See Fig. 22.1 (b).)

By setting the relevant parameter, the workpiece coordinate system can also be employed as the programming coordinate system. In this case, as the table turns, the position and direction of the workpiece fixed on the table change with respect to the programming coordinate system. It is therefore necessary to take into account the rotation of the table when specifying the end point. In this case, too, when a straight line is specified, the tool center point moves along a straight path with respect to the workpiece as instructed.

Fig. 22.1 (c) illustrates how linear interpolation is accomplished with a composite type machine, showing the relationship between the case when a table-fixed coordinate system is used as the programming coordinate system and the case when the workpiece coordinate system is used. If linear interpolation is specified in this function mode, speed control is exerted in such a way that the tool center point moves at a specified speed with respect to the workpiece.

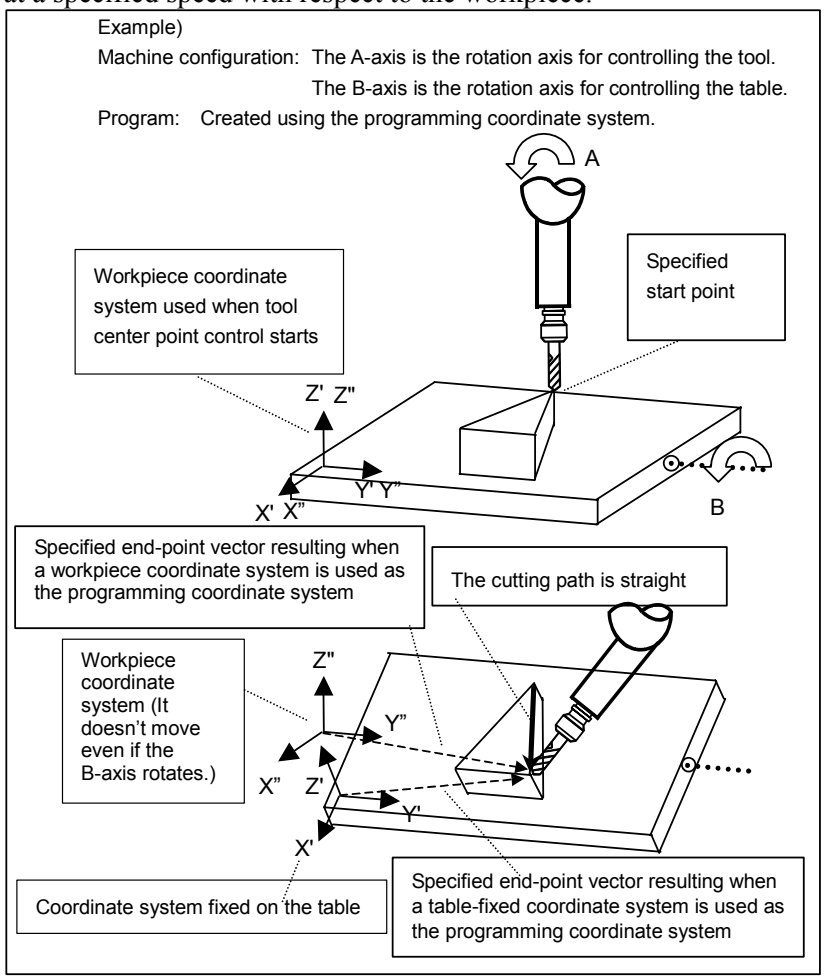


Fig. 22.1 (c) Linear interpolation with a composite type machine

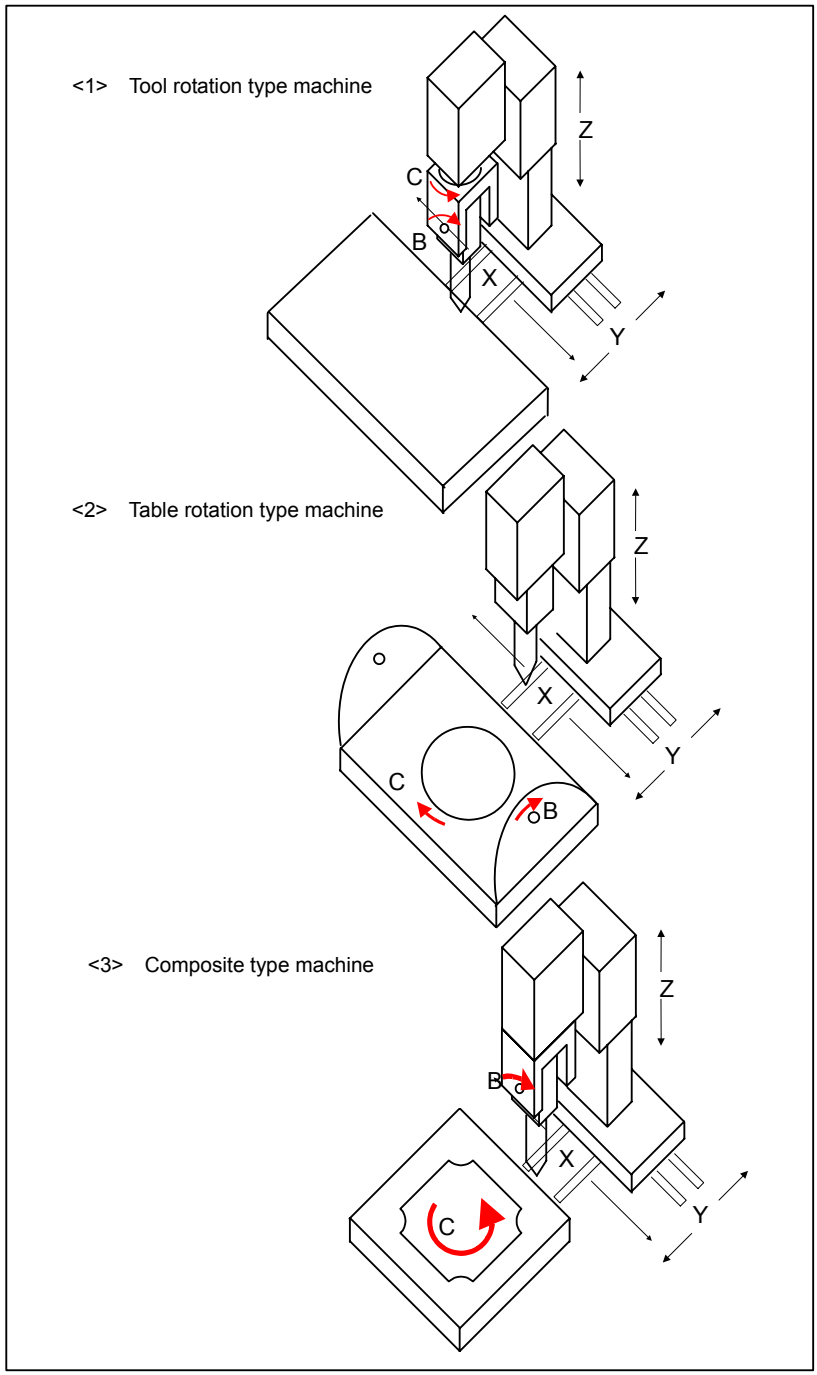


Fig. 22.1 (d) Three types of 5-axis machine

Even if the rotary axis that controls the tool does not intersect the one that controls the table, this function can still be used.

There are two types, as described below, one of which is used depending on how the direction of the tool axis is specified.

(1) Type 1
The block end point of the rotary axes is specified (e.g. A, B, C).
The CNC performs tool length compensation by the specified amount in the tool axis direction that is calculated from the specified position of the rotary axes and exerts control so that the tip of the tool moves along the specified path.

(2) Type 2
The direction of the tool axis (I, J, K) at the block end point, as seen from the coordinate system fixed on the table, is specified, instead of the position of the rotary axes.
The CNC calculates an end point of the rotary axes where the tool will face the specified direction, performs tool length compensation by the specified amount in the tool axis direction that is calculated from the position of the rotary axes, and exerts control so that the tip of the tool moves along the specified path.

Format

- Positioning and linear interpolation for tool center point control (type 1)

G43.4 IP_ α_ β_ H_ ; Starts tool center point control (type 1).
IP_ α_ β_ ;
:

IP : In the case of an absolute programming, the coordinate value of the end point of the tool tip movement
In the case of an incremental programming, the amount of the tool tip movement

α, β : In the case of an absolute programming, the coordinate value of the end point of the rotary axes
In the case of an incremental programming, the amount of the rotary axis movement

H : Tool offset number

Movement to the position specified by the G43.4 block (startup) does not constitute tool center point control. Only tool length compensation is performed.
As for rotary axes, either table rotation axes or tool rotation axes are specified.

While performing interpolation for the rotary axes, the CNC controls the control points so that the tool center point moves along a straight line with respect to the table (workpiece). The end of the tool center point comes to the point specified on the programming coordinate system.

- Positioning and linear interpolation for tool center point control (type 2)

G43.5 IP_ H_ Q_ ; Starts tool center point control (type 2).
IP_ I_ J_ K_ ;
:

IP	: In the case of an absolute programming, the coordinate value of the end point of the tool tip movement In the case of an incremental programming, the amount of the tool tip movement
I, J, K	: Tool axis direction at the block end point as seen from the programming coordinate system
H	: Tool offset number
Q	: Inclination angle of the tool (in degrees)

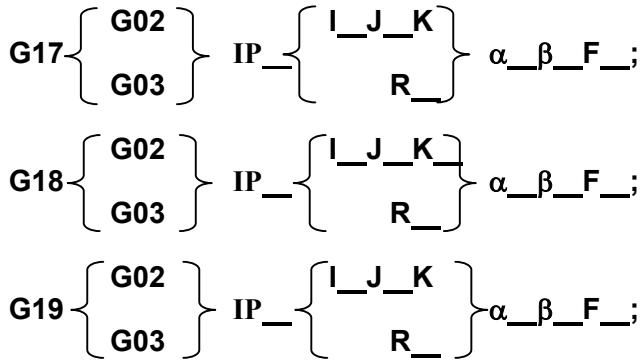
Movement to the position specified by the G43.5 block does not constitute tool center point control. Only tool length compensation is performed.
No rotary axes are specified. Instead, the direction of the tool end point is specified as I, J, K, as seen from the programming coordinate system (the one fixed on the table when G43.5 is specified).
With a tool rotation type machine, I, J, K can be specified using the G43.5 block. In the case of a table rotation type or composite type machine, however, these cannot be specified. Specifying them with a table rotation type or composite type machine causes alarm PS5421.

While performing interpolation for the rotary axes, the CNC controls the control points so that the tool center point moves along a straight line with respect to the table (workpiece). The end of the tool center point comes to the point specified on the programming coordinate system.

⚠ CAUTION
1 If one or two of the I, J, and K values are omitted, the omitted value or values are considered to be 0.
2 In a block in which I, J, and K are all omitted, the compensation vector of the preceding block is used.
3 This block can be used only when the programming coordinate system is fixed on the table (when the bit 5 (WKP) of parameter No.19696 is set to 0). Specifying G43.5 when bit 5 (WKP) of parameter No.19696 is set to 1 causes alarm PS5459.
4 Type 2 cannot be used when there is only one rotary axis or when any hypothetical axis is used. Specifying G43.5 in such a case causes alarm PS5459.
5 When using the rotary axis rollover function or the rotary axis control function, set parameter No.1260 (amount of movement per rotation of the rotary axis) to 360 degrees.

- Circular interpolation for tool center point control (type 1)

G43.4 IP_ H_ ; Starts tool center point control (type 1).



:

G17 : X-Y plane of the table coordinate system
G18 : Z-X plane of the table coordinate system
G19 : Y-Z plane of the table coordinate system
G02 : Clockwise (CW) circular interpolation
G03 : Counterclockwise (CCW) circular interpolation
IP : In the case of an absolute programming, the coordinate value of the end point of the tool tip movement
In the case of an incremental programming, the amount of the tool tip movement
(This pertains only to two axes on the plane.)
I, J, K : Specify the distance between the start point in the rotary axis position of the block start point and the center of the arc, as seen from the programming coordinate system.
R : Arc radius R > 0: The center angle of the arc is less than 180°.
R < 0: The center angle of the arc is more than 180°.
α, β : In the case of an absolute programming, the coordinate value of the end point of the rotary axis
In the case of an incremental programming, the amount of the rotary axis movement
F : Specified speed (speed in the tangent direction of the arc as seen from the table coordinate system)
H : Tool offset number

Movement to the position specified by the G43.4 block does not constitute tool center point control. Only tool length compensation is performed. While performing interpolation for the rotary axes, the CNC controls the control points so that the tool center point moves along an arc with respect to the table (workpiece). The end of the tool center point comes to the point specified on the programming coordinate system.

⚠ **CAUTION**
Any command that does not move the tool center point with respect to the workpiece (one that moves the rotary axes only) must be executed in G00 or G01 mode.

- Circular interpolation for tool center point control (type 2)

G43.5 IP_ H_ Q_ ; Starts tool center point control (type 2).

G17 { **G02** / **G03** } **IP__I__J__K__ R__ F__;**

G18 { **G02** / **G03** } **IP__I__J__K__ R__ F__;**

G19 { **G02** / **G03** } **IP__I__J__K__ R__ F__;**

:

G17 : X-Y plane of the table coordinate system
G18 : Z-X plane of the table coordinate system
G19 : Y-Z plane of the table coordinate system
G02 : Clockwise (CW) circular interpolation
G03 : Counterclockwise (CCW) circular interpolation
IP : In the case of an absolute programming, the coordinate value of the end point of the tool tip movement
In the case of an incremental programming, the amount of the tool tip movement
(This pertains only to two axes on the plane.)
I, J, K : Tool axis direction at the block end point as seen from the programming coordinate system
R : Arc radius R > 0: The center angle of the arc is less than 180°.
R < 0: The center angle of the arc is more than 180°.
F : Specified speed (speed in the tangent direction of the arc as seen from the table coordinate system)
H : Tool offset number
Q : Inclination angle of the tool (in degrees)

Movement to the position specified by the G43.5 block does not constitute tool center point control. Only tool length compensation is performed.
No rotary axes are specified. Instead, the direction of the tool end point is specified as I, J, K, as seen from the programming coordinate system (the one fixed on the table when G43.5 is specified).
With a tool rotation type machine, I, J, K can be specified using the G43.5 block. In the case of a table rotation type or composite type machine, however, these cannot be specified. Specifying them with a table rotation type or composite type machine causes alarm PS5421.
While performing interpolation for the rotary axes, the CNC controls the control points so that the tool center point moves along an arc with respect to the table (workpiece). The end of the tool center point comes to the point specified on the programming coordinate system.

⚠ CAUTION

1 Only arc radius R can be specified. (The distance from the start point to the center of the arc cannot be specified using I, J, and K.)
2 A round circle (the start point and end point are the same) cannot be specified.
Any command that does not move the tool center point with respect to the workpiece (one that moves the rotary axes only) must be executed in G00 or G01 mode.
3 See the CAUTION box for tool center point control (type 2).

- Helical interpolation for tool center point control (type 1)

G43.4 IP_ H ; Starts tool center point control (type 1).

G17 {**G02**, **G03**} **IP__** {**I__J__K__**, **R__**} **α__β__γ__F__;**

G18 {**G02**, **G03**} **IP__** {**I__J__K__**, **R__**} **α__β__γ__F__;**

G19 {**G02**, **G03**} **IP__** {**I__J__K__**, **R__**} **α__β__γ__F__;**

:

G17 : X-Y plane of the table coordinate system
G18 : Z-X plane of the table coordinate system
G19 : Y-Z plane of the table coordinate system
G02 : Clockwise (CW) circular interpolation
G03 : Counterclockwise (CCW) circular interpolation
IP : In the case of an absolute programming, the coordinate value of the end point of the tool tip movement
In the case of an incremental programming, the amount of the tool tip movement
(This pertains only to two axes on the plane.)
I, J, K : Specify the distance between the start point in the rotary axis position of the block start point and the center of the arc, as seen from the programming coordinate system.
R : Arc radius R > 0: The center angle of the arc is less than 180°.
R < 0: The center angle of the arc is more than 180°.
α, β : In the case of an absolute programming, the coordinate value of the end point of the rotary axis
In the case of an incremental programming, the amount of the rotary axis movement
γ : In the case of an absolute programming, the coordinate value of the end point of the tool tip movement
In the case of an incremental programming, the amount of the tool tip movement
(This pertains only to one of the linear axes subject to tool center point control which does not exist on the plane. Linear interpolation is performed simultaneously with circular interpolation.)
F : Specified speed (speed in the tangent direction of the arc as seen from the table coordinate system)
H : Tool offset number

Movement to the position specified by the G43.4 block does not constitute tool center point control. Only tool length compensation is performed.

Because the specified speed is usually the speed in the tangent direction of the arc, the speed of the linear axis, when seen from the table coordinate system, is: $F \times \frac{\text{Length of the linear axis}}{\text{Length of the arc}}$.

Depending on bit 5 (HTG) of parameter No.1403, the specified speed varies as shown in the following table.

Bit 5 (HTG) of parameter No.1403	
0	**1**
Tangential speed of the arc	Synthetic speed of the linear axis speed and tangential speed

While performing interpolation for the rotary axes, the CNC controls the control points so that the tool center point moves helically with respect to the table (workpiece). The end of the tool center point comes to the point specified on the programming coordinate system.

⚠ CAUTION
Any command that does not move the tool center point with respect to the workpiece (one that moves the rotary axes only) must be executed in G00 or G01 mode.

- Helical interpolation for tool center point control (type 2)

G43.5 IP_ H_ Q_; Starts tool center point control (type 2).

G17 { **G02** / **G03** } **IP__I__J__K__ R__γ__F__;**

G18 { **G02** / **G03** } **IP__I__J__K__ R__γ__F__;**

G19 { **G02** / **G03** } **IP__I__J__K__ R__γ__F__;**

:

G17 : X-Y plane of the table coordinate system
G18 : Z-X plane of the table coordinate system
G19 : Y-Z plane of the table coordinate system
G02 : Clockwise (CW) circular interpolation
G03 : Counterclockwise (CCW) circular interpolation
IP : In the case of an absolute programming, the coordinate value of the end point of the tool tip movement
In the case of an incremental programming, the amount of the tool tip movement
(This pertains only to two axes on the plane.)
I, J, K : Tool axis direction at the block end point as seen from the programming coordinate system
R : Arc radius R > 0: The center angle of the arc is less than 180°.
R < 0: The center angle of the arc is more than 180°.
γ : In the case of an absolute programming, the coordinate value of the end point of the tool tip movement
In the case of an incremental programming, the amount of the tool tip movement
(This pertains only to one of the linear axes subject to tool center point control which does not exist on the plane. Linear interpolation is performed simultaneously with circular interpolation.)
F : Specified speed (speed in the tangent direction of the arc as seen from the table coordinate system)
H : Tool offset number
Q : Inclination angle of the tool (in degrees)

Movement to the position specified by the G43.5 block does not constitute tool center point control. Only tool length compensation is performed.
Because the specified speed is the speed in the tangent direction of the arc, the speed of the linear axis, when seen from the table coordinate system, is: $F \times \frac{\text{Length of the linear axis}}{\text{Length of the arc}}$.

Depending on bit 5 (HTG) of parameter No.1403, the specified speed varies as shown in the following table.

Bit 5 (HTG) of parameter No.1403	
0	**1**
Tangential speed of the arc	Synthetic speed of the linear axis speed and tangential speed

No rotary axes are specified. Instead, the direction of the tool end point is specified as I, J, K, as seen from the programming coordinate system (the one fixed on the table when G43.5 is specified).
With a tool rotation type machine, I, J, K can be specified using the G43.5 block. In the case of a table rotation type or composite type machine, however, these cannot be specified. Specifying them with a table rotation type or composite type machine causes alarm PS5421.
While performing interpolation for the rotary axes, the CNC controls the control points so that the tool center point moves helically with respect to the table (workpiece). The end of the tool center point comes to the point specified on the programming coordinate system.

⚠ CAUTION
1 Only arc radius R can be specified. (The distance from the start point to the center of the arc cannot be specified using I, J, and K.)
2 A round circle cannot be specified.
3 Any command that does not move the tool center point with respect to the workpiece (one that moves the rotary axes only) must be executed in G00 or G01 mode.
4 See the CAUTION box for tool center point control (type 2).

- Tool center point control cancellation command

G49 IP_ α_ β_ ; Cancels tool center point control.

IP : In the case of an absolute programming, the coordinate value of the end point of the tool control point movement
In the case of an incremental programming, the amount of the tool control point movement

α, β : In the case of an absolute programming, the coordinate value of the end point of the rotary axes
In the case of an incremental programming, the amount of the rotary axis movement

The cancellation block for tool center point control is the one that controls buffering.

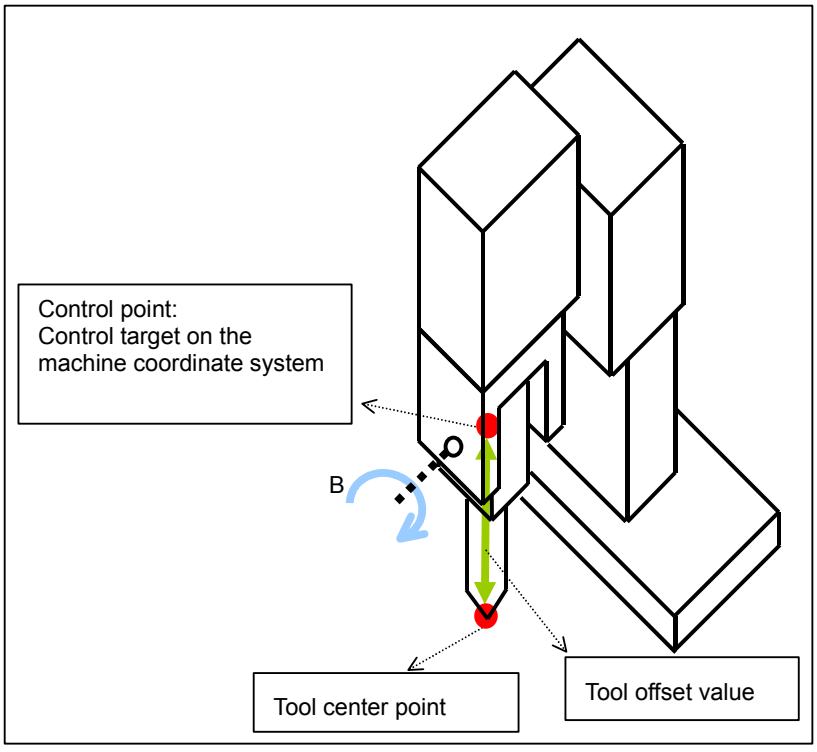
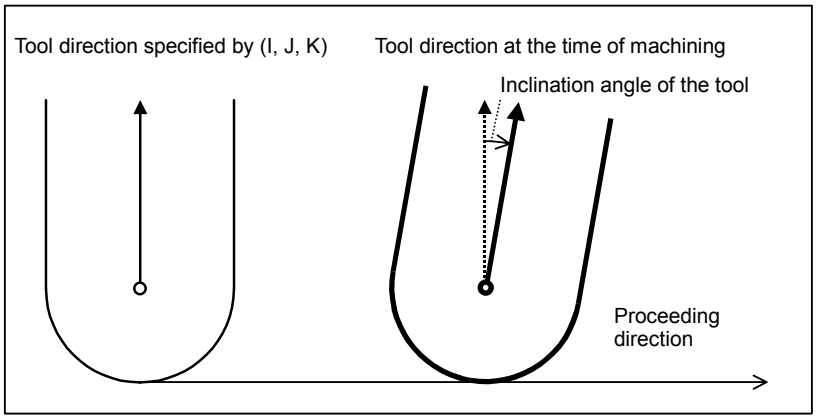


Fig. 22.1 (e) Control point and tool center point

⚠ CAUTION
The G49 command must be executed in G00 or G01 mode.

- Inclination angle of the tool

In the case of tool center point control of type 2, the inclination angle of the tool can be specified using address Q of G43.5. The inclination angle of the tool represents how inclined the tool direction is toward the proceeding direction with respect to the direction specified by (I, J, K) at the time of machining on the plane produced by the tool direction specified by (I, J, K) and the proceeding direction on the programming coordinate system. (See the figure below.)
Overviewly, the normal direction on the machining plane is specified by (I, J, K). If the tool direction needs to be inclined toward the proceeding direction with respect to the normal direction at the time of machining, perform compensation using Q.
If the direction specified by (I, J, K) matches the direction in which the tool moves at the time of machining, commanding Q is unnecessary.



Example)

To incline the tool by two degrees toward the proceeding direction at the time of machining, enter the following command:
G43.5 I_ J_ K_ H_ Q2.0

Explanation

- When a coordinate system fixed on the table is used as the programming coordinate system

The programming coordinate system is used for tool center point control. When the G43.4 or G43.5 command is specified with bit 5 (WKP) of parameter No.19696 set to 0, the workpiece coordinate system that is fixed on the table at that point of time becomes the programming coordinate system. Thereafter, the programming coordinate system rotates as the table turns around.
It does not rotate with the tool head.
X, Y, and Z mentioned hereinafter are assumed to be commanded on the programming coordinate system.
When the G43.4 or G43.5 command is specified or when the rotary axis of the table has moved in a block preceding G43.4 or G43.5, the angle of the table's rotary axis represents the initial state of the programming coordinate system.
In the case of type 2, the tool direction seen from the coordinate system that is fixed on the table is specified by I, J, K.
In the descriptions that follow, the table-fixed coordinate system is represented by X', Y', and Z'.

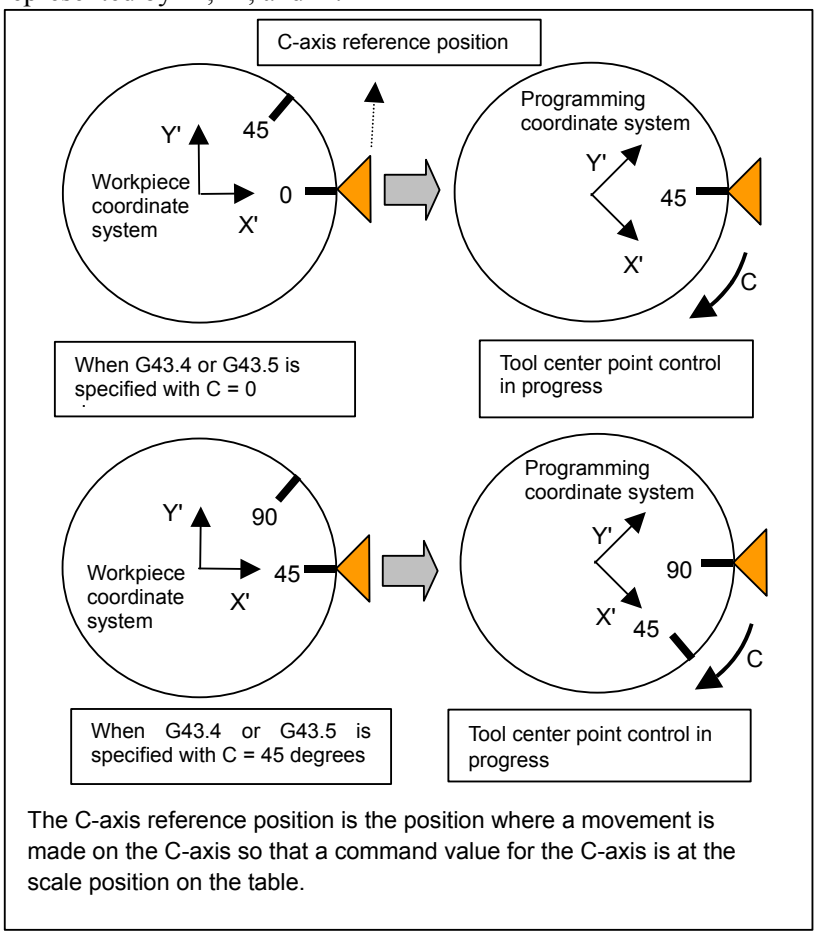


Fig. 22.1 (f) Programming coordinate system fixed on the table

- When the workpiece coordinate system is used as the programming coordinate system

When the G43.4 command is specified with bit 5 (WKP) of parameter No.19696 set to 1, the workpiece coordinate system that is in use at that point of time becomes the programming coordinate system.
In this case, the programming coordinate system does not rotate as the table turns around but remains fixed on the workpiece coordinate system.
Hereinafter, when X, Y, and Z are commanded, the tool moves along a straight line toward the table (workpiece). For X, Y, Z, specify the end point location after the rotation of the table, as seen from the programming coordinate system.
Type 2 cannot be used. Specifying G43.5 when bit 5 (WKP) of parameter No.19696 set to 1 causes alarm PS5459.
In the descriptions that follow, the coordinate values of the workpiece coordinate system used as the programming coordinate system are represented by X", Y", and Z".

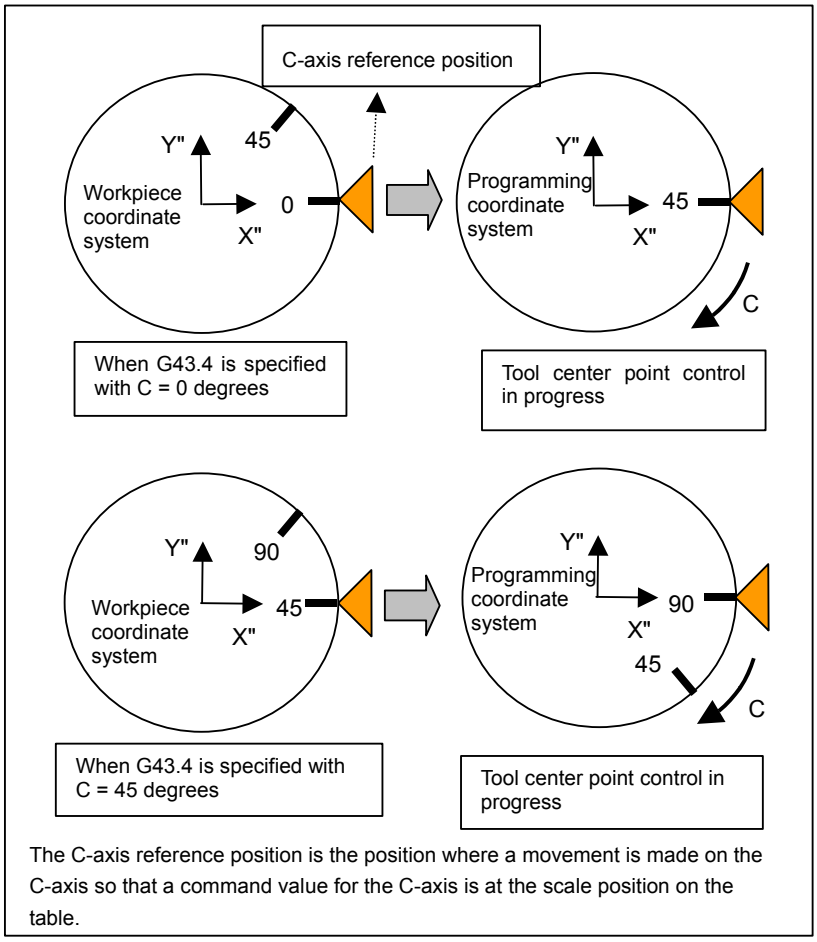


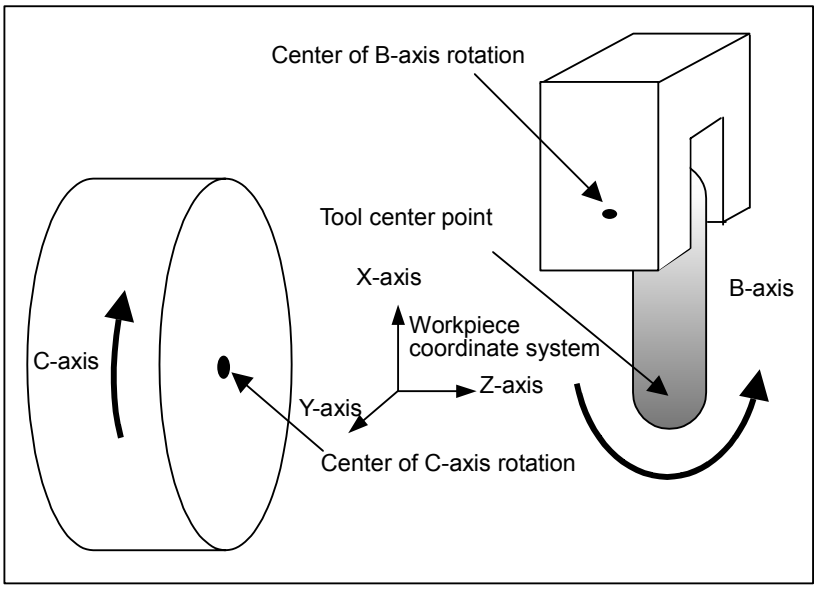
Fig. 22.1 (g) Programming coordinate system identical to the workpiece coordinate system

- Notes on performing circular interpolation and helical interpolation when using the workpiece coordinate system as the programming coordinate system

- The start point, end point, and center of an arc change as the table rotation axis rotates.
- I, J, K commands the vector of the block start point to the center of the arc from the start point in the rotary axis position.
- Note the following:
  <1> Only a table rotation axis normal to a selected plane can be rotated during circular interpolation.
  <2> During circular interpolation, those table rotation axes not normal to a selected plane must continue to be at the same position as when tool center point control is started.
  If <1> or <2> is not satisfied, the alarm PS5421 is issued.
  No restriction is imposed on rotation on a tool rotation axis.

- Program examples

- In the case of a composite type machine
  Descriptions are based on the following machine configuration.



When the G17 (X-Y plane) command is executed

After the G43.4 command, the X-Y plane is selected using the G17 command and circular interpolation is performed by rotating the C-axis (table rotation axis) (including those cases where the C-axis moves before the G43.4 command). → This case corresponds to <1> and allows circular interpolation.

Example)

```
:
(G01 C90. ;)
G43.4 H1 ;
G17 G02 IP IR   B10. C20. ;
:
```

IP: Coordinates of the end point
IR: Arc radius

When the G18 (Z-X plane) or G19 (Y-Z plane) command is executed

After the G43.4 command, the Z-X plane is selected using the G18 command and circular interpolation is performed without rotating the C-axis (including those cases where the C-axis moves before the G43.4 command). → This case corresponds to <2> and allows circular interpolation.

The same is also true when the G19 command is used.

Example)

```
:
(G01 C90. ;)
G43.4 H1 ;
G18 G02 IP IR ;
:
```

After the G43.4 command, the Z-X plane is selected using the G18 command and the C-axis is rotated during circular interpolation . → Alarm (violation of <2>)

The same is also true when the G19 command is used.

Example)

```
:
G43.4 H1 ;
G18 G02 IP IR C20. ;
:
```

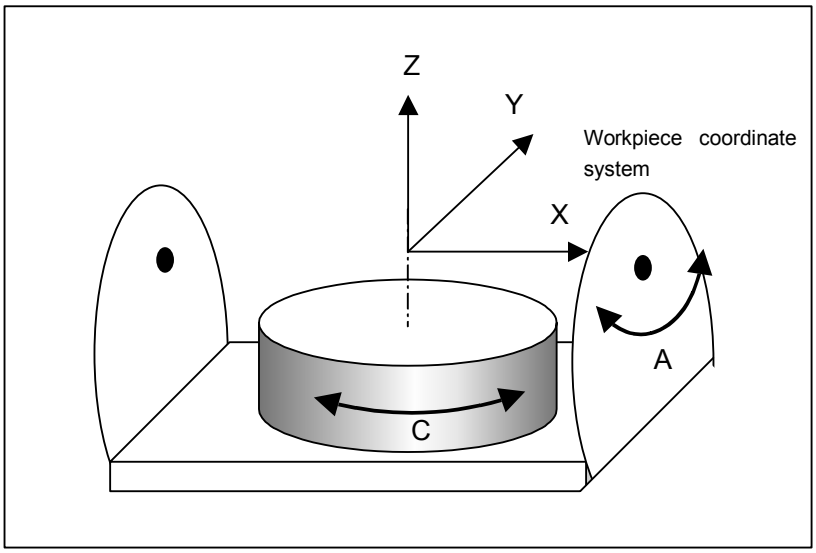
After the G43.4 command, the Z-X plane is selected using the G18 command and circular interpolation is performed after rotating the C-axis. → Alarm (violation of <2>)

The same is also true when the G19 command is used.

Example)

```
:
G43.4 H1 ;
G01 C10.
G18 G02 IP IR ;
:
```

- In the case of a table rotation type machine
  Descriptions are based on the following machine configuration.
  A table rotation type machine can be considered equivalent to a composite type machine if any of its two table rotation axes does not move.



When the G17 (X-Y plane) command is executed
The master axis (A-axis) moves before the G43.4 command and, after the G43.4 command, circular interpolation is performed without rotating the A- or C-axis (including those cases where the C-axis moves before the G43.4 command). → This case corresponds to <2> and allows circular interpolation.
Example)

```
    :
G01 A90. (C10.);
G43.4 H1 ;
G17 G02 IP IR ;
    :
```

The master axis (A-axis) moves before the G43.4 command and, after the G43.4 command, circular interpolation is performed using the G17 (X-Y plane) command by rotating the C-axis, or the C-axis is rotated during circular interpolation. → Alarm (violation of <2>)
Example)

```
:
G01 A90.;
G43.4 H1 ;
G01 C10. ;
G17 G02 IP IR ;
:
:
G01 A90.;
G43.4 H1 ;
G17 G02 IP IR C10.;
:
```

After the G43.4 command, the A-axis is moved and circular interpolation is performed using the G17 command (X-Y plane). → Alarm (violation of <2>)
Example)

```
:
G43.4 H1 ;
G01 A10. ;
G17 G02 IP IR C10.;
:
```

G19 (Y-Z plane) command is executed
The G43.4 command is executed after moving the A-axis and circular interpolation is performed while rotating the A-axis using the G19 (Y-Z plane) command. → This case corresponds to <1> and allows circular interpolation.
Example)

```
:
G01 A90. ;
G43.4 H1 ;
G19 G02 IP IR A10. ;
:
```

After the G43.4 command, the C-axis is rotated and circular interpolation is performed using the G19 (Y-Z plane) command. → Alarm (violation of <2>)
Example)

```
:
G43.4 H1 ;
G01 C10. ;
G19 G02 IP IR ;
:
```

When the G18 (Z-X plane) command is executed

The G43.4 command is executed after moving the A- and C-axes, and circular interpolation is performed using the G18 (Z-X plane) command without moving any rotary axis. → This case corresponds to <2> and allows circular interpolation.

Example)

```
:
G01 A90. C10. ;
G43.4 H1 ;
G18 G02 IP IR;
:
```

After the G43.4 command, circular interpolation is performed using the G18 command (Z-X plane) by moving any of the rotary axes. → Alarm (violation of <2>)

Example)

```
:
G43.4 H1 ;
G01 A10. (C10.)
G18 G02 IP IR;
:
```

- Tool center point control command

During tool center point control, the command specifies the location of each block end point as seen from the programming coordinate system.

The program specifies the tool center point.

As for the rotary axis, the command specifies the coordinate values of each block end point in the case of type 1 or the tool direction at each block end point in the case of type 2.

The feedrate is specified by the tangential speed relative to the workpiece (the tool's relative speed as opposed to the workpiece), which is represented by F.

- Commands that can be specified during tool center point control

The commands that can be specified during tool center point control are linear interpolation (G01), positioning (G00), circular interpolation (G02, G03), and helical interpolation (G02, G03).

When linear interpolation (G01) is specified during tool center point control, speed control is exerted so that the tool center point moves at the specified speed.

The circular interpolation command (G02, G03) controls the tangential speed of the arc path along which the tool center point moves.

The helical interpolation command (G02, G03) controls the tangential speed of the arc path along which the tool center point moves or a synthetic speed including that of the helical axis. (This is dependent on the setting of bit 5 (HTG) of parameter No.1403.)

As the actual speed, the speed at the control point is shown.

NOTE

When the tool center point control mode is entered, acceleration/deceleration before look ahead interpolation is enabled automatically. Be sure to specify the following parameters:

(1) Bit 1 (LRP) of parameter No.1401=1 :
Linear rapid traverse
(2) Bit 5 (FRP) of parameter No.19501=1 :
Acceleration/deceleration before interpolation is used for rapid traverse.
(3) Parameter No.1671 :
Acceleration before interpolation for rapid traverse.
(4) Parameter No.1672 :
Change time for bell-shaped acceleration before interpolation for rapid traverse.
(5) Parameter No.1660 :
Maximum permissible acceleration for acceleration/deceleration before interpolation

If they are not specified, alarm PS5420 is generated.

- Rotary axis command

If a command is specified during tool center point control that prohibits the tool center point from moving with respect to the workpiece, the maximum cutting speed (parameter No.1430) is assumed as the feedrate of the rotary axis when bit 6 (RFC) of parameter No.19696 is 0, and the speed specified by F is assumed when bit 6 (RFC) of parameter No.19696 is 1.

The rotary axis command cannot be specified during tool center point control of type 2. Specifying the command with type 2 causes alarm PS5421.

- The moving distance of the rotary axis is long compared to that of the linear axis

If the moving distance of the rotary axis is long compared to that of the linear axis, the rotary axis moves faster so that the tool center point moves at the specified speed, possibly resulting in the tool center point traveling on an inadequate path.

In such a case, it is possible to exert control to slow down the speed and ensure that the tool center point travels on the specified path, by setting bit 6 (CRS) of parameter No.19746 to 1. When using this control, specify in parameter No.19751 (for rapid traverse) and parameter No.19752 (for cutting feed) the extent of deviation of the path at which the speed is to be slowed down (maximum allowable deviation of the path). When 0 is specified, the least input increment is regarded as the maximum allowable deviation of the path.

- Tool behavior at startup and cancellation

When tool center point control is started (G43.4/G43.5) or canceled (G49), the tool moves by a tool offset value.
Compensation vector calculation is performed only at the end of a block.

- Current position display during tool center point control

During tool center point control, the position of the control point (rotation center of the tool rotation axis) is displayed as the machine coordinate.
When bit 5 (WKP) of parameter No.19696 is 0, whether to use absolute or relative coordinates can be selected using bit 2 (DET) of parameter No.19608.
If bit 2 (DET) of parameter No.19608 is 0, the position of the tool center point on the programming coordinate system is displayed.
If bit 2 (DET) of parameter No.19608 is 1, the position of the tool center point in the workpiece coordinate system is displayed.

- Tool offset

If tool offsets are used based on tool numbers, tool center point control is carried out using the tool length compensation value corresponding to the relevant tool number (T code).
If tool life management is used, tool center point control is carried out using the tool length compensation value corresponding to the tool in use.

- Function for placing priority on the path rather than on speed

If, in a block specifying a movement on a rotary axis, a high speed (speed command or rapid traverse rate) is specified or the travel distance on the rotary axis is larger than the travel distance of the tool center point, an excessive path deviation can occur in the middle of the block when a movement is made by the specified speed.
In such a case, priority can be placed on the path instead of the speed so that the speed can be automatically lowered to prevent a deviation from the path.

This function is enabled by setting bit 6 (CRS) of parameter No. 19746 to 1.
Set a maximum allowable path deviation in parameter No. 19751 (for rapid traverse) or in parameter No. 19751 (for cutting feed).
When 0 is set as a maximum allowable path deviation, the specification of the minimum unit of setting data of the parameter is assumed.
When a negative value is set, the function for placing priority on the path is disabled.

When a judgment is made to lower the speed, the speed of the entire block is lowered.
As a result of speed reduction, the path error may be smaller than a parameter-set value due to a calculation error.

- Canned cycle during tool center point control

When bit 5 (TFA) of parameter No. 5105 is set to 1, if a canned cycle being subjected to tool center point control is performed when the position of the rotation axis that determines the tool axis direction is not $\pm 90° \times n$ (n=0, 1, 2...) on the workpiece coordinate system, alarm PS5424 (invalid tool axis direction) occurs.

> ⚠ **CAUTION**
> A canned cycle being subjected to tool center point control operates in the direction of the program coordinate system. It does not always operates in the current tool axis direction.
> Be extremely careful about programming.

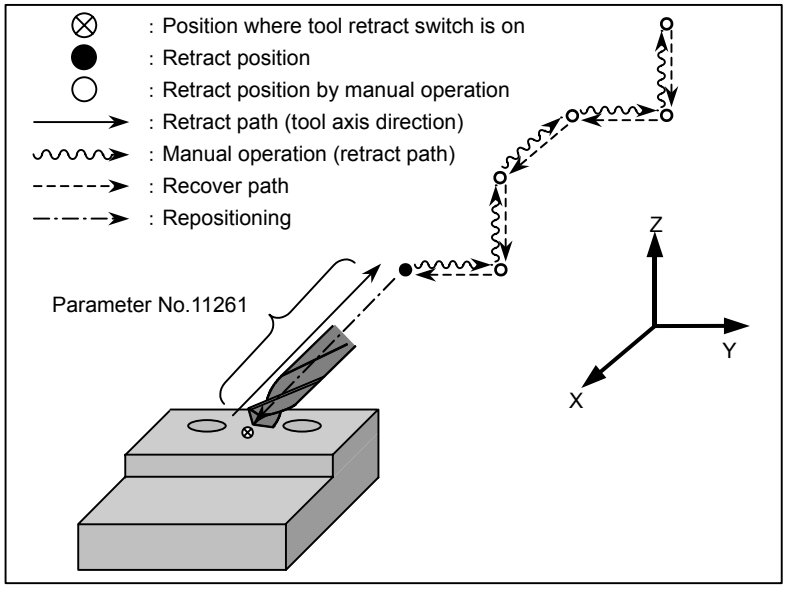
- Tool retract and recover during tool center point control

A tool retract and recover operation can be performed during tool center point control. A retract operation is performed in the tool axis direction by setting a tool axis direction retract amount in parameter No. 11261.
This function is enabled on a machine of head rotation type.
For tool retract and recover operation, refer to "TOOL RETRACT AND RECOVER" in Part for Operation.

Tool axis direction retract operation is enabled under the following conditions:

1. The G10.6 command is singly executed (no axis address is specified in the block specifying G10.6).
2. Bit 2 (RPS) of parameter No. 7040 is set to 1.
3. Tool center point control or workpiece setting error compensation are being exercised. (However, bit 0 (RCM) of parameter No. 11200 is set to 1 to perform compensation in the tool direction.)
4. A nonzero value is set in parameter No. 11261.

If the tool retract signal TRESC is set to 1 under the conditions above, a retract operation is performed in the tool axis direction. Parameter No. 11261 is used to specify a retract amount. A retract operation is performed using the value of parameter No. 11261 as an incremental retract amount.



If the tool retract signal TRESC is set to 1 during tool center point control when the tool axis direction retract conditions mentioned above are not satisfied, a retract operation of the conventional type is performed. In such a case, specify the tool center point retract position in a G10.6 block.

<table>
<tr><th colspan="3">G10.6 alone</th><th>G10.6 X_ Y_ Z_</th></tr>
<tr><td colspan="2">Bit 2 (RPS) of parameter No.7040 = 1</td><td>= 0</td><td rowspan="3">Conventional retract</td></tr>
<tr><td>Parameter No.11261 ≠ 0.0</td><td>= 0.0</td><td rowspan="2">Retract operation is not performed.</td></tr>
<tr><td>Tool axis direction retract</td><td>Conventional retract (Parameter No. 7041)</td></tr>
</table>

- Angle of the rotary axis for type 2 (when the movement range is not specified)

When the direction of the tool is specified by I, J, K, Q for type 2, more than two pairs of "computed angles" of the rotary axes usually exist.
The "computed angle" is the candidate angle at which the rotary axis is to be controlled in the specified tool axis direction.
The "output angle" is determined from the "computed angle" based on the "output judgment conditions" described below.

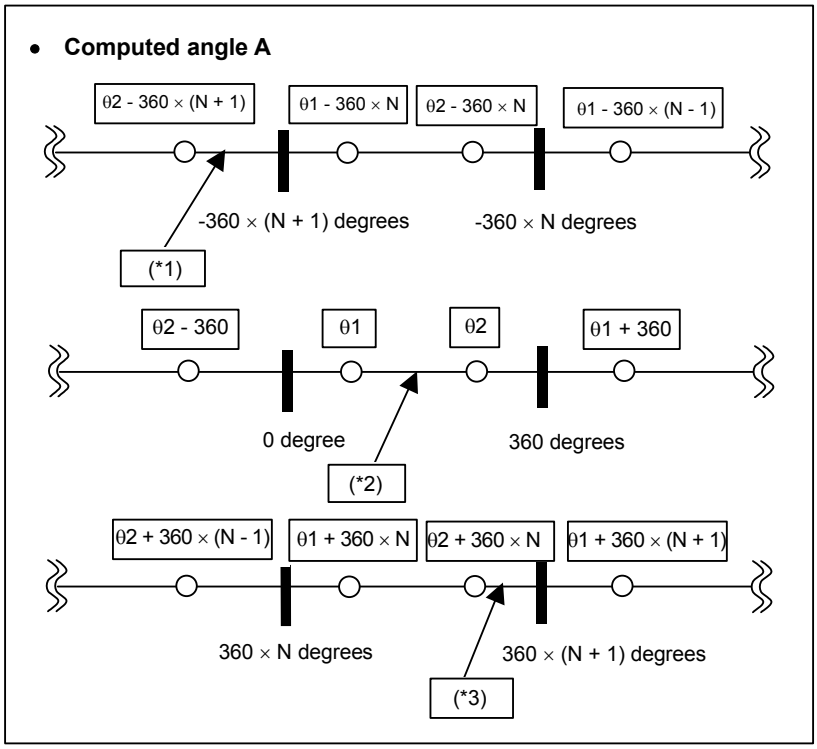
The following descriptions assume that there is no movement range specification (parameter No.19741 - No.19744 = 0).

"Output judgment conditions"

Tool rotation type or table rotation type machine
<1> The "output angles" are represented by the computed rotary axis angle pair whose master axis (first rotary axis) moving angle is smaller. ↓ ↓ When the master axis moving angle is the same ↓
<2> The "output angles" are represented by the computed rotary axis angle pair whose slave axis (second rotary axis) moving angle is smaller. ↓ ↓ When the slave axis moving angle is the same ↓
<3> The "output angles" are represented by the computed rotary axis angle pair whose master axis (first rotary axis) angle is nearer to 0 degree (multiple of 360 degrees). ↓ ↓ When the master axis angle is equally near to 0 degree ↓
<4> The "output angles" are represented by the computed rotary axis angle pair whose slave axis (second rotary axis) angle is nearer to 0 degree (multiple of 360 degrees).
Composite type machine
<1> The "output angles" are represented by the computed rotary axis angle pair whose table (second rotary axis) moving angle is smaller. ↓ ↓ When the table moving angle is the same ↓
<2> The "output angles" are represented by the computed rotary axis angle pair whose tool (first rotary axis) moving angle is smaller. ↓ ↓ When the tool moving angle is the same ↓
<3> The "output angles" are represented by the computed rotary axis angle pair whose table (second rotary axis) angle is nearer to 0 degree (multiple of 360 degrees). ↓ ↓ When the table angle is equally near to 0 degree ↓
<4> The "output angles" are represented by the computed rotary axis angle pair whose tool (first rotary axis) angle is nearer to 0 degree (multiple of 360 degrees).

The process of judging whether the moving angle is smaller or larger as the output judgement condition is called "movement judgement."
When bit 5 (PRI) of parameter No.19608 is 1, the movement judgements for the first rotary axis and second rotary axis are made in reverse order.

The "movement judgement" process is explained below.
When the "computed angle" is within the range between 0 and 360 degrees, it is called the "basic computed angle."
Usually, two pairs of "basic computed angles" exist.
For example, assume that a tool rotation type or table rotation type machine has rotary axis A (master) and rotary axis B (slave) and that there are two pairs of basic computed angles as follows:
(A θ1 degree; B ϕ1 degree)
(A θ2 degrees; B ϕ2 degrees) where θ1 ≤ θ2.
The "computed angle" is obtained from either of the following expressions: "basic computed angle" + 360 degrees × N or "basic computed angle" - 360 degrees × N.
The current position of rotary axis A (master) is PA, and that of rotary axis B (slave) is 0 degree.
Based on the PA angle, the "movement judgement" process is done as follows (when bit 5 (PRI) of parameter No.19608 is 0).



"Movement judgment"

When the PA angle is (*1):
The output angle is: (A θ2 - 360 × (N + 1) degrees; B ϕ2 degrees).
Namely, θ2 - 360 × (N + 1) degrees is adopted that is nearer to the computed angle of A, and ϕ2, which is the same group as θ2, is adopted as the computed angle of B.

When the PA angle is (*2):
The output angle is: (A θ1 degrees; B ϕ1 degrees).
Namely, θ1 degrees is adopted that is nearer to the computed angle of A, and ϕ1, which is the same group as θ1, is adopted as the computed angle of B.

When the PA angle is (*3):
The output angle is: (A θ + 360 × N degrees; B ϕ2 degrees).
Namely, θ2 + 360 × N degrees is adopted that is nearer to the computed angle of A, and ϕ2, which is the same group as θ2. is adopted as the computed angle of B.

When the moving angle of rotary axis A (master) is the same, a "movement judgement" is made for rotary axis B (slave) according to the "output judgment conditions."
If the "output angle" of rotary axis A is determined by the "movement judgement" for rotary axis A, the computed angle representing the "smaller moving angle" is adopted as the "output angle" of rotary axis B.
Similarly, if the "output angle" of rotary axis B is determined by the "movement judgement" for rotary axis B, the computed angle representing the "smaller moving angle" is adopted as the "output angle" of rotary axis A.

The "output angle" is explained below using a tool rotation type machine as an example.
This example illustrates a machine having a "BC type tool axis Z."

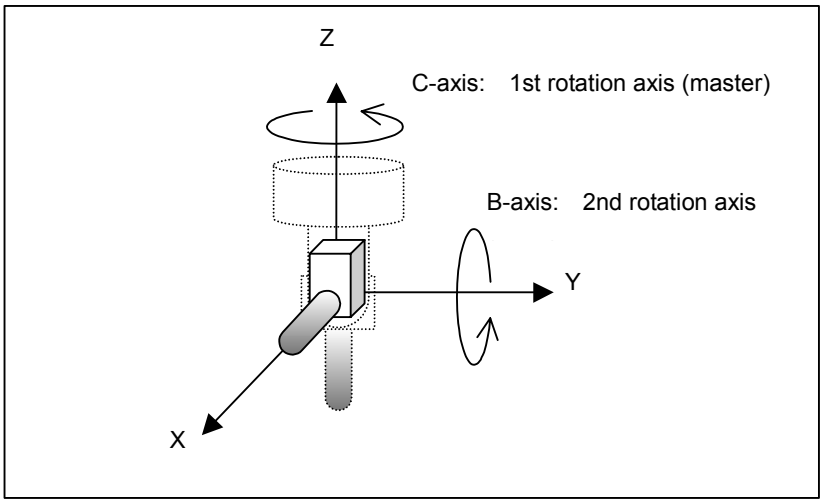


Fig. 22.1 (h) BC type tool axis Z

The following two pairs of "computed basic angles" exist that direct the tool axis toward the + X axis direction.
(B 90 degrees; C 180 degrees)
(B 270 degrees; C 0 degree)

<1> When the current rotary axis angles are (B -70 degrees; C 30 degrees)
The "output angles" are (B -90 degrees; C 0 degree).
0 degree is adopted because it is nearer to the current position (30 degrees) of the C-axis that is the master axis. For the B-axis, 270 degrees is adopted which is the same group. However, this is changed to -90 degrees (270 degrees - 360 degrees) which is the nearest to the current position of the B-axis (-70 degrees).

<2> When the current rotary axis angles are (B 80 degrees; C 500 degrees)
The "output angles" are (B 90 degrees; C 540 degrees).
540 degrees (180 degrees + 360 degrees) is adopted because it is nearer to the current position (500 degrees) of the C-axis that is the master axis. For the B-axis, 90 degrees is adopted which is the same group.

<3> When the current rotary axis angles are (B 60 degrees; C 90 degrees)
The "output angles" are (B 90 degrees; C 180 degrees).
Since the two candidates are equally near to the current position (90 degrees) of the C-axis that is the master axis, a judgment is made based on the current position of the B-axis. 90 degrees is adopted because it is nearer to the current position (60 degrees) of the B-axis that is the slave axis. For the C-axis, 180 degrees is adopted which is the same group.

<4> When the current rotary axis angles are (B 180 degrees; C 90 degrees)
The "output angles" are (B 270 degrees; C 0 degree).
Since the two candidates are equally near to the current position (90 degrees) of the C-axis that is the master axis, a judgment is made based on the current position of the B-axis. In this case, however, the two candidates are also equally near to the current position of the B-axis (180 degrees). Therefore, the candidate is adopted in which the C-axis (master axis) is nearer to 0 degree.
That is, the pair is adopted whose C axis angle is 0 degree and whose B axis angle is 270 degrees.

When the slave axis angle is 0 degree, the direction of the tool axis becomes fixed regardless of the master axis angle.
In that case, the master axis does not move from the current angle.

An explanation is shown below using a machine having a "BC type tool axis Z" as an example.

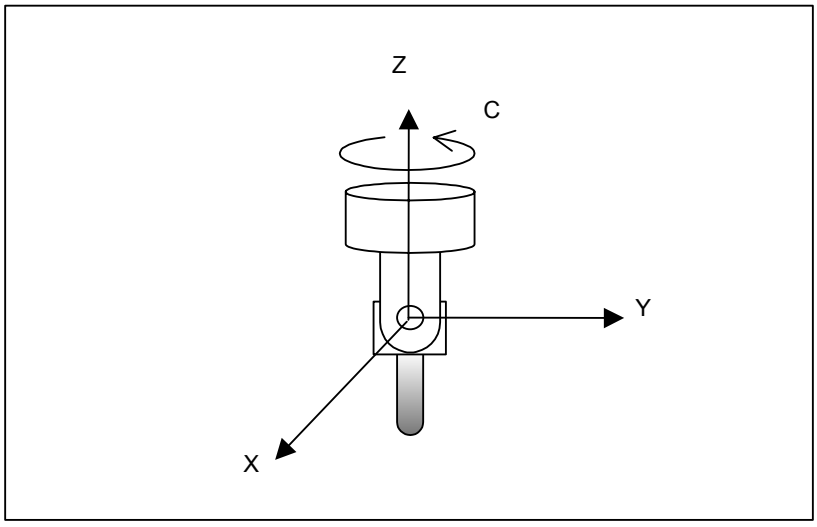


Fig. 22.1 (i) BC type tool axis Z

When the current rotary axis angles are (B 45 degrees; C 90 degrees), the "output angles" are (B 0 degree; C 90 degrees).

- Angle of the rotary axis for type 2 (when the movement range is specified)

If the upper and lower limits of the movement range of the rotary axis is specified using parameters No.19741 to No.19744, the rotary axis will move only within the specified range when the direction is specified using I, J, K, Q command for type 2 control.

Although the procedure for determining the angles is the same as that used "when the movement range is not specified," the "output angles" need to be selected from **those computed angles that are within the specified movement range for both axes**.

"Output judgment conditions"

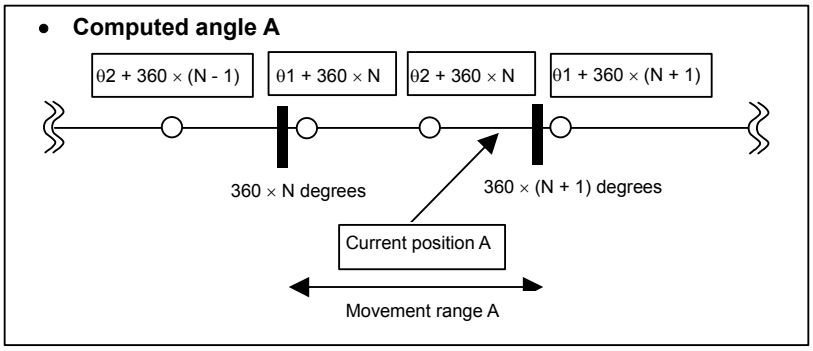
Tool rotation type or table rotation type machine
<1> Of the angle pairs whose master and slave axis angles are both within the specified movement range, the rotary axis angle pair whose master axis (first rotary axis) moving angle is smaller represents the "output angles."
↓
↓ When the master axis moving angle is the same
↓
<2> The "output angles" are represented by the computed rotary axis angle pair whose slave axis (second rotary axis) moving angle is smaller.
↓
↓ When the slave axis moving angle is the same
↓
<3> The "output angles" are represented by the computed rotary axis angle pair whose master axis (first rotary axis) angle is nearer to 0 degree (multiple of 360 degrees).
↓
↓ When the master axis angle is equally near to 0 degree
↓
<4> The "output angles" are represented by the computed rotary axis angle pair whose slave axis (second rotary axis) angle is nearer to 0 degree (multiple of 360 degrees).
Composite type machine
<1> Of the angle pairs whose master and slave axis angles are both within the specified movement range, the rotary axis angle pair whose table (second rotary axis) moving angle is smaller represents the "output angles."
↓
↓ When the table moving angle is the same
↓
<2> The "output angles" are represented by the computed rotary axis angle pair whose tool (first rotary axis) moving angle is smaller.
↓
↓ When the tool moving angle is the same
↓
<3> The "output angles" are represented by the computed rotary axis angle pair whose table (second rotary axis) angle is nearer to 0 degree (multiple of 360 degrees).
↓
↓ When the table angle is equally near to 0 degree
↓
<4> The "output angles" are represented by the computed rotary axis angle pair whose tool (first rotary axis) angle is nearer to 0 degree (multiple of 360 degrees).

When bit 5 (PRI) of parameter No.19608 is 1, the movement judgements for the first rotary axis and second rotary axis are made in reverse order.

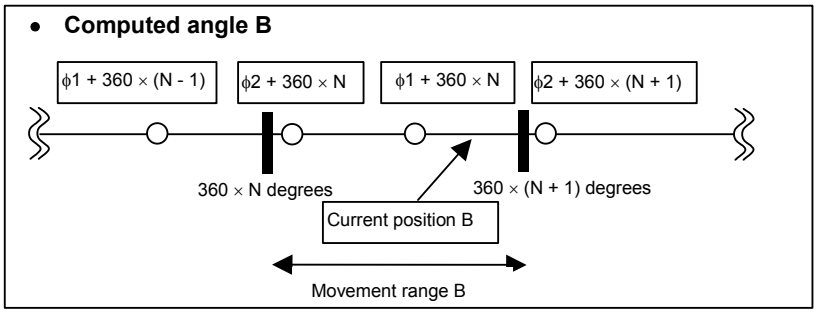
⚠ CAUTION

1 If the lower limit of the movement range is larger than the upper limit, alarm PS5459 occurs when G43.5 is specified.
2 If no "computed angle" is found within the movement range because the range is too small, alarm PS5459 occurs.
3 If 0 is set for both parameters that specify the upper and lower limits of the movement range, the tool operates assuming that there is no range specification.
4 When the rotary axis rollover function or rotary axis control function is used (in which case, set parameter No.1260 (amount of movement per rotation of the rotary axis) to 360 degrees), the tool does not move beyond 0 degree (360 degrees) (does not take the shortcut) if the movement range is set between 0 and 360 degrees. Also, do not specify a negative value or a value larger than 360 degrees for the movement range.

An example of the "movement judgement" process is given below.
Assume that a tool rotation type or table rotation type machine has rotary axis A (master) and rotary axis B (slave) and that there are two pairs of basic computed angles as follows:
(A θ1 degree; B ϕ1 degree)
(A θ2 degrees; B ϕ2 degrees) where $\theta 1 \leq \theta 2$.
The "computed angle" is obtained from either of the following expressions: "basic computed angle" + 360 degrees × N or "basic computed angle" - 360 degrees × N.
Assume that the current positions and movement ranges of rotary axis A (master) and rotary axis B (slave) are as shown in the following figure.



"Computed angle of rotary axis A and its current position and movement range"



"Computed angle of rotary axis B and its current position and movement range"

When the two axes have a positional relationship as shown in the figure, the output angle of rotary axis A is (θ2 + 360 × N) degrees and that of rotary axis B is (ϕ2 + 360 × N) degrees (when bit 5 (PRI) of parameter No.19608 is set to 0).
More concretely, from the computed angles obtained for rotary axis A, the nearest angle within the movement range, i.e. θ2 + 360 × N degrees, is first adopted. Then, from the computed angles obtained for rotary axis B, the angle belonging to the same group as θ2, i.e. ϕ2 + 360 × N, is adopted.

Note that, in this example, the output angles and moving direction differ depending on whether the movement range is specified or not (0 to 360 degrees), even if N is set to 0 and coordinates are rounded to 0 to 360 degrees.
Namely, if the movement range is not specified, θ1 + 360 degrees nearest to the current position is adopted as the computed angle for rotary axis A and, from the computed angles belonging to the same group as θ1, ϕ1 degrees nearest to the current position is adopted as the computed angle for rotary axis B. Rotary axis A moves in the plus direction. As its coordinate is rounded to 360 degrees, rotary axis A reaches θ1 degrees while moving in the plus direction.
By contrast, when the movement range is set to 0 to 360 degrees, the output angles are (A θ2 degrees; B ϕ2 degrees). Neither rotary axis A nor B moves in a way that it exceeds 0 degree (360 degrees).

Operation examples

- In the case of a tool rotation type machine

Explanations are given below assuming a machine configuration in which a tool rotation axis that turns around the Y-axis is located beneath another tool rotation axis that turns around the Z-axis. (See Fig. 22.1 (j).)

If linear interpolation is specified for the X-, Y-, and Z-axes when a workpiece coordinate system is used as the programming coordinate system, control is exerted in such a way that the tool center point moves along a specified straight line with respect to the table (workpiece) as the tool rotates.
Also, speed control is exerted so that the tool center point moves at the specified speed with respect to the table (workpiece).

In the case of a machine having two tool rotation axes, the table does not rotate with respect to the workpiece coordinate system even if the rotary axes move. Therefore, the programming coordinate system always matches the workpiece coordinate system, regardless of whether bit 5 (WKP) of parameter No.19696 is set to 0 or 1.

Examples)
For type 1:

```
O100 (Sample Program1) ;
N1 G00 G90 B0 C0 ;
N2 G55 ;                          Prepares the programming coordinate
                                  system.
N3 G43.4 H01 ;                    Starts tool center point control.
                                  H01 is the tool compensation number.
N4 G00 X200.0 Y150.0 Z20.0 ;      Moves to the start point.
N5 G01 X5.0 Y5.0 Z5.0 C60.0 B45.0 F500 ; Linear interpolation
N6 G49;                           Cancels tool center point control.
N7 M30;
```

For type 2:

```
O100 (Sample Program1) ;
N1 G00 G90 B0 C0 ;
N2 G55 ;                  Prepares the programming coordinate system.
N3 G43.5 H01 ;            Starts tool center point control.
                          H01 is the tool compensation number.
N4 G00 X200.0 Y150.0 Z20.0 ; Moves to the start point.
N5 G01 X5.0 Y5.0 Z5.0 I1.0 J1.732 K2.0 F500 ; Linear interpolation
N6 G49;                   Cancels tool center point control.
N7 M30;
```

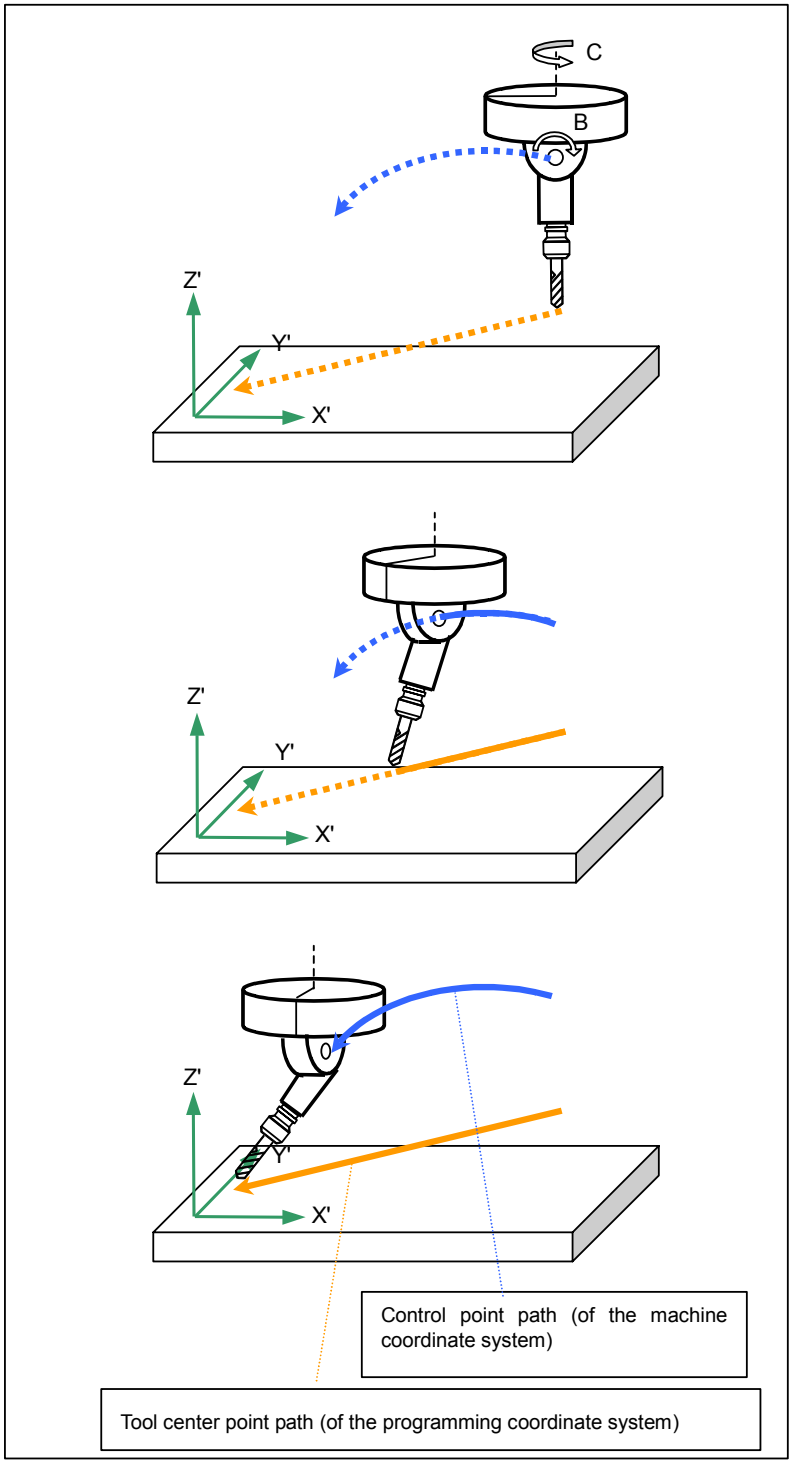


Fig. 22.1 (j) Example for a tool rotation type machine

- In the case of a table rotation type machine

Explanations are given below assuming a machine configuration (trunnion) in which a rotation table that turns around the Y-axis is located above another table rotation axis that turns around the X-axis. (See Fig. 22.1 (k).)

If linear interpolation is specified for the X-, Y-, and Z-axes on the programming coordinate system and if the rotary axis that moves the rotation table is specified (in the case of type 1) or the tool direction is specified (in the case of type 2), control is exerted in such a way that the tool center point moves along a specified straight line with respect to the table (workpiece) as the rotation table rotates.
Also, speed control is exerted so that the tool center point moves at the specified speed with respect to the table (workpiece).

Example)
When type 1 is selected and the coordinate system fixed on the table is used as the programming coordinate system
(Bit 5 (WKP) of parameter No.19696 = 0):

```
O200 (Sample Program2) ;
N1 G00 G90 A0 B0 ;
N2 G55 ;                          Prepares the programming coordinate
                                  system.
N3 G43.4 H01 ;                    Starts tool center point control.
                                  H01 is the tool compensation number.
N4 G00 X20.0 Y100.0 Z0 ;          Moves to the start point.
N5 G01 X10.0 Y20.0 Z30.0 A60.0 B45.0 F500 ; Linear interpolation
N6 G49;                           Cancels tool center point control.
N7 M30;
```

When type 1 is selected and the workpiece coordinate system is used as the programming coordinate system
([Bit 5 (WKP) of parameter No.19696 = 1):

```
O200 (Sample Program2) ;
N1 G00 G90 A0 B0 ;
N2 G55 ;                          Prepares the programming coordinate
                                  system.
N3 G43.4 H01 ;                    Starts tool center point control.
                                  H01 is the tool compensation number.
N4 G00 X20.0 Y100.0 Z0 ;          Moves to the start point.
N5 G01 X7.574 Y47.247 Z83.052 A60.0 B45.0 F500 ;
                                  Linear interpolation
N6 G49;                           Cancels tool center point control.
N7 M30;
```

For type 2 (when the coordinate system fixed on the table is used as the programming coordinate system (only when bit 5 (WKP) of parameter No.19696 is set to 0)):

```
O200 (Sample Program2) ;
N1 G00 G90 A0 B0 ;
N2 G55 ;                          Prepares the programming coordinate
                                  system.
N3 G43.5 H01 ;                    Starts tool center point control.
                                  H01 is the tool compensation number.
N4 G00 X20.0 Y100.0 Z0 ;          Moves to the start point.
N5 G01 X10.0 Y20.0 Z30.0 I-1.0 J2.449 K1.0 F500 ;
                                  Linear interpolation
N6 G49;                           Cancels tool center point control.
N7 M30;
```

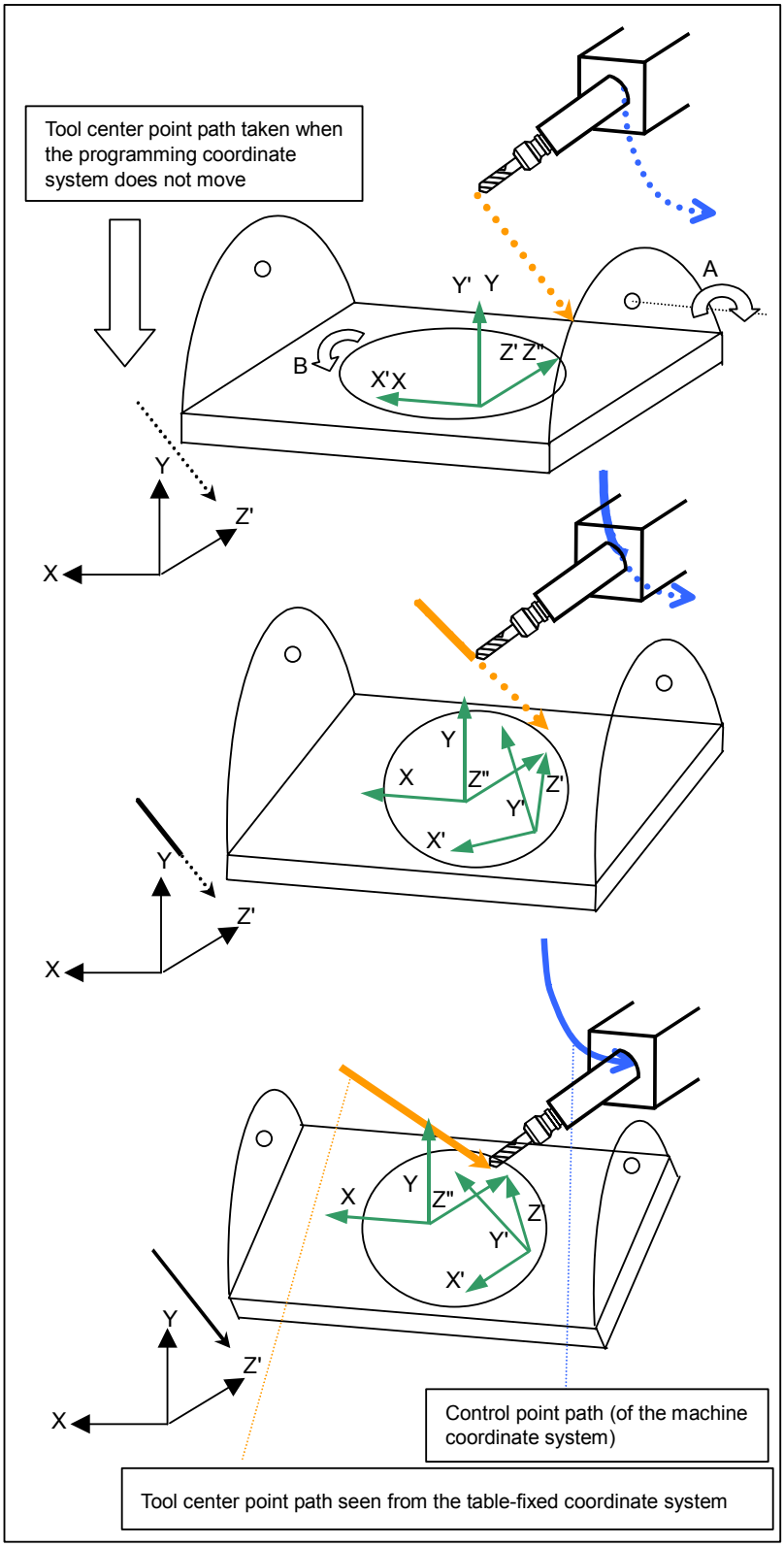


Fig. 22.1 (k) Example for a table rotation type machine

- In the case of a composite type machine

Explanations are given below assuming a composite type machine configuration that has one table rotation axis (which turns around the X-axis) and one tool rotation axis (which turns around the Y-axis). (See Fig. 22.1 (l).)

If linear interpolation is specified for the X-, Y-, and Z-axes on the programming coordinate system and if the rotary axis that moves the rotation table and the tool rotation axis are specified (in the case of type 1) or the tool direction is specified (in the case of type 2), control is exerted in such a way that the tool center point moves along a specified straight line with respect to the table (workpiece) as the rotation table and tool rotate.
Also, speed control is exerted so that the tool center point moves at the specified speed with respect to the table (workpiece).

Example)
When type 1 is selected and the coordinate system fixed on the table is used as the programming coordinate system
(Bit 5 (WKP) of parameter No.19696 = 0):

```
O300 (Sample Program3) ;
N1 G00 G90 A0 B0 ;
N2 G55 ;                          Prepares the programming coordinate
                                  system.
N3 G43.4 H01 ;                    Starts tool center point control.
                                  H01 is the tool compensation number.
N4 G00 X200.0 Y150.0 Z20.0 ;      Moves to the start point.
N5 G01 X5.0 Y5.0 Z5.0 A60.0 B45.0 F500;
                                  Linear interpolation
N6 G49 ;                          Cancels tool center point control.
N7 M30;
```

When type 1 is selected and the workpiece coordinate system is used as the programming coordinate system
(Bit 5 (WKP) of parameter No.19696 = 1):

```
O300 (Sample Program3) ;
N1 G00 G90 A0 B0 ;
N2 G55 ;                          Prepares the programming coordinate
                                  system.
N3 G43.4 H01 ;                    Starts tool center point control.
                                  H01 is the tool compensation number.
N4 G00 X200.0 Y150.0 Z20.0 ;      Moves to the start point.
N5 G01 X5.0 Y48.170 Z-79.772 A60.0 B45.0 F500;
                                  Linear interpolation
N6 G49 ;                          Cancels tool center point control.
N7 M30;
```

For type 2 (when the coordinate system fixed on the table is used as the programming coordinate system (only when bit 5 (WKP) of parameter No.19696 is set to 0)):

```
O300 (Sample Program3) ;
N1 G00 G90 A0 B0 ;
N2 G55 ;                                  Prepares the programming coordinate
                                          system.
N3 G43.5 H01 ;                            Starts tool center point control.
                                          H01 is the tool compensation number.
N4 G00 X200.0 Y150.0 Z20.0 ; Moves to the start point.
N5 G01 X5.0 Y5.0 Z5.0 I2.0 J1.732 K1.0 F500;
                                          Linear interpolation
N6 G49 ;                                  Cancels tool center point control.
N7 M30;
```

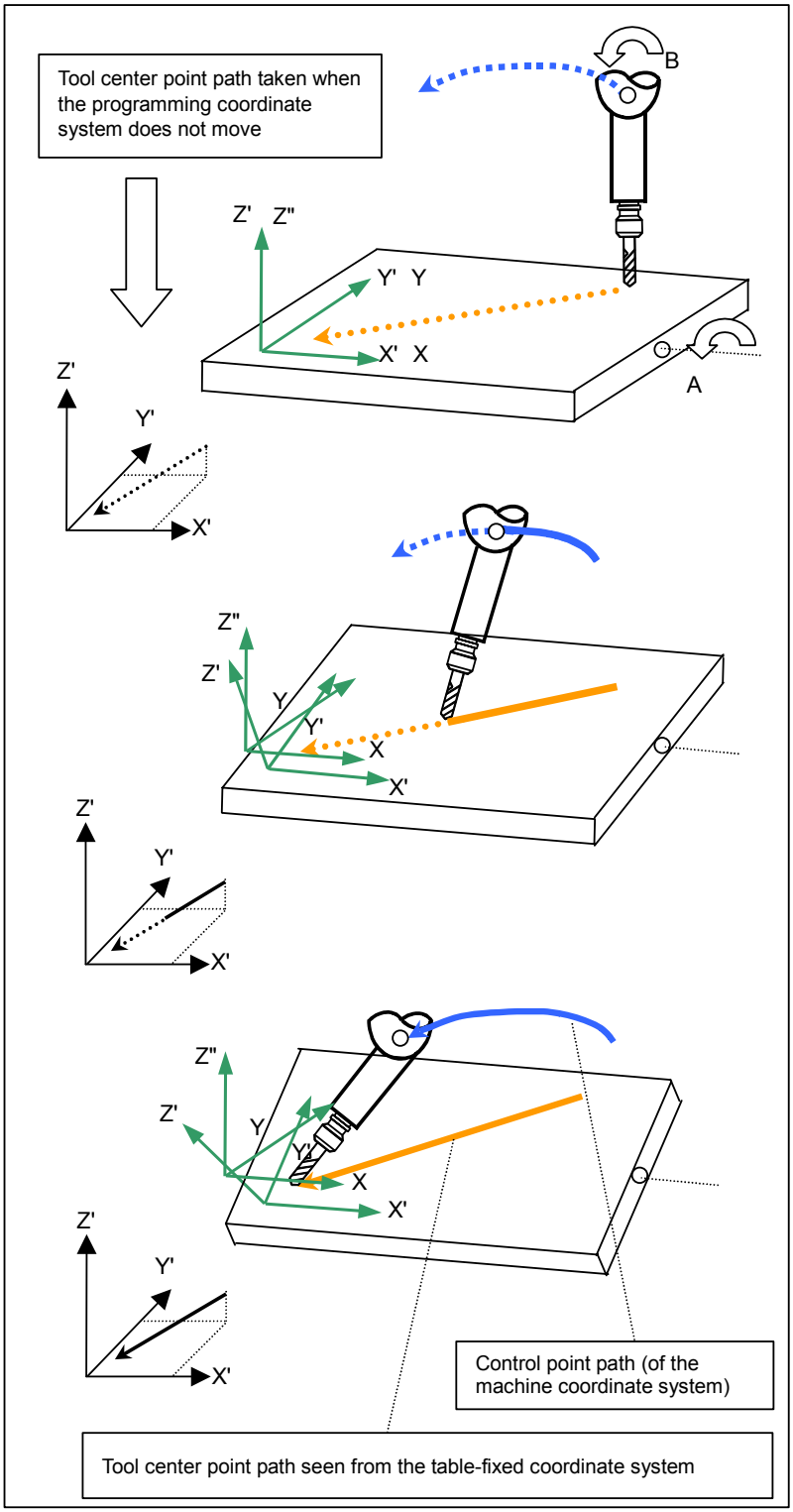


Fig. 22.1 (l) Example for a composite type machine

- When linear interpolation is performed during tool center point control

Examples are given below in which each 100-mm-long side of an equilateral triangle is cut at B-axis angles of 0, 30 to 60, and 60 degrees, respectively.

Example)
When type 1 is selected and the table-fixed coordinate system is used as the programming coordinate system:

```
O400 (Sample Program4) ;
N10 G55 ;                                  Prepares the programming
                                           coordinate system.
N20 G90 X50.0 Y-70.0 Z300.0 B0 C0 ;        Moves to the initial position.
N30 G01 G43.4 H01 Z20.0 F500. ;            Starts tool center point control.
                                           Moves to the approaching
                                           position.
                                           H01 is the tool compensation
                                           number.
N40 X28.868 Y-50.0 Z10.0 B30.0 ;           The Z-axis height on the
                                           machining plane is 10.0.
N50 Y50.0 ;
N60 B45.0 C120.0 ;
N70 X-57.735 Y0 B60.0 C180.0 ;             Moves X and Y while operating
                                           both B- and C-axes.
N80 C240.0 ;
N90 X28.868 Y-50.0 ;
N100 X50.0 Y-70.0 Z20.0 B0 C360.0 ;        X, Y, and Z are approaching
                                           positions.
                                           The rotary axes move to their
                                           original positions.
N110 G49 Z300.0 ;                          Cancels tool center point
                                           control.
                                           Moves the Z-axis to its initial
                                           position.
N120 M30;
```

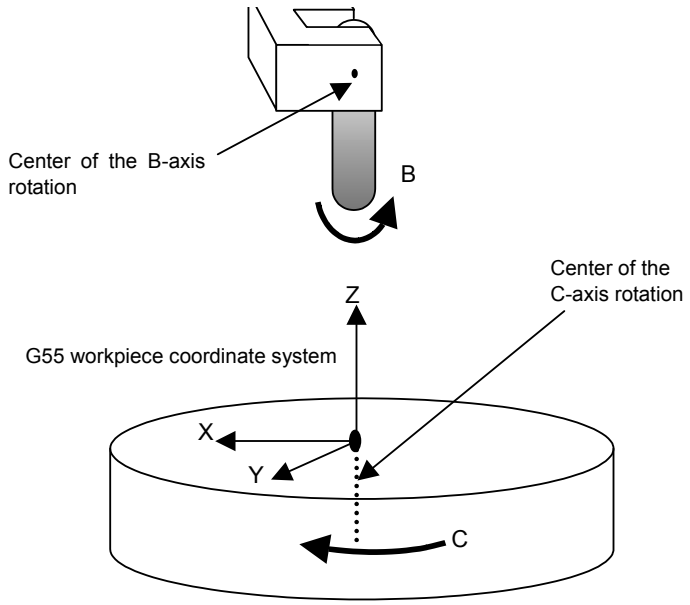
When type 1 is selected and the workpiece coordinate system is used as the programming coordinate system
(Note that the values of N60 to N90 are different from those specified in the preceding example.):

```
O400 (Sample Program4) ;
N10 G55 ;                                  Prepares the programming
                                           coordinate system.
N20 G90 X50.0 Y-70.0 Z300.0 B0 C0 ;        Moves to the initial position.
N30 G01 G43.4 H01 Z20.0 F500. ;            Starts tool center point control.
                                           Moves to the approaching
                                           position.
                                           H01 is the tool compensation
                                           number.
N40 X28.868 Y-50.0 Z10.0 B30.0 ;           The Z-axis height on the
                                           machining plane is 10.0.
N50 Y50.0 ;
N60 Y-50.0 B45.0 C120.0 ;
N70 X57.735 Y0 B60.0 C180.0 ;              Moves X and Y while operating
                                           both B- and C-axes.
N80 X28.868 Y-50.0 C240.0 ;
N90 Y50.0 ;
N100 X50.0 Y-70.0 Z20.0 B0 C360.0 ;        X, Y, and Z are approaching
                                           positions.
                                           The rotary axes move to their
                                           original positions.
N110 G49 Z300.0 ;                          Cancels tool center point
                                           control.
                                           Moves the Z-axis to its initial
                                           position.
N120 M30;
```

When type 2 is selected and the table-fixed coordinate system is used as the programming coordinate system:

```
O400 (Sample Program4) ;
N10 G55 ;                                  Prepares the programming
                                           coordinate system.
N20 G90 X50.0 Y-70.0 Z300.0 B0 C0 ;        Moves to the initial position.
N30 G01 G43.5 H01 Z20.0 F500. ;            Starts tool center point control.
                                           Moves to the approaching
                                           position.
                                           H01 is the tool compensation
                                           number.
N40 X28.868 Y-50.0 Z10.0 I1.0 K1.732 ;     The Z-axis height on the
                                           machining plane is 10.0.
N50 Y50.0 ;
N60 I-0.5 J0.866 K1.0 ;
N70 X-57.735 Y0 I-1.732 K1.0 ;             Moves X and Y while operating
                                           both B- and C-axes.
N80 I-0.866 J1.5 K1.0 ;
N90 X28.868 Y-50.0 ;
N100 X50.0 Y-70.0 Z20.0 K1.0 ;             X, Y, and Z are approaching
                                           positions.
                                           The rotary axes move to their
                                           original positions.
N110 G49 Z300.0 ;                          Cancels tool center point
                                           control.
                                           Moves the Z-axis to its initial
                                           position.
N120 M30;
```

- **Composite type machine (tool rotation axis = B-axis; table rotation axis = C-axis; tool axis = Z direction)**



Machine configuration for the example

The following figure illustrates the position of the workpiece, as well as the position of the tool head (relative to the workpiece), as seen from the table-fixed programming coordinate system in the +Z direction.

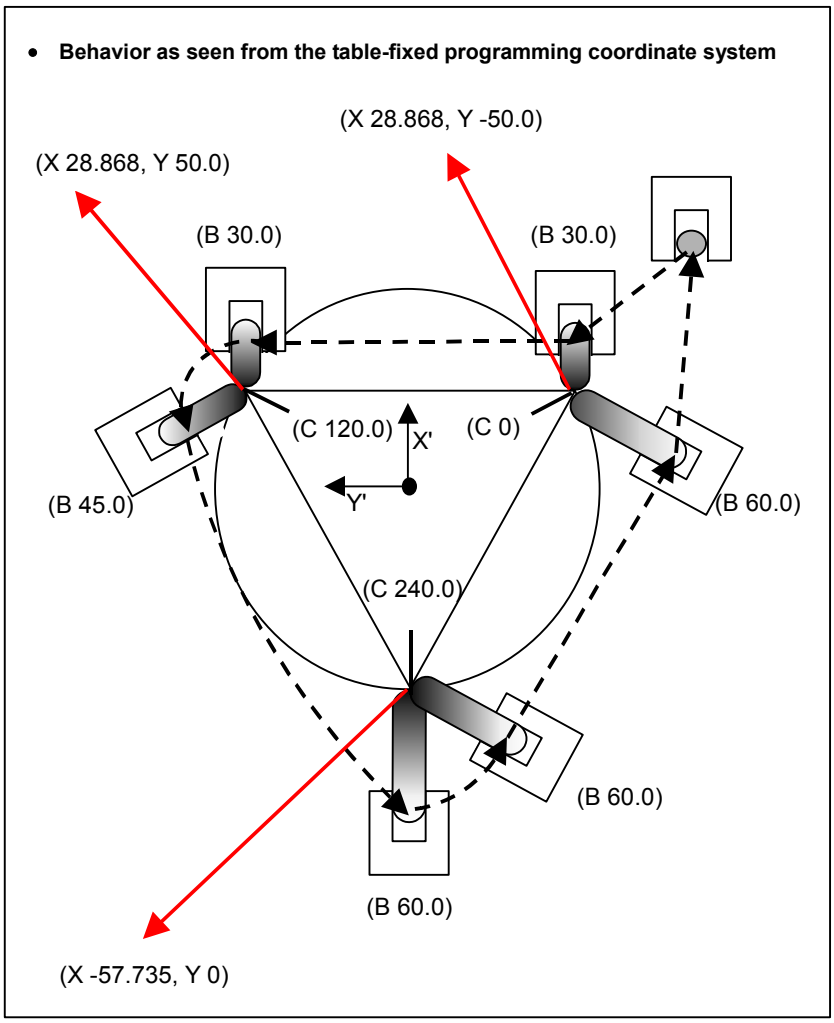
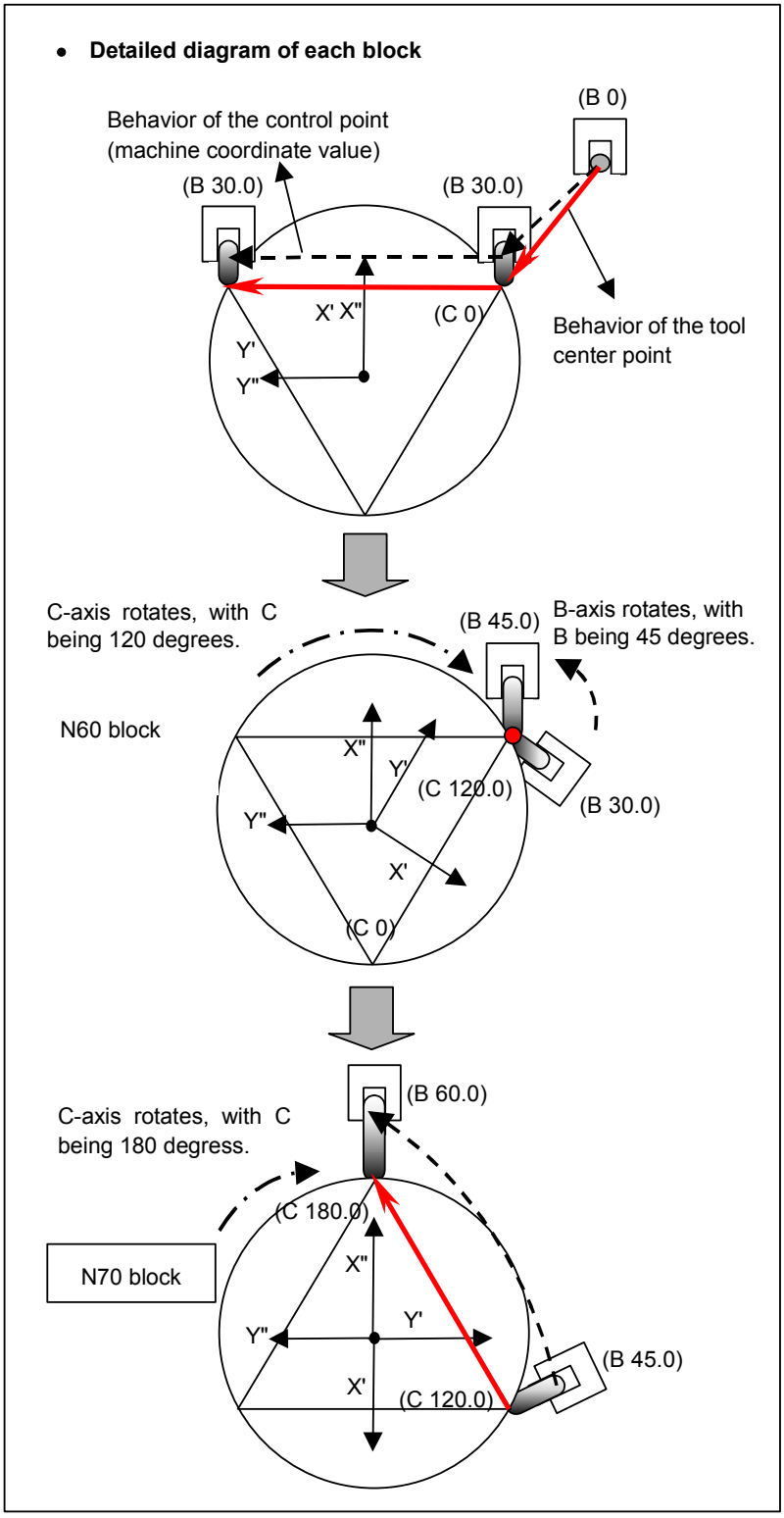
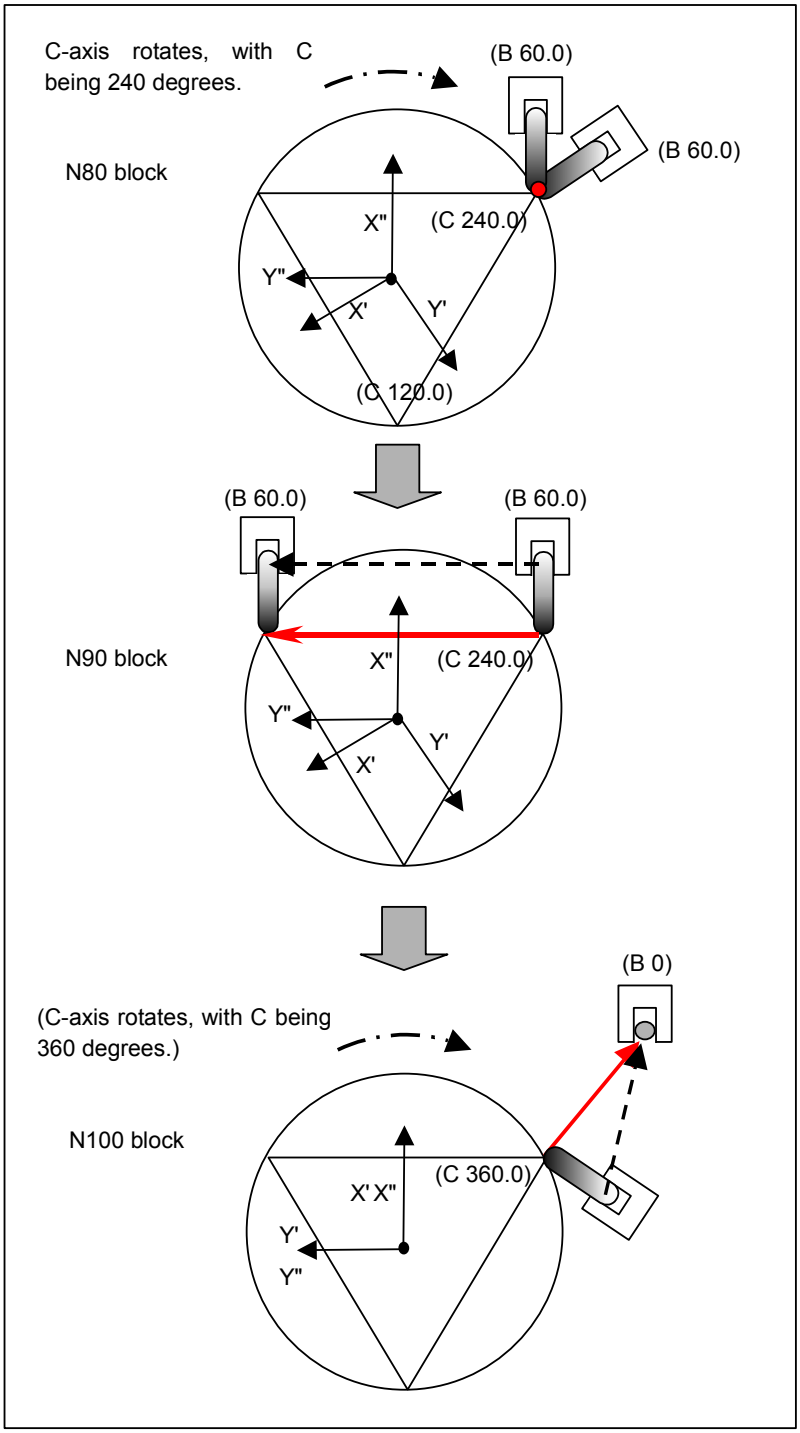


Illustration of the example (1)



Detailed diagram of each block (1)



Detailed diagram of each block (2)

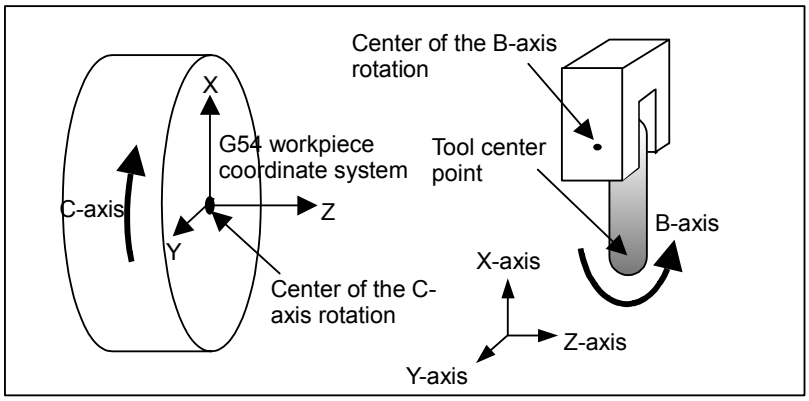
- When circular interpolation is performed during tool center point control

In this example, one of the three sides of an equilateral triangle, each being 100 mm long side, is specified as a straight line and the other two are specified as arcs, and each side is cut at B-axis angles of -60, -45 to -30, and -30 degrees, respectively. (For the X-axis, its radius is specified.)

Example)

The program sample given below assumes that the table-fixed coordinate system is used as the programming coordinate system. When the workpiece coordinate system is used as the programming coordinate system (when bit 5 (WKP) of parameter No.19696 is set to 1), the same shape can be obtained by specifying the values shown in parentheses.

```
N001 T0000 ;                          Cancels the tool offset.
N002 G54 ;                            Selects the workpiece coordinate system.
N003 G90 X50.0 Y-70.0 Z300.0 B-90.0 C0.0 ;   Moves to the initial position.
                                      The tool tip faces the Z- direction at B-axis
                                      angle of -90 degrees.
N020 G01 G43.4 H01                    Starts tool center point control.
N021 Z20.0 ;                          Moves to the approaching position.
N022 X28.868 Y-50.0 Z10.0 B-60.0 ;    The Z-axis height on the machining
                                      plane is 10.0.
N031 Y50.0 ;
N032 B-45.0 C90.0 ;
(N032 X50.0 Y-28.868 B-45.0 C90.0 ;)
N033 G17 G03 X-57.735 Y0.0 J-100.0 B-30.0 C150.0 ;
                                      Moves X and Y while operating both B-
                                      and C-axes.
(N033 G17 G03 X50.0 Y28.868 I-100.0 B-30.0 C150.0 ;)
N034 G01 B-30.0 C210.0 ;
(N034 G01 X50.0 Y-28.867 B-30.0 C210.0 ;)
N035 G03 X28.868 Y-50.0 I86.603 J50.0 C270.0 ;
(N035 G03 X50.0 Y28.868 I-100.0 C270.0 ;)
N041 G01 X50.0 Y-70.0 Z20.0 B-90.0 C360.0 ;
                                      X, Y, and Z are approaching positions.
                                      The rotary axes move to their original
                                      positions.
N050 G49 ;                            Cancels tool center point control.
N051 Z300.0 ;                         Moves the Z-axis to its initial position.
```



Machine configuration for the circular interpolation example

The following figure illustrates the relative positional relationship between the workpiece and the tool head and their positions, as seen from the +Z direction. The X and Y coordinate values shown in the figure are those on the table-fixed programming coordinate system (which rotates as the C-axis turns).

- **Behavior as seen from the table-fixed programming coordinate system**

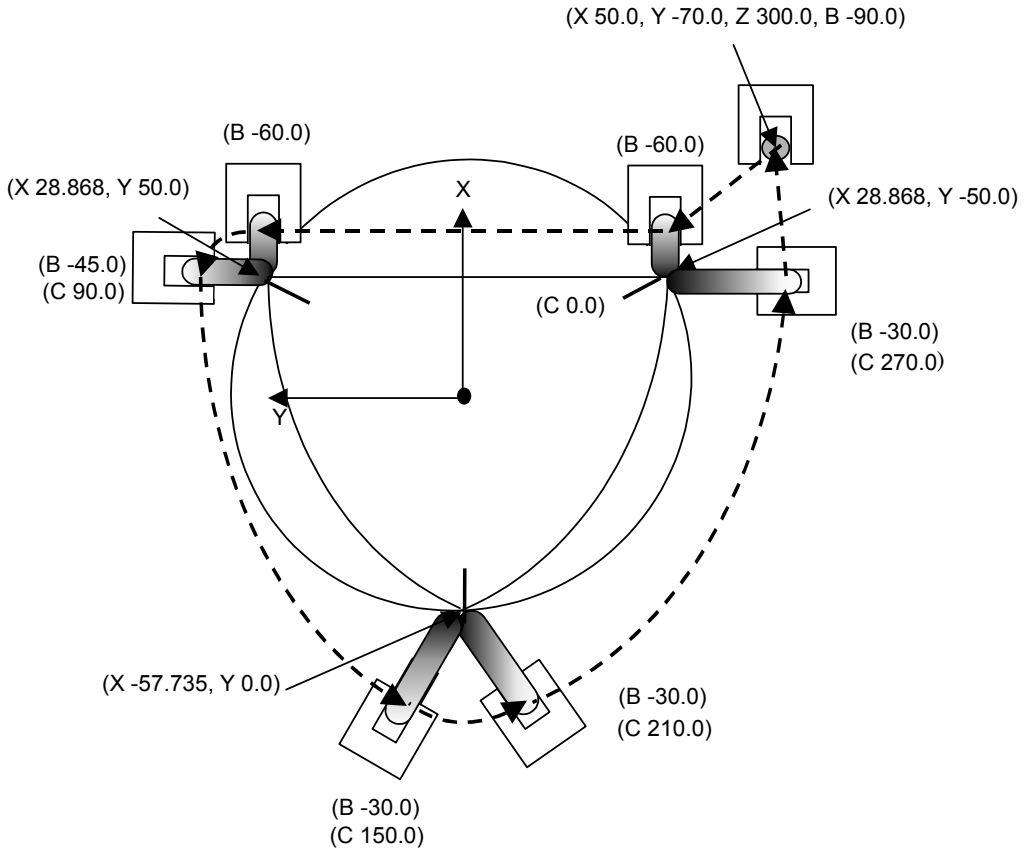
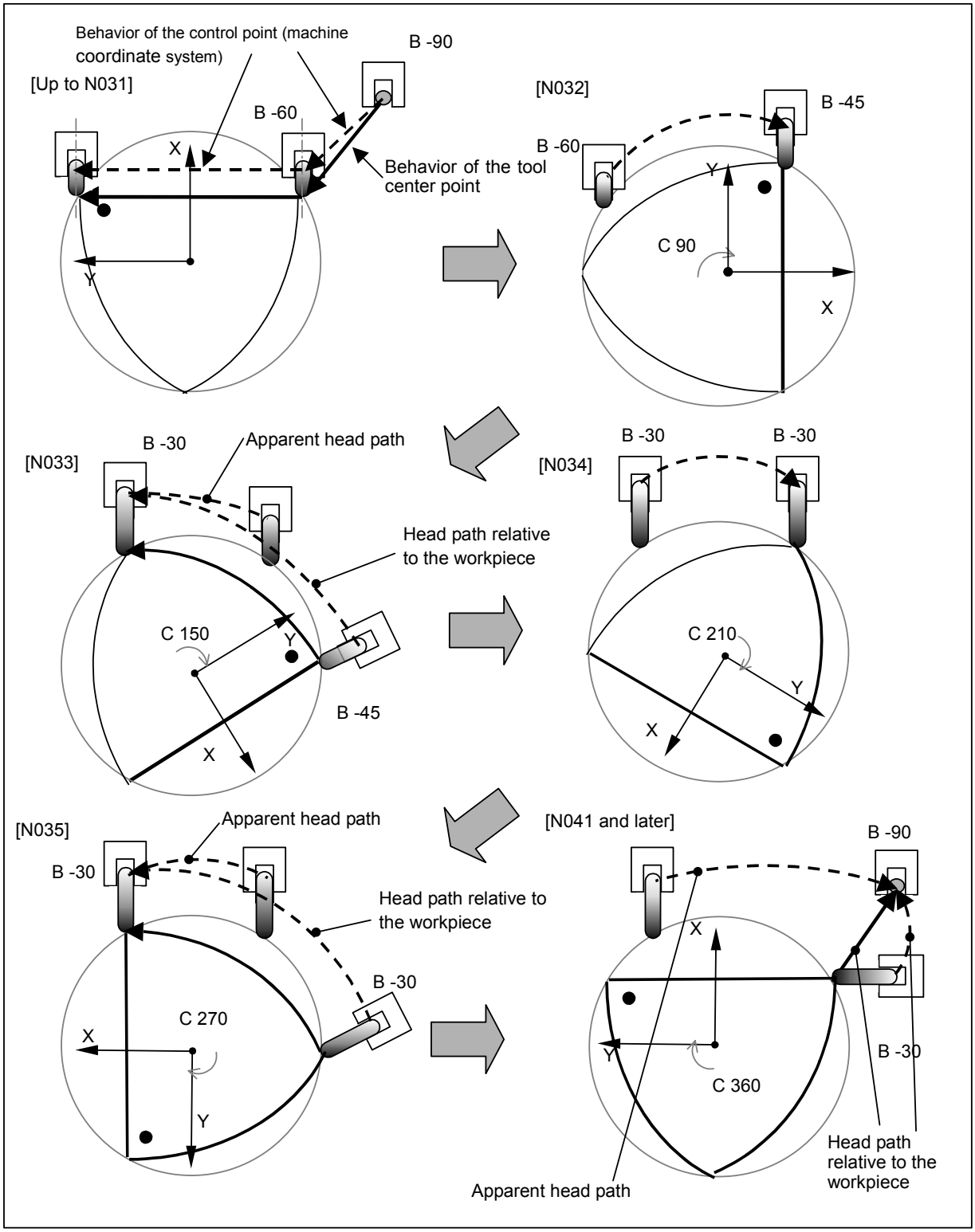


Illustration of the example



Detailed diagram of each block

Limitation

- Manual intervention

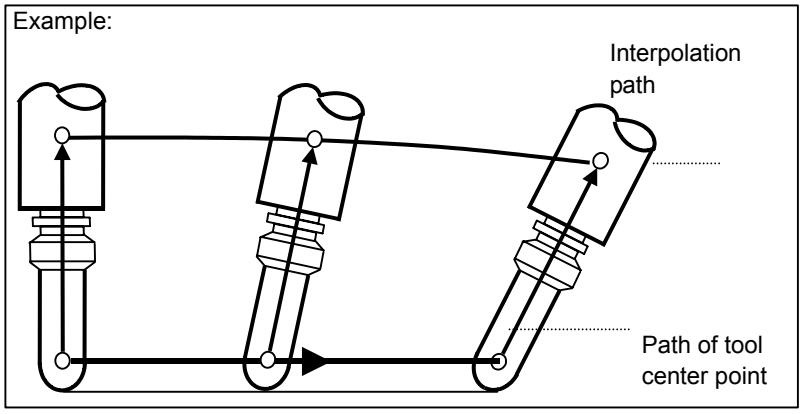
Manual intervention is impossible during tool center point control.
If manual intervention is performed, alarm PS5421 is issued when automatic operation is started later.

- Hypothetical axis of a table rotation axis

When a table rotation axis is set as a hypothetical axis, tool center point control is performed on the assumption that the table rotation axis is at 0 degrees.

- Look-ahead pre-interpolation acceleration/deceleration

When the tool center point control mode is set, look-ahead acceleration/deceleration before interpolation is automatically enabled. Set the parameter for look-ahead acceleration/deceleration before interpolation. If the parameter is not set correctly, alarm PS5420 is issued.



In the above example, the feedrate is controlled so that the tool center point moves at a specified feedrate, so the feedrate on the interpolation path is equal to or higher than the specified feedrate. At this time, look-ahead acceleration/deceleration before interpolation allows the feedrate to be clamped so that the feedrate on the interpolation path does not exceed the maximum cutting feedrate or maximum rapid traverse rate.

- Deceleration at a corner

Under tool center point control, the controlled point may move along a curve even if a straight-line command is issued. Some commands may cause a corner movement.
When AI contour control is not used, a shock can be applied to the machine at a corner.
So, it is recommended to use AI contour control as well and set an allowable speed difference at a corner (parameter No. 1783) and an allowable acceleration rate (parameter No. 1660, No. 1737).

- 3-dimensional cutter compensation

3-dimensional cutter compensation and tool center point control can be used at the same time.
For restrictions in such a case, see "Restrictions" of 3-dimensional cutter compensation.

- Programmable mirror image

Note the following points when making a programmable mirror image:

- In the case of tool center point control of type 1
  Mirroring the linear axis alone does not create a mirror image for the rotary axis. To make the direction of the tool symmetrical, it is necessary to make a mirror image for the rotary axis as well.
- In the case of tool center point control of type 2
  When bit 6 (MIR) of parameter No.19608 is set to 0, mirroring the linear axis alone does not create a mirror image for I, J, K. Note that the inclination direction specified by Q is the movement direction of the tool center point after mirroring.
  When bit 6 (MIR) of parameter No.19608 is set to 1, mirroring the linear axis automatically creates a mirror image for I, J, K.
  A mirror image cannot be put on the rotary axis directly.

- Scaling

Do not apply scaling to a rotary axis.

- Machine coordinate system command

During tool center point control, the machine coordinate system command (G53) cannot be used.
However, G53 can be specified singly to suppress look-ahead operation.

- Stored limit check before move

During tool center point control, stored limit check before move is disabled.

- Type 2

When a type 2 command (G43.5 command) is specified on a machine of table rotation type or composite type (parameter No. 19680 = 12 or 21), the table coordinate system needs to be set as the programming coordinate system (by setting bit 5 (WKP) of parameter No. 19696 to 0). If type 2 is specified with a workpiece coordinate system set as the programming coordinate system, alarm PS5459 is issued.

- Specifiable G codes

The G codes that can be specified in the tool center point control mode are listed below.
Specifying a G code other than these codes results in alarm PS5421.

- Positioning (G00)
- Linear interpolation (G01)
- Circular interpolation / helical interpolation (G02/G03)
- Dwell (G04)

- Exact stop (G09)
- Programmable data input (G10)
- Tool retract and recover (G10.6)
- Programmable data input mode cancel (G11)
- Plane selection (G17/G18/G19)
- Tool radius compensation cancel (G40)
- 3-dimensional cutter compensation (G41.2/G42.2/G41.4/G42.4/G41.5/G42.5)
- Tool length compensation cancel (G49)
- Scaling (G50/G51)
- Programmable mirror image (G50.1/G51.1)
- Machine coordinate system command (G53) specified singly
- Cutting mode(G64)
- Macro call (G65)
- Macro modal call A (G66)
- Macro modal call B (G66.1)
- Macro modal call A/B cancel (G67)
- Absolute programming (G90)
- Incremental programming (G91)
- Canned cycle : return to initial level (G98)
- Canned cycle : return to R point level (G99)

- Modal G codes that allow specification of tool center point control

Tool center point control can be specified in the modal G code states listed below.
Tool center point control cannot be specified in a modal state other than these.

- Modal G codes included in "Specifiable G codes" described previously
- Polar coordinate interpolation mode cancel (G13.1)
- Polar coordinates command cancel (G15)
- Input in inch (G20 (G70))
- Input in mm (G21 (G71))
- Polygon turning cancel (G50.2)
- Workpiece coordinate system setting (G54 to G59)
- Canned cycle cancel (G80)
- Constant surface speed control cancel (G97)

M

- Coordinate system rotation cancel or 3-dimensional coordinate conversion mode off (G69)
- Feed per minute (G94)
- Polar coordinate interpolation mode cancel (G113)

T

- Mirror image for double turret off/balanced cutting mode cancel (G69)
- Coordinate system rotation cancel or 3-dimensional coordinate conversion mode off (G69.1)
- Feed per minute (G98 (G94))

- Specification of axes not relating to tool center point control

Axes not relating to tool center point control cannot be specified. If such an axis is specified, alarm PS5421 is issued.

- Linear axes under tool center point control

The basic three axes set in parameter No. 1022 are regarded as the three linear axes for tool center point control. Axes parallel to the basic three axes cannot be used as the linear axes for tool center point control.
If the basic three axes are not set in parameter No. 1022, alarm PS5459 is issued.

T

- Extended tool selection

When tool center point control is used with a lathe system, the tool change method needs to be ATC type (bit 3 (TCT) of parameter No. 5040 = 1).
In this case, only the value set in "Z" on the tool compensation memory screen is used as a tool length compensation amount. Set values such as "X", "R", and "Y" axis offset values are not used.

22.2 TOOL POSTURE CONTROL

Overview

Under tool center point control, the tool tip moves along a specified path even when the tool direction relative to the workpiece changes. Usually, however, the two rotary axes are controlled independently of each other, and the posture of the tool is not controlled.
Under tool posture control, the tool posture is controlled every moment for operation on the plane formed by the tool postures (tool length compensation vectors) at the start point and end point of a block.
This control method is suitable for machining a plane with the side of a tool by using tool center point control.

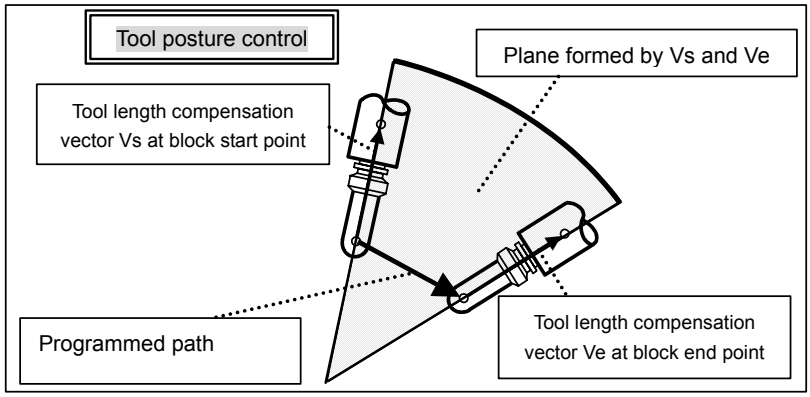


Fig. 22.2 (a) Tool posture control

This function is optional. The tool center point control option is also needed.

Format

Format
G43.4 IP_ α_ β_ H_ P_ ; Tool center point control (type 1) **G43.5 IP_ H_ Q_ P_ ; Tool center point control (type 2)**
Description of symbols
IP : For an absolute command, the coordinates of travel end point of the tool center point For an incremental command, the travel distance of the tool center point
α, β : For an absolute command, the coordinates of the end point of the rotation axis For an incremental command, the travel distance of the rotation axis
H : Tool offset number
Q : Tool tilted angle (degree)
P : Tool posture control disabled (P0) or tool posture control enabled (P1) However, bit 0 (TPC) of setting parameter No. 19604 can be used to select the behavior taken when the address P command is not present.

Tool posture control is enabled when P1 is specified or disabled when P0 is specified in the G43.4 and G43.5 blocks. The behavior taken when the address P command is not present can be selected by bit 0 (TPC) of parameter No. 19604. The setting of tool posture control selected in the G43.4 and G43.5 blocks is kept until the tool center point control cancellation command (G49) is specified.

Example:

```
O0010
…
N10 G90 G00 X30.0 Y-30.0 Z0.0;
N20 G43.5 H1 P1;
N30 G42.6 D2;
N40 G01 X20.0 Y-20.0 I-1 J1 K2 F_;
N50 Y20.0 I-1 J-1 K2;
N60 X-20.0 I1 J-1 K2;
N70 Y-20.0 I1 J1 K2;
N80 X20.0 I-1 J1 K2;
…
G40;
G49;
M30;
```

In N20 to N80 in program O0010 to the left, the sides of a quadrangular pyramid trapezoid as shown below can be machined by the side of the tool using tool posture control. The following figure illustrates tool movement by N50. The command in N30 is the three-dimensional cutter compensation command.

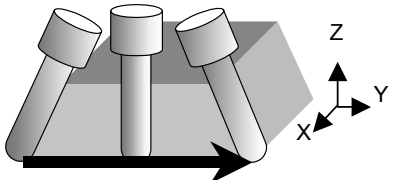


Fig. 22.2 (b) Example of tool posture control

Explanation

In tool posture control, the two rotation axes are controlled so that the momentary tool posture (tool length compensation vector) satisfies the following conditions (Fig. 22.2 (c)).

The momentary tool posture (tool length compensation vector) is present on the plane configured by the tool length compensation vector at the block start point and that at the block end point.

The angle which the tool length compensation vector at the block start point makes with the momentary tool length compensation vector is the angle obtained by proportionally dividing the angle which the tool length compensation vector at the block start point makes with the tool length compensation vector at the block end point.

Tool posture control is enabled when P1 is specified or disabled when P0 is specified in the tool center point control start command (G43.4/G43.5) blocks. Bit 0 (TPC) of setting parameter No. 19604 can be used to select whether tool posture control is enabled or disabled when the address P command is not present. The setting of tool posture control selected in the G43.4 and G43.5 blocks is kept until the tool center point control cancellation command (G49) is specified.

This function can be used for positioning (G00) and linear interpolation (G01) during tool center point control.

Other notes and restrictions are the same as those for tool center point control.

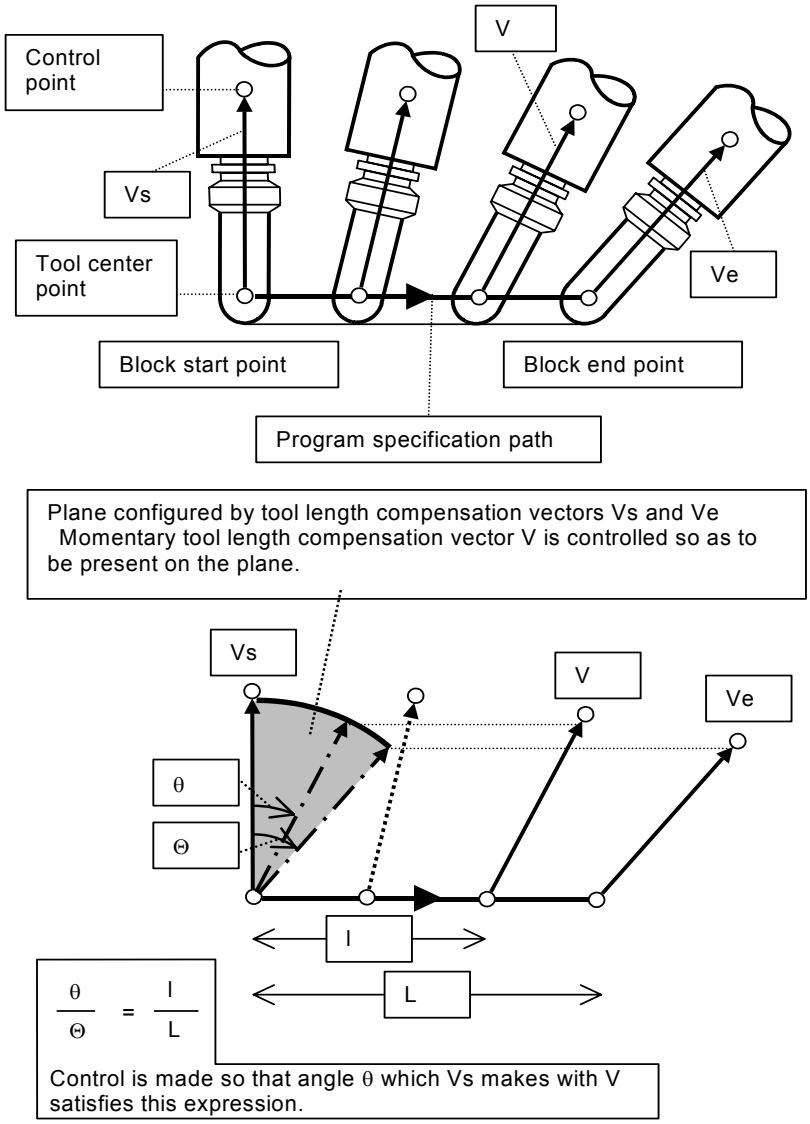


Fig. 22.2 (c) Tool posture control

Singular point

- "Tool-side rotary axis" and "workpiece-side rotary axis"

When tool center point control type 2 is specified, programming needs to be performed considering a singular point and singular point posture. The "tool-side rotary axis" and "workpiece-side rotary axis" used in the descriptions of a singular point and singular point posture later are explained below.

On a 5-axis machine that has two rotary axes, one rotary axis has a function for tilting the tool toward a workpiece. This rotary axis is referred to as a "tool-side rotary axis".

The other rotary axis is referred to as a "workpiece-side rotary axis".

The table below indicates the axes that represent these rotary axes for each mechanical unit type.

Table 22.2 (a) "Tool-side rotary axis" and "workpiece-side rotary axis"

Mechanical unit type (parameter No. 19680)	Tool-side rotary axis	Workpiece-side rotary axis
Tool rotation type (2)	Slave axis	Master axis
Table rotation type (12)	Master axis	Slave axis
Composite type (21)	Tool rotary axis	Table rotary axis

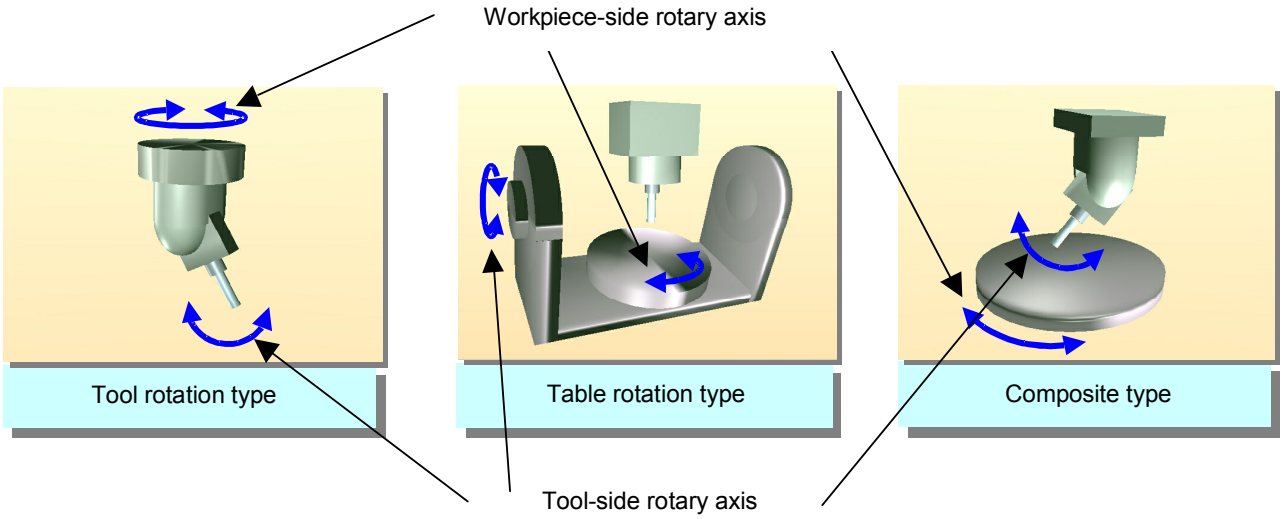


Fig. 22.2 (d) "Tool-side rotary axis" and "workpiece-side rotary axis"

- Singular point, singular point posture

A tool posture is uniquely determined when the angular displacements of the two rotary axes are determined. However, a combination of the angular displacements of the two rotary axes for determining a particular tool posture is not uniquely determined usually.

In particular, a tool posture with the angular displacement of the workpiece-side rotary axis being arbitrary is referred to as a "singular point posture", and the angular displacement of the tool-side rotary axis that causes the tool posture to assume a singular point posture is referred to as the "singular point (or singular point angular displacement)". When the angular displacement of the tool-side rotary axis is at a singular point, the center of the workpiece-side

rotary axis and the tool posture (tool orientation) are parallel with each other.

Example)

Suppose that a tool rotation type machine is used, the master axis is the C-axis (about the Z-axis), the slave axis is the B-axis (about the Y-axis), and the reference tool axis direction is the Z-axis direction. In this case, singular point B = ..., 0°, 180°, ..., and a singular point posture (the same direction as the Z-axis direction or the direction opposite to the Z-axis direction) results at an arbitrary C-axis angular displacement.

For example, when the B-axis angular displacement is B = 0 (singular point) as shown in Fig. 22.2 (e), the same tool posture (singular point posture) is assumed at an arbitrary C-axis angular displacement.

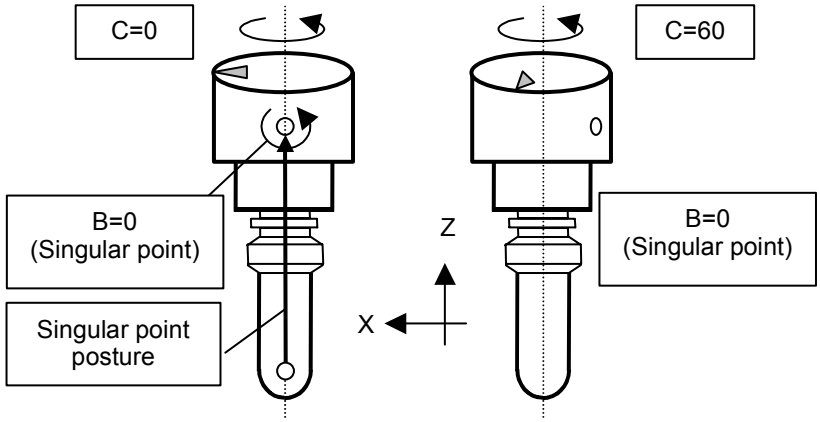
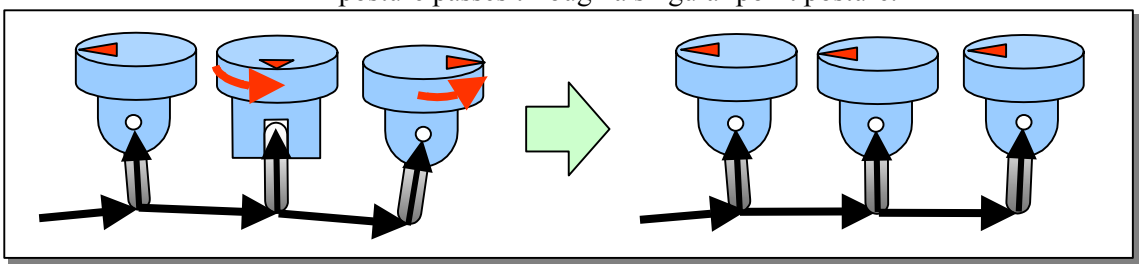


Fig. 22.2 (e) Singular point and singular point posture

There is no singular point and singular point posture on some types of machines as in the case where the reference tool axis direction is tilted (parameter No. 19698 and No. 19699) or a rotary axis is an angular axis (parameter No. 19682, No. 19683, No. 19687, and No. 19688).

- When the tool posture is close to a singular point posture

When tool posture control is exercised on a machine that has a singular point, and the tool posture becomes close to a singular point posture during execution of a block, the operation of the workpiece-side rotary axis can become so large, resulting in unstable machine operation.
In such a case, machine operation can be made stabler, the machining time can be reduced, and the surface roughness can be improved by changing the tool posture at the block end point so that the tool posture passes through a singular point posture.



A large and abrupt movement is made on the workpiece-side rotary axis when the tool posture passes through a singular point close posture.

A large and abrupt movement on the workpiece-side rotary axis can be avoided by passing through a singular point posture. By passing through a singular point posture, the tool posture at the block end is changed within the allowable range.

Fig. 22.2 (f) Changing the tool posture at a block end point

When tool posture control is used, and the angle between a tool posture and singular point posture is equal to or less than the value of (parameter No. 19738), the tool posture is determined to be close to a singular point posture. The tool posture at this time is referred to as a "singular point close posture". When tool center point control is specified (G43.5) based on type 2, which specifies the tool posture at the block end point with tool axis direction (I, J, K), each block is checked to see if a singular point close posture is assumed. If a singular point close posture is assumed in a block, the tool posture at the end point of the block is changed (the rotary axis angular displacement is changed) to pass through a singular point posture. If the end points before and after the workpiece-side rotary axis is changed vary largely from each other as the result of changing the tool posture at the block end point, the change in the tool posture at the block end point can be suppressed.
However, if the value of parameter No. 19738 is 0, the tool posture at the block end point is not changed.
If the tool posture at the start point or end point of a block is a singular point posture or the tool posture becomes a singular point posture in the middle of a block, the tool posture is not changed at the block end point.

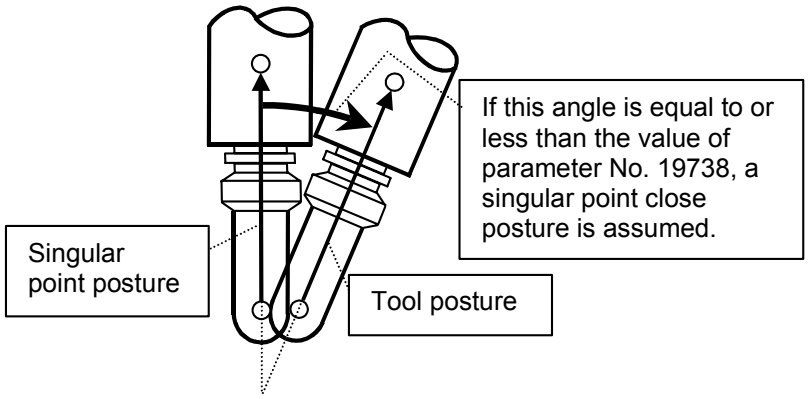


Fig. 22.2 (g) Singular point close posture

If a singular point close posture is assumed during execution of a block, change the angular displacements of the rotary axes at the end point of the block calculated from the tool posture specified by (I, J, K) so that the tool posture passes through a singular point posture in the block as indicated below.

Tool-side rotary axis:

Turned reversely relative to the singular point angular displacement

Example:

When the end point angular displacement before a change is 60°, and the singular point angular displacement is 20°, the end point angular displacement after a change is -20°.

Workpiece-side rotary axis:

Same as the angular displacement at the start point

Example:

When the end point angular displacement after a change is 30°, and the start point angular displacement is -150°, the end point angular displacement after a change is -150°.

However, if both of the difference between the end point angular displacement before a change and (end point angular displacement after a change + 180°) and the difference between the end point angular displacement before a change and (end point angular displacement after a change - 180°) are equal to or greater than the value of parameter No. 19739, the program is executed according to the setting of bit 3 (NPC) of parameter No. 19696, suppressing the change, or alarm PS5421 is issued.

This function is used to avoid unstable machine operation in a block where a singular point close posture is assumed as shown in Fig. 22.2 (f).

Notes

- When the reference tool axis or a rotary axis is inclined

When tool center point control is used, the inclination of the reference tool axis can be set in parameter No. 19698 and No. 19699, and the inclination of a rotary axis can be set in parameter No. 19682, No. 19683, No. 19687 and No. 19688.

On a machine with the reference tool axis or a rotary axis being inclined, all tool postures cannot be assumed. Some tool postures can never be assumed by setting the angular displacements of the two rotary axes in any way. If the tool posture passes through any of those impossible tool postures during program execution when tool posture control is enabled, alarm PS0432 is issued before execution of the block.

Example:

On a tool rotation type machine where the master axis is the C-axis (about the Z-axis), the slave axis is the B-axis (about the Y-axis), and the reference tool axis direction is inclined relative to the positive Z-axis direction (a non-zero value is set in parameter No. 19698 and No. 19699), no tool posture can be assumed in the positive or negative direction of the Z-axis. So, if a block specifies such a tool posture during tool posture control, alarm PS0432 is issued before execution of the block.

```
O0010
…
G43.5 H1 P1;
…
N10 X_ Y_ Z_ I1 J0 K2;
N20 X_ Y_ Z_ I-1 J0 K2;
…
G49;
M30;
```

If O0010 is executed on the machine in the example above, the tool posture assumes the positive Z-axis direction during N20, so alarm PS0432 is issued before execution of N20.

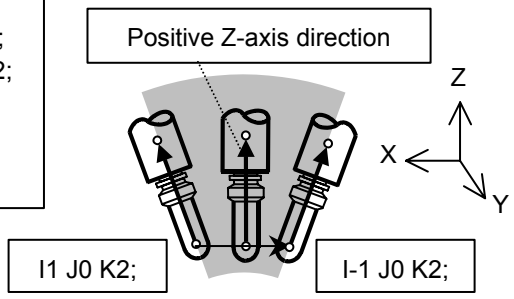


Fig. 22.2 (h) Example of impossible tool posture

- When the operation range of rotary axes is specified

When tool center point control is used, the operation range of rotary axes can be set in parameter No. 19741 to No. 19744.
If either of the rotary axes exceeds the set operation range during program execution when tool posture control is enabled, alarm DS0029 is issued.
If tool posture control is disabled, rotary axis operation range specification based on parameter No. 19741 to No. 19744 is enabled only when the tool center point control type 2 command (G43.5) is specified. If tool posture control is enabled, the check mentioned above can be made even when the tool center point control type 1 command (G43.4) is specified.
If the operation range of the first rotary axis is not specified, set 0 in both of parameter No. 19741 and No. 19742.
If the operation range of the second rotary axis is not specified, set 0 in both of parameter No. 19743 and No. 19744.

- When tool center point control type 1 is used

Tool posture control can also be exercised by using the tool center point control type 1 command (G43.4), which specifies the tool posture at the end point of a block with the angular displacements of the two rotary axes. However, when tool posture control is exercised using the tool posture at the end point of a block calculated from the rotary axis angular displacement specified by the program, alarm PS5421 is issued before execution of the block if the program-specified rotary axis angular displacement does not result.
Example:

Suppose that a tool rotation type machine where the master axis is the C-axis (about the Z-axis), the slave axis is the B-axis (about the Y-axis), the reference tool axis direction is the Z-axis direction, and the reference tool axis and rotary axes are not inclined is used. If the rotary axis angular displacements at the end point of a block are specified as (B, C) = (-90°, -90°) (with the tool posture being in the positive Y-axis direction) when the rotary axis angular displacements at the start point of the block are (B, C) = (90°, 0°) (with the tool posture being in the positive X-axis direction), tool posture control needs to maintain the tool posture in parallel with the XY plane. In this state, however, the specified end point (B, C) = (-90°, -90°) cannot be reached, so that alarm PS5421 is issued.
If (B, C) = (90°, 90°) for representing the same tool posture in the positive Y-axis direction at the end point of a block is specified, tool posture control is possible.

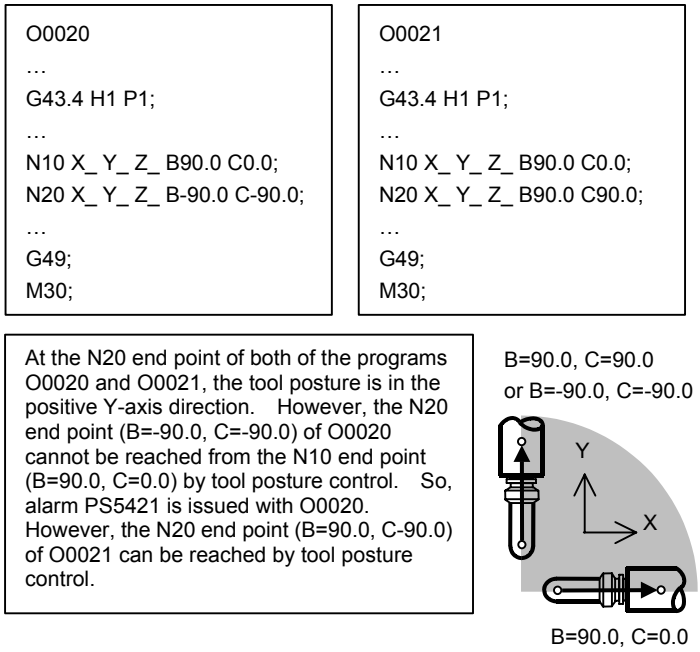


Fig. 22.2 (i) Example where tool posture control is impossible when type 1 is specified

- When the tool posture directions at the start point and end point of a block match each other

If the direction of the tool posture at the start point of a block matches the direction of the tool posture at the end point of the block (same direction or opposite direction), no plane can be determined by these tool postures. So, tool posture control in that block is temporarily disabled. However, whether to enable/disable tool posture control is kept as modal information. So, tool posture control temporarily disabled for the cause mentioned above is enabled again in the subsequent blocks.

For other notes, those described for tool center point control are applicable.

Limitation

This function is enabled for positioning (G00) and linear interpolation (G01).

For other restrictions, those described for tool center point control are applicable.

22.3 TILTED WORKING PLANE COMMAND

22.3.1 Tilted Working Plane Command

Overview

Programming for creating holes, pockets, and other figures in a datum plane tilted with respect to the workpiece would be easy if commands can be specified in a coordinate system fixed to this plane (called a feature coordinate system). This function enables commands to be specified in the feature coordinate system. The feature coordinate system is defined in the workpiece coordinate system.
For explanations about the relationship between the feature coordinate system and workpiece coordinate system, see Fig. 22.3 (a).

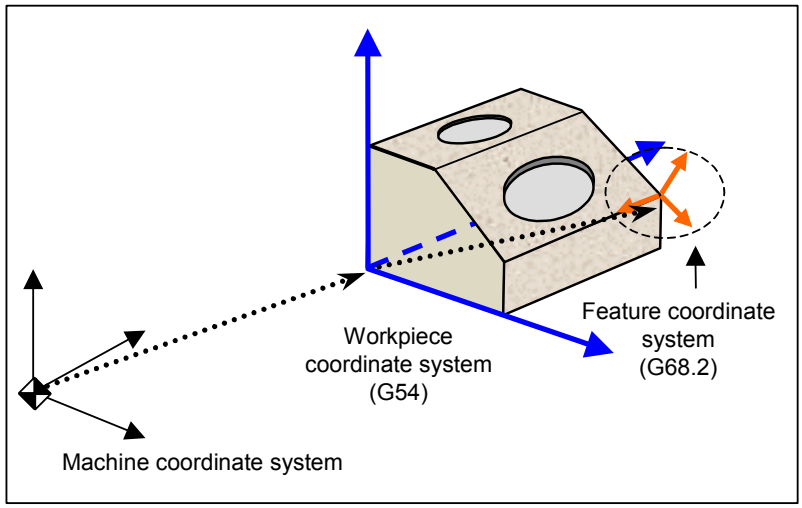


Fig. 22.3 (a) Feature coordinate system

The G68.2 command causes the programming coordinate system to switch to the feature coordinate system. The commands in all subsequent blocks are assumed to be specified in the feature coordinate system until G69 appears.
If G68.2 specifies the relationship between the feature coordinate system and the workpiece coordinate system, G53.1 automatically specifies the +Z direction of the feature coordinate system as the tool axis direction even if no angle is specified for the rotary axis. (See Fig. 22.3 (c).)
For explanations about the tool axis direction, see Fig. 22.3 (b).

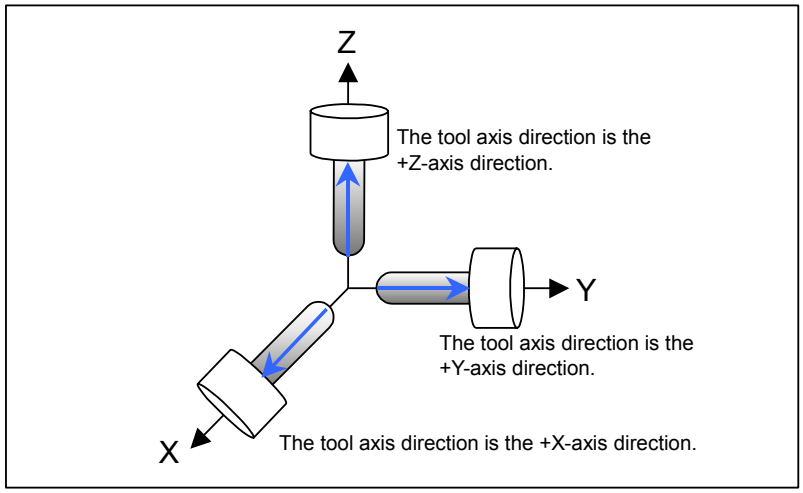


Fig. 22.3 (b) Tool axis direction

This function regards the direction normal to the machining plane as the +Z-axis direction of the feature coordinate system. After the G53.1 command, the tool is controlled so that it remains perpendicular to the machining plane.

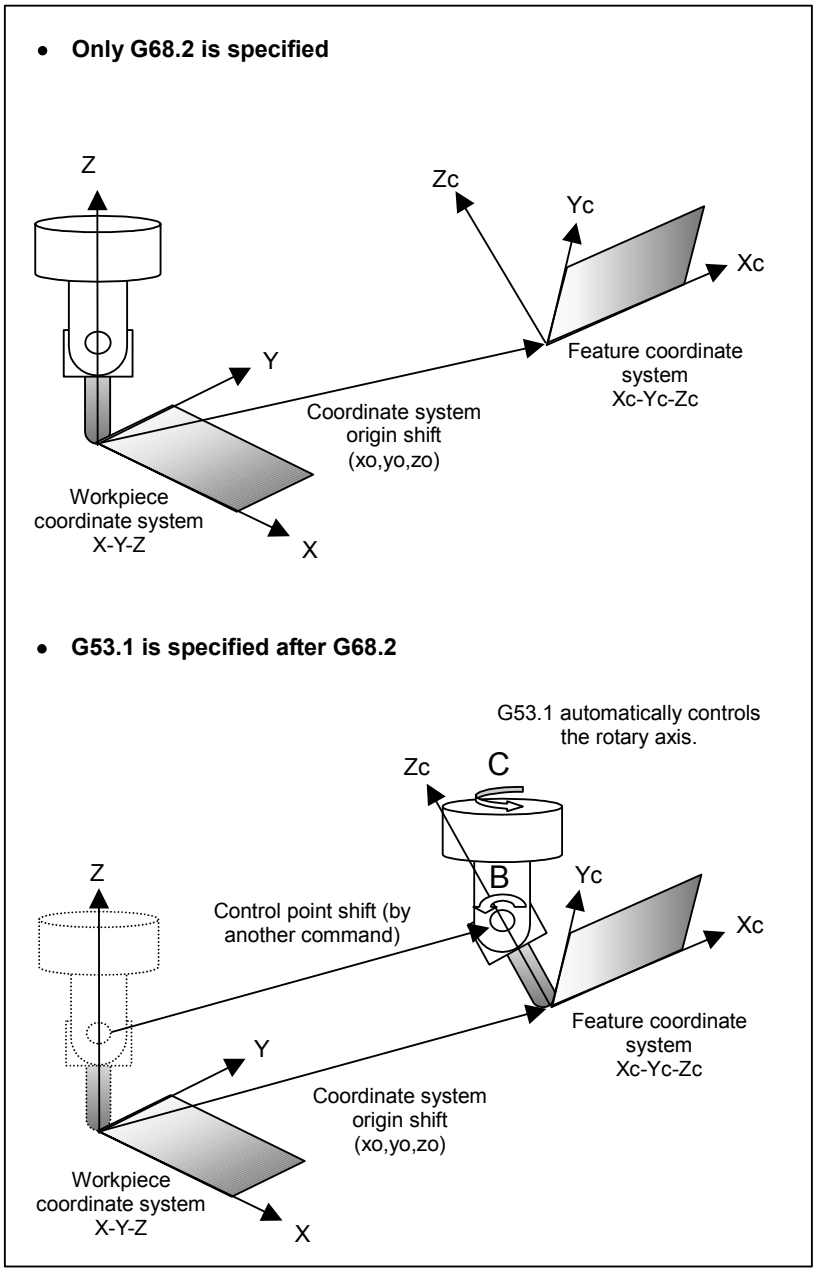


Fig. 22.3 (c) G68.2 and G53.1 commands

This function is applicable to the following machine configurations. (See Fig. 22.3 (d).)

<1> Tool rotation type machine controlled with two tool rotation axes

<2> Table rotation type machine controlled with two table rotation axes

<3> Composite type machine controlled with one tool rotation axis and one rotary axis

The function can also be used for a machine configuration in which the rotary axis for controlling the tool does not intersect the rotary axis for controlling the table.

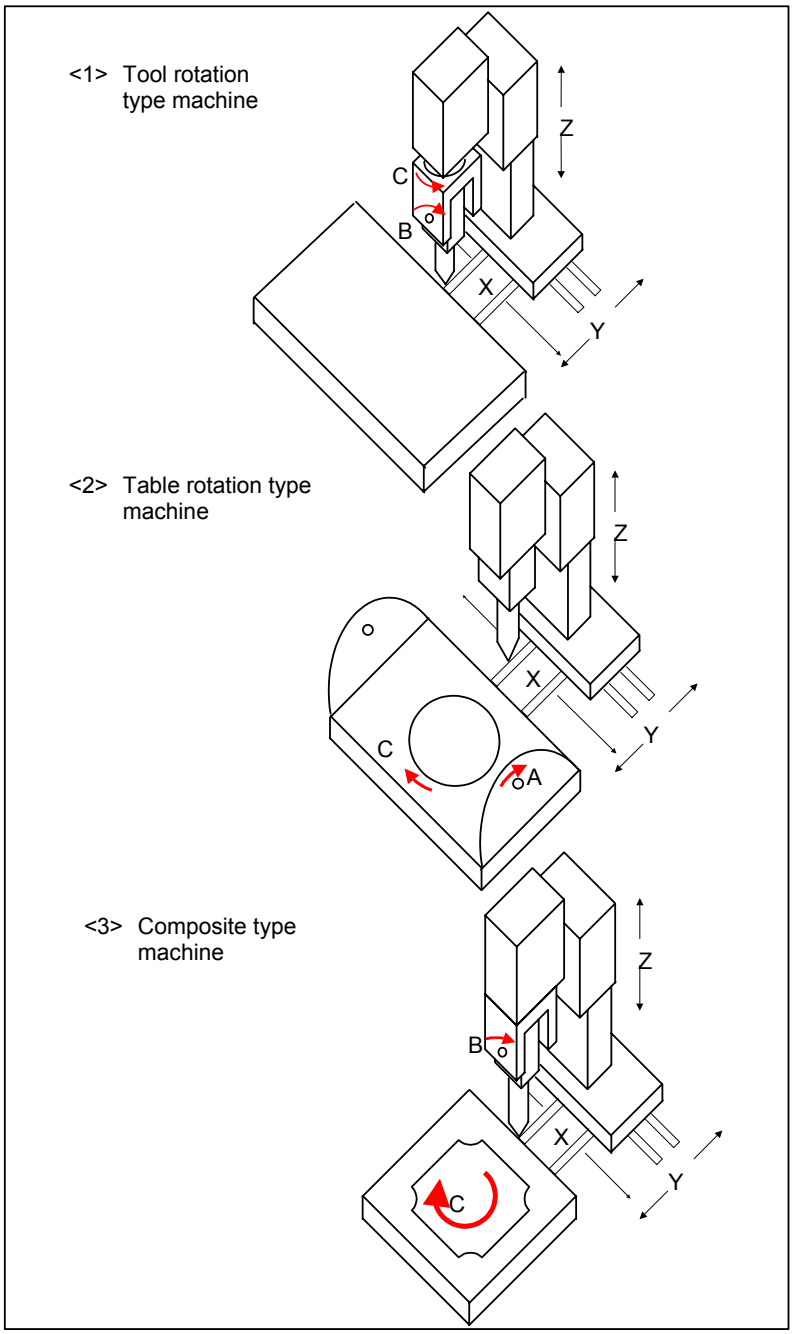


Fig. 22.3 (d) Three types of 5-axis machine

Format

- Tilted working plane command (G68.2)

M

G68.2 X x_0 Y y_0 Z z_0 Iα Jβ Kγ ;
Tilted working plane command
G69 ; Cancels the tilted working plane command.

X, Y, Z : Feature coordinate system origin
The axes specified here are the three axes of the feature coordinate system. Specify the three basic axes or parallel axes set by parameter No. 1022. When specification is omitted, the X, Y, and Z of the basic three axes are assumed to be 0.
I, J, K : Euler's angle for determining the orientation of the feature coordinate system

T

G68.2 X x_0 Y y_0 Z z_0 Iα Jβ Kγ ;
Tilted working plane command
G69.1 ; Cancels the tilted working plane command.

X, Y, Z : Feature coordinate system origin
The axes specified here are the three axes of the feature coordinate system. Specify the three basic axes or parallel axes set by parameter No. 1022. When specification is omitted, the X, Y, and Z of the basic three axes are assumed to be 0.
I, J, K : Euler's angle for determining the orientation of the feature coordinate system

- Tool axis direction control (G53.1)

G53.1 ; Controls the tool axis direction.

⚠ **CAUTION**
1 G53.1 must be specified in a block after the block that contains G68.2.
An alarm occurs if G53.1 is specified without G68.2 being specified in a preceding block.
2 G53.1 must be specified in a block in which there is no other command.
3 The rotary axis moves at the maximum rapid traverse federate in the case of rapid traverse and at the specified federate in the case of cutting feed.

Explanation

- Coordinate conversion using an Euler's angle

Coordinate conversion by rotation is assumed to be performed around the workpiece coordinate system origin.
Let the coordinate system obtained by rotating the workpiece coordinate system around the Z-axis by an angle of α degrees be coordinate system 1. Similarly, let the coordinate system obtained by rotating coordinate system 1 around the X-axis by an angle of β be coordinate system 2. The feature coordinate system is the coordinate system obtained by shifting the coordinate system that is obtained by rotating coordinate system 2 around the Z-axis through an angle of γ degrees from the workpiece coordinate system origin by (Xo, Yo, Zo). Fig. 22.3 (e) shows the relationship between the workpiece coordinate system and the feature coordinate system.
The figure also gives examples of displacement on the X-Y plane.

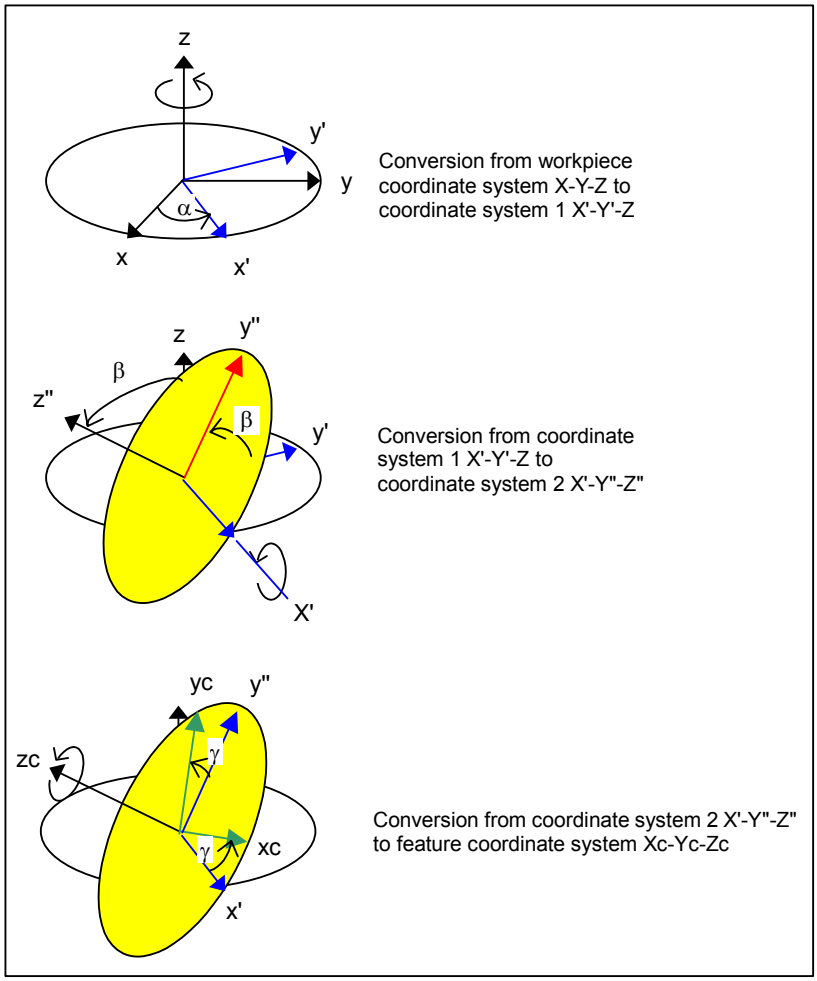


Fig. 22.3 (e) Coordinate conversion using an Euler's angle

- I0 J0 K0 command

When I0 J0 K0 is specified as an Euler's angle, the alarm (PS5457) (G68.2 format error) usually occurs. When bit 1 (ATW) of parameter No. 13451 is set to 1, the feature coordinate system with a tilted angle of 0 degree is used.

- Constant surface speed control

Constant surface speed control is exercised by using, as the reference, the machine axis specified in address P in a G96 block or the machine axis (not in the feature coordinate system but in the actually operating workpiece coordinate system) set in parameter No. 3770.

- Tool rotation type machine

The following paragraphs describe several cases of the tool rotation type machine operation.

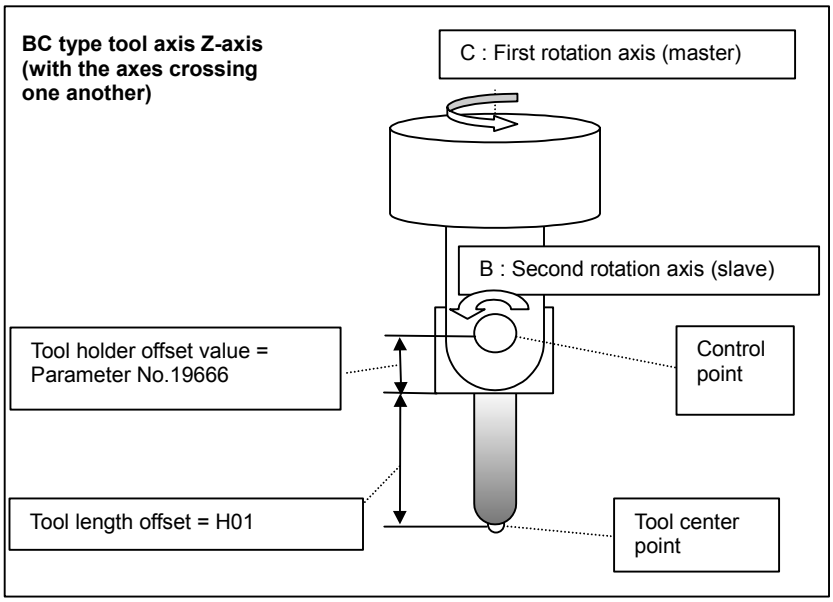
Operation description 1: When G43 (tool length compensation) is specified for a machine with its axes crossing one another

The G53.1 command, when specified after the G68.2 command, automatically controls the rotary axis in such a way that the tool axis will be oriented in the +Z direction of the feature coordinate system.

Example)

```
O100 (Sample Program1) ;
N1 G55 ;
N2 G90 G01 X0 Y0 Z30.0 F1000 ;
N3 G68.2 X100.0 Y100.0 Z50.0 I30.0 J15.0 K20.0 ;
N4 G01 X0 Y0 Z30.0 F1000 ;
N5 G53.1 ;
N6 G43 H01 X0 Y0 Z0 ;
N7 . . .
```

In this example, the "BC type tool axis Z-axis" is used as the machine configuration.
In addition, the tool axis, tool rotation axis B, and tool rotation axis C cross one another.



Block N3 : Defines a feature coordinate system in the workpiece coordinate system.
Block N4 : Shifts the control point to point Z30.0 in the feature coordinate system.
Block N5 : Exerts automatic control over the rotary axes.
Block N6 : Performs tool length compensation in the feature coordinate system.
The tool center point is shifted to the origin of the feature coordinate system.

Fig. 22.3 (f) shows the behavior of the machine when it runs sample program 1.

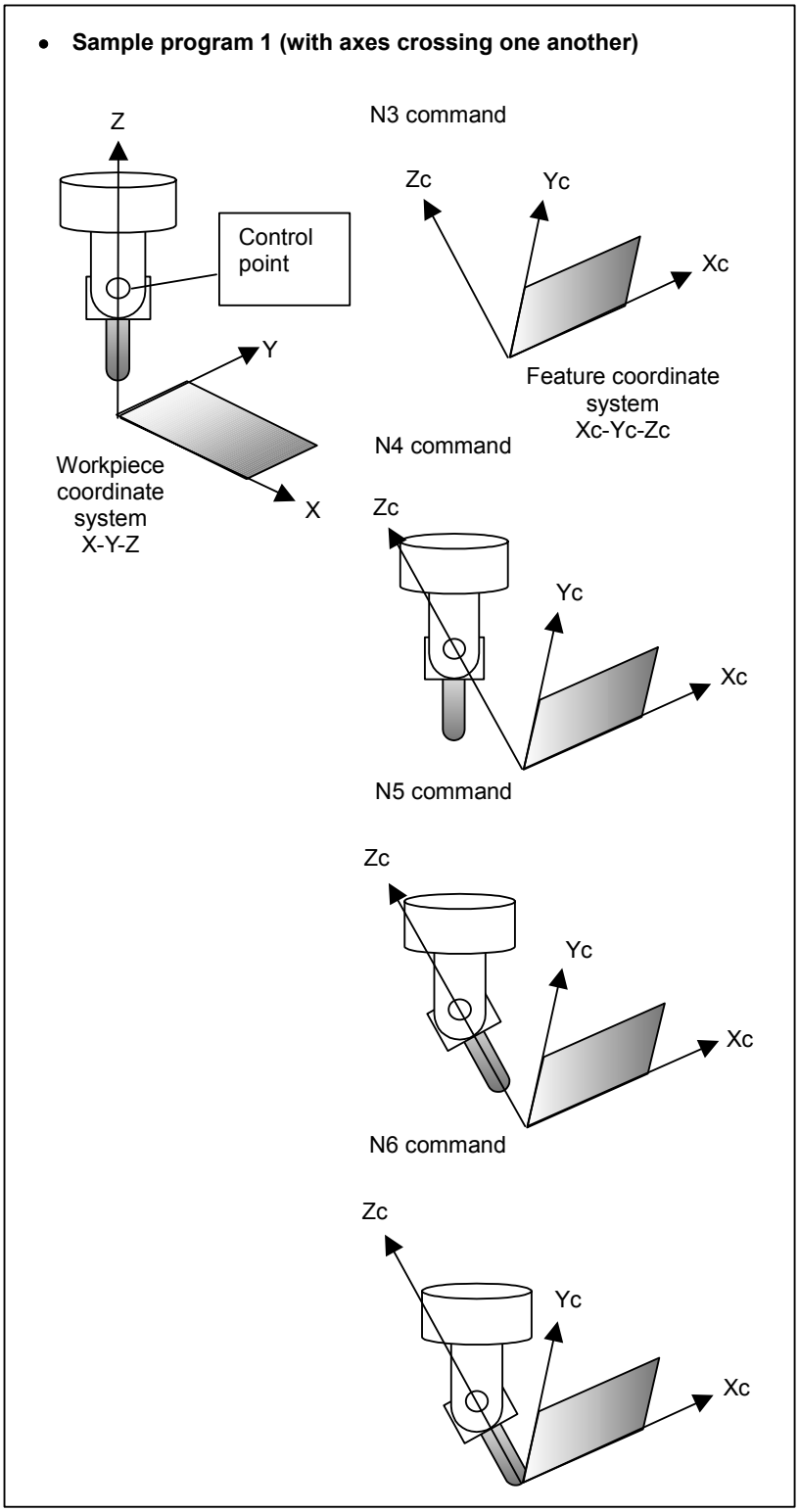
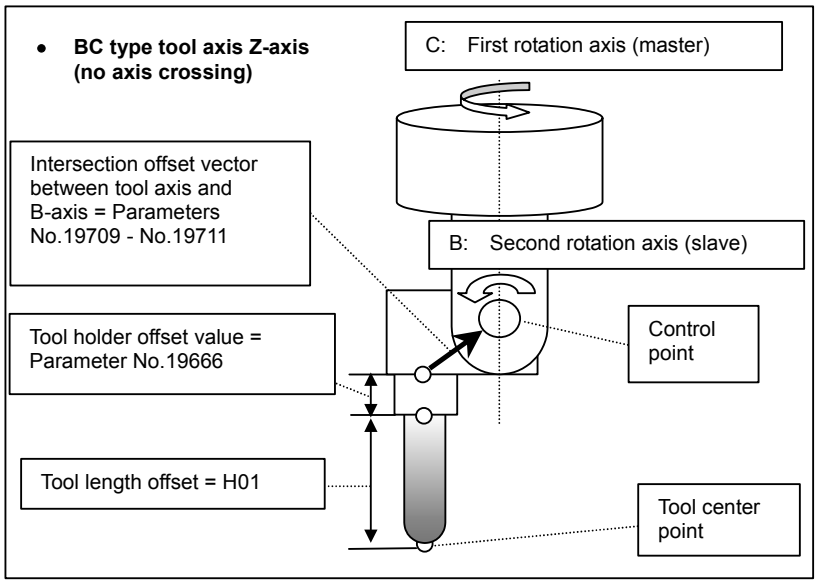


Fig. 22.3 (f) Tool axis direction control 1

Operation description 2: When G43 (tool length compensation) is specified for a machine with no axis crossing

Here is the case where no axis of the machine crosses any other axis.
It is assumed that sample program 1 is used.

In this example, the "BC type tool axis Z-axis" is used as the machine configuration.
It is assumed, however, that the tool axis does not cross the B-axis while the B-axis and C-axis cross each other.



Block N4 : Shifts the control point to point Z30.0 in the feature coordinate system.
Block N5 : Exerts automatic control over the rotary axes.
Block N6 : An intersection offset vector between the tool axis and the B-axis with automatic control for rotary axes taken into consideration is output in the feature coordinate system.
Performs tool length compensation in the feature coordinate system.
The tool center point is shifted to the origin of the feature coordinate system.

This is also true when the B-axis does not cross the C-axis.
For explanations about the offset to be applied when the B-axis does not cross the C-axis, see the descriptions about parameters No.19712, No.19713, and No.19714.

Fig. 22.3 (g) shows the behavior of the machine when it runs sample program 1.

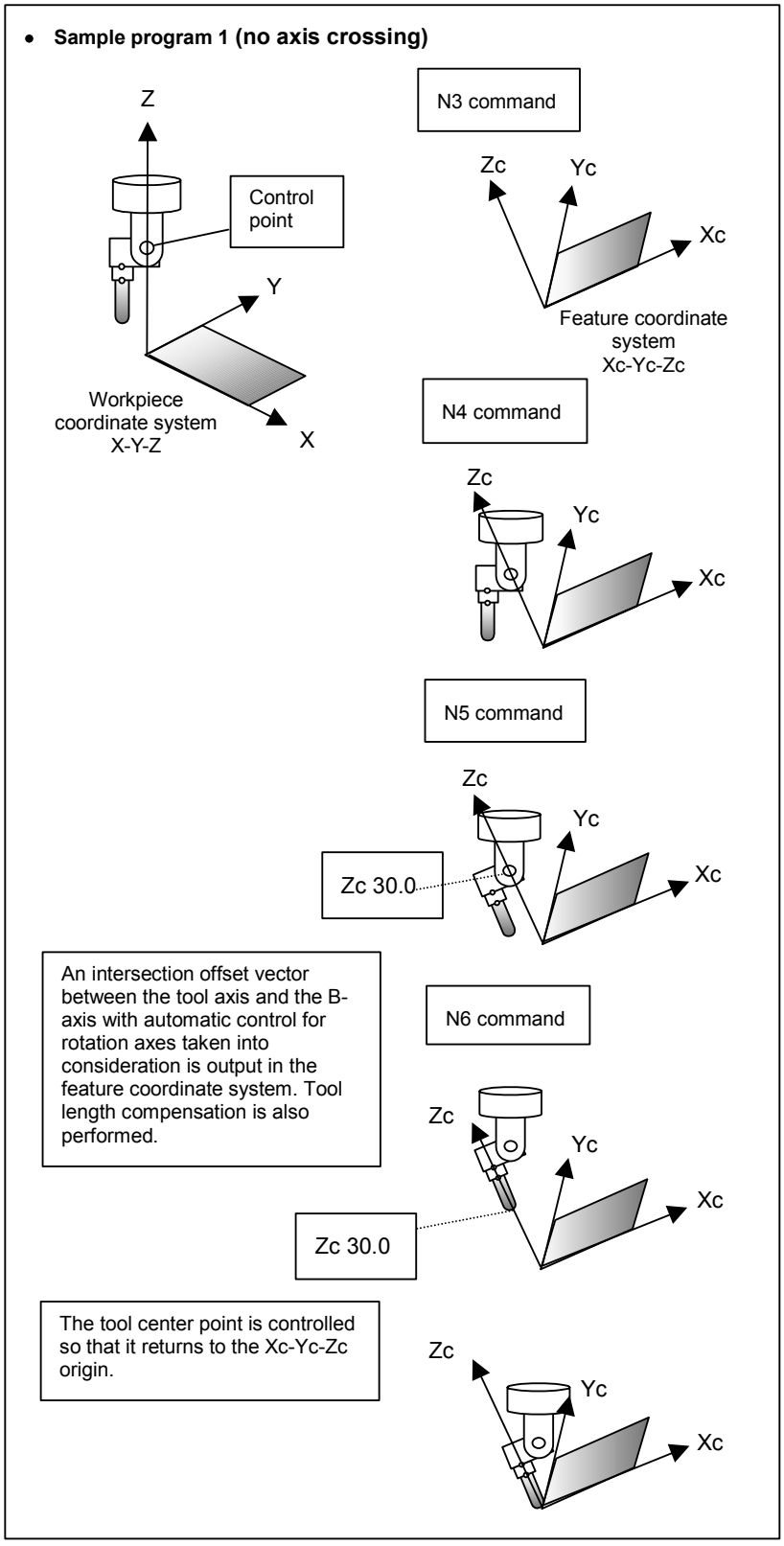


Fig. 22.3 (g) Tool axis direction control 2

Operation description 3: When no G43 (tool length compensation) command is specified or if no G53.1 (tool axis direction control) command is specified

Sample program 2 of O200 is equivalent to sample program 1 except that sample program 2 has no tool length compensation command (G43).
Example)

```
O200 (Sample Program2) ;
N1 G55 ;
N2 G90 G01 X0 Y0 Z30.0 F1000 ;
N3 G68.2 X100.0 Y100.0 Z50.0 I30.0 J15.0 K20.0 ;
N4 G01 X0 Y0 Z0 F1000 ;
N5 G53.1 ;
N6 . . . ;
```

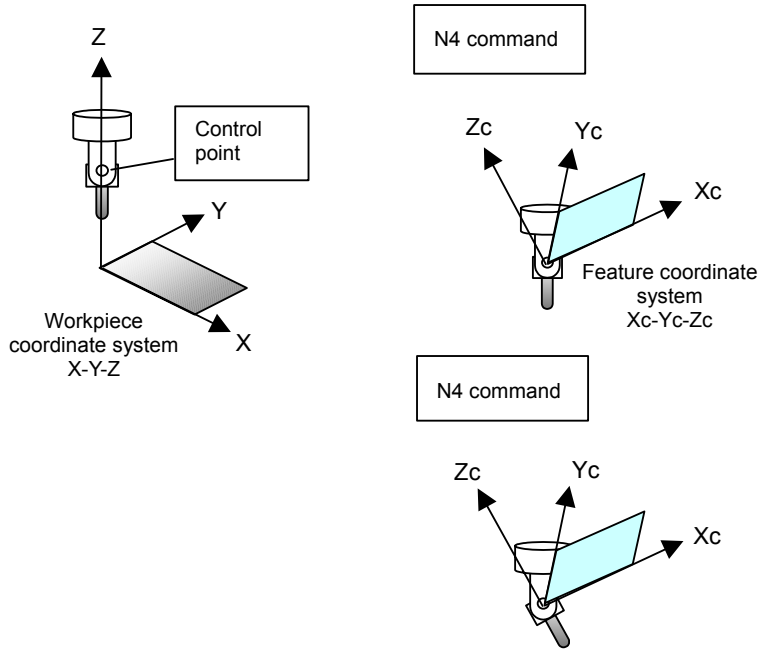
In this example, the "BC type tool axis Z-axis" is used as the machine configuration.
The case in which the axes cross one another and the case in which no axis crosses any other axis are described.
Fig. 22.3 (h) shows the behavior of the machine when it runs sample program 2.

Sample program 3 of O300 is equivalent to sample program 1 except that sample program 3 has no tool axis direction control command (G53.1).
Example)

```
O300 (Sample Program3) ;
N1 G55 ;
N2 G90 G01 X0 Y0 Z30.0 F1000 ;
N3 G68.2 X100.0 Y100.0 Z50.0 I30.0 J15.0 K20.0 ;
N4 G01 X0 Y0 Z0 F1000 ;
N5 G43 H01 ;
N6 . . . ;
```

In this example, the "BC type tool axis Z-axis" is used as the machine configuration.
The case in which the axes cross one another and the case in which no axis crosses any other axis are described.
Tool length compensation is applied in the +Z-axis direction of the feature coordinate system.
Fig. 22.3 (i) shows the behavior of the machine when it runs sample program 3.

- **Sample program 2 (with axes crossing one another)**



- **Sample program 2 (no axis crossing)**

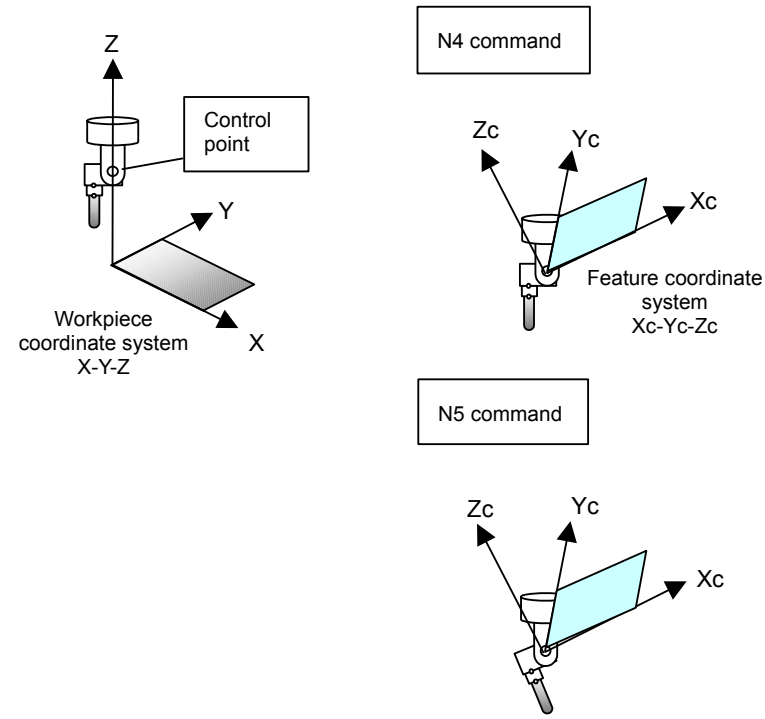
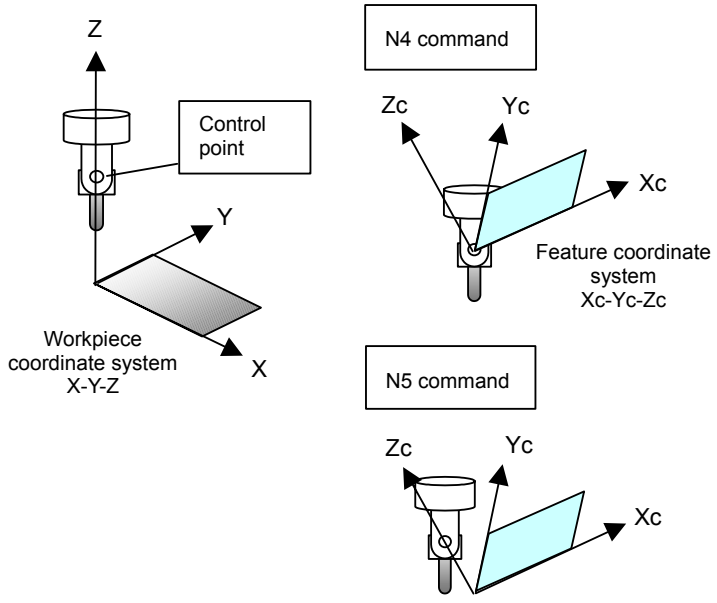


Fig. 22.3 (h) When the tool length compensation command is not specified

- **Sample program 3 (with axes crossing one another)**



- **Sample program 3 (no axis crossing)**

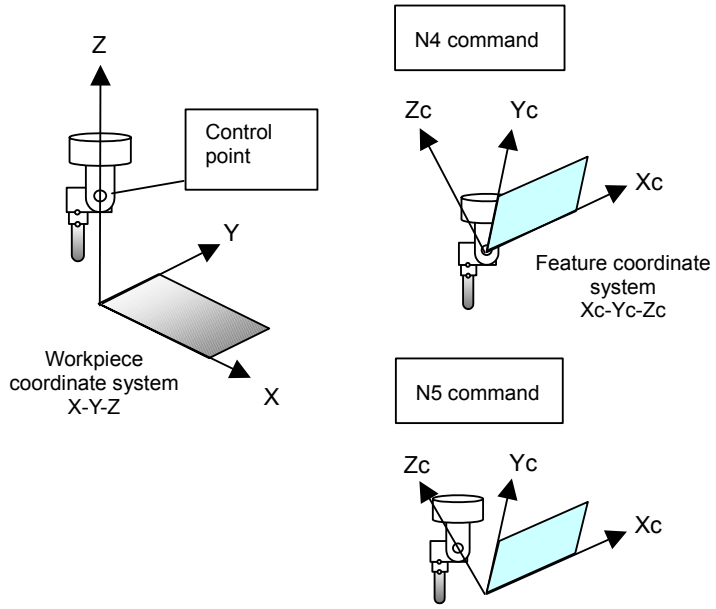


Fig. 22.3 (i) When the tool axis direction control command is not specified

- Composite type machine Basic operation

This function is also available for a composite type machine in which the tool head rotates on the tool rotation axis and the table rotates on the table rotation axis.
The feature coordinate system Xc-Yc-Zc is set in the workpiece coordinate system based on the coordinate system origin shift (xo, yo, zo) and the Euler's angle.
Given the A-axis and B-axis shown in Fig. 22.3 (j), control is performed in such a way that the A-axis rotates until Zc comes in the X-Z plane and the B-axis is controlled so that the tool axis is oriented toward the +Z-axis direction of the feature coordinate system.

- **Tool axis direction control for a composite type machine**

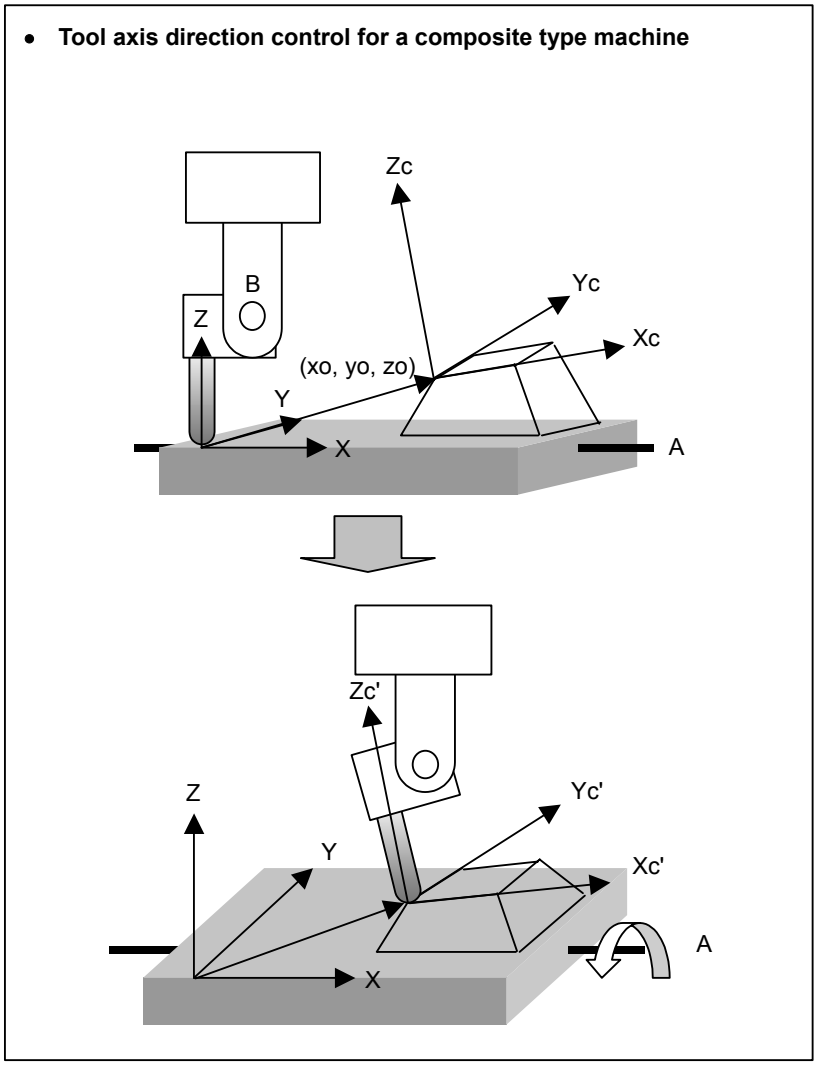


Fig. 22.3 (j) Composite type machine

- Feature coordinate system with the table rotated by G53.1 (tool axis direction control)

The composite type machine shown in Fig. 22.3 (j) is explained as an example.
If the table rotates by the tool axis direction control command (G53.1), the feature coordinate system (called the first feature coordinate system), which is set in the workpiece coordinate system by the tilted working plane command (G68.2), rotates as much as the table rotates.
The feature coordinate system that has rotated is called the second feature coordinate system.
Once G53.1 is specified, the subsequent machining commands are assumed to be specified in the second feature coordinate system. (See Fig. 22.3 (k).)
In the composite type machine, the specified feature coordinate system (the first feature coordinate system) may differ from the feature coordinate system to be used for machining (the second feature coordinate system).

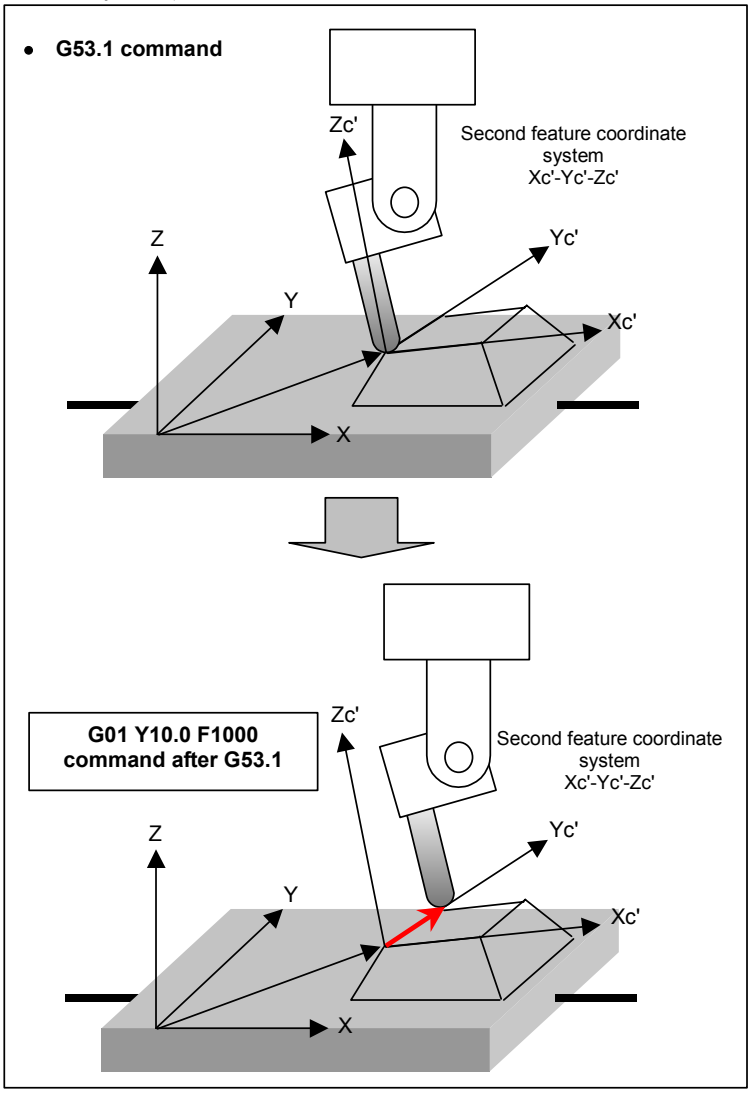


Fig. 22.3 (k) Resetting of the feature coordinate system

- Rotation direction of the table rotation axis

The composite type machine shown in Fig. 22.3 (j) is explained as an example.

Set parameter No.19684 to 1 if the rotation direction of the rotation table corresponding to the positive-direction move command is clockwise when viewed from the positive direction of the rotation center axis on which the table rotation axis rotates. If the rotation direction is counterclockwise, set parameter No.19684 to 0.

Let's take sample program 4 of O400 as an example, where the movement of the table is specified by G53.1.

If parameter No.19684 is set to 1, control is performed in such a way that the table is rotated to A-45.0.

If parameter No.19684 is set to 0, control is performed in such a way that the table is rotated to A45.0.

Example)

```
O400 (Sample Program4) ;
N1 G68.2 X100.0 Y100.0 Z0 I180.0 J45.0 K0 ;
N2 G53.1 ;
N3 . . . ;
```

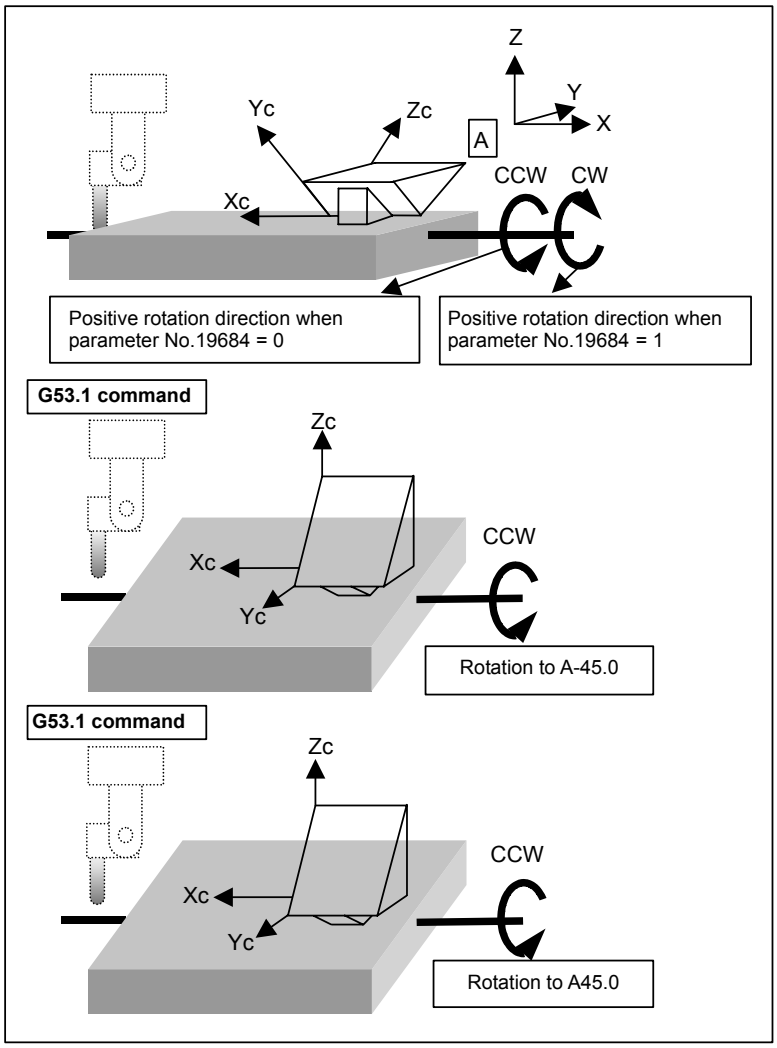


Fig. 22.3 (l) Rotation direction of the table rotation axis

- Table rotation type machine Basic operation

This function is also usable for a table rotation type machine with two table rotation axes.
The feature coordinate system Xc-Yc-Zc is set in the workpiece coordinate system based on the coordinate system origin shift (xo, yo, zo) and the Euler's angle.
Given the A-axis and C-axis shown in Fig. 22.3 (m), the A-axis and C-axis rotate until Zc comes in the X-Z plane and the tool axis is directed toward the +Z-axis direction of the feature coordinate system.

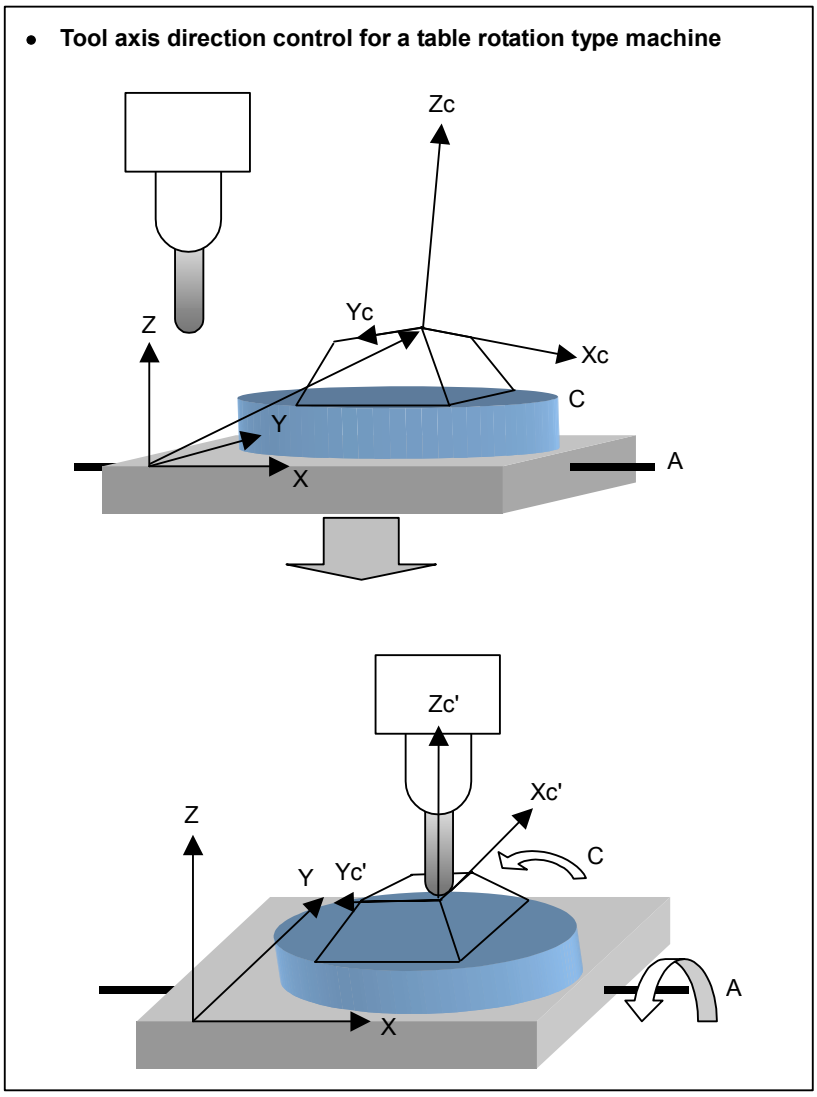


Fig. 22.3 (m) Table rotation type machine

- Feature coordinate system with the table rotated by G53.1 (tool axis direction control)

The table rotation type machine shown in Fig. 22.3 (m) is explained as an example.
If the table rotates by the tool axis direction control command (G53.1), the feature coordinate system (called the first feature coordinate system), which is set in the workpiece coordinate system by the tilted working plane command (G68.2), rotates as much as the table rotates.
The feature coordinate system that has rotated is called the second feature coordinate system.
Once G53.1 is specified, the subsequent machining commands are assumed to be specified in the second feature coordinate system. (See Fig. 22.3 (n).)
In the table rotation type machine, the specified feature coordinate system (the first feature coordinate system) may differ from the feature coordinate system to be used for machining (the second feature coordinate system).

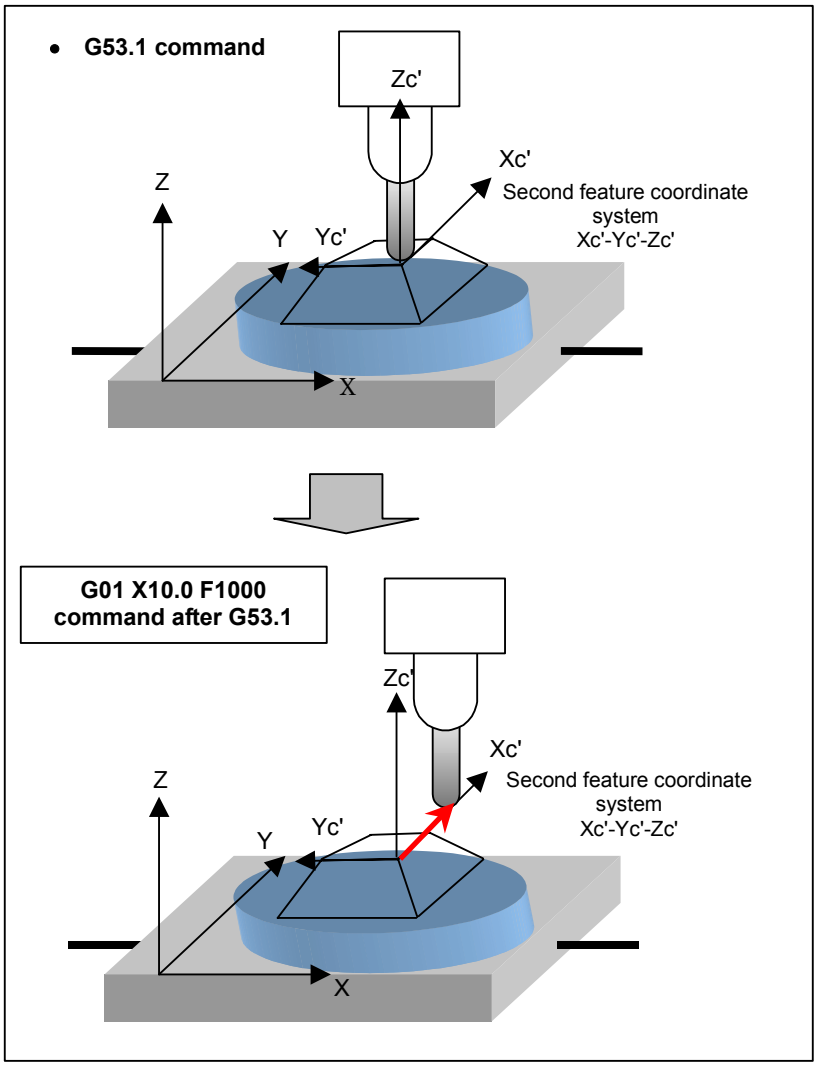


Fig. 22.3 (n) Resetting of the feature coordinate system

- Angle of the rotary axis

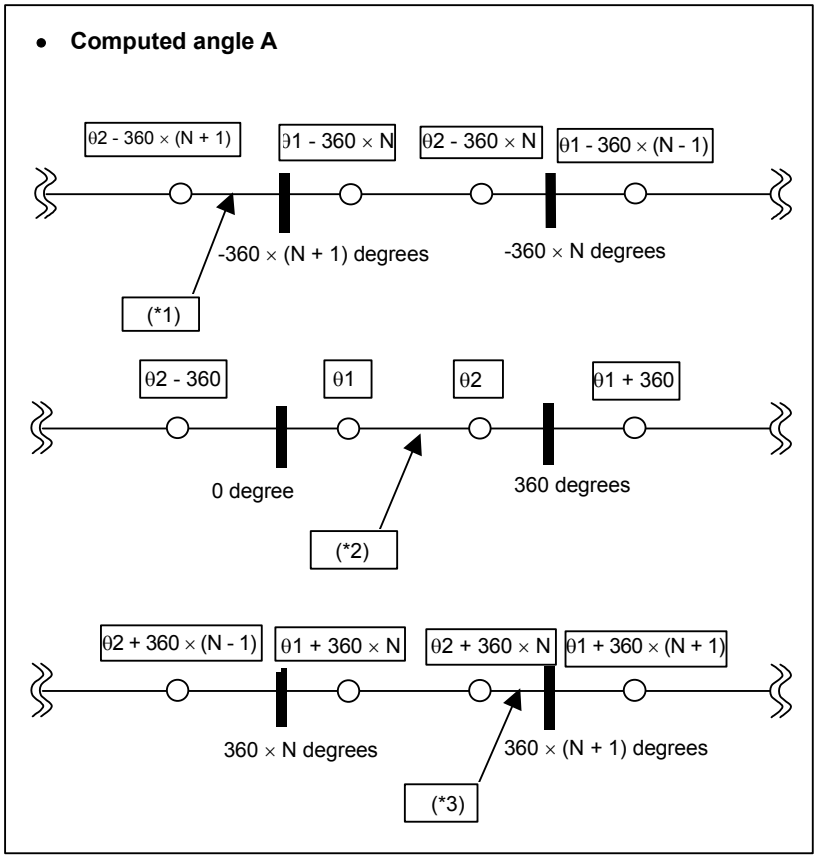
When tool axis direction control (G53.1) has been performed, more than two pairs of "computed angles" of the rotary axes usually exist.
The "computed angle" is the candidate angle at which the rotary axis is to be controlled in the tool axis direction specified by G53.1.
The "output angle" is determined from the "computed angle" based on the "output judgment conditions" described below.
When the upper and lower limits of the movement range of rotary axes are specified by parameters No. 19741 to No. 19744 at this time, a calculated angle that makes the two axes fall within the movement range is selected.

"Output judgment conditions"

Tool rotation type or table rotation type machine
<1> The "output angles" are represented by the computed rotary axis angle pair whose master axis (first rotary axis) moving angle is smaller. ↓ ↓ When the master axis moving angle is the same ↓
<2> The "output angles" are represented by the computed rotary axis angle pair whose slave axis (second rotary axis) moving angle is smaller. ↓ ↓ When the slave axis moving angle is the same ↓
<3> The "output angles" are represented by the computed rotary axis angle pair whose master axis (first rotary axis) angle is nearer to 0 degree (multiple of 360 degrees). ↓ ↓ When the master axis angle is equally near to 0 degree ↓
<4> The "output angles" are represented by the computed rotary axis angle pair whose slave axis (second rotary axis) angle is nearer to 0 degree (multiple of 360 degrees).
Composite type machine
<1> The "output angles" are represented by the computed rotary axis angle pair whose table (second rotary axis) moving angle is smaller. ↓ ↓ When the table moving angle is the same ↓
<2> The "output angles" are represented by the computed rotary axis angle pair whose tool (first rotary axis) moving angle is smaller. ↓ ↓ When the tool moving angle is the same ↓
<3> The "output angles" are represented by the computed rotary axis angle pair whose table (second rotary axis) angle is nearer to 0 degree (multiple of 360 degrees). ↓ ↓ When the master axis angle is equally near to 0 degree ↓
<4> The "output angles" are represented by the computed rotary axis angle pair whose tool (first rotary axis) angle is nearer to 0 degree (multiple of 360 degrees).

The process of judging whether the moving angle is smaller or larger as the output judgement condition is called "movement judgement."

The "movement judgement" process is explained below.
When the "computed angle" is within the range between 0 and 360 degrees, it is called the "basic computed angle."
Usually, two pairs of "basic computed angles" exist.
For example, assume that the machine has rotary axis A (master) and rotary axis B (slave) and that there are two pairs of basic computed angles as follows:
(A θ1 degree; B ϕ1 degree)
(A θ2 degrees; B ϕ2 degrees) where θ1 ≤ θ2 and ϕ1 ≤ ϕ2.
The "computed angle" is obtained from either of the following expressions: "basic computed angle" + 360 degrees × N or "basic computed angle" - 360 degrees × N.
The current position of rotary axis A (master) is PA, and that of rotary axis B (slave) is 0 degree.
Based on the PA angle, the "movement judgement" process is done as follows.



"Movement judgment"

When the PA angle is (*1):
The output angle is: (A θ2 - 360 × (N + 1) degrees; B ϕ2 degrees).
Namely, θ2 - 360 × (N + 1) degrees is adopted that is nearer to the computed angle of A, and ϕ2, which is the same group as θ2, is adopted as the computed angle of B.

When the PA angle is (*2):
The output angle is: (A θ1 degrees; B ϕ1 degrees).
Namely, θ1 degrees is adopted that is nearer to the computed angle of A, and ϕ1, which is the same group as θ1, is adopted as the computed angle of B.

When the PA angle is (*3):
The output angle is: (A θ2 + 360 × N degrees; B ϕ2 degrees).
Namely, θ2 + 360 × N degrees is adopted that is nearer to the computed angle of A, and ϕ2, which is the same group as θ2. is adopted as the computed angle of B.

When the moving angle of rotary axis A (master) is the same, a "movement judgement" is made for rotary axis B (slave) according to the "output judgment conditions."
If the "output angle" of rotary axis A is determined by the "movement judgement" for rotary axis A, the computed angle representing the "smaller moving angle" is adopted as the "output angle" of rotary axis B.
Similarly, if the "output angle" of rotary axis B is determined by the "movement judgement" for rotary axis B, the computed angle representing the "smaller moving angle" is adopted as the "output angle" of rotary axis A.

⚠ CAUTION
1 To use the rotary axis roll-over function, set parameter No. 1260 (amount of rotary axis movement per rotation) to 360 degrees.
2 A stroke limit before movement is applied to the rotary axis subject to tool axis direction control.
3 If the setting of the lower limit (parameters No. 19742 and No. 19744) is greater than that of the upper limit (parameters No. 19741 and No. 19743), alarm PS5459 is issued.
4 If there is no calculated angle that falls within the movement range because the movement range is too small, alarm PS5459 is issued.
5 When the parameters that specify the upper limit and lower limit of the movement range are set to 0, no movement range is assumed to be specified.

The "output angle" is explained below using a tool rotation type machine as an example.
This example illustrates a machine having a "BC type tool axis Z."

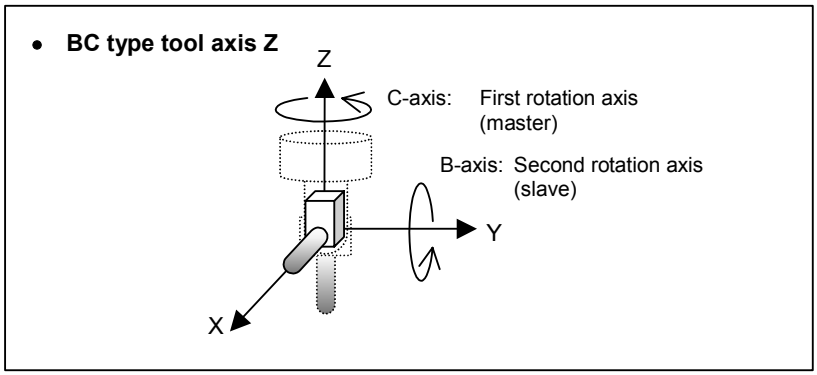


Fig. 22.3 (o) BC type tool axis Z

The following two pairs of "computed basic angles" exist that direct the tool axis toward the +X-axis direction.
(B 90 degrees; C 180 degrees)
(B 270 degrees; C 0 degree)

<1> When the current rotary axis angles are (B -70 degrees; C 30 degrees)
The "output angles" are (B -90 degrees; C 0 degree).
0 degree is adopted because it is nearer to the current position (30 degrees) of the C-axis that is the master axis. For the B-axis, 270 degrees is adopted which is the same group. However, this is changed to -90 degrees (270 degrees - 360 degrees) which is the nearest to the current position of the B-axis (-70 degrees).

<2> When the current rotary axis angles are (B 80 degrees; C 500 degrees)
The "output angles" are (B 90 degrees; C 540 degrees).
540 degrees (180 degrees + 360 degrees) is adopted because it is nearer to the current position (500 degrees) of the C-axis that is the master axis. For the B-axis, 90 degrees is adopted which is the same group.

<3> When the current rotary axis angles are (B 60 degrees; C 90 degrees)
The "output angles" are (B 90 degrees; C 180 degrees).
Since the two candidates are equally near to the current position (90 degrees) of the C-axis that is the master axis, a judgment is made based on the current position of the B-axis. 90 degrees is adopted because it is nearer to the current position (60 degrees) of the B-axis that is the slave axis. For the C-axis, 180 degrees is adopted which is the same group.

<4> When the current rotary axis angles are (B 180 degrees; C 90 degrees)
The "output angles" are (B 270 degrees; C 0 degree).
Since the two candidates are equally near to the current position (90 degrees) of the C-axis that is the master axis, a judgment is made based on the current position of the B-axis. In this case, however, the two candidates are also equally near to the current position of the B-axis (180 degrees). Therefore, the candidate is adopted in which the C-axis (master axis) is nearer to 0 degree.
That is, the pair is adopted whose C-axis angle is 0 degree and whose B-axis angle is 270 degrees.

When the slave axis angle is 0 degree, the direction of the tool axis becomes fixed regardless of the master axis angle.
In that case, the master axis does not move from the current angle.

An explanation is shown below using a machine having a "BC type tool axis Z" as an example.

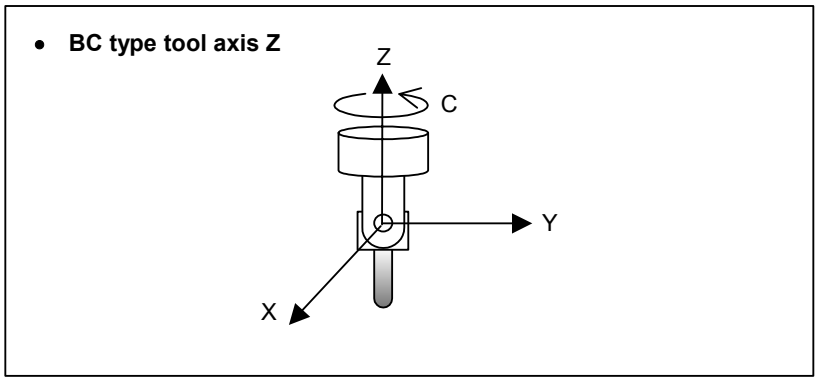


Fig. 22.3 (p) BC type tool axis Z

When the current rotary axis angles are (B 45 degrees; C 90 degrees), the "output angles" are (B 0 degree; C 90 degrees).

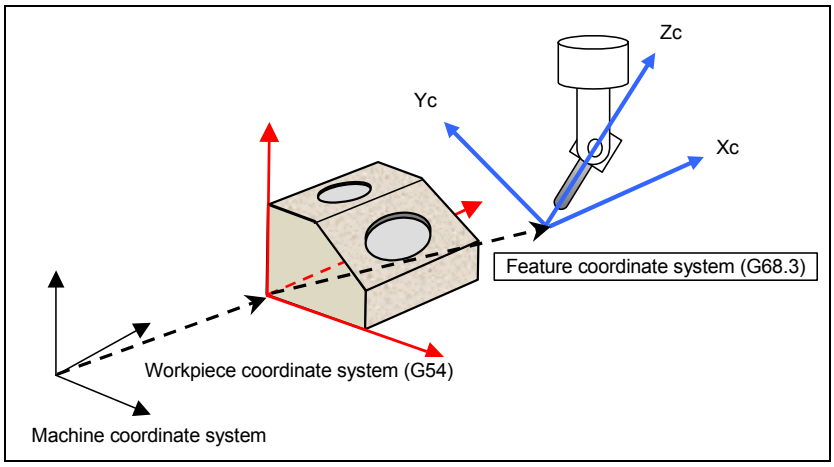
- Specification for rotary axes

The tilted working plane command assumes that positioning of a rotary axis is performed by tool axis direction control (G53.1) and then machining is performed without the rotary axis being moved.
When movement is made with the rotary axis specified directly, the movement of the rotary axis is not considered in the movement of the linear axis. Do not specify the movement of the rotary axis during execution of the tilted working plane command to ensure correct machining.

22.3.2 Tilted Working Plane Command by Tool Axis Direction

Overview

By specifying G68.3, a coordinate system (feature coordinate system) where the tool axis direction is the +Z-axis direction can be automatically specified. When a feature coordinate system is used, a program for cutting a hole or pocket in a plane tilted relative to the workpiece coordinate system can be made simpler.
This function can automatically generate a feature coordinate system that is normal to the tool direction.

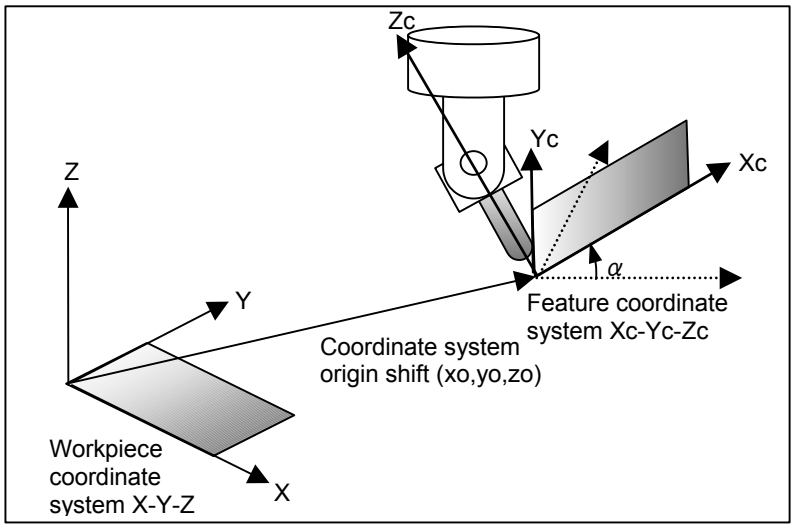


Feature coordinate system

When G68.3 is specified in a block, the coordinate system for programming is changed to a feature coordinate system. All commands after the block are regarded as commands in the feature coordinate system until G69 is specified.

Format

Format
G68.3 X x_0 Y y_0 Z z_0 Rα ; Tilted working plane command
G69 ; Cancel tilted working plane command (M series).
G69.1; Cancel tilted working plane command (T series).
Explanation of symbols
X,Y,Z : Origin of a feature coordinate system (absolute) By default, the current position is used as the origin of the feature coordinate system.
R : Angular displacement about the Z-axis in the feature coordinate system. The default is 0°.



G68.3 command

Explanation

- Feature coordinate system

By specifying G68.3, a feature coordinate system with the tool axis direction being the +Z-axis direction can be created. The tool axis direction means the tool axis direction based on the rotation axis position reached by automatic operation or manual operation.

- Origin of a feature coordinate system

The origin of a feature coordinate system must be specified using an absolute command.
Even in the incremental command mode, the specified origin is regarded as an absolute position.
When 1 address or 2 addresses are omitted in X, Y, Z, alarm PS5457 is issued.
When the origin of a feature coordinate system is not specified, the position when G68.3 is specified is used as the origin.

- Determination of a feature coordinate system

When G68.3 is specified, the tool axis direction vector (T) represents the +Z direction (Zc) of the feature coordinate system.

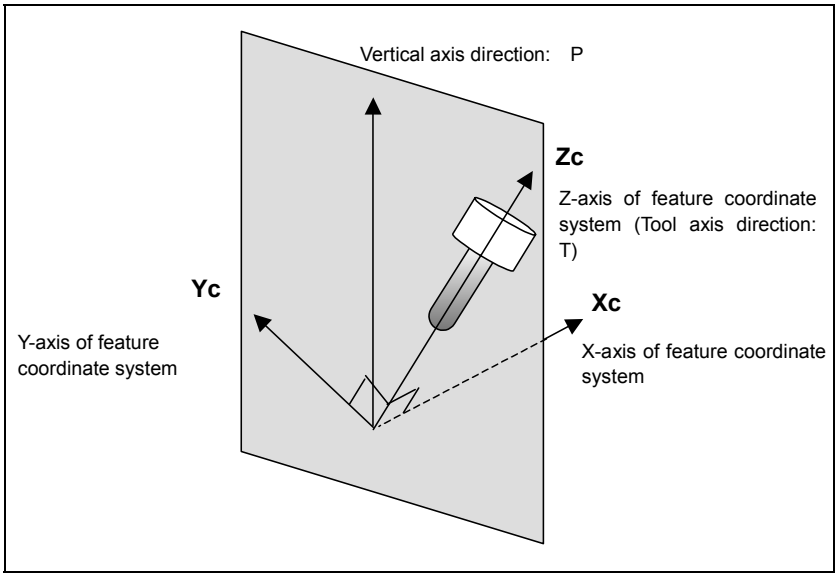
The vector normal to a plane formed by the +Z direction (Zc) of the feature coordinate system and the vertical axis direction vector (P) (parameter No. 12321) represents the +X direction (Xc) of the feature coordinate system.

Expression: $Xc = P \times Zc$

The vector normal to the +Z direction (Zc) of the feature coordinate system and the +X direction (Xc) of the feature coordinate system represents the +Y direction (Yc) of the feature coordinate system.

Expression: $Yc = Zc \times Xc$

When R is commanded, a coordinate system rotated by R around Zc from the above-mentioned coordinate system is the feature coordinate system.



Determination of a feature coordinate system

When the tool axis direction vector (T) is parallel with the vertical axis direction vector (P) (parameter No. 12321) (when the angle between the vector (T) and the vector (P) is equal to or less than the value of parameter No. 12322), the feature coordinate system Xc-Yc-Zc is as indicated below. By specifying angular displacement R, a feature coordinate system rotated about the Z-axis of this coordinate system can be specified.

Parameter No. 12321	Z-axis of feature coordinate system Zc	X-axis of feature coordinate system Xc	Y-axis of feature coordinate system Yc
1	+X direction	+Y direction	+Z direction
2	+Y direction	+Z direction	+X direction
3	+Z direction	+X direction	+Y direction

When 0 is set in parameter No. 12321, the vertical axis direction is the reference tool axis direction (parameter No. 19697).
If a value other than 0 through 3 is set in parameter No. 12321, the PS5459 alarm is issued.

> ⚠ **CAUTION**
> Tool axis direction is Z-axis direction of feature coordinate system regardless of the reference tool axis direction (parameter No. 19697).

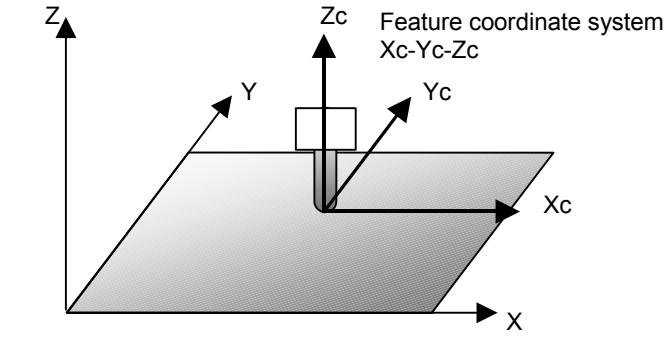
- Angular displacement R

Angular displacement R is positive when a rotation is made clockwise viewed in the Z-axis direction of the feature coordinate system. The range of angular displacement R is: $0.0° \leq R \leq 360.0°$.

- Machine of table rotation type

On a machine of table rotation type, the tool direction remains unchanged. So, a feature coordinate system based on the reference tool axis direction (parameter No. 19697) is set. However, the origin specification of the feature coordinate system and angular displacement R about the Z-axis are valid.

Example: When reference tool axis direction represents +Z-axis direction (with 3 set in parameter No. 19697)



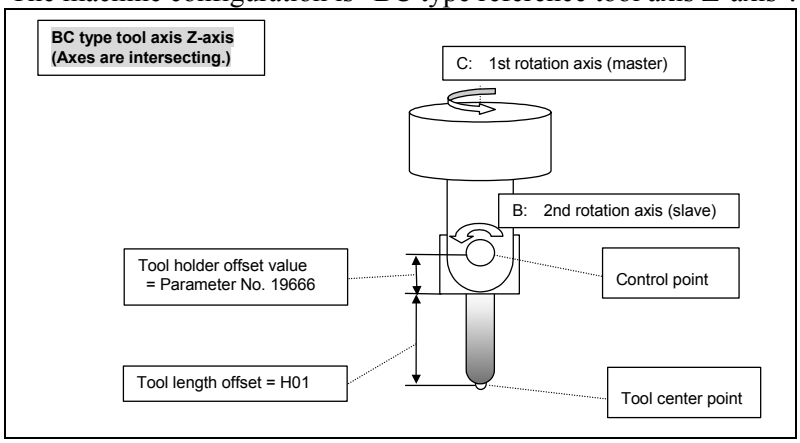
- Use in combination with tool length compensation

The G68.3 command can be specified even during tool length compensation.

> **NOTE**
> Do not specify G68.3 during workpiece setting error compensation. Otherwise, alarm PS0437 is issued.

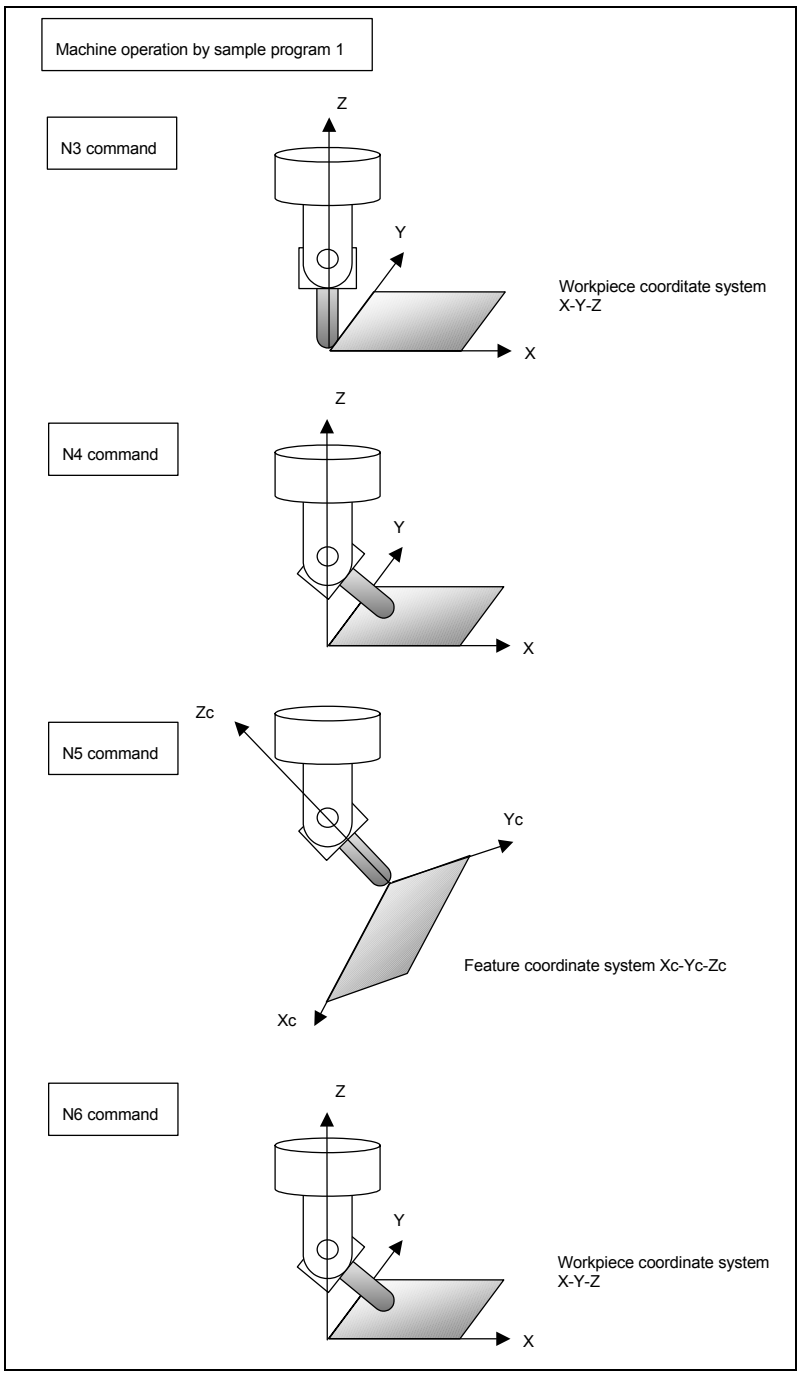
- Example of operation

An example of operation on a machine of tool rotation type is given below.
The machine configuration is "BC type reference tool axis Z-axis".



Sample program 1

```
O0100 ;
N1 G55 ;
N2 G90 G01 X0Y0Z50.0 F1000 ;
N3 G43 H01 X0 Y0 Z0 ;
N4 B-45.0 ;
N5 G68.3 ;
:
N6 G69 ;
:
```



N3 block: Performs tool length compensation in the workpiece coordinate system.
The tool center point moves to the origin of the workpiece coordinate system.

N4 block: Tilts the tool.

N5 block: Sets a feature coordinate system where the tool axis direction is the Z-axis direction and the tool center point is placed at the origin.

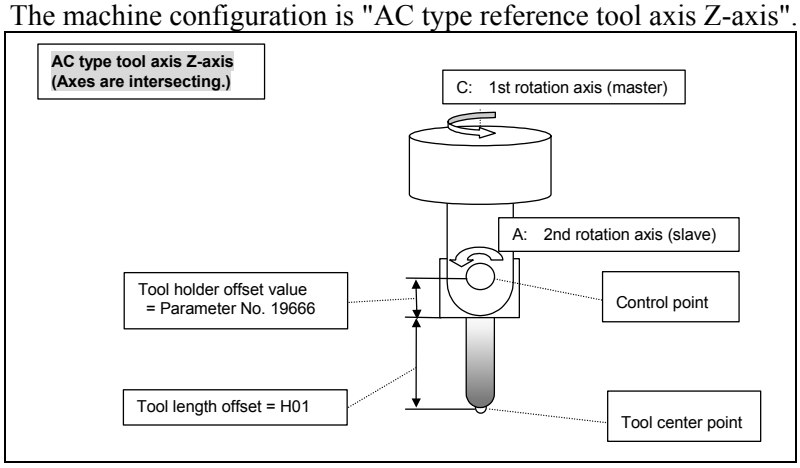
N6 block: Cancels the feature coordinate system to return to the workpiece coordinate system.

- Multiple G68.3

After the tool axis direction is changed in G68.3 mode, by specifying G68.3, a new feature coordinate system where the tool axis direction is the +Z-axis direction can be specified.

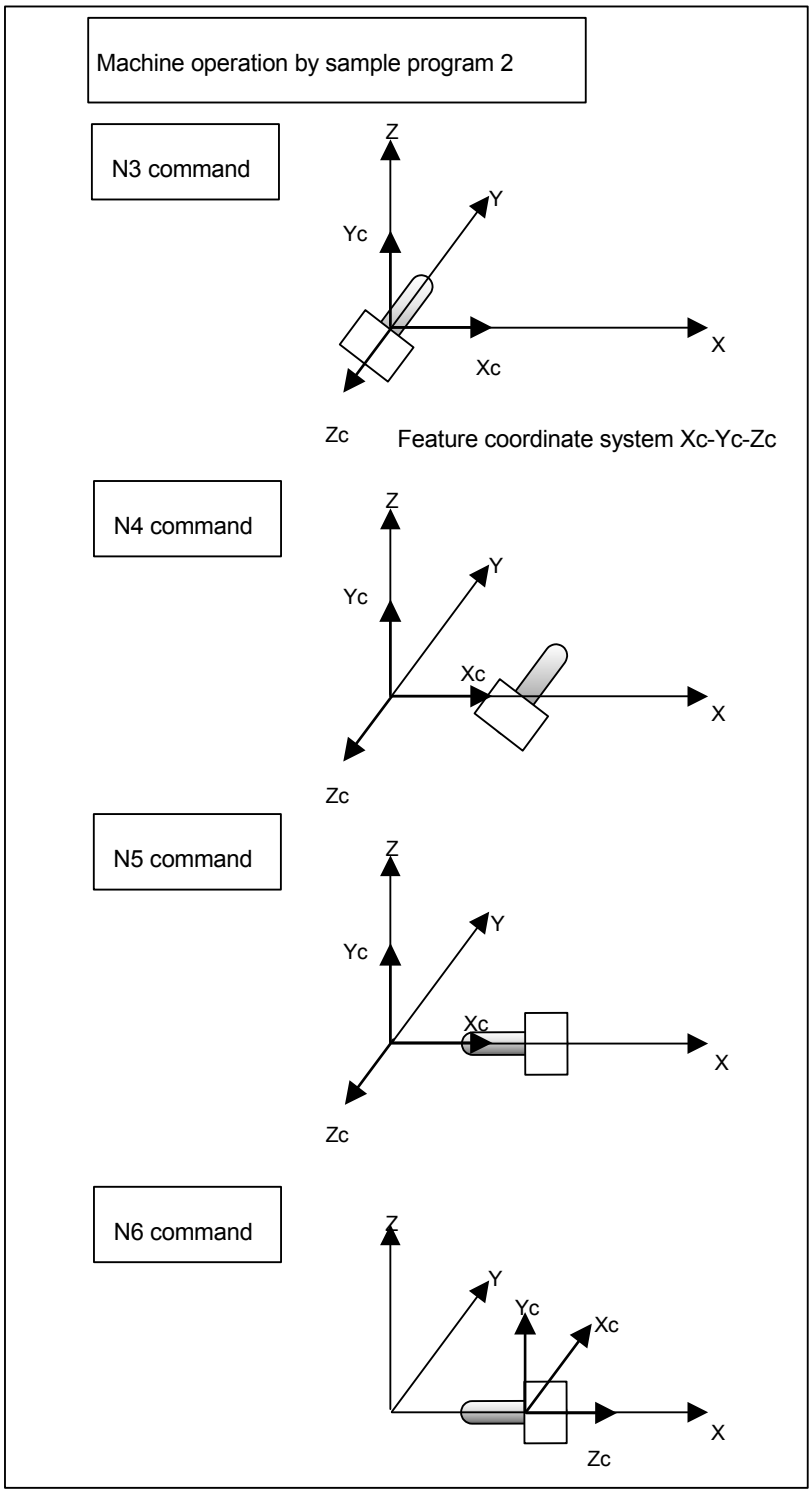
Example of operation

An example of operation on a machine of tool rotation type is given below.
The machine configuration is "AC type reference tool axis Z-axis".



Sample program 2

```
O0200 ;
N1 G55 ;
N2 G01 A90.0 F1000 ;
N3 G68.3 X0 Y0 Z0 R0;
:
N4 X10.0 Y0 Z0 ;
N5 C90.0;
N6 G68.3 X10.0 Y0 Z0 ;
:
N7 G69 ;
:
```



N3 block: Sets a feature coordinate system according to the tool direction.
N4 block: Specifies coordinates in the feature coordinate system.
N5 block: Changes the tool direction.
N6 block: Sets a feature coordinate system according to the tool direction.

22.3.3 Tilted Working Plane command with Guidance

Overview

With the conventional tilted working plane command, a tilted working plane can be specified based on Eulerian angle and tool axis direction. This function enables a tilted working plane to be specified based on the following methods:

- Tilted working plane command based on roll-pitch-yaw
- Tilted working plane command based on three points
- Tilted working plane command based on two vectors
- Tilted working plane command based on projection angles

The user can select most suitable commands for various types of machining.

With the guidance screen function, a tilted working plane command block can be created by entering tilted working plane data on the screen. A complicate tilted working plane command block can be created easily.

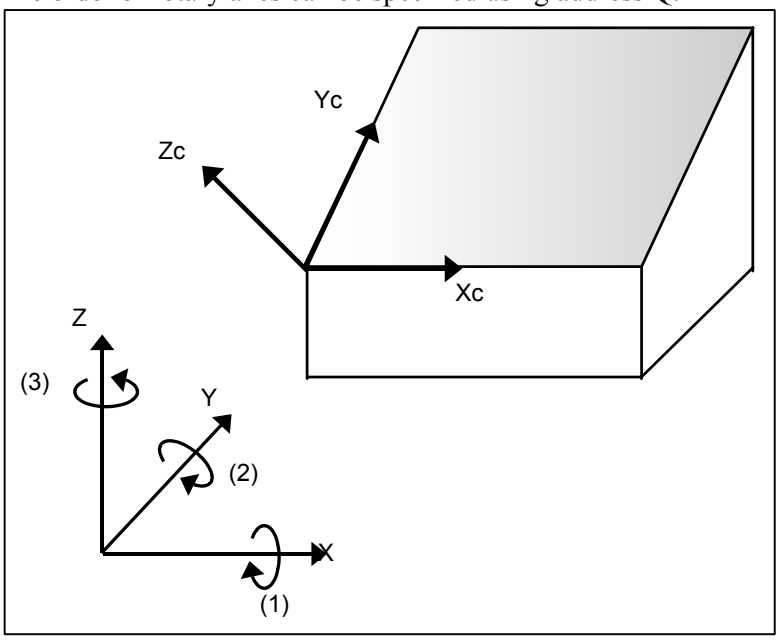
The guidance screen function can be enabled by setting bit 1 (GGD) of parameter No. 11304 to 1.

The tilted working plane command with guidance is an optional function.

22.3.3.1 Tilted working plane command based on roll-pitch-yaw

Overview

With the tilted working plane command, coordinate system conversion by rotation about the X-axis, Y-axis, and Z-axis of a workpiece coordinate system in this order can be used (roll-pitch-yaw).
The order of rotary axes can be specified using address Q.



Format

Format
G68.2 P1 Qq X_ Y_ Z_ Iα Jβ Kγ; Tilted working plane command **G69 ;** Cancel tilted working plane command (M series). **G69.1;** Cancel tilted working plane command (T series).
Explanation of symbols
Q : Order in which axes are rotated X_ Y_ Z_ : Origin of a feature coordinate system I : Angle of rotation about the X-axis (roll angle) J : Angle of rotation about the Y-axis (pitch angle) K : Angle of rotation about the Z-axis (yaw angle)

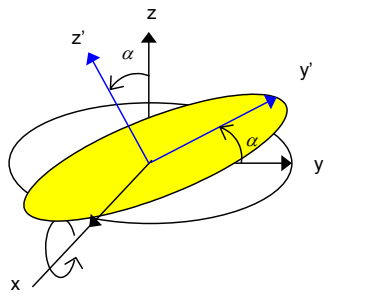
The values of address Q and the order in which axes are rotated are shown below.

	First rotation axis	Second rotation axis	Third rotation axis
Q 123	X axis	Y axis	Z axis
Q 132	X axis	Z axis	Y axis
Q 213	Y axis	X axis	Z axis
Q 231	Y axis	Z axis	X axis
Q 312	Z axis	X axis	Y axis
Q 321	Z axis	Y axis	X axis

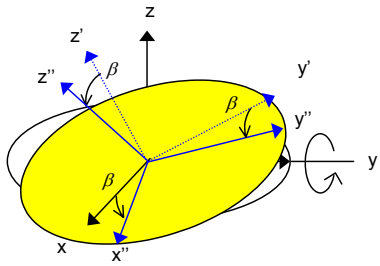
⚠ CAUTION
1 When address Q is omitted, the X-axis, Y-axis, and Z-axis are rotated in this order. (equivalent to Q123)
2 When address Q is set to a value other than the above, alarm PS5457 is issued.

Explanation

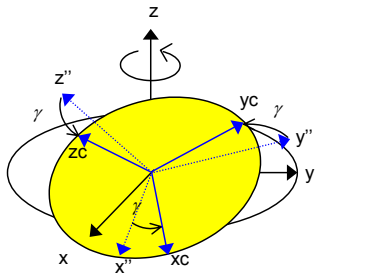
Suppose that the coordinate system is rotated about (1) the X-axis, (2) the Y-axis, and (3) the Z-axis in this order.
A "workpiece coordinate system" rotated by angle α about the X-axis is "coordinate system 1".
"Coordinate system 1" rotated by angle β about the Y-axis is "coordinate system 2". "Coordinate system 2" rotated by angle γ about the Z-axis then shifted by (X_0,Y_0,Z_0) from the workpiece coordinate system origin is a "feature coordinate system".



Conversion from the workpiece coordinate system X-Y-Z to coordinate system 1 X'-Y'-Z'



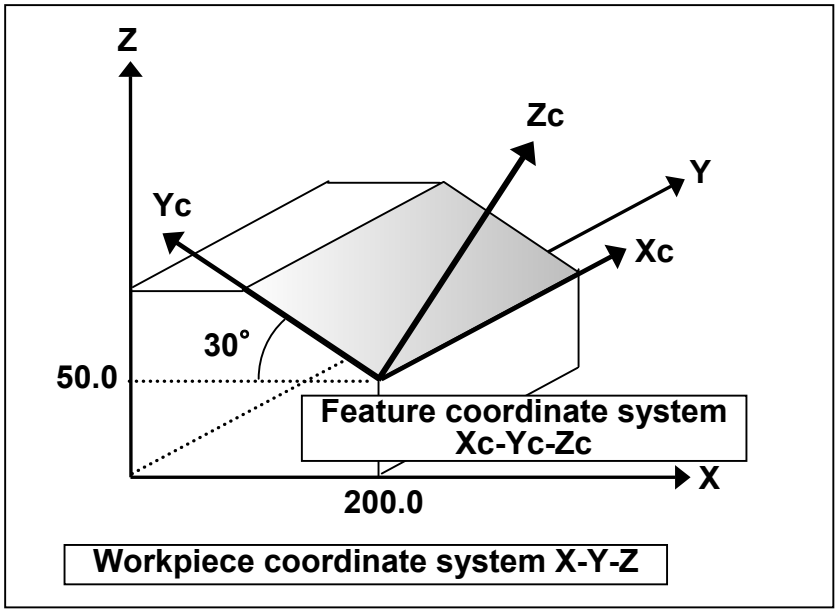
Conversion from coordinate system 1 X'-Y'-Z' to coordinate system 2 X"-Y"-Z"



Conversion from coordinate system 2 X"-Y"-Z" to the feature coordinate system Xc-Yc-Zc

Example

The example of a program when feature coordinate system like the figure below is used is shown below.



- Feature coordinate system origin : (200.0, 0, 50.0)
- Order in which axes are rotated : X axis → Y axis → Z axis
- Angle of rotation about the X-axis : 30 degrees
- Angle of rotation about the Y-axis : 0 degree
- Angle of rotation about the Z-axis : 90 degrees

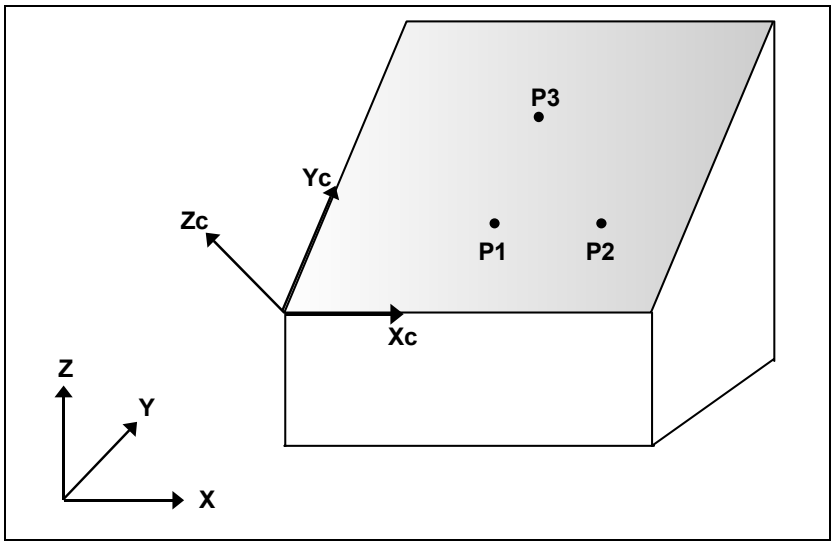
Example of a program

```
G68.2 P1 Q123 X200.0 Y0 Z50.0 I30.0 J0 K90.0 ;
G53.1 ;
  :
```

22.3.3.2 Tilted working plane command based on three points

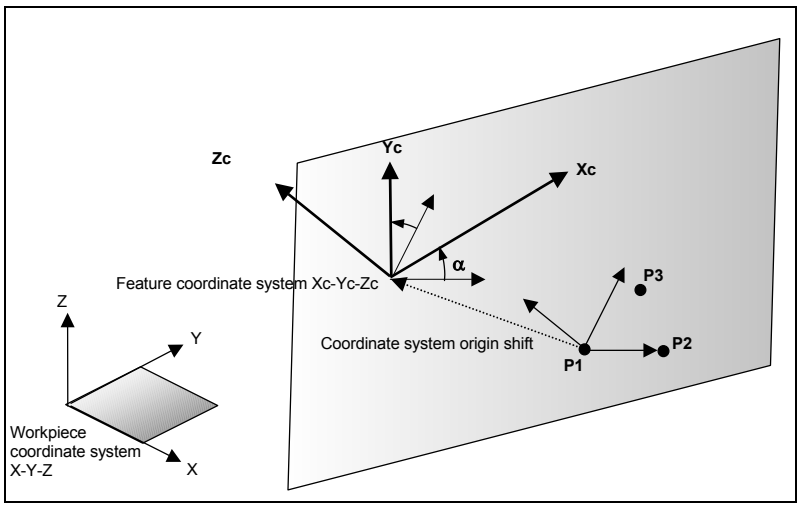
Overview

With the tilted working plane command, a tilted working plane can be specified by specifying three points in a feature coordinate system.



Format

Format
G68.2 P2 Q0 X $\underline{x_0}$ Y $\underline{y_0}$ Z $\underline{z_0}$ Rα ; **G68.2 P2 Q1 X $\underline{x_1}$ Y $\underline{y_1}$ Z $\underline{z_1}$;** **G68.2 P2 Q2 X $\underline{x_2}$ Y $\underline{y_2}$ Z $\underline{z_2}$;** **G68.2 P2 Q3 X $\underline{x_3}$ Y $\underline{y_3}$ Z $\underline{z_3}$; Tilted working plane command** **G69 ; Cancel tilted working plane command (M series).** **G69.1; Cancel tilted working plane command (T series).**
Explanation of symbols
Q0 X x_0 Y y_0 Z z_0 : Amount of shift from the first point to the origin of a feature coordinate system By default, this value is 0. Q1 X x_1 Y y_1 Z z_1 : First point. (origin of a feature coordinate system) Q2 X x_2 Y y_2 Z z_2 : Second point. Q3 X x_3 Y y_3 Z z_3 : Third point. R : Angle of rotation about the Z-axis of a feature coordinate system By default, this value is 0. Any block in the G68.2 P2 command can be specified.



⚠ CAUTION

1 Three G68.2P2 commands (Q1, Q2, and Q3) determine a tilted plane. If the G68.2P2 commands are interrupted, alarm PS5457 is issued.

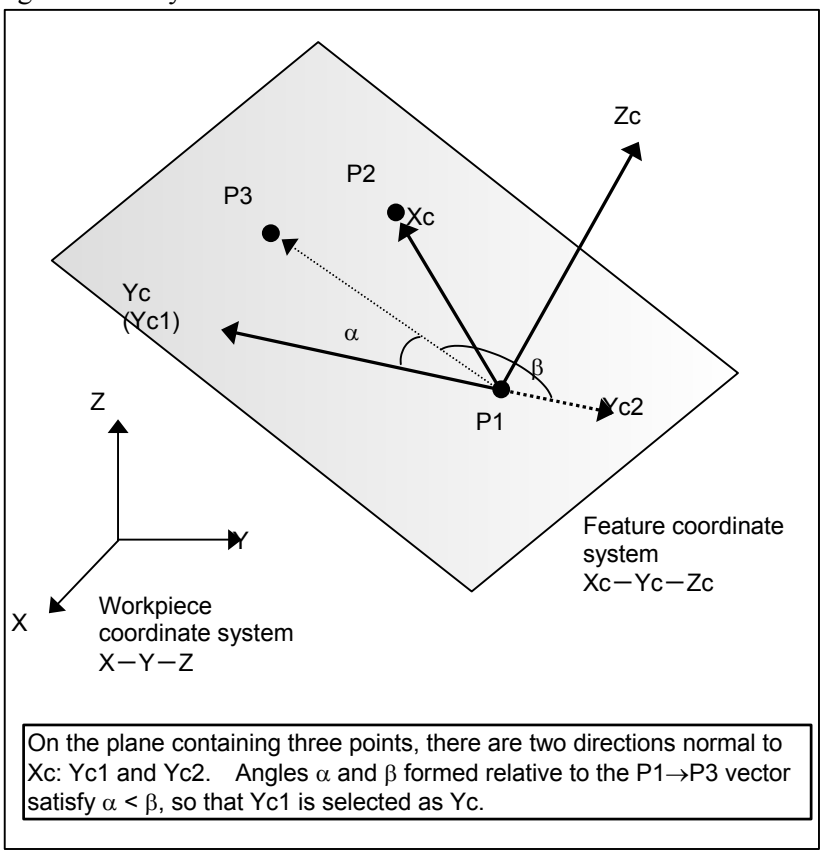
2 When any of the following conditions is met or address Q is set to a value other than the above, alarm PS5457 is issued.

- When two or more points are duplicate (the plane is not determined.)
- When three points are placed on a line (the plane is not determined.)
- When the distance between a line passing two of the three points and the remaining point is smaller than the setting of parameter No. 11220 (the plane is unstable.)

Explanation

- Determination of a feature coordinate system

Three entered points are named P1, P2, and P3 in the order of entry.
The P1-to-P2 direction is defined as the X-axis of a feature coordinate system. Among the directions that are on the plane containing the three points and are normal to the X-axis of the feature coordinate system, the direction that makes a smaller angle with the P1→P3 vector is defined as the Y-axis of the feature coordinate system. The Z-axis of the feature coordinate system is defined according to the right-handed system.



On the plane containing three points, there are two directions normal to Xc: Yc1 and Yc2. Angles α and β formed relative to the P1→P3 vector satisfy $\alpha < \beta$, so that Yc1 is selected as Yc.

- Origin of the feature coordinate system

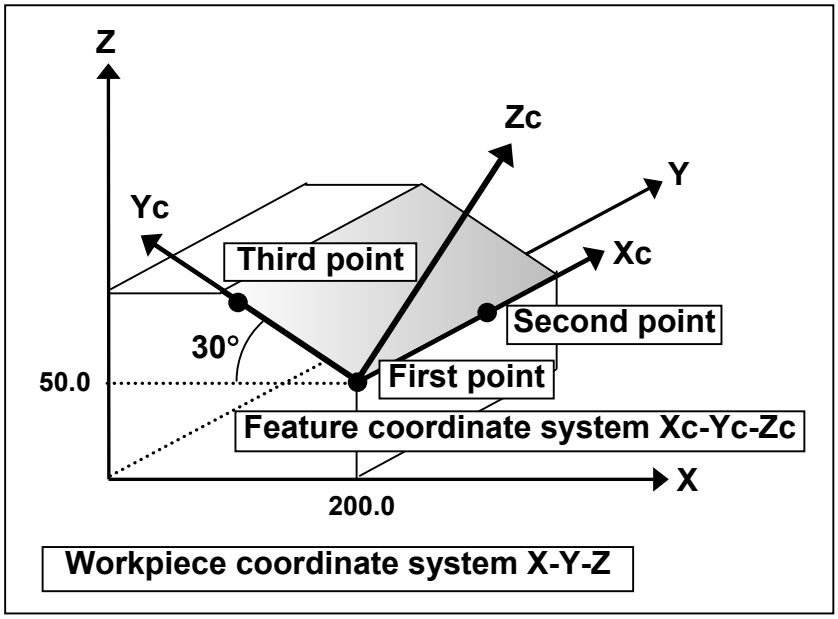
The origin of the feature coordinate system is the first specified point P1.
By setting an origin shift amount (G68.2 P2 Q0 X_Y_Z_), the origin of the feature coordinate system is shifted by (X,Y,Z) from P1. Specify (X,Y,Z) in the feature coordinate system.

- Angular displacement R

The angular displacement R is positive for clockwise rotation viewed in the Z-axis direction in the feature coordinate system.

Example

The example of a program when feature coordinate system like the figure below is used is shown below.



- First point (origin of a feature coordinate system)
  : (200.0, 0, 50.0)
- Second point : (200.0, 100.0, 50.0)
- Third point : (26.795, 0, 150.0)

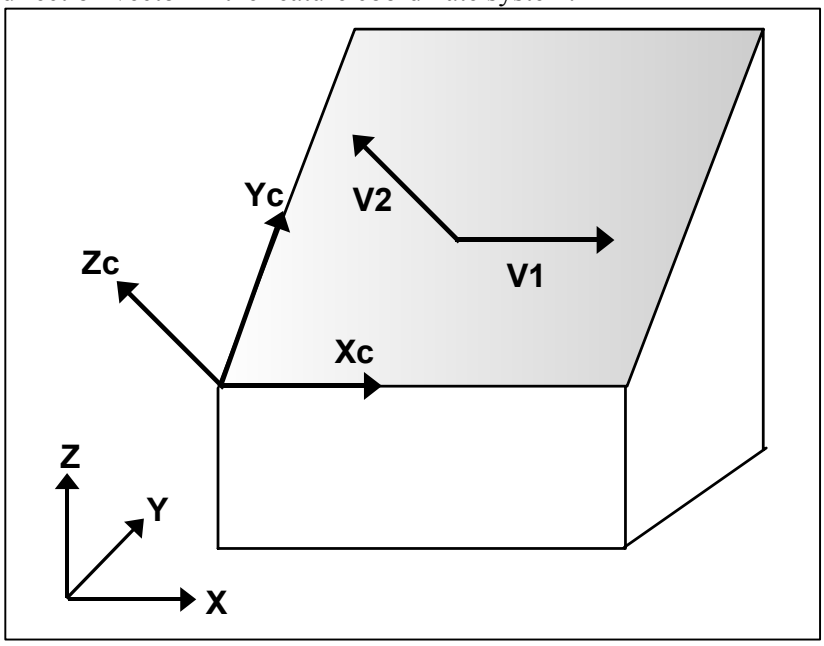
Example of a program

```
G68.2 P2 Q1 X200.0 Y0 Z50.0 ;
G68.2 P2 Q2 X200.0 Y100.0 Z50.0 ;
G68.2 P2 Q3 X26.795 Y0 Z150.0 ;
G53.1 ;
 . . .
```

22.3.3.3 Tilted working plane command based on two vectors

Overview

With the tilted working plane command, a tilted working plane can be specified by specifying an X-axis direction vector and a Z-axis direction vector in the feature coordinate system.



Format

Format
G68.2 P3 Q1 X_ Y_ Z_ Iα_1 Jβ_1 Kγ_1 ; **G68.2 P3 Q2 Iα_2 Jβ_2 Kγ_2 ; Tilted working plane command** **G69 ; Cancel tilted working plane command (M series).** **G69.1; Cancel tilted working plane command (T series).**
Explanation of symbols
X_ Y_ Z_ : Origin of a feature coordinate system (specifiable in the Q1 block)
Q1 Iα_1 Jβ_1 Kγ_1 : Feature coordinate system X-axis direction relative to the workpiece coordinate system (first vector)
Q2 Iα_2 Jβ_2 Kγ_2 : Feature coordinate system Z-axis direction relative to the workpiece coordinate system (second vector)

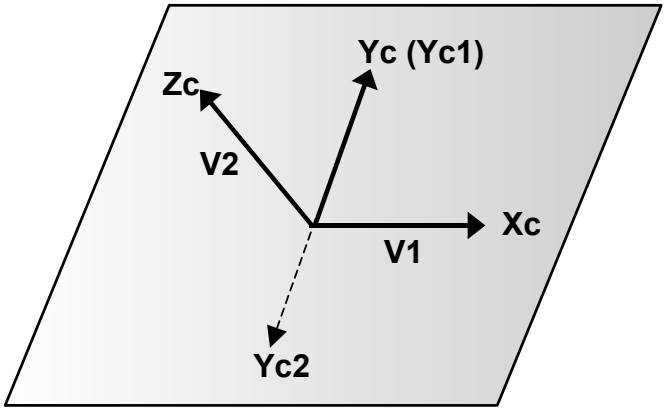
⚠ CAUTION

1 The two G68.2P3 commands (Q1 and Q2) determine a tilted plane. The G68.2P3 commands are interrupted, alarm PS5457 is issued.
2 If the angle between the two vectors is 5 degrees or more off the 90 degrees, alarm PS5457 is issued.
3 If (I, J, K) is set to a 0 vector, alarm PS5457 is issued.

Explanation

- Determination of a feature coordinate system

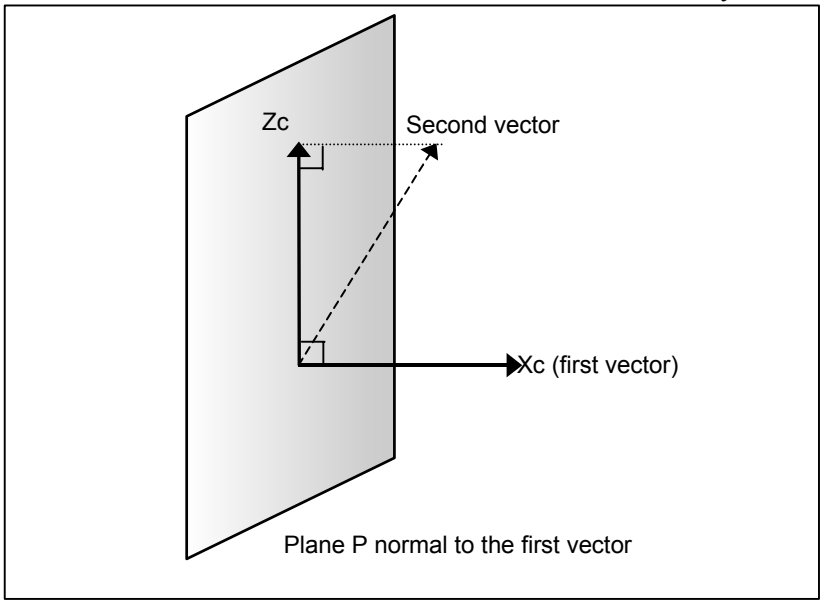
The first vector is defined as the X-axis of the feature coordinate system, and the second vector is defined as the Z-axis of the feature coordinate system. The Y-axis of the feature coordinate system is defined according to the right-handed system.



There are two vectors normal to the Xc-axis and Zc-axis. However, Yc1 is defined as the Yc-axis of the feature coordinate system according to the right-handed system.

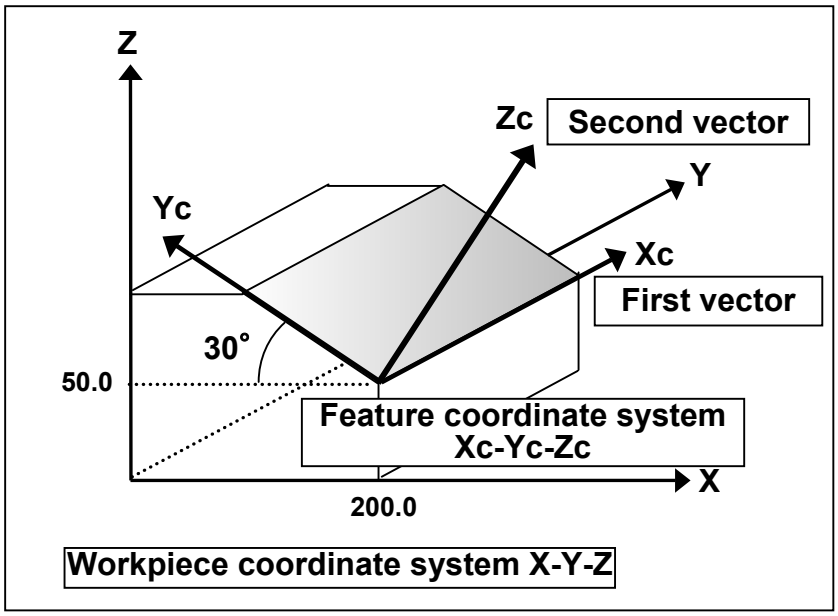
- When the first and second vectors are not normal

When the first vector and second vector are not normal, the orthographical vector from the second vector to plane P normal to the first vector is defined as the Z-axis of the feature coordinate system.



Example

The example of a program when feature coordinate system like the figure below is used is shown below.



Origin of a feature coordinate system : (200.0, 0, 50.0)
X-axis direction of a feature coordinate system : (0, 1.0, 0)
(First vector)
Z-axis direction of a feature coordinate system : (100.0, 0, 173.205)
(Second vector)

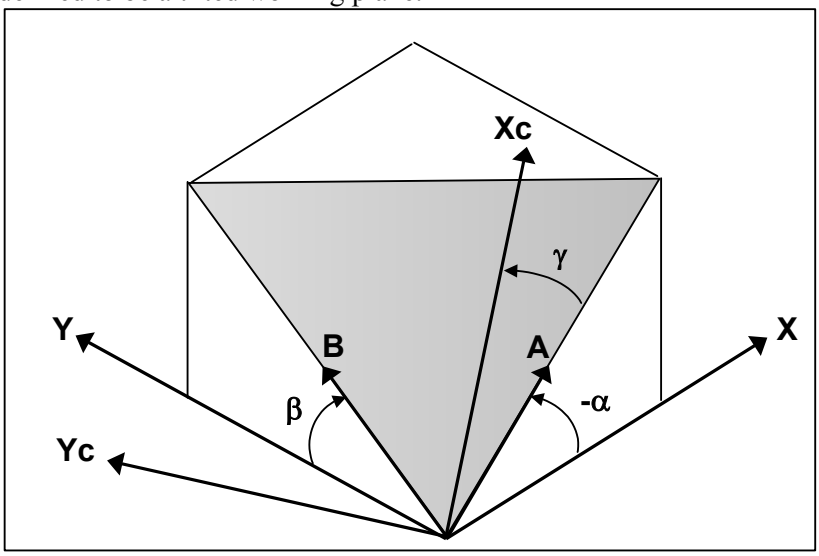
Example of a program

```
G68.2 P3 Q1 X200.0 Y0 Z50.0 I0 J1.0 K0 ;
G68.2 P3 Q2 I100.0 J0 K173.205 ;
G53.1 ;
 . . .
```

22.3.3.4 Tilted working plane command based on projection angles

Overview

With the tilted working plane command, a tilted working plane can be specified based on projection angles.
A plane determined by vector A and vector B produced by rotating the X-axis vector and Y-axis vector of the workpiece coordinate system is defined to be a tilted working plane.



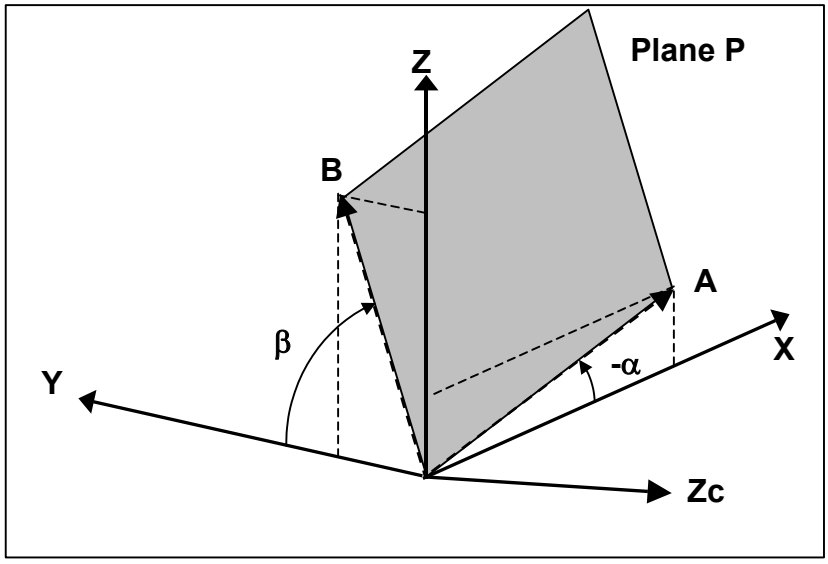
Format

Format
G68.2 P4 X_ Y_ Z_ Iα Jβ Kγ; Tilted working plane command **G69 ; Cancel tilted working plane command (M series).** **G69.1; Cancel tilted working plane command (T series).**
Explanation of symbols
X_ Y_ Z_ : Origin of a feature coordinate system α : Angle by which the X-axis vector is rotated about the Y-axis of the workpiece coordinate system β : Angle by which the Y-axis vector is rotated about the X-axis of the workpiece coordinate system γ : Angle of rotation about the Z-axis of the feature coordinate system

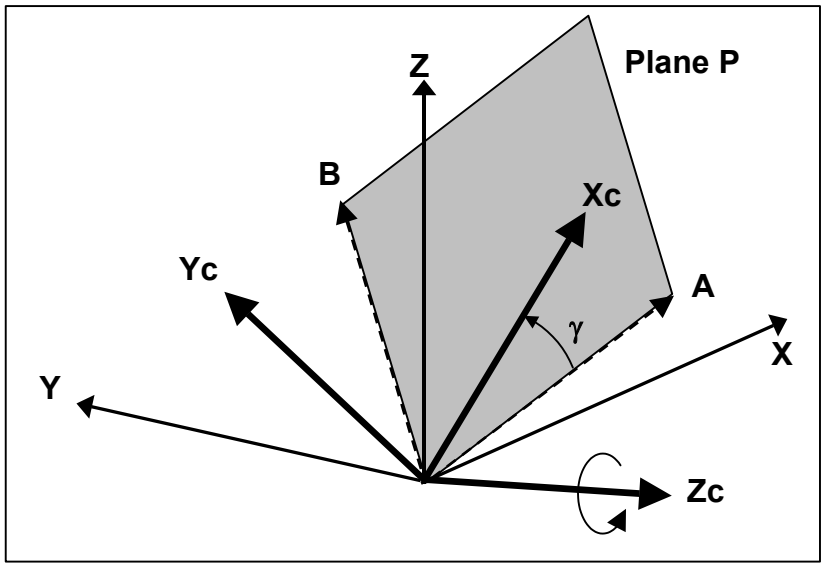
Explanation

- Determination of a feature coordinate system

The X-axis direction vector of the workpiece coordinate system rotated by α about the Y-axis of the workpiece coordinate system is defined as vector A. The Y-axis direction vector of the workpiece coordinate system rotated by β about the X-axis of the workpiece coordinate system is defined to be vector B.
The direction normal to plane P containing vector A and vector B (direction of the outer product of A × B) is defined to be the Z-axis direction of the feature coordinate system.
Vector A rotated by γ about the Z-axis of the feature coordinate system is defined to be the X-axis direction of the feature coordinate system. The Y-axis of the feature coordinate system is defined according to the right-handed system.



By the third command angle α and second command angle β, the Z-axis of the feature coordinate system are determined.

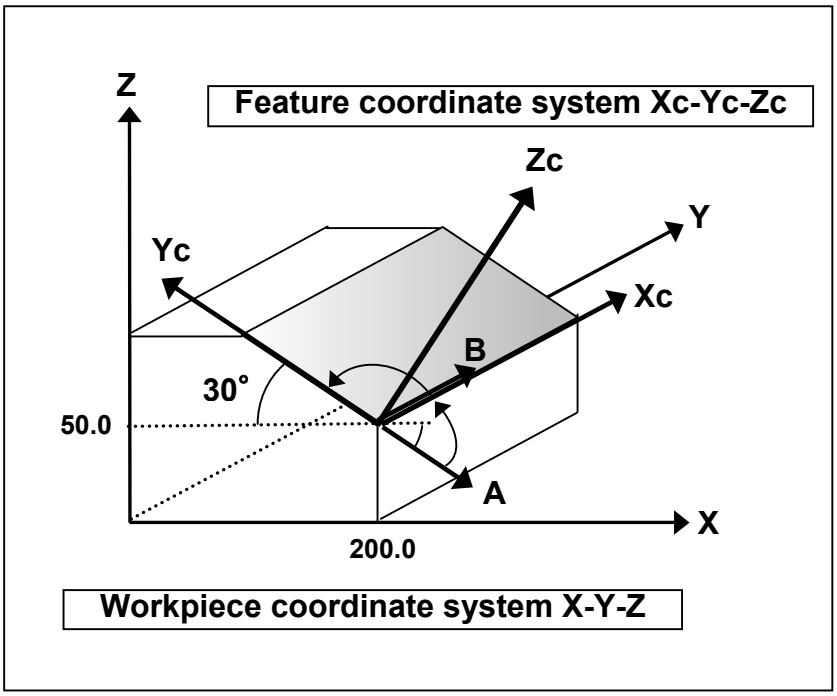


By the third command angle γ, the X-axis and Y-axis of the feature coordinate system are determined.

NOTE
When vector A and vector B are considered to be parallel with each other (when the angle formed by the two vectors is smaller than 1°), alarm PS5457 is issued.

Example

The example of a program when feature coordinate system like the figure below is used is shown below.



- Origin of a feature coordinate system : (200.0, 0, 50.0)
- Angle by which the X-axis vector is rotated about the Y-axis of the workpiece coordinate system : 30 degrees
- Angle by which the Y-axis vector is rotated about the X-axis of the workpiece coordinate system : 0 degree
- Angle of rotation about the Z-axis of the feature coordinate system : 90 degrees

Example of a program

```
G68.2 P4 X200.0 Y0 Z50.0 I30.0 J0 K90.0 ;
G53.1 ;
  :
```

22.3.3.5 Absolute multiple command

By additionally specifying G68.2 in the tilted working plane command mode, a feature coordinate system produced by additionally applying coordinate system conversion to the workpiece coordinate system can be set. The workpiece coordinate system is resumed by specifying G69.
This function is enabled by setting bit 0 (MTW) of parameter No. 11221.

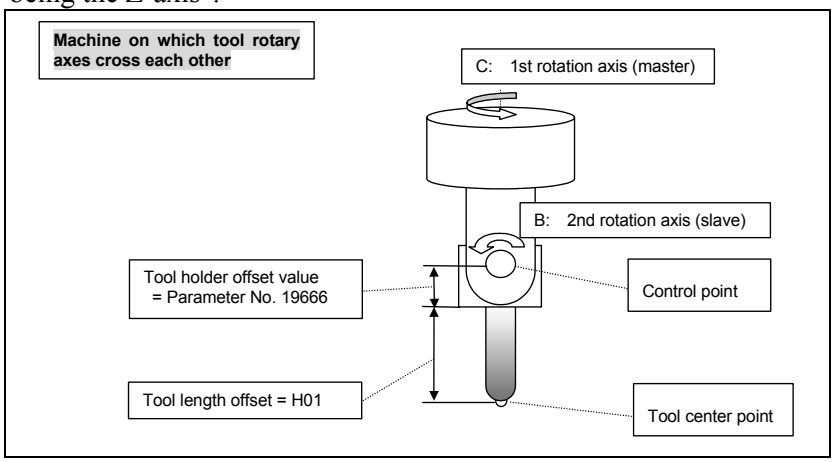
Format

The format of the tilted working plane command (G68.2) is applicable.
Specify the origin of a feature coordinate system in the workpiece coordinate system.

NOTE
Before specifying G68.2, cancel tool length compensation and tool radius compensation. If G68.2 is specified during tool length compensation or tool radius compensation, alarm PS5462 is issued.

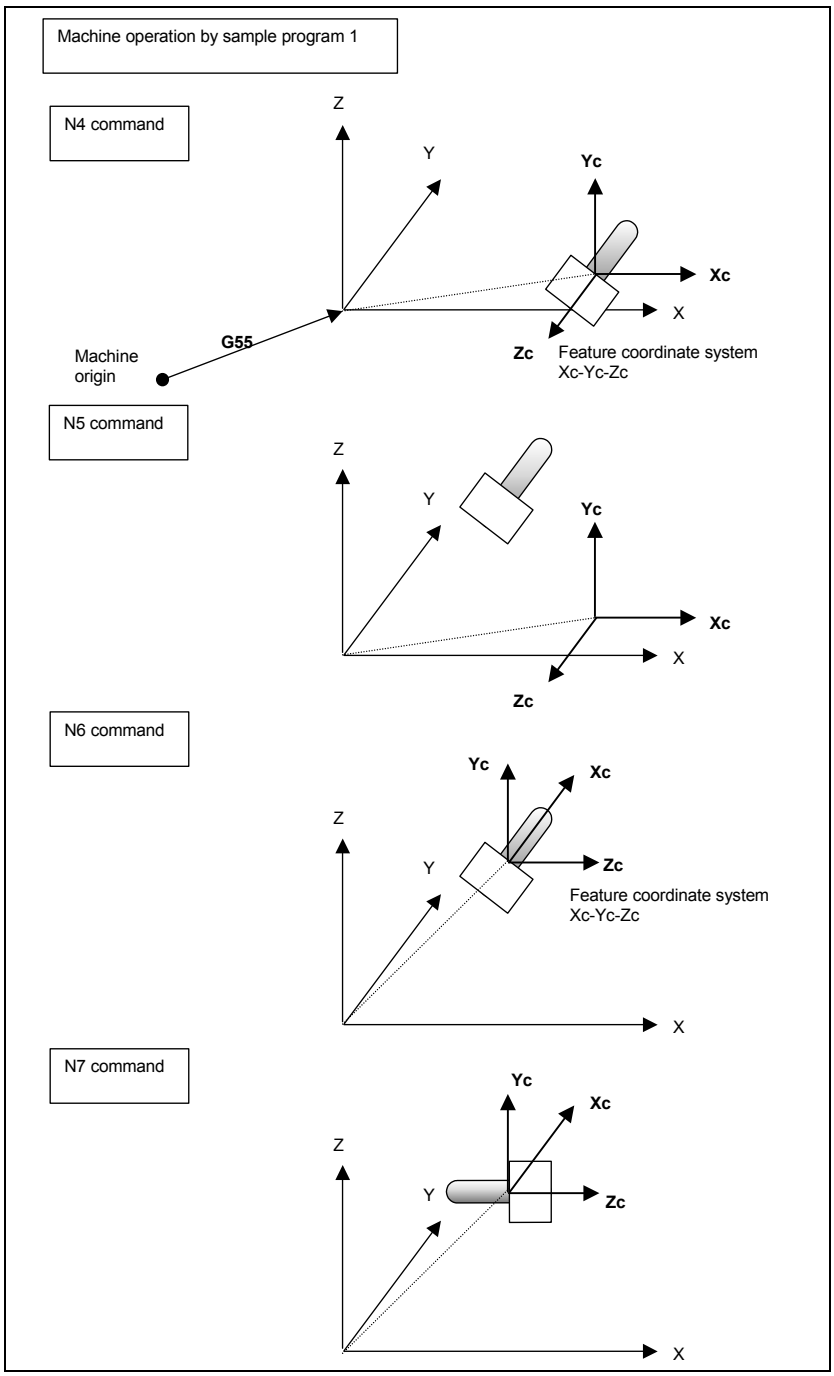
Example of operation

An example of operation on a tool rotation type machine is explained below.
The machine configuration is "BC type with the reference tool axis being the Z-axis".



Sample program 2

```
O0100 ;
N1 G55 ;
N2 G90 G01 X20.0 Y5.0 Z0 F1000 ;
N3 G68.2 X20.0 Y5.0 Z0 I0 J90.0 K0 ;
N4 G53.1 ;
  :
N5 X-15.0 Y0 Z-15.0 ;
N6 G68.2 X5.0 Y20.0 Z0 I90.0 J90.0 K0 ;
N7 G53.1;
  :
N8 G69 ;
  :
```



N4 block: Rotates the tool in the Z-axis direction in the feature coordinate system.
N5 block: Specifies coordinates in the feature coordinate system.
N6 block: Sets a new feature coordinate system.
N7 block: Rotates the tool in the Z-axis direction in the new feature coordinate system.

22.3.3.6 Incremental multiple command

By specifying G68.4, coordinate system conversion can be applied to the currently set feature coordinate system.
This function is enabled by setting bit 0 (MTW) of parameter No. 11221.

Format

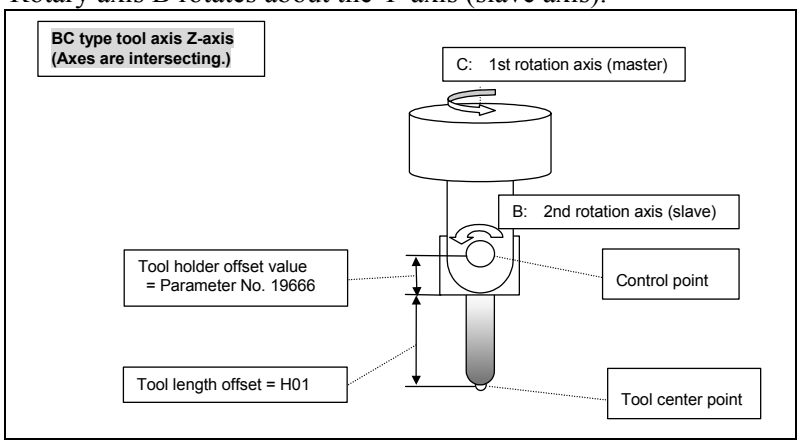
The format of the tilted working plane command (G68.2) is applicable.
Specify the origin of a feature coordinate system in the immediately preceding feature coordinate system.

Specification method	Incremental multiple command
Eulerian angle	G68.4
Roll-pitch-yaw	G68.4 P1
Three points	G68.4 P2
Two vectors	G68.4 P3
Projection angles	G68.4 P4

NOTE
1 Before specifying G68.4, cancel tool length compensation and tool radius compensation. If G68.4 is specified during tool length compensation or tool radius compensation, alarm PS5462 is issued.
2 Do not specify G68.4 during workpiece setting error compensation. Otherwise, alarm PS0437 is issued.

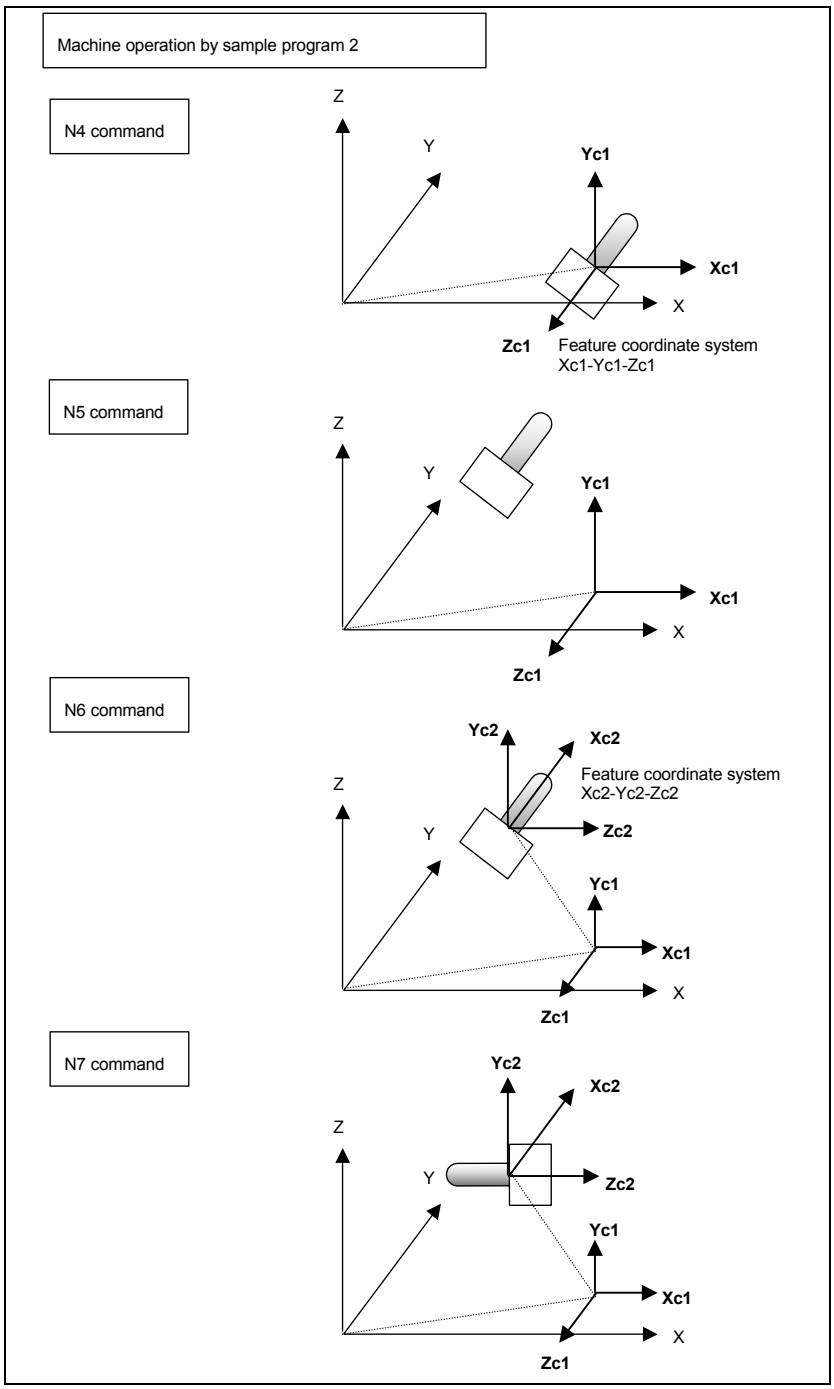
Example of operation

An example of operation on a tool rotation type machine is explained below.
Rotary axis C rotates about the Z-axis (master axis).
Rotary axis B rotates about the Y-axis (slave axis).



Sample program 3

```
O0200 ;
N1 G55 ;
N2 G90 G01 X20.0 Y5.0 Z0 F1000 ;
N3 G68.2 X20.0 Y5.0 Z0 I0 J90.0 K0 ;
N4 G53.1 ;
  :
N5 X-15.0 Y0 Z-15.0 ;
N6 G68.4 X-15.0 Y0 Z-15.0 I90.0 J90.0 K-90.0 ;
N7 G53.1;
  :
N8 G69 ;
  :
```



N4 block: Rotates the tool in the Z-axis direction in the feature coordinate system.
N5 block: Specifies coordinates in the feature coordinate system.
N6 block: Applies coordinate system conversion to the feature coordinate system based on Eulerian angle to set a new feature coordinate system.
N7 block: Rotates the tool in the Z-axis direction in the new feature coordinate system.

Limitation

- Basic restrictions

The restrictions for this function are similar to those for the three-dimensional coordinate system conversion function.

- Increment system

The same increment system must be used for the basic three axes used by this function.

- Rapid traverse command

The rapid traverse command must specify linear rapid traverse (Bit 1 (LRP) of parameter No.1401= 1).

- Feature coordinate system and three-dimensional coordinate system conversion

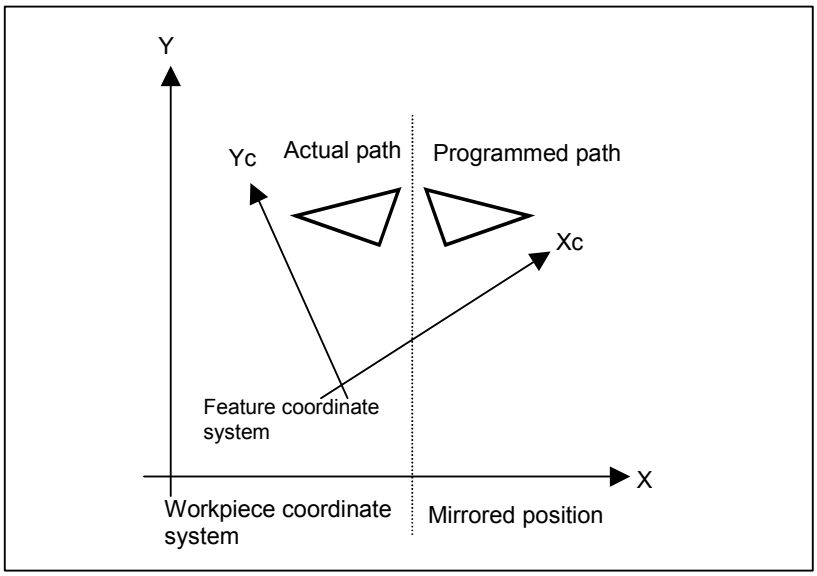
An alarm occurs if an attempt is made to set a feature coordinate system in another feature coordinate system.
An alarm also occurs if an attempt is made to set a new coordinate system by performing three-dimensional coordinate system conversion in a feature coordinate system.

- Positioning in the machine coordinate system

Positioning commands in the machine coordinate system, such as G28, G30, and G53, operate in the machine coordinate system rather than in the feature coordinate system.

- External mirror image

If an attempt is made to use this function and the external mirror image function simultaneously, this function takes effect before the external mirror image function.



- Relationships with other modal commands

G41, G42, and G40 (tool radius compensation), G43 and G49 (tool length compensation), G51.1 and G50.1 (programmable mirror image), and canned cycle commands must have nesting relationships with G68.2.
In other words, first issue G68.2 when the modes mentioned above are off, then turn the modes on and off, and then issue G69.

- Parallel axis control

When a parking signal is applied to an axis during parallel axis control, conversion to a feature coordinate system occurs for another axis if a move command is issued for that another axis. For this reason, an axis may move even if a parking signal has been applied to it.

- Specifiable G codes

The G codes that can be specified in the tilted working plane command mode are listed below.
Specifying a G code other than these codes results in alarm PS5462.

- Positioning (G00)
- Linear interpolation (G01)
- Circular interpolation / helical interpolation (G02/G03)
- Dwell (G04)
- Programmable data input (G10)
- Programmable data input mode cancel (G11)
- Plane selection (G17/G18/G19)
- Automatic return to reference position (G28)
- Movement from reference position (G29)
- 2nd, 3rd and 4th reference position return (G30)
- Skip function (G31)
- Tool radius compensation : cancel (G40)
- Tool radius · tool nose radius compensation / 3-dimensional cutter compensation (G41/G42)
- Tool length compensation + (G43)
- Tool length compensation - (G44)
- Tool length compensation cancel (G49)
- Machine coordinate system setting (G53)
- Tool axis direction control (G53.1)
- Macro call (G65)
- Macro modal call A (G66)
- Macro modal call B (G66.1)
- Macro modal call A/B cancel (G67)
- Absolute programming (G90)
- Incremental programming (G91)
- Canned cycle for drilling (G73, G74, G76, G80 to G89)
- Constant surface speed control (G96)
- Constant surface speed control cancel (G97)
- Canned cycle : return to initial level (G98)
- Canned cycle : return to R point level (G99)

M

- Programmable mirror image (G50.1/G51.1)
- Coordinate system rotation cancel or 3-dimensional coordinate conversion mode off (G69)
- Feed per minute (G94)
- Feed per revolution (G94)

T

- Coordinate system rotation cancel or 3-dimensional coordinate conversion mode off (G69.1)
- Feed per minute (G98 (G94))
- Feed per revolution (G99 (G94))

- Modal G codes that allow specification of a tilted working plane command

A tilted working plane command can be specified in the modal G code states listed below.
In a modal state other than the following modal G codes, specifying the tilted working plane command results in alarm PS5462:

- Modal G codes included in "Specifiable G codes" described previously
  (However, G43/G44 and G41/G42 are excluded.)
- Polar coordinate interpolation mode cancel (G13.1)
- Polar coordinates command cancel (G15)
- Input in inch (G20 (G70))
- Input in mm (G21 (G71))
- Stored stroke check function (G22/G23)
- Scaling cancel (G50)
- Polygon turning cancel (G50.2)
- Workpiece coordinate system 1 selection (G54 to G59)
- Workpiece setting error compensation (G54.4)
- Exact stop mode (G61)
- Automatic corner override (G62)
- Tapping mode (G63)
- Cutting mode (G64)
- Inverse time feed (G93)
- Constant surface speed control cancel (G97)

M

- Polar coordinate interpolation mode cancel (G113)

T

- Programmable mirror image cancel (G50.1)
- Mirror image for double turret off/balanced cutting mode cancel (G69)

22.4 INCLINED ROTARY AXIS CONTROL

Overview

The conventional tilted working plane command / tool center point control function / 3-dimensional cutter compensation / three-dimensional manual feed can be used only for those machines whose tool rotation axis or table rotation axis is parallel to the basic axis of the basic coordinate system. (See Fig. 22.4 (a).)

Inclined rotary axis control is intended to allow the tilted working plane command / tool center point control function / 3-dimensional cutter compensation / three-dimensional manual handle feed to be applied even when the tool rotation axis or table rotation axis is inclined with respect to the X-Z plane, a Y-Z plane or a Z-X plane of the machine coordinate system.

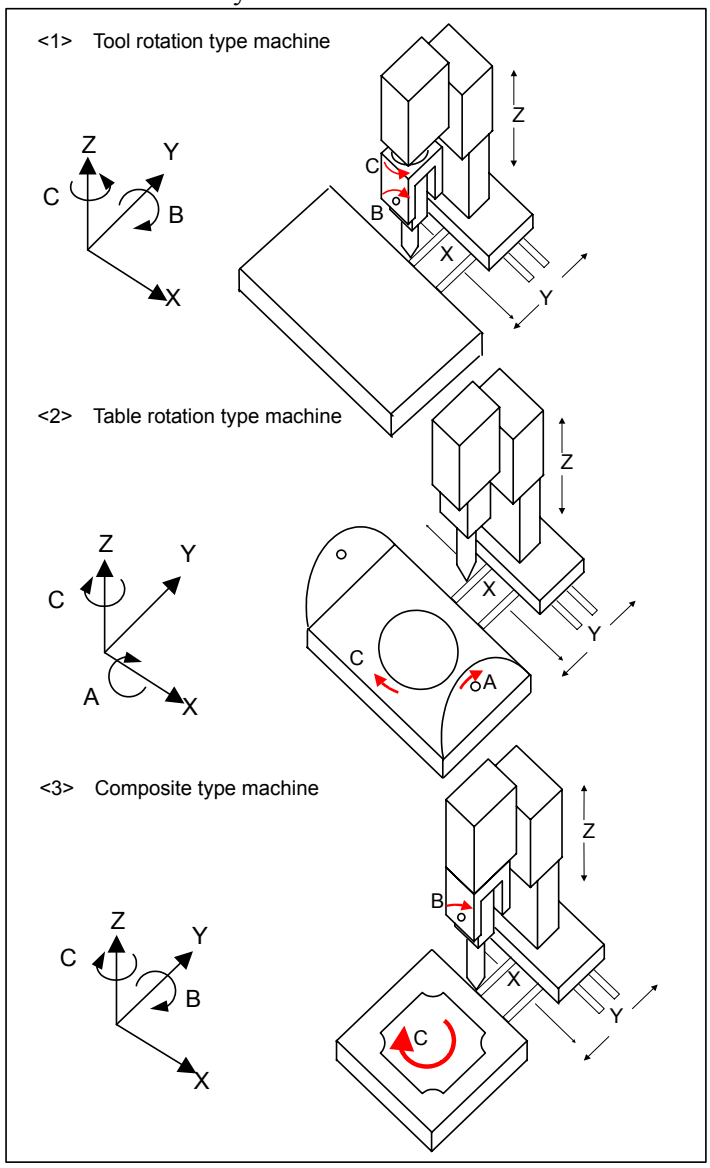


Fig. 22.4 (a) Three types of 5-axis machine

An example of a tool rotation type machine is explained below. (See Fig. 22.4 (b).)
The machine shown in Fig. 22.4 (b) has rotary axis B (master) that turns around the Y-axis and rotary axis C (slave) whose Y-axis is inclined at an angle of 45 degrees on the Y-Z plane.
The tilted working plane command / tool center point control function / 3-dimensional cutter compensation / three-dimensional manual feed can be used even for the machine configuration shown in Fig. 22.4 (b).

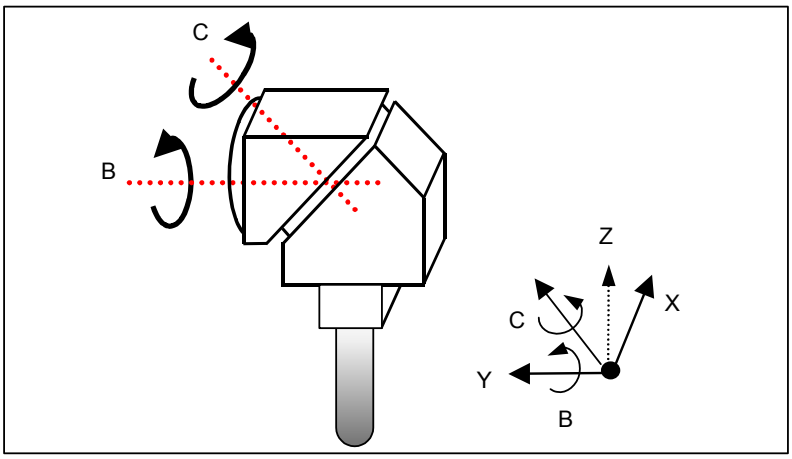


Fig. 22.4 (b) Tool rotation type machine

An example of a table rotation type machine is explained below. (See Fig. 22.4 (c).)
The machine shown in Fig. 22.4 (c) has rotary axis B (master) whose Y-axis is inclined at an angle of -45 degrees on the Y-Z plane and rotary axis C (slave) that turns around the Z-axis.
The tilted working plane command / tool center point control function / 3-dimensional cutter compensation / three-dimensional manual feed can be used even for the machine configuration shown in Fig. 22.4 (c).

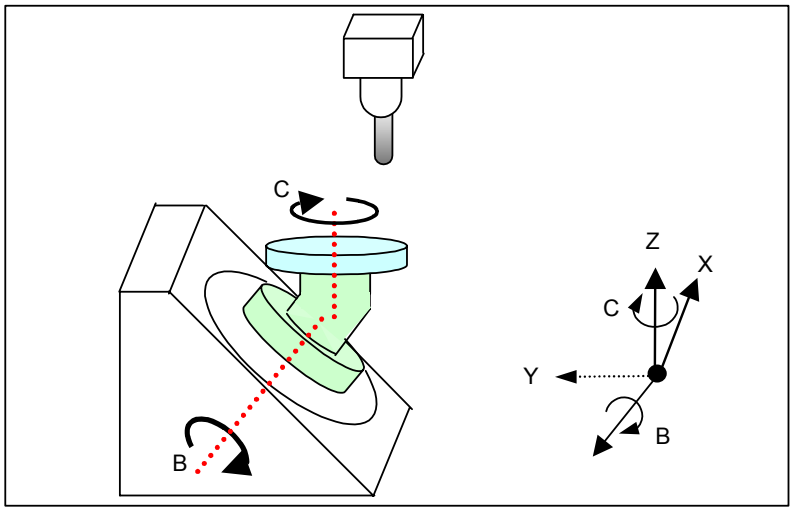


Fig. 22.4 (c) Table rotation type machine

An example of a composite type machine is explained below. (See Fig. 22.4 (d).)
The machine shown in Fig. 22.4 (d) has table rotation axis B whose Y-axis is inclined at an angle of -45 degrees on the Y-Z plane and tool rotation axis C that turns around the Z-axis.
The tilted working plane command / tool center point control function / 3-dimensional cutter compensation / three-dimensional manual feed can be used even for the machine configuration shown in Fig. 22.4 (d).

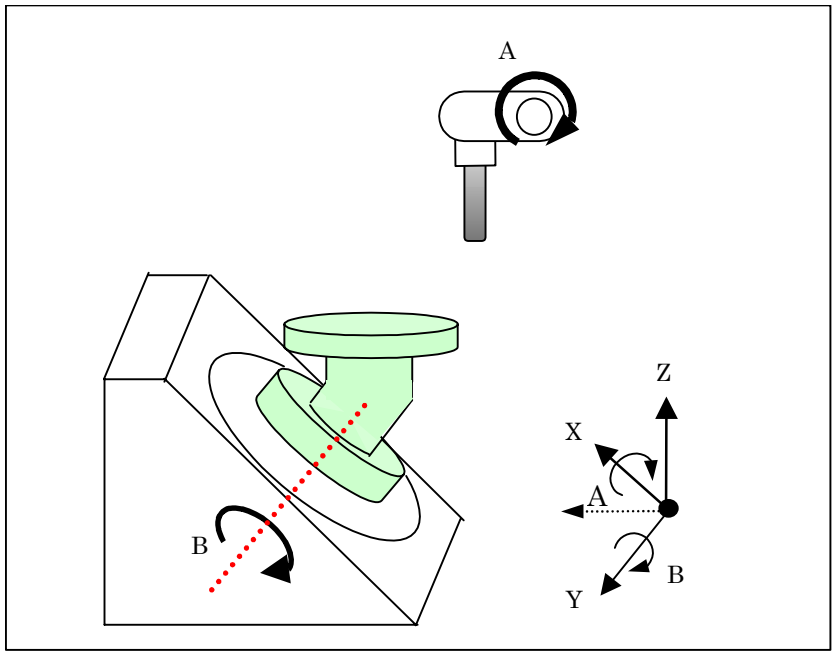


Fig. 22.4 (d) Composite type machine

Format and operation

The operation of the tilted working plane command / tool center point control function / 3-dimensional cutter compensation / three-dimensional manual feed during the inclined rotary axis control is similar to the operation performed when the inclined rotary axis control is not in use.
For details, see the description of each function.

Limitation

The operation of the tilted working plane command / tool center point control function / 3-dimensional cutter compensation / three-dimensional manual feed during the inclined rotary axis control is similar to the operation performed when the inclined rotary axis control is not in use.
For details, see the description of each function.

22.5 3-DIMENSIONAL CUTTER COMPENSATION

Overview

For machines having multiple rotary axes for freely controlling the orientation of a tool axis, this function calculates a tool vector from the positions of these rotary axes. The function then calculates a compensation vector in a plane (compensation plane) perpendicular to the tool vector and performs 3-dimensional cutter compensation.
The compensation is performed on the block end point.

- Machine configuration

This function is applicable to the following machine configurations:

<1> Tool rotation type machine controlled with two tool rotation axes

<2> Table rotation type machine controlled with two table rotation axes

<3> Composite type machine controlled with one tool rotation axis and one table rotation axis

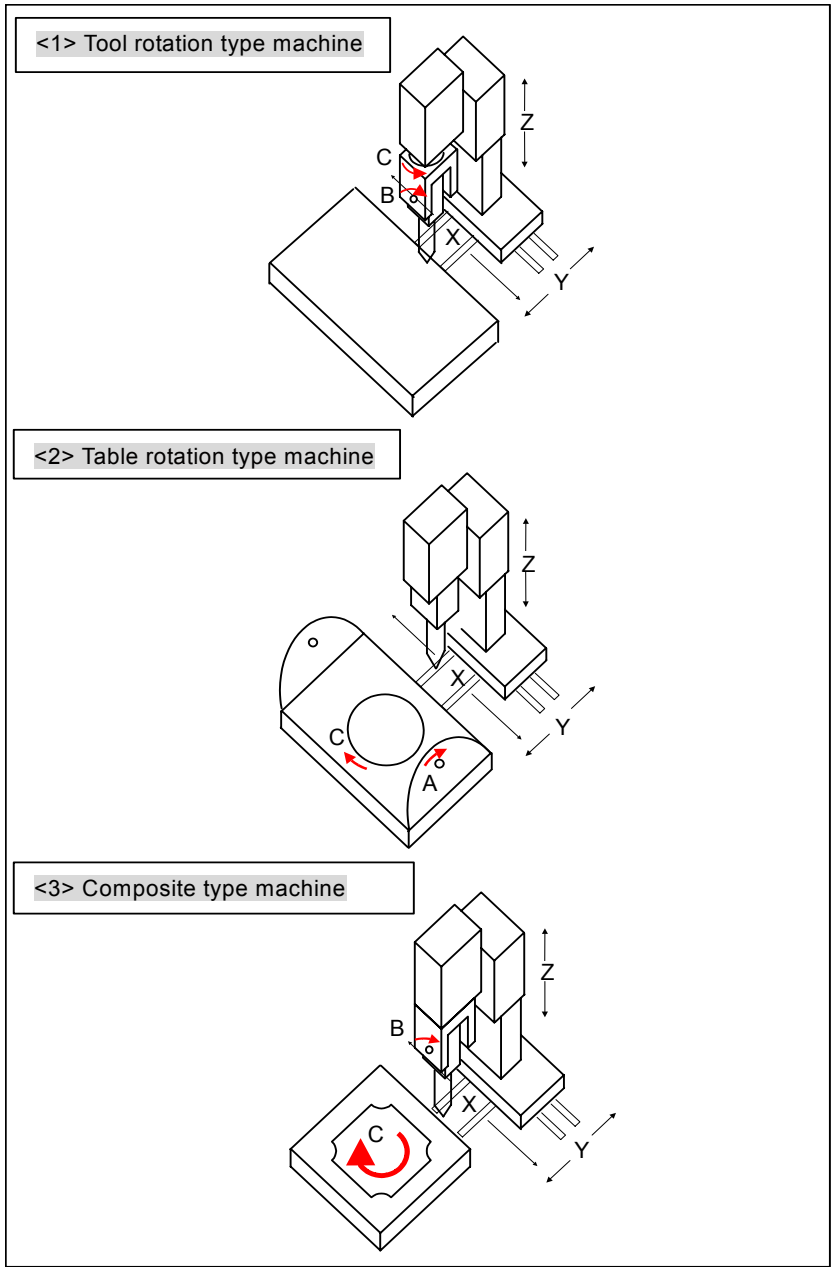


Fig. 22.5 (a) Three types of machine configuration

The coordinate system in which to execute a program for 3-dimensional cutter compensation is called a programming coordinate system.
If, in a 5-axis machine having a table rotation axis, 3-dimensional cutter compensation (tool side offset) is to be performed, the table coordinate system (coordinate system fixed to the table) can be selected as a programming coordinate system.
Alternatively, the workpiece coordinate system fixed to the machine coordinate system can be selected as a programming coordinate system.
There are two types of 3-dimensional cutter compensation (tool side offset), which differ in the method of specifying the tool axis direction.

(1) Type 1

Specify the block end position (for example, A, B, C) of the rotation axis. The CNC performs cutter compensation on the plane perpendicular to the tool axis direction calculated from the specified position on the rotation axis.

(2) Type 2

Instead of specifying a position on the rotation axis, specify the tool axis direction (I, J, K) at the block end point as viewed from the table coordinate system. The CNC calculates the end position on the rotation axis so that the tool points to the specified direction at the end point, and performs cutter compensation on the plane perpendicular to the tool axis direction calculated from the position on the rotation axis.

By using type 2, it is possible to perform the same machining with the same program regardless of the machine configuration of the 5-axis machine (tool rotation type, table rotation type, or composite type).

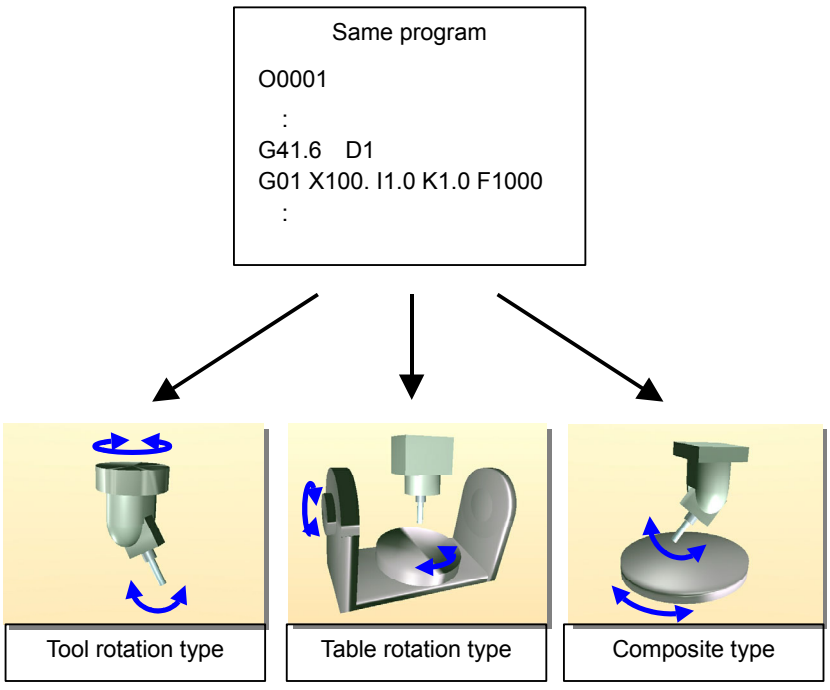


Fig. 22.5 (b) Type 2 Program

22.5.1 Cutter Compensation in Tool Rotation Type Machine

Overview

In a 5-axis machine having two tool rotation axes as shown in Fig. 22.5.1 (a), this function can perform cutter compensation.
Shown below is a 5-axis machine that has tool rotation axis B on the Y-axis and tool rotation axis C on the Z-axis.
This machine configuration is used as a sample configuration in the following explanation unless otherwise noted.

Cutter compensation in tool rotation machines is classified into two types according to the way of machining: tool side offset and leading edge offset.

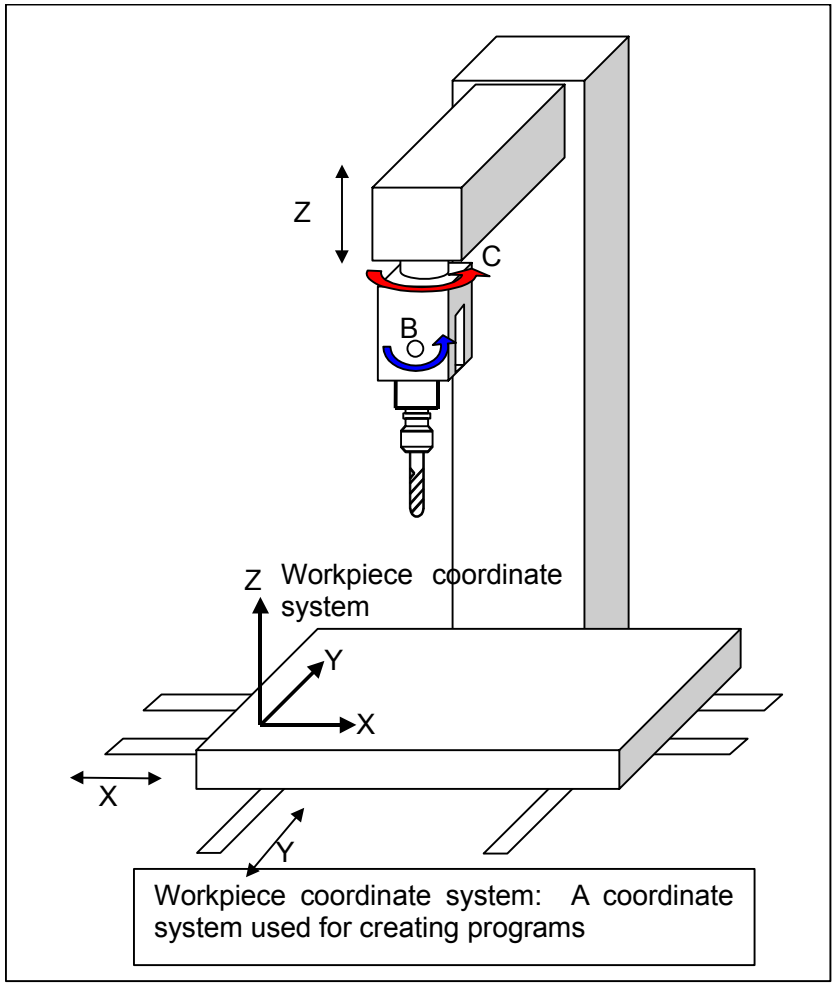


Fig. 22.5.1 (a) Machine having two tool rotation axes

22.5.1.1 Tool side offset

Overview

This type of cutter compensation performs three-dimensional compensation in a plane (compensation plane) perpendicular to the tool vector.

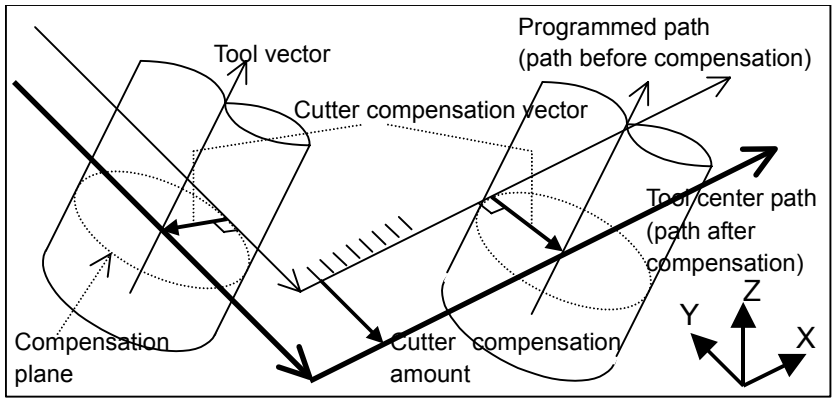


Fig. 22.5.1.1 (a) Tool side offset

Format

- Tool side offset (type 1)

G41.2 (or G42.2) IP_ D_ ;
IP_ ;
:
G41.2: Cutter compensation left (group 07)
G42.2: Cutter compensation right (group 07)
IP_: Value specified for axis movement (including rotary axis)
D_: Code specifying the cutter compensation amount (1 to 3 digits)

- Tool side offset (type 2)

G41.6 (or G42.6) IP_ I_ J_ K_ D_ Q_ ;
IP_ I_ J_ K_ ;
:
G41.6: Cutter compensation left (group 07)
G42.6: Cutter compensation right (group 07)
IP_: Value specified for axis movement (without rotary axis)
D_: Code specifying the cutter compensation amount (1 to 3 digits)
Q_: Tool's angle of gradient (in degrees)
I_ J_ K_: Tool axis direction at the block end point as viewed from the programming coordinate system

For type 2, do not specify a rotation axis but specify the direction at the tool end point as viewed from the programming coordinate system (workpiece coordinate system), with I, J, and K. Specifying a rotation axis causes alarm PS5460 to be generated.

NOTE
When a movement perpendicular to the next movement (bit 1 (SUV) of parameter No. 5003 is set to 1) is specified as the operation performed at the time of startup or cancellation, a move command such as X_ Y_ Z_ must not be specified in the G41.2 and G42.2 block.

The following are the notes on type 2.

NOTE
1 If one or two of I, J, and K are omitted, the omitted ones of I, J, and K are assumed to be 0.
2 In a block in which all of I, J, and K are omitted, the values of I, J, and K in the previous block are used.
3 If there is only one rotation axis (a hypothetical axis is used), type 2 cannot be used. In this case, if an attempt is made to issue G41.6/G42.6, alarm PS5460 is generated.
4 If using the rotation axis rollover function or rotary axis control function, specify 360 degrees in parameter No. 1260 (mount of travel per rotation about the rotation axis).
5 Only tool side offset provides type 2 commands. Leading edge offset, described later, does not provide type 2 commands.

- Canceling the tool side offset

G40 IP_ ;
G40: Cancellation of cutter compensation (group 07)
IP_: Value specified for axis movement

Explanation

- Tool's angle of gradient in type 2

For type 2 of 3-dimensional cutter compensation, the tool's angle of gradient can be specified with address Q in a G41.6/G42.6 command block. The tool's angle of gradient refers to the angle by which the tool direction to be assumed when machining is actually performed is inclined from the direction specified with (I, J, K) toward the traveling direction on the plane formed by the tool direction specified with (I, J, K) and the traveling direction in the programming coordinate system. (See Fig. 22.5.1.1 (b).)

Because in general, the normal direction of the machining surface is specified with (I, J, K), if it is desired to incline the tool direction to be assumed when machining is actually performed from the normal direction toward the traveling direction, correction may be performed with a Q command.

If the direction specified with (I, J, K) matches the tool direction to be assumed when machining is actually performed, no Q command is necessary.

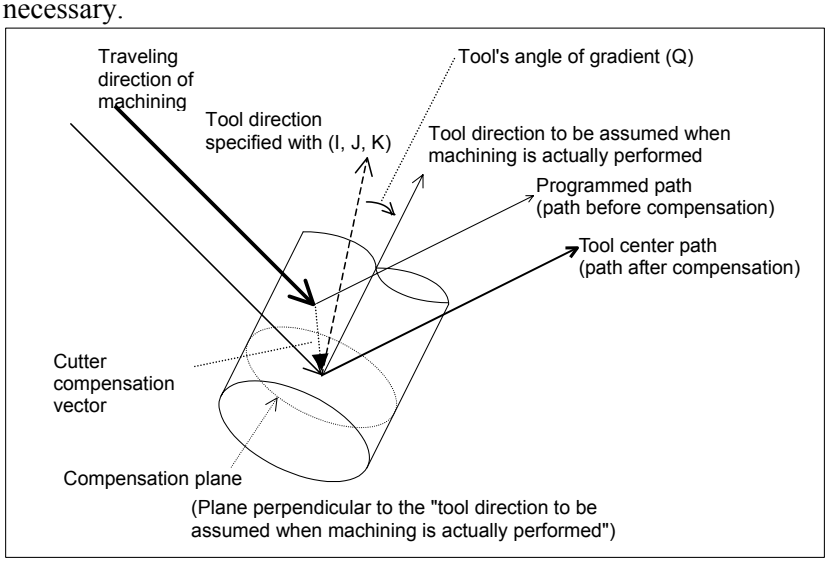


Fig. 22.5.1.1 (b) Tool's angle of gradient in type 2

(Example)

To perform machining by inclining the tool's traveling direction twice, issue a command such as the following:

G41.6 I_ J_ K_ H_ Q2.0

- Operation at startup and cancellation

<1> Type A

The tool is moved in the same way as for cutter compensation as shown below.

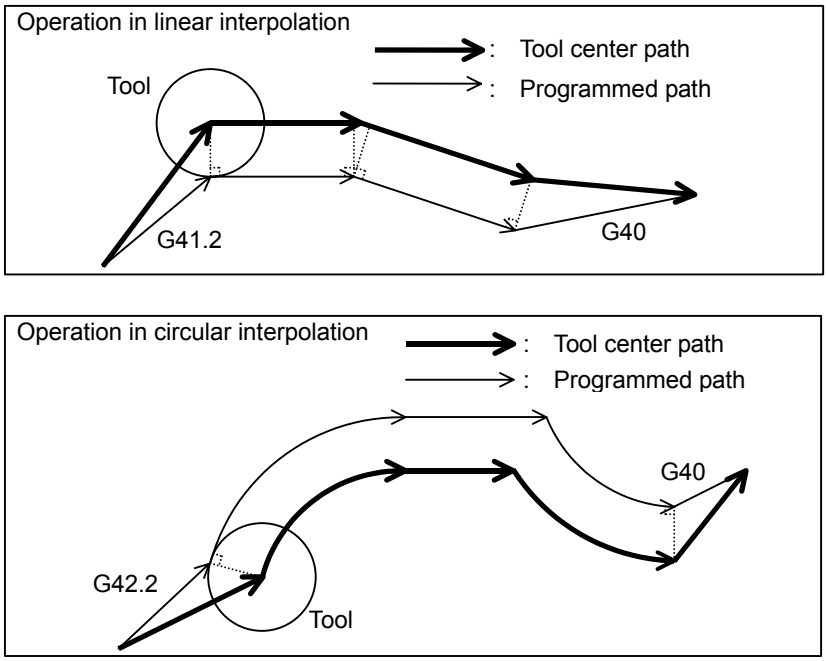


Fig. 22.5.1.1 (c) Operation at startup and cancellation (type A)

When the G40 block contains no movement command, type C cancellation is performed (see <3> Type C below).

<2> Type B

The tool is moved in the same way as for cutter compensation as shown below.

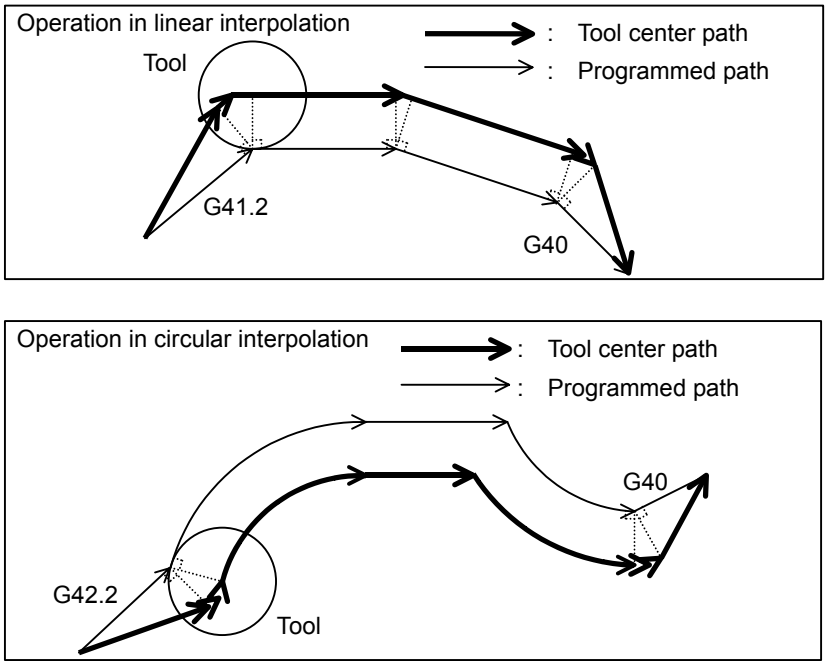


Fig. 22.5.1.1 (d) Operation at startup and cancellation (type B)

When the G40 block contains no movement command, type C cancellation is performed (see <3> Type C below).

<3> Type C
When G41.2, G42.2, or G40 is specified as shown below, a linear block specifying movement by the amount of cutter compensation in the direction orthogonal to the movement direction of the block following startup or the block preceding cancellation is inserted.

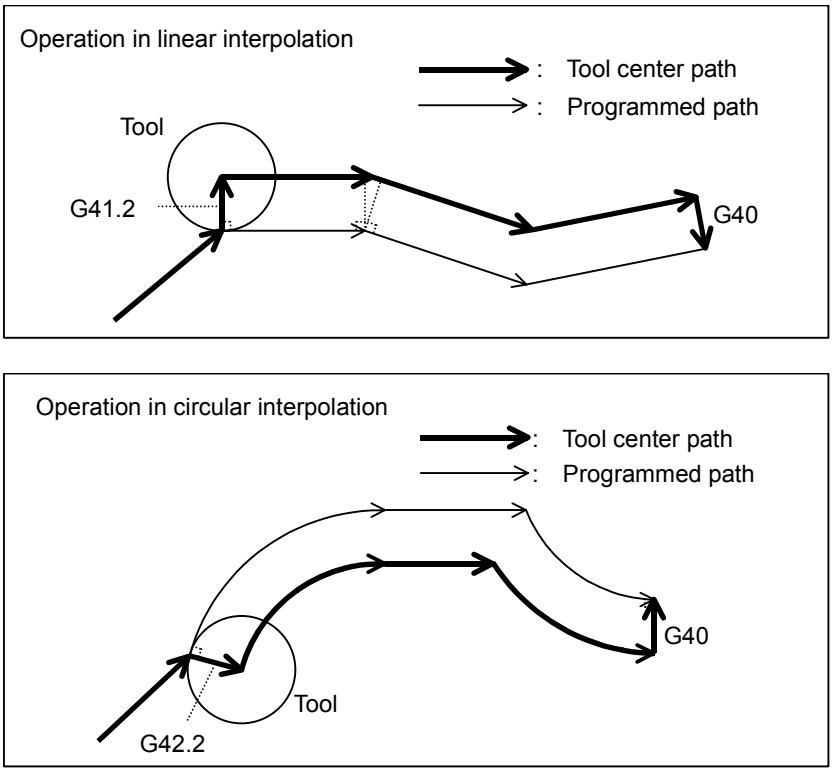


Fig. 22.5.1.1 (e) Operation at startup and cancellation (type C)

NOTE
For type C (bit 1 (SUV) of parameter No. 5003 is 1), the following conditions need to be satisfied during startup and cancellation.
1 A block specifying G40, G41.2, or G42.2 must be in the G00 or G01 mode.
2 A block specifying G40, G41.2, or G42.2 must contain no move command.
3 The block next to a block specifying G41.2 or G42.2 must contain move command G00, G01, G02, or G03.

- Operation during compensation

Operations such as change of the offset direction and offset value, retention of a vector, and interference checks are performed in the same way as for cutter compensation. However, G39 (corner rounding) cannot be specified. So, note the following:

<1> When the tool center path goes outside the programmed path at a corner, a linear movement takes place at the corner without inserting an arc. When the tool center path goes inside the programmed path, nothing is inserted.

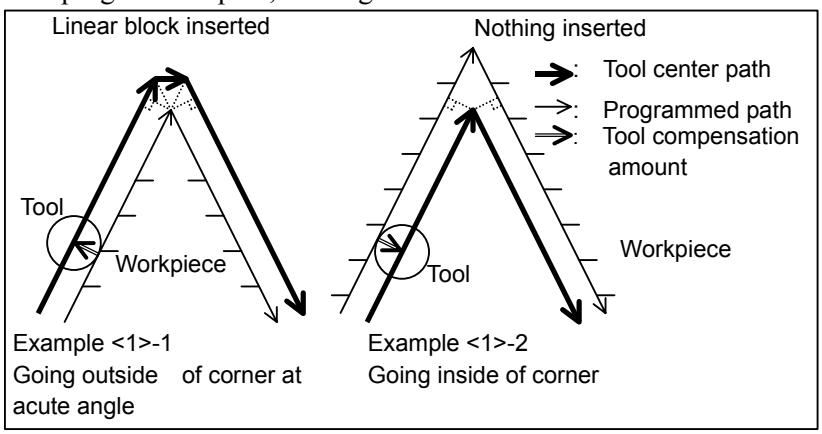


Fig. 22.5.1.1 (f) Operation during compensation <1>-1 and <1>-2

In the above examples, the terms "inside" and "outside" denote how the tool center path is positioned with respect to the programmed path. In the figure below, example <1>-3 shows the same relationship between the tool center path and programmed path as example <1>-1 and indicates that the tool center path is outside the programmed path; example <1>-4 shows the same relationship as example <1>-2 and indicates that the tool center path is inside the programmed path.

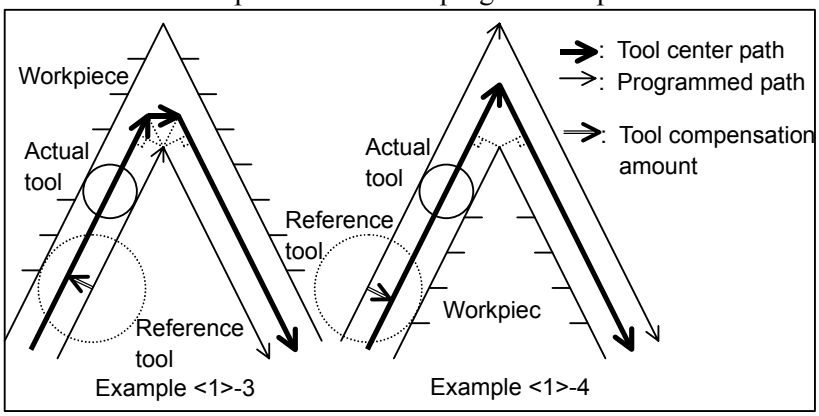


Fig. 22.5.1.1 (g) Operation during compensation <1>-3 and <1>-4

<2> When the tool moves at a corner, the feedrate of the previous block is used if the corner is positioned before a single-block stop point; if the corner is after a single-block stop point, the feedrate of the next block is used.

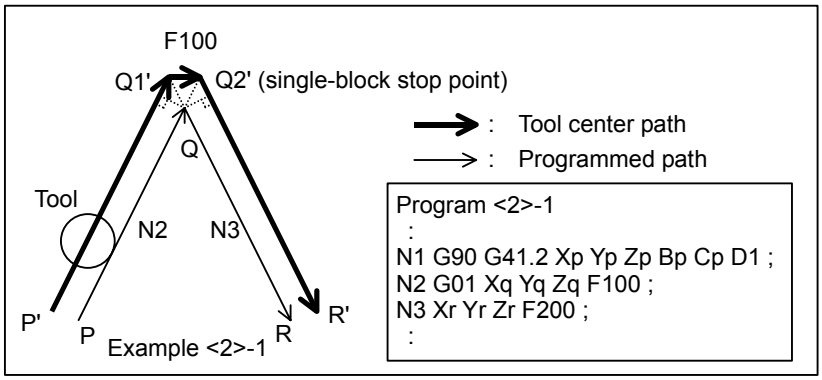


Fig. 22.5.1.1 (h) Operation during compensation <2>

In the above example, the single-block stop point of N2 is Q2', so the feedrates along paths P'-Q1' and Q1'-Q2' are both F100.

<3> When a command that makes the tool retrace the path of the previous block is specified, the tool path can match the locus of the previous block by changing the G code to change the offset direction. If the G code is left unchanged, the operation shown in example <3>-2 results:

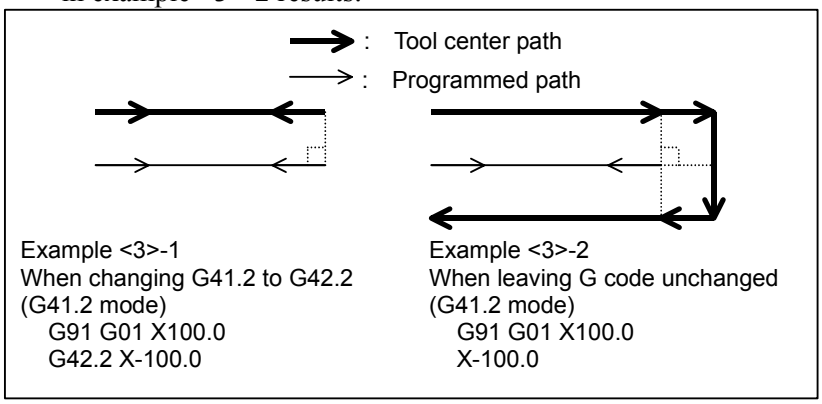


Fig. 22.5.1.1 (i) Operation during compensation <3>

- Interference check when the compensation plane changes

An interference check is made when the compensation plane (a plane perpendicular to the tool vector) has changed.

Example:

If the following program is executed, an alarm PS0041 (INTERFERENCE IN CUTTER COMPENSATION) is issued at N4:

```
O100 F3000
N1 G90 G00 X0 Y0 Z0 A-46 C180
N2 G41.2 D1
N3 G01 X100
N4 Y-200 Z-200
N5 A45
N6 Y-400 Z0
N7 X0
N8 Y-200 Z-200
N9 A-46
N10 Y0 Z0
N11 G40
M30
```

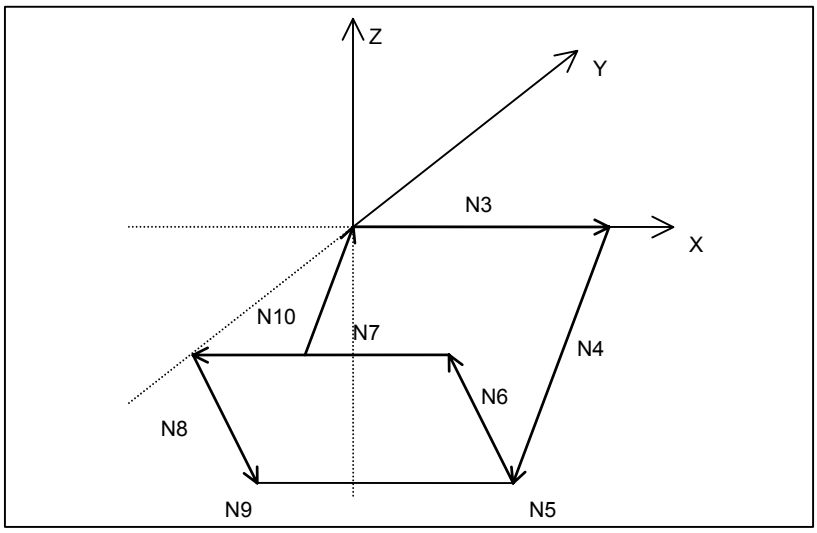


Fig. 22.5.1.1 (j) Conceptual diagram

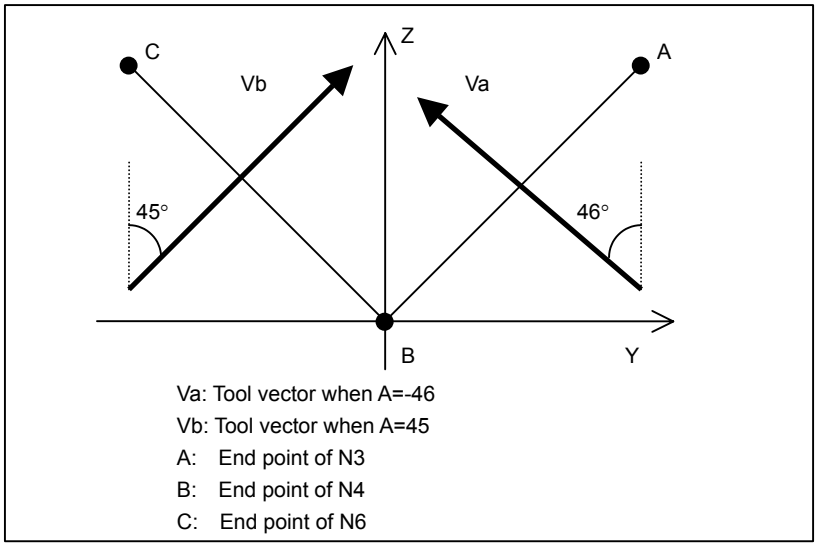


Fig. 22.5.1.1 (k) Tool vector

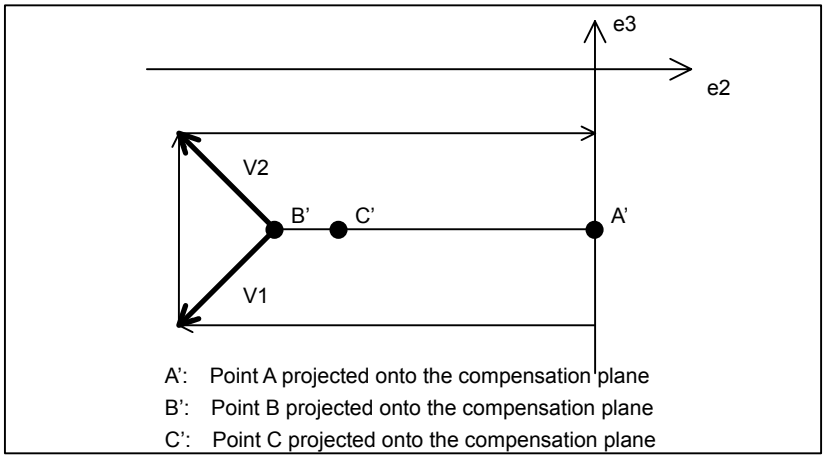


Fig. 22.5.1.1 (l) Compensation vector at the end point (point B) of N4 (in the compensation plane)

The movement direction of A'B' is opposite to that of B'C', so two compensation vectors V1 and V2 are produced at point B' (the end point of N4). In such a case, there is a possibility of overcutting, so an alarm PS0041 is issued at N4.

<1> Conditions for issuing the interference alarm

Suppose that a move command for a rotary axis makes the tool vector change significantly from one block to another. In this case, an interference alarm is assumed because compensation vectors are regarded as being generated in wrong directions when the path angle difference in the compensation plane is large, even though the angle difference between the directions of compensation vectors to be generated by those blocks is small.

Here, the compensation plane is perpendicular to the tool direction (Va in the figure below) of the first of the two blocks.

Specifically, the conditions listed below are used for issuing the alarm.

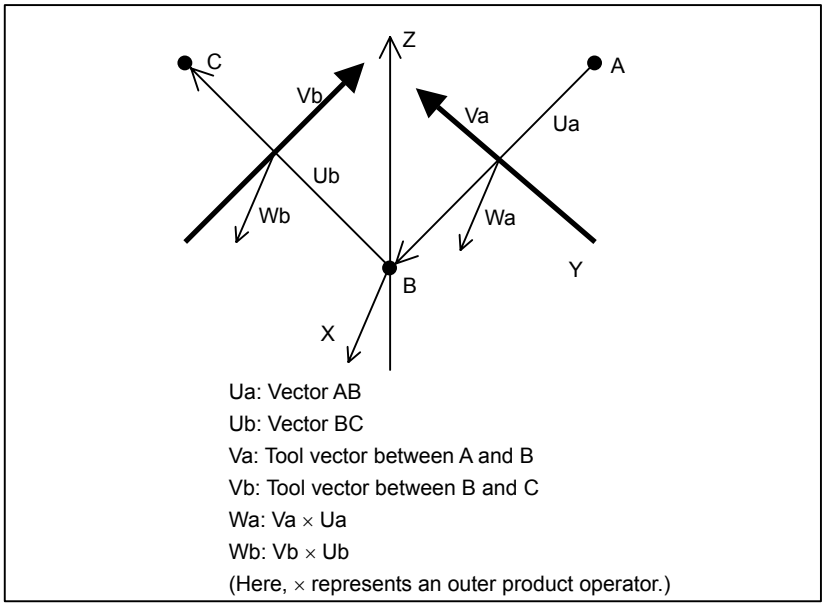


Fig. 22.5.1.1 (m) Conceptual diagram

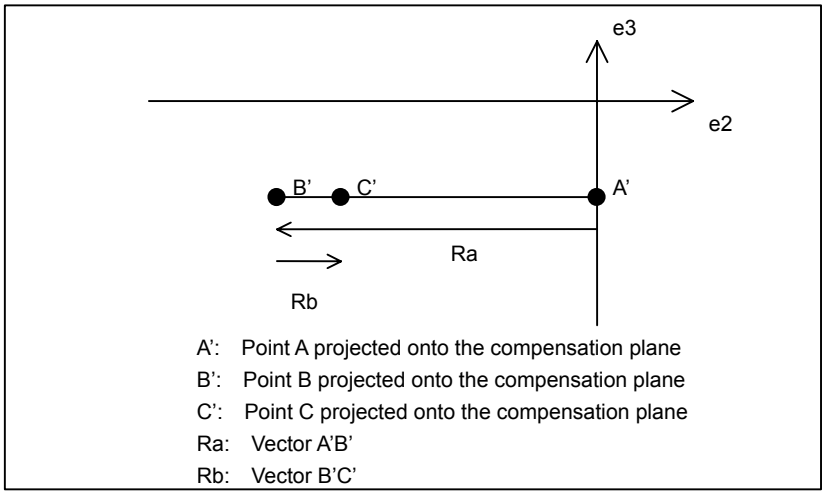


Fig. 22.5.1.1 (n) Programmed path before and after the end point (point B) of N4 (in the compensation plane)

When all the following conditions are satisfied, an alarm PS0041) is issued:

(1) The tool vector changes significantly.
   α: Angle for determination set in parameter No. 19635 (The default is 45°.)
   $(Va,Vb) \leq \cos(\alpha)$(where, (Va,Vb) means an inner product.)

(2) The difference between the directions of the compensation vectors to be generated is small.
   Wa: Direction of a compensation vector to be generated by block AB
   Wb: Direction of a compensation vector to be generated by block BC
   $Wa = Va \times Ua$
   $Wb = Vb \times Ub$
   $(Wa,Wb) \geq 0$

(3) The path angle difference in the compensation plane is large.
(Ra,Rb) < 0

<2> Suppressing the issue of the alarm with a Q command
By inserting a Q command into a block that resulted in the alarm, the issue of the alarm can be suppressed.

(1) Q1 command
By inserting a Q1 command, a perpendicular vector is generated.
Example: N4 Y-200 Z-200 Q1

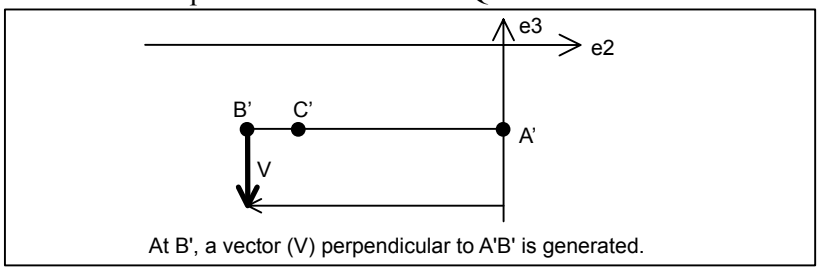


Fig. 22.5.1.1 (o) Q1 command

A perpendicular vector can also be generated by specifying G41.2 or G42.1 in the next block as follows:
Example: N6 G41.2 Y-400 Z0

(2) Q2 command
With a program specifying a linear-to-linear connection, up to two compensation vectors are generated. In this case, the second vector is deleted by inserting a Q2 command.
The Q2 command has no effect on circular interpolation.
Example: N4 Y-200 Z-200 Q2

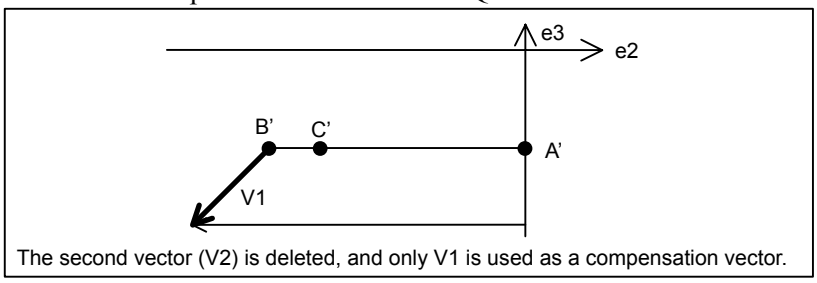


Fig. 22.5.1.1 (p) Q2 command

(3) Q3 command
By inserting a Q3 command, the issue of the alarm can be suppressed.
Example: N4 Y-200 Z-200 Q3

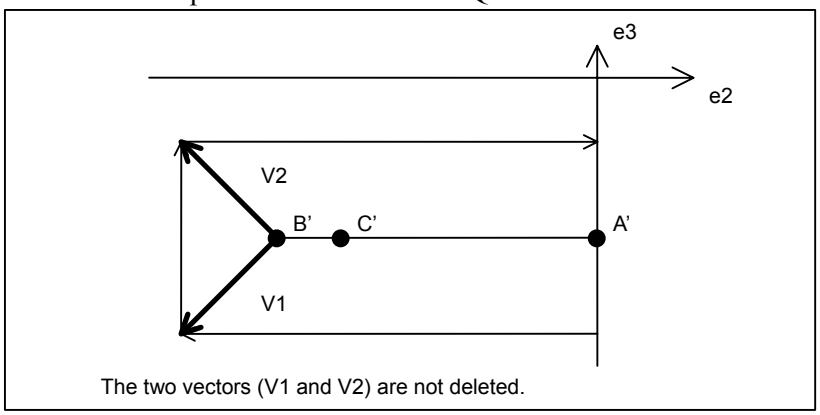


Fig. 22.5.1.1 (q) Q3 command

- Others

When the tool movement changes linear to circular (helical), circular (helical) to linear, or circular (helical) to circular (helical), the start, end, and center points of a circular (helical) movement are projected on the compensation plane that is perpendicular to the tool axis, and a compensation vector is calculated in the plane. The obtained vector is added to the originally specified position to create a position to be specified. Then, the tool is moved linearly or circularly (helically) to the created position.

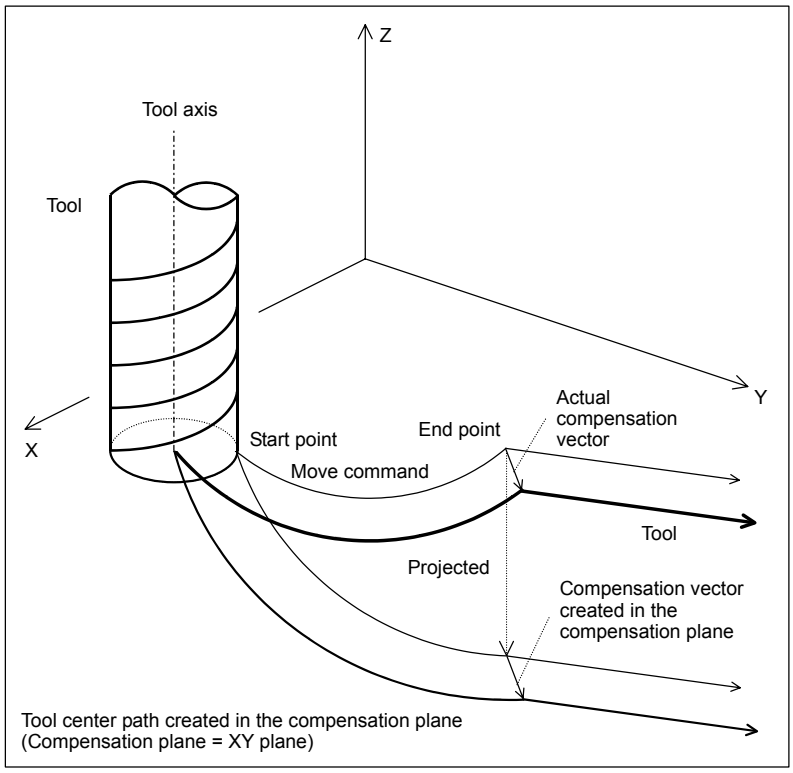


Fig. 22.5.1.1 (r) Operation during compensation

- Angle of the rotary axis for type 2 (when the movement range is not specified)

When the direction of the tool is specified by I, J, K, Q for type 2, more than two pairs of "computed angles" of the rotary axes usually exist.
The "computed angle" is the candidate angle at which the rotary axis is to be controlled in the specified tool axis direction.
The "output angle" is determined from the "computed angle" based on the "output judgment conditions" described below.
The following descriptions assume that there is no movement range specification (parameter No.19741 - No.19744 = 0).

Tool rotation type or table rotation type machine
<1> The "output angles" are represented by the computed rotary axis angle pair whose master axis (first rotary axis) moving angle is smaller. ↓ ↓ When the master axis moving angle is the same ↓
<2> The "output angles" are represented by the computed rotary axis angle pair whose slave axis (second rotary axis) moving angle is smaller. ↓ ↓ When the slave axis moving angle is the same ↓
<3> The "output angles" are represented by the computed rotary axis angle pair whose master axis (first rotary axis) angle is nearer to 0 degree (multiple of 360 degrees). ↓ ↓ When the master axis angle is equally near to 0 degree ↓
<4> The "output angles" are represented by the computed rotary axis angle pair whose slave axis (second rotary axis) angle is nearer to 0 degree (multiple of 360 degrees).
Composite type machine
<1> The "output angles" are represented by the computed rotary axis angle pair whose table (second rotary axis) moving angle is smaller. ↓ ↓ When the table moving angle is the same ↓
<2> The "output angles" are represented by the computed rotary axis angle pair whose tool (first rotary axis) moving angle is smaller. ↓ ↓ When the tool moving angle is the same ↓
<3> The "output angles" are represented by the computed rotary axis angle pair whose table (second rotary axis) angle is nearer to 0 degree (multiple of 360 degrees). ↓ ↓ When the table angle is equally near to 0 degree ↓
<4> The "output angles" are represented by the computed rotary axis angle pair whose tool (first rotary axis) angle is nearer to 0 degree (multiple of 360 degrees).

Fig. 22.5.1.1 (s) Output judgment conditions

The process of judging whether the moving angle is smaller or larger as the output judgement condition is called "movement judgement."
When bit 5 (PRI) of parameter No.19608 is 1, the movement judgements for the first rotary axis and second rotary axis are made in reverse order.

The "movement judgement" process is explained below.
When the "computed angle" is within the range between 0 and 360 degrees, it is called the "basic computed angle."
Usually, two pairs of "basic computed angles" exist.
For example, assume that a tool rotation type or table rotation type machine has rotary axis A (master) and rotary axis B (slave) and that there are two pairs of basic computed angles as follows:
(A θ1 degree; B ϕ1 degree)
(A θ2 degrees; B ϕ2 degrees) where θ1 ≤ θ2.
The "computed angle" is obtained from either of the following expressions: "basic computed angle" + 360 degrees × N or "basic computed angle" - 360 degrees × N.
The current position of rotary axis A (master) is PA, and that of rotary axis B (slave) is 0 degree.
Based on the PA angle, the "movement judgement" process is done as follows (when bit 5 (PRI) of parameter No.19608 is 0).

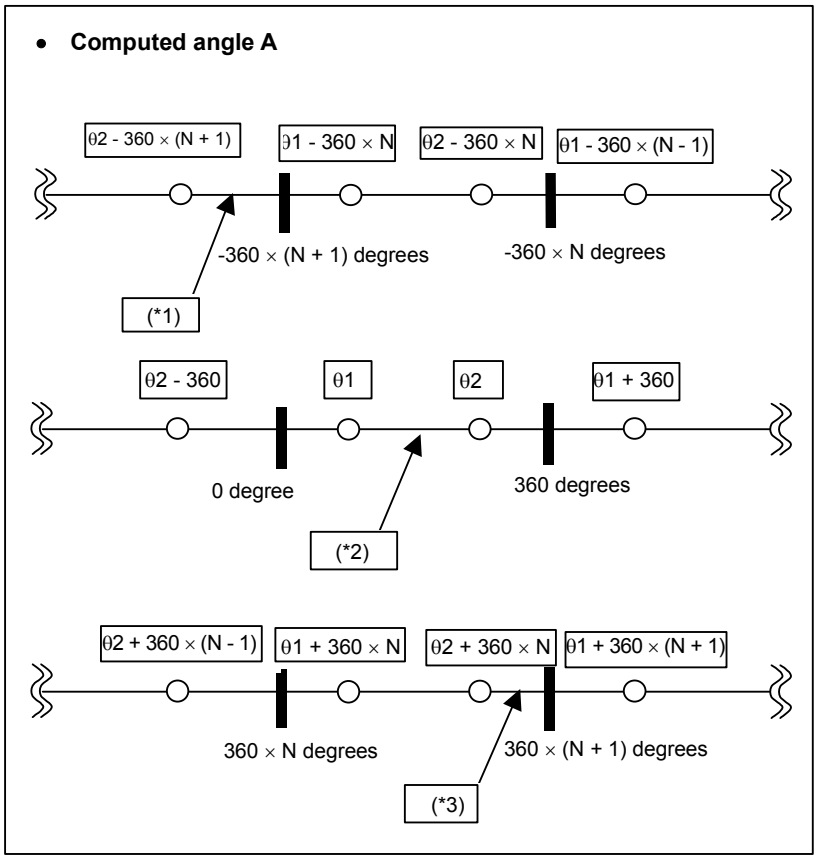


Fig. 22.5.1.1 (t) Movement judgment

When the PA angle is (*1):
The output angle is: (A θ2 - 360 × (N + 1) degrees; B ϕ2 degrees).
Namely, θ2 - 360 × (N + 1) degrees is adopted that is nearer to the computed angle of A, and ϕ2, which is the same group as θ2, is adopted as the computed angle of B.

When the PA angle is (*2):
The output angle is: (A θ1 degrees; B ϕ1 degrees).
Namely, θ1 degrees is adopted that is nearer to the computed angle of A, and ϕ1, which is the same group as θ1, is adopted as the computed angle of B.

When the PA angle is (*3):
The output angle is: (A θ2 + 360 × N degrees; B ϕ2 degrees).
Namely, θ2 + 360 × N degrees is adopted that is nearer to the computed angle of A, and ϕ2, which is the same group as θ2, is adopted as the computed angle of B.

When the moving angle of rotary axis A (master) is the same, a "movement judgement" is made for rotary axis B (slave) according to the "output judgment conditions."
If the "output angle" of rotary axis A is determined by the "movement judgement" for rotary axis A, the computed angle representing the "smaller moving angle" is adopted as the "output angle" of rotary axis B.
Similarly, if the "output angle" of rotary axis B is determined by the "movement judgement" for rotary axis B, the computed angle representing the "smaller moving angle" is adopted as the "output angle" of rotary axis A.

The "output angle" is explained below using a tool rotation type machine as an example.
This example illustrates a machine having a "BC type tool axis Z."

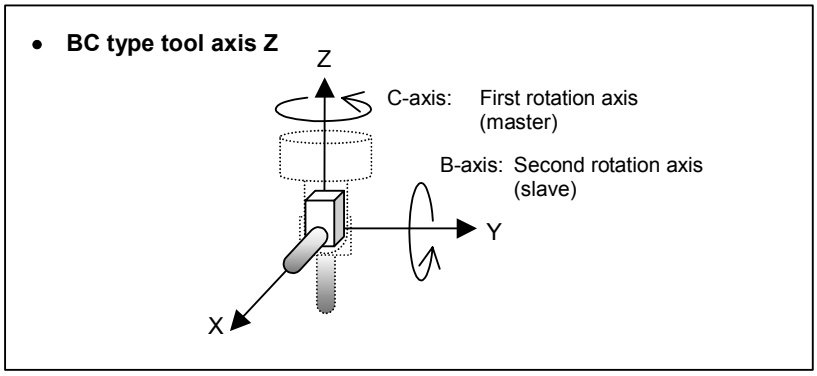


Fig. 22.5.1.1 (u) BC type tool axis Z

The following two pairs of "computed basic angles" exist that direct the tool axis toward the +X-axis direction.
(B 90 degrees; C 180 degrees)
(B 270 degrees; C 0 degree)

<1> When the current rotary axis angles are (B -70 degrees; C 30 degrees)
The "output angles" are (B -90 degrees; C 0 degree).
0 degree is adopted because it is nearer to the current position (30 degrees) of the C-axis that is the master axis. For the B-axis, 270 degrees is adopted which is the same group. However, this is changed to -90 degrees (270 degrees - 360 degrees) which is the nearest to the current position of the B-axis (-70 degrees).

<2> When the current rotary axis angles are (B 80 degrees; C 500 degrees)
The "output angles" are (B 90 degrees; C 540 degrees).
540 degrees (180 degrees + 360 degrees) is adopted because it is nearer to the current position (500 degrees) of the C-axis that is the master axis. For the B-axis, 90 degrees is adopted which is the same group.

<3> When the current rotary axis angles are (B 60 degrees; C 90 degrees)
The "output angles" are (B 90 degrees; C 180 degrees).
Since the two candidates are equally near to the current position (90 degrees) of the C-axis that is the master axis, a judgment is made based on the current position of the B-axis. 90 degrees is adopted because it is nearer to the current position (60 degrees) of the B-axis that is the slave axis. For the C-axis, 180 degrees is adopted which is the same group.

<4> When the current rotary axis angles are (B 180 degrees; C 90 degrees)
The "output angles" are (B 270 degrees; C 0 degree).
Since the two candidates are equally near to the current position (90 degrees) of the C-axis that is the master axis, a judgment is made based on the current position of the B-axis. In this case, however, the two candidates are also equally near to the current position of the B-axis (180 degrees). Therefore, the candidate is adopted in which the C-axis (master axis) is nearer to 0 degree.
That is, the pair is adopted whose C-axis angle is 0 degree and whose B-axis angle is 270 degrees.
When the slave axis angle is 0 degree, the direction of the tool axis becomes fixed regardless of the master axis angle.

In that case, the master axis does not move from the current angle.

An explanation is shown below using a machine having a "BC type tool axis Z" as an example.

- **BC type tool axis Z**

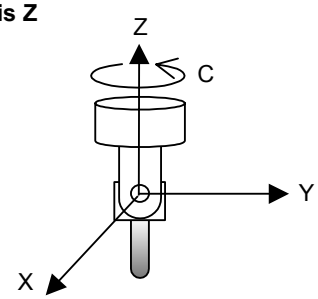


Fig. 22.5.1.1 (v) BC type tool axis Z

When the current rotary axis angles are (B 45 degrees; C 90 degrees), the "output angles" are (B 0 degree; C 90 degrees).

- Angle of the rotary axis for type 2 (when the movement range is specified)

If the upper and lower limits of the movement range of the rotary axis is specified using parameters No.19741 to No.19744, the rotary axis will move only within the specified range when the direction is specified using I, J, K, Q command for type 2 control.

Although the procedure for determining the angles is the same as that used "when the movement range is not specified," the "output angles" need to be selected from **those computed angles that are within the specified movement range for both axes**.

Tool rotation type or table rotation type machine
<1> Of the angle pairs whose master and slave axis angles are both within the specified movement range, the rotary axis angle pair whose master axis (first rotary axis) moving angle is smaller represents the "output angles."
↓
↓ When the master axis moving angle is the same
↓
<2> The "output angles" are represented by the computed rotary axis angle pair whose slave axis (second rotary axis) moving angle is smaller.
↓
↓ When the slave axis moving angle is the same
↓
<3> The "output angles" are represented by the computed rotary axis angle pair whose master axis (first rotary axis) angle is nearer to 0 degree (multiple of 360 degrees).
↓
↓ When the master axis angle is equally near to 0 degree
↓
<4> The "output angles" are represented by the computed rotary axis angle pair whose slave axis (second rotary axis) angle is nearer to 0 degree (multiple of 360 degrees).
Composite type machine
<1> Of the angle pairs whose master and slave axis angles are both within the specified movement range, the rotary axis angle pair whose table (second rotary axis) moving angle is smaller represents the "output angles."
↓
↓ When the table moving angle is the same
↓
<2> The "output angles" are represented by the computed rotary axis angle pair whose tool (first rotary axis) moving angle is smaller.
↓
↓ When the tool moving angle is the same
↓
<3> The "output angles" are represented by the computed rotary axis angle pair whose table (second rotary axis) angle is nearer to 0 degree (multiple of 360 degrees).
↓
↓ When the table angle is equally near to 0 degree
↓
<4> The "output angles" are represented by the computed rotary axis angle pair whose tool (first rotary axis) angle is nearer to 0 degree (multiple of 360 degrees).

Fig. 22.5.1.1 (w) Output judgment conditions

When bit 5 (PRI) of parameter No.19608 is 1, the movement judgements for the first rotary axis and second rotary axis are made in reverse order.

⚠ CAUTION

1 If the lower limit of the movement range is larger than the upper limit, alarm PS5459 occurs when G43.5 is specified.
2 If no "computed angle" is found within the movement range because the range is too small, alarm PS5459 occurs.
3 If 0 is set for both parameters that specify the upper and lower limits of the movement range, the tool operates assuming that there is no range specification.
4 When the rotary axis rollover function or rotary axis control function is used (in which case, set parameter No.1260 (amount of movement per rotation of the rotary axis) to 360 degrees), the tool does not move beyond 0 degree (360 degrees) (does not take the shortcut) if the movement range is set between 0 and 360 degrees. Also, do not specify a negative value or a value larger than 360 degrees for the movement range.

An example of the "movement judgement" process is given below.
Assume that a tool rotation type or table rotation type machine has rotary axis A (master) and rotary axis B (slave) and that there are two pairs of basic computed angles as follows:
(A θ1 degree; B ϕ1 degree)
(A θ2 degrees; B ϕ2 degrees) where θ1 ≤ θ2.
The "computed angle" is obtained from either of the following expressions: "basic computed angle" + 360 degrees × N or "basic computed angle" - 360 degrees × N.
Assume that the current positions and movement ranges of rotary axis A (master) and rotary axis B (slave) are as shown in Fig. 22.5.1.1 (x), Fig. 22.5.1.1 (y).

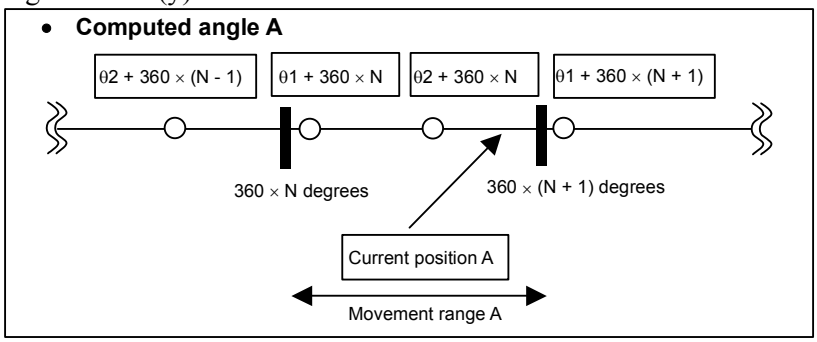


Fig. 22.5.1.1 (x) Computed angle of rotary axis A and its current position and movement range

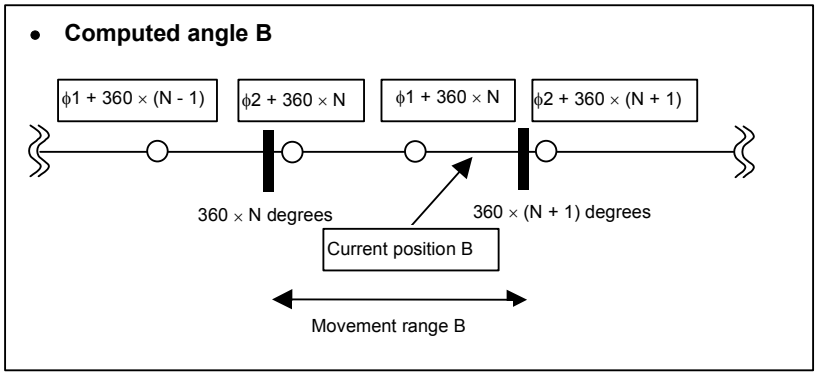


Fig. 22.5.1.1 (y) Computed angle of rotary axis B and its current position and movement range

When the two axes have a positional relationship as shown in the figure, the output angle of rotary axis A is (θ2 + 360 × N) degrees and that of rotary axis B is (ϕ2 + 360 × N) degrees (when bit 5 (PRI) of parameter No.19608 is set to 0).
More concretely, from the computed angles obtained for rotary axis A, the nearest angle within the movement range, i.e. θ2 + 360 × N degrees, is first adopted. Then, from the computed angles obtained for rotary axis B, the angle belonging to the same group as θ2, i.e. ϕ2 + 360 × N, is adopted.

Note that, in this example, the output angles and moving direction differ depending on whether the movement range is specified or not (0 to 360 degrees), even if N is set to 0 and coordinates are rounded to 0 to 360 degrees.
Namely, if the movement range is not specified, θ1 + 360 degrees nearest to the current position is adopted as the computed angle for rotary axis A and, from the computed angles belonging to the same group as θ1, ϕ1 degrees nearest to the current position is adopted as the computed angle for rotary axis B. Rotary axis A moves in the plus direction. As its coordinate is rounded to 360 degrees, rotary axis A reaches θ1 degrees while moving in the plus direction.
By contrast, when the movement range is set to 0 to 360 degrees, the output angles are (A θ2 degrees; B ϕ2 degrees). Neither rotary axis A nor B moves in a way that it exceeds 0 degree (360 degrees).

22.5.1.2 Leading edge offset

Overview

Leading edge offset is a type of cutter compensation used when a workpiece is machined with the edge of a tool. The tool is automatically shifted by the amount of cutter compensation on the line where a plane formed by a tool vector and tool movement direction meets a plane perpendicular to the tool axis direction.

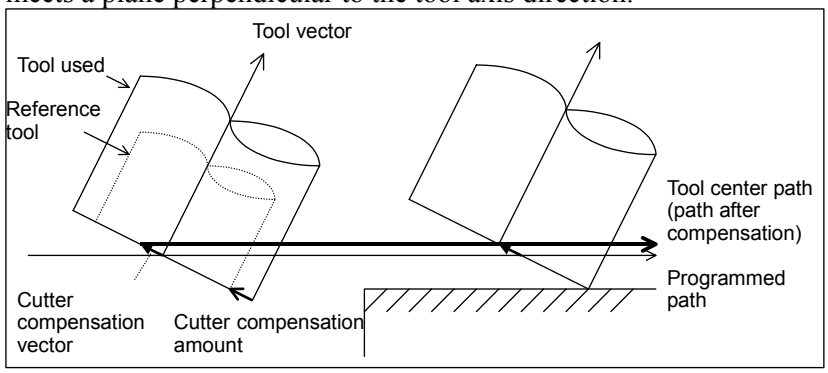


Fig. 22.5.1.2 (a) Leading edge offset

Format

- Leading edge offset

```
G41.3 D_ ;
```

- Canceling the leading edge offset

```
G40 ;
```

NOTE

1 G41.3 can be specified only in the G00 and G01 modes. In a block containing G41.3 or G40, only addresses D, O, and N can be specified.
2 The block that follows a block containing a G41.3 command must contain a move command. In the block after G41.3, however, a tool movement in the same direction as the tool axis direction or the opposite direction cannot be specified.
3 No modal G code that belongs to the same group as G00 and G01 can be specified in the G41.3 mode. If such a modal G code is specified, alarm PS5460 is issued.
4 Leading edge offset does not provide type 2 commands. It does not allow the tool direction to be specified with the I, J, and K commands.

Explanation

- Operation at startup and cancellation

The operation performed at leading edge offset startup and cancellation does not vary. When G41.3 is specified, the tool is moved by the amount of compensation (Vc) in the plane formed by the movement vector (V_M) of the block after the G41.3 block and the tool vector (V_T) obtained at the time of G41.3 specification. The tool movement is perpendicular to the tool vector. When G40 is specified, the tool is moved to cancel V_C. The following illustrates how the compensation is performed:

<1> When the tool vector is inclined in the tool movement direction

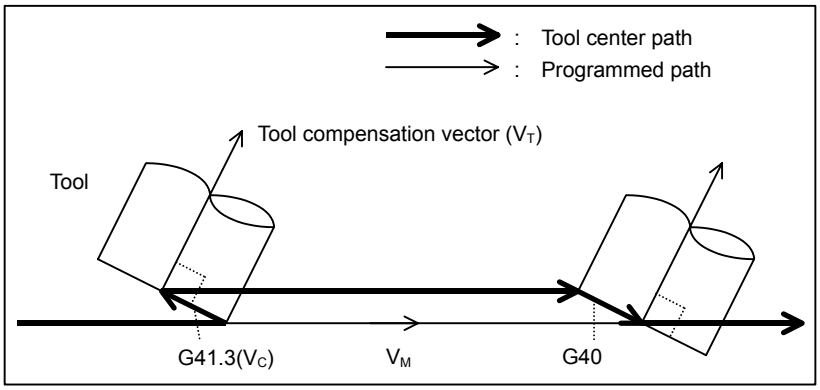


Fig. 22.5.1.2 (b) When the tool vector is inclined in the tool movement direction

<2> When the tool vector is inclined in the direction opposite to the tool movement direction

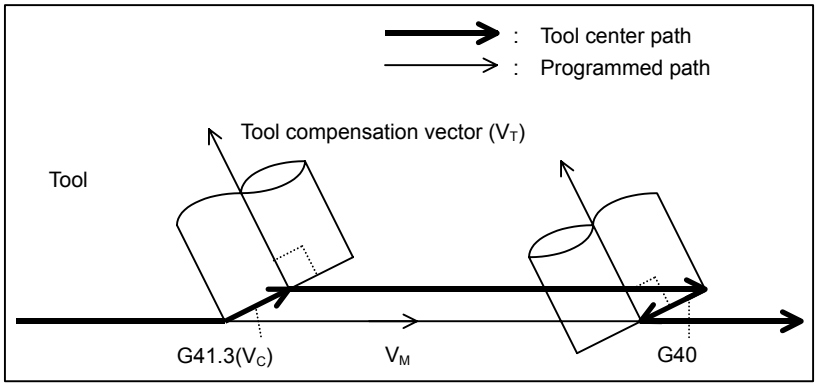


Fig. 22.5.1.2 (c) When the tool vector is inclined in the direction opposite to the tool movement direction

- Operation during compensation

The tool center moves so that a compensation vector (VC) perpendicular to the tool vector (VT) is created in the plane formed by the tool vector (VT) at the end point of each block and the movement vector (VM) of the next block.

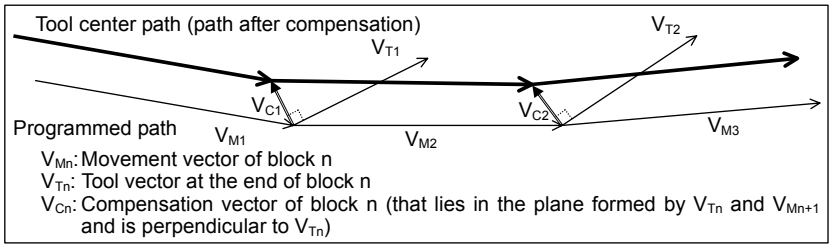


Fig. 22.5.1.2 (d) Operation during compensation

If a G code or M code that suppresses buffering is specified in the compensation mode, however, the compensation vector created immediately before the specification is maintained.

When a block specifying no movement (including a block containing a move command for a rotary axis only) is specified in the compensation mode, the movement vector of the block after the block specifying no movement is used to create a compensation vector as shown below.

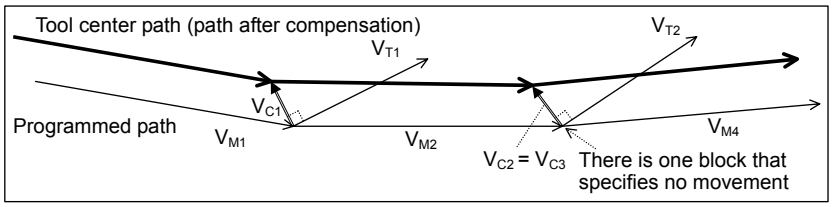


Fig. 22.5.1.2 (e) When there is one block that specifies no movement

If block 3 specifies no movement, the compensation vector of block 2 (VC2) is created in a plane formed by the movement vector of block 4 (VM4) and the tool vector (VT2) at the end of block 2. VC2 is perpendicular to VT2.

> ⚠ **CAUTION**
> If two or more successive blocks specify no movement, the previously created compensation vector is maintained. However, such specification should be avoided.

- Block immediately before the offset cancel command (G40)

In the block immediately before the offset cancel command (G40), a compensation vector is created from the movement vector of that block and the tool vector at the end point of the block as shown below.

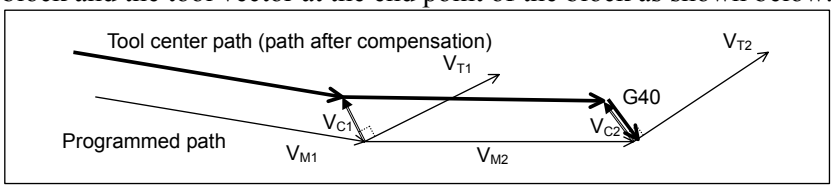


Fig. 22.5.1.2 (f) Block immediately before G40

The compensation vector (VC2) of block 2 is created in a plane formed by the tool vector (VT2) at the end point of block 2 and the movement vector (VM2) of block 2. VC2 is perpendicular to VT2.

- Compensation performed when θ is approximately 0°, 90°, or 180°

When the included angle θ between VMn+1 and VTn is regarded as 0°, 180°, or 90°, the compensation vector is created in a different way. So, when creating a program, note the following points:

<1> Setting a variation range for determining θ to be 0°, 180°, or 90°
When the included angle (θ) between the tool vector (VTn) and movement vector (VMn+1) becomes approximately 0°, 180°, or 90°, the system regards θ as 0°, 180°, or 90°, respectively, then creates a compensation vector which is different from the normal compensation vector. The variation range used for determining θ to be 0°, 180°, and 90° is set in parameter No. 19631. For example, let the angle set in this parameter be Δθ. Then, the system regards θ as follows:

(1) If $0 \le \theta \le \Delta\theta$, θ is regarded as 0°.

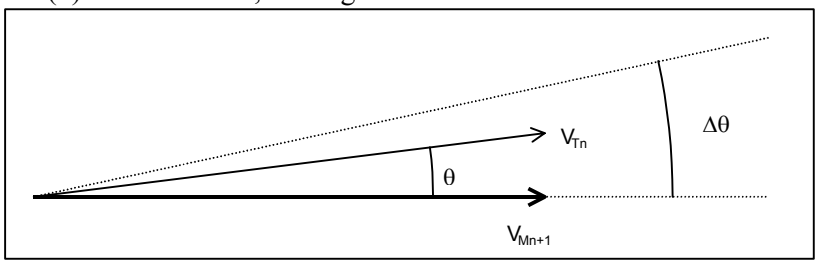


Fig. 22.5.1.2 (g) Determination of θ = 0°

(2) If $(180 - \Delta\theta) \le \theta \le 180$, θ is regarded as 180°.

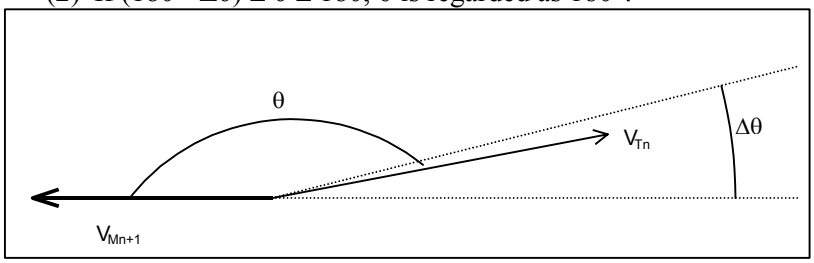


Fig. 22.5.1.2 (h) Determination of θ = 180°

(3) If $(90 - \Delta\theta) \le \theta \le (90 + \Delta\theta)$, θ is regarded as 90°.

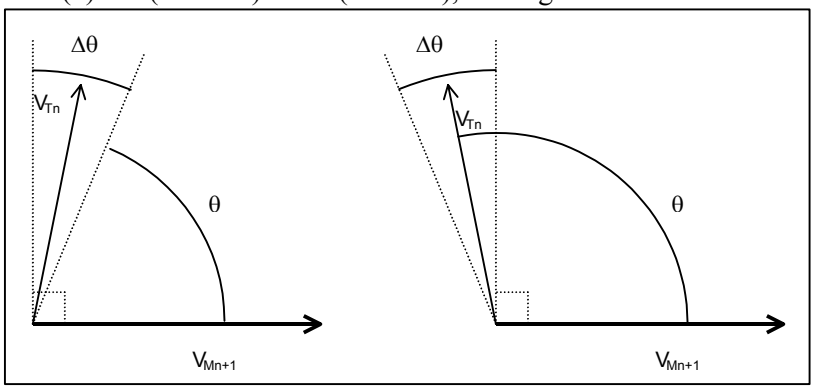


Fig. 22.5.1.2 (i) Determination of θ = 90°

<2> Compensation vector when θ is regarded as 0° or 180°

At startup (when G41.3 is specified), alarm PS5408 is issued.

This means that the tool vector of a block and the movement vector of the next block must not point in the same direction or in opposite directions at startup.

At other than startup, the previously created compensation vector is maintained without change.

If the included angles between VT2 and VM3, VT3 and VM4, and VT4 and VM5 are regarded as 0°, compensation vector VC1 of block 1 is maintained as compensation vectors VC2, VC3, and VC4 of blocks 2, 3, and 4, respectively.

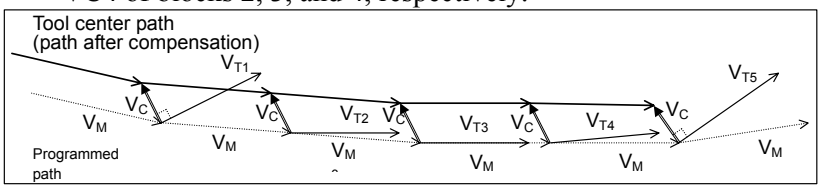


Fig. 22.5.1.2 (j) When θ = 0° is determined

If the included angles between VT2 and VM3, VT3 and VM4, and VT4 and VM5 are regarded as 180°, compensation vector VC1 of block 1 is maintained as compensation vectors VC2, VC3, and VC4 of blocks 2, 3, and 4, respectively.

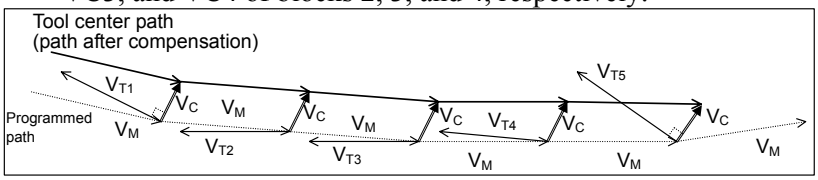


Fig. 22.5.1.2 (k) When θ = 180° is determined

<3> Compensation vector when θ is regarded as 90°

If the previous compensation vector (VCn-1) points in the opposite direction ((VMn × VTn-1) × VTn-1 direction) to VMn with respect to VTn-1, the current compensation vector (VCn) is created so that it also points in the (VMn+1 × VTn) × VTn direction.

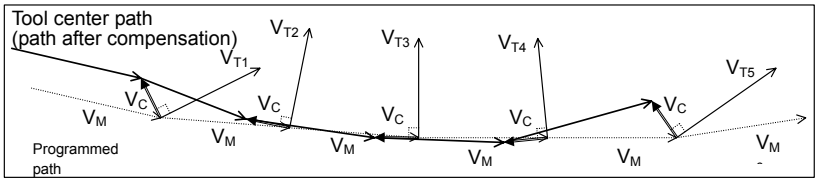


Fig. 22.5.1.2 (l) When θ = 90° is determined 1

If the previous compensation vector (VCn-1) points in the same direction (-(VMn × VTn-1) × VTn-1 direction) as VMn with respect to VTn-1, the current compensation vector (VCn) is created so that it also points in the -(VMn+1 × VTn) × VTn direction.

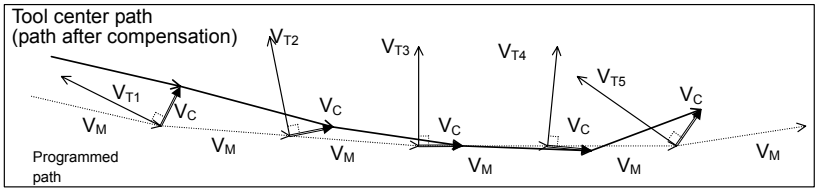


Fig. 22.5.1.2 (m) When θ = 90° is determined 2

22.5.1.3 Tool tip position (cutting point) command

Overview

For machines having a rotary axis for rotating a tool, this function performs 3-dimensional cutter compensation at the tool tip position if a programmed point is specified with a pivot point.
When this function is used, the programmed point (pivot point) is converted into a tool tip position (cutting point) and a vector of 3-dimensional cutter compensation is calculated for the position obtained by the conversion. Then, the programmed point (pivot point) is compensated for with the vector of 3-dimensional cutter compensation.
If the tool side offset (G41.2/G42.2) of 3-dimensional cutter compensation is performed, the operation of this function is as follows:

(1) If parameter No. 19632 is 0
The vector of 3-dimensional cutter compensation is calculated at the programmed point (pivot point).
(2) If parameter No. 19632 is not 0 (this function)
The vector of 3-dimensional cutter compensation is calculated at the tool tip position (cutting point).

Explanation

- Operation explanation

This function calculates a vector at the tool tip position for the 3-dimensional cutter compensation function as described below.

(1) Convert the programmed coordinates from a programmed point (pivot point) to a tool tip position (cutting point). Parameter No. 19632 is used to store the distance from the programmed point (pivot point) to the tool tip position (cutting point).
(2) Calculate a vector of 3-dimensional cutter compensation at the tool tip position (cutting point).
(3) Add the cutter compensation vector to the programmed point (pivot point).

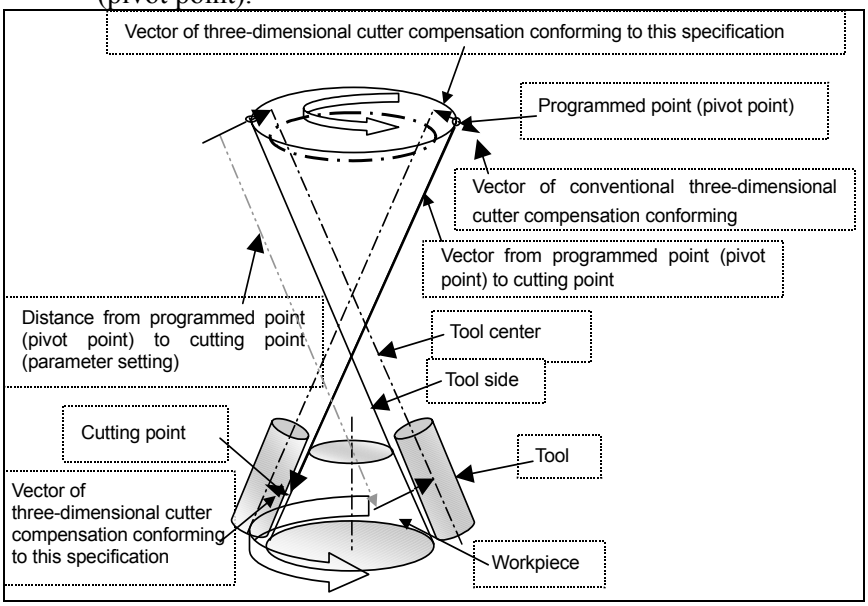


Fig. 22.5.1.3 (a) Basic operation (for G42.2)

- Operation example

For a machine configuration in which the tool axis direction is along the Z-axis and the rotary axes are the B and C axes (Fig. 22.5.1.3 (b))

LC: Parameter (No. 19632) specifying the distance from the programmed point (pivot point) to the tool tip position (cutting point)

b: Specified B-axis value, c: Specified C-axis value

Q = (Qx,Qy,Qz): Programmed point (pivot point)

P, R: Programmed points (pivot points) in the preceding and succeeding blocks

QT = (QTx,QTy,QTz): Tool position (tool tip position (cutting point)) resulting from conversion

PT, RT: Tool positions (tool tip positions (cutting positions)) in the preceding and succeeding blocks

Then,

<1> Convert programmed points (pivot points) P, Q, and R to tool tip positions (cutting points) PT, QT, and RT.

$QTx = LC \times \sin(b) \times \cos(c) + Qx$

$QTy = LC \times \sin(b) \times \sin(c) + Qy$

$QTz \quad LC \times \cos(b) + Qz$

(The same applies to PT and RT.)

<2> Calculate vector VD of 3-dimensional cutter compensation from tool tip positions (cutting points) PT, QT, and RT and tool gradient VT.

<3> Add cutter vector VD to programmed point (pivot point) Q and set the result as the end point position.

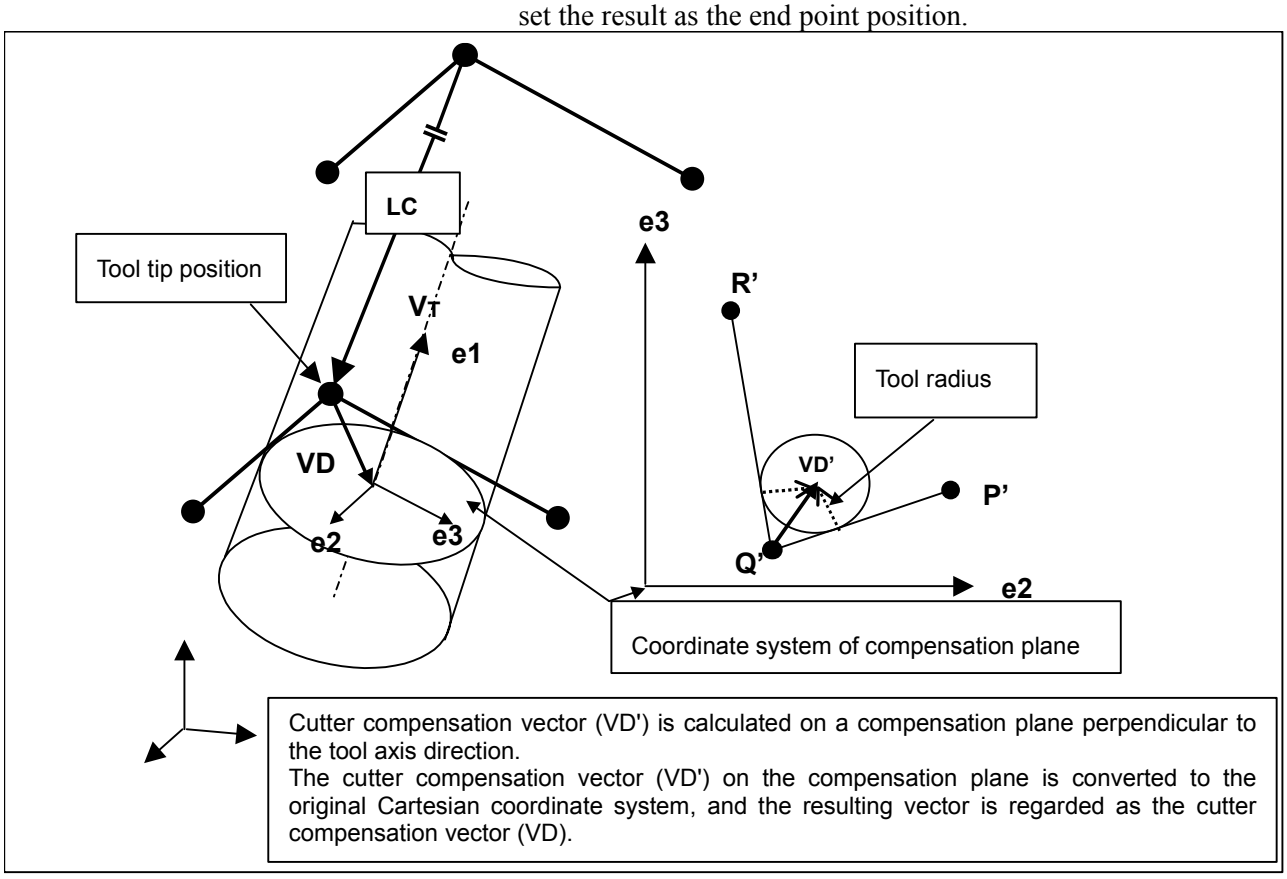


Fig. 22.5.1.3 (b) Operation

NOTE
1 This function is disabled for leading edge offset.
2 With a command for a rotary axis only, this function does not calculate a cutter compensation vector.
3 This function cannot be used in the three-dimensional coordinate system conversion mode.
4 In addition to the cautions given here, the cautions on the 3-dimensional cutter compensation function apply to this function.

22.5.2 Cutter Compensation in Table Rotation Type Machine

Overview

Cutter compensation can be performed for a 5-axis machine having a rotary table as shown in Fig. 22.5.2 (a).
Shown below is a 5-axis machine that has table rotation axis A on the X-axis and table rotation axis B on the Y-axis.
This machine configuration is used as a sample configuration in the following explanation unless otherwise noted:

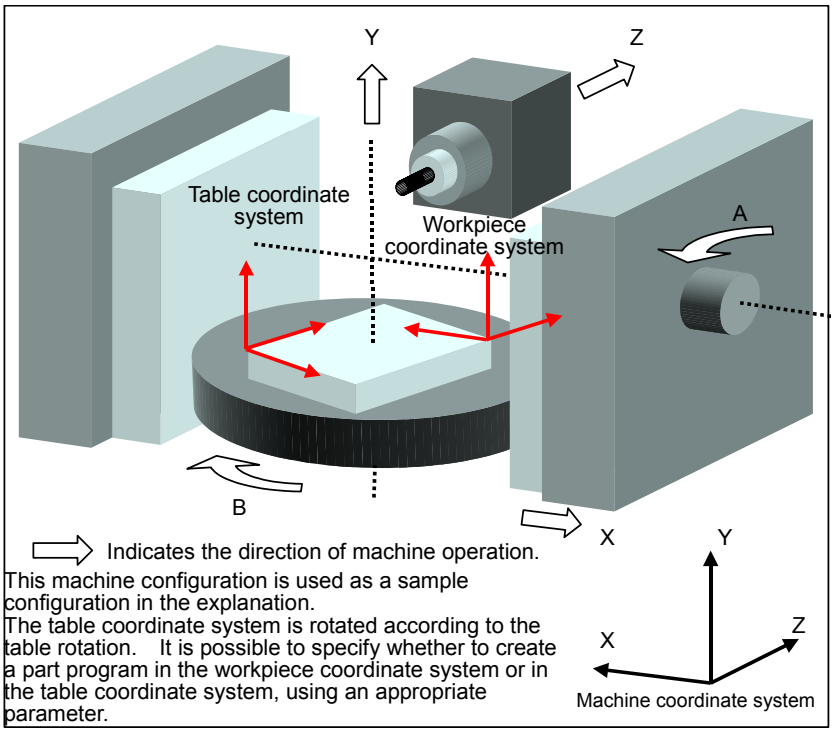


Fig. 22.5.2 (a) Machine having a rotary table

Format

- Startup (start of cutter compensation) (type 1)

When bit 1 (SPG) of parameter No. 19607 is 0

G41.2 (or G42.2) IP_ D_ ;

G41.2: Cutter compensation left (group 07)

G42.2: Cutter compensation right (group 07)

IP_: Value specified for axis moving as viewed from the programming coordinate system (including rotary axis)

D_: Code specifying the cutter compensation amount (1 to 3 digits)

When bit 1 (SPG) of parameter No. 19607 is 1

G41.4 (or G42.4) IP_ D_ ;

G41.4: Cutter compensation left (group 07)

G42.4: Cutter compensation right (group 07)

IP_: Value specified for axis moving as viewed from the programming coordinate system (including rotary axis)

D_: Code specifying the cutter compensation amount (1 to 3 digits)

NOTE

1 In a table rotation type machine (parameter No. 19680 = 12), if an attempt is made to issue G41.4 or G42.4 with bit 1 (SPG) of parameter No. 19607 equal to 0, alarm PS0010 is generated.
2 In a table rotation type machine, if an attempt is made to issue G41.2 or G42.2 with bit 1 (SPG) of parameter No. 19607 equal to 1, alarm PS5460 is generated.
3 In machine not of the table rotation type machine, if an attempt is made to issue G41.4 or G42.4 with bit 1 (SPG) of parameter No. 19607 equal to 1, alarm PS5460 is generated.

- Startup (start of cutter compensation) (type 2)

G41.6(or G42.6) IP_ D_ Q_ ;
IP_ I_ J_ K_ ;
:

G41.6: Cutter compensation left (group 07)
G42.6: Cutter compensation right (group 07)
IP_: Value specified for axis moving as viewed from the programming coordinate system (including rotary axis)
D_: Code specifying the cutter compensation amount (1 to 3 digits)
Q_: Tool's angle of gradient (in degrees)
I_ J_ K_: Tool axis direction at the block end point as viewed from the programming coordinate system

For type 2, do not specify a rotation axis but specify the direction at the tool end point as viewed from the programming coordinate system (table coordinate system), with I, J, and K. Specifying a rotation axis causes alarm PS5460 to be generated.

In a tool rotation type machine, I, J, and K can be specified in a G41.6/G42.6 command block; in a table rotation type machine, however, they cannot. If an attempt is made to specify them, alarm PS5460 is generated.
The following are the notes on type 2.

NOTE
1 If one or two of I, J, and K are omitted, the omitted ones of I, J, and K are assumed to be 0.
2 In a block in which all of I, J, and K are omitted, the values of I, J, and K in the previous block are used.
3 If there is only one rotation axis (a hypothetical axis is used), type 2 cannot be used. In this case, if an attempt is made to issue G41.6/G42.6, alarm PS5460 is generated.
4 If using the rotation axis rollover function or rotary axis control function, specify 360 degrees in parameter No. 1260 (mount of travel per rotation about the rotation axis).
5 They can be used only with the settings that select the table coordinate system as a programming coordinate system (bit 5 (WKP) of parameter No.19696 = 0 and bit 4 (TBP) of parameter No.19746 = 1). If an attempt is made to issue G41.6/G42.6 with the settings that select the workpiece coordinate system as a programming coordinate system, alarm PS5460 is generated.

- Canceling the cutter compensation

G40 IP_ ;
G40: Cutter compensation cancellation (group 07)
IP_: Value specified for axis movement

- Selecting an offset plane

When bit 1 (PTD) of parameter No. 19746 is 1, compensation is performed on the selected plane, on the assumption that the tool is pointing to the direction perpendicular to that plane.

Offset plane	Plane selection command	IP_
XpYp	G17 ;	Xp_Yp_
ZpXp	G18 ;	Xp_Zp_
YpZp	G19 ;	Yp_Zp_

(Example: In the case of Fig. 22.5.2 (a), the XpYp plane is selected.)
The two axes of a selected plane must be included in the three basic axes (the axes for which parameter No. 1022 is set to 1 to 3).

When bit 1 (PTD) of parameter No. 19746 is 0, compensation is performed on the plane perpendicular to the tool direction specified with parameters Nos. 19697, 19698, and 19699, regardless of which plane is selected.

NOTE
This function is enabled for tool side offset only. If leading edge offset is specified, alarm PS5460 is generated.

Explanation

- Tool's angle of gradient in type 2

For type 2 of 3-dimensional cutter compensation, the tool's angle of gradient can be specified with address Q in a G41.6/G42.6 command block. The tool's angle of gradient refers to the angle by which the tool direction to be assumed when machining is actually performed is inclined from the direction specified with (I, J, K) toward the traveling direction on the plane formed by the tool direction specified with (I, J, K) and the traveling direction in the programming coordinate system. (See Fig. 22.5.2 (b).)

Because in general, the normal direction of the machining surface is specified with (I, J, K), if it is desired to incline the tool direction to be assumed when machining is actually performed from the normal direction toward the traveling direction, correction may be performed with a Q command.

If the direction specified with (I, J, K) matches the tool direction to be assumed when machining is actually performed, no Q command is necessary.

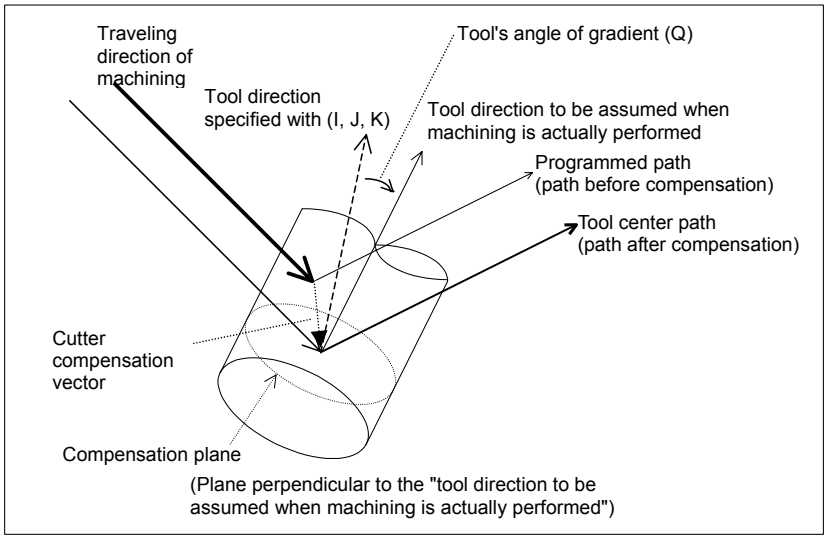


Fig. 22.5.2 (b) Tool's angle of gradient in type 2

- Cutter compensation

The function for 3-dimensional cutter compensation in a table rotation type machine basically performs operations in conformance with 3-dimensional cutter compensation in a tool rotation type machine. The operations different from those of 3-dimensional cutter compensation in a tool rotation type machine are mainly described below.

For the specifications and cautions not mentioned here, see the description of 3-dimensional cutter compensation in a tool rotation type machine.

- Startup

When cutter compensation for the rotary table is specified (G41.2 or G42.2, G41.4 or G42.4, a dimension word other than 0 in the offset plane, or a D code other than D0) in the offset cancel mode, the CNC enters the offset mode.
Startup is specified with positioning (G00) or linear interpolation (G01).

NOTE
If a command such as circular interpolation (G02 or G03) and involute interpolation (G02.2 or G03.2) is specified at startup, alarm PS0034 is issued.

- Commands in the offset mode

In the offset mode, compensation is performed for positioning (G00) and linear interpolation (G01).

NOTE
If a command such as circular interpolation (G02 or G03) and involute interpolation (G02.2 or G03.2) is specified in the mode for 3-dimensional cutter compensation in a table rotation type machine, alarm PS5460 is issued.

- Offset mode cancellation

If a block satisfying either of the following conditions is executed in the offset mode, the CNC enters the offset cancel mode:

1 G40 is specified.
2 0 is specified as a code specifying the cutter compensation amount (D code).

When offset cancellation is to be performed, neither circular (G02 or G03) nor involute (G02.2 or G03.2) command can be specified. If offset cancellation is specified in such a mode, alarm PS0034 is issued.

- If selecting the table coordinate system as a programming coordinate system

If bit 4 (TBP) of parameter No. 19746 is 1 and bit 5 (WKP) of parameter No. 19696 is 0, specifying 3-dimensional cutter compensation causes the table coordinate system to be selected as a programming coordinate system. The table coordinate system refers to the workpiece coordinate system fixed to the table when 3-dimensional cutter compensation.
In the blocks subsequent to the one in which 3-dimensional cutter compensation is specified, the table coordinate system rotates with the rotation of the table.
A linear axis (X, Y, Z) command is assumed to be issued on the table coordinate system.
Specifying linear interpolation causes cutter compensation to be performed on the linear interpolation command on the table coordinate system.
The table coordinate system does not rotate with the rotation of the tool head.

The cancel (G40) block becomes a block that suppresses buffering.

It is possible to switch between absolute and relative coordinate displays, using bit 2 (DET) of parameter No. 19608 as follows:

- When DET is 0, the position in the table coordinate system is displayed.
- When DET is 1, the position in the workpiece coordinate system is displayed.

Note, however, the distance to go is always that in the programming coordinate system.

NOTE

1 If the 3-dimensional cutter compensation mode is entered when the table coordinate system is used as the programming coordinate system, look-ahead acceleration/deceleration before interpolation is automatically enabled. Be sure to specify the following parameters:
   (1) Bit 1 (LRP) of parameter No.1401 = 1:
   Linear rapid traverse
   (2) Bit 5 (FRP) of parameter No.19501 = 1:
   Acceleration/deceleration before interpolation is used for rapid traverse.
   (3) Parameter No. 1671:
   Acceleration of acceleration/deceleration before interpolation for rapid traverse
   (4) Parameter No. 1672:
   Change time for bell-shaped acceleration/deceleration before interpolation for rapid traverse
   (5) Parameter No. 1660:
   Maximum permissible acceleration for acceleration/deceleration before interpolation
   If they are not specified, alarm PS5463 is generated.

NOTE
2 When table rotation axis movement is specified in the start block of 3-dimensional cutter compensation, after the movement is completed, the workpiece coordinate system is fixed to the table and assumed to be a table coordinate system.
3 In the mode of 3-dimensional cutter compensation, do not change the workpiece coordinate system or change the workpiece offset value. If an attempt is made to specify workpiece coordinate system selection (G54 to G59), alarm PS5460 is generated.

- If selecting the workpiece coordinate system as a programming coordinate system

If bit 4 (TBP) of parameter No. 19746 is 0 or if bit 4 (TBP) of parameter No. 19746 is 1 and bit 5 (WKP) of parameter No. 19696 is 1, the programming coordinate system does not rotate with the rotation of the table, being fixed to the workpiece coordinate system.

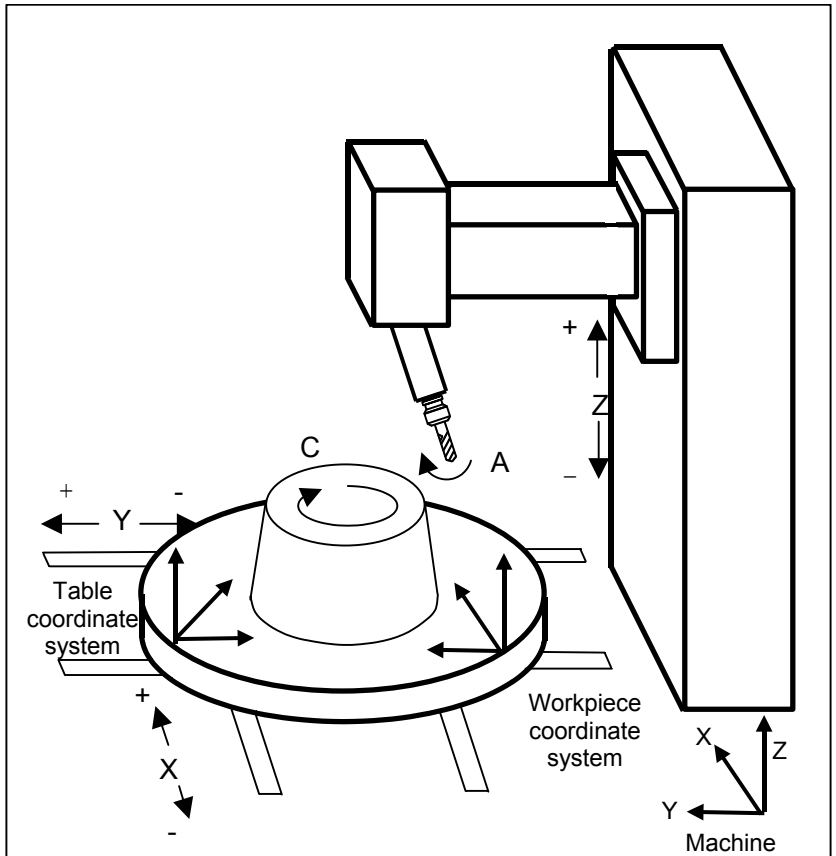
- Angle of the rotation axis in type 2

For an explanation of how the rotation axis end point is determined when the tool direction is specified with an IJKQ command in type 2, see the explanations of cutter compensation in a tool rotation type machine, "Angle of the rotation axis in type 2 (if the operating range is not specified)" and "Angle of the rotation axis in type 2 (if the operating range is specified)".

22.5.3 Cutter Compensation in Composite Type Machine

Overview

This function can perform 3-dimensional cutter compensation in a 5-axis machine having a rotary table and a tool axis as shown in Fig. 22.5.3 (a).
Shown below is a 5-axis machine that has tool axis A on the X-axis (the tool axis direction is along the Z-axis) and table rotation axis B on the Y-axis.
This machine configuration is used as a sample configuration in the following explanation unless otherwise noted:



The table coordinate system is rotated according to the table rotation. It is possible to specify whether to create a part program in the workpiece coordinate system or in the table coordinate system, using an appropriate parameter.

Fig. 22.5.3 (a) Machine having a tool rotation axis and table rotation axis

Format

- Startup (start of cutter compensation) (type 1)

When bit 1 (SPG) of parameter No. 19607 is 0

G41.2 (or G42.2) IP_ D_ ;

G41.2: Cutter compensation left (group 07)

G42.2: Cutter compensation right (group 07)

IP_: Value specified for axis moving as viewed from the programming coordinate system (including rotary axis)

D_: Code specifying the cutter compensation amount (1 to 3 digits)

When bit 1 (SPG) of parameter No. 19607 is 1

G41.5 (or G42.5) IP_ D_ ;

G41.5: Cutter compensation left (group 07)

G42.5: Cutter compensation right (group 07)

IP_: Value specified for axis moving as viewed from the programming coordinate system (including rotary axis)

D_: Code specifying the cutter compensation amount (1 to 3 digits)

NOTE

1 In a composite type machine (parameter No. 19680 = 21), if an attempt is made to issue G41.5 or G42.5 with bit 1 (SPG) of parameter No. 19607 equal to 0, alarm PS0010 is generated.

2 In a composite type machine, if an attempt is made to issue G41.2 or G42.2 with bit 1 (SPG) of parameter No. 19607 equal to 1, alarm PS5460 is generated.

3 In machine not of the composite type machine, if an attempt is made to issue G41.5 or G42.5 with bit 1 (SPG) of parameter No. 19607 equal to 1, alarm PS5460 is generated.

- Startup (start of cutter compensation) (type 2)

G41.6 (or G42.6) IP_ D_ Q_ ;
IP_ I_ J_ K_ ;
:

G41.6: Cutter compensation left (group 07)

G42.6: Cutter compensation right (group 07)

IP_: Value specified for axis moving as viewed from the programming coordinate system (including rotary axis)

D_: Code specifying the cutter compensation amount (1 to 3 digits)

Q_: Tool's angle of gradient (in degrees)

I_ J_ K_: Tool axis direction at the block end point as viewed from the programming coordinate system

For type 2, do not specify a rotation axis but specify the direction at the tool end point as viewed from the programming coordinate system (table coordinate system), with I, J, and K. Specifying a rotation axis causes alarm PS5460 to be generated.

In a tool rotation type machine, I, J, and K can be specified in a G41.6/G42.6 command block; in a composite type machine, however, they cannot. If an attempt is made to specify them, alarm PS5460 is generated.
The following are the notes on type 2.

NOTE
1 If one or two of I, J, and K are omitted, the omitted ones of I, J, and K are assumed to be 0.
2 In a block in which all of I, J, and K are omitted, the values of I, J, and K in the previous block are used.
3 If there is only one rotation axis (a hypothetical axis is used), type 2 cannot be used. In this case, if an attempt is made to issue G41.6/G42.6, alarm PS5460 is generated.
4 If using the rotation axis rollover function or rotary axis control function, specify 360 degrees in parameter No. 1260 (mount of travel per rotation about the rotation axis).
5 They can be used only with the settings that select the table coordinate system as a programming coordinate system (bit 5 (WKP) of parameter No.19696 = 0 and bit 4 (TBP) of parameter No.19746 = 1). If an attempt is made to issue G41.6/G42.6 with the settings that select the workpiece coordinate system as a programming coordinate system, alarm PS5460 is generated.

- Canceling the cutter compensation

G40 IP_ ;
G40: Cutter compensation cancellation (group 07)
IP_: Value specified for axis movement

NOTE
This function is enabled for tool side offset only. If leading edge offset is specified, alarm PS5460 is generated.

Explanation

- Tool's angle of gradient in type 2

For type 2 of 3-dimensional cutter compensation, the tool's angle of gradient can be specified with address Q in a G41.6/G42.6 command block. The tool's angle of gradient refers to the angle by which the tool direction to be assumed when machining is actually performed is inclined from the direction specified with (I, J, K) toward the traveling direction on the plane formed by the tool direction specified with (I, J, K) and the traveling direction in the programming coordinate system. (See Fig. 22.5.2 (b).)
Because in general, the normal direction of the machining surface is specified with (I, J, K), if it is desired to incline the tool direction to be assumed when machining is actually performed from the normal direction toward the traveling direction, correction may be performed with a Q command.
If the direction specified with (I, J, K) matches the tool direction to be assumed when machining is actually performed, no Q command is necessary.

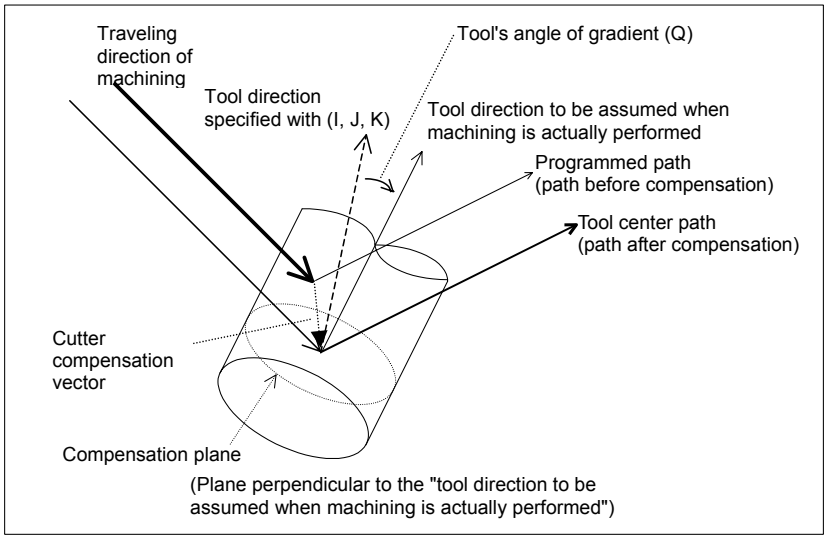


Fig. 22.5.3 (b) Tool's angle of gradient in type 2

- Cutter compensation

The function for 3-dimensional cutter compensation in a composite type machine basically performs operations in conformance with 3-dimensional cutter compensation in a tool rotation type machine. The operations different from those of 3-dimensional cutter compensation in a tool rotation type machine are mainly described below.
For the specifications and cautions not mentioned here, see the description of 3-dimensional cutter compensation in a tool rotation type machine.

- Startup

When 3-dimensional cutter compensation in a composite type machine (G41.2 or G42.2, G41.5 or G42.5, or a D code other than D0) is specified in the offset cancel mode, the CNC enters the offset mode. Startup is specified with positioning (G00) or linear interpolation (G01).

NOTE
If a command such as circular interpolation (G02 or G03) and involute interpolation (G02.2 or G03.2) is specified at startup, alarm PS0034 is issued.

- Commands in the offset mode

In the offset mode, compensation is performed for positioning (G00) and linear interpolation (G01).

NOTE
If a command such as circular interpolation (G02 or G03) and involute interpolation (G02.2 or G03.2) is specified in the mode for 3-dimensional cutter compensation in a composite type machine, alarm PS5460 is issued.

- Offset mode cancellation

If a block satisfying either of the following conditions is executed in the offset mode, the CNC enters the offset cancel mode:

1 G40 is specified.
2 0 is specified as a code specifying the cutter compensation amount (D code).

When offset cancellation is to be performed, neither circular (G02 or G03) nor involute (G02.2 or G03.2) command can be specified. If offset cancellation is specified in such a mode, alarm PS0034 is issued.

NOTE
1 This function cannot be used in the three-dimensional coordinate system conversion mode.
2 In addition to the cautions given here, the cautions on the 3-dimensional cutter compensation function apply to this function.

- If selecting the table coordinate system as a programming coordinate system

If bit 4 (TBP) of parameter No.19746 is 1 and bit 5 (WKP) of parameter No.19696 is 0, specifying 3-dimensional cutter compensation causes the table coordinate system to be selected as a programming coordinate system. The table coordinate system refers to the workpiece coordinate system fixed to the table when 3-dimensional cutter compensation.
In the blocks subsequent to the one in which 3-dimensional cutter compensation is specified, the table coordinate system rotates with the rotation of the table.
A linear axis (X, Y, Z) command is assumed to be issued on the table coordinate system.
Specifying linear interpolation causes cutter compensation to be performed on the linear interpolation command on the table coordinate system.
The table coordinate system does not rotate with the rotation of the tool head.

The cancel (G40) block becomes a block that suppresses buffering.

It is possible to switch between absolute and relative coordinate displays, using bit 2 (DET) of parameter No.19608 as follows:

- When DET is 0, the position in the table coordinate system is displayed.
- When DET is 1, the position in the workpiece coordinate system is displayed.

Note, however, the distance to go is always that in the programming coordinate system.

NOTE

1 If the 3-dimensional cutter compensation mode is entered when the table coordinate system is used as the programming coordinate system, look-ahead acceleration/deceleration before interpolation is automatically enabled. Be sure to specify the following parameters:
   (1) Bit 1 (LRP) of parameter No.1401 = 1:
   Linear rapid traverse
   (2) Bit 5 (FRP) of parameter No.19501 = 1:
   Acceleration/deceleration before interpolation is used for rapid traverse.
   (3) Parameter No. 1671:
   Acceleration of acceleration/deceleration before interpolation for rapid traverse
   (4) Parameter No. 1672:
   Change time for bell-shaped acceleration/deceleration before interpolation for rapid traverse
   (5) Parameter No. 1660:
   Maximum permissible acceleration for acceleration/deceleration before interpolation
   If they are not specified, alarm PS5463 is generated.

NOTE
2 When table rotation axis movement is specified in the start block of 3-dimensional cutter compensation, after the movement is completed, the workpiece coordinate system is fixed to the table and assumed to be a table coordinate system.
3 In the mode of 3-dimensional cutter compensation, do not change the workpiece coordinate system or change the workpiece offset value. If an attempt is made to specify workpiece coordinate system selection (G54 to G59), alarm PS5460 is generated.

- If selecting the workpiece coordinate system as a programming coordinate system

If bit 4 (TBP) of parameter No.19746 is 0 or if bit 4 (TBP) of parameter No.19746 is 1 and bit 5 (WKP) of parameter No.19696 is 1, the programming coordinate system does not rotate with the rotation of the table, being fixed to the workpiece coordinate system.

- Angle of the rotation axis in type 2

For an explanation of how the rotation axis end point is determined when the tool direction is specified with an IJKQ command in type 2, see the explanations of cutter compensation in a tool rotation type machine, "Angle of the rotation axis in type 2 (if the operating range is not specified)" and "Angle of the rotation axis in type 2 (if the operating range is specified)".

22.5.4 Interference Check and Interference Avoidance

Overview

By setting bit 1 (NI5) of parameter No. 19608 to 1, this function performs an interference check on the plane (compensation plane) perpendicular to the tool axis direction regardless of the machine configuration.
If bit 5 (CAV) of parameter No. 19607 is set to 1, a vector is generated to avoid interference on the same plane.

Explanation

- For a tool rotation type machine

An interference check is performed, as well as interference avoidance, with the tool path as projected from the workpiece coordinate system (X-Y-Z) onto the compensation plane (X'-Y'-Z') and a compensation vector.

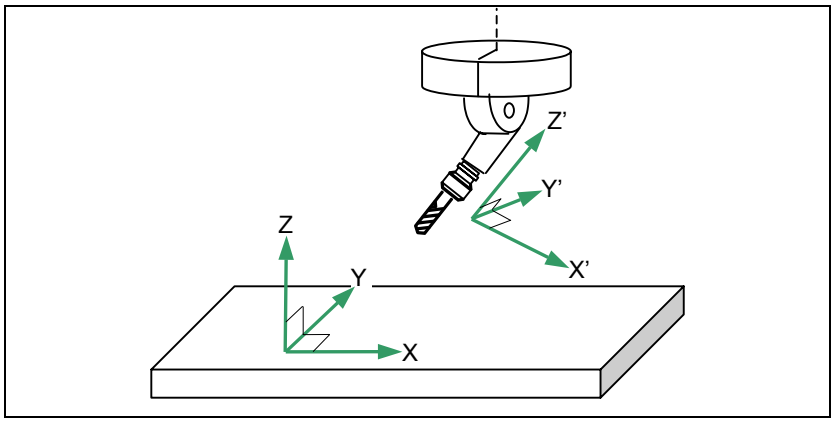


Fig. 22.5.4 (a) Tool rotation type

- For a table rotation type machine

An interference check is performed, as well as interference avoidance, with the tool path as converted from the workpiece coordinate system (X-Y-Z) into the table coordinate system (X'-Y'-Z') and a compensation vector.

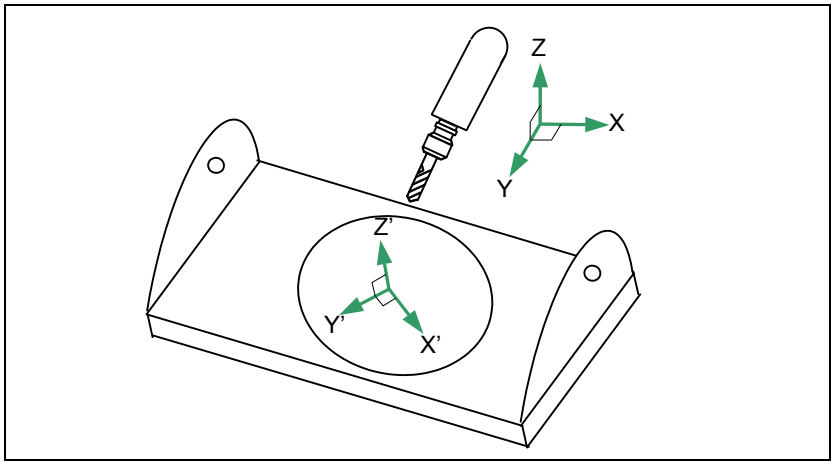


Fig. 22.5.4 (b) Table rotation type

- For a composite type machine

An interference check is performed, as well as interference avoidance, with the tool path as projected from the workpiece coordinate system (X-Y-Z) onto the table coordinate system (X'-Y'-Z') and then onto the compensation plane perpendicular to the tool axis direction (X"-Y"-Z") and a compensation vector.

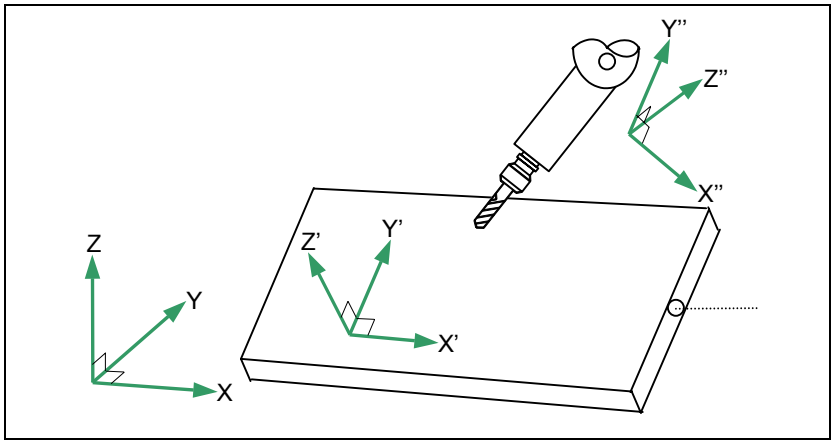


Fig. 22.5.4 (c) Composite type

- Interference avoidance

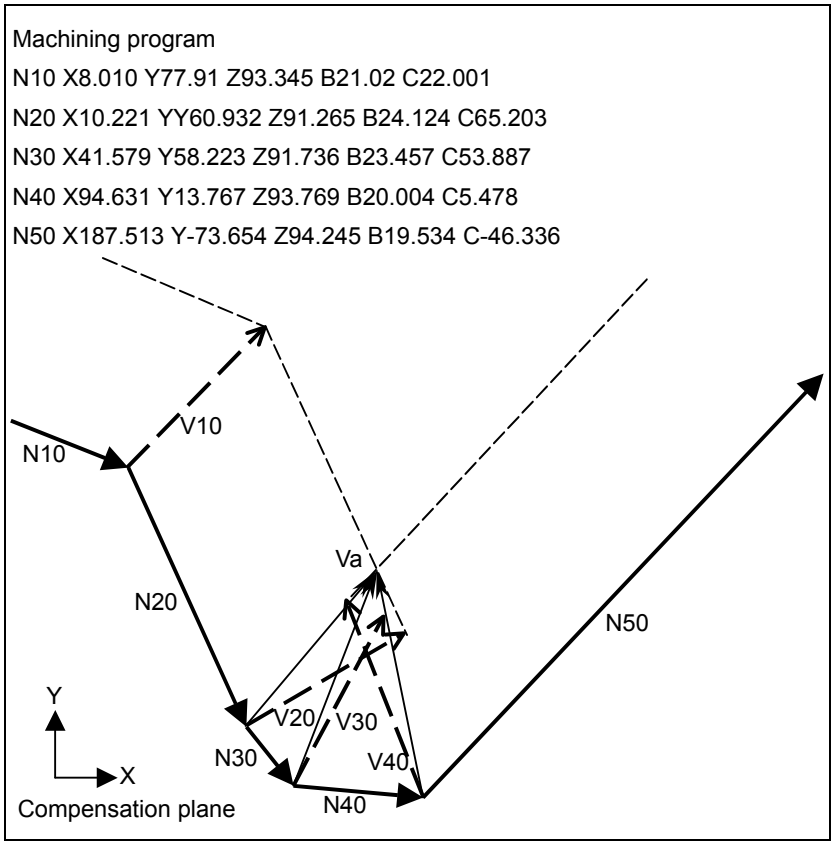


Fig. 22.5.4 (d) Example of interference avoidance

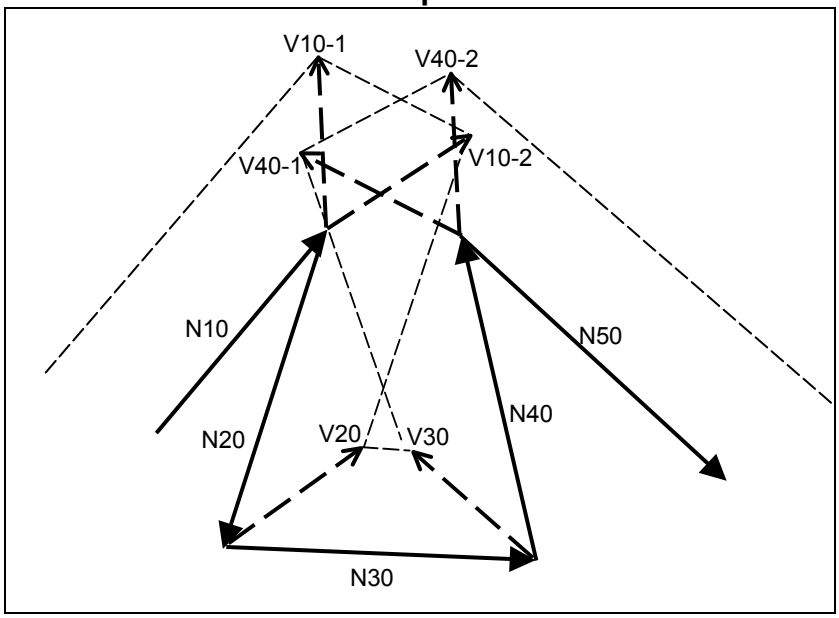
Fig. 22.4.4(d) shows a tool path in the workpiece coordinate system as projected onto the compensation plane. This is an example when the parameter No.19626 is set to 4.
At the start of the execution of the N10 block, the system looks at N20 to N50 ahead and generates V20 to V40.
Then, because the movement direction of N30 greatly differs from the direction from V20 to V30, V20 and V30 are considered interferences and eliminated. Similarly, because the movement direction from N30 to N40 greatly differs from the movement direction from V20 to V40, V40 is also considered an interference and eliminated.
Then, the interference avoidance vector Va is generated between N20 and N50 and uses it instead of V20, V30, and V40. At this time, because N20 and N50 generally do not intersect, the plane perpendicular to the tool axis direction at the N20 end point is regarded as a compensation plane, N20 and N50 are projected onto this plane, and Va is determined by calculating the intersection point.

NOTE
Strictly speaking, if the tool axis direction at the N20 end point differs from the tool axis direction at the N50 start point, correct intersection point calculation is not possible. For this reason, the maximum permissible angle by which the tool axis directions in the two blocks used to determine the interference avoidance vector may differ can be set in parameter No. 19636, and if the change in tool axis direction is within the range, an approximate interference avoidance vector is calculated.
If the maximum angle is exceeded, the compensation vector is determined but interference avoidance is not performed.

- If interference avoidance is not possible

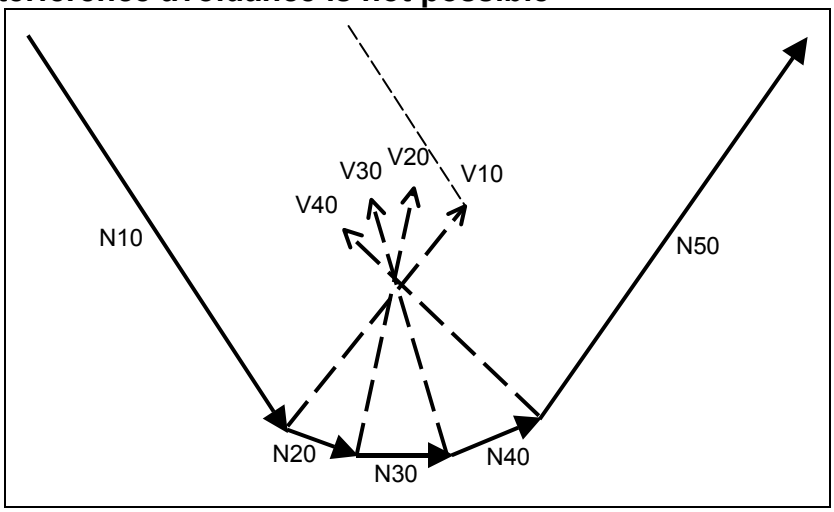
If there are three consecutive interfering blocks, no interference vector can be generated.

Example 1 in which interference avoidance is not possible



N20 to N40 interfere, so that no interference avoidance vector can be generated. Too much cutting results.

Example 2 in which interference avoidance is not possible



N10 to N40 interfere, so that no interference avoidance vector can be generated. V10 causes an interference alarm.

22.5.5 Restrictions

22.5.5.1 Restrictions common to machine configurations

- Corner rounding (G39)

In the mode for 3-dimensional cutter compensation, G39 cannot be specified. Specifying G39 causes an alarm.

- Reset

Whenever a reset is made in the mode for 3-dimensional cutter compensation (G41.2, G42.2, G41.4, G42.4, G41.5, or G42.5), the cancel mode (G40) is entered.

- AI contour control I and II

To set the AI contour control I or II mode, the corresponding G code must be specified. Specifying 3-dimensional cutter compensation does not automatically place the CNC in the AI contour control I or II mode.

- Restricted commands

In the mode for 3-dimensional cutter compensation, the following functions are available, but their statuses cannot be changed:

- Inch/metric input (If an attempt is made to change the status by using G20 or G21, alarm PS5000 is issued.)
- Mirror image (The signal status cannot be changed.)
- One-digit F code feed (The feedrate cannot be changed by using the manual handle.)

- When a rotary axis is moved significantly

In 3-dimensional cutter compensation, compensation is performed at the block end point.
Even when the tool direction relative to the workpiece changes momentarily as a rotary axis moves, compensation is not performed momentarily during execution of the block.
Accordingly, when a rotary axis is moved significantly, carry out programming to divide a block into small movement blocks to prevent the machining precision from being reduced.

- Operation at cancellation

In a block for canceling 3-dimensional cutter compensation, compensation accompanying movement is cancelled regardless of whether the movement command is present.

- Unavailable commands

In the mode for 3-dimensional cutter compensation, the functions listed below cannot be specified. Specifying any of these functions results in an alarm.

- Hypothetical axis interpolation G07
- Circular interpolation G07.1
- Polar coordinate interpolation G12.1, G13.1
- Polar coordinate command G15, G16
- Reference position return check G27
- Reference position return G28, G29, G30
- Skip G31
- Threading G33
- Automatic tool length measurement G37
- Normal direction control G40.1, G41.1, G42.1
- Tool radius/tool nose radius compensation G41, G42, G39
- 3-dimensional cutter compensation G41
- Wheel wear compensation G41
- Tool offset G45, G46, G47, G48
- Local coordinate system G52
- Machine coordinate system G53
- Workpiece coordinate system setting G54-G59, G54.1
- Rotary table dynamic fixture offset G54.2
- Single direction positioning G60
- Automatic corner override G62
- Tapping mode G63
- Three-dimensional coordinate system conversion G68, G69
- Tilted working plane command G68.2, G69
- Figure copy G72.1, G72.2
- Canned cycle G73-G79, G80, G81-G89, G98, G99
- Electric gear box G80, G81
- Hobbling machine function G80, G81
- External operation function G81
- Chopping G81.1
- Small hole peck drilling cycle G83
- Workpiece coordinate system setting G92
- Workpiece coordinate system presetting G92.1
- Feed per revolution G95
- Constant surface speed control G96, G97

- Unavailable functions

If the following function is specified in the 3-dimensional cutter compensation mode, a warning message is issued:

- MDI interruption

If one of the following functions is specified in the 3-dimensional cutter compensation mode, a PS alarm is issued:

- Manual interruption operation
- Tool retract and recover

In the 3-dimensional cutter compensation mode, the following functions cannot be used:

- Twin table control
- Flexible synchronous control
- Sequence number comparison and stop (It is not possible to cause a stop by sequence number in the 3-dimensional cutter compensation mode.)
- Index table indexing
- Retrace function (Programs using the 3-dimensional cutter compensation mode must not use the retrace function.)
- Rotary axis control
- Manual handle interruption
- External deceleration (External deceleration is not performed.)
- Angular axis control

Combinations of other NC commands may be restricted. Refer to the manual on each function.

22.5.5.2 Restriction on tool rotation type

- Unavailable commands (leading edge offset)

In the G41.3 mode, the following commands cannot be specified:

- G functions of group 01 other than G00 and G01

- Use with tool center point control

If 3-dimensional cutter compensation is specified before tool center point control when 3-dimensional cutter compensation and tool center point control are specified together, tool center point control must be canceled before 3-dimensional cutter compensation is canceled (Example 1). Contrariwise, if tool center point control is specified before 3-dimensional cutter compensation, 3-dimensional cutter compensation must be canceled before tool center point control is canceled (Example 2).

```
Example 1

G41.2  D1
 :
G43.4  H1
 :
G49
 :
G40
```

```
Example 2

G43.4  H1
 :
G41.2  D1
 :
G40
 :
G49
```

If the specification sequence of G40 and G49 is reversed, alarm PS5460 is issued.

When 3-dimensional cutter compensation is specified before tool center point control, the block for canceling tool center point control suppresses buffering. Note that, as a result, the block before the G49 block generates a compensation vector for 3-dimensional cutter compensation, which is perpendicular to the movement.
For each of 3-dimensional cutter compensation and tool center point control, two commands, type 1 and type 2, are provided. Be sure to specify commands of the same type. If commands of different types are specified, alarm PS5460 is issued.
When the tool is tilted by address Q in type 2 command specification, if Q is specified in both of the block for starting tool center point control and the block for starting 3-dimensional cutter compensation, the Q command specified earlier becomes valid.

- Combination with circular interpolation

When circular interpolation is specified in a block, if a rotary axis is moved in the block, the previous block, or the next block, a cutter compensation vector may be created in the direction orthogonal to the plane. In this case, helical interpolation is performed, so the helical interpolation option is required.

22.5.5.3 Restriction on machine configurations having table rotation axes (table rotation type and composite type)

- Unavailable commands

For machines having table rotation axes, the following commands cannot be specified during 3-dimensional cutter compensation:

- G functions of group 01 other than G00 and G01

- Use with tool center point control

When the workpiece coordinate system is used as the programming coordinate system, the same restrictions as for the tool rotation type apply. See "Use with tool center point control" in the restrictions on the tool rotation type described previously.
When the table coordinate system is used as the programming coordinate system, the restrictions to apply will be explained below.

- Type 2

When a type 2 command (G41.6/G42.6 command) is executed, the table coordinate system needs to be used as the programming coordinate system (bit 4 (TBP) of parameter No. 19746 = 1 and bit 5 (WKP) of parameter No. 19696 = 0). If a type 2 command (G41.6/G42.6 command) is executed with the workpiece coordinate system used as the programming coordinate system, alarm PS5459 is issued.

- Restrictions when the table coordinate system is used as the programming coordinate system

Use with tool center point control

When 3-dimensional cutter compensation and tool center point control are specified together, tool center point control must be specified before cutter compensation. In addition, tool center point control must be canceled after 3-dimensional cutter compensation is canceled.
(Example 1)
If tool center point control is specified in the 3-dimensional cutter compensation mode (Example 2), or if tool center point control is canceled without canceling 3-dimensional cutter compensation (Example 3), alarm PS5460 is issued.

Example 1 (Correct specification)	Example 2 (Specification resulting in alarm)	Example 3 (Specification resulting in alarm)
G43.4 H1 : G41.2 D1 : G40 : G49	G41.2 D1 : G43.4 H1 :	G43.4 H1 : G41.2 D1 : G49

If the setting of the programming coordinate system differs between 3-dimensional cutter compensation and tool center point control, specifying both functions together results in alarm PS5460. (See the following table:)

	TBP=0		TBP=1	
	WKP=0	WKP=1	WKP=0	WKP=1
Tool center point control	Table coordinate system	Workpiece coordinate system	Table coordinate system	Workpiece coordinate system
3-dimensional cutter compensation	Workpiece coordinate system			
Both specified together	Alarm PS5460			

Programming coordinate system determined by bit 4 (TBP) of parameter No. 19746 and bit 5 (WKP) of parameter No. 19696

The table rotation axis position at the start of tool center point control and the table rotation axis position at the start of 3-dimensional cutter compensation must always match.
Each programming coordinate system matches the workpiece coordinate system used when each function is started.
If these functions are started when their table rotation axis positions differ, the programming coordinate system for tool center point control and the programming coordinate system for 3-dimensional cutter compensation do not match, which results in alarm PS5460.
Example: When the A-axis is the table rotation axis:

(Correct example)

```
G90 G00 A0.0
G43.4 H1
G01 Z100.0 F1000.
G41.2 D1    ← After G43.4 is specified, G41.2 is specified
              without A-axis movement.
:
```

(Wrong example)

```
G90 G00 A0.0
G43.4 H1
G01 Z100.0 A30.0 F1000.
G41.2 D1    ← After G43.4 is specified, A-axis movement
              is made, then G41.2 is specified.
:
```

* Furthermore, no A-axis command can be included in the block specifying G41.2.

For each of 3-dimensional cutter compensation and tool center point control, two commands, type 1 and type 2, are provided. Be sure to specify commands of the same type. If commands of different types are specified, alarm PS5460 is issued.
When the tool is tilted by address Q in type 2 command specification, if Q is specified in both of the block for starting tool center point control and the block for starting 3-dimensional cutter compensation, the Q command specified earlier becomes valid.

Look-ahead acceleration/deceleration before interpolation

If the 3-dimensional cutter compensation mode is entered, look-ahead acceleration/deceleration before interpolation is automatically enabled. Set the look-ahead acceleration/deceleration parameter. If the parameter is incorrectly set, an alarm is issued.

Deceleration at a corner

Under 3-dimensional cutter compensation, the controlled point may move along a curve even if a straight-line command is issued. Some commands may cause a corner movement.
When AI contour control is not used, there is a possibility that the machine is shocked at a corner.
Accordingly, it is recommended that AI contour control be used and the permissible acceleration (parameter No. 1737) and permissible speed difference at a corner (parameter No. 1783) be set.

Specifiable G codes

When the table coordinate system is used as the programming coordinate system, the G codes that can be specified in the 3-dimensional cutter compensation mode are listed below.
Specifying a G code other than these codes results in alarm PS5460:

- Positioning (G00)
- Linear interpolation (G01)
- Dwell (G04)
- Exact stop (G09)
- Programmable data input (G10)
- Programmable data input mode cancel (G11)
- Plane selection (G17/G18/G19)
- Cutter compensation : cancel (G40)
- Tool length compensation cancel (G49)
- Programmable mirror image (G50.1/G51.1)
- Macro call (G65)
- Macro modal call A (G66)
- Macro modal call B (G66.1)
- Macro modal call A/B cancel (G67)
- Absolute programming (G90)
- Incremental programming (G91)

Modal G codes that allow specification of 3-dimensional cutter compensation

When the table coordinate system is used as the programming coordinate system, 3-dimensional cutter compensation can be specified in the modal G code states listed below.
In a modal state other than the following modal G codes, specifying tool center point control results in alarm PS5421.

- Modal G codes included in "Specifiable G codes" described previously
- Polar coordinate interpolation mode cancel (G13.1)
- Polar coordinates command cancel (G15)
- Input in inch (G20 (G70))
- Input in mm (G21 (G71))
- Tool center point control (G43.4/G43.5)
- Polygon turning cancel (G50.2)
- Workpiece coordinate system 1 selection (G54 to G59)
- Canned cycle cancel (G80)
- Constant surface speed control cancel (G97)
- Canned cycle : return to initial level (G98)
- Canned cycle : return to R point level (G99)

M

- Coordinate system rotation start or 3-dimensional coordinate conversion mode on (G69)
- Feed per minute (G94)
- Polar coordinate interpolation mode cancel (G113)

T

- Mirror image for double turret off/balanced cutting mode cancel (G69)
- Coordinate system rotation cancel or 3-dimensional coordinate conversion mode off (G69.1)
- Feed per minute (G98 (G94))

Specification of axes not relating to 3-dimensional cutter compensation

Axes not relating to 3-dimensional cutter compensation cannot be specified. If such an axis is specified, alarm PS5460 is issued.

22.5.6 Examples

O100 is a sample program.
This is an example in which each side of a square is cut at an angle of 30 degrees on the B-axis in a composite type machine.
Programs 1 to 3 all perform the same machining.

Program 1: Type 1 and the table coordinate system is selected as a programming coordinate system

```
O100（Sample Program1）；
N10 G55 ;                                Preparations for the programming coordinate system
N20 G90 X0 Y0 Z300.0 B0 C0 ;             Movement to the initial position
N30 G01 G43.4 H01 Z40.0 F500. ;          Start of tool center point control
                                         H01 is a tool length compensation number
N40 G41.2 D01;                           Start of cutter compensation
                                         D01 is a tool cutter compensation number
N50 X50.0 Y50.0 Z20.0 B30.0 C45.0 ;      The Z-axis height on the machining plane is 20.0.
N60 X-50.0 C135.0 ;
N70 X-100.0 Y-100.0 C225.0 ;
N80 X100.0 C315.0 ;
N90 X50.0 Y50.0 C405.0 ;
N100 X0 Y0 Z40.0 B0 C360.0 ;
N110 G40 ;                               Cutter compensation cancellation
N120 G49 Z300.0 ;                        Tool center point control cancellation
                                         Movement to the initial position on the Z-axis
N130 M30;
```

Program 2: Type 1 and the workpiece coordinate system is selected as a programming coordinate system
(Note that the values specified in N50 to N90 differ from those in program 1.)

```
O100（Sample Program2）；
N10 G55 ;                                Preparations for the programming coordinate system
N20 G90 X0 Y0 Z300.0 B0 C0 ;             Movement to the initial position
N30 G01 G43.4 H01 Z40.0 F500. ;          Start of tool center point control
                                         H01 is a tool length compensation number
N40 G41.2 D01;                           Start of cutter compensation
                                         D01 is a tool cutter compensation number
N50 X70.711 Y0 Z20.0 B30.0 C45.0 ;       The Z-axis height on the machining plane is 20.0.
N60 C135.0 ;
N70 X141.421 C225.0 ;
N80 C315.0 ;
N90 X70.711 C405.0 ;
N100 X0 Y0 Z40.0 B0 C360.0 ;
N110 G40 ;                               Cutter compensation cancellation
N120 G49 Z300.0 ;                        Tool center point control cancellation
                                         Movement to the initial position on the Z-axis
N130 M30;
```

Program 3: When the type 2 is used:
(The table coordinate system is selected as a programming coordinate system)

```
O100（Sample Program3）;
N10 G55 ;                                          Preparations for the programming coordinate system
N20 G90 X0 Y0 Z300.0 B0 C0 ;                       Movement to the initial position
N30 G01 G43.5 H01 Z40.0 F500. ;                    Start of tool center point control
                                                   H01 is a tool length compensation number
N40 G41.6 D01;                                     Start of cutter compensation
                                                   D01 is a tool cutter compensation number
N50 X50.0 Y50.0 Z20.0 I35.355 J35.355 K86.603 ;    The Z-axis height on the machining plane is 20.
N60 X-50.0 I-35.355 J35.355 K86.603 ;
N70 X-100.0 Y-100.0 I-35.355 J-35.355 K86.603 ;
N80 X100.0 I35.355 J-35.355 K86.603 ;
N90 X50.0 Y50.0 I35.355 J35.355 K86.603 ;
N100 X0 Y0 Z40.0 K1.0 ;
N110 G40 ;                                         Cutter compensation cancellation
N120 G49 Z300.0 ;                                  Tool center point control cancellation
                                                   Movement to the initial position on the Z-axis
N130 M30;
```

By using type 2 as in program 3, the same program can be used with machines with different configurations, regardless of whether the machine configuration is the tool rotation type, table rotation type, or composite type.

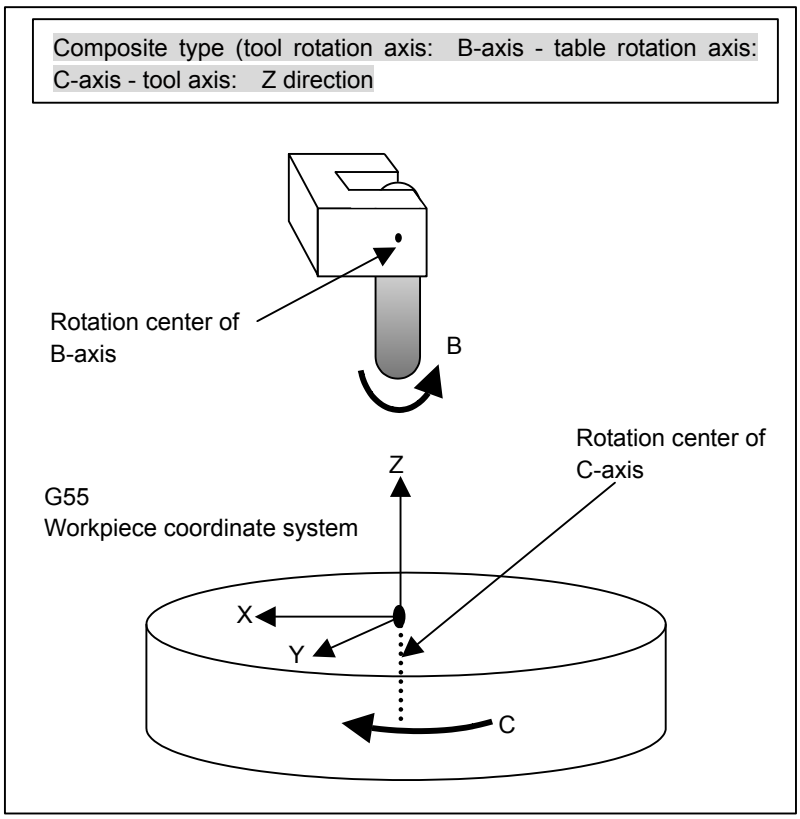


Fig. 22.5.6 (a) Machine configuration for example

Fig. 22.5.6 (b) shows the attitudes of the workpiece (object to be machined) and the tool head (relative to the workpiece (object to be machined)) as viewed in the positive Z direction of the programming coordinate system fixed to the table (table coordinate system).

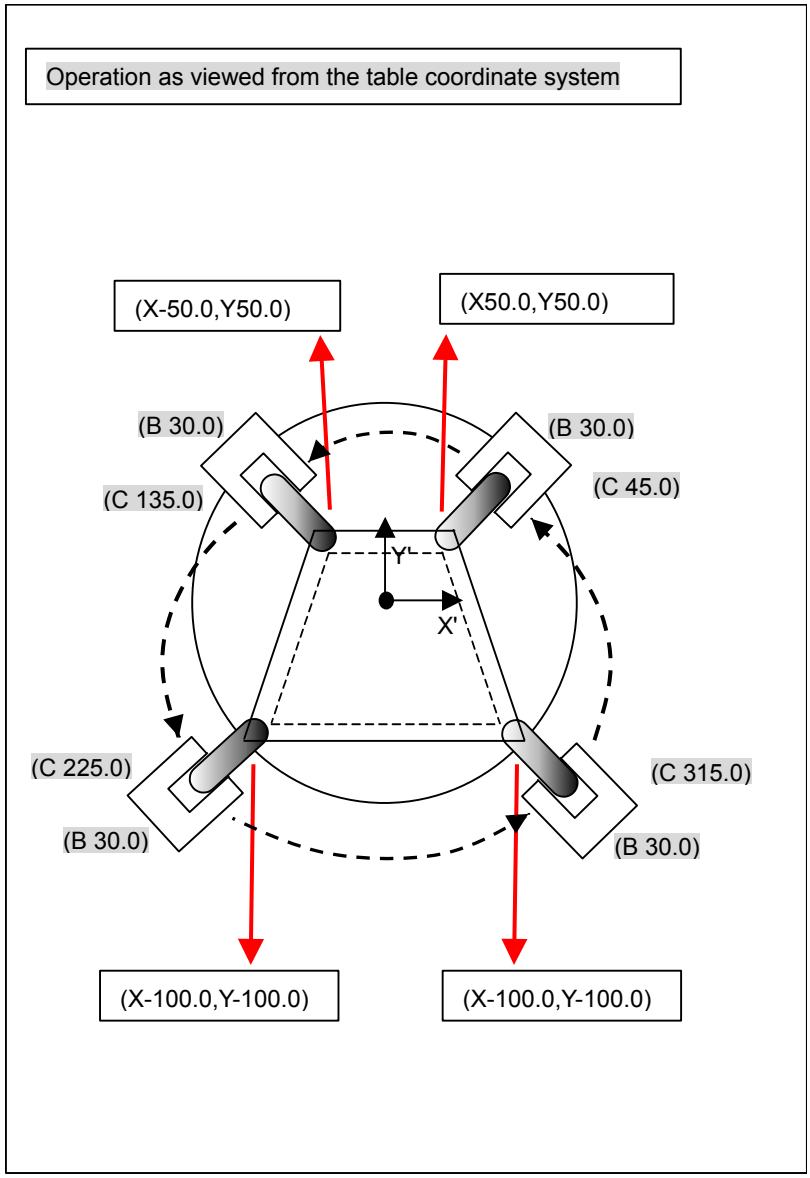


Fig. 22.5.6 (b) Illustration of example

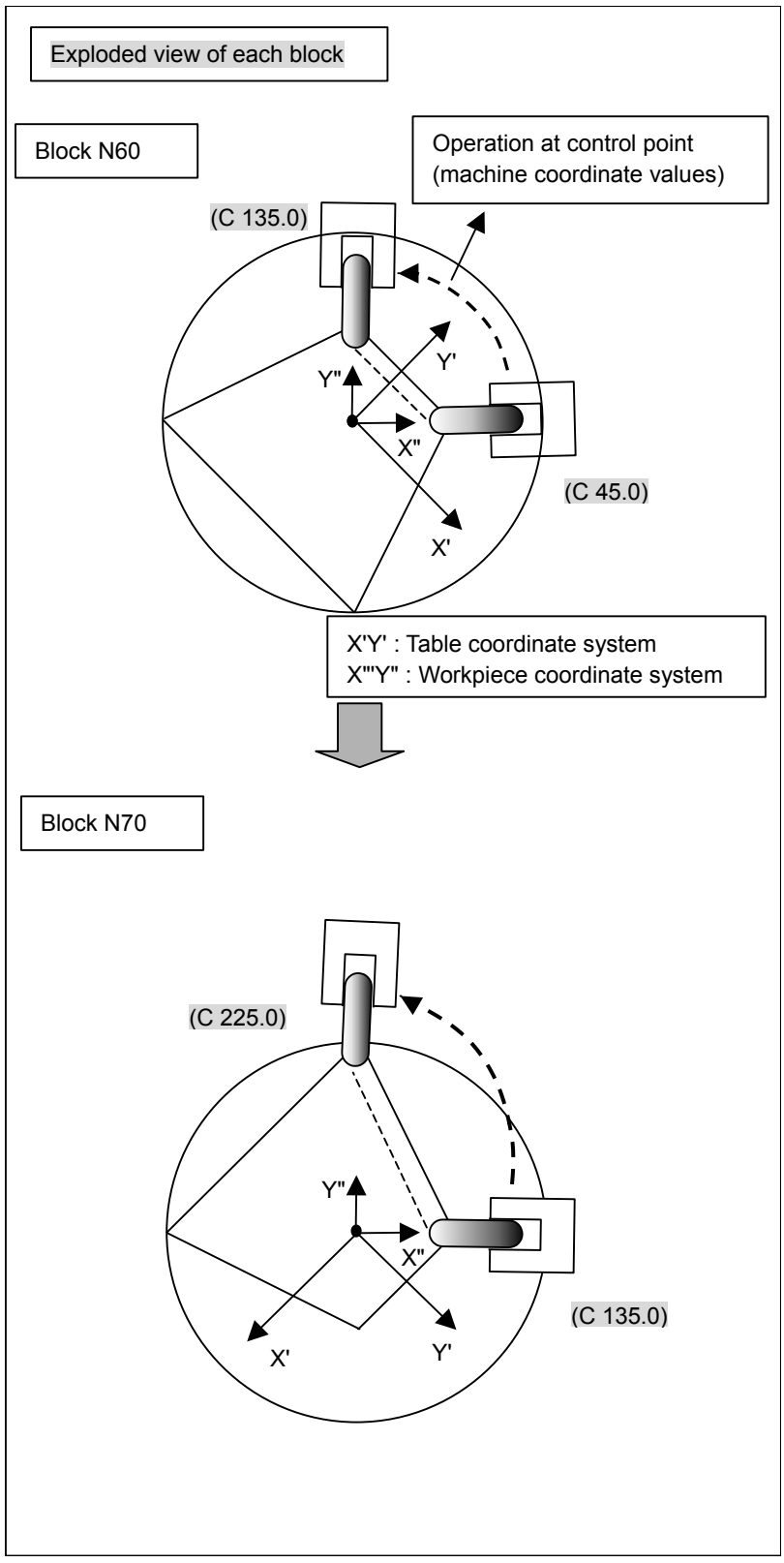


Fig. 22.5.6 (c) Exploded View of Each Block (1)

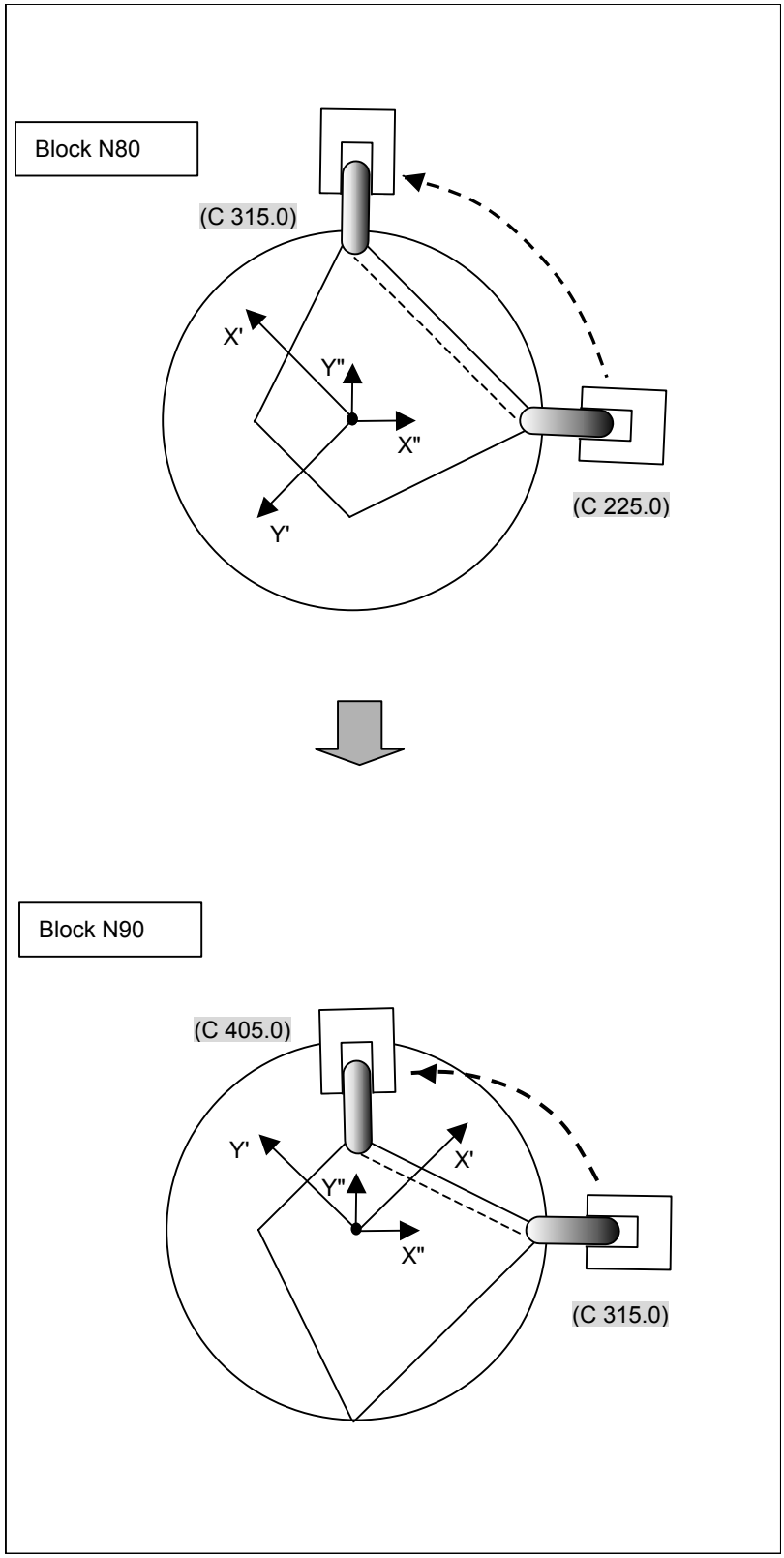


Fig. 22.5.6 (d) Exploded View of Each Block (2)

23 MUITI-PATH CONTROL FUNCTION

Chapter 23, "MUITI-PATH CONTROL FUNCTION", consists of the following sections:

23.1 OVERVIEW 964
23.2 WAITING FUNCTION FOR PATHS 966
23.3 COMMON MEMORY BETWEEN EACH PATH.....972
23.4 SPINDLE CONTROL BETWEEN EACH PATH.....974
23.5 SYNCHRONOUS/COMPOSITE/SUPERIMPOSED CONTROL 975

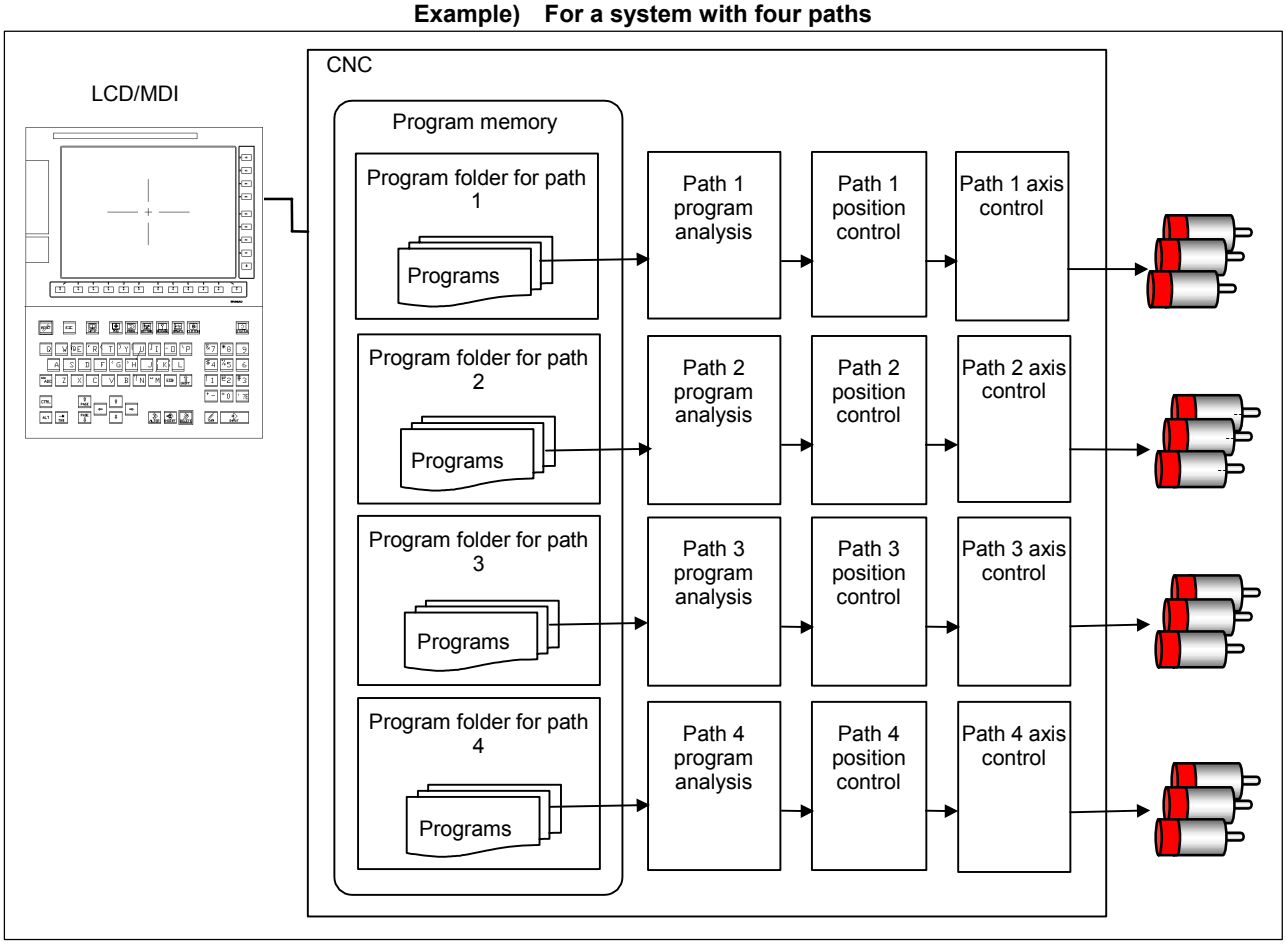
23.1 OVERVIEW

The multi-path control function is designed to enable 10 independent simultaneous machining with up to 10 paths (10-path control). This function is applicable to lathes and automatic lathes which perform cutting simultaneously with multiple tool posts, combined machine tools which perform turning and milling simultaneously with multiple paths, and machines which require additional control paths such as a loader control path.

For multi-path simultaneous machining, each machining program is stored in a folder in program memory for each path. When automatic operation is to be performed, each path is activated after selecting a program for machining with path 1 and programs for machining with paths 2 to 10 from the programs stored in the respective folders in program memory. Then the programs selected for the tool posts are executed independently at the same time. When tool post 1 and tool post 2 need to wait for each other during machining, the waiting function is available.
Other available functions specific to multi-path control include interference check for each path, balance cut, synchronous control, composite control, spindle control between each path, and common memory between each path.

Just one LCD/MDI is provided for the all paths. Before operation and display on the LCD/MDI, the path selection signal is used to switch between the paths.

Example) For a system with four paths



23.2 WAITING FUNCTION FOR PATHS

Overview

Control based on M codes is used to cause one path to wait for the other during machining. When an M code for waiting is specified in a block for one path during automatic operation, the other path waits for the same M code to be specified before staring the execution of the next block.
A range of M codes used as M codes for waiting is to be set in the parameters (Nos. 8110 and 8111) beforehand. Waiting can be ignored using a signal.

Format

M m (Pp) ;
m: Number of an M code for waiting
p: (1) In the binary value specification mode, specify the sum of the binary values corresponding to the numbers of paths which are to wait for one another.
(2) In the path number specification mode, specify the numbers of all paths that are to wait for one another in combination.

Explanation

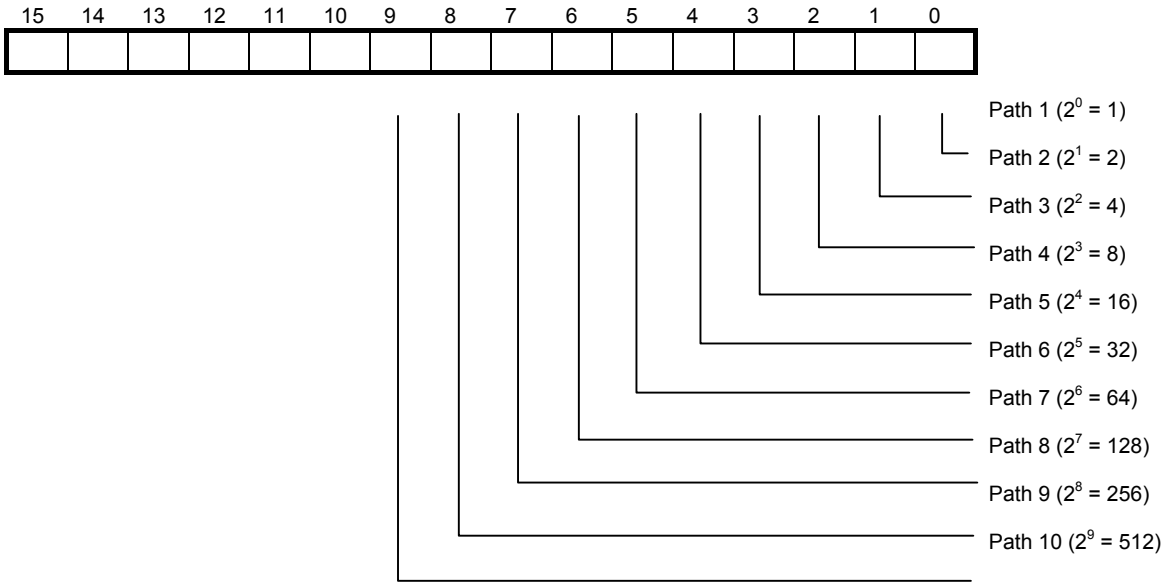
The following two methods for specifying paths which are to wait for one another at address P in the same block in which an M code for waiting is specified are available and can be selected using bit 1 (MWP) of parameter No. 8103. One method is to specify the paths with the sum of their corresponding binary values (waiting set for three or more paths specified with binary values). The other is to specify them with their path numbers in combination (waiting specified with path numbers).
When address P is not specified, paths 1 and 2 wait for each other (waiting set for two paths). Always specify address P in a single block.

- Waiting specified with binary values

When bit 1 (MWP) of parameter No. 8103 is set to 0, the value specified at address P is assumed to be obtained using binary values. The following table lists the path numbers and corresponding binary values.

Path number	Binary value (decimal number)
1	1
2	2
3	4
4	8
5	16
6	32
7	64
8	128
9	256
10	512

The bit position of each path in binary representation is shown below.



To make all of paths 1, 2, and 3 wait for one another, the P value is obtained as follows:

Binary value of path 1	1 (0000 0000 0000 0001)
Binary value of path 2	2 (0000 0000 0000 0010)
Binary value of path 3	4 (0000 0000 0000 0100)
Sum	7 (0000 0000 0000 0111)

All of the three paths can be made to wait for one another by specifying P7 together with an M code for waiting.
To make all of paths 1, 3, 5, 7, and 9 wait for one another, the P value is obtained as follows:

Binary value of path 1	1 (0000 0000 0000 0001)
Binary value of path 3	4 (0000 0000 0000 0100)
Binary value of path 5	16 (0000 0000 0001 0000)
Binary value of path 7	64 (0000 0000 0100 0000)
Binary value of path 9	256 (0000 0001 0000 0000)
Sum	341 (0000 0001 0101 0101)

All of the five paths can be made to wait for one another by specifying P341 together with an M code for waiting.

- Waiting specified with a combination of path numbers

When bit 1 (MWP) of parameter No. 8103 is set to 1, the value specified at address P is assumed to be a combination of path numbers. The following table lists the path numbers and corresponding values.

Path number	Value (decimal number)
1	1
2	2
3	3
4	4
5	5
6	6
7	7
8	8
9	9
10	0

To make all of paths 1, 2, and 3 wait for one another, the P value is a number consisting of 1, 2, and 3.

Example) P123

There are no restrictions on the order in which the numeric characters are specified, and the following six possible values can be specified:

P123, P132, P213, P231, P312, P321

Path numbers specified in combination in different orders for different paths are effective as long as the numbers of the relevant paths are specified.

Example) The following are treated as the same P value and these paths can be made to wait for one another:
M200P123 for path 1, M200P231 for path 2, and M200P321 for path 3

To make all of paths 1, 3, 5, 7, and 9 wait for one another, the P value is a number consisting of 1, 3, 5, 7, and 9.

Example) P13579

- Waiting for path 10

To make path 10 and another path wait for each other, specify a value of 0 for the combination.
If a number begins with 0, 0 cannot be recognized. Specify 0 in the second or subsequent digit from the left.

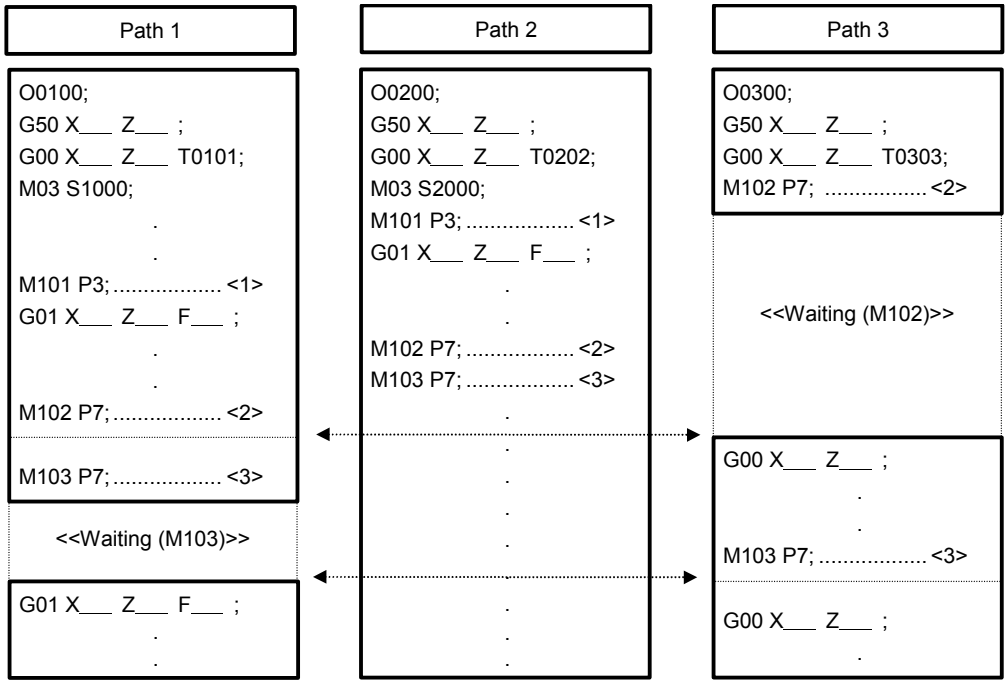
Incorrect example) P013579
Correct example) P103579

In the incorrect example, the P value is assumed to be the same as P13579. Path 10 cannot be made to wait for other paths and other paths cannot be made to wait for path 10.

Example

- When the value specified at P is obtained using binary values

Assume that the waiting ignore signal for path 2 (G1063.7 for a system with three or more paths) is set to 1 and M101 to M103 (parameter No. 8110 = 101 and parameter No. 8111 = 103) are set as M codes for waiting. In this case, programs O100, O200, and O300 for individual paths are executed as follows:



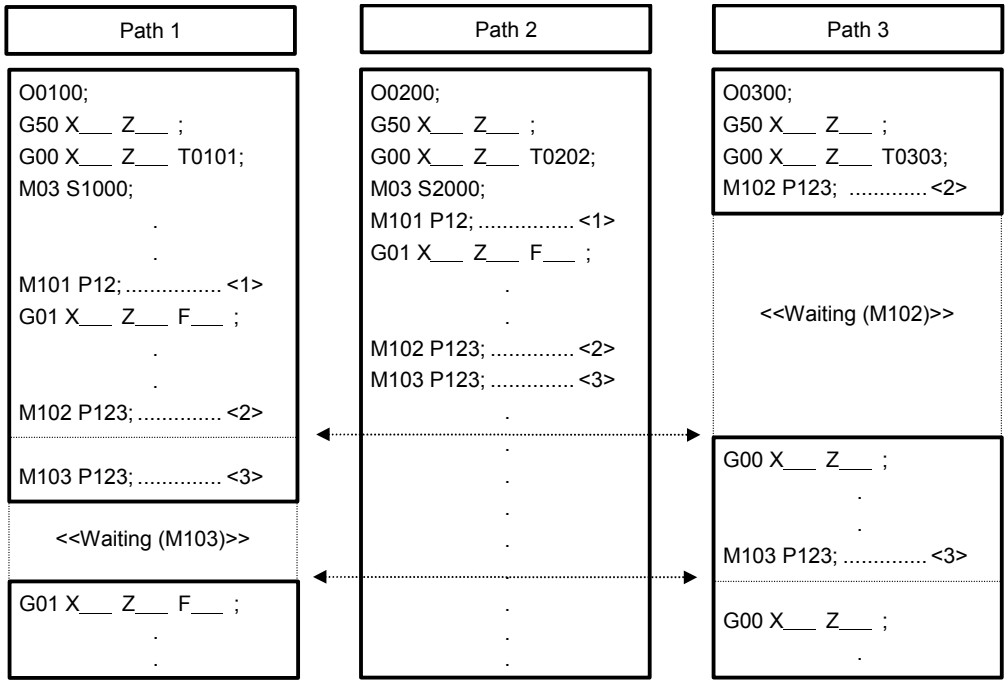
<1> M101 P3; (making paths 1 and 2 wait for each other)
When the waiting ignore signal for path 2 is set to 0, paths 1 and 2 wait for each other. Because the signal is set to 1, however, paths 1 and 2 ignore the M code for waiting and immediately execute the next block.

<2> M102 P7; (making paths 1, 2, and 3 wait for one another)
In this example, path 3 waits for processing on paths 1 and 2 to terminate. Because the waiting ignore signal for path 2 is set to 1, however, path 3 assumes that waiting is completed when processing on path 1 terminates, and executes the next block.

<3> M103 P7; (making paths 1, 2, and 3 wait for one another)
In this example, paths 1 and 2 wait for processing on path 3 to terminate. Because the waiting ignore signal for path 2 is set to 1, however, path 2 does not wait for processing on path 3 to terminate and executes the next block, but path 1 waits for path 3.

- When the value specified at P is obtained using path numbers in combination

Assume that the waiting ignore signal for path 2 (G1063.7 for a system with three or more paths) is set to 1 and M101 to M103 (parameter No. 8110 = 101 and parameter No. 8111 = 103) are set as M codes for waiting. In this case, programs O100, O200, and O300 for individual paths are executed as follows:



<1> M101 P12; (making paths 1 and 2 wait for each other)
When the waiting ignore signal for path 2 is set to 0, paths 1 and 2 wait for each other. Because the signal is set to 1, however, paths 1 and 2 ignore the M code for waiting and immediately execute the next block.

<2> M102 P123; (making paths 1, 2, and 3 wait for one another)
In this example, path 3 waits for processing on paths 1 and 2 to terminate. Because the waiting ignore signal for path 2 is set to 1, however, path 3 assumes that waiting is completed when processing on path 1 terminates, and executes the next block.

<3> M103 P123; (making paths 1, 2, and 3 wait for one another)
In this example, paths 1 and 2 wait for processing on path 3 to terminate. Because the waiting ignore signal for path 2 is set to 1, however, path 2 does not wait for processing on path 3 to terminate and executes the next block, but path 1 waits for path 3.

⚠ CAUTION

1 An M code for waiting must always be specified in a single block.
2 Unlike other M codes, the M code for waiting is not output to the PMC.
3 If the operation of a single path is required, the M code for waiting need not be deleted. By using the signal to specify that waiting be ignored (NOWT for the system with two paths, NMWT for the system with three or more paths), the M code for waiting in a machining program can be ignored. For details, refer to the manual supplied by the machine tool builder.
4 If using a waiting M code in block multiple-M code command mode, be sure to specify it as the first M code.

23.3 COMMON MEMORY BETWEEN EACH PATH

Overview

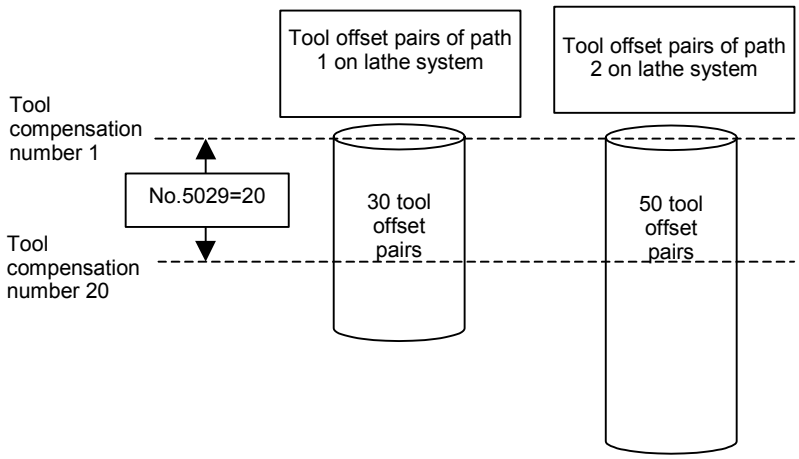
In a multi-path system, this function enables data within the specified range to be accessed as data common to all paths. The data includes tool compensation memory and custom macro common variables.

Explanation

The path common memory function enables the following operations.

- Tool compensation memory

Part or all of tool compensation memory for individual paths can be used as common data by setting parameter No. 5029.

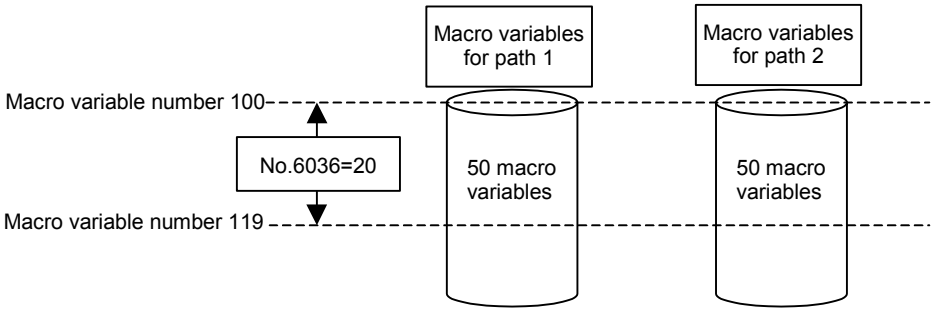


NOTE

1 When a combined system including a machining center system and lathe system is used, data is made common according to the same path control type.
2 The same unit for tool compensation (bits 0 to 3 of parameter No. 5042) must be set for each of the machining center and lathe systems.
3 Set a value less than the number of tool compensation values for each path for parameter No. 5029.
4 If the value set for parameter No. 5029 is greater than the number of tool compensation values for each path, the minimum number of tool compensation values for individual paths is assumed.
5 For details, refer to the relevant manual of the machine tool builder.

- Custom macro common variables

All or part of custom macro common variables #100 to #149 (, #199, or #499) and #500 to #599 (or #999) can be used as common data by setting parameters No. 6036 (#100 to #149 (, #199, or #499)) and 6037 (#500 to #599 (or #999)).



NOTE
1 If the value of parameter No. 6036 or 6037 exceeds the maximum number of macro common variables, the maximum number of macro common variables is assumed.
2 Common variables #150 to #199, #150 to #499, and #600 to #999 are optional functions.

23.4 SPINDLE CONTROL BETWEEN EACH PATH

Overview

This function allows a workpiece attached to one spindle to be machined simultaneously with two tool posts and each of two workpieces attached to each of two spindles to be machined simultaneously with each of two tool posts.

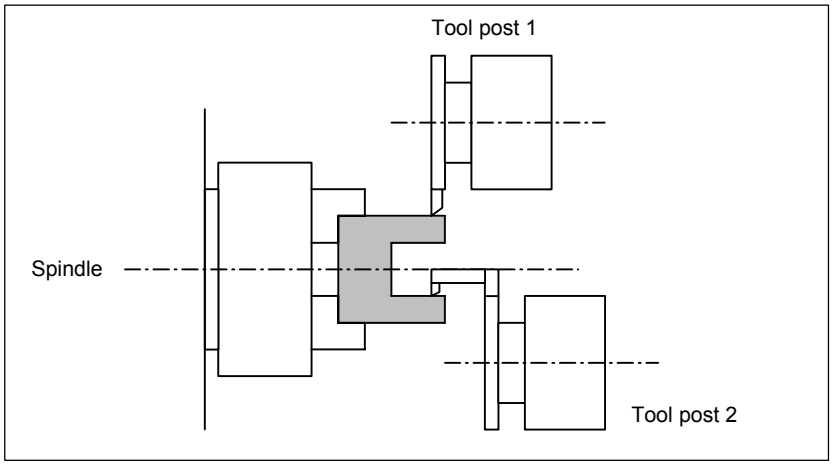


Fig. 23.4 (a) Application to a lathe with one spindle and two tool posts

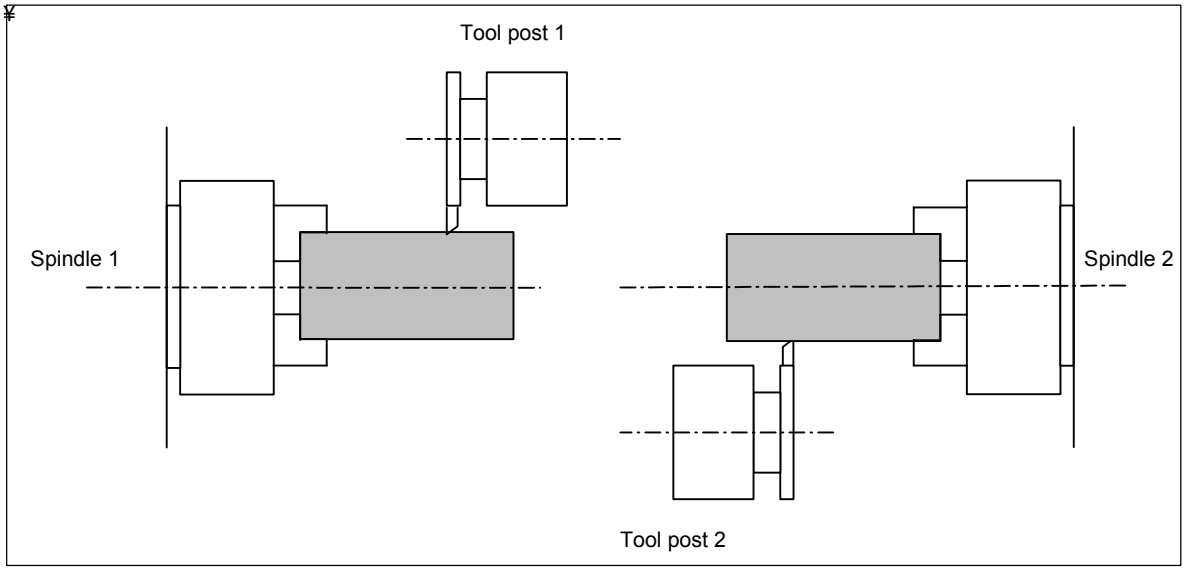


Fig. 1.1 (b) Application to a lathe with two spindles and two tool posts

The spindle belonging to each path can generally be controlled by programmed commands for the path. With path spindle command selection signals, programmed commands for any path can control the spindle belonging to any path.

NOTE
For the method of spindle command selection, refer to the relevant manual of the machine tool builder.

23.5 SYNCHRONOUS/COMPOSITE/SUPERIMPOSED CONTROL

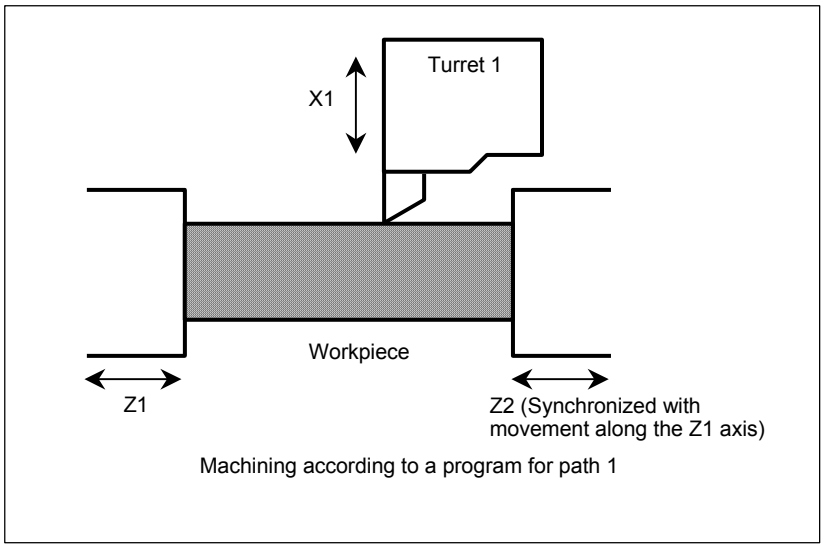
Overview

In multi-path control, the synchronous control function, composite control function, and superimposed control function enable synchronous control, composite control, and superimposed control in a single path system or between multi-path systems.

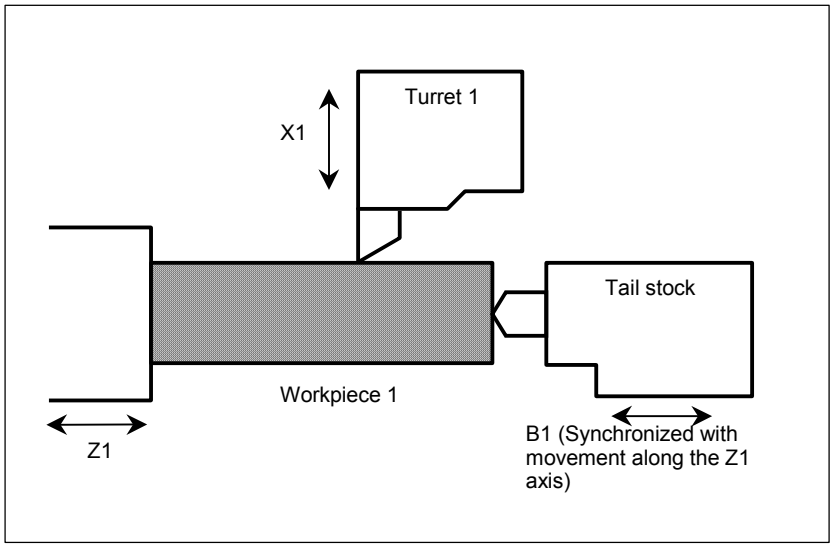
Explanation

- Synchronous control

- Synchronizes movement along an axis of one system with that along an axis of the other path.
  Example)
  Synchronizing movement along the Z1 (master) and Z2 (slave) axes (in the case of turning)



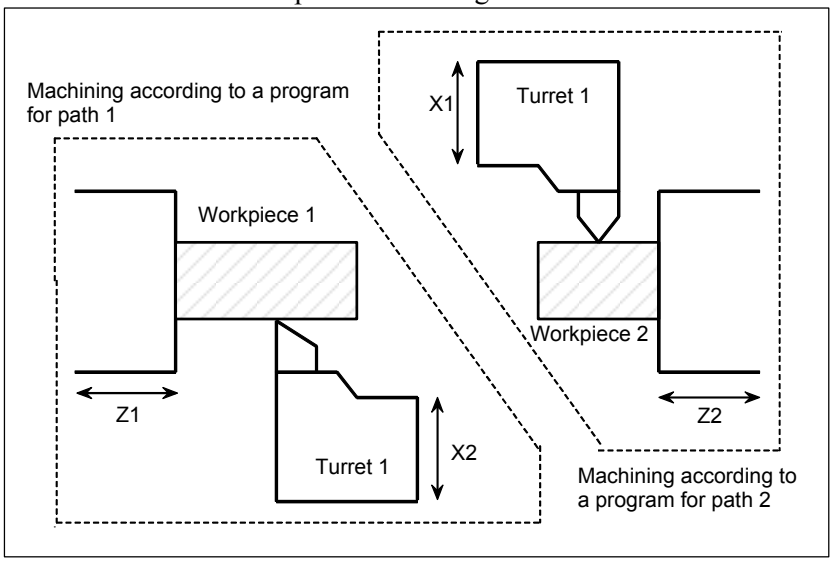
- Synchronizes movement along an axis of one path with that along another axis of the same path.
  Example)
  Synchronizing movement along the Z1 (master) and B1 (slave) axes (in the case of turning)



- Composite control

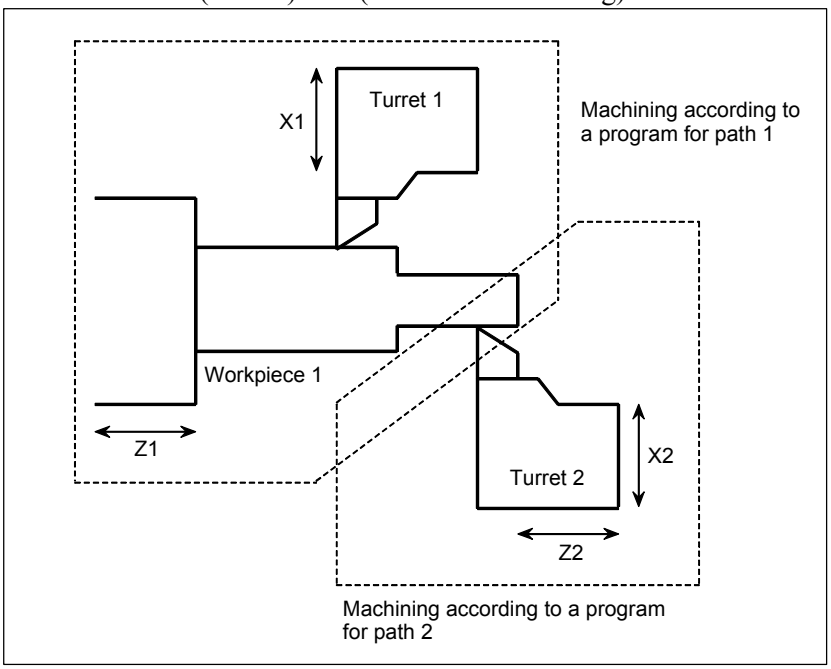
- Exchanges the move commands for different axes of different path.

  Example)
  Exchanging the commands for the X1 and X2 axes (in the case of turning)
  → Upon the execution of a command programmed for path 1, movement is performed along the X2 and Z1 axes.
  Upon the execution of a command programmed for path 2, movement is performed along the X1 and Z2 axes.



- Superimposed control

- Provides a move command of an axis for a different axis in another path.
  Example)
  Providing the Z2 (slave) axis with a move command specified for the Z1 (master) axis (in the case of turning)



NOTE
The method used to specify synchronous, composite, or superimposed control varies with the machine tool builder. For details, refer to the manual supplied by the machine tool builder.

III. OPERATION

1 GENERAL

Chapter 1, "GENERAL", consists of the following sections:

1.1 MANUAL OPERATION....982
1.2 TOOL MOVEMENT BY PROGRAMING - AUTOMATIC OPERATION....984
1.3 AUTOMATIC OPERATION....986
1.4 TESTING A PROGRAM....988
1.5 EDITING A PROGRAM....990
1.6 DISPLAYING AND SETTING DATA....991
1.7 DISPLAY....994
1.8 ADJUSTMENT OF THE BRIGHTNESS OF THE MONOCHROME LCD....997

1.1 MANUAL OPERATION

Explanation

- Manual reference position return

The CNC machine tool has a position used to determine the machine position.
This position is called the reference position, where the tool is replaced or the coordinate are set. Ordinarily, after the power is turned on, the tool is moved to the reference position.
Manual reference position return is to move the tool to the reference position using switches and pushbuttons located on the operator's panel. (See Section III-3.1)

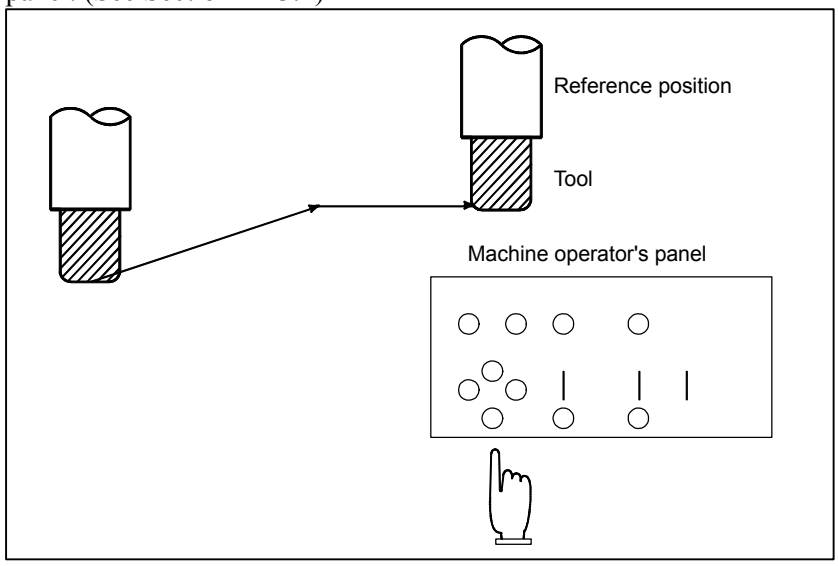


Fig. 1.1 (a) Manual reference position return

The tool can be moved to the reference position also with program commands.
This operation is called automatic reference position return (See Section II-6).

- The tool movement by manual operation

Using machine operator's panel switches, pushbuttons, or the manual handle, the tool can be moved along each axis.

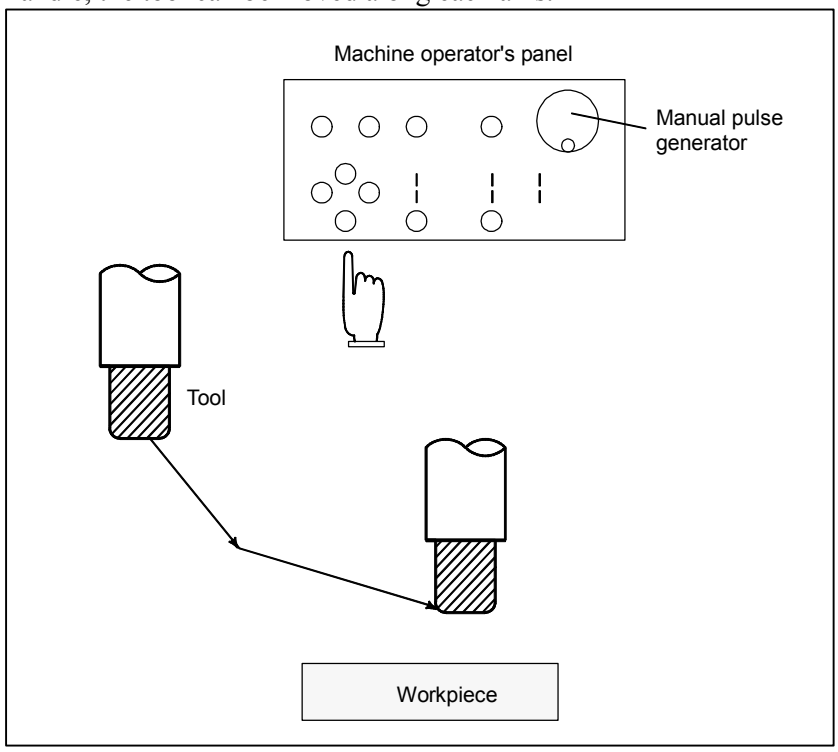


Fig. 1.1 (b) The tool movement by manual operation

The tool can be moved in the following ways:

(i) Jog feed (See Section III-3.2)
The tool moves continuously while a pushbutton remains pressed.

(ii) Incremental feed (See Section III-3.3)
The tool moves by the predetermined distance each time a button is pressed.

(iii) Manual handle feed (See Section III-3.4)
By rotating the manual handle, the tool moves by the distance corresponding to the degree of handle rotation.

1.2 TOOL MOVEMENT BY PROGRAMING - AUTOMATIC OPERATION

Automatic operation is to operate the machine according to the created program. It includes memory, MDI and DNC operations. (See Section III-4).

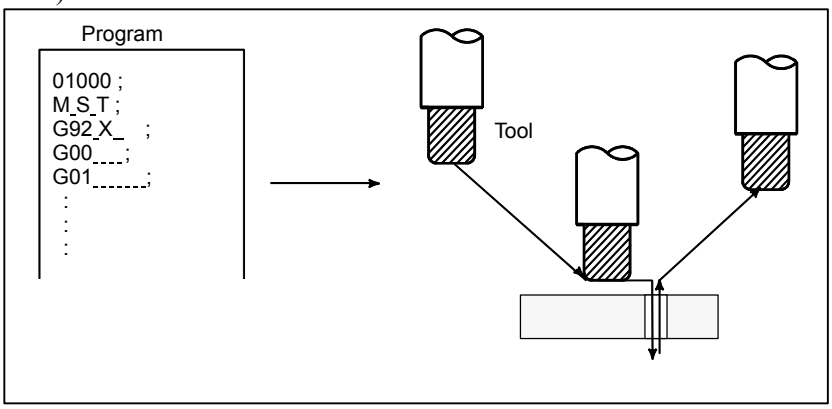


Fig. 1.2 (a) Tool Movement by programming

Explanation

- Memory operation

After the program is once registered in memory of CNC, the machine can be run according to the program instructions. This operation is called memory operation.

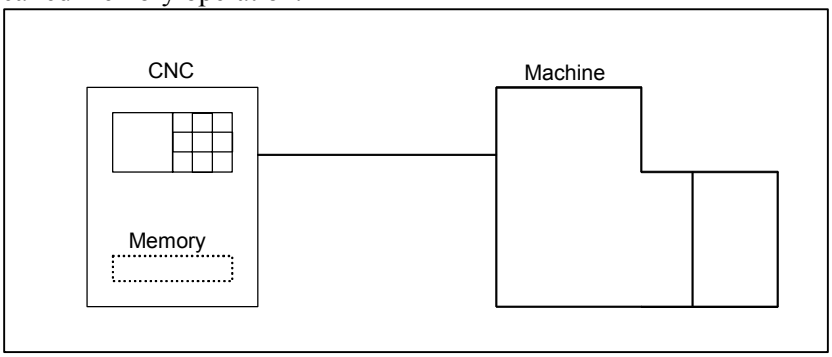


Fig. 1.2 (b) Memory operation

- MDI operation

After the program is entered, as an command group, from the MDI keyboard, the machine can be run according to the program. This operation is called MDI operation.

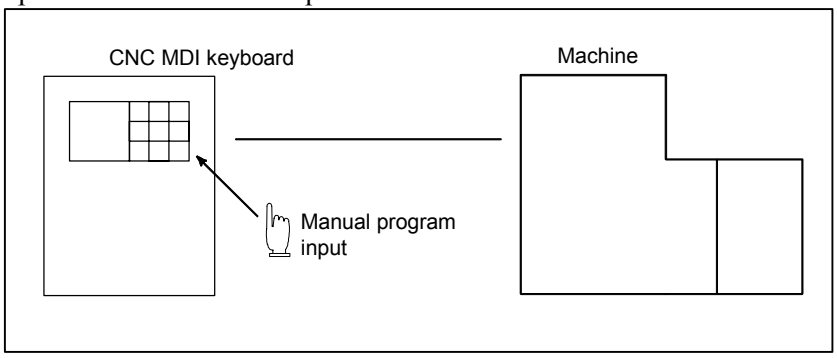


Fig. 1.2 (c) MDI operation

- DNC operation

In this mode of operation, the program is not registered in the CNC memory. It is read from the external input/output devices instead. This is called DNC operation.

1.3 AUTOMATIC OPERATION

Explanation

- Program selection

Select the program used for the workpiece. Ordinarily, one program is prepared for one workpiece. If two or more programs are in memory, select the program to be used, by searching the program number (Section III-9.3).

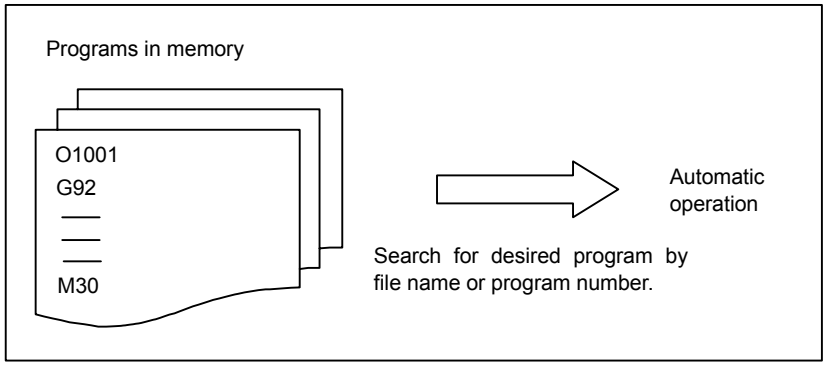


Fig. 1.3 (a) Program selection for automatic operation

- Start and stop

Pressing the cycle start pushbutton causes automatic operation to start. By pressing the feed hold or reset pushbutton, automatic operation pauses or stops. By specifying the program stop or program termination command in the program, the running will stop during automatic operation. When one process machining is completed, automatic operation stops. (See Section III-4)

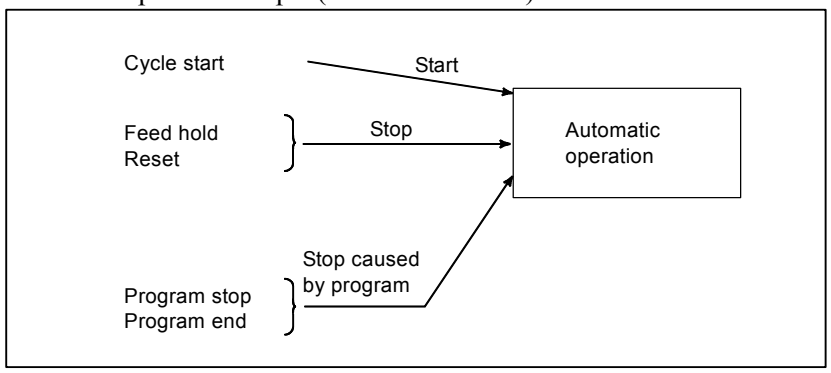


Fig. 1.3 (b) Start and stop for automatic operation

- Handle interruption

While automatic operation is being executed, tool movement can overlap automatic operation by rotating the manual handle. (See Section III-4.4)

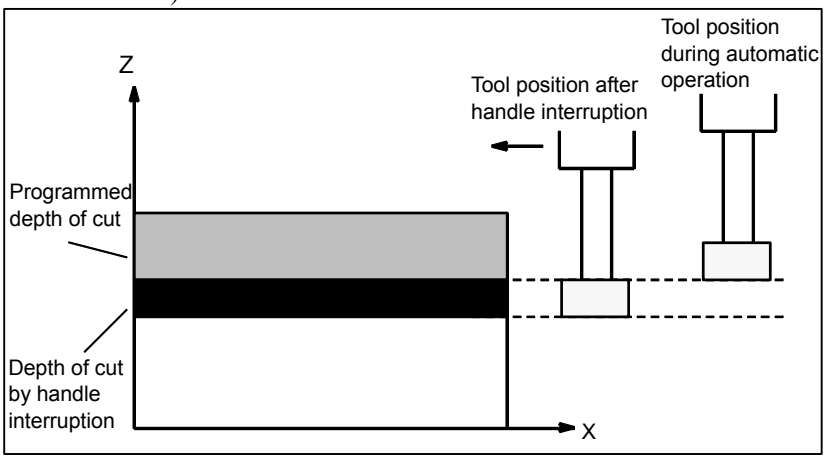


Fig. 1.3 (c) Handle interruption for automatic operation

1.4 TESTING A PROGRAM

Before machining is started, the automatic running check can be executed.
It checks whether the created program can operate the machine as desired.
This check can be accomplished by running the machine actually or viewing the position display change (without running the machine) (See Section III-5).

1.4.1 Check by Running the Machine

Explanation

- Dry run

Remove the workpiece, check only movement of the tool. Select the tool movement rate using the dial on the operator's panel. (See Section III-5.4)

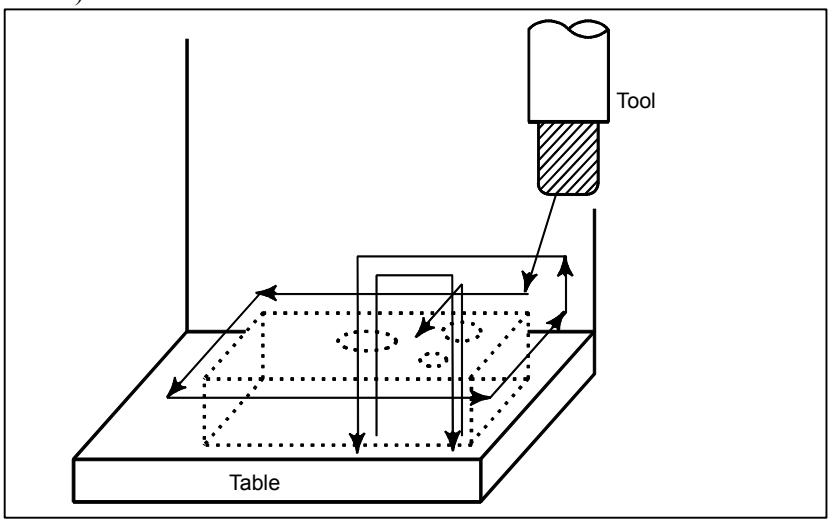


Fig. 1.4.1 (a) Dry run

- Feedrate override

Check the program by changing the feedrate specified in the program. (See Section III-5.2)

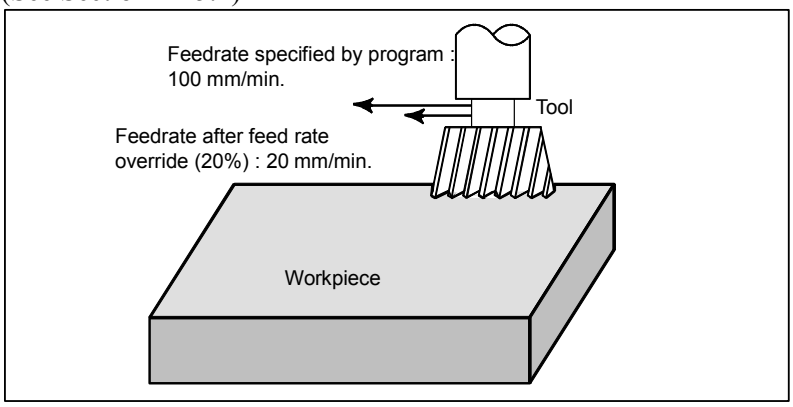


Fig. 1.4.1 (b) Feedrate override

- Single block

When the cycle start pushbutton is pressed, the tool executes one operation then stops. By pressing the cycle start again, the tool executes the next operation then stops. The program is checked in this manner. (See Section III-5.5)

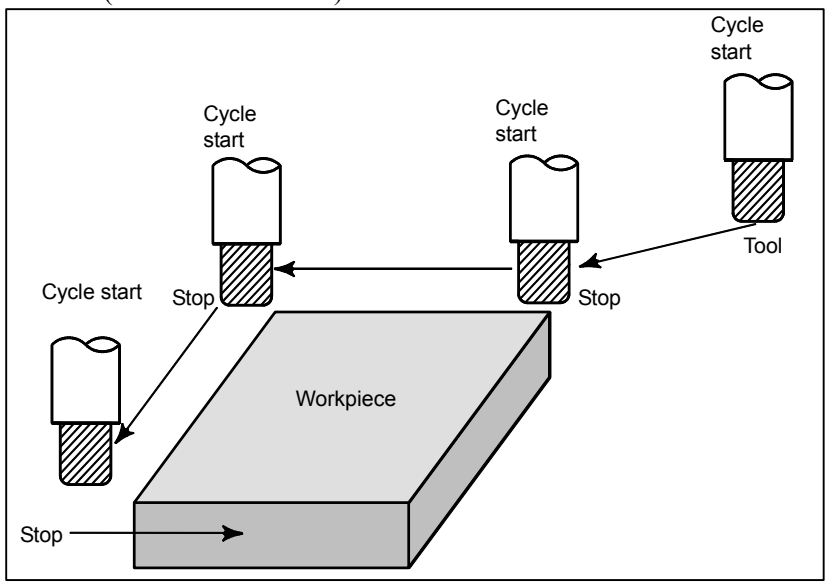


Fig. 1.4.1 (c) Single Block

1.4.2 How to View the Position Display Change without Running the Machine

Explanation

- Machine Lock

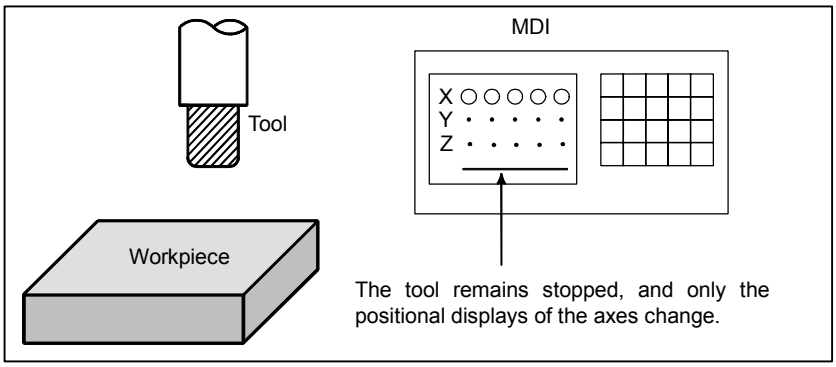


Fig. 1.4.2 (a) Machine Lock

- Auxiliary function lock

When automatic running is placed into the auxiliary function lock mode during the machine lock mode (See Sections III-5.1), all auxiliary functions (spindle rotation, tool replacement, coolant on/off, etc.) (See Section III-5.1) are disabled.

1.5 EDITING A PROGRAM

After a created program is once registered in memory, it can be corrected or modified from the MDI panel (See Section III-10).
This operation can be executed using the program edit function.

1.6 DISPLAYING AND SETTING DATA

The operator can display or change a value stored in CNC internal memory by key operation on the MDI screen (See III-12).

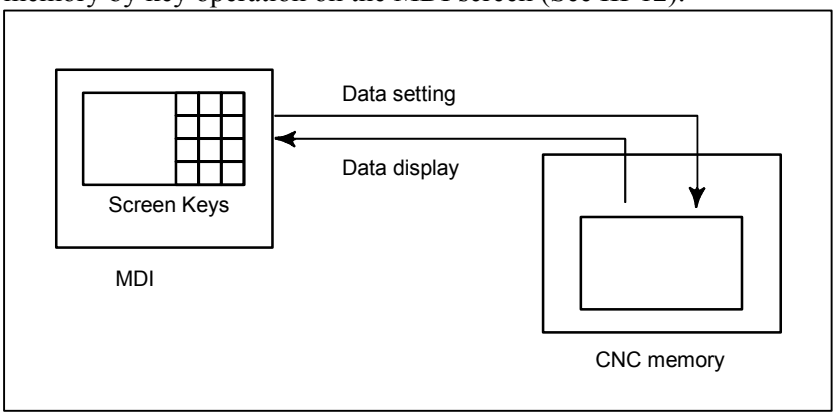


Fig. 1.6 (a) Displaying and setting data

Explanation

- Offset value

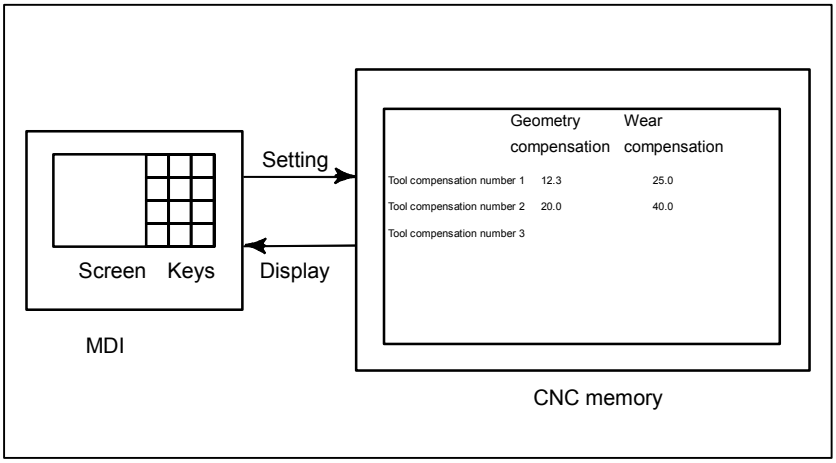


Fig. 1.6 (b) Displaying and Setting Offset Values

The tool has the tool dimension (length, diameter). When a workpiece is machined, the tool movement value depends on the tool dimensions. By setting tool dimension data in CNC memory beforehand, CNC automatically generates tool routes that permit any tool to cut the workpiece specified by the program. Tool dimension data is called the offset value.

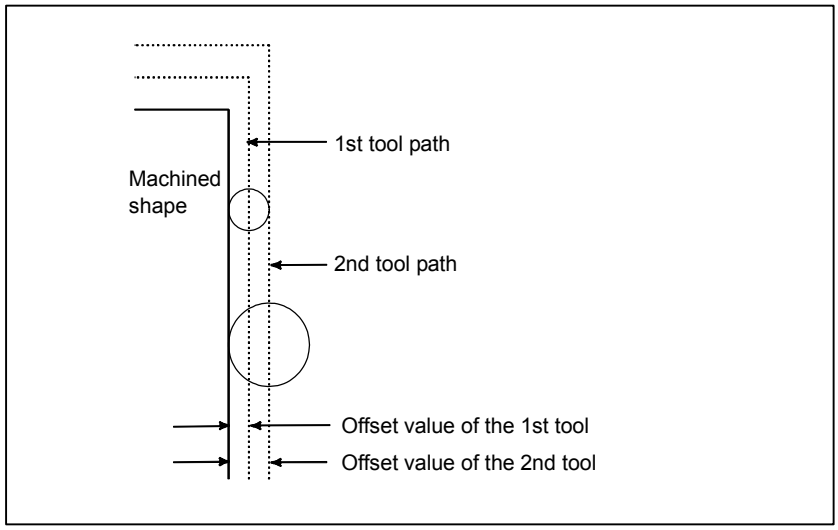


Fig. 1.6 (c) Offset value

- Displaying and setting operator's setting data

Apart from parameters, there is data that is set by the operator in operation. This data causes machine characteristics to change.
For example, the following data can be set:

- Inch/Metric switching
- Selection of I/O devices
- Mirror image cutting on/off

The above data is called setting data (See Section III-12.3.1).

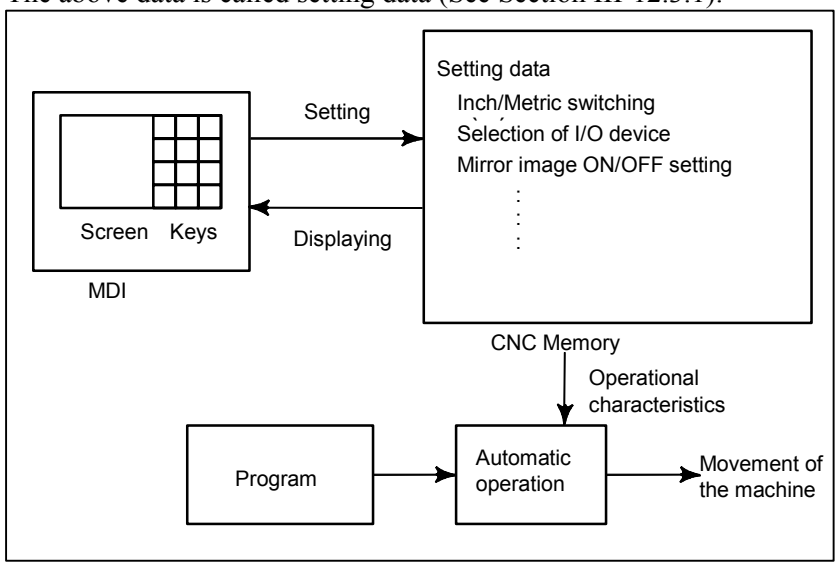


Fig. 1.6 (d) Displaying and setting operator's setting data

- Displaying and setting parameters

The CNC functions have versatility in order to take action in characteristics of various machines.
For example, CNC can specify the following:

- Rapid traverse rate of each axis
- Whether increment system is based on metric system or inch system.
- How to set command multiply/detect multiply (CMR/DMR)

Data to make the above specification is called parameters (See Section III-12.4.1).
Parameters differ depending on machine tool.

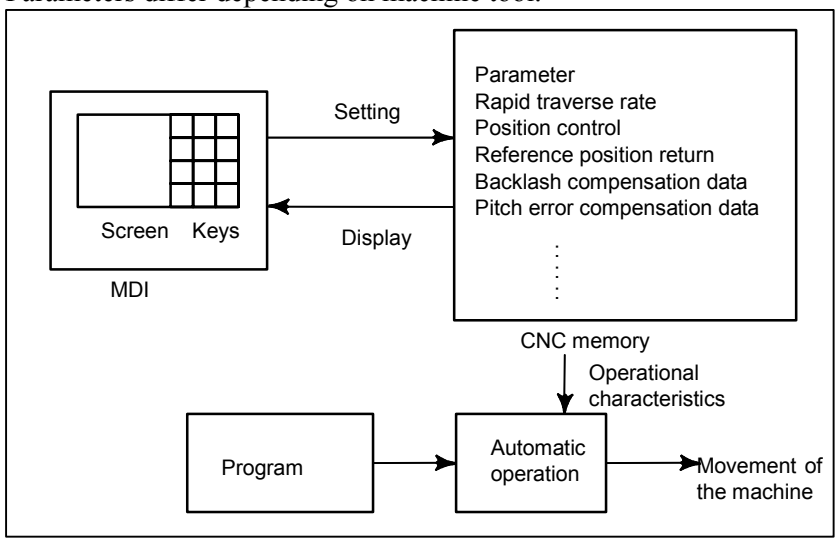


Fig. 1.6 (e) Displaying and setting parameters

- Data protection key

A key called the data protection key can be defined. It is used to prevent part programs, offset values, parameters, and setting data from being registered, modified, or deleted erroneously (See Section III-12).

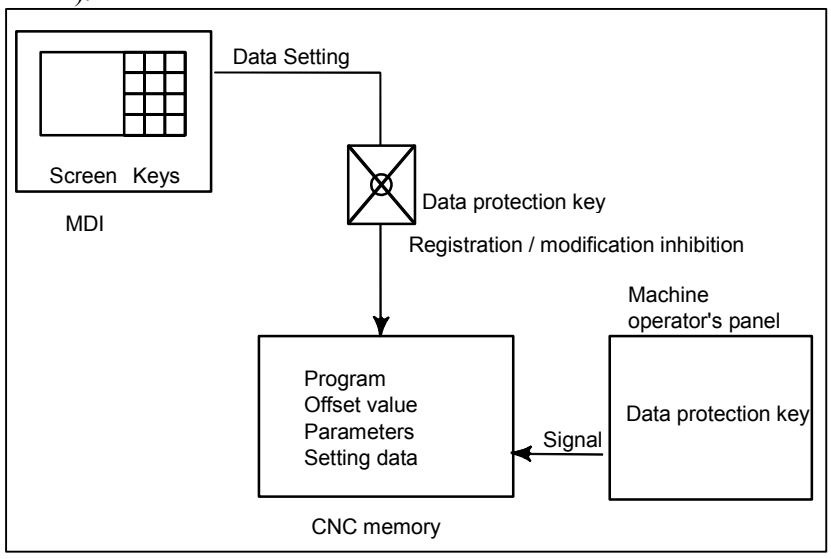


Fig. 1.6 (f) Data protection key

1.7 DISPLAY

1.7.1 Program Display

The contents of the currently active program are displayed.
(See Section III-12.2.1)

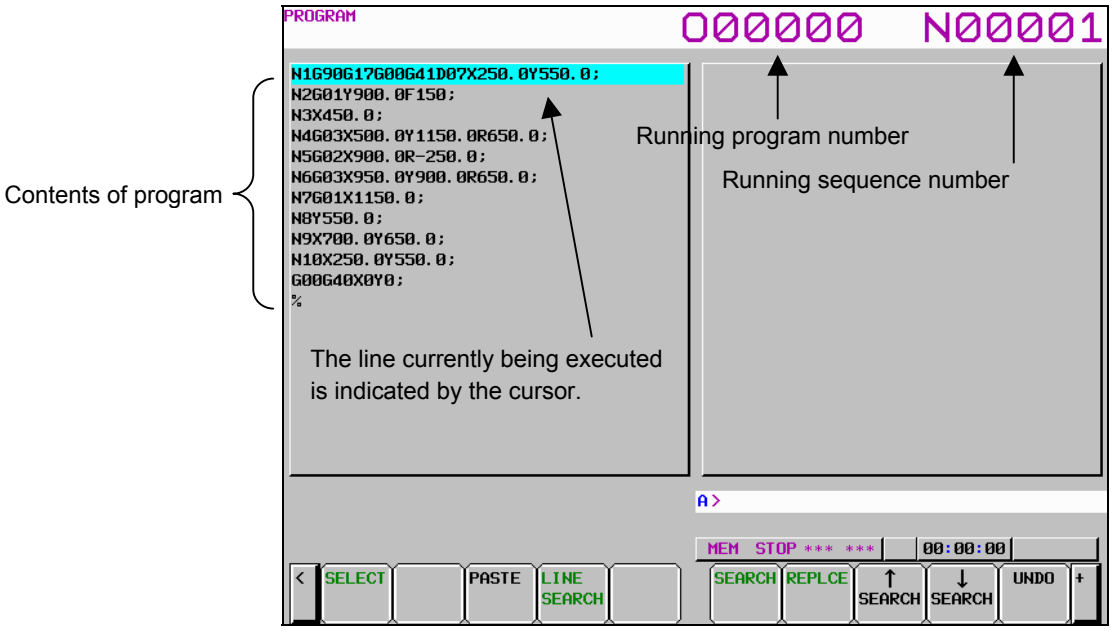


Fig. 1.7.1 (a)

A list of the programs held in the currently selected folder is displayed.

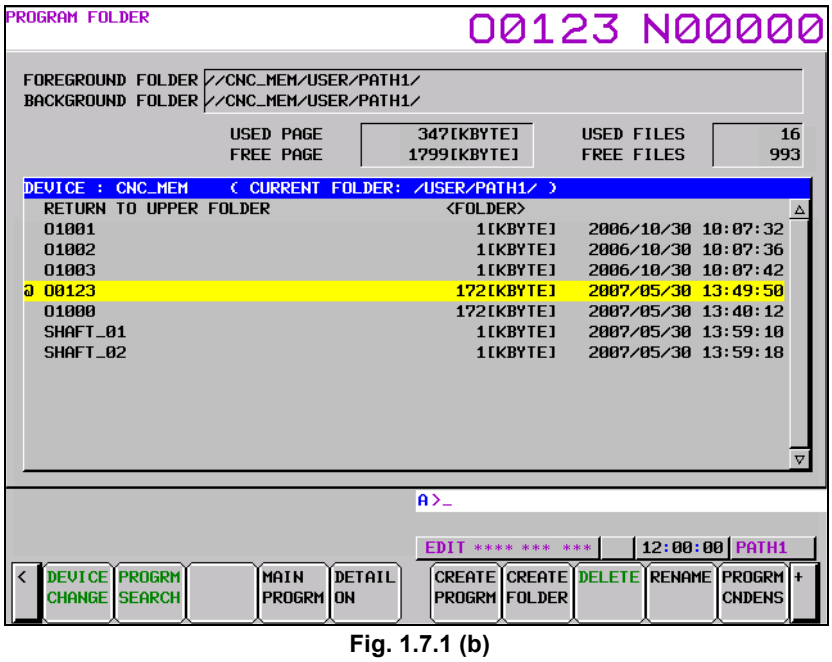


Fig. 1.7.1 (b)

1.7.2 Current Position Display

The current position of the tool is displayed with the coordinate values.
Moreover, the distance from the current position to a target point can be displayed as a remaining travel distance.
(See Section III-12.1.1, 12.1.2, 12.1.3.)

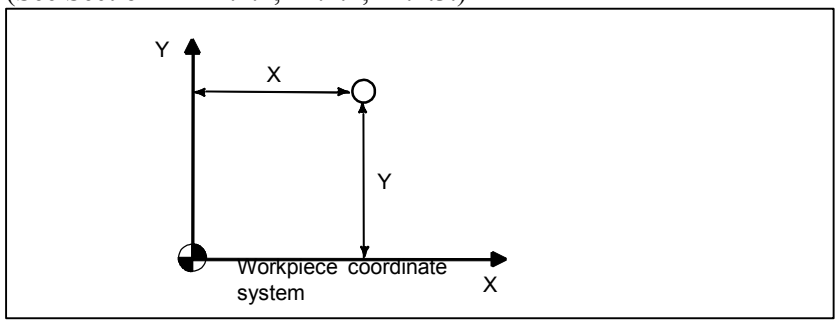


Fig. 1.7.2 (a)

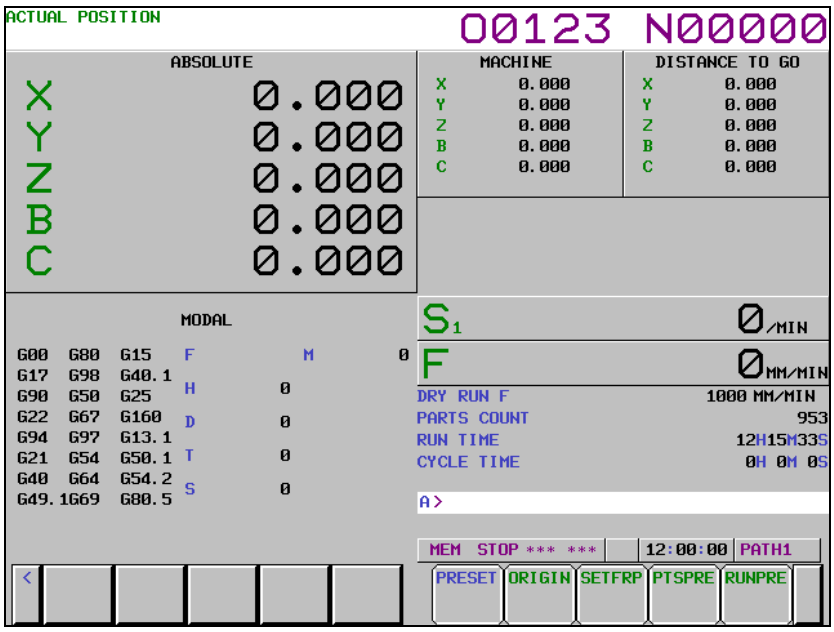


Fig. 1.7.2 (b)

1.7.3 Alarm Display

When a trouble occurs during operation, error code and alarm message are displayed on the screen. (See Section III-7.1.)
See APPENDIX G for the list of error codes and their meanings.

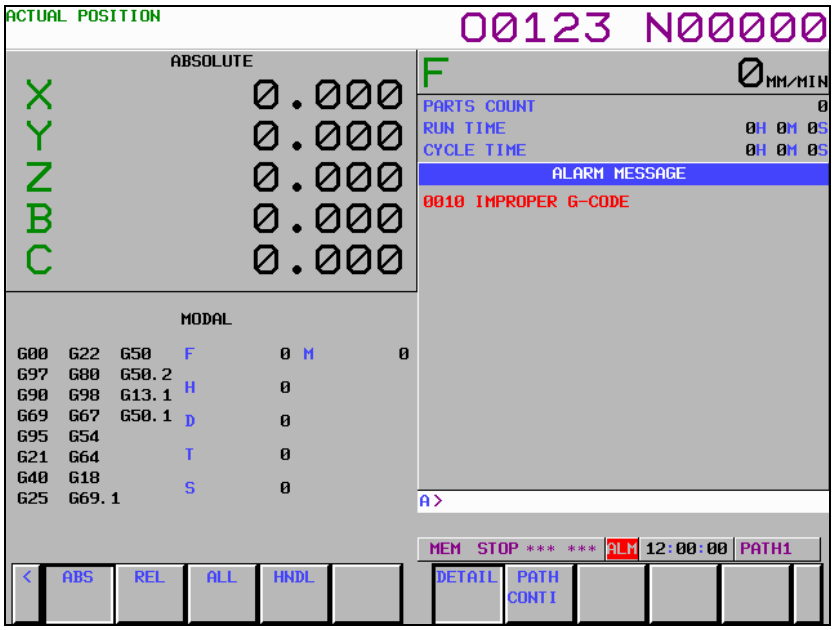


Fig. 1.7.3 (a)

1.7.4 Parts Count Display, Run Time Display

The position display screen displays a run time, cycle time, and parts count. (See Section lll-12.3.3.)

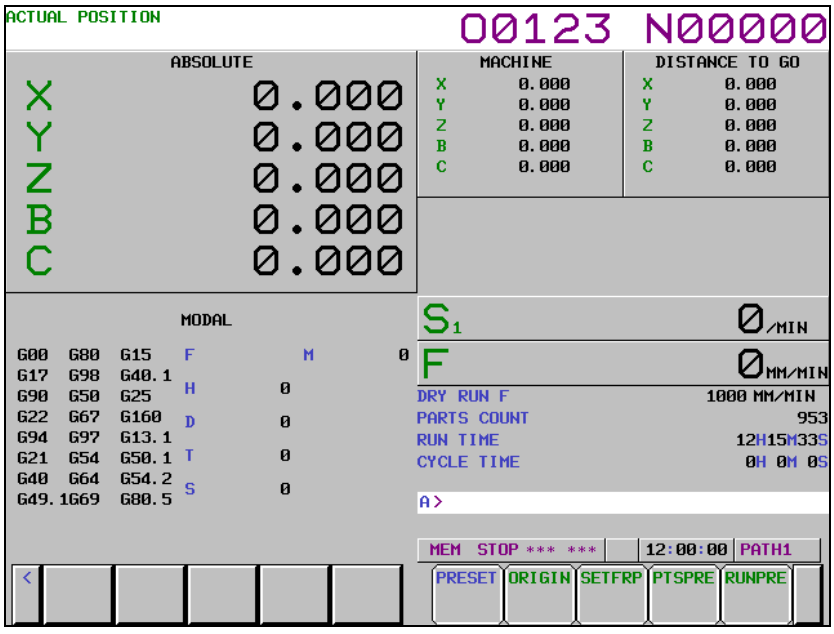


Fig. 1.7.4 (a)

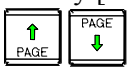
1.8 ADJUSTMENT OF THE BRIGHTNESS OF THE MONOCHROME LCD

The brightness of the monochrome LCD can be adjusted.

- Procedure

To adjust the brightness of the monochrome LCD, follow the procedure below on the SETTING (HANDY) screen.

1 Press function key .

2 Press soft key [SETING] to display the SETTING DATA screen. Press  multiple times until the SETTING (HANDY) screen appears.

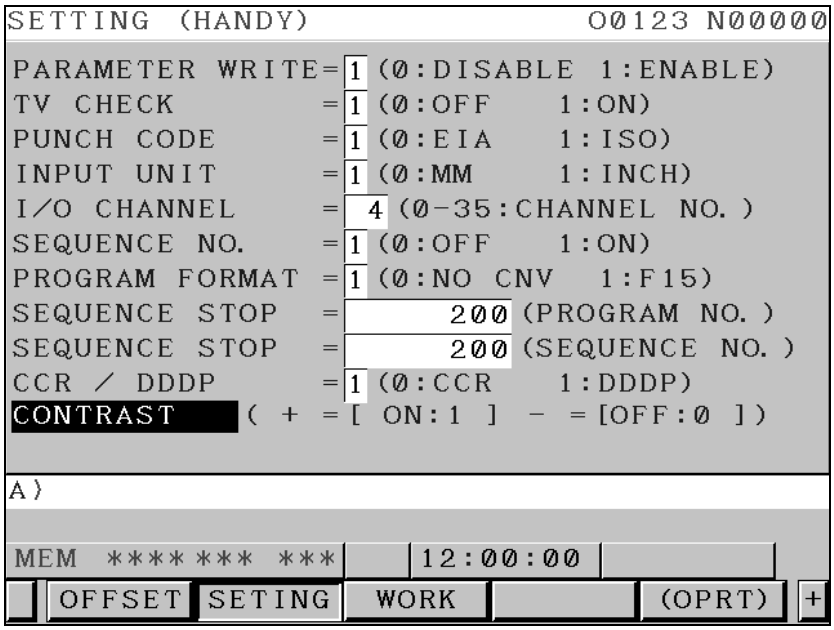


Fig. 1.8 (a) Adjustment of the brightness of the SETTING screen

3 Move the cursor to the CONTRAST item.

4 When soft key [(OPRT)] is pressed, soft keys [ON:1] and [OFF:0] are displayed.
Each time soft key [ON:1] is pressed, the brightness of the screen is increased.
Each time soft key [OFF:0] is pressed, the brightness of the screen is decreased.

NOTE
1 The new brightness of the screen is saved across power-off.
2 This setting item appears only when the monochrome LCD is connected.

2 OPERATIONAL DEVICES

As operational devices, setting and display devices attached to the CNC, and machine operator's panels are available.
For machine operator's panels, refer to the relevant manual of the machine tool builder.

Chapter 2, "OPERATIONAL DEVICES", consists of the following sections:

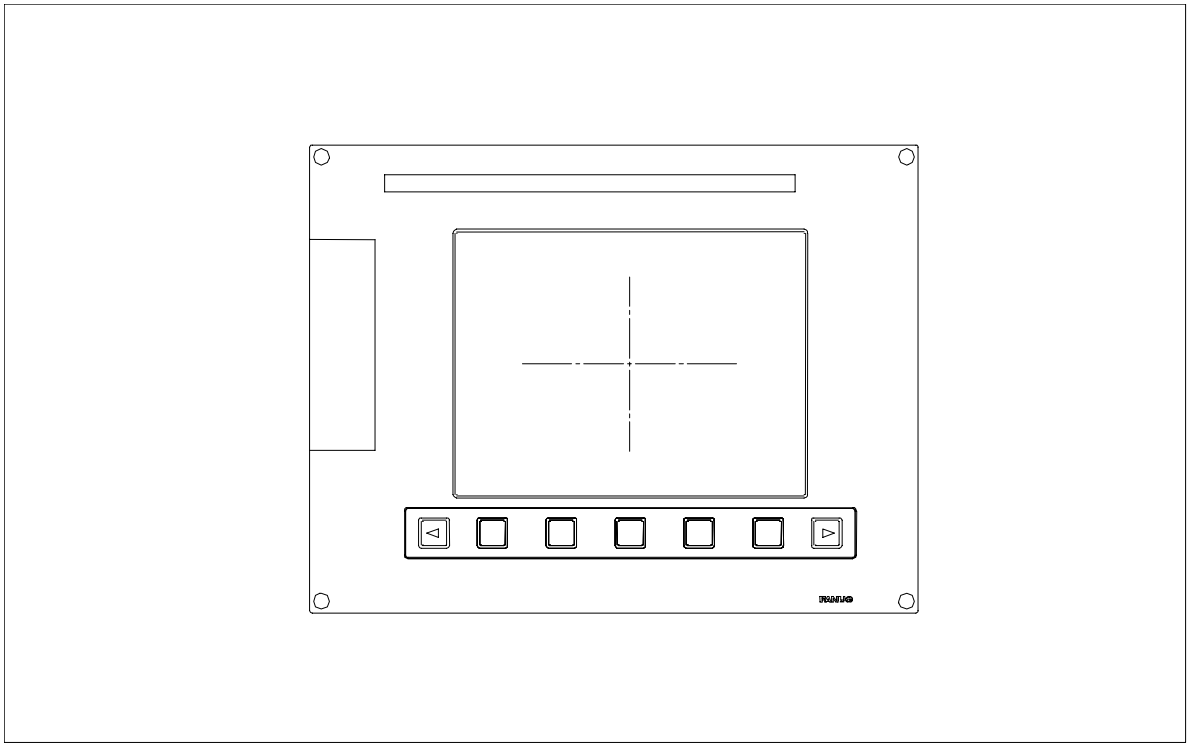
2.1 SETTING AND DISPLAY UNITS 999
2.2 OPERATIONAL DEVICES 1006
2.3 FUNCTION KEYS AND SOFT KEYS 1009
2.4 EXTERNAL I/O DEVICES.....1025
2.5 POWER ON/OFF.....1027

2.1 SETTING AND DISPLAY UNITS

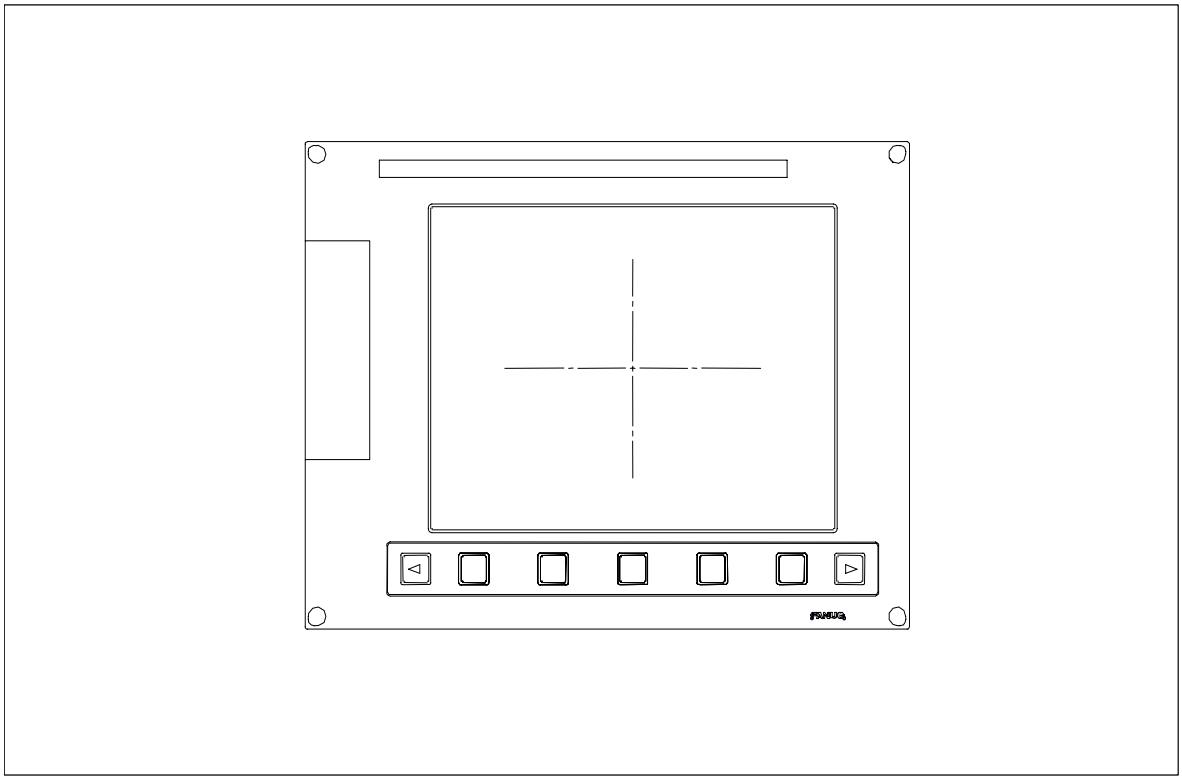
The setting and display units are shown in Subsections 2.1.1 to 2.1.8 of Part III.

7.2" LCD CNC Display Panel III-2.1.1
8.4" LCD CNC Display Panel III-2.1.2
10.4" LCD CNC Display Panel III-2.1.3
12.1" LCD CNC Display Panel III-2.1.4
15" LCD CNC Display Panel III-2.1.5
Standard MDI Unit (ONG Key) III-2.1.6
Standard MDI Unit (QWERTY Key) III-2.1.7
Small MDI Unit (ONG Key) III-2.1.8

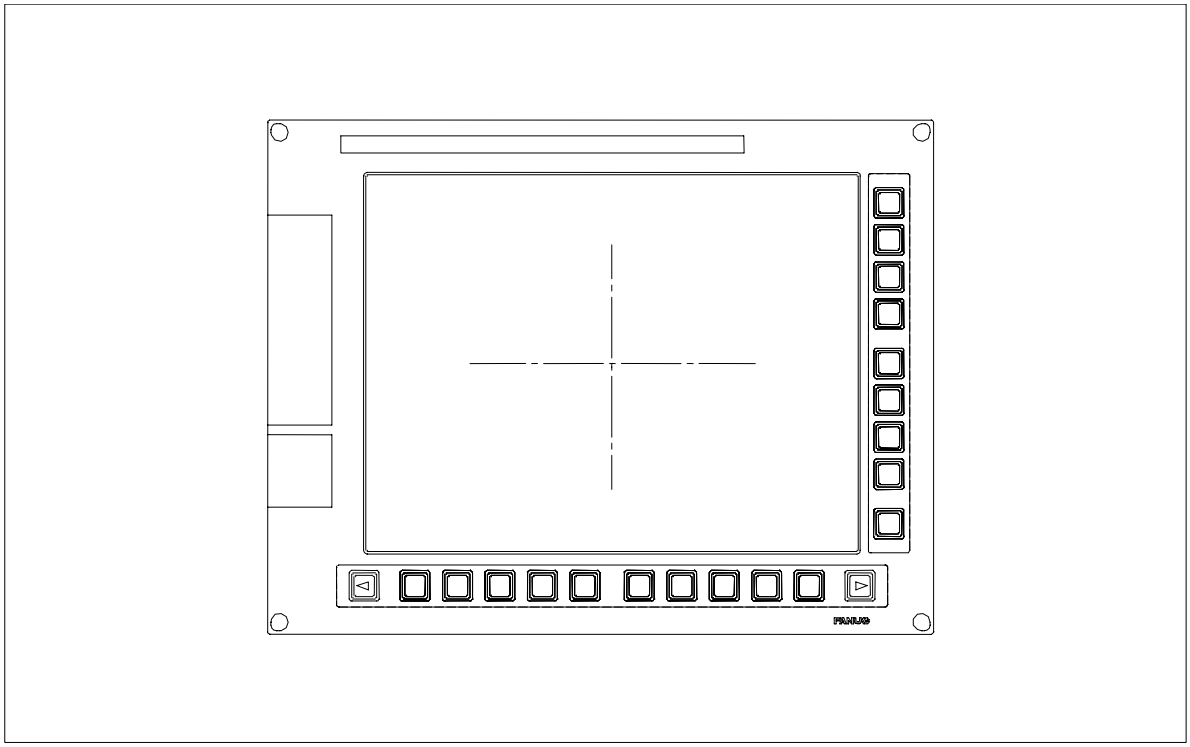
2.1.1 7.2" LCD CNC Display Panel



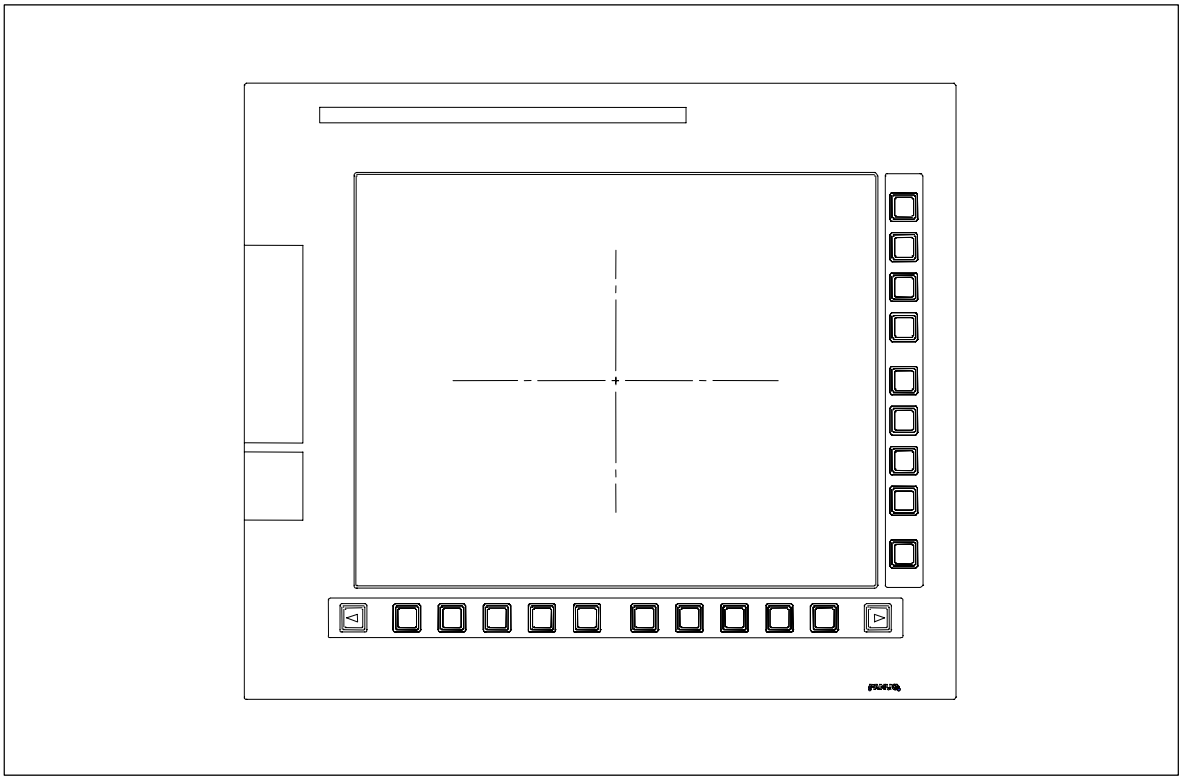
2.1.2 8.4" LCD CNC Display Panel



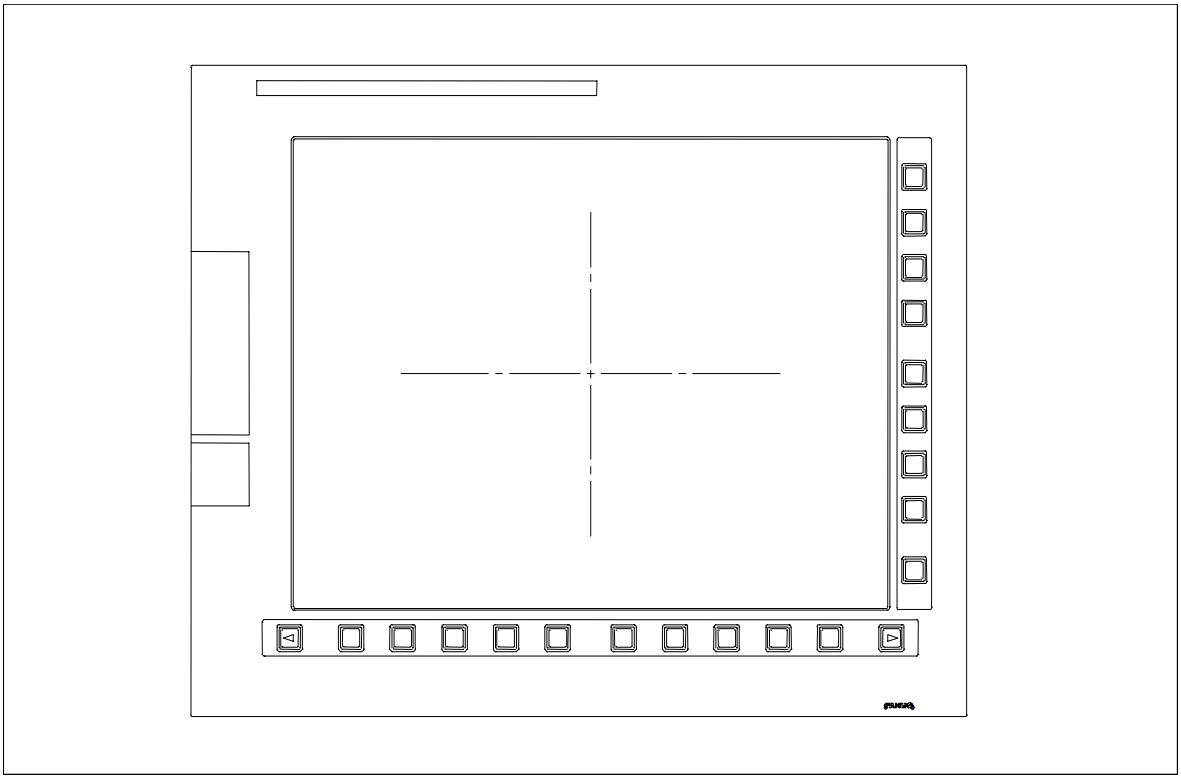
2.1.3 10.4" LCD CNC Display Panel



2.1.4 12.1" LCD CNC Display Panel

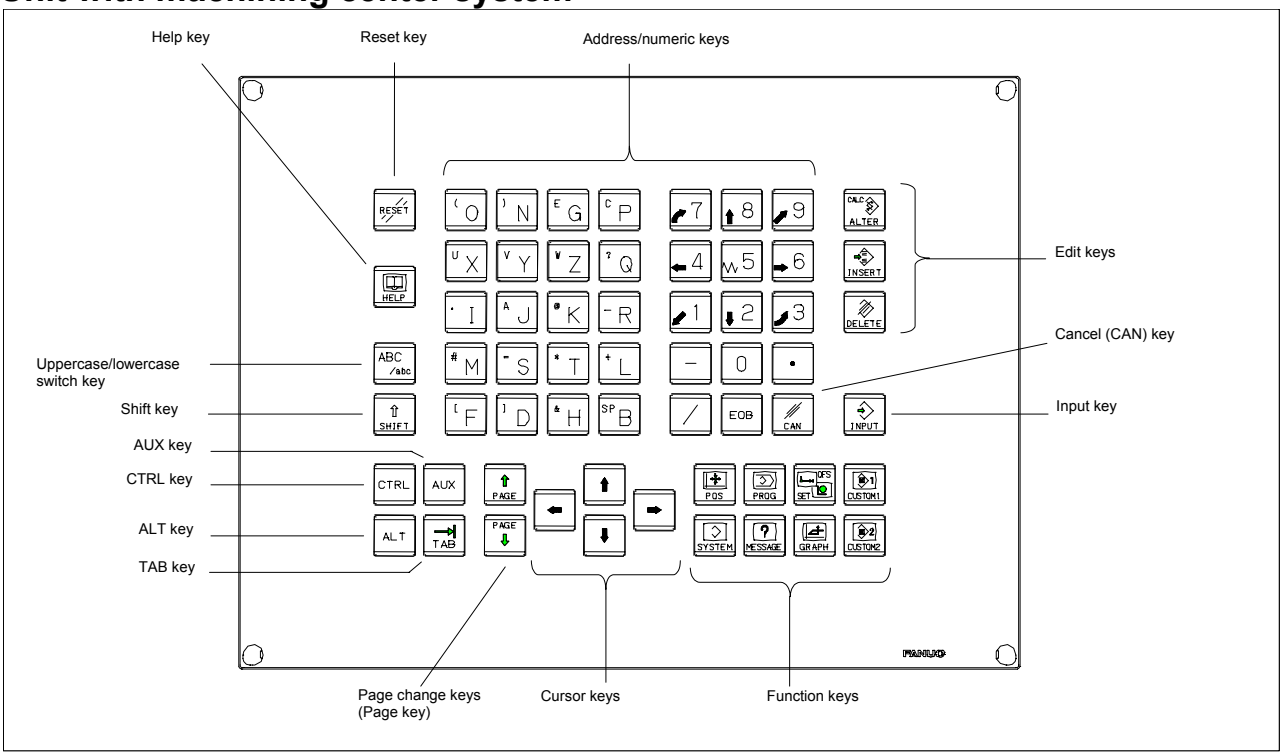


2.1.5 15" LCD CNC Display Panel

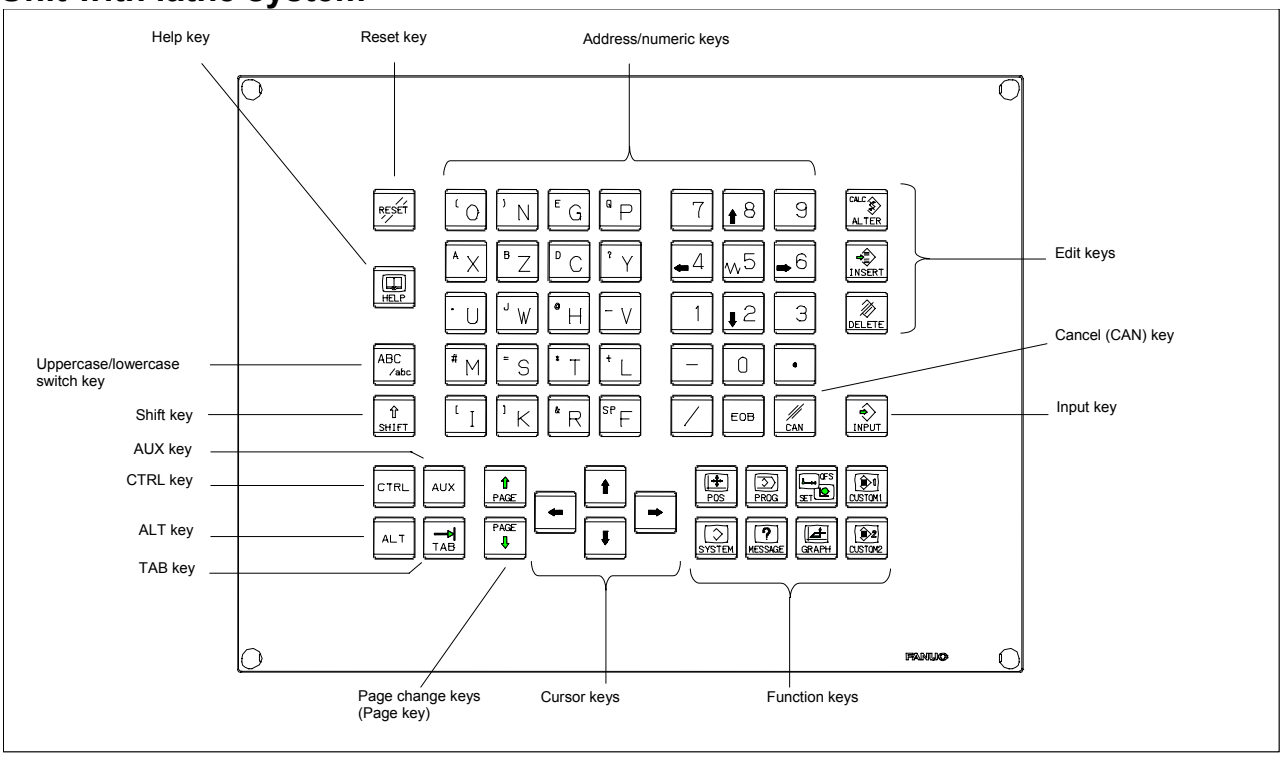


2.1.6 Standard MDI Unit (ONG Key)

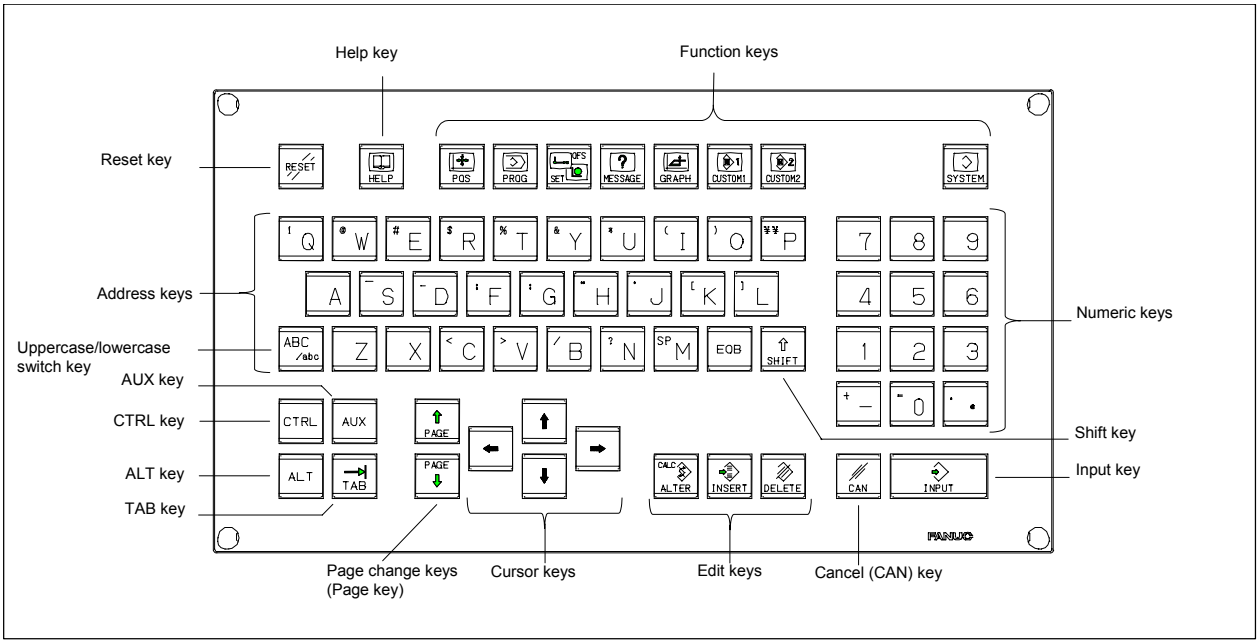
Unit with machining center system



Unit with lathe system

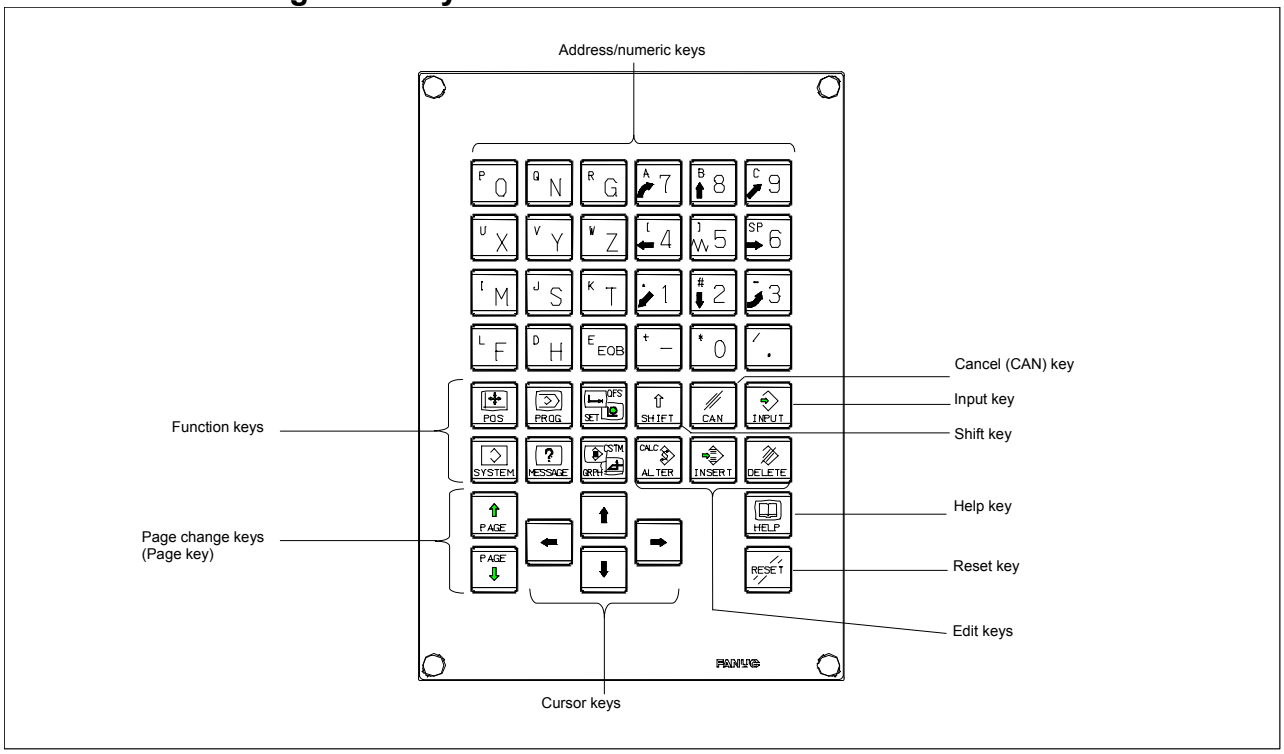


2.1.7 Standard MDI Unit (QWERTY Key)

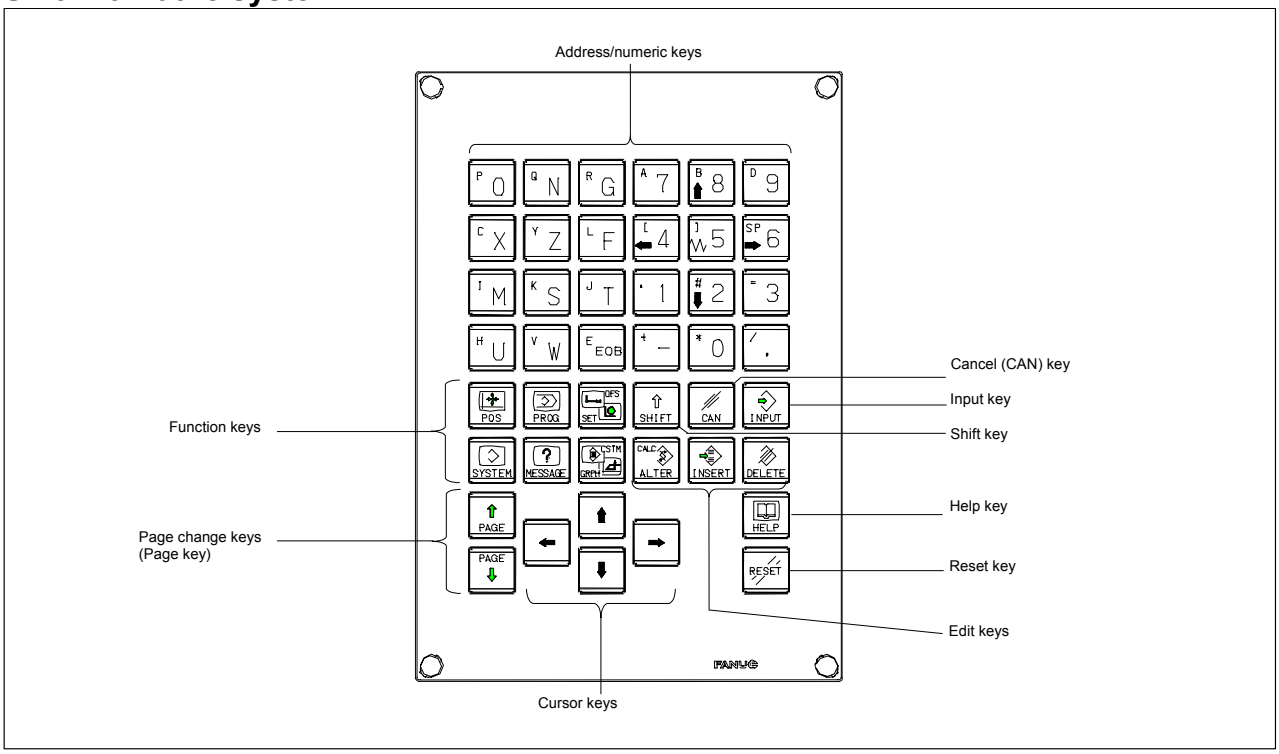


2.1.8 Small MDI Unit (ONG Key)

Unit with machining center system



Unit with lathe system



2.2 OPERATIONAL DEVICES

Table 2.2 (a) Explanation of the MDI keyboard

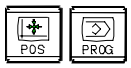
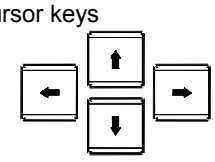
Number	Name	Explanation
1	RESET key	Press this key to reset the CNC, to cancel an alarm, etc.
2	HELP key	Press this key to use the help function when uncertain about the operation of an MDI key (help function).
3	Soft keys	The soft keys have various functions, according to the Applications. The soft key functions are displayed on the display unit.
4	Address and numeric keys	Press these keys to input alphabetic, numeric, and other characters.
5	SHIFT key	Some keys have two characters on their keytop. Pressing the <SHIFT> key switches the characters. Special character ^ is displayed on the screen when a character indicated at the bottom right corner on the keytop can be entered.
6	INPUT key	When an address or a numerical key is pressed, the data is input to the buffer, and it is displayed on the screen. To copy the data in the key input buffer to the offset register, etc., press the key. This key is equivalent to the [INPUT] key of the soft keys, and either can be pressed to produce the same result. This key is also used to move to a folder on the program folder screen.
7	CANCEL (CAN) key	Press this key to delete the last character or symbol input to the key input buffer. Example) When the key input buffer displays >N001X100Z_ and the cancel key is pressed, Z is canceled and >N001X100_ is displayed.
8	Edit keys	Press these keys when editing the program. : ALTER : INSERT : IDELETE
9	Function keys 	Press theses keys to switch display screens for each function. See III-2.3 for details of the function keys.
10	Cursor keys 	There are four different cursor move keys. : This key is used to move the cursor to the right or in the forward direction. The cursor is moved in short units in the forward direction. : This key is used to move the cursor to the left or in the reverse direction. The cursor is moved in short units in the reverse direction. : This key is used to move the cursor in a downward or forward direction. The cursor is moved in large units in the forward direction. : This key is used to move the cursor in an upward or reverse direction. The cursor is moved in large units in the reverse direction.

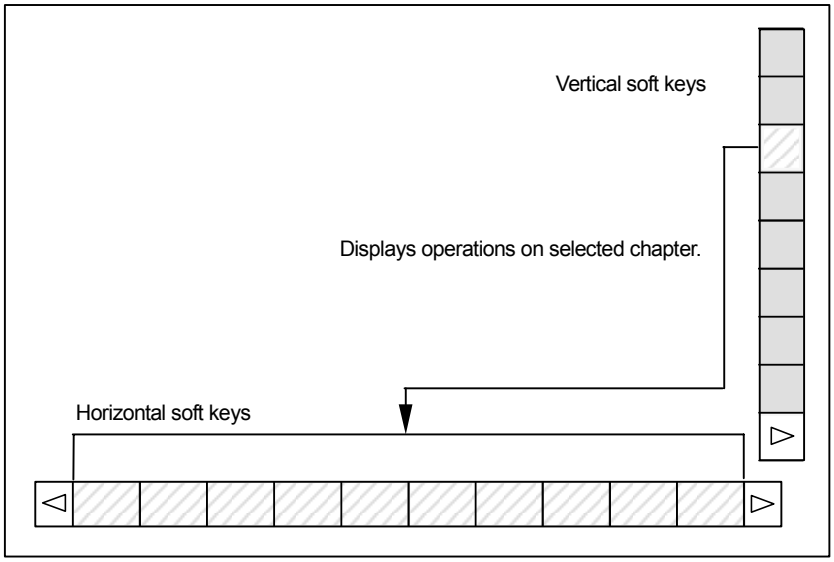
Table 2.2 (a) Explanation of the MDI keyboard

Number	Name	Explanation
11	Page change keys (Page keys)	Two kinds of page change keys are described below. : This key is used to changeover the page on the screen in the forward direction. : This key is used to changeover the page on the screen in the reverse direction.
12	Uppercase/lowercase switch key	Press this key to switch between uppercase and lowercase when entering alphabetic characters.
13	PC function key	These keys are used with the personal computer function of the 300*i*/300*i*s, 310*i*/310*i*s, 320*i*/320*i*s.

Explanation

- 15” LCD/MDI soft key configuration

The 15” LCD panel has 12 soft keys horizontally and 9 soft keys vertically.
As shown below, the 8 vertical soft keys and the lowermost key are used as chapter selection soft keys. By pressing each of these keys, the screen (chapter) belonging to each function can be selected. The horizontal 12 soft keys are used to perform operations on the screen selected with a vertical soft key.
For an LCD display with a touch panel, touch a soft key on the screen to select it.



In this manual, the descriptions below assume a 10.4" LCD display panel with 12 soft keys.

- Key operation with multi-path control

In the multi-path control, be sure to select the tool post for which data is specified, using the path selection switch on the machine operator's panel. Then, perform keyboard operation, such as displaying or specifying various data items, and editing a program.

2.3 FUNCTION KEYS AND SOFT KEYS

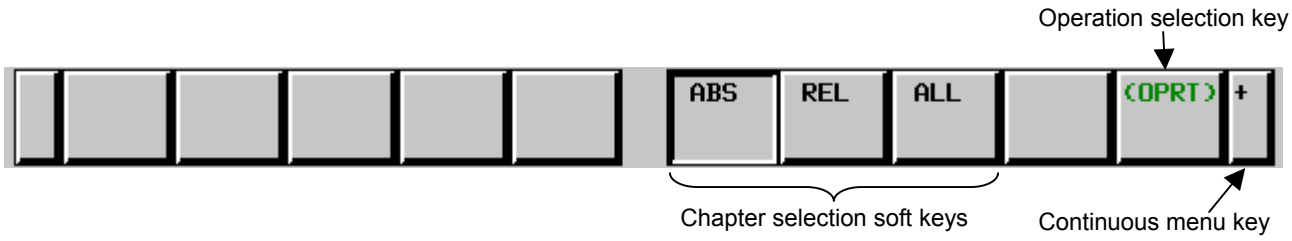
The function keys are used to select the type of screen (function) to be displayed. When a soft key (section select soft key) is pressed immediately after a function key, the screen (section) corresponding to the selected function can be selected.

2.3.1 General Screen Operations

- Procedure

1 By pressing a function key on the MDI panel, the chapter selection soft keys that belong to the function are displayed.

Example 1)

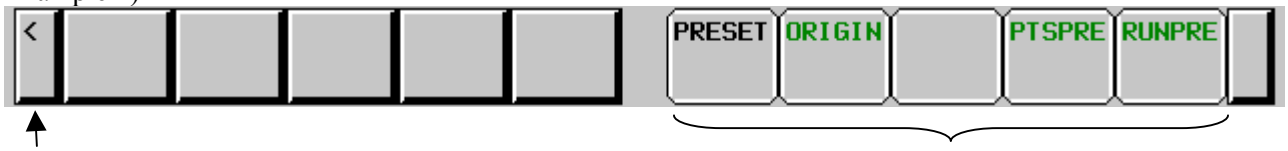


2 When one of the chapter selection soft keys is pressed, the screen of the chapter is displayed. If the soft key of a desired chapter is not displayed, press the continuous menu key.
In a chapter, a further choice may be made from multiple chapters.

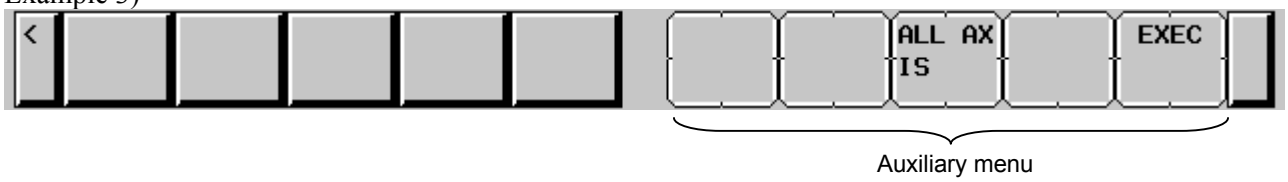
3 When the screen of a desired chapter is displayed, press the operation selection key to display operations to be performed.

4 Select a desired operation with the operation selection soft key.
Depending on the operation to be executed, an auxiliary menu of soft keys is displayed. Perform an operation according to the indications on the auxiliary menu.

Example 2)



Example 3)



5 To return to the display of chapter selection soft keys, press the return menu key.

A general screen display procedure is provided above.
The actual display procedure varies from one screen to another.
For details, see each description of operation.

- Button design change depending on soft key state

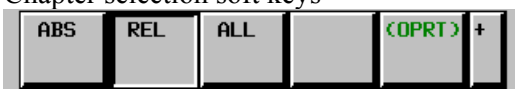
The soft keys assume one of the following states, depending on the selection target:

- Chapter selection soft keys
- Operation selection soft keys
- Auxiliary menu of operation selection soft keys

Depending on the state, the button images of the soft keys change. From the button images, which state the soft keys are assuming can be known.

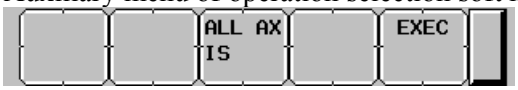
Example)

- Chapter selection soft keys



- Operation selection soft keys

- Auxiliary menu of operation selection soft keys



2.3.2 Function Keys

Function keys are provided to select the type of screen to be displayed. The following function keys are provided on the MDI panel:

Press this key to display the position screen.

Press this key to display the program screen.

Press this key to display the offset/setting screen.

Press this key to display the system screen.

Press this key to display the message screen.

Press this key to display the graphics screen.

Press this key to display the custom screen 1 (conversational macro screen or C language executor screen).

Press this key to display the custom screen 2 (conversational macro screen or C language executor screen).

2.3.3 Soft Keys

By pressing a soft key after a function key, the corresponding screen of the function can be displayed.

The chapter selection soft keys of each function are described below. The horizontal four keys on the right-hand side are assigned to chapter selection soft keys. When multiple pages are used for chapter selection soft keys, [+] is displayed on the continuous menu key (rightmost soft key). Press the continuous menu key to switch between chapter selection soft keys.

NOTE
1 Press function keys to switch between screens that are used frequently.
2 Some soft keys are not displayed depending on the option configuration.

If position indications are provided on the left half of the screen when a key other than the function key

is pressed, the left half of the soft keys is displayed as follows at all times:

Position display screen

The chapter selection soft keys that belong to the function key

and the function of each screen are described below.

Page 1

(1)	(2)	(3)	(4)	(5)	
ABS	REL	ALL	HNDL	(OPRT)	+

Page 2

(6)	(7)	(8)	(9)	(10)	
MONI	3-D MANUAL			(OPRT)	+

Table 2.3.3 (a) Position display screen

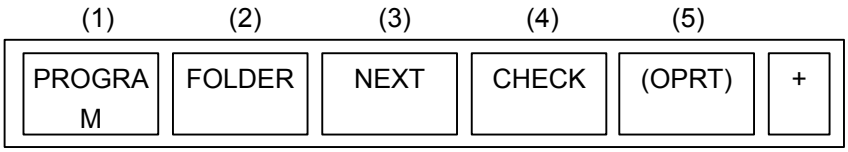
No.	Chapter menu	Description
(1)	ABS	Selects the absolute coordinate display screen.
(2)	REL	Selects the relative coordinate display screen.
(3)	ALL	Selects the overall coordinate display screen.
(4)	HNDL	Selects the operation screen for manual handle operation.
(6)	MONI	Selects the screen for displaying the servo axis load meter, serial spindle load meter, and speedometer.
(7)	3-D MANUAL	Displays a handle pulse interrupt amount in three-dimensional manual feed.

Program screen

The chapter selection soft keys that belong to the function key

and the function of each screen are described below.

Page 1



Page 2

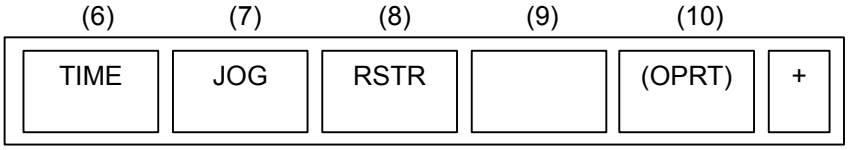


Table 2.3.3 (b) Program

No.	Chapter menu	Description
(1)	PROGRAM	Selects the screen for displaying a list of part programs currently registered.
(2)	FOLDER	Selects the screen for displaying a list of part programs currently registered.
(3)	NEXT	Selects the screen for displaying the command values of the block currently executed and the next block to be executed among the command values.
(4)	CHECK	Selects the screen for displaying programs, position data, modal information, and so forth simultaneously.
(6)	TIME	Selects the screen for displaying executed program operation time.
(7)	JOG	Selects the screen for executing, in the JOG mode, data specified in the program format from the MDI.
(8)	RSTR	Selects the operation screen for restarting an interrupted program operation.

Offset/setting screen

The chapter selection soft keys that belong to the function key

and the function of each screen are described below.

	(1)	(2)	(3)	(4)	(5)	
Page 1	OFFSET	SETTING	WORK		(OPRT)	+

	(6)	(7)	(8)	(9)	(10)	
Page 2	MACRO		OPR	TOOL MANAGER	(OPRT)	+

	(11)	(12)	(13)	(14)	(15)	
Page 3	OFST.2	W.SHFT	GEOM.2		(OPRT)	+

	(16)	(17)	(18)	(19)	(20)	
Page 4		PR-LV	EXTEND OFFSET		(OPRT)	+

	(21)	(22)	(23)	(24)	(25)	
Page 5	CHUCK TAIL	LANG.	PROTECT	GUARD	(OPRT)	+

	(26)	(27)	(28)	(29)	(30)	
Page 6				TOOL LIFE	(OPRT)	+

	(31)	(32)	(33)	(34)	(35)	
Page 7	WORK SET ER				(OPRT)	+

Table 2.3.3 (c) Offset

No.	Chapter menu	Description
(1)	OFFSET	Selects the screen for setting tool offset values.
(2)	SETTING	Selects the screen for setting the setting parameters.
(3)	WORK	Selects the screen for setting a workpiece coordinate system offset.
(6)	MACRO	Selects the screen for setting macro variables.
(8)	OPR	Selects the screen for operating some operation switches on the machine operator's panel as soft switches.
(9)	TOOL MANAGER	Selects the screen for setting data related to tool management.
(11)	OFST.2	Selects the screen for setting a Y-axis offset.
(12)	W.SHFT	Selects the screen for setting a workpiece coordinate system shift value.
(13)	GEOM.2	Selects the screen for setting a second geometry offset.
(17)	PR-LV	Selects the screen for setting a precision level.
(18)	EXTEND OFFSET	Selects the screen for setting the offsets of the fourth and fifth axes.
(21)	CHUCK TAIL	Selects the chuck tail stock barrier screen.
(22)	LANG.	Selects the screen for setting a display language.
(23)	PROTECT	Selects the screen for setting data protection.
(24)	GUARD	Selects the screen for setting wrong operation prevention.
(29)	TOOL LIFE	Selects the screen for operations and setting related to tool life management.
(31)	WRK ERR COMP	Selects the screen for setting errors related to workpiece mounting position.

System screen

The chapter selection soft keys that belong to the function key

and the function of each screen are described below.

Page 1

(1)	(2)	(3)	(4)	(5)	
PARAM	DGNOS	SERVO GUIDEM	SYSTEM	(OPRT)	+

Page 2

(6)	(7)	(8)	(9)	(10)	
MEMORY	PITCH	SERVO PARAM	SP.SET	(OPRT)	+

Page 3

(11)	(12)	(13)	(14)	(15)	
PMC MAINTE	PMC LADDER	PMC CONFIG			+

Page 4

(16)	(17)	(18)	(19)	(20)	
MCNG TUNING		ALL IO	OPEHIS	(OPRT)	+

Page 5

(21)	(22)	(23)	(24)	(25)	
COLOR	MAINTE	M-INFO	W. DGNS	(OPRT)	+

Page 6

(26)	(27)	(28)	(29)	(30)	
	FSSB	PRMTUN	P.MATE MGR.	(OPRT)	+

Page 7

(31)	(32)	(33)	(34)	(35)	
EMBED PORT	PCMCIA LAN	ETHNET BOARD	PROFI MASTER	(OPRT)	+

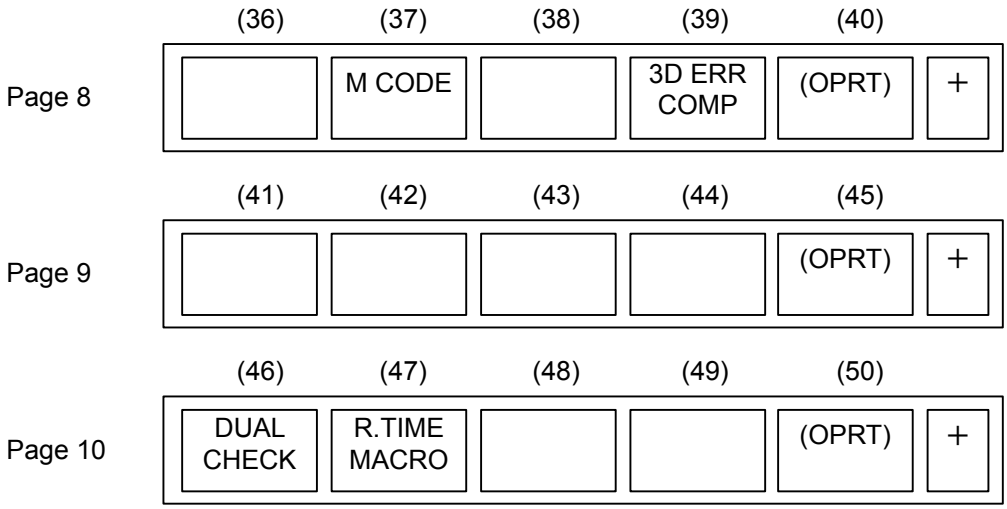


Table 2.3.3 (d) System

No.	Chapter menu	Description
(1)	PARAM	Selects the screen for setting parameters.
(2)	DGNOS	Selects the screen for displaying CNC state.
(3)	SERVO GUIDEM	Selects the screen for displaying the servo guide mate.
(4)	SYSTEM	Selects the screen for displaying the current system status.
(6)	MEMORY	Selects the screen for displaying the contents of memory.
(7)	PITCH	Selects the screen for setting pith error compensation.
(8)	SERVO PARAM	Selects the screen for setting the servo-related parameters.
(9)	SP.SET	Selects the screen for spindle-related setting.
(11)	PMC MAINTE	Selects the screen related to PMC maintenance such as PMC signal state monitoring and tracing, and PMC parameter display/editing.
(12)	PMC LADDER	Selects the screen related to ladder display/editing.
(13)	PMC CONFIG	Displays the screen for displaying/editing data other than ladders that makes up a sequence program and for setting the PMC function.
(16)	MCNG TUNING	Displays the screen for setting the parameter set for emphasis on speed (LV1) or emphasis on precision (LV10).
(18)	ALL IO	Selects the screen for data input to and output from the memory card.
(19)	OPEHIS	Selects the screen for displaying the history of operations performed by the operator and issued alarms.
(21)	COLOR	Selects the screen for setting colors to be used on the screen.
(22)	MAINTE	Selects the screen for setting maintenance items to be managed periodically.
(23)	M-INFO	Selects the screen for displaying information about maintenance performed.

No.	Chapter menu	Description
(24)	W.DGNS	Selects the screen for displaying data such as servo positional deviation values, torque values, machine signals, and so forth as graphs.
(27)	FSSB	Selects the screen for making settings related to the high-speed serial servo bus (FSSB: Fanuc Serial Servo Bus).
(28)	PRMTUN	Selects the screen for setting parameters necessary for start-up and tuning.
(31)	EMBED PORT	Selects the screen for making settings related to the embedded Ethernet (embedded port).
(32)	PCMCIA LAN	Selects the screen for making settings related to the embedded Ethernet (PCMCIA Ethernet card).
(33)	ETHNET BOARD	Selects the screen for making settings related to the fast Ethernet/fast data server.
(34)	PROFI MASTER	Selects the screen for making settings related to the profibus master function.
(37)	M CODE	Selects the screen for setting an M code group.

Message screen

The chapter selection soft keys that belong to the function key

and the function of each screen are described below.

Page 1

(1)	(2)	(3)	(4)	(5)	
ALARM	MSG	HISTRY	MSGHIS	(OPRT)	+

Page 2

(6)	(7)	(8)	(9)	(10)	
EMBED LOG	PCMCIA LOG	BOARD LOG		(OPRT)	+

Table 2.3.3 (e) Message

No.	Chapter menu	Description
(1)	ALARM	Selects the alarm message screen.
(2)	MSG	Selects the operator message screen.
(3)	HISTRY	Selects the screen for displaying the details of alarms issued so far.
(4)	MSGHIS	Selects the external operator message screen.
(6)	EMBED LOG	Selects the screen for displaying error messages related to the embedded Ethernet (embedded port).
(7)	PCMCIA LOG	Selects the screen for displaying error messages related to the embedded Ethernet (PCMCIA Ethernet card).
(8)	BOARD LOG	Selects the screen for displaying error messages related to the fast Ethernet/fast data server.

Graphic screen

The chapter selection soft keys that belong to the function key

and the function of each screen are described below.

When the graphic display function is enabled:

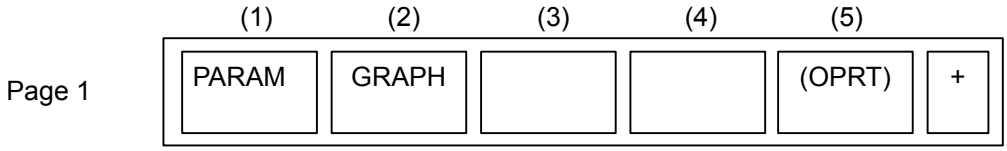


Table 2.3.3 (f) Graphic

No.	Chapter menu	Description
(1)	PARAM	Selects the screen for setting graphic parameters.
(2)	GRAPH	Selects the screen for graphically displaying the tool path.

When the dynamic graphic display function is enabled:

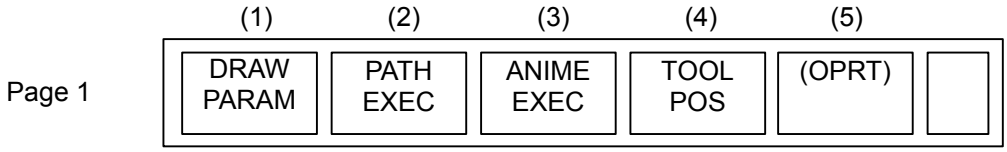


Table 2.3.3 (g) Graphic

No.	Chapter menu	Description
(1)	DRAW PARAM	Selects the screen for setting drawing parameters.
(2)	PATH EXEC	Selects the screen for drawing tool paths.
(3)	ANIME EXEC	Selects the screen for drawing animation.
(4)	TOOL POS	Selects the screen for displaying tool positions.

2.3.4 Key Input and Input Buffer

When an address and a numeric key are pressed, the character corresponding to that key is input once into the key input buffer. The contents of the key input buffer is displayed at the bottom of the screen.
In order to indicate that it is key input data, a ">" symbol is displayed immediately in front of it. A "_" is displayed at the end of the key input data indicating the input position of the next character.

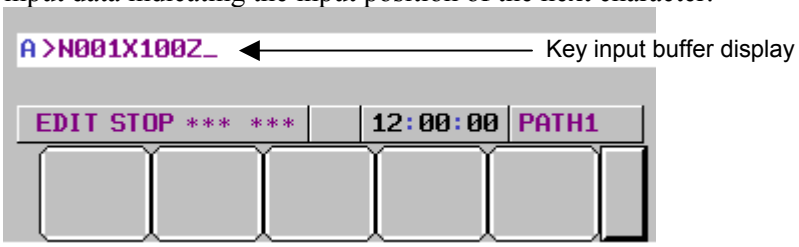


Fig. 2.3.4 (a) Key input buffer display

- Switching between upper and lower key characters

To input the lower character of the keys that have two characters inscribed on them, first press the SHIFT key and then the key in question.
When the SHIFT key is pressed, "_" indicating the next character input position changes to "^". Now lowercase characters can be entered (shift state).
When a character is input in shift status the shift status is canceled. Furthermore, if the SHIFT key is pressed in shift status, the shift status is canceled.
It is possible to input up to 128 characters at a time in the key input buffer.
Press the CAN key to cancel a character or symbol input in the key input buffer.

(Example)
When the key input buffer displays
>N001X100Z_
and the cancel CAN key is pressed, Z is canceled and
>N001X100_
is displayed.

- Switching between uppercase and lowercase alphabetic characters

When entering alphabetic characters, the user can switch between uppercase and lowercase.

By pressing the uppercase/lowercase switch key

, the display of the key input buffer changes to enable uppercase or lowercase alphabetic characters to be entered as described below.

Example)

A> A B C _
State enabling uppercase input

a> a b c _
State enabling lowercase input

2.3.5 Warning Messages

After a character or number has been input from the MDI panel, a data check is executed when

key or a soft key is pressed. In the case of incorrect input data or the wrong operation a flashing warning message will be displayed on the status display line.

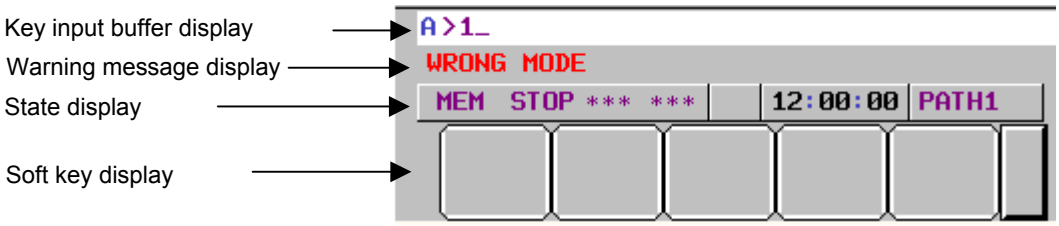


Fig. 2.3.5 (a) Warning message display

Table 2.3.5 (a) Warning Messages

Warning message	Content
FORMAT ERROR	The format is incorrect.
WRITE PROTECT	Key input is invalid because of data protect key or the parameter is not write enabled.
DATA IS OUT OF RANGE	The input value exceeds the permitted range.
TOO MANY DIGITS	The input value exceeds the permitted number of digits.
WRONG MODE	Parameter input is not possible in any mode other than MDI mode.
EDIT REJECTED	It is not possible to edit in the current CNC status.

2.4 EXTERNAL I/O DEVICES

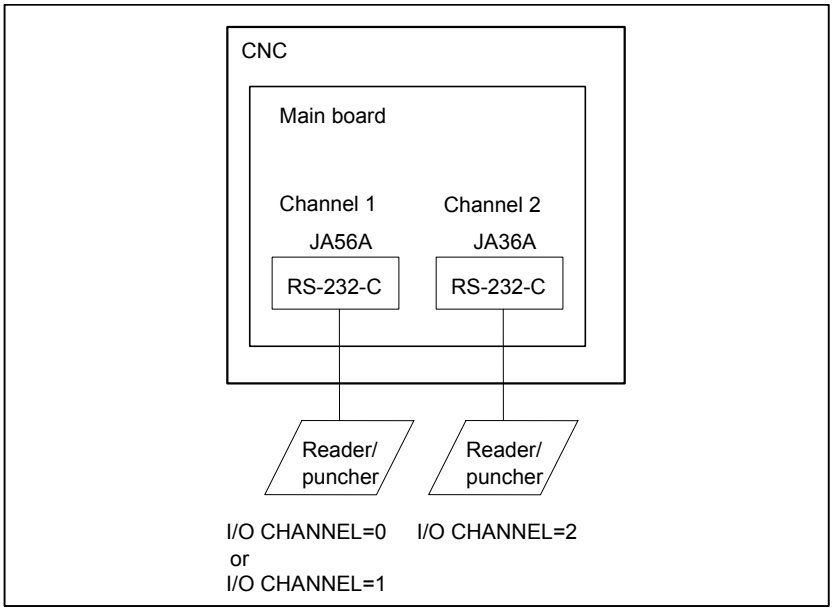
External I/O devices such as a memory card are available.
By using an external I/O device such as a memory card, the following data can be input or output:

1. Programs
2. Offset data
3. Parameters
4. Custom macro common variables

For how to input or output data and input data from or output it to a memory card, see III-8.

- Parameter setting

Before an external input/output device can be used, parameters must be set as follows.



This CNC has a total of two channels of reader/puncher interfaces. It also has a memory card interface. The input/output device to be used is specified by setting the channel (interface) connected to that device in setting parameter I/O CHANNEL.
The specified data, such as a baud rate and the number of stop bits, of an input/output device connected to a specific channel must be set in parameters for that channel in advance. (These settings are not required for the memory card interface.)
For channel 1, two combinations of parameters to specify the input/output device data are provided.
The following shows the interrelation between the reader/puncher interface parameters for the channels.

0020 I/O CHANNEL or foreground input

Set channels to be used for data input/output.

I/O CHANNEL (0 to 5)
=0 : Channel 1
=1 : Channel 1
=2 : Channel 2
=3 : Channel 3
:
:
:

Input/output to and from the memory card interface, etc. is also possible.

When IO4 is set

0021 Foreground output

0022 Background input

0023 Background input

The channel setting is the same as No.0020.

Input/output channel number (parameter No.0020)

↓

I/O CHANNEL=0 (Channel 1)
- 0101 Stop bit and other data
- 0102 Number specified for the input/output device
- 0103 Baud rate

I/O CHANNEL=1 (Channel 1)
- 0111 Stop bit and other data
- 0112 Number specified for the input/output device
- 0113 Baud rate

I/O CHANNEL=2 (Channel 2)
- 0121 Stop bit and other data
- 0122 Number specified for the input/output device
- 0123 Baud rate

:
:
:

I/O CHANNEL=5

2.5 POWER ON/OFF

2.5.1 Turning on the Power

Procedure of turning on the power

Procedure

1 Check that the appearance of the CNC machine tool is normal. (For example, check that front door and rear door are closed.)
2 Turn on the power according to the manual issued by the machine tool builder.
3 After the power is turned on, check that the position screen is displayed. An alarm screen is displayed if an alarm occurs upon power-on.

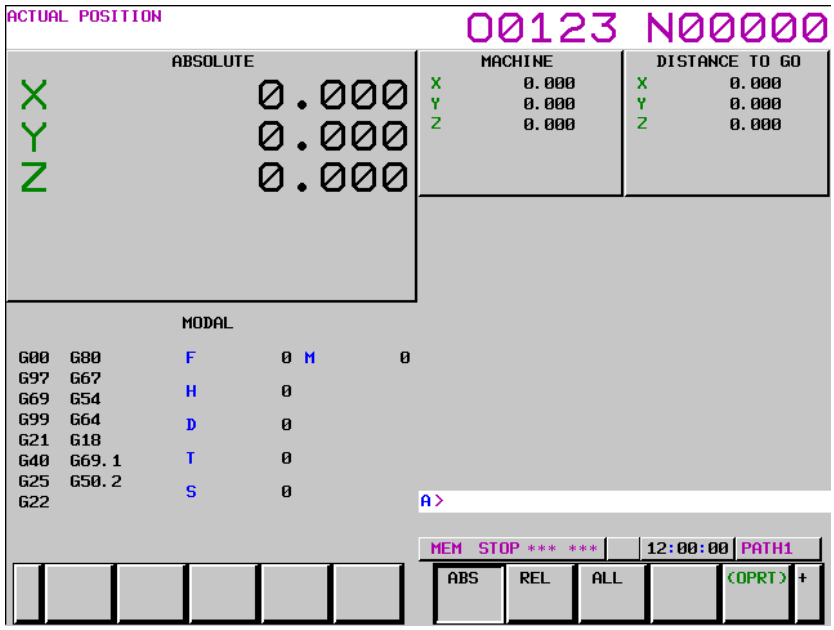


Fig. 2.5.1 (a) Position screen (for machining center system)

4 Check that the fan motor is rotating.

> ⚠ **WARNING**
> Until the positional or alarm screen is displayed at the power on, do not touch them. Some keys are used for the maintenance or special operation purpose. When they are pressed, unexpected operation may be caused.

2.5.2 Power Disconnection

Procedure of power disconnection

Procedure

1 Check that the LED indicating the cycle start is off on the operator's panel.
2 Check that all movable parts of the CNC machine tool is stopping.
3 If an external input/output device such as the Handy File is connected to the CNC, turn off the external input/output device.
4 Continue to press the <POWER OFF> button for about 5 seconds.
5 Refer to the machine tool builder's manual for turning off the power to the machine.

3 MANUAL OPERATION

MANUAL OPERATION are twelve kinds as follows :

3.1 MANUAL REFERENCE POSITION RETURN.....1030
3.2 JOG FEED (JOG).....1032
3.3 INCREMENTAL FEED.....1034
3.4 MANUAL HANDLE FEED.....1036
3.5 MANUAL ABSOLUTE ON AND OFF.....1040
3.6 MANUAL LINEAR/CIRCULAR INTERPOLATION.....1046
3.7 RIGID TAPPING BY MANUAL HANDLE.....1052
3.8 MANUAL NUMERICAL COMMAND.....1055
3.9 THREE-DIMENSIONAL MANUAL FEED.....1064
3.10 DISTANCE CODED LINEAR SCALE INTERFACE.....1080
3.11 LINEAR SCALE WITH DISTANCE-CODED REFERENCE MARKS (SERIAL).....1087
3.12 MANUAL HANDLE RETRACE.....1093
3.13 AUXILIARY FUNCTION OUTPUT BLOCK REVERSE MOVEMENT FOR MANUAL HANDLE RETRACE.....1108

3.1 MANUAL REFERENCE POSITION RETURN

The tool is returned to the reference position as follows :
The tool is moved in the direction specified in bit 5 (ZMI) of parameter No.1006 for each axis with the reference position return switch on the machine operator's panel. The tool moves to the deceleration point at the rapid traverse rate, then moves to the reference position at the FL speed. The rapid traverse rate and FL speed are specified in parameters Nos. 1424,1421, and 1425.
Four step rapid traverse override is effective during rapid traverse.
When the tool has returned to the reference position, the reference position return completion LED goes on. The tool generally moves along only a single axis, but can move along three axes simultaneously when specified so in bit 0 (JAX) of parameter No.1002.

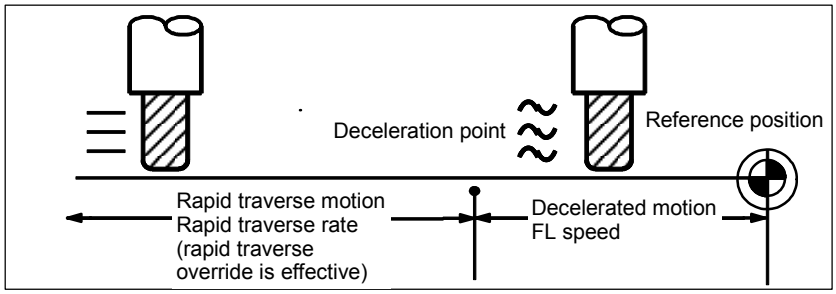


Fig. 3.1 (a) Manual reference position return

Procedure for manual reference position return

Procedure

1 Press the reference position return switch, one of the mode selection switches.
2 To decrease the feedrate, press a rapid traverse override switch.
3 Press the feed axis and direction selection switch corresponding to the axis and direction for reference position return. Continue pressing the switch until the tool returns to the reference position. The tool can be moved along three axes simultaneously when specified so in an appropriate parameter setting. The tool moves to the deceleration point at the rapid traverse rate, then moves to the reference position at the FL speed set in a parameter. When the tool has returned to the reference position, the reference position return completion LED goes on.
4 Perform the same operations for other axes, if necessary.

The above is an example. Refer to the appropriate manual provided by the machine tool builder for the actual operations.

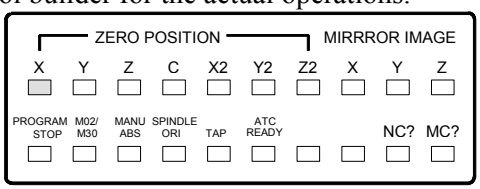


Fig. 3.1 (b)

Explanation

- Automatically setting the coordinate system

Bit 0 (ZPR) of parameter No.1201 is used for automatically setting the coordinate system. When ZPR is set, the coordinate system is automatically determined when manual reference position return is performed.

When α, β and γ are set in parameter 1250, the workpiece coordinate system is determined so that reference point on the tool holder or the position of the tip of the reference tool is X= α, Y = β, Z = γ when reference position return is performed. This has the same effect as specifying the following command for reference position return:

G92XαYβZγ;

However, when options of the workpiece coordinate system is selected, it is not able to use.

Limitation

- Moving the tool again

Once the reference position return completion LED lights at the completion of reference position return, the tool does not move unless the reference position return switch is turned off.

- Reference position return completion LED

The reference position return completion LED is extinguished by either of the following operations:

- Moving from the reference position.
- Entering an emergency stop state.

- The distance to return to reference position

For the distance (Not in the deceleration condition) to return the tool to the reference position, refer to the manual issued by the machine tool builder.

3.2 JOG FEED (JOG)

In the jog mode, pressing a feed axis and direction selection switch on the machine operator's panel continuously moves the tool along the selected axis in the selected direction.
The jog feedrate is specified in a parameter No.1423.
The jog feedrate can be adjusted with the jog feedrate override dial.
Pressing the rapid traverse switch moves the tool at the rapid traverse feedrate No. 1424 regardless of the position of the jog feedrate override dial. This function is called the manual rapid traverse.
Manual operation is allowed for one axis at a time. 3 axes can be selected at a time by bit 0 (JAX) of parameter No.1002.

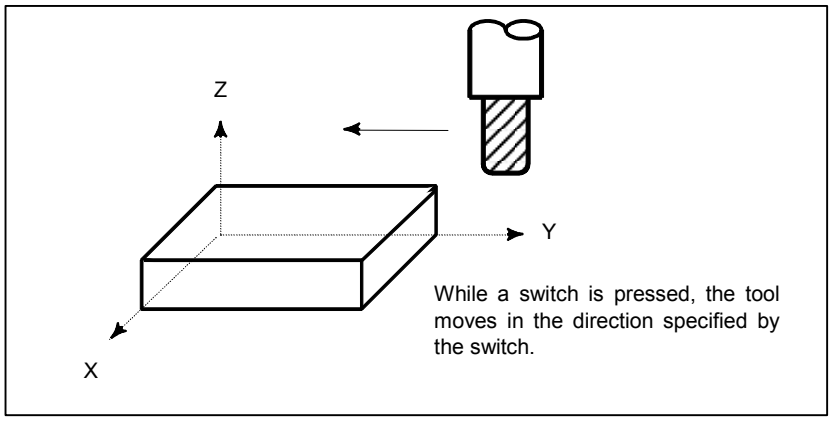


Fig. 3.2 (a) Jog Feed (JOG)

Procedure for JOG feed

Procedure

1 Press the jog switch, one of the mode selection switches.
2 Press the feed axis and direction selection switch corresponding to the axis and the direction where the tool is to be moved. While the switch is pressed, the tool moves at the feedrate specified in the parameter No. 1423. The tool stops when the switch is released.
3 The jog feedrate can be adjusted with the jog feedrate override dial.
4 Pressing the rapid traverse switch while pressing a feed axis and direction selection switch moves the tool at the rapid traverse rate while the rapid traverse switch is pressed. Rapid traverse override by the rapid traverse override switches is effective during rapid traverse.

The above is an example. Refer to the appropriate manual provided by the machine tool builder for the actual operations.

Explanation

- Manual per revolution feed

The manual per revolution feed is enabled for jog feed by setting bit 4 (JRV) of parameter No.1402.

During the manual per revolution feed, the tool is jogged at the feedrate that is obtained by multiplying the spindle speed by the feed amount per revolution, which is calculated by multiplying the jog feedrate override value by the feed amount per revolution specified with the parameter (No. 1423).

During manual per revolution feed, the tool is jogged at the following feedrate:

Feed distance per rotation of the spindle (mm/rev) (specified with parameter No. 1423) × JOG feedrate override × actual spindle speed (rev/min).

Limitation

- Acceleration/deceleration for rapid traverse

Feedrate, time constant and method of automatic acceleration/ deceleration for manual rapid traverse are the same as G00 in programmed command.

- Change of modes

Changing the mode to the jog mode while pressing a feed axis and direction selection switch does not enable jog feed. To enable jog feed, enter the jog mode at first, then press a feed axis and direction selection switch.

- Rapid traverse prior to reference position return

If reference position return is not performed after power-on, pushing rapid traverse button does not actuate the rapid traverse but the remains at the JOG feedrate. This function can be disabled by setting bit 0 (RPD) of parameter No.1401.

3.3 INCREMENTAL FEED

In the incremental (INC) mode, pressing a feed axis and direction selection switch on the machine operator's panel moves the tool one step along the selected axis in the selected direction. The minimum distance the tool is moved is the least input increment. Each step can be 10, 100, or 1000 times the least input increment.
With using bit 2 (HNT) of parameter No. 7103, each step can be additionally 10 times the lest input increment.
The feedrate set in parameter No. 1423 is applied.
By using the manual feedrate override signal, the feedrate can be increased or decreased.
The tool can also be moved at the rapid traverse rate by using the manual rapid traverse selection signal, independent of the manual feedrate override signal.

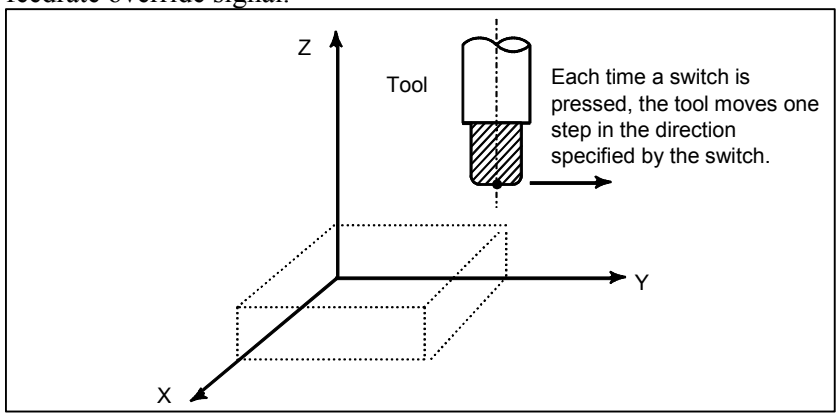


Fig. 3.3 (a) Incremental feed

Procedure for incremental feed

Procedure

1 Press the INC switch, one of the mode selection switches.
2 Select the distance to be moved for each step with the magnification dial.
3 Press the feed axis and direction selection switch corresponding to the axis and direction the tool is to be moved. Each time a switch is pressed, the tool moves one step. The feedrate is the same as the jog feedrate.
4 Pressing the rapid traverse switch while pressing a feed axis and direction selection switch moves the tool at the rapid traverse rate.
Rapid traverse override by the rapid traverse override switch is effective during rapid traverse.

The above is an example. Refer to the appropriate manual provided by the machine tool builder for the actual operations.

Explanation

- Travel distance specified with a diameter

T

The distance the tool travels along the X-axis can be specified with a diameter.

3.4 MANUAL HANDLE FEED

In the handle mode, the tool can be minutely moved by rotating the manual pulse generator on the machine operator's panel. Select the axis along which the tool is to be moved with the handle feed axis selection switches.
The minimum distance the tool is moved when the manual pulse generator is rotated by one graduation is equal to the least input increment. One of four types of magnifiers selected with the manual handle feed move distance select signal can be applied. With bit 2 (HNT) of parameter No. 7103, the minimum distance can be further 10 times greater.
The number of manual pulse generators depends on the option as follows:

- Control on 1 manual handle: Up to 1
- Control on 2 or 3 manual handles:
  Up to 3 (Up to three manual pulse generators can be moved at a time.)
- Control on 2 to 5 manual handles:
  Up to 5 (Up to five manual pulse generators can be moved at a time.)

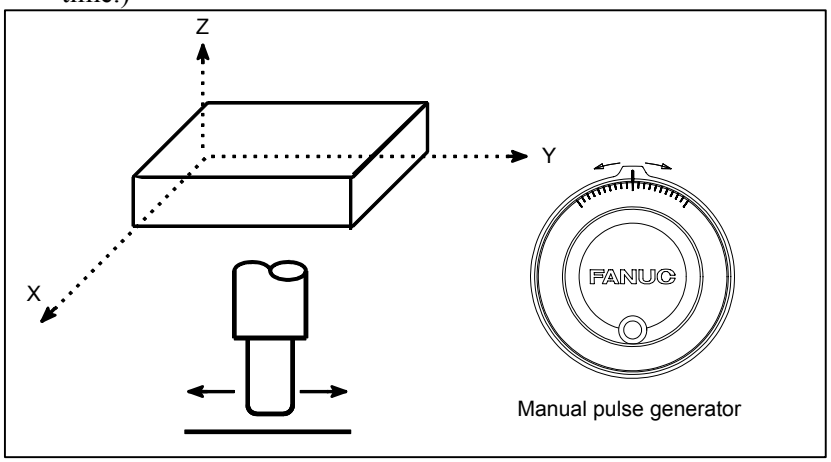


Fig. 3.4 (a) Manual handle feed

Procedure for manual handle feed

Procedure

1 Press the handle switch, one of the mode selection switches.
2 Select the axis along which the tool is to be moved by pressing a handle feed axis selection switch.
3 Select the magnification for the distance the tool is to be moved by pressing a handle feed magnification switch. The minimum distance the tool is moved when the manual pulse generator is rotated by one graduation is equal to the least input increment.
4 Move the tool along the selected axis by rotating the handle. Rotating the handle 360 degrees moves the tool the distance equivalent to 100 graduations.

The above is an example. Refer to the appropriate manual provided by the machine tool builder for the actual operations.

Explanation

- Availability of manual pulse generator in Jog mode (JHD)

Bit 0 (JHD) of parameter No.7100 enables or disables the manual handle feed in the JOG mode.
When bit 0 (JHD) of parameter No.7100 is set 1,both manual handle feed and incremental feed are enabled.

- Manual handle feed in TEACH IN JOG mode (THD)

By setting bit 1 (THD) of parameter No. 7100, manual handle feed in TEACH IN JOG mode can be enabled or disabled.

- When manual handle feed exceeding the rapid traverse rate is specified

The amount of pulses exceeding the rapid traverse rate can be saved by CNC as B. And amount of pulses B will be output as pulses C.

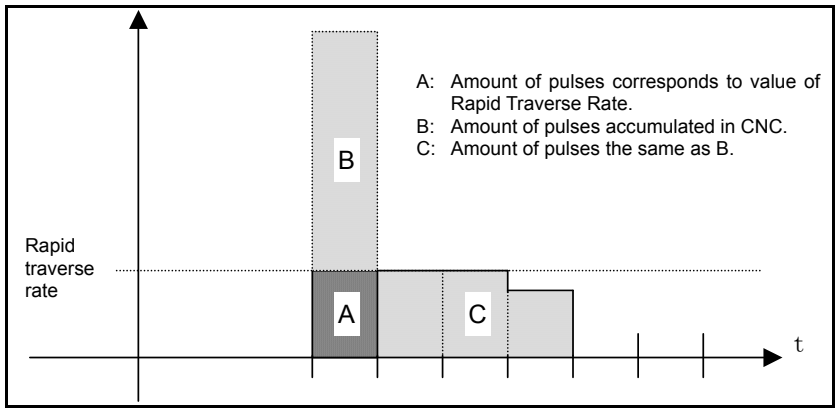


Fig. 3.4 (b) Amount of pulses output by CNC in Manual Handle Feed

Amount of pulses B is calculated in 2 cases as following:

In case of
1) Parameter No.7117 = 0
   The feedrate is clamped at the Rapid Traverse Rate and generated pulses exceeding the Rapid Traverse Rate are ignored (B=0).

In case of
2) Parameter No.7117 > 0
   The feedrate is clamped as the Rapid Traverse Rate, but the pulses exceeding the Rapid Traverse Rate is not ignored. Amount of pulses accumulated in CNC is calculated as following. (Although the rotation of manual pulse generator is stopped, if there is pulses accumulated in CNC, it will be output and the tool will move as long as amount of it.)

   Magnification set by MP1, MP2<Gn019.4,5> is m, value of parameter No.7117 is n.

   n < m: Clamping is set performed at value of parameter No.7117.

n ≥ m: Amount A+B, shown in figure, which's value is multiple of m and small than n. As a result, clamping is performed as an integral multiple of the selected magnification.

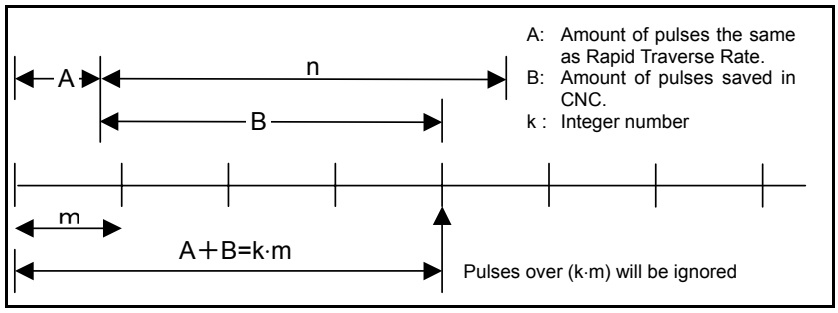


Fig. 3.4 (c) Amount of pulses exceeding the Rapid Traverse Rate (n ≥ m)

NOTE
Due to change of mode, clamping can be performed not as an integral multiple of the selected magnification.
The distance the tool moves may not match the graduations on the manual pulse generator.

- Upper feedrate limit in manual handle feed

The upper feedrate limit depends on the input signal (maximum manual handle feedrate switch signal HNDLF) from the PMC as follows:

- When HNDLF is set to 0, the feedrate is clamped to the manual rapid traverse rate (parameter No. 1424).
- When HNDLF is set to 1, the feedrate is clamped to the feedrate set in parameter No. 1434.

- Movement direction of an axis to the rotation of MPG (HNGx)

Bit 0 (HNGx) of parameter No 7102 switches the direction of MPG in which the tool moves along an axis, corresponding to the direction in which the handle of the manual pulse generator is rotated.

This parameter is valid only for the following functions:

- Manual handle feed
- Manual handle interruption

Limitation

⚠ WARNING
Rotating the handle quickly with a large magnification such as ×100 moves the tool too fast. The feedrate is clamped at the rapid traverse feedrate.

NOTE

Rotate the manual pulse generator at a rate of five rotations per second or lower. If the manual pulse generator is rotated at a rate higher than five rotations per second, the tool may not stop immediately after the handle is no longer rotated or the distance the tool moves may not match the graduations on the manual pulse generator.

3.5 MANUAL ABSOLUTE ON AND OFF

Whether the distance the tool is moved by manual operation is added to the coordinates can be selected by turning the manual absolute switch on or off on the machine operator's panel. When the switch is turned on, the distance the tool is moved by manual operation is added to the coordinates. When the switch is turned off, the distance the tool is moved by manual operation is not added to the coordinates.

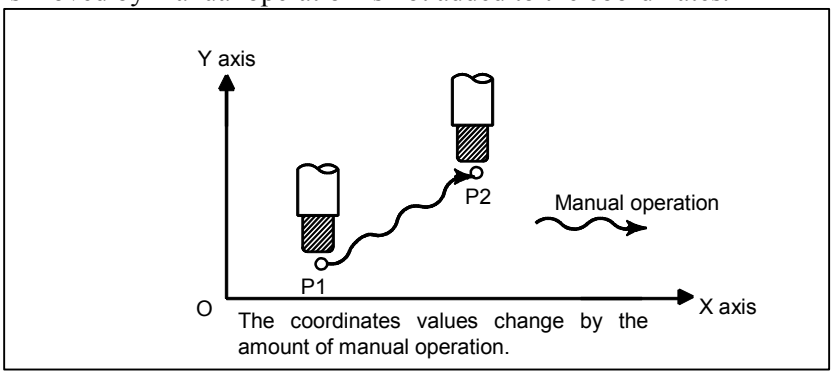


Fig. 3.5 (a) Coordinates with the switch ON

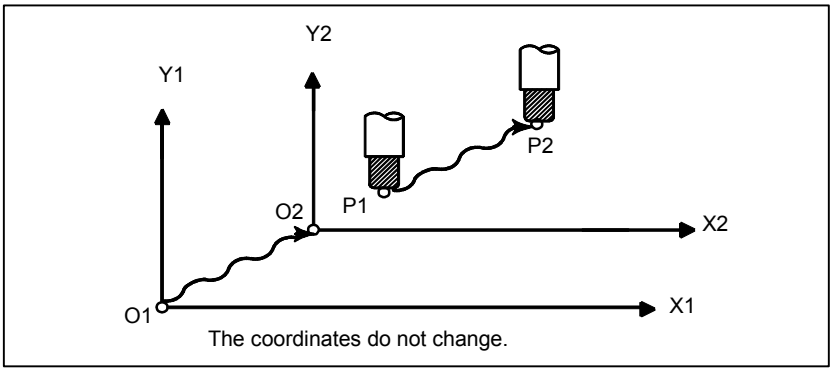


Fig. 3.5 (b) Coordinates with the switch OFF

Explanation

The following describes the relation between manual operation and coordinates when the manual absolute switch is turned on or off, using a program example.

```
G01G90 X100.0Y100.0F010 ;        <1>
       X200.0Y150.0      ;        <2>
       X300.0Y200.0      ;        <3>
```

Fig. 3.5 (c) Program example

The subsequent figures use the following notation:

→ Movement of the tool when the switch is on

- - - → Movement of the tool when the switch is off

The coordinates after manual operation include the distance the tool is moved by the manual operation. When the switch is off, therefore, subtract the distance the tool is moved by the manual operation.

- Manual operation after the end of block

Coordinates when block <1> has been executed after manual operation (X-axis +20.0, Y-axis +100.0) at the end of movement of block <2>.

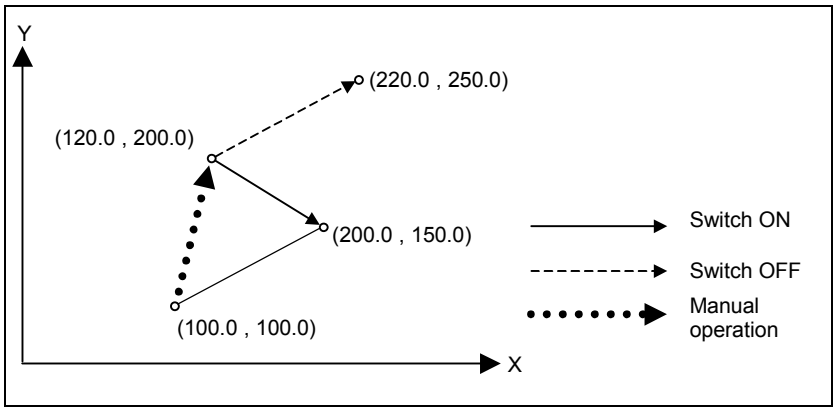


Fig. 3.5 (d) Manual operation after the end of block

- Manual operation after a feed hold

Coordinates when the feed hold button is pressed while block <2> is being executed, manual operation (Y-axis + 75.0) is performed, and the cycle start button is pressed and released.

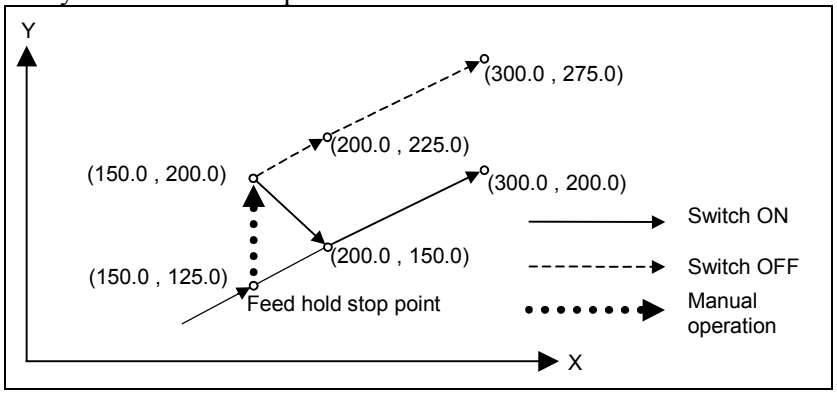


Fig. 3.5 (e) Manual operation after a feed hold

- When reset after a manual operation following a feed hold

Coordinates when the feed hold button is pressed while block <2> is being executed, manual operation (Y-axis +75.0) is performed, the control unit is reset with the RESET button, and block <2> is read again.

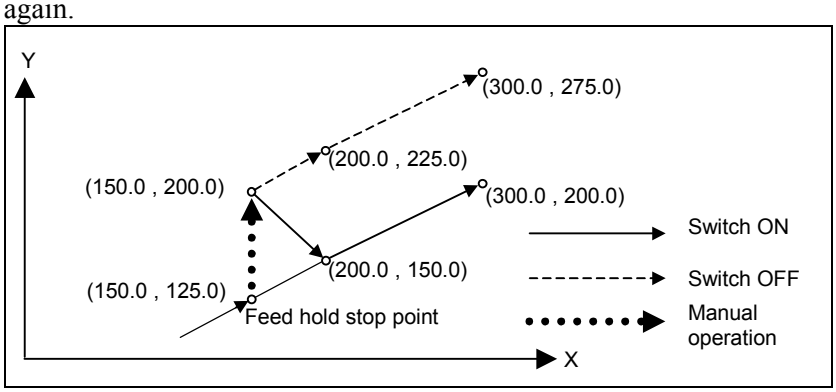


Fig. 3.5 (f) When reset after a manual operation following a feed hold

- When a movement command in the next block is only one axis

When there is only one axis in the following command, only the commanded axis returns.

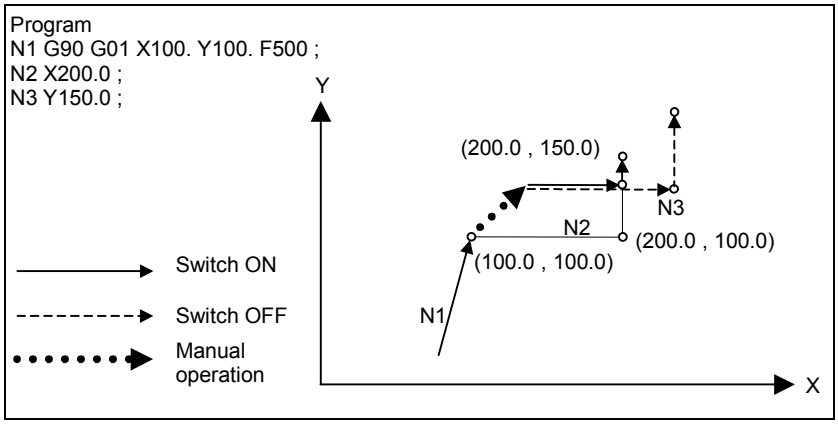


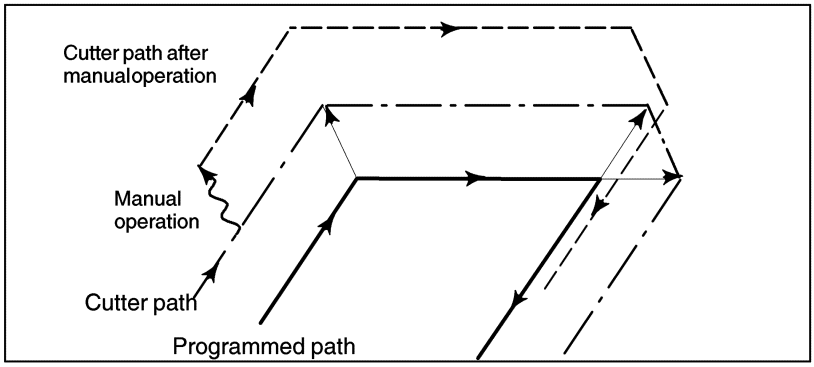
Fig. 3.5 (g) When a movement command in the next block is only one axis

- When the next move block is an incremental

When the following commands are incremental commands, operation is the same as when the switch is OFF.

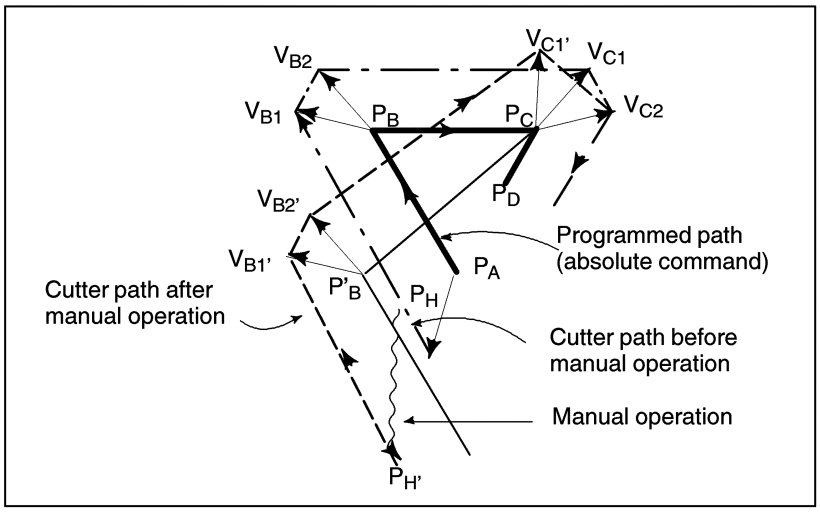
- Manual operation during cutter or tool nose radius compensation

- When the switch is OFF
  After manual operation is performed with the switch OFF during cutter or tool nose radius compensation, automatic operation is restarted then the tool moves parallel to the movement that would have been performed if manual movement had not been performed.
  The amount of separation equals to the amount that was performed manually.

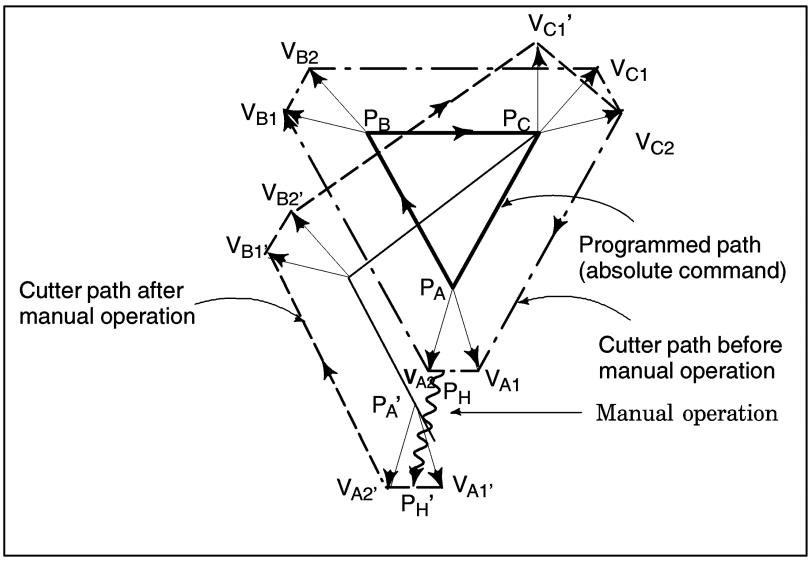


- When the switch is ON during cutter or tool nose radius compensation
  Operation of the machine upon return to automatic operation after manual intervention with the switch is ON during execution with an absolute command program in the cutter or tool nose radius compensation mode will be described. The vector created from the remaining part of the current block and the beginning of the next block is shifted in parallel. A new vector is created based on the next block, the block following the next block and the amount of manual movement. This also applies when manual operation is performed during cornering.

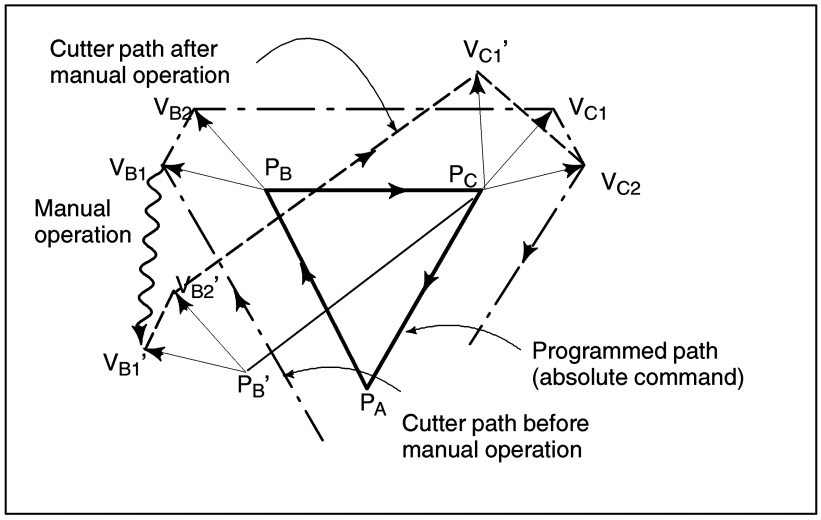
- Manual operation performed in other than cornering
  Assume that the feed hold was applied at point P_H while moving from P_A to P_B of programmed path P_A, P_B, and P_C and that the tool was manually moved to PH'. The block end point P_B moves to the point P_B' by the amount of manual movement, and vectors V_{B1} and V_{B2} at P_B also move to V_{B1}' and V_{B2}'. Vectors V_{C1} and V_{C2} between the next two blocks P_B - P_C and P_C - P_D are discarded and new vectors V_{C1}' and V_{C2}' (V_{C2}' = V_{C2} in this example) are produced from the relation between P_B' - P_C and P_C - P_D. However, since V_{B2}' is not a newly calculated vector, correct offset is not performed at block P_B' - P_C. Offset is correctly performed after P_C.



- Manual operation during cornering
  This is an example when manual operation is performed during cornering. V_{A2}', V_{B1}', and V_{B2}' are vectors moved in parallel with V_{A2}, V_{B1} and V_{B2} by the amount of manual movement. The new vectors are calculated from V_{C1} and V_{C2}. Then correct cutter or tool nose radius compensation is performed for the blocks following P_C.



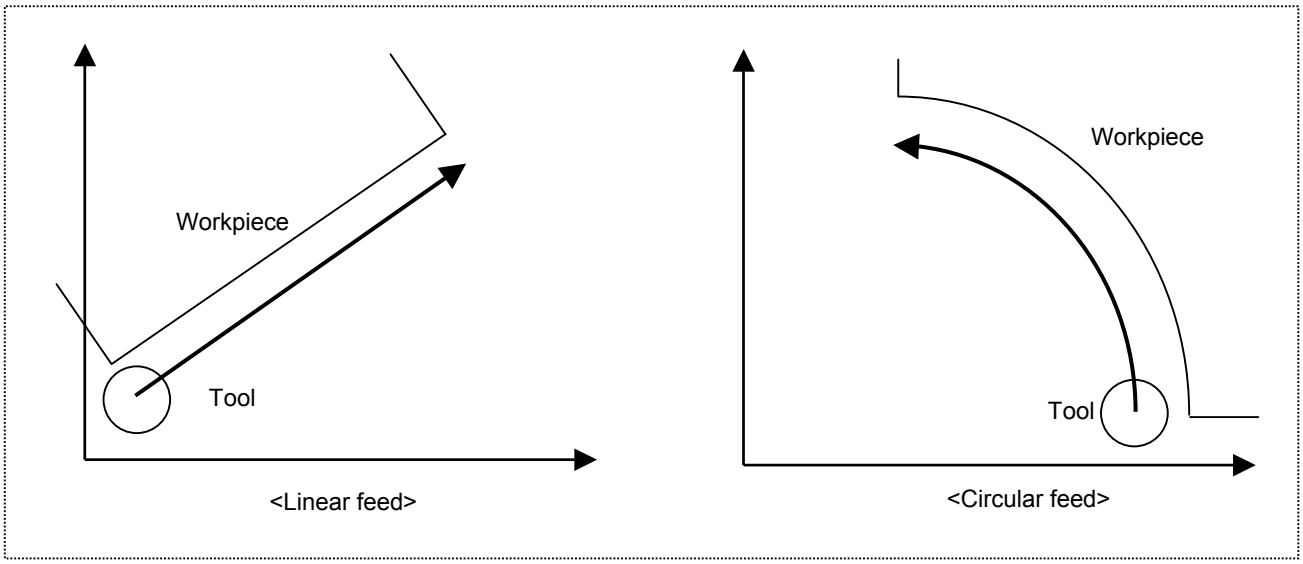
- Manual operation after single block stop
  Manual operation was performed when execution of a block was terminated by single block stop.
  Vectors V_{B1} and V_{B2} are shifted by the amount of manual operation. Sub-sequent processing is the same as case a described above. An MDI operation can also be intervened as well as manual operation. The movement is the same as that by manual operation.


Cutter path after
manual operation
V_{C1}'
V_{B2}
V_{C1}
V_{B1}
P_B
P_C
V_{C2}
Manual
operation
V_{B2}'
V_{B1}'
P_B'
P_A
Programmed path
(absolute command)
Cutter path before
manual operation

3.6 MANUAL LINEAR/CIRCULAR INTERPOLATION

In manual handle feed or jog feed, the following types of feed operations are possible along with the conventional feed operation with simultaneous single-axis control (for X, Y, Z, or other axis).

- Feed along a tilted straight line in the XY, YZ, or ZX plane based on simultaneous 2-axis control (linear feed)
- Feed along a circle in the XY, YZ, or ZX plane based on simultaneous 2-axis control (circular feed)



NOTE
1 Two control axes should be included in the three standard axes.
2 In the following, only the case of the XY plane is described. If any other plane is used, replace XY with YZ or ZX. (Refer to the table below.)

XY plane	YZ plane	ZX plane
X	Y	Z
Y	Z	X

Explanations

Procedure

1 For manual handle feed, select the manual handle feed mode. For jog feed, select the jog feed mode.
2 For manual handle feed, use the handle feed axis selection switch to select the feed axis (simultaneous 1-axis feed in the X-, Y-, or Z-axis or simultaneous 2-axes feed of the X- and Y-axes along a specified line or circle) used when the manual handle is turned.
For jog feed, use the feed axis direction selection switch to select the feed axis and its direction. The tool moves at the feedrate (jog feedrate) set in parameter No. 1423 in the specified axis direction or along the specified line or circle while the feed axis and its direction are selected.
3 For the manual handle, the tool moves in the specified axis direction when the manual handle is turned. The feedrate depends on the rotation speed of the manual handle. The amount of movement when the manual handle is turned for one pulse can be selected by the manual handle feed movement scale switch.
For jog feed, the manual feedrate override dial can be used to override the feedrate.

The above procedure is an example. For actual operation, refer to the manual provided by the machine tool builder.

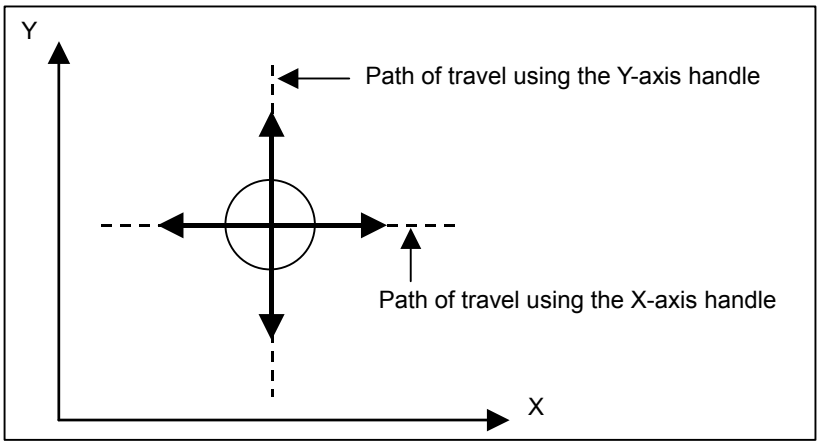
- Line/circle definition

A line or circle definition is not required for axis direction feed. For linear or circular feed, however, a line or circle must be defined. (For example, in the case of circular feed, the center and radius of the circle must be specified.)
Refer to the manual provided by the machine tool builder.

- Manual handle feed

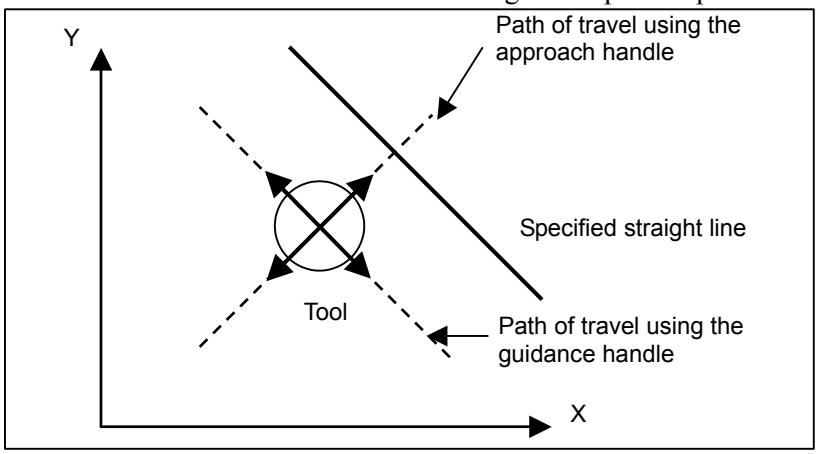
In manual handle feed, the tool can be moved along a specified axis (X-axis, Y-axis, Z-axis, or N^{th}-axis), along a rotated straight line (linear feed), or along a circle (circular feed).

(1) Feed along a specified axis (simultaneous 1-axis control)
A single manual handle operation can move the tool along the desired axis, such as X-axis, Y-axis, or Z-axis, on a simultaneous 1-axis control basis.



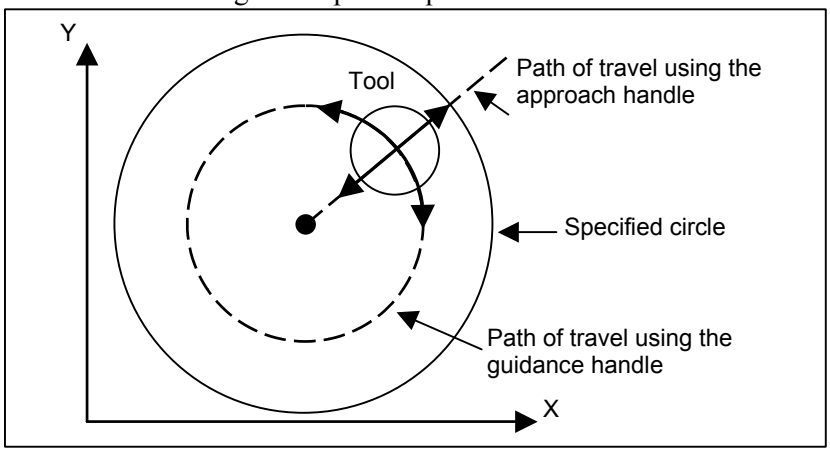
Feed along a specified axis

(2) Linear feed (simultaneous 2-axis control)
A single manual handle operation can move the tool in parallel to the straight line on a simltaneous 2-axis control basis. This manual handle is referred to as the "guidance handle". Also, the tool can be moved at right angles to a specified straight line on a simultaneous 2-axis control basis. This manual handle is referred to as the "approach handle". When the guidance handle or approach handle is turned clockwise or counterclockwise, the tool travels forward or backward along the respective path.



Linear feed

(3) Circular feed (simultaneous 2-axis control)
A single manual handle operation can move the tool from the current position along a concentric circle that has the same center as a specified circle on a simultaneous 2-axis control basis. This manual handle is referred to as the guidance handle. Also, a single manual handle operation can move the tool along a straight line normal to a specified circle on simultaneous 2-axis control. This manual handle is refrred to as the approach handle. When the guidance handle or approach handle is turned clockwise or counterclockwise, the tool travels forward or backward along the respective path.



Circular feed

- Feedrate for manual handle feed

The feedrate depends on the speed at which a manual handle is turned. The distance to be traveled by the tool (along a tangent in the case of linear or circular feed) per manual handle pulse can be selected using the manual handle feed travel distance magnification switch.
The approach handle clamp feedrate can be set using parameter No. 7160. The guidance handle clamp feedrate can be set using parameter No. 7161.

- Manual handle selection

Up to five manual handles to be connected. For information on specifying the usage of the manual handle connected to each interface (for feed in the axis direction, use as a guidance handle, or use as an approach handle), refer to the manual issued by the machine tool builder.

- Direction of movement using manual handles

When the tool is moved along a straight line or circle using the guidance handle or approach handle, the direction of the tool movement (for example, whether to make a clockwise or counterclockwise movement along a circle) when the handle is turned clockwise or counterclockwise can be specified.
Refer to the manual provided by the machine tool builder.

- Jog feed

In jog feed, the tool can be moved along a specified axis (X-axis, Y-axis, Z-axis, etc.), along a rotated straight line (linear feed), or along a circle (circular feed).

(1) Feed along a specified axis (simultaneous 1-axis control)
While a feed axis and its direction are specified with the feed axis direction selection switch, the tool moves in the specified axis direction at the feedrate specified in parameter No. 1423. The feedrate can be overriden using the manual feedrate override dial.

(2) Linear feed (simultaneous 2-axis control)
By defining a straight line, the tool can be moved as follows:
- While a feed axis and its direction are selected using the feed axis direction selection switch, the tool moves along a straight line parallel to the specified straight line on a simultaneous 2-axis control basis.
- While a feed axis and its direction are selected using the feed axis direction selection switch, the tool moves along a straight line perpendicular to the specified staight line on a simultaneous 2-axis control basis.

The feedrate in the tangential direction is specified in parameter No. 1410.
The feedrate can be overridden using the manual feedrate override dial.

(3) Circular feed (simultaneous 2-axis control)
By defining a circle, the tool can be moved as follows:
- While a feed axis and its direction are selected using the feed axis direction selection switch, the tool moves from the current position along the concentric circle that has the same center as the specified circle.
- While a feed axis and its direction are selected using the feed axis direction selection switch, the tool moves along a straight line normal to the specified circle.

The feedrate in the tangential direction is specified in parameter No. 1410. The feedrate can be overridden using the manual feedrate override dial.

- Manual handle feed in JOG mode

Even in JOG mode, manual handle feed can be enabled using bit 0 (JHD) of parameter No. 7100. In this case, however, manual handle feed is enabled only when the tool is not moved using jog feed.

Limitations

- Mirror image

The mirror image function is not available during the manual operation. (The manual operation can be executed when the mirror image switch is off and the mirror image setting is off.)

NOTE
If the tool is operated with the mirror image function activated, the normal operation will not be guaranteed.

- Rotation direction

When this function is effective, bit 0 (HNGx) of parameter No. 7102 is invalid which is used to change the axis movement direction with respect to the rotation direction of the manual pulse generator. (The rotation direction of the manual pulse generator is always the same as the axis movement direction; that is, the operation is the same as when bit 0 of parameter No. 7102 is set to "0".)

- The amount of the shift for manual handle feed

When this function is effective, parameters No. 12350 and No. 12351 used to determine the magnification of manual handle feed for each axis are invalid and the values of parameter No. 7113 and No. 7114 are used. (Parameters No. 7131 and No. 7132 are used for the second manual pulse generator, parameters No. 7133 and No. 7134 for the third, parameters No. 7135 and No. 7136 for the fourth, and parameters No. 7137 and No. 7138 for fifth.)

- Incremental feed

This function is invalid for the incremental feed.

- Synchronous/composite control

If the axis is set as the synchronous/composite control axis, this function is invalid.

3.7 RIGID TAPPING BY MANUAL HANDLE

For execution of rigid tapping, set rigid mode, then switch to handle mode and move the tapping axis with a manual handle. For rigid tapping, refer to Section 4.4 in Part II of the User's Manual (T series) or Section 5.2 in Part II of the User's Manual (M series) and the relevant manual of the machine tool builder.

Rigid tapping by manual handle

Procedure

1 Stop the spindle and servo axes, then set MDI mode by pressing the MDI switch among the mode selection switches.
2 Enter and execute the following program:
Example 1) M series / T series (G code systems B, C)
```
M29 S100 ;
G91 G84 Z0 F1000 ;
```
Example 2) T series (G code system A)
```
M29 S100 ;
G84 W0 F1000 ;
```
The program above is required to determine a screw lead and set rigid tapping mode. In this program, a tapping axis must always be specified. Specify a value that does not operate the tapping axis. With the M series, specify G84 (G74). With the T series, specify G84 (G88).

> ⚠ **WARNING**
> In this MDI programming, never specify commands to position the tool at a drilling position and at point R. Otherwise, the tool moves along an axis.

3 When the entered program is executed, rigid tapping mode is set.
4 After rigid mode is set upon completion of MDI program execution, switch to the handle mode by pressing the handle switch among the mode selection switches.

> ⚠ **CAUTION**
> At this time, never press the reset key. Otherwise, rigid mode is canceled.

5 To perform rigid tapping, select a tapping axis with the handle feed axis select switch, and move the tapping axis with the manual handle.

Explanation

- Manual rigid tapping

Manual rigid tapping is enabled by parameter HRG (No. 5203#0) to 1.

- Cancellation of rigid mode

To cancel rigid mode, specify G80 as same the normal rigid tapping. When the reset key is pressed, rigid mode is canceled, but the canned cycle is not canceled.
When the rigid mode switch is to be set to off for rigid mode cancellation (when parameter CRG (No. 5200#2) is set to 0), the G80 command ends after the rigid mode switch is set to off.

- Spindle rotation direction

The rotation direction of the spindle is determined by a specified tapping cycle G code and the setting of parameter HRM (No. 5203#1). For example, when the HRM parameter is set to 0 in G84 mode, the spindle makes forward rotations as the tapping axis moves in the minus direction. (When the tapping axis moves in the plus direction, the spindle makes reverse rotations.)

- Arbitrary tapping axis

With the M series, an arbitrary tapping axis can be selected by setting bit 0 (FXY) of parameter No. 5101 to 1. With the T series, an arbitrary tapping axis can be selected not by using bit 0 (FXY) of parameter No. 5101 (but by using bit 0 (FXY) of parameter No. 5101 when the FS15 format is specified). In this case, specify a G code for plane selection and the address of the tapping axis when setting the rigid tapping mode in the MDI mode.

- G84 command/same block specifying both M29 and G84

As the MDI program command for setting the rigid mode, G84 can be specified as a rigid tapping G code (by setting bit 0 (G84) of parameter No. 5200 to 1), or M29 and G84 can be specified in the same block.
Example 1: G91 G84 Z0 F1000 S100 ;
Example 2: G91 G84 Z0 F1000 M29 S100 ;

- Specification of manual handle feed faster than the rapid traverse rate

Set parameter (No.7117) to 0 so that when manual handle feed is specified which is faster than the rapid traverse rate, the handle pulses beyond the rapid traverse rate are ignored.

- Series 15 format command

By setting bit 1 (FCV) of parameter No. 0001 to 1, the Series15 format can be used for specification.
Example 1: G91 G84.2(G84.3) Z0 F1000 S100; (M series)
Example 2: G84.2 W0 F1000 S100; (T series : G code systems A)
Example 3: G91 G84.2 Z0 F1000 S100; (T series : G code system B, C)

- Acceleration/deceleration type

When manual rigid tapping is executed, the acceleration/deceleration type and acceleration/deceleration time constant set in the rigid tapping parameters are valid.
The same settings are valid also for extraction.

- In the case of multi-spindle operation

In the case of multi-spindle operation, a spindle can be selected by specifying a P command as well as an S command.
Example : When the 2nd spindle is selected:
M29 S100 **P2**;
G91 G84 Z0 F1000 ;

Limitation

- Excessive error check

In manual rigid tapping, only an excessive error during movement is checked.

- Tool axis direction handle feed

Tool axis direction handle feed is disabled.

- Extraction override

In manual rigid tapping, the extraction override function is disabled, and the use of an acceleration/deceleration time constant for extraction is disabled.

- Number of repeats

In MDI programming, never specify K0 and L0, which are used to specify that the number of repeats is 0 and to disable the execution of a G84 block. If K0 or L0 is specified, rigid mode cannot be set.

- Positioning of the tool to a drilling position

When positioning the tool to a drilling position, select the X-axis or Y-axis with the axis select switch in handle mode. Never use the method of positioning to a drilling position in MDI mode or MEM mode. The method can operate the tapping axis.

- Three-dimensional rigid tapping

Three-dimensional rigid tapping cannot be used in rigid tapping using the manual handle.

- Interpolation type rigid tapping

Interpolation type rigid tapping cannot be used in rigid tapping using the manual handle. To perform rigid tapping using the manual handle, select the conventional type of rigid tapping with bit 3 (CHR) of parameter No. 5202.

3.8 MANUAL NUMERICAL COMMAND

The manual numerical command function allows data programmed through the MDI to be executed in jog mode. Whenever the system is ready for jog feed, a manual numerical command can be executed. The following eight functions are supported:

(1) Positioning (G00)
(2) Linear interpolation (G01)
(3) Automatic reference position return (G28)
(4) 2nd/3rd/4th reference position return (G30)
(5) M codes (miscellaneous functions)
(6) S codes (spindle speed functions)
(7) T codes (tool functions)
(8) B codes (second auxiliary functions)

By setting the following parameters, the commands for axial motion and the M, S, T, and B functions can be disabled:

(1) Positioning (G00): .....Parameter JAXx (No. 7010#0)
(2) Linear interpolation (G01):.....Parameter JAXx (No. 7010#0)
(3) Automatic reference position return (G28):
.....Parameter JAXx (No. 7010#0)
(4) 2nd/3rd/4th reference position return (G30):
.....Parameter JAXx (No. 7010#0)
(5) M codes (miscellaneous functions): Parameter JMF (No. 7002#0)
(6) S codes (spindle speed functions):.... Parameter JSF (No. 7002#1)
(7) T codes (tool functions):..... Parameter JSF (No. 7002#2)
(8) B codes (second auxiliary functions):Parameter JBF (No. 7002#3)

Procedure Manual numerical command

Procedure

1 Press the jog switch (one of the mode selection switches).
2 Press function key

.
3 Press soft key [JOG] on the screen. The following manual numerical command screen is displayed.

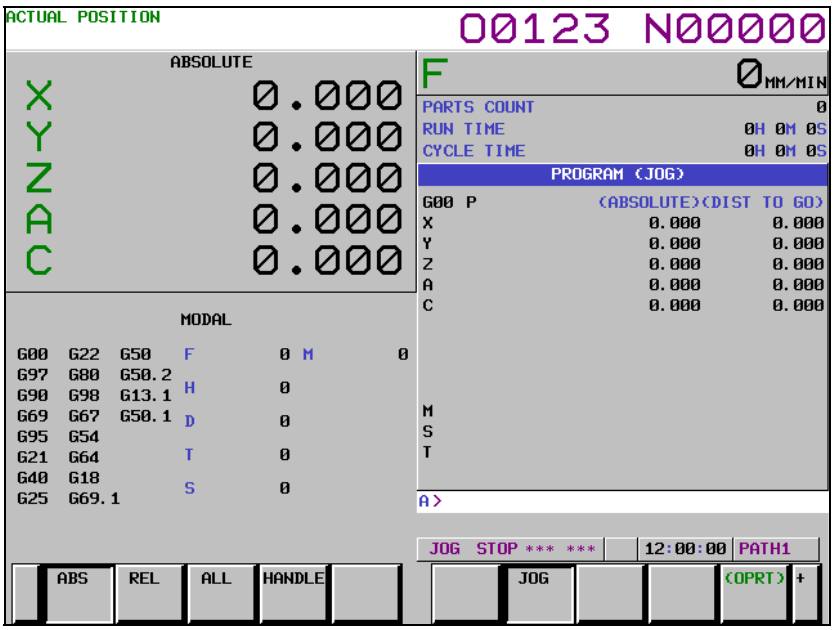


Fig. 3.8 (a) Manual numerical command screen

The remaining portion of the axis information currently not shown on the screen can be displayed by pressing the PAGE ↑ or PAGE ↓ key.

NOTE
1 The actual feedrate (F) and the actual spindle speed (S) are displayed only for 9” window.
2 If two or more identical axis names exist for the same path, they are not displayed.

4 Enter the required commands by using address keys and numeric keys on the MDI panel, then press soft key [INPUT] or the

key to set the entered data.

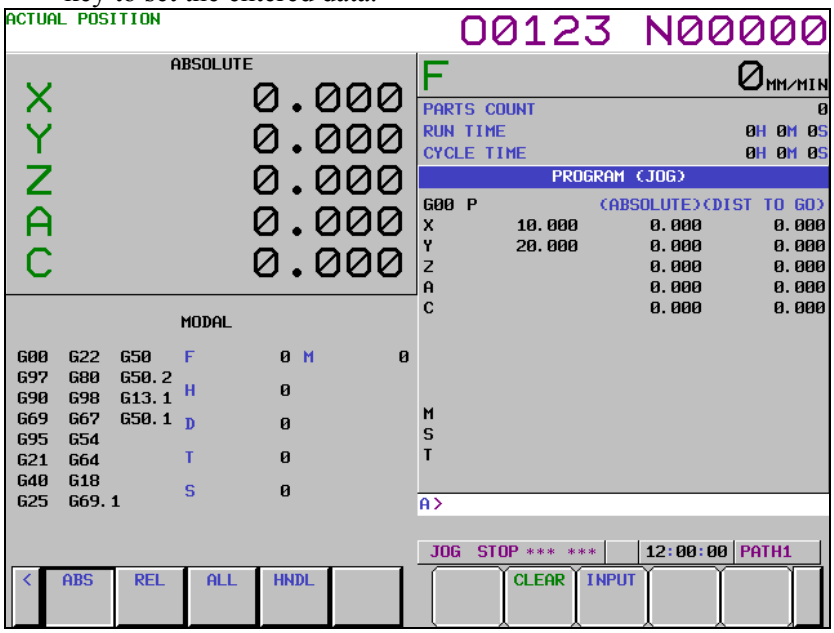


Fig. 3.8 (b) Example of inputting numerical value

The following data can be set:

1. G00: ..... Positioning
2. G01: ..... Linear interpolation
3. G28: ..... Automatic reference position return
4. G30: ..... 2nd/3rd/4th reference position return
5. M codes: ..... Miscellaneous functions
6. S codes:..... Spindle speed functions
7. T codes:..... Tool functions
8. B codes: ..... Second auxiliary functions

The set data is maintained even when the screen or mode is changed.

NOTE
When an alarm state exists, data cannot be set.

5 Press the cycle start switch on the machine operator's panel to start command execution. The status is indicated as "MSTR." The automatic operation signal, STL, can be turned on by setting bit 2 (JST) of parameter No.7001.

NOTE
If the cycle start switch is pressed while an alarm state exists, a "START REJECTED (ALARM EXISTED)" warning is generated, and the entered data cannot be executed.

6 Upon the completion of execution, the "MSTR" status indication is cleared from the screen, and automatic operation signal STL is turned off. The set data is cleared entirely. G codes are set to G00 or G01 according to the setting of parameter G01 (No.3402#0).

Explanation

- Positioning

An amount of travel is given as a numeric value, preceded by an address such as X, Y, or Z. This is always regarded as being an incremental command, regardless of whether G90 or G91 is specified.

	Manual rapid traverse selection switch	
	Off	**On**
Feedrate (parameter)	Jog feedrate for each axis (No.1423)	Rapid traverse rate for each axis (No.1420)
Automatic acceleration/ deceleration (parameter)	Exponential acceleration/ deceleration in jog feed for each axis (No.1624)	Linear acceleration/deceleration in rapid traverse for each axis (No.1620)
Override	Manual feed override	Rapid traverse override

NOTE
1 When the manual rapid traverse switch is off, the feedrate is clamped so that the axis-by-axis manual rapid traverse rate (parameter No. 1424) is not exceeded.
2 The tool cannot move if linear interpolation type positioning (where the tool path is linear) has been performed by setting bit 1 (LRP) of parameter No. 1401.

- Linear interpolation (G01)

An amount of travel is given as a numeric value, preceded by an address such as X, Y, or Z. This is always regarded as being an incremental command, regardless of whether G90 or G91 is specified. Axial movements are always performed in incremental mode even during scaling or polar coordinate interpolation. In addition, movement is always performed in feed per minute mode regardless of the specification of G94 or G95.

Feedrate (parameter)	Dry run feedrate (No. 1410)
Automatic acceleration/deceleration (parameter)	Exponential acceleration/deceleration in cutting feed for each axis (No.1622)
Override	Manual feed override

NOTE
Since the feedrate is always set to the dry run feedrate, regardless of the setting of the dry run switch, the feedrate cannot be specified using F. The feedrate is clamped such that the maximum cutting feedrate, set in parameter No. 1430, is not exceeded.

- Automatic reference position return (G28)

The tool returns directly to the reference position without passing through any intermediate points, regardless of the specified amount of travel. For axes for which no move command is specified, however, a return operation is not performed.

Feedrate (parameter)	Rapid traverse rate (No. 1420)
Automatic acceleration/deceleration (parameter)	Linear acceleration/deceleration in rapid traverse for each axis (No. 1620)
Override	Rapid traverse override

- 2nd, 3rd, or 4th reference position return (G30)

The tool returns directly to the 2nd, 3rd, or 4th reference position without passing through any intermediate points, regardless of the specified amount of travel. To select a reference position, specify 2, 3, or 4 in address P. If address P is omitted, a return to the second reference position is performed.

Feedrate (parameter)	Rapid traverse rate (No. 1420)
Automatic acceleration/deceleration (parameter)	Linear acceleration/deceleration in rapid traverse for each axis (No. 1620)
Override	Rapid traverse override

NOTE
The function for 3rd/4th reference position return is optional.
1 When the option is not selected
If "P2" is not specified in address P, the alarm PS0046 is generated and the function cannot be executed.
2 When the option is selected
If "P2," "P3," or "P4" is not specified in address P, the alarm PS0046 is generated and the function cannot be executed.

- M codes (Auxiliary functions)

After address M, specify a numeric value of no more than the number of digits specified by parameter No. 3030. When M98 or M99 is specified, it is executed but not output to the PMC.

NOTE
Neither subprogram calls nor custom macro calls can be performed using M codes.

- S codes (spindle speed functions)

After address S, specify a numeric value of no more than the number of digits specified by parameter No. 3031.

NOTE
Subprogram calls cannot be performed using S codes.

- T codes (tool functions)

After address T, specify a numeric value of no more than the number of digits specified by parameter No. 3032.

NOTE
1 Subprogram calls cannot be performed using T codes.
2 In lathe systems, tool compensation by T code commands is not performed.

- B codes (second auxiliary functions)

After address B, specify a numeric value of no more than the number of digits specified by parameter No. 3033.

NOTE
1 B codes can be renamed "U," "V," "W," "A," or "C" by setting parameter No. 3460. If the new name is the same as an axis name address, "B" is used. Note that "U," "V," and "W" can be used for the T codes only when the G codes are B or C.
2 Subprogram calls cannot be performed using B codes.

- Data input

(1) When addresses and numeric values of a command are typed, then soft key [INPUT] is pressed, the entered data is set. In this case, the input unit is either the least input increment or calculator-type input format, according to the setting of bit 0 (DPI) of parameter No. 3401.

The

key on the MDI panel can be used instead of soft key [INPUT].

(2) Commands can be typed successively.
(3) Key entry is disabled during execution.

If soft key [INPUT] or the key on the MDI panel is pressed during execution, an "EXECUTION/MODE SWITCHING IN PROGRESS" warning is output.

(4) If input data contains an error, the following warnings may appear:

Warning	Description
FORMAT ERROR	- A G code other than G00, G01, and G28 has been entered. - An address other than those displayed on the manual numerical command screen has been entered. A value that exceeds the following limitations has been entered. - Address G:..... 2 digits - Address P:..... 1 digit - Axis address:..... 9 digits - M, S, T, B: ..... The parameter-set number of digits

NOTE
Even when the memory protection key is set, key input can nevertheless be performed.

- Erasing data

(1) When soft key [CLEAR] is pressed, followed by soft key [EXEC], all the set data is cleared. In this case, however, the G codes are set to G00 or G01, depending on the setting of bit 0 (G01) of parameter No. 3402.

Data can also be cleared by pressing the key on the MDI panel.

(2) If soft key [CLEAR] is pressed during execution, an "EXECUTION/MODE SWITCHING IN PROGRESS" warning is output.

- Scrolling pages

If the whole axis information cannot be displayed on a single page, you can scroll pages by pressing the or key.

- Halting execution

If one of the following occurs during execution, execution is halted, and the data is cleared in the same way as when soft key [CLEAR] is pressed. The remaining distance to be traveled is canceled.

(1) When a feed hold is applied
(2) When the mode is changed to other than jog feed mode
(3) When an alarm is generated
(4) When a reset or emergency stop is applied

The M, S, T, and B functions remain effective even upon the occurrence of the above events, with the exception of (4).

- Modal information

Modal G codes and addresses used in automatic operation or MDI operation are not affected by the execution of commands specified using the manual numerical command function.

- Jog feed

When the tool is moved along an axis using a feed axis and direction selection switch on the manual numerical command screen, the remaining amount of travel is always shown as "0".

- Disabling the M, S, T, and B functions

By setting bit 0 to bit 3 (JMF, JSF, JTF, and JBF) of parameter No. 7002, the M, S, T, and B functions can be disabled. If a disabled function is specified, the warning "THIS COMMAND CANNOT BE EXECUTED." is issued.

Limitation

- M, S, T, and B functions

While automatic operation is halted, manual numerical commands can be executed. In the following cases, however, a "START REJECTED (ALREADY EXECUTING)" warning is output, and command execution is disabled.

(1) When an M, S, T, or B function is already being executed, a manual numerical command containing an M, S, T, or B function cannot be executed.
(2) When an M, S, T, or B function is already being executed, and that function alone is specified or a block specifying that function also contains another function (such as a move command or dwell function) which has already been completed, a manual numerical command cannot be executed.

- Jog feed

When a manual numerical command is specified while the tool is being moved along an axis by using a feed axis and direction selection switch, the axial movement is interrupted, and the manual numerical command is executed. Therefore, the tool cannot be moved along an axis by using a feed axis and direction selection switch during execution of a manual numerical command.

- Mirror image

A mirror image cannot be produced for the direction of a specified axial movement.

- REF mode

The manual numerical command screen is not displayed in the REF mode.

- Indexing of the index table and chopping

Commands cannot be specified for an axis along which operation is being performed during indexing or chopping.
If such an axis is specified for execution, a "THIS COMMAND CAN NOT EXECUTE" warning is output.

- Functions that cannot be specified

Commands cannot be specified for axes that operate using any of the following functions.

- Index table indexing
- Chopping
- Spindle positioning
- Polygon turning
- Synchronous, composite, and superimposed control

If the commands are executed for any such axis, a "THIS COMMAND CAN NOT EXECUTE" warning is generated.

- Functions that cannot be used

Commands cannot be specified for the following functions.

- Extended axis name
- Extended spindle name
- Address P command for multi spindle
- Cs contour control function

3.9 THREE-DIMENSIONAL MANUAL FEED

This function enables the use of the following functions.

- Three-dimensional manual feed
  - Tool axis direction handle feed/tool axis direction JOG feed/tool axis direction incremental feed
  - Tool axis right-angle direction handle feed/tool axis right-angle direction JOG feed/tool axis right-angle direction incremental feed
  - Tool tip center rotation handle feed/tool tip center rotation JOG feed/tool tip center rotation incremental feed
  - Table vertical direction handle feed/table vertical direction JOG feed/table vertical direction incremental feed
  - Table horizontal direction handle feed/table horizontal direction JOG feed/table horizontal direction incremental feed

  A handle interrupt can be generated for each handle feed. Handle interrupts work according to the corresponding handle feed specifications described hereinafter unless otherwise noted.

- Screen display functions
  - Display of the coordinate of the tool tip
  - Display of pulse values
  - Display of the amount of machine axes movement

NOTE
1 To execute three-dimensional handle feed requires the manual handle feed option. Also, to generate three-dimensional handle interrupts requires the manual handle interrupt option.
2 A three-dimensional handle interrupt must not be generated when a rotation axis command is being executed during automatic operation.
3 Three-dimensional manual feed is disabled when the manual reference position return mode is selected.

3.9.1 Tool Axis Direction Handle Feed / Tool Axis Direction JOG Feed / Tool Axis Direction Incremental Feed

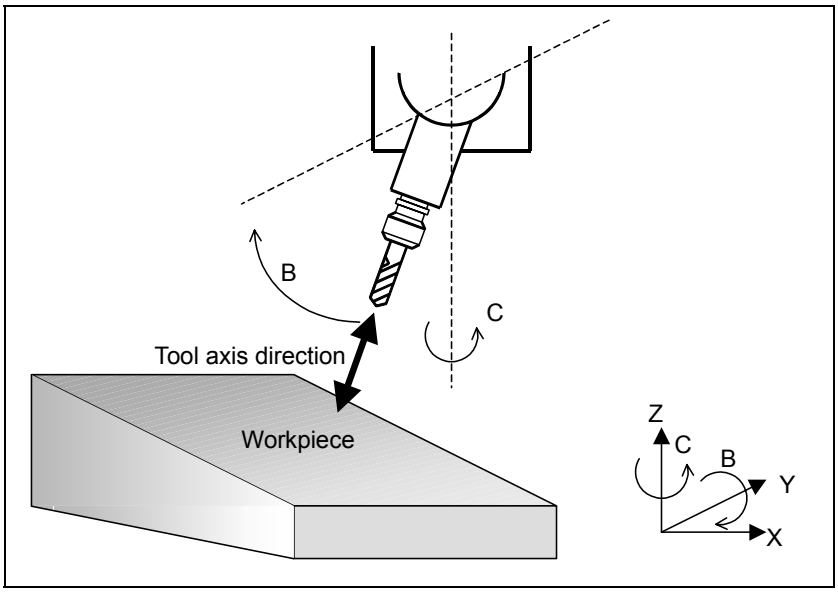
Overview

In the tool axis direction handle feed, tool axis direction JOG feed, and tool axis direction incremental feed, the tool or table is moved in the tool axis direction.

Explanation

- Tool axis direction

The tool axis direction that is taken when all the rotation axes for controlling the tool are at an angle of 0 degree is specified in parameters No.19697, No.19698, and No.19699. As the rotation axes for controlling the tool rotate, the tool axis direction changes according to the rotation axis angle.



- Tool axis direction feed in the tilted working plane command mode

If bit 0 (TWD) of parameter No. 12320 is set to 1, the feed direction of the tool axis direction feed in the tilted working plane command mode is assumed to be the Z direction in the feature coordinate system of the tilted working plane command.

- Tool axis direction handle feed

The tool axis direction handle feed is enabled when the following four conditions are satisfied:

<1> Handle mode is selected.
<2> The tool axis direction feed mode signal (ALNGH) is set to "1" and the table base signal (TB_BASE) is set to "0".
<3> The state of the first manual handle feed axis selection signals (HS1A - HS1E) to make the tool axis direction handle feed mode effective is set in parameter No.12310.
<4> The value of parameter No.12310 matches the first manual handle feed axis selection signals (HS1A - HS1E).

Amount of movement

When the manual pulse generator is rotated, the tool is moved in the tool axis direction by the amount of rotation.

Feedrate clamp

The feedrate is clamped so that the speed of each moving axis dose not exceed the manual rapid traverse rate (parameter No.1424). Handle pulses generated while the clamp feedrate is exceeded are ignored.

- Tool axis direction JOG feed/tool axis direction incremental feed

The tool axis direction JOG feed or tool axis direction incremental feed is enabled when the following three conditions are satisfied:

<1> JOG mode or incremental feed mode is selected.

<2> The tool axis direction feed mode signal (ALNGH) is set to "1" and the table base signal (TB_BASE) is set to "0".

<3> The feed axis direction selection signal (+Jn, -Jn (where n = 1 to the number of controlled axes)) is set to "1" for the axis corresponding to the direction specified by parameter No.19697. (Even when the tool axis direction is slant because of the settings of parameters No.19698 and No.19699, the signal that activates the tool axis direction JOG feed or tool axis direction incremental feed is determined by parameter No.19697 only.)

Ex.) No.19697 = 3 (+Z-axis direction); Z-axis is the 3rd axis.

- +J3 : Tool axis direction +
- -J3 : Tool axis direction -

Feedrate

The feedrate is the dry run rate (parameter No.1410). The manual feedrate override feature is available.

If bit 2 (JFR) of parameter No. 12320 is set to 1, the feedrate of a rotation axis is the jog feedrate of the axis to be rotated (parameter No. 1423). The manual feedrate override feature is available.

Feedrate clamp

The feedrate is clamped so that the speed of each moving axis dose not exceed the manual rapid traverse rate (parameter No.1424).

3.9.2 Tool Axis Right-Angle Direction Handle Feed / Tool Axis Right-Angle Direction JOG Feed / Tool Axis Right-Angle Direction Incremental Feed

Overview

In the tool axis right-angle direction handle feed, tool axis direction JOG feed, or tool axis direction incremental feed, the tool or table is moved in the tool axis direction.
If bit 1 (FLL) of parameter No. 12320 is set to 1, the tool or table is moved in the latitude or longitude direction determined by the tool axis direction vector.

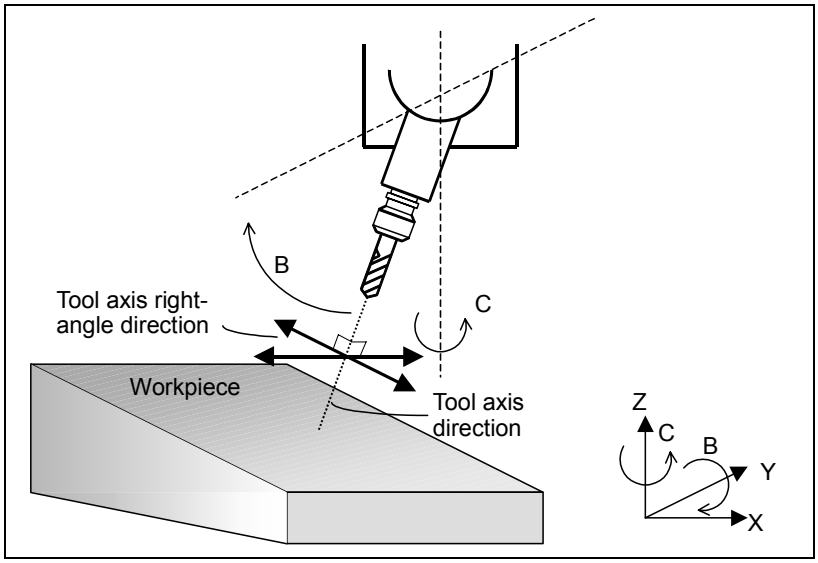
Explanation

- Tool axis right-angle direction

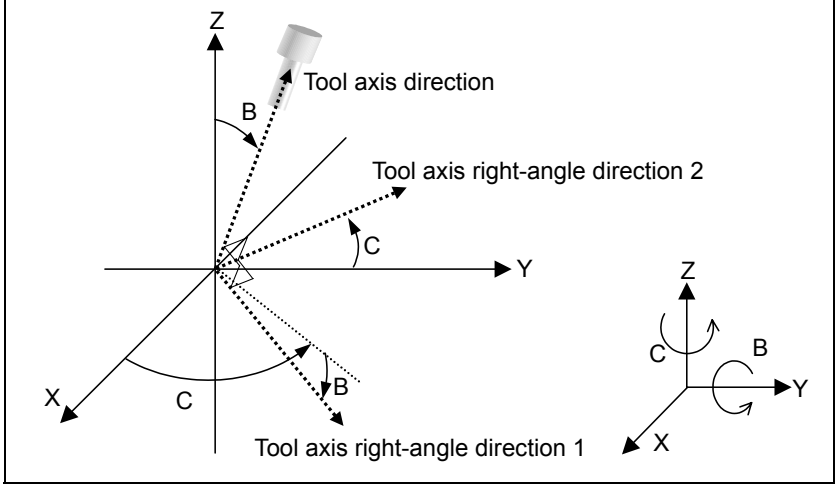
There are two tool axis right-angle directions, which are perpendicular to the tool axis direction (see the previous section).

Parameter No.19697	Tool axis right-angle direction 1	Tool axis right-angle direction 2
1 (The reference tool direction is +X.)	+Y direction	+Z direction
2 (The reference tool direction is +Y.)	+Z direction	+X direction
3 (The reference tool direction is +Z.)	+X direction	+Y direction

This table shows the tool axis right-angle directions that may be taken when the angles of all the rotation axes for controlling the tool are 0 degree and when parameters No.19698 and No.19699 are both set to 0.
When the reference tool axis direction is inclined based on the settings of parameters No.19698 and No.19699, the tool axis right-angle direction is also inclined as much.
As the rotation axes for controlling the tool rotate, the tool axis right-angle direction changes according to the rotation axis angle.



(Example) When the tool rotation axes are B-axis and C-axis and the tool axis direction is the Z-axis direction



- Latitude and longitude directions

When bit 1 (FLL) of parameter No. 12320 is set to 1, the feed direction is defined as follows:

Let a vector perpendicular to a plane formed by the tool axis direction vector ($\vec{T}$) and normal axis direction vector ($\vec{P}$) (parameter No. 12321) be the tool axis right-angle direction 1 (longitude direction) vector ($\vec{R}1$). When tool axis right-angle direction 1 is selected, a movement in the positive direction means a movement in this vector direction, and a movement in the negative direction means a movement in the direction opposite to the vector direction. (Longitude direction feed)

Equation: $\vec{R}1 = \vec{P} \times \vec{T}$

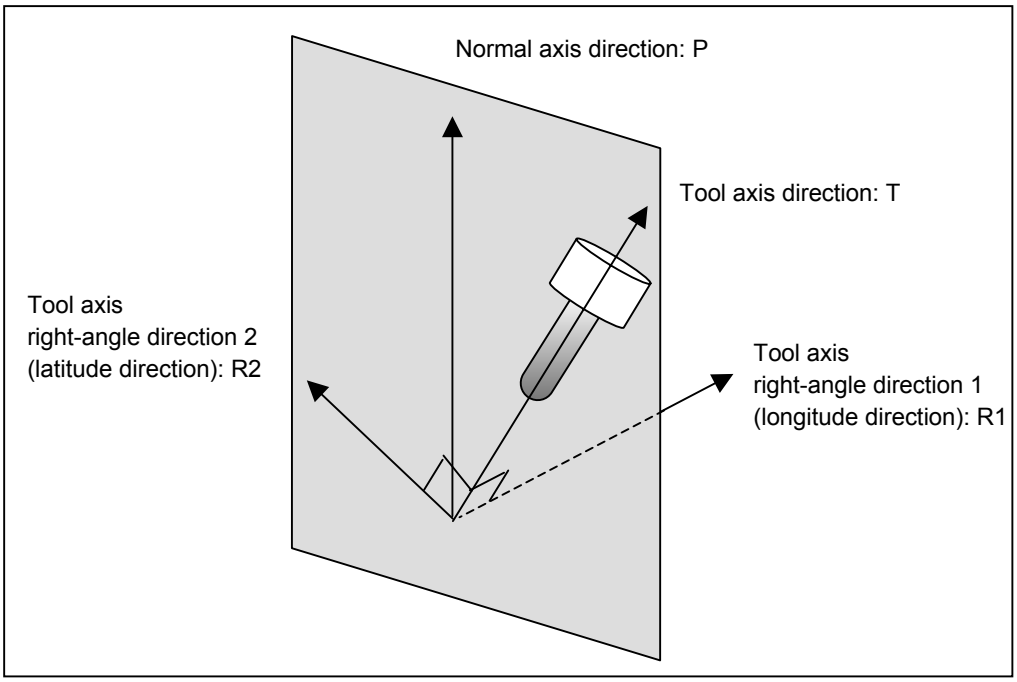
Let a vector perpendicular to the tool axis direction vector ($\vec{T}$) and tool axis right-angle direction 1 (longitude direction) vector ($\vec{R}1$) be the tool axis right-angle direction 2 (latitude direction) vector ($\vec{R}2$). When tool axis right-angle direction 2 is selected, a movement in the positive direction means a movement in this vector direction, and a movement in the negative direction means a movement in the direction opposite to the vector direction. (Latitude direction)

Equation: $\vec{R}2 = \vec{T} \times \vec{R}1$

When the tool axis direction vector ($\vec{T}$) is parallel to the normal axis direction vector ($\vec{P}$) (parameter No. 12321) (when the angle between them is not greater than the setting of parameter No. 12322), tool axis right-angle direction 1 and tool axis right-angle direction 2 are assumed as follows:

Parameter No.12321	Normal axis direction	Tool axis right-angle direction 1	Tool axis right-angle direction 2
1	+X direction	+Y direction	+Z direction
2	+Y direction	+Z direction	+X direction
3	+Z direction	+X direction	+Y direction

If 0 is set in parameter No. 12321, the normal axis direction is set to the reference tool axis direction (parameter No. 19697).
If a value other than 0 to 3 is specified in parameter No. 12321, alarm PS5459 is issued.



- Tool axis right-angle direction feed in the tilted working plane command mode

If bit 0 (TWD) of parameter No. 12320 is set to 1, the feed direction of the tool axis right-angle direction feed in the tilted working plane command mode is defined as follows:

Tool axis right-angle direction 1: X direction in the feature coordinate system of the tilted working plane command
Tool axis right-angle direction 2: Y direction in the feature coordinate system of the tilted working plane command

- Tool axis right-angle direction handle feed

The tool axis right-angle direction handle feed is enabled when the following four conditions are satisfied:

<1> Handle mode is selected.
<2> The tool axis right-angle direction feed mode signal (RGHTH) is set to "1" and the table base signal (TB_BASE) is set to "0".
<3> The state of the first manual handle feed axis selection signals (HS1A - HS1E) to make the tool axis right-angle direction handle feed mode effective is set in parameter No.12311 or No.12312.
<4> The value of parameter No.12311 or No.12312 matches the first manual handle feed axis selection signals (HS1A - HS1E).

Amount of movement

When the manual pulse generator is rotated, the tool is moved in the tool axis right-angle direction by the amount of rotation.

Feedrate clamp

The feedrate is clamped so that the speed of each moving axis dose not exceed the manual rapid traverse rate (parameter No.1424). Handle pulses generated while the clamp feedrate is exceeded are ignored.

- Tool axis right-angle direction JOG feed/tool axis right-angle direction incremental feed

The tool axis right-angle direction JOG feed or tool axis right-angle direction incremental feed is enabled when the following three conditions are satisfied:

<1> JOG mode or incremental feed mode is selected.

<2> The tool axis right-angle direction feed mode signal (RGHTH) is set to "1" and the table base signal (TB_BASE) is set to "0".

<3> The feed axis direction selection signal (+Jn, -Jn (where n = 1 to the number of controlled axes)) is set to "1" for the axis corresponding to the direction that is perpendicular to the direction specified by parameter No.19697. (Even when the tool axis direction is slant because of the settings of parameters No.19698 and No.19699, the signal that activates the tool axis right-angle direction JOG feed or tool axis right-angle direction incremental feed is determined by parameter No.19697 only.)

Ex.) No.19697=3 (+Z-axis direction); X-, Y-, and Z-axes are the 1st, 2nd, and 3rd axes respectively.

- +J1 : Tool axis right-angle direction 1 +
- -J1 : Tool axis right-angle direction 1 -
- +J2 : Tool axis right-angle direction 2 +
- -J2 : Tool axis right-angle direction 2 -

Feedrate

The feedrate is the dry run rate (parameter No.1410). The manual feedrate override feature is available.

If bit 2 (JFR) of parameter No. 12320 is set to 1, the feedrate is the jog feedrate (parameter No. 1423) for a driven feed axis direction selection signal. The manual feedrate override feature is available.

Feedrate clamp

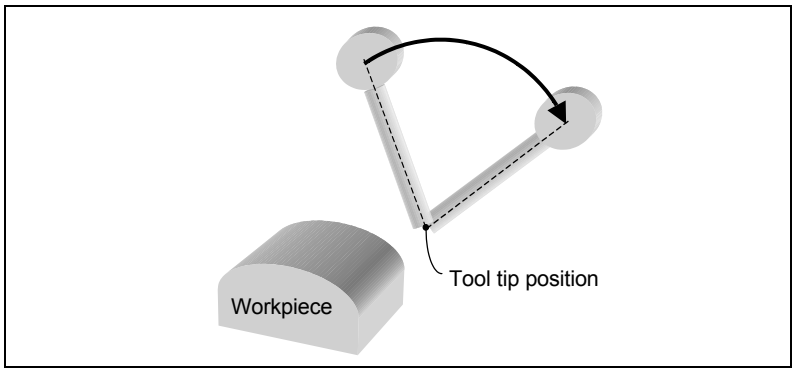
The feedrate is clamped so that the speed of each moving axis dose not exceed the manual rapid traverse rate (parameter No.1424).

3.9.3 Tool Tip Center Rotation Handle Feed / Tool Tip Center Rotation JOG Feed / Tool Tip Center Rotation Incremental Feed

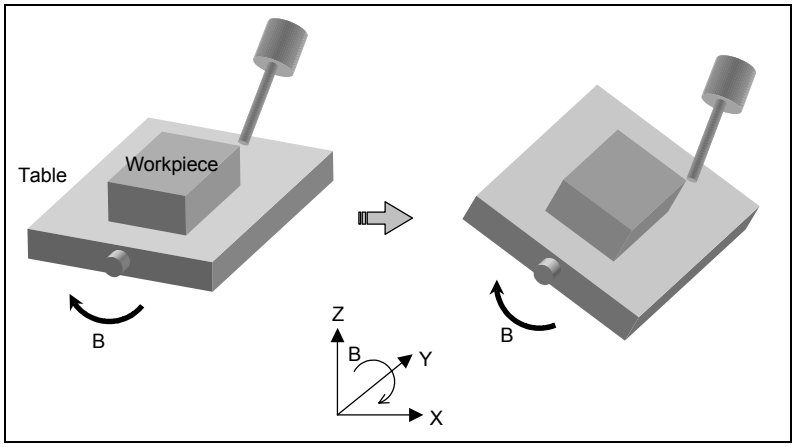
Overview

In the tool tip center rotation handle feed, tool tip center rotation JOG feed, and tool tip center rotation incremental feed, when a rotary axis is rotated by manual feed, the linear axes (X, Y, and Z axes) are moved so that turning the rotation axis does not change the relative relationship between the tool tip position and the workpiece (table).

- The following figure shows an example where the tool is rotated on the rotation axis. In this case, the linear axes are moved so that the position of the tool tip is not moved with respect to the workpiece.



- The following figure shows an example where the table is rotated on the rotation axis. As in the previous case, the linear axes are moved so that the position of the tool tip is not moved with respect to the workpiece (table).



- Tool tip center rotation handle feed

The tool tip center rotation handle feed is enabled when the following four conditions are satisfied:

<1> Handle mode is selected.
<2> The tool tip center rotation feed mode signal (RNDH) is set to "1".
<3> The state of the first manual handle feed axis selection signals (HS1A - HS1E) to make the tool tip center rotation handle feed mode effective is set in parameter No.12313 or No.12314.
<4> The value of parameter No.12313 or No.12314 matches the first manual handle feed axis selection signals (HS1A - HS1E).

Amount of movement

When the manual pulse generator is rotated, the rotation axis is moved by the amount of rotation. The linear axes (X, Y, and Z axes) are moved so that turning the rotation axis does not change the relative relationship between the tool tip position and the workpiece.

Feedrate clamp

The feedrate is clamped so that the synthetic speed of the linear axes (in the tangential direction) does not exceed the manual rapid traverse rate (parameter No.1424) (of any moving linear axis). The feedrate is also clamped so that the speed of the rotation axis does not exceed the manual rapid traverse rate (parameter No.1424) (of that particular axis). Handle pulses generated while the clamp feedrate is exceeded are ignored.

- Tool tip center rotation JOG feed/tool tip center rotation incremental feed

The tool tip center rotation JOG feed or tool tip center rotation incremental feed is enabled when the following three conditions are satisfied:

<1> JOG mode or incremental feed mode is selected.
<2> The tool tip center rotation feed mode signal (RNDH) is set to "1".
<3> The feed axis direction selection signal (+Jn, -Jn (where n = 1 to the number of controlled axes)) is set to "1" for the rotation axis to be rotated.
Ex.) When the B-axis (4th axis) is rotated
- +J4 : Tool tip center rotation feed +
- -J4 : Tool tip center rotation feed -

Feedrate

Control is exerted so that the synthetic speed of the linear axes (in the tangential direction) is the dry run rate (parameter No.1410). The manual feedrate override feature is available.
If bit 2 (JFR) of parameter No. 12320 is set to 1, the feedrate of a rotation axis is the jog feedrate of the axis to be rotated (parameter No. 1423). The manual feedrate override feature is available.

Feedrate clamp

The feedrate is clamped so that the synthetic speed of the linear axes (in the tangential direction) does not exceed the manual rapid traverse rate (parameter No.1424) (of any moving linear axis). The feedrate is also clamped so that the speed of the rotation axis does not exceed the manual rapid traverse rate (parameter No.1424) (of that particular axis).

- Selection of the tool length offset value

The tool length in three-dimensional manual feed is determined as explained below. Table 3.9.3 (a))
If bit 2 (LOD) of parameter No. 19746 is set to 0, the value set in parameter No. 12318 is assumed to be the tool length.
If the LOD parameter is set to 1, and the tool length offset function is performed, the offset data specified for the tool length offset is assumed to be the tool length.
If the LOD parameter is set to 1, and the tool length offset function is not performed, the tool length is determined as follows. If bit 3 (LOZ) of parameter No. 19746 is set to 0, the value set in parameter No. 12318 is assumed to be the tool length in three-dimensional manual feed; if LOZ is set to 1, the tool length is assumed to be 0.

Table 3.9.3 (a) Tool length offset value in three-dimensional manual feed

<table>
<tr><th colspan="2" rowspan="3"></th><th colspan="3">Parameter No. 19746#2(LOD)</th></tr>
<tr><th rowspan="2">= 0</th><th colspan="2">= 1</th></tr>
<tr><th>Tool length offset enabled</th><th>Tool length offset canceled</th></tr>
<tr><td rowspan="2">Parameter No. 19746#3 (LOZ)</td><td>= 0</td><td rowspan="2">Parameter No. 12318</td><td rowspan="2">Offset data</td><td>Parameter No. 12318</td></tr>
<tr><td>= 1</td><td>0</td></tr>
</table>

The tool length offset function is enabled when the following two conditions are both satisfied:

- The tool length offset function listed below is enabled (modal code of group 8 except G49)
  - G43／G44 : Tool length compensation
  - G43.4／G43.5 : Tool center point control
- The H/D code is other than 0.

If bit 6 (CLR) of parameter No. 3402 is set to 0 not to clear the tool length offset vector, G codes of group 8, and H codes at the time of a reset, the tool length offset status is maintained when a reset is made in the tool length offset mode.

3.9.4 Table Vertical Direction Handle Feed / Table Vertical Direction JOG Feed / Table Vertical Direction Incremental Feed

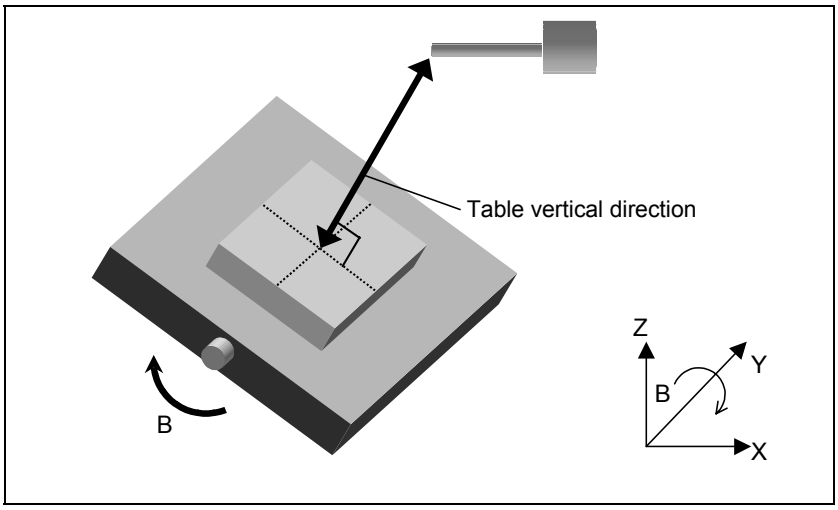
Overview

In the table vertical direction handle feed, table vertical direction JOG feed, and table vertical direction incremental feed, the tool is moved in the table vertical direction.

Explanation

- Table vertical direction

The table vertical direction is a direction vertical to the table. It is equal to the tool axis direction specified in parameter No.19697 when all of the rotation axes for controlling the table are at a an angle of 0 degree. When the rotation axes for controlling the table rotate, the table vertical direction changes according to the rotation axis angle.



- Table-based vertical direction feed in the tilted working plane command mode

If bit 0 (TWD) of parameter No. 12320 is set to 1, the feed direction of the table-based vertical direction feed in the tilted working plane command mode is assumed to be the Z direction in the feature coordinate system of the tilted working plane command.

- Table vertical direction handle feed

The table vertical direction handle feed is enabled when the following four conditions are satisfied:

<1> Handle mode is selected.
<2> Both the tool axis direction feed mode signal (ALNGH) and the table base signal (TB_BASE) are set to "1".
<3> The state of the first manual handle feed axis selection signals (HS1A - HS1E) to make the table vertical handle feed mode effective is set in parameter No.12310.
<4> The value of parameter No.12310 matches the first manual handle feed axis selection signals (HS1A - HS1E).

Amount of movement

When the manual pulse generator is rotated, the tool is moved in the table vertical direction by the amount of rotation.

Feedrate clamp

The feedrate is clamped so that the speed of each moving axis dose not exceed the manual rapid traverse rate (parameter No.1424). Handle pulses generated while the clamp feedrate is exceeded are ignored.

- Table vertical direction JOG feed/table vertical direction incremental feed

The table vertical direction JOG feed or table vertical direction incremental feed is enabled when the following three conditions are satisfied:

<1> JOG mode or incremental feed mode is selected.

<2> Both the tool axis direction feed mode signal (ALNGH) and the table base signal (TB_BASE) are set to "1".

<3> The feed axis direction selection signal [+Jn,-Jn (where n = 1 to the number of controlled axes)] is set to "1" for the axis corresponding to the direction specified by parameter No.19697.

Ex.) No.19697 = 3 (+Z-axis direction); Z-axis is the 3rd axis.

- +J3 : Table vertical direction +
- -J3 : Table vertical direction -

Feedrate

The feedrate is the dry run rate (parameter No.1410). The manual feedrate override feature is available.

If bit 2 (JFR) of parameter No. 12320 is set to 1, the feedrate is the jog feedrate (parameter No. 1423) for a driven feed axis direction selection signal. The manual feedrate override feature is available.

Feedrate clamp

The feedrate is clamped so that the speed of each moving axis dose not exceed the manual rapid traverse rate (parameter No.1424).

3.9.5 Table Horizontal Direction Handle Feed / Table Horizontal Direction JOG Feed / Table Horizontal Direction Incremental Feed

Overview

In the table horizontal direction handle feed, table horizontal direction JOG feed, and table horizontal direction incremental feed, the tool is moved in the table horizontal direction.
If bit 1 (FLL) of parameter No. 12320 is set to 1, the tool or table is moved in the latitude or longitude direction determined by the table-based vertical direction vector.

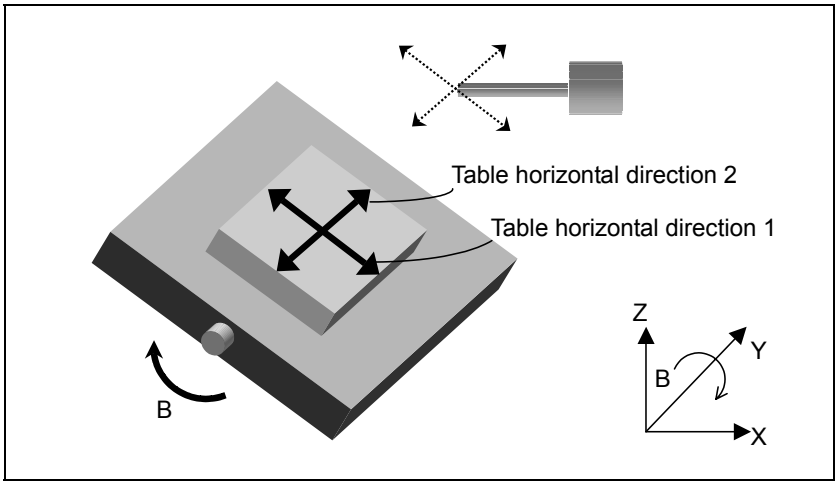
Explanation

- Table horizontal direction

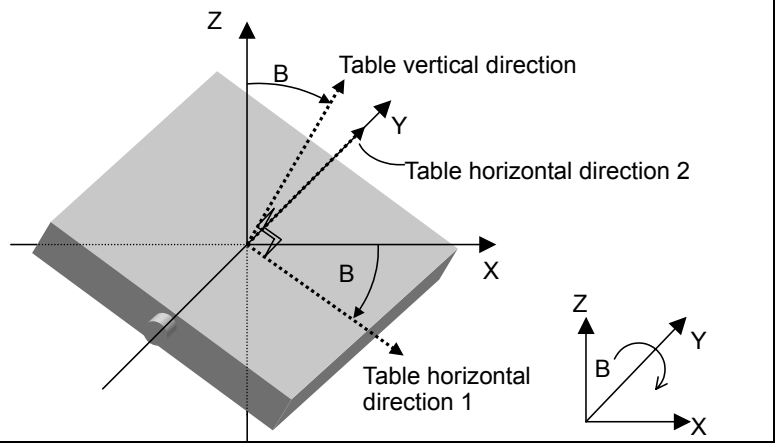
There are two table horizontal directions, which are perpendicular to the table vertical direction (see the previous section).

Parameter No.19697	Table horizontal direction 1	Table horizontal direction 2
1 (The reference tool direction is +X.)	+Y direction	+Z direction
2 (The reference tool direction is +Y.)	+Z direction	+X direction
3 (The reference tool direction is +Z.)	+X direction	+Y direction

This table shows the table horizontal directions that may be taken when the angles of all the rotation axes for controlling the table are 0 degree.
As the rotation axes for controlling the table rotate, the table horizontal direction changes according to the rotation axis angle.



(Example) When the table rotation axis is the B-axis, and the table vertical direction is the Z-axis direction



- Latitude and longitude directions

When bit 1 (FLL) of parameter No. 12320 is set to 1, the feed direction is defined as follows:

Let a vector perpendicular to a plane formed by the table-based vertical direction vector ($\vec{T}$) and normal axis direction vector ($\vec{P}$) (parameter No. 12321) be the table-based horizontal direction 1 (longitude direction) vector ($\vec{R1}$). When tool axis right-angle direction 1 is selected, a movement in the positive direction means a movement in this vector direction, and a movement in the negative direction means a movement in the direction opposite to the vector direction. (Longitude direction feed)

Equation: $\vec{R1} = \vec{P} \times \vec{T}$

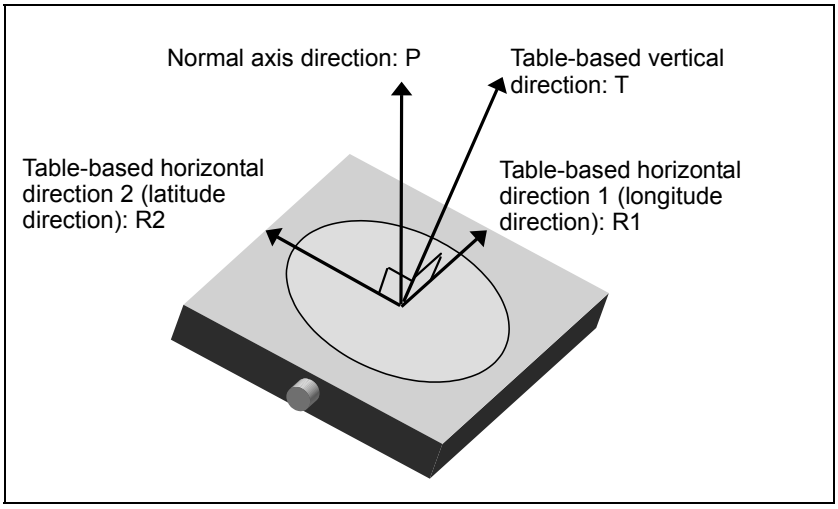
Let a vector perpendicular to the table-based vertical direction vector ($\vec{T}$) and table-based horizontal direction 1 (longitude direction) vector ($\vec{R1}$) be the table-based horizontal direction 2 (latitude direction) vector ($\vec{R2}$). When tool axis right-angle direction 2 is selected, a movement in the positive direction means a movement in this vector direction, and a movement in the negative direction means a movement in the direction opposite to the vector direction. (Latitude direction)

Equation: $\vec{R2} = \vec{T} \times \vec{R1}$

When table-based vertical direction vector ($\vec{T}$) is parallel to the normal axis direction vector ($\vec{P}$) (parameter No. 12321) (when the angle between them is not greater than the setting of parameter No. 12322), tool axis right-angle direction 1 and tool axis right-angle direction 2 are assumed as follows:

Parameter No. 12321	Normal axis direction	Table-based horizontal direction 1	Table-based horizontal direction 2
1	+X direction	+Y direction	+Z direction
2	+Y direction	+Z direction	+X direction
3	+Z direction	+X direction	+Y direction

If 0 is set in parameter No. 12321, the normal axis direction is set to the tool axis direction.
If a value other than 0 to 3 is specified in parameter No. 12321, alarm PS5459 is issued.



- Table-based horizontal direction feed in the tilted working plane command mode

If bit 0 (TWD) of parameter No. 12320 is set to 1, the feed direction of the table-based horizontal direction feed in the tilted working plane command mode is defined as follows:

Table-based horizontal direction 1: X direction in the feature coordinate system of the tilted working plane command
Table-based horizontal direction 2: Y direction in the feature coordinate system of the tilted working plane command

- Table horizontal direction handle feed

The table horizontal direction handle feed is enabled when the following four conditions are satisfied:

<1> Handle mode is selected.
<2> Both the tool axis right-angle direction feed mode signal (RGHTH) and the table base signal (TB_BASE) are set to 1.
<3> The state of the first manual handle feed axis selection signals (HS1A - HS1E) to make the table horizontal direction handle feed mode effective is set in parameter No.12311 or No.12312.
<4> The value of parameter No.12311 or No.12312 matches the first manual handle feed axis selection signals (HS1A - HS1E).

Amount of movement

When the manual pulse generator is rotated, the tool is moved in the table horizontal direction by the amount of rotation.

Feedrate clamp

The feedrate is clamped so that the speed of each moving axis dose not exceed the manual rapid traverse rate (parameter No.1424). Handle pulses generated while the clamp feedrate is exceeded are ignored.

- Table horizontal direction JOG feed/table horizontal direction incremental feed

The table horizontal direction JOG feed or table horizontal direction incremental feed is enabled when the following three conditions are satisfied:

<1> JOG mode or incremental feed mode is selected.

<2> Both the tool axis right-angle direction feed mode signal (RGHTH) and the table base signal (TB_BASE) are set to "1".

<3> The feed axis direction selection signal (+Jn, -Jn (where n = 1 to the number of controlled axes)) is set to "1" for the axis corresponding to the direction that is perpendicular to the direction specified by parameter No.19697.

Ex.) No.19697 = 3 (+Z-axis direction); X-, Y-, and Z-axes are the 1st, 2nd, and 3rd axes respectively.

- +J1 : Table horizontal direction 1 +
- -J1 : Table horizontal direction 1 -
- +J2 : Table horizontal direction 2 +
- -J2 : Table horizontal direction 2 -

Feedrate

The feedrate is the dry run rate (parameter No.1410). The manual feedrate override feature is available.

If bit 2 (JFR) of parameter No. 12320 is set to 1, the feedrate is the jog feedrate (parameter No. 1423) of a driven feed axis direction selection signal. The manual feedrate override feature is available.

Feedrate clamp

The feedrate is clamped so that the speed of each moving axis dose not exceed the manual rapid traverse rate (parameter No.1424).

Note

1 To perform a three-dimensional handle feed, the manual handle feed option is required. To perform three-dimensional handle interrupt, the manual handle interrupt option is required.

2 When a three-dimensional handle interrupt is performed, rotation axis command execution must not be in progress in automatic operation.

3 When the manual reference position return mode is selected, three-dimensional manual feed is not enabled.

4 When the offset value specified for the tool length offset function is used for tool center point rotation feed (when bit 2 (LOD) of parameter No. 19746 is set to 1), the controlled point should generally be shifted. (Set bit 5 (SVC) of parameter No. 19665 to 1.)
In this case, specify the tool length with a radius value.

3.10 DISTANCE CODED LINEAR SCALE INTERFACE

Overview

The interval of each reference marks of distance coded linear scale are variable. Accordingly, if the interval is determined, the absolute position can be determined. The CNC measures the interval of reference marks by axis moving of short distance and determines the absolute position. Consequently the reference position can be established without moving to reference position.

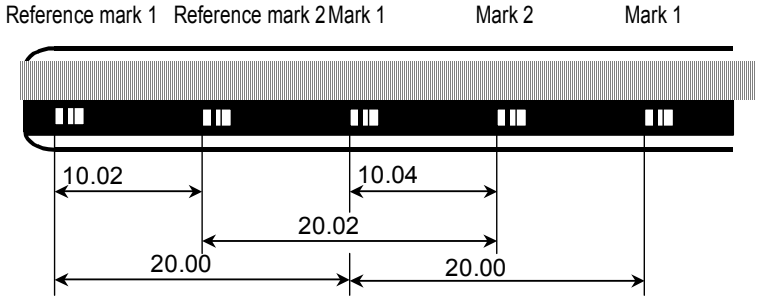


Fig. 3.10 (a) Example of distance coded linear scale

This is an optional function.

3.10.1 Procedure for Reference Position Establishment

Procedure

(1) Select the JOG mode, and set the manual reference position return selection signal ZRN to "1".
(2) Set a direction selection signal(+J1,-J1,+J2,-J2,…) for a target axis.
(3) The axis is fed at a constant low speed (reference position return FL feedrate specified by parameter (No.1425) setting).
(4) When a reference mark is detected, the axis stops, then the axis is fed at a constant low speed again.
(5) Above (4) is executed repeatedly until two, three or four reference marks are detected. And absolute position is determined and reference position establishment signal (ZRF1,ZRF2,ZRF3, …) turns to "1".
(A number of reference marks is determined by bit 2 (DC2x) and 1 (DC4x) of parameter No.1802.)

Even when the direction selection signal (+J1, -J1, +J2, -J2,...) is set to "0" while steps (2) to (5) are being performed, the feedrate operation does not stop, and the operation for establishing a reference position is carried out continuously.

The timing chart for this procedures is given below.

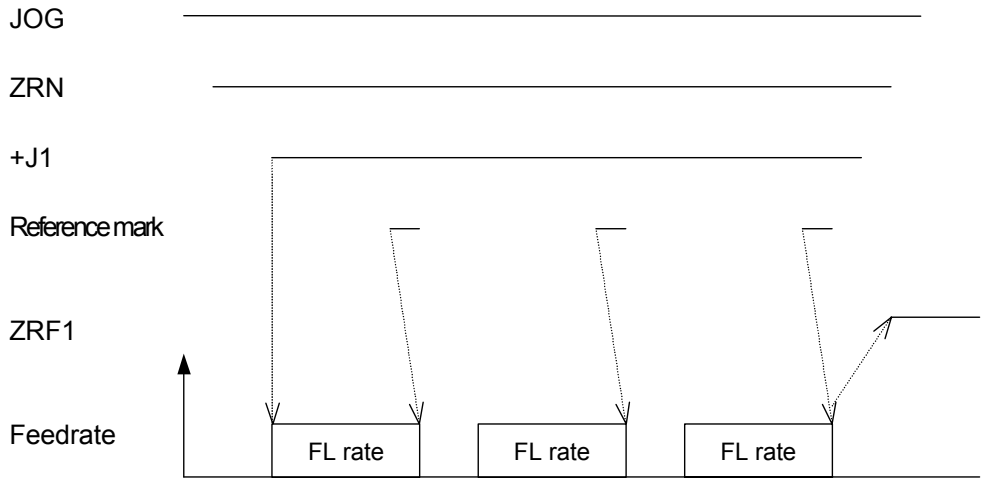


Fig. 3.10.1 (a) Timing chart for reference position establishment

- Procedure for establishing a reference position through automatic operation

If an automatic reference position return (G28) is specified before a reference position is not established, steps (3) to (5) above are performed automatically.
After the reference position is established, the automatic reference position return is performed.

- Stopping the operation for establishing a reference position

The operation for establishing a reference position is stopped if any of the following operations is performed in steps (3) to (5), described above.

- Reset
- Setting the feed axis direction selection signal (+J1, -J1, +J2, -J2, etc.) to 0

If any of the following operations is performed during the operation of automatic reference position return (G28) before a reference position is not established, the operation for establishing a reference position stops:

- Reset
- Performing feed hold during movement from an intermediate position

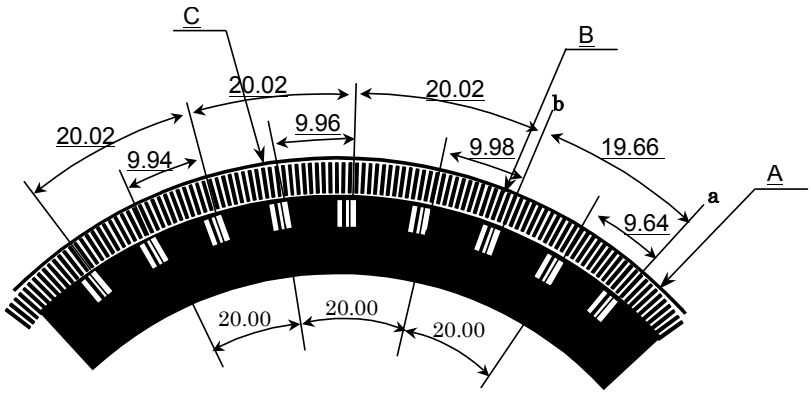
If the operation for establishing a reference position is stopped by an operation other than a reset, the operation for establishing a reference position must be reset and resumed.

3.10.2 Reference Position Return

(1) When the reference position is not established and the axis moved by turning the feed axis direction signal (+J1,-J1,+J2,-J2,...) to "1" in REF mode, the reference position establishment procedure is executed.
(2) When the reference position is already established and the axis is moved by turning the feed axis direction signal (+J1,-J1,+J2,-J2,...) to "1" in REF mode, the axis is moved to the reference point without executing the reference position establishment procedure.
(3) When the reference position is not established and the reference position return command (G28) is executed, the reference position establishment procedure is executed. The next movement the axis depends on the setting of parameter RFS (No.1818#0).
(4) When the reference position is already established and the reference position command (G28) is executed, the movement of the axis depends on the setting of bit 1 (RF2) of parameter No.1818.

3.10.3 Distance Coded Rotary Encoder

In case of setting a rotary axis, if bit 3 (DCRx) of parameter No.1815 is set, the setting axis is regarded as being equipped with a distance coded rotary encoder.
In case of distance coded rotary encoder, the marker interval may be different from parameter setting value. (a-b section of the following figure) When the reference point return is executed through this section, it is not able to establish the reference point. Therefore, in case of distance coded rotary encoder, if the reference point return is started for B point from A point of below figure, the reference point is not established yet at B point. The reference point return is re-started for C point. The reference point return procedure is finished at C point.



- When the reference point return procedure is executed, the coordinate value are rounded in 0 to 360 degree, even if Machine coordinate values are linear axis type.

- In case of distance coded rotary encoder, only the measurement by three points or four points is possible. (bit 2 (DC2) of parameter No.1802 is disregarded as 0.)

3.10.4 Axis Synchronization Control

Requirements when this function is used with axis synchronization control axes

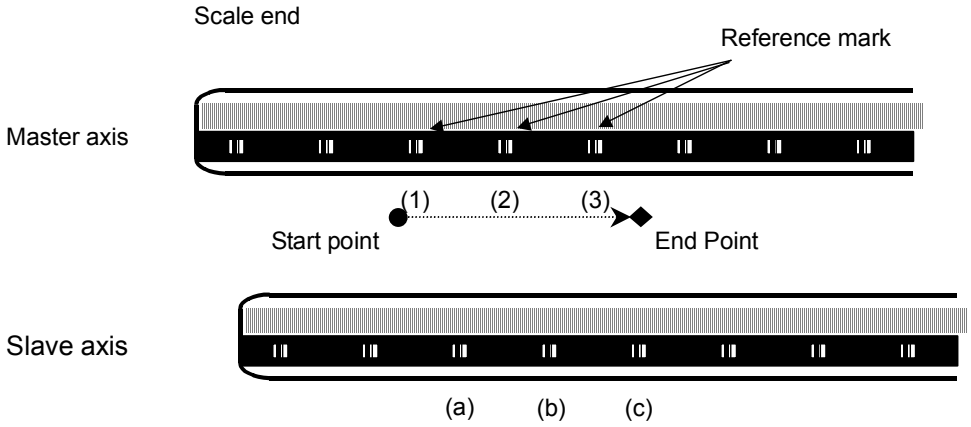
When this function is used with axis synchronization control axes, the distance coded linear scale used for the master axis and that used for the slave axis must have reference marks placed at identical intervals. (Set identical values in parameters Nos. 1821 and 1882 for both the master and slave axes.)
This function does not work unless the use of this function is specified for both the master and slave axes (bit 2 (DCL) of parameter No. 1815 is 1).
Also, in all parameters related to this function, except parameter No. 1883, 1884 (distance from the scale zero point to reference position 1, 2), set identical values for both the master and slave axes.
If a parameter value for the master axis differs from the corresponding parameter value for the slave axis, alarm SV1051 is issued.

NOTE
When this function is used with axis synchronization control axes for which the operation mode is switched between synchronization operation and normal operation, this function is enabled only if the synchronization select signal (SYNC1, SYNC2, ...<Gn138>) is 1. (During establishment of a reference position, the synchronization select signal status must be maintained.)

Reference position establishment with axis synchronization control axes

With axis synchronization control axes, a reference position is established as follows. When a reference mark for the master or slave axis is detected, a stop takes place temporarily. Then, a feed operation is performed again at the reference position return FL feedrate. This sequence is repeated until a reference mark is detected three or four times for both the master and slave axes. Then the absolute position is calculated for both the master and slave axes, and the reference position establish signals ZRF1, ZRF2, ... <F120> are set to 1.
After the reference position has been established by the above operation, a synchronization error is corrected. (Checking for excessive synchronization error alarm 2 is made even during reference position establishment.)

(Example of 3 points measurement system)



In the above example, the following sequence is executed.

a. When the reference mark (1) of the master axis is detected, both master axis and slave axis stop.
b. Both the axes begin to move again at a reference position return FL feedrate.
c. When the reference mark (a) of the slave axis is detected, both axes stop again.
d. Both the axes begin to move again at FL feedrate.
e. Both axes repeat the operation until all point ((2) -> (b) ->(3) -> (c)) are detected.
f. When the slave axis detects the third reference mark (c), both the axes end the reference position establishment.

NOTE
In case of this function is used with axis synchronization control axes, if the value of parameters Nos. 1883 and 1884 for both the master and slave axes is 0, the reference position is not established. Also, the reference position establish signals ZRF1, ZRF2, ... <F120> are set to 0.

3.10.5 Axis Control by PMC

In PMC axis control, if the reference position return command (axis control command code 05H) is issued for an axis having a distance coded linear scale, reference position return is performed according to the reference position return sequence for the distance coded linear scale. Specifically, the following operations take place:

Before reference position establishment	The reference position is established by detecting two, three or four reference marks. Movement to the reference position is not performed.
After reference position establishment	Positioning at the reference position is performed.

3.10.6 Angular Axis Control

There are the following limitations when the angular axis control is used.

(a) It is necessary to use the linear scale with the distance coded reference mark for both the perpendicular axis and the angular axis.

(b) When the reference point of the perpendicular axis is established, it is necessary to establish the reference point of the angular axis previously. When the reference point of the angular axis is not previously established, the alarm DS0020 is generated.

(c) During the reference point establishment operation of the angular axis, the command in the perpendicular axis is invalid in the manual reference point return.

3.10.7 Note

(1) In the case of the actual interval of reference marks is different from parameter setting value, the alarm DS1449 occurs.

(2) This function is disabled if any of the following conditions is satisfied:

- Either parameter No.1821 (mark-1 interval) or parameter No.1882 (mark-2 interval) is set to 0.
- The setting of parameter No.1821 is greater than or equal to the setting of parameter No.1882.
- The difference between the settings made for parameters 1821 and 1882 is greater than or equal to twice either setting.
- The absolute-position detection function is enabled.(Bit 5 (APCx) of parameter No.1815 is set to 1.)

(3) A difference of parameters Nos.1821 and 1882 must be more than 4.

Example)

When the scale, which is that mark1 interval is 20.000mm and mark2 interval is 20.004mm, is used on IS-B machine :

When the detection unit of 0.001mm is selected, parameter No.1821 and No.1882 must be set "20000" and "20004", and the difference of them is "4".

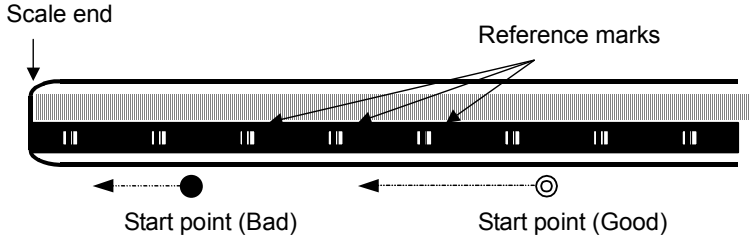
To use such a scale, please adjust the detection unit by modification of parameters No.1820 (CMR) and Nos.2084/2085(flexible feed gear) to make the difference of parameters Nos.1821 and 1882 more than 4 as following examples.

(a) Set the detection unit=0.0001mm, and set No.1821=200000, No.1882=200040

(b) Set the detection unit=0.0005mm, and set No.1821=40000, No.1882=40008

NOTE
When the detection unit is changed, parameters relating to the detection unit (such as the effective area and positional deviation limit) must also be changed accordingly.

(4) In this procedure, the axis does not stop until two, three or four reference marks are detected. If this procedure is started at the position near the scale end, CNC can not detect three or four reference marks and the axis does not stop until over travel alarm occurs. Please care to start at the position that has enough distance from scale end.



(5) When the axis used this function, the following function can not be used.
- Absolute position detection (absolute pulse coder)
- Three-dimensional error correction

(6) If axial movement is made in the direction opposite to that of reference position return, the movement is reversed to the direction of reference position return after three or four reference marks have been detected. Steps (3) to (5) of the basic procedure for establishing a reference position are carried out to establish the reference position.

(7) Straightness compensation function
When the reference point establishment of moving axis is executed after the establishment of compensation axis, the compensation axis is moved by straightness compensation amount when the reference point of moving axis is established.

(8) The reference point establishment is not performed during synchronous control is activated.

(9) The reference point establishment is not performed during composite control is activated.

(10) The reference point establishment is not performed during superimposed control is activated.

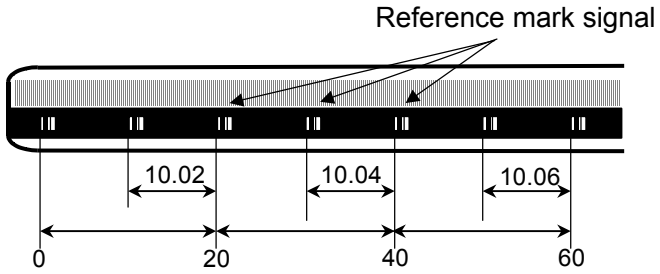
3.11 LINEAR SCALE WITH DISTANCE-CODED REFERENCE MARKS (SERIAL)

Overview

By using High-resolution serial output circuit for the linear scale with distance-coded reference marks (serial), the CNC measures the interval of referenced mark by axis moving of short distance and determines the absolute position.
This function enables high-speed high-precision detection by using High-resolution serial output circuit.
It is available that using maximum stroke 30 meters length.

Explanation

The linear scale with distance-coded reference marks (serial) is combined the irregular reference marked linear scale with the High-resolution serial output circuit, it can detect the accurate position.

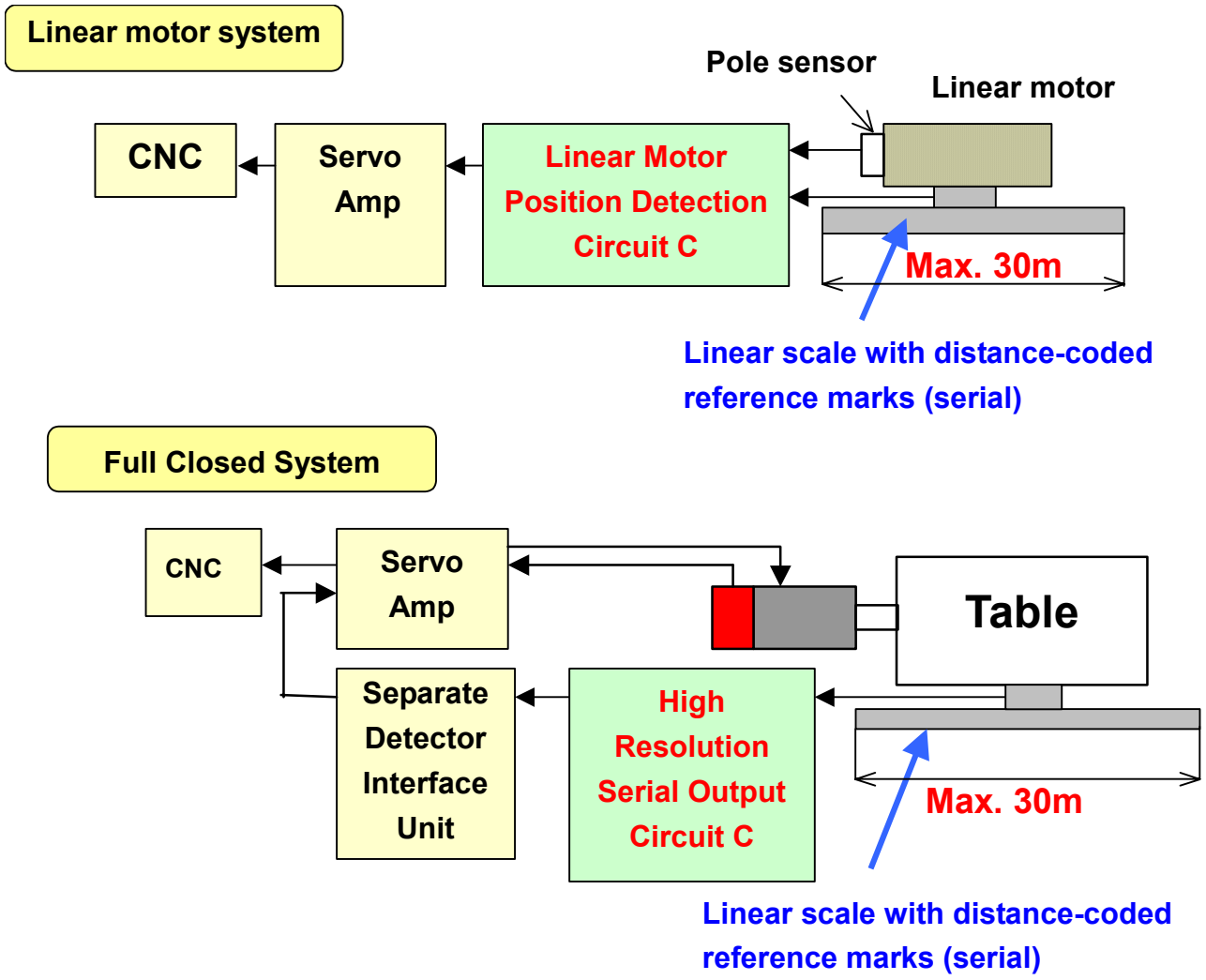


The CNC measures the interval of referenced mark by axis moving of short distance and determines the absolute position, because of the interval of each reference mark is different with regular interval.
It is not necessary that the axis is moved to the reference position for establishment of reference position.

This function enables high-speed high-precision detection by using High-resolution serial output circuit.
It is available that using maximum stroke 30 meters length.

- Connection

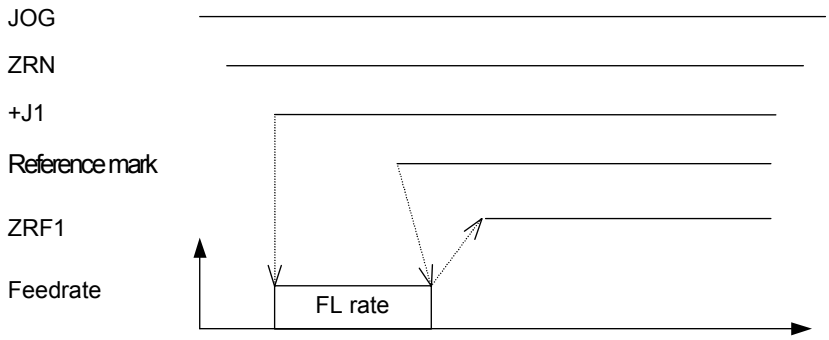
It is available under linear motor system and full closed system.



- Procedure for reference position establishment through manual operation

(1) Select the JOG mode, and set the manual reference position return selection signal ZRN to "1".
(2) Set a direction selection signal(+J1,-J1,+J2,-J2,…) for a target axis.
(3) The axis is fed at a constant low speed (reference position return FL feedrate specified by parameter No.1425 setting).
(4) When the absolute position of linear scale with distance-coded reference marks (serial) is detected, the axis stops. Then the absolute position of CNC is calculated and reference position establishment signal (ZRF1,ZRF2,ZRF3,…) turns to "1".

The timing chart for this procedures is given below.



- Procedure for reference position establishment through automatic operation

If an automatic reference position return (G28) is specified before a reference position is not established, steps (3) to (4) above are performed automatically.
After the reference position is established, the automatic reference position return is performed by setting of parameter RFS No.1818#0.

- Stopping the operation for establishing a reference position

The operation for establishing a reference position is stopped if any of the following operations is performed in steps (3) to (4), described above.

- Reset
- Setting the feed axis direction selection signal (+J1, -J1, +J2, -J2, etc.) to 0
- Setting the Servo off signals (SVF1, SVF2, etc.) to 1

If any of the following operations is performed during the operation of automatic reference position return (G28) before a reference position is not established, the operation for establishing a reference position stops:

- Reset
- Performing feed hold during movement from an intermediate position
- Setting the Servo off signals (SVF1, SVF2, etc.) to 1

If the operation for establishing a reference position is stopped by an operation other than a reset, the operation for establishing a reference position must be reset and resumed.

- Establishing a reference position and moving to the reference position

By following operation, establishing a reference position and moving to the reference position is performed.

	Moving through manual operation in REF mode	Moving through automatic operation by automatic reference position return (G28)
The reference position is not established.	Establishing the reference position	Firstly, moving to the intermediate position, and establishing the reference position. Secondly, whether moving to the reference position or not is performed by setting bit 0 (RFS) of parameter No.1818.
The reference position is established.	Moving to the reference position	Whether moving to the intermediate position and the reference position or not is performed by setting bit 1 (RF2) of parameter No.1818.

- Feed axis synchronization control

In case of using the axis synchronization control, please confirm the following items.

- When this function is used with axis synchronization control axes, the linear scale with distance-coded reference marks (serial) used for the master axis and that used for the slave axis must have reference marks placed at identical intervals.
- The master axis scale and the slave axis scale should be installed in parallel direction. (The zero positions should be faced the same direction.)
- To the parameters, which relate to this function (except No.1883, No.1884), the same value must be set for the master axis and for the slave axis.
- The linear scale with distance-coded reference marks (serial) should be applied for the master axis and the slave axis.
  If either of the master axis or the slave axis is not the linear scale with distance-coded reference marks (serial), alarm DS0018 occurs when reference position establishment is tried.
- During operating the establishment of reference position, the state of signal for selecting synchronized axis(SYNCn<Gn138> or SYNCJn<Gn140>) should be kept.

Procedure for reference position establishment by axis synchronization control is as follows.

- Both of axes (master axis and slave axis) are fed on the reference position return FL feedrate until distance coded scales of both axes detect the absolute position.
- Then absolute position of both axes are calculated and Reference Position Establishment Signals (ZRF1,ZRF2,...) turn to "1".

- Angular axis control

In case of using the angular axis control, please confirm the following items.

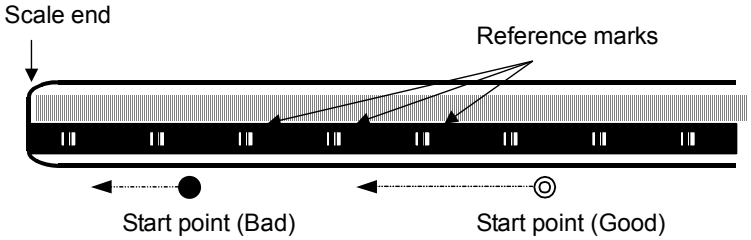
- It is necessary to use the linear scale with distance-coded reference marks (serial) for both the perpendicular axis and the angular axis.
  If not, the alarm DS0019 occurs when reference position establishment is commanded.
- When the reference point establishment of angular and perpendicular axes are tried, please set bit 2 (AZR) of parameter No. 8200 to '0' and input signal NOZAGC <G063.5> to '0'.
  If not, the alarm DS0019 occurs when reference position establishment is commanded.
- When the reference point of the perpendicular axis is established, it is necessary to establish the reference point of the angular axis previously. When the reference point of the angular axis is not previously established, the alarm DS0020 occurs.
- On angular axis control, if you use automatic setting of parameter No.1883,1884 on reference point establishment (bit 2 (DATx) of parameter No.1819 = 1), please establish reference point of perpendicular axis after reference point establishment and return of angular axis.

In manual reference position return, the perpendicular axis cannot be specified while the angular axis reference point is being established. The perpendicular axis, if specified, is ignored.

⚠ CAUTION

1 When the Linear scale with distance-coded reference marks (serial) is used, please set bit 3 (SDCx) of parameter No.1818 to 1.

2 On the Linear scale with distance-coded reference marks (serial), the axis does not stop until three reference marks are detected. If this procedure is started at the position near the scale end, CNC can not detect three reference marks and the axis does not stop until over travel alarm occurs. Please care to start at the position that has enough distance from scale end.
And if establishment of reference position is failed, the establishment is retried. Then axis does not stop until still more three reference marks are detected. So please set the maximum move amount (detection unit : parameter No.14010) not to reach the scale end.



3 On flexible synchronization control mode, reference position can't be established.

4 Straightness compensation function
When the reference point establishment of moving axis is executed after the establishment of compensation axis, the compensation axis is moved by straightness compensation amount when the reference point of moving axis is established.

5 It is not available to use this function and the temporary absolute coordinate setting together.

6 The reference point cannot be established during twin table synchronous operation or single operation.

7 Angular axis control cannot be performed together with synchronous control, composite control, or superposition control.

3.12 MANUAL HANDLE RETRACE

Overview

In this function, the program can be executed both forward and backward with a manual handle (manual pulse generator) under automatic operation.
Therefore, errors of a program, interference, and so on can be checked easily by working a machine actually.

- Checking mode

In this mode, the program can be executed forward and backward and the program can be checked.
To change to the checking mode, it is necessary to change the mode to the memory mode (MEM mode), and the checking mode signal MMOD<Gn067.2> is set to "1". This function makes the data to execute the program backward when the program is executed forward in the checking mode.
To work a machine synchronizing with a pulse generated by a manual handle in the checking mode, the manual handle check signal MCHK <Gn067.3> is set to "1" in addition to the above-mentioned. As a result, it becomes possible to check the program with a manual handle.

NOTE
During the checking mode, it is not possible to change the parameter and offset.

- Forward movement with a manual handle

The "forward movement" is that the program is executed forward by turning a manual handle in the positive direction (when the manual handle check signal is set to "1".) or in no relation to rotation of a manual handle (when the manual handle check signal is set to "0".).
When the manual handle check signal is set to "1", the execution speed of the program is proportional to the number of rotations of a manual handle. The program is executed forward rapidly when a manual handle is turned to the positive direction rapidly. And, the program is executed forward slowly when a manual handle is turned to the positive direction slowly. The distance magnification traveled per pulse from manual handle can be switched as same as a usual manual handle feed function.
When the manual handle check signal is set to "0", the execution of the program is controlled as same as an automatic operation.

- Backward movement

The "backward movement " is that the program executed forward once is executed backward by turning a manual handle in the negative direction.
The program can be executed backward only for the block executed forward. And, the number of blocks for it is about 190 blocks. This block number changes by the content of the commanded program.
The program is executed backward rapidly when a manual handle is turned to the negative direction rapidly. And the program is executed backward slowly when a manual handle is turned to the negative direction slowly. The distance traveled per pulse from manual handle can switch magnification as well as a usual manual handle feed.

NOTE
In the system which exceeds 4 paths and 20 axes, the maximum number of blocks which can move backward becomes about 97 blocks.

Explanation

- Control by the manual handle

Program execution start

The checking mode signal MMOD<Gn067.2> is set to "1" in the memory mode (MEM mode) in order to change the checking mode. Then, the program execution is begun by turning ST signal from "1" to "0".
If the manual handle check signal MCHK <Gn067.3> is set to "1" at this time, the execution of the program is controlled by a manual handle. The program is executed synchronizing with rotation of a manual handle.
When a manual handle check signal MCHK<Gn067.3> is set to "0", it is controlled as usual execution.
When check mode signal MMOD<Gn067.2> is set to "1" during the operation of the program, it is enabled a check mode from the block that next buffering is done.
That is, even if check mode signal set to "1", check mode is not always enabled at once.
When check mode is enabled, check mode confirmation signal MMMOD<Fn091.3> is set to "1".

NOTE
After the signal MMOD is turned to "0" during the execution of the program, the program cannot be executed forward and backward.

Control with the manual handle

The value of the parameter No.6410 and the scale factors decide the moving speed of the machine by one pulse generated by a manual handle.
When a manual handle is turned, the actual movement speed of the machine is as follows.
[Feedrate command value] × [Number of the handle pulse per a second]
× [Handle magnification] × ([Parameter setting value]/100) × (8/1000) (mm/min or inch/min)
Example) When feedrate command value is 30mm/min, handle magnification is 100, parameter No.6410 is set to 1 and manual pulse generator is rotated at 100 pulse/rev, the feedrate of axis is decided as follow.
[Feedrate]=30[mm/min] × 100[pulse/s] × 100 × (1/100) × (8/1000)[s]
=24[mm/min]

When the feedrate exceeds the override 100% feedrate by turning a manual handle rapidly, the feedrate is clamped at the speed of override 100%. That is, if the pulse of the following formula exceeds "1", the feedrate is clamped.
[Number of the handle pulse per a second]
× [Handle magnification] × ([Parameter setting value]/100) × (8/1000)
The rapid traverse feedrate is clamped at 10%. However, the feedrate of the rapid traverse is clamped at 100% when bit 0 (HDRPD) of parameter No.6400 is set to "1".
And if parameter No.6405 is set to an optional value, it can be clamped to override by nearly optional value.
When the parameter No.6405 is set to larger value than "100", it is clamped to nearly 100%.
When parameter No.6405 is set to "0", the setting of bit 0 (RPO) of parameter No.6400 becomes valid.
The single block signal and the feed hold signal are effective in the checking mode. When the execution of a program is stopped by the single block stop or the feed hold stop, it is necessary to turn ST signal from "1" to "0" in order to restart the program.
In the block with the movement and the block of dwell, the execution speed of the program can be controlled by turning a manual handle. As for the block of neither movement nor dwell such as the block of only address M, S, T, and F, the program advances to the following block even if a manual handle does not turn.
The rotation of the spindle does not synchronize with a pulse of a manual handle. During the checking mode, the spindle rotates at the specified rotation speed. As for the feed per revolution, a program is executed at the feedrate which was converted from the rotation speed of the spindle to the corresponding feed per minute inside CNC.

NOTE
The manual handle used by this function is always the first. The 2nd and 3rd manual handle cannot be used in this function.

Forward movement and backward movement with a manual handle

The program is executed forward when a manual handle is turned to the positive direction. And, the program is executed backward when a manual handle is turned to the negative direction.
The program is executed backward as soon as a manual handle is turned to the negative direction in executing the program forward.
When a manual handle keeps being turned in a negative direction, the program is executed backward and the execution stops in the block of O number. Then, if a manual handle is turned to the positive direction, the program is executed forward again.
Even if a manual handle controls the program execution, the program is executed forward in no relation to a pulse generated by a manual handle on setting the manual handle check signal to "0".

Program end

When the block of M2 or M30 is executed, the manual handle retrace ends. It is not possible to execute the program backward from the block of M2 or M30.
When the execution of the program ends, RESET signal must be set to "1", and the checking mode signal and the manual handle check signal must be set to "0".
In 2 path control system, FIN signal must not be set to "1" when the block of M2 or M30 is executed in only one of paths. After the block of M2 or M30 has been executed in both paths, FIN signal is set to "1". (Except for the block of waiting M code is commanded before M2 or M30 in both paths.)

- Notice of the operation

Dry-run can not operate during the checking mode. Dry-run signal must be set to "0".
Automatic operation starts immediately with the feedrate commanded by the program, when the checking mode signal or the synchronous operation with handle signal is turned off during executing the program in the checking mode.
The edit of the program and the change of the parameter and the offset must not be done.

- Backward movement of each code

All modal information of G, T, S-code is memorized in executing the program forward. And, the memorized data of the modal G, T, S-code are used in executing the program backward.
As for M-codes, they are grouped and the modal information is managed by parameter No.6411 to 6490. Therefore, M-code can be executed backward according to the information. As for the modal information of the M-code, a change in each group is memorized in the execution data.
As for the codes except for G, M, S, and T, the same code is output between forward movement and backward movement.

- G-code

If G-code that changes modal information is commanded in backward movement, the modal information of previous block is executed.
Example)

```
N1G99;
N2G01X_F_;
N3X_Z_;
N4G98; ..... backward movement starts from this block
N5X_Y_Z_;
```

If backward movement starts from N4 block, the modal information is changed from G98 to G99 and G99 is executed from N3.
G-code with a movement is traced along the route opposite to forward movement.
G-code that can be command in executing the program backward is as follows.
The other G-codes cannot be command in executing the program backward.
The G-codes in the G-code system B and C also can be used.

Lathe system

G00	G01	G02	G03	G04	G22	G23
G25	G26	G28	G30	G40	G41	G42
G50	G53	G65	G70	G71	G72	G73
G75	G80	G83	G85	G87	G89	G90
G94	G96	G97	G98	G99		

(G-code system A)

Machining center system

G00	G01	G02	G03	G04	G22	G23
G25	G26	G28	G30	G40	G41	G42
G43	G44	G49	G53	G65	G73	G76
G80	G81	G82	G83	G85	G86	G87
G88	G89	G82	G94	G95	G96	G97

NOTE
1 In Small-Holl Pecking Drilling Cycle(G83), backward movement is prohibited.
2 In forward movement of Boring Cycle(G88), the sequence of actions at bottom of holl is shown as follows (dwell -> stop of spindle motor -> hold state). But in backward movement, that is (rotation of spindle -> hold state -> dwell after restart).

- M-code

If there is M-code in the same group is commanded in previous blocks, modal information of the M-code, commanded at the last in previous blocks, is output.
If no M-code is commanded in previous blocks, the M-code set to the first parameter in the same M-code group is output.
If M-code is not set to group M-code in parameter, the same M-code is output in backward movement.
If the parameter RVN(6400#5) is set to "1", the backward movement is prohibited when the M-code, which is not set to group M-code, is commanded in backward movement.

NOTE
When setting the parameter RVN, backward movement prohibition is enabled except the M-code which was set in the grouping but backward movement can be enabled for the following M-code exceptionally.
1. Subprogram Call by M98/M99.
2. Subprogram Call using an M code
3. Macro Call using an M code
4. Waiting M code
5. M0

Example) Output of M-codes that are set to groups by parameters in backward movement
Setting of parameters:
No.6400#2=1, #3=0 (5 M-codes/group and 16 groups)
No.6411=100
No.6412=101
No.6413=102
No.6414=103
No.6415=104
} Group A
No.6416=200
No.6417=201
No.6418=202
No.6419=203
No.6420=204
} Group B

Program O10 is executed in forward movement from N1 to N15 and backward movement is executed from N15. In backward movement, the output of M-codes is shown as next table.

	Forward movement	Backward movement
O10;		
N1G4X1.;		
N2M101;	M101	M100 (*1)
N3G4X1.;		
N4M204;	M204	M200 (*1)
N5G4X1.;		
N6M104;	M104	M101 (*2)
N7G4X1.;		

	Forward movement	Backward movement
N8M300;	M300	M300
N9G4X1.;		
N10M200;	M200	M204 (*1)
N11G4X1.;		
N12M0;	M0	M0 (*3)
N13G4X1.;		
N14M102;	M102	M104 (*2)
N15G4X1.;	Backward movement starts from this block	
M2;		

*1 No M-code in the same group is commanded before this block, so the M-code, which is set in the 1st. parameter of the same group, is output.

*2 M-code in the same group is commanded before this block, so the M-code, which is commanded at the last before this block, is output.

*3 M-code is not set to group M-code, so the same M-code is output.

- S and T-code

A modal value of the previous block is output.
When movement command and S-code or T-code is commanded in the same block, the timing of the output of the S-code and T-code is different. Because, the timing where S-code and T-code are output at the forward movement is different from that at the backward movement. By setting bit 7 (STO) of parameter No.6401 to "1", the timing of the output of S and T code at the forward movement is the same as the one at the backward movement.

Example)

T-code output timing at the backward movement

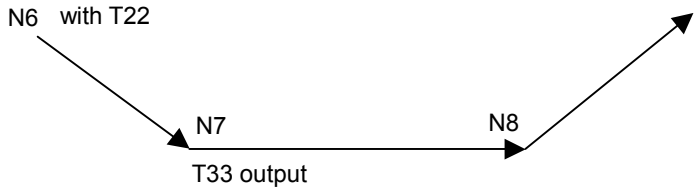
T-code is output as follows when the program proceeds backward after the forward movement to N8 block.

	Forward movement	Backward movement	
		Parameter STO=0	Parameter STO=1
O1000;			
N1G98G00X0Z0;			Default T output
N2G00X-10.T11;	T11 output	Default T output	
N3G00X100.;			T11 output
N4G00X10.Z20.T22;	T22 output	T11 output	No T code output
N5G00X30.Z30.;			
N6G00X-10.Z-20.;			T22 output
N7G00X50.Z40.T33;	T33 output	T22 output	T33 output
N8G04X5.;		(Backward start)	(Backward start)
M30;			

The “Default T” means a T-code status at N1 block in forward movement. If the status is T0, “T0” signal is output as “Default T” in the backward movement.

The timing of T-code output of N7 and N8 in O1000 shown in the example above is as follows.

Forward movement :



Backward movement (When parameter STO is set to “0”) :

Backward movement (When parameter STO is set to “1”) :

- Direction change prohibition

The direction change prohibition is a state not changing the direction where the program is executed.
In the state, even if the rotating direction of a manual handle is reversed, the reversed rotation is ignored. A manual handle must be rotate in the same direction as present direction for removing this state.
The direction change prohibition can be confirmed by output signal MNCHG<Fn091.1>.

It becomes the change prohibition state under the following condition.

- While the block with the code waiting for FIN is executing
- After a block has done and until the next block begins to operate
- During thread cutting
- Modal G code of G68.1, G68 and G51.2
- The block with the axis that ends movement earlier in the block with G02 or nonlinear type position (G00) etc.

- Backward movement prohibition

The backward movement prohibition is a state that the program cannot be executed from a certain block backward. In this state, the negative rotation of a manual handle is ignored, and the only positive rotation is effective. The program must be executed forward by rotating a manual handle in the positive direction for removing this state.
The backward movement prohibition can be confirmed by output signal MRVSP<Fn091.2>.
If the following blocks are executed in backward movement, backward movement is prohibited.

- Program number block of main program (except subprogram and macro program)
- Over the maximum number of the blocks for reverse movement
- The block including backward movement prohibition G-code (which is not described in the paragraph "G-code")
- The block which is executed while in modal including backward movement prohibition G-code (which is not described in the paragraph "G-code")

- Status display

In manual handle retrace, the status of manual handle retrace is displayed on clock display of CNC state display line. This status display is displayed during the execution of manual handle retrace. The clock is displayed usually.

When the all condition is filled, "M.H.RTR." is displayed on clock display of CNC state display line. This status is displayed by the color of color number 3 (INPUT KEY, O/N NO. and STATUS are the same color). The screen display is as shown in Fig.3.12(a). When the following conditions are not full, the clock is displayed.

- When bit 2 (CHS) of parameter No.6401 is set to "0":
  1) Software option of handle manual retrace is enabled.
  2) Status display disable/enable bit 6 (HST) of parameter No.6401 is set to "1".
  3) Check mode confirmation signal MMMOD<Fn091.3> is set to "1".

- When bit 2 (CHS) of parameter No.6401 is set to "1":
  1) Software option of handle manual retrace is enabled.
  2) Status display disable/enable bit 6 (HST) of parameter No.6401 is set to "1".
  3) Cycle start signal STL<Fn000.5> is set to "1".
  4) Check mode signal MMOD<Gn067.2> is set to "1".
  5) Manual handle check signal MCHK<Gn067.3> is set to "1".

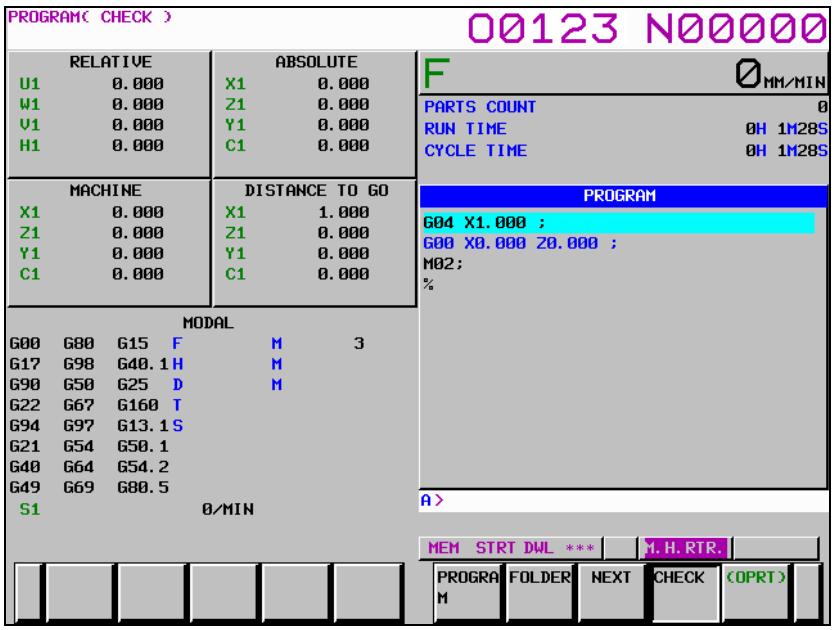


Fig. 3.12 (a) "M.H.RTR." status display

Besides, when reverse movement prohibition signal MRVSP<Fn091.2> is set to "1",
the "NO RVRS." is displayed. This status is displayed by blinking/reversing in the color of color number 1 (ALARM is the same color). The screen display is as shown in Fig.3.12(b). When reverse movement prohibition signal MRVSP<Fn091.2> is set to "0", the " M.H.RTR." is displayed again.

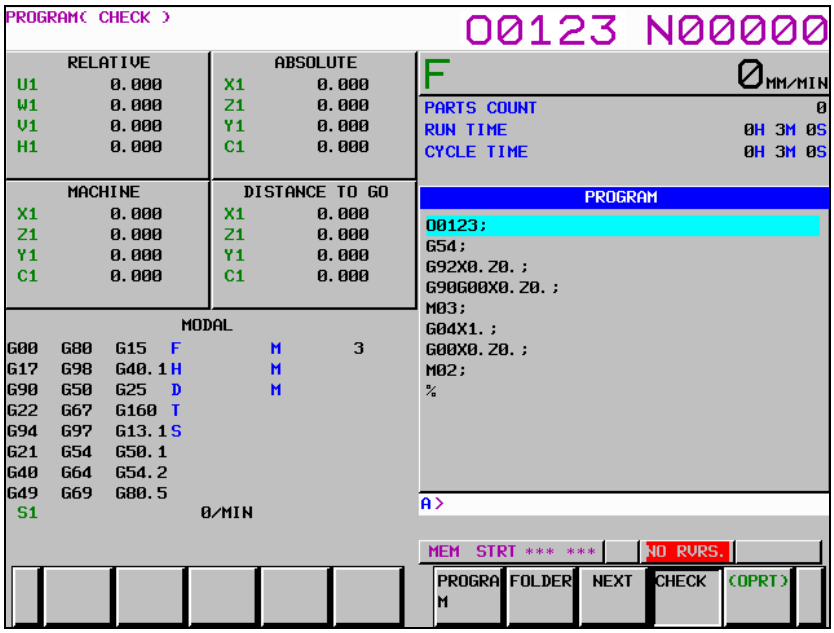


Fig. 3.12 (b) " NO RVRS." status display

Besides, when direction change prohibition signal MNCHG<F0091.1> is set to "1" and the direction of program's execution is changed by manual handle, this status display changes from "M.H.RTR." to "NO.CHAG.".

This status is displayed by blinking/reversing in the color of color number 3 (INPUT KEY, O/N NO. and STATUS are the same color) . The screen display is shown as Fig.3.12(c). When the program is executed in the direction as the same as before by manual handle or direction change prohibition signal MNCHG<Fn091.1> is set to "0", the " M.H.RTR." is displayed again.
Moreover, when parameter FWD(No.6400#1) is set to "1" and the program is executed to change direction by manual handle, this status display changes from "M.H.RTR." to "NO.CHAG.".

NOTE
When the improvement of direction change movement in auxiliary function output block is enabled, the state of direction change prohibition signal MNCHG<Fn091.1> is set to "1" and direction change is possible. Therefore please note that the status display of the direction change prohibition is displayed though it is possible to direction change.

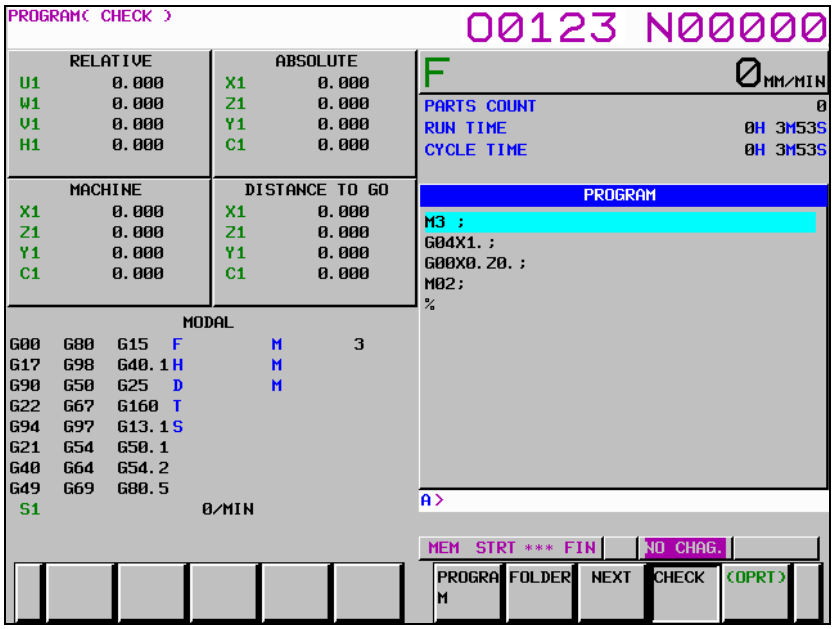


Fig. 3.12 (c) "NO.CHAG." status display

Limitation

- Movement in automatic operation by DNC operation mode(RMT)

In the automatic operation by DNC operation mode(RMT), the backward movement is prohibited though the forward movement is enable.

- Movement in subprogram operation by external subprogram call

In M198 or M-code for subprogram operation by external subprogram call (parameter No.6030), the backward movement is prohibited though the forward movement is enable.

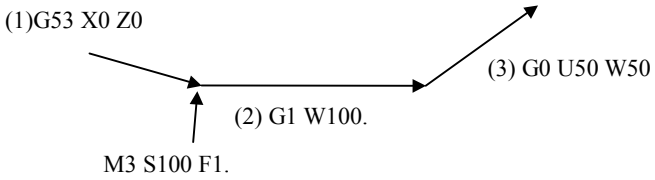
- Movement command and M,S,T-code

When M,S,T-codes and movement commands are in the same block, the timing outputting codes changes between in forward movement and backward movement. Therefore, M, S, T-codes should be commanded in backward movement after confirming that "DEN" signal is set to "1".

Example of executing the following programs with lathe system

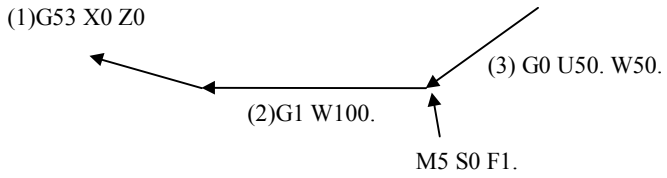
```
O0001 ;
M5 S0 F0 ;
G53 X0 Z0 ;..... (1)
G1 W100 M3 S100 F1. ; ..... (2)
G0 U50. W50. ;..... (3)
M2 ;
```

[Forward movement]



The block of (2) moves with M3 S100 F1.

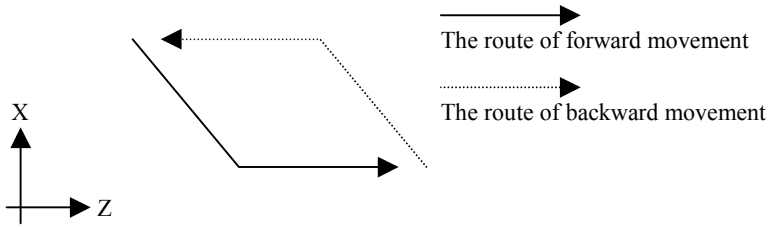
[Backward movement]



The block of (2) moves with M5 S0 F1.

- Non linear interpolation type positioning

In the non-linear interpolation type positioning, the route is different between forward movement and backward movement.



Please use the interpolation type positioning to ward off danger. (Set the bit 1 (LRP) of parameter No.1401 to "1")

- Threading in forward movement

Threading (G32,G76,G84,G88,G92) is always executed at 100% override speed. That is to say, a pulse generated by a manual handle is ignored in executing a threading block. In thread cutting cycle, the pulse is invalid at the time actually cutting thread, but the one are valid in the other movements.

- Macro

In macro statement, the setting, operation, and so on of the macro variable is executed in only first forward movement. That is to say, the setting, operation, and so on of the macro variable is never executed in the block executing them once.

- Axis control by PMC

The movement of axis control by PMC cannot be controlled by this function.

- Multiple path simultaneous check in the multi-path system

When using the manual handle retrace function at the same time in multiple paths, the timing of block operation may slightly differ between these paths due to the repetition of forward and backward motion or differences in the rotation speed of the manual handle. To synchronize block operation between the paths, use the wait M code.

- Check of path unit by multi-path system

In the multi-path system, the program check of an arbitrary path is possible. In the path not to check, please select the mode excluding the MEM mode. Even if bit 4 (HMP) of parameter No.6400 is set to "1", it is possible to execute the check of forward movement, direction change and backward movement.
If waiting M code exist in the program, please set No-wait signal of two path NOWT <Gn063.1>, or No-wait signal of three path NMWT<Gn063.7> to "1".

- Modal display

In the backward movement with manual handle, the modal display is updated according to the operation condition of the program.

- Modal information

In the backward movement with manual handle, the state of modal information is updated according to the operation condition of the program.

- Change in operation mode

When you change to EDIT mode during the checking mode, the backward movement and the re-forward movement cannot be executed in the blocks which have been already executed.

- ON/OFF of Manual Handle Retrace mode

When check mode signal MMOD<Gn067.2> is set to "0" and handle available signal in checking mode MCHK<Gn067.3> is set to "0", the check mode might not be turned off at once.
Basically, in the middle of block, the check mode doesn't switch from ON to OFF or from OFF to ON. After the block is ended, the check mode switches from ON to OFF or from OFF to ON.

- AI contour control

If the checking mode signal MMOD<Gn067.2> is "1", AI contour control mode(G05.1Q1) is canceled. If AI contour control is canceled, forward movement and backward movement can be executed.
Furthermore, if the signal MMOD<Gn067.2> is set to "1" in AI contour control mode, the checking mode is not effective until AI contour control mode is invalid(G05.1Q0).

- Multi Spindle

During the backward movement, both TYPE-A and TYPE-B multi spindle control may not be operated exactly.

- Path Table Operation

In path table operation, the backward movement is prohibited. Furthermore, in forward movement, regardless of manual handle, the motion is performed as the command of path table operation.

- Execution of measurement G-code with the speed of override 100%

When bit 6 (MGO) of parameter No.6400 is set to "1", a handle pulse is invalid and it is always executed at 100% override. When bit 6 (MGO) of parameter No.6400 is set to "0", this function is invalid and a handle pulse is valid.
In the multi-path system, this function is not effective in the execution of another path and handle pulse is valid in another path. The measurement G code to which this function is effective is as follows.

1) G31 for skip
2) G31, G31P1, G31P2, G31P3, G31P4, G04, G04Q1, G04Q2, G04Q3 and G04Q4 for multistage skip
3) G31P99 and G31P98 for torque limit skip

When the measurement by G36 and G37 (G37.1 and G37.2 when bit 3(G36) of parameter No.3405 is set to "1") of automatic tool compensation is executed, a handle pulse is invalid and it is always executed at 100% override regardless of setting bit 6 (MGO) of parameter No.6400. At the rapid traverse before measurement, handle pulse is valid.

When bit 7 (SKF) of parameter No.6200 is set to "0" and bit 2 (SFN) of parameter No.6207 is set to "0", handle pulse at G31 is invalid and it is always executed at 100% override regardless of setting bit 6 (MGO) of parameter No.6400.

When bit 7 (MG4) of parameter No.6400 is set to "1" and the software option of multistage skip is enabled and the setting of parameter from No.6202 to No.6206 is enabled, the backward movement prohibition is enabled in G04 block for multistage skip.
The G code to which this function is effective is as follows.

1) G04, G04Q1, G04Q2, G04Q03 and G04Q4 for multistage skip

- Relation to another function

This function cannot coexist with the following functions

- Retrace
- Learning control for parts cutting
- Preview repetitive control
- Learning control

3.13 AUXILIARY FUNCTION OUTPUT BLOCK REVERSE MOVEMENT FOR MANUAL HANDLE RETRACE

Overview

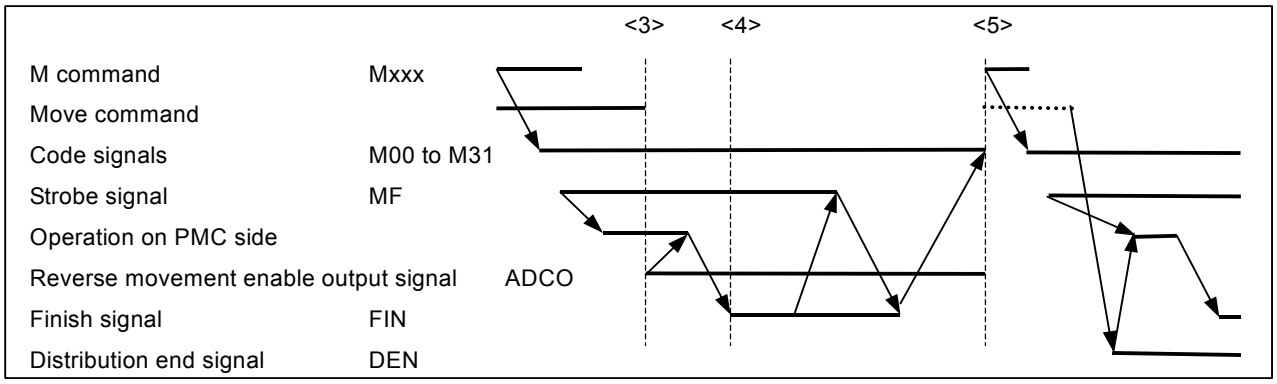
This function enables reverse movement during manual handle retrace even if a move command and an auxiliary function (M/S/T/B code) are specified in the same block.
This is an optional function.

Explanation

Reverse movement is enabled for an auxiliary function output block if all the conditions and sequences described below are met.

<1> The function is effective when bit 0 (ADC) of parameter No. 6401 (auxiliary function output block reverse movement enable parameter) is set to "1".
<2> The function is effective when a move command and an auxiliary function (M/S/T/B code) are specified in the same block.
<3> If the rotation of the manual pulse generator is reversed when all the conditions mentioned in <1> and <2> are met, the NC sets auxiliary function output block reverse movement enable output signal ADCO<Fn091.5> to "1".
<4> When auxiliary function output block reverse movement enable output signal ADCO<Fn091.5> is set to "1", the PMC performs the corresponding operation with a ladder language program. After that, set finish signal FIN<Gn004.3> to "1". This enables the NC to move the axis in reverse direction (move it backward or move it forward again).
<5> When finish signal FIN<Gn004.3> is set to "1", the NC sets ADCO<Fn091.5> to "0" and starts to move the axis in reverse direction (move it backward or move it forward again).
As for parameter-grouped M codes or S or T codes, those of the previous one block are output. As for M codes that are not grouped by parameter or B codes, the same codes as for the forward movement are also output for the reverse movement.

The figure below presents a timing chart depicting this process. The chart shows an example of M codes.
This example describes how an auxiliary function is executed after waiting for the move command to complete (distribution end signal DEN) following the reverse movement.
It is also possible to execute an auxiliary function without waiting for the move command to complete after the reverse movement.



Notes

NOTE

1 When a single auxiliary function is specified individually, carry out the regular auxiliary function completion sequence. Reverse movement becomes possible after moving to the next (or previous) block.
2 If a move command and an auxiliary function are specified in the same block and if no travel distance remains after movement, carry out the regular auxiliary function completion sequence. Reverse movement becomes possible after moving to the next (or previous) block.
3 To enable reverse movement using this function when an M code other than grouped M codes and a move command are specified in the same block, set bit 5 (RVN) of parameter No. 6400 to "0".
4 This function is not supported for M02 or M30.
5 If an auxiliary function is specified in a block having no move command, such as G70, carry out the regular auxiliary function completion sequence. Reverse movement becomes possible after moving to the next (or previous) block.
6 This function is optional.

4 AUTOMATIC OPERATION

Programmed operation of a CNC machine tool is referred to as automatic operation.
This chapter explains the following types of automatic operation:

4.1 MEMORY OPERATION....1111
Operation by executing a program registered in CNC memory
4.2 MDI OPERATION....1114
Operation by executing a program entered from the MDI panel
4.3 DNC OPERATION....1118
Function for executing a program while reading the program from an input device connected to the reader/puncher interface or a memory card.
4.4 SCHEDULE OPERATION....1121
Function for executing a program while reading the program from an input device connected to the reader/puncher interface or a memory card according to a schedule.
4.5 EXTERNAL SUBPROGRAM CALL (M198)....1124
Function for calling and executing subprograms (files) registered in an external input/output device during memory operation
4.6 MANUAL HANDLE INTERRUPTION....1127
Function for performing manual feed during movement executed by automatic operation
4.7 MIRROR IMAGE....1134
Function for enabling mirror-image movement along an axis during automatic operation
4.8 PROGRAM RESTART....1136
Restarting a program for automatic operation from an intermediate point
4.9 TOOL RETRACT AND RECOVER....1149
Function for retracting the tool to a desired position
4.10 RETRACE....1161
Function for executing a program in the reverse direction.
4.11 ACTIVE BLOCK CANCEL....1172
Function for stopping operation and canceling the remaining movement by inputting a signal during automatic operation.

4.1 MEMORY OPERATION

Programs are registered in memory in advance. When one of these programs is selected and the cycle start switch on the machine operator's panel is pressed, automatic operation starts, and the cycle start LED goes on.
When the feed hold switch on the machine operator's panel is pressed during automatic operation, automatic operation is stopped temporarily. When the cycle start switch is pressed again, automatic operation is restarted.
When the

key on the MDI panel is pressed, automatic operation terminates and the reset state is entered.
For the multi-path control, the programs for the multiple paths can be executed simultaneously so the multiple paths can operate independently at the same time.
The following procedure is given as an example. For actual operation, refer to the manual supplied by the machine tool builder.

Memory operation

Procedure

1 Press the MEMORY mode selection switch.
2 Select a program from the registered programs. To do this, follow the steps below.
   2-1 Press

key to display the program screen.
   2-2 Press address

key.
   2-3 Enter a program number using the numeric keys.
   2-4 Press the [O SRH] soft key.
3 For the multi-path control, select the path to be operated with the path selection switch on the machine operator's panel.
4 Press the cycle start switch on the machine operator's panel.
   Automatic operation starts, and the cycle start LED goes on. When automatic operation terminates, the cycle start LED goes off.
5 To stop or cancel memory operation midway through, follow the steps below.
   a. Stopping memory operation
      Press the feed hold switch on the machine operator's panel. The feed hold LED goes on and the cycle start LED goes off. The machine responds as follows:
      (i) When the machine was moving, feed operation decelerates and stops.
      (ii) When dwell was being performed, dwell is stopped.
      (iii) When M, S, or T was being executed, the operation is stopped after M, S, or T is finished.
      When the cycle start switch on the machine operator's panel is pressed while the feed hold LED is on, machine operation restarts.
   b. Terminating memory operation

Press the RESET key on the MDI panel.

Automatic operation is terminated and the reset state is entered.
When a reset is applied during movement, movement decelerates then stops.

Explanation

- Memory operation

After memory operation is started, the following are executed:

(1) A one-block command is read from the specified program.
(2) The block command is decoded.
(3) The command execution is started.
(4) The command in the next block is read.
(5) Buffering is executed. That is, the command is decoded to allow immediate execution.
(6) Immediately after the preceding block is executed, execution of the next block can be started. This is because buffering has been executed.
(7) Hereafter, memory operation can be executed by repeating the steps (4) to (6).

- Stopping and terminating memory operation

Memory operation can be stopped using one of two methods: Specify a stop command, or press a key on the machine operator's panel.

- The stop commands include M00 (program stop), M01 (optional stop), and M02 and M30 (program end).
- There are two keys to stop memory operation: The feed hold key and reset key.

- Program stop (M00)

Memory operation is stopped after a block containing M00 is executed. When the program is stopped, all existing modal information remains unchanged as in single block operation. The memory operation can be restarted by pressing the cycle start button. Operation may vary depending on the machine tool builder. Refer to the manual supplied by the machine tool builder.

- Optional stop (M01)

Similarly to M00, memory operation is stopped after a block containing M01 is executed. This code is only effective when the Optional Stop switch on the machine operator's panel is set to ON. Operation may vary depending on the machine tool builder. Refer to the manual supplied by the machine tool builder.

- Program end (M02, M30)

When M02 or M30 (specified at the end of the main program) is read, memory operation is terminated and the reset state is entered.
In some machines, M30 returns control to the top of the program. For details, refer to the manual supplied by the machine tool builder.

- Feed hold

When Feed Hold button on the operator's panel is pressed during memory operation, the tool decelerates to a stop at a time.

- Reset

Automatic operation can be stopped and the system can be made to the reset state by using

key on the MDI panel or external reset signal. When reset operation is applied to the system during a tool moving status, the motion is slowed down then stops.

- Optional block skip

When the optional block skip switch on the machine operator's panel is turned on, blocks containing a slash (/) are ignored.

- Cycle start for the multi-path control

For the multi-path control, a cycle start switch is provided for each path. This allows the operator to activate a single paths to operate them at the same time in memory operation or MDI operation. In general, select the path to be operated with the path selection switch on the machine operator's panel and then press the cycle start button to activate the selected path. (The procedure may vary with the machine tool builder. Refer to the appropriate manual issued by the machine tool builder.)

4.2 MDI OPERATION

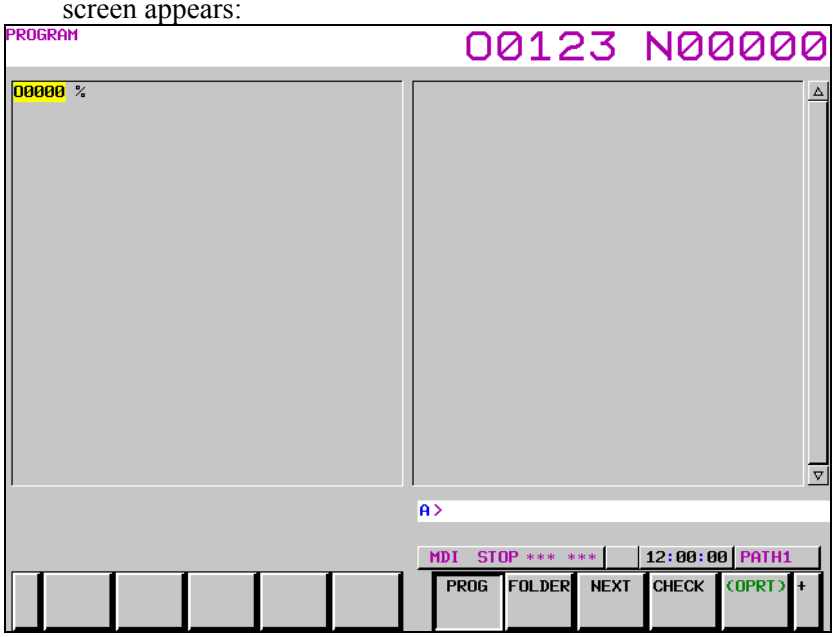
In the MDI mode, a program consisting of up to 511 characters can be created in the same format as normal programs and executed from the MDI panel.
MDI operation is used for simple test operations.
The following procedure is given as an example. For actual operation, refer to the manual supplied by the machine tool builder.

MDI Operation

Procedure

1 Select the MDI mode.
For the multi-path control, select the path for which a program is to be created with the path selection switch. Create a separate program for each path.

2 Press the PROG key to select the program screen. The following screen appears:



MDI program screen

At this time, program number “O0000” is inserted automatically.

3 Prepare a program to be executed by an operation similar to normal program editing. M99 specified in the last block can return control to the beginning of the program after operation ends. Word insertion, modification, deletion, word search, address search, and program search are available for programs created in the MDI mode.

4 To entirely erase a program created in MDI mode, use one of the following methods:

a. Enter address [P 0] key, then press the [DELETE] key.

b. Alternatively, press the [RESET] key. In this case, set parameter MCL (No. 3203#7) to 1 in advance.

5 To execute a program, set the cursor on the head of the program. Push Cycle Start button on the operator's panel. By this action, the prepared program will start.
(For the multi-path control, select the path to be operated with the path selection switch on the machine operator's panel beforehand.) When the program end (M02, M30) or ER(%) is executed, the prepared program will be automatically erased and the operation will end.
By command of M99, control returns to the head of the prepared program.

6 To stop or terminate MDI operation in midway through, follow the steps below.

a. Stopping MDI operation
Press the feed hold switch on the machine operator's panel. The feed hold LED goes on and the cycle start LED goes off. The machine responds as follows:
(i) When the machine was moving, feed operation decelerates and stops.
(ii) When dwell was being performed, dwell is stopped.
(iii) When M, S, or T was being executed, the operation is stopped after M, S, or T is finished.
When the cycle start switch on the machine operator's panel is pressed, machine operation restarts.

b. Terminating MDI operation
Press the [RESET] key.
Automatic operation is terminated and the reset state is entered.
When a reset is applied during movement, movement decelerates then stops.

Explanation

The previous explanation of how to execute and stop memory operation also applies to MDI operation, except that in MDI operation, M30 does not return control to the beginning of the program (M99 performs this function).

- Erasing the program

Programs prepared in the MDI mode will be erased in the following cases:

- In MDI operation, if M02, M30 or ER(%) is executed.
- When bit 6 (MER) of parameter No. 3203 is set to 1, and the last block of the program is executed in single block operation

NOTE
In the two cases above, program erasure can be prevented by setting bit 6 (MKP) of parameter No. 3204 to 1.

- In MEM mode, if memory operation is performed.
- In EDIT mode, if any editing is performed.
- When the and keys are pressed.
- Upon reset when bit 7 (MCL) of parameter No.3203 is set to 1

NOTE
Upon reset when the parameter MCL = 0, the cursor moves to the end of the program.

- Restart

If a program is not executed even once after the program is input, the program is executed from the beginning, regardless of where the cursor is placed. However, a program is executed starting at the beginning of the block where the cursor is place, if the program is stopped for a reason such as single block operation after restart of an MDI operation then is restarted after an editing operation.

⚠ **CAUTION**
When an MDI program is restarted, the program is executed starting at the beginning of the block where the cursor is placed, regardless of the cursor position in the block.
(Example)
When the cursor is placed on G90
:
G91 X100.0 G90Y200.0 Z300.0 ;
:
The program is executed starting at the beginning (namely, G91) of this block. So, the tool moves by 100.0 along the X-axis in the incremental programming, and moves to 200.0 and 300.0 along the Y-axis and Z-axis, respectively, in the absolute programming.

- Editing a program during MDI operation

A program can be edited during MDI operation. By setting bit 5 (MIE) of parameter No. 3203 to 1, editing can be disabled. However, even when bit 5 (MIE) of parameter No. 3203 is set to 1, editing can be enabled by resetting the operation.

- Absolute/incremental command

When bit 4 (MAB) of parameter No. 3401 is set to 1, the absolute/incremental programming of MDI operation does not depend on G90/G91. In this case, the incremental programming is set when bit 5 (ABS) of parameter No. 3401 is set to 0, and the absolute programming is set when bit 5 (ABS) of parameter No. 3401 is set to 1.

<table>
<tr><th>Parameter MAB (No.3401#4)=0</th><th colspan="2">Parameter MAB (No.3401#4)=1</th></tr>
<tr><td rowspan="2">Absolute mode operation with G90 command, and incremental programming operation with G91 command</td><td>Parameter ABS (No.3401#5)=0</td><td>Parameter ABS (No.3401#5)=1</td></tr>
<tr><td>Incremental mode operation at all times, independent of G90/G91 command</td><td>Absolute mode operation at all times, independent of G90/G91 command</td></tr>
</table>

NOTE
When G code system A is used on a lathe system, the parameters MAB and ABS are invalid.

Limitation

- Program registration

Programs created in MDI mode cannot be registered.

- Number of characters in a program

A created program can consist of up to 511 characters including "O0000" automatically inserted.

- Subprogram nesting

The subprogram call command (M98) can be described in a program created in MDI mode. That is, programs that are registered in memory through MDI operation can be called and executed. The level of subprogram call nesting is the same as in MEM operation.

- Macro call

When the custom macro function is enabled, a macro program can be created and executed even in the MDI mode. Moreover, a macro program can be called for execution.

NOTE
The GOTO statement, WHILE statement, and DO statement cannot be executed in a program created in the MDI mode. An alarm PS0377 is issued.
When a program including those statements is to be executed, register the program in the program memory then call the program for execution.

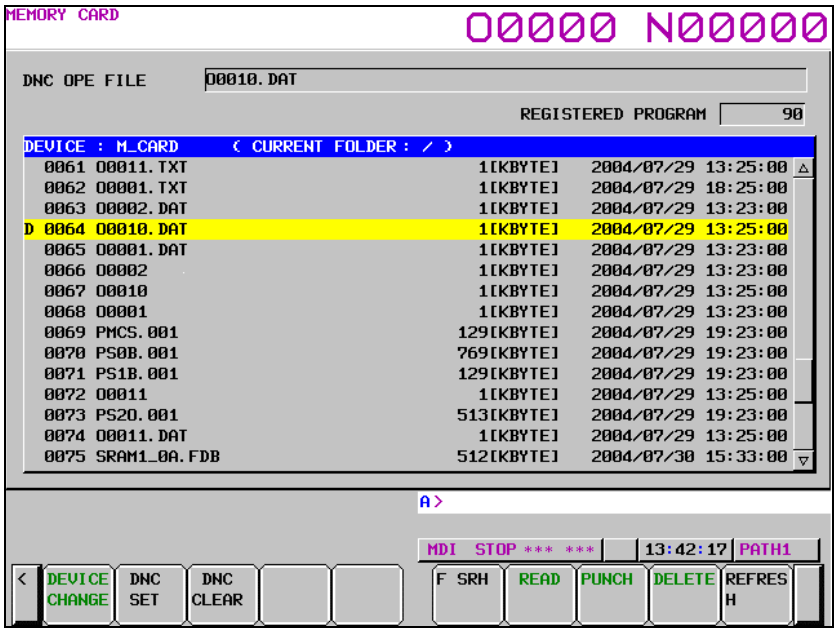
4.3 DNC OPERATION

By activating automatic operation during the DNC operation mode (RMT), it is possible to perform machining (DNC operation) while a program is being read in via reader/puncher interface.
To use the DNC operation function, it is necessary to set the parameters related to the reader/punch interface in advance.
The procedure described below is just an example. For actual operation, refer to the relevant manual of the machine tool builder.

DNC operation

Procedure

1 Press the REMOTE switch on the machine operator's panel to enter the RMT mode.
2 Select the program to be executed.
- Selecting a DNC operation file
  On the memory card (or floppy cassette) list screen, move the cursor to the file to be subjected to DNC operation and press "DNC SET" to select the file to be subjected to DNC operation. (The selected file is marked with "D".)
- Releasing a DNC operation file
  On the memory card (or floppy cassette) list screen, move the cursor to the file for which DNC operation is released and press "DNC CLEAR" to release the DNC operation file. (The "D" mark for the file is removed.)



3 Press the cycle start switch to execute the selected file. For details on the REMOTE switch, refer to the manual provided by the machine tool builder.

4 During DNC operation, executed programs are listed on the program check screen and program screen.

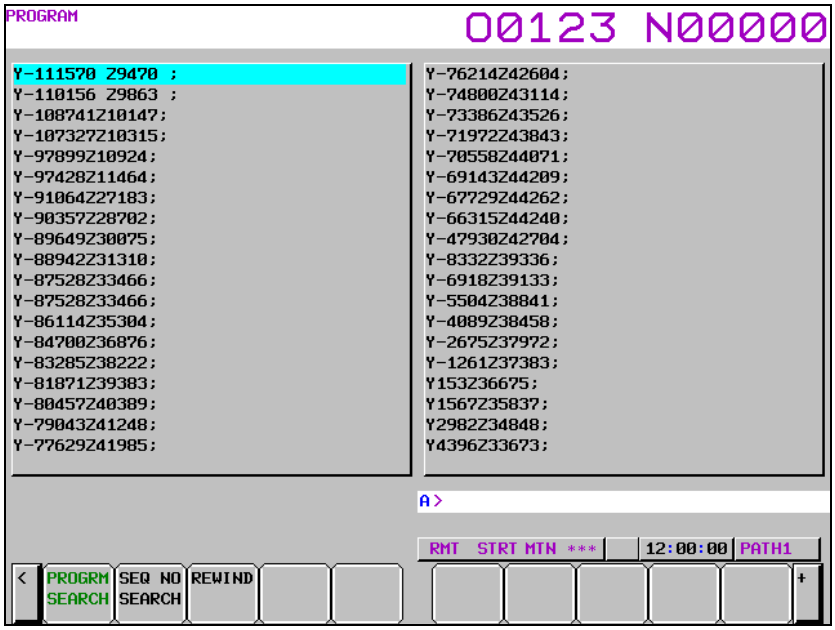


Fig. 4.3 (a) PROGRAM screen

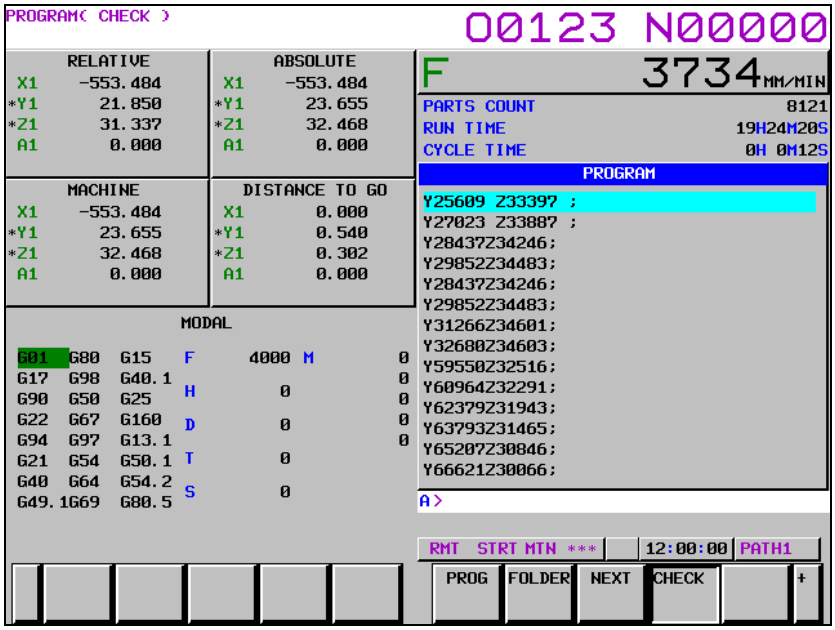


Fig. 4.3 (b) PROGRAM CHECK screen

NOTE
1 Before selecting a DNC operation file, be sure to release all schedule data. DNC operation and schedule operation cannot be specified at the same time.
2 A DNC operation file cannot be released during DNC operation.
3 To switch between devices when DNC settings are made, release the settings and then make the settings again.

Explanation

During DNC operation, subprograms and macro programs stored in memory can be called.

Limitation

- M198 (command for calling a program from within an external input/output unit)

In DNC operation, M198 cannot be executed. If M198 is executed, alarm PS0210 is issued.

- Custom macro

In DNC operation, custom macros can be specified, but no repeat instruction and branch instruction can be programmed. If a repeat instruction or branch instruction is executed, alarm PS0123 is issued.

- M99

For returning from a subprogram or macro program to the calling program during DNC operation, the specification of a return command (M99P...) with a sequence number specified is not allowed.

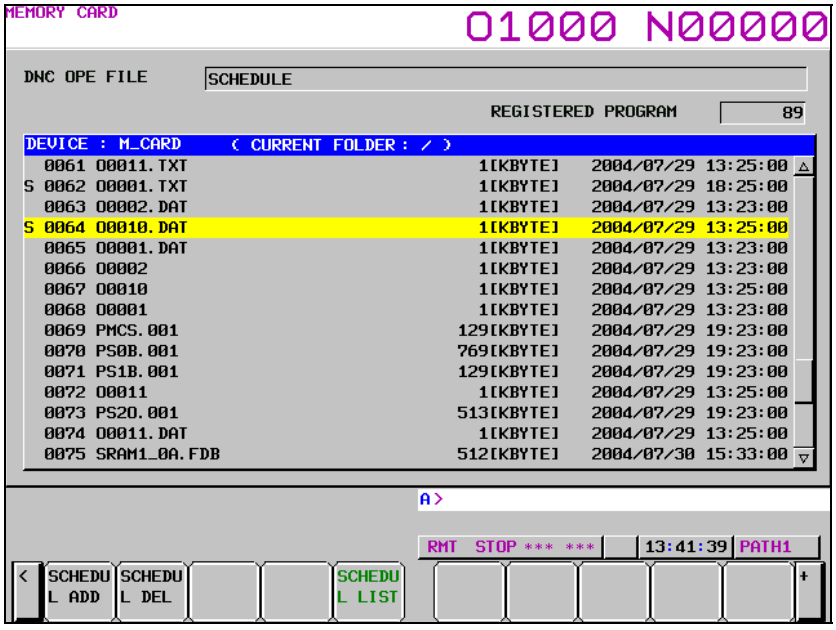
4.4 SCHEDULE OPERATION

To perform schedule operation, select files (programs) registered in a memory card and specify the sequence of execution and the repetition count of each program.

Schedule operation

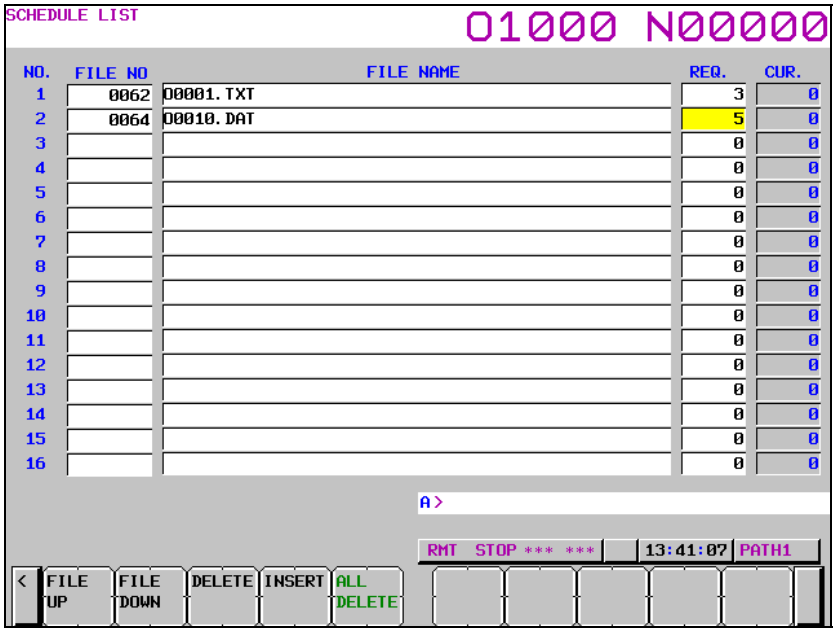
Procedure

1 Press the REMOTE switch on the machine operator's panel to enter the RMT mode.
2 Select the program to be subjected to schedule operation.
- Setting a schedule
  Move the cursor to the file to be subjected to schedule operation and press the soft key [SCHEDUL ADD]. The files registered as schedule data are marked with "S" on the left sides of their file names. The files are subjected to schedule operation in the order in which the files are selected. When settings are made this way, the repetition count of execution of each file is set to 1. To change the repetition count or execution order or check the schedule, press the soft key [SCHEDUL LIST] to display the schedule list screen.



[SCHEDUL ADD]
Adds the file at the cursor position as schedule data.
[SCHEDUL DEL]
Deletes the file at the cursor position from schedule data when the file is registered as schedule data.
[SCHEDUL LIST]
Lists the settings of schedule data to edit the repetition count or the like (see the next page for details).

- Editing a schedule
  To edit schedule data, press the soft key [SCHEDUL LIST] to display the schedule list screen, on which schedule data can be edited.



[FILE UP]
Moves the file at the cursor position up one line and moves the replaced file down one line.
[FILE DOWN]
Moves the file at the cursor position down one line and moves the replaced file up one line.
[DELETE]
Deletes the file at the cursor position and moves the files below the cursor up one line.
[INSERT]
Moves the files below the cursor down one line.
[ALL DELETE]
Deletes all records.

3 Press the cycle start switch to execute the selected files. For details on the REMOTE switch, refer to the manual provided by the machine tool builder.

NOTE
1 Before setting schedule operation, release DNC operation files in the MDI mode. DNC operation and schedule operation cannot be specified at the same time.
2 Before starting schedule operation, confirm that schedule data is set correctly on the schedule list screen.
3 Schedule data cannot be changed or edited during schedule operation. Before changing schedule data, make a reset to stop operation.

Restrictions

- Repetition count

The maximum repetition count during schedule operation is 9999. When a negative value is specified, an endless loop (LOOP display) is assumed. The file for which 0 is set is skipped and processing proceeds to the next file.

- Number of registered files

The maximum number of programs that can be registered as schedule setting data is 20.

- Selectable files

The files to be selected as schedule setting data must be registered in the same directory. (Files in the different directories cannot be selected.)

- M code

Even if a code other than M02 and M30 in the execution program is executed, the current count on the schedule execution status screen is not increased.

- Floppy disk directory display during execution of a file

During schedule operation, directories in a floppy disk cannot be displayed in a background edit.

- Intervention during automatic operation

Intervention in schedule operation cannot be performed during automatic operation.

- During multipath control

The scheduling function cannot be used by multiple paths at the same time.

4.5 EXTERNAL SUBPROGRAM CALL (M198)

During memory operation, you can call and execute a subprogram registered in an external device (such as a Memory Card, Handy File, or Data Server) connected to the CNC.

Format

M198 Pxxxxxxxx Lyyyyyyyy ;

↑ ↑

Pxxxxxxxx : Program number (or file number)
Lyyyyyyyy : Number of repetitive calls
When address L is omitted, the number of repetitive calls is assumed to be 1.

FS16 compatible command format
(The following command format is valid only when a 4-digit program number is used.)

M198 P<u>xxxx</u> <u>yyyy</u> ;

↑ ↑

<u>xxxx</u> : Number of repetitive calls
<u>yyyy</u> : Program number (or file number)
When the number of repetitive calls is omitted, it is assumed to be 1.

Explanation

M code M198 specifies an external subprogram call. You can also call an external subprogram using an M code set in parameter No. 6030. (When an M code other than M198 is set as an M code for calling an external subprogram, M198 is executed as a normal M code.)

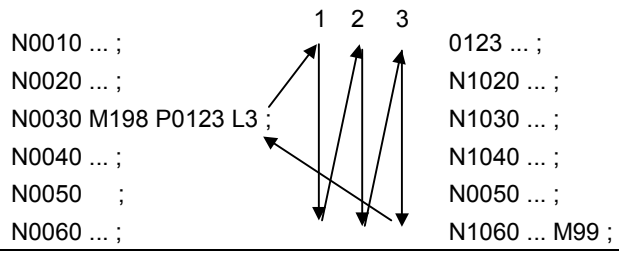
Specify a program number (file number) registered in an external device at address P. If the specified program number (file number) is not registered in the connected external device, an alarm (PS1079) is issued.

Example)

M198 P0123 L3;

This command specifies that the subprogram having external subprogram number O0123 is to be called three times repeatedly.

The subprogram is called from the main program and executed as follows:



- Program number call

You can also specify a subprogram call with its program number instead of the file number by the setting of bit 2 (SBP) of parameter No. 3404.

NOTE

1 An external subprogram call can be specified only during program operation in the MEM mode and cannot be specified in the MDI mode.

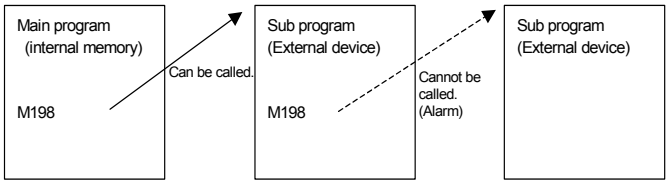
2 An external subprogram call is available for the following external devices:

External device name	Program number call	File number call
Handy File	Available	Available
FLOPPY CASSTTE	Available	Available
Memory Card	Available	Unavailable
Data Server	Available	Unavailable

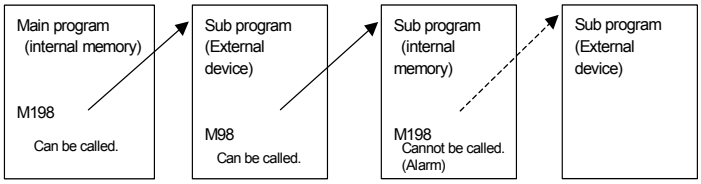
3 To perform a subprogram call using a Memory Card as the external device, set bit 7 (MNC) of parameter No. 138 to 1 and I/O channel (parameter No. 0020) to 4.
A program number call is always enabled regardless of the setting of bit 2 (SBP) of parameter No. 3404.

NOTE

4 An external device subprogram call cannot be performed from a subprogram called using another external device subprogram call. (An alarm (PS1080) is issued.)



5 A subprogram registered in internal memory can be called from a subprogram called using an external device subprogram call. From the called subprogram in internal memory, another external device subprogram call cannot be performed. (An alarm (PS1080) is issued.)



6 A call using the external device subprogram call function is counted as one level of subprogram nesting.

7 In a multipath system, an external device subprogram call cannot be performed simultaneously from multiple paths.

4.6 MANUAL HANDLE INTERRUPTION

By rotating the manual pulse generator in the automatic operation mode (manual data input, DNC operation, or memory operation) or in the memory editing mode, handle feed can be superimposed on movement by automatic operation. A handle interruption axis is selected using the manual handle interruption axis selection signal.
The minimum unit of travel distance per scale division is the least input increment. One of four types of magnifiers selected with MP1 and MP2 <G019.4 and 5> can be applied. With bit 3 (HIT) of parameter No. 7103, the minimum unit of travel distance can be further 10 times greater. A handle feed magnifier is selected using the manual handle feed amount selection signal. (See "MANUAL HANDLE FEED".)

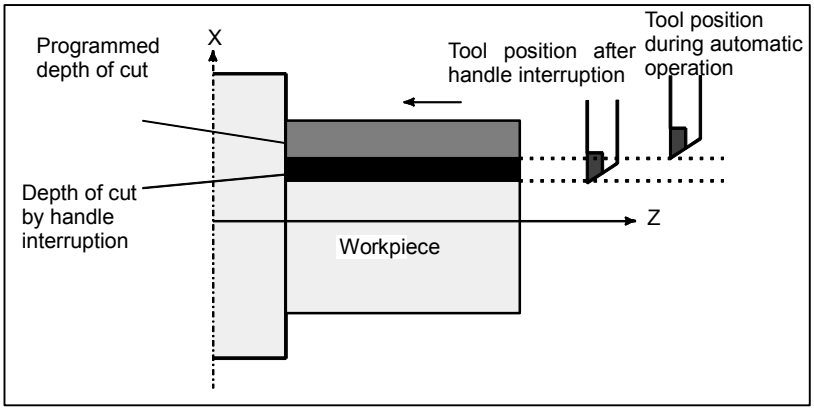


Fig. 4.6(a) Manual handle interruption

> ⚠ **WARNING**
> The travel distance per scale division by manual handle interruption is the least input increment as with manual handle feed. With a machine of metric input and inch output, for example, the travel distance per 254 scale divisions is 0.01 inch. With a machine of inch input and metric output, the travel distance per 100 scale divisions is 0.254 mm.

Explanation

- Interruption operation

1 When the handle interruption axis selection signal for a handle interruption axis is set to 1 in the automatic operation mode (manual data input, DNC operation, or memory operation) or in the memory editing mode, manual handle interruption can be performed by rotating the handle of the manual pulse generator.

NOTE
Even when the feedrate override signal sets 0%, manual handle interruption can be accepted.

2 For the method of selecting a manual handle interruption axis, refer to the relevant manual of the machine tool builder.

3 The feedrate during manual handle interruption is the sum of feedrate used for automatic operation and the feedrate used for movement by manual handle interruption. However, the feedrate during manual handle interruption is controlled so that it does not exceed the maximum allowable cutting feedrate for the axis.

Example
Suppose that the maximum allowable cutting feedrate for an axis is 5 m/min, and that a movement is made in the + direction at 2 m/min along the axis. In this case, manual handle interruption can be accepted even when the manual pulse generator is rotated up to a speed equivalent to 3 m/min. Manual handle interruption by rotation in one direction can be accepted even when the manual pulse generator is rotated to a speed equivalent to 7 m/min.
If the manual pulse generator is rotated to a speed beyond the upper limits, those pulses from the manual pulse generator that correspond to the excess are lost, resulting in a mismatch between the scale mark of the manual pulse generator and the actually interrupted travel distance.

4 For a magnifier for manual handle interruption, refer to the relevant manual of the machine tool builder.

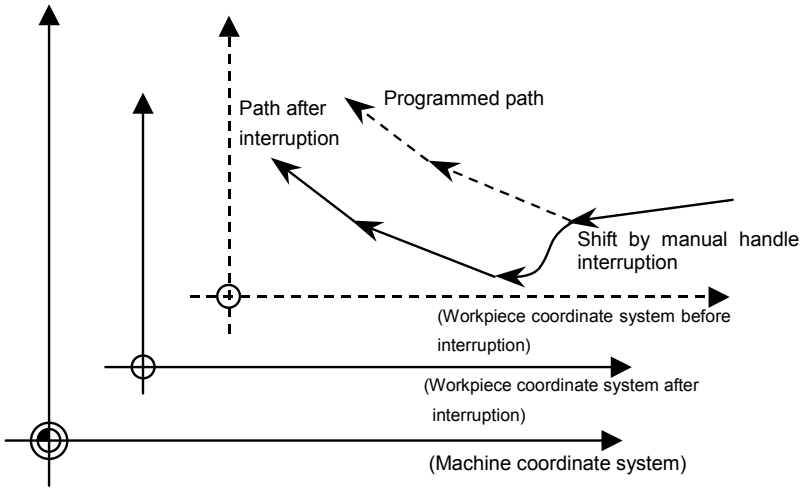
5 If the travel direction is reversed as a result of manual handle interruption, backlash compensation is performed. Pitch error compensation is performed for the position after interruption.

6 In manual handle interruption, only acceleration/deceleration for cutting feed is enabled. By setting bit 0 (MNJ) of parameter No. 1606 to 1, acceleration/deceleration for both of cutting feed and jog feed can be applied to manual handle interruption.

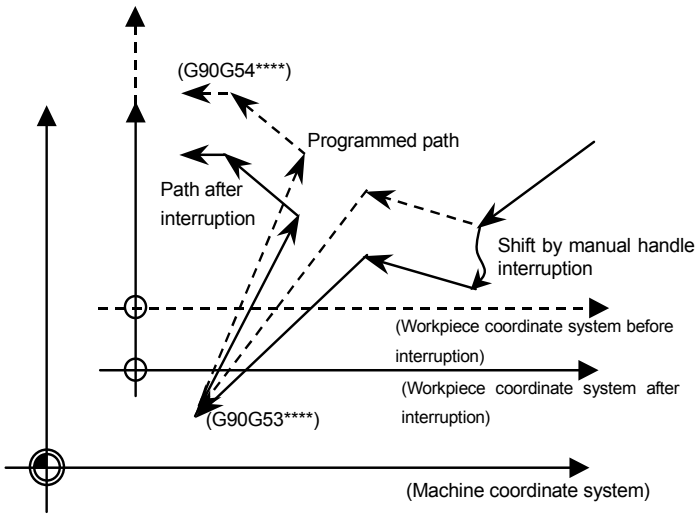
- Manual handle interruption and coordinate system

1 The amount of manual handle interruption shifts the workpiece coordinate systems and the local coordinate system. So, the machine moves, but the coordinates in the workpiece coordinate systems and the local coordinate system remain unchanged.
Regardless of which coordinate system is selected, all workpiece coordinate systems and the local coordinate system shift by the same amount.

- Absolute coordinates
  → Remain unchanged by handle interruption.
- Relative coordinates
  → Change by the amount of handle interruption.
- Machine coordinates
  → Change by the amount of handle interruption.



2 Even when manual handle interruption is performed, the machine coordinate system remains unchanged. The absolute command (G53) in the machine coordinate system is not affected by manual handle interruption.

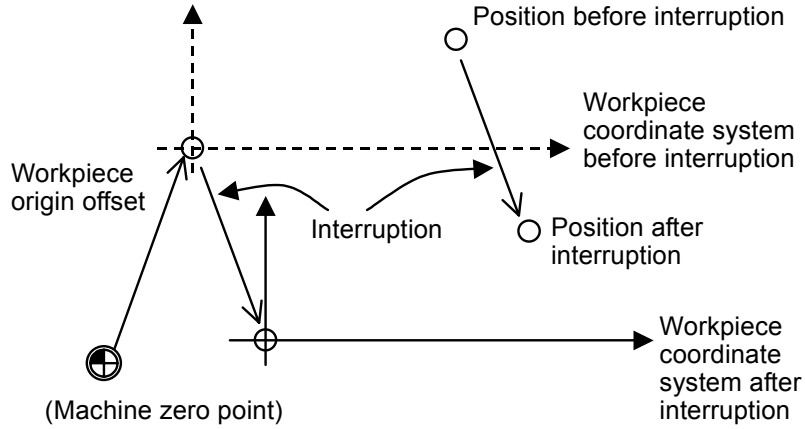


3 In automatic reference position return (G28), the end point (reference position) is not affected by manual handle interruption. However, the midpoint is in the workpiece coordinate system, so that the position shifted by the amount of interruption becomes the midpoint.

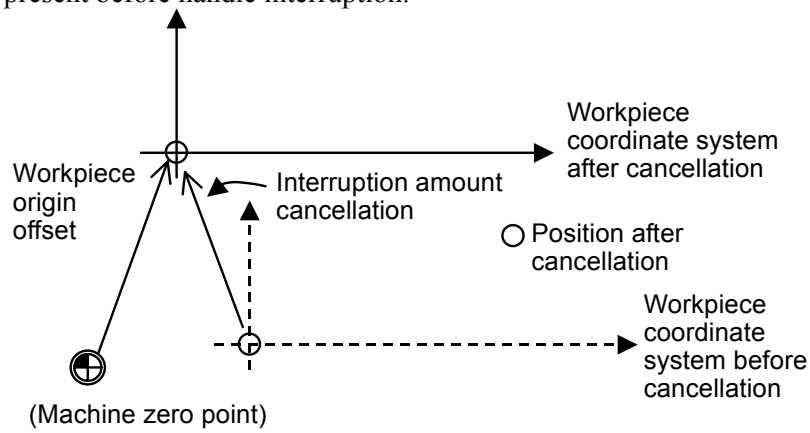
- Cancellation of the amount of interruption

Operation by which the workpiece coordinate system shifted by manual handle interruption from the machine coordinate system is returned to the original workpiece coordinate system is referred to cancellation of the amount of interruption.
When the amount of interruption is canceled, the workpiece coordinate system is shifted by the amount of manual handle interruption, and the amount of interruption is reflected in the absolute coordinates.

Interruption shifts the workpiece coordinate system from the machine coordinate system.



By cancellation, the workpiece coordinate system returns to the state present before handle interruption.



In the following cases, the amount of interruption is canceled:

- When a reset is made (when bit 1 (RTH) of parameter No. 7103 is set to 1)
- When emergency stop state is canceled (when bit 1 (RTH) of parameter No. 7103 is set to 1)

- When a manual reference position return operation is performed (when G28 is specified before a reference position is established)
- When a reference position is set without dogs
- When the workpiece coordinate system is preset

NOTE
When the amount of interruption is cleared using soft keys, only the indication of the amount of interruption becomes 0, and the workpiece coordinate system remains unchanged.

- Clearing the amount of interruption by soft keys

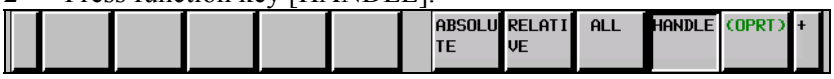
Clearing of the amount of interruption means that the indication of the amount of interruption by manual handle interruption is set to 0. The workpiece coordinate system does not change.
"Clearing all axes" or "Clearing any axis" is performed on the path for which the amount of manual handle interruption is indicated.
When bit 3 (HLC) of parameter No. 7100 is enabled, soft key [INTRPT CANCEL], which is used for this operation, appears. When HLC is disabled, soft key [INTRPT CANCEL] does not appear.

To select "Clearing all axes" or "Clearing any axis", follow the procedure below.

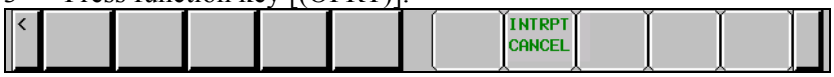
1 Press function key

on the MDI panel.

2 Press function key [HANDLE].



3 Press function key [(OPRT)].



4 To prepare for "Clearing all axes" or "Clearing any axis", press soft key [INTRPT CANCEL].

To prepare for "Clearing all axes" or "Clearing any axis", perform one of the following.

- Clearing all axes
  Press [INTRPT CANCEL] and then press [ALL AXIS].

- Clearing any axis
  - Enter the axis name and then press [INTRPT CANCEL].
  - Press [INTRPT CANCEL], enter the axis name, and press [EXEC].

  If an incorrect axis name is entered, a warning message stating "FORMAT ERROR" appears.

- Relation with other functions

The following table indicates the relation between other functions and the movement by handle interruption.

Table 4.6(a) Relation between other functions and the movement by handle interruption

Signals	Relation
Machine lock	Machine lock is effective. When machine lock is on, no movement is made due to handle interruption.
Interlock	Interlock is effective. When interlock is on, no movement is made due to handle interruption.
Mirror image	Mirror image is not effective. Interrupt functions on the plus direction by plus direction command, even if this signal turns on.

- Position display

The following table shows the relation between various position display data and the movement by handle interruption.

Table4.6(b) relation between various position display data and the movement by handle interruption

Signals	Relation
Absolute coordinate value	Handle interruption does not change absolute coordinates.
Relative coordinate value	Relative coordinates are changed by the travel distance specified by handle interruption.
Machine coordinate value	Machine coordinates are changed by the travel distance specified by handle interruption.

- Travel distance display

Press the function key

, then press the chapter selection soft key [HNDL]. The move amount by the handle interruption is displayed. The following 4 kinds of data are displayed concurrently.

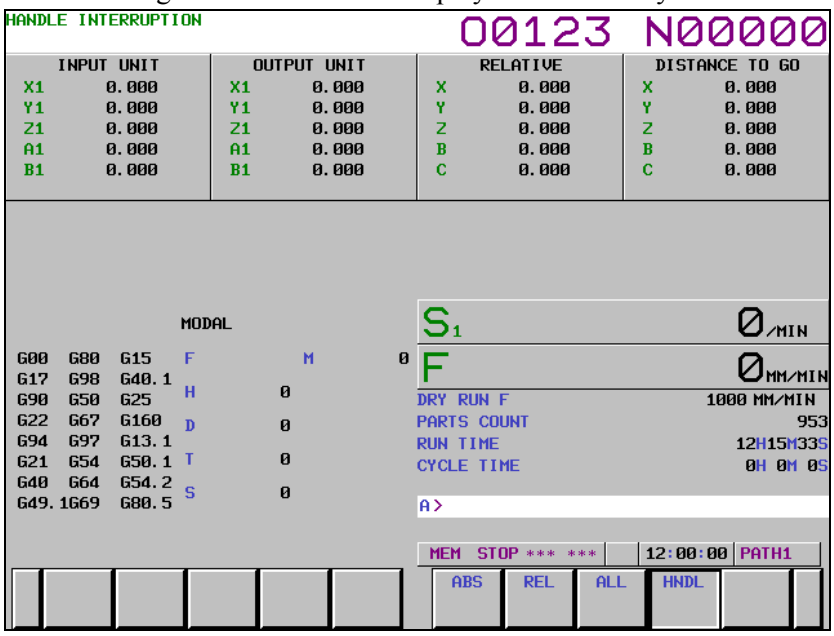


Fig. 4.6 (b)

(a) INPUT UNIT:
Handle interruption move amount in input unit system
Indicates the travel distance specified by handle interruption according to the least input increment.

(b) OUTPUT UNIT :
Handle interruption move amount in output unit system
Indicates the travel distance specified by handle interruption according to the least command increment.

(c) RELATIVE:
Position in relative coordinate system
These values have no effect on the travel distance specified by handle interruption.

(d) DISTANCE TO GO:
The remaining travel distance in the current block has no effect on the travel distance specified by handle interruption.

The handle interruption move amount is cleared when the manual reference position return ends every axis.

- Display for five-axis systems or better

Systems having five or more axes provide the same display as the overall position display. See III-12.1.3.

Note

NOTE
1 In a manual operation mode such as the jog feed mode, manual handle feed mode, or TEACH IN HANDLE mode, handle interruption cannot be performed.
2 During a machine lock or interlock, handle interruption does not cause movement.
3 Manual handle interruption is disabled for the axis in any of the following states.
  - Follow-up state
  - PMC axis control state
4 Manual handle interruption cannot be performed for the axis specified in the G00 mode.

4.7 MIRROR IMAGE

During automatic operation, the mirror image function can be used for movement along an axis. To use this function, set the mirror image switch to ON on the machine operator's panel, or set the mirror image setting to ON from the MDI panel.

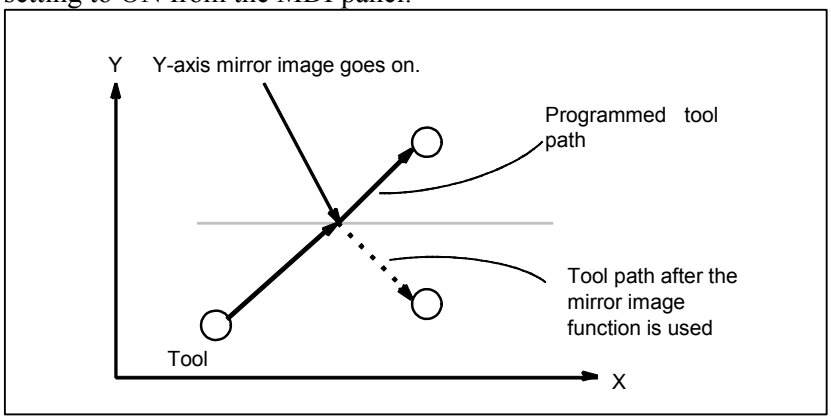


Fig. 4.7 (a) Mirror image

Procedure for mirror image

Procedure

The following procedure is given as an example. For actual operation, refer to the manual supplied by the machine tool builder.

1 Press the single block switch to stop automatic operation.
When the mirror image function is used from the beginning of operation, this step is omitted.

2 Press the mirror image switch for the target axis on the machine operator's panel.
Alternatively, turn on the mirror image setting by following the steps below:
2-1 Set the MDI mode.
2-2 Press the function key .

2-3 Press the [SETING] soft key for chapter selection to display the setting screen.

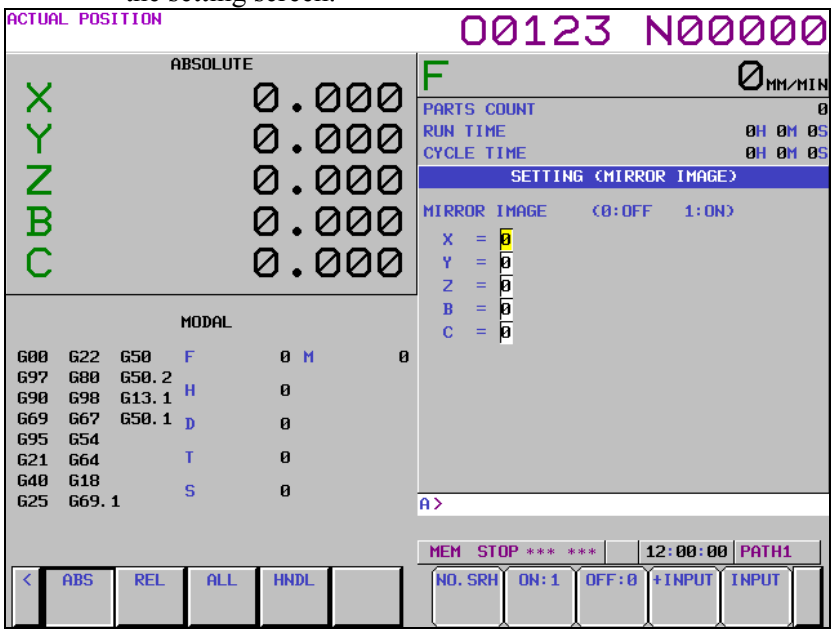


Fig. 4.7 (b) Setting screen

2-4 Move the cursor to the mirror image setting position, then set the target axis to 1.

3 Enter an automatic operation mode (memory mode or MDI mode), then press the cycle start button to start automatic operation.

Explanation

- The mirror image function can also be turned on and off by setting bit 0 (MIRx) of parameter No.0012 to 1 or 0.
- For the mirror image switches, refer to the manual supplied by the machine tool builder.

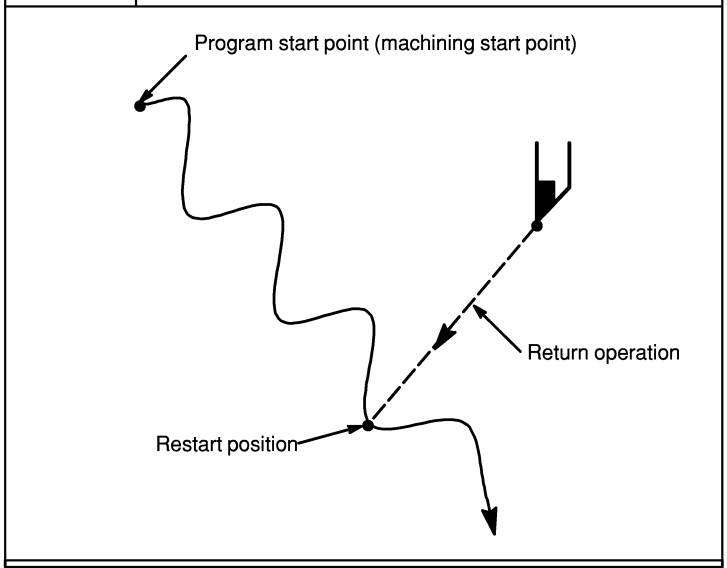
Limitation

The direction of movement during manual operation, the direction of movement from an intermediate point to the reference position during automatic reference position return (G28), the direction of approach during single direction positioning (G60), and the shift direction in a boring cycle (G76, G87) cannot be reversed.

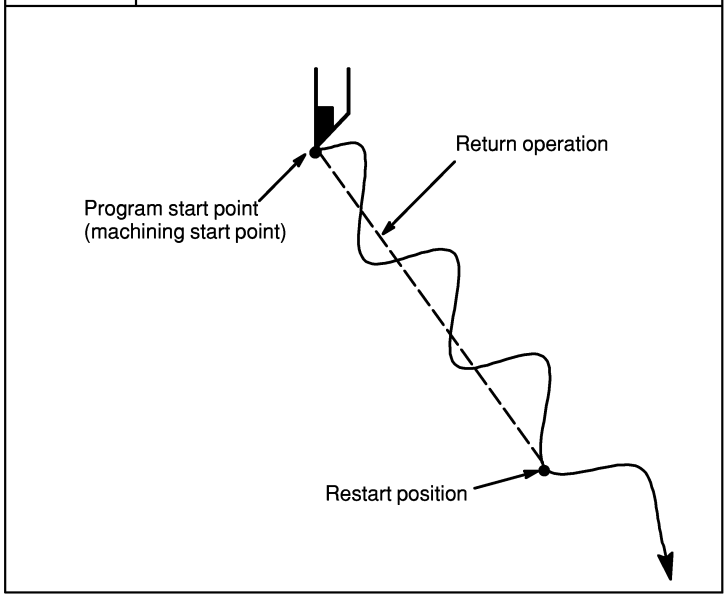
4.8 PROGRAM RESTART

This function specifies Sequence No. of a block to be restarted when a tool is broken down or when it is desired to restart machining operation after a day off, and restarts the machining operation from that block. It can also be used as a high-speed program check function.
There are two restart methods: the P-type method and Q-type method.

P TYPE	Operation can be restarted anywhere. This restart method is used when operation is stopped because of a broken tool.



Q TYPE	Before operation can be restarted, the machine must be moved to the programmed start point (machining start point)



Procedure for program restart by specifying a sequence number

Procedure 1

[P TYPE]

1 Retract the tool and replace it with a new one. When necessary, change the offset. (Go to step 2.)

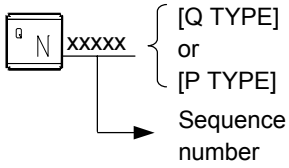
[Q TYPE]

1 When power is turned ON or emergency stop is released, perform all necessary operations at that time, including the reference position return.
2 Move the machine manually to the program starting point (machining start point), and keep the modal data and coordinate system in the same conditions as at the machining start.
3 If necessary, modify the offset amount. (Go to step 2.)

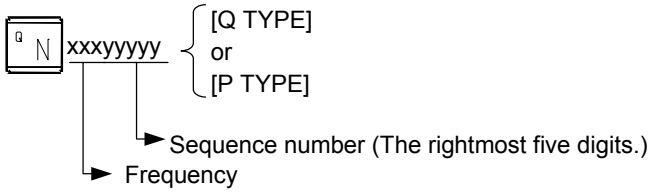
Procedure 2

[COMMON TO P TYPE / Q TYPE]

1 Turn the program restart switch on the machine operator's panel ON.
2 Press PROG key to display the desired program.
3 Find the program head. Press RESET key.
4 Enter the sequence number of the block to be restarted, then press the [P TYPE] or [Q TYPE] soft key.



If the same sequence number appears more than once, the location of the target block must be specified. Specify a frequency and a sequence number.



5 The sequence number is searched for, and the program restart screen appears on the LCD display.

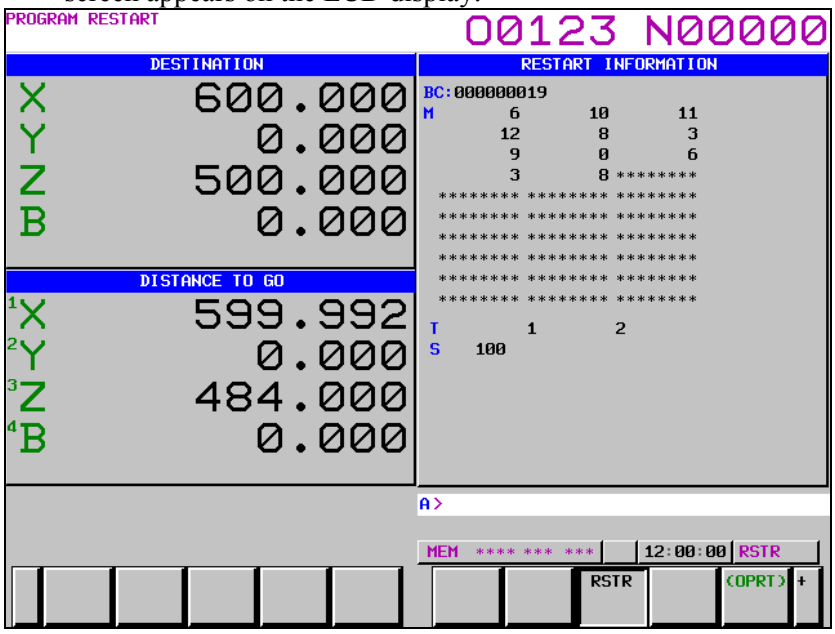


Fig. 4.8 (a) Program restart screen

DESTINATION shows the position at which machining is to restart.
DISTANCE TO GO shows the distance from the current tool position to the position where machining is to restart. A number to the left of each axis name indicates the order of axes (determined by parameter setting) along which the tool moves to the restart position.
The coordinates and amount of travel for restarting the program can be displayed for up to four axes. If your system supports five or more axes, pressing the [RSTR] soft key again displays the data for the fifth and subsequent axes.

M : Up to 35 most recently specified M codes. The maximum number of displayed M codes differs depending on the size of the display.
With 15-inch or 10.4-inch LCD/MDI panel :
Up to 30 M codes
With 7.2/8.4-inch LCD/MDI panel : Up to 6 M codes

T : Two most recently specified T codes
S : Most recently specified S code
B : Most recently specified B code

Codes are displayed in the order in which they are specified. All codes are cleared by a program restart command or cycle start in the reset state.

6 Turn the program re-start switch OFF. At this time, the figure at the left side of axis name DISTANCE TO GO blinks.

7 Check the screen for the M, S, T, and B codes to be executed. If they are found, enter the MDI mode, then execute the M, S, T, and B functions. After execution, restore the previous mode. These codes are not displayed on the program restart screen.

8 Check that the distance indicated under DISTANCE TO GO is correct. Also check whether there is the possibility that the tool might hit a workpiece or other objects when it moves to the machining restart position. If such a possibility exists, move the tool manually to a position from which the tool can move to the machining restart position without encountering any obstacles.

9 Press the cycle start button. The tool moves to the machining restart position at the dry run feedrate sequentially along axes in the order specified by parameter No. 7310 settings. Machining is then restarted.

Procedure for program restart by specifying a block Number

Procedure 1

[P TYPE]

1 Retract the tool and replace it with a new one. When necessary, change the offset. (Go to step 2.)

[Q TYPE]

1 When power is turned ON or emergency stop is released, perform all necessary operations at that time, including the reference position return.

2 Move the machine manually to the program starting point (machining start point), and keep the modal data and coordinate system in the same conditions as at the machining start.

3 If necessary, modify the offset amount. (Go to step 2.)

Procedure 2

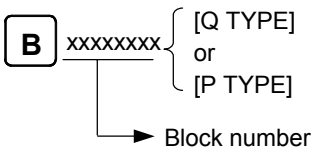
[COMMON TO P TYPE / Q TYPE]

1 Turn the program restart switch on the machine operator's panel ON.

2 Press key to display the desired program.

3 Find the program head. Press key.

4 Enter the number of the block to be restarted then press the [P TYPE] or [Q TYPE] soft key. The block number cannot exceed eight digits.



5 The block number is searched for, and the program restart screen appears on the LCD display.

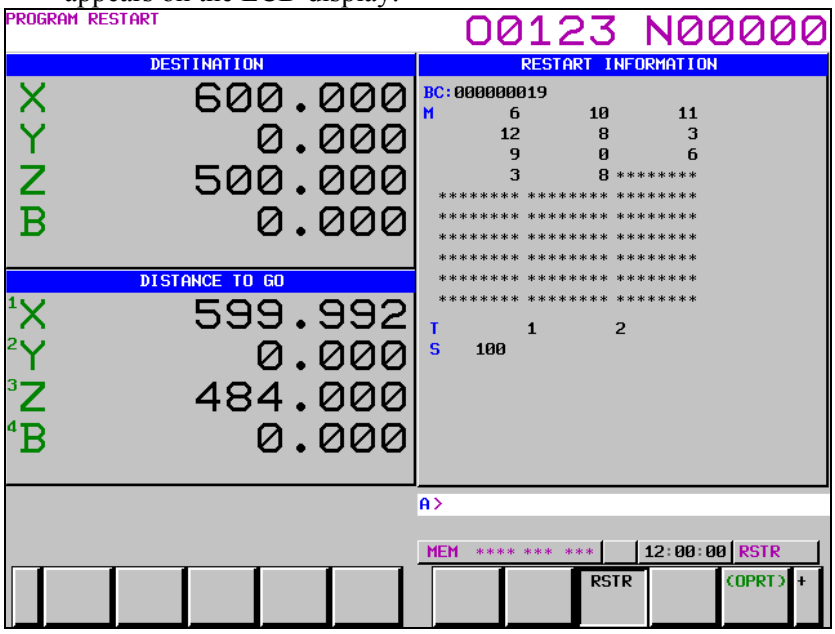


Fig. 4.8 (b) Program restart screen

DESTINATION shows the position at which machining is to restart.
DISTANCE TO GO shows the distance from the current tool position to the position where machining is to restart. A number to the left of each axis name indicates the order of axes (determined by parameter setting) along which the tool moves to the restart position.
The coordinates and amount of travel for restarting the program can be displayed for up to four axes. If your system supports five or more axes, pressing the [RSTR] soft key again displays the data for the fifth and subsequent axes.

M : Up to 35 most recently specified M codes. The maximum number of displayed M codes differs depending on the size of the display.
With 15-inch or 10.4-inch LCD/MDI panel :
Up to 30 M codes
With 9.5-inch LCD/MDI panel : Up to 6 M codes

T : Two most recently specified T codes

S : Most recently specified S code

B : Most recently specified B code

Codes are displayed in the order in which they are specified. All codes are cleared by a program restart command or cycle start in the reset state.

6 Turn the program re-start switch OFF. At this time, the figure at the left side of axis name DISTANCE TO GO blinks.

7 Check the screen for the M, S, T, and B codes to be executed. If they are found, enter the MDI mode, then execute the M, S, T, and B functions. After execution, restore the previous mode. These codes are not displayed on the program restart screen.

8 Check that the distance indicated under DISTANCE TO GO is correct. Also check whether there is the possibility that the tool might hit a workpiece or other objects when it moves to the machining restart position. If such a possibility exists, move the tool manually to a position from which the tool can move to the machining restart position without encountering any obstacles.

9 Press the cycle start button. The tool moves to the machining restart position at the dry run feedrate sequentially along axes in the order specified by parameter No. 7310 settings. Machining is then restarted.

Outputting the M, S, T, and B codes for program restart

After the block to be restarted is searched for, you can perform the following operations:

1 Before the tool is moved to the machining restart position

<1> The most recently specified M, S, T, and B codes can automatically be output to the PMC.
The most recently specified S code is output as the maximum spindle speed when the S code is specified in the block containing G92 or as the specified spindle speed in other cases. As the most recently specified S code, only one S code is displayed on the program restart screen regardless of whether the S code is specified in the block containing G92.

<2> While the block to be restarted is being searched for, all sampled M codes and most recently specified S, T, and B codes can automatically be output to the PMC. Up to 35 M codes can be sampled. If the number of sampled M codes exceeds 35, the 35 most recently specified M codes are output to the PMC.

Switch between operations <1> and <2> using bit 6 (MOA) of parameter No. 7300.

2 Before the tool reaches the machining restart position
On the program restart screen, you can specify M, S, T, and B codes from the MDI panel in the MEM or RMT mode without changing the mode.

Outputting the most recently specified M, S, T, and B codes

When bit 7 (MOP) of parameter No. 7300 is set to 1, pressing the cycle start switch after searching for the block to be restarted automatically outputs the most recently specified M, S, T, and B codes to the PMC before the tool is moved to the machining restart position.
In the single block stop status, after the most recently specified M, S, T, and B codes are output, pressing the cycle start switch again moves the tool to the machining restart position.

Outputting all M codes and most recently specified S, T, and B codes

When bit 6 (MOA) of parameter No. 7300 is set to 1, pressing the cycle start switch after searching for the block to be restarted automatically outputs all M codes and most recently specified S, T, and B codes to the PMC before the tool is moved to the machining restart position.

(Example)

When M10, M11, M12, M13, M14, T0101, S1000, and B10 are sampled, a program is executed in the format shown below before the tool is moved to the machining restart position:

```
M10 T0101 S1000 B10 ;
M11 ;
M12 ;
M13 ;
M14 ;
```

Outputting M, S, T, and B codes on the program restart screen

When bit 7 (MOP) of parameter No. 7300 is set to 1, you can specify M, S, T, and B codes from the MDI panel in the MEM or RMT mode without changing the mode after searching for the block to be restarted until the tool reaches the machining restart position.

Procedure

1 When the block to be restarted is searched for using the program restart function, the program restart screen appears. When bit 7 (MOP) of parameter No. 7300 is set to 1, operation soft keys [OVERSTORE], [ERASE], and [INPUT] are displayed.

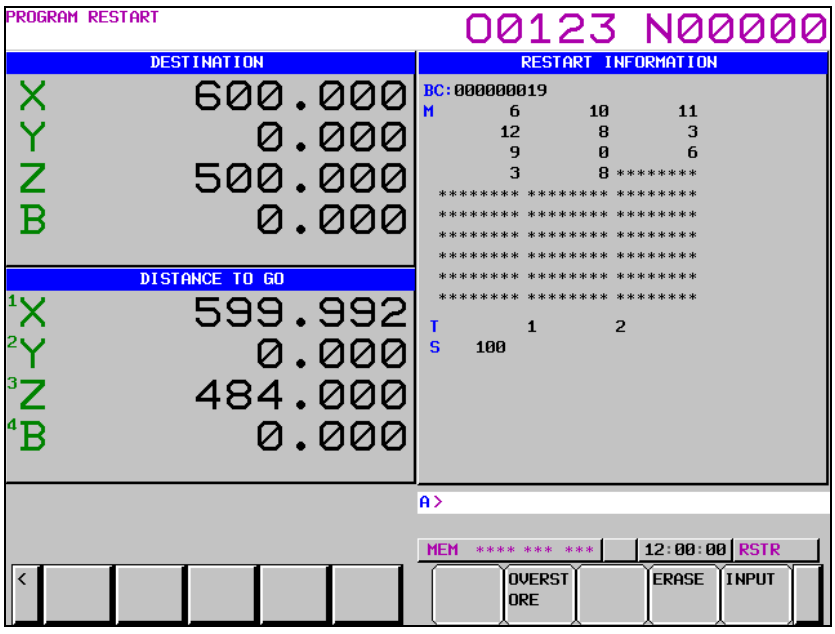


Fig. 4.8 (c) Program restart screen (outputting M, S, T, and B codes)

2 Before the tool reaches the machining restart position, pressing soft key [OVERSTORE] selects the over store mode. In the over store mode, data can be entered in the M, S, T, and B fields displayed in the (OVERSTORE) section.
To select the over store mode while the tool is moving to the machining restart position, hold restart operation by feed hold and press soft key [OVERSTORE].
Enter M, S, T, and B codes to be output in the (OVERSTORE) section from the MDI panel.
(Example)
To enter M10, S1000, T101, and B20 in the (OVERSTORE) section:
<1> Enter M 1 0 from the MDI panel.
<2> Press the [INPUT] key.
You can also enter the S, T, and B codes by performing steps <1> and <2>.

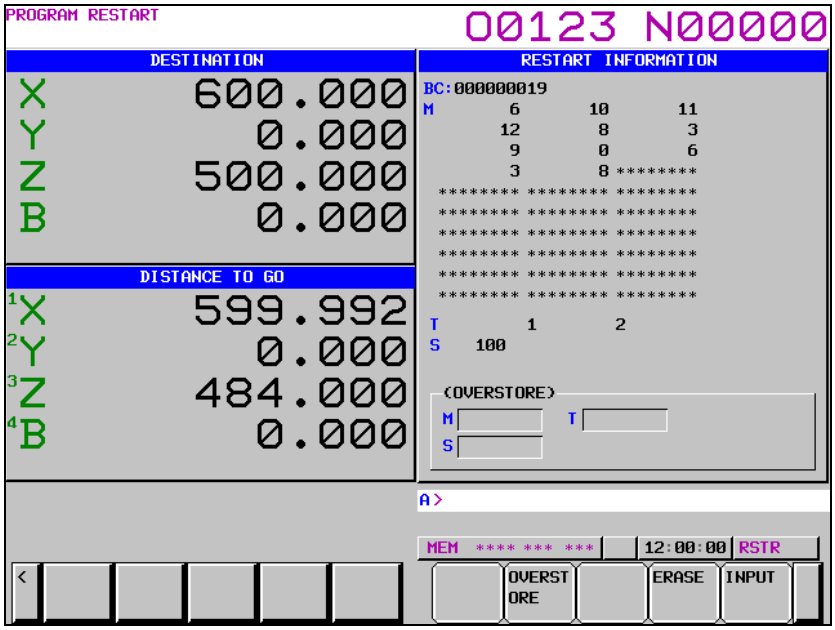


Fig. 4.8 (d) Program restart screen when M, S, T, and B codes are output

3 When values have been entered in the (OVERSTORE) section, pressing the cycle start switch outputs each code in the (OVERSTORE) section. The values in the (OVERSTORE) section are cleared.
4 To clear the values entered in the (OVERSTORE) section as M, S, T, and B codes, press soft key [ERASE]. All entered values are cleared.
5 Pressing soft key [OVERSTORE] again in the over store mode cancels the mode. Pressing the reset key also cancels the over store mode.
6 To continue with restart operation, cancel the over store mode and press the cycle start switch.

⚠ CAUTION

1 The M, S, T, and B codes specified in the over store mode are not displayed on the program restart screen.
2 In the over store mode, changing the operation mode to other than the MEM or RMT mode does not cancel the over store mode. In this case, no values can be entered in the (OVERSTORE) section.
3 In lathe systems, do not specify a T code in the over store mode. If a T code is specified, it is not executed.

Explanation

- Block number

When the CNC is stopped, the number of executed blocks is displayed on the program screen or program restart screen. The operator can specify the number of the block from which the program is to be restarted, by referencing the number displayed on the screen. The displayed number indicates the number of the block that was executed most recently. For example, to restart the program from the block at which execution stopped, specify the displayed number, plus one.
The number of blocks is counted from the start of machining, assuming one NC line of a CNC program to be one block.

(Example 1)

CNC Program	Number of blocks
O 0001 ;	1
G90 G92 X0 Y0 Z0 ;	2
G01 X100. F100 ;	3
G03 X01 -50. F50 ;	4
M30 ;	5

(Example 2)

CNC Program	Number of blocks
O 0001 ;	1
G90 G92 X0 Y0 Z0 ;	2
G90 G00 Z100. ;	3
G81 X100. Y0. Z120. R-80. F50. ;	4
#1=#1+1 ;	4
#2=#2+1 ;	4
#3=#3+1 ;	4
G00 X0 Z0 ;	5
M30 ;	6

Macro statements are not counted as blocks.

- Storing / clearing the block number

The block number is held in memory while no power is supplied. The number can be cleared by cycle start in the reset state.

- Block number when a program is halted or stopped

The program screen usually displays the number of the block currently being executed. When the execution of a block is completed, the CNC is reset, or the program is executed in single-block stop mode, the program screen displays the number of the program that was executed most recently. When a CNC program is halted or stopped by feed hold, reset, or single-block stop, the following block numbers are displayed:

Feed hold : Block being executed
Reset : Block executed most recently
Single-block stop : Block executed most recently

For example, when the CNC is reset during the execution of block 10, the displayed block number changes from 10 to 9.

- MDI intervention

When MDI intervention is performed while the program is stopped by single-block stop, the CNC commands used for intervention are not counted as a block.

- Block number exceeding eight digits

When the block number displayed on the program screen exceeds eight digits, the block number is reset to 0 and counting continues.

Limitation

- P type Restart

In the following conditions, P type restart cannot be performed:

- Automatic operation has not been performed since power-on.
- Automatic operation has not been performed since emergency stop was released.
- Automatic operation has not been performed since the coordinate system was changed or shifted (change of the external workpiece origin offset value).

The block that can be restored properly by P type program restart is a block for which coordinate system setup or change was performed most recently before machining was interrupted.

- Restart block

The block where the program is to restart is not necessarily be the block at which the program was interrupted. You can restart the program from any block. For P-type restart, however, the block where the program is to restart must use the same coordinate system as when program execution was interrupted.

- Single block

When the single-block operation is enabled at the time of a movement to the restart point, a single-block stop occurs each time an axis operation takes place. In this case, no MDI operation is allowed.

- Manual intervention

During movement to the restart point, manual intervention is allowed for an axis for which a return operation has not yet been performed. However, manual operations do not cause any movement along axes for which a return operation has already been completed.

- MDI

When the search operation has ended, no move command can be specified by MDI before axis movement.

- Reset

Do not perform a reset operation during the time from the start of the search operation of the restart sequence until machining is restarted.
If a reset operation is performed, the restart steps must be performed again from the beginning.

- Feed hold

If a feed hold operation is performed during the search, the restart steps must be performed again from the beginning.

- Manual absolute

Every manual operation must be performed with the manual absolute mode turned on regardless of whether the manual operation is performed before or after machining.

- Reference position return

Unless an absolute position detector (absolute pulse coder) is provided, be sure to perform reference position return after power-up, then perform restart operation.

- Program restart switch

When the program restart switch is on, pressing the cycle start switch does not start operation.

- Blocks specifying a macro statement, macro call, and subprogram call

Blocks specifying a macro statement, macro call, and subprogram call are not searched for even when they have a sequence number. In such a case, search for a block previously preceding such a block.

- Interruption type custom macro

During movement to the machining restart point at a dry run feedrate, no interruption type custom macro can be started. If an interruption type custom macro is started, alarm DS024 is issued.

- Commands that prevent program restart

Program restart cannot be performed for blocks placed in the following modes:

- Cs contouring control
- Polygon turning (G50.2)
- Threading (G32,G33), Circular threading (G35,G36), Threading cycle (G92), Multiple repetitive threading cycle (G76)
- Polar coordinate interpolation (G12.1)
- Balance cutting (G68)
- Rigid tapping

If any of the following commands is included between the beginning of a program and the block where the program is to restart, program restart cannot be performed:

- Workpiece coordinate system preset (G92.1,G50.3)
- Commands for enabling and disabling synchronous/composite control and superimposed control
- Commands for enabling and disabling axis synchronous control

- M, S, and T commands not usable in over store mode

The M, S, and T functions listed below, unlike the other M, S, and T functions, have special meanings within the CNC. These M, S, and T commands cannot be specified from the over store screen. To specify these commands, cancel the over store mode, and execute them in MDI operation.

Example:

- Spindle positioning
- Rigid tapping
- Tool management

⚠ WARNING

As a rule, the tool cannot be returned to a correct position under the following conditions.
Special care must be taken in the following cases since none of them cause an alarm:

- Manual operation is performed when the manual absolute mode is OFF.
- Manual operation is performed when the machine is locked.
- When the mirror image is used. However, P type return is possible for a block that switched between ON and OFF most recently or a subsequent block. In this case, the mirror image signal status present when the program was interrupted must be maintained.
- When no coordinate system is set up at the beginning of a program in which main commands are executed in the incremental mode.
- When manual operation is performed in the course of axis movement for returning operation.
- When the program restart is commanded for a block between the block for skip cutting and subsequent absolute command block.
- When program restart is specified in the machine lock state, then the machine lock is canceled.
- When program restart specified for an intermediate block for a multiple repetitive canned cycle
- In general, when a coordinate system is set up, changed, or shifted after the search operation ends, the tool cannot be returned to a correct position.

⚠ CAUTION

Keep the following in mind when restarting a program including macro variables.

- Common variable
  When the program is restarted, the previous values are inherited as common variables without being preset automatically. Before restarting the program, initialize the appropriate variables to the original values used at start of the previous automatic operation.
- DI/DO
  At restart of the program, DI can be read by a system variable, but DO cannot be output.
- Clock
  When the program is being restarted, the clock time can be obtained by a system variable, but the time cannot be preset.
- Tool offset and workpiece origin offset
  When the program is being restarted, the offset can be read by a system variable, but change of the offset is allowed only for the Q type.

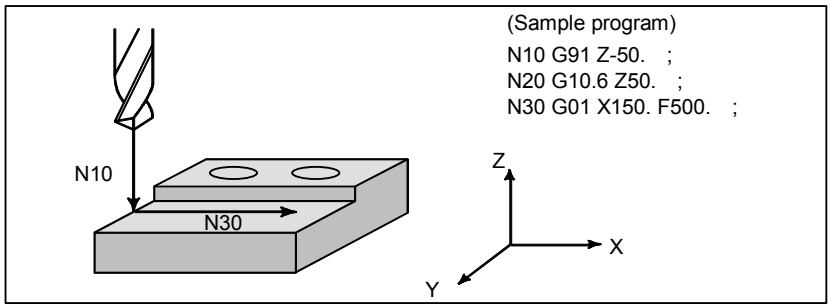
4.9 TOOL RETRACT AND RECOVER

The tool can be retracted from a workpiece to replace the tool, if damaged during machining, or to check the status of machining. Then, the tool can be returned to restart machining efficiently.

Procedure for tool retract and recover

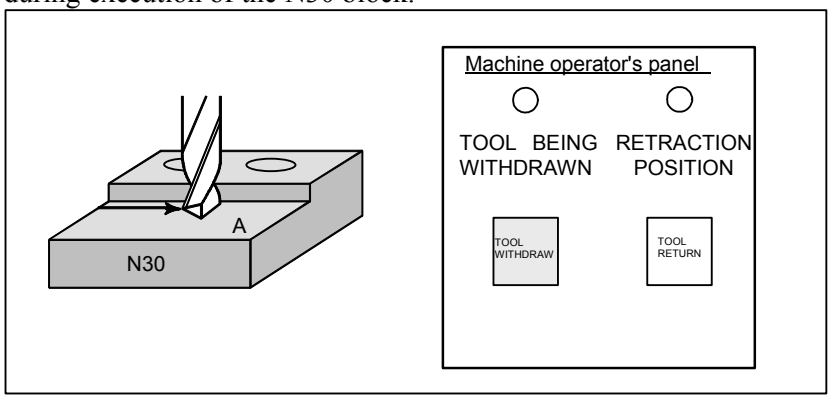
Procedure1 - Programming

Specify a retraction axis and distance in command G10.6IP_ beforehand. In the sample program below, the N20 block specifies that the Z-axis is the retraction axis and the retraction distance is to be 50 mm.

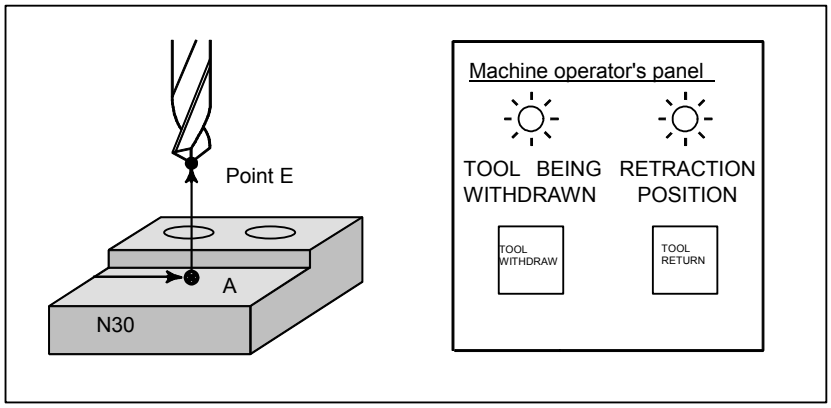


Procedure 2 - Retract

Suppose that the TOOL WITHDRAW switch on the machine operator's panel is turned on when the tool is positioned at point A during execution of the N30 block.



Next, the tool withdrawal mode is set and the TOOL BEING WITHDRAWN LED goes on. At this time, automatic operation is temporarily halted. The tool is then retracted by the programmed distance. If point A is the end point of the block, retraction is performed after automatic operation is stopped. Retraction is based on linear interpolation. The dry run feedrate is used for retraction. Upon completion of retraction, the RETRACT POSITION LED on the operator's panel goes on.

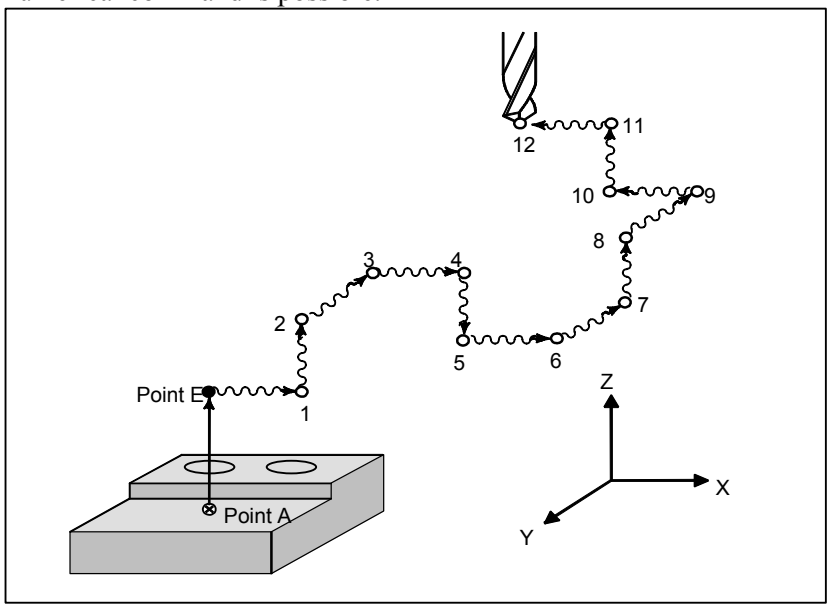


During retraction, the screen displays PTRR and STRT.

- PTRR blinks in the field for indicating states such as the program editing status.
- STRT is displayed in the automatic operation status field.
- MTN is displayed in the field for indicating status such as movement along an axis.

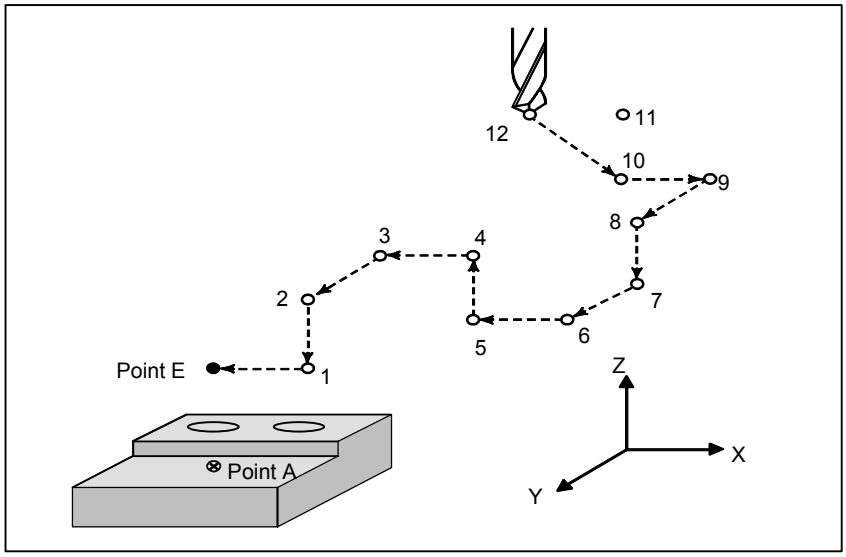
Procedure 3 - Withdrawal

Set the manual operation mode, then withdraw the tool. For manual operation, either jog feed incremental feed, handle feed, or manual numerical command is possible.



Procedure 4 - Return

After withdrawing the tool and any additional operation such as replacing the tool, move the tool back to the previous retraction position. To return the tool to the retraction position, return the mode to automatic operation mode, then turn the TOOL RETURN switch on the operator's panel on then off again. The tool returns to the retraction position at the dry run feedrate, regardless of whether the dry run switch is on or off.
When the tool has returned to the retraction position, the RETRACTION POSITION LED comes on.

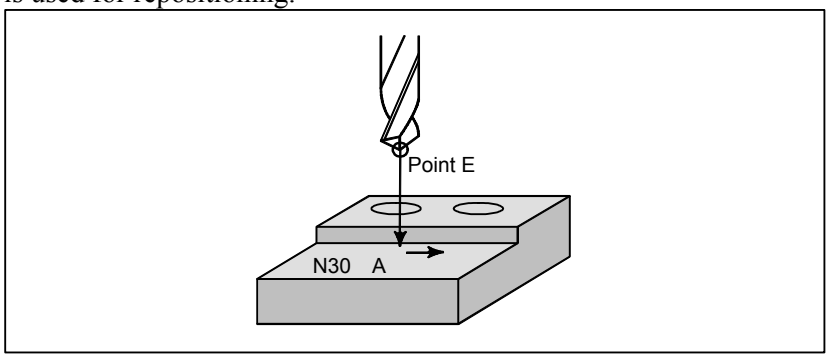


During return operation, the screen displays PTRR and MSTR.

- PTRR blinks in the field for indicating states such as program editing status.
- MSTR is displayed in the automatic operation status field.
- MTN is displayed in the field for indicating states such as movement along an axis.

Procedure 5 - Repositioning

While the tool is at the retraction position (point E in the figure below) and the RETRACTION POSITION LED is on, press the cycle start switch. The tool is then repositioned at the point where retraction was started (i.e. where the TOOL WITHDRAW switch was turned on). Repositioning is based on linear interpolation. The dry run feedrate is used for repositioning.



Upon completion of repositioning, the tool withdraw mode is cancelled, and the TOOL BEING WITHDRAWN LED goes off and restart N30.

Limitation

1 If the origin, presetting, workpiece origin offset value (or external workpiece origin offset value), or workpiece coordinate shift amount (for a lathe system) is changed after the retraction position is specified with G10.6 in the absolute mode, the change is not reflected in the retraction position. After such changes are made or the workpiece origin offset value (or external workpiece origin offset value) or workpiece coordinate shift amount (for a lathe system) is changed, respectively the retraction position with G10.6.

2 When retracting the tool manually in the tool withdrawal mode, do not use the machine lock, mirror-image, or scaling function.

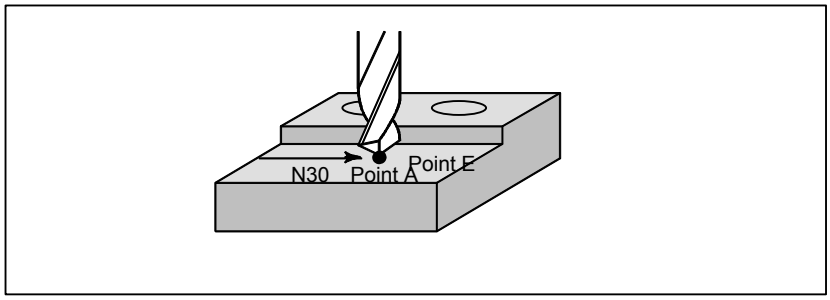
> ⚠ **WARNING**
> The retraction axis and retraction distance specified in G10.6 must be changed in an appropriate block according to the figure being machined. Be very careful when specifying the retraction distance; an incorrect retraction distance may damage the workpiece, machine, or tool.

4.9.1 Retract

Explanation

- When no retraction distance is specified

If no retraction distance or direction required for retraction are specified, retraction is not performed when the TOOL WITHDRAW switch on the operator's panel is turned on. Instead, the block being executed in automatic operation is interrupted (automatic operation is held or stopped). In this state, the tool can be withdrawn and returned.



- Retraction from the automatic operation hold or stop state

When the single block switch is turned on during automatic operation, or the TOOL WITHDRAW switch is turned on after the automatic operation hold or stop state is set by feed hold: Retraction is performed, then the automatic operation hold or stop state is set again.

- Stopping retraction

During retraction, feed hold operation is ignored. However, reset operation is enabled (retraction is stopped at reset). When an alarm is issued during retraction, the retraction is stopped immediately.

- Repositioning immediately after retraction

After retraction is completed, tool repositioning can be started without performing the withdraw and return operations.

4.9.2 Withdrawal

Explanation

- Axis selection

To move the tool along an axis, select the corresponding axis selection signal. Never specify axis selection signals for two or more axes at a time.

- Path memorization

When the tool is moved in manual operation along an axis, the control unit memorizes up to ten paths of movements. If the tool is stopped after being moved along a selected axis and is then moved along another selected axis, the position where this switch takes place is memorized. After ten paths have been memorized, the control unit does not memorize any additional switching points.

- Reset

Upon reset, memorized position data is lost and the tool withdraw mode is cancelled.

NOTE
If an attempt is made to move the tool simultaneously along two axes using the manual numeric command in the tool withdrawal mode, an alarm (PS0015) is issued.

4.9.3 Return

Explanation

- Return path

When there are more than ten return paths, the tool first moves to the tenth position, then to the ninth position, then to the eighth position, and so forth until the retraction position is reached.

- Single block

The single block switch is enabled during return operation. If the single block switch is turned off, continuous return operation is performed. If the single block switch is turned off, the tool stops at each memorized position. In this case, return operation can be resumed by turning the TOOL RETURN switch on then off again.

- Interruption of return operation

When an alarm is issued during return operation, return operation stops.

- Feed hold

The feed hold function is enabled during return operation.

4.9.4 Repositioning

Explanation

- Feed hold

The feed hold function is disabled during repositioning.

- Operation after completion of repositioning

The operation after completion of repositioning depends on the automatic operation state present when the TOOL WITHDRAW switch is turned on.

1 When automatic operation is being started
After completion of repositioning, the interrupted execution of the block is resumed.

2 When automatic operation is held or stopped
After completion of repositioning, the tool stops once at the repositioned point, then the original automatic operation hold or stop state is set. When the cycle start switch is pressed, automatic operation is resumed.

4.9.5 Tool Retract and Recover for Threading

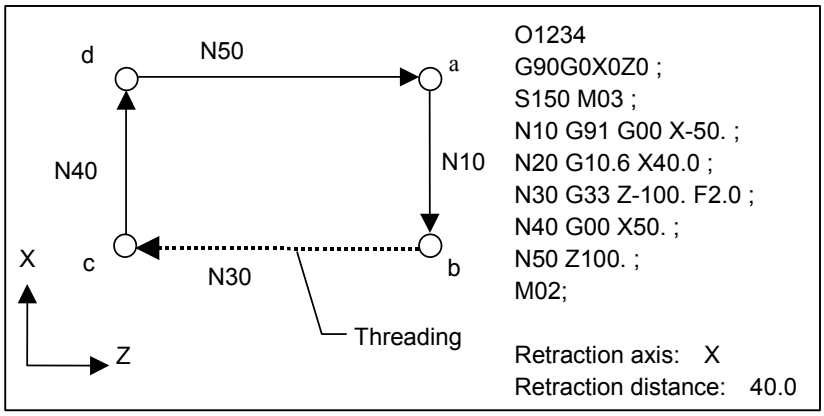
Explanation

- Differences between ordinary tool retract and recover and tool retract and recover for threading

1 During retraction, chamfering is performed between the specified retraction axis and threading axis.
2 After retraction, one block that does not specify threading is executed and the tool stops.
3 When the major axis for threading is specified as the retraction axis, retraction is not performed by turning the TOOL WITHDRAW switch on. In this case, after a block that does not specify threading is executed, an alarm PS0429 is issued and the tool stops.
4 As repositioning, the tool is returned to the position specified in the first block that does not specify threading.

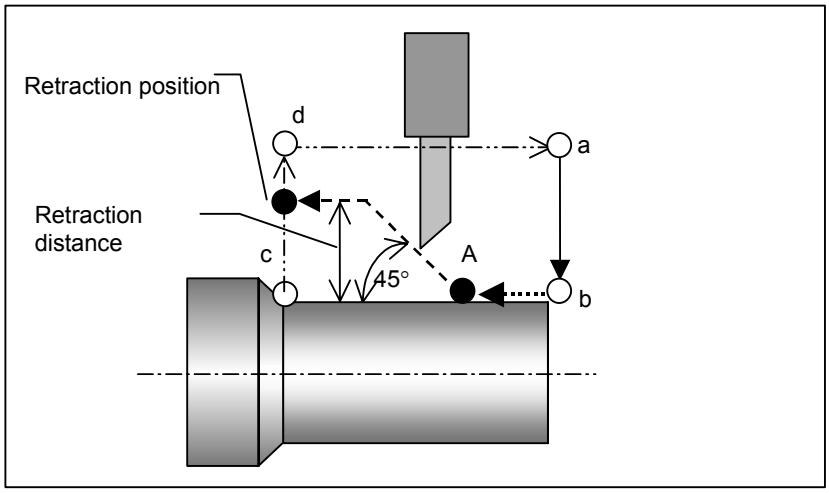
- Operation procedure

1 Specify a retraction axis and retraction distance in command "G10.6IP- -;".



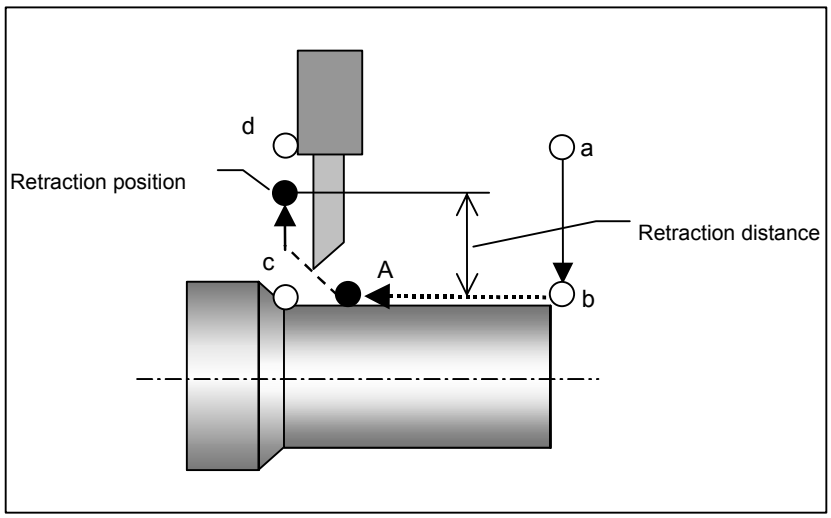
2 Turn the TOOL WITHDRAW switch during the execution of a threading command block.
3 The tool withdrawal mode is set and retraction is performed. Chamfering at 45 degrees is performed between the retraction axis and major axis for threading using the retraction distance as the chamfering amount during retraction.
Details of retraction differ depending on whether the remaining travel distance for the threading command is smaller than the retraction distance when the TOOL WITHDRAW switch is turned on as follows:

(1) When remaining travel distance for threading ≥ retraction distance



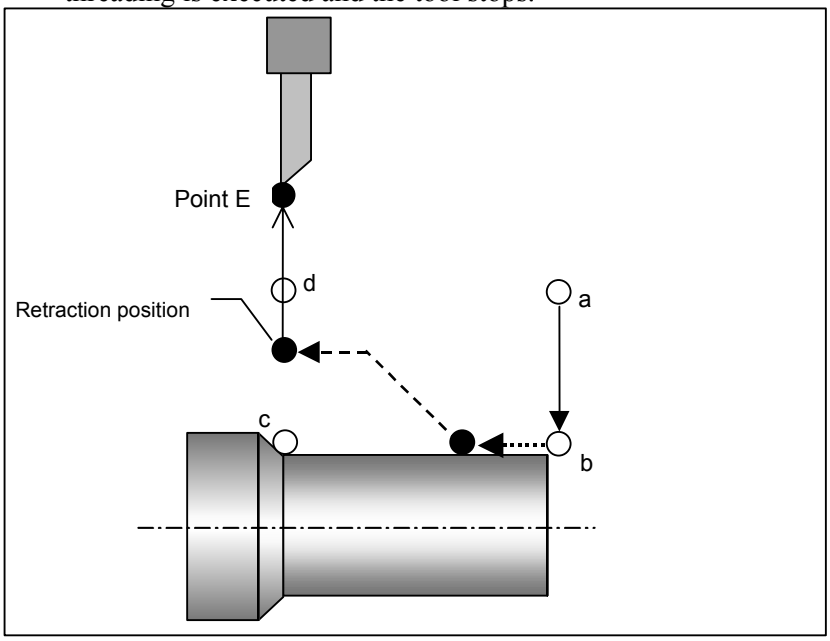
When the position where 45-degree chamfering by the retraction distance ends does not exceed the threading end position (c), the tool moves to the threading end position after the termination of chamfering.

(2) When the remaining travel distance for threading < retraction distance



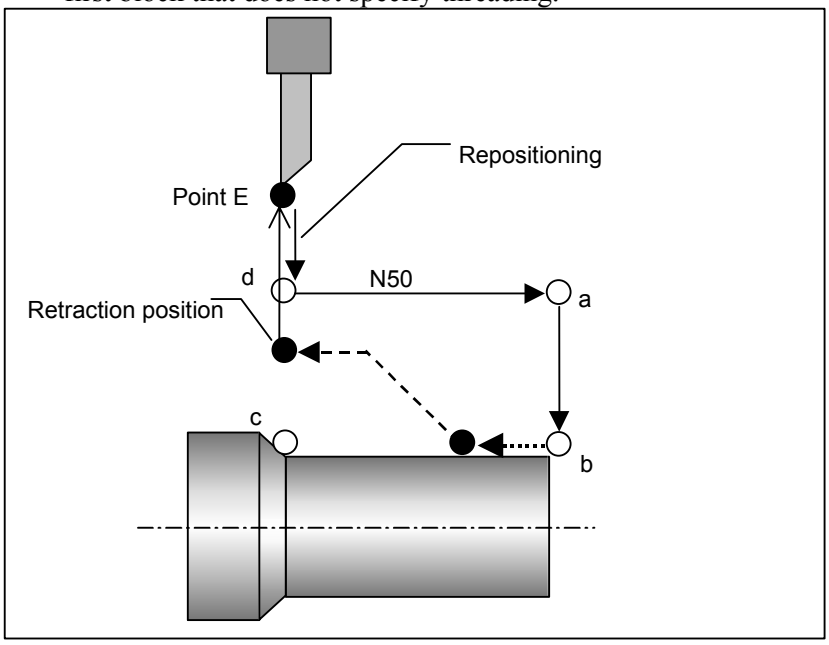
When the position where 45-degree chamfering by the retraction distance ends exceeds the threading end position (c), the tool moves to the retraction position along the retraction axis after it reaches the threading end position.

4 After retraction is completed, the next block that does not specify threading is executed and the tool stops.



In this example, “X50.0” is specified in the first block that does not specify threading in the incremental mode, the tool moves to point E and stops.
If the major axis for threading is specified as the retraction axis, the block that does not specify threading is executed without performing retraction, an alarm PS0429 is issued, and the tool stops.

5 As repositioning, the tool returns to the position specified in the first block that does not specify threading.



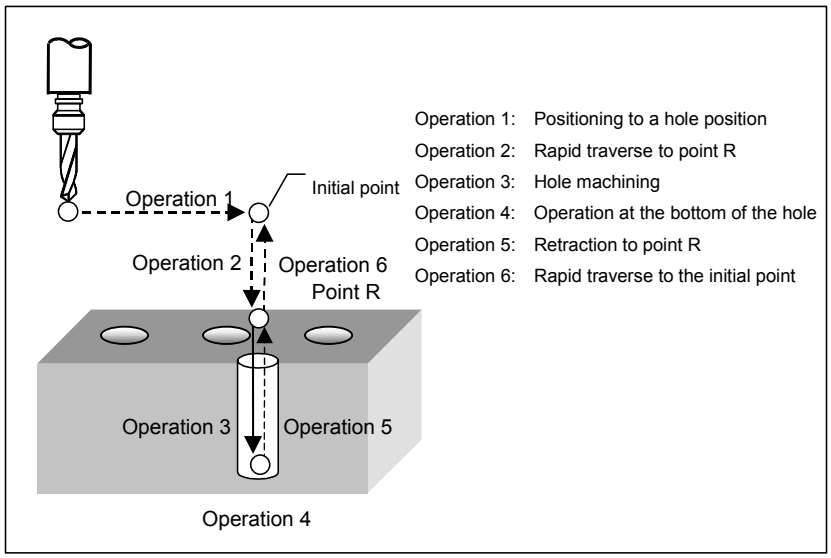
In this example, the repositioning position is point d. Automatic operation after repositioning starts at the N50 block.

4.9.6 Operation Procedure for a Canned Cycle for Drilling

Explanation

- Retract

When the TOOL WITHDRAW switch is turned on during a canned cycle for drilling (abbreviated as a canned cycle below), retraction is performed depending on the cycle operation being executed at that time.



1 During operation 1, the tool moves by the retraction distance specified in G10.6 in the same way as for ordinary retraction.
2 During operation 2, the tool stops operation 2, moves to the initial point, and stops.
3 During operation 3, the tool stops operation 3, executes cycle operations 4, 5, and 6 from that position, and stops at the initial point.
4 During operation 4, 5, or 6, the tool continues the operation and stops at the initial point.
When the TOOL WITHDRAW switch is turned on during operation 2 to 6, the tool does not move according to the retraction specified in G10.6. After the TOOL WITHDRAW switch is turned on and the tool moves to the initial point, however, the tool withdrawal mode is set.
When the second or subsequent canned cycle is being executed and the TOOL WITHDRAW switch is turned on during operation 2 to 6, the retraction position differs depending on G98 (return to initial level) or G99 (return to point R level).
   - G98 (return to initial level): The tool moves to the initial level.
   - G99 (return to point R level):The tool moves to the point R level.
5 During operation 2 to 6, the tool also moves to the initial point and stops when the TOOL WITHDRAW switch is turned on without the G10.6 command specified.

- Repositioning

When the tool is at the retraction position and the cycle start switch is pressed, repositioning is performed for the canned cycle.

1 Repositioning performed when the TOOL WITHDRAW switch is turned on during operation 1
After the completion of repositioning, automatic operation is resumed in the same way as for ordinary repositioning.

2 Repositioning performed when the TOOL WITHDRAW switch is turned on during operation 2
The canned cycle is reexecuted from operation 2.

3 Repositioning performed when the TOOL WITHDRAW switch is turned on during operation 3
The canned cycle is reexecuted from operation 2.

4 Repositioning performed when the TOOL WITHDRAW switch is turned on during operation 4, 5, or 6
The canned cycle is reexecuted for the same hole position from operation 2.

4.10 RETRACE

M

Overview

The tool can retrace the path along which the tool has moved so far (reverse execution). Furthermore, the tool can move along the retraced path in the forward direction (forward reexecution). After forward reexecution is performed until the tool reaches the position at which reverse execution started, machining is continued as programmed.

Procedure

- Forward execution → reverse execution

To perform forward execution of a program, set the "REVERSE" switch on the machine operator's panel to off, then perform a cycle start operation. If the "REVERSE" switch on the machine operator's panel is set to on, reverse execution or the end of reverse execution results.

To perform reverse execution of a program, use one of the following three methods:

1) Set the "REVERSE" switch on the machine operator's panel to on during forward execution of a block.
2) Perform a single block stop operation during forward execution, then set the "REVERSE" switch on the machine operator's panel to on.
3) Perform a feed hold stop operation during forward execution, then set the "REVERSE" switch on the machine operator's panel to on.

When method 1) is used, reverse execution starts after the end of the block being executed (after execution up to the single block stop position). Reverse execution does not start as soon as the "REVERSE" switch on the machine operator's panel is set to on.

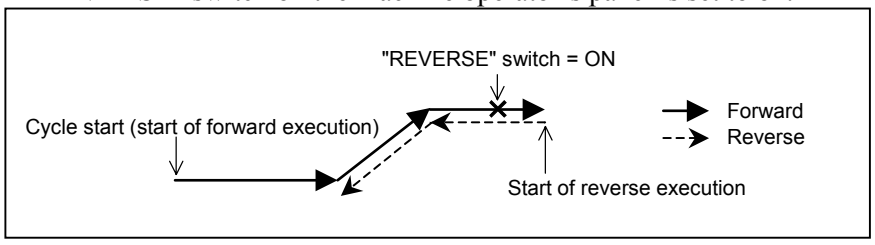


Fig. 4.10 (a)

If functions for gas cutting machine are enabled, however, only when linear interpolation (G01), circular interpolation (G02, G03), or skip function (G31) is specified, reverse execution starts at the position at which the "REVERSE" switch is set to on.

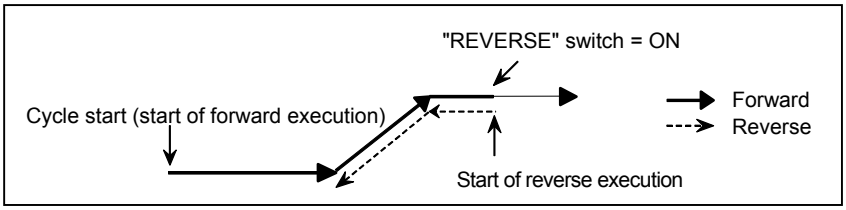


Fig. 4.10 (b)

When method 2) is used, performing a cycle start operation starts reverse execution from the position at which a single block stop takes place.

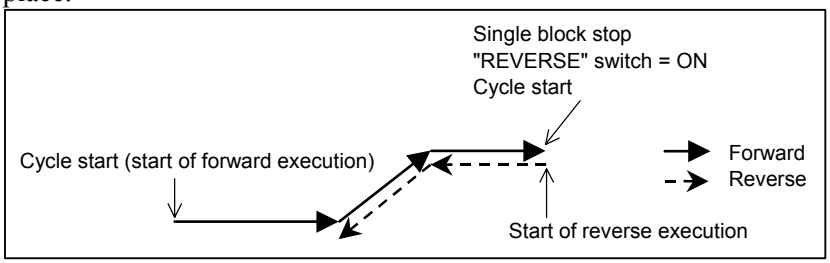


Fig. 4.10 (c)

When method 3) is used, performing a cycle start operation starts reverse execution from the position at which a feed hold stop takes place.

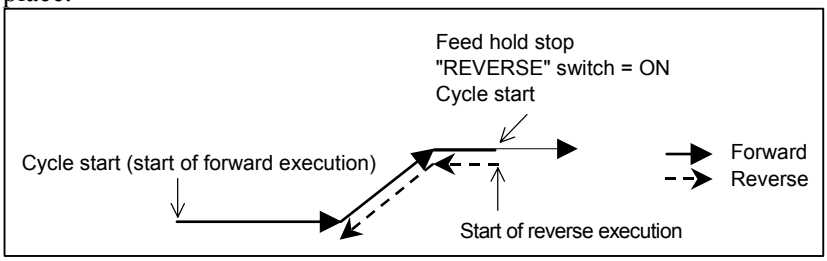


Fig. 4.10 (d)

- Reverse execution → forward reexecution

To perform forward reexecution of a program, use one of the following three methods:

1) Set the “REVERSE” switch on the machine operator’s panel to off during reverse execution of a block.
2) Set the “REVERSE” switch on the machine operator’s panel to off after a single block stop takes place during reverse execution.
3) Set the “REVERSE” switch on the machine operator’s panel to off after a feed hold stop takes place during reverse execution.

When method 1) is used, forward reexecution starts after the block being executed ends (after execution up to the position at which a single block stop takes place). Forward reexecution does not start as soon as the “REVERSE” switch on the machine operator’s panel is set to off.

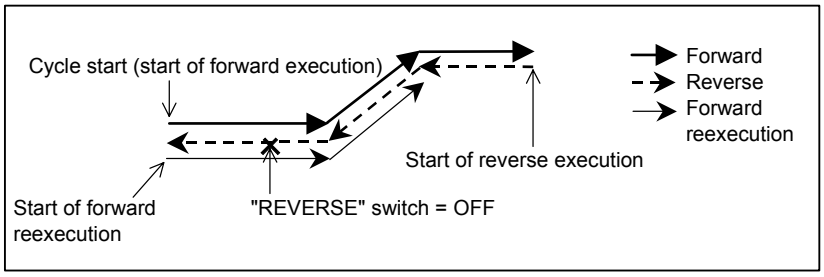


Fig. 4.10 (e)

If functions for gas cutting machine are enabled, however, only when linear interpolation (G01), circular interpolation (G02, G03), or skip function (G31) is specified, forward reexecution starts at the position at which the “REVERSE” switch is set to 0.

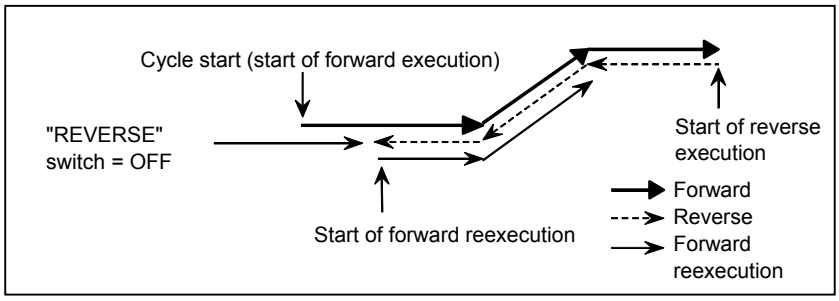


Fig. 4.10 (f)

When method 2) is used, performing a cycle start operation starts forward reexecution from the position at which a single block stop takes place.

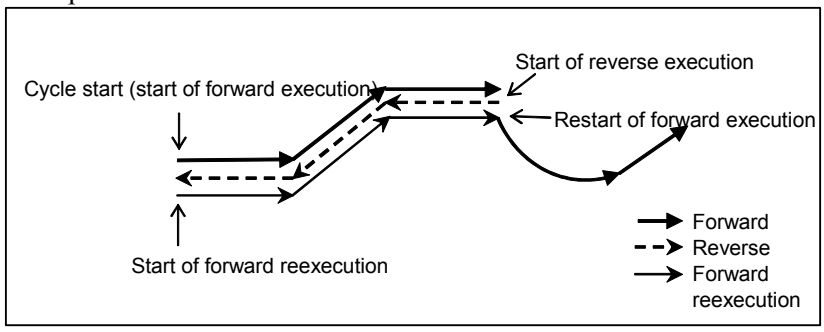


Fig. 4.10 (g)

When method 3) is used, performing a cycle start operation starts forward reexecution from the position at which a feed hold stop takes place.

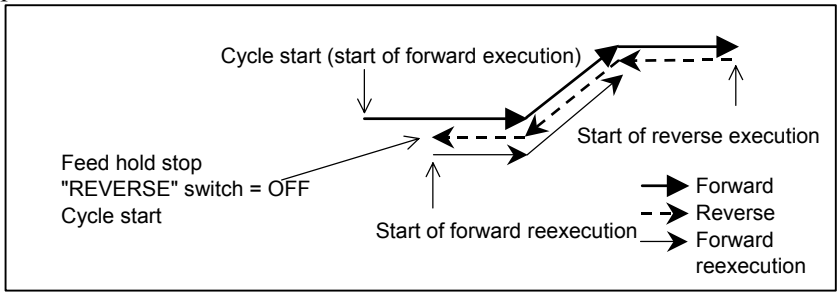


Fig. 4.10 (h)

- Reverse execution → end of reverse execution → forward reexecution

When a block to be executed is no longer present during reverse execution (when reverse execution has been performed up to the block where forward execution started, or when forward execution has not yet been performed), the reverse execution end state is entered, and operation stops.

Even when a cycle start operation is performed while the "REVERSE" switch on the machine operator's panel is held on, operation is not performed, and the reverse execution end state is maintained. Forward reexecution (or forward execution) is started by setting the "REVERSE" switch on the machine operator's panel to off then performing a cycle start operation.

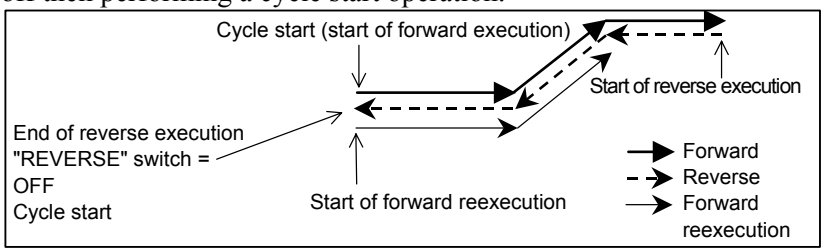


Fig. 4.10 (i)

- Forward reexecution → forward execution

After forward reexecution is performed up to the block at which reverse execution started, forward execution starts automatically, and commands are read from the program again and executed. No particular operation is required.

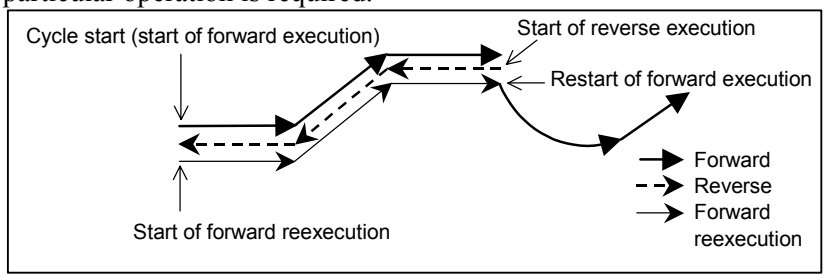


Fig. 4.10 (j)

If reverse execution was performed after feed hold stop, forward reexecution ends when the feed hold stop position is reached, then forward execution is performed. Also if single block operation was performed, forward reexecution ends at the single block stop position.

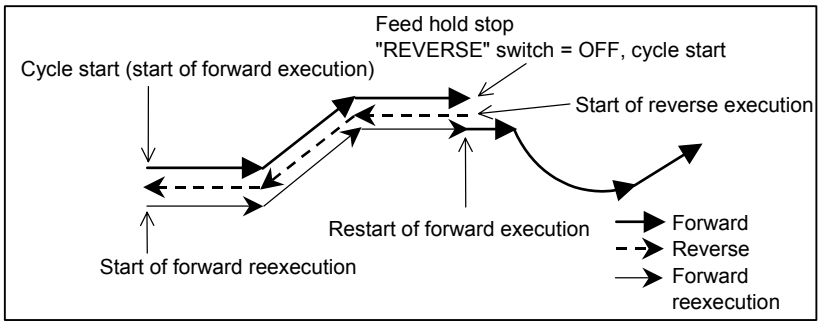


Fig. 4.10 (k)

Explanation

- Reverse execution and forward execution

Usually in automatic operation, a program is executed in the programmed order. This is called forward execution. This function allows a program executed by forward execution to be executed in the reverse direction. This is called reverse execution. Reverse execution allows the tool to retrace the path along which the tool has moved during forward execution.
Reverse execution of a program can be performed only for blocks that have been executed by forward execution.
Furthermore, in single block operation, reverse execution can also be performed on a block-by-block basis .

- Forward reexecution

Blocks that have been executed by reverse execution can be reexecuted in the forward direction up to the block from which reverse execution started. This is called forward reexecution. Forward reexecution allows the tool to retrace the same tool path as in forward execution until the position at which reverse execution started is reached.
After the block from which reverse execution started is reached, the program is executed again in the programmed order (forward execution).
Furthermore, in single block operation, forward reexecution can also be performed on a block-by-block basis.

- End of reverse execution

When a block to be executed is no longer present during reverse execution (when stored blocks have all been executed during reverse execution, or when forward execution has not yet been performed), operation stops. This is called the end of reverse execution.

- Status indication

During reverse execution, characters "RVRS" blink on the screen. During forward reexecution, characters "RTRY" blink to indicate that forward reexecution is in progress. The "RTRY" indication is kept blinking until the block at which reverse execution started is reached and normal operation starts (until forward execution is restarted).
When a block to be executed is no longer present during reverse execution, or if an attempt is made to perform reverse execution for a block that cannot be executed by reverse execution, characters "RVED" blink, notifying the user that reverse execution can no longer be performed.

- Number of blocks that can be executed by reverse execution

Up to about 100 blocks can be executed by reverse execution. Depending on the specified program, the maximum number of executable blocks may decrease.

- Reset

A reset operation (the reset button on the MDI panel, the external reset signal, or the reset & rewind signal) clears the blocks stored for reverse execution.

- Feedrate

A feedrate to be applied during reverse execution can be specified in parameter No. 1414. If this parameter is set to 0, the feedrate in reverse execution is assumed to be the same as that in forward execution. Rapid traverse, however, is performed always at the rapid traverse rate, regardless of the setting of this parameter.
The feedrate in forward reexecution is always the same as that in forward execution.
In reverse execution or forward reexecution, feedrate override, rapid traverse override, and dry run are allowed.

- Start of reverse execution or forward reexecution after the end of a block

In a block for rapid traverse (G00), linear interpolation (G01), circular interpolation (G02, G03), dwelling (G04), skip cutting (G31), or an auxiliary function in an automatic operation mode (memory operation, part program operation, or MDI operation), reverse execution or forward reexecution can be started. However, reverse execution and forward reexecution do not start as soon as the reverse execution signal status is changed. When the block has ended, that is, after a movement, dwelling, or an auxiliary function is completed, reverse execution or forward reexecution starts.

- Start of reverse execution or forward reexecution after feed hold stop

When a feed hold stop operation is performed during execution of rapid traverse (G00), linear interpolation (G01), circular interpolation (G02, G03), or skip cutting (G31), then the reverse execution signal status is changed and operation is restarted, reverse execution or forward reexecution can be started immediately from the stop position. This cannot be performed when dwelling (G04) or an auxiliary function is being executed.

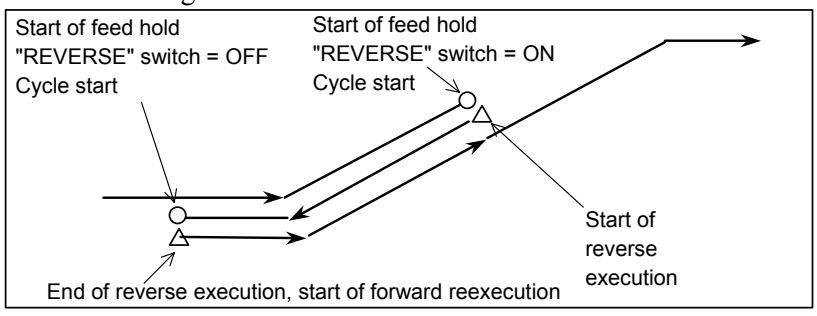


Fig. 4.10 (l)

When reverse execution is started after feed hold stop operation, the part from the start point of that block to the feed hold stop position is stored as one block. Therefore, when forward reexecution is performed with the single block switch set to 1, a single block stop takes place as soon as the position at which reverse execution started is reached.

- Start of reverse execution or forward reexecution after single block stop

After a single block stop takes place, reverse execution or forward reexecution can be started immediately when the reverse execution signal status is changed and restart operation is performed.

Limitation

- Blocks that cannot be executed by reverse execution

In the modes listed below, reverse execution cannot be performed.
When one of these commands appears during reverse execution, reverse execution ends immediately and "RVED" is displayed.

- Involutes interpolation (G02.2, G03.2)
- Exponential interpolation (G02.3, G03.3)
- Three-dimensional circular interpolation (G02.4, G03.4)
- NURBS interpolation (G06.2)
- Cylindrical interpolation (G07.1,G107)
- Polar coordinate interpolation (G12.1, G13.1,G112,G113)
- Polar coordinate command (G16)
- Functions related thread cutting (G33,G34,G35,G36)
- 3-dimensional cutter compensation (G41.2, G41.3, G42.2, G42.3)
- Tool length compensation in tool axis direction (G43.1)
- Tool center point control (G43.4, G43.5)
- Polygon turning (G50.2, G51.2)
- Tool axis direction control (G53.1)
- Single direction positioning (G60)
- Tapping mode (G63)
- Tapping cycle (G84,G74)
- Rigid tapping cycle (G84,G74,G84.2,G84.3)
- Fine boring ccyle (G76)
- Back boring ccyle (G87)

It is impossible to perform reverse execution for blocks specifying the commands listed below. If one of these commands appears during reverse execution, reverse execution ends immediately and "RVED" is displayed.
Some of these commands turn a mode on and off. It is possible to start reverse execution and perform forward reexecution in a mode set by such a command. However, if a block that turns the mode on or off is reached during reverse execution, the reverse execution ends at that block, and "RVED" is displayed.

- Functions related AI contour control (G05,G05.1,G08)
- HRV3 on/off (G05.4)
- Hypothetical axis interpolation (G07)
- Inch/metric conversion (G20, G21)
- Stored stroke check on/off (G22, G23)
- Functions related referece position return (G27, G28, G29, G30)
- Floating referece position return (G30.1)
- 3-dimensional coordinate system conversion (G68, G69)
- Feature coordinate system (G68.2)

- Figure copying (G72.1,G72.2)
- Chopping (G81.1)
- Index table indexing
- Cs contouring control
- Spindle positioning

- Manual intervention

To execute a program in the reverse direction after a feed hold stop or single block stop, when manual intervention is performed after the stop, make a return to the original position and then turn on the reverse signal. Movement by manual intervention is ignored during reverse execution and forward reexecution.

- Single block stop position

A block that is internally generated by the control unit is also treated as one block during reverse execution.

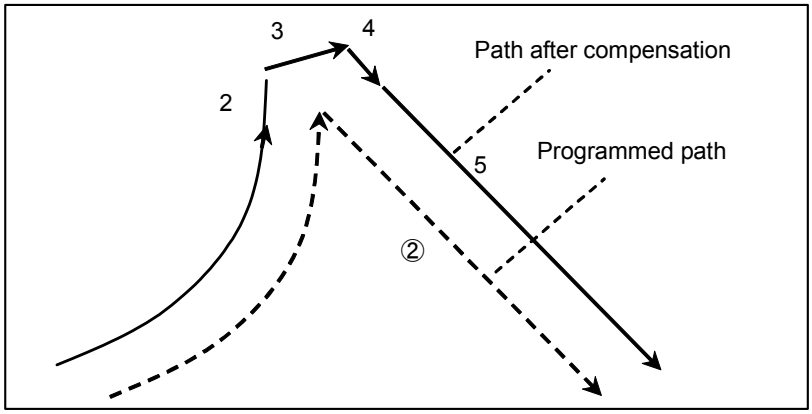


Fig. 4.10 (m) Path when cutter compensation is applied

In the above example, the program specifies two blocks, but in actual operation, move commands for five blocks are generated.
In such a case, positions at which a single block stop takes place may differ between forward execution and reverse execution.

- Positioning (G00)

When non-linear type positioning is performed (bit 1 (LRP) of parameter No. 1401 is set to 0), the tool path in reverse execution and that in forward execution do not match. The tool path in forward reexecution is the same as that in forward execution.
When linear type positioning is performed (bit 1 (LRP) of parameter No. 1401 is set to 1), the tool path in reverse execution is the same as that in forward execution.

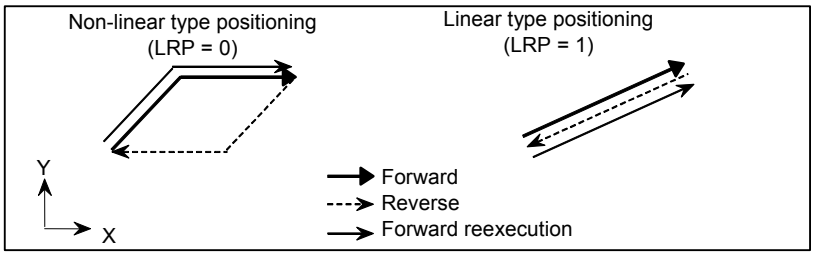


Fig. 4.10 (n)

- Dwell command (G04)

During reverse execution or forward reexecution, the dwell command (G04) is executed in the same way as in normal operation.

- Programmable data input (G10)

Tool compensation values, parameters, pitch error data, workpiece origin offsets, and tool life management values set or modified by programmable data input (G10) are ignored during reverse execution and forward reexecution.

- Skip function (G31) and automatic tool length compensation (G37)

The skip signal and the automatic tool length measurement signal are ignored during reverse execution and forward reexecution. During reverse execution and forward reexecution, the tool moves along the path that the tool has actually passed during forward execution.

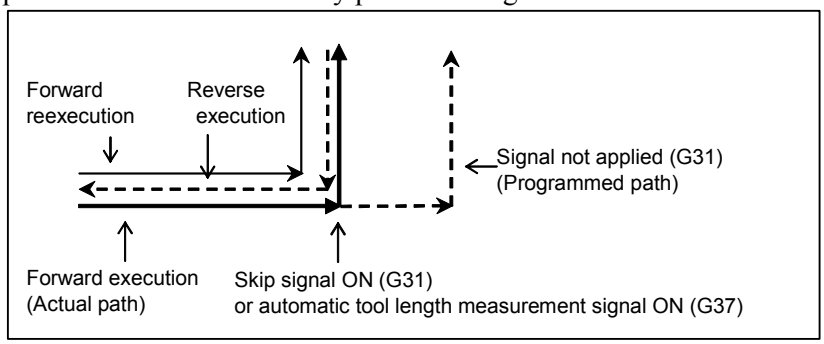


Fig. 4.10 (o)

- Setup of a coordinate system (G92, G54 to G59, G54.1P_, G52, and G92.1)

When setup of a coordinate system (G92, G54 to G59, G54.1P_, and G52) is specified during reverse execution, the indicated current position may differ from the position indicated during forward execution. However, the actual machine position does not differ.

- Mirror image

When a block to which a mirror image is applied by programmable mirror image (G50.1, G51.1) is executed during reverse execution, the tool moves along the actual path resulting from the application of a mirror image in the reverse direction.

When a mirror image is applied to a block by setting or a machine signal, the block with the mirror image not applied is stored. Mirror image application by setting or a machine signal is enabled also during reverse execution and forward reexecution. Therefore, during reverse execution and forward reexecution, the mirror image by setting data or machine signal must be turned on and off so that this on/off status and the on/off status during forward execution match.

- Changing offsets

Even when cutter compensation data or tool length offsets are changed during reverse execution or forward reexecution, the change in compensation or offset data does not become valid until forward reexecution ends and normal operation starts. Until then, the tool moves with the offset data that was applied when the block was executed for the first time during forward execution.

- Interrupt type custom macro

(1) Do not initiate any interrupt during reverse execution.
(2) Do not execute an interrupted block and the interrupt program in reverse execution.

- Tool management function

The tool life is not counted during reverse execution and forward reexecution.

- Inverse time feed (G93)

If a nonzero value is set as the feedrate to be applied during reverse execution in parameter No. 1414, a block that moves the tool by inverse time feed during forward execution is executed at the parameter-set feedrate (feed per minute) during reverse execution.
If the feedrate during reverse execution (parameter No. 1414) is not set (= 0), the same feedrate as applied during forward execution is used.

- Maximum spindle speed clamp (G92Sxxxx)

Clamping at a maximum spindle speed specified during reverse execution becomes valid. This means that if G92Sxxxx appears during reverse execution, the spindle speed is clamped at Sxxxx in the subsequent reverse execution. As a result, the clamp speed may differ between reverse execution and forward execution even when the same block is executed. The spindle speed is clamped when the G96 mode is set.

- Auxiliary functions

M, S, T, and the second auxiliary function (B function) are output directly also during reverse execution and forward reexecution.
When specified together with a move command in the same block, M, S, T, and the second auxiliary function (B function) are output with the move command at the same time during forward execution, reverse execution, and forward reexecution. Therefore, the output positions of M, S, T, and the second auxiliary function (B function) during reverse execution differ from those during forward execution and forward reexecution.

- Custom macro operation

Custom macro operations are ignored during reverse execution and forward reexecution.

- Tool retract and recover function

For retract operation and repositioning operation by the tool retract and recover function, reverse execution cannot be performed. Retract operation and repositioning operation are ignored during reverse execution and forward reexecution.

- AI contour control

During reverse execution and forward reexecution, the feedrate clamp function by acceleration under AI contour control is disabled.

- Display

During reverse execution and forward reexecution, the modal display and the display of the currently executed program are not updated; information obtained at the start of reverse execution is maintained.

Warning

⚠ WARNING

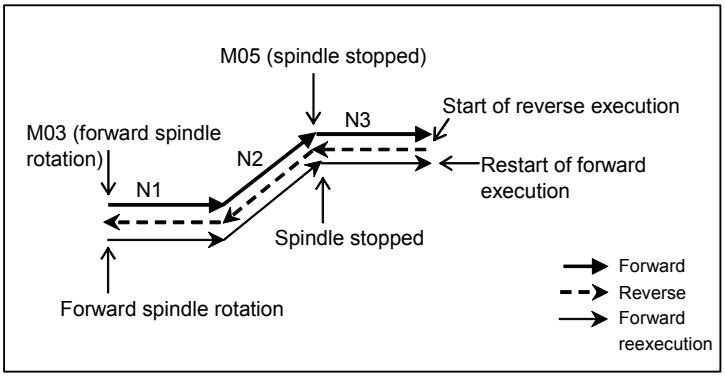
1 Auxiliary functions are output directly even during reverse execution and forward reexecution. Accordingly, the execution status of an auxiliary function during forward execution may be reversed during reverse execution.

Example:

When forward rotation of the spindle (M03) and stop (M05) are specified

When N3 is executed during reverse execution, M05 is output. So, when N2 and N1 are executed during reverse execution, operation is performed with the spindle stopped.

When N1 is executed during forward reexecution, M03 is output. So, when N1 and N2 are executed during forward reexecution, operation is performed with the spindle rotating in the forward direction.



2 To perform reverse execution after a feed hold stop or single block stop operation, be sure to restore the original position if manual intervention has been performed after the stop , then set the "REVERSE" switch to on. Movements made by manual intervention are ignored during reverse execution and forward reexecution. (The same operation as in the manual absolute off state takes place.)

4.11 ACTIVE BLOCK CANCEL FUNCTION

The automatic operation can be stopped by inputting block cancellation signal BCAN<Gn297.0> while automatic operation. After an automatic operation becomes stop, automatic operation signal OP <Fn000.7> and cycle start lamp signal STL <Fn000.5> becomes "0".
All modal information is preserved.

At this time,

(1) If the machine is moving, it stops while decelerating. And the distance remaining is cleared.
(2) If the dwell is executing, it is canceled.
(3) If auxiliary function, spindle speed, tool function, the second auxiliary function is executing, the each strobe signal MF, SF, TF, BF becomes "0" and the block is ended.
If the high speed M/S/T/B interface (bit 7 (MHI) of parameter No.3001 = 1) is used, the block is ended without the each strobe signal MF, SF, TF and BF changing.
Therefore, after completion signal MFIN / SFIN / TFIN / BFIN is matched in the same logical level as the strobe signal, it is necessary to restart the operation.

An automatic operation is restarted from the next block of the canceled block by cycle start.

Active block cancel function

Explanation

Operation when operation restarts

G90/G91

When the operation is restarted, the operation of the restart is decided depending on the command in the next block of the canceled block.
When the next block is absolute command (G90), the operation is restarted toward the terminal from the stopping position to the next block. When the next block is incremental command (G91), the incremental operation is restarted from the stopping position.

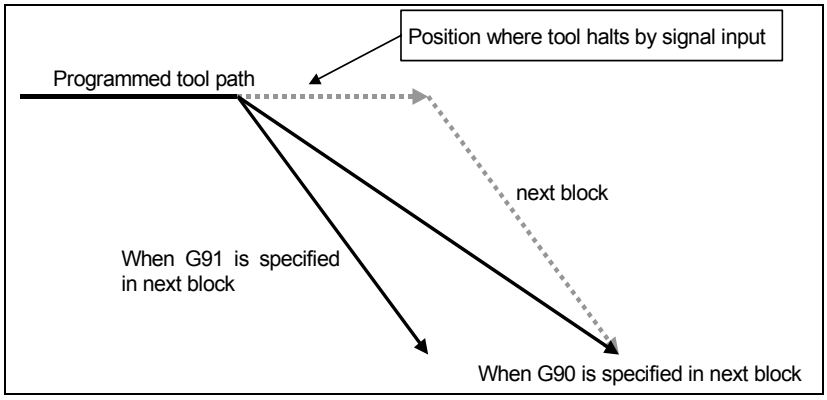


Fig. 4.11 (a) Absolute / Incremental

⚠ CAUTION

1 Because the tool becomes a start from the canceled position when an automatic operation restarts, the route of an actual movement is different from original program route. Therefore, there is a possibility that the tool route and the work piece interfere at the restart. Please do the restart operation after confirming neither the tool nor work piece interfere.

2 If the part program is incremental command, the terminal of the next block shifts cancellation amount when automatic operation restarts. This amount of the gap keeps remaining until the absolute command is executed next. Therefore, this function cannot be used to check the block terminal by the program made by the incremental command.

Tool radius / tool nose radius compensation

The offset mode is temporarily canceled at that time when canceling in the tool radius / tool nose radius compensation mode. And, the start-up operation is done at the restart.
For example, when canceling in N20 of the undermentioned program, the movement of restarting is the start-up operation toward the position commanded with N30 in the figure below.

Example

```
N10 G90 G01 G42 X_ D_ S_ F_ ;
N20 X_ ;
N30 X_ Y_;
N40 X_ ;
```

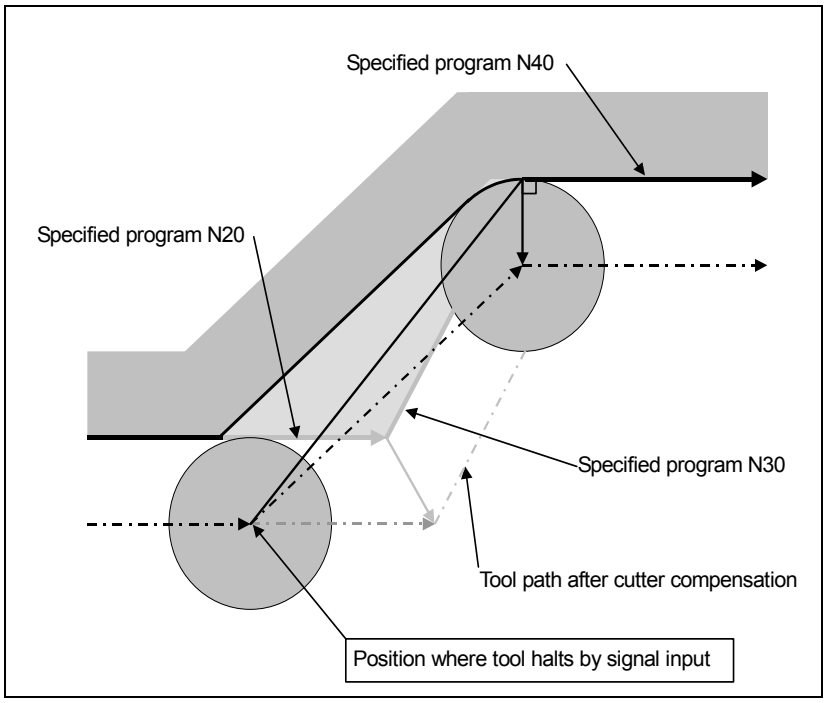


Fig. 4.11 (b) Tool radius / tool nose radius compensation

Canned cycle for drilling

When a block is canceled during a canned cycle for drilling, that canned cycle is canceled. Program execution is restarted from the next block.

Example1
```
N10 G91 G84 X_ Y_ Z_ R_ K3 ;
N20 G90 G00 X_ Y_ Z_ ;
```

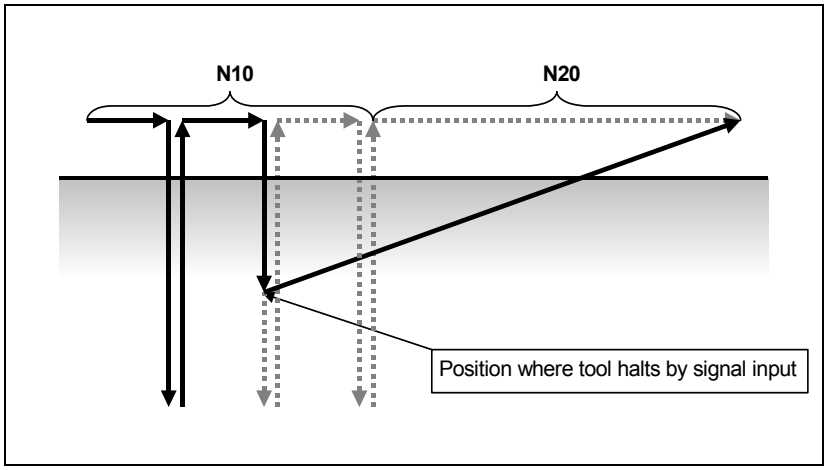


Fig. 4.11 (c)Canned cycle for drilling 1

Example 2
```
N10 G90 G84 X_ Y_ Z_ R_ ;
N20 X_ Y_ ;
N30 X_ Y_ ;
N40 X_ Y_ ;
N50 G00 X_ Y_ Z_ ;
```

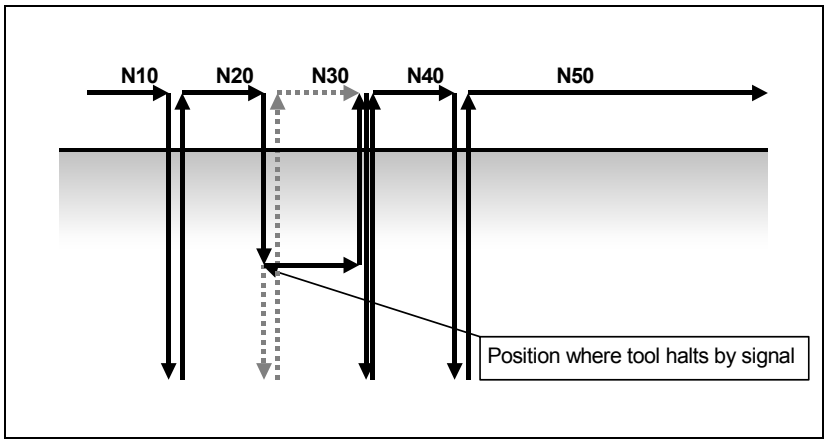


Fig. 4.11 (d) Canned cycle for drilling 2

At the restart, the operation is executed from the next N30 block when the operation is canceled while executing in N20 cycle.
At this time, because the G84 mode is maintained, the tool positions in R point after moving to the X-Y position commanded with N30 and starts the hole opening operation.

Notes

⚠ CAUTION

1 Please confirm neither the tool route nor work piece interference when the operation is restarted.
2 When an automatic operation is restarted, the route of an actual movement is different from an original route because it is a start from the canceled position. Therefore, there is a possibility that the tool route and the work piece interference at the restart.
3 When reset or the emergency stops is executed during the stopping by this function, this function becomes invalid. In a word, maintained various data and modal data are changed to the clear or reset states by the reset operation.
4 This function is effective only while automatic operation. (OP=1 and STL=1)
5 If the part program is incremental command, the terminal of the next block shifts cancellation amount when automatic operation restarts. This amount of the gap keeps remaining until the next absolute command is executed. Therefore, this function cannot be used to check the block terminal by the program made by the incremental command.
6 This function is not available during AI contour control mode. Even if the cancel signal is input, it is ignored.

5 TEST OPERATION

The following functions are used to check before actual machining whether the machine operates as specified by the created program.

5.1 MACHINE LOCK AND AUXILIARY FUNCTION LOCK 1178
5.2 FEEDRATE OVERRIDE....1180
5.3 RAPID TRAVERSE OVERRIDE....1181
5.4 DWELL/AUXILIARY FUNCTION TIME OVERRIDE....1182
5.5 DRY RUN....1184
5.6 SINGLE BLOCK....1185
5.7 HIGH SPEED PROGRAM CHECK FUNCTION....1187

5.1 MACHINE LOCK AND AUXILIARY FUNCTION LOCK

To display the change in the position without moving the tool, use machine lock.
There are two types of machine lock: all-axis machine lock, which stops the movement along all axes, and specified-axis machine lock, which stops the movement along specified axes only. In addition, auxiliary function lock, which disables M, S, T, and B (2nd auxiliary function) commands, is available for checking a program together with machine lock.

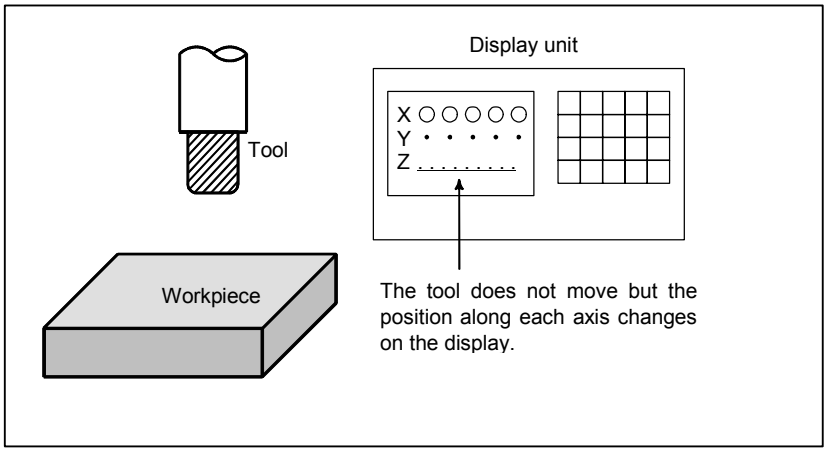


Fig. 5.1 (a) Machine lock

Machine lock and auxiliary function lock

Procedure

- Machine Lock

Press the machine lock switch on the operator's panel. The tool does not move but the position along each axis changes on the display as if the tool were moving.
Some machines have a machine lock switch for each axis. On such machines, press the machine lock switches for the axes along which the tool is to be stopped. Refer to the appropriate manual provided by the machine tool builder for machine lock.

> ⚠ **WARNING**
> The positional relationship between the workpiece coordinates and machine coordinates may differ before and after automatic operation using machine lock. In such a case, specify the workpiece coordinate system by using a coordinate setting command or by performing manual reference position return.

- Auxiliary function lock

Press the auxiliary function lock switch on the operator's panel. M, S, T, and B codes are disabled and not executed. Refer to the appropriate manual provided by the machine tool builder for auxiliary function lock.

Limitation

- M, S, T, B command by only machine lock

M, S, T and B commands are executed in the machine lock state.

- Reference position return under machine lock

When a G27, G28, or G30 command is issued in the machine lock state, the command is accepted but the tool does not move to the reference position and the reference position return LED does not go on.

- M codes not locked by auxiliary function lock

M00, M01, M02, M30, M98, M99, and M198 (subprogram call function) commands are executed even in the auxiliary function lock state. M codes for calling a subprogram (parameters No. 6071 to 6079) and those for calling a custom macro (parameters No. 6080 to 6089) are also executed.

5.2 FEEDRATE OVERRIDE

A programmed feedrate can be reduced or increased by a percentage (%) selected by the override dial. This feature is used to check a program.
For example, when a feedrate of 100 mm/min is specified in the program, setting the override dial to 50% moves the tool at 50 mm/min.

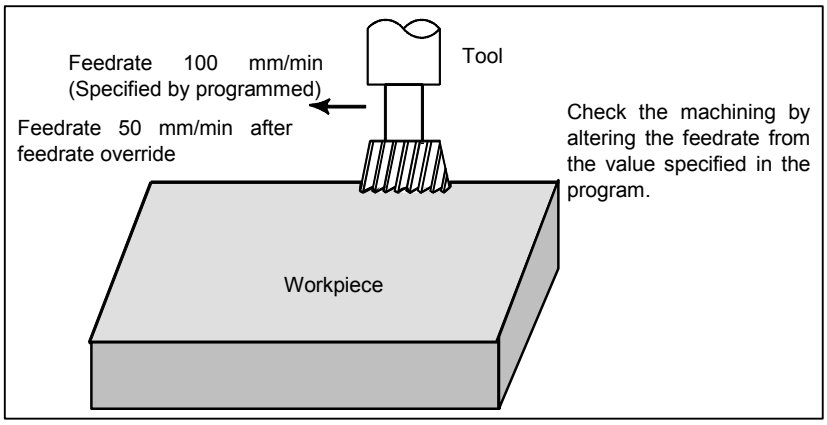


Fig. 5.2 (a) Feedrate override

Feedrate override

Procedure

Set the feedrate override dial to the desired percentage (%) on the machine operator's panel, before or during automatic operation.
On some machines, the same dial is used for the feedrate override dial and jog feedrate dial. Refer to the appropriate manual provided by the machine tool builder for feedrate override.

Limitation

- Override range

The override that can be specified ranges from 0 to 254%. For individual machines, the range depends on the specifications of the machine tool builder.

- Override during thread

During the threading process, the override setting is ignored; it is always regarded as 100% during the process.

5.3 RAPID TRAVERSE OVERRIDE

An override of four steps (F0, 25%, 50%, and 100%) can be applied to the rapid traverse rate. F0 is set by a parameter No. 1421.

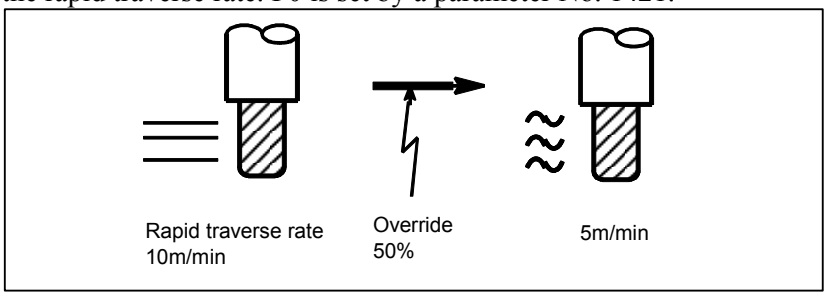


Fig. 5.3 (a) Rapid traverse override

Rapid traverse override

Procedure

Select one of the four feedrates with the rapid traverse override switch during rapid traverse. Refer to the appropriate manual provided by the machine tool builder for rapid traverse override.

Explanation

The following types of rapid traverse are available. Rapid traverse override can be applied for each of them.

(1) Rapid traverse by G00
(2) Rapid traverse during a canned cycle
(3) Rapid traverse in G27, G28, G29, G30, G53
(4) Manual rapid traverse
(5) Rapid traverse of manual reference position return

5.4 DWELL/AUXILIARY FUNCTION TIME OVERRIDE

An override can be applied to dwell and the auxiliary function (M/S/T/B) in increments of 1% in the range from 0 to 100%.
For the auxiliary function, however, if the override is less than 100%, the next block is started after a lapse of actual processing time plus shortage time.

Dwell/auxiliary function time override

Procedure

Press the 1% rapid traverse override selection switch and turn the 1% rapid traverse override switch to select the desired percentage.
For details on 1% rapid traverse override, refer to the manual provided by the machine tool builder.

Explanation

- Dwell

An override can be applied to dwell.
[Example]
Override and dwell time when G04 P10000; is executed

Override	Dwell time
25%	40.0 sec.
50%	20.0 sec.
75%	13.3 sec.
100%	10.0 sec.

NOTE
An override is disabled for dwell per revolution.

- AUXILIARY functions (M/S/T/B)

An override can be applied in the time from when M/S/T/B is sent until the FIN signal is turned off (or until MFIN, SFIN, TFIN, and BFIN are changed in the case of the high speed M/S/T/B). The time required to communicate with PMC is included.
When the FIN signal is turned off and then shortage time elapses, the next block is started.
[Example]
When the following program is executed with the override set to 60%

```
N10 M03 ;
N20 · · · · · ;
```

When the processing time of the N10 block is 80 msec, the N20 block is started approx. 56 msec after completion of the processing.

Shortage time = 80 msec × 100/60 - 80 msec = 53.333 ≅ 56 msec

NOTE
The time count is incremented every 4 msec or 8 msec and the fraction is rounded up. The counting cycle of time may depend on the system.

The 1% rapid traverse override signal reads the value upon completion of the auxiliary function. Accordingly, the value is not ensured if the override value changes between before and after the reading.

When the movement command and auxiliary function are specified in the same block:

(1) When the movement command is completed first
- An override is enabled.

(2) When the auxiliary command is completed first
- An override is disabled.

When auxiliary functions are specified in the same block, the shortage time is calculated using the auxiliary function with the longest processing time.

NOTE
1 This function is disabled for dwell by PMC axis control (04H) and the auxiliary functions (12H, 14H, 15H).
2 For the dwell, when the 1% rapid traverse override signal is 0%, a transition to the next block is not performed.
3 The time when a feed hold is caused in the auxiliary function during shortage time is not included in the shortage time.
4 For the auxiliary function, when the 1% rapid traverse override signal is 0%, the setting of parameter No. 3018 is used as the override value. When the setting of the parameter is 0, it is assumed to be 10%. While the 1% rapid traverse override signal is 0%, a transition to the next block is not performed.
5 For auxiliary functions M02 and M30, bit 7 (OVM) of parameter No. 3002 can be used to enable or disable an override.

5.5 DRY RUN

The tool is moved at the feedrate specified by a parameter regardless of the feedrate specified in the program. This function is used for checking the movement of the tool under the state that the workpiece is removed from the table.

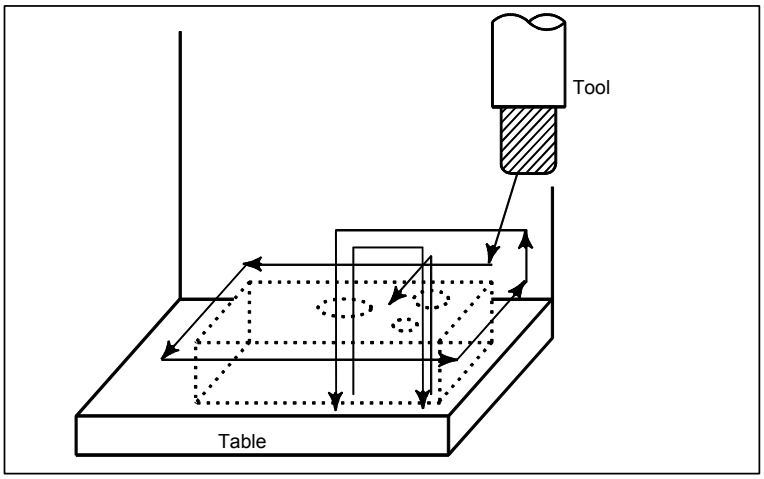


Fig. 5.5 (a) Dry run

Dry run

Procedure

Press the dry run switch on the machine operator's panel during automatic operation.
The tool moves at the feedrate specified in a parameter. The rapid traverse switch can also be used for changing the feedrate.
Refer to the appropriate manual provided by the machine tool builder for dry run.

Explanation

- Dry run feedrate

The dry run feedrate changes as shown in the table below according to the rapid traverse switch and parameters.

Table 5.5 (a)

Rapid traverse switch	Program command	
	Rapid traverse	Cutting feed
ON	Rapid traverse rate	Dry run feedrate × Jvmax(*2)
OFF	Dry run feedrate × JV, or rapid traverse rate (*1)	Dry run feedrate × JV(*2)

Max. cutting feedrate ..... Setting by parameter No.1430
Rapid traverse rate ..... Setting by parameter No.1420
Dry run feedrate ..... Setting by parameter No.1410

(*1) Dry run feedrate × JV when parameter RDR (No. 1401#6) is 1.
Rapid traverse rate when parameter RDR is 0.
JV Jog feedrate override

(*2) Clamped to the maximum cutting feedrate
Jvmax Maximum value of jog feedrate override

5.6 SINGLE BLOCK

Pressing the single block switch starts the single block mode. When the cycle start button is pressed in the single block mode, the tool stops after a single block in the program is executed. Check the program in the single block mode by executing the program block by block.

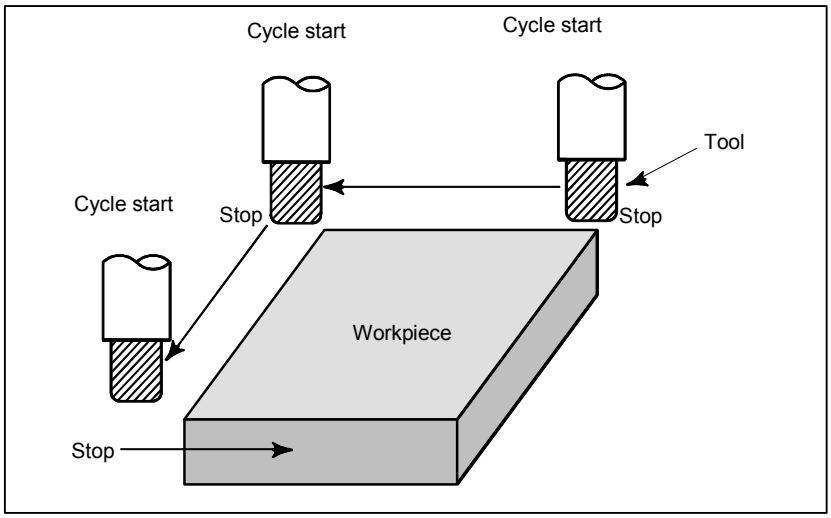


Fig. 5.6 (a) Single block

Single block

Procedure

1 Press the single block switch on the machine operator's panel.
The execution of the program is stopped after the current block is executed.

2 Press the cycle start button to execute the next block. The tool stops after the block is executed.
Refer to the appropriate manual provided by the machine tool builder for single block execution.

Explanation

- Reference position return and single block

If G28, G29, and G30 are issued, the single block function is effective at the intermediate point.

- Single block during a canned cycle

In a canned cycle, the single block stop points are the end of <1>, <2>, and <6> shown below. When the single block stop is made after the point <1> or <2>, the feed hold LED lights.

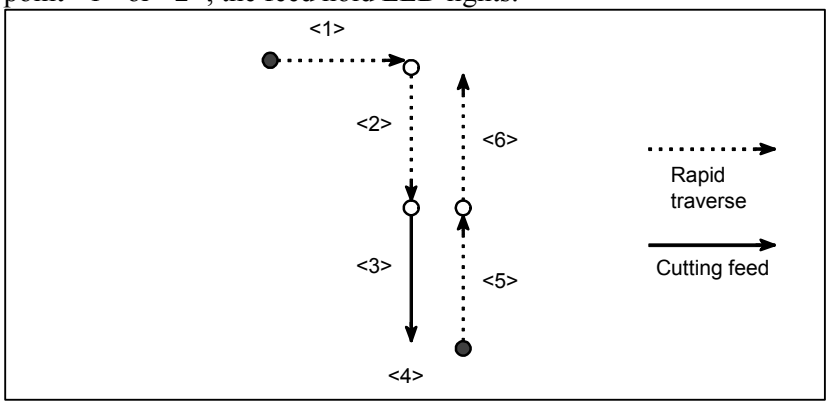


Fig. 5.6 (b) Single block during canned cycle

- Subprogram call and single block

Single block stop is not performed in a block containing M98P_;. M99; or G65.

However, single block stop is even performed in a block with M98P_ or M99 command, if the block contains an address other than O, N, P, L.

5.7 HIGH SPEED PROGRAM CHECK FUNCTION

When the cycle start button is pressed with the high speed program check mode set to be enabled, the program syntax and stroke limit are checked without axis movement.
A program check is performed at the maximum speed that can be specified by the CNC, regardless of the specified speed.

NOTE
This function is optional.

High speed program check

Procedure

1 Perform position of the machine at the start position of the machining program in advance.
Set the workpiece coordinate system or tool compensation to the same state as in actual machining.
2 Reset the CNC and press the high speed program check switch on the machine operator's panel to enter the high speed program check mode. In the high speed program check mode, "PRG-CHK" blinks at the lower-left of the CNC screen.
3 Press the cycle start button to execute the program without axis movement. The feedrate is the maximum speed that can be specified by the CNC, regardless of the specified speed.
The program syntax and stroke limit are checked during execution of the program.
4 Upon completion of the check, reset the CNC and release the high speed program check mode with the switch on the machine operator's panel.

Explanation

- Stroke Limit Check

Following the stroke limit check is executed without axes movements during the high speed program check mode.

- Stored stroke check 1
- Stored stroke check 2, 3
- Stroke limit check before move
- Rotation area interference check

If an alarm about the stroke limit occurs in the high speed program check mode, clear the alarm as described below.
(1) Change the mode to other than MEM or DNC.
(2) Reset the CNC.

⚠ WARNING

1 If the machine coordinate system is not established, a stroke limit check is not performed correctly. When a reference position return is completed after power-on, perform a high speed program check.
2 When the width of the prohibited area is small in a stored stroke limit 2 and 3 check and rotation area interference check, if a command to intersect the prohibited area in one block is issued, the stroke limit may not be detected. This means that a program not causing a stroke check alarm in a high speed program check may cause an alarm in actual operation.

- Program Check Feedrate

During the high speed program check mode, the feedrate of program execution becomes the maximum feedrate of CNC system regardless of the specified feedrate.
The feedrate clamp, override and dry run are not effective.
The execution speed of circular interpolation, involute interpolation, spiral/conical interpolation, and NURBS interpolation can be selected by bit 7 (PGF) of parameter No. 1490 from the maximum speed and dry run speed.

NOTE

1 The execution time of dwell is the same as a normal operation.
2 The PS011 alarm occurs when there is no F command in a normal operation. However, it is executed at maximum speed without the alarm in the high speed program check mode even if there is no F command.

- Workpiece Coordinate System Preset

Upon completion of the high speed program check mode, the workpiece coordinate system is automatically preset. This enables program operation without making a reference position return upon completion of a program check. For details, refer to the manual provided by the machine tool builder.

> ⚠ **WARNING**
> When the coordinate system is set before the high speed program check mode is turned on by the work coordinate system setting G92 (machining center system and G code system B or C of the lathe system), G50 (G code system A of the lathe system) and the local coordinate system setting G52, their setting values are deleted by the execution of the work coordinate system preset.

- Machine Coordinate Display For Program Check

During the high speed program check mode, it is possible to change the display of machine coordinates from the actual machine coordinates to the machine coordinates for the program check by setting bit 7 (PGM) of parameter No.11320.
And it is possible to change the display of machine coordinate for each axis from the actual machine coordinate to the machine coordinate for the program check by setting bit 4 (PGA) of parameter No.3115.

- Tool life

If tool life management command or tool management command is specified during the high speed program check mode, tool life is not counted.

- Unavailable command

Following functions can not be specified during the high speed program check mode. If the following functions are specified, the alarm PS5364 occurs.

- High speed cycle cutting
- Chopping
  As for the chopping by the input signal:
  The alarm PS5364 occurs when the program check is started during the chopping by the input signal.
  The alarm PS5364 occurs when the chopping by the input signal is started at the program check execution.
- Simple electronic gear box

･ Commands not covered by a high speed program check

Auxiliary functions and other commands are not covered by a high speed program check. For details, refer to the manual provided by the machine tool builder.

6 SAFETY FUNCTIONS

To immediately stop the machine for safety, press the Emergency stop button. To prevent the tool from exceeding the stroke ends, Overtravel check and Stored stroke check are available. This chapter describes emergency stop, overtravel check, and stored stroke check.

Chapter 6, "SAFETY FUNCTIONS", consists of the following sections:

6.1 EMERGENCY STOP 1191
6.2 OVERTRAVEL 1192
6.3 STORED STROKE CHECK 1193
6.4 STROKE LIMIT CHECK BEFORE MOVE.....1198
6.5 WRONG OPERATION PREVENTION FUNCTIONS.....1201

6.1 EMERGENCY STOP

If you press Emergency Stop button on the machine operator's panel, the machine movement stops in a moment.

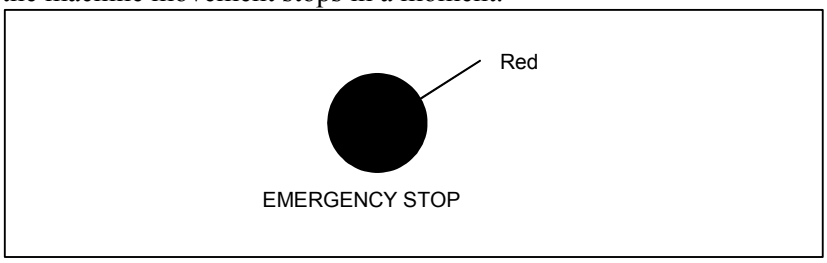


Fig. 6.1 (a) Emergency stop

This button is locked when it is pressed. Although it varies with the machine tool builder, the button can usually be unlocked by twisting it.

Explanation

EMERGENCY STOP interrupts the current to the motor.
Causes of trouble must be removed before the button is released.

6.2 OVERTRAVEL

When the tool tries to move beyond the stroke end set by the machine tool limit switch, the tool decelerates and stops because of working the limit switch and an OVER TRAVEL is displayed.

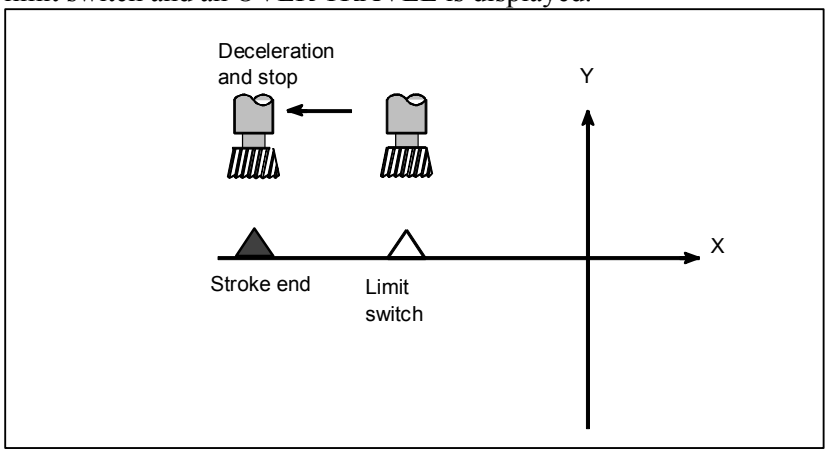


Fig. 6.2 (a) Overtravel

Explanation

- Overtravel during automatic operation

When the tool touches a limit switch along an axis during automatic operation, the tool is decelerated and stopped along all axes and an overtravel alarm is displayed.

- Overtravel during manual operation

In manual operation, the tool is decelerated and stopped only along the axis for which the tool has touched a limit switch. The tool still moves along the other axes.

- Releasing overtravel

Press the reset button to reset the alarm after moving the tool to the safety direction by manual operation. For details on operation, refer to the operator's manual of the machine tool builder.

Alarm

Table6.2 (a)

Alarm No.	Message	Description
OT0506	+ OVERTRAVEL (HARD)	The stroke limit switch in the positive direction was triggered. This alarm is generated when the machine reaches the stroke end. When this alarm is not generated, feed of all axes is stopped during automatic operation. During manual operation, only the feed of the axis on which the alarm occurred is stopped.
OT0507	- OVERTRAVEL (HARD)	The stroke limit switch in the negative direction was triggered. This alarm is generated when the machine reaches the stroke end. When this alarm is not generated, feed of all axes is stopped during automatic operation. During manual operation, only the feed of the axis on which the alarm occurred is stopped.

6.3 STORED STROKE CHECK

Three areas which the tool cannot enter can be specified with stored stroke check 1, stored stroke check 2, and stored stroke check 3.

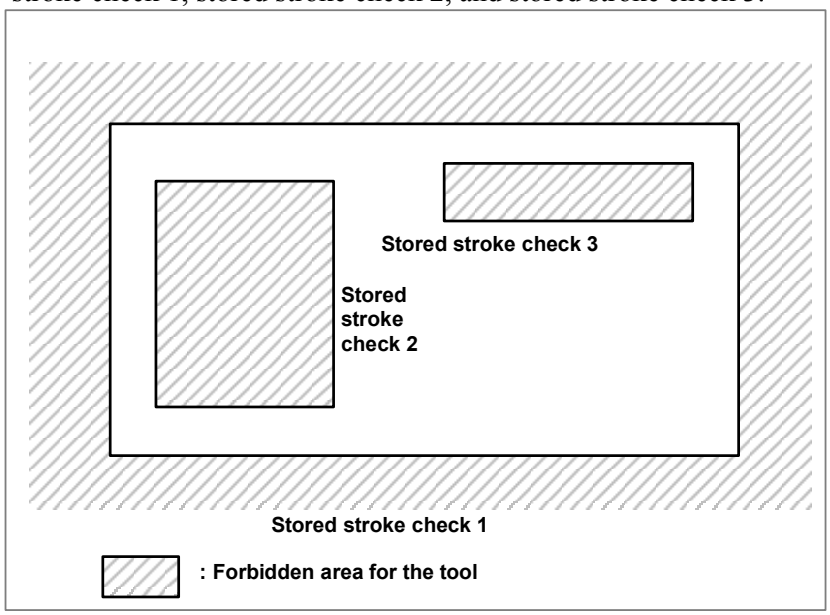


Fig. 6.3 (a) Stroke check

The following shows the areas which the tool cannot enter for each stored stroke check.

- Stored stroke check 1: Outside
- Stored stroke check 2: Outside or inside (switchable)
- Stored stroke check 3: Inside

When the tool moves into the forbidden area, an alarm is displayed and the tool is decelerated and stopped.
When the tool enters a forbidden area and an alarm is generated, the tool can be moved in the reverse direction from which the tool came.
The stored stroke check 2 and 3 functions are optional.

Explanation

- Stored stroke check 1

Parameters (Nos. 1320, 1321 or Nos. 1326, 1327) set boundary. Outside the area of the set limits is a forbidden area. The machine tool builder usually sets this area as the maximum stroke.
When the tool enters a forbidden area and an alarm is generated, the tool can be moved in the reverse direction from which the tool came.
At this time, a signal (overtravel alarm signal) can be output to the PMC if bit 6 (OTS) of parameter No. 1301 is set to 1. In addition, when the tool enters the forbidden area during manual operation, the signal (overtravel alarm signal) can be output to the PMC without generating the alarm by setting bit 1 (NAL) of parameter No. 1300 to 1. With this parameter setting, the alarm is generated when the tool enters the forbidden area during automatic operation.

> ⚠ **CAUTION**
> 1 If the two points for specifying a forbidden area are identical, all areas are handled as forbidden areas for stored stroke check 1.
> 2 The size of a forbidden area must be set carefully. If the size is set incorrectly, the stroke becomes infinite.

- Stored stroke check 2

Parameters (Nos. 1322, 1323) or commands set these boundaries. Inside or outside the area of the limit can be set as the forbidden area. Parameter OUT (No. 1300#0) selects either inside or outside as the forbidden area.
In case of program command a G22 command forbids the tool to enter the forbidden area, and a G23 command permits the tool to enter the forbidden area.
Each of G22; and G23; should be commanded independently of another commands in a block.
The command below creates or changes the forbidden area:

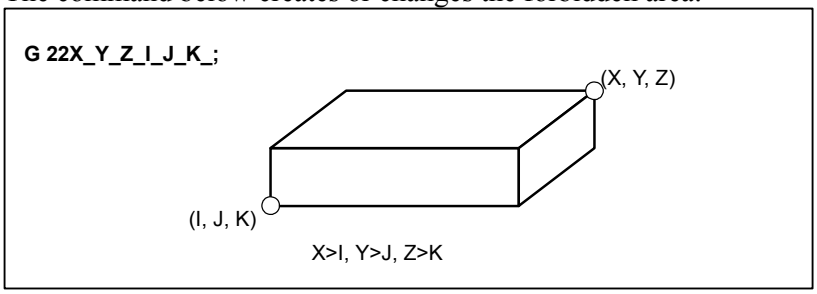


Fig. 6.3 (b) Creating or changing the forbidden area using a program

When setting the area by parameters, points A and B in the figure below must be set.

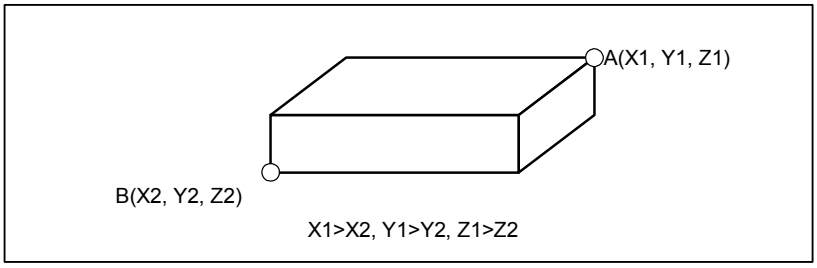


Fig. 6.3 (c) Creating or changing the forbidden area using a parameters

The values X1, Y1, Z1, X2, Y2, and Z2, which are set by parameters No. 1322 and No. 1323, must be specified by the distance from the machine coordinate system (machine unit). The values X, Y, Z, I, J, and K, which are set by a G22 command, must be specified by the distance in the least input increment (input unit).
Values set by a program are then converted in the machine increment and the values are set as the parameters.

- Stored stroke check 3

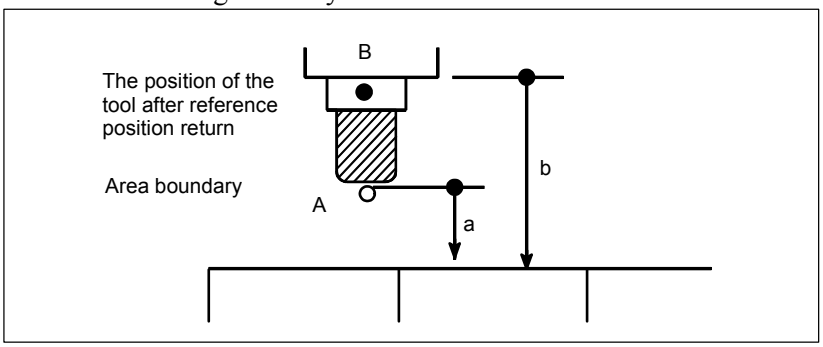
Set the boundary with parameters No. 1324 and 1325. The area inside the boundary becomes the forbidden area. The values X1, Y1, Z1, X2, Y2, and Z2 must be set as coordinates (machine unit) in the machine coordinate system.

> ⚠ **CAUTION**
> 1 If the two points for specifying a forbidden area are identical, all areas are handled as movable areas for stored stroke check 2/3.
> 2 Even if the two points for specifying a forbidden area are erroneously set, the rectangular parallelepiped having the points as vertices is assumed as a boundary.
> 3 Since an axis without the reference position return function has no forbidden areas, there are no alarms about forbidden areas for the axis.

- Checkpoints in the forbidden area

The parameter setting or programmed value (XYZIJK) depends on which part of the tool or tool holder is checked for entering the forbidden area.
If point A (the top of the tool) is checked in Fig. 6.3(d), the distance "a" should be set as the data for the stored stroke limit function. If point B (the tool chuck) is checked, the distance "b" must be set. When checking the tool tip (like point A), and if the tool length varies for each tool, setting the forbidden area for the longest tool requires no re-setting and results in safe operation.

- For machining center system



- For lathe system

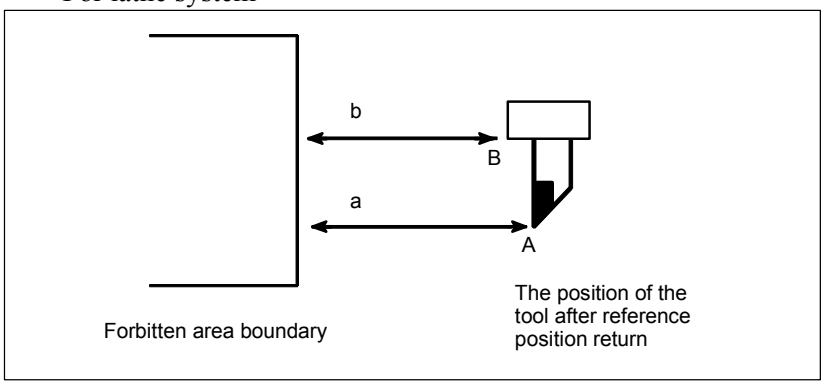


Fig. 6.3 (d) Setting the forbidden area

- Forbidden area overlapping

Area can be set in piles.

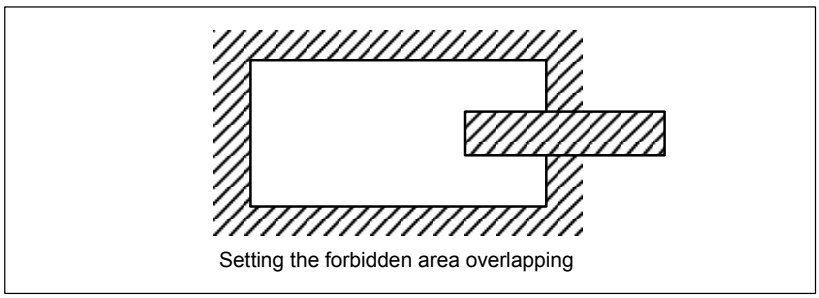


Fig. 6.3 (e) Setting the forbidden area overlapping

Unnecessary limits should be set beyond the machine stroke.

- Condition under which each check is enabled

Each check becomes effective after the power is turned on and manual reference position return or automatic reference position return by G28 has been performed.
After the power is turned on, if the reference position is in the forbidden area of each limit, an alarm is generated immediately. (Only in G22 mode for stored stroke check 2).

- Releasing the alarms

If the enters a forbidden area and an alarm is generated, the tool can be moved only in the backward direction. To cancel the alarm, move the tool backward until it is outside the forbidden area and reset the system. When the alarm is canceled, the tool can be moved both backward and forward.

- Change from G23 to G22 in a forbidden area

When G23 is switched to G22 in the forbidden area, the following results.

<1> When the forbidden area is inside, an alarm is informed in the next move.

<2> When the forbidden area is outside, an alarm is informed immediately.

- Timing for displaying an alarm

In stored stroke check 1/2/3, parameter BFA (bit 7 of No. 1300) selects whether an alarm is displayed immediately before the tool enters the forbidden area or immediately after the tool has entered the forbidden area.

Alarm

Number	Message	Description
OT0500	+ OVERTRAVEL (SOFT 1)	A movement in the positive direction exceeded stored stroke check 1.
OT0501	- OVERTRAVEL (SOFT 1)	A movement in the negative direction exceeded stored stroke check 1.
OT0502	+ OVERTRAVEL (SOFT 2)	A movement in the positive direction exceeded stored stroke check 2.
OT0503	- OVERTRAVEL (SOFT 2)	A movement in the negative direction exceeded stored stroke check 2.
OT0504	+ OVERTRAVEL (SOFT 3)	A movement in the positive direction exceeded stored stroke check 3.
OT0505	- OVERTRAVEL (SOFT 3)	A movement in the negative direction exceeded stored stroke check 3.

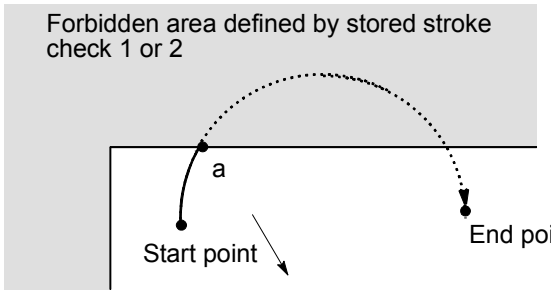
6.4 STROKE LIMIT CHECK BEFORE MOVE

During automatic operation, before the movement specified by a given block is started, whether the tool enters the forbidden area defined by stored stroke check 1, 2, or 3 is checked by determining the position of the end point from the current position of the machine and a specified amount of travel. If the tool is found to enter the forbidden area defined by a stored stroke limit, the tool is stopped immediately upon the start of movement for that block, and an alarm is displayed.

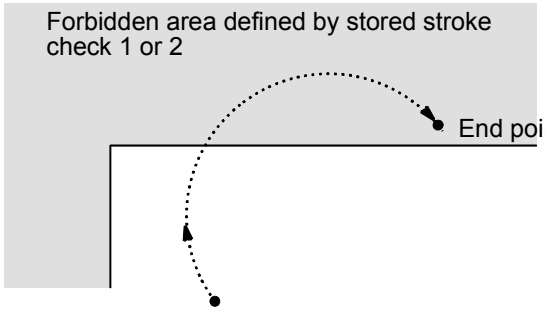
⚠ WARNING

Whether the coordinates of the end point, reached as a result of traversing the distance specified in each block, are in a forbidden area is checked. In this case, the path followed by a move command is not checked. However, if the tool enters the forbidden area defined by stored stroke check 1, 2, or 3, an alarm is issued. (See the examples below.)

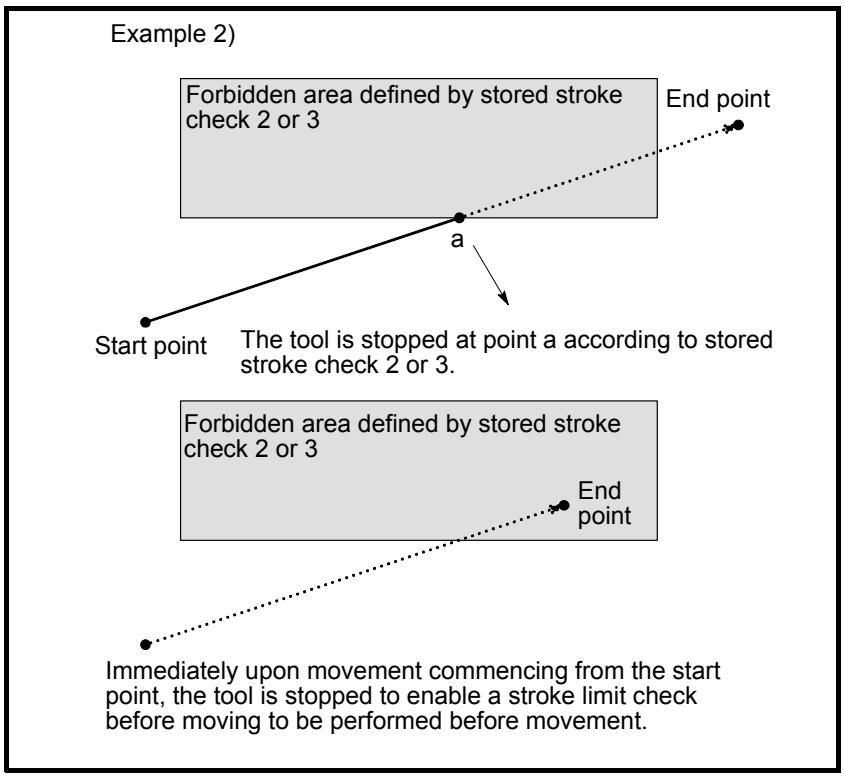
Example 1)



The tool is stopped at point a according to stored stroke check 1 or 2.



Immediately upon movement commencing from the start point, the tool is stopped to enable a stroke limit check before moving to be performed before movement.



Explanation

When a stroke limit check before moving is performed, whether to check the movement performed by a G31 (skip) block and G37 (automatic tool length measurement) block can be determined using (parameter NPC (No. 1301#2)).

Limitation

- Machine lock

If machine lock is applied at the start of movement, no stroke limit check made before movement is performed.

- G23

When stored stroke check 2 is disabled (G23 mode), no check is made to determine whether the tool enters the forbidden area defined by stored stroke check 2.

- Program restart

When a program is restarted, an alarm is issued if the restart position is within a forbidden area.

- Manual intervention following a feed hold stop

When the execution of a block is restarted after manual intervention following a feed hold stop, no alarm is issued even if the end point following a manual intervention is within a forbidden area.

- A block consisting of multiple operations

If a block consisting of multiple operations (such as a canned cycle and exponential interpolation) is executed, an alarm is issued at the start point of any operation whose end point falls within a inhibited area.

- Cylindrical interpolation mode

In cylindrical interpolation mode, no check is made.

- Polar coordinate interpolation mode

In polar coordinate interpolation mode, no check is made.

- Three-dimensional coordinate system conversion

In three-dimensional coordinate system conversion mode, no check is made.

- PMC axis control

No check is made for a movement based on PMC axis control.

Alarm

Table 6.4 (a) Alarm

Number	Message	Description
OT0510	+ OVERTRAVEL (PRE-CHECK)	The block end point was found in the + side stroke limit prohibition area during a stroke check before movement. Modify the program.
OT0511	- OVERTRAVEL (PRE-CHECK)	The block end point was found in the - side stroke limit prohibition area during a stroke check before movement. Modify the program.

6.5 WRONG OPERATION PREVENTION FUNCTIONS

An improper tool offset setting or an improper operation of the machine can result in the workpiece being cut inadequately or the tool being damaged. Also, if data is lost due to an operation mistake, it takes extra time to recover from the mistake.
The operation confirmation functions described below are meant to prevent the operator from performing any unintended operation (hereinafter referred to as an improper operation).

1 Functions that are used when data is set
- Data check to verify that the offset data is within the valid setting range
- Incremental input operation confirmation
- Prohibition of the absolute input by the soft key to prevent any improper absolute or incremental input operation
- Confirmation of any operation of deleting the program or all data
- Confirmation of a data update during the data setting process

2 Functions that are used when the program is executed
- Highlighting of updated modal information
- Display of the executed block status prior to the program execution
- Display of the axis status, such as the mirror image function enabled or the interlock function enabled
- Check for starting from the middle of the program
- Data check to verify that the offset data is within the effective setting range
- Maximum incremental value check

6.5.1 Functions that are Used When Data is Set

The following functions are provided to prevent improper operations when data is set.

- Input data range check
- Confirmation of incremental input
- Prohibition of the absolute input by the soft key
- Confirmation of the deletion of the program
- Confirmation of the deletion of all data
- Confirmation of a data update during the data setting process

Set these functions on the operation confirmation function setting screen. For the input data range check, set a valid input data range, e.g. the upper and lower limits, for each input screen. For the other functions, specify whether to enable or disable them.
For information about how to display the individual setting screens, how to manipulate them, and other details, see the item “Operation confirmation setting screen” that describes the operation procedures.

6.5.1.1 Input data range check

This function allows an effective data range to be set and checks whether the input data is within the set range.

Input data range check

Explanation

- Outline of the input data range check

This function allows an effective data range to be set for the data of each input screen listed later and checks whether the input data is within the set range. If the input data is out of the effective data range, the warning message "DATA IS OUT OF RANGE" is displayed and the data is rejected.

For example, assume that the effective data range for a certain tool offset number is set to -200. to 200, and that you are going to input 100.[INPUT]. Even if you inadvertently press the 0 key one more time, resulting in 1000.[INPUT], the input of 1000. is not accepted.
The function detects a setting mistake and prevents the program from running with invalid data.

- Input screens for which this function is effective

- Tool compensation
- Workpiece origin offset

T

- Y-axis tool offset
- Workpiece shift

- Settings

To enable this function, set an effective data range for each input screen on the operation confirmation function setting screen. For information about how to display the individual setting screens, how to set data ranges, and other details, see the items that describe the setting of the data ranges.
If the set data range is invalid, no data input is accepted. Correct the data range setting, and then input data.

- Disabling the function

The input data range check is disabled if you make any of the following settings on the operation confirmation function setting screen.

- Both the upper and lower limit values for the tool offset number or workpiece coordinate system are 0.
- The upper and lower limit values for each offset are identical.

- Messages displayed during the input data range check

When the cursor moves into an input field of an input screen, one of the following messages and warning messages is displayed. No message is displayed when the input data range check is disabled.

When the set effective data range is valid

Message list 1		
Input data status	**Message**	**Color**
The data in the input field is within the range.	Input range xxx - xxx	Black
The data in the input field is out of the range.	Input range xxx - xxx	Red

xxx: Upper and lower limit values

When the set effective data range is invalid

Message list 2		
Range check status	**Message**	**Color**
Tool offset number overlap	NG SETTING (OFFSET NUM OVERLAP)	Red
Workpiece coordinate system overlap	NG SETTING (WORK COORD VAL OVERLAP)	Red
Invalid upper and lower limit values	NG SETTING (U-LMT AND L-LMT ILLEGAL)	Red

The message "NG SETTING (U-LMT AND L-LMT ILLEGAL)" is displayed in the following cases:

- The upper and lower limit values are reversed.
- The values are not effective (e.g., more pairs of offset numbers than allowed are set).
- Either of the tool offset numbers is 0.

- Range check for data changed by G10 or system variable

If the data changed by G10 or system variable is out of the effective data range, the alarm PS0334 "OFFSET DATA OUT OF RANGE" is displayed.

6.5.1.2 Confirmation of incremental input

This function displays a confirmation message when you attempt to input an incremental value by using the [+INPUT] soft key.

Confirmation of incremental input

Explanation

- Outline of the confirmation of incremental input

This function displays a confirmation message when you attempt to input an incremental value by using the [+INPUT] soft key in any of the input screens listed below. It lets you confirm whether you really want to change data or not before making that change.

For example, when you set 5.[+INPUT] for 10., the message "15. INPUT OK?" is displayed.
The function prevents improper absolute or incremental input operations.

NOTE
This function cannot be used to input two or more values consecutively by delimiting them by semicolons (;).

- Input screens for which this function is effective

- Tool compensation
- Workpiece origin offset
- Settings
- Parameter
- Pitch error compensation

M

- Chopping

T

- Workpiece shift
- Y-axis tool offset
- Second tool geometry offset
- Chuck tail stock barrier
- Tool geometry data

- Settings

In the operation confirmation function setting screen, check or uncheck the "INCREMENTAL INPUT" box to enable or disable this function. For information about how to display the setting screen, how to set the function, and other details, see the item “Operation confirmation setting” that describes the setting of the operation confirmation function.

6.5.1.3 Prohibition of the absolute input by the soft key

This function prohibits the absolute input using the [INPUT] soft key.

Prohibition of the absolute input by the soft key

Explanation

- Outline of the prohibition of the absolute input by the soft key

This function prohibits the absolute input by the [INPUT] soft key in the input screens listed later.
It prevents improper absolute or incremental input operations by requiring that the absolute input be made using the

MDI key and that the incremental input be made using the [+INPUT] soft key.

- Input screens for which this function is effective

- Tool compensation
- Workpiece origin offset

T

- Y-axis tool offset
- Workpiece shift

- Settings

In the operation confirmation function setting screen, check or uncheck the "DISABLED SOFTKEY[INPUT] IN" box to enable or disable this function. For information about how to display the setting screen, how to set the function, and other details, see the item “Operation confirmation setting” that describes the setting of the operation confirmation function.

6.5.1.4 Confirmation of the deletion of the program

This function displays the confirmation message "DELETE PROGRAM ?" when you attempt to delete the program.

Confirmation of the deletion of the program

Explanation

- Outline of the confirmation of the deletion of the program

When you attempt to delete the program, this function displays the confirmation message "DELETE PROGRAM ?" It lets you confirm whether you really want to delete the program or not before executing the deletion.
The function prevents the program from being deleted due to an improper operation.

- Settings

In the operation confirmation function setting screen, check or uncheck the "PROGRAM DELETE" box to enable or disable this function. For information about how to display the setting screen, how to set the function, and other details, see the item “Operation confirmation setting” that describes the setting of the operation confirmation function.

6.5.1.5 Confirmation of the deletion of all data

This function displays the confirmation message "DELETE ALL DATA?" when you attempt to delete all data.

Confirmation of the deletion of all data

Explanation

- Outline of the confirmation of the deletion of all data

When you attempt to delete all data on the input screen described later, this function displays the confirmation message "DELETE ALL DATA?". It lets you confirm whether you really want to delete all data or not before executing the deletion.
The function prevents all data from being deleted due to an improper operation.

- Input screens for which this function is effective

T

- Tool compensation
- Y-axis tool offset

- Settings

In the operation confirmation function setting screen, check or uncheck the "ALL DATA DELETE" box to enable or disable this function. For information about how to display the setting screen, how to set the function, and other details, see the item “Operation confirmation setting” that describes the setting of the operation confirmation function.

6.5.1.6 Confirmation of a data update during the data setting process

This function displays the [CAN] and [EXEC] soft keys for confirmation when you attempt to update the data of an input screen during the data setting process.

Confirmation of a data update during the data setting process

Explanation

- Outline of the confirmation of a data update during the data setting process

When you input data in input screen during the data setting process, this function displays the [CAN] and [EXEC] soft keys for confirmation. It lets you confirm whether you really want to update the data or not before executing the update.
The function prevents set values from being lost due to an improper operation.

If you input data using the [+INPUT] soft key when the confirmation of incremental input is enabled, a message is displayed to confirm the incremental input.

- Settings

In the operation confirmation function setting screen, check or uncheck the "INPUT IN SETTING" box to enable or disable this function. For information about how to display the setting screen, how to set the function, and other details, see the item “Operation confirmation setting” that describes the setting of the operation confirmation function.

6.5.2 Functions that are Used when the Program is Executed

Overview

The following functions are provided to prevent improper operations when the program is executed.

- Display of updated modal information
- Start check signal
- Axis status display
- Confirmation of the start from a middle block
- Data range check
- Maximum incremental value check

In the operation confirmation function setting screen, specify whether to enable or disable these functions individually.
For information about how to display the setting screen, how to manipulate it, and other details, see the item “Operation confirmation setting screen” that describes the operation procedures.

6.5.2.1 Display of updated modal information

This function allows modal information updated by the NC command or RESET to be highlighted in the modal information display for the current block.

Display of updated modal information

Explanation

- Outline of the display of updated modal information

This function allows modal information updated by the NC command or RESET to be highlighted in the modal information display for the current block.
For example, when a absolute command has been changed to an incremental command or when the workpiece coordinate system has been initialized by RESET, the function displays the changed part of the data in an easy-to-recognize manner, in order to prevent improper operations during the execution of the program.

- Settings

In the operation confirmation function setting screen, check or uncheck the "UPDATE MODAL HIGHLIGHT DISPLAY" box to enable or disable this function. For information about how to display the setting screen, how to set the function, and other details, see the item “Operation confirmation setting” that describes the setting of the operation confirmation function.

6.5.2.2 Start check signal

This function displays the remaining amount of travel and modal information of the block to be executed and puts the program to a temporary halt before the program is executed.

Start check signal

Explanation

- Outline of the start check signal

When a cycle start is made with the start check signal STCHK <G0408 #0> set to 1, the function displays the remaining amount of travel and modal information of the block to be executed and puts the program to a temporary halt. Making the cycle start again resumes the execution of the program.
The function lets you check the status of the block before executing it, thus helping to prevent improper operations at the time of execution.
Using this function in combination with the updated modal information display function described in the preceding subsection makes it easier to check the status of the block to be executed.

- Settings

This function does not require any setting on the operation confirmation function setting screen.

6.5.2.3 Axis status display

This function displays the axis status to the left of the axis name in the coordinate display screen.

Axis status display

Explanation

- Outline of the axis status display

This function displays the axis status to the left of the axis name in the display of the machine coordinates, absolute coordinates, relative coordinates, and remaining travel amounts.
For example, when the mirror image function is enabled for the X1 axis, the absolute coordinates are displayed as follows.

```
ABSOLUTE
M  X1     10.000
   Y1     10.000
   Z1      0.000
```

By displaying the axis status as shown above, the function prevents improper operations at the time of execution.

- Axis status indication

The axis status is indicated as follows. These indications are listed in order of priority.

AXIS DETACH	: D
INTERLOCK	: I
MACHINE LOCK	: L
SERVO OFF	: S
AXIS MOVING	: *
MIRROR IMAGE	: M

- Settings

In the operation confirmation function setting screen, check or uncheck the "AXIS STATUS DISPLAY" box to enable or disable this function. For information about how to display the setting screen, how to set the function, and other details, see the item “Operation confirmation setting” that describes the setting of the operation confirmation function.

6.5.2.4 Confirmation of the start from a middle block

This function displays a confirmation message when you attempt to execute a memory operation with the cursor placed on a block in the middle of the program.

Confirmation of the start from a middle block

Explanation

- Outline of the confirmation of the start from a middle block

This function displays the confirmation message "START FROM MIDDLE OF PROG (START/RESET)" when you attempt to execute a memory operation with the cursor placed on a block in the middle of the program. It lets you confirm whether you really want to start execution from that block or not before executing the program.
The function prevents you from inadvertently making a cycle start from a block in the middle of the program.

- Settings

In the operation confirmation function setting screen, check or uncheck the "START FROM MIDDLE OF PROGRAM" box to enable or disable this function. For information about how to display the setting screen, how to set the function, and other details, see the item “Operation confirmation setting” that describes the setting of the operation confirmation function.

6.5.2.5 Data range check

This function lets you set an effective data range and check whether the data to be used for execution is within the set range.

Data range check

Explanation

- Outline of the data range check

This function lets you set an effective data range for each data item listed later and check whether the data to be used for execution is within the set range. If the data is out of the effective range, the alarm PS0334 "OFFSET DATA OUT OF EFFECTIVE RANGE" is displayed.
The function detects data setting mistakes and prevents the program from running with invalid data.

- Data for which this function is effective

- Tool compensation
- Workpiece origin offset

T

- Y-axis tool offset
- Workpiece shift

NOTE
To use this function, you need to set each effective data range correctly. For information about how to set the data ranges, see the item “Effective value range for each data”.

6.5.2.6 Maximum incremental value check

This function checks the maximum incremental value specified for each axis by the NC command.

Maximum incremental value check

Explanation

- Outline of the maximum incremental value check

When the maximum incremental value is specified by the NC command described later, make sure that the absolute value of the travel distance by the incremental command does not exceed the specified value. If the specified value is exceeded, the alarm PS0337 "EXCESS MAXIMUM INCREMENTAL VALUE" is displayed.
A maximum incremental value can be specified on a per-axis basis and remains effective until 0 is set or the value is reset.
For example, when AI contour control is used, the function checks whether the amount of movement between blocks is kept to the specified value or less. Through this process, it detects erroneous program settings and prevents the program from running with invalid data.

- Format

The format of the NC command used to specify the maximum incremental value is as follows.

G91.1 IP_ ;
IP_ ; Maximum incremental value

To cancel the maximum incremental value check, set 0.

6.5.3 Setting Screen

This section describes how to display the operation confirmation function setting screen and how to set the individual data items on this screen.

The operation confirmation function setting screen allows you to set the following items:

- Enabling or disabling each operation confirmation function
- Effective value range for the tool offset
- Effective value range for the workpiece origin offset

T

- Effective value range for the Y-axis tool offset
- Effective value range for the work shift

6.5.3.1 Operation confirmation function setting screen

This screen displays the enable/disable setting status of the following operation confirmation functions and lets you change their settings. (Hereinafter, the screen is referred to as the operation confirmation function setting screen.)

- Confirmation of incremental input
- Prohibition of the absolute input by the soft key
- Confirmation of the deletion of the program
- Confirmation of the deletion of all data
- Confirmation of a data update during the data setting process
- Display of updated modal information
- Axis status display
- Confirmation of the start from a middle block

Displaying and setting the operation confirmation function setting screen

Procedure

1 Press the function key.

2 Press the continuous menu key at the right edge of the screen several times until the [GUARD] soft key is displayed.

3 Click the [GUARD] soft key. The setting screen that was displayed last with relation to any operation confirmation function is displayed (the operation confirmation function setting screen is the first such screen that appears after the system is restarted).

4 If any screen other than the operation confirmation function setting screen is displayed, click the [GUARD] soft key. The operation confirmation function setting screen is displayed.

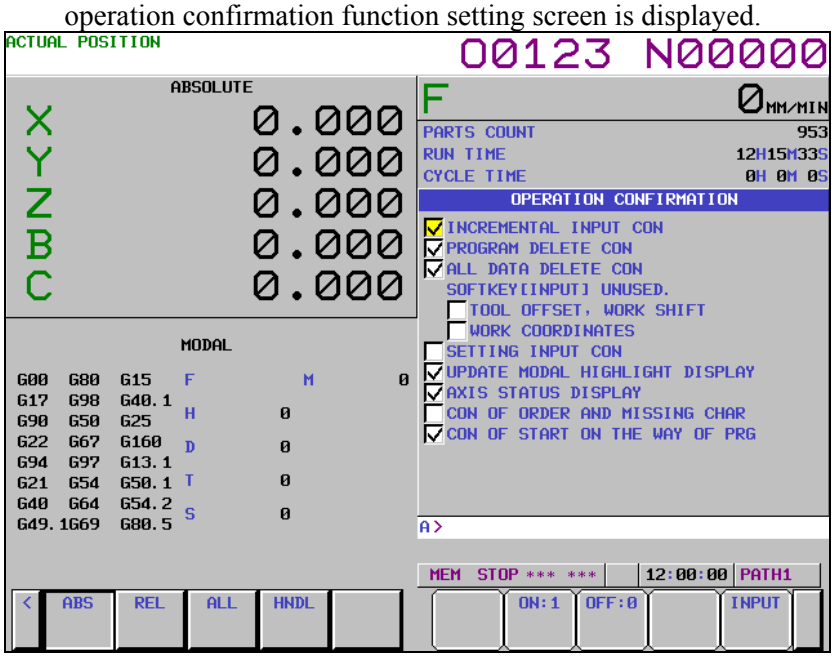


Fig. 6.5.3.1 (a) Operation confirmation function setting screen

5 In the operation confirmation function setting screen, the check box of each enabled function is checked (✓). Move the cursor to the check box of the item you want to set, by pressing the [↑], [↓], [←], and [→] keys.

6 Click the operation soft key [ON:1] or [OFF:0]. When you click the [ON:1] soft key, a check mark (✓) appears in the corresponding check box, indicating that the function is enabled. When you click the [OFF:0] soft key, the check mark disappears from the check box, indicating that the function is disabled.

Explanation

- Items to be set

The following table shows what is displayed for each item to be set and the corresponding functions.

Displayed item	Corresponding function
INCREMENTAL INPUT	Confirmation of incremental input
DISABLED SOFTKEY[INPUT] IN TOOL OFFSET, WORK SHIFT	Prohibition of the absolute input by the soft key (tool offset, Y-axis tool offset (lathe system), and work shift (lathe system))
DISABLED SOFTKEY[INPUT] IN WORK COORDINATES	Prohibition of the absolute input by the soft key (workpiece origin offset)
PROGRAM DELETE	Confirmation of the deletion of the program
ALL DATA DELETE	Confirmation of the deletion of all data
INPUT IN SETTING	Confirmation of a data update during the data setting process
UPDATE MODAL HIGHLIGHT DISPLAY	Display of updated modal information
AXIS STATUS DISPLAY	Axis status display
START FROM MIDDLE OF PROGRAM	Confirmation of the start from a middle block

6.5.3.2 Tool offset range setting screen

This screen displays the setting status of tool offset effective data ranges and lets you change their settings. (Hereinafter, the screen is referred to as the tool offset range setting screen.)
Up to 20 pairs of numbers can be specified to identify tool offset number ranges, and an effective offset value range can be defined for each of these 20 pairs.

Displaying and setting the tool offset range setting screen

Procedure

1 Press the function key.

2 Press the continuous menu key at the right edge of the screen several times until the [GUARD] soft key is displayed.

3 Click the [GUARD] soft key. The setting screen that was displayed last with relation to any operation confirmation function is displayed (the operation confirmation function setting screen is the first such screen that appears after the system is restarted).

4 If any screen other than the tool offset range setting screen is displayed, click the [OFFSET] soft key. The tool offset range setting screen is displayed. What is displayed in this screen differs depending on the system configuration described later.

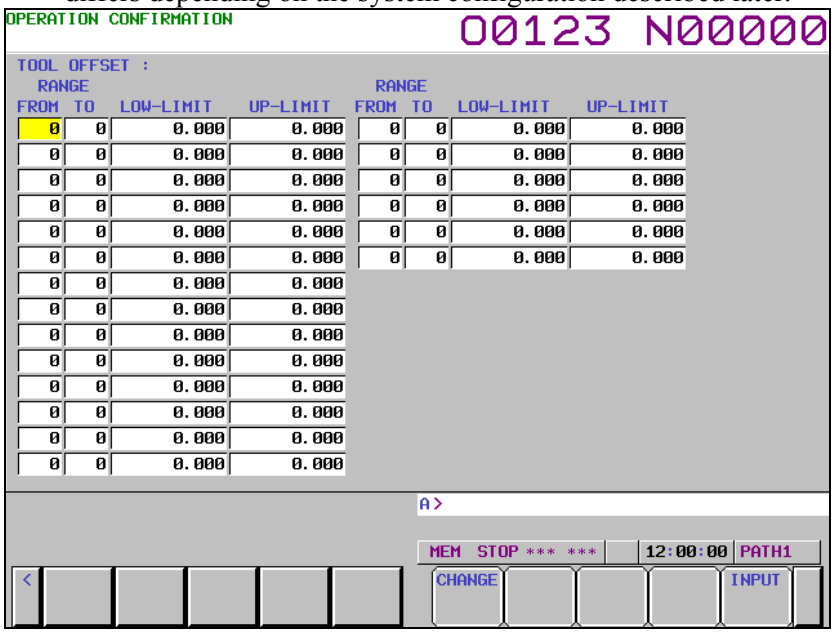


Fig. 6.5.3.2 (a) Tool offset range setting screen

5 Move the cursor to the item you want to set, by using the and keys, , , , and keys, or the [SWITCH] soft key.

6 Press the MDI key, enter necessary data, and then click the [INPUT] soft key.

If the set effective data range is invalid for any of the reasons listed below, the input data range check is not performed normally and the input data is rejected.

- There is a tool offset number overlap.
- The upper and lower limit values are reversed.
- The values are not effective (e.g., more pairs of offset numbers than allowed are set).
- Either of the tool offset numbers is 0.

Also, the input data range check is invalidated in the following cases.

- Both the upper and lower limit values for the tool offset number are 0.
- The upper and lower offset limit values are identical.

Explanation

- System configuration

What to set differs for each of the following system configurations:

M

- Tool offset memory A
- Tool offset memory B
- Tool offset memory C

T

- Without geometry and wear offset
- With geometry and wear offset

M

- What to set with tool offset memory A

With tool offset memory A, an effective data range is specified using the following four items.

Displayed item		What to set
RANGE	FROM	Specify a tool offset number range.
	TO	
-	LOW-LIMIT	Specify a valid tool offset value range in connection with a specified tool offset number range.
	UP-LIMIT	

- What to set with tool offset memory B

With tool offset memory A, an effective data range is specified using the following six items.

Displayed item		What to set
RANGE	FROM	Specify a tool offset number range.
	TO	
GEOM	LOW-LIMIT	Specify a valid tool offset value range for geometry in connection with a specified tool offset number range.
	UP-LIMIT	
WEAR	LOW-LIMIT	Specify a valid tool offset value range for wear in connection with a specified tool offset number range.
	UP-LIMIT	

- What to set with tool offset memory C

With tool offset memory C, an effective data range is specified using the following ten items.

Displayed item			What to set
RANGE		FROM	Specify a tool offset number range.
		TO	
GEOM	LENGTH	LOW-LIMIT	Specify a valid tool offset value range for geometry length in connection with a specified tool offset number range.
		UP-LIMIT	
	RADIUS	LOW-LIMIT	Specify a valid tool offset value range for geometry radius in connection with a specified tool offset number range.
		UP-LIMIT	
WEAR	LENGTH	LOW-LIMIT	Specify a valid tool offset value range for wear length in connection with a specified tool offset number range.
		UP-LIMIT	
	RADIUS	LOW-LIMIT	Specify a valid tool offset value range for wear radius in connection with a specified tool offset number range.
		UP-LIMIT	

In the case of this configuration, all the information needed to set an input data range cannot be displayed in a single screen page. Set the information while switching pages using the [SWITCH] soft key. The screen provides an indication that lets you know which part of the information is currently displayed.

T

- What to set without geometry/wear offset

Without geometry/wear offset, an effective data range is specified using the following eight items.

Displayed item		What to set
RANGE	FROM	Specify a tool offset number range.
	TO	
X	LOW-LIMIT	Specify a valid tool offset value range for the X-axis in connection with a specified tool offset number range.
	UP-LIMIT	
Z	LOW-LIMIT	Specify a valid tool offset value range for the Z-axis in connection with a specified tool offset number range.
	UP-LIMIT	
RADIUS	LOW-LIMIT	Specify a valid tool offset value range for tool-nose radius in connection with a specified tool offset number range.
	UP-LIMIT	

NOTE
The radius-related items are not displayed if the cutter or tool nose radius compensation is not used.

- What to set with geometry/wear offset

With geometry/wear offset, an effective data range is specified using the following 12 items.

Displayed item			What to set
RANGE		FROM	Specify a tool offset number range.
		TO	
GEOM	X	LOW-LIMIT	Specify a valid tool offset value range for the geometry X-axis in connection with a specified tool offset number range.
		UP-LIMIT	
	Z	LOW-LIMIT	Specify a valid tool offset value range for the geometry Z-axis in connection with a specified tool offset number range.
		UP-LIMIT	
	RADIUS	LOW-LIMIT	Specify a valid tool offset value range for geometry tool-nose radius in connection with a specified tool offset number range.
		UP-LIMIT	
WEAR	X	LOW-LIMIT	Specify a valid tool offset value range for the wear X-axis in connection with a specified tool offset number range.
		UP-LIMIT	
	Z	LOW-LIMIT	Specify a valid tool offset value range for the wear Z-axis in connection with a specified tool offset number range.
		UP-LIMIT	
	RADIUS	LOW-LIMIT	Specify a valid tool offset value range for wear tool-nose radius in connection with a specified tool offset number range.
		UP-LIMIT	

In the case of this system, all the information needed to set an input data range cannot be displayed in a single screen page. Set the information while switching pages using the [SWITCH] soft key. The screen provides an indication that lets you know which part of the information is currently displayed.

NOTE
The radius-related items are not displayed if the tool nose radius compensation option is not displayed.

- Example of setting an input data range

For example, suppose that the following values are set with offset memory A.

FROM : TO	LOW-LIMIT : UP-LIMIT
1 : 20	0.000 : 100.000

In this case, the tool offset input screen accepts only offset values from 0.000 to 100.000 for offset numbers 1 to 20.
If you attempt to input any other value, the warning message "DATA IS OUT OF RANGE" is displayed.

6.5.3.3 Workpiece origin offset range setting screen

This screen displays the setting status of workpiece origin offset and external workpiece origin offset effective data ranges and lets you change their settings. (Hereinafter, the screen is referred to as the workpiece origin offset range setting screen.)

Up to six pairs of values can be specified to identify workpiece coordinate ranges for the workpiece origin offset, and an effective offset value range can be defined for each of the axes of these six pairs. As for the external workpiece origin offset, an effective offset value range can be specified for each axis.

Displaying and setting the workpiece origin offset range setting screen

Procedure

1 Press the function key.

2 Press the continuous menu key at the right edge of the screen several times until the [GUARD] soft key is displayed.

3 Click the [GUARD] soft key. The setting screen that was displayed last with relation to any operation confirmation function is displayed (the operation confirmation function setting screen is the first such screen that appears after the system is restarted).

4 If any screen other than the workpiece origin offset range setting screen is displayed, click the [WORK] soft key. The workpiece origin offset range setting screen is displayed.

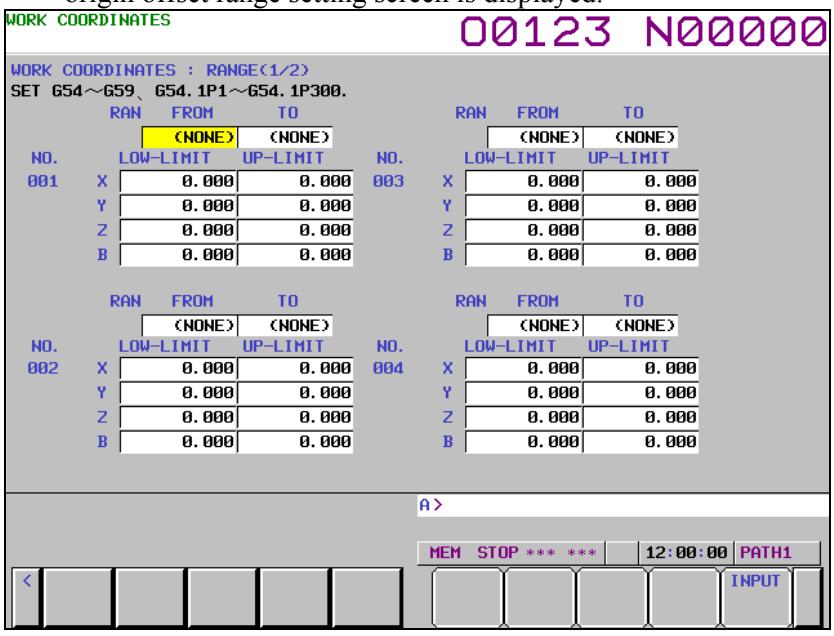


Fig. 6.5.3.3 (a) Workpiece origin offset range setting screen

5 Move the cursor to the item you want to set, by using the PAGE and PAGE keys, , , , and keys, or the [SWITCH] soft key.
6 Press the MDI key, enter necessary data, and then click the [INPUT] soft key.

If the set effective data range is invalid for any of the reasons listed below, the input data range check is not performed normally and the input data is rejected.

- There is a workpiece coordinate overlap.
- The upper and lower limit values are reversed.
- The values are not effective (e.g., an invalid workpiece coordinate system is set).
- The upper limit value is set for the workpiece coordinate system when 0 is set for the lower limit value.

Also, the input data range check is invalidated in the following cases.

- Both the upper and lower limit values for the workpiece coordinate system are 0.
- The upper and lower limit values for each offset are identical.

Explanation

- What to set for the workpiece origin offset

For the workpiece origin offset, an effective data range is specified using the following four items.

Displayed item		What to set
RANGE	FROM	Specify a workpiece coordinate system range.
	TO	
AXIS NAME	LOW-LIMIT	Specify a valid offset value range in connection with a specified workpiece coordinate system range.
	UP-LIMIT	

- What to set for the external workpiece origin offset

For the external workpiece origin offset, an effective data range is specified using the following two items.

Displayed item		What to set
AXIS NAME	LOW-LIMIT	Specify a valid external workpiece origin offset value range on each axis.
	UP-LIMIT	

6.5.3.4 Y-axis tool offset range setting screen

T

In the case of a lathe system, this screen displays the setting status of Y-axis tool offset effective data ranges and lets you change their settings. (Hereinafter, the screen is referred to as the Y-axis tool offset range setting screen.)
Up to four pairs of values can be specified to identify Y-axis tool offset number ranges, and an effective offset value range can be defined for each of these four pairs.

Displaying and setting the Y-axis tool offset range setting screen

Procedure

1 Press the function key.

2 Press the continuous menu key ▷ at the right edge of the screen several times until the [GUARD] soft key is displayed.

3 Click the [GUARD] soft key. The setting screen that was displayed last with relation to any operation confirmation function is displayed (the operation confirmation function setting screen is the first such screen that appears after the system is restarted).

4 If any screen other than the Y-axis tool offset range setting screen is displayed, click the [OFST.2] soft key. The Y-axis tool offset range setting screen is displayed. What is displayed in this screen differs depending on such factors as whether tool geometry/wear offsets are present.

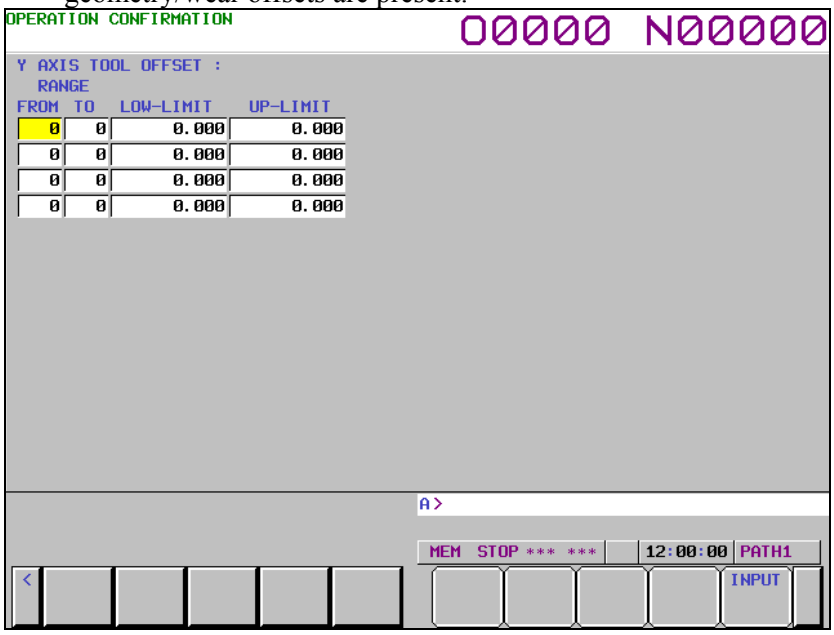
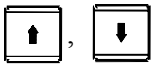


Fig. 6.5.3.4 (a) Y-axis tool offset range setting screen

5 Move the cursor to the item you want to set, by using the

and

keys,  , and keys, or the [SWITCH] soft key.

6 Press the MDI key, enter necessary data, and then click the [INPUT] soft key.

If the set effective data range is invalid for any of the reasons listed below, the input data range check is not performed normally and the input data is rejected.

- There is a tool offset number overlap.
- The upper and lower limit values are reversed.
- The values are not effective (e.g., more pairs of offset numbers than allowed are set).
- Either of the tool offset numbers is 0.

Also, the input data range check is invalidated in the following cases.

- Both the upper and lower limit values for the tool offset number are 0.
- The upper and lower offset limit values are identical.

Explanation

- What to set without geometry/wear offsets

Without geometry/wear offsets, an effective data range is specified using the following four items.

Displayed item		What to set
RANGE	FROM	Specify a Y-axis tool offset number range.
	TO	
-	LOW-LIMIT	Specify a valid tool offset value range in connection with a specified Y-axis tool offset number range.
	UP-LIMIT	

- What to set with geometry/wear offsets

With geometry/wear offsets, an effective data range is specified using the following six items.

Displayed item		What to set
RANGE	FROM	Specify a Y-axis tool offset number range.
	TO	
GEOM	LOW-LIMIT	Specify a valid tool offset value range for geometry in connection with a specified Y-axis tool offset number range.
	UP-LIMIT	
WEAR	LOW-LIMIT	Specify a valid tool offset value range for wear in connection with a specified Y-axis tool offset number range.
	UP-LIMIT	

6.5.3.5 Workpiece shift range setting screen

T

In the case of a lathe system, this screen displays the setting status of shift effective data ranges of workpiece shift coordinate systems and lets you change their settings. (Hereinafter, the screen is referred to as the workpiece shift range setting screen.)
An offset value range can be specified for each axis.

Displaying and setting workpiece shift input ranges

Procedure

1 Press the OFS SET function key.

2 Press the continuous menu key ▷ at the right edge of the screen several times until the [GUARD] soft key is displayed.

3 Click the [GUARD] soft key. The setting screen that was displayed last with relation to any operation confirmation function is displayed (the operation confirmation function setting screen is the first such screen that appears after the system is restarted).

4 If any screen other than the workpiece shift range setting screen is displayed, click the [WORK SHIFT] soft key. The workpiece shift range setting screen is displayed.

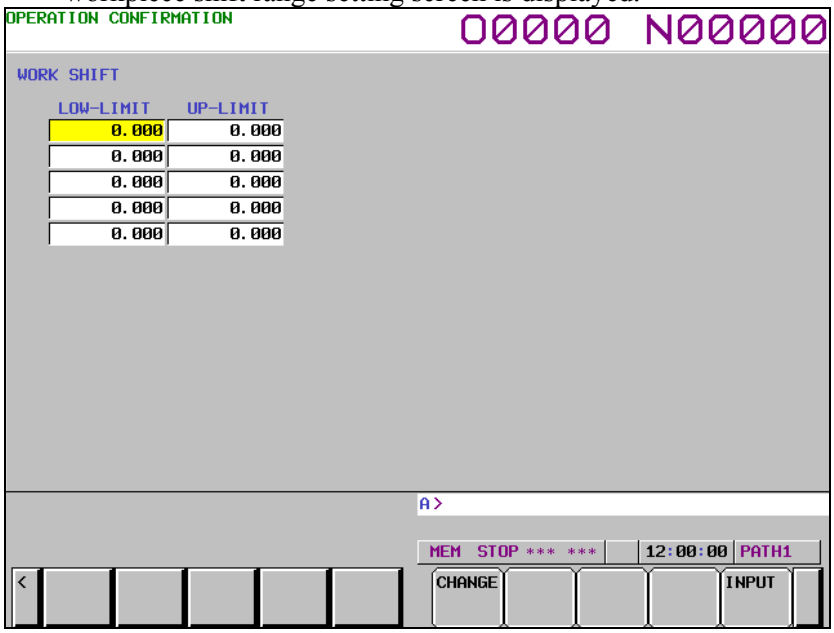


Fig. 6.5.3.5 (a) Workpiece shift range setting screen

5 Move the cursor to the item you want to set, by using the PAGE ↑ and PAGE ↓ keys, ↑, ↓, ←, and → keys, or the [SWITCH] soft key.

6 Press the MDI key, enter necessary data, and then click the [INPUT] soft key.

If the set effective data range is invalid for any of the reasons listed below, the input data range check is not performed normally and the input data is rejected.

- The upper and lower limit values are reversed.

Also, the input data range check is invalidated in the following cases.

- The upper and lower offset limit values are identical.

Explanation

- What to set for the workpiece shift

For the workpiece shift, an effective data range is specified using the following two items.

Displayed item		What to set
AXIS NAME	LOW-LIMIT	Specify a valid workpiece shift coordinate system shift value range on each axis.
	UP-LIMIT	

7 ALARM AND SELF-DIAGNOSIS FUNCTIONS

When an alarm occurs, the corresponding alarm screen appears to indicate the cause of the alarm. The causes of alarms are classified by error codes and number. Up to 60 previous alarms can be stored and displayed on the screen (alarm history display).
The system may sometimes seem to be at a halt, although no alarm is displayed. In this case, the system may be performing some processing. The state of the system can be checked using the self-diagnosis function.

7.1 ALARM DISPLAY

Explanation

- Alarm screen

When an alarm is issued, the display changes to the alarm screen. Two alarm screens "DETAIL" and "ALL PATH" are provided. You can choose one of the screens by pressing the corresponding soft key.

- Detail screen
  Alarm information for the currently selected path is displayed.

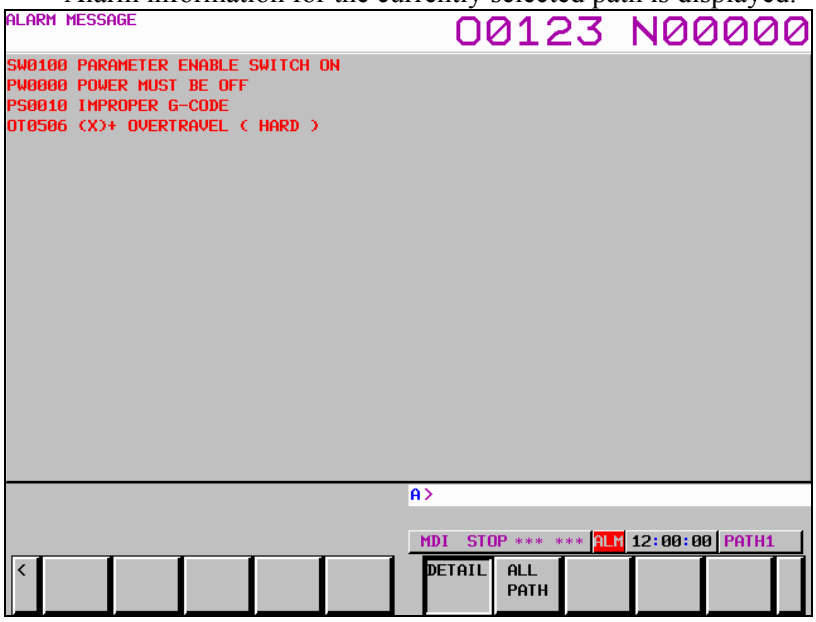


Fig. 7.1 (a) Alarm detail screen

- All path screen
  Alarm information for all paths is displayed sequentially from path 1.

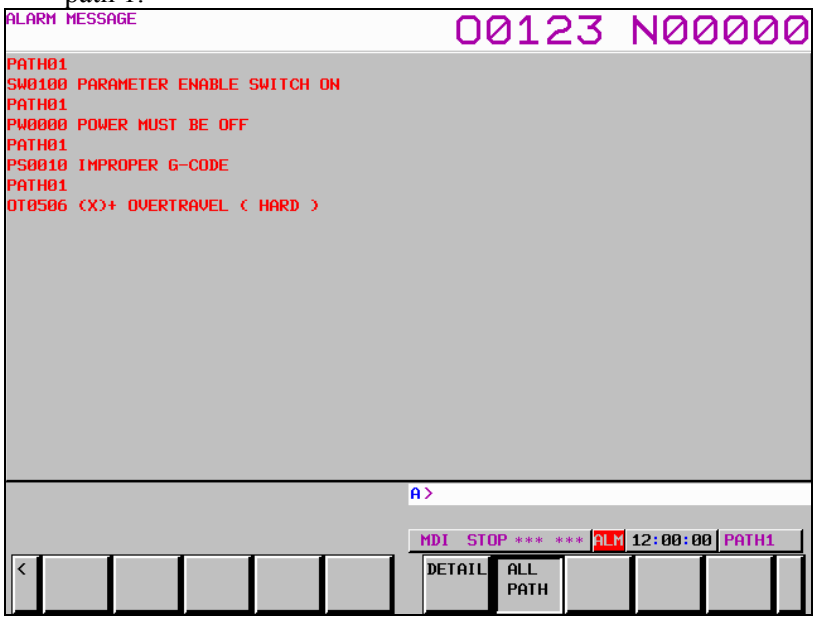


Fig. 7.1 (b) All path screen

- Displaying an alarm screen

ALM is sometimes indicated in the bottom part of the screen display without displaying an alarm screen.

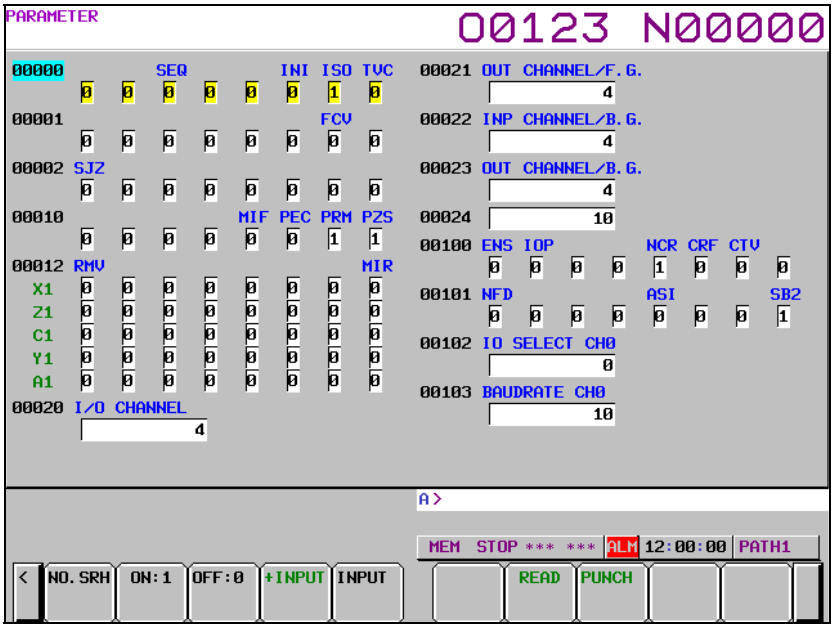


Fig. 7.1 (c) Parameter screen

In this case, display the alarm screen by following the steps below.

1 Press the function key .

2 Press the soft key [ALARM].

3 Pressing the [ALARM] soft key changes the screen display to the "DETAIL" screen (or the alarm screen selected previously), and the [DETAIL] and [ALL PATH] soft keys appears.

- Pressing the [DETAIL] soft key displays the "DETAIL" screen.
- Pressing the [ALL PATH] soft key displays the "ALL PATH" screen.

If the number of paths is 1, pressing the [ALARM] soft key displays the "DETAIL" screen, but the [ALARM] soft key indication remains unchanged.

4 You can change pages by using the page key.

- Releasing alarm

The cause of an alarm can be determined from the error code, number, and associated message. To release the alarm, generally correct the cause, then press the reset key.

- Error code and number

The type of an alarm is indicated by an error code and number.
Example: PS0010, SV0004, etc.
For details, see Appendix G, "ALARMS".

7.2 ALARM HISTORY DISPLAY

Up to 60 alarms (in 10 screen pages) issued by the CNC including the latest alarm are stored and displayed on the screen.
The display procedure is explained below.

Alarm history display

Procedure

1 Press the function key

.

2 Press the soft key [HISTRY].
An alarm history is displayed.
The following information is displayed:
<1> Date and time of alarm issuance
<2> Alarm type
<3> Alarm number
<4> Alarm message (sometimes not displayed depending on the alarm)
<5> Page No.

3 You can change pages by using the page key.

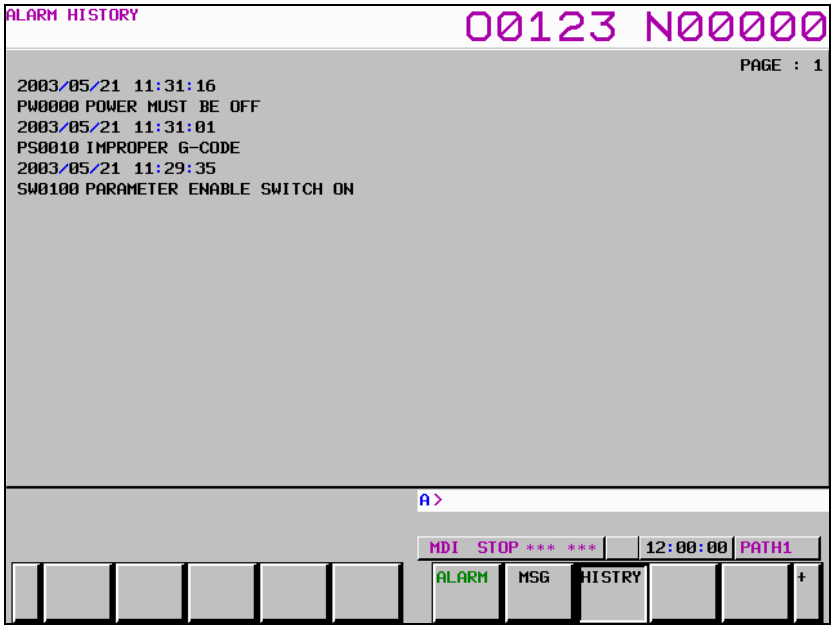


Fig. 7.2 (a) Alarm history screen

7.3 CHECKING BY DIAGNOSTIC DISPLAY

The system may sometimes seem to be at a halt, although no alarm has occurred. In this case, the system may be performing some processing. Diagnostic display can be used to check the system status.

Procedure for Diagnostic display

Procedure

1 Press the function key

.

2 Press the soft key [DGNOS].

3 The diagnosis screen has more than 1 pages. Select the screen by the following operation.
   (1) Change the page by the page change key.
   (2) Method by soft key
      - Key input the number of the diagnosis data to be displayed.
      - Press the soft key [NO.SRH].

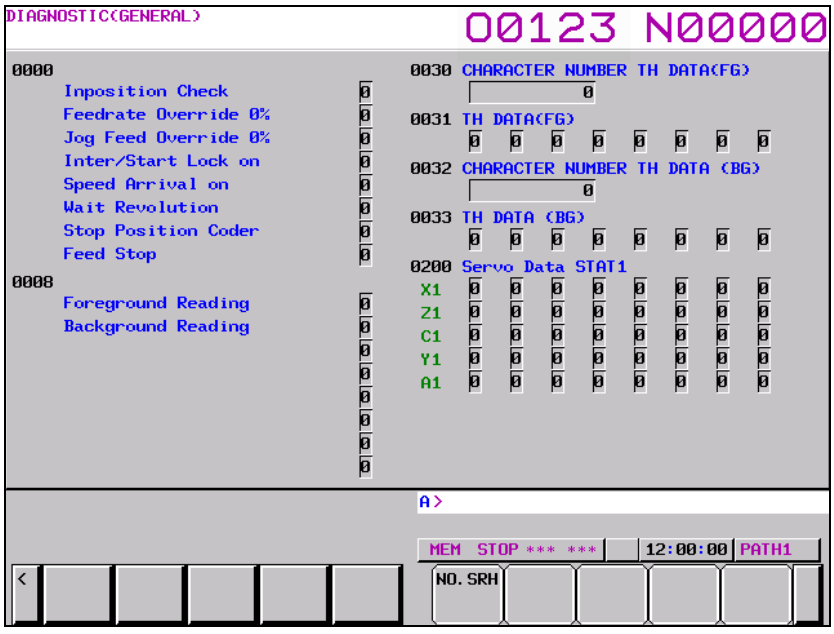


Fig. 7.3 (a) Diagnostic display

7.4 RETURN FROM THE ALARM SCREEN

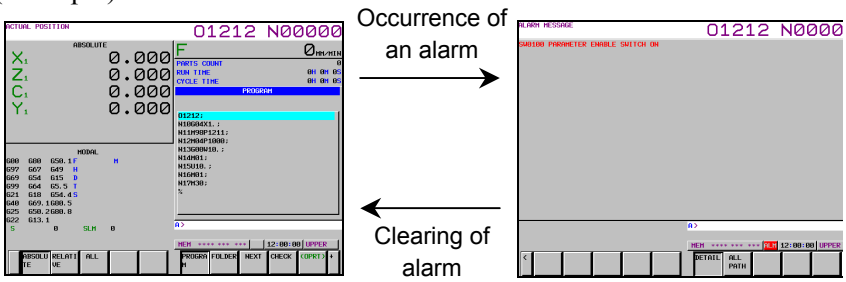
7.4.1 Return from the Alarm Screen

When alarms are cleared or function key is pressed on the alarm screen, the screen displayed before the alarm screen appears.
To enable this function, set bit 4 (ADC) of parameter No. 11302 is set to 1.

Switching between screens when alarms are cleared

When all alarms are cleared on the alarm screen, the screen displayed before the alarm screen appears again.
When the alarm screen was displayed automatically due to occurrence of an alarm, the screen displayed immediately before the alarm appears again.
When the alarm screen was displayed by pressing function key during occurrence of an alarm, the screen displayed immediately before the alarm appears again.

(Example)

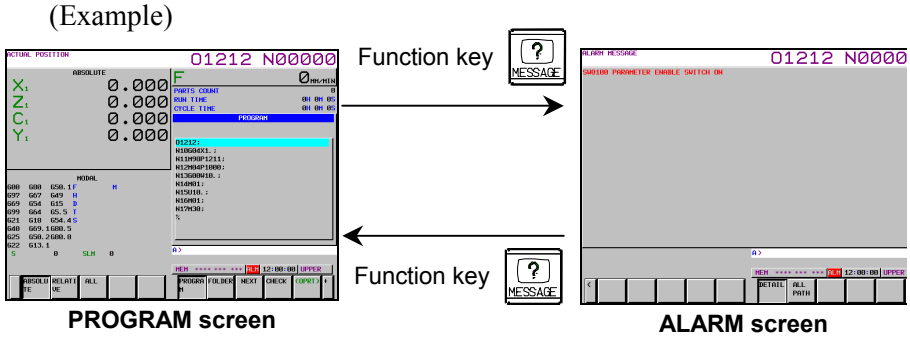


NOTE
Even if alarms are cleared when the alarm screen is not displayed, the current screen is not changed.

Switching between screens by the function key

When function key MESSAGE is pressed on the alarm screen, the screen displayed before the alarm screen appears. Press function key MESSAGE to switch to the alarm screen for checking for alarms and then press function key MESSAGE to return to the previous screen.

(Example)



If function key MESSAGE is pressed when the alarm screen was displayed automatically due to occurrence of an alarm, the screen displayed before the alarm screen appears again.

Restrictions

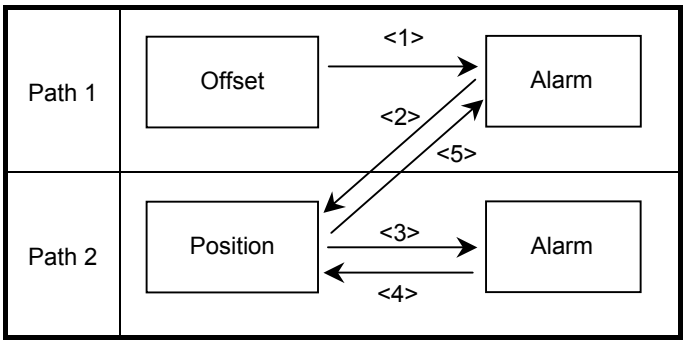
- Switching to the interactive macro screen is not performed.
- The screens to which the alarm screen can be switched are only the screen selected by the chapter selection soft key.

7.4.2 Relationship with Other Functions

Relationship between the screen switching function and a return from the alarm screen during switching between paths

(1) When bit 5 (PSC) of parameter No. 3208 is set to 0, if paths are switched by the path switching signal or switching to the loader, the screen last selected in the path appears again.
At this time, even if a return from the alarm screen to the previous screen is performed in one path, a return is not performed in the other path and the alarm screen remains displayed.

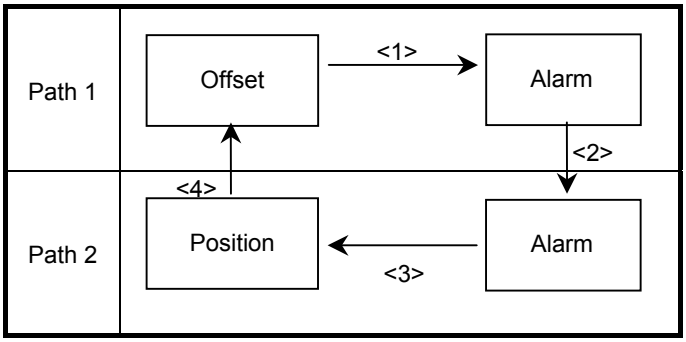
(Example)



<1> When the message key is pressed on the offset screen of path 1, the alarm screen (path 1) appears.
<2> When switching to path 2 is performed on the alarm screen of path 1, the position screen of path 2 appears (when the screen last displayed in path 2 is the position screen).
<3> When the message key is pressed on the position screen of path 2, the alarm screen (path 2) appears.
<4> When the alarm is cleared or the message key is pressed on the alarm screen of path 2, a return to the position screen (path 2) is performed.
<5> When switching to path 1 is performed, the alarm screen (path 1) appears.

(2) When PSC of parameter No. 3208 is set to 1, if paths are switched by the path switching signal or switching to the loader, the screen displayed immediately before the path switching appears again.
At this time, if a return from the alarm screen to the previous screen is performed in one path, the screen of the path in which a return was performed appears in the other path.

(Example)



<1> When the message key is pressed on the offset screen of path 1, the alarm screen (path 1) appears.
<2> When switching to path 2 is performed on the alarm screen of path 1, the alarm screen (path 2) appears.
<3> When the alarm is cleared on the alarm screen of path 2, the offset screen (path 2) appears.
<5> When switching to path 1 is performed, the offset screen of path 1 appears.

8 DATA INPUT/OUTPUT

Information stored in external I/O devices can be read into the CNC, and information can be written into external I/O devices.
External I/O devices include memory cards that can be mounted to the memory card interface located on the left side of the display unit and personal computers and data servers that can be connected via embedded Ethernet.
The following types of data can be input and output.

Data type	Default file name
Program	ALL-PROG.TXT
Offset data	TOOLOFST.TXT
Parameter	CNC-PARA.TXT
Pitch error compensation data	PITCH.TXT
Custom macro common variable	MACRO.TXT
Workpiece coordinate system data	EXT_WKZ.TXT
Operation history data	OPRT_HIS.TXT
Maintenance information	MAINTINF.TXT
Periodic maintenance data (periodic maintenance:status screen)	MAINTENA.TXT
Tool management data (tool management function)	TOOL_MNG.TXT
Magazine data (tool management function)	MAGAZINE.TXT
Tool life status name (tool management function)	STATUS.TXT
Customize data name (tool management function)	CUSTOMIZ.TXT
Second geometry tool offset	SEC_GEOM.TXT
Three-dimensional error compensation data	COMP3D.TXT
Rotary table dynamic fixture offset	FIXTURE.TXT
Customize data displayed as tool management data (tool management expansion)	DISPCSTM.TXT
Spindle waiting position name (tool management expansion)	POSNAME.TXT
Decimal point position data of customize data (tool management expansion)	POINTPOS.TXT
Real time custom macro variable	RTMMACRO.TXT
System configuration data	SYS-CONF.TXT
PMC signal protect	DIDOENBL.TXT
Tool geometry data (tool management function for oversize tools)	TOOLGEOM.TXT
Cause data (tool management tool attachment/detachment function)	TLCAUSE.TXT
Attachment history data (tool management tool attachment/detachment function)	TLATAHIS.TXT
Detachment history data (tool management tool attachment/detachment function)	TLDTAHIS.TXT
Servo/spindle information	SV_SP_ID.TXT
Machine system name data (periodic maintenance:machine system screen)	MAINTEMC.TXT
Servo waveform diagnosis	WAVE-DGN.TXT
Tool geometry data (check for tool post interference)	TOOL-FRM.TXT
P code variable	PCODE.TXT

Data type	Default file name
Learning control data (learning control for parts cutting)	PRT_LN.TXT
Learning control data (learning control)	LEARN.TXT
High precision pitch error compensation	UPEC.TXT

The above types of data can be input and output on the screens for displaying and setting those types of data.
If NC data such as programs and parameters is to be written to a memory card, and if a file with the same name already exists, it is possible to select whether to overwrite the existing file or cancel writing with appropriate operation.
This function is enabled by setting bit 1 (COW) of parameter No. 11308.

The external I/O device set in NC parameter No. 0020 is selected. See the table below for details.

Correspondence between settings and input/output units	
Setting	**Description**
0,1	RS-232-C serial port 1
2	RS-232-C serial port 2
4	Memory card interface
5	Data server interface
9	Embedded Ethernet interface

NOTE
External I/O devices can handle file names of up to 12 characters each. NC programs on the memory of the CNC can each be assigned any file name of up to 32 characters, but the name is restricted to up to 12 characters in the input/output to and from external I/O devices.

8.1 OVERWRITING FILES ON A MEMORY CARD

Screen display

When an attempt is made to output NC data to a memory card, and if the specified file name or the default file name is the same as an existing file name on the memory card, a confirmation message "OVERWRITE?" appears. (Bit 1 (COW) of parameter No. 11308 = 1)

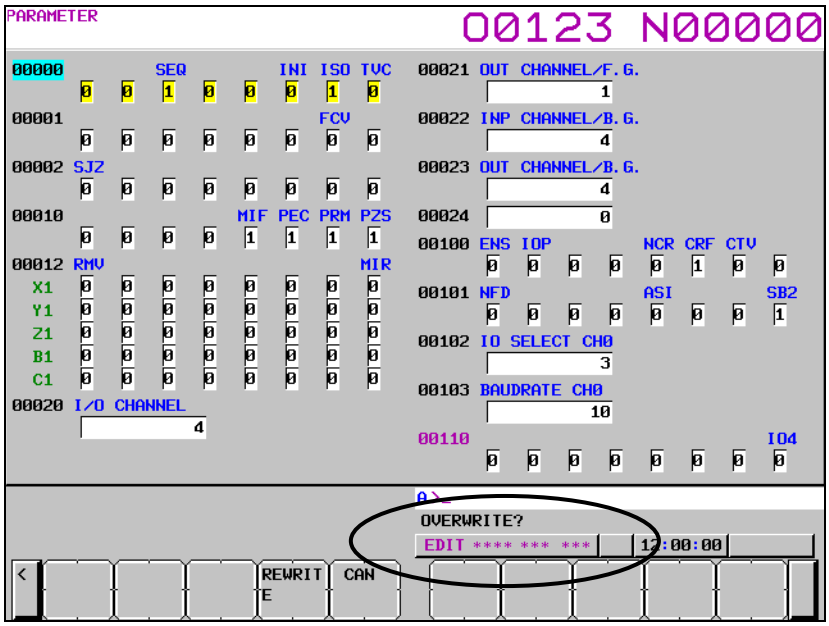


Fig. 8.1 (a) Screen display example

Procedure

On the output screen for the desired function, perform the following operation.

1 Press the soft key [PUNCH].
2 Press the soft key [EXEC]. If a file with the same name does not exist on the memory card, the file is output in this step.
3 If a file with the same name exists on the memory card, soft keys [REWRITE] and [CAN] appear.
Pressing the soft key [REWRITE] causes the file to be overwritten.
Pressing the soft key [CAN] causes output to be canceled.

Example) Output from the parameter screen (for 7.2/8.4/10.4-inch display unit)

1 Press the function key SYSTEM.
2 Press the soft key [PARAMETER].
3 Enter the EDIT mode or state emergency stop.
4 Press the soft key [(OPRT)].
5 For 7.2/8.4-inch display unit, press the continuous menu key ▷.

6 Press the soft key [PUNCH]. The soft key display switches from the one in Fig. 8.1 (b) to the one in Fig. 8.1 (c).
7 Press the soft key [EXEC]. Because no file name is specified, the file is output with a file name of CNC-PARA.TXT, but if a file with the same name exists on the memory card, the soft key display switches from the one in Fig. 8.1 (c) to the one in Fig. 8.1 (d), with a confirmation message appearing.
If a file with the same name does not exist on the memory card, the file is output directly.
8 Pressing the soft key [REWRITE] causes the file to be overwritten.
Pressing the soft key [CAN] causes output to be canceled. If wishing to output the file after changing the file name, specify a file name after step 6, and perform step 7 again.

Fig. 8.1 (b) Soft key display before [PUNCH] is pressed

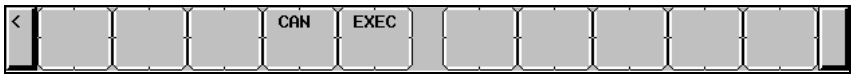


Fig. 8.1 (c) Soft key display after [PUNCH] is pressed

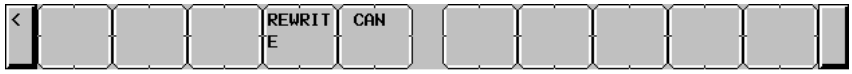


Fig. 8.1 (d) Soft key display after [EXEC] is pressed

Example) Output from the parameter screen (for 15-inch display unit)

1 Press the function key

.
2 Press the vertical soft key [PARAMETER].
3 Enter the EDIT mode or state emergency stop.
4 Press the horizontal soft key [PUNCH]. The horizontal soft key display switches from the one shown in Fig. 8.1 (e) to the one in Fig. 8.1 (f).
5 Press the horizontal soft key [EXEC]. Because no file name is specified, the file is output with a file name of CNC-PARA.TXT, but if a file with the same name exists on the memory card, the horizontal soft key display switches from the one in Fig. 8.1 (f) to the one in Fig. 8.1 (g), with a confirmation message appearing.
If a file with the same name does not exist on the memory card, the file is output directly.
6 Pressing the horizontal soft key [REWRITE] causes the file to be overwritten.
Pressing the horizontal soft key [CAN] causes output to be canceled. If wishing to output the file after changing the file name, specify a file name after step 4, and perform step 5 again.

Fig. 8.1 (e) Soft key display before [PUNCH] is pressed

Fig. 8.1 (f) Soft key display after [PUNCH] is pressed

Fig. 8.1 (g) Soft key display after [EXEC] is pressed

> ⚠ **CAUTION**
> Even if soft key [REWRITE] is pressed, warning message "OVER WRITE FAILED" is issued and output is canceled in the following cases:
> - The file to be overwritten is for read only.
> - The memory card is pulled out while the confirmation message appears.

8.2 INPUT/OUTPUT ON EACH SCREEN

This section explains how to input and output data of the following types to and from each operation screen: program, parameter, offset, pitch error compensation, three-dimensional error compensation, macro variable, workpiece coordinate system data, operation history, and tool management.

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8.2.1 Inputting and Outputting a Program

8.2.1.1 Inputting a program

The following explains how to input a program from an external device to the memory of the CNC by using the program editing screen or program folder screen.

Inputting a program (for 7.2/8.4/10.4-inch display unit)

Procedure

1 Make sure the input device is ready for reading.
2 Press the function key

to display the program editing screen or program folder screen.
3 Press the EDIT switch on the machine operator's panel or enter state emergency stop.
4 Press the soft key [(OPRT)].
5 Press the continuous menu key until soft key [READAD] appears.
Press the soft key [READ].
6 Type the name of the file that you want to input.
Press the soft key [F SET]. To specify the program name to input, type the program name and press the soft key [P SET].
If the input file name is omitted, default input file name "ALL-PROG.TXT" is assumed.
For an explanation of the operations to be performed if an input file name [F SET] and an input program name [P SET] are omitted, see the table below.
7 Press the soft key [EXEC].
This starts reading the program, and "INPUT" blinks in the lower right part of the screen. When the read operation ends, the "INPUT" indication disappears.
To cancel the input of the program, press the soft key [CAN].

Once read, the program is registered in the currently selected foreground folder.

NOTE
Foreground folders are displayed on the program folder screen. For an explanation of foreground folders, see II-12.1.3, and for how to change foreground folders, see III-11.6

[F SET]	[P SET]	Input file name	Input program	Input program name
BLANK	BLANK	ALL-PROG.TXT	All programs in ALL-PROG.TXT	File name at the time the file is saved
BLANK	INPUT	ALL-PROG.TXT	First program in ALL-PROG.TXT	Program name set with [P SET]
INPUT	BLANK	File name set with [F SET]	All programs in the file specified with [F SET]	File name at the time the file is saved
INPUT	INPUT	File name set with [F SET]	First program in the file specified with [F SET]	Program name set with [P SET]

Inputting a program (for 15-inch display unit)

Procedure

1 Make sure the input device is ready for reading.
2 Press the function key .
3 Press the vertical soft key [PROGRAM] or [FOLDER] to display programs or a program directory.
4 Press the EDIT switch on the machine operator's panel or enter state emergency stop.
5 Press the continuous menu key until horizontal soft key [READ] appears.
Press the horizontal soft key [READ].
6 Type the name of the file that you want to input.
Press the horizontal soft key [F SET]. To specify the program name to input, type the program name and press the horizontal soft key [P SET].
If the input file name is omitted, default input file name "ALL-PROG.TXT" is assumed.
For an explanation of the operations to be performed if an input file name [F SET] and an input program name [P SET] are omitted, see the table below.
7 Press the horizontal soft key [EXEC].
This starts reading the program, and "INPUT" blinks in the lower right part of the screen. When the read operation ends, the "INPUT" indication disappears.
To cancel the input of the program, press the horizontal soft key [CAN].

Once read, the program is registered in the currently selected foreground folder.

NOTE
Foreground folders are displayed on the program folder screen. For an explanation of foreground folders, see II-12.1.3, and for how to change foreground folders, see III-11.6

[F SET]	[P SET]	Input file name	Input program	Input program name
BLANK	BLANK	ALL-PROG.TXT	All programs in ALL-PROG.TXT	File name at the time the file is saved
BLANK	INPUT	ALL-PROG.TXT	First program in ALL-PROG.TXT	Program name set with [P SET]
INPUT	BLANK	File name set with [F SET]	All programs in the file specified with [F SET]	File name at the time the file is saved
INPUT	INPUT	File name set with [F SET]	First program in the file specified with [F SET]	Program name set with [P SET]

8.2.1.2 Outputting a program

A program stored in the memory of the CNC unit is output to an external device.

Outputting a program (for 7.2/8.4/10.4-inch display unit)

Procedure

1 Make sure the output device is ready for output.
2 Press the function key

to display the program editing screen or program folder screen.
3 Press the EDIT switch on the machine operator's panel or enter state emergency stop.
4 Press the soft key [(OPRT)].
5 Press the continuous menu key until soft key [PUNCH] appears.
Press the soft key [PUNCH].
6 Type the program number to output and press the soft key [P SET]. To specify an output file name, type the output file name and press the soft key [F SET].
If, in this step, no output file name or no program name is specified, all programs in the foreground folders are output to "ALL-PROG.TXT".
For an explanation of the operations to be performed if an output file name [F SET] and an output program name [P SET] are omitted, see the table below.
7 Press the soft key [EXEC].
This starts outputting the program, and "OUTPUT" blinks in the lower right part of the screen. When the read operation ends, the "OUTPUT" indication disappears.
To cancel the output, press the soft key [CAN].

NOTE
Foreground folders are displayed on the program folder screen. For an explanation of foreground folders, see II-12.1.3, and for how to change foreground folders, see III-11.6

[F SET]	[P SET]	Output file name	Output program
BLANK	BLANK or (O-9999)	ALL-PROG.TXT	All programs in the foreground folders displayed in the program folder
BLANK	INPUT	Program name set with [P SET]	Program in the NC that is set with [P SET]
INPUT	BLANK or (O-9999)	File name set with [F SET]	All programs in the foreground folders displayed in the program folder
INPUT	INPUT	File name set with [F SET]	Program in the NC that is set with [P SET]

Outputting a program (for 15-inch display unit)

Procedure

1 Make sure the output device is ready for output.
2 Press the function key .
3 Press the vertical soft key [PROGRAM] or [FOLDER] to display programs or a program directory.
4 Press the EDIT switch on the machine operator's panel or enter state emergency stop.
5 Press the continuous menu key until horizontal soft key [PUNCH] appears.
Press the horizontal soft key [PUNCH].
6 Type the program number to output and press the horizontal soft key [P SET]. To specify an output file name, type the output file name and press the horizontal soft key [F SET].
If, in this step, no output file name or no program name is specified, all programs in the foreground folders are output to "ALL-PROG.TXT".
For an explanation of the operations to be performed if an output file name [F SET] and an output program name [P SET] are omitted, see the table below.
7 Press the horizontal soft key [EXEC].
This starts outputting the program, and "OUTPUT" blinks in the lower right part of the screen. When the read operation ends, the "OUTPUT" indication disappears.
To cancel the output, press the horizontal soft key [CAN].

NOTE
Foreground folders are displayed on the program folder screen. For an explanation of foreground folders, see II-12.1.3, and for how to change foreground folders, see III-11.6

[F SET]	[P SET]	Output file name	Output program
BLANK	BLANK or (O-9999)	ALL-PROG.TXT	All programs in the foreground folders displayed in the program folder
BLANK	INPUT	Program name set with [P SET]	Program in the NC that is set with [P SET]
INPUT	BLANK or (O-9999)	File name set with [F SET]	All programs in the foreground folders displayed in the program folder
INPUT	INPUT	File name set with [F SET]	Program in the NC that is set with [P SET]

8.2.2 Inputting and Outputting Parameters

8.2.2.1 Inputting parameters

Parameters are loaded into the memory of the CNC unit from an external device. The input format is the same as the output format. When a parameter is loaded which has the same data number as a parameter already registered in the memory, the loaded parameter replaces the existing parameter.

Inputting parameters (for 7.2/8.4/10.4-inch display unit)

Procedure

1 Make sure the input device is ready for reading.
2 Press the function key .
3 Press the continuous menu key until soft key [SETTING] appears.
Press the soft key [SETTING].
4 Press the MDI switch on the machine operator's panel or enter state emergency stop.
5 Enter 1 in response to the prompt for "PARAMETER WRITE" in setting data.
Alarm SW0100 appears.
6 Press the function key .
7 Press the soft key [PARAMETER], then the parameter screen appears.
8 Press the EDIT switch on the machine operator's panel or enter state emergency stop.
9 Press the soft key [(OPRT)].
10 Press the continuous menu key until soft key [READ] appears.
Press the soft key [READ].
11 Type the name of the file that you want to input.
If the input file name is omitted, default input file name "CNC-PARA.TXT" is assumed.
12 Press the soft key [EXEC].
This starts reading the parameter, and "INPUT" blinks in the lower right part of the screen. When the read operation ends, the "INPUT" indication disappears.
To cancel the input of the program, press the soft key [CAN].
13 Press the function key .
14 Press the soft key [SETTING].
15 Press the MDI switch on the machine operator's panel or enter state emergency stop.
16 Enter 0 in response to the prompt for "PARAMETER WRITE" in setting data.
17 Turn the power of the CNC on again.

Inputting parameters (for 15-inch display unit)

Procedure

1 Make sure the input device is ready for reading.
2 Press the function key .
3 Press the vertical soft key [NEXT PAGE] until vertical soft key [SETTING] appears.
Press the vertical soft key [SETTING].
4 Press the MDI switch on the machine operator's panel or enter state emergency stop.
5 Enter 1 in response to the prompt for "PARAMETER WRITE" in setting data.
Alarm SW0100 appears.
6 Press the function key .
7 Press the vertical soft key [NEXT PAGE] until vertical soft key [PARAMETER] appears.
Press the vertical soft key [PARAMETER], then the parameter screen appears.
8 Press the EDIT switch on the machine operator's panel or enter state emergency stop.
9 Press the horizontal soft key [READ].
10 Type the name of the file that you want to input.
If the input file name is omitted, default input file name "CNC-PARA.TXT" is assumed.
11 Press the horizontal soft key [EXEC].
This starts reading the parameter, and "INPUT" blinks in the lower right part of the screen. When the read operation ends, the "INPUT" indication disappears.
To cancel the input of the program, press the horizontal soft key [CAN].
12 Press the function key .
13 Press the vertical soft key [SETTING].
14 Press the MDI switch on the machine operator's panel or enter state emergency stop.
15 Enter 0 in response to the prompt for "PARAMETER WRITE" in setting data.
16 Turn the power of the CNC on again.

8.2.2.2 Outputting parameters

All parameters are output in a defined output format from the memory of the CNC to an external device.

Outputting parameters (for 7.2/8.4/10.4-inch display unit)

Procedure

1 Make sure the output device is ready for output.
2 Press the function key .
3 Press the soft key [PARAMETER], then the parameter screen appears.
4 Press the EDIT switch on the machine operator's panel or enter state emergency stop.
5 Press the soft key [(OPRT)].
6 Press the continuous menu key until soft key [PUNCH] appears.
Press the soft key [PUNCH].
7 Type the file name that you want to output.
If the file name is omitted, default file name "CNC-PARA.TXT" is assumed.
8 Press the soft key [EXEC].
This starts outputting the parameter, and "OUTPUT" blinks in the lower right part of the screen. When the read operation ends, the "OUTPUT" indication disappears.
To cancel the output, press the soft key [CAN].

Outputting parameters (for 15-inch display unit)

Procedure

1 Make sure the output device is ready for output.
2 Press the function key .
3 Press the vertical soft key [NEXT PAGE] until vertical soft key [PARAMETER] appears.
Press the vertical soft key [PARAMETER], then the parameter screen appears.
4 Press the EDIT switch on the machine operator's panel or enter state emergency stop.
5 Press the horizontal soft key [PUNCH].
6 Type the file name that you want to output.
If the file name is omitted, default file name "CNC-PARA.TXT" is assumed.
7 Press the horizontal soft key [EXEC].
This starts outputting the parameter, and "OUTPUT" blinks in the lower right part of the screen. When the read operation ends, the "OUTPUT" indication disappears.
To cancel the output, press the horizontal soft key [CAN].

Explanation

- Suppressing output of parameters set to 0

When bit 1 (PRM) of parameter No. 0010 is set to 1, and soft key [EXEC] is pressed, the following parameters are not output:

	Other than axis type	Axis type
Bit type	Parameter for which all bits are set to 0.	Parameter for an axis for which all bits are set to 0.
Value type	Parameter whose value is 0.	Parameter for an axis for which the value is 0.

8.2.3 Inputting and Outputting Offset Data

8.2.3.1 Inputting offset data

Offset data is loaded into the memory of the CNC from an external device. The input format is the same as for offset value output. When an offset value is loaded which has the same offset number as an offset number already registered in the memory, the loaded offset data replaces existing data.

Inputting offset data (for 7.2/8.4/10.4-inch display unit)

Procedure

1 Make sure the input device is ready for reading.
2 Press the function key .
3 Press the continuous menu key until soft key [OFFSET] appears.
Press the soft key [OFFSET].
4 Press the EDIT switch on the machine operator's panel or enter state emergency stop.
5 Press the soft key [(OPRT)].
6 Press the continuous menu key until soft key [READ] appears.
Press the soft key [READ].
7 Type the name of the file that you want to input.
If the input file name is omitted, default input file name "TOOLOFST.TXT" is assumed.
8 Press the soft key [EXEC].
This starts reading the offset data, and "INPUT" blinks in the lower right part of the screen. When the read operation ends, the "INPUT" indication disappears.
To cancel the input of the program, press the soft key [CAN].

Inputting offset data (for 15-inch display unit)

Procedure

1 Make sure the input device is ready for reading.
2 Press the function key .
3 Press the vertical soft key [NEXT PAGE] until vertical soft key [OFFSET] appears.
Press the vertical soft key [OFFSET].
4 Press the EDIT switch on the machine operator's panel or enter state emergency stop.
5 Press the horizontal soft key [READ].
6 Type the name of the file that you want to input.
If the input file name is omitted, default input file name "TOOLOFST.TXT" is assumed.
7 Press the horizontal soft key [EXEC].
This starts reading the offset data, and "INPUT" blinks in the lower right part of the screen. When the read operation ends, the "INPUT" indication disappears.
To cancel the input of the program, press the horizontal soft key [CAN].

8.2.3.2 Outputting offset data

All offset data is output in a defined output format from the memory of the CNC to an external device.

Outputting offset data (for 7.2/8.4/10.4-inch display unit)

Procedure

1 Make sure the output device is ready for output.
2 Press the function key .
3 Press the continuous menu key until soft key [OFFSET] appears.
Press the soft key [OFFSET].
4 Press the EDIT switch on the machine operator's panel or enter state emergency stop.
5 Press the soft key [(OPRT)].
6 Press the continuous menu key until soft key [PUNCH] appears.
Press the soft key [PUNCH].
7 Type the file name that you want to output.
If the file name is omitted, default file name "TOOLOFST.TXT" is assumed.
8 Press the soft key [EXEC].
This starts outputting the offset data, and "OUTPUT" blinks in the lower right part of the screen. When the read operation ends, the "OUTPUT" indication disappears.
To cancel the output, press the soft key [CAN].

Outputting offset data (for 15-inch display unit)

Procedure

1 Make sure the output device is ready for output.
2 Press the function key .
3 Press the vertical soft key [NEXT PAGE] until vertical soft key [OFFSET] appears.
Press the vertical soft key [OFFSET].
4 Press the EDIT switch on the machine operator's panel or enter state emergency stop.
5 Press the horizontal soft key [PUNCH].
6 Type the file name that you want to output.
If the file name is omitted, default file name "TOOLOFST.TXT" is assumed.
7 Press the horizontal soft key [EXEC].
This starts outputting the offset data, and "OUTPUT" blinks in the lower right part of the screen. When the read operation ends, the "OUTPUT" indication disappears.
To cancel the output, press the horizontal soft key [CAN].

Explanation

- Output format

Output format is as follows:

M

- Tool compensation memory A

```
%
G10 G90 P01 R_ Q_
G10 G90 P02 R_ Q_
 ...
G10 G90 P_ R_
%
```

Q_ : Virtual tool nose number (TIP). Not output when the virtual tool nose direction is not used.
P_ : Tool offset number (1 to the number of tool compensation pairs)
R_ : Tool compensation data. Output with a decimal point in the input unit used at output.

- Tool compensation memory B

```
%
G10 G90 L10 P01 R_ Q_
G10 G90 L11 P01 R_
G10 G90 L10 P02 R_ Q_
 ...
G10 G90 L11 P_ R_
%
```

L10 : Geometric compensation amount
L11 : Wear compensation amount
Q_, P_, and R_ have the same meanings as for tool compensation memory A.

- Tool compensation memory C

```
%
G10 G90 L10 P01 R_ Q_
G10 G90 L11 P01 R_
G10 G90 L12 P01 R_
G10 G90 L13 P01 R_
G10 G90 L10 P02 R_ Q_
 ...
G10 G90 L12 P_ R_
G10 G90 L13 P_ R_
%
```

L10 : Geometry compensation amount corresponding to the H code
L11 : Wear compensation amount corresponding to the H code
L12 : Geometry compensation amount corresponding to the D code
L13 : Wear compensation amount corresponding to the D code
Q_, P_, and R_ have the same meanings as for tool compensation memory A.

T

The tool compensation amount and tool nose radius compensation amount are output in the following format.

```
%
G10 P01 X_ Z_ R_ Q_ Y_
G10 P02 X_ Z_ R_ Q_ Y_
...
G10 P__ X_ Z_ R_ Q_ Y_
G10 P10001 X_ Z_ R_ Y_
G10 P10002 X_ Z_ R_ Y_
...
G10 P100__ X_ Z_ R_ Y_
%
```

P_: Tool compensation number (1 to the number of tool compensation pairs)
Tool offset number: Specification of the tool compensation amount and tool wear compensation amount
10000 + tool offset number: Specification of the tool geometry compensation amount

X_: Tool compensation data (X). Output with a decimal point in the input unit used at output.

Z_: Tool compensation data (Z). Same as X_.

R_: Tool nose radius offset amount (R). The data format is the same as for X_.
When tool nose radius compensation is not provided, this item is not output.

Q_: Virtual tool nose number (TIP). When tool nose radius compensation is not provided, this item is not output.

Y_ : Tool compensation data (Y). The data format is the same as for X_.
When no Y-axis offset is provided, this item is not output.

The second geometry tool offset data is output in the following format.

```
%
G10 P20001 X_ Z_ Y_
G10 P20002 X_ Z_ Y_

G10 P200__ X_ Z_ Y_
%
```

P_ : Tool compensation number (1 to the number of tool compensation pairs)
Tool offset number: Specification of the tool compensation amount and tool wear compensation amount
20000 + tool offset number: Specification of the second tool geometry compensation amount
The other addresses are the same as for the tool compensation amount.

NOTE
The input format and output format do not depend on the G-code system A/B/C.

8.2.4 Inputting and Outputting Pitch Error Compensation Data

8.2.4.1 Inputting pitch error compensation data

Pitch error compensation data are loaded into the memory of the CNC from an external device. The input format is the same as the output format. When a pitch error compensation data is loaded which has the corresponding data number as a pitch error compensation data already registered in the memory, the loaded data replaces the existing data.

Inputting pitch error compensation data (for 7.2/8.4/10.4-inch display unit)

Procedure

1 Make sure the input device is ready for reading.
2 Press the function key .
3 Press the continuous menu key until soft key [SETTING] appears.
Press the soft key [SETTING].
4 Press the MDI switch on the machine operator's panel or enter state emergency stop.
5 Enter 1 in response to the prompt for "PARAMETER WRITE" in setting data.
Alarm SW0100 appears.
6 Press the function key .
7 Press the continuous menu key until soft key [PITCH ERROR] appears.
Press the soft key [PITCH ERROR].
8 Press the EDIT switch on the machine operator's panel or enter state emergency stop.
9 Press the soft key [(OPRT)].
10 Press the continuous menu key until soft key [READ] appears.
Press the soft key [READ].
11 Type the name of the file that you want to input.
If the input file name is omitted, default input file name "PITCH.TXT" is assumed.
12 Press the soft key [EXEC].
This starts reading the pitch error compensation data, and "INPUT" blinks in the lower right part of the screen. When the read operation ends, the "INPUT" indication disappears.
To cancel the input of the program, press the soft key [CAN].
13 Press the function key .
14 Press the soft key [SETTING].
15 Press the MDI switch on the machine operator's panel or enter state emergency stop.
16 Enter 0 in response to the prompt for "PARAMETER WRITE" in setting data.
17 Turn the power of the CNC on again.

Inputting pitch error compensation data (for 15-inch display unit)

Procedure

1 Make sure the input device is ready for reading.
2 Press the function key .
3 Press the vertical soft key [NEXT PAGE] until vertical soft key [SETTING] appears.
Press the vertical soft key [SETTING].
4 Press the MDI switch on the machine operator's panel or enter state emergency stop.
5 Enter 1 in response to the prompt for "PARAMETER WRITE" in setting data.
Alarm SW0100 appears.
6 Press the function key .
7 Press the vertical soft key [NEXT PAGE] until vertical soft key [PITCH ERROR] appears.
Press the vertical soft key [PITCH ERROR].
8 Press the EDIT switch on the machine operator's panel or enter state emergency stop.
9 Press the horizontal soft key [READ].
10 Type the name of the file that you want to input.
If the input file name is omitted, default input file name "PITCH.TXT" is assumed.
11 Press the horizontal soft key [EXEC].
This starts reading the pitch error compensation data, and "INPUT" blinks in the lower right part of the screen. When the read operation ends, the "INPUT" indication disappears.
To cancel the input of the program, press the horizontal soft key [CAN].
12 Press the function key .
13 Press the vertical soft key [SETTING].
14 Press the MDI switch on the machine operator's panel or enter state emergency stop.
15 Enter 0 in response to the prompt for "PARAMETER WRITE" in setting data.
16 Turn the power of the CNC on again.

8.2.4.2 Outputting pitch error compensation data

All pitch error compensation data are output in a defined output format from the memory of the CNC to an external device.

Outputting pitch error compensation data (for 7.2/8.4/10.4-inch display unit)

Procedure

1 Make sure the output device is ready for output.
2 Press the function key SYSTEM.
3 Press the continuous menu key ▷ until soft key [PITCH ERROR] appears.
Press the soft key [PITCH ERROR].
4 Press the EDIT switch on the machine operator's panel or enter state emergency stop.
5 Press the soft key [(OPRT)].
6 Press the continuous menu key ▷ until soft key [PUNCH] appears.
Press the soft key [PUNCH].
7 Type the file name that you want to output.
If the file name is omitted, default file name "PITCH.TXT" is assumed.
8 Press the soft key [EXEC].
This starts outputting the pitch error compensation data, and "OUTPUT" blinks in the lower right part of the screen. When the read operation ends, the "OUTPUT" indication disappears.
To cancel the output, press the soft key [CAN].

Outputting pitch error compensation data (for 15-inch display unit)

Procedure

1 Make sure the output device is ready for output.
2 Press the function key SYSTEM.
3 Press the vertical soft key [NEXT PAGE] until vertical soft key [SETTING] appears.
Press the vertical soft key [PITCH ERROR].
4 Press the EDIT switch on the machine operator's panel or enter state emergency stop.
5 Press the horizontal soft key [PUNCH].
6 Type the file name that you want to output.
If the file name is omitted, default file name "PITCH.TXT" is assumed.
7 Press the horizontal soft key [EXEC].
This starts outputting the pitch error compensation data, and "OUTPUT" blinks in the lower right part of the screen. When the read operation ends, the "OUTPUT" indication disappears.
To cancel the output, press the horizontal soft key [CAN].

8.2.4.3 Input/output format of pitch error compensation data

Pitch error compensation data is input and output in the following input and output formats.

- Keywords

The following alphabets are used as keywords.
The numeric value following each keyword has the meaning listed below:

Keyword	Meaning of the following numeric value
N	Pitch error compensation data number + 10000
Q	Data identification (1 : Parameter data, 0 : Pitch error compensation data)
P	Pitch error compensation data value

- Format

Pitch error compensation data is output in the following format:

N	*****	Q0	P	****	;

The 5-digit numeric value following N indicates a pitch error compensation data number to which a value of 10000 is added.
Q0 indicates pitch error compensation data
The numeric value following P indicates the value (integer value) of pitch error compensation data between -128 to 127.
The semicolon (;) indicates the end of block (LF in the ISO code or CR in the EIA code).

Example

```
N10001Q0P100;
   Pitch error compensation data number    1
   Pitch error compensation data value     100
```

- Beginning and end of a record

A pitch error compensation data record begins with % and ends with %.

Example

```
% ;          .....Beginning of record
N10000Q0P10;
N10001Q0P100;
   :
N11279Q0P0;
%            ..... End of record
```

When parameters and pitch error compensation data are integrated into one file, % is added to the beginning and end of the file.

8.2.5 Inputting and Outputting Three-dimensional Error Compensation Data

8.2.5.1 Inputting three-dimensional error compensation data

Three-dimensional error compensation data are loaded into the memory of the CNC from an external device. The input format is the same as the output format. When a three-dimensional error compensation data is loaded which has the corresponding data number as a three-dimensional error compensation data already registered in the memory, the loaded data replaces the existing data.

Inputting three-dimensional error compensation data (for 7.2/8.4/10.4-inch display unit)

Procedure

1 Make sure the input device is ready for reading.
2 Press the function key .
3 Press the continuous menu key until soft key [SETTING] appears.
Press the soft key [SETTING].
4 Press the MDI switch on the machine operator's panel or enter state emergency stop.
5 Enter 1 in response to the prompt for "PARAMETER WRITE" in setting data.
Alarm SW0100 appears.
6 Press the function key .
7 Press the continuous menu key until soft key [3D ERR COMP] appears.
Press the soft key [3D ERR COMP].
8 Press the EDIT switch on the machine operator's panel or enter state emergency stop.
9 Press the soft key [(OPRT)].
10 Press the continuous menu key until soft key [READ] appears.
Press the soft key [READ].
11 Type the name of the file that you want to input.
If the input file name is omitted, default input file name "COMP3D.TXT" is assumed.
12 Press the soft key [EXEC].
This starts reading the three-dimensional error compensation data, and "INPUT" blinks in the lower right part of the screen. When the read operation ends, the "INPUT" indication disappears.
To cancel the input of the program, press the soft key [CAN].
13 Press the function key .
14 Press the soft key [SETTING].
15 Press the MDI switch on the machine operator's panel or enter state emergency stop.

16 Enter 0 in response to the prompt for “PARAMETER WRITE” in setting data.
17 Turn the power of the CNC on again.

Inputting three-dimensional error compensation data (for 15-inch display unit)

Procedure

1 Make sure the input device is ready for reading.
2 Press the function key .
3 Press the vertical soft key [NEXT PAGE] until vertical soft key [SETTING] appears.
Press the vertical soft key [SETTING].
4 Press the MDI switch on the machine operator’s panel or enter state emergency stop.
5 Enter 1 in response to the prompt for “PARAMETER WRITE” in setting data.
Alarm SW0100 appears.
6 Press the function key .
7 Press the vertical soft key [NEXT PAGE] until vertical soft key [3D ERR COMP] appears.
Press the vertical soft key [3D ERR COMP].
8 Press the EDIT switch on the machine operator’s panel or enter state emergency stop.
9 Press the horizontal soft key [READ].
10 Type the name of the file that you want to input.
If the input file name is omitted, default input file name “COMP3D.TXT” is assumed.
11 Press the horizontal soft key [EXEC].
This starts reading the three-dimensional error compensation data, and “INPUT” blinks in the lower right part of the screen. When the read operation ends, the “INPUT” indication disappears.
To cancel the input of the program, press the horizontal soft key [CAN].
12 Press the function key .
13 Press the vertical soft key [SETTING].
14 Press the MDI switch on the machine operator’s panel or enter state emergency stop.
15 Enter 0 in response to the prompt for “PARAMETER WRITE” in setting data.
16 Turn the power of the CNC on again.

8.2.5.2 Outputting three-dimensional error compensation data

All three-dimensional error compensation data are output in a defined output format from the memory of the CNC to an external device.

Outputting three-dimensional error compensation data (for 7.2/8.4/10.4-inch display unit)

Procedure

1 Make sure the output device is ready for output.
2 Press the function key .
3 Press the continuous menu key until soft key [3D ERR COMP] appears.
Press the soft key [3D ERR COMP].
4 Press the EDIT switch on the machine operator's panel or enter state emergency stop.
5 Press the soft key [(OPRT)].
6 Press the continuous menu key until soft key [PUNCH] appears.
Press the soft key [PUNCH].
7 Type the file name that you want to output.
If the file name is omitted, default file name "COMP3D.TXT" is assumed.
8 Press the soft key [EXEC].
This starts outputting the three-dimensional error compensation data, and "OUTPUT" blinks in the lower right part of the screen. When the read operation ends, the "OUTPUT" indication disappears.
To cancel the output, press the soft key [CAN].

Outputting three-dimensional error compensation data (for 15-inch display unit)

Procedure

1 Make sure the output device is ready for output.
2 Press the function key .
3 Press the vertical soft key [NEXT PAGE] until vertical soft key [3D ERR COMP] appears.
Press the vertical soft key [3D ERR COMP].
4 Press the EDIT switch on the machine operator's panel or enter state emergency stop.
5 Press the horizontal soft key [PUNCH].
6 Type the file name that you want to output.
If the file name is omitted, default file name "COMP3D.TXT" is assumed.
7 Press the horizontal soft key [EXEC].
This starts outputting the three-dimensional error compensation data, and "OUTPUT" blinks in the lower right part of the screen. When the read operation ends, the "OUTPUT" indication disappears.
To cancel the output, press the horizontal soft key [CAN].

8.2.5.3 Input/output format of three-dimensional error compensation data

Three-dimensional error compensation data is input and output in the following input and output formats.

- Keywords

The following alphabets are used as keywords.
The numeric value following each keyword has the meaning listed below:

Keyword	Meaning of the following numeric value
N	Data number (compensation point number + 100000) as represented with six digits
A1	1st compensation axis
A2	2nd compensation axis
A3	3rd compensation axis
P	Compensation data (-128 to 127)

- Format

Three-dimensional error compensation data is output in the following format:

N	*****	A1	P	****	A2	P	****	A3	P	****	;

The 6-digit numeric value following N indicates a three-dimensional error compensation data number to which a value of 100000 is added.
The numeric value following P indicates the value (integer value) of three-dimensional error compensation data between -128 to 127.
The semicolon (;) indicates the end of block (LF in the ISO code or CR in the EIA code).

Example

N100001A1P100A2P110A3P120;

Three-dimensional error compensation data number: 1
Error compensation data value for the 1st compensation axis: 100
Error compensation data value for the 2nd compensation axis: 110
Error compensation data value for the 3rd compensation axis: 120

- Beginning and end of a record

A three-dimensional error compensation data record begins with % and ends with %.

Example

```
%  ;          .....Beginning of record
N100001 A1 P1 A2 P2 A3 P3 ;
N100002 A1 P0 A2 P0 A3 P-3 ;
   :
N115625 A1 P1 A2 P1 A3 P0 ;
%             ..... End of record
```

- Input of compensation data using G10

Compensation data can be changed from a machining program, using the programmable parameter input function.

The command format is as follows:

```
%
G10 L51 ;
N_ P_ R_ ;
N_ P_ R_ ;
:
G11 ;
%
```

G10 L51: Three-dimensional error compensation data input mode
G11 : Cancellation of three-dimensional error compensation data input mode
N : Compensation point number (1-15625)
P : Compensation axis number (1-3)
R : Compensation data (-128-127)

NOTE

1 To input compensation data using G10, the option of the programmable parameter input function is required.
2 In three-dimensional error compensation data input mode, no other NC statements can be issued.
3 The decimal point cannot be used in address N, P, and R.

8.2.6 Inputting and Outputting Custom Macro Common Variables

8.2.6.1 Inputting custom macro common variables

The value of a custom macro common variable is loaded into the memory of the CNC from an external device. The same format used to output custom macro common variables is used for input.

Inputting custom macro common variables (for 7.2/8.4/10.4-inch display unit)

Procedure

1 Make sure the input device is ready for reading.
2 Press the function key .
3 Press the continuous menu key until soft key [MACRO] appears.
Press the soft key [MACRO].
4 Press the EDIT switch on the machine operator's panel or enter state emergency stop.
5 Press the soft key [(OPRT)].
6 Press the continuous menu key until soft key [READ] appears.
Press the soft key [READ].
7 Type the name of the file that you want to input.
If the input file name is omitted, default input file name "MACRO.TXT" is assumed.
8 Press the soft key [EXEC].
This starts reading the custom macro common variables, and "INPUT" blinks in the lower right part of the screen. When the read operation ends, the "INPUT" indication disappears.
To cancel the input of the program, press the soft key [CAN].

Inputting custom macro common variables (for 15-inch display unit)

Procedure

1 Make sure the input device is ready for reading.
2 Press the function key .
3 Press the vertical soft key [NEXT PAGE] until vertical soft key [MACRO] appears.
Press the vertical soft key [MACRO].
4 Press the EDIT switch on the machine operator's panel or enter state emergency stop.
5 Press the horizontal soft key [READ].
6 Type the name of the file that you want to input.
If the input file name is omitted, default input file name "MACRO.TXT" is assumed.
7 Press the horizontal soft key [EXEC].
This starts reading the custom macro common variables, and "INPUT" blinks in the lower right part of the screen. When the read operation ends, the "INPUT" indication disappears.
To cancel the input of the program, press the horizontal soft key [CAN].

Explanation

- Common variables

The common variables (#500 to #549) can be input and output.
(When the option for addition of common variable is specified, values from #500 to #999 can be input and output.) #100 to #149 can be input when bit 3 (PV5) of parameter No. 6001 is set to 1. (When the option for addition of common variable is specified, values from #100 to #199 can be input and output.)

8.2.6.2 Outputting custom macro common variables

Custom macro common variables stored in the memory of the CNC can be output in a defined output format to an external device.

Outputting custom macro common variables (for 7.2/8.4/10.4-inch display unit)

Procedure

1 Make sure the output device is ready for output.
2 Press the function key .
3 Press the continuous menu key until soft key [MACRO] appears.
Press the soft key [MACRO].
4 Press the EDIT switch on the machine operator's panel or enter state emergency stop.
5 Press the soft key [(OPRT)].
6 Press the continuous menu key until soft key [PUNCH] appears.
Press the soft key [PUNCH].
7 Type the file name that you want to output.
If the file name is omitted, default file name "MACRO.TXT" is assumed.
8 Press the soft key [EXEC].
This starts outputting the custom macro common variables, and "OUTPUT" blinks in the lower right part of the screen. When the read operation ends, the "OUTPUT" indication disappears.
To cancel the output, press the soft key [CAN].

Outputting custom macro common variables (for 15-inch display unit)

Procedure

1 Make sure the output device is ready for output.
2 Press the function key .
3 Press the vertical soft key [NEXT PAGE] until vertical soft key [MACRO] appears.
Press the vertical soft key [MACRO].
4 Press the EDIT switch on the machine operator's panel or enter state emergency stop.
5 Press the horizontal soft key [PUNCH].
6 Type the file name that you want to output.
If the file name is omitted, default file name "MACRO.TXT" is assumed.
7 Press the horizontal soft key [EXEC].
This starts outputting the custom macro common variables, and "OUTPUT" blinks in the lower right part of the screen. When the read operation ends, the "OUTPUT" indication disappears.
To cancel the output, press the horizontal soft key [CAN].

Explanation

- Output format

The output format is as follows:
The values of custom macro variables are output in a bit-image hexadecimal representation of double-precision floating-point type data.

```
%
G10L85P200(0000000000000000)
G10L85P200(0000000000000000)
G10L85P200(FFFFFFFFFFFFFFFF)
 :
G10L85P500(4024000000000000)
G10L85P501(4021000000000000)
G10L85P502(0000000000000000)
 :
SETVN500[ABC,DEF]
SETVN501[GHI,JKL]
SETVN502[MNO,PQR]
 :
M02
%
```

NOTE
The conventional custom macro statement program format cannot be used for output.

By setting bit 0 (MCO) of parameter No. 6019, it is possible to output macro variable numbers and variable data values as comments after normally output data.
The output comments do not affect data input.

- Common variable

The common variables (#500 to #549) can be input and output.
(When the option for addition of common variable is specified, values from #500 to #999 can be input and output.) #100 to #149 can be output when bit 3 (PV5) of parameter No. 6001 is set to 1. (When the option for addition of common variable is specified, values from #100 to #199 can be input and output.)

8.2.7 Inputting and Outputting Workpiece Coordinates System Data

8.2.7.1 Inputting workpiece coordinate system data

Coordinate system variable data is loaded into the memory of the CNC from an external device. The input format is the same as the output format. When coordinate system variable data with a data number corresponding to existing coordinate system variable data registered in the memory is loaded, the loaded coordinate system variable data replaces the existing coordinate system variable data.

Inputting workpiece coordinate system data (for 7.2/8.4/10.4-inch display unit)

Procedure

1 Make sure the input device is ready for reading.
2 Press the function key .
3 Press the continuous menu key ▷ until soft key [WORK] appears.
Press the soft key [WORK].
4 Press the EDIT switch on the machine operator's panel or enter state emergency stop.
5 Press the soft key [(OPRT)].
6 Press the continuous menu key ▷ until soft key [READ] appears.
Press the soft key [READ].
7 Type the name of the file that you want to input.
If the input file name is omitted, default input file name "EXT_WKZ.TXT" is assumed.
8 Press the soft key [EXEC].
This starts reading the workpiece coordinate system data, and "INPUT" blinks in the lower right part of the screen. When the read operation ends, the "INPUT" indication disappears.
To cancel the input of the program, press the soft key [CAN].

Inputting workpiece coordinate system data (for 15-inch display unit)

Procedure

1 Make sure the input device is ready for reading.
2 Press the function key .
3 Press the vertical soft key [NEXT PAGE] until vertical soft key [WORK] appears.
Press veertical soft key [WORK].
4 Press the EDIT switch on the machine operator's panel or enter state emergency stop.
5 Press the horizontal soft key [READ].
6 Type the name of the file that you want to input.
If the input file name is omitted, default input file name "EXT_WKZ.TXT" is assumed.
7 Press the horizontal soft key [EXEC].
This starts reading the workpiece coordinate system data, and "INPUT" blinks in the lower right part of the screen. When the read operation ends, the "INPUT" indication disappears.
To cancel the input of the program, press the horizontal soft key [CAN].

8.2.7.2 Outputting workpiece coordinate system data

All coordinate system variable data is output in the output format from the memory of the CNC to an external device.

Outputting workpiece coordinate system data (for 7.2/8.4/10.4-inch display unit)

Procedure

1 Make sure the output device is ready for output.
2 Press the function key .
3 Press the continuous menu key until soft key [WORK] appears.
Press the soft key [WORK].
4 Press the EDIT switch on the machine operator's panel or enter state emergency stop.
5 Press the soft key [(OPRT)].
6 Press the continuous menu key until soft key [PUNCH] appears.
Press the soft key [PUNCH].
7 Type the file name that you want to output.
If the file name is omitted, default file name "EXT_WKZ.TXT" is assumed.
8 Press the soft key [EXEC].
This starts outputting the workpiece coordinate system data, and "OUTPUT" blinks in the lower right part of the screen. When the read operation ends, the "OUTPUT" indication disappears.
To cancel the output, press the soft key [CAN].

Outputting workpiece coordinate system data (for 15-inch display unit)

Procedure

1 Make sure the output device is ready for output.
2 Press the function key .
3 Press the vertical soft key [NEXT PAGE] until vertical soft key [WORK] appears.
Press veertical soft key [WORK].
4 Press the EDIT switch on the machine operator's panel or enter state emergency stop.
5 Press the horizontal soft key [PUNCH].
6 Type the file name that you want to output.
If the file name is omitted, default file name "EXT_WKZ.TXT" is assumed.
7 Press the horizontal soft key [EXEC].
This starts outputting the workpiece coordinate system data, and "OUTPUT" blinks in the lower right part of the screen. When the read operation ends, the "OUTPUT" indication disappears.
To cancel the output, press the horizontal soft key [CAN].

8.2.8 Inputting and Outputting Operation History Data

Only output operation is permitted on operation history data.
The output data is in text format. So, to reference the output data you must use an application that can handle text files on the personal computer.

8.2.8.1 Outputting operation history data

All operation history data is output in the output format form the memory of the CNC to an external device.

Outputting operation history data (for 7.2/8.4/10.4-inch display unit)

Procedure

1 Make sure the output device is ready for output.
2 Press the function key

.
3 Press the continuous menu key until soft key [OPERAT HISTRY] appears.
Press the soft key [OPERAT HISTRY].
4 Press the EDIT switch on the machine operator's panel or enter state emergency stop.
5 Press the soft key [(OPRT)].
6 Press the soft key [PUNCH].
7 Type the file name that you want to output.
If the file name is omitted, default file name "OPRT_HIS.TXT" is assumed.
8 Press the soft key [EXEC].
This starts outputting the operation history data, and "OUTPUT" blinks in the lower right part of the screen. When the read operation ends, the "OUTPUT" indication disappears.
To cancel the output, press the soft key [CAN].

Outputting operation history data (for 15-inch display unit)

Procedure

1 Make sure the output device is ready for output.
2 Press the function key

.
3 Press the vertical soft key [NEXT PAGE] until vertical soft key [OPERAT HISTRY] appears.
Press the vertical soft key [OPERAT HISTRY].
4 Press the EDIT switch on the machine operator's panel or enter state emergency stop.
5 Press the horizontal soft key [PUNCH].
6 Type the file name that you want to output.
If the file name is omitted, default file name "OPRT_HIS.TXT" is assumed.
7 Press the horizontal soft key [EXEC].
This starts outputting the operation history data, and "OUTPUT" blinks in the lower right part of the screen. When the read operation ends, the "OUTPUT" indication disappears.
To cancel the output, press the horizontal soft key [CAN].

8.2.9 Inputting and Outputting Tool Management Data

NOTE
1 For multi-path systems, place all paths in the EDIT mode before performing input and output operations.
2 The format used is the same as the registration format of the G10 format.

8.2.9.1 Inputting tool management data

Tool management data is loaded into the memory of the CNC from an external device. The input format is the same as the output format. When tool management data with a data number corresponding to existing tool management data registered in the memory is loaded, the loaded tool management data replaces the existing tool management data.

Inputting tool management data (for 7.2/8.4/10.4-inch display unit)

Procedure

1 Make sure the input device is ready for reading.
2 Press the function key to display the tool management screen or magazine screen.
3 Press the EDIT switch on the machine operator's panel.
4 Press the soft key [(OPRT)].
5 Press the soft key [READ].
6 Press the soft key [TOOL].
7 Type the name of the file that you want to input.
If the input file name is omitted, default input file name "TOOL_MNG.TXT" is assumed.
8 Press the soft key [EXEC].
This starts reading the tool management data, and "INPUT" blinks in the lower right part of the screen. When the read operation ends, the "INPUT" indication disappears.
To cancel the input of the program, press the soft key [CAN].

NOTE
When using oversize tool support of the tool management function, keep the following in mind.
- If a target tool is registered in a cartridge and interferes with other tools in registration or modification of tool geometry data of the tool management data, PS alarm 5360 is issued. (The data is not input.)
- When restoring backup data in the state where all data related to the tool management function in the NC is cleared, restore tool geometry data, tool management data, and cartridge management table data in this order.

Inputting tool management data (for 15-inch display unit)

Procedure

1 Make sure the input device is ready for reading.
2 Press the function key .
3 Press the vertical soft key [NEXT PAGE] until vertical soft key [TOOL MANAGER] appears.
Press the vertical soft key [TOOL MANAGER].
4 Press the vertical soft key [MAGAZINE] or [TOOL].
5 Press the EDIT switch on the machine operator's panel.
6 Press the horizontal soft key [READ].
7 Press the horizontal soft key [TOOL].
8 Type the name of the file that you want to input.
If the input file name is omitted, default input file name "TOOL_MNG.TXT" is assumed.
9 Press the horizontal soft key [EXEC].
This starts reading the tool management data, and "INPUT" blinks in the lower right part of the screen. When the read operation ends, the "INPUT" indication disappears.
To cancel the input of the program, press the horizontal soft key [CAN].

NOTE
When using oversize tool support of the tool management function, keep the following in mind.
- If a target tool is registered in a cartridge and interferes with other tools in registration or modification of tool geometry data of the tool management data, PS alarm 5360 is issued. (The data is not input.)
- When restoring backup data in the state where all data related to the tool management function in the NC is cleared, restore tool geometry data, tool management data, and cartridge management table data in this order.

8.2.9.2 Outputting tool management data

All tool management data is output in the output format from the memory of the CNC to an external device.

Outputting tool management data (for 7.2/8.4/10.4-inch display unit)

Procedure

1 Make sure the output device is ready for output.
2 Press the function key to display the tool management screen or magazine screen.
3 Press the EDIT switch on the machine operator's panel.
4 Press the soft key [(OPRT)].
5 Press the soft key [PUNCH].
6 Press the soft key [TOOL].
7 Type the file name that you want to output.
If the output file name is omitted, default output file name "TOOL_MNG.TXT" is assumed.
8 Press the soft key [EXEC].
This starts outputting the tool management data, and "OUTPUT" blinks in the lower right part of the screen. When the read operation ends, the "OUTPUT" indication disappears.
To cancel the output, press the soft key [CAN].

Outputting tool management data (for 15-inch display unit)

Procedure

1 Make sure the output device is ready for output.
2 Press the function key .
3 Press the vertical soft key [NEXT PAGE] until vertical soft key [TOOL MANAGER] appears.
Press the vertical soft key [TOOL MANAGER].
4 Press the vertical soft key [MAGAZINE] or [TOOL].
5 Press the EDIT switch on the machine operator's panel.
6 Press the horizontal soft key [PUNCH].
7 Press the horizontal soft key [TOOL].
8 Type the file name that you want to output.
If the output file name is omitted, default output file name "TOOL_MNG.TXT" is assumed.
9 Press the horizontal soft key [EXEC].
This starts outputting the tool management data, and "OUTPUT" blinks in the lower right part of the screen. When the read operation ends, the "OUTPUT" indication disappears.
To cancel the output, press the horizontal soft key [CAN].

8.2.9.3 Inputting magazine data

Magazine data is loaded into the memory of the CNC from an external device. The input format is the same as the output format. When magazine data with a data number corresponding to existing magazine data registered in the memory is loaded, the loaded magazine data replaces the existing magazine data.

Inputting magazine data (for 7.2/8.4/10.4-inch display unit)

Procedure

1 Make sure the input device is ready for reading.
2 Press the function key to display the tool management screen or magazine screen.
3 Press the EDIT switch on the machine operator's panel.
4 Press the soft key [(OPRT)].
5 Press the soft key [READ].
6 Press the soft key [MAGAZINE].
7 Type the name of the file that you want to input.
If the input file name is omitted, default input file name "MAGAZINE.TXT" is assumed.
8 Press the soft key [EXEC].
This starts reading the magazine data, and "INPUT" blinks in the lower right part of the screen. When the read operation ends, the "INPUT" indication disappears.
To cancel the input of the program, press the soft key [CAN].

NOTE
When using oversize tool support of the tool management function, keep the following in mind.
- If a tool interferes with other tools in registration in or modification to the cartridge management table, PS alarm 5360 is issued. (The data is not input.)
- When restoring backup data in the state where all data related to the tool management function in the NC is cleared, restore tool geometry data, tool management data, and cartridge management table data in this order.

Inputting magazine data (for 15-inch display unit)

Procedure

1 Make sure the input device is ready for reading.
2 Press the function key .
3 Press the vertical soft key [NEXT PAGE] until vertical soft key [TOOL MANAGER] appears.
Press the vertical soft key [TOOL MANAGER].
4 Press the vertical soft key [MAGAZINE] or [TOOL].
5 Press the EDIT switch on the machine operator's panel.
6 Press the horizontal soft key [READ].
7 Press the horizontal soft key [MAGAZINE].
8 Type the name of the file that you want to input.
If the input file name is omitted, default input file name "MAGAZINE.TXT" is assumed.
9 Press the horizontal soft key [EXEC].
This starts reading the magazine data, and "INPUT" blinks in the lower right part of the screen. When the read operation ends, the "INPUT" indication disappears.
To cancel the input of the program, press the horizontal soft key [CAN].

NOTE
When using oversize tool support of the tool management function, keep the following in mind.
- If a tool interferes with other tools in registration in or modification to the cartridge management table, PS alarm 5360 is issued. (The data is not input.)
- When restoring backup data in the state where all data related to the tool management function in the NC is cleared, restore tool geometry data, tool management data, and cartridge management table data in this order.

8.2.9.4 Outputting magazine data

All magazine data is output in the output format from the memory of the CNC to an external device.

Outputting magazine data (for 7.2/8.4/10.4-inch display unit)

Procedure

1 Make sure the output device is ready for output.
2 Press the function key to display the tool management screen or magazine screen.
3 Press the EDIT switch on the machine operator's panel.
4 Press the soft key [(OPRT)].
5 Press the soft key [PUNCH].
6 Press the soft key [MAGAZINE].
7 Type the file name that you want to output.
If the file name is omitted, default file name "MAGAZINE.TXT" is assumed.
8 Press the soft key [EXEC].
This starts outputting the magazine data, and "OUTPUT" blinks in the lower right part of the screen. When the read operation ends, the "OUTPUT" indication disappears.
To cancel the output, press the soft key [CAN].

Outputting magazine data (for 15-inch display unit)

Procedure

1 Make sure the output device is ready for output.
2 Press the function key .
3 Press the vertical soft key [NEXT PAGE] until vertical soft key [TOOL MANAGER] appears.
Press the vertical soft key [TOOL MANAGER].
4 Press the vertical soft key [MAGAZINE] or [TOOL].
5 Press the EDIT switch on the machine operator's panel.
6 Press the horizontal soft key [PUNCH].
7 Press the horizontal soft key [MAGAZINE].
8 Type the file name that you want to output.
If the file name is omitted, default file name "MAGAZINE.TXT" is assumed.
9 Press the horizontal soft key [EXEC].
This starts outputting the magazine data, and "OUTPUT" blinks in the lower right part of the screen. When the read operation ends, the "OUTPUT" indication disappears.
To cancel the output, press the horizontal soft key [CAN].

8.2.9.5 Inputting tool life status name data

Tool life status name data is loaded into the memory of the CNC from an external device. The input format is the same as the output format. When tool life status name data with a data number corresponding to existing tool life status name data registered in the memory is loaded, the loaded tool life status name data replaces the existing tool life status name data.

Inputting tool life status name data (for 7.2/8.4/10.4-inch display unit)

Procedure

1 Make sure the input device is ready for reading.
2 Press the function key to display the tool management screen or magazine screen.
3 Press the EDIT switch on the machine operator's panel.
4 Press the soft key [(OPRT)].
5 Press the soft key [READ].
6 Press the soft key [STATUS].
7 Type the name of the file that you want to input.
If the input file name is omitted, default input file name "STATUS.TXT" is assumed.
8 Press the soft key [EXEC].
This starts reading the tool life status name data, and "INPUT" blinks in the lower right part of the screen. When the read operation ends, the "INPUT" indication disappears.
To cancel the input of the program, press the soft key [CAN].

Inputting tool life status name data (for 15-inch display unit)

Procedure

1 Make sure the input device is ready for reading.
2 Press the function key .
3 Press the vertical soft key [NEXT PAGE] until vertical soft key [TOOL MANAGER] appears.
Press the vertical soft key [TOOL MANAGER].
4 Press the vertical soft key [MAGAZINE] or [TOOL].
5 Press the EDIT switch on the machine operator's panel.
6 Press the horizontal soft key [READ].
7 Press the horizontal soft key [STATUS].
8 Type the name of the file that you want to input.
If the input file name is omitted, default input file name "STATUS.TXT" is assumed.
9 Press the horizontal soft key [EXEC].
This starts reading the tool life status name data, and "INPUT" blinks in the lower right part of the screen. When the read operation ends, the "INPUT" indication disappears.
To cancel the input of the program, press the horizontal soft key [CAN].

8.2.9.6 Outputting tool life status name data

All tool life status name data is output in the output format from the memory of the CNC to an external device.

Outputting tool life status name data (for 7.2/8.4/10.4-inch display unit)

Procedure

1 Make sure the output device is ready for output.
2 Press the function key to display the tool management screen or magazine screen.
3 Press the EDIT switch on the machine operator's panel.
4 Press the soft key [(OPRT)].
5 Press the soft key [PUNCH].
6 Press the soft key [STATUS].
7 Type the file name that you want to output.
If the file name is omitted, default file name "STATUS.TXT" is assumed.
8 Press the soft key [EXEC].
This starts outputting the tool life status name data, and "OUTPUT" blinks in the lower right part of the screen. When the read operation ends, the "OUTPUT" indication disappears.
To cancel the output, press the soft key [CAN].

Outputting tool life status name data (for 15-inch display unit)

Procedure

1 Make sure the output device is ready for output.
2 Press the function key .
3 Press the vertical soft key [NEXT PAGE] until vertical soft key [TOOL MANAGER] appears.
Press the vertical soft key [TOOL MANAGER].
4 Press the vertical soft key [MAGAZINE] or [TOOL].
5 Press the EDIT switch on the machine operator's panel.
6 Press the horizontal soft key [PUNCH].
7 Press the horizontal soft key [STATUS].
8 Type the file name that you want to output.
If the file name is omitted, default file name "STATUS.TXT" is assumed.
9 Press the horizontal soft key [EXEC].
This starts outputting the tool life status name data, and "OUTPUT" blinks in the lower right part of the screen. When the read operation ends, the "OUTPUT" indication disappears.
To cancel the output, press the horizontal soft key [CAN].

8.2.9.7 Inputting name data of customize data

Name data of customize data is loaded into the memory of the CNC from an external device. The input format is the same as the output format. When name data of customize data with a data number corresponding to existing name data of customize data registered in the memory is loaded, the loaded name data of customize data replaces the existing data.

Inputting name data of customize data (for 7.2/8.4/10.4-inch display unit)

Procedure

1 Make sure the input device is ready for reading.
2 Press the function key to display the tool management screen or magazine screen.
3 Press the EDIT switch on the machine operator's panel.
4 Press the soft key [(OPRT)].
5 Press the soft key [READ].
6 Press the soft key [CUSTOM].
7 Type the name of the file that you want to input.
If the input file name is omitted, default input file name "CUSTOMIZ.TXT" is assumed.
8 Press the soft key [EXEC].
This starts reading the name data of customize data, and "INPUT" blinks in the lower right part of the screen. When the read operation ends, the "INPUT" indication disappears.
To cancel the input of the program, press the soft key [CAN].

Inputting name data of customize data (for 15-inch display unit)

Procedure

1 Make sure the input device is ready for reading.
2 Press the function key .
3 Press the vertical soft key [NEXT PAGE] until vertical soft key [TOOL MANAGER] appears.
Press the vertical soft key [TOOL MANAGER].
4 Press the vertical soft key [MAGAZINE] or [TOOL].
5 Press the EDIT switch on the machine operator's panel.
6 Press the horizontal soft key [READ].
7 Press the horizontal soft key [CUSTOM].
8 Type the name of the file that you want to input.
If the input file name is omitted, default input file name "CUSTOMIZ.TXT" is assumed.
9 Press the horizontal soft key [EXEC].
This starts reading the name data of customize data, and "INPUT" blinks in the lower right part of the screen. When the read operation ends, the "INPUT" indication disappears.
To cancel the input of the program, press the horizontal soft key [CAN].

8.2.9.8 Outputting name data of customize data

All name data of customize data is output in the output format from the memory of the CNC to an external device.

Outputting name data of customize data (for 7.2/8.4/10.4-inch display unit)

Procedure

1 Make sure the output device is ready for output.
2 Press the function key to display the tool management screen or magazine screen.
3 Press the EDIT switch on the machine operator's panel.
4 Press the soft key [(OPRT)].
5 Press the soft key [PUNCH].
6 Press the soft key [CUSTOM].
7 Type the file name that you want to output.
If the file name is omitted, default file name "CUSTOMIZ.TXT" is assumed.
8 Press the soft key [EXEC].
This starts outputting the name data of customize data, and "OUTPUT" blinks in the lower right part of the screen. When the read operation ends, the "OUTPUT" indication disappears.
To cancel the output, press the soft key [CAN].

Outputting name data of customize data (for 15-inch display unit)

Procedure

1 Make sure the output device is ready for output.
2 Press the function key .
3 Press the vertical soft key [NEXT PAGE] until vertical soft key [TOOL MANAGER] appears.
Press the vertical soft key [TOOL MANAGER].
4 Press the vertical soft key [MAGAZINE] or [TOOL].
5 Press the EDIT switch on the machine operator's panel.
6 Press the horizontal soft key [PUNCH].
7 Press the horizontal soft key [CUSTOM].
8 Type the file name that you want to output.
If the file name is omitted, default file name "CUSTOMIZ.TXT" is assumed.
9 Press the horizontal soft key [EXEC].
This starts outputting the name data of customize data, and "OUTPUT" blinks in the lower right part of the screen. When the read operation ends, the "OUTPUT" indication disappears.
To cancel the output, press the horizontal soft key [CAN].

8.2.9.9 Inputting customize data displayed as tool management data

Customize data displayed as tool management data is loaded into the memory of the CNC from an external device. The input format is the same as the output format. When customize data displayed as tool management data with a data number corresponding to existing customize data displayed as tool management data registered in the memory is loaded, the loaded customize data replaces the existing data.

Inputting customize data displayed as tool management data (for 7.2/8.4/10.4-inch display unit)

Procedure

1 Make sure the input device is ready for reading.
2 Press the function key

to display the tool management screen, magazine screen, or each tool data screen.
3 Press the EDIT switch on the machine operator's panel.
4 Press the soft key [(OPRT)].
5 Press the soft key [READ].
6 Press the soft key [DSP-LAYOUT].
7 Type the name of the file that you want to input.
If the input file name is omitted, default input file name "DISPCSTM.TXT" is assumed.
8 Press the soft key [EXEC].
This starts reading the customize data displayed as tool management data, and "INPUT" blinks in the lower right part of the screen. When the read operation ends, the "INPUT" indication disappears.
To cancel the input of the program, press the soft key [CAN].

Inputting customize data displayed as tool management data (for 15-inch display unit)

Procedure

1 Make sure the input device is ready for reading.
2 Press the function key .
3 Press the vertical soft key [NEXT PAGE] until vertical soft key [TOOL MANAGER] appears.
Press the vertical soft key [TOOL MANAGER].
4 Press the vertical soft key [MAGAZINE], [EACH TOOL], or [TOOL].
5 Press the EDIT switch on the machine operator's panel.
6 Press the horizontal soft key [READ].
7 Press the horizontal soft key [DSP-LAYOUT].
8 Type the name of the file that you want to input.
If the input file name is omitted, default input file name "DISPCSTM.TXT" is assumed.
9 Press the horizontal soft key [EXEC].
This starts reading the customize data displayed as tool management data, and "INPUT" blinks in the lower right part of the screen. When the read operation ends, the "INPUT" indication disappears.
To cancel the input of the program, press the horizontal soft key [CAN].

8.2.9.10 Outputting customize data displayed as tool management data

Customize data displayed as tool management data is output from the memory of the CNC to an external device in the output format.

Outputting customize data displayed as tool management data (for 7.2/8.4/10.4-inch display unit)

Procedure

1 Make sure the output device is ready for output.
2 Press the function key to display the tool management screen, magazine screen, or each tool data screen.
3 Press the EDIT switch on the machine operator's panel.
4 Press the soft key [(OPRT)].
5 Press the soft key [PUNCH].
6 Press the soft key [DSP-LAYOUT].
7 Type the file name that you want to output.
If the file name is omitted, default file name "DISPCSTM.TXT" is assumed.
8 Press the soft key [EXEC].
This starts outputting the customize data displayed as tool management data, and "OUTPUT" blinks in the lower right part of the screen. When the read operation ends, the "OUTPUT" indication disappears.
To cancel the output, press the soft key [CAN].

Outputting customize data displayed as tool management data (for 15-inch display unit)

Procedure

1 Make sure the output device is ready for output.
2 Press the function key .
3 Press the vertical soft key [NEXT PAGE] until vertical soft key [TOOL MANAGER] appears.
Press the vertical soft key [TOOL MANAGER].
4 Press the vertical soft key [MAGAZINE], [EACH TOOL], or [TOOL].
5 Press the EDIT switch on the machine operator's panel.
6 Press the horizontal soft key [PUNCH].
7 Press the horizontal soft key [DSP-LAYOUT].
8 Type the file name that you want to output.
If the file name is omitted, default file name "DISPCSTM.TXT" is assumed.
9 Press the horizontal soft key [EXEC].
This starts outputting the customize data displayed as tool management data, and "OUTPUT" blinks in the lower right part of the screen. When the read operation ends, the "OUTPUT" indication disappears.
To cancel the output, press the horizontal soft key [CAN].

8.2.9.11 Inputting spindle waiting position name data

Spindle waiting position name data is loaded into the memory of the CNC from an external device. The input format is the same as the output format. When spindle waiting position name data with a data number corresponding to existing spindle waiting position name data registered in the memory is loaded, the loaded spindle waiting position name data replaces the existing data.

Inputting spindle waiting position name data (for 7.2/8.4/10.4-inch display unit)

Procedure

1 Make sure the input device is ready for reading.
2 Press the function key to display the tool management screen, magazine screen, or each tool data screen.
3 Press the EDIT switch on the machine operator's panel.
4 Press the soft key [(OPRT)].
5 Press the soft key [READ].
6 Press the soft key [SPWP-NAME].
7 Type the name of the file that you want to input.
If the input file name is omitted, default input file name "POSNAME.TXT" is assumed.
8 Press the soft key [EXEC].
This starts reading the spindle waiting position name data, and "INPUT" blinks in the lower right part of the screen. When the read operation ends, the "INPUT" indication disappears.
To cancel the input of the program, press the soft key [CAN].

Inputting spindle waiting position name data (for 15-inch display unit)

Procedure

1 Make sure the input device is ready for reading.
2 Press the function key .
3 Press the vertical soft key [NEXT PAGE] until vertical soft key [TOOL MANAGER] appears.
Press the vertical soft key [TOOL MANAGER].
4 Press the vertical soft key [MAGAZINE], [EACH TOOL], or [TOOL].
5 Press the EDIT switch on the machine operator's panel.
6 Press the horizontal soft key [READ].
7 Press the horizontal soft key [SPWP-NAME].
8 Type the name of the file that you want to input.
If the input file name is omitted, default input file name "POSNAME.TXT" is assumed.
9 Press the horizontal soft key [EXEC].
This starts reading the spindle waiting position name data, and "INPUT" blinks in the lower right part of the screen. When the read operation ends, the "INPUT" indication disappears.
To cancel the input of the program, press the horizontal soft key [CAN].

8.2.9.12 Outputting spindle waiting position name data

Spindle waiting position name data is output from the memory of the CNC to an external device in the output format.

Outputting spindle waiting position name data (for 7.2/8.4/10.4-inch display unit)

Procedure

1 Make sure the output device is ready for output.
2 Press the function key to display the tool management screen, magazine screen, or each tool data screen.
3 Press the EDIT switch on the machine operator's panel.
4 Press the soft key [(OPRT)].
5 Press the soft key [PUNCH].
6 Press the soft key [SPWP-NAME].
7 Type the file name that you want to output.
If the file name is omitted, default file name "POSNAME.TXT" is assumed.
8 Press the soft key [EXEC].
This starts outputting the spindle waiting position name data display, and "OUTPUT" blinks in the lower right part of the screen. When the read operation ends, the "OUTPUT" indication disappears.
To cancel the output, press the soft key [CAN].

Outputting spindle waiting position name data (for 15-inch display unit)

Procedure

1 Make sure the output device is ready for output.
2 Press the function key .
3 Press the vertical soft key [NEXT PAGE] until vertical soft key [TOOL MANAGER] appears.
Press the vertical soft key [TOOL MANAGER].
4 Press the vertical soft key [MAGAZINE], [EACH TOOL], or [TOOL].
5 Press the EDIT switch on the machine operator's panel.
6 Press the horizontal soft key [PUNCH].
7 Press the horizontal soft key [SPWP-NAME].
8 Type the file name that you want to output.
If the file name is omitted, default file name "POSNAME.TXT" is assumed.
9 Press the horizontal soft key [EXEC].
This starts outputting the spindle waiting position name data display, and "OUTPUT" blinks in the lower right part of the screen. When the read operation ends, the "OUTPUT" indication disappears.
To cancel the output, press the horizontal soft key [CAN].

8.2.9.13 Inputting decimal point position data of customize data

Decimal point position data of customize data is loaded into the memory of the CNC from an external device. The input format is the same as the output format. When decimal point position data of customize data with a data number corresponding to existing decimal point position data of customize data registered in the memory is loaded, the loaded decimal point position data of customize data replaces the existing data.

Inputting decimal point position data of customize data (for 7.2/8.4/10.4-inch display unit)

Procedure

1 Make sure the input device is ready for reading.
2 Press the function key

to display the tool management screen, magazine screen, or each tool data screen.
3 Press the EDIT switch on the machine operator's panel.
4 Press the soft key [(OPRT)].
5 Press the soft key [READ].
6 Press the soft key [CSTM-DPOINT].
7 Type the name of the file that you want to input.
If the input file name is omitted, default input file name "POINTPOS.TXT" is assumed.
8 Press the soft key [EXEC].
This starts reading the decimal point position data of customize data, and "INPUT" blinks in the lower right part of the screen. When the read operation ends, the "INPUT" indication disappears.
To cancel the input of the program, press the soft key [CAN].

Inputting decimal point position data of customize data (for 15-inch display unit)

Procedure

1 Make sure the input device is ready for reading.
2 Press the function key .
3 Press the vertical soft key [NEXT PAGE] until vertical soft key [TOOL MANAGER] appears.
Press the vertical soft key [TOOL MANAGER].
4 Press the vertical soft key [MAGAZINE], [EACH TOOL], or [TOOL].
5 Press the EDIT switch on the machine operator's panel.
6 Press the horizontal soft key [READ].
7 Press the horizontal soft key [CSTM-DPOINT].
8 Type the name of the file that you want to input.
If the input file name is omitted, default input file name "POINTPOS.TXT" is assumed.
9 Press the horizontal soft key [EXEC].
This starts reading the decimal point position data of customize data, and "INPUT" blinks in the lower right part of the screen. When the read operation ends, the "INPUT" indication disappears.
To cancel the input of the program, press the horizontal soft key [CAN].

8.2.9.14 Outputting decimal point position data of customize data

Decimal point position data of customize data is output from the memory of the CNC to an external device in the output format.

Outputting decimal point position data of customize data (for 7.2/8.4/10.4-inch display unit)

Procedure

1 Make sure the output device is ready for output.
2 Press the function key to display the tool management screen, magazine screen, or each tool data screen.
3 Press the EDIT switch on the machine operator's panel.
4 Press the soft key [(OPRT)].
5 Press the soft key [PUNCH].
6 Press the soft key [CSTM-DPOINT].
7 Type the file name that you want to output.
If the file name is omitted, default file name "POINTPOS.TXT" is assumed.
8 Press the soft key [EXEC].
This starts outputting the decimal point position data of customize data, and "OUTPUT" blinks in the lower right part of the screen. When the read operation ends, the "OUTPUT" indication disappears.
To cancel the output, press the soft key [CAN].

Outputting decimal point position data of customize data (for 15-inch display unit)

Procedure

1 Make sure the output device is ready for output.
2 Press the function key .
3 Press the vertical soft key [NEXT PAGE] until vertical soft key [TOOL MANAGER] appears.
Press the vertical soft key [TOOL MANAGER].
4 Press the vertical soft key [MAGAZINE], [EACH TOOL], or [TOOL].
5 Press the EDIT switch on the machine operator's panel.
6 Press the horizontal soft key [PUNCH].
7 Press the horizontal soft key [CSTM-DPOINT].
8 Type the file name that you want to output.
If the file name is omitted, default file name "POINTPOS.TXT" is assumed.
9 Press the horizontal soft key [EXEC].
This starts outputting the decimal point position data of customize data, and "OUTPUT" blinks in the lower right part of the screen. When the read operation ends, the "OUTPUT" indication disappears.
To cancel the output, press the horizontal soft key [CAN].

8.2.9.15 Inputting tool geometry data

Tool geometry data is loaded into the memory of the CNC from an external device. The input format is the same as the output format. When tool geometry data with a data number corresponding to existing tool geometry data registered in the memory is loaded, the loaded tool geometry data replaces the existing data.

Inputting tool geometry data (for 7.2/8.4/10.4-inch display unit)

Procedure

1 Make sure the input device is ready for reading.
2 Press the function key to display the tool management screen, magazine screen, each tool data screen, or tool geometry data screen.
3 Press the EDIT switch on the machine operator's panel.
4 Press the soft key [(OPRT)].
5 Press the soft key [READ].
6 Press the soft key [TOOL- GEOM].
7 Type the name of the file that you want to input.
If the input file name is omitted, default input file name "TOOLGEOM.TXT" is assumed.
8 Press the soft key [EXEC].
This starts reading the tool geometry data, and "INPUT" blinks in the lower right part of the screen. When the read operation ends, the "INPUT" indication disappears.
To cancel the input of the program, press the soft key [CAN].

NOTE
1 If the tool with a number of tool geometry data to be changed is registered to the magazine when an attempt is made to change the tool geometry data, an alarm PS5360 is issued. (The data is not input.)
2 After data related to the tool management functions in the NC has all been cleared, restore backup data in the following order: Tool geometry data, tool management data, and magazine management table.

Inputting tool geometry data (for 15-inch display unit)

Procedure

1 Make sure the input device is ready for reading.
2 Press the function key .
3 Press the vertical soft key [NEXT PAGE] until vertical soft key [TOOL MANAGER] appears.
Press the vertical soft key [TOOL MANAGER].
4 Press the vertical soft key [MAGAZINE], [EACH TOOL], [TOOL], or [TOOL- GEOM].
5 Press the EDIT switch on the machine operator's panel.
6 Press the horizontal soft key [READ].
7 Press the horizontal soft key [TOOL- GEOM].
8 Type the name of the file that you want to input.
If the input file name is omitted, default input file name "TOOLGEOM.TXT" is assumed.
9 Press the horizontal soft key [EXEC].
This starts reading the tool geometry data, and "INPUT" blinks in the lower right part of the screen. When the read operation ends, the "INPUT" indication disappears.
To cancel the input of the program, press the horizontal soft key [CAN].

NOTE
1 If the tool with a number of tool geometry data to be changed is registered to the magazine when an attempt is made to change the tool geometry data, an alarm PS5360 is issued. (The data is not input.)
2 After data related to the tool management functions in the NC has all been cleared, restore backup data in the following order: Tool geometry data, tool management data, and magazine management table.

8.2.9.16 Outputting tool geometry data

Tool geometry data is output from the memory of the CNC to an external device in the output format.

Outputting tool geometry data (for 7.2/8.4/10.4-inch display unit)

Procedure

1 Make sure the output device is ready for output.
2 Press the function key to display the tool management screen, magazine screen, each tool data screen, or tool geometry data screen.
3 Press the EDIT switch on the machine operator's panel.
4 Press the soft key [(OPRT)].
5 Press the soft key [PUNCH].
6 Press the soft key [TOOL- GEOM].
7 Type the file name that you want to output.
If the output file name is omitted, default output file name "TOOLGEOM.TXT" is assumed.
8 Press the soft key [EXEC].
This starts outputting the tool geometry data, and "OUTPUT" blinks in the lower right part of the screen. When the read operation ends, the "OUTPUT" indication disappears.
To cancel the output, press the soft key [CAN].

Outputting tool geometry data (for 15-inch display unit)

Procedure

1 Make sure the output device is ready for output.
2 Press the function key .
3 Press the vertical soft key [NEXT PAGE] until vertical soft key [TOOL MANAGER] appears.
Press the vertical soft key [TOOL MANAGER].
4 Press the vertical soft key [MAGAZINE], [EACH TOOL], [TOOL], or [TOOL- GEOM].
5 Press the EDIT switch on the machine operator's panel.
6 Press the horizontal soft key [PUNCH].
7 Press the horizontal soft key [TOOL- GEOM].
8 Type the file name that you want to output.
If the output file name is omitted, default output file name "TOOLGEOM.TXT" is assumed.
9 Press the horizontal soft key [EXEC].
This starts outputting the tool geometry data, and "OUTPUT" blinks in the lower right part of the screen. When the read operation ends, the "OUTPUT" indication disappears.
To cancel the output, press the horizontal soft key [CAN].

8.3 INPUT/OUTPUT ON THE ALL IO SCREEN

Just by using the ALL IO screen, you can input and output programs, parameters, offset data, pitch error compensation data, macro variables, workpiece coordinate system data, operation history data, and tool management data.

NOTE
The ALL IO screen can be operated only if a memory card interface is selected as an external I/O device.

The following explains how to display the ALL IO screen:

Displaying the ALL IO screen (for 7.2/8.4/10.4-inch display unit)

Procedure

1 Press the function key .
2 Press the continuous menu key several times.
3 Press the soft key [ALL IO] to display the ALL IO screen.

The subsequent steps to select data from the ALL IO screen will be explained for each type of data.

Displaying the ALL IO screen (for 15-inch display unit)

Procedure

1 Press the function key .
2 Press the vertical soft key [NEXT PAGE] until vertical soft key [ALL I/O] appears.
3 Press the vertical soft key [ALL I/O] to display the ALL IO screen.

The subsequent steps to select data from the ALL IO screen will be explained for each type of data.

Configuration of this section

Section 8.3, "INPUT/OUTPUT ON THE ALL IO SCREEN", consists of the following subsections:

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8.3.1 Inputting/Outputting a Program

A program can be input and output using the ALL IO screen.

Inputting a program (for 7.2/8.4/10.4-inch display unit)

Procedure

1 Press the soft key [PROGRAM] on the ALL IO screen.
2 Press the EDIT switch on the machine operator's panel or enter state emergency stop.
3 Press the soft key [(OPRT)].
4 Press the soft key [N READ].
5 Set the name of the file that you want to input.
Type a file name, and press the soft key [F NAME].
If the input file name is omitted, default input file name "ALL-PROG.TXT" is assumed.
See the table below for details.
6 Set the program number to be used after the input.
Type a program number, and press the soft key [P SET].
If no program number is specified here, the program number in the file is adopted as is.
7 Press the soft key [EXEC].
This starts reading the program, and "INPUT" blinks in the lower right part of the screen. When the read operation ends, the "INPUT" indication disappears.
To cancel the input of the program, press the soft key [CAN].

[F SET]	[P SET]	Input file name	Input program	Input program name
BLANK	BLANK	ALL-PROG.TXT	All programs in ALL-PROG.TXT	File name at the time the file is saved
BLANK	INPUT	ALL-PROG.TXT	First program in ALL-PROG.TXT	Program name set with [P SET]
INPUT	BLANK	File name set with [F NAME]	All programs in the file specified with [F NAME]	File name at the time the file is saved
INPUT	INPUT	File name set with [F NAME]	First program in the file specified with [F NAME]	Program name set with [P SET]

Inputting a program (for 15-inch display unit)

Procedure

1 On the ALL IO screen, press the vertical soft key [NEXT PAGE] until vertical soft key [PROGRAM] appears.
Press the vertical soft key [PROGRAM].
2 Press the EDIT switch on the machine operator's panel or enter state emergency stop.
3 Press the horizontal soft key [N READ].
4 Set the name of the file that you want to input.
Type a file name, and press the horizontal soft key [F NAME].
If the input file name is omitted, default input file name "ALL-PROG.TXT" is assumed.
See the table below for details.
5 Set the program number to be used after the input.
Type a program number, and press the horizontal soft key [P SET].
If no program number is specified here, the program number in the file is adopted as is.
6 Press the horizontal soft key [EXEC].
This starts reading the program, and "INPUT" blinks in the lower right part of the screen. When the read operation ends, the "INPUT" indication disappears.
To cancel the input of the program, press the horizontal soft key [CAN].

[F SET]	[P SET]	Input file name	Input program	Input program name
BLANK	BLANK	ALL-PROG.TXT	All programs in ALL-PROG.TXT	File name at the time the file is saved
BLANK	INPUT	ALL-PROG.TXT	First program in ALL-PROG.TXT	Program name set with [P SET]
INPUT	BLANK	File name set with [F NAME]	All programs in the file specified with [F NAME]	File name at the time the file is saved
INPUT	INPUT	File name set with [F NAME]	First program in the file specified with [F NAME]	Program name set with [P SET]

Outputting a program (for 7.2/8.4/10.4-inch display unit)

Procedure

1 Press the soft key [PROGRAM] on the ALL IO screen.
2 Press the EDIT switch on the machine operator's panel or enter state emergency stop.
3 Press the soft key [(OPRT)].
4 Press the soft key [PUNCH].
5 Set the program that you want to output.
Type a program number, and press the soft key [P SET].
If O-9999 is typed, all programs in the memory are output.
6 Set the file name to be output.
Type a file name, and press the soft key [F NAME].
When no file name is set, the output file name is assumed to be "O-number" if a single program number is specified; if O-9999 is specified, the output file name is assumed to be "ALL-PROG.TXT".
See the table below for details.
(For an explanation of foreground folders, see II-12.1.3, and for how to change foreground folders, see III-11.6.)
7 Press the soft key [EXEC].
This starts outputting the program, and "OUTPUT" blinks in the lower right part of the screen. When the read operation ends, the "OUTPUT" indication disappears.
To cancel the output, press the soft key [CAN].

[F SET]	[P SET]	Output file name	Output program
BLANK	BLANK or (O-9999)	ALL-PROG.TXT	All programs in the foreground folders displayed in the program folder
BLANK	INPUT	Program name set with [P SET]	Program in the NC that is set with [P SET]
INPUT	BLANK or (O-9999)	File name set with [F NAME]	All programs in the foreground folders displayed in the program folder
INPUT	INPUT	File name set with [F NAME]	Program in the NC that is set with [P SET]

Outputting a program (for 15-inch display unit)

Procedure

1 On the ALL IO screen, press the vertical soft key [NEXT PAGE] until vertical soft key [PROGRAM] appears.
Press the vertical soft key [PROGRAM].
2 Press the EDIT switch on the machine operator's panel or enter state emergency stop.
3 Press the horizontal soft key [PUNCH].
4 Set the program that you want to output.
Type a program number, and press the horizontal soft key [P SET].
If O-9999 is typed, all programs in the memory are output.
5 Set the file name to be output.
Type a file name, and press the horizontal soft key [F NAME].
When no file name is set, the output file name is assumed to be "O-number" if a single program number is specified; if O-9999 is specified, the output file name is assumed to be "ALL-PROG.TXT".
See the table below for details.
(For an explanation of foreground folders, see II-12.1.3, and for how to change foreground folders, see III-11.6.)
6 Press the horizontal soft key [EXEC].
This starts outputting the program, and "OUTPUT" blinks in the lower right part of the screen. When the read operation ends, the "OUTPUT" indication disappears.
To cancel the output, press the horizontal soft key [CAN].

[F SET]	[P SET]	Output file name	Output program
BLANK	BLANK or (O-9999)	ALL-PROG.TXT	All programs in the foreground folders displayed in the program folder
BLANK	INPUT	Program name set with [P SET]	Program in the NC that is set with [P SET]
INPUT	BLANK or (O-9999)	File name set with [F NAME]	All programs in the foreground folders displayed in the program folder
INPUT	INPUT	File name set with [F NAME]	Program in the NC that is set with [P SET]

8.3.2 Inputting and Outputting Parameters

Parameters can be input and output using the ALL IO screen.

Inputting parameters (for 7.2/8.4/10.4-inch display unit)

Procedure

1 Press the function key .
2 Press the continuous menu key until soft key [SETTING] appears.
Press the soft key [SETTING].
3 Press the MDI switch on the machine operator's panel or enter state emergency stop.
4 Enter 1 in response to the prompt for "PARAMETER WRITE" in setting data.Alarm SW0100 appears.
5 Press the soft key [PARAMETER] on the ALL IO screen.
6 Press the EDIT switch on the machine operator's panel or enter state emergency stop.
7 Press the soft key [(OPRT)].
8 Press the soft key [N READ].
9 Set the name of the file that you want to input.
Type a file name, and press the soft key [F NAME].
If the input file name is omitted, default input file name "CNC-PARA.TXT" is assumed.
10 Press the soft key [EXEC].
This starts reading the parameter, and "INPUT" blinks in the lower right part of the screen. When the read operation ends, the "INPUT" indication disappears.
To cancel the input of the program, press the soft key [CAN].
11 Press the function key .
12 Press the soft key [SETTING].
13 Press the MDI switch on the machine operator's panel or enter state emergency stop.
14 Enter 0 in response to the prompt for "PARAMETER WRITE" in setting data.
15 Turn the power of the CNC on again.

Inputting parameters (for 15-inch display unit)

Procedure

1 Press the function key .
2 Press the vertical soft key [NEXT PAGE] until vertical soft key [SETTING] appears.
Press the vertical soft key [SETTING].
3 Press the MDI switch on the machine operator's panel or enter state emergency stop.
4 Enter 1 in response to the prompt for "PARAMETER WRITE" in setting data.Alarm SW0100 appears.
5 On the ALL IO screen, press the vertical soft key [NEXT PAGE] until vertical soft key [PARAMETER] appears.
Press the vertical soft key [PARAMETER].
6 Press the EDIT switch on the machine operator's panel or enter state emergency stop.
7 Press the horizontal soft key [N READ].
8 Set the name of the file that you want to input.
Type a file name, and press the horizontal soft key [F NAME].
If the input file name is omitted, default input file name "CNC-PARA.TXT" is assumed.
9 Press the horizontal soft key [EXEC].
This starts reading the parameter, and "INPUT" blinks in the lower right part of the screen. When the read operation ends, the "INPUT" indication disappears.
To cancel the input of the program, press the horizontal soft key [CAN].
10 Press the function key .
11 Press the vertical soft key [SETTING].
12 Press the MDI switch on the machine operator's panel or enter state emergency stop.
13 Enter 0 in response to the prompt for "PARAMETER WRITE" in setting data.
14 Turn the power of the CNC on again.

Outputting parameters (for 7.2/8.4/10.4-inch display unit)

Procedure

1 Press the soft key [PARAMETER] on the ALL IO screen.
2 Press the EDIT switch on the machine operator's panel or enter state emergency stop.
3 Press the soft key [(OPRT)].
4 Press the soft key [PUNCH].
5 Set the file name to be output.
Type a file name, and press the soft key [F NAME].
If the file name is omitted, default file name "CNC-PARA.TXT" is assumed.
6 Press the soft key [EXEC].
This starts outputting the parameter, and "OUTPUT" blinks in the lower right part of the screen. When the read operation ends, the "OUTPUT" indication disappears.
To cancel the output, press the soft key [CAN].

Outputting parameters (for 15-inch display unit)

Procedure

1 On the ALL IO screen, press the vertical soft key [NEXT PAGE] until vertical soft key [PARAMETER] appears.
Press the vertical soft key [PARAMETER].
2 Press the EDIT switch on the machine operator's panel or enter state emergency stop.
3 Press the horizontal soft key [PUNCH].
4 Set the file name to be output.
Type a file name, and press the horizontal soft key [F NAME].
If the file name is omitted, default file name "CNC-PARA.TXT" is assumed.
5 Press the horizontal soft key [EXEC].
This starts outputting the parameter, and "OUTPUT" blinks in the lower right part of the screen. When the read operation ends, the "OUTPUT" indication disappears.
To cancel the output, press the horizontal soft key [CAN].

8.3.3 Inputting and Outputting Offset Data

Offset data can be input and output using the ALL IO screen.

Inputting offset data (for 7.2/8.4/10.4-inch display unit)

Procedure

1 Press the soft key [OFFSET] on the ALL IO screen.
2 Press the EDIT switch on the machine operator's panel or enter state emergency stop.
3 Press the soft key [(OPRT)].
4 Press the soft key [N READ].
5 Set the name of the file that you want to input.
Type a file name, and press the soft key [F NAME].
If the input file name is omitted, default input file name "TOOLOFST.TXT" is assumed.
6 Press the soft key [EXEC].
This starts reading the offset data, and "INPUT" blinks in the lower right part of the screen. When the read operation ends, the "INPUT" indication disappears.
To cancel the input of the program, press the soft key [CAN].

Inputting offset data (for 15-inch display unit)

Procedure

1 On the ALL IO screen, press the vertical soft key [NEXT PAGE] until vertical soft key [OFFSET] appears.
Press the vertical soft key [OFFSET].
2 Press the EDIT switch on the machine operator's panel or enter state emergency stop.
3 Press the horizontal soft key [N READ].
4 Set the name of the file that you want to input.
Type a file name, and press the horizontal soft key [F NAME].
If the input file name is omitted, default input file name "TOOLOFST.TXT" is assumed.
5 Press the horizontal soft key [EXEC].
This starts reading the offset data, and "INPUT" blinks in the lower right part of the screen. When the read operation ends, the "INPUT" indication disappears.
To cancel the input of the program, press the horizontal soft key [CAN].

Outputting offset data (for 7.2/8.4/10.4-inch display unit)

Procedure

1 Press the soft key [OFFSET] on the ALL IO screen.
2 Press the EDIT switch on the machine operator's panel or enter state emergency stop.
3 Press the soft key [(OPRT)].
4 Press the soft key [PUNCH].
5 Set the file name to be output.
Type a file name, and press the soft key [F NAME].
If the file name is omitted, default file name "TOOLOFST.TXT" is assumed.
6 Press the soft key [EXEC].
This starts outputting the offset data, and "OUTPUT" blinks in the lower right part of the screen. When the read operation ends, the "OUTPUT" indication disappears.
To cancel the output, press the soft key [CAN].

Outputting offset data (for 15-inch display unit)

Procedure

1 On the ALL IO screen, press the vertical soft key [NEXT PAGE] until vertical soft key [OFFSET] appears.
Press the vertical soft key [OFFSET].
2 Press the EDIT switch on the machine operator's panel or enter state emergency stop.
3 Press the horizontal soft key [PUNCH].
4 Set the file name to be output.
Type a file name, and press the horizontal soft key [F NAME].
If the file name is omitted, default file name "TOOLOFST.TXT" is assumed.
5 Press the horizontal soft key [EXEC].
This starts outputting the offset data, and "OUTPUT" blinks in the lower right part of the screen. When the read operation ends, the "OUTPUT" indication disappears.
To cancel the output, press the horizontal soft key [CAN].

8.3.4 Inputting/Outputting Pitch Error Compensation Data

Pitch error compensation data can be input and output using the ALL IO screen.

Inputting pitch error compensation data (for 7.2/8.4/10.4-inch display unit)

Procedure

1 Press the function key .
2 Press the continuous menu key until soft key [SETTING] appears.
Press the soft key [SETTING].
3 Press the MDI switch on the machine operator's panel or enter state emergency stop.
4 Enter 1 in response to the prompt for "PARAMETER WRITE" in setting data.Alarm SW0100 appears.
5 Press the soft key [PITCH] on the ALL IO screen.
6 Press the EDIT switch on the machine operator's panel or enter state emergency stop.
7 Press the soft key [(OPRT)].
8 Press the soft key [N READ].
9 Set the name of the file that you want to input.
Type a file name, and press the soft key [F NAME].
If the input file name is omitted, default input file name "PITCH.TXT" is assumed.
10 Press the soft key [EXEC].
This starts reading the pitch error compensation data, and "INPUT" blinks in the lower right part of the screen. When the read operation ends, the "INPUT" indication disappears.
To cancel the input of the program, press the soft key [CAN].
11 Press the function key .
12 Press the soft key [SETTING].
13 Press the MDI switch on the machine operator's panel or enter state emergency stop.
14 Enter 0 in response to the prompt for "PARAMETER WRITE" in setting data.
15 Turn the power of the CNC on again.

Inputting pitch error compensation data (for 15-inch display unit)

Procedure

1 Press the function key .
2 Press the vertical soft key [NEXT PAGE] until vertical soft key [SETTING] appears.
Press the vertical soft key [SETTING].
3 Press the MDI switch on the machine operator's panel or enter state emergency stop.
4 Enter 1 in response to the prompt for "PARAMETER WRITE" in setting data.Alarm SW0100 appears.
5 On the ALL IO screen, press the vertical soft key [NEXT PAGE] until vertical soft key [PITCH ERROR] appears.
Press the vertical soft key [PITCH ERROR].
6 Press the EDIT switch on the machine operator's panel or enter state emergency stop.
7 Press the horizontal soft key [N READ].
8 Set the name of the file that you want to input.
Type a file name, and press the horizontal soft key [F NAME].
If the input file name is omitted, default input file name "PITCH.TXT" is assumed.
9 Press the horizontal soft key [EXEC].
This starts reading the pitch error compensation data, and "INPUT" blinks in the lower right part of the screen. When the read operation ends, the "INPUT" indication disappears.
To cancel the input of the program, press the horizontal soft key [CAN].
10 Press the function key .
11 Press the vertical soft key [SETTING].
12 Press the MDI switch on the machine operator's panel or enter state emergency stop.
13 Enter 0 in response to the prompt for "PARAMETER WRITE" in setting data.
14 Turn the power of the CNC on again.

Outputting pitch error compensation data (for 7.2/8.4/10.4-inch display unit)

Procedure

1 Press the soft key [PITCH] on the ALL IO screen.
2 Press the EDIT switch on the machine operator's panel or enter state emergency stop.
3 Press the soft key [(OPRT)].
4 Press the soft key [PUNCH].
5 Set the file name to be output.
Type a file name, and press the soft key [F NAME].
If the file name is omitted, default file name "PITCH.TXT" is assumed.
6 Press the soft key [EXEC].
This starts outputting the pitch error compensation data, and "OUTPUT" blinks in the lower right part of the screen. When the read operation ends, the "OUTPUT" indication disappears.
To cancel the output, press the soft key [CAN].

Outputting pitch error compensation data (for 15-inch display unit)

Procedure

1 On the ALL IO screen, press the vertical soft key [NEXT PAGE] until vertical soft key [PITCH ERROR] appears.
Press the vertical soft key [PITCH ERROR].
2 Press the EDIT switch on the machine operator's panel or enter state emergency stop.
3 Press the horizontal soft key [PUNCH].
4 Set the file name to be output.
Type a file name, and press the horizontal soft key [F NAME].
If the file name is omitted, default file name "PITCH.TXT" is assumed.
5 Press the horizontal soft key [EXEC].
This starts outputting the pitch error compensation data, and "OUTPUT" blinks in the lower right part of the screen. When the read operation ends, the "OUTPUT" indication disappears.
To cancel the output, press the horizontal soft key [CAN].

8.3.5 Inputting/Outputting Custom Macro Common Variables

Custom macro common variables can be input and output using the ALL IO screen.

Inputting custom macro common variables (for 7.2/8.4/10.4-inch display unit)

Procedure

1 Press the soft key [MACRO] on the ALL IO screen.
2 Press the EDIT switch on the machine operator's panel or enter state emergency stop.
3 Press the soft key [(OPRT)].
4 Press the soft key [N READ].
5 Set the name of the file that you want to input.
Type a file name, and press the soft key [F NAME].
If the input file name is omitted, default input file name "MACRO.TXT" is assumed.
6 Press the soft key [EXEC].
This starts reading the custom macro common variables, and "INPUT" blinks in the lower right part of the screen. When the read operation ends, the "INPUT" indication disappears.
To cancel the input of the program, press the soft key [CAN].

Inputting custom macro common variables (for 15-inch display unit)

Procedure

1 On the ALL IO screen, press the vertical soft key [NEXT PAGE] until vertical soft key [MACRO] appears.
Press the vertical soft key [MACRO].
2 Press the EDIT switch on the machine operator's panel or enter state emergency stop.
3 Press the horizontal soft key [N READ].
4 Set the name of the file that you want to input.
Type a file name, and press the horizontal soft key [F NAME].
If the input file name is omitted, default input file name "MACRO.TXT" is assumed.
5 Press the horizontal soft key [EXEC].
This starts reading the custom macro common variables, and "INPUT" blinks in the lower right part of the screen. When the read operation ends, the "INPUT" indication disappears.
To cancel the input of the program, press the horizontal soft key [CAN].

Outputting custom macro common variables (for 7.2/8.4/10.4-inch display unit)

Procedure

1 Press the soft key [MACRO] on the ALL IO screen.
2 Press the EDIT switch on the machine operator's panel or enter state emergency stop.
3 Press the soft key [(OPRT)].
4 Press the soft key [PUNCH].
5 Set the file name to be output.
Type a file name, and press the soft key [F NAME].
If the file name is omitted, default file name "MACRO.TXT" is assumed.
6 Press the soft key [EXEC].
This starts outputting the custom macro common variables, and "OUTPUT" blinks in the lower right part of the screen. When the read operation ends, the "OUTPUT" indication disappears.
To cancel the output, press the soft key [CAN].

Outputting custom macro common variables (for 15-inch display unit)

Procedure

1 On the ALL IO screen, press the vertical soft key [NEXT PAGE] until vertical soft key [MACRO] appears.
Press the vertical soft key [MACRO].
2 Press the EDIT switch on the machine operator's panel or enter state emergency stop.
3 Press the horizontal soft key [PUNCH].
4 Set the file name to be output.
Type a file name, and press the horizontal soft key [F NAME].
If the file name is omitted, default file name "MACRO.TXT" is assumed.
5 Press the horizontal soft key [EXEC].
This starts outputting the custom macro common variables, and "OUTPUT" blinks in the lower right part of the screen. When the read operation ends, the "OUTPUT" indication disappears.
To cancel the output, press the horizontal soft key [CAN].

8.3.6 Inputting and Outputting Workpiece Coordinates System Data

Workpiece coordinates system data can be input and output using the ALL IO screen.

Inputting workpiece coordinate system data (for 7.2/8.4/10.4-inch display unit)

Procedure

1 Press the soft key [WORK] on the ALL IO screen.
2 Press the EDIT switch on the machine operator's panel or enter state emergency stop.
3 Press the soft key [(OPRT)].
4 Press the soft key [N READ].
5 Set the name of the file that you want to input.
Type a file name, and press the soft key [F NAME].
If the input file name is omitted, default input file name "EXT_WKZ.TXT" is assumed.
6 Press the soft key [EXEC].
This starts reading the workpiece coordinate system data, and "INPUT" blinks in the lower right part of the screen. When the read operation ends, the "INPUT" indication disappears.
To cancel the input of the program, press the soft key [CAN].

Inputting workpiece coordinate system data (for 15-inch display unit)

Procedure

1 On the ALL IO screen, press the vertical soft key [NEXT PAGE] until vertical soft key [WORK] appears.
Press veertical soft key [WORK].
2 Press the EDIT switch on the machine operator's panel or enter state emergency stop.
3 Press the horizontal soft key [N READ].
4 Set the name of the file that you want to input.
Type a file name, and press the horizontal soft key [F NAME].
If the input file name is omitted, default input file name "EXT_WKZ.TXT" is assumed.
5 Press the horizontal soft key [EXEC].
This starts reading the workpiece coordinate system data, and "INPUT" blinks in the lower right part of the screen. When the read operation ends, the "INPUT" indication disappears.
To cancel the input of the program, press the horizontal soft key [CAN].

Outputting workpiece coordinate system data (for 7.2/8.4/10.4-inch display unit)

Procedure

1 Press the soft key [WORK] on the ALL IO screen.
2 Press the EDIT switch on the machine operator's panel or enter state emergency stop.
3 Press the soft key [(OPRT)].
4 Press the soft key [PUNCH].
5 Set the file name to be output.
Type a file name, and press the soft key [F NAME].
If the file name is omitted, default file name "EXT_WKZ.TXT" is assumed.
6 Press the soft key [EXEC].
This starts outputting the workpiece coordinate system data, and "OUTPUT" blinks in the lower right part of the screen. When the read operation ends, the "OUTPUT" indication disappears.
To cancel the output, press the soft key [CAN].

Outputting workpiece coordinate system data (for 15-inch display unit)

Procedure

1 On the ALL IO screen, press the vertical soft key [NEXT PAGE] until vertical soft key [WORK] appears.
Press veertical soft key [WORK].
2 Press the EDIT switch on the machine operator's panel or enter state emergency stop.
3 Press the horizontal soft key [PUNCH].
4 Set the file name to be output.
Type a file name, and press the horizontal soft key [F NAME].
If the file name is omitted, default file name "EXT_WKZ.TXT" is assumed.
5 Press the horizontal soft key [EXEC].
This starts outputting the workpiece coordinate system data, and "OUTPUT" blinks in the lower right part of the screen. When the read operation ends, the "OUTPUT" indication disappears.
To cancel the output, press the horizontal soft key [CAN].

8.3.7 Inputting and Outputting Operation History Data

Operation history data can be output on the ALL IO screen (with function key

for 15-inch display unit).

Only output operation is permitted for operation history data.
The output data is of text format. So, to reference the output data you must use an application that can handle text files on the personal computer.

Outputting operation history data (for 7.2/8.4/10.4-inch display unit)

Procedure

1 Press the soft key [OPERAT HISTRY] on the ALL IO screen.
2 Press the EDIT switch on the machine operator's panel or enter state emergency stop.
3 Press the soft key [(OPRT)].
4 Press the soft key [PUNCH].
5 Set the file name to be output.
Type a file name, and press the soft key [F NAME].
If the file name is omitted, default file name "OPRT_HIS.TXT" is assumed.
6 Press the soft key [EXEC].
This starts outputting the operation history data, and "OUTPUT" blinks in the lower right part of the screen. When the read operation ends, the "OUTPUT" indication disappears.
To cancel the output, press the soft key [CAN].

Outputting operation history data (for 15-inch display unit)

Procedure

1 Press the function key

.
2 On the ALL IO screen, press the vertical soft key [NEXT PAGE] until vertical soft key [OPERAT HISTRY] appears.
Press the vertical soft key [OPERAT HISTRY].
3 Press the EDIT switch on the machine operator's panel or enter state emergency stop.
4 Press the horizontal soft key [PUNCH].
5 Set the file name to be output.
If the file name is omitted, default file name "OPRT_HIS.TXT" is assumed.
6 Press the horizontal soft key [EXEC].
This starts outputting the operation history data, and "OUTPUT" blinks in the lower right part of the screen. When the read operation ends, the "OUTPUT" indication disappears.To cancel the output, press the horizontal soft key [CAN].

8.3.8 Inputting and Outputting Tool Management Data

Tool management data can be input and output using the ALL IO screen.

NOTE
1 For multi-path systems, place all paths in the EDIT mode before performing input and output operations.
2 The format used is the same as the registration format of the G10 format.

Inputting tool management data (for 7.2/8.4/10.4-inch display unit)

Procedure

1 Press the soft key [TOOL] on the ALL IO screen.
2 Press the EDIT switch on the machine operator's panel.
3 Press the soft key [(OPRT)].
4 Press the soft key [N READ].
5 Set the name of the file that you want to input.
Type a file name, and press the soft key [F NAME].
If the input file name is omitted, default input file name "TOOL_MNG.TXT" is assumed.
6 Press the soft key [EXEC].
This starts reading the tool management data, and "INPUT" blinks in the lower right part of the screen. When the read operation ends, the "INPUT" indication disappears.
To cancel the input of the program, press the soft key [CAN].

Inputting tool management data (for 15-inch display unit)

Procedure

1 On the ALL IO screen, press the vertical soft key [NEXT PAGE] until vertical soft key [TOOL] appears.
Press the vertical soft key [TOOL].
2 Press the EDIT switch on the machine operator's panel.
3 Press the horizontal soft key [N READ].
4 Set the name of the file that you want to input.
Type a file name, and press the horizontal soft key [F NAME].
If the input file name is omitted, default input file name "TOOL_MNG.TXT" is assumed.
5 Press the horizontal soft key [EXEC].
This starts reading the tool management data, and "INPUT" blinks in the lower right part of the screen. When the read operation ends, the "INPUT" indication disappears.
To cancel the input of the program, press the horizontal soft key [CAN].

Outputting tool management data (for 7.2/8.4/10.4-inch display unit)

Procedure

1 Press the soft key [TOOL] on the ALL IO screen.
2 Press the EDIT switch on the machine operator's panel.
3 Press the soft key [(OPRT)].
4 Press the soft key [PUNCH].
5 Set the file name to be output.
Type a file name, and press the soft key [F NAME].
If the output file name is omitted, default output file name "TOOL_MNG.TXT" is assumed.
6 Press the soft key [EXEC].
This starts outputting the tool management data, and "OUTPUT" blinks in the lower right part of the screen. When the read operation ends, the "OUTPUT" indication disappears.
To cancel the output, press the soft key [CAN].

Outputting tool management data (for 15-inch display unit)

Procedure

1 On the ALL IO screen, press the vertical soft key [NEXT PAGE] until vertical soft key [TOOL] appears.
Press the vertical soft key [TOOL].
2 Press the EDIT switch on the machine operator's panel.
3 Press the horizontal soft key [PUNCH].
4 Set the file name to be output.
Type a file name, and press the horizontal soft key [F NAME].
If the output file name is omitted, default output file name "TOOL_MNG.TXT" is assumed.
5 Press the horizontal soft key [EXEC].
This starts outputting the tool management data, and "OUTPUT" blinks in the lower right part of the screen. When the read operation ends, the "OUTPUT" indication disappears.
To cancel the output, press the horizontal soft key [CAN].

Inputting magazine data (for 7.2/8.4/10.4-inch display unit)

Procedure

1 Press the soft key [MAGAZINE] on the ALL IO screen.
2 Press the EDIT switch on the machine operator's panel.
3 Press the soft key [(OPRT)].
4 Press the soft key [N READ].
5 Set the name of the file that you want to input.
Type a file name, and press the soft key [F NAME].
If the input file name is omitted, default input file name "MAGAZINE.TXT" is assumed.
6 Press the soft key [EXEC].
This starts reading the magazine data, and "INPUT" blinks in the lower right part of the screen. When the read operation ends, the "INPUT" indication disappears.
To cancel the input of the program, press the soft key [CAN].

Inputting magazine data (for 15-inch display unit)

Procedure

1 On the ALL IO screen, press the vertical soft key [NEXT PAGE] until vertical soft key [MAGAZINE] appears.
Press the vertical soft key [MAGAZINE].
2 Press the EDIT switch on the machine operator's panel.
3 Press the horizontal soft key [N READ].
4 Set the name of the file that you want to input.
Type a file name, and press the horizontal soft key [F NAME].
If the input file name is omitted, default input file name "MAGAZINE.TXT" is assumed.
5 Press the horizontal soft key [EXEC].
This starts reading the magazine data, and "INPUT" blinks in the lower right part of the screen. When the read operation ends, the "INPUT" indication disappears.
To cancel the input of the program, press the horizontal soft key [CAN].

Outputting magazine data (for 7.2/8.4/10.4-inch display unit)

Procedure

1 Press the soft key [MAGAZINE] on the ALL IO screen.
2 Press the EDIT switch on the machine operator's panel.
3 Press the soft key [(OPRT)].
4 Press the soft key [PUNCH].
5 Set the file name to be output.
Type a file name, and press the soft key [F NAME].
If the file name is omitted, default file name "MAGAZINE.TXT" is assumed.
6 Press the soft key [EXEC].
This starts outputting the magazine data, and "OUTPUT" blinks in the lower right part of the screen. When the read operation ends, the "OUTPUT" indication disappears.
To cancel the output, press the soft key [CAN].

Outputting magazine data (for 15-inch display unit)

Procedure

1 On the ALL IO screen, press the vertical soft key [NEXT PAGE] until vertical soft key [MAGAZINE] appears.
Press the vertical soft key [MAGAZINE].
2 Press the EDIT switch on the machine operator's panel.
3 Press the horizontal soft key [PUNCH].
4 Set the file name to be output.
Type a file name, and press the horizontal soft key [F NAME].
If the file name is omitted, default file name "MAGAZINE.TXT" is assumed.
5 Press the horizontal soft key [EXEC].
This starts outputting the magazine data, and "OUTPUT" blinks in the lower right part of the screen. When the read operation ends, the "OUTPUT" indication disappears.
To cancel the output, press the horizontal soft key [CAN].

Inputting tool life status name data (for 7.2/8.4/10.4-inch display unit)

Procedure

1 Press the soft key [STATUS] on the ALL IO screen.
2 Press the EDIT switch on the machine operator's panel.
3 Press the soft key [(OPRT)].
4 Press the soft key [N READ].
5 Set the name of the file that you want to input.
Type a file name, and press the soft key [F NAME].
If the input file name is omitted, default input file name "STATUS.TXT" is assumed.
6 Press the soft key [EXEC].
This starts reading the tool life status name data, and "INPUT" blinks in the lower right part of the screen. When the read operation ends, the "INPUT" indication disappears.
To cancel the input of the program, press the soft key [CAN].

Inputting tool life status name data (for 15-inch display unit)

Procedure

1 On the ALL IO screen, press the vertical soft key [NEXT PAGE] until vertical soft key [STATUS] appears.
Press the vertical soft key [STATUS].
2 Press the EDIT switch on the machine operator's panel.
3 Press the horizontal soft key [N READ].
4 Set the name of the file that you want to input.
Type a file name, and press the horizontal soft key [F NAME].
If the input file name is omitted, default input file name "STATUS.TXT" is assumed.
5 Press the horizontal soft key [EXEC].
This starts reading the tool life status name data, and "INPUT" blinks in the lower right part of the screen. When the read operation ends, the "INPUT" indication disappears.
To cancel the input of the program, press the horizontal soft key [CAN].

Outputting tool life status name data (for 7.2/8.4/10.4-inch display unit)

Procedure

1 Press the soft key [STATUS] on the ALL IO screen.
2 Press the EDIT switch on the machine operator's panel.
3 Press the soft key [(OPRT)].
4 Press the soft key [PUNCH].
5 Set the file name to be output.
Type a file name, and press the soft key [F NAME].
If the file name is omitted, default file name "STATUS.TXT" is assumed.
6 Press the soft key [EXEC].
This starts outputting the tool life status name data, and "OUTPUT" blinks in the lower right part of the screen. When the read operation ends, the "OUTPUT" indication disappears.
To cancel the output, press the soft key [CAN].

Outputting tool life status name data (for 15-inch display unit)

Procedure

1 On the ALL IO screen, press the vertical soft key [NEXT PAGE] until vertical soft key [STATUS] appears.
Press the vertical soft key [STATUS].
2 Press the EDIT switch on the machine operator's panel.
3 Press the horizontal soft key [PUNCH].
4 Set the file name to be output.
Type a file name, and press the horizontal soft key [F NAME].
If the file name is omitted, default file name "STATUS.TXT" is assumed.
5 Press the horizontal soft key [EXEC].
This starts outputting the tool life status name data, and "OUTPUT" blinks in the lower right part of the screen. When the read operation ends, the "OUTPUT" indication disappears.
To cancel the output, press the horizontal soft key [CAN].

Inputting name data of customize data (for 7.2/8.4/10.4-inch display unit)

Procedure

1 Press the soft key [CUSTOM] on the ALL IO screen.
2 Press the EDIT switch on the machine operator's panel.
3 Press the soft key [(OPRT)].
4 Press the soft key [N READ].
5 Set the name of the file that you want to input.
Type a file name, and press the soft key [F NAME].
If the input file name is omitted, default input file name "CUSTOMIZ.TXT" is assumed.
6 Press the soft key [EXEC].
This starts reading the name data of customize data, and "INPUT" blinks in the lower right part of the screen. When the read operation ends, the "INPUT" indication disappears.
To cancel the input of the program, press the soft key [CAN].

Inputting name data of customize data (for 15-inch display unit)

Procedure

1 On the ALL IO screen, press the vertical soft key [NEXT PAGE] until vertical soft key [CUSTOM] appears.
Press the vertical soft key [CUSTOM].
2 Press the EDIT switch on the machine operator's panel.
3 Press the horizontal soft key [N READ].
4 Set the name of the file that you want to input.
Type a file name, and press the horizontal soft key [F NAME].
If the input file name is omitted, default input file name "CUSTOMIZ.TXT" is assumed.
5 Press the horizontal soft key [EXEC].
This starts reading the name data of customize data, and "INPUT" blinks in the lower right part of the screen. When the read operation ends, the "INPUT" indication disappears.
To cancel the input of the program, press the horizontal soft key [CAN].

Outputting name data of customize data (for 7.2/8.4/10.4-inch display unit)

Procedure

1 Press the soft key [CUSTOM] on the ALL IO screen.
2 Press the EDIT switch on the machine operator's panel.
3 Press the soft key [(OPRT)].
4 Press the soft key [PUNCH].
5 Set the file name to be output.
Type a file name, and press the soft key [F NAME].
If the file name is omitted, default file name "CUSTOMIZ.TXT" is assumed.
6 Press the soft key [EXEC].
This starts outputting the name data of customize data, and "OUTPUT" blinks in the lower right part of the screen. When the read operation ends, the "OUTPUT" indication disappears.
To cancel the output, press the soft key [CAN].

Outputting name data of customize data (for 15-inch display unit)

Procedure

1 On the ALL IO screen, press the vertical soft key [NEXT PAGE] until vertical soft key [CUSTOM] appears.
Press the vertical soft key [CUSTOM].
2 Press the EDIT switch on the machine operator's panel.
3 Press the horizontal soft key [PUNCH].
4 Set the file name to be output.
Type a file name, and press the horizontal soft key [F NAME].
If the file name is omitted, default file name "CUSTOMIZ.TXT" is assumed.
5 Press the horizontal soft key [EXEC].
This starts outputting the name data of customize data, and "OUTPUT" blinks in the lower right part of the screen. When the read operation ends, the "OUTPUT" indication disappears.
To cancel the output, press the horizontal soft key [CAN].

8.3.9 File Format and Error Messages

Explanation

- File format

All files that are read from and written to an external device are of text format. The format is described below.
A file starts with % or LF, followed by the actual data. A file always ends with %. In a read operation, data between the first % and the next LF is skipped. Each block ends with an LF, not a semicolon (;).

- LF: 0A (hexadecimal) of ASCII code
- When a file containing lowercase letters, kana characters, and several special characters (such as $, \, and !) is read, those letters and characters are ignored.

Example)

```
%
O0001(MEMORY CARD SAMPLE FILE)
G17 G49 G97
G92 X-11.3 Y2.33
 :
 :
M30
%
```

- ASCII code is used for input/output, regardless of the setting parameter (ISO/EIA).
- Bit 3 (NCR) of parameter No. 0100 can be used to specify whether the end of block code (EOB) is output as "LF" only, or as "LF, CR, CR."

Limitation

- Memory card specification

Use memory cards that comply with PCMCIA Ver2.0 or JEIDA Ver4.1.

- Attribute memory

Memory cards that have no attribute memory or contain no device information in the attribute memory cannot be used.

- Flash ROM card

Flash ROM cards can be used for reading only.

8.4 EMBEDDED ETHERNET OPERATIONS

8.4.1 FTP File Transfer Function

The operation of the FTP file transfer function is described below.

Host file list display (for 7.2/8.4/10.4-inch display unit)

A list of the files held on the host computer is displayed.

Procedure

1 Press the function key

.

2 Press the soft key [FOLDER]. The program list screen appears. (If the soft key does not appear, press the continuous menu key .)

3 Press the soft key [(OPRT)].

4 Press the soft key [DEVICE CHANGE].

5 Press the soft key [EMB ETHER], and the EMBEDDED ETHERNET HOST FILE LIST screen appears.

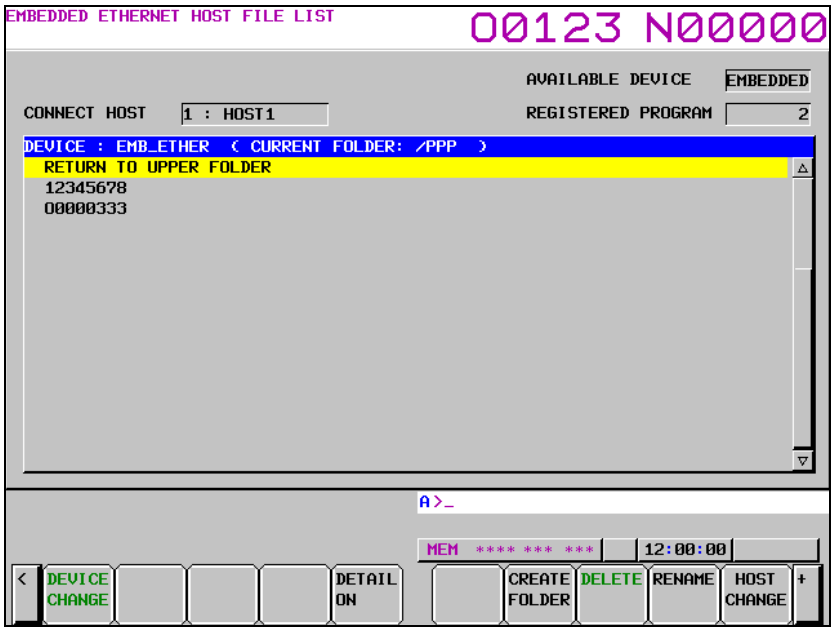
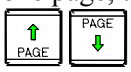


Fig. 8.4.1 (a) Embedded Ethernet host file list screen

NOTE
1 When using the FTP file transfer function, check that the valid device is the embedded Ethernet port. The two conditions below determine a connection destination on the host file list screen:
(1)Check that the valid device is the embedded Ethernet port. Make a selection in "DEVICE SELECTION" on the Ethernet setting screen.
(2)A host computer can be selected from connection destinations 1, 2, and 3. Make a selection according to soft key [HOST CHANGE] described later.
2 File names including kanji, hiragana, and katakana characters cannot be displayed correctly.

6 When a list of files is larger than one page, the screen display can be switched using the page keys 

.

7 Press the soft key [REFRESH] to update the screen display.

8 Press the soft key [DETAIL OFF] to display the host file list (file names only) screen. Press the soft key [DETAIL ON] to display the host file list (detail) screen.

9 When selecting a folder with cursor and pressing the MDI key

, the selected folder is changed to the new current folder.

Host file list display (for 15-inch display unit)

A list of the files held on the host computer is displayed.

Procedure

1 Press the function key PROG.
2 Press the vertical soft key [FOLDER]. The program list screen appears.
3 Press the continuous menu key ▷ until horizontal soft key [DEVICE CHANGE] appears.
Press the horizontal soft key [DEVICE CHANGE].
4 Press the horizontal soft key [EMB ETHER], and the EMBEDDED ETHERNET HOST FILE LIST screen appears.

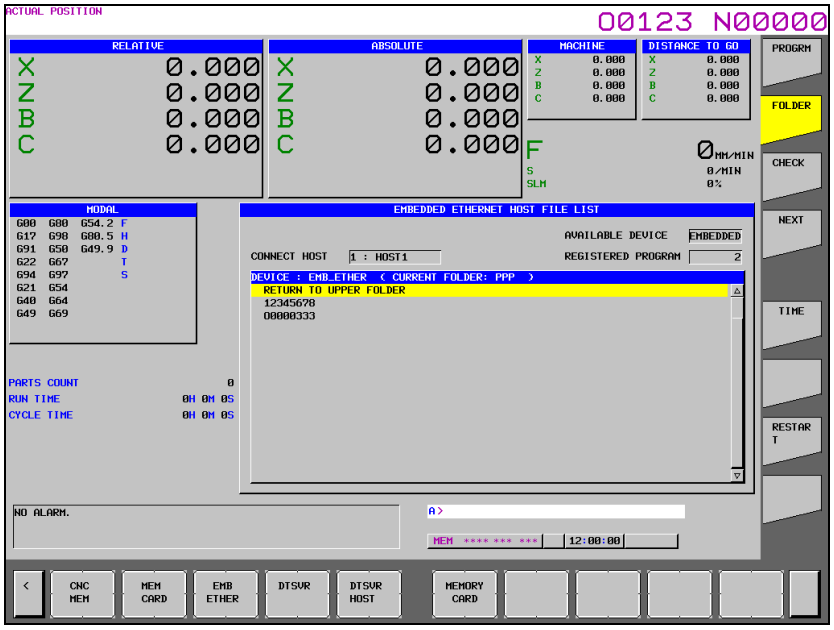
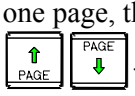


Fig. 8.4.1 (b) Embedded Ethernet host file list screen

NOTE
1 When using the FTP file transfer function, check that the valid device is the embedded Ethernet port. The two conditions below determine a connection destination on the host file list screen:
(1) Check that the valid device is the embedded Ethernet port. Make a selection in "DEVICE SELECTION" on the Ethernet setting screen.
(2) A host computer can be selected from connection destinations 1, 2, and 3. Make a selection according to soft key [HOST CHANGE] described later.
2 File names including kanji, hiragana, and katakana characters cannot be displayed correctly.

5 When a list of files is larger than one page, the screen display can be switched using the page keys  .

6 Press the horizontal soft key [REFRESH] to update the screen display.

7 Press the horizontal soft key [DETAIL OFF] to display the host file list (file names only) screen. Press the horizontal soft key [DETAIL ON] to display the host file list (detail) screen.

8 When selecting a folder with cursor and pressing the MDI key , the selected folder is changed to the new current folder.

Display item (common to 7.2/8.4/10.4/15-inch display units)

AVAILABLE DEVICE

"EMBEDDED" or "PCMCIA" is displayed.

CONNECT HOST

The currently selected connection host number and host name are displayed.

REGISTERED PROGRAM

The number of files registered in the work folder of the connected host is displayed. Up to 8 digits can be displayed.

CURRENT FOLDER

The current folder name of the connected host is displayed.
If the folder-path is long compared with the display-item, characters: "…" and only the first and last ten letters of the folder name are displayed.

FILE LIST

There is no distinction between file names and folder names.
Although the maximum number of displayed characters is 127, characters are displayed as far as they can be displayed in one line.

Operation list (common to 7.2/8.4/10.4/15-inch display units)

DETAIL ON, DETAIL OFF

The screen display can be switched between the display of file names only and the display of details.

REFRESH

Display data can be updated.

READ

A file can be input from the host computer to the part program storage memory of the CNC. This item is displayed only when 9 is set as the I/O device number of the CNC.

PUNCH

A file can be output from the part program storage memory of the CNC to the host computer. This item is displayed only when 9 is set as the I/O device number of the CNC.

DELETE

A folder or file can be deleted from the host computer.

RENAME

A file or folder on the host computer can be renamed.

DELETE FOLDER

A folder can be deleted from the host computer.

CREATE FOLDER

A folder can be created on the host computer.

HOST CHANGE

The connected host can be changed.

NC program input (for 7.2/8.4/10.4-inch display unit)

A file (NC program) stored on the host computer can be input into the part program storage memory of the CNC.

Procedure

1 Display the EMBEDDED ETHERNET HOST FILE LIST screen.
2 Press the EDIT switch on the machine operator's panel.
3 Press the soft key [(OPRT)].
4 Press the continuous menu key ▷ until soft key [READ] appears.
Press the soft key [READ].
5 Position the cursor on the file to input, and press [F GET] or type the file name to input.
Press the soft key [F SET].
If the input file name is omitted, default input file name "ALL-PROG.TXT" is assumed.
6 Type the program name, and press the soft key [P SET].
For an explanation of the operations to be performed if an input file name [F SET] and an input program name [P SET] are omitted, see the table below.
7 Press the soft key [EXEC].
This starts reading the program, and "INPUT" blinks in the lower right part of the screen. When the read operation ends, the "INPUT" indication disappears.
To cancel the input of the program, press the soft key [CAN].

[F SET]	[P SET]	Input file name	Input program	Input program name
BLANK	BLANK	ALL-PROG.TXT	All programs in ALL-PROG.TXT	File name at the time the file is saved
BLANK	INPUT	ALL-PROG.TXT	First program in ALL-PROG.TXT	Program name set with [P SET]
INPUT	BLANK	File name set with [F SET]	All programs in the file specified with [F SET]	File name at the time the file is saved
INPUT	INPUT	File name set with [F SET]	First program in the file specified with [F SET]	Program name set with [P SET]

NC program input (for 15-inch display unit)

Procedure

A file (NC program) stored on the host computer can be input into the part program storage memory of the CNC.

1 Display the EMBEDDED ETHERNET HOST FILE LIST screen.
2 Press the EDIT switch on the machine operator's panel.
3 Press the continuous menu key ▷ until horizontal soft key [READ] appears.
Press the horizontal soft key [READ].
4 Position the cursor on the file to input, and press [F GET] or type the file name to input.
Press the horizontal soft key [F SET].
If the input file name is omitted, default input file name "ALL-PROG.TXT" is assumed.
5 Type the program name, and press the horizontal soft key [P SET].
For an explanation of the operations to be performed if an input file name [F SET] and an input program name [P SET] are omitted, see the table below.
6 Press the horizontal soft key [EXEC].
This starts reading the program, and "INPUT" blinks in the lower right part of the screen. When the read operation ends, the "INPUT" indication disappears.
To cancel the input of the program, press the horizontal soft key [CAN].

[F SET]	[P SET]	Input file name	Input program	Input program name
BLANK	BLANK	ALL-PROG.TXT	All programs in ALL-PROG.TXT	File name at the time the file is saved
BLANK	INPUT	ALL-PROG.TXT	First program in ALL-PROG.TXT	Program name set with [P SET]
INPUT	BLANK	File name set with [F SET]	All programs in the file specified with [F SET]	File name at the time the file is saved
INPUT	INPUT	File name set with [F SET]	First program in the file specified with [F SET]	Program name set with [P SET]

NC program output (for 7.2/8.4/10.4-inch display unit)

A file (NC program) stored in the part program storage memory of the CNC can be output to the host computer.

Procedure

1 Display the EMBEDDED ETHERNET HOST FILE LIST screen.
2 Press the EDIT switch on the machine operator's panel.
3 Press the soft key [(OPRT)].
4 Press the continuous menu key ▷ until soft key [PUNCH] appears.
Press the soft key [PUNCH].
5 Type the program name to output, and press the soft key [P SET].
To specify an output file name, type the output file name and press the soft key [F SET].
If, in this step, no output file name or no program name is specified, all programs in the foreground folders are output to "ALL-PROG.TXT".
For an explanation of the operations to be performed if an output file name [F SET] and an output program name [P SET] are omitted, see the table below.
(For an explanation of foreground folders, see II-12.1.3, and for how to change foreground folders, see III-11.6)
6 Press the soft key [EXEC].
This starts outputting the program, and "OUTPUT" blinks in the lower right part of the screen. When the read operation ends, the "OUTPUT" indication disappears.
To cancel the output of the program, press the soft key [CAN].

[F SET]	[P SET]	Output file name	Output program
BLANK	BLANK or (O-9999)	ALL-PROG.TXT	All programs in the foreground folders displayed in the program folder
BLANK	INPUT	Program name set with [P SET]	Program in the NC that is set with [P SET]
INPUT	BLANK or (O-9999)	File name set with [F SET]	All programs in the foreground folders displayed in the program folder
INPUT	INPUT	File name set with [F SET]	Program in the NC that is set with [P SET]

NC program output (for 15-inch display unit)

A file (NC program) stored in the part program storage memory of the CNC can be output to the host computer.

Procedure

1 Display the EMBEDDED ETHERNET HOST FILE LIST screen.
2 Press the EDIT switch on the machine operator's panel.
3 Press the continuous menu key ▷ until horizontal soft key [PUNCH] appears.
Press the horizontal soft key [PUNCH].
4 Type the program name to output, and press the horizontal soft key [P SET]. To specify an output file name, type the output file name and press the horizontal soft key [F SET].
If, in this step, no output file name or no program name is specified, all programs in the foreground folders are output to "ALL-PROG.TXT".
For an explanation of the operations to be performed if an output file name [F SET] and an output program name [P SET] are omitted, see the table below.
(For an explanation of foreground folders, see II-12.1.3, and for how to change foreground folders, see III-11.6)
5 Press the horizontal soft key [EXEC].
This starts outputting the program, and "OUTPUT" blinks in the lower right part of the screen. When the read operation ends, the "OUTPUT" indication disappears.
To cancel the output, press the horizontal soft key [CAN].

[F SET]	[P SET]	Output file name	Output program
BLANK	BLANK or (O-9999)	ALL-PROG.TXT	All programs in the foreground folders displayed in the program folder
BLANK	INPUT	Program name set with [P SET]	Program in the NC that is set with [P SET]
INPUT	BLANK or (O-9999)	File name set with [F SET]	All programs in the foreground folders displayed in the program folder
INPUT	INPUT	File name set with [F SET]	Program in the NC that is set with [P SET]

8.5 SCREEN HARD COPY FUNCTION

Overview

This function converts screen information displayed on the CNC into bit map format data and output it to a memory card. Once output, bit map format data can be displayed and edited on a personal computer.

Explanation

- Start/cancellation methods

The screen hard copy function is started by pressing and holding down key

for five seconds. The function can be canceled by pressing key

.

- Acquisition and output of screen data

When started, the screen hard copy function starts acquiring screen data. As soon as it has acquired it, the function outputs bit map format data to the memory card inserted into the LCD unit. While screen data is being acquired, the screen remains stationary for a few seconds. Acquired screen data can be output from the memory card screen. Also, while screen data is being output, "OUTPUT" blinks in the status display.

- Screen data file names

Bit map format screen data files created by this function are assigned the names below, starting with the one created after the power is turned on.
"HDCPY000.BMP" (name of the first file output to the memory card after the power is turned on)
"HDCPY001.BMP" (name of the second file output to the memory card after the power is turned on)
:
:
"HDCPY999.BMP" (name of the 1000th file output to the memory card after the power is turned on)
If, after a file with "HDCPY999.BMP" is output, the screen hard copy function is executed, the file name returns to "HDCPY000.BMP". If, however, a file with the same file name as the one to be output when the screen hard copy function is executed exists on the memory card, alarm SR1973 is issued. If the capacity of the memory card is exceeded, alarm SR1962 is issued.
In either case, screen data is not output, so that either the existing file needs to be deleted or the memory card needs to be replaced with a new one.

Limitation

- Screens whose hard copies cannot be made

Hard copies of the BOOT screen, the IPL screen, and the system alarm screen cannot be made.

- Foreground I/O devices

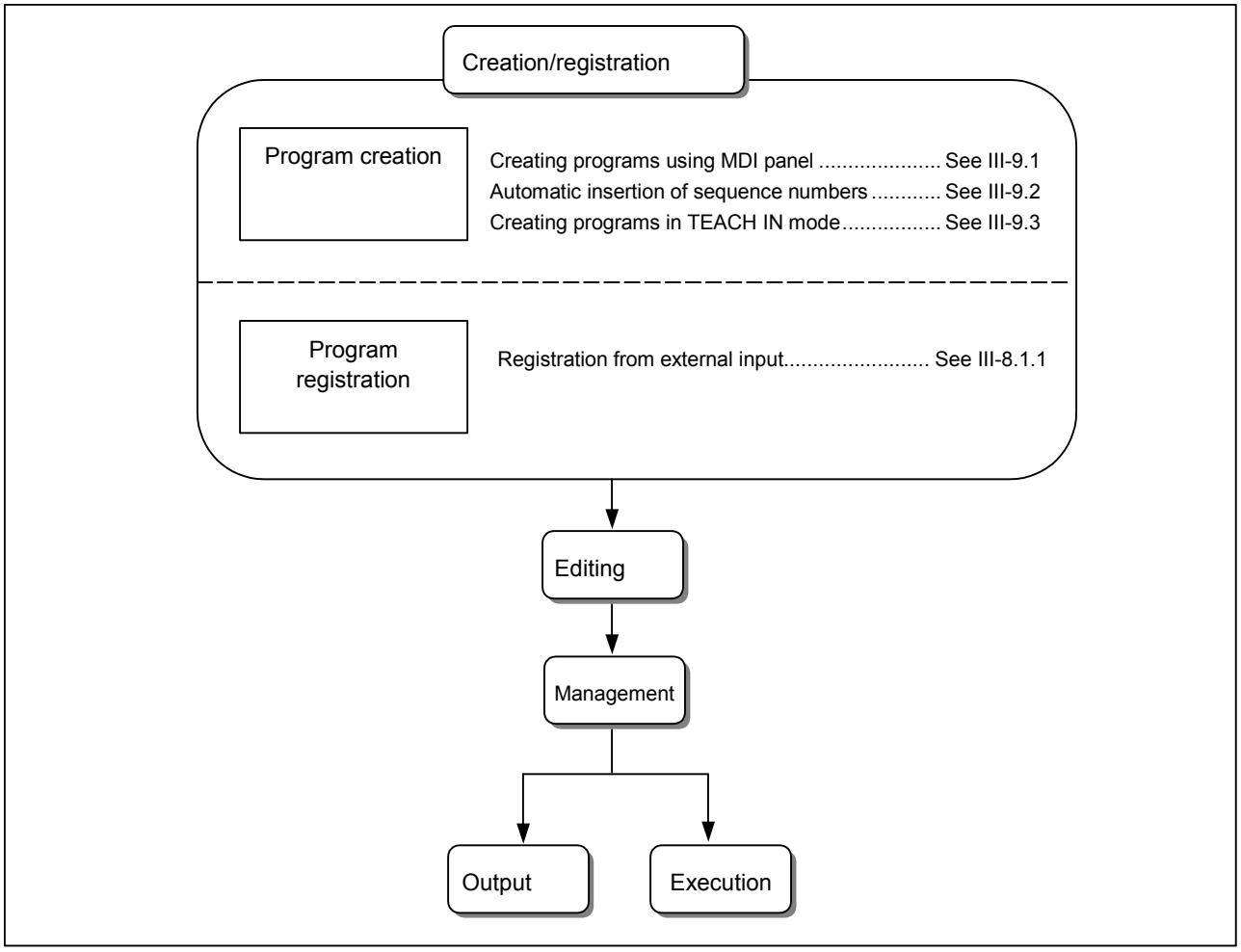
During DNC operation, for example, screen data cannot be output while a foreground I/O device is in use.

- Canceling the hard copy function

If the hard copy function is canceled before a hard copy is completed, an incomplete bit map file of data that has been output is created.

9 CREATING PROGRAMS

This chapter explains how to create programs by MDI of the CNC.
This chapter also explains automatic insertion of sequence numbers and how to create programs in TEACH IN mode.



9.1 CREATING PROGRAMS USING THE MDI PANEL

Programs can be created in the EDIT mode using the program editing functions described in III-10.

Procedure for Creating Programs Using the MDI Panel

1 Enter the EDIT mode.
2 Press the PROG key.
3 Press address key P 0 and enter the program number.
4 Press the INSERT key.
5 Create a program using the program editing functions described in III-10.

Explanation

- Comments in a program

Comments can be written in a program using the control in/out codes.
Example) O0001 (TEST PROGRAM) ;
M08 (COOLANT ON) ;

- When the INSERT key is pressed after the control-out code "(", comments, and control-in code ")" have been typed, the typed comments are registered.
- When the INSERT key is pressed midway through comments, to enter the rest of comments later, the data typed before the INSERT key is pressed may not be correctly registered (not entered, modified, or lost) because the data is subject to an entry check which is performed in normal editing.
- Control-out code "(" or Control-in code ")" cannot be registered by itself.

9.2 AUTOMATIC INSERTION OF SEQUENCE NUMBERS

Sequence numbers can be automatically inserted in each block when a program is created using the MDI keys in the EDIT mode.
Set the increment for sequence numbers in parameter No. 3216.

Procedure for automatic insertion of sequence numbers

Procedure

1 Enter 1 in response to the prompt for "SEQUENCE NO." in setting data. (see III-12.3.1).
2 Enter the EDIT mode.
3 Press

key to display the program screen.
4 Search for or register the number of a program to be edited and move the cursor to the EOB (;) of the block after which automatic insertion of sequence numbers is started.
When a program number is registered and an EOB (;) is entered with the

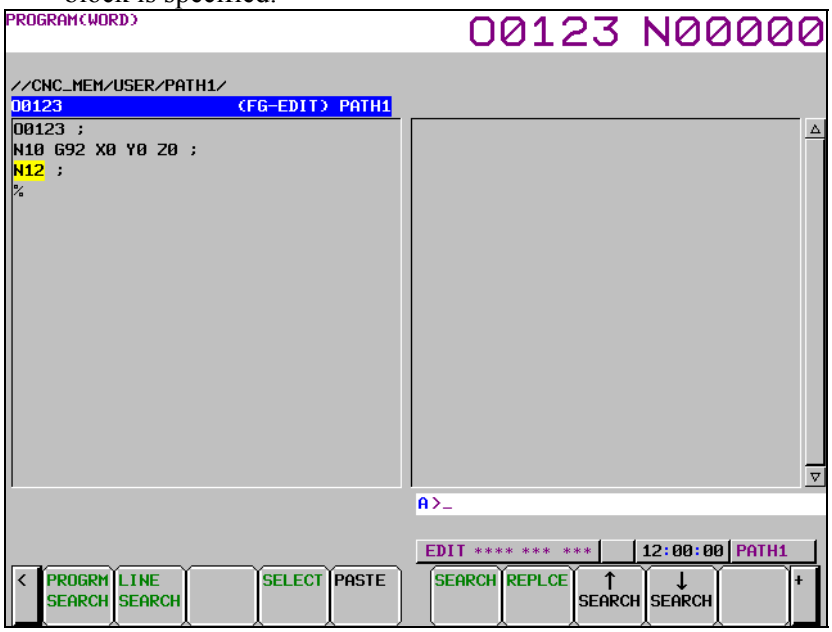
key, sequence numbers are automatically inserted starting with 0. Change the initial value, if required, according to step 10, then skip to step 7.
5 Press address key

and enter the initial value of N.
6 Press

key.
7 Enter each word of a block.
8 Press

key.

9 Press INSERT key. The EOB is registered in memory and sequence numbers are automatically inserted. For example, if the initial value of N is 10 and the parameter for the increment is set to 2, N12 inserted and displayed below the line where a new block is specified.



10 In the example above, if N12 is not necessary in the next block, pressing the DELETE key after N12 is displayed deletes N12.
If wishing to insert N100, not N12, into the next block, type N100 immediately after N12 is displayed, and press CALC ALTER. This causes N100 to be registered and the initial value to be changed to 100.

9.3 CREATING PROGRAMS IN TEACH IN MODE (PLAYBACK)

In the TEACH IN JOG or TEACH IN HANDLE mode, you can create a program while inserting the coordinate of the current position along each axis in the absolute coordinate system when the tool is moved by manual operation into the program.
You can input the words other than axis names in the same way as in the EDIT mode.

Program screen in the TEACH IN JOG mode

Displayed items

In the TEACH IN JOG or TEACH IN HANDLE MODE, the following program screen is displayed.
On the left of the screen, the coordinates of the current position in the absolute and relative coordinate systems are displayed; on the right of the screen, the contents of a program are displayed. You can create a program while checking the current position by manual operation.

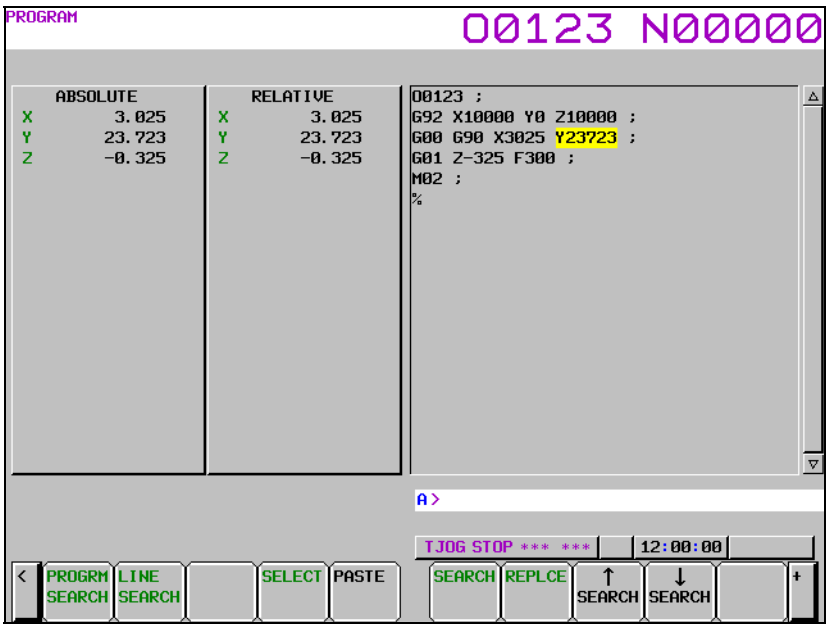


Fig. 9.3 (a) Program screen in the TEACH IN JOG mode

Inputting the coordinates of the current position

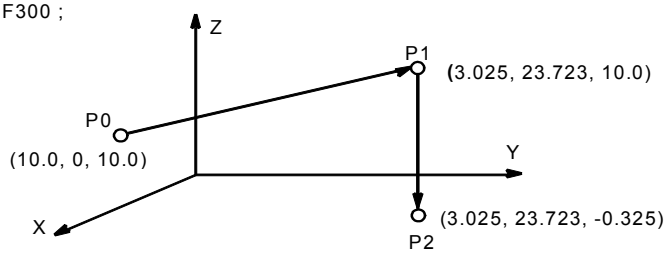
You can use the following procedure to insert the coordinate of the current position along each axis in the absolute coordinate system:

1 Select the TEACH IN JOG mode or TEACH IN HANDLE mode.
2 Press PROG key to display the program screen. Search for or register the number of a program to be edited and move the cursor to the position where the current position along each axis is to be inserted.
3 Move the tool to the desired position with jog or handle.
4 Key in the axis name of an axis along which you want to insert the coordinate of the current position.
5 Press the INSERT key. Then, the coordinate of the current position along the specified axis is inserted in the program.

(Example) X10.521 Coordinate of the current position
X10521 Data inserted in the program

Example

```
O1234 ;
G92 X10000 Y0 Z10000 ;
G00 G90 X3025 Y23723 ;
G01 Z-325 F300 ;
M02 ;
```



1 Select the TEACH IN HANDLE mode.
2 Make positioning at position P0 by the manual pulse generator.
3 Select the program screen.
4 Enter program number O1234 as follows:

O 1 2 3 4 INSERT

This operation input program number O1234 in memory.
Next, press the following keys:

EOB INSERT

An EOB (;) is entered after program number O1234.

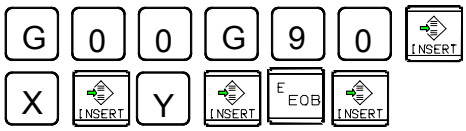
5 Enter the P0 machine position for data of the first block as follows:

G 9 2 INSERT X INSERT Y INSERT
Z INSERT EOB INSERT

This operation registers G92X10000Y0Z10000; in program.

6 Position the tool at P1 with the manual pulse generator.

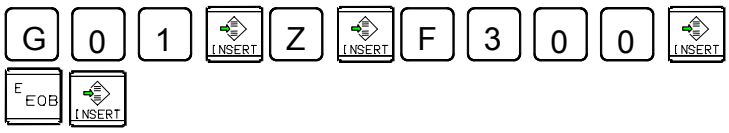
7 Enter the P1 machine position for data of the second block as follows:



This operation input G00G90X3025Z23723; in program.

8 Position the tool at P2 with the manual pulse generator.

9 Enter the P2 machine position for data of the third block as follows:



This operation input G01Z -325F300; in program.

10 Input M02; in program as follows:

This completes the registration of the sample program.

Explanation

- Registering a position with compensation

When an axis name and a numeric value are keyed in and the

key is pressed, the value keyed in is added to the absolute coordinate of the current position and the coordinate is inserted. This operation allows insertion of a corrected value for the absolute coordinate position.

When an extended axis name is used and the name ends with a numeric value, insert an equal sign (=) between the axis name and numeric value.

- Registering commands other than position commands

Commands to be entered before and after a position command must be entered before and after the machine position is registered, by using the same operation as program editing in EDIT mode.

- Calculator-type input

When the calculator-type input format is disabled (bit 0 (DPI) of parameter No. 3401 is set to 0), the coordinate of the current position is inserted into the program in least input increments. When the calculator-type input format is enabled (the bit is set to 1), the coordinate is inserted with a decimal point.

(Example)

Coordinate of the current position X10.521

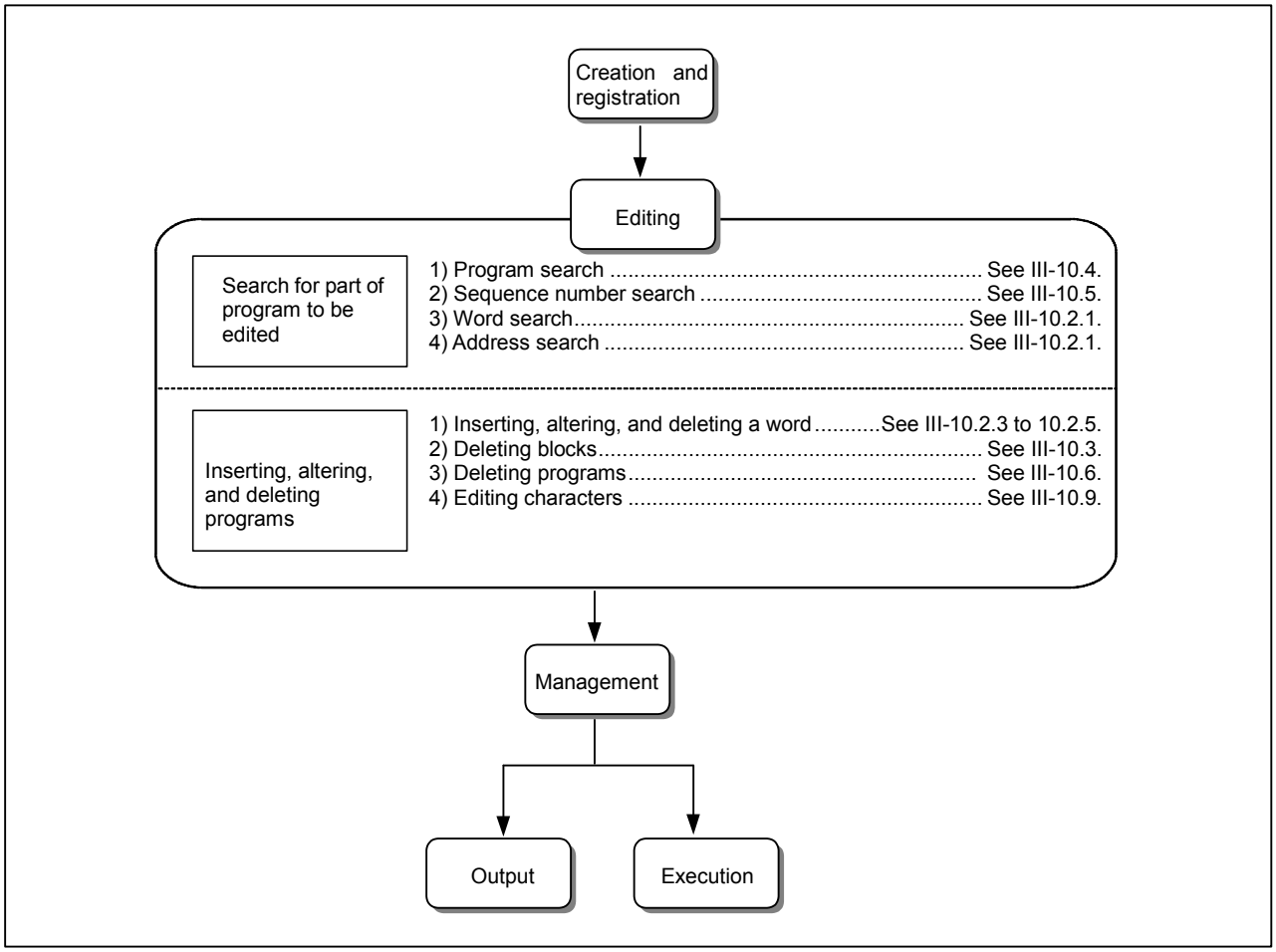
At this time, the X-axis coordinate is inserted into the program as follows:

When the calculator-type input format is disabled X10521

When the calculator-type input format is enabled X10.521

10 EDITING PROGRAMS

This chapter describes how to edit programs registered in the CNC. Editing includes the insertion, modification, and deletion of words. Editing also includes deletion of the entire program and automatic insertion of sequence numbers. In addition, PC-like program text copy and move operations are possible. This chapter also describes program search, sequence number search, word search, and address search, which are performed before editing the program.



10.1 EDIT DISABLE ATTRIBUTE

Before a program can be edited, the edit disable attribute must be removed.
The edit disable attribute can be set for each program and folder.
Programs with the edit disable attribute and programs in folders with the edit disable attribute cannot be edited.

Procedure for removing the edit disable attribute

1 Select EDIT mode.
2 Press the function key

.
3 Press the chapter selection soft key [FOLDER].
4 Press the soft key [DETAIL ON].
5 Move the cursor to the program or folder from which you want to remove the edit disable attribute.
6 Press the soft key [CHANGE ATTR].
7 Press the soft key [EDIT ENABLE].
8 Press the soft key [END].

CAUTION
1 After completing editing, set the edit disable attribute as necessary.
2 To set the edit disable attribute, follow the same procedure as for removing the attribute.
In the step 7, press the soft key [EDIT DISABL].

10.2 INSERTING, ALTERING AND DELETING A WORD

This section outlines the procedure for inserting, altering, and deleting a word in a program registered in memory.

Procedure for inserting, altering and deleting a word

1 Select EDIT mode.

2 Press the function key

.

3 Select a program to be edited.
If a program to be edited is selected, perform the operation 4.
If a program to be edited is not selected, search for the program number.

4 Move the cursor to the position to be edited, with one of the methods below.
- Scan method
- Word search method
- Address search method

For each method, refer to III-10.2.1.

5 Perform an operation such as altering, inserting, or deleting a word.

Explanation

- Concept of word and editing unit

A word is an address followed by a number. With a custom macro, the concept of word is ambiguous.
So the editing unit is considered here.
The editing unit is a unit subject to alteration or deletion in one operation. In one scan operation, the cursor indicates the start of an editing unit.
An insertion is made after an editing unit.
Definition of editing unit
- Program portion from an address to immediately before the next address
- An address is an alphabet, IF, WHILE, GOTO, END, DO=, or ; (EOB).

According to this definition, a word is an editing unit.
The word "word," when used in the description of editing, means an editing unit according to the precise definition.

⚠ WARNING
When a change, insertion, or deletion was performed on data of a program by pausing machining with the single block stop, feed hold, or other operations during execution of a program, be sure to return the cursor to the original position before restarting the program. To execute the program with the cursor positioned at another position, be sure to make a reset.
Otherwise, the program may not be executed as expected from the program shown on the screen after machining restarts.

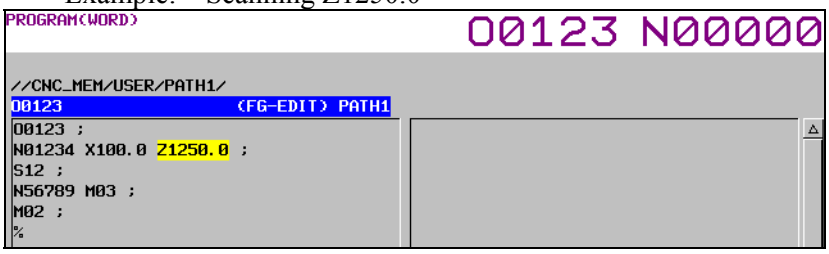
10.2.1 Word Search

A word can be searched for by merely moving the cursor through the text (scanning), by word search, or by address search.

Procedure for scanning a program

1 Press the cursor key [→].
The cursor moves forward word by word; the cursor is displayed at a selected word.

2 Press the cursor key [←].
The cursor moves backward word by word; the cursor is displayed at a selected word.
Example: Scanning Z1250.0



3 Holding down the cursor key [→] or [←] scans words continuously.

4 The first word of the next block is searched for when the cursor key [↓] is pressed.

5 The first word of the previous block is searched for when the cursor key [↑] is pressed.

6 Holding down the cursor key [↓] or [↑] moves the cursor to the head of a block continuously.

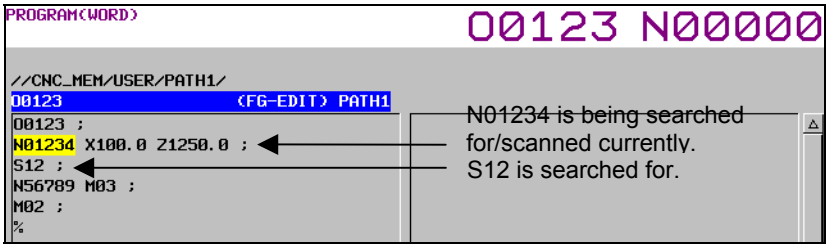
7 Pressing the page key [PAGE ↓] displays the next page and searches for the first word of the page.

8 Pressing the page key [↑ PAGE] displays the previous page and searches for the first word of the page.

9 Holding down the page key [PAGE ↓] or [↑ PAGE] displays one page after another.

Procedure for searching a word

Example of searching for S12



1 Press soft key [SEARCH].

2 Key in address [S].

3 Key in [1] [2].

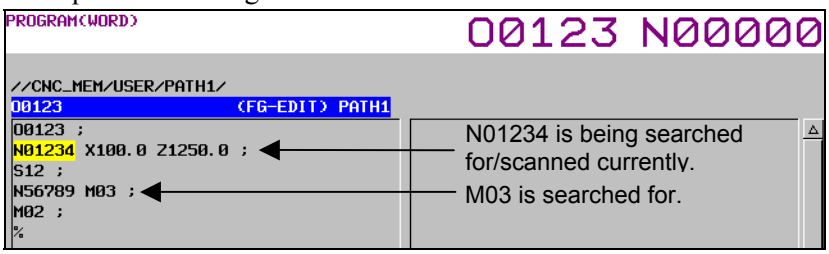
- S12 cannot be searched for if only S1 is keyed in.
- S09 cannot be searched for by keying in only S9. To search for S09, be sure to key in S09.

4 Pressing soft key [↓ SEARCH] starts searching forward from the cursor position. Pressing [↑ SEARCH] starts searching backward.

5 To search for the same word successively, press [↓ SEARCH] or [↑ SEARCH].

Procedure for searching an address

Example of searching for M03



1 Pressing the [SEARCH] key

2 Key in address [M].

3 Pressing soft key [↓ SEARCH] starts searching forward from the cursor position. Pressing [↑ SEARCH] starts searching backward.

4 To search for the same word successively, press [↓ SEARCH] or [↑ SEARCH].

10.2.2 Heading a Program

The cursor can be jumped to the top of a program. This function is called heading the program pointer. This section describes the four methods for heading the program pointer.

Procedure for heading a program

Method 1

1 Press RESET key when the program screen is selected in EDIT mode.
When the cursor has returned to the start of the program, the contents of the program are displayed from its start on the screen.

Method 2

1 Search for the program number.
When the program screen is selected in MEMORY or EDIT mode, enter a program number or program name.
When entering a program number, press address key O then type the program number.
2 Press soft key [PROGRM SEARCH].

Method 3

1 Select the program screen or program check screen in MEMORY mode.
2 Press soft key [(OPRT)].
3 Press soft key [REWIND].

Method 4

1 Select the program screen in EDIT mode.
2 Press soft key [(OPRT)].
3 Press soft key [LINE SEARCH].
4 Press soft key [TOP].

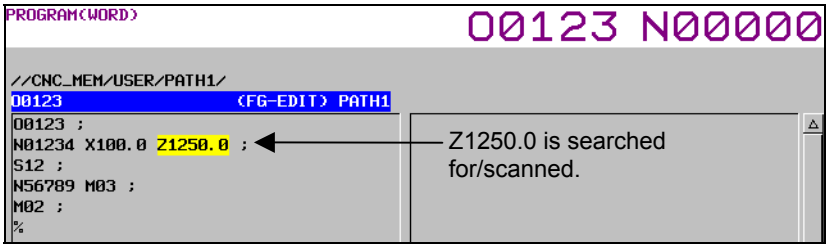
10.2.3 Inserting a Word

Procedure for inserting a word

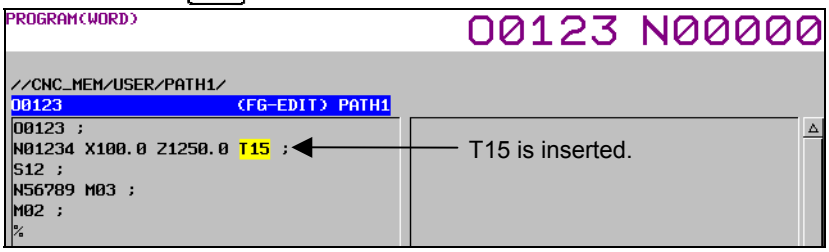
1 Search for or scan the word immediately before a word to be inserted.
2 Key in an address to be inserted.
3 Key in data.
4 Press the INSERT key.

Example of Inserting T15

1 Search for or scan Z1250.



2 Key in T 1 5 .
3 Press the INSERT key.



10.2.4 Altering a Word

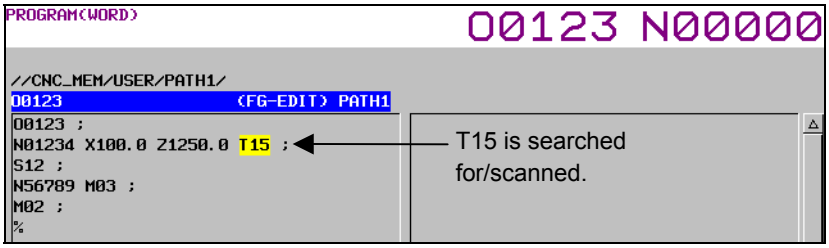
Procedure for altering a word

1 Search for or scan a word to be altered.
2 Key in an address to be inserted.
3 Key in data.
4 Press the

key.

Example of changing T15 to M15

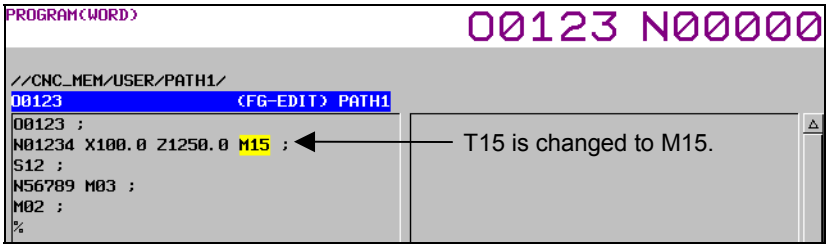
1 Search for or scan T15.



2 Key in M 1 5 .

3 Press the

key.



10.2.5 Deleting a Word

Procedure for deleting a word

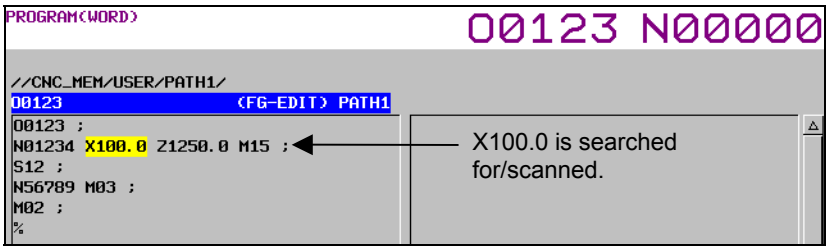
1 Search for or scan a word to be deleted.

2 Press the

key.

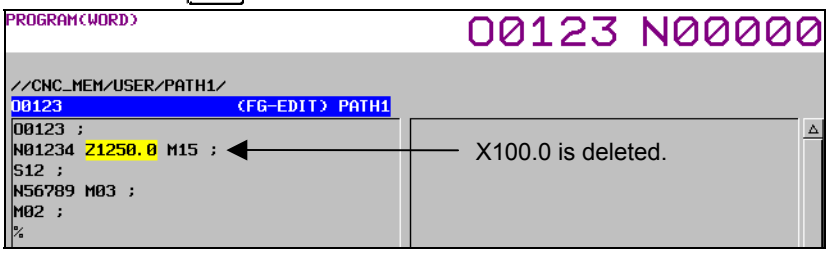
Example of deleting X100.0

1 Search for or scan X100.0.



2 Press the

key.



10.3 DELETING BLOCKS

A block or blocks can be deleted in a program.

10.3.1 Deleting a Block

The portion from the current word position to the next EOB is deleted. The cursor is then placed in the word next to the deleted EOB.

Procedure for deleting a block

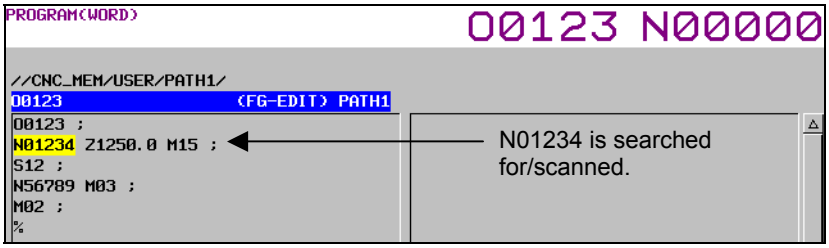
1 Search for or scan address N for a block to be deleted.

2 Press the EOB key.

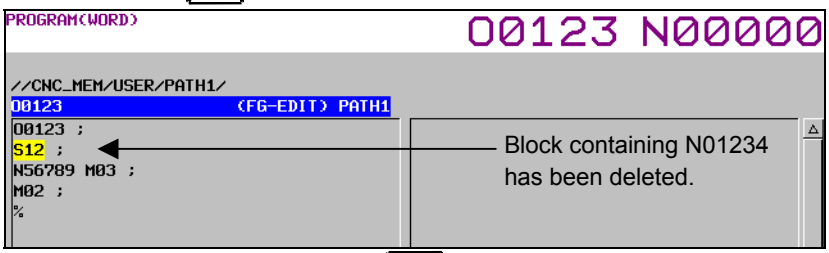
3 Press the editing key DELETE.

Example of deleting a block of N01234

1 Search for or scan N01234.



2 Press the EOB key.



3 Press the editing key DELETE.

10.3.2 Deleting Multiple Blocks

The several blocks in the forward direction from the current word position up to the EOB of the farthest of those blocks are deleted. The cursor is then placed in the word next to the deleted EOB.

Procedure for deleting blocks

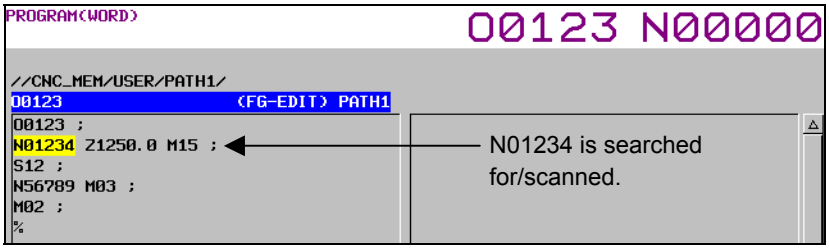
1 Search for or scan a word in the first block of a portion to be deleted.

2 Press the EOB key as many times as the number of blocks that you want to delete.

3 Press the editing key DELETE.

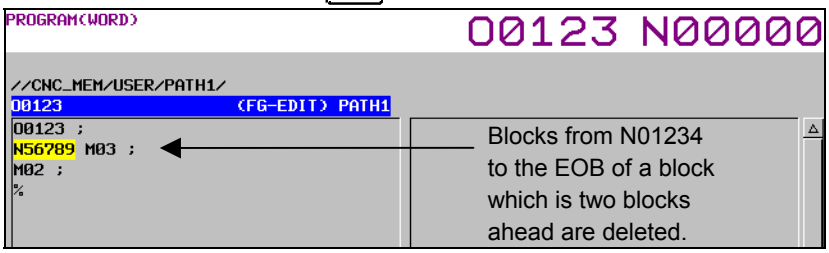
Example of deleting blocks from N01234 to the EOB of a block which is two blocks ahead

1 Search for or scan N01234.



2 Press EOB EOB.

3 Press the editing key DELETE.



10.4 PROGRAM SEARCH

When memory holds multiple programs, a program can be searched for. There are four methods as follows.

Procedure for program search

Method 1

1 Select EDIT or MEMORY mode.
2 Press function key [PROG] to display the program screen.
3 Enter a program number or program name. When entering a program number, press address key [P 0] then type the program number.
4 Press soft key [PROGRM SEARCH]. Alternatively, press the cursor key [↓] or [↑].
5 Upon completion of search operation, the program number searched for is displayed in the upper-right corner of the screen
If the program is not found , alarm PS0071 occurs.

Method 2

1 Select EDIT or MEMORY mode.
2 Press function key [PROG] to display the program screen.
3 Press soft key [PROGRM SEARCH].
4 Then, press soft key [PREV PROGRM] or [NEXT PROGRM].
Pressing [PREV PROGRM] searches for the previous program on the folder, and pressing [NEXT PROGRM] searches for the next program.

Method 3

This method searches for the program number (0001 to 0031) corresponding to a signal on the machine tool side to start automatic operation. Refer to the relevant manual prepared by the machine tool builder for detailed information on operation.

1 Select MEMORY mode.
2 Set the reset state.
- The reset state is the state where the LED for indicating that automatic operation is in progress is off. (Refer to the relevant manual of the machine tool builder.)

3 Set the program number selection signal on the machine tool side to a number from 01 to 31.
4 Press the cycle start button.
- When the signal on the machine tool side represents 00, program number search operation is not performed.
- If the program corresponding to a signal on the machine tool side is not registered, alarm DS0059 is raised.

Method 4

When bit 4 (PGS) of parameter No. 11308 is 1, search for an O number program by specifying digits only.

1 Select EDIT or MEMORY mode.

2 Press function key to display the program screen.

3 Enter the digits of a program number. Address key need not be pressed before the digits.

4 Press soft key [PROGRM SEARCH]. Alternatively, press the cursor key or .

Example

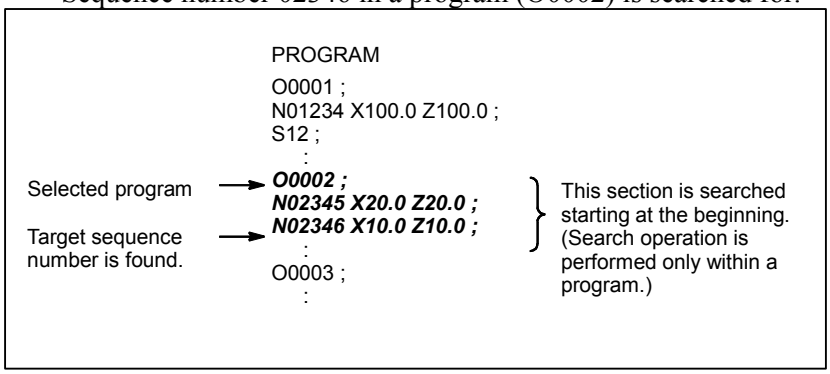
- When bit 4 (PGS) of parameter No. 11308 is 1
  Specify "1000" → Search for program O1000.
  Specify "ABC" → Search for program ABC.
- When bit 4 (PGS) of parameter No. 11308 is 0
  Specify "1000" → Search for program 1000.
  Specify "ABC" → Search for program ABC.

When bit 4 (PGS) of parameter No. 11308 is 1, it is not possible to search for a program with any name consisting of digits only.

10.5 SEQUENCE NUMBER SEARCH

Sequence number search operation is usually used to search for a sequence number in the middle of a program so that execution can be started or restarted at the block of the sequence number.

Example)
Sequence number 02346 in a program (O0002) is searched for.



Procedure for sequence number search

Procedure

1 Select MEMORY mode.
2 Press function key .
3 If the program contains a sequence number to be searched for, perform the operations 4 to 7 below.
If the program does not contain a sequence number to be searched for, select the program number of the program that contains the sequence number to be searched for.
4 Press address key .
5 Key in a sequence number to be searched for.
6 Press soft key [N SRH].
7 Upon completion of search operation, the sequence number searched for is displayed in the upper-right corner of the screen.
If the specified sequence number is not found in the program currently selected, alarm PS0060 occurs.

Explanation

- Operation during Search

Those blocks that are skipped do not affect the CNC. This means that the data in the skipped blocks such as coordinates and M, S, and T codes does not alter the CNC coordinates and modal values.
So, in the first block where execution is to be started or restarted by using a sequence number search command, be sure to enter required M, S, and T codes and coordinates. A block searched for by sequence number search usually represents a point of shifting from one process to another. When a block in the middle of a process must be searched for to restart execution at the block, specify M, S, and T codes, G codes, coordinates, and so forth as required from the MDI after closely checking the machine tool and NC states at that point.

- Checking during search

During search operation, the following checks are made:

- Optional block skip

Limitation

- Searching in sub-program

During sequence number search operation, M98Pxxxx (subprogram call) is not executed. So an alarm PS0060 is raised if an attempt is made to search for a sequence number in a subprogram called by the program currently selected.

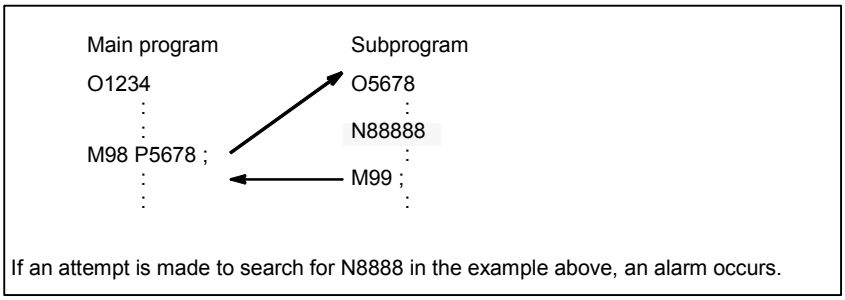


Fig. 10.5 (a)

10.6 DELETING PROGRAMS

Programs registered in memory can be deleted, either one program by one program or all at once.

10.6.1 Deleting One Program

A single program in the folder containing the program currently being edited can be deleted.

Procedure for deleting one program

1 Select the EDIT mode.
2 Press function key to display the program screen.
3 Press address key .
4 Key in a desired program number.
5 Press the editing key . The program with the entered program number is deleted.

10.6.2 Deleting All Programs

All programs in the folder containing the program currently being edited can be deleted.

Procedure for deleting all programs

1 Select the EDIT mode.
2 Press function key to display the program screen.
3 Press address key .
4 Key in -9999.
5 Press editing key to delete all programs.

10.7 EDITING OF CUSTOM MACROS

Unlike ordinary programs, custom macro programs are modified, inserted, or deleted based on editing units.
Custom macro words can be entered in abbreviated form.
Comments can be entered in a program.
Refer to the III-9.1 for the comments of a program.

Explanation

- Editing unit

When editing a custom macro already entered, the user can move the cursor to each editing unit that starts with any of the following characters and symbols:

(a) Address
(b) # located at the start of the left side of a substitution statement
(c) /, (,=, and ;
(d) First character of IF, WHILE, GOTO, END, DO, POPEN, BPRNT, DPRNT and PCLOS

On the screen, a blank is placed before each of the above characters and symbols.

Example) Head positions where the cursor is placed

```
N001 X-#100 ;
#1 =123 ;
N002 /2 X[12/#3] ;
N003 X-SQRT[#3/3*[#4+1]] ;
N004 X-#2 Z#1 ;
N005 #5 =1+2-#10 ;
IF[#1NE0] GOTO10 ;
WHILE[#2LE5] DO1 ;
#[200+#2] =#2*10 ;
#2 =#2+1 ;
END1 ;
```

- Abbreviations of custom macro word

When a custom macro word is altered or inserted, the first two characters or more can replace the entire word.
Namely,

WHILE → WH	GOTO → GO	XOR → XO	AND → AN
SIN → SI	ASIN → AS	COS → CO	ACOS → AC
TAN → TA	ATAN → AT	SQRT → SQ	ABS → AB
BCD → BC	BIN → BI	FIX → FI	FUP → FU
ROUND → RO	END → EN	POPEN → PO	BPRNT → BP
DPRNT → DP	PCLOS → PC	EXP → EX	THEN → TH

Example) Keying in

```
WH [AB [#2 ] LE RO [#3 ] ] has the same effect as
WHILE  [ABS  [#2 ] LE ROUND [#3 ] ]
```

The program is also displayed in this way.

10.8 CURSOR MOVEMENT LIMITATIONS ON PROGRAM EDITING

If a program is to be edited while it is stopped or halted, limitations are imposed so that already executed blocks and buffered blocks cannot be edited. Modifiable blocks are those at and below the cursor position at the time the system switches to the edit mode (EDIT). In the remainder of this manual, the area of editable blocks is called the edit-enabled area, while the area of executed blocks and buffered blocks is called the edit-disabled area.
These limitations are applied when the automatic operation status signal <F0000.7> is ON and when a single block execution of MDI operation is edited, and are also applied to TEACH IN HNDL or TEACH IN JOG mode when the playback option is enabled.
This function is enabled when bit 3 (FPD) of parameter No. 11308 is 1.

Explanation

- States where cursor movement limitations are imposed

- EDIT mode while automatic operation is stopped or halted
- MDI mode and when a single block is set
- TEACH IN HNDL or TEACH IN JOG mode (only when the playback function is enabled)

- Cursor movement

Operations that cause the cursor to move to blocks above the edit mode start position are ignored.

- ↑
- ←
- PAGE UP
- CAN（character editing）

- Search and replacement

Operations that cause the cursor to move to the edit-disabled area are disabled.

- Top line search: Results in a search of the top line of the edit-enabled block.
- Specified-line search (character editing): Made through the edit-enabled area.
  If, however, an attempt is made to search through the edit-disabled area, warning "PROGRAM IN OPERATION" is issued, and the operation is disabled.
- Upward search: Made through the edit-enabled area.
  If attempted at the beginning of the edit-enabled block, this operation is ignored.
- Upward search (replacement): Made through the edit-enabled area.
  If attempted at the beginning of the edit-enabled block, this operation is ignored.

- Upward candidate search: Made through the edit-enabled area. If attempted at the beginning of the edit-enabled block, this operation is ignored.
- Character string + ↑ (search): Typing the character string to search for and pressing the ↑ key searches for the character string. This search is made through the edit-enabled area. If attempted at the beginning of the edit-enabled block, this operation is ignored.
- Replace all: This operation is disabled.

- Reset

The program execution status is canceled, and the cursor movement limitations are also canceled.

- Program search

If an attempt is made to search for the program currently used for operation, warning "PROGRAM IN OPERATION" is issued, and the operation is disabled.

- Time insertion

When the machining time stamp function is enabled, soft key [INSERT TIME] does not appear in the state in which the cursor movement limitations are imposed.

10.9 PASSWORD FUNCTION

The password function locks bit 4 (NE9) of parameter No. 3202, which protects programs with program Nos. O9000 to O9999 and programs and folders having the edit/display disable attribute, according to the settings in two parameters, PASSWORD (parameter No. 3210) and KEYWD (parameter No. 3211). In the locked state, parameter NE9 cannot be set to 0. In this case, the protection of programs with program Nos. O9000 to O9999 and programs and folders having the edit/display disable attribute cannot be released unless a correct keyword is set.
A locked state means that the value set in the parameter PASSWD differs from the value set in the parameter KEYWD. The values set in these parameters are not displayed. The locked state is released when the value already set in the parameter PASSWD is also set in parameter KEYWD. When 0 is displayed in parameter PASSWD, parameter PASSWD is not set.

Procedure for locking and unlocking

Locking

1 Set the MDI mode.
2 Enable parameter writing (III-12.3.1). At this time, alarm PS0100 is issued on the CNC.
3 Set parameter No. 3210 (PASSWD). At this time, the locked state is set.
4 Disable parameter writing.
5 Press the RESET key to release the alarm state.

Unlocking

1 Set the MDI mode.
2 Enable parameter writing (III-12.3.1). At this time, alarm SW0100 is issued on the CNC.
3 In parameter No. 3211 (KEYWD), set the same value as set in parameter No. 3210 (PASSWD) for locking. At this time, the locked state is released.
4 Set bit 4 (NE9) of parameter No.3202 to 0.
5 Disable parameter writing.
6 Press the RESET key to release the alarm state.
7 Subprograms from program Nos. 9000 to 9999 can now be edited.

Explanation

- Setting parameter PASSWD

The locked state is set when a value is set in the parameter PASSWD. However, note that parameter PASSWD can be set only when the locked state is not set (when PASSWD = 0, or PASSWD = KEYWD). If an attempt is made to set parameter PASSWD in other cases, a warning is given to indicate that writing is disabled. When the locked state is set (when PASSWD ≠ 0 and PASSWD ≠ KEYWD), parameter NE9 is automatically set to 1. If an attempt is made to set NE9 to 0, a warning is given to indicate that writing is disabled.

- Changing parameter PASSWD

Parameter PASSWD can be changed when the locked state is released (when PASSWD = 0, or PASSWD = KEYWD). After step 3 in the procedure for unlocking, a new value can be set in the parameter PASSWD. From that time on, this new value must be set in parameter KEYWD to release the locked state.

- Setting 0 in parameter PASSWD

When 0 is set in the parameter PASSWD, the number 0 is displayed, and the password function is disabled. In other words, the password function can be disabled by either not setting parameter PASSWD at all, or by setting 0 in parameter PASSWD after step 3 of the procedure for unlocking. To ensure that the locked state is not entered, care must be taken not to set a value other than 0 in parameter PASSWD.

- Re-locking

After the locked state has been released, it can be set again by setting a different value in parameter PASSWD, or by turning the power to the NC off then on again to reset parameter KEYWD.

⚠ CAUTION

1 Once the locked state is set, parameter NE9 cannot be set to 0 and parameter PASSWD cannot be changed until the locked state is released or the memory all-clear operation is performed. Special care must be taken in setting parameter PASSWD.
2 The edit/display disable attribute cannot be set unless parameter PASSWD is set.
3 In the locked state, programs with the edit/display disable attribute are treated as follows:
   - The presence of the programs is hidden. This means that these programs are not displayed on screens such as the program folder screen. These programs cannot be edited either.
   - These programs cannot be selected as a main program. They can be called as subprograms.

⚠ CAUTION

4 In the unlocked state, programs with the edit/display disable attribute are treated in the same manner as ordinary programs.
5 The programs in a folder having the edit/display disable attribute are also treated as described in Caution 3 and 4 above.
6 In the locked state, the folders in a folder having the edit/display disable attribute are treated as follows:
   - The presence of the folders is hidden. This means that these folders are not displayed on screens such as the program folder screen.
7 In the unlocked state, the folders in a folder having the edit/display disable attribute are treated in the same manner as ordinary folders.

10.10 EDITING PROGRAM CHARACTERS

This section describes how to edit programs registered in the CNC.
Editing operations include character insertion, modification, deletion, and replacement.
While program word editing is performed by recognizing program words, program character editing is performed on a character-by-character basis.

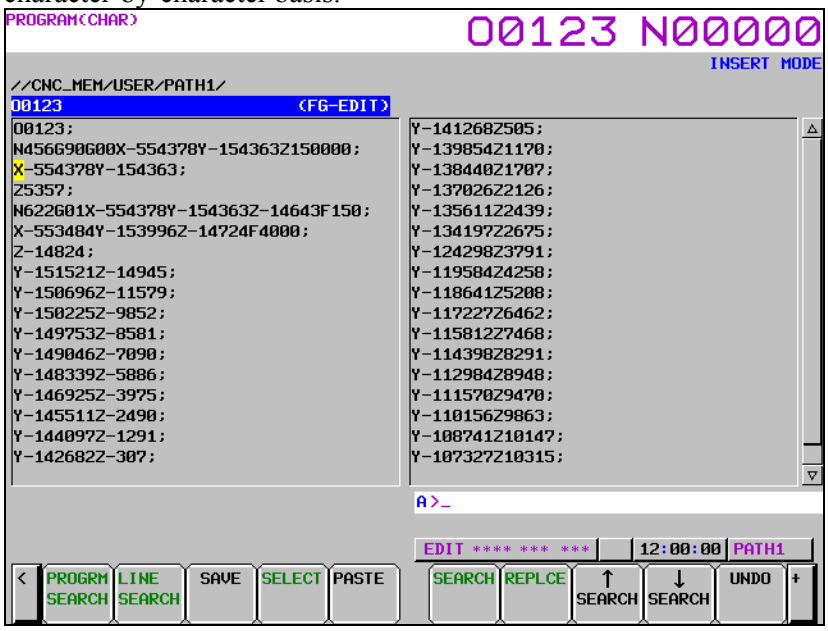


Fig. 10.10 (a) Character editing

Explanation

- Edit unit

Character editing is performed on a character-by-character basis. Select between character editing and word editing according to what to edit.

- Line

A range delimited by an EOB is specified as a line.
In character editing, edited data is saved on a line-by-line basis.
When one program line contains many characters, it extends over multiple lines on the screen, but these lines are counted as one program line.

- Line splitting

When the cursor is on a line during line editing, pressing edit key

causes the line to be split into two, one before the cursor and the other after the cursor.
Pressing edit key

immediately after the line is split causes the lines to return to a single one.
Alternatively, moving the cursor to the EOB of the first line resulting from the split and pressing edit key

causes the lines to return to a single one.

- Line merging

If the EOB of a line is deleted, the line is combined with the next line to form a single line.

- Maximum number of characters in one line

One line can consist of up to 140 characters.

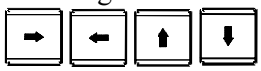
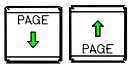
- Line number

The number of lines are counted starting with the starting line of a program, which is counted as the first line. Even when a line wraps around to the next and subsequent lines, these lines are counted as a single line.

- Clipboard

The clipboard is an area to store characters when a cut or copy operation is performed. The area has a capacity for holding approximately 4000 characters.
The characters that are cut or copied by one cut or copy operation are stored. The most recently stored character string can be used by the paste function.
The information stored by the paste operation is kept unchanged until another cut or copy operation is performed.
The characters stored in the clipboard are maintained until the power to the CNC is turned off unless the clipboard is updated again.

- Undo function

The function for undoing operations performed in program editing undoes operations sequentially, starting with the one performed last. It can undo only those operations that update character strings.
A single undo operation undoes a single input or deletion operation.
If data updated with replace operations is to be undone, a single undo operation undoes the replacement of a single item.
Operations of the cursor keys  and the page keys  cannot be undone with this function because they do not update character strings.

- Example of the undo function

1 Suppose that the following character string is present before modifications:

N110AX[#AXIS3]=100.0;

2 Edit key is pressed five times.

X[#AXIS3]=100.0;

3 Soft key [UNDO] is pressed five times. Then, the original character string is restored as follows:

AX[#AXIS3]=100.0;
0AX[#AXIS3]=100.0;
10AX[#AXIS3]=100.0;
110AX[#AXIS3]=100.0;
N110AX[#AXIS3]=100.0;

- Input modes for editing

Input modes for program editing include insert mode and overwrite mode.
To switch between the input modes, press soft key [INPUT MODE].
Initially, insert mode is set.

- Insert mode
  In insert mode, an entered character is inserted between the current cursor position and the preceding character position.

  1234567890

  When the cursor is placed at 6, and X is entered, the following results:

  12345X67890

- Overwrite mode
  In overwrite mode, the character at the cursor position is replaced by an entered character.

  1234567890

  When the cursor is placed at 6, and X is entered, the following results:

  12345X7890

- Restrictions on editing

O numbers and file names cannot be edited.
EOR (%) cannot be deleted.

- Line editing and automatic saving

When a line is edited, the line is displayed in the update color, blue (which can be changed with color settings), to indicate that the line has been edited. When the line is displayed in the update color, the edit has not yet been saved.
When the cursor is moved to a line preceding or succeeding the edited line, the edit is automatically saved, and the line display changes from the update color to the normal color.
Alternatively, pressing soft key [SAVE] enables the edit to be saved.

- Function for automatically saving an edited line vs. function for undoing editing operations (UNDO function)

When an edit operation is undone, the line is displayed in the update color, and the edit is not saved any longer. If the cursor moves to another line when the operation is undone, the edit is automatically saved.

10.10.1 Available Keys

The available keys are as follows:

- Cursor keys

Cursor keys , , , and move the cursor.

- Editing key

Deletes the character at the cursor position.

- Editing key

Deletes the character immediately before the cursor position.
When the cursor is at the beginning of a line, the character at the end of the previous line is deleted.

- Editing key

Causes a line change.

- Page change keys

Pages are changed using page key

or

Page key

moves to the next page.

When page key

is pressed on the last page, the cursor moves to the last character position of the last line (the % position).

Page key

moves to the previous page.

When page key

is pressed on the first page, the cursor moves to the first character position of the first line.

- Character keys

Characters are entered using these keys.

10.10.2 Input Mode

Input modes include insert mode and overwrite mode.

Changing input mode

To switch between the input modes, use soft key [INPUT MODE].
Pressing soft key [INPUT MODE] causes insert mode and overwrite mode to be selected alternately.
The current mode is indicated in the lower right part of the editing screen.

10.10.3 Line Number Display

This function is used to display a program with line numbers. Pressing soft key [LINE NUMBER] displays a program with line numbers. Pressing soft key [LINE NUMBER] again causes the line numbers to disappear.

10.10.4 Search

A program is searched for a character string.

Search

Procedure

1 Press soft key [SEARCH].
2 A search character string input area is displayed. Enter the character string to search for. To cancel the search, press soft key [CANCEL].
3 Upward search operation
Pressing soft key [UP] causes the program to be searched in the upward direction (toward the top), starting at the current cursor position.
When a search character string is found, the cursor moves to the character string.
Pressing soft key [UP] again causes the program to be searched for the next candidate.
4 Downward search operation
Pressing soft key [DOWN] causes the program to be searched in the downward direction (toward the bottom), starting at the current cursor position.
When a search character string is found, the cursor moves to the character string.
Pressing soft key [DOWN] again causes the program to be searched for the next candidate.

10.10.5 Replacement

A character string in a program is replaced with a specified character string.

Replacement

Procedure

1 Press soft key [REPLCE].

2 Search and replacement character string input areas are displayed.
Enter a search character string (character string to be replaced) and a replacement character string (character string to replace the former character string).
To move between the search character string input area and the replacement character string input area, use the cursor keys and .
If a search character string is input, but no replacement character string is input, the former character string will be deleted.
To cancel the replacement, press soft key [CANCEL].

3 Replacement operations
Replacement operations include an operation for searching for a search character string and an operation for actually replacing it.
To perform an operation for searching for a search string, use soft key [↑ SEARCH] or [↓ SEARCH].
To perform an operation for actually replacing it, use [REPLCE] or [REPLCE ALL].

- Soft key [↑ SEARCH]
  Searches the program in the upward direction (toward the top), starting at the current cursor position.
  When a search character string is found, the cursor moves to the character string.
  Pressing soft key [↑ SEARCH] again causes the program to be searched for the next candidate.
- Soft key [↓ SEARCH]
  Searches the program in the downward direction (toward the bottom), starting at the current cursor position.
  When a search character string is found, the cursor moves to the character string.
  Pressing soft key [↓ SEARCH] again causes the program to be searched for the next candidate.
- Soft key [REPLCE]
  Replaces the character string found with a search operation with a replacement character string.
- Soft key [REPLCE ALL]
  Replace all instances of the search character string in the program text with the replacement character string.
  Pressing this soft key causes message "ARE YOU SURE YOU WANT TO EXECUTE?" to appear and also soft key [YES] and soft key [NO] to appear. Pressing soft key [YES] causes all replacements to be executed.
  Pressing soft key [NO] cancels the replacement operation, without executing the replacements.

10.10.6 Reversing Edit Operations (Undo Function)

Edit operations performed on a program can be undone sequentially starting with the one performed last.

Reversing edit operations (undo function)

Procedure

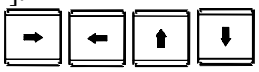
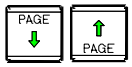
1 Pressing soft key [UNDO] once undoes a single operation.
When there are no longer any operation to be undone, pressing this soft key has no effect.

10.10.7 Copy

A selected character string can be copied to the clipboard.

Copying

Procedure

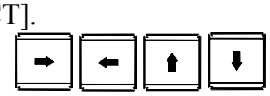
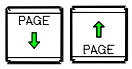
1 Move the cursor to the beginning of the character string to select.
2 Press soft key [SELECT].
3 Using the cursor keys  and the page keys  , move the cursor to the end of the character string to select.
At this time, the background of the selected character string is displayed in the selection color (cursor color).
4 Press soft key [COPY] to copy the selected character string to the clipboard.
To cancel the copying, press soft key [CANCEL].

10.10.8 Cut

A selected character string can be deleted.
When deleted, the character string is copied to the clipboard.

Cut

Procedure

1 Move the cursor to the beginning of the character string to select.
2 Press soft key [SELECT].
3 Using the cursor keys  and the page keys  , move the cursor to the end of the character string to select.
At this time, the background of the selected character string is displayed in the selection color (cursor color).
4 Press soft key [CUT] to delete the selected character string.
To cancel the deletion, pres soft key [CANCEL].

10.10.9 Paste

The character string stored in the clipboard can be inserted at the current cursor position.
After pasted, the character string remains stored in the clipboard, so that the character string can be pasted as many times as necessary.

Paste

Procedure

1 Move the cursor to the position at which you want to paste a character string.
2 Press the soft key [PASTE].

10.10.10 Saving

Moving the cursor to a line preceding or succeeding the edited line causes the edit to be automatically saved, but it is possible to save the edit by pressing soft key [SAVE], if necessary. For a program consisting of only a single line, in particular, it is necessary to press soft key [SAVE] to save the edit.

Saving

Procedure

1 Press the soft key [SAVE].

10.10.11 Creation

A new program can be created.

Creation

Procedure

1 Press soft key [NEW PROGRM].
2 A program name input area appears.
3 Enter the name of a program to be created.
4 Press the soft key [EXECUTE]. This creates a new program and displays the editing screen.
To cancel the creation of a new program, press soft key [CANCEL].

10.10.12 Line Search

The cursor can be moved to a specified line.
The cursor can be moved to a line with a specified line number, to the first line of the program, and to the last line of the program.

Line search

Procedure

Movement to a line with a specified line number

1 Press soft key [LINE SEARCH].
2 A line number input area appears.
3 Enter a line number. The first line is regarded as having a line number of 1.
4 Press soft key [LINE NUMBER], and the cursor moves to the specified line.

Movement to the first line

1 Press soft key [LINE SEARCH].
2 Press soft key [TOP], and the cursor moves to the first line of the program.

Movement to the last line

1 Press soft key [LINE SEARCH].
2 Press soft key [BOTTOM], and the cursor moves to the last line of the program.

10.11 PROGRAM COPY FUNCTION

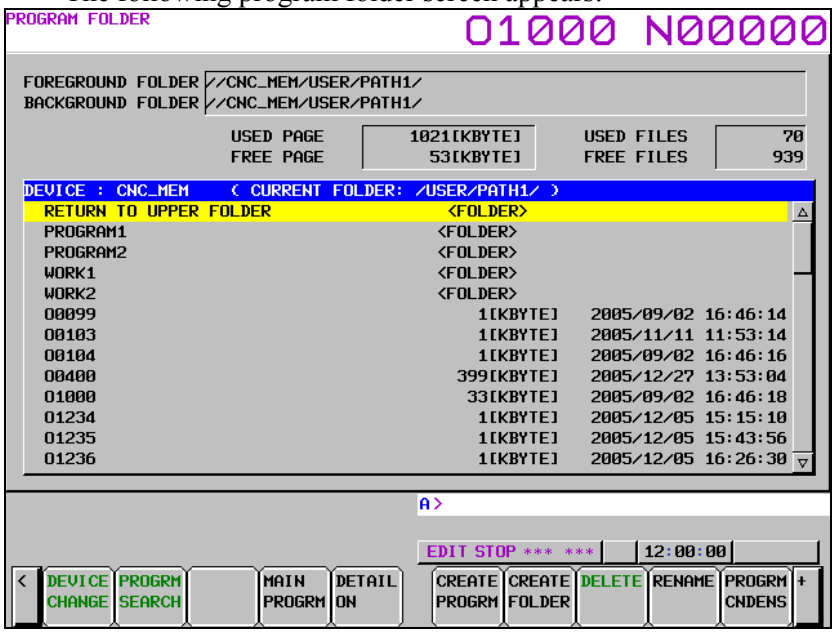
A program is copied or moved between folders.

Procedure for copying a program compact

Procedure

1 Press function key PROG.

2 Press chapter selection soft key [FOLDER].
The following program folder screen appears:



3 Press the soft key [(OPRT)].

4 Move the cursor to a folder that contains the program that you want to copy or move, and press soft key [CHANGE FOLDER].

5 Press the continuous menu key ▷ until soft key [SELECT START] appears.
Press the soft key [SELECT START].

6 Move the cursor to the program that you want to select, and press soft key [SELECT]. The background color of the selected program becomes light blue, indicating the selected state.

7 Press the soft key [SELECT END]. The selected program is determined.

8 Press soft key [CHANGE FOLDER], and move the cursor to the folder to which you want to copy or move the program.

9 Press the soft key [COPY]. The selected program is copied. If just one program is selected, pressing soft key [COPY] after typing a program name performs a copy operation with the entered name.

10 To move the program, press soft key [MOVE]. The selected file is then moved. If just one program is selected, pressing soft key [MOVE] after typing a program name performs a move operation with the entered name.

Explanation

Operations are accepted only when the data protection key is set to ON.

If the program storage capacity on the copy destination side is insufficient, the copy operation is not accepted.

The currently selected program is highlighted. Multiple programs can be selected in the same folder. Each time [SELECT] is pressed, the program currently indicated by the cursor is selected.

A selected program can be deselected by pressing [SELECT] again or by pressing [CANCEL].

If the same program name is already present in the copy or move destination folder, "REWRITE: FILE NAME" is displayed, allowing you to determine whether to overwrite the existing program by pressing soft key [NO] or [REWRITE]. Pressing [REWRITE] performs overwriting, and pressing [NO] does not perform the copy or move operation of that program.

Pressing soft key [DELETE] deletes all the files currently selected.

A program can be neither copied nor moved to the same folder as the selected folder.
When only one program is selected, and a program name is already entered, however, the program can be copied or moved within the same folder.

NOTE
Once a copy or move operation starts, it cannot be canceled. So, start the operation very carefully.

10.11.1 Copying and Moving Files between Devices

Overview

Files can be copied and moved between different devices.

There are two procedures available on the program folder screen.

<1> File copying and moving (device change)
Select a source file first, and then select a destination folder by first changing the device and then selecting a folder, decide on the destination folder, and execute copy or move.

<2> File copying and moving with split display
On 10.4/15-inch LCDs, the program folder screen can be split. By using this screen split function, it is possible to perform copy or move operations while checking the file to be copied or moved on the screen on which the folder containing the source file and the destination folder are displayed at the same time.

File copying and moving (device change)

Procedure

1 Select EDIT mode.
2 Press function key

to display the program folder screen.
3 Press soft key [(OPRT)].
4 Press the continuous menu key until [SELECT START] appears. Press soft key [SELECT START].
5 Use the cursor keys as appropriate to position the cursor on the desired file.
6 Press soft key [SELECT] to select the file.
7 After the desired file has been selected, press soft key [SELECT END].
8 To select a destination device, press soft key [DEVICE CNAHGE].
9 Select the desired device, using the soft key.
10 In the selected device, move to a destination folder.
11 To copy the selected file to the destination folder, press soft key [COPY], and to move the file to the folder, press soft key [MOVE].

File copying and moving with split display

Procedure 1

- Procedure for selecting a file first and then a destination folder

1 Select EDIT mode.
2 Press function key PROG to display the program folder screen.
3 Press soft key [(OPRT)].
4 Press the continuous menu key ▷ until [MULTI LIST] appears. Press soft key [MULTI LIST].
5 In either of the folder displays, select the folder containing the source file.
6 Press soft key [SELECT START].
7 Use the cursor keys as appropriate to position the cursor on the desired file.
8 Press soft key [SELECT] to select the file.
9 Use the SHIFT + cursor key combination, select the other folder display.
10 To select a destination device, press soft key [DEVICE CNAHGE].
11 Select the desired device, using the soft key.
12 In the selected device, move to a destination folder.
13 To copy the selected file to the destination folder, press soft key [COPY], and to move the file to the folder, press soft key [MOVE].

Procedure 2

- Procedure for selecting a destination folder first and then a file

1 Select EDIT mode.
2 Press function key PROG to display the program folder screen.
3 Press soft key [(OPRT)].
4 Press the continuous menu key ▷ until [MULTI LIST] appears. Press soft key［MULTI LIST］.
5 In either of the folder displays, select the folder containing the source file, use the SHIFT + cursor key combination to switch to the other folder display, and on this folder display, select a destination folder.
6 Use the SHIFT + cursor key combination to switch to the folder display in which the source folder has been selected.
7 Press the continuous menu key ▷ until [SELECT START] appears.
Press soft key [SELECT START].
8 Use the cursor keys as appropriate to position the cursor on the desired file.
9 Press soft key [SELECT] to select the file.
10 After the file has been selected, press soft key [COPY] to copy the selected file to the folder selected in the other folder display and press soft key [MOVE] to move the file.

Explanation

- Multiple selectable files

Only multiple files that are in the same folder can be selected at the same time with soft key [SELECT]. Files in other devices and folders cannot be selected at the same time.
Up to ten files can be selected at the same time.

- Changing the file name during copying and movement

If only one file is selected for copying or moving, the file can be copied or moved by changing its file name.
After selecting a file, enter a file name and press soft key [COPY] or [MOVE]. This causes the file to be copied or moved with the entered file name.

- Copying and moving a file to the same folder

It is not possible to copy and move a selected file to the folder that contains the file.
If an attempt is made to do so, warning "SAMEFILE NOTCOPY" is issued.
If, however, only one file is selected, and is to be copied or moved by changing its file name, it can be copied or moved to the same folder.

- Overwriting files

When copying or moving is attempted, if a file with the same name exists in the destination folder, message "REWRITE: File name" flashes below the key-in buffer. If the file with the same name is to be overwritten, press soft key [REWRITE] and otherwise, press soft key [NO].

- File search when selecting a file

After soft key [SELECT START] is pressed, it is possible to search for a file through a folder using soft key [SEARCH] when selecting the file to copy or move.
Enter the file name to search for first and then press soft key [SEARCH]. If a file with the entered file name exists, the cursor is positioned on the file. If no file with the entered file name exists, warning "SPECIFIED PROGRAM NOT FOUND" appears.
Soft key [SEARCH] simply moves the cursor to the program found on the directory screen. It is not possible to switch operations, using bit 5 (PES) of parameter No. 11302.

- Display of a selected file name

When a file is selected with soft key [SELECT], the color of the background of the file name display is changed. The color of the background of the file name display remains unchanged until the file is deselected.

- Deselecting a file

When a file is selected with soft key [SELECT], the selected file can be deselected by positioning the cursor on the file and pressing soft key [SELECT] again. All selected files are deselected when any of the following operations is performed.

- Pressing of soft key [CANCEL] after performing a select operation.
- Completion of a move operation.
- Change of the source folder during selection.
- Termination of a select operation with soft key [SELECT END] and pressing of soft key [SELECT START] to enter the start of selection state.

NOTE
When bit 6 (RTC) of parameter No. 11318 is 1, all selected files are also deselected upon completion of a copy operation.

Limitation

- Mode for performing copy and move operations

To copy and move files between different devices or data servers, enter EDIT mode. In a multipath system, place all paths in EDIT mode. If an attempt to perform an operation in a mode other than EDIT mode, warning "CHANGE TO EDIT MODE" is issued.

- Restrictions on file operations in various devices

Those devices between which file copy and move operations can be performed are as follows:

- CNC MEM
- MEM CARD (binary format)
  (Note that a folder on a MEM CARD can be selected as a source folder from which a file is to be read.)
- Data server

Between devices other than those above, copy or move operations cannot be performed.
It is not possible to copy and move to devices other than those above or to a MEM CARD or to move from a MEM CARD.
If an attempt is made to copy or move, warning "CAN NOT COPY/MOVE" is issued.

- Folder copy and move operations

Folder copy or move operations cannot be performed.
If an attempt is made to select a folder as an object to copy or move, warning "DISABLE TO SELECT" is issued.

Note

NOTE
Once a copy or move operation starts, it cannot be canceled. Thus, check the files and folder subject to the operation carefully before starting it.

10.12 KEYS AND PROGRAM ENCRYPTION

Overview

Program contents can be protected by setting parameters for encryption and for the program security range.

Explanation

1 Security with a password and a security range
When the password and security range parameters are specified, the display, editing, and input/output operations are disabled for the programs within the security range.
This prevents the custom macro programs created by the machine tool builder from being accidentally changed or deleted by the end user. This also provides security because the contents of the programs are not displayed.

2 Input/output of encrypted programs
The programs in the security range can be encrypted before being output. Once encrypted, the programs cannot be decrypted. In addition, encrypted programs can be input directly.

- Locking/unlocking

When the programs in the security range are secured, the program memory is said to be locked.
If they are not secured, the program memory is said to be unlocked.

- Parameter

- Parameter PASSWORD (No.3220)
  Set the password required to lock the program memory. A password other than 0 can be set. The password is not displayed. A password can be set when no password is set (PASSWORD = 0) or the program memory is unlocked.
- Parameter KEY (No.3221)
  When KEY is set equal to PASSWORD, the program memory is unlocked. The value set for the parameter is never displayed. Each time the CNC is started, 0 is set for the parameter. This means that whenever the CNC is started, a lock is provided if a password is set (PASSWORD ≠ 0).
- Minimum value (in parameter No. 3222) and maximum value (in parameter No. 3223) of the program security range
  Set the desired security range. The value set for MINIMUM must not be greater than that set for MAXIMUM. The programs in the range of MINIMUM to MAXIMUM are secured. If MINIMUM is set to 0, it is regarded as being 9000. If MAXIMUM is set to 0, it is regarded as being 9999.
  A range can be set when no password is set or the program memory is unlocked.

NOTE
1 For security, the values set for PASSWORD and KEY are not displayed. For the same reason, PASSWORD, MINIMUM, and MAXIMUM can be specified only when no password is set or the program memory is unlocked. Set a password, taking great care to avoid a situation where the program memory cannot be unlocked because a password is set incorrectly.
2 The [+INPUT] key used to specify PASSWORD and KEY behave in the same way as the [INPUT] soft key.

Example: When 99 is set for KEY, pressing [1] and [+INPUT] changes the value to.

3 These four parameters cannot be output externally. Also, even when these parameters are input by the parameter read operation, they are ignored.

- Inputting/outputting and collating programs

When a program is encrypted, a password is output. The password is used to load the program.
The following output operations are enabled for programs outside the security range if the locked state is present or for programs within the security range if the unlocked state is present:

Outputting all programs

Locked/unlocked	Results
Locked	All the programs outside the security range are output in the normal way.
Unlocked	All the programs within the security range are encrypted and output.
Password not set	All the programs in the program memory are output in the normal way.

Outputting a single program

Locked/unlocked	Results
Locked	If a program is outside the security range, it is output in the normal way. If it is within the security range, a warning "PROTECTED" is issued.
Unlocked	If a program is outside the security range, it is output in the normal way. If it is within the security range, it is encrypted and output.
Password not set	The program is output in a normal way.

Outputting specified multiple programs

Locked/unlocked	Results
Locked	When all of the specified programs fall outside the protected range, they are output as usual. When all of the specified programs are within the security range, warning message "PROTECTED" is issued. When some of the specified program numbers are outside the security range, and the others are within the security range, only the programs outside the security range are output normally. When the programs to be output are not within the specified range, warning message "THE WRONG DATA IS USED" is issued.
Unlocked	When all of the specified programs fall outside the protected range, they are output as usual. When all the specified programs fall within the protected range, they are output as encoded programs. When some of the specified program numbers are outside the security range, and the others are within the security range, only the programs within the security range are encoded and output. When the programs to be output are not within the specified range, warning message "THE WRONG DATA IS USED" is issued.
Password not set	The program is output in a normal way.

Inputting an un-encrypted program

Locked/unlocked	Results
Locked	When the program to be read is outside the security range, it is input normally. When the program to be read is within the security range, warning message "PROTECTED" is issued.
Unlocked, or password not set	The program is input.

Inputting an encrypted program

Password set in the system and password of the program	Results
Password set in the system	Warning “WRITE PROTECT” is issued.
Password for the program	When the program is within the security range, it is input normally. When the program is outside the security range, warning message "WRITE PROTECTED" is issued.

Password set in the system and password of the program	Results
Password not set in the system	The program is input. The PSW in the file is set for parameter No. 3220. This applies if the program is in the security range. If the program is outside the security range, warning message "PROTECTED" is issued.

Collating a program with an encrypted program

In the unlocked state, the following takes place:

Password set in the system and password of the program	Results
Password set in the system	Alarm SR0075 “PROTECT” is issued.
Password set in the system = Password of the program, or password not set in the system	The program is collated.

In the locked state, program collation cannot be performed.

NOTE
To encrypt programs, set parameter ISO (bit 1 of No.0000) to 1 (to specify that the punch code is ISO).

- Program display

1 On the program folder screen, all program numbers are displayed together with comments.
2 In the locked state, the programs within the security range are not displayed on the program screen. In the unlocked state, the programs within the security range are displayed in the same way as normal programs.

NOTE
Immediately after switched from the unlocked state to the locked state or from the locked state to the unlocked state, programs are displayed in the previous state. They are displayed in the new state when a program search or other operation is performed.

- Editing and deleting programs

When the program memory is locked, the programs within the security range cannot be edited or deleted. When the program memory is locked, an attempt to delete all programs results in only those programs outside the security range being deleted.
In the locked state, the programs in the security range cannot be selected as a main program. If an attempt is made to select such a program on the program folder screen as a main program, warning message "SETTING A MAIN PROGRAM FAILED" is issued.

- Searching for programs

In the locked state, a program search is performed within the protected range as described below.

1 When no program number is specified, programs within the protected range are skipped.
2 When an attempt is made to search for a program within the security range by specifying a program number, the search operation is ignored, resulting in warning message "Program is protected".

10.13 SIMULTANEOUS EDITING OF MULTIPATH PROGRAMS

Simultaneous editing of multipath programs enables simultaneous editing of programs for multiple paths on a single screen.
This function is enabled when bit 0 (DHD) of parameter No. 3106 is 1.

Explanation

- Procedure

1 Set the paths subject to simultaneous editing, with multipath simultaneous display parameters Nos. 13131 and 13132.
(For details of the parameters, refer to the Parameter Manual (B-63950EN).)
2 Place the paths subject to simultaneous editing in EDIT mode.
3 Press function key

.
4 Press soft key [PROGRAM] to display the program editing screen.

- Screen display

Figure 10.13 (a) to (c) show examples of performing simultaneous editing of multipath programs.
Above each program is a status line on which three items of information are displayed: the program name, "FG-EDIT", which indicates that the program is being edited in the foreground, and the path name.
For the program currently being edited, its status line is displayed in reverse video.

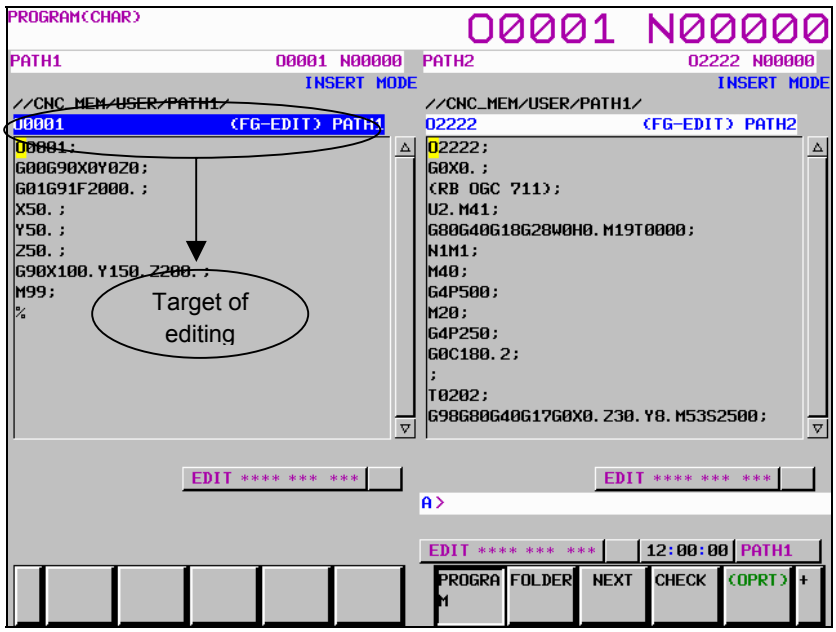


Figure 10.13 (a) Simultaneous editing of multipath programs screen (10.4-inch LCD)

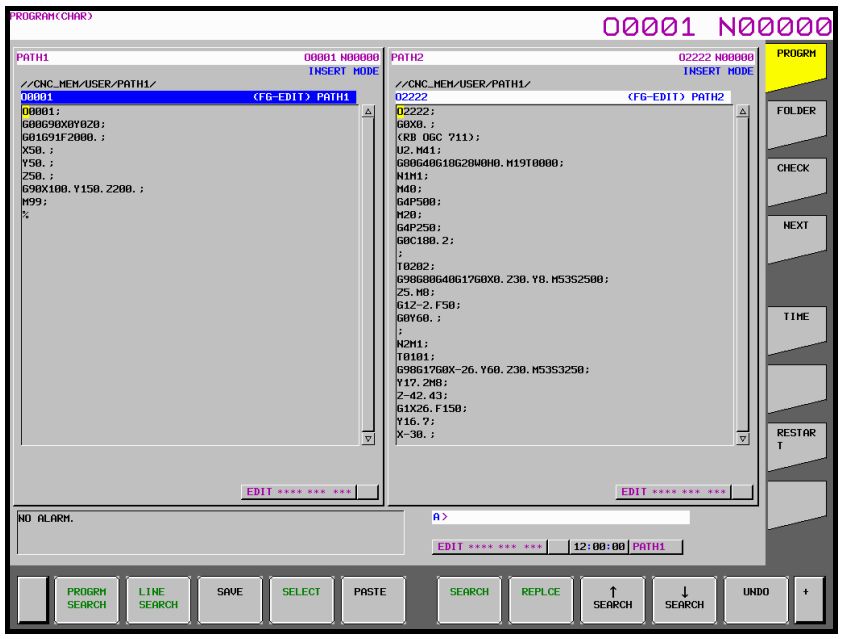


Figure 10.13 (b) Simultaneous editing of multipath programs screen (15-inch LCD)

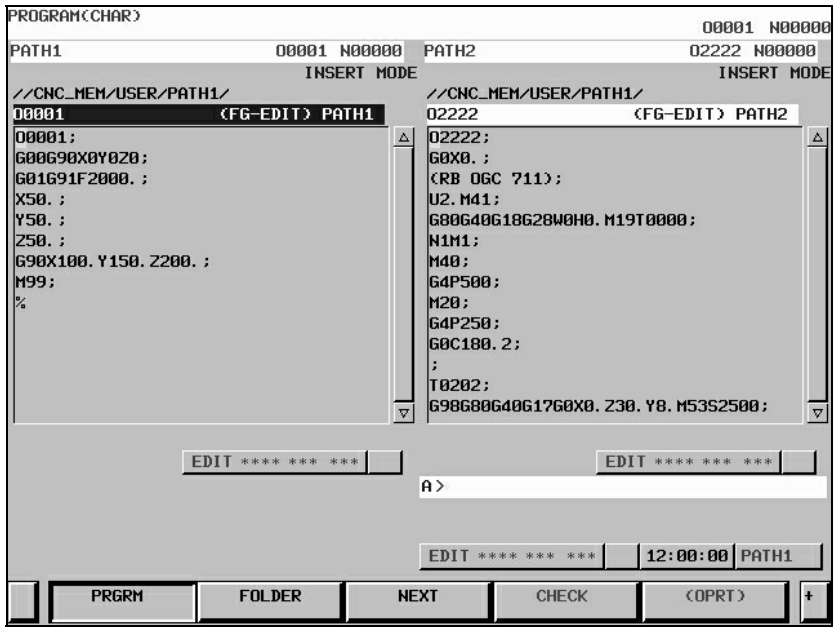


Figure 10.13 (c) Simultaneous editing of multipath programs screen (7.2-inch LCD)

- Modes

When the paths to be displayed simultaneously are in either EDIT or MEM mode, the multipath programs are displayed simultaneously on the program screen. Selecting the path for which EDIT mode is selected enables program editing to be performed.
Figure 10.13 (d) shows an example in which both EDIT and MEM modes are selected.
For path 1, MEM mode (execution status screen) is selected, and for path 2, EDIT mode (editing screen) is selected.

Soft keys are switched according to the mode for the currently selected path.

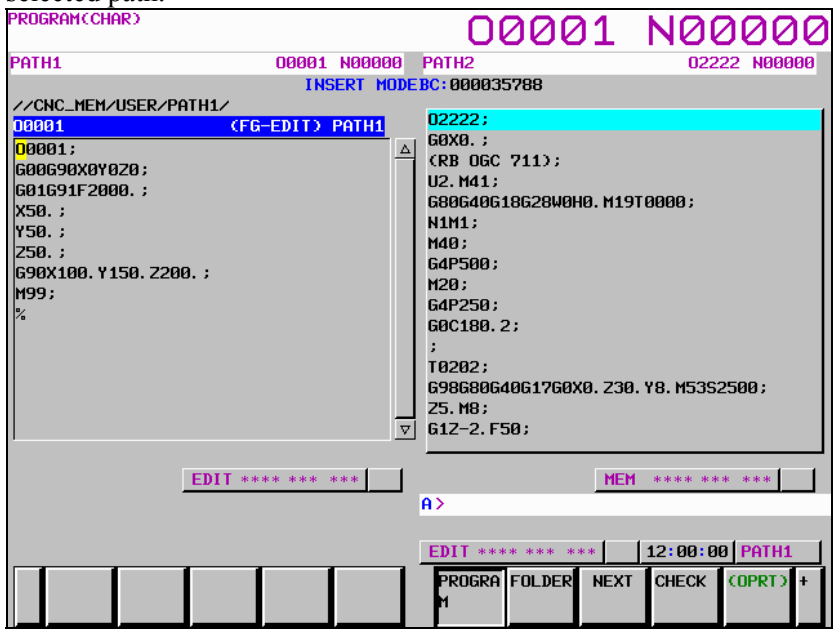


Figure 10.13 (d) Screen on which both MEM and EDIT modes are selected

- Switching the path subject to editing

The path selected with the path selection signal is subject to editing.

- Maximum number of paths that can be subject to editing simultaneously

The maximum number of paths that can be subject to editing simultaneously on each LCD is as given in the table below.

LCD size	Maximum number of paths that can be subject to editing simultaneously
7.2, 8.4, or 10.4-inch	3
15-inch	4

- Conditions under which simultaneous editing is not usable

Simultaneous editing of multipath programs is disabled in the following cases:

- The program screen is selected as a small screen.
- The paths to be displayed simultaneously include a path for which a mode other than EDIT and MEM modes is selected.
- The virtual MDI key function is enabled.

If, in the state of simultaneous editing of multipath programs, background editing is started, background editing is performed on all screens. Simultaneous editing of multipath programs and background editing cannot be used at the same time.

- Simultaneous editing on 7.2/8.4-inch LCDs

When simultaneous editing is performed on 7.2/8.4-inch LCDs, the characters get smaller.

The number of characters per path in the editing area is as follows:

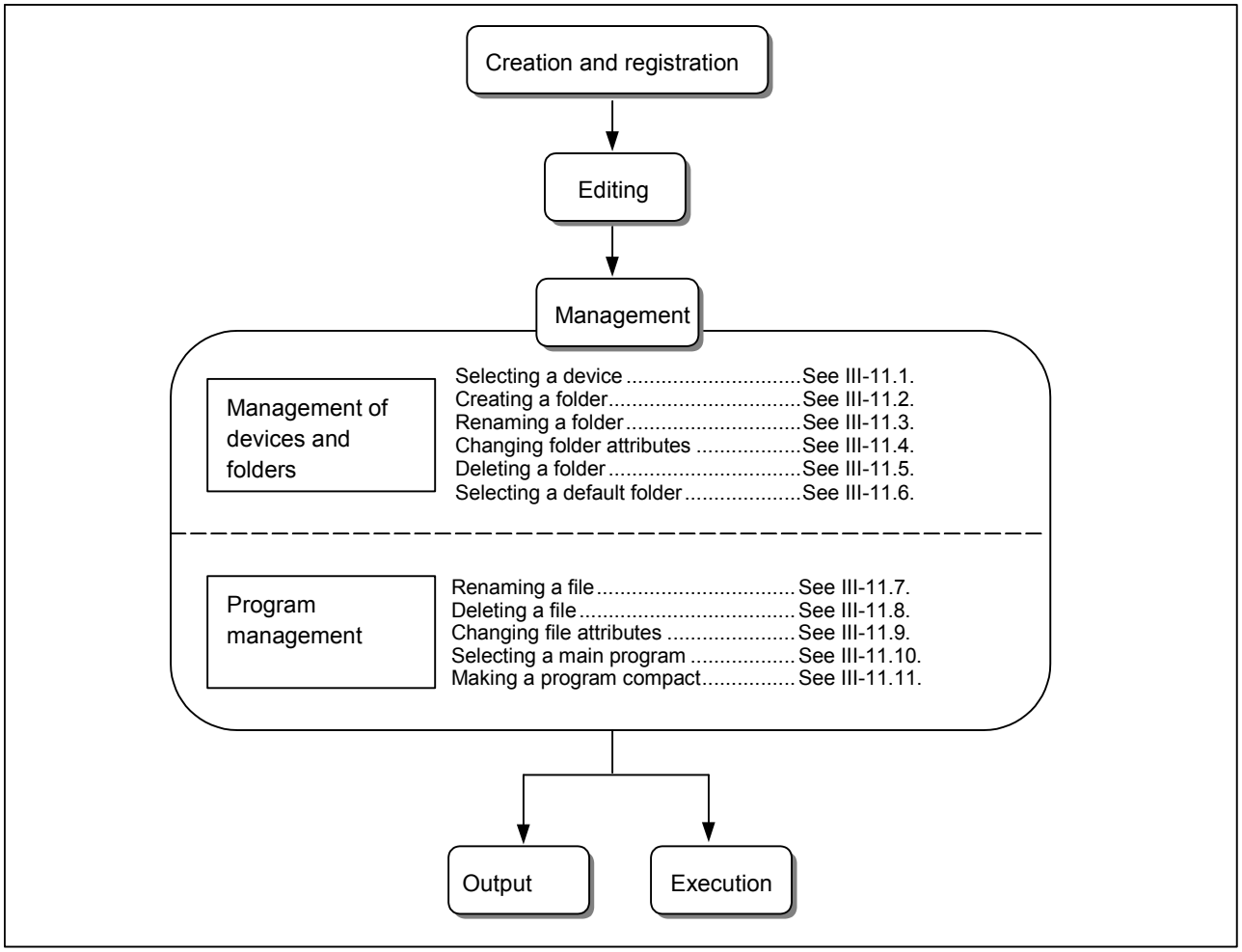
- Display of 38 columns and 10 rows for non-simultaneous display
- Display of 74 columns and 15 rows for 1-path simultaneous display
- Display of 35 columns and 14 rows for 2-path simultaneous display
- Display of 22 columns and 14 rows for 3-path simultaneous display

11 PROGRAM MANAGEMENT

Program management functions are classified into the following two types:

- Functions for folders
- Functions for programs

Functions for folders include creation, deletion, change of names and attributes, and so on.
Functions for programs include main program selection, deletion, change of names and attributes, program compaction, and so on.



11.1 SELECTING A DEVICE

When the fast data server function (option) is provided, a program storage device can be selected. This section explains the selection procedure.

Procedure for selecting a device

1 Press the function key

.
2 Press the soft key [FOLDER].
3 Press the soft key [(OPRT)].
4 Press the soft key [DEVICECHANGE].
5 Press the soft key for the desired device.

11.1.1 Selecting a Memory Card Program as a Device

Overview

By selecting a memory card including a program storage file (named "FANUCPRG.BIN") as a device, memory operation can be performed with the program in the program storage file selected as the main program.
In addition, the content of a program storage file can be displayed on the program list screen or a program in a program storage file can be edited on the program edit screen.
A program storage file can be created using a memory card program tool (A08B-9010-J700#ZZ11) on a commercially available personal computer. To be used, the program storage file must be written to a memory card formatted in FAT16 format.
(A program held in a program storage file is hereinafter referred to as a memory card program. Moreover, a memory card storing a program storage file is referred to as a program storage memory card.)

Procedure for selecting a device

1 Press the function key

.
2 Press the soft key [FOLDER].
3 Press the soft key [(OPRT)].
4 Press the soft key [DEVICECHANGE].
5 Press the soft key [MEMCARD].

NOTE
1 A FAT16-formatted memory card containing the program storage file FANUCPRG.BIN is recognized as a program storage memory card.
2 For a program storage memory card containing more than 63 folders and programs, the option for extending the number of memory card program registrations is required. The option for extending the number of memory card program registrations is applicable to a program storage memory card containing a maximum of 1000 folders and programs.

Procedure for removing a device

When a program storage memory card is replaced or a memory card is used for normal usage such as data input/output, clear the recognition of the program storage memory card with removal operation.

1 Press the function key

.
2 Press the soft key [FOLDER].
3 Press the soft key [(OPRT)].
4 Press the soft key [DEVICECHANGE].
5 Press the soft key [DETACH].

NOTE
1 This soft key appears when the CNC recognizes a program storage memory card during device change operation.
2 This operation is enabled only in EDIT mode or MEM mode.
When a memory card program is selected in the main programs of multiple paths in a multipath control system, set the modes of all paths to EDIT mode or MEM mode.
3 When the default folder is a folder in a program storage file, it is changed to "//CNC_MEM/" by a removal operation.
4 When the main program is a memory card program, the main program enters the unselected state by a removal operation.

Explanation

- About operation

A memory card program can be selected as a main program to perform memory operation.
Memory operation has the following features:

- Subprogram call nesting is allowed.
- Macro program call nesting is allowed.
- In a custom macro, a control command using a GOTO statement/WHILE statement can be specified.
- With the T series, a multiple repetitive canned cycle for turning can be specified.

NOTE
To use the functions for calling macros, using custom macros, and using a multiple repetitive canned cycle for turning, the options corresponding to the respective functions are required.

- Selection as a main program

As a main program to be automatically executed in the memory mode, a memory card program can be selected.

- Sub program (call using M98/G72.1/G72.2)
- Macro program (call using G65/G66/G66.1/M96)

The following subprogram/macro program held in the same folder as containing the main program is called:

- Sub program call (M98)
- Macro call (Simple macro call G65 / Modal call G66,G66.1)
- Macro interrupt (M96)
- Figure copying (G72.1, G72.2)

If the program cannot be found in the same folder as the main program, the following folder is searched:

- Common program folder on the CNC_MEM device (CNC program storage memory)

NOTE
To use the functions for macro call, macro interrupt, and figure copying, the options corresponding to the respective functions are required.

- Sub program (call using M code/S code/T code/particular address/the second auxiliary function)
- Macro program (call using G code/M code)

The following subprogram/macro program calls a program from the CNC_MEM device (CNC program storage memory):

- Subprogram call using M code/S code/T code/particular addresses/the second auxiliary function
- Macro call using G code/M code

The folders set as search targets are searched in the following order, and the first found program is called:

1 Common program folder among the initial folders
2 MTB-dedicated folder 2 among the initial folders
3 MTB-dedicated folder 1 among the initial folders
4 System folder among the initial folders

Search target folders are set in parameter No. 3457.

NOTE
For a memory card program, subprogram call using M code/S code/T code/particular addresses/the second auxiliary function or macro call using G code/M code can be specified. However, a program on the CNC_MEM device (CNC program storage memory) is called.

- External program number search / External workpiece number search

A program on a program storage memory card can be searched for with the external program number search function or external workpiece number search function.

Limitation

For a memory card program, M198 cannot be specified. Moreover, no memory card program can be called from a program on the CNC_MEM device (CNC program storage memory) by specifying M198.

When a setting is made to enable an external device subprogram call from a memory card (M198) or DNC operation from a memory card (bit 7 (MNC) of parameter No. 0138 = 1), the content of program storage file cannot be displayed during automatic operation.

When a program storage memory card is selected, the memory card cannot be used for the ordinary purposes listed below. To use a memory card in such a case, perform a "removal" operation to cancel the recognition of the program storage memory card.

- ALL I/O screen
  Display of the contents of a memory card, and reading/punching data to and from a memory card
- PMC data I/O screen
  Display of the contents of a memory card, and reading/punching to and from a memory card
- Program directory screen
  Reading/punching program data to and from a memory card
- External device subprogram call (M198) operation
  Subprogram call (M198) with a memory card set as an external device
- DNC operation
  DNC operation from a memory card

⚠ CAUTION

1 Do not remove the memory card when a program that specifies a write to the memory card is being edited. The data can be destructed.
2 If an editing operation is completed, the results of editing are preserved even when the power to the CNC is turned off.

> ⚠ **CAUTION**
> 3 When removing the memory card, be sure to perform a "removal" operation. If the memory card is removed without performing a "removal" operation and an attempt is made to access the memory card, the alarm (SR1964) or alarm (IO1030) is issued.
> If the card is removed inadvertently, insert the card again and perform a "removal" operation.
> When an alarm is issued, perform the following operation:
> - When the alarm (SR1964) is issued
> Reset the alarm after performing a "removal" operation.
> - When the alarm (IO1030) is issued
> The alarm can be reset only by turning off the power to the CNC.
>
> 4 There are cases in which when a memory card is replaced with another, the CNC cannot detect the replacement. Thus, it is risky to replace a memory card without performing a "removal" operation, and this should never be attempted.

- Operation of creattion, edition, and management of a program

When “memory card program as a device” is selected, operation of creattion, edition, and management of a program is below:

Item	Usable
Creation of a program	Unusable
Edition prohibition attribute	Unusable
Inserting, alteration, and deletion a Word	Usable
Deletion of a block	Usable
Program search	Usable
Sequence number search	Usable
Deletion of a program	Unusable
Editing a customa macro	Usable
Password function	Unusable
Editing a program character	Usable
Program copy function	Unusable
Key and program encryption	Unusable
Selecting a device	Usable
Creation of a folder	Unusable
Renaming of a folder	Unusable
Changing the attribute of a folder	Unusable
Deletion of a folder	Unusable
Selecting a default folder	Usable
Renaming of a file	Unusable
Deletion of a file	Unusable
Changing the attribute of a file	Unusable
Selecting a main program	Usable
Making a program compact	Unusable
Input/output of program	Unusable

11.2 CREATING A FOLDER

This section explains the procedure for creating a folder.

Procedure for creating a folder

1 Select EDIT mode.
2 Press the function key .
3 Move to the folder in which you want to create a folder.
Use the cursor keys and to move among folders.
After selecting the folder, press the key.
4 Press the soft key [(OPRT)].
5 Type folder name.
6 Press the soft key [CREATE FOLDER].

NOTE
1 Each folder name must be unique within the same folder.
2 Each time a folder is created, the number of programs that can be registered decreases by one.
3 Depending on the operation status and protection status, a folder cannot sometimes be created.

11.3 RENAMING A FOLDER

This section explains the procedure for renaming a folder.

Procedure for renaming a folder

1 Select EDIT mode.
2 Press the function key

.
3 Press the soft key [FOLDER].
4 Select the folder that you want to rename.
To select a folder, use the cursor keys and .
5 Press the soft key [(OPRT)].
6 Type new folder name.
7 Press the soft key [RENAME].

NOTE
1 You cannot rename the initial folders.
2 Each folder name must be unique within the same folder.
3 Depending on the operation status and protection status, a folder cannot sometimes be renamed.

11.4 CHANGING FOLDER ATTRIBUTES

This section explains the procedure for changing the attribute of a folder (edit disable or edit/display disable).

Procedure for changing folder attributes

1 Select EDIT mode.
2 Press the function key .
3 Press the soft key [FOLDER].
4 Select the folder of which attribute is to be changed.
To select a folder, use the cursor keys and .
5 Press the soft key [(OPRT)].
6 Press the soft key [DETAILON].
7 Press the soft key [CHANGE ATTR].
- To disable editing, press the soft key [EDIT DISABL].
- To enable editing, press the soft key [EDIT ENABLE].
- To disable editing and display, press the soft key [DISP DISABL].
- To enable editing and display, press the soft key [DISP ENABLE].

NOTE
1 Depending on the operation status and protection status, the attribute of a folder cannot sometimes be changed.
2 When the edit disable attribute is set for a folder, editing of folders and files in that folder is disabled.
3 When the edit/display disable attribute is set for a folder, editing and display of folders and files in that folder is disabled, and these folders and files are not displayed.
4 The items that can be set vary depending on the status of parameters and so on.

11.5 DELETING A FOLDER

This section explains the procedure for deleting a folder.

Procedure for deleting a folder

1 Select EDIT mode.
2 Press the function key .
3 Press the soft key [FOLDER].
4 Select the folder that you want to delete.
To select a folder, use the cursor keys and .
5 Press the soft key [(OPRT)].
6 Press the soft key [DELETE].
- To perform the deletion, press the soft key [EXEC].
- To cancel the deletion, press the soft key [CANCEL].

NOTE
1 You cannot delete the initial folders.
2 A folder cannot be deleted unless the folder is empty.
(An empty folder means that the folder contains neither folders nor files.)
3 If a folder contains a folder or file having the edit/display disable attribute, the folder may seem to be empty when displayed, but the folder is not actually empty, so it cannot be deleted.
4 Depending on the operation status and protection status, a folder cannot sometimes be deleted.

11.6 SELECTING A DEFAULT FOLDER

This section explains the procedure for selecting a foreground or background default folder.

Procedure for selecting a default folder

1 Select EDIT mode.
2 Press the function key .
3 Press the soft key [FOLDER].
4 Move to the folder that you want to select as the default folder.
Use the cursor keys and to move among folders.
After selecting the folder, press the key.
5 Press the soft key [(OPRT)].
6
- To select the foreground, press the soft key [FORE CHANGE].
- To select the background, press the soft key [BACK CHANGE].

NOTE
1 When the foreground or background default folder is not set, the path folder, which is an initial folder, is assumed.
2 The settings for the foreground and background default folders are stored in the default folder setting file.
3 When a program file, program folder, or program folder management file is cleared, the default folder setting file is also cleared at the same time.

11.7 RENAMING A FILE

This section explains the procedure for renaming a file.

Procedure for renaming a file

1 Select EDIT mode.
2 Press the function key .
3 Press the soft key [FOLDER].
4 Move to the folder containing the file that you want to rename.
Use the cursor keys and to move among folders.
After selecting the folder, press the key.
5 Select the file that you want to rename.
To select a file, use the cursor keys and .
6 Press the soft key [(OPRT)].
7 Type new file name.
8 Press the soft key [RENAME].

NOTE
1 Each file name must be unique within the same file.
2 When the assigned file name cannot be treated as a program number, the program in the file is restricted as follows:
- Specification by program number is impossible. (Such as a subprogram call)
- Information output by program number is impossible.

3 Depending on the operation status and protection status, a file cannot sometimes be renamed.

11.8 DELETING A FILE

This section explains the procedure for deleting a file.

Procedure for deleting a file

1 Select EDIT mode.
2 Press the function key .
3 Press the soft key [FOLDER].
4 Move to the folder containing the file that you want to delete.
Use the cursor keys and to move among folders.
After selecting the folder, press the key.
5 Select the file that you want to delete.
To select a file, use the cursor keys and .
6 Press the soft key [(OPRT)].
7 Press the soft key [DELETE].
- To perform the deletion, press the soft key [EXEC].
- To cancel the deletion, press the soft key [CANCEL].

NOTE
Depending on the operation status and protection status, a file cannot sometimes be deleted.

11.9 CHANGING FILE ATTRIBUTES

This section explains the procedure for changing the attribute of a file (edit disable, edit/display disable, encoding, or protection of data at eight levels).

Procedure for selecting the attribute of a file

1 Select EDIT mode.
2 Press the function key

.
3 Press the soft key [FOLDER].
4 Move to the folder containing the file of which attribute is to be changed.
Use the cursor keys and to move among folders.
After selecting the folder, press the

key.
5 Select the file of which attribute is to be changed.
To select a file, use the cursor keys and .
6 Press the soft key [(OPRT)].
7 Press the soft key [DETAILON].
8 Press the continuous menu key until the desired soft key appears.
9
- To disable editing, press the soft key [EDIT DISABL].
- To enable editing, press the soft key [EDIT ENABLE].
- To disable editing and display, press the soft key [DISP DISABL].
- To enable editing/display, press the soft key [DISP ENABLE].
- To set encoding, press the soft key [ENCODESET].
- To cancel encoding, press the soft key [ENCODE RESET].
- To change the change protection level, type a change protection level, then press the soft key [CHANGELEVEL].
- To change the output protection level, type an output protection level, then press soft key [OUT LEVEL].

NOTE
1 Depending on the operation status and protection status, a file cannot sometimes be deleted.
2 The items that can be set vary depending on the status of options, parameters, and so on.

11.10 SELECTING A MAIN PROGRAM

This section explains the procedure for selecting a main program.

Procedure for selecting a main program

1 Select EDIT mode.
2 Press the function key PROG.
3 Press the soft key [FOLDER].
4 Move to the folder containing the file that you want to use as a main program.
Use the cursor keys ↓ and ↑ to move among folders.
After selecting the folder, press the INPUT key.
5 Select the file that you want to use as the main program.
To select a file, use the cursor keys ↓ and ↑.
6 Press the soft key [(OPRT)].
7 Press the soft key [MAIN PROGRM].

NOTE
Depending on the operation status and protection status, the main program cannot sometimes be selected.

11.11 MAKING A PROGRAM COMPACT

This section explains the procedure for making a program compact.

Procedure for making a program compact

1 Select EDIT mode.
2 Press the function key .
3 Press the soft key [FOLDER].
4 Move to the folder containing the file of the program that you want to make compact.
Use the cursor keys and to move among folders.
After selecting the folder, press the key.
5 Select the file of the program that you want to make compact.
To select a file, use the cursor keys and .
6 Press the soft key [(OPRT)].
7 Press the soft key [PROGRMCNDENS].

NOTE
1 Depending on the operation status and protection status, a program cannot sometimes be made compact.
2 nly programs on the CNC_MEM device can be made compact.

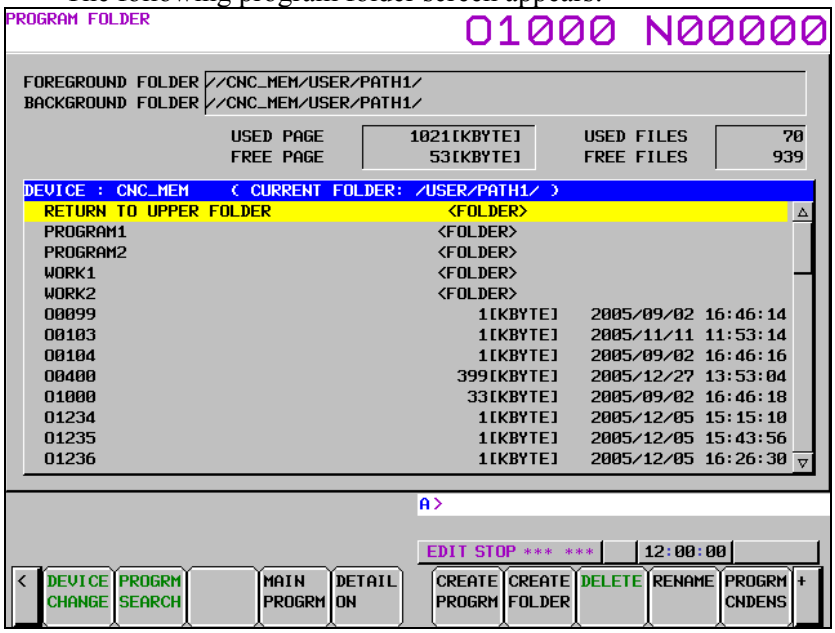
11.12 PROGRAM COPY FUNCTION

A program is copied or moved from folder to folder.

Procedure for copying a program compact

1 Press function key PROG.

2 Press chapter selection soft key [FOLDER].
The following program folder screen appears:



3 Press the soft key [(OPRT)].

4 Display the folder from which the program to be copied or moved. Move the cursor to the folder (indicated by <FOLDER>) on the screen and press INPUT to move to the folder.

5 Press the continuous menu key ▷ until soft key [SELECT START] appears.
Press the soft key [SELECT START].

6 Move the cursor to the program that you want to select, and press soft key [SELECT]. The background color of the selected program changes to blue to indicate that the program is selected. To select multiple programs, repeat this procedure.

7 Press the soft key [SELECT END]. The selected program is determined.

8 Display the folder to which the program is to be copied or moved. Move the cursor to the folder (indicated by <FOLDER>) on the screen and press INPUT to move to the folder.

9 Press soft key [COPY] in the folder to which the selected program is copied to copy the program. If just one program is selected, pressing soft key [COPY] after typing a program name performs a copy operation with the entered name.
10 To move the program, press soft key [MOVE]. The selected file is then moved. If just one program is selected, pressing soft key [MOVE] after typing a program name performs a move operation with the entered name.

Explanation

Operations are accepted only when the data protection key is set to ON.

If the program storage capacity on the copy destination side is insufficient, the copy operation is not accepted.

A selected program can be deselected by pressing [SELECT] again or by pressing [CANCEL].

If a program with the same name is present in the copy or move destination, soft keys [NO] and [REWRITE] are displayed together with the "REWRITE: FILE NAME" message to ask whether to overwrite the existing program. Pressing [REWRITE] performs overwriting, and pressing [NO] does not perform the copy or move operation of that program.

Pressing soft key [DELETE] deletes all the files currently selected.

A program can be neither copied nor moved to the same folder as the selected folder.
However, when only one program is selected, if the name of the program is entered, the program is copied or moved with the specified name to the same folder.

NOTE
Once a copy or move operation starts, it cannot be canceled. So, start the operation very carefully.

12 SETTING AND DISPLAYING DATA

To operate a CNC machine tool, various data must be set on the MDI panel for the CNC. The operator can monitor the state of operation with data displayed during operation.
This chapter describes how to display and set data for each function.

Chapter 12, "SETTING AND DISPLAYING DATA", consists of the following sections:

12.1 SCREENS DISPLAYED BY FUNCTION KEY 1429
12.2 SCREENS DISPLAYED BY FUNCTION KEY 1465
12.3 SCREENS DISPLAYED BY FUNCTION KEY 1571
12.4 SCREENS DISPLAYED BY FUNCTION KEY 1721
12.5 SCREENS DISPLAYED BY FUNCTION KEY 1881
12.6 DISPLAYING THE PROGRAM NUMBER/NAME, SEQUENCE NUMBER, AND STATUS, AND WARNING MESSAGES FOR DATA SETTING OR INPUT/OUTPUT OPERATION.....1882
12.7 SCREEN ERASURE FUNCTION AND AUTOMATIC SCREEN ERASURE FUNCTION.....1892
12.8 LOAD METER SCREEN 1894

Explanation

- Screen transition chart

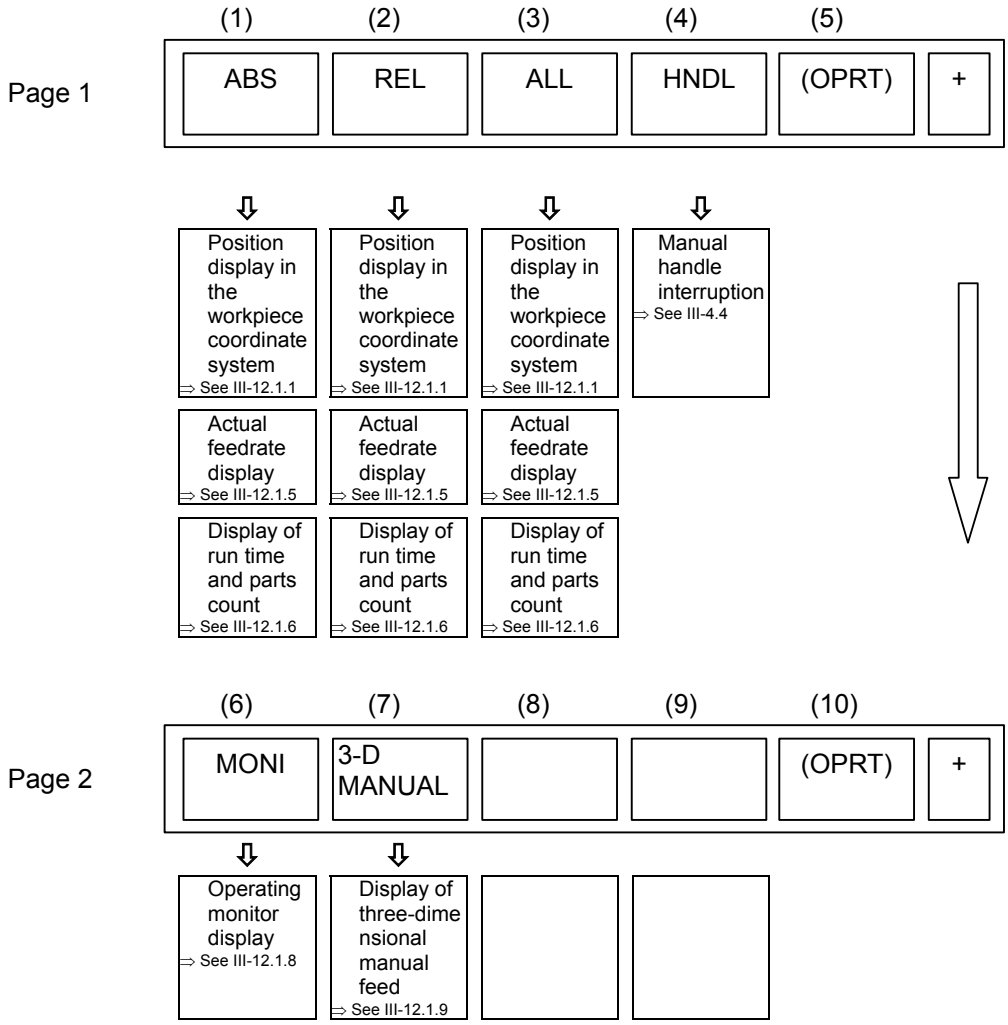
The screen transition for when each function key on the MDI panel is pressed is shown below. The subsections referenced for each screen are also shown. See the appropriate subsection for details of each screen and the setting procedure on the screen. See other chapters for screens not described in this chapter.

See Chapter 7 for the screen that appears when function key is pressed. In general, function key or is prepared by the machine tool builder and used for macros. Refer to the manual issued by the machine tool builder for the screen that appears when function key or is pressed.

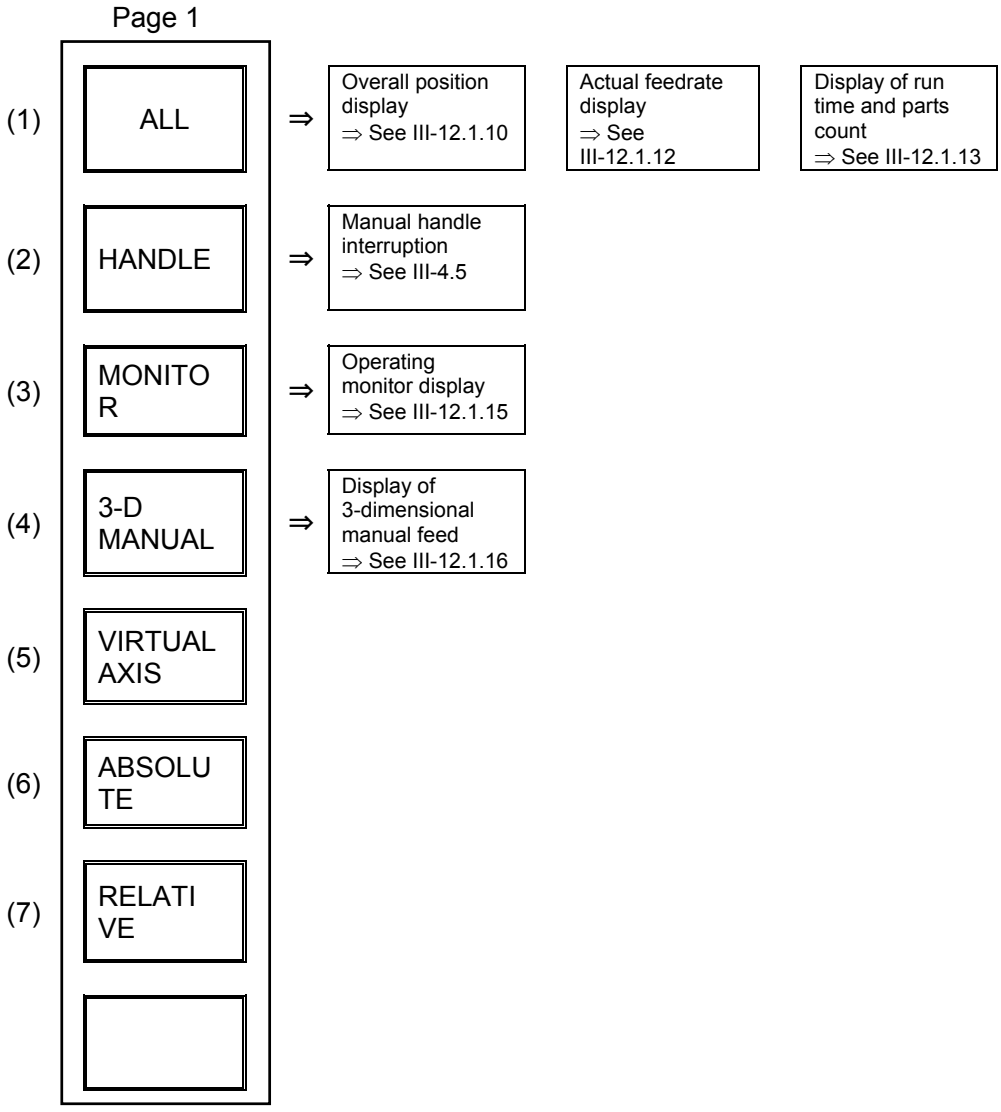
- Data protection key

The machine may have a data protection key to protect part programs, tool compensation values, setting data, and custom macro variables. Refer to the manual issued by the machine tool builder for where the data protection key is located and how to use it.

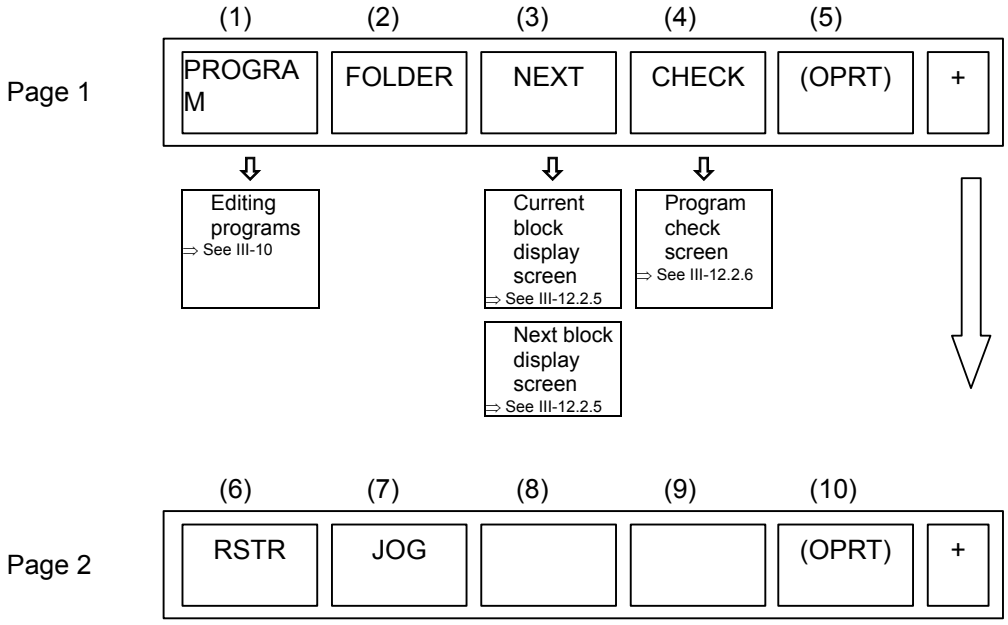
Screen displayed when the function key POS is pressed (for 7.2/8.4/10.4-inch display unit)



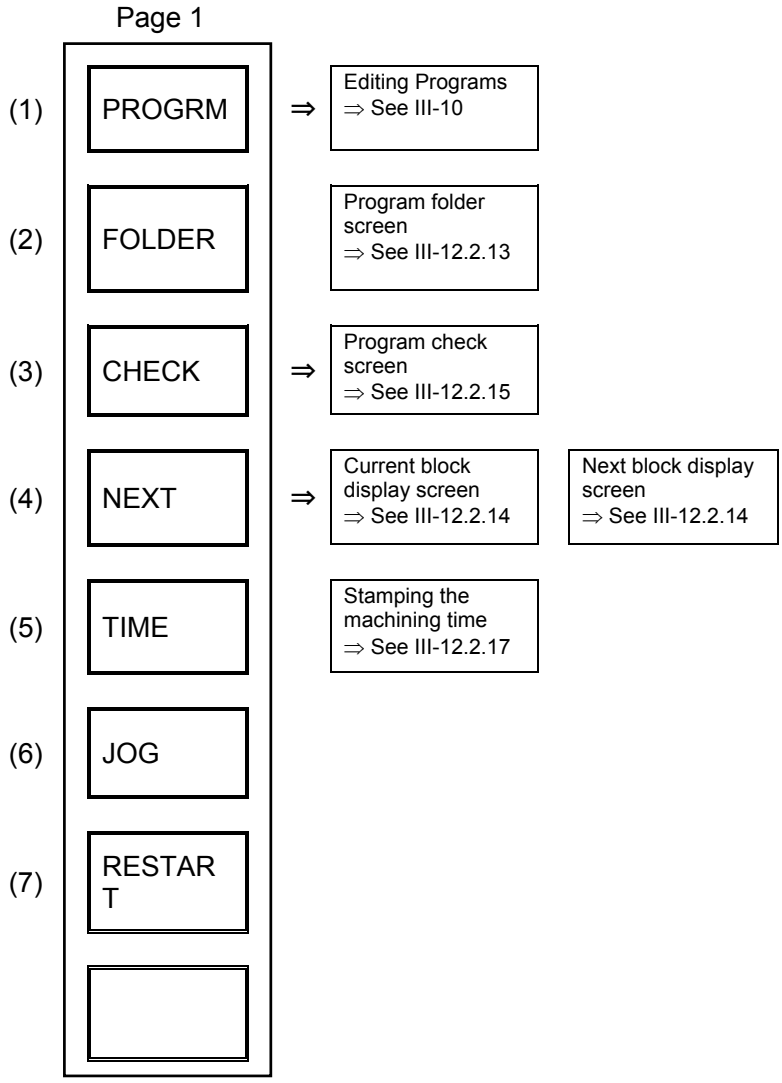
Screen displayed when the function key POS is pressed (for 15-inch display unit)



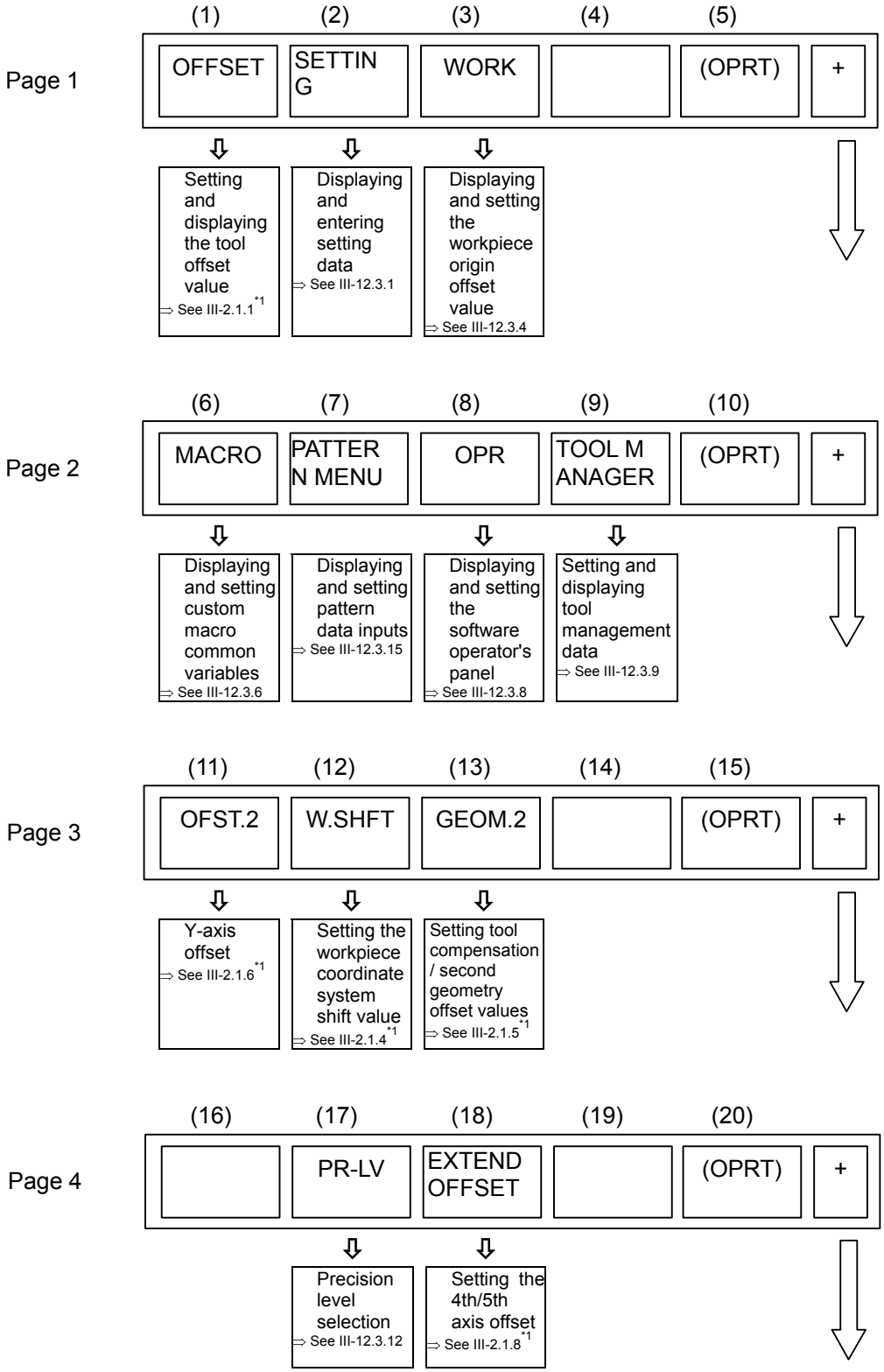
Screen displayed when the function key PROG is pressed (for 7.2/8.4/10.4-inch display unit)



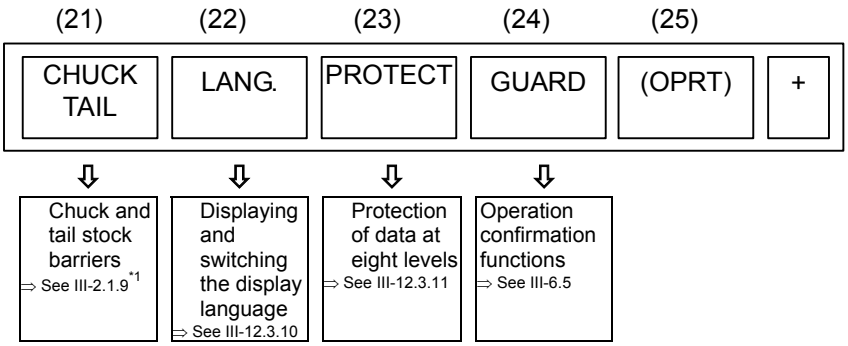
Screen displayed when the function key PROG is pressed (for 15-inch display unit)



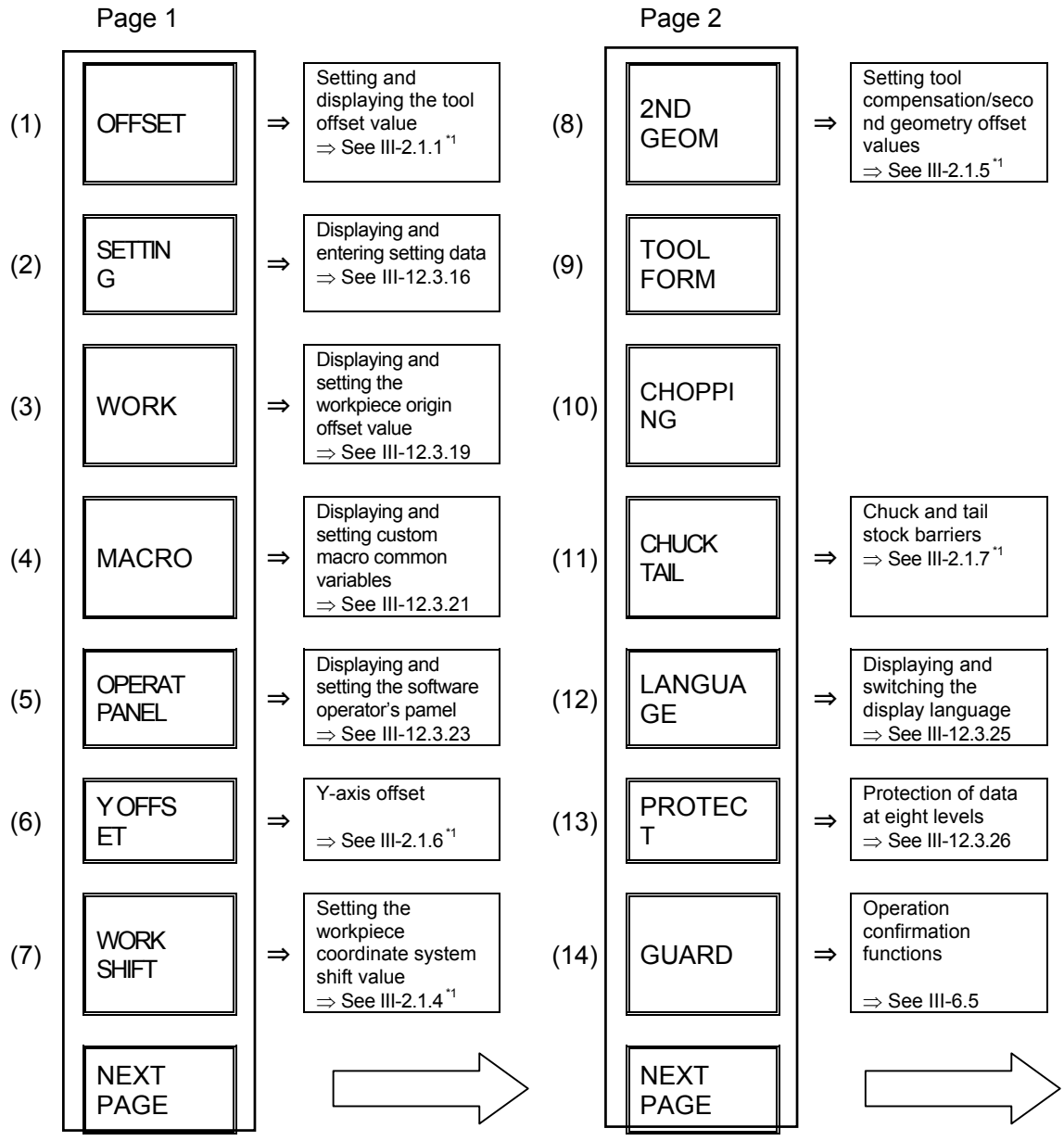
Screen displayed when the function key is pressed (for 7.2/8.4/10.4-inch display unit)

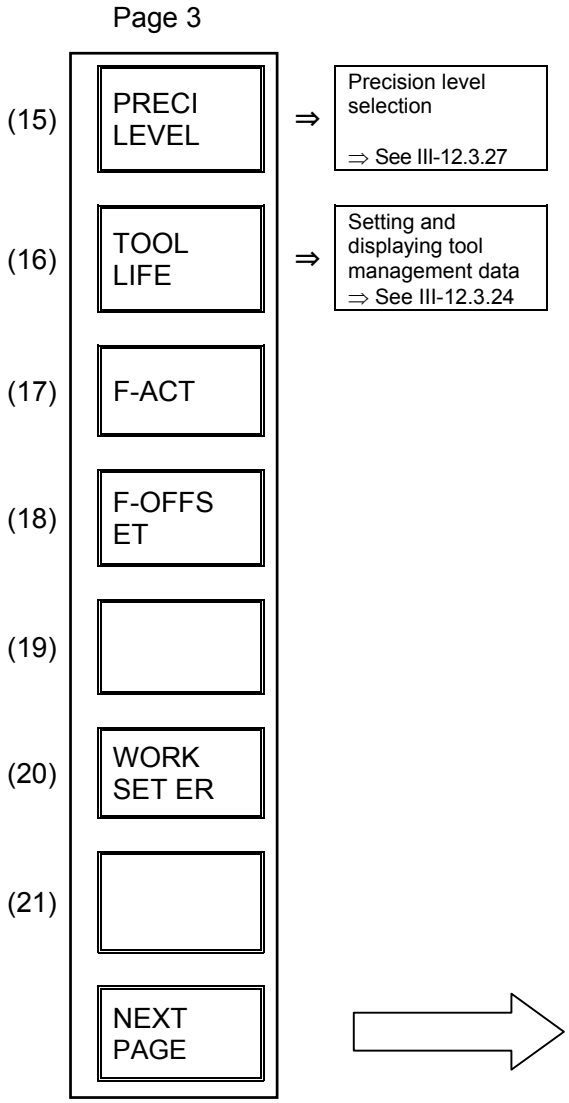


Page 5

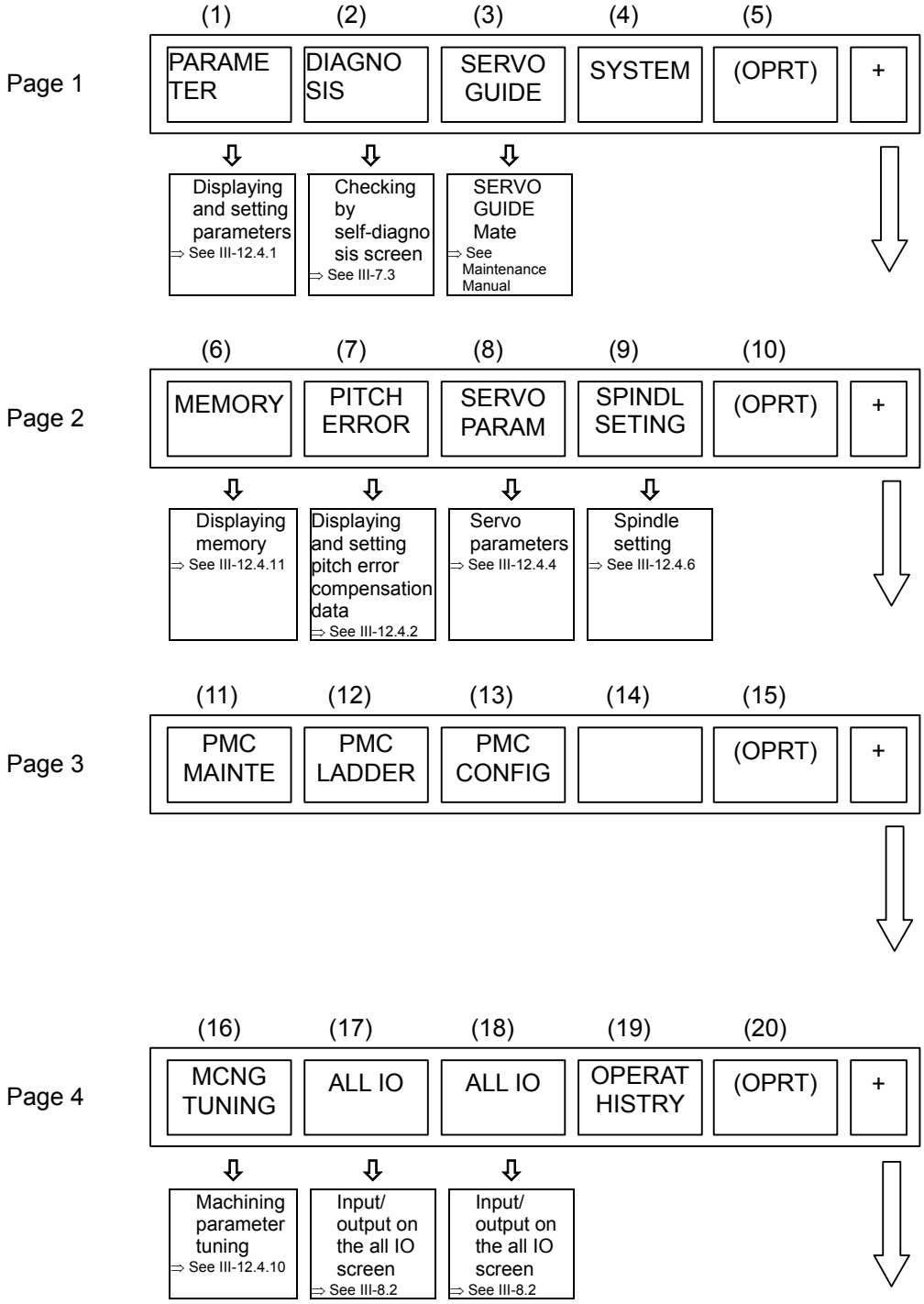

(21)
(22)
(23)
(24)
(25)
CHUCK TAIL
LANG.
PROTECT
GUARD
(OPRT)
+
Chuck and tail stock barriers
⇨ See III-2.1.9 *1
Displaying and switching the display language
⇨ See III-12.3.10
Protection of data at eight levels
⇨ See III-12.3.11
Operation confirmation functions
⇨ See III-6.5

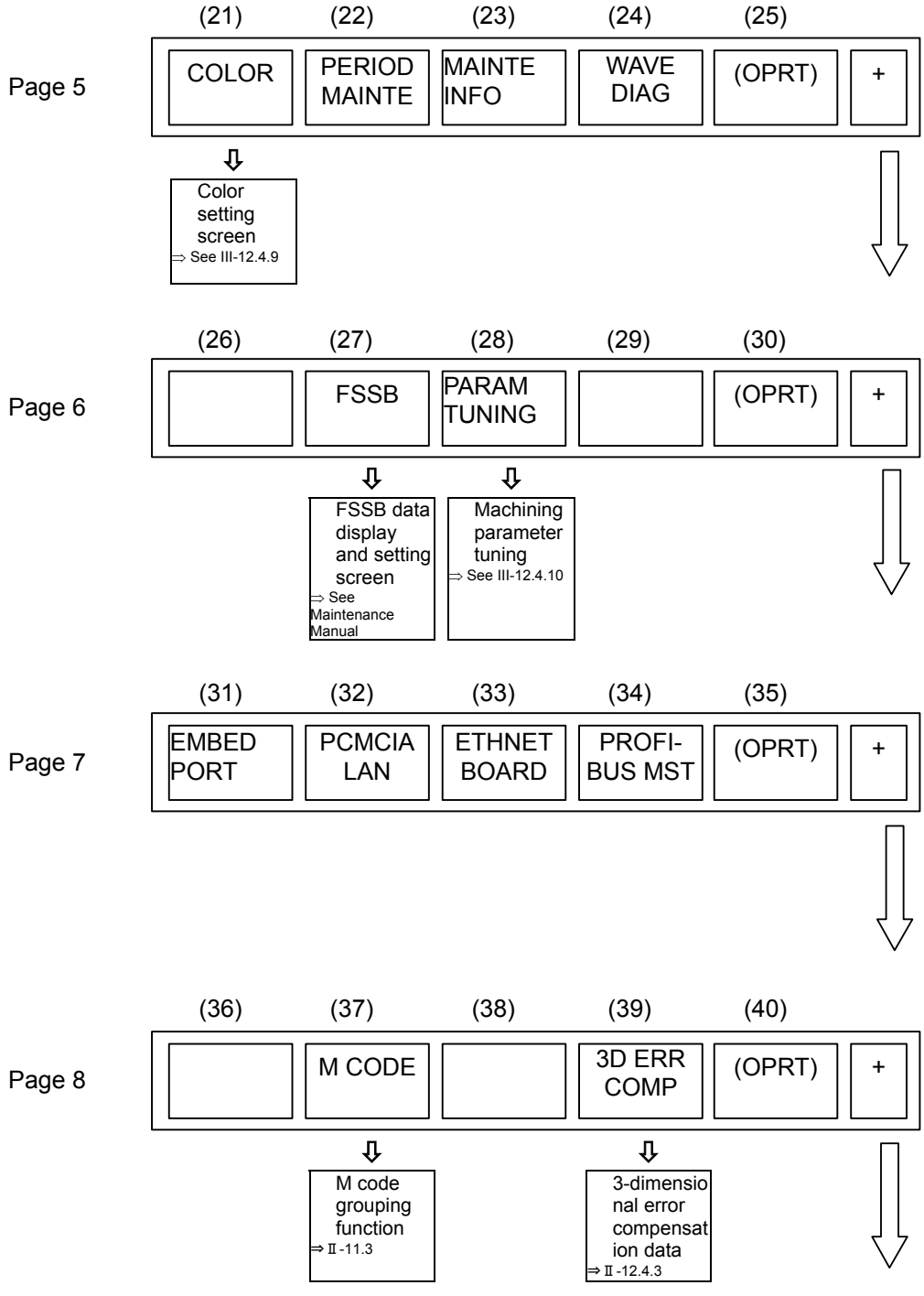
Screen displayed when the function key is pressed (for 15-inch display unit)

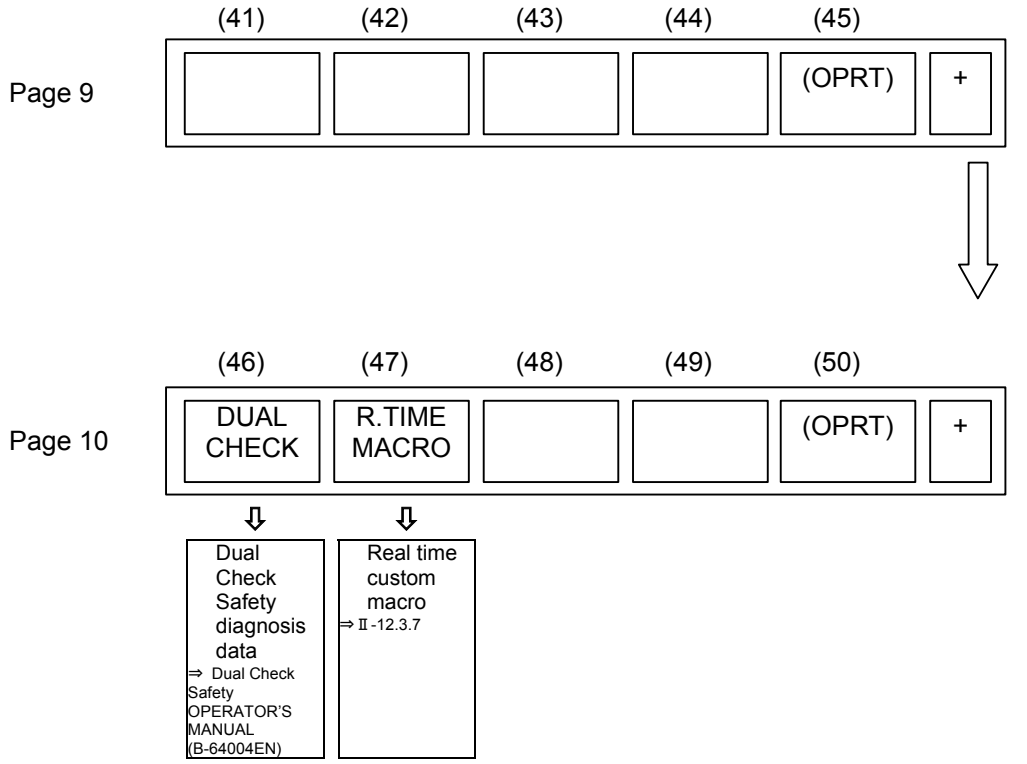



Page 3
(15)
PRECI LEVEL
⇒
Precision level selection
⇒ See III-12.3.27
(16)
TOOL LIFE
⇒
Setting and displaying tool management data
⇒ See III-12.3.24
(17)
F-ACT
(18)
F-OFFS ET
(19)
(20)
WORK SET ER
(21)
NEXT PAGE

Screen displayed when the function key SYSTEM is pressed (for 7.2/8.4/10.4-inch display unit)



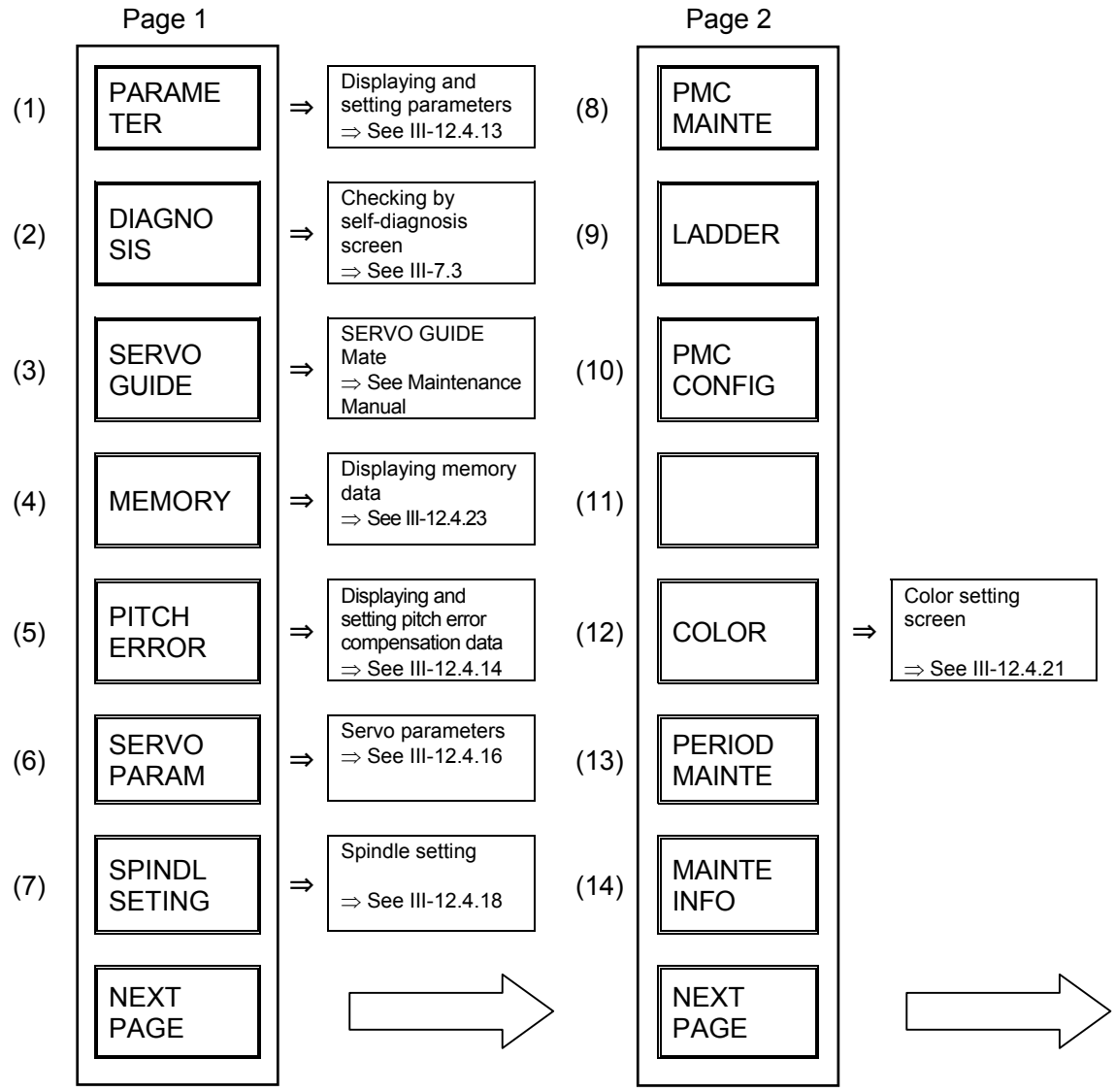

Page 5
(21)
(22)
(23)
(24)
(25)
COLOR
PERIOD MAINTE
MAINTE INFO
WAVE DIAG
(OPRT)
+
Color setting screen
⇨ See III-12.4.9
Page 6
(26)
(27)
(28)
(29)
(30)
FSSB
PARAM TUNING
(OPRT)
+
FSSB data display and setting screen
⇨ See Maintenance Manual
Machining parameter tuning
⇨ See III-12.4.10
Page 7
(31)
(32)
(33)
(34)
(35)
EMBED PORT
PCMCIA LAN
ETHNET BOARD
PROFI-BUS MST
(OPRT)
+
Page 8
(36)
(37)
(38)
(39)
(40)
M CODE
3D ERR COMP
(OPRT)
+
M code grouping function
⇨ Ⅱ-11.3
3-dimensional error compensation data
⇨ Ⅱ-12.4.3

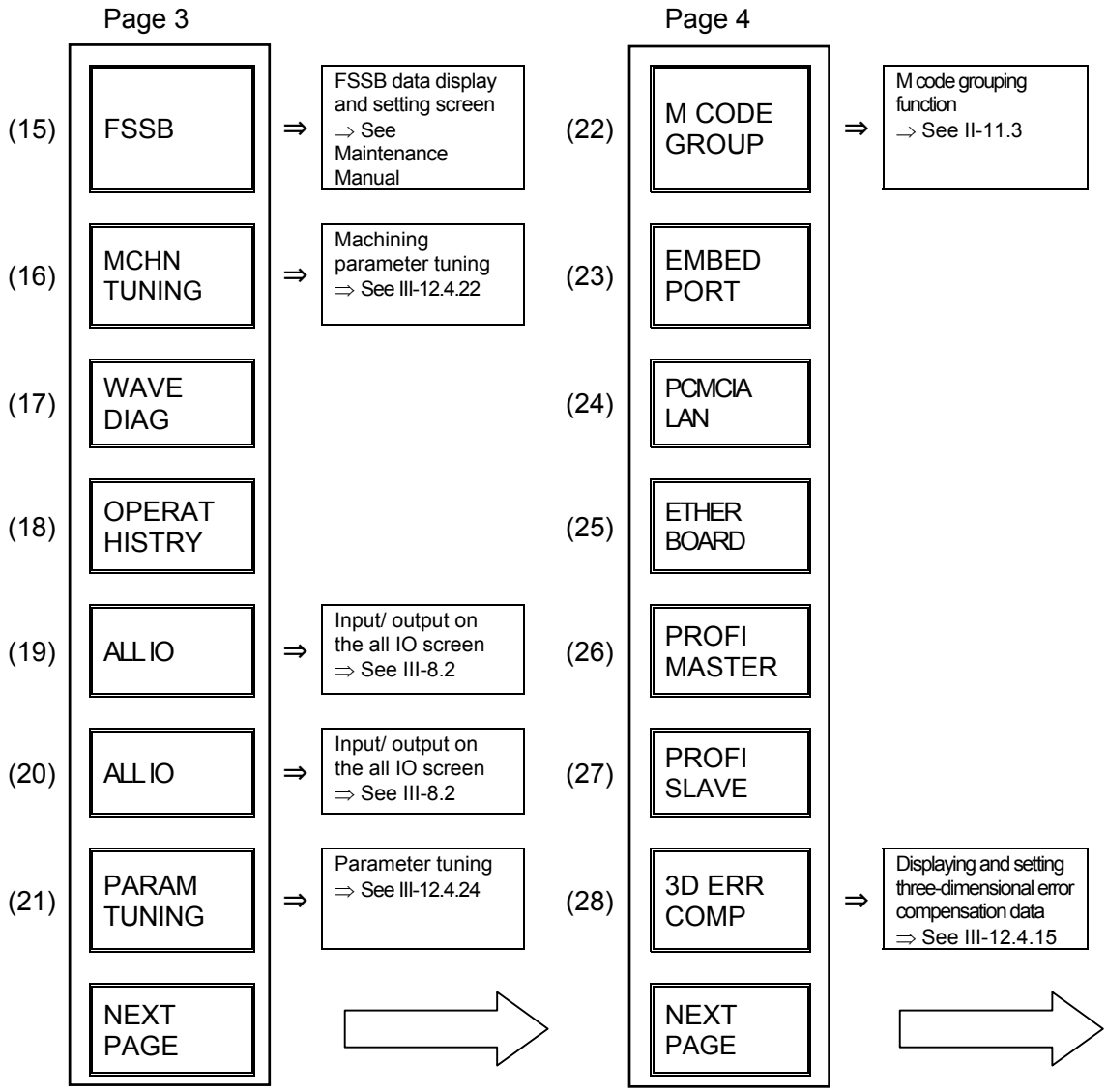


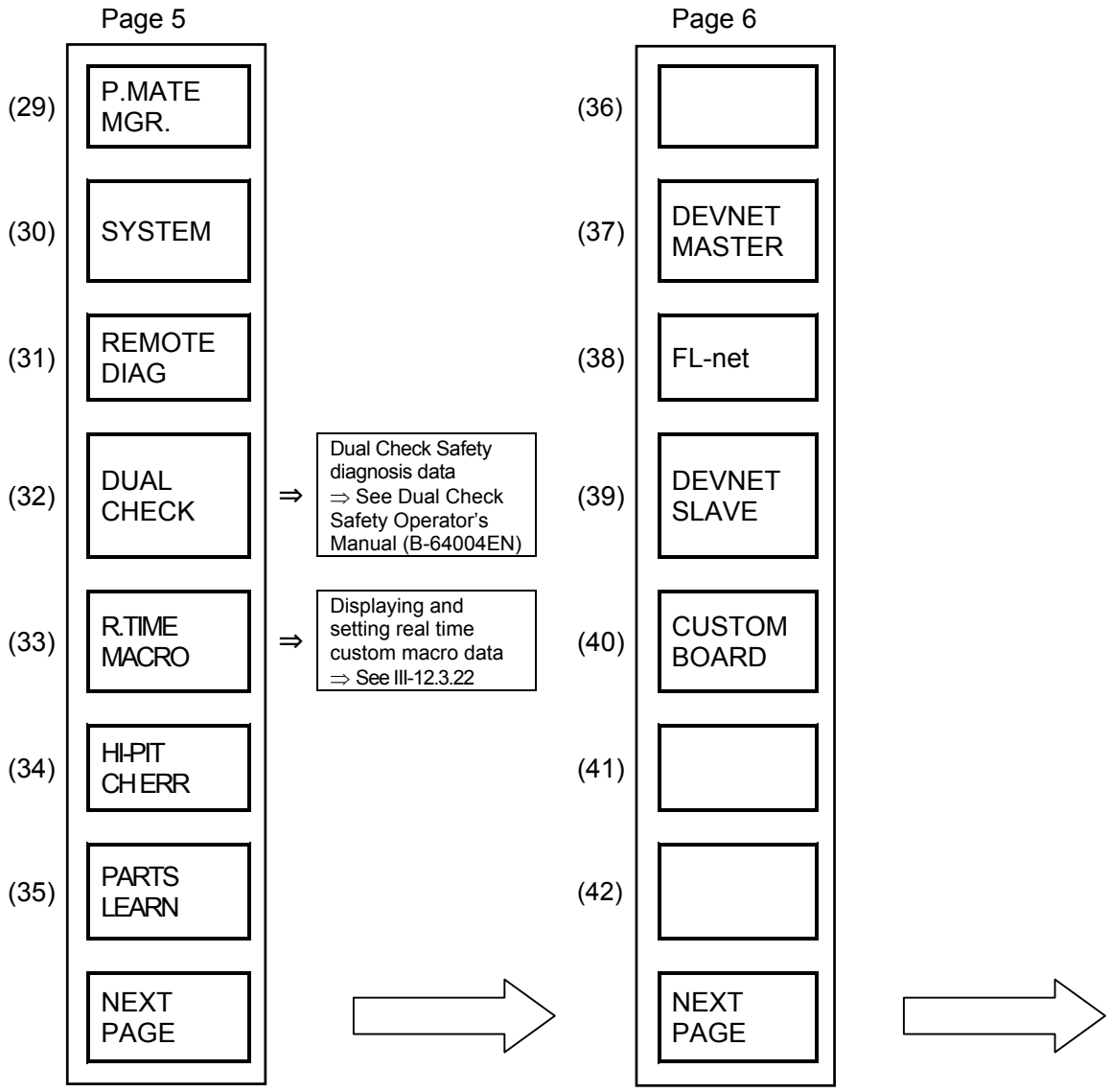
NOTE

For information about a dedicated screen for each path control type in the lathe system/machining center system, refer to the manuals:
*1: User's manual (T series) (B-63944EN-1)
*2: User's manual (M series) (B-63944EN-2)

Screen displayed when the function key SYSTEM is pressed (for 15-inch display unit)




Page 3
(15)
FSSB
⇒
FSSB data display and setting screen
⇒ See Maintenance Manual
(16)
MCHN TUNING
⇒
Machining parameter tuning
⇒ See III-12.4.22
(17)
WAVE DIAG
(18)
OPERAT HISTRY
(19)
ALL IO
⇒
Input/ output on the all IO screen
⇒ See III-8.2
(20)
ALL IO
⇒
Input/ output on the all IO screen
⇒ See III-8.2
(21)
PARAM TUNING
⇒
Parameter tuning
⇒ See III-12.4.24
NEXT PAGE
Page 4
(22)
M CODE GROUP
⇒
M code grouping function
⇒ See II-11.3
(23)
EMBED PORT
(24)
PCMCIA LAN
(25)
ETHER BOARD
(26)
PROFI MASTER
(27)
PROFI SLAVE
(28)
3D ERR COMP
⇒
Displaying and setting three-dimensional error compensation data
⇒ See III-12.4.15
NEXT PAGE



NOTE

For information about a dedicated screen for each path control type in the lathe system/machining center system, refer to the manuals:
*1: User's manual (T series) (B-63944EN-1)
*2: User's manual (M series) (B-63944EN-2)

12.1 SCREENS DISPLAYED BY FUNCTION KEY

Section 12.1, "SCREENS DISPLAYED BY FUNCTION KEY ", consists of the following subsections:

------ Screens of a 7.2/8.4/10.4-inch display unit ------
12.1.1 Position Display in the Workpiece Coordinate System 1431
12.1.2 Position Display in the Relative Coordinate System 1433
12.1.3 Overall Position Display 1436
12.1.4 Workpiece Coordinate System Preset 1438
12.1.5 Actual Feedrate Display 1439
12.1.6 Display of Run Time and Parts Count 1441
12.1.7 Setting the Floating Reference Position 1443
12.1.8 Operating Monitor Display 1444
12.1.9 Display of Three-dimensional Manual Feed (Tool Tip Coordinates, Number of Pulses, Machine Axis Move Amount) 1447
------ Screens of a 15-inch display unit ------
12.1.10 Overall Position Display (15-inch Display Unit).....1451
12.1.11 Workpiece Coordinate System Preset (15-Inch Display Unit) 1454
12.1.12 Actual Feedrate Display (15-inch Display Unit) 1455
12.1.13 Display of Run Time and Parts Count (15-Inch Display Unit) 1457
12.1.14 Setting the Floating Reference Position (15-Inch Display Unit).....1459
12.1.15 Operating Monitor Display (15-inch Display Unit).....1460
12.1.16 Display of Three-dimensional Manual Feed (Tool Tip Coordinates, Number of Pulses, Machine Axis Move Amount) (15-inch Display Unit).....1463

Screens of a 7.2/8.4/10.4-inch display unit

Press function key to display the current position of the tool.

The following three screens are used to display the current position of the tool:

- Current position display screen for the workpiece coordinate system.
- Current position display screen for the relative coordinate system.
- Current overall position display screen.

The above screens can also display the feedrate, run time, and the number of parts. In addition, a floating reference position can be set on these screens.

Function key can also be used to display the load on the servo motor and spindle motor and the rotation speed of the spindle motor (operating monitor display).

Function key can also be used to display the screen for displaying the distance moved by handle interruption. See III- 4.6 for details on this screen.

On any of the position display screens, the status (such as D, I, L, S, *, or M) of an axis is indicated on the left of the axis name to prevent wrong operations. See "Axis status display" in III-6.5, "WRONG OPERATION PREVENTION FUNCTIONS" for details.

12.1.1 Position Display in the Workpiece Coordinate System

Displays the current position of the tool in the workpiece coordinate system. The current position changes as the tool moves. The least input increment is used as the unit for numeric values. The title at the top of the screen indicates that absolute coordinates are used.

Display procedure for the current position screen in the workpiece coordinate

Procedure

1 Press function key

.

2 Press soft key [ABSOLUTE].

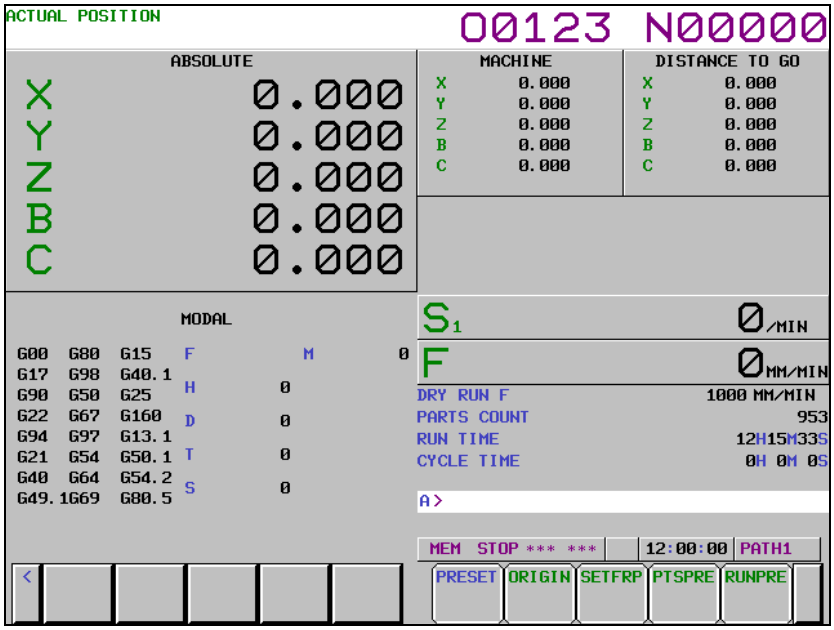


Fig. 12.1.1 (a) Current position (absolute) screen (M series)(10.4-inch)

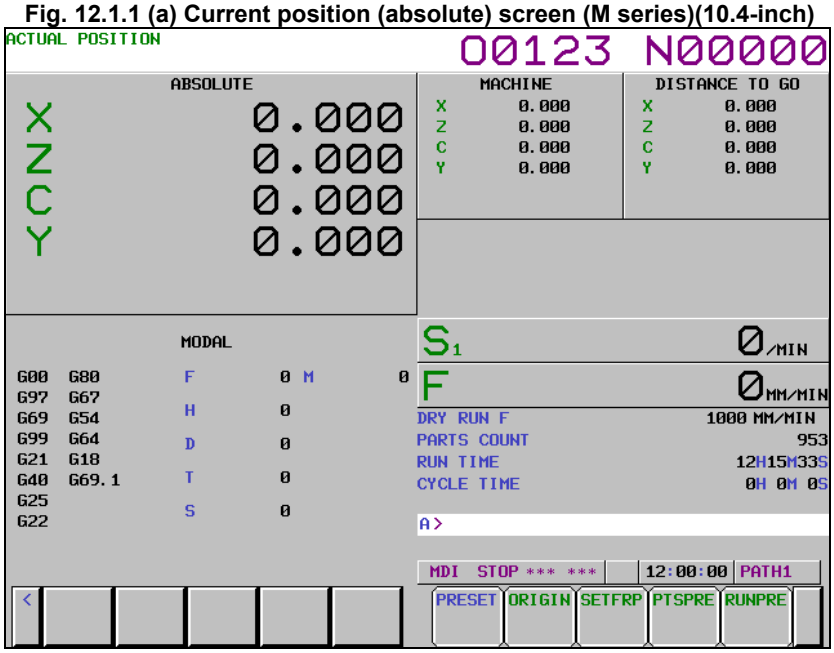


Fig. 12.1.1 (b) Current position (absolute) screen (T series)(10.4-inch)

Explanation

- Display including compensation values

M

Bits 6 (DAL) and 7 (DAC) of parameter No. 3104 can be used to select whether the displayed values include tool length compensation and cutter compensation.

T

Bit 1 (DAP) parameter No. 3129 and bit 7 (DAC) of parameter No. 3104 can be used to select whether the displayed values include tool offset and tool nose radius compensation.

12.1.2 Position Display in the Relative Coordinate System

Displays the current position of the tool in a relative coordinate system based on the coordinates (see Explanation) set by the operator. The current position changes as the tool moves. The increment system is used as the unit for numeric values.
The title at the top of the screen indicates that relative coordinates are used.

Display procedure for the current position screen with the relative coordinate system

Procedure

1 Press function key

.

2 Press soft key [RELATIVE].

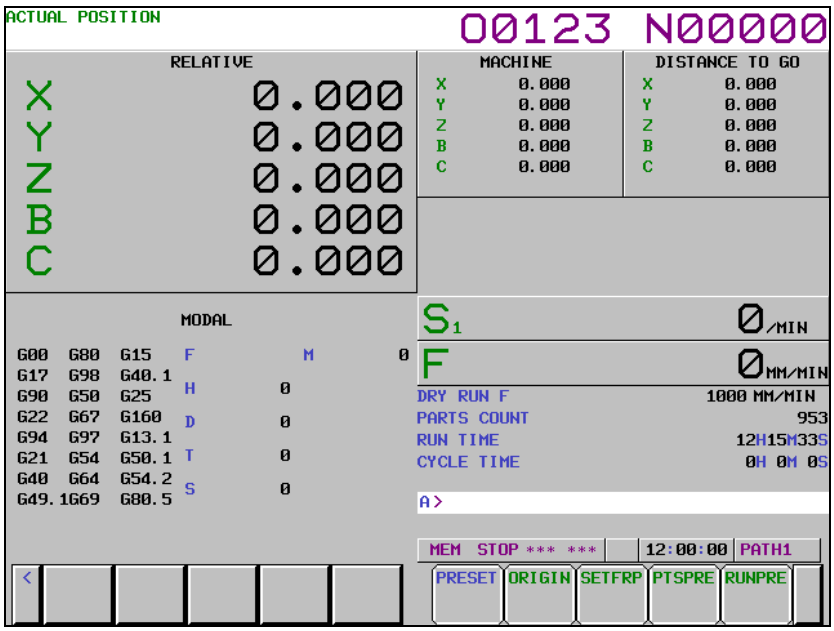


Fig. 12.1.2 (a) Current position (relative) screen (M series)(10.4-inch)

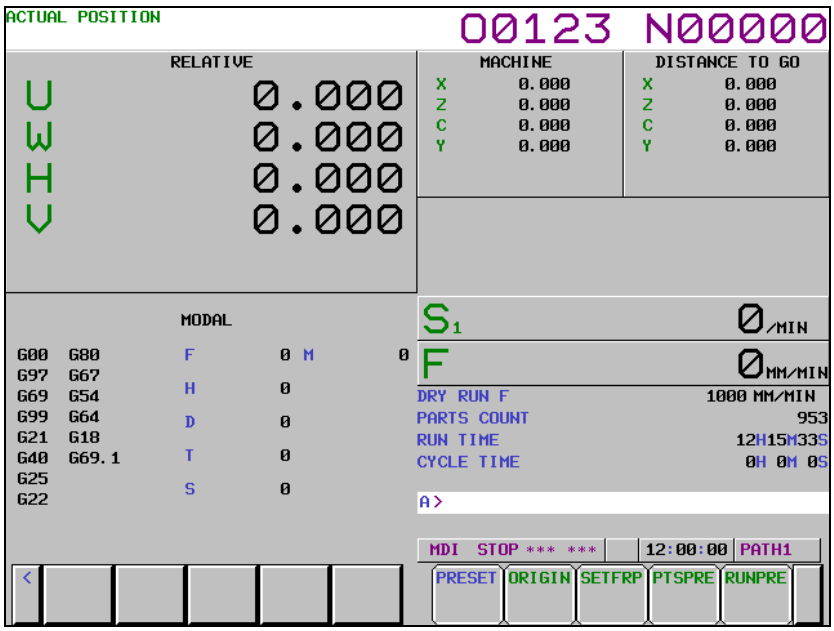


Fig. 12.1.2 (b) Current position (relative) screen (T series) (10.4-inch)

See Explanation for the procedure for setting the coordinates.

Explanation

- Setting the relative coordinates

The current position of the tool in the relative coordinate system can be reset to 0 or preset to a specified value as follows:

Procedure to set the axis coordinate to a specified value

Procedure

1 To reset the coordinate to 0, press soft key [ORGIN].
Key in an axis name to be reset (such as X or Y), then press soft key [EXEC].
2 For presetting to a specified value, key in an axis name to be preset and a preset value (such as X100.000), then press soft key [PRESET].

- Display including compensation values

M

Bits 4 (DRL) and 5 (DRC) of parameter No. 3104 can be used to select whether the displayed values include tool length compensation and cutter compensation.

T

Bit 0 (DRP) parameter No. 3129 and bit 5 (DRC) of parameter No. 3104 can be used to select whether the displayed values include tool offset and tool nose radius compensation.

- Presetting by setting a coordinate system

M

Bit 3 (PPD) of parameter No. 3104 can be used to specify whether the position indication values in the absolute coordinate system are preset as those in the relative coordinate system during coordinate system setting or manual reference position return by the G92 command.

T

Bit 3 (PPD) of parameter No. 3104 can be used to specify whether the position indication values in the absolute coordinate system are preset as those in the relative coordinate system during coordinate system setting or manual reference position return by the G50 command (for G code system A) or G92 command (for G code system B or C).

12.1.3 Overall Position Display

Displays the following positions on a screen : Current positions of the tool in the workpiece coordinate system, relative coordinate system, and machine coordinate system, and the remaining distance.
The relative coordinates can also be set on this screen. See III-12.1.2 for the procedure.

Procedure for displaying overall position display screen

Procedure

1 Press function key

.

2 Press soft key [ALL].

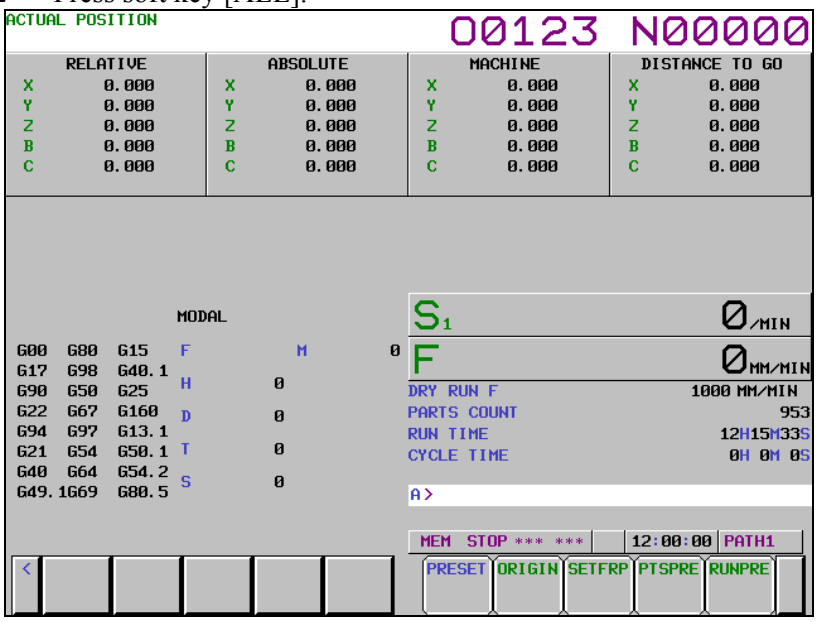


Fig. 12.1.3 (a) Current position (overall) screen (M series) (10.4-inch)

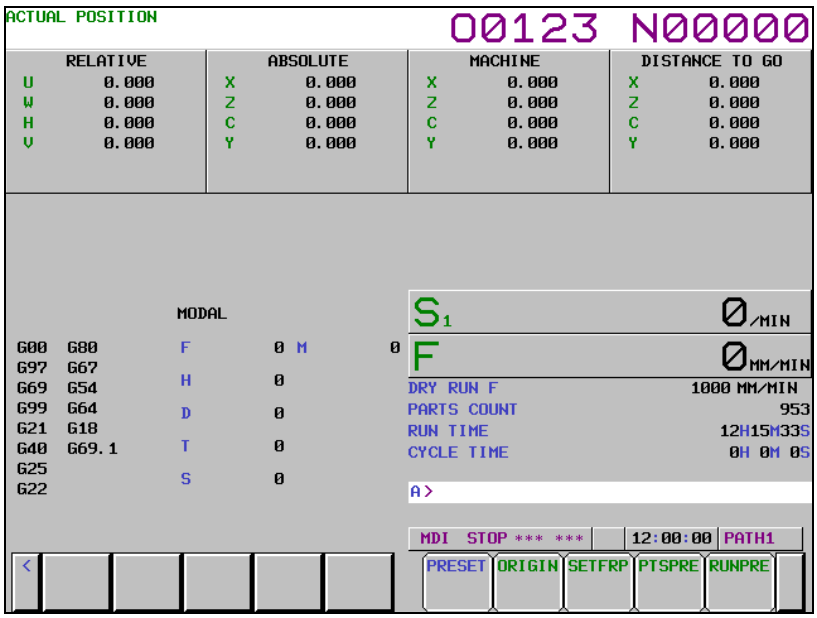


Fig. 12.1.3 (b) Current position (overall) screen (T series) (10.4-inch)

Explanation

- Coordinate display

The current positions of the tool in the following coordinate systems are displayed at the same time:

- Current position in the relative coordinate system (relative coordinate)
- Current position in the workpiece coordinate system (absolute coordinate)
- Current position in the machine coordinate system (machine coordinate)
- Distance to go (distance to go)

- Distance to go

The distance remaining is displayed in the MEMORY or MDI mode. The distance the tool is yet to be moved in the current block is displayed.

- Machine coordinate system

The least command increment is used as the unit for values displayed in the machine coordinate system. However, the least input increment can be used by setting bit 0 (MCN) of parameter No. 3104.

- Resetting the relative coordinates

The total position display screen also supports the resetting of the relative coordinates to 0 or presetting of them to specified values. See the procedure for resetting the relative coordinates described in Subsection III-12.1.2

12.1.4 Workpiece Coordinate System Preset

If a workpiece coordinate system has been shifted with manual intervention or any other operation, an MDI operation can be performed to preset the system to a workpiece coordinate system that is offset by a workpiece origin offset, from the machine zero point that has been set before the shifting.
A command (G92.1) can be programmed to preset a workpiece coordinate system. (See II-7.2.4.)

Procedure for the workpiece coordinate system preset

Procedure

1 Press function key

.

2 Press soft key [(OPRT)].

3 Enter the axis name (

,

, ...) and

.

4 Press soft key [PRESET].

Explanation

- Operation mode

This function can be executed when the reset state or automatic operation stop state is entered, regardless of the operation mode.

- Presetting relative coordinates

As with absolute coordinates, bit 3 (PPD) of parameter No. 3104 is used to specify whether to preset relative coordinates (RELATIVE).

12.1.5 Actual Feedrate Display

The actual feedrate on the machine (per minute) can be displayed on a current position display screen or program check screen.
On the 12 soft keys display unit, the actual feedrate is always displayed.

Display procedure for the actual feedrate on the current position display screen

Procedure

1 Press the function key POS to display a current position display screen. At the location indicated by ⬚, an actual feedrate is displayed.

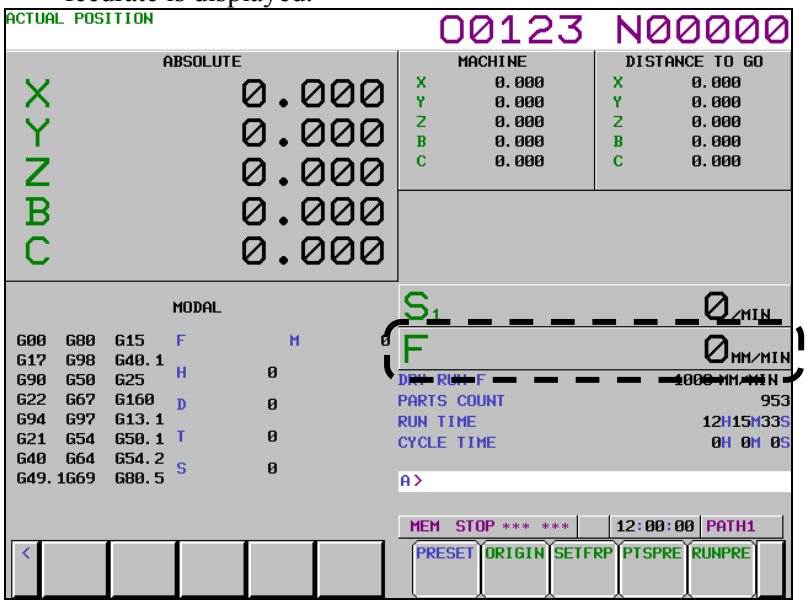


Fig. 12.1.5 (a) Current position (absolute) screen (M series) (10.4-inch)

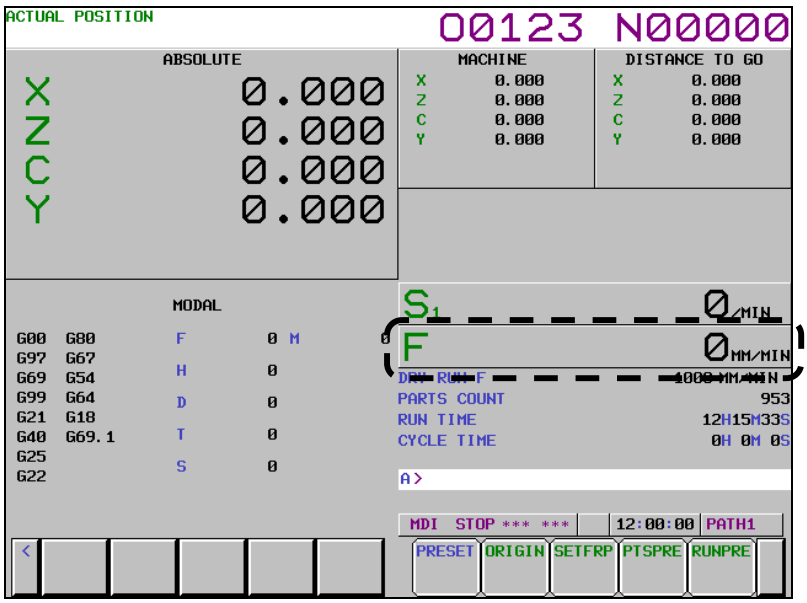


Fig. 12.1.5 (b) Current position (absolute) screen (T series) (10.4-inch)

The actual feedrate is displayed in units of millimeter/min or inch/min (depending on the specified least input increment) under the display of the current position.

Explanation

- Actual feedrate value

The actual rate is calculated by the following expression:

$$Fact = \sqrt{\sum_{i=1}^{n} (fi)^2}$$

n : Number of axes
fi : Cutting feed rate in the tangential direction of each axis or rapid traverse rate
Fact : Actual feedrate displayed

The display unit:

mm/min (metric input).
inch/min (Inch input, two digits below the decimal point are displayed.)

- Actual feedrate display of feed per revolution

In the case of feed per revolution and thread cutting, the actual feedrate displayed is the feed per minute rather than feed per revolution.

- Actual feedrate display of rotary axis

In the case of movement of rotary axis, the speed is displayed in units of deg/min but is displayed on the screen in units of input system at that time. For example, when the rotary axis moves at 50 deg/min, the following is displayed: 50 mm/min (in metric input) or 0.50 inch/min (in inch input).

- Actual feedrate display on the other screen

The program check screen also displays the actual feedrate.

12.1.6 Display of Run Time and Parts Count

The run time, cycle time, and the number of machined parts are displayed on the current position display screens.

Procedure for displaying run time and parts count on the current position display screen

Procedure

1 Press the function key POS to display a current position display screen. At the location indicated by ⬚, a run time and parts count are displayed.

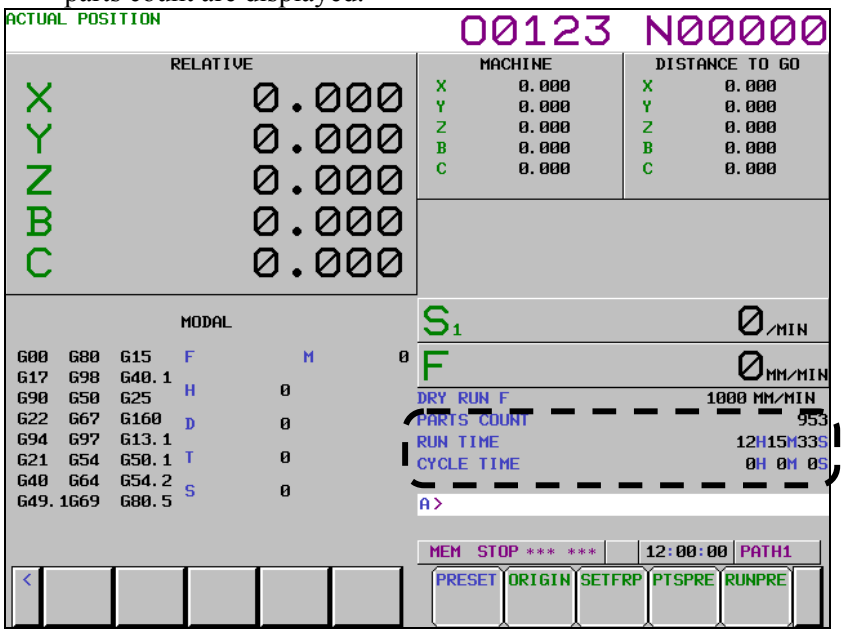


Fig. 12.1.6 (a) Current position (relative) screen (M series) (10.4-inch)

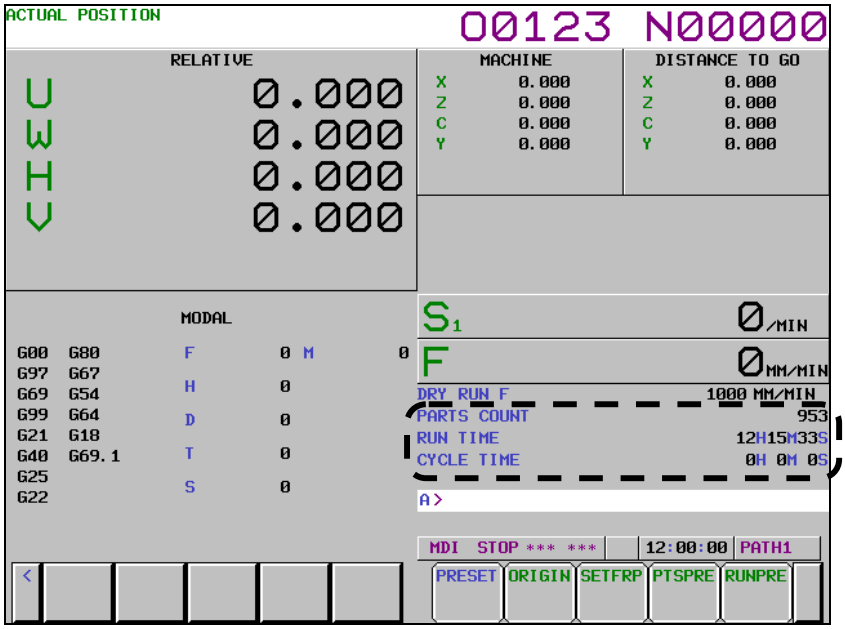


Fig. 12.1.6 (b) Current position (relative) screen (T series) (10.4-inch)

The number of machined parts (PART COUNT), run time (RUN TIME), and cycle time (CYCLE TIME) are displayed under the current position.

Explanation

- PART COUNT

Indicates the number of machined parts. The number is incremented each time M02, M30, or an M code specified by parameter No. 6710 is executed.

- RUN TIME

Indicates the total run time during automatic operation, excluding the stop and feed hold time.

- CYCLE TIME

Indicates the run time of one automatic operation, excluding the stop and feed hold time. This is automatically preset to 0 when a cycle start is performed at reset state. It is preset to 0 even when power is removed.

- Display on the other screen

Details of the run time and the number of machined parts are displayed on the setting screen. See III-12.3.3.

- Parameter setting

The number of machined parts and run time cannot be set on current position display screens.
The number of machined parts and run time cannot be set on current position display screens. They can be set by parameters Nos. 6711, 6751, and 6752 or on the setting screen.

- Incrementing the number of machined parts

Bit 0 (PCM) of parameter No. 6700 is used to specify whether the number of machined parts is incremented each time M02, M30, or an M code specified by parameter No. 6710 is executed, or only each time an M code specified by parameter No. 6710 is executed.

12.1.7 Setting the Floating Reference Position

To perform floating reference position return with a G30.1 command, the floating reference position must be set beforehand.

Procedure for setting the floating reference position

Procedure

1 Press function key to display a screen used for displaying the current position.
2 Move the tool to the floating reference position by jogging.
3 Press soft key [(OPRT)].
4 Press soft key [SET FRP].
5 To register the floating reference positions for all axes, press soft key [ALL AXIS].
To register the floating reference position of a specific axis, enter the name of the axis (, , etc.), then press soft key [AXISNAME]. Two or more names can be entered consecutively (e.g.,) [AXIS NAME]).
The above operation stores the floating reference position. It can be checked with parameter (No. 1244).
6 In step 4, the floating reference position along a specified axis can also be stored by entering the axis name (such as or) and pressing soft key [SET FRP].

Explanation

- Presetting the relative coordinate system

When bit 3 (FPC) of parameter No. 1201 is set to 1, the floating reference position can be registered, while the relative position display can be preset to 0.

12.1.8 Operating Monitor Display

The load meter for a servo axis can be displayed. Also, the load meter and speed meter for a serial spindle can be displayed.
To enable this function, bit 5 (OPM) of parameter No. 3111 must be set to 1.

Procedure for displaying the operating monitor

Procedure

1 Press function key

.

2 Press continuous menu key
.

3 Press soft key [MONITOR].

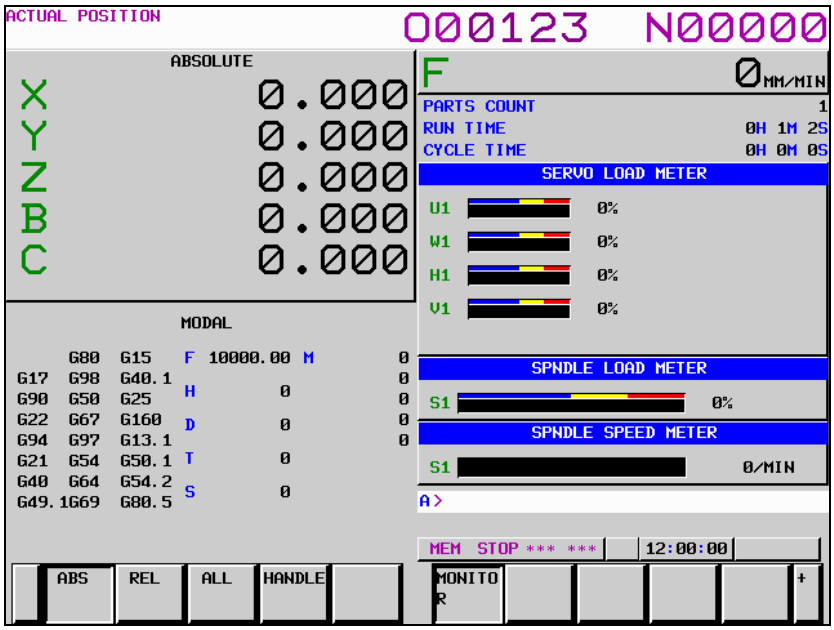


Fig. 12.1.8 (a) Operating monitor (M series) (10.4-inch)

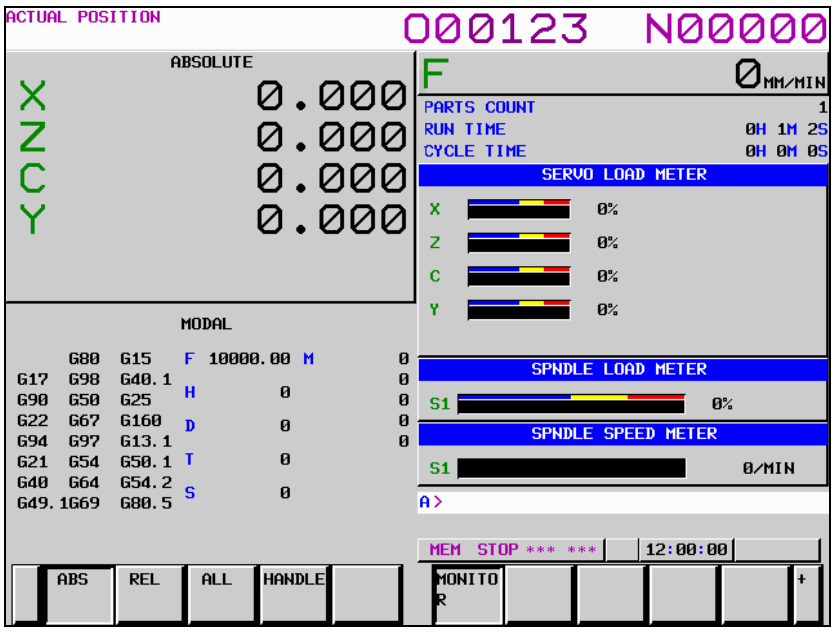


Fig. 12.1.8 (b) Operating monitor (T series) (10.4-inch)

Explanation

- Display of the servo axes

Servo axis load meters as many as the maximum number of controlled axes of the path can be displayed. One screen displays load meters for up to five axes at a time. By pressing the soft key [MONITOR], load meters for the 6th axis and up are displayed.

- Display of the spindle axes

When serial spindles are used, the reading on the load meter and speedometer can be displayed only for the main serial spindle.

- Unit of graph

The bar graph for the load meter shows load up to 200% (only a value is displayed for load exceeding 200%).
The bar graph for the speedometer shows the ratio of the current spindle speed to the maximum spindle speed (100%).

- Load meter

The reading on the load meter depends on servo parameter No. 2086 and spindle parameter No. 4127.

- Speedometer

Although the speedometer normally indicates the speed of the spindle motor, it can also be used to indicate the speed of the spindle by setting bit 6 (OPS) of parameter No. 3111 to 1.
The spindle speed to be displayed during operation monitoring is calculated from the speed of the spindle motor (see the formula below).
The spindle speed can therefore be displayed, during operation monitoring, even when no position coder is used.
To display the correct spindle speed, however, the maximum spindle speed for each gear (spindle speed at each gear ratio when the spindle motor rotates at the maximum speed) must be set in parameters No. 3741 to No.3744.
The input of the clutch and gear signals for the first serial spindle is used to determine the gear which is currently selected. Control the input of the CTH1A and CTH2A signals according to the gear selection, by referring to the table below.
(Formula for calculating the spindle speed to be displayed)

$$\text{Spindle speed displayed during operation monitoring} = \frac{\text{Speed of spindle motor}}{\text{Maximum speed of spindle motor}} \times \text{Maximum spindle speed with the gear being used}$$

The following table lists the correspondence between clutch and gear selection signals CTH1A and CTH2A, used to determine the gear being used, and parameters:

CTH1A	CTH2A	Parameter	Serial spindle specification
0	0	=No.3741 (Maximum spindle speed with gear 1)	HIGH
0	1	=No.3742 (Maximum spindle speed with gear 2)	MEDIUM HIGH
1	0	=No.3743 (Maximum spindle speed with gear 3)	MEDIUM LOW
1	1	=No.3744 (Maximum spindle speed with gear 4)	LOW

The speed of the spindle motor and spindle can be displayed, during operation monitoring, only for the first serial spindle and the spindle switching axis for the first serial spindle. It cannot be displayed for the second spindle.

- Color of graph

If the value of a load meter exceeds 100%, the bar graph turns purple.

12.1.9 Display of Three-dimensional Manual Feed (Tool Tip Coordinates, Number of Pulses, Machine Axis Move Amount)

The absolute coordinates of the tool tip, the number of pulses, and a machine axis move amount based on three-dimensional manual feed are displayed.

Displaying the screen for three-dimensional manual feed

Procedure

1 Press function key

.

2 Press the continuous menu key several times to display the soft key [3-D MANUAL].

3 Press the soft key [3-D MANUAL] to display the screen of three-dimensional manual feed.

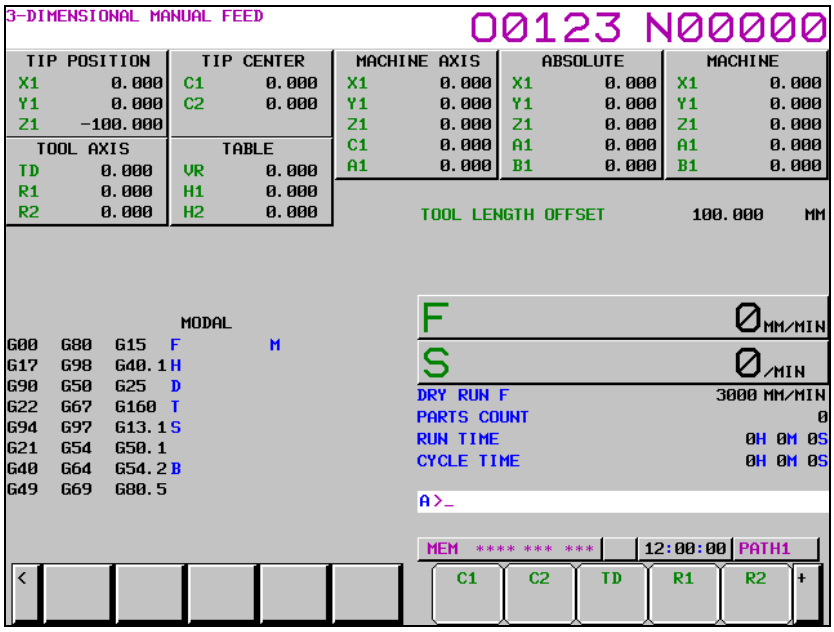


Fig. 12.1.9 (a) Three-dimensional manual feed screen(10.4-inch)

Explanation

- Tool tip position

The addresses of the three basic machine configuration axes for performing three-dimensional manual feed and the current position of the tool tip are displayed.

- Tool axis reference (number of pulses)

TD

The amount of travel in the tool axis direction in tool axis direction handle feed, tool axis direction jog feed, or tool axis direction incremental feed is displayed.
The unit is the least input increment of the axis in the direction specified by parameter No. 19697.

R1

The amount of travel in the first axis direction in tool axis right-angle direction handle feed, tool axis right-angle direction jog feed, or tool axis right-angle direction incremental feed is displayed.
The unit is the least input increment of the axis in the first axis direction normal to the direction specified by parameter No. 19697.

R2

The amount of travel in the second axis direction in tool axis right-angle direction handle feed, tool axis right-angle direction jog feed, or tool axis normal direction incremental feed is displayed.
The unit is the least input increment of the axis in the second axis direction normal to the direction specified by parameter No. 19697.

- Tool tip center (number of pulses)

C1

The angular displacement in tool tip center rotation handle feed, tool tip center rotation jog feed, or tool tip center rotation incremental feed for rotation of the first rotary axis is displayed. The unit is the least input increment of the first rotary axis.

C2

The angular displacement in tool tip center rotation handle feed, tool tip center rotation jog feed, or tool tip center rotation incremental feed for rotation of the second rotary axis is displayed. The unit is the least input increment of the second rotary axis.

- Table reference (number of pulses)

VR

The amount of travel in the table reference vertical direction in table reference vertical direction handle feed, table reference vertical direction jog feed, or table reference vertical direction incremental feed is displayed.
The unit is the least input increment of the axis in the direction specified by parameter No. 19697.

H1

The amount of travel in the first axis direction in table reference horizontal direction handle feed, table reference horizontal direction jog feed, or table reference horizontal direction incremental feed is displayed.
The unit is the least input increment of the axis in the first axis direction normal to the direction specified by parameter No. 19697.

H2

The amount of travel in the second axis direction in table reference horizontal direction handle feed, table reference horizontal direction jog feed, or table reference horizontal direction incremental feed is displayed.
The unit is the least input increment of the axis in the second axis direction normal to the direction specified by parameter No. 19697.

- Amounts of machine axis travel

The addresses of machine configuration axes used for three-dimensional manual feed and the sum of the amount of travel of each axis used for three-dimensional manual feed are displayed.
The values of the basic three axes (X-axis, Y-axis, and Z-axis), the first rotary axis, and the second rotary axis are displayed in this order.
For the definition of the first rotary axis and second rotary axis, see the description of parameter No. 19680.
When bit 0 (CLR) of parameter No. 13113 is set to 1, the displayed data is cleared by a reset.

- Absolute coordinates, machine coordinates

The absolute coordinates and machine coordinates of all axes are displayed. If too many axes are involved for display on one screen, the remaining axes can be displayed by pressing the soft key [3-D MANUAL] for page feed.

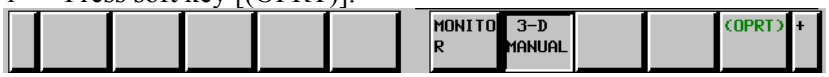
- F (feedrate)

- When bit 3 (CFD) of parameter No. 13113 is set to 0
  The composite feedrate at a control point on a linear axis or rotary axis is displayed.
- When bit 3 (CFD) of parameter No. 13113 is set to 1
  The feedrate of the tool tip is displayed.

Operation

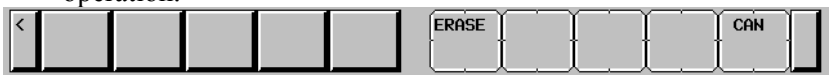
The display of the number of pulses can be cleared by soft key operations.

1 Press soft key [(OPRT)].



2 Select the soft key corresponding to a function subject to clearing of the amount of travel.
Pressing the rightmost soft key displays the second page.

3 Press soft key [ERASE] to clear the amount of travel of the specified function. Press soft key [CAN] to cancel erase operation.



Screens of a 15-inch display unit

Press function key to display the current overall position display screen that shows the current position of the tool.
The above screen can also display the feedrate, run time, and the number of parts. In addition, a floating reference position can be set on this screen.

Function key can also be used to display the load on the servo motor and spindle motor and the rotation speed of the spindle motor (operating monitor display).

Function key can also be used to display the screen for displaying the distance moved by handle interruption. See III-4.6 for details on this screen.

12.1.10 Overall Position Display (15-inch Display Unit)

Displays the following positions on a screen : Current positions of the tool in the workpiece coordinate system, relative coordinate system, and machine coordinate system, and the remaining distance.
The relative coordinates can also be set on this screen. (Refer to Explanation.)

Procedure for displaying overall position display screen

Procedure

1 Press function key

.

2 Press vertical soft key [ALL].

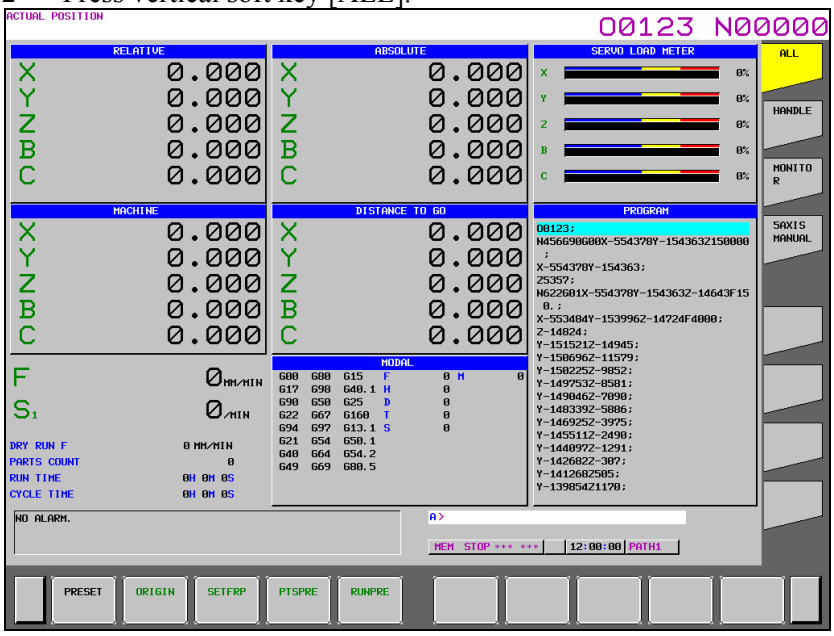


Fig. 12.1.10 (a) Current position screen (M series) (15-inch)

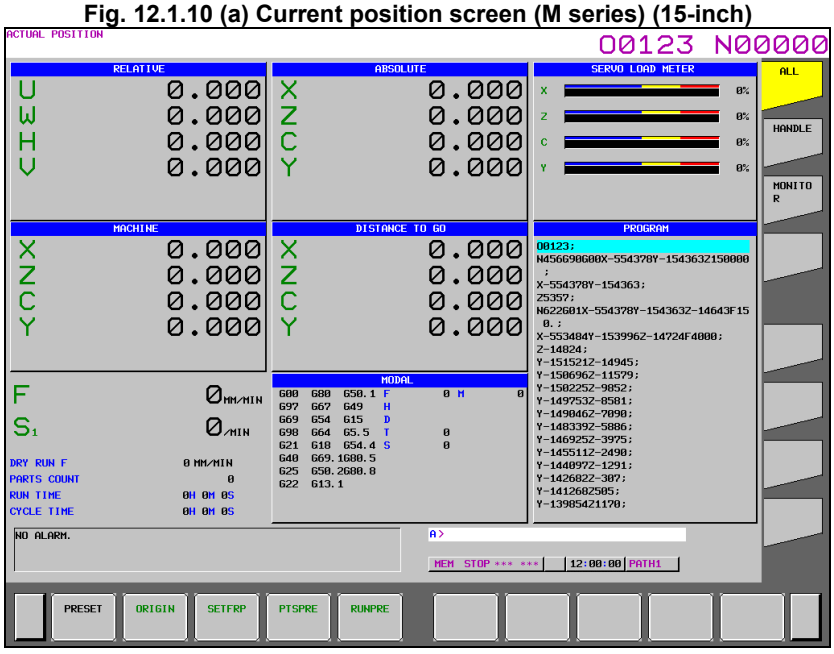


Fig. 12.1.10 (b) Current position screen (T series) (15-inch)

Explanation

- Coordinate display

The current positions of the tool in the following coordinate systems are displayed at the same time:

- Current position in the relative coordinate system (relative coordinate)
- Current position in the work coordinate system (absolute coordinate)
- Current position in the machine coordinate system (machine coordinate)
- Distance to go (distance to go)

- Distance to go

The distance remaining is displayed in the MEMORY or MDI mode. The distance the tool is yet to be moved in the current block is displayed.

- Machine coordinate system

The least command increment is used as the unit for values displayed in the machine coordinate system. However, the least input increment can be used by setting bit 0 (MCN) of parameter No. 3104.

- Setting the relative coordinates

The current position of the tool in the relative coordinate system can be reset to 0 or preset to a specified value as follows:

Procedure to set the axis coordinate to a specified value

Procedure

1 To reset the coordinate to 0, press horizontal soft key [ORGIN]. Key in an axis name to be reset (such as X or Y), then press soft key [EXEC].
2 For presetting to a specified value, key in an axis name to be preset and a preset value (such as X100.000), then press horizontal soft key [PRESET].

- Display including compensation values

M

The following parameters can be used to determine whether the displayed values include tool length compensation and cutter compensation.

- Workpiece coordinate system: Bits 6 (DAL) and 7 (DAC) of parameter No. 3104
- Relative coordinate system: Bits 4 (DRL) and 5 (DRC) of parameter No. 3104

T

The following parameters can be used to determine whether the displayed values include tool offset and tool nose radius compensation.

- Workpiece coordinate system: Bit 1 (DAP) of parameter No. 3129 and bit 7 (DAC) of parameter No. 3104
- Relative coordinate system: Bit 0 (DRP) of parameter No. 3129 and bit 5 (DRC) of parameter No. 3104

- Presetting by setting a coordinate system

M

Bit 3 (PPD) of parameter No. 3104 can be used to specify whether the position indication values in the absolute coordinate system are preset as those in the relative coordinate system during coordinate system setting or manual reference position return by the G92 command.

T

Bit 3 (PPD) of parameter No. 3104 can be used to specify whether the position indication values in the absolute coordinate system are preset as those in the relative coordinate system during coordinate system setting or manual reference position return by the G50 command (for G code system A) or G92 command (for G code system B or C).

12.1.11 Workpiece Coordinate System Preset (15-inch Display Unit)

If a workpiece coordinate system has been shifted with manual intervention or any other operation, an MDI operation can be performed to preset the system to a workpiece coordinate system that is offset by a workpiece origin offset, from the machine zero point that has been set before the shifting.
A command (G92.1) can be programmed to preset a workpiece coordinate system. (See II-7.2.4.)

Procedure for the workpiece coordinate system preset

Procedure

1 Press function key

.

2 Press vertical soft key [ALL].

3 Enter the axis name (

,

, ...) and

.

4 Press soft key [PRESET].

Explanation

- Operation mode

This function can be executed when the reset state or automatic operation stop state is entered, regardless of the operation mode.

- Presetting relative coordinates

As with absolute coordinates, bit 3 (PPD) of parameter No. 3104 is used to specify whether to preset relative coordinates (RELATIVE).

12.1.12 Actual Feedrate Display (15-inch Display Unit)

The actual feedrate on the machine (per minute) can be displayed on a current position display screen or program check screen.

Display procedure for the actual feedrate on the current position display screen

Procedure

1 Press the function key

to display a current position display screen. At the location indicated by , an actual feedrate is displayed.

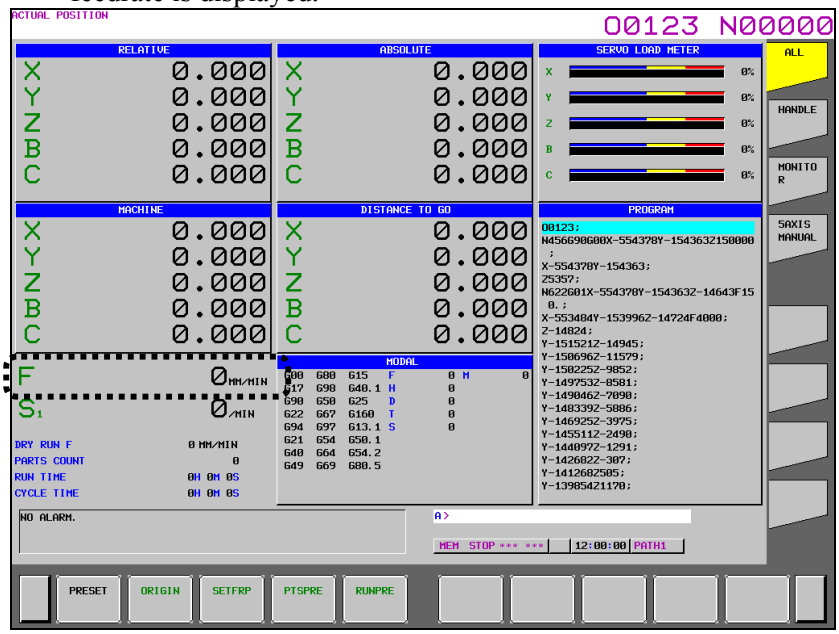


Fig. 12.1.12 (a) Current position screen (M series) (15-inch)

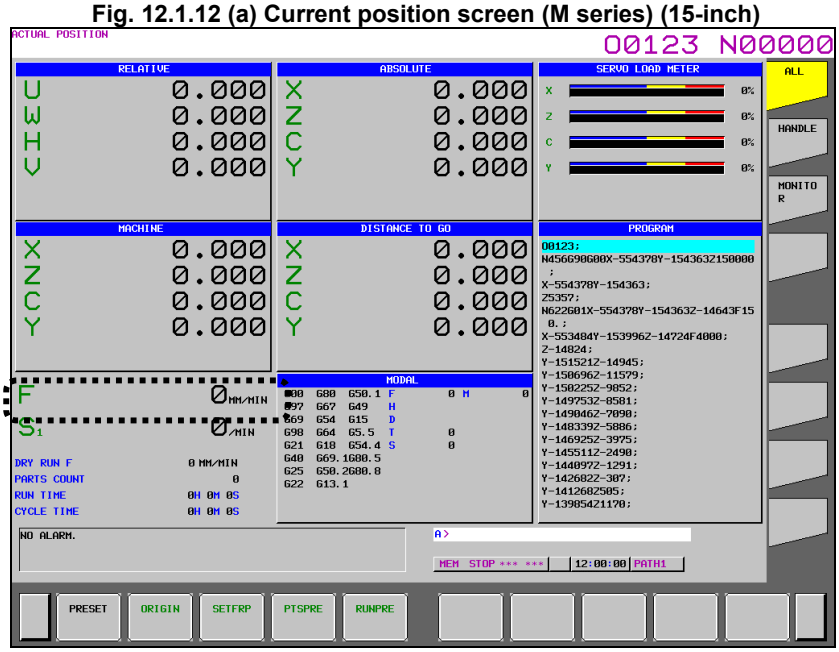


Fig. 12.1.12 (b) Current position screen (T series) (15-inch)

The actual feedrate is displayed in units of millimeter/min or inch/min (depending on the specified least input increment) under the display of the current position.

Explanation

- Actual feedrate value

The actual rate is calculated by the following expression:

$$Fact = \sqrt{\sum_{i=1}^{n} (fi)^2}$$

n : Number of axes
fi : Cutting feed rate in the tangential direction of each axis or rapid traverse rate
Fact : Actual feedrate displayed

The display unit:

mm/min (metric input).
inch/min (Inch input, two digits below the decimal point are displayed.)

- Actual feedrate display of feed per revolution

In the case of feed per revolution and thread cutting, the actual feedrate displayed is the feed per minute rather than feed per revolution.

- Actual feedrate display of rotary axis

In the case of movement of rotary axis, the speed is displayed in units of deg/min but is displayed on the screen in units of input system at that time. For example, when the rotary axis moves at 50 deg/min, the following is displayed: 50 mm/min (in metric input) or 0.50 inch/min (in inch input).

- Actual feedrate display on the other screen

The program check screen also displays the actual feedrate.

12.1.13 Display of Run Time and Parts Count (15-inch Display Unit)

The run time, cycle time, and the number of machined parts are displayed on the current position display screens.

Procedure for displaying run time and parts count on the current position display screen

Procedure

1 Press the function key

to display a current position display screen. At the location indicated by , a run time and parts count are displayed.

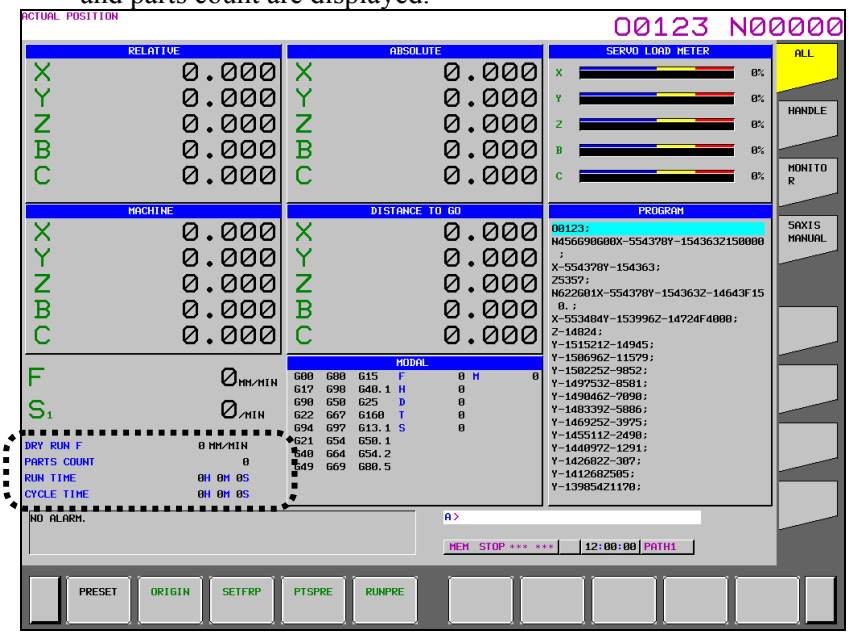


Fig. 12.1.13 (a) Current position screen (M series) (15-inch)

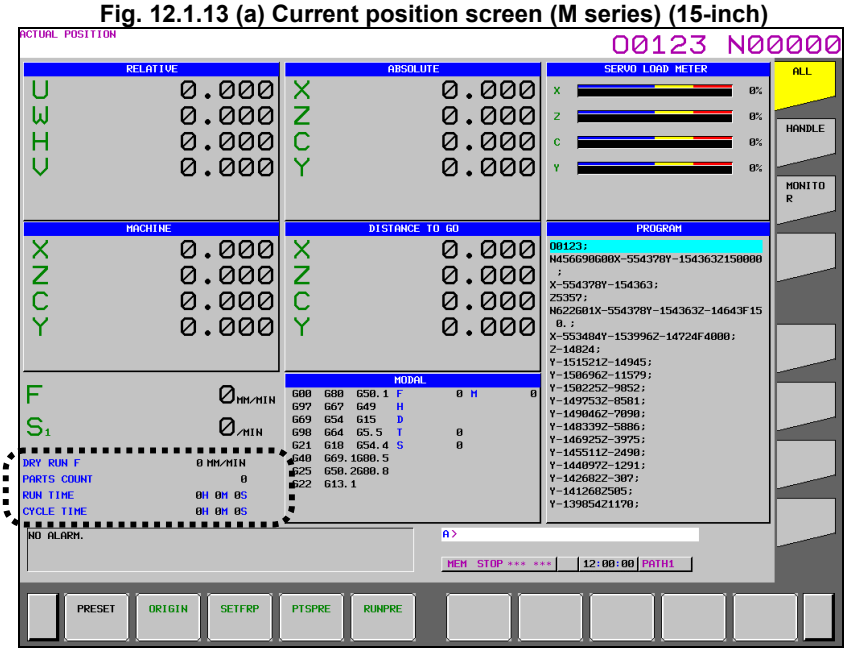


Fig. 12.1.13 (b) Current position screen (T series) (15-inch)

The number of machined parts (PART COUNT), run time (RUN TIME), and cycle time (CYCLE TIME) are displayed under the current position.

Explanation

- PART COUNT

Indicates the number of machined parts. The number is incremented each time M02, M30, or an M code specified by parameter No. 6710 is executed.

- RUN TIME

Indicates the total run time during automatic operation, excluding the stop and feed hold time.

- CYCLE TIME

Indicates the run time of one automatic operation, excluding the stop and feed hold time. This is automatically preset to 0 when a cycle start is performed at reset state. It is preset to 0 even when power is removed.

- Display on the other screen

Details of the run time and the number of machined parts are displayed on the setting screen. See III-12.3.18.

- Parameter setting

The number of machined parts and run time cannot be set on current position display screens.
The number of machined parts and run time cannot be set on current position display screens. They can be set by parameters Nos. 6711, 6751, and 6752 or on the setting screen.

- Incrementing the number of machined parts

Bit 0 (PCM) of parameter No. 6700 is used to specify whether the number of machined parts is incremented each time M02, M30, or an M code specified by parameter No. 6710 is executed, or only each time an M code specified by parameter No. 6710 is executed.

12.1.14 Setting the Floating Reference Position (15-inch Display Unit)

To perform floating reference position return with a G30.1 command, the floating reference position must be set beforehand.

Procedure for setting the floating reference position

Procedure

1 Press function key to display a screen used for displaying the current position.
2 Move the tool to the floating reference position by jogging.
3 Press horizontal soft key [SET FRP].
4 To register the floating reference positions for all axes, press soft key [ALL AXIS].
To register the floating reference position of a specific axis, enter the name of the axis (, , etc.), then press horizontal soft key [AXIS NAME]. Two or more names can be entered consecutively (e.g.,) [AXIS NAME]).
The above operation stores the floating reference position. It can be checked with parameter (No. 1244).
5 In step 3, the floating reference position along a specified axis can also be stored by entering the axis name (such as or) and pressing horizontal soft key [SET FRP].

Explanation

- Presetting the relative coordinate system

When bit 3 (FPC) of parameter No. 1201 is set to 1, the floating reference position can be registered, while the relative position display can be preset to 0.

12.1.15 Operating Monitor Display (15-inch Display Unit)

The load meter for each servo axis can be displayed. Also, the load meter and speed meter for a serial spindle can be displayed.
To enable this function, bit 5 (OPM) of parameter No. 3111 must be set to 1.

Procedure for displaying the operating monitor

Procedure

1 Press function key

.

2 Press vertical soft key [MONITOR].

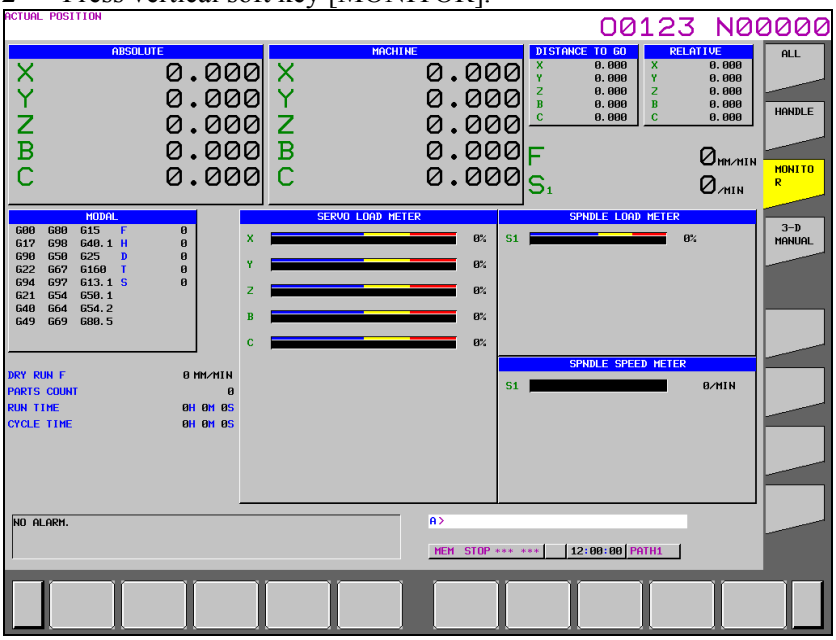


Fig. 12.1.15 (a) Operating monitor (M series) (15-inch)

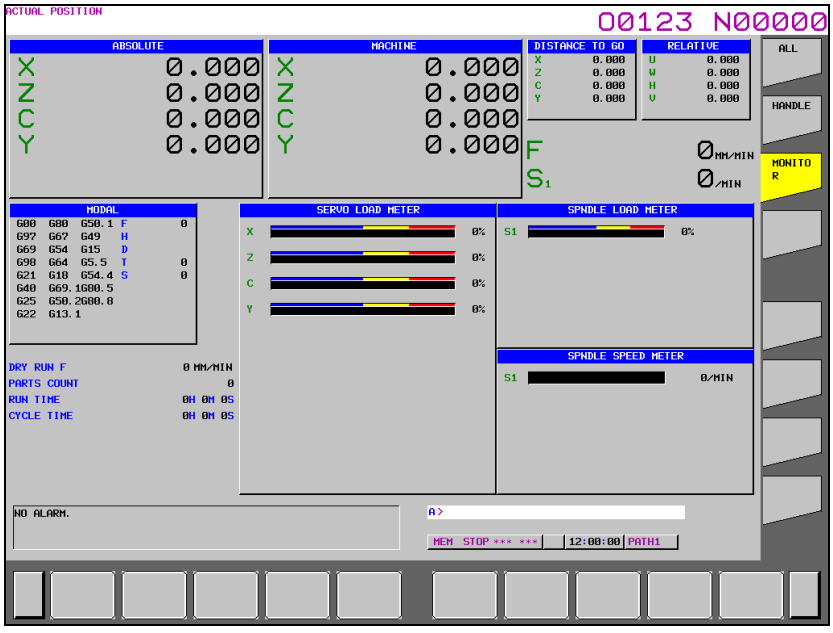


Fig. 12.1.15 (b) Operating monitor (T series) (15-inch)

Explanation

- Display of the servo axes

Servo axis load meters as many as the maximum number of controlled axes of the path can be displayed. One screen displays load meters for up to five axes at a time. By pressing the vertical soft key [MONITOR], load meters for the 6th axis and up are displayed.

- Display of the spindle axes

When serial spindles are used, the reading on the load meter and speedometer can be displayed only for the main serial spindle.

- Unit of graph

The bar graph for the load meter shows load up to 200% (only a value is displayed for load exceeding 200%).
The bar graph for the speedometer shows the ratio of the current spindle speed to the maximum spindle speed (100%).

- Load meter

The reading on the load meter depends on servo parameter No. 2086 and spindle parameter No. 4127.

- Speedometer

Although the speedometer normally indicates the speed of the spindle motor, it can also be used to indicate the speed of the spindle by setting bit 6 (OPS) of parameter No. 3111 to 1.
The spindle speed to be displayed during operation monitoring is calculated from the speed of the spindle motor (see the formula below).
The spindle speed can therefore be displayed, during operation monitoring, even when no position coder is used.
To display the correct spindle speed, however, the maximum spindle speed for each gear (spindle speed at each gear ratio when the spindle motor rotates at the maximum speed) must be set in parameters No. 3741 to No.3744.
The input of the clutch and gear signals for the first serial spindle is used to determine the gear which is currently selected. Control the input of the CTH1A and CTH2A signals according to the gear selection, by referring to the table below.
(Formula for calculating the spindle speed to be displayed)

$$\text{Spindle speed displayed during operation monitoring} = \frac{\text{Speed of spindle motor}}{\text{Maximum speed of spindle motor}} \times \text{Maximum spindle speed with the gear being used}$$

The following table lists the correspondence between clutch and gear selection signals CTH1A and CTH2A, used to determine the gear being used, and parameters:

CTH1A	CTH2A	Parameter	Serial spindle specification
0	0	=No.3741 (Maximum spindle speed with gear 1)	HIGH
0	1	=No.3742 (Maximum spindle speed with gear 2)	MEDIUM HIGH
1	0	=No.3743 (Maximum spindle speed with gear 3)	MEDIUM LOW
1	1	=No.3744 (Maximum spindle speed with gear 4)	LOW

The speed of the spindle motor and spindle can be displayed, during operation monitoring, only for the first serial spindle and the spindle switching axis for the first serial spindle. It cannot be displayed for the second spindle.

- Color of graph

If the value of a load meter exceeds 100%, the bar graph turns red.

12.1.16 Display of Three-dimensional Manual Feed (Tool Tip Coordinates, Number of Pulses, Machine Axis Move Amount) (15-inch Display Unit)

The absolute coordinates of the tool tip, the number of pulses, and a machine axis move amount based on three-dimensional manual feed are displayed.

Displaying the screen for three-dimensional manual feed

Procedure

1 Press function key

.

2 Press the vertical soft key [3-D MANUAL] to display the screen of three-dimensional manual feed.

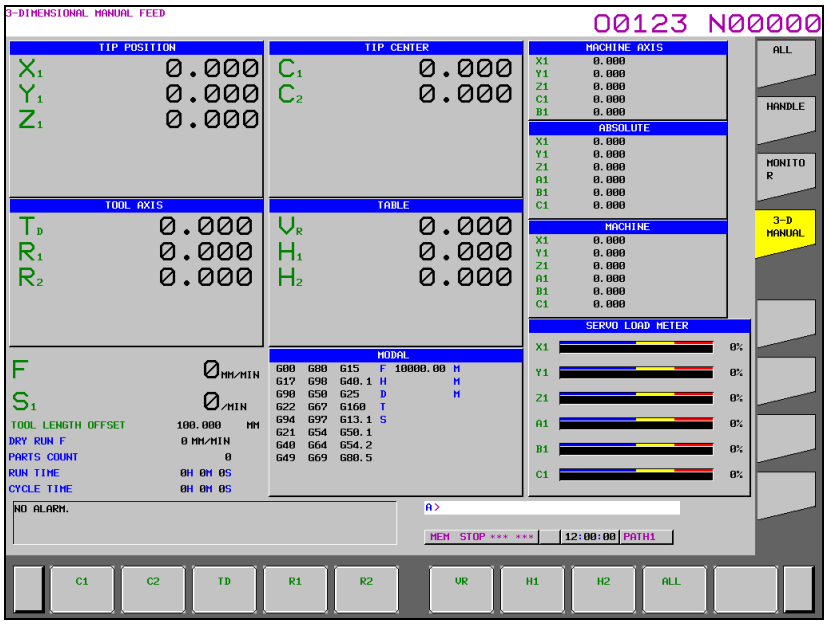


Fig. 12.1.16 (a) Three-dimensional manual feed screen(15-inch)

Explanation

- Tool tip position

The addresses of the three basic machine configuration axes for performing three-dimensional manual feed and the current position of the tool tip are displayed.

- Tool axis reference (number of pulses)

TD

The amount of travel in the tool axis direction in tool axis direction handle feed, tool axis direction jog feed, or tool axis direction incremental feed is displayed.
The unit is the least input increment of the axis in the direction specified by parameter No. 19697.

R1

The amount of travel in the first axis direction in tool axis right-angle direction handle feed, tool axis right-angle direction jog feed, or tool axis right-angle direction incremental feed is displayed.
The unit is the least input increment of the axis in the first axis direction normal to the direction specified by parameter No. 19697.

R2

The amount of travel in the second axis direction in tool axis right-angle direction handle feed, tool axis right-angle direction jog feed, or tool axis normal direction incremental feed is displayed.
The unit is the least input increment of the axis in the second axis direction normal to the direction specified by parameter No. 19697.

- Tool tip center (number of pulses)

C1

The angular displacement in tool tip center rotation handle feed, tool tip center rotation jog feed, or tool tip center rotation incremental feed for rotation of the first rotary axis is displayed. The unit is the least input increment of the first rotary axis. For the definition of the first rotary axis, see the description of parameter No. 19680.

C2

The angular displacement in tool tip center rotation handle feed, tool tip center rotation jog feed, or tool tip center rotation incremental feed for rotation of the second rotary axis is displayed. The unit is the least input increment of the second rotary axis. For the definition of the second rotary axis, see the description of parameter No. 19680.

- Table reference (number of pulses)

VR

The amount of travel in the table reference vertical direction in table reference vertical direction handle feed, table reference vertical direction jog feed, or table reference vertical direction incremental feed is displayed.
The unit is the least input increment of the axis in the direction specified by parameter No. 19697.

H1

The amount of travel in the first axis direction in table reference horizontal direction handle feed, table reference horizontal direction jog feed, or table reference horizontal direction incremental feed is displayed.
The unit is the least input increment of the axis in the first axis direction normal to the direction specified by parameter No. 19697.

H2

The amount of travel in the second axis direction in table reference horizontal direction handle feed, table reference horizontal direction jog feed, or table reference horizontal direction incremental feed is displayed.
The unit is the least input increment of the axis in the second axis direction normal to the direction specified by parameter No. 19697.

- Amounts of machine axis travel

The addresses of machine configuration axes used for three-dimensional manual feed and the sum of the amount of travel of each axis used for three-dimensional manual feed are displayed.
The values of the basic three axes (X-axis, Y-axis, and Z-axis), the first rotary axis, and the second rotary axis are displayed in this order.
For the definition of the first rotary axis and second rotary axis, see the description of parameter No. 19680.
When bit 0 (CLR) of parameter No. 13113 is set to 1, the displayed data is cleared by a reset.

- Absolute coordinates, machine coordinates

The absolute coordinates and machine coordinates of all axes are displayed. If too many axes are involved for display on one screen, the remaining axes can be displayed by pressing the vertical soft key [3-D MANUAL] for page feed.

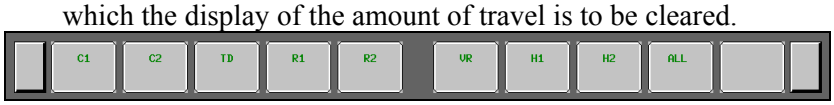
- F (feedrate)

- When bit 3 (CFD) of parameter No. 13113 is set to 0
  The composite feedrate at a control point on a linear axis or rotary axis is displayed.
- When bit 3 (CFD) of parameter No. 13113 is set to 1
  The feedrate of the tool tip is displayed.

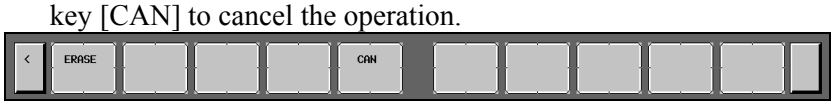
Operation

The display of the number of pulses can be cleared to 0 with horizontal soft keys.

1 Select the horizontal soft key corresponding to a function for which the display of the amount of travel is to be cleared.



2 Press horizontal soft key [ERASE] to clear the display of the amount of travel of the specified function, or press horizontal soft key [CAN] to cancel the operation.



12.2 SCREENS DISPLAYED BY FUNCTION KEY

Section 12.2, "SCREENS DISPLAYED BY FUNCTION KEY ", consists of the following subsections:

———— Screens of a 7.2/8.4/10.4-inch display unit
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———— Screens of a 15-inch display unit
12.2.10 Program Contents Display (15-inch Display Unit).....1521
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12.2.18 Screen for Assistance in Entering Tilted Working Plane Command (15-inch Display Unit).....1550

Screens of a 7.2/8.4/10.4-inch display unit

This section describes the screens displayed by pressing function key . The screens include a program editing screen, program folder list display screen, and screens for displaying the command states of the program currently being executed.

1. Program screen
2. Program folder screen
3. Next block display screen
4. Program check screen

On the program screen, you edit the program that is currently selected, and display the block that is currently executed during program operation. In MDI mode, you also edit an MDI operation program, and display the block that is currently executed.

12.2.1 Program Contents Display

Displays the program currently being executed in MEMORY mode.

Displaying the program being executed

Procedure

1 Press function key

to display the program screen.

2 Press chapter selection soft key [PROGRAM].
The cursor is positioned at the block currently being executed.

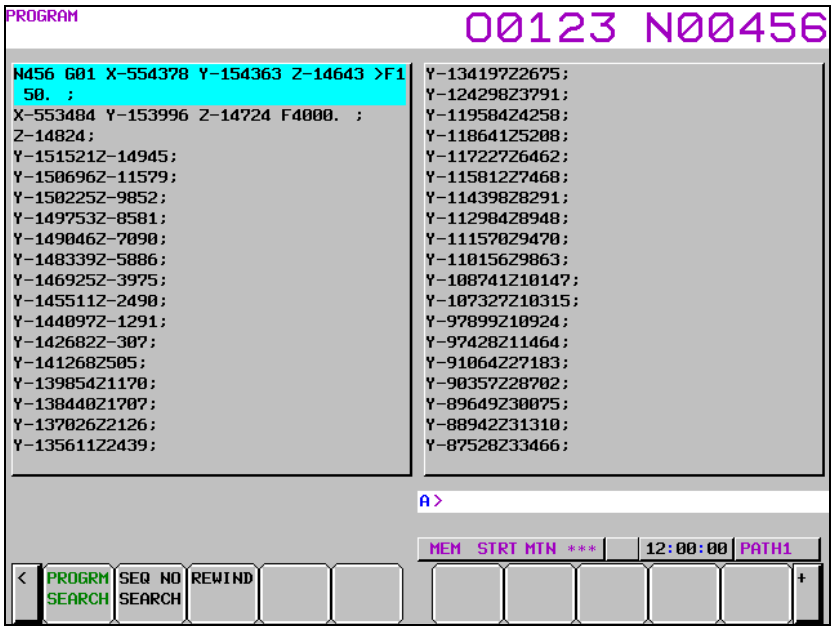


Fig. 12.2.1 (a) Screen for displaying the program being executed (10.4-inch)

12.2.1.1 Displaying the executed block

Overview

When the program being executed is displayed, one executed block can be displayed. This function can be enabled by setting bit 3 (FPD) of parameter No. 11308.

The program screen shows the following:

- Executed block
- Block being executed
- Look-ahead block(*)
- Non-look-ahead block

(*) For details on look-ahead, see the explanation "Program look-ahead" below.

The executed block is displayed on the top of the screen, and ">" is shown in front of the block being executed, across which the cursor is placed. Below the block being executed, look-ahead blocks are displayed. Below them, non-look-ahead blocks are displayed.
There is a dividing line between the look-ahead blocks and the non-look-ahead blocks.

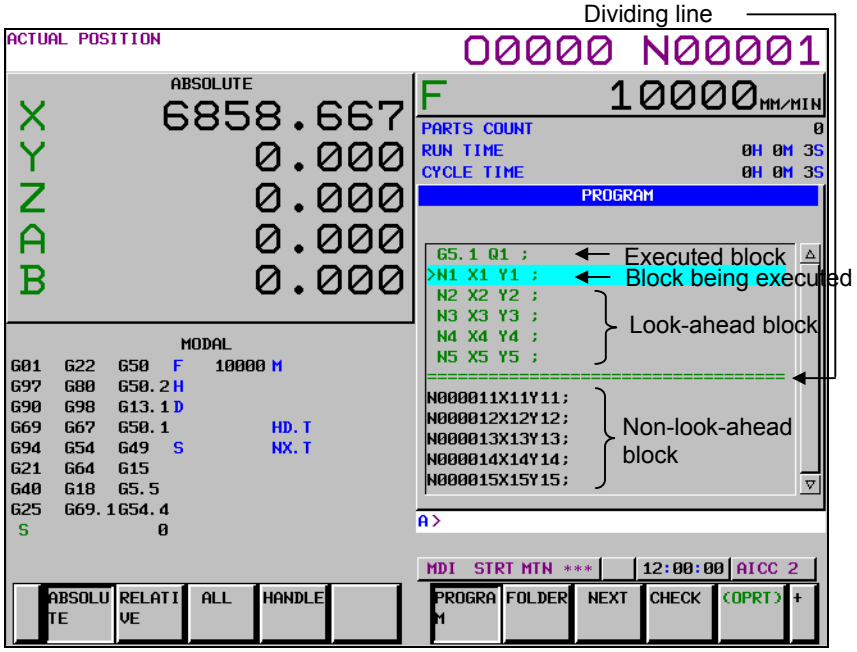


Fig. 12.2.1.1 (b) Screen for displaying the program being executed (MDI mode)

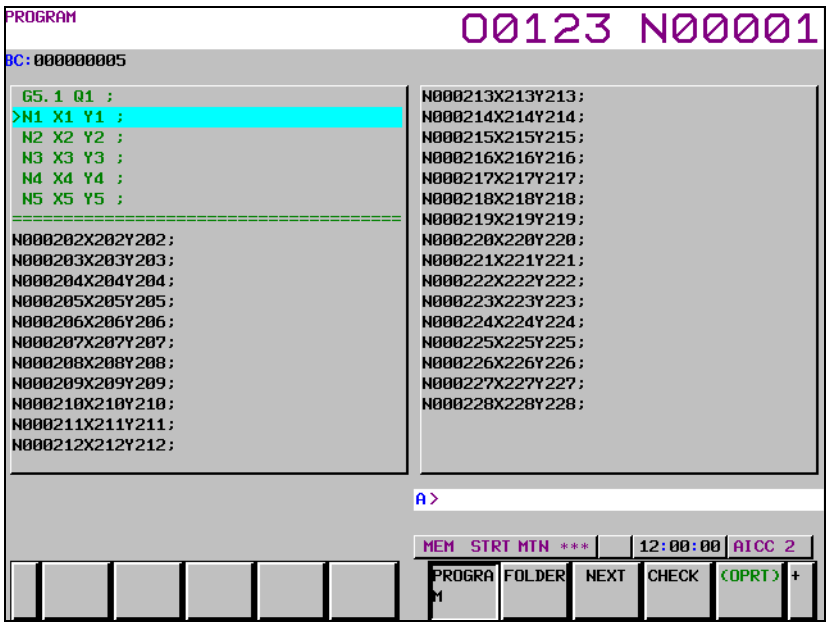


Fig. 12.2.1.1 (c) Screen for displaying the program being executed (MEM mode)

Explanation

- Program look-ahead

When an automatic operation cycle starts, a program is executed as described below.

<1> The command of one block is read from the specified program.
<2> The read command is analyzed and converted to executable data.
<3> The command of the block is executed.

During execution of the command, steps <1> and <2> are performed for the next block in advance. Upon completion of the current block, execution of the next block can be started immediately. This processing is referred to as look-ahead or buffering.

The number of look-ahead blocks depends on the processing status or parameters.

12.2.2 Editing a Program

A program can be edited in the EDIT mode.
Two modes of editing are available. One mode is word editing, which performs word-by-word editing. The other is character editing, which performs character-by-character editing.
For program creation and editing operation, see Chapter III-9, “CREATING PROGRAMS” and Chapter III-10, “EDITING PROGRAMS”.

Displaying the program editing screen

Procedure

1 Press function key

to display the program screen.

2 Press chapter selection soft key [PROGRAM].

- Word editing

Editing operations such as text insertion, modification, and deletion, and cursor movements are performed on a word-by-word basis.

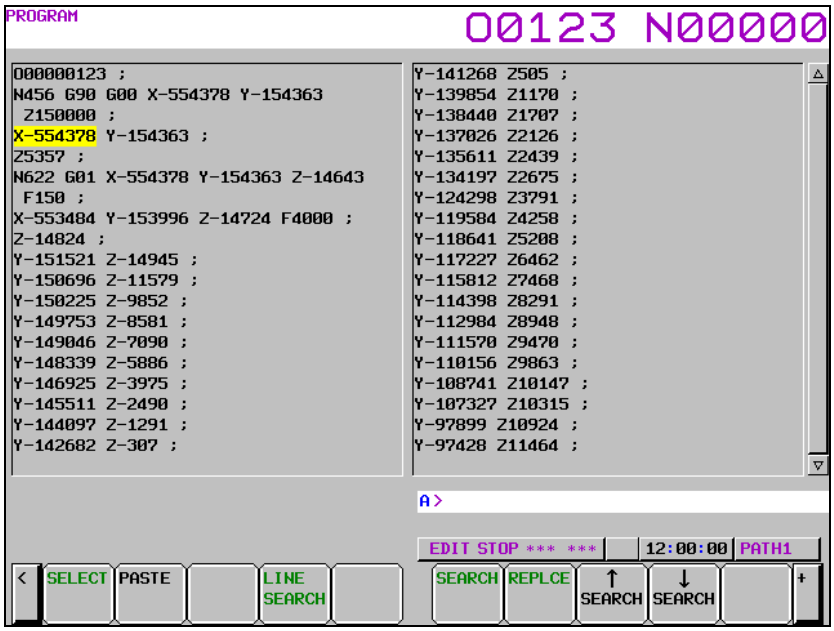


Fig. 12.2.2 (a) Program word editing screen (10.4-inch)

- Character editing

Program editing operations and cursor movements are performed on a character-by-character basis as with a general text editor.
Text is input directly to the cursor position instead of using the key input buffer.

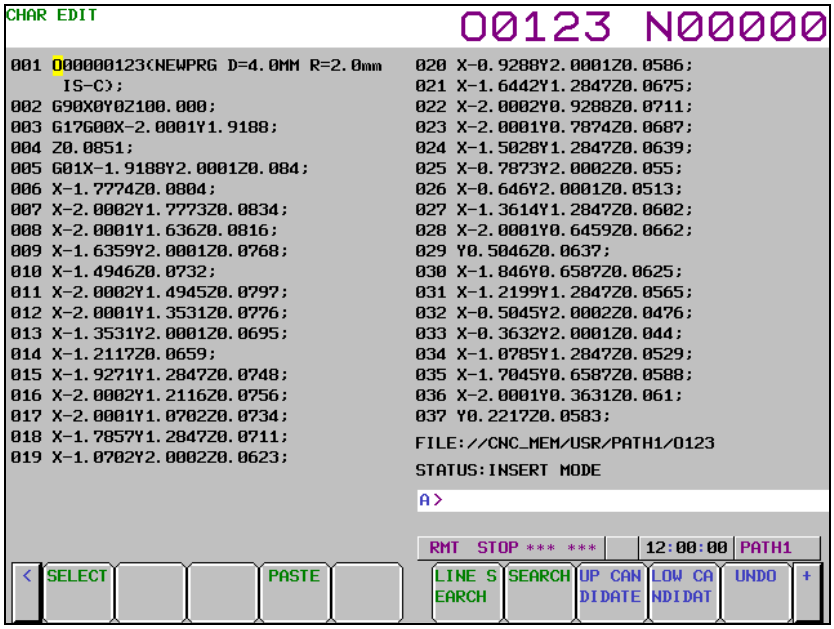


Fig. 12.2.2 (b) Program character editing screen (10.4-inch)

Switching between program editing modes

You can switch between word editing and character editing with soft keys.

Procedure

1 Press function key

to display the program screen.
2 Press chapter selection soft key [PROGRAM].
3 Press operation soft key [(OPRT)].
4 Pressing operation soft key [CHANGE EDITOR] switches the editing mode between word editing and character editing.

12.2.3 Program Screen for MDI Operation

During MDI operation or editing of an MDI operation program in the MDI mode, the program currently being executed mode is displayed. For MDI operation, see Section III-4.2, “MDI Operation”.

Procedure for displaying the program screen for MDI operation

Procedure

1 Press function key

.

2 Press chapter selection soft key [PROGRAM].
The program input from the MDI is displayed.

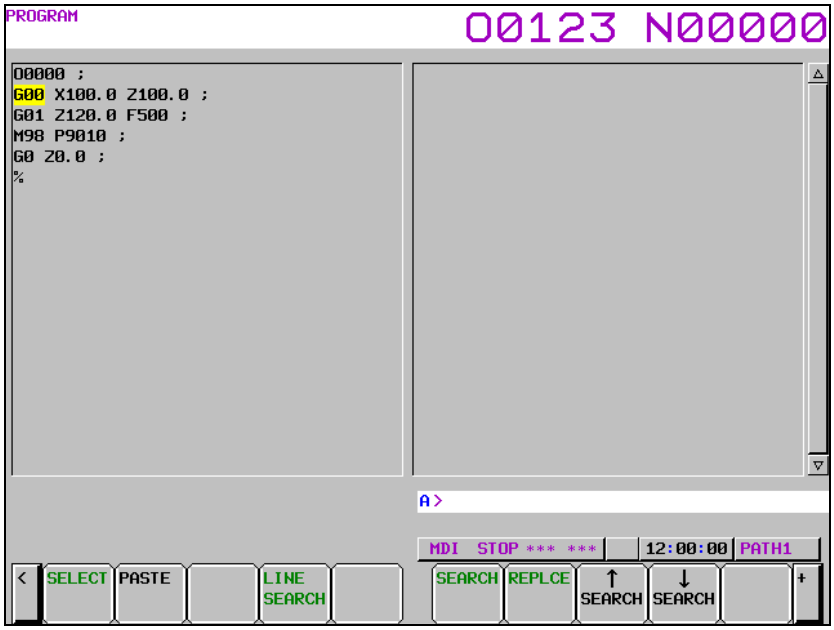


Fig. 12.2.3 (a) MDI operation program screen (10.4-inch)

12.2.4 Program Folder Screen

A list of programs registered in the program memory is displayed.
For the program folder screen, see Chapter III-11, “PROGRAM MANAGEMENT”.

Displaying the program folder screen

Procedure

1 Press function key

.

2 Press chapter selection soft key [FOLDER].

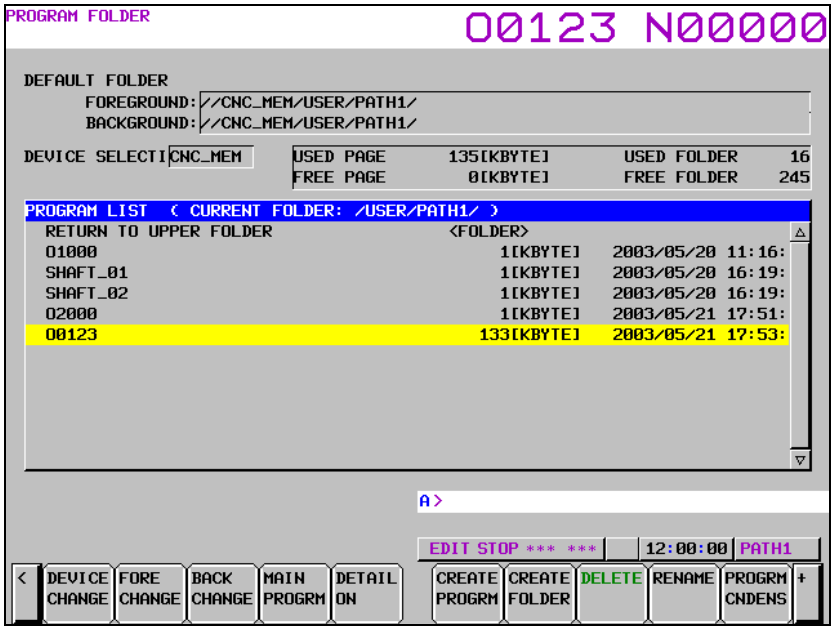


Fig. 12.2.4 (a) Program folder screen (10.4-inch)

12.2.4.1 Split display on the program folder screen

Overview

On the program folder screen, the folder information display can be split into two folder information views, upper and lower, as shown in the figure below. This function is enabled by setting bit 1 (MLD) of parameter No. 11318 to 1.

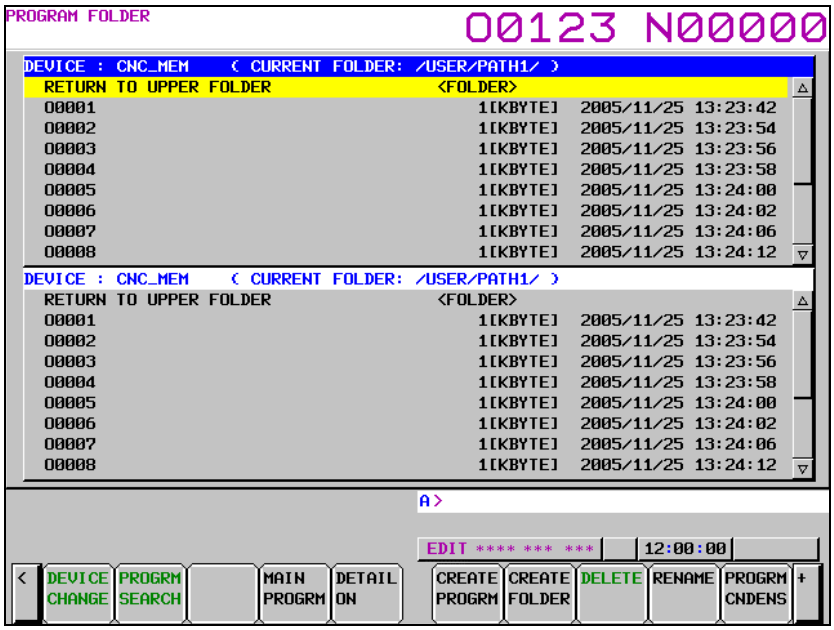
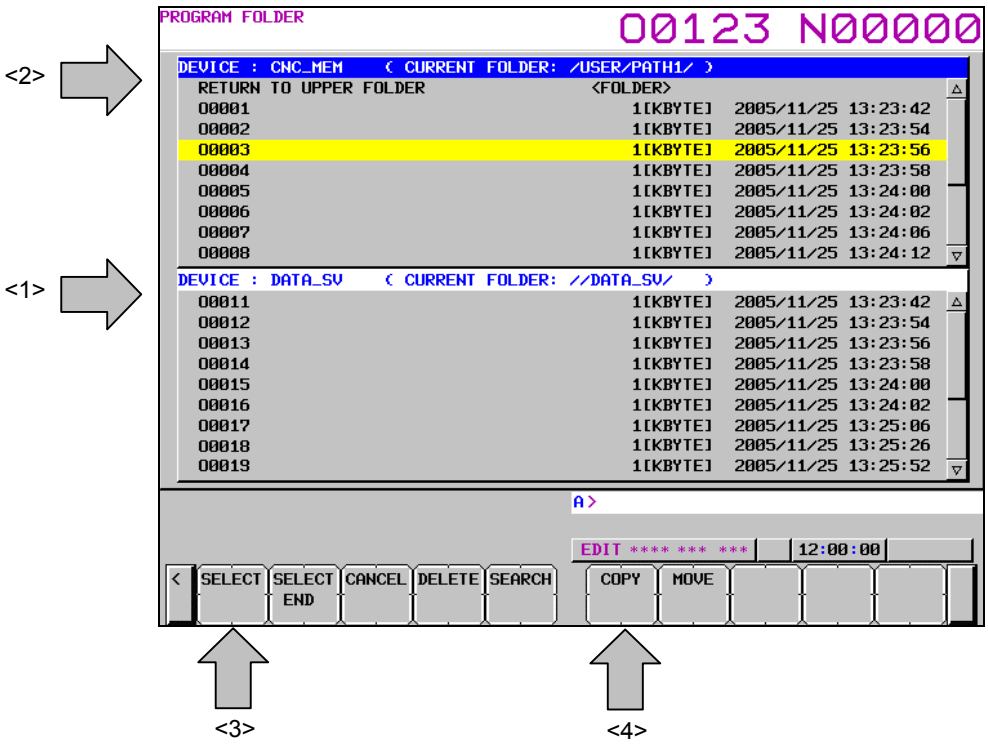


Fig. 12.2.4.1 (a) Program folder screen (split-display screen)(10.4-inch)

On the split folder display, file information about different sets of devices or folders can be shown in different views at the same time, so you can copy or move files easily.

Example of copying a file from the CNC MEM to the Data Server



Procedure

<1> Change to a destination folder on the Data Server.
<2> Change to a folder on the CNC_MEM that contains a file you want to copy.
<3> Select the file.
<4> Copy the file.

Procedure for switching to split display

Procedure

1 Press function key

to display the program folder screen.

2 Press soft key [(OPRT)].

3 Press continuous menu key twice.
Soft key [MULTI LIST] appears.

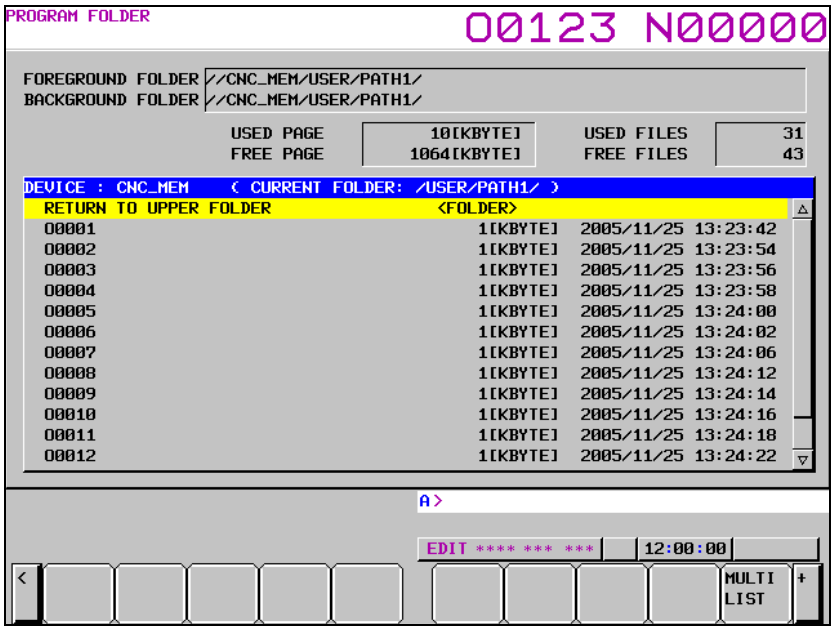


Fig. 12.2.4.1 (b) Program folder screen (normal screen)(10.4-inch)

4 Press soft key [MULTI LIST].
The folder information display is split into two folder views, upper and lower, which show the same folder information. Immediately after the splitting, the upper folder view becomes active for operations. At the same time, soft key [SINGLE LIST] appears.

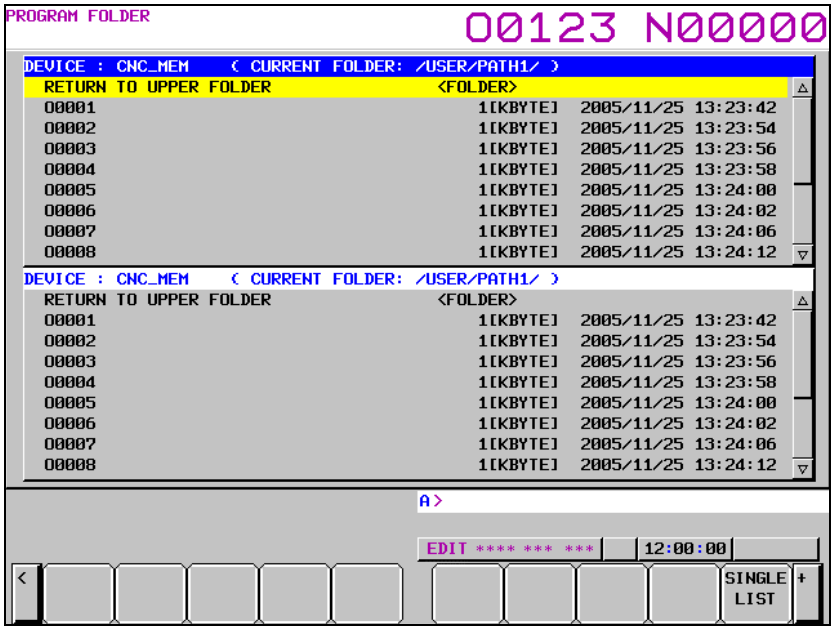


Fig. 12.2.4.1 (c) Program folder screen (split-display screen)(10.4-inch)

5 Pressing soft key [SINGLE LIST] takes you back to the original program folder display.
Then, the folders are shown which have been selected in the active view for operations on the split display.

Procedure for switching between active folder views for file operations on the split display

On the split folder display, you can switch between active folder views for file operations as described below.
In the active folder view for file operations, the cursor is displayed on the file list.

Procedure

1 When the upper folder view is active for file operations and you press MDI key and cursor key , the lower folder view becomes active for file operations.
2 When the lower folder view is active for file operations and you press MDI key and cursor key , the upper folder view becomes active for file operations.

NOTE

1 In step 1, even if you press MDI key and cursor key when the upper folder view is active for file operations, the lower folder view becomes active for file operations.
2 In step 2, even if you press MDI key and cursor key when the lower folder view is active for file operations, the upper folder view becomes active for file operations.

Explanation

- File operations on the split display

The following operations are allowed on the split display.

<1> You can select and display devices or folders individually in each folder view.

<2> You can copy or move files between folders selected in different folder views.

(For details, see the section on operations of copying/moving between devices.)

In the selected folder view, you can perform the same operations as on the normal screen.

- Path switching

When path switching is performed while the split display is provided for a multi-path system, the folder view in which the CNC MEM is selected as a device is changed to a view of the last selected folder for each path on the CNC MEM.

- Device information display

The device information as shown in the normal program folder screen (portion in a dotted box in the figure below) is not shown on the split display (Fig. 12.2.4.1 (c)).

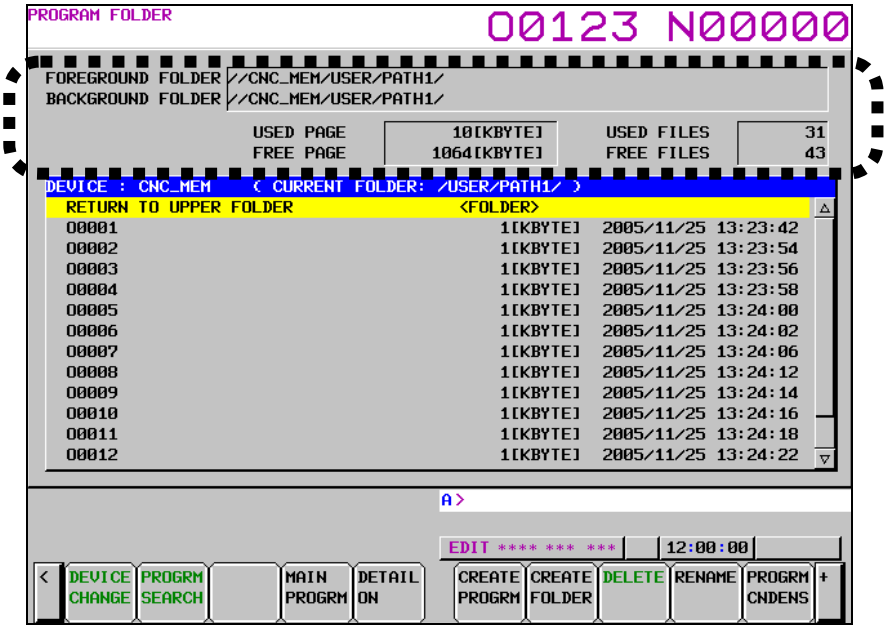


Fig. 12.2.4.1 (d) Program folder screen (normal screen)(10.4-inch)

You should see the device information on the normal screen.

Limitation

- Device that can be selected in both views at the same time on the split display

The same device of any of the following types can be selected in both views at the same time on the split display.

- CNC MEM
- MEM CARD(binary format)
- Data Server

For other types, the same device cannot be selected for display in the folder views at the same time.

If any other type of device is selected when split display starts, the upper and lower folder views cannot show the same sets of folders due to the above limitation. Therefore, the lower folder view shows the folder selected as a foreground folder.

- Display units that support the split display

The split display is supported on 10.4-inch and 15-inch display units.

12.2.5 Next Block Display Screen

Displays the block currently being executed and the block to be executed next.

Procedure for displaying the next block display screen

Procedure

1 Press function key

.

2 Press chapter selection soft key [NEXT].
The G codes, addresses, command values specified in the block currently being executed and the next block are displayed.

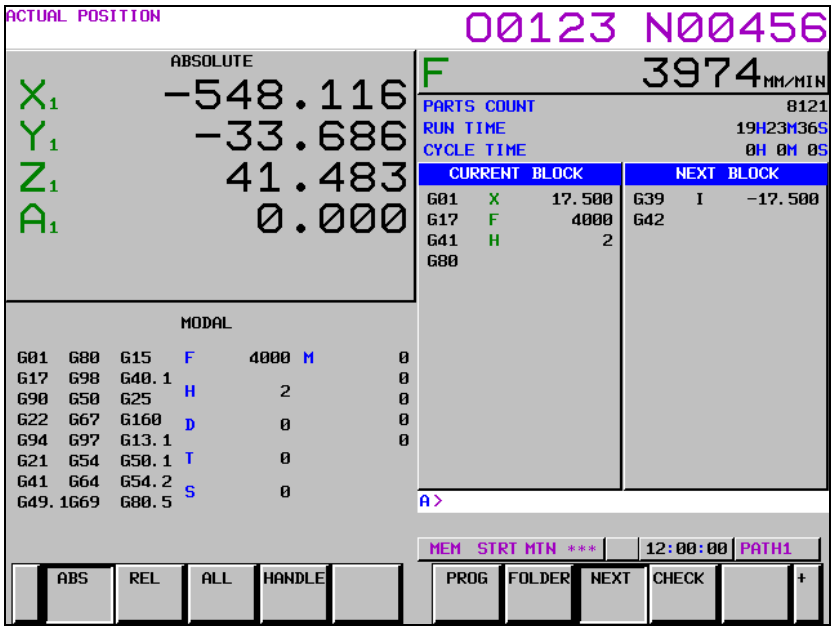


Fig. 12.2.5 (a) Next block display screen (10.4-inch)

12.2.6 Program Check Screen

Displays the program currently being executed, current position of the tool, and modal data.

Procedure for displaying the program check screen

Procedure

1 Press function key

.

2 Press chapter selection soft key [CHECK].
The program currently being executed, current position of the tool, and modal data are displayed.

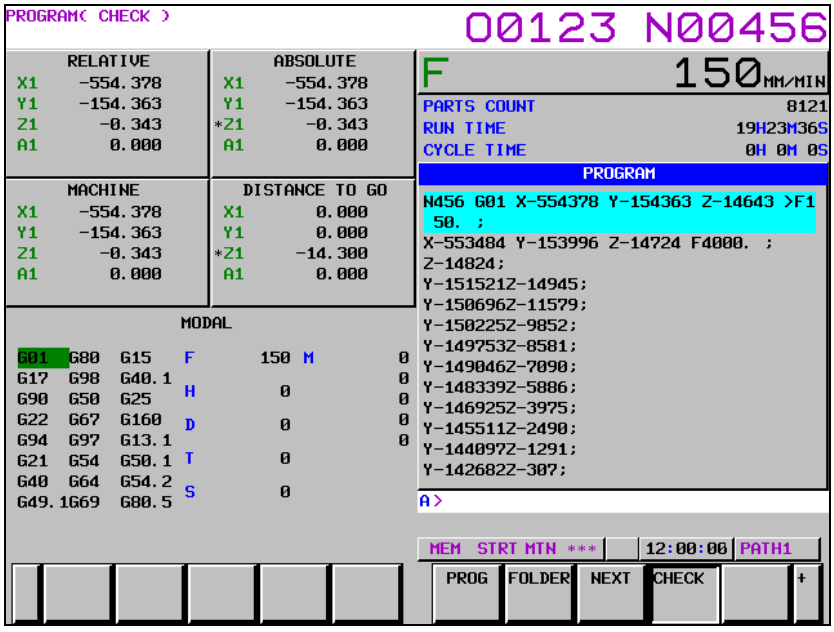


Fig. 12.2.6 (a) Program check screen (10.4-inch)

Explanation

- Program display

The program currently being executed is displayed.
The block being executed is displayed in reverse video.

- Current position display

The current position in the relative coordinate system, workpiece coordinate system, and machine coordinate system, and the remaining amount of travel are displayed.

- Modal G codes

Up to 24 modal G codes are displayed.

12.2.7 Background Editing

Editing one program during execution of another program is referred to as background editing. You can perform the same edit operations in the background as those in normal editing (foreground editing).

On a 10.4-inch or 15-inch display, you can perform background editing for the programs displayed in multiple tiled windows. You can switch from a program displayed in one of the tiled windows to another to perform copy, paste, and other edit operations, which allows you to edit programs efficiently.
You can edit two programs simultaneously on a 10.4-inch display or four programs on a 15-inch display.

Function

- Background editing

A program other than the currently selected program can be edited. Background editing can be performed in any mode.

- Program selected in the foreground

If the program selected in the foreground is specified as a program to be edited in the background, background editing is started in the read-only mode. The text at an arbitrary position of the program currently being executed can be checked.

- Switching from the program directory screen to the editing screen

By selecting a program from the program folder screen, background editing can be started.
On the program folder screen, move the cursor to a program to be edited in the background, then press the MDI key

. The screen display switches to the background editing screen.
Background editing can be started without entering a program name.

- Background editing on a multi-path system

Background editing does not depend on the path.
Even if the path to be operated is changed using the path selection signal, the background editing currently being executed is just continued.

Display

When background editing starts, the ordinary program editing screen switches to the background editing screen.
When two or more programs are edited in the background, the screen is split to display these programs. For a 10.4-inch display, you can edit two programs simultaneously; for a 15-inch display, you can edit four programs simultaneously.

- Word editing

Fig. 12.2.7 (a) shows background word editing performed simultaneously for two programs (right and left programs). On the status line at the top of the window for each program, the program name and "BG-EDIT" (indicating background editing in progress) are displayed.
The status line of the program being edited is displayed in reverse video.

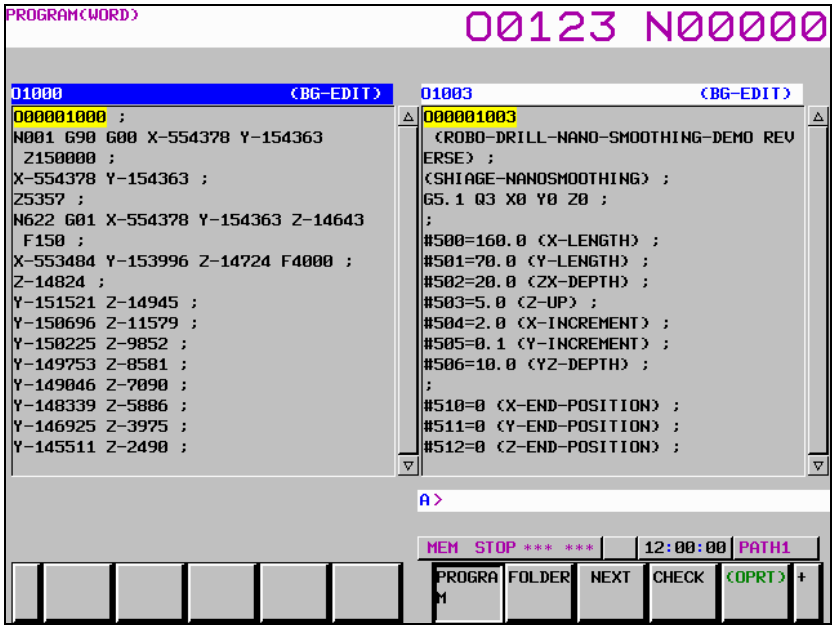


Fig. 12.2.7 (a) Background editing screen (word editing) (10.4-inch)

- Character editing

Fig. 12.2.7 (b) shows background character editing performed simultaneously for two programs (right and left programs). Similarly to word editing, at the top of the window for each program, the status line is displayed. In addition, the current input mode (INSERT MODE or OVERWRITE MODE) is displayed at the upper right of the screen for character editing.
The status line of the program being edited is displayed in reverse video.

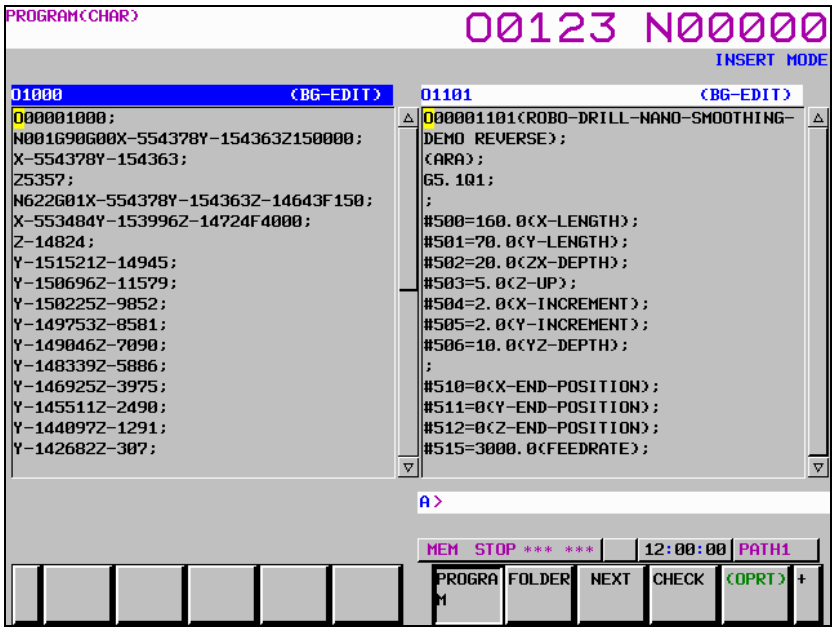


Fig. 12.2.7 (b) Background editing screen (character editing) (10.4-inch)

- Editing status

The following items are displayed on the status line and program editing area according to the background editing status.

Editing status	Displayed items
No program selected	(BG-EDIT) “NO PROGRAM” is displayed in the editing area.
Program opened	program-name + (BG-EDIT)
Read-only program opened	program-name + (BG:READ ONLY) The contents of the program are displayed in green.

Starting background editing from the editing screen

Procedure

Method 1

1 Press function key .
2 Press soft key [PROGRAM].
3 Press soft key [(OPRT)], then soft key [BG EDIT].
4 Press soft key [PROGRM SEARCH] to select a program to be edited.

Method 2 (only for word editing)

1 Press function key .
2 Press soft key [PROGRAM].
3 Key in the name of a program to be edited in the background.
4 Press soft key [BG EDIT].

When background editing has already started and editing start operation is performed again, the program being edited and newly specified program are edited simultaneously.

Explanation

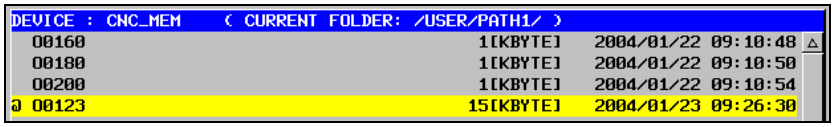
When a program name is input in the key-in buffer, background editing for the program starts. When the specified program is not found, a new program is created and background editing starts.
When background editing is started with no program name specified, new background editing starts in the no program state. Perform a program search or create a new program.
When character editing is selected, background editing starts in the no program state first. Perform a program search or create a new program.

Starting background editing from the program folder screen

By selecting a program from the program folder screen, background editing can be started. The cursor is used to select a program. You do not need to enter a program name.

Procedure

1 Press function key PROG.
2 Press soft key [FOLDER].
3 Move the cursor to a program to be edited.
4 Press the MDI key INPUT. The screen display is switched to the background editing screen.



Background editing operation

- Editing operation

The same editing operations as performed in the foreground can be performed.

- Switching from a program to another for editing

To switch from a program to another for editing when editing multiple programs simultaneously in the background, press SHIFT key, then ← or →.

- Editing operation in the read-only mode

Cursor movement and page switching are possible. However, program editing is disabled.

- Copying and pasting text between programs

Text in a program selected and copied during program editing is kept held after the program to be edited is changed. You can paste text in a program onto another program.

Background editing end operation

Background editing can be ended using the procedure described below.
The procedure for ending background editing of one program and that for ending all background editing of multiple programs are shown below.

- Ending editing of one program

1 Press function key .
2 Press soft key [PROGRAM].
3 Select a program for which to end editing.
4 Press soft key [(OPRT)], then soft key [BG END].

- Ending editing of all programs

1 Press function key .
2 Press soft key [PROGRAM].
3 Press soft key [(OPRT)], then soft key [BG ALL END].

To return to ordinary foreground editing, end all background editing. If at least one program is being edited in the background, you cannot return to foreground editing.

12.2.8 Stamping the Machining Time

The execution times of the most recently executed ten programs can be displayed in hours, minutes, and seconds.
The calculated machining time can be inserted as a comment of the program to check the machining time on the program folder screen.

Procedure for stamping the machining time

Procedure

- Displaying the machining time

1 Press function key PROG.
2 Press the continuous menu key ▷ several times to display soft key [TIME].
3 Press soft key [TIME]. The machining time display screen appears.

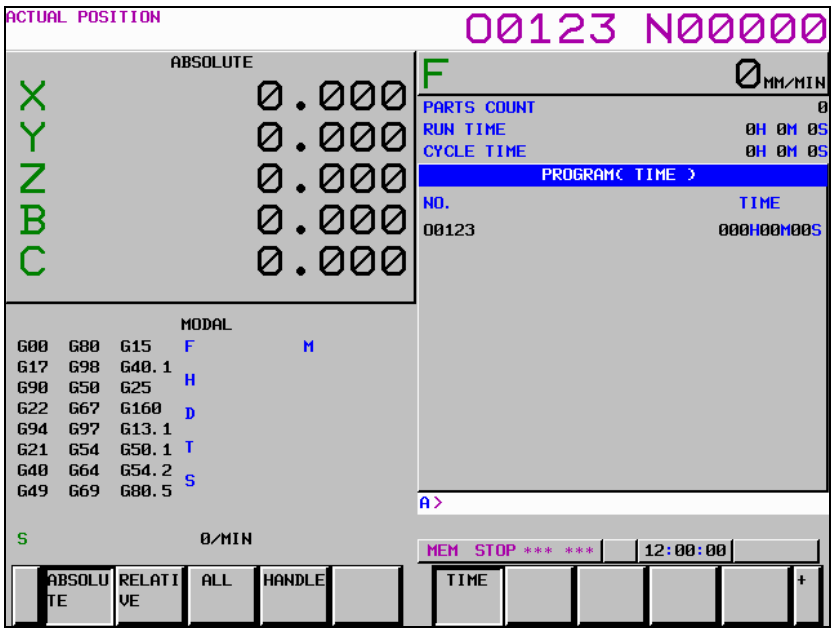


Fig. 12.2.8 (a) Machining time display screen (10.4-inch)

- Calculating the machining time

1 Select the memory operation mode, then press the RESET key.
2 Select the program screen, then select a program of which machining time you want to calculate.
3 Execute the program to perform actual machining.
4 When the RESET key is pressed or, M02 or M30 is executed, the machining time count operation stops. When the machining time display screen is selected, the program number of the stopped main program and its machining time are displayed.

5 The following figure shows the screen when the machining times of the ten main programs O0020, O0040, …, and O0200 are displayed and the screen when the machining time of O0220 is newly calculated after that.

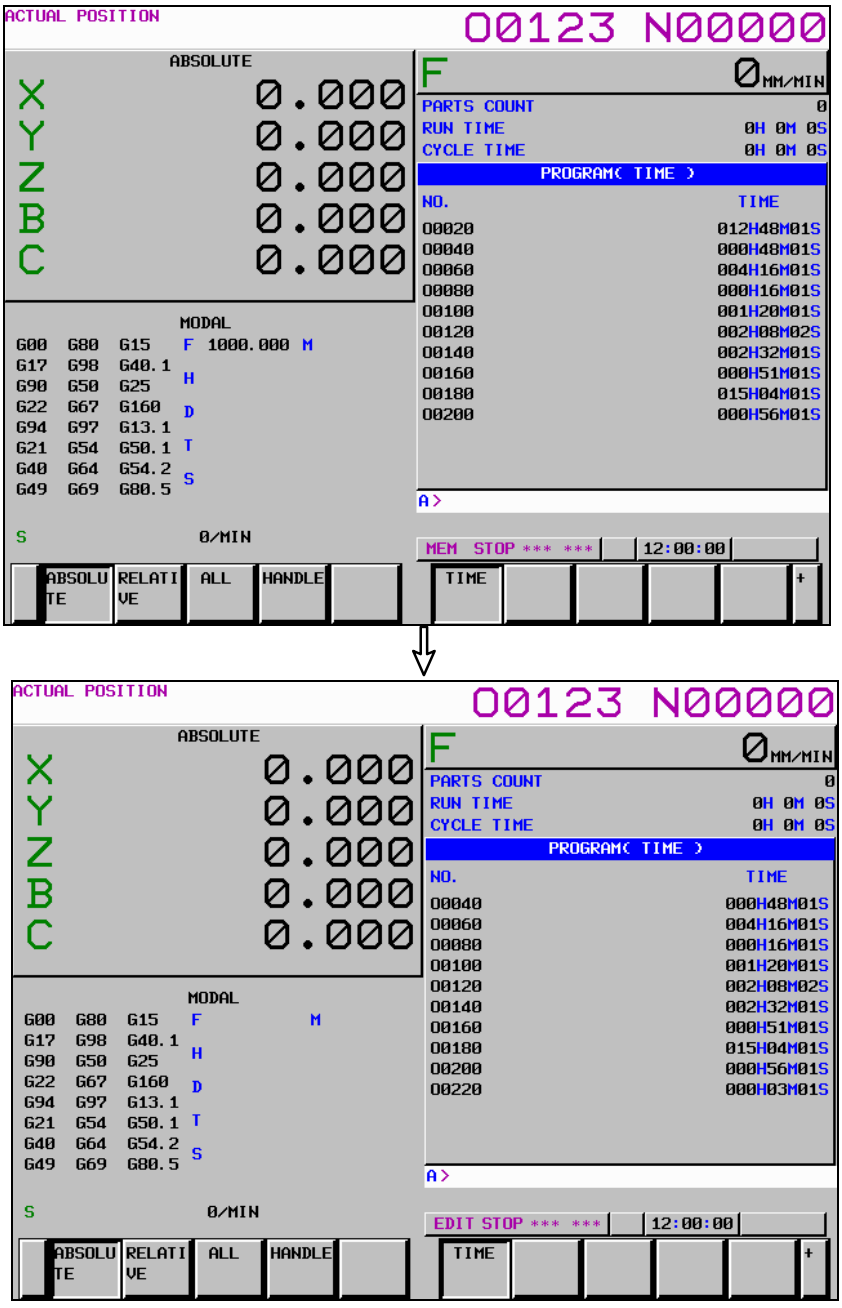


Fig. 12.2.8 (b) Stamping the machining time (10.4-inch)

Procedure for inserting the machining time on the program screen

Procedure

You can display the machining time of a program as a comment of the program. The procedure is shown below:

1 To insert the calculated machining time of a program as a comment, the machining time of the program must be displayed on the machining time display screen. Before stamping the machining time of the program, check that the program number is displayed on the machining time display screen.
2 Set the edit mode or background editing state and select the program screen. Then, select a program of which machining time you want to insert.
3 Suppose that the machining time of O0100 is displayed on the machining time display screen.
Press soft key [(OPRT)] to display the operation soft keys. Then, press the continuous menu key ▷ several times to display soft key [INSERT TIME]. Press soft key [INSERT TIME]. The beginning of the program is displayed and the machining time of the program is inserted after the program name.

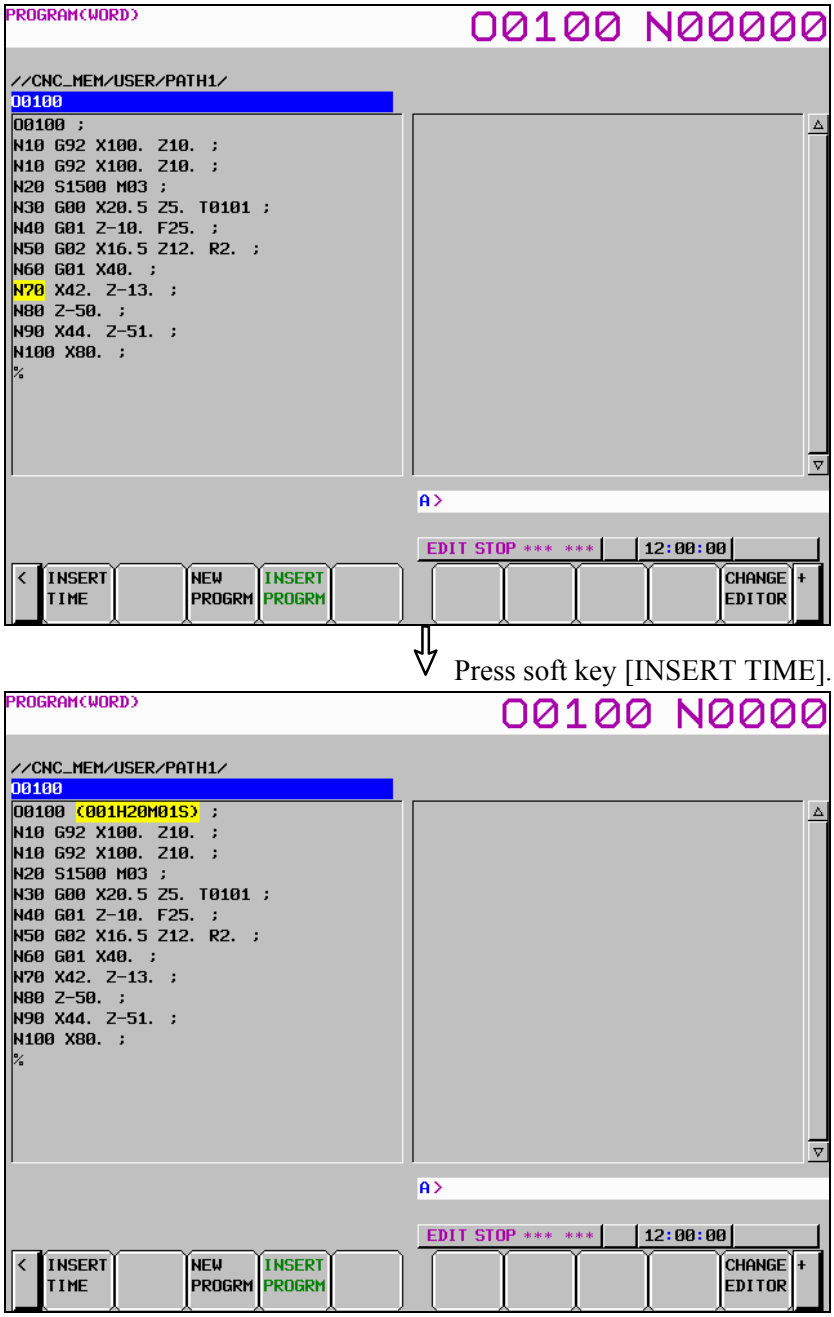


Fig. 12.2.8 (c) Program screen (10.4-inch)

4 If a comment is written in the block containing the program number of a program of which machining time is to be inserted, the machining time is inserted after the comment.

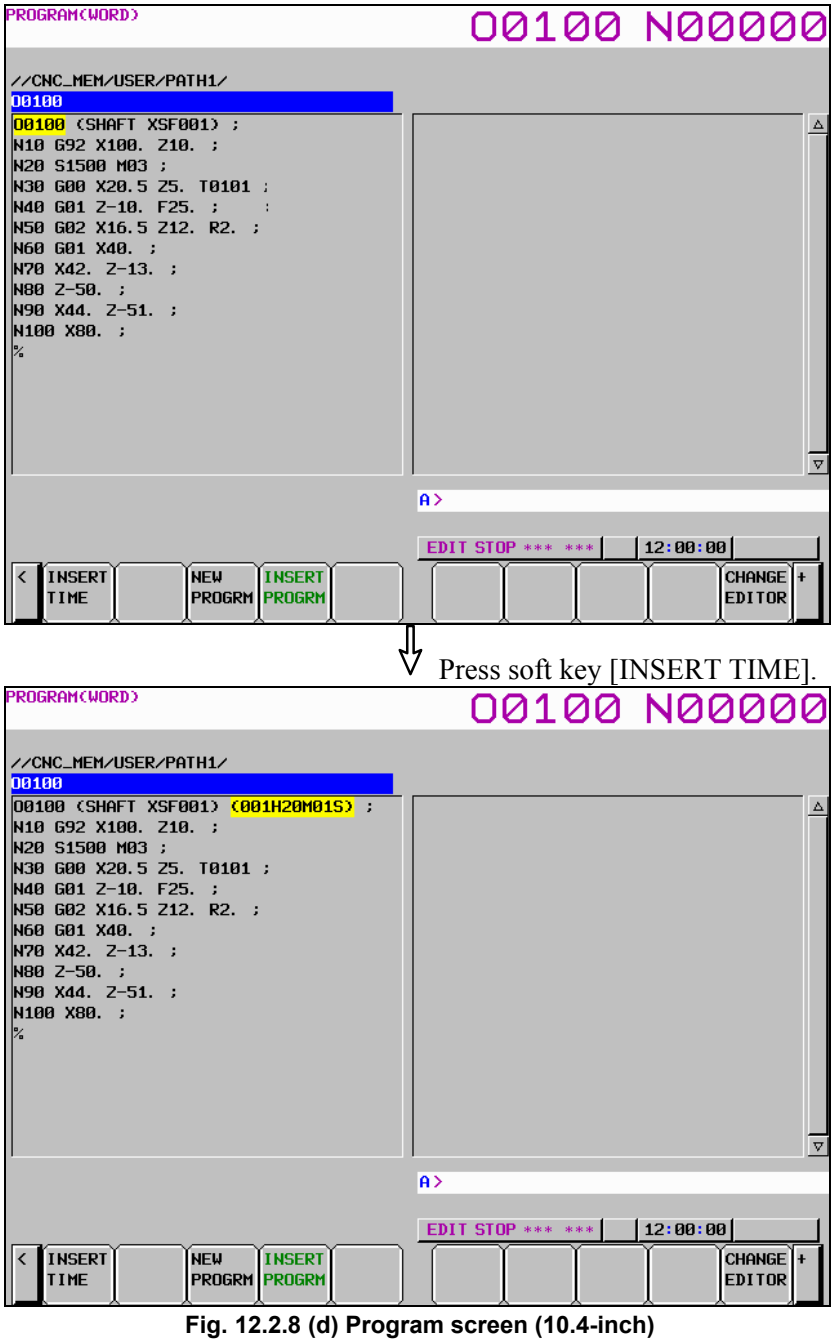


Fig. 12.2.8 (d) Program screen (10.4-inch)

Display on the program folder screen

The machining time of a program inserted in the program as a comment is displayed after the existing comment of the program on the program folder screen.

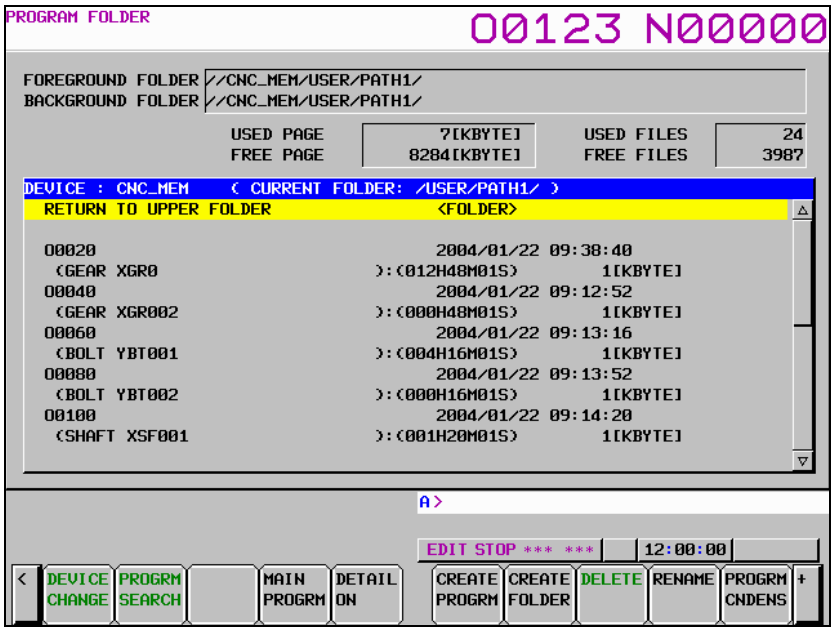


Fig. 12.2.8 (e) Program folder screen (10.4-inch)

Explanation

- Machining time

The machining time is counted from the initial start after a reset in the memory operation mode to the next reset. If a reset is not performed during operation, the machining time is counted from the start to M02 or M30. The time during which operation is held is not counted, but the time used to wait for the completion of the M, S, T, and/or B functions is counted, however.

- Stamping the machining time

You can insert (stamp) the displayed machining time in a program stored in memory as a comment. The machining time is inserted as a comment after the program number.

- Program folder

You can display the machining time inserted after a program number on the program folder screen. This operation lets you know the machining time of each program to use the time as useful reference data when planning processes on site.

Limitation

- Alarm

The execution of a program may be held by an alarm during the machining time count operation. In this case, the machining time is counted until the alarm is released by a reset.

- M02

It may be specified that M02 does not reset the CNC, but returns completion signal FIN to the CNC to restart the program from the beginning successively (bit 5 (M02) of parameter No. 3404 is set to 0). In this case, when M02 returns completion signal FIN, the machining time count operation stops.

- Stamping the machining time

When an attempt is made to stamp the machining time of a program in the program, the machining time of the program may not be displayed on the machining time display screen. In this case, the machining time cannot be inserted in the program even if soft key [INSERT TIME] is pressed.

- Correcting the machining time

If an incorrect machining time is calculated (such as when a reset is made during the execution of a program), reexecute the program to calculate the correct machining time. The same program number may be displayed in two or more lines on the machining time display screen. In this case, the latest calculated machining time is inserted in the program.

- States of the stamped machining time

In the following states, the stamped machining time is displayed on the program folder screen as shown below.

1 When the comment of a program is longer than 16 characters
The 17th and subsequent characters of the comment are discarded and the machining time display field is left blank.

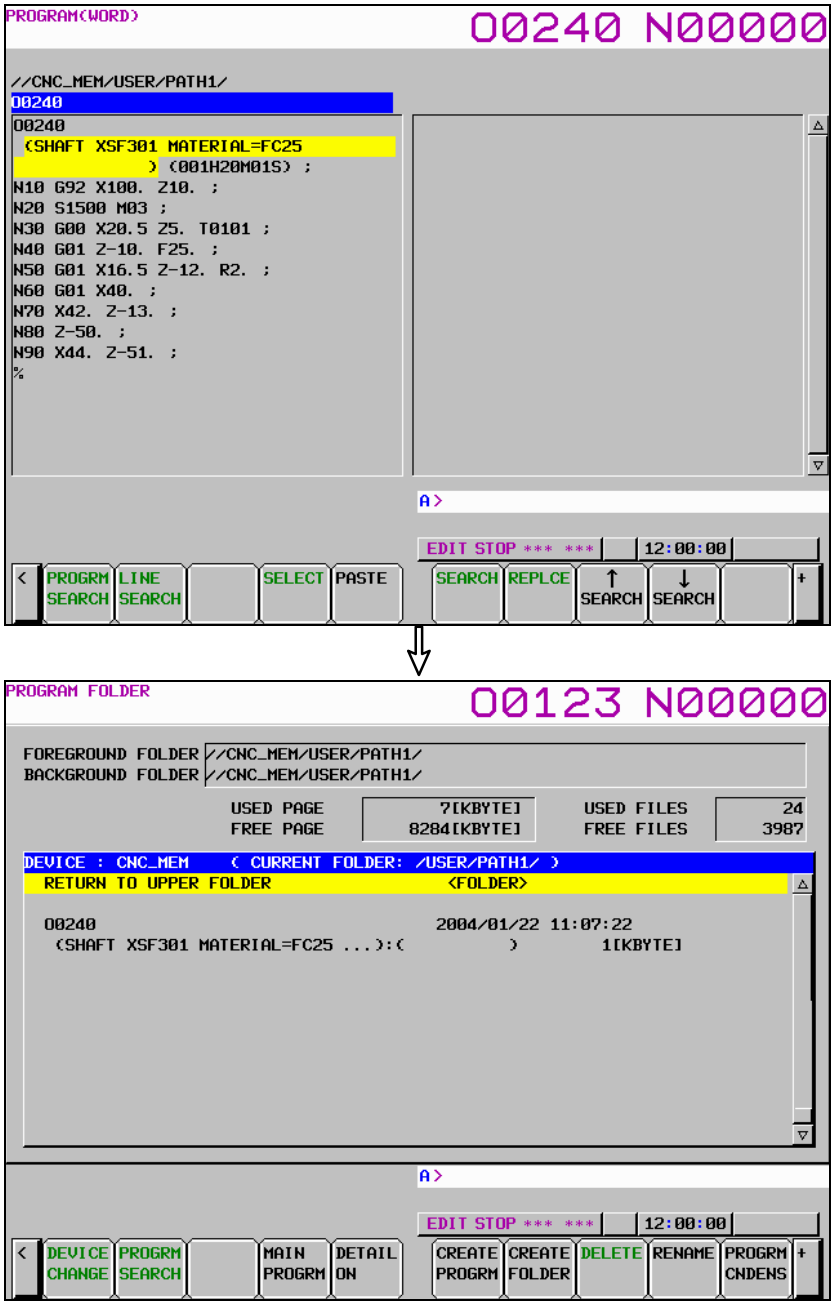


Fig. 12.2.8 (f) 1. When the comment of a program is longer than 16 characters (10.4-inch)

2 When two or more machining times are stamped
The first machining time is displayed.

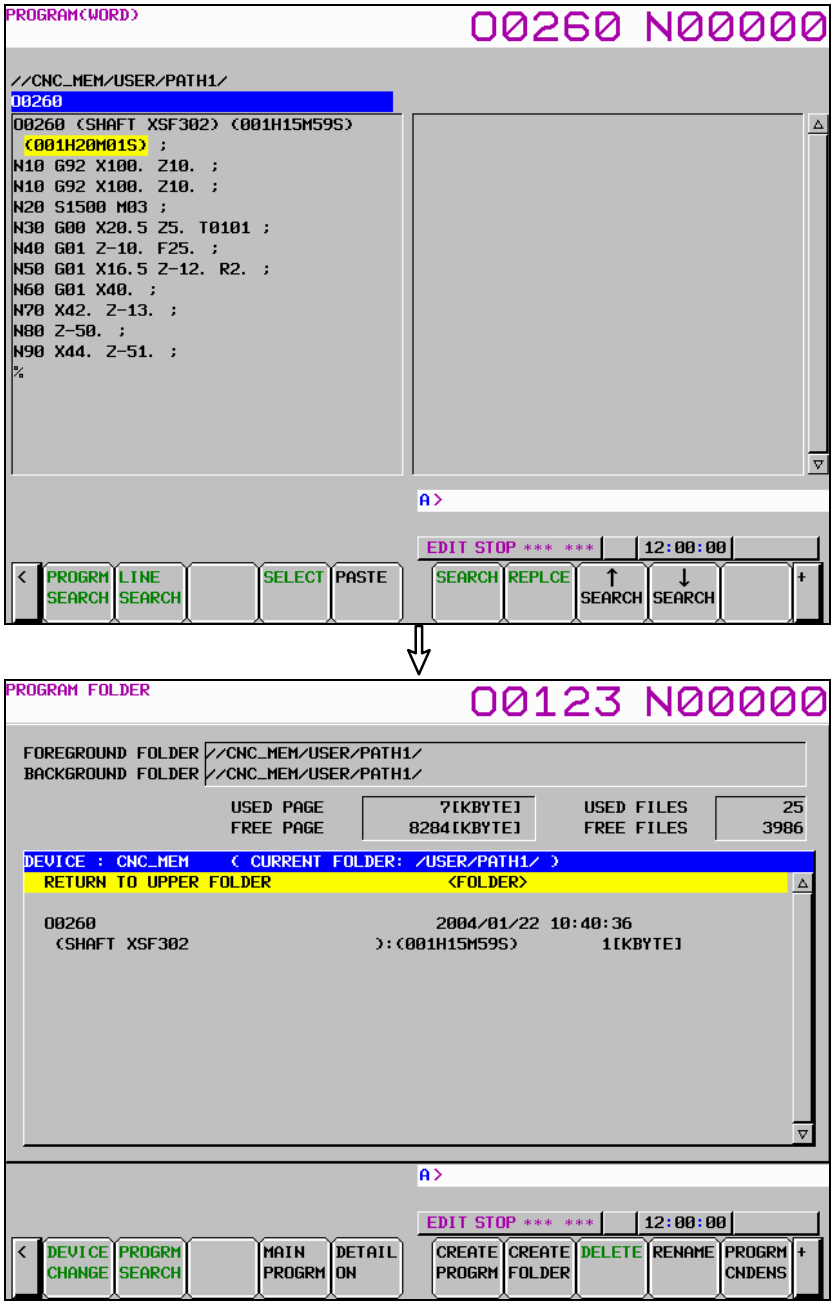


Fig. 12.2.8 (g) When two or more machining times are stamped (10.4-inch)

3 When the format of an inserted machining time is not “hhhHmmMssS” (H following a 3-digit number, M following a 2-digit number, and S following a 2-digit number, in this order) The machining time display field is left blank.

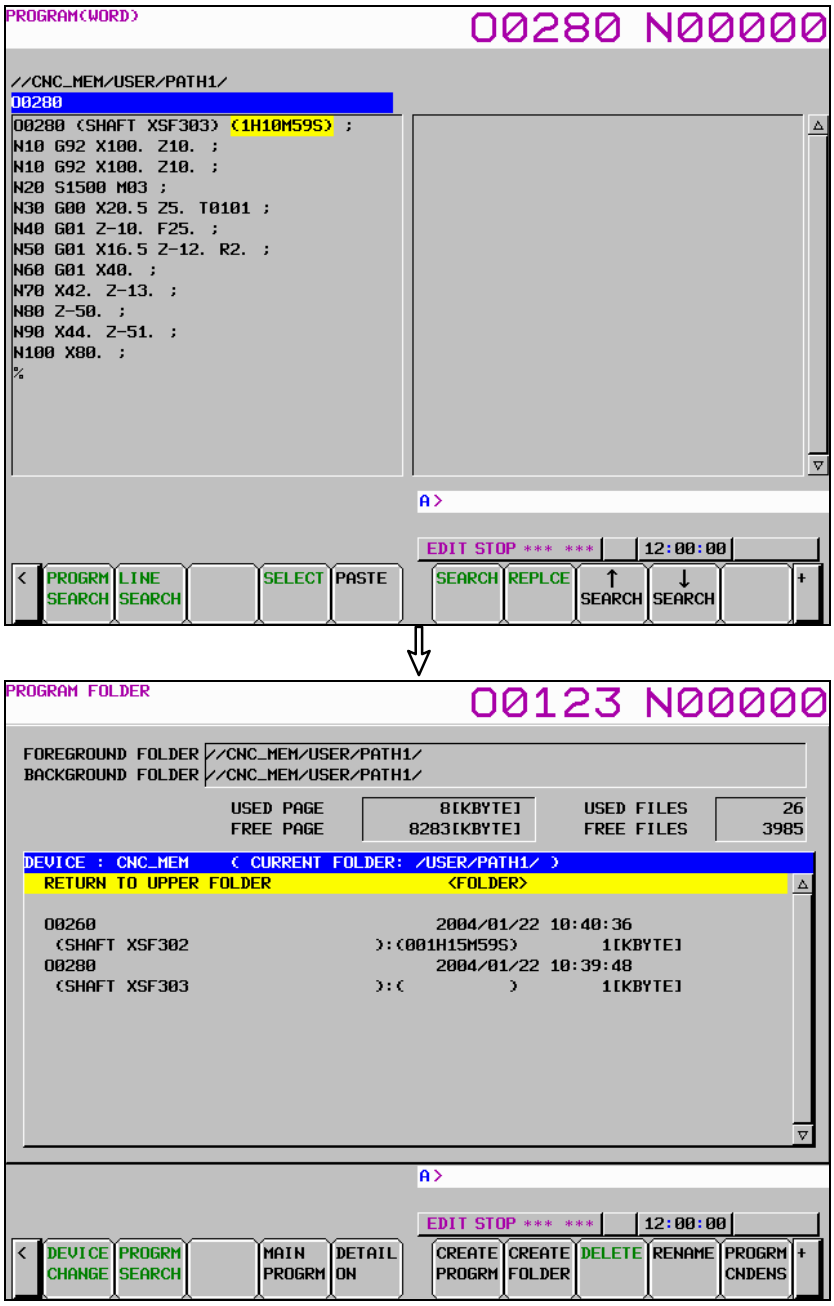


Fig. 12.2.8 (h) When the format of an inserted machining time is not “hhhHmmMssS” (H following a 3-digit number, M following a 2-digit number, and S following a 2-digit number, in this order) (10.4-inch)

12.2.9 Screen for Assistance in Entering Tilted Working Plane Command

The screens for assistance in entering tilted working plane commands (hereinafter referred to as guidance screens) include a command type selection screen and a tilted working plane data setting screen. The command type selection screen is used to select a tilted working plane command. The tilted working plane data setting screen is used to set specified tilted working plane data required for the selected command. By making settings and performing operations on these guidance screens, a tilted working plane command block can be created.
The created block is reflected as a new insertion to a program being edited or as a modification to an existing block.
This function can be enabled by setting bit 1 (GGD) of parameter No. 11304 to 1.

Creation of a new block

The following describes the procedure for creating a tilted working plane command block on guidance screens and for inserting the block to a program being edited on a program editing screen.

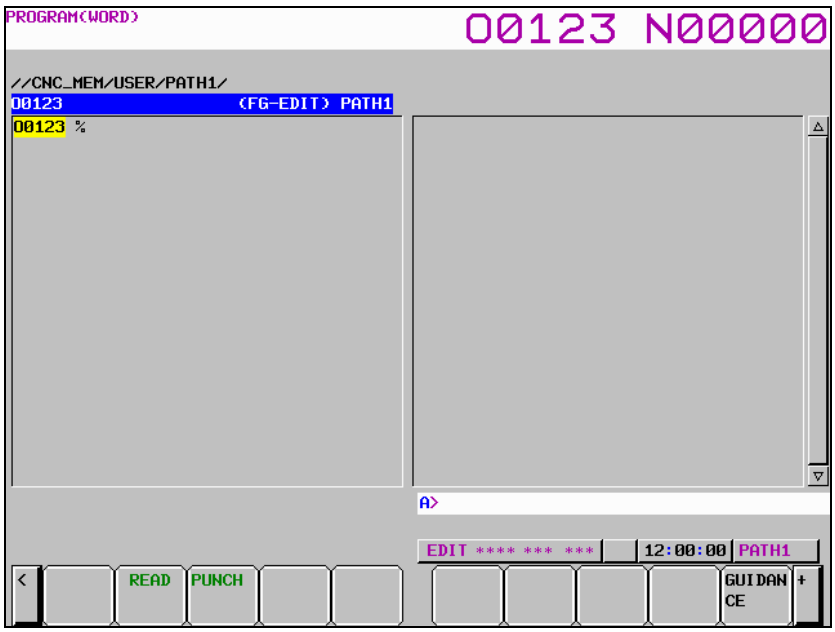
1 On a program editing screen, display a program to which you want to insert a tilted working plane command block.
The foreground editing screen, background editing screen, or MDI editing screen should be displayed.

- Displaying the foreground editing screen
  <1> Select the EDIT mode.
  <2> Press function key .
  <3> Press soft key [PROGRAM].

- Displaying the background editing screen
  <1> Press function key .
  <2> Press soft key [FOLDER].
  <3> Press any of the cursor keys to move the cursor to a program to be edited in the background.
  <4> Press key.

- Displaying the MDI editing screen
  <1> Select the MDI mode.
  <2> Press function key .
  <3> Press soft key [PROGRAM].

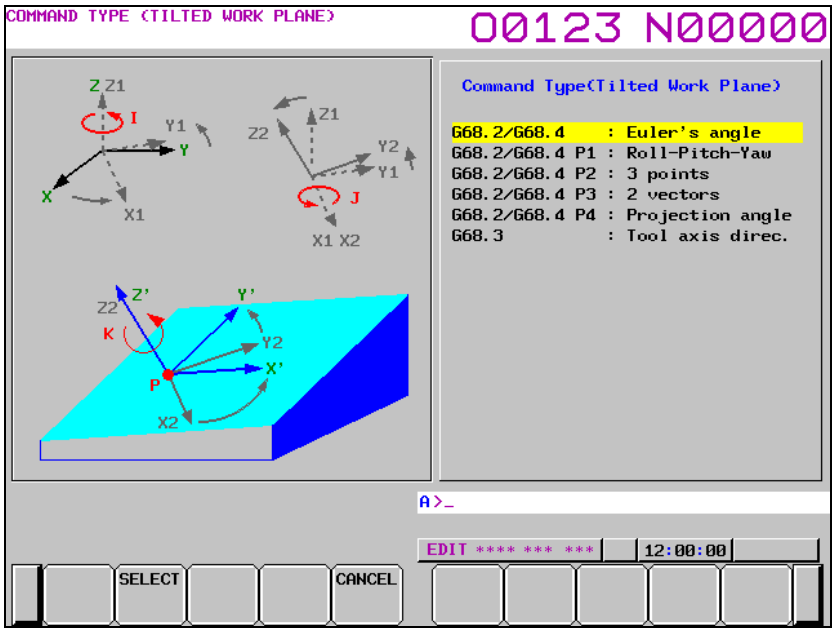
The program editing screen is displayed.



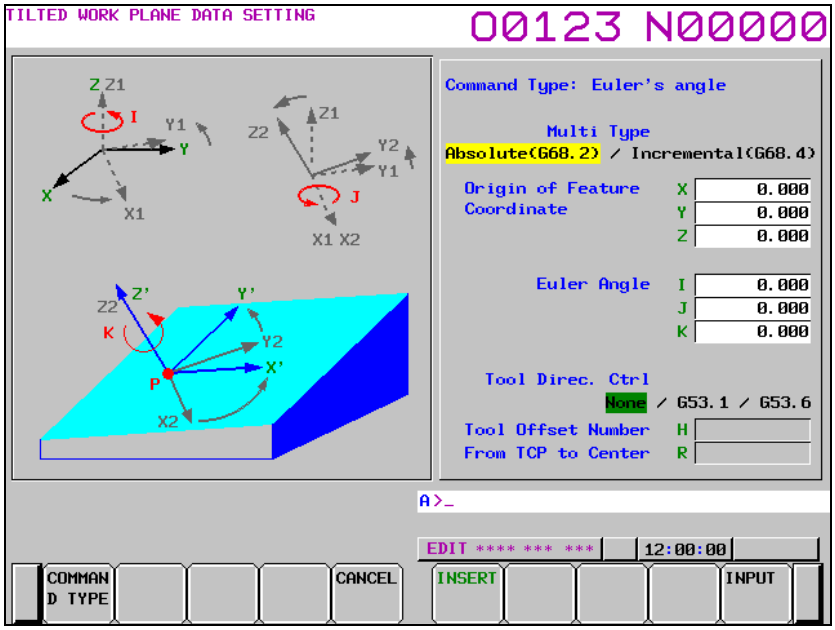
2 Press any of the cursor keys to move the cursor to a position where you want to insert a block.
Note that a block created on the guidance screens is inserted after the block at the cursor position. (If the block at the cursor position includes a tilted working plane command, the existing block is modified. See "Modification to an existing block" below.)

3 Press soft key [(OPRT)].

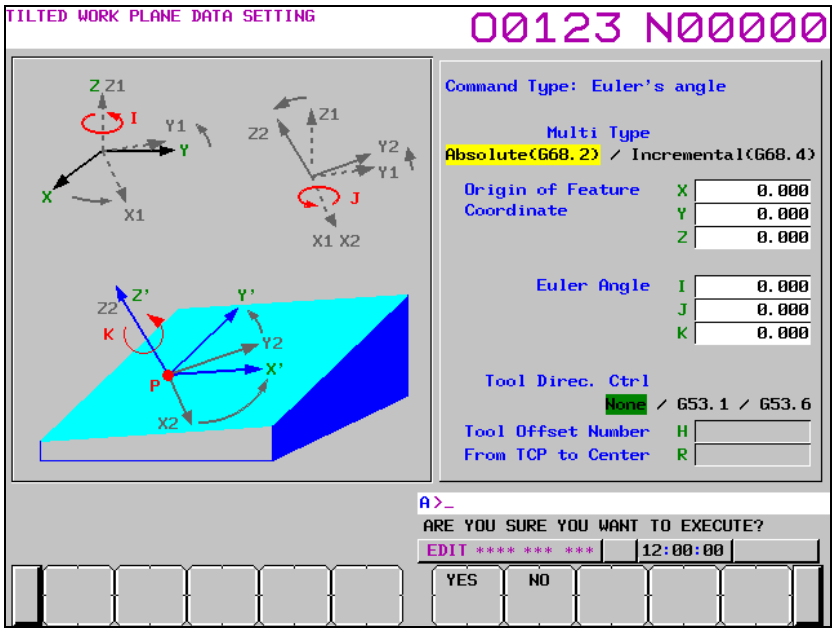
4 Press continuous menu key ▷ several times, and then press soft key [GUIDANCE].
The command type selection screen is displayed.



5 Select a command type with any of the cursor keys, and then press soft key [SELECT].
The tilted working plane data setting screen is displayed.



6 Enter command data for the setting items.
7 Press soft key [INSERT].

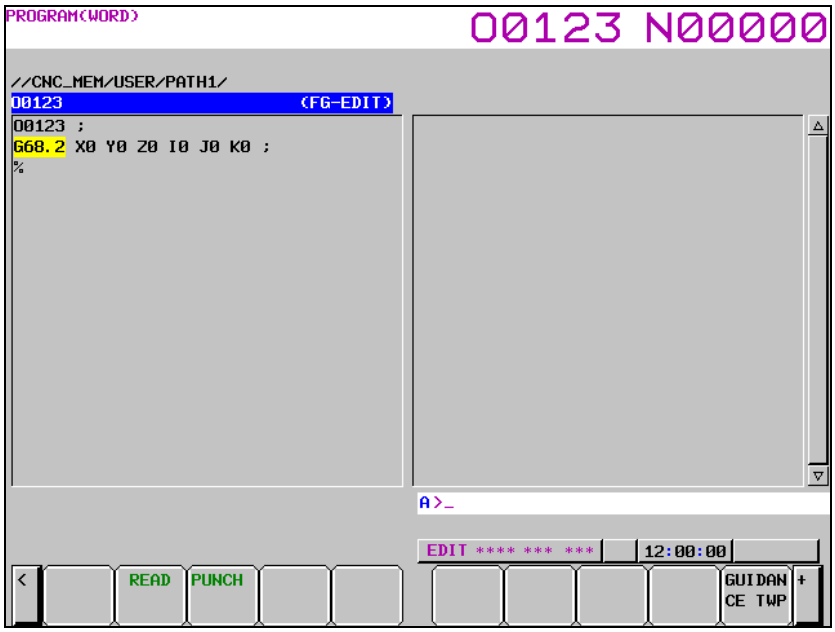


8 Press soft key [YES].
This takes you back to the program editing screen, where the new block is inserted after the block at the cursor position.

Modification to an existing block

The following describes the procedure for replacing a block in a program being edited on a program editing screen, with a tilted working plane command block created on a guidance screen.

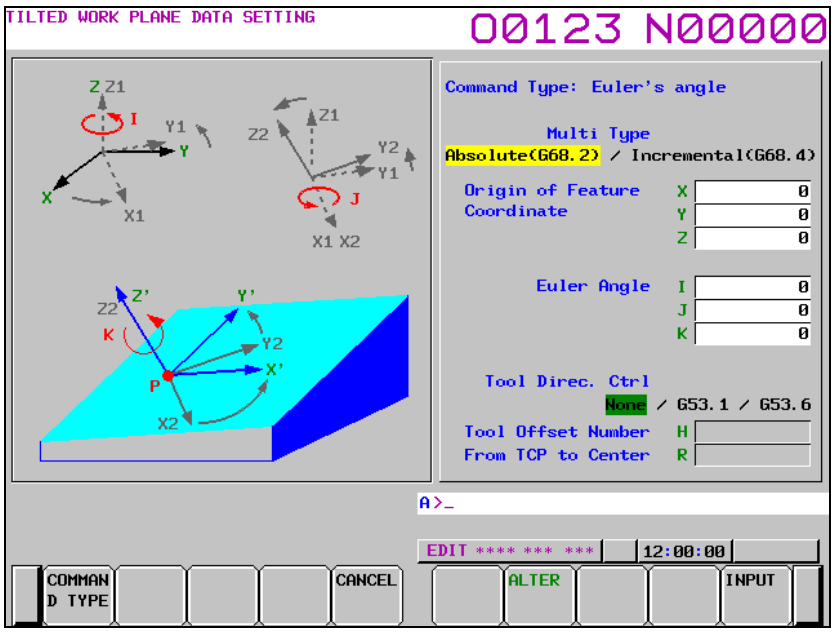
1 On a program editing screen, display a program to be edited.
(For the procedure for displaying a program editing screen, see step 1 in "Creation of a new block".)
The program editing screen is displayed.



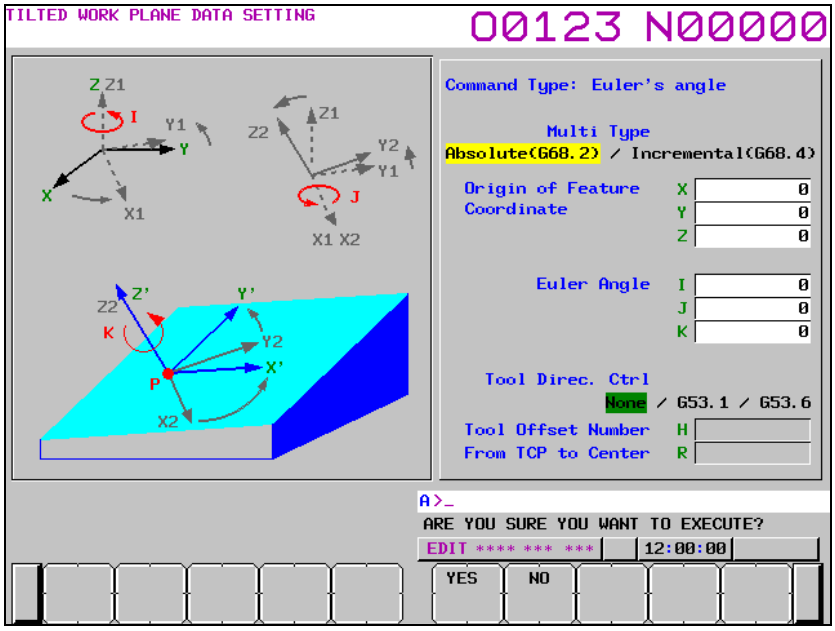
2 Press any of the cursor keys to move the cursor to a block to be modified.
For a command extending over more than one block, move the cursor to the first block.

3 Press soft key [(OPRT)].

4 Press continuous menu key ▷ several times, and then press soft key [GUIDANCE TWP]. The tilted working plane data setting screen is displayed.



5 Enter command data for setting items to be modified.
6 Press soft key [ALTER].



7 Press soft key [YES].
This takes you back to the program editing screen, where the block at the cursor position is replaced.

Guidance screen cancellation

Pressing soft key [CANCEL] on a guidance screen takes you back to the program editing screen. At this time, the data that has been set on the guidance screen is discarded.

NOTE

1 In addition to the above operation, the following operations also cancel a guidance screen. The data that has been set on the guidance screen is discarded.
- When bit 7 (CPG) of parameter No. 11302 is 1 (setting for automatically switching between program-related screens according to CNC mode), the CNC mode is changed.
- When a guidance screen is displayed from the foreground editing screen, the CNC mode is changed to a mode other than EDIT, TJOG, or THND.
- When a guidance screen is displayed from the MDI editing screen, the CNC mode is changed to a mode other than MDI.
- When a guidance screen is displayed on a 15-inch display unit, the screen is switched by a vertical soft key.
- The screen is switched by an MDI key.
- The screen is switched by a path select signal.
- An event that causes screen switching has occurred, including occurrence of an alarm, display of an operator message, or signal-based display of a screen (such as a tool compensation screen, workpiece shift screen, workpiece coordinate system setting screen, or C language executor screen).

2 If MDI key

is pressed after switching from a guidance screen to another screen, a program editing screen is displayed, instead of the guidance screen.

Notes

- Conditions under which soft key [GUIDANCE TWP] is displayed
  Soft key [GUIDANCE TWP] is displayed on a program editing screen under the following conditions:

  1 Foreground editing screen
    - The CNC mode is EDIT, TJOG, or THND.
    - The editing and display are not prohibited for a program to be edited.
  2 Background editing screen
    - A program to be edited is not in reference mode.
    - The editing and display are not prohibited for a program to be edited.
  3 MDI editing screen
    - The CNC mode is MDI.

- Screen displayed when soft key [GUIDANCE TWP] is pressed
  When soft key [GUIDANCE TWP] is pressed, which guidance screen is displayed is automatically determined according to the conditions described below.
  1 When a tilted working plane command is not included in a block at the cursor position on a program editing screen
    The command type selection screen is displayed. A block created on the guidance screens is inserted after the block at the cursor position on the program editing screen.
  2 When a tilted working plane command is included in a block at the cursor position on a program editing screen
    The tilted working plane data setting screen is displayed, which shows the data for the tilted working plane command in the block at the cursor position on the program editing screen. When a block of a tool axis direction control command exists immediately after the tilted working plane command, the command data for the block is also shown. The tilted working plane command in the block at the cursor position on the program editing screen is replaced with a block created on the guidance screen.

NOTE
If the CNC is in the reset state or emergency stop state when soft key [GUIDANCE TWP] is pressed on the foreground editing screen or MDI editing screen, the warning "PROGRAM READ FAILED" appears, and the operation cannot be continued. (Soft keys other than [CANCEL] are not displayed.) Press soft key [CANCEL] to return to the program editing screen, and then press soft key [GUIDANCE TWP] again.

12.2.9.1 Command type selection screen

The command type selection screen is used to select the type of a tilted working plane command you want to insert into a program to be edited. One of the following command types can be selected:

- G68.2 / G68.4 (Euler's angle)
- G68.2 / G68.4 P1 (Roll-Pitch-Yaw angle)
- G68.2 / G68.4 P2 (3 points specification)
- G68.2 / G68.4 P3 (2 vectors specification)
- G68.2 / G68.4 P4 (Projection angle)
- G68.3 (Tool Axis Direction)

G68.2 is an absolute command, and G68.4 is an incremental command.

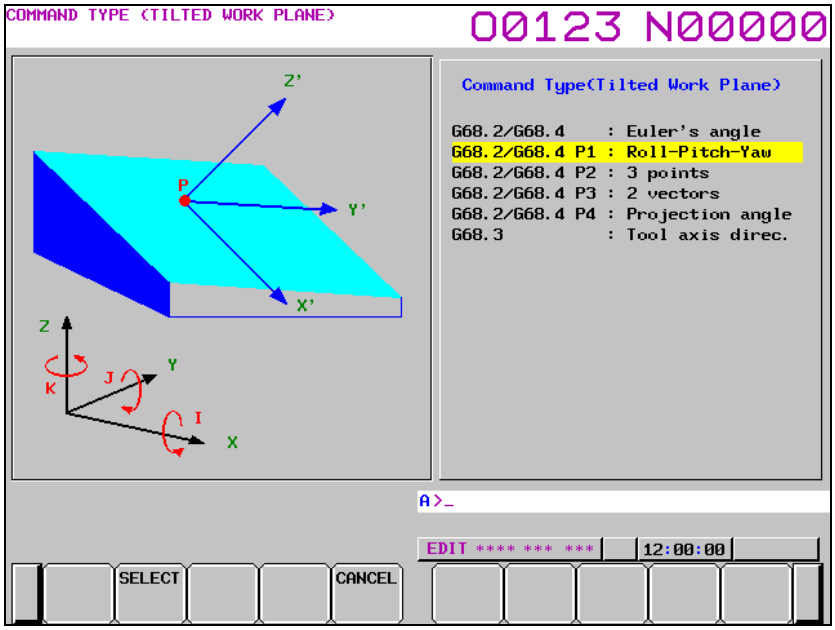


Fig. 12.2.9.1 (a) Command type selection screen (10.4-inch)

Selection of a command type

1 Press cursor key ↑ or ↓ to move the cursor to a command type you want to select. As the cursor is moved, the figure corresponding to the command type at the cursor position is displayed.

2 When soft key [SELECT] is pressed, the command type at the cursor position is accepted, and the tilted working plane data setting screen is displayed.

NOTE

If the warning "PROGRAM READ FAILED" appears when the command type selection screen is displayed, the operation cannot be continued. (Soft keys other than [CANCEL] are not displayed.) Press soft key [CANCEL] to return to the program editing screen, and then press soft key [GUIDANCE TWP] again.

12.2.9.2 Tilted working plane data setting screen

The tilted working plane data setting screen is used to set specified tilted working plane data required for a tilted working plane command of the type that has been selected on the command type selection screen or has been selected when soft key [GUIDANCE TWP] is pressed.
Different types of tilted working plane data setting screen are provided for different command types. For details of each type of tilted working plane data setting screen, see "Details of the tilted working plane data setting screen".

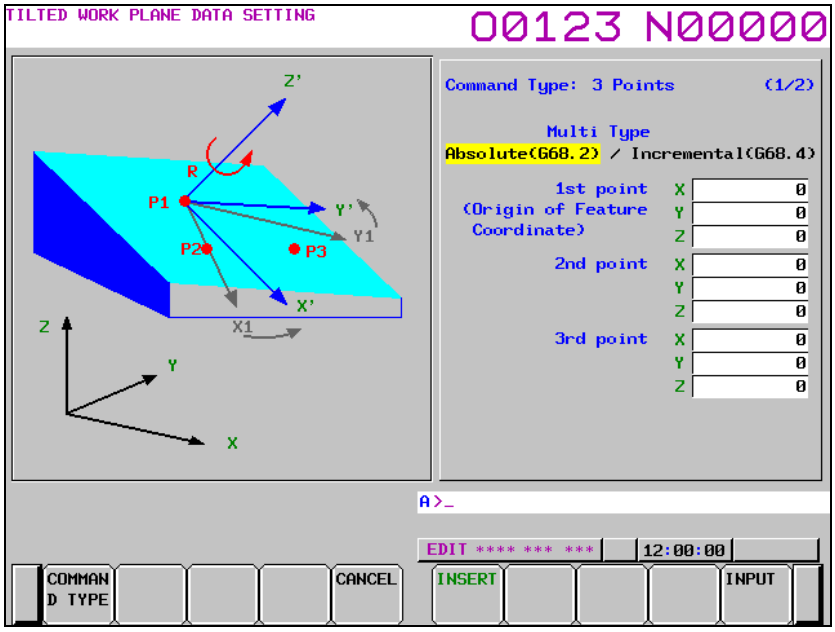


Fig. 12.2.9.2 (a) Tilted working plane data setting screen—3 points specification(10.4-inch)

Display of the tilted working plane data setting screen

When a new block is created, the initial values are set in the setting and selection items.
When an existing block is modified, the command data for the block at the cursor position on the program editing screen is reflected in the setting and selection items.

NOTE
1 For existing block modification, if the guidance screen is displayed when the cursor is placed in the middle of multiple blocks for a command, the parameters for the block(s) before the cursor are not reflected in the setting and selection items, but only the parameters for the block(s) after the cursor are reflected. If no data is to be reflected in a setting or selection item, the initial value is set in the item.
2 For existing block modification, only the commands with parameter-specified axis names are reflected in the setting items on the screen. (The values specified in data for commands with incorrect axis names are not reflected in the setting items.)

Command data input

- Item for which to enter a value

  Press cursor key or to move the cursor to an item you want to set.

  Enter a value, and then press the key or soft key [INPUT].

  Example)

  When the origin of a feature coordinate system is set as shown above, addresses X, Y, and Z are specified as follows:

  G68.2 X0.001 Y0.01 Z1000 ...

  For two items "Tool Offset Number" and "From TCP to center", the state with no command can be set by deleting the specified values according to the following procedure.

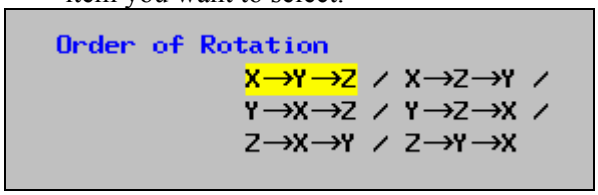
  1 Press cursor key or to move the cursor to the item "Tool Offset Number" or "From TCP to Center".

  2 Press or soft key [INPUT] without entering anything.

- Item to be selected from a list

  1 Press cursor key [↑] or [↓] to move the cursor to an item you want to set.

  2 Press cursor key [←] or [→] to move the cursor to an item you want to select.



Example)
Order of rotation for Roll-Pitch Yaw angle

Insertion of a block

If the guidance screen is displayed when the block at the cursor position on the program editing screen does not include a tilted working plane command, soft key [INSERT] is displayed on the tilted working plane data setting screen. An operation for inserting a tilted working plane command block is described below.
For warnings that may be issued at the time of block insertion, see "Limitation".

1 Press soft key [INSERT].
The confirmation message "ARE YOU SURE YOU WANT TO EXECUTE?" is displayed, and soft keys [YES] and [NO] are displayed.

2 Press soft key [YES].
A block is created from the command type and command data and then is inserted after the block at the cursor position in the program to be edited. After block insertion, the guidance screen is closed and the program editing screen is displayed.
Pressing soft key [NO] takes you back to the tilted working plane data setting screen.

Example)

G00 X0.;
↓
G00 X0.;
G68.2 P2 Q0...
G68.2 P2 Q1...
G68.2 P2 Q2...
G68.2 P2 Q3...

When the guidance screen is displayed and the 3-point specification is selected as a command type for block insertion, a created block is inserted after the block at the cursor position.

Replacement of a block

If the guidance screen is displayed when the block at the cursor position on the program editing screen includes a tilted working plane command, soft key [ALTER] is displayed on the tilted working plane data setting screen. An operation for replacing a tilted working plane command block is described below.
For warnings that may be issued at the time of block replacement, see "Limitation".

1 Press soft key [ALTER].
The confirmation message "ARE YOU SURE YOU WANT TO EXECUTE?" is displayed, and soft keys [YES] and [NO] are displayed.
2 Press soft key [YES].
A block is created from the command type and command data and then replaces the tilted working plane command in the block at the cursor position on the program editing screen. After block replacement, the guidance screen is closed and the program editing screen is displayed.
Pressing soft key [NO] takes you back to the tilted working plane data setting screen.

NOTE
1 If a block to be replaced includes a command other than a tilted working plane command, the command is deleted during block replacement. However, only the sequence number at the beginning is preserved.
2 If the guidance screen is displayed when the cursor is placed in the middle of multiple blocks for a command, the block(s) before the cursor are not replaced. These blocks remain unchanged after replacement.
3 If soft key [GUIDANCE TWP] is pressed when the cursor is placed on a block of a tool axis direction control command, the command type selection screen is displayed in new insertion mode. A block created on the guidance screens is inserted after the block of the tool axis direction control command.

Limitation

The following lists the warnings that may be issued at the time of block insertion or replacement.
If a warning is displayed, return to the program editing screen with soft key [CANCEL] and press soft key [GUIDANCE TWP] again, or eliminate the cause of the warning and retry the operation.

Warning	Description
“PROGRAM WRITE FAILED”	• The guidance screen was displayed from the foreground editing screen or MDI editing screen, and a block insertion or replacement operation was performed when the CNC was in the reset state or emergency stop state.
”PROGRAM CANNOT BE WRITTEN”	• A block insertion or replacement operation was performed after a program to be edited was updated due to program downloading by an external application while the guidance screen was displayed. • A block insertion or replacement operation was performed after the start of the main program was located by a reset while the guidance screen was displayed.
“WRITE PROTECT”	• A block insertion or replacement operation was performed when the editing or display was prohibited for a program to be edited. • A block insertion or replacement operation was performed when the memory protection signal (KEY signal) for program registration or editing was off. • A block insertion or replacement operation was performed when the operation level of the 8-level data protection function was lower than the protection level of a part program editing operation.

Display of the command type selection screen

When soft key [COMMAND TYPE] is pressed, the command type selection screen is displayed. When the command type is changed on the command type selection screen, the values set on the tilted working plane data setting screen are cleared.
When a tilted working plane command is included in the block at the cursor position on the program editing screen, you can display the guidance screen, change the command type on the command type selection screen, and then perform block replacement. In this case, the block at the cursor position on the program editing screen is replaced with a block created from the command type and command data that have been set.

12.2.9.3 Details of the tilted working plane data setting screen

The following six tilted working plane commands are supported.
For details of the commands, see II-22.3, "TILTED WORKING PLANE COMMANDS".

- G68.2 / G68.4 (Euler's angle)
- G68.2 / G68.4 P1 (Roll-Pitch-Yaw angle)
- G68.2 / G68.4 P2 (3 points specification)
- G68.2 / G68.4 P3 (2 vectors specification)
- G68.2 / G68.4 P4 (Projection angle)
- G68.3 (Tool Axis Direction)

G68.2 is an absolute command, and G68.4 is an incremental command.

G68.2 / G68.4(Euler's angle)

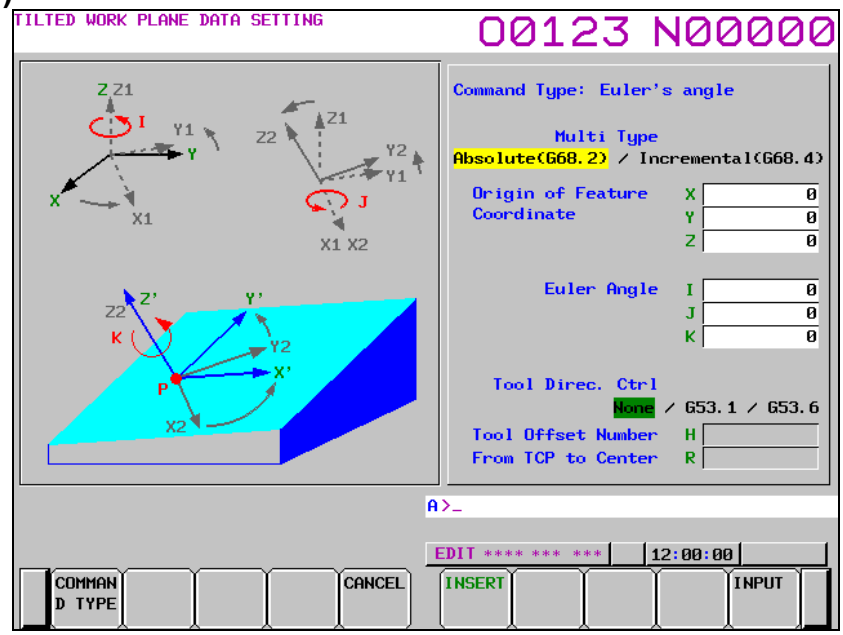


Fig. 12.2.9.3 (a) Tilted working plane data setting screen-Euler's angle(10.4-inch)

- Multi Type
  Absolute:
  It is assumed that values of specified data are in a workpiece coordinate system, regardless of whether tilted working plane command mode is set.
  Incremental:
  It is assumed that values of specified data are in a feature coordinate system if tilted working plane command mode is already set.
- Origin of Feature Coordinate
  Specify the origin (X, Y, and Z of point P) of a feature coordinate system as coordinates in a workpiece coordinate system (for the absolute type) or the current feature coordinate system (for the incremental type).

- Euler's angle

  I : Specify an angle of rotation around the Z-axis of a workpiece coordinate system (for the absolute type) or the current feature coordinate system (for the incremental type). This rotation determines coordinate system 1 (X1-Y1-Z1) from the workpiece coordinate system (for the absolute type) or the current feature coordinate system (for the incremental type) (X-Y-Z).

  J : Specify an angle of rotation around the X-axis of coordinate system 1. This rotation determines coordinate system 2 (X2-Y2-Z2) from coordinate system 1 (X1-Y1-Z1).

  K : Specify an angle of rotation around the Z-axis of coordinate system 2. After this rotation, a feature coordinate system is obtained by shifting the origin of the workpiece coordinate system by the coordinates specified in "Origin of Feature Coordinate".

G68.2 / G68.4(Roll-Pitch-Yaw angle)

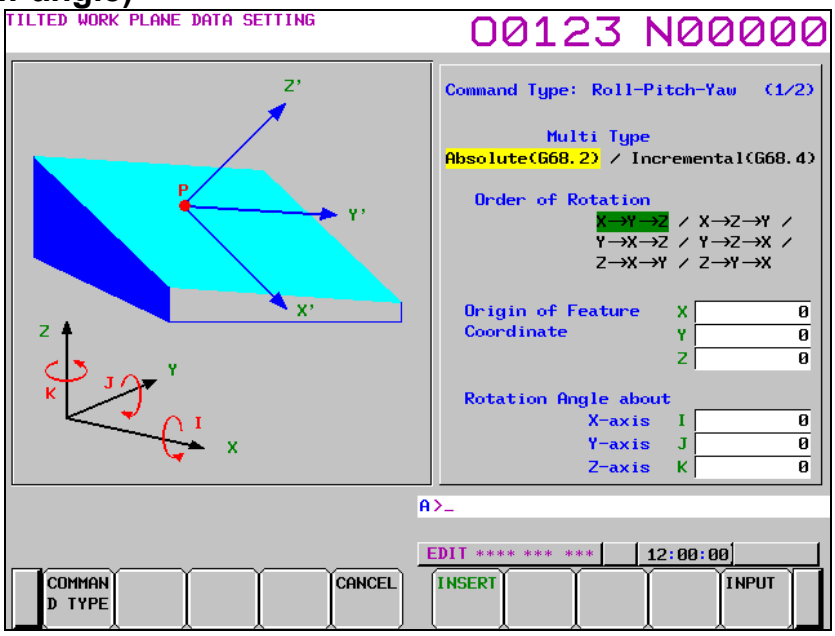


Fig. 12.2.9.3 (b) Tilted working plane data setting screen- Roll-Pitch-Yaw angle(10.4-inch)

- Multi Type

  Absolute:

  It is assumed that values of specified data are in a workpiece coordinate system, regardless of whether tilted working plane command mode is set.

  Incremental:

  It is assumed that values of specified data are in a feature coordinate system if tilted working plane command mode is already set.

- Order of Rotation
  Select an order in which the X-axis, Y-axis, and Z-axis are rotated in a workpiece coordinate system (for the absolute type) or the current feature coordinate system (for the incremental type). The selectable rotation orders are as follows:

	1st rotation axis	2nd rotation axis	3rd rotation axis
X→Y→Z	X-axis	Y-axis	Z-axis
X→Z→Y	X-axis	Z-axis	Y-axis
Y→X→Z	Y-axis	X-axis	Z-axis
Y→Z→X	Y-axis	Z-axis	X-axis
Z→X→Y	Z-axis	X-axis	Y-axis
Z→Y→X	Z-axis	Y-axis	X-axis

- Origin of Feature Coordinate
  Specify the origin (X, Y, and Z of point P) of a feature coordinate system as coordinates in a workpiece coordinate system (for the absolute type) or the current feature coordinate system (for the incremental type).

- Rotation Angle about X-axis
  Specify an angle of rotation around the X-axis of a workpiece coordinate system (for the absolute type) or the current feature coordinate system (for the incremental type).

- Rotation Angle about Y-axis
  Specify an angle of rotation around the Y-axis of a workpiece coordinate system (for the absolute type) or the current feature coordinate system (for the incremental type).

- Rotation Angle about Z-axis
  Specify an angle of rotation around the Z-axis of a workpiece coordinate system (for the absolute type) or the current feature coordinate system (for the incremental type).

G68.2 / G68.4(3 points specification)

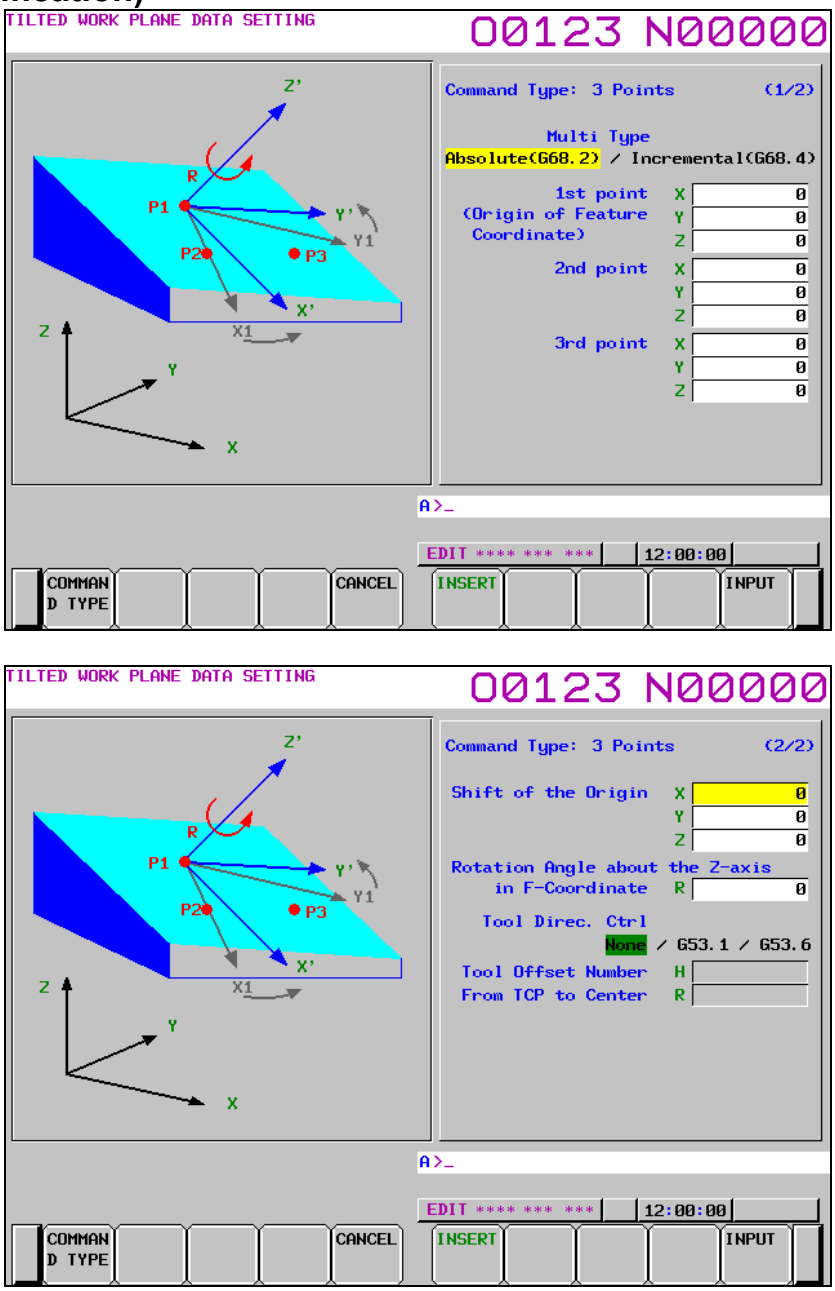


Fig. 12.2.9.3 (c) Tilted working plane data setting screen- 3 points specification(10.4-inch)

- Multi Type

  Absolute:

  It is assumed that values of specified data are in a workpiece coordinate system, regardless of whether tilted working plane command mode is set.

  Incremental:

  It is assumed that values of specified data are in a feature coordinate system if tilted working plane command mode is already set.

- Shift of the Origin
  Specify, in a feature coordinate system, an amount of shift from the feature coordinate system origin specified for the 1st point (point P1).

- Rotation Angle about the Z-axis in F-Coordinate
  Specify an angle of rotation around the Z-axis of a feature coordinate system. The direction of rotation angle R is positive when a rotation is made clockwise as viewed in the Z-axis direction of the feature coordinate system.

- 1st point (Origin of Feature Coordinate)
  Specify the origin (X, Y, and Z of point P1) of a feature coordinate system as coordinates in a workpiece coordinate system (for the absolute type) or the current feature coordinate system (for the incremental type).

- 2nd point
  Specify the coordinates (X, Y, and Z of point P2) of the 2nd point as coordinates in a workpiece coordinate system (for the absolute type) or the current feature coordinate system (for the incremental type). The 1st point and 2nd point determine the X-axis of the feature coordinate system.

- 3rd point
  Specify the coordinates (X, Y, and Z of point P3) of the 3rd point as coordinates in a workpiece coordinate system (for the absolute type) or the current feature coordinate system (for the incremental type). Among the directions normal to the X-axis, a direction with a smaller angle relative to the P1 → P3 vector is the Y-axis of the feature coordinate system.

G68.2 / G68.4(2 vectors specification)

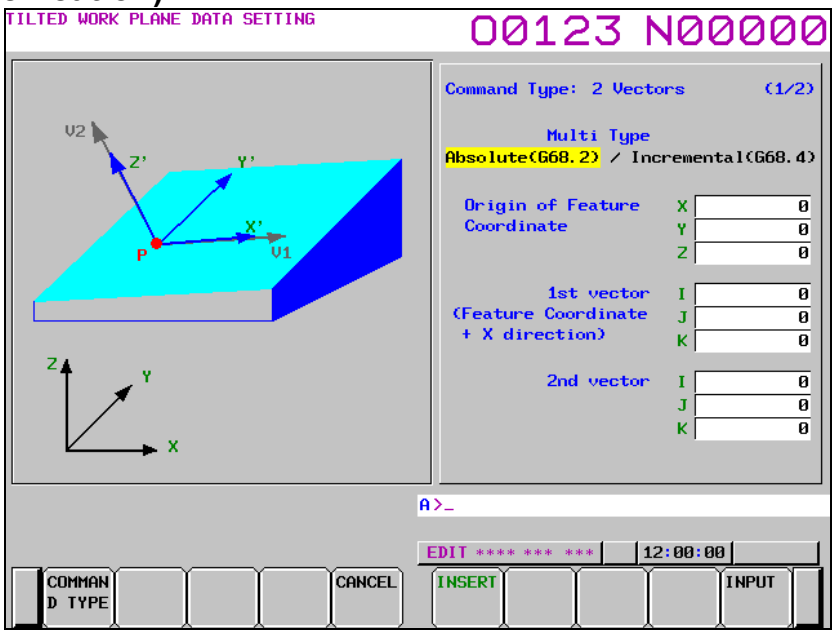


Fig. 12.2.9.3 (d) Tilted working plane data setting screen-2 vectors specification(10.4-inch)

- Multi Type
  Absolute:
  It is assumed that values of specified data are in a workpiece coordinate system, regardless of whether tilted working plane command mode is set.
  Incremental:
  It is assumed that values of specified data are in a feature coordinate system if tilted working plane command mode is already set.

- Origin of Feature Coordinate
  Specify the origin (X, Y, and Z of point P) of a feature coordinate system as coordinates in a workpiece coordinate system (for the absolute type) or the current feature coordinate system (for the incremental type).

- V1 (X') vector
  Specify the X-axis direction vector of a feature coordinate system as values in a workpiece coordinate system (for the absolute type) or the current feature coordinate system (for the incremental type).

- V2 (Z') vector
  Specify the Z-axis direction vector of a feature coordinate system as values in a workpiece coordinate system (for the absolute type) or the current feature coordinate system (for the incremental type).

G68.2 / G68.4(Projection angle)

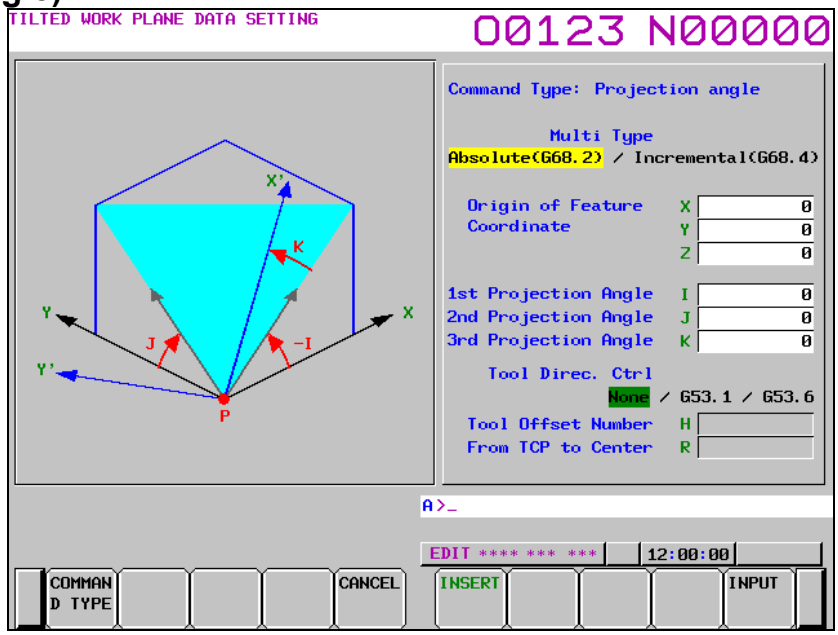


Fig. 12.2.9.3 (e) Tilted working plane data setting screen-Projection angle(10.4-inch)

- Multi Type
  Absolute:
  It is assumed that values of specified data are in a workpiece coordinate system, regardless of whether tilted working plane command mode is set.
  Incremental:
  It is assumed that values of specified data are in a feature coordinate system if tilted working plane command mode is already set.

- Origin of Feature Coordinate
  Specify the origin (X, Y, and Z of point P) of a feature coordinate system as coordinates in a workpiece coordinate system (for the absolute type) or the current feature coordinate system (for the incremental type).

- Projection angle
  I : Specify a projection angle relative to the X-axis of a workpiece coordinate system (for the absolute type) or the current feature coordinate system (for the incremental type).
  J : Specify a projection angle relative to the Y-axis of a workpiece coordinate system (for the absolute type) or the current feature coordinate system (for the incremental type).
  K : Specify an angle of rotation around the Z-axis of a feature coordinate system.

G68.3(Tool Axis Direction)

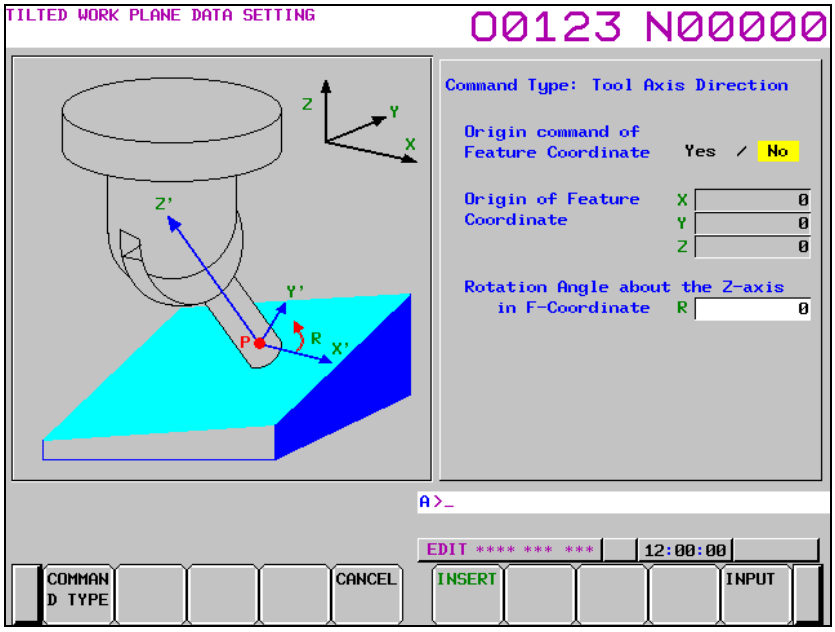


Fig. 12.2.9.3 (f) Tilted working plane data setting screen-Tool Axis Direction(10.4-inch)
(When "No" is selected in "Origin command of Feature Coordinate")

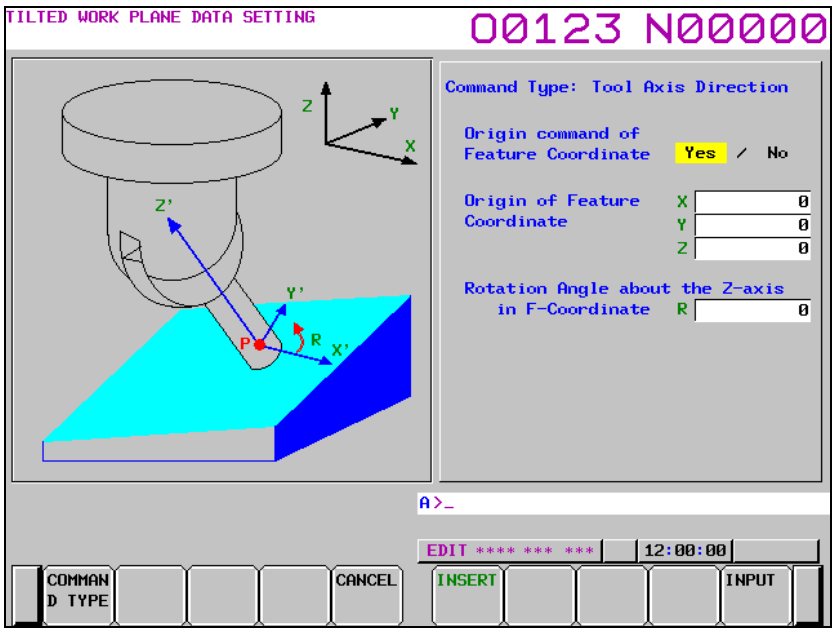


Fig. 12.2.9.3 (g) Tilted working plane data setting screen-Tool Axis Direction(10.4-inch)
(When "Yes" is selected in "Origin command of Feature Coordinate")

- Origin command of Feature Coordinate
  Select whether to specify a feature coordinate system origin.
  Yes : A feature coordinate system origin is specified.
  No : A feature coordinate system origin is not specified.

- Origin of Feature Coordinate
  Specify the origin (X, Y, and Z of point P) of a feature coordinate system as coordinates in a workpiece coordinate system.
  This setting cannot be made if "No" is selected in "Origin command of Feature Coordinate".

- Rotation Angle about the Z-axis in F-Coordinate
  Specify an angle of rotation around the Z-axis of a feature coordinate system. The direction of rotation angle R is positive when a rotation is made clockwise as viewed in the Z-axis direction of the feature coordinate system.

12.2.9.4 Limitation

This function is supported on 10.4-inch and 15-inch display units. This function is not displayed on 7.2-inch and 8.4-inch display units.

Screens of a 15-inch display unit

This section describes the screens displayed by pressing function key

. The screens include a program editing screen, program folder list display screen, and screens for displaying the command states of the program currently being executed.

1. Program screen
2. Program folder screen
3. Next block display screen
4. Program check screen

On the program screen, you edit the program that is currently selected, and display the block that is currently executed during program operation. In MDI mode, you also edit an MDI operation program, and display the block that is currently executed.

12.2.10 Program Contents Display (15-inch Display Unit)

Displays the program currently being executed in MEMORY mode.

Displaying the program being executed

Procedure

1 Press function key

to display the program screen.

2 Press vertical soft key [PROGRAM].
The cursor is positioned at the block currently being executed.

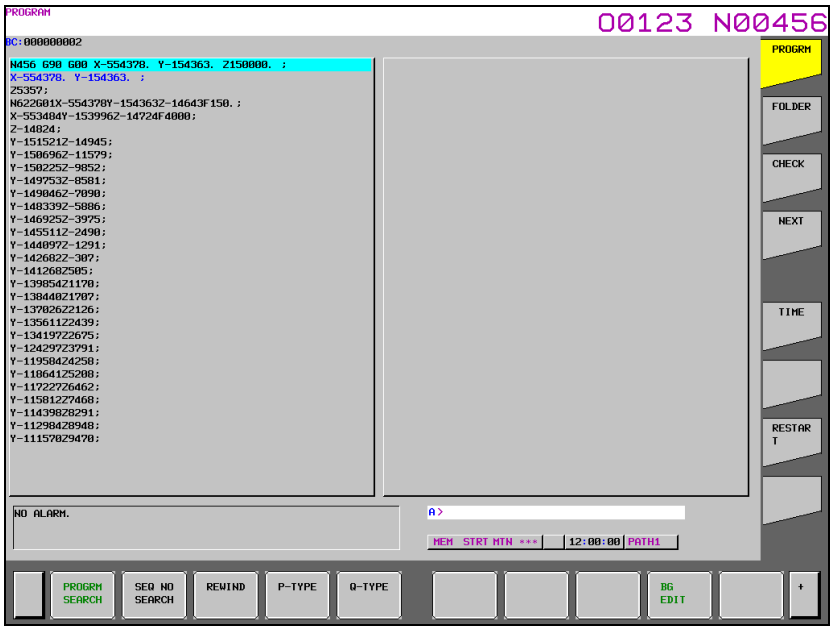


Fig. 12.2.10 (a) Screen for displaying the program being executed (15-inch)

12.2.10.1 Displaying the executed block

For an explanation of how to display the executed block, see Subsection 12.2.1.1, "Displaying the executed block".

12.2.11 Editing a Program (15-inch Display Unit)

A program can be edited in the EDIT mode.
Two modes of editing are available. One mode is word editing, which performs word-by-word editing. The other is character editing, which performs character-by-character editing.
For program creation and editing operation, see Chapter III-9, “CREATING PROGRAMS “ and Chapter III-10, “EDITING PROGRAMS”.

Displaying the program editing screen

Procedure

1 Press function key

to display the program screen.

2 Press vertical soft key [PROGRAM].

- Word editing

Editing operations such as text insertion, modification, and deletion, and cursor movements are performed on a word-by-word basis.

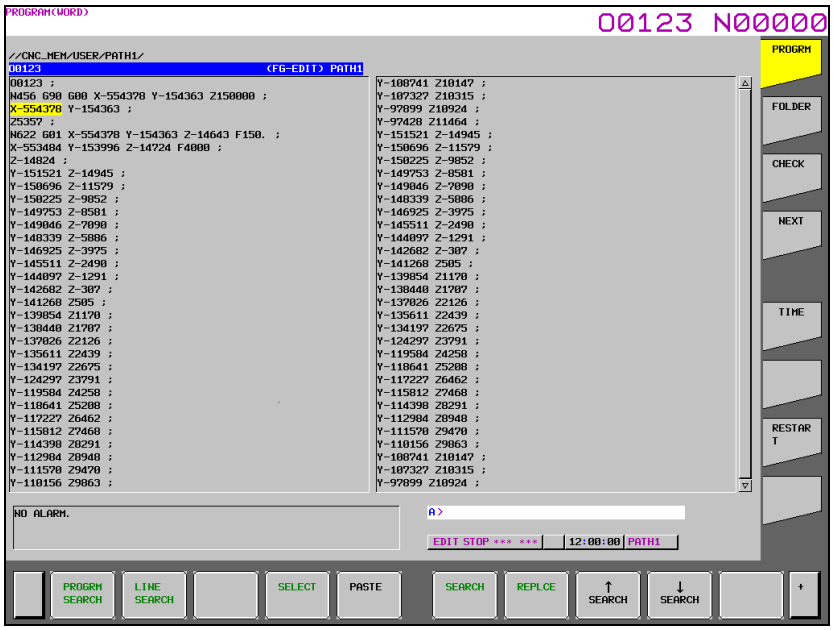


Fig. 12.2.11 (a) Program word editing screen (15-inch)

- Character editing

Program editing operations and cursor movements are performed on a character-by-character basis as with a general text editor.
Text is input directly to the cursor position instead of using the key input buffer.

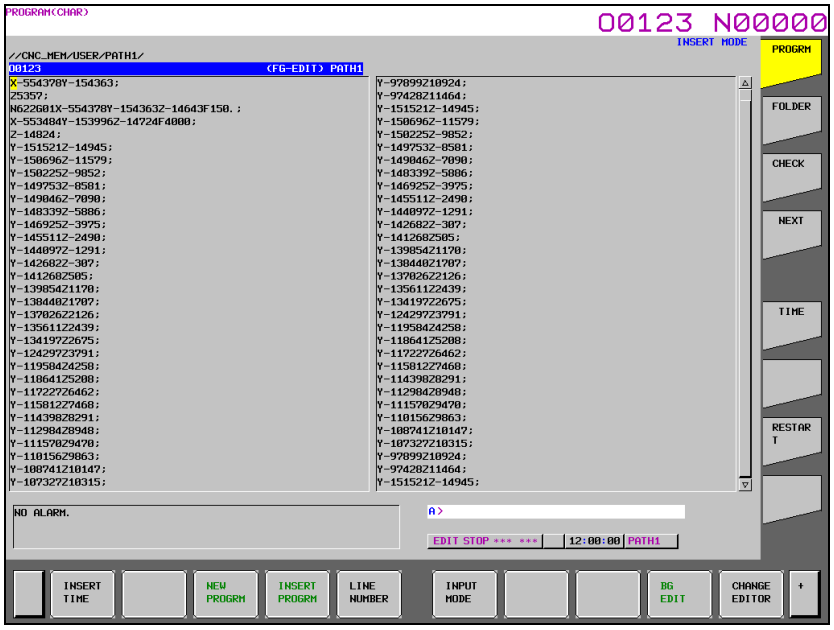


Fig. 12.2.11 (b) Program character editing screen (15-inch)

Switching between program editing modes

You can switch between word editing and character editing with horizontal soft keys.

Procedure

1 Press function key to display the program screen.
2 Press vertical soft key [PROGRAM].
3 Press continuous menu key to display horizontal soft key [CHANGE EDITOR].
4 Pressing horizontal soft key [CHANGE EDITOR] switches the editing mode between word editing and character editing.

12.2.12 Program Screen for MDI Operation (15-inch Display Unit)

During MDI operation or editing of an MDI operation program in the MDI mode, the program currently being executed mode is displayed. For MDI operation, see Section III-4.2, “MDI Operation”.

Procedure for displaying the program screen for MDI operation

Procedure

1 Press function key

.

2 Press vertical soft key [PROGRAM].
The program input from the MDI is displayed.

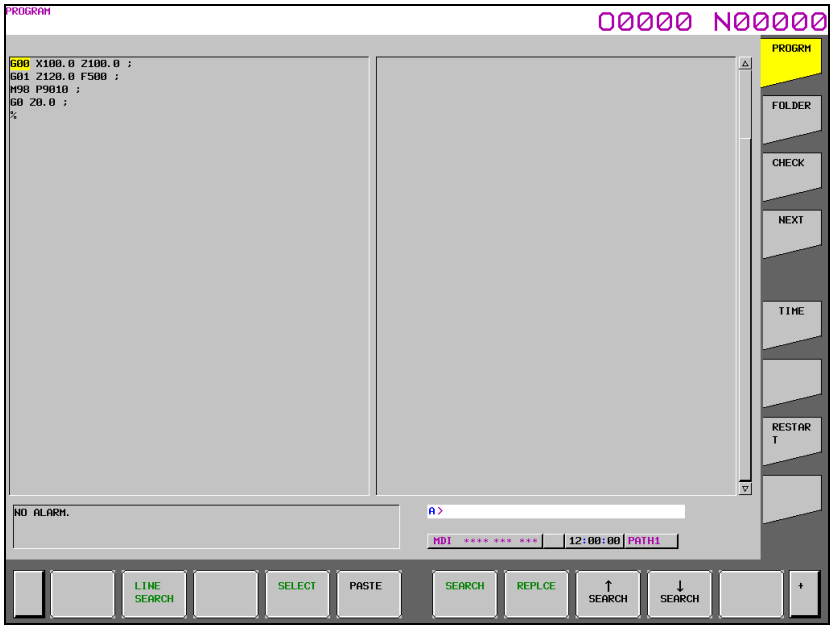


Fig. 12.2.12 (a) MDI operation program screen (15-inch)

12.2.13 Program Folder Screen (15-inch Display Unit)

A list of programs registered in the program memory is displayed.
For the program folder screen, see Chapter III-11, “PROGRAM MANAGEMENT”.

Displaying the program folder screen

Procedure

1 Press function key

.

2 Press vertical soft key [FOLDER].

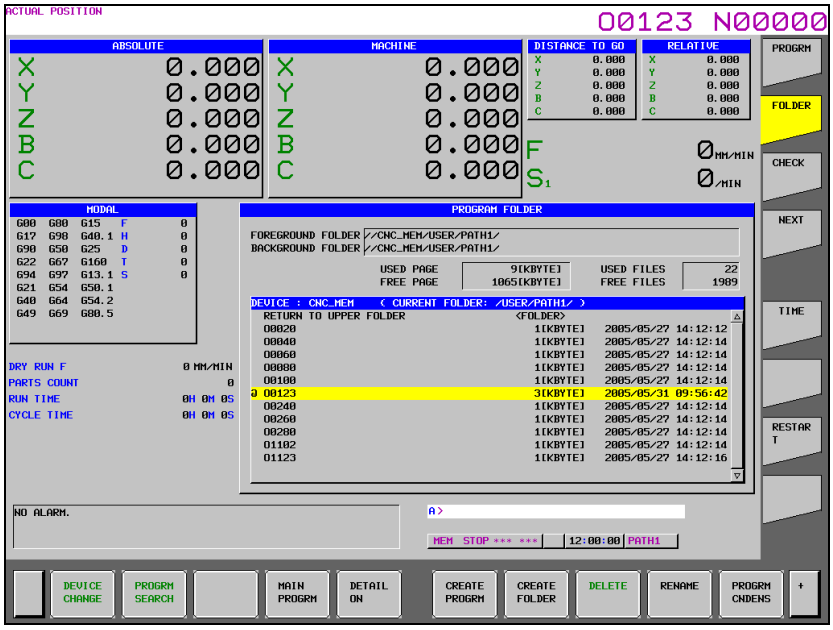


Fig. 12.2.13 (a) Program folder screen (15-inch)

12.2.13.1 Split display on the program folder screen

Overview

On the program folder screen, the folder information display can be split into two folder information views, upper and lower, as shown in the figure below.

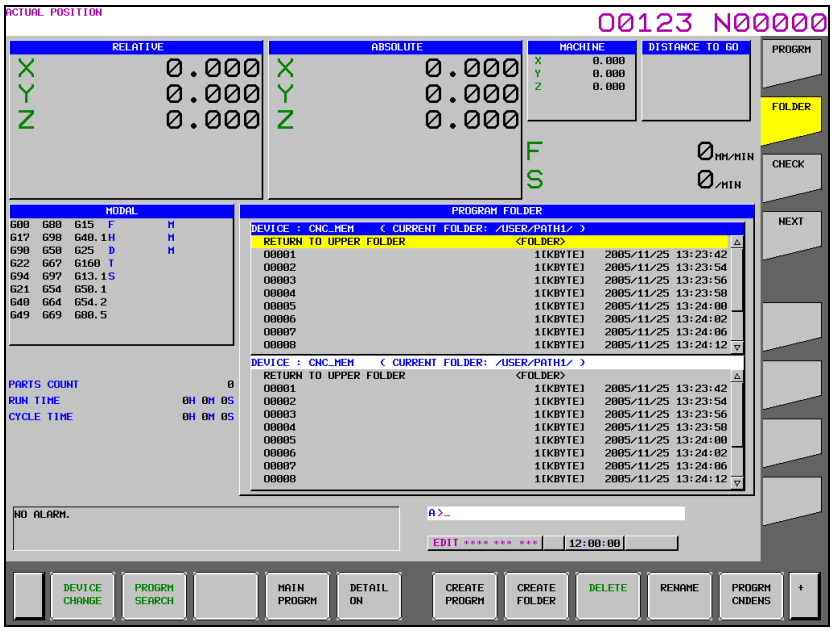
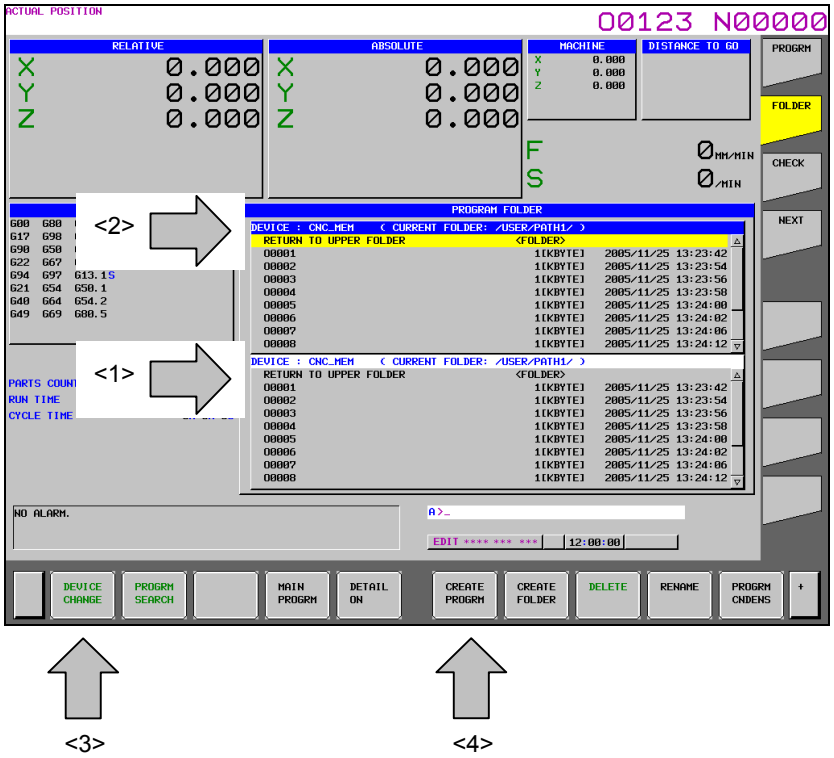


Fig. 12.2.13.1 (a) Program folder screen (split-display screen)(15-inch)

On the split folder display, file information about different sets of devices or folders can be shown in different views at the same time, so you can copy or move files easily.

Example of copying a file from the CNC MEM to the Data Server



Procedure

<1> Change to a destination folder on the Data Server.
<2> Change to a folder on the CNC_MEM that contains a file you want to copy.
<3> Select the file.
<4> Copy the file.

Procedure for switching to split display

Procedure

1 Press function key PROG to display the program folder screen.
2 Press horizontal soft key [(OPRT)].
3 Press continuous menu key twice.
Horizontal soft key [MULTI LIST] appears.

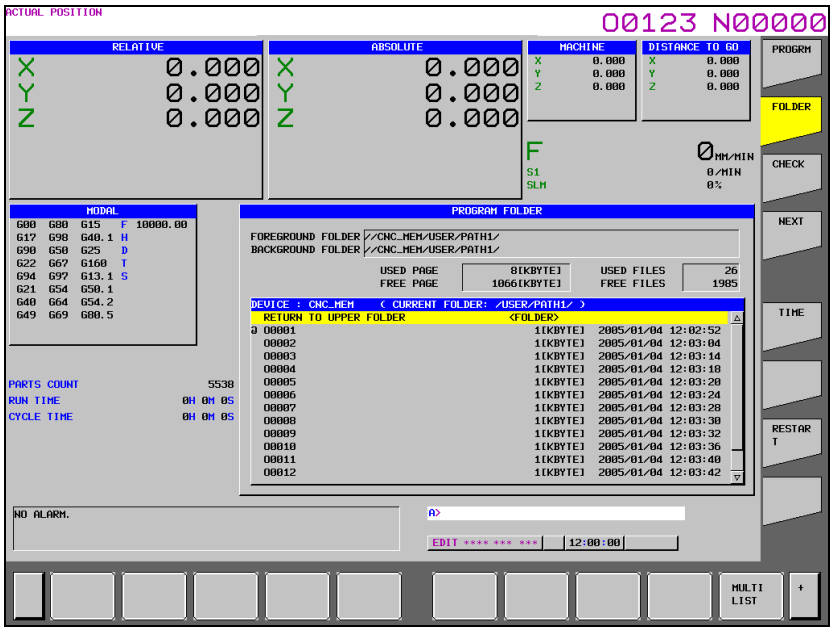


Fig. 12.2.13.1 (b) Program folder screen normal screen)(15-inch)

4 Press horizontal soft key [MULTI LIST].
The folder information display is split into two folder views, upper and lower, which show the same folder information. Immediately after the splitting, the upper folder view becomes active for operations. At the same time, horizontal soft key [SINGLE LIST] appears.

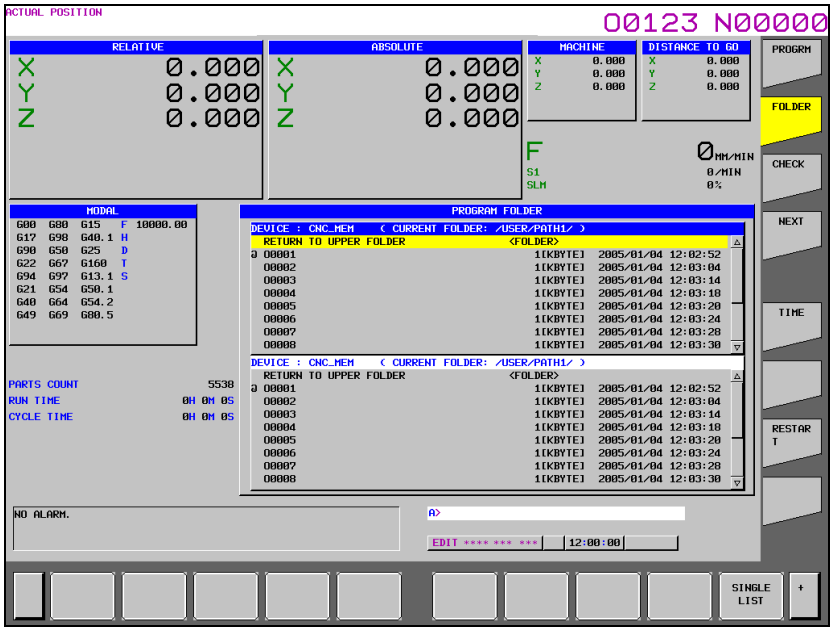


Fig. 12.2.13.1 (c) Program folder screen (split-display screen)(15-inch)

5 Pressing horizontal soft key [SINGLE LIST] takes you back to the original program folder display.
Then, the folders are shown which have been selected in the active view for operations on the split display.

Procedure for switching between active folder views for file operations on the split display

On the split folder display, you can switch between active folder views for file operations as described below.
In the active folder view for file operations, the cursor is displayed on the file list.

Procedure

1 When the upper folder view is active for file operations and you press MDI key [SHIFT] and cursor key [↓], the lower folder view becomes active for file operations.
2 When the lower folder view is active for file operations and you press MDI key [SHIFT] and cursor key [↑], the upper folder view becomes active for file operations.

NOTE
1 In step 1, even if you press MDI key [SHIFT] and cursor key [↑] when the upper folder view is active for file operations, the lower folder view becomes active for file operations.
2 In step 2, even if you press MDI key [SHIFT] and cursor key [↓] when the lower folder view is active for file operations, the upper folder view becomes active for file operations.

Explanation

- File operations on the split display

The following operations are allowed on the split display.
<1> You can select and display devices or folders individually in each folder view.
<2> You can copy or move files between folders selected in different folder views.
(For details, see the section on operations of copying/moving between devices.)
In the selected folder view, you can perform the same operations as on the normal screen.

- Path switching

When path switching is performed while the split display is provided for a multi-path system, the folder view in which the CNC MEM is selected as a device is changed to a view of the last selected folder for each path on the CNC MEM.

- Device information display

The device information as shown in the normal program folder screen (portion in a dotted box in the figure below) is not shown on the split display (Fig. 12.2.13.1 (c)).

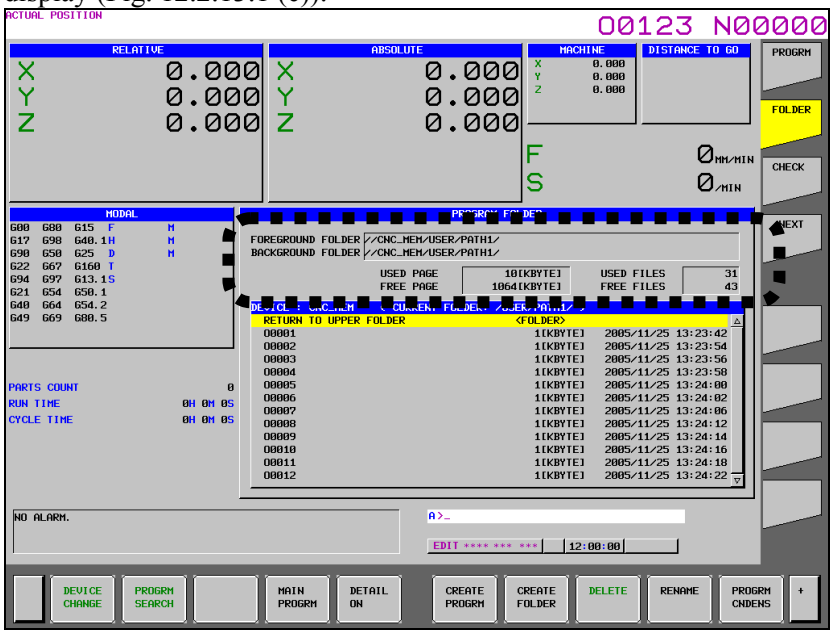


Fig. 12.2.13.1 (d) Program folder screen (normal screen)(15-inch)

You should see the device information on the normal screen.

Limitation

- Device that can be selected in both views at the same time on the split display

The same device of any of the following types can be selected in both views at the same time on the split display.

- CNC MEM
- MEM CARD(binary format)
- Data Server

For other types, the same device cannot be selected for display in the folder views at the same time.

If any other type of device is selected when split display starts, the upper and lower folder views cannot show the same sets of folders due to the above limitation. Therefore, the lower folder view shows the folder selected as a foreground folder.

- Display units that support the split display

The split display is supported on 10.4-inch and 15-inch display units.

12.2.14 Next Block Display Screen (15-inch Display Unit)

Displays the block currently being executed and the block to be executed next.

Procedure for displaying the next block display screen

Procedure

1 Press function key PROG.

2 Press vertical soft key [NEXT].
The G codes, addresses, command values specified in the block currently being executed and the next block are displayed.

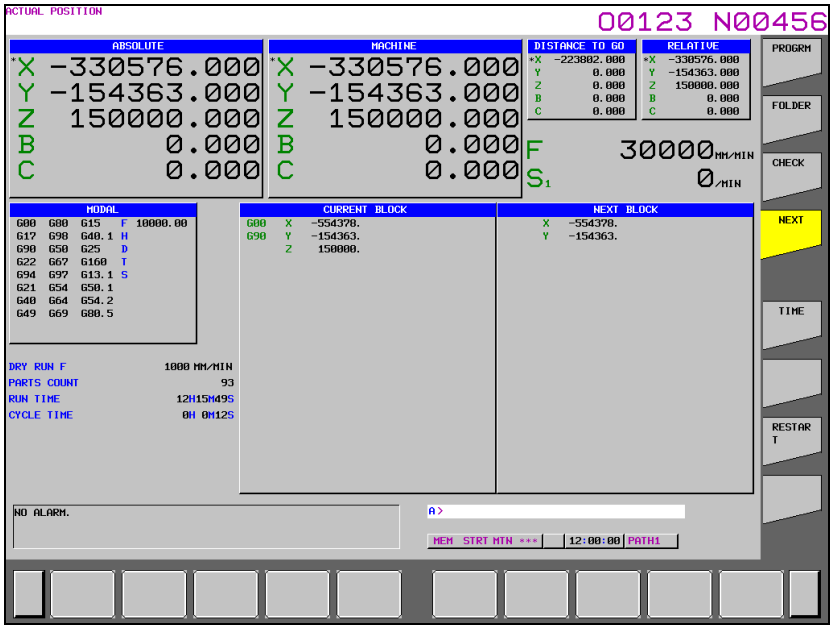


Fig. 12.2.14 (a) Next block display screen (15-inch)

12.2.15 Program Check Screen (15-inch Display Unit)

Displays the program currently being executed, current position of the tool, and modal data.

Procedure for displaying the program check screen

Procedure

1 Press function key

.

2 Press vertical soft key [CHECK].
The program currently being executed, current position of the tool, and modal data are displayed.

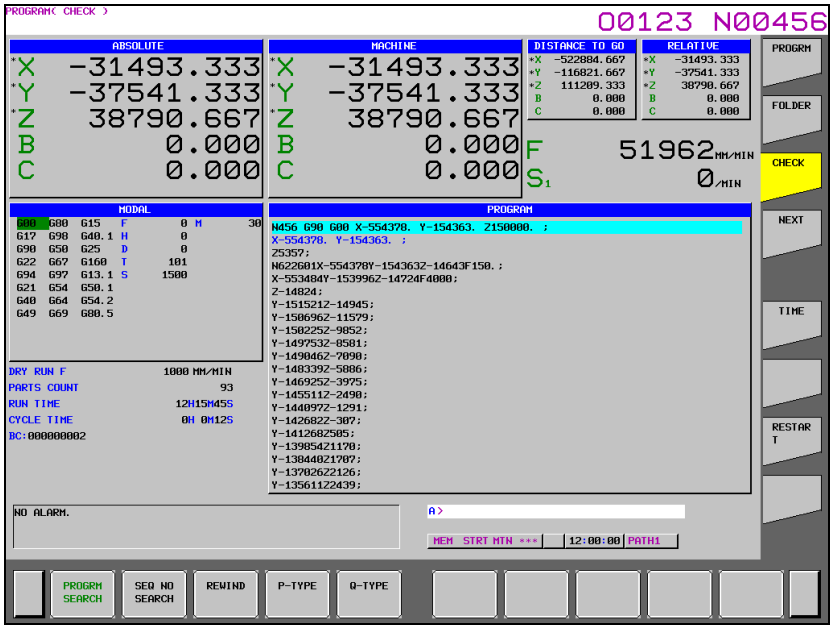


Fig. 12.2.15 (a) Program check screen (15-inch)

Explanation

- Program display

The program currently being executed is displayed.
The block being executed is displayed in reverse video.

- Current position display

The current position in the relative coordinate system, workpiece coordinate system, and machine coordinate system, and the remaining amount of travel are displayed.

- Modal G codes

Up to 24 modal G codes are displayed.

12.2.16 Background Editing (15-inch Display Unit)

Editing one program during execution of another program is referred to as background editing. You can perform the same edit operations in the background as those in normal editing (foreground editing).

On a 10.4-inch or 15-inch display, you can perform background editing for the programs displayed in multiple tiled windows. You can switch from a program displayed in one of the tiled windows to another to perform copy, paste, and other edit operations, which allows you to edit programs efficiently.
You can edit two programs simultaneously on a 10.4-inch display or four programs on a 15-inch display.

Function

- Background editing

A program other than the currently selected program can be edited. Background editing can be performed in any mode.

- Program selected in the foreground

If the program selected in the foreground is specified as a program to be edited in the background, background editing is started in the read-only mode. The text at an arbitrary position of the program currently being executed can be checked.

- Switching from the program directory screen to the editing screen

By selecting a program from the program folder screen, background editing can be started.
On the program folder screen, move the cursor to a program to be edited in the background, then press the MDI key

. The screen display switches to the background editing screen.
Background editing can be started without entering a program name.

- Background editing on a multi-path system

Background editing does not depend on the path.
Even if the path to be operated is changed using the path selection signal, the background editing currently being executed is just continued.

Display

When background editing starts, the ordinary program editing screen switches to the background editing screen.
When two or more programs are edited in the background, the screen is split to display these programs. For a 10.4-inch display, you can edit two programs simultaneously; for a 15-inch display, you can edit four programs simultaneously.

- Word editing

Fig. 12.2.16 (a) shows background word editing performed simultaneously for two programs (right and left programs). On the status line at the top of the window for each program, the program name and "BG-EDIT" (indicating background editing in progress) are displayed.
The status line of the program being edited is displayed in reverse video.

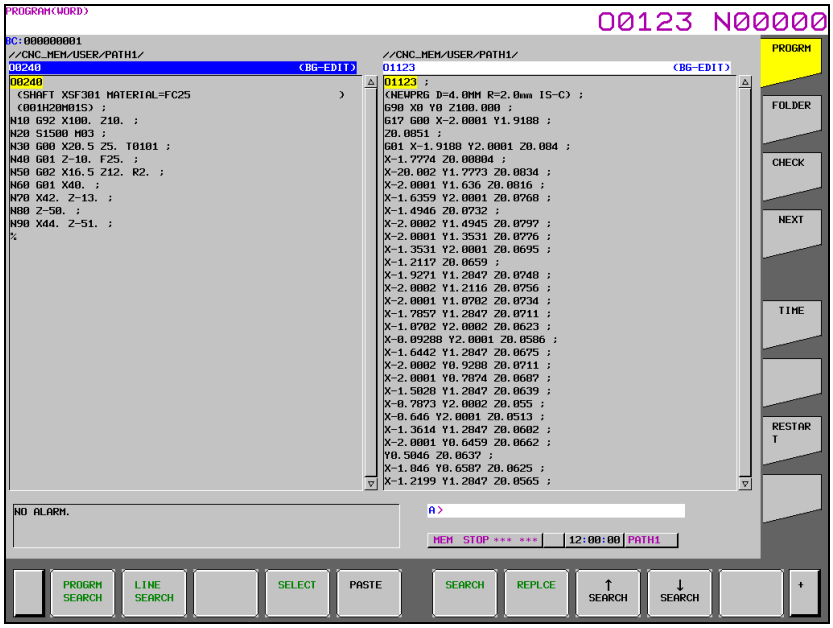


Fig. 12.2.16 (a) Background editing screen (word editing) (15-inch)

- Character editing

Fig. 12.2.16 (b) shows background character editing performed simultaneously for two programs (right and left programs). Similarly to word editing, at the top of the window for each program, the status line is displayed. In addition, the current input mode (INSERT MODE or OVERWRITE MODE) is displayed at the upper right of the screen for character editing.
The status line of the program being edited is displayed in reverse video.

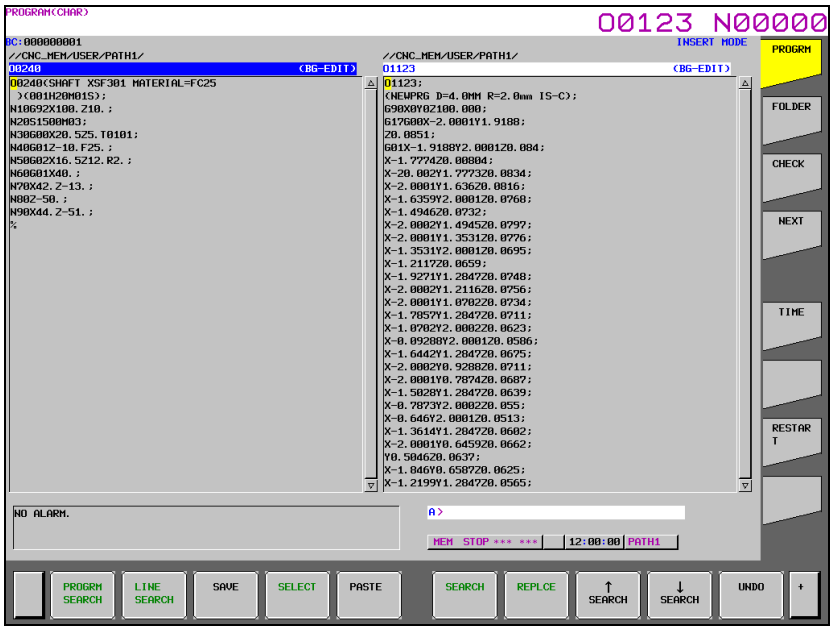


Fig. 12.2.16 (b) Background editing screen (character editing) (15-inch)

- Editing status

The following items are displayed on the status line and program editing area according to the background editing status.

Editing status	Displayed items
No program selected	(BG-EDIT) “NO PROGRAM” is displayed in the editing area.
Program opened	program-name + (BG-EDIT)
Read-only program opened	program-name + (BG:READ ONLY) The contents of the program are displayed in green.

Starting background editing from the editing screen

Procedure

Method (only for word editing)

1 Press function key

.
2 Press vertical soft key [PROGRAM].
3 Key in the name of a program to be edited in the background.
4 Press horizontal soft key [BG EDIT].

When background editing has already started and editing start operation is performed again, the program being edited and newly specified program are edited simultaneously.

Explanation

When a program name is input in the key-in buffer, background editing for the program starts. When the specified program is not found, a new program is created and background editing starts.
When background editing is started with no program name specified, new background editing starts in the no program state. Perform a program search or create a new program.
When character editing is selected, background editing starts in the no program state first. Perform a program search or create a new program.

Starting background editing from the program folder screen

By selecting a program from the program folder screen, background editing can be started. The cursor is used to select a program. You do not need to enter a program name.

Procedure

1 Press function key

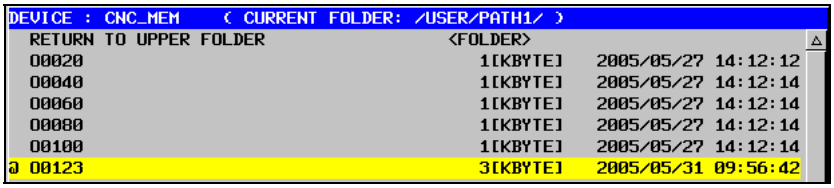
.

2 Press vertical soft key [FOLDER].

3 Move the cursor to a program to be edited.

4 Press the MDI key

. The screen display is switched to the background editing screen.



Background editing operation

- Editing operation

The same editing operations as performed in the foreground can be performed.

- Switching from a program to another for editing

To switch from a program to another for editing when editing multiple programs simultaneously in the background, press

key, then or .

- Editing operation in the read-only mode

Cursor movement and page switching are possible. However, program editing is disabled.

- Copying and pasting text between programs

Text in a program selected and copied during program editing is kept held after the program to be edited is changed. You can paste text in a program onto another program.

Background editing end operation

Background editing can be ended using the procedure described below.
The procedure for ending background editing of one program and that for ending all background editing of multiple programs are shown below.

- Ending editing of one program

1 Press function key

.
2 Press vertical soft key [PROGRAM].
3 Select a program for which to end editing.
4 Press horizontal soft key [BG END].

- Ending editing of all programs

1 Press function key

.
2 Press vertical soft key [PROGRAM].
3 Press horizontal soft key [BG ALL END].

To return to ordinary foreground editing, end all background editing. If at least one program is being edited in the background, you cannot return to foreground editing.

12.2.17 Stamping the Machining Time (15-inch Display Unit)

The execution times of the most recently executed ten programs can be displayed in hours, minutes, and seconds.
The calculated machining time can be inserted as a comment of the program to check the machining time on the program folder screen.

Procedure for stamping the machining time

Procedure

- Displaying the machining time

1 Press function key

.

2 Press vertical soft key [TIME]. The machining time display screen appears.

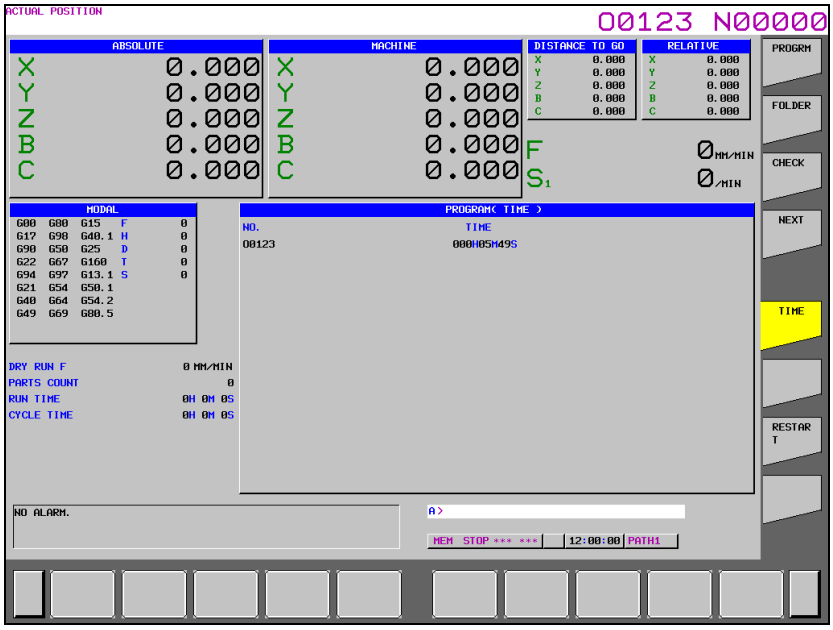


Fig. 12.2.17 (a) Machining time display screen (15-inch)

- Calculating the machining time

1 Select the memory operation mode, then press the

key.

2 Select the program screen, then select a program of which machining time you want to calculate.

3 Execute the program to perform actual machining.

4 When the

key is pressed or, M02 or M30 is executed, the machining time count operation stops. When the machining time display screen is selected, the program number of the stopped main program and its machining time are displayed.

5 The following figure shows the screen when the machining times of the ten main programs O0020, O0040, …, and O0200 are displayed and the screen when the machining time of O0220 is newly calculated after that.

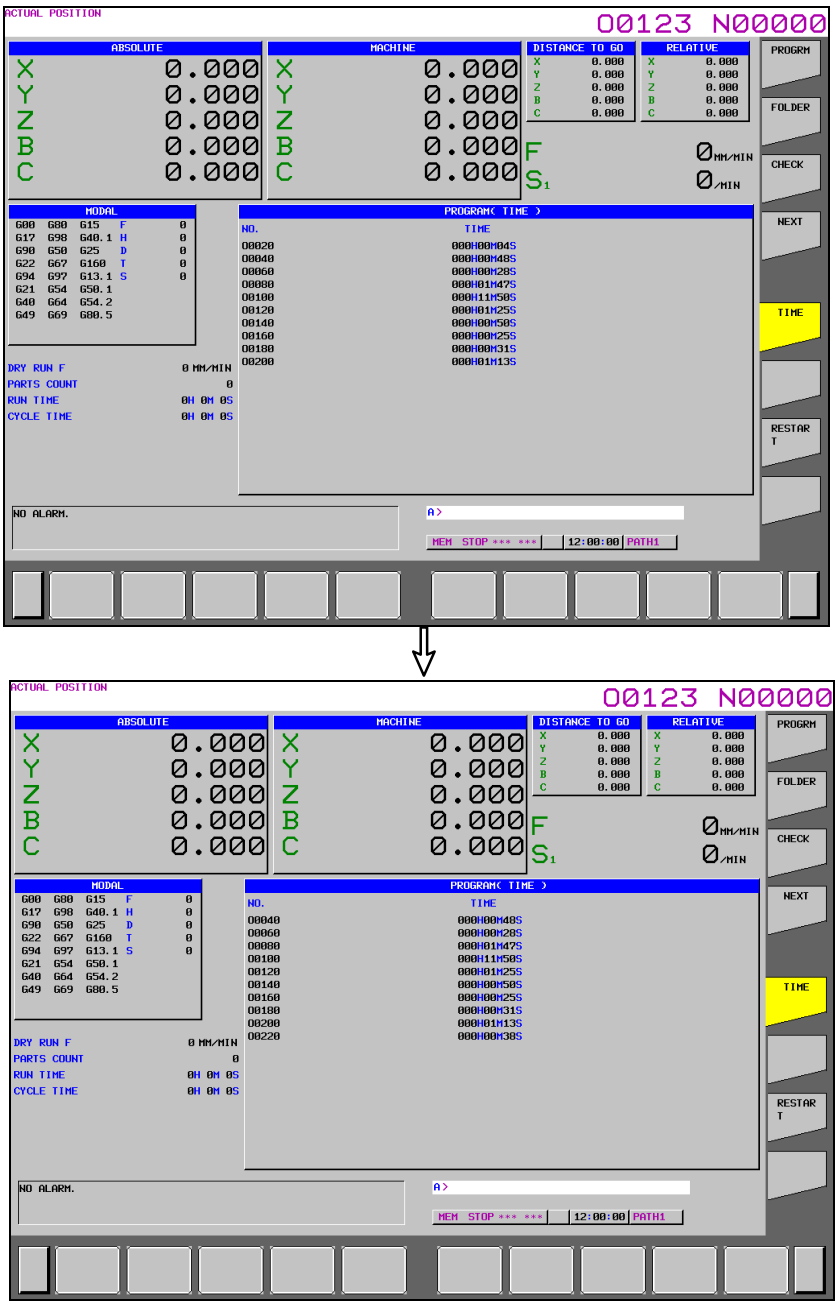


Fig. 12.2.17 (b) Stamping the machining time (15-inch)

Procedure for inserting the machining time on the program screen

Procedure

You can display the machining time of a program as a comment of the program. The procedure is shown below:

1 To insert the calculated machining time of a program as a comment, the machining time of the program must be displayed on the machining time display screen. Before stamping the machining time of the program, check that the program number is displayed on the machining time display screen.
2 Set the edit mode or background editing state and select the program screen. Then, select a program of which machining time you want to insert.
3 For example, the machining time of O0100 is displayed on the machining time display screen. Press continuous menu key ▷ until horizontal soft key [INSERT TIME] appears. When horizontal soft key [INSERT TIME] is pressed, the start of the program is located, and the machining time of the program is inserted after the program name.

Press soft key [INSERT TIME].

Fig. 12.2.17 (c) Program screen (15-inch)

4 If a comment is written in the block containing the program number of a program of which machining time is to be inserted, the machining time is inserted after the comment.

Press soft key [INSERT TIME].

Fig. 12.2.17 (d) Program screen (15-inch)

Display on the program folder screen

The machining time of a program inserted in the program as a comment is displayed after the existing comment of the program on the program folder screen.

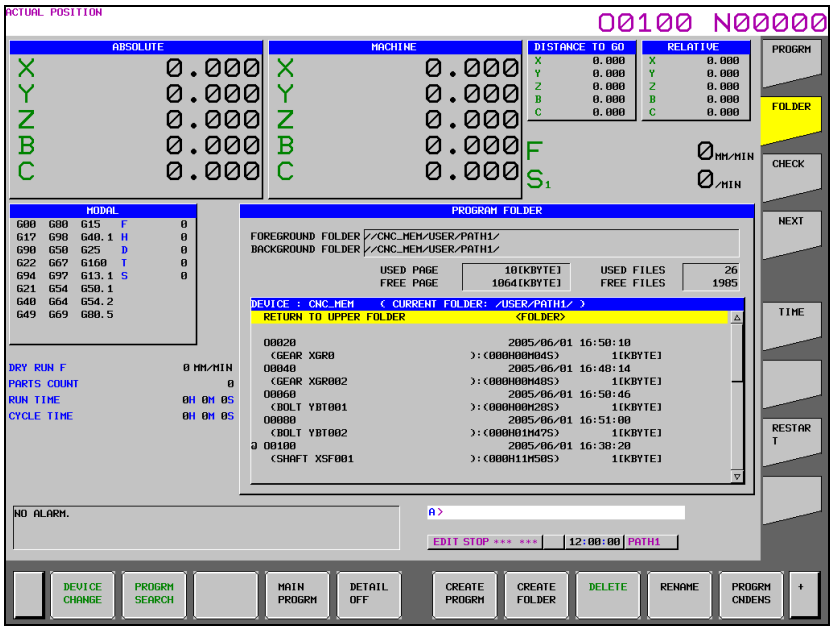


Fig. 12.2.17 (e) Program folder screen (15-inch)

Explanation

- Machining time

The machining time is counted from the initial start after a reset in the memory operation mode to the next reset. If a reset is not performed during operation, the machining time is counted from the start to M02 or M30. The time during which operation is held is not counted, but the time used to wait for the completion of the M, S, T, and/or B functions is counted, however.

- Stamping the machining time

You can insert (stamp) the displayed machining time in a program stored in memory as a comment. The machining time is inserted as a comment after the program number.

- Program folder

You can display the machining time inserted after a program number on the program folder screen. This operation lets you know the machining time of each program to use the time as useful reference data when planning processes on site.

Limitation

- Alarm

The execution of a program may be held by an alarm during the machining time count operation. In this case, the machining time is counted until the alarm is released by a reset.

- M02

It may be specified that M02 does not reset the CNC, but returns completion signal FIN to the CNC to restart the program from the beginning successively (bit 5 (M02) of parameter No. 3404 is set to 0). In this case, when M02 returns completion signal FIN, the machining time count operation stops.

- Stamping the machining time

When an attempt is made to stamp the machining time of a program in the program, the machining time of the program may not be displayed on the machining time display screen. In this case, the machining time cannot be inserted in the program even if horizontal soft key [INSERT TIME] is pressed.

- Correcting the machining time

If an incorrect machining time is calculated (such as when a reset is made during the execution of a program), reexecute the program to calculate the correct machining time. The same program number may be displayed in two or more lines on the machining time display screen. In this case, the latest calculated machining time is inserted in the program.

- States of the stamped machining time

In the following states, the stamped machining time is displayed on the program folder screen as shown below.

1 When the comment of a program is longer than 16 characters
The 17th and subsequent characters of the comment are discarded and the machining time display field is left blank.

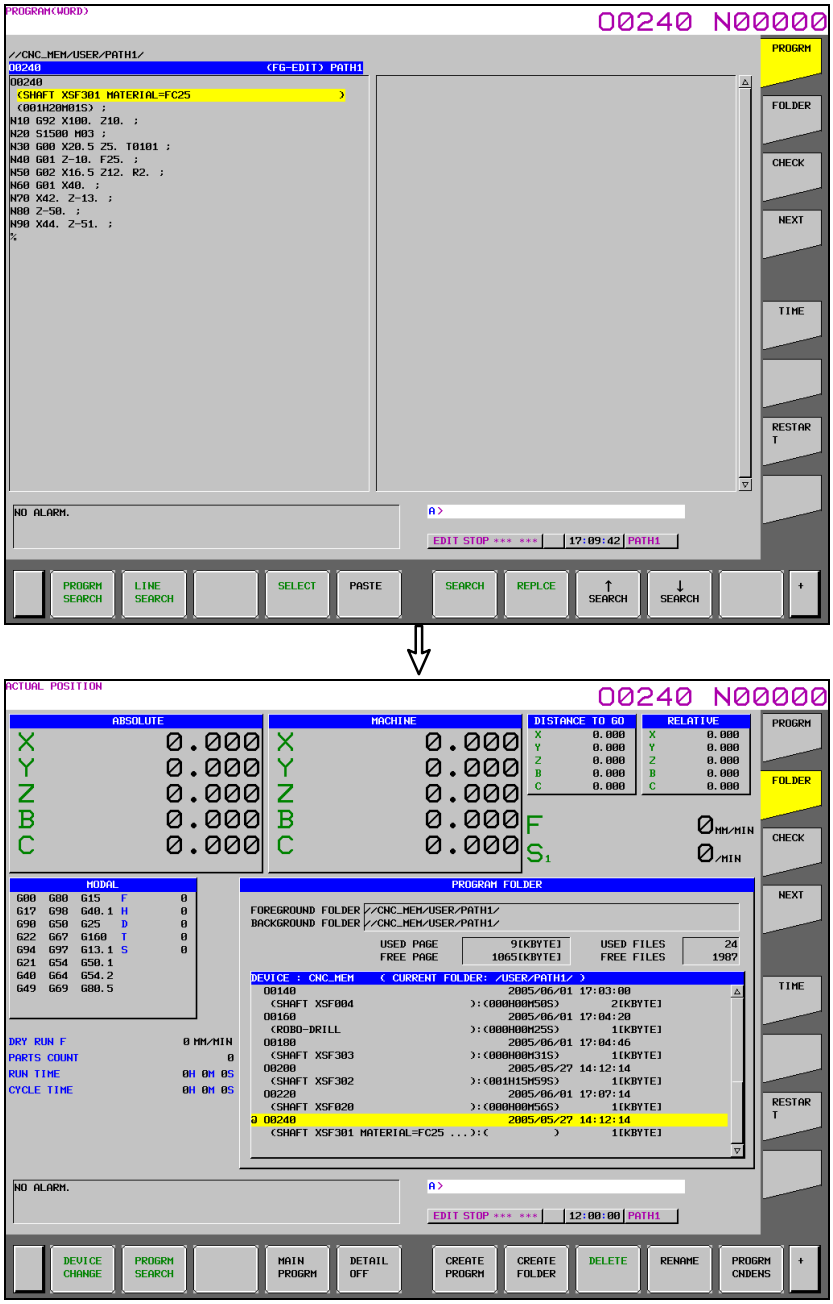


Fig. 12.2.17 (f) When the comment of a program is longer than 16 characters (15-inch)

2 When two or more machining times are stamped
The first machining time is displayed.

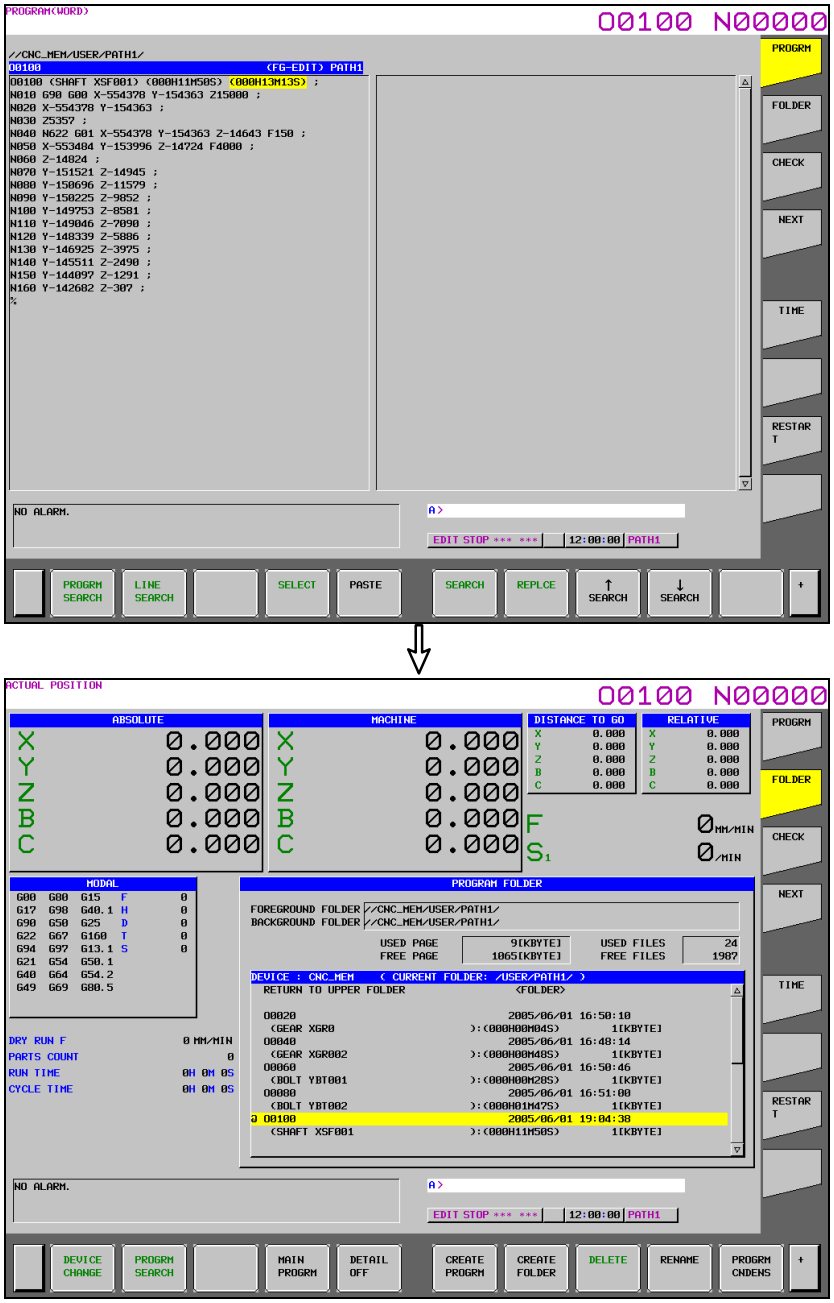


Fig. 12.2.17 (g) When two or more machining times are stamped (15-inch)

3 When the format of an inserted machining time is not “hhhHmmMssS” (H following a 3-digit number, M following a 2-digit number, and S following a 2-digit number, in this order)
The machining time display field is left blank.

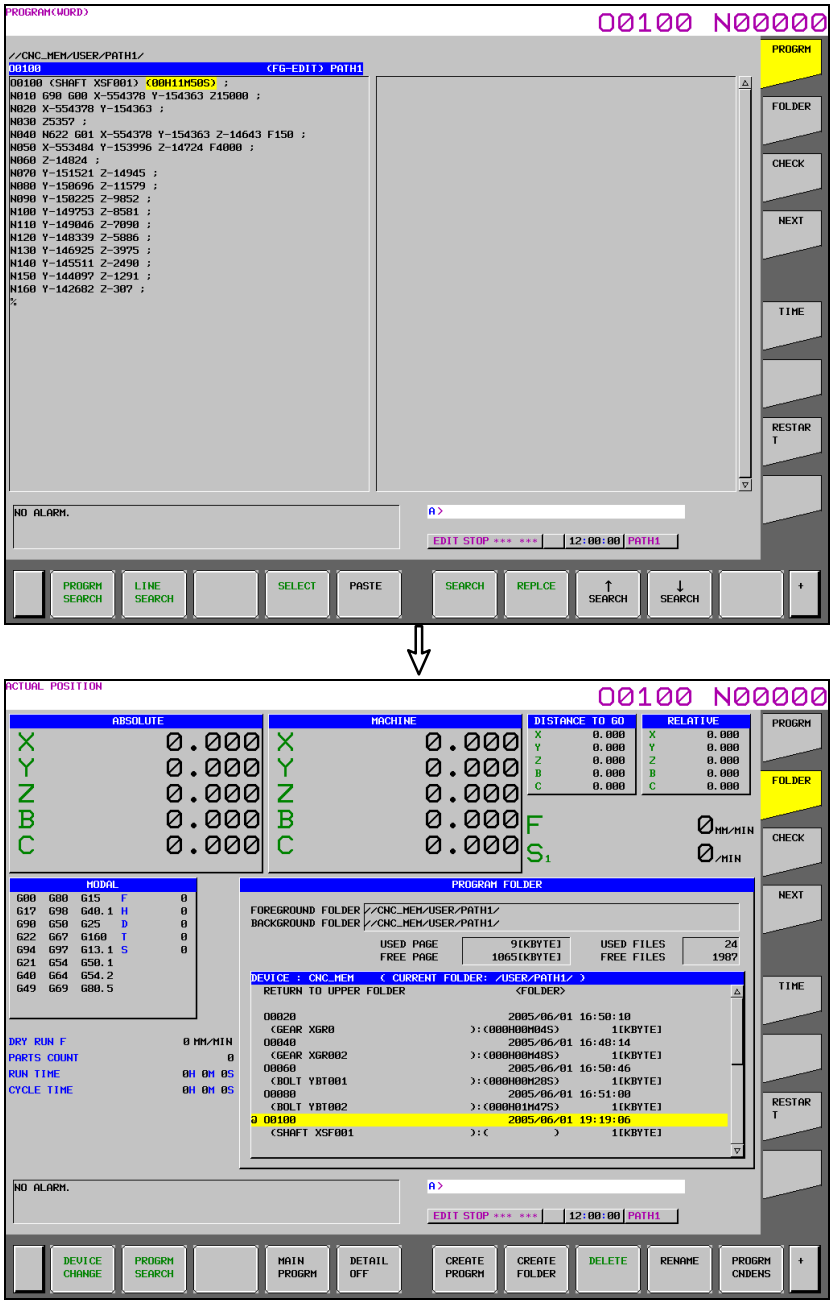


Fig. 12.2.17 (h) When the format of an inserted machining time is not “hhhHmmMssS” (H following a 3-digit number, M following a 2-digit number, and S following a 2-digit number, in this order) (15-inch)

12.2.18 Screen for Assistance in Entering Tilted Working Plane Command (15-inch Display Unit)

The screens for assistance in entering tilted working plane commands (hereinafter referred to as guidance screens) include a command type selection screen and a tilted working plane data setting screen. The command type selection screen is used to select a tilted working plane command. The tilted working plane data setting screen is used to set specified tilted working plane data required for the selected command. By making settings and performing operations on these guidance screens, a tilted working plane command block can be created.
The created block is reflected as a new insertion to a program being edited or as a modification to an existing block.
This function can be enabled by setting bit 1 (GGD) of parameter No. 11304 to 1.

Creation of a new block

The following describes the procedure for creating a tilted working plane command block on guidance screens and for inserting the block to a program being edited on a program editing screen.

1 On a program editing screen, display a program to which you want to insert a tilted working plane command block.
The foreground editing screen, background editing screen, or MDI editing screen should be displayed.

- Displaying the foreground editing screen
  <1> Select the EDIT mode.
  <2> Press function key

.
  <3> Press vertical soft key [PROGRAM].

- Displaying the background editing screen
  <1> Press function key

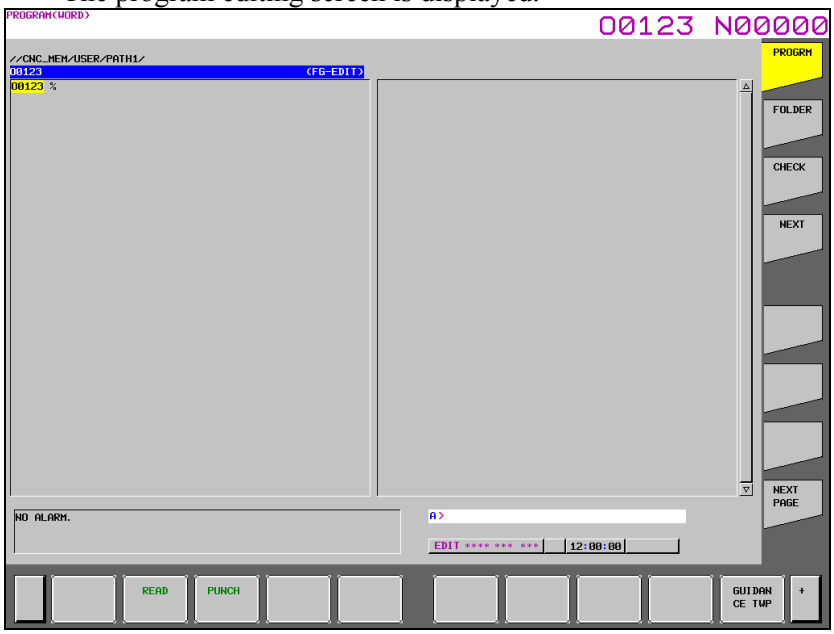
.
  <2> Press soft key [FOLDER].
  <3> Press any of the cursor keys to move the cursor to a program to be edited in the background.
  <4> Press

key.

- Displaying the MDI editing screen
  <1> Select the MDI mode.
  <2> Press function key

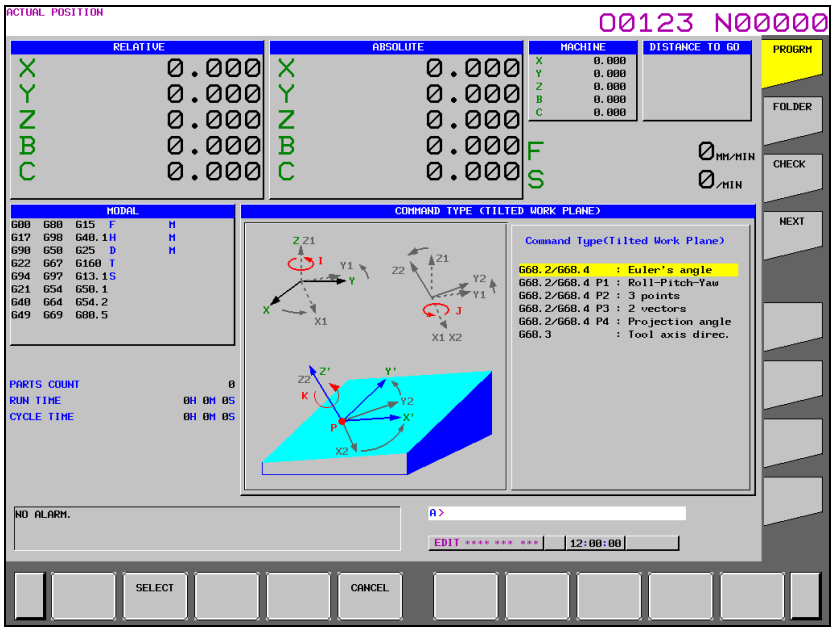
.
  <3> Press vertical soft key [PROGRAM].

The program editing screen is displayed.

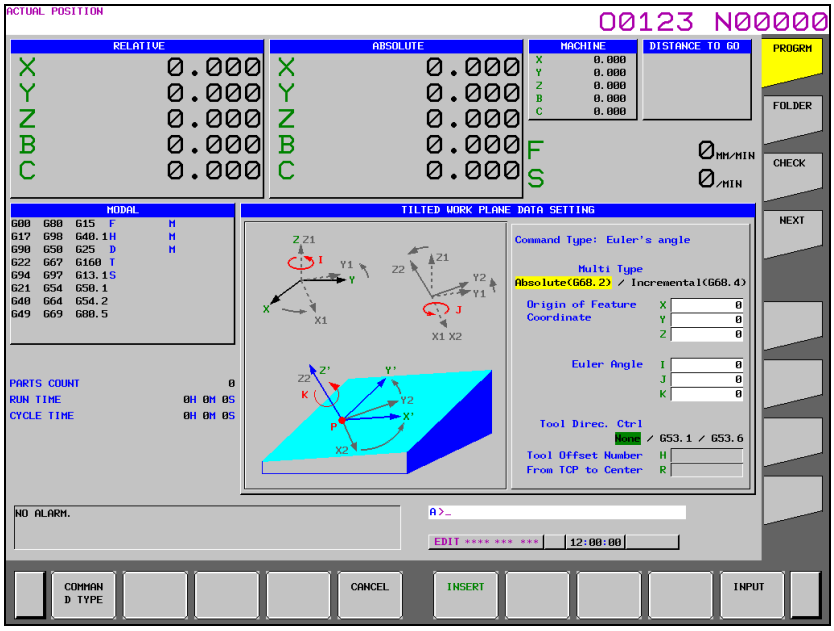


2 Press any of the cursor keys to move the cursor to a position where you want to insert a block.
Note that a block created on the guidance screens is inserted after the block at the cursor position. (If the block at the cursor position includes a tilted working plane command, the existing block is modified. See "Modification to an existing block" below.)

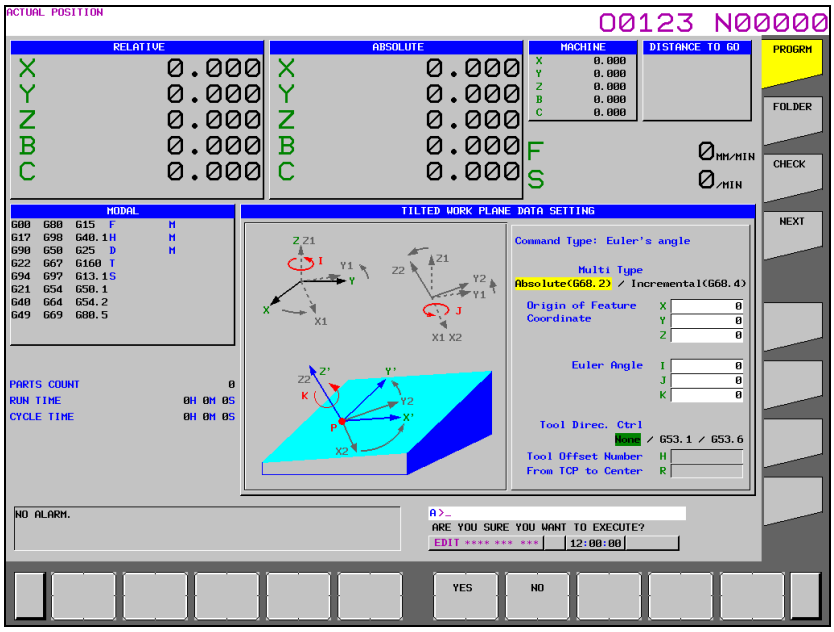
3 Press continuous menu key ▷ several times, and then press horizontal soft key [GUIDANCE TWP].
The command type selection screen is displayed.



4 Select a command type with any of the cursor keys, and then press horizontal soft key [SELECT].
The tilted working plane data setting screen is displayed.



5 Enter command data for the setting items.
6 Press horizontal soft key [INSERT].

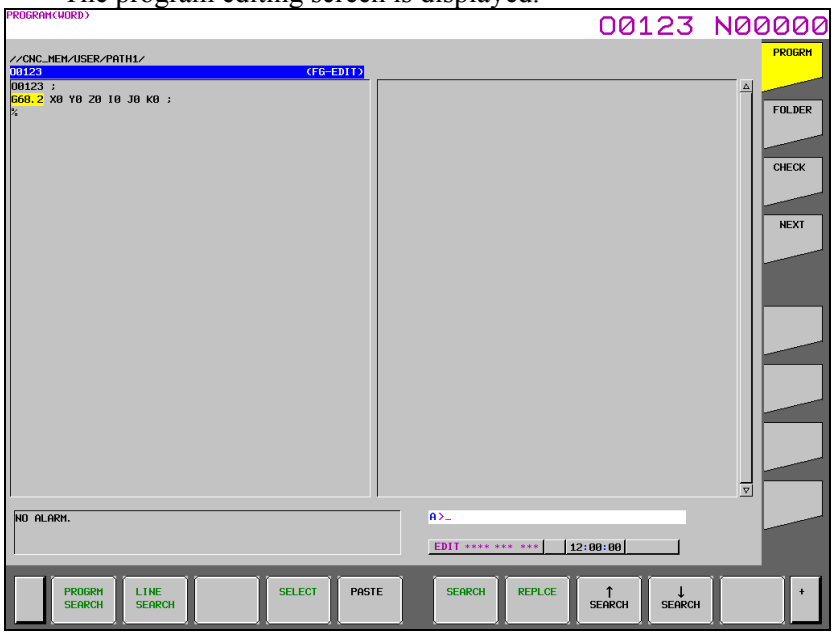


7 Press horizontal soft key [YES].
This takes you back to the program editing screen, where the new block is inserted after the block at the cursor position.

Modification to an existing block

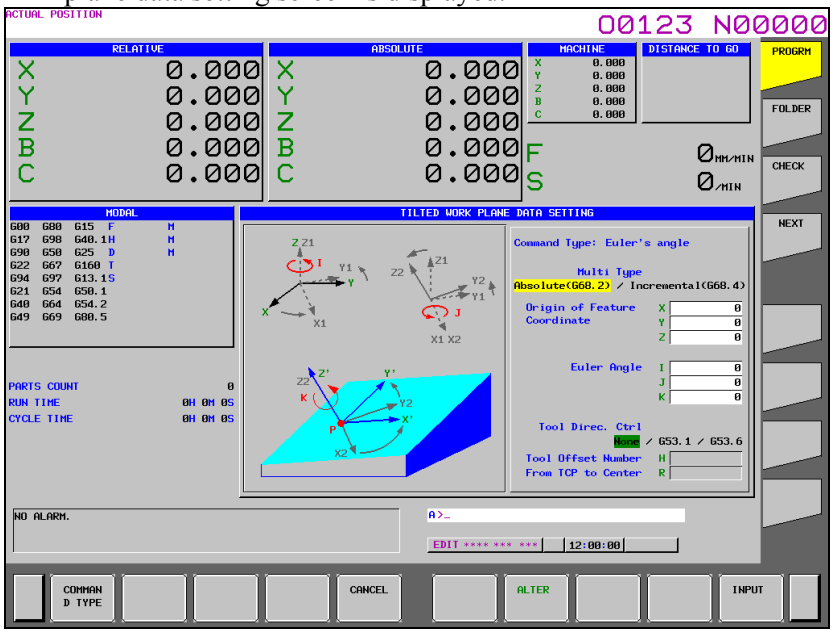
The following describes the procedure for replacing a block in a program being edited on a program editing screen, with a tilted working plane command block created on a guidance screen.

1 On a program editing screen, display a program to be edited.
(For the procedure for displaying a program editing screen, see step 1 in "Creation of a new block".)
The program editing screen is displayed.



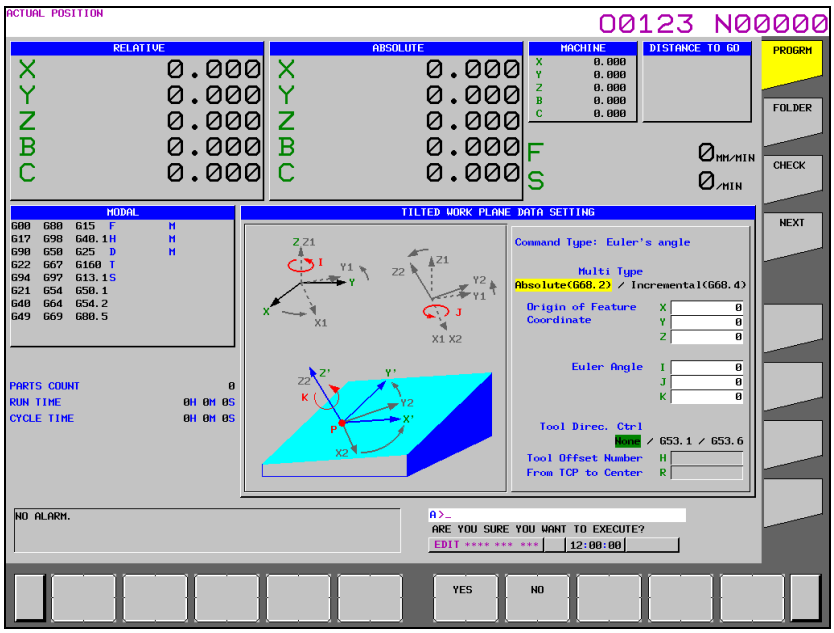
2 Press any of the cursor keys to move the cursor to a block to be modified.
For a command extending over more than one block, move the cursor to the first block.

3 Press continuous menu key ▷ several times, and then press horizontal soft key [GUIDANCE TWP]. The tilted working plane data setting screen is displayed.



4 Enter command data for setting items to be modified.

5 Press horizontal soft key [ALTER].



6 Press horizontal soft key [YES].
This takes you back to the program editing screen, where the block at the cursor position is replaced.

Guidance screen cancellation

Pressing horizontal soft key [CANCEL] on a guidance screen takes you back to the program editing screen. At this time, the data that has been set on the guidance screen is discarded.

NOTE

1 In addition to the above operation, the following operations also cancel a guidance screen. The data that has been set on the guidance screen is discarded.
- When bit 7 (CPG) of parameter No. 11302 is 1 (setting for automatically switching between program-related screens according to CNC mode), the CNC mode is changed.
- When a guidance screen is displayed from the foreground editing screen, the CNC mode is changed to a mode other than EDIT, TJOG, or THND.
- When a guidance screen is displayed from the MDI editing screen, the CNC mode is changed to a mode other than MDI.
- When a guidance screen is displayed on a 15-inch display unit, the screen is switched by a vertical soft key.
- The screen is switched by an MDI key.
- The screen is switched by a path select signal.
- An event that causes screen switching has occurred, including occurrence of an alarm, display of an operator message, or signal-based display of a screen (such as a tool compensation screen, workpiece shift screen, workpiece coordinate system setting screen, or C language executor screen).

2 If MDI key

is pressed after switching from a guidance screen to another screen, a program editing screen is displayed, instead of the guidance screen.

Notes

- Conditions under which horizontal soft key [GUIDANCE TWP] is displayed

  Horizontal soft key [GUIDANCE TWP] is displayed on a program editing screen under the following conditions:

  1 Foreground editing screen
  - The CNC mode is EDIT, TJOG, or THND.
  - The editing and display are not prohibited for a program to be edited.

2 Background editing screen
- A program to be edited is not in reference mode.
- The editing and display are not prohibited for a program to be edited.

3 MDI editing screen
- The CNC mode is MDI.

- Screen displayed when horizontal soft key [GUIDANCE TWP] is pressed

  When horizontal soft key [GUIDANCE TWP] is pressed, which guidance screen is displayed is automatically determined according to the conditions described below.

  1 When a tilted working plane command is not included in a block at the cursor position on a program editing screen

    The command type selection screen is displayed. A block created on the guidance screens is inserted after the block at the cursor position on the program editing screen.

  2 When a tilted working plane command is included in a block at the cursor position on a program editing screen

    The tilted working plane data setting screen is displayed, which shows the data for the tilted working plane command in the block at the cursor position on the program editing screen. When a block of a tool axis direction control command exists immediately after the tilted working plane command, the command data for the block is also shown. The tilted working plane command in the block at the cursor position on the program editing screen is replaced with a block created on the guidance screen.

NOTE

If the CNC is in the reset state or emergency stop state when horizontal soft key [GUIDANCE TWP] is pressed on the foreground editing screen or MDI editing screen, the warning "PROGRAM READ FAILED" appears, and the operation cannot be continued. (Horizontal soft keys other than [CANCEL] are not displayed.) Press horizontal soft key [CANCEL] to return to the program editing screen, and then press horizontal soft key [GUIDANCE TWP] again.

12.2.18.1 Command type selection screen

The command type selection screen is used to select the type of a tilted working plane command you want to insert into a program to be edited. One of the following command types can be selected:

- G68.2 / G68.4 (Euler's angle)
- G68.2 / G68.4 P1 (Roll-Pitch-Yaw angle)
- G68.2 / G68.4 P2 (3 points specification)
- G68.2 / G68.4 P3 (2 vectors specification)
- G68.2 / G68.4 P4 (Projection angle)
- G68.3 (Tool Axis Direction)

G68.2 is an absolute command, and G68.4 is an incremental command.

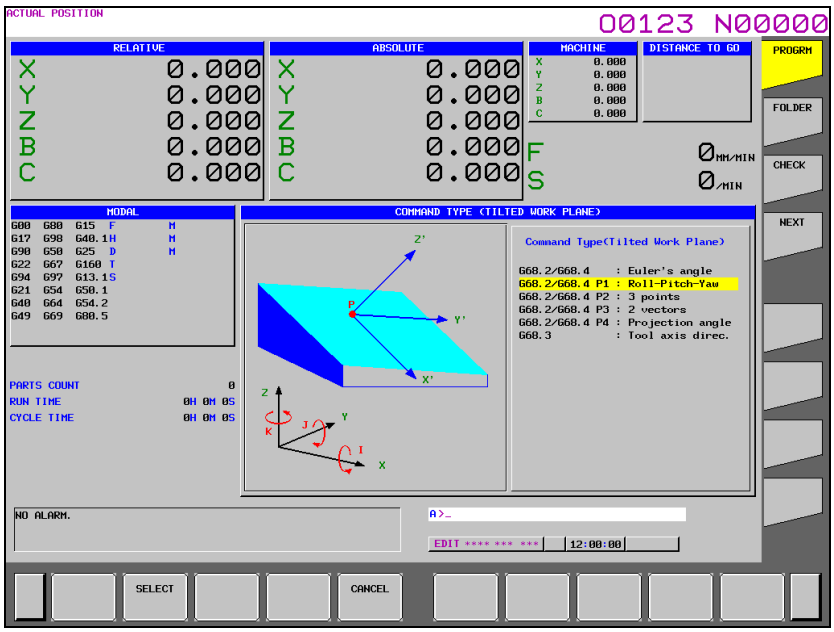


Fig. 12.2.18.1 (a) Command type selection screen (15-inch)

Selection of a command type

1 Press cursor key ↑ or ↓ to move the cursor to a command type you want to select. As the cursor is moved, the figure corresponding to the command type at the cursor position is displayed.

2 When horizontal soft key [SELECT] is pressed, the command type at the cursor position is accepted, and the tilted working plane data setting screen is displayed.

NOTE

If the warning "PROGRAM READ FAILED" appears when the command type selection screen is displayed, the operation cannot be continued. (Horizontal soft keys other than [CANCEL] are not displayed.) Press horizontal soft key [CANCEL] to return to the program editing screen, and then press horizontal soft key [GUIDANCE TWP] again.

12.2.18.2 Tilted working plane data setting screen

The tilted working plane data setting screen is used to set specified tilted working plane data required for a tilted working plane command of the type that has been selected on the command type selection screen or has been selected when horizontal soft key [GUIDANCE TWP] is pressed.
Different types of tilted working plane data setting screen are provided for different command types. For details of each type of tilted working plane data setting screen, see "Details of the tilted working plane data setting screen".

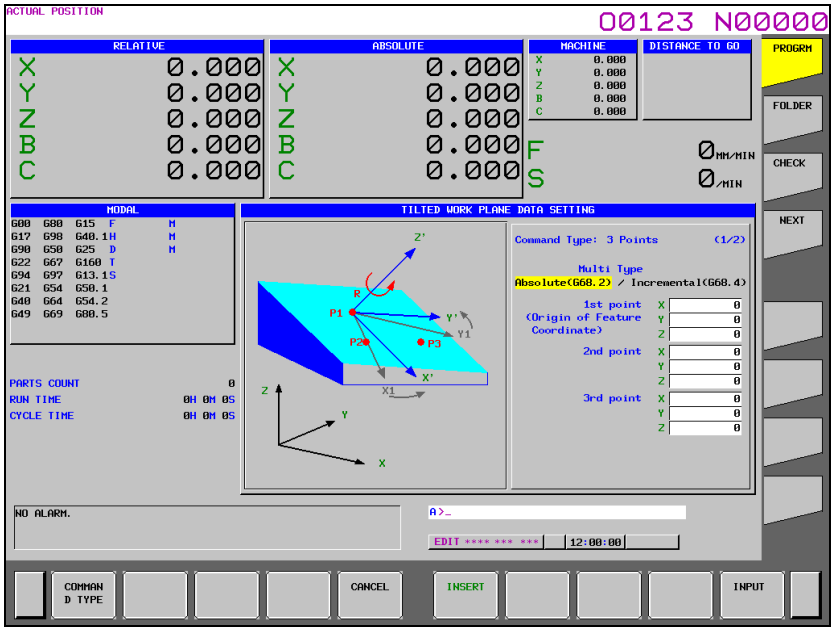


Fig. 12.2.18.2 (a) Tilted working plane data setting screen−3 points specification(15-inch)

Display of the tilted working plane data setting screen

When a new block is created, the initial values are set in the setting and selection items.
When an existing block is modified, the command data for the block at the cursor position on the program editing screen is reflected in the setting and selection items.

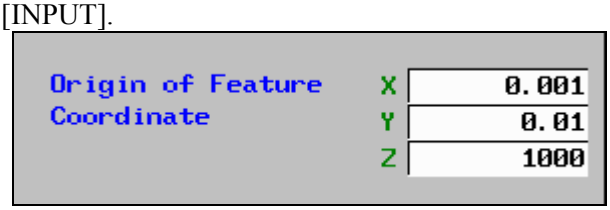
NOTE
1 For existing block modification, if the guidance screen is displayed when the cursor is placed in the middle of multiple blocks for a command, the parameters for the block(s) before the cursor are not reflected in the setting and selection items, but only the parameters for the block(s) after the cursor are reflected. If no data is to be reflected in a setting or selection item, the initial value is set in the item.
2 For existing block modification, only the commands with parameter-specified axis names are reflected in the setting items on the screen. (The values specified in data for commands with incorrect axis names are not reflected in the setting items.)

Command data input

- Item for which to enter a value

  Press cursor key or to move the cursor to an item you want to set.

  Enter a value, and then press the key or horizontal soft key [INPUT].



  Example)

  When the origin of a feature coordinate system is set as shown above, addresses X, Y, and Z are specified as follows:

  G68.2 X0.001 Y0.01 Z1000 ...

  For two items "Tool Offset Number" and "From TCP to center", the state with no command can be set by deleting the specified values according to the following procedure.

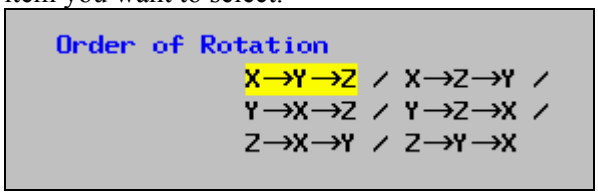
  1 Press cursor key or to move the cursor to the item "Tool Offset Number" or "From TCP to Center".

  2 Press or horizontal soft key [INPUT] without entering anything.

- Item to be selected from a list

  1 Press cursor key [↑] or [↓] to move the cursor to an item you want to set.

  2 Press cursor key [←] or [→] to move the cursor to an item you want to select.



Example)
Order of rotation for Roll-Pitch Yaw angle

Insertion of a block

If the guidance screen is displayed when the block at the cursor position on the program editing screen does not include a tilted working plane command, horizontal soft key [INSERT] is displayed on the tilted working plane data setting screen. An operation for inserting a tilted working plane command block is described below.
For warnings that may be issued at the time of block insertion, see "Limitation".

1 Press horizontal soft key [INSERT].
The confirmation message "ARE YOU SURE YOU WANT TO EXECUTE?" is displayed, and horizontal soft keys [YES] and [NO] are displayed.

2 Press horizontal soft key [YES].
A block is created from the command type and command data and then is inserted after the block at the cursor position in the program to be edited. After block insertion, the guidance screen is closed and the program editing screen is displayed.
Pressing horizontal soft key [NO] takes you back to the tilted working plane data setting screen.

Example)

```
G00 X0.;
   ↓
G00 X0.;
G68.2 P2 Q0...
G68.2 P2 Q1...
G68.2 P2 Q2...
G68.2 P2 Q3...
```

When the guidance screen is displayed and the 3-point specification is selected as a command type for block insertion, a created block is inserted after the block at the cursor position.

Replacement of a block

If the guidance screen is displayed when the block at the cursor position on the program editing screen includes a tilted working plane command, horizontal soft key [ALTER] is displayed on the tilted working plane data setting screen. An operation for replacing a tilted working plane command block is described below.
For warnings that may be issued at the time of block replacement, see "Limitation".

1 Press horizontal soft key [ALTER].
The confirmation message "ARE YOU SURE YOU WANT TO EXECUTE?" is displayed, and horizontal soft keys [YES] and [NO] are displayed.
2 Press horizontal soft key [YES].
A block is created from the command type and command data and then replaces the tilted working plane command in the block at the cursor position on the program editing screen. After block replacement, the guidance screen is closed and the program editing screen is displayed.
Pressing horizontal soft key [NO] takes you back to the tilted working plane data setting screen.

NOTE
1 If a block to be replaced includes a command other than a tilted working plane command, the command is deleted during block replacement. However, only the sequence number at the beginning is preserved.
2 If the guidance screen is displayed when the cursor is placed in the middle of multiple blocks for a command, the block(s) before the cursor are not replaced. These blocks remain unchanged after replacement.
3 If horizontal soft key [GUIDANCE TWP] is pressed when the cursor is placed on a block of a tool axis direction control command, the command type selection screen is displayed in new insertion mode. A block created on the guidance screens is inserted after the block of the tool axis direction control command.

Limitation

The following lists the warnings that may be issued at the time of block insertion or replacement.
If a warning is displayed, return to the program editing screen with horizontal soft key [CANCEL] and press horizontal soft key [GUIDANCE TWP] again, or eliminate the cause of the warning and retry the operation.

Warning	Description
“PROGRAM WRITE FAILED”	• The guidance screen was displayed from the foreground editing screen or MDI editing screen, and a block insertion or replacement operation was performed when the CNC was in the reset state or emergency stop state.
”PROGRAM CANNOT BE WRITTEN”	• A block insertion or replacement operation was performed after a program to be edited was updated due to program downloading by an external application while the guidance screen was displayed. • A block insertion or replacement operation was performed after the start of the main program was located by a reset while the guidance screen was displayed.
“WRITE PROTECT”	• A block insertion or replacement operation was performed when the editing or display was prohibited for a program to be edited. • A block insertion or replacement operation was performed when the memory protection signal (KEY signal) for program registration or editing was off. • A block insertion or replacement operation was performed when the operation level of the 8-level data protection function was lower than the protection level of a part program editing operation.

Display of the command type selection screen

When horizontal soft key [COMMAND TYPE] is pressed, the command type selection screen is displayed. When the command type is changed on the command type selection screen, the values set on the tilted working plane data setting screen are cleared.
When a tilted working plane command is included in the block at the cursor position on the program editing screen, you can display the guidance screen, change the command type on the command type selection screen, and then perform block replacement. In this case, the block at the cursor position on the program editing screen is replaced with a block created from the command type and command data that have been set.

12.2.18.3 Details of the tilted working plane data setting screen

The following six tilted working plane commands are supported.
For details of the commands, see II-22.3, "TILTED WORKING PLANE COMMANDS".

- G68.2 / G68.4 (Euler's angle)
- G68.2 / G68.4 P1 (Roll-Pitch-Yaw angle)
- G68.2 / G68.4 P2 (3 points specification)
- G68.2 / G68.4 P3 (2 vectors specification)
- G68.2 / G68.4 P4 (Projection angle)
- G68.3 (Tool Axis Direction)

G68.2 is an absolute command, and G68.4 is an incremental command.

G68.2 / G68.4(Euler's angle)

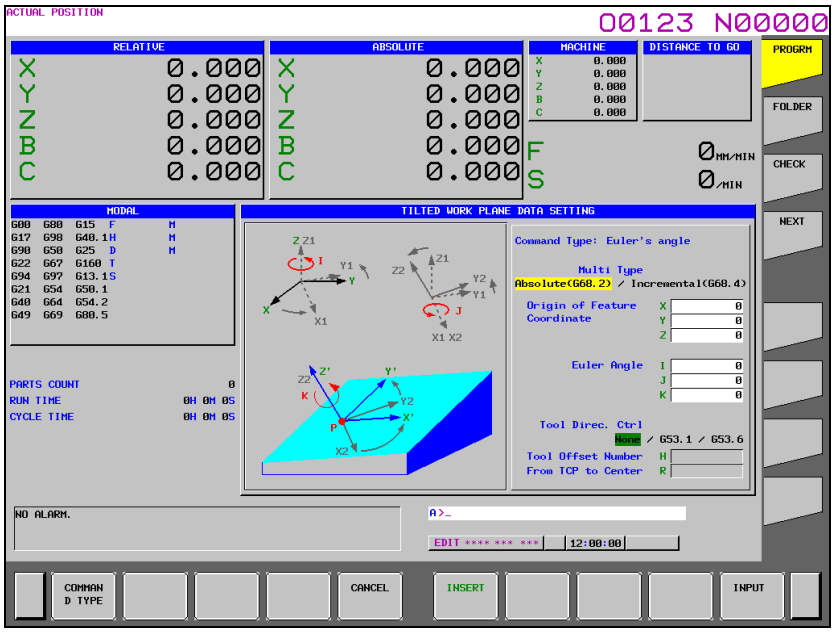


Fig. 12.2.18.3 (a) Tilted working plane data setting screen-Euler's angle(15-inch)

- Multi Type
  Absolute:
  It is assumed that values of specified data are in a workpiece coordinate system, regardless of whether tilted working plane command mode is set.
  Incremental:
  It is assumed that values of specified data are in a feature coordinate system if tilted working plane command mode is already set.

- Origin of Feature Coordinate
  Specify the origin (X, Y, and Z of point P) of a feature coordinate system as coordinates in a workpiece coordinate system (for the absolute type) or the current feature coordinate system (for the incremental type).

- Euler's angle

  I : Specify an angle of rotation around the Z-axis of a workpiece coordinate system (for the absolute type) or the current feature coordinate system (for the incremental type). This rotation determines coordinate system 1 (X1-Y1-Z1) from the workpiece coordinate system (for the absolute type) or the current feature coordinate system (for the incremental type) (X-Y-Z).

  J : Specify an angle of rotation around the X-axis of coordinate system 1. This rotation determines coordinate system 2 (X2-Y2-Z2) from coordinate system 1 (X1-Y1-Z1).

  K : Specify an angle of rotation around the Z-axis of coordinate system 2. After this rotation, a feature coordinate system is obtained by shifting the origin of the workpiece coordinate system by the coordinates specified in "Origin of Feature Coordinate".

G68.2 / G68.4(Roll-Pitch-Yaw angle)

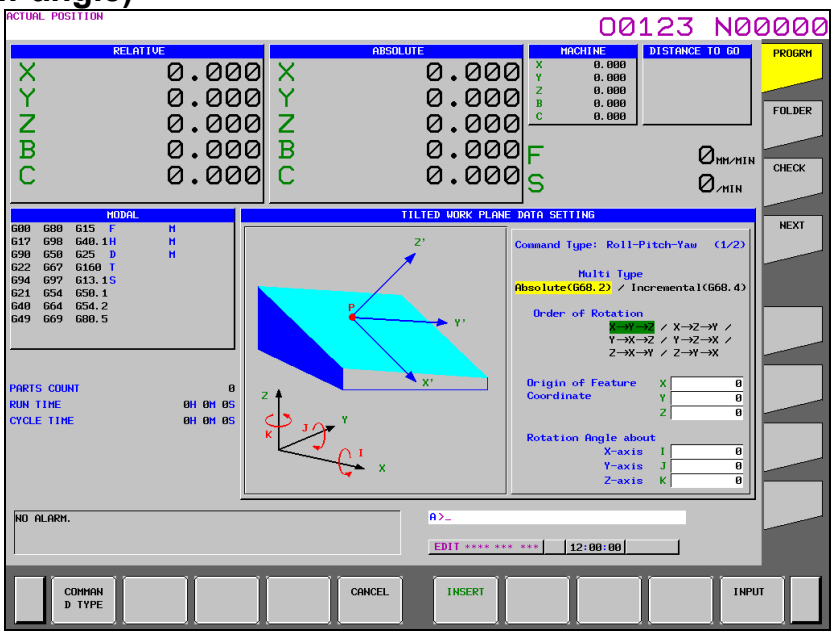


Fig. 12.2.18.3 (b) Tilted working plane data setting screen-Roll-Pitch-Yaw angle(15-inch)

- Multi Type

  Absolute:

  It is assumed that values of specified data are in a workpiece coordinate system, regardless of whether tilted working plane command mode is set.

  Incremental:

  It is assumed that values of specified data are in a feature coordinate system if tilted working plane command mode is already set.

- Order of Rotation
  Select an order in which the X-axis, Y-axis, and Z-axis are rotated in a workpiece coordinate system (for the absolute type) or the current feature coordinate system (for the incremental type). The selectable rotation orders are as follows:

	1st rotation axis	2nd rotation axis	3rd rotation axis
X→Y→Z	X-axis	Y-axis	Z-axis
X→Z→Y	X-axis	Z-axis	Y-axis
Y→X→Z	Y-axis	X-axis	Z-axis
Y→Z→X	Y-axis	Z-axis	X-axis
Z→X→Y	Z-axis	X-axis	Y-axis
Z→Y→X	Z-axis	Y-axis	X-axis

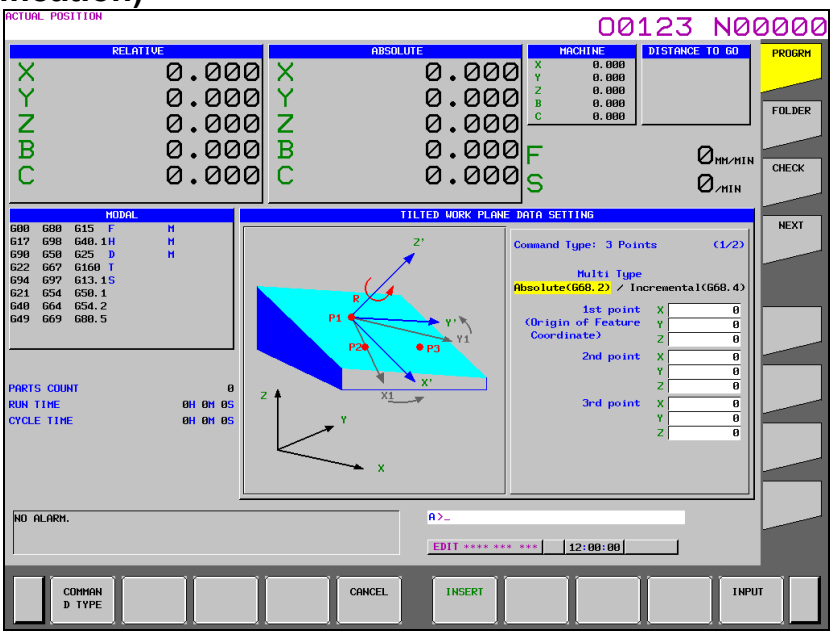
- Origin of Feature Coordinate
  Specify the origin (X, Y, and Z of point P) of a feature coordinate system as coordinates in a workpiece coordinate system (for the absolute type) or the current feature coordinate system (for the incremental type).

- Rotation Angle about X-axis
  Specify an angle of rotation around the X-axis of a workpiece coordinate system (for the absolute type) or the current feature coordinate system (for the incremental type).

- Rotation Angle about Y-axis
  Specify an angle of rotation around the Y-axis of a workpiece coordinate system (for the absolute type) or the current feature coordinate system (for the incremental type).

- Rotation Angle about Z-axis
  Specify an angle of rotation around the Z-axis of a workpiece coordinate system (for the absolute type) or the current feature coordinate system (for the incremental type).

G68.2 / G68.4(3 points specification)



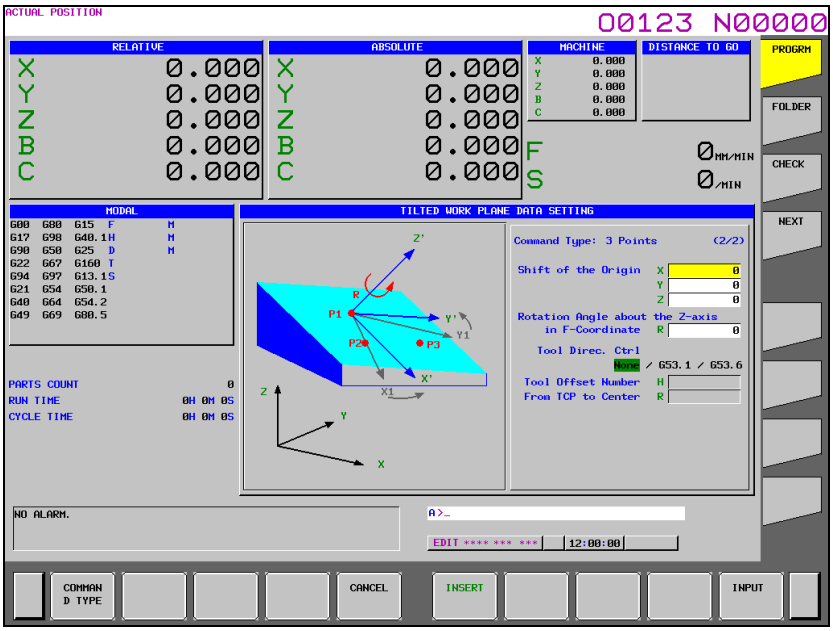


Fig. 12.2.18.3 (c) Tilted working plane data setting screen- 3 points specification(15-inch)

- Multi Type

  Absolute:

  It is assumed that values of specified data are in a workpiece coordinate system, regardless of whether tilted working plane command mode is set.

  Incremental:

  It is assumed that values of specified data are in a feature coordinate system if tilted working plane command mode is already set.

- Shift of the Origin
  Specify, in a feature coordinate system, an amount of shift from the feature coordinate system origin specified for the 1st point (point P1).

- Rotation Angle about the Z-axis in F-Coordinate
  Specify an angle of rotation around the Z-axis of a feature coordinate system. The direction of rotation angle R is positive when a rotation is made clockwise as viewed in the Z-axis direction of the feature coordinate system.

- 1st point (Origin of Feature Coordinate)
  Specify the origin (X, Y, and Z of point P1) of a feature coordinate system as coordinates in a workpiece coordinate system (for the absolute type) or the current feature coordinate system (for the incremental type).

- 2nd point
  Specify the coordinates (X, Y, and Z of point P2) of the 2nd point as coordinates in a workpiece coordinate system (for the absolute type) or the current feature coordinate system (for the incremental type). The 1st point and 2nd point determine the X-axis of the feature coordinate system.

- 3rd point
  Specify the coordinates (X, Y, and Z of point P3) of the 3rd point as coordinates in a workpiece coordinate system (for the absolute type) or the current feature coordinate system (for the incremental type). Among the directions normal to the X-axis, a direction with a smaller angle relative to the P1 → P3 vector is the Y-axis of the feature coordinate system.

G68.2 / G68.4(2 vectors specification)

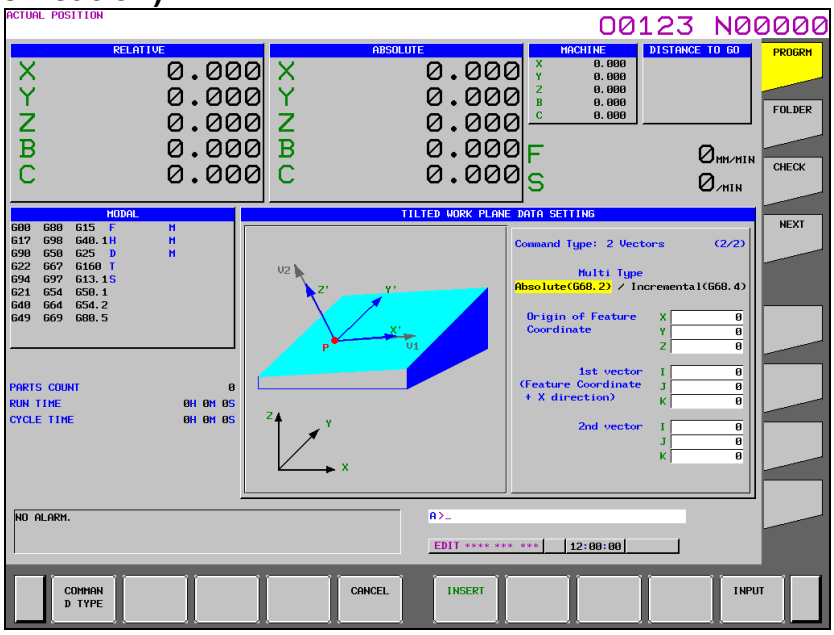


Fig. 12.2.18.3 (d) Tilted working plane data setting screen-2 vectors specification(15-inch)

- Multi Type
  Absolute:
  It is assumed that values of specified data are in a workpiece coordinate system, regardless of whether tilted working plane command mode is set.
  Incremental:
  It is assumed that values of specified data are in a feature coordinate system if tilted working plane command mode is already set.

- Origin of Feature Coordinate
  Specify the origin (X, Y, and Z of point P) of a feature coordinate system as coordinates in a workpiece coordinate system (for the absolute type) or the current feature coordinate system (for the incremental type).

- V1 (X') vector
  Specify the X-axis direction vector of a feature coordinate system as values in a workpiece coordinate system (for the absolute type) or the current feature coordinate system (for the incremental type).

- V2 (Z') vector
  Specify the Z-axis direction vector of a feature coordinate system as values in a workpiece coordinate system (for the absolute type) or the current feature coordinate system (for the incremental type).

G68.2 / G68.4(Projection angle)

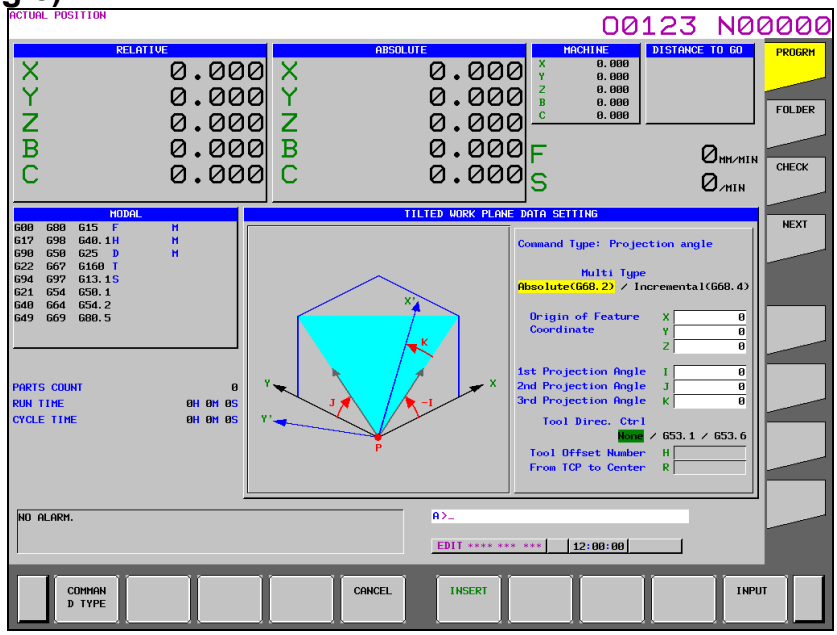


Fig. 12.2.18.3 (e) Tilted working plane data setting screen-Projection angle(15-inch)

- Multi Type
  Absolute:
  It is assumed that values of specified data are in a workpiece coordinate system, regardless of whether tilted working plane command mode is set.
  Incremental:
  It is assumed that values of specified data are in a feature coordinate system if tilted working plane command mode is already set.

- Origin of Feature Coordinate
  Specify the origin (X, Y, and Z of point P) of a feature coordinate system as coordinates in a workpiece coordinate system (for the absolute type) or the current feature coordinate system (for the incremental type).

- Projection angle
  I : Specify a projection angle relative to the X-axis of a workpiece coordinate system (for the absolute type) or the current feature coordinate system (for the incremental type).
  J : Specify a projection angle relative to the Y-axis of a workpiece coordinate system (for the absolute type) or the current feature coordinate system (for the incremental type).
  K : Specify an angle of rotation around the Z-axis of a feature coordinate system.

G68.3(Tool Axis Direction)

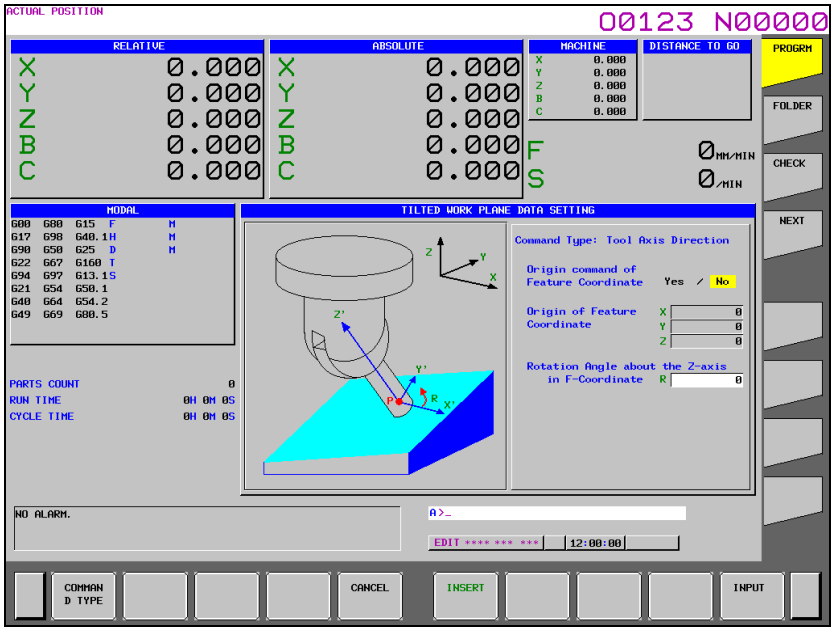


Fig. 12.2.18.3 (f) Tilted working plane data setting screen-Tool Axis Direction(15-inch)
(When "No" is selected in "Origin command of Feature Coordinate")

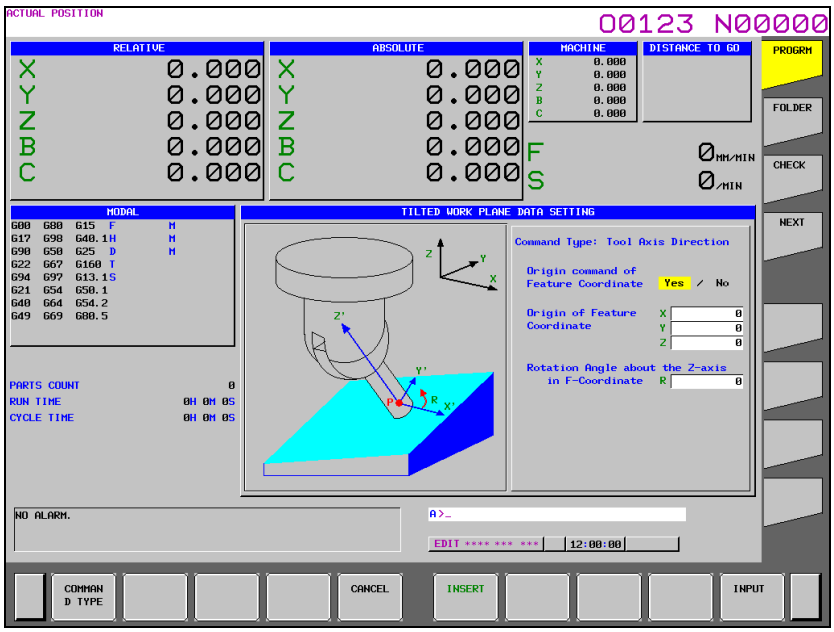


Fig. 12.2.18.3 (g) Tilted working plane data setting screen-Tool Axis Direction(15-inch)
(When "Yes" is selected in "Origin command of Feature Coordinate")

- Origin command of Feature Coordinate
  Select whether to specify a feature coordinate system origin.
  Yes : A feature coordinate system origin is specified.
  No : A feature coordinate system origin is not specified.

- Origin of Feature Coordinate
  Specify the origin (X, Y, and Z of point P) of a feature coordinate system as coordinates in a workpiece coordinate system.
  This setting cannot be made if "No" is selected in "Origin command of Feature Coordinate".

- Rotation Angle about the Z-axis in F-Coordinate
  Specify an angle of rotation around the Z-axis of a feature coordinate system. The direction of rotation angle R is positive when a rotation is made clockwise as viewed in the Z-axis direction of the feature coordinate system.

12.2.18.4 Limitation

This function is supported on 10.4-inch and 15-inch display units. This function is not displayed on 7.2-inch and 8.4-inch display units.

12.3 SCREENS DISPLAYED BY FUNCTION KEY

Section 12.3, "SCREENS DISPLAYED BY FUNCTION KEY ", consists of the following subsections:

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12.3.3 Displaying and Setting Run Time, Parts Count, and Time 1578
12.3.4 Displaying and Setting the Workpiece Origin Offset Value....1581
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12.3.8 Displaying and Setting the Software Operator's Panel....1588
12.3.9 Setting and Displaying Tool Management Data....1591
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12.3.30 Displaying and Setting Pattern Data Inputs (15-inch Display Unit) 1718

Screens of a 7.2/8.4/10.4-inch display unit

Press function key

to display or set tool compensation values and other data.
This section describes how to display or set the following data:

1. Tool compensation value
2. Settings
3. Sequence number comparison and stop
4. Run time and part count
5. Workpiece origin offset value
6. Custom macro common variables
7. Software operator’s panel
8. Tool management data
9. Display language switching
10. Protection of data at eight levels
11. Precision level selection
12. Tool life management data
13. Workpiece setting error compensation data
14. Pattern data input

The software operator’s panel, display language switching, precision level selection, and pattern data input depend on the specifications of the machine tool builder. See the manual issued by the machine tool builder for details.

12.3.1 Displaying and Entering Setting Data

Data such as the TV check flag and punch code is set on the setting data screen. On this screen, the operator can also enable/disable parameter writing, enable/disable the automatic insertion of sequence numbers in program editing, and perform settings for the sequence number comparison and stop function.
See III-9.2 for automatic insertion of sequence numbers.
See III-12.3.2 for the sequence number comparison and stop function.
This subsection describes how to set data.

Procedure for setting the setting data

Procedure

1 Select the MDI mode.

2 Press function key .

3 Press soft key [SETTING] to display the setting data screen.
This screen consists of several pages.

Press page key or until the desired screen is displayed.

An example of the setting data screen is shown below.

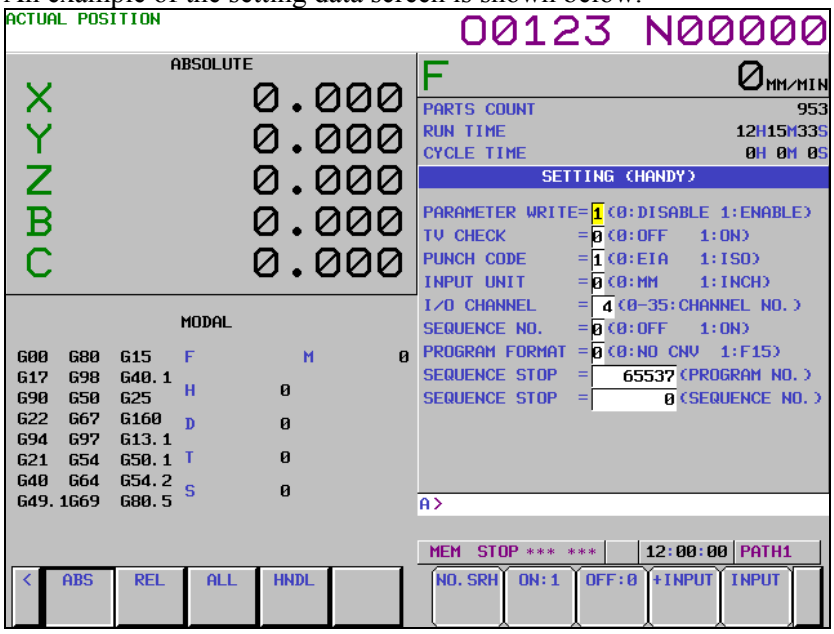


Fig. 12.3.1 (a) SETTING (HANDY) screen (10.4-inch)

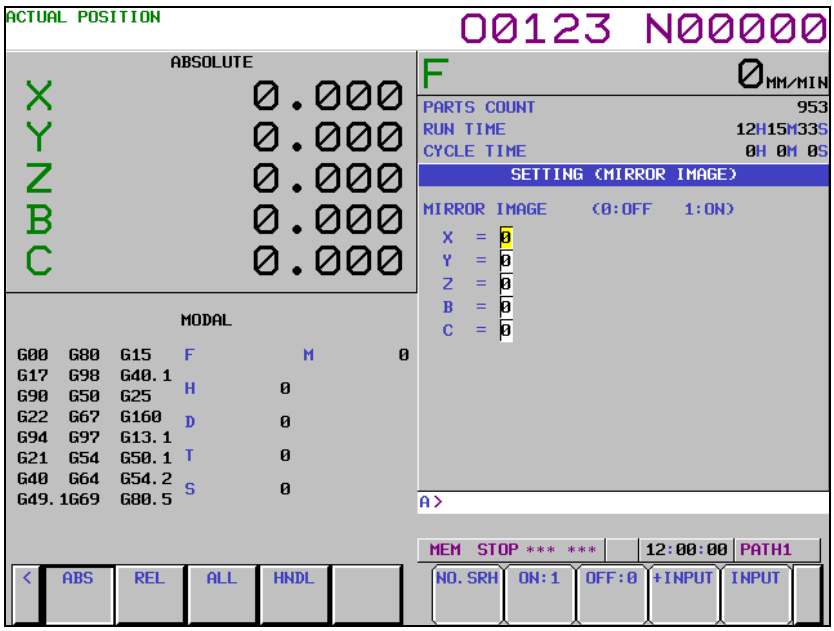
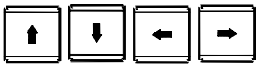


Fig. 12.3.1 (b) SETTING (MIRROR IMAGE) screen (10.4-inch)

4 Move the cursor to the item to be changed by pressing cursor keys  .

5 Enter a new value and press soft key [INPUT].

Explanation

- PARAMETER WRITE

Setting whether parameter writing is enabled or disabled.
0 : Disabled
1 : Enabled

- TV CHECK

Setting to perform TV check.
0 : No TV check
1 : Perform TV check

- PUNCH CODE

Setting code when data is output through reader/puncher interface.
0 : EIA code output
1 : ISO code output

- INPUT UNIT

Setting a program input unit, inch or metric system
0 : Metric
1 : Inch

- I/O CHANNEL

Using channel of reader/puncher interface.
0 : Channel 0
1 : Channel 1
2 : Channel 2

- SEQUENCE NO.

Setting of whether to perform automatic insertion of the sequence number or not at program edit in the EDIT mode.
0 : Does not perform automatic sequence number insertion.
1 : Perform automatic sequence number insertion.

- PROGRAM FORMAT

Setting of whether to use the Series 15 format.
0: Uses the standard format.
1: Uses the Series 15 format.
For the Series 15 format, refer to Chapter II-6 in the User's Manual (T series) or Chapter II-7 of Part II in the User's Manual (M series).

- SEQUENCE STOP

Setting the sequence number with which the operation stops for the sequence number comparison and stop function and the number of the program to which the sequence number belongs

- MIRROR IMAGE

Setting of mirror image ON/OFF for each axes.
0 : Mirror image off
1 : Mirror image on

- Others

Page key

or

can also be pressed to display the SETTING (TIMER) screen. See III-12.3.3 for this screen.

12.3.2 Sequence Number Comparison and Stop

If a block containing a specified sequence number appears in the program being executed, operation enters single block mode after the block is executed.

Procedure for sequence number comparison and stop

Procedure

1 Select the MDI mode.
2 Press function key .
3 Press chapter selection soft key [SETTING].
4 Press page key or several times until the following screen is displayed.

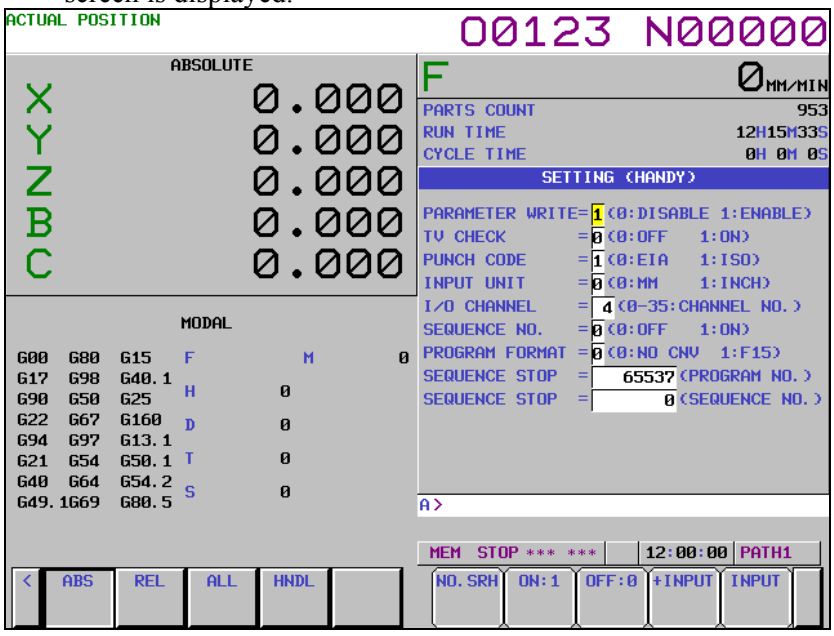


Fig. 12.3.2 (a) SETTING (HANDY) screen (10.4-inch)

5 Enter in (PROGRAM NO.) for SEQUENCE STOP the number (1 to 99999999) of the program containing the sequence number with which operation stops.
6 Enter in (SEQUENCE NO.) for SEQUENCE STOP (with eight or less digits) the sequence number with which operation is stopped.
7 When automatic operation is executed, operation enters single block mode at the block containing the sequence number which has been set.

Explanation

- Sequence number after the program is executed

After the specified sequence number is found during the execution of the program, the sequence number settting for sequence number compensation and stop becomes “-1”.

- Exceptional blocks

If the predetermined sequence number is found in a block in which all commands are those to be processed within the CNC control unit, the execution does not stop at that block.

```
[Example]  N1 #1=1 ;
N2 IF[#1 EQ 1]GOTO 08 ;
N3 GOTO 09 ;
N4 M98 P1000 ;
N5 M99 ;
```

In the example shown above, if the predetermined sequence number is found, the execution of the program does not stop.

- Stop in the canned cycle

If the predetermined sequence number is found in a block which has a canned cycle command, the execution of the program stops after the return operation is completed.

- When the same sequence number is found several times in the program

If the predetermined sequence number appears twice or more in a program, the execution of the program stops after the block in which the predetermined sequence number is found for the first time is executed.

- Block to be repeated a specified number of times

If the predetermined sequence number is found in a block which is to be executed repeatedly, the execution of the program stops after the block is executed specified times.

12.3.3 Displaying and Setting Run Time, Parts Count, and Time

Various run times, the total number of machined parts, number of parts required, and number of machined parts can be displayed.
This data can be set by parameters or on this screen (except for the total number of machined parts and the time during which the power is on, which can be set only by parameters).
This screen can also display the clock time. The time can be set on the screen.

Procedure for Displaying and Setting Run Time, Parts Count and Time

Procedure

1 Select the MDI mode.
2 Press function key .
3 Press chapter selection soft key [SETTING].
4 Press page key or several times until the following screen is displayed.

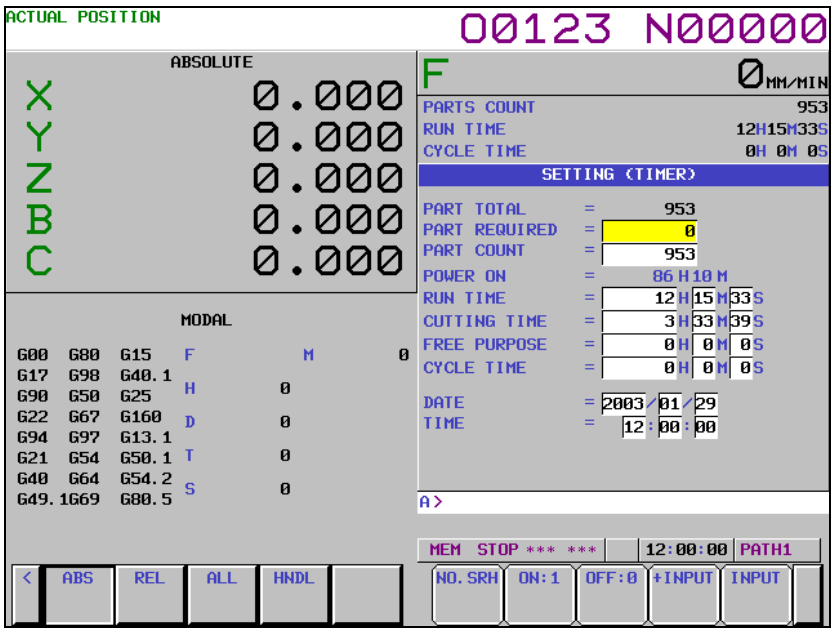


Fig. 12.3.3 (a) SETTING (TIMER) screen (10.4-inch)

5 To set the number of parts required, move the cursor to PARTS REQUIRED and enter the number of parts to be machined.
6 To set the clock, move the cursor to DATE or TIME, enter a new date or time, then press soft key [INPUT].

Explanation

- PARTS TOTAL

This value is incremented by one when M02, M30, or an M code specified by parameter No. 6710 is executed. This value cannot be set on this screen. Set the value in parameter No. 6712.

- PARTS REQUIRED

It is used for setting the number of machined parts required.
When the “0” is set to it, there is no limitation to the number of parts.
Also, its setting can be made by the parameter No 6713.

- PARTS COUNT

This value is incremented by one when M02, M30, or an M code specified by parameter No. 6710 is executed. The value can also be set by parameter No. 6711. In general, this value is reset when it reaches the number of parts required. Refer to the manual issued by the machine tool builder for details.

- POWER ON

Displays the total time which the power is on. This value cannot be set on this screen but can be preset in parameter No. 6750.

- RUN TIME

Indicates the total run time during automatic operation, excluding the stop and feed hold time.
This value can be preset in parameter No. 6751 or No. 6752.

- CUTTING TIME

Displays the total time taken by cutting that involves cutting feed such as linear interpolation (G01) and circular interpolation (G02 or G03). This value can be preset in parameter No. 6753 or No. 6754.

- FREE PURPOSE

This value can be used, for example, as the total time during which coolant flows. Refer to the manual issued by the machine tool builder for details.

- CYCLE TIME

Indicates the run time of one automatic operation, excluding the stop and feed hold time. This is automatically preset to 0 when a cycle start is performed at reset state. It is preset to 0 even when power is removed.

- DATA and TIME

Displays the current date and time. The date and time can be set on this screen.

- Usage

When the command of M02 or M30 is executed, the total number of machined parts and the number of machined parts are incremented by one. Therefore, create the program so that M02 or M30 is executed every time the processing of one part is completed. Furthermore, if an M code set to the parameter No. 6710 is executed, counting is made in the similar manner. Also, it is possible to disable counting even if M02 or M30 is executed (bit 0 (PCM) of parameter No. 6700 is set to 1). For details, see the manual issued by machine tool builders.

Limitation

- Run time and part count settings

Negative value cannot be set. Also, the setting of "M" and "S" of run time is valid from 0 to 59.
Negative value may not be set to the total number of machined parts.

- Time settings

Neither negative value nor the value exceeding the value in the following table can be set.

Table 12.3.3 (a)

Item	Maximum value	Item	Maximum value
Year	2096	Hour	23
Month	12	Minute	59
Day	31	Second	59

12.3.4 Displaying and Setting the Workpiece Origin Offset Value

Displays the workpiece origin offset for each workpiece coordinate system (G54 to G59, G54.1 P1 to G54.1 P48 and G54.1 P1 to G54.1 P300) and external workpiece origin offset.
The workpiece origin offset value and external workpiece origin offset value can be set on this screen.

Procedure for displaying and setting the workpiece origin offset value

Procedure

1 Press function key .

2 Press chapter selection soft key [WORK].
The workpiece coordinate system setting screen is displayed.

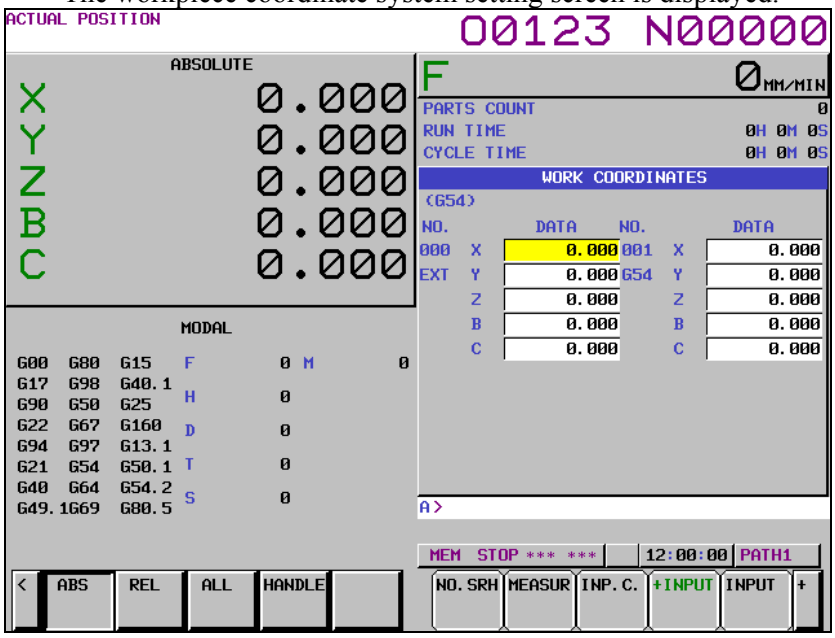


Fig. 12.3.4 (a) WORK COORDINATES screen (10.4-inch)

3 The screen for displaying the workpiece origin offset values consists of two or more pages.
Display a desired page in either of the following two ways:

- Press the page key or .
- Enter the workpiece coordinate system number (0 : external workpiece origin offset, 1 to 6: workpiece coordinate systems G54 to G59, P1 to P48 : workpiece coordinate systems G54.1 P1 to G54.1 P48, P1 to P300 : workpiece coordinate systems G54.1 P1 to G54.1 P300) and press operation selection soft key [NO.SRH].

4 Turn off the data protection key to enable writing.

5 Move the cursor to the workpiece origin offset to be changed.

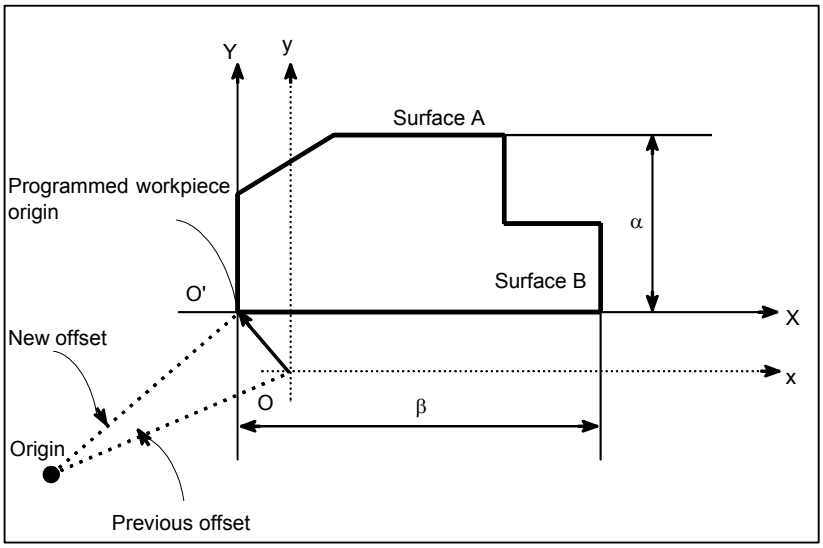
6 Enter a desired value by pressing numeric keys, then press soft key [INPUT]. The entered value is specified in the workpiece origin offset value. Or, by entering a desired value with numeric keys and pressing soft key [+INPUT], the entered value can be added to the previous offset value.
7 Repeat steps 5 and 6 to change other offset values.
8 Turn on the data protection key to disable writing.

12.3.5 Direct Input of Workpiece Origin Offset Value Measured

This function is used to compensate for the difference between the programmed workpiece coordinate system and the actual workpiece coordinate system. The measured offset for the origin of the workpiece coordinate system can be input on the screen such that the command values match the actual dimensions.
Selecting the new coordinate system matches the programmed coordinate system with the actual coordinate system.

Procedure for direct input of workpiece origin offset value measured

Procedure



1 When the workpiece is shaped as shown above, position the reference tool manually until it touches surface A of the workpiece.
2 Retract the tool without changing the Y coordinate.
3 Measure distance α between surface A and the programmed origin of the workpiece coordinate system as shown above.
4 Press function key .

5 To display the WORK COORDINATES screen, press the chapter selection soft key [WORK].

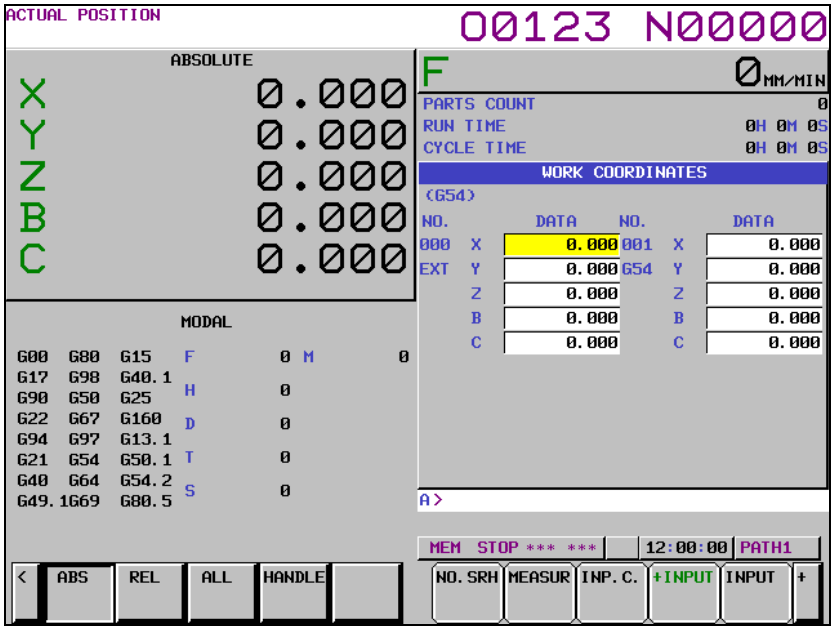


Fig. 12.3.5 (a) WORK COORDINATES screen (10.4-inch)

6 Position the cursor to the workpiece origin offset value to be set.
7 Press the address key for the axis along which the offset is to be set (Y-axis in this example).
8 Enter the measured value (α) then press soft key [MEASUR].
9 Move the reference tool manually until it touches surface B of the workpiece.
10 Retract the tool without changing the X coordinate.
11 Enter the measured value of distance (β) to surface B as an X value as with steps 7 and 8.

Limitation

- Consecutive input

Offsets for two or more axes cannot be input at the same time.

- During program execution

This function cannot be used while a program is being executed.

12.3.6 Displaying and Setting Custom Macro Common Variables

Displays common variables (#100 to #149 or #100 to #199, and #500 to #531 or #500 to #999) on the screen.
The values for variables can be set on this screen.
Relative coordinates can also be set to variables.

Procedure for displaying and setting custom macro common variables

Procedure

1 Press function key .

2 Press the continuous menu key , chapter selection soft key [MACRO], and soft key [(OPRT)] in the stated order.
The following screen is displayed.

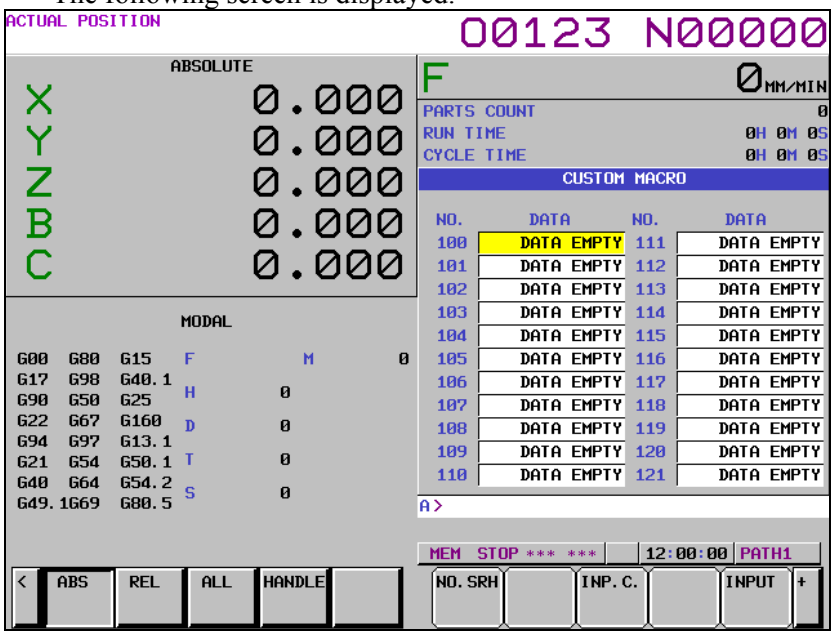


Fig. 12.3.6 (a) CUSTOM MACRO screen (10.4-inch)

3 Move the cursor to the variable number to set using either of the following methods:
- Enter the variable number and press soft key [NO.SRH].
- Move the cursor to the variable number to set by pressing page keys and/or and cursor keys , , , and/or .

4 Enter data with numeric keys and press soft key [INPUT].

5 To set a relative coordinate in a variable, press address key , , or , then press soft key [INP.C.].

6 To set a blank in a variable, just press soft key [INPUT]. The value field for the variable becomes blank.

Explanation

If the value of a variable produced by an operation is not displayable, an indication below is provided.
When the significant number of digits is 12 (with bit 0 (F16) of parameter No. 6008 set to 0):

Variable value range	Variable value indication
0 < Variable value < +0.00000000001	+Underflow
0 > Variable value > -0.00000000001	-Underflow
Variable value > 999999999999	+Overflow
Variable value < -999999999999	-Overflow

When the significant number of digits is 8 (with bit 0 (F16) of parameter No. 6008 set to 1):

Variable value range	Variable value indication
0 < Variable value < +0.0000001	+Underflow
0 > Variable value > -0.0000001	-Underflow
Variable value > 99999999	+Overflow
Variable value < -99999999	-Overflow

12.3.7 Displaying and Setting Real Time Custom Macro Data

Real time macro variables (RTM variables) are dedicated to real time custom macros.
RTM variables are divided into temporary real time macro variables (temporary RTM variables) and permanent real time macro variables (permanent RTM variables).
The values of permanent RTM variables are kept stored after the power is turned off.
Temporary RTM variables are cleared to 0 when the power is turned off.
System variables (DI/DO variables) dedicated to real time custom macros are used to read and write PMC interface signals. Data can be read and written in bit and byte units. Before writing a signal, release the protection of the corresponding variable on the PMC signal protect screen.

Displaying and setting real time custom macro variables

Procedure

1 Press function key SYSTEM.

2 Press the continuous menu key ▷ several times, then press chapter selection soft key [R.TIME MACRO]. The following screen appears:

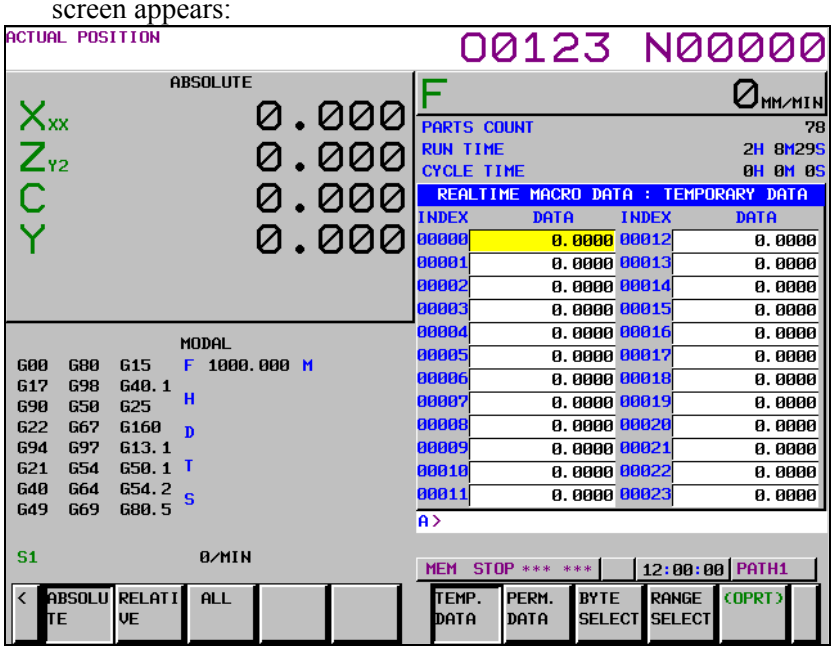


Fig. 12.3.7 (a) REAL TIME MACRO screen (10.4-inch)

3 To display or set real time custom macro variables of which values are not stored at power-off, press soft key [TEMP. DATA].

4 To display or set real time custom macro variables of which values are stored at power-off, press soft key [PERM. DATA].

5 Move the cursor to the number of a real time custom macro variable you want to set using either of the following methods:
- Enter the number of a real time custom macro variable and press soft key [NO. SRH].
- Move the cursor to the number of a real time custom macro variable you want to set by pressing page keys PAGE and/or PAGE and cursor keys ↑, ↓, ←, and/or →.

6 Enter data.

Displaying and setting DI/DO variables

Procedure

For setting in byte units:

1 Press function key SYSTEM.

2 Press the continuous menu key ▷ several times, then press chapter selection soft key [R.TIME MACRO]. The following screen appears:

3 Press soft key [BYTE SELECT]. The following screen appears:

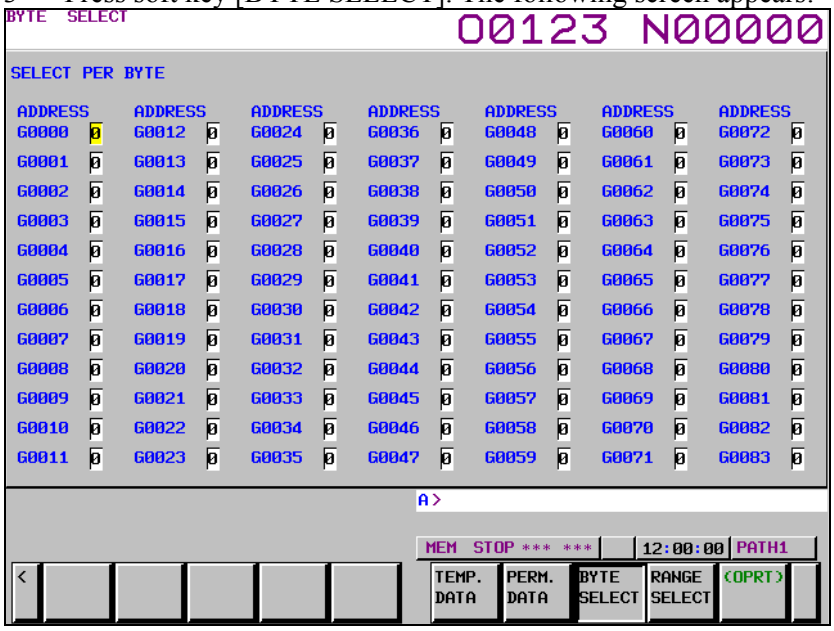


Fig. 12.3.7 (b) BYTE SELECT screen (10.4-inch)

4 Move the cursor to the number of a DI/DO variable you want to set using either of the following methods:
- Enter the number and press soft key [NO. SRH].
- Move the cursor to a desired number by pressing page keys PAGE and/or PAGE and cursor keys ↑, ↓, ←, and/or →.

5 Enter data.

12.3.8 Displaying and Setting the Software Operator's Panel

Operations on the MDI panel can substitute for the functions of switches on the machine operator's panel. This means that a mode selection, jog feed override selection, and so forth can be made with operations on the MDI panel, eliminating the need to use the corresponding switches on the machine operator's panel.
Jog feed can be performed using numeric keys.

Procedure for displaying and setting the software operator's panel

Procedure

1 Press function key .

2 Press the continuous menu key , then press chapter selection soft key [OPR].

3 The screen consists of several pages. Press page key or until the desired screen is displayed.

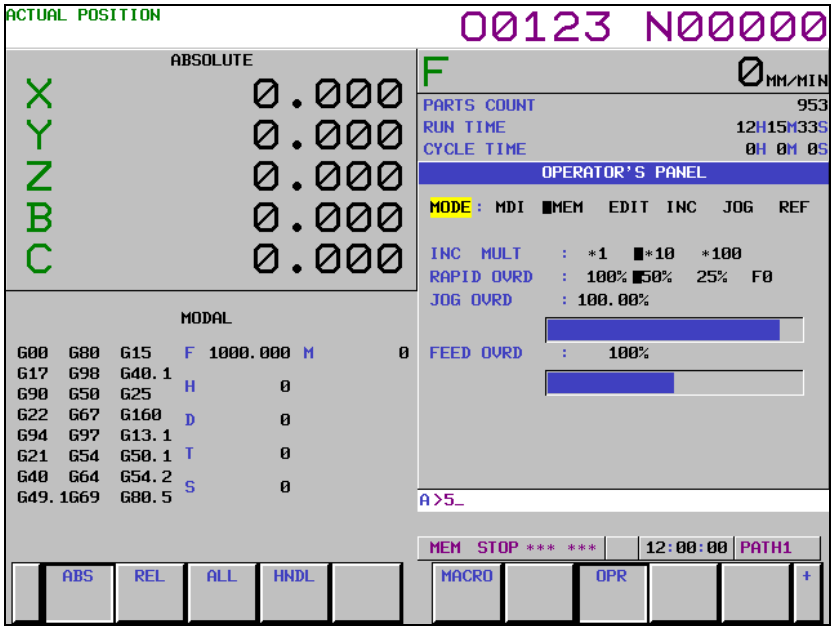


Fig. 12.3.8 (a) Without the manual handle feed function (10.4-inch)

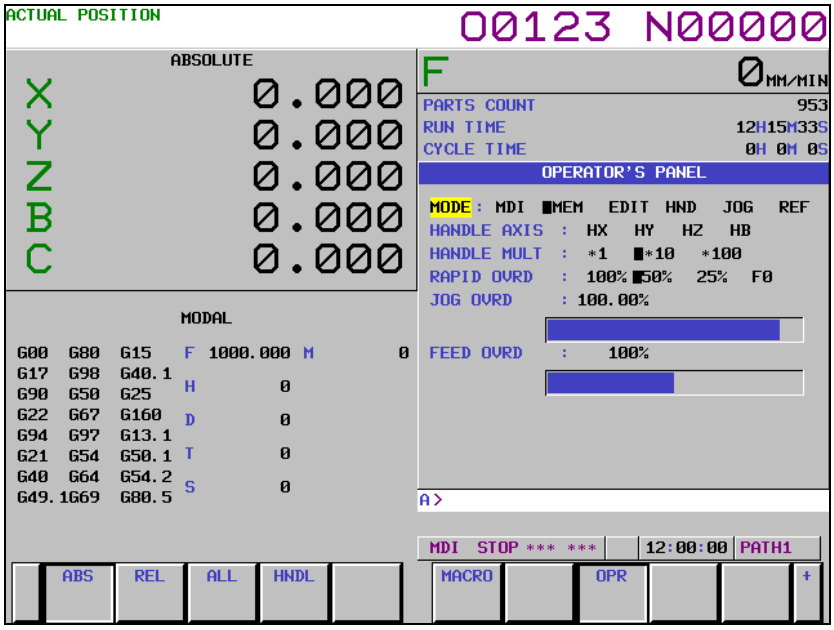


Fig. 12.3.8 (b) With the manual handle feed function (10.4-inch)

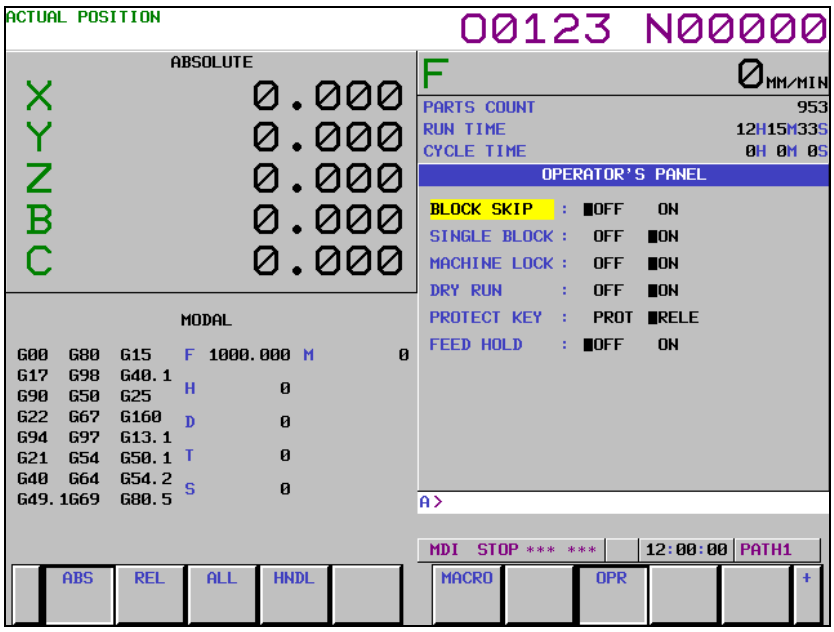


Fig. 12.3.8 (c) (10.4-inch)

4 Move the cursor to the desired switch by pressing cursor key ↑ or ↓.

5 Push the cursor key ← or → to match the mark ■ to an arbitrary position and set the desired condition.

6 Press one of the following arrow keys to perform jog feed. Press the [5] key together with an arrow key to perform jog rapid traverse.

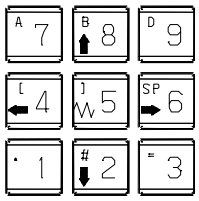


Fig. 12.3.8 (c) MDI arrow keys

Explanation

- Valid operations

The valid operations on the software operator's panel are shown below. Whether to use each group can be chosen using parameter No. 7200. Those groups that are not used are not displayed on the software operator's panel.

Group1 : Mode selection
Group2 : Selection of jog feed axis, Manual rapid traverse
Group3 : Selection of manual pulse generator feed axis, selection of manual pulse magnification
Group4 : Jog federate, federate override, rapid traverse override
Group5 : Optional block skip, single block, machine lock, dry run
Group6 : Protect key
Group7 : Feed hold

- Screens on which jog feed is valid

When the LCD indicates other than the software operator's panel screen and self-diagnosis screen, jog feed is not conducted even if the arrow key is pushed.

- Jog feed and arrow keys

The feed axis and direction corresponding to the arrow keys can be set with parameters (Nos. 7210 to 7217).

- General purpose switches

For the meanings of these switches, refer to the manual issued by machine tool builder.

12.3.9 Setting and Displaying Tool Management Data

The tool management function totally manages tool information including tool offsets and tool life information.
This function provides a magazine screen and tool management screen.
This subsection explains how to set these screens.

12.3.9.1 Displaying and setting magazine screen

1 Press function key .

2 Press chapter selection soft key [TOOL MANAGER].
Alternatively, press function key several times until the tool management screen appears.
Press soft key [MAGAZINE].

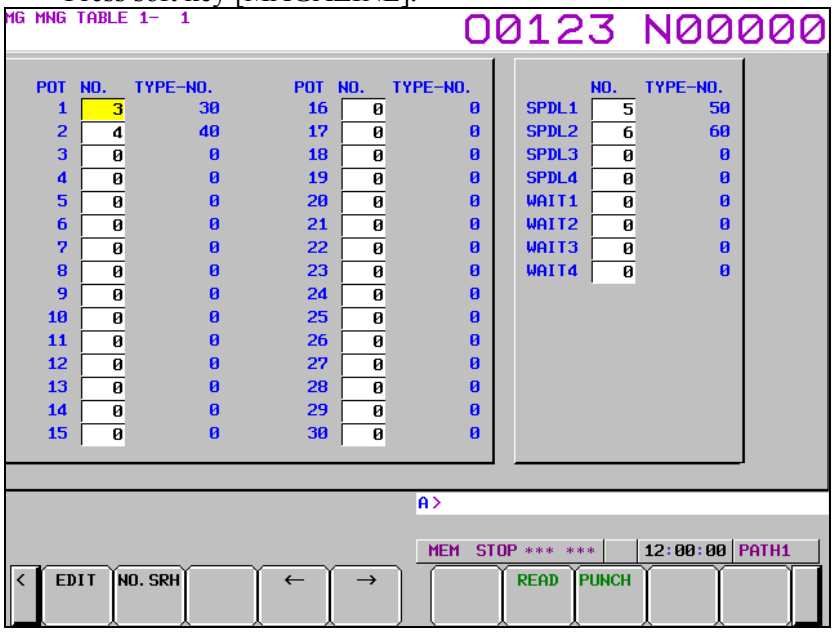


Fig. 12.3.9.1 (a) Tool management magazine screen (10.4-inch)

3 By using the page keys, cursor keys, and soft keys [←] and [→], move the cursor to the pot No. position of the magazine for which you want to set or modify data.
Alternatively, type the number of the tool management data you wan to set or modify, and press soft key [NO. SRH].

4 Press soft key [EDIT].

5 To set the tool management data number of a pot, type the tool management data number, then press soft key [INPUT].
To delete the tool management data number set for a pot, follow the steps below.
<1> Press soft key [ERASE].
<2> Press soft key [CURDAT]. To delete the tool management data numbers registered for all pots, press soft key [ALL].
<3> Press soft key [EXEC]. Alternatively, type 0.

6 To end the edit operation, press soft key [EXIT].
This returns the screen display to the conventional tool management screen.

Explanation

- Another method

Magazine data can be input/output also by using external I/O devices. See III-8, “DATA I/O”.

- Displayed data

Pot : As many pots as specified in parameter No. 13222, 13227, 13232, or 13237 (settable for each magazine) are displayed. The starting pot number can be set for each magazine in parameter No. 13223, 13228, 13233, or 13238.

NO. : Tool management data numbers are displayed.
In the following cases, tools cannot be set in magazines:
1. A tool management data number beyond the range from 0 to (the number of valid pairs in parameter No. 13220) is set.
2. Tool management data is invalid (bit 1 of tool information is 0).
3. The tool management data number to be set is already set for another pot.

Type No. : Tool type numbers corresponding to tool management data numbers are displayed.
Values cannot be set on this screen.

Spindle : The tool management data numbers and tool type numbers of spindle positions are displayed.

Wait : The tool management data numbers and tool type numbers of wait positions are displayed.

- Edit mode

To edit data, press soft key [EDIT] to enter tool management data edit mode.

- Tool management data number

Tool management data numbers can optionally be extended to 64, 240, or 1000 pairs. In addition, the number of tools used can be set in parameter No. 13220.

- Display of the spindle positions and wait positions

The spindle position and wait position can be set for each path.

12.3.9.2 Displaying and setting tool management screen

Procedure

1 Press function key .

2 Press chapter selection soft key [TOOL MANAGER].
Alternatively, press function key several times until the tool management screen appears.

3 Press soft key [TOOL MANAGER].

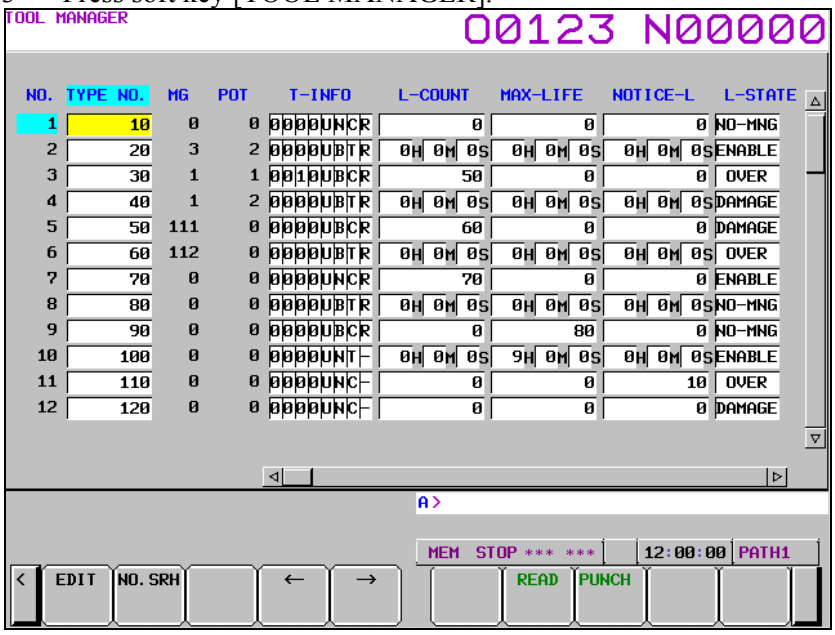


Fig. 12.3.9.2 (a) Tool management data screen (10.4-inch)

4 By using the page keys, cursor keys, and soft keys [←] and [→], move the cursor to the position of the tool information of the tool number for which you want to set or modify data.
Alternatively, type the tool number of the data you want to set or modify, and press soft key [NO. SRH].

5 Press soft key [EDIT].

6 To set tool data, type a desired value, then press soft key [INPUT].
To delete tool information, follow the steps below.
<1> In step 4, move the cursor to the tool information you want to erase.
<2> Press soft key [ERASE].
<3> Press soft key [CURDAT]. To delete entire tool information, press soft key [ALL].
<4> Press soft key [EXEC].

7 To end the edit operation, press soft key [EXIT].
This returns the screen display to the conventional tool management screen.

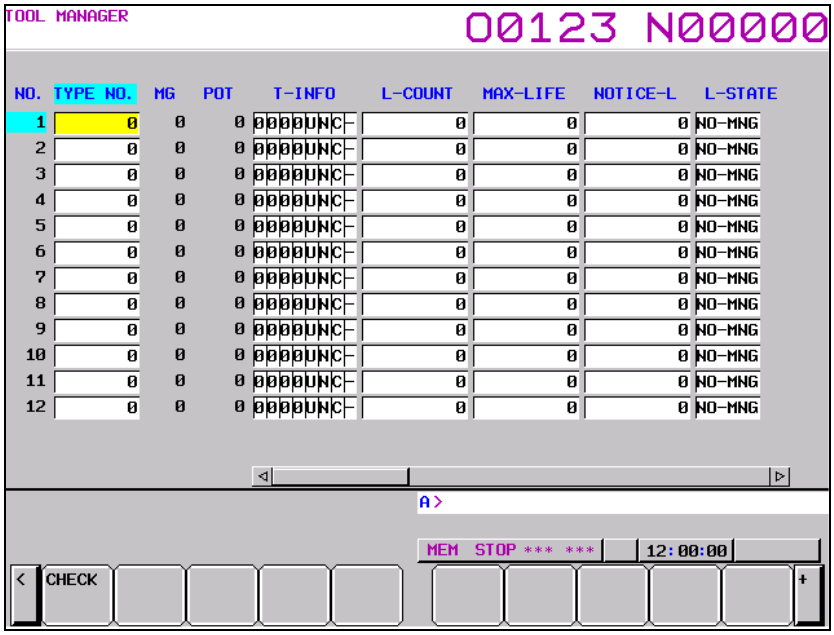


Fig. 12.3.9.2 (b) Tool management data screen (check function) (10.4-inch)

8 When soft key [CHECK] is pressed, if there are tools with the same number but with different count types (count and time), the cursor moves to the tool type number of the smallest tool management number in the tool type numbers and a warning message appears.
The warning messages issued by the check function are shown below.
<1> For the same type of tools with different count types
L-COUNT TYPE MISMATCH:XXXXXXXX
<2> For the same type of tools with the same count type
L-COUNT TYPE MATCH

Explanation

- Another method

Tool management data, customize data, and names set for tool states can be input/output also by using external I/O devices.
See III-8, “DATA I/O”.

- Displayed information

- Life information

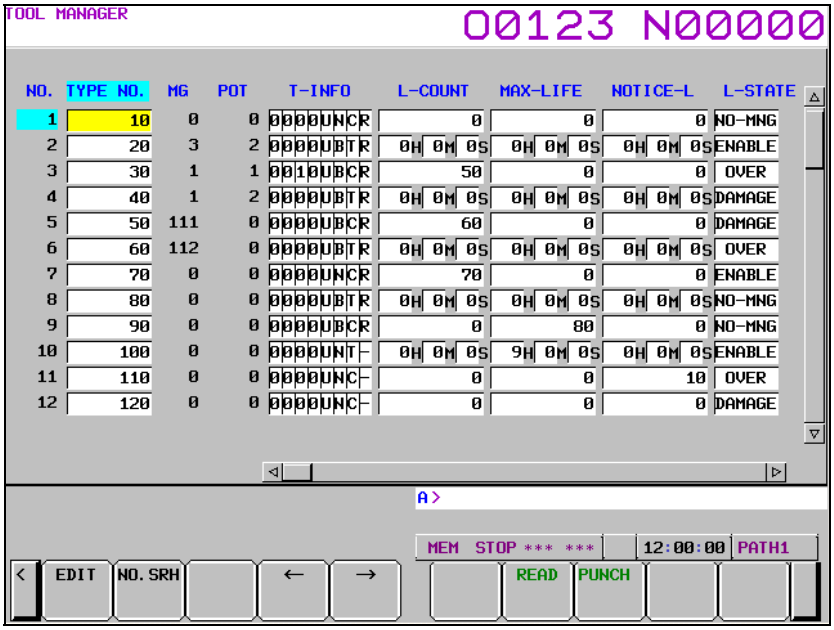


Fig. 12.3.9.2 (c) Tool management data life status screen(10.4-inch)

NO. : Tool management data numbers are displayed. These numbers can be displayed but cannot be set. The tool management data number of edited data is kept blinking until tool management data edit mode ends. Tool management data numbers set or deleted on the magazine screen also blink on the tool management screen.

Type No. : Tool type numbers are displayed.
A value from 0 to 99,999,999 can be set.

MG : The magazine number assigned to each tool is indicated. These numbers can be displayed but cannot be set.

Pot : The pot number assigned to each tool is indicated. These numbers can be displayed but cannot be set.

T-INFO : The following five types of information are displayed sequentially from the right:

- Tool management data: Valid (R)/invalid (-))
- Life count type: Time (T)/count (C)
- Tool type: Oversize tool (B)/normal tool (N)
- Data access: Locked (L)/unlocked (U)
- When a tool is not under life management:
  Included in the tools to be searched (1)/not included in the tools to be searched (0)

NOTE
1 The tool types and data access information vary depending on the specifications defined by the machine tool builder.
2 The same type of tools must have the same life count type.

L-COUNT : The number of use times/use period of time of each tool is indicated.
Up to 99,999,999 times or 999 hours 59 minutes 59 seconds can be set.
MAX-LIFE : The maximum life value of each tool is indicated.
Up to 99,999,999 times or 999 hours 59 minutes 59 seconds can be set.
NOTICE-L : Noticed life value of a tool
Up to 99,999,999 times or 999 hours 59 minutes 59 seconds can be set.
L-STATE : Current tool state
One of the four states, including invalid (0), present (1, 2), not present (3), and broken (4), is indicated. The numbers in parentheses are data values used when these states are input in MDI.

- Spindle speed/feedrate

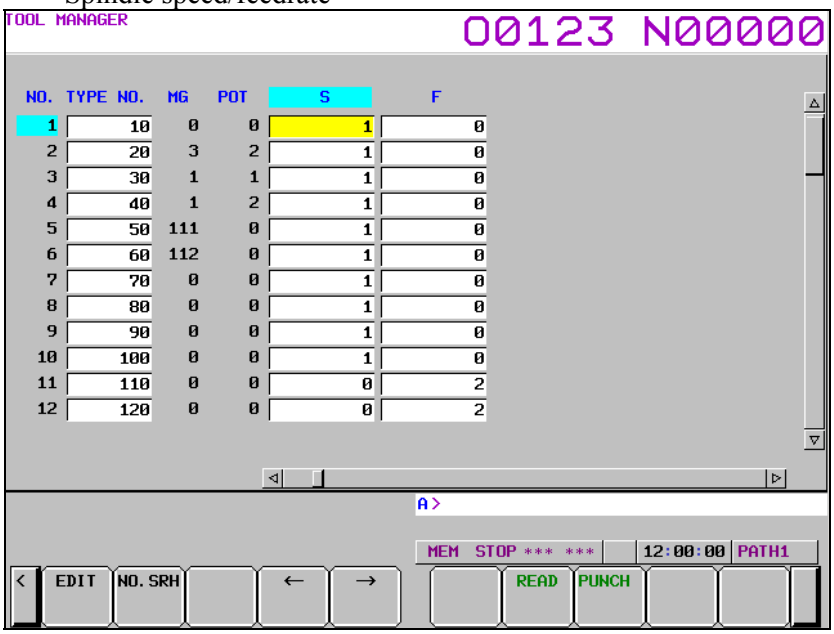


Fig. 12.3.9.2 (c) Tool management data spindle speed/feedrate screen (10.4-inch)

S : Spindle speed.
A value from 0 to 99,999 can be set.
F : Feedrate.
A value from 0 to 99,999,999 can be set.

- Tool offset information

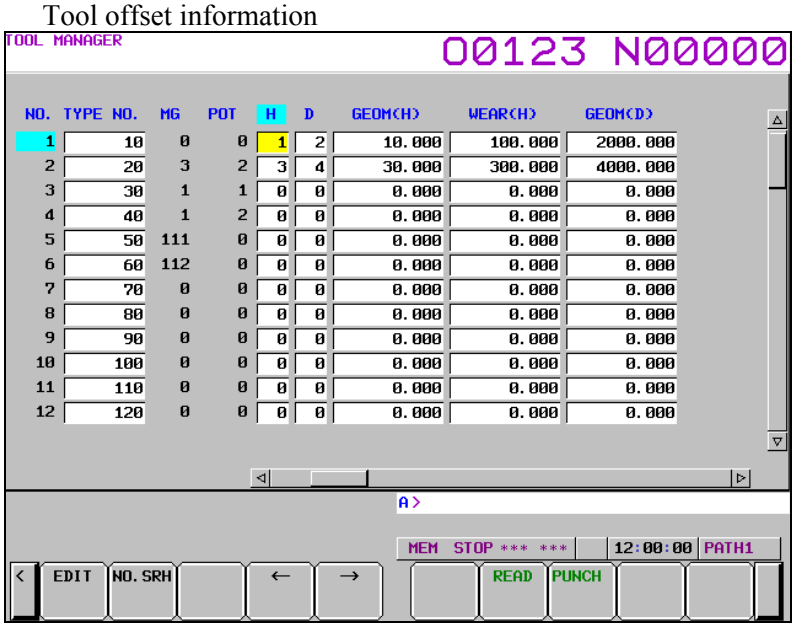


Fig. 12.3.9.2 (d) Tool management data tool offset screen (10.4-inch)

H : Tool length compensation number (for machining center systems only).
A value from 0 to 999 can be set.

D : Cutter compensation number (for machining center systems only).
A value from 0 to 999 can be set.

TG : Tool geometry compensation number (for lathe systems only).
A value from 0 to 999 can be set.

TW : Tool wear compensation number (for lathe systems only).
A value from 0 to 999 can be set.

The displayed offset data is determined depending on the option configuration and parameter settings (bits 1(DCR), 2(DOY), 3(DOB), 4(DO2), 6(DOT), and 7(DOM) of parameter No. 13202).

- Customize information

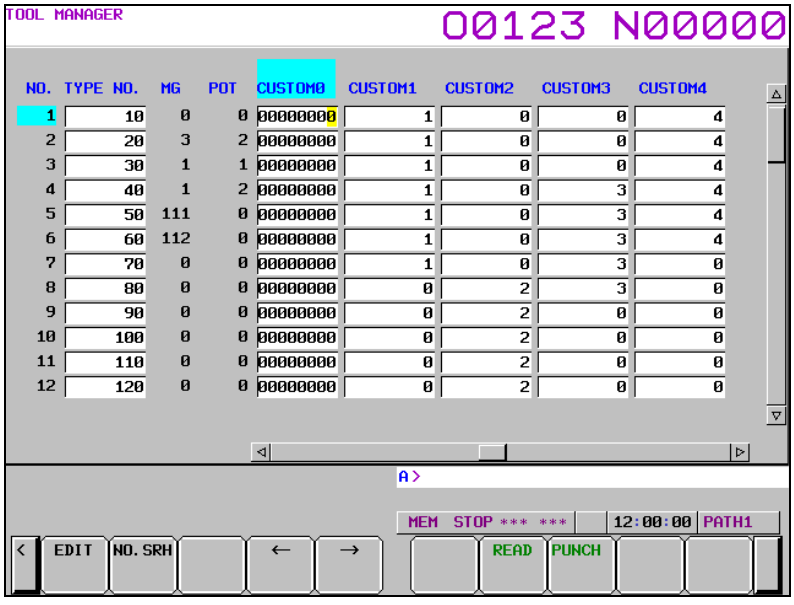


Fig. 12.3.9.2 (e) Tool management data customize data screen (10.4-inch)

Customize 0 : Bit-type customize information.
For each bit, 1 or 0 can be input.
Customize 1 to 4 : Customize information. Any value from -99,999,999 to 99,999,999 can be set.
Customize 5 to 20 : Customize information. These items are displayed only when customize data extension option (5 to 20) of the tool management function is enabled. Any value from -99,999,999 to 99,999,999 can be set.
Customize 21 to 40 : Customize information. These items are displayed only when customize data extension option (5 to 40) of the tool management function is enabled. Any value from -99,999,999 to 99,999,999 can be set.

- Tool management data number

Tool management data numbers can optionally be extended to 64, 240, or 1000 pairs. The number of tools used can be set in parameter No. 13220.

- Edit mode

To edit data, press soft key [EDIT] to enter tool management data edit mode.

- Life count override

When the period of time is set as the tool life count type, the life count can be overridden if bit 2 (LFV) of parameter No. 6801 is set to 1. Specify an override value by using a switch on the machine operator's panel.

Example) When cutting is performed for 10 minutes with an override of 0.1, one minute is counted in the tool life counter.

- Tool management extension function

When tool management extension functions are enabled, you can use the following functions in addition to the tool management functions:

- A value with a decimal point can be set as customize data.
  The maximum number of decimal places can be set to 7 by G10 data input and file read.
- Various types of tool management data can be protected using KEY signals.
  An attempt to enter a value to a protected data causes a warning message "WRITE PROTECT."
- Tool life count period selection

Bit 5 of tool information is used to make a life count period selection.

Item		Description
Data length		1 byte (flag data)
#5	REV	0: A life count period of 1 sec is used. 1: A life count period of 8 msec is used.

Range of count is as follows.

1sec : 0 to 3,599,999 seconds (999 hours 59 minutes 59 seconds)
8msec : 0 to 3,599,992 ms (59 minutes 59 seconds 992 milliseconds)

12.3.9.3 Each tool data screen

Each tool data screen

Procedure

1 Press function key .

2 Press chapter selection soft key [TOOL MANAGER].
Alternatively, press function key several times until the tool management screen appears.

3 Press soft key [EACH TOOL]. The each tool data screen appears.

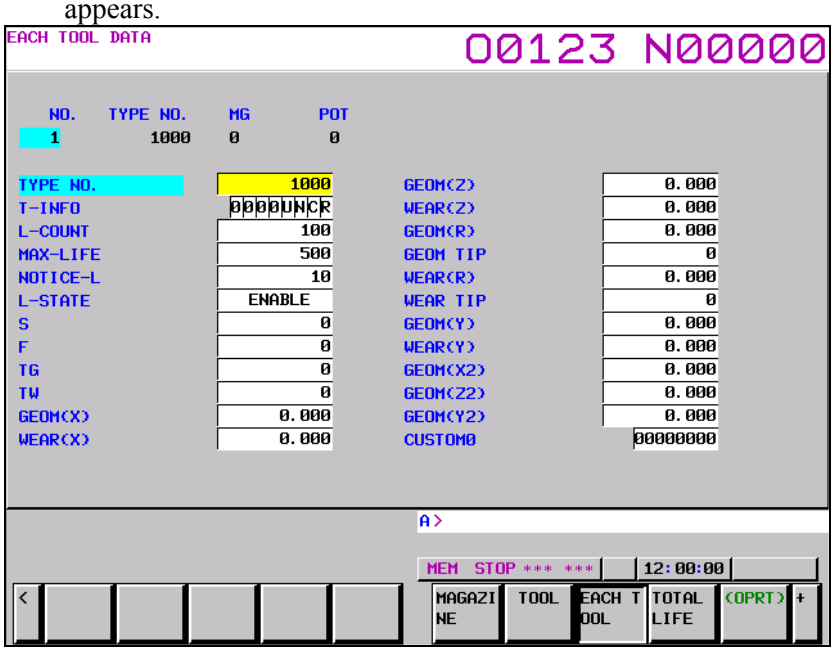


Fig. 12.3.9.3 (a) Each tool data screen (10.4-inch)

Explanation

- Header

The following four data items are displayed: NO., TYPE NO., MG, and POT.
When the data table of a tool extends over two or more pages, the same header is displayed on these pages.

- Data table

The data table shows data items related to a tool at a time.
The data items are displayed from the upper left to the lower left, the upper right, and the lower right in ascending order of screen display position numbers specified using the customize function. The number of digits displayed for one data item is fixed to 11.
Up to 24 data items are displayed on one page. (12 data items × 2 columns)
When more than 24 data items are set for a tool, the 25th and subsequent data items are displayed on the next page. (Up to three pages)

When a data item is set as a screen element of the tool management data screen twice or more using the tool management data display customize function (one of the tool management extension functions), only the data item with the smaller display position number is displayed. When no value is set for a data item, the field for the data item is not displayed and the next data item is displayed.

Key operations

- MDI key operations

- Numeral keys Inputs a numeric value.
-

Displays the previous page for the same tool.
-

Displays the next page for the same tool.
-
Moves the cursor to the previous data item.

  When the cursor is on the top row of the data table, it moves to the bottom row of the left column. When the cursor is on the first data item, it moves to the last data item.
-
Moves the cursor to the next data item.

  When the cursor is on the bottom row of the data table, it moves to the top row of the right column. When the cursor is on the last data item, it moves to the first data item.
-
Moves the cursor left on the screen.

  When the cursor is on the left column of the data table, it moves to the right column on the row immediately above. When the cursor is on the first data item, it moves to the last data item.
-
Moves the cursor right on the screen.

  When the cursor is on the right column of the data table, it moves to the left column on the row immediately below. When the cursor is on the last data item, it moves to the first data item.

- Soft key operations

- Soft key [EDIT]
  Enters the management data edit mode.
- Soft key [NO.SRH]
  Displays data with a tool management data number when pressed after the tool management data number is entered.
- Soft key [PREV.TOOL]
  Returns to the previous tool management data number.
- Soft key [NEXT.TOOL]
  Proceeds to the next tool management number.

Operation in the management data edit mode

To edit data, press soft key [EDIT] to enter the management data edit mode.

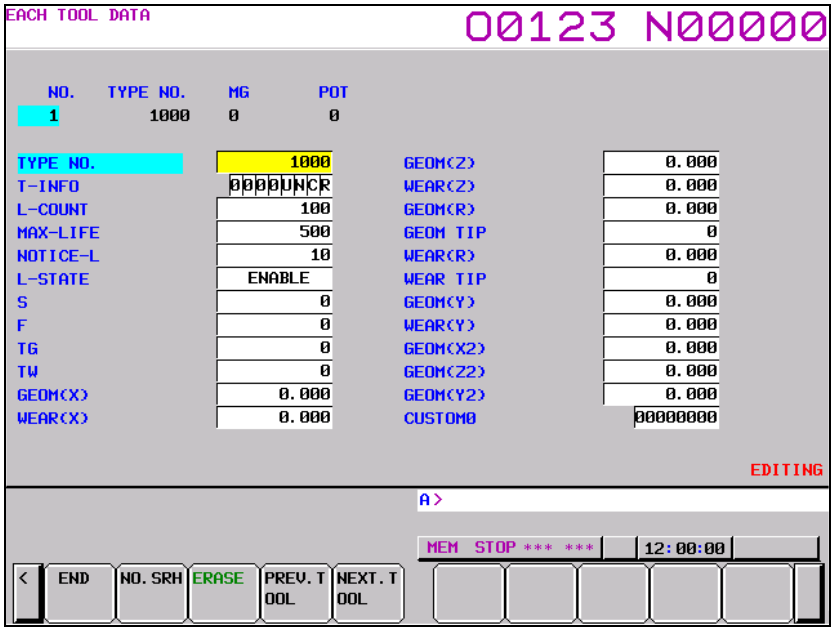


Fig. 12.3.9.3 (b) Each tool data screen (10.4-inch)

In the management data edit mode, “EDITING” is displayed at the lower right of the screen.
In addition to the above key operations, the following key operations are available in the management data edit mode:

-

  Actually writes a value entered using numeric keys.
- Soft key [END]
  Ends the management data edit mode.
  Operations for editing tool information are the same as those on the tool management data screen.

12.3.9.4 Displaying the total life of tools of the same type

Total life data screen

Procedure

1 Press function key .

2 Press chapter selection soft key [TOOL MANAGER].
Alternatively, press key several times until the tool management screen appears.

3 Press soft key [TOTAL LIFE]. The total life data screen appears.

4 Press soft key [(OPRT)], then soft key [CHANGE] to switch the display between total life data of the count counting type tools and that of the time counting type tools.

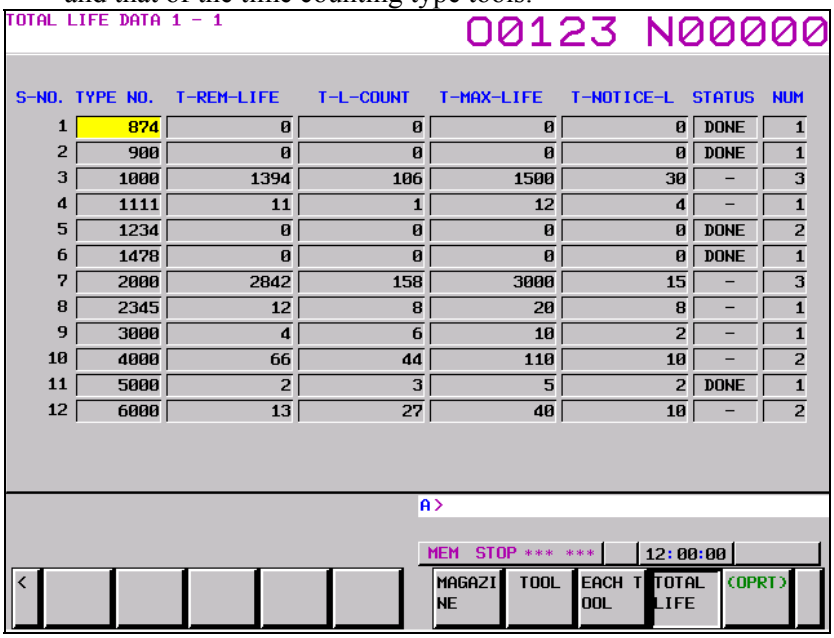


Fig. 12.3.9.4 (a) Count display (10.4-inch)

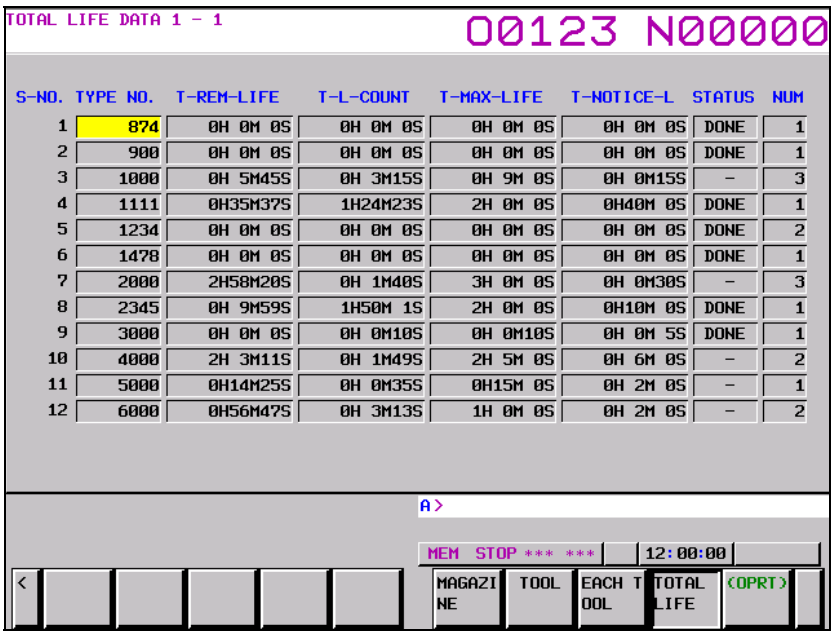


Fig. 12.3.9.4 (b) Time display (10.4-inch)

Displayed information

S-NO. : Sequential number of each tool type
TYPE NO. : Tool type number
T-REM-LIFE : Total of remaining life values of tools with the same tool type number
T-L-COUNT : Total of used counts/times of tools with the same tool type number
T-MAX-LIFE : Total of maximum life values of tools with the same tool type number
NUM : Number of tools with the same tool type number
T-NOTICE-L : Total of notice life values of tools with the same tool type number when the tool life arrival signal is to be output for each tool type
STATUS : Whether the tool life arrival signal has been output when the signal is to be output for each tool type
Either of the two states (UNDONE and DONE) is displayed.

When bit 3 (ETE) of parameter No. 13200 is set to 0 and bit 2 (TRT) of parameter No. 13200 is set to 1, the tool life arrival signal is to be output for each tool type number and T-NOTICE-L and STATUS are displayed.

Key operations

- MDI key operations

Displays the previous page. The cursor moves to the last data item on that page.

Displays the next page. The cursor moves to the first data item on that page.

Moves the cursor up on the screen.

Moves the cursor down on the screen.

Disabled.

Disabled.

- Soft key operations

Soft key [DETAIL]
Displays the detailed life data screen.
Soft key [UPDATE]
Updates data displayed on the total life data screen.
Soft key [CHANGE]
Switch the display between the count and time types.
Soft key [T-TYPENO. SRH]
Moves the cursor to a tool type number when pressed after the number is entered.
Soft key [T-ASCE -SORT]
Sorts data in ascending order of tool type numbers.
Soft key [T-DESC -SORT]
Sorts data in descending order of tool type numbers.
Soft key [R-ASCE -SORT]
Sorts data in ascending order of remaining life values.
Soft key [R-DESC -SORT]
Sorts data in descending order of remaining life values.

NOTE

1 After soft key [T-ASCE -SORT], [T-DESC -SORT], [R-ASCE -SORT], or [R-DESC -SORT] is pressed, the cursor is positioned at the top of page 1 of the total life data screen.
2 When the power is turned on, data of the count counting type is displayed in ascending order of tool type numbers. When the display type is changed or data is sorted in a different order, the status is kept.
3 If soft key [DETAIL] is pressed when no data is displayed on the total life data screen, the warning message “NO DETAILED LIFE DATA SCREEN” is output.
4 If an unregistered tool type is specified and soft key [T-TYPENO. SRH] is pressed, the warning message “UNREGISTERED NUMBER” is output.

Detailed life data screen

Procedure

1 Press function key .
2 Press chapter selection soft key [TOOL MANAGER].
Alternatively, press several times until the tool management screen appears.
3 Press soft key [TOTAL LIFE]. The total life data screen appears.
4 Press soft key [(OPRT)], then soft key [DETAIL]. The detailed life data screen appears.

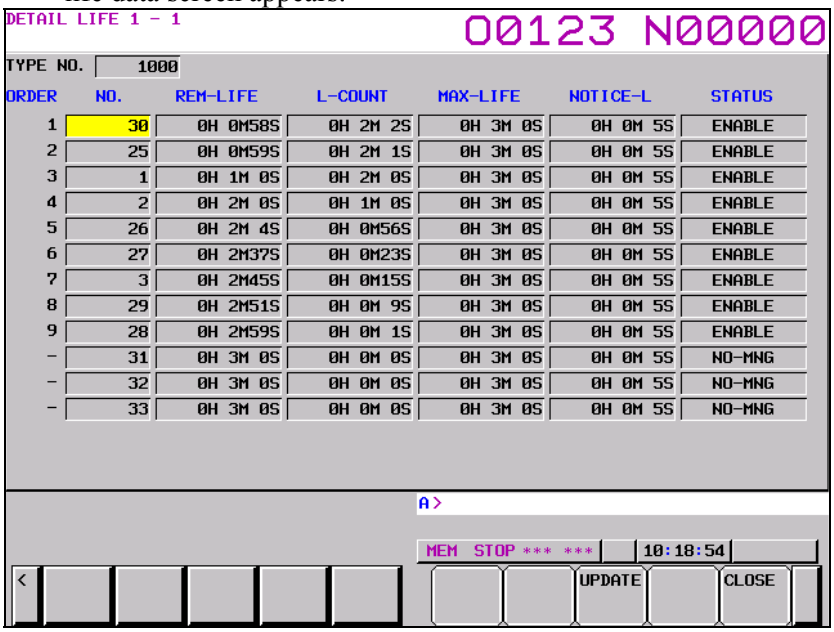


Fig. 12.3.9.4 (c) Detailed life data screen (10.4-inch)

- Displayed information

TYPE NO. : Tool type number
ORDER : Sequential number in ascending order of remaining life times or the order in which the customize data is set.
NO. : Tool management data number
REM-LIFE : Remaining life value obtained by subtracting the life count value from the maximum life value
L-COUNT : Total used count or time of the tool
MAX-LIFE : Maximum life value of the tool
NOTICE-L : Notice life value of the tool
STATUS : Life status of the tool
One of the four states (NO-MNG (0), ENABLE (1, 2), NONE (3), and DAMAGED (4)) is displayed.

Key operations

- MDI key operations

Displays the previous page.

Displays the next page.

Moves the cursor up on the screen. The cursor moves to the last data item on that page.

Moves the cursor down on the screen. The cursor moves to the first data item on that page.

Disabled.

Disabled.

- Soft key operations

Soft key [UPDATA]

Updates data displayed on the detailed life data screen.
The cursor is positioned at the beginning of page 1 after this soft key is pressed.

Soft key [CLOSE]

Closes the detailed life data screen and returns to the total life data screen.

NOTE

1 When soft key [CLOSE] is pressed and the total life data screen is displayed again, the cursor on the total life screen data is positioned as follows.
2 When the tools with the tool type number displayed when the detailed data screen is closed are registered as tool management data, the cursor on the total life screen is positioned at that tool type number.
3 When the tools with the tool type number displayed when the detailed data screen is closed are not registered as tool management data, the cursor on the total life data screen is positioned at the first total life data.

12.3.9.5 Tool geometry data screen

Tool geometry data screen

Procedure

1 Press function key .

2 Press chapter selection soft key [TOOL MANAGER].
Alternatively, press several times until the tool management screen appears.

3 Press continuous menu key , then soft key [TOOL-GEOM].
The tool geometry data screen appears.

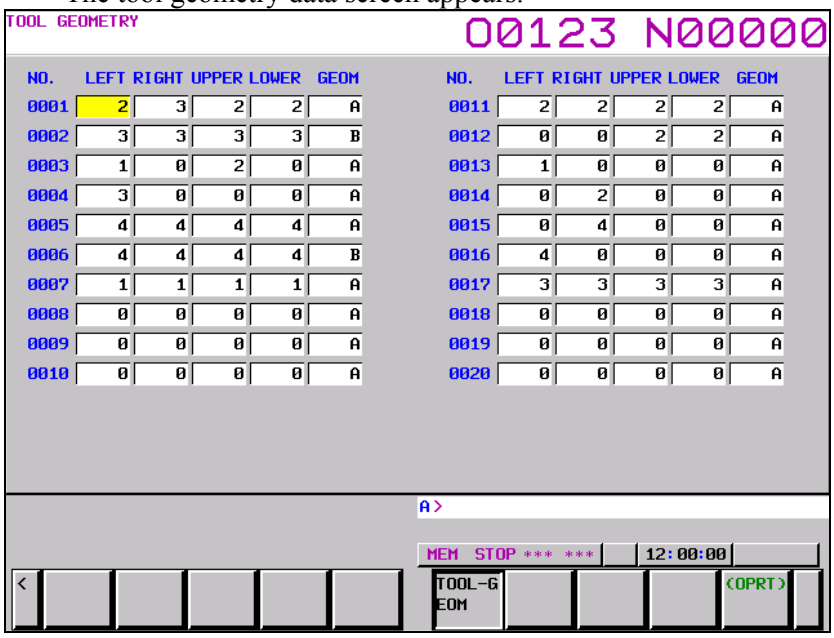


Fig. 12.3.9.5 (a) Tool geometry data screen (10.4-inch)

- Displayed item

NO. : Tool geometry number
Up to 20 numbers can be displayed.

LEFT : Sets the number of pots on the left of the reference pot that are to be occupied.
A value between 0 and 4 can be set.

RIGHT : Sets the number of pots on the right of the reference pot that are to be occupied.
A value between 0 and 4 can be set.

UPPER : Sets the number of pots above the reference pot that are to be occupied.
A value between 0 and 4 can be set. (Use this item when the magazine is of the matrix type.)

LOWER : Sets the number of pots under the reference pot that are to be occupied.
A value between 0 and 4 can be set. (Use this item when the magazine is of the matrix type.)

Key operations

- Operations in the standard mode

MDI key operations

Numeral keys Inputs a numeric value.

Moves the cursor up on the screen.

Moves the cursor down on the screen.

Moves the cursor left on the screen.

Moves the cursor right on the screen.

Soft key operations

Soft key [NO.SRH]
Moves the cursor on a tool geometry number when pressed after the number is entered.
Soft key [EDIT]
Enters the data edit mode.
Soft key [READ]
Reads data related to the tool management functions.
This key is available only in the standard mode.
Put the NC in the EDIT mode.
Soft key [PUNCH]
Punches data related to the tool management functions.
This key is available only in the standard mode.
Put the NC in the EDIT mode.
In the management data edit mode, the following key operations are available in addition to the above key operations.

- Operations in the edit mode

To edit data, press soft key [EDIT] to enter the edit mode. In the edit mode, “EDITING” is displayed at the lower right of the screen.

MDI key operation

Actually writes a value entered using numeric keys.

Soft key operation

Soft key [END]
Ends the data edit mode.

Example

Set the edit mode. When the tool geometry with tool geometry number 1 occupies 1 pot in the left direction, 0.5 pots in the right direction, and 1.5 pots in the down direction, set data as shown in the figure below:

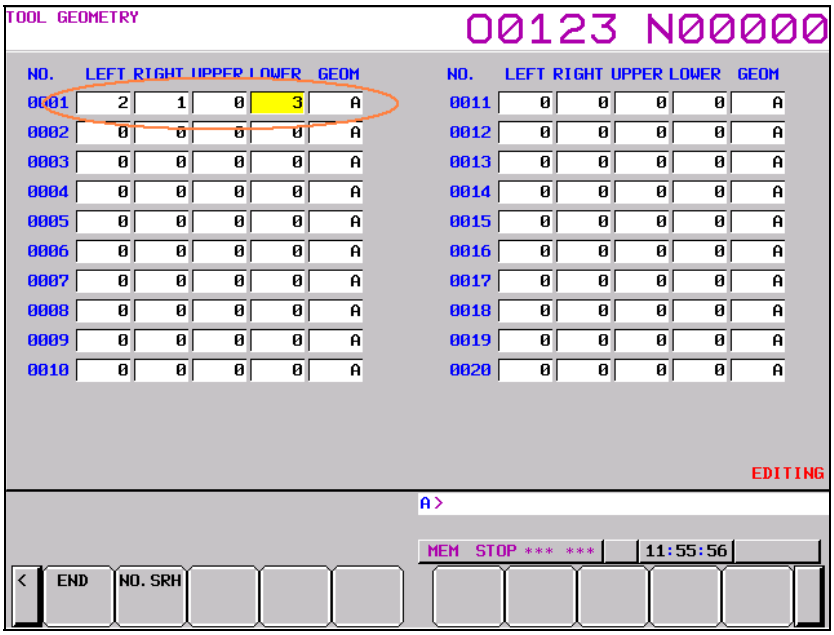


Fig. 12.3.9.5 (b) Example of setting data on the tool geometry data screen (10.4-inch)

- Display of occupied pots in the magazine management table

Each pot occupied by a tool stored in another pot is indicated with an asterisk (*).

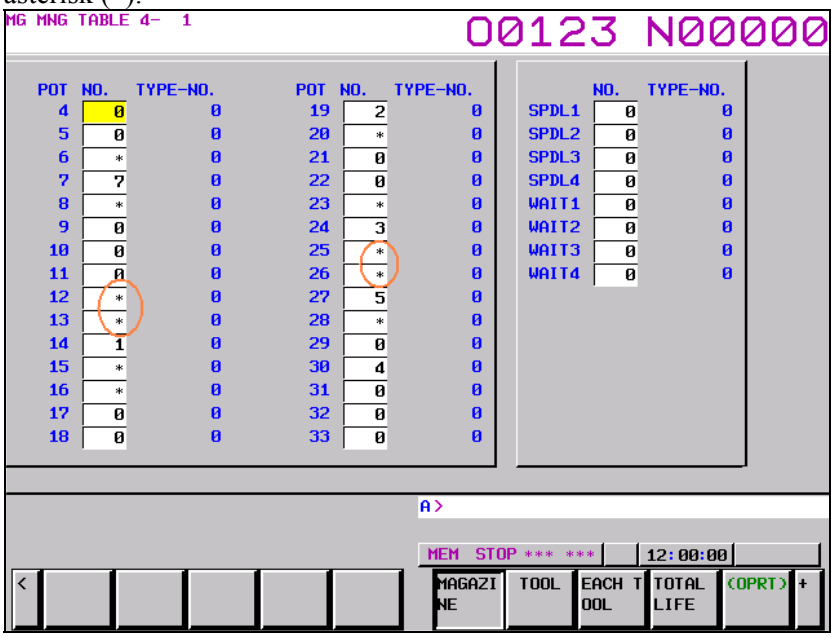


Fig. 12.3.9.5 © Magazine management table (10.4-inch)

If a tool to be registered for a magazine is determined to interfere with another tool, the warning message “TOOL INTERFERENCE CHECK ERROR:xxxx,xxxx” is displayed. xxxx indicates the tool number of each of the two tools. If a tool is determined to interfere with multiple tools, only the first tool detected by the NC interference check is displayed.
If a tool is determined to interfere with the frame of a magazine, the tool number of the tool to be stored and “FRAME” are displayed.

- Search for an empty pot for a oversize tool

In the management data edit mode, pressing soft key [(OPRT)], then continuous menu key ▷ displays the soft keys for searching for an empty pot for a oversize tool.

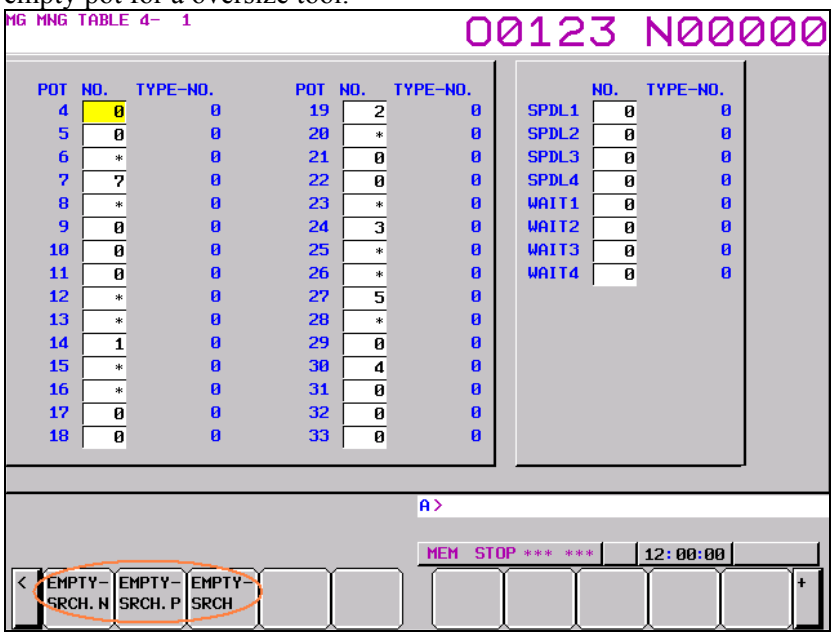


Fig. 12.3.9.5 (c) Searching for an empty pot for a oversize tool (10.4-inch)

Enter the tool geometry number in the key-in buffer and press a search soft key. The cursor moves to an empty pot fit for the geometry.
EMPTY-SRCH.N : Searches for an empty pot forward.
EMPTY-SRCH.P : Searches for an empty pot backward.
EMPTY-SRCH : Searches for the pot nearest to the current position.

- Tool management screen

You can use bit 2 of tool information to switch between a oversize tool and normal tool. For a oversize tool, set a tool geometry number fit for the tool.

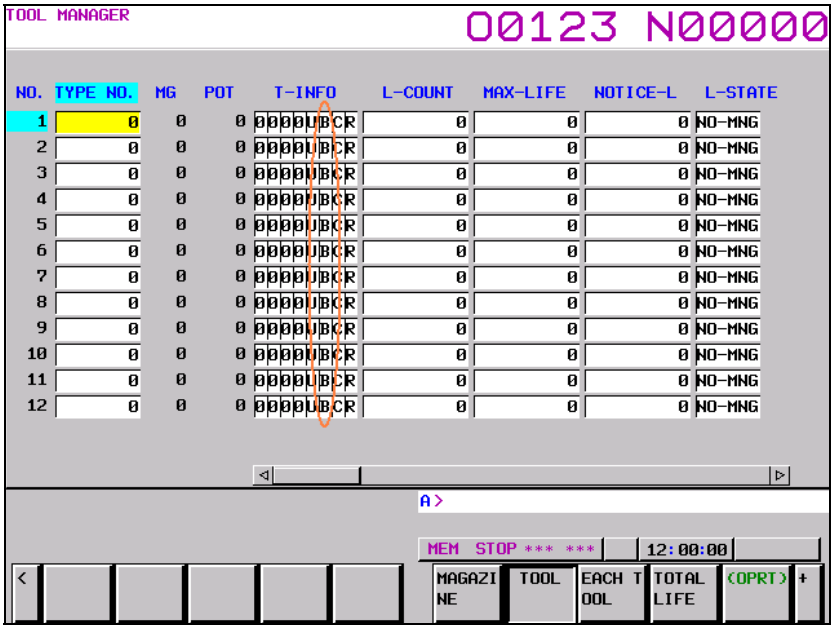


Fig. 12.3.9.5 (d) Bit for switching between a normal tool and oversize tool (10.4-inch)

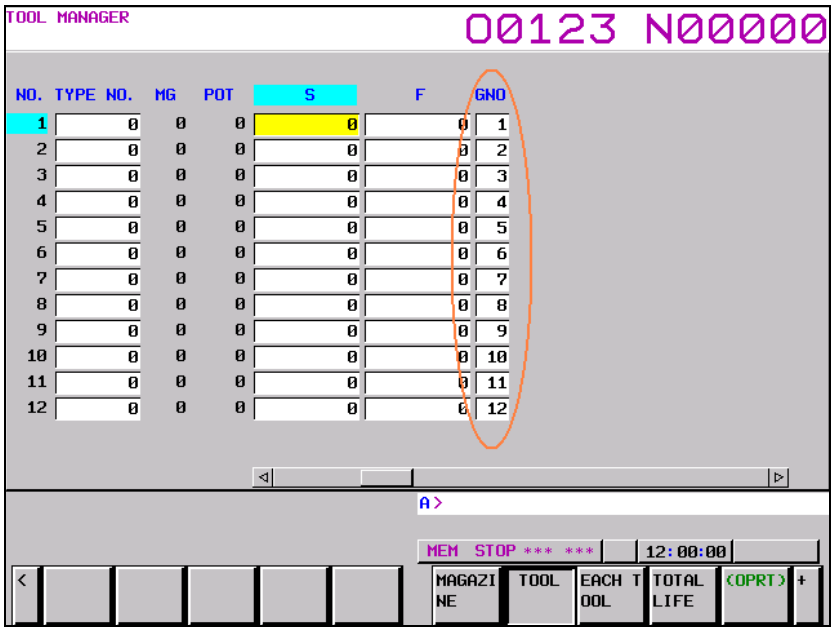


Fig. 12.3.9.5 (e) Tool geometry number (10.4-inch)

12.3.10 Displaying and Switching the Display Language

The language used for display can be switched to another language.
A display language can be set using a parameter. However, by modifying the setting of the display language on this screen, the display language can be switched without turning off then on the power.

Displaying and setting the display language

Procedure

1 Press function key .

2 Press the continuous menu key several times.

3 Press soft key [LANGUAGE] to display the language screen.

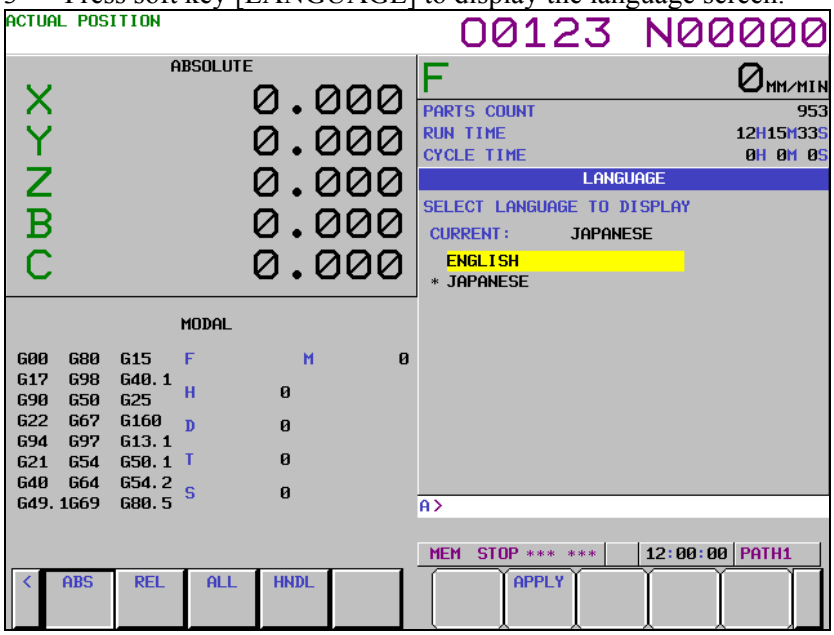


Fig. 12.3.10 (a) LANGUAGE screen (10.4-inch)

4 Press page key or , then press cursor keys , to move the cursor to a desired display language.

5 Press operation soft key [APPLY]. The display language is switched to the selected language. The language specified on this screen continues to be used if the power is turned off then back on.

Explanation

- Language switching

The language screen can be displayed if bit 0 (NLC) of parameter No. 3280 is set to 0.

- Selectable languages

The display languages selectable on this screen are as follows:

1. English
2. Japanese
3. German
4. French
5. Chinese (Traditional)
6. Chinese (Simplified)
7. Italian
8. Korean
9. Spanish
10. Dutch
11. Danish
12. Portuguese
13. Polish
14. Hungarian
15. Swedish
16. Czech
17. Russian
18. Turkish

Among the languages listed above, English and other usable languages are displayed on the screen as a list of switchable languages.

Limitation

- Language parameter modification on the parameter screen

Which language to use for display is specified with parameter No. 3281. This parameter can be modified using the parameter screen as well. However, if a modification is made on the parameter screen, the new setting is not reflected until “APPLY” operation is performed on the language screen or the power is turned on again. If an invalid value is set in parameter No. 3281 on the parameter screen, the screen is displayed in English after the power is turned on again.

12.3.11 Protection of Data at Eight Levels

You can set eight CNC and PMC operation levels and one of eight protection levels for each type of CNC and PMC data.
When an attempt is made to change CNC and PMC data or output it to an external unit, the operation level is compared with the protection level to determine whether to allow the change or external output.

12.3.11.1 Operation level setting

You can set eight CNC and PMC operation levels.

Displaying and setting the operation level setting screen

Procedure

1 Press function key .

2 Press the continuous menu key several times until soft key [PROTECT] is displayed.

3 Press soft key [PROTECT].
The operation level setting screen shown below is displayed.

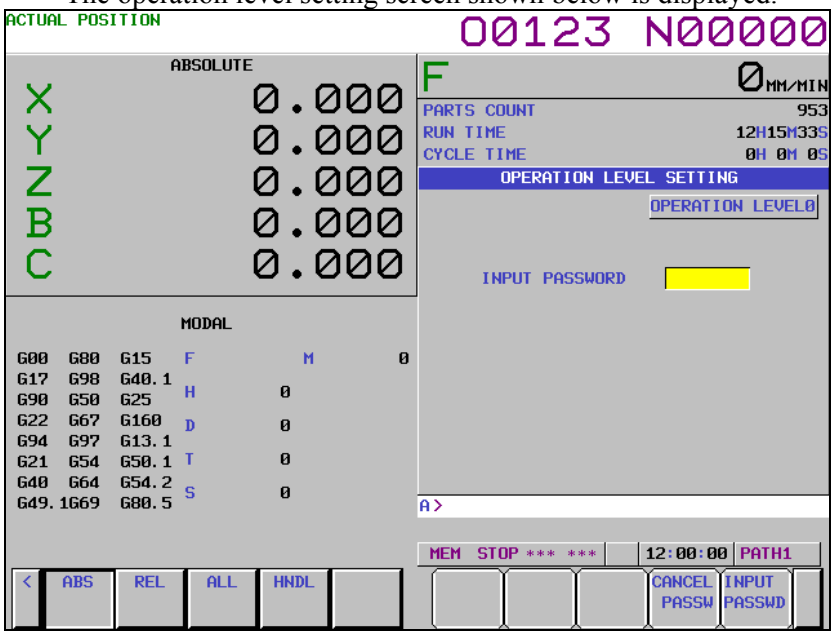


Fig. 12.3.11.1 (a) Operation level setting screen (10.4-inch)

4 Key in the password for an operation level to be set/modified, then press soft key [INPUT PASSWD].

5 To return the operation level to 0, 1, 2, or 3, press soft key [CANCEL PASSWD].

Explanation

- Operation level setting

To select operation level 0 to 3, use the corresponding memory protection key signal.
To select operation level 4 to 7, use the corresponding password.

Table 12.3.11.1 (a) Operation level setting

Operation level	Setting	Sample grouping
7 (high)	Password	-
6	Password	MTB
5	Password	Dealer and integrator
4	Password	End user
3	Memory protection key signal	User level (level 1)
2	Memory protection key signal	User level (level 2)
1	Memory protection key signal	User level (level 3)
0 (low)	Memory protection key signal	User level (level 4)

When operation level 4 to 7 is set, the operation level remains unchanged until the password is cleared.
(The operation level also remains unchanged if the power is turned off.)
Operation level 7 is reserved for CNC and PMC maintenance.

NOTE
When a password is being entered, an asterisk (*) is displayed instead of each entered character.

12.3.11.2 Password modification

The current operation level is displayed.
The password for each of operation levels 4 to 7 can be modified.

Displaying and setting the password modification screen

Procedure

1 Press function key .
2 Press the continuous menu key several times until soft key [PROTECT] is displayed.
3 Press soft key [PROTECT].
4 Press soft key [PASSWORD].
The PASSWORD CHANGE screen shown below is displayed.

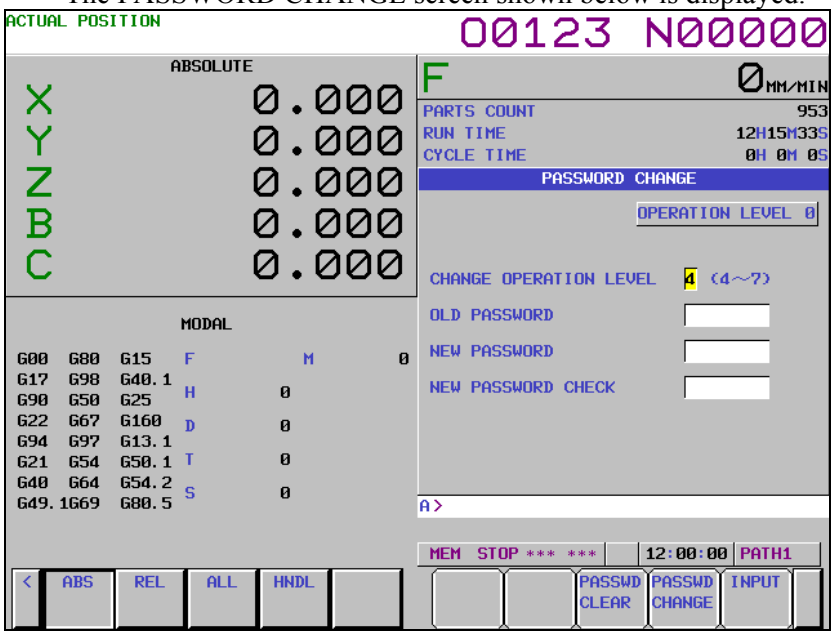


Fig. 12.3.11.2 (a) PASSWORD CHANGE screen (10.4-inch)

5 Key in an operation level whose password is to be modified, then press soft key [INPUT].
6 Key in the current password for the operation level whose password is to be modified, then press soft key [INPUT].
7 Key in a new password, then press soft key [INPUT].
8 Key in the new password again for confirmation, then press soft key [INPUT].
9 Press soft key [PASSWDCHANGE].
10 To clear the password, press soft key [PASSWDCLEAR].

Explanation

Up to eight characters (only uppercase alphabetic characters and numeric characters) can be input.

NOTE

1 For a password, consisting of three to eight characters, the following characters are available:
   - Uppercase alphabetic characters
   - Numeric characters

2 When a password is being entered, an asterisk (*) is displayed instead of each entered character.

3 Whether a password can be changed at the current operation level is determined as follows:
   - Password of an operation level higher than the current operation level
     Cannot be changed.
   - Password of the current operation level
     Can be changed.
   - Password of an operation level lower than the current operation level
     Can be changed (only to the initial password).

4 The set password is not displayed.
   Be careful not to forget the password.

12.3.11.3 Protection level setting

The current operation level is displayed.
The change protection level and output protection level of each data item are displayed.
The change protection level and output protection level of each data item can be changed.

Confirmation based on protection level setting

Procedure

1 Press function key .
2 Press the continuous menu key several times until soft key [PROTECT] is displayed.
3 Press soft key [PROTECT].
4 Press soft key [DATA LEVEL] to change the protection level of CNC data or press soft key [PMC LEVEL] to change the protection level of PMC data.

The following protection level change screen is displayed.

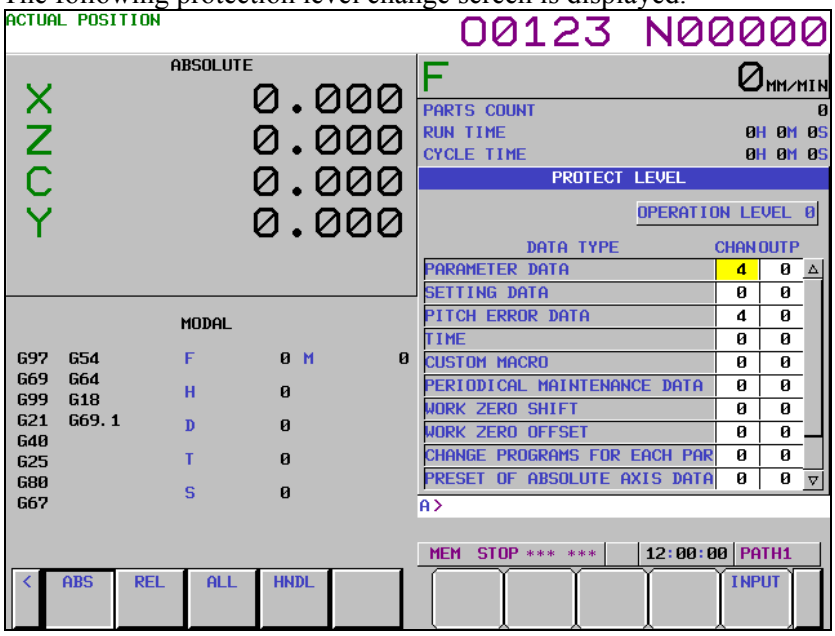


Fig. 12.3.11.3 (a) Protection level change screen (10.4-inch)

5 Move the cursor to the change level or output level of a desired data item.
6 Key in a new desired level, then press soft key [INPUT].

NOTE
When the protection level of PMC data is set, soft key [SWITCH PMC] is used to switch between PMC paths to be set, for multipath PMC.

Explanation

When the protection level of a data item is higher than the current operation level, the protection level of the data item cannot be changed.
The protection level of a data item cannot be changed to a protection level higher than the current operation level.
For each of the following types of data, you can set a data protection level. There are the following two types of data protection levels:

- Change protection level

Sets the protection level used when data is changed.

- Output protection level

Sets the protection level used when data is output (punched out) to an external unit.

As a protection level, you can set a value of 0 (low) to 7 (high).

Table 12.3.11.3 (a) Protection level of each type of data

Type of data	Initial protection level	
	Change	Output
Custom macro variable data <CUSTOM MACRO> (including variable data dedicated to the macro executor)	0	0
Periodical maintenance data <PERIODICAL MAINTENANCE DATA>	0	0
Tool management data <TOOL LIFE TIME DATA>	0	0
Tool offset data <TOOL OFFSET DATA> (For each type when tool geometry compensation and tool wear compensation are treated differently)	0	0
Clock data <TIME>	0	0
Workpiece origin shift amount data <WORK ZERO SHIFT>	0	0
Workpiece origin offset data <WORK ZERO OFFSET>	0	0
Ethernet setting data <ETHER NET SETTING>	0	0
Fixture offset data <FIXTURE OFFSET DATA>	0	0
Parameter data <PARAMETER DATA>	4	0
Settings <SETTING DATA>	0	0
Pitch error compensation data <PITCH ERROR DATA> (including three-dimensional error compensation data)	4	0
Parameter data for Power Mate CNC manager function <PARAMETER DATA FOR POWER MATE CNC MANAGER>	0	0
Each part program <PROGRAMS FOR EACH PARTS>	0	0
Part program editing operation <CHANGE PROGRAMS FOR EACH PARTS>	0	0
Absolute coordinate preset operation <PRESET OF ABSOLUTE AXIS DATA>	0	0

Table 12.3.11.3 (b) Protection level of PMC data

Type of data	Initial protection level	
	Change	Output
Composition parameter	0	0
Setting (online)	0	0
Setting (each path)	0	0
Sequence program	0	0
PMC parameter	0	0
Timer	0	0
Counter	0	0
Keep relay	0	0
Keep relay (system)	0	0
Data table	0	0
Data table control	0	0
PMC momory	0	0

NOTE

1 For some types of data, the output function is not provided.
2 When the protection level of data is higher than the current operation level, the protection level cannot be changed.
3 The protection level of data cannot be changed to a level higher than the current operation level.
4 Settable types of data increase or decrease, depending on the option configuration.
5 For details on the protection level of PMC data, refer to “PMC Programming Manual (B-63983EN)”.
6 Data related to tool information on the EACH TOOL DATA screen for the tool management function is protected by protecting tool offset data.
7 The type of tool offset data put in effect varies depending on the tool compensation value memory used.
8 To change the protection level for each part program, do so on the PROGRAM FOLDERscreen rather than on the PROTECT LEVEL screen.
9 Part program editing includes program editing for the MDI mode.
10 Presetting absolute coordinates causes preset workpiece coordinate system values to be protected.
11 During tool offset data input/output, if any tool offset data type is not allowed to be changed or output to the outside, it is processed as follows:
- Input : Any data type other than those whose change is not allowed is changed.
- Output : Any data type other than those whose change is not allowed is output.

12.3.11.4 Setting the change protection level and output protection level of a program

The display/operations indicated below can be performed from the directory screen.
The change protection level and output protection level of each part program are displayed.
The change protection level and output protection level of each part program can be changed.

Setting the change protection level and output protection level of a program

Procedure

1 Press function key .
2 Press soft key [FOLDER].
The program directory screen shown below is displayed.

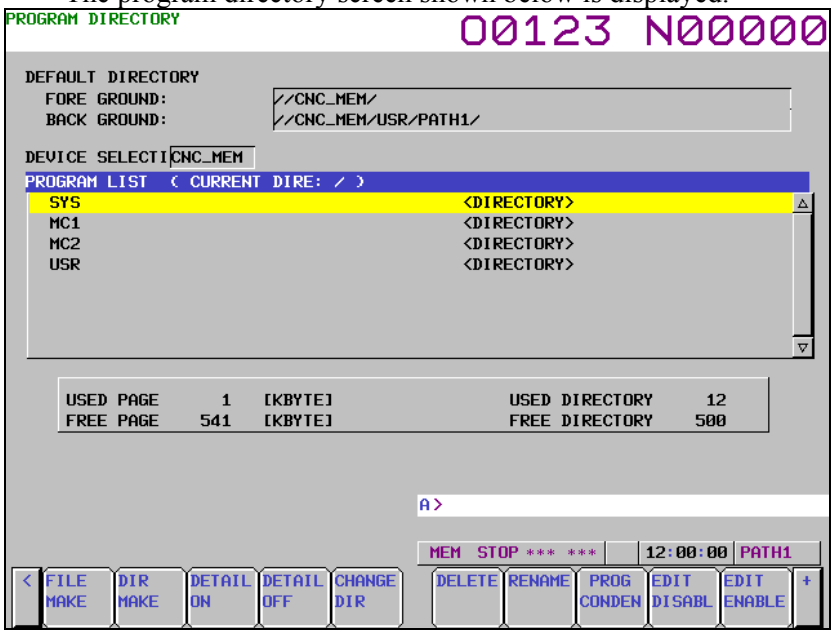


Fig. 12.3.11.4 (a) Program directory screen (10.4-inch)

3 Press the continuous menu key several times until [DETAIL ON] is displayed.
4 Press soft key [DETAIL ON]. The screen display switches to the detail display screen.
5 Move the cursor to a desired program.
6 Press the continuous menu key several times until [CHANGE LEVEL] is displayed.
7 Key in a new desired level, then press soft key [CHANGE LEVEL].
8 To change the output protection level, key in a new desired level, then press soft key [OUT LEVEL].

Explanation

The change protection level (0 to 7) and output protection level (0 to 7) are displayed as “CHANGE PROTECTION LEVEL VALUE/OUTPUT PROTECTION LEVEL”.

NOTE
1 When the protection level of data is higher than the current operation level, the protection level cannot be changed.
2 The protection level of data cannot be changed to a level higher than the current operation level.
3 A protection level can be set only for those part programs that are held on the “CNC_MEM” device.

12.3.12 Precision Level Selection

An intermediate precision level between the parameters for emphasis on velocity (precision level 1) and the parameters for emphasis on precision (precision level 10) set on the machining parameter tuning screen can be selected. As shown in the figure below, the levels are proportionally linear, and an intermediate level can be selected so that optimal parameters can be automatically calculated to perform machining.

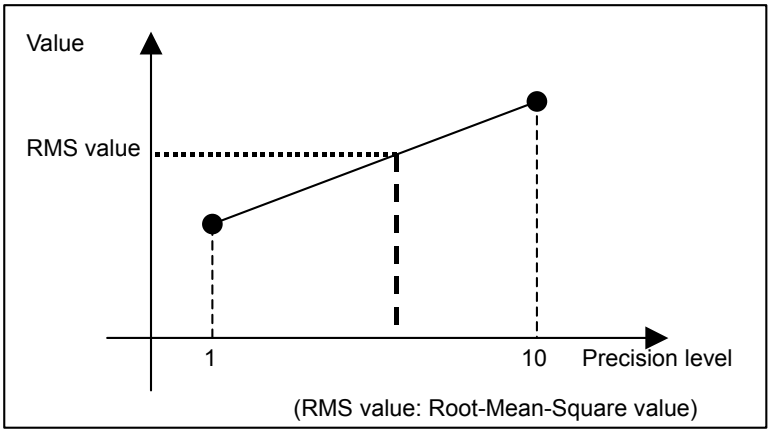


Fig. 12.3.12 (a) Image of "level"

Procedure for precision level selection

1 Select the MDI mode.

2 Press function key .

3 Press soft key [PR-LEV].

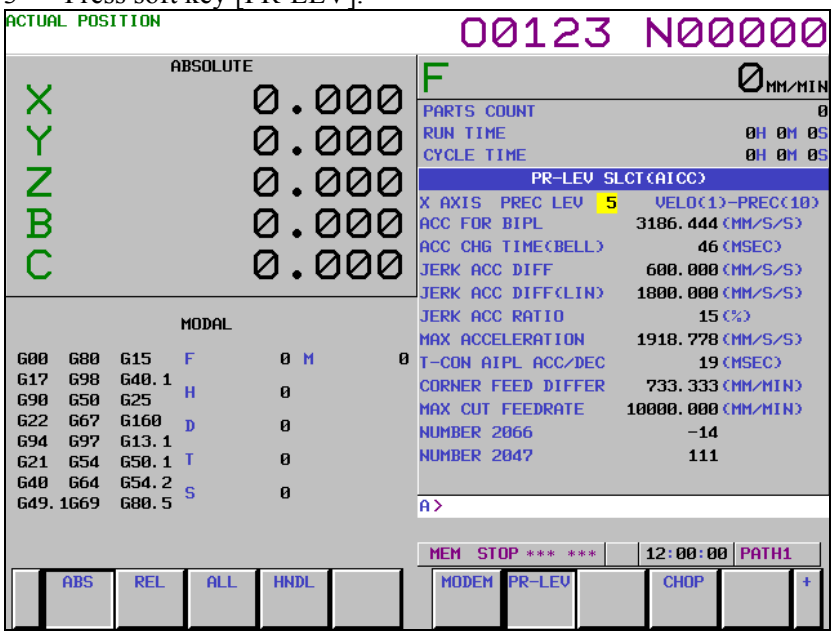


Fig. 12.3.12 (b) Precision level selection screen (10.4-inch)

4 To change the precision level, key in a desired precision level (1 to 10), then press the

key on the MDI panel.

5 When the precision level is changed, a RMS value is obtained from the velocity-emphasized parameter set and precision-emphasized parameter set for parameter modification.
For the modified parameters, see the description of the machining parameter tuning.

6 If there is an axis in addition to the currently displayed axes, press page key

or

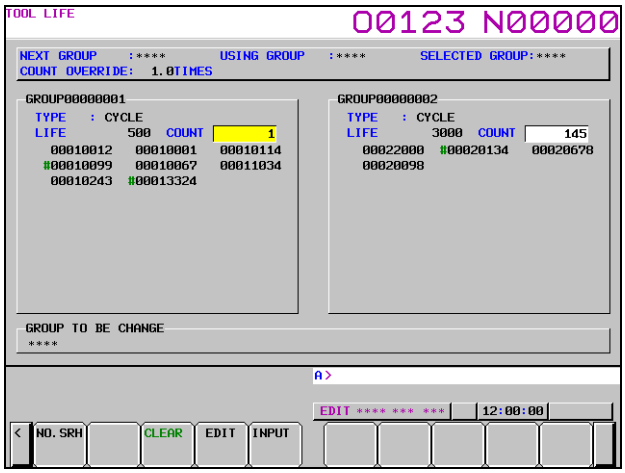
several times to display the screen for the axis.

12.3.13 Displaying and Setting Tool Life Management Data

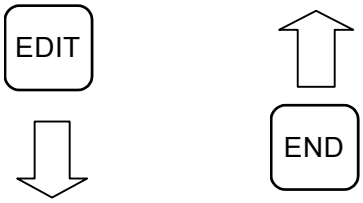
Displaying tool life management data on a screen enables the current status of tool life management to be grasped. Also on the screen, tool life management data can be edited. The screen is either:

- Tool life management (list screen) or
- Tool life management (group editing screen)

Overview

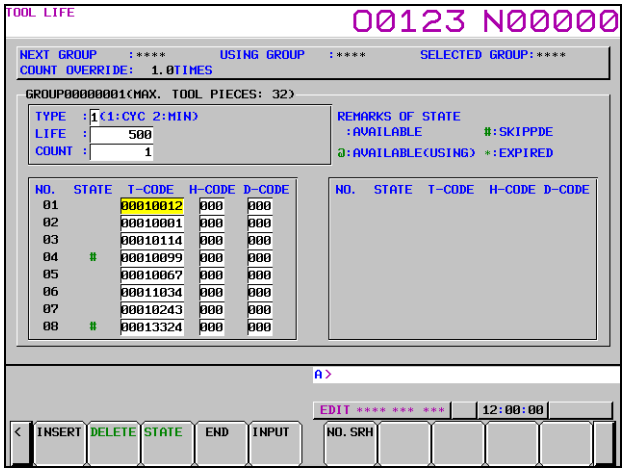


List screen



Tool life management (list screen)
Displayed items:
- NEXT GROUP
- SELECTED GROUP
- GROUP NO.
- LIFE
- TOOL MANAGEMENT STATUS
- GROUP TO BE CHANGE

Functions:
- Searching for groups
- Clearing execution data
- Setting the tool life counter



Group editing screen

Tool life management (group editing screen)
Displayed items:
- NEXT GROUP
- SELECTED GROUP
- MAX.TOOL PIECES
- LIFE
- T-CODE
- H-CODE and D-CODE

Functions:
- Setting tool data
- Setting a tool status
- Selecting too skip
- Setting a life count type, life value, and tool life counter

Pressing horizontal soft key [EDIT] or [END] switches from the list screen to the group editing screen or vice versa.

M

If the tool life management B function is enabled (bit 4 (LFB) of parameter No. 6805 = 1), the following parameters can be used to display arbitrary groups and remaining set values.

- If bit 5 (TGN) of parameter No. 6802 = 1
  An arbitrary group is displayed on the list and group editing screens.
- If bit 3 (GRP) of parameter No. 6802 = 1
  Remaining set values are displayed on the group editing screen. The group editing screen always displays H and D codes.

T

The T series is provided with two tool change types, turret type and ATC type. For the T series, displays vary depending on which tool change type is in use.

If the turret type is in use (bit 3 (TCT) of parameter No. 5040 = 0)

- Neither the H code nor the D code is used; neither is displayed.
- No arbitrary group can be used; no arbitrary group is displayed.

If the ATC type is in use (bit 3 (TCT) of parameter No. 5040 = 1)

- The D code is displayed on the group editing screen.
- If the tool life management B function is enabled (bit 4 (LFB) of parameter No. 6805 = 1) and bit 5 (TGN) of parameter No. 6802 = 1, arbitrary groups are displayed on the list and group editing screens.

If the tool life management B function is enabled (bit 4 (LFB) of parameter No.6805 = 1) and bit 3 (GRP) or parameter No. 6802 = 1, remaining set values are displayed on the group editing screen.

12.3.13.1 Tool life management (list screen)

This screen can display the life management status of all tools in tool groups and whether the life of the tool groups has expired. It also enables you to set tool life counters and clear execution data.

Displays on the list screen

Procedure

1 Press function key .

2 Press the continuous menu key several times to display soft key [TOOL LIFE].

3 Press soft key [TOOL LIFE].

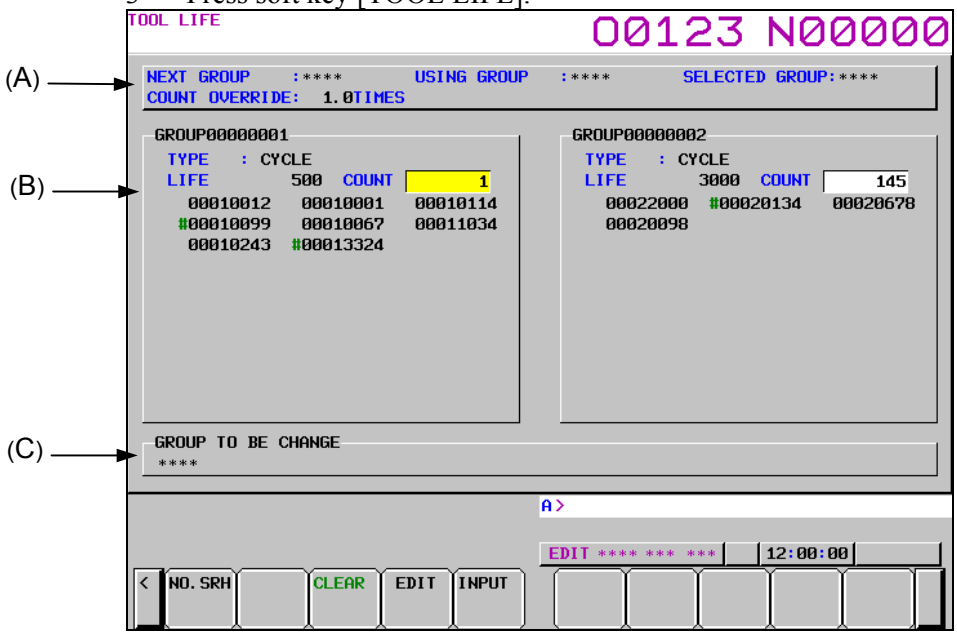


Fig. 12.3.13.1 (a) Displaying tool life management (list screen) (10.4-inch)

- Contents of (A)

(A) displays tool group numbers and an override value. If there is no tool group to display, ****, instead of tool group numbers, is displayed.

NEXT GROUP:

Tool group number for which life counting is started by the next M06 command.

USING GROUP:

Tool group number for which life counting is currently under way.

SELECTED GROUP:

Tool group number for which life counting is currently under way or life counting has been performed most recently.

COUNT OVERRIDE:

"1.0TIMES" is displayed if the tool life counter override signal is disabled (bit 2 (LFV) of parameter No. 6801 = 0).

NOTE
If arbitrary group numbers are enabled, NEXT GROUP, USING GROUP, and SELECTED GROUP are each represented with an arbitrary group number rather than the tool group number.

- Contents of (B)

(B) displays the set life value, the current content of the tool life counter, and the registered tool numbers (in the order they are used) for each tool group. If the life count type is specification by duration, the measurement unit used in displaying and specifying set life values and tool life counter values is selected according to the setting of bit 0 (FCO) of parameter No. 6805 as listed below.

Bit 0 (FCO) of parameter No. 6805	0	1
Measurement unit used in displaying and specifying set life values and tool life counter values	1 minute	0.1 minutes

The following table lists the prefixes used with tool numbers.

Tool status	Tool in use	Tool not in use
Life remaining	@	No indication
Skip	#	#
Life expired	*	*

NOTE
1 The tool life counter indicates the count value for the tool indicated with @.
2 If bit 3 (EMD) of parameter No. 6801 = 0, a tool number remains prefixed with @ even if the life of the tool has expired until another tool is selected.
3 If bit 3 (EMD) of parameter No. 6801 =1, the following differences can occur depending on the type of the life counter in use.
- If the life counter type is specification by duration, the prefix is changed to "*" (life expired) as soon as the tool life expires.
- If the life counter type is specification by count, the counter is incremented by one at the end of a program (such as M02 or M30). So, the prefix is not changed to "*" (life expired) even if the tool life counter matches the life value. The symbol "*" (life expired) appears when the tool life count is incremented after the CNC is reset.

4 If bit 2 (ETE) of parameter No. 6804 = 1, the symbol "*" indicating the expiration of the life of the last tool appears on the tool life management screen when the life counter for the last tool of the tool group of interest matches the life value. This way, the information about the last tool in the FOCAS2 or PMC window indicates that the life of the tool has expired if the tool change signal TLCH <Fn064.0> is 1.

If arbitrary group numbers are enabled, an arbitrary group number is displayed in the parentheses beside the tool group number. If no arbitrary group number is specified, "********" is displayed instead.

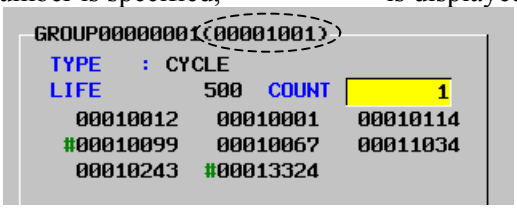


Fig. 12.3.13.1 (b) Displaying arbitrary group numbers

Arbitrary group numbers are enabled by setting the following parameters.

M

- The tool life management B function is enabled (bit 4 (LFB) of parameter No. 6805 = 1).
- Arbitrary group numbers are enabled (bit 5 (TGN) of parameter No. 6802 = 1).

T

- The current tool change type is ATC (bit 3 (TCT) of parameter No. 5040 = 1).
- The tool life management B function is enabled (bit 4 (LFB) of parameter No. 6805 = 1).
- Arbitrary group numbers are enabled (bit 5 (TGN) of parameter No. 6802 = 1).

- Contents of (C)

(C) displays tool group numbers for which a tool change signal has been issued.
If there are so many tool group numbers that all the numbers cannot be displayed, some are omitted, and ">>" is displayed instead.
If there is no tool group number in need of change, "****" is displayed.

NOTE
If arbitrary group numbers are enabled, a tool group number in need of change is represented with an arbitrary group number rather than the tool group number.

Setting data on the list screen

Tool life management data can be specified in the reset state (both the OP and RST signals are “0”). However, setting bit 1 (TCI) of parameter No. 6804 to 1 enables tool life management data to be specified even during automatic operation (the OP signal is “1”).

NOTE
As for USING GROUP or NEXT GROUP settings:
1) During automatic operation (OP signal = “1” and bit 1 (TCI) of parameter No. 6804 = 1), only the tool life counter can be changed.
2) In the reset state (OP signal = “0” and RST signal “0”), the following editing operation stops life management because it cannot be continued.
- Clearing execution data

Procedure

- Setting the tool life counter

The tool life counter can be set with a value, using the following methods.

Method 1

1 Place the cursor on the tool life counter for a desired tool group.
2 Enter the value from the keypad.
3 Press soft key [INPUT].

Method 2

1 Place the cursor on the tool life counter for a desired tool group.
2 Enter the value from the keypad.
3 Press

key.

- Clearing execution data

All existing execution data for a tool group selected by the cursor can be cleared as follows:

1 Place the cursor on the tool group whose execution data you want to clear.
2 Press soft key [ERASE].
3 Press soft key [EXEC].

NOTE
Setting bit 4 (GRS) of parameter No. 6800 to 1 enables execution data for all registered tool groups to be cleared.

- Selecting tool groups

Tool groups can be selected using the following methods.
Method 1

1 Enter a tool group number from the keypad.
2 Press soft key [NO.SRH].

NOTE
If arbitrary group numbers are enabled, a tool group is selected by searching for an arbitrary group number rather than the tool group number.

Method 2

1 Press page key or to display desired groups.
2 Press cursor key or to move the cursor to the desired group at either the left or right.

- Switching to the group editing screen

Switch to tool life management (group editing screen).

1 Move the cursor to the tool group you want to edit:
2 Press soft key [EDIT].

12.3.13.2 Tool life management (group editing screen)

On this screen, it is possible to edit tool life management data (such as tool life value, tool life counter, and tool data) for the tool group of interest.

Displaying the group editing screen

Procedure

1 Place the cursor on the tool group you want to edit from the list screen.
2 Press soft key [(OPRT)].
3 Press soft key [EDIT].

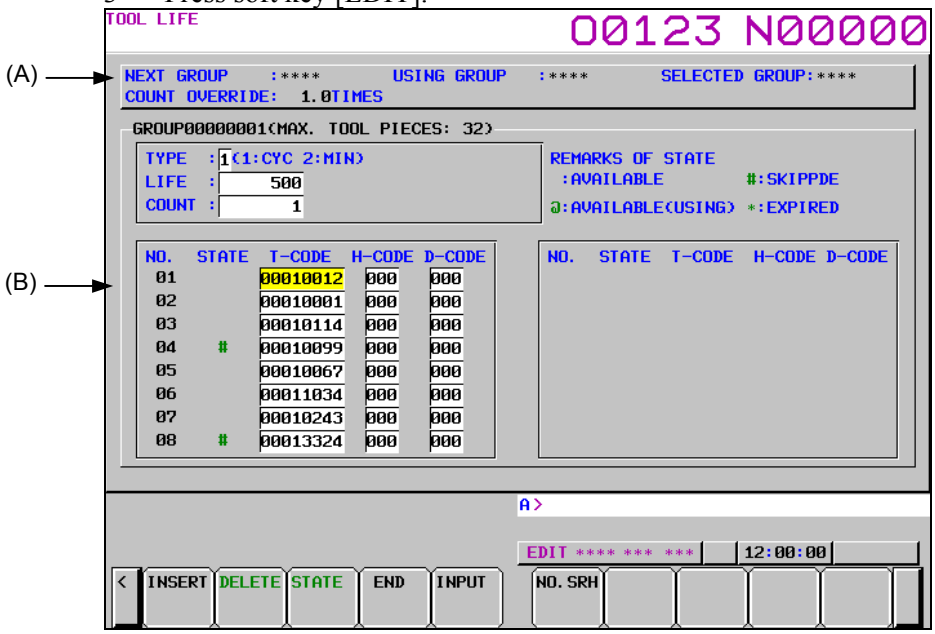


Fig. 12.3.13.2 (a) Displaying tool life management (group editing screen) (10.4-inch)

NOTE
If no tool is registered with a tool group, none of a life count type, a life value, and a tool life counter value is displayed for the tool group.

- Contents of (A)

Like the counterpart of the list screen, (A) of the editing screen displays the next tool group number and override value. If there is no appropriate tool group, **** is displayed instead.

NEXT GROUP:
Tool group for which life counting is started by the next M06 command.

USING GROUP:
Tool group number for which life counting is currently under way.

SELECTED GROUP:
Tool group number for which life counting is currently under way or life counting has been performed most recently.
COUNT OVERRIDE:
"1.0TIMES" is displayed if the tool life counter override signal is disabled (bit 2 (LFV) of parameter No. 6801 = 0).

NOTE
If arbitrary group numbers are enabled, NEXT GROUP, USING GROUP, and SELECTED GROUP are each represented with an arbitrary group number rather than the tool group number.

- Contents of (B)

(B) details tool life management data related to a selected tool group as follows:

TYPE :

1	Specification by count
2	Specification by duration

LIFE : Tool life value
COUNT : Tool life counter

STATE :

Tool status	Tool in use	Tool not in use
Life remaining	@	No indication
Skip	#	#
Life expired	*	*

T-CODE : Tool number

M

H-CODE : Tool length compensation specification code
D-CODE : Cutter compensation specification code

T

H-CODE : No display.
D-CODE : Tool offset value specification code if ATC type is in use (bit 3 (TCT) of parameter No. 5040 = 1)

NOTE

1 The tool life counter indicates the count value for the tool indicated with @.

2 If bit 3 (EMD) of parameter No. 6801 = 0, a tool number remains prefixed with @ even if the life of the tool has expired until another tool is selected.

3 If bit 3 (EMD) of parameter No. 6801 = 1, the following differences can occur depending on the type of the life counter in use.

- If the life counter type is specification by duration, the prefix is changed to "*" (life expired) as soon as the tool life expires.
- If the life counter type is specification by count, the counter is incremented by one at the end of a program (such as M02 or M30). So, the prefix is not changed to "*" (life expired) even if the tool life counter matches the life value. The symbol "*" (life expired) appears when the tool life count is incremented after the CNC is reset.

4 If bit 2 (ETE) of parameter No. 6804 = 1, the symbol "*" indicating the expiration of the life of the last tool appears on the tool life management screen when the life counter for the last tool of the tool group of interest matches the life value. This way, the information about the last tool in the FOCAS2 or PMC window indicates that the life of the tool has expired if the tool change signal TLCH <Fn064.0> is 1.

Displaying arbitrary groups and remaining set values

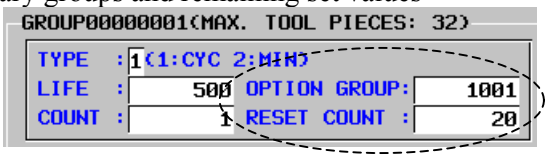


Fig. 12.3.13.2 (b) Displaying arbitrary group numbers and remaining set values on the editing screen

M

If the tool life management B function is enabled (bit 4 (LFB) of parameter No. 6805 = 1) and the following parameters are enabled, arbitrary groups and remaining set values are displayed.

OPTION GROUP:

Arbitrary group number
(if bit 5 (TGN) of parameter No. 6802 = 1)

REST COUNT:

Remaining set value used until a new tool is selected
(if bit 3 (GRP) of parameter No. 6802 = 1)

T

If the tool life management B function is enabled (bit 4 (LFB) of parameter No. 6805 = 1)
Turret type (bit 3 (TCT) of parameter No. 5040 = 0)

- OPTION GROUP:
  No display.
- REST COUNT:
  Remaining set value used until a new tool is selected
  (if bit 3 (GRP) of parameter No. 6802 = 1)

ATC type (bit 3 (TCT) of parameter No. 5040 = 1)

- OPTION GROUP:
  Arbitrary group number
  (if bit 5 (TGN) of parameter No. 6802 = 1)
- REST COUNT:
  Remaining set value used until a new tool is selected
  (if bit 3 (GRP) of parameter No. 6802 = 1)

Setting data on the group editing screen

Tool life management data can be specified in the reset state (both the OP and RST signals are "0"). However, setting bit 1 (TCI) of parameter No. 6804 to "1" enables tool life management data to be specified even during automatic operation (the OP signal is "1").
The available editing operations are listed below.

M

Items that can be edited	Mode
Setting a life count type, life value, tool life counter, and tool data (T code, H code, and D code)	All modes
Adding tool numbers (T code)	MDI
Deleting all tool group data at a time	MDI
Deleting tool data (status, T code, H code, and D code)	MDI
Selecting tool skip	MDI
Specifying to clear tool data (life re-set)	MDI

T

Items that can be edited	Mode
Setting a life count type, life value, tool life counter, and tool data (T code and D code) However, the D code can be edited only if the ATC type is in use (bit 3 (TCT) of parameter No. 5040 = 1).	All modes
Adding tool numbers (T code)	MDI
Deleting all tool group data at a time	MDI
Deleting tool data (status, T code, and D code)	MDI
Selecting tool skip	MDI
Specifying to clear tool data (life re-set)	MDI

M
T

If no tool is registered with a tool group, none of a life count type, a tool life value, and a tool life counter value can be set for the tool group. First of all, add a tool number (T code).

NOTE

1 As for USING GROUP or NEXT GROUP editing:
   <1> During automatic operation (OP signal = “1” and bit 1 (TCI) of parameter No. 6804 = 1), only the tool life counter can be changed.
   <2> In the reset state (OP signal = “0” and RST signal “0”), the following editing operation stops life management because it cannot be continued.
      - Adding tool numbers (T code)
      - Deleting all tool group data at a time
      - Deleting tool data (status, T code, H code, and D code)
2 The following editing operations may set the tool change signal to ”1”.
   - Selecting tool skip for the last tool.
   - Deleting tool numbers, resulting in any tool other than those whose life has expired or who have been skipped being not found in the tool group of interest.
3 The following editing operations may reset the tool change signal to ”0”.
   - Adding tool numbers, leading to tools whose life has not expired being set in the tool group of interest.
   - Selecting tool clear.

Procedure

- Setting a life count type, tool life value, tool life counter, tool data, arbitrary group number, and remaining set value

Setting a life count type, tool life value, tool life counter, tool data, arbitrary group number, and remaining set value

Method 1

1 Place the cursor on the desired item.
2 Enter a value from the keypad.
3 Press soft key [INPUT].

Method 2

1 Place the cursor on the desired item.
2 Enter a value from the keypad.
3 Press

.

NOTE
1 Changing a tool life value or tool life counter does not affect the tool status or tool change signal.
2 Changing a life count type causes the tool life value and too life counter to be reset to 0.

M

If the tool life management B function is enabled (bit 4 (LFB) of parameter No. 6805 = 1) and the following parameters are enabled, arbitrary groups and remaining set values can be specified

OPTION GROUP:
Arbitrary group number
(if bit 5 (TGN) of parameter No. 6802 = 1)

REST COUNT:
Remaining set value used until a new tool is selected
(if bit 3 (GRP) of parameter No. 6802 = 1)

T

If the tool life management B function is enabled (bit 4 (LFB) of parameter No. 6805 = 1)

Turret type (bit 3 (TCT) of parameter No. 5040 = 0)

- OPTION GROUP:
  Cannot be set.
- REST COUNT:
  It is possible to set a remaining set value to be used until a new tool is selected (if bit 3 (GRP) of parameter No. 6802 = 1).

ATC type (bit 3 (TCT) of parameter No. 5040 = 1)

- OPTION GROUP:
  Arbitrary group numbers can be set (if bit 5 (TGN) of parameter No. 6802 = 1).
- REST COUNT:
  It is possible to set a remaining set value to be used until a new tool is selected (if bit 3 (GRP) of parameter No. 6802 = 1).

- Adding tool numbers

Tool numbers can be added to a tool group as follows:

1 Select the MDI mode.
2 Place the cursor on the tool data (T code, H code, or D code) just before a tool number to be added.
3 Enter the tool number from the key pad.
4 Press soft key [INSERT].

(Example)

Adding tool number 1550 between numbers 1 and 2 (for the M series)

1 Move the cursor to the data for number 1, enter "1550", and press [INSERT].

NO.	STATE	T-CODE	H-CODE	D-CODE
01		00010012	000	000
02		00010001	000	000
03		00010114	000	000

2 The entered T code 1550 is inserted in the position of number 2. The H and D codes are reset to 0.

NO.	STATE	T-CODE	H-CODE	D-CODE
01		00010012	000	000
02		00001550	000	000
03		00010001	000	000

- Deleting all tool group data at a time

All tool group data can be deleted at a time as follows:

1 Select the MDI mode.
2 Place the cursor on the tool data (T code, H code, or D code) just Select the tool group from which all tool data you want to delete at a time.
3 Press soft key [DELETE].
4 Press soft key [GROUP].
5 Press soft key [EXEC].

- Deleting tool data

Tool data can be deleted from a tool group as follows:

1 Select the MDI mode.
2 Place the cursor on the tool data (T code, H code, or D code) you want to delete.
3 Press soft key [DELETE].
4 Press soft key [<CURS>].

NOTE
1 Deleting all tools from a tool group is equivalent to deleting the tool group itself.
2 Deleting a tool indicated with @ (in use) results in @ being moved to the previous tool whose life has expired or which has been skipped.

- Selecting tool skip

Tool data can be placed in a skip state as follows:

1 Select the MDI mode.
2 Place the cursor on the tool data (T code, H code, or D code) for a tool you want to skip.
3 Press soft key [STATE].
4 Press soft key [SKIP].

- Specifying to clear tool data (life re-set)

Tool data state can be cleared as follows:

1 Select the MDI mode.
2 Place the cursor on the tool data (T code, H code, or D code) for a tool you want to clear.
3 Press soft key [STATE].
4 Press soft key [CLEAR].

- Selecting a tool group

A tool group can be selected as follows:

Method 1

1 Enter a tool group number from the keypad.
2 Press soft key [NO.SRH].

NOTE
If arbitrary group numbers are enabled, a tool group is selected by searching for an arbitrary group number rather than the tool group number.

Method 2

1 Press page key

or

to display the target tool group.

- Switching to the list screen

The tool life management (list screen) can be resumed as follows:

1 Press soft key [END].

12.3.14 Displaying and Setting Workpiece Setting Error Compensation Data

An amount of error used in workpiece setting error compensation can be set on the workpiece setting error screen.
The workpiece setting error screen is displayed as follows:

1 Press function key .

2 Press chapter selection soft key [WORD SET ER].
The workpiece setting error screen appears.

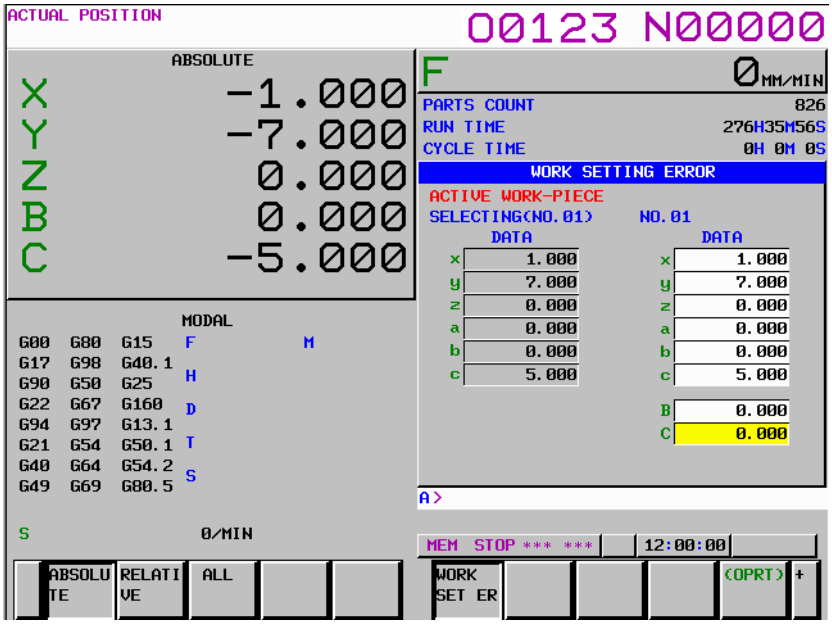


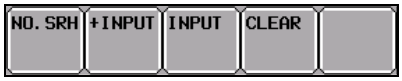
Fig. 12.3.14 (a) Workpiece setting error screen (10.4-inch)

Up to 7 sets (No. 01 to No. 07) of different amounts of workpiece setting error can be specified. In addition, it is possible to specify an amount of error (No. 00 (COMMON)) to be added to an error in the 7 sets in common.
The X-direction error Δx, Y-direction error Δy, and Z-direction error Δz are specified, respectively, in the x, y, and z fields. The rotational errors Δa, Δb, and Δc are specified, respectively, in the a, b, and c fields.
Table rotation axis positions 1 and 2 are specified in the fields under the a, b, and c fields (C and B in the figure).
Rotation axis names specified in parameter Nos. 19681 and 19686 are displayed in the axis name fields.
Table rotation axis positions 1 and 2 are specified for rotation axes. No position must be specified for axes (including virtual axes) other than table rotation axes. So, no such item is displayed for setting.

Table rotation axis position setting items are displayed under the following conditions.

- It must have been correctly specified which axis is a table rotation axis.
  If parameter No. 19680 = 12 (table rotation type), axis numbers must have been specified correctly in parameter Nos. 19681 and 19686.
  If parameter No. 19680 = 21 (mixed type), an axis number must have been specified correctly in parameter No. 19686.
- Bit 0 of parameter No. 1006 must be 1 (rotation axis) for the axes specified above.

The following soft key can be used in data setting.



- After a workpiece setting error number you want to display is entered, pressing [NO.SRH] causes a setting screen for the target workpiece setting error to appear.
- After a number is entered, pressing soft key [+INPUT] causes the number to be added.
- After a number is entered, pressing soft key [INPUT] causes the number to be set.
- Pressing soft key [ERASE] causes the next soft key to appear.

- After a workpiece setting error number you want to delete is entered, pressing soft key [NO.] causes the data corresponding to the error number to be deleted.
- Pressing soft key [ALL] causes the data corresponding to all workpiece setting error numbers to be deleted.

12.3.15 Displaying and Setting Pattern Data Inputs

Described below are a method for displaying machining menus (pattern menus) created by machine tool builders and a method for setting them. The descriptions are based on an example. For actual pattern menus and pattern data, refer to manuals released by respective machine tool builders.

Displaying pattern data and pattern menus

Given below is a procedure for displaying pattern menus.

1 Press function key .

2 Press the continuous menu key .

3 Press soft key [PATTERN MENU].
The pattern menu screen shown below appears.

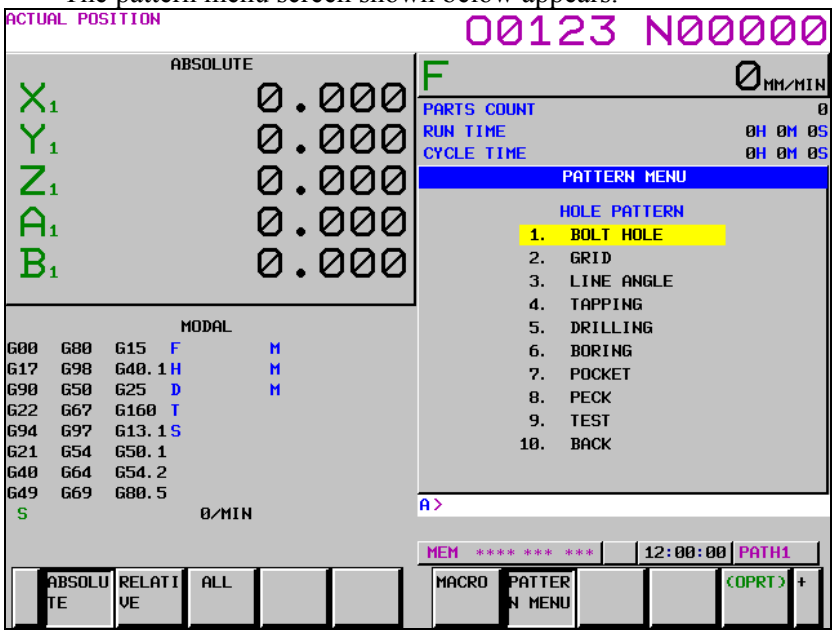


Fig. 12.3.15 (a) Pattern menu screen (10.4-inch)

On this screen, a pattern to be used can be selected.
The following two methods can be used to select patterns.

- Using the cursor
  Move the cursor to a pattern name you want to select, using cursor key ↑ or ↓, and then press soft key [SELECT] or [INPUT].
- Specifying a pattern number
  Enter a number displayed at the left side of a pattern name, and press soft key [SELECT] or [INPUT].

The custom macro screen (pattern data screen) shown below appears.

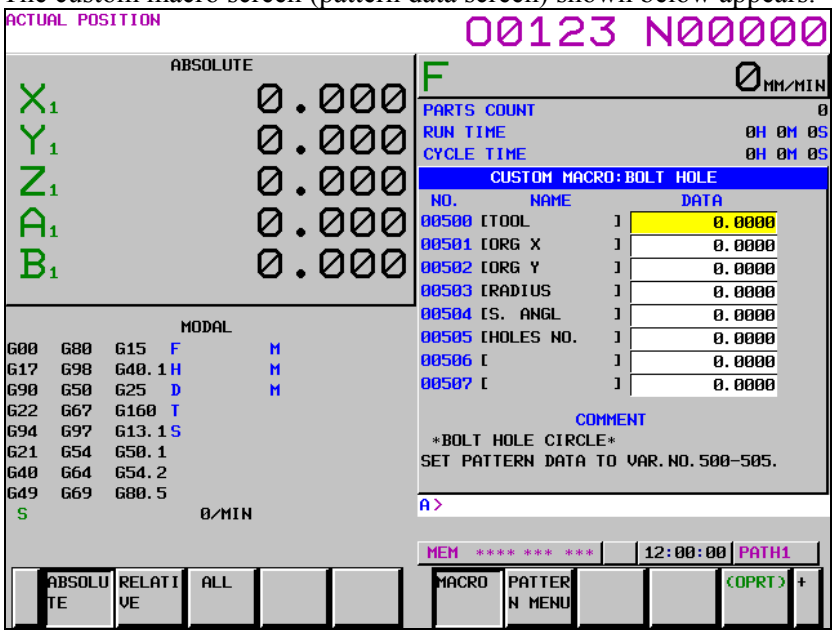


Fig. 12.3.15 (b) Custom macro screen (pattern data) (10.4-inch)

Enter the necessary pattern data, and press [INPUT].

After entering all necessary data, select the MEMORY mode and press the cycle start button. Machining begins.

Explanation

- Explanations about the pattern menu screen

HOLE PATTERN

An arbitrary character string consisting of 12 or less characters can be displayed as a menu title.

BOLT HOLE

An arbitrary character string consisting of 10 or less characters can be displayed as a pattern name.

Machine tool builders should program menu title and pattern name character strings, using custom macros, and save them in program memory.

- Explanations about the custom macro screen (pattern data screen)

BOLT HOLE

An arbitrary character string consisting of 12 or less characters can be displayed as a pattern data title.

TOOL.

An arbitrary character string consisting of 10 or less characters can be displayed as a variable name.
BOLT HOLE
The comment text consists of nine 12-character blocks, or the comment can be up to 12 lines (10.4-inch display) or 8 lines (7.2 or 8.4-inch display) with one block counted as one line.

Machine tool builders should program variable name and comment text character strings, using custom macros, and save them in program memory.

Screens of a 15-inch display unit

12.3.16 Displaying and Entering Setting Data (15-inch Display Unit)

Data such as the TV check flag and punch code is set on the setting data screen. On this screen, the operator can also enable/disable parameter writing, enable/disable the automatic insertion of sequence numbers in program editing, and perform settings for the sequence number comparison and stop function.
See III-9.2 for automatic insertion of sequence numbers.
See III-12.3.17 for the sequence number comparison and stop function.
This subsection describes how to set data.

Procedure for setting the setting data

Procedure

1 Select the MDI mode.

2 Press function key

.

3 Press vertical soft key [SETTING] to display the setting data screen.
This screen consists of several pages.
Press page key

or

until the desired screen is displayed.
An example of the setting data screen is shown below.

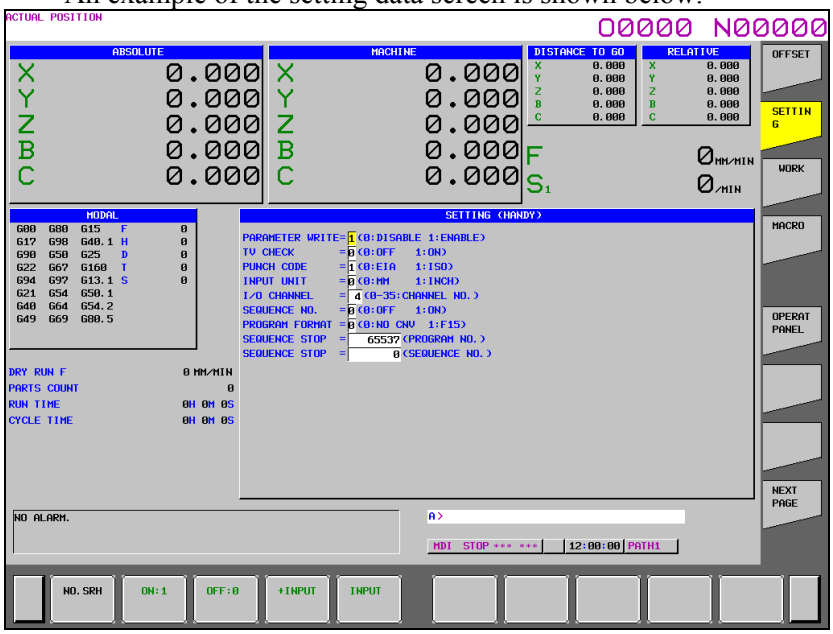


Fig. 12.3.16 (a) SETTING (HANDY) screen (15-inch)

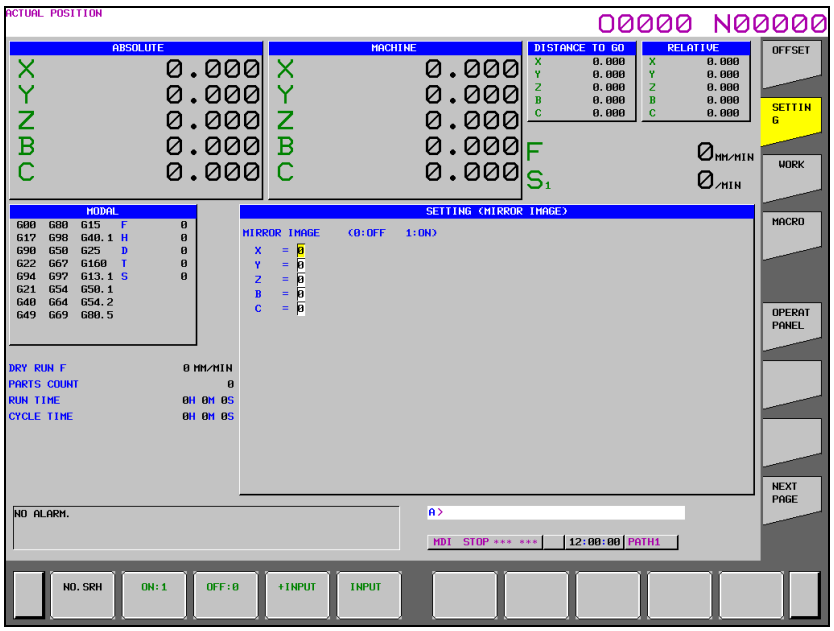
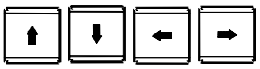


Fig. 12.3.16 (b) SETTING (MIRROR IMAGE) screen (15-inch)

4 Move the cursor to the item to be changed by pressing cursor keys  .

5 Enter a new value and press horizontal soft key [INPUT].

Explanation

- PARAMETER WRITE

Setting whether parameter writing is enabled or disabled.
0 : Disabled
1 : Enabled

- TV CHECK

Setting to perform TV check.
0 : No TV check
1 : Perform TV check

- PUNCH CODE

Setting code when data is output through reader/puncher interface.
0 : EIA code output
1 : ISO code output

- INPUT UNIT

Setting a program input unit, inch or metric system
0 : Metric
1 : Inch

- I/O CHANNEL

Using channel of reader/puncher interface.
0 : Channel 0
1 : Channel 1
2 : Channel 2

- SEQUENCE NO.

Setting of whether to perform automatic insertion of the sequence number or not at program edit in the EDIT mode.
0 : Does not perform automatic sequence number insertion.
1 : Perform automatic sequence number insertion.

- PROGRAM FORMAT

Setting of whether to use the Series 15 format.
0: Uses the standard format.
1: Uses the Series 15 format.
For the Series 15 format, refer to Chapter II-6 in the User's Manual (T series) or Chapter II-7 of Part II in the User's Manual (M series).

- SEQUENCE STOP

Setting the sequence number with which the operation stops for the sequence number comparison and stop function and the number of the program to which the sequence number belongs

- MIRROR IMAGE

Setting of mirror image ON/OFF for each axes.
0 : Mirror image off
1 : Mirror image on

- Others

Page key

or

can also be pressed to display the SETTING (TIMER) screen. See III-12.3.18 for this screen.

12.3.17 Sequence Number Comparison and Stop (15-inch Display Unit)

If a block containing a specified sequence number appears in the program being executed, operation enters single block mode after the block is executed.

Procedure for sequence number comparison and stop

Procedure

1 Select the MDI mode.

2 Press function key .

3 Press vertical soft key [SETTING].

4 Press page key or several times until the following screen is displayed.

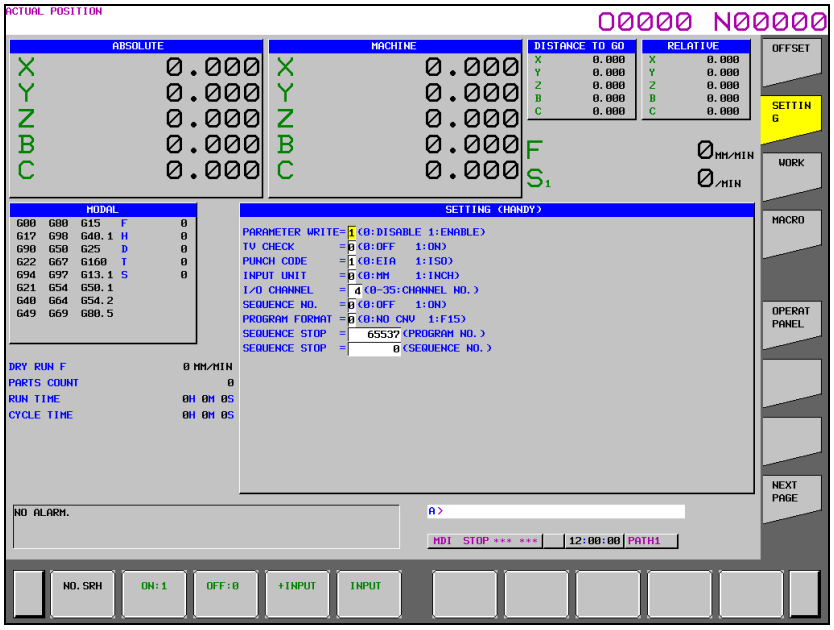


Fig. 12.3.17 (a) SETTING (HANDY) screen (15-inch)

5 Enter in (PROGRAM NO.) for SEQUENCE STOP the number (1 to 99999999) of the program containing the sequence number with which operation stops.

6 Enter in (SEQUENCE NO.) for SEQUENCE STOP (with eight or less digits) the sequence number with which operation is stopped.

7 When automatic operation is executed, operation enters single block mode at the block containing the sequence number which has been set.

Explanation

- Sequence number after the program is executed

After the specified sequence number is found during the execution of the program, the sequence number set for sequence number compensation and stop is decremented by one.

- Exceptional blocks

If the predetermined sequence number is found in a block in which all commands are those to be processed within the CNC control unit, the execution does not stop at that block.

```
[Example]  N1 #1=1 ;
           N2 IF[#1 EQ 1]GOTO 08 ;
           N3 GOTO 09 ;
           N4 M98 P1000 ;
           N5 M99 ;
```

In the example shown above, if the predetermined sequence number is found, the execution of the program does not stop.

- Stop in the canned cycle

If the predetermined sequence number is found in a block which has a canned cycle command, the execution of the program stops after the return operation is completed.

- When the same sequence number is found several times in the program

If the predetermined sequence number appears twice or more in a program, the execution of the program stops after the block in which the predetermined sequence number is found for the first time is executed.

- Block to be repeated a specified number of times

If the predetermined sequence number is found in a block which is to be executed repeatedly, the execution of the program stops after the block is executed specified times.

12.3.18 Displaying and Setting Run Time, Parts Count, and Time (15-inch Display Unit)

Various run times, the total number of machined parts, number of parts required, and number of machined parts can be displayed.
This data can be set by parameters or on this screen (except for the total number of machined parts and the time during which the power is on, which can be set only by parameters).
This screen can also display the clock time. The time can be set on the screen.

Procedure for Displaying and Setting Run Time, Parts Count and Time

Procedure

1 Select the MDI mode.
2 Press function key .
3 Press vertical soft key [SETTING].
4 Press page key or several times until the following screen is displayed.

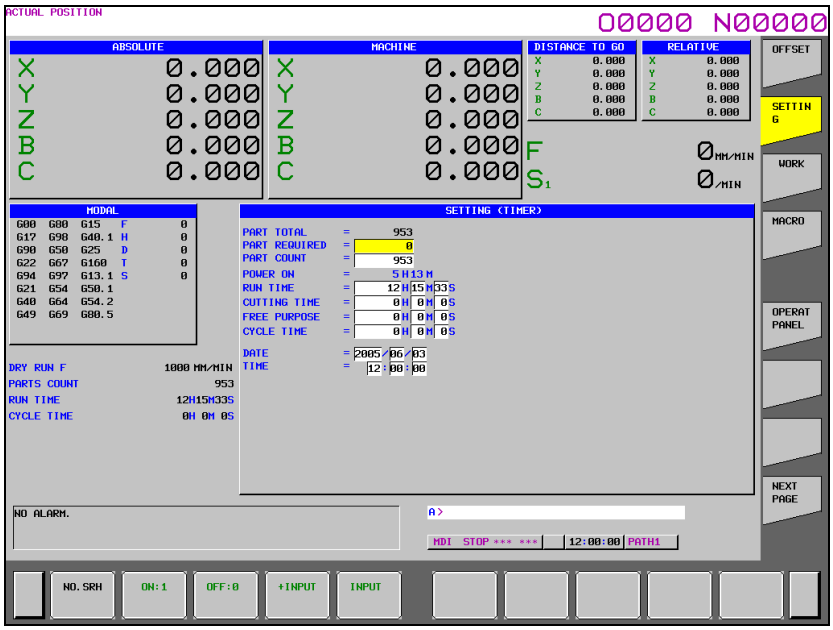


Fig. 12.3.18 (a) SETTING (TIMER) screen (15-inch)

5 To set the number of parts required, move the cursor to PARTS REQUIRED and enter the number of parts to be machined.
6 To set the clock, move the cursor to DATE or TIME, enter a new date or time, then press horizontal soft key [INPUT].

Explanation

- PARTS TOTAL

This value is incremented by one when M02, M30, or an M code specified by parameter No. 6710 is executed. This value cannot be set on this screen. Set the value in parameter No. 6712.

- PARTS REQUIRED

It is used for setting the number of machined parts required.
When the "0" is set to it, there is no limitation to the number of parts.
Also, its setting can be made by the parameter No 6713.

- PARTS COUNT

This value is incremented by one when M02, M30, or an M code specified by parameter No. 6710 is executed. The value can also be set by parameter No. 6711. In general, this value is reset when it reaches the number of parts required. Refer to the manual issued by the machine tool builder for details.

- POWER ON

Displays the total time which the power is on. This value cannot be set on this screen but can be preset in parameter No. 6750.

- RUN TIME

Indicates the total run time during automatic operation, excluding the stop and feed hold time.
This value can be preset in parameter No. 6751 or No. 6752.

- CUTTING TIME

Displays the total time taken by cutting that involves cutting feed such as linear interpolation (G01) and circular interpolation (G02 or G03). This value can be preset in parameter No. 6753 or No. 6754.

- FREE PURPOSE

This value can be used, for example, as the total time during which coolant flows. Refer to the manual issued by the machine tool builder for details.

- CYCLE TIME

Indicates the run time of one automatic operation, excluding the stop and feed hold time. This is automatically preset to 0 when a cycle start is performed at reset state. It is preset to 0 even when power is removed.

- DATA and TIME

Displays the current date and time. The date and time can be set on this screen.

- Usage

When the command of M02 or M30 is executed, the total number of machined parts and the number of machined parts are incremented by one. Therefore, create the program so that M02 or M30 is executed every time the processing of one part is completed. Furthermore, if an M code set to the parameter No. 6710 is executed, counting is made in the similar manner. Also, it is possible to disable counting even if M02 or M30 is executed (bit 0 (PCM) of parameter No. 6700 is set to 1). For details, see the manual issued by machine tool builders.

Limitation

- Run time and part count settings

Negative value cannot be set. Also, the setting of "M" and "S" of run time is valid from 0 to 59.
Negative value may not be set to the total number of machined parts.

- Time settings

Neither negative value nor the value exceeding the value in the following table can be set.

Table 12.3.18 (a)

Item	Maximum value	Item	Maximum value
Year	2096	Hour	23
Month	12	Minute	59
Day	31	Second	59

12.3.19 Displaying and Setting the Workpiece Origin Offset Value (15-inch Display Unit)

Displays the workpiece origin offset for each workpiece coordinate system (G54 to G59, G54.1 P1 to G54.1 P48 and G54.1 P1 to G54.1 P300) and external workpiece origin offset.
The workpiece origin offset value and external workpiece origin offset value can be set on this screen.

Procedure for displaying and setting the workpiece origin offset value

Procedure

1 Press function key .

2 Press vertical soft key [WORK].
The workpiece coordinate system setting screen is displayed.

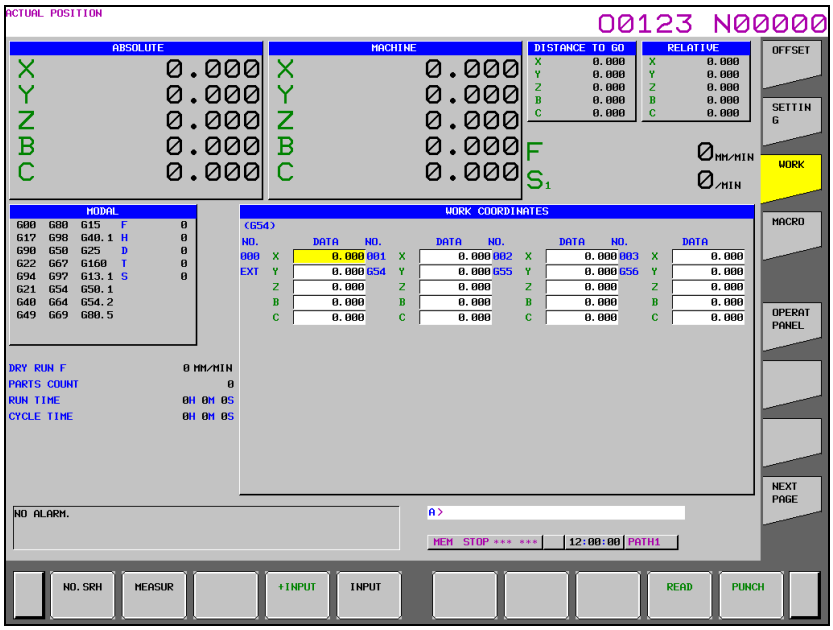


Fig. 12.3.19 (a) WORK COORDINATES screen (15-inch)

3 The screen for displaying the workpiece origin offset values consists of two or more pages.
Display a desired page in either of the following two ways:

- Press the page key or .
- Enter the workpiece coordinate system number (0 : external workpiece origin offset, 1 to 6: workpiece coordinate systems G54 to G59, P1 to P48 : workpiece coordinate systems G54.1 P1 to G54.1 P48, P1 to P300 : workpiece coordinate systems G54.1 P1 to G54.1 P300) and press horizontal soft key [NO.SRH].

4 Turn off the data protection key to enable writing.

5 Move the cursor to the workpiece origin offset to be changed.

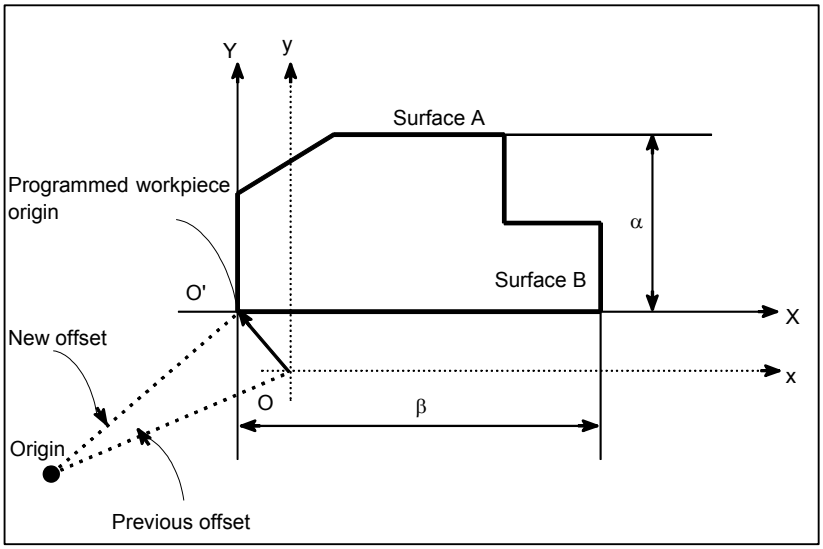
6 Enter a desired value by pressing numeric keys, then press horizontal soft key [INPUT]. The entered value is specified in the workpiece origin offset value. Or, by entering a desired value with numeric keys and pressing horizontal soft key [+INPUT], the entered value can be added to the previous offset value.
7 Repeat steps 5 and 6 to change other offset values.
8 Turn on the data protection key to disable writing.

12.3.20 Direct Input of Workpiece Origin Offset Value Measured (15-inch Display Unit)

This function is used to compensate for the difference between the programmed workpiece coordinate system and the actual workpiece coordinate system. The measured offset for the origin of the workpiece coordinate system can be input on the screen such that the command values match the actual dimensions.
Selecting the new coordinate system matches the programmed coordinate system with the actual coordinate system.

Procedure for direct input of workpiece origin offset value measured

Procedure



1 When the workpiece is shaped as shown above, position the reference tool manually until it touches surface A of the workpiece.
2 Retract the tool without changing the Y coordinate.
3 Measure distance α between surface A and the programmed origin of the workpiece coordinate system as shown above.
4 Press function key .

5 To display the WORK COORDINATES screen, press vertical soft key [WORK].

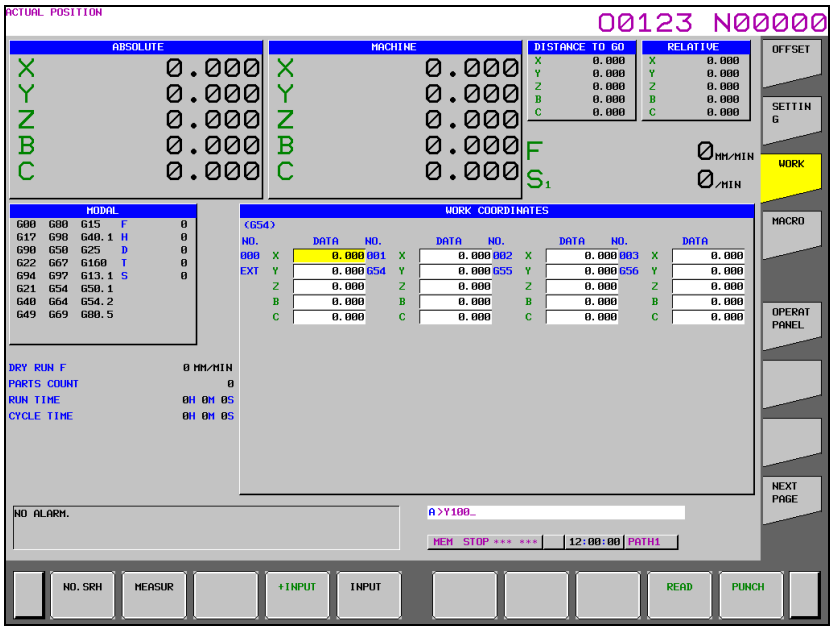


Fig. 12.3.20 (a) WORK COORDINATES screen (15-inch)

6 Position the cursor to the workpiece origin offset value to be set.
7 Press the address key for the axis along which the offset is to be set (Y-axis in this example).
8 Enter the measured value (α) then press horizontal soft key [MEASUR].
9 Move the reference tool manually until it touches surface B of the workpiece.
10 Retract the tool without changing the X coordinate.
11 Enter the measured value of distance (β) to surface B as an X value as with steps 7 and 8.

Limitation

- Consecutive input

Offsets for two or more axes cannot be input at the same time.

- During program execution

This function cannot be used while a program is being executed.

12.3.21 Displaying and Setting Custom Macro Common Variables (15-inch Display Unit)

Displays common variables (#100 to #149 or #100 to #199, and #500 to #531 or #500 to #999) on the screen.
The values for variables can be set on this screen.
Relative coordinates can also be set to variables.

Procedure for displaying and setting custom macro common variables

Procedure

1 Press function key .

2 Press vertical soft key [MACRO].
The following screen is displayed.

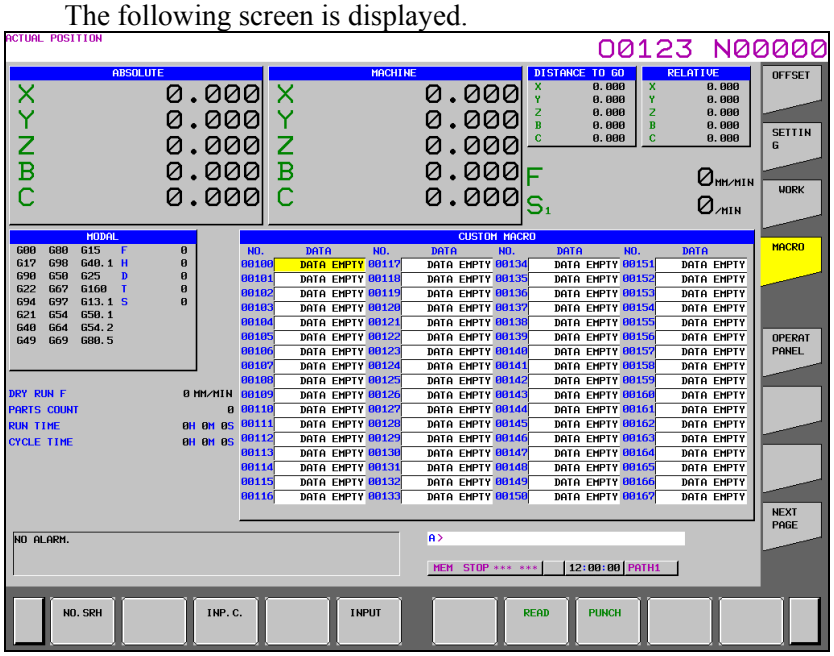


Fig. 12.3.21 (a) CUSTOM MACRO screen (15-inch)

3 Move the cursor to the variable number to set using either of the following methods:

- Enter the variable number and press horizontal soft key [NO.SRH].
- Move the cursor to the variable number to set by pressing page keys and/or and cursor keys , , , and/or .

4 Enter data with numeric keys and press horizontal soft key [INPUT].

5 To set a relative coordinate in a variable, press address key , , or , then press horizontal soft key [INP.C.].

6 To set a blank in a variable, just press horizontal soft key [INPUT]. The value field for the variable becomes blank.

Explanation

If the value of a variable produced by an operation is not displayable, an indication below is provided.
When the significant number of digits is 12 (with bit 0 (F16) of parameter No. 6008 set to 0):

Variable value range	Variable value indication
0 < Variable value < +0.00000000001	+Underflow
0 > Variable value > -0.00000000001	-Underflow
Variable value > 999999999999	+Overflow
Variable value < -999999999999	-Overflow

When the significant number of digits is 8 (with bit 0 (F16) of parameter No. 6008 set to 1):

Variable value range	Variable value indication
0 < Variable value < +0.0000001	+Underflow
0 > Variable value > -0.0000001	-Underflow
Variable value > 99999999	+Overflow
Variable value < -99999999	-Overflow

12.3.22 Displaying and Setting Real Time Custom Macro Data (15-inch Display Unit)

Real time macro variables (RTM variables) are dedicated to real time custom macros.
RTM variables are divided into temporary real time macro variables (temporary RTM variables) and permanent real time macro variables (permanent RTM variables).
The values of permanent RTM variables are kept stored after the power is turned off.
Temporary RTM variables are cleared to 0 when the power is turned off.

System variables (DI/DO variables) dedicated to real time custom macros are used to read and write PMC interface signals. Data can be read and written in bit and byte units. Before writing a signal, release the protection of the corresponding variable on the PMC signal protect screen.

Displaying and setting real time custom macro variables

Procedure

1 Press function key SYSTEM.

2 Press vertical soft key [NEXT PAGE] several times and then vertical soft key [R.TIME MACRO]. The screen shown below appears.

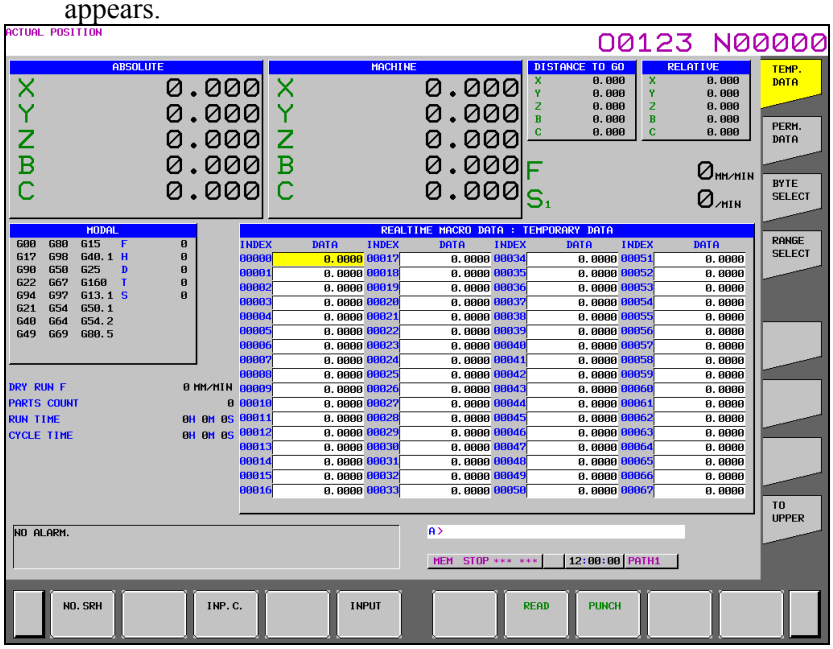


Fig. 12.3.22 (a) REAL TIME MACRO screen (15-inch)

3 To display or set real time custom macro variables of which values are not stored at power-off, press vertical soft key [TEMP. DATA].

4 To display or set real time custom macro variables of which values are stored at power-off, press vertical soft key [PERM. DATA].

5 Move the cursor to the number of a real time custom macro variable you want to set using either of the following methods:

- Enter the number of a real time custom macro variable and press horizontal soft soft key [NO. SRH].
- Move the cursor to the number of a real time custom macro variable you want to set by pressing page keys

and/or

and cursor keys , , , and/or .

6 Enter data.

Displaying and setting DI/DO variables

Procedure

For setting in byte units:

1 Press function key

.

2 Press vertical soft key [NEXT PAGE] several times and then vertical soft key [R.TIME MACRO]. The screen shown below appears.

3 Press vertical soft key [BYTE SELECT]. The following screen appears:

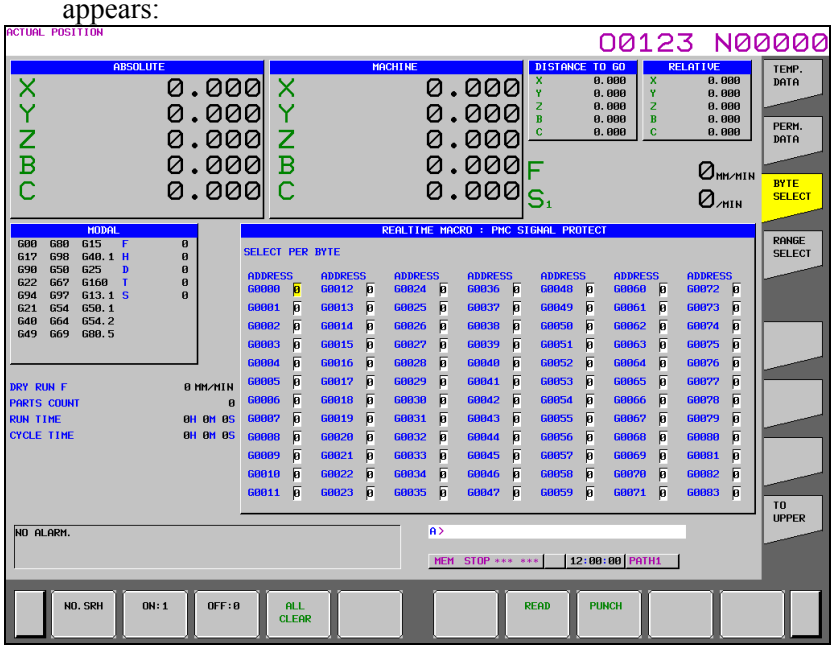


Fig. 12.3.22 (b) BYTE SELECT screen (15-inch)

4 Move the cursor to the number of a DI/DO variable you want to set using either of the following methods:
- Enter the number and press horizontal soft key [NO. SRH].
- Move the cursor to a desired number by pressing page keys

and/or

and cursor keys , , , and/or .

5 Enter data.

12.3.23 Displaying and Setting the Software Operator's Panel (15-inch Display Unit)

Operations on the MDI panel can substitute for the functions of switches on the machine operator's panel. This means that a mode selection, jog feed override selection, and so forth can be made with operations on the MDI panel, eliminating the need to use the corresponding switches on the machine operator's panel.
Jog feed can be performed using numeric keys.

Procedure for displaying and setting the software operator's panel

Procedure

1 Press function key

.

2 Press vertical soft key [NEXT PAGE] and then vertical soft key [OPERAT PANEL].

3 The screen consists of several pages. Press page key

or

until the desired screen is displayed.

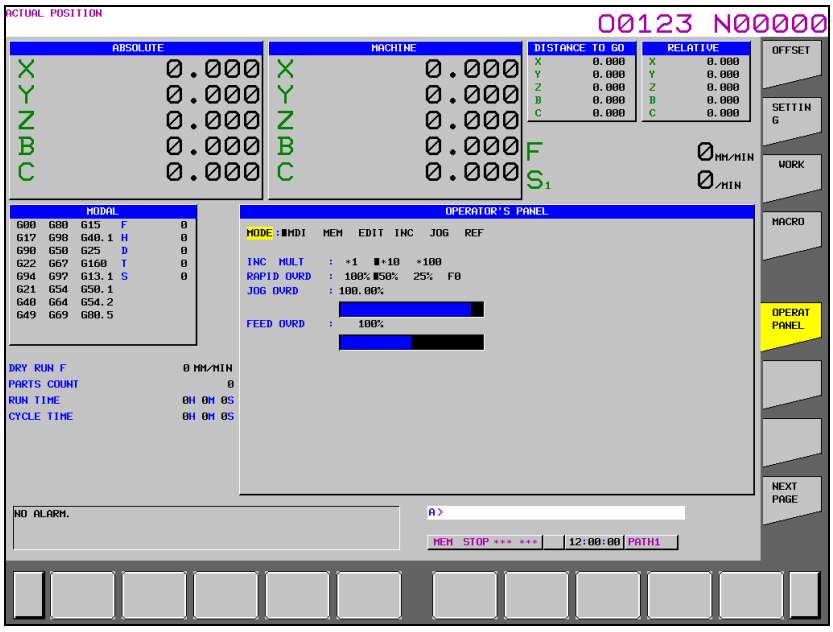


Fig. 12.3.23 (a) Without the manual handle feed function (15-inch)

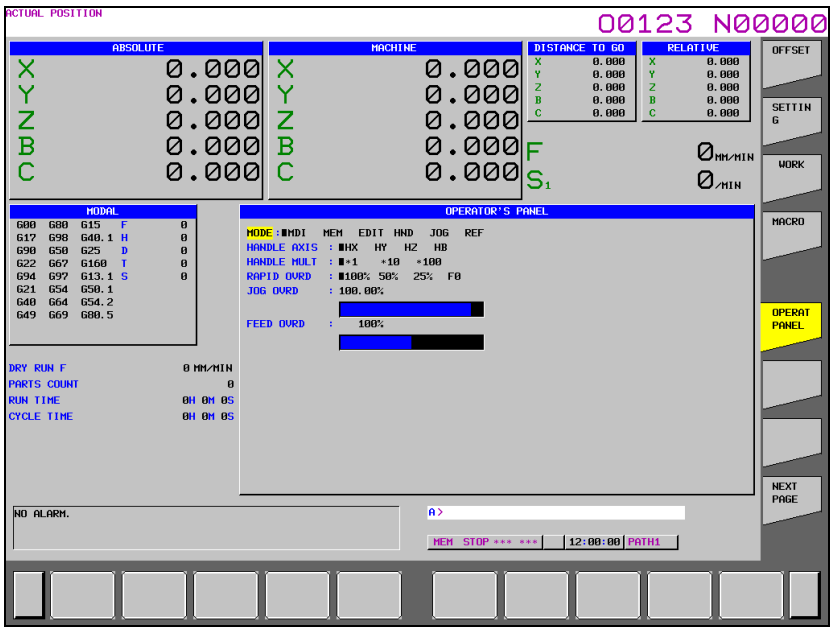


Fig. 12.3.23 (b) With the manual handle feed function (15-inch)

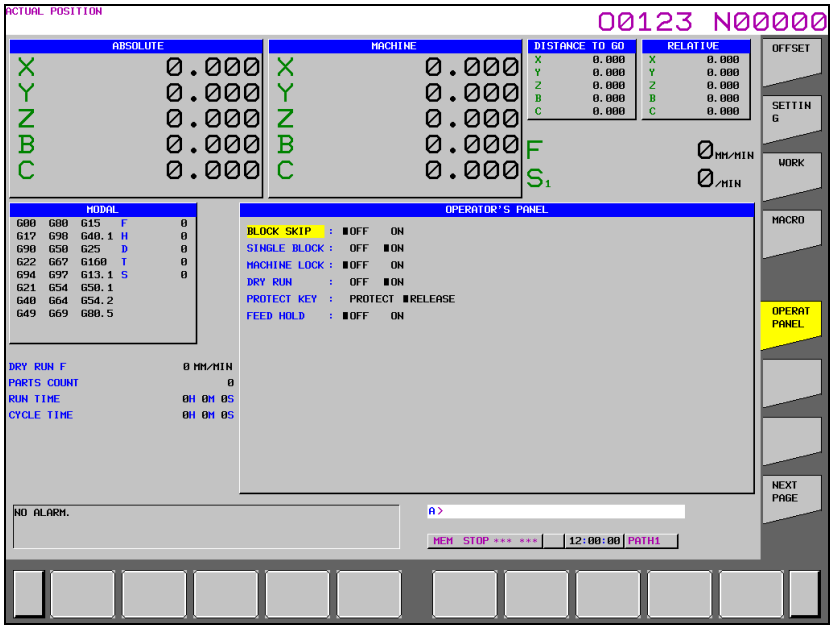


Fig. 12.3.23 (c) (15-inch)

4 Move the cursor to the desired switch by pressing cursor key [↑] or [↓].

5 Push the cursor key [←] or [→] to match the mark ■ to an arbitrary position and set the desired condition.

6 Press one of the following arrow keys to perform jog feed. Press the [J 5] key together with an arrow key to perform jog rapid traverse.

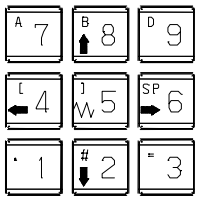


Fig. 12.3.23 (d) MDI arrow keys

Explanation

- Valid operations

The valid operations on the software operator's panel are shown below. Whether to use each group can be chosen using parameter No. 7200. Those groups that are not used are not displayed on the software operator's panel.

Group1 : Mode selection
Group2 : Selection of jog feed axis, Manual rapid traverse
Group3 : Selection of manual pulse generator feed axis, selection of manual pulse magnification
Group4 : Jog federate, federate override, rapid traverse override
Group5 : Optional block skip, single block, machine lock, dry run
Group6 : Protect key
Group7 : Feed hold

- Screens on which jog feed is valid

When the LCD indicates other than the software operator's panel screen and self-diagnosis screen, jog feed is not conducted even if the arrow key is pushed.

- Jog feed and arrow keys

The feed axis and direction corresponding to the arrow keys can be set with parameters (Nos. 7210 to 7217).

- General purpose switches

For the meanings of these switches, refer to the manual issued by machine tool builder.

12.3.24 Setting and Displaying Tool Management Data (15-inch Display Unit)

The tool management function totally manages tool information including tool offsets and tool life information.
This function provides a magazine screen and tool management screen. This subsection explains how to set these screens.

12.3.24.1 Displaying and setting magazine screen (15-inch display unit)

Procedure

1 Press function key .

2 Press vertical soft key [NEXT PAGE] several times and then vertical soft key [TOOL MANAGER]. The screen shown below appears.

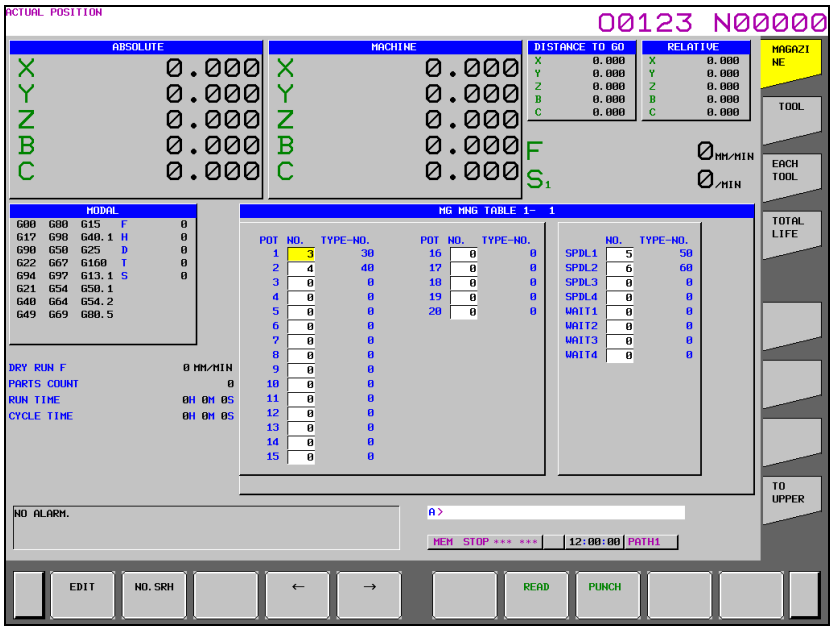


Fig. 12.3.24.1 (a)) Tool management magazine screen (15-inch)

3 By using the page keys, cursor keys, and horizontal soft keys [←] and [→], move the cursor to the pot No. position of the magazine for which you want to set or modify data.
Alternatively, type the number of the tool management data you wan to set or modify, and press horizontal soft key [NO. SRH].

4 Press horizontal soft key [EDIT].

5 To set the tool management data number of a pot, type the tool management data number, then press MDI key .

To delete the tool management data number set for a pot, follow the steps below.

<1> Press horizontal soft key [ERASE].
<2> Press horizontal soft key [CURDAT]. To delete the tool management data numbers registered for all pots, press horizontal soft key [ALL].
<3> Press horizontal soft key [EXEC]. Alternatively, type 0.

6 To end the edit operation, press horizontal soft key [EXIT].
This returns the screen display to the conventional tool management screen.

Explanation

- Another method

Magazine data can be input/output also by using external I/O devices. See III-8, "DATA I/O".

- Displayed data

Pot : As many pots as specified in parameter No. 13222, 13227, 13232, or 13237 (settable for each magazine) are displayed. The starting pot number can be set for each magazine in parameter No. 13223, 13228, 13233, or 13238.

NO. : Tool management data numbers are displayed.
In the following cases, tools cannot be set in magazines:
1. A tool management data number beyond the range from 0 to (the number of valid pairs in parameter No. 13220) is set.
2. Tool management data is invalid (bit 1 of tool information is 0).
3. The tool management data number to be set is already set for another pot.

Type No. : Tool type numbers corresponding to tool management data numbers are displayed.
Values cannot be set on this screen.

Spindle : The tool management data numbers and tool type numbers of spindle positions are displayed.

Wait : The tool management data numbers and tool type numbers of wait positions are displayed.

- Edit mode

To edit data, press horizontal soft key [EDIT] to enter tool management data edit mode.

- Tool management data number

Tool management data numbers can optionally be extended to 64, 240, or 1000 pairs. In addition, the number of tools used can be set in parameter No. 13220.

- Display of the spindle positions and wait positions

The spindle position and wait position can be set for each path.

12.3.24.2 Displaying and setting tool management screen (15-inch display unit)

Procedure

1 Press function key .

2 Press vertical soft key [NEXT PAGE] several times and then vertical soft key [TOOL MANAGER].

3 Press vertical soft key [TOOL].

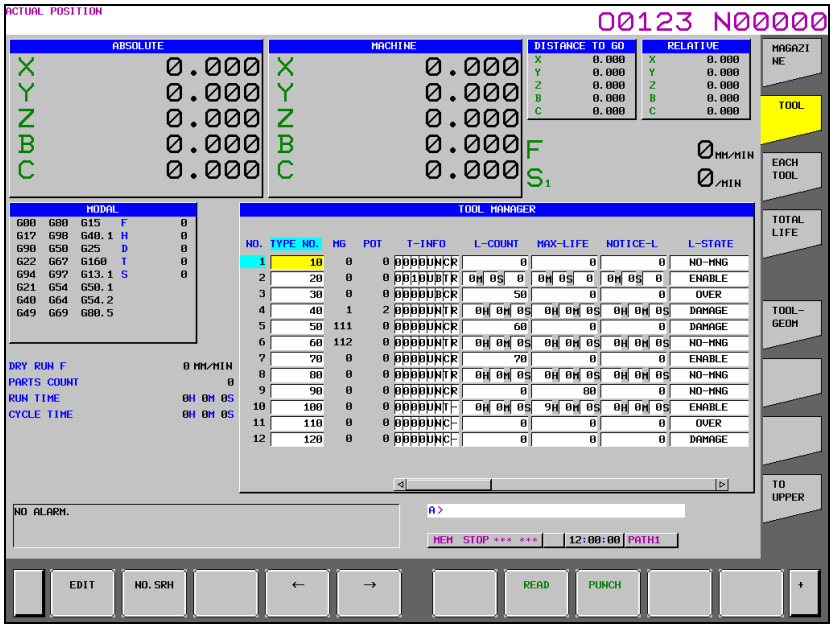


Fig. 12.3.24.2 (a) Tool management data screen (15-inch)

4 By using the page keys, cursor keys, and horizontal soft keys [←] and [→], move the cursor to the position of the tool information of the tool number for which you want to set or modify data.
Alternatively, type the tool number of the data you want to set or modify, and press horizontal soft key [NO. SRH].

5 Press horizontal soft key [EDIT].

6 To set tool data, type a desired value, then press MDI key .

To delete tool information, follow the steps below.

<1> In step 4, move the cursor to the tool information you want to erase.

<2> Press horizontal soft key [ERASE].

<3> Press horizontal soft key [CURDAT]. To delete entire tool information, press horizontal soft key [ALL].

<4> Press horizontal soft key [EXEC].

7 To end the edit operation, press horizontal soft key [EXIT].
This returns the screen display to the conventional tool management screen.

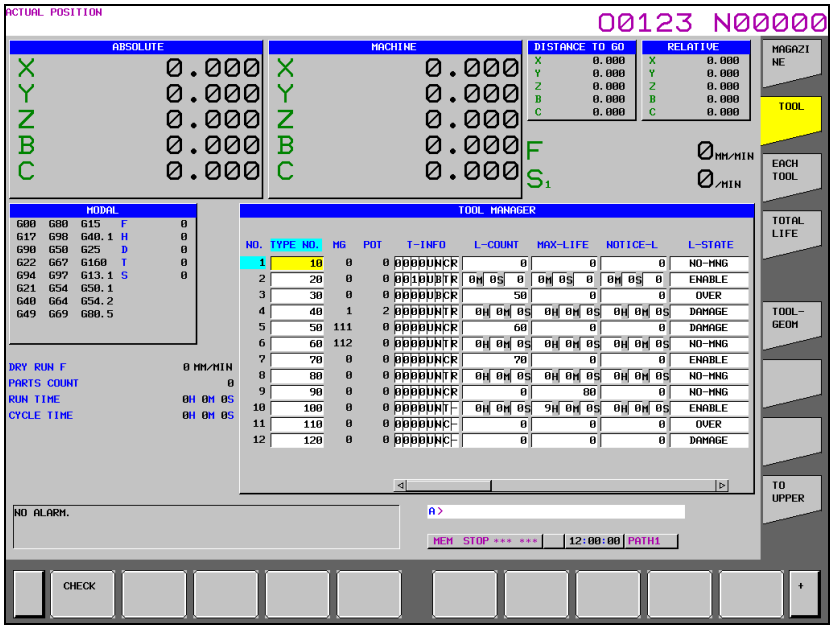


Fig. 12.3.24.2 (b) Tool management data screen (check function) (15-inch)

8 When horizontal soft key [CHECK] is pressed, if there are tools with the same number but with different count types (count and time), the cursor moves to the tool type number of the smallest tool management number in the tool type numbers and a warning message appears.
The warning messages issued by the check function are shown below.

<1> For the same type of tools with different count types
L-COUNT TYPE MISMATCH:XXXXXXXX

<2> For the same type of tools with the same count type
L-COUNT TYPE MATCH

Explanation

- Another method

Tool management data, customize data, and names set for tool states can be input/output also by using external I/O devices.
See III-8, "DATA I/O".

- Displayed information

- Life information

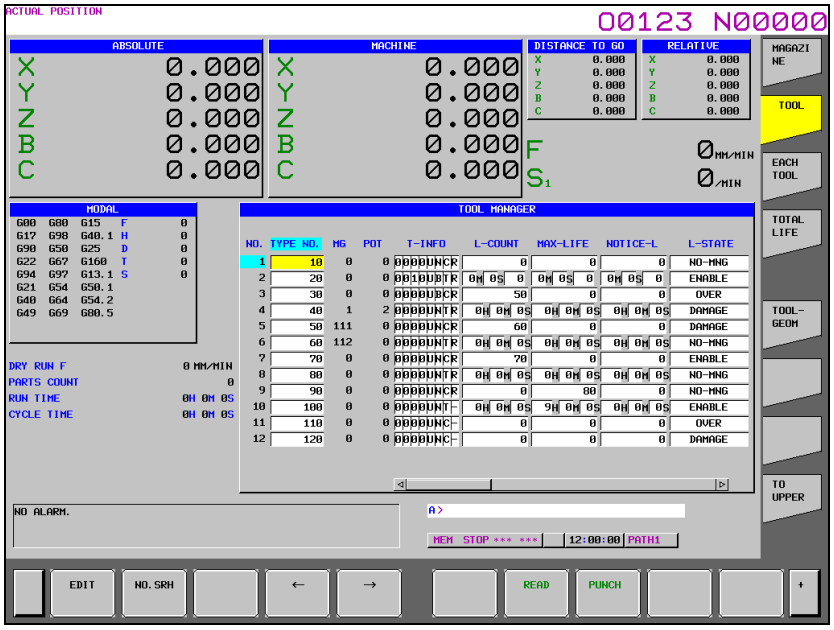


Fig. 12.3.24.2 (c) Tool management data life status screen(15-inch)

NO. : Tool management data numbers are displayed. These numbers can be displayed but cannot be set. The tool management data number of edited data is kept blinking until tool management data edit mode ends. Tool management data numbers set or deleted on the magazine screen also blink on the tool management screen.

Type No. : Tool type numbers are displayed.
A value from 0 to 99,999,999 can be set.

MG : The magazine number assigned to each tool is indicated. These numbers can be displayed but cannot be set.

Pot : The pot number assigned to each tool is indicated. These numbers can be displayed but cannot be set.

Tool information: The following five types of information are displayed sequentially from the right:

- Tool management data: Valid (R)/invalid (-))
- Life count type: Time (T)/count (C)
- Tool type: Oversize tool (B)/ normal tool (N)
- Data access: Locked (L)/unlocked (U)
- When a tool is not under life management: Included in the tools to be searched (1)/not included in the tools to be searched (0)

NOTE
1 The tool types and data access information vary depending on the specifications defined by the machine tool builder.
2 The same type of tools must have the same life count type.

Life counter: The number of use times/use period of time of each tool is indicated.
Up to 99,999,999 times or 999 hours 59 minutes 59 seconds can be set.

Maximum life: The maximum life value of each tool is indicated.
Up to 99,999,999 times or 999 hours 59 minutes 59 seconds can be set.

Noticed life: Noticed life value of a tool
Up to 99,999,999 times or 999 hours 59 minutes 59 seconds can be set.

Life state: Current tool state
One of the four states, including invalid (0), present (1, 2), not present (3), and broken (4), is indicated. The numbers in parentheses are data values used when these states are input in MDI.

- Spindle speed/feedrate

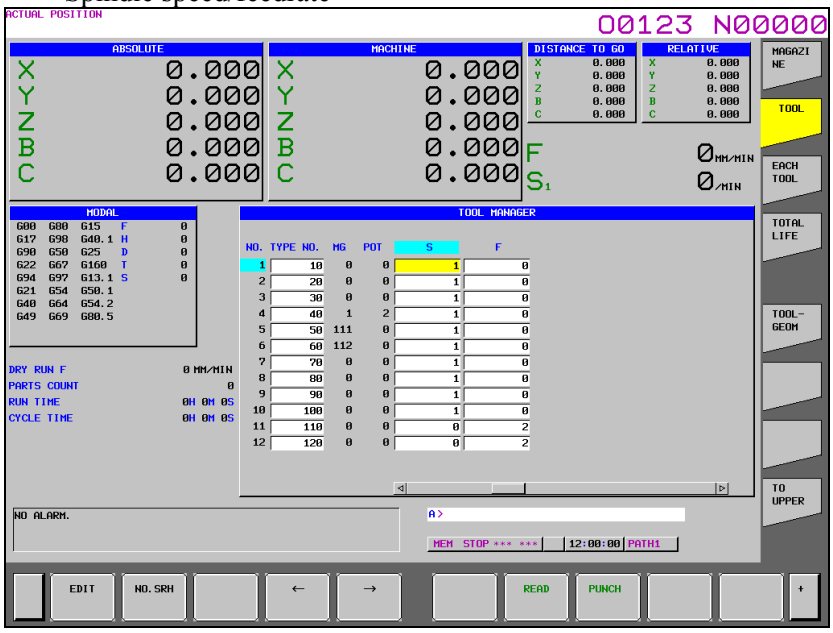


Fig. 12.3.24.2 (d) Tool management data spindle speed/feedrate screen (15-inch)

S : Spindle speed.
A value from 0 to 99,999 can be set.

F : Feedrate.
A value from 0 to 99,999,999 can be set.

- Tool offset information

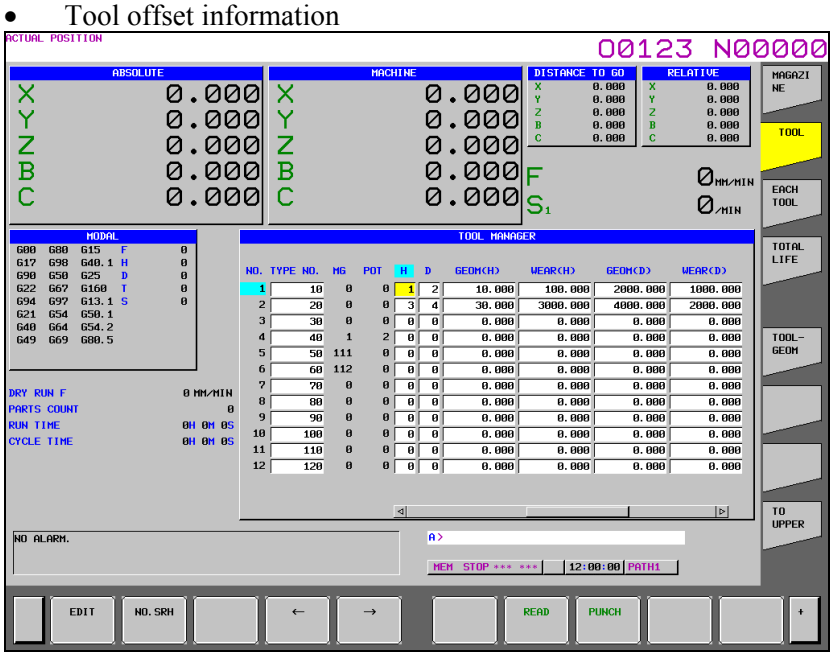


Fig. 12.3.24.2 (e) Tool management data tool offset screen (15-inch)

H : Tool length compensation number (for machining center systems only).
A value from 0 to 999 can be set.

D : Cutter compensation number (for machining center systems only).
A value from 0 to 999 can be set.

TG : Tool geometry compensation number (for lathe systems only).
A value from 0 to 999 can be set.

TW : Tool wear compensation number (for lathe systems only).
A value from 0 to 999 can be set.

The displayed offset data is determined depending on the option configuration and parameter settings (bits 1(DCR), 2(DOY), 3(DOB), 4(DO2), 6(DOT), and 7(DOM) of parameter No. 13202).

- Customize information

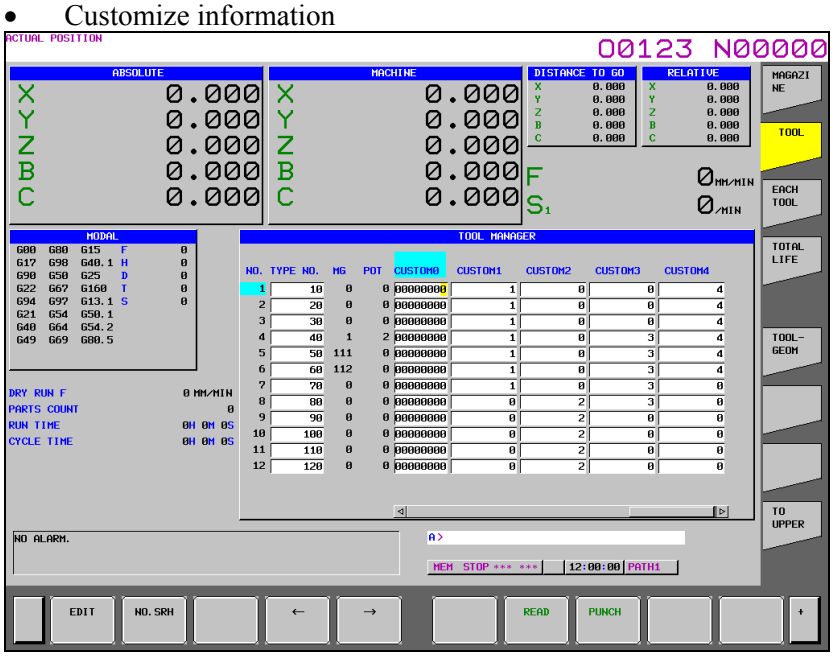


Fig. 12.3.24.2 (f) Tool management data customize data screen (15-inch)

Customize 0 : Bit-type customize information.
For each bit, 1 or 0 can be input.

Customize 1 to 4 : Customize information. Any value from -99,999,999 to 99,999,999 can be set.

Customize 5 to 20 : Customize information. These items are displayed only when customize data extension option (5 to 20) of the tool management function is enabled. Any value from -99,999,999 to 99,999,999 can be set.

Customize 21 to 40 : Customize information. These items are displayed only when customize data extension option (5 to 40) of the tool management function is enabled. Any value from -99,999,999 to 99,999,999 can be set.

- Tool management data number

Tool management data numbers can optionally be extended to 64, 240, or 1000 pairs. The number of tools used can be set in parameter No. 13220.

- Edit mode

To edit data, press horizontal soft key [EDIT] to enter tool management data edit mode.

- Life count override

When the period of time is set as the tool life count type, the life count can be overridden if bit 2 (LFV) of parameter No. 6801 is set to 1. Specify an override value by using a switch on the machine operator's panel.

Example) When cutting is performed for 10 minutes with an override of 0.1, one minute is counted in the tool life counter.

- Tool management extension function

When tool management extension functions are enabled, you can use the following functions in addition to the tool management functions:

- A value with a decimal point can be set as customize data.
  The maximum number of decimal places can be set to 7 by G10 data input and file read.
- Various types of tool management data can be protected using KEY signals.
  An attempt to enter a value to a protected data causes a warning message "WRITE PROTECT."
- Tool life count period selection
  Bit 5 of tool information is used to make a life count period selection.

Item		Description
Data length		1 byte (flag data)
#5	REV	0: A life count period of 1 sec is used. 1: A life count period of 8 msec is used.

Range of count is as follows.

1sec : 0 to 3,599,999 seconds (999 hours 59 minutes 59 seconds)
8msec : 0 to 3,599,992 ms (59 minutes 59 seconds 992 milliseconds)

12.3.24.3 Each tool data screen (15-inch display unit)

Each tool data screen

Procedure

1 Press function key .

2 Press vertical soft key [NEXT PAGE] several times and then vertical soft key [TOOL MANAGER].

3 Press vertical soft key [EACH TOOL]. The each tool data screen appears.

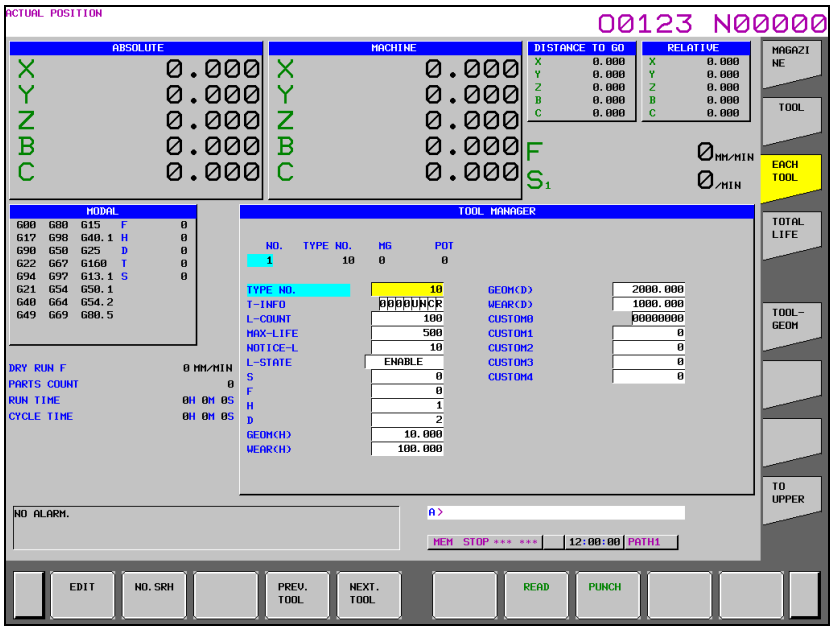


Fig. 12.3.24.3 (a) Each tool data screen (15-inch)

Explanation

- Header

The following four data items are displayed: NO., TYPE NO., MG, and POT.
When the data table of a tool extends over two or more pages, the same header is displayed on these pages.

- Data table

The data table shows data items related to a tool at a time.
The data items are displayed from the upper left to the lower left, the upper right, and the lower right in ascending order of screen display position numbers specified using the customize function. The number of digits displayed for one data item is fixed to 11.
Up to 24 data items are displayed on one page. (12 data items × 2 columns)
When more than 24 data items are set for a tool, the 25th and subsequent data items are displayed on the next page. (Up to three pages)

When a data item is set as a screen element of the tool management data screen twice or more using the tool management data display customize function (one of the tool management extension functions), only the data item with the smaller display position number is displayed. When no value is set for a data item, the field for the data item is not displayed and the next data item is displayed.

Key operations

- MDI key operations

- Numeral keys Inputs a numeric value.
-

Displays the previous page for the same tool.
-

Displays the next page for the same tool.
-
Moves the cursor to the previous data item.

  When the cursor is on the top row of the data table, it moves to the bottom row of the left column. When the cursor is on the first data item, it moves to the last data item.
-
Moves the cursor to the next data item.

  When the cursor is on the bottom row of the data table, it moves to the top row of the right column. When the cursor is on the last data item, it moves to the first data item.
-
Moves the cursor left on the screen.

  When the cursor is on the left column of the data table, it moves to the right column on the row immediately above. When the cursor is on the first data item, it moves to the last data item.
-
Moves the cursor right on the screen.

  When the cursor is on the right column of the data table, it moves to the left column on the row immediately below. When the cursor is on the last data item, it moves to the first data item.

- Soft key operations

Horizontal soft key [EDIT]
Enters the management data edit mode.
Horizontal soft key [NO.SRH]
Displays data with a tool management data number when pressed after the tool management data number is entered.
Horizontal soft key [PREV.TOOL]
Returns to the previous tool management data number.
Horizontal soft key [NEXT.TOOL]
Proceeds to the next tool management number.
Horizontal soft key [READ]
Reads data related to the tool management function.
Can be used only in the standard mode.
Requires placing the NC in the EDIT mode.

Horizontal soft key [PUNCH]
Writes data related to the tool management function.
Can be used only in the standard mode.
Requires placing the NC in the EDIT mode.

Operation in the management data edit mode

To edit data, press horizontal soft key [EDIT] to enter the management data edit mode.

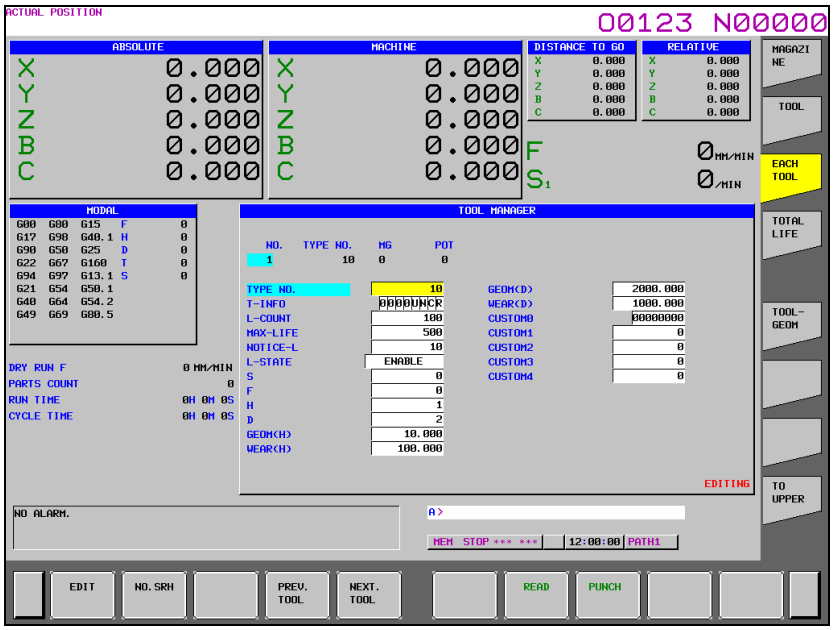


Fig. 12.3.24.3 (b) Each tool data editing screen (15-inch)

In the management data edit mode, “EDITING” is displayed at the lower right of the screen.
In the management data edit mode, the following key operations, in addition to those mentioned above except [READ] and [PUNCH], can be performed.

-

  Actually writes a value entered using numeric keys.
- Horizontal soft key [END]
  Ends the management data edit mode.

Operations for editing tool information are the same as those on the tool management data screen.

12.3.24.4 Displaying the total life of tools of the same type (15-inch display unit)

Total life data screen

Procedure

1 Press function key .

2 Press vertical soft key [NEXT PAGE] several times and then vertical soft key [TOOL MANAGER].

3 Press vertical soft key [TOTAL LIFE]. The total life data screen appears.

4 Using horizontal soft key [CHANGE] can switch total tool life data displays between specification by count and specification by duration.

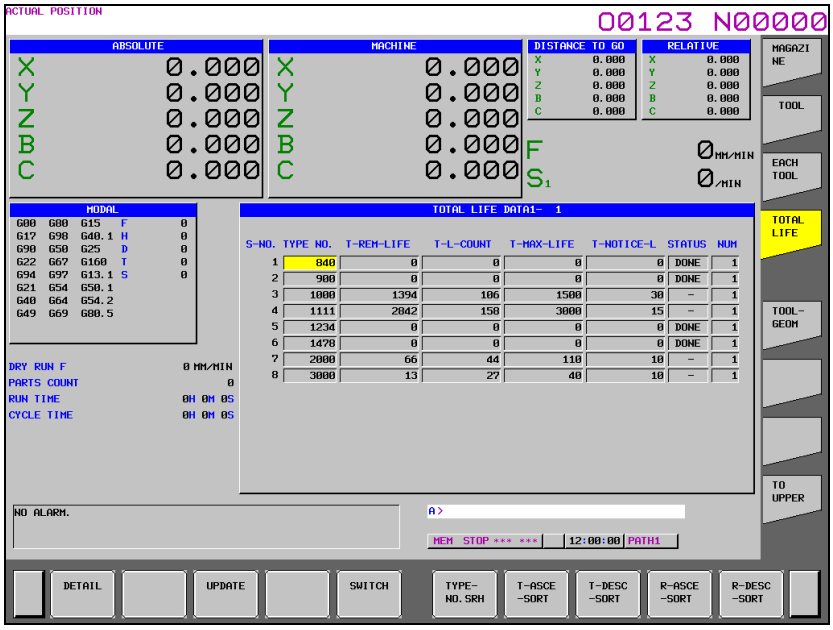


Fig. 12.3.24.4 (a) Count display (15-inch)

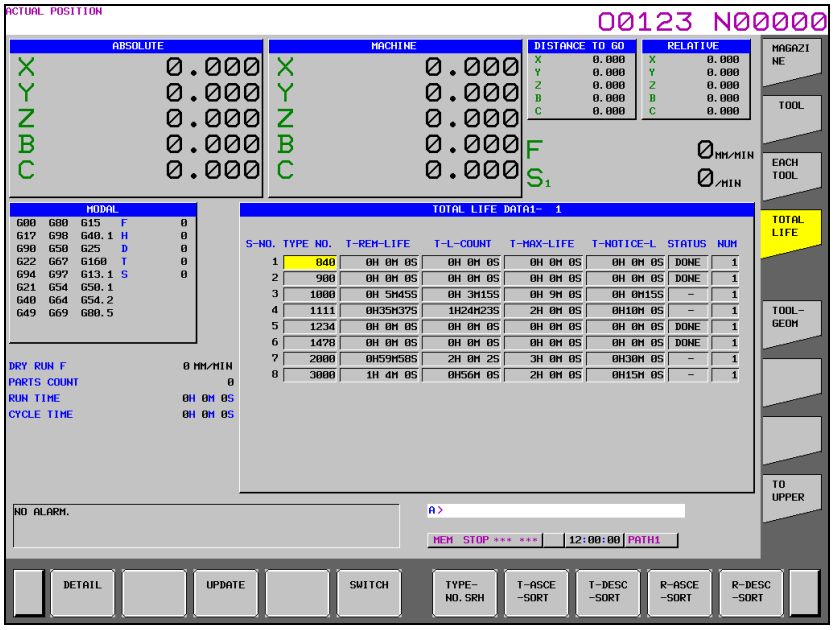


Fig. 12.3.24.4 (b) Time display (15-inch)

Displayed information

S-NO. : Sequential number of each tool type

TYPE NO. : Tool type number

T-REM-LIFE : Total of remaining life values of tools with the same tool type number

T-L-COUNT : Total of used counts/times of tools with the same tool type number

T-MAX-LIFE : Total of maximum life values of tools with the same tool type number

T-NOTICE-L : Total of notice life values of tools with the same tool type number when the tool life arrival signal is to be output for each tool type

STATUS : Whether the tool life arrival signal has been output when the signal is to be output for each tool type
Either of the two states (UNDONE and DONE) is displayed.

NUM : Number of tools with the same tool type number

When bit 3 (ETE) of parameter No. 13200 is set to 0 and bit 2 (TRT) of parameter No. 13200 is set to 1, the tool life arrival signal is to be output for each tool type number and T-NOTICE-L and STATUS are displayed.

Key operations

- MDI key operations

Displays the previous page. The cursor moves to the last data item on that page.

Displays the next page. The cursor moves to the first data item on that page.

Moves the cursor up on the screen.

Moves the cursor down on the screen.

Disabled.

Disabled.

- Soft key operations

Horizontal soft key [DETAIL]
Displays the detailed life data screen.
Horizontal soft key [UPDATE]
Updates data displayed on the total life data screen.
Horizontal soft key [CHANGE]
Switch the display between the count and time types.
Horizontal soft key [T-TYPENO. SRH]
Moves the cursor to a tool type number when pressed after the number is entered.
Horizontal soft key [T-ASCE -SORT]
Sorts data in ascending order of tool type numbers.
Horizontal soft key [T-DESC -SORT]
Sorts data in descending order of tool type numbers.
Horizontal soft key [R-ASCE -SORT]
Sorts data in ascending order of remaining life values.
Horizontal soft key [R-DESC -SORT]
Sorts data in descending order of remaining life values.

NOTE

1 After horizontal soft key [T-ASCE -SORT], [T-DESC -SORT], [R-ASCE -SORT], or [R-DESC -SORT] is pressed, the cursor is positioned at the top of page 1 of the total life data screen.
2 When the power is turned on, data of the count counting type is displayed in ascending order of tool type numbers. When the display type is changed or data is sorted in a different order, the status is kept.
3 If horizontal soft key [DETAIL] is pressed when no data is displayed on the total life data screen, the warning message “NO DETAILED LIFE DATA SCREEN” is output.
4 If an unregistered tool type is specified and horizontal soft key [T-TYPENO. SRH] is pressed, the warning message “UNREGISTERED NUMBER” is output.

Detailed life data screen

Procedure

1 Press function key .
2 Press vertical soft key [NEXT PAGE] several times and then vertical soft key [TOOL MANAGER].
3 Press vertical soft key [TOTAL LIFE]. The total life data screen appears.
4 Selecting horizontal soft key [DETAIL] causes the detailed life data screen to appear.

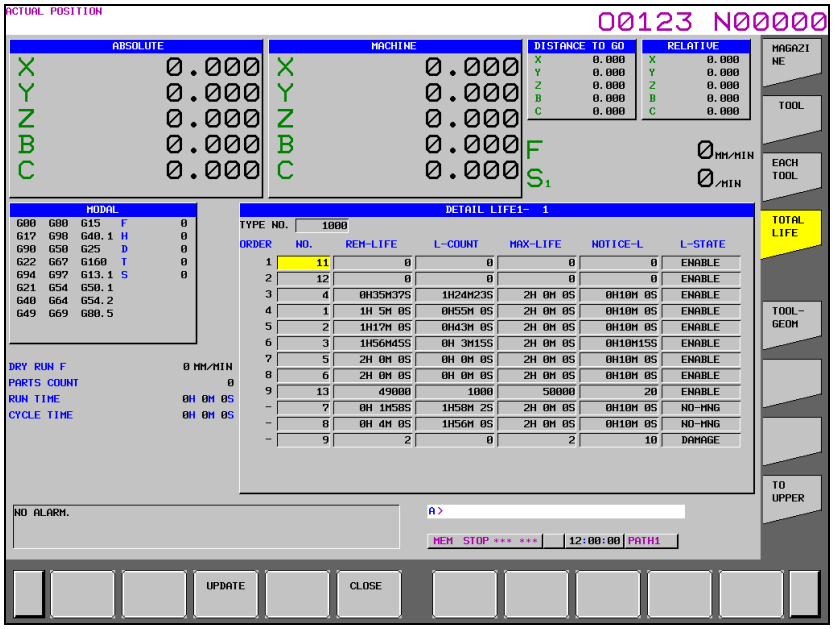


Fig. 12.3.24.4 (c) Detailed life data screen (15-inch)

- Displayed information

TYPE NO. : Tool type number
ORDER : Sequential number in ascending order of remaining life times or the order in which the customize data is set.
NO. : Tool management data number
REM-LIFE : Remaining life value obtained by subtracting the life count value from the maximum life value
L-COUNT : Total used count or time of the tool
MAX-LIFE : Maximum life value of the tool
NOTICE-L : Notice life value of the tool
STATUS : Life status of the tool
One of the four states (NO-MNG (0), ENABLE (1, 2), NONE (3), and DAMAGED (4)) is displayed.

Key operations

- MDI key operations

Displays the previous page.

Displays the next page.

Moves the cursor up on the screen. The cursor moves to the last data item on that page.

Moves the cursor down on the screen. The cursor moves to the first data item on that page.

Disabled.

Disabled.

- Soft key operations

Horizontal soft key [UPDATA]

Updates data displayed on the detailed life data screen.
The cursor is positioned at the beginning of page 1 after this soft key is pressed.

Horizontal soft key [CLOSE]

Closes the detailed life data screen and returns to the total life data screen.

NOTE

1 When horizontal soft key [CLOSE] is pressed and the total life data screen is displayed again, the cursor on the total life screen data is positioned as follows.
2 When the tools with the tool type number displayed when the detailed data screen is closed are registered as tool management data, the cursor on the total life screen is positioned at that tool type number.
3 When the tools with the tool type number displayed when the detailed data screen is closed are not registered as tool management data, the cursor on the total life data screen is positioned at the first total life data.

12.3.24.5 Tool geometry data screen (15-inch display unit)

Tool geometry data screen

Procedure

1 Press function key .

2 Press vertical soft key [NEXT PAGE] several times and then vertical soft key [TOOL MANAGER].

3 Selecting vertical soft key [TOOL-GEOM] causes the tool geometry data screen to appear.

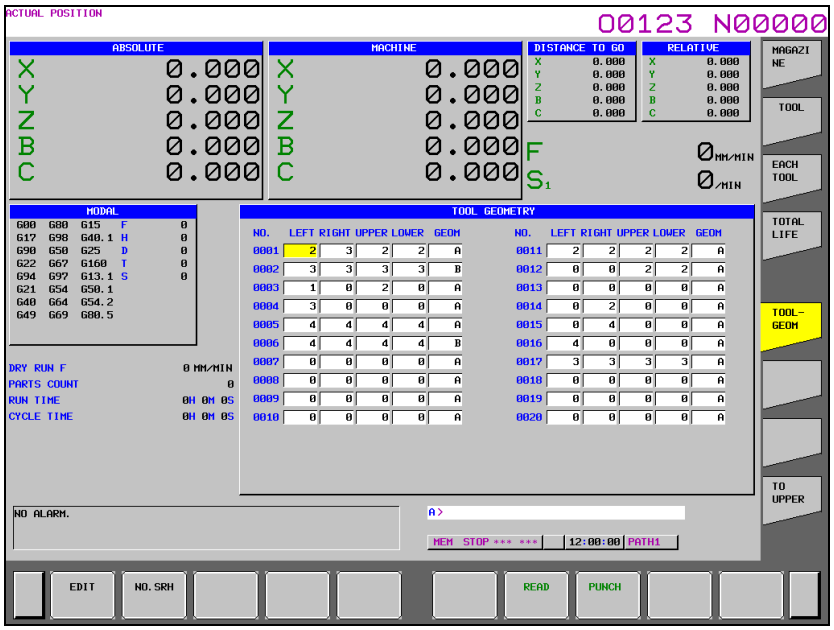


Fig. 12.3.24.5 (a) Tool geometry data screen (15-inch)

- Displayed item

NO. : Tool geometry number
Up to 20 numbers can be displayed.

LEFT : Sets the number of pots on the left of the reference pot that are to be occupied.
A value between 0 and 4 can be set.

RIGHT : Sets the number of pots on the right of the reference pot that are to be occupied.
A value between 0 and 4 can be set.

UPPER : Sets the number of pots above the reference pot that are to be occupied.
A value between 0 and 4 can be set. (Use this item when the magazine is of the matrix type.)

LOWER : Sets the number of pots under the reference pot that are to be occupied.
A value between 0 and 4 can be set. (Use this item when the magazine is of the matrix type.)

Key operations

- Operations in the standard mode

MDI key operations

Numeral keys Inputs a numeric value.

Moves the cursor up on the screen.

Moves the cursor down on the screen.

Moves the cursor left on the screen.

Moves the cursor right on the screen.

Soft key operations

Horizontal soft key [NO.SRH]
Moves the cursor on a tool geometry number when pressed after the number is entered.
Horizontal soft key [EDIT]
Enters the data edit mode.
Horizontal soft key [READ]
Reads data related to the tool management functions.
This key is available only in the standard mode.
Put the NC in the EDIT mode.
Horizontal soft key [PUNCH]
Punches data related to the tool management functions.
This key is available only in the standard mode.
Put the NC in the EDIT mode.
In the management data edit mode, the following key operations, in addition to [NO.SRH] mentioned above, can be performed.

- Operations in the edit mode

To edit data, press horizontal soft key [EDIT] to enter the edit mode. In the edit mode, “EDITING” is displayed at the lower right of the screen.

MDI key operation

Actually writes a value entered using numeric keys.

Soft key operation

Horizontal soft key [END]
Ends the data edit mode.

Example

Set the edit mode. When the tool geometry with tool geometry number 1 occupies 1 pot in the left direction, 0.5 pots in the right direction, and 1.5 pots in the down direction, set data as shown in the figure below:

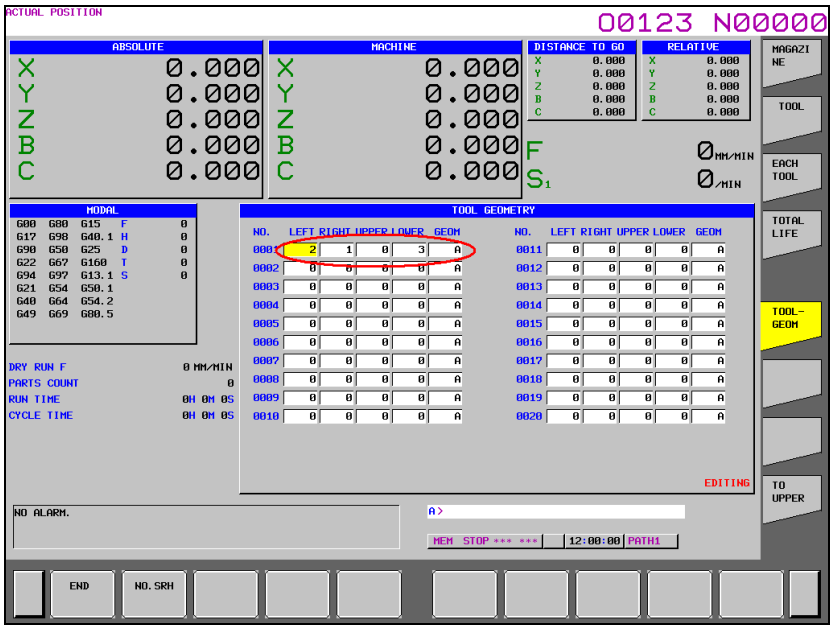


Fig. 12.3.24.5 (b) Example of setting data on the tool geometry data screen (15-inch)

- Display of occupied pots in the magazine management table

Each pot occupied by a tool stored in another pot is indicated with an asterisk (*).

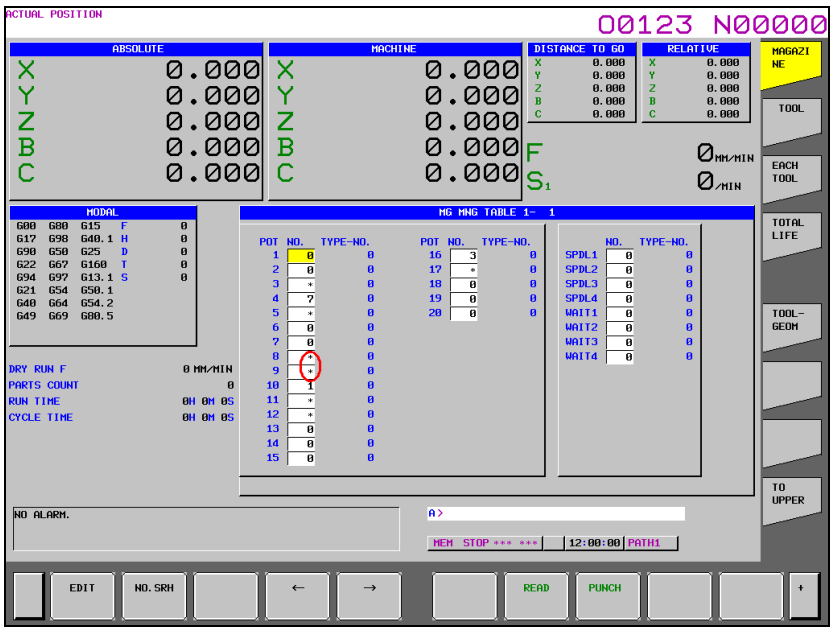


Fig. 12.3.24.5 (c) Magazine management table (15-inch)

If a tool to be registered for a magazine is determined to interfere with another tool, the warning message “TOOL INTERFERENCE CHECK ERROR:xxxx,xxxx” is displayed. xxxx indicates the tool number of each of the two tools. If a tool is determined to interfere with multiple tools, only the first tool detected by the NC interference check is displayed.
If a tool is determined to interfere with the frame of a magazine, the tool number of the tool to be stored and “FRAME” are displayed.

- Search for an empty pot for a oversize tool

During the management data editing, a horizontal soft key for searching for empty oversize tool pots is displayed.

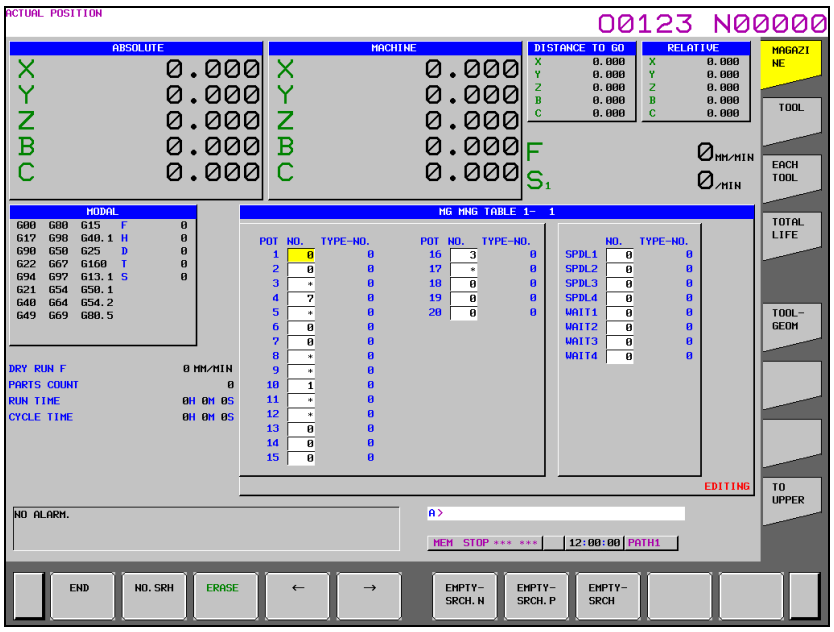


Fig. 12.3.24.5 (d) Searching for an empty pot for a oversize tool (15-inch)

Enter the tool geometry number in the key-in buffer and press a search horizontal soft key. The cursor moves to an empty pot fit for the geometry.

EMPTY-SRCH.N: Searches for an empty pot forward.
EMPTY-SRCH.P: Searches for an empty pot backward.
EMPTY-SRCH: Searches for the pot nearest to the current position.

- Tool management screen

You can use bit 2 of tool information to switch between a oversize tool and normal tool. For a oversize tool, set a tool geometry number fit for the tool.

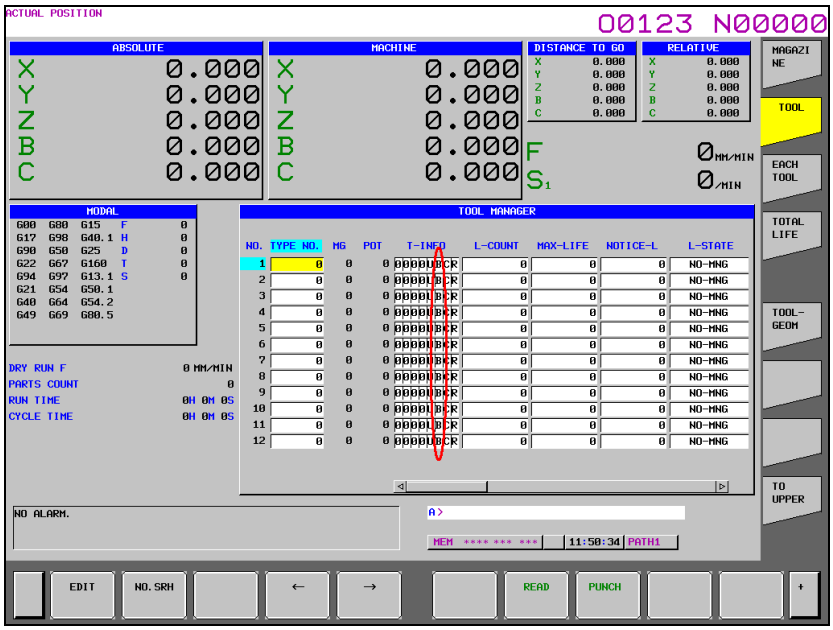


Fig. 12.3.24.5 (e) Bit for switching between a normal tool and oversize tool (15-inch)

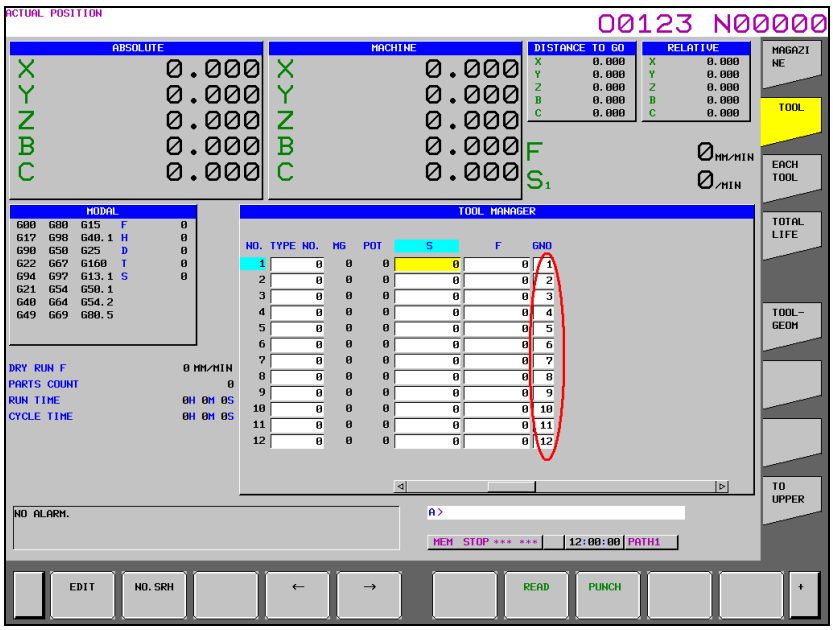


Fig. 12.3.24.5 (f) Tool geometry number (15-inch)

12.3.25 Displaying and Switching the Display Language (15-inch Display Unit)

The language used for display can be switched to another language.
A display language can be set using a parameter. However, by modifying the setting of the display language on this screen, the display language can be switched without turning off then on the power.

Displaying and setting the display language

Procedure

1 Press function key .

2 Press vertical soft key [NEXT PAGE] several times to display soft key [LANGUAGE].

3 Press vertical soft key [LANGUAGE] to display the language screen.

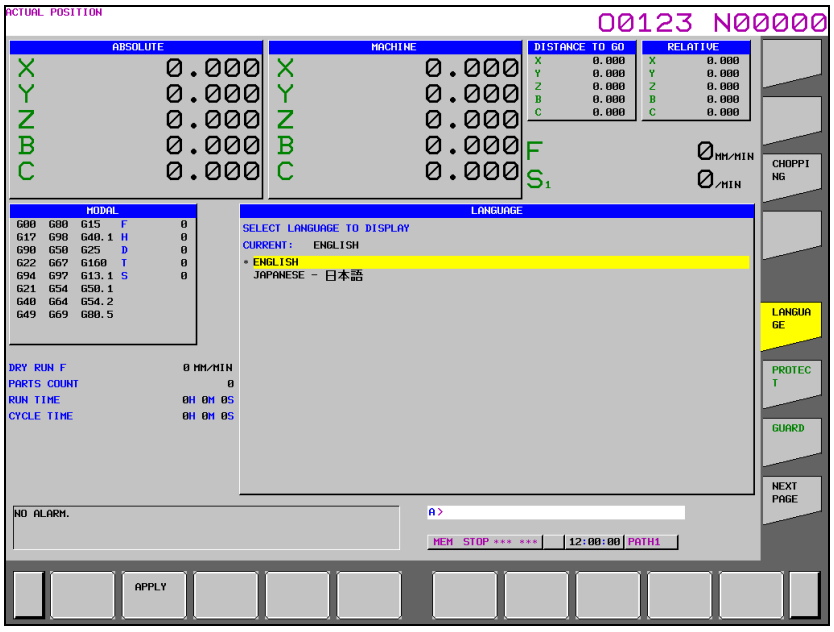


Fig. 12.3.25 (a) LANGUAGE screen (15-inch)

4 Press page key or , then press cursor keys , to move the cursor to a desired display language.

5 Press horizontal soft key [APPLY]. The display language is switched to the selected language. The language specified on this screen continues to be used if the power is turned off then back on.

Explanation

- Language switching

The language screen can be displayed if bit 0 (NLC) of parameter No. 3280 is set to 0.

- Selectable languages

The display languages selectable on this screen are as follows:

1. English
2. Japanese
3. German
4. French
5. Chinese (Traditional)
6. Chinese (Simplified)
7. Italian
8. Korean
9. Spanish
10. Dutch
11. Danish
12. Portuguese
13. Polish
14. Hungarian
15. Swedish
16. Czech
17. Russian
18. Turkish

Among the languages listed above, English and other usable languages are displayed on the screen as a list of switchable languages.

Limitation

- Language parameter modification on the parameter screen

Which language to be used for display is specified with parameter No. 3281. This parameter can be modified using the parameter screen as well. However, if a modification is made on the parameter screen, the new setting is not reflected until "APPLY" operation is performed on the language screen or the power is turned on again. If an invalid value is set in parameter No. 3281 on the parameter screen, the screen is displayed in English after the power is turned on again.

12.3.26 Protection of Data at Eight Levels (15-inch Display Unit)

You can set eight CNC and PMC operation levels and one of eight protection levels for each type of CNC and PMC data.
When an attempt is made to change CNC and PMC data or output it to an external unit, the operation level is compared with the protection level to determine whether to allow the change or external output.

12.3.26.1 Operation level setting (15-inch display unit)

You can set eight CNC and PMC operation levels.

Displaying and setting the operation level setting screen

Procedure

1 Press function key .

2 Press vertical soft key [NEXT PAGE] several times to display soft key [PROTECT].

3 Press vertical soft key [PROTECT].
The operation level setting screen shown below is displayed.

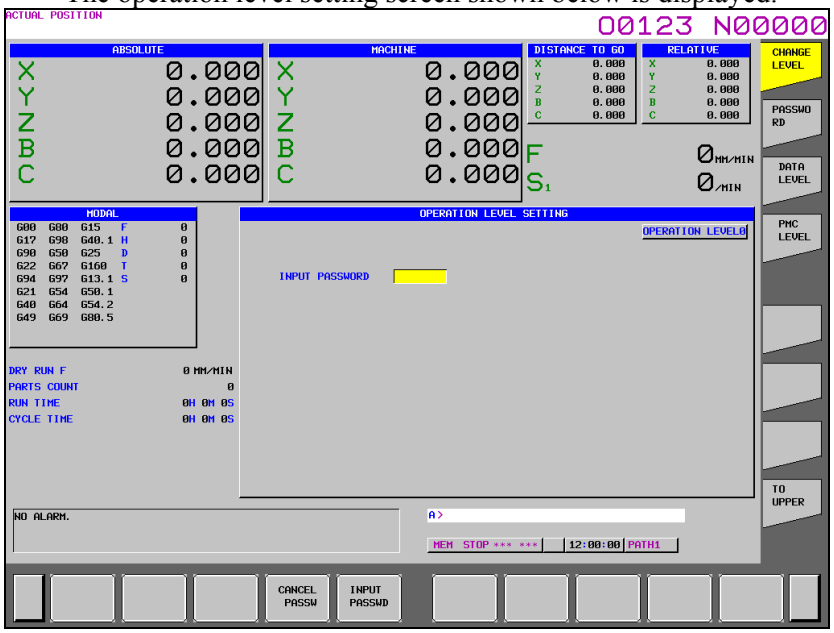


Fig. 12.3.26.1 (a) Operation level setting screen (15-inch)

4 Key in the password for an operation level to be set/modified, then press horizontal soft key [INPUT PASSWD].

5 To return the operation level to 0, 1, 2, or 3, press horizontal soft key [CANCEL PASSWD].

Explanation

- Operation level setting

To select operation level 0 to 3, use the corresponding memory protection key signal.
To select operation level 4 to 7, use the corresponding password.

Table 12.3.26.1 (a) Operation level setting

Operation level	Setting	Sample grouping
7 (high)	Password	-
6	Password	MTB
5	Password	Dealer and integrator
4	Password	End user
3	Memory protection key signal	User level (level 1)
2	Memory protection key signal	User level (level 2)
1	Memory protection key signal	User level (level 3)
0 (low)	Memory protection key signal	User level (level 4)

When operation level 4 to 7 is set, the operation level remains unchanged until the password is cleared.
(The operation level also remains unchanged if the power is turned off.)
Operation level 7 is reserved for CNC and PMC maintenance.

NOTE
When a password is being entered, an asterisk (*) is displayed instead of each entered character.

12.3.26.2 Password modification (15-inch display unit)

The current operation level is displayed.
The password for each of operation levels 4 to 7 can be modified.

Displaying and setting the password modification screen

Procedure

1 Press function key .
2 Press vertical soft key [NEXT PAGE] several times to display soft key [PROTECT].
3 Press vertical soft key [PROTECT].
4 Press vertical soft key [PASSWORD].
The PASSWORD CHANGE screen shown below is displayed.

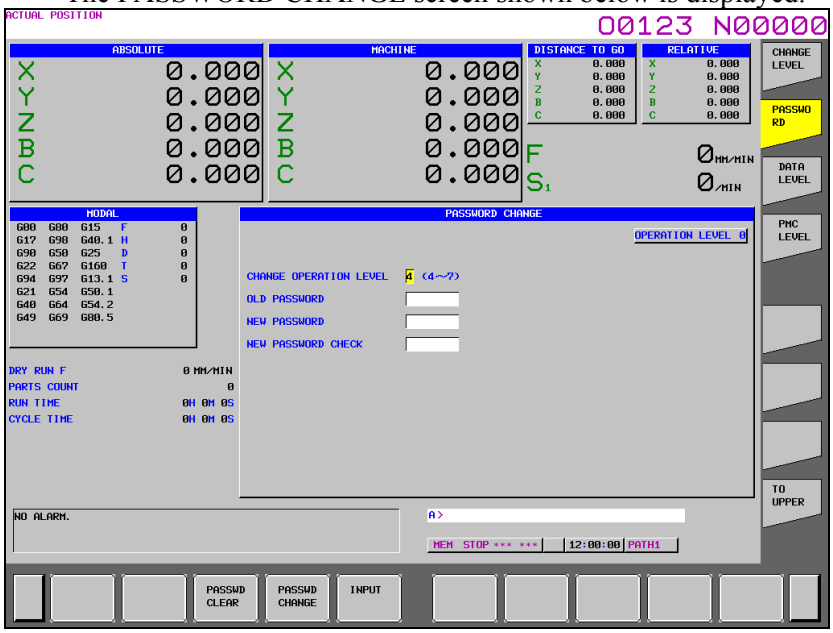


Fig. 12.3.26.2 (a) PASSWORD CHANGE screen (15-inch)

5 Key in an operation level whose password is to be modified, then press horizontal soft key [INPUT].
6 Key in the current password for the operation level whose password is to be modified, then press horizontal soft key [INPUT].
7 Key in a new password, then press horizontal soft key [INPUT].
8 Key in the new password again for confirmation, then press horizontal soft key [INPUT].
9 Press horizontal soft key [PASSWD CHANGE].
10 To clear the password, press horizontal soft key [PASSWD CLEAR].

Explanation

Up to eight characters (only uppercase alphabetic characters and numeric characters) can be input.

NOTE

1 For a password, consisting of three to eight characters, the following characters are available:
   - Uppercase alphabetic characters
   - Numeric characters

2 When a password is being entered, an asterisk (*) is displayed instead of each entered character.

3 Whether a password can be changed at the current operation level is determined as follows:
   - Password of an operation level higher than the current operation level
     Cannot be changed.
   - Password of the current operation level
     Can be changed.
   - Password of an operation level lower than the current operation level
     Can be changed (only to the initial password).

4 The set password is not displayed.
   Be careful not to forget the password.

12.3.26.3 Protection level setting (15-inch display unit)

The current operation level is displayed.
The change protection level and output protection level of each data item are displayed.
The change protection level and output protection level of each data item can be changed.

Confirmation based on protection level setting

Procedure

1 Press function key .

2 Press vertical soft key [NEXT PAGE] several times to display soft key [PROTECT].

3 Press vertical soft key [PROTECT].

4 Press vertical soft key [DATA LEVEL] to change the protection level of CNC data or press vertical soft key [PMC LEVEL] to change the protection level of PMC data.
The following protection level change screen is displayed.

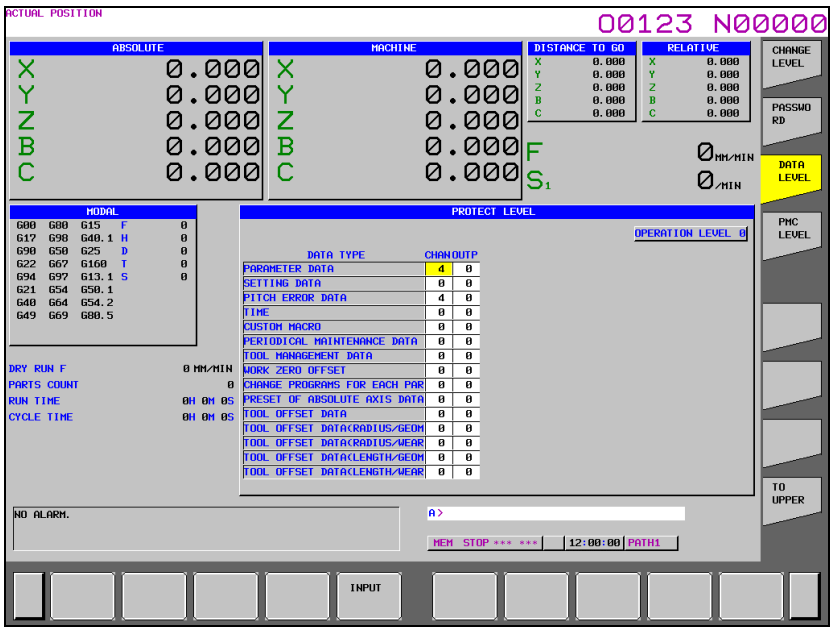


Fig. 12.3.26.3 (a) Protection level change screen (15-inch)

5 Move the cursor to the change level or output level of a desired data item.

6 Key in a new desired level, then press horizontal soft key [INPUT].

NOTE
When the protection level of PMC data is set, soft key [SWITCH PMC] is used to switch between PMC paths to be set, for multipath PMC.

Explanation

When the protection level of a data item is higher than the current operation level, the protection level of the data item cannot be changed.
The protection level of a data item cannot be changed to a protection level higher than the current operation level.
For each of the following types of data, you can set a data protection level. There are the following two types of data protection levels:

- Change protection level
  Sets the protection level used when data is changed.
- Output protection level
  Sets the protection level used when data is output (punched out) to an external unit.

As a protection level, you can set a value of 0 (low) to 7 (high).

Table 12.3.26.3 (a) Protection level of each type of data

Type of data	Initial protection level	
	Change	Output
Custom macro variable data <CUSTOM MACRO> (including variable data dedicated to the macro executor)	0	0
Periodical maintenance data <PERIODICAL MAINTENANCE DATA>	0	0
Tool management data <"TOOL LIFE TIME DATA>	0	0
Tool offset data <TOOL OFFSET DATA> (For each type when tool geometry compensation and tool wear compensation are treated differently)	0	0
Clock data <TIME>	0	0
Workpiece origin shift amount data <WORK ZERO SHIFT>	0	0
Workpiece origin offset data <WORK ZERO OFFSET>	0	0
Ethernet setting data <ETHER NET SETTING>	0	0
Fixture offset data <FIXTURE OFFSET DATA>	0	0
Parameter data <PARAMETER DATA>	4	0
Settings <SETTING DATA>	0	0
Pitch error compensation data <PITCH ERROR DATA> (including three-dimensional error compensation data)	4	0
Parameter data for Power Mate CNC manager function <PARAMETER DATA FOR POWER MATE CNC MANAGER>	0	0
Each part program <PROGRAMS FOR EACH PARTS>	0	0
Part program editing operation <CHANGE PROGRAMS FOR EACH PARTS>	0	0
Absolute coordinate preset operation <PRESET OF ABSOLUTE AXIS DATA>	0	0

Table 12.3.26.3 (b) Protection level of PMC data

Type of data	Initial protection level	
	Change	Output
Composition parameter	0	0
Setting (online)	0	0
Setting (each path)	0	0
Sequence program	0	0
PMC parameter	0	0
Timer	0	0
Counter	0	0
Keep relay	0	0
Keep relay (system)	0	0
Data table	0	0
Data table control	0	0
PMC momory	0	0

NOTE

1 For some types of data, the output function is not provided.
2 When the protection level of data is higher than the current operation level, the protection level cannot be changed.
3 The protection level of data cannot be changed to a level higher than the current operation level.
4 Settable types of data increase or decrease, depending on the option configuration.
5 For details on the protection level of PMC data, refer to “PMC Programming Manual (B-63983EN)”.
6 Data related to tool information on the EACH TOOL DATA screen for the tool management function is protected by protecting tool offset data.
7 The type of tool offset data put in effect varies depending on the tool compensation value memory used.
8 To change the protection level for each part program, do so on the PROGRAM FOLDERscreen rather than on the PROTECT LEVEL screen.
9 Part program editing includes program editing for the MDI mode.
10 Presetting absolute coordinates causes preset workpiece coordinate system values to be protected.
11 During tool offset data input/output, if any tool offset data type is not allowed to be changed or output to the outside, it is processed as follows:
- Input : Any data type other than those whose change is not allowed is changed.
- Output : Any data type other than those whose change is not allowed is output.

12.3.26.4 Setting the change protection level and output protection level of a program (15-inch display unit)

The display/operations indicated below can be performed from the directory screen.
The change protection level and output protection level of each part program are displayed.
The change protection level and output protection level of each part program can be changed.

Setting the change protection level and output protection level of a program

Procedure

1 Press function key PROG.
2 Press vertical soft key [FOLDER].

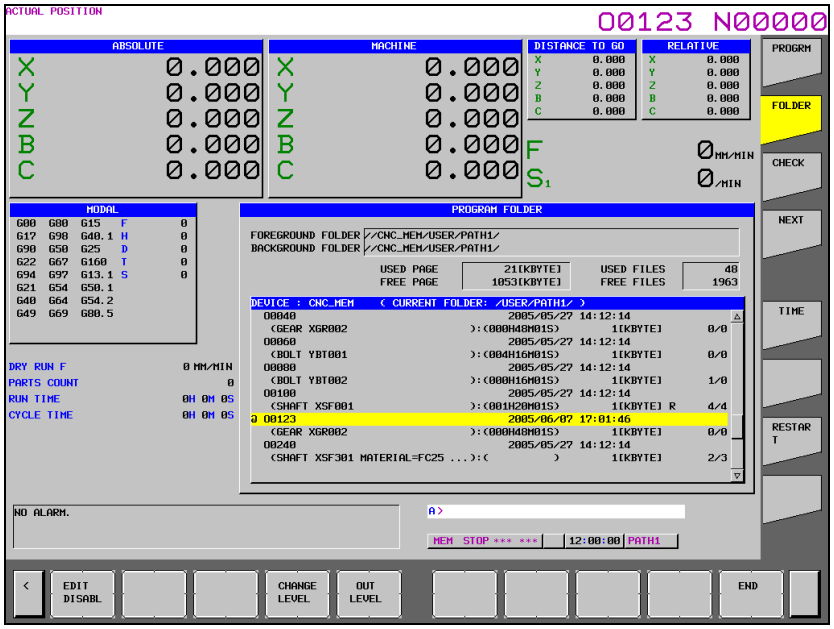


Fig. 12.3.26.4 (a) Program directory screen (15-inch)

3 Press horizontal soft key [DETAIL ON] to switch to detail displays.
4 Move the cursor to a desired program.
5 Press the continuous menu key to display horizontal soft key [CHANGE ATTR].
6 Press horizontal soft key [CHANGE ATTR].
7 Key in a new desired level, then press horizontal soft key [CHANGE LEVEL].
8 To change the output protection level, key in a new desired level, then press horizontal soft key [OUT LEVEL].

Explanation

The change protection level (0 to 7) and output protection level (0 to 7) are displayed as "CHANGE PROTECTION LEVEL VALUE/OUTPUT PROTECTION LEVEL".

NOTE
1 When the protection level of data is higher than the current operation level, the protection level cannot be changed.
2 The protection level of data cannot be changed to a level higher than the current operation level.
3 A protection level can be set only for those part programs that are held on the "CNC_MEM" device.

12.3.27 Precision Level Selection (15-inch Display Unit)

An intermediate precision level between the parameters for emphasis on velocity (precision level 1) and the parameters for emphasis on precision (precision level 10) set on the machining parameter tuning screen can be selected. As shown in the figure below, the levels are proportionally linear, and an intermediate level can be selected so that optimal parameters can be automatically calculated to perform machining.

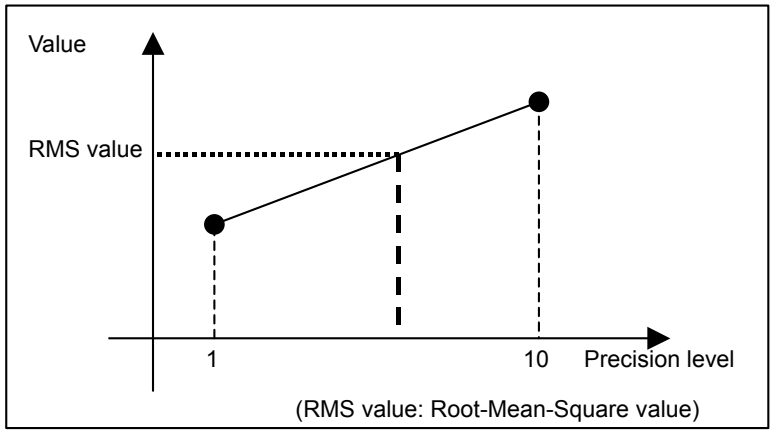


Fig. 12.3.27 (a) Image of "level"

Procedure for precision level selection

1 Select the MDI mode.

2 Press function key .

3 Press vertical soft key [NEXT PAGE] several times to display vertical soft key [PRECI LEVEL].

4 Press vertical soft key [PRECI LEVEL].

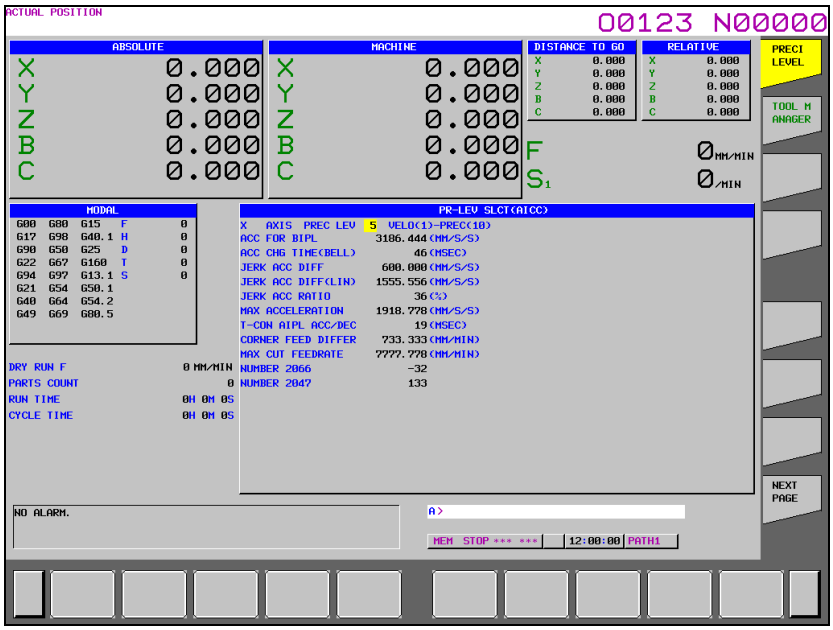


Fig. 12.3.27 (b) Precision level selection screen

5 To change the precision level, key in a desired precision level (1 to 10), then press the

key on the MDI panel.

6 When the precision level is changed, a RMS value is obtained from the velocity-emphasized parameter set and precision-emphasized parameter set for parameter modification. For the modified parameters, see the description of the machining parameter tuning.

7 If there is an axis in addition to the currently displayed axes, press page key

or

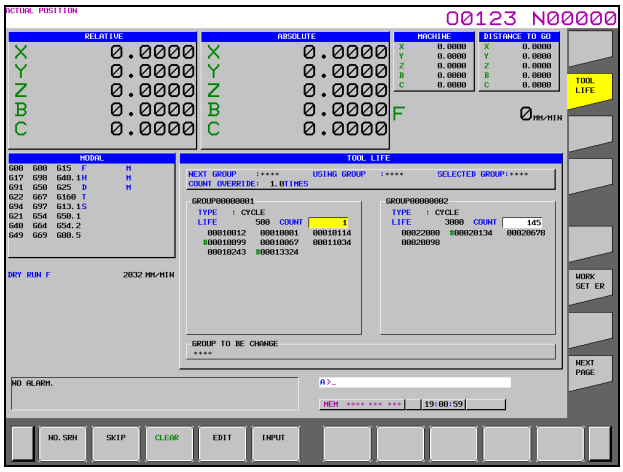
several times to display the screen for the axis.

12.3.28 Displaying and Setting Tool Life Management Data (15-inch Display Unit)

Displaying tool life management data on a screen enables the current status of tool life management to be grasped. Also on the screen, tool life management data can be edited. The screen is either:

- Tool life management (list screen) or
- Tool life management (group editing screen)

Overview



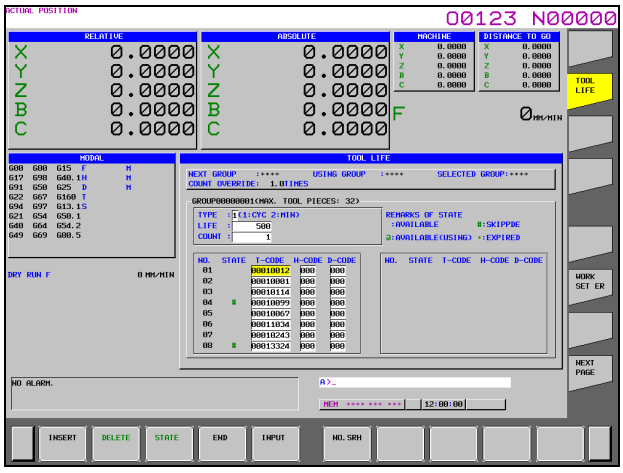
List screen

EDIT

END

Tool life management (list screen)
Displayed items:
- NEXT GROUP
- SELECTED GROUP
- GROUP NO.
- LIFE
- TOOL MANAGEMENT STATUS
- GROUP TO BE CHANGE

Functions:
- Searching for groups
- Clearing execution data
- Setting the tool life counter



Group editing screen

Tool life management (group editing screen)
Displayed items:
- NEXT GROUP
- SELECTED GROUP
- MAX.TOOL PIECES
- LIFE
- T-CODE
- H-CODE and D-CODE

Functions:
- Setting tool data
- Setting a tool status
- Selecting too skip
- Setting a life count type, life value, and tool life counter

Pressing horizontal soft key [EDIT] or [END] switches from the list screen to the group editing screen or vice versa.

M

If the tool life management B function is enabled (bit 4 (LFB) of parameter No. 6805 = 1), the following parameters can be used to display arbitrary groups and remaining set values.

- If bit 5 (TGN) of parameter No. 6802 = 1
  An arbitrary group is displayed on the list and group editing screens.
- If bit 3 (GRP) of parameter No. 6802 = 1
  Remaining set values are displayed on the group editing screen. The group editing screen always displays H and D codes.

T

The T series is provided with two tool change types, turret type and ATC type. For the T series, displays vary depending on which tool change type is in use.

If the turret type is in use (bit 3 (TCT) of parameter No. 5040 = 0)

- Neither the H code nor the D code is used; neither is displayed.
- No arbitrary group can be used; no arbitrary group is displayed.

If the ATC type is in use (bit 3 (TCT) of parameter No. 5040 = 1)

- The D code is displayed on the group editing screen.
- If the tool life management B function is enabled (bit 4 (LFB) of parameter No. 6805 = 1) and bit 5 (TGN) of parameter No. 6802 = 1, arbitrary groups are displayed on the list and group editing screens.

If the tool life management B function is enabled (bit 4 (LFB) of parameter No.6805 = 1) and bit 3 (GRP) or parameter No. 6802 = 1, remaining set values are displayed on the group editing screen.

12.3.28.1 Tool life management (list screen) (15-inch display unit)

This screen can display the life management status of all tools in tool groups and whether the life of the tool groups has expired. It also enables you to set tool life counters and clear execution data.

Displays on the list screen

Procedure

1 Press function key .

2 Press vertical soft key [NEXT PAGE] several times to display soft key [TOOL LIFE].

3 Press vertical soft key [TOOL LIFE].

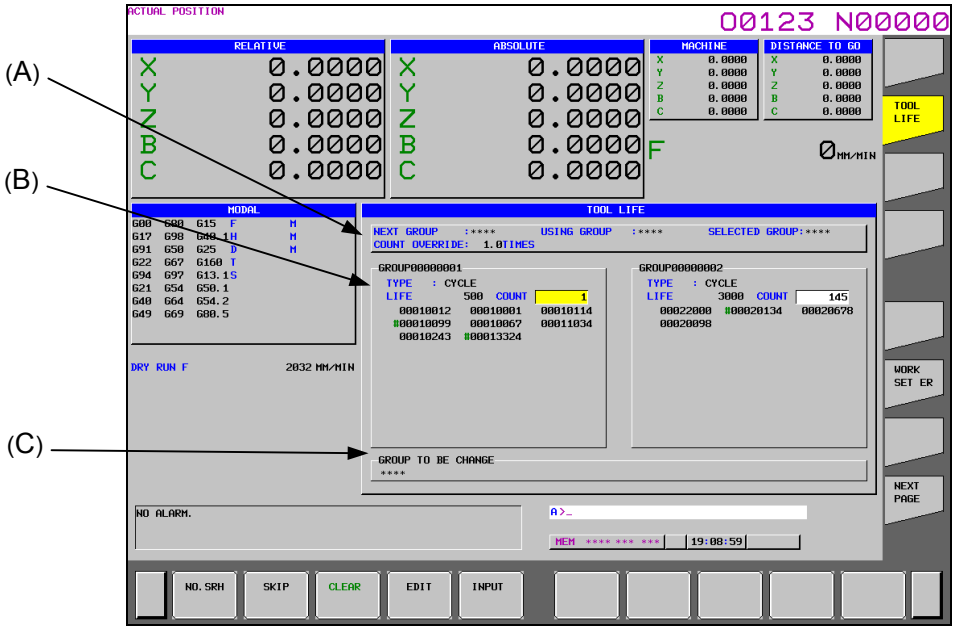


Fig. 12.3.28.1 (a) Displaying tool life management (list screen) (15-inch)

- Contents of (A)

(A) displays tool group numbers and an override value. If there is no tool group to display, ****, instead of tool group numbers, is displayed.

NEXT GROUP:

Tool group number for which life counting is started by the next M06 command.

USING GROUP:

Tool group number for which life counting is currently under way.

SELECTED GROUP:

Tool group number for which life counting is currently under way or life counting has been performed most recently.

COUNT OVERRIDE:

“1.0TIMES” is displayed if the tool life counter override signal is disabled (bit 2 (LFV) of parameter No. 6801 = 0).

NOTE
If arbitrary group numbers are enabled, NEXT GROUP, USING GROUP, and SELECTED GROUP are each represented with an arbitrary group number rather than the tool group number.

- Contents of (B)

(B) displays the set life value, the current content of the tool life counter, and the registered tool numbers (in the order they are used) for each tool group. If the life count type is specification by duration, the measurement unit used in displaying and specifying set life values and tool life counter values is selected according to the setting of bit 0 (FCO) of parameter No. 6805 as listed below.

Bit 0 (FCO) of parameter No. 6805	0	1
Measurement unit used in displaying and specifying set life values and tool life counter values	1 minute	0.1 minutes

The following table lists the prefixes used with tool numbers.

Tool status	Tool in use	Tool not in use
Life remaining	@	No indication
Skip	#	#
Life expired	*	*

NOTE
1 The tool life counter indicates the count value for the tool indicated with @.
2 If bit 3 (EMD) of parameter No. 6801 = 0, a tool number remains prefixed with @ even if the life of the tool has expired until another tool is selected.
3 If bit 3 (EMD) of parameter No. 6801 =1, the following differences can occur depending on the type of the life counter in use.
- If the life counter type is specification by duration, the prefix is changed to "*" (life expired) as soon as the tool life expires.
- If the life counter type is specification by count, the counter is incremented by one at the end of a program (such as M02 or M30). So, the prefix is not changed to "*" (life expired) even if the tool life counter matches the life value. The symbol "*" (life expired) appears when the tool life count is incremented after the CNC is reset.

4 If bit 2 (ETE) of parameter No. 6804 = 1, the symbol "*" indicating the expiration of the life of the last tool appears on the tool life management screen when the life counter for the last tool of the tool group of interest matches the life value. This way, the information about the last tool in the FOCAS2 or PMC window indicates that the life of the tool has expired if the tool change signal TLCH <Fn064.0> is 1.

If arbitrary group numbers are enabled, an arbitrary group number is displayed in the parentheses beside the tool group number. If no arbitrary group number is specified, "********" is displayed instead.

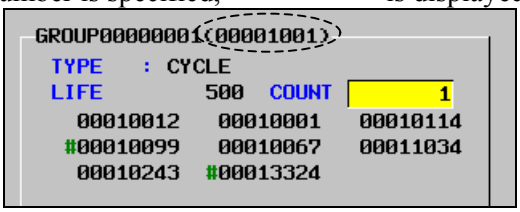


Fig. 12.3.28.1 (b) Displaying arbitrary group numbers

Arbitrary group numbers are enabled by setting the following parameters.

M

- The tool life management B function is enabled (bit 4 (LFB) of parameter No. 6805 = 1).
- Arbitrary group numbers are enabled (bit 5 (TGN) of parameter No. 6802 = 1).

T

- The current tool change type is ATC (bit 3 (TCT) of parameter No. 5040 = 1).
- The tool life management B function is enabled (bit 4 (LFB) of parameter No. 6805 = 1).
- Arbitrary group numbers are enabled (bit 5 (TGN) of parameter No. 6802 = 1).

- Contents of (C)

(C) displays tool group numbers for which a tool change signal has been issued.
If there are so many tool group numbers that all the numbers cannot be displayed, some are omitted, and "》" is displayed instead.
If there is no tool group number in need of change, "****" is displayed.

NOTE
If arbitrary group numbers are enabled, a tool group number in need of change is represented with an arbitrary group number rather than the tool group number.

Setting data on the list screen

Tool life management data can be specified in the reset state (both the OP and RST signals are "0"). However, setting bit 1 (TCI) of parameter No. 6804 to 1 enables tool life management data to be specified even during automatic operation (the OP signal is "1").

NOTE
As for USING GROUP or NEXT GROUP settings:
1) During automatic operation (OP signal = “1” and bit 1 (TCI) of parameter No. 6804 = 1), only the tool life counter can be changed.
2) In the reset state (OP signal = “0” and RST signal “0”), the following editing operation stops life management because it cannot be continued.
- Clearing execution data

Procedure

- Setting the tool life counter

The tool life counter can be set with a value, using the following methods.

Method 1

1 Place the cursor on the tool life counter for a desired tool group.
2 Enter the value from the keypad.
3 Press horizontal soft key [INPUT].

Method 2

1 Place the cursor on the tool life counter for a desired tool group.
2 Enter the value from the keypad.
3 Press key.

- Clearing execution data

All existing execution data for a tool group selected by the cursor can be cleared as follows:

1 Place the cursor on the tool group whose execution data you want to clear.
2 Press horizontal soft key [ERASE].
3 Press horizontal soft key [EXEC].

NOTE
Setting bit 4 (GRS) of parameter No. 6800 to 1 enables execution data for all registered tool groups to be cleared.

- Selecting tool groups

Tool groups can be selected using the following methods.

Method 1

1 Enter a tool group number from the keypad.
2 Press soft key [NO.SRH].

NOTE
If arbitrary group numbers are enabled, a tool group is selected by searching for an arbitrary group number rather than the tool group number.

Method 2

1 Press page key or to display desired groups.

2 Press cursor key or to move the cursor to the desired group at either the left or right.

- Switching to the group editing screen

Switch to tool life management (group editing screen).

1 Move the cursor to the tool group you want to edit:

2 Press horizontal soft key [EDIT].

12.3.28.2 Tool life management (group editing screen) (15-inch display unit)

On this screen, it is possible to edit tool life management data (such as tool life value, tool life counter, and tool data) for the tool group of interest.

Displaying the group editing screen

Procedure

1. Place the cursor on the tool group you want to edit from the list screen.
2. Press horizontal soft key [EDIT].

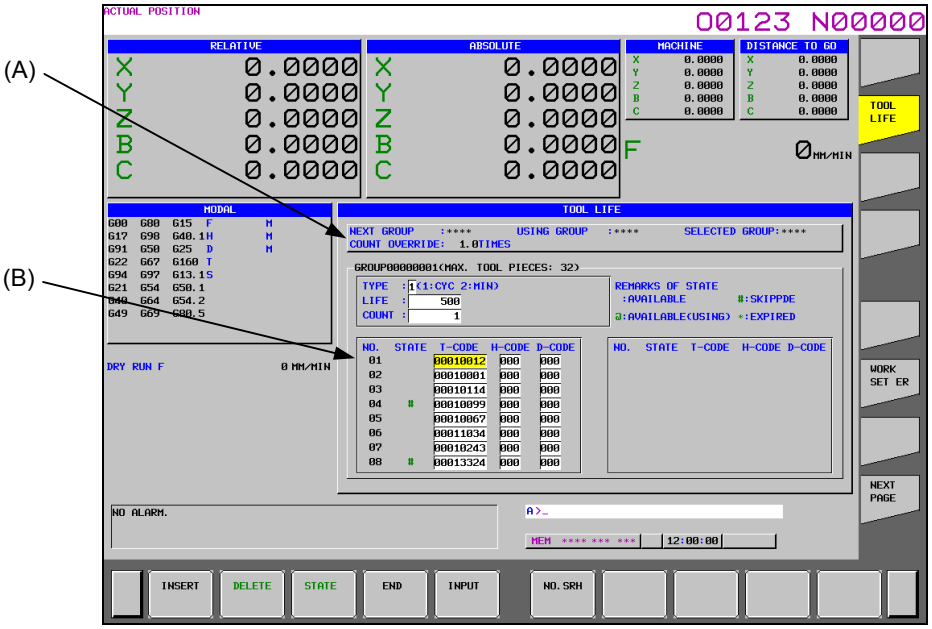


Fig. 12.3.28.2 (a) Displaying tool life management (group editing screen) (15-inch)

NOTE
If no tool is registered with a tool group, none of a life count type, a life value, and a tool life counter value is displayed for the tool group.

- Contents of (A)

Like the counterpart of the list screen, (A) of the editing screen displays the next tool group number and override value. If there is no appropriate tool group, **** is displayed instead.

NEXT GROUP:
Tool group for which life counting is started by the next M06 command.

USING GROUP:
Tool group number for which life counting is currently under way.

SELECTED GROUP:
Tool group number for which life counting is currently under way or life counting has been performed most recently.
COUNT OVERRIDE:
"1.0TIMES" is displayed if the tool life counter override signal is disabled (bit 2 (LFV) of parameter No. 6801 = 0).

NOTE
If arbitrary group numbers are enabled, NEXT GROUP, USING GROUP, and SELECTED GROUP are each represented with an arbitrary group number rather than the tool group number.

- Contents of (B)

(B) details tool life management data related to a selected tool group as follows:

TYPE :

1	Specification by count
2	Specification by duration

LIFE : Tool life value
COUNT : Tool life counter

STATE :

Tool status	Tool in use	Tool not in use
Life remaining	@	No indication
Skip	#	#
Life expired	*	*

T-CODE : Tool number

M

H-CODE : Tool length compensation specification code
D-CODE : Cutter compensation specification code

T

H-CODE : No display.
D-CODE : Tool offset value specification code
if ATC type is in use (bit 3 (TCT) of parameter No. 5040 = 1)

NOTE

1 The tool life counter indicates the count value for the tool indicated with @.

2 If bit 3 (EMD) of parameter No. 6801 = 0, a tool number remains prefixed with @ even if the life of the tool has expired until another tool is selected.

3 If bit 3 (EMD) of parameter No. 6801 = 1, the following differences can occur depending on the type of the life counter in use.

- If the life counter type is specification by duration, the prefix is changed to "*" (life expired) as soon as the tool life expires.
- If the life counter type is specification by count, the counter is incremented by one at the end of a program (such as M02 or M30). So, the prefix is not changed to "*" (life expired) even if the tool life counter matches the life value. The symbol "*" (life expired) appears when the tool life count is incremented after the CNC is reset.

4 If bit 2 (ETE) of parameter No. 6804 = 1, the symbol "*" indicating the expiration of the life of the last tool appears on the tool life management screen when the life counter for the last tool of the tool group of interest matches the life value. This way, the information about the last tool in the FOCAS2 or PMC window indicates that the life of the tool has expired if the tool change signal TLCH <Fn064.0> is 1.

Displaying arbitrary groups and remaining set values

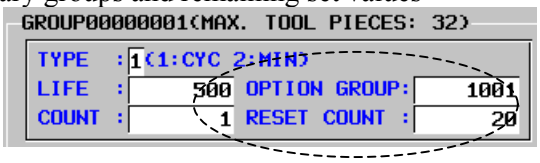


Fig. 12.3.28.2 (b) Displaying arbitrary group numbers and remaining set values on the editing screen

M

If the tool life management B function is enabled (bit 4 (LFB) of parameter No. 6805 = 1) and the following parameters are enabled, arbitrary groups and remaining set values are displayed.

OPTION GROUP:

Arbitrary group number
(if bit 5 (TGN) of parameter No. 6802 = 1)

REST COUNT:

Remaining set value used until a new tool is selected
(if bit 3 (GRP) of parameter No. 6802 = 1)

T

If the tool life management B function is enabled (bit 4 (LFB) of parameter No. 6805 = 1)
Turret type (bit 3 (TCT) of parameter No. 5040 = 0)

- OPTION GROUP:
  No display.
- REST COUNT:
  Remaining set value used until a new tool is selected
  (if bit 3 (GRP) of parameter No. 6802 = 1)

ATC type (bit 3 (TCT) of parameter No. 5040 = 1)

- OPTION GROUP:
  Arbitrary group number
  (if bit 5 (TGN) of parameter No. 6802 = 1)
- REST COUNT:
  Remaining set value used until a new tool is selected
  (if bit 3 (GRP) of parameter No. 6802 = 1)

Setting data on the group editing screen

Tool life management data can be specified in the reset state (both the OP and RST signals are "0"). However, setting bit 1 (TCI) of parameter No. 6804 to "1" enables tool life management data to be specified even during automatic operation (the OP signal is "1").
The available editing operations are listed below.

M

Items that can be edited	Mode
Setting a life count type, life value, tool life counter, and tool data (T code, H code, and D code)	All modes
Adding tool numbers (T code)	MDI
Deleting all tool group data at a time	MDI
Deleting tool data (status, T code, H code, and D code)	MDI
Selecting tool skip	MDI
Specifying to clear tool data (life re-set)	MDI

T

Items that can be edited	Mode
Setting a life count type, life value, tool life counter, and tool data (T code and D code) However, the D code can be edited only if the ATC type is in use (bit 3 (TCT) of parameter No. 5040 = 1).	All modes
Adding tool numbers (T code)	MDI
Deleting all tool group data at a time	MDI
Deleting tool data (status, T code, and D code)	MDI
Selecting tool skip	MDI
Specifying to clear tool data (life re-set)	MDI

M
T

If no tool is registered with a tool group, none of a life count type, a tool life value, and a tool life counter value can be set for the tool group. First of all, add a tool number (T code).

NOTE

1 As for USING GROUP or NEXT GROUP editing:
   <1> During automatic operation (OP signal = “1” and bit 1 (TCI) of parameter No. 6804 = 1), only the tool life counter can be changed.
   <2> In the reset state (OP signal = “0” and RST signal “0”), the following editing operation stops life management because it cannot be continued.
      - Adding tool numbers (T code)
      - Deleting all tool group data at a time
      - Deleting tool data (status, T code, H code, and D code)
2 The following editing operations may set the tool change signal to ”1”.
   - Selecting tool skip for the last tool.
   - Deleting tool numbers, resulting in any tool other than those whose life has expired or who have been skipped being not found in the tool group of interest.
3 The following editing operations may reset the tool change signal to ”0”.
   - Adding tool numbers, leading to tools whose life has not expired being set in the tool group of interest.
   - Selecting tool clear.

Procedure

- Setting a life count type, tool life value, tool life counter, tool data, arbitrary group number, and remaining set value

Setting a life count type, tool life value, tool life counter, tool data, arbitrary group number, and remaining set value

Method 1

1 Place the cursor on the desired item.
2 Enter a value from the keypad.
3 Press horizontal soft key [INPUT].

Method 2

1 Place the cursor on the desired item.
2 Enter a value from the keypad.
3 Press

.

NOTE
1 Changing a tool life value or tool life counter does not affect the tool status or tool change signal.
2 Changing a life count type causes the tool life value and too life counter to be reset to 0.

M

If the tool life management B function is enabled (bit 4 (LFB) of parameter No. 6805 = 1) and the following parameters are enabled, arbitrary groups and remaining set values can be specified

OPTION GROUP:
Arbitrary group number
(if bit 5 (TGN) of parameter No. 6802 = 1)

REST COUNT:
Remaining set value used until a new tool is selected
(if bit 3 (GRP) of parameter No. 6802 = 1)

T

If the tool life management B function is enabled (bit 4 (LFB) of parameter No. 6805 = 1)

Turret type (bit 3 (TCT) of parameter No. 5040 = 0)

- OPTION GROUP:
  Cannot be set.
- REST COUNT:
  It is possible to set a remaining set value to be used until a new tool is selected (if bit 3 (GRP) of parameter No. 6802 = 1).

ATC type (bit 3 (TCT) of parameter No. 5040 = 1)

- OPTION GROUP:
  Arbitrary group numbers can be set (if bit 5 (TGN) of parameter No. 6802 = 1).
- REST COUNT:
  It is possible to set a remaining set value to be used until a new tool is selected (if bit 3 (GRP) of parameter No. 6802 = 1).

- Adding tool numbers

Tool numbers can be added to a tool group as follows:

1 Select the MDI mode.
2 Place the cursor on the tool data (T code, H code, or D code) just before a tool number to be added.
3 Enter the tool number from the key pad.
4 Press horizontal soft key [INSERT].

(Example)

Adding tool number 1550 between numbers 1 and 2 (for the M series)

1 Move the cursor to the data for number 1, enter “1550”, and press [INSERT].

NO.	STATE	T-CODE	H-CODE	D-CODE
01		00010012	000	000
02		00010001	000	000
03		00010114	000	000

2 The entered T code 1550 is inserted in the position of number 2. The H and D codes are reset to 0.

NO.	STATE	T-CODE	H-CODE	D-CODE
01		00010012	000	000
02		00001550	000	000
03		00010001	000	000

- Deleting all tool group data at a time

All tool group data can be deleted at a time as follows:

1 Select the MDI mode.
2 Place the cursor on the tool data (T code, H code, or D code) just Select the tool group from which all tool data you want to delete at a time.
3 Press horizontal soft key [DELETE].
4 Press horizontal soft key [GROUP].
5 Press horizontal soft key [EXEC].

- Deleting tool data

Tool data can be deleted from a tool group as follows:

1 Select the MDI mode.
2 Place the cursor on the tool data (T code, H code, or D code) you want to delete.
3 Press horizontal ssoft key [DELETE].
4 Press horizontal ssoft key [<CURS>].

NOTE
1 Deleting all tools from a tool group is equivalent to deleting the tool group itself.
2 Deleting a tool indicated with @ (in use) results in @ being moved to the previous tool whose life has expired or which has been skipped.

- Selecting tool skip

Tool data can be placed in a skip state as follows:

1 Select the MDI mode.
2 Place the cursor on the tool data (T code, H code, or D code) for a tool you want to skip.
3 Press horizontal soft key [STATE].
4 Press horizontal soft key [SKIP].

- Specifying to clear tool data (life re-set)

Tool data state can be cleared as follows:

1 Select the MDI mode.
2 Place the cursor on the tool data (T code, H code, or D code) for a tool you want to clear.
3 Press horizontal soft key [STATE].
4 Press horizontal soft key [CLEAR].

- Selecting a tool group

A tool group can be selected as follows:

Method 1

1 Enter a tool group number from the keypad.
2 Press horizontal soft key [NO.SRH].

NOTE
If arbitrary group numbers are enabled, a tool group is selected by searching for an arbitrary group number rather than the tool group number.

Method 2

1 Press page key or to display the target tool group.

- Switching to the list screen

The tool life management (list screen) can be resumed as follows:

1 Press horizontal soft key [END].

12.3.29 Displaying and Setting Workpiece Setting Error Compensation Data (15-inch Display Unit)

An amount of error used in workpiece setting error compensation can be set on the workpiece setting error screen.
The workpiece setting error screen is displayed as follows:

1 Press function key .

2 Press vertical soft key [WORK SET ER].
The workpiece setting error screen appears.

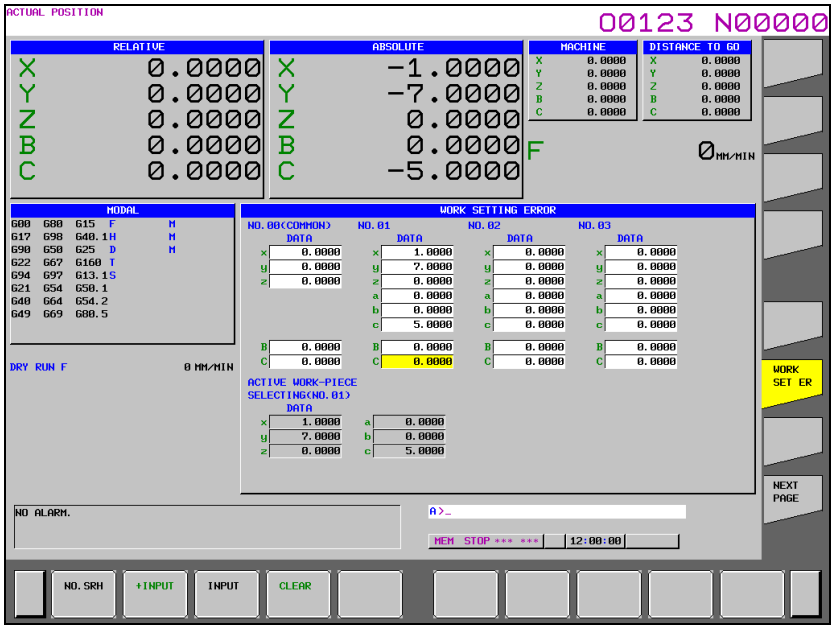


Fig. 12.3.29 (a) Workpiece setting error screen (15-inch)

Up to 7 sets (No. 01 to No. 07) of different amounts of workpiece setting error can be specified. In addition, it is possible to specify an amount of error (No. 00 (COMMON)) to be added to an error in the 7 sets in common.
The X-direction error Δx, Y-direction error Δy, and Z-direction error Δz are specified, respectively, in the x, y, and z fields. The rotational errors Δa, Δb, and Δc are specified, respectively, in the a, b, and c fields.
Table rotation axis positions 1 and 2 are specified in the fields under the a, b, and c fields (C and B in the figure).
Rotation axis names specified in parameter Nos. 19681 and 19686 are displayed in the axis name fields.
Table rotation axis positions 1 and 2 are specified for rotation axes. No position must be specified for axes (including virtual axes) other than table rotation axes. So, no such item is displayed for setting.
Table rotation axis position setting items are displayed under the following conditions.

- It must have been correctly specified which axis is a table rotation axis.
  If parameter No. 19680 = 12 (table rotation type), axis numbers must have been specified correctly in parameter Nos. 19681 and 19686.
  If parameter No. 19680 = 21 (mixed type), an axis number must have been specified correctly in parameter No. 19686.
- Bit 0 of parameter No. 1006 must be 1 (rotation axis) for the axes specified above.

The following soft key can be used in data setting.

- After a workpiece setting error number you want to display is entered, pressing horizontal soft key [NO.SRH] causes a setting screen for the target workpiece setting error to appear.
- After a number is entered, pressing horizontal soft key [+INPUT] causes the number to be added.
- After a number is entered, pressing horizontal soft key [INPUT] causes the number to be set.
- Pressing horizontal soft key [CLEAR] causes the next soft key to appear.

- After a workpiece setting error number to be deleted is entered, pressing horizontal soft key [NO.] deletes the data corresponding to the workpiece setting error number.
- Pressing horizontal soft key [ALL] deletes data for all workpiece setting error numbers.

12.3.30 Displaying and Setting Pattern Data Inputs (15-inch Display Unit)

Described below are a method for displaying machining menus (pattern menus) created by machine tool builders and a method for setting them. The descriptions are based on an example. For actual pattern menus and pattern data, refer to manuals released by respective machine tool builders.

Displaying pattern data and pattern menus

Given below is a procedure for displaying pattern menus.

1 Press function key .

2 Press vertical soft key [NEXT PAGE] several times.

3 Press vertical soft key [PATTERN MENU].
The pattern menu screen shown below appears.

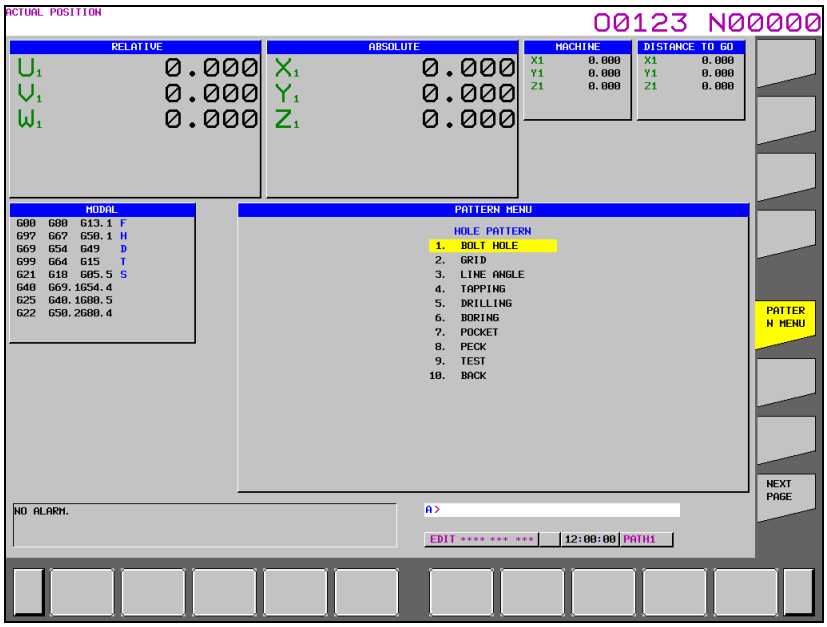


Fig. 12.3.30 (a) Pattern menu screen (15-inch)

On this screen, a pattern to be used can be selected.
The following two methods can be used to select patterns.

- Using the cursor
  Move the cursor to a pattern name you want to select, using cursor key ↑ or ↓, and then press soft key [SELECT] or [INPUT].
- Specifying a pattern number
  Enter a number displayed at the left side of a pattern name, and press soft key [SELECT] or [INPUT].

The custom macro screen (pattern data screen) shown below appears.

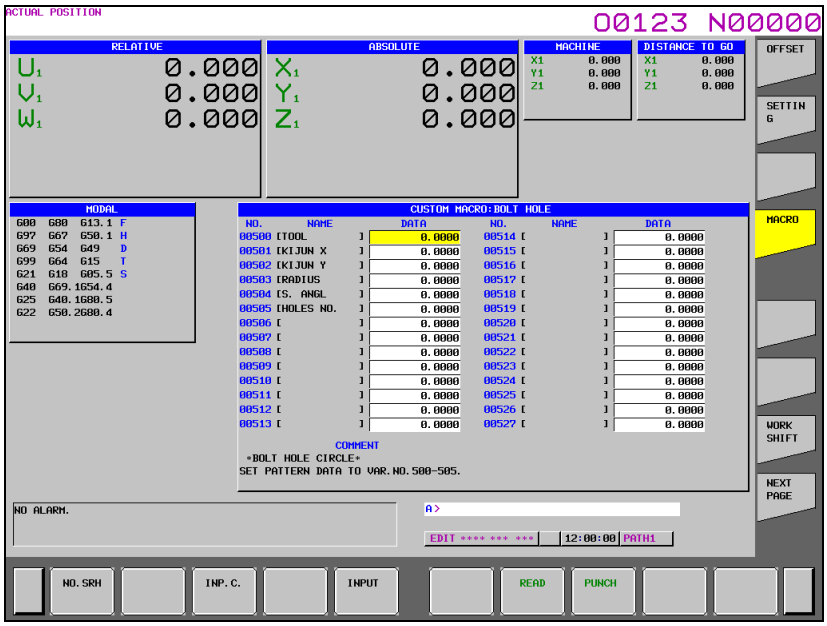


Fig. 12.3.30 (b) Custom macro screen (pattern data) (15-inch)

Enter the necessary pattern data, and press [INPUT].

After entering all necessary data, select the MEMORY mode and press the cycle start button. Machining begins.

Explanation

- Explanations about the pattern menu screen

HOLE PATTERN

An arbitrary character string consisting of 12 or less characters can be displayed as a menu title.

BOLT HOLE

An arbitrary character string consisting of 10 or less characters can be displayed as a pattern name.

Machine tool builders should program menu title and pattern name character strings, using custom macros, and save them in program memory.

- Explanations about the custom macro screen (pattern data screen)

BOLT HOLE

An arbitrary character string consisting of 12 or less characters can be displayed as a pattern data title.

TOOL.

An arbitrary character string consisting of 10 or less characters can be displayed as a variable name.
BOLT HOLE
The comment text consists of nine 12-character blocks, or the comment can be up to 12 lines (10.4-inch display) or 8 lines (7.2 or 8.4-inch display) with one block counted as one line.

Machine tool builders should program variable name and comment text character strings, using custom macros, and save them in program memory.

12.4 SCREENS DISPLAYED BY FUNCTION KEY SYSTEM

When the CNC and machine are connected, parameters must be set to determine the specifications and functions of the machine in order to fully utilize the characteristics of the servo motor or other parts.
This chapter describes how to set parameters on the MDI panel. Parameters can also be set with external input/output devices such as the memory card (see III-8).
In addition, pitch error compensation data used for improving the precision in positioning with the ball screw on the machine can be set or displayed by the operations under function key SYSTEM.

See III-7 for the diagnosis screens displayed by pressing function key SYSTEM.

Section 12.4, "SCREENS DISPLAYED BY FUNCTION KEY SYSTEM", consists of the following subsections:

———— Screens of a 7.2/8.4/10.4-inch display unit
12.4.1 Displaying and Setting Parameters 1723
12.4.2 Displaying and Setting Pitch Error Compensation Data....1726
12.4.3 Displaying and Setting Three-Dimensional Error Compensation Data 1729
12.4.4 Servo Parameters.....1733
12.4.5 Servo Tuning.....1734
12.4.6 Spindle Setting 1735
12.4.7 Spindle Tuning.....1736
12.4.8 Spindle Monitor 1737
12.4.9 Color Setting Screen 1738
12.4.10 Machining Parameter Tuning.....1741
12.4.11 Displaying Memory Data.....1749
12.4.12 Parameter Tuning Screen 1751
12.4.13 Periodic Maintenance Screen.....1768
12.4.14 System Configuration Screen.....1776
12.4.15 Overview of the History Function 1779
———— Screens of a 15-inch display unit
12.4.16 Displaying and Setting Parameters (15-inch Display Unit)1802
12.4.17 Displaying and Setting Pitch Error Compensation Data (15-inch Display Unit) 1805
12.4.18 Displaying and Setting Three-Dimensional Error Compensation Data (15-inch Display Unit).....1808
12.4.19 Servo Parameters (15-inch Display Unit) 1812
12.4.20 Servo Tuning (15-inch Display Unit) 1813
12.4.21 Spindle Setting (15-inch Display Unit).....1814
12.4.22 Spindle Tuning (15-inch Display Unit) 1815
12.4.23 Spindle Monitor (15-inch Display Unit).....1816
12.4.24 Color Setting Screen (15-inch Display Unit).....1817
12.4.25 Machining Parameter Tuning (15-inch Display Unit) 1820

12.4.25 Machining Parameter Tuning (15-inch Display Unit) 1820
12.4.26 Displaying Memory Data (15-inch Display Unit) 1828
12.4.27 Parameter Tuning Screen (15-inch Display Unit).....1830
12.4.28 Periodic Maintenance Screen (15-inch Display Unit) 1847
12.4.29 System Configuration Screen (15-inch Display Unit) 1855
12.4.30 Overview of the History Function (15-inch Display Unit) 1858

Screens of a 7.2/8.4/10.4-inch display unit

12.4.1 Displaying and Setting Parameters

When the CNC and machine are connected, parameters are set to determine the specifications and functions of the machine in order to fully utilize the characteristics of the servo motor. The setting of parameters depends on the machine. Refer to the parameter list prepared by the machine tool builder.
Normally, the user need not change parameter setting.

Procedure for displaying and setting parameters

Procedure

1 Set 1 for PARAMETER WRITE to enable writing. See the procedure for enabling/disabling parameter writing described below.

2 Press function key [SYSTEM].

3 Press chapter selection soft key [PARAM] to display the parameter screen.

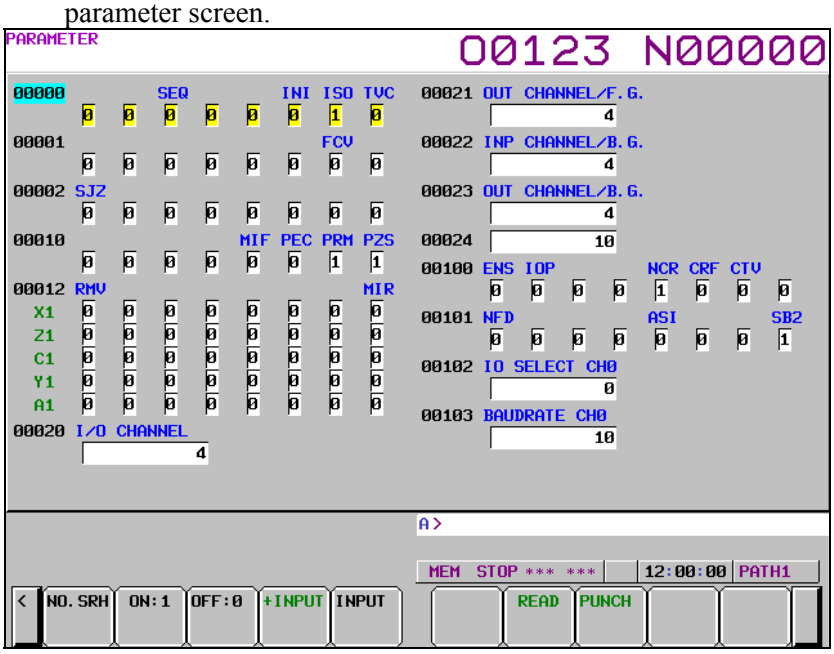


Fig. 12.4.1 (a) PARAMETER screen (10.4-inch)

4 Move the cursor to the parameter number to be set or displayed in either of the following ways:

- Enter the parameter number and press soft key [NO.SRH] .
- Move the cursor to the parameter number using the page keys, [PAGE↑] and [PAGE↓], and cursor keys, [↑], [↓], [←], and [→].

5 To set the parameter, enter a new value with numeric keys and press soft key [INPUT]. The parameter is set to the entered value and the value is displayed.

6 Set 0 for PARAMETER WRITE to disable writing.

Procedure for enabling/displaying parameter writing

Procedure

1 Select the MDI mode or enter state emergency stop.

2 Press function key .

3 Press soft key [SETTING] to display the setting screen.

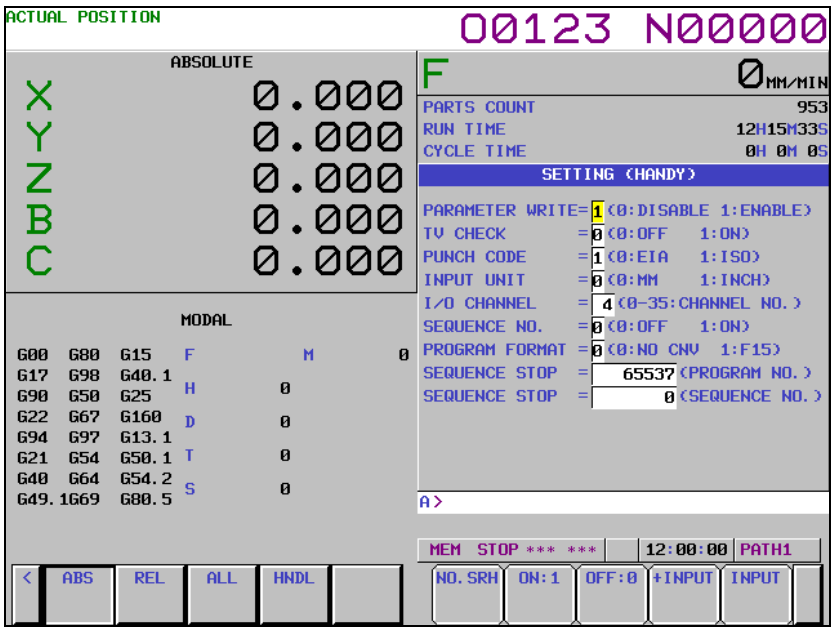


Fig. 12.4.1 (b) SETTING screen (10.4-inch)

4 Move the cursor to PARAMETER WRITE using cursor keys.

5 Press soft key [(OPRT)], then press [ON:1] to enable parameter writing.
At this time, the CNC enters the alarm state SW0100.

6 After setting parameters, return to the setting screen. Move the cursor to PARAMETER WRITE and press soft key [(OPRT)] , then press [OFF:0].

7 Press the key to release the alarm condition.
When an alarm PW0000 occurred, however, it is not released unless the power is turned off and back on.

Explanation

- Setting parameters with external input/output devices

See III-8 for setting parameters with external input/output devices such as the memory card.

- Parameters that require turning off the power

Some parameters are not effective until the power is turned off and on again after they are set. Setting such parameters causes alarm PW0000. In this case, turn off the power, then turn it on again.

- Parameter list

Refer to the Parameter Manual (B-63950EN) for the parameter list.

- Setting data

Some parameters can be set on the setting screen if the parameter list indicates "Setting entry is acceptable". Setting 1 for PARAMETER WRITE is not necessary when these parameters are set on the setting screen.

12.4.2 Displaying and Setting Pitch Error Compensation Data

If pitch error compensation data is specified, pitch errors of each axis can be compensated in detection unit per axis.

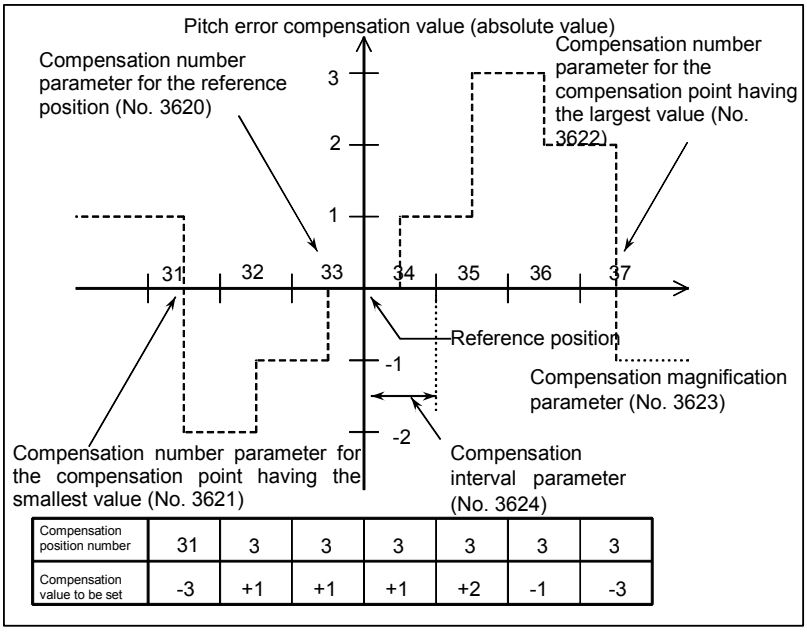
Pitch error compensation data is set for each compensation point at the intervals specified for each axis. The origin of compensation is the reference position to which the tool is returned.

The pitch error compensation data is set according to the characteristics of the machine connected to the NC. The content of this data varies according to the machine model. If it is changed, the machine accuracy is reduced. In principle, the end user must not alter this data.

Pitch error compensation data can be set with external devices such as the memory card (see Chapter III-8). Compensation data can also be written directly with the MDI panel.

The following parameters must be set for pitch error compensation. Set the pitch error compensation value for each pitch error compensation point number set by these parameters.

In the following example, 33 is set for the pitch error compensation point at the reference position.



Compensation position number	31	3	3	3	3	3	3
Compensation value to be set	-3	+1	+1	+1	+2	-1	-3

- Number of the pitch error compensation point at the reference position (for each axis): Parameter No. 3620
- Number of the pitch error compensation point having the smallest value (for each axis): Parameter No. 3621
- Number of the pitch error compensation point having the largest value (for each axis): Parameter No. 3622
- Pitch error compensation magnification (for each axis): Parameter No. 3623
- Interval of the pitch error compensation points (for each axis): Parameter No. 3624

- Travel distance per revolution of pitch error compensation of the rotary axis type (for each axis): Parameter No. 3625

- Bi-directional pitch error compensation

The bi-directional pitch error compensation function allows independent pitch error compensation in different travel directions. (When the movement is reversed, compensation is automatically carried out as in a backlash.)
To use this function, specify pitch error compensation for each travel direction, that is, separately for the positive and negative directions of a movement.
When using bi-directional pitch error compensation (setting bit 0 (BDPx) of parameter No. 3605 to 1), specify the following parameters in addition to the pitch error compensation parameter.

- Number of the pitch error compensation point at the negative end (for travel in the positive direction, for each axis): Parameter No. 3621
- Number of the pitch error compensation point at the positive end (for travel in the positive direction, for each axis): Parameter No. 3622
- Number of the pitch error compensation point at the negative end (for travel in the negative direction, for each axis): Parameter No. 3626
- Pitch error compensation in the reference position when moving to the reference position from opposite to the reference position return direction (for each axis): Parameter No. 3627

Procedure for displaying and setting the pitch error compensation data

Procedure

1 Set the following parameters:
- Number of the pitch error compensation point at the reference position (for each axis): Parameter No. 3620
- Number of the pitch error compensation point having the smallest value (for each axis): Parameter No. 3621
- Number of the pitch error compensation point having the largest value (for each axis): Parameter No. 3622
- Pitch error compensation magnification (for each axis): Parameter No. 3623
- Interval of the pitch error compensation points (for each axis): Parameter No. 3624
- Travel distance per revolution of pitch error compensation of the rotary axis type (for each axis): Parameter No. 3625

When using bi-directional pitch error compensation (setting bit 0 (BDPx) of parameter No. 3605 to 1), specify the following parameters in addition to the pitch error compensation parameter.
- Number of the pitch error compensation point at the negative end (for travel in the positive direction, for each axis): Parameter No. 3621

- Number of the pitch error compensation point at the positive end (for travel in the positive direction, for each axis): Parameter No. 3622
- Number of the pitch error compensation point at the negative end (for travel in the negative direction, for each axis): Parameter No. 3626
- Pitch error compensation in the reference position when moving to the reference position from opposite to the reference position return direction (for each axis): Parameter No. 3627

2 Press function key [SYSTEM].

3 Press the continuous menu key [▷], then press chapter selection soft key [PITCH ERROR]. The following screen is displayed:

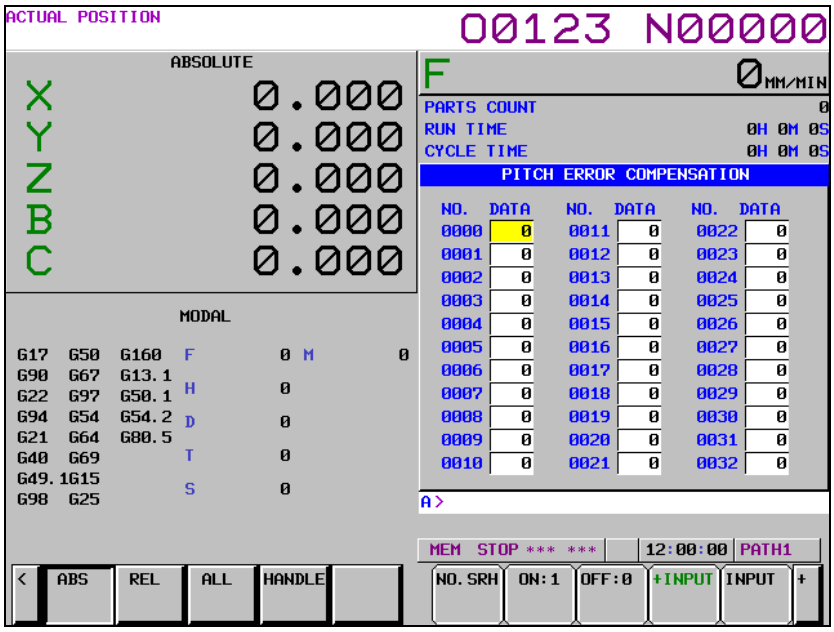


Fig. 12.4.2 (a) PITCH ERROR COMPENSATION screen (10.4-inch)

4 Move the cursor to the compensation point number to be set in either of the following ways:

- Enter the compensation point number and press the soft key [NO.SRH].
- Move the cursor to the compensation point number using the page keys, [PAGE↑] and [PAGE↓], and cursor keys, [↑], [↓], [←], and [→].

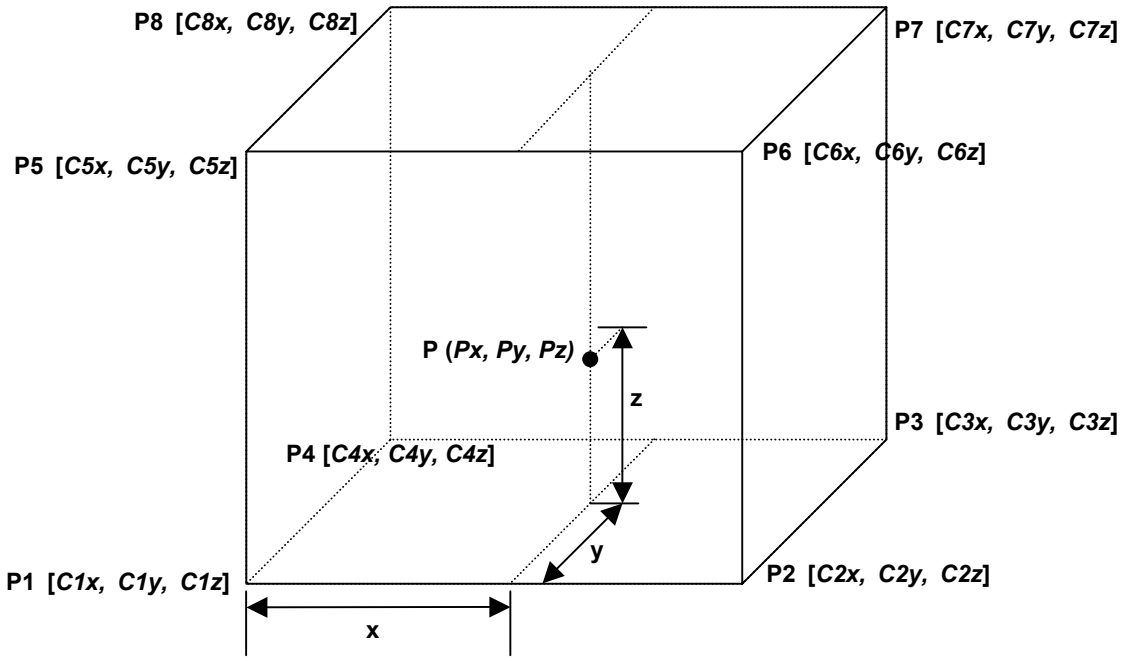
5 Enter a value with numeric keys and press soft key [INPUT].

12.4.3 Displaying and Setting Three-Dimensional Error Compensation Data

In ordinary pitch error compensation, compensation is applied to a specified compensation axis (single axis) by using its position information. For example, pitch error compensation is applied to X-axis by using the position information of X-axis.
Three-dimensional error compensation is a function that adjusts the current position by calculating compensation data (for three axes) from the compensation amounts at surrounding compensation points (eight points) on the basis of the interior division ratio in the compensation area (rectangular parallelepiped) containing the current position on up to three compensation axes.

- Calculation of compensation

Three-dimensional error compensation is calculated as follows.



Suppose three compensation axes to be X, Y, and Z (three basic axes) and the coordinates of the current position be P (Px, Py, Pz). consider a compensation space (rectangular parallelepiped) containing P. Let its vertexes be P1, P2, …, and P8 and the compensation values for the individual axes at the individual vertexes be Cnx, Cny, and Cnz (where n is a number between 1 and 8).
Let the interior division ratio on X-axis at P be x. Here, x is standardized in the range of 0 to 1 as follows:

$$x = \frac{|Px - P1x|}{|P2x - P1x|}$$

P1x and P2x are the X coordinates of P1 and P2. The interior division ratios on Y and Z-axes are determined in the same way.

The compensation amount Cx for X-axis at P is determined as follows:

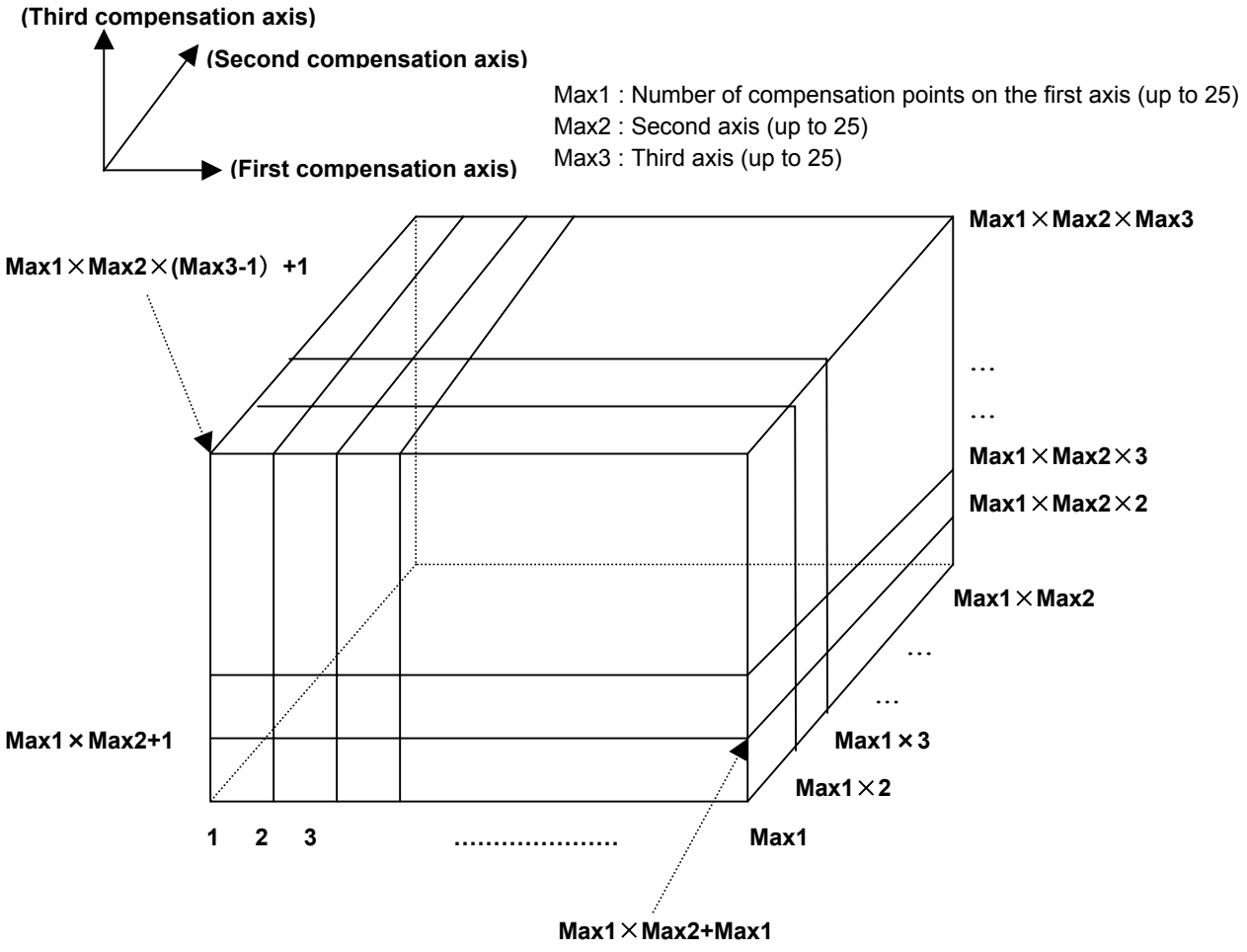
$$\begin{aligned} Cx = {} & C1x \times (1-x) \times (1-y) \times (1-z) + C2x \times x \times (1-y) \times (1-z) \\ & + C3x \times x \times y \times (1-z) + C4x \times (1-x) \times y \times (1-z) \\ & + C5x \times (1-x) \times (1-y) \times z + C6x \times x \times (1-y) \times z \\ & + C7x \times x \times y \times z + C8x \times (1-x) \times y \times z \end{aligned}$$

The compensation amount Cy and Cz on Y and Z-axes are determined in the same way.

The actual compensation amounts are the calculated compensation amounts multiplied by the compensation magnifications (Parameter No.10809 to 10811).

- Number of compensation points

Up to 15625 compensation points (up to 25 points per axis) can be set. The numbers of compensation points on the individual axes are set for parameters No.10803 to 10805. The ordering of the compensation point numbers in the compensation space is as follows.



Displaying and setting three-dimensional error compensation data

Procedure

1 Set the following parameters:
- First compensation axis for three-dimensional error compensation : Parameter No. 10800
- Second compensation axis for three-dimensional error compensation : Parameter No. 10801
- Third compensation axis for three-dimensional error compensation : Parameter No. 10802
- Number of compensation points for three-dimensional error compensation (first compensation axis)
  Parameter No. 10803
- Number of compensation points for three-dimensional error compensation (second compensation axis)
  Parameter No. 10804
- Number of compensation points for three-dimensional error compensation (third compensation axis)
  Parameter No. 10805
- Compensation point number of the reference position for three-dimensional error compensation (first compensation axis) : Parameter No. 10806
- Compensation point number of the reference position for three-dimensional error compensation (second compensation axis) : Parameter No. 10807
- Compensation point number of the reference position for three-dimensional error compensation (third compensation axis) : Parameter No. 10808
- Magnification for three-dimensional error compensation (first compensation axis) : Parameter No. 10809
- Magnification for three-dimensional error compensation (second compensation axis) : Parameter No. 10810
- Magnification for three-dimensional error compensation (third compensation axis) : Parameter No. 10811
- Compensation interval for three-dimensional error compensation (first compensation axis)
  Parameter No. 10812
- Compensation interval for three-dimensional error compensation (second compensation axis)
  Parameter No. 10813
- Compensation interval for three-dimensional error compensation (third compensation axis)
  Parameter No. 10814

2 Press function key

.

3 Press the continuous menu key ▷ several times, then press chapter selection soft key [3D ERR COMP]. The following screen appears:

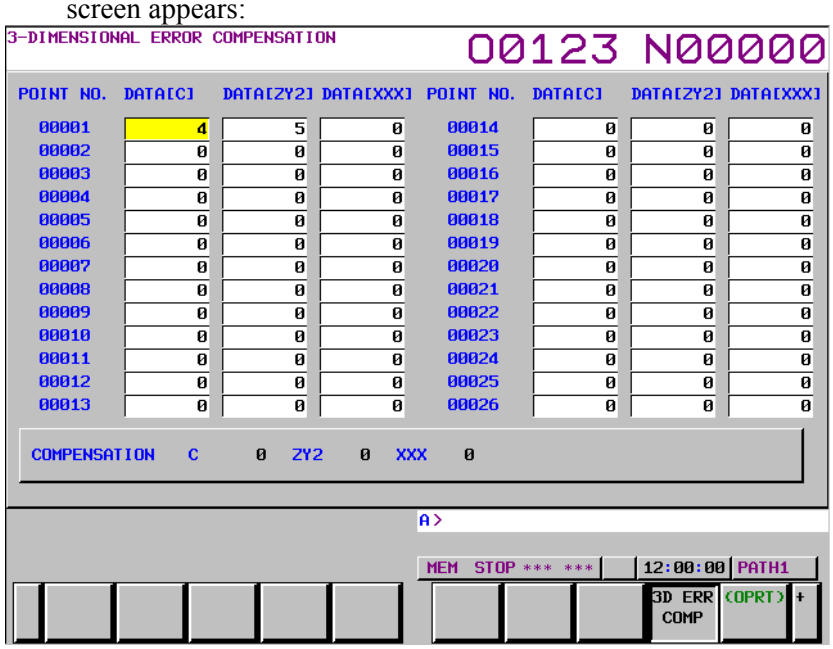


Fig. 12.4.3 (a) 3-DIMENSIONAL ERROR COMPENSATION screen (10.4-inch)

4 Move the cursor to the position of the compensation point number you want to set using either of the following methods:

- Select the MDI mode.
- Set bit 0 (PWE) of parameter No. 8900 to 1.
- Enter a compensation point number and press soft key [NO. SRH].
- Move the cursor to the compensation point number or compensation axis you want to set by pressing page keys PAGE↑ and/or PAGE↓, and cursor keys ↑, ↓, ←, and →.

5 Enter compensation data in detection units. The valid data range is from -128 to 127.

12.4.4 Servo Parameters

This subsection describes the initialization of digital servo parameters performed, for example, at the time of field tuning of the machine tool.

Procedure for servo parameter setting

Procedure

1 Turn on the power in the emergency stop state.
2 Set bit 0 (SVS) of parameter No. 3111 to 1 to display servo setting and tuning screens.
3 Press function key SYSTEM, continuous menu key ▷, then soft key [SERVO PARAM].
4 Press soft key [SERVO PARAM] to select the servo parameter screen.
The following screen appears:

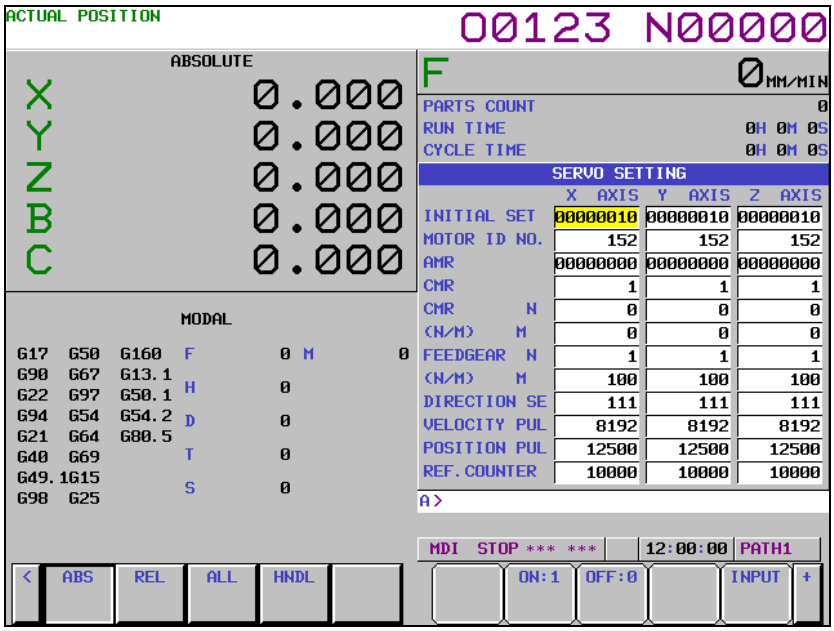


Fig. 12.4.4 (a) Servo parameter setting screen (10.4-inch)

5 With the page keys and cursor keys, move the cursor to the position of data to be set or modified.
6 Key in a desired value then press soft key [INPUT].

12.4.5 Servo Tuning

Data related to servo tuning is displayed and set.

Procedure for servo tuning

Procedure

1 Turn on the power in the emergency stop state.
2 Set bit 0 (SVS) of parameter No. 3111 to 1 to display servo setting and tuning screens.
3 Press function key SYSTEM, continuous menu key ▷, then soft key [SERVO PARAM].
4 Press soft key [SERVO TUNE] to select the servo tuning screen.
5 Use the cursor and page keys to enter necessary data for initial setting. The following screen appears:

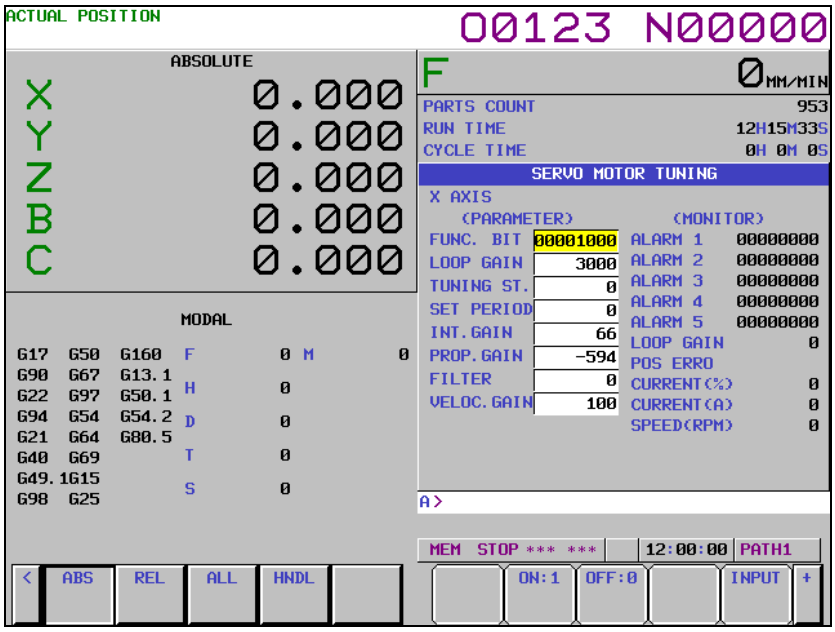


Fig. 12.4.5 (a) Servo tuning screen (10.4-inch)

6 With the page keys and cursor keys, move the cursor to the position of data to be set or modified.
7 Key in a desired value then press soft key [INPUT].

12.4.6 Spindle Setting

Parameters related to spindles are set and displayed. In addition to the parameters, related data can be displayed. Screens for spindle setting, spindle tuning, and spindle monitoring are provided.

Setting spindle parameters

Procedure

1 Set bit 1 (SPS) of parameter No. 3111 to 1 to display spindle setting and tuning screens.

2 Press function key SYSTEM, continuous menu key ▷, then soft key [SPINDL SETING].

3 Press soft key [SPINDL SETING] to select the spindle setting screen.
The following screen appears:

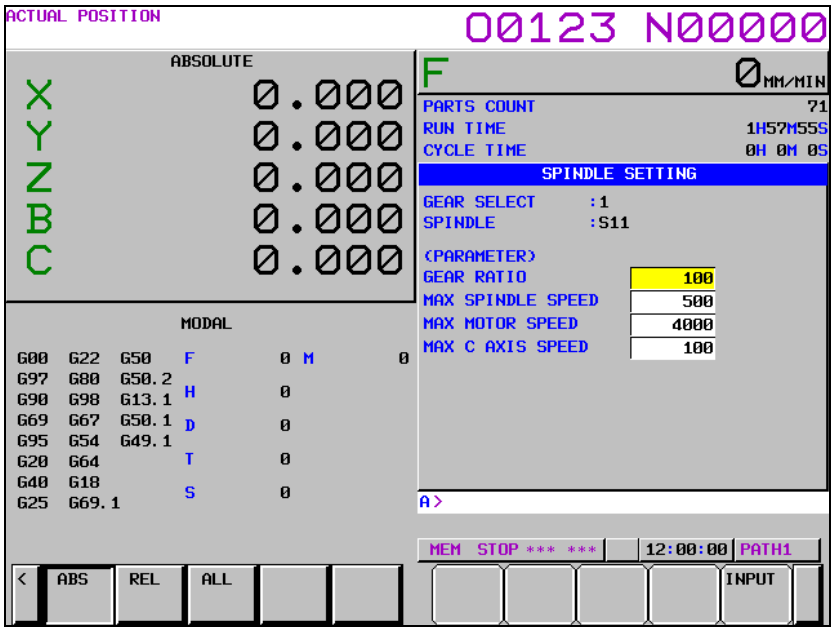


Fig. 12.4.6 (a) Spindle setting screen (10.4-inch)

4 With the page keys and cursor keys, move the cursor to the position of data to be set or modified.

5 Key in a desired value then press soft key [INPUT].

12.4.7 Spindle Tuning

Spindle tuning data is displayed and set.

Setting for spindle tuning

Procedure

1 Set bit 1 (SPS) of parameter No. 3111 to 1 to display spindle setting and tuning screens.
2 Press function key SYSTEM, continuous menu key ▷, then soft key [SPINDL SETING].
3 Press soft key [SPINDL TUNE] to select the spindle tune screen.
4 The following screen appears:

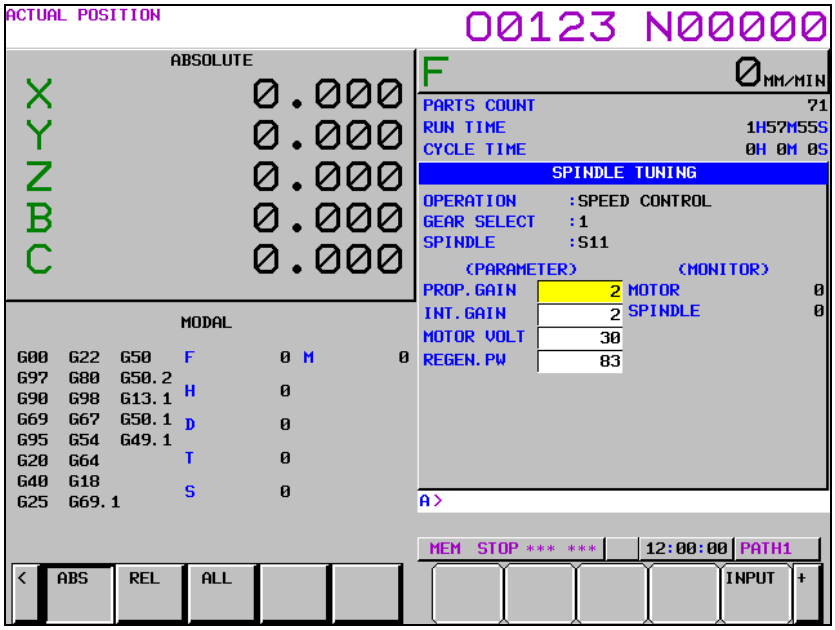


Fig. 12.4.7 (a) Spindle tuning screen (10.4-inch)

5 With the page keys and cursor keys, move the cursor to the position of data to be set or modified.
6 Key in a desired value then press soft key [INPUT].

12.4.8 Spindle Monitor

Spindle-related data is displayed.

Displaying the spindle monitor

Procedure

1 Set bit 1 (SPS) of parameter No. 3111 to 1 to display spindle setting and tuning screens.
2 Press function key SYSTEM, continuous menu key ▷, then soft key [SPINDL SETING].
3 Press soft key [SP.MON] to select the spindle monitor screen.
4 The following screen appears:

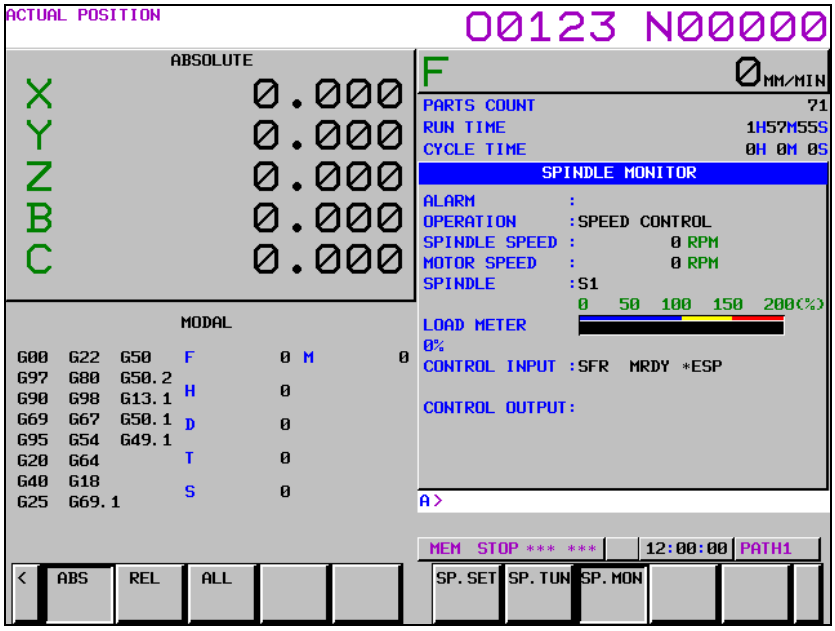


Fig. 12.4.8 (a) Spindle monitor screen (10.4-inch)

12.4.9 Color Setting Screen

Screen colors can be set on the color setting screen.

Displaying the color setting screen

Procedure

1 Press function key SYSTEM.
2 Press the continuous menu key ▷ several times to display soft key [COLOR].
3 Press soft key [COLOR] to display the color setting screen.

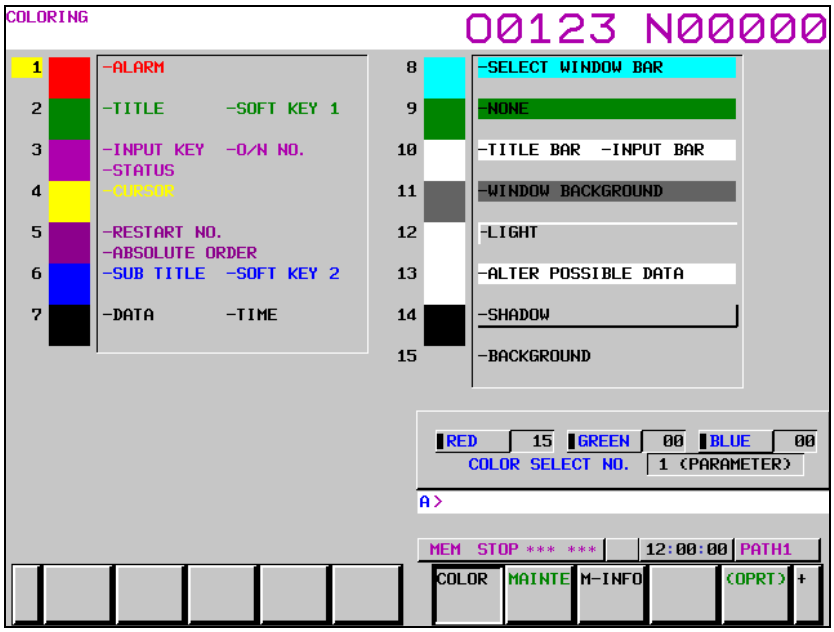
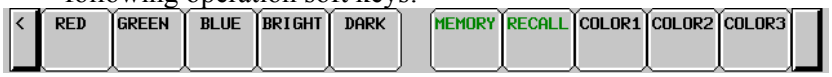


Fig. 12.4.9 (a) Color setting screen (10.4-inch)

Procedure for operating the color setting screen

- Modifying the color (color palette values)

1 Press soft key [(OPRT)]. The soft key display changes to the following operation soft keys:



2 Move the cursor to a color number whose color palette values are to be modified.
The current color palette value set for each primary color is displayed.

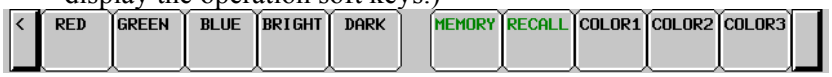
3 Select a primary color whose setting is to be modified, with the corresponding operation soft key [RED], [GREEN], or [BLUE].
More than one primary color can be selected at the same time.
Each time operation soft key [RED], [GREEN], or [BLUE] is pressed, the operation soft key toggles between selection and deselection.
(When operation soft keys [RED], [GREEN], and [BLUE] are not displayed, press the rightmost soft key to display the operation soft keys.)

4 Select operation soft key [BRIGHT] or [DARK] to modify the brightness of the selected prime color(s).

- Storing the color (color palette values)

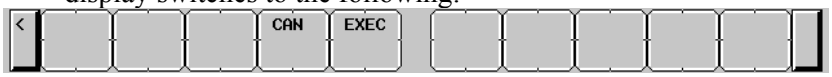
The set color palette values can be stored.

1 Press operation soft key [COLOR1], [COLOR2], or [COLOR3] to select a storage area.
(When operation soft keys [COLOR1], [COLOR2], and [COLOR3] are not displayed, press the rightmost soft key to display the operation soft keys.)



COLOR1 Standard color data parameters Nos. 6581 to 6595
COLOR2 Parameters Nos. 10421 to 10435
COLOR3 Parameters Nos. 10461 to 10475

2 Press operation soft key [MEMORY]. The operation soft key display switches to the following:

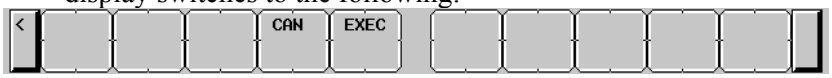


3 Pressing operation soft key [EXEC] stores the current color palette settings in the selected area.
Pressing operation soft key [CAN] or the leftmost key does not store the current color palette settings in the selected area.

- Calling the color (color palette values)

1 Press operation soft key [COLOR1], [COLOR2], or [COLOR3] to select a storage area where color palette values are stored.
(When operation soft keys [COLOR1], [COLOR2], and [COLOR3] are not displayed, press the rightmost soft key to display the operation soft keys.)

2 Press operation soft key [RECALL]. The operation soft key display switches to the following:



3 Pressing operation soft key [EXEC] calls the color palette values from the selected area to enable the color to be modified. This operation is invalid if no color palette values are stored.
Pressing operation soft key [CAN] or the leftmost key does not call the color palette values from the selected area.

NOTE
1 Immediately after the power is turned on, the settings of COLOR1 (parameters) are used for display. If nothing is stored for COLOR1, the default colors are used for display.
2 Do not modify the color setting data parameters directly by MDI key input. When modifying the standard color data, be sure to perform a storage operation on the color setting screen.

12.4.10 Machining Parameter Tuning

In AI contour control, by setting a velocity-emphasized parameter set and precision-emphasized parameter set and setting the precision level matching a machining condition such as rough machining or finish machining on the precision level setting screen or by programming, the parameters suitable for the condition can be automatically calculated to perform machining.
On this screen, the parameter sets for emphasis on velocity (precision level 1) and emphasis on precision (precision level 10) can be set.
Set the following parameters:

- Acceleration rate of acceleration/deceleration before interpolation
- Acceleration change time (bell-shaped)
- Allowable acceleration change value for each axis in velocity control based on acceleration change under jerk control
- Allowable acceleration change value for each axis in acceleration change under jerk control in successive linear interpolation operations
- Ratio of the change time of the rate of change of acceleration in smooth bell-shaped acceleration/deceleration before interpolation
- Allowable acceleration rate
- Acceleration rate of acceleration/deceleration after interpolation
- Corner speed difference
- Maximum feedrate
- Items that can be set freely (2 items)

For details of each parameter, see the descriptions of AI contour control and jerk control.
By setting bit 0 (MPR) of parameter No. 13601 to 1, this screen can be hidden.
For the method of setting a precision level, see the description of the precision level selection screen in Subsection 11.2.14.

Procedure for machining parameter tuning

Procedure

1 Set the MDI mode.
2 Press function key

.
3 Press soft key [M-TUN] to display the machining parameter tuning screen.

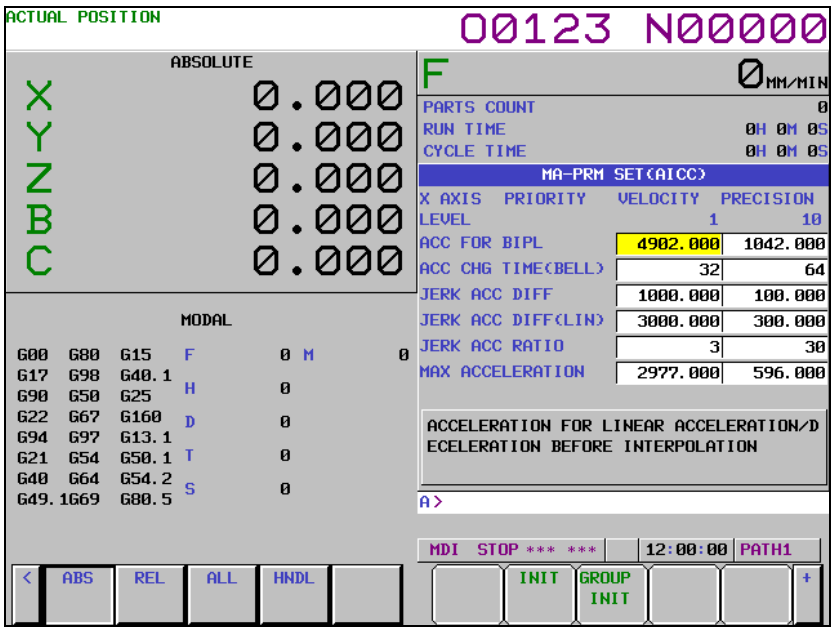


Fig. 12.4.10 (a) Machining parameter tuning screen (10.4-inch)

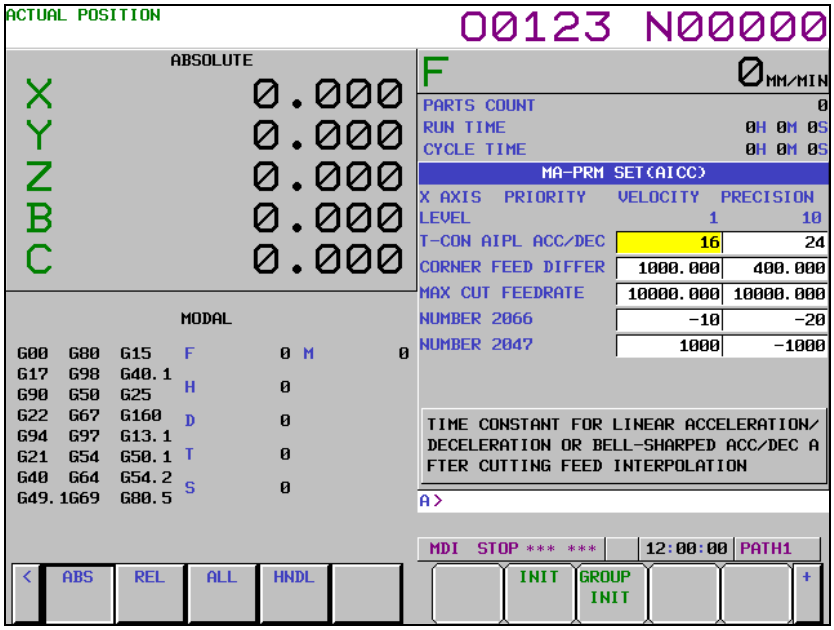


Fig. 12.4.10 (b) Machining parameter tuning screen (10.4-inch)

4 Move the cursor to the position of a parameter to be set, as follows:

Press page key PAGE or PAGE, and cursor keys ↑, ↓, ←, and /or → to move the cursor to the parameter.

5 Key in desired data then press the INPUT key on the MDI panel.

6 When data is input, a RMS value is found according to the precision level parameters. (The precision level can be changed on the precision level selection screen or parameter setting screen.) If a RMS value calculation fails, a warning (indicating that automatic setting failed) is displayed.

7 Repeat steps 4 and 5 until all machining parameters are set.
8 In addition to the setting method described above, a parameter setting method using soft keys is available. Pressing soft key [INIT] displays the standard value (recommended by FANUC) of the item selected by the cursor in the key input buffer. Pressing soft key [EXEC] initializes the item to the standard value. Pressing soft key [GROUP INIT] initializes all items of a group (emphasis on velocity or emphasis on precision) selected by the cursor to the standard values.

The table below indicates the initial settings.

Table 12.4.10 (a) Initial settings

Setting item	AI contour control		Unit
	Emphasis on velocity (LV1)	Emphasis on precision (LV10)	
Acceleration rate of acceleration/deceleration before interpolation <ACC FOR BIPL>	4902.000	1042.000	mm/sec^2
Acceleration change time (bell-shaped) <ACC CHG TIME(BELL)>	32	64	msec
Allowable acceleration change value <JERK ACC DIFF>	0	0	mm/sec^2
Allowable acceleration change value in successive linear interpolation operations <JERK ACC DIFF(LIN)>	0	0	mm/sec^2
Ratio of the change time of the jerk control <JERK ACC RATIO>	0	0	%
Allowable acceleration rate <MAX AACELERATION>	2977.000	596.000	mm/sec^2
Time constant for acceleration/deceleration after interpolation <T-CON AIPL ACC/DEC>	24	24	msec
Corner speed difference <CORNER FEED DIFFER>	1000	400	mm/min
Maximum cutting speed <MAX CUT FEEDRATE>	10000	10000	mm/min

Explanation

- Look-ahead acceleration/deceleration before interpolation

Set an acceleration rate for a linear portion in look-ahead acceleration/deceleration before interpolation.
Unit of data: mm/sec^2, inch/sec^2, deg/sec^2 (machine unit)

The parameter set on the machining parameter tuning screen is reflected in the following parameters:
Parameter No. 13610 (velocity-emphasized parameter)
Parameter No. 13611 (precision-emphasized parameter)

Moreover, the following parameter is also set according to the precision level:
Parameter No. 1660: Maximum allowable acceleration rate for each axis in acceleration/deceleration before interpolation

- Acceleration change time (bell-shaped)

Set a time constant for a bell-shaped portion in acceleration/ deceleration before look-ahead interpolation.
Unit of data: ms

The parameter set on the machining parameter tuning screen is reflected in the following parameters:
Parameter No. 13612 (velocity-emphasized parameter)
Parameter No. 13613 (precision-emphasized parameter)

Moreover, the following parameter is also set according to the precision level:
Parameter No. 1772: Time constant for bell-shaped look-ahead acceleration/deceleration before interpolation of constant acceleration time type

> ⚠ **CAUTION**
> A set time constant is applied to all axes. So, a modification made to this item changes the settings for all axes.

- Allowable acceleration change value in velocity control based on acceleration change under jerk control

Unit of data: mm/sec^2, $inch/sec^2$, deg/sec^2 (machine unit)

Set an allowable acceleration change value per ms for each axis in velocity control based on acceleration change under jerk control.

The parameter set on the machining parameter tuning screen is reflected in the following parameters:
Parameter No. 13614 (velocity-emphasized parameter)
Parameter No. 13615 (precision-emphasized parameter)

Moreover, the following parameter is also set according to the precision level:
Parameter No. 1788: Allowable acceleration change value for each axis in velocity control based on acceleration change under jerk control

> **NOTE**
> This setting item is displayed only when the jerk control function is enabled.

- Allowable acceleration change value for each axis in velocity control based on acceleration change under jerk control in successive linear interpolation operations

Unit of data: mm/sec^2, inch/sec^2, deg/sec^2 (machine unit)

Set an allowable acceleration change value per ms for each axis in velocity control based on acceleration change under jerk control in successive linear interpolation operations.

The parameter set on the machining parameter tuning screen is reflected in the following parameters:
Parameter No. 13616 (velocity-emphasized parameter)
Parameter No. 13617 (precision-emphasized parameter)

Moreover, the following parameter is also set according to the precision level:
Parameter No. 1789: Allowable acceleration change value for each axis in velocity control based on acceleration change under jerk control in successive linear interpolation operations

⚠ CAUTION
1 For an axis with 0 set in this parameter, the parameters (allowable acceleration change value in velocity control based on acceleration change under jerk control: No. 13614, No.13615) are valid.
2 For an axis with 0 set in the parameter (allowable acceleration change value in velocity control based on acceleration change under jerk control: No. 13614, No.13615), velocity control based on acceleration change is disabled, so that this parameter has no effect.

NOTE
This setting item is displayed only when the jerk control function is enabled.

- Ratio of the change time of the jerk control in smooth bell-shaped acceleration/deceleration before interpolation

Unit of data: %

Set the ratio (in %) of the change time of jerk control to the change time of acceleration in smooth bell-shaped acceleration/deceleration before interpolation.

The parameter set on the machining parameter tuning screen is reflected in the following parameters:
Parameter No. 13618 (velocity-emphasized parameter)
Parameter No. 13619 (precision-emphasized parameter)

Moreover, the following parameter is also set according to the precision level:
Parameter No. 1790: Ratio of the change time of the jerk control in smooth bell-shaped acceleration/deceleration before interpolation

NOTE
This setting item is displayed only when the jerk control function is enabled.

- Allowable acceleration rate

Set an allowable acceleration rate in acceleration-based speed determination.
Unit of data: mm/sec^2, $inch/sec^2$, deg/sec^2 (machine unit)

The parameter set on the machining parameter tuning screen is reflected in the following parameters:
Parameter No. 13620 (velocity-emphasized parameter)
Parameter No. 13621 (precision-emphasized parameter)

Moreover, the following parameter is also set according to the precision level:
Parameter No. 1735: Allowable acceleration rate for each axis applicable to the deceleration function based on acceleration in circular interpolation
Parameter No. 1737: Allowable acceleration rate for each axis applicable to the deceleration function based on acceleration in AI contour control

⚠ CAUTION
When bit 0 (MCR) of parameter No. 13600 is set to 1, the deceleration function based on acceleration in circular interpolation is not set.

- Time constant for acceleration/deceleration after interpolation

Set a time constant for acceleration/deceleration after interpolation.
Unit of data: ms

The parameter set on the machining parameter tuning screen is reflected in the following parameters:
Parameter No. 13622 (velocity-emphasized parameter)
Parameter No. 13623 (precision-emphasized parameter)

Moreover, the following parameter is also set according to the precision level:
Parameter No. 1769: Time constant for acceleration/deceleration after cutting feed interpolation

- Corner speed difference

Set an allowable corner speed difference used for speed determination.
Unit of data: mm/sec, inch/sec, deg/sec (machine unit)

The parameter set on the machining parameter tuning screen is reflected in the following parameters:
Parameter No. 13624 (velocity-emphasized parameter)
Parameter No. 13625 (precision-emphasized parameter)

Moreover, the following parameter is also set according to the precision level:
Parameter No. 1783: Allowable speed difference for each axis in automatic corner deceleration based on speed difference

- Maximum cutting speed

Set a maximum cutting speed for each axis.
Unit of data: mm/sec, inch/sec, deg/sec (machine unit)

The parameter set on the machining parameter tuning screen is reflected in the following parameters:
Parameter No. 13626 (velocity-emphasized parameter)
Parameter No. 13627 (precision-emphasized parameter)

Moreover, the following parameter is also set from the precision level:
Parameter No. 1432: Maximum cutting feedrate for each axis in the AI contour control mode

- Arbitrary items

Two arbitrary parameters can be registered. Each item can correspond to a CNC parameter or servo parameter. A parameter number corresponding to each item is to be specified with parameters.

As indicated below, set the parameters for corresponding parameter numbers, velocity-emphasized parameters (precision level 1), and precision-emphasized parameters (precision level 10).

Table 12.4.10 (b) Parameters related to arbitrary items

	Corresponding parameter number	Setting of velocity-emphasized (precision level 1) value	Setting of precision-emphasized (precision level 10) value
Arbitrary item 1	No.13628	No.13630	No.13632
Arbitrary item 2	No.13629	No.13631	No.13633

- Display
  Tuning target parameter numbers are displayed.

> ⚠ **CAUTION**
> As arbitrary items, the numbers of the following parameters cannot be specified:
> - Bit parameter
> - Spindle parameters (Parameter Nos. 4000 to 4799)
> - Real-type parameter
> - Power-off parameter
> - Nonexistent parameter

12.4.11 Displaying Memory Data

The contents of the CNC memory can be displayed starting at a specified address.

Displaying memory data

Procedure

1 Set bit 0 (MEM) of parameter No. 8950 to 1 to display the memory contents display screen.

2 Press function key [SYSTEM].

3 Press the continuous menu key [▷], then press chapter selection soft key [MEMORY].
The following screen appears:

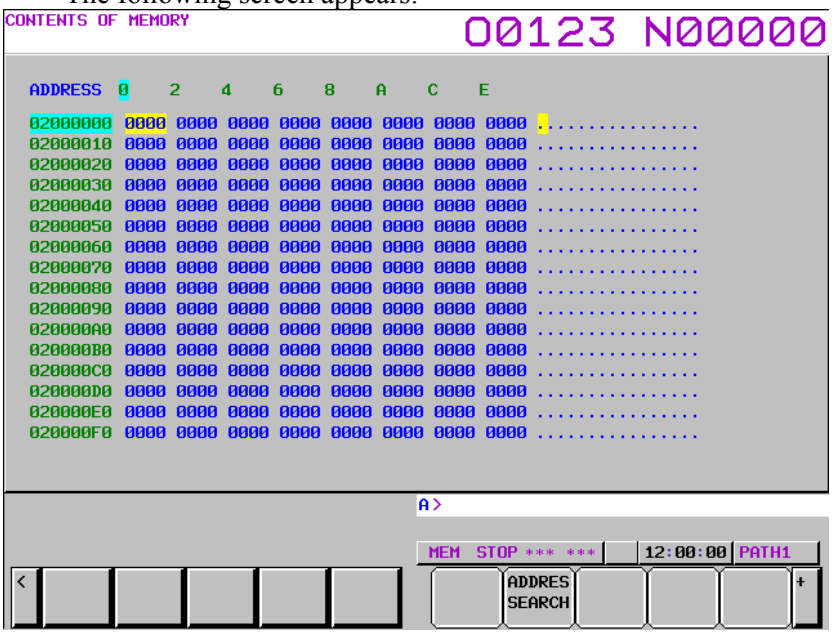


Fig. 12.4.11 (a) Memory contents display screen (10.4-inch)

4 Key in a desired address (hexadecimal) then press soft key [ADDRES SEARCH]. Starting at the specified address, 256-byte data is displayed.
(Example: When you input 100000 then press [ADDRES SEARCH], data is displayed starting at 100000H.)

5 Display data can be switched by using page keys [PAGE↑] and [PAGE↓] and cursor keys [↑], [↓], [←], and/or [→].

6 By pressing soft key [BYTE], [WORD], [LONG], or [DOUBLE], select a data type to be displayed.

Explanation

A memory data display format can be selected from the following four options:

Byte display (1 byte in hexadecimal)
Word display (2 bytes in hexadecimal)
Long display (4 bytes in hexadecimal)
Double display (8 bytes in decimal: Double precision floating-point display)

One screen displays 256-byte memory data.

NOTE
1 When an address is input, "H" for representing "hexadecimal" need not be specified at the end of the address. If H is added, a warning is displayed to indicate that the format is invalid.
2 When word display is selected as the display format, an input address is rounded to a multiple of 2 bytes. When long display or double display is selected, an input address is rounded to a multiple of 4 bytes.

⚠ WARNING
1 If a memory address that must not be accessed in address search is input, a system alarm is issued. When making an address search, check that the address is accessible and that the address is input correctly.
2 This function is designed for maintenance, and must not be used by general users.

12.4.12 Parameter Tuning Screen

The parameter tuning screen is a screen for parameter setting and tuning designed to achieve the following:

1 The minimum required parameters that must be set when the machine is started up are collectively displayed to facilitate start-up of the machine.
2 The servo tuning screen, spindle tuning screen, and machining parameter tuning screen are displayed for smooth tuning.

The parameter tuning screen consists of a menu screen and several setting screens.

12.4.12.1 Displaying the menu screen and selecting a menu item

The parameter tuning menu screen displays the following items:

[START UP]
- SYSTEM SETTING
- AXIS SETTING
- FSSB (AMP)
- FSSB (AXIS)
- SERVO SETTING
- SPINDLE SETTING
- MISCELLANY

[TUNING]
- SERVO TUNING
- SPINDLE TUNING
- AICC TUNING

On the parameter tuning menu screen, one of the displayed items can be selected to display the corresponding screen. From each setting screen, you can return to the menu screen by performing a soft key operation.

NOTE
1 Some items may not be displayed, depending on the system configuration.
2 When bit 0 (SVS) of parameter No. 3111 is set to 0, "SERVO SETTING" and "SERVO TUNING" are not displayed. When bit 1 (SPS) of parameter No. 3111 is set to 0, "SPINDLE TUNING" is not displayed.

Displaying the menu screen and selecting a setting screen

Procedure

1 Set the MDI mode.
2 Switch the setting of "PARAMETER WRITE" to "ENABLED". For details, see the procedure for "PARAMETER WRITE" in Subsection III-12.4.1.
3 Press function key SYSTEM.
4 Press the continuous menu key ▷ several times.
5 Press soft key [PARAM TUNING] to display the menu screen for parameter tuning .

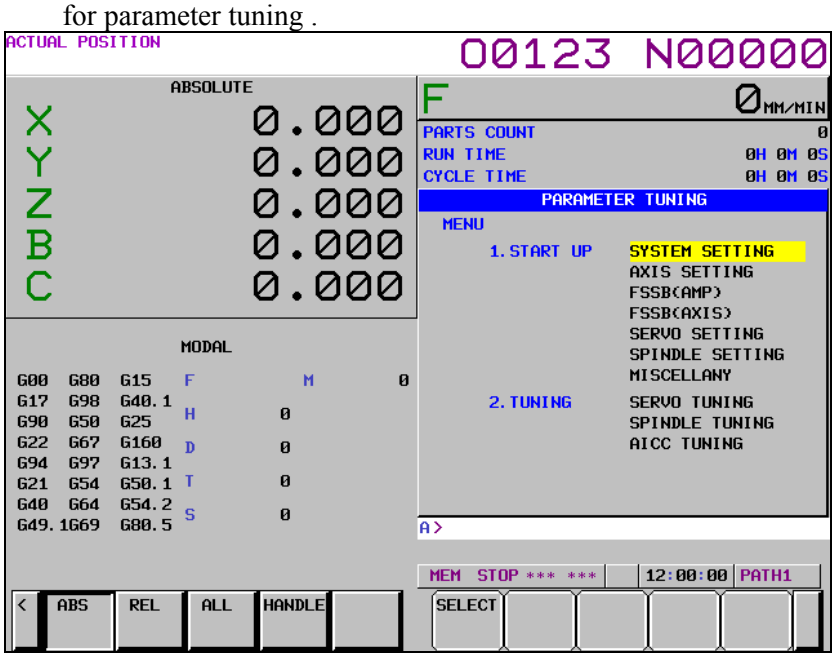


Fig. 12.4.12.1 (a) Menu screen for parameter tuning (10.4-inch)

6 Move the cursor to a desired item by pressing cursor key ↑ or ↓.
7 Press soft key [SELECT]. The screen display switches to the selected screen.

Returning to the menu screen

Procedure

1 Press soft key [SELECT] on the parameter tuning menu screen described in Subsection III-12.4.12.1. The screen and soft keys shown below are displayed. (The screen below is displayed when "AXIS SETTING" is selected.)

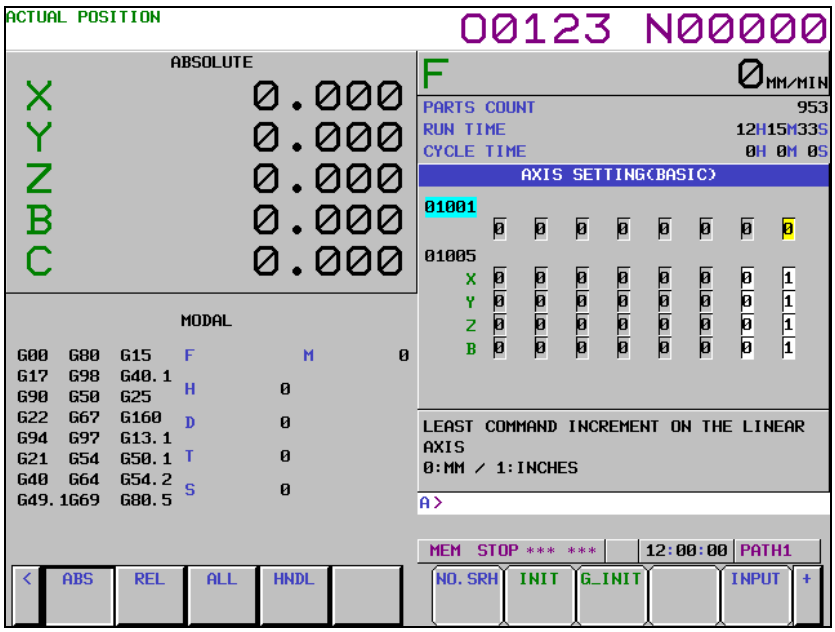


Fig. 12.4.12.1 (b) Axis setting screen (10.4-inch)

2 Press the continuous menu key ▷ several times.

3 Press soft key [MENU].
The screen display returns to the parameter tuning menu screen.

4 Upon completion of parameter setting, switch the setting of "PARAMETER WRITE" to "DISABLED".

NOTE
Some setting screens can also be displayed by a chapter selection soft key. If a screen is selected using a chapter selection soft key, however, you cannot return to the parameter tuning menu screen.

Explanation

- Items displayed with [START UP]

The items of [START UP] indicate the screens for setting the minimum required parameters for starting up the machine.

Table 12.4.12.1 (a) Items displayed with [START UP]

Display item	Description
SYSTEM SETTING	Screen for setting CNC parameters related to the entire system configuration
AXIS SETTING	Screen for setting CNC parameters related to axes, coordinates, feedrate, and acceleration/deceleration
FSSB (AMP)	FSSB amplifier setting screen
FSSB (AXIS)	FSSB axis setting screen
SERVO SETTING	Servo setting screen
SPINDLE SETTING	Screen for setting spindle-related parameters
MISCELLANY	Screen for setting parameters related to the allowable number of M code digits and whether to display the servo setting and spindle tuning screens

NOTE
Some items may not be displayed, depending on the system configuration.

- Items displayed with [TUNING]

The items of [TUNING] indicate the screens for servo, spindle, and high-speed high-precision machining tuning.

Fig. 12.4.12.1 (b) Items displayed with [TUNING]

Display item	Description
SERVO TUNING	Servo tuning screen
SPINDLE TUNING	Spindle tuning screen
AICC TUNING	Machining parameter tuning screen

NOTE
1 Some items may not be displayed, depending on the system configuration.
2 When bit 0 (SVS) of parameter No. 3111 is set to 0, "SERVO SETTING" and "SERVO TUNING" are not displayed. When bit 1 (SPS) of parameter No. 3111 is set to 0, "SPINDLE TUNING" is not displayed.

12.4.12.2 Parameter tuning screen (system setting)

This screen enables the parameters related to the entire system configuration to be displayed and modified. The parameters can be initialized to the standard values (recommended by FANUC).

Display and setting

Procedure

1 Move the cursor to [SYSTEM SETTING] by pressing cursor key or on the parameter tuning menu screen described in Subsection III-12.4.12.1.

2 Press soft key [SELECT]. The screen display switches to the screen and soft keys shown below.

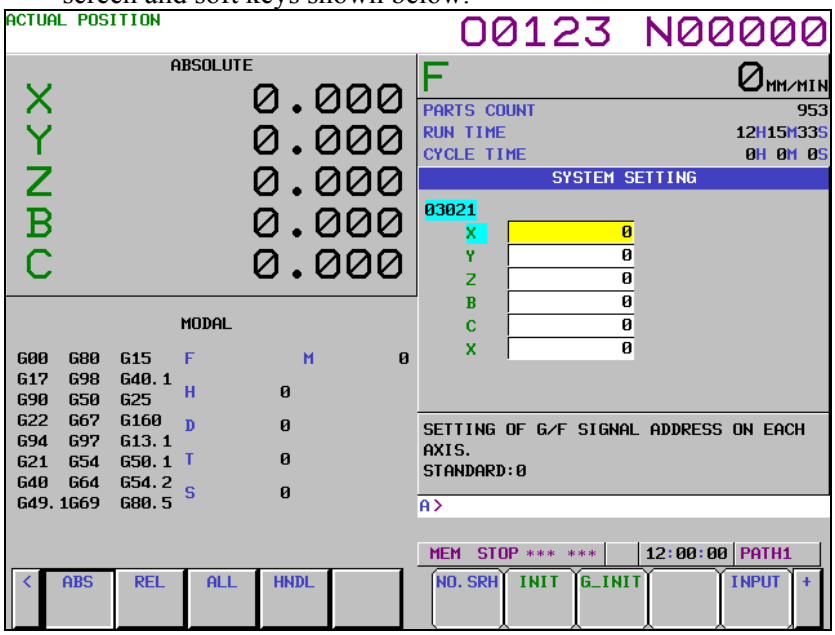


Fig. 12.4.12.2 (a) Parameter tuning screen (system setting) (10.4-inch)

3 Move the cursor to a parameter number to be set or displayed, according to one of the methods below.

- Enter the parameter number and press soft key [NO.SRH] .
- Move the cursor to the desired parameter number by pressing page key or , and cursor keys , , , and/or .
  A brief description of the parameter where the cursor is placed is provided at the bottom of the screen. However, no description is provided when the cursor is placed on multiple bits for bit parameters.

4 Input desired data then press the key on the MDI panel to set the parameter.

5 Press soft key [INIT]. The standard value (recommended by FANUC) for the item selected by the cursor is displayed in the key input buffer. Pressing soft key [EXEC] in this state initializes the item to the standard value.
6 Press soft key [G_INIT]. A message asking whether to set the group standard values is displayed on the screen. Pressing soft key [EXEC] in this state inputs all of the standard values of the group.

NOTE
1 If the cursor is placed on a parameter that has no standard value assigned, no standard value is input even when [INIT] is pressed.
2 When the cursor is placed on multiple bits for bit parameters, the multiple bits can be input simultaneously. When [INIT] is pressed in this state, the key input buffer displays the standard values for the bits where the cursor is placed. If a bit has no standard value assigned, "*" is displayed for the bit, and no value is input for the bit.
3 When [G_INIT] is pressed, those parameters that have no standard values assigned are not initialized.

12.4.12.3 Parameter tuning screen (axis setting)

This screen enables the CNC parameters related to axes, coordinates, feedrate, and acceleration/deceleration to be displayed and set. The parameters displayed can be divided into four groups:

(Basic) group :

The parameters related to basic settings are displayed.

(Coordinate) group :

The parameters related to coordinates are displayed.

(Feedrate) group :

The parameters related to feedrate are displayed.

(Acceleration/deceleration) group:

The parameters related to acceleration/deceleration are displayed.

The parameters can be initialized to the standard values (recommended by FANUC). For the display and setting procedure, see the description of the parameter tuning screen (system setting) in Subsection III-12.4.12.2.

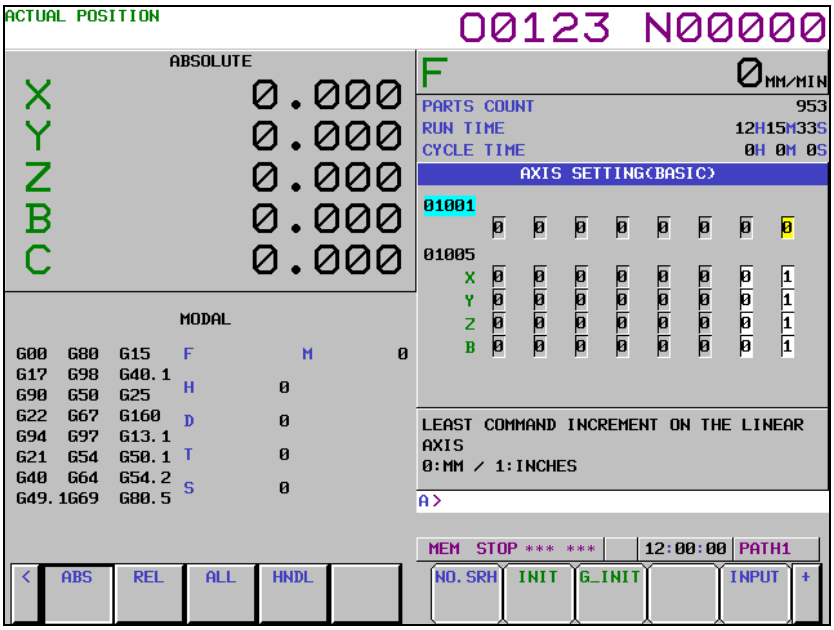


Fig. 12.4.12.3 (a) Parameter tuning screen (axis setting) (10.4-inch)

12.4.12.4 Displaying and setting the FSSB amplifier setting screen

From the parameter tuning screen, the FSSB amplifier setting screen can be displayed. For details of the FSSB amplifier setting screen, see the description of the FSSB amplifier setting screen in Subsection 1.4.4 in the Connection Manual (Function) (B-63943EN-1).

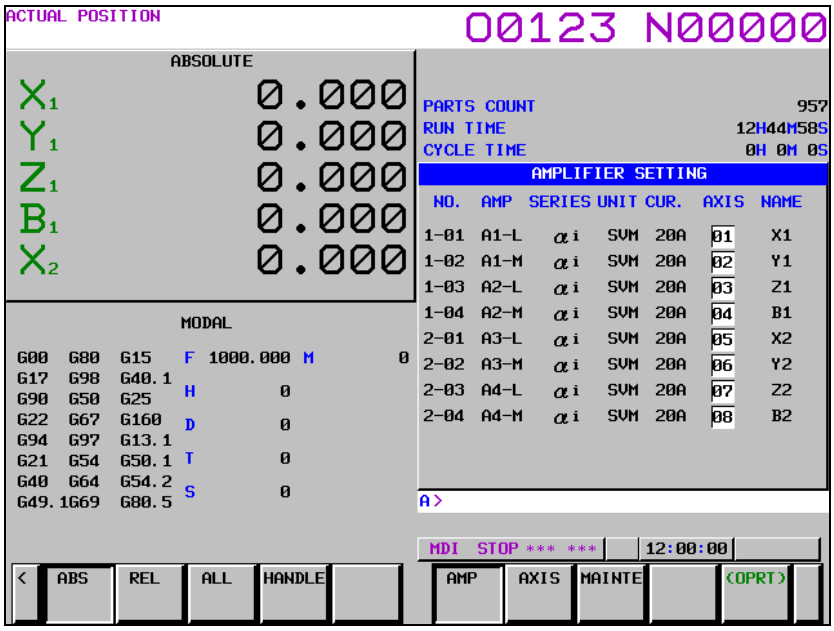


Fig. 12.4.12.4 (a) FSSB amplifier setting screen (10.4-inch)

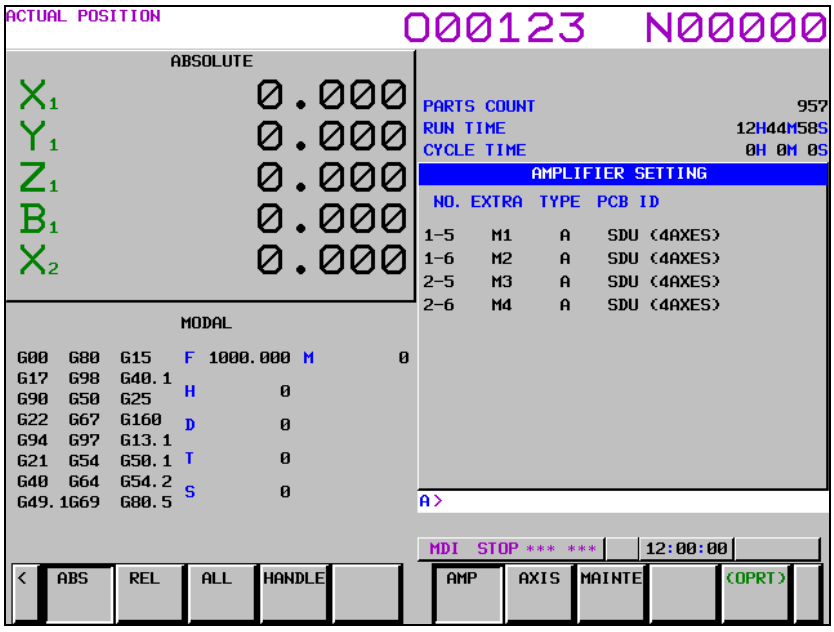


Fig. 12.4.12.4 (b) FSSB amplifier setting screen 2 (10.4-inch)

12.4.12.5 Displaying and setting the FSSB axis setting screen

From the parameter tuning screen, the FSSB axis setting screen can be displayed. For details of the FSSB axis setting screen, see the description of the FSSB axis setting screen Subsection 1.4.4 in the Connection Manual (Function) (B-63943EN-1).

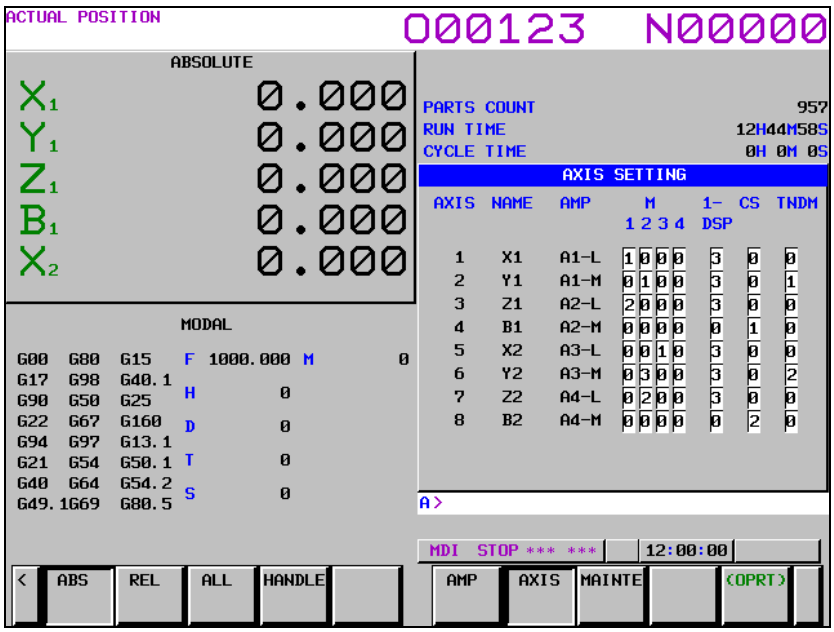


Fig. 12.4.12.5 (a) FSSB axis setting screen (10.4-inch)

12.4.12.6 Displaying and setting the servo setting screen

From the parameter tuning screen, the servo setting screen can be displayed. For details of the servo setting screen, see the description of the servo setting screen in Subsection III-12.4.4.

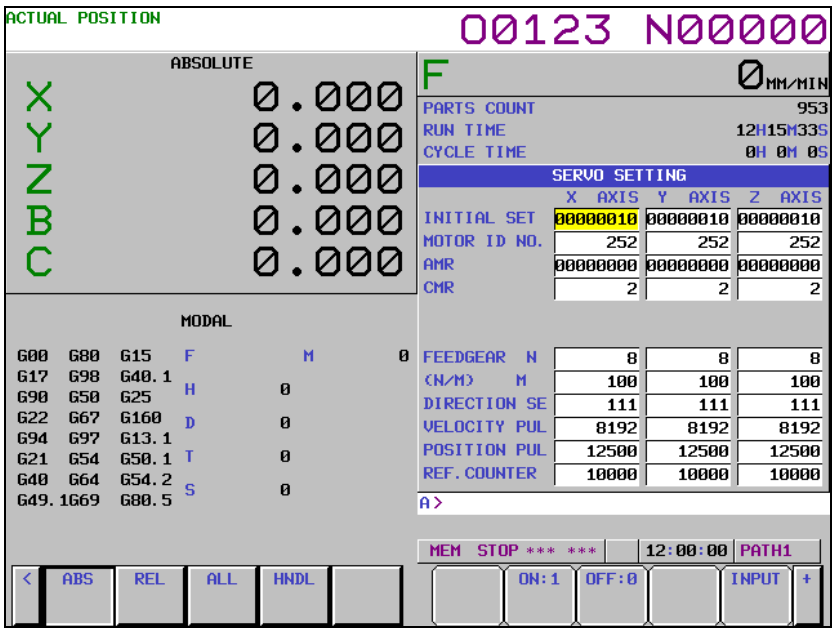


Fig. 12.4.12.6 (a) Servo setting screen (10.4-inch)

12.4.12.7 Parameter tuning screen (spindle setting)

The spindle-related parameters can be displayed and modified.
For the display and setting procedure, see the description of the parameter tuning screen (system setting) in Subsection III-12.4.12.2.

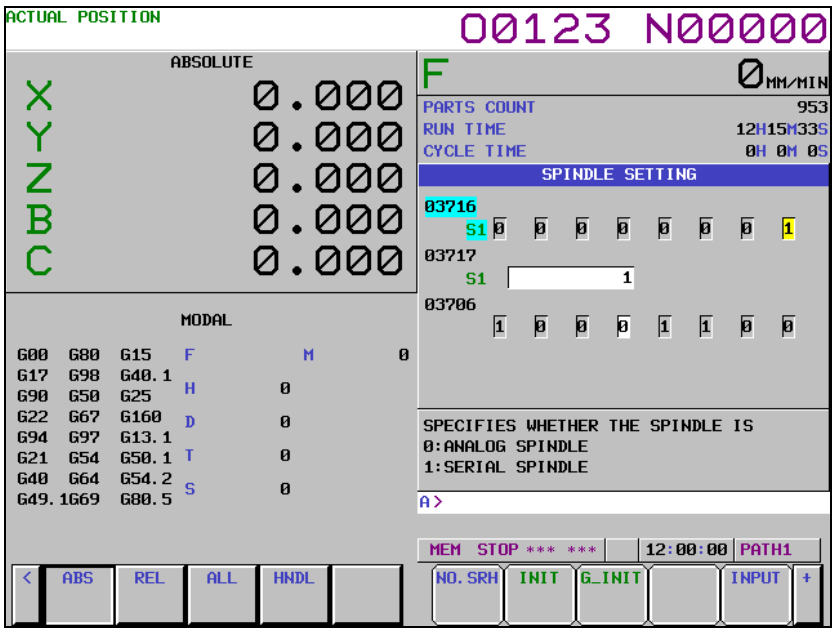


Fig. 12.4.12.7 (a) Parameter tuning screen (spindle setting) (10.4-inch)

12.4.12.8 Parameter tuning screen (miscellaneous settings)

The parameters related to the allowable number of M code digits and whether to display the servo setting and spindle tuning screens can be displayed and modified. Moreover, the parameters can be initialized to the standard values (recommended by FANUC).
For the display and setting procedure, see the description of the parameter tuning screen (system setting) in Subsection III-12.4.12.2.

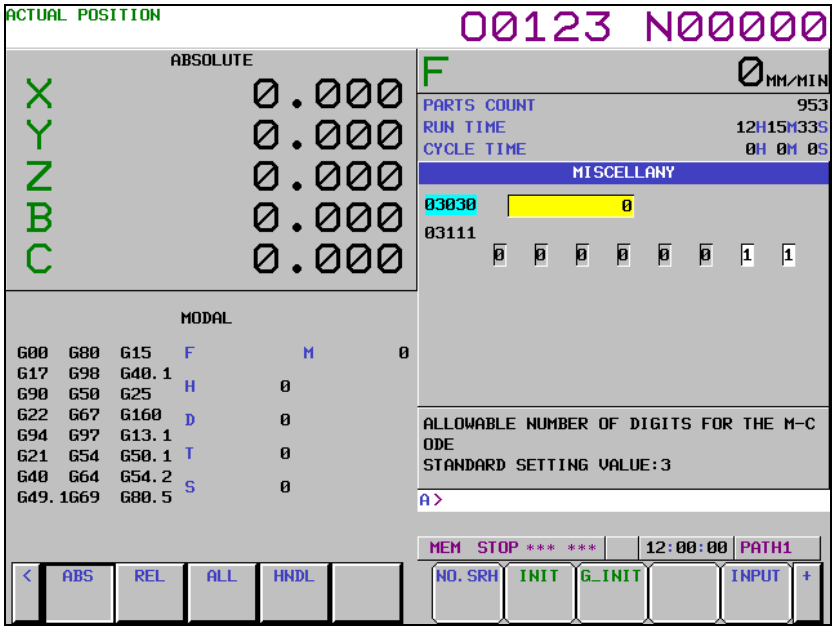


Fig. 12.4.12.8 (a) Parameter tuning screen (Miscellaneous setting) (10.4-inch)

12.4.12.9 Displaying and setting the servo tuning screen

From the parameter tuning screen, the servo tuning screen can be displayed. For details of the servo tuning screen, see the description of the servo tuning screen in Subsection III-12.4.5.

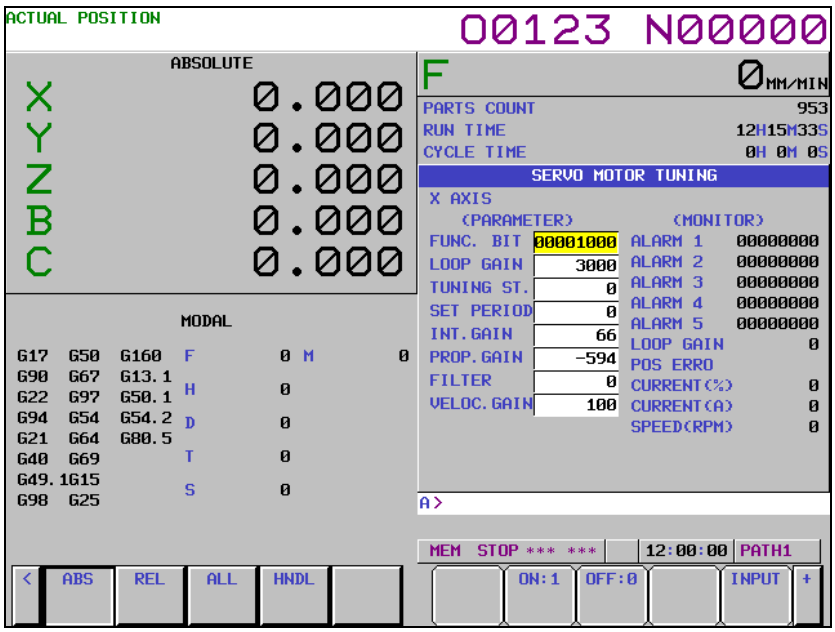


Fig. 12.4.12.9 (a) Servo tuning screen (10.4-inch)

12.4.12.10 Displaying and setting the spindle tuning screen

From the parameter tuning screen, the spindle tuning screen can be displayed. For details of the spindle tuning screen, see the description of the spindle tuning screen in Subsection III-12.4.7.

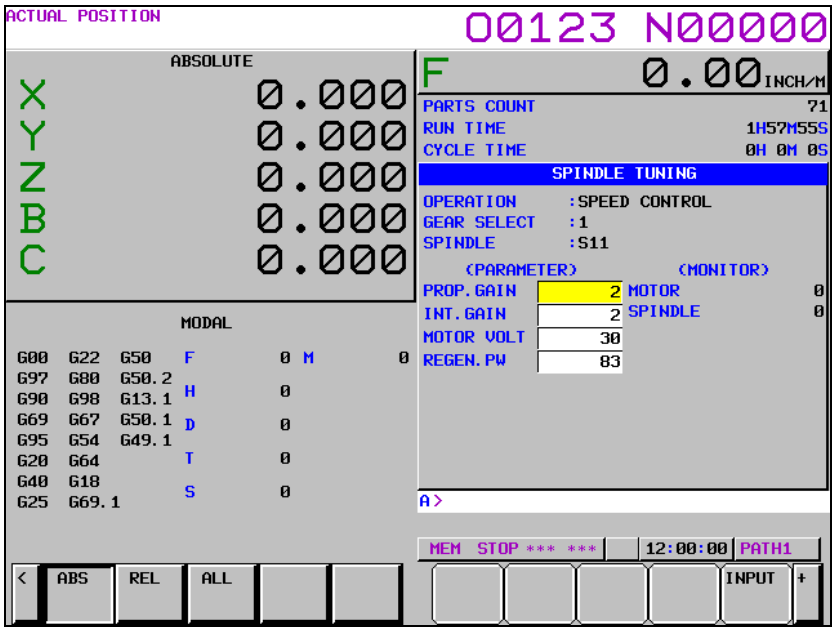


Fig. 12.4.12.10 (a) Spindle tuning screen (10.4-inch)

12.4.12.11 Displaying and setting the machining parameter tuning screen

From the parameter tuning screen, the machining parameter tuning screen can be displayed. For details of the machining parameter tuning screen, see the description of the machining parameter tuning screen in Subsection III-12.4.10.

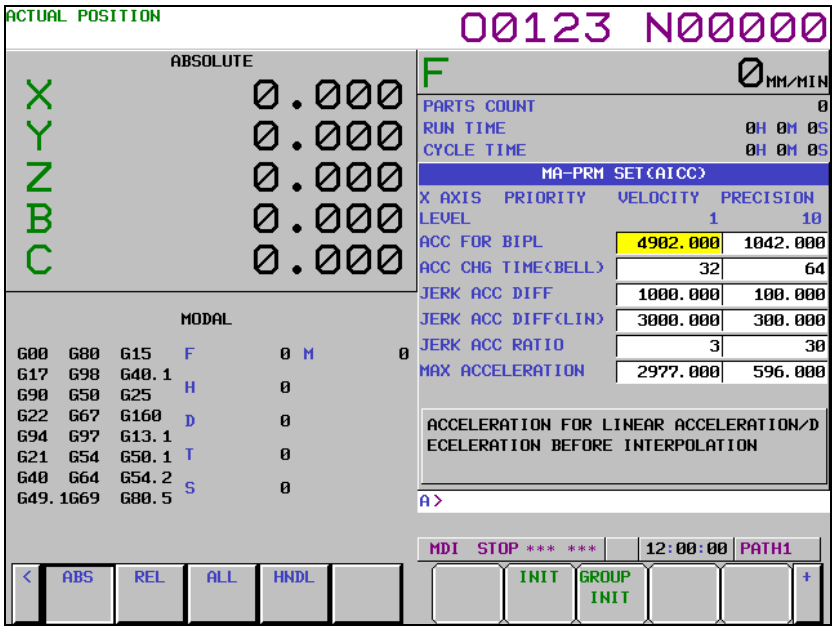


Fig. 12.4.12.11 (a) Machining parameter tuning screen (10.4-inch)

Explanation

- Parameters displayed for parameter tuning

Table 12.4.12 (a) Parameters displayed for parameter tuning (1)

Menu	Group	Parameter No.	Name	Brief description	Standard setting
SYSTEM SETTING	System setting	981		Sets the path of each axis.	
		982		Sets the path of each spindle.	
		983		Sets the T series/M series of each path. 0:T series/1:M series	
		984#0	LCP	Sets the attribute of each path. 0:Normal/1:Loader control	*1
		3021		Sets the G/F signal address of each axis.	*2
		3022		Sets the G/F signal address of each spindle.	*3
		3006#0	GDC	The deceleration signal for reference position return is: 0:X signal/1:G signal	1
		3008#2	XSGx	The signal address allocated to address X is: 0:Fixed/1:Set by the parameter	1
		3013		Address allocated to the deceleration signal for reference position return	*4
		3014		Bit position allocated to the deceleration signal for reference position return	*5

*1 : The value 1 is set for the paths as many as the number of loader paths starting from the greatest path number. For path 1, the value 0 is set at all times.
Example) When the number of loader paths is 3 in a 10-path system:
The value 1 is set for paths 8 to 10. The value 0 is set for others.

*2 : When intra-path axis number ≤ 8, (path number - 1)*10+(intra-path axis number - 1)
When intra-path axis number ≥ 9, no standard value is available.
Example) When path 1 has 9 axes, and path 2 has 3 axes:
0,1,...,7,(none) for axes of path 1, 10,11,12 for axes of path 2

*3 : When intra-path spindle number ≤ 4, (path number - 1)*10+(intra-path spindle number - 1)
When intra-path spindle number ≥ 5, no standard value is available.
Example) When path 1 has 5 spindles, and path 2 has 1 spindle:
0,1,...,4,(none) for spindles of path 1, 10 for spindle of path 2

*4 : When (path number ≤ 3) and (intra-path axis number ≤ 8)
Axes of path 1: 9
Axes of path 2: 7
Axes of path 3: 10
Other axes: No standard value is available.
Example) When path 1 has 9 axes, and path 2 has 3 axes:
9,9,9,9,9,9,9,9,(none) for axes of path 1, 7,7,7 for axes of path 2

*5 : When (path number ≤ 3) and (intra-path axis number ≤ 8)
(Intra-path axis number - 1)
Other axes: No standard value is available.
Example) When path 1 has 9 axes, and path 2 has 3 axes:
0,1,...,7,(none) for axes of path 1, 0,1,2 for axes of path 2

Table 12.4.12 (b) Parameters displayed for parameter tuning (2)

Menu	Group	Parameter No.	Name	Brief description	Standard setting
SPINDLE SETTING	Spindle setting	3716#0	A/Ss	Sets the type of spindle motor: 0:Analaog/1:Serial.	
		3717		Sets a motor number to be assigned to each spindle.	
		3706#4	GTT	Specifies a spindle gear selection method. 0:M type/1:T type	
		3718		Sets a suffix to be added to spindle speed display on a screen such as the position display screen.	
		3735		Sets the minimum clamp speed of the spindle motor.	
		3736		Sets the maximum clamp speed of the spindle motor.	
		3741		Sets the maximum spindle speed for gear 1.	
		3742		Sets the maximum spindle speed for gear 2.	
		3743		Sets the maximum spindle speed for gear 3.	
		3744		Sets the maximum spindle speed for gear 4.	
		3772		Sets the maximum speed of the spindle. When 0 is set, the speed is not clamped.	
		4133		Sets the motor model code of the serial spindle. (This setting is not required for an analog spindle.)	
		4019#7	***	The parameters for the serial spindle are: 0:Not automatically set/1:Automatically set (This setting is not required for an analog spindle.)	

Table 12.4.12 (c) Parameters displayed for parameter tuning (3)

Menu	Group	Parameter No.	Name	Brief description	Standard setting
AXIS SETTING	Basic	1001#0	INM	Least command increment on linear axes: 0:Metric (millimeter machine) 1:Inch (inch machine)	
		1005#0	ZRNx	When an automatic operation (other than G28) is executed before reference position return: 0:Alarm is issued (PS224) 1:No alarm is issued.	0
		1005#1	DLZx	Reference position return without dogs: 0:Disabled 1:Enabled	
		1006#0	ROTx	Setting of linear axes or rotary axis: 0:Linear axis 1:Rotation axis	
		1006#3	DIAx	Setting of the amount of travel: 0:Radius specification 1: Diameter specification	
		1006#5	ZMIx	Reference position return direction: 0:Plus direction 1:Minus direction	
		1008#0	ROAx	Rotation axis roll-over function: 0:Disabled 1:Enabled	1
		1008#2	RRLx	With the amount of travel per revolution, relative coordinates are: 0:Not rounded 1:Rounded	1
		1013#0	ISAx	Sets the least input increment and least command increment: 0:IS-B 1:IS-A	
		1013#1	ISCx	Sets the least input increment and least command increment: 0:IS-B 1:IS-C	
		1013#2	ISDx	Sets the least input increment and least command increment: 0:IS-B 1:IS-D	
		1013#3	ISEx	Sets the least input increment and least command increment: 0:IS-B 1:IS-E	
		1020		Program name	*1
		1022		Sets each axis in the basic coordinate system.	*2
		1023		Servo axis number	
		1815#1	OPTx	A separate pulse coder is: 0:Not used 1:Used	
		1815#4	APZx	The correspondence between machine positions and absolute-position detector positions is: 0:Not established 1:Established	
		1815#5	APCx	The position detector used is: 0:Other than an absolute-position detector 1:Absolute-position detector	
		1825		Servo loop gain	
		1826		Effective area	
		1828		Positional deviation limit during travel	
		1829		Positional deviation limit during stop	500

*1 : For the M series: 88(X),89(Y),90(Z) sequentially starting with the first axis (no standard value for the fourth axis and up)
For the T series: 88(X),90(Z) (no standard value for the third axis and up)

*2 : For the M series: 1,2,3 sequentially starting with the first axis (no standard value for the fourth axis and up)
For the T series: 1,3 (no standard value for the third axis and up)

Table 12.4.12 (d) Parameters displayed for parameter tuning (4)

Menu	Group	Parameter No.	Name	Brief description	Standard setting
AXIS SETTING	Coordinate	1240		Machine coordinate of the first reference position	
		1241		Machine coordinate of the second reference position	
		1260		Amount of travel per revolution of a rotary axis	360.000
		1320		Coordinate of the boundary of stored stroke check 1 in the positive direction	
		1321		Coordinate of the boundary of stored stroke check 1 in the negative direction	
	Feedrate	1401#6	RDR	For a rapid traverse command, dry run is: 0:Disabled 1:Enabled	0
		1410		Dry run feedrate	
		1420		Rapid traverse rate	
		1421		Rapid traverse override F0 rate	
		1423		Jog feedrate	
		1424		Manual rapid traverse rate	
		1425		FL feedrate for reference position return	
		1428		Reference position return feedrate	
		1430		Maximum cutting feedrate	
	Acceleration/ deceleration	1610#0	CTLx	Acceleration/deceleration for cutting feed is: 0:Exponential acceleration/deceleration 1:Linear acceleration/deceleration after interpolation	
		1610#4	JGLx	Acceleration/deceleration for jog feed is: 0:Exponential acceleration/deceleration 1:Same as acceleration/deceleration for cutting feed (The settings of bit 1 (CTBx) and bit 0 (CTLx) of parameter No. 1610 are followed.)	
		1620		Time constant for linear acceleration/deceleration for rapid traverse	
		1622		Time constant for acceleration/deceleration for cutting feed	
		1623		FL feedrate for acceleration/deceleration after interpolation for cutting feed	
		1624		Time constant for acceleration/deceleration for jog feed	
		1625		FL feedrate for exponential acceleration/deceleration for jog feed	
MISCELLANY	MISC	3030		Allowable number of digits of an M code	3
		3111#0	SVS	The servo setting screen is: 0:Not displayed 1:Displayed	1
		3111#1	SPS	The spindle setting screen is: 0:Not displayed 1:Displayed	1

12.4.13 Periodic Maintenance Screen

Periodic maintenance screens are used for managing consumables (such as the backlight of a LCD unit and backup batteries). By setting the name of a consumable, its life time, and the method for counting down the life time, the life of the consumable can be counted down in the appropriate method set for the consumable, and the remaining life time can be displayed.
With these screens, the user can easily manage consumables that require periodic replacement.

Explanation

There are four periodic maintenance screens: the status screen, the setting screen, the machine menu screen, and the NC menu screen.

- Status screen:
  Item names, remaining times, and count statuses are displayed, and item names are set.
- Setting screen:
  Life times, remaining times, and count types (count-down method) are set.
- Machine menu screen:
  The names of consumables in the machine can be registered.
- NC menu screen:
  The names of consumables in the NC are already registered.

Using periodic maintenance screens

<1> Referencing a periodic maintenance screen
Display the status screen. For the meanings of items on the status screen, see **Status screen**.

<2> Adding the name of a new consumable item to a periodic maintenance screen or editing an existing consumable item on the screen
The name of a consumable item can be added or edited on the machine menu screen. For details, see **Machine menu screen**.

<3> Adding or editing a life time, remaining time, and a method of counting down the remaining time for a consumable on a periodic maintenance screen
A life time and remaining time can be added or edited on the setting screen. For details, see **Setting screen**.

<4> Newly displaying the item name and remaining time of a consumable on a periodic maintenance screen.
1 Setting an item name
Select the item name of a consumable to be displayed from the machine menu screen or NC menu screen, or enter the name using the MDI keys. For the procedure, see **Item name** in **Status screen**.

2 Setting a life time, remaining time, and count type
Select the life time, remaining time, and count type of a consumable to be displayed from the setting screen. For the procedure, see **Remaining time** in **Status screen**.

Procedure for displaying a periodic maintenance screen

1 Press function key SYSTEM.
2 Press the continuous menu key ▷ several times to display [PERIOD MAINTE].
3 Press soft key [PERIOD MAINTE] to display the periodic maintenance screen.

Status screen

When soft key [STATUS] is pressed, the status screen is displayed. The status screen shows the item names, count statuses, and remaining times of managed consumables.

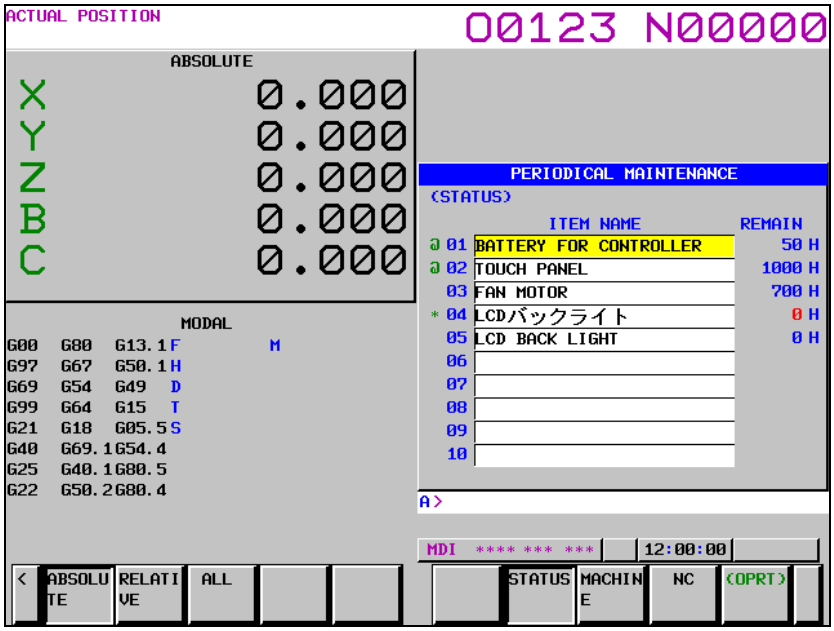


Fig. 12.4.13 (a) Status screen

- Item name

As the item name, set the name of a consumable to be managed by periodic maintenance. To set an item name, select a name from the machine menu screen or NC menu screen, or directly enter the name using the MDI keys.

Setting an item name from a menu screen

1 On the status screen, move the cursor to a target item name, press soft key [(OPRT)], then press soft key [ENTRY].
2 Press soft key [MACHINE] or [NC] to display the machine menu screen or NC menu screen.
3 Move the cursor to an existing item name on the menu screen, press soft key [(OPRT)], and press soft key [SELECT] then [EXEC] successively.

4 The screen display returns to the status screen, and the item name selected on the menu screen is added to the status screen.

Initially, there is no item name set on the machine menu screen, so item names must be registered in advance. For the registration method, see the description of the procedure for registering item names on the machine menu screen.

Setting an item name using the MDI keys

1 Press soft key [(OPRT)].
2 Type the alphanumeric characters to be input, then press soft key [INPUT].
3 The entered item name is registered on the status screen.

When soft key [+INPUT] is pressed instead of soft key [INPUT], the entered characters can be added to an existing item name.
When typing 2-byte characters, type "*" before and after the character codes. The character codes must conform to FANUC codes. (See Appendix G, "FANUC 2-BYTE CHARACTER CODE TABLE".)
An item name to be registered must be up to 24 characters long if it consists of alphanumeric characters only; or it must be up to 12 characters long if it consists of 2-byte characters only.
Example: To register "LCD バックライト", enter the following:
>LCD＊110E10F410CC114010B610FE＊_

NOTE
1 An asterisk "*" is used as a control code, so it cannot be used in item names. In addition, characters "[", "]", "(", and ")" cannot be used in item names.
2 When an item name consisting of both alphanumeric characters and 2-byte characters is registered, the warning "DATA OUT OF RANGE" may be output.

Deleting an item name

To delete a registered item name, move the cursor to the item name, press soft key [ERASE], then press soft key [EXEC].
When an item name is deleted, its life time, remaining time, and count type are deleted at the same time.

- Remaining time

As the remaining time, the period of time left until the time for replacement is reached by count down operation is displayed.
If the percentage of the remaining time to the life time has reached to the value (%) specified by parameter No. 8911, or smaller, the remaining time is displayed in red.
Even after the life time has expired, count down operation continues.

Setting the remaining time

1 On the status screen, place the cursor on an item for which the remaining time is to be set (the item name must have been set in advance).
2 Press soft key [(OPRT)], then press soft key [CHANGE].
3 The screen display changes to the setting screen.
4 Set the life time, remaining time, and count type. For the setting method and other information, see **Setting screen**.

NOTE
On the status screen, the remaining time and life time cannot be set.
These items must be set on the setting screen.

- Count status

The count status is indicated on the left side of the item number as follows:

Indication	Count status
Blank	Counting stopped
@	Counting in progress
*	Life expired

Setting screen

On the setting screen, the life time, remaining time, and count type of a managed consumable are set.

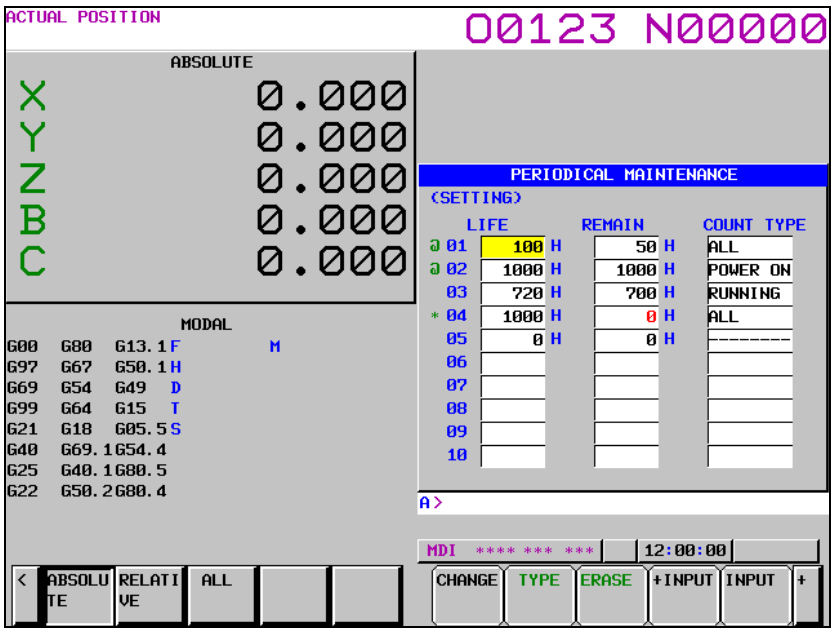


Fig. 12.4.13 (b) Setting screen

Display procedure

1 When the status screen is displayed, press soft key [(OPRT)].
2 Press soft key [CHANGE].

- Life time

Set the life time of a consumable.
Move the cursor to an existing item, type a life time, then press soft key [INPUT] (or the

key). The life time is then set, and the same value is set also as the remaining time.
At this time, "------" is indicated in the count type field.
When soft key [+INPUT] is pressed, the entered value can be added to the life time already set. The same value as the added value is added also to the remaining time.
A value ranging from 0 to 65535 (in hours) can be set.

NOTE
1 If a setting operation is attempted when the item name is not registered, the warning "EDIT REJECTED" is issued.
2 If a value exceeding the valid range is entered, the warning "DATA IS OUT OF RANGE" is issued.
3 If soft key [ERASE] or [TYPE] is pressed, the warning "EDIT REJECTED" is issued.

- Remaining time

The period of time left until the time for replacement is reached by count down operation is indicated.
If the percentage of the remaining time to the life time has reached the value (%) specified by parameter No. 8911, or smaller, the remaining time is displayed in red.
Even after the life time has expired, count down operation continues.
Move the cursor to the remaining time of a target registered number, type a remaining time, then press soft key [INPUT] (or the

key). The remaining time is then set.
When soft key [+INPUT] is pressed, the entered value can be added to the remaining time already set.
A value ranging from 0 to (life time) can be set.
When soft key [ERASE] then soft key [EXEC] are pressed, the same value as the life time is set.

NOTE
1 If a setting operation is attempted when the item name or life time is not registered, the warning “EDIT REJECTED” is issued.
2 When a value exceeding the valid range is entered, the warning “DATA IS OUT OF RANGE” is issued.
3 If soft key [TYPE] is pressed, the warning “EDIT REJECTED” is issued.

- Count type

As the count type, select the way of counting.
Place the cursor on the count type of a target registration number, then press soft key [TYPE]. Count types are displayed as soft keys as shown below. Select one of these soft keys, then press soft key [EXEC].

Soft key	Meaning	Indication
[NO CNT]	Counting is not performed (stopped)	— — — — — —
[ALL]	Counting is performed at all times	All times
[POWER ON]	Counting is performed when power is on.	When power is on
[RUN]	Counting is performed when operation is in progress.	When operation is in progress
[CUT]	Counting is performed when cutting is being performed.	When cutting is being performed

NOTE
1 If a setting operation is attempted when the item name or life time is not registered, the warning “EDIT REJECTED” is issued.
2 Soft keys [INPUT] and [+INPUT] have no effect.
3 When counting is performed at all times, a 24-hour error is generated in leap year.
4 If soft key [ERASE] is pressed, the warning “EDIT REJECTED” is issued.

Machine menu screen

On the machine menu screen, the names of consumables of the machine are registered. From this screen, item names can be added to the status screen. For the method of addition to the status screen, see the description of the status screen.

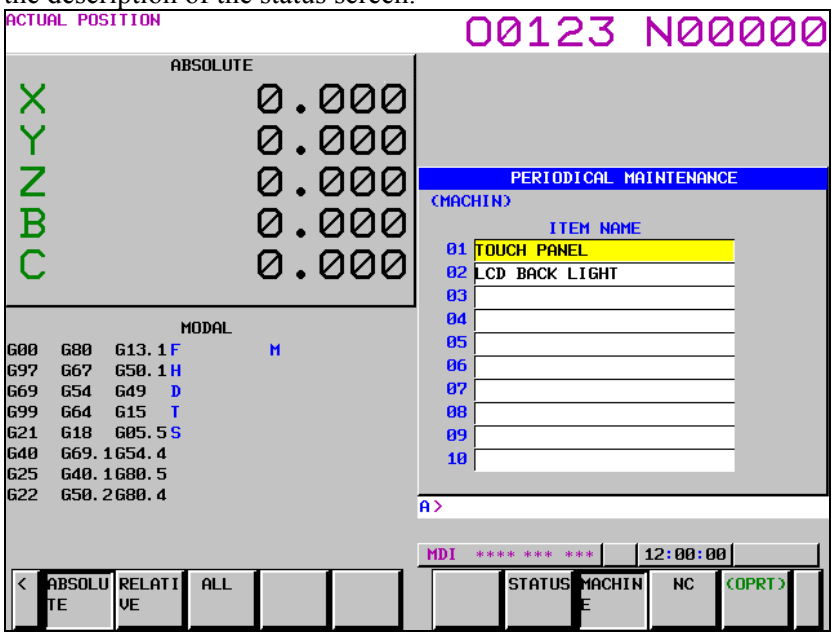


Fig. 12.4.13 (c) Machine menu screen

- Displaying the screen

1 When the status screen is displayed, press soft key [MACHINE].

On the machine menu screen, item names can be registered using one of the following two methods:

- Registration from a program
- Registration using the MDI keys

- Registration from a program

By executing a program having the following format, an item name can be registered in the menu of the machine:

Format

G10 L61 Px [n]

X Registration number

n Item name, format:

[alphanumeric-characters*2-byte-characters*alphanumeric-characters]

- Registration using the MDI keys

An item name can be registered on the menu screen of the machine by typing the item name in the format shown below on the machine menu screen, then pressing soft key [INPUT] (or the

key).

When soft key [+INPUT] is pressed, the typed characters can be added to an already registered item name.

Format

Alphanumeric-characters*2-byte-characters*alphanumeric-characters

Two-byte codes must conform to FANUC codes. (See Appendix G, "FANUC 2-BYTE CHARACTER CODE TABLE".)
When typing 2-byte characters, type an asterisk "*" before and after the character codes. An item name to be registered must be up to 24 characters long if it consists of alphanumeric characters only; or it must be up to 12 characters long if it consists of 2-byte characters only.

Example:

To register "LCD バックライト", enter the following:

>LCD＊110E10F410CC114010B610FE＊_

NOTE
1 An asterisk "*" is used as a control code, so it cannot be used in item names. In addition, characters "[", "]", "(", and ")" cannot be used in item names.
2 When an item name consisting of both alphanumeric characters and 2-byte characters is registered, the warning "DATA IS OUT OF RANGE" may be output.
3 When a blank item name is selected on the machine screen, the warning "EDIT REJECTED" is issued.

To delete a registered item name, move the cursor to the item name, press soft key [ERASE], then press soft key [EXEC].

From this screen, an item name can be registered on the status screen. For the method of registration to the status screen, see the description of the status screen.

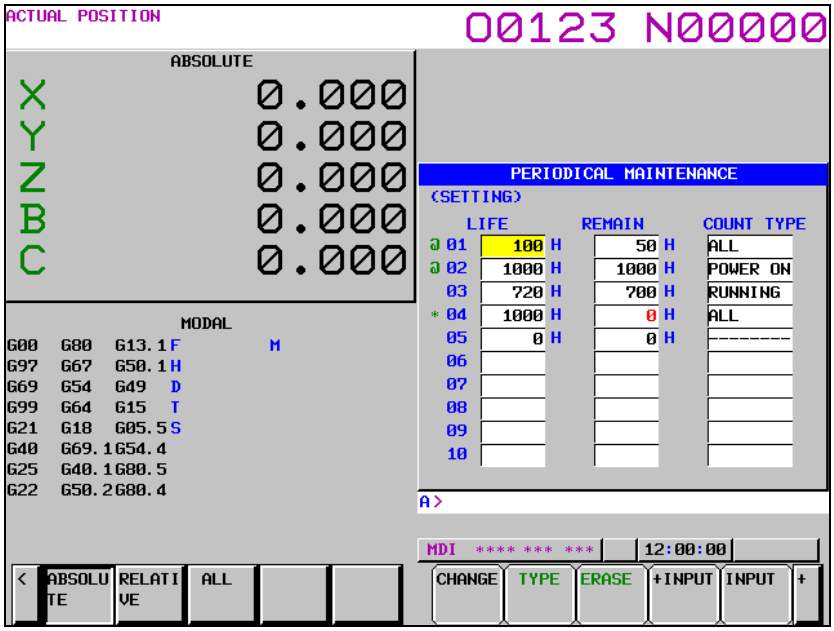


Fig. 12.4.13 (d) NC menu screen

- Displaying the screen

1 When the status screen is displayed, press soft key [NC].

NOTE
On the NC screen, the registration, deletion, and I/O of item names cannot be performed.
When a blank item name is selected, a blank is set.

12.4.14 System Configuration Screen

The system configuration screen provides information about the types of installed hardware and software.

Procedure for displaying the screen

Procedure

1 Press the SYSTEM key to display a screen that shows parameters and other information.
2 Press soft key [SYSTEM]. The system configuration screen is displayed.

There are two types of system configuration screens: the hardware configuration screen and the software configuration screen. The screen display can switch between these screens by using PAGE ↑ and PAGE ↓. When soft key [SERVO INFO] or [SPINDLE INFO] is pressed, information about the connected servo system or spindles is displayed.

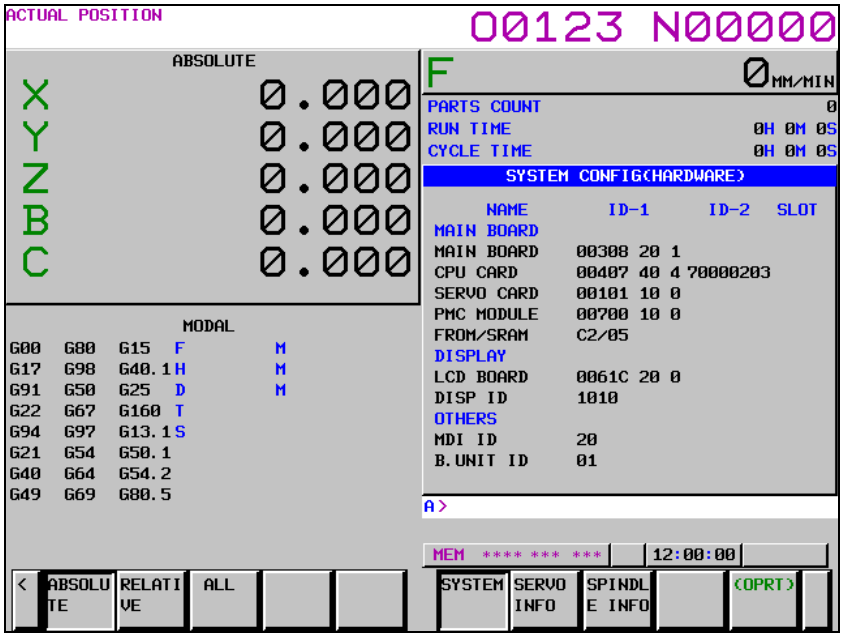


Fig. 12.4.14 (a) System configuration screen

Hardware configuration screen

This screen shows the names and IDs of the hardware used by the NC.

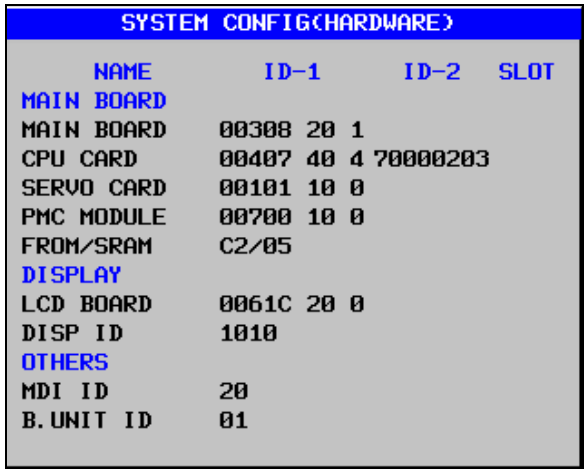


Fig. 12.4.14 (b) Hardware configuration screen

Software configuration screen

This screen shows the names and series/editions of the software used by the NC.

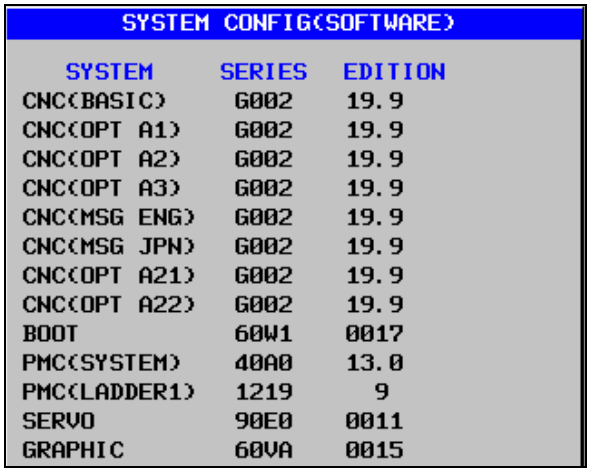


Fig. 12.4.14 (c) Software configuration screen

Servo information screen

When a servo system is connected to the NC, ID information of the connected servo devices (servo motors and servo amplifier modules) can be displayed on the NC.

Displaying the screen

1 When the system configuration screen is displayed, press soft key [SERVO INFO].
2 The servo information screen is displayed.

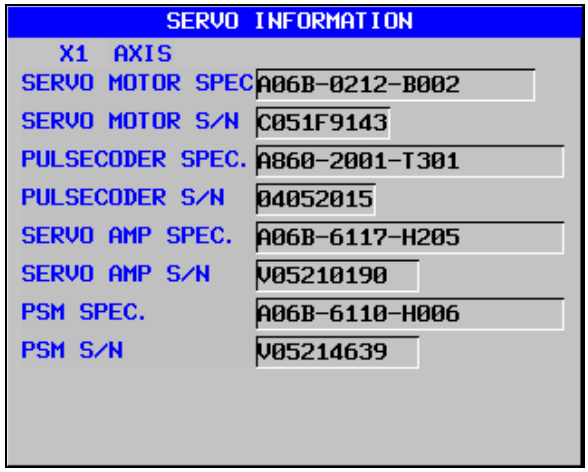


Fig. 12.4.14 (d) Servo information screen

Spindle information screen

When a spindle system is connected to the NC, the ID information of the connected spindle devices (spindle motors and spindle amplifier modules) can be displayed on the NC.

Displaying the screen

1 When the system configuration screen is displayed, press soft key [SPINDLE INFO].
2 The spindle information screen is displayed.

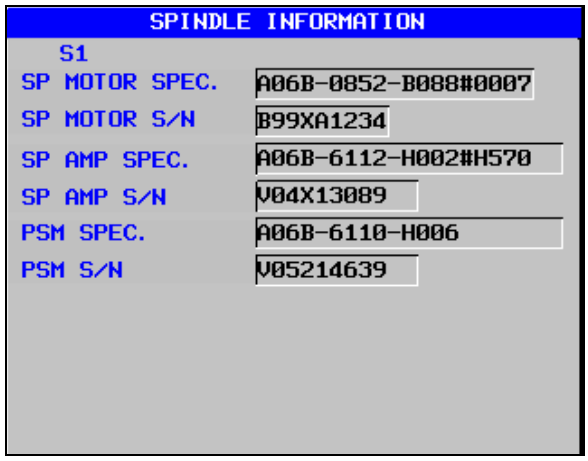


Fig. 12.4.14 (e) Spindle information screen

12.4.15 Overview of the History Function

The history function makes it possible to record a history of operations performed by the operator, alarms and external operator messages issued, and other history data, check the history, and output history data.

1 Conditions for recording history
   a Display screen
   A history of screens other than the operation history screen is recorded.
   b Signal selection
   Up to 60 I/O signals can be selected so that they can be recorded in history data.
   c Parameter setting
   With parameters, it is possible to separately set whether to record an MDI key operation history and an external operator message history, add external alarms and messages, and record data modification histories such as operation histories of parameters, tool offsets, workpiece offsets (workpiece shift amounts), custom macro common variables, and I/O signals.
   d Storage capacity
   When the storage capacity becomes full, data is deleted in order from the oldest data.Up to about 8000 items of history data can be recorded if the data consists of history data of MDI key operations only.

2 Data output
   All history data stored can be output to external input/output devices. (See Subsection 12.4.15.5, "Outputting all history data".)

3 Erasure of history data
   If bit 7 (EKE) of parameter No. 3195 is set to 1, soft key [CLEAR] is displayed, allowing the user to erase history data.

	#7	#6	#5	#4	#3	#2	#1	#0
3195	EKE							

[Data type] Bit

 # 7 EKE Soft key [ALL CLEAR] for clearing all history data is:
0: Not displayed.
1: Displayed.

NOTE

1 History data remains even after the power is turned off.
Memory clear operation, however, erases history data as well.
2 Set the time and date correctly on the setting screen.
3 All history data including data of alarms, external operator messages, operations, and I/O signals is integrated and maintained in a single storage area. When the storage capacity becomes full, history data is erased in order from the oldest data regardless of the history type.
When history data is erased, all history data is erased, so care should be taken.
It is impossible to erase only a specific history.

12.4.15.1 Alarm history

From all history data recorded, only alarm history is extracted and displayed on the screen. Note that when the amount of history data exceeds the storage capacity, history data is automatically erased in order from the oldest history data.

Fig. 12.4.15.1 (a) Alarm history screen

Screen display

Issued alarms are displayed sequentially from the latest alarm.
The following information is displayed for each alarm:

- Path name (only when multipath operation is performed)
- Time and date of alarm occurrence
- Type and number of the alarm
- Alarm message

NOTE
To record also external alarm and macro alarm messages as history data, set bit 3 (EAH) of parameter No. 3112 to 1, and at the same time set bit 7 (HAL) of parameter No. 3196 to 0.

When a name such as a path name, axis name, or spindle name has been changed after the issuance of an alarm, a newly assigned name is displayed on the alarm history screen.
When modal O data that is a program name is output, only the first five characters are output.

Procedure

1 Press function key [SYSTEM] to display a screen of parameters and so on.
2 Press return menu key [◁].
3 Press continuous menu key [▷] several times until soft key [HISTRY] is displayed.
4 Press soft key [HISTRY]. The alarm history screen is then displayed.
5 The screen display can be changed to the previous page and the next page by using page keys [PAGE↑] and [PAGE↓].

Erasing history data from the alarm history screen

Procedure

1 Display the alarm history screen.
2 Press soft key [(OPRT)].
3 Press soft key [CLEAR]. All history data is then erased.

NOTE
When history data is erased, not only alarm history data but also other history data such as external operator message history data and operation history data is erased. It is impossible to erase only a specific history.

Display of external alarms and macro alarms

When an external alarm or macro alarm is issued, it becomes possible to record its message as well as the alarm number in the alarm history if the parameter shown below is set. As the message of an external alarm or macro alarm, the first 32 characters for the external alarm or the first 26 characters for the macro alarm are recorded.

	#7	#6	#5	#4	#3	#2	#1	#0
3112					EAH			

[Data type] Bit

 # 3 EAH Messages of the external alarm/macro alarm in alarm or operation history:
0: Not recorded
1: Recorded

NOTE
This parameter is valid when bit 7 (HAL) of parameter No. 3196 is set to 0.

	#7	#6	#5	#4	#3	#2	#1	#0
3196	HAL							

[Data type] Bit

 # 7 HAL When an alarm is issued, additional information (modal data, absolute coordinates, and machine coordinates present at the issuance of the alarm) is:

0: Recorded in the operation history and alarm history.
1: Not recorded in the operation history and alarm history.

NOTE

1 History data such as alarm history data, operation history data, external operator message history data, and data modification history data is stored in the same storage area. Therefore, alarm history data may be erased when the storage capacity is exceeded by other history data.

2 The maximum number of history data items that can be stored is as follows, where the history data is assumed to include alarm data only:
When bit 7 (HAL) of parameter No. 3196 is set to 1, up to about 4000 history data items can be stored.
When bit 7 (HAL) of parameter No. 3196 is set to 0, modal information and coordinates observed when an alarm is issued are also recorded, so the maximum number of history data items that can be stored varies from system to system. (Example: For a 5-axis system, up to about 280 data items can be stored; for a 10-axis system, up to about 200 data items can be stored.)
When bit 3 (EAH) of parameter No. 3112 is set to 1, however, macro and external alarm messages are also stored, so the number of data items that can be stored further decreases.

12.4.15.2 External operator message history

From all history data recorded, only external operator message history and macro message history are extracted and displayed on the screen. When the amount of history data exceeds the storage capacity, history data is automatically erased in order from the oldest history data.

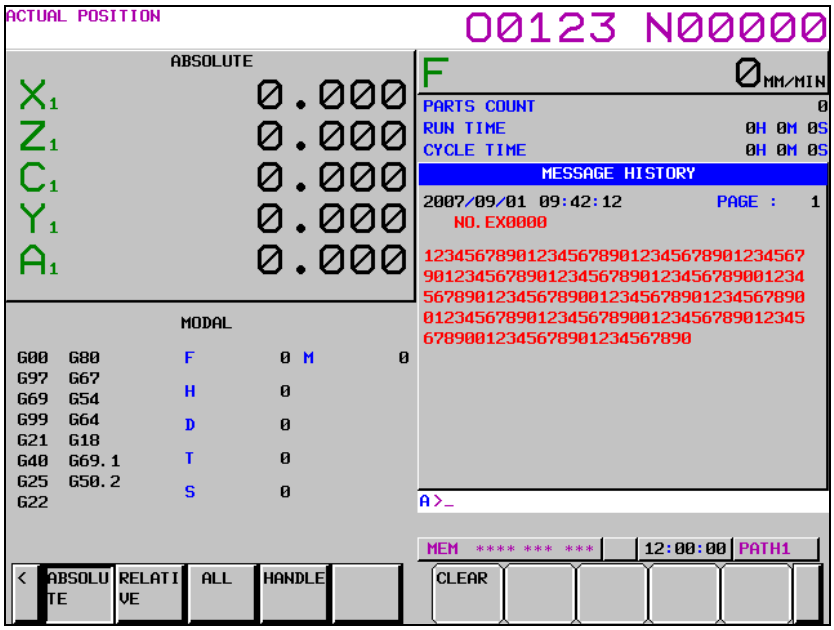


Fig. 12.4.15.2 (a) External operator message history screen

Screen display

Procedure

1 Press function key MESSAGE.

2 Press continuous menu key ▷ several times until soft key [MESSAGE HISTRY] is displayed.

3 Press soft key [MESSAGE HISTRY]. The external operator message history screen is displayed.

4 The screen display can be changed to the previous page and the next page by using page keys PAGE ↑ and PAGE ↓.

NOTE

1 History data such as external operator message history data, operation history data, alarm history data, and data modification history data is stored in the same storage area. Therefore, external operator message history data may be erased when the storage capacity is exceeded by other history data.

2 The maximum number of history data items that can be stored is about 150 if the history data includes external operator message data only.

Erasing history data from the external operator message history screen

Procedure

1 Display the external operator message history screen.
2 Press soft key [(OPRT)].
3 Press soft key [CLEAR]. All history data is then erased.

NOTE
When history data is erased, not only external operator message history data but also other history data such as operation history data and alarm history data is erased. It is impossible to erase only a specific history.

Parameter setting

	#7	#6	#5	#4	#3	#2	#1	#0
3112						OMH		

[Data type] Bit

 # 2 OMH The external operator message history screen is:
0: Not displayed.
1: Displayed.

	#7	#6	#5	#4	#3	#2	#1	#0
3196		HOM						

[Data type] Bit

 # 6 HOM A history of external operator messages and macro messages is:
0: Recorded.
1: Not recorded.

12.4.15.3 Operation history

This function displays a history of the operator's key operations and signal operations made when a failure occurred or an alarm was issued, and also information about alarms.
The following data is recorded:

a Operation history
  i MDI key operations made by the operator
  ii I/O signal (X,Y,G,F) on/off switching
b Alarm history
  i Alarms issued
  ii Modal information in a block executed and coordinates observed when an alarm was issued
  (Not displayed on the screen)
c Data modification history
  i Modification of tool offset data
  (When bit 0 (HTO) of parameter No. 3196 is set to 1)
  ii Modification of workpiece offset data/extended workpiece offset data/workpiece shift (T series)
  (When bit 1 (HWO) of parameter No. 3196 is set to 1)
  iii Modification of parameters
  (When bit 2 (HPM) of parameter No. 3196 is set to 1)
  iv Modification of custom macro common variable data
  (When bit 3 (HMV) of parameter No. 3196 is set to 1)
d External operator message history and macro message history
(When bit 6 (HOM) of parameter No. 3196 is set to 0)
e Time stamp (time and date)

With some exceptions, history data of the operation history and alarm history can be viewed on the operation history screen. (The data modification history, external operator message history, and alarm messages are not displayed.)
All recorded history data can be output to external input/output devices.

NOTE
Up to about 8000 items of history data can be recorded if the data includes history data of MDI key operations only. However, because history data varies in size, the maximum number of history data items that can be recorded is not fixed.

Parameter setting

	#7	#6	#5	#4	#3	#2	#1	#0
3106				OPH				

[Data type] Bit

 # 4 OPH The operation history screen is:
0: Not displayed.
1: Displayed.

3122	Time interval used to record time data in operation history

[Input type] Parameter input
[Data type] Word path
[Unit of data] min
[Valid data range] 0 to 1440
When history data is recorded within a set time period, the time for each set time period is recorded in the history data.
When 0 is set, the specification of a time period of 10 minutes is assumed.
If no data is recorded within a set time period, the time for that period is not recorded.

NOTE
The same value must be set in this parameter for all paths.

	#7	#6	#5	#4	#3	#2	#1	#0
3195	EKE	HDE	HKE					

[Input type] Parameter input
[Data type] Common to the bit system

 # 5 HKE A key operation history is:
0: Recorded.
1: Not recorded.

 # 6 HDE A DI/DO history is:
0: Recorded.
1: Not recorded.

 # 7 EKE Soft key [ALL CLEAR] for clearing all history data is:
0: Not displayed.
1: Displayed.

	#7	#6	#5	#4	#3	#2	#1	#0
3196	HAL	HOM			HMV	HPM	HWO	HTO

[Data type] Bit

 # 0 HTO A modification history of tool offset data is:
0: Not recorded.
1: Recorded.

 # 1 HWO A modification history of workpiece offset data/extended workpiece offset data/workpiece shift (T series) is:
0: Not recorded.
1: Recorded.

 # 2 HPM A modification history of parameters is:
0: Not recorded.
1: Recorded.

 # 3 HMV A modification history of custom macro common variables is:
0: Not recorded.
1: Recorded.

 # 6 HOM A history of external operator messages and macro messages is:
0: Recorded.
1: Not recorded.

 # 7 HAL When an alarm is issued, additional information (modal data, absolute coordinates, and machine coordinates present at the issuance of the alarm) is:
0: Recorded in the operation history and alarm history.
1: Not recorded in the operation history and alarm history.
To record as many alarm history items as possible, rather than detailed alarm information, set 1. The numbers of ten G code modal groups to be recorded are set in parameter Nos. 12990 to 12999.

12990	(1st) G code modal group to be recorded in the history when an alarm is issued

[Input type] Parameter input
[Data type] Byte path
[Valid data range] 1 to the maximum number of G code groups
Set the number of a G code modal group to be recorded in the alarm history and operation history when an alarm is issued.
* If a value beyond the valid data range is set, the status of group 01 is recorded.

12991 **(2nd) G code modal group to be recorded in the history when an alarm is issued**

[Input type] Parameter input
[Data type] Byte path
[Valid data range] 1 to the maximum number of G code groups
Set the number of a G code modal group to be recorded in the alarm history and operation history when an alarm is issued.
* If a value beyond the valid data range is set, the status of group 02 is recorded.

12992 **(3rd) G code modal group to be recorded in the history when an alarm is issued**

[Input type] Parameter input
[Data type] Byte path
[Valid data range] 1 to the maximum number of G code groups
Set the number of a G code modal group to be recorded in the alarm history and operation history when an alarm is issued.
* If a value beyond the valid data range is set, the status of group 03 is recorded.

12993 **(4th) G code modal group to be recorded in the history when an alarm is issued**

[Input type] Parameter input
[Data type] Byte path
[Valid data range] 1 to the maximum number of G code groups
Set the number of a G code modal group to be recorded in the alarm history and operation history when an alarm is issued.
* If a value beyond the valid data range is set, the status of group 04 is recorded.

12994 **(5th) G code modal group to be recorded in the history when an alarm is issued**

[Input type] Parameter input
[Data type] Byte path
[Valid data range] 1 to the maximum number of G code groups
Set the number of a G code modal group to be recorded in the alarm history and operation history when an alarm is issued.
* If a value beyond the valid data range is set, the status of group 05 is recorded.

12995 **(6th) G code modal group to be recorded in the history when an alarm is issued**

[Input type] Parameter input
[Data type] Byte path
[Valid data range] 1 to the maximum number of G code groups
Set the number of a G code modal group to be recorded in the alarm history and operation history when an alarm is issued.
* If a value beyond the valid data range is set, the status of group 06 is recorded.

12996 **(7th) G code modal group to be recorded in the history when an alarm is issued**

[Input type] Parameter input
[Data type] Byte path
[Valid data range] 1 to the maximum number of G code groups
Set the number of a G code modal group to be recorded in the alarm history and operation history when an alarm is issued.
* If a value beyond the valid data range is set, the status of group 07 is recorded.

12997 **(8th) G code modal group to be recorded in the history when an alarm is issued**

[Input type] Parameter input
[Data type] Byte path
[Valid data range] 1 to the maximum number of G code groups
Set the number of a G code modal group to be recorded in the alarm history and operation history when an alarm is issued.
* If a value beyond the valid data range is set, the status of group 08 is recorded.

12998 **(9th) G code modal group to be recorded in the history when an alarm is issued**

[Input type] Parameter input
[Data type] Byte path
[Valid data range] 1 to the maximum number of G code groups
Set the number of a G code modal group to be recorded in the alarm history and operation history when an alarm is issued.
* If a value beyond the valid data range is set, the status of group 09 is recorded.

12999 **(10tht) G code modal group to be recorded in the history when an alarm is issued**

[Input type] Parameter input
[Data type] Byte path
[Valid data range] 1 to the maximum number of G code groups
Set the number of a G code modal group to be recorded in the alarm history and operation history when an alarm is issued.
* If a value beyond the valid data range is set, the status of group 10 is recorded.

Screen display

Procedure

1 Press function key [SYSTEM].

2 Press continuous menu key [▷] several times until soft key [OPERAT HISTRY] is displayed.

3 Press soft key [OPERAT HISTRY], then press newly displayed soft key [OPERAT HISTRY]. The operation history screen is then displayed.

4 To display the previous page and next page of the operation history, use page keys [PAGE↑] and [PAGE↓].

To display a part across two pages, use the cursor keys [←] and [→]. The screen display is shifted by half page.

(With the 7.2- and 8.4-inch display devices, the screen display is shifted by one column.)

By pressing soft key [(OPRT)] on the operation history screen, it becomes possible to perform the following soft key operations:

a [TOP] displays the starting page (the oldest data).

b [BOTTOM] displays the end page (the latest data).

c [NO.SRH] displays specified operation history data.

(Example) Specifying 50 then [NO.SRH] displays the 50th data.

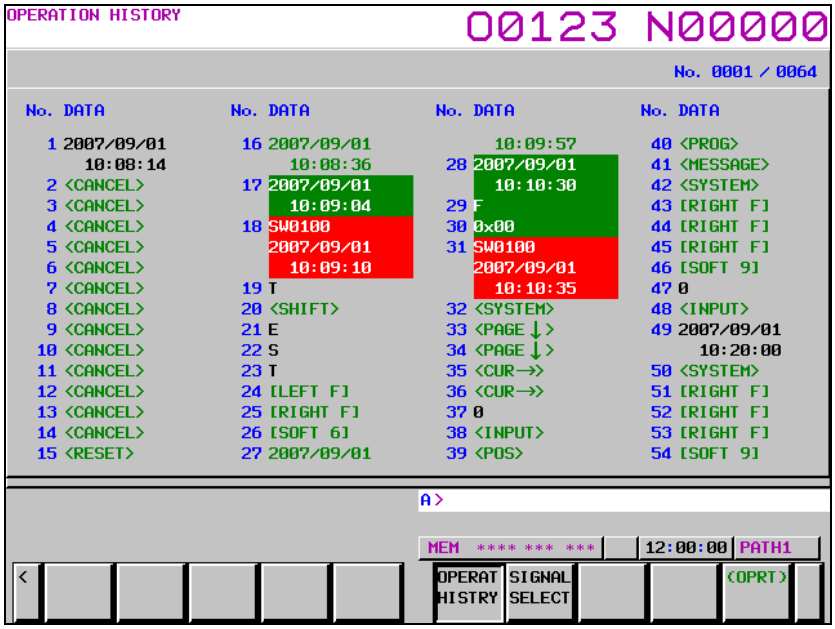


Fig. 12.4.15.3 (a) Operation history screen

Displayed information

1 Serial number and display start history number/total number of history data items

A serial number is indicated on the left side of each recorded history data item. A smaller serial number indicates an older data item.

In the upper right part of the screen, the display start history number and the total number of history data items are indicated. The total number of history data items does not include history data items not displayed on the screen.

2 Data

- MDI key

  When bit 5 (HKE) of parameter No. 3195 is set to 0, key operations are recorded.

  A key operation is indicated following a path number (for example, "1_[LEFT F]", and "2_[LEFT F]"). (When only one path is used, the path number is not indicated.)

  P_ used in "P_[LEFT F]", for example, indicates a key operation made from the outside.

  i Address keys and numeric keys

  Characters such as A to Z, 0 to 9, ;, +, and - are indicated directly.

  These characters are displayed in black.

  ii Function menu keys, operation menu keys, and soft keys

  These keys are enclosed in brackets [] (for example, "[LEFT F]", "[SOFT 1]" to "[SOFT 10]", and "[RIGHT F]"). These keys are displayed with green characters.

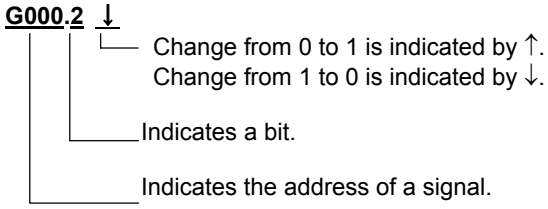
  iii Function keys, page keys, cursor keys, and so on

  These keys are enclosed in angle brackets <> (for example, "<POS>", "<SYSTEM>", "<PAGE ↑>", "<CUR →>", "<RESET>", and "<INPUT>"). These keys are displayed with green characters.

  iv Power-on key

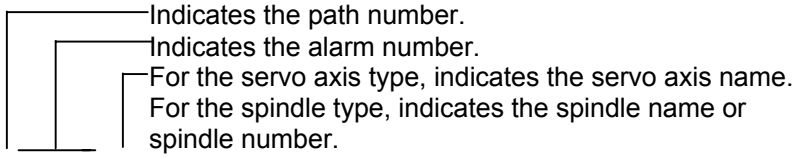
  This key is displayed with white characters in the green background.

- I/O signals
  When bit 6 (HDE) of parameter No. 3195 is set to 0, I/O signals specified on the operation history signal selection screen are recorded. Recorded signals are indicated on a bit-by-bit basis with information about the signal address and a change in bit. These signals are displayed with purple characters.



NOTE
1 When more than one bit at the same address changes at the same time, the changes of these bits are treated as one history data item.
2 A signal fluctuation for less than 4 msec is not recorded as history data.

- Alarms
  Alarm numbers and the issuance times are displayed on the operation history screen.
  This alarm information is displayed with white characters in the red background.



1_OT0506 (XA1)
2004/01/11 Indicates the date and time of issuance in two lines.
11:22:33

When a path name, axis name, or spindle name was changed after the issuance of an alarm, the alarm is indicated with a newly assigned name.

- Time and date
  The time and date of the following are displayed in two lines:
  i The time and date of power-on. These are displayed with white characters in the green background.
  ii The time and date of power-off. These are displayed with green characters.
  iii Date on which the date changed. This is displayed with black characters.
  iv Date and time at regular intervals of the period set in parameter No. 3122. These are displayed with black characters.

v Date and time when history data was erased. These are displayed with black characters.

NOTE
1 When times are recorded at regular intervals, if there is no data to record within an interval, the time is not recorded. (When the date is changed, the time is recorded on the assumption that there is data to record.)
2 When the date and time in the CNC system are changed, the date and time at which the change is made may be recorded as the date and time in iii or iv above.

History data not displayed on the screen

In addition to history data of MDI keys, I/O signal status, alarms issued, external operator messages (not displayed on the operation history screen), and time stamps, data described below can be recorded with time. Such history data cannot be displayed on the screen but can be output to external input/output devices. (See Subsection 12.4.15.5, "Outputting all history data".)

1 Detailed data at alarm issuance
If bit 7 (HAL) of parameter No. 3196 is set to 0, 10 modal G codes, auxiliary function codes D, E, F, H, M, N, O, S, and T, absolute coordinates, and machine coordinates in the block being executed when an alarm was issued are recorded together with the alarm number and the time of alarm issuance.
The group numbers of the 10 modal G codes to be recorded are set in parameter Nos. 12990 to 12999. If these parameters are not specified, modal G codes of groups 01 to 10 are recorded.

NOTE
1 For modal O data that is a program name, only the first five characters are recorded.
2 To record as many items as possible instead of recording detailed data at alarm issuance, set bit 7 (HAL) of parameter No. 3196 to 1.

2 External alarm messages and macro alarm messages
If bit 3 (EAH) of parameter No. 3112 is set to 1, external alarm messages and macro alarm messages can also be recorded as history data. However, only the first 64 characters of each message are recorded.

NOTE
To record also external alarm and macro alarm messages as history data, set bit 3 (EAH) of parameter No. 3112 to 1, and at the same time set bit 7 (HAL) of parameter No. 3196 to 0.

3 Modification of tool offset data
If bit 0 (HTO) of parameter No. 3196 is set to 1, when tool offset data is modified, the number and type of the tool offset are recorded as well as the tool offset data before modification, the tool offset data after modification, and the time of modification.

4 Modification of workpiece offset/extended workpiece offset/workpiece shift (T series) data
If bit 1 (HWO) of parameter No. 3196 is set to 1, when workpiece offset data is modified, the number of the modified workpiece offset is recorded as well as the workpiece offset data before modification, the workpiece offset data after modification, and the time of modification. Similar data is recorded also when extended workpiece offsets and workpiece shift amounts (T series) are modified.

5 Modification of parameters
If bit 2 (HPM) of parameter No. 3196 is set to 1, when a parameter is modified, the number and type (axis type, spindle type, path type, or machine group type) of the parameter are recorded as well as the parameter data before modification, the parameter data after modification, and the time of modification.

NOTE
Modifications made at power-on and modifications of passwords and keys are not recorded as history data.

6 Modification of custom macro common variables (#100 to #999)
If bit 3 (HMV) of parameter No. 3196 is set to 1, when a custom macro common variable is modified, the number of the common variable is recorded as well as the common variable value before modification, the common variable value after modification, and the time of modification.

Erasing history data from the operation history screen

Procedure

1 Display the operation history screen.
2 Press soft key [(OPRT)].
3 Press soft key [ALL CLEAR].
4 Press soft key [EXEC]. All history data is erased.

NOTE
An erasure operation erases not only operation history data but also other history data such as external operator message history data and alarm history data. It is impossible to erase only a specific history.

12.4.15.4 Selecting operation history signals

I/O signals to be recorded as history data can be selected. Up to 60 signals can be set.

Setting data

1 Press function key SYSTEM.
2 Press continuous menu key ▷ several times until soft key [OPERAT HISTRY] is displayed.
3 Press soft key [OPERAT HISTRY].
4 Press soft key [SIGNAL SELECT] to display the operation history signal selection screen.
5 Press soft key [(OPRT)].
6 Move the cursor to a desired position by using cursor keys ↑ and ↓.
7 Type a signal type (X, G, F, or Y) and an address, then press INPUT.
Example:
When G0004 is entered then INPUT is pressed:
The entered signal address, G0004, is set in ADDRESS, and initial value 00000000 is set in SIGNAL.
8 Select the bits to be recorded in the history.
To change the status of all bits of the specified signal address, place the cursor in all bits so that the bits are highlighted (for example, "00000000"), then press soft key [ON:1] or [OFF:0]. Then, the bits are set to 11111111 or 00000000.
To change the status of only a certain bit, move the cursor to the bit by using cursor keys ← and →, and press [ON:1] or [OFF:0]. The selected bit is then set to 1 or 0.
9 Up to 60 addresses can be set to select signals. Addresses need not necessarily be set sequentially starting from No. 1.

NOTE
1 While the operation history signal selection screen is being displayed, recording of history data is not performed.
2 Only X, Y, G, and F can be set for I/O signals.
For data not set, "********" is displayed.
3 Even when an address is set, history data is not recorded if all bits are set to 0.
4 When the ON/OFF width of an input signal is less than 4 msec, recording of history data is not performed.
Also,there are some signals that are not recorded.
5 When many signals are selected, the processing speed may lower.

Clearing data individually

1 Display the operation history signal selection screen.
2 Move the cursor to the data to be cleared.
3 Press soft key [DELETE].
4 Press soft key [EXEC].

Clearing all data

1 Display the operation history signal selection screen.
2 Press soft key [ALLDEL].
3 Press soft key [EXEC].

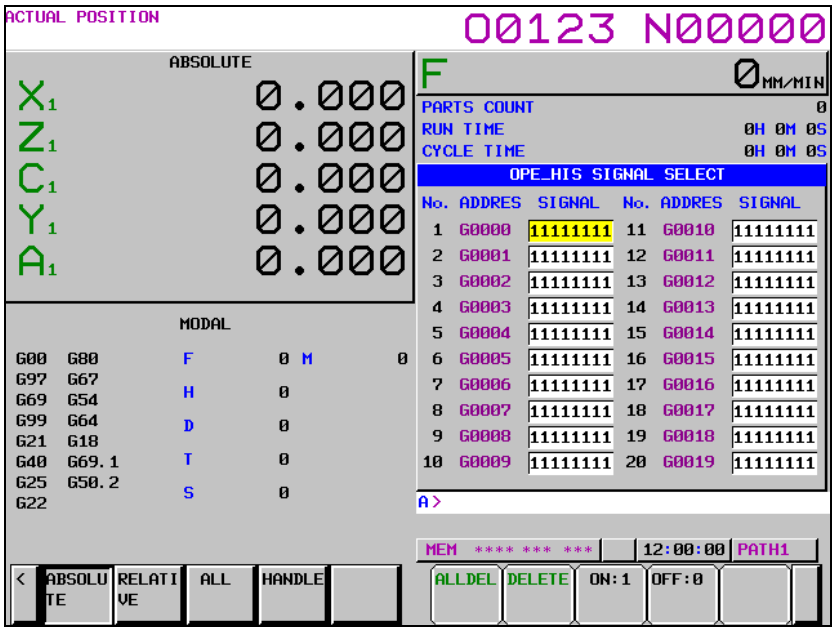


Fig. 12.4.15.4 (a) Operation history signal selection screen

12.4.15.5 Outputting all history data

All history data can be output to external input/output devices.
It is impossible, however, to output history data individually.

Procedure

1 Make an output device ready for output.
2 Set the EDIT mode.
3 Press function key

.
4 Press continuous menu key several times until soft key [OPERAT HISTRY] is displayed.
5 Press soft key [OPERAT HISTRY], then press newly displayed soft key [OPERAT HISTRY]. The operation history screen is displayed.
6 Press soft key [(OPRT)].
7 Press soft key [PUNCH].
8 Enter a file name, and press soft key [EXEC].
When soft key [EXEC] is pressed without entering a file name, the output file name is assumed to be OPRT_HIS.TXT.

Output format

History data is output as an ASCII file in the following format:

1 MDI keys
After "MDI", "path-number_", "key-data", and "input-time" are output in this order.
(Key data input at power-on is indicated as "Power on MDI".)
<Example>
MDI 01_A 12:23:34
MDI 02_<CAN> 12:23:34
MDI 02_[SOFT HF1] 12:23:35
MDI P_<RESET> 12:34:56
Power on MDI 01_<RESET> 12:34:56

2 I/O signals
After "DI/DO", "PMC-number_", "signal-address_bit-status", and "time-of-change" are output in this order.
<Example>
DI/DO 1_F0002.2_on 12:34:56
DI/DO 1_ G0043.0_off G0043.1_off 12:35:00
(For multiple bits at the same address)

3 Alarms
After "Alarm", "path-number_", "type", "alarm-number", "G-code-modal-data", "modal-data-other-than-G-code", "absolute-coordinate-value" and "machine-coordinate-value" for each axis, and "date-and-time-of-alarm-issuance" are output in this order.
An asterisk "*" is output before the modal data that was specified in a block executed when the alarm was issued.

<Example>

- Alarm 01_SR01973
  *G0. G97. G69. G99. G21. G50.2 G25. G13.1 B0. D0. E0. *F100. H0. M10. *N123. Test_ S1000. T1010.
  X1 ABS 197.999 MCN 197.999 Y1 ABS -199806.00 MCN -199806.00 Z1 ABS 297.009 MCN 0.123 C1 ABS 10395.999 MCN 0.000 at 2007/09/01 19:03:28
- Alarm 02_ OT00506(ZA2)
  *G1. G17. G90. G22. G94. G20. *G42. G49. G80. G12.1 B0. *D12. E0. *F100. H34. M0. *N123. O123 S0. T0.
  X2 ABS 123.999 MCN 234.000 Y2 ABS -123.00 MCN -234.00 ZA2 ABS 1234.567 MCN –1234.567 at 2007/09/01 12:34:56
- When additional information is not to be recorded when an alarm is issued (bit 7 (HAL) of parameter No. 3196 is set to 1), only "path-number_", "alarm-number", and "date-and-time-of-alarm-issuance" are output.
  Alarm 01_OT00506(XC1) at 2007/09/01 22:08:32
  Alarm 02_SW00100 at 2007/09/01 19:07:52
- When external alarm/macro alarm messages are to be recorded (bit 7 (HAL) of parameter No. 3196 is set to 0) and bit 3 (EAH) of parameter No. 3112 is set to 1, the messages are also output.
  Alarm 01_MC00001 Message ATC ALARM
  G0. G97. G69. G99. G21. G40. G25. G22. G80. D0. E0. F0. H0. M0. N0. O9999 S0. T0.
  X2 ABS 10.000 MCN 0.000Y2 ABS 123.000 MCN 0.000Z2 ABS 0.000 MCN 0.000 at 2007/09/01 10:06:43

4 External operator messages
After "EXT_Message", "message-number", "message", and "date-and-time-of-issuance" are output in this order.

<Example>
EXT_Message 01234 OIL PRESSURE DECREASE at 2007/09/01 2:38:43

5 Modification of tool offset data
After "Tool Offset", "path-number_", "type", "offset-number", "offset-data-before-modification", "offset-data-after-modification", and "time-of-modification" are output in this order. The following types are provided:

Common to M/T	:	G=Geometric compensation	W=Wear compensation
M series	:	H=Tool length compensation	D=Cutter compensation
T series	:	R=Tool-nose radius compensation	T=Tool nose direction

X2,Z2,Y2=Second geometric compensation

<Example>
Tool Offset 01_X0002 0.000 → 1 at 12:15:43
Tool Offset 02_XW0001 -9999.999 → 9999.999 at 12:15:46
Tool Offset 01_RG0032 0.000 → 0.003 at 12:15:52
Tool Offset 02_T0001 5. → 2. at 19:34:11
Tool Offset 02_W0123 -10.000 → 123.456 at 10:28:58
Tool Offset 01_HG0456 0.000 → 999.999 at 11:37:40
Tool Offset 01_ 0064 12.340 → 12.569 at 11:39:42

6 Modification of workpiece offset/extended workpiece offset/workpiece shift (T series) data
After "Work Offset", "EXT Work Offset", or "Work Shift", "path-number_(axis-name)", "type", "offset-number", "offset-data-before-modification", "offset-data-after-modification", and "time-of-modification" are output in this order.
<Example>
Work Offset 01_G55(XA1) 15.000 → 0.007 at 09:23:03
Work Offset 02_EXT(Z2) 0.000 → 300.003 at 09:22:50
EXT Work Offset 02_G54.1P300 (Y2) 123.456 → 9999.999 at 12:15:46
Work Shift (X) 02_999999.999 → 999999.999 at 10:22:37

7 Modification of parameters
After "Parameter", "type", "parameter-number", "parameter-before-modification", "parameter-after-modification", and "time-of-modification" are output in this order.
The following types are provided:
Path type : L is added before the path number.
Axis type : A is added before the axis number.
Spindle type : S is added before the spindle number.
Machine group type : Indicated as the machine type. T is added before the machine group number.
Others : No type is output.
<Example>
Paramater N03112 00000100 → 00001100 at 11:18:40
Paramater Path type N01410 L02 0.000 → 1000.000 at 18:58:48
Paramater Axis type N01423 A04(B2) 0.000 → 10000.000 at 18:58:48
Paramater Spindle type N04011 S1(S) 10011010 → 10011010 at 18:58:53
Paramater Machine type N06310 T01 0 → -32768 at 19:21:13

8 Modification of custom macro common variables (#100 to #999)
After "Macro variable", "path-number_", "#variable-number", "common-variable-value-before-modification", "common-variable-value-after-modification", and "time-of-modification" are output in this order.
Variable values are output in the data format M × (10**(-E)).
<Example>

- When #149 on the first path was modified from <empty> to 12.345
  Macro valiable 01_#149 Empty → 123450000*(10**-7) at 15:02:35
- When #549 on the second path was modified from -12.345 to 123456789012
  Macro valiable 02_#549 -123450000*(10**-7) →123456789*(10**3) at 15:03:27

9 Date and time

Power on at 2007/09/01 17:11:17	(Date and time of power-on)
Power off at 2007/09/01 17:49:17	(Date and time of power-off)
Date 2007/09/01 00:00:00	(Record indicating a change in date)
Time stamp at 2007/09/01 15:51:00	(Record at regular intervals)
Data delete at 2007/09/01 10:56:18	(Date and time when history data was deleted)

Screens of a 15-inch display unit

12.4.16 Displaying and Setting Parameters (15-inch Display Unit)

When the CNC and machine are connected, parameters are set to determine the specifications and functions of the machine in order to fully utilize the characteristics of the servo motor. The setting of parameters depends on the machine. Refer to the parameter list prepared by the machine tool builder.
Normally, the user need not change parameter setting.

Procedure for displaying and setting parameters

Procedure

1 Set 1 for PARAMETER WRITE to enable writing. See the procedure for enabling/disabling parameter writing described below.

2 Press function key [SYSTEM].

3 Press vertical soft key [PARAM] to display the parameter screen.

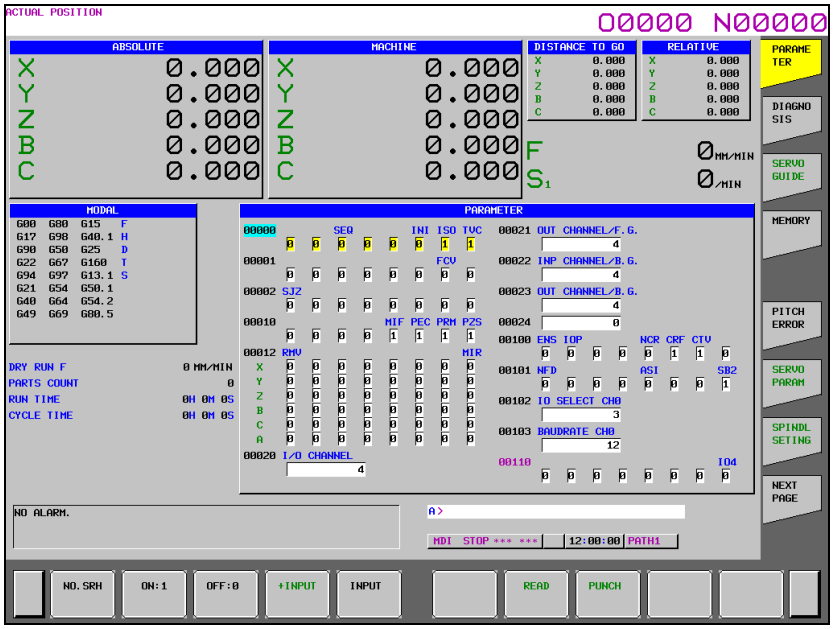


Fig. 12.4.16 (a) PARAMETER screen (15-inch)

4 Move the cursor to the parameter number to be set or displayed in either of the following ways:

- Enter the parameter number and press horizontal soft key [NO.SRH] .
- Move the cursor to the parameter number using the page keys, [PAGE↑] and [PAGE↓], and cursor keys, [↑], [↓], [←], and [→].

5 To set the parameter, enter a new value with numeric keys and press horizontal soft key [INPUT]. The parameter is set to the entered value and the value is displayed.
6 Set 0 for PARAMETER WRITE to disable writing.

Procedure for enabling/displaying parameter writing

Procedure

1 Select the MDI mode or enter state emergency stop.
2 Press function key .
3 Press vertical soft key [SETTING] to display the setting screen.

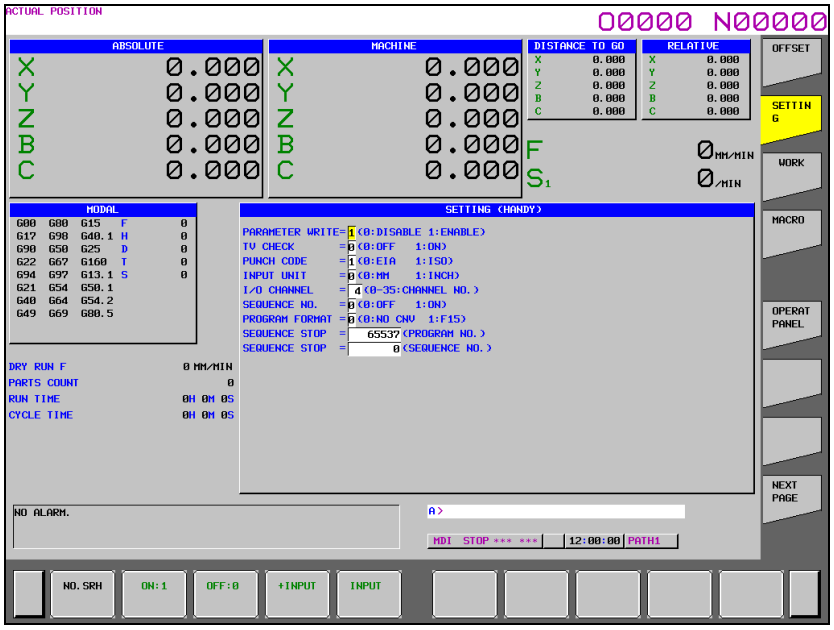


Fig. 12.4.16 (b) SETTING screen (15-inch)

4 Move the cursor to PARAMETER WRITE using cursor keys.
5 Press horizontal soft key [ON:1] to enable parameter writing.
At this time, the CNC enters the alarm state SW0100.
6 After completing parameter setting, move the cursor to PARAMETER WRITE on the setting screen again, and press horizontal soft key [OFF:0].
7 Depress the key to release the alarm condition.

When an alarm PW0000 occurred, however, it is not released unless the power is turned off and back on.

Explanation

- Setting parameters with external input/output devices

See III-8 for setting parameters with external input/output devices such as the memory card.

- Parameters that require turning off the power

Some parameters are not effective until the power is turned off and on again after they are set. Setting such parameters causes alarm PW0000. In this case, turn off the power, then turn it on again.

- Parameter list

Refer to the Parameter Manual (B-63950EN) for the parameter list.

- Setting data

Some parameters can be set on the setting screen if the parameter list indicates "Setting entry is acceptable". Setting 1 for PARAMETER WRITE is not necessary when three parameters are set on the setting screen.

12.4.17 Displaying and Setting Pitch Error Compensation Data (15-inch Display Unit)

If pitch error compensation data is specified, pitch errors of each axis can be compensated in detection unit per axis.

Pitch error compensation data is set for each compensation point at the intervals specified for each axis. The origin of compensation is the reference position to which the tool is returned.

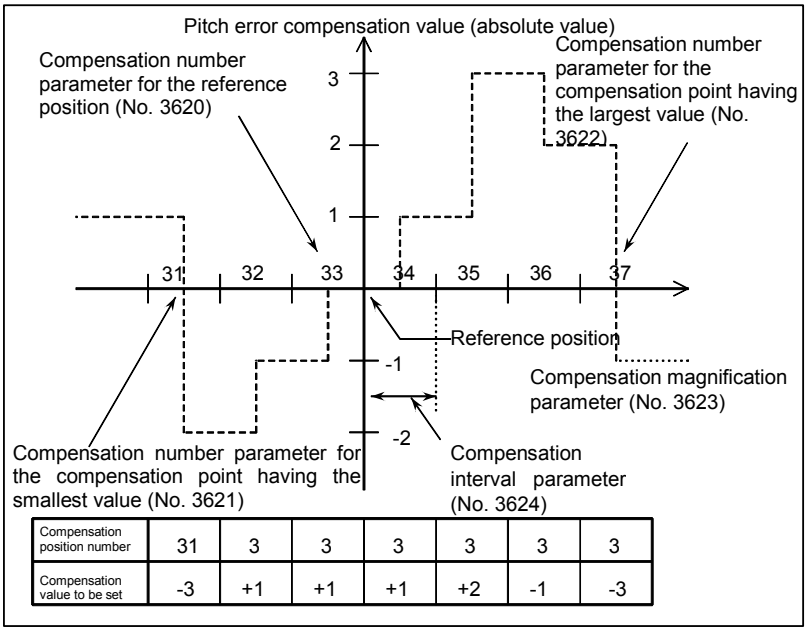
The pitch error compensation data is set according to the characteristics of the machine connected to the NC. The content of this data varies according to the machine model. If it is changed, the machine accuracy is reduced.

In principle, the end user must not alter this data.

Pitch error compensation data can be set with external devices such as the memory card (see Chapter III-8). Compensation data can also be written directly with the MDI panel.

The following parameters must be set for pitch error compensation. Set the pitch error compensation value for each pitch error compensation point number set by these parameters.

In the following example, 33 is set for the pitch error compensation point at the reference position.



Compensation position number	31	3	3	3	3	3	3
Compensation value to be set	-3	+1	+1	+1	+2	-1	-3

- Number of the pitch error compensation point at the reference position (for each axis): Parameter No. 3620
- Number of the pitch error compensation point having the smallest value (for each axis): Parameter No. 3621
- Number of the pitch error compensation point having the largest value (for each axis): Parameter No. 3622
- Pitch error compensation magnification (for each axis): Parameter No. 3623
- Interval of the pitch error compensation points (for each axis): Parameter No. 3624

- Travel distance per revolution of pitch error compensation of the rotary axis type (for each axis): Parameter No. 3625

- Bi-directional pitch error compensation

The bi-directional pitch error compensation function allows independent pitch error compensation in different travel directions. (When the movement is reversed, compensation is automatically carried out as in a backlash.)
To use this function, specify pitch error compensation for each travel direction, that is, separately for the positive and negative directions of a movement.
When using bi-directional pitch error compensation (setting bit 0 (BDPx) of parameter No. 3605 to 1), specify the following parameters in addition to the pitch error compensation parameter.

- Number of the pitch error compensation point at the negative end (for travel in the positive direction, for each axis): Parameter No. 3621
- Number of the pitch error compensation point at the positive end (for travel in the positive direction, for each axis): Parameter No. 3622
- Number of the pitch error compensation point at the negative end (for travel in the negative direction, for each axis): Parameter No. 3626
- Pitch error compensation in the reference position when moving to the reference position from opposite to the reference position return direction (for each axis): Parameter No. 3627

Procedure for displaying and setting the pitch error compensation data

Procedure

1 Set the following parameters:
- Number of the pitch error compensation point at the reference position (for each axis): Parameter No. 3620
- Number of the pitch error compensation point having the smallest value (for each axis): Parameter No. 3621
- Number of the pitch error compensation point having the largest value (for each axis): Parameter No. 3622
- Pitch error compensation magnification (for each axis): Parameter No. 3623
- Interval of the pitch error compensation points (for each axis): Parameter No. 3624
- Travel distance per revolution of pitch error compensation of the rotary axis type (for each axis): Parameter No. 3625

When using bi-directional pitch error compensation (setting bit 0 (BDPx) of parameter No. 3605 to 1), specify the following parameters in addition to the pitch error compensation parameter.
- Number of the pitch error compensation point at the negative end (for travel in the positive direction, for each axis): Parameter No. 3621

- Number of the pitch error compensation point at the positive end (for travel in the positive direction, for each axis): Parameter No. 3622
- Number of the pitch error compensation point at the negative end (for travel in the negative direction, for each axis): Parameter No. 3626
- Pitch error compensation in the reference position when moving to the reference position from opposite to the reference position return direction (for each axis): Parameter No. 3627

2 Press function key SYSTEM.

3 Press vertical soft key [PITCH ERROR]. The following screen is displayed:

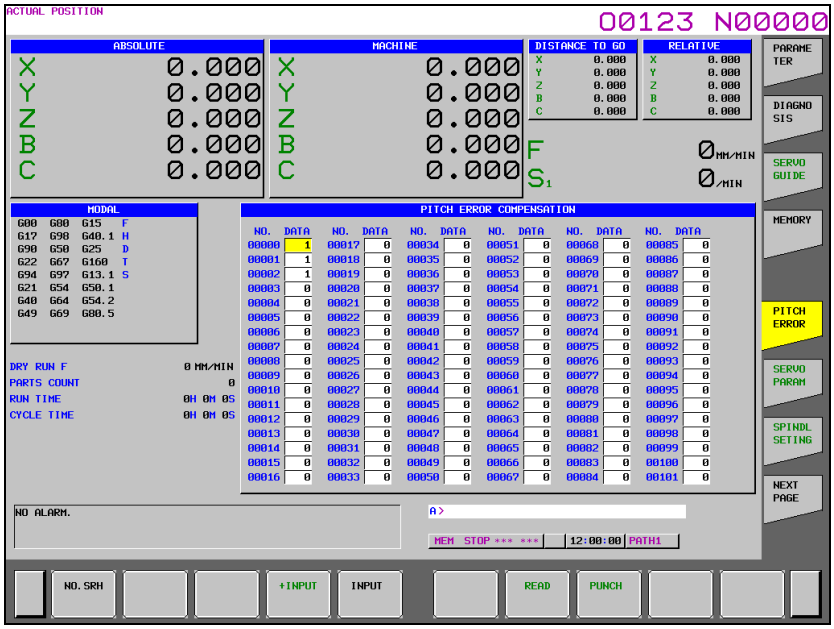


Fig. 12.4.17 (a) PITCH ERROR COMPENSATION screen (15-inch)

4 Move the cursor to the compensation point number to be set in either of the following ways:

- Enter the compensation point number and press the horizontal soft key [NO.SRH].
- Move the cursor to the compensation point number using the page keys, PAGE and PAGE, and cursor keys, ↑, ↓, ←, and →.

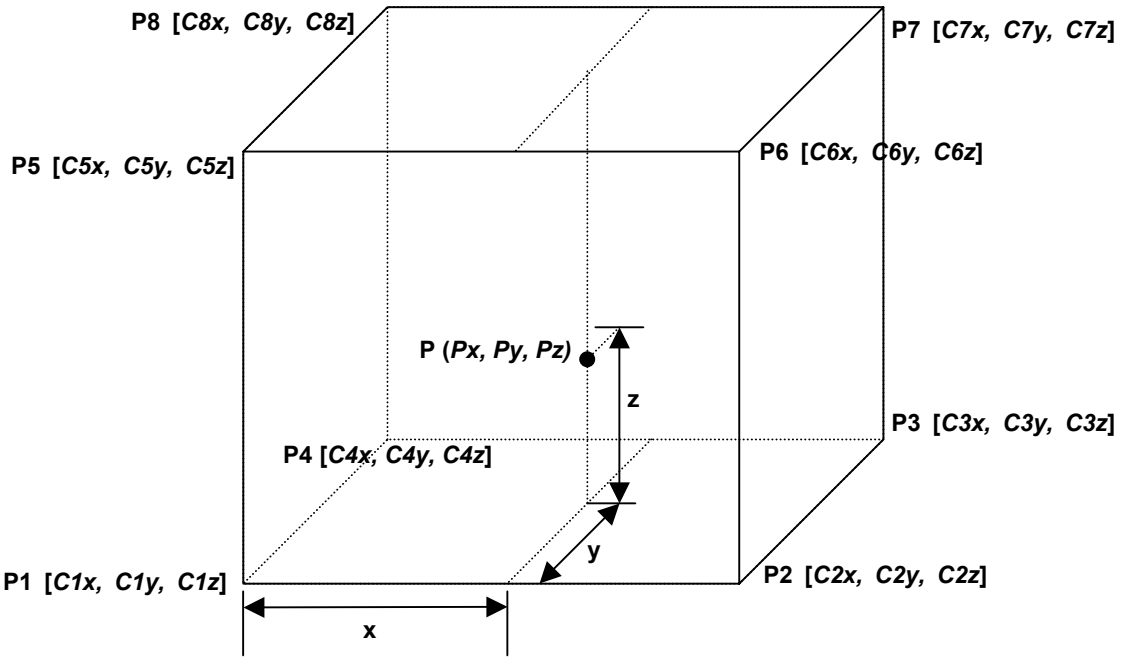
5 Enter a value with numeric keys and press horizontal soft key [INPUT].

12.4.18 Displaying and Setting Three-Dimensional Error Compensation Data (15-inch Display Unit)

In ordinary pitch error compensation, compensation is applied to a specified compensation axis (single axis) by using its position information. For example, pitch error compensation is applied to X-axis by using the position information of X-axis.
Three-dimensional error compensation is a function that adjusts the current position by calculating compensation data (for three axes) from the compensation amounts at surrounding compensation points (eight points) on the basis of the interior division ratio in the compensation area (rectangular parallelepiped) containing the current position on up to three compensation axes.

- Calculation of compensation

Three-dimensional error compensation is calculated as follows.



Let three compensation axes be X, Y, and Z (three basic axes) and the coordinates of the current position be P (Px, Py, Pz). consider a compensation space (rectangular parallelepiped) containing P. Let its vertexes be P1, P2, …, and P8 and the compensation values for the individual axes at the individual vertexes be Cnx, Cny, and Cnz (where n is a number between 1 and 8).
Let the interior division ratio on X-axis at P be x. Here, x is standardized in the range of 0 to 1 as follows:

$$x = \frac{|Px - P1x|}{|P2x - P1x|}$$

P1x and P2x are the X coordinates of P1 and P2. The interior division ratios on Y and Z-axes are determined in the same way.

The compensation amount Cx for X-axis at P is determined as follows:

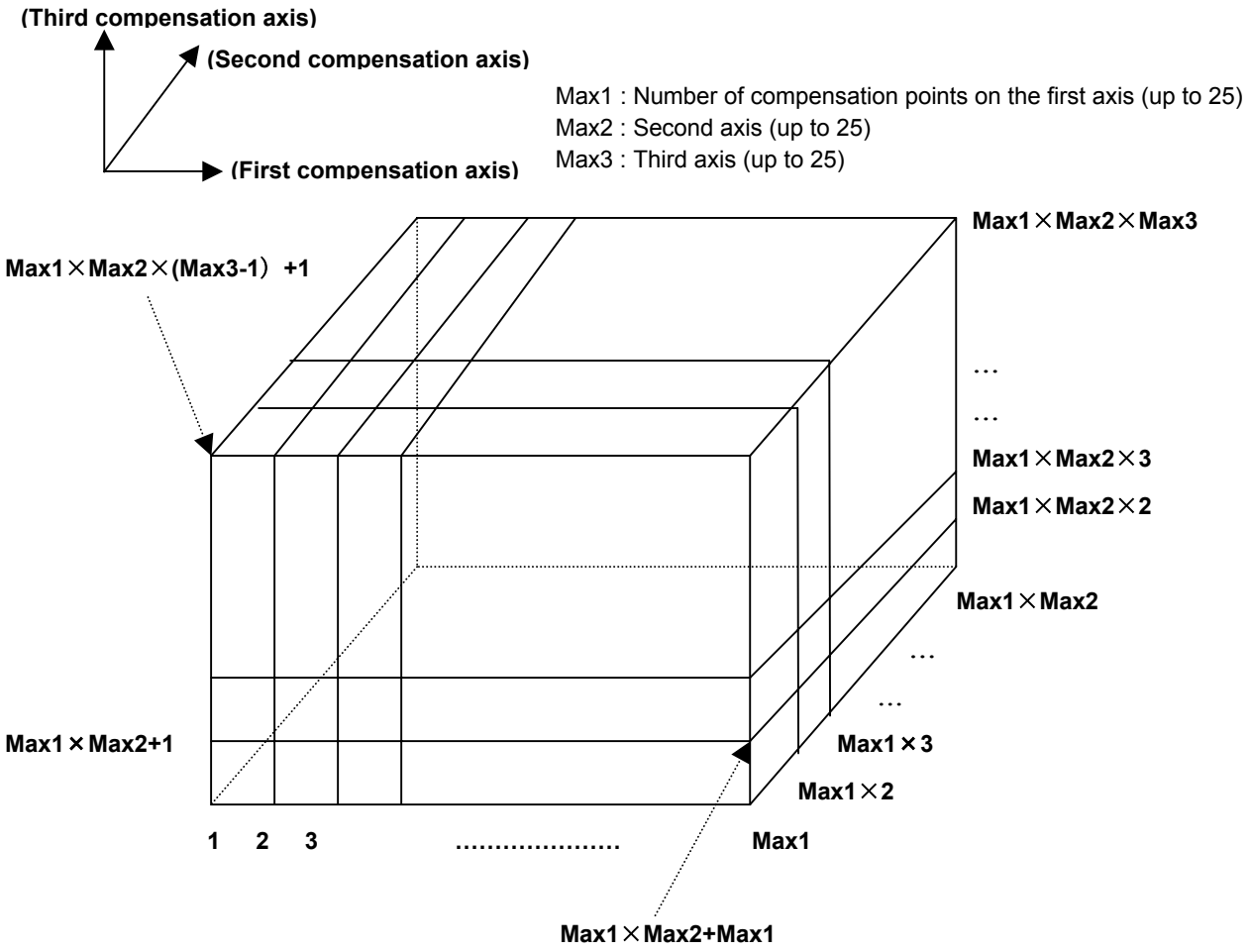
$$\begin{aligned} Cx = {} & C1x \times (1-x) \times (1-y) \times (1-z) + C2x \times x \times (1-y) \times (1-z) \\ & + C3x \times x \times y \times (1-z) + C4x \times (1-x) \times y \times (1-z) \\ & + C5x \times (1-x) \times (1-y) \times z + C6x \times x \times (1-y) \times z \\ & + C7x \times x \times y \times z + C8x \times (1-x) \times y \times z \end{aligned}$$

The compensation amount Cy and Cz on Y and Z-axes are determined in the same way.

The actual compensation amounts are the calculated compensation amounts multiplied by the compensation magnifications (Parameter No.10809 to 10811).

- Number of compensation points

Up to 15625 compensation points (up to 25 points per axis) can be set. The numbers of compensation points on the individual axes are set for parameters No.10803 to 10805. The ordering of the compensation point numbers in the compensation space is as follows.



Displaying and setting three-dimensional error compensation data

Procedure

1 Set the following parameters:
- First compensation axis for three-dimensional error compensation : Parameter No. 10800
- Second compensation axis for three-dimensional error compensation : Parameter No. 10801
- Third compensation axis for three-dimensional error compensation : Parameter No. 10802
- Number of compensation points for three-dimensional error compensation (first compensation axis)
  Parameter No. 10803
- Number of compensation points for three-dimensional error compensation (second compensation axis)
  Parameter No. 10804
- Number of compensation points for three-dimensional error compensation (third compensation axis)
  Parameter No. 10805
- Compensation point number of the reference position for three-dimensional error compensation (first compensation axis) : Parameter No. 10806
- Compensation point number of the reference position for three-dimensional error compensation (second compensation axis) : Parameter No. 10807
- Compensation point number of the reference position for three-dimensional error compensation (third compensation axis) : Parameter No. 10808
- Magnification for three-dimensional error compensation (first compensation axis) : Parameter No. 10809
- Magnification for three-dimensional error compensation (second compensation axis) : Parameter No. 10810
- Magnification for three-dimensional error compensation (third compensation axis) : Parameter No. 10811
- Compensation interval for three-dimensional error compensation (first compensation axis)
  Parameter No. 10812
- Compensation interval for three-dimensional error compensation (second compensation axis)
  Parameter No. 10813
- Compensation interval for three-dimensional error compensation (third compensation axis)
  Parameter No. 10814

2 Press function key

.

3 Press vertical soft key [NEXT PAGE] several times, then press vertical soft key [3D ERR COMP]. The following screen appears:

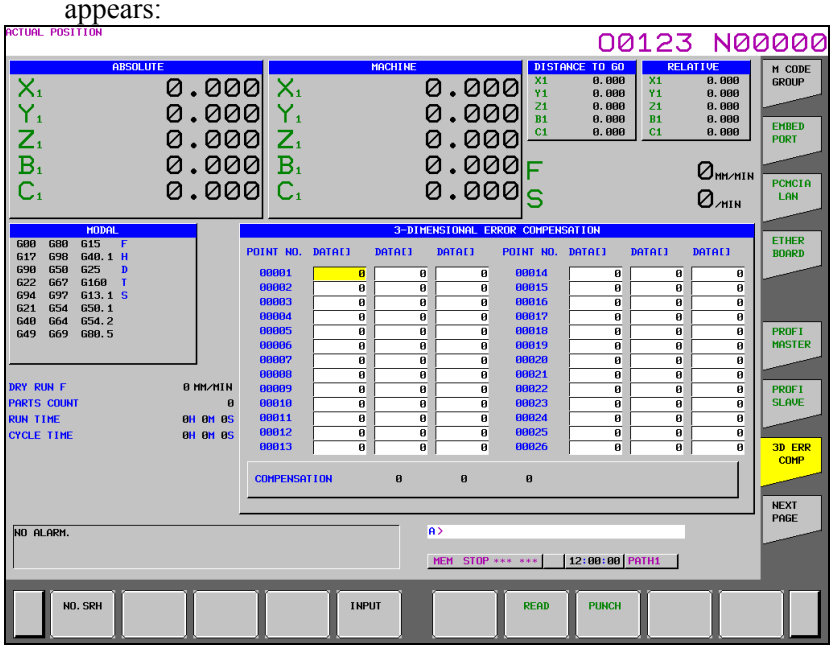


Fig. 12.4.18 (a) 3-DIMENSIONAL ERROR COMPENSATION screen (15-inch)

4 Move the cursor to the position of the compensation point number you want to set using either of the following methods:

- Select the MDI mode.
- Set bit 0 (PWE) of parameter No. 8900 to 1.
- Enter a compensation point number and press horizontal soft key [NO. SRH].
- Move the cursor to the compensation point number or compensation axis you want to set by pressing page keys PAGE ↑ and/or PAGE ↓, and cursor keys ↑, ↓, ←, and →.

5 Enter compensation data in detection units. The valid data range is from -128 to 127.

12.4.19 Servo Parameters (15-inch Display Unit)

This subsection describes the initialization of digital servo parameters performed, for example, at the time of field tuning of the machine tool.

Procedure for servo parameter setting

Procedure

1 Turn on the power in the emergency stop state.
2 Set bit 0 (SVS) of parameter No. 3111 to 1 to display servo setting and tuning screens.
3 Press function key SYSTEM then vertical soft key [SERVO PARAM]. The following screen appears:

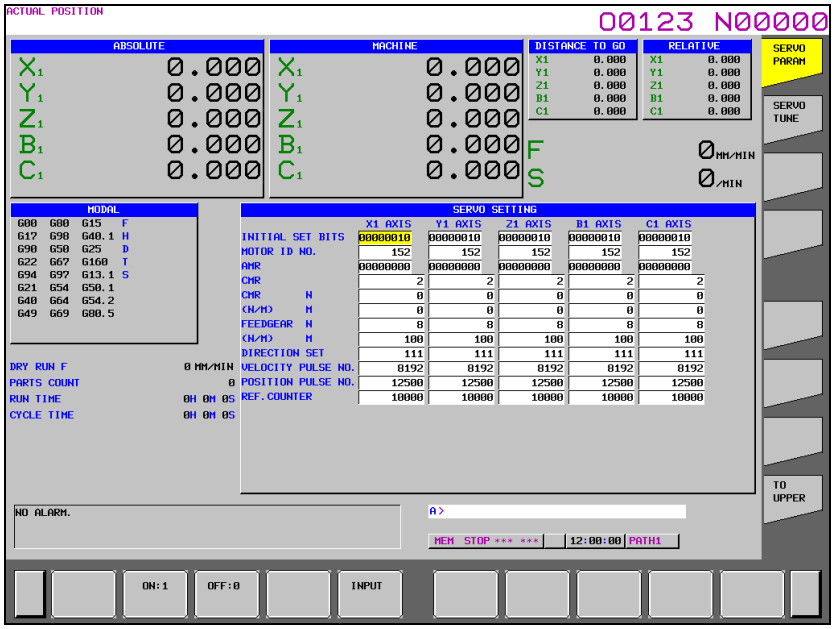


Fig. 12.4.19 (a) Servo parameter setting screen (15-inch)

4 With the page keys and cursor keys, move the cursor to the position of data to be set or modified.
5 Key in a desired value then press horizontal soft key [INPUT].

12.4.20 Servo Tuning (15-inch Display Unit)

Data related to servo tuning is displayed and set.

Procedure for servo tuning

Procedure

1 Turn on the power in the emergency stop state.
2 Set bit 0 (SVS) of parameter No. 3111 to 1 to display servo setting and tuning screens.
3 Press function key SYSTEM then vertical soft key [SERVO PARAM].
4 Press vertical soft key [SERVO TUNE] to select the servo tuning screen. The following screen appears:

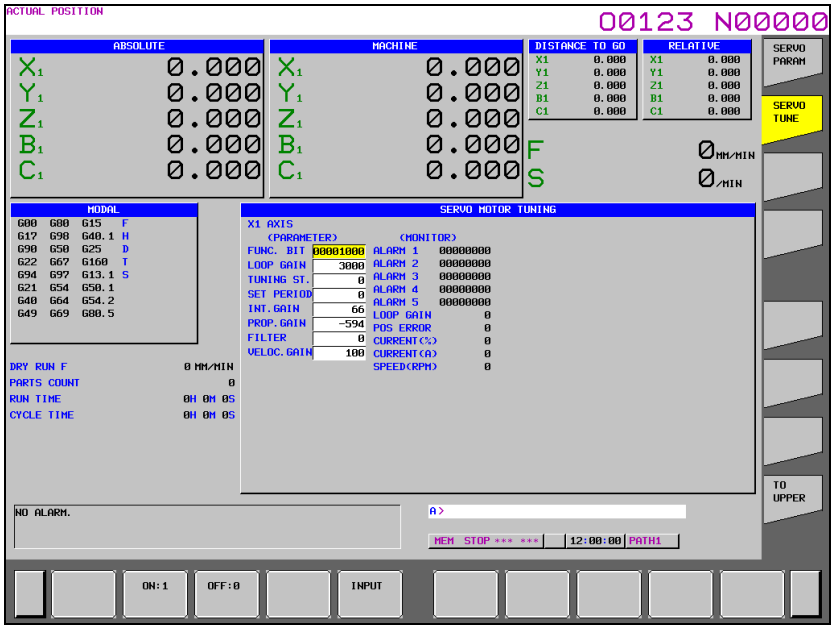


Fig. 12.4.20 (a) Servo tuning screen (15-inch)

5 With the page keys and cursor keys, move the cursor to the position of data to be set or modified.
6 Key in a desired value then press horizontal soft key [INPUT].

12.4.21 Spindle Setting (15-inch Display Unit)

Parameters related to spindles are set and displayed. In addition to the parameters, related data can be displayed. Screens for spindle setting, spindle tuning, and spindle monitoring are provided.

Setting spindle parameters

Procedure

1 Set bit 1 (SPS) of parameter No. 3111 to 1 to display spindle setting and tuning screens.

2 Press function key

then vertical soft key [SPINDL SETING].

3 Press vertical soft key [SPINDL SETING] to select the spindle setting screen.
The following screen appears:

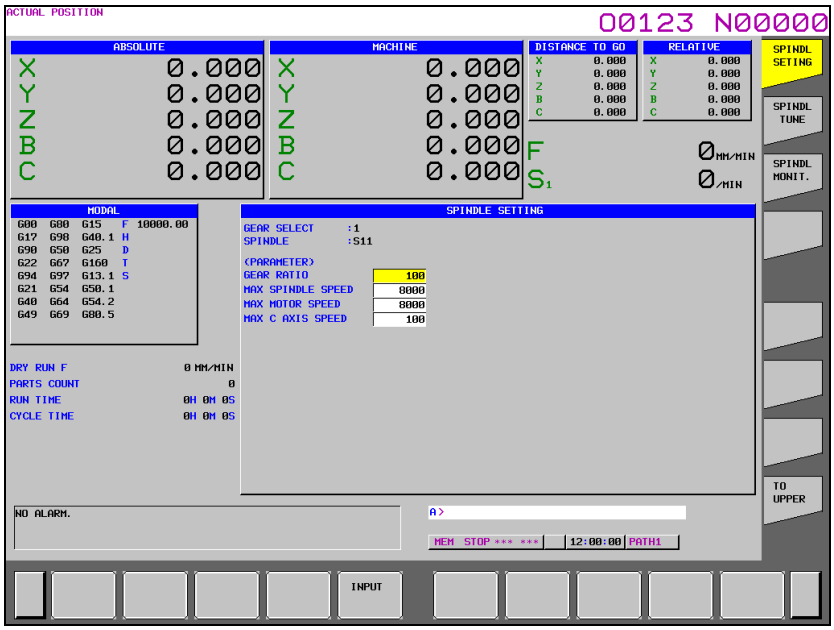


Fig. 12.4.21 (a) Spindle setting screen (15-inch)

4 With the page keys and cursor keys, move the cursor to the position of data to be set or modified.

5 Key in a desired value then press horizontal soft key [INPUT].

12.4.22 Spindle Tuning (15-inch Display Unit)

Spindle tuning data is displayed and set.

Setting for spindle tuning

Procedure

1 Set bit 1 (SPS) of parameter No. 3111 to 1 to display spindle setting and tuning screens.

2 Press function key SYSTEM then vertical soft key [SPINDL SETING].

3 Press vertical soft key [SPINDL TUNE] to select the spindle tuning screen.
The following screen appears:

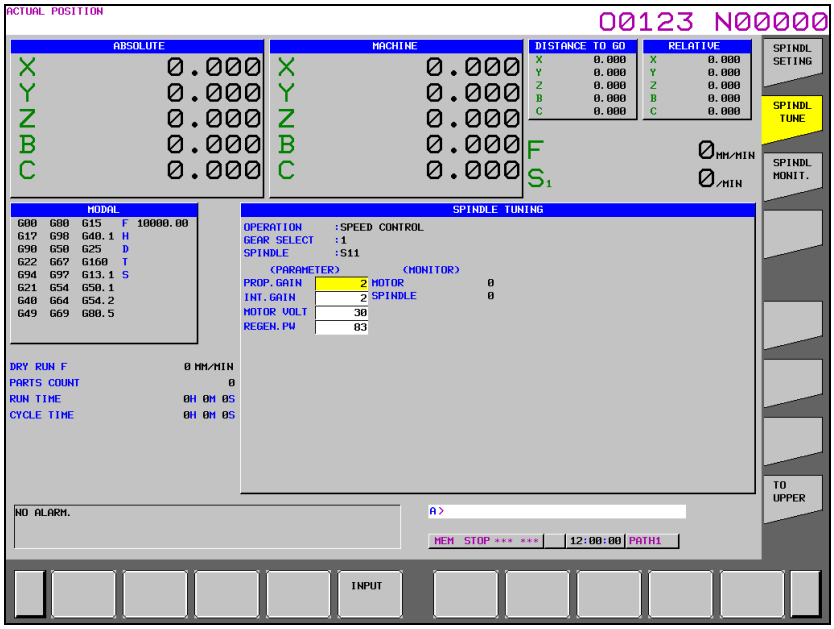


Fig. 12.4.22 (a) Spindle tuning screen (15-inch)

4 With the page keys and cursor keys, move the cursor to the position of data to be set or modified.

5 Key in a desired value then press horizontal soft key [INPUT].

12.4.23 Spindle Monitor (15-inch Display Unit)

Spindle-related data is displayed.

Displaying the spindle monitor

Procedure

1 Set bit 1 (SPS) of parameter No. 3111 to 1 to display spindle setting and tuning screens.

2 Press function key

then vertical soft key [SPINDL SETING].

3 Press vertical soft key [SPINDL MONIT.] to select the spindle monitor screen.
The following screen appears:

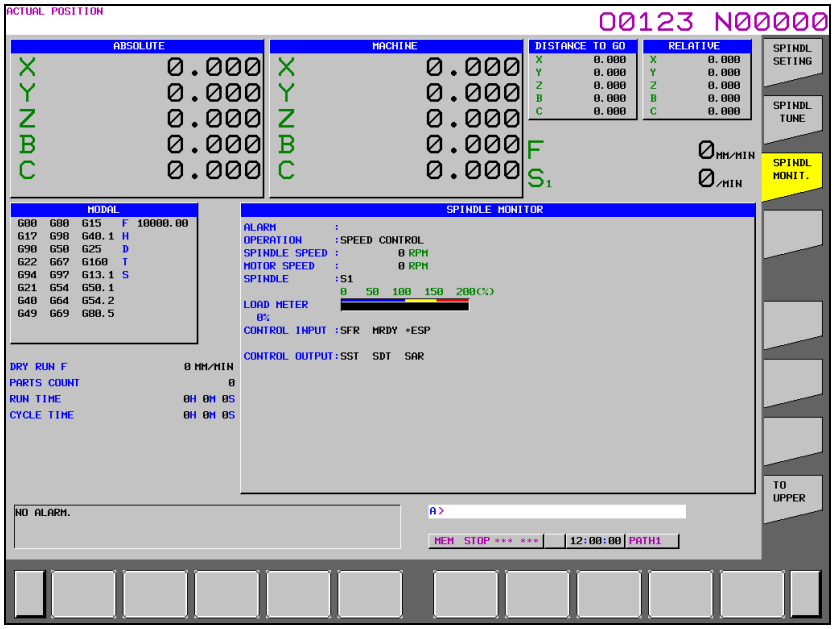


Fig. 12.4.23 (a) Spindle monitor screen (15-inch)

12.4.24 Color Setting Screen (15-inch Display Unit)

Screen colors can be set on the color setting screen.

Displaying the color setting screen

Procedure

1 Press function key SYSTEM.
2 Press vertical soft key [NEXT PAGE] several times to display vertical soft key [COLOR].
3 Press vertical soft key [COLOR] to display the color setting screen.

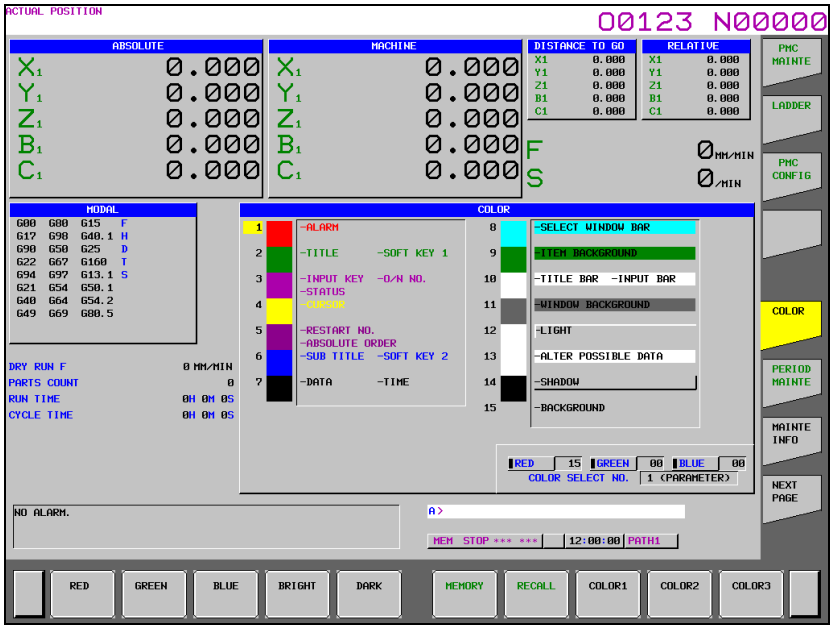


Fig. 12.4.24 (a) Color setting screen (15-inch)

Procedure for operating the color setting screen

- Modifying the color (color palette values)

1 The following horizontal soft keys are displayed:

2 Move the cursor to a color number whose color palette values are to be modified.
The current color palette value set for each primary color is displayed.

3 Select a primary color whose setting is to be modified, with the corresponding horizontal soft key [RED], [GREEN], or [BLUE].
More than one primary color can be selected at the same time.
Each time horizontal soft key [RED], [GREEN], or [BLUE] is pressed, the operation soft key toggles between selection and deselection.

4 Select horizontal soft key [BRIGHT] or [DARK] to modify the brightness of the selected prime color(s).

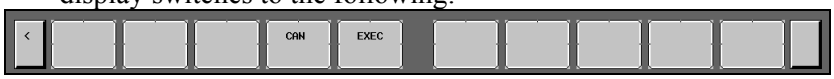
- Storing the color (color palette values)

The set color palette values can be stored.

1 Press horizontal soft key [COLOR1], [COLOR2], or [COLOR3] to select a storage area.

COLOR1 Standard color data parameters (Nos. 6581 to 6595)
COLOR2 Parameters (Nos. 10421 to 10435)
COLOR3 Parameters (Nos. 10461 to 10475)

2 Press horizontal soft key [MEMORY]. The horizontal soft key display switches to the following:

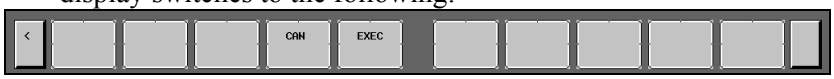


3 Pressing horizontal soft key [EXEC] stores the current color palette settings in the selected area.
Pressing horizontal soft key [CAN] or the leftmost key does not store the current color palette settings in the selected area.

- Calling the color (color palette values)

1 Press horizontal soft key [COLOR1], [COLOR2], or [COLOR3] to select a storage area where color palette values are stored.

2 Press horizontal soft key [RECALL]. The horizontal soft key display switches to the following:



3 Pressing horizontal soft key [EXEC] calls the color palette values from the selected area to enable the color to be modified. This operation is invalid if no color palette values are stored.
Pressing horizontal soft key [CAN] or the leftmost key does not call the color palette values from the selected area.

NOTE
1 Immediately after the power is turned on, the settings of COLOR1 (parameters) are used for display. If nothing is stored for COLOR1, the default colors are used for display.
2 Do not modify the color setting data parameters directly by MDI key input. When modifying the standard color data, be sure to perform a storage operation on the color setting screen.

12.4.25 Machining Parameter Tuning (15-inch Display Unit)

In AI contour control, by setting a velocity-emphasized parameter set and precision-emphasized parameter set and setting the precision level matching a machining condition such as rough machining or finish machining on the precision level setting screen or by programming, the parameters suitable for the condition can be automatically calculated to perform machining.
On this screen, the parameter sets for emphasis on velocity (precision level 1) and emphasis on precision (precision level 10) can be set.
Set the following parameters:

- Acceleration rate of acceleration/deceleration before interpolation
- Acceleration change time (bell-shaped)
- Allowable acceleration change value for each axis in velocity control based on acceleration change under jerk control
- Allowable acceleration change value for each axis in acceleration change under jerk control in successive linear interpolation operations
- Ratio of the change time of the rate of change of acceleration in smooth bell-shaped acceleration/deceleration before interpolation
- Allowable acceleration rate
- Acceleration rate of acceleration/deceleration after interpolation
- Corner speed difference
- Maximum feedrate
- Items that can be set freely (2 items)

For details of each parameter, see the descriptions of AI contour control and jerk control.
By setting bit 0 (MPR) of parameter No. 13601 to 1, this screen can be hidden.
For the method of setting a precision level, see the description of the precision level selection screen in Subsection 12.3.27.

Procedure for machining parameter tuning

Procedure

1 Set the MDI mode.

2 Press function key [SYSTEM].

3 Press vertical soft key [NEXT PAGE] several times to display vertical soft key [MCHN TUNING]. The following screen appears:

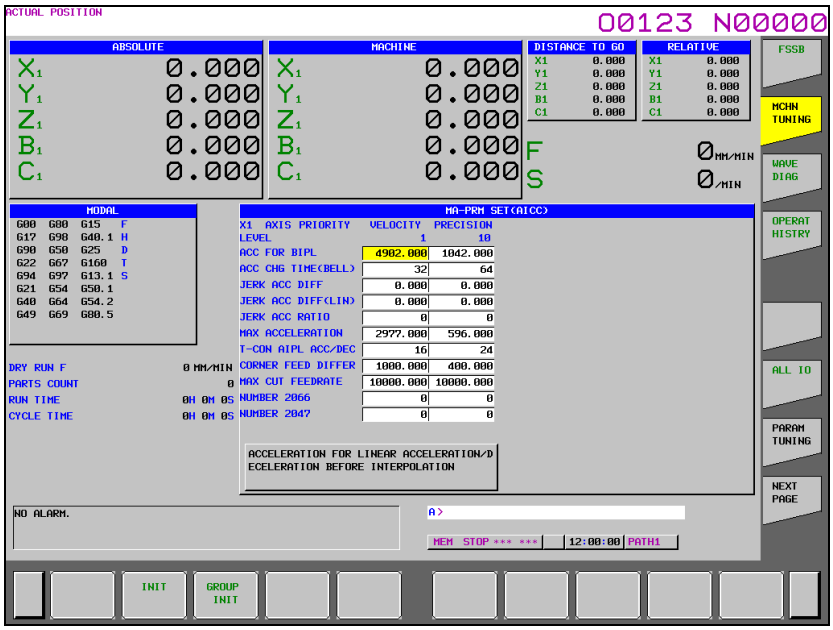


Fig. 12.4.25 (a) Machining parameter tuning screen (15-inch)

4 Move the cursor to the position of a parameter to be set, as follows:

Press page key [PAGE] or [PAGE], and cursor keys [↑], [↓], [←] and /or [→] to move the cursor to the parameter.

5 Key in desired data then press the [INPUT] key on the MDI panel.

6 When data is input, a RMS value is found according to the precision level parameters. (The precision level can be changed on the precision level selection screen or parameter setting screen.) If a RMS value calculation fails, a warning (indicating that automatic setting failed) is displayed.

7 Repeat steps 4 and 5 until all machining parameters are set.

8 In addition to the setting method described above, a parameter setting method using horizontal soft keys is available. Pressing horizontal soft key [INIT] displays the standard value (recommended by FANUC) of the item selected by the cursor in the key input buffer. Pressing horizontal soft key [EXEC] initializes the item to the standard value. Pressing horizontal soft key [GROUP INIT] initializes all items of a group (emphasis on velocity or emphasis on precision) selected by the cursor to the standard values.

The table below indicates the initial settings.

Table 12.4.25 (a) Initial settings

Setting item	AI contour control		Unit
	Emphasis on velocity (LV1)	Emphasis on precision (LV10)	
Acceleration rate of acceleration/deceleration before interpolation <ACC FOR BIPL>	4902.000	1042.000	mm/sec^2
Acceleration change time (bell-shaped) <ACC CHG TIME(BELL)>	32	64	msec
Allowable acceleration change value <JERK ACC DIFF>	0	0	mm/sec^2
Allowable acceleration change value in successive linear interpolation operations <JERK ACC DIFF(LIN)>	0	0	mm/sec^2
Ratio of the change time of the jerk control <JERK ACC RATIO>	0	0	%
Allowable acceleration rate <MAX AACELERATION>	2977.000	596.000	mm/sec^2
Time constant for acceleration/deceleration after interpolation <T-CON AIPL ACC/DEC>	24	24	msec
Corner speed difference <CORNER FEED DIFFER>	1000	400	mm/min
Maximum cutting speed <MAX CUT FEEDRATE>	10000	10000	mm/min

Explanation

- Look-ahead acceleration/deceleration before interpolation

Set an acceleration rate for a linear portion in look-ahead acceleration/deceleration before interpolation.
Unit of data: mm/sec^2, inch/sec^2, deg/sec^2 (machine unit)

The parameter set on the machining parameter tuning screen is reflected in the following parameters:
Parameter No. 13610 (velocity-emphasized parameter)
Parameter No. 13611 (precision-emphasized parameter)

Moreover, the following parameter is also set according to the precision level:
Parameter No. 1660: Maximum allowable acceleration rate for each axis in acceleration/deceleration before interpolation

- Acceleration change time (bell-shaped)

Set a time constant for a bell-shaped portion in acceleration/ deceleration before look-ahead interpolation.
Unit of data: ms

The parameter set on the machining parameter tuning screen is reflected in the following parameters:
Parameter No. 13612 (velocity-emphasized parameter)
Parameter No. 13613 (precision-emphasized parameter)

When bit 7 of parameter No. 13600 is set to 1, the parameter set on the machining parameter tuning screen is reflected in the following parameters for which the valid data range is extended to the range from 0 to 200:
Parameter No. 13662 (velocity-emphasized parameter)
Parameter No. 13663 (precision-emphasized parameter)

Moreover, the following parameter is also set according to the precision level:
Parameter No. 1772: Time constant for bell-shaped look-ahead acceleration/deceleration before interpolation of constant acceleration time type

> ⚠ **CAUTION**
> A set time constant is applied to all axes. So, a modification made to this item changes the settings for all axes.

- Allowable acceleration change value in velocity control based on acceleration change under jerk control

Unit of data: mm/sec^2, inch/sec^2, deg/sec^2 (machine unit)

Set an allowable acceleration change value per ms for each axis in velocity control based on acceleration change under jerk control.

The parameter set on the machining parameter tuning screen is reflected in the following parameters:
Parameter No. 13614 (velocity-emphasized parameter)
Parameter No. 13615 (precision-emphasized parameter)

Moreover, the following parameter is also set according to the precision level:
Parameter No. 1788: Allowable acceleration change value for each axis in velocity control based on acceleration change under jerk control

> **NOTE**
> This setting item is displayed only when the jerk control function is enabled.

- Allowable acceleration change value for each axis in velocity control based on acceleration change under jerk control in successive linear interpolation operations

Unit of data: mm/sec^2, inch/sec^2, deg/sec^2 (machine unit)

Set an allowable acceleration change value per ms for each axis in velocity control based on acceleration change under jerk control in successive linear interpolation operations.

The parameter set on the machining parameter tuning screen is reflected in the following parameters:
Parameter No. 13616 (velocity-emphasized parameter)
Parameter No. 13617 (precision-emphasized parameter)

Moreover, the following parameter is also set according to the precision level:
Parameter No. 1789: Allowable acceleration change value for each axis in velocity control based on acceleration change under jerk control in successive linear interpolation operations

⚠ CAUTION
1 For an axis with 0 set in this parameter, the parameters (allowable acceleration change value in velocity control based on acceleration change under jerk control: No. 13614, No.13615) are valid.
2 For an axis with 0 set in the parameter (allowable acceleration change value in velocity control based on acceleration change under jerk control: No. 13614, No.13615), velocity control based on acceleration change is disabled, so that this parameter has no effect.

NOTE
This setting item is displayed only when the jerk control function is enabled.

- Ratio of the change time of the jerk control in smooth bell-shaped acceleration/deceleration before interpolation

Unit of data: %

Set the ratio (in %) of the change time of jerk control to the change time of acceleration in smooth bell-shaped acceleration/deceleration before interpolation.

The parameter set on the machining parameter tuning screen is reflected in the following parameters:
Parameter No. 13618 (velocity-emphasized parameter)
Parameter No. 13619 (precision-emphasized parameter)

Moreover, the following parameter is also set according to the precision level:
Parameter No. 1790: Ratio of the change time of the jerk control in smooth bell-shaped acceleration/deceleration before interpolation

NOTE
This setting item is displayed only when the jerk control function is enabled.

- Allowable acceleration rate

Set an allowable acceleration rate in acceleration-based speed determination.
Unit of data: mm/sec^2, $inch/sec^2$, deg/sec^2 (machine unit)

The parameter set on the machining parameter tuning screen is reflected in the following parameters:
Parameter No. 13620 (velocity-emphasized parameter)
Parameter No. 13621 (precision-emphasized parameter)

Moreover, the following parameter is also set according to the precision level:
Parameter No. 1735: Allowable acceleration rate for each axis applicable to the deceleration function based on acceleration in circular interpolation
Parameter No. 1737: Allowable acceleration rate for each axis applicable to the deceleration function based on acceleration in AI contour control

⚠ CAUTION
When bit 0 (MCR) of parameter No. 13600 is set to 1, the deceleration function based on acceleration in circular interpolation is not set.

- Time constant for acceleration/deceleration after interpolation

Set a time constant for acceleration/deceleration after interpolation.
Unit of data: ms

The parameter set on the machining parameter tuning screen is reflected in the following parameters:
Parameter No. 13622 (velocity-emphasized parameter)
Parameter No. 13623 (precision-emphasized parameter)

Moreover, the following parameter is also set according to the precision level:
Parameter No. 1769: Time constant for acceleration/deceleration after cutting feed interpolation

- Corner speed difference

Set an allowable corner speed difference used for speed determination.
Unit of data: mm/sec, inch/sec, deg/sec (machine unit)

The parameter set on the machining parameter tuning screen is reflected in the following parameters:
Parameter No. 13624 (velocity-emphasized parameter)
Parameter No. 13625 (precision-emphasized parameter)

Moreover, the following parameter is also set according to the precision level:
Parameter No. 1783: Allowable speed difference for each axis in automatic corner deceleration based on speed difference

- Maximum cutting speed

Set a maximum cutting speed for each axis.
Unit of data: mm/sec, inch/sec, deg/sec (machine unit)

The parameter set on the machining parameter tuning screen is reflected in the following parameters:
Parameter No. 13626 (velocity-emphasized parameter)
Parameter No. 13627 (precision-emphasized parameter)

Moreover, the following parameter is also set from the precision level:
Parameter No. 1432: Maximum cutting feedrate for each axis in the AI contour control mode

- Arbitrary items

Two arbitrary parameters can be registered. Each item can correspond to a CNC parameter or servo parameter. A parameter number corresponding to each item is to be specified with parameters.

As indicated below, set the parameters for corresponding parameter numbers, velocity-emphasized parameters (precision level 1), and precision-emphasized parameters (precision level 10).

Table 12.4.25 (a) Parameters related to arbitrary items

	Corresponding parameter number	Setting of velocity-emphasized (precision level 1) value	Setting of precision-emphasized (precision level 10) value
Arbitrary item 1	No.13628	No.13630	No.13632
Arbitrary item 2	No.13629	No.13631	No.13633

- Display
  Tuning target parameter numbers are displayed.

> ⚠ **CAUTION**
> As arbitrary items, the numbers of the following parameters cannot be specified:
> - Bit parameter
> - Spindle parameters (Parameter Nos. 4000 to 4799)
> - Real-type parameter
> - Power-off parameter
> - Nonexistent parameter

12.4.26 Displaying Memory Data (15-inch Display Unit)

The contents of the CNC memory can be displayed starting at a specified address.

Displaying memory data

Procedure

1 Set bit 0 (MEM) of parameter No. 8950 to 1 to display the memory contents display screen.

2 Press function key SYSTEM.

3 Press vertical soft key [MEMORY].
The following screen appears:

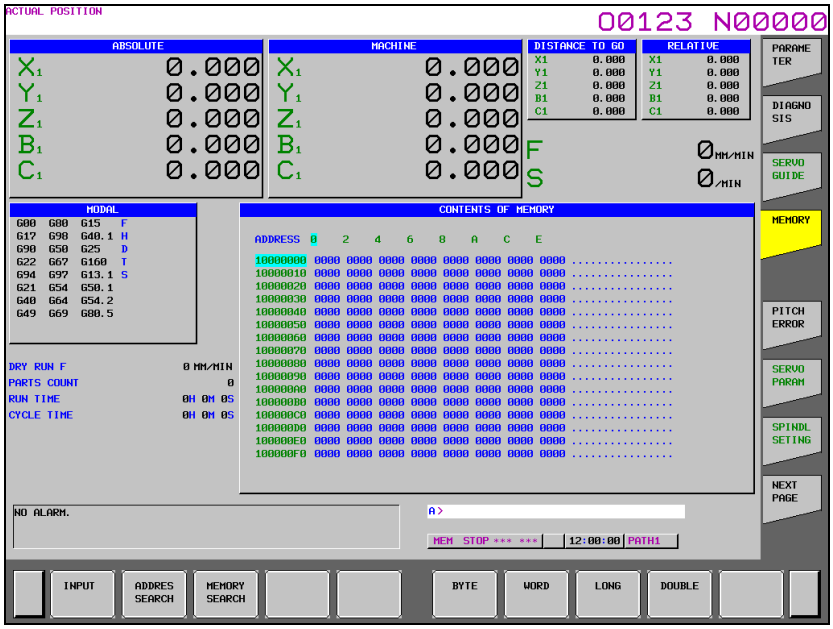


Fig. 12.4.26 (a) Memory contents display screen (15-inch)

4 Key in a desired address (hexadecimal) then press horizontal soft key [ADDRES SEARCH]. Starting at the specified address, 256-byte data is displayed.
(Example: When you input 100000 then press [ADDRES SEARCH], data is displayed starting at 100000H.)

5 Display data can be switched by using page keys PAGE and PAGE and cursor keys ↑, ↓, ←, and/or →.

6 By pressing horizontal soft key [BYTE], [WORD], [LONG], or [DOUBLE], select a data type to be displayed.

Explanation

A memory data display format can be selected from the following four options:

Byte display (1 byte in hexadecimal)
Word display (2 bytes in hexadecimal)
Long display (4 bytes in hexadecimal)
Double display (8 bytes in decimal: Double precision floating-point display)

One screen displays 256-byte memory data.

NOTE
1 When an address is input, "H" for representing "hexadecimal" need not be specified at the end of the address. If H is added, a warning is displayed to indicate that the format is invalid.
2 When word display is selected as the display format, an input address is rounded to a multiple of 2 bytes. When long display or double display is selected, an input address is rounded to a multiple of 4 bytes.

⚠ WARNING
1 If a memory address that must not be accessed in address search is input, a system alarm is issued. When making an address search, check that the address is accessible and that the address is input correctly.
2 This function is designed for maintenance, and must not be used by general users.

12.4.27 Parameter Tuning Screen (15-inch Display Unit)

The parameter tuning screen is a screen for parameter setting and tuning designed to achieve the following:

1 The minimum required parameters that must be set when the machine is started up are collectively displayed to facilitate start-up of the machine.
2 The servo tuning screen, spindle tuning screen, and machining parameter tuning screen are displayed for smooth tuning.

The parameter tuning screen consists of a menu screen and several setting screens.

12.4.27.1 Displaying the menu screen and selecting a menu item (15-inch display unit)

The parameter tuning menu screen displays the following items:

[START UP]

- SYSTEM SETTING
- AXIS SETTING
- FSSB (AMP)
- FSSB (AXIS)
- SERVO SETTING
- SPINDLE SETTING
- MISCELLANY

[TUNING]

- SERVO TUNING
- SPINDLE TUNING
- AICC TUNING

On the parameter tuning menu screen, one of the displayed items can be selected to display the corresponding screen. From each setting screen, you can return to the menu screen by performing a horizontal soft key operation.

NOTE

1 Some items may not be displayed, depending on the system configuration.
2 When bit 0 (SVS) of parameter No. 3111 is set to 0, "SERVO SETTING" and "SERVO TUNING" are not displayed. When bit 1 (SPS) of parameter No. 3111 is set to 0, "SPINDLE TUNING" is not displayed.

Displaying the menu screen and selecting a setting screen

Procedure

1 Set the MDI mode.
2 Switch the setting of "PARAMETER WRITE" to "ENABLED". For details, see the procedure for "PARAMETER WRITE" in Subsection III-12.3.16.
3 Press function key

.
4 Press vertical soft key [NEXT PAGE] several times then press vertical soft key [PARAM TUNING]. The following screen appears:

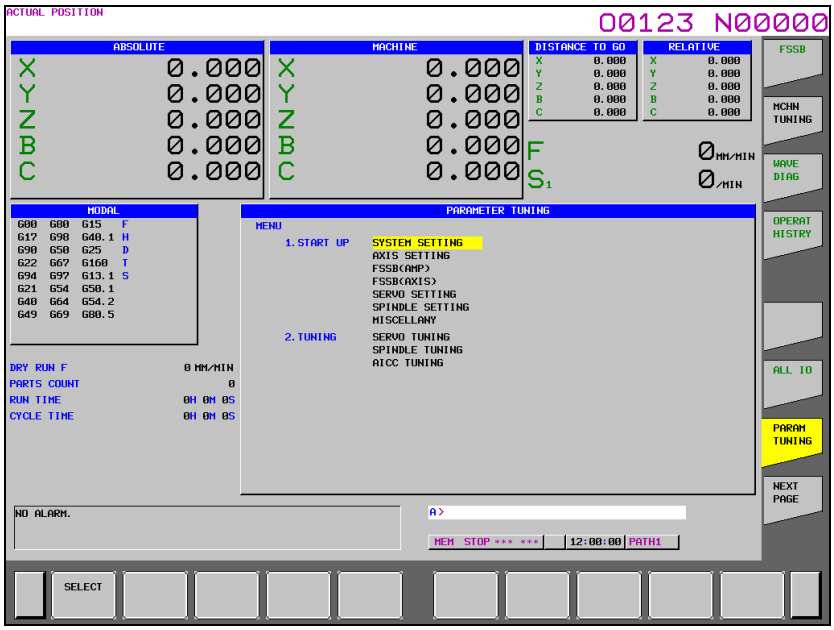


Fig. 12.4.27.1 (a) Parameter tuning menu screen (15-inch)

5 Move the cursor to a desired item by pressing cursor key
or
.
6 Press horizontal soft key [SELECT]. The screen display switches to the selected screen.

Returning to the menu screen

Procedure

1 Press horizontal soft key [SELECT] on the parameter tuning menu screen described in Subsection III-12.4.27.1. The screen and soft keys shown below are displayed. (The screen below is displayed when "AXIS SETTING" is selected.)

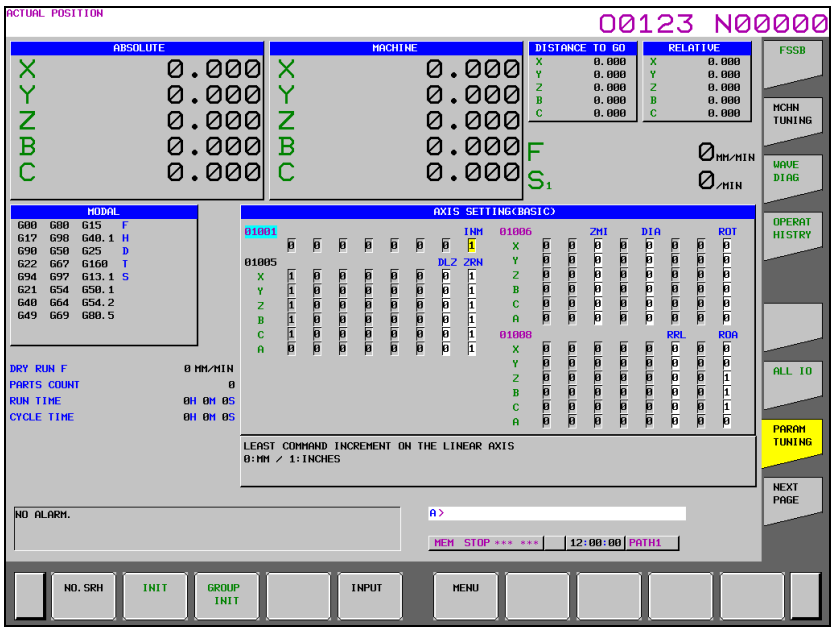


Fig. 12.4.27.1 (b) Axis setting screen (15-inch)

2 Press horizontal soft key [MENU].
The screen display returns to the parameter tuning menu screen.

3 Upon completion of parameter setting, switch the setting of "PARAMETER WRITE" to "DISABLED".

NOTE
Part of the setting screens can be displayed also by using a vertical soft key for chapter selection. When these screens are displayed using the vertical soft key for chapter selection, it becomes impossible to return to the parameter tuning menu screen.

Explanation

- Items displayed with [START UP]

The items of [START UP] indicate the screens for setting the minimum required parameters for starting up the machine.

Table 12.4.27.1 (a) Items displayed with [START UP]

Display item	Description
SYSTEM SETTING	Screen for setting CNC parameters related to the entire system configuration
AXIS SETTING	Screen for setting CNC parameters related to axes, coordinates, feedrate, and acceleration/deceleration
FSSB (AMP)	FSSB amplifier setting screen
FSSB (AXIS)	FSSB axis setting screen
SERVO SETTING	Servo setting screen
SPINDLE SETTING	Screen for setting spindle-related parameters
MISCELLANY	Screen for setting parameters related to the allowable number of M code digits and whether to display the servo setting and spindle tuning screens

NOTE
Some items may not be displayed, depending on the system configuration.

- Items displayed with [TUNING]

The items of [TUNING] indicate the screens for servo, spindle, and high-speed high-precision machining tuning.

Fig. 12.4.27.1 (b) Items displayed with [TUNING]

Display item	Description
SERVO TUNING	Servo tuning screen
SPINDLE TUNING	Spindle tuning screen
AICC TUNING	Machining parameter tuning screen

NOTE
1 Some items may not be displayed, depending on the system configuration.
2 When bit 0 (SVS) of parameter No. 3111 is set to 0, "SERVO SETTING" and "SERVO TUNING" are not displayed. When bit 1 (SPS) of parameter No. 3111 is set to 0, "SPINDLE TUNING" is not displayed.

12.4.27.2 Parameter tuning screen (system setting) (15-inch display unit)

This screen enables the parameters related to the entire system configuration to be displayed and modified. The parameters can be initialized to the standard values (recommended by FANUC).

Display and setting

Procedure

1 Move the cursor to [SYSTEM SETTING] by pressing cursor key or on the parameter tuning menu screen described in Subsection III-12.4.27.1.

2 Press horizontal soft key [SELECT]. The screen display switches to the screen and horizontal soft keys shown below.

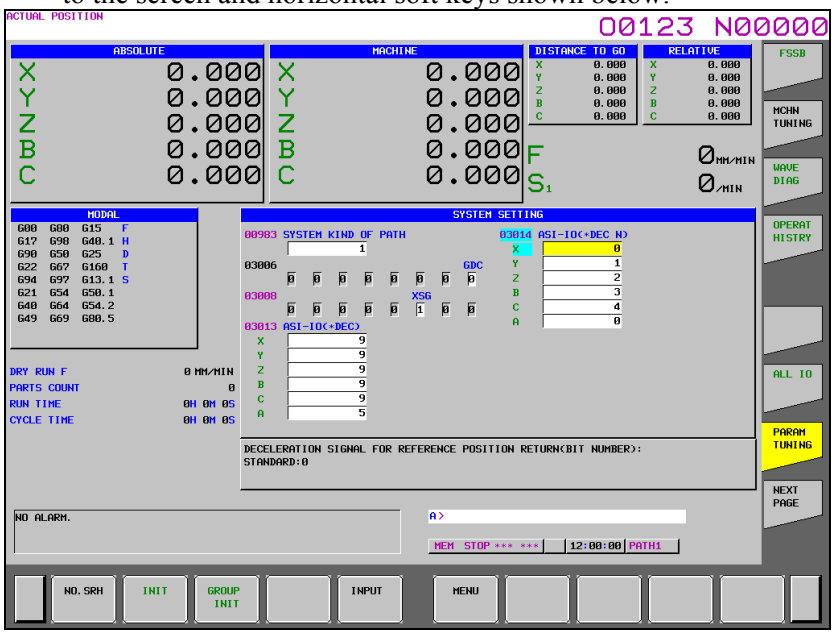


Fig. 12.4.27.2 (a) Parameter tuning screen (system setting) (15-inch)

3 Move the cursor to a parameter number to be set or displayed, according to one of the methods below.

- Input the desired parameter number, then press horizontal soft key [NO.SRH].
- Move the cursor to the desired parameter number by pressing page key or , and cursor keys , , , and/or .

  A brief description of the parameter where the cursor is placed is provided at the bottom of the screen. However, no description is provided when the cursor is placed on multiple bits for bit parameters.

4 Input desired data then press the key on the MDI panel to set the parameter.

5 Press horizontal soft key [INIT]. The standard value (recommended by FANUC) for the item selected by the cursor is displayed in the key input buffer. Pressing horizontal soft key [EXEC] in this state initializes the item to the standard value.
6 Press horizontal soft key [GROUP INIT]. A message asking whether to set the group standard values is displayed on the screen. Pressing horizontal soft key [EXEC] in this state inputs all of the standard values of the group.

NOTE
1 If the cursor is placed on a parameter that has no standard value assigned, no standard value is input even when [INIT] is pressed.
2 When the cursor is placed on multiple bits for bit parameters, the multiple bits can be input simultaneously. When [INIT] is pressed in this state, the key input buffer displays the standard values for the bits where the cursor is placed. If a bit has no standard value assigned, "*" is displayed for the bit, and no value is input for the bit.
3 When [GROUP INIT] is pressed, those parameters that have no standard values assigned are not initialized.

12.4.27.3 Parameter tuning screen (axis setting) (15-inch display unit)

This screen enables the CNC parameters related to axes, coordinates, feedrate, and acceleration/deceleration to be displayed and set. The parameters displayed can be divided into four groups:

(Basic) group :

The parameters related to basic settings are displayed.

(Coordinate) group :

The parameters related to coordinates are displayed.

(Feedrate) group :

The parameters related to feedrate are displayed.

(Acceleration/deceleration) group:

The parameters related to acceleration/deceleration are displayed.

The parameters can be initialized to the standard values (recommended by FANUC). For the display and setting procedure, see the description of the parameter tuning screen (axis setting) in Subsection III-12.4.27.2.

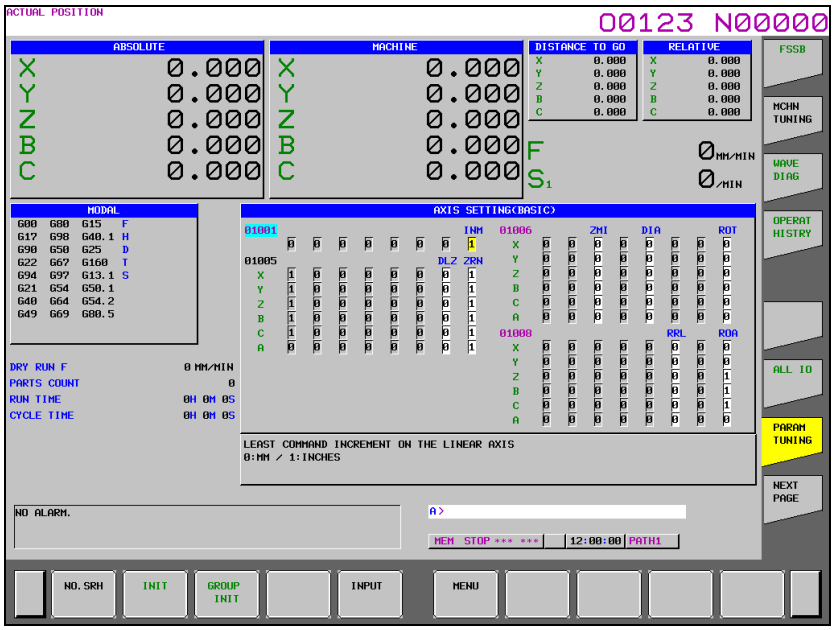


Fig. 12.4.27.3 (a) Parameter tuning screen (axis setting) (15-inch)

12.4.27.4 Displaying and setting the FSSB amplifier setting screen (15-inch display unit)

From the parameter tuning screen, the FSSB amplifier setting screen can be displayed. For details of the FSSB amplifier setting screen, see the description of the FSSB amplifier setting screen in Subsection 1.4.4 in the Connection Manual (Function) (B-63943EN-1).

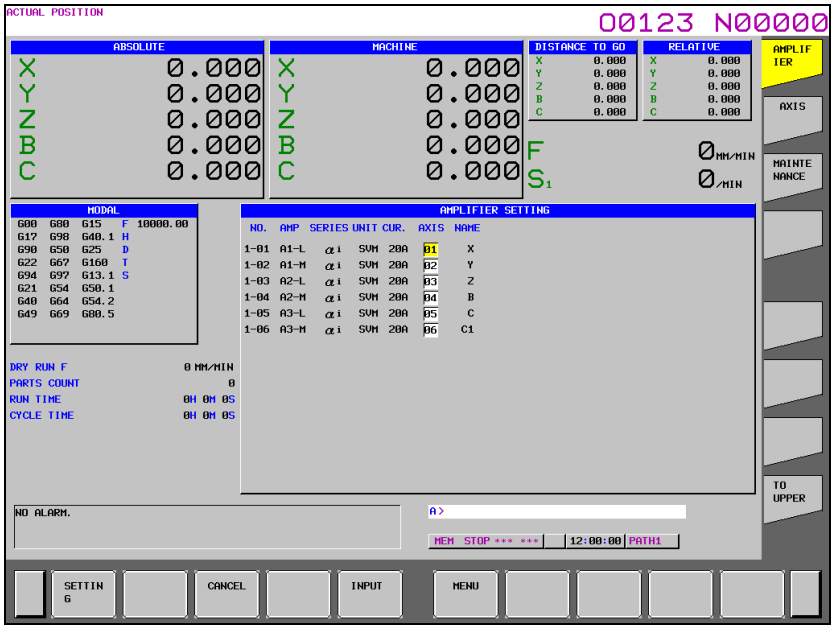


Fig. 12.4.27.4 (a) FSSB amplifier setting screen (15-inch)

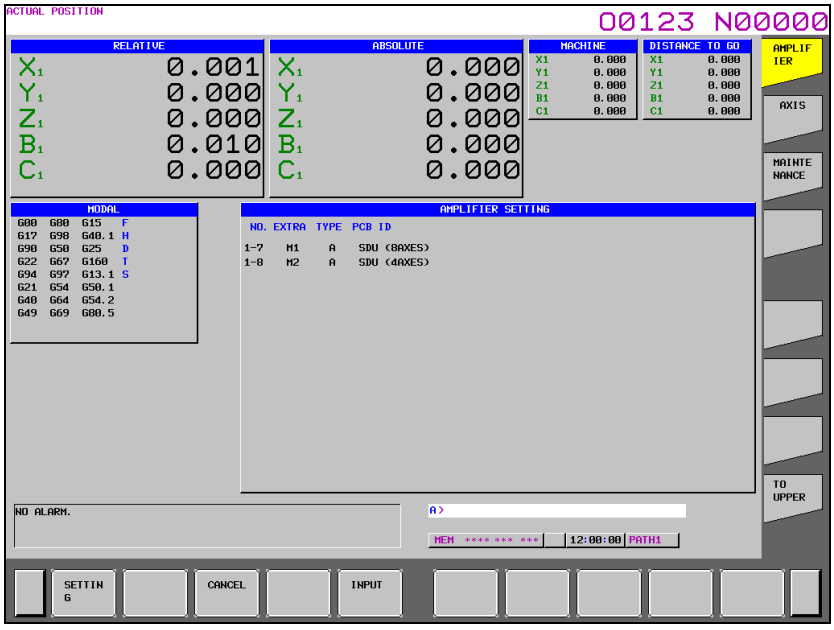


Fig. 12.4.27.4 (b) FSSB amplifier setting screen 2 (15-inch)

12.4.27.5 Displaying and setting the FSSB axis setting screen (15-inch display unit)

From the parameter tuning screen, the FSSB axis setting screen can be displayed. For details of the FSSB axis setting screen, see the description of the FSSB axis setting screen Subsection 1.4.4 in the Connection Manual (Function) (B-63943EN-1).

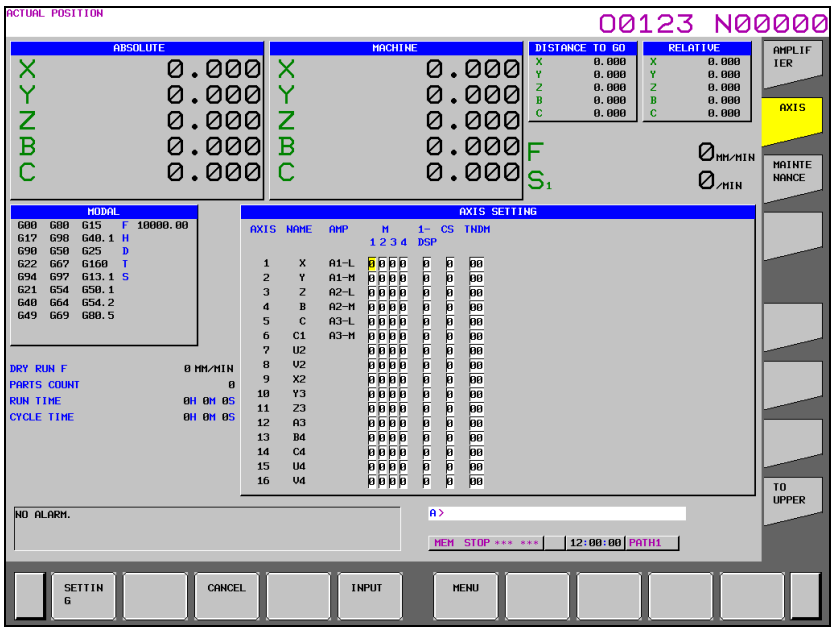


Fig. 12.4.27.5 (a) FSSB axis setting screen (15-inch)

12.4.27.6 Displaying and setting the servo setting screen (15-inch display unit)

From the parameter tuning screen, the servo setting screen can be displayed. For details of the servo setting screen, see the description of the servo setting screen in Subsection III-12.4.19.

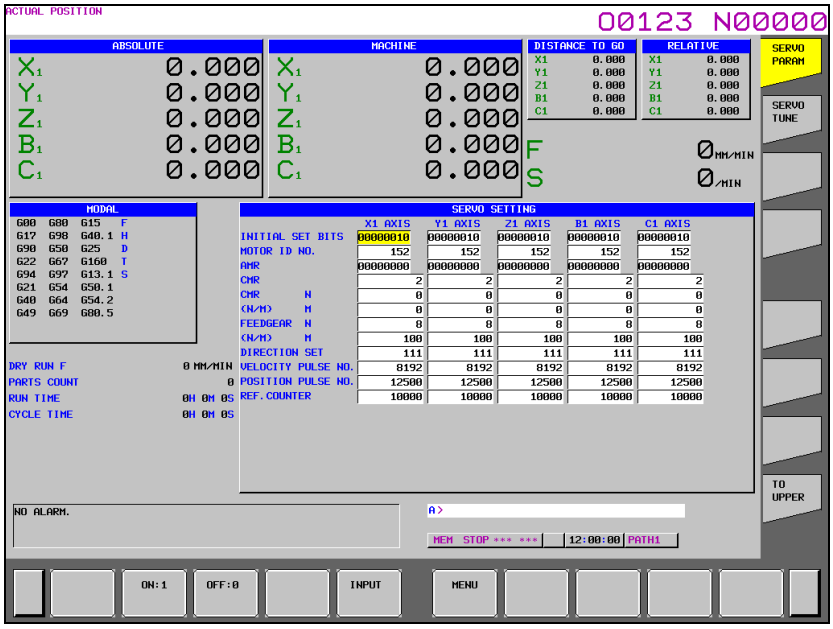


Fig. 12.4.27.6 (a) Servo setting screen (15-inch)

12.4.27.7 Parameter tuning screen (spindle setting) (15-inch display unit)

The spindle-related parameters can be displayed and modified.
For the display and setting procedure, see the description of the parameter tuning screen (system setting) in Subsection III-12.4.27.2.

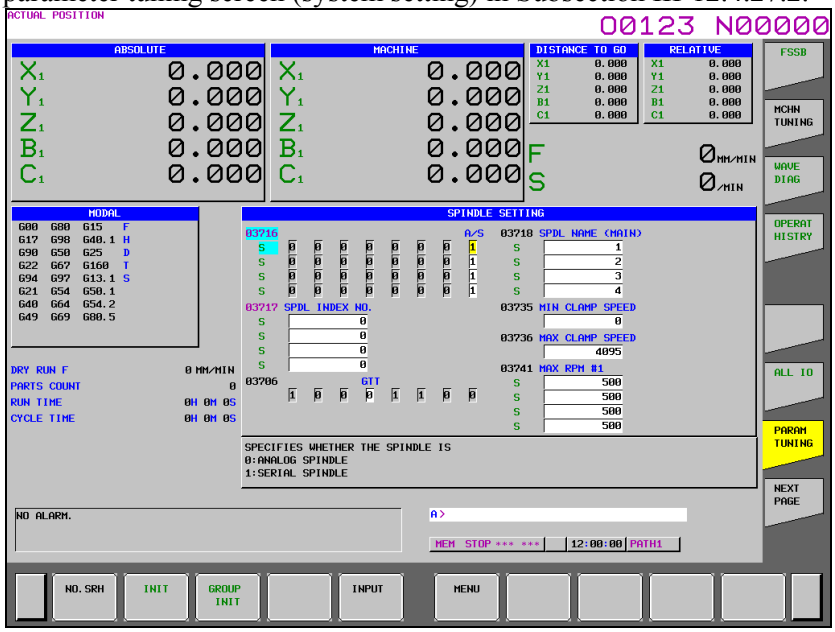


Fig. 12.4.27.7 (a) Parameter tuning screen (spindle setting) (15-inch)

12.4.27.8 Parameter tuning screen (miscellaneous settings) (15-inch display unit)

The parameters related to the allowable number of M code digits and whether to display the servo setting and spindle tuning screens can be displayed and modified. Moreover, the parameters can be initialized to the standard values (recommended by FANUC).
For the display and setting procedure, see the description of the parameter tuning screen (system setting) in Subsection III-12.4.27.2.

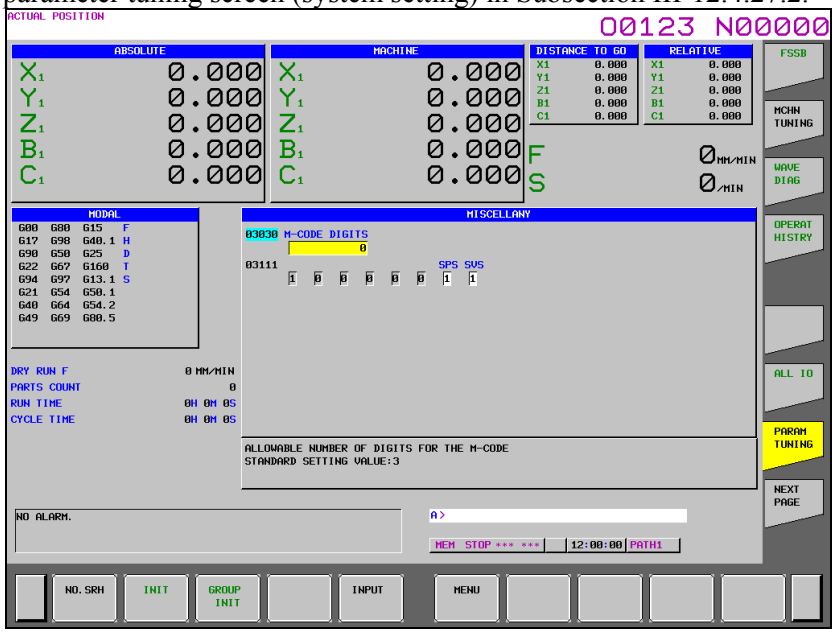


Fig. 12.4.27.8 (a) Parameter tuning screen (Miscellaneous setting) (15-inch)

12.4.27.9 Displaying and setting the servo tuning screen (15-inch display unit)

From the parameter tuning screen, the servo tuning screen can be displayed. For details of the servo tuning screen, see the description of the servo tuning screen in Subsection III-12.4.20.

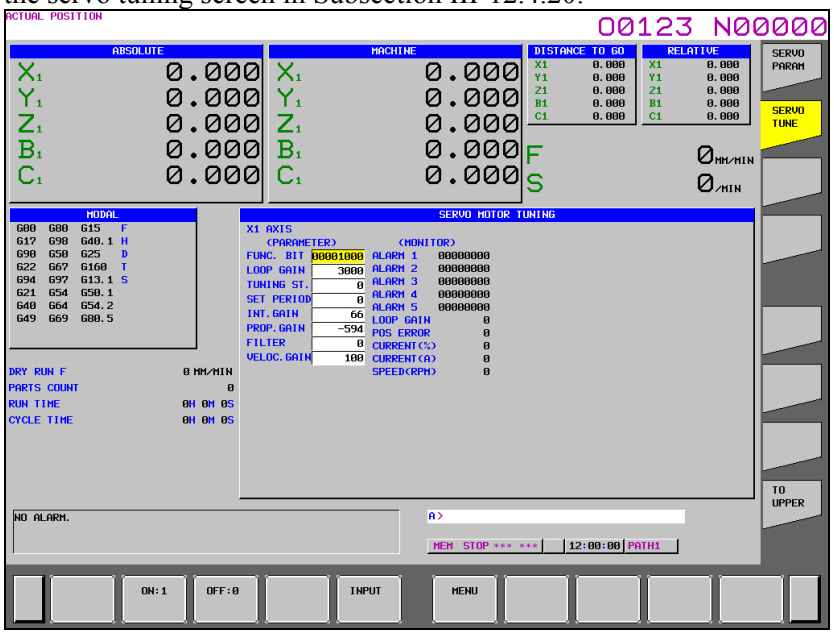


Fig. 12.4.27.9 (a) Servo tuning screen (15-inch)

12.4.27.10 Displaying and setting the spindle tuning screen (15-inch display unit)

From the parameter tuning screen, the spindle tuning screen can be displayed. For details of the spindle tuning screen, see the description of the spindle tuning screen in Subsection III-12.4.22.

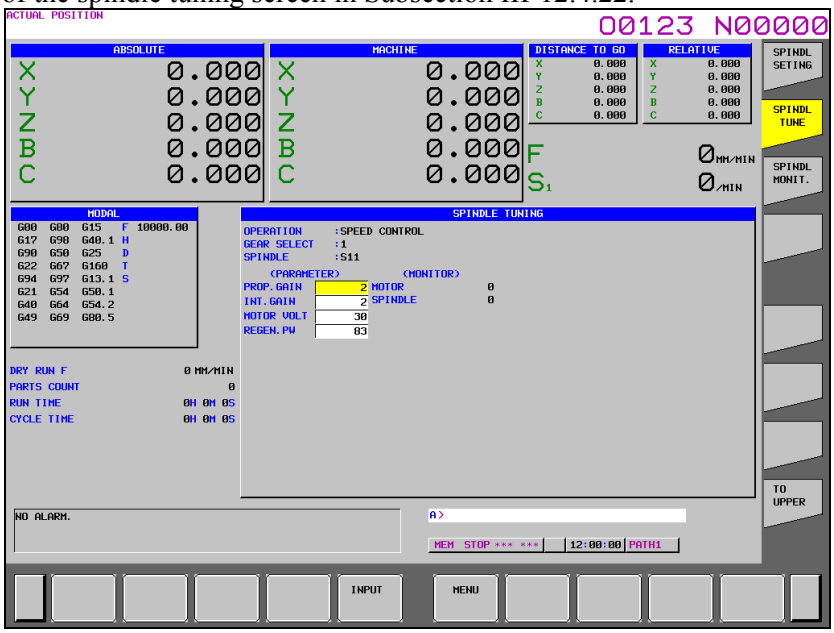


Fig. 12.4.27.10 (a) Spindle tuning screen (15-inch)

12.4.27.11 Displaying and setting the machining parameter tuning screen (15-inch display unit)

From the parameter tuning screen, the machining parameter tuning screen can be displayed. For details of the machining parameter tuning screen, see the description of the machining parameter tuning screen in Subsection III-12.4.25.

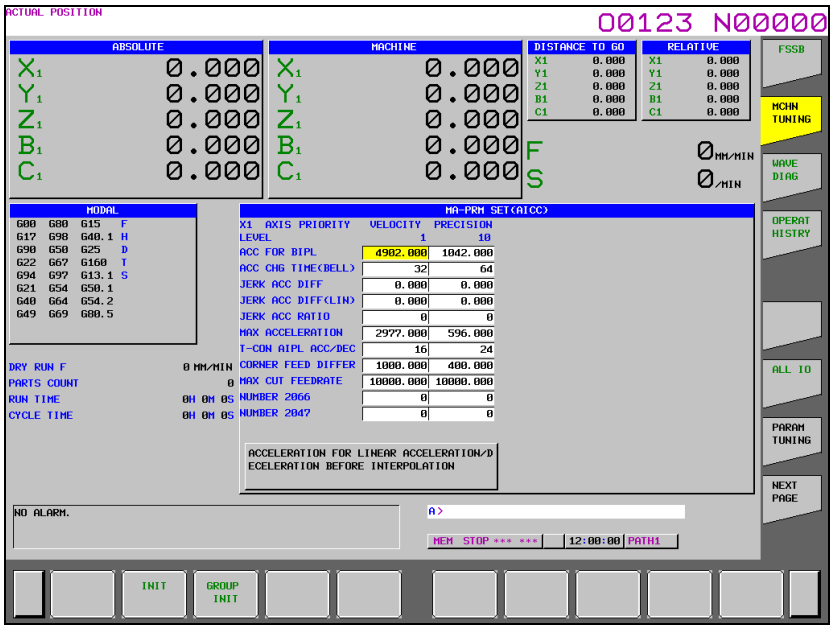


Fig. 12.4.27.11 (a) Machining parameter tuning screen (15-inch)

Explanation

- Parameters displayed for parameter tuning

Table 12.4.27 (a) Parameters displayed for parameter tuning (1)

Menu	Group	Parameter No.	Name	Brief description	Standard setting
SYSTEM SETTING	System setting	981		Sets the path of each axis.	
		982		Sets the path of each spindle.	
		983		Sets the T series/M series of each path. 0:T series/1:M series	
		984#0	LCP	Sets the attribute of each path. 0:Normal/1:Loader control	*1
		3021		Sets the G/F signal address of each axis.	*2
		3022		Sets the G/F signal address of each spindle.	*3
		3006#0	GDC	The deceleration signal for reference position return is: 0:X signal/1:G signal	1
		3008#2	XSGx	The signal address allocated to address X is: 0:Fixed/1:Set by the parameter	1
		3013		Address allocated to the deceleration signal for reference position return	*4
		3014		Bit position allocated to the deceleration signal for reference position return	*5

*1 : The value 1 is set for the paths as many as the number of loader paths starting from the greatest path number. For path 1, the value 0 is set at all times.
Example) When the number of loader paths is 3 in a 10-path system: The value 1 is set for paths 8 to 10. The value 0 is set for others.

*2 : When intra-path axis number ≤ 8, (path number - 1)*10+(intra-path axis number - 1)
When intra-path axis number ≥ 9, no standard value is available.
Example) When path 1 has 9 axes, and path 2 has 3 axes:
0,1,...,7,(none) for axes of path 1, 10,11,12 for axes of path 2

*3 : When intra-path spindle number ≤ 4, (path number - 1)*10+(intra-path spindle number - 1)
When intra-path spindle number ≥ 5, no standard value is available.
Example) When path 1 has 5 spindles, and path 2 has 1 spindle: 0,1,...,4,(none) for spindles of path 1, 10 for spindle of path 2

*4 : When (path number ≤ 3) and (intra-path axis number ≤ 8)
Axes of path 1: 9
Axes of path 2: 7
Axes of path 3: 10
Other axes: No standard value is available.
Example) When path 1 has 9 axes, and path 2 has 3 axes: 9,9,9,9,9,9,9,9,(none) for axes of path 1, 7,7,7 for axes of path 2

*5 : When (path number ≤ 3) and (intra-path axis number ≤ 8)
(Intra-path axis number - 1)
Other axes: No standard value is available.
Example) When path 1 has 9 axes, and path 2 has 3 axes: 0,1,...,7,(none) for axes of path 1, 0,1,2 for axes of path 2

Table 12.4.27 (b) Parameters displayed for parameter tuning (2)

Menu	Group	Parameter No.	Name	Brief description	Standard setting
SPINDLE SETTING	Spindle setting	3716#0	A/Ss	Sets the type of spindle motor: 0:Analaog/1:Serial.	
		3717		Sets a motor number to be assigned to each spindle.	
		3706#4	GTT	Specifies a spindle gear selection method. 0:M type/1:T type	
		3718		Sets a suffix to be added to spindle speed display on a screen such as the position display screen.	
		3735		Sets the minimum clamp speed of the spindle motor.	
		3736		Sets the maximum clamp speed of the spindle motor.	
		3741		Sets the maximum spindle speed for gear 1.	
		3742		Sets the maximum spindle speed for gear 2.	
		3743		Sets the maximum spindle speed for gear 3.	
		3744		Sets the maximum spindle speed for gear 4.	
		3772		Sets the maximum speed of the spindle. When 0 is set, the speed is not clamped.	
		4133		Sets the motor model code of the serial spindle. (This setting is not required for an analog spindle.)	
		4019#7	***	The parameters for the serial spindle are: 0:Not automatically set/1:Automatically set (This setting is not required for an analog spindle.)	

Table 12.4.27 (c) Parameters displayed for parameter tuning (3)

Menu	Group	Parameter No.	Name	Brief description	Standard setting
AXIS SETTING	Basic	1001#0	INM	Least command increment on linear axes: 0:Metric (millimeter machine) 1:Inch (inch machine)	
		1005#0	ZRNx	When an automatic operation (other than G28) is executed before reference position return: 0:Alarm is issued (PS0224) 1:No alarm is issued.	0
		1005#1	DLZx	Reference position return without dogs: 0:Disabled 1:Enabled	
		1006#0	ROTx	Setting of linear axes or rotary axis: 0:Linear axis 1:Rotation axis	
		1006#3	DIAx	Setting of the amount of travel: 0:Radius specification 1: Diameter specification	
		1006#5	ZMIx	Reference position return direction: 0:Plus direction 1:Minus direction	
		1008#0	ROAx	Rotation axis roll-over function: 0:Disabled 1:Enabled	1
		1008#2	RRLx	With the amount of travel per revolution, relative coordinates are: 0:Not rounded 1:Rounded	1
		1013#0	ISAx	Sets the least input increment and least command increment: 0:IS-B 1:IS-A	
		1013#1	ISCx	Sets the least input increment and least command increment: 0:IS-B 1:IS-C	
		1013#2	ISDx	Sets the least input increment and least command increment: 0:IS-B 1:IS-D	
		1013#3	ISEx	Sets the least input increment and least command increment: 0:IS-B 1:IS-E	
		1020		Program name	*1
		1022		Sets each axis in the basic coordinate system.	*2
		1023		Servo axis number	
		1815#1	OPTx	A separate pulse coder is: 0:Not used 1:Used	
		1815#4	APZx	The correspondence between machine positions and absolute-position detector positions is: 0:Not established 1:Established	
		1815#5	APCx	The position detector used is: 0:Other than an absolute-position detector 1:Absolute-position detector	
		1825		Servo loop gain	
		1826		Effective area	
		1828		Positional deviation limit during travel	
		1829		Positional deviation limit during stop	500

*1 : For the M series: 88(X),89(Y),90(Z) sequentially starting with the first axis (no standard value for the fourth axis and up)
For the T series: 88(X),90(Z) (no standard value for the third axis and up)

*2 : For the M series: 1,2,3 sequentially starting with the first axis (no standard value for the fourth axis and up)
For the T series: 1,3 (no standard value for the third axis and up)

Table 12.4.27 (d) Parameters displayed for parameter tuning (4)

Menu	Group	Parameter No.	Name	Brief description	Standard setting
AXIS SETTING	Coordinate	1240		Machine coordinate of the first reference position	
		1241		Machine coordinate of the second reference position	
		1260		Amount of travel per revolution of a rotary axis	360.000
		1320		Coordinate of the boundary of stored stroke check 1 in the positive direction	
		1321		Coordinate of the boundary of stored stroke check 1 in the negative direction	
	Feedrate	1401#6	RDR	For a rapid traverse command, dry run is: 0:Disabled 1:Enabled	0
		1410		Dry run feedrate	
		1420		Rapid traverse rate	
		1421		Rapid traverse override F0 rate	
		1423		Jog feedrate	
		1424		Manual rapid traverse rate	
		1425		FL feedrate for reference position return	
		1428		Reference position return feedrate	
		1430		Maximum cutting feedrate	
	Acceleration/ deceleration	1610#0	CTLx	Acceleration/deceleration for cutting feed is: 0:Exponential acceleration/deceleration 1:Linear acceleration/deceleration after interpolation	
		1610#4	JGLx	Acceleration/deceleration for jog feed is: 0:Exponential acceleration/deceleration 1:Same as acceleration/deceleration for cutting feed (The settings of bit 1 (CTBx) and bit 0 (CTLx) of parameter No. 1610 are followed.)	
		1620		Time constant for linear acceleration/deceleration for rapid traverse	
		1622		Time constant for acceleration/deceleration for cutting feed	
		1623		FL feedrate for acceleration/deceleration after interpolation for cutting feed	
		1624		Time constant for acceleration/deceleration for jog feed	
		1625		FL feedrate for exponential acceleration/deceleration for jog feed	
MISCELLANY	MISC	3030		Allowable number of digits of an M code	3
		3111#0	SVS	The servo setting screen is: 0:Not displayed 1:Displayed	1
		3111#1	SPS	The spindle setting screen is: 0:Not displayed 1:Displayed	1

12.4.28 Periodic Maintenance Screen (15-inch Display Unit)

Periodic maintenance screens are used for managing consumables (such as the backlight of a LCD unit and backup batteries). By setting the name of a consumable, its life time, and the method for counting down the life time, the life of the consumable can be counted down in the appropriate method set for the consumable, and the remaining life time can be displayed.
With these screens, the user can easily manage consumables that require periodic replacement.

Explanation

There are four periodic maintenance screens: the status screen, the setting screen, the machine menu screen, and the NC menu screen.

- Status screen:
  Item names, remaining times, and count statuses are displayed, and item names are set.
- Setting screen:
  Life times, remaining times, and count types (count-down method) are set.
- Machine menu screen:
  The names of consumables in the machine can be registered.
- NC menu screen:
  The names of consumables in the NC are already registered.

Using periodic maintenance screens

<1> Referencing a periodic maintenance screen
Display the status screen. For the meanings of items on the status screen, see **Status screen**.

<2> Adding the name of a new consumable item to a periodic maintenance screen or editing an existing consumable item on the screen
The name of a consumable item can be added or edited on the machine menu screen. For details, see **Machine menu screen**.

<3> Adding or editing a life time, remaining time, and a method of counting down the remaining time for a consumable on a periodic maintenance screen
A life time and remaining time can be added or edited on the setting screen. For details, see **Setting screen**.

<4> Newly displaying the item name and remaining time of a consumable on a periodic maintenance screen.
1 Setting an item name
Select the item name of a consumable to be displayed from the machine menu screen or NC menu screen, or enter the name using the MDI keys. For the procedure, see **Item name** in **Status screen**.

2 Setting a life time, remaining time, and count type
Select the life time, remaining time, and count type of a consumable to be displayed from the setting screen. For the procedure, see **Remaining time** in **Status screen**.

Procedure for displaying a periodic maintenance screen

1 Press function key SYSTEM.
2 Press vertical soft key [NEXT PAGE] several times to display [PERIOD MAINTE].
3 Press vertical soft key [PERIOD MAINTE] to display the periodic maintenance screen.

Status screen

When vertical soft key [STATUS] is pressed, the status screen is displayed. The status screen shows the item names, count statuses, and remaining times of managed consumables.

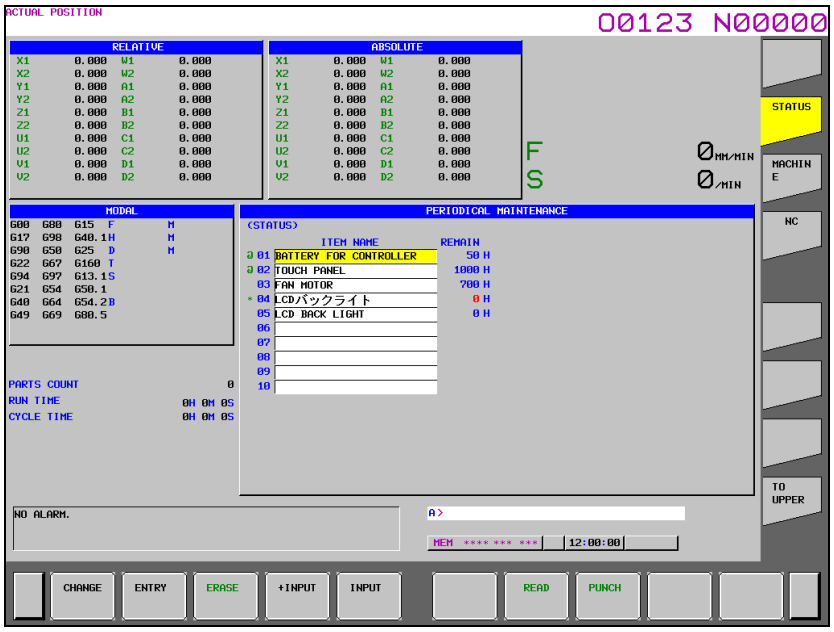


Fig. 12.4.28 (a) Status screen

- Item name

As the item name, set the name of a consumable to be managed by periodic maintenance. To set an item name, select a name from the machine menu screen or NC menu screen, or directly enter the name using the MDI keys.

Setting an item name from a menu screen

1 On the status screen, move the cursor to the name of the item to be set, and press horizontal soft key [ENTRY].
2 Press vertical soft key [MACHINE] or [NC] to display the machine menu screen or NC menu screen.
3 Move the cursor to an item name registered on the menu screen, and press horizontal soft key [SELECT] then [EXEC] successively.

4 The screen display returns to the status screen, and the item name selected on the menu screen is added to the status screen.

Initially, there is no item name set on the machine menu screen, so item names must be registered in advance. For the registration method, see the description of the procedure for registering item names on the machine menu screen.

Setting an item name using the MDI keys

1 Type the alphanumeric characters to be input, then press horizontal soft key [INPUT].
2 The entered item name is registered on the status screen.

When horizontal soft key [+INPUT] is pressed instead of horizontal soft key [INPUT], the entered characters can be added to an existing item name.
When typing 2-byte characters, type "*" before and after the character codes. The character codes must conform to FANUC codes. (See Appendix G, "FANUC 2-BYTE CHARACTER CODE TABLE".)
An item name to be registered must be up to 24 characters long if it consists of alphanumeric characters only; or it must be up to 12 characters long if it consists of 2-byte characters only.
Example: To register "LCD バックライト", enter the following:
>LCD＊110E10F410CC114010B610FE＊_

NOTE
1 An asterisk "*" is used as a control code, so it cannot be used in item names. In addition, characters "[", "]", "(", and ")" cannot be used in item names.
2 When an item name consisting of both alphanumeric characters and 2-byte characters is registered, the warning "DATA OUT OF RANGE" may be output.

Deleting an item name

To delete a registered item name, move the cursor to the item name, press horizontal soft key [ERASE], then press horizontal soft key [EXEC].
When an item name is deleted, its life time, remaining time, and count type are deleted at the same time.

- Remaining time

As the remaining time, the period of time left until the time for replacement is reached by count down operation is displayed.
If the percentage of the remaining time to the life time has reached to the value (%) specified by parameter No. 8911, or smaller, the remaining time is displayed in red.
Even after the life time has expired, count down operation continues.

Setting the remaining time

1 On the status screen, place the cursor on an item for which the remaining time is to be set (the item name must have been set in advance).
2 Press horizontal soft key [CHANGE].
3 The screen display changes to the setting screen.
4 Set the life time, remaining time, and count type. For the setting method and other information, see **Setting screen**.

NOTE
On the status screen, the remaining time and life time cannot be set.
These items must be set on the setting screen.

- Count status

The count status is indicated on the left side of the item number as follows:

Indication	Count status
Blank	Counting stopped
@	Counting in progress
*	Life expired

Setting screen

On the setting screen, the life time, remaining time, and count type of a managed consumable are set.

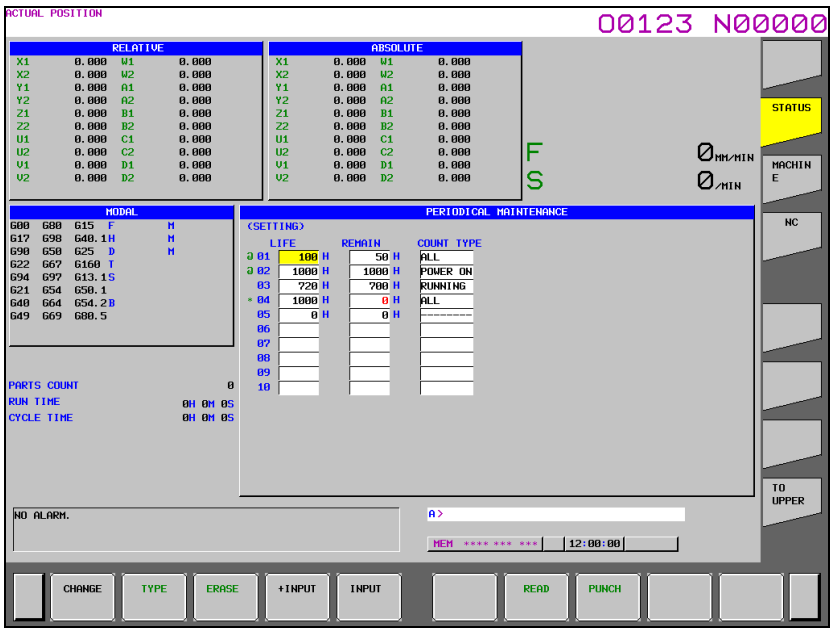


Fig. 12.4.28 (b) Setting screen

Display procedure

1 Press horizontal soft key [CHANGE].

- Life time

Set the life time of a consumable.
Move the cursor to an existing item, type a life time, then press horizontal soft key [INPUT] (or the

key). The life time is then set, and the same value is set also as the remaining time.
At this time, "------" is indicated in the count type field.
When horizontal soft key [+INPUT] is pressed, the entered value can be added to the life time already set. The same value as the added value is added also to the remaining time.
A value ranging from 0 to 65535 (in hours) can be set.

NOTE
1 If a setting operation is attempted when the item name is not registered, the warning "EDIT REJECTED" is issued.
2 If a value exceeding the valid range is entered, the warning "DATA IS OUT OF RANGE" is issued.
3 If soft key [ERASE] or [TYPE] is pressed, the warning "EDIT REJECTED" is issued.

- Remaining time

The period of time left until the time for replacement is reached by count down operation is indicated.
If the percentage of the remaining time to the life time has reached the value (%) specified by parameter No. 8911, or smaller, the remaining time is displayed in red.
Even after the life time has expired, count down operation continues.
Move the cursor to the remaining time of a target registered number, type a remaining time, then press horizontal soft key [INPUT] (or the

key). The remaining time is then set.
When horizontal soft key [+INPUT] is pressed, the entered value can be added to the remaining time already set.
A value ranging from 0 to (life time) can be set.
When horizontal soft key [ERASE] then horizontal soft key [EXEC] are pressed, the same value as the life time is set.

NOTE
1 If a setting operation is attempted when the item name or life time is not registered, the warning “EDIT REJECTED” is issued.
2 When a value exceeding the valid range is entered, the warning “DATA IS OUT OF RANGE” is issued.
3 If horizontal soft key [TYPE] is pressed, the warning “EDIT REJECTED” is issued.

- Count type

As the count type, select the way of counting.
Place the cursor on the count type of a target registration number, then press horizontal soft key [TYPE]. Count types are displayed as soft keys as shown below. Select one of these soft keys, then press horizontal soft key [EXEC].

Soft key	Meaning	Indication
[NO CNT]	Counting is not performed (stopped)	— — — — — —
[ALL]	Counting is performed at all times	All times
[POWER ON]	Counting is performed when power is on.	When power is on
[RUN]	Counting is performed when operation is in progress.	When operation is in progress
[CUT]	Counting is performed when cutting is being performed.	When cutting is being performed

NOTE
1 If a setting operation is attempted when the item name or life time is not registered, the warning “EDIT REJECTED” is issued.
2 Horizontal soft keys [INPUT] and [+INPUT] have no effect.
3 When counting is performed at all times, a 24-hour error is generated in leap year.
4 If horizontal soft key [ERASE] is pressed, the warning “EDIT REJECTED” is issued.

Machine menu screen

On the machine menu screen, the names of consumables of the machine are registered. From this screen, item names can be added to the status screen. For the method of addition to the status screen, see the description of the status screen.

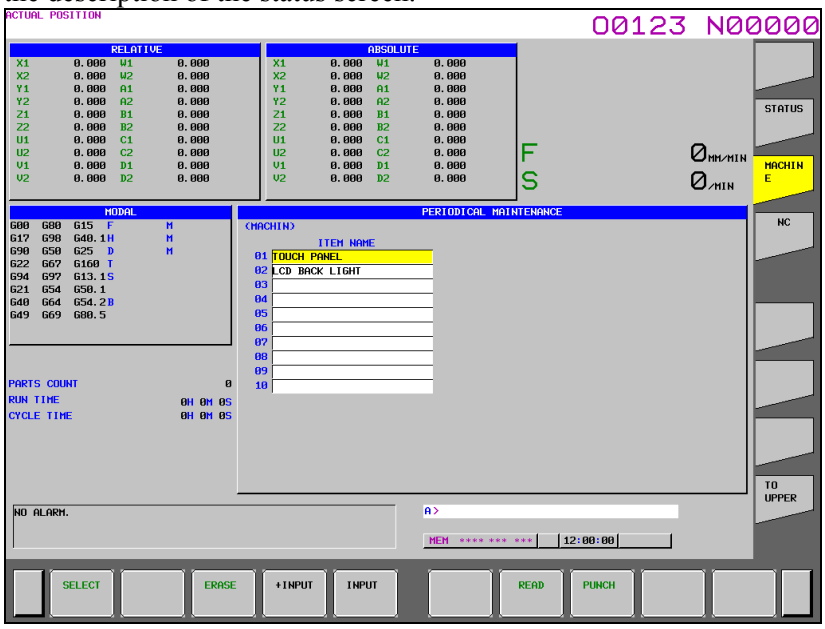


Fig. 12.4.28 (c) Machine menu screen

- Displaying the screen

1 When the status screen is displayed, press vertical soft key [MACHINE].

On the machine menu screen, item names can be registered using one of the following two methods:

- Registration from a program
- Registration using the MDI keys

- Registration from a program

By executing a program having the following format, an item name can be registered in the menu of the machine:

Format

G10 L61 Px [n]

X Registration number

n Item name, format:

[alphanumeric-characters*2-byte-characters*alphanumeric-characters]

- Registration using the MDI keys

An item name can be registered on the menu screen of the machine by typing the item name in the format shown below on the machine menu screen, then pressing horizontal soft key [INPUT] (or the

key).

When horizontal soft key [+INPUT] is pressed, the typed characters can be added to an already registered item name.

Format

Alphanumeric-characters*2-byte-characters*alphanumeric-characters

Two-byte codes must conform to FANUC codes. (See Appendix G, "FANUC 2-BYTE CHARACTER CODE TABLE".)
When typing 2-byte characters, type an asterisk "*" before and after the character codes. An item name to be registered must be up to 24 characters long if it consists of alphanumeric characters only; or it must be up to 12 characters long if it consists of 2-byte characters only.

Example:

To register "LCD バックライト", enter the following:

>LCD＊110E10F410CC114010B610FE＊_

NOTE
1 An asterisk "*" is used as a control code, so it cannot be used in item names. In addition, characters "[", "]", "(", and ")" cannot be used in item names.
2 When an item name consisting of both alphanumeric characters and 2-byte characters is registered, the warning “DATA IS OUT OF RANGE” may be output.
3 When a blank item name is selected on the machine screen, the warning “EDIT REJECTED” is issued.

To delete a registered item name, move the cursor to the item name, press horizontal soft key [ERASE], then press horizontal soft key [EXEC].

From this screen, an item name can be registered on the status screen. For the method of registration to the status screen, see the description of the status screen.

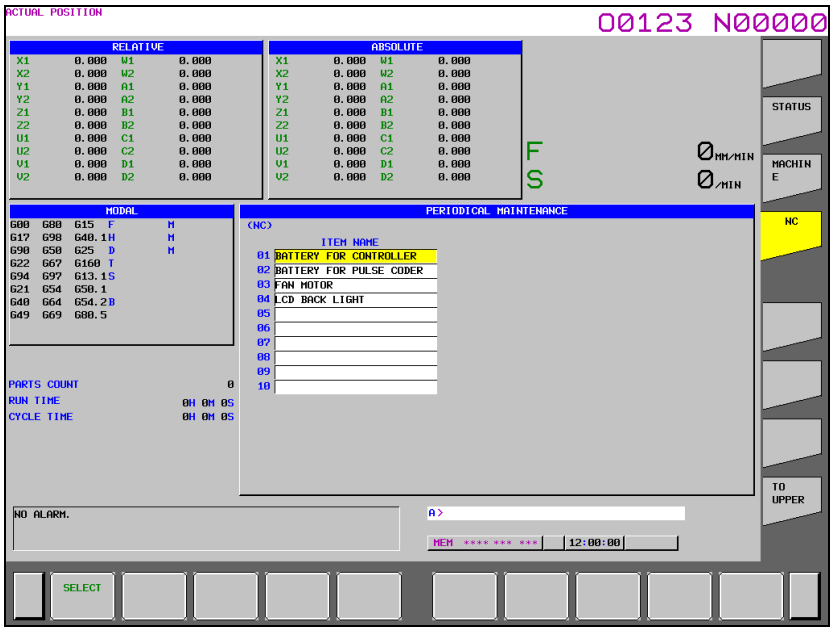


Fig. 12.4.28 (a) NC menu screen

- Displaying the screen

1 When the status screen is displayed, press soft key [NC].

NOTE
On the NC screen, the registration, deletion, and I/O of item names cannot be performed.
When a blank item name is selected, a blank is set.

12.4.29 System Configuration Screen (15-inch Display Unit)

The system configuration screen provides information about the types of installed hardware and software.

Procedure for displaying the screen

Procedure

1 Press the SYSTEM key, then press vertical soft key [NEXT PAGE] several times until vertical soft key [SYSTEM] is displayed.
2 Press vertical soft key [SYSTEM]. The system configuration screen is displayed.

There are two types of system configuration screens: the hardware configuration screen and the software configuration screen. The screen display can switch between these screens by using PAGE ↑ and PAGE ↓.

When vertical soft key [SERVO INFO] or [SPINDLE INFO] is pressed, information about the connected servo system or spindles is displayed.

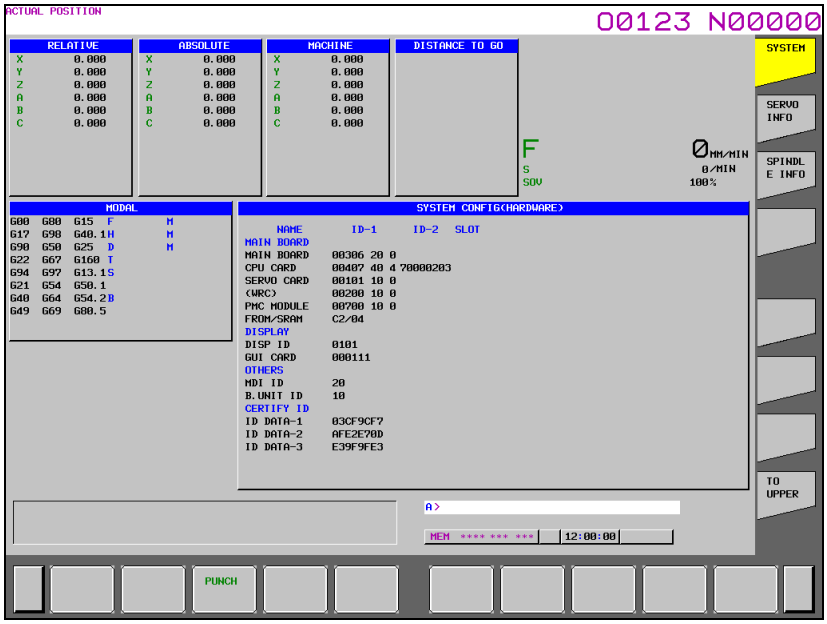


Fig. 12.4.29 (a) System configuration screen

Hardware configuration screen

This screen shows the names and IDs of the hardware used by the NC.

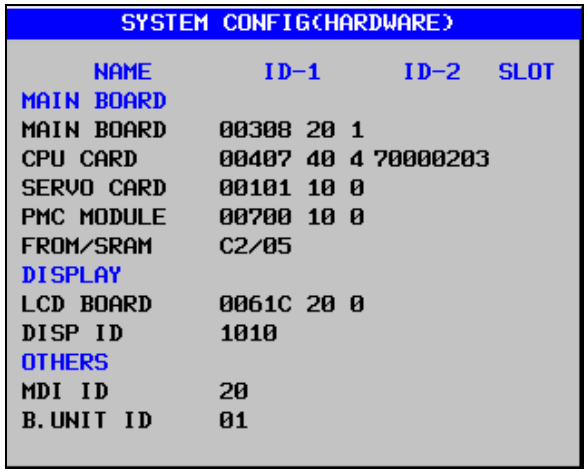


Fig. 12.4.29 (b) Hardware configuration screen

Software configuration screen

This screen shows the names and series/editions of the software used by the NC.

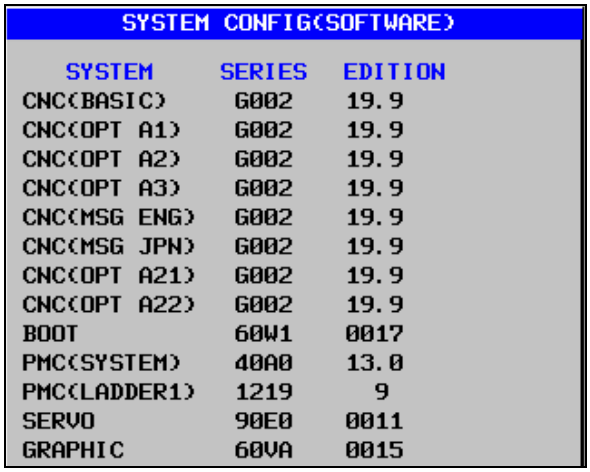


Fig. 12.4.29 (c) Software configuration screen

Servo information screen

When a servo system is connected to the NC, ID information of the connected servo devices (servo motors and servo amplifier modules) can be displayed on the NC.

Displaying the screen

1 When the system configuration screen is displayed, press vertical soft key [SERVO INFO].
2 The servo information screen is displayed.

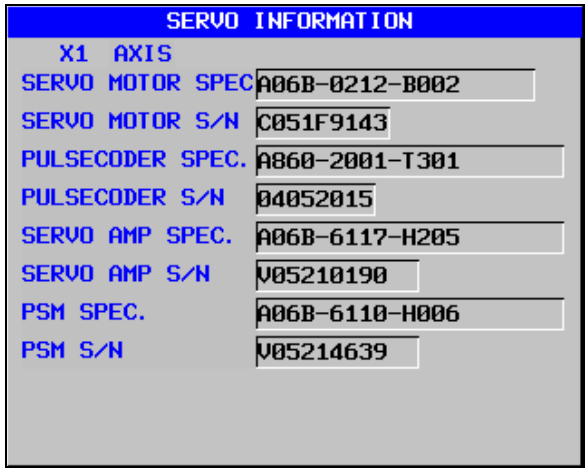


Fig. 12.4.29 (d) Servo information screen

Spindle information screen

When a spindle system is connected to the NC, the ID information of the connected spindle devices (spindle motors and spindle amplifier modules) can be displayed on the NC.

Displaying the screen

1 When the system configuration screen is displayed, press vertical soft key [SPINDLE INFO].
2 The spindle information screen is displayed.

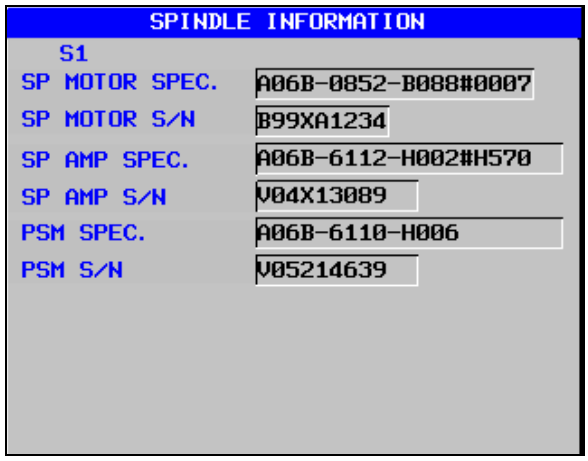


Fig. 12.4.29 (e) Spindle information screen

12.4.30 Overview of the History Function (15-inch Display Unit)

The history function makes it possible to record a history of operations performed by the operator, alarms and external operator messages issued, and other history data, check the history, and output history data.

1 Conditions for recording history
   a Display screen
   A history of screens other than the operation history screen is recorded.
   b Signal selection
   Up to 60 I/O signals can be selected so that they can be recorded in history data.
   c Parameter setting
   With parameters, it is possible to separately set whether to record an MDI key operation history and an external operator message history, add external alarms and messages, and record data modification histories such as operation histories of parameters, tool offsets, workpiece offsets (workpiece shift amounts), custom macro common variables, and I/O signals.
   d Storage capacity
   When the storage capacity becomes full, data is deleted in order from the oldest data.Up to about 8000 items of history data can be recorded if the data consists of history data of MDI key operations only.

2 Data output
   All history data stored can be output to external input/output devices. (See Subsection 12.4.30.5, "Outputting all history data".)

3 Erasure of history data
   If bit 7 (EKE) of parameter No. 3195 is set to 1, soft key [CLEAR] is displayed, allowing the user to erase history data.

	#7	#6	#5	#4	#3	#2	#1	#0
3195	EKE							

[Data type] Bit

 # 7 EKE Soft key [ALL CLEAR] for clearing all history data is:
0: Not displayed.
1: Displayed.

NOTE

1 History data remains even after the power is turned off.
Memory clear operation, however, erases history data as well.
2 Set the time and date correctly on the setting screen.
3 All history data including data of alarms, external operator messages, operations, and I/O signals is integrated and maintained in a single storage area. When the storage capacity becomes full, history data is erased in order from the oldest data regardless of the history type.
When history data is erased, all history data is erased, so care should be taken.
It is impossible to erase only a specific history.

12.4.30.1 Alarm history

From all history data recorded, only alarm history is extracted and displayed on the screen. Note that when the amount of history data exceeds the storage capacity, history data is automatically erased in order from the oldest history data.

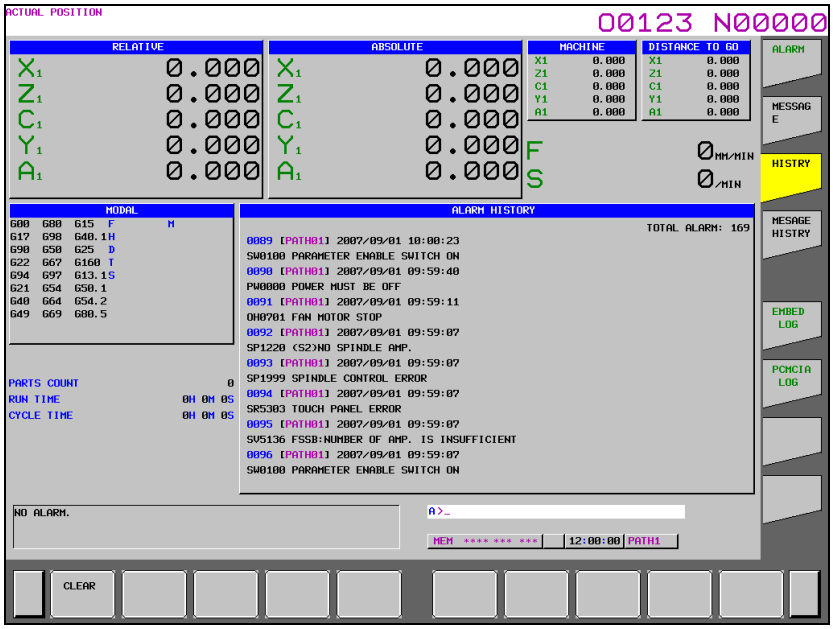


Fig. 12.4.30.1 (a) Alarm history screen

Screen display

Issued alarms are displayed sequentially from the latest alarm.
The following information is displayed for each alarm:

- Path name (only when multipath operation is performed)
- Time and date of alarm occurrence
- Type and number of the alarm
- Alarm message

NOTE
To record also external alarm and macro alarm messages as history data, set bit 3 (EAH) of parameter No. 3112 to 1, and at the same time set bit 7 (HAL) of parameter No. 3196 to 0.

When a name such as a path name, axis name, or spindle name has been changed after the issuance of an alarm, a newly assigned name is displayed on the alarm history screen.
When modal O data that is a program name is output, only the first five characters are output.

Procedure

1 Press function key to display a screen of parameters and so on.
2 Press vertical soft key [HISTRY] to display the alarm history screen.
3 The screen display can be changed to the previous page and the next page by using page keys and .

Erasing history data from the alarm history screen

Procedure

1 Display the alarm history screen.
2 Press horizontal soft key [CLEAR]. All history data is then erased.

NOTE
When history data is erased, not only alarm history data but also other history data such as external operator message history data and operation history data is erased. It is impossible to erase only a specific history.

Display of external alarms and macro alarms

When an external alarm or macro alarm is issued, it becomes possible to record its message as well as the alarm number in the alarm history if the parameter shown below is set. As the message of an external alarm or macro alarm, the first 32 characters for the external alarm or the first 26 characters for the macro alarm are recorded.

	#7	#6	#5	#4	#3	#2	#1	#0
3112					EAH			

[Data type] Bit

 # 3 EAH Messages of the external alarm/macro alarm in alarm or operation history:
0: Not recorded
1: Recorded

NOTE
This parameter is valid when bit 7 (HAL) of parameter No. 3196 is set to 0.

	#7	#6	#5	#4	#3	#2	#1	#0
3196	HAL							

[Data type] Bit

 # 7 HAL When an alarm is issued, additional information (modal data, absolute coordinates, and machine coordinates present at the issuance of the alarm) is:

0: Recorded in the operation history and alarm history.
1: Not recorded in the operation history and alarm history.

NOTE

1 History data such as alarm history data, operation history data, external operator message history data, and data modification history data is stored in the same storage area. Therefore, alarm history data may be erased when the storage capacity is exceeded by other history data.

2 The maximum number of history data items that can be stored is as follows, where the history data is assumed to include alarm data only:
When bit 7 (HAL) of parameter No. 3196 is set to 1, up to about 4000 history data items can be stored.
When bit 7 (HAL) of parameter No. 3196 is set to 0, modal information and coordinates observed when an alarm is issued are also recorded, so the maximum number of history data items that can be stored varies from system to system. (Example: For a 5-axis system, up to about 280 data items can be stored; for a 10-axis system, up to about 200 data items can be stored.)
When bit 3 (EAH) of parameter No. 3112 is set to 1, however, macro and external alarm messages are also stored, so the number of data items that can be stored further decreases.

12.4.30.2 External operator message history

From all history data recorded, only external operator message history and macro message history are extracted and displayed on the screen. When the amount of history data exceeds the storage capacity, history data is automatically erased in order from the oldest history data.

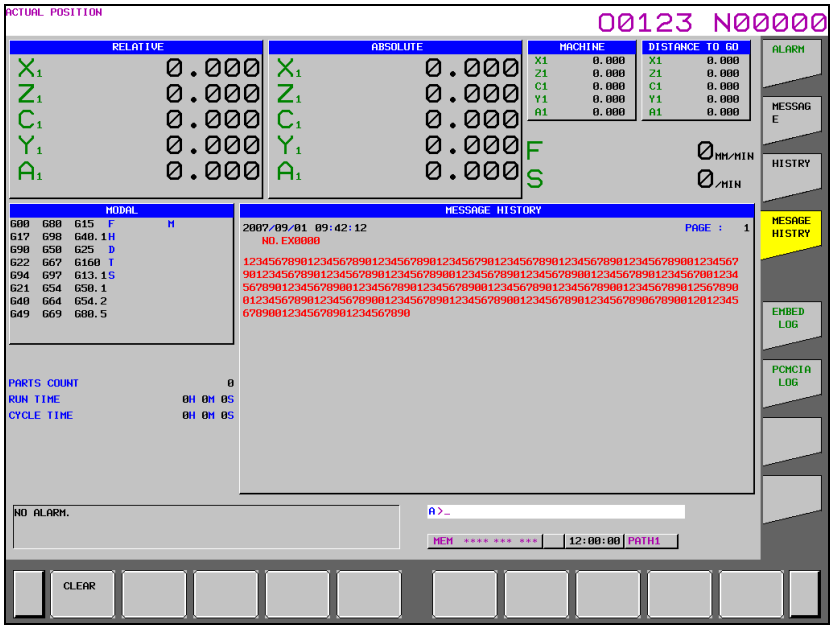


Fig. 12.4.30.2 (a) External operator message history screen

Screen display

Procedure

1 Press function key .

2 Press vertical soft key [MESAGE HISTRY] to display the external operator message history screen.

3 The screen display can be changed to the previous page and the next page by using page keys and .

NOTE

1 History data such as external operator message history data, operation history data, alarm history data, and data modification history data is stored in the same storage area. Therefore, external operator message history data may be erased when the storage capacity is exceeded by other history data.

2 The maximum number of history data items that can be stored is about 150 if the history data includes external operator message data only.

Erasing history data from the external operator message history screen

Procedure

1 Display the external operator message history screen.
2 Press horizontal soft key [CLEAR]. All history data is then erased.

NOTE
When history data is erased, not only external operator message history data but also other history data such as operation history data and alarm history data is erased. It is impossible to erase only a specific history.

Parameter setting

	#7	#6	#5	#4	#3	#2	#1	#0
3112						OMH		

[Data type] Bit

 # 2 OMH The external operator message history screen is:
0: Not displayed.
1: Displayed.

	#7	#6	#5	#4	#3	#2	#1	#0
3196		HOM						

[Data type] Bit

 # 6 HOM A history of external operator messages and macro messages is:
0: Recorded.
1: Not recorded.

12.4.30.3 Operation history

This function displays a history of the operator's key operations and signal operations made when a failure occurred or an alarm was issued, and also information about alarms.
The following data is recorded:

a Operation history
  i MDI key operations made by the operator
  ii I/O signal (X,Y,G,F) on/off switching
b Alarm history
  i Alarms issued
  ii Modal information in a block executed and coordinates observed when an alarm was issued
  (Not displayed on the screen)
c Data modification history
  i Modification of tool offset data
  (When bit 0 (HTO) of parameter No. 3196 is set to 1)
  ii Modification of workpiece offset data/extended workpiece offset data/workpiece shift (T series)
  (When bit 1 (HWO) of parameter No. 3196 is set to 1)
  iii Modification of parameters
  (When bit 2 (HPM) of parameter No. 3196 is set to 1)
  iv Modification of custom macro common variable data
  (When bit 3 (HMV) of parameter No. 3196 is set to 1)
d External operator message history and macro message history
(When bit 6 (HOM) of parameter No. 3196 is set to 0)
e Time stamp (time and date)

With some exceptions, history data of the operation history and alarm history can be viewed on the operation history screen. (The data modification history, external operator message history, and alarm messages are not displayed.)
All recorded history data can be output to external input/output devices.

NOTE
Up to about 8000 items of history data can be recorded if the data includes history data of MDI key operations only. However, because history data varies in size, the maximum number of history data items that can be recorded is not fixed.

Parameter setting

	#7	#6	#5	#4	#3	#2	#1	#0
3106				OPH				

[Data type] Bit

 # 4 OPH The operation history screen is:
0: Not displayed.
1: Displayed.

3122	Time interval used to record time data in operation history

[Input type] Parameter input
[Data type] Word path
[Unit of data] min
[Valid data range] 0 to 1440
When history data is recorded within a set time period, the time for each set time period is recorded in the history data.
When 0 is set, the specification of a time period of 10 minutes is assumed.
If no data is recorded within a set time period, the time for that period is not recorded.

NOTE
The same value must be set in this parameter for all paths.

	#7	#6	#5	#4	#3	#2	#1	#0
3195	EKE	HDE	HKE					

[Input type] Parameter input
[Data type] Common to the bit system

 # 5 HKE A key operation history is:
0: Recorded.
1: Not recorded.

 # 6 HDE A DI/DO history is:
0: Recorded.
1: Not recorded.

 # 7 EKE Soft key [ALL CLEAR] for clearing all history data is:
0: Not displayed.
1: Displayed.

	#7	#6	#5	#4	#3	#2	#1	#0
3196	HAL	HOM			HMV	HPM	HWO	HTO

[Data type] Bit

 # 0 HTO A modification history of tool offset data is:
0: Not recorded.
1: Recorded.

 # 1 HWO A modification history of workpiece offset data/extended workpiece offset data/workpiece shift (T series) is:
0: Not recorded.
1: Recorded.

 # 2 HPM A modification history of parameters is:
0: Not recorded.
1: Recorded.

 # 3 HMV A modification history of custom macro common variables is:
0: Not recorded.
1: Recorded.

 # 6 HOM A history of external operator messages and macro messages is:
0: Recorded.
1: Not recorded.

 # 7 HAL When an alarm is issued, additional information (modal data, absolute coordinates, and machine coordinates present at the issuance of the alarm) is:
0: Recorded in the operation history and alarm history.
1: Not recorded in the operation history and alarm history.
To record as many alarm history items as possible, rather than detailed alarm information, set 1. The numbers of ten G code modal groups to be recorded are set in parameter Nos. 12990 to 12999.

12990	(1st) G code modal group to be recorded in the history when an alarm is issued

[Input type] Parameter input
[Data type] Byte path
[Valid data range] 1 to the maximum number of G code groups
Set the number of a G code modal group to be recorded in the alarm history and operation history when an alarm is issued.
* If a value beyond the valid data range is set, the status of group 01 is recorded.

12991 **(2nd) G code modal group to be recorded in the history when an alarm is issued**

[Input type] Parameter input
[Data type] Byte path
[Valid data range] 1 to the maximum number of G code groups
Set the number of a G code modal group to be recorded in the alarm history and operation history when an alarm is issued.
* If a value beyond the valid data range is set, the status of group 02 is recorded.

12992 **(3rd) G code modal group to be recorded in the history when an alarm is issued**

[Input type] Parameter input
[Data type] Byte path
[Valid data range] 1 to the maximum number of G code groups
Set the number of a G code modal group to be recorded in the alarm history and operation history when an alarm is issued.
* If a value beyond the valid data range is set, the status of group 03 is recorded.

12993 **(4th) G code modal group to be recorded in the history when an alarm is issued**

[Input type] Parameter input
[Data type] Byte path
[Valid data range] 1 to the maximum number of G code groups
Set the number of a G code modal group to be recorded in the alarm history and operation history when an alarm is issued.
* If a value beyond the valid data range is set, the status of group 04 is recorded.

12994 **(5th) G code modal group to be recorded in the history when an alarm is issued**

[Input type] Parameter input
[Data type] Byte path
[Valid data range] 1 to the maximum number of G code groups
Set the number of a G code modal group to be recorded in the alarm history and operation history when an alarm is issued.
* If a value beyond the valid data range is set, the status of group 05 is recorded.

12995 **(6th) G code modal group to be recorded in the history when an alarm is issued**

[Input type] Parameter input
[Data type] Byte path
[Valid data range] 1 to the maximum number of G code groups
Set the number of a G code modal group to be recorded in the alarm history and operation history when an alarm is issued.
* If a value beyond the valid data range is set, the status of group 06 is recorded.

12996 **(7th) G code modal group to be recorded in the history when an alarm is issued**

[Input type] Parameter input
[Data type] Byte path
[Valid data range] 1 to the maximum number of G code groups
Set the number of a G code modal group to be recorded in the alarm history and operation history when an alarm is issued.
* If a value beyond the valid data range is set, the status of group 07 is recorded.

12997 **(8th) G code modal group to be recorded in the history when an alarm is issued**

[Input type] Parameter input
[Data type] Byte path
[Valid data range] 1 to the maximum number of G code groups
Set the number of a G code modal group to be recorded in the alarm history and operation history when an alarm is issued.
* If a value beyond the valid data range is set, the status of group 08 is recorded.

12998 **(9th) G code modal group to be recorded in the history when an alarm is issued**

[Input type] Parameter input
[Data type] Byte path
[Valid data range] 1 to the maximum number of G code groups
Set the number of a G code modal group to be recorded in the alarm history and operation history when an alarm is issued.
* If a value beyond the valid data range is set, the status of group 09 is recorded.

12999 **(10tht) G code modal group to be recorded in the history when an alarm is issued**

[Input type] Parameter input
[Data type] Byte path
[Valid data range] 1 to the maximum number of G code groups
Set the number of a G code modal group to be recorded in the alarm history and operation history when an alarm is issued.
* If a value beyond the valid data range is set, the status of group 10 is recorded.

Screen display

Procedure

1 Press function key SYSTEM.

2 Press vertical soft key [NEXT PAGE] several times until soft key [OPERAT HISTRY] is displayed.

3 Press vertical soft key [OPERAT HISTRY], then press newly displayed vertical soft key [OPERAT HISTRY]. The operation history screen is then displayed.

4 To display the previous page and next page of the operation history, use page keys PAGE ↑ and PAGE ↓.

To display a part across two pages, use the cursor keys ← and →. The screen display is shifted by half page.

(With the 7.2- and 8.4-inch display devices, the screen display is shifted by one column.)

When the operation history screen is displayed, the following horizontal soft key operations can be performed:

a [TOP] displays the starting page (the oldest data).

b [BOTTOM] displays the end page (the latest data).

c [NO.SRH] displays specified operation history data.
(Example) Specifying 50 then [NO.SRH] displays the 50th data.

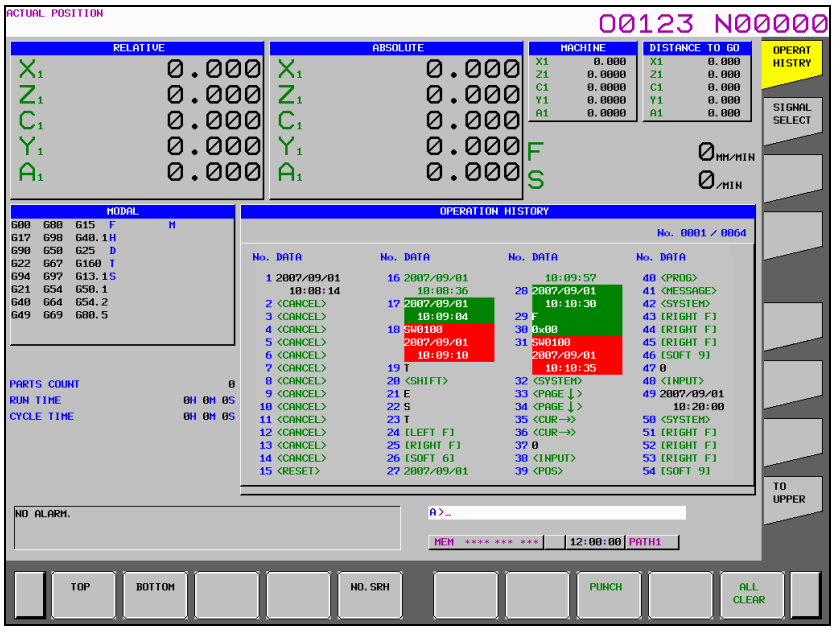


Fig. 12.4.30.3 (a) Operation history screen

Displayed information

1 Serial number and display start history number/total number of history data items

A serial number is indicated on the left side of each recorded history data item. A smaller serial number indicates an older data item.

In the upper right part of the screen, the display start history number and the total number of history data items are indicated. The total number of history data items does not include history data items not displayed on the screen.

2 Data

- MDI key

  When bit 5 (HKE) of parameter No. 3195 is set to 0, key operations are recorded.

  A key operation is indicated following a path number (for example, "1_[LEFT F]", and "2_[LEFT F]"). (When only one path is used, the path number is not indicated.)

  P_ used in "P_[LEFT F]", for example, indicates a key operation made from the outside.

  i Address keys and numeric keys

    Characters such as A to Z, 0 to 9, ;, +, and - are indicated directly.

    These characters are displayed in black.

  ii Function menu keys, operation menu keys, and soft keys

    These keys are enclosed in brackets [] (for example, "[LEFT F]", "[SOFT 1]" to "[SOFT 10]", and "[RIGHT F]"). These keys are displayed with green characters.

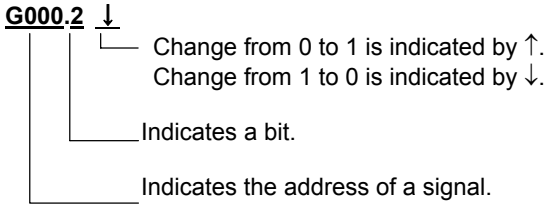
  iii Function keys, page keys, cursor keys, and so on

    These keys are enclosed in angle brackets <> (for example, "<POS>", "<SYSTEM>", "<PAGE ↑>", "<CUR →>", "<RESET>", and "<INPUT>"). These keys are displayed with green characters.

  iv Power-on key

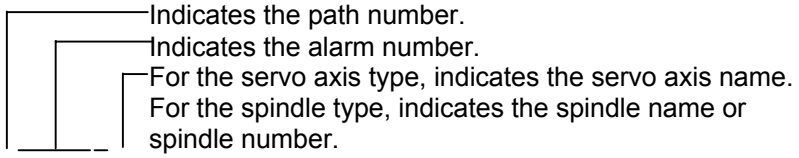
    This key is displayed with white characters in the green background.

- I/O signals
  When bit 6 (HDE) of parameter No. 3195 is set to 0, I/O signals specified on the operation history signal selection screen are recorded. Recorded signals are indicated on a bit-by-bit basis with information about the signal address and a change in bit. These signals are displayed with purple characters.



NOTE
1 When more than one bit at the same address changes at the same time, the changes of these bits are treated as one history data item.
2 A signal fluctuation for less than 4 msec is not recorded as history data.

- Alarms
  Alarm numbers and the issuance times are displayed on the operation history screen.
  This alarm information is displayed with white characters in the red background.



1_OT0506（X A1)
2004/01/11 Indicates the date and time of issuance in two lines.
11:22:33

When a path name, axis name, or spindle name was changed after the issuance of an alarm, the alarm is indicated with a newly assigned name.

- Time and date
  The time and date of the following are displayed in two lines:
  i The time and date of power-on. These are displayed with white characters in the green background.
  ii The time and date of power-off. These are displayed with green characters.
  iii Date on which the date changed. This is displayed with black characters.
  iv Date and time at regular intervals of the period set in parameter No. 3122. These are displayed with black characters.

v Date and time when history data was erased. These are displayed with black characters.

NOTE
1 When times are recorded at regular intervals, if there is no data to record within an interval, the time is not recorded. (When the date is changed, the time is recorded on the assumption that there is data to record.)
2 When the date and time in the CNC system are changed, the date and time at which the change is made may be recorded as the date and time in iii or iv above.

History data not displayed on the screen

In addition to history data of MDI keys, I/O signal status, alarms issued, external operator messages (not displayed on the operation history screen), and time stamps, data described below can be recorded with time. Such history data cannot be displayed on the screen but can be output to external input/output devices. (See Subsection 12.4.30.5, "Outputting all history data".)

1 Detailed data at alarm issuance
If bit 7 (HAL) of parameter No. 3196 is set to 0, 10 modal G codes, auxiliary function codes D, E, F, H, M, N, O, S, and T, absolute coordinates, and machine coordinates in the block being executed when an alarm was issued are recorded together with the alarm number and the time of alarm issuance.
The group numbers of the 10 modal G codes to be recorded are set in parameter Nos. 12990 to 12999. If these parameters are not specified, modal G codes of groups 01 to 10 are recorded.

NOTE
1 For modal O data that is a program name, only the first five characters are recorded.
2 To record as many items as possible instead of recording detailed data at alarm issuance, set bit 7 (HAL) of parameter No. 3196 to 1.

2 External alarm messages and macro alarm messages
If bit 3 (EAH) of parameter No. 3112 is set to 1, external alarm messages and macro alarm messages can also be recorded as history data. However, only the first 64 characters of each message are recorded.

NOTE
To record also external alarm and macro alarm messages as history data, set bit 3 (EAH) of parameter No. 3112 to 1, and at the same time set bit 7 (HAL) of parameter No. 3196 to 0.

3 Modification of tool offset data
If bit 0 (HTO) of parameter No. 3196 is set to 1, when tool offset data is modified, the number and type of the tool offset are recorded as well as the tool offset data before modification, the tool offset data after modification, and the time of modification.

4 Modification of workpiece offset/extended workpiece offset/workpiece shift (T series) data
If bit 1 (HWO) of parameter No. 3196 is set to 1, when workpiece offset data is modified, the number of the modified workpiece offset is recorded as well as the workpiece offset data before modification, the workpiece offset data after modification, and the time of modification. Similar data is recorded also when extended workpiece offsets and workpiece shift amounts (T series) are modified.

5 Modification of parameters
If bit 2 (HPM) of parameter No. 3196 is set to 1, when a parameter is modified, the number and type (axis type, spindle type, path type, or machine group type) of the parameter are recorded as well as the parameter data before modification, the parameter data after modification, and the time of modification.

NOTE
Modifications made at power-on and modifications of passwords and keys are not recorded as history data.

6 Modification of custom macro common variables (#100 to #999)
If bit 3 (HMV) of parameter No. 3196 is set to 1, when a custom macro common variable is modified, the number of the common variable is recorded as well as the common variable value before modification, the common variable value after modification, and the time of modification.

Erasing history data from the operation history screen

Procedure

1 Display the operation history screen.
2 Press horizontal soft key [ALL CLEAR].
3 Press horizontal soft key [EXEC]. All history data is erased.

NOTE
An erasure operation erases not only operation history data but also other history data such as external operator message history data and alarm history data. It is impossible to erase only a specific history.

12.4.30.4 Selecting operation history signals

I/O signals to be recorded as history data can be selected. Up to 60 signals can be set.

Setting data

1 Press function key SYSTEM.
2 Press vertical soft key [NEXT PAGE] several times until vertical soft key [OPERAT HISTRY] is displayed.
3 Press vertical soft key [OPERAT HISTRY].
4 Press vertical soft key [SIGNAL SELECT] to display the operation history signal selection screen.
5 Move the cursor to a desired position by using cursor keys ↑ and ↓.
6 Type a signal type (X, G, F, or Y) and an address, then press INPUT.
Example:
When G0004 is entered then INPUT is pressed:
The entered signal address, G0004, is set in ADDRESS, and initial value 00000000 is set in SIGNAL.
7 Select the bits to be recorded in the history.
To change the status of all bits of the specified signal address, place the cursor in all bits so that the bits are highlighted (for example, "00000000"), then press horizontal soft key [ON:1] or [OFF:0]. Then, the bits are set to 11111111 or 00000000.
To change the status of only a certain bit, move the cursor to the bit by using cursor keys ← and →, and press horizontal soft key [ON:1] or [OFF:0]. The selected bit is then set to 1 or 0.
8 Up to 60 addresses can be set to select signals. Addresses need not necessarily be set sequentially starting from No. 1.

NOTE
1 While the operation history signal selection screen is being displayed, recording of history data is not performed.
2 Only X, Y, G, and F can be set for I/O signals.
For data not set, "********" is displayed.
3 Even when an address is set, history data is not recorded if all bits are set to 0.
4 When the ON/OFF width of an input signal is less than 4 msec, recording of history data is not performed.
Also,there are some signals that are not recorded.
5 When many signals are selected, the processing speed may lower.

Clearing data individually

1 Display the operation history signal selection screen.
2 Move the cursor to the data to be cleared.
3 Press horizontal soft key [DELETE].
4 Press horizontal soft key [EXEC].

Clearing all data

1 Display the operation history signal selection screen.
2 Press horizontal soft key [ALLDEL].
3 Press horizontal soft key [EXEC].

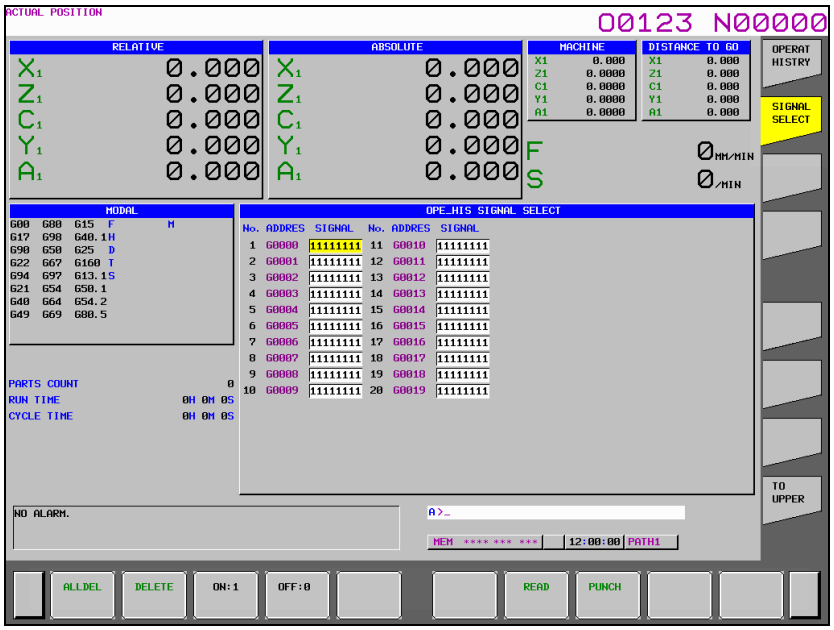


Fig. 12.4.30.4 (a) Operation history signal selection screen

12.4.30.5 Outputting all history data

All history data can be output to external input/output devices.
It is impossible, however, to output history data individually.

Procedure

1 Make an output device ready for output.
2 Set the EDIT mode.
3 Press function key

.
4 Press vertical soft key [NEXT PAGE] several times until vertical soft key [OPERAT HISTRY] is displayed.
5 Press vertical soft key [OPERAT HISTRY], then press newly displayed vertical soft key [OPERAT HISTRY]. The operation history screen is displayed.
7 Press horizontal soft key [PUNCH].
8 Enter a file name, and press horizontal soft key [EXEC].
When horizontal soft key [EXEC] is pressed without entering a file name, the output file name is assumed to be OPRT_HIS.TXT.

Output format

History data is output as an ASCII file in the following format:

1 MDI keys
After "MDI", "path-number_", "key-data", and "input-time" are output in this order.
(Key data input at power-on is indicated as "Power on MDI".)
<Example>
MDI 01_A 12:23:34
MDI 02_<CAN> 12:23:34
MDI 02_[SOFT HF1] 12:23:35
MDI P_<RESET> 12:34:56
Power on MDI 01_<RESET> 12:34:56

2 I/O signals
After "DI/DO", "PMC-number_", "signal-address_bit-status", and "time-of-change" are output in this order.
<Example>
DI/DO 1_F0002.2_on 12:34:56
DI/DO 1_ G0043.0_off G0043.1_off 12:35:00
(For multiple bits at the same address)

3 Alarms
After "Alarm", "path-number_", "type", "alarm-number", "G-code-modal-data", "modal-data-other-than-G-code", "absolute-coordinate-value" and "machine-coordinate-value" for each axis, and "date-and-time-of-alarm-issuance" are output in this order.
An asterisk "*" is output before the modal data that was specified in a block executed when the alarm was issued.

<Example>

- Alarm 01_SR01973
  *G0. G97. G69. G99. G21. G50.2 G25. G13.1 B0. D0. E0. *F100. H0. M10. *N123. Test_ S1000. T1010.
  X1 ABS 197.999 MCN 197.999 Y1 ABS -199806.00 MCN -199806.00 Z1 ABS 297.009 MCN 0.123 C1 ABS 10395.999 MCN 0.000 at 2007/09/01 19:03:28
- Alarm 02_ OT00506(ZA2)
  *G1. G17. G90. G22. G94. G20. *G42. G49. G80. G12.1 B0. *D12. E0. *F100. H34. M0. *N123. O123 S0. T0.
  X2 ABS 123.999 MCN 234.000 Y2 ABS -123.00 MCN -234.00 ZA2 ABS 1234.567 MCN –1234.567 at 2007/09/01 12:34:56
- When additional information is not to be recorded when an alarm is issued (bit 7 (HAL) of parameter No. 3196 is set to 1), only "path-number_", "alarm-number", and "date-and-time-of-alarm-issuance" are output.
  Alarm 01_OT00506(XC1) at 2007/09/01 22:08:32
  Alarm 02_SW00100 at 2007/09/01 19:07:52
- When external alarm/macro alarm messages are to be recorded (bit 7 (HAL) of parameter No. 3196 is set to 0) and bit 3 (EAH) of parameter No. 3112 is set to 1, the messages are also output.
  Alarm 01_MC00001 Message ATC ALARM
  G0. G97. G69. G99. G21. G40. G25. G22. G80. D0. E0. F0. H0. M0. N0. O9999 S0. T0.
  X2 ABS 10.000 MCN 0.000Y2 ABS 123.000 MCN 0.000Z2 ABS 0.000 MCN 0.000 at 2007/09/01 10:06:43

4 External operator messages
After "EXT_Message", "message-number", "message", and "date-and-time-of-issuance" are output in this order.

<Example>
EXT_Message 01234 OIL PRESSURE DECREASE at 2007/09/01 2:38:43

5 Modification of tool offset data
After "Tool Offset", "path-number_", "type", "offset-number", "offset-data-before-modification", "offset-data-after-modification", and "time-of-modification" are output in this order. The following types are provided:

Common to M/T	:	G=Geometric compensation	W=Wear compensation
M series	:	H=Tool length compensation	D=Cutter compensation
T series	:	R=Tool-nose radius compensation	T=Tool nose direction

X2,Z2,Y2=Second geometric compensation

<Example>
Tool Offset 01_X0002 0.000 → 1 at 12:15:43
Tool Offset 02_XW0001 -9999.999 → 9999.999 at 12:15:46
Tool Offset 01_RG0032 0.000 → 0.003 at 12:15:52
Tool Offset 02_T0001 5. → 2. at 19:34:11
Tool Offset 02_W0123 -10.000 → 123.456 at 10:28:58
Tool Offset 01_HG0456 0.000 → 999.999 at 11:37:40
Tool Offset 01_ 0064 12.340 → 12.569 at 11:39:42

6 Modification of workpiece offset/extended workpiece offset/workpiece shift (T series) data
After "Work Offset", "EXT Work Offset", or "Work Shift", "path-number_(axis-name)", "type", "offset-number", "offset-data-before-modification", "offset-data-after-modification", and "time-of-modification" are output in this order.
<Example>
Work Offset 01_G55(XA1) 15.000 → 0.007 at 09:23:03
Work Offset 02_EXT(Z2) 0.000 → 300.003 at 09:22:50
EXT Work Offset 02_G54.1P300 (Y2) 123.456 → 9999.999 at 12:15:46
Work Shift (X) 02_999999.999 → 999999.999 at 10:22:37

7 Modification of parameters
After "Parameter", "type", "parameter-number", "parameter-before-modification", "parameter-after-modification", and "time-of-modification" are output in this order.
The following types are provided:
Path type : L is added before the path number.
Axis type : A is added before the axis number.
Spindle type : S is added before the spindle number.
Machine group type : Indicated as the machine type. T is added before the machine group number.
Others : No type is output.
<Example>
Paramater N03112 00000100 → 00001100 at 11:18:40
Paramater Path type N01410 L02 0.000 → 1000.000 at 18:58:48
Paramater Axis type N01423 A04(B2) 0.000 → 10000.000 at 18:58:48
Paramater Spindle type N04011 S1(S) 10011010 → 10011010 at 18:58:53
Paramater Machine type N06310 T01 0 → -32768 at 19:21:13

8 Modification of custom macro common variables (#100 to #999)
After "Macro variable", "path-number_", "#variable-number", "common-variable-value-before-modification", "common-variable-value-after-modification", and "time-of-modification" are output in this order.
Variable values are output in the data format M × (10**(-E)).
<Example>

- When #149 on the first path was modified from <empty> to 12.345
  Macro valiable 01_#149 Empty → 123450000*(10**-7) at 15:02:35
- When #549 on the second path was modified from -12.345 to 123456789012
  Macro valiable 02_#549 -123450000*(10**-7) →123456789*(10**3) at 15:03:27

9 Date and time

Power on at 2007/09/01 17:11:17	(Date and time of power-on)
Power off at 2007/09/01 17:49:17	(Date and time of power-off)
Date 2007/09/01 00:00:00	(Record indicating a change in date)
Time stamp at 2007/09/01 15:51:00	(Record at regular intervals)
Data delete at 2007/09/01 10:56:18	(Date and time when history data was deleted)

12.5 SCREENS DISPLAYED BY FUNCTION KEY

By pressing the function key

, data such as alarms, and alarm history data can be displayed.
For information relating to alarm display, see Section III-7.1. For information relating to alarm history display, see Section III-7.2.

12.6 DISPLAYING THE PROGRAM NUMBER/NAME, SEQUENCE NUMBER, AND STATUS, AND WARNING MESSAGES FOR DATA SETTING OR INPUT/OUTPUT OPERATION

The program number, program name, sequence number, and current CNC status are always displayed on the screen except when the power is turned on, a system alarm occurs.
If data setting or the input/output operation is incorrect, the CNC does not accept the operation and displays a warning message.
This section describes the display of the program number, sequence number, and status, and warning messages displayed for incorrect data setting or input/output operation.

Screens of a 10.4-inch display unit

12.6.1 Displaying the Program Number, Program Name, and Sequence Number

The number and name of the program currently selected or currently executed and the current sequence number are indicated in the upper right part as shown below.

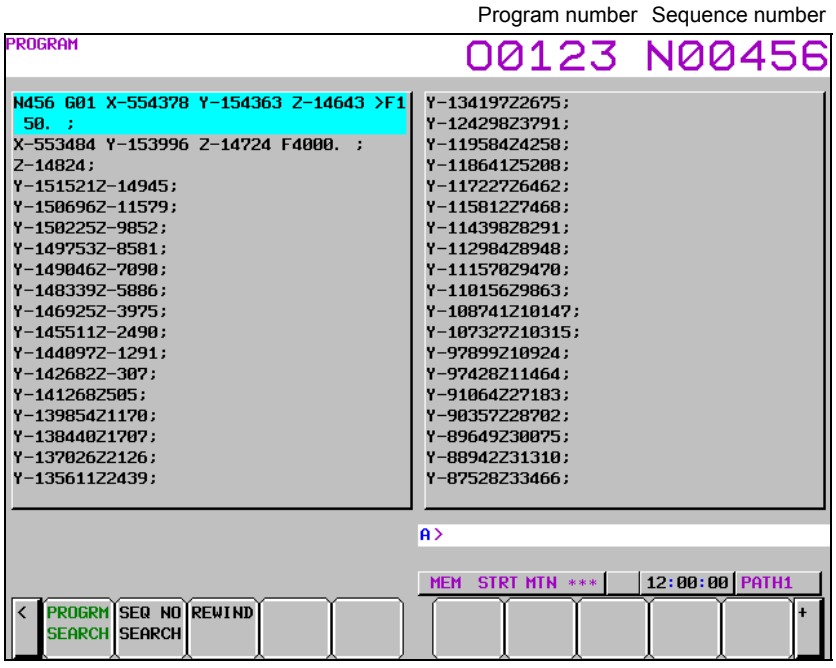


Fig. 12.6.1 (a) Program number and sequence number (10.4-inch)

In the EDIT mode, the number and name of the program currently edited in the foreground are indicated.

12.6.2 Displaying the Status and Warning for Data Setting or Input/Output Operation

The current mode, automatic operation state, alarm state, and program editing state are displayed on the next to last line on the screen allowing the operator to readily understand the operation condition of the system. If data setting or the input/output operation is incorrect, the CNC does not accept the operation and a warning message is displayed on the next to last line of the screen. This prevents invalid data setting and input/output errors.

Explanation

- Description of each display

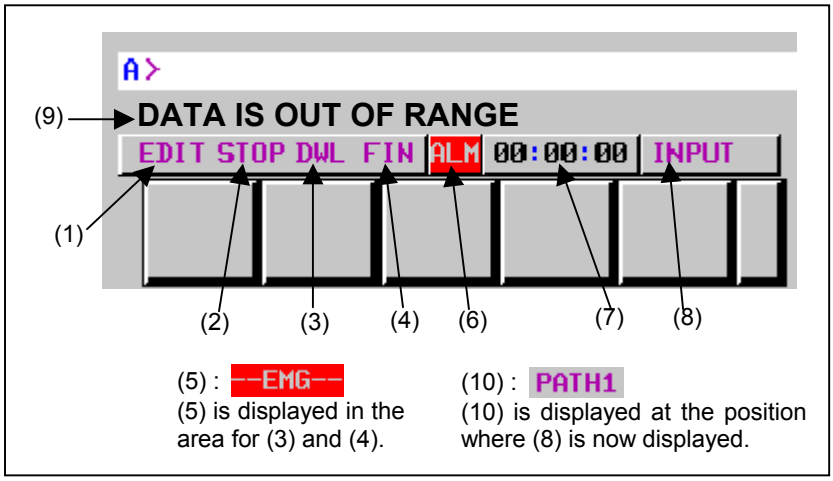


Fig. 12.6.2 (a) Positions of status indications

(1) Current mode

MDI	:	Manual data input, MDI operation
MEM	:	Automatic operation (memory operation)
RMT	:	Automatic operation (DNC operation, or such like)
EDIT	:	Memory editing
HND	:	Manual handle feed
JOG	:	Jog feed
INC	:	Manual incremental feed
REF	:	Manual reference position return
****	:	Mode other than the above

(2) Automatic operation status

****	:	Reset (When the power is turned on or the state in which program execution has terminated and automatic operation has terminated.)
STOP	:	Automatic operation stop (The state in which one block has been executed and automatic operation is stopped.)
HOLD	:	Feed hold (The state in which execution of one block has been interrupted and automatic operation is stopped.)
STRT	:	Automatic operation start-up (The state in which the system operates automatically)

MSTR : Manual numerical command start state (The state in which a manual numerical command is being executed) Alternatively, tool retract and recover operation state (The state in which a recover operation and repositioning operation are being performed)

(3) Axis moving status/dwell status

MTN : Indicates that the axis is moving.
DWL : Indicates the dwell state.
*** : Indicates a state other than the above.

(4) State in which an auxiliary function is being executed

FIN : Indicates the state in which an auxiliary function is being executed. (Waiting for the complete signal from the PMC)
*** : Indicates a state other than the above.

(5) Emergency stop or reset status

--EMG-- : Indicates emergency stop.(Blinks in reversed display.)
-RESET- : Indicates that the reset signal is being received.

(6) Alarm status

ALM : Indicates that an alarm is issued. (Blinks in reversed display.)
BAT : Indicates that the voltage of the lithium battery (the backup battery of the CNC) has decreased. (Blinks in reversed display.)
APC : Indicates that the voltage of the backup battery of the absolute pulse coder has decreased. (Blinks in reversed display.)
FAN : Indicates that the rotation speed of the fan has decreased. (Blinks in reversed display.)
Space : Indicates a state other than the above.

(7) Current time

hh : mm : ss - Hours, minutes, and seconds

(8) Program editing status

INPUT : Indicates that data is being input.
OUTPUT : Indicates that data is being output.
SEARCH : Indicates that a search is being performed.
EDIT : Indicates that another editing operation is being performed (insertion, modification, etc.)
LSK : Indicates that labels are skipped when data is input.
RSTR : Indicates that the program is being restarted
COMPARE : Indicates that a data comparison is being made.
OFST : Indicates that the tool length compensation amount measurement mode is set (for the machining center system) or that the tool length compensation amount write mode is set (for the lathe system).
WOFS : Indicates that the workpiece origin offset amount measurement mode is set.

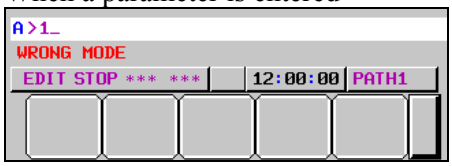
AICC1	:	Indicates that operation is being performed in the AI contour control I mode.
AICC2	:	Indicates that operation is being performed in the AI contour control II mode.
MEM-CHK	:	Indicates that a program memory check is being made.
WSFT	:	Indicates that the workpiece shift amount write mode is set.
LEN	:	Indicates that the active offset value change mode (tool length offset value of the M series) is set.
RAD	:	Indicates that the active offset value change mode (tool radius compensation amount of the M series) is set.
WZR	:	Indicates that the active offset value change mode (workpiece origin offset value) is set.
TOFS	:	Indicates that the active offset value change mode (tool offset value of the T series) is set.
OFSX	:	Indicates that the active offset value change mode (X-axis tool offset value of the T series) is set.
OFSZ	:	Indicates that the active offset value change mode (Z-axis tool offset value of the T series) is set.
OFSY	:	Indicates that the active offset value change mode (Y-axis tool offset value of the T series).
TCP	:	Indicates that operation is being performed in the tool center point control.
TWP	:	Indicates that operation is being performed in the tilted working plane command mode.
Space	:	Indicates that no editing operation is being performed.

(9) Warning for data setting or input/output operation

When invalid data is entered (wrong format, value out of range, etc.), when input is disabled (wrong mode, write disabled, etc.), or when input/output operation is incorrect (wrong mode, etc.), a warning message is displayed. In this case, the CNC does not accept the setting or input/output operation (retry the operation according to the message).

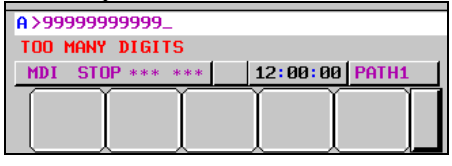
Example 1)

When a parameter is entered



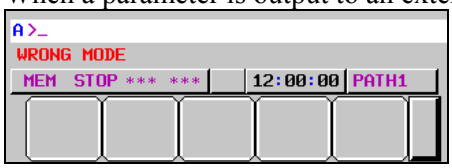
Example 2)

When a parameter is entered



Example 3)

When a parameter is output to an external input/output device



(10) Tool post name

The number of a path whose status is indicated is displayed.

PATH1 : Indicates that the status being indicated is for path 1.
Other names can be used depending on the settings of parameters 3141 to 3147. The tool post name is displayed at the position where (8) is now displayed. While the program is edited, (8) is displayed.

Screens of a 15-inch display unit

12.6.3 Displaying the Program Number, Program Name, and Sequence Number (15-inch Display Unit)

The number and name of the program currently selected or currently executed and the current sequence number are indicated in the upper right part as shown below.

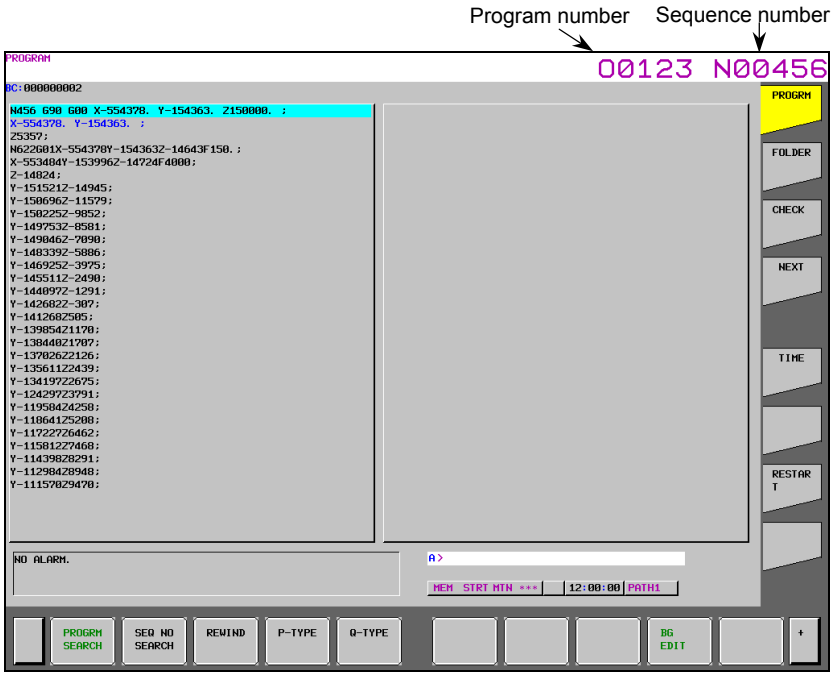


Fig. 12.6.3 (a) Program number and sequence number (15-inch)

In the EDIT mode, the number and name of the program currently edited in the foreground are indicated.

12.6.4 Displaying the Status and Warning for Data Setting or Input/Output Operation (15-inch Display Unit)

The current mode, automatic operation state, alarm state, and program editing state are displayed on the next to last line on the screen allowing the operator to readily understand the operation condition of the system. If data setting or the input/output operation is incorrect, the CNC does not accept the operation and a warning message is displayed on the next to last line of the screen. This prevents invalid data setting and input/output errors.

Explanation

- Description of each display

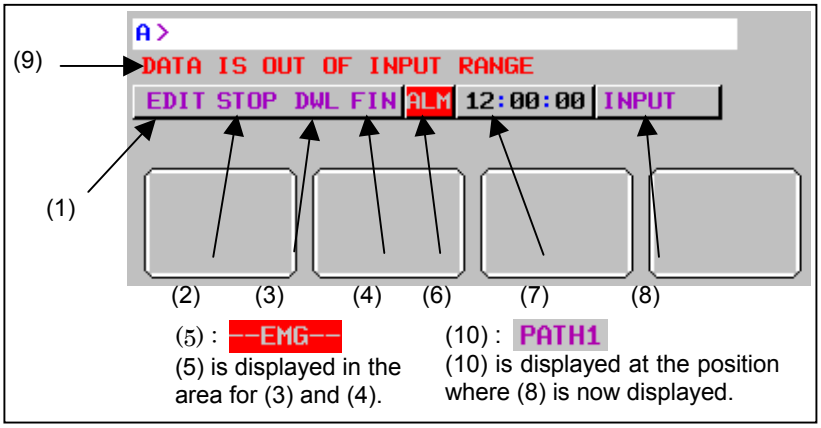


Fig. 12.6.4 (a) Positions of status indications

(1) Current mode

MDI	:	Manual data input, MDI operation
MEM	:	Automatic operation (memory operation)
RMT	:	Automatic operation (DNC operation, or such like)
EDIT	:	Memory editing
HND	:	Manual handle feed
JOG	:	Jog feed
INC	:	Manual incremental feed
REF	:	Manual reference position return
****	:	Mode other than the above

(2) Automatic operation status

****	:	Reset (When the power is turned on or the state in which program execution has terminated and automatic operation has terminated.)
STOP	:	Automatic operation stop (The state in which one block has been executed and automatic operation is stopped.)
HOLD	:	Feed hold (The state in which execution of one block has been interrupted and automatic operation is stopped.)
STRT	:	Automatic operation start-up (The state in which the system operates automatically)

MSTR : Manual numerical command start state (The state in which a manual numerical command is being executed) Alternatively, tool retract and recover operation state (The state in which a recover operation and repositioning operation are being performed)

(3) Axis moving status/dwell status

MTN : Indicates that the axis is moving.
DWL : Indicates the dwell state.
*** : Indicates a state other than the above.

(4) State in which an auxiliary function is being executed

FIN : Indicates the state in which an auxiliary function is being executed. (Waiting for the complete signal from the PMC)
*** : Indicates a state other than the above.

(5) Emergency stop or reset status

-EMG- : Indicates emergency stop.(Blinks in reversed display.)
-RESET- : Indicates that the reset signal is being received.

(6) Alarm status

ALM : Indicates that an alarm is issued. (Blinks in reversed display.)
BAT : Indicates that the voltage of the lithium battery (the backup battery of the CNC) has decreased. (Blinks in reversed display.)
APC : Indicates that the voltage of the backup battery of the absolute pulse coder has decreased. (Blinks in reversed display.)
FAN : Indicates that the rotation speed of the fan has decreased. (Blinks in reversed display.)
Space : Indicates a state other than the above.

(7) Current time

hh : mm : ss - Hours, minutes, and seconds

(8) Program editing status

INPUT : Indicates that data is being input.
OUTPUT : Indicates that data is being output.
SEARCH : Indicates that a search is being performed.
EDIT : Indicates that another editing operation is being performed (insertion, modification, etc.)
LSK : Indicates that labels are skipped when data is input.
RSTR : Indicates that the program is being restarted
COMPARE : Indicates that a data comparison is being made.
OFST : Indicates that the tool length compensation amount measurement mode is set (for the machining center system) or that the tool length compensation amount write mode is set (for the lathe system).
WOFS : Indicates that the workpiece origin offset amount measurement mode is set.

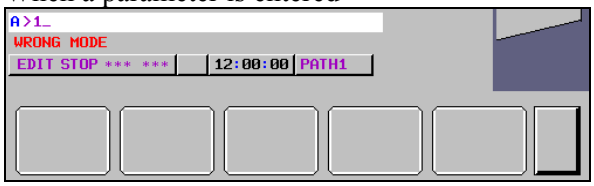
AICC1	:	Indicates that operation is being performed in the AI contour control I mode.
AICC2	:	Indicates that operation is being performed in the AI contour control II mode.
MEM-CHK	:	Indicates that a program memory check is being made.
WSFT	:	Indicates that the workpiece shift amount write mode is set.
LEN	:	Indicates that the active offset value change mode (tool length offset value of the M series) is set.
RAD	:	Indicates that the active offset value change mode (tool radius compensation amount of the M series) is set.
WZR	:	Indicates that the active offset value change mode (workpiece origin offset value) is set.
TOFS	:	Indicates that the active offset value change mode (tool offset value of the T series) is set.
OFSX	:	Indicates that the active offset value change mode (X-axis tool offset value of the T series) is set.
OFSZ	:	Indicates that the active offset value change mode (Z-axis tool offset value of the T series) is set.
OFSY	:	Indicates that the active offset value change mode (Y-axis tool offset value of the T series).
TCP	:	Indicates that operation is being performed in the tool center point control.
TWP	:	Indicates that operation is being performed in the tilted working plane command mode.
Space	:	Indicates that no editing operation is being performed.

(9) Warning for data setting or input/output operation

When invalid data is entered (wrong format, value out of range, etc.), when input is disabled (wrong mode, write disabled, etc.), or when input/output operation is incorrect (wrong mode, etc.), a warning message is displayed. In this case, the CNC does not accept the setting or input/output operation (retry the operation according to the message).

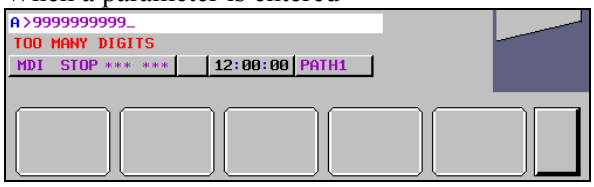
Example 1)

When a parameter is entered



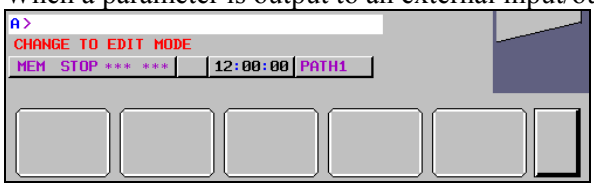
Example 2)

When a parameter is entered



Example 3)

When a parameter is output to an external input/output device



(10) Tool post name

The number of a path whose status is indicated is displayed.

PATH1 : Indicates that the status being indicated is for path 1.
Other names can be used depending on the settings of parameters 3141 to 3147. The tool post name is displayed at the position where (8) is now displayed. While the program is edited, (8) is displayed.

12.7 SCREEN ERASURE FUNCTION AND AUTOMATIC SCREEN ERASURE FUNCTION

Overview

Keeping the same characters displayed in the same positions on the screen for a long time will shorten the life of the LCD.
To prevent this, the CNC screen can be erased. The screen erasure function allows the user to perform a key operation to erase the screen. The automatic screen erasure function erases the screen automatically when there has been no key operation for a parameter-set period of time.

Screen erasure function

When 0 is set in parameter No. 3123, the CNC screen can be erased by pressing the

key and any function key (such as

or

) at the same time. The CNC screen can be displayed again by pressing any function key.

Automatic screen erasure function

When there has been no key operation for a time (in minutes) set in parameter No. 3123, the CNC screen is erased automatically. The CNC screen is displayed again by pressing a key.

- Screen erasure by the automatic screen erasure function

If the following conditions are all satisfied for the time (in minutes) set in parameter No. 3123, the CNC screen is erased.
Conditions for automatically erasing the CNC screen

- Parameter No. 3123 ≠ 0
- None of the following key operations is performed.
  MDI keys
  Soft keys
  External key input
- No alarm is issued.

- Redisplay of the screen by the automatic screen erasure function

If one of the following conditions is satisfied when the CNC screen is off, the CNC screen is displayed again:
Conditions for redisplaying the CNC screen

- One of the following key operations is performed.
  MDI keys
  Soft keys
  External key input
- An alarm is issued.

- Screen erasure by using the CAN key + function key

When a non-zero value is set in parameter No. 3123, the screen is not erased with the CAN key and a function key.

- Set time

Only the time set in parameter No. 3123 for path 1 is valid.

- Alarm in another path

When an alarm is issued in any of the paths, the screen is not erased.

Parameter

3123	Time required before a screen saver is activated

[Input type] Setting input
[Data type] Byte path
[Unit of data] min
[Valid data range] 0 to 127

After a time (in minutes) set in parameter No. 3123 passes without key operation, the NC screen is erased automatically. Pressing a key causes the NC screen to reappear.

NOTE
1 Setting 0 disables automatic screen erasure.
2 This function cannot be used together with manual screen erasure. If 1 or a larger value is set in this parameter, manual screen erasure is disabled.

12.8 LOAD METER SCREEN

Overview

The servo and spindle load meters can be displayed in place of the modal code display part and the remaining travel distance part of the current position display on the program check screen.
This function is enabled only when the 10.4-inch display unit is used.

12.8.1 Single-Path Display

Screen layout

In the single-path display mode, the servo or spindle load meter is displayed in place of the remaining travel distance display of the total position display.

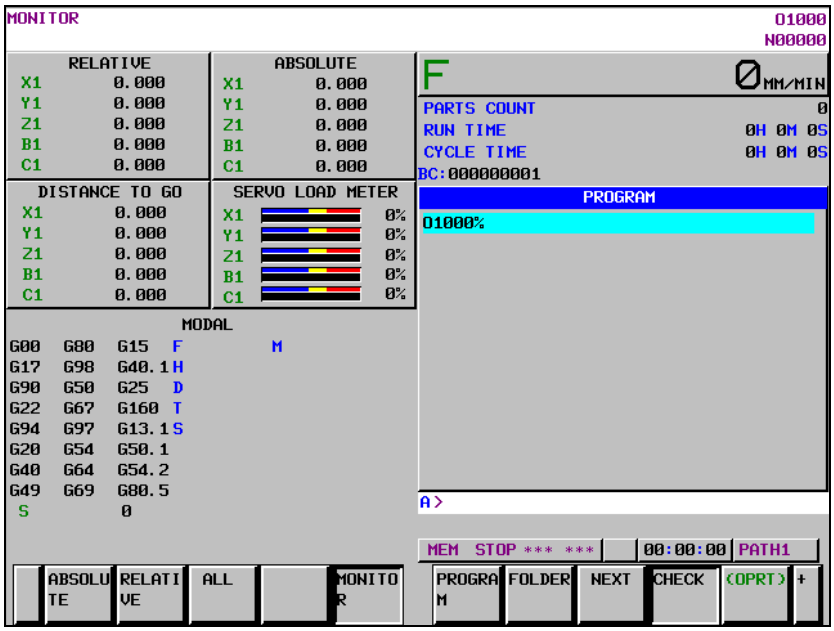


Fig. 12.8.1.(a) Servo load meter

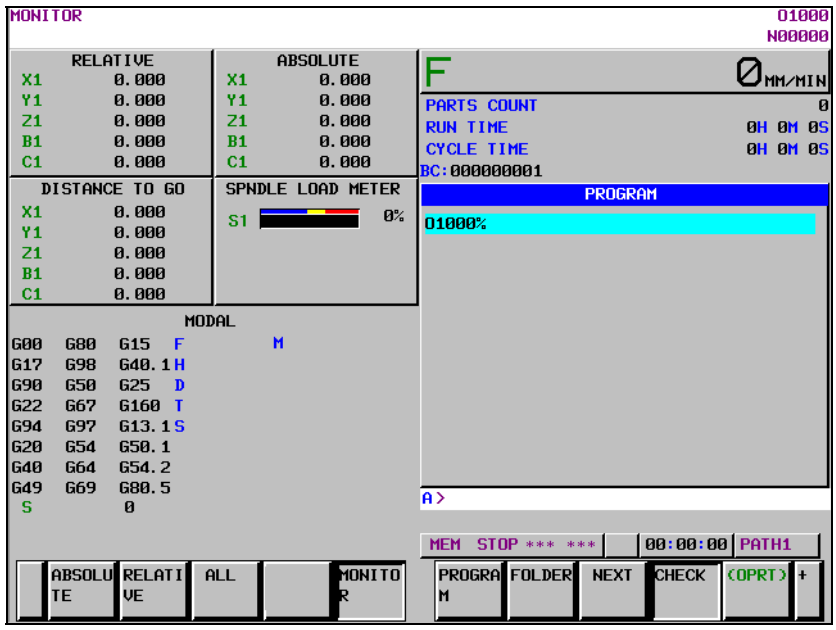
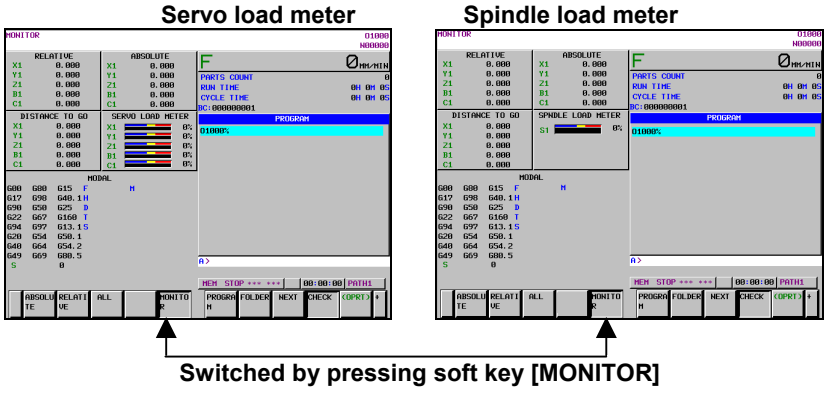


Fig. 12.8.1 (b) Spindle load meter

Screen switching

The servo load meter and spindle load meter are displayed by pressing soft key [MONITOR] on the left side of the screen. Initially the servo load meter is displayed. Each time soft key [MONITOR] is pressed, the servo load meter and spindle load meter alternate.



12.8.2 Two-Path Display and Three-Path Display

Screen layout

In the two- or three-path display mode, the modal information display part on the program check screen can be replaced by the servo or spindle load meter.

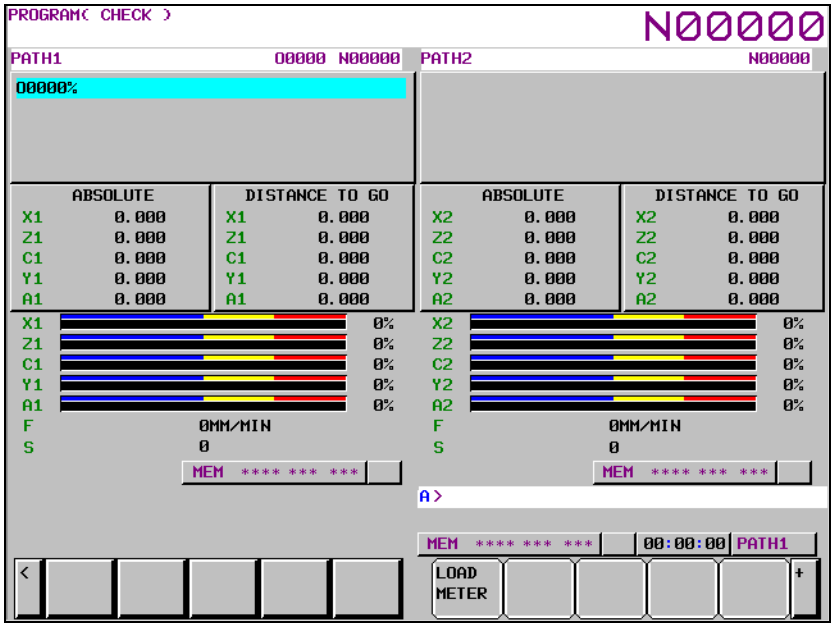


Fig. 12.4.30.5 (a) Servo load meter screen

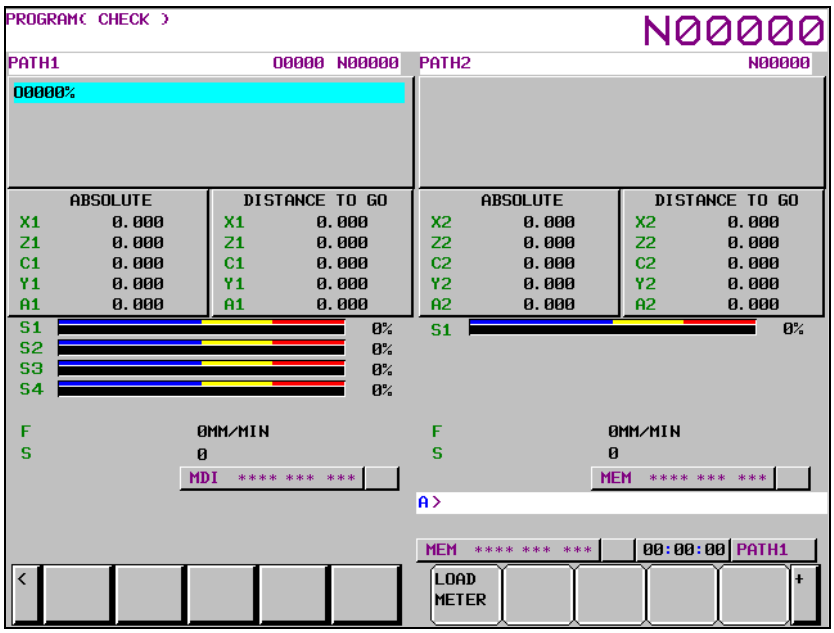
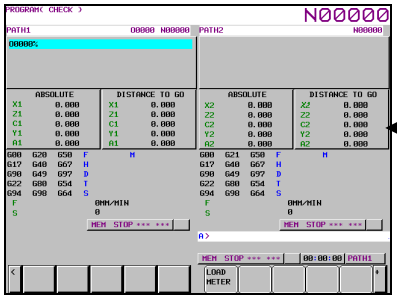


Fig. 12.4.30.5 (a) Spindle load meter screen

Screen switching

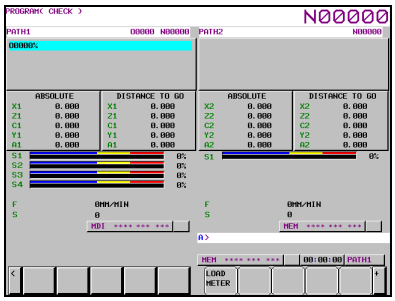
By pressing soft key [LOAD METER], the screen display can be switched among the modal information display, servo load meter display, and spindle load meter display.

Modal information



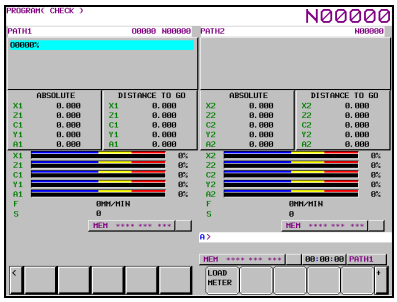
Press soft key [LOAD METER].

Spindle load meter



Press soft key [LOAD METER].

Servo load meter



Press soft key [LOAD METER].

Parameter

	#7	#6	#5	#4	#3	#2	#1	#0
3192	PLD							

[Input type] Parameter input
[Data type] Bit

 # 7 PLD When the current position is indicated for a path, and when the program check screen is displayed in a two- or three-path system, the function for displaying servo load meters and spindle load meters is :
0: Disabled.
1: Enabled.

13140	First character in spindle load meter display
13141	Second character in spindle load meter display

[Input type] Setting input
[Data type] Byte spindle
[Valid data range] These parameters set character codes to set the name of each spindle that appears in spindle load meter display. Any character string consisting of numeric characters, alphabetical characters, katakana characters, and symbols with a maximum length of two characters can be displayed as a spindle name.
If 0 is set, the following is displayed:

1st spindle S1
2nd spindle S2
3rd spindle S3
4th spindle S4

13 GRAPHIC FUNCTION

Chapter 13, "GRAPHIC FUNCTION", consists of the following sections:

13.1 GRAPHIC DISPLAY.....1900
13.2 DYNAMIC GRAPHIC DISPLAY.....1913

13.1 GRAPHIC DISPLAY

The graphic display functions enable drawing of the tool path of the program currently used for machining.
These functions are intended to display the movement of the tool during automatic operation or during manual operation.
This enables the operator to check the progress of machining and the current tool position.
These functions include the following:

- The current tool position in the workpiece coordinate system is displayed.
- Graphic coordinates can be set freely.
- Rapid traverse and cutting feed can be drawn using a different color for each.
- The values of F, S, and T in the program during drawing are displayed.
- Graphic enlargement or reduction is possible.

Graphic display procedure

Explanation

Press the function key

then press the [GRAPH] soft key to display the tool path graphic screen.

- Tool path graphic screen

The tool path graphic screen consists of three major areas:

- Graphic area for drawing a tool path
- Area for displaying machining information such as tool position information
- Area for displaying a graphic coordinate system

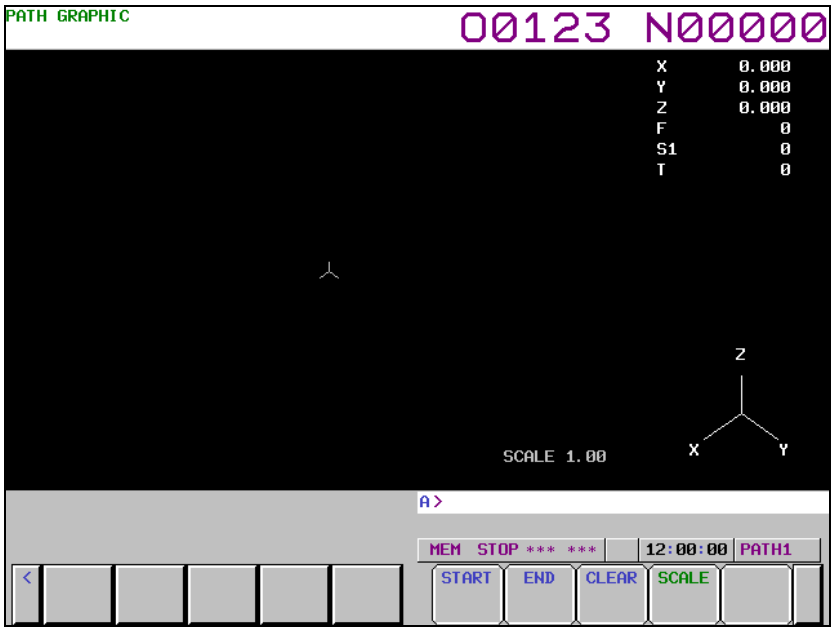


Fig. 13.1 (a) Tool path graphic screen (M series)

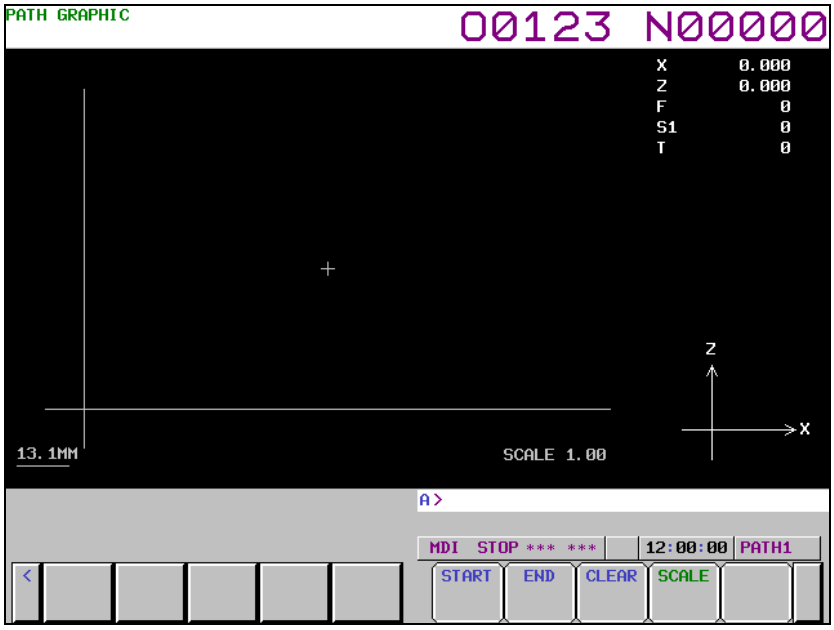


Fig. 13.1 (b) Tool path graphic screen (T series)

- Tool path

In a graphic coordinate system set by the graphic parameters described later, a tool path in the workpiece coordinate system is drawn.
Even when the tool position changes discontinuously for a cause such as the setting of the origin and the switching of the workpiece coordinate system, drawing is performed assuming that the tool has moved.
A tool path is continuously drawn even when the screen display is changed to another screen.

- Machining information

On the right side of the screen, the positions (along only three axes used for drawing) in the workpiece coordinate system, feedrate (F), spindle speed (S), and tool number (T) are displayed.

NOTE
Up to three graphic axes are used with the M series, and up to two graphic axes are used with the T series.

- Graphic coordinate system

On the lower-right portion of the screen, the coordinate axes and axis names of the graphic coordinate system are displayed.

Graphic parameter screen

Explanation

Press the function key GRAPH then press the [PARAM] soft key to display the tool path graphic screen.
On the graphic parameter screen, make settings necessary for drawing a tool path.
The graphic parameter screen consists of three pages.

M

- Graphic parameter screen page 1

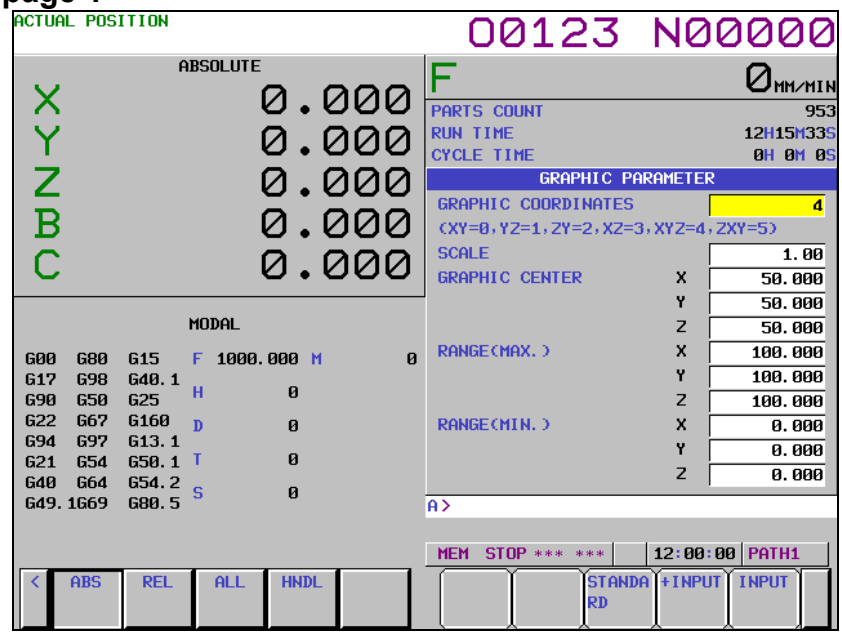


Fig. 13.1 (c) Graphic parameter screen page 1

On graphic parameter screen page 1, a graphic coordinate system, graphic range, and so forth are set.
In the setting of a graphic coordinate system, the coordinate axes and axis names of the set coordinate system are graphically displayed. When a three-dimensional coordinate system is displayed, a rotation angle is also displayed.
A graphic range is set using one of two methods. One method sets a graphic scale and graphic center coordinates. The other sets the maximum values and minimum values of a graphic range.

- Graphic parameter screen page 2

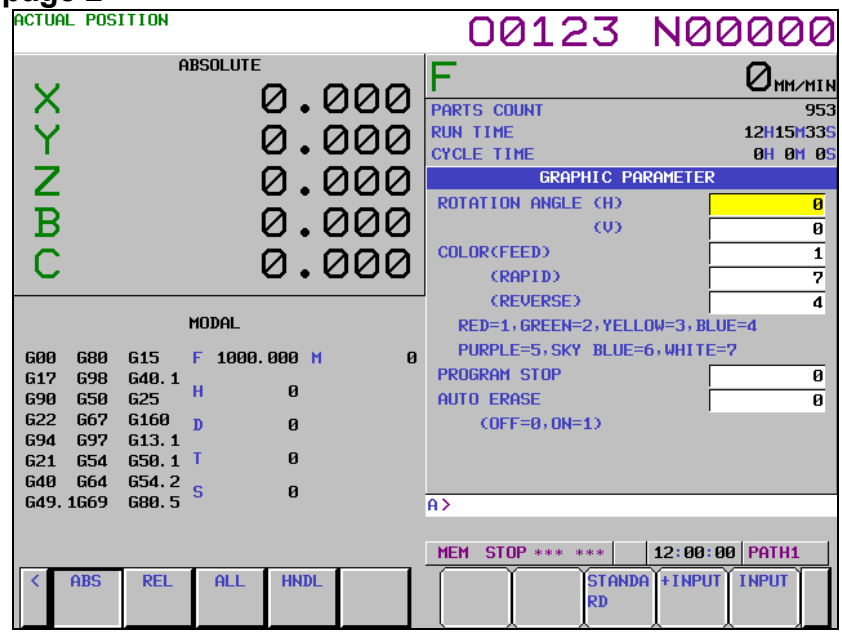


Fig. 13.1 (d) Graphic parameter screen page 2

On graphic parameter screen page 2, graphic colors, rotation angles, and whether to perform automatic erase operation are set.

- Graphic parameter screen page 3

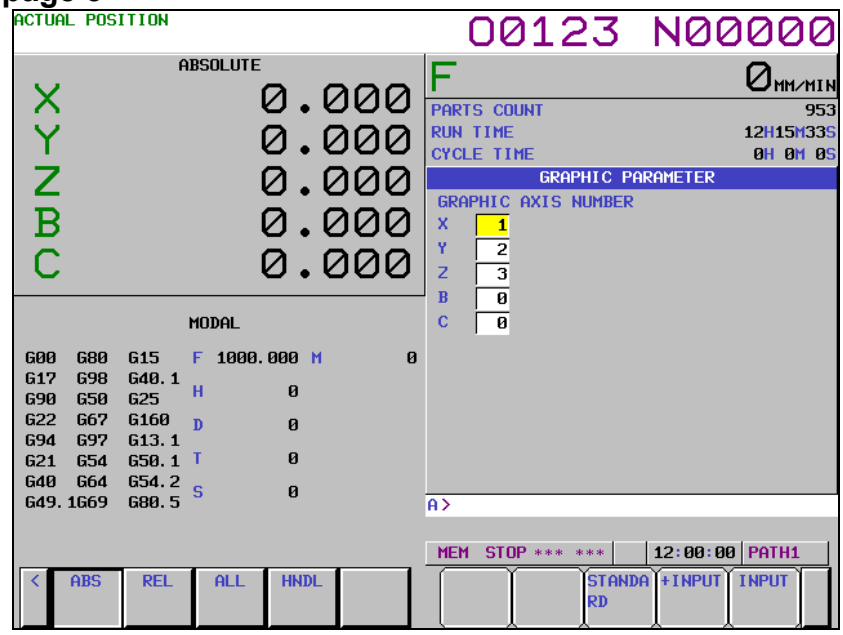


Fig. 13.1 (e) Graphic parameter screen page 3

On graphic parameter screen page 3, coordinate axes to be used for drawing are set.

T

- Graphic parameter screen page 1

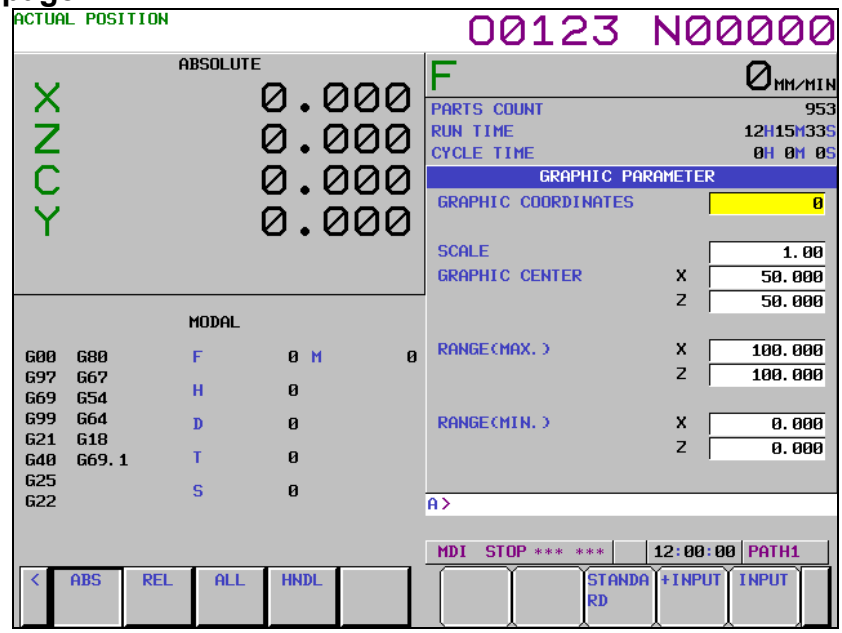


Fig. 13.1 (f) Graphic parameter screen page 1

On graphic parameter screen page 1, a graphic coordinate system, graphic range, and so forth are set.
In the setting of a graphic coordinate system, the coordinate axes and axis names of the set coordinate system are graphically displayed. When a three-dimensional coordinate system is displayed, a rotation angle is also displayed.
A graphic range is set using one of two methods. One method sets a graphic scale and graphic center coordinates. The other sets the maximum values and minimum values of a graphic range.

- Graphic parameter screen page 2

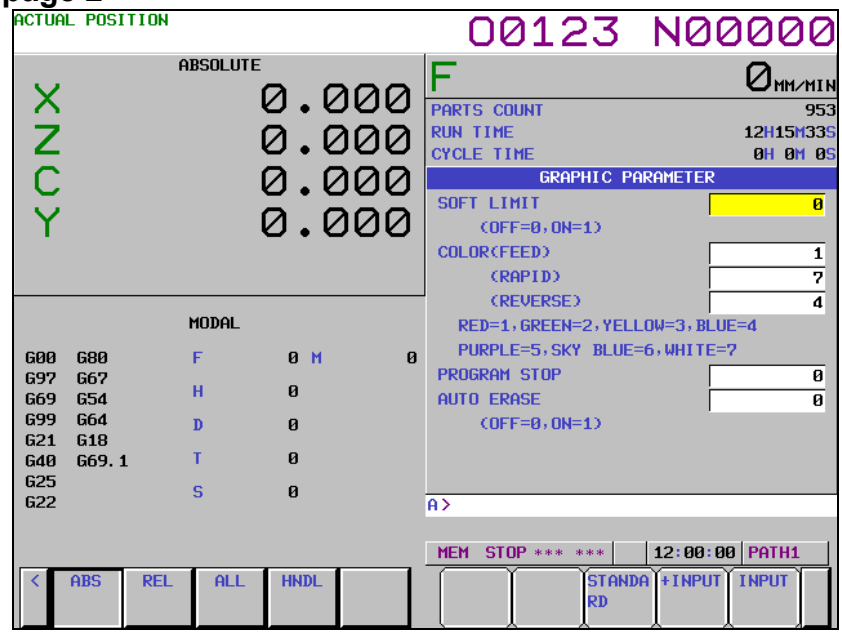


Fig. 13.1 (g) Graphic parameter screen page 2 (T series)

On graphic parameter screen page 2, graphic colors and whether to perform automatic erase operation are set.

- Graphic parameter screen page 3

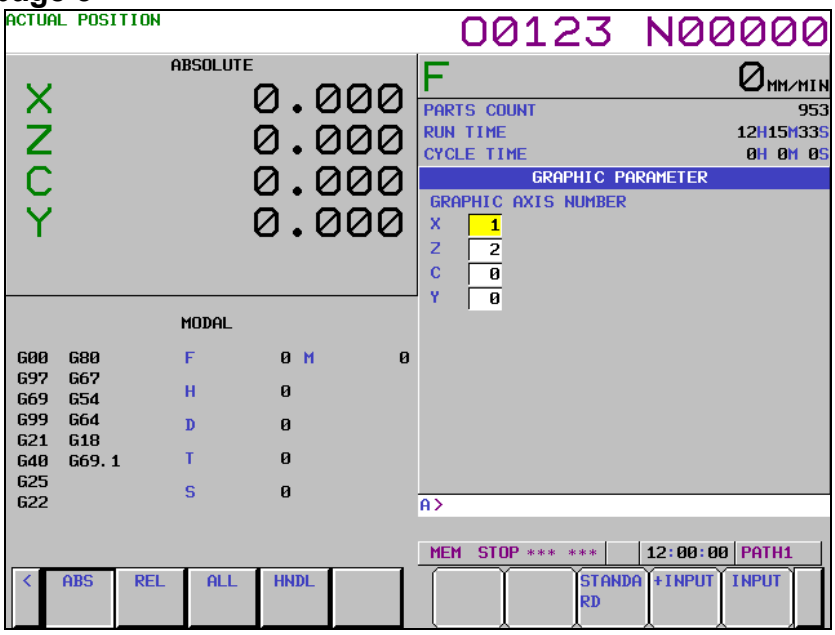


Fig. 13.1 (h) Graphic parameter screen page 3

On graphic parameter screen page 3, coordinate axes to be used for drawing are set.

Graphic parameter setting

Explanation

For tool path drawing, a graphic coordinate system, tool path graphic colors, and graphic range need to be set on the graphic parameter screen.
The graphic parameters to be set on the graphic parameter screen are described below.
When a value is set for a parameter, the parameter value becomes immediately effective. If a tool path is already drawn, the tool path is cleared when new parameter values are set.

- Graphic coordinate system

Select a desired graphic coordinate system for tool path drawing then set the corresponding number.

M

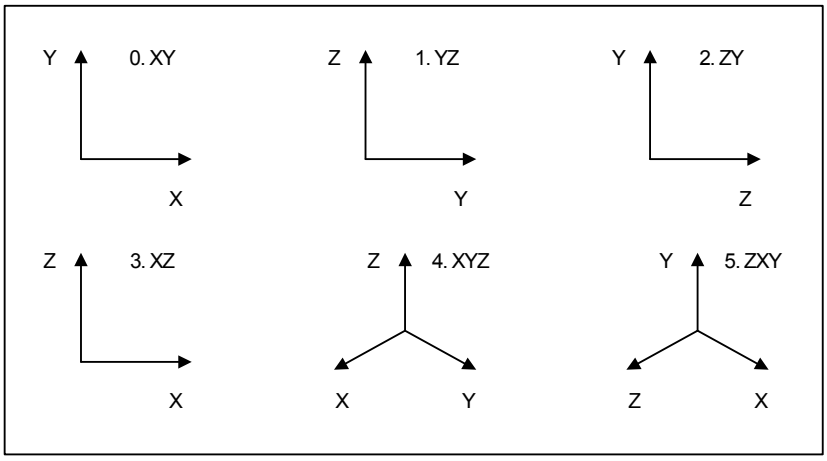


Fig. 13.1 (i) Graphic coordinate system (M series)

T

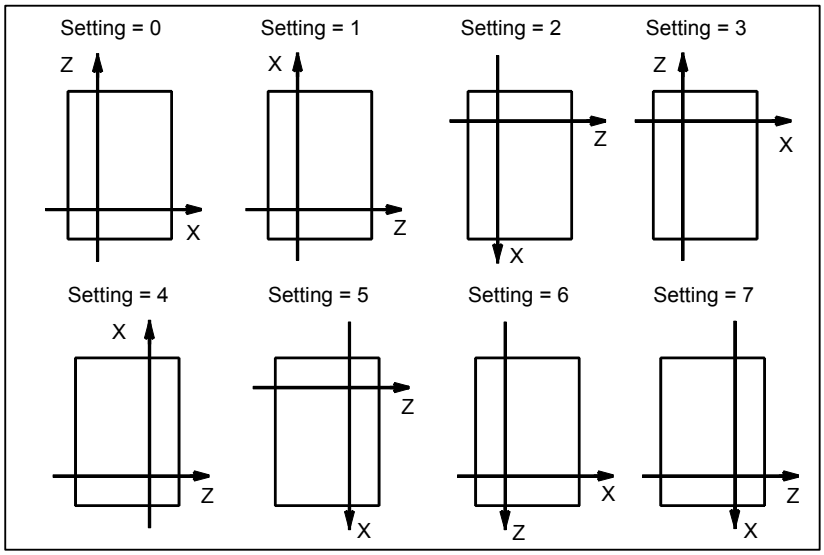


Fig. 13.1 (j) Graphic coordinate system (T series)

M

- Horizontal rotation angle

When a three-dimensional graphic coordinate system such as 4.XYZ or 5.ZXY is selected, the coordinate system can be rotated with the horizontal plane used as the rotation plane. Set a rotation angle from -360° to +360°.

The initial rotation angle can be set with parameter No. 24831, and the actual rotation angle is equal to the "initial rotation angle + specified rotation angle".

In Fig. 13.1 (k) below, the graphic coordinate system XYZ is converted to X"Y"Z" by the following settings:

Initial rotation angle: 180°
Horizontal rotation angle: 30°

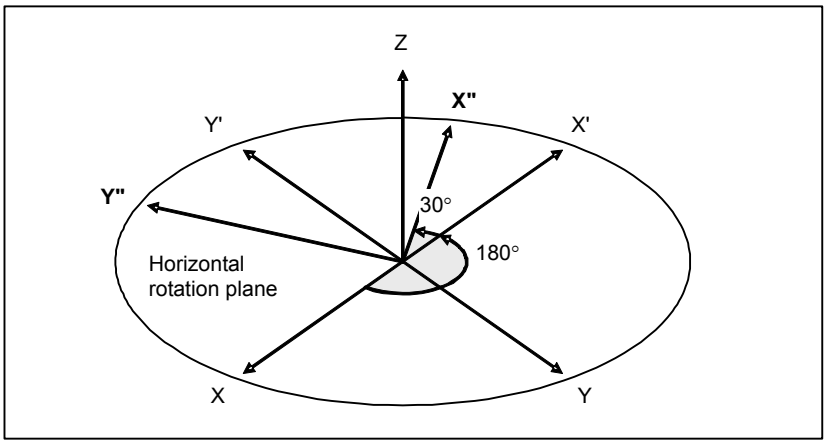


Fig. 13.1 (k) Coordinate system rotation in horizontal direction

- Vertical rotation angle

When a three-dimensional coordinate system such as 4.XYZ or 5.ZXY is selected, the coordinate system can be rotated with an axis on the horizontal plane specified as a vertical rotation axis. Set a rotation angle from -360° to +360°.

A vertical rotation axis can be set with the angle with the horizontal axis of the screen on the horizontal plane. This angle can be set with parameter No. 24832.

In Fig. 13.1 (l) below, the graphic coordinate system XYZ is converted to X'Y'Z' by the following settings:

Rotation angle of the vertical rotation axis: 65°
Vertical rotation angle: 20°

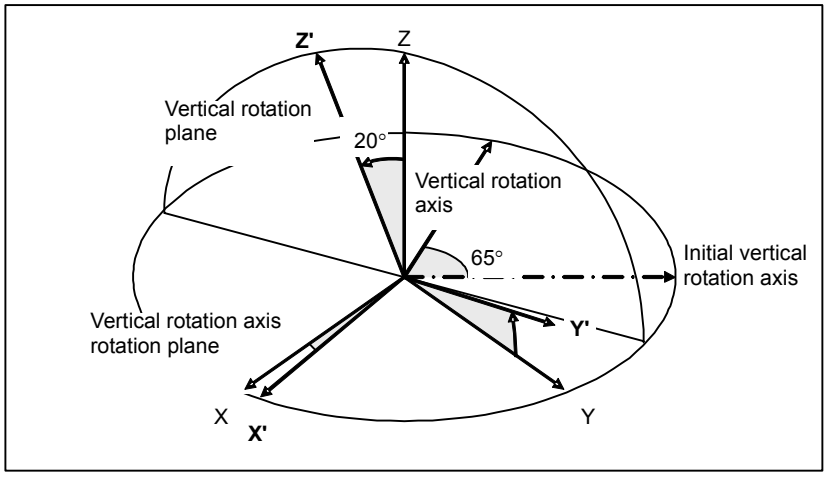


Fig. 13.1 (l) Coordinate system rotation in vertical direction

- Graphic color

Set a graphic color number for a tool path for each of cutting feed and rapid traverse.

1: Red 2: Green 3: Yellow 4: Blue
5: Purple 6: Sky blue 7: White

- Graphic range setting

Set a graphic range so that a tool path can be drawn in the tool path graphic area. Two methods are available:

1. Method that sets graphic center coordinates and a scale
2. Method that sets the maximum values and minimum values of a graphic range

Whether method 1 or method 2 is used is determined by which parameters were set most recently. A set graphic range is preserved even if the power is turned off.

1. Method that sets graphic center coordinates and a scale

Set the coordinates of the center of a tool path graphic area in the workpiece coordinate system. Then, set a scale used to contain the graphic range in the tool path graphic area.
Set a scale value from 0.01 to 100 (magnification).
As a smaller scale is set, a tool path can be drawn in a wider range. As a larger scale is set, a tool path is drawn by enlarging the tool path around the graphic center coordinates.

2. Method that sets the maximum values and minimum values of a graphic range

Set the maximum coordinates and minimum coordinates of a desired graphic range in the workpiece coordinate system. Drawing is performed so that the specified entire range is contained in the tool path graphic area.

From set maximum values and minimum values, graphic center coordinates and a sale are automatically calculated to update the graphic center coordinates and scale on the graphic parameter screen.

When a scale is automatically determined, the scale is clamped to within the range 0.01 to 100. Moreover, a maximum value must be greater than the corresponding minimum value.

NOTE
When the maximum values and minimum values of a graphic range are set, the graphic center coordinates and scale are automatically updated. However, when the graphic center coordinates and scale are changed, the maximum values and minimum values of the graphic range are not automatically updated.

- Automatic erasure

Before drawing is started, the previous drawing can be erased automatically.

1: Immediately before drawing is started, the previous drawing is erased automatically.
0: The previous drawing is not erased automatically.

- Graphic axis number

Set which controlled axis to be assigned to which graphic axis.
For each controlled axis, set one of the following graphic axis numbers:

First graphic axis:	1
Second graphic axis:	2
Third graphic axis:	3
Axis not used for drawing:	0

NOTE
1 When 0 is set for all controlled axes, it is assumed that 1, 2, and 3 are set sequentially for the first to the third controlled axes.
2 With the T series, a tool path is drawn along the first and second graphic axes. No tool path is drawn along the third graphic axis.

Operation for graphic parameter setting

Operation

- Moving the cursor

The cursor can be moved to a desired parameter by the page key or and the cursor key , , , or .
With the cursor keys, however, you cannot move from page 1 or 2 to page 3.

- Input of settings (absolute input)

Method 1

(1) Key in a value to be set.
(2) Press the [INPUT] soft key.

Method 2

(1) Key in a value to be set.
(2) Press the key.

- Input of settings (incremental input)

Method 1

(1) Key in a value to be incremented to or decremented from the current setting.
(2) Press the [+INPUT] soft key.

Procedure for tool path drawing

Procedure

- Start of drawing

(1) Display the tool path graphic screen.
(2) Press the [START] soft key.
The state that enables the movement of the tool in automatic operation or manual operation to be drawn is set.
Afterwards, tool path drawing continues even if another screen is displayed.
(3) Start automatic operation or manual operation.

- End of drawing

(1) Display the tool path graphic screen.
(2) Press the [END] soft key.
Tool path drawing ends.

- Erasure of the drawing

(1) Press the [END] soft key to end drawing.
(2) Press the [CLEAR] soft key. The tool path drawn so far is erased.

NOTE
1 Set the machine lock state to perform drawing only without moving the tool.
2 When the feedrate is high, the tool path may not be drawn correctly. In such a case, decrease the feedrate by performing, for example, a dry run.

Enlarged/reduced display

On the tool path graphic screen, you can move the center position of the tool path drawing or enlarge the tool path drawing while viewing the drawn tool path.
If any of these operations is executed, the tool path already drawn is cleared.

- Procedure for changing the graphic range by setting a graphic center and magnification

The center position of drawing can be moved. At the same time, the scale can also be changed. So, the tool path can be enlarged or reduced at a desired new center position.

(1) Press the [SCALE] soft key then the [CENTER] soft key.
A yellow cursor appears at the center of the screen, and the soft key display is changed.
(2) Move the yellow cursor to a new graphic center position by using the cursor key [↑], [↓], [←], or [→].
(3) When changing the scale, key in a value from 0.01 to 100 (magnification) then press the [INPUT] soft key. An input value is displayed at "SCALE" in the lower-right corner of the screen.
When you press the [+INPUT] soft key, the current magnification is incremented by an input value.
(4) Press the [EXEC] soft key to end the operation.
After this step, the setting for graphic movement is effective to enable drawing with the new setting.

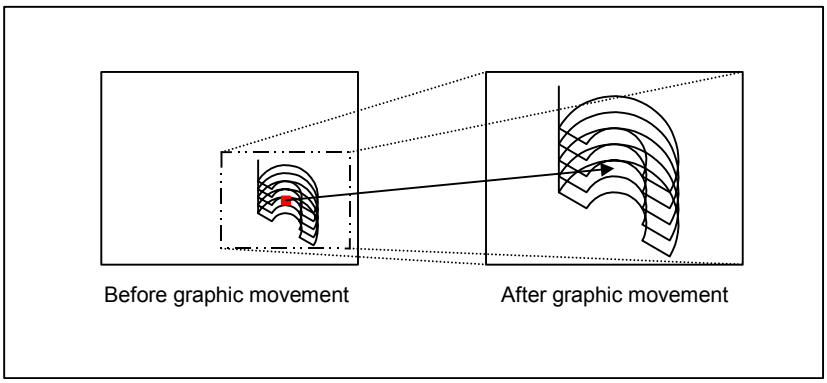


Fig. 13.1 (m) Graphic movement (magnification = 2.00)

- Procedure for changing the graphic range with a rectangle

A tool path can be drawn by enlarging a specified rectangular area.

(1) Press the [SCALE] soft key then the [RECTANGLE] soft key.
Two cursors, one in red and the other in yellow, appear at the center of the screen, and the soft key display is changed.

(2) Move the yellow cursor by using the cursor key [↑], [↓], [←], or [→]. The cursor to be moved can be switched by pressing the [CURSOR UP/DOWN] soft key.
Move the two cursors to the diagonal points of a new rectangular graphic range. A tool path is drawn next time so that the drawn tool path is contained in this rectangular range.

(3) Press the [EXEC] soft key to end the operation.
After this step, the setting of graphic enlargement is effective to enable drawing with the new setting.

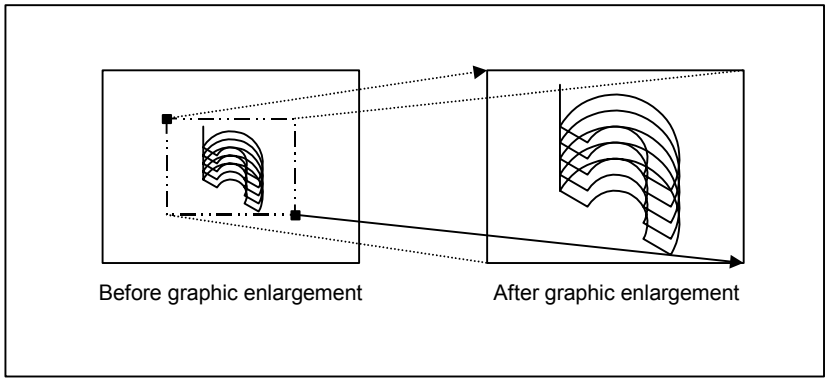


Fig. 13.1 (n) Graphic enlargement

NOTE
1 To stop an enlargement/reduction operation, press the [CANCEL] soft key.
2 Even if you perform an enlargement/reduction operation, the tool path already drawn on the screen is neither moved nor enlarged. The setting for enlargement/reduction becomes effective starting with tool path drawing after you press the [EXEC] soft key.

13.2 DYNAMIC GRAPHIC DISPLAY

Overview

The dynamic graphic display function has two features:

- Path Drawing
  The path of coordinates specified in a program is drawn on the screen.
  By displaying a travel path on the screen, the path can be checked easily before performing machining actually.

- Animation
  The figure of a workpiece to be machined by a programmed tool movement is drawn.
  By drawing the three-dimensional figure of a workpiece to be machined in an animation-like way, the intermediate machining process and final figure can be grasped easily.

When machining is performed according to a program, this function can draw a tool path with another program.
This function performs drawing much faster than the graphic display function based on automatic operation, so that a program check can be made more swiftly.
This function is distinguished from automatic program operation as indicated by the following terminology:

Automatic operation	Operation performed for actual machining
Background operation	Virtual operation performed for drawing

13.2.1 Path Drawing

Overview

The following tool path drawing screens are used to make various settings and execute drawing:

- GRAPHIC PARAMETER (DYNAMIC GRAPHIC) screen
  This screen is used to set data needed for tool path drawing.

- PATH GRAPHIC screen
  This screen is used for tool path drawing.

- PATH GRAPHIC (TOOL POSITION) screen
  This screen enables the position of the tool during operation to be checked on the drawn tool path.

13.2.1.1 GRAPHIC PARAMETER (DYNAMIC GRAPHIC) screen

This screen is used to set graphic parameters needed for tool path drawing.
Data set using this screen is made valid by displaying the PATH GRAPHIC screen or executing drawing. If a tool path is already drawn, the path is erased.
Graphic parameter data once set is preserved even if the power is turned off.

Graphic Parameter (Dynamic Graphic) Screen Procedure

Procedure

1 Press the function key

(or

when a small MDI unit is used) to display the GRAPHIC PARAMETER (DYNAMIC GRAPHIC) screen.

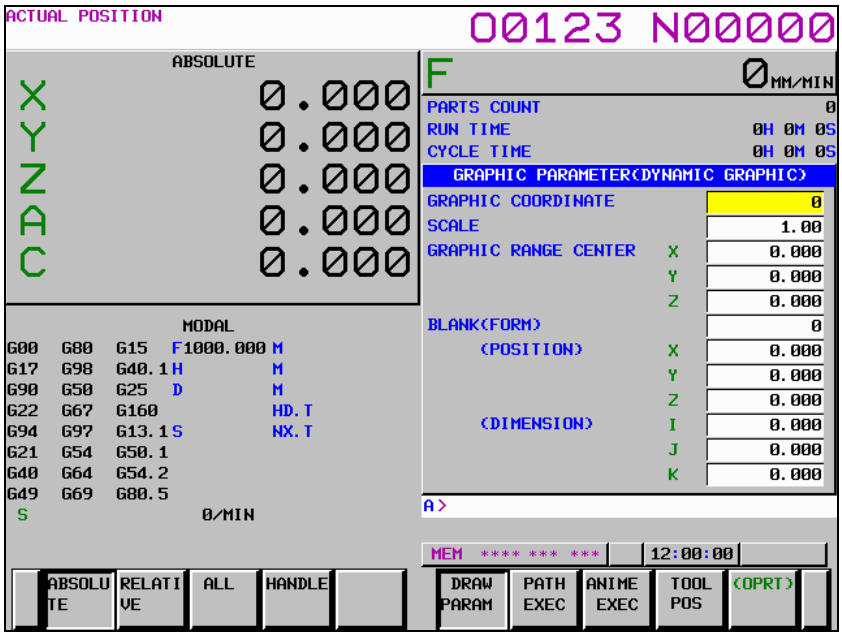


Fig. 13.2.1.1 (a) GRAPHIC PARAMETER screen (first page)

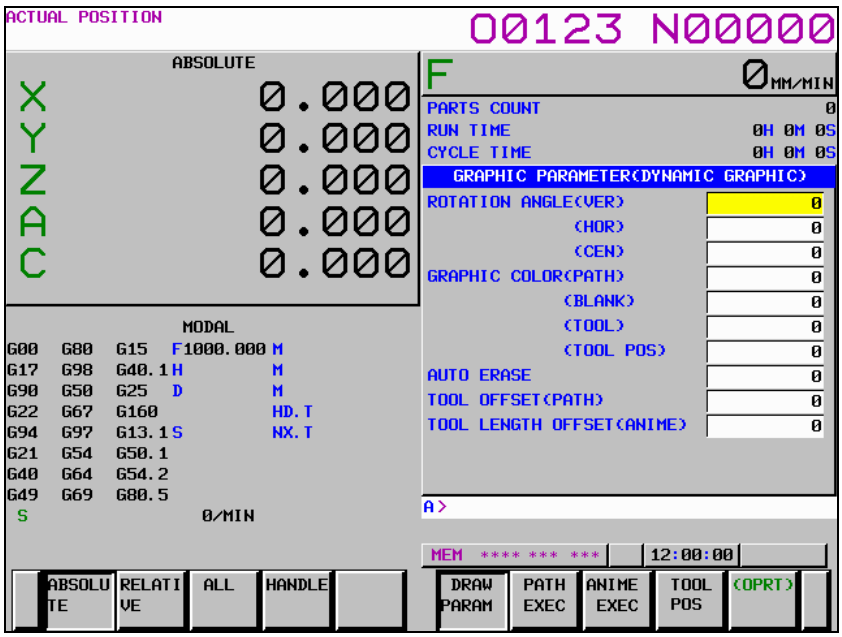


Fig. 13.2.1.1 (b) GRAPHIC PARAMETER screen (second page)

2 Two screens are used for the GRAPHIC PARAMETER screen. Use the MDI page keys to switch between the screens for display of a desired setting item.

3 Move the cursor to a desired setting item with the MDI cursor keys then key in a desired value with the numeric keys.

4 For direct input of the value set in step 3, press the INPUT key or the [INPUT] soft key.
For incremental input of the value set in step 3, press the [+INPUT] soft key.

For each setting item, see the explanation.

Explanation

The setting items on the GRAPHIC PARAMETER screen are described below.

- Graphic coordinate system

Select a graphic coordinate system for drawing from the following and set its number.

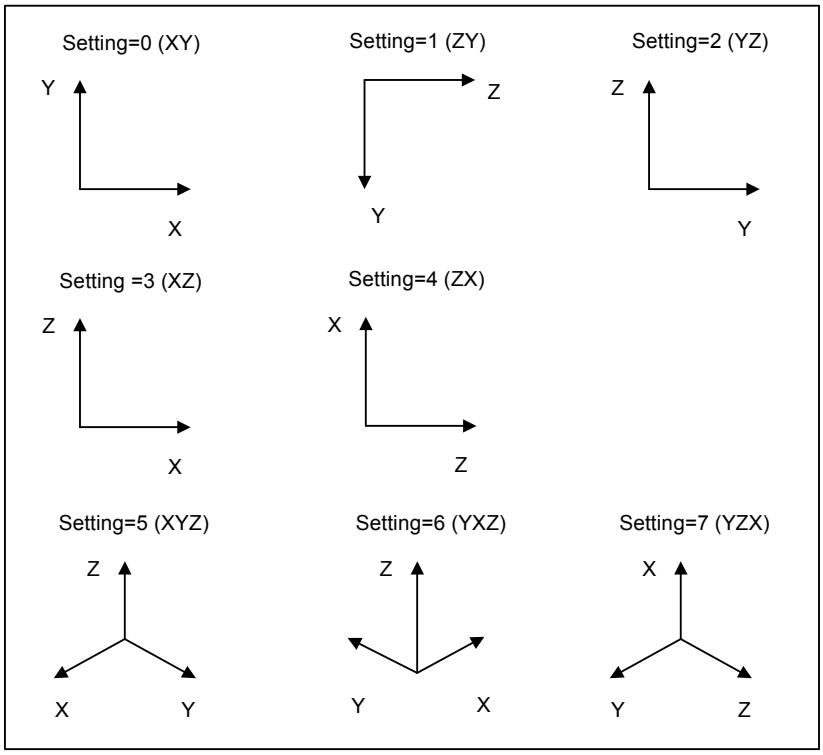


Fig. 13.2.1.1 (c) Graphic coordinate system

- Scale

Set the scale factor for drawing in the range of 0.01 to 100.00 (times). With a small scale factor, it is possible to draw within a wide range. With a large scale factor, it is possible to draw in the vicinity of the graphic center being enlarged.

- Graphic range center

To specify the center coordinates of a graphic range, set a coordinate on each axis in the workpiece coordinate system of the program executed for drawing.

NOTE
Set a coordinate on each axis in the machine coordinate system in case of bit 3 (BGM) of parameter No.11329=1.

- Blank figure

With a drawing program, set the figure, position, and dimensions of a blank to be machined.
The graphic range where the blank figure is included in the drawing area is automatically decided by the value set here.

NOTE
1 With a lathe system, no rectangular parallelepiped can be set.
2 With a lathe system, the Y-axis, one of the basic three axes, may not exist. In such a case, the Y-axis for the reference position of a blank need not be set.
3 When a figure or dimension is set, a scale value and screen center coordinates are automatically set as the initial graphic range values so that the blank figure is located within the graphic range on the screen. Until a modification is made to the blank figure, dimension or the setting of the graphic coordinate system, the graphic range is determined by the set scale and screen center coordinates.

Figure

Select a type of blank figure from the following and set the corresponding value:

Setting	Figure
0	Column or cylinder (parallel with the Z-axis)
1	Rectangular parallelepiped

Position

Set the reference position of a blank with coordinates (X,Y,Z) in the workpiece coordinate system.

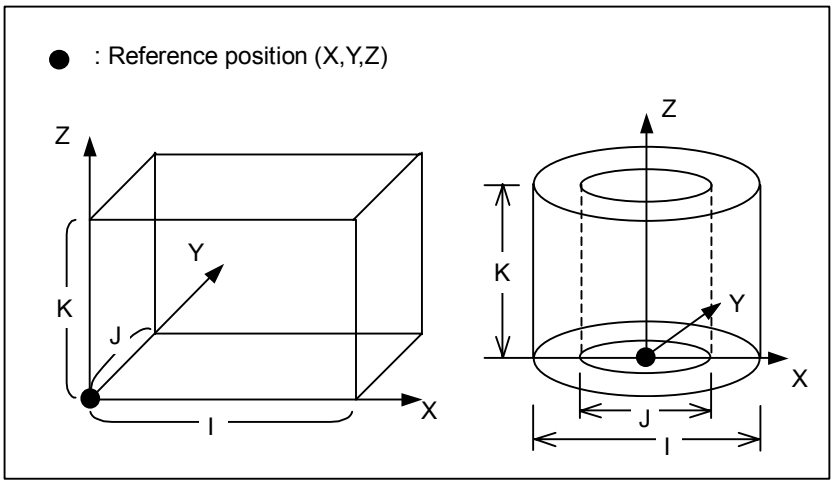
NOTE
Set the reference position of a blank with coordinates (X,Y,Z) in the machine coordinate system in case of bit 3 (BGM) of parameter No.11329=1.

Dimensions

Set the dimensions of each type of blank figure as indicated below.

Type of blank figure	Dimension I	Dimension J	Dimension K
Rectangular parallelepiped	Length in X-axis direction	Length in Y-axis direction	Length in Z-axis direction
Column	Diameter of column	0	Length of column
Cylinder	Diameter of outer circle of cylinder	Diameter of inner circle of cylinder	Length of cylinder

The relationship between the position and dimensions of a blank is shown below.

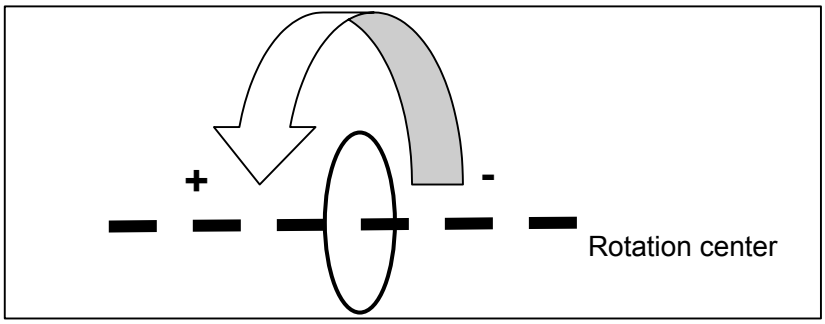


- Rotation angle

Set a rotating angle of the graphic coordinate system that centers on the graphic range center. The rotating angle is a range of -360°-+360°. Set a rotating angle as a reference position (position of the rotating angle 0°) on the indicated direction of each graphic coordinate system.

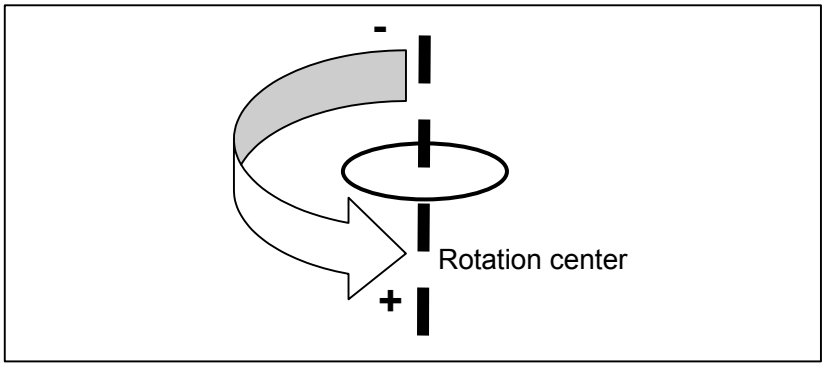
Vertical plane rotation angle

Set a rotating angle at the horizontal direction center in front of the screen.
The rotation direction is as follows.



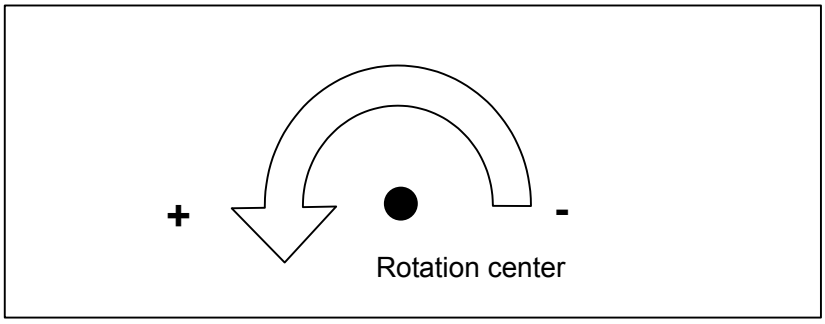
Horizontal plane rotation angle

Set a rotating angle at the vertical direction center in front of the screen.
The rotation direction is as follows.



Screen center rotation angle

Set a rotating angle at the vertical direction center of the screen plane.
The rotation direction is as follows.



- Graphic color

Set colors to be used for tool path drawing.
The colors that can be set are indicated below together with their setting values:

Graphic color	White	Red	Green	Yellow	Blue	Purple	Light blue	White
Setting value	0	1	2	3	4	5	6	7

Path

Set colors to be used for a drawn tool path.

Tool position

Set a graphic cursor color to be used for PATH GRAPHIC (TOOL POSITION) screen.

- Auto erase

When drawing is started with the [START] soft key from a state where drawing is not being executed or is not stopped temporarily, the previously drawn path can be erased.
Setting 0: The previously drawn path is not erased.
1: The previously drawn path is erased.

- Tool offset（Path）

For tool path drawing, whether to enable or disable the tool offset function (tool length compensation, cutter/tool-noise radius compensation) can be selected.
Setting 0: The tool offset function is enabled for drawing.
1: The tool offset function is disabled for drawing.

13.2.1.2 PATH GRAPHIC screen

The PATH GRAPHIC screen is used to draw a tool path. The following operations can be performed:

- Starting/ending tool path drawing
- Rewind of a drawing target program
- Erasing a drawn tool path

The screen is composed of the following items:

(1) Drawing area
(2) Status display of Background operation
(3) Program name and sequence number for drawing execution
(4) Current coordinates (absolute coordinates)
(5) Feedrate and M/S/T/D code instruction information
(6) Graphic coordinate system

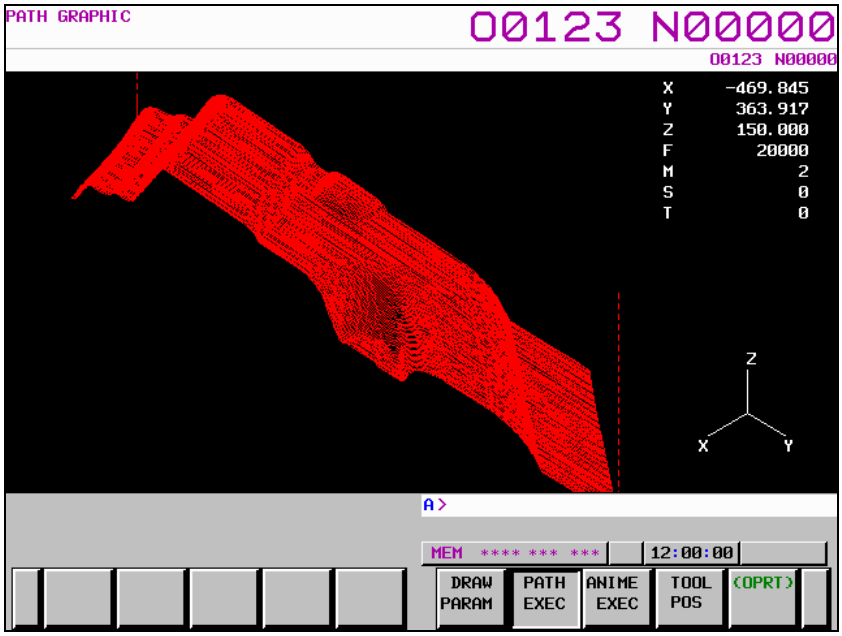


Fig. 13.2.1.2 (a) PATH GRAPHIC screen

Path Graphic Screen Procedure

Procedure

1 Press the function key

(or

when a small MDI unit is used) to display the GRAPHIC PARAMETER (DYNAMIC GRAPHIC) screen.

2 Press the [PATH EXEC] soft key. The PATH GRAPHIC screen is displayed.

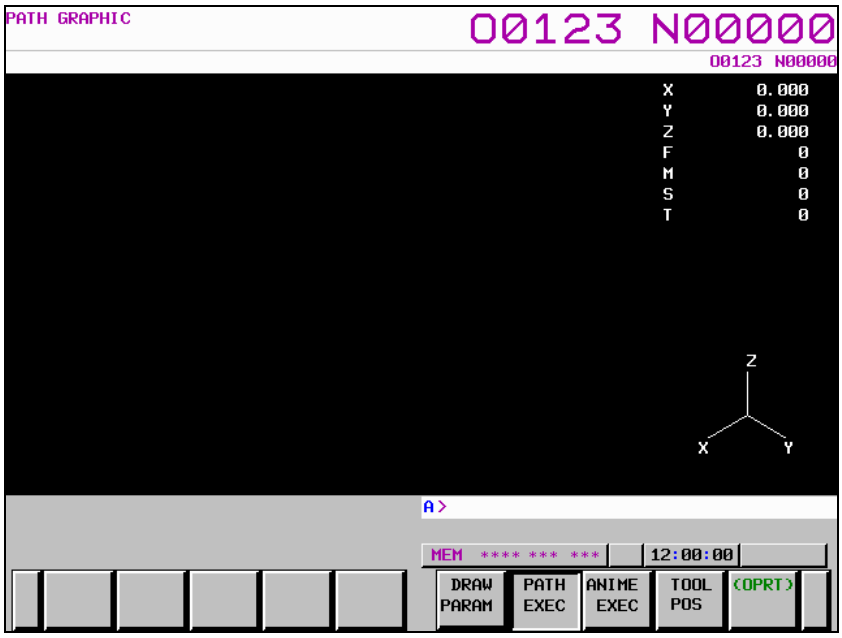


Fig. 13.2.1.2 (b) PATH GRAPHIC screen

3 Press the [(OPRT)] soft key. The soft keys for tool path drawing are displayed.

Fig. 13.2.1.2 (c) PATH GRAPHIC screen (operation)

4 Press the continuous menu key to display the soft keys for enlarging/reducing/moving the graphic range.

Fig. 13.2.1.2 (d) PATH GRAPHIC screen (enlarging/reducing/moving the graphic range)

5 Press the [COORDINATES] soft key to display the soft keys for changing the graphic coordinate system.

Fig. 13.2.1.2 (e) PATH GRAPHIC screen (changing the graphic coordinate system)

6 Press the [ROTATION] soft key to display the soft keys for rotating the graphic coordinate system.

Fig. 13.2.1.2 (f) PATH GRAPHIC screen (rotating the graphic coordinate system)

For the operation of each soft key, see the explanation.

Explanation

- Graphic program selection

Press the [PRGRAM SELECT] soft key to select a drawing target program.
Pressing the [PRGRAM SELECT] soft key displays the program list screen.
On the program list screen, move the cursor to a desired program then press the [DRAW SELECT] soft key.

Fig. 13.2.1.2 (g) Program list screen ([DRAW SELECT] soft key)

Pressing the [DRAW SELECT] soft key selects a drawing target program and switches the screen display to the PATH GRAPHIC screen.

The file whose name is prefixed by a "#" mark on the program list screen is selected as the drawing target program. And the mark of "+" is displayed in the file that has been selected as a program for drawing in other path systems.

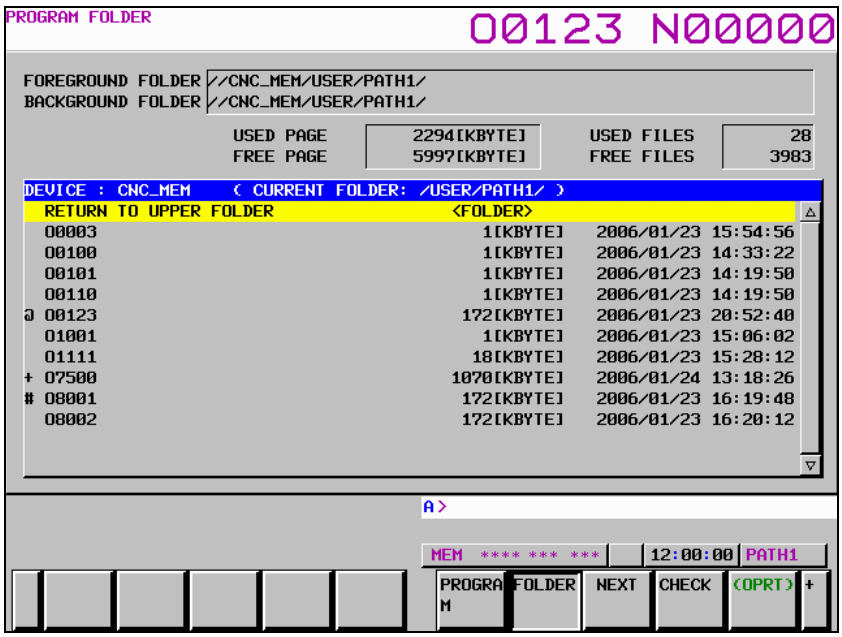


Fig. 13.2.1.2 (h) Program list screen (drawing target program selection state)

NOTE
The file that can be selected as the drawing target program is only a file that can be selected as the main program.

- Rewind of a drawing target program

If the execution of drawing of a selected program has ended or is stopped halfway, press the [REWIND] soft key to restart drawing from the beginning of the program.

- Starting / Stopping drawing

To draw the tool path of a program selected for drawing, press one of the following soft keys displayed by step 3 mentioned above:

- [START] soft key
  This soft key starts drawing from the head of the program.
- [1BLOCK] soft key
  This soft key executes the program for drawing and stops temporarily on a block-by-block basis as with ordinary single block stop operation.

If a program is executed with a soft key above, the soft key display is changed to the following:

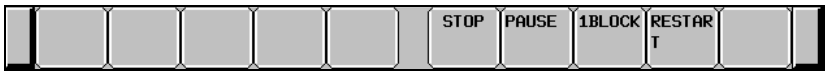


Fig. 13.2.1.2 (i) Soft keys displayed during drawing execution

The operations of these soft keys are as follows:

- [STOP] soft key
  This soft key terminates the execution of the drawing target program to stop drawing.
- [PAUSE] soft key
  This soft key temporarily stops the execution of the drawing target program to stop drawing temporarily.
- [1BLOCK] soft key
  This soft key executes the program for drawing and stops drawing temporarily on a block-by-block basis as with ordinary single block stop operation.
- [RESTART] soft key
  If the [RESTART] soft key is pressed in the stop state set by the [PAUSE] soft key or the [1BLOCK] soft key, the execution of drawing can be restarted from the block where drawing is stopped.

NOTE
When the new path was drawn by the operation of starting drawing without erasing of the old path before the operation, it is impossible to redraw the old path by each operation of enlarging/reducing/moving the graphic range and changing/rotating the graphic coordinate system.

The state of drawing is indicated as follows:

DRAWING: Indicates that drawing is being executed.

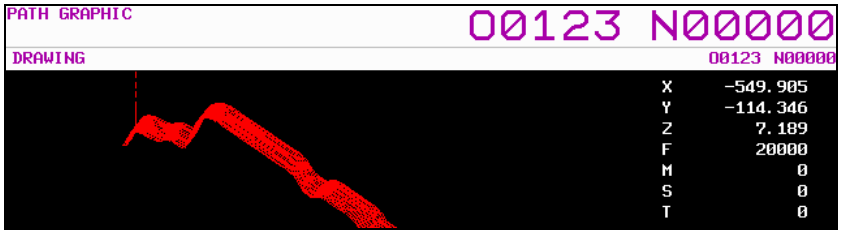


Fig. 13.2.1.2 (j) State indication during drawing

STOP: Indicates that drawing is temporarily stopped.

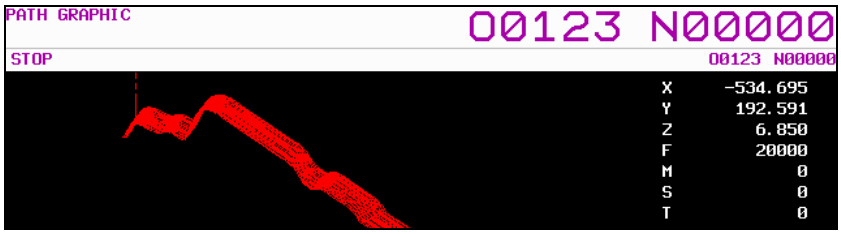


Fig. 13.2.1.2 (k) State indication during temporary stop

ALM: Indicates that occurring of alarm in the Background operation.

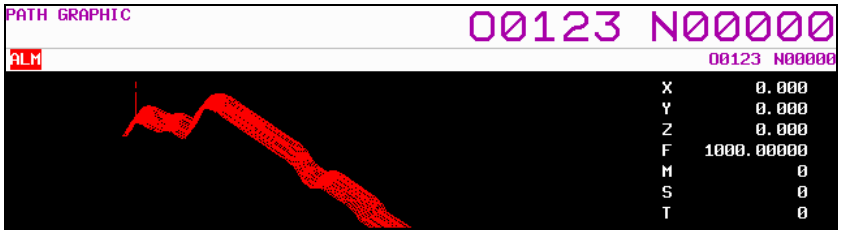


Fig. 13.2.1.2 (l) State indication during occurring of alarm

- End of drawing

When M02 or M30 is executed, the program executed for drawing terminates drawing.
Upon program termination, the soft key display returns to the soft keys (Fig. 13.2.1.2 (c)) displayed before drawing is started.

- Erasing a drawn tool path

Press the [ERASE] soft key to erase a drawn tool path.

NOTE
1 If the screen display is switched or the path is switched during tool path drawing, the background operation is stopped to end drawing.
2 A tool path once drawn is erased when the screen display is switched or the path is switched.

- Enlarging/reducing the graphic range

The following soft keys displayed by step 4 are used:

- [LARGE] soft key
  This soft key increases the scale to enlarge the graphic range.
- [SMALL] soft key
  This soft key decreases the scale to reduce the graphic range.
- [AUTO] soft key
  When the blank figure specified in a graphic parameter is input, this soft key automatically scales the graphic range so that the blank figure is contained within the graphic range.
  If no blank figure is input, this soft key has no effect.

NOTE
1 Set the unit of scale for one enlargement/reduction operation in parameter No. 14713.
2 An enlargement/reduction scale used here is set in the graphic parameter for scale.

- Moving the graphic range

The following soft keys displayed by step 4 are used:

- [←MOVE] soft key
  This soft key moves the graphic range to left.
- [MOVE→] soft key
  This soft key moves the graphic range to right.
- [↑MOVE] soft key
  This soft key moves the graphic range upward.
- [↓MOVE] soft key
  This soft key moves the graphic range downward.
- [CENTER] soft key
  This soft key returns the graphic range to the original position.

NOTE
1 Set the travel increment made by one horizontal move operation in parameter No. 14714.
2 Set the travel increment made by one vertical move operation in parameter No. 14715.
3 The graphic range modified here is not set in the graphic parameter for graphic range center.

- Changing the graphic coordinate system

The following soft keys displayed by step 5 are used.
A graphic coordinate system selected here is the same one as set in the graphic parameter for the graphic coordinate system.

- [XY] soft key
  This soft key selects the graphic coordinate system of XY (with a setting of 0).
- [ZY] soft key
  This soft key selects the graphic coordinate system of ZY (with a setting of 1).
- [YZ] soft key
  This soft key selects the graphic coordinate system of YZ (with a setting of 2).
- [XZ] soft key
  This soft key selects the graphic coordinate system of XZ (with a setting of 3).
- [ZX] soft key
  This soft key selects the graphic coordinate system of ZX (with a setting of 4).
- [XYZ] soft key
  This soft key selects the graphic coordinate system of XYZ (with a setting of 5).
- [YXZ] soft key
  This soft key selects the graphic coordinate system of YXZ (with a setting of 6).
- [YZX] soft key
  This soft key selects the graphic coordinate system of YZX (with a setting of 7).
- [OK] soft key
  This soft key changes the current graphic coordinate system to the graphic coordinate system selected by one of the soft keys above.
- [CANCEL] soft key
  This soft key cancels the graphic coordinate system selected by one of the soft keys above to return to the original graphic coordinate system.

NOTE
The graphic coordinate system selected here is set in the graphic parameter for the graphic coordinate system.

- Rotating the graphic coordinate system

The following soft keys displayed by step 6 are used.

- [↑] soft key
  This soft key rotates the graphic coordinate system upward.
- [↓] soft key
  This soft key rotates the graphic coordinate system downward.
- [←] soft key
  This soft key rotates the graphic coordinate system to left.
- [→] soft key
  This soft key rotates the graphic coordinate system to right.
- [CW] soft key
  This soft key rotates the graphic coordinate system clockwise.
- [CCW] soft key
  This soft key rotates the graphic coordinate system counterclockwise.
- [OK] soft key
  This soft key changes the rotation angle of the current graphic coordinate system to the one set by one of the soft keys above.
- [CANCEL] soft key
  This soft key cancels the rotation of the graphic coordinate system performed by one of the soft keys above to return to the original graphic coordinate system.

NOTE
1 Set the travel increment made by one rotation operation in parameter No. 14716.
2 The rotation angle of the graphic coordinate system set here is not set in the graphic parameter for rotation angle.

13.2.1.3 PATH GRAPHIC (TOOL POSITION) screen

The tool position during operation can be checked by displaying the cursor for indicating the tool position during operation on the tool path drawn on the PATH GRAPHIC (EXECUTION) screen.

The screen is composed of the following items:
(1) Drawing area
(2) Current coordinates (absolute coordinates)
(3) Feedrate and M/S/T/D code instruction information
(4) Graphic coordinate system
(5) Cursor to show tool position

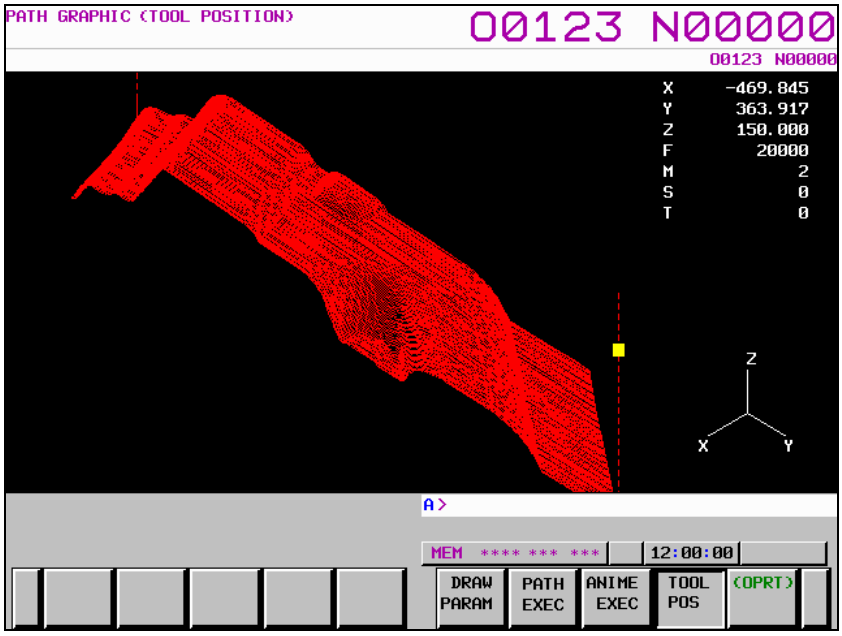


Fig. 13.2.1.3 (a) PATH GRAPHIC (TOOL POSITION) screen

Path Graphic (Tool Position) Screen Procedure

Procedure

1 Press the function key

(or

when a small MDI unit is used) to display the PATH GRAPHIC (PARAMETER) screen.

2 Press the [TOOL POS] soft key.
The screen display changes to the PATH GRAPHIC (TOOL POSITION) screen and the cursor indicating the tool position is displayed.

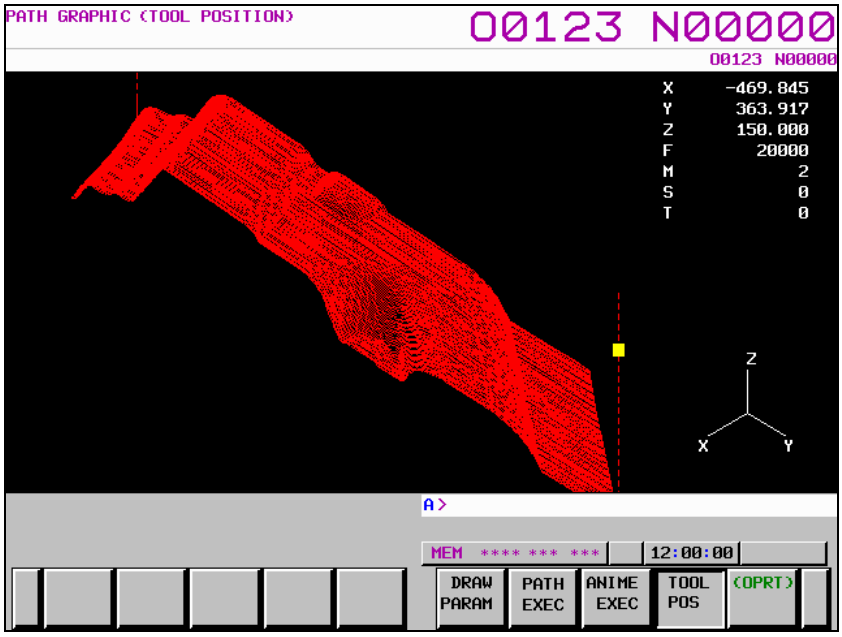


Fig. 13.2.1.3 (b) PATH GRAPHIC (TOOL POSITION) screen

For the method of checking the current tool position, see the explanation.
Pressing a soft key other than the [TOOL POS] soft key displays the corresponding screen.

Explanation

Use the following procedure to check the tool position during operation on the PATH GRAPHIC (TOOL POSITION) screen:

(1) Select a drawing target program for operation.
(2) Draw the tool path of the selected program on the PATH GRAPHIC screen.
(3) After the end of drawing, switch the screen display to the PATH GRAPHIC (TOOL POSITION) screen.
(4) Start automatic operation with the program executed for tool path drawing.

After automatic operation is started, the blinking cursor moves along the drawn tool path as the tool moves. The cursor blinking rate is fast when the tool is moving, and the cursor blinking rate is slow when the tool is stopped.

The following items displayed on the screen are provided for the program under automatic operation:

- Current coordinates
- Feedrate and M/S/T/D code specification information

NOTE
1 A tool path drawn by setting the tool offset parameter to 1 (to disable the tool offset function) is different from the actual tool path. In this case, the cursor indicating the tool position may not move along the drawn tool path.
2 The tool path cannot be drawn correctly if a command not supporting drawing or a command that specifies operation different from operation performed during drawing is specified or if an operation dependent on the operation state or setting on the machine side is performed. So, the tool path in actual operation may differ from a drawn tool path. In this case, the cursor indicating the tool position does not move along the drawn tool path.
3 When the graphic parameter of the graphic coordinate system, scale, graphic range center, blank figure / position / dimensions and rotation angle is changed, the drawn tool path drawn is erased.
Therefore, please draw the tool path again on the PATH GRAPHIC screen to display the tool position by setting the changed graphic parameter.

13.2.2 Animation

Overview

For animation drawing, make necessary settings and perform operations for drawing execution on the following screens:

- GRAPHIC PARAMETER (DYNAMIC GRAPHIC) screen
  On this screen, data required to execute animation drawing can be set.
- ANIMATION GRAPHIC screen
  This screen is used for animation drawing.
  On this screen, the graphic range can be reduced/enlarged, and the graphic coordinate system can be rotated.

13.2.2.1 GRAPHIC PARAMETER (DYNAMIC GRAPHIC) SCREEN

This screen is used to set graphic parameters needed for animation drawing.
Data set using this screen is made valid by displaying the ANIMATION GRAPHIC screen or executing drawing.
Graphic parameter data once set is preserved even if the power is turned off.

Graphic Parameter (Dynamic Graphic) Screen Procedure

Procedure

1 Press the function key

(or

when a small MDI unit is used) to display the GRAPHIC PARAMETER (DYNAMIC GRAPHIC) screen.

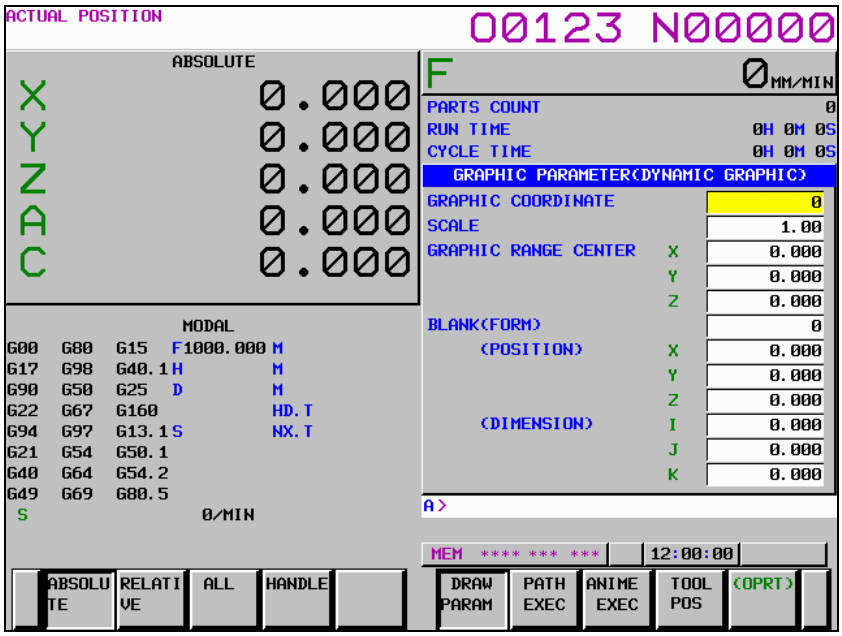


Fig. 13.2.2.1 (a) GRAPHIC PARAMETER screen (first page)

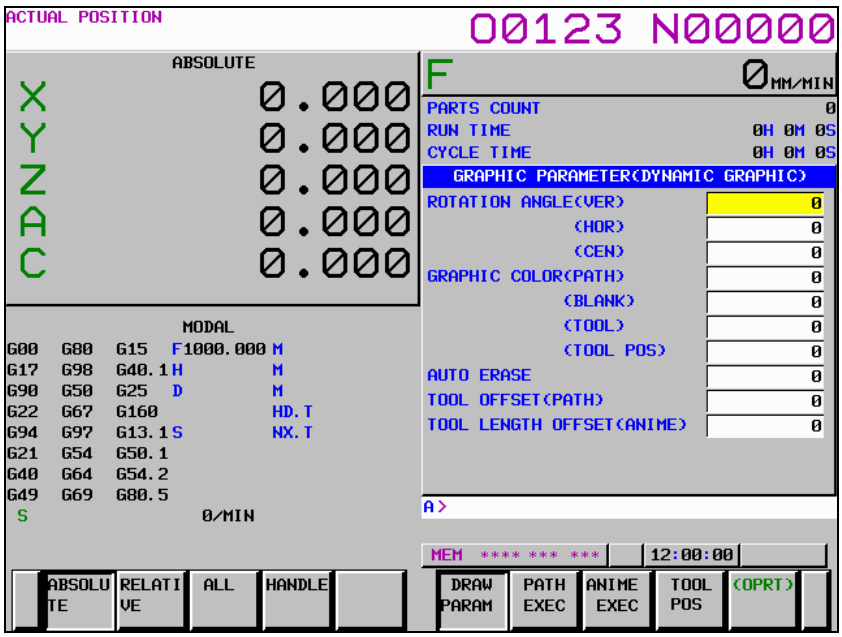


Fig. 13.2.2.1 (b) GRAPHIC PARAMETER screen (second page)

2 Two screens are used for the GRAPHIC PARAMETER screen. Use the MDI page keys to switch between the screens for display of a desired setting item.

3 Move the cursor to a desired setting item with the MDI cursor keys then key in a desired value with the numeric keys.

4 For direct input of the value set in step 3, press the

key or the [INPUT] soft key.
For incremental input of the value set in step 3, press the [+INPUT] soft key.

For each setting item, see the explanation.

Explanation

The setting items on the GRAPHIC PARAMETER screen are described below.
However, the graphic parameters listed below are shared for tool path drawing. So, refer to the explanation of the GRAPHIC PARAMETER screen for tool path drawing.

- Graphic coordinate system
- Scale
- Graphic range center
- Rotation angles (vertical plane, horizontal plane, screen center)

- Blank figure

With a drawing program, set the figure, position, and dimensions of a blank to be machined.

NOTE
1 With a lathe system, no rectangular parallelepiped can be set.
2 With a lathe system, the Y-axis, one of the basic three axes, may not exist. In such a case, the Y-axis for the reference position of a blank need not be set.
3 When a figure or dimension is set, a scale value and screen center coordinates are automatically set as the initial graphic range values so that the blank figure is located within the graphic range on the screen. Until a modification is made to the blank figure, dimension or the setting of the graphic coordinate system, the graphic range is determined by the set scale and screen center coordinates.

Figure

Select a type of blank figure from the following and set the corresponding value:

Setting	Figure
0	Column or cylinder (parallel with the Z-axis)
1	Rectangular parallelepiped

Position

Set the reference position of a blank with coordinates (X,Y,Z) in the workpiece coordinate system.

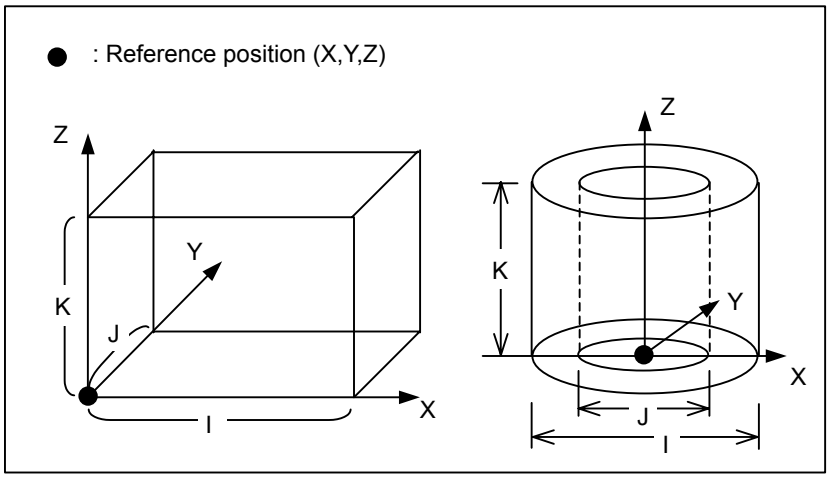
NOTE
Set the reference position of a blank with coordinates (X,Y,Z) in the machine coordinate system in case of bit 3 (BGM) of parameter No.11329=1.

Dimensions

Set the dimensions of each type of blank figure as indicated below.

Type of blank figure	Dimension I	Dimension J	Dimension K
Rectangular parallelepiped	Length in X-axis direction	Length in Y-axis direction	Length in Z-axis direction
Column	Diameter of column	0	Length of column
Cylinder	Diameter of outer circle of cylinder	Diameter of inner circle of cylinder	Length of cylinder

The relationship between the position and dimensions of a blank is shown below.



- Graphic color

Set colors to be used for animation drawing.
The colors that can be set are indicated below together with their setting values:

Graphic color	White	Red	Green	Yellow	Blue	Purple	Light blue	White
Setting value	0	1	2	3	4	5	6	7

Blank

Set the color of a blank to be drawn in animation drawing.

Tool

Set the color of a tool to be drawn in animation drawing.

- Tool length offset (Anime)

For animation drawing, whether to enable or disable the tool length offset can be selected.

Setting 0: The tool length offset is disabled for drawing.
1: The tool length offset is enabled for drawing.

NOTE
In animation drawing, the cutter/tool-noise radius compensation is always enabled for drawing.

13.2.2.2 ANIMATION GRAPHIC screen

The ANIMATION GRAPHIC screen is used to draw a animation. The following operations can be performed:

- Starting/ending animation drawing
- Rewind of a drawing target program
- Initializing a blank
- Enlarging/reducing/moving the graphic range
- Rotating the graphic coordinate system

The screen configuration is as follows:

(1) Drawing area
(2) Status display of Background operation
(3) Program name and sequence number for drawing execution
(4) Current coordinates (absolute coordinates)
(5) Feedrate and M/S/T/D code instruction information
(6) Graphic coordinate system
(7) Material position, Material dimension, Scale, Graphic range center of graphic Parameter
(8) Tool name

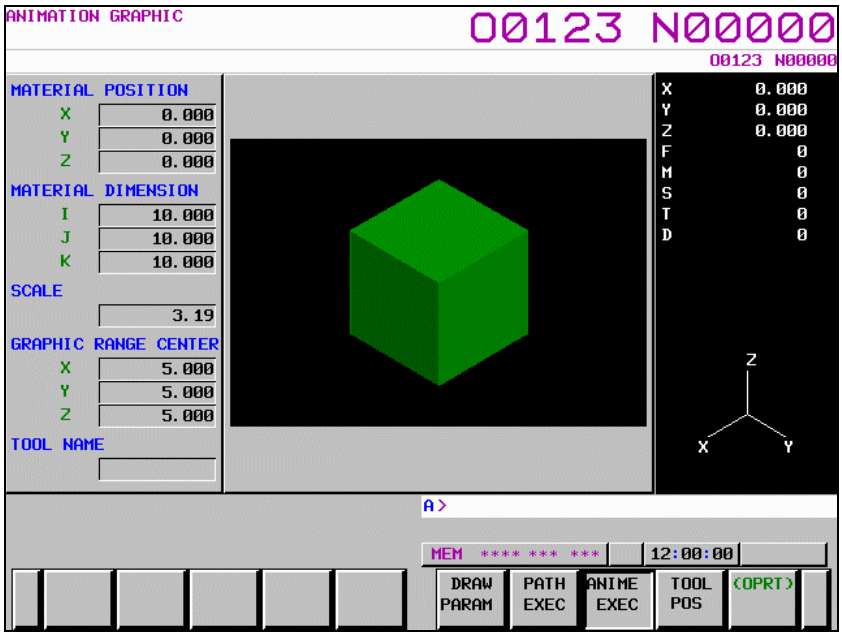


Fig. 13.2.2.2 (a) ANIMATION GRAPHIC screen

Animation Graphic Screen Procedure

Procedure

1 Press the function key

(or

when a small MDI unit is used) to display the GRAPHIC PARAMETER (DYNAMIC GRAPHIC) screen.

2 Press the [ANIME EXEC] soft key. The ANIMATION GRAPHIC screen is displayed.

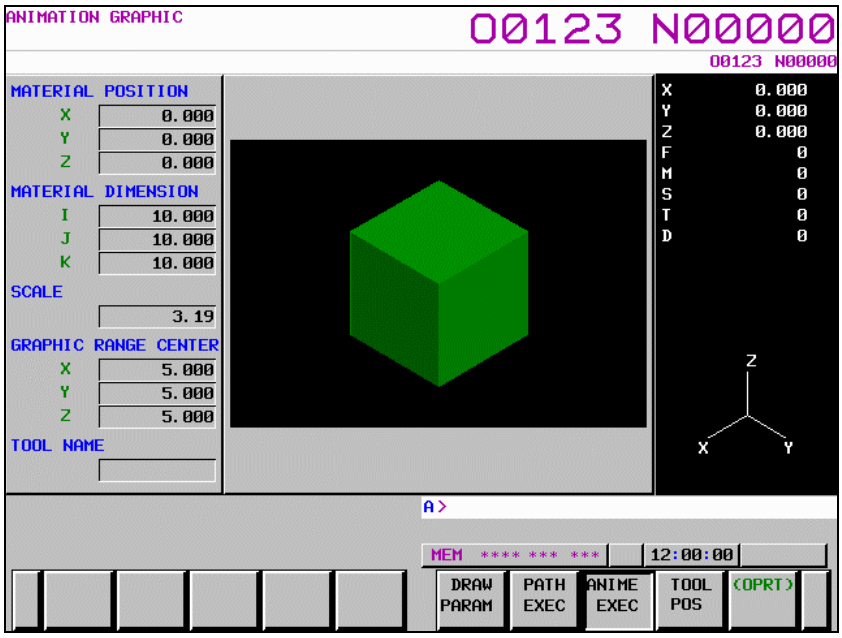


Fig. 13.2.2.2 (b) ANIMATION GRAPHIC screen

3 Press the [(OPRT)] soft key. The soft keys for tool path drawing are displayed.

Fig. 13.2.2.2 (c) ANIMATION GRAPHIC screen (operation)

4 Press the continuous menu key to display the soft keys for enlarging/reducing/moving the graphic range.

Fig. 13.2.2.2 (d) ANIMATION GRAPHIC screen (enlarging/reducing/moving the graphic range)

5 Press the [COORDINATES] soft key to display the soft keys for changing the graphic coordinate system.

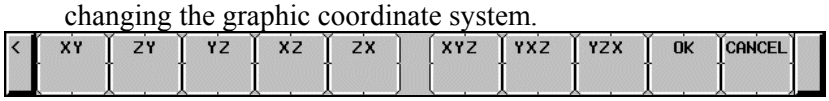


Fig. 13.2.2.2 (e) ANIMATION GRAPHIC screen (changing the graphic coordinate system)

6 Press the [ROTATION] soft key to display the soft keys for rotating the graphic coordinate system.

Fig. 13.2.2.2 (f) ANIMATION GRAPHIC screen (rotating the graphic coordinate system)

For the operation of each soft key, see the explanation.

Explanation

The operations listed below are the same operations as for the tool path drawing screen. See the explanation of the tool path drawing screen.

- Graphic program selection
- Rewind of a drawing target program
- Starting / Stopping of drawing
- End of drawing

- Initializing a blank

Press the [INIT.] soft key to initialize and return a drawn machining profile to the original blank figure.

NOTE
1 Blank initialization can also be performed by any of the following operations:
  - Start of drawing
  - Changing of the graphic coordinate system and graphic range by performing enlargement/reduction/movement/rotation operations
  - Change of screen
2 A machining profile once drawn for a blank is erased when the screen or the path is switched.

- Enlarging/reducing the graphic range

The following soft keys displayed by step 4 are used:

- [LARGE] soft key
  This soft key increases the scale to enlarge the graphic range.
- [SMALL] soft key
  This soft key decreases the scale to reduce the graphic range.
- [AUTO] soft key
  When the blank figure specified in a graphic parameter is input, this soft key automatically scales the graphic range so that the blank figure is contained within the graphic range.
  If no blank figure is input, this soft key has no effect.

NOTE
1 Set the unit of scale for one enlargement/reduction operation in parameter No. 14713.
2 An enlargement/reduction scale used here is set in the graphic parameter for scale.

- Moving the graphic range

The following soft keys displayed by step 4 are used:

- [←MOVE] soft key
  This soft key moves the graphic range to left.
- [MOVE→] soft key
  This soft key moves the graphic range to right.
- [↑MOVE] soft key
  This soft key moves the graphic range upward.
- [↓MOVE] soft key
  This soft key moves the graphic range downward.
- [CENTER] soft key
  This soft key returns the graphic range to the original position.

NOTE
1 Set the travel increment made by one horizontal move operation in parameter No. 14714.
2 Set the travel increment made by one vertical move operation in parameter No. 14715.
3 The graphic range modified here is not set in the graphic parameter for graphic range center.

- Changing the graphic coordinate system

The following soft keys displayed by step 5 are used.
A graphic coordinate system selected here is the same one as set in the graphic parameter for the graphic coordinate system.

- [XY] soft key
  This soft key selects the graphic coordinate system of XY (with a setting of 0).
- [ZY] soft key
  This soft key selects the graphic coordinate system of ZY (with a setting of 1).
- [YZ] soft key
  This soft key selects the graphic coordinate system of YZ (with a setting of 2).
- [XZ] soft key
  This soft key selects the graphic coordinate system of XZ (with a setting of 3).
- [ZX] soft key
  This soft key selects the graphic coordinate system of ZX (with a setting of 4).
- [XYZ] soft key
  This soft key selects the graphic coordinate system of XYZ (with a setting of 5).

- [YXZ] soft key
  This soft key selects the graphic coordinate system of YXZ (with a setting of 6).
- [YZX] soft key
  This soft key selects the graphic coordinate system of YZX (with a setting of 7).
- [OK] soft key
  This soft key changes the current graphic coordinate system to the graphic coordinate system selected by one of the soft keys above.
- [CANCEL] soft key
  This soft key cancels the graphic coordinate system selected by one of the soft keys above to return to the original graphic coordinate system.

NOTE
The graphic coordinate system selected here is set in the graphic parameter for the graphic coordinate system.

- Rotating the graphic coordinate system

The following soft keys displayed by step 6 are used.

- [↑] soft key
  This soft key rotates the graphic coordinate system upward.
- [↓] soft key
  This soft key rotates the graphic coordinate system downward.
- [←] soft key
  This soft key rotates the graphic coordinate system to left.
- [→] soft key
  This soft key rotates the graphic coordinate system to right.
- [CW] soft key
  This soft key rotates the graphic coordinate system clockwise.
- [CCW] soft key
  This soft key rotates the graphic coordinate system counterclockwise.
- [OK] soft key
  This soft key changes the rotation angle of the current graphic coordinate system to the one set by one of the soft keys above.
- [CANCEL] soft key
  This soft key cancels the rotation of the graphic coordinate system performed by one of the soft keys above to return to the original graphic coordinate system.

NOTE
1 Set the travel increment made by one rotation operation in parameter No. 14716.
2 The rotation angle of the graphic coordinate system set here is not set in the graphic parameter for rotation angle.

- Drawing of a tool

In animation drawing, not only a blank figure but also a tool figure is drawn in three dimensions.

NOTE
The following function is necessary to draw in the tool figure.
- Tool geometry size data 100/300 pairs

Figure data

For tool figure drawing, the following data is used:

- Tool geometry size data 100/300 pairs
- Tool compensation data (tool-nose radius compensation value, virtual tool tip direction, cutter compensation value)
- Parameters

The table below indicates the drawable types of tools and figure data of each type of tool.

Table 13.2.2.2 (a) List of drawable tools (for turning)

Tool name	Tool geometry size data	Tool compensation	Parameter No.	
General-purpose tool	Setting		Tip position	No.27350#0
	Cutting edge angle		Cutting edge length	No.27351
	Tool angle		Holder length	No.27352
			Holder width	No.27353
			Holder length 2	No.27354
			Holder width 2	No.27355
Threading tool	Setting		Tip position	No.27356#0
	Tool angle		Cutting edge width	No.27357
			Holder length	No.27358
			Holder width	No.27359
Grooving tool	Setting		Tip position	No.27360#0
	Tip width		Holder length	No.27361
	Tip length		Holder width	No.27362
Round-nose tool	Setting	Tool-nose radius	Tip position	No.27363#0
	Tip length	Virtual tool tip	Holder length	No.27364
			Holder width	No.27365
Point nose straight tool	Setting		Tip position	No.27366#0
	Cutting edge angle		Cutting edge length	No.27367
	Tool angle		Holder length	No.27368
			Holder width	No.27369
			Holder length 2	No.27370
			Holder width 2	No.27371
Versatile tool	Setting		Tip position	No.27384#0
	Tip width		Holder length	No.27385
	Tip length		Holder width	No.27386

Table 13.2.2.2 (b) List of drawable tools (for milling)

Tool name	Tool geometry size data	Cutter compensation	Parameter No.	
Drill	Setting Tool angle	Diameter	Tip length	No.27372
counter sink tool	Setting Cutter diameter	Small diameter	Tool angle Tip length Cutter length Shank length Shank diameter	No.27375 No.27376 No.27377 No.27378 No.27379
Flat end mill	Setting	Diameter	Tip length	No.27373
Ball end mill	Setting	Diameter	Tip length	No.27380
Tap	Setting	Diameter	Tip length	No.27374
Reamer	Setting	Diameter	Tip length	No.27381
Boring tool	Setting	Diameter	Tip length	No.27382
Face mill	Setting	Diameter	Tip length	No.27383

NOTE
In lathes system, the round-nose tool and milling tools can be drawn only when the cutter compensation/tool-nose radius compensation function is provided.

Timing of drawing

If a code indicated below is specified during drawing execution, a tool figure is drawn according to the tool geometry size data that has the same number as the code.

- With a lathe system
  T code command
  If three numbers, namely, a tool number, tool geometry offset number, and tool wear offset number, are specified in a T code, the tool geometry size data that has the same number as the tool geometry offset number is obtained.

  NOTE
  When the tool management function is provided, a tool offset is specified with a D code. So, the tool geometry size data that has the offset number indicated by the D code is obtained.

- With a machining center system
  D code command

If the tool geometry size data corresponding to a specified number does not exist or the tool geometry size data is not set correctly, tool drawing is disabled with the warning "ILLEGAL SETTING OF TOOL FIGURE DATA".

13.2.3 Programmable Data Input (G10) for Blank Figure Drawing Parameters

Overview

Each of the drawing parameters for a blank figure, a position, and dimensions that are for the automatic scaling of the drawing areas of tool path drawing and animation drawing with the dynamic graphic display function can be set with an NC program, using the programmable data input command (G10 command).

Format

G10 L90 P_ IP_ I_ J_ K_ ;

P_ : Blank figure
IP_ : Addresses of the basic three axes and the reference position of a blank
I_,J_,K_ : Dimensions of a blank

NOTE
1 This command is enabled only during the execution of drawing. It is ignored during normal automatic operation.
2 This command is a one-shot G code.
3 This command must be specified in a single block.

Explanation

- Blank figure (P_)

Specify the type of a blank figure with either of the following settings for shapes.

Setting	Figure
0	Column or cylinder (parallel with the Z-axis)
1	Rectangular parallelepiped

The specified value is set in parameter No. 11343.

- Reference position of a blank (IP_)

Specify the reference position of a blank with coordinates (X,Y,Z) in the workpiece coordinate system.

NOTE
Specify the reference position of a blank with coordinates (X,Y,Z) in the machine coordinate system in case of bit 3 (BGM) of parameter No.11329=1.

The specified value is set in parameter No. 11344.

- Dimensions of a blank (I_,J_,K_)

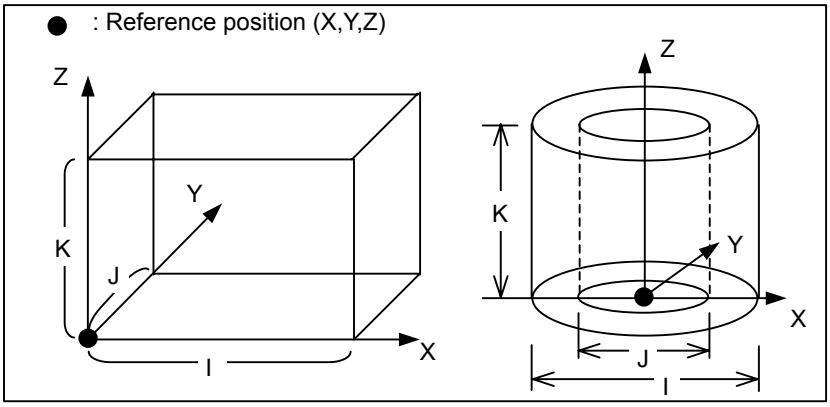
For the shape of each blank, specify the dimensions of the blank as follows:

Blank figure	Address I	Address J	Address K
Rectangular parallelepiped	Length in X-axis direction	Length in Y-axis direction	Length in Z-axis direction
Column	Diameter of column	0	Length of column
Cylinder	Diameter of outer circle of cylinder	Diameter of inner circle of cylinder	Length of cylinder

The specified value are set in parameter No.11345 (address I), parameter No.11346 (address J), and parameter No.11347 (address K).

- Relationship between the position and dimensions of a blank

The relationship between the position and dimensions of a blank is shown in the figure below.



- Operation to be performed when this command is issued

When this command is executed in the background operation, the specified values are set in the drawing parameters for the blank figure, reference position, and dimensions that correspond to the specified arguments, and the drawing area is automatically set again with the values set.
This causes this command and subsequent commands to perform drawing in the new drawing area.

NOTE

Before this command, drawing is performed as described below.

- The tool path is erased.
- In animation drawing, the blank figure is initialized.
- The drawing start position and the tool figure become ineffective.

Thus, issue this command at the start of a drawing program.

13.2.4 Warning Messages

Warning message	Content
START REJECTED	This program cannot be drawn.
NO PROGRAM SELECTED	No drawing target program is selected.
UNAVAILABLE COMMAND IS IN DRAWING	An NC statement/macro statement that can not execute drawing is instructed.
ILLEGAL SETTING OF GRAPHIC PARAMETERS	Graphic parameters are not set correctly.
ILLEGAL SETTING OF TOOL FIGURE DATA	Tool figure data are not set correctly, or tool figure data are not exist.

13.2.5 Note

NOTE

1 The coordinates used in drawing are absolute coordinates.
Therefore, even if the coordinate system is changed while it is drawing, it draws in the coordinate system when having begun to draw.

2 The drawing target axes are the basic three axes.
The basic three axes are the X-axis, Y-axis, and Z-axis on the basic coordinate system set with parameter No.1022.

3 In a drawing program having a sequence of contiguous small blocks, drawing requires longer processing time, possibly causing machining time to become longer than actual machining time.

4 When drawing is executed, data is treated as described below.

(1) Parameters
The same parameters as for automatic operation are used. However, parameters cannot be rewritten with a command such as the G10 command. If an attempt is made to rewrite a parameter with programmable parameter input (G10L52), the command is ignored.
However, by setting bit 7 (GST) of parameter No.11329, drawing can be temporarily stopped with the warning "UNAVAILABLE COMMAND IS IN DRAWING".

(2) Tool compensation value/workpiece origin offset/extended workpiece origin offset/macro variable/fixture offset data, etc.
For each of drawing execution and automatic operation, separate data is used. At the start of drawing, data for drawing is produced by copying data for automatic operation. In subsequent operations, the data for drawing and the data for automatic operation are treated independently of each other. So, even if data is rewritten with a command such as G10, drawing execution and automatic operation do not affect each other. Note, however, that data rewritten in drawing execution is not reflected in the data for automatic operation but is deleted.

(3) Tool life management data, tool number offset data, three-dimensional error correction data
These data items are not used for drawing execution. So, if an attempt is made to rewrite data with G10, the command is ignored.
However, by setting bit 7 (GST) of parameter No.11329, drawing can be temporarily stopped with the warning "UNAVAILABLE COMMAND IS IN DRAWING".

13.2.6 Restrictions

- Simultaneous drawing with multiple paths

This function does not support simultaneous tool path drawing by program execution with multiple paths.
For example, the tool paths of programs executed simultaneously with multiple paths on a 2-path or 3-path lathe or a combined machine tool cannot be drawn. This function supports only drawing executed singly with a single path.

- Simultaneous multipath display

The screens of this function do not support simultaneous multipath display based on the setting of parameter No. 13131 and No. 13132.

- Functions that operate differently in drawing execution and automatic operation

The operations of the following functions in drawing execution differ from the operations in automatic operation:

1. Operations that differ, depending on the custom macro
   1) Interface signal
      #1000 to #1035 are assumed to be 0 at all times.
   2) Message output
      A message to be output with #3006 is not displayed but is ignored.
   3) clock
      #3001 and #3002 are ignored. So, note that drawing is not continued, for example, if the following is specified:
      #3001=0 ;
      WHILE [#3001 LE 100] DO1 ;
      END1 ;
   4) Mirror image
      #3007 is assumed to be 0 at all times.
   5) Program-restarted state
      #3008 is assumed to be 0 at all times.
   6) External output command
      BPRNT, DPRNT, POPEN, and PCLOS are ignored.

2 Functions that perform partly different operations
   1) When G28 (automatic reference position return) is specified, up to the intermediate point is drawn.
   2) When G29 (automatic return from the reference position) is specified, drawing is performed from the intermediate point.
   3) When G27 (reference position return check) is specified, no reference position return check is made.
   4) No stored stroke limit check is made.
   5) When G31 (skip function), or G31.1, G31.2, or G31.3 (multistage skip function) is specified, drawing is performed up to the specified position, regardless of the skip signal.
   6) When G60 (single direction positioning) is specified, drawing is performed directly up to the specified position at all times even if the positioning direction is opposite.

3. Functions that perform different operations
   If the following functions are specified, the operations described below result:
   1) G02.2/G03.2 (involute interpolation)
      Circular interpolation is performed.
   2) G02.3/G03.3 (exponential interpolation)
      Linear interpolation is performed only with linear axes.
   3) G07 (virtual axis interpolation)
      Circular interpolation is performed.
   4) G07.1 (cylindrical interpolation)
      Linear interpolation is performed only with linear axes.
   5) G12.1 (polar coordinate interpolation)
      Linear interpolation is performed only with linear axes.

- Functions that do not support drawing

The following functions are ignored in drawing execution:

1) G04 (dwell)
2) G20, G21 (inch/metric switch)
3) Auxiliary function (M, S, T, B)
4) G22, G23 (stored stroke limit check on/off)
5) G10.6 (tool retract data setting)
6) G81.1 (chopping)
7) G25/G26 (spindle variation detection on/off)
8) G10 (programmable data input)

NOTE
If G10 (programmable data input) is specified, drawing can be temporarily stopped with the warning "UNAVAILABLE COMMAND IS IN DRAWING" by setting bit 7 (GST) of parameter No.11329.

The following functions are warning in drawing execution:

1) G02.4/G03.4 (three-dimensional circular interpolation)
2) M198 (external subprogram call)

- Functions that support drawing

The operations performed with the following program commands can be drawn:

1) G00 (Positioning)
2) G01 (Linear interpolation)
3) G02/G03 (Circular interpolation)
   However, helical interpolation with the instruction axis other than basic three axes cannot be drawn.
4) G17/G18/G19 (Plane selection)
5) G33/G34 (Threading)
   However, it is drawn as linear interpolation.
6) G35/G36 (Circular threading)
   However, it is drawn as circular interpolation.
7) G40/G41/G42 (Cutter compensation/cancel)
8) G52 (Local coordinate system)

9) G53 (Machine coordinate system selection)
10) G54 to G59 (Workpiece coordinate system selection)
11) G54.1 (Extended workpiece coordinate system selection)
12) G65 (Macro call)
13) G68/G69 (Coordinate system rotation, three-dimensional coordinate system conversion/cancel)
14) G90/G91 (Absolute/incremental command)
15) G92 (Workpiece coordinate system change)
16) G92.1 (Workpiece coordinate system preset)
17) G94/G95 (Feed per minute/Feed per rotation)
18) G96/G97 (Constant surface speed control/cancel)
19) M98 (Subprogram call)
20) G90/G92/G94 (Canned cycle, only Lathes system)
21) G71/G72/G73/G74/G75/G76/G70 (Multiple repetitive cycle, only Lathes system)
22) G83/G84G85/G87/G87/G88/G80 (Canned cycle for drilling, only Lathes system)
23) G73/G74/G76/G81/G82/G83/G84/G85/G86/G87/G88/G89/G80 (Canned cycle for drilling, only Machining center system)

NOTE
1 When a lathe system is used, refer to the G codes above as codes of G code system B.
2 It is possible to draw with the G68 (Coordinate system rotation, three-dimensional coordinate system conversion) instruction only in the tool path drawing.
And, the display of coordinates when instructing in G68 is a coordinate value on the workpiece coordinate system.
3 In threading cycle of canned cycle and multiple repetitive cycle, movement of thread chamfering is not drawn.
4 In animation drawing, shape by the movement of back boring cycle command is different from actual shape.
5 In animation drawing, movement to shift amount at the bottom of a hole in the fine boring cycle and back boring cycle command is not drawn by setting bit 1 (ABC) of parameter No.11349=0.

- Operations that do not support drawing

The following operations cannot be drawn:
1) Backward movement based on the retrace function
2) Forward movement/backward movement/resumed forward movement based on the manual handle retrace function
3) Movement made with skip signal input for a skip command or multistage skip command
4) Movement on an axis based on PMC axis control

5) Movement on an axis based on real-time custom macro specification
6) Operation based on manual interrupt, manual handle interrupt, etc.
7) Operation based on synchronous control, mixture control, and superimposed control
8) Operation based on mirror image of facing tool posts
9) Operation based on the 5-axis machining function
10) Operation based on the high-speed, high-precision function (AI contour control I, AI contour control II and high-speed processing)
11) Operation based on nano smoothing or NURBS interpolation
12) Operation based on the program coordinate system switching function
13) Operation based on the offset memory switching function
14) Operation based on Cs contour control
15) Operation based on Spindle positioning.

- Dynamic switching of diameter/radius specification in the background operation

It is possible to perform drawing by switching the amount-of-travel setting on each axis that is to be specified in the drawing program to a diameter or radius value, using programmable switching of diameter/radius specification (G10.9 command) of the function of dynamic switching of diameter/radius specification.

When the G10.9 command of the function of dynamic switching of diameter/radius specification is enabled (bit 5 (PGD) of parameter No. 3400 = 1), drawing is performed by internally switching the diameter/radius specification for the move command for each axis in the background operation, as follows:

- If the specified address value is 0 (radius specification) → Drawing is performed with the radius specification.
- If the specified address value is 1 (diameter specification) → Drawing is performed with the diameter specification.

NOTE
1 For the G10.9 command in the background operation, the diameter/radius specification switching status signal of the function of dynamic switching of diameter/radius specification is not output.
2 The diameter/radius specification switching signal of the function of dynamic switching of diameter/radius specification is ineffective to the background operation.
3 The switching due to the G10.9 command in the background operation is canceled if the background operation is stopped.
4 Using this command requires the function of dynamic switching of diameter/radius specification.

- Data items and commands that can be switched
  For the following data item and command, switching of diameter/radius specification is performed according to the switched specification:
  - Move command from a program
  - Current position display

- Data items that cannot be switched
  For the following data items, switching of diameter/radius specification is not performed because they follow the setting of bit 3 (DIAx) of parameter No. 1006.
  - Parameter
  - Offset
  - Workpiece coordinate system

  In custom macros, coordinates that can be read with the system variables indicating position information, listed below.
  #5001 to #5020 and #100001 to #100050: End point position in the immediately preceding block (workpiece coordinate system)
  #5021 to #5040 and #100051 to #100100: Specified current position (machine coordinate system)
  #5041 to #5060 and #100101 and #100150: Specified current position (workpiece coordinate system)
  #5061 to #5080 and #100151 to #100200: Skip position (workpiece coordinate system)

NOTE
For offsets, the settings of bit 1 (ORC) and bit 2 (ODI) of parameter No. 5004 take precedence.

Screen center coordinates and values of blank (position) drawing parameters are handled with radius specification regardless of the setting of parameter bit DIAx.

- Operation of the P-CODE workpiece number search function in the background operation

When drawing is started with the setting of bit 3 (WNS) of parameter No. 11349, the execution macro specified in system variable #8610 is executed before the drawing program.

- Calling conditions and operations
  a) Bit 6 (PWSR) of compilation parameter No. 9002 is set to 1.
  b) The program number of an execution macro is set in system variable #8610 before the start of drawing.

  If the conditions above are met, pressing soft key [START] or [1BLOCK] to execute drawing causes the following operations to be performed:

(1) The execution macro specified with system variable #8610 is called.
If bit 4 (P98) of compilation parameter No. 9163 is 0, the execution macro is called with an operation equivalent to a simple call (G65), and if P98 is 1, it is called with an operation equivalent to a subprogram call (M98).
(2) When the execution macro terminates, the drawing program is executed.

NOTE

1 This function calls an execution macro if the conditions described in a) and b) are met, regardless of the program selected. Thus, if using this function to perform drawing with an execution macro, modify the program of the execution macro so that macro statements, auxiliary functions, and other commands in the execution macro that are not necessary for drawing are not executed, by referring to the system variable below.

System variable	Value	Execution state
#3010	1	Normal state (if drawing is not executed)
	-1	Drawing execution

2 System variable #8610, referenced during the execution of drawing, is the same as that used during automatic operation. If, therefore, the value of the variable is changed to execute drawing, it is necessary to return the variable to the original value before automatic operation is performed. Also, automatic operation is performed with the variable value at the time drawing is started and, therefore, caution is required so that the value set is not changed during automatic operation.

- Nesting and local variables
Nesting is accumulated in an execution macro. An execution macro is called with an operation equivalent to a simple call (G65) or a subprogram call (M98) and, therefore, other execution macros can be called from the called execution macro with all calling methods.
In the case of an operation equivalent to a simple call (G65), the level of local variables is changed to that of the drawing program. That is, the drawing program do not inherit the local variables used in the execution macro.
In the case of an operation equivalent to a subprogram call (M98), the local variables at the same level as the drawing program are used. That is, the drawing program inherits the local variables used in the execution macro.

- Drawing start position

In tool path drawing, if G92, G52, or G92.1 (machining center system) or G50, G52, or G50.3 (lathe system) is specified at the start of a drawing target program, the position specified with the G code is the drawing start position. If none of these G codes is specified, the end point of the first moving command is the drawing start position.
In animation drawing, the current position when the tool figure is displayed is the drawing start position.

- Use of this function with other functions

When this function is specified, the following functions cannot be used:

- Graphic display
- Manual Guide *i*

- VGA display based on the C language executor

If VGA display based on the C language executor is used, the VGA window is erased when the screen display is switched to a screen of this function.
So, when using the VGA window, determine the screen of this function by screen number and close the VGA window before switching the screen display.

- Use of the CNC screen display function

The following restriction exists when the screen of dynamic graphic display function is displayed by the CNC screen display function.

1) The drawing screen is not displayed in the CNC screen display function by way of via embedded Ethernet.
   Please use the CNC screen display function by way of the FAST Ethernet board.
2) The drawing screen is not displayed in the CNC screen dual display function.
3) Please do not start / shut down the CNC screen display function when you display the drawing screen. Please start / shut down the CNC screen display function after it switches to other screens.

- Display unit

This function can be used with a 10.4-inch display unit.
On the screens of this function, the vertical soft keys displayed on a 10.4-inch display unit are not displayed.
To use the vertical soft keys, switch the screen display to another screen.

- Use of the Screen hard copy function

If the screen is copied while executing drawing by screen hard copy function, the drawing stops temporarily. The drawing can be restarted by pushing [RESTART] soft key if necessary after the copy operation is finished.

14 VIRTUAL MDI KEY FUNCTION

Chapter 14, "VIRTUAL MDI KEY FUNCTION", consists of the following sections:

14.1 VIRTUAL MDI KEY 1957

14.1 VIRTUAL MDI KEY

Overview

This function is used to perform program editing and changing of various data using the keyboard displayed on the LCD with a touch panel.

Screen on which a CNC screen is displayed in the upper left 1/4 area

Explanation

A CNC screen equivalent to that displayed on 7.2/8.4-inch LCDs is displayed in the upper left area of the screen, and the keyboard is displayed in the remaining area.

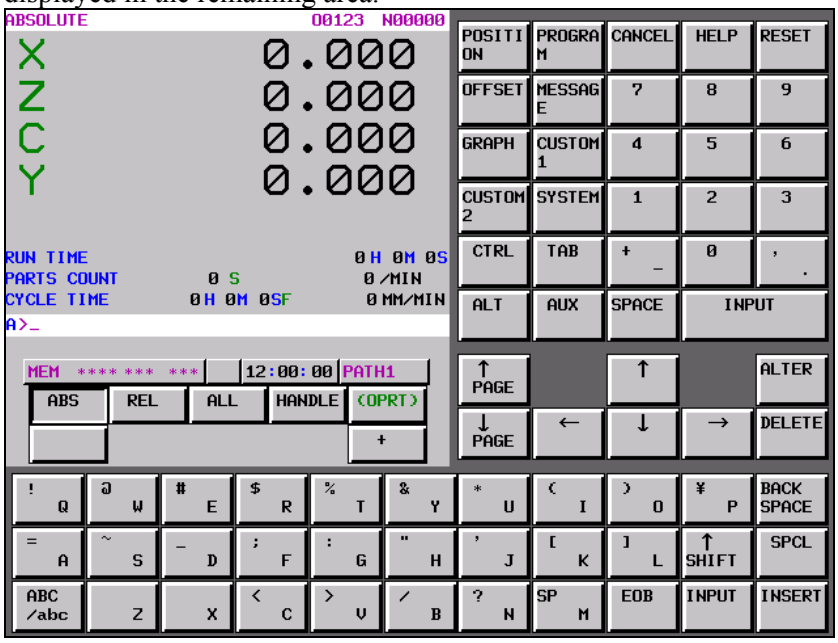


Figure 14.1 (a) Screen on which a CNC screen is displayed in the upper left area

Operation

- Input key

The display "INPUT" on the virtual MDI keyboard is equivalent to the input key.

- Cancel key

The displays "BACK SPACE" and "CAN" on the virtual MDI keyboard are equivalent to the cancel key.

- Shift key

The display "SHIFT" ↑ on the virtual MDI keyboard is equivalent to the shift key.
Pressing the "SHIFT" key once places the system in the shift state, and pressing the "SHIFT" key again or another key releases the system from the state.
Pressing a key in the shift state causes the character indicated in the upper left corner of the key to be entered.

- Simultaneous pressing of two keys

The operation to be performed for pressing two key simultaneously, such as the "CAN" and "RESET" keys to erase alarm PS100, is as follows:

(1) Press the "SPCL" key. The "SPCL" key is held down, and places the system in SPCL mode.
(2) Press the keys to be pressed simultaneously, one at a time.
(3) Press the "INPUT" key. Of the keys pressed in SPCL mode, the last two are assumed to have been pressed simultaneously. The system is released from SPCL mode, and the SPCL key, which has been held down, returns to the normal state.

Example: "SPCL" → "CAN" → "RESET" → "INPUT"

NOTE
1 In SPCL mode, other keys are disabled until the "SPCL" or "INPUT" key is pressed.
2 Pressing the "SPCL" key in "SPCL" mode causes all keys pressed in SPCL mode to be disabled. The system is released from SPCL mode, and the SPCL key, which has been held down, returns to the normal state.

Screen on which a CNC screen is displayed in the entire area

Explanation

A CNC screen equivalent to that displayed on 7.2/8.4-inch LCDs is displayed in the entire area. Function keys and the ON/OFF button for virtual keys are displayed at the bottom of the screen.

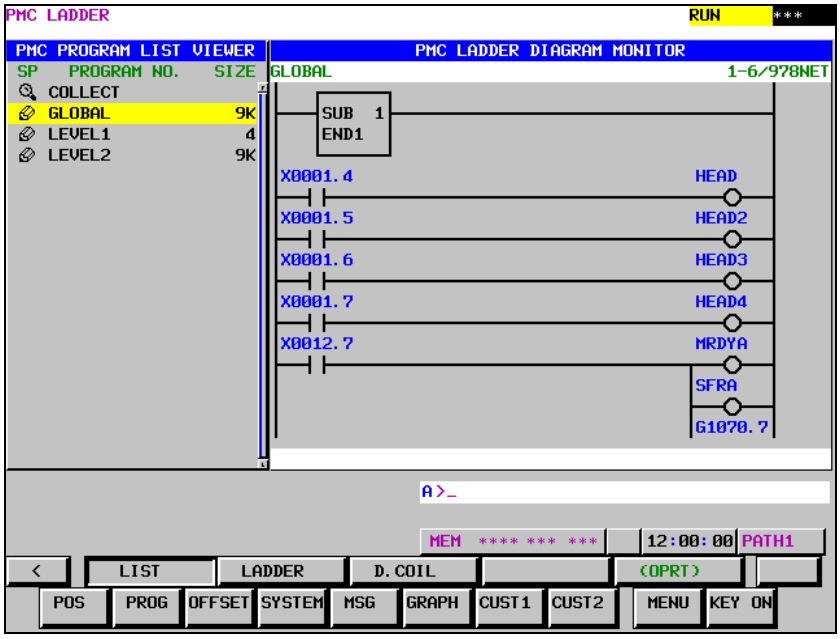
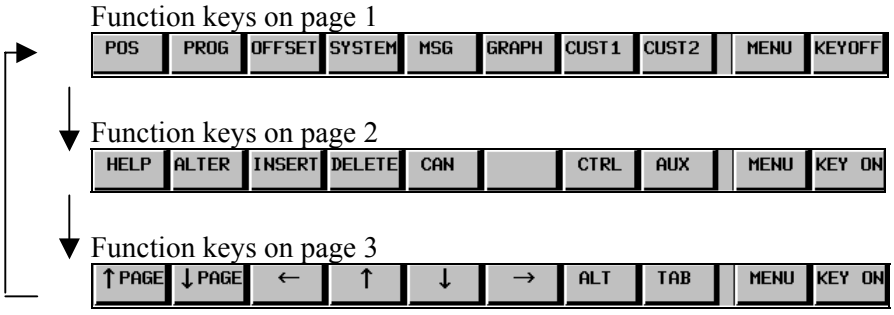


Figure 14.1 (b) Screen on which a CNC screen is displayed in the entire area

Operation

- Function key page switching

Pressing "MENU" located near the lower right corner of the screen switches the screen to page 1, page 2, page 3, and back to page 1 in this order.



- Display of virtual keys

Pressing "KEY ON" located at the lower right corner of the screen displays virtual MDI keys. The character string on the key top changes to "KEYOFF".
Pressing "KEYOFF" hides virtual MDI keys.

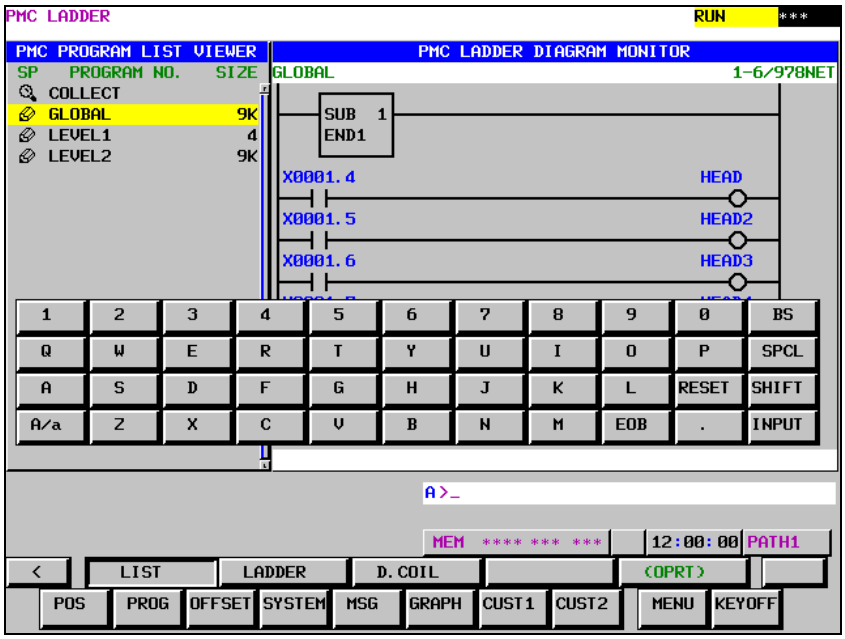


Figure 14.1 (c) State in which virtual keys are ON

- Input key

The display "INPUT" on the virtual MDI keyboard is equivalent to the input key.

- Cancel key

The displays "BS" and "CAN" on the virtual MDI keyboard are equivalent to the cancel key.

- Shift key

The display "SHIFT" on the virtual MDI keyboard is equivalent to the shift key.
The characters on the key tops change to those that can be entered in the shift state.

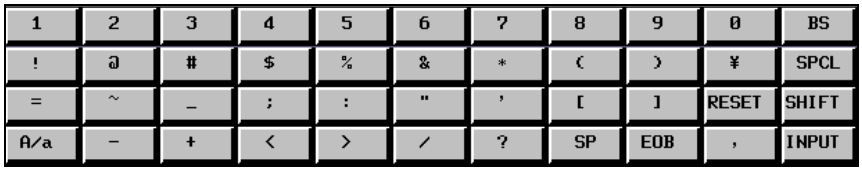


Figure 14.1 (d) Key tops in the shift state

- Simultaneous pressing of two keys

The operation to be performed for pressing two key simultaneously, such as the "CAN" and "RESET" keys to erase alarm PS100, is as follows:

(1) Press the "SPCL" key. The "SPCL" key is held down, and places the system in SPCL mode.
(2) Press the keys to be pressed simultaneously, one at a time.
(3) Press the "INPUT" key.
Example: "SPCL" → "CAN" → "RESET" → "INPUT"

NOTE
1 In SPCL mode, other keys are disabled until the "SPCL" or "INPUT" key is pressed.
2 Pressing the "SPCL" key in "SPCL" mode causes all keys pressed in SPCL mode to be disabled. The system is released from SPCL mode, and the SPCL key, which has been held down, returns to the normal state.

14.1.1 Limitations

- Display of VGA windows on the C language executor

This function uses one VGA window, so that the number of VGA windows that can be used on the C language executor is reduced by one.

- LCDs

LCDs on which this function can be used are 10.4-inch LCDs with a touch panel.

IV. MAINTENANCE

1 ROUTINE MAINTENANCE

This chapter describes routine maintenance work that the operator can perform when using the CNC.

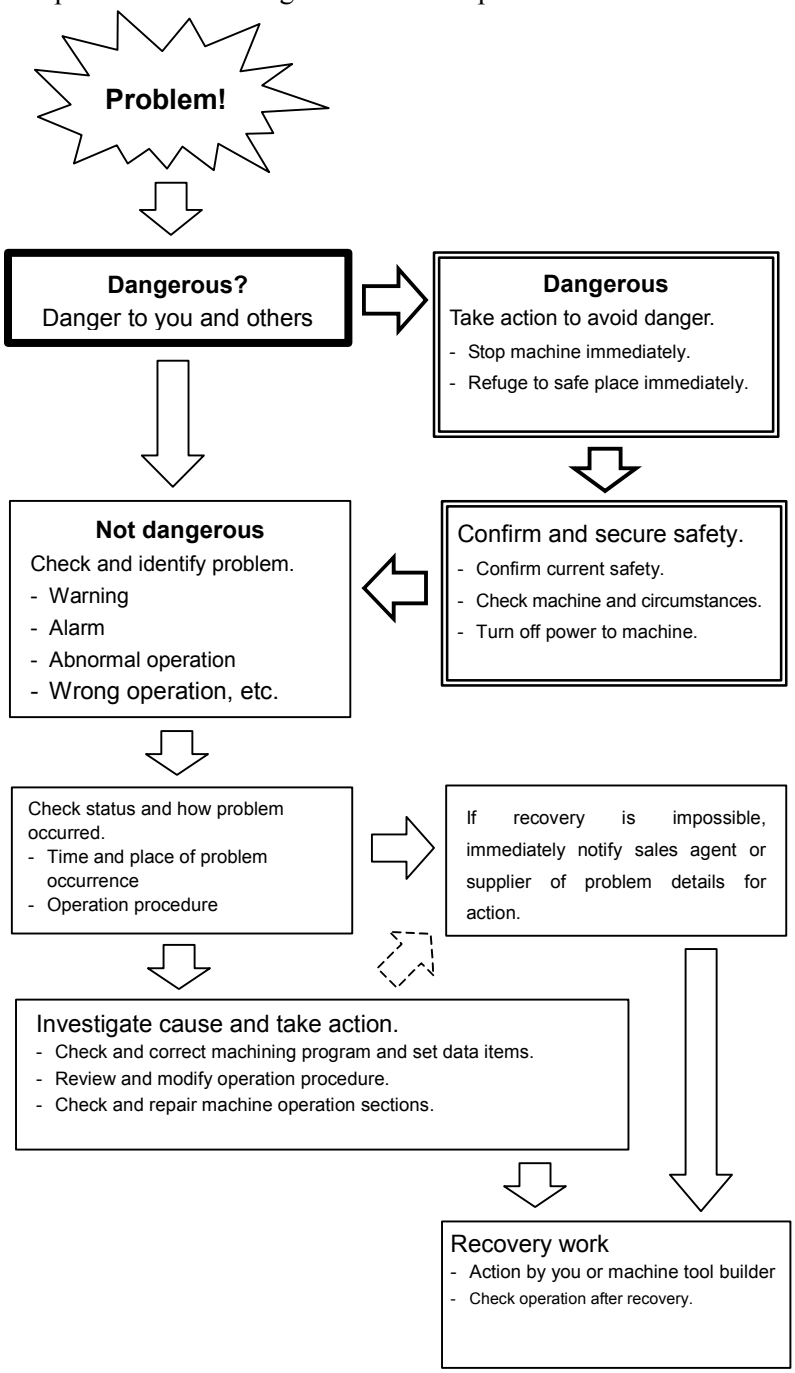
> ⚠ **WARNING**
> Only those persons who have been educated for maintenance and safety may perform maintenance work not described in this chapter.

Chapter 1, "ROUTINE MAINTENANCE", consists of the following sections:

1.1 ACTION TO BE TAKEN WHEN A PROBLEM OCCURRED 1964
1.2 BACKING UP VARIOUS DATA ITEMS 1965
1.3 METHOD OF REPLACING BATTERY 1967

1.1 ACTION TO BE TAKEN WHEN A PROBLEM OCCURRED

If an unexpected operation occurs or an alarm or warning is output when the CNC and machine are used, the problem needs to be solved quickly. For this purpose, the status of the problem must be identified correctly, and a proper action must be taken.
The procedure for taking an action for a problem is shown below.



For details of investigation and action on problems arising from the CNC, refer to "TROUBLESHOOTING PROCEDURE" in the Maintenance Manual (B-63945EN) issued by FANUC.

1.2 BACKING UP VARIOUS DATA ITEMS

With the CNC, various data items such as offset data and system parameters are stored in the SRAM of the control unit and are protected by a backup battery. However, an accident can erase the data. By storing the data at another location (outside the CNC), the data, when lost, can be restored.
So, when the machine is started up or data is updated, for example, the data should be backed up (stored outside the CNC).

- Data backup operation

The data items listed below should be backed up. For the method of data output operation, see the chapter of "DATA INPUT/OUTPUT" in this manual.

<1> System parameters
→ See III-8.1.2.
<2> Machining programs
→ See III-8.1.1.
<3> Tool offset data
→ See III-8.1.3.
<4> PMC data
→ See PMC PROGRAMMING MANUAL (B-63983EN).
<5> Pitch error compensation data (when the pitch error compensation function is selected.)
→ See III-8.1.4.
<6> Three-dimensional error compensation (when the three-dimensional error compensation function is selected.)
→ See III-8.1.5.
<7> Custom macro variables (when the custom macro function is selected.)
→ See III-8.1.6.
<8> Workpiece coordinate system setting data (when the workpiece coordinate system function is selected.)
→ See III-8.1.7.
<9> Tool management data (when the tool management function is selected.)
→ See III-8.1.9.

It is recommended that recording media (such as floppy disks and memory cards) daily used with the machine be used to store data. Stored data should be managed properly so that the data can be restored quickly if a problem occurs.

- Data restoration work

In order to restore lost data to the state of the stored data, input the data backed up according to the previous item into the CNC. For the method of data input operation, see the chapter of "DATA INPUT/OUTPUT" in this manual.

> ⚠ **WARNING**
> After inputting stored data, do not start an operation immediately. Instead, check that the data is input correctly and that settings are made to meet a desired operation.
> If an operation is executed without making this check, the machine and workpiece can be damaged and personal injury can occur due to an unexpected machine movement. Use sufficient care.

> ⚠ **CAUTION**
> Before recovery of the following data items, consult with the machine tool builder of the machine used:
> - System parameters
> - PMC data
> - Macro programs and custom macro variables
> - Pitch error compensation values

> **NOTE**
> The method of recovery described in this section is intended just to restore the state of the backed up data, and does not guarantee recovery of the state that was present when the data was lost.

1.3 METHOD OF REPLACING BATTERY

This chapter describes how to replace the CNC backup battery and absolute Pulsecoder battery. This section consists of the following subsections::

1.3.1 Replacing Battery for LCD-mounted Type CNC Control Unit 1968
1.3.2 Replacing the Battery for Stand-alone Type CNC Control Unit 1971
1.3.3 Battery in the PANEL *i* (3 VDC) 1973
1.3.4 Battery for Absolute Pulse coders 1975

Battery for memory backup

Offset data, and system parameters are stored in SRAM in the control unit. The power to the SRAM is backed up by a lithium battery mounted on the front panel of the control unit. Therefore, the above data is not lost even if the main battery fails. The backup battery is installed in the control unit prior to being shipped from the factory. This battery can provide backup for the memory contents for about a year.

When the battery voltage falls, alarm message "BAT" blinks on the LCD display and the battery alarm signal is output to the PMC. When this alarm is displayed, replace the battery as soon as possible. In general, the battery can be replaced within one or two weeks of the alarm first being issued. This, however, depends on the system configuration.

If the battery voltage subsequently drops further, backup of memory can no longer be provided. Turning on the power to the control unit in this state causes system alarm to be issued because the contents of memory are lost. Replace the battery, clear the entire memory, then reenter the data.

Replace the memory backup battery while the control unit is brought off.

The following two kinds of batteries can be used.

- Lithium battery, incorporated into the CNC control unit.
- Two alkaline dry cells (size D) in an external battery case.

NOTE
A lithium battery is installed as standard at the factory.

1.3.1 Replacing Battery for LCD-mounted Type CNC Control Unit

When using a lithium battery
- Replacement procedure

When a lithium battery is used

Prepare a new lithium battery (ordering code: A02B-0200-K102 (FANUC specification: A98L-0031-0012)).

<1> Turn on the power to the CNC. After about 30 seconds, turn off the power.

<2> Remove the old battery from the back of the CNC control unit.
First, unplug the connector by yanking the battery cable, then take the battery out of its case.
The battery case of a control unit without option slots is located at the back of the unit as shown in the figure below. The battery case of a control unit with option slots is located next to the fan on the top of the unit.

<3> Insert a new battery and reconnect the connector.

<4> Clamp the battery cable as shown in Fig. 1.3.1 (c).

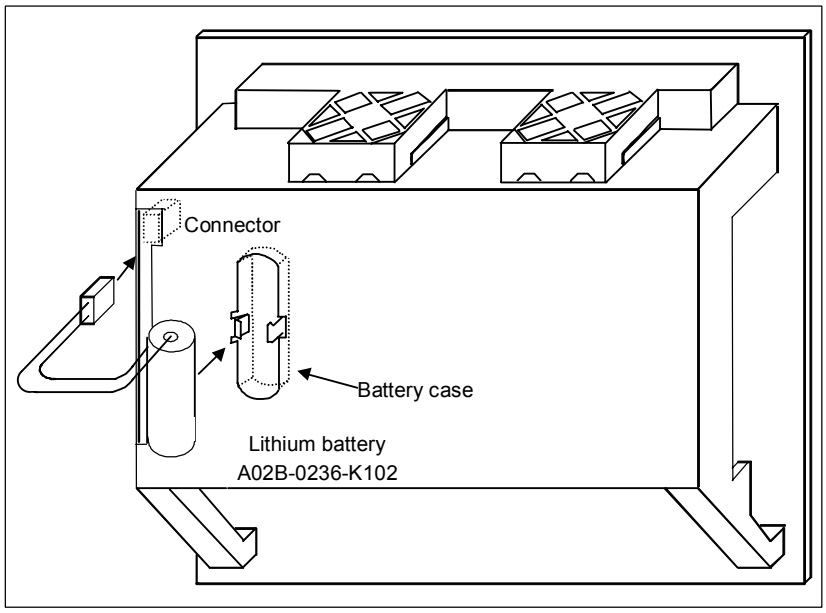


Fig. 1.3.1 (a) Unit without option slots

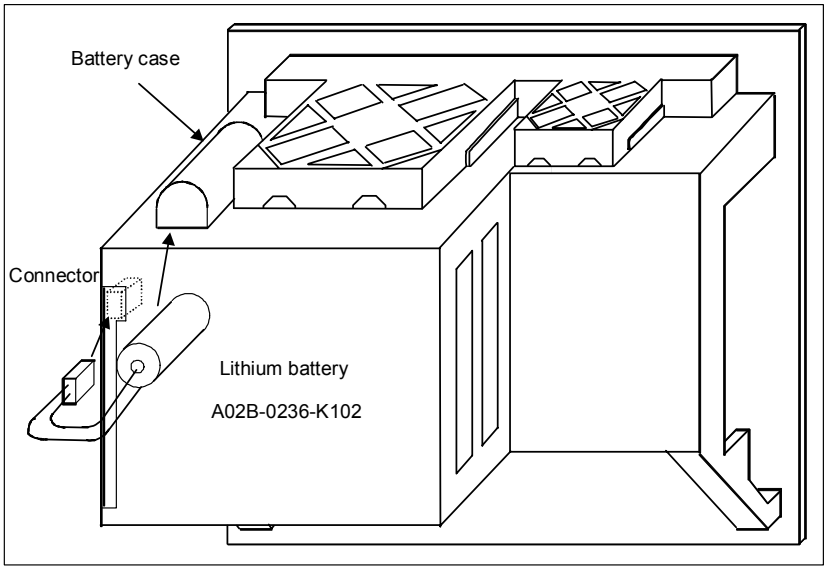


Fig. 1.3.1 (b) Unit with option slots

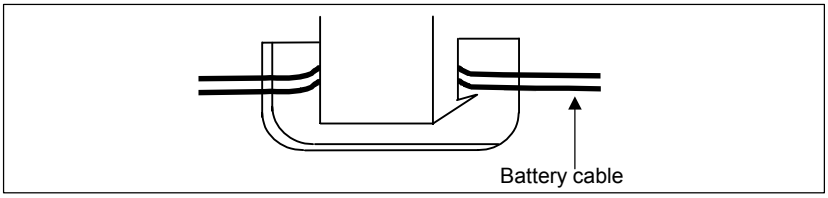


Fig. 1.3.1 (c) Clamping the battery cable

> ⚠ **WARNING**
> Using other than the recommended battery may result in the battery exploding. Replace the battery only with the specified battery (A02B-0200-K102).

> ⚠ **CAUTION**
> Steps <1> to <3> should be completed within 30 minutes.
> Do not leave the control unit without a battery for any longer than the specified period. Otherwise, the contents of memory may be lost.
> If steps <1> to <3> may not be completed within 30 minutes, save all contents of the SRAM to the memory card beforehand. Thus, if the contents of the SRAM are lost, the contents can be restored easily.
> For the method of operation, refer to Maintenance Manual.

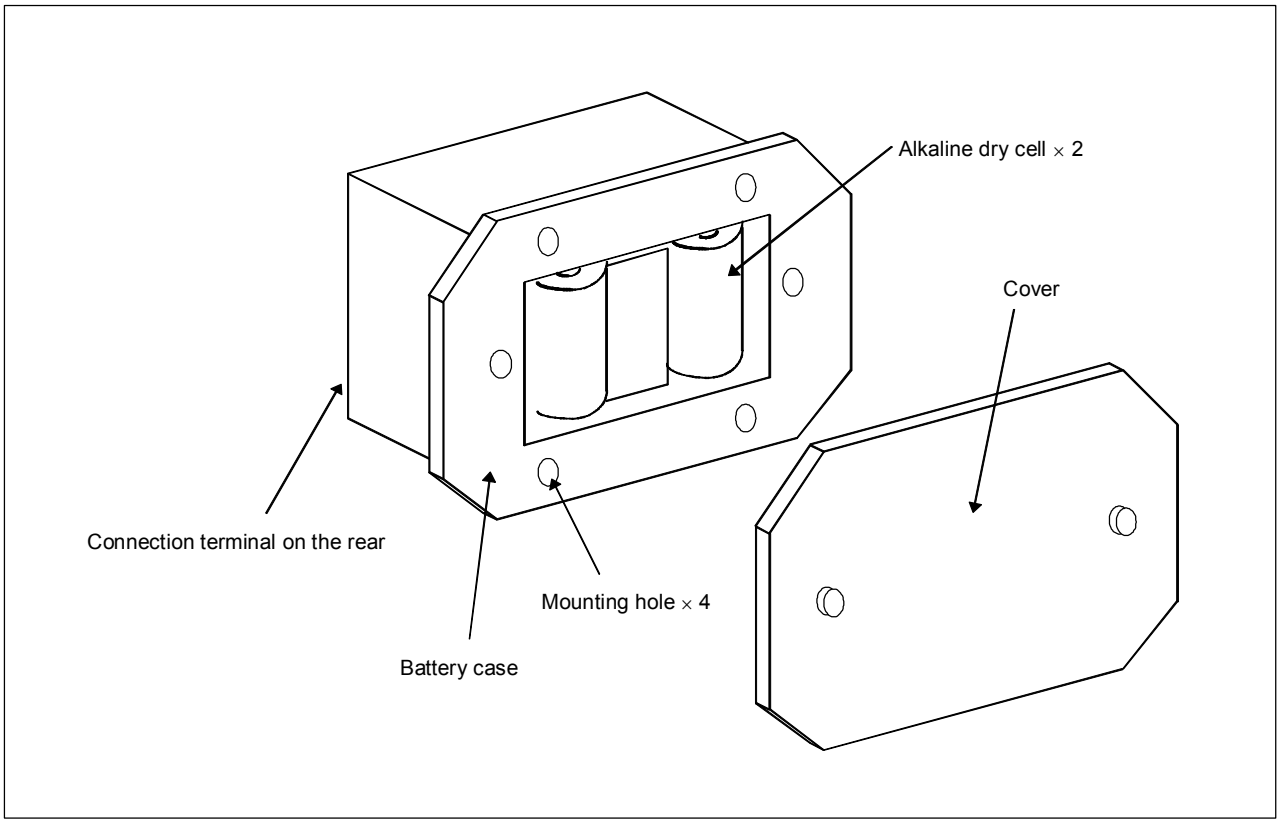
When discarding a battery, observe the applicable ordinances or other rules of your local government. Also, cover the terminals of the battery with vinyl tape or the like to prevent a short-circuit.

When using commercial alkaline dry cells (size D)

- Replacement procedure

<1> Prepare two alkaline dry cells (size D) commercially available.
<2> Turn on the power to the control unit.
<3> Remove the battery case cover.
<4> Replace the cells, paying careful attention to their orientation.
<5> Reinstall the cover onto the battery case.

> ⚠ **CAUTION**
> When replacing the alkaline dry cells while the power is off, use the same procedure as that for lithium battery replacement described above.



1.3.2 Replacing the Battery for Stand-alone Type CNC Control Unit

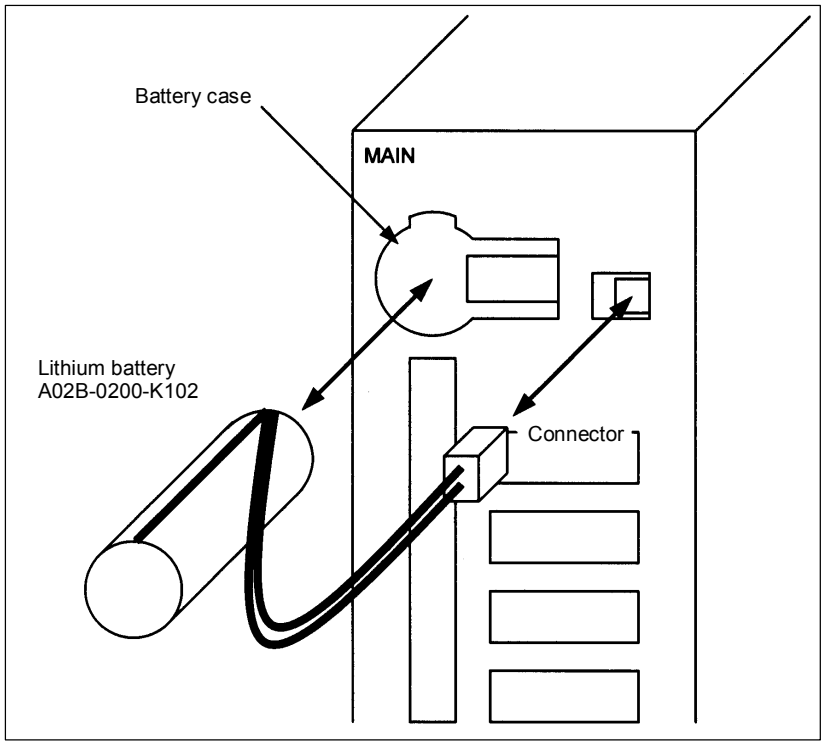
When using a lithium battery
- Replacing the battery

If a lithium battery is used, have A02B-0200-K102 (FANUC internal code: A98L-0031-0012) handy.

<1> Turn the CNC on. About 30 seconds later, turn the CNC off.

<2> Remove the battery from the top area of the CNC unit.
First, unplug the connector by yanking the battery cable, then take the battery out of the case.
The battery case is provided in the top area of the face plate of the main board.

<3> Replace the battery, then connect the connector.



⚠ **WARNING**
The incorrect mounting of the battery may cause an explosion. Avoid using any battery other than the one specified here (A02B-0200-K102).

> ⚠ **CAUTION**
> Complete steps <1> to <3> within 30 minutes.
> If the battery is left removed for a long time, the memory would lose the contents.
> If there is a danger that the replacement cannot be completed within 30 minutes, save the whole contents of the SRAM to a memory card. The contents of the memory can be easily restored with the memory card in case the memory loses the contents.

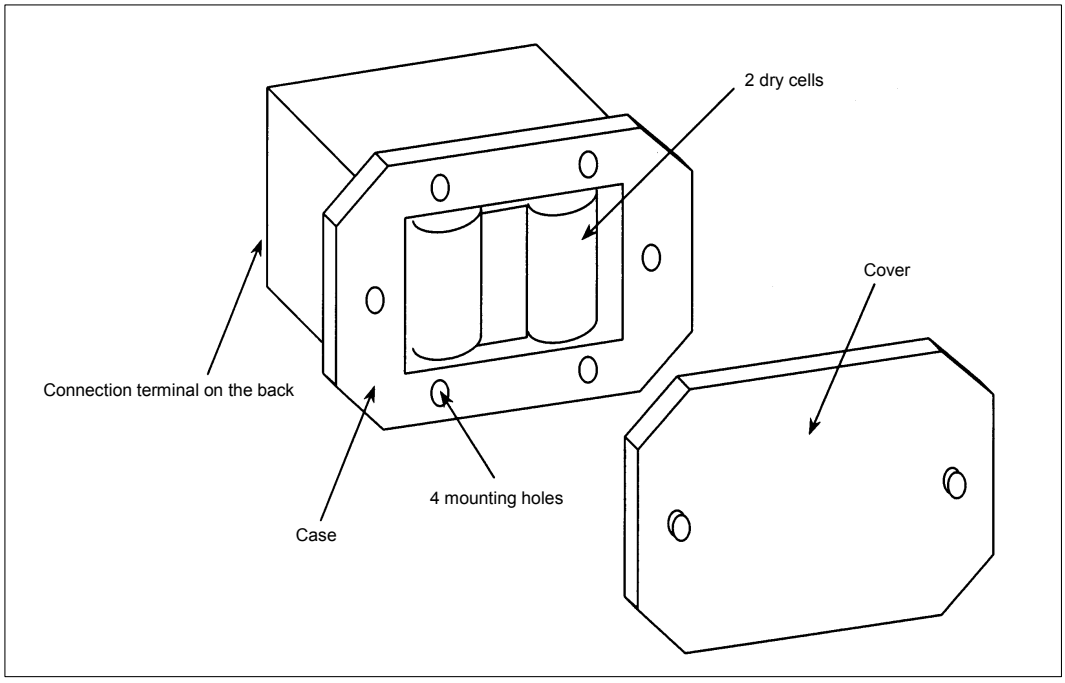
Discard the dead battery, observing appropriate municipal rules and regulations. When discarding the battery, insulate the terminal with a tape so that no short-circuit would occur.

When using commercial D-size alkaline dry cells

- Replacing the battery

<1> Have commercial D-size alkaline dry cells handy.
<2> Turn the CNC on.
<3> Remove the cover from the battery case.
<4> Replace the old dry cells with new ones. Mount the dry cells in a correct orientation.
<5> Replace the cover on the battery case.

> ⚠ **CAUTION**
> In the power-off state, the battery should be replaced as in the case of the lithium battery, which is descried above.



1.3.3 Battery in the PANEL *i* (3 VDC)

A lithium battery is used to back up BIOS data in the PANEL *i*. This battery is factory-set in the PANEL *i*. This battery has sufficient capacity to retain BIOS data for one year.
When the battery voltage becomes low, the following message appears on the self test screen displayed at turn-on, and the self test pauses.

CMOS Battery Failure

If this message appears, replace the battery as soon as possible (within one week). FANUC recommends that the battery be replaced once per year regardless of whether a battery alarm is issued.

- Replacing the battery

(1) To guard against the possible loss or destruction of BIOS parameters, write down the BIOS parameter values.
(2) Obtain a new lithium battery (A02B-0200-K102).
(3) After power has been supplied for at least five seconds, turn off the power to PANEL *i*. Remove the intelligent terminal from the panel so that replacement work can be done from the rear of the intelligent terminal.
(4) Detach the connector of the lithium battery, and remove the battery from the battery holder.
(5) Attach the connector, and place the battery in the battery holder.
(6) Install PANEL *i* again.
(7) Turn on the power, and check that the BIOS parameters are maintained (BIOS setup is not activated forcibly).

Between removing an old battery and inserting new battery, no more than five minutes must be allowed to elapse.

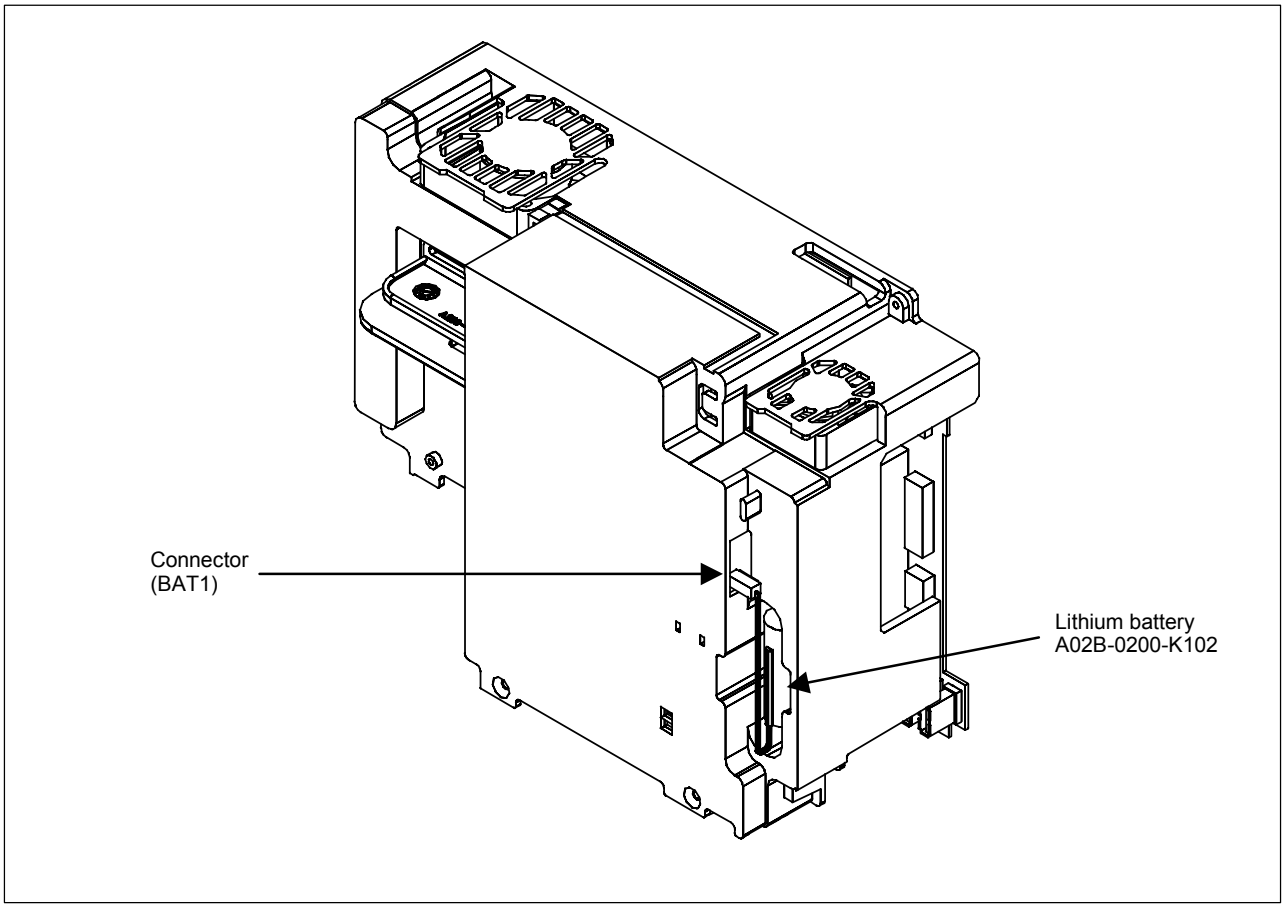


Fig. 1.3.3 (a) Lithium battery connection for PANEL *i*

1.3.4 Battery for Absolute Pulsecoders

(1) When the absolute pulsecoder battery voltage falls, "APC" blinks in the status display on the screen. When the battery voltage further falls, alarms DS0306 to 0308 are issued.
(2) When alarm DS0307 (alarm indicating the voltage of the battery becomes low) occurs, replace the battery as soon as possible. In general, the battery should be replaced within one or two weeks, however, this depends on the number of Pulsecoders used.
(3) If the voltage of the battery becomes any lower, alarm DS0306 (battery zero alarm) occurs. In this case, the current positions for the Pulsecoders can no longer be maintained.
In this state, alarm DS0300 (reference position return request alarm) occurs. After replacing the batteries, be sure to perform manual reference position return.
(4) The service life of the batteries is about two years if they are used in a six-axis configuration with α*i*/β*i* series servo motors and one year if they are used in a six-axis configuration with α/β series servo motors. FANUC recommends that you replace the batteries periodically according to the battery service life.
(5) To connect the battery, use the battery case or incorporate the battery into the servo amplifier. Note that the attachment method of the battery depends on the connection method and the type of servo amplifier.

- Replacing batteries

To prevent absolute position information in absolute Pulsecoders from being lost, turn on the machine power before replacing the battery. The replacement procedure is described below. (Note: The turning-on step is not required when the α*i* series servo motor or β*i* series servo motor (β*i*S 0.4 to β*i*S 22) is used.)

(1) Turn the servo unit (machine) on.
(2) Place the machine in the emergency stop state.
(3) Confirm that servo motors are not active.
(4) Make sure the DC link charge LED is off.
(5) Remove the old battery and then attach a new one.
(6) Now, replacement has been completed. The system power can be turned off.

NOTE

The absolute Pulsecoder of the servo motor $\alpha i/\alpha i$s series or βiS (βiS 0.4 to βiS 22) series is incorporated with a backup capacitor as standard. This backup capacitor enables an absolute position detection to be continued for about 10 minutes. Therefore, no manual reference position return need be performed if the time during which servo amplifier power is kept off for battery replacement is within 10 minutes. If battery replacement takes 10 minutes or more, the power must remain turned on.

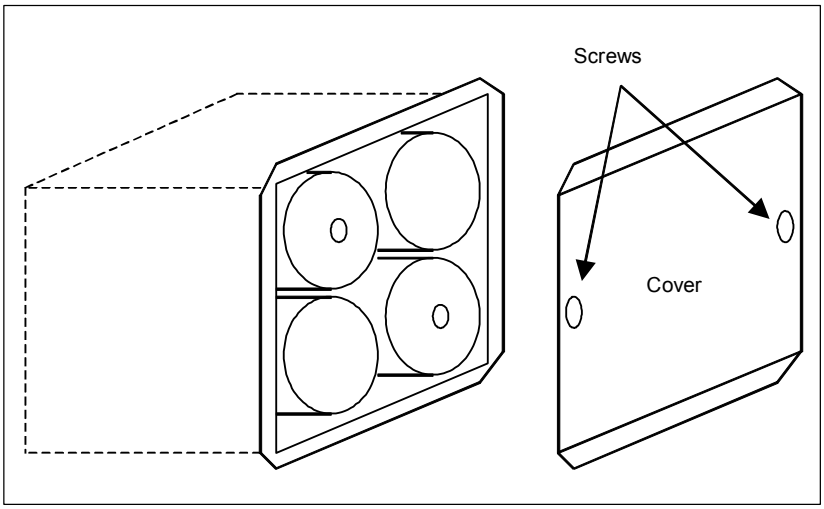
⚠ **WARNING**

1 When replacing the battery, be careful not to touch bare metal parts in the panel. In particular, be careful not to touch any high-voltage circuits due to the electric shock hazard.
2 Before replacing the battery, make sure the DC link charge LED is off. Otherwise, an electric shock may be received.
3 Be sure to use the specified battery. If another type of battery is used, it may overheat, blow out, or catch fire.
4 Install the battery with correct polarity. If the battery is installed with incorrect polarity, it may overheat, blow out, or catch fire. Or, absolute position information in absolute Pulsecoders may be lost.
5 During attachment of the battery, insert the factory-attached protection socket into the CX5X or CX5Y connector, whichever is not used. If the +6 V pin and 0 V pin are short-circuited, the battery may overheat, blow out, or catch fire. Or, absolute position information in absolute Pulsecoders may be lost.

- Replacing D-size alkaline dry cells in the battery case

Replace four D-size alkaline batteries (A06B-6050-K061) in the battery case installed in the machine.

(1) Have four D-size alkaline batteries on hand.
(2) Loosen the screws on the battery case. Remove the cover.
(3) Replace the alkaline batteries in the case. Pay careful attention to the polarity of the alkaline batteries.
(4) Attach the cover.



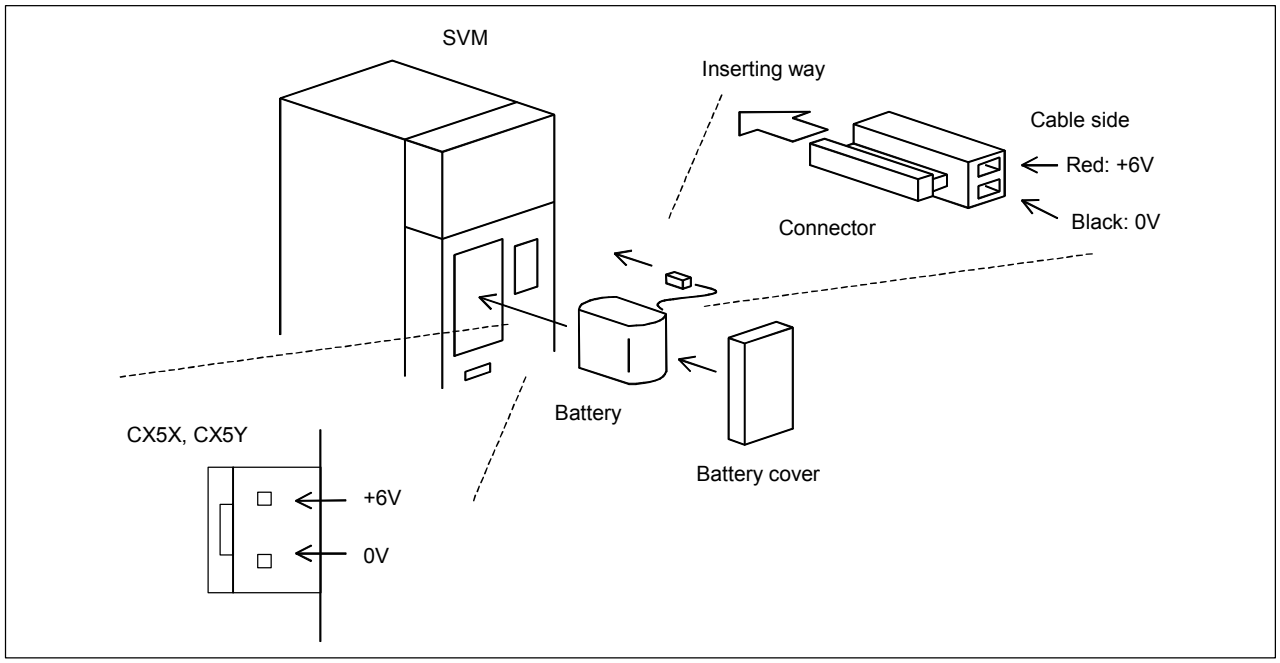
> ⚠ **WARNING**
> Install the battery with correct polarity. If the battery is installed with incorrect polarity, it may overheat, blow out, or catch fire. Or, absolute position information in absolute Pulsecoders may be lost.

- Attaching the built-in battery (αi series servo amplifier)

Attach the lithium battery (A06B-6073-K001) to the servo amplifier.

[Attachment procedure]

(1) Remove a battery cover from the servo amplifier.
(2) Attach the battery as shown below.
(3) Re-attach the cover.
(4) Connect the connector of battery with CX5X of the servo amplifier.



⚠ CAUTION

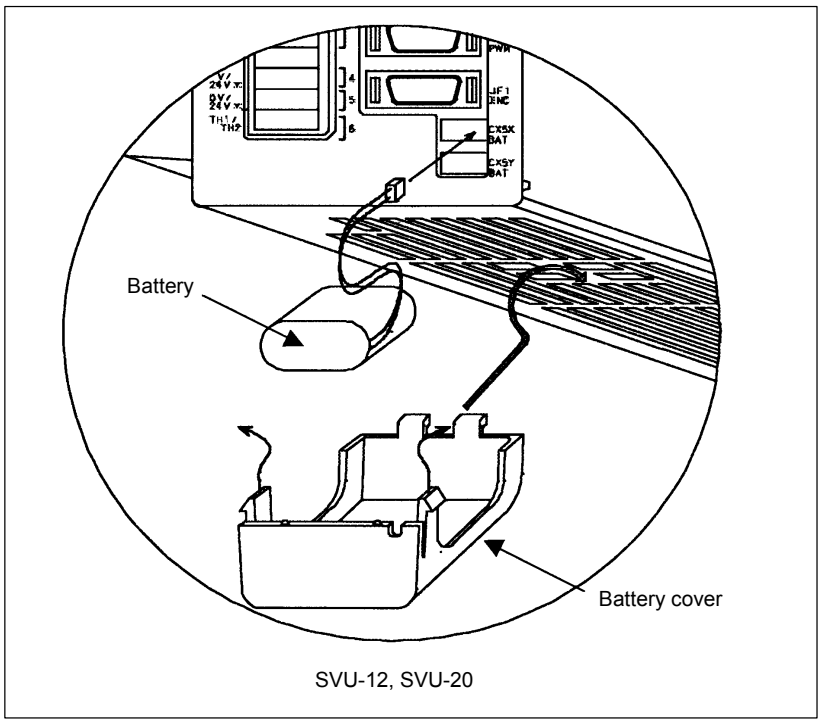
Attaching the battery from the cable outlet applies tension to the cable. Therefore, attach the cable from another place to prevent the cable from being stretched. If this cable is connected on a stretch condition, a bad conductivity may be occurred.

- Attaching the built-in battery (β series servo amplifier)

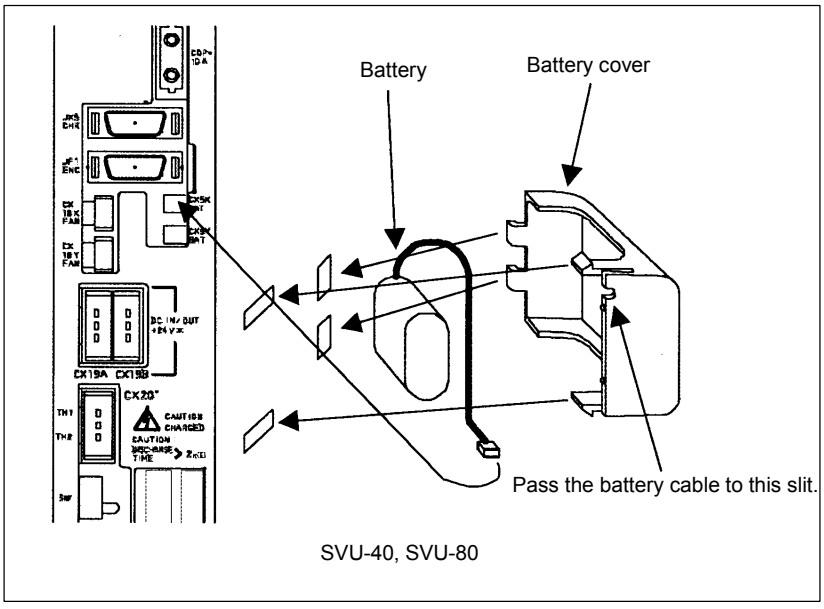
Attach the lithium battery (A06B-6093-K001) to the servo amplifier.

[Attachment procedure]

(1) In case of SVU-12 or SVU-20, remove the battery cover under the servo amplifier grasping its left and right sides. In case of SVU-40 or SVU-80, remove the cover attached on right side of the servo amplifier grasping its upper and lower sides.
(2) Remove the connector of battery. (Connector CX5X or CX5Y)
(3) Replace the battery and connect the connector.
(4) Mount the battery cover.



SVU-12, SVU-20



SVU-40, SVU-80

> ⚠ **CAUTION**
> 1 The connector of the battery can be connected with either of CX5X and CX5Y.
> 2 Attaching the battery from the cable outlet applies tension to the cable. Therefore, attach the cable from another place to prevent the cable from being stretched. If this cable is connected on a stretch condition, a bad conductivity may be occurred.

- Used batteries

Old batteries should be disposed as "INDUSTRIAL WASTES" according to the regulations of the country or autonomy where your machine has been installed.

APPENDIX

A PARAMETERS

This manual describes all parameters indicated in this manual.
For those parameters that are not indicated in this manual and other parameters, refer to the parameter manual.

NOTE

A parameter that is valid with only one of the path control types for the lathe system (T series) and machining center system (M series) is indicated in the upper or lower row as described below. A blank represents an unusable parameter.

[Example 1]

The parameter HTG is common to the T series and M series, and RTV and ROC are parameters used with the T series only.

	#7	#6	#5	#4	#3	#2	#1	#0	
1403	RTV		HTG	ROC					T series
			HTG						M series

[Example 2]

The following parameter is used with the M series only:

1411		T series
	Cutting feedrate	M series

A.1 DESCRIPTION OF PARAMETERS

	#7	#6	#5	#4	#3	#2	#1	#0
0000							ISO	TVC

[Input type] Setting input
[Data type] Bit path

 # 0 TVC TV check
0: Not performed
1: Performed

 # 1 ISO Code used for data output
0: EIA code
1: ISO code

NOTE
ASCII code is used at all times for output to the memory card.

	#7	#6	#5	#4	#3	#2	#1	#0
0001							FCV	

[Input type] Setting input
[Data type] Bit path

 # 1 FCV Program format
0: Series 16 standard format
1: Series 15 format

NOTE
1 Programs created in the Series 15 program format can be used for operation on the following functions:
  1 Subprogram call M98
  2 Thread cutting with equal leads G32 (T series)
  3 Canned cycle G90, G92, G94 (T series)
  4 Multiple repetitive canned cycle G71 to G76 (T series)
  5 Drilling canned cycle
    G83.1, G80 to G89 (T series)
    G73, G74, G76, G80 to G89(M series)
2 When the program format used in the Series 15 is used for this CNC, some limits may add. Refer to the User's Manual.

	#7	#6	#5	#4	#3	#2	#1	#0
0010							PRM	

[Input type] Setting input
[Data type] Bit path

 # 1 PRM When parameters are output, the parameters whose values are 0 are:
0: Output.
1: Not output.

	#7	#6	#5	#4	#3	#2	#1	#0
0012								MIRx

[Input type] Setting input
[Data type] Bit axis

 # 0 MIRx Mirror image for each axis
0: Mirror image is off. (Normal)
1: Mirror image is on. (Mirror)

0020	**I/O CHANNEL : Input/output device selection, or interface number for a foreground input device**

[Input type] Setting input
[Data type] Byte
[Valid data range] 0 to 5

The CNC has the following interfaces for transferring data to and from an external input/output device and the host computer:

Input/output device interface (RS-232-C serial ports 1 and 2)
Memory card interface
Data server interface
Embedded Ethernet interface

By setting bit 0 (IO4) of parameter No. 0110, data input/output can be controlled separately. When IO4 is not set, data input/output is performed using the channel set in parameter No. 0020. When IO4 is set, a channel can be assigned to each of foreground input, foreground output, background input, and background output.

In these parameters, specify the interface connected to each input/output device to and from which data is to be transferred. See the table below for these settings.

Correspondence between settings and input/output devices

Setting	Description
0,1	RS-232-C serial port 1
2	RS-232-C serial port 2
4	Memory card interface
5	Data server interface
9	Embedded Ethernet interface

	#7	#6	#5	#4	#3	#2	#1	#0
0100					NCR		CTV	

[Input type] Setting input
[Data type] Bit

 # 1 CTV Character counting for TV check in the comment section of a program.
0: Performed
1: Not performed

 # 3 NCR Output of the end of block (EOB) in ISO code
0: LF, CR, CR are output.
1: Only LF is output.

	#7	#6	#5	#4	#3	#2	#1	#0
0138	MNC							

[Input type] Parameter input
[Data type] Bit

 # 7 MNC DNC operation from the memory card and external device subprogram call from the memory card are:
0: Not performed.
1: Performed.

0983	Path control type of each path

NOTE
When this parameter is set, the power must be turned off before operation is continued.

[Input type] Parameter input
[Data type] Byte path
[Valid data range] 0 to 1
Set the path control type of each path.
The following two path control types are available:
T series (lathe system) : 0
M series (machining system) : 1

	#7	#6	#5	#4	#3	#2	#1	#0
0984								LCP

[Input type] Parameter input
[Data type] Bit path

NOTE
When this parameter is set, the power must be turned off before operation is continued.

 # 0 LCP Set whether the path is a loader control path.
0: The path is not a loader control path.
1: The path is a loader control path.

	#7	#6	#5	#4	#3	#2	#1	#0
1001								INM

[Input type] Parameter input
[Data type] Bit path

NOTE
When this parameter is set, the power must be turned off before operation is continued.

 # 0 INM Least command increment on the linear axis
0: In mm (metric system machine)
1: In inches (inch system machine)

	#7	#6	#5	#4	#3	#2	#1	#0
1002	IDG			XIK	AZR			JAX

[Input type] Parameter input
[Data type] Bit path

 # 0 JAX Number of axes controlled simultaneously in jog feed, manual rapid traverse and manual reference position return
0: 1 axis
1: 3 axes

 # 3 AZR When no reference position is set, the G28 command causes:

0: Reference position return using deceleration dogs (as during manual reference position return) to be executed.

1: Alarm (PS0304) "G28 was specified when no reference position is set" to be displayed.

> **NOTE**
> When reference position return without dogs is specified, (when bit 1 (DLZ) of parameter No.1005 is set to 1) the G28 command specified before a reference position is set causes an alarm PS0304 to be issued, regardless of the setting of AZR.

 # 4 XIK When LRP, bit 1 of parameter No.1401, is set to 0, namely, when positioning is performed using non-linear type positioning, if an interlock is applied to the machine along one of axes in positioning,

0: The machine stops moving along the axis for which the interlock is applied and continues to move along the other axes.

1: The machine stops moving along all the axes.

 # 7 IDG When the reference position is set without dogs, automatic setting of the IDGx parameter (bit 0 of parameter No.1012) to prevent the reference position from being set again is:

0: Not performed.

1: Performed.

> **NOTE**
> When this parameter is set to 0, bit 0 (IDGx) of parameter No. 1012 is invalid.

	#7	#6	#5	#4	#3	#2	#1	#0
1004	IPR							

[Input type] Parameter input
[Data type] Bit path

 # 7 IPR When a number with no decimal point is specified, the least input increment of each axis is:

0: Not 10 times greater than the least command increment

1: 10 times greater than the least command increment

When the increment system is IS-A, and bit 0 (DPI) of parameter No. 3401 is set to 1 (fixed-point format), the least input increment cannot be 10 times greater than the least command increment.

	#7	#6	#5	#4	#3	#2	#1	#0
1005			EDMx	EDPx				ZRNx

[Input type] Parameter input
[Data type] Bit axis

 # 0 ZRNx If a move command other than G28 is specified by automatic operation when no reference position return is performed yet after the power is turned on:

0: The alarm (PS0224) "PERFORM REFERENCE POSITION RETURN." is issued.
1: Operation is performed without issuing an alarm.

NOTE
1 The state in which a reference position has not been established refers to the following state:
- When an absolute position detector is not used and reference position return has not been performed even once after power-up
- When an absolute position detector is used and the association of the machine position with the position detected with the absolute position detector has not been completed (See the description of bit 4 (APZx) of parameter No. 1815.)

2 When the Cs axis coordinates are to be set up, set ZRN to 0.

 # 4 EDPx In cutting feed, an external deceleration signal in the + direction for each axis is:

0: Invalid
1: Valid

 # 5 EDMx In cutting feed, an external deceleration signal in the - direction for each axis is:

0: Invalid
1: Valid

	#7	#6	#5	#4	#3	#2	#1	#0
1006			ZMIx		DIAx		ROSx	ROTx

[Input type] Parameter input
[Data type] Bit axis

NOTE
When at least one of these parameters is set, the power must be turned off before operation is continued.

ROTx, ROSx Setting linear or rotation axis.

ROSx	ROTx	Meaning
0	0	Linear axis (1)Inch/metric conversion is done. (2)All coordinate values are linear axis type. (Is not rounded in 0 to 360°) (3)Stored pitch error compensation is linear axis type (Refer to parameter No.3624)
0	1	Rotation axis (A type) (1)Inch/metric conversion is not done. Machine coordinate values are rounded in 0 to 360°. Absolute coordinate values are rounded or not rounded by parameter No.1008#0(ROAx) and #2(RRLx). (2)Stored pitch error compensation is the rotation type. (Refer to parameter No.3624) (3)Automatic reference position return (G28, G30) is done in the reference position return direction and the move amount does not exceed one rotation.
1	1	Rotation axis (B type) (1)Inch/metric conversion, absolute coordinate values and relative coordinate values are not done. (2)Machine coordinate values, absolute coordinate values and relative coordinate values are linear axis type. (Is not rounded in 0 to 360°). (3)Stored pitch error compensation is linear axis type (Refer to parameter No.3624) (4)Cannot be used with the rotation axis roll over function and the index table indexing function (M series)
Except for the above.		Setting is invalid (unused)

 # 3 DIAx The move command for each axis is based on:
0: Radius specification
1: Diameter specification

 # 5 ZMIx The direction of manual reference position return is:
0: + direction
1: - direction

	#7	#6	#5	#4	#3	#2	#1	#0
1007			G90x		RAAx			

[Input type] Parameter input
[Data type] Bit axis

 # 3 RAAx Rotary axis control is:
0: Not performed.
1: Performed.
When an absolute command is specified, the rotary axis control function determines the direction of rotation from the sign of the command value and determines an end coordinate from the absolute value of the command value.

NOTE
RAA is valid when bit 0 (ROA) of parameter No. 1008 is set to 1 and bit 1 (RAB) of parameter No. 1008 is set to 0.
To use this function, the option for rotary axis control is required.

 # 5 G90x A command for a rotary axis control (RAAx = 1) is:
0: Regarded as an absolute/incremental command according to the G90/G91 mode setting.
1: Regarded as an absolute command at all times.

	#7	#6	#5	#4	#3	#2	#1	#0
1008						RRLx	RABx	ROAx

[Input type] Parameter input
[Data type] Bit axis

NOTE
When at least one of these parameters is set, the power must be turned off before operation is continued.

 # 0 ROAx The roll-over function of a rotation axis is
0: Invalid
1: Valid

NOTE
ROAx specifies the function only for a rotation axis (for which bit 0 (ROTx) of parameter No.1006, is set to 1)

 # 1 RABx In the absolute commands, the axis rotates in the direction
0: In which the distance to the target is shorter.
1: Specified by the sign of command value.

> **NOTE**
> RABx is valid only when ROAx is 1.

 # 2 RRLx Relative coordinates are
0: Not rounded by the amount of the shift per one rotation
1: Rounded by the amount of the shift per one rotation

> **NOTE**
> 1 RRLx is valid only when ROAx is 1.
> 2 Assign the amount of the shift per one rotation in parameter No.1260.

	#7	#6	#5	#4	#3	#2	#1	#0
1013					ISEx	ISDx	ISCx	ISAx

[Input type] Parameter input
[Data type] Bit axis

> **NOTE**
> When at least one of these parameters is set, the power must be turned off before operation is continued.

 # 0 ISA
 # 1 ISC
 # 2 ISD
 # 3 ISE Increment system of each axis

Increment system	#3 ISE	#2 ISD	#1 ISC	#0 ISA
IS-A	0	0	0	1
IS-B	0	0	0	0
IS-C	0	0	1	0
IS-D	0	1	0	0
IS-E	1	0	0	0

	#7	#6	#5	#4	#3	#2	#1	#0
1015	DWT							

[Input type] Parameter input
[Data type] Bit path

 # 7 DWT When time for dwell per second is specified by P, the increment system:
0: Depends on the increment system
1: Does not depend on the increment system (1 ms)

1020	Program axis name for each axis

[Input type] Parameter input
[Data type] Byte axis
[Valid data range] 67,85 to 90

An axis name (axis name 1: parameter No. 1020) can be arbitrarily selected from 'A', 'B', 'C', 'U', 'V', 'W', 'X', 'Y', and 'Z'. (When G code system A is used with the lathe system, however, 'U', 'V', and 'W' are not selectable.) When bit 0 (EEA) of parameter No. 1000 is set to 1, the length of an axis name can be extended to three characters by setting axis name 2 (parameter No. 1025) and axis name 3 (parameter No. 1026) (extended axis name).
For axis names 2 and 3, a character from '0' to '9' and 'A' to 'Z' of ASCII code can be arbitrarily selected. However, the setting of axis name 3 for each axis is invalid if axis name 2 is not set. Moreover, if a character from '0' to '9' is set as axis name 2, do not use a character from 'A' to 'Z' as axis name 3.

(Tip) ASCII code

Axis name	X	Y	Z	A	B	C	U	V	W
Setting	88	89	90	65	66	67	85	86	87

When G code system A is used with the lathe system, and the character 'X','Y','Z', or 'C' is used as axis name 1 of an axis, a command with 'U','V','W', or 'H' specified for axis name 1 represents an incremental command for the axis.

NOTE
1 When a multiple repetitive canned cycle for turning is used, no character other than 'X','Y', and 'Z' can be used as the address of the axis.
2 An address other than addresses 'A', 'B', and 'C' cannot be used as the address of a rotary axis used for the function for tool length compensation in a specified direction or the tool center point control function.
3 When the custom macro function is enabled, the same extended axis name as a reserved word cannot be used. Such an extended axis name is regarded as a reserved word.
Because of reserved words of custom macros, extended axis names that start with the following two characters cannot be used:
AB,AC,AD,AN,AS,AT,AX,BC,BI,BP,CA,CL,CO,US,WH, WR,XO
4 In a macro call, no extended axis name can be used as an argument.

1022	Setting of each axis in the basic coordinate system

[Input type] Parameter input
[Data type] Byte axis
[Valid data range] 0 to 7

To determine a plane for circular interpolation, cutter compensation, and so forth (G17: Xp-Yp plane, G18: Zp-Xp plane, G19: Yp-Zp plane) and a 3-dimensional tool compensation space (XpYpZp), specify which of the basic three axes (X, Y, and Z) is used for each control axis, or a parallel axis of which basic axis is used for each control axis.

A basic axis (X, Y, or Z) can be specified only for one control axis.

Two or more control axes can be set as parallel axes for the same basic axis.

Setting	Meaning
0	Rotation axis (Neither the basic three axes nor a parallel axis)
1	X axis of the basic three axes
2	Y axis of the basic three axes
3	Z axis of the basic three axes
5	Axis parallel to the X axis
6	Axis parallel to the Y axis
7	Axis parallel to the Z axis

In general, the increment system and diameter/radius specification of an axis set as a parallel axis are to be set in the same way as for the basic three axes.

1023	Number of the servo axis for each axis

NOTE
When this parameter is set, the power must be turned off before operation is continued.

[Input type] Parameter input
[Data type] Byte axis
[Valid data range] 0 to Number of controlled axes

Set the servo axis for each control axis.
Usually set to same number as the control axis number.
The control axis number is the order number that is used for setting the axis-type parameters or axis-type machine signals

- With an axis for which Cs contour control/spindle positioning is to be performed, set -(spindle number) as the servo axis number.
  Example)
  When exercising Cs contour control on the fourth controlled axis by using the first spindle, set -1.
- For tandem controlled axes or electronic gear box (EGB) controlled axes, two axes need to be specified as one pair. So, make a setting as described below.
  Tandem axis:
  For a master axis, set an odd (1, 3, 5, 7, ...) servo axis number. For a slave axis to be paired, set a value obtained by adding 1 to the value set for the master axis.
  EGB axis:
  For a slave axis, set an odd (1, 3, 5, 7, ...) servo axis number. For a dummy axis to be paired, set a value obtained by adding 1 to the value set for the slave axis.

1025	Program axis name 2 for each axis

1026	Program axis name 3 for each axis

[Input type] Parameter input
[Data type] Byte axis
[Valid data range] 48 to 57, 65 to 90

When axis name extension is enabled (when bit 0 (EEA) of parameter No. 1000 is set to 1), the length of an axis name can be extended to a maximum of three characters by setting axis name 2 and axis name 3.

NOTE
If program axis name 2 is not set, program axis name 3 is invalid.

1031	Reference axis

[Input type] Parameter input
[Data type] Byte path
[Valid data range] 1 to Number of controlled axes

The unit of some parameters common to all axes such as those for dry run feedrate and single-digit F1 feedrate may vary according to the increment system. An increment system can be selected by a parameter on an axis-by-axis basis. So, the unit of those parameters is to match the increment system of a reference axis. Set which axis to use as a reference axis.

Among the basic three axes, the axis with the finest increment system is generally selected as a reference axis.

	#7	#6	#5	#4	#3	#2	#1	#0
1201					FPC	ZCL		ZPR

[Input type] Parameter input
[Data type] Bit path

 # 0 ZPR Automatic setting of a coordinate system when the manual reference position return is performed

0: Not set automatically
1: Set automatically

NOTE
ZPR is valid while a workpiece coordinate system function is not provided. If a workpiece coordinate system function is provided, making a manual reference position return always causes the workpiece coordinate system to be established on the basis of the workpiece zero point offset (parameters No. 1220 to No. 1226), irrespective of this parameter setting.

 # 2 ZCL Local coordinate system when the manual reference position return is performed

0: The local coordinate system is not canceled.
1: The local coordinate system is canceled.

NOTE
ZCL is valid when the workpiece coordinate system option is specified. In order to use the local coordinate system (G52), the workpiece coordinate system option is required.

 # 3 FPC When a floating reference position is set with a soft key, the relative position indication is:

0: Not preset to 0 (The relative position indication remains unchanged.)
1: Preset to 0.

	#7	#6	#5	#4	#3	#2	#1	#0
1202						G92		

[Input type] Parameter input
[Data type] Bit path

 # 2 G92 When the CNC has commands G52 to G59 specifying workpiece coordinate systems (optional function), if the G command for setting a coordinate system (G92 for M series, G50 for T series (or the G92 command in G command system B or C)) is specified,

0: G command is executed and no alarm is issued.
1: G command is not executed and an alarm (PS0010) is issued.

1240	Coordinate value of the reference position in the machine coordinate system

NOTE
When this parameter is set, the power must be turned off before operation is continued.

[Input type] Parameter input
[Data type] Real axis
[Unit of data] mm, inch, degree (machine unit)
[Minimum unit of data] Depend on the increment system of the applied axis
[Valid data range] 9 digit of minimum unit of data (refer to standard parameter setting table (A))
(When the increment system is IS-B, -999999.999 to +999999.999)
Set the coordinate values of the reference position in the machine coordinate system.

1241	Coordinate value of the second reference position in the machine coordinate system
1242	Coordinate value of the third reference position in the machine coordinate system
1243	Coordinate value of the fourth reference position in the machine coordinate system

[Input type] Parameter input
[Data type] Real axis
[Unit of data] mm, inch, degree (machine unit)
[Minimum unit of data] Depend on the increment system of the applied axis
[Valid data range] 9 digit of minimum unit of data (refer to standard parameter setting table (A))
(When the increment system is IS-B, -999999.999 to +999999.999)
Set the coordinate values of the second to fourth reference positions in the machine coordinate system.

1244	Coordinate value of the floating reference position in the machine coordinate system

[Input type] Parameter input
[Data type] Real axis
[Unit of data] mm, inch, degree (machine unit)
[Minimum unit of data] Depend on the increment system of the applied axis
[Valid data range] 9 digit of minimum unit of data (refer to standard parameter setting table (A))
(When the increment system is IS-B, -999999.999 to +999999.999)
Set the coordinate values of the floating reference position in the machine coordinate system.

1250	Coordinate system of the reference position used when automatic coordinate system setting is performed

[Input type] Parameter input
[Data type] Real axis
[Unit of data] mm, inch, degree (input unit)
[Minimum unit of data] Depend on the increment system of the applied axis
[Valid data range] 9 digit of minimum unit of data (refer to standard parameter setting table (A))
(When the increment system is IS-B, -999999.999 to +999999.999)
Set the coordinate system of the reference position on each axis to be used for setting a coordinate system automatically.

1260	The shift amount per one rotation of a rotation axis

NOTE
When this parameter is set, the power must be turned off before operation is continued.

[Input type] Parameter input
[Data type] Real axis
[Unit of data] Degree
[Minimum unit of data] Depend on the increment system of the applied axis
[Valid data range] 0 or positive 9 digit of minimum unit of data (refer to the standard parameter setting table (B))
(When the increment system is IS-B, 0.0 to +999999.999)
Set the shift amount per one rotation of a rotation axis.
For the rotation axis used for cylindrical interpolation, set the standard value.

	#7	#6	#5	#4	#3	#2	#1	#0
1300	BFA						NAL	OUT

[Input type] Setting input
[Data type] Bit path

 # 0 OUT The area inside or outside of the stored stroke check 2 is set as an inhibition area
0: Inside
1: Outside

 # 1 NAL When the tool enters the inhibition area of stored stroke limit 1:
0: The overtravel alarm signal is not output.
1: The overtravel alarm signal is output, and the tool is decelerated to a stop.
If manual operation is in progress at this time, the alarm is not output.

NOTE
When this parameter is set to 1, the alarm is issued if the tool enters stored stroke limit 1 during automatic operation.

 # 7 BFA When the stored stroke check 1, 2, or 3 alarm is issued, an interference alarm is issued with the inter-path interference check function (T series), or a chuck/tail stock barrier (T series) alarm is issued:
0: The tool stops after entering the prohibited area.
1: The tool stops before the prohibited area.

	#7	#6	#5	#4	#3	#2	#1	#0
1301		OTS				NPC		

[Input type] Setting input
[Data type] Bit path

 # 2 NPC As part of the stroke limit check performed before movement, the movement specified in G31 (skip) and G37 (automatic tool length measurement) blocks is:
0: Checked
1: Not checked

NOTE
This parameter is valid only when the option for stroke limit check before movement is selected.

 # 6 OTS When the overtravel alarm is issued:
0: The overtravel alarm signal is not output to the PMC.
1: The overtravel alarm signal is output to the PMC.

1320	Coordinate value I of stored stroke check 1 in the positive direction on each axis
1321	Coordinate value I of stored stroke check 1 in the negative direction on each axis

[Input type] Parameter input
[Data type] Real axis
[Unit of data] mm, inch, degree (machine unit)
[Minimum unit of data] Depend on the increment system of the applied axis
[Valid data range] 9 digit of minimum unit of data (refer to standard parameter setting table (A))
(When the increment system is IS-B, -999999.999 to +999999.999)
Set the coordinate value of stored stroke check 1 on each axis in the + or - direction in the machine coordinate system.

NOTE
1 Specify diameter values for any axes for which diameter programming is specified.
2 The area outside the area set by parameter No. 1320 and No. 1321 is a prohibited area.

1322	Coordinate value of stored stroke check 2 in the positive direction on each axis

1323	Coordinate value of stored stroke check 2 in the negative direction on each axis

[Input type] Setting input
[Data type] Real axis
[Unit of data] mm, inch, degree (machine unit)
[Minimum unit of data] Depend on the increment system of the applied axis
[Valid data range] 9 digit of minimum unit of data (refer to standard parameter setting table (A))
(When the increment system is IS-B, -999999.999 to +999999.999)
Set the coordinate value of stored stroke check 2 on each axis in the + or - direction in the machine coordinate system.

NOTE
1 Specify diameter values for any axes for which diameter programming is specified.
2 Whether the inside area or outside area is a prohibited area is set using bit 0 (OUT) of parameter No. 1300.

1324	Coordinate value of stored stroke check 3 in the positive direction on each axis

1325	Coordinate value of stored stroke check 3 in the negative direction on each axis

[Input type] Setting input
[Data type] Real axis
[Unit of data] mm, inch, degree (machine unit)
[Minimum unit of data] Depend on the increment system of the applied axis
[Valid data range] 9 digit of minimum unit of data (refer to standard parameter setting table (A))
(When the increment system is IS-B, -999999.999 to +999999.999)
Set the coordinate value of stored stroke check 3 on each axis in the + or - direction in the machine coordinate system.

NOTE
1 Specify diameter values for any axes for which diameter programming is specified.
2 The area inside the area set by parameter No. 1324 and No. 1325 is a prohibited area.

1326	Coordinate value II of stored stroke check 1 in the negative direction on each axis
1327	Coordinate value II of stored stroke check 1 in the negative direction on each axis

[Input type] Parameter input
[Data type] Real axis
[Unit of data] mm, inch, degree (machine unit)
[Minimum unit of data] Depend on the increment system of the applied axis
[Valid data range] 9 digit of minimum unit of data (refer to standard parameter setting table (A))
(When the increment system is IS-B, -999999.999 to +999999.999)
Set the coordinate value of stored stroke check 1 on each axis in the + or - direction in the machine coordinate system.
When the stored stroke check switch signal EXLM is set to 1, or the stored stroke check switch signal for each axis direction +EXLx is set to 1, parameter No. 1326 and No. 1327 are used for stroke check instead of parameter No.1320 and No. 1321.

NOTE
1 Specify diameter values for any axes for which diameter programming is specified.
2 The area outside the area set by parameter No. 1326 and No. 1327 is a prohibited area.
3 The EXLM signal is valid only when bit 2 (LMS) of parameter No. 1300 is set to 1.
4 The +EXLx signal is valid only when bit 0 (DLM) of parameter No. 1301 is set to 1.

	#7	#6	#5	#4	#3	#2	#1	#0
1401				RF0			LRP	RPD

[Input type] Parameter input
[Data type] Bit path

 # 0 RPD Manual rapid traverse during the period from power-on time to the completion of the reference position return.
0: Disabled (Jog feed is performed.)
1: Enabled

 # 1 LRP Positioning (G00)
0: Positioning is performed with non-linear type positioning so that the tool moves along each axis independently at rapid traverse.
1: Positioning is performed with linear interpolation so that the tool moves in a straight line.
When using three-dimensional coordinate conversion, set this parameter to 1.

 # 4 RF0 When cutting feedrate override is 0% during rapid traverse,
0: The machine tool does not stop moving.
1: The machine tool stops moving.

	#7	#6	#5	#4	#3	#2	#1	#0
1402				JRV				NPC

[Input type] Parameter input
[Data type] Bit path

 # 0 NPC Feed per revolution without the position coder (function for converting feed per revolution F to feed per minute F in the feed per revolution mode (G95)) is:
0: Not used
1: Used

 # 4 JRV Jog feed or incremental feed is
0: Performed at feed per minute.
1: Performed at feed per rotation.

NOTE
1 Specify a feedrate in parameter No.1423.
2 For the machining center system, the option for threading/synchronous feed is required.

	#7	#6	#5	#4	#3	#2	#1	#0
1403			HTG					

[Input type] Parameter input
[Data type] Bit path

 # 5 HTG The feedrate for helical interpolation/helical involute interpolation/ three-dimensional circular interpolation is:
0: Specified using the feedrate along the tangent to an arc/involute curve/three-dimensional arc
1: Specified using the feedrate along axes including a linear axis (specified axes other than the circular interpolation axis in the case of three-dimensional circular interpolation)

	#7	#6	#5	#4	#3	#2	#1	#0
1404						FM3		

[Input type] Parameter input
[Data type] Bit path

 # 2 FM3 The increment system of an F command without a decimal point in feed per minute is:
0: 1 mm/min (0.01 inch/min for inch input)
1: 0.001 mm/min (0.00001 inch/min for inch input)

	#7	#6	#5	#4	#3	#2	#1	#0
1405								
							FR3	

[Input type] Parameter input
[Data type] Bit path

 # 1 FR3 The increment system of an F command without a decimal point in feed per revolution is:
0: 0.01 mm/rev (0.0001 inch/rev for inch input)
1: 0.001 mm/rev (0.00001 inch/rev for inch input)

	#7	#6	#5	#4	#3	#2	#1	#0
1408								RFDx

[Input type] Parameter input
[Data type] Bit axis

 # 0 RFDx Feedrate control on a rotation axis is exercised using:
0: Conventional method
1: Method that specifies a feedrate on the virtual circle of the rotation axis

1410	Dry run rate

[Input type] Parameter input
[Data type] Real path
[Unit of data] mm/min, inch/min, degree/min (machine unit)
[Minimum unit of data] Depend on the increment system of the reference axis
[Valid data range] Refer to the standard parameter setting table (C)
(When the increment system is IS-B, 0.0 to +999000.0)
Set the dry run rate at the 100% position on the jog feedrate specification dial. The unit of data depends on the increment system of the reference axis.

1414	
	Feedrate for retrace

[Input type] Parameter input
[Data type] Real path
[Unit of data] mm/min, inch/min, degree/min (machine unit)
[Minimum unit of data] Depend on the increment system of the reference axis
[Valid data range] Refer to the standard parameter setting table (C)
(When the increment system is IS-B, 0.0 to +999000.0)
Set a cutting feedrate for retrace operation. When 0 is set, a retrace operation is performed at a programmed feedrate.

1420	Rapid traverse rate for each axis

[Input type] Parameter input
[Data type] Real axis
[Unit of data] mm/min, inch/min, degree/min (machine unit)
[Minimum unit of data] Depend on the increment system of the applied axis
[Valid data range] Refer to the standard parameter setting table (C)
(When the increment system is IS-B, 0.0 to +999000.0)
Set the rapid traverse rate when the rapid traverse override is 100% for each axis.

1421	F0 rate of rapid traverse override for each axis

[Input type] Parameter input
[Data type] Real axis
[Unit of data] mm/min, inch/min, degree/min (machine unit)
[Minimum unit of data] Depend on the increment system of the applied axis
[Valid data range] Refer to the standard parameter setting table (C)
(When the increment system is IS-B, 0.0 to +999000.0)
Set the F0 rate of the rapid traverse override for each axis.

1423	Feedrate in manual continuous feed (jog feed) for each axis

[Input type] Parameter input
[Data type] Real axis
[Unit of data] mm/min, inch/min, degree/min (machine unit)
[Minimum unit of data] Depend on the increment system of the applied axis
[Valid data range] Refer to the standard parameter setting table (C)
(When the increment system is IS-B, 0.0 to +999000.0)

(1) When JRV, bit 4 of parameter No.1402, is set to 0 (feed per minute), specify a jog feedrate (feed per minute) under an override of 100%.
(2) When JRV, bit 4 of parameter No.1402, is set to 1 (feed per revolution), specify a jog feedrate (feed per revolution) under an override of 100%.

NOTE
This parameter is clamped to the axis-by-axis manual rapid traverse rate (parameter No. 1424).

1424	Manual rapid traverse rate for each axis

[Input type] Parameter input
[Data type] Real axis
[Unit of data] mm/min, inch/min, degree/min (machine unit)
[Minimum unit of data] Depend on the increment system of the applied axis
[Valid data range] Refer to the standard parameter setting table (C)
(When the increment system is IS-B, 0.0 to +999000.0)
Set the rate of manual rapid traverse when the rapid traverse override is 100% for each axis.

NOTE
1 If 0 is set, the rate set in parameter 1420 (rapid traverse rate for each axis) is assumed.
2 When manual rapid traverse is selected (bit 0 (RPD) of parameter No. 1401 is set to 1), manual feed is performed at the feedrate set in this parameter, regardless of the setting of bit 4 (JRV) of parameter No. 1402.

1425	FL rate of the reference position return for each axis

[Input type] Parameter input
[Data type] Real axis
[Unit of data] mm/min, inch/min, degree/min (machine unit)
[Minimum unit of data] Depend on the increment system of the applied axis
[Valid data range] Refer to the standard parameter setting table (C)
(When the increment system is IS-B, 0.0 to +999000.0)
Set feedrate (FL rate) after deceleration when the reference position return is performed for each axis.

1427	External deceleration rate of rapid traverse for each axis

[Input type] Parameter input
[Data type] Real axis
[Unit of data] mm/min, inch/min, degree/min (machine unit)
[Minimum unit of data] Depend on the increment system of the applied axis
[Valid data range] Refer to the standard parameter setting table (C)
(When the increment system is IS-B, 0.0 to +999000.0)
Set the external deceleration rate of rapid traverse for each axis.

1428	Reference position return feedrate for each axis

[Input type] Parameter input
[Data type] Real axis
[Unit of data] mm/min, inch/min, degree/min (machine unit)
[Minimum unit of data] Depend on the increment system of the applied axis
[Valid data range] Refer to the standard parameter setting table (C)
(When the increment system is IS-B, 0.0 to +999000.0)

This parameter sets a rapid traverse rate for reference position return operation using deceleration dogs, or for reference position return operation before a reference position is set.
This parameter is also used to set a feedrate for the rapid traverse command (G00) in automatic operation before a reference position is set.

NOTE

1 To this feedrate setting (100%), a rapid traverse override (F0, 25, 50, or 100%) is applicable.
2 For automatic return after completion of reference position return and machine coordinate system establishment, the normal rapid traverse rate is used.
3 As a manual rapid traverse rate before machine coordinate system establishment by reference position return, the jog feedrate or manual rapid traverse rate can be selected with bit 0 (RPD) of parameter No. 1401.

	Before coordinate system establishment	After coordinate system establishment
Automatic reference position return (G28)	No.1428	No.1420
Automatic rapid traverse (G00)	No.1428	No.1420
Manual reference position return *1	No.1428	No.1428 *3
Manual rapid traverse	No.1423 *2	No.1424

4 When parameter No. 1428 is set to 0, the following parameter-set feedrates are applied.

	Before coordinate system establishment	After coordinate system establishment
Automatic reference position return (G28)	No.1420	No.1420
Automatic rapid traverse (G00)	No.1420	No.1420
Manual reference position return *1	No.1424	No.1424 *3
Manual rapid traverse	No.1423 *2	No.1424

1420: rapid traverse rate
1423: Jog feedrate
1424: Manual rapid traverse rate
*1 : By using bit 2 (JZR) of parameter No. 1401, the jog feedrate can be used for manual reference position return at all times.
*2 : When bit 0 (RPD) of parameter No. 1401 is set to 1, the setting of parameter No. 1424 is used.
*3 : When rapid traverse is used for reference position return without dogs or manual reference position return after reference position establishment, regardless of the deceleration dog, the feedrate for manual reference position return based on these functions is used (the setting of bit 1 (DLF) of parameter No. 1404 is followed).

1430	Maximum cutting feedrate for each axis

[Input type] Parameter input
[Data type] Real axis
[Unit of data] mm/min, inch/min, degree/min (machine unit)
[Minimum unit of data] Depend on the increment system of the applied axis
[Valid data range] Refer to the standard parameter setting table (C)
(When the increment system is IS-B, 0.0 to +999000.0)
Specify the maximum cutting feedrate for each axis.

1432	Maximum cutting feedrate for all axes in the acceleration/deceleration before interpolation

[Input type] Parameter input
[Data type] Real axis
[Unit of data] mm/min, inch/min, degree/min (machine unit)
[Minimum unit of data] Depend on the increment system of the applied axis
[Valid data range] Refer to the standard parameter setting table (C)
(When the increment system is IS-B, 0.0 to +999000.0)
Set a maximum cutting feedrate for each axis in the acceleration/deceleration before interpolation mode such as AI contour control. When the acceleration/deceleration before interpolation mode is not set, the maximum cutting feedrate set in parameter No. 1430 is used.

1434	Maximum manual handle feedrate for each axis

[Input type] Parameter input
[Data type] Real axis
[Unit of data] mm/min, inch/min, degree/min (machine unit)
[Minimum unit of data] Depend on the increment system of the applied axis
[Valid data range] Refer to the standard parameter setting table (C)
(When the increment system is IS-B, 0.0 to +999000.0)
Set a maximum manual handle feedrate for each axis in case of maximum manual handle feedrate switch signal HNDLF<Gn023.3>=1.

1441	External deceleration rate setting 2 for each axis in rapid traverse

[Input type] Parameter input
[Data type] Real axis
[Unit of data] mm/min, inch/min, degree/min (machine unit)
[Minimum unit of data] Depend on the increment system of the applied axis
[Valid data range] Refer to the standard parameter setting table (C)
(When the increment system is IS-B, 0.0 to +999000.0)
Set external deceleration rate 2 for each axis in rapid traverse.

1444	External deceleration rate setting 3 for each axis in rapid traverse

[Input type] Parameter input
[Data type] Real axis
[Unit of data] mm/min, inch/min, degree/min (machine unit)
[Minimum unit of data] Depend on the increment system of the applied axis
[Valid data range] Refer to the standard parameter setting table (C)
(When the increment system is IS-B, 0.0 to +999000.0)
Set external deceleration rate 3 for each axis in rapid traverse.

1450	Change of feedrate for one graduation on the manual pulse generator during one-digit F feed code

[Input type] Parameter input
[Data type] Byte path
[Valid data range] 1 to 127
Set the constant that determines the change in feedrate as the manual pulse generator is rotated one graduation during one-digit F feed code.

$$\Delta F = \frac{F \max i}{100n} \quad \text{(where, i=1 or 2)}$$

In the above equation, set n. That is, the number of revolutions of the manual pulse generator, required to reach feedrate Fmaxi is obtained. Fmaxi refers to the upper limit of the feedrate for a one-digit F code feed command, and set it in parameters No. 1460 or No. 1461.
Fmax1: Upper limit of the feedrate for F1 to F4 (parameter No. 1460)
Fmax2: Upper limit of the feedrate for F5 to F9 (parameter No. 1461)

1451	Feedrate for F1
to	to
1459	Feedrate for F9

[Input type] Setting input
[Data type] Real path
[Unit of data] mm/min, inch/min, degree/min (machine unit)
[Minimum unit of data] Depend on the increment system of the reference axis
[Valid data range] Refer to the standard parameter setting table (C)
(When the increment system is IS-B, 0.0 to +999000.0)
These parameters set the feedrates for one-digit F code feed commands F1 to F9. When a one-digit F code feed command is specified, and the feedrate is changed by turning the manual pulse generator, the parameter-set value also changes accordingly.

1460	
	Upper limit of feedrate for F1 to F4

1461	
	Upper limit of feedrate for F5 to F9

[Input type] Parameter input
[Data type] Real path
[Unit of data] mm/min, inch/min, degree/min (machine unit)
[Minimum unit of data] Depend on the increment system of the reference axis
[Valid data range] Refer to the standard parameter setting table (C)
(When the increment system is IS-B, 0.0 to +999000.0)
Set the upper limit of feedrate for the one-digit F code feed command. As the feedrate increases by turning the manual pulse generator, the feedrate is clamped when it reaches the upper limit set. If a one-digit F feed command F1 to F4 is executed, the upper limit is that set in parameter No. 1460. If a one-digit F code feed command F5 to F9 is executed, the upper limit is that set in parameter No. 1461.

1465	**Radius of a virtual circle when a feedrate is specified on the virtual circle of a rotation axis**

[Input type] Parameter input
[Data type] Real axis
[Unit of data] mm, inch (input unit)
[Minimum unit of data] Depend on the increment system of the applied axis
[Valid data range] Refer to the standard parameter setting table (B)
Set the radius of a virtual circle when a feedrate on the virtual circle of a rotation axis is specified.
If 0 is set for a rotation axis, the axis is excluded from feedrate calculation.
If the input unit is the inch, enter a value in inches.
The data is then converted to a millimeter value and displayed.

	#7	#6	#5	#4	#3	#2	#1	#0
1490	PGF							

[Data type] Bit path

 # 7 PGF The feedrate specified for circular interpolation, involute interpolation, spiral/conical interpolation, and NURBS interpolation in the high-speed program check mode is:

0: The dry run feedrate.
At this time, manual feedrate override signals *JV0 to *JV15 <Gn010 to Gn011> can be used.

1: The maximum feedrate specified by the CNC.

> ⚠ **CAUTION**
> If this parameter is set to 1, feedrate clamp, override, and dry run for circular interpolation, involute interpolation, spiral/conical interpolation, and NURBS interpolation are disabled. If a movement around a stroke limit is specified, therefore, a stroke limit check cannot sometimes be made correctly.

	#7	#6	#5	#4	#3	#2	#1	#0
1601			NCI					

[Input type] Parameter input
[Data type] Bit path

 # 5 NCI An in-position check:

0: Confirms that the specified feedrate becomes 0 (the acceleration/deceleration delay becomes 0) at deceleration time and that the machine position has reached a specified position (the servo positional deviation is within the in-position width set by parameter No. 1826).

1: Confirms only that the specified feedrate becomes 0 (the acceleration/deceleration delay becomes 0) at deceleration time.

	#7	#6	#5	#4	#3	#2	#1	#0
1604								SHP

[Input type] Parameter input
[Data type] Bit path

 # 0 SHP When automatic operation is started, the state equivalent to the specification of G5.1Q1 for AI contour control is:

0: Not set
1: Set

Upon reset, the state where G5.1Q1 is specified is set.

	#7	#6	#5	#4	#3	#2	#1	#0
1606								MNJx

[Input type] Parameter input
[Data type] Bit axis

 # 0 MNJx In manual handle interrupt or automatic manual simultaneous operation (interrupt type):
0: Only cutting feed acceleration/deceleration is enabled, and jog feed acceleration/deceleration is disabled.
1: Both cutting feed acceleration/deceleration and jog feed acceleration/deceleration are applied.

	#7	#6	#5	#4	#3	#2	#1	#0
1610							CTBx	CTLx

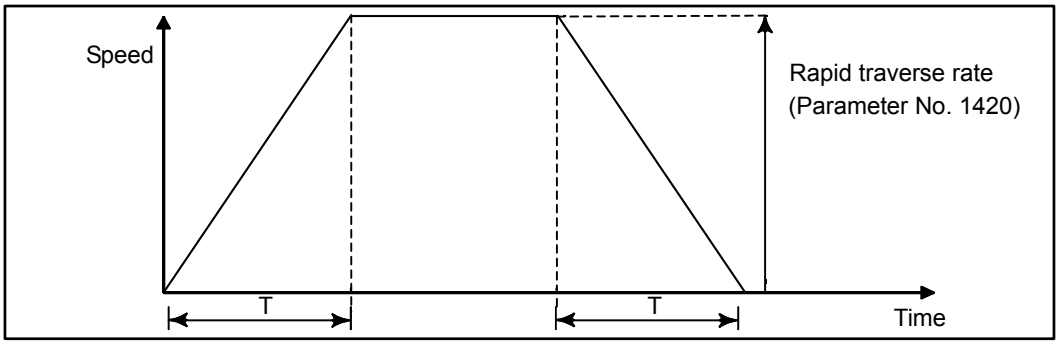
[Input type] Parameter input
[Data type] Bit axis

 # 0 CTLx Acceleration/deceleration in cutting feed or dry run
0: Exponential acceleration/deceleration is applied.
1: Linear acceleration/deceleration after interpolation is applied.

 # 1 CTBx Acceleration/deceleration in cutting feed or dry run
0: Exponential acceleration/deceleration or linear acceleration/ deceleration is applied.
(depending on the setting in CTLx, bit 0 of parameter No.1610)
1: Bell-shaped acceleration/deceleration is applied.

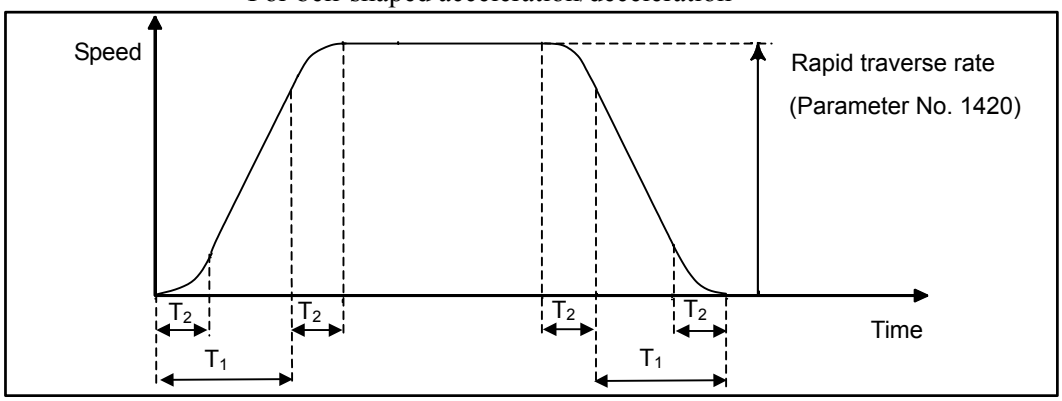
1620	**Time constant T or T1 used for linear acceleration/deceleration or bell-shaped acceleration/deceleration in rapid traverse for each axis**

[Input type] Parameter input
[Data type] Word axis
[Unit of data] msec
[Valid data range] 0 to 4000
Specify a time constant used for acceleration/deceleration in rapid traverse.
[Example]
For linear acceleration/deceleration



T : Setting of parameter No. 1620

For bell-shaped acceleration/deceleration



T_1 : Setting of parameter No. 1620
T_2 : Setting of parameter No. 1621
(However, $T_1 \geq T_2$ must be satisfied.)
Total acceleration (deceleration) time : $T_1 + T_2$
Time for linear portion : $T_1 - T_2$
Time for curve portion : $T_2 \times 2$

1622 **Time constant of acceleration/deceleration in cutting feed for each axis**

[Input type] Parameter input
[Data type] Word axis
[Unit of data] msec
[Valid data range] 0 to 4000

Set the time constant used for exponential acceleration/deceleration in cutting feed, bell-shaped acceleration/deceleration after interpolation or linear acceleration/deceleration after interpolation in cutting feed for each axis. Which type to use is selected with bits 1(CTBx) and 0(CTLx) of parameter No.1610. Except for special applications, the same time constant must be set for all axes in this parameter. If the time constants set for the axes differ from each other, proper straight lines and arcs cannot be obtained.

1624 **Time constant of acceleration/deceleration in jog feed for each axis.**

[Input type] Parameter input
[Data type] Word axis
[Unit of data] msec
[Valid data range] 0 to 4000

Set the time constant used for acceleration/deceleration in jog feed for each axis.

1660	Maximum allowable acceleration rate in acceleration/deceleration before interpolation for each axis

[Input type] Parameter input
[Data type] Real axis
[Unit of data] mm/sec^2, inch/sec^2, degree/sec^2 (machine unit)
[Minimum unit of data] Depend on the increment system of the applied axis
[Valid data range] Refer to the standard parameter setting table (D)
(When the machine system is metric system, 0.0 to +100000.0. When the machine system is inch system, machine, 0.0 to +10000.0.)
Set a maximum allowable acceleration rate in acceleration/ deceleration before interpolation for each axis.
If a value greater than 100000.0 is set, the value is clamped to 100000.0.
If 0 is set, the specification of 100000.0 is assumed. If 0 is set for all axes, however, acceleration/deceleration before interpolation is not performed.

If a maximum allowable acceleration rate set for one axis is greater than a maximum allowable acceleration rate set for another axis by a factor or 2 or more, the feedrate at a corner where the direction of travel abruptly changes can decrease temporarily.

1671	Maximum allowable acceleration rate in acceleration/deceleration before interpolation for linear rapid traverse for each axis, or maximum allowable reference acceleration rate in optimum torque acceleration/deceleration

[Input type] Parameter input
[Data type] Real axis
[Unit of data] mm/sec^2, inch/sec^2, degree/sec^2 (machine unit)
[Minimum unit of data] Depend on the increment system of the applied axis
[Valid data range] Refer to the standard parameter setting table (D)
(When the machine system is metric system, 0.0 to +100000.0. When the machine system is inch system, machine, 0.0 to +10000.0.)

(1) Set a maximum allowable acceleration rate in acceleration/ deceleration before interpolation for linear rapid traverse.
If a value greater than 100000.0, the value is clamped to 100000.0.
If 0 is set, the specification of the following is assumed:
1000.0 mm/sec^2
100.0 inch/sec^2
100.0 degrees/sec^2
If 0 is specified for all axes, however, acceleration/deceleration before interpolation is not performed.

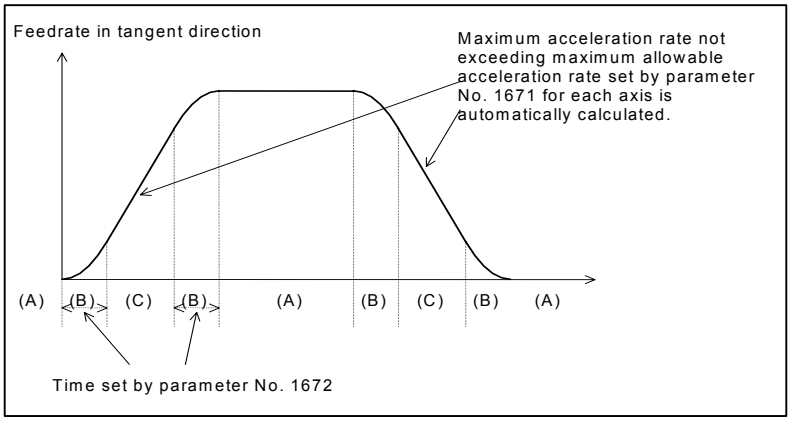
(2) Maximum allowable reference acceleration rate in optimum torque acceleration/deceleration

1672	**Acceleration change time of bell-shaped acceleration/deceleration before interpolation for linear rapid traverse, or acceleration change time of bell-shaped acceleration/deceleration in optimum torque acceleration/deceleration**

[Input type] Parameter input
[Data type] 2-word path
[Unit of data] msec
[Valid data range] 0 to 200

(1) Set an acceleration change time of bell-shaped acceleration/ deceleration for linear rapid traverse (time for changing from the state of constant feedrate (A) to the state of constant acceleration/deceleration (C) at the acceleration rate calculated from the acceleration rate set in parameter No. 1671: time of (B) in the figure below).

(2) Set an acceleration change time of bell-shaped acceleration/ deceleration in optimum torque acceleration/deceleration (time for changing from the state of constant feedrate (A) to the state of acceleration/deceleration (C) at the acceleration rate calculated from optimum torque acceleration/deceleration: time of (B) in the figure below).



1710	Minimum deceleration ratio (MDR) for inner circular cutting feedrate change by automatic corner override

[Input type] Parameter input
[Data type] Byte path
[Unit of data] %
[Valid data range] 0 to 100

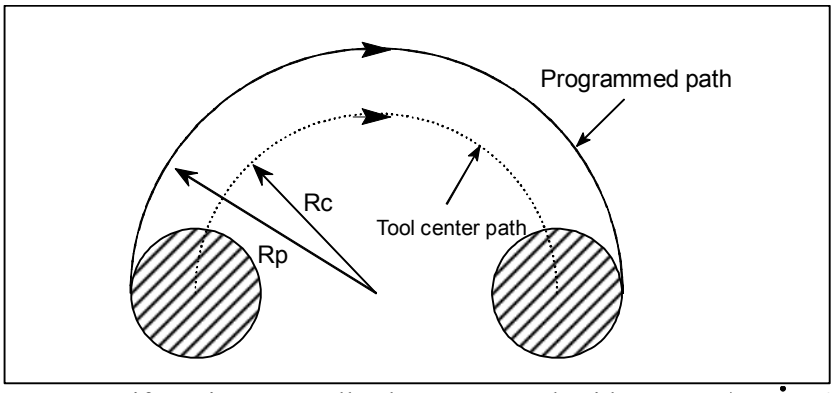
Set a minimum deceleration ratio (MDR) for an inner circular cutting feedrate change by automatic corner override.

In the case of circular cutting offset inward, the actual feedrate is determined by a specified feedrate (F) as follows:

$$F \times \frac{Rc}{Rp}$$

(Rc:Radius of tool center path
Rp:Programmed radius)

Thus, the feedrate along the programmed path satisfies the specified value of F.



However, if Rc is too small when compared with Rp, Rc/Rp $\fallingdotseq$ 0 results to stop the tool. So, a minimum deceleration ratio (MDR) is set, and the feedrate of the tool is set to F×(MDR) when Rc/Rp ≤ MDR.

1711	Inner determination angle (θp) for inner corner override

[Input type] Parameter input
[Data type] Real path
[Unit of data] deg
[Minimum unit of data] Depend on the increment system of the reference axis
[Valid data range] 2 to 178

Set an inner determination angle for inner corner override in automatic corner overriding.

1712	Override value for inner corner override

[Input type] Parameter input
[Data type] Byte path
[Unit of data] %
[Valid data range] 1 to 100

Set an inner corner override value in automatic corner overriding.

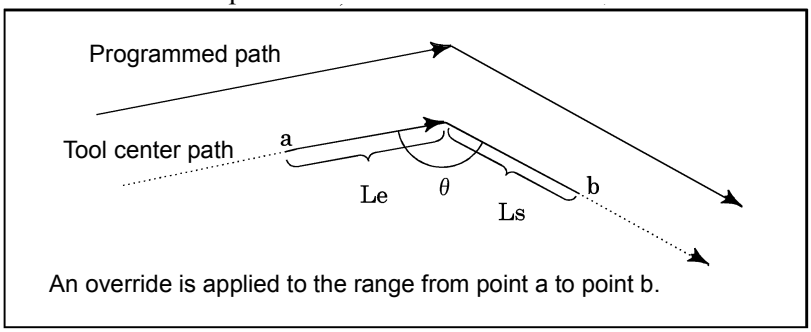
1713	Start distance (Le) for inner corner override

[Input type] Setting input
[Data type] Real path
[Unit of data] mm, inch (input unit)
[Minimum unit of data] Depend on the increment system of the reference axis
[Valid data range] 9 digit of minimum unit of data (refer to standard parameter setting table (A))
(When the increment system is IS-B, -999999.999 to +999999.999)
Set a start distance for inner corner override in automatic corner overriding.

1714	End distance (Ls) for inner corner override

[Input type] Setting input
[Data type] Real path
[Unit of data] mm, inch (input unit)
[Minimum unit of data] Depend on the increment system of the reference axis
[Valid data range] 9 digit of minimum unit of data (refer to standard parameter setting table (A))
(When the increment system is IS-B, -999999.999 to +999999.999)
Set an end distance for inner corner override in automatic corner overriding.

When $\theta \leq \theta p$, an inner corner is assumed. (Parameter No. 1711 is used to set θp.)
When a corner is determined to be an inner corner, an override is applied to the feedrate in the range of Le in the previous block from the intersection of the corner and in the range of Ls in the next block from the intersection of the corner.
Distances Le and Ls represent linear distances from the intersection of a corner to points on the tool center path.
Le and Ls are set in parameter No. 1713 and No. 1714.



1732	Minimum allowable feedrate for the deceleration function based on acceleration in circular interpolation

[Input type] Parameter input
[Data type] Real path
[Unit of data] mm/min, inch/min, degree/min (machine unit)
[Minimum unit of data] Depend on the increment system of the reference axis
[Valid data range] Refer to the standard parameter setting table (C)
(When the increment system is IS-B, 0.0 to +999000.0)

With the deceleration function based on acceleration in circular interpolation, an optimum feedrate is automatically calculated so that acceleration produced by changing the move direction in circular interpolation does not exceed the maximum allowable acceleration rate specified in parameter No. 1735.
If the radius of an arc is very small, a calculated feedrate may become too low.
In such a case, the feedrate is prevented from decreasing below the value specified in this parameter.

NOTE
During involute interpolation, the minimum allowable feedrate of "clamping of acceleration near a basic circle" in involute interpolation automatic feedrate control is used.

1735	Maximum allowable acceleration rate for the deceleration function based on acceleration in circular interpolation for each axis

[Input type] Parameter input
[Data type] Real axis
[Unit of data] mm/sec^2, $inch/sec^2$, $degree/sec^2$ (machine unit)
[Minimum unit of data] Depend on the increment system of the applied axis
[Valid data range] Refer to the standard parameter setting table (D)
(When the machine system is metric system, 0.0 to +100000.0. When the machine system is inch system, machine, 0.0 to +10000.0.)

Set a maximum allowable acceleration rate for the deceleration function based on acceleration in circular interpolation.
Feedrate is controlled so that acceleration produced by changing the move direction in circular interpolation does not exceed the value specified in this parameter.
For an axis with 0 set in this parameter, the deceleration function based on acceleration is disabled.
If a different value is set in this parameter for each axis, a feedrate is determined from the smaller of the acceleration rates specified for the two circular axes.

NOTE
During involute interpolation, the minimum allowable feedrate of "clamping of acceleration near a basic circle" in involute interpolation automatic feedrate control is used.

1737	Maximum allowable acceleration rate for the deceleration function based on acceleration in AI contour control for each axis

[Input type] Parameter input
[Data type] Real axis
[Unit of data] mm/sec^2, inch/sec^2, degree/sec^2 (machine unit)
[Minimum unit of data] Depend on the increment system of the applied axis
[Valid data range] Refer to the standard parameter setting table (D)
(When the machine system is metric system, 0.0 to +100000.0. When the machine system is inch system, machine, 0.0 to +10000.0.)
Set a maximum allowable acceleration rate produced by changing the tool move direction.
For an axis with 0 set in this parameter, the deceleration function based on acceleration is disabled. If 0 is set for all axes, the deceleration function based on acceleration is not performed.
In circular interpolation, however, the deceleration function based on feedrate control using acceleration in circular interpolation (parameter No. 1735) is enabled.

1738	Minimum allowable feedrate for the deceleration function based on acceleration in AI contour control

[Input type] Parameter input
[Data type] Real path
[Unit of data] mm/min, inch/min, degree/min (machine unit)
[Minimum unit of data] Depend on the increment system of the reference axis
[Valid data range] Refer to the standard parameter setting table (C)
(When the increment system is IS-B, 0.0 to +999000.0)
With the deceleration function based on acceleration in AI contour control, a feedrate most suitable for a desired figure is automatically calculated.
Depending on the figure, however, the calculated feedrate may become too low.
In such a case, the feedrate is prevented from decreasing below the value specified in this parameter.
If overriding using the deceleration function based on cutting load is enabled, a feedrate lower than the minimum allowable feedrate may be used.

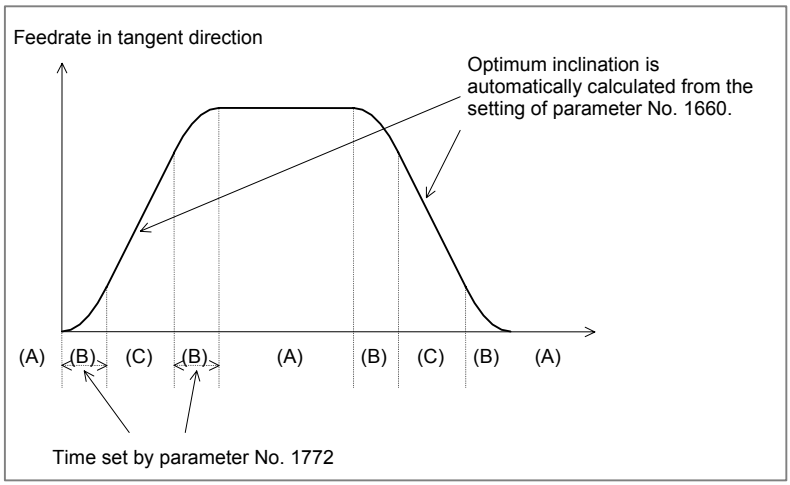
1769	Time constant for acceleration/deceleration after cutting feed interpolation in the acceleration/deceleration before interpolation mode

[Input type] Parameter input
[Data type] Word axis
[Unit of data] msec
[Valid data range] 0 to 4000
In the acceleration/deceleration before interpolation mode as in AI contour control, not the ordinary time constant (parameter No. 1622) but the value of this parameter is used.
Be sure to specify the same time constant value for all axes except for a special application. If different values are set, correct linear and circular figures cannot be obtained.

1772	Acceleration change time of bell-shaped acceleration/deceleration before interpolation

[Input type] Parameter input
[Data type] 2-word path
[Unit of data] msec
[Valid data range] 0 to 200

Set an acceleration change time of bell-shaped acceleration/ deceleration before interpolation (time for changing from the state of constant feedrate (A) to the state of constant acceleration/deceleration (C) at the acceleration rate calculated from the acceleration rate set in parameter No. 1660: time of (B) in the figure below).



1783	Maximum allowable feedrate difference for feedrate determination based on corner feedrate difference

[Input type] Parameter input
[Data type] Real axis
[Unit of data] mm/min, inch/min, degree/min (machine unit)
[Minimum unit of data] Depend on the increment system of the applied axis
[Valid data range] Refer to the standard parameter setting table (C)
(When the increment system is IS-B, 0.0 to +999000.0)

If a feedrate component change for each axis exceeding the value set in this parameter occurs at the joint of blocks, the feedrate determination function based on corner feedrate difference finds a feedrate not exceeding the set value and performs deceleration by using acceleration/deceleration before interpolation. Thus, a shock to the machine and machining error at a corner can be reduced.

1788	Maximum allowable acceleration change rate in feedrate determination based on acceleration change for each axis

[Input type] Parameter input
[Data type] Real axis
[Unit of data] mm/sec^2, inch/sec^2, degree/sec^2 (machine unit)
[Minimum unit of data] Depend on the increment system of the applied axis
[Valid data range] Refer to the standard parameter setting table (D)
(When the machine system is metric system, 0.0 to +100000.0. When the machine system is inch system, machine, 0.0 to +10000.0.)
Set a maximum allowable acceleration change rate for each axis in feedrate control based on acceleration change under control on the rate of change of acceleration.
For an axis with 0 set in this parameter, feedrate control based on acceleration change is disabled.
If 0 is set for all axes, feedrate control based on acceleration change is not exercised.

1789	Maximum allowable acceleration change rate in feedrate determination based on acceleration change for each axis (linear interpolation)

[Input type] Parameter input
[Data type] Real axis
[Unit of data] mm/sec^2, inch/sec^2, degree/sec^2 (machine unit)
[Minimum unit of data] Depend on the increment system of the applied axis
[Valid data range] Refer to the standard parameter setting table (D)
(When the machine system is metric system, 0.0 to +100000.0. When the machine system is inch system, machine, 0.0 to +10000.0.)
Set a maximum allowable acceleration change rate for each axis in feedrate control based on acceleration change under control on the rate of change of acceleration in successive linear interpolation operations.
In feedrate control based on acceleration change at a corner between linear interpolation operations, the maximum allowable acceleration change rate not set in parameter No. 1788 but set in this parameter is valid.
For an axis with 0 set in this parameter, the maximum allowable acceleration change rate set in parameter No. 1788 is valid.
Feedrate control based on acceleration change is disabled for an axis with 0 set in parameter No. 1788, so that the setting of this parameter for such an axis is ignored.

1790	Ratio of change time of the rate of change of acceleration in smooth bell-shaped acceleration/deceleration before interpolation

[Input type] Parameter input
[Data type] Byte path
[Unit of data] %
[Valid data range] 0 to 50

Set the ratio of the change time of the rate of change of acceleration to the change time of acceleration(*1) by percentage (%) in smooth bell-shaped acceleration/deceleration before look-ahead interpolation.
If 0 is set in this parameter or a value not within the valid data range is specified in this parameter, smooth bell-shaped acceleration/ deceleration before look-ahead interpolation is not performed.

(*1) Parameter No. 1772 for acceleration/deceleration before look-ahead interpolation (cutting feed).
Parameter No. 1672 for acceleration/deceleration before interpolation in linear rapid traverse, or for optimum torque acceleration/ deceleration.

	#7	#6	#5	#4	#3	#2	#1	#0
1802						DC2x	DC4x	

[Input type] Parameter input
[Data type] Bit axis

 # 1 DC4x When the reference position is established on the linear scale with reference marks:
0: An absolute position is established by detecting three reference marks.
1: An absolute position is established by detecting four reference marks.

 # 2 DC2x Reference position establishment operation for a linear scale with reference marks is performed as follows:
0: The setting of bit 1 (DC4) of parameter No. 1802 is followed.
1: An absolute position is established by detecting two reference marks.

NOTE
1 When this parameter is set to 1, specify the direction of the scale zero point by setting bit 4 (SCP) of parameter No. 1817.
2 When a rotary encoder with absolute address reference marks is used, this parameter is invalid. Even when this parameter is set to 1, the setting of bit 1 (DC4) of parameter No. 1802 is followed.

	#7	#6	#5	#4	#3	#2	#1	#0
1815			APCx	APZx	DCRx		OPTx	

[Input type] Parameter input
[Data type] Bit axis

NOTE
When at least one of these parameters is set, the power must be turned off before operation is continued.

 # 1 OPTx Position detector
0: A separate pulse coder is not used.
1: A separate pulse coder is used.

NOTE
Set this parameter to 1 when using a linear scale with reference marks or a linear scale with an absolute address zero point (full-closed system).

 # 3 DCRx As a scale with absolute address reference marks:
0: A rotary encoder with absolute address reference marks is not used.
1: A rotary encoder with absolute address reference marks is used.

NOTE
When using a rotary encoder with absolute address reference marks, set also bit 2 (DCLx) of parameter No. 1815 to 1.

 # 4 APZx Machine position and position on absolute position detector when the absolute position detector is used
0: Not corresponding
1: Corresponding
When an absolute position detector is used, after primary adjustment is performed or after the absolute position detector is replaced, this parameter must be set to 0, power must be turned off and on, then manual reference position return must be performed. This completes the positional correspondence between the machine position and the position on the absolute position detector, and sets this parameter to 1 automatically.

 # 5 APCx Position detector
0: Other than absolute position detector
1: Absolute position detector (absolute pulse coder)

	#7	#6	#5	#4	#3	#2	#1	#0
1817		TANx						

[Input type] Parameter input
[Data type] Bit axis

NOTE
When this parameter is set, the power must be turned off before operation is continued..

 # 6 TANx Tandem control
0: Not used
1: Used

NOTE
Set this parameter to both master axis and slave axis.

	#7	#6	#5	#4	#3	#2	#1	#0
1818					SDC		RF2x	RFSx

[Input type] Parameter input
[Data type] Bit axis

 # 0 RFSx If G28 is specified for an axis for which a reference position is not established (ZRF = 0) when a linear scale with an absolute address zero point or a linear scale with absolute address reference marks is used:
0: A movement is made to the reference position after reference position establishment operation.
1: No movement is made after reference position establishment operation, but the operation is completed.

NOTE
This parameter disables movement based on the G28 command to a reference position. So, use this parameter only in special cases.

 # 1 RF2x If G28 is specified for an axis for which a reference position is already established (ZRF = 1) when a linear scale with an absolute address zero point or a linear scale with absolute address reference marks is used:
0: A movement is made to the reference position.
1: No movement is made to the intermediate position and reference position, but the operation is completed.

NOTE
This parameter disables movement based on the G28 command to a reference position. So, use this parameter only in special cases.

 # 3 SDCx A linear scale with an absolute address zero point is:
0: Not used.
1: Used.

	#7	#6	#5	#4	#3	#2	#1	#0
1819						DATx		

[Input type] Parameter input
[Data type] Bit axis

 # 2 DATx When a linear scale with an absolute address zero point or a linear scale with absolute address reference marks is used, the automatic setting of parameter No. 1883 and No. 1884 at manual reference position return time is:
0: Not performed.
1: Performed.

The automatic setting procedure is as follows:
<1> Set an appropriate value in parameter No. 1815, No. 1821, and No. 1882.
<2> Position the machine at the reference position by manual operation.
<3> Set this parameter to 1.
<4> Perform a manual reference position return operation. Upon completion of manual reference position return operation, parameter No. 1883 and No. 1884 are set, and this parameter is automatically set to 0.

1820	Command multiplier for each axis (CMR)

NOTE
When this parameter is set, the power must be turned off before operation is continued.

[Input type] Parameter input
[Data type] Byte axis
[Valid data range] See below :
Set a command multiplier indicating the ratio of the least command increment to the detection unit for each axis.
Least command increment = detection unit × command multiplier

Relationship between the increment system and the least command increment
(1) T series

			Least input increment	Least command increment
IS-B	Millimeter machine	Millimeter input	0.001 mm (diameter specification)	0.0005 mm
			0.001 mm (radius specification)	0.001 mm
		Inch input	0.0001 inch (diameter specification)	0.0005 mm
			0.0001 inch (radius specification)	0.001 mm
	Inch machine	Millimeter input	0.001 mm (diameter specification)	0.00005 inch
			0.001 mm (radius specification)	0.0001 inch
		Inch input	0.0001 inch (diameter specification)	0.00005 inch
			0.0001 inch (radius specification)	0.0001 inch
	Rotation axis		0.001 deg	0.001 deg

			Least input increment	Least command increment
IS-C	Millimeter machine	Millimeter input	0.0001 mm (diameter specification)	0.00005 mm
			0.0001 mm (radius specification)	0.0001 mm
		Inch input	0.00001 inch (diameter specification)	0.00005 mm
			0.00001 inch (radius specification)	0.0001 mm
	Inch machine	Millimeter input	0.0001 mm (diameter specification)	0.000005 inch
			0.0001 mm (radius specification)	0.00001 inch
		Inch input	0.00001 inch (diameter specification)	0.000005 inch
			0.00001 inch (radius specification)	0.00001 inch
	Rotation axis		0.0001 deg	0.0001 deg

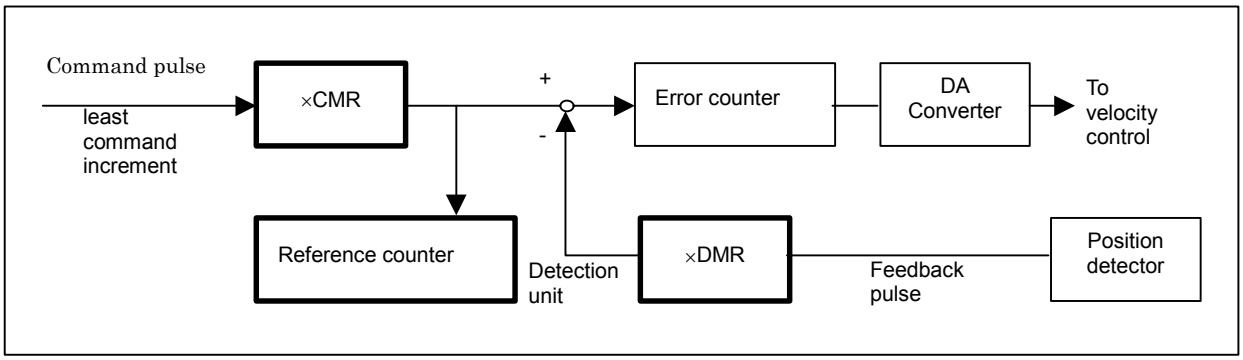
			Least input increment	Least command increment
IS-D	Millimeter machine	Millimeter input	0.00001 mm (diameter specification)	0.000005 mm
			0.00001 mm (radius specification)	0.00001 mm
		Inch input	0.000001 inch (diameter specification)	0.000005 mm
			0.000001 inch (radius specification)	0.00001 mm
	Inch machine	Millimeter input	0.00001 mm (diameter specification)	0.0000005 inch
			0.00001 mm (radius specification)	0.000001 inch
		Inch input	0.000001 inch (diameter specification)	0.0000005 inch
			0.000001 inch (radius specification)	0.000001 inch
	Rotation axis		0.00001 deg	0.00001 deg

			Least input increment	Least command increment
IS-E	Millimeter machine	Millimeter input	0.000001 mm (diameter specification)	0.0000005 mm
			0.000001 mm (radius specification)	0.000001 mm
		Inch input	0.0000001 inch (diameter specification)	0.0000005 mm
			0.0000001 inch (radius specification)	0.000001 mm
	Inch machine	Millimeter input	0.000001 mm (diameter specification)	0.00000005 inch
			0.000001 mm (radius specification)	0.0000001 inch
		Inch input	0.0000001 inch (diameter specification)	0.00000005 inch
			0.0000001 inch (radius specification)	0.0000001 inch
	Rotation axis		0.000001 deg	0.000001 deg

(2) M series

Increment system	Least input increment and least command increment					
	IS-A	IS-B	IS-C	IS-D	IS-E	Unit
Millimeter machine	0.01	0.001	0.0001	0.00001	0.000001	mm
Millimeter input	0.001	0.0001	0.00001	0.000001	0.0000001	inch
Rotation axis	0.01	0.001	0.0001	0.00001	0.000001	deg

Setting command multiply (CMR), detection multiply (DMR), and the capacity of the reference counter



Set CMR and DMR so that the pulse weight of + input (command from the CNC) into the error counter matches the pulse weight of -input (feedback from the position detector).
[Least command increment]/CMR=[Detection unit]=
[Feedback pulse unit]/DMR

[Least command increment]:
Minimum unit of commands issued from the CNC to the machine

[Detection unit]: Minimum unit for machine position detection

The unit of feedback pulses varies, depending on the type of detector.
[Feedback pulse unit]=[Amount of travel per rotation of the pulse coder]/[Number of pulses per rotation of the pulse coder]

As the size of the reference counter, specify the grid interval for the reference position return in the grid method.
[Size of the reference counter]=[Grid interval]/[Detection unit]
[Grid interval]=[Amount of travel per rotation of the pulse coder]

The setting of a command multiplier is as follows:
(1) When command multiplier is 1 to 1/27
Set value = 1 / command multiplier + 100
Valid data range : 101 to 127
(2) When command multiply is 0.5 to 48
Set value = 2 × command multiplier
Valid data range : 1 to 96

NOTE
If a feedrate exceeding the feedrate found by the expression below is used, an incorrect travel amount may result or a servo alarm may be issued. Be sure to use a feedrate not exceeding the feedrate found by the following expression:
Fmax[mm/min] = 196602 × 10^4 × least command increment / CMR

1821	Reference counter size for each axis

NOTE
When this parameter is set, the power must be turned off before operation is continued.

[Input type] Parameter input
[Data type] 2-word axis
[Unit of data] Detection unit
[Valid data range] 0 to 999999999
Set a reference counter size.
As a reference counter size, specify a grid interval for reference position return based on the grid method.
When a value less than 0 is set, the specification of 10000 is assumed.
When a linear scale with absolute address reference marks is used, set the interval of mark 1.

1828	Positioning deviation limit for each axis in movement

[Input type] Parameter input
[Data type] 2-word axis
[Unit of data] Detection unit
[Valid data range] 0 to 99999999

Set the positioning deviation limit in movement for each axis.
If the positioning deviation exceeds the positioning deviation limit during movement, a servo alarm (SV0411) is generated, and operation is stopped immediately (as in emergency stop).
Generally, set the positioning deviation for rapid traverse plus some margin in this parameter.

1829	Positioning deviation limit for each axis in the stopped state

[Input type] Parameter input
[Data type] 2-word axis
[Unit of data] Detection unit
[Valid data range] 0 to 99999999

Set the positioning deviation limit in the stopped state for each axis.
If, in the stopped state, the positioning deviation exceeds the positioning deviation limit set for stopped state, a servo alarm (SV0410) is generated, and operation is stopped immediately (as in emergency stop).

1838	Position deviarion limit for each axis in moving during safety check

NOTE
When this parameter is set, the power must be turned off before operation is continued.

[Input type] Paramete input
[Data type] 2-word axis
[Unit of data] Detection unit
[Valid data range] 0 to 99999999

Position deviation limit for each axis during moving for safety check of Dual Check Safety function is specified.
If position deviation of a moving axis exceeds position deviation limit while Safety Check is carried out (Safety Monitoring Request "*VLDVx" =0), a servo alarm SV0475 or SV1071 is generated and axes are stopped immediately like emergency stop state..
In Dual Check Safety function, positin deviation is always checked by CNC and Servo. In case that Safety Check is carried out (Safety Monitoring Request "*VLDVx" =0), the servo alarm SV0475 or SV1071 is generated when each CPU finds out that the deviation exceeds position deviation limit in moving state.

1841	Position deviation limit of each axis in moving state during other than Dual Check Safety monitoring (for Dual Check Safety Function)

NOTE
When this parameter is set, the power must be turned off before operation is continued.

[Input type] Parameter input
[Data type] 2 word axis
[Unit of data] Detection unit
[Valid data range] 0 to 99999999

Set the positioning deviation limit in moving state for each axis for Dual Check Safety function, in case that Safety Check is not carried out (Safety Monitoring Request "*VLDVx"=1).
In case that Safety Check is not carried out (Safety Monitoring Request "*VLDVx" =1), servo alarm (SV0475,SV1071) is generated and operation is stopped immediately (as in emergency stop), when each CPU finds out that the deviation exceeds position deviation limit in moving state.
If the value of this parameter is "0", the parameter No.1828 is used for the value of deviation limit in moving state.
In case that Safety Check is carried out (Safety Monitoring Request "*VLDVx" =0), the parameter No.1838 is used for the value of deviation limit in moving state.

1851	Backlash compensating value for each axis

[Input type] Parameter input
[Data type] Word axis
[Unit of data] Detection unit
[Valid data range] -9999 to 9999

Set the backlash compensating value for each axis.
When the machine moves in a direction opposite to the reference position return direction after the power is turned on, the first backlash compensation is performed.

1882	Interval of mark 2 of a linear scale with absolute address reference marks

NOTE
When this parameter is set, the power must be turned off before operation is continued.

[Input type] Parameter input
[Data type] 2-word axis
[Unit of data] Detection unit
[Valid data range] 0 to 999999999

Set the interval of mark 2 of a linear scale with absolute address reference marks.

1883	Distance 1 from the scale zero point to reference position (linear scale with absolute address reference marks) or distance 1 from the base point to reference position (linear scale with an absolute address zero point)

NOTE
When this parameter is set, the power must be turned off before operation is continued.

[Input type] Parameter input
[Data type] 2-word axis
[Unit of data] Detection unit
[Valid data range] -999999999 to 999999999

1884	Distance 2 from the scale zero point to reference position (linear scale with absolute address reference marks) or distance 2 from the base point to reference position (linear scale with an absolute address zero point)

NOTE
When this parameter is set, the power must be turned off before operation is continued.

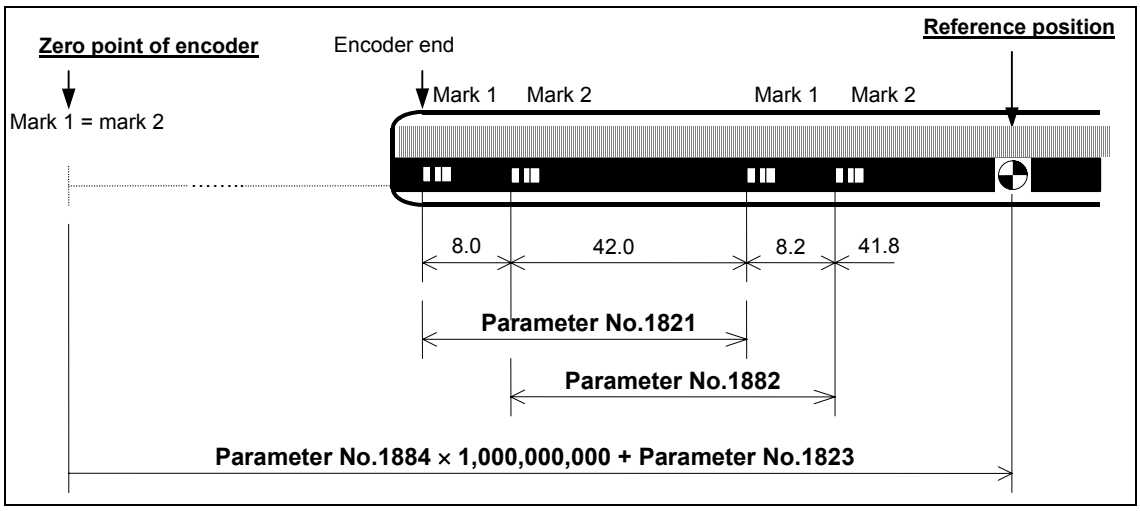
[Input type] Parameter input
[Data type] 2-word axis
[Unit of data] Detection unit
[Valid data range] -999 to 999

When a linear scale with absolute address reference marks is used, set the distance from the scale zero point to reference position in parameter Nos. 1883 and 1884).

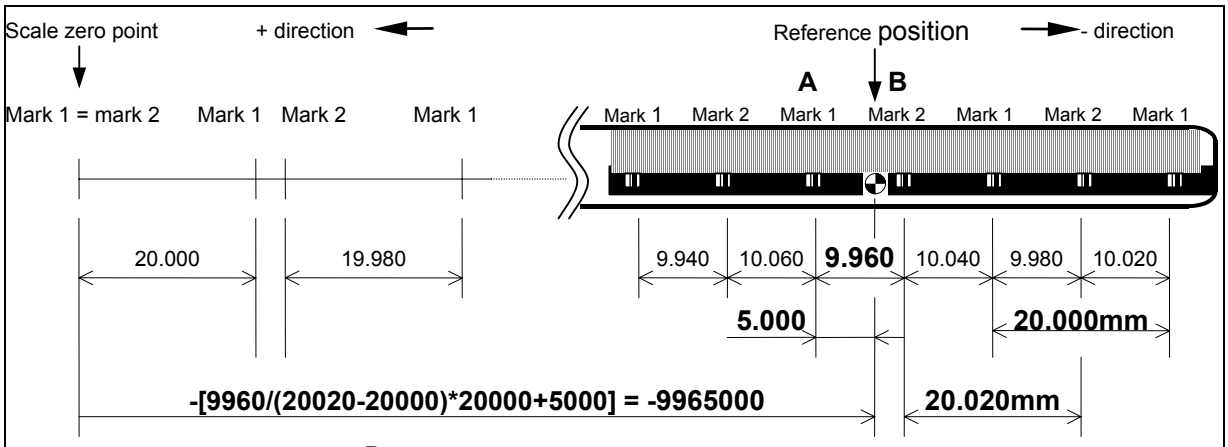
Distance from the zero point to the reference position of a linear scale
= No. 1884 × 1,000,000,000 + No. 1883

The scale zero point represents a point where mark 1 and mark 2 match. Usually, this point is a virtual point that does not physically exist on the scale. (See the figure below.)

If the reference position is placed in the + direction when viewed from the scale zero point, set a positive value. If the reference position is placed in the - direction when viewed from the scale zero point, set a negative value.



[Example of parameter settings] When an encoder as shown below is used with an IS-B, millimeter machine:



Parameters

No.1821 (interval of mark 1) = 20000
No.1882 (interval of mark 2) = 20020
No.1883 (reference position) = position of point A + 5.000
= distance between A and B/(mark 2 – mark 1) × mark 1 + 5000
= 9960/(20020-20000) × 20000 + 5000
= 9965000
= -9965000 (the reference position is on the negative side)

[Setting parameter No. 1883]

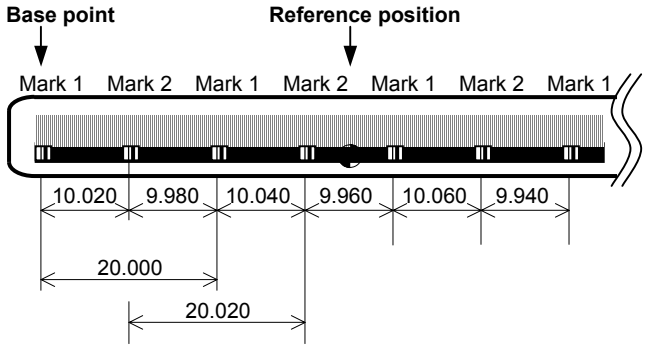
When it is difficult to measure the distance from the scale zero point to the reference position (parameter No. 1883), the method described below can be used to find the distance.

<1> Set parameter No. 1815 to enable this function.
Set an appropriate value in parameter No. 1821 and No. 1882.
Set 0 in parameter No. 1240.
Set 0 in parameter No. 1883 and No. 1884.

<2> At an appropriate position, establish a reference position.
(As a result, the machine coordinate represents the distance from the scale zero point to the current position.)

<3> By jog feed or handle feed, place the machine at the accurate reference position.
<4> In parameter No. 1883, set the machine coordinate of that time converted to the detection unit (machine coordinate × CMR).
<5> If necessary, set parameter No. 1240.

When a linear scale with an absolute address zero point is used, set the distance from the base point to the reference position in parameter Nos. 1883 and 1884. The base point is a point at a scale end as shown below.



If the reference position is located in the positive direction when viewed from the base point, set a positive value; if the reference position is located in the negative direction, set a negative value. Set the value by following the steps explained below.

<1> Set bit 1 (OPT) of parameter No. 1815 , bit 2 (DCL) of parameter No. 1815, and bit 3 (SDC) of parameter No. 1818 to enable this function.
Set 0 in parameter No. 1240.
Set 0 in parameter No. 1883 and No. 1884.
<2> At an appropriate position, establish a reference position.
(Consequently, the machine coordinate value indicates the distance from the base point to current position.)
<3> By jog feed or handle feed, place the machine at the accurate reference position.
<4> In parameters Nos. 1883 and 1884, set the machine coordinate of that time converted to the detection unit (machine coordinate × CMR).
If necessary, set parameter No. 1240.

NOTE
1 Set parameter Nos. 1883 and 1884 so that the distance from the scale zero point (for a linear scale with absolute address reference marks) or the base point (for a linear scale with an absolute address zero point) to the reference position is within the range from -999,999,999,999 to +999,999,999,999. If a value beyond this range is set, an alarm (PS 5325) is issued.
2 The scale area on the scale cannot be extended across the scale zero point or base point. Make parameter settings not to cause the scale area to extend beyond the scale zero point or base point.

	#7	#6	#5	#4	#3	#2	#1	#0
1902							ASE	FMD

[Input type] Parameter input
[Data type] Bit

NOTE
When at least one of these parameters is set, the power must be turned off before operation is continued.

 # 0 FMD The FSSB setting mode is:
0: Automatic setting mode.
(When the relationship between an axis and amplifier is defined on the FSSB setting screen, parameter Nos. 1023, 1905, 1936 to 1939, and 14340 to 14407 (plus parameter Nos. 14408 to 14425 and 14444 to 14459 if an additional axis board is attached) are automatically set.
1: Manual setting 2 mode.
(Parameter Nos. 1023, 1905, 1936 to 1939 and 14340 to 14407 (plus parameter Nos. 14408 to 14425 and 14444 to 14459 if an additional axis board is attached) are to be manually set.)

 # 1 ASE When automatic setting mode is selected for FSSB setting (when the FMD parameter (bit 0 of parameter No.1902) is set to 0), automatic setting is:
0: Not completed.
1: Completed.
This bit is automatically set to 1 upon the completion of automatic setting.

	#7	#6	#5	#4	#3	#2	#1	#0
1905	PM2	PM1				PM4	PM3	

[Input type] Parameter input
[Data type] Bit axis

NOTE
When at least one of these parameters is set, the power must be turned off before operation is continued.

 # 1 PM3 The third separate detector interface unit is:
0: Not used.
1: Used.

 # 2 PM4 The fourth separate detector interface unit is:
0: Not used.
1: Used.

 # 6 **PM1** The first separate detector interface unit is:
0: Not used.
1: Used.

 # 7 **PM2** The second separate detector interface unit is:
0: Not used.
1: Used.

NOTE
When automatic setting mode is selected for FSSB setting (when the parameter FMD (No.1902#0) is set to 0), this parameter is automatically set when input is performed with the FSSB setting screen. When manual setting 2 mode is selected for FSSB setting (when the parameter FMD (No.1902#0) is set to 1), this parameter must be set directly. When a separate detector interface unit is used, a connector number must be set in the corresponding parameter (No.1936, No.1937, No.1938, or No.1939).

1936	Connector number of the first separate detector interface unit
1937	Connector number of the second separate detector interface unit
1938	Connector number of the third separate detector interface unit
1939	Connector number of the fourth separate detector interface unit

NOTE
When these parameters are set, the power must be turned off before operation is continued.

[Input type] Parameter input
[Data type] Byte axis
[Valid data range] 0 to 7
Set the connector numbers corresponding to connectors to be connected when the separate detector interface unit set by bit 1, 2, 6, or 7 of parameter No. 1905 is used. The values to be set are indicated below.
Within one separate detector interface unit, use connector numbers sequentially. No intermediate number may be omitted.

Correspondence between connectors and connector numbers	
Connector	**Connector number**
JF101	0
JF102	1
JF103	2
JF104	3
JF105	4
JF106	5

Correspondence between connectors and connector numbers	
Connector	**Connector number**
JF107	6
JF108	7

Example of setting)

Controlled axis	Separate detector connection destination				Parameter setting				
	Connectors for 1st unit	Connectors for 2nd unit	Connectors for 3rd unit	Connectors for 4th unit	No. 1936	No. 1937	No. 1938	No. 1939	No.1905 (#7,#6,#2,#1)
X1	JF101	-	-	-	0	-	-	-	0,1,0,0
Y1	-	JF102	-	-	-	1	-	-	1,0,0,0
Z1	-	-	JF102	-	-	-	1	-	0,0,0,1
X2	-	JF101	-	-	-	0	-	-	1,0,0,0
Y2	-	-	-	JF101	-	-	-	0	0,0,1,0
Z2	-	-	-	-	-	-	-	-	0,0,0,0
A1	-	-	JF101	-	-	-	0	-	0,0,0,1
B1	-	-	-	JF102	-	-	-	1	0,0,1,0
C1	-	JF104	-	-	-	3	-	-	1,0,0,0
A2	JF102	-	-	-	1	-	-	-	0,1,0,0
B2	-	JF103	-	-	-	2	-	-	1,0,0,0
C2	-	-	-	JF103	-	-	-	2	0,0,1,0

NOTE

When automatic setting mode is selected for FSSB setting (when the parameter FMD (No.1902#0) is set to 0), these parameters are automatically set when input is performed with the FSSB setting screen. When manual setting 2 mode is selected for FSSB setting (when the parameter FMD (No.1902#0) is set to 1), these parameters must be set directly.

Parameters No.2000 to 2999 are for digital servo, The following parameters are not explained in this manual. Refer to FANUC AC SERVO MOTOR αi series PARAMETER MANUAL (B-65270EN)

	#7	#6	#5	#4	#3	#2	#1	#0
2011	XIAx							

[Input type] Parameter input
[Data type] Bit axis

 # 7 XIAx Temporary absolute coordinate setting is:
0: Not used.
1: Used.

NOTE
1 When temporary absolute coordinate setting is used, bit 1 (OPTx) of parameter No. 1815, bit 5 (APCx) of parameter No. 1815, parameter No. 1874, and parameter No. 1875 must be set.
2 The setting of this parameter becomes effective after the power is turned off then back on.

2031	Torque command difference threshold of torque difference alarm

[Input type] Parameter input
[Data type] Word axis
[Valid data range] 0 to 14564
If the absolute value of the torque command difference between two axes exceeds the value set in this parameter, an alarm is issued.
Set the same value for two axes that are placed under axis synchronous control.
The servo axis numbers of the synchronized master axis and slave axis must be assigned so that an odd number is assigned to the master axis and the next axis number is assigned to the slave axis. Examples are (1,2) and (3,4).

	#7	#6	#5	#4	#3	#2	#1	#0
3002	OVM	POV						

[Input type] Parameter input
[Data type] Bit path

 # 6 POV Dwell/Auxiliary function time override function is:
0: Invalid.
1: Valid.

 # 7 OVM In Dwell/Auxiliary function time override function, override function for M02,M30 is:
0: Invalid.
1: Valid.

	#7	#6	#5	#4	#3	#2	#1	#0
3008						XSG		

[Input type] Parameter input
[Data type] Bit path

NOTE
When this parameter is set, the power must be turned off before operation is continued.

 # 2 XSG A signal assigned to an X address is:
0: Fixed at the address.
1: Able to be reassigned to an arbitrary X address.

NOTE
When this parameter is set to 1, set parameter No. 3013, No. 3014, No. 3012, and No. 3019. If parameter No. 3013 and No. 3014 are not set, the deceleration signal for reference position return is assigned to bit 0 of X0000. If parameter No. 3012 and No. 3019 are not set, the skip signal, the PMC axis control skip signal, the measurement position arrival signal, the manual feed interlock signal for each axis direction, and the tool compensation value write signal are assigned to X0000.

3012	Skip signal assignment address

NOTE
When this parameter is set, the power must be turned off before operation is continued.

[Input type] Parameter input
[Data type] Word path
[Valid data range] 0 to 727
Set an X address to which the skip signal (SKIPn) is to be assigned.

NOTE
This parameter is valid when bit 2 (XSG) of parameter No. 3008 is set to 1.
Depending on the option configuration of the I/O Link, the actually usable X addresses are:
X0 to X127, X200 to X327, X400 to X527, X600 to X727

3013	X address to which the deceleration signal for reference position return is assigned

NOTE
When this parameter is set, the power must be turned off before operation is continued.

[Input type] Parameter input
[Data type] Word axis
[Valid data range] 0 to 727
Set an address to which the deceleration signal (*DECn) for reference position return for each axis is to be assigned.

NOTE
This parameter is valid when bit 2 (XSG) of parameter No. 3008 is set to 1.
Depending on the option configuration of the I/O Link, the actually usable X addresses are:
X0 to X127, X200 to X327, X400 to X527, X600 to X727

3018	Percentage assumed when the 1% rapid traverse override signal indicates 0% during execution of an auxiliary function

[Input type] Parameter input
[Data type] Byte path
[Unit of data] %
[Valid data range] 0 to 100
When the 1% rapid traverse override signal indicates 0% in the dwell/auxiliary function time override function, this parameter sets the percentage for calculating an insufficient time.
0% is assumed to be 10%.

3019	Address to which the PMC axis control skip signal and the measurement position arrival signal are assigned

NOTE
When this parameter is set, the power must be turned off before operation is continued.

[Input type] Parameter input
[Data type] Word path
[Valid data range] 0 to 727
Set an X address to which the PMC axis control skip signal ESKIP and the measurement position arrival signals (XAE, YAE, and ZAE (M series) or XAE and ZAE (T series)) are to be assigned.

Example 1. When No.3012 is set to 5 and No.3019 is set to 6

When XSG (bit 2 of parameter No. 3008) is 1, the PMC axis control skip signal, and measurement position arrival signal are allocated to X0006 and the skip signal is allocated to X0005.

	#7	#6	#5	#4	#3	#2	#1	#0	
X005	SKIP	SKIP6	SKIP5	SKIP4	SKIP3	SKIP2	SKIP8	SKIP7	(T series)
	SKIP	SKIP6	SKIP5	SKIP4	SKIP3	SKIP2	SKIP8	SKIP7	(M series)

	#7	#6	#5	#4	#3	#2	#1	#0	
X006		ESKIP	-MIT2	+MIT2	-MIT1	+MIT1	ZAE	XAE	(T series)
		ESKIP				ZAE	YAE	XAE	(M series)

Example 2. When No.3012 is set to 5 and No.3019 is set to 5

When XSG (bit 2 of parameter No. 3008) is 1, the PMC axis control skip signal, measurement position arrival signal, and skip signal are allocated to X0005.

	#7	#6	#5	#4	#3	#2	#1	#0	
X005	SKIP	ESKIP	-MIT2	+MIT2	-MIT1	+MIT1	ZAE	XAE	(T series)
		SKIP6	SKIP5	SKIP4	SKIP3	SKIP2	SKIP8	SKIP7	

	#7	#6	#5	#4	#3	#2	#1	#0	
	SKIP	ESKIP	SKIP5	SKIP4	SKIP3	ZAE	YAE	XAE	(M series)
		SKIP6				SKIP2	SKIP8	SKIP7	

NOTE
This parameter is valid when bit 2 (XSG) of parameter No. 3008 is set to 1.
Depending on the option configuration of the I/O Link, the actually usable X addresses are:
X0 to X127, X200 to X327, X400 to X527, X600 to X727

3021	Address to which an axis signal is assigned

NOTE
When this parameter is set, the power must be turned off before operation is continued.

[Input type] Parameter input
[Data type] Byte axis
[Valid data range] 0 to 7, 10 to 17, 20 to 27, ... , 90 to 97
For each axis of the CNC, set a PMC interface address.
Set a value according to the tables below.

Value of parameter No. 3021 (the second digit)

Setting value	Input signal address	Output signal address
0	G0000 to G0767	F0000 to F0767
1	G1000 to G1767	F1000 to F1767
	:	
9	G9000 to G9767	F9000 to F9767

Value of parameter No. 3021 (the first digit)

Setting value	Input signal address	Output signal address
0	#0	#0
1	#1	#1
	:	
7	#7	#7

[Example of setting]

Axis number	No.3021	Signal allocation
1	0	+J1<G0100#0>, -J1<G0102#0>, ZP1<F0090#0>, ...
2	1	+J2<G0100#1>, -J2<G0102#1>, ZP2<F0090#1>, ...
3	2	+J3<G0100#2>, -J3<G0102#2>, ZP3<F0090#2>, ...
4	10	+J4<G1100#0>, -J4<G1102#0>, ZP4<F1090#0>, ...
5	11	+J5<G1100#1>, -J5<G1102#1>, ZP5<F1090#1>, ...

If eight or less axes are used per path, the following signal allocation results when 0 is set for all axes:

Axis 1 of path 1 = Setting equivalent to 0
Axis 2 of path 1 = Setting equivalent to 1
:
Axis 1 of path 2 = Setting equivalent to 10
:

NOTE
Set this parameter when more than eight axes are used per path.
The valid data range varies, depending on the NC system type.

3022 **Address to which a spindle signal is assigned**

NOTE
When this parameter is set, the power must be turned off before operation is continued.

[Input type] Parameter input
[Data type] Byte spindle
[Valid data range] 0to3,10to13,20to23, ... ,90to93

For each axis of the CNC, set a PMC interface address.
Set a value according to the tables below.

Value of parameter No. 3022 (the second digit)

Setting value	Input signal address	Output signal address
0	G0000toG0767	F0000toF0767
1	G1000toG1767	F1000toF1767
	:	
9	G9000toG9767	F9000toF9767

Value of parameter No. 3022 (the first digit)

Setting value	Input signal address	Output signal address
0	Bit position A	Bit position A
1	Bit position B	Bit position B
2	Bit position C	Bit position C
3	Bit position D	Bit position D

(The bit positions A, B, C, and D vary, depending on the type of signal.)

[Example of setting]

Spindle number	No.3022	Signal allocation
1	0	TLMLA<G0070.0>, TLMHA<G0070.1>, ALMA<F0045.0>, ...
2	1	TLMLB<G0074.0>, TLMHB<G0074.1>, ALMB<F0049.0>, ...
3	10	TLMLA<G1070.0>, TLMHA<G1070.1>, ALMA<F1045.0>, ...
4	11	TLMLB<G1074.0>, TLMHB<G1074.1>, ALMB<F1049.0>, ...

If four or less axes are used per path, the following signal allocation results when 0 is set for all axes:

Axis 1 of path 1 = Setting equivalent to 0
Axis 2 of path 1 = Setting equivalent to 1
:
Axis 1 of path 2 = Setting equivalent to 10
:

NOTE
The valid data range varies, depending on the system software.

3030 **Allowable number of digits for the M code**

3031 **Allowable number of digits for the S code**

3032 **Allowable number of digits for the T code**

[Input type] Parameter input
[Data type] Byte path
[Valid data range] 1 to 8

Set the allowable numbers of digits for the M, S, and T codes.
When 0 is set, the allowable number of digits is assumed to be 8.

3033	Allowable number of digits for the B code (second auxiliary function)

[Input type] Parameter input
[Data type] Byte path
[Valid data range] 1 to 8

Set the allowable number of digits for the second auxiliary function.
When 0 is set, the allowable number of digits is assumed to be 8.
To enable a decimal point to be specified, bit 0 (AUP) of parameter No. 3450 must be set to 1. In this case, the allowable number of digits set in this parameter includes the number of decimal places.
If a value exceeding the allowable number of digits is specified, the alarm PS0003 is issued.

	#7	#6	#5	#4	#3	#2	#1	#0
3104	DAC		DRC		PPD			MCN
	DAC	DAL	DRC	DRL	PPD			MCN

[Input type] Parameter input
[Data type] Bit path

 # 0 MCN Machine position

0: Regardless of whether input is made in mm or inches, the machine position is displayed in mm for millimeter machines, or in inches for inch machines.
1: When input is made in mm, the machine position is displayed in mm, and when input is made in inches, the machine position is displayed in inches accordingly.

 # 3 PPD Relative position display when a coordinate system is set

0: Not preset
1: Preset

NOTE
If any of the following is executed when PPD is set to 1, the relative position display is preset to the same value as the absolute position display:
(1) Manual reference position return
(2) Coordinate system setting based on G92 (G50 for G code system A on the lathe system)
(3) Workpiece coordinate system presetting based on G92.1 (G50.3 for G code system A on the lath system)
(4) When a T code for the lathe system is specified.

 # 4 DRL Relative position

0: The actual position displayed takes into account tool length offset.
1: The programmed position displayed does not take into account tool length offset.

 # 5 DRC When relative positions are displayed:
0: Values not excluding the amount of travel based on tool radius · tool nose radius compensation are displayed.
1: Values excluding the amount of travel based on tool radius · tool nose radius compensation (programmed positions) are displayed.

 # 6 DAL Absolute position
0: The actual position displayed takes into account tool length offset.
1: The programmed position displayed does not take into account tool length offset.

NOTE
In lathe systems, whether to exclude a tool offset when displaying the absolute position is determined by the setting of bit 1 (DAP) of parameter No. 3129.

 # 7 DAC When an absolute position are displayed:
0: Values not excluding the amount of travel based on tool radius · tool nose radius compensation are displayed.
1: Values excluding the amount of travel based on tool radius · tool nose radius compensation (programmed positions) are displayed.

	#7	#6	#5	#4	#3	#2	#1	#0
3106		DAK						

[Input type] Setting input
[Data type] Bit

 # 6 DAK Specifies whether to display coordinates in the program coordinate system or workpiece coordinate system as absolute coordinates when the three-dimensional coordinate conversion mode, the tilted working plane command mode or the workpiece setting error compensation mode is set.
0: Display coordinates in the program coordinate system.
1: Display coordinates in the workpiece coordinate system.

	#7	#6	#5	#4	#3	#2	#1	#0
3111		OPS	OPM				SPS	SVS

[Input type] Setting input
[Data type] Bit path

 # 0 SVS Servo setting screen and servo tuning screen
0: Not displayed
1: Displayed

 # 1 SPS Spindle tuning screen
0: Not displayed
1: Displayed

 # 5 OPM Operating monitor
0: Not displayed
1: Displayed

 # 6 OPS The speedometer on the operating monitor screen indicates:
0: Spindle motor speed
1: Spindle speed

	#7	#6	#5	#4	#3	#2	#1	#0
3115							NDAx	NDPx

[Input type] Parameter input
[Data type] Bit axis

 # 0 NDPx The current position is:
0: Displayed.
1: Not displayed.

NOTE
When using the electric gear box (EGB) function, set 1 for the EGB dummy axis to disable current position display.

 # 1 NDAx The current position and the amount of the movement to be made in absolute and relative coordinates are:
0: Displayed.
1: Not displayed.

	#7	#6	#5	#4	#3	#2	#1	#0
3129							DAP	DRP

[Input type] Parameter input
[Data type] Bit path

 # 0 DRP For relative coordinate display:
0: The actual position considering a tool offset (tool movement) is displayed.
1: The programmed position excluding a tool offset (tool movement) is displayed.

 # 1 DAP For absolute coordinate display:

0: The actual position considering a tool offset (tool movement) is displayed.

1: The programmed position excluding a tool offset (tool movement) is displayed.

NOTE
In machining center systems, whether to exclude the tool length offset when displaying the absolute position is determined according to the setting of bit 6 (DAL) of parameter No. 3104.

3131	Subscript of axis name

[Input type] Parameter input
[Data type] Byte axis
[Valid data range] 0 to 9, 65 to 90

In order to distinguish axes under parallel operation, synchronization control, and tandem control, specify a subscript for each axis name.

Setting value	Meaning
0	Each axis is set as an axis other than a parallel axis, synchronization control axis, and tandem control axis.
1 to 9	A set value is used as a subscript.
65 to 90	A set letter (ASCII code) is used as a subscript.

Example) When the axis name is X, a subscript is added as indicated below.

Setting value	Axis name displayed on a screen such as the position display screen
0	X
1	X1
77	XM
83	XS

If a multi-path system is used, no extended axis name is used within a path, and no subscript is set for the axis names, then the path number is automatically used as the subscript for the axis names. To disable the display of axis name subscripts, set a blank (32) of ASCII code in the parameter for specifying an axis name subscript.

NOTE
If even one axis in a path uses an extended axis name when bit 2 (EAS) of parameter No. 11308 is set to 0, subscripts cannot be used for axis names in the path.

3141	Path name (1st character)
3142	Path name (2nd character)
3143	Path name (3rd character)
3144	Path name (4th character)
3145	Path name (5th character)
3146	Path name (6th character)
3147	Path name (7th character)

[Input type] Parameter input
[Data type] Byte path
[Valid data range] See the character-code correspondence table.
Specify a path name with codes.
Any character string consisting of alphanumeric characters, katakana characters, and special characters with a maximum length of seven characters can be displayed as a series name.

NOTE
1 For characters and codes, see Appendix A, “CHARACTER CODE LIST”.
2 When 0 is set in parameter No. 3141, PATH1(,PATH2...) are displayed as path names.

	#7	#6	#5	#4	#3	#2	#1	#0
3201		NPE						

[Input type] Parameter input
[Data type] Bit path

 # 6 NPE With an M02, M30, or M99 block, program registration is assumed to be:
0: Completed
1: Not completed

	#7	#6	#5	#4	#3	#2	#1	#0
3202				NE9				NE8

[Input type] Parameter input
[Data type] Bit path

 # 0 NE8 Editing of subprograms with program numbers 8000 to 8999
0: Not inhibited
1: Inhibited
When this parameter is set to 1, the following editing operations are disabled:
(1) Program deletion (Even when deletion of all programs is specified, programs with program numbers 8000 to 8999 are not deleted.)
(2) Program output (Even when outputting all programs is specified, programs with program numbers 8000 to 8999 are not output.)
(3) Program number search
(4) Program editing of registered programs
(5) Program registration
(6) Program collation
(7) Displaying programs

NOTE
This parameter setting does not affect the following programs:
(1) Programs on the Data Server
(2) Programs for running and editing memory card programs on a memory card

 # 4 NE9 Editing of subprograms with program numbers 9000 to 9999
0: Not inhibited
1: Inhibited
When this parameter is set to 1, the following editing operations are disabled:
(1) Program deletion (Even when deletion of all programs is specified, programs with program numbers 9000 to 9999 are not deleted.)
(2) Program output (Even when outputting all programs is specified, programs with program numbers 9000 to 9999 are not output.)
(3) Program number search
(4) Program editing of registered programs
(5) Program registration
(6) Program collation
(7) Displaying programs

NOTE
This parameter setting does not affect the following programs:
(1) Programs on the Data Server
(2) Programs for running and editing memory card programs on a memory card

	#7	#6	#5	#4	#3	#2	#1	#0
3203	MCL	MER	MZE					

[Input type] Parameter input
[Data type] Bit path

 # 5 MZE After MDI operation is started, program editing during operation is:
0: Enabled
1: Disabled

 # 6 MER When the last block of a program has been executed at single block operation in the MDI mode, the executed block is:
0: Not deleted
1: Deleted

NOTE
When MER is set to 0, the program is deleted if the end-of-record mark (%) is read and executed. (The mark % is automatically inserted at the end of a program.)

 # 7 MCL Whether a program prepared in the MDI mode is cleared by reset
0: Not deleted
1: Deleted

	#7	#6	#5	#4	#3	#2	#1	#0
3204		MKP						

[Input type] Parameter input
[Data type] Bit path

 # 6 MKP When M02, M30, or EOR(%) is executed during MDI operation, the created MDI program is:
0: Erased automatically.
1: Not erased automatically.

NOTE
If the bit 6 (MER) of parameter No. 3203 is 1, executing the last block provides a choice of whether to automatically erase a created program.

	#7	#6	#5	#4	#3	#2	#1	#0
3207			VRN					

[Input type] Parameter input
[Data type] Bit

 # 5 VRN On the custom macro variable screen, the variable names of common variables #500 to #549 are:
0: Not displayed.
1: Displayed.

3210	Program protection (PSW)

[Input type] Parameter input
[Data type] 2-word
[Valid data range] 0 to 99999999

This parameter sets a password for protecting program Nos. 9000 to 9999. When a value other than zero is set in this parameter and this value differs from the keyword set in parameter No.3211, bit 4 (NE9) of parameter No.3202 for protecting program Nos. 9000 to 9999 is automatically set to 1.
This disables the editing of program Nos. 9000 to 9999. Until the value set as the password is set as a keyword, NE9 cannot be set to 0 and the password cannot be modified.

NOTE
1 The state where password ≠ 0 and password ≠ keyword is referred to as the locked state. When an attempt is made to modify the password by MDI input operation in this state, the warning message "WRITE PROTECTED" is displayed to indicate that the password cannot be modified. When an attempt is made to modify the password with G10 (programmable parameter input), alarm (PS0231) is issued.
2 When the value of the password is not 0, the parameter screen does not display the password. Care must be taken in setting a password.

3211	Program protection key (KEY)

[Input type] Parameter input
[Data type] 2-word
[Valid data range] 0 to 99999999
When the value set as the password (set in parameter No.3210) is set in this parameter, the locked state is released and the user can now modify the password and the value set in bit 4 (NE9) of parameter No.3202.

NOTE
The value set in this parameter is not displayed. When the power is turned off, this parameter is set to 0.

3220	Password (PSW)

[Input type] Locked parameter
[Data type] 2-word
[Valid data range] 0 to 99999999
This parameter sets a password (PSW). When a value other than 0 is set, a password is set. When a password is set, a blank is displayed in this parameter, and the state (locked state) where an operation such as program editing is locked is set. When password (PSW) = 0, namely, in the normal state, or when password (PSW) = keyword (KEY), namely, in the unlock state, this parameter can be set.

3221	Keyword (KEY)

[Input type] Locked parameter
[Data type] 2-word
[Valid data range] 0 to 99999999
When the same value as the password (PSW) is set in this parameter, the lock is released (unlock state). The value set in this parameter is not displayed.
The value of this parameter is initialized to 0 automatically when the power is turned on. So, if the power is turned off in the unlock state then is turned on again, the lock state is automatically set.

3222	Program protection range (minimum value) (PMIN)
3223	Program protection range (maximum value) (PMAX)

[Input type] Locked parameter
[Data type] 2-word
[Valid data range] 0 to 99999999

The programs in a range set here can be locked. Set the minimum program number and maximum program number of a desired range.
Set these parameters to satisfy PMAX > PMIN.
These parameters can be set when password (PSW) = 0, namely, in the normal state, or when password (PSW) = keyword (KEY), namely in the unlock state.
Example)
Parameter No.3222 = 7000
Parameter No.3223 = 8499
When the values above are set, the programs from O7000 to O8499 can be locked.
When PMIN = 0, the specification of PMIN = 9000 is assumed. When PMAX = 0, the specification of PMAX = 9999 is assumed. So, when these parameters are set to the defaults, the programs from O9000 to O9999 are locked.

NOTE
1 Parameter No. 3220 to No. 3223 are neither punched nor read.
2 Parameter No. 3220 to No. 3223 are not cleared even when a parameter file clear operation is performed in the IPL state.
3 The values of a password (PSW) and keyword (KEY) are not displayed. When password (PSW) = 0, 0 is displayed in parameter No. 3220 to indicate that the normal state is set.
4 When a password (PSW) or keyword (KEY) is set, [+INPUT] has the same effect as [INPUT]. For example, if the input operation "1[+INPUT]" is performed when 99 is set in the keyword (KEY) parameter, "1" is set.

	#7	#6	#5	#4	#3	#2	#1	#0
3233							PDM	PCE

[Input type] Parameter input
[Data type] Bit

 # 0 PCE Program editing is:
0: Performed in the word edit mode.
1: Performed in the character edit mode.

 # 1 PDM On the Data Server file list screen:
0: M198 operation folders and DNC operation files can be set.
1: Folders in the Data Server can be set as the foreground folder and background folder.

NOTE
When an M198 external subprogram call or DNC operation is performed on the Data Server, set this bit to 0.
For the foreground and background folders, refer to Chapter 12, "PROGRAM MANAGEMENT", in Part III in the user's manual (B-63944EN).

	#7	#6	#5	#4	#3	#2	#1	#0
3280								NLC

[Input type] Parameter input
[Data type] Bit

 # 0 NLC Dynamic display language switching is:
0: Enabled.
1: Disabled.
When dynamic display language switching is disabled, the language setting screen is not displayed. In this case, change the setting of parameter No. 3281 on the parameter screen then turn on the power again to switch the display language.

3281	Display language

[Input type] Parameter input
[Data type] Byte
[Valid data range] 0 to 17
Select a display language from the following:
0 : English
1 : Japanese
2 : German
3 : French
4 : Chinese(traditional characters)
5 : Italian
6 : Korean
7 : Spanish
8 : Dutch
9 : Danish
10 : Portuguese
11 : Polish
12 : Hungarian
13 : Swedish
14 : Czech
15 : Chinese(simplified characters)
16 : Russian
17 : Turkish
If a number not indicated above is set, English is selected.

	#7	#6	#5	#4	#3	#2	#1	#0
3400			PGD				MGC	

[Input type] Parameter input
[Data type] Bit path

 # 1 MGC When a single block specifies multiple M commands, an M code group check is:
0: Made.
1: Not made.

 # 5 PGD The G10.9 command (programmable diameter/radius specification switching) is:
0: Disabled.
1: Enabled.

NOTE
1 The option for the diameter and radius switching function is required.
2 When the G10.9 command is enabled by this parameter, signal-based diameter/radius switching is disabled.

	#7	#6	#5	#4	#3	#2	#1	#0
3401	GSC	GSB	ABS	MAB				DPI
			ABS	MAB				DPI

[Input type] Parameter input
[Data type] Bit path

 # 0 DPI When a decimal point is omitted in an address that can include a decimal point
0: The least input increment is assumed. (Normal decimal point input)
1: The unit of mm, inches, degree, or second is assumed. (Pocket calculator type decimal point input)

 # 4 MAB Switching between the absolute and incremental commands in MDI operation
0: Performed by G90 or G91
1: Depending on the setting of bit 5 (ABS) of parameter No.3401

NOTE
When G code system A of the lathe system is used, this parameter is invalid.

 # 5 ABS Program command in MDI operation
0: Assumed as an incremental command
1: Assumed as an absolute command

> **NOTE**
> ABS is valid when bit 4 (MAB) of parameter No.3401 is set to 1. When G code system A of the lathe system is used, this parameter is invalid.

 # 6 GSB The G code system is set.
 # 7 GSC

GSC	GSB	G code
0	0	G code system A
0	1	G code system B
1	0	G code system C

> **NOTE**
> G code system B and G code system C are optional functions. When no option is selected, G code system A is used, regardless of the setting of these parameters.

	#7	#6	#5	#4	#3	#2	#1	#0
3402	G23	CLR		FPM	G91			G01
	G23	CLR			G91	G19	G18	G01

[Input type] Parameter input
[Data type] Bit path

 # 0 G01 G01 Mode entered when the power is turned on or when the control is cleared
0: G00 mode (positioning)
1: G01 mode (linear interpolation)

 # 1 G18 Plane selected when power is turned on or when the control is cleared
0: G17 mode (plane XY)
1: G18 mode (plane ZX)

 # 2 G19 Plane selected when power is turned on or when the control is cleared
0: The setting of bit 1 (G18) of parameter No.3402 is followed.
1: G19 mode (plane YZ)
When this bit is set to 1, set bit 1 (G18) of parameter No.3402 to 0.

 # 3 G91 When the power is turned on or when the control is cleared
0: G90 mode (absolute command)
1: G91 mode (incremental command)

 # 4 FPM At power-on time or in the cleared state:
0: G99 or G95 mode (feed per revolution) is set.
1: G98 or G94 mode (feed per minute) is set.

 # 6 CLR Reset button on the MDI panel, external reset signal, reset and rewind signal, and emergency stop signal
0: Cause reset state.
1: Cause clear state.
For the reset and clear states, refer to Appendix in the User's Manual.

 # 7 G23 When the power is turned on
0: G22 mode (stored stroke check on)
1: G23 mode (stored stroke check off)

	#7	#6	#5	#4	#3	#2	#1	#0
3404	M3B		M02	M30		SBP		

[Input type] Parameter input
[Data type] Bit path

 # 2 SBP In an external device subprogram call, the address P format is based on:
0: File number specification
1: Program number specification

NOTE
In memory card operation, the program number specification format is used, regardless of the setting of this parameter.

 # 4 M30 When M30 is specified in a memory operation:
0: M30 is sent to the machine, and the head of the program is automatically searched for. So, when the ready signal FIN is returned and a reset or reset and rewind operation is not performed, the program is executed, starting from the beginning.
1: M30 is sent to the machine, but the head of the program is not searched for. (The head of the program is searched for by the reset and rewind signal.)

 # 5 M02 When M02 is specified in memory operation
0: M02 is sent to the machine, and the head of the program is automatically searched for. So, when the end signal FIN is returned and a reset or reset and rewind operation is not performed, the program is executed, starting from the beginning.
1: M02 is sent to the machine, but the head of the program is not searched for. (The head of the program is searched for by the reset and rewind signal.)

 # 7 M3B The number of M codes that can be specified in one block
0: One
1: Up to three

	#7	#6	#5	#4	#3	#2	#1	#0
3405				CCR	G36		DWL	AUX
							DWL	AUX

[Input type] Parameter input
[Data type] Bit path

 # 0 AUX When the second auxiliary function is specified in the calculator-type decimal point input format or with a decimal point, the multiplication factor for a value output (onto the code signal) relative to a specified value is such that:
0: The same multiplication factor is used for both of metric input and inch input.
1: A multiplication factor used for inch input is 10 times greater than that used for metric input.

When the second auxiliary function is specified in the calculator-type decimal point input format or with a decimal point, the value output onto the code signal is a specified value multiplied by a value indicated below.

Increment system		Parameter AUX=0	Parameter AUX=1
Metric input system	IS-A for reference axis	100 times	100 times
	IS-B for reference axis	1000 times	1000 times
	IS-C for reference axis	10000 times	10000 times
	IS-D for reference axis	100000 times	100000 times
	IS-E for reference axis	1000000 times	1000000 times
Inch input system	IS-A for reference axis	100 times	1000 times
	IS-B for reference axis	1000 times	10000 times
	IS-C for reference axis	10000 times	100000 times
	IS-D for reference axis	100000 times	1000000 times
	IS-E for reference axis	1000000 times	10000000 times

 # 1 DWL The dwell time (G04) is:
0: Always dwell per second.
1: Dwell per second in the feed per minute mode (G94), or dwell per rotation in the feed per rotation mode (G95).

 # 3 G36 As a G code to be used with the automatic tool length measurement function (M series)/automatic tool offset function (T series) is:
0: G36 (T series only)/G37 is used.
1: G37.1/G37.2/G37.3 is used.

NOTE
If it is necessary to perform circular threading (counterclockwise), set this parameter to 1.

 # 4 CCR Addresses used for chamfering

0: Address is “I”, “J”, or “K”.
In direct drawing dimension programming, addresses ",C", ",R", and ",A" (with comma) are used in stead of "C", "R", and "A".

1: Address is “C”.
Addresses used for direct drawing dimension programming are "C", "R", and "A" without comma.

NOTE
If this bit (CCR) is set to 0, the function for changing the compensation direction by specifying I, J, or K in a G01 block in the tool radius · tool nose radius compensation mode cannot be used.
If this bit (CCR) is set to 1 when address C is used as an axis name, the chamfer function cannot be used.

	#7	#6	#5	#4	#3	#2	#1	#0
3406	C07	C06	C05	C04	C03	C02	C01	
3407	C15	C14	C13	C12	C11	C10	C09	C08
3408	C23	C22	C21	C20	C19	C18	C17	C16
3409		C30	C29	C28	C27	C26	C25	C24

[Input type] Parameter input
[Data type] Bit

C01 to C30 If bit 6 (CLR) of parameter No.3402 is set to 1, set a group of G codes to be placed in the cleared state when the CNC is reset by the RESET key of the MDI panel, the external reset signal, the reset & rewind signal, or the emergency stop signal.
The table below indicates the correspondence between bits and G code groups
The setting of a bit has the following meaning:

0: Places the G code group in the cleared state.
1: Does not place G code group in the cleared state.

Parameter	G code group
C01	01
C02	02
C03	03
:	:
C30	30

3410	Tolerance of arc radius

[Input type] Setting input
[Data type] Real path
[Unit of data] mm, inch (input unit)
[Minimum unit of data] Depend on the increment system of the reference axis
[Valid data range] 0 or positive 9 digit of minimum unit of data (refer to the standard parameter setting table (B))
(When the increment system is IS-B, 0.0 to +999999.999)
When a circular interpolation command is executed, the tolerance for the radius between the start point and the end point is set.

3411	M code preventing buffering 1
3412	M code preventing buffering 2
:	
3420	M code preventing buffering 10

[Input type] Parameter input
[Data type] 2-word path
[Valid data range] 0 to 999999999
Set M codes that prevent buffering the following blocks. If processing directed by an M code must be performed by the machine without buffering the following block, specify the M code.
M00, M01, M02, and M30 always prevent buffering even when they are not specified in these parameters.

3421	Range specification 1 of M codes that do not perform buffering (lower limit)
3422	Range specification 1 of M codes that do not perform buffering (upper limit)
3423	Range specification 2 of M codes that do not perform buffering (lower limit)
3424	Range specification 2 of M codes that do not perform buffering (upper limit)
3425	Range specification 3 of M codes that do not perform buffering (lower limit)
3426	Range specification 3 of M codes that do not perform buffering (upper limit)
3427	Range specification 4 of M codes that do not perform buffering (lower limit)
3428	Range specification 4 of M codes that do not perform buffering (upper limit)
3429	Range specification 5 of M codes that do not perform buffering (lower limit)
3430	Range specification 5 of M codes that do not perform buffering (upper limit)
3431	Range specification 6 of M codes that do not perform buffering (lower limit)

3432	Range specification 6 of M codes that do not perform buffering (upper limit)

[Input type] Parameter input
[Data type] 2-word path
[Valid data range] 3 to 99999999

When a specified M code is within the range specified with parameter Nos.3421 and 3422, 3423 and 3424, 3425 and 3426, 3427 and 3428, 3429 and 3430, or 3431 and 3432, buffering for the next block is not performed until the execution of the block is completed.

NOTE
M00, M01, M02, and M30 are M codes that do not perform buffering, regardless of parameter setting. M98, M99, M codes for calling subprograms, and M codes for calling custom macros are M codes that performs buffering, regardless of parameter setting.

3441	Start number of M codes for which an M code group can be set (1)
3442	Start number of M codes for which an M code group can be set (2)
3443	Start number of M codes for which an M code group can be set (3)
3444	Start number of M codes for which an M code group can be set (4)

[Input type] Parameter input
[Data type] 2-word path
[Valid data range] 0, 100to99999999

Code numbers 0 to 99 on the M code group setting screen correspond to M00 to M99. When adding M codes after the first 100 M codes, specify a start M code number in these parameters. Thus, up to 400 M codes can be added to the M code group setting screen in groups of 100 M codes starting with the set value. When 0 is set, no M codes are added to the M code group setting screen.
When setting these parameters, follow the setting condition described below. If the condition is not satisfied, no M codes are added to the M code group setting screen as in the case where 0 is set.
(Setting condition)

The settings of parameters (1) to (4) (excluding the setting of 0) must satisfy:
99 < (1), (1)+99 < (2), (2)+99 < (39, (3) +99 < (4)

	#7	#6	#5	#4	#3	#2	#1	#0
3450	BDX							AUP

[Input type] Parameter input
[Data type] Bit path

 # 0 AUP The second auxiliary function specified in the calculator-type decimal point input format, with a decimal point, or with a negative value is:

0: Disabled.
1: Enabled.

If the second auxiliary function is specified after setting this bit to 0, the following operation results:

1. When a value is specified without a decimal point
   A specified value is output onto the code signal without modification, regardless of the setting of the calculator-type decimal point input format (with bit 0 (DPI) of parameter No. 3401).
2. When a value is specified with a decimal point
   The alarm (PS0007) is issued.
3. When a negative value is specified
   The alarm (PS0006) is issued.

 # 7 BDX When ASCII code is called using the same address as the address for the second auxiliary function (specified by parameter No.3460), this parameter prevents the argument unit used when the option for the second auxiliary function is selected from differing from the argument unit used when the same option is not selected.

0: When bit 0 (AUP) of parameter No.3450 is set to 1, the argument unit differs, depending on whether the option for the second auxiliary function is selected or not.
1: The same argument unit is used. (The unit applied when the option for the second auxiliary function is selected is used.)

[Example]

A setting is made so that address B is used to call O9004, and the program O1 below is executed with parameter No.3460 = 66.

```
O1      O9004
B2      #500 = #146
M30     M99
```

When the increment system is IS-B, and metric input is used, #500 assumes a value indicated in the table below.

Bit 0 (DPI) of parameter No. 3401	Bit 0 (AUP) of parameter No. 3450	BDX=0		BDX=1
		Without the second auxiliary function option	With the second auxiliary function option	
0	0	2.000	2.000	2.000
	1	2.000	0.002	0.002
1	0	2.000	2.000	2.000
	1	2.000	2.000	2.000

	#7	#6	#5	#4	#3	#2	#1	#0
3451								
								GQS

[Input type] Parameter input
[Data type] Bit path

 # 0 GQS When threading is specified, the threading start angle shift function (Q) is:
0: Disabled.
1: Enabled.

	#7	#6	#5	#4	#3	#2	#1	#0
3452	EAP							

[Input type] Parameter input
[Data type] Bit path

 # 7 EAP When bit 0 (ADX) of parameter No.3455 is set to 1, calculator-type decimal point input at a macro calling argument address is:
0: Enabled.
1: Disabled.

NOTE
This parameter is valid when bit 0 (DPI) of parameter No.3401 is set to 0.

	#7	#6	#5	#4	#3	#2	#1	#0
3455								AXDx

[Input type] Parameter input
[Data type] Bit axis

 # 0 AXDx If a decimal point is omitted for an axis address with which a decimal point can be used, the value is determined:

0: In accordance with the least input increment. (Normal decimal point input)
1: In millimeters, inches, or seconds. (calculator-type decimal point input)

NOTE
This parameter specifies the calculator-type decimal point input function for each axis.
For the same axis name, be sure to make the same setting.

	#7	#6	#5	#4	#3	#2	#1	#0
3457	SCF	SCC			SYS	MC1	MC2	LIB

[Input type] Parameter input
[Data type] Bit path

NOTE
1 The parameters LIB, MC2, MC1, and SYS are used to set a search folder for the following subprogram/macro calls:
- Subprogram call based on an M code
- Subprogram call based on a particular address
- Subprogram call based on a second auxiliary function code
- Macro call based on a G code
- Macro call based on an M code
- Macro call based on a T code
- One-touch macro call

2 The parameter SCF is used to set whether to add a search folder for the following subprogram/macro calls:
- Subprogram call based on M98
- Figure copy based on G72.1/G72.2
- Macro call based on G65/G66/G66.1
- Macro interrupt based on M96

 # 0 LIB The common program directory "//CNC_MEM/USER/LIBRARY/" of the initial directories is:

0: Set as a search directory.
1: Not set as a search directory.

 # 1 **MC2** MTB dedicated directory 2 "//CNC_MEM/MTB2/" of the initial directories is:
0: Set as a search directory.
1: Not set as a search directory.

 # 2 **MC1** MTB dedicated directory 1 "//CNC_MEM/MTB1/" of the initial directories is:
0: Set as a search directory.
1: Not set as a search directory.

 # 3 **SYS** The system directory "//CNC_MEM/SYSTEM/" of the initial directories is:
0: Set as a search directory.
1: Not set as a search directory.

 # 6 **SCC** The same folder as the main program is added to the top of the search order as a search folder for the following each subprogram call and macro call.

- Subprogram call by M code
- Subprogram call by ASCII code
- Subprogram call by the second auxiliary function code
- Macro call by Scode
- Macro call by T code
- Macro call by G code
- Macro call by M code
- One-touch macro call

The same folder as the main program is:
0: Not added in the search order.
1: Added in the search order.
When a search folder is added, a search is made in the following order:
0) Folder only for embedded macro (With the embedded macro-function.)
1) Folder where the main program is stored
2) Common program folder, which is an initial folder (LIBRARY)
3) MTB-dedicated folder 2, which is an initial folder (MTB2)
4) MTB-dedicated folder 1, which is an initial folder (MTB1)
5) System folder, which is an initial folder (SYSTEM)
The folders of 2) through 5) can be excluded from search target folders by setting the bits 0 (LIB), 1 (MC2), 2 (MC1), and 3 (SYS) of parameter No.3457.

 # 7 SCF A search folder is:
0: Not added.
1: Added.

When a search folder is added, a search is made in the following order:
0) Folder only for embedded macro (With the embedded macro-function.)
1) Folder where the main program is stored
2) Common program folder, which is an initial folder (LIBRARY)
3) MTB-dedicated folder 2, which is an initial folder (MTB2)
4) MTB-dedicated folder 1, which is an initial folder (MTB1)
5) System folder, which is an initial folder (SYSTEM)

The folders of 3) through 5) can be excluded from search target folders by setting the bits 1 (MC2), 2 (MC1), and 3 (SYS) of parameter No.3457.

3460 **Second auxiliary function specification address**

[Input type] Parameter input
[Data type] Byte path
[Valid data range] 65to67, 85to87
Specify which of A, B, C, U, V, and W is to be used as the address for specifying the second auxiliary function. If an address used as an axis name is specified, the second auxiliary function is disabled.

Name	A	B	C	U	V	W
Setting value	65	66	67	85	86	87

Address B is assumed when a value other than the above is set.
However, the name U, V, or W can be used with the T series only when G code system B or C is used. When a value from 85 to 87 is specified with G code system A, the specification address for the second auxiliary function is B.

3471 **Allowable difference between the specified end position and the end position obtained from the increase/decrease and frequency in spiral interpolation or conic interpolation**

[Input type] Parameter input
[Data type] Real axis
[Unit of data] mm, inch (input unit)
[Minimum unit of data] Depend on the increment system of the applied axis
[Valid data range] 0 or positive 9 digit of minimum unit of data (refer to the standard parameter setting table (B))
(When the increment system is IS-B, 0.0 to +999999.999)
This parameter sets the maximum allowable difference (absolute value) between the specified end position and the end position obtained from the increase/decrease and frequency in spiral or conic interpolation.

3472	Minimum radius needed to maintain the actual speed in spiral or conic interpolation

[Input type] Parameter input
[Data type] Real path
[Unit of data] mm, inch (input unit)
[Minimum unit of data] Depend on the increment system of the reference axis
[Valid data range] (For IS-B and millimeter machines, 1.0 to 999999.999; for inch machines, 1.0 to 99999.9999)
If this parameter value is 0 or a value outside the valid data range, the minimum value of the range is assumed.
In spiral interpolation and conic interpolation, the speed is generally held constant. In an area near the center, the spiral radius decreases, resulting in an extremely high angular velocity. To prevent this, once the spiral radius has reached the parameter-set value, the angular velocity subsequently remains constant. As a result, the actual speed decreases.

	#7	#6	#5	#4	#3	#2	#1	#0
3605								BDPx

[Input type] Parameter input
[Data type] Bit axis

NOTE
When this parameter is set, the power must be turned off before operation is continued..

 # 0 BDPx Both-direction pitch error compensation is:
0: Not used.
1: Used.

3620	Number of the pitch error compensation position for the reference position for each axis

NOTE
When this parameter is set, the power must be turned off before operation is continued.

[Input type] Parameter input
[Data type] Word axis
[Valid data range] 0 to 1535
Set the number of the pitch error compensation position for the reference position for each axis.

3621	Number of the pitch error compensation position at extremely negative position for each axis

NOTE
When this parameter is set, the power must be turned off before operation is continued.

[Input type] Parameter input
[Data type] Word axis
[Valid data range] 0 to 1535
Set the number of the pitch error compensation position at the extremely negative position for each axis.

3622	Number of the pitch error compensation position at extremely positive position for each axis

NOTE
When this parameter is set, the power must be turned off before operation is continued.

[Input type] Parameter input
[Data type] Word axis
[Valid data range] 0 to 1535
Set the number of the pitch error compensation position at the extremely positive position for each axis.
This value must be larger than set value of parameter (No.3620).

3623 **Magnification for pitch error compensation for each axis**

NOTE
When this parameter is set, the power must be turned off before operation is continued.

[Input type] Parameter input
[Data type] Byte axis
[Valid data range] 0 to 100
Set the magnification for pitch error compensation for each axis.
If the magnification is set to 1, the same unit as the detection unit is used for the compensation data.
If 0 is set, compensation is not performed.

3624 **Interval between pitch error compensation positions for each axis**

NOTE
When this parameter is set, the power must be turned off before operation is continued.

[Input type] Parameter input
[Data type] Real axis
[Unit of data] mm, inch, degree (machine unit)
[Minimum unit of data] Depend on the increment system of the applied axis
[Valid data range] See the description below.
The pitch error compensation positions are arranged with equal spacing. The space between two adjacent positions is set for each axis.
The minimum interval between pitch error compensation positions is limited and obtained from the following equation:
Minimum interval between pitch error compensation positions = maximum feedrate/7500
Unit : mm, inch, deg or mm/min, inch/min, deg/min
Example:
When the maximum feedrate is 15000 mm/min, the minimum interval between pitch error compensation positions is 2 mm.

3625 **Travel distance per revolution in pitch error compensation of rotation axis type**

NOTE
When this parameter is set, the power must be turned off before operation is continued.

[Input type] Parameter input
[Data type] Real axis
[Unit of data] mm, inch, degree (machine unit)
[Minimum unit of data] Depend on the increment system of the applied axis
[Valid data range] See the description below.

If the pitch error compensation of rotation axis type is performed (bit 1 (ROSx) of parameter No.1006 is set to 0 and bit 0 (ROTx) of parameter No.1006 is set to 1), set the travel distance per revolution. The travel distance per revolution does not have to be 360 degrees, and a cycle of pitch error compensation of rotation axis type can be set.

However, the travel distance per revolution, compensation interval, and number of compensation points must satisfy the following condition:

(Travel distance per revolution)
= (Compensation interval) × (Number of compensation points)

The compensation at each compensation point must be set so that the total compensation per revolution equals 0.

NOTE
If 0 is set, the travel distance per revolution becomes 360 degrees.

3626 **Number of the both-direction pitch error compensation position at extremely negative position (for movement in the negative direction)**

NOTE
When this parameter is set, the power must be turned off before operation is continued.

[Input type] Parameter input
[Data type] Word axis
[Valid data range] 0 to 1023, 3000 to 4023

When using both-direction pitch error compensation, set the number of compensation point at the farthest end in the negative direction for a movement in the negative direction.

NOTE
1 For a movement in the positive direction, set the compensation point number at the farthest end in the negative direction in parameter No.3621.
2 A set of compensation data items for a single axis should not be set to lie astride 1023 to 3000.

3627	Pitch error compensation at reference position when a movement to the reference position is made from the direction opposite to the direction of reference position return

NOTE
When this parameter is set, the power must be turned off before operation is continued.

[Input type] Parameter input
[Data type] Word axis
[Unit of data] Detection unit
[Valid data range] -32768 to 32767

Set the absolute value of pitch error compensation at reference position when a movement to the reference position is made from the negative direction if the direction of reference position return (bit 5 (ZMI) of parameter No.1006) is positive or from the positive direction if the direction of reference position return is negative.

	#7	#6	#5	#4	#3	#2	#1	#0
3700							NRF	CRF

[Input type] Parameter input
[Data type] Bit path

 # 0 CRF Reference position setting at an arbitrary position under Cs contour control is:
0: Not used.
1: Used.

NOTE
When this function is used, an attempt to specify G00 for a Cs contour control axis without performing a reference position return operation even once after switching the serial spindle to the Cs contour control mode results in the alarm (PS0303) even if bit 1 (NRF) of parameter No. 3700 is set to 0. Be sure to perform a reference position return operation by specifying G28.

 # 1 NRF With the first move command (G00) after switching the series spindle to Cs contour control mode:
0: A reference position return operation is once performed then positioning is performed.
1: A normal positioning operation is performed.

	#7	#6	#5	#4	#3	#2	#1	#0
3702							EMS	

[Input type] Parameter input
[Data type] Bit path

 # 1 EMS The multi-spindle control function is:
0: Used.
1: Not used.

	#7	#6	#5	#4	#3	#2	#1	#0
3716								A/Ss

[Input type] Parameter input
[Data type] Bit spindle

NOTE
When this parameter is set, the power must be turned off before operation is continued.

 # 0 A/Ss Spindle motor type is :
0: Analog spindle.
1: Serial spindle.

NOTE
1 When an analog spindle is used, the option for spindle analog output is required.
2 When a serial spindle is used, the option for spindle serial output is required.
3 The option for the number of controlled spindles needs to be specified.

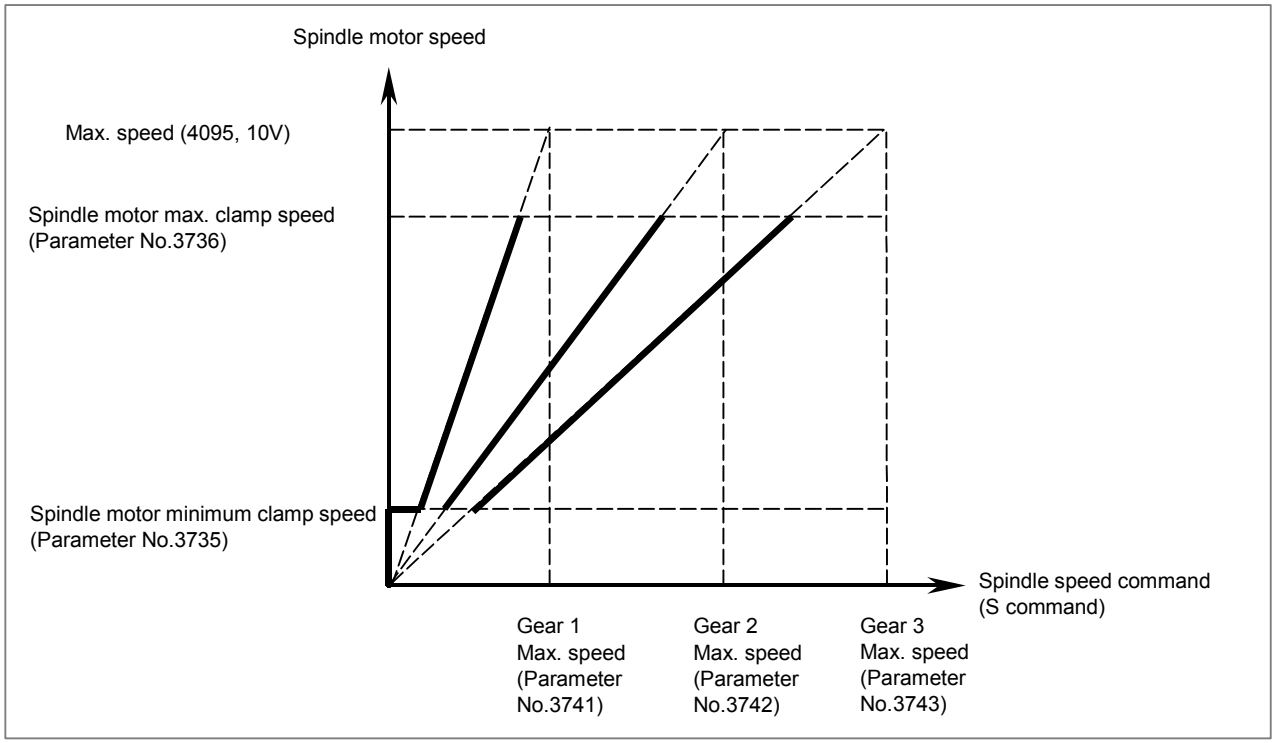
3717	Motor number to each spindle

NOTE
When this parameter is set, the power must be turned off before operation is continued.

[Input type] Parameter input
[Data type] Byte spindle
[Valid data range] 0 to Maximum number of controlled axes
Set a spindle amplifier number to be assigned to each spindle.
0: No spindle amplifier is connected.
1: Spindle motor connected to amplifier number 1 is used.
2: Spindle motor connected to amplifier number 2 is used.
to
n: Spindle motor connected to amplifier number n is used.

3741	Maximum spindle speed for gear 1
3742	Maximum spindle speed for gear 2
3743	Maximum spindle speed for gear 3
3744	Maximum spindle speed for gear 4

[Input type] Parameter input
[Data type] 2-word spindle
[Unit of data] min^{-1}
[Valid data range] 0 to 99999999
Set the maximum spindle speed corresponding to each gear.



3770	Axis as the calculation reference in constant surface speed control

[Input type] Parameter input
[Data type] Byte path
[Valid data range] 0 to Number of controlled axes
Set the axis as the calculation reference in constant surface speed control.

NOTE
When 0 is set, constant surface speed control is always applied to the X-axis. In this case, specifying P in a G96 block has no effect on the constant surface speed control.

3781	P code for selecting the spindle in multi-spindle control

[Input type] Parameter input
[Data type] Word spindle
[Valid data range] 0 to 32767

If bit 3 (MPP) of parameter No. 3703 is set to 1, set the P code to select each spindle under multi-spindle control. Specify the P code in a block containing the S command.

Example)

If the P code value for selecting the second spindle is set to 3,
S1000 P3;
causes the second spindle to rotate at S1000.

NOTE

1 This parameter is valid if bit 3 (MPP) of parameter No. 3703 is set to 1.
2 If this parameter is set to 0, the corresponding spindle cannot be selected by a P code.
3 Under multipath control, the P code specified here is valid for each path.
For instance, if the P code to select the first spindle of path 2 is set to 21, specifying S1000 P21; in path 1 causes the first spindle of path 2 to be rotated at S1000.
4 Identical P code values cannot be used for different spindles. (Identical P code values cannot be used even if the paths are different.)
5 When this parameter is used (when bit 3 (MPP) of parameter No. 3703 is set to 1), the spindle command selection signal is invalid.
6 To use this parameter, the multi-spindle control function is needed.

Parameters Nos. 4000 to 4799 below are basically used with the serial spindle amplifier. For details of these parameters, refer to either of the following manuals and other related documents, depending on the spindle that is actually connected.

- FANUC AC SPINDLE MOTOR αi series Parameter Manual (B-65280EN)

	#7	#6	#5	#4	#3	#2	#1	#0
4900								FLRs

[Input type] Parameter input
[Data type] Bit spindle

 # 0 FLRs When the spindle speed fluctuation detection function is used, the unit of an allowable ratio (q) and fluctuation ratio (r) set by parameter No. 4911 and No. 4912 is:
0: 1%
1: 0.1%

4911	Allowable speed ratio (q) used to assume that the spindle has reached a specified speed

[Input type] Parameter input
[Data type] Word spindle
[Unit of data] 1%, 0.1%
[Valid data range] 1 to 100, 1 to 1000
When the spindle speed fluctuation detection function is used, set an allowable speed ratio (q) used to assume that the spindle has reached a specified speed.

NOTE
The unit of data is determined by bit 0 (FLR) of parameter No. 4900.

4912	Spindle variation ratio (r) for not issuing a spindle speed fluctuation detection alarm

[Input type] Parameter input
[Data type] Word spindle
[Unit of data] 1%, 0.1%
[Valid data range] 1 to 100, 1 to 1000
When the spindle speed fluctuation detection function is used, set a spindle fluctuation ratio (r) for not issuing an alarm.

NOTE
The unit of data is determined by bit 0 (FLR) of parameter No. 4900.

4913	Spindle speed fluctuation width (i) for not issuing a spindle speed fluctuation detection alarm

[Input type] Parameter input
[Data type] 2-word spindle
[Unit of data] min^{-1}
[Valid data range] 0 to 99999
When the spindle speed fluctuation detection function is used, set an allowable fluctuation width (i) for not issuing an alarm.

4914	Time (p) from the change of a specified speed until spindle speed fluctuation detection is started

[Input type] Parameter input
[Data type] 2-word spindle
[Unit of data] msec
[Valid data range] 0 to 999999
When the spindle speed fluctuation detection function is used, set a time (p) from the change of a specified speed until spindle speed fluctuation detection is started. In other words, spindle speed fluctuation detection is not performed until a set time has elapsed after a specified speed is changed. However, when the actual spindle speed is assumed to have reached a specified value within a set time (p), spindle speed fluctuation detection is started.

	#7	#6	#5	#4	#3	#2	#1	#0
4950						ISZs	IDMs	IORs

[Input type] Parameter input
[Data type] Bit spindle

 # 0 IORs Resetting the system in the spindle positioning mode
0: Does not release the mode.
1: Releases the mode

 # 1 IDMs The direction of spindle positioning (half-fixed angle positioning based on M codes) is:
0: Plus direction.
1: Minus direction.

 # 2 ISZs When an M code for switching to the spindle positioning mode is specified for spindle positioning:
0: The spindle is switched to the spindle positioning mode, and spindle orientation operation is performed.
1: Only the switching of the spindle to the spindle positioning mode is performed. (Spindle orientation operation is not performed.)

4960	M code specifying the spindle orientation

[Input type] Parameter input
[Data type] 2-word spindle
[Valid data range] 6 to 97
Set an M code for switching to the spindle positioning mode.

NOTE
1 Do not set an M code that duplicates other M codes used for spindle positioning.
2 Do not set an M code used with other functions (such as M00-05, 30, 98, and 99, and M codes for calling subprograms).

4961	M code releasing the spindle positioning mode

[Input type] Parameter input
[Data type] 2-word spindle
[Valid data range] 6 to 97
Set an M code for canceling the spindle positioning mode on the spindle positioning axis.

NOTE
1 Do not set an M code that duplicates other M codes used for spindle positioning.
2 Do not set an M code used with other functions (such as M00-05, 30, 98, and 99, and M codes for calling subprograms).

4962 **M code for specifying a spindle positioning angle**

[Input type] Parameter input
[Data type] 2-word spindle
[Valid data range] 6 to 9999999

Two methods are available for specifying spindle positioning. One method uses axis address for arbitrary-angle positioning. The other use an M code for half-fixed angle positioning. This parameter sets an M code for the latter method.

In this parameter, set an M code to be used for half-fixed angle positioning based on M codes.

Six M code from Mα to M(α+5) are used for half-fixed angle positioning, when α is the value of this parameter.

- When the number of M codes is set in parameter No. 4964, let α be the value set in parameter No. 4962, and let β be the value set in parameter No. 4964. Then, β M codes from Mα to M(α+β-1) are used as M codes for half-fixed angle positioning based on M codes.

The table below indicates the relationship between the M codes and positioning angles.

M code	Positioning angle	Example: Positioning angle when θ = 30°
Mα	θ	30°
M(α+1)	2θ	60°
M(α+2)	3θ	90°
M(α+3)	4θ	120°
M(α+4)	5θ	150°
M(α+5)	6θ	180°
:	:	:
M(α+β-1)	β×θ	β×30°

β represents the number of M codes set in parameter No. 4964.
(When parameter No. 4964 is set to 0, β = 6.)
θ represents the basic angular displacement set in parameter No.4963.

NOTE
1 Do not set an M code that duplicates other M codes used for spindle positioning.
2 Do not set an M code used with other functions (such as M00-05, 30, 98, and 99, and M codes for calling subprograms).

4963 **Basic angle for half-fixed angle positioning**

[Input type] Parameter input
[Data type] Real spindle
[Unit of data] Degree
[Minimum unit of data] Depend on the increment system of the applied axis
[Valid data range] 0 to 60

This parameter sets a basic angular displacement used for half-fixed angle positioning using M codes.

4964	Number of M codes for specifying a spindle positioning angle

[Input type] Parameter input
[Data type] 2-word spindle
[Valid data range] 0 to 255

This parameter sets the number of M codes used for Half-fixed angle positioning using M codes.
As many M codes as the number specified in this parameter, starting with the M code specified in parameter No.4962, are used to specify half-fixed angle positioning.
Let α be the value of parameter No.4962, and let β be the value of parameter No.4964. That is, M codes from Mα to M(α+β-1) are used for half-fixed angle positioning.
Setting this parameter to 0 has the same effect as setting 6. That is, M code from Mα to M(α+5) are used for half-fixed angle positioning.

NOTE
1 Make sure that M codes from Mα to M (α+β-1) do not duplicate other M codes.
2 Do not set an M code that duplicates other M codes used for spindle positioning.
3 Do not set an M code used with other functions (such as M00-05, 30, 98, and 99, and M codes for calling subprograms).

	#7	#6	#5	#4	#3	#2	#1	#0
5001		EVO						
		EVO			TAL		TLB	TLC

[Input type] Parameter input
[Data type] Bit path

 # 0 TLC

 # 1 TLB These bits are used to select a tool length compensation type.

Type	TLB	TLC
Tool length compensation A	0	0
Tool length compensation B	1	0
Tool length compensation C	-	1

The axis to which cutter compensation is applied varies from type to type as described below.
Tool length compensation A :
Z-axis at all times
Tool length compensation B :
Axis perpendicular to a specified plane (G17/G18/G19)
Tool length compensation C :
Axis specified in a block that specifies G43/G44

 # 3 TAL Tool length compensation C
0: Generates an alarm when two or more axes are offset
1: Not generate an alarm even if two or more axes are offset

 # 6 EVO If a tool compensation value modification is made for tool length compensation A or tool length compensation B in the offset mode (G43 or G44):

0: The new value becomes valid in a block where G43, G44, or an H code is specified next.

1: The new value becomes valid in a block where buffering is performed next.

	#7	#6	#5	#4	#3	#2	#1	#0
5002						LWT	LGN	

[Input type] Parameter input
[Data type] Bit path

 # 1 LGN Geometry offset number of tool offset

0: Is the same as wear offset number

1: Specifies the geometry offset number by the tool selection number

NOTE
This parameter is valid when the option for tool geometry compensation or tool wear compensation is specified.

 # 2 LWT Tool wear compensation is performed by:

0: Moving the tool.

1: Shifting the coordinate system.

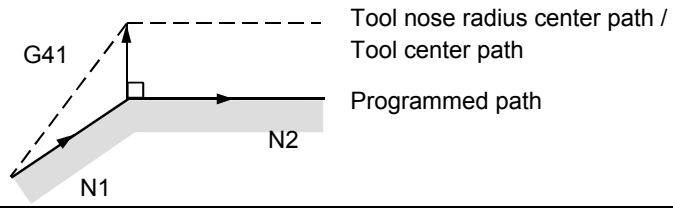
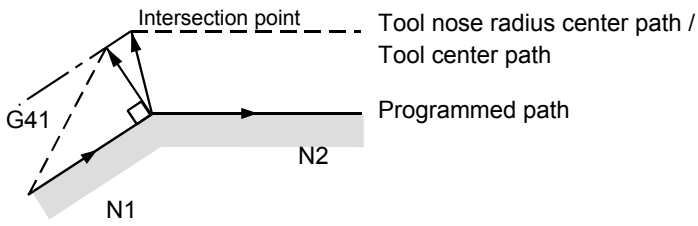
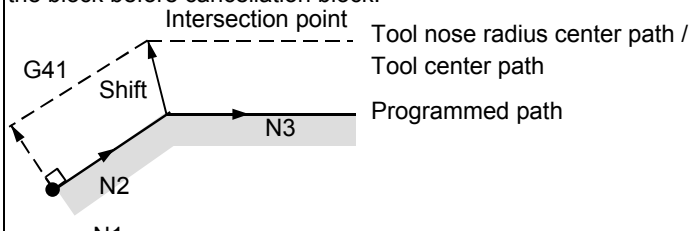
NOTE
This parameter is valid when the option for tool geometry compensation or tool wear compensation is specified.

	#7	#6	#5	#4	#3	#2	#1	#0
5003							SUV	SUP

[Input type] Parameter input
[Data type] Bit path

 # 0 SUP

 # 1 SUV These bits are used to specify the type of startup/cancellation of tool radius · tool nose radius compensation.

SUV	SUP	Type	Operation
0	0	Type A	A compensation vector perpendicular to the block next to the startup block or the block preceding the cancellation block is output. 
0	1	Type B	A compensation vector perpendicular to the startup block or cancellation block and an intersection vector are output. 
1	0 1	Type C	When the startup block or cancellation block specifies no movement operation, the tool is shifted by the cutter compensation amount in a direction perpendicular to the block next to the startup or the block before cancellation block.  When the block specifies movement operation, the type is set according to the SUP setting; if SUP is 0, type A is set, and if SUP is 1, type B is set.

NOTE
When SUV,SUP = 0,1 (type B), an operation equivalent to that of FS16*i*-T is performed.

	#7	#6	#5	#4	#3	#2	#1	#0
5004							ORC	
						ODI		

[Input type] Parameter input
[Data type] Bit path

 # 1 ORC The setting of a tool offset value is corrected as:
0: Diameter value
1: Radius value

NOTE
This parameter is valid only for an axis based on diameter specification. For an axis based on radius specification, specify a radius value, regardless of the setting of this parameter.

 # 2 ODI The setting of a tool radius · tool nose radius compensation value is corrected as:
0: Radius value
1: Diameter value

	#7	#6	#5	#4	#3	#2	#1	#0
5008				MCR				

[Input type] Parameter input
[Data type] Bit path

 # 4 MCR If G41/G42 (tool radius · tool nose radius compensation) is specified in the MDI mode, an alarm is:
0: Not raised.
1: Raised. (alarm PS5257)

5028	Number of digits of an offset number used with a T code command

[Input type] Parameter input
[Data type] Byte path
[Valid data range] 0 to 3

Specify the number of digits of a T code portion that is used for a tool offset number (wear offset number when the tool geometry/wear compensation function is used).
When 0 is set, the number of digits is determined by the number of tool compensation values.
When the number of tool compensation values is 1 to 9: Lower 1 digit
When the number of tool compensation values is 10 to 99: Lower 2 digits
When the number of tool compensation values is 100 to 999: Lower 3 digits
Example :

When an offset number is specified using the lower 2 digits of a T code, set 2 in parameter No. 5028.
Txxxxxx yy
xxxxxx : Tool selection
yy : Tool offset number

NOTE
A value longer than the setting of parameter No. 3032 (allowable number of digits of a T code) cannot be set.

5029	Number of tool compensation value memories common to paths

NOTE
When this parameter is set, the power must be turned off before operation is continued.

[Input type] Parameter input
[Data type] Word
[Valid data range] 0 to 999

When using memories common to paths, set the number of common tool compensation values in this parameter.
Ensure that the setting of this parameter does not exceed the number of tool compensation values set for each path (parameter No. 5024).
[Example 1]
When parameter No. 5029 = 10, parameter No. 5024 (path 1) = 15, and parameter No. 5024 (path 2) = 30 in a 2-path system, tool compensation numbers 1 to 10 of all paths are made common.
[Example 2]
When parameter No. 5029 = 20 and the other conditions are the same as for Example 1, tool compensation numbers 1 to 15 are made common.

NOTE
1 When a multi-path system involving the machining center system and lathe system is used, memories are made common in each system.
2 In each of the machining center system and lathe system, the same unit of tool compensation values needs to be used.
3 Ensure that the setting of parameter No. 5029 does not exceed the number of tool compensation values for each path (parameter No. 5024). If the setting of parameter No. 5029 exceeds the number of compensation values of a path, the least of the numbers of compensation values in all paths is made common.
4 When 0 or a negative value is set, memories common to paths are not used.

	#7	#6	#5	#4	#3	#2	#1	#0
5040					TCT			OWD

[Input type] Parameter input
[Data type] Bit path

 # 0 OWD In radius programming (bit 1 (ORC) of parameter No. 5004 is set to 1),

0: Tool offset values of both geometry compensation and wear compensation are specified by radius.

1: Tool offset value of geometry compensation is specified by radius and tool offset value of wear compensation is specified by diameter, for an axis of diameter programming.

NOTE
This parameter is valid when the option for tool geometry compensation or tool wear compensation is specified.

 # 3 TCT The tool change method is based on:

0: Turret rotation. (Tool change operation is performed with a T command only.)
With a T command, an auxiliary function and tool offset operation are performed.

1: Automatic tool changer (ATC).
(Tool change operation is performed with an M command (such as M06)).
With a T command, an auxiliary function only is performed.

This parameter is valid with a lathe system only.

⚠ **WARNING**
Before changing the setting of this parameter, cancel the offset. If the setting is changed while the offset is applied, the subsequent offset operation may not be performed correctly.

	#7	#6	#5	#4	#3	#2	#1	#0
5042					OFE	OFD	OFC	OFA

[Input type] Parameter input
[Data type] Bit path

NOTE
When at least one of these parameters is set, the power must be turned off before operation is continued.

 # 0 OFA
 # 1 OFC
 # 2 OFD
 # 3 OFE These bits are used to specify the increment system and valid data range of a tool offset value.

For metric input

OFE	OFD	OFC	OFA	Unit	Valid data range
0	0	0	1	0.01mm	±9999.99mm
0	0	0	0	0.001mm	±9999.999mm
0	0	1	0	0.0001mm	±9999.9999mm
0	1	0	0	0.00001mm	±9999.99999mm
1	0	0	0	0.000001mm	±999.999999mm

For inch input

OFE	OFD	OFC	OFA	Unit	Valid data range
0	0	0	1	0.001inch	±999.999inch
0	0	0	0	0.0001inch	±999.9999inch
0	0	1	0	0.00001inch	±999.99999inch
0	1	0	0	0.000001inch	±999.999999inch
1	0	0	0	0.0000001inch	±99.9999999inch

	Axis number for which Y-axis offset is used
5043	

[Input type] Parameter input
[Data type] Byte path
[Valid data range] 0 to 24
Set the number of an axis for which the tool offset is corrected.
If 0 or a value beyond the valid data range is set, the Y-axis offset is applied to the Y-axis of the basic three axes. If setting is made for the X- or Z-axis of the basic three axes, the standard tool offset for the X- or Z-axis is not used, and only the Y-axis offset is used.

5071	Number of first axis for grinding –wheel wear compensation
5072	Number of second axis for grinding–wheel wear compensation

[Input Type] Parameter Input
[Data Type] Byte path
[Valid data Range] 1 to number of controlled axis
This parameter specifies the controlled axis numbers of the first and second axis for which grinding-wheel wear compensation is applied.

5081	1st-axis coordinate value of compensation center 1 in grinding-wheel wear compensation
5082	2nd-axis coordinate value of compensation center 1 in grinding-wheel wear compensation
5083	1st-axis coordinate value of compensation center 2 in grinding-wheel wear compensation
5084	2nd-axis coordinate value of compensation center 2 in grinding-wheel wear compensation
5085	1st-axis coordinate value of compensation center 3 in grinding-wheel wear compensation
5086	2nd-axis coordinate value of compensation center 3 in grinding-wheel wear compensation

[Input type] Parameter input
[Data type] Real path
[Unit of data] mm, inch (input unit)
[Minimum unit of data] Depend on the increment system of the reference axis
[Valid data range] 9 digit of minimum unit of data (refer to standard parameter setting table (A))
(When the increment system is IS-B, -999999.999 to +999999.999)
Set the coordinate value (in the workpiece coordinate system) of the center of compensation in grinding-wheel wear compensation.

	#7	#6	#5	#4	#3	#2	#1	#0
5101								FXY

[Input type] Parameter input
[Data type] Bit path

 # 0 FXY The drilling axis in the drilling canned cycle, or cutting axis in the grinding canned cycle is:

0: In case of the Drilling canned cycle:
Z-axis at all times.
In case of the Grinding canned cycle:
- For the Lathe system
Z-axis at all times.
- For the Machining Center system
G75,G77 command :Y-axis
G78,G79 command :Z-axis

1: Axis selected by the program

NOTE
In the case of the T series, this parameter is valid only for the drilling canned cycle in the Series 15 format.

	#7	#6	#5	#4	#3	#2	#1	#0
5105			TFA					

[Input type] Parameter input
[Data type] Bit path

 # 5 TFA During tool center point control or tool length compensation in tool axis direction, canned cycles:

0: Cannot be used.
1: Can be used. However, an alarm (PS5424) is issued if the position of the rotation axis is not ±90° × n (n=0, 1, 2, ...) in the workpiece coordinate system.

	#7	#6	#5	#4	#3	#2	#1	#0
5106								GFX

[Input type] Parameter input
[Data type] Bit path

NOTE
When this parameter is set, the power must be turned off before operation is continued.

 # 0 GFX When the options of multiple respective canned cycle and grinding canned cycle are both specified, G71/G72/G73/G74 commands are:

0: Multiple respective canned cycle.
1: Grinding canned cycle.

5176	Grinding axis number in Traverse Grinding Cycle(G71) Grinding axis number in Plunge Grinding Cycle(G75)

[Input type] Parameter input
[Data type] Byte path
[Valid data range] 0 to Number of controlled axes
For the Lathe system:
Set the Grinding axis number of Traverse Grinding Cycle(G71).
For the Machining Center system:
Set the Grinding axis number of Plunge Grinding Cycle(G75).

NOTE
The axis number except for the cutting axis can be specified. When the axis number which is same to cutting axis is specified, PS0456 alarm is issued at the time of execution. The Grinding Cycle is executed when this parameter value is 0, PS0456 alarm is also issued.

5177	Grinding axis number of Traverse direct constant-size Grinding cycle(G72) Grinding axis number of Direct Constant Dimension Plunge Grinding Cycle(G77)

[Input type] Parameter input
[Data type] Byte path
[Valid data range] 0 to Number of controlled axes
For the Lathe system:
Set the Grinding axis number of Traverse direct constant-size Grinding cycle(G72).
For the Machining Center system:
Set the Grinding axis number of Direct Constant Dimension Plunge Grinding Cycle (G77).

NOTE
The axis number except for the cutting axis can be specified. When the axis number which is same to cutting axis is specified, PS0456 alarm is issued at the time of execution. The Grinding Cycle is executed when this parameter value is 0, PS0456 alarm is also issued.

5178	Grinding axis number of Oscillation Grinding Cycle(G73)
	Grinding axis number of Continuous feed surface grinding cycle(G78)

[Input type] Parameter input
[Data type] Byte path
[Valid data range] 0 to Number of controlled axes
For the Lathe system:
Set the Grinding axis number of Oscillation Grinding Cycle(G73).
For the Machining Center system:
Set the Grinding axis number of Continuous feed surface grinding cycle(G78).

NOTE
The axis number except for the cutting axis can be specified. When the axis number which is same to cutting axis is specified, PS0456 alarm is issued at the time of execution. The Grinding Cycle is executed when this parameter value is 0, PS0456 alarm is also issued.

5179	Grinding axis number of Oscillation Direct Fixed Dimension Grinding Cycle(G74)
	Grinding axis number of Intermittent feed surface grinding cycle(G79)

[Input type] Parameter input
[Data type] Byte path
[Valid data range] 0 to Number of controlled axes
For the Lathe system:
Set the Grinding axis number of Oscillation Direct Fixed Dimension Grinding Cycle(G74).
For the Machining Center system:
Set the Grinding axis number of Intermittent feed surface grinding cycle(G79).

NOTE
The axis number except for the cutting axis can be specified. When the axis number which is same to cutting axis is specified, PS0456 alarm is issued at the time of execution. The Grinding Cycle is executed when this parameter value is 0, PS0456 alarm is also issued.

5180	
	Axis number of dressing axis in Plunge grinding cycle(G75)

[Input type] Parameter input
[Data type] Byte path
[Valid data range] 0 to Number of controlled axes
Set the axis number of dressing axis in Plunge grinding cycle(G75).

NOTE
The axis number except for the cutting axis or grinding axis can be specified. When the axis number which is same to cutting axis or grinding axis is specified, PS0456 alarm is issued at the time of execution. The Grinding Cycle is executed when this parameter value is 0 and address "L" is specified in NC program, the PS0456 alarm is also issued.

5181	
	Axis number of dressing axis in Direct constant dimension plunge grinding cycle(G75)

[Input type] Parameter input
[Data type] Byte path
[Valid data range] 0 to Number of controlled axes
Set the axis number of dressing axis in Direct constant dimension plunge grinding cycle(G75).

NOTE
The axis number except for the cutting axis or grinding axis can be specified. When the axis number which is same to cutting axis or grinding axis is specified, PS0456 alarm is issued at the time of execution. The Grinding Cycle is executed when this parameter value is 0 and address "L" is specified in NC program, the PS0456 alarm is also issued.

5182	Axis number of dressing axis in Continuous feed surface grinding cycle(G78)

[Input type] Parameter input
[Data type] Byte path
[Valid data range] 0 to Number of controlled axes
Set the axis number of dressing axis in Continuous feed surface grinding cycle(G78).

NOTE
The axis number except for the cutting axis or grinding axis can be specified. When the axis number which is same to cutting axis or grinding axis is specified, PS0456 alarm is issued at the time of execution. The Grinding Cycle is executed when this parameter value is 0 and address "L" is specified in NC program, the PS0456 alarm is also issued.

5183	Axis number of dressing axis in Intermittent feed surface grinding cycle(G79)

[Input type] Parameter input
[Data type] Byte path
[Valid data range] 0 to Number of controlled axes
Set the axis number of dressing axis in Intermittent feed surface grinding cycle(G79).

NOTE
The axis number except for the cutting axis or grinding axis can be specified. When the axis number which is same to cutting axis or grinding axis is specified, PS0456 alarm is issued at the time of execution. The Grinding Cycle is executed when this parameter value is 0 and address "L" is specified in NC program, the PS0456 alarm is also issued.

	#7	#6	#5	#4	#3	#2	#1	#0
5200						CRG		G84

[Input type] Parameter input
[Data type] Bit path

 # 0 G84 Method for specifying rigid tapping:
0: An M code specifying the rigid tapping mode is specified prior to the issue of the G84 (or G74) command. (See parameter No.5210).
1: An M code specifying the rigid tapping mode is not used. (G84 cannot be used as a G code for the tapping cycle; G74 cannot be used for the reverse tapping cycle.)

 # 2 CRG Rigid mode when a rigid mode cancel command is specified (G80, G01 group G code, reset, etc.) :
0: Canceled after rigid tapping signal RGTAP is set to "0".
1: Canceled before rigid tapping signal RGTAP is set to "0".

	#7	#6	#5	#4	#3	#2	#1	#0
5202					CHR			

[Input type] Parameter input
[Data type] Bit path

NOTE
When this parameter is set, the power must be turned off before operation is continued..

 # 3 CHR Rigid tapping is:
0: Normal rigid tapping.
1: Interpolation type rigid tapping.

NOTE
1 The rigid tapping function and interpolation type rigid tapping function cannot be used together within a path.
2 Before the interpolation type rigid tapping function can be used, the Cs contour control function is required. If interpolation type rigid tapping is specified when the Cs contour control function is disabled, an alarm (PS1223) is issued.
3 The interpolation type rigid tapping function cannot be used in a path that has a spindle positioning axis. If interpolation type rigid tapping is specified for a path that has a spindle positioning axis, an alarm (PS1223) is issued.

	#7	#6	#5	#4	#3	#2	#1	#0
5203							HRM	HRG

[Input type] Parameter input
[Data type] Bit path

 # 0 HRG Rigid tapping by the manual handle is:
0: Disabled.
1: Enabled.

 # 1 HRM When the tapping axis moves in the negative direction during rigid tapping controlled by the manual handle, the direction in which the spindle rotates is determined as follows:
0: In G84 mode, the spindle rotates in a normal direction. In G74 mode, the spindle rotates in reverse.
1: In G84 mode, the spindle rotates in reverse. In G74 mode, the spindle rotates in a normal direction.

5241	Maximum spindle speed in rigid tapping (first gear)
5242	Maximum spindle speed in rigid tapping (second gear)
5243	Maximum spindle speed in rigid tapping (third gear)
5244	Maximum spindle speed in rigid tapping (fourth gear)

[Input type] Parameter input
[Data type] 2-word spindle
[Unit of data] min^{-1}
[Valid data range] 0 to 9999
Spindle position coder gear ratio
1 : 1 0 to 7400
1 : 2 0 to 9999
1 : 4 0 to 9999
1 : 8 0 to 9999
Each of these parameters is used to set a maximum spindle speed for each gear in rigid tapping.
Set the same value for both parameter No.5241 and parameter No.5243 for a one-stage gear system. For a two-stage gear system, set the same value as set in parameter No. 5242 in parameter No. 5243. Otherwise, alarm PS0200 will be issued. This applies to the M series.

	#7	#6	#5	#4	#3	#2	#1	#0
5400	SCR	XSC				D3R		

[Input type] Parameter input
[Data type] Bit path

 # 2 D3R When Reset is done by reset operation or reset signal from PMC, three-dimensional coordinate system conversion mode, tilted working plane command mode and workpiece setting error compensation mode is:
0: Canceled.
1: Not canceled.

 # 6 XSC The setting of a scaling magnification (axis-by-axis scaling) is:
0: Disabled.
1: Enabled.

 # 7 SCR Scaling (G51) magnification unit:
0: 0.00001 times (1/100,000)
1: 0.001 times

	#7	#6	#5	#4	#3	#2	#1	#0
5401								SCLx

[Input type] Parameter input
[Data type] Bit axis

 # 0 SCLx Scaling on this axis:
0: Invalidated
1: Validated

5411	Scaling (G51) magnification

[Input type] Setting input
[Data type] 2-word path
[Unit of data] 0.001 or 0.00001 times (Selected using SCR, #7 of parameter No.5400)
[Valid data range] 1to999999999
This parameter sets a scaling magnification when axis-by-axis scaling is disabled (with bit 6 (XSC) of parameter No. 5400 set to 0). If no scaling magnification (P) is specified in the program, the setting of this parameter is used as a scaling magnification.

NOTE
When bit 7 (SCR) of parameter No.5400 is set to 1, the valid data range is 1 to 9999999.

5412	Rapid traverse rate for canned cycle for drilling in three-dimensional coordinate conversion mode

[Input type] Parameter input
[Data type] Real path
[Unit of data] mm/min, inch/min, degree/min (machine unit)
[Minimum unit of data] Depend on the increment system of the reference axis
[Valid data range] Refer to the standard parameter setting table (C)
(When the increment system is IS-B, 0.0 to +999000.0)
This parameter sets a rapid traverse rate for canned cycle for drilling in the three-dimensional coordinate conversion mode, the tilted working plane command mode, or the workpiece setting error compensation mode.

5421	Scaling magnification for each axis

[Input type] Setting input
[Data type] 2-word axis
[Unit of data] 0.001 or 0.00001 times (Selected using SCR, #7 of parameter No.5400)
[Valid data range] -999999999 to –1, 1 to 999999999
This parameter sets a scaling magnification for each axis when axis-by-axis scaling is enabled (with bit 6 (XSC) of parameter No. 5400 set to 1). For the first spindle to the third spindle (X-axis to Z-axis), the setting of this parameter is used as a scaling magnification if scaling magnifications (I, J, K) are not specified in the program.

NOTE
When bit 7 (SCR) of parameter No.5400 is set to 1, the valid data ranges are -9999999 to -1 and 1 to 9999999.

	#7	#6	#5	#4	#3	#2	#1	#0
5431								MDL

[Input type] Parameter input
[Data type] Bit path

NOTE
When this parameter is set, the power must be turned off before operation is continued..

 # 0 MDL The G60 code (one-direction positioning) is:
0: One-shot G code (group 00).
1: Modal G code (group 01).

5440	Positioning direction and overrun distance in single directional positioning

[Input type] Parameter input
[Data type] Real axis
[Unit of data] mm, inch, degree (machine unit)
[Minimum unit of data] Depend on the increment system of the applied axis
[Valid data range] Refer to the standard parameter setting table (A)
(When the increment system is IS-B, -999999.999 to +999999.999)
This parameter sets the positioning direction and overrun distance in single directional positioning (G60) for each axis. The positioning direction is specified using a setting data sign, and the overrun distance using a value set here.
Overrun distance>0: The positioning direction is positive (+).
Overrun distance<0: The positioning direction is negative (*).
Overrun distance=0: Single directional positioning is not performed.

	#7	#6	#5	#4	#3	#2	#1	#0
5450						PLS		

[Input type] Parameter input
[Data type] Bit path

 # 2 PLS The polar coordinate interpolation shift function is:
0: Not used.
1: Used.
This enables machining using the workpiece coordinate system with a desired point which is not the center of the rotation axis set as the origin of the coordinate system in polar coordinate interpolation.

5460	Axis (linear axis) specification for polar coordinate interpolation

[Input type] Parameter input
[Data type] Byte path
[Valid data range] 1 to number of controlled axes
This parameter sets control axis numbers of linear axis to execute polar interpolation.

5461	Axis (rotation axis) specification for polar coordinate interpolation

[Input type] Parameter input
[Data type] Byte path
[Valid data range] 1 to number of controlled axes
This parameter sets control axis numbers of rotation axis to execute polar interpolation.

5463	Automatic override tolerance ratio for polar coordinate interpolation

[Input type] Parameter input
[Data type] Byte path
[Unit of data] %
[Valid data range] 0 to 100
Typical setting: 90% (treated as 90% when set to 0)
Set the tolerance ratio of the fastest cutting feedrate to the speed of the rotation axis during automatic override of polar coordinate interpolation.

5464	Compensation for error on hypothetical axis of polar coordinate interpolation

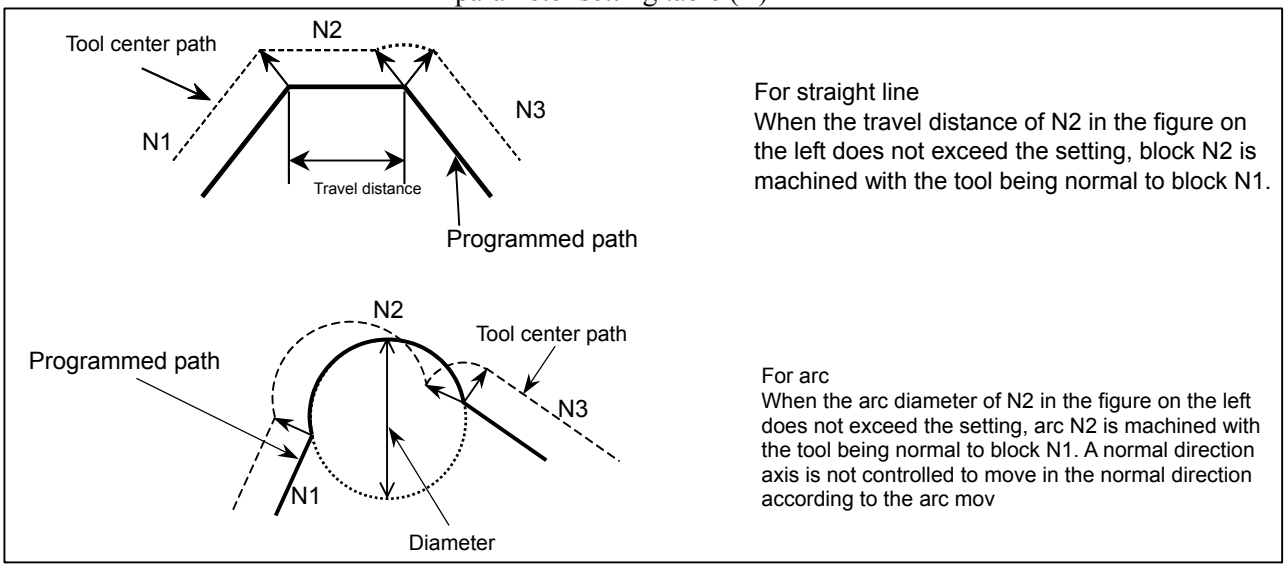
[Input type] Parameter input
[Data type] Byte path
[Unit of data] mm, inch (input unit)
[Minimum unit of data] Depend on the increment system of the reference axis
[Valid data range] 9 digit of minimum unit of data (refer to standard parameter setting table (A))
(For IS-B, -999999.999 to +999999.999)
This parameter is used to set the error if the center of the rotation axis on which polar coordinate interpolation is performed is not on the X-axis.
If the setting of the parameter is 0, regular polar coordinate interpolation is performed.

5481	
	Feedrate of rotation of the normal direction controlled axis

[Input type] Parameter input
[Data type] Real axis
[Unit of data] deg/min
[Minimum unit of data] Depend on the increment system of the applied axis
[Valid data range] Refer to the standard parameter setting table (C)
This parameter sets the feedrate of the movement along the normal direction controlled axis that is inserted at the start point of a block during normal direction control.

5483	Limit value of movement that is executed at the normal direction angle of a preceding block

[Input type] Parameter input
[Data type] Real path
[Unit of data] mm, inch (input unit)
[Minimum unit of data] Depend on the increment system of the reference axis
[Valid data range] 0 or positive 9 digit of minimum unit of data (refer to standard parameter setting table (B)



	#7	#6	#5	#4	#3	#2	#1	#0
5630								SPN

[Input type] Parameter input
[Data type] Bit path

 # 0 SPN The amount of linear axis division (span value) in exponential interpolation is:
0: Specified with parameter No.5643.
1: Specified using address K in a block containing G02.3/G03.3. When address K is not specified, the value set with parameter No.5643 is used.

5641	Linear axis number subject to exponential interpolation

[Input type] Parameter input
[Data type] Byte path
[Valid data range] 1 to number of controlled axes
This parameter sets the ordinal number, among the controlled axes, for the linear axis to which exponential interpolation is applied.

5642	Rotation axis number subject exponential interpolation

[Input type] Parameter input
[Data type] Byte path
[Valid data range] 1 to number of controlled axes
This parameter sets the ordinal number, among the controlled axes, for the rotation axis to which exponential interpolation is applied.

5643	Amount of linear axis division (span value) in exponential interpolation

[Input type] Setting input
[Data type] Real path
[Unit of data] mm, inch (machine unit)
[Minimum unit of data] Depend on the increment system of the reference axis
[Valid data range] 0 to 999999999
This parameter sets an amount of linear axis division in exponential interpolation when bit 0 (SPN) of parameter No.5630 is set to 0 or when address K is not specified.

	#7	#6	#5	#4	#3	#2	#1	#0
6000			SBM	HGO			MGO	G67
			SBM	HGO	V15		MGO	G67

[Input type] Parameter input
[Data type] Bit path

 # 0 G67 If the macro modal call cancel command (G67) is specified when the macro modal call mode (G66/G66.1) is not set:
0: Alarm PS0122 is issued.
1: The specification of G67 is ignored.

 # 1 MGO When a GOTO statement for specifying custom macro control is executed, a high-speed branch to 20 sequence numbers executed from the start of the program is:
0: A high-speed branch is not caused to n sequence numbers from the start of the executed program.
1: A high-speed branch is caused to n sequence numbers from the start of the program.

 # 3 V15 As system variable numbers for tool offset:
0 : The standard system variable numbers for the Series 16 are used.
1 : The same system variable numbers as those used for the Series 15 are used.

The tables below indicate the system variables for tool offset numbers 1 to 999. The values for tool offset numbers 1 to 200 can be read from or assigned to the system variables in parentheses.

(1) Tool offset memory A

	System variable number	
	V15 = 0	V15 = 1
Wear offset value	#10001 to #10999 (#2001 to #2200)	#10001 to #10999 (#2001 to #2200)

(2) Tool offset memory B

	System variable number	
	V15 = 0	V15 = 1
Geometry offset value	#11001 to #11999 (#2201 to #2400)	#10001 to #10999 (#2001 to #2200)
Wear offset value	#10001 to #10999 (#2001 to #2200)	#11001 to #11999 (#2201 to #2400)

(3) Tool offset memory C

		System variable number	
		V15 = 0	V15 = 1
Tool length offset	Geometry offset value	#11001 to #11999 (#2201 to #2400)	#10001 to #10999 (#2001 to #2200)
	Wear offset value	#10001 to #10999 (#2001 to #2200)	#11001 to #11999 (#2201 to #2400)
Tool radius offset	Geometry offset value	#13001 to #13999	#12001 to #12999
	Wear offset value	#12001 to #12999	#13001 to #13999

 # 4 HGO When a GOTO statement in a custom macro control command is executed, a high-speed branch to the 30 sequence numbers immediately before the executed statement is:
0: Not made.
1: Made.

 # 5 SBM Custom macro statement
0: Not stop the single block
1: Stops the single block
If you want to disable the single blocks in custom macro statements using system variable #3003, set this parameter to 0. If this parameter is set to 1, the single blocks in custom macro statements cannot be disabled using system variable #3003. To control single blocks in custom macro statements using system variable #3003, use bit 7 (SBV) of parameter No. 6000.

	#7	#6	#5	#4	#3	#2	#1	#0
6001		CCV	TCS	CRO	PV5		PRT	MIF

[Input type] Parameter input
[Data type] Bit path

 # 0 MIF The custom macro interface signals are based on:
0: Standard specification.
(The signals UI000 to UI015, UO000 to UO015, and UO100 to UO131 are used.)
1: Extended specification.
(The signals UI000 to UI031, UI100 to UI131, UI200 to UI231, UI300 to UI331, UO000 to UO031, UO100 to UO131, UO200 to UO231, and UO300 to UO331 are used.)

 # 1 PRT Reading zero when data is output using a DPRINT command
0: Outputs a space
1: Outputs no data

 # 3 PV5 Custom macro common variables:
0: #500 to #549 are output. (Note)
1: #100 to #149 and #500 to 549 are output. (Note)

NOTE
Output variables are as follows according to the combination of added options.

When the parameter PV5=0

		Option "Addition of custom macro common variables"	
		Non	**Yes**
Option "Embedded macro"	**Non**	#500 to #549	#500 to #999
	Yes	#500 to #549	#500 to #999

When the parameter PV5=1

		Option "Addition of custom macro common variables"	
		Non	**Yes**
Option "Embedded macro"	**Non**	#100 to #149 and #500 to #549	#100 to #199 and #500 to #999
	Yes	#100 to #149, #200 to #499 and #500 to #549	#100 to #199, #200 to #499 and #500 to #999

 # 4 CRO ISO code in BPRWT or DPRNT command
0: Outputs only "LF" after data is output
1: Outputs "LF" and "CR" after data is output

 # 5 TCS Custom macro (subprogram)
0: Not called using a T code
1: Called using a T code

 # 6 CCV Common variables #100 to #149(NOTE) cleared by power-off are:
0: Cleared to <null> by reset
1: Not cleared by reset

NOTE
Cleared variables are as follows according to the combination of added options.

		Option "Addition of custom macro common variables"	
		Non	**Yes**
Option "Embedded macro"	**Non**	#100 to #149	#100 to #199
	Yes	#100 to #149 and #200 to #499	#100 to #199 and #200 to #499

	#7	#6	#5	#4	#3	#2	#1	#0
6003			MSB	MPR	TSE	MIN		

[Input type] Parameter input
[Data type] Bit path

NOTE
When at least one of these parameters is set, the power must be turned off before operation is continued.

 # 2 MIN Custom macro interrupt
0: Performed by interrupting an in-execution block (Custom macro interrupt type I)
1: Performed after an in-execution block is completed (Custom macro interrupt type II)

 # 3 TSE Custom macro interrupt signal UINT
0: Edge trigger method (Rising edge)
1: Status trigger method

 # 4 MPR Custom macro interrupt valid/invalid M code
0: M96/M97
1: M code set using parameters (Nos. 6033 and 6034)

 # 5 MSB Interrupt program
0: Uses a dedicated local variable (Macro-type interrupt)
1: Uses the same local variable as in the main program (Subprogram- type interrupt)

	#7	#6	#5	#4	#3	#2	#1	#0
6004						VHD		NAT
			D15					NAT

[Input type] Parameter input
[Data type] Bit path

 # 0 NAT The results of the custom macro functions ATAN (with 2 arguments) and ASIN are specified as follows:
0: The result of ATAN is 0 to 360.0.
The result of ASIN is 270.0 to 0 to 90.0.
1: The result of ATAN is -180.0 to 0 to 180.0.
The result of ASIN is -90.0 to 0 to 90.0.

 # 2 VHD With system variables #5121 to #5140:
0: The tool offset value (geometry offset value) in the block currently being executed is read. (This parameter is valid only when tool geometry/tool wear compensation memories are available.)
1: An interrupt travel distance based on manual handle interrupt is read.

 # 5 D15 When tool compensation memory C is used, for reading or writing tool offset values (for up to offset number 200) for D code (tool radius), the same system variables, #2401 through #2800, as Series 15 are:

0: Not used.
1: Used.

When bit 3 (V15) of parameter No. 6000 is set to 1

D code				
Compensation number	Geometry		Wear	
	Variable number	Variable name	Variable number	Variable name
1	#2401	[#_OFSDG[1]]	#2601	[#_OFSDW[1]]
2	#2402	[#_OFSDG[2]]	#2602	[#_OFSDW[2]]
3	#2403	[#_OFSDG[3]]	#2603	[#_OFSDW[3]]
:	:	:	:	:
199	#2599	[#_OFSDG[199]]	#2799	[#_OFSDW[199]]
200	#2600	[#_OFSDG[200]]	#2800	[#_OFSDW[200]]

	#7	#6	#5	#4	#3	#2	#1	#0
6005								SQC

[Input type] Parameter input
[Data type] Bit path

 # 0 SQC In the subprogram call function, a subprogram sequence number call is:

0: Not used.
1: Used.

	#7	#6	#5	#4	#3	#2	#1	#0
6007				CVA	MGE	BCS	SCS	DPG

[Input type] Parameter input
[Data type] Bit path

 # 0 DPG Specifies whether to allow G codes with a decimal point to be called.

0: Do not allow.
1: Allow.

 # 1 SCS Specifies whether to call subprograms with S codes.

0: Do not call with S codes.
1: Call with S codes.

 # 2 BCS Specifies whether to call subprograms with the second auxiliary function codes.

0: Do not call with the second auxiliary function codes.
1: Call with the second auxiliary function codes.

 # 3 MGE Specifies whether a G code modal call is made after movement or for each block.

0: Make a call for each block (equivalent to G66.1).
1: Make a call after movement (equivalent to G66).

 # 4 CVA The format for macro call arguments is specified as follows:

0: Arguments are passed in NC format without modifications.
1: Arguments are converted to macro format then passed.

Example)

When G65 P_ X10 ; is specified, the value in local variable #24 in the calling program is set as follows:

Command	CVA=0	CVA=1
#24	0.01	0.01
ADP[#24]	10.0	0.01

NOTE
External operations are the same unless the ADP function is used.

	#7	#6	#5	#4	#3	#2	#1	#0
6008	IJK	GMP		ISO			MCA	F16

[Input type] Parameter input
[Data type] Bit path

 # 0 F16 The precision of operation is based on:

0: New specification.
1: FS16*i* compatible specification.

 # 1 MCA A macro alarm specification based on system variable #3000 is selected as follows:

0: An alarm number obtained by adding 3000 to a value assigned to variable #3000 and the corresponding message are displayed. (A value from 0 to 200 can be assigned to variable #3000.)
1: A value assigned to variable #3000 and the corresponding message are displayed. (A value from 0 to 4095 can be assigned to variable #3000.)

(Example)

Execution of #3000=1 (ALARM MESSAGE);
When bit 1 (MCA) of parameter No. 6008 is set to 0:
The alarm screen displays "MC 3001 ALARM MESSAGE".
When bit 1 (MCA) of parameter No. 6008 is set to 1:
The alarm screen displays "MC0001 ALARM MESSAGE".

 # 4 ISO

0: When the EIA code is used, the bit patterns of codes specified instead of [,], #, *, =, ?, @, &, and _ are set in parameter No. 6010 to No. 6018.
1: When the ISO/ASCII code is used, the bit patters of codes specified instead of [,], #, *, =, ?, @, &, and _ are set in parameter No. 6010 to No. 6018.

 # 6 GMP The calling of M, S, T, a second auxiliary function code, or a particular code during the calling of a G code, and the calling of a G code during the calling of M, S, T, a second auxiliary function code, or particular code are:

0: Not allowed. (They are executed as an ordinary G, M, S, T, second auxiliary function code, and NC address.)
1: Allowed.

#7 IJK For addresses I, J, and K specified as arguments:

0: Argument specification I or II is automatically determined.
1: Argument specification I is always used.

Example
When K_J_I_ is specified:

- When this parameter is set to 0:
  Argument specification II is used and K=#6, J=#8, and I=#10 are specified.
- When this parameter is set to1:
  Argument specification I is used and I=#4, J=#5, and K=#6 are specified regardless of the specification order.
  (Argument specification II cannot be used.)

	#7	#6	#5	#4	#3	#2	#1	#0
6010	*7	*6	*5	*4	*3	*2	*1	*0
	#7	#6	#5	#4	#3	#2	#1	#0
6011	=7	=6	=5	=4	=3	=2	=1	=0
	#7	#6	#5	#4	#3	#2	#1	#0
6012	#7	#6	#5	#4	#3	#2	#1	#0
	#7	#6	#5	#4	#3	#2	#1	#0
6013	[7	[6	[5	[4	[3	[2	[1	[0
	#7	#6	#5	#4	#3	#2	#1	#0
6014	]7	]6	]5	]4	]3	]2	]1	]0
	#7	#6	#5	#4	#3	#2	#1	#0
6015	?7	?6	?5	?4	?3	?2	?1	?0
	#7	#6	#5	#4	#3	#2	#1	#0
6016	@7	@6	@5	@4	@3	@2	@1	@0
	#7	#6	#5	#4	#3	#2	#1	#0
6017	&7	&6	&5	&4	&3	&2	&1	&0
	#7	#6	#5	#4	#3	#2	#1	#0
6018	_7	_6	_5	_4	_3	_2	_1	_0

[Input type] Parameter input
[Data type] Bit path

*0 to *7 : The bit pattern of the EIA or ISO/ASCII code indicating * is set.
=0 to =7 : The bit pattern of the EIA or ISO/ASCII code indicating = is set.
#0 to #7 : The bit pattern of the EIA or ISO/ASCII code indicating # is set.
[0 to [7 : The bit pattern of the EIA or ISO/ASCII code indicating [is set.
]0 to]7 : The bit pattern of the EIA or ISO/ASCII code indicating] is set.
?0 to ?7 : The bit pattern of the EIA or ISO/ASCII code indicating ? is set.
@0 to @7 : The bit pattern of the EIA or ISO/ASCII code indicating @ is set.
&0 to &7 : The bit pattern of the EIA or ISO/ASCII code indicating & is set.
_0 to _7 : The bit pattern of the EIA or ISO/ASCII code indicating _ is set.
0: A corresponding bit is 0.
1: A corresponding bit is 1.

	#7	#6	#5	#4	#3	#2	#1	#0
6019								MCO

[Input type] Parameter input
[Data type] Bit

#0 MCO When data is output, the decimal number value of the macro variable data is
0: Not output as a comment.
1: Output at the same time as a comment.

After the number, data, and the variable name of the macro variable are output when data output operation is performed the variable number and the value of the macro variable data in decimal number are output as a comment.

NOTE
1 Output data by this parameter is "Comment", and this is ignored at the time of reading.
2 Accuracy of the output data of the comment is up to 15 digits. The range of output data are nine digits above decimalpoint and eight digits below decimal point. "± OVER FLOW" is output instead of a value when the total digits number is more than 16 and the digit number above the decimal point is ten or more. When the number of digits below the decimal point becomes nine digits or more, the ninth place of the decimal point is rounded off and output. Moreover, the seventh place or the eighth place of the decimal point is rounded off and output when the total digits number is more than 16 and the digit number above decimal point is nine or eight.
3 The output becomes "EMPTY" when displayed, the macro variable data is "DATA EMPTY".

6030	M code to execute external device subprogram calls

[Input type] Setting input
[Data type] 2-word path
[Valid data range] 0 to 99999999
Set the M code to execute external device subprogram calls. When 0 is set, M198 is used. M01, M02, M30, M98, and M99 cannot be used to execute external device subprogram calls. When a negative number, 1, 2, 30, 98, or 99 is set for this parameter, M198 is used to execute external device subprogram calls.

6031	Start number of common variables to be protected among the common variables (#500 to #999)
6032	End number of common variables to be protected among the common variables (#500 to #999)

[Input type] Parameter input
[Data type] Word path
[Valid data range] 500 to 999

Among the common variables (#500 to #999), the range of common variables specified by this parameter can be protected (by setting their attributes to read-only). If a write attempt (on the left side) is made, an alarm is issued.

NOTE
Set 0 in both parameter No. 6031 and No. 6032 not to protect common variables.

6033	M code that validates a custom macro interrupt
6034	M code that invalidates a custom macro interrupt

[Input type] Parameter input
[Data type] 2-word path
[Valid data range] 03 to 99999999 (excluding 30, 98 and 99)

These parameters can be used when bit 4 (MPR) of parameter No.6003, is 1. M96 is used as a valid M code and M97 is used as an invalid M code when MPR is 0, irrespective of the state of this parameter.

6036	Number of custom macro variables common to tool path (for #100 to #199 (#499))

[Input type] Parameter input
[Data type] Word
[Valid data range] 0 to 400

When the memory common to paths is used, this parameter sets the number of custom macro common variables to be shared (custom macro variables common to paths). Common variables #100 to #199 (up to #499 in a system with the embedded macro option) may be shared. Ensure that the maximum number of usable macro common variables is not exceeded.

Example

When 20 is set in parameter No. 6036
#100 to #119: Shared by all paths
#120 to #149: Used by each path independently

NOTE

1 To use up to #199, the option for adding custom macro common variables is required.
2 To use up to #499, the embedded macro option is required.
3 When 0 or a negative value is set, the memory common to paths is not used.
4 When the option for embedded macro is effective and the option for addition of custom macro common variables is not effective, #150 to #199 can not be used but this parameter should be set the number which includes #150 to #199.

6037 **Number of custom macro variables common to tool path (for #500 to #999)**

[Input type] Parameter input
[Data type] Word
[Valid data range] 0 to 500

When the memory common to paths is used, this parameter sets the number of custom macro common variables to be shared (custom macro variables common to paths). Common variables #500 to #999 may be shared. Ensure that the maximum number of usable macro common variables is not exceeded.

Example
When 50 is set in parameter No. 6037
#500 to #549: Shared by all paths
#550 to #599: Used by each path independently

NOTE
1 To use up to #999, the option for adding custom macro common variables is required.
2 When 0 or a negative value is set, the memory common to paths is not used.

6038 **Start G code used to call a custom macro**

[Input type] Parameter input
[Data type] Word path
[Valid data range] -9999 to 9999

6039 **Start program number of a custom macro called by G code**

[Input type] Parameter input
[Data type] 2-word path
[Valid data range] 1 to 9999

6040	Number of G codes used to call custom macros

[Input type] Parameter input
[Data type] Word path
[Valid data range] 0 to 255

Set this parameter to define multiple custom macro calls using G codes at a time. With G codes as many as the value set in parameter No. 6040 starting with the G code set in parameter No. 6038, the custom macros of program numbers as many as the value set in parameter No. 6040 starting with the program number set in parameter No. 6039 can be called. Set 0 in parameter No. 6040 to disable this mode of calling.

If a negative value is set in parameter No. 6038, the modal call mode is entered. Whether the modal call is equivalent to G66 or G66.1 depends on bit 3 (MGE) of parameter No. 6007.

Example)

When parameter No. 6038 = 900, parameter No. 6039 = 1000, and parameter No. 6040 = 100 are set, a set of 100 custom macro calls (simple calls) is defined as follows:

G900 → O1000
G901 → O1001
G902 → O1002
:
G999 → O1099

When the setting of parameter No. 6038 is changed to -900, the same set of custom macro calls (modal calls) is defined.

NOTE
1 When the following conditions are satisfied, all calls using these parameters are disabled:
  1) When a value not within the specifiable range is set in each parameter
  2) (Value of parameter No.6039 + value of parameter No.6040 - 1) > 9999
2 The specification of a mixture of simple calls and modal calls is not allowed.
3 If a range of G codes set by these parameters duplicate G codes specified in parameter No.6050 to No.6059, the calls specified by parameter No.6050 to 6059 are made preferentially.

6041 **Start G code with a decimal point used to call a custom macro**

[Input type] Parameter input
[Data type] Word path
[Valid data range] -999 to 999

6042 **Start program number of a custom macro called by G code with a decimal point**

[Input type] Parameter input
[Data type] 2-word path
[Valid data range] 1 to 9999

6043 **Number of G codes with a decimal point used to call custom macros**

[Input type] Parameter input
[Data type] Word path
[Valid data range] 0 to 255

Set this parameter to define multiple custom macro calls using G codes with a decimal point at a time. With G codes with a decimal point as many as the value set in parameter No. 6043 starting with the G code with a decimal point set in parameter No. 6041, the custom macros of program numbers as many as the value set in parameter No. 6043 starting with the program number set in parameter No. 6042 can be called. Set 0 in parameter No. 6043 to disable this mode of calling.
If a negative value is set in parameter No. 6041, the modal call mode is entered. Whether the modal call is equivalent to G66 or G66.1 depends on bit 3 (MGE) of parameter No. 6007.

Example)

When parameter No. 6041 = 900, parameter No. 6042 = 2000, and parameter No. 6043 = 100 are set, a set of 100 custom macro calls (simple calls) is defined as follows:
G90.0 → O2000
G90.1 → O2001
G90.2 → O2002
:
G99.9 → O2099
When the setting of parameter No. 6041 is changed to -900, the same set of custom macro calls (modal calls) is defined.

NOTE
1 When the following conditions are satisfied, all calls using these parameters are disabled:
  1) When a value not within the specifiable range is set in each parameter
  2) (Value of parameter No.6042 + value of parameter No.6043 - 1) > 9999
  3) When bit 0 (DPG) of parameter No. 6007 = 0 (to disable calls using G codes with a decimal point)
2 The specification of a mixture of simple calls and modal calls is not allowed.
3 If a range of G codes set by these parameters duplicate G codes specified in parameter No.6060 to No.6069, the calls specified by parameter No.6060 to 6069 are made preferentially.

6044	Start M code used to call a subprogram

[Input type] Parameter input
[Data type] 2-word path
[Valid data range] 3 to 99999999

6045	Start program number of a subprogram called by M code

[Input type] Parameter input
[Data type] 2-word path
[Valid data range] 1 to 9999

6046	Number of M codes used to call subprograms (number of subprograms called by M codes)

[Input type] Parameter input
[Data type] 2-word path
[Valid data range] 0 to 32767

Set this parameter to define multiple subprogram calls using M codes at a time. With M codes as many as the value set in parameter No. 6046 starting with the M code set in parameter No. 6044, the subprograms of program numbers as many as the value set in parameter No. 6046 starting with the program number set in 6045 can be called. Set 0 in parameter No. 6046 to disable this mode of calling.

Example)

When parameter No. 6044 = 80000000, parameter No. 6045 = 3000, and parameter No. 6046 = 100 are set, a set of 100 subprogram calls is defined as follows:

M80000000 → O3000
M80000001 → O3001
M80000002 → O3002
:
M80000099 → O3099

NOTE

1 When the following conditions are satisfied, all calls using these parameters are disabled:
   1) When a value not within the specifiable range is set in each parameter
   2) (Value of parameter No. 6045 + value of parameter No. 6046 - 1) > 9999
2 If a range of M codes set by these parameters duplicate M codes specified in parameter No. 6071 to No. 6079, the calls specified by parameter No. 6071 to 6079 are made preferentially.

6047	Start M code used to call a custom macro

[Input type] Parameter input
[Data type] 2-word path
[Valid data range] 3 to 99999999

6048	Start program number of a custom macro called by M code

[Input type] Parameter input
[Data type] 2-word path
[Valid data range] 1 to 9999

6049	Number of M codes used to call custom macros (number of custom macros called by M codes)

[Input type] Parameter input
[Data type] 2-word path
[Valid data range] 0 to 32767

Set this parameter to define multiple custom macro calls using M codes at a time. With M codes as many as the value set in parameter No. 6049 starting with the M code set in parameter No. 6047, the custom macros of program numbers as many as the value set in parameter No. 6049 starting with the program number set in parameter No. 6048 can be called. Set 0 in parameter No. 6049 to disable this mode of calling.

Example)

When parameter No. 6047 = 90000000, parameter No. 6048 = 4000, and parameter No. 6049 = 100 are set, a set of 100 custom macro calls (simple calls) is defined as follows:

M90000000 → O4000
M90000001 → O4001
M90000002 → O4002
:
M90000099 → O4099

NOTE

1 When the following conditions are satisfied, all calls using these parameters are disabled:
   1) When a value not within the specifiable range is set in each parameter
   2) (Value of parameter No. 6048 + value of parameter No. 6049 - 1) > 9999
2 If a range of M codes set by these parameters duplicate M codes specified in parameter No. 6080 through No. 6089, the calls specified by parameter No. 6080 through 6089 are made preferentially.

6050	G code that calls the custom macro of program number 9010
6051	G code that calls the custom macro of program number 9011
6052	G code that calls the custom macro of program number 9012
6053	G code that calls the custom macro of program number 9013
6054	G code that calls the custom macro of program number 9014
6055	G code that calls the custom macro of program number 9015
6056	G code that calls the custom macro of program number 9016
6057	G code that calls the custom macro of program number 9017
6058	G code that calls the custom macro of program number 9018
6059	G code that calls the custom macro of program number 9019

[Input type] Parameter input
[Data type] Word path
[Valid data range] (-9999 to 9999 : excluding 0, 5, 65, 66 and 67)

Set the G codes used to call the custom macros of program numbers 9010 to 9019. However, note that when a negative value is set in this parameter, it becomes a modal call. For example, if this parameter is set to -11, the modal call mode is entered by G11.
Whether the modal call is equivalent to G66 or G66.1 depends on bit 3 (MGE) of parameter No. 6007.

6060	G code with a decimal point used to call the custom macro of program number 9040
6061	G code with a decimal point used to call the custom macro of program number 9041
6062	G code with a decimal point used to call the custom macro of program number 9042
6063	G code with a decimal point used to call the custom macro of program number 9043
6064	G code with a decimal point used to call the custom macro of program number 9044
6065	G code with a decimal point used to call the custom macro of program number 9045
6066	G code with a decimal point used to call the custom macro of program number 9046
6067	G code with a decimal point used to call the custom macro of program number 9047
6068	G code with a decimal point used to call the custom macro of program number 9048
6069	G code with a decimal point used to call the custom macro of program number 9049

[Input type] Parameter input
[Data type] Word path
[Valid data range] -999 to 999

Set the G codes used to call the custom macros of program numbers 9040 to 9049. However, note that when a negative value is set in this parameter, it becomes a modal call. For example, if this parameter is set to -11, the modal call mode is entered by G1.1.

Whether the modal call is equivalent to G66 or G66.1 depends on bit 3 (MGE) of parameter No. 6007. Set G codes in the format Gm.n. The value expressed by (m×10+n) is set in the parameter. The values m and n must satisfy the following relationships: $0 \leq m \leq 99$, $0 \leq n \leq 9$.

NOTE
Parameter Nos. 6060 to 6069 are valid when bit 0 (DPG) of parameter No. 6007 is set to 1.

6071	M code used to call the subprogram of program number 9001
6072	M code used to call the subprogram of program number 9002
6073	M code used to call the subprogram of program number 9003
6074	M code used to call the subprogram of program number 9004
6075	M code used to call the subprogram of program number 9005
6076	M code used to call the subprogram of program number 9006
6077	M code used to call the subprogram of program number 9007
6078	M code used to call the subprogram of program number 9008
6079	M code used to call the subprogram of program number 9009

[Input type] Parameter input
[Data type] 2-word path
[Valid data range] 3 to 99999999 (excluding 30, 98 and 99)

These parameters set the M codes that call the subprograms of program numbers 9001 to 9009.

NOTE
If the same M code is set in these parameters, the younger number is called preferentially. For example, if 100 is set in parameter No. 6071 and 6072, and programs O9001 and O9002 both exist, O9001 is called when M100 is specified.

6080	M code used to call the custom macro of program number 9020
6081	M code used to call the custom macro of program number 9021
6082	M code used to call the custom macro of program number 9022
6083	M code used to call the custom macro of program number 9023
6084	M code used to call the custom macro of program number 9024
6085	M code used to call the custom macro of program number 9025
6086	M code used to call the custom macro of program number 9026
6087	M code used to call the custom macro of program number 9027
6088	M code used to call the custom macro of program number 9028
6089	M code used to call the custom macro of program number 9029

[Input type] Parameter input
[Data type] 2-word path
[Valid data range] 3 to 99999999 (excluding 30, 98 and 99)
Set the M codes used to call the custom macros of program numbers 9020 to 9029. The simple call mode is set.

NOTE
1 If the same M code is set in these parameters, the younger number is called preferentially. For example, if 200 is set in parameter No. 6081 and No. 6082, and programs O9021 and O9022 both exist, O9021 is called when M200 is specified.
2 If the same M code is set in a parameter (No. 6071 to No. 6079) used to call subprograms and in a parameter (No. 6080 to No. 6089) used to call custom macros, a custom macro is called preferentially. For example, if 300 is set in parameter No. 6071 and No. 6081, and programs O9001 and O9021 both exist, O9021 is called when M300 is specified.

6090	ASCII code that calls the subprogram of program number 9004

6091	ASCII code that calls the subprogram of program number 9005

[Input type] Parameter input
[Data type] Byte path
[Valid data range] 65(A:41H) to 90(Z:5AH)
These parameters set the ASCII codes that call subprograms in decimal.
The settable addresses are indicated below.

Address	Parameter setting value	T series	M series
A	65	O	O
B	66	O	O
D	68	X	O
F	70	O	O
H	72	O	O
I	73	O	O
J	74	O	O
K	75	O	O
L	76	O	O
M	77	O	O
P	80	O	O
Q	81	O	O
R	82	O	O
S	83	O	O
T	84	O	O
V	86	X	O
X	88	X	O
Y	89	X	O
Z	90	X	O

NOTE
1 When address L is set, the number of repeats cannot be specified.
2 Set 0 when no subprogram is called.

	#7	#6	#5	#4	#3	#2	#1	#0
6200	SKF	SRE	SLS	HSS			SK0	GSK

[Input type] Parameter input
[Data type] Bit path

 # 0 GSK As a skip signal, the skip signal SKIPP is:
0: Invalid.
1: Valid.

 # 1 SK0 This parameter specifies whether the skip signal is made valid under the state of the skip signal SKIP and the multistage skip signals SKIP2 to SKIP8.
0: Skip signal is valid when these signals are 1.
1: Skip signal is valid when these signals are 0.

 # 4 HSS

0: The skip function does not use high-speed skip signals while skip signals are input. (The conventional skip signal is used.)
1: The step skip function uses high-speed skip signals while skip signals are input.

 # 5 SLS

0: The multi-step skip function does not use high-speed skip signals while skip signals are input. (The conventional skip signal is used.)
1: The multi-step skip function uses high-speed skip signals while skip signals are input.

NOTE
The skip signals (SKIP and SKIP2 to SKIP8) are valid regardless of the setting of this parameter. They can also be disabled using bit 4 (IGX) of parameter No. 6201.

 # 6 SRE When a high-speed skip signal or high-speed measurement position arrival signal is used:

0: The signal is assumed to be input on the rising edge (contact open → close).
1: The signal is assumed to be input on the falling edge (contact close → open).

 # 7 SKF Dry run, override, and automatic acceleration/deceleration for G31 skip command

0: Disabled
1: Enabled

	#7	#6	#5	#4	#3	#2	#1	#0
6201	SKPXE		CSE	IGX		TSE	SEB	

[Input type] Parameter input
[Data type] Bit path

 # 1 SEB When a skip signal or measurement position arrival signal goes on while the skip function, or the automatic tool length measurement (M series) or automatic tool compensation (T series) is used, the accumulated pulses and positional deviation due to acceleration/deceleration are:

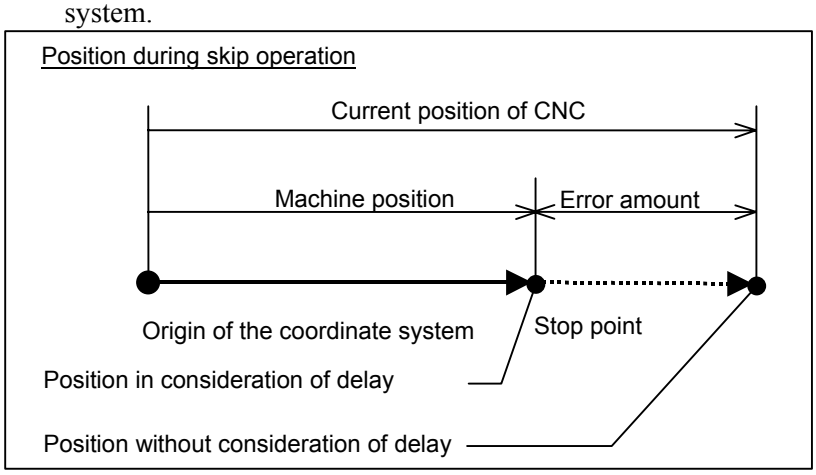
0: Ignored.
1: Considered and compensated.

The accumulated pulses and positional deviation due to actual acceleration/deceleration when the skip signal or measurement position arrival signal goes on are considered to obtain the position at which the signal is input.

 # 2 TSE When the torque limit skip function (G31 P98/99) is used, the skip position held in a system variable (#5061 to #5080) is:

0: Position that is offset considering the delay (positional deviation) incurred by the servo system.

1: Position that does not reflect the delay incurred by the servo system.



 # 4 IGX When the high-speed skip function is used, SKIP, SKIPP, and SKIP2 to SKIP8 are:

0: Enabled as skip signals.

1: Disabled as skip signals.

 # 5 CSE For the continuous high-speed skip command, high-speed skip signals are:

0: Effective at either a rising or falling edge (depending on the setting of bit 6 (SRE) of parameter No. 6200).

1: Effective at both the rising and falling edges.

 # 7 SKPXE For the skip function (G31), the skip signal SKIP is:

0: Enabled.

1: Disabled.

Whether the skip signals are enabled or disabled

Parameter	IGX (No.6201#4)	GSK (No.6200#0)	SKPXE (No.6201#7)	Skip signal SKIPP	Skip signal SKIP	Multistage skip signals SKIP2-SKIP8
Setting	0	0	0	Disabled	Enabled	Enabled
	0	1	0	Enabled	Enabled	Enabled
	0	0	1	Disabled	Disabled	Enabled
	0	1	1	Enabled	Disabled	Enabled
	1	0	0	Disabled	Disabled	Disabled
	1	1	0	Disabled	Disabled	Disabled
	1	0	1	Disabled	Disabled	Disabled
	1	1	1	Disabled	Disabled	Disabled

Bit 4 (IGX) of parameter No. 6201 is valid for the skip function using high-speed skip signals (when bit 4 (HSS) of parameter No. 6200 is set to 1) or for the multistage skip function using high-speed skip signals (when bit 5 (SLS) of parameter No. 6200 is set to 1).
To use multistage skip signals, the multistage skip function option is required.

	#7	#6	#5	#4	#3	#2	#1	#0
6202	1S8	1S7	1S6	1S5	1S4	1S3	1S2	1S1

[Input type] Parameter input
[Data type] Bit path

1S1 to 1S8 These parameters specify whether to enable or disable each high-speed skip signal when the G31 skip command is issued.
The following table shows the correspondence between the bits, input signals, and commands.
The settings of the bits have the following meaning :
0: The high-speed skip signal corresponding to a bit is disabled.
1: The high-speed skip signal corresponding to a bit is enabled.

Parameter	High-speed skip signals	Parameter	High-speed skip signals
1S1	HDI0	1S5	HDI4
1S2	HDI1	1S6	HDI5
1S3	HDI2	1S7	HDI6
1S4	HDI3	1S8	HDI7

NOTE
Do not specify the same signal simultaneously for different paths.

	#7	#6	#5	#4	#3	#2	#1	#0
6203	2S8	2S7	2S6	2S5	2S4	2S3	2S2	2S1
6204	3S8	3S7	3S6	3S5	3S4	3S3	3S2	3S1
6205	4S8	4S7	4S6	4S5	4S4	4S3	4S2	4S1
6206	DS8	DS7	DS6	DS5	DS4	DS3	DS2	DS1

[Input type] Parameter input
[Data type] Bit path

1S1to1S8, 2S1to2S8, 3S1to3S8, 4S1to4S8, DS1toDS8

Specify which skip signal is enabled when the skip command (G31, or G31P1 to G31P4) and the dwell command (G04, G04Q1 to G04Q4) are issued with the multi-step skip function.
The following table shows the correspondence between the bits, input signals, and commands.
The setting of the bits have the following meaning :
0: The skip signal corresponding to a bit is invalid.
1: The skip signal corresponding to a bit is enabled.

Multi-step skip function					
Command / Input signal	G31 G31P1 G04Q1	G31P2 G04Q2	G31P3 G04Q3	G31P4 G04Q4	G04
SKIP/HDI0	1S1	2S1	3S1	4S1	DS1
SKIP2/HDI1	1S2	2S2	3S2	4S2	DS2
SKIP3/HDI2	1S3	2S3	3S3	4S3	DS3
SKIP4/HDI3	1S4	2S4	3S4	4S4	DS4
SKIP5/HDI4	1S5	2S5	3S5	4S5	DS5
SKIP6/HDI5	1S6	2S6	3S6	4S6	DS6
SKIP7/HDI6	1S7	2S7	3S7	4S7	DS7
SKIP8/HDI7	1S8	2S8	3S8	4S8	DS8

NOTE
HDI0 to HDI7 are high-speed skip signals. Do not specify the same signal simultaneously for different paths.

When bit 0 (GSK) of parameter No. 6200 is set to 1, commands to be skipped can be selected by setting the following parameter:

Commands skipped by SKIPP signal <G006#6>

Parameter	Command skipped
When bit 0 (1S1) of parameter No. 6202 is set to 1	G31P1,G04Q1
When bit 0 (2S1) of parameter No. 6203 is set to 1	G31P2,G04Q2
When bit 0 (3S1) of parameter No. 6204 is set to 1	G31P3,G04Q3
When bit 0 (4S1) of parameter No. 6205 is set to 1	G31P4,G04Q4
When bit 6 (DS1) of parameter No. 6206 is set to 1	G04,G04Q1,G04Q2,G04Q3,G04Q4

	#7	#6	#5	#4	#3	#2	#1	#0
6207						SFN	SFP	

[Input type] Parameter input
[Data type] Bit path

 # 1 SFP The feedrate used when the skip function (G31) is being executed is:
0: Feedrate of a programmed F code.
1: Feedrate set in parameter No. 6281.

NOTE
For the multi-stage skip function and high-speed skip, see the description of bit 2 (SFN) of parameter No. 6207.

 # 2 SFN The feedrate used when the skip function based on high-speed skip signals (with bit 1 (HSS) of parameter No. 6200 set to 1) or the multi-skip function is being executed is:
0: Feedrate of a programmed F code.
1: Feedrate set in a parameter from parameter No. 6282 to No. 6285.

NOTE
For not the multistage skip function, but the skip function using no high-speed skip signals (when bit 4 (HSS) of parameter No. 6200 is set to 0), see the description of bit 1 (SFP) of parameter No. 6207.

	#7	#6	#5	#4	#3	#2	#1	#0
6208	9S8	9S7	9S6	9S5	9S4	9S3	9S2	9S1

[Input type] Parameter input
[Data type] Bit path

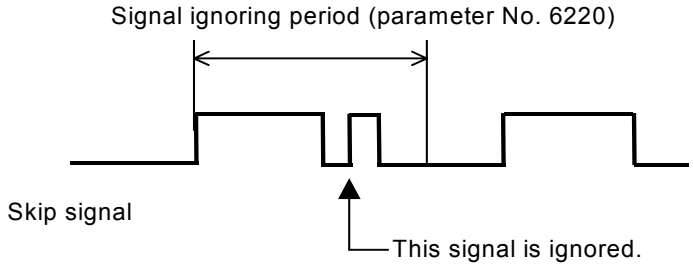
9S1 to 9S8 Specify which high-speed skip signal is enabled for the continuous high-speed skip command G31P90 or the EGB skip command G31.8.
The settings of each bit have the following meaning:
0: The high-speed skip signal corresponding to the bit is disabled.
1: The high-speed skip signal corresponding to the bit is enabled.
The bits correspond to signals as follows:

Parameter	High-speed skip signal	Parameter	High-speed skip signal
9S1	HDI0	9S5	HDI4
9S2	HDI1	9S6	HDI5
9S3	HDI2	9S7	HDI6
9S4	HDI3	9S8	HDI7

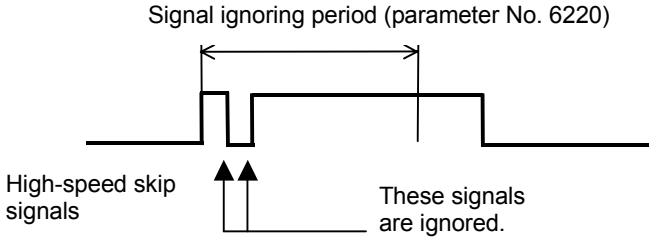
6220	Period during which skip signal input is ignored for the continuous high-speed skip function and EGB axis skip function

[Input type] Parameter input
[Data type] Byte path
[Unit of data] 8msec
[Valid data range] 3 to 127(× 8msec)

This parameter specifies the period from when a skip signal is input to when the next skip signal can be input for the continuous high-speed skip function and EGB axis skip function. This parameter is used to ignore chattering in skip signals.
If a value that falls outside the valid range is specified, the setting is assumed to be 24 msec.



When high-speed skip signals are used and bit 5 (CSE) of parameter No. 6201 is set to 1, signals are handled as follows:



6221	Torque limit dead zone time for a torque limit skip command

[Input type] Parameter input
[Data type] 2-word axis
[Unit of data] msec
[Valid data range] 0 to 65535
The torque limit skip arrival signal is ignored for a set period of time.
If G31P98 is specified, skip operation is not performed for a set period of time after the torque limit skip arrival signal is set to 1.
If G31P99 is specified, skip operation is not performed for a set period of time after the torque limit skip arrival signal is set to 1.
However, if a skip signal is input, skip operation is performed, regardless of the period of time set in this parameter.

6254	ε value on the X axis during automatic tool compensation (T series) ε value during automatic tool length measurement (M series) (for the XAE1 and GAE1 signals)

6255	ε value on the Z axis during automatic tool compensation (T series) ε value during automatic tool length measurement (M series) (for the XAE2 and GAE2 signals)

[Input type] Parameter input
[Data type] 2-word path
[Unit of data] mm, inch, deg (machine unit)
[Minimum unit of data] Depend on the increment system of the applied axis
[Valid data range] 9 digit of minimum unit of data (refer to standard parameter setting table (A))
(When the increment system is IS-B, -999999.999 to +999999.999)
These parameters set the relevant ε value during automatic tool compensation (T series) or automatic tool length measurement (M series).

NOTE
1 For the M series, when the setting of parameter No. 6252 or 6253 is 0, the setting of parameter No. 6251 is used.
2 Set a radius value regardless of whether diameter or radius programming is specified.

6281	Feedrate for the skip function (G31)

[Input type] Parameter input
[Data type] Real path
[Unit of data] mm/min, inch/min, degree/min (machine unit)
[Minimum unit of data] Depend on the increment system of the reference axis
[Valid data range] Refer to the standard parameter setting table (C)
(When the increment system is IS-B, 0.0 to +999000.0)
This parameter sets a feedrate for the skip function (G31). This parameter is valid when bit 1 (SFP) of parameter No. 6207 is set to 1.

NOTE
For the multi-stage skip function and high-speed skip, see the description of parameter No. 6282 to No. 6285.

6282	Feedrate for the skip function (G31, G31 P1)
6283	Feedrate for the skip function (G31 P2)
6284	Feedrate for the skip function (G31 P3)
6285	Feedrate for the skip function (G31 P4)

[Input type] Parameter input
[Data type] Real path
[Unit of data] mm/min, inch/min, degree/min (machine unit)
[Minimum unit of data] Depend on the increment system of the reference axis
[Valid data range] Refer to the standard parameter setting table (C)
(When the increment system is IS-B, 0.0 to +999000.0)
Each of these parameters sets a feedrate for each skip function G code. These parameters are valid when bit 2 (SFN) of parameter No. 6207 is set to 1.

6287	Positional deviation limit in torque limit skip

[Input type] Parameter input
[Data type] 2-word axis
[Unit of data] Detection unit
[Valid data range] 0 to 327670
This parameter sets a positional deviation limit for each axis imposed when torque limit skip is specified. When the actual positional deviation exceeds the positional deviation limit, the alarm (SV0004) is issued and an immediate stop takes place.

	#7	#6	#5	#4	#3	#2	#1	#0
6400	MG4	MGO	RVN	HMP	MC8	MC5	FWD	RPO

[Input type] Parameter input
[Data type] Bit path

 # 0 RPO With the manual handle retrace function, the rapid traverse rate is clamped, assuming that:
0: An override of 10% is used.
1: An override of 100% is used.

 # 1 FWD With the manual handle retrace function, program execution can be performed:
0: In both forward and backward directions.
1: In the forward direction only. Execution in the backward direction is not permitted.

 # 2 MC5
 # 3 MC8

These parameters set the number of M code groups and the number of M codes per group.
(See explanations of parameters Nos. 6411 to 6490.)

MC5	MC8	M code group setting
0	0	Standard (20 groups of four)
1	0	16 groups of five
0	1	10 groups of eight

When 16 groups of five are used, the meanings of parameters are changed as follows:
Group A No.6411(1) to No.6415(5)
Group B No.6416(1) to No.6420(5)
:
Group P No.6486(1) to No.6490(5)
When 10 groups of eight are used, they are changed as follows:
Group A No.6411(1) to No.6418(8)
Group B No.6419(1) to No.6426(8)
:
Group J No.6483(1) to No.6490(8)

 # 4 HMP When inversion or backward movement is inhibited in other paths:
0: Inversion or backward movement is not inhibited for the currently executed path.
1: Inversion or backward movement is inhibited also for the currently executed path.

 # 5 RVN When the manual handle retrace function is used, M codes other than grouped M codes:
0: Do not disable backward movement.
1: Disable backward movement.

When this parameter is set to 1, M codes other than grouped M codes disable backward movement in general. Exceptionally, however, the following M codes allow backward movement:
1. Subprogram call based on M98/M99
2. Subprogram call based on an M code
3. Macro call based on an M code
4. Waiting M code
5. M0

 # 6 MGO When the manual handle retrace function is used, handle pulses during execution of a G code related to measurement are:
0: Valid.
1: Invalid. A speed with an override of 100% is used for execution at all times.

 # 7 MG4 In the manual handle retrace function, for blocks for which multi-step skip G04 is enabled (when the multi-step skip software option is used, and the settings of parameter Nos. 6202 to 6206 are valid):
0: Backward movement is not prohibited.
1: Backward movement is prohibited.

	#7	#6	#5	#4	#3	#2	#1	#0
6401	STO	HST				CHS		

[Input type] Parameter input
[Data type] Bit path

 # 2 CHS In manual handle retrace:
0: The status is displayed if the following conditions are all satisfied:
(1) The manual handle retrace software option is used.
(2) Bit 6 (HST) of parameter No. 6401, which specifies whether to enable or disable status display, is set to 1.
(3) Check mode output signal MMMOD<Fn091.3> is set to 1.
1: The status is displayed if the following conditions are all satisfied:
(1) The manual handle retrace software option is used.
(2) Bit 6 (HST) of parameter No. 6401, which specifies whether to enable or disable status display, is set to 1.
(3) Cycle start lamp signal STL<Fn000.5> is set to 1.
(4) Check mode input signal MMOD<Gn067.2> is set to 1.
(5) Handle input signal MCHK<Gn067.3> is set to 1 in the check mode.

 # 6 HST When the manual handle retrace function is used, the time display field on the status display line of the CNC screen:
0: Does not display status.
1: Displays status.

 # 7 STO In the manual handle retrace function, the timing for outputting an S code and T code during backward movement is:
0: Different from the timing during forward movement:
1: The same as during forward movement.

6405	Override value (equivalence) for clamping the rapid traverse rate used with the manual handle retrace function

[Input type] Parameter input
[Data type] Word path
[Unit of data] %
[Valid data range] 0 to 100
This parameter sets an override value (equivalence) for clamping the rapid traverse rate used with the manual handle retrace function. If a value greater than 100 is set in parameter (No.6405), the rapid traverse rate is clamped to an override of 100%. This function is invalid if 0 is set in parameter (No.6405). In this case, the setting of bit 0 (RPO) of parameter No. 6400 is used.

6410	Travel distance per pulse generated from the manual pulse generator

[Input type] Parameter input
[Data type] Word path
[Unit of data] %
[Valid data range] 0 to 100

Set the travel distance per pulse generated from the manual pulse generator in terms of the override value.
The distance traveled by the machine when the manual handle is actually turned can be found by the following expression:
[Specified speed] × [Handle magnification] × ([Setting of this parameter]/100) × (8/60000) (mm or inch)
[Example]
When a specified feedrate is 30mm/min, the manual handle magnification is 100, and parameter No. 6410 is set to 1, the travel distance per pulse generated from the manual pulse generator is calculated as follows:
[Travel distance per pulse]
=30[mm/min] × 100 × (1/100) × (8/60000)[min]= 0.004mm

6411	M code of group A in manual handle retrace (1)
to	
6414	M code of group A in manual handle retrace (4)
6415	M code of group B in manual handle retrace (1)
to	
6418	M code of group B in manual handle retrace (4)
6419	M code of group C in manual handle retrace (1)
to	
6422	M code of group C in manual handle retrace (4)
6423	M code of group D in manual handle retrace (1)
to	
6426	M code of group D in manual handle retrace (4)
6427	M code of group E in manual handle retrace (1)
to	
6430	M code of group E in manual handle retrace (4)
6431	M code of group F in manual handle retrace (1)
to	
6434	M code of group F in manual handle retrace (4)
6435	M code of group G in manual handle retrace (1)
to	
6438	M code of group G in manual handle retrace (4)

6439	M code of group H in manual handle retrace (1)
to	
6442	M code of group H in manual handle retrace (4)
6443	M code of group I in manual handle retrace (1)
to	
6446	M code of group I in manual handle retrace (4)
6447	M code of group J in manual handle retrace (1)
to	
6450	M code of group J in manual handle retrace (4)
6451	M code of group K in manual handle retrace (1)
to	
6454	M code of group K in manual handle retrace (4)
6455	M code of group L in manual handle retrace (1)
to	
6458	M code of group L in manual handle retrace (4)
6459	M code of group M in manual handle retrace (1)
to	
6462	M code of group M in manual handle retrace (4)
6463	M code of group N in manual handle retrace (1)
to	
6466	M code of group N in manual handle retrace (4)
6467	M code of group O in manual handle retrace (1)
to	
6470	M code of group O in manual handle retrace (4)
6471	M code of group P in manual handle retrace (1)
to	
6474	M code of group P in manual handle retrace (4)
6475	M code of group O in manual handle retrace (1)
to	
6478	M code of group Q in manual handle retrace (4)
6479	M code of group R in manual handle retrace (1)
to	
6482	M code of group R in manual handle retrace (4)
6483	M code of group S in manual handle retrace (1)
to	
6486	M code of group S in manual handle retrace (4)

6487	M code of group T in manual handle retrace (1)
to	
6490	M code of group T in manual handle retrace (4)

[Input type] Parameter input
[Data type] 2-word path
[Valid data range] 0 to 9999

Set a group of M codes output during backward movement.
For backward movement for an M code, the modal M code in the same group set by the parameter is output.
The first M code in each group is set as the default.
When the number of M codes in a group is 3 or less, set the parameter corresponding to an unused M code to 0.
For backward movement for "M0", "M0" is output regardless of which M code is set for the parameter. "0" set in the parameter is ignored.
For an M code which is not set in any group by any of the above parameters, the M code for forward movement is output.
With these parameters, an M code in the same group can be output in backward movement only when the M code is the first M code in each block. When a block contains two or more M codes, the same M codes as output in forward movement are output as a second M code and up.

NOTE
The above explanation of M code groups applies to the standard settings. The number of M codes in each group and the number of M code groups vary depending on the settings of bit 2 (MC5) and bit 3 (MC8) of parameter No. 6400.

6581	RGB value of color palette 1
to	
6595	RGB value of color palette 15

[Input type] Parameter input
[Data type] 2-word
[Valid data range] 0 to 151515

Each of these parameters sets the RGB value of each color palette by specifying a 6-digit number as described below.
rrggbb: 6-digit number (rr: red data, gg: green data, bb: blue data)
The valid data range of each color is 0 to 15 (same as the tone levels on the color setting screen). When a number equal to or greater than 16 is specified, the specification of 15 is assumed.
Example)

When the tone level of a color is: red:1 green:2, blue:3, set 10203 in the parameter.

	#7	#6	#5	#4	#3	#2	#1	#0
6700								PCM

[Input type] Parameter input
[Data type] Bit path

 # 0 PCM M code that counts the total number of machined parts and the number of machined parts
0: M02, or M30, or an M code specified by parameter No.6710
1: Only M code specified by parameter No.6710

6710	M code that counts the number of machined parts

[Input type] Parameter input
[Data type] 2-word path
[Valid data range] 0 to 999999999
The total number of machined parts and the number of machined parts are counted (+1) when the M code set is executed.

NOTE
The setting of 0 is invalid (no count operation is performed with M00.) Moreover, M98, M99, M198 (external device subprogram calling), and M codes used for subprogram calling and macro calling cannot be set as M codes for count-up operation. (Even when such an M code is set, count-up operation is not performed, ignoring the M code.)

6711	Number of machined parts

[Input type] Setting input
[Data type] 2-word path
[Valid data range] 0 to 999999999
The number of machined parts is counted (+1) together with the total number of machined parts when the M02, M30, or a M code specified by parameter No.6710 is executed.

NOTE
The number of parts is not counted for M02, M03, when bit 0 (PCM) of parameter No. 6700 is set to 1.

6712	Total number of machined parts

[Input type] Setting input
[Data type] 2-word path
[Valid data range] 0 to 999999999
This parameter sets the total number of machined parts.
The total number of machined parts is counted (+1) when M02, M30, or an M code specified by parameter No.6710 is executed.

NOTE
The number of parts is not counted for M02, M30, when bit 0 (PCM) of parameter No. 6700 is set to 1.

6713	Number of required parts

[Input type] Setting input
[Data type] 2-word path
[Valid data range] 0 to 999999999
This parameter sets the number of required machined parts.
Required parts finish signal PRTSF <F0062.7> is output to PMC when the number of machined parts reaches the number of required parts. The number of parts is regarded as infinity when the number of required parts is zero. The PRTSF signal is then not output.

6750	Integrated value of power-on period

[Input type] Parameter input
[Data type] 2-word path
[Unit of data] min
[Valid data range] 0 to 999999999
This parameter displays the integrated value of power-on period.

6751	Operation time (integrated value of time during automatic operation) 1

[Input type] Setting input
[Data type] 2-word path
[Unit of data] msec
[Valid data range] 0 to 59999
For details, see the description of parameter No. 6752.

6752	Operation time (integrated value of time during automatic operation) 2

[Input type] Setting input
[Data type] 2-word path
[Unit of data] min
[Valid data range] 0 to 999999999
This parameter displays the integrated value of time during automatic operation (neither stop nor hold time included).
The actual time accumulated during operation is the sum of this parameter No. 6751 and parameter No. 6752.

6753	Integrated value of cutting time 1

[Input type] Setting input
[Data type] 2-word path
[Unit of data] msec
[Valid data range] 0 to 59999
For details, see the description of parameter No. 6754.

6754	Integrated value of cutting time 2

[Input type] Setting input
[Data type] 2-word path
[Unit of data] min
[Valid data range] 0 to 999999999
This parameter displays the integrated value of a cutting time that is performed in cutting feed such as linear interpolation (G01) and circular interpolation (G02 or G03).
The actual time accumulated during cutting is the sum of this parameter No. 6753 and parameter No. 6754.

	#7	#6	#5	#4	#3	#2	#1	#0
6800	M6T	IGI	SNG	GRS	SIG	LTM	GS2	GS1

[Input type] Parameter input
[Data type] Bit path

 # 0 GS1
 # 1 GS2 For the maximum number of groups set in parameter No. 6813, up to four tools can be registered per group. The combination of the number of registrable groups and the number of tools per group can be changed by setting GS1 and GS2.

GS2	GS1	Number of groups	Number of tools
0	0	1/8 of maximum number of groups (No.6813)	32
0	1	1/4 of maximum number of groups (No.6813)	16
1	0	1/2 of maximum number of groups (No.6813)	8
1	1	Maximum number of groups (No.6813)	4

NOTE
After changing these parameters, set data again by using G10 L3 ;(registration after deletion of data of all groups).

 # 2 LTM The tool life count is specified by:
0: Count.
1: Duration.

> **NOTE**
> After changing this parameter,set data again by using G10 L3 ;(registration after deletion of data of all groups).

 # 3 SIG When a tool is skipped by a signal, the group number is:
0: Not input by the tool group number selection signals.
1: Input by the tool group number selection signals.

> **NOTE**
> When this parameter is set to 0, a tool of the currently used group is skipped.

 # 4 GRS When the tool change reset signal (TLRST) is input:
0: If the life of the group specified by the tool group number selection signals has expired, the execution data of the group is cleared.
1: The execution data of all registered groups is cleared.
If this parameter is set to 1, the execution data of all registered groups is cleared also when the clear operation to clear execution data is performed on the tool life management list screen.

 # 5 SNG When the tool skip signal (TLSKP) is input while a tool not controlled by the tool life management function is being used:
0: A tool of the most recently used group or a specified group (bit 3 (SIG) of parameter No. 6800) is skipped.
1: The tool skip signal is ignored.

 # 6 IGI Tool back numbers are:
0: Not ignored.
1: Ignored.

 # 7 M6T A T code specified in the same block as M06 is:
0: Assumed to be a back number.
1: Assumed to be a command specifying the next tool group.

	#7	#6	#5	#4	#3	#2	#1	#0
6801	M6E				EMD	LVF	TSM	
	M6E				EMD	LVF		

NOTE
The use of this parameter varies depending on whether the tool management function or tool life management function is used.

[Input type] Parameter input
[Data type] Bit path

 # 1 TSM In the tool life management function, life counting is performed as follows when more than one offset is specified:
0: Counting is performed for each tool number.
1: Counting is performed for each tool.

 # 2 LVF When the life value is counted by duration in the tool management function or tool life management function, tool life count override signals *TLV0 to *TLV9 <G049.0 to G050.1> are:
0: Not used.
1: Used.

 # 3 EMD In the tool life management function, the mark "*" indicating that the life has expired is displayed when:
0: The next tool is used.
1: The life has just expired.

NOTE
If this parameter is set to 0, the "@" mark (indicating that the tool is in use) is kept displayed unless the next tool whose life has not expired is used. If this parameter is set to 1, marks are displayed in different ways depending on the life count type.
If the life count type is the duration specification type, the "*" mark (indicating that the life has expired) appears when the life has expired. If the life count type is the count specification type, one count is not assumed until the end of the program (M02, M30, and so on). Therefore, even when the life value and the tool life counter value match, the "*" mark (life has expired) does not appear. The "*" mark (life has expired) appears when the tool is used again by a tool group command (T code) or tool change command (M06) issued after the CNC is reset.

 # 7 M6E When a T code is specified in the same block as M06:

0: The T code is treated as a back number or the group number to be selected next.
Which number is assumed depends on the setting of bit 7 (M6T) of parameter No. 6800.

1: Life counting for the tool group starts immediately.

	#7	#6	#5	#4	#3	#2	#1	#0
6802	RMT	TSK	TGN	ARL	GRP	E17	TCO	T99

[Input type] Parameter input
[Data type] Bit path

 # 0 T99 When M99 of the main program is executed, and there is a the life was expired tool group:

0: The tool change signal is not output.

1: The tool change signal is output, and the automatic operation becomes a stopped state..

If the life count is specified by use count and this parameter 1, the tool change signal TLCH <Fn064.0> is output and the automatic operation becomes a stopped state if the life of at least one tool group has expired when the M99 command is specified.
If the life count type is the duration specification type, the automatic operation becomes a stopped state if the life of at least one tool group has expired when the M99 command is specified.

M

If the life count is specified by use count, after the M99 command is specified, a tool group command (T code) selects, from a specified group, a tool whose life has not expired, and the next tool change command (M06) increments the tool life counter by one.

T

If the life count is specified by use count, when a tool group command (T code) is specified after the M99 command is specified, a tool whose life has not expired is selected from a specified group, and the tool life counter is incremented by one.
When the tool change type is the ATC type (bit 3 (TCT) of parameter No. 5040 = 1), the same specifications as for the M series apply.

#1 TCO
#2 E17

Specifies whether to allow the FOCAS2 or PMC window function to write tool information of a group being used or a group to be used next during automatic operation (the OP signal is set to "1").

Condition			6802#1(TCO)		
			0	1	
				6802#2(E17)	
				1	0
During automatic operation	Group being used or to be used next	Tool being used	×	□	○
		Tool not being used	×	○	○
	Group neither being used nor to be used next		○	○	○
Not during automatic operation			○	○	○

○: Tool information can be written from FOCAS2 and PMC window.
×: Tool information cannot be written from FOCAS2 and PMC window.
When an attempt is made to write tool information from PMC window, completion code 13 (REJECT ALARM) is returned.
□: Tool information cannot be cleared.

NOTE
When tool information of a tool being used (marked with "@") in the group being used or to be used next or tool information of the most recently used tool (marked with "@") in a group that is neither the group being used nor the group to be used next is cleared, the life counter is reset to 0.
It is possible to modify tool information of a tool in the group to be used next. However, because tool selection is already completed, the selected tool does not change even when the tool information is modified.
This parameter has no influence on modifications to tool information by edit operations from the tool life management screen.

 # 3 GRP Management data of tool life arrival notice signal TLCHB is:
0: Managed using the remaining life value set in parameter No. 6844 and 6845.
1: Managed using the remaining life value set in tool life management data.

NOTE
When the tool life arrival notice signal function is used, bit 4 (LFB) of parameter No. 6805 must be set to 1 to enable the tool life management B function.

 # 4 ARL Tool life arrival notice signal TLCHB of tool life management is:
0: Output for each tool.
1: Output for the last tool of a group.
This parameter is valid only when bit 3 (GRP) of parameter No. 6802 is set to 1.

 # 5 TGN In the tool life management function, the optional group number function is:
0: Not used.
1: Used.

> **NOTE**
> When the optional group number function is used, bit 4 (LFB) of parameter No. 6805 must be set to 1 to enable the tool life management B function.
> In lathe systems, the optional group number function can be used if the tool change type is the ATC type (bit 3 (TCT) of parameter No. 5040 = 1).

 # 6 TSK If the count type in tool life management is the duration type, then when the last tool of a group is skipped by a signal:
0: The count value for the last tool equals the life value.
1: The count value for the last tool remains unchanged.

 # 7 RMT Tool life arrival notice signal TLCHB is turned on and off as follows:
0: The signal is turned on if the remaining life value (the life value minus the life counter value) is smaller than or equal to the remaining life setting. The signal is turned off if the remaining life value (the life value minus the life counter value) is greater than the remaining life setting.
1: The signal is turned on if the remaining life value (the life value minus the life counter value) is equal to the remaining life setting. The signal is turned off if the remaining life value (the life value minus the life counter value) is not equal to the remaining life setting.

> **NOTE**
> When using the life count override feature, set bit 7 (RMT) of parameter No. 6802 to 0. When the life count is specified by duration, the unit used for determining the result of comparison between the remaining life and the remaining life setting varies depending on the life count interval (bit 0 (FCO) of parameter No. 6805). If the life is counted every second, the comparison is made in units of 1 minute; if the life is counted every 0.1 second, the comparison is made in units of 0.1 minute.

	#7	#6	#5	#4	#3	#2	#1	#0
6804		LFI				ETE	TCI	

[Input type] Parameter input
[Data type] Bit path

 # 1 TCI During automatic operation (the OP signal is "1"), editing of tool life data is:
0: Disabled.
1: Enabled.

NOTE
When this parameter is set to 1, tool life data can be edited even during automatic operation (the OP signal is "1"). If the target group for editing is the group being used or the group to be used next, however, only presetting of the life counter is permitted, and other data cannot be modified.

 # 2 ETE In the tool life management screen, the mark of the tool at the life was expired of the final tool in the group :
0: depends on setting parameter EMD (No.6801#3).
1: is "*" mark.
If bit 2 (ETE) of parameter No. 6804 is set to 1, when the life counter of the final tool in the group becomes equal to the life value, display mark "*" in the final tool of the tool life management screen.
When tool change signal TLCH<Fn064.0> is "1", the state of the life was expired of the tool can be read by reading tool information on the final tool in FOCAS2 or the PMC window.

 # 6 LFI In tool life management, counting of the life of a selected tool is:
0: Enabled.
1: Enabled or disabled according to the status of tool life counting disable signal LFCIV<G048.2>.

	#7	#6	#5	#4	#3	#2	#1	#0
6805	TAD	TRU	TRS	LFB			FGL	FCO

[Input type] Parameter input
[Data type] Bit path

 # 0 FCO If the life count type is the duration specification type, the life is counted as follows:
0: Every second.
1: Every 0.1 second.
According to the setting of this parameter, the increment system of life values and tool life counter values displayed on the tool life management screen is set as follows:

Parameter FCO	0	1
Increment system for display and setting of life values and life counter values	1-minute increments	0.1-minute increments

NOTE
After changing the setting of this parameter, set data again by using G10L3;(registration after deletion of data of all groups).

 # 1 FGL If the life count type is the duration specification type, life data registered by G10 is:
0: In one-minute increments.
1: In 0.1-second increments.

 # 4 LFB The tool life management B function is:
0: Disabled.
1: Enabled.
When the tool life management B function is enabled, the following functions can be used:
<1> Tool life value extension (count specification: 99999999 times, duration specification: 100000 minutes)
<2> Optional group number function
<3> Tool life arrival notice function
In lathe systems, if the tool change type is the ATC type (bit 3 (TCT) of parameter No. 5040 = 1), the optional group number function can be used.

 # 5 TRS Tool change reset signal TLRST is valid when reset signal RST is not "1" and:

0: The reset state (automatic operation signal OP is "0") is observed.

1: The reset state (automatic operation signal OP is "0"), automatic operation stop state (The STL and SPL signals are "0" and the OP signal is "1"), or the automatic operation pause state (the STL signal is "0" and the SPL signal is "1") is observed. The TLRST signal, however, is invalid when the automatic operation stop state, automatic operation pause state, and automatic operation start state (the STL signal is "1") is observed during execution of a data setting command (G10L3).

 # 6 TRU When the life count type is the duration specification type, and the life is counted every second (bit 0 (FCO) of parameter No. 6805 is set to 0):

0: Cutting time less than one second is discarded and is not counted.

1: Cutting time less than one second is rounded up and is counted as one second.

NOTE
If the life is counted every 0.1 second (bit 0 (FCO) of parameter No. 6805 is set to 1), cutting time less than 0.1 second is always rounded up and is counted as 0.1 second.

 # 7 TAD With tool change type D (bit 7 (M6E) of parameter No. 6801 is set to 1), when a block specifying M06 contains no T command:

0: An alarm PS0153 is issued.

1: No alarm is issued.

6810	Tool life management ignore number

[Input type] Parameter input
[Data type] 2-word path
[Valid data range] 0 to 99999999

This parameter sets the tool life management ignore number.
When the value specified in a T code exceeds the value set in this parameter, the value obtained by subtracting the parameter-set value from the T code value is assumed to be the tool group number for tool life management.

6811	Tool life count restart M code

[Input type] Parameter input
[Data type] Byte path
[Valid data range] 0 to 127 (except 01, 02, 30, 98, and 99)
When 0 is specified, it is ignored.
When the life is specified by count, the tool change signal (TLCH) is output if the life of at least one tool group has expired when the tool life count restart M code is issued.
The T code (tool life management group command) specified after the tool life count restart M code selects a tool whose life has not expired from a specified group, and the next M06 command increments the tool life counter by one.
When the life is specified by duration, specifying the tool life count restart M code causes nothing. When 0 is set in this parameter, the tool life count restart M code is invalid. When the data of M code exceeds 127 values, set 0 in parameter No.6811, and set the value of M code in parameter No.13221. The data range of parameter No.13221 is from 0 to 255.

NOTE
The use of this parameter varies depending on whether the tool management function or tool life management function is used.

6813	Maximum number of groups in tool life management

NOTE
After this parameter has been set, the power must be turned off then back on for the setting to become effective.

[Input type] Parameter input
[Data type] Word path
[Unit of data] Group
[Valid data range] 0, 8, 16 to 256
This parameter sets the maximum number of groups to be used for each path. As the maximum number of groups, set a multiple of eight. When the tool life management function is not used, 0 must be set. Set this parameter so that the total number of groups in all paths does not exceed the total number of groups in the entire system (256 groups).

NOTE
When the power is turned on, all tool life management file data is initialized. So, tool life management data must be set for all paths that use tool life management.

6844	Remaining tool life (use count)

[Input type] Parameter input
[Data type] Word path
[Valid data range] 0 to 9999
This parameter sets a remaining tool life (use count) used to output the tool life arrival notice signal when the tool life is specified by use count. If a value greater than the tool life value or 0 is set in this parameter, the tool life arrival notice signal is not output.

6845	Remaining tool life (use duration)

[Input type] Parameter input
[Data type] 2-word path
[Unit of data] min
[Valid data range] Not greater than the tool life value
This parameter sets the remaining tool life (use duration) used to output the tool life arrival notice signal when the tool life is specified by use duration. If a value greater than the tool life value or 0 is specified in this parameter, the tool life arrival notice signal is not output.

NOTE
When the life is counted every 0.1 second (bit 0 (FCO) of parameter No. 6805 = 1), the parameter value is in 0.1-minute increments.

6930	Maximum value of the operating range of the 1-st position switch (PSW101)
6931	Maximum value of the operating range of the 2-nd position switch (PSW102)
to	to
6945	Maximum value of the operating range of the 16-th position switch (PSW116)

[Input type] Parameter input
[Data type] Real path
[Unit of data] mm, inch, degree (machine unit)
[Minimum unit of data] Depend on the increment system of the reference axis
[Valid data range] 9 digit of minimum unit of data (refer to standard parameter setting table (A))
(When the increment system is IS-B, -999999.999 to +999999.999)
Set the maximum value of the operating range of the first to sixteenth position switches.

NOTE
1 For a diameter-specified axis, use radius values to specify the parameters used to set the maximum and minimum values of an operating range.
2 The position switch function is enabled upon completion of reference position return.

6950	Minimum value of the operating range of the 1-st position switch (PSW201)
6951	Minimum value of the operating range of the 2-nd position switch (PSW202)
to	to
6965	Minimum value of the operating range of the 16-th position switch (PSW216)

[Input type] Parameter input
[Data type] Real path
[Unit of data] mm, inch, degree (machine unit)
[Minimum unit of data] Depend on the increment system of the reference axis
[Valid data range] 9 digit of minimum unit of data (refer to standard parameter setting table (A))
(When the increment system is IS-B, -999999.999 to +999999.999)
Set the minimum value of the operating range of the first to sixteenth position switches.

NOTE
1 For a diameter-specified axis, use radius values to specify the parameters used to set the maximum and minimum values of an operating range.
2 The position switch function is enabled upon completion of reference position return.

	#7	#6	#5	#4	#3	#2	#1	#0
7001						JST		

[Input type] Parameter input
[Data type] Bit path

 # 2 JST In manual numerical specification, the STL signal indicating that automatic operation is being started is:
0: Not output.
1: Output.

	#7	#6	#5	#4	#3	#2	#1	#0
7002					JBF	JTF	JSF	JMF

[Input type] Parameter input
[Data type] Bit path

 # 0 JMF In manual numerical specification, M function specification is:
0: Allowed.
1: Not allowed.

 # 1 JSF In manual numerical specification, S function specification is:
0: Allowed.
1: Not allowed.

 # 2 JTF In manual numerical specification, T function specification is:
0: Allowed.
1: Not allowed.

 # 3 JBF In manual numerical specification, B function specification is:
0: Allowed.
1: Not allowed.

	#7	#6	#5	#4	#3	#2	#1	#0
7040						RPS		

[Input type] Parameter input
[Data type] Bit path

 # 2 RPS When the tool retract signal TRESC is set to 1 after G10.6 is specified alone:
0: The tool is not retracted.
1: The tool is retracted with the value set for parameter No. 7041 or 1261 used as the incremental retraction distance.

7041	Retraction distance in tool retract and return

[Input type] Setting input
[Data type] Real axis
[Unit of data] mm, inch, deg (input unit)
[Minimum unit of data] Depend on the increment system of the applied axis
[Valid data range] 9 digit of minimum unit of data (refer to standard parameter setting table (A))
(When the increment system is IS-B, -999999.999 to +999999.999)
This parameter sets the retraction distance used when G10.6 is specified alone for the tool retract and return. The tool is retracted by the distance set for this parameter in the incremental mode. This data is valid only when bit 2 (RPS) of parameter No. 7040 is set to 1.
During tool center point control and workpiece setting error compensation, however, this parameter is valid if parameter No. 11261 is set to 0.

	#7	#6	#5	#4	#3	#2	#1	#0
7055					BCG			

[Input type] Parameter input
[Data type] Bit path

 # 3 BCG The pre-interpolation bell-shaped acceleration/deceleration time constant change function is:
0: Disabled.
1: Enabled.

7066	Acceleration/deceleration reference speed for the time constant change function of bell-shaped acceleration/deceleration before interpolation

[Input type] Setting input
[Data type] Real path
[Unit of data] mm/min, inch/min, degree/min (input unit)
[Minimum unit of data] Depend on the increment system of the reference axis
[Valid data range] Refer to the standard parameter setting table (C)
(When the increment system is IS-B, 0.0 to +999000.0)
This parameter is used when the time constant change function of bell-shaped acceleration/deceleration before interpolation is used.

	#7	#6	#5	#4	#3	#2	#1	#0
7100					HCL		THD	JHD

[Input type] Parameter input
[Data type] Bit path

 # 0 JHD Manual handle feed in JOG feed mode or incremental feed in the manual handle feed
0: Invalid
1: Valid

 # 1 THD In the TEACH IN JOG mode, the manual pulse generator is:
0: Disabled.
1: Enabled.

 # 3 HCL The clearing of handle interruption amount display by soft key [CAN] operation is:
0: Disabled.
1: Enabled.

	#7	#6	#5	#4	#3	#2	#1	#0
7102								HNGx

[Input type] Parameter input
[Data type] Bit axis

 # 0 HNGx Axis movement direction for rotation direction of manual pulse generator
0: Same in direction
1: Reverse in direction

	#7	#6	#5	#4	#3	#2	#1	#0
7103					HIT	HNT	RTH	

[Input type] Parameter input
[Data type] Bit path

 # 1 RTH By a reset or emergency stop, the amount of manual handle interruption is:
0: Not canceled.
1: Canceled.

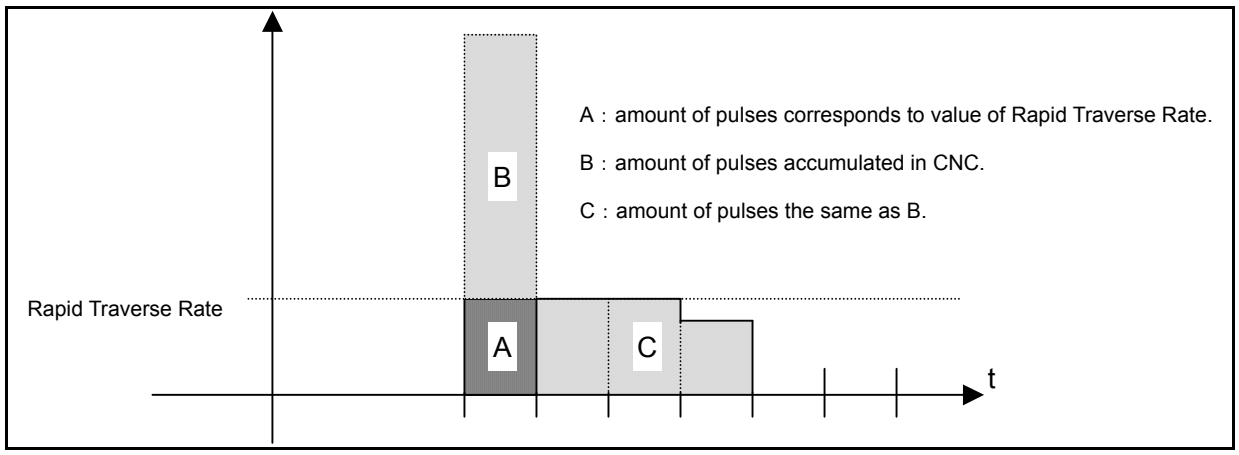
 # 2 HNT When compared with the travel distance magnification selected by the manual handle feed travel distance selection signals (incremental feed signals) (MP1, MP2), the travel distance magnification for incremental feed/manual handle feed is:
0: Same.
1: 10 times greater.

 # 3 HIT When compared with the travel distance magnification selected by the manual handle feed travel distance selection signals (incremental feed signals (MP1, MP2), the travel distance magnification for manual handle interrupt is:
0: Same.
1: 10 times greater.

7117	**Allowable number of pulses that can be accumulated during manual handle feed**

[Input type] Parameter input
[Data type] 2-word path
[Unit of data] Pulse
[Valid data range] 0 to 999999999
This parameter sets the number of pulses from the manual pulse generator that exceed the rapid traverse rate and can be accumulated without being discarded if manual handle feed faster than the rapid traverse rate is specified.

The amount of pulses exceeding the rapid traverse rate can be saved by CNC as B. And amount of pulses B will be exported as pulses C.



Amount of pulses exported by CNC in Manual Handle Feed

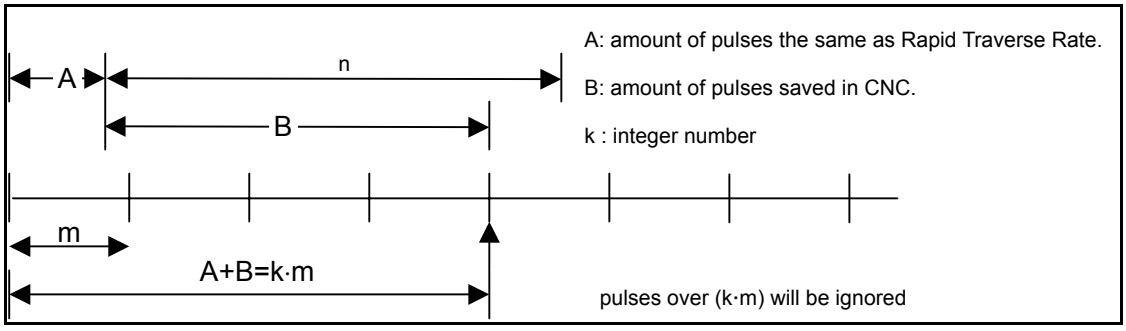
Amount of pulses B is calculated in 2 cases as following:

In case of
1) Parameter No.7117 = 0
   The feedrate is clamped at the Rapid Traverse Rate and generated pulses exceeding the Rapid Traverse Rate are ignored (B=0)

In case of
2) Parameter No.7117 > 0
   The feedrate is clamped as the Rapid Traverse Rate, but the pulses exceeding the Rapid Traverse Rate is not ignored. Amount of pulses accumulated in CNC is calculated as following. (Although stopping the rotation of manual pulse generator, if there is pulses accumulated in CNC, it will be exported and the tool will move as long as amount of it.)

   Magnification set by MP1, MP2<Gn019.4,.5> is m, value of parameter No.7117 is n.

   n < m: Clamping is set performed at value of parameter No.7117.
   n ≥ m: Amount A+B, showed in figure, which's value is multiple of m and small than n. As a result, clamping is performed as an integral multiple of the selected magnification.



Amount of pulses exceeding the Rapid Traverse Rate (n ≥ m)

NOTE
Due to change of mode, clamping can be performed not as an integral multiple of the selected magnification.
The distance the tool moves may not match the graduations on the manual pulse generator.

7160	Approach handle clamp feedrate

[Input type] Parameter input
[Data type] Real path
[Unit of data] mm/min, inch/min, degree/min (machine unit)
[Minimum unit of data] Depend on the increment system of the reference axis
[Valid data range] Refer to standard parameter setting table (C)
Approach handle clamp feedrate is set.
When the setting is 0, the feedrate is not clamped.

7161	Guidance handle clamp feedrate

[Input type] Parameter input
[Data type] Real path
[Unit of data] mm/min, inch/min, degree/min (machine unit)
[Minimum unit of data] Depend on the increment system of the reference axis
[Valid data range] Refer to standard parameter setting table (C)
Guidance handle clamp feedrate is set.
When the setting is 0, the feedrate is not clamped.

	#7	#6	#5	#4	#3	#2	#1	#0
7200		OP7	OP6	OP5	OP4	OP3	OP2	OP1

[Input type] Parameter input
[Data type] Bit path

NOTE
When at least one of these parameters is set, the power must be turned off before operation is continued.

 # 0 OP1 Mode selection on software operator's panel
0: Not performed
1: Performed

 # 1 OP2 JOG feed axis select and manual rapid traverse select on software operator's panel
0: Not performed
1: Performed

 # 2 **OP3** Manual pulse generator's axis select and manual pulse generator's magnification select on software operator's panel
0: Not performed
1: Performed

 # 3 **OP4** JOG feedrate override select, feedrate override select, and rapid traverse override select on software operator's panel
0: Not performed
1: Performed

 # 4 **OP5** Optional block skip select, single block select, machine lock select, and dry run select on software operator's panel
0: Not performed
1: Performed

 # 5 **OP6** Protect key on software operator's panel
0: Not performed
1: Performed

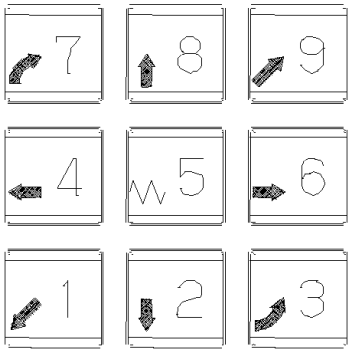
 # 6 **OP7** Feed hold on software operator's panel
0: Not performed
1: Performed

7210	Jog-movement axis and its direction on software operator's panel "↑"
7211	Jog-movement axis and its direction on software operator's panel "↓"
7212	Jog-movement axis and its direction on software operator's panel "→"
7213	Jog-movement axis and its direction on software operator's panel "←"
7214	Jog-movement axis and its direction on software operator's panel "↙"
7215	Jog-movement axis and its direction on software operator's panel "↗"
7216	Jog-movement axis and its direction on software operator's panel "↖"
7217	Jog-movement axis and its direction on software operator's panel "↗"

[Input type] Parameter input
[Data type] Byte path
[Valid data range] 0 to 8

On software operator's panel, set a feed axis corresponding to an arrow key on the MDI panel when jog feed is performed.

Arrow keys on MDI panel



Setting value	Feed axis and direction
0	Not moved
1	First axis, positive direction
2	First axis, negative direction
3	Second axis, positive direction
4	Second axis, negative direction
5	Third axis, positive direction
6	Third axis, negative direction
7	Fourth axis, positive direction
8	Fourth axis, negative direction

Example)

Under X, Y, and Z axis configuration, to set arrow keys to feed the axes in the direction specified as follows, set the parameters to the values given below. [8↑] to the positive direction of the Z axis, [2↓] to the negative direction of the Z axis, [6→] to the positive direction of the X axis [4←] to the negative direction of the X axis, [1↙] to the positive direction of the Y axis, [9↗] to the negative direction of the Y axis

Parameter No.7210 = 5 (Z axis, positive direction)
Parameter No.7211 = 6 (Z axis, negative direction)
Parameter No.7212 = 1 (X axis, positive direction)
Parameter No.7213 = 2 (X axis, negative direction)
Parameter No.7214 = 3 (Y axis, positive direction)
Parameter No.7215 = 4 (Y axis, negative direction)
Parameter No.7216 = 0 (Not used)
Parameter No.7217 = 0 (Not used)

	#7	#6	#5	#4	#3	#2	#1	#0
7300	MOU	MOA						

[Input type] Parameter input
[Data type] Bit path

 # 6 MOA In program restart operation, before movement to a machining restart point:
0: The last M, S, T, and B codes are output.
1: All M codes and the last S, T, and B codes are output.
This parameter is enabled when the MOU parameter is set to 1.

 # 7 MOU In program restart operation, before movement to a machining restart point after restart block search:
0: The M, S, T, and B codes are not output.
1: The last M, S, T, and B codes are output.

7310	Ordinal number of an axis along which a movement is made in dry run after program restart

[Input type] Setting input
[Data type] Byte axis
[Valid data range] 1 to (Number of controlled axes)
This parameter sets the ordinal number of an axis along which a movement is made in dry run after the program is restarted.

	#7	#6	#5	#4	#3	#2	#1	#0
7501	IPC	IT2	IT1	IT0				

[Input type] Parameter input
[Data type] Bit path

 # 4 IT0
 # 5 IT1
 # 6 IT2

IT2	IT1	IT0	Interpolation of high-speed cutting G05 data (ms)
0	0	0	8
0	0	1	2
0	1	0	4
0	1	1	1
1	0	0	16
1	1	1	0.5

 # 7 IPC While high-speed cutting (G05) is being executed using cycle machining data, monitoring to see whether distribution processing is stopped or not is:
0: Not performed.
1: Performed. (If distribution processing is stopped in the middle, an alarm (PS0179) is issued after distribution is completed.)

	#7	#6	#5	#4	#3	#2	#1	#0
7502			LC2	LC1				

[Input type] Parameter input
[Data type] Bit path

 # 4 LC1
 # 5 LC2

LC2	LC1	End timing of servo learning function during high-speed cycle cutting retract function
0	0	Disables the servo learning function, after which retract operation starts.
0	1	Disables the servo learning function upon the completion of retract operation.
1	0	Disables the servo learning function upon the completion of a retract cycle.

	#7	#6	#5	#4	#3	#2	#1	#0
7503				HCT				

[Input type] Parameter input
[Data type] Bit path

 # 4 HCT The variable number for starting storage of high-speed cycle machining distribution data is:
0: Specified by 1/10.
1: Specified by 1/100.

	#7	#6	#5	#4	#3	#2	#1	#0
7505							HUN	

[Input type] Parameter input
[Data type] Bit axis

NOTE
When this parameter bit is set, the power must be turned off before operation is continued.

 # 1 HUN During high-speed cutting, the increment system of distribution data is:
0: 1 pulse.
1: 10 pulses.

NOTE
If data distributed at one time is longer than one word because of the least input increment and the maximum movement feedrate, this parameter is used. If bit parameter HUN is set to 1 for an axis, high-speed cycle machining data distributed to that axis is internally multiplied by 10 by the CNC and then input. Therefore, when using HUN for an axis, set a tenth of high-speed cycle machining data to be distributed to that axis.

7510	Number of controlled axes for high-speed cycle machining

[Input type] Parameter input
[Data type] Word path
[Valid data range] 1 to number of controlled axes
Set the number of axes to be controlled when G05 is specified to perform high-speed cycle machining.

7514	Retract direction and retract feedrate in high-speed cycle machining retract operation

[Input type] Parameter input
[Data type] Real axis
[Valid data range] Refer to standard parameter setting table (C)
This parameter sets a retract direction and retract feedrate along each axis in a high-speed cycle machining retract operation. The superimposed feedrate for each axis is specified, and the retract direction is specified with a sign.

7515	Number of retract operation distributions in a high-speed cycle machining retract operation

[Input type] Parameter input
[Data type] 2-word path
[Valid data range] 0 to 99999999

This parameter sets the number of retract operation distributions in a high-speed cycle machining retract operation.
When the cycle currently being executed ends before the number of distributions specified in this parameter are performed, retract operation is terminated. When 0 is specified in this parameter, the number of retract operation distributions is assumed to be infinite. In this case, retract operation is performed until the cycle currently being executed ends.

	#7	#6	#5	#4	#3	#2	#1	#0
7600	PLZ							

[Input type] Parameter input
[Data type] Bit path

 # 7 PLZ Reference position return based on a G28 command on the tool rotation axis for polygon turning is:
0: Performed in the same sequence as manual reference position return.
1: Performed by positioning using the rapid traverse rate.
The synchronous axis returns to the reference position in the same sequence as the manual reference position return when no return-to-reference position is performed after the power is turned on.

7610	Control axis number of tool rotation axis for polygon turning

NOTE
When this parameter is set, the power must be turned off before operation is continued.

[Input type] Parameter input
[Data type] Byte path
[Valid data range] 1 to number of controlled axes

This parameter sets the control axis number of a rotation tool axis used for polygon turning.
However, when a G51.2 command is executed by setting 0 in this parameter, operation stops with the alarm (PS0314).

7640	Master axis in spindle-spindle polygon turning

[Input type] Parameter input
[Data type] Byte path
[Valid data range] 0 to Maximum number of controlled axes (Within a path)
This parameter sets the master axis in spindle-spindle polygon turning.

NOTE
1 Spindle-spindle polygon turning is enabled only for serial spindles.
2 When any one of parameter No. 7640 and No. 7641 is set to 0, polygon turning is performed using the first spindle (master axis) and the second spindle (polygon synchronous axis) in the path to which the parameter belongs.
3 When a spindle other than the first serial spindle is used as a master axis, the multi-spindle control option is required to specify an S command for the master axis.
4 When the PMC window function or G10 command is used to rewrite this parameter, rewrite this parameter before the block specifying the spindle-spindle polygon command G51.2. When the PMC window function is used to rewrite this parameter in the block immediately before G51.2, specify the rewriting of this parameter by using an M code (parameter No. 3411 and up) without buffering.

7641	Polygon synchronous axis in spindle-spindle polygon turning

[Input type] Parameter input
[Data type] Byte path
[Valid data range] 0 to Maximum number of controlled axes (Within a path)
This parameter sets the polygon synchronous (slave) axis in spindle-spindle polygon turning.

NOTE
1 Spindle-spindle polygon turning is enabled only for serial spindles.
2 When any one of parameter No. 7640 and No. 7641 is set to 0, polygon turning is performed using the first spindle (master axis) and the second spindle (polygon synchronous axis) in the path to which the parameter belongs.
3 When a spindle other than the first serial spindle is used as a master axis, the multi-spindle control option is required to specify an S command for the master axis.
4 When the PMC window function or G10 command is used to rewrite this parameter, rewrite this parameter before the block specifying the spindle-spindle polygon command G51.2. When the PMC window function is used to rewrite this parameter in the block immediately before G51.2, specify the rewriting of this parameter by using an M code (parameter No. 3411 and up) without buffering.

7642	Master axis in spindle-spindle polygon turning (spindle number common to the system)

[Input type] Parameter input
[Data type] Byte path
[Valid data range] 0 to Maximum number of controlled axes (Common to the system)
This parameter sets the master axis in spindle-spindle polygon turning.

NOTE
1 Spindle-spindle polygon turning is enabled only for serial spindles.
2 This parameter is invalid if either parameter No. 7642 or No.7643 is set to 0. In this case, the settings of parameter No. 7640 and No.7641 are valid.
3 When a spindle other than the first serial spindle is used as a master axis, the multi-spindle control option is required to specify an S command for the master axis.
4 When the PMC window function or G10 command is used to rewrite this parameter, rewrite this parameter before the block specifying the spindle-spindle polygon command G51.2. When the PMC window function is used to rewrite this parameter in the block immediately before G51.2, specify the rewriting of this parameter by using an M code (parameter No. 3411 and up) without buffering.
5 A spindle number common to the system is to be set in this parameter. When using this parameter, set 0 in parameter No. 7640 and No. 7641.

7643	Polygon synchronous axis in spindle-spindle polygon turning (spindle number common to the system)

[Input type] Parameter input
[Data type] Byte path
[Valid data range] 0 to Maximum number of controlled axes (Common to the system)
This parameter sets the polygon synchronous (slave) axis in spindle-spindle polygon turning.

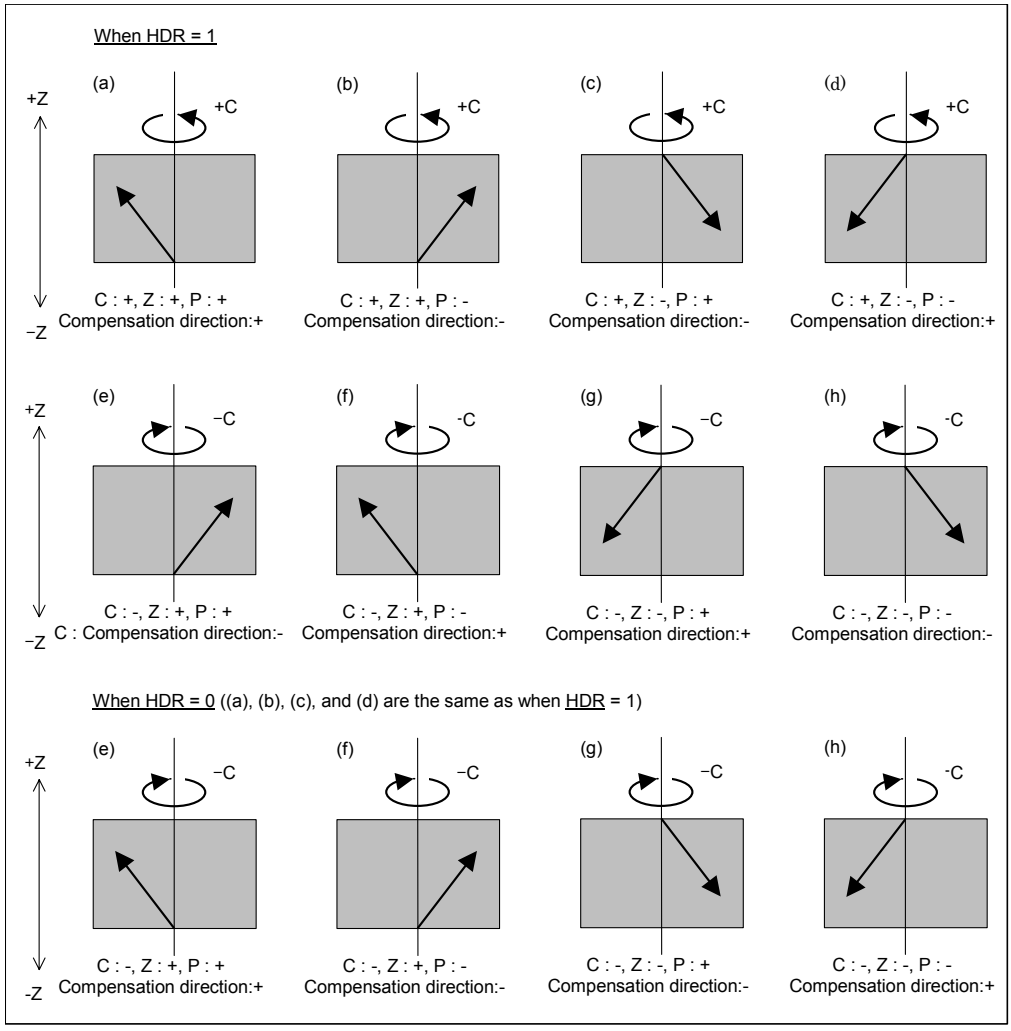
NOTE
1 Spindle-spindle polygon turning is enabled only for serial spindles.
2 This parameter is invalid if either parameter No. 7642 or No.7643 is set to 0. In this case, the settings of parameter No. 7640 and No.7641 are valid.
3 When a spindle other than the first serial spindle is used as a master axis, the multi-spindle control option is required to specify an S command for the master axis.
4 When the PMC window function or G10 command is used to rewrite this parameter, rewrite this parameter before the block specifying the spindle-spindle polygon command G51.2. When the PMC window function is used to rewrite this parameter in the block immediately before G51.2, specify the rewriting of this parameter by using an M code (parameter No. 3411 and up) without buffering.
5 A spindle number common to the system is to be set in this parameter. When using this parameter, set 0 in parameter No. 7640 and No. 7641.

	#7	#6	#5	#4	#3	#2	#1	#0
7700						HDR		HBR

[Input type] Parameter input
[Data type] Bit path

 # 0 HBR When the electronic gear box (EGB) function is used, performing a reset:
0: Cancels the synchronous mode (G81 or G81.5).
1: Does not cancel the synchronous mode. The mode is canceled only by the G80 or G80.5 command.

 # 2 HDR Direction of helical gear compensation (usually, set 1.)
(Example) To cut a left-twisted herical gear when the direction of rotation about the C-axis is the negative (-) direction:
0: Set a negative (-) value in P.
1: Set a positive (+) value in P.



	#7	#6	#5	#4	#3	#2	#1	#0
7701					LZR			

[Input type] Parameter input
[Data type] Bit path

 # 3 LZR When L (number of hob threads) = 0 is specified at the start of EGB synchronization (G81):
0: Synchronization is started, assuming that L = 1 is specified.
1: Synchronization is not started, assuming that L = 0 is specified. However, helical gear compensation is performed.

	#7	#6	#5	#4	#3	#2	#1	#0
7702	PHD	PHS			ART			TDP

[Input type] Parameter input
[Data type] Bit path

 # 0 TDP The specifiable number of teeth, T, of the electronic gear box (G81) is:
0: 1 to 1000
1: 0.1 to 100 (1/10 of a specified value)

NOTE
In either case, a value from 1 to 1000 can be specified.

 # 3 ART The retract function executed when an alarm is issued is:
0: Disabled.
1: Enabled.
When an alarm is issued, a retract operation is performed with a set feedrate and travel distance (parameter Nos. 7740 and 7741).

NOTE
If a servo alarm is issued for other than the axis along which a retract operation is performed, the servo activating current is maintained until the retract operation is completed.

 # 6 PHS When the G81/G80 block contains no R command:
0: Acceleration/deceleration is not performed at the start or cancellation of EGB synchronization.
1: Acceleration/deceleration is performed at the start or cancellation of EGB synchronization. After acceleration at the start of synchronization, phase synchronization is automatically performed.

 # 7 PHD The direction of movement for automatic phase synchronization is:
0: Positive (+).
1: Negative (-).

	#7	#6	#5	#4	#3	#2	#1	#0
7703						ARO	ARE	ERV

[Input type] Parameter input
[Data type] Bit path

 # 0 ERV During EGB synchronization (G81), feed per revolution is performed for:
0: Feedback pulses.
1: Pulses converted to the speed for the workpiece axis.

 # 1 ARE The retract function executed when an alarm is issued retracts the tool during:
0: EGB synchronization or automatic operation (automatic operation signal OP = 1).
1: EGB synchronization.

 # 2 ARO The retract function executed when an alarm is issued retracts the tool during:
0: EGB synchronization.
1: EGB synchronization and automatic operation (automatic operation signal OP = 1).

NOTE
This parameter is valid when bit 1 (ARE) of parameter No. 7703 is set to 1.

The following table lists the parameter settings and corresponding operation.

ARE	ARO	Operation
1	0	During EGB synchronization
1	1	During EGB synchronization and automatic operation
0	0	During EGB synchronization or automatic operation
0	1	

NOTE
1 Parameters ARE and ARO are valid when bit 3 (ART) of parameter No. 7702 is set to 1 (when the retract function executed when an alarm is issued).
2 This parameter is valid when bit 1 (ARE) of parameter No. 7703 is set to 1.

7710	Axis number of an axis to be synchronized using the method of command specification for a hobbing machine

NOTE
When this parameter is set, the power must be turned off before operation is continued.

[Input type] Parameter input
[Data type] 2-word path
[Valid data range] 1 to Number of controlled axes

When there are several groups of axes to be synchronized (the axes for which bit 0 (SYNMOD) of parameter No. 2011 is set to 1), an axis for which to start synchronization is specified using the following command (for a hobbing machine):

G81 T t L $\pm l$;

t: Spindle speed ($1 \leq t \leq 1000$)

l: Number of synchronized axis rotations ($1 \leq l \leq 21$)

Synchronization between the spindle and a specified axis is established with the ratio of $\pm l$ rotations about the synchronized axis to t spindle rotations.

t and l correspond to the number of teeth and the number of threads on the hobbing machine, respectively.

Above command is issued without setting this parameter when there are several groups of axes to be synchronized, the alarm (PS1593) is issued.

When only one group of axes is to be synchronized, this parameter is ignored.

	#7	#6	#5	#4	#3	#2	#1	#0
7731					ECN			EFX

[Input type] Parameter input
[Data type] Bit path

 # 0 EFX As the EGB command:
0: G80 and G81 are used.
1: G80.4 and G81.4 are used.

NOTE
When this parameter is set to 0, no drilling canned cycle can be used.

 # 3 ECN When the automatic phase synchronization function for the electronic gear box is disabled, during EGB synchronization, the G81 or G81.5 command:
0: Cannot be issued again. (The alarm (PS1595) is issued.)
1: Can be issued again.

7740	Feedrate during retraction

[Input type] Parameter input
[Data type] Real axis
[Unit of data] mm/min, inch/min, degree/min (machine unit)
[Minimum unit of data] Depend on the increment system of the applied axis
[Valid data range] Refer to the standard parameter setting table (C)
(When the increment system is IS-B, 0.0 to +999000.0)
This parameter sets the feedrate during retraction for each axis.

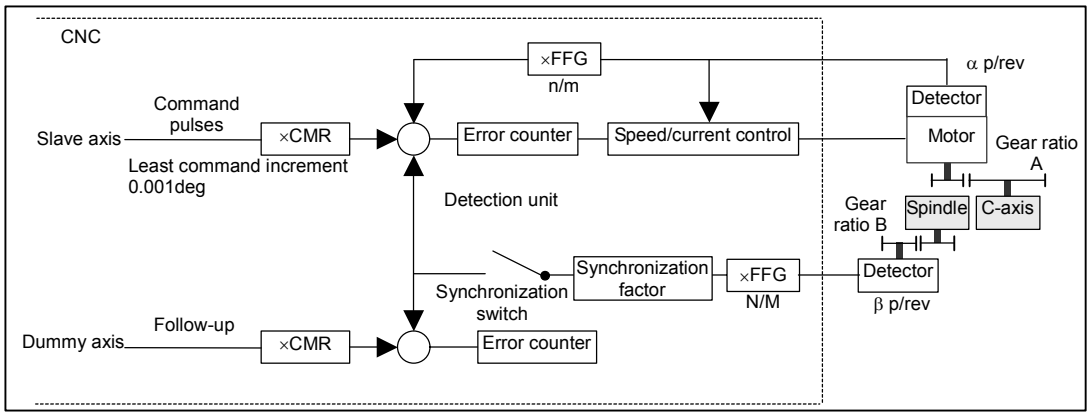
7772	Number of position detector pulses per rotation about the tool axis

[Input type] Parameter input
[Data type] 2-word path
[Valid data range] 1 to 999999999
This parameter sets the number of pulses per rotation about the tool axis (on the spindle side), for the position detector.
For an A/B phase detector, set this parameter with four pulses equaling one A/B phase cycle.

7773	Number of position detector pulses per rotation about the workpiece axis

[Input type] Parameter input
[Data type] 2-word path
[Valid data range] 1 to 999999999
This parameter sets the number of pulses per rotation about the workpiece axis (on the slave side), for the position detector.
Set the number of pulses output by the detection unit.
Set parameters Nos. 7772 and 7773 when using the G81 EGB synchronization command.

[Example 1] When the EGB master axis is the spindle and the EGB slave axis is the C-axis



Gear ratio of the spindle to the detector B:
1/1 (The spindle and detector are directly connected to each other.)
Number of detector pulses per spindle rotation β: 80,000 pulses/rev
(Calculated for four pulses for one A/B phase cycle)

FFG N/M of the EGB dummy axis: 1/1
Gear ratio of the C-axis A: 1/36 (One rotation about the C-axis to 36 motor rotations)
Number of detector pulses per C-axis rotation α: 1,000,000 pulses/rev
C-axis CMR: 1
C-axis FFG n/m: 1/100

In this case, the number of pulses per spindle rotation is:
$80000 \times 1/1 = 80000$
Therefore, set 80000 for parameter No. 7772.
The number of pulses per C-axis rotation in the detection unit is:
$1000000 \div 1/36 \times 1/100 = 360000$
Therefore, set 360000 for parameter No. 7773.

[Example 2] When the gear ratio of the spindle to the detector B is 2/3 for the above example (When the detector rotates twice for three spindle rotations)
In this case, the number of pulses per spindle rotation is:

$$80000 \times \frac{2}{3} = \frac{160000}{3}$$

160000 cannot be divided by 3 without a remainder. In this case, change the setting of parameter No. 7773 so that the ratio of the settings of parameters Nos. 7772 and 7773 indicates the value you want to set.

$$\frac{\text{No.7772}}{\text{No.7773}} = \frac{160000 \diagup 3}{360000} = \frac{160000}{360000 \times 3} = \frac{160000}{1080000}$$

Therefore, set 160000 for parameter No. 7772 and 1080000 for parameter No. 7773.
As described above, all the settings of parameters Nos. 7772 and 7773 have to do is to indicate the ratio correctly. So, you can reduce the fraction indicated by the settings. For example, you may set 16 for parameter No. 7772 and 108 for parameter No. 7773 for this case.

7776 **Feedrate during automatic phase synchronization for the workpiece axis**

[Input type] Parameter input
[Data type] Real path
[Unit of data] deg/min
[Minimum unit of data] Depend on the increment system of the applied axis
[Valid data range] Refer to the standard parameter setting table (C)
(When the increment system is IS-B, 0.0 to +999000.0)
This parameter sets the feedrate during automatic phase synchronization for the workpiece axis.
When this parameter is set to 0, the rapid traverse rate (parameter No. 1420) is used as the feedrate during automatic phase synchronization.

7777	Angle shifted from the spindle position (one-rotation signal position) which the workpiece axis uses as the reference of phase synchronization

[Input type] Parameter input
[Data type] Real path
[Unit of data] deg
[Minimum unit of data] Depend on the increment system of the applied axis
[Valid data range] 9 digit of minimum unit of data (refer to standard parameter setting table (A))
(When the increment system is IS-B, -999999.999 to +999999.999)
This parameter sets the angle shifted from the spindle position (one-rotation signal position) which the workpiece axis uses as the reference of phase synchronization.

7778	Acceleration for acceleration/deceleration for the workpiece axis

[Input type] Parameter input
[Data type] Real axis
[Unit of data] deg/sec^2
[Minimum unit of data] Depend on the increment system of the applied axis
[Valid data range] Refer to the standard parameter setting table (D)
(For a millimeter machine, 0.0 to +100000.0, for an inch machine, 0.0 to +10000.0)
This parameter sets an acceleration for acceleration/deceleration for the workpiece axis.

NOTE
1 In the FS16*i* series, acceleration/deceleration for automatic phase matching is set by specifying a feedrate and a time constant in parameter Nos. 2135 and 2136 separately; in the FS30i series, acceleration/deceleration is directly set in parameter No. 7778.
2 If this parameter is set to 0, specifying G81 results in an alarm (PS1598).

7782	Number of pulses from the position detector per EGB master axis rotation

[Input type] Parameter input
[Data type] 2-word axis
[Valid data range] 1 to 999999999
For a slave axis, set the number of pulses generated from the position detector per EGB master axis rotation.
For an A/B phase detector, set this parameter with four pulses equaling one A/B phase cycle.

7783	Number of pulses from the position detector per EGB slave axis rotation

[Input type] Parameter input
[Data type] 2-word axis
[Unit of data] Detection unit
[Valid data range] 1 to 999999999

For a slave axis, set the number of pulses generated from the position detector per EGB slave axis rotation.
Set the number of pulses output by the detection unit.
Set this parameter when using the G81.5 EGB synchronization command.
The method for setting parameters Nos. 7782 and 7783 is the same as for parameters Nos. 7772 and 7773. For the method, see the description of parameters Nos. 7772 and 7773.

The ratio of the number of pulses for the master slave to that of pulses for the slave axis may be valid, but the settings of the parameters may not indicate the actual number of pulses. For example, the number of pulses may not be able to be divided without a remainder for the reason of the master and slave axis gear ratios as described in example 2. In this case, the following methods cannot be used for the G81.5 command:

G81.5 T_ C_ ; When the speed is specified for the master axis and the travel distance is specified for the slave axis
G81.5 P_ C0 L_ ; When the number of pulses is specified for the master axis and the speed is specified for the slave axis

	#7	#6	#5	#4	#3	#2	#1	#0
8001					RDE	OVE		MLE

[Input type] Parameter input
[Data type] Bit path

 # 0 MLE Whether all axis machine lock signal MLK is valid for PMC-controlled axes
0: Valid
1: Invalid
The axis-by-axis machine lock signal MLKx depends on the setting of bit 1 of parameter No. 8006.

 # 2 OVE Signals related to dry run and override used in PMC axis control
0: Same signals as those used for the CNC
1: Signals specific to the PMC
The signals used depend on the settings of these parameter bits as indicated below.

Signals	No.8001#2=0 (same signals as those used for the CNC)		No.8001#2=1 (signals specific to the PMC)	
Feedrate override signals	*FV0 to *FV7	G012	*EFV0 to *EFV7	G151
Override cancellation signal	OVC	G006.	EOVC	G150.5
Rapid traverse override signals	ROV1,2	G014.0, .1	EROV1,2	G150.0, .1
Dry run signal	DRN	G46.7	EDRN	G150.7
Rapid traverse selection signal	RT	G19.7	ERT	G150.6

(The signal addresses at PMC selection time are for the first group.)

 # 3 RDE Whether dry run is valid for rapid traverse in PMC axis control
0: Invalid
1: Valid

	#7	#6	#5	#4	#3	#2	#1	#0
8002	FR2	FR1	PF2	PF1	F10			RPD

[Input type] Parameter input
[Data type] Bit path

 # 0 RPD Rapid traverse rate for PMC-controlled axes
0: Feedrate specified with parameter No.1420
1: Feedrate specified with the feedrate data in an axis control command by PMC

 # 3 F10 Least increment for the feedrate for cutting feed (per minute) in PMC axis control
The following settings are applied when bit 4 (PF1) of parameter No. 8002 is set to 0 and bit 5 (PF2) of parameter No. 8002 is set to 0.

	F10	IS-A	IS-B	IS-C	IS-D	IS-E
Millimeter input (mm/min)	0	10	1	0.1	0.01	0.001
	1	100	10	1	0.1	0.01
Inch input (inch/min)	0	0.1	0.01	0.001	0.0001	0.00001
	1	1	0.1	0.01	0.001	0.0001

 # 4 PF1
 # 5 PF2 Set the feedrate unit of cutting feedrate (feed per minute) for an axis controlled by the PMC.

Bit 5 (PF2) of parameter No. 8002	Bit 4 (PF1) of parameter No. 8002	Feedrate unit
0	0	1 / 1
0	1	1 / 10
1	0	1 / 100
1	1	1 / 1000

 # 6 FR1
 # 7 FR2 Set the feedrate unit for cutting feedrate (feed per rotation) for an axis controlled by the PMC.

Bit 7 (FR2) of parameter No. 8002	Bit 6 (FR1) of parameter No. 8002	Millimeter input (mm/rev)	Inch input (inch/rev)
0	0	0.0001	0.000001
1	1		
0	1	0.001	0.00001
1	0	0.01	0.0001

	#7	#6	#5	#4	#3	#2	#1	#0
8004		NCI						

[Input type] Parameter input
[Data type] Bit path

 # 6 NCI In axis control by the PMC, a position check at the time of deceleration is:
0: Performed.
1: Not performed.

	#7	#6	#5	#4	#3	#2	#1	#0
8005								EDC

[Input type] Setting input
[Data type] Bit path

 # 0 EDC In axis control by the PMC, an external deceleration function is:
0: Disabled.
1: Enabled.

	#7	#6	#5	#4	#3	#2	#1	#0
8006		EZR		EFD				

[Input type] Parameter input
[Data type] Bit path

 # 4 EFD When cutting feed (feed per minute) is used in PMC axis control, the specification unit of feedrate data is:
0: Unchanged (1 times).
1: 100 times greater.

NOTE
When this parameter is set to 1, bit 3 (F10) of parameter No. 8002 is invalid.

 # 6 EZR In PMC axis control, bit 0 (ZRNx) of parameter No. 1005 is:

0: Invalid.
With a PMC controlled axis, the alarm (PS0224) is not issued.

1: Valid.
A reference position return state check is made on a PMC controlled axis as with an NC axis according to the setting of bit 0 (ZRNx) of parameter No. 1005.

	#7	#6	#5	#4	#3	#2	#1	#0
8008								EMRx

[Input type] Parameter input
[Data type] Bit axis

 # 0 EMRx When a PMC axis control command is issued in mirror image state, the mirror image is:

0: Not considered.
1: Considered.

This parameter is valid in the mirror image mode set with the mirror image signals MI1 to MI8 <G106.0 to .7> set to 1 or bit 0 (MIRx) of parameter No. 12 set to 1.
If a movement is made along the same axis by doubly specifying a command with the CNC and PMC axis control when this parameter is set to 0, and the mirror image mode is set, a coordinate shift can occur afterwards. So, do not attempt to make such a movement.

8010	Selection of the DI/DO group for each axis controlled by the PMC

[Input type] Parameter input
[Data type] Byte axis
[Valid data range] 1 to 40
Specify the DI/DO group to be used to specify a command for each PMC-controlled axis.

For addresses of the fifth group and up, 1000 is added in steps of 4 groups.
For example:
The start address of the 10th group is G2154.
The start address of the 25th axis is G6142.

P8010	Description
1	DI/DO 1st group (G142 to G153) is used.
2	DI/DO 2nd group (G154 to G165) is used.
3	DI/DO 3rd group (G166 to G177) is used.
4	DI/DO 4th group (G178 to G189) is used.
5	DI/DO 5th group (G1142 to G1153) is used.
6	DI/DO 6th group (G1154 to G1165) is used.
:	:
13	DI/DO 13th group (G3142 to G3153) is used.
:	:
20	DI/DO 20th group (G4178 to G4189) is used.
21	DI/DO 21st group (G5142toG5153) is used.
:	:

P8010	Description
29	DI/DO 29th group (G7142toG7153) is used.
:	:
35	DI/DO 35th group (G8166toG8177) is used.
36	DI/DO 36th group (G8178toG8189) is used.
37	DI/DO 37th group (G9142toG9153) is used.
38	DI/DO 38th group (G9154toG9165) is used.
39	DI/DO 39th group (G9166toG9177) is used.
40	DI/DO 40th group (G9178toG9189) is used.

NOTE
When a value other than the above is set, the axis is not controlled by the PMC.

	#7	#6	#5	#4	#3	#2	#1	#0
8011								XRT

[Input type] Parameter input
[Data type] Bit axis

 # 0 XRT The axis that uses the group specified by parameter No. 8010 is:
0: Not controlled by the real time custom macro.
1: Controlled by the real time custom macro.

NOTE
1 This parameter is invalid for the axis for which 0 or a value outside the range is set by parameter No. 8010
2 When multiple axes are assigned to the same group by parameter No. 8010, these axes cannot be controlled by the real time custom macro. When multiple axes are assigned to the same group, be sure to set this bit to 0.
3 When this parameter (No. 8011) is all 0s, the axis is used for PMC axis control.

8030	Time constant for exponential acceleration/deceleration in cutting feed or continuous feed under PMC axis control

[Input type] Parameter input
[Data type] 2-word axis
[Unit of data] msec
[Valid data range] 0 to 4000
For each axis, this parameter sets a time constant for exponential acceleration/deceleration in cutting feed or continuous feed under PMC axis control.

NOTE
When 0 is set in this parameter, the value set in parameter No. 1622 is used.
The value set in parameter No. 1622 is used also for linear acceleration/deceleration after cutting interpolation.

	#7	#6	#5	#4	#3	#2	#1	#0
8103							MWP	MWT

[Input type] Parameter input
[Data type] Bit

NOTE
When at least one of these parameters is set, the power must be turned off before operation is continued.

 # 0 MWT As the signal interface for the waiting M code:
0: The path individual signal interface is used.
1: The path common signal interface is used.
This parameter can be selected only when 2-path control is used.

 # 1 MWP To specify a P command for the waiting M code/balance cut:
0: A binary value is used as conventionally done.
1: A path number combination is used.

8110	Waiting M code range (minimum value)

8111	Waiting M code range (maximum value)

[Input type] Parameter input
[Data type] 2-word
[Valid data range] 0 ,100to99999999
A range of M code values can be set by specifying a minimum waiting M coder value (parameter No. 8110) and a maximum waiting M code value (parameter No. 8111).
(parameter No. 8110) ≤ (waiting M code) ≤ (parameter No. 8111)
Set 0 in these parameters when the waiting M code is not used.

	#7	#6	#5	#4	#3	#2	#1	#0
8162						PKUx		

[Input type] Parameter input
[Data type] Bit axis

 # 2 PKUx In the parking state,
0: The absolute, relative, and machine coordinates are not updated.
1: The absolute and relative coordinates are updated. The machine coordinates are not updated.

NOTE
1 With an axis for which polar coordinate interpolation is specified, set this parameter to 1. If this parameter is set to 0, a coordinate shift can occur when a single block stop or feed hold is performed in the polar coordinate interpolation mode.
2 With an axis that is set to function as a synchronous master axis and synchronous slave axis at the same time (with bit 1 (SYWx) of parameter No. 8167), set this parameter to 1.
3 With an axis specified in the three-dimensional coordinate conversion mode, set this parameter to 1. If this parameter is set to 0, the alarm (PS0367) is issued.

	#7	#6	#5	#4	#3	#2	#1	#0
8163	NUMx							

[Input type] Parameter input
[Data type] Bit axis

 # 7 NUMx When neither synchronous control nor composite control is applied, a move command for the axis is:
0: Not disabled.
1: Disabled.

NOTE
If a move command is specified for an axis with NUMx set to 1 when neither synchronous control nor composite control is applied, alarm PS0353 is issued.

8180 **Master axis with which an axis is synchronized under synchronous control**

[Input type] Parameter input
[Data type] Word axis
[Valid data range] 101, 102, 103, . . . , (path number)*100+(intra-path relative axis number) (101, 102, 103, . . . , 201, 202, 203, . . . , 1001, 1002, 1003, . . .)

This parameter sets the path number and intra-path relative axis number of the master axis with which each axis is synchronized. When zero is specified, the axis does not become a slave axis and is not synchronized with another axis. When an identical number is specified in two or more parameters, one master axis has two or more slave axes.

8183 **Composite control axis of the other path in composite control for each axis**

[Input type] Parameter input
[Data type] Word axis
[Valid data range] 101, 102, 103, . . . , (path number)*100+(intra-path relative axis number) (101, 102, 103, . . . , 201, 202, 203, . . . , 1001, 1002, 1003, . . .)

This parameter sets with which axis each axis is to be placed under composite control. When zero is specified, control of the axis is not replaced under composite control. An identical number can be specified in two or more parameters, but composite control cannot be exercised for all of them at a time.

NOTE
When the two-path interface is used (bit 1 (MIX) of parameter No. 8166 is set to 1), set this parameter for path 2.

8186	Master axis under superimposed control

[Input type] Parameter input
[Data type] Word axis
[Valid data range] 101, 102, 103, . . . , (path number)*100+(intra-path relative axis number) (101, 102, 103, . . . , 201, 202, 203, . . . , 1001, 1002, 1003, . . .)

This parameter sets the path number and intra-path relative axis number of a superimposed master axis for each axis when superimposed control is exercised. When zero is specified, the axis does not become a slave axis under superimposed control and the move pulse of another axis is not superimposed.

An identical number can be specified in two or more parameters to exercise superimposed control simultaneously. This means that superimposed control with one master axis and multiple slave axes is possible.

A slave axis may function as the master axis of another axis to allow three-generation superimposed control: parent (master axis) - child (slave axis/master axis) - grandchild (slave axis).

In this case, a movement along the child is made by its travel distance plus the travel distance of the parent, and a movement along the grandchild is made by its travel distance plus the travel distance of the child plus the travel distance of the parent.

Example of the relationship of parent (X1 of path 1) - child (X2 of path 2) - grandchild (X3 of path 3):
The travel distance of X1 is superimposed on X2, and the travel distances of X1 and X2 are further superimposed on X3.

Parameter No. 8186x of path 2 = 101
Parameter No. 8186x of path 3 = 201

	#7	#6	#5	#4	#3	#2	#1	#0
8200						AZR		AAC

[Input type] Parameter input
[Data type] Bit path

NOTE
When at least one of these parameters is set, the power must be turned off before operation is continued.

 # 0 AAC

0: Does not perform angular axis control.
1: Performs inclined axis control.

 # 2 AZR

0: The machine tool is moved along the Cartesian axis during manual reference position return along the slanted axis under angular axis control.
1: The machine tool is not moved along the Cartesian axis during manual reference position return along the slanted axis under angular axis control.

	#7	#6	#5	#4	#3	#2	#1	#0
8201		A53				AO3	AO2	AOT

[Input type] Parameter input
[Data type] Bit path

NOTE
When at least one of these parameters is set, the power must be turned off before operation is continued.

 # 0 AOT Stored stroke limit 1 under angular axis control is handled as:
0: Value in the slanted coordinate system.
1: Value in the Cartesian coordinate system.

 # 1 AO2 Stored stroke limit 2 under angular axis control is handled as:
0: Value in the slanted coordinate system.
1: Value in the Cartesian coordinate system.

 # 2 AO3 Stored stroke limit 3 under angular axis control is handled as:
0: Value in the slanted coordinate system.
1: Value in the Cartesian coordinate system.

 # 6 A53 So far, if a slanted axis is singly specified by a machine coordinate command (G53) in angular axis control, this parameter set to 0 specifies that "compensation is applied to the Cartesian axis", and this parameter set to 1 specifies that "a movement is made along the slanted axis only". However, the specification has been changed so that "a movement is made along the slanted axis only", regardless of whether this parameter is set to 0 or 1.

8210	Slant angle of a slanted axis in angular axis control

[Input type] Parameter input
[Data type] Real path
[Unit of data] Degree
[Minimum unit of data] Depend on the increment system of the applied axis
[Valid data range] -180.000 to 180.000. However, angular axis control is disabled in the ranges -95.000 to -85.000 and 85.000 to 95.000 (in the case of IS-B).

8211	Axis number of a slanted axis subject to angular axis control

8212	Axis number of a Cartesian axis subject to slanted axis control

NOTE
When these parameters are set, the power must be turned off before operation is continued.

[Input type] Parameter input
[Data type] Word path
[Valid data range] 1 to number of controlled axes

When angular axis control is to be applied to an arbitrary axis, these parameters set the axis numbers of a slanted axis and Cartesian axis. If 0 is set in either of the two parameters, the same number is set in the two parameters, or a number other than the controlled axis numbers is set in either of the two parameters, a slanted axis and Cartesian axis are selected as indicated in the following table:

	Slanted axis	Cartesian axis
M series	Y-axis (axis with 2 set in parameter No. 1022) of the basic three axes	Z-axis (axis with 3 set in parameter No. 1022) of the basic three axes
T series	X-axis (axis with 1 set in parameter No. 1022) of the basic three axes	Z-axis (axis with 3 set in parameter No. 1022) of the basic three axes

	#7	#6	#5	#4	#3	#2	#1	#0
8301				SYA				

[Input type] Parameter input
[Data type] Bit path

 # 4 SYA In the servo-off state in axis synchronous control, the limit of the difference between the positioning deviation of the master axis and that of the slave axis is:
0: Checked.
1: Not checked.

	#7	#6	#5	#4	#3	#2	#1	#0
8302	SMA							

[Input type] Parameter input
[Data type] Bit path

NOTE
When this parameter is set, the power must be turned off before operation is continued.

 # 7 SMA When an absolute position detector is attached, and bit 4 (APZ) of parameter No. 1815 for an axis in synchronous operation is set to OFF, APZ of the pairing axis in synchronous operation is:
0: Not set to OFF.
1: Set to OFF.

	#7	#6	#5	#4	#3	#2	#1	#0
8303	SOF					SAF	ATS	ATE

[Input type] Parameter input
[Data type] Bit axis

NOTE
When at least one of these parameters is set, the power must be turned off before operation is continued.

 # 0 ATE In axis synchronous control, automatic setting for grid positioning is:
0: Disabled
1: Enabled
Set this parameter with a slave axis.

 # 1 ATS In axis synchronous control, automatic setting for grid positioning is:
0: Not started
1: Started
Set this parameter with a slave axis.

NOTE
When starting automatic setting for grid positioning, set ATS to 1. Upon the completion of setting, ATS is automatically set to 0.

 # 2 SAF In axis synchronous control, a movement along a slave axis is:
0: Not added to actual feedrate display.
1: Added to actual feedrate display.
Set this parameter with a slave axis.

 # 7 SOF In axis synchronous control, the synchronization establishment function based on machine coordinates is:
0: Disabled.
1: Enabled.
Set this parameter with a slave axis.
When using synchronization error compensation, set this parameter to 0.

	#7	#6	#5	#4	#3	#2	#1	#0
8304	SYE	SMS	SCA	MVB	CLP	ADJ		SSA

[Input type] Parameter input
[Data type] Bit axis

 # 0 SSA When the one-direction synchronization establishment function under axis synchronous control is used:
0: The axis with a larger machine coordinate is used as the reference.
1: The axis with a smaller machine coordinate is used as the reference.

NOTE
1 When at least one of these parameters is set, the power must be turned off before operation is continued.
2 Set this parameter (SSA) to the same value for both the master and slave axes.

 # 2 ADJ In axis synchronous control, this parameter specifies an axis along which a movement is made in the modification mode.
0: A movement is not made in the modification mode along the axis.
1: A movement is made in the modification mode along the axis.
When this parameter is set to 1, the modification mode is set.
Along an axis with this parameter set to 1, a movement is made by a move command for the master axis.
Set this parameter for one of the master and slave axes.
When there are multiple slave axes for one master axis, set this parameter to 1 for an axis with which a synchronization error excessive alarm is issued for recovery. If an alarm is issued with multiple axes, modify this parameter after recovery of one axis to recover another axis.

 # 3 CLP In axis feed synchronous control, synchronization error compensation is:
0: Disabled.
1: Enabled.
Set this parameter with a slave axis.

 # 4 MVB In the modification mode, a move command in a direction that increases a synchronization error is:
0: Ignored.
1: Valid.
When there are multiple slave axes for one master axis, an attempt to reduce the synchronous error of a slave axis by a movement along the master axis can increase the synchronization error of another slave axis. If this parameter is set to 0 in such a case, a movement can be made in neither direction along the master axis. In this case, set bit 2 (ADJ) of parameter No. 8304 to make a movement along a slave axis to perform a corrective operation.

 # 5 SCA In axis synchronous control:
0: Synchronous operation is performed when the axis synchronous control manual feed selection signal SYNCJ or the axis synchronous control selection signal SYNC for slave axes is set to 1.
1: Synchronous operation is performed at all times.
Set this parameter with a slave axis.

 # 6 SMS The synchronization error smooth suppress function is:
0: Disabled.
1: Enabled.
Set this parameter with a slave axis.

 # 7 SYE When external machine coordinate system shift is specified by external data input/output for the master axis in synchronous control, the slave axis is:
0: Not shifted.
1: Shifted by the same amount as specified for the master axis.
Set this parameter for the slave axis.
This function is disabled during normal operation.

	#7	#6	#5	#4	#3	#2	#1	#0
8305							SSE	SSO

[Input type] Parameter input
[Data type] Bit path

 # 0 SSO The uni-directional synchronization function in axis synchronous control is:
0: Disabled.
1: Enabled.

 # 1 SSE After emergency stop, the uni-directional synchronization function in axis synchronous control is:
0: Disabled.
1: Enabled.

8311	Axis number of master axis in axis synchronous control

NOTE
When this parameter is set, the power must be turned off before operation is continued.

[Input type] Parameter input
[Data type] Byte axis
[Valid data range] 0 to Number of controlled axes

Select a master axis in axis synchronous control. In the parameter for the slave axis, set the axis number of the master axis.

Example 1)

When one set of axis synchronous control is used:
When the master axis is the first axis (X-axis), and the slave axis is the third axis (Z-axis), set parameter No. 8311 as follows:
Parameter No.8311 X (first axis) = 0
Parameter No.8311 Y (second axis) = 0
Parameter No.8311 Z (third axis) = 1
Parameter No.8311 A (fourth axis) = 0

Example 2)

When three sets of axis synchronous control is used:
When the master axes are the first axis, second axis, and third axis, and the slave axes are the sixth axis, fifth axis, and fourth axis, set parameter No. 8311 as follows:
Parameter No.8311 X (first axis) = 0
Parameter No.8311 Y (second axis) = 0
Parameter No.8311 Z (third axis) = 0
Parameter No.8311 A (fourth axis) = 3
Parameter No.8311 B (fifth axis) = 2
Parameter No.8311 C (sixth axis) = 1

8312	Enabling/disabling mirror image in axis synchronous control

[Input type] Parameter input
[Data type] Word axis
[Valid data range] -127 to 128

This parameter sets mirror image for the slave axis. When 100 or a more value is set with this parameter, the mirror image function is applied to synchronous control. Set this parameter to the slave axis.

Example)

For reverse synchronization with the master axis being the third axis and the slave axis being the fourth axis, set parameter No. 8312 as follows:

Parameter No.8312 X (first axis) = 0
Parameter No.8312 Y (second axis) = 0
Parameter No.8312 Z (third axis) = 0
Parameter No.8312 A (fourth axis) = 100

NOTE
In synchronous operation with mirror image applied, synchronization error compensation, synchronization establishment, synchronization error checking, and modification mode cannot be used.

8314	Maximum allowable error in synchronization error check based on machine coordinates

[Input type] Parameter input
[Data type] Real axis
[Unit of data] mm, inch, degree (machine unit)
[Minimum unit of data] Depend on the increment system of the applied axis
[Valid data range] 0 or positive 9 digit of minimum unit of data (refer to the standard parameter setting table (B))
(When the increment system is IS-B, 0.0 to +999999.999)

This parameter sets a maximum allowable error in a synchronization error check based on machine coordinates. When the error between the master and slave axes in machine coordinates exceeds the value set in this parameter, the machine stops with the servo alarm (SV0005).
Set this parameter with a slave axis.

NOTE
Set 0 in this parameter when a synchronization error check is not made.

8323 **Limit in positional deviation check in axis synchronous control**

[Input type] Parameter input
[Data type] 2-word axis
[Unit of data] Detection unit
[Valid data range] 0 to 999999999

This parameter sets the maximum allowable difference between the master axis and slave axis position deviations. When the absolute value of a positional deviation difference exceeds the value set in this parameter in axis synchronous control, the alarm (DS0001) is issued.
Set this parameter with a slave axis. If 0 is specified in this parameter, no position deviation difference check is made.

8325 **Maximum compensation value in synchronization establishment based on machine coordinates**

[Input type] Parameter input
[Data type] Real axis
[Unit of data] mm, inch, degree (machine unit)
[Minimum unit of data] Depend on the increment system of the applied axis
[Valid data range] 0 or positive 9 digit of minimum unit of data (refer to the standard parameter setting table (B))
(When the increment system is IS-B, 0.0 to +999999.999)

This parameter sets the maximum compensation value for synchronization. When a compensation value exceeding the value set in this parameter is detected, the servo alarm (SV0001) is issued, and the synchronization establishment is not performed.
Specify a slave axis for this parameter. To enable this parameter, set the parameter SOF (bit 7 of parameter No.8303) to 1. When 0 is set in this parameter, synchronization establishment is not performed.

8326 **Difference between master axis and slave axis reference counters**

[Input type] Parameter input
[Data type] 2-word axis
[Unit of data] Detection unit
[Valid data range] 0 to 999999999

The difference between the master axis reference counter and slave axis reference counter (master axis and slave axis grid shift) is automatically set when automatic setting for grid positioning is performed. Then, the difference is transferred together with an ordinary grid shift value to the servo system when the power is turned on. This parameter is set with a slave axis.

8327	Torque difference alarm detection timer

[Input type] Parameter input
[Data type] 2-word axis
[Unit of data] msec
[Valid data range] 0 to 4000

This parameter sets a time from the servo preparation completion signal, SA <F000.6>, being set to 1 until torque difference alarm detection is started in axis synchronous control.
When 0 is set in this parameter, the specification of 512 msec is assumed.
Set this parameter with a slave axis.

8330	Multiplier for a maximum allowable synchronization error immediately after power-up

NOTE
When this parameter is set, the power must be turned off before operation is continued.

[Input type] Parameter input
[Data type] Word path
[Valid data range] 1 to 100

Until synchronization establishment is completed immediately after power-up, synchronization error excessive alarm 2 is checked using the maximum allowable error (parameter No. 8332) multiplied by the value set in this parameter.
If the result produced by multiplying the value of parameter No. 8332 by the value of this parameter exceeds 32767, the value is clamped to 32767.

8331	Maximum allowable synchronization error for synchronization error excessive alarm 1

[Input type] Parameter input
[Data type] 2-word axis
[Unit of data] Detection unit
[Valid data range] 1 to 32767

This parameter sets a maximum allowable synchronization error for synchronization error excessive alarm 1.
Set this parameter with a slave axis.

8332	Maximum allowable synchronization error for synchronization error excessive alarm 2

NOTE
When this parameter is set, the power must be turned off before operation is continued.

[Input type] Parameter input
[Data type] 2-word axis
[Unit of data] Detection unit
[Valid data range] 1 to 32767
This parameter sets a maximum allowable synchronization error for synchronization error excessive alarm 2.
Set this parameter with a slave axis.

8333	Synchronization error zero width for each axis

[Input type] Parameter input
[Data type] 2-word axis
[Unit of data] Detection unit
[Valid data range] 1 to 32767
When a synchronization error below the value set in this parameter is detected, synchronization error compensation is not performed.
Set this parameter with a slave axis.

8334	Synchronization error compensation gain for each axis

[Input type] Parameter input
[Data type] Word axis
[Valid data range] 1 to 1024
This parameter sets a synchronization error compensation gain.
Compensation pulses found by the following expression are output for the slave axis:

Compensation pulses = Synchronization error × (Ci/1024)

Ci: Compensation gain

Set this parameter with a slave axis.

8335	Synchronization error zero width 2 for each axis

[Input type] Parameter input
[Data type] 2-word axis
[Unit of data] Detection unit
[Valid data range] 0 to 32767
This parameter sets synchronization error zero width 2 for synchronization error smooth suppression.
Set this parameter with a slave axis.

NOTE
Set a value less than the value set in parameter No. 8333.

8336	Synchronization error compensation gain 2 for each axis

[Input type] Parameter input
[Data type] Word axis
[Valid data range] 0 to 1024
This parameter sets synchronization error compensation gain 2 for synchronization error smooth suppression.
Set this parameter with a slave axis.

NOTE
Set a value less than the value set in parameter No. 8334.

8337	M code for turning off synchronization in axis synchronous control

[Input type] Parameter input
[Data type] 2-word path
[Valid data range] 1 to 999999999
This parameter specifies an M code for switching from synchronous operation to normal operation.
The M code set in this parameter is not buffered.

8338	M code for turning on synchronization in axis synchronous control

[Input type] Parameter input
[Data type] 2-word path
[Valid data range] 1 to 999999999
This parameter specifies an M code for switching from normal operation to synchronous operation.
The M code set in this parameter is not buffered.

	#7	#6	#5	#4	#3	#2	#1	#0
8451	NOF			ZAG				

[Input type] Setting input
[Data type] Bit path

 # 4 ZAG The deceleration function based on cutting load in AI contour control (deceleration based on Z-axis fall angle) is:
0: Not performed.
1: Performed.
When this parameter is set to 1, be sure to set parameter Nos. 8456, 8457, and 8458.

 # 7 NOF In AI contour control, an F command is:
0: Not ignored.
1: Ignored.
When this parameter is set to 1, the specification of the maximum allowable feedrate set in parameter No. 8465 is assumed.

8456	Override for range 2 that is applied during deceleration according to the cutting load in AI contour control
8457	Override for range 3 that is applied during deceleration according to the cutting load in AI contour control
8458	Override for range 4 that is applied during deceleration according to the cutting load in AI contour control

[Input type] Setting input
[Data type] Word path
[Unit of data] %
[Valid data range] 1 to 100

For the function of decelerating according to the cutting load in AI contour control, the override set in a parameter can be applied according to the angle at which the tool moves downward along the Z-axis. The feedrate obtained according to other conditions is multiplied by the override for the range containing angle θ at which the tool moves downward.

However, when bit 1 (ZG2) of parameter No. 19515 is set to 0, no parameter is available to range 1, and 100% is applied at all times. When bit 1 (ZG2) of parameter No. 19515 is set to 1, set an override value for range 1 in parameter No. 19516.

Range 1 $0° \leq \theta < 30°$
Range 2 $30° \leq \theta < 45°$
Range 3 $45° \leq \theta < 60°$
Range 4 $60° \leq \theta \leq 90°$

8465	Maximum allowable feedrate for AI contour control

[Input type] Setting input
[Data type] Real path
[Unit of data] mm/min, inch/min, degree/min (input unit)
[Minimum unit of data] Depend on the increment system of the reference axis
[Valid data range] Refer to the standard parameter setting table (C)
(When the increment system is IS-B, 0.0 to +999000.0)

This parameter sets the maximum allowable feedrate for contour control.

If a feedrate higher than the setting of this parameter is specified in the AI contour control mode, the feedrate is clamped to that set in this parameter.

If this parameter is set to 0, no clamping is performed.

When bit 7 (NOF) of parameter No. 8451 is set to 1, the tool moves, assuming that the feedrate set in this parameter is specified. If 0 is set in this parameter at this time, a movement is made at the specified feedrate.

8486	Maximum travel distance of a block where smooth interpolation or Nano smoothing is applied

[Input type] Setting input
[Data type] Real path
[Unit of data] mm, inch (input unit)
[Minimum unit of data] Depend on the increment system of the reference axis
[Valid data range] 9 digit of minimum unit of data (refer to standard parameter setting table (A))
(When the increment system is IS-B, -999999.999 to +999999.999)
This parameter specifies a block length used as a reference to decide whether to apply smooth interpolation or Nano smoothing. If the line specified in a block is longer than the value set in the parameter, smooth interpolation or Nano smoothing is not applied to that block.

8487	Angle at which smooth interpolation or Nano smoothing is turned off

[Input type] Setting input
[Data type] Real path
[Unit of data] Degree
[Minimum unit of data] Depend on the increment system of the reference axis
[Valid data range] 0 to 90
This parameter sets the angle used to determine whether to apply smooth interpolation or Nano smoothing.
At a point having a difference in angle greater than this setting, smooth interpolation or Nano smoothing is turned off.

8490	Minimum travel distance of a block where smooth interpolation or Nano smoothing is applied

[Input type] Setting input
[Data type] Real path
[Unit of data] mm, inch (input unit)
[Minimum unit of data] Depend on the increment system of the reference axis
[Valid data range] 9 digit of minimum unit of data (refer to standard parameter setting table (A))
(When the increment system is IS-B, -999999.999 to +999999.999)
This parameter sets a block length used to determine whether to apply smooth interpolation or Nano smoothing.
If the line specified in a block is shorter than the value set in this parameter, smooth interpolation or Nano smoothing is not applied to that block.

	#7	#6	#5	#4	#3	#2	#1	#0
8900								PWE

[Input type] Setting input
[Data type] Bit

 # 0 PWE The setting, from an external device and MDI panel, of those parameters that cannot be set by setting input is:
0: Disabled.
1: Enabled.

10461	RGB value of color palette 1 for text for color set 3
10462	RGB value of color palette 2 for text for color set 3
:	
10475	RGB value of color palette 15 for text for color set 3

[Input type] Parameter input
[Data type] 2-word
[Valid data range] 0 to 151515
Each of these parameters sets the RGB value of each color palette for text by specifying a 6-digit number as described below.
rrggbb: 6-digit number (rr: red data, gg: green data, bb: blue data)
The valid data range of each color is 0 to 15 (same as the tone levels on the color setting screen). When a number equal to or greater than 16 is specified, the specification of 15 is assumed.
Example)
When the tone level of a color is: red:1 green:2, blue:3, set 10203 in the parameter.

10800	First compensation axis for three-dimensional error compensation
10801	Second compensation axis for three-dimensional error compensation
10802	Third compensation axis for three-dimensional error compensation

NOTE
When these parameters are set, the power must be turned off before operation is continued.

[Input type] Parameter input
[Data type] Byte path
[Valid data range] 1 to Number of controlled axes
These parameters set three compensation axes for applying three-dimensional error compensation.

10803	Number of compensation points for three-dimensional error compensation (first compensation axis)
10804	Number of compensation points for three-dimensional error compensation (second compensation axis)
10805	Number of compensation points for three-dimensional error compensation (third compensation axis)

NOTE
When these parameters are set, the power must be turned off before operation is continued.

[Input type] Parameter input
[Data type] Byte path
[Valid data range] 2 to 25
These parameters set the number of compensation points for each axis for three-dimensional error compensation.

10806	Compensation point number of the reference position for three-dimensional error compensation (first compensation axis)
10807	Compensation point number of the reference position for three-dimensional error compensation (second compensation axis)
10808	Compensation point number of the reference position for three-dimensional error compensation (third compensation axis)

NOTE
When these parameters are set, the power must be turned off before operation is continued.

[Input type] Parameter input
[Data type] Byte path
[Valid data range] 1 to number of compensation points
These parameters set the compensation point number of the reference position for each axis for three-dimensional error compensation.

10809	Magnification for three-dimensional error compensation (first compensation axis)
10810	Magnification for three-dimensional error compensation (second compensation axis)
10811	Magnification for three-dimensional error compensation (third compensation axis)

NOTE
When these parameters are set, the power must be turned off before operation is continued.

[Input type] Parameter input
[Data type] Byte path
[Valid data range] 1 to 100
These parameters set the magnification for each axis for three-dimensional error compensation.

10812	Compensation interval for three-dimensional error compensation (first compensation axis)
10813	Compensation interval for three-dimensional error compensation (second compensation axis)
10814	Compensation interval for three-dimensional error compensation (third compensation axis)

NOTE
When these parameters are set, the power must be turned off before operation is continued.

[Input type] Parameter input
[Data type] Real path
[Unit of data] mm, inch (machine unit)
[Minimum unit of data] Depend on the increment system of the reference axis
[Valid data range] 9 digit of minimum unit of data (refer to standard parameter setting table (A))
(When the increment system is IS-B, -999999.999 to +999999.999)
These parameters set the compensation interval for each axis for three-dimensional error compensation.

	#7	#6	#5	#4	#3	#2	#1	#0
11005								SIC

[Input type] Parameter input
[Data type] Bit

 # 0 SIC Spindle indexing is:
0: Performed based on absolute coordinates.
1: Performed based on machine coordinates.

11090	Path number with which the rotation of each spindle is specified

[Input type] Parameter input
[Data type] Byte spindle
[Valid data range] 0 to 10
When a path is specified for spindle commands, this parameter sets a path number with which the rotation of a spindle can be specified.
0: Spindle commands can be issued from all paths.
1 to 10: Spindle commands can be issued from a set path.

NOTE
1 This parameter is valid when SPSP<Gn536.7> is set to 1.
2 If the setting is illegal, an alarm (PS5305) is issued when a spindle command is issued from any one of the paths.
3 This setting does not apply to spindle commands using the spindle select signals (SWS1 to SWS4<Gn027.0 to .2, Gn026.3>).

	#7	#6	#5	#4	#3	#2	#1	#0
11200								RCM

[Input type] Setting input
[Data type] Bit path

 # 0 RCM When workpiece setting error compensation is performed with a 5-axis machine, tool direction compensation (compensation for a rotation axis) is:
0: Not performed.
1: Performed.

NOTE
When this parameter is set to 0, machining cannot be performed precisely. For 5-axis machines, usually set this parameter to 1.

11201	The number of decimal places of rotation direction errors in workpiece setting error compensation

[Input type] Setting input
[Data type] Byte path
[Valid data range] 0 to 8

This parameter sets the number of decimal places of rotation direction errors in workpiece setting error compensation.

Parameter No. 11201	1	2	3	4
Least input increment (deg)	0.1	0.01	0.001	0.0001
Maximum settable value (deg)	±99,999,999.9	±9999,999.99	±999,999.999	±99,999.9999

Parameter No. 11201	5	6	7	8
Least input increment (deg)	0.00001	0.000001	0.0000001	0.00000001
Maximum settable value (deg)	±9,999.99999	±999.999999	±99.9999999	±9.99999999

Note, however, that a value from 1 to 8 can be specified in this parameter.
If a value not within the specifiable range is specified in this parameter, the least input increment of the reference axis is followed.

Unit system of reference axis	IS-A	IS-B	IS-C	IS-D	IS-E
Least input increment (deg)	0.01	0.001	0.0001	0.00001	0.000001
Maximum settable value (deg)	±999,999.99	±999,999.999	±99,999.9999	±9,999.99999	±999.999999

11202	Override for maximum cutting feedrate during workpiece setting error compensation mode

[Input type] Parameter input
[Data type] Byte path
[Unit of data] %
[Valid data range] 0 to 100

When the parameter RCM(No.11200#0)=1 in order to enable tool direction compensation, during workpiece setting error compensation or during tool center point control with three-dimensional coordinate conversion or with tilted working plane command, the override for maximum cutting feedrate (Parameter No.1432) is set.
This override is done only to rotary axes.
In the case of 0 of this parameter, the override is regarded as 100%.

11203	Override for rapid traverse during workpiece setting error compensation mode

[Input type] Parameter input
[Data type] Byte path
[Unit of data] %
[Valid data range] 0 to 100

When the parameter RCM (No.11200#0)=1 in order to enable tool direction compensation, during workpiece setting error compensation or during tool center point control with three-dimensional coordinate conversion or with tilted working plane command, the override for rapid traverse (Parameter No.1420) is set.
This override is done only to rotary axes.
In the case of 0 of this parameter, the override is regarded as 100%.

11204	Angle to decide singular posture (for Workpiece setting error compensation)

[Input type] Parameter input
[Data type] Real path
[Unit of data] Degree
[Minimum unit of data] Depend on the increment system of the reference axis
[Valid data range] 9 digit of minimum unit of data (refer to standard parameter setting table (A)
(When the increment system is IS-B, -999999.999 to +999999.999)

When Tool direction is compensated(Parameter RCM(No.11200#0)=1), in the case that Tool center point control is active during Workpiece setting error compensation, 3-dimension coordinates system conversion or Tilted working plane command, rotary axes are compensated. Then, compensation may be different when the tool is in singular posture.
When the angle between the tool posture and the singular posture is less than this parameter, the compensation is done regarding tool posture as singular posture.

11220	Minimum distance used for determining a plane when a tilted working plane command with three points is specified

[Input type] Parameter input
[Data type] Real axis
[Unit of data] mm, inch (input unit)
[Minimum unit of data] Depend on the increment system of the applied axis
[Valid data range] 0 or positive 9 digit of minimum unit of data (refer to the standard parameter setting table (B))
(When the increment system is IS-B, 0.0 to +999999.999)

When a tilted working plane command with three points is specified, if the distance (used for determining a place) between a straight line passing two points and the remaining one point is short, the plane is unstable. In this parameter, set the minimum distance used for determining a plane. If the distance is shorter than the value set in this parameter, an alarm (PS5457) is issued.

	#7	#6	#5	#4	#3	#2	#1	#0
11221								MTW

[Input type] Parameter input
[Data type] Bit path

 # 0 MTW Multiple tilted working plane commands are:
0: Not used.
1: Used.

	#7	#6	#5	#4	#3	#2	#1	#0
11222	PDM						CIM	NIM

[Input type] Parameter input
[Data type] Bit path

 # 0 NIM Automatic conversion of a coordinate system by an inch/metric conversion command (G20 or G21) is:
0: Not performed.
1: Performed.

 # 1 CIM When an inch/metric conversion command (G20 or G21) is specified, if the workpiece coordinate system is shifted by the shift amount as described below:
0: An alarm (PS1298) is issued.
1: Clearing is performed.
If bit 0 (NIM) of parameter No. 11222 is set to 1, or if bit 2 (IRF) of parameter No. 14000 is set to 1, this parameter clears the following:
- Manual intervention made when the manual absolute signal is off
- Issuance of a move command with the machine locked
- Movement by handle interrupt
- Operation with a mirror image
- Shifting of a workpiece coordinate system when a local coordinate system or workpiece coordinate system is set up

 # 7 PDM When switching between diameter and radius specification is made with the function for dynamic switching of diameter/radius specification, coordinates in the machine coordinate system select command (G53) are:
0: Switched between diameter and radius specification.
1: Set according to the setting of bit 3 (DIAx) of parameter No. 1006.

	#7	#6	#5	#4	#3	#2	#1	#0
11240								FAE

[Input type] Parameter input
[Data type] Bit path

 # 0 FAE During positioning when the AI contour control mode is canceled, the optimum torque acceleration/deceleration is:
0: Disabled.
1: Enabled.

11261	The amount of a retract operation in the tool axis direction during tool retract and return

[Input type] Setting input
[Data type] Real axis
[Unit of data] mm, inch, degree (input unit)
[Minimum unit of data] Depend on the increment system of the reference axis
[Valid data range] 9 digit of minimum unit of data (refer to standard parameter setting table (A))
(When the increment system is IS-B, -999999.999 to +999999.999)
This parameter sets the amount of a retract operation in the tool axis direction when G10.6 is specified alone during tool retraction and return (during tool center point control and workpiece setting error compensation). The retract operation is performed by using the value set in this parameter in an incremental manner. This data is valid only when bit 2 (RPS) of parameter No. 7040 is set to 1. If 0 is set in this parameter, a retract operation in the tool axis direction is not performed, but a retract operation is performed according to the setting of parameter No. 7041.

	#7	#6	#5	#4	#3	#2	#1	#0
11304							GGD	

[Input type] Parameter input
[Data type] Bit

NOTE
When this parameter is set, the power must be turned off before operation is continued..

 # 1 GGD The G code guidance screen is:
0: Not displayed.
1: Displayed.

11305	Maximum number of simultaneously displayed axes

NOTE
When this parameter is set, the power must be turned off before operation is continued.

[Input type] Parameter input
[Data type] Byte path
[Valid data range] 0 to 2
By setting this parameter, the maximum number of simultaneously displayed axes at the current position can be changed. A value set in this parameter corresponds to the maximum number of simultaneously displayed axis as follows:

Max. number of simultaneously displayed axes	5	10	20
Setting	0	1	2

A value other than 1 and 2 is assumed to be 0.

11307	Display sequence of the coordinates in current position display

NOTE
When this parameter is set, the power must be turned off before operation is continued.

[Input type] Parameter input
[Data type] Byte path
[Valid data range] 0 to 5

This parameter sets the display sequence of the coordinates of a position displayed on the following screens:

10.4- and 15-inch display units
- Total position display screen
- Total position display on each screen

7.2- and 8.4-inch display units
- Total position display screen

When the maximum number of simultaneously displayed axes is set to 20 (when 2 is set in parameter No. 11305), two sets of coordinates are displayed simultaneously as the current position display on each screen.

When the first set is displayed, switching to the second set can be made by pressing

then pressing the chapter selection soft key being selected.

When the above operation is performed again, the displayed set changes to the first set.
The display sequence of coordinates corresponds to the parameter setting as follows:

Setting \ Display sequence of coordinates	1	2	3	4
0	Relative coordinates	Absolute coordinates	Machine coordinates	Remaining travel distance
1	Relative coordinates	Machine coordinates	Absolute coordinates	Remaining travel distance
2	Relative coordinates	Remaining travel distance	Absolute coordinates	Machine coordinates
3	Absolute coordinates	Machine coordinates	Relative coordinates	Remaining travel distance
4	Absolute coordinates	Remaining travel distance	Relative coordinates	Machine coordinates
5	Machine coordinates	Remaining travel distance	Relative coordinates	Absolute coordinates

If the setting is beyond the valid data range, 0 is assumed.
When the multipath simultaneous display function is enabled (parameter No. 13131 is set to a nonzero value, and parameter No. 13132 is set to 1 or a greater value), this parameter becomes invalid.

	#7	#6	#5	#4	#3	#2	#1	#0
11308							COW	

[Input type] parameter input
[Data type] Bit

#1 COW When the file of specified name already exists on memory card,
0: It is not overwritten
Alarm (SR1973 FILE ALREADY EXIST) is generated.
1: It is overwritten.
Because the confirmation message is displayed before overwriting even if COW is 1, overwriting can be canceled.

NOTE
When the overwritten file is read only attribute, it is not possible to overwrite even if parameter COW (No.11308#1)=1.

	#7	#6	#5	#4	#3	#2	#1	#0
11329	GST				BGM			

[Input type] Parameter input
[Data type] Bit path

#3 BGM Coordinates used by the dynamic graphic display function are:
0: Absolute coordinates.
1: Machine coordinates.

#7 GST When drawing cannot be performed for a command with the dynamic graphic display function:
0: The command is ignored, and drawing continues without stopping drawing.
1: Drawing stops.

11343	Blank figure in dynamic graphic display

[Input type] Parameter input
[Data type] Byte path
[Valid data range] 0 to 1
This parameter sets the type of a blank figure in dynamic graphic display.

Setting	Figure
0	Cylinder or hollow cylinder (parallel to the Z-axis)
1	Rectangular parallelepiped

11344	Blank reference position in dynamic graphic display

[Input type] Parameter input
[Data type] Real axis
[Unit of data] mm,inch (input unit)
[Minimum unit of data] Depend on the increment system of the applied axis
[Valid data range] 9 digit of minimum unit of data (refer to standard parameter setting table (A))
(When the increment system is IS-B, -999999.999 to +999999.999)
This parameter sets the reference position of a blank in the dynamic graphic display function by using coordinate values in the workpiece coordinate system.

NOTE
If bit 3 (BGM) of parameter No. 11329 is set to 1, set coordinate values in the machine coordinate system.

11345	Blank dimension I in dynamic graphic display

11346	Blank dimension J in dynamic graphic display

11347	Blank dimension K in dynamic graphic display

[Input type] Parameter input
[Data type] Real axis
[Unit of data] mm,inch (input unit)
[Minimum unit of data] Depend on the increment system of the reference axis
[Valid data range] 0 or positive 9 digit of minimum unit of data (refer to the standard parameter setting table (B))
(When the increment system is IS-B, 0.000 to +999999.999)
These parameters set the dimensions of a blank in the dynamic graphic display function according to the blank figure as follows:

Blank type	Dimension I	Dimension J	Dimension K
Cylinder	Column diameter	0	Column length
Hollow cylinder	Diameter of outer circle of cylinder	Diameter of inner circle of cylinder	Cylinder length
Rectangular prism	Length in X-axis direction	Length in Y-axis direction	Length in Z-axis direction

	#7	#6	#5	#4	#3	#2	#1	#0
11349					WNS		ABC	

[Input type] Parameter input
[Data type] Bit

 # 1 ABC In animated simulation in the dynamic graphic display function, when a fine boring cycle or back boring cycle, which is a hole machining canned cycle, is performed, the movement for a shift at the hole bottom is:
0: Not drawn.
1: Drawn.

 # 3 WNS In the dynamic graphic display function, P-CODE workpiece number search is:
0: Disabled.
1: Enabled.

NOTE
A macro executor option, or a macro executor and C executor options are required.

	#7	#6	#5	#4	#3	#2	#1	#0
11353								SEK

[Input type] Parameter input
[Data type] Bit

#0 SEK When the power is turned on, or when the clear state is present, sequence numbers are:
0: Not maintained.
1: Maintained.

NOTE
During a subprogram call, the sequence number of the subprogram is maintained.

	#7	#6	#5	#4	#3	#2	#1	#0
11630								FRD

[Data type] Bit path

 # 0 FRD The minimum command unit of the rotation angles of coordinate rotation and three-dimensional coordinate system conversion is:
0: 0.001 degree.
1: 0.00001 degree. (1/100,000)

12310	States of the manual handle feed axis selection signals when tool axis direction handle feed/interrupt and table-based vertical direction handle feed/interrupt are performed

[Input type] Parameter input
[Data type] Byte path
[Valid data range] 1 to 24

This parameter sets the states of the manual handle feed axis selection signal (HS1A to HS1E for the first manual handle) or the manual handle interrupt axis selection signal (HS1IA to HS1IE for the first manual handle) to perform tool axis direction handle feed/interrupt and table-based vertical direction handle feed/interrupt.
The handle for which the signal states are set is determined by parameter No. 12323.

<Table of correspondence with the manual handle feed axis selection signals>
If parameter No. 12323 is set to 1, the states of the manual handle feed axis selection signals or manual handle interrupt axis selection signals for the first manual handle in the three-dimensional manual feed (handle feed) mode and corresponding parameter settings are listed in the table below. When the first manual handle pulse generator is turned after setting the signals corresponding to the value set in the parameter, operation is performed in the specified mode.

HS1E (HS1IE)	HS1D (HS1ID)	HS1C (HS1IC)	HS1B (HS1IB)	HS1A (HS1IA)	Parameter (No.12310)
0	0	0	0	1	1
0	0	0	1	0	2
0	0	0	1	1	3
0	0	1	0	0	4
0	0	1	0	1	5
0	0	1	1	0	6
0	0	1	1	1	7
0	1	0	0	0	8
0	1	0	0	1	9
0	1	0	1	0	10
0	1	0	1	1	11
0	1	1	0	0	12
0	1	1	0	1	13
0	1	1	1	0	14
0	1	1	1	1	15
1	0	0	0	0	16
1	0	0	0	1	17
1	0	0	1	0	18
1	0	0	1	1	19
1	0	1	0	0	20
1	0	1	0	1	21
1	0	1	1	0	22
1	0	1	1	1	23
1	1	0	0	0	24

If parameter No. 12323 is set to 2 to 5, replace "1" in HS1A to HS1E and HS1IA to HS1IE above with 2 to 5.

12311	States of the manual handle feed axis selection signals when a movement is made in the first axis direction in tool axis normal direction handle feed/interrupt and table-based horizontal direction handle feed/interrupt

[Input type] Parameter input
[Data type] Byte path
[Valid data range] 1 to 24

This parameter sets the states of the manual handle feed axis selection signals (HS1A to HS1E for the first manual handle) or the manual handle interrupt axis selection signal (HS1IA to HS1IE for the first manual handle) when a movement is made in the first axis direction. (For settings, see "Table of correspondence with the manual handle feed axis selection signals" in the description of parameter No. 12310.)
The handle for which the signal states are set is determined by parameter No. 12323.

The table below indicates the relationships of tool axis directions, first axis directions, and second axis directions.

Parameter No.19697	Tool axis directions	First axis directions	Second axis directions
1	X	Y	Z
2	Y	Z	X
3	Z	X	Y

Note, however, that the table above indicates the directions applicable when the angles of all rotation axes are set to 0.
In tool axis direction/tool axis normal direction feed (not table-based), the directions indicated above assume that 0 is set in parameter No. 19698 and No. 19699. When a rotation axis has made a turn or a nonzero value is set in these parameters in tool axis direction/tool axis normal direction feed, the relevant directions are inclined accordingly.

12312	States of the manual handle feed axis selection signals when a movement is made in the second axis direction in tool axis normal direction handle feed/interrupt and table-based horizontal direction handle feed/interrupt

[Input type] Parameter input
[Data type] Byte path
[Valid data range] 1 to 24

This parameter sets the states of the manual handle feed axis selection signals (HS1A to HS1E for the first manual handle) or the manual handle interrupt axis selection signals (HS1IA to HS1IE for the first manual handle) when a movement is made in the second axis direction. (For settings, see "Table of correspondence with the manual handle feed axis selection signals" in the description of parameter No. 12310.)
The handle for which the signal states are set is determined by parameter No. 12323.

12313	States of the manual handle feed axis selection signals when the first rotation axis is turned in tool tip center rotation handle feed/interrupt

[Input type] Parameter input
[Data type] Byte path
[Valid data range] 1 to 24

This parameter sets the states of the manual handle feed axis selection signals (HS1A to HS1E for the first manual handle) or the manual handle interrupt axis selection signals (HS1IA to HS1IE for the first manual handle) when the first rotation axis is turned in tool tip center rotation handle feed or interrupt. (For settings, see "Table of correspondence with the manual handle feed axis selection signals" in the description of parameter No. 12310.)
The handle for which the signal states are set is determined by parameter No. 12323.

12314	States of the manual handle feed axis selection signals when the second rotation axis is turned in tool tip center rotation handle feed/interrupt

[Input type] Parameter input
[Data type] Byte path
[Valid data range] 1 to 24

This parameter sets the states of the manual handle feed axis selection signals (HS1A to HS1E for the first manual handle) or the manual handle interrupt axis selection signals (HS1IA to HS1IE for the first manual handle) when the second rotation axis is turned in tool tip center rotation handle feed or interrupt. (For settings, see "Table of correspondence with the manual handle feed axis selection signals" in the description of parameter No. 12310.)
The handle for which the signal states are set is determined by parameter No. 12323.

12318	Tool length in three-dimensional machining manual feed

[Input type] Setting input
[Data type] Real path
[Unit of data] mm, inch (machine unit)
[Minimum unit of data] Depend on the increment system of the reference axis
[Valid data range] 9 digit of minimum unit of data (refer to standard parameter setting table (A))
(When the increment system is IS-B, -999999.999 to +999999.999)

This parameter sets a tool length when tool tip center rotation feed is performed with the three-dimensional machining manual feed function and when the three-dimensional machining manual feed screen is displayed.

NOTE
Specify a radius value to set this parameter.

	#7	#6	#5	#4	#3	#2	#1	#0
12320						JFR	FLL	TWD

[Input type] Parameter input
[Data type] Bit path

 # 0 TWD The directions of three-dimensional machining manual feed (other than tool tip center rotation feed) when the tilted working plane command is issued are:
0: Same as those not in the tilted working plane command. That is, the directions are:
Tool axis normal direction 1 (table-based horizontal direction 1)
Tool axis normal direction 2 (table-based horizontal direction 2)
Tool axis direction (table-based vertical direction)
1: X, Y, and Z directions in the feature coordinate system.

 # 1 FLL The directions of tool axis normal direction feed or table-based horizontal direction feed in the three-dimensional machining manual feed mode are:
0: Tool axis normal direction 1 (table-based horizontal direction 1) and tool axis normal direction 2 (table-based horizontal direction 2).
1: Longitude direction and latitude direction.

Parameter FLL (No.12320#1)	Parameter TWD (No.12320#0)	Directions of three-dimensional machining manual feed
0	0	Conventional directions
0	1	When the tilted working plane command is issued: X, Y, and Z directions in the feature coordinate system When the command is not issued: Conventional directions
1	0	Longitude direction and latitude direction
1	1	When the tilted working plane command is issued: X, Y, and Z directions in the feature coordinate system When the command is not issued: Longitude direction and latitude direction

 # 2 JFR As the feeedrate of three-dimensional machining manual feed (jog feed or incremental feed) :
0: The dry run rate (parameter No. 1410) is used.
1: The jog feedrate (parameter No. 1423) is used.

12321	Normal axis direction

[Input type] Parameter input
[Data type] Byte path
[Valid data range] 0 to 3
When a tilted working plane command (G68.3) is issued to perform three-dimensional machining manual feed in the latitude direction, longitude direction, and tool axis direction, this parameter sets an axis parallel to the normal direction.
1 : Positive (+) X-axis direction
2 : Positive (+) Y-axis direction
3 : Positive (+) Z-axis direction
0 : Reference tool axis direction (parameter No. 19697)

12322	Angle used to determine whether to assume the tool axis direction to be parallel to the normal direction (parameter No. 12321)

[Input type] Parameter input
[Data type] Real path
[Unit of data] deg
[Minimum unit of data] Depend on the increment system of the reference axis
[Valid data range] 0 to 90

When a tilted working plane command (G68.3) is issued to perform three-dimensional machining manual feed in the latitude direction, longitude direction, and tool axis direction, if the angle between the tool axis direction and normal direction (parameter No. 12321) is too small, the tool axis direction is assumed to be parallel to the normal direction (parameter No. 12321). This parameter sets the maximum angle at which the tool axis direction is assumed to be parallel to the normal direction.
When this parameter is set to 0 or a value outside the valid range, it is set to 1 degree.

12323	Number of a manual handle used for three-dimensional machining manual feed

[Input type] Setting input
[Data type] Byte path
[Valid data range] 0 to 5

When three-dimensional machining manual feed (handle feed) is performed, set the number of the manual handle to be used.
When the second or third manual handle is used for three-dimensional machining manual feed, the option for manual handle feed with 2/3 handles is required.
When the fourth or fifth manual handle is used for three-dimensional machining manual feed, the option for manual handle feed with 4/5 handles is required.
If 0 or the number of an unavailable handle is set, the first handle is assumed.

12600	Identification Number for synchronous, composite and superimposed control with program command

[Input type] Parameter input
[Data type] Word axis
[Valid data range] 0,1 to 32767

Set identification numbers that can be specified with P,Q addresses.
The axis whose identification number is "0" cannot become under synchronous /composite /superimposed control by CNC program.
The same identification number cannot be set to two or more axes through all paths.
When the same identification number is set, PS alarm (PS5339) occurs at G50.4/G50.5/G50.6/G51.4/G51.5/G51.6 block.

	#7	#6	#5	#4	#3	#2	#1	#0
13113					CFD			CLR

[Input type] Parameter input
[Data type] Bit path

 # 0 CLR Upon reset, the display of a travel distance by three-dimensional machining manual feed is:
0: Not cleared.
1: Cleared.

 # 3 CFD As feedrate F, the three-dimensional machining manual feed screen displays:
0: Composite feedrate at the linear axis/rotation axis control point.
1: Feedrate at the tool tip.

	#7	#6	#5	#4	#3	#2	#1	#0
13200					ETE	TRT		

[Input type] Parameter input
[Data type] Bit path

 # 2 TRT As the remaining lifetime value for outputting the tool life arrival notice signal:
0: The remaining lifetime of the last tool is used.
1: The sum of the remaining lifetimes of the tools with the same type number is used.

NOTE
This parameter is valid when bit 3 (ETE) of parameter No. 13200 is set to 0 (arrival notice for each type number).

 # 3 ETE The tool life arrival notice signal is output:
0: For each tool type.
1: For each tool.

	#7	#6	#5	#4	#3	#2	#1	#0
13201							TDN	TDC

[Input type] Parameter input
[Data type] Bit

NOTE
When at least one of these parameters is set, the power must be turned off before operation is continued.

 # 0 TDC The function of customizing the tool management data screen of the tool management function is:
0: Disabled.
1: Enabled.

 # 1 TDN On the tool management function screen, the character string for indicating the tool life status can contain:
0: Up to 6 characters.
1: Up to 12 characters.

	#7	#6	#5	#4	#3	#2	#1	#0
13202	DOM	DOT		DO2	DOB	DOY	DCR	

[Input type] Parameter input
[Data type] Bit

 # 1 DCR On the tool management function screen, tool nose radius compensation data is:
0: Displayed.
1: Not displayed.

NOTE
This parameter is valid when the machine control type is the lathe system or compound system.

 # 2 DOY On the tool management function screen, Y-axis offset data is:
0: Displayed.
1: Not displayed.

NOTE
This parameter is valid when the machine control type is the lathe system or compound system.

 # 3 DOB On the tool management function screen, B-axis offset data is:
0: Displayed.
1: Not displayed.

NOTE
This parameter is valid when the machine control type is the lathe system or compound system.

 # 4 DO2 On the tool management function screen, the second geometry tool offset data is:
0: Displayed.
1: Not displayed.

NOTE
This parameter is valid when the machine control type is the lathe system or compound system.

 # 6 DOT On the tool management function screen, the tool offset data (X, Z) of the T series is:
0: Displayed.
1: Not displayed.

NOTE
This parameter is valid when the machine control type is the lathe system or compound system.

 # 7 DOM On the tool management function screen, the tool offset data of the M series is:
0: Displayed.
1: Not displayed.

NOTE
This parameter is valid when the machine control type is the machining center system or compound system.

	#7	#6	#5	#4	#3	#2	#1	#0
13204								TDL

[Input type] Parameter input
[Data type] Bit

 # 0 TDL The protection function for tool management data using a key is:
0: Disabled.
1: Enabled.

13220	Number of valid tools in tool management data

NOTE
When this parameter is set, the power must be turned off before operation is continued.

[Input type] Parameter input
[Data type] Word
[Valid data range] 0 to 64 (Extended to 240 or 1000 by the addition of an option)
This parameter sets the number of valid tools in tool management data.

13221	M code for tool life count restart

[Input type] Parameter input
[Data type] Word path
[Valid data range] 0 to 65535
When 0 is set in this parameter, this parameter is ignored.
When an M code for tool life count restart is specified, the counting of the life of the tool attached at the spindle position is started. When the type for counting the number of use times is selected, the target of life counting is switched to the tool attached at the spindle position, and the life count is incremented by 1.
When the type for counting time is selected, the target of life counting is switched to the tool attached at the spindle position but no other operations are performed. If the tool attached at the spindle position is not a tool under tool life management, no operation is performed.
The M code set in parameter No. 6811 waits for FIN. However, the M code set in this parameter does not wait for FIN.
The M code set in parameter No. 13221 must not be specified in a block where another auxiliary function is specified.
The M code set in parameter No. 13221 does not wait for FIN. So, do not use the M code for other purposes.

NOTE
The use of this parameter varies depending on whether it is used by the tool management function or tool life management function.

13222	Number of data items in the first cartridge

NOTE
When this parameter is set, the power must be turned off before operation is continued.

[Input type] Parameter input
[Data type] Word
[Valid data range] 1 to 64 (Extended to 240 or 1000 by the addition of an option)
This parameter sets the number of data items used with the first cartridge.

13223	Start pot number of the first cartridge

NOTE
When this parameter is set, the power must be turned off before operation is continued.

[Input type] Parameter input
[Data type] Word
[Valid data range] 1 to 9999
This parameter sets the start pot number to be used with the first cartridge. Pot numbers starting with the value set in this parameter and sequentially incremented by 1 are assigned to all data items.

13227	Number of data items in the second cartridge

NOTE
When this parameter is set, the power must be turned off before operation is continued.

[Input type] Parameter input
[Data type] Word
[Valid data range] 1 to 64(Extended to 240 or 1000 by the addition of an option)
This parameter sets the number of data items used with the second cartridge.

13228	Start pot number of the second cartridge

NOTE
When this parameter is set, the power must be turned off before operation is continued.

[Input type] Parameter input
[Data type] Word
[Valid data range] 1 to 9999
This parameter sets the start pot number to be used with the second cartridge. Pot numbers starting with the value set in this parameter and sequentially incremented by 1 are assigned to all data items.

13232	Number of data items in the third cartridge

NOTE
When this parameter is set, the power must be turned off before operation is continued.

[Input type] Parameter input
[Data type] Word
[Valid data range] 1 to 64(Extended to 240 or 1000 by the addition of an option)
This parameter sets the number of data items used with the third cartridge.

13233	Start pot number of the third cartridge

NOTE
When this parameter is set, the power must be turned off before operation is continued.

[Input type] Parameter input
[Data type] Word
[Valid data range] 1 to 9999
This parameter sets the start pot number to be used with the third cartridge. Pot numbers starting with the value set in this parameter and sequentially incremented by 1 are assigned to all data items.

13237	Number of data items in the fourth cartridge

NOTE
When this parameter is set, the power must be turned off before operation is continued.

[Input type] Parameter input
[Data type] Word
[Valid data range] 1 to 64(Extended to 240 or 1000 by the addition of an option)
This parameter sets the number of data items used with the fourth cartridge.

13238	Start pot number of the fourth cartridge

NOTE
When this parameter is set, the power must be turned off before operation is continued.

[Input type] Parameter input
[Data type] Word
[Valid data range] 1 to 9999
This parameter sets the start pot number to be used with the fourth cartridge. Pot numbers starting with the value set in this parameter and sequentially incremented by 1 are assigned to all data items.

13252	M code for specifying a particular tool

[Input type] Parameter input
[Data type] Word path
[Valid data range] 0 to 65535
This parameter sets not a tool type number but an M code for directly specifying the T code of a particular tool.

13265	Number for selecting a spindle position offset number with tool life or H code for using the tool length offset in tool life management with tool life management

[Input type] Parameter input
[Data type] 2-word path
[Valid data range] 0 to 999

- With tool management functions
  This parameters sets an H/D code for selecting an offset number registered in the data of the tool attached at the spindle position.
  When 0 is set, an ordinary used code such as H99/D99 is used. When a value other than 0 is set, H99/D99 no longer has a particular meaning. So, when H99/D99 is specified in this case, the specification of offset number 99 is assumed.
  With the T series, address D only is used to specify a tool number and offset number, so that a restriction is imposed on the number of digits. So, the valid data range of this parameter varies according the number of digits of an offset number.
  When the number of digits of an offset number is 1: to 9
  When the number of digits of an offset number is 2: to 99
  When the number of digits of an offset number is 3: to 999
- With tool life management functions
  Usually, when H99 is specified, tool length offset is enabled by the H code of the tool being used. By setting any H code in this parameter, the H code instead of H99 can be used. If 0 is specified, H99 is assumed.
  A value ranging from 0 to 9999 can be set.

13266	D code for enabling cutter compensation in tool life management

[Input type] Parameter input
[Data type] 2-word path
[Valid data range] 0 to 9999
Usually in tool life management, specifying D99 allows the D code of the tool being used to enable cutter compensation. By setting any D code in this parameter, the D code instead of D99 can be used. If 0 is set, D99 is assumed.
A value ranging from 0 to 9999 can be set.

	#7	#6	#5	#4	#3	#2	#1	#0
13451							ATW	

[Input type] Parameter input
[Data type] Bit path

 # 1 ATW When I, J, and K are all set to 0 in a block that specifies a feature coordinate system setup command (G68.2), which is a tilted working plane command:
0: An alarm (PS5457) is issued.
1: A feature coordinate system with a tilted plane angle of 0 degrees is assumed for operation.

	#7	#6	#5	#4	#3	#2	#1	#0
13600	MSA							MCR

[Input type] Parameter input
[Data type] Bit path

 # 0 MCR When an allowable acceleration rate adjustment is made with the machining condition selection function (machining parameter adjustment screen, precision level selection screen), parameter No. 1735 for the deceleration function based on acceleration in circular interpolation is:
0: Modified.
1: Not modified.

 # 7 MSA When the machining condition selection function is used, the acceleration rate change time (bell-shaped) (LV1, LV10) is:
0: Set using parameter Nos. 13612 and 13613.
1: Set using parameter Nos. 13662 and 13663.

	#7	#6	#5	#4	#3	#2	#1	#0
13601								MPR

[Input type] Parameter input
[Data type] Bit

NOTE
When this parameter is set, the power must be turned off before operation is continued.

 # 0 MPR The machining parameter adjustment screen is:
0: Displayed.
1: Not displayed.
Even when this parameter is set to 1, the precision level selection screen is displayed.

13610	Acceleration rate for acceleration/deceleration before look-ahead interpolation in AI contour control (precision level 1)
13611	Acceleration rate for acceleration/deceleration before look-ahead interpolation in AI contour control (precision level 10)

[Input type] Parameter input
[Data type] Real axis
[Unit of data] mm/sec/sec, inch/sec/sec, degree/sec/sec (machine unit)
[Minimum unit of data] Depend on the increment system of the applied axis
[Valid data range] Refer to the standard parameter setting table (D)
(When the machine system is metric system, 0.0 to +100000.0. When the machine system is inch system, machine, 0.0 to +10000.0)
Each of these parameters sets an acceleration rate for acceleration/ deceleration before interpolation in AI contour control. Set a value (precision level 1) with emphasis placed on speed, and a value (precision level 10) with emphasis on precision.

13612	Acceleration rate change time (bell-shaped) when AI contour control is used (precision level 1)
13613	Acceleration rate change time (bell-shaped) when AI contour control is used (precision level 10)

[Input type] Parameter input
[Data type] Byte path
[Unit of data] msec
[Valid data range] 0 to 127
Each of these parameters sets an acceleration rate change time (bell-shaped) in AI contour control. Set a value (precision level 1) with emphasis placed on speed, and a value (precision level 10) with emphasis on precision.

13614	**Allowable acceleration rate change amount for each axis in speed control based on acceleration rate change under control on the rate of change of acceleration (precision level 1)**
13615	**Allowable acceleration rate change amount for each axis in speed control based on acceleration rate change under control on the rate of change of acceleration (precision level 10)**

[Input type] Parameter input
[Data type] Real axis
[Unit of data] mm/sec/sec, inch/sec/sec, degree/sec/sec (machine unit)
[Minimum unit of data] Depend on the increment system of the applied axis
[Valid data range] Refer to the standard parameter setting table (D)
(When the machine system is metric system, 0.0 to +100000.0. When the machine system is inch system, machine, 0.0 to +10000.0)
Each of these parameters sets an allowable acceleration rate change amount per 1 ms for each axis in speed control based on acceleration rate change under control on the rate of change of acceleration during AI contour control.
Set a value (precision level 1) with emphasis placed on speed, and a value (precision level 10) with emphasis on precision.

13616	**Allowable acceleration rate change amount for each axis in speed control based on acceleration rate change under control on the rate of change of acceleration in successive linear interpolation operations (precision level 1)**
13617	**Allowable acceleration rate change amount for each axis in speed control based on acceleration rate change under control on the rate of change of acceleration in successive linear interpolation operations (precision level 10)**

[Input type] Parameter input
[Data type] Real axis
[Unit of data] mm/sec/sec, inch/sec/sec, degree/sec/sec (machine unit)
[Minimum unit of data] Depend on the increment system of the applied axis
[Valid data range] Refer to the standard parameter setting table (D)
(When the machine system is metric system, 0.0 to +100000.0. When the machine system is inch system, machine, 0.0 to +10000.0)
Each of these parameters sets an allowable acceleration rate change amount per 1 ms for each axis in speed control based on acceleration rate change under control on the rate of change of acceleration in successive linear interpolation operations during AI contour control.
Set a value (precision level 1) with emphasis placed on speed, and a value (precision level 10) with emphasis on precision.

NOTE
1 For an axis with 0 set in this parameter, parameter No. 13614 and No. 13615 (allowable acceleration rate change amount in speed control based on acceleration rate change under control on the rate of change of acceleration) are valid.
2 For an axis with 0 set in parameter No. 13614 and No. 13615 (allowable acceleration rate change amount in speed control based on acceleration rate change under control on the rate of change of acceleration), speed control based on acceleration rate change is disabled, so that the specification of this parameter has no effect.

13618	**Rate of change time of the rate of change of acceleration in smooth bell-shaped acceleration/deceleration before interpolation when AI contour control is used (precision level 1)**
13619	**Rate of change time of the rate of change of acceleration in smooth bell-shaped acceleration/deceleration before interpolation when AI contour control is used (precision level 10)**

[Input type] Parameter input
[Data type] Byte path
[Unit of data] %
[Valid data range] 0 to 50

Each of these parameters sets the rate (percentage) of the change time of the rate of change of acceleration to the change time of acceleration rate change in smooth bell-shaped acceleration/deceleration before look-ahead interpolation during AI contour control.
Set a value (precision level 1) with emphasis placed on speed, and a value (precision level 10) with emphasis on precision.

NOTE
When 0 or a value not within the valid data range is set in this parameter, smooth bell-shaped acceleration/deceleration before look-ahead interpolation is not performed.

13620	Allowable acceleration rate when AI contour control is used (precision level 1)

13621	Allowable acceleration rate when AI contour control is used (precision level 10)

[Input type] Parameter input
[Data type] Real axis
[Unit of data] mm/sec/sec, inch/sec/sec, degree/sec/sec (machine unit)
[Minimum unit of data] Depend on the increment system of the applied axis
[Valid data range] Refer to the standard parameter setting table (D)
(When the machine system is metric system, 0.0 to +100000.0. When the machine system is inch system, machine, 0.0 to +10000.0)
Each of these parameters sets an allowable acceleration rate in AI contour control. Set a value (precision level 1) with emphasis placed on speed, and a value (precision level 10) with emphasis on precision.

13622	Time constant for acceleration/deceleration after interpolation when AI contour control is used (precision level 1)

13623	Time constant for acceleration/deceleration after interpolation when AI contour control is used (precision level 10)

[Input type] Parameter input
[Data type] Word axis
[Unit of data] msec
[Valid data range] 1 to 512
Each of these parameters sets a time constant for acceleration/ deceleration after interpolation when AI contour control is used. Set a value (precision level 1) with emphasis placed on speed, and a value (precision level 10) with emphasis on precision.

13624	Corner speed difference when AI contour control is used (precision level 1)

13625	Corner speed difference when AI contour control is used (precision level 10)

[Input type] Parameter input
[Data type] Real axis
[Unit of data] mm/min, inch/min, degree/min (machine unit)
[Minimum unit of data] Depend on the increment system of the applied axis
[Valid data range] Refer to the standard parameter setting table (C)
(When the increment system is IS-B, 0.0 to +999000.0)
Each of these parameters sets an allowable speed difference for speed determination based on corner speed difference in AI contour control. Set a value (precision level 1) with emphasis placed on speed, and a value (precision level 10) with emphasis on precision.

13626	Maximum cutting speed when AI contour control is used (precision level 1)

13627	Maximum cutting speed when AI contour control is used (precision level 10)

[Input type] Parameter input
[Data type] Real axis
[Unit of data] mm/min, inch/min, degree/min (machine unit)
[Minimum unit of data] Depend on the increment system of the applied axis
[Valid data range] Refer to the standard parameter setting table (C)
(When the increment system is IS-B, 0.0 to +999000.0)
Each of these parameters sets a maximum cutting speed in AI contour control. Set a value (precision level 1) with emphasis placed on speed, and a value (precision level 10) with emphasis on precision.

13628	Parameter number corresponding to arbitrary item 1 when AI contour control is used

13629	Parameter number corresponding to arbitrary item 2 when AI contour control is used

NOTE
When these parameters are set, the power must be turned off before operation is continued.

[Input type] Parameter input
[Data type] 2-word path
[Valid data range] 1 to 65535
These parameters set the parameter numbers corresponding to arbitrary items 1 and 2.

NOTE
The parameter numbers corresponding to the following cannot be specified:
- Bit parameters
- Spindle parameters (No. 4000 to No. 4799)
- Parameters of real number type
- Parameters that require power-off (for which the alarm (PW0000) is issued)
- Nonexistent parameters

13630	Value with emphasis on speed (precision level 1) of the parameter corresponding to arbitrary item 1 when AI contour control is used
13631	Value with emphasis on speed (precision level 1) of the parameter corresponding to arbitrary item 2 when AI contour control is used
13632	Value with emphasis on speed (precision level 10) of the parameter corresponding to arbitrary item 1 when AI contour control is used
13633	Value with emphasis on speed (precision level 10) of the parameter corresponding to arbitrary item 2 when AI contour control is used

[Input type] Parameter input
[Data type] 2-word axis
[Unit of data] Depend on the type of parameter for an arbitrary item
[Valid data range] Depend on the type of parameter for an arbitrary item

Each of these parameters sets a value with emphasis placed on speed or precision for a parameter.

13662	Acceleration rate change time (bell-shaped) when AI contour control is used (precision level 1), range extended
13663	Acceleration rate change time (bell-shaped) when AI contour control is used (precision level 10), range extended

[Input type] Parameter input
[Data type] 2-word path
[Unit of data] msec
[Valid data range] 0 to 200

Each of these parameters sets an acceleration rate change time (bell-shaped) in AI contour control. Set a value (precision 1) with emphasis placed on speed, and a value (precision level 10) with emphasis on precision.

	#7	#6	#5	#4	#3	#2	#1	#0
14000						IRF		

[Input type] Parameter input
[Data type] Bit axis

 # 2 IRF An inch-metric switch command (G20, G21) at the reference position is:
0: Disabled.
1: Enabled.

When this function is enabled for an axis, if an attempt to switch between the inch and metric unit is made although the tool is not at the reference position on that axis, an alarm (PS5362) is issued, and switching between the inch and metric unit is canceled.
Be sure to move the tool to the reference position by, for example, specifying G28 before switching between the inch and metric unit.

14010	Maximum allowable travel distance when the reference position is established for a linear scale with an absolute address reference position

[Input type] Parameter input
[Data type] 2-word axis
[Unit of data] Detection unit
[Valid data range] 0 to 99999999

This parameter sets the maximum allowable travel distance at the FL feedrate when the reference position is established for a linear scale with an absolute address reference position. When the travel distance exceeds the setting of this parameter, the alarm (PS5326) (SCALE WITH REFERENCE POSITION: REFERENCE POSITION ESTABLISHMENT FAILED) is issued. When this parameter is set to 0, the maximum allowable travel distance is not checked.

14340	ATR value corresponding to slave 01 on FSSB line 1
14341	ATR value corresponding to slave 02 on FSSB line 1
:	
14357	ATR value corresponding to slave 18 on FSSB line 1

NOTE
When these parameters are set, the power must be turned off before operation is continued.

[Input type] Parameter input
[Data type] Byte
[Valid data range] 0 to 23, 64, -56, -96

Each of these parameters sets the value (ATR value) of the address translation table corresponding to each of slave 1 to slave 18 on FSSB line 1 (first optical connector).
The slave is a generic term for servo amplifiers and separate detector interface units connected via an FSSB optical cable to the CNC. Numbers 1 to 18 are assigned to slaves, with younger numbers sequentially assigned to slaves closer to the CNC.
A 2-axis amplifier consists of two slaves, and a 3-axis amplifier consists of three slaves. In each of these parameters, set a value as described below, depending on whether the slave is an amplifier, separate detector, or nonexistent.

- When the slave is an amplifier:
  Set a value obtained by subtracting 1 from the setting of parameter No. 1023 for the axis to which the amplifier is assigned.
- When the slave is a separate detector interface unit:
  Set 64 for the first separate detector interface unit (connected near the CNC), and set -56 for the second separate detector interface unit (connected far from the CNC).
- When the slave is nonexistent:
  Set -96.

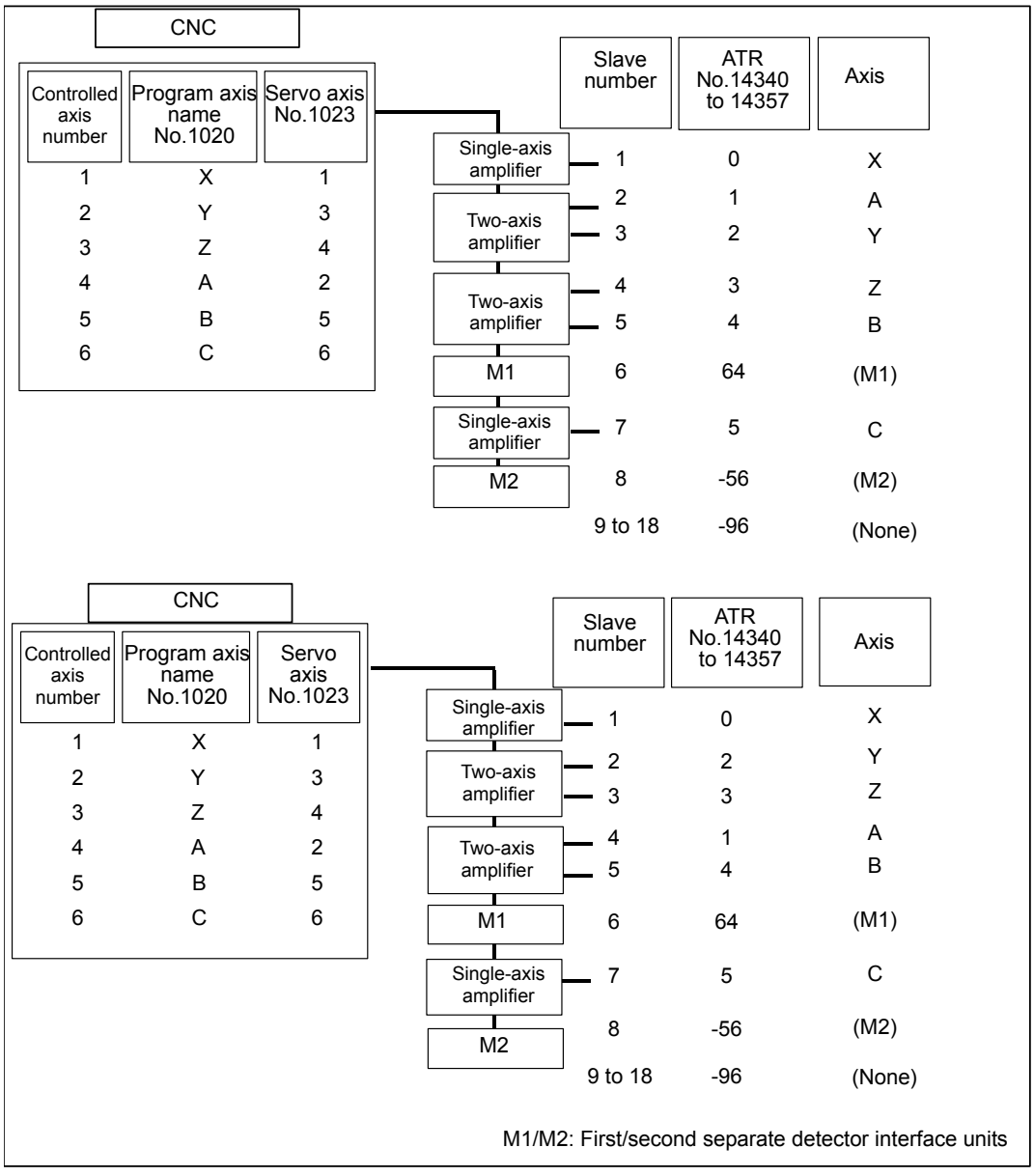
NOTE

1 When the electric gear box (EGB) function is used
Although an amplifier is not actually required for an EGB dummy axis, set this parameter with assuming that a dummy amplifier is connected. That is, as the address conversion table value for a nonexistent slave, set the value obtained by subtracting 1 from the setting of parameter No. 1023 for the EGB dummy axis, instead of -96.

2 When the FSSB is set to the automatic setting mode (when the parameter FMD (No.1902#0) is set to 0), parameter Nos. 14340 to 14357 are automatically set as data is input on the FSSB setting screen. When the manual setting 2 mode is set (when the parameter FMD (No.1902#0) is set to 1), be sure to directly set values in parameter Nos. 14340 to 14357.

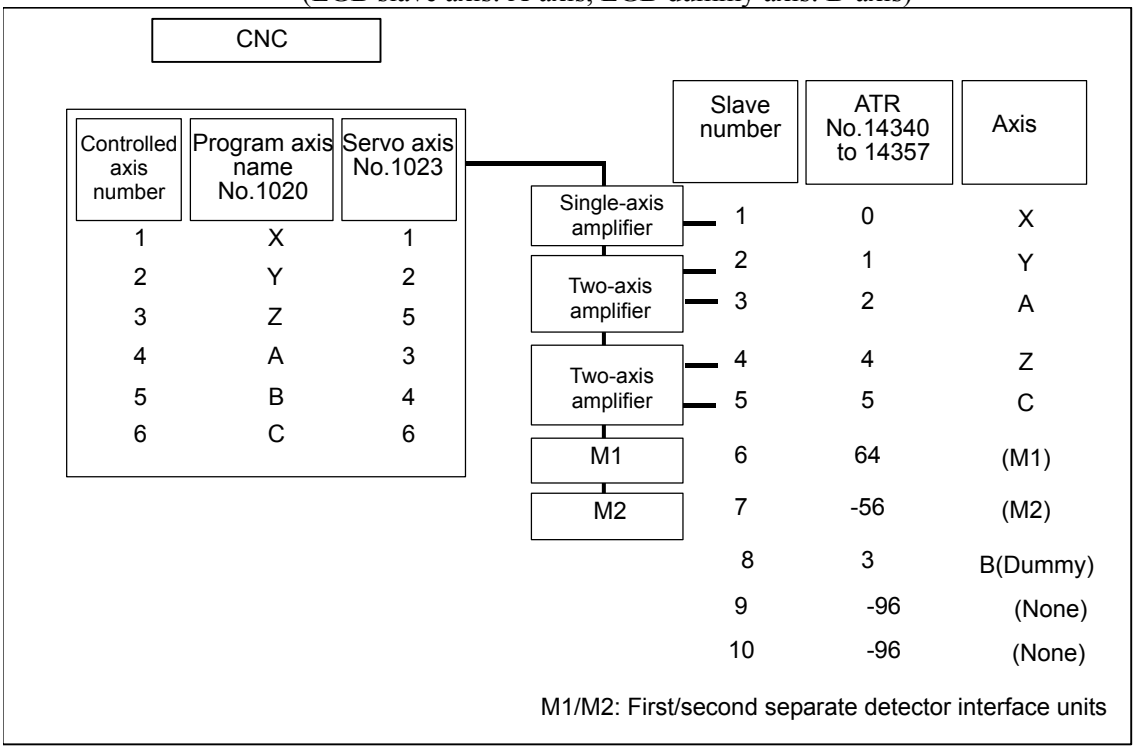
Example of axis configuration and parameter settings

- Example 1



- Example 2

Example of axis configuration and parameter settings when the electronic gear box (EGB) function is used
(EGB slave axis: A-axis, EGB dummy axis: B-axis)



14358	ATR value corresponding to slave 01 on FSSB line 2
14359	ATR value corresponding to slave 02 on FSSB line 2
:	
14375	ATR value corresponding to slave 18 on FSSB line 2

NOTE
When these parameters are set, the power must be turned off before operation is continued.

[Input type] Parameter input
[Data type] Byte
[Valid data range] 0 to 23, 80, -40, -96

Each of these parameters sets the value (ATR value) of the address translation table corresponding to each of slave 1 to slave 18 on FSSB line 2 (second optical connector).

- When the slave is an amplifier:
  Set a value obtained by subtracting 1 from the setting of parameter No. 1023 for the axis to which the amplifier is assigned.
- When the slave is a separate detector interface unit:
  Set 80 for the third separate detector interface unit (connected near the CNC), and set -40 for the fourth separate detector interface unit (connected far from the CNC).
- When the slave is nonexistent:
  Set -96.

NOTE
1 Set these parameters only when a servo axis control card with two optical connectors (FSSB lines) is used.
2 When the FSSB is set to the automatic setting mode (when the parameter FMD (No.1902#0) is set to 0), parameter Nos. 14358 to 14375 are automatically set as data is input on the FSSB setting screen. When the manual setting 2 mode is set (when the parameter FMD (No.1902#0) is set to 1), be sure to directly set values in parameter Nos. 14358 to 14375.

14376	ATR value corresponding to connector 1 on the first separate detector interface unit
14377	ATR value corresponding to connector 2 on the first separate detector interface unit
:	
14383	ATR value corresponding to connector 8 on the first separate detector interface unit
14384	ATR value corresponding to connector 1 on the second separate detector interface unit
:	
14391	ATR value corresponding to connector 8 on the second separate detector interface unit
14392	ATR value corresponding to connector 1 on the third separate detector interface unit
:	
14399	ATR value corresponding to connector 8 on the third separate detector interface unit
14400	ATR value corresponding to connector 1 on the fourth separate detector interface unit
:	
14407	ATR value corresponding to connector 8 on the fourth separate detector interface unit

NOTE
When these parameters are set, the power must be turned off before operation is continued.

[Input type] Parameter input
[Data type] Byte
[Valid data range] 0 to 32

Each of these parameters sets the value (ATR value) of the address translation table corresponding to each connector on a separate detector interface unit.
The first and second separate detector interface units are connected to FSSB line 1, and the third and fourth separate detector interface units are connected to FSSB line 2.
In each of these parameters, set a value obtained by subtracting 1 from the setting of parameter No. 1023 for the axis connected to a connector on a separate detector interface unit.
When there are axes for which settings are made to use a separate detector interface unit (bit 6 (PM1x) of parameter No. 1905 is set to 1, bit 7 (PM2x) of parameter No. 1905 is set to 1, bit 1 (PM3x) of parameter No. 1905 is set to 1, or bit 2 (PM4x) of parameter No. 1905 is set to 1), set 32 for connectors not used.

NOTE
When the FSSB is set to the automatic setting mode (when the parameter FMD (No.1902#0) is set to 0), parameter Nos. 14376 to 14407 are automatically set as data is input on the FSSB setting screen. When the manual setting 2 mode is set (when the parameter FMD (No.1902#0) is set to 1), be sure to directly set values in parameter Nos. 14376 to 14407.

14713	Unit of magnification by which enlargement and reduction is performed with the dynamic graphic display function

[Input type] Parameter input
[Data type] Word
[Valid data range] 0 to 255
This parameter sets the unit of magnification by which enlargement and reduction is performed with the dynamic graphic display function.
Unit of magnification = 64 / setting
If 0 is set, 64 is assumed.

14714	Unit of horizontal movement when a movement is made with the dynamic graphic display function

[Input type] Parameter input
[Data type] Word
[Valid data range] 0 to 255
This parameter sets the unit of horizontal movement (in dots) applied when a movement is made with the dynamic graphic display function.
If 0 is set, 64 is assumed.

14715	Unit of vertical movement when a movement is made with the dynamic graphic display function

[Input type] Parameter input
[Data type] Word
[Valid data range] 0 to 255
This parameter sets the unit of vertical movement (in dots) applied when a movement is made with the dynamic graphic display function.
If 0 is set, 35 is assumed.

14716	Unit of rotation angle when rotation is performed with the dynamic graphic display function

[Input type] Parameter input
[Data type] Word
[Valid data range] 0 to 255
This parameter sets the unit (in degrees) of a rotation angle by which the drawing coordinate system is rotated with the dynamic graphic display function.
If 0 is set, 10 is assumed.

18060	M code that prohibits backward movement

[Input type] Parameter input
[Data type] Word path
[Valid data range] 1 to 999
When an M code that prohibits backward movement is specified during backward movement, backward movement of blocks before the M code is prohibited. In this case, backward movement prohibition signal MRVSP<Fn091.2> is output.

The M code that prohibits backward movement is not output as an M code. As the M code that prohibits backward movement, set an M code that is not used by auxiliary functions and macros.

18065	M code that prohibits backward movement 1

[Input type] Parameter input
[Data type] Word path
[Valid data range] 1 to 999
When an M code that prohibits backward movement is specified during backward movement, backward movement of blocks before the M code is prohibited. In this case, backward movement prohibition signal MRVSP<Fn091.2> is output.

18066	M code that prohibits backward movement 2

[Input type] Parameter input
[Data type] Word path
[Valid data range] 1 to 999
When an M code that prohibits backward movement is specified during backward movement, backward movement of blocks before the M code is prohibited. In this case, backward movement prohibition signal MRVSP<Fn091.2> is output.

	#7	#6	#5	#4	#3	#2	#1	#0
19500		FNW						

[Input type] Parameter input
[Data type] Bit path

 # 6 FNW When the feedrate is determined according to the feedrate difference and acceleration in AI contour control:

0: The maximum feedrate at which the allowable feedrate difference and acceleration for each axis are not exceeded is used.

1: The maximum feedrate at which the allowable feedrate difference and acceleration for each axis are not exceeded is used. The feedrate is determined so that the decreased feedrate is constant regardless of the move direction when the profile is the same.

A feedrate is determined to satisfy the condition that the allowable feedrate difference and allowable acceleration rate of each axis are not exceeded, and also to ensure that a constant deceleration rate is applied to the same figure regardless of the direction of movement.

	#7	#6	#5	#4	#3	#2	#1	#0
19501			FRP					

[Input type] Parameter input
[Data type] Bit path

 # 5 FRP Linear rapid traverse is:

0: Acceleration/deceleration after interpolation

1: Acceleration/deceleration before interpolation

Set a maximum allowable acceleration rate for each axis in parameter No. 1671.

When using bell-shaped acceleration/deceleration before interpolation, set an acceleration rate change time in parameter No. 1672.

When this parameter is set to 1, acceleration/deceleration before interpolation is also applied to rapid traverse if all conditions below are satisfied. At this time, acceleration/deceleration after interpolation is not applied.

- Bit 1 (LRP) of parameter No. 1401 is set to 1: Linear interpolation type positioning
- A value other than 0 is set in parameter No. 1671 for an axis.
- The AI contour control mode is set.

If all of these conditions are not satisfied, acceleration/deceleration after interpolation is applied.

	#7	#6	#5	#4	#3	#2	#1	#0
19503				ZOL				HPF

[Input type] Parameter input
[Data type] Bit path

 # 0 HPF When a feedrate is determined based on acceleration in AI contour control, smooth feedrate control is:
0: Not used.
1: Used.

 # 4 ZOL The deceleration function based on cutting load in AI contour control (deceleration based on Z-axis fall angle) is:
0: Enabled for all commands.
1: Enabled for linear interpolation commands only.

	#7	#6	#5	#4	#3	#2	#1	#0
19515							ZG2	

[Input type] Parameter input
[Data type] Bit path

 # 1 ZG2 When the deceleration function based on cutting load in AI contour control (deceleration based on Z-axis fall angle) is used:
0: Stepwise override values are applied.
1: Inclined override values are applied.
This parameter is valid only when bit 4 (ZAG) of parameter No. 8451 is set to 1.
When this parameter is set to 1, be sure to set parameter Nos. 19516, 8456, 8457, and 8458.

19516	Override for area 1 in deceleration based on cutting load in AI contour control

[Input type] Parameter input
[Data type] Word path
[Unit of data] %
[Valid data range] 1 to 100
This parameter sets an override value for area 1 when the deceleration function based on cutting load in AI contour control is used.
This parameter is valid only when bit 1 (ZG2) of parameter No. 19515 is set to 1.

	#7	#6	#5	#4	#3	#2	#1	#0
19530		CYS						

[Input type] Parameter input
[Data type] Bit path

 # 6 CYS Specifies whether when the cylindrical interpolation cutting point compensation function is used, cutting point compensation is performed between blocks or together with a block movement if the cutting point compensation value is less than the setting of parameter No. 19534.

0: Performed between blocks.
1: Performed together with a block movement if the cutting point compensation value is less than the setting of parameter No. 19534.

19534	Limit for changing cylindrical interpolation cutting point compensation in a single block

[Input type] Parameter input
[Data type] Real path
[Unit of data] mm, inch (input unit)
[Minimum unit of data] Depend on the increment system of the reference axis
[Valid data range] 1 to 999999999

The following operation is performed, depending on the setting of parameter No.19530:

(1) Bit 6 (CYS) of parameter No. 19530) is set to 0
If the amount of cylindrical interpolation cutting point compensation is smaller than the value set in this parameter, cylindrical interpolation cutting point compensation is not performed. Instead, this ignored amount of cylindrical interpolation cutting point compensation is added to the next amount of cylindrical interpolation cutting point compensation to determine whether to perform cylindrical interpolation cutting point compensation.

(2) Bit 6 (CYS) of parameter No. 19530) is set to 1
If the amount of cylindrical interpolation cutting point compensation is smaller than the value set in this parameter, cylindrical interpolation cutting point compensation is performed together with the movement of the specified block.

NOTE
Set this parameter as follows:
Setting > (setting for a rotation axis in parameter No. 1422) × 4/3 where 4/3 is a constant for internal processing.

19535	Limit of travel distance moved with the cylindrical interpolation cutting point compensation in the previous block unchanged.

[Input type] Parameter input
[Data type] Real path
[Unit of data] mm, inch (input unit)
[Minimum unit of data] Depend on the increment system of the reference axis
[Valid data range] 1 to 999999999

The following operation is performed, depending on the type of interpolation:

(1) For linear interpolation
If the travel distance in a specified block is smaller than the value set in this parameter, machining is performed without changing the cylindrical interpolation cutting point compensation in the previous block.

(2) For circular interpolation
If the diameter of a specified arc is smaller than the value set in this parameter, machining is performed without changing the cylindrical interpolation cutting point compensation in the previous block. Cylindrical interpolation cutting point compensation is not performed according to a circular movement.

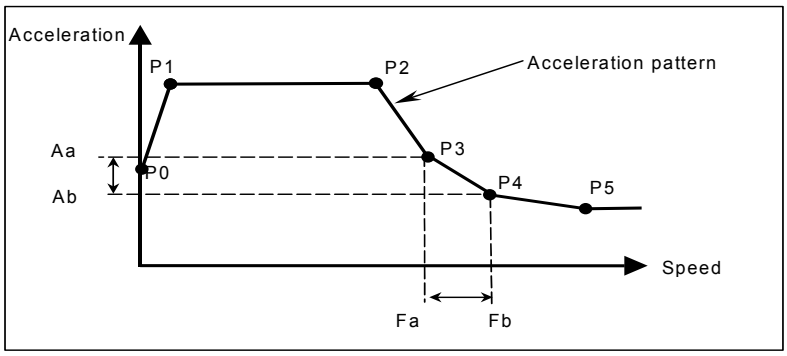
	#7	#6	#5	#4	#3	#2	#1	#0
19540								FAP

[Input type] Parameter input
[Data type] Bit path

 # 0 FAP Optimal torque acceleration/deceleration is:
0: Disabled.
1: Enabled.

When the linear positioning parameters, namely bit 1 (LRP) of parameter No. 1401 and bit 0 (FAP) of parameter No. 19540, are set to 1, and a value other than 0 is set in reference acceleration parameter (No. 1671) for an axis, the acceleration/deceleration for rapid traverse becomes optimal torque acceleration/deceleration in the mode for acceleration/deceleration before look-ahead interpolation (or the AI contour control mode). Optimal torque acceleration/ deceleration is controlled according to parameter-set restricted acceleration curve data.

Setting an acceleration pattern



Set the speed at each of the acceleration setting points (P0 to P5) in a corresponding parameter, then in parameters for each axis, set acceleration rates applicable in the following four cases at these speeds: when a movement in the positive direction is accelerated, when a movement in the positive direction is decelerated, when a movement in the negative direction is accelerated, and when a movement in the negative direction is decelerated.
The line connecting the acceleration setting points is the acceleration pattern.
The acceleration rate for each axis is calculated. For example, between speeds Fa to Fb in the above figure, the acceleration rates corresponding to these speeds, Aa to Ab, are used for calculation.
The tangent acceleration is controlled so that it does not exceed the calculated acceleration rate for each axis.

> ⚠ **CAUTION**
> When an acceleration pattern is set, setting a high acceleration rate immediately after a speed of 0 can cause an impact on the machine, so it is not desirable. Therefore, **be sure to apply a relatively low acceleration rate at a speed of 0** as shown in the above figure.

19541	Optimal torque acceleration/deceleration (speed at P1)
19542	Optimal torque acceleration/deceleration (speed at P2)
19543	Optimal torque acceleration/deceleration (speed at P3)
19544	Optimal torque acceleration/deceleration (speed at P4)

[Input type] Parameter input
[Data type] Word axis
[Unit of data] 0.01%
[Valid data range] 0 to 10000

The speeds at acceleration setting points P1 to P4 are to be set with speed parameters Nos. 19541 to 19544 as ratios to the rapid traverse speed (parameter No. 1420). The speed at P0 is 0, and the speed at P5 is the rapid traverse rate specified with parameter (No. 1420). Any acceleration setting point for which the speed parameter (one of Nos. 19541 to 19544) is set to 0 will be skipped.

19545	Optimal torque acceleration/deceleration (acceleration at P0 during movement in + direction and acceleration)
19546	Optimal torque acceleration/deceleration (acceleration at P1 during movement in + direction and acceleration)
19547	Optimal torque acceleration/deceleration (acceleration at P2 during movement in + direction and acceleration)
19548	Optimal torque acceleration/deceleration (acceleration at P3 during movement in + direction and acceleration)
19549	Optimal torque acceleration/deceleration (acceleration at P4 during movement in + direction and acceleration)
19550	Optimal torque acceleration/deceleration (acceleration at P5 during movement in + direction and acceleration)
19551	Optimal torque acceleration/deceleration (acceleration at P0 during movement in - direction and acceleration)
19552	Optimal torque acceleration/deceleration (acceleration at P1 during movement in - direction and acceleration)
19553	Optimal torque acceleration/deceleration (acceleration at P2 during movement in - direction and acceleration)
19554	Optimal torque acceleration/deceleration (acceleration at P3 during movement in - direction and acceleration)
19555	Optimal torque acceleration/deceleration (acceleration at P4 during movement in - direction and acceleration)

19556	Optimal torque acceleration/deceleration (acceleration at P5 during movement in - direction and acceleration)
19557	Optimal torque acceleration/deceleration (acceleration at P0 during movement in + direction and deceleration)
19558	Optimal torque acceleration/deceleration (acceleration at P1 during movement in + direction and deceleration)
19559	Optimal torque acceleration/deceleration (acceleration at P2 during movement in + direction and deceleration)
19560	Optimal torque acceleration/deceleration (acceleration at P3 during movement in + direction and deceleration)
19561	Optimal torque acceleration/deceleration (acceleration at P4 during movement in + direction and deceleration)
19562	Optimal torque acceleration/deceleration (acceleration at P5 during movement in + direction and deceleration)
19563	Optimal torque acceleration/deceleration (acceleration at P0 during movement in - direction and deceleration)
19564	Optimal torque acceleration/deceleration (acceleration at P1 during movement in - direction and deceleration)
19565	Optimal torque acceleration/deceleration (acceleration at P2 during movement in - direction and deceleration)
19566	Optimal torque acceleration/deceleration (acceleration at P3 during movement in - direction and deceleration)
19567	Optimal torque acceleration/deceleration (acceleration at P3 during movement in - direction and deceleration)
19568	Optimal torque acceleration/deceleration (acceleration at P5 during movement in - direction and deceleration)

[Input type] Parameter input
[Data type] Word axis
[Unit of data] 0.01%
[Valid data range] 0 to 32767

For each travel direction and each acceleration/deceleration operation, set the allowable acceleration rate at each of the acceleration setting points (P0 to P5). As an allowable acceleration rate, set a ratio to the value set in the reference acceleration parameter (No. 1671). When 0 is set, the specification of 100% is assumed.

19581	Tolerance smoothing for nano smoothing

[Input type] Setting input
[Data type] Real path
[Unit of data] mm, inch, degree (input unit)
[Minimum unit of data] Depend on the increment system of the applied axis
[Valid data range] 0 or positive 9 digit of minimum unit of data (refer to the standard parameter setting table (B))
(When the increment system is IS-B, 0.0 to +999999.999)
This parameter sets a tolerance value for a program created using miniature line segments in nano smoothing.
When 0 is set in this parameter, a minimum amount of travel in the increment system is regarded as a tolerance value.

19582	Minimum amount of travel of a block that makes a decision based on an angular difference between blocks for nano smoothing

[Input type] Setting input
[Data type] Real path
[Unit of data] mm, inch, degree (input unit)
[Minimum unit of data] Depend on the increment system of the reference axis
[Valid data range] 0 or positive 9 digit of minimum unit of data (refer to the standard parameter setting table (B))
(When the increment system is IS-B, 0.0 to +999999.999)
This parameter sets the minimum amount of travel of a block that makes a decision based on an angular difference between blocks for nano smoothing. A block that specifies an amount of travel less than the value set in this parameter makes no decision based on an angular difference.
When 0 is set in this parameter, a decision based on an angular difference is made with all blocks.
A value greater than the value set in parameter No. 8490 for making a decision based on the minimum travel distance of a block must be set.

19587	Tolerance of rotary axes for nano smoothing 2

[Input type] Setting input
[Data type] Real axis
[Unit of data] degree (input unit)
[Minimum unit of data] Depend on the increment system of the reference axis
[Valid data range] 0 or positive 9 digit of minimum unit of data (refer to the standard parameter setting table (B))
(When the increment system is IS-B, 0.0 - +999999.999)
This parameter sets the tolerance of rotation axes in a program created using small line segments in nano smoothing 2. This parameter is valid only for the rotation axes specified in nano smoothing 2.
When 0 is set in this parameter, a minimum amount of travel in the increment system is regarded as a tolerance value.

	#7	#6	#5	#4	#3	#2	#1	#0
19604								TPC

[Input type] Setting input
[Data type] Bit path

 # 5 TPC In the case that there is no address P at the start of tool center point control (G43.4/G43.5), tool posture control
0: Does not work.
1: Works.

	#7	#6	#5	#4	#3	#2	#1	#0
19605								NSC

[Input type] Parameter input
[Data type] Bit path

 # 0 NSC For the machine type that has no rotation axis for rotating the tool (when parameter No. 19680 is set to 12 to specify the table rotation type), control point shifting in the tilted working plane command is:
0: Enabled.
Set bit 4 (SPR) and bit 5 (SVC) of parameter No. 19665.
1: Disabled.

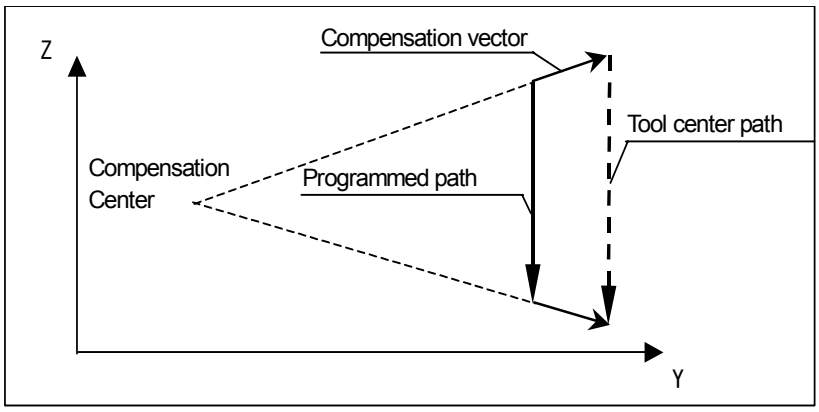
	#7	#6	#5	#4	#3	#2	#1	#0
19607			CAV				SPG	
			CAV		WCD		SPG	

[Input type] Parameter input
[Data type] Bit path

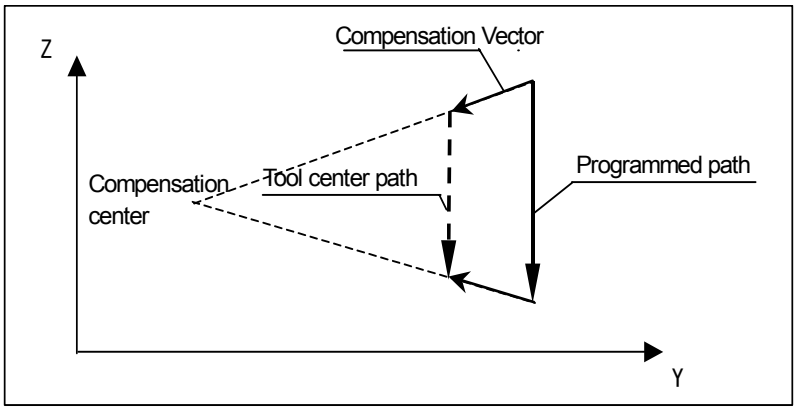
 # 1 SPG To apply 3-dimensional tool compensation to a machine having a table rotation axis, as the G code to be specified:
0: G41.2/G42.2 is used regardless of the machine type.
1: G41.4/G42.4 is used for a table rotation type machine; G41.5/G42.5 for a mixed type machine.

 # 3 WCD This parameter specify a direction of compensation vector by a sign of offset value in geinding-wheel wear compensation

		Offset vale by D code	
		Minus	Plus
19607#3	0	Direction from compensation center to command end position.	Direction from command end position to compensation center
	1	Direction from command end position to compensation center	Direction from compensation center to command end position.



Direction from compensation center to command end position



Direction from command end position to compensation center

 # 5 CAV When an interference check finds that interference (overcutting) occurred:

0: Machining stops with the alarm (PS0041).
(Interference check alarm function)

1: Machining is continued by changing the tool path to prevent interference (overcutting) from occurring. (Interference check avoidance function)

For the interference check method, see the descriptions of bit 1 (CNC) of parameter No. 5008 and bit 3 (CNV) of parameter No. 5008.

	#7	#6	#5	#4	#3	#2	#1	#0
19608		MIR	PRI			DET	NI5	

[Input type] Parameter input
[Data type] Bit path

#1 NI5 The interference check in 3-dimensional cutter compensation is performed by:

0: Projecting a look-ahead command position onto a plane perpendicular to the tool axis direction of a block for which compensation is planned.
The interference check avoidance function cannot be used.

1: Projecting a look-ahead command position onto a plane that is always perpendicular to the tool axis direction. When a table rotation axis is present, checking is made in the table coordinate system.
The interference check avoidance function can be used.

 # 2 DET When the programming coordinate system is fastened to the table in tool center point control or 3-dimensional tool compensation, the relative position and absolute position of a specified path are:

0: Displayed in the programming coordinate system (fastened to the table).

1: Displayed in the workpiece coordinate system (not fastened to the table).

 # 5 PRI Among multiple end point candidates that exist when a movement is made on a rotation axis by a command such as I, J, and K when a slanted surface machining command is specified under tool center point control (type 2) or 3-dimensional tool compensation (type 2):

0: A combination in which the master (first rotation axis) makes a smaller angular movement is selected for a machine of tool rotation type or table rotation type. A combination in which the table (second rotation axis) makes a smaller angular movement is selected for a machine of composite type.

1: A combination in which the slave (second rotation axis) makes a smaller angular movement is selected for a machine of tool rotation type or table rotation type. A combination in which the tool (first rotation axis) makes a smaller angular movement is selected for a machine of composite type.

 # 6 MIR When programmable mirror image is applied to a linear axis in tool center point control (type 2) or 3-dimensional tool compensation (type 2), mirror image is:

0: Not applied to a specified I, J, or K command

1: Applied to a specified I, J, or K command.

19631	Variation for determining an angle when the leading-edge offset function is performed

[Input type] Parameter input
[Data type] Real axis
[Unit of data] degree
[Minimum unit of data] Depend on the increment system of the reference axis
[Valid data range] 9 digit of minimum unit of data (refer to standard parameter setting table (A))
(When the increment system is IS-B, -999999.999 to +999999.999)

In the leading-edge offset function is performed, this parameter sets the range of variations applicable when the included angle of the tool direction vector (VT) and the advancing direction vector (VM) is determined to be 0°, 180°, or 90°.

For example, let the included angle between VT and VM be θ ($0 \leq \theta \leq 180$) and the angle set in this parameter be $\Delta\theta$. θ is then determined as follows:

If $0 \leq \theta \leq \Delta\theta$	$\theta = 0°$
$(180-\Delta\theta) \leq \theta \leq 180$	$\theta = 180°$
$(90-\Delta\theta) \leq \theta \leq (90+\Delta\theta)$	$\theta = 90°$

Normally, set a value around 1.0.

19632	Distance from a programmed point (pivot point) to tool center point position (cutting point)

[Input type] Parameter input
[Data type] Real axis
[Unit of data] mm, inch (input unit)
[Minimum unit of data] Depend on the increment system of the reference axis
[Valid data range] 9 digit of minimum unit of data (refer to standard parameter setting table (A))
(When the increment system is IS-B, -999999.999 to +999999.999)

This parameter sets the distance from a programmed point to the actual cutting point to perform vector calculation for 3-dimensional cutter compensation at the tool center point.

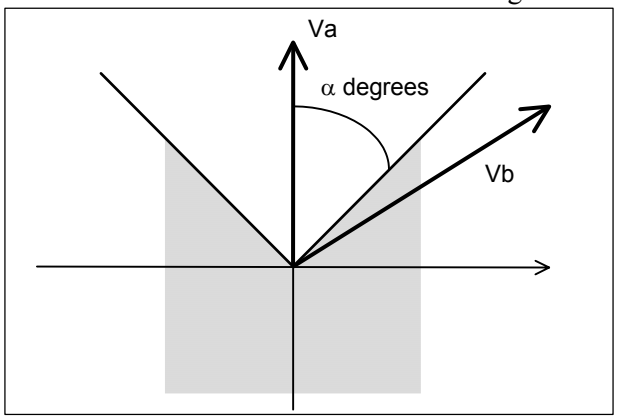
When this parameter is set to 0, the tool center point is not supported by the 3-dimensional cutter compensation function.

NOTE
This parameter can be rewritten only before the 3-dimensional cutter compensation mode is entered.

19635	Angle used as a criterion for the interference check in 3-dimensional cutter compensation

[Input type] Setting input
[Data type] Real axis
[Unit of data] degree
[Minimum unit of data] Depend on the increment system of the reference axis
[Valid data range] 9 digit of minimum unit of data (refer to standard parameter setting table (A))
(When the increment system is IS-B, -999999.999 to +999999.999)
In 3-dimensional cutter compensation, if the difference in angle between two tool direction vectors is equal to or greater than the setting of this parameter, the tool direction is determined to have been changed.
When 0 is set, 45 degrees is assumed.

Let two tool direction vectors be Va and Vb. Then, if the difference in angle is α degrees or greater as shown in the figure below, the tool direction vector is determined to have changed.



19636	Angle used to determine whether to execute the interference check/avoidance function of 3-dimensional tool compensation machining

[Input type] Setting input
[Data type] Real path
[Unit of data] deg
[Minimum unit of data] Depend on the increment system of the reference axis
[Valid data range] 9 digit of minimum unit of data (refer to standard parameter setting table (A))
The interference check/avoidance function of 3-dimensional tool compensation machining is executed when the angle difference between the tool direction vectors for the target two points is less than the setting.

This parameter is valid when bit 1 (NI5) of parameter No. 19608 is set to 1. When the setting is 0, the angle is assumed to be 10.0 degrees.

19658	Angular displacement of a rotation axis

[Input type] Parameter input
[Data type] Real axis
[Unit of data] deg
[Minimum unit of data] Depend on the increment system of the applied axis
[Valid data range] 9 digit of minimum unit of data (refer to standard parameter setting table (B))
(When the increment system is IS-B, -999999.999 to +999999.999)
This parameter sets the coordinate of a rotation axis, among the rotation axes determining the tool axis direction, which is not controlled by the CNC for the tool axis direction tool length compensation function. Whether this parameter is valid or invalid is determined by the setting of bit 1 (RAP) of parameter No. 19650.

19659	Offset value for the angular displacement of a rotation axis

[Input type] Parameter input
[Data type] Real axis
[Unit of data] deg
[Minimum unit of data] Depend on the increment system of the applied axis
[Valid data range] 9 digit of minimum unit of data (refer to standard parameter setting table (A))
(When the increment system is IS-B, -999999.999 to +999999.999)
An offset can be applied to the angular displacement for the tool axis direction tool length compensation function to compensate for the move direction.

19660	Origin offset value of a rotation axis

[Input type] Parameter input
[Data type] Real axis
[Unit of data] deg
[Minimum unit of data] Depend on the increment system of the applied axis
[Valid data range] 9 digit of minimum unit of data (refer to standard parameter setting table (A))
(When the increment system is IS-B, -999999.999 to +999999.999)
This parameter sets an angular displacement shifted from the origin for a rotation axis for the tool axis direction tool length compensation function.

19661	Rotation center compensation vector in tool axis direction tool length compensation

[Input type] Parameter input
[Data type] Real axis
[Unit of data] mm, inch (machine unit)
[Minimum unit of data] Depend on the increment system of the applied axis
[Valid data range] 9 digit of minimum unit of data (refer to standard parameter setting table (A))
(When the increment system is IS-B, -999999.999 to +999999.999)
This parameter sets the vector from the first rotation axis center to the second rotation axis center for the tool axis direction tool length compensation function.

19662	Spindle center compensation vector in tool axis direction tool length compensation

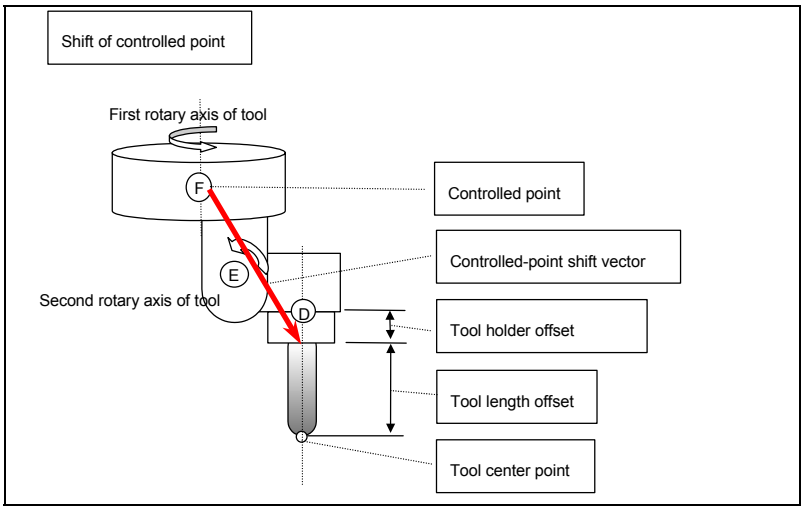
[Input type] Parameter input
[Data type] Real axis
[Unit of data] mm, inch (machine unit)
[Minimum unit of data] Depend on the increment system of the applied axis
[Valid data range] 9 digit of minimum unit of data (refer to standard parameter setting table (A))
(When the increment system is IS-B, -999999.999 to +999999.999)
This parameter sets the compensation vector for the spindle center for the tool axis direction tool length compensation function.

	#7	#6	#5	#4	#3	#2	#1	#0
19665			SVC	SPR				

[Input type] Parameter input
[Data type] Bit path

 # 4 SPR The controlled point is shifted by:
0: Automatic calculation.
1: Using parameter No. 19667.

Bit 5 (SVC) of parameter No. 19665	Bit 4 (SPR) of parameter No. 19665	Shift of controlled point
0	-	Shift is not performed as not done conventionally.
1	0	The controlled point is shifted according to the result of the following automatic calculation: - (Intersection offset vector between the tool axis and the first rotation axis of the tool + intersection offset vector between the second and first rotation axes of the tool + tool holder offset (parameter No. 19666)) (See the figure below.)
1	1	The controlled point is shifted. As the shift vector, the vector set in parameter No. 19667 is used.



[Controlled-point shift vector when automatically calculated]

 # 5 SVC The controlled point is:
0: Not shifted.
1: Shifted.
The method of shifting is specified by bit 4 (SPR) of parameter No. 19665.

NOTE
When the machine has no rotation axis for rotating the tool (when parameter No. 19680 is set to 12 to specify the table rotation type), the controlled point is not shifted regardless of the setting of this parameter.

19666	Tool holder offset value

[Input type] Parameter input
[Data type] Real path
[Unit of data] mm, inch (machine unit)
[Minimum unit of data] Depend on the increment system of the reference axis
[Valid data range] 9 digit of minimum unit of data (refer to standard parameter setting table (A))
(When the increment system is IS-B, -999999.999 to +999999.999)
When the tool axis direction tool length compensation function, tool tip center rotation handle feed/interrupt, and the display of the tool center point position are performed, specify the offset for the machine-specific section from the rotation center of the rotation axis to the tool mounting position (the tool holder offset value) in tool length compensation during tool center point control and tilted working plane command mode (after G53.1). For the tool axis direction tool length compensation function, the tool holder offset function can be enabled or disabled by setting bit 7 (ETH) of parameter No. 19665.

NOTE
Set a radius value.

19667	Controlled-point shift vector

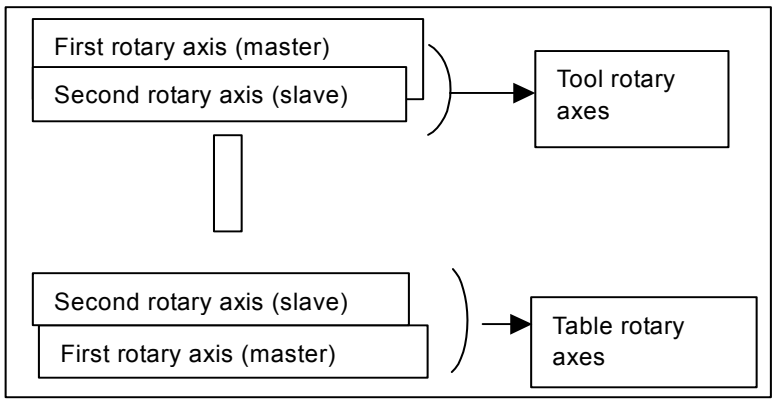
[Input type] Parameter input
[Data type] Real axis
[Unit of data] mm, inch (machine unit)
[Minimum unit of data] Depend on the increment system of the applied axis
[Valid data range] 9 digit of minimum unit of data (refer to standard parameter setting table (A))
(When the increment system is IS-B, -999999.999 to +999999.999)
Set the shift vector for the controlled point. This value becomes valid when bit 5 (SVC) of parameter No. 19665 is set to 1, and bit 4 (SPR) of parameter No. 19665 is set to 1.

NOTE
Set a radius value.

19680	Mechanical unit type

[Input type] Parameter input
[Data type] Byte path
[Valid data range] 0 to 21
Specify the type of the mechanical unit.

Parameter No. 19680	Mechanical unit type	Controlled rotation axis	Master and slave
0		Mechanism having no rotation axis	
2	Tool rotation type	Two rotation axes of the tool	The first rotation axis is the master, and the second rotation axis is the slave.
12	Table rotation type	Two rotation axes of the table	The first rotation axis is the master, and the second rotation axis is the slave.
21	Mixed type	One rotation axis of the tool + one rotation axis of the table	The first rotation axis is the tool rotation axis, and the second rotation axis is the table rotation axis.



NOTE
A hypothetical axis is also counted as a controlled rotary axis.
<Hypothetical axis>
In some cases, it is convenient to use an imaginary rotary axis whose angle is fixed to a certain value. For example, suppose that a tool is mounted in a tilted manner through an attachment. In such a case, the rotary axis considered hypothetically is a hypothetical axis. Bits 0 (IA1) and 1 (IA2) of parameter No. 19696 determine whether each rotary axis is an ordinary rotary axis or a hypothetical axis.

19681	Controlled-axis number for the first rotation axis

[Input type] Parameter input
[Data type] Byte path
[Valid data range] 0 to Number of controlled axes

Set the controlled-axis number for the first rotation axis.
For a hypothetical axis (when bit 0 (IA1) of parameter No. 19696 is 1), set 0.

Example)

Assuming that the axis configuration in path 1 is X,Y,Z,B,C and the axis configuration in path 2 is X,Z,C,Y,B, set the parameter to 5 in path 1 and to 3 in path 2 if C is the first rotation axis in each path.

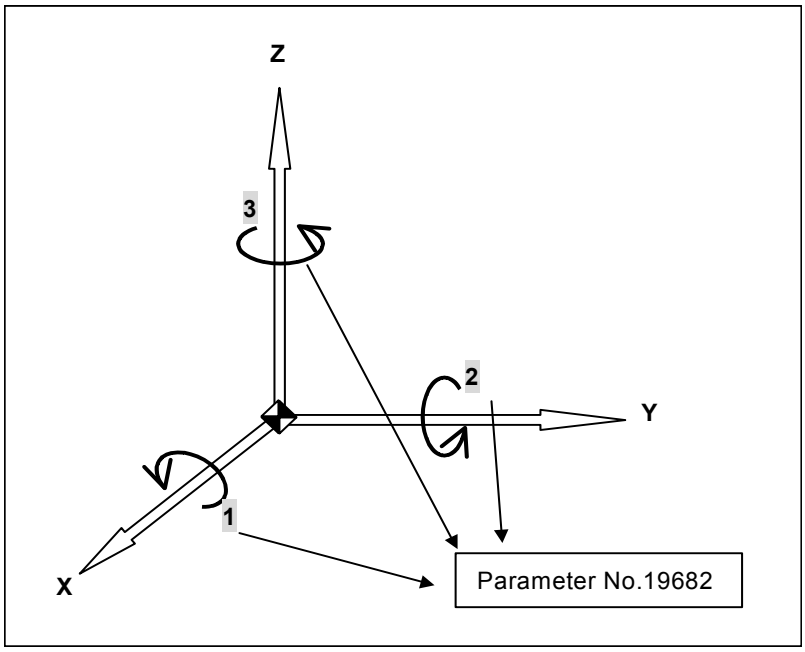
19682	Axis direction of the first rotation axis

[Input type] Parameter input
[Data type] Byte path
[Valid data range] 0 to 6

Specify the axis direction of the first rotation axis.

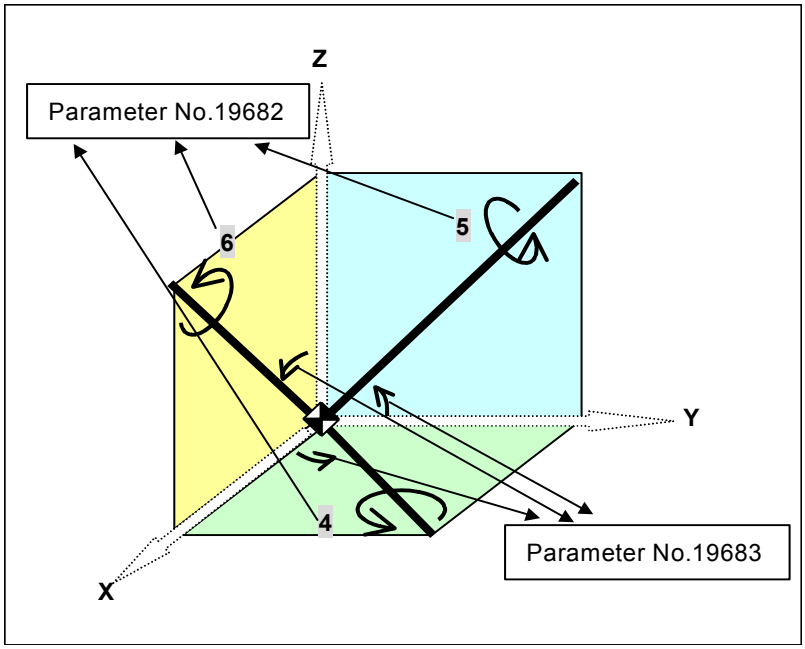
1: On X-axis
2: On Y-axis
3: On Z-axis
4: On an axis tilted a certain angle from the X-axis from the positive X-axis to positive Y-axis
5: On an axis tilted a certain angle from the Y-axis from the positive Y-axis to positive Z-axis
6: On an axis tilted a certain angle from the Z-axis from the positive Z-axis to positive X-axis

(A value 4 to 6 is to be set when the inclined rotation axis control function is used.)



19683	Inclination angle when the first rotation axis is an inclined axis

[Input type] Parameter input
[Data type] Real path
[Unit of data] Degree
[Minimum unit of data] The increment system of the reference axis is to be followed.
[Valid data range] Nine digits of the least input increment (see standard parameter setting table (A).)
(-999999.999 to +999999.999 for IS-B)
When a value 1 to 3 is set in parameter No. 19682, set 0 degrees.
When a value 4 to 6 is set in parameter No.19682, specify the inclination angle.



19684	Rotation direction of the first rotation axis

[Input type] Parameter input
[Data type] Byte path
[Valid data range] 0 to 1
Set the direction in which the first rotation axis rotates as a mechanical motion when a positive move command is issued.
0: Clockwise direction as viewed from the negative to positive direction of the axis specified in parameter No. 19682 (right-hand thread rotation)
1: Counterclockwise direction as viewed from the negative to positive direction of the axis specified in parameter No. 19682 (left-hand thread rotation)
Normally, 0 is set for a tool rotation axis, and 1 is set for a table rotation axis.

19685	Rotation angle when the first rotation axis is a hypothetical axis

[Input type] Parameter input
[Data type] Real path
[Unit of data] Degree
[Minimum unit of data] Depend on the increment system of the reference axis
[Valid data range] 9 digit of minimum unit of data (refer to standard parameter setting table (A))
(When the increment system is IS-B, -999999.999 to +999999.999)
When the first rotation axis is a hypothetical axis (bit 0 (IA1) of parameter No. 19696 is 1), set the rotation angle.

19686	Controlled-axis number for the second rotation axis

[Input type] Parameter input
[Data type] Byte path
[Valid data range] 0 to Number of controlled axes
Set the controlled-axis number for the second rotation axis.
For a hypothetical axis (bit 1 (IA2) of parameter No. 19696 is 1), set 0.
Example)
Assuming that the axis configuration in path 1 is X,Y,Z,B,C and the axis configuration in path 2 is X,Z,C,Y,B, set the parameter to 4 in path 1 and to 5 in path 2 if B is the second rotation axis in each path.

19687	Axis direction of the second rotation axis

[Input type] Parameter input
[Data type] Byte path
[Valid data range] 0 to 6
Specify the axis direction of the second rotation axis.
1: On X-axis
2: On Y-axis
3: On Z-axis
4: On an axis tilted a certain angle from the X-axis from the positive X-axis to positive Y-axis
5: On an axis tilted a certain angle from the Y-axis from the positive Y-axis to positive Z-axis
6: On an axis tilted a certain angle from the Z-axis from the positive Z-axis to positive X-axis
(A value 4 to 6 is to be set when the inclined rotation axis control function is used.)
When the second rotation axis is the slave axis, the direction when the master axis is at 0 degrees must be set.

19688	Inclination angle when the second rotation axis is inclined

[Input type] Parameter input
[Data type] Real path
[Unit of data] Degree
[Minimum unit of data] Depend on the increment system of the reference axis
[Valid data range] 9 digit of minimum unit of data (refer to standard parameter setting
(When the increment system is IS-B, -999999.999 to +999999.999)
If parameter No. 19687 is set to a value 1 to 3, set 0 degrees.
If parameter No. 19687 is set to a value 4 to 6, set the inclination angle.

19689	Rotation direction of the second rotation axis

[Input type] Parameter input
[Data type] Byte path
[Valid data range] 0 to 1
Set the direction in which the second rotation axis rotates as a mechanical motion when a positive move command is issued.
0: Clockwise direction as viewed from the negative to positive direction of the axis specified in parameter No. 19687 (right-hand thread rotation)
1: Counterclockwise direction as viewed from the negative to positive direction of the axis specified in parameter No. 19687 (left-hand thread rotation)
Normally, 0 is set for a tool rotation axis, and 1 is set for a table rotation axis.

19690	Rotation angle when the second rotation axis is a hypothetical axis

[Input type] Parameter input
[Data type] Real path
[Unit of data] Degree
[Minimum unit of data] Depend on the increment system of the reference axis
[Valid data range] 9 digit of minimum unit of data (refer to standard parameter setting table (A))
(When the increment system is IS-B, -999999.999 to +999999.999)
When the second rotation axis is a hypothetical axis (bit 1 (IA2) of parameter No. 19696 is 1), set the rotation angle.

	#7	#6	#5	#4	#3	#2	#1	#0
19696		RFC	WKP		NPC		IA2	IA1

[Input type] Parameter input
[Data type] Bit path

 # 0 IA1

0: The first rotation axis is an ordinary rotation axis.
1: The first rotation axis is a hypothetical axis.

If IA1 is 1, set 0 as the controlled-axis number for the first rotation axis (parameter No. 19681).
Also, set parameter Nos. 19682 to 19685 on the assumption that there is a rotation axis.

 # 1 IA2

0: The second rotation axis is an ordinary rotation axis.
1: The second rotation axis is a hypothetical axis.

If IA2 is 1, set 0 as the controlled-axis number for the second rotation axis (parameter No. 19686).
Also, set parameter Nos. 19687 to 19690 on the assumption that there is a rotation axis.

 # 3 NPC In tool posture control for tool center point control type II, when the change of tool posture at the block end is not done with the parameters (No.19738, 19739), even if the tool posture passes the singular posture,

0: Program is executed without the change of tool posture.
1: The alarm PS5421 occurs.

 # 5 WKP For a 5-axis machine having a table rotation axis, as the programming coordinate system for tool center point control or 3-dimensional tool compensation machining:

0: The table coordinate system (coordinate system fixed on the rotary table) is used.
1: The workpiece coordinate system is used.

NOTE
For 3-dimensional tool compensation machining, the setting of this parameter is used only when bit 4 (TBP) of parameter No. 19746 is set to 1.

 # 6 RFC In tool center point control, when a command that does not move the tool center point with respect to the workpiece is issued, the feedrate of the rotation axis is:

0: The maximum cutting feedrate (parameter No. 1422).
1: A specified feedrate.

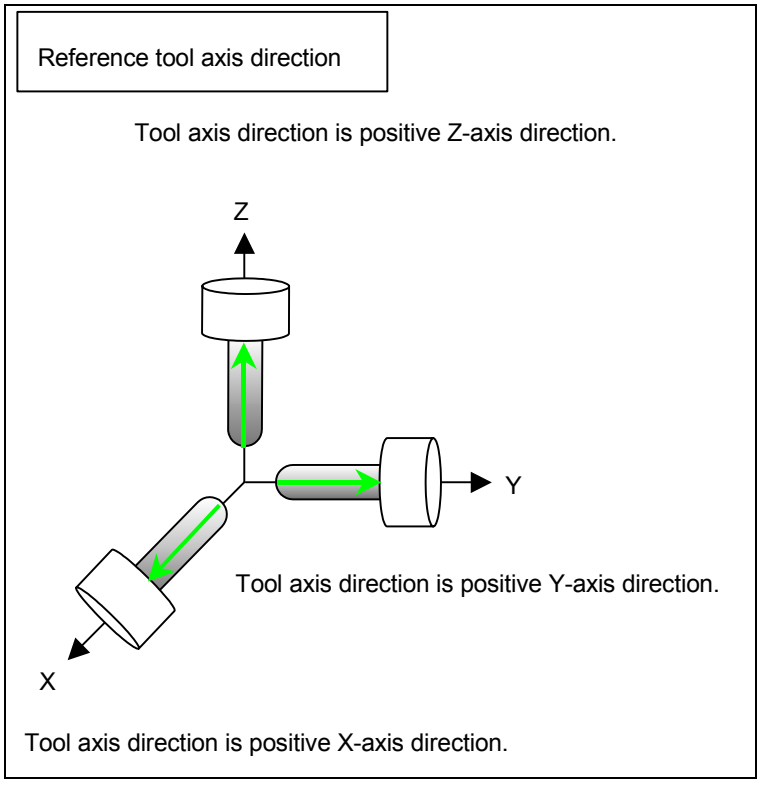
19697	Reference tool axis direction

[Input type] Parameter input
[Data type] Byte path
[Valid data range] 0 to 3

Set the tool axis direction in the machine coordinate system when the rotation axes for controlling the tool are all at 0 degrees. Also, set the tool axis direction in the machine coordinate system in a mechanism in which only the rotation axes for controlling the table are present (there is no rotation axis for controlling the tool).

1: Positive X-axis direction
2: Positive Y-axis direction
3: Positive Z-axis direction

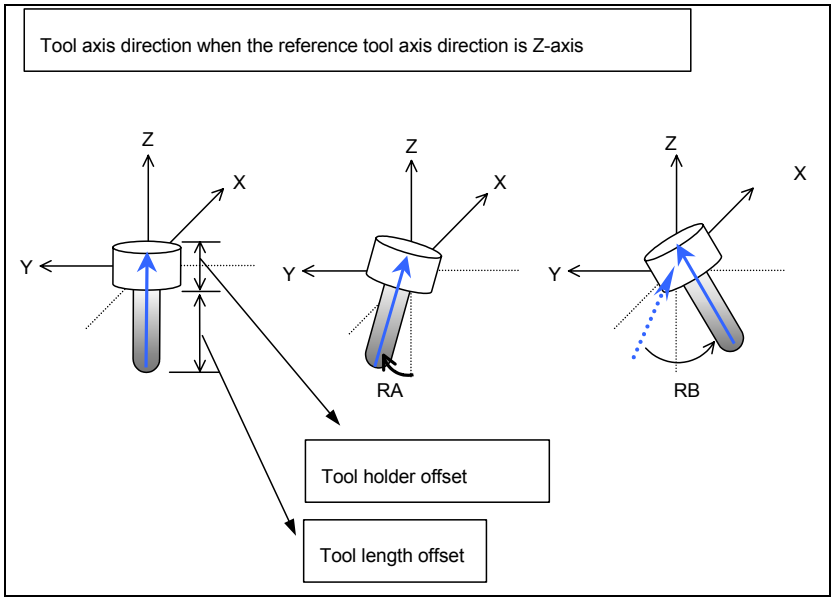
When the reference tool axis direction is neither the X-, Y-, nor Z-axis direction, set the reference direction in this parameter, then set appropriate angles as the reference angle RA and reference angle RB (parameter Nos. 19698 and 19699).



19698	Angle when the reference tool axis direction is tilted (reference angle RA)

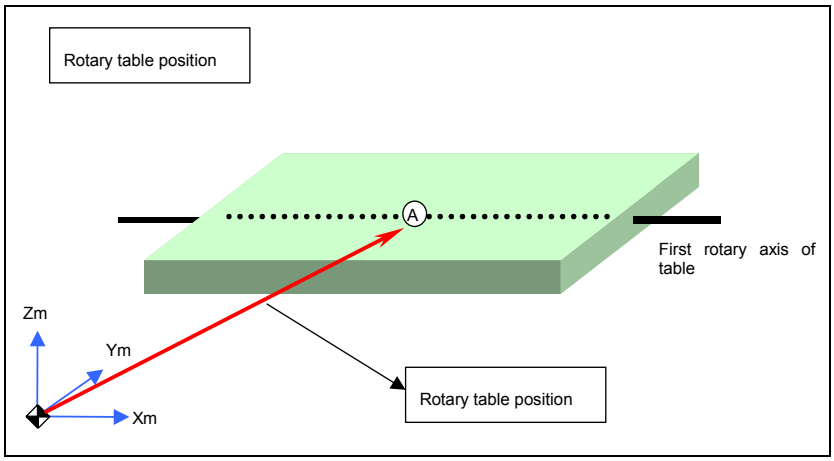
19699	Angle when the reference tool axis direction is tilted (reference angle RB)

[Input type] Parameter input
[Data type] Real path
[Unit of data] Degree
[Minimum unit of data] Depend on the increment system of the reference axis
[Valid data range] 9 digit of minimum unit of data (refer to standard parameter setting
(When the increment system is IS-B, -999999.999 to +999999.999)
When the reference tool axis direction (parameter No. 19697) is set to 1, the tool axis is tilted the RA degrees on the Z-axis from the positive X-axis direction to positive Y-axis direction, then the tool axis is tilted the RB degrees on the X-axis from the positive Y-axis direction to positive Z-axis direction.
When the reference tool axis direction (parameter No. 19697) is set to 2, the tool axis is tilted the RA degrees on the X-axis from the positive Y-axis direction to positive Z-axis direction, then the tool axis is tilted the RB degrees on the Y-axis from the positive Z-axis direction to positive X-axis direction.
When the reference tool axis direction (parameter No. 19697) is set to 3, the tool axis is tilted the RA degrees on the Y-axis from the positive Z-axis direction to positive X-axis direction, then the tool axis is tilted the RB degrees on the Z-axis from the positive X-axis direction to positive Y-axis direction.



19700	Rotary table position (X-axis of the basic three axes)
19701	Rotary table position (Y-axis of the basic three axes)
19702	Rotary table position (Z-axis of the basic three axes)

[Input type] Parameter input
[Data type] Real path
[Unit of data] mm, inch (machine unit)
[Minimum unit of data] Depend on the increment system of the applied axis
[Valid data range] 9 digit of minimum unit of data (refer to standard parameter setting table (A))
(When the increment system is IS-B, -999999.999 to +999999.999)
Set these parameters when parameter No. 19680 is set to 12 or 21. The vector from the origin of the machine coordinate system to point A on the first rotation axis of the table is set as the rotary table position in the machine coordinate system.

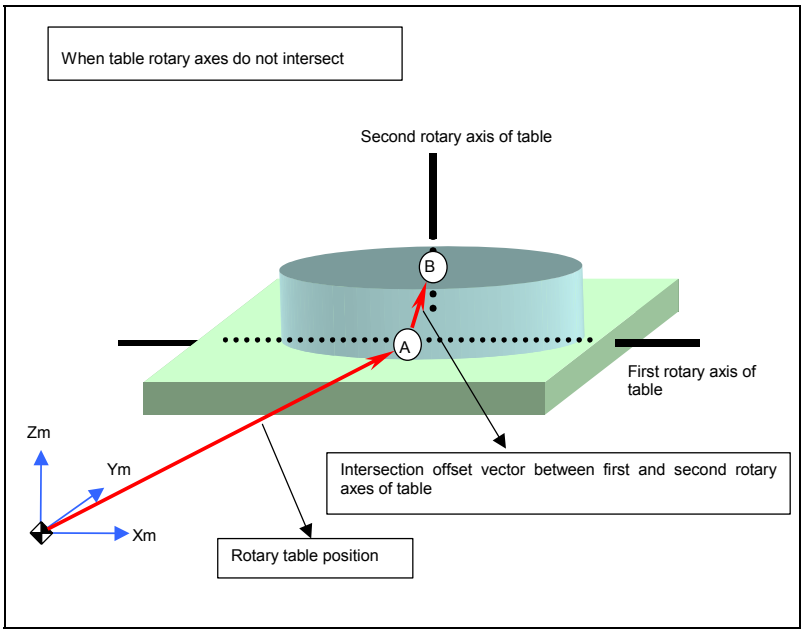


NOTE
As point A, set a position that is easy to measure on the first rotary axis of the table.
Set a radius value.
If the rotary table is moved along the X-, Y-, or Z-axis or all of these axes, set the position of the rotary table when the machine coordinates of the X-, Y-, and Z-axes are all set to 0.

19703	**Intersection offset vector between the first and second rotation axes of the table (X-axis of the basic three axes)**
19704	**Intersection offset vector between the first and second rotation axes of the table (Y-axis of the basic three axes)**
19705	**Intersection offset vector between the first and second rotation axes of the table (Z-axis of the basic three axes)**

[Input type] Parameter input
[Data type] Real path
[Unit of data] mm, inch (machine unit)
[Minimum unit of data] Depend on the increment system of the applied axis
[Valid data range] 9 digit of minimum unit of data (refer to standard parameter setting table (A))
(When the increment system is IS-B, -999999.999 to +999999.999)

Set these parameters when the first rotation axis and second rotation axis of the table do not intersect. These parameters are valid when parameter No. 19680 is set to 12. When the rotation axes for controlling the table are all at 0 degrees, the vector from point A to point B on the second rotation axis of the table is set as the intersection offset vector in the machine coordinate system.

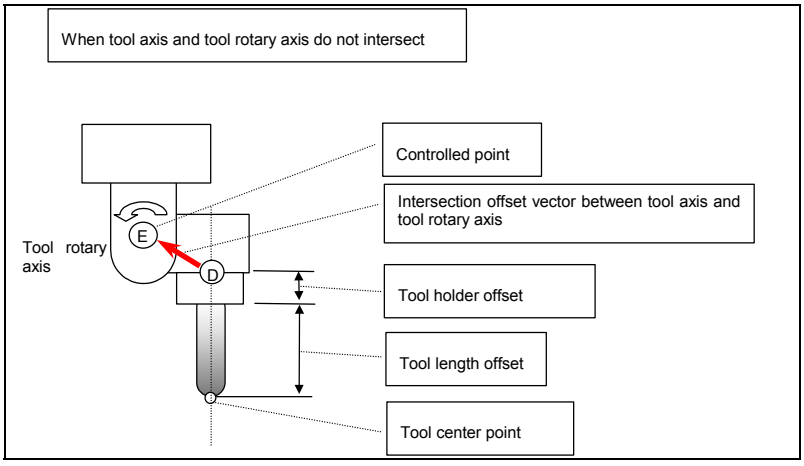


NOTE
As point B, set a position that is easy to measure on the second rotary axis of the table.
Set a radius value.

19709	**Intersection offset vector between the tool axis and tool rotation axis (X-axis of the basic three axes)**
19710	**Intersection offset vector between the tool axis and tool rotation axis (Y-axis of the basic three axes)**
19711	**Intersection offset vector between the tool axis and tool rotation axis (Z-axis of the basic three axes)**

[Input type] Parameter input
[Data type] Real path
[Unit of data] mm, inch (machine unit)
[Minimum unit of data] Depend on the increment system of the applied axis
[Valid data range] 9 digit of minimum unit of data (refer to standard parameter setting table (A))
(When the increment system is IS-B, -999999.999 to +999999.999)

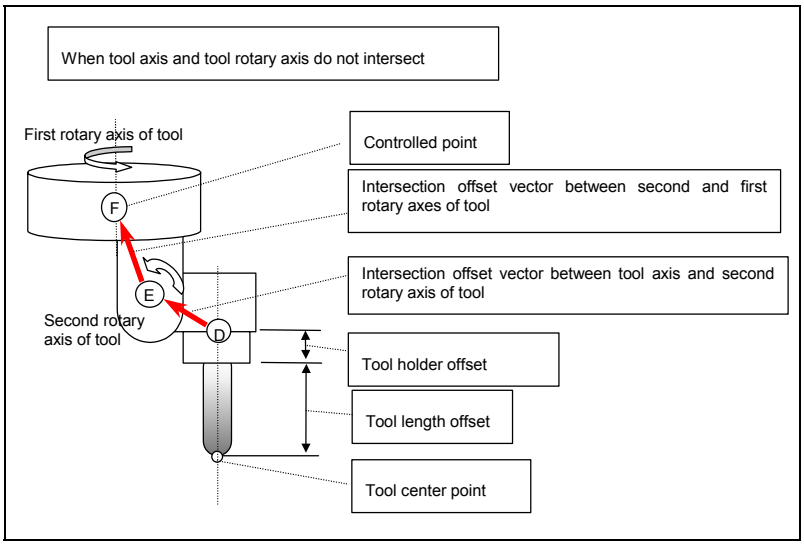
Set these parameters when the tool axis and tool rotation axis do not intersect.
These parameters are valid when parameter No. 19680 is set to 2 or 21.
If parameter No. 19680 is 21, set the vector from point D on the tool axis to point E determined on the tool rotation axis as the intersection offset vector in the machine coordinate system when the rotation axes for controlling the tool are all at 0 degrees.
If parameter No. 19680 is 2, set the vector from point D on the tool axis to point E determined on the second rotation axis of the tool as the intersection offset vector in the machine coordinate system when the rotation axes for controlling the tool are all at 0 degrees.



NOTE
Point D is determined by adding the tool length offset and tool holder offset (parameter No. 19666) to the tool tip. As point E, set a position that is easy to measure.
Set a radius value.

19712	**Intersection offset vector between the second and first rotation axes of the tool (X-axis of the basic three axes)**
19713	**Intersection offset vector between the second and first rotation axes of the tool (Y-axis of the basic three axes)**
19714	**Intersection offset vector between the second and first rotation axes of the tool (Z-axis of the basic three axes)**

[Input type] Parameter input
[Data type] Real path
[Unit of data] mm, inch (machine unit)
[Minimum unit of data] Depend on the increment system of the applied axis
[Valid data range] 9 digit of minimum unit of data (refer to standard parameter setting table (A))
(When the increment system is IS-B, -999999.999 to +999999.999)
Set these parameters when the rotation axes of the tool do not intersect.
These parameters are valid when parameter No. 19680 is set to 2.
Set the vector from point E on the second rotation axis of the tool to point F on the first rotation axis of the tool as the intersection offset vector in the machine coordinate system when the rotation axes for controlling the tool are all at 0 degrees.



NOTE
As point F, set a position that is easy to measure.
Set a radius value.

19738	Angle to check if Tool posture is near Singular posture or not

[Input type] Parameter input
[Data type] Real path
[Unit of data] degree
[Minimum unit of data] Depend on the increment system of the reference axis
[Valid data range] 9 digit of minimum unit of data (refer to standard parameter setting table(A))
(When the increment system is IS-B, -999999.999 to +999999.999)
In the case of tool posture control for tool center point control (type II), when the difference(angle) between a tool posture and the singular posture is less than this parameter, the tool posture is regarded as 'near the singular posture'. Then, the tool posture at the block end is changed so that the tool posture passes the singular posture during the block.
If this parameter is 0.0, the tool posture at the block end is not changed.

19739	Angle to decide that the tool posture at block end is not changed

[Input type] Parameter input
[Data type] Real path
[Unit of data] degree
[Minimum unit of data] Depend on the increment system of the reference axis
[Valid data range] 9 digit of minimum unit of data (refer to standard parameter setting table(A))
(When the increment system is IS-B, -999999.999 to +999999.999)
When an appropriate value is set in parameter No. 19738 in tool posture control for tool center point control (type II), a tool posture near the singular point may occur during the execution of a block. If this happens, change the tool posture at the end point so that the singular posture is passed within the block. With respect to the angle of the rotation axis nearer to the workpiece before and after the tool posture is changed (the rotation axis is the master axis when tehe tool turns, the slave axis when the table turns, or the table roration axis when the rotation is of the mixed type), however, the tool posture must not be changed if both the difference between the angle after the change and (angle before the change + 180 degrees) and the difference between the angle after the change and (angle before the change − 180 degrees) are equal to or greater than the value set in the parameter,

19741 **Upper limit of the movement range of the first rotation axis**

[Input type] Parameter input
[Data type] Real path
[Unit of data] Degree
[Minimum unit of data] Depend on the increment system of the reference axis
[Valid data range] 9 digit of minimum unit of data (refer to standard parameter setting
(When the increment system is IS-B, -999999.999 to +999999.999)
This parameter sets the upper limit of the movement range of the first rotation axis in tool center point control (type 2), 3-dimensional cutter compensation (type 2), or tool axis direction control of the tilted working plane command (G53.1). When the movement range of the first rotation axis is not specified, this parameter and parameter No. 19742 must both be set to 0.

19742 **Lower limit of the movement range of the first rotation axis**

[Input type] Parameter input
[Data type] Real path
[Unit of data] Degree
[Minimum unit of data] Depend on the increment system of the reference axis
[Valid data range] 9 digit of minimum unit of data (refer to standard parameter setting
(When the increment system is IS-B, -999999.999 to +999999.999)
This parameter sets the lower limit of the movement range of the first rotation axis in tool center point control (type 2), 3-dimensional cutter compensation (type 2), or tool axis direction control of the tilted working plane command (G53.1). When the movement range of the first rotation axis is not specified, this parameter and parameter No. 19741 must both be set to 0.

19743 **Upper limit of the movement range of the second rotation axis**

[Input type] Parameter input
[Data type] Real path
[Unit of data] Degree
[Minimum unit of data] Depend on the increment system of the reference axis
[Valid data range] 9 digit of minimum unit of data (refer to standard parameter setting
(When the increment system is IS-B, -999999.999 to +999999.999)
This parameter sets the upper limit of the movement range of the second rotation axis in tool center point control (type 2), 3-dimensional cutter compensation (type 2), or tool axis direction control of the tilted working plane command (G53.1). When the movement range of the second rotation axis is not specified, this parameter and parameter No. 19744 must both be set to 0.

19744	Lower limit of the movement range of the second rotation axis

[Input type] Parameter input
[Data type] Real path
[Unit of data] Degree
[Minimum unit of data] Depend on the increment system of the reference axis
[Valid data range] 9 digit of minimum unit of data (refer to standard parameter setting
(When the increment system is IS-B, -999999.999 to +999999.999)
This parameter sets the lower limit of the movement range of the second rotation axis in tool center point control (type 2), 3-dimensional cutter compensation (type 2), or tool axis direction control of the tilted working plane command (G53.1). When the movement range of the second rotation axis is not specified, this parameter and parameter No. 19743 must both be set to 0.

	#7	#6	#5	#4	#3	#2	#1	#0
19746		CRS		TBP	LOZ	LOD	PTD	

[Input type] Parameter input
[Data type] Bit path

 # 1 PTD When 3-dimensional cutter compensation is performed for a table rotation type machine, the direction of the tool is:
0: Specified by parameter Nos. 19697, 19698, and 19699.
1: Specified as a direction perpendicular to the plane specified by G17, G18, or G19.

 # 2 LOD As the tool length for three-dimensional machining manual feed:
0: The value of parameter No. 12318 is used.
1: The tool length currently used for tool length compensation is used.

 # 3 LOZ When bit 2 (LOD) of parameter No. 19746 is set to 1 and tool length compensation is not applied, as the tool length for three-dimensional machining manual feed:
0: The value of parameter No. 12318 is used.
1: 0 is used.

 # 4 TBP For a 5-axis machine having a table rotation axis, as the programming coordinate system for 3-dimensional tool compensation machining:
0: The workpiece coordinate system is used.
1: The setting of bit 5 (WKP) of parameter No. 19696 is used.

 # 6 CRS In tool center point control, when the deviation from the path during movement at the specified cutting feedrate or rapid traverse rate is determined to exceed the limit:

0: The feedrate or rapid traverse rate is not decreased.

1: The feedrate or rapid traverse rate is controlled so that the limit of the deviation from the path set in the parameter for the cutting feed or rapid traverse is not exceeded.

When this parameter is set to 1:

In the rapid traverse mode, the rapid traverse rate is decreased so that the deviation from the path does not exceed the limit specified in parameter No. 19751.

In the cutting feed mode, the cutting feedrate is decreased so that the deviation from the path does not exceed the limit specified in parameter No. 19752.

19751	Limit of the deviation from the path (for rapid traverse)

[Input type] Parameter input
[Data type] Real path
[Unit of data] mm, inch (machine unit)
[Minimum unit of data] Depend on the increment system of the reference axis
[Valid data range] 9 digit of minimum unit of data (refer to standard parameter setting table (A))
(When the increment system is IS-B, -999999.999 to +999999.999)

This parameter sets the limit of the deviation from the path in the rapid traverse mode in tool center point control.

If the tool moves at the specified rate, the deviation from the path may exceed the value specified in this parameter. In this case, the rate is decreased so that the tool moves along the path.

This parameter is valid when bit 6 (CRS) of parameter No. 19746 is set to 1.

When 0 is set, the least input increment is assumed to be the limit of the deviation from the path.

If a negative value is set, the rapid traverse rate is not decreased.

NOTE
The error generated after the rate is decreased may be smaller than the value set in this parameter depending on the calculation error.

19752	Limit of the deviation from the path (for cutting feed)

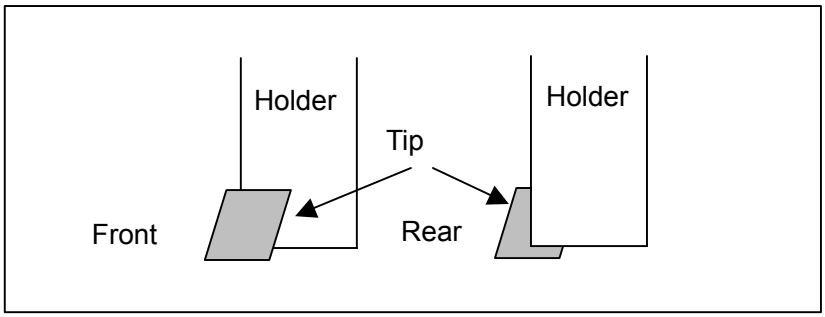
[Input type] Parameter input
[Data type] Real path
[Unit of data] mm, inch (machine unit)
[Minimum unit of data] Depend on the increment system of the reference axis
[Valid data range] 9 digit of minimum unit of data (refer to standard parameter setting table (A))
(When the increment system is IS-B, -999999.999 to +999999.999)
This parameter sets the limit of the deviation from the path in the cutting feed mode in tool center point control.
If the tool moves at the specified rate, the deviation from the path may exceed the value specified in this parameter. In this case, the rate is decreased so that the tool moves along the path.
This parameter is valid when bit 6 (CRS) of parameter No. 19746 is set to 1.
When 0 is set, the least input increment is assumed to be the limit of the deviation from the path.
If a negative value is set, the cutting feedrate is not decreased.

NOTE
The error generated after the rate is decreased may be smaller than the value set in this parameter depending on the calculation error.

	#7	#6	#5	#4	#3	#2	#1	#0
27350								GTP

[Input type] Parameter input
[Data type] Bit

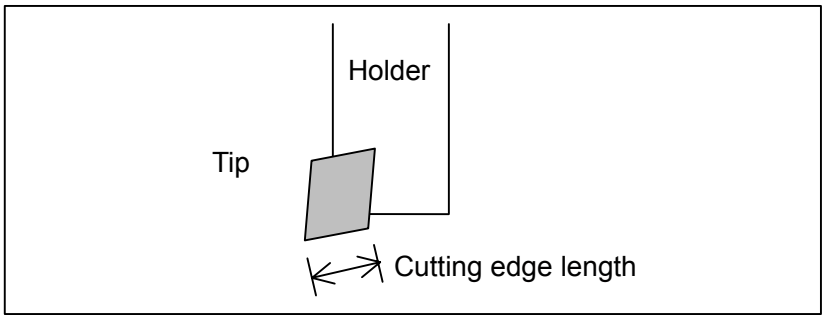
 # 0 GTP When a general-purpose tool is drawn in animated simulation, the tip is:
0: Positioned on the front.
1: Positioned on the rear.



27351	Cutting edge length applied when a general-purpose tool is drawn in animated simulation

[Input type] Parameter input
[Data type] 2-word
[Unit of data] 0.001mm (metric input), 0.0001inch (inch input)
[Valid data range] 0 or larger

This parameter sets the cutting edge length applied when a general-purpose tool is drawn in animated simulation.

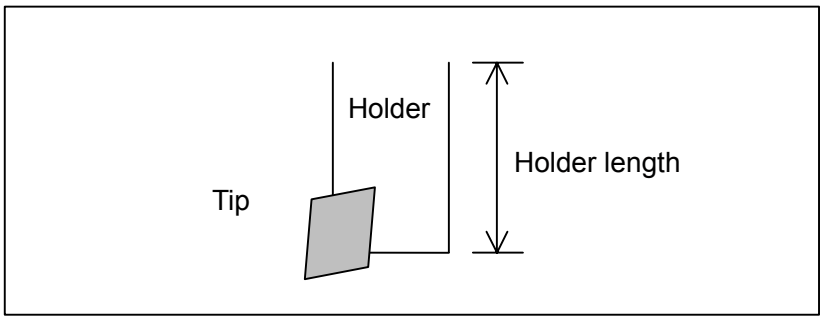


When 0 is set, 12 mm for metric input or 0.4724 inch for inch input is assumed.

27352	Holder length applied when a general-purpose tool is drawn in animated simulation

[Input type] Parameter input
[Data type] 2-word
[Unit of data] 0.001mm (metric input), 0.0001inch (inch input)
[Valid data range] 0 or larger

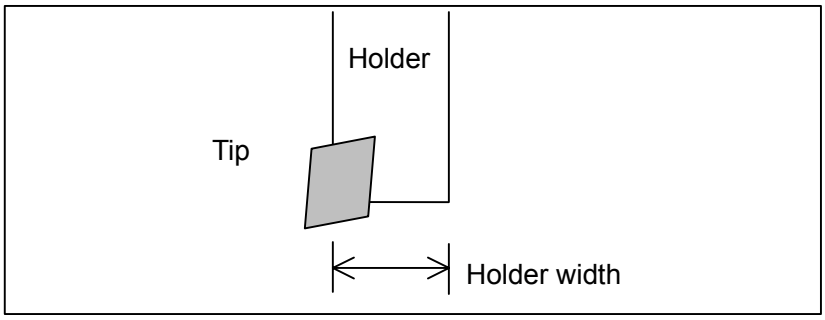
This parameter sets the holder length applied when a general-purpose tool is drawn in animated simulation.



When 0 is set, 50 mm for metric input or 1.9685 inch for inch input is assumed.

27353	Holder width applied when a general-purpose tool is drawn in animated simulation

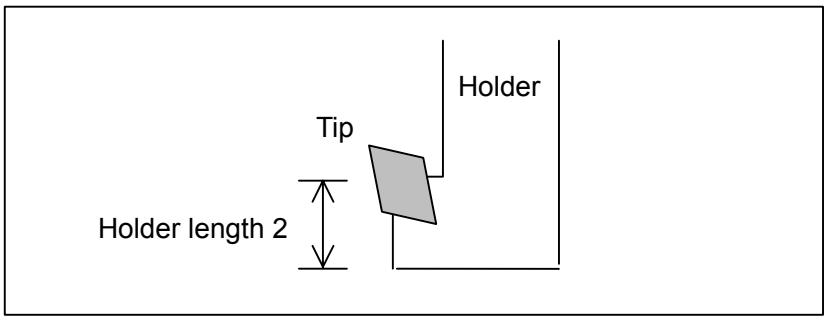
[Input type] Parameter input
[Data type] 2-word
[Unit of data] 0.001mm (metric input), 0.0001inch (inch input)
[Valid data range] 0 or larger
This parameter sets the holder width applied when a general-purpose tool is drawn in animated simulation.



When 0 is set, 14 mm for metric input or 0.5512 inch for inch input is assumed.

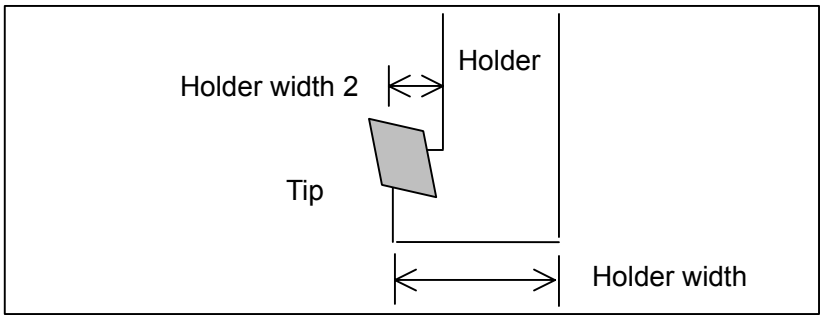
27354	Holder length 2 applied when a general-purpose tool is drawn in animated simulation

[Input type] Parameter input
[Data type] 2-word
[Unit of data] 0.001mm (metric input), 0.0001inch (inch input)
[Valid data range] 0 or larger
This parameter sets the holder length 2 applied when a general-purpose tool is drawn in animated simulation.



27355	Holder width 2 applied when a general-purpose tool is drawn in animated simulation

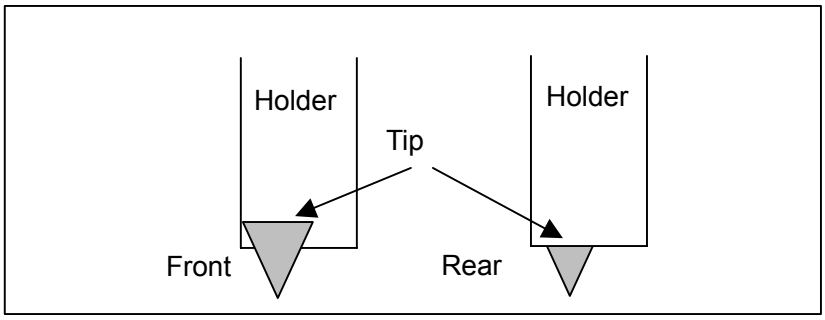
[Input type] Parameter input
[Data type] 2-word
[Unit of data] 0.001mm (metric input), 0.0001inch (inch input)
[Valid data range] 0 or larger
This parameter sets the holder width 2 applied when a general-purpose tool is drawn in animated simulation.



	#7	#6	#5	#4	#3	#2	#1	#0
27356								TTP

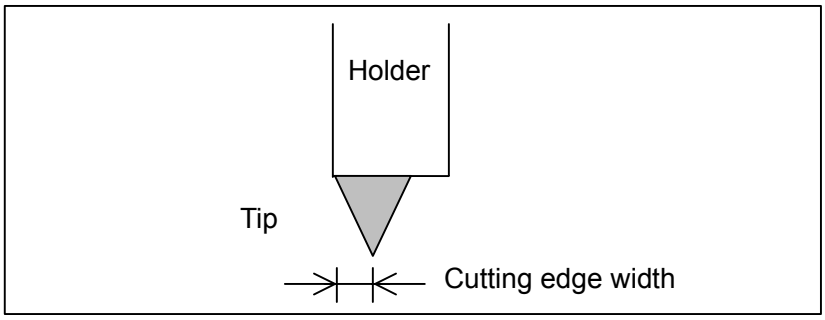
[Input type] Parameter input
[Data type] Bit

 # 0 TTP When a threading tool is drawn in animated simulation, the tip is:
0: Positioned on the front.
1: Positioned on the rear.



27357	Cutting edge width applied when a threading tool is drawn in animated simulation

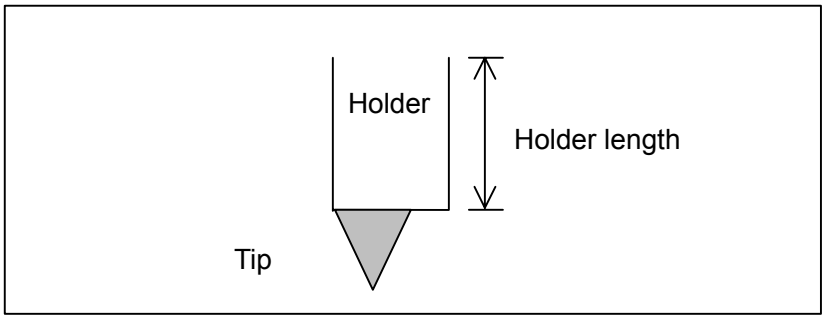
[Input type] Parameter input
[Data type] 2-word
[Unit of data] 0.001mm (metric input), 0.0001inch (inch input)
[Valid data range] 0 or larger
This parameter sets the cutting edge width applied when a threading tool is drawn in animated simulation.



When 0 is set, 3 mm for metric input or 0.11811 inch for inch input is assumed.

27358	Holder length applied when a threading tool is drawn in animated simulation

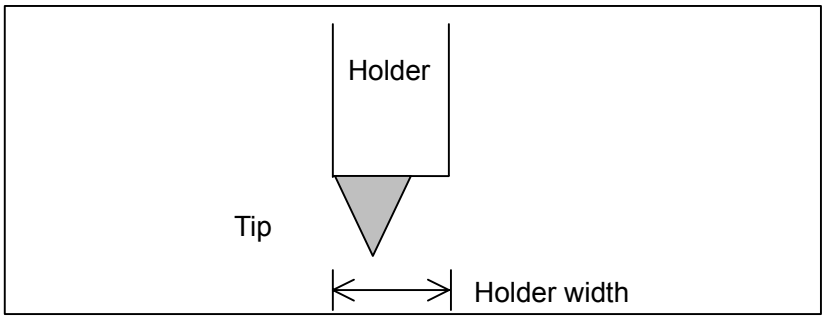
[Input type] Parameter input
[Data type] 2-word
[Unit of data] 0.001mm (metric input), 0.0001inch (inch input)
[Valid data range] 0 or larger
This parameter sets the holder length applied when a threading tool is drawn in animated simulation.



When 0 is set, 50 mm for metric input or 1.9685 inch for inch input is assumed.

27359	Holder width applied when a threading tool is drawn in animated simulation

[Input type] Parameter input
[Data type] 2-word
[Unit of data] 0.001mm (metric input), 0.0001inch (inch input)
[Valid data range] 0 or larger
This parameter sets the holder width applied when a threading tool is drawn in animated simulation.

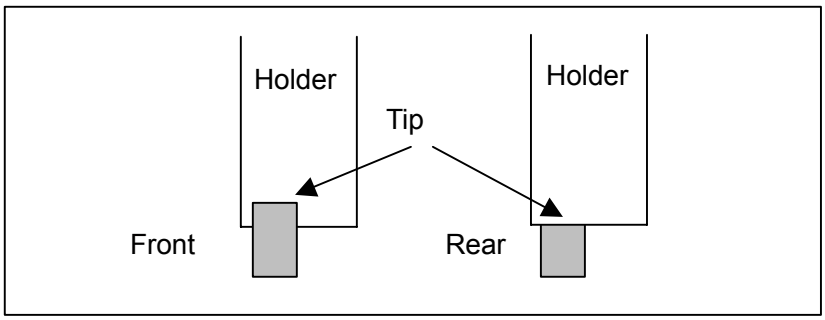


When 0 is set, 14 mm for metric input or 0.5512 inch for inch input is assumed.

	#7	#6	#5	#4	#3	#2	#1	#0
27360								GVP

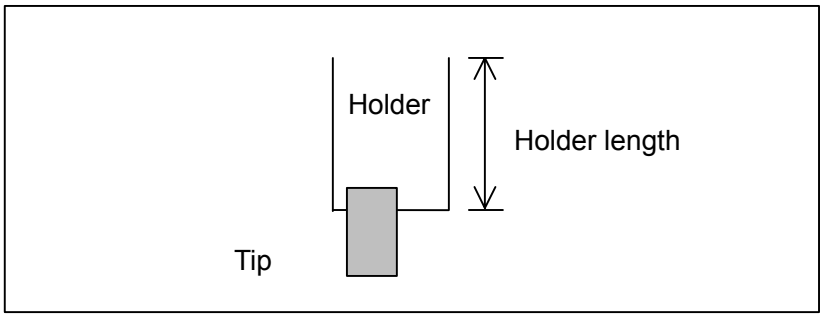
[Input type] Parameter input
[Data type] Bit

 # 0 GVP When a groove cutting tool is drawn in animated simulation, the tip is:
0: Positioned on the front.
1: Positioned on the rear.



27361	Holder length applied when a groove cutting tool is drawn in animated simulation

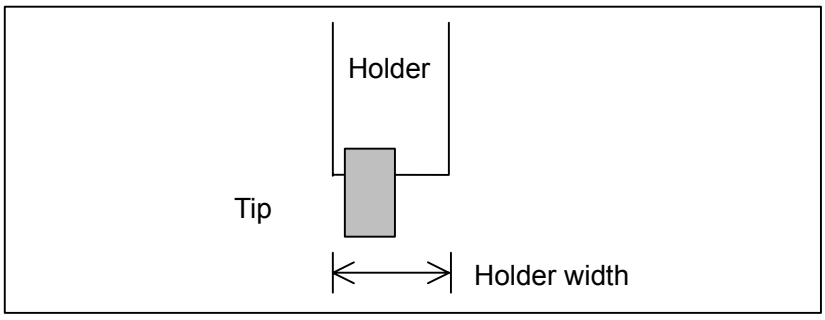
[Input type] Parameter input
[Data type] 2-word
[Unit of data] 0.001mm (metric input), 0.0001inch (inch input)
[Valid data range] 0 or larger
This parameter sets the holder length applied when a groove cutting tool is drawn in animated simulation.



When 0 is set, 50 mm for metric input or 1.9685 inch for inch input is assumed.

27362	Holder width applied when a groove cutting tool is drawn in animated simulation

[Input type] Parameter input
[Data type] 2-word
[Unit of data] 0.001mm (metric input), 0.0001inch (inch input)
[Valid data range] 0 or larger
This parameter sets the holder width applied when a groove cutting tool is drawn in animated simulation.

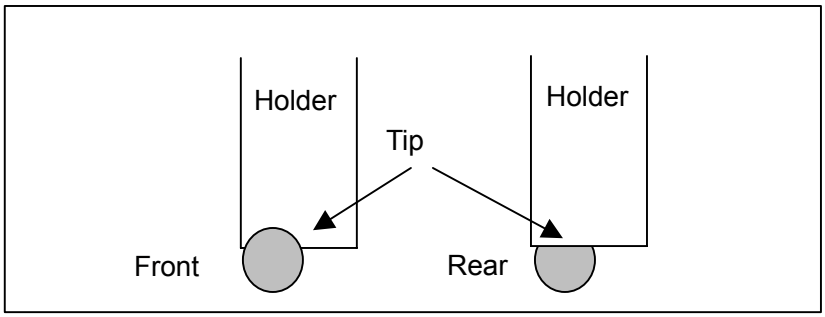


When 0 is set, 14 mm for metric input or 0.5512 inch for inch input is assumed.

	#7	#6	#5	#4	#3	#2	#1	#0
27363								BTP

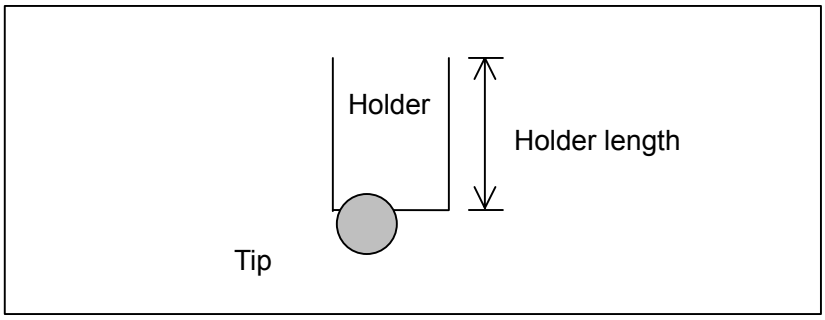
[Input type] Parameter input
[Data type] Bit

 # 0 BTP When a round-nose tool is drawn in animated simulation, the tip is:
0: Positioned on the front.
1: Positioned on the rear.



27364	Holder length applied when a round-nose tool is drawn in animated simulation

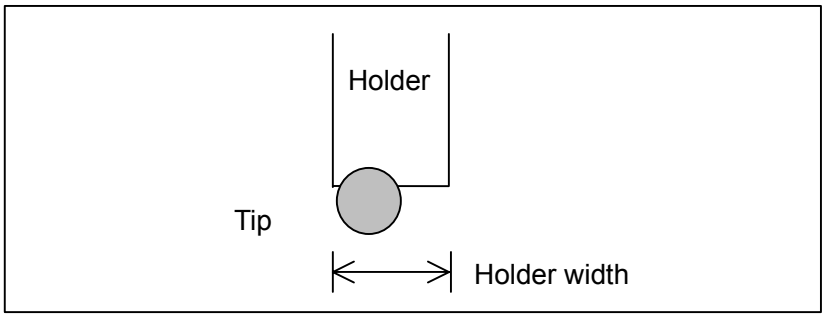
[Input type] Parameter input
[Data type] 2-word
[Unit of data] 0.001mm (metric input), 0.0001inch (inch input)
[Valid data range] 0 or larger
This parameter sets the holder length applied when a round-nose tool is drawn in animated simulation.



When 0 is set, 50 mm for metric input or 1.9685 inch for inch input is assumed.

27365	Holder width applied when a round-nose tool is drawn in animated simulation

[Input type] Parameter input
[Data type] 2-word
[Unit of data] 0.001mm (metric input), 0.0001inch (inch input)
[Valid data range] 0 or larger
This parameter sets the holder width applied when a round-nose tool is drawn in animated simulation.

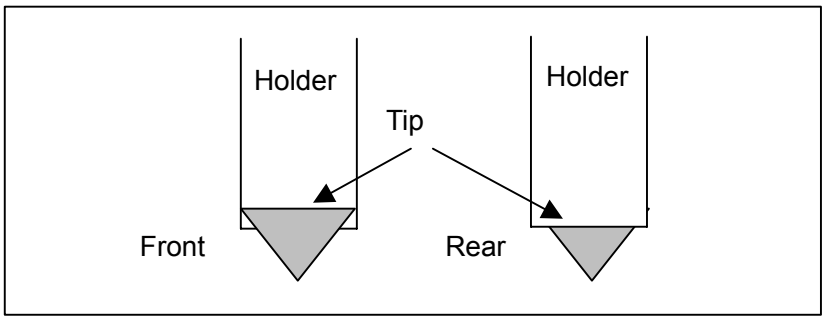


When 0 is set, 14 mm for metric input or 0.5512 inch for inch input is assumed.

	#7	#6	#5	#4	#3	#2	#1	#0
27366								STP

[Input type] Parameter input
[Data type] Bit

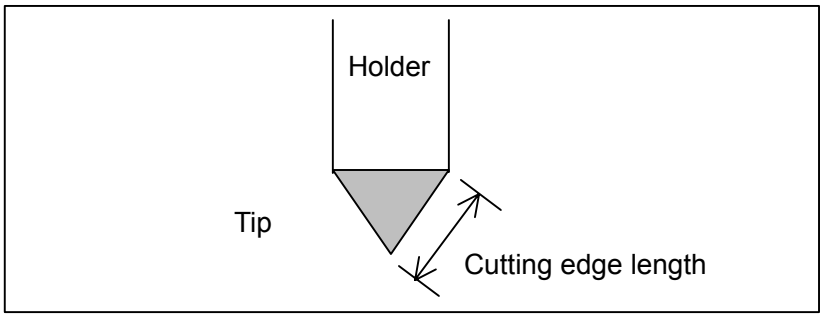
 # 0 STP When a point nose straight tool is drawn in animated simulation, the tip is:
0: Positioned on the front.
1: Positioned on the rear.



27367	Cutting edge length applied when a point nose straight tool is drawn in animated simulation

[Input type] Parameter input
[Data type] 2-word
[Unit of data] 0.001mm (metric input), 0.0001inch (inch input)
[Valid data range] 0 or larger

This parameter sets the cutting edge length applied when a point nose straight tool is drawn in animated simulation.

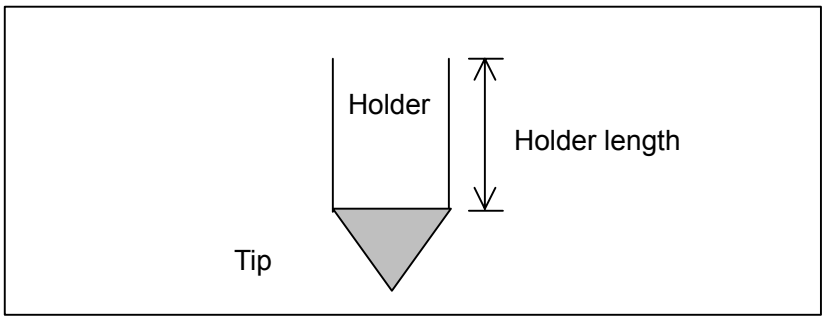


When 0 is set, 12 mm for metric input or 0.4724 inch for inch input is assumed.

27368	Holder length applied when a point nose straight tool is drawn in animated simulation

[Input type] Parameter input
[Data type] 2-word
[Unit of data] 0.001mm (metric input), 0.0001inch (inch input)
[Valid data range] 0 or larger

This parameter sets the holder length applied when a point nose straight tool is drawn in animated simulation.

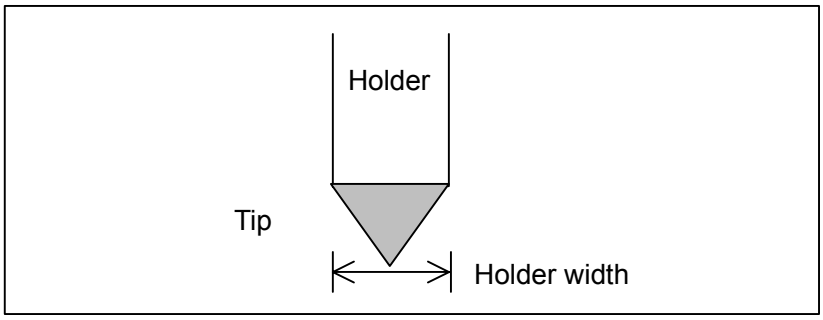


When 0 is set, 50 mm for metric input or 1.9685 inch for inch input is assumed.

27369	Holder width applied when a point nose straight tool is drawn in animated simulation

[Input type] Parameter input
[Data type] 2-word
[Unit of data] 0.001mm (metric input), 0.0001inch (inch input)
[Valid data range] 0 or larger

This parameter sets the holder width applied when a point nose straight tool is drawn in animated simulation.

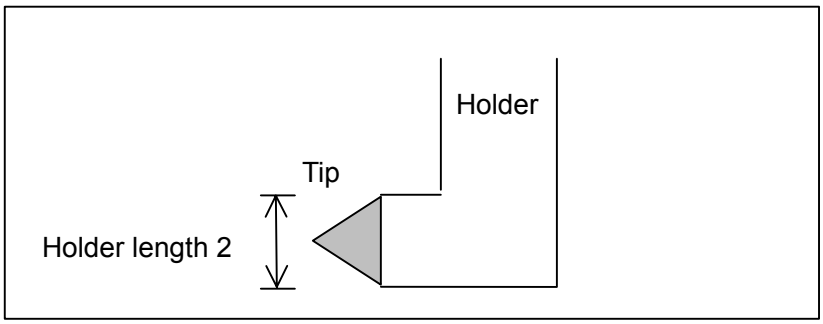


When 0 is set, 14 mm for metric input or 0.5512 inch for inch input is assumed.

27370	Holder length 2 applied when a point nose straight tool is drawn in animated simulation

[Input type] Parameter input
[Data type] 2-word
[Unit of data] 0.001mm (metric input), 0.0001inch (inch input)
[Valid data range] 0 or larger

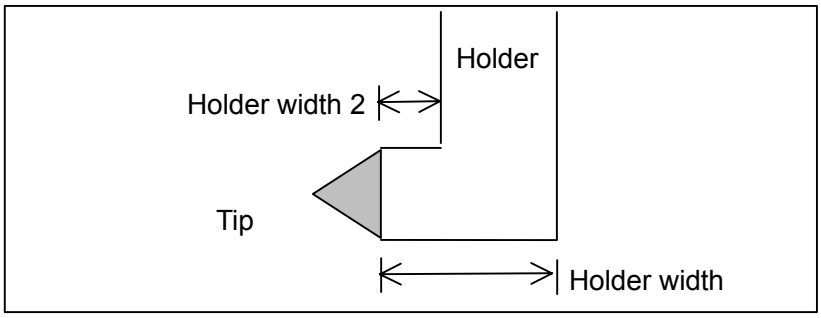
This parameter sets the holder length 2 applied when a point nose straight tool is drawn in animated simulation.



27371	Holder width 2 applied when a point nose straight tool is drawn in animated simulation

[Input type] Parameter input
[Data type] 2-word
[Unit of data] 0.001mm (metric input), 0.0001inch (inch input)
[Valid data range] 0 or larger

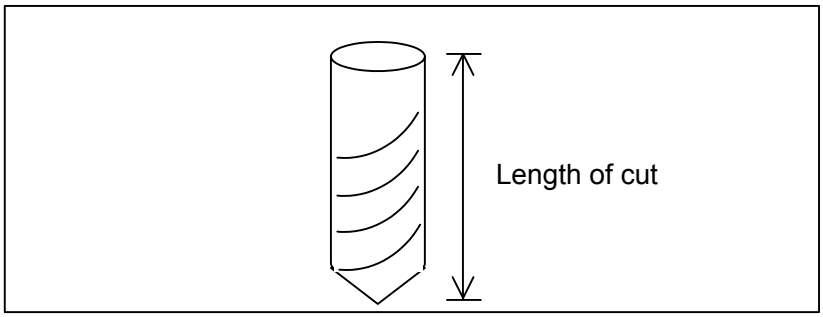
This parameter sets the holder width 2 applied when a point nose straight tool is drawn in animated simulation.



27372	Length of cut applied when a drill tool is drawn in animated simulation

[Input type] Parameter input
[Data type] 2-word
[Unit of data] 0.001mm (metric input), 0.0001inch (inch input)
[Valid data range] 0 or larger

This parameter sets the length of cut applied when a drill tool is drawn in animated simulation.

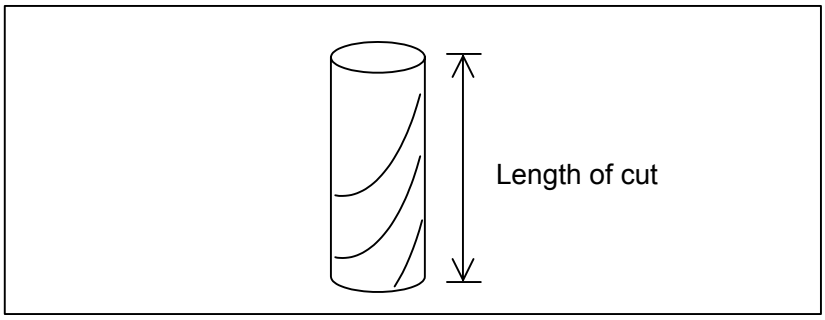


When 0 is set, 50 mm for metric input or 1.9685 inch for inch input is assumed.

27373	Length of cut applied when a flat end milling cutter is drawn in animated simulation

[Input type] Parameter input
[Data type] 2-word
[Unit of data] 0.001mm (metric input), 0.0001inch (inch input)
[Valid data range] 0 or larger

This parameter sets the length of cut applied when a flat end milling cutter is drawn in animated simulation.

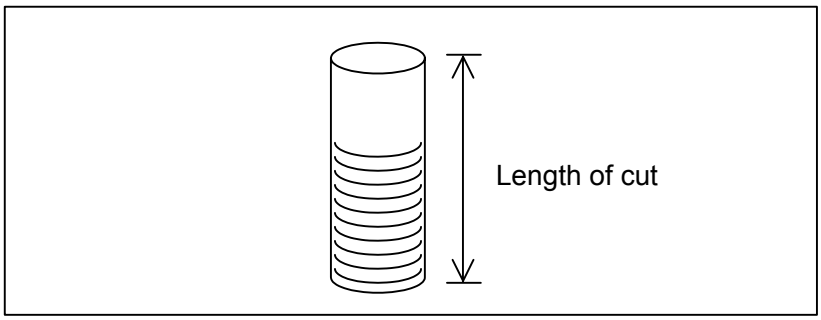


When 0 is set, 50 mm for metric input or 1.9685 inch for inch input is assumed.

27374	Length of cut applied when a tapping tool is drawn in animated simulation

[Input type] Parameter input
[Data type] 2-word
[Unit of data] 0.001mm (metric input), 0.0001inch (inch input)
[Valid data range] 0 or larger

This parameter sets the length of cut applied when a tapping tool is drawn in animated simulation.

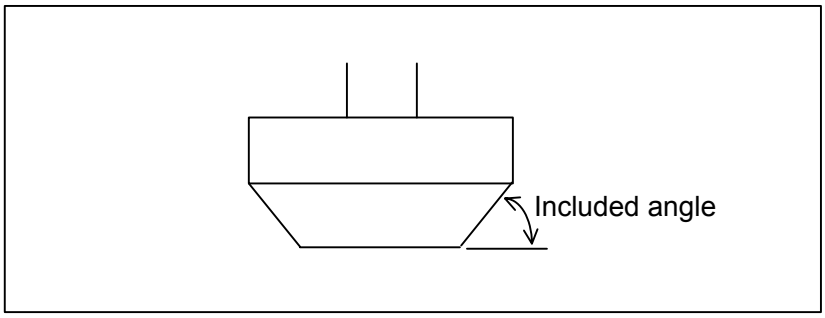


When 0 is set, 50 mm for metric input or 1.9685 inch for inch input is assumed.

27375	Included angle applied when a chamfering tool is drawn in animated simulation

[Input type] Parameter input
[Data type] 2-word
[Unit of data] degree
[Valid data range] 0 to 90

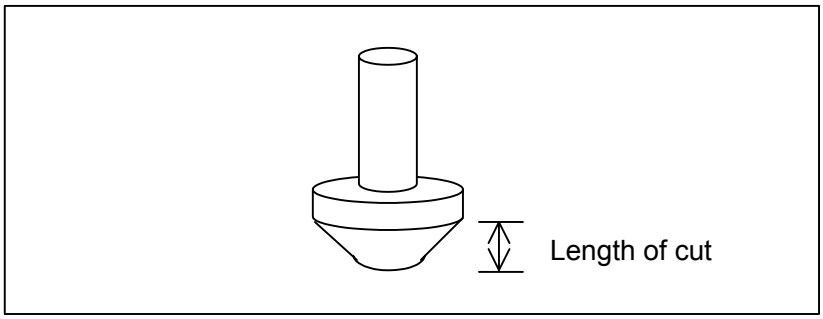
This parameter sets the included angle applied when a a chamfering tool is drawn in animated simulation.



27376	Length of cut applied when a chamfering tool is drawn in animated simulation

[Input type] Parameter input
[Data type] 2-word
[Unit of data] 0.001mm (metric input), 0.0001inch (inch input)
[Valid data range] 0 or larger

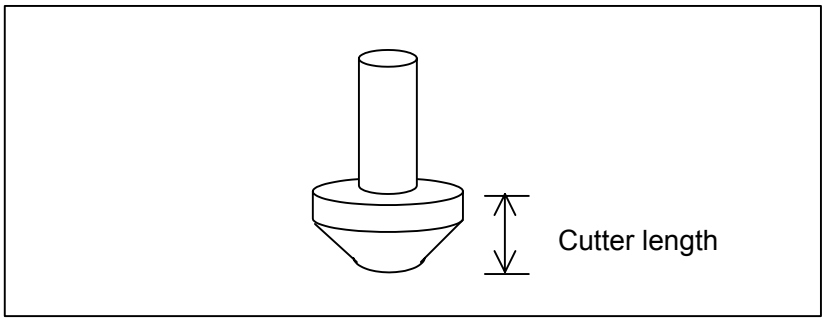
This parameter sets the length of cut applied when a chamfering tool is drawn in animated simulation.



When 0 is set, 26 mm for metric input or 1.0236 inch for inch input is assumed.

27377	Cutter length applied when a chamfering tool is drawn in animated simulation

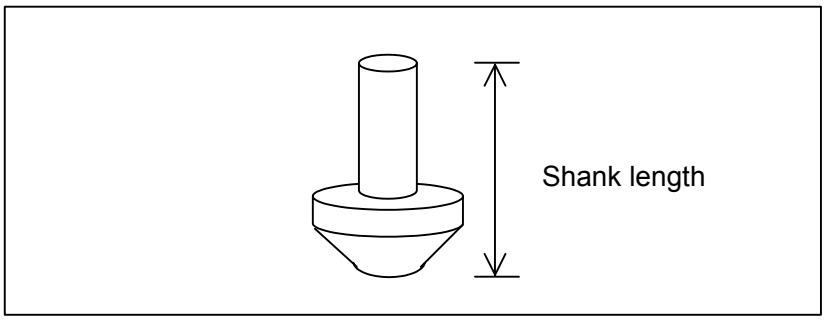
[Input type] Parameter input
[Data type] 2-word
[Unit of data] 0.001mm (metric input), 0.0001inch (inch input)
[Valid data range] 0 or larger
This parameter sets the cutter length applied when a chamfering tool is drawn in animated simulation.



When 0 is set, 50 mm for metric input or 1.9685 inch for inch input is assumed.

27378	Shank length applied when a chamfering tool is drawn in animated simulation

[Input type] Parameter input
[Data type] 2-word
[Unit of data] 0.001mm (metric input), 0.0001inch (inch input)
[Valid data range] 0 or larger
This parameter sets the shank length applied when a chamfering tool is drawn in animated simulation.

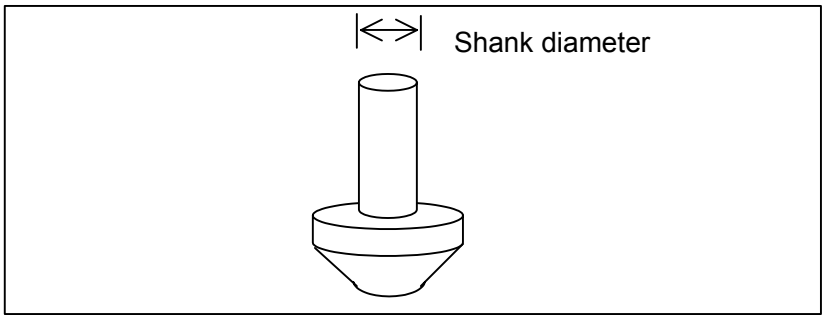


When 0 is set, 130 mm for metric input or 5.1181 inch for inch input is assumed.

27379	Shank diameter applied when a chamfering tool is drawn in animated simulation

[Input type] Parameter input
[Data type] 2-word
[Unit of data] 0.001mm (metric input), 0.0001inch (inch input)
[Valid data range] 0 or larger

This parameter sets the shank diameter applied when a chamfering tool is drawn in animated simulation.

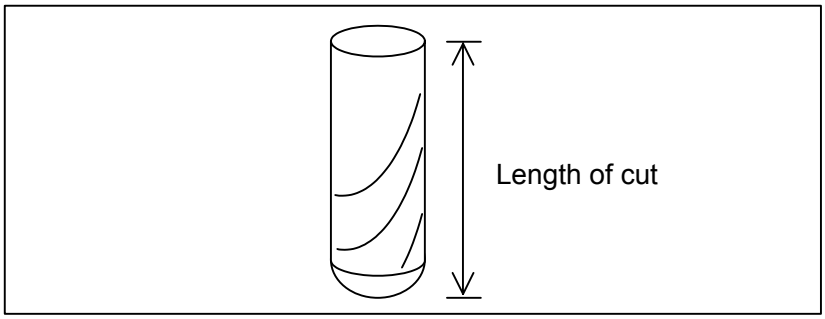


When 0 is set, 32 mm for metric input or 1.2598 inch for inch input is assumed.

27380	Length of cut applied when a ball end mill is drawn in animated simulation

[Input type] Parameter input
[Data type] 2-word
[Unit of data] 0.001mm (metric input), 0.0001inch (inch input)
[Valid data range] 0 or larger

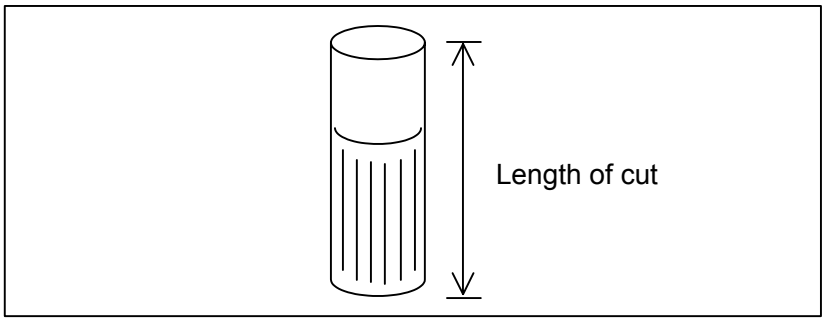
This parameter sets the length of cut applied when a ball end mill is drawn in animated simulation.



When 0 is set, 50 mm for metric input or 1.9685 inch for inch input is assumed.

27381 **Length of cut applied when a reamer is drawn in animated simulation**

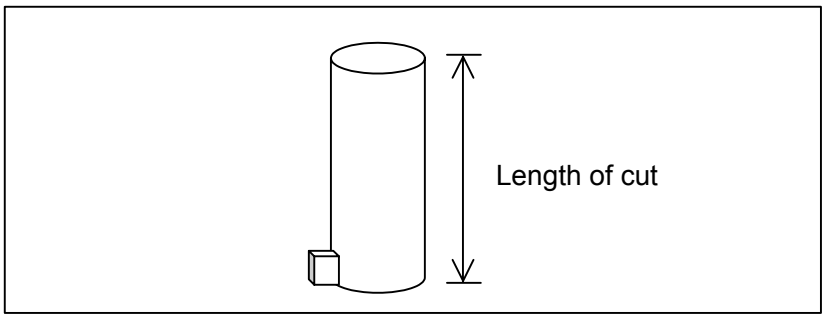
[Input type] Parameter input
[Data type] 2-word
[Unit of data] 0.001mm (metric input), 0.0001inch (inch input)
[Valid data range] 0 or larger
This parameter sets the length of cut applied when a reamer is drawn in animated simulation.



When 0 is set, 50 mm for metric input or 1.9685 inch for inch input is assumed.

27382 **Length of cut applied when a boring tool is drawn in animated simulation**

[Input type] Parameter input
[Data type] 2-word
[Unit of data] 0.001mm (metric input), 0.0001inch (inch input)
[Valid data range] 0 or larger
This parameter sets the length of cut applied when a boring tool is drawn in animated simulation.

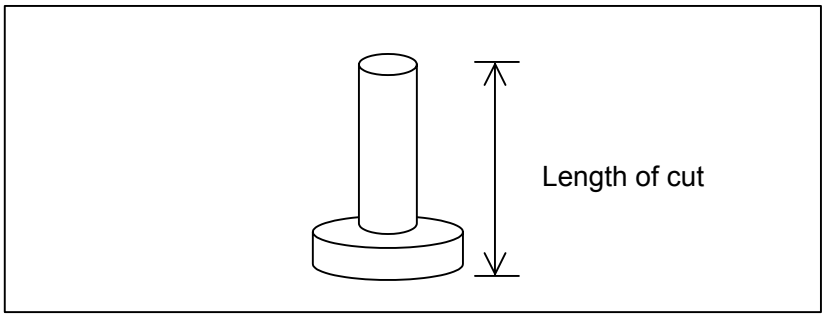


When 0 is set, 50 mm for metric input or 1.9685 inch for inch input is assumed.

27383	Length of cut applied when a face milling cutter is drawn in animated simulation

[Input type] Parameter input
[Data type] 2-word
[Unit of data] 0.001mm (metric input), 0.0001inch (inch input)
[Valid data range] 0 or larger

This parameter sets the length of cut applied when a face milling cutter is drawn in animated simulation.

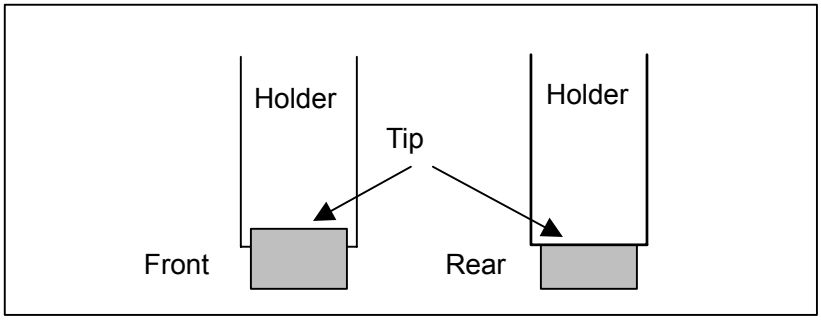


When 0 is set, 63 mm for metric input or 2.4803 inch for inch input is assumed.

	#7	#6	#5	#4	#3	#2	#1	#0
27384								VRP

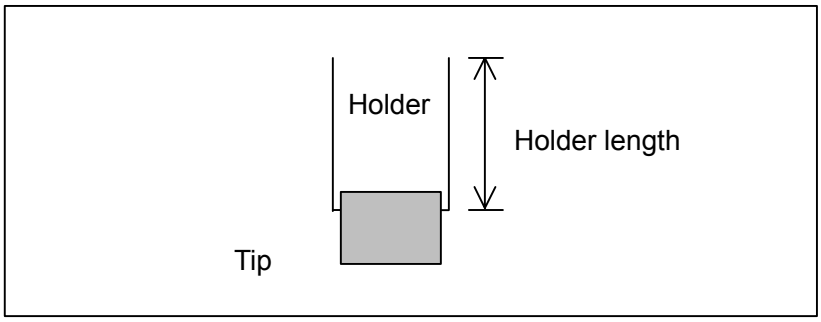
[Input type] Parameter input
[Data type] Bit

 # 0 VRP When a multifunction tool is drawn in animated simulation, the tip is:
0: Positioned on the front.
1: Positioned on the rear.



27385	Holder length applied when a multifunction tool is drawn in animated simulation

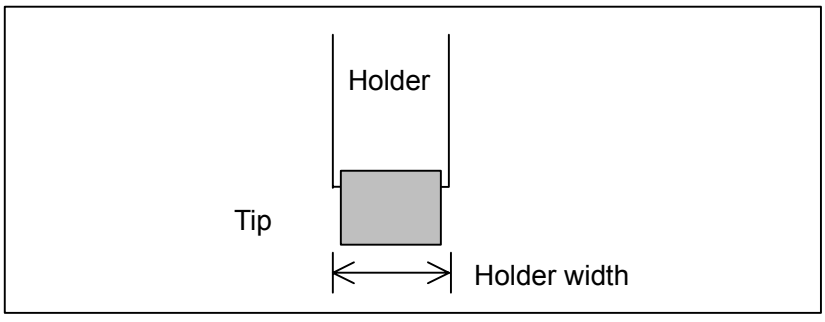
[Input type] Parameter input
[Data type] 2-word
[Unit of data] 0.001mm (metric input), 0.0001inch (inch input)
[Valid data range] 0 or larger
This parameter sets the holder length applied when a multifunction tool is drawn in animated simulation.



When 0 is set, 50 mm for metric input or 1.9685 inch for inch input is assumed.

27386	Holder width applied when a multifunction tool is drawn in animated simulation

[Input type] Parameter input
[Data type] 2-word
[Unit of data] 0.001mm (metric input), 0.0001inch (inch input)
[Valid data range] 0 or larger
This parameter sets the holder width applied when a multifunction tool is drawn in animated simulation.



When 0 is set, 14 mm for metric input or 0.5512 inch for inch input is assumed.

A.2 DATA TYPE

Parameters are classified by data type as follows:

Data type	Valid data range	Remarks
Bit	0 or 1	
Bit machine group		
Bit path		
Bit axis		
Bit spindle		
Byte	-128 to 127 0 to 255	Some parameters handle these types of data as unsigned data.
Byte machine group		
Byte path		
Byte axis		
Byte spindle		
Word	-32768 to 32767 0 to 65535	Some parameters handle these types of data as unsigned data.
Word machine group		
Word path		
Word axis		
Word spindle		
2-word	0 to ±999999999	Some parameters handle these types of data as unsigned data.
2-word machine group		
2-word path		
2-word axis		
2-word spindle		
Real	See the standard parameter setting tables.	
Real machine group		
Real path		
Real axis		
Real spindle		

NOTE

1 Each of the parameters of the bit, bit machine group, bit path, bit axis, and bit spindle types consists of 8 bits for one data number (parameters with eight different meanings).
2 For machine group types, parameters corresponding to the maximum number of machine groups are present, so that independent data can be set for each machine group.
3 For path types, parameters corresponding to the maximum number of paths are present, so that independent data can be set for each path.
4 For axis types, parameters corresponding to the maximum number of control axes are present, so that independent data can be set for each control axis.
5 For spindle types, parameters corresponding to the maximum number of spindles are present, so that independent data can be set for each spindle axis.
6 The valid data range for each data type indicates a general range. The range varies according to the parameters. For the valid data range of a specific parameter, see the explanation of the parameter.

A.3 STANDARD PARAMETER SETTING TABLES

This section defines the standard minimum data units and valid data ranges of the CNC parameters of the real type, real machine group type, real path type, real axis type, and real spindle type. The data type and unit of data of each parameter conform to the specifications of each function.

NOTE
1 Values are rounded up or down to the nearest multiples of the minimum data unit.
2 A valid data range means data input limits, and may differ from values representing actual performance.
3 For information on the ranges of commands to the CNC, refer to Appendix D, "Range of Command Value."

(A) Length and angle parameters (type 1)

Unit of data	Increment	Minimum	Valid data range
mm deg.	IS-A	0.01	-999999.99 to +999999.99
	IS-B	0.001	-999999.999 to +999999.999
	IS-C	0.0001	-99999.9999 to +99999.9999
	IS-D	0.00001	-9999.99999 to +9999.99999
	IS-E	0.000001	-999.999999 to +999.999999
inch	IS-A	0.001	-99999.999 to +99999.999
	IS-B	0.0001	-99999.9999 to +99999.9999
	IS-C	0.00001	-9999.99999 to +9999.99999
	IS-D	0.000001	-999.999999 to +999.999999
	IS-E	0.0000001	-99.9999999 to +99.9999999

(B) Length and angle parameters (type 2)

Unit of data	Increment system	Minimum data unit	Valid data range
mm deg.	IS-A	0.01	0.00 to +999999.99
	IS-B	0.001	0.000 to +999999.999
	IS-C	0.0001	0.0000 to +99999.9999
	IS-D	0.00001	0.00000 to +9999.99999
	IS-E	0.000001	0.000000 to +999.999999
inch	IS-A	0.001	0.000 to +99999.999
	IS-B	0.0001	0.0000 to +99999.9999
	IS-C	0.00001	0.00000 to +9999.99999
	IS-D	0.000001	0.000000 to +999.999999
	IS-E	0.0000001	0.0000000 to +99.9999999

(C) Velocity and angular velocity parameters

Unit of data	Increment system	Minimum data unit	Valid data range
mm/min degree/min	IS-A	0.01	0.00 to +999000.00
	IS-B	0.001	0.000 to +999000.000
	IS-C	0.0001	0.0000 to +99999.9999
	IS-D	0.00001	0.00000 to +9999.99999
	IS-E	0.000001	0.000000 to +999.999999
inch/min	IS-A	0.001	0.000 to +96000.000
	IS-B	0.0001	0.0000 to +9600.0000
	IS-C	0.00001	0.00000 to +4000.00000
	IS-D	0.000001	0.000000 to +400.000000
	IS-E	0.0000001	0.0000000 to +40.0000000

(D)Acceleration and angular acceleration parameters

Unit of data	Increment system	Minimum data unit	Valid data range
mm/sec^2 $deg./sec^2$	IS-A	0.01	0.00 to +999999.99
	IS-B	0.001	0.000 to +999999.999
	IS-C	0.0001	0.0000 to +99999.9999
	IS-D	0.00001	0.00000 to +9999.99999
	IS-E	0.000001	0.000000 to +999.999999
$inch/sec^2$	IS-A	0.001	0.000 to +99999.999
	IS-B	0.0001	0.0000 to +99999.9999
	IS-C	0.00001	0.00000 to +9999.99999
	IS-D	0.000001	0.000000 to +999.999999
	IS-E	0.0000001	0.0000000 to +99.9999999

B PROGRAM CODE LIST

Character name	ISO code		EIA code		Custom macro		Usable as file name
	Character	Code (hexadecimal)	Character	Code (hexadecimal)	without custom macro	with custom macro	
Number 0	0	30	0	20			*
Number 1	1	B1	1	01			*
Number 2	2	B2	2	02			*
Number 3	3	33	3	13			*
Number 4	4	B4	4	04			*
Number 5	5	35	5	15			*
Number 6	6	36	6	16			*
Number 7	7	B7	7	07			*
Number 8	8	B8	8	08			*
Number 9	9	39	9	19			*
Address A	A	41	a	61			*
Address B	B	42	b	62			*
Address C	C	C3	c	73			*
Address D	D	44	d	64			*
Address E	E	C5	e	75			*
Address F	F	C6	f	76			*
Address G	G	47	g	67			*
Address H	H	48	h	68			*
Address I	I	C9	i	79			*
Address J	J	CA	j	51			*
Address K	K	4B	k	52			*
Address L	L	CC	l	43			*
Address M	M	4D	m	54			*
Address N	N	4E	n	45			*
Address O	O	CF	o	46			*
Address P	P	50	p	57			*
Address Q	Q	D1	q	58			*
Address R	R	D2	r	49			*
Address S	S	53	s	32			*
Address T	T	D4	t	23			*
Address U	U	55	u	34			*
Address V	V	56	v	25			*
Address W	W	D7	w	26			*
Address X	X	D8	x	37			*
Address Y	Y	59	y	38			*
Address Z	Z	5A	z	29			*
Delete	DEL	FF	Del	7F	×	×	
Back space	BS	88	BS	2A	×	×	
Tabulator	HT	09	Tab	2E	×	×	
End of block	LF or NL	0A	CR or EOB	80			
Carriage return	CR	8D			×	×	

Character name	ISO code		EIA code		Custom macro		Usable as file name
	Character	Code (hexadecimal)	Character	Code (hexadecimal)	without custom macro	with custom macro	
Space	SP	A0	SP	10	□	□	
Absolute rewind stop	%	A5	ER	0B			
Control out (start of comment)	(	28	(2-4-5)	1A			
Control in (end of comment)	)	A9	(2-4-7)	4A			
Plus sign	+	2B	+	70			*
Minus sign	-	2D	-	40			*
Colon (address O)	:	3A					
Optional block skip	/	AF	/	31			
Period (decimal point)	.	2E	.	6B			*
Sharp	#	A3	Parameter (No.6012)				
Dollar sign	$	24			□	□	
Ampersand	&	A6	&	0E			
Apostrophe	’	27			□	□	
Asterisk	*	AA	Parameter (No.6010)				
Comma	,	AC	,	3B			
Semicolon	;	FB			□	□	
Left angle bracket	<	2C					
Equal sign	=	BD	Parameter (No.6011)				
Right angle bracket	>	BE					
Question mark	?	3F			△	○	
Commercial at mark	@	C0			△	△	
Quotation mark	“	22			□	□	
Left square bracket	[	DB	Parameter (No.6013)		△		
Right square bracket	]	DD	Parameter (No.6014)		△		
Underscore	_	6F	Parameter (No.6018)		△	△	*
Lowercase letter a	a	E1			△	△	*
Lowercase letter b	b	E2			△	△	*
Lowercase letter c	c	63			△	△	*
Lowercase letter d	d	E4			△	△	*
Lowercase letter e	e	65			△	△	*
Lowercase letter f	f	66			△	△	*
Lowercase letter g	g	E7			△	△	*
Lowercase letter h	h	E8			△	△	*
Lowercase letter i	i	69			△	△	*
Lowercase letter j	j	6A			△	△	*
Lowercase letter k	k	EB			△	△	*
Lowercase letter l	l	6C			△	△	*
Lowercase letter m	m	ED			△	△	*
Lowercase letter n	n	EE			△	△	*
Lowercase letter o	o	6F			△	△	*
Lowercase letter p	p	F0			△	△	*

Character name	ISO code		EIA code		Custom macro		Usable as file name
	Character	Code (hexadecimal)	Character	Code (hexadecimal)	without custom macro	with custom macro	
Lowercase letter q	q	71			△	△	*
Lowercase letter r	r	72			△	△	*
Lowercase letter s	s	F3			△	△	*
Lowercase letter t	t	74			△	△	*
Lowercase letter u	u	F5			△	△	*
Lowercase letter v	v	F6			△	△	*
Lowercase letter w	w	77			△	△	*
Lowercase letter x	x	78			△	△	*
Lowercase letter y	y	F9			△	△	*
Lowercase letter z	z	FA			△	△	*

NOTE

1 The symbols used in the "Custom macro" column have the following meanings.
(Space) : The character will be registered in memory and has a specific meaning. It is used incorrectly in a statement other than a comment, an alarm occurs.
× : The character will not be registered in memory and will be ignored.
△ : The character will be registered in memory, but will be ignored during program execution. The character that has "*" in its "Usable as file name" field and is used as a file name is not ignored.
○ : The character will be registered in memory. If it is used in a statement other than a comment, an alarm occurs.
□ : If it is used in a statement other than a comment, the character will not be registered in memory. If it is used in a comment, it will be registered in memory.

2 The symbol used in the column "Usable as file name" has the following meaning:
* : Capable of being coded between "<" and ">" as a file name.

3 Codes not in this table are ignored if their parity is correct.

4 Codes with incorrect parity cause the TH alarm. But they are ignored without generating the TH alarm when they are in the comment section.

C LIST OF FUNCTIONS AND PROGRAM FORMAT

With some functions, the format used for specification on the machining center system differs from the format used for specification on the lathe system. Moreover, some functions are used for only one of the control types for the machining center system and lathe system. Some functions cannot be added as options depending on the model. For details of command formats, see the relevant sections or subsections.

In the list, the following symbols are used:

- For machining center
  x : 1st basic axis (X),
  y : 2nd basic axis (Y),
  z : 3rd basic axis (Z)
- For lathe
  x : 1st basic axis (X),
  z : 2nd basic axis (Z),
  coded using G code system A
- IP_ : presents a combination of arbitrary axis addresses using X,Y,Z,A,B and C (such as X_ Y_ Z_).

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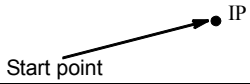
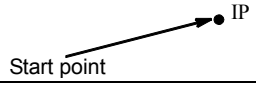
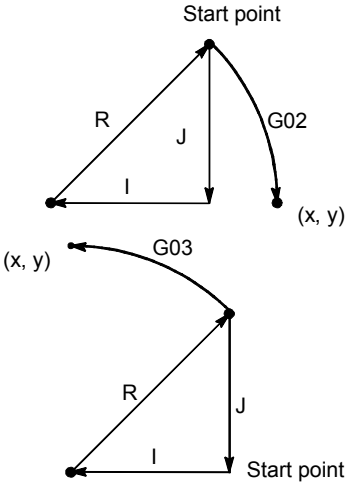
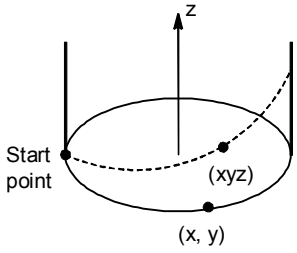
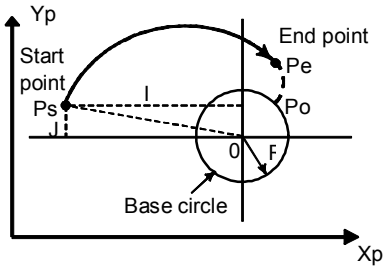
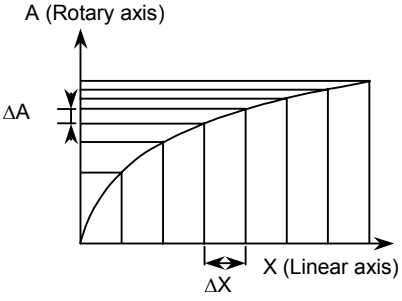
Functions	Illustration	Program format
Positioning (G00)		G00 IP_ ;
Linear interpolation (G01)		G01 IP_ F_;
Circular interpolation (G02, G03)		• **For machining center** G17 {G02 / G03} X_ Y_ {R_ / I_ J_} F_ ; G18 {G02 / G03} X_ Z_ {R_ / I_ K_} F_ ; G19 {G02 / G03} Y_ Z_ {R_ / J_ K_} F_ ; • **For lathe** {G02 / G03} X_ Z_ {R_ / I_ K_} F_ ;
Helical interpolation (G02, G03)	 In case of G03 on X-Y plane	G17 {G02 / G03} X_ Y_ {R_ / I_ J_} α_ F_ ; G18 {G02 / G03} X_ Z_ {R_ / I_ K_} α_ F_ ; G19 {G02 / G03} Y_ Z_ {R_ / J_ K_} α_ F_ ; α: Arbitrary address except the circular interpolation axis
Involute interpolation (G02.2, G03.2)	 (In case of X-Y plane)	G17 {G02.2 / G03.2} Xp_ Yp_ I_ J_ R_ F_ ; G18 {G02.2 / G03.2} Zp_Xp_ K_ I_ R_ F_ ; G19 {G02.2 / G03.2} Yp_ Zp_ J_ K_ R_ F_ ;
Exponential interpolation (G02.3, G03.3)	 Relationship between X-axis and A-axis	Positive rotation G02.3 X_ Y_ Z_ I_ J_ K_ R_ F_ Q_; Negative rotation G03.3 X_ Y_ Z_ I_ J_ K_ R_ F_ Q_;

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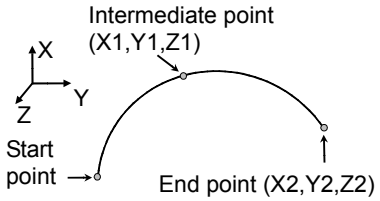
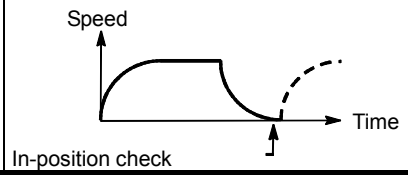
Functions	Illustration	Program format
Three-dimensional circular interpolation (G02.4, G03.4)		G02.4 X_{X1} Y_{Y1} Z_{Z1} $\alpha_{\alpha1}$ $\beta_{\beta1}$; First block (mid-point of the arc) X_{X2} Y_{Y2} Z_{Z2} $\alpha_{\alpha2}$ $\beta_{\beta2}$; Second block (end point of the arc) α, β : Arbitrary axes other than the 3-dimensional circular interpolation axis (up to two axes) G03.4 can also be specified instead of G02.4.
Dwell (G04)		G04 { X_ / P_ } ;
AI contour control (G05)		G05 P10000 ; AI contour control start G05 P0 ; AI contour control end
AI contour control (G05.1)		G05.1 Q1 ; AI contour control mode on G05.1 Q0 ; AI contour control mode off
Nano smoothing (G05.1)		G05.1 Q3 IP0 ; Nano smoothing mode on G05.1 Q0 ; Nano smoothing mode off
Smooth interpolation (G05.1)		G05.1 Q2 ; Smooth interpolation mode on G05.1 Q0 ; Smooth interpolation mode off
NURBS interpolation (G06.2)		G06.2[P_] K_ IP_ [R_] [F_]; NURBS interpolation mode on P : Rank of NURBS curve IP : Control point R : Weight K : Knot F : Feedrate
Hypothetical axis interpolation (G07)		G07 IP0 ; Hypothetical axis setting G07 IP1 ; Hypothetical axis cancel
Cylindrical interpolation (G07.1)		G07 IP_ r_; Cylindrical interpolation mode r : Cylinder radius G07 IP 0; Cylindrical interpolation mode cancel
AI contour control (Advanced preview control) (G08)		G08 P1 ; AI contour control mode on G08 P0 ; AI contour control mode off
Exact stop (G09)		G09 { G01 / G02 / G03 } IP_ ;

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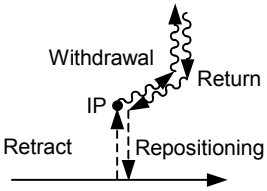
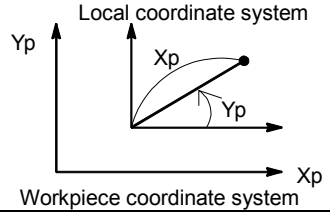
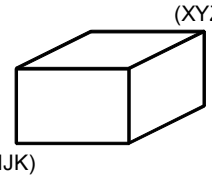
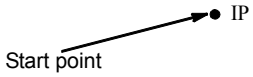
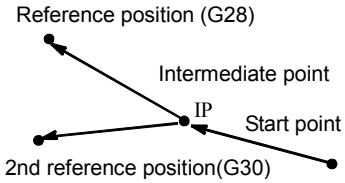
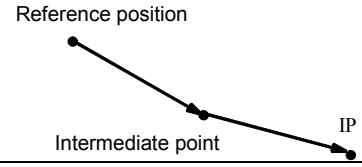
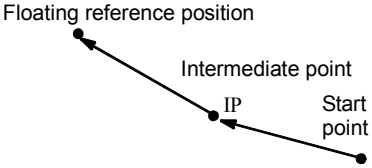
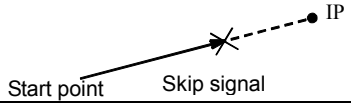
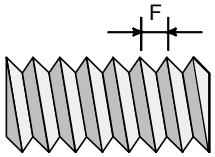
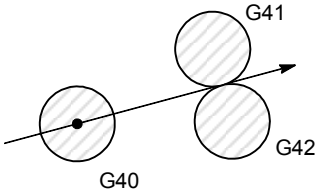
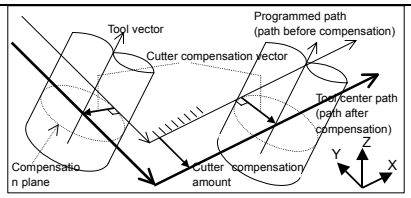
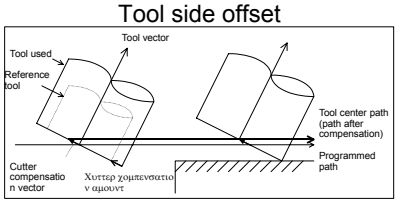
Functions	Illustration	Program format
Programmable data input (G10)		• **For machining center** Tool compensation memory A G10 L01 P_ R_ ; Tool compensation memory B G10 L10 P_ R_ ; (Geometry offset amount) G10 L11 P_ R_ ; (Wear offset amount) Tool compensation memory C G10 L10 P_ R_ ; (Geometry offset amount/H) G10 L11 P_ R_ ; (Wear offset amount/H) G10 L12 P_ R_ ; (Geometry offset amount/D) G10 L13 P_ R_ ; (Wear offset amount/D) • **For lathe** Geometry offset amount G10 P_ X_ Z_ R_ Q_ ; P = 10000 + Geometry offset number Wear offset amount G10 P_ X_ Z_ C_ Q_ ; P = Wear offset number
Tool retract and recover (G10.6)		G10.6 IP_ ; Specify the amount of retraction G10.6 (as a single block containing no other commands) ; Cancel the amount of retraction
Polar coordinate interpolation (G12.1, G13.1)		G12.1 ; Polar coordinate interpolation mode on G13.1 ; Polar coordinate interpolation cancel
Polar coordinate command (G15, G16)		G17 G16 Xp_ Yp_ . . . ; G18 G16 Zp_ Xp_ . . . ; G19 G16 Yp_ Zp_ . . . ; G15 ; Cancel
Plane selection (G17, G18, G19)		G17 ; Xp Yp-plane selection G18 ; Zp Xp-plane selection G19 ; Yp Zp-plane selection
Inch/metric conversion (G20, G21)		Inch input G20 ; Metric input G21 ;
Stored stroke check (G22, 23)		G22 X_ Y_ Z_ I_ J_ K_ ; G23 ; Cancel
Reference position return check (G27)		G27 IP_ ;

Functions	Illustration	Program format
Reference position return (G28) 2nd Reference position return (G30)		G28 IP_ ; G30 IP_ ;
Movement from reference position (G29)		G29 IP_ ;
Floating reference position return (G30.1)		G30.1 IP_ ;
Skip function (G31)		G31 IP_ F_ ;
Threading (G33) Threading (G32)		• **For machining center** G33 IP_ F_ ; F : Lead • **For lathe** Equal lead threading G32 P_ F_ ;
Tool radius/tool nose radius compensation, 3-dimensional cutter compensation (G38, G39, G40 to G42)		• **For machining center** {G17, G18, G19} {G41, G42} D_ ; D : Tool compensation number G40 : Cancel
Tool radius/tool nose radius compensation (G40 to G42) (G38, G39)		• **For lathe only** {G41, G42} IP_ ; G40 : Cancel
3-dimensional cutter compensation (G41.2, G42.2) (G41.3)	 Tool side offset  Leading edge offset	{G41.2, G42.2} IP_ D_ ; Tool side offset G40 IP_ ; Cancellation of cutter compensation G41.3 D_ ; Leading edge offset G40 ; Canceling the leading edge offset

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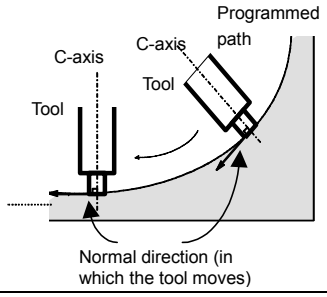
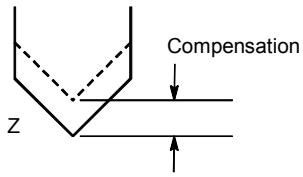
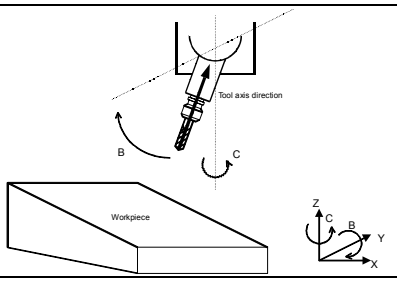
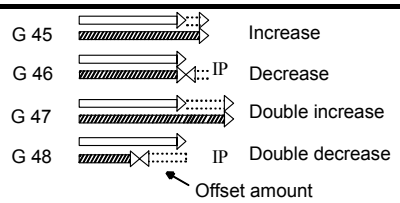
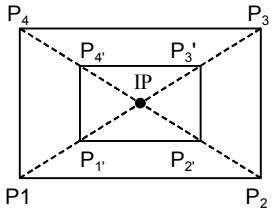
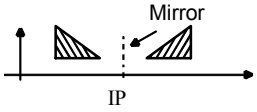
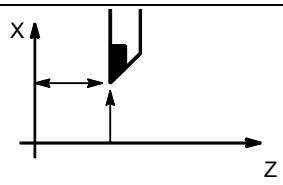
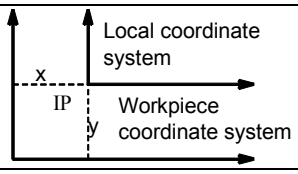
Functions	Illustration	Program format
Normal direction control (G40.1, G41.1, G42.1)		G41.1 ; Normal direction control on : right G42.1 ; Normal direction control on : left G40.1 ; Normal direction control cancel
Tool length compensation (G43, G44, G49)		{G43, G44} Z_ H_ ; {G43, G44} H_ ; H : Tool compensation number G49 : Cancel
Tool length compensation in tool axis direction (G43.1)		G43.1 H_ ; Tool length compensation in tool axis direction H : Compensation number G49 ; Compensation cancel
Tool center point control (TYPE1) (G43.4)		G43.4 IP α β H ; Tool center point control (TYPE1) start IP α β ; IP : In the case of an absolute command, the coordinate value of the end point of the tool tip movement In the case of an incremental command, the amount of the tool tip movement α, β : In the case of an absolute command, the coordinate value of the end point of the rotation axes In the case of an incremental command, the amount of the rotation axis movement H : Tool compensation number
Tool center point control (TYPE2) (G43.5)		G43.5 IP_ H_ Q_ ; Tool center point control (TYPE2) start IP_ I_ J_ K_ ; IP : In the case of an absolute command, the coordinate value of the end point of the tool tip movement In the case of an incremental command, the amount of the tool tip movement I,J,K : Tool axis direction at the block end point as seen from the programming coordinate system H : Tool compensation number Q : Inclination angle of the tool (in degrees)
Tool offset (G43.7)		• **For lathe only**

Functions	Illustration	Program format
Tool offset (G45 to G48)		• **For machining center** {G45 G46 G47 G48} IP_ D_ ; D : Tool offset number
Scaling (G50, G51)		• **For machining center** G51 X_ Y_ Z_ {P_ / I_ J_ K_} ; P, I, J, K : Scaling magnification X, Y, Z : Control position of scaling G50 : Cancel • **For lathe only** Enabled when G code system B/C is used
Programmable mirror image (G50.1, G51.1)		G51.1 IP_ ; G50.1 ; . . . Cancel
Synchronous, composite, and superimposed control by program command (G50.4, G51.4, G50.5, G51.5, G50.6, G51.6)		G51.4 P_Q_(L_) ; Start synchronous control (L_ can be omitted.) G50.4 Q_ ; Cancel synchronous control. P : Number to identify synchronous master axis Q :Number to identify synchronous slave axis L : Parking start command G51.5 P_Q_ ; Start composite control G50.5 P_Q_ ; Cancel composite control P : Number to identify composite axis 1 Q : Number to identify composite axis 2 G51.6 P_Q_ ; Start superimposed control G50.6 Q_ ; Cancel superimposed control P : Number to identify superimposed master axis Q :Number to identify superimposed slave axis
Coordinate system setting Maximum spindle speed clamp (G50)		• **For lathe only** G50 IP_ ; (Coordinate system setting) G50 S_ ; (Maximum spindle speed clamp)
Local coordinate system setting (G52)		G52 IP_ ;
Command in machine coordinate system (G53)		G53 IP_ ;
Tool axis direction control (G53.1)		G53.1 ; Tool axis direction control

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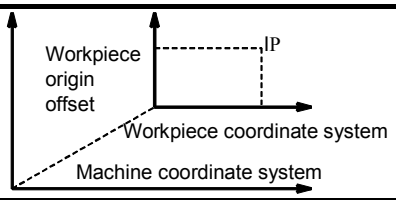
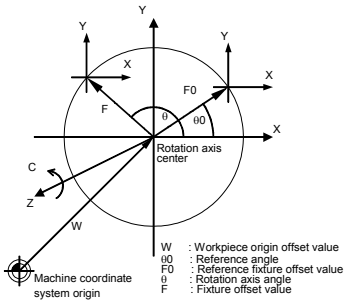
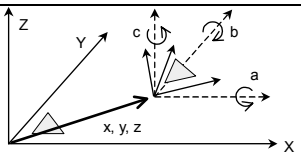
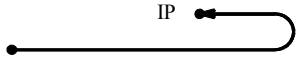
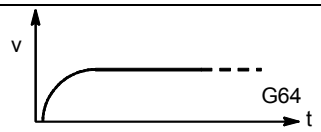
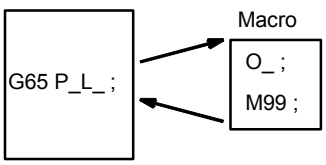
Functions	Illustration	Program format
Selection of workpiece coordinate system (G54 to G59)		{G54 : G59} IP_ ;
Rotary table dynamic fixture offset (G54.2)		• **For machining center** G54.2 P_ ; Fixture offset P : Reference fixture offset value number G54.2 P0 ; Offset cancel
Workpiece setting error compensation (G54.4)	 x : X-axis direction error y : Y-axis direction error z : Z-axis direction error a : Rotation direction error (about the X-axis) b : Rotation direction error (about the Y-axis) c : Rotation direction error (about the Z-axis)	G54.4 P_ ; Workpiece setting error compensation P : Workpiece setting error number G54.4 P0 ; Workpiece setting error compensation cancel
Single direction positioning (G60)		G60 IP_ ;
Cutting mode (G64) Exact stop mode (G61) Tapping mode (G63)		G64_ ; Cutting mode G61_ ; Exact stop mode G63_ ; Tapping mode
Automatic corner override (G62)		G62_ ; Automatic corner override
Custom macro (G65, G66, G66.1, G67)		One-shot call G65 P_ L_ <Argument assignment> ; P : Program number L : Number of repetition Modal call G66 P_ L_ <Argument assignment> ; Call after the move command G66.1 P_ L_ <Argument assignment> ; Each block call G67 ; Cancel

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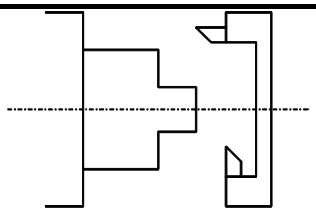
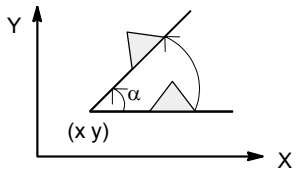
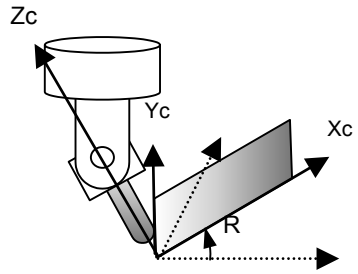
Functions	Illustration	Program format
Mirror image for double turret (G68, G69)		• **For lathe only** G68 : Mirror image for double turret G69 : Mirror image cancel
Coordinate system rotation, Three-dimensional coordinate conversion (G68, G69) (G68.1, G69.1)	 In case of X-Y plane	• **For machining center** G68 { G17 X_ Y_ / G18 Z_ X_ / G19 Y_ Z_ } R α ; G69 ; Cancel • **For lathe** G68.1 { G17 X_ Y_ / G18 Z_ X_ / G19 Y_ Z_ } R α ; G69.1 ; Cancel
Tilted working plane command (G68.2)		G68.2 X_ Y_ Z_ I_ J_ K_ ; Tilted working plane command G69 ; Tilted working plane command cancel X, Y, Z : Feature coordinate system origin I, J, K : Euler angles for determining the orientation of the feature coordinate system
Tilted working plane command with guidance (G68.2, G68.4)		G68.2/G68.4 P_ X_ Y_ Z_ I_ J_ K_ ; G68.2 : Tilted working plane command G68.4 : Tilted working plane command (incremental multiplexed command) G69 ; Tilted working plane command cancel Without P : Euler's angle P1 : Roll, pitch, yaw P2 : Three points P3 : Two vectors P4 : Projection angle
Angular surface command by tool axis direction (G68.3)	 Feature coordinate system Xc-Yc-Zc	G68.3 X_ Y_ Z_ R_ ; G69 ; Cancel X,Y, Z : Feature coordinate system origin R : Feature coordinate system rotation angle about the Z-axis

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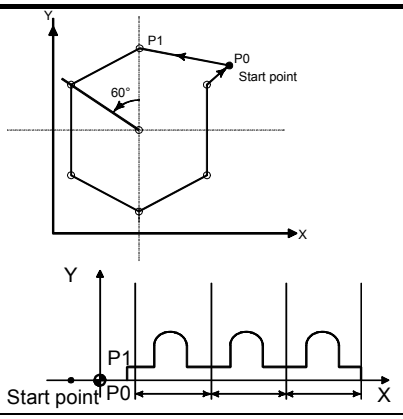
Functions	Illustration	Program format
Figure copy (G72.1, G72.2)		Rotational copy {(G17)/(G18)/(G19)} G72.1 P_ L_ {X_ Y_/Z_ X_/Y_ Z_} R_ ; Linear copy {(G17)/(G18)/(G19)} G72.2 P_ L_ {I_ J_/K_ I_/J_ K_} ;
Canned cycle for drilling (G73, G74, G80 to G89)		• For machining center G80 ; Cancel G73 G74 G76 G81 : G89 } X_Y_Z_P_Q_R_F_K_ ;
Canned cycle (G71 to G76) (G90, G92, G94)		**• For lathe only** N_ G70P_ Q_ ; G71U_ R_ ; G71P_ Q_ U_ W_ F_ S_ T_ ; G72W_ R_ ; G72P_ Q_ U_ W_ F_ S_ T_ ; G73U_ W_ R_ ; G73P_ Q_ U_ W_ F_ S_ T_ ; G74R_ ; G74X(u)_ Z(w)_ P_ Q_ R_ F_ ; G75R_ ; G75X(u)_ Z(w)_ P_ Q_ R_ F_ ; G76P_ Q_ R_ ; G76X(u)_ Z(w)_ P_ Q_ R_ F_ ; {G90/G92} X_ Z_ I_ F_ ; G94X_ Z_ K_ F_ ;
Canned grinding cycle (for grinding machine) (G71 to G75, G77 to G79)		**• For machining center** G75 I_ J_ K_ α_ R_ F_ P_ L_ ; G77 I_ J_ K_ α_ R_ F_ P_ L_ ; G78 I_ (J_) K_ α_ F_ P_ L_ ; G79 I_ J_ K_ α_ R_ F_ P_ L_ ; α : Arbitrary axis address of the grinding axis **• For lathe** G71 A_ B_ W_ U_ I_ K_ H_ ; G72 P_ A_ B_ W_ U_ I_ K_ H_ ; G73 A_ (B_) W_ U_ K_ H_ ; G74 P_ A_ (B_) W_ U_ K_ H_ ;

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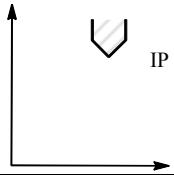
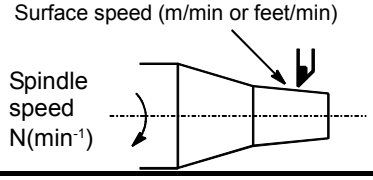
Functions	Illustration	Program format
Absolute/incremental programming (G90/G91)		• **For machining center** G90_ ; Absolute programming G91_ ; Incremental programming : G90_ G91_ ; Programming in both modes • **For lathe** X_ Z_ C_ : Absolute programming U_ W_ H_ : Incremental programming Distinguished by an address specified in combined use with a G function such as G00 and G01.
Maximum incremental command value check (G91.1)		G91.1 IP_ ; IP_ ; Maximum incremental value Set 0 to cancel maximum incremental value check.
Change of workpiece coordinate system (G92) Maximum spindle speed clamp (G92)		• **For machining center** G92 IP_ ; Change of workpiece coordinate system G92 S_ ; Constant surface speed control : Maximum spindle speed clamp
Workpiece coordinate system preset (G92.1)		• **For machining center** G92.1 IP 0 ;
Inverse time feed (G93)		G93 ; Inverse time setting mode
Feed per minute, Feed per revolution (G94, G95) (G98, G99)	mm/min inch/min mm/rev inch/rev	• **For machining center** G94 F_ ; Feed per minute G95 F_ ; Feed per revolution • **For lathe** G98 F_ ; Feed per minute G99 F_ ; Feed per revolution
Constant surface speed control (G96, G97)		G96 S_ ; Constant surface speed control on (surface speed specification) G97 S_ ; Constant surface speed control off (spindle speed specification)

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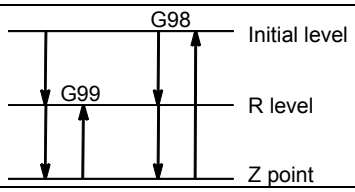
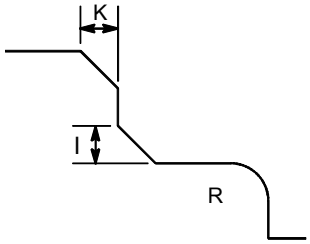
Functions	Illustration	Program format
Speed display function of a milling tool with servo motor (G96.1,G96.2,G96.3,G96.4)		G96.1 P_R_; The next block starts operating upon completion of spindle indexing (the SV speed control mode is off). G96.2 P_R_; The next block starts operating without waiting for spindle indexing to complete. G96.3 P_; The next block starts operating after confirming the completion of spindle indexing (the SV speed control mode is off). G96.4 P_; The SV speed control mode is on.
Initial point return / R point return (G98, G99)		• **For machining center** G98_ ; G99_ ;
Optional chamfering/corner R Chamfering/corner R		• **For machining center** ,C_ : Chamfering ,R_ : Corner R • **For lathe only** X_ {C±K, R_} P_; Z_ {C±K, R_} P_;
In-feed control (for grinding machine) (G160, G161)		• **For machining center** G161 R_ ; Figure program (G01, G02, G03) G160 ;

D RANGE OF COMMAND VALUE

Linear axis

- In case of millimeter input, feed screw is millimeter

	Increment system				
	IS-A	**IS-B**	**IS-C**	**IS-D**	**IS-E**
Least input increment (mm)	0.01	0.001	0.0001	0.00001	0.000001
Least command increment (mm)	0.01	0.001	0.0001	0.00001	0.000001
Max. programmable dimension (mm)	±999,999.99	±999,999.999	±99,999.9999	±9,999.99999	±999.999999
Max. rapid traverse (mm/min)[*1]	999,000	999,000	100,000	10,000	1,000
Feedrate range (mm/min)[*1]	0.01 to 999,000	0.01 to 999,000	0.0001 to 100,000	0.00001 to 10,000	0.000001 to 1,000
Incremental feed (mm/step)	0.01 0.1 1.0 10.0	0.001 0.01 0.1 1.0	0.0001 0.001 0.01 0.1	0.00001 0.0001 0.001 0.01	0.000001 0.00001 0.0001 0.001
Tool compensation amount (mm)[*2]	0 to ±9,999.99	0 to ±9,999.999	0 to ±9,999.9999	0 to ±9,999.99999	0 to ±999.999999
Backlash compensation amount (pulses)[*3]	0 to ±9,999	0 to ±9,999	0 to ±9,999	0 to ±9,999	0 to ±9,999
Dwell (sec)[*4]	0 to 999,999.99	0 to 999,999.999	0 to 99,999.9999	0 to 9,999.99999	0 to 999.999999

- In case of inch input, feed screw is millimeter

	Increment system				
	IS-A	**IS-B**	**IS-C**	**IS-D**	**IS-E**
Least input increment (inch)	0.001	0.0001	0.00001	0.000001	0.0000001
Least command increment (inch)	0.001	0.0001	0.00001	0.000001	0.0000001
Max. programmable dimension (inch)	±39,370.078	±39,370.0787	±3,937.00787	±393.700787	±39.3700787
Max. rapid traverse (mm/min)[*1]	999,000	999,000	100,000	10,000	1,000
Feedrate range (inch/min)[*1]	0.001 to 96,000	0.0001 to 9,600	0.00001 to 4,000	0.000001 to 400	0.0000001 to 40
Incremental feed (inch/step)	0.001 0.01 0.1 1.0	0.0001 0.001 0.01 0.1	0.00001 0.0001 0.001 0.01	0.000001 0.00001 0.0001 0.001	0.0000001 0.000001 0.00001 0.0001
Tool compensation amount (inch)[*2]	0 to ±999.999	0 to ±999.9999	0 to ±999.99999	0 to ±999.999999	0 to ±99.9999999
Backlash compensation amount (pulses)[*3]	0 to ±9,999	0 to ±9,999	0 to ±9,999	0 to ±9,999	0 to ±9,999
Dwell (sec)[*4]	0 to 999,999.99	0 to 999,999.999	0 to 99,999.9999	0 to 9,999.99999	0 to 999.999999

- In case of inch input, feed screw is inch

	Increment system				
	IS-A	**IS-B**	**IS-C**	**IS-D**	**IS-E**
Least input increment (inch)	0.001	0.0001	0.00001	0.000001	0.0000001
Least command increment (inch)	0.001	0.0001	0.00001	0.000001	0.0000001
Max. programmable dimension (inch)	±99,999.999	±99,999.9999	±9,999.99999	±999.999999	±99.9999999
Max. rapid traverse (inch/min)[*1]	96,000	9,600	4,000	400	40
Feedrate range (inch/min)[*1]	0.001 to 96,000	0.0001 to 9,600	0.00001 to 4,000	0.000001 to 400	0.0000001 to 40
Incremental feed (inch/step)	0.001 0.01 0.1 1.0	0.0001 0.001 0.01 0.1	0.00001 0.0001 0.001 0.01	0.000001 0.00001 0.0001 0.001	0.0000001 0.000001 0.00001 0.0001
Tool compensation amount (inch)[*4]	0 to ±999.999	0 to ±999.9999	0 to ±999.99999	0 to ±999.999999	0 to ±99.9999999
Backlash compensation amount (pulses)[*3]	0 to ±9,999	0 to ±9,999	0 to ±9,999	0 to ±9,999	0 to ±9,999
Dwell (sec)[*4]	0 to 999,999.99	0 to 999,999.999	0 to 99,999.9999	0 to 9,999.99999	0 to 999.999999

- In case of millimeter input, feed screw is inch

	Increment system				
	IS-A	**IS-B**	**IS-C**	**IS-D**	**IS-E**
Least input increment (mm)	0.01	0.001	0.0001	0.00001	0.000001
Least command increment (mm)	0.01	0.001	0.0001	0.00001	0.000001
Max. programmable dimension (mm)	±999,999.99	±999,999.999	±99,999.9999	±9,999.99999	±999.999999
Max. rapid traverse (inch/min)[*1]	96,000	9,600	4,000	400	40
Feedrate range (mm/min)[*1]	0.01 to 999,000	0.001 to 999,000	0.0001 to 100,000	0.00001 to 10,000	0.000001 to 1,000
Incremental feed (mm/step)	0.01 0.1 1.0 10.0	0.001 0.01 0.1 1.0	0.0001 0.001 0.01 0.1	0.00001 0.0001 0.001 0.01	0.000001 0.00001 0.0001 0.001
Tool compensation amount (mm)[*2]	0 to ±9,999.99	0 to ±9,999.999	0 to ±9,999.9999	0 to ±9,999.99999	0 to ±999.999999
Backlash compensation amount (pulses)[*3]	0 to ±9,999	0 to ±9,999	0 to ±9,999	0 to ±9,999	0 to ±9,999
Dwell (sec)[*4]	0 to 999,999.99	0 to 999,999.999	0 to 99,999.9999	0 to 9,999.99999	0 to 999.999999

- Rotary axis

	Increment system				
	IS-A	**IS-B**	**IS-C**	**IS-D**	**IS-E**
Least input increment (deg)	0.01	0.001	0.0001	0.00001	0.000001
Least command increment (deg)	0.01	0.001	0.0001	0.00001	0.000001
Max. programmable dimension (deg)	±999,999.99	±999,999.999	±99,999.9999	±9,999.99999	±999.999999
Max. rapid traverse (deg/min)*1	999,000	999,000	100,000	10,000	1,000
Feedrate range (deg/min)*1	0.01 to 999,000	0.001 to 999,000	0.0001 to 100,000	0.00001 to 10,000	0.000001 to 1,000
Incremental feed (deg/step)	0.01 0.1 1.0 10.0	0.001 0.01 0.1 1.0	0.0001 0.001 0.01 0.1	0.00001 0.0001 0.001 0.01	0.000001 0.00001 0.0001 0.001
Tool compensation amount (deg)*2	0 to ±9,999.99	0 to ±9,999.999	0 to ±9,999.9999	0 to ±9,999.99999	0 to ±999.999999
Backlash compensation amount (pulses)*3	0 to ±9,999	0 to ±9,999	0 to ±9,999	0 to ±9,999	0 to ±9,999
Dwell (sec)*4	0 to 999,999.99	0 to 999,999.999	0 to 99,999.9999	0 to 9,999.99999	0 to 999.999999

NOTE

*1 The feedrate range shown above are limitations depending on CNC interpolation capacity. As a whole system, limitations depending on servo system must also be considered.

*2 If the mode of input is switched between inch input and metric input, the maximum compensation value that can be set at inch input time is (maximum compensation value) × 1/25.4. If a value exceeding this value is specified at inch input time, the compensation value is not converted to a metric value correctly when the mode of input is switched to metric input.

*3 The unit is the detection unit.

*4 Depends on the increment system of the axis at in address X.

E NOMOGRAPHS

Appendix E, "NOMOGRAPHS", consists of the following sections:

E.1 INCORRECT THREADED LENGTH.....2302
E.2 SIMPLE CALCULATION OF INCORRECT THREAD LENGTH.....2304
E.3 TOOL PATH AT CORNER.....2306
E.4 RADIUS DIRECTION ERROR AT CIRCLE CUTTING.....2309

E.1 INCORRECT THREADED LENGTH

The leads of a thread are generally incorrect in δ_1 and δ_2, as shown in Fig. E.1 (a), due to automatic acceleration and deceleration.

Thus distance allowances must be made to the extent of δ_1 and δ_2 in the program.

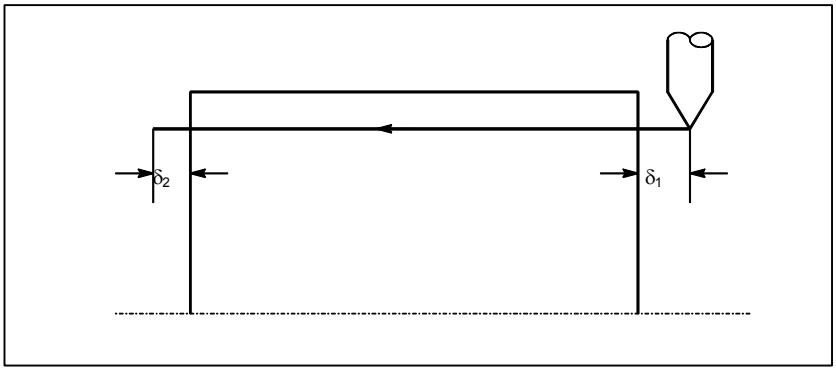


Fig. E.1 (a) Incorrect thread position

Explanation

- How to determine δ_2

$\delta_2 = T_1 V$ (mm). . . . (1)

$$V = \frac{1}{60} RL$$

T_1 : Time constant of servo system (sec)
V : Threading speed (mm/sec)
R : Spindle speed (min^{-1})
L : Thread feed (mm)
Time constant T_1 (sec) of the servo system: Usually 0.033 s.

- How to determine δ_1

$$\delta_1 = \{t - T_1 + T_1 \exp(-\frac{t}{T_1})\} V \quad \ldots\ldots (2)$$

$$a = \exp(-\frac{t}{T_1}) \quad \ldots\ldots (3)$$

T_1 : Time constant of servo system (sec)
V : Threading speed (mm/sec)
Time constant T_1 (sec) of the servo system: Usually 0.033 s.

The lead at the beginning of thread cutting is shorter than the specified lead L, and the allowable lead error is ΔL. Then as follows.

$$a = \frac{\Delta L}{L}$$

When the value of "a" is determined, the time lapse until the thread accuracy is attained. The time "t" is substituted in (2) to determine δ_1: Constants V and T_1 are determined in the same way as for δ_2. Since the calculation of δ_1 is rather complex, a nomography is provided on the following pages.

- How to use nomograph

First specify the class and the lead of a thread. The thread accuracy, a, will be obtained at (1), and depending on the time constant of cutting feed acceleration/ deceleration, the δ_1 value when V = 10mm/s will be obtained at (2). Then, depending on the speed of thread cutting, δ_1 for speed other than 10mm/s can be obtained at (3).

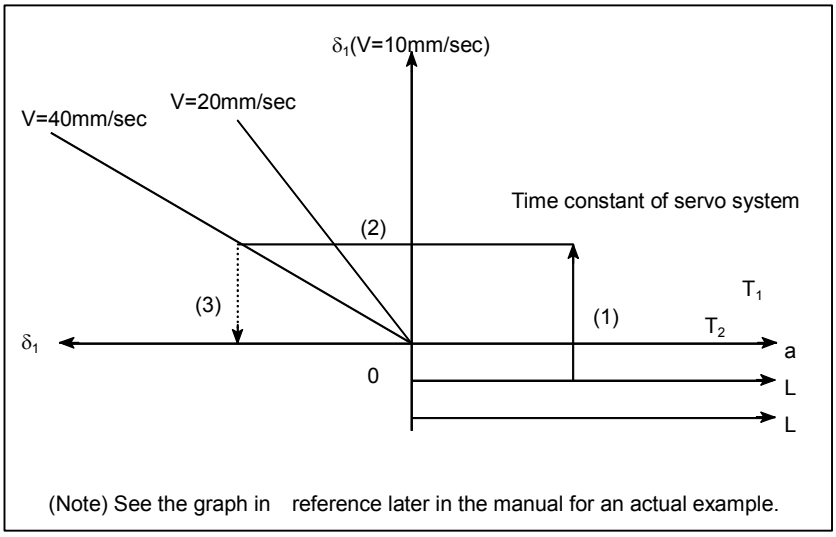


Fig. E.1 (b) Nomograph

NOTE
The equations for δ_1, and δ_2 are for when the acceleration/ deceleration time constant for cutting feed is 0.

E.2 SIMPLE CALCULATION OF INCORRECT THREAD LENGTH

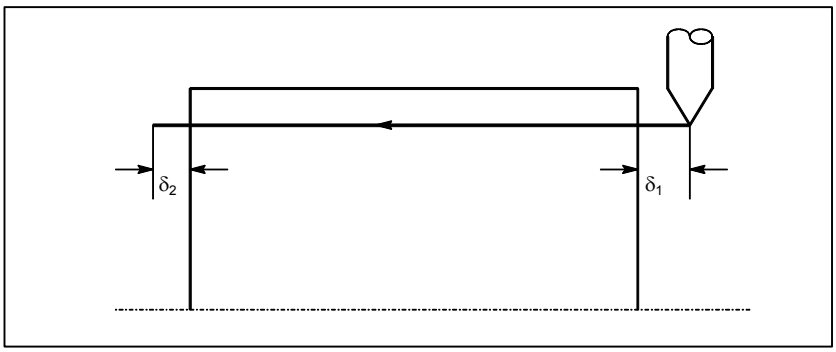


Fig. E.2 (a) Incorrect threaded portion

Explanation

- How to determine δ_2

$$a=\frac{\Delta L}{L}(\text{mm})$$

R : Spindle speed (min^{-1})
L : Thread lead (mm)
* When time constant T_1 of the servo system is 0.033 s.

- How to determine δ_1

$$\delta_1 = \frac{LR}{1800*}(- 1 - \ln a) \ (\text{mm})$$
$$= \delta_2(- 1 - \ln a)(\text{mm})$$

R : Spindle speed (min^{-1})
L : Thread lead (mm)
* When time constant T_1 of the servo system is 0.033 s.
Following a is a permited value of thread.

a	-1-lna
0.005	4.298
0.01	3.605
0.015	3.200
0.02	2.912

Example

R=350rpm
L=1mm
a=0.01
then

$$\delta_2 = \frac{350 \times 1}{1800} = 0.194(\text{mm})$$
$$\delta_1 = \delta_2 \times 3.605 = 0.701(\text{mm})$$

Reference

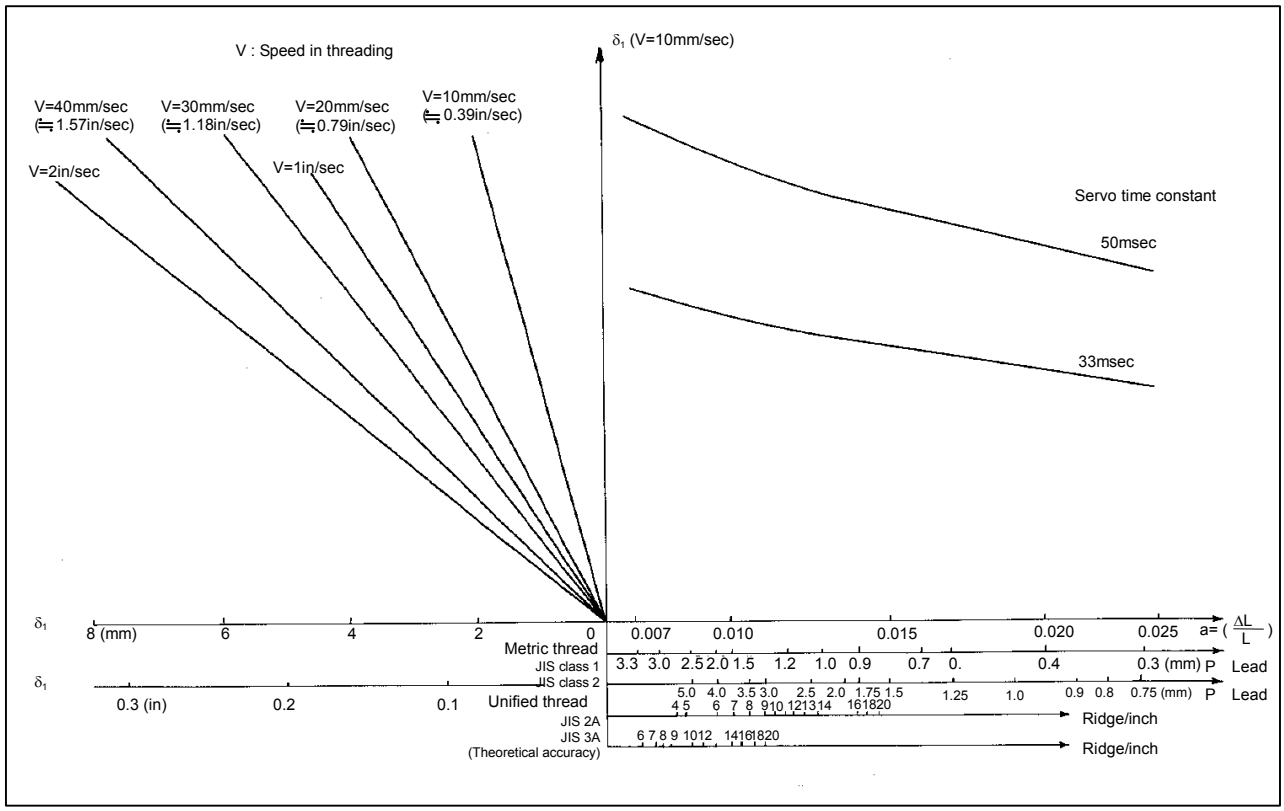


Fig. E.2 (b) Nomograph for obtaining approach distance δ_1

E.3 TOOL PATH AT CORNER

When servo system delay (by exponential acceleration/deceleration at cutting or caused by the positioning system when a servo motor is used) is accompanied by cornering, a slight deviation is produced between the tool path (tool center path) and the programmed path as shown in Fig. E.3 (a).

Time constant T_1 of the exponential acceleration/deceleration is fixed to 0.

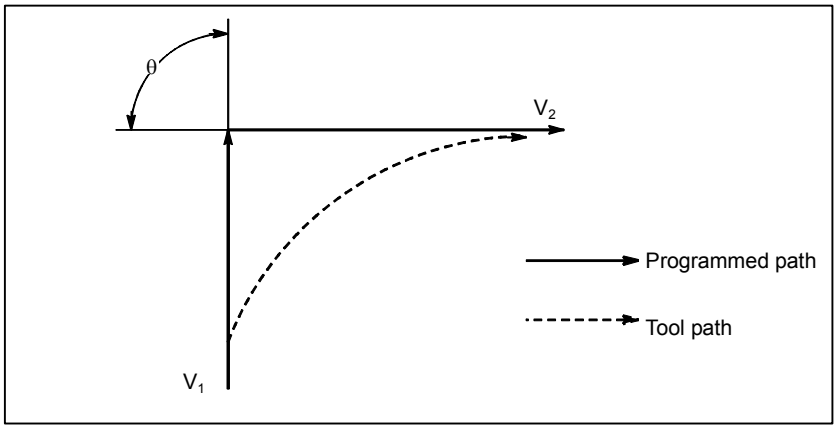


Fig. E.3 (a) Slight deviation between the tool path and the programmed path

This tool path is determined by the following parameters:

- Feedrate (V_1, V_2)
- Corner angle (θ)
- Exponential acceleration / deceleration time constant (T_1) at cutting ($T_1 = 0$)
- Presence or absence of buffer register.

The above parameters are used to theoretically analyze the tool path and above tool path is drawn with the parameter which is set as an example.

When actually programming, the above items must be considered and programming must be performed carefully so that the shape of the workpiece is within the desired precision.

In other words, when the shape of the workpiece is not within the theoretical precision, the commands of the next block must not be read until the specified feedrate becomes zero. The dwell function is then used to stop the machine for the appropriate period.

Explanation

- Analysis

The tool path shown in Fig. E.3 (b) is analyzed based on the following conditions:

- Feedrate is constant at both blocks before and after cornering.
- The controller has a buffer register. (The error differs with the reading speed of the tape reader, number of characters of the next block, etc.)

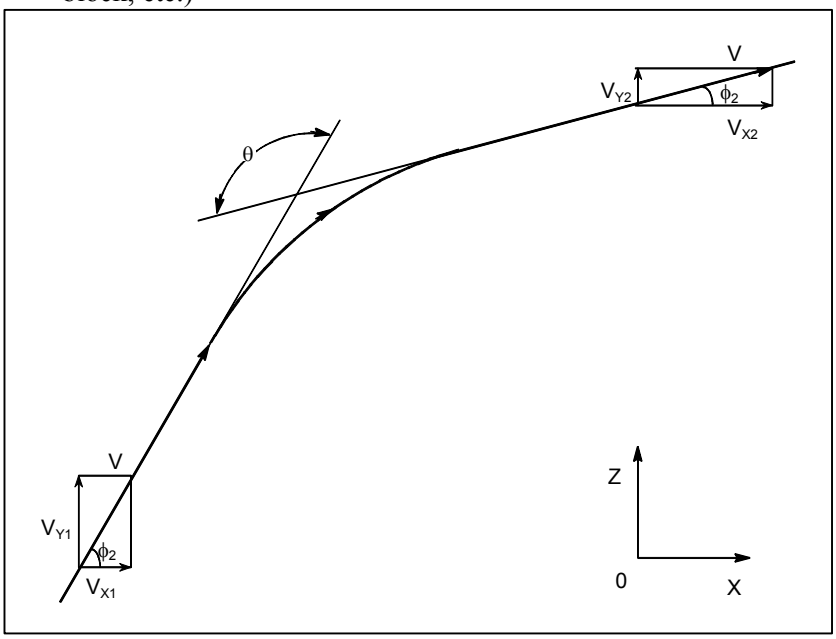


Fig. E.3 (b) Example of tool path

- Description of conditions and symbols

$V_{X1} = V\cos\phi_1$
$V_{Y1} = V\sin\phi_1$
$V_{X2} = V\cos\phi_2$
$V_{Y2} = V\sin\phi 2$

V : Feedrate at both blocks before and after cornering
V_{X1} : X-axis component of feedrate of preceding block
V_{Y1} : Y-axis component of feedrate of preceding block
V_{X2} : X-axis component of feedrate of following block
V_{Y2} : Y-axis component of feedrate of following block
θ : Corner angle
ϕ_1 : Angle formed by specified path direction of preceding block and X-axis
ϕ_2 : Angle formed by specified path direction of following block and X-axis

- Initial value calculation

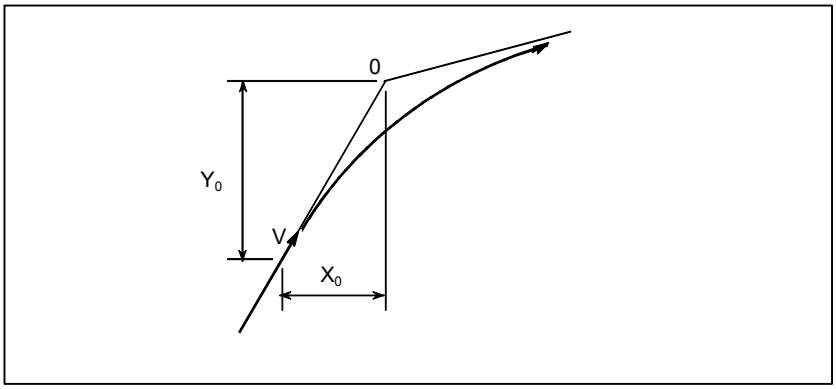


Fig. E.3 (c) Initial value

The initial value when cornering begins, that is, the X and Y coordinates at the end of command distribution by the controller, is determined by the feedrate and the positioning system time constant of the servo motor.

$$X_0 = V_{X1}(T_1 + T_2)$$

$$Y_0 = V_{Y1}(T_1 + T_2)$$

T_1: Exponential acceleration/deceleration time constant. (T=0)
T_2: Time constant of positioning system (Inverse of position loop gain)

- Analysis of corner tool path

The equations below represent the feedrate for the corner section in X-axis direction and Y-axis direction.

$$V_X^{(t)} = (V_{X2} - V_{X1})[1 - \frac{V_{X1}}{T_1 - T_2}\{T_1 \exp(-\frac{t}{T_1}) - T_2 \exp(-\frac{t}{T_2})\} + V_{X1}]$$

$$= V_{X2}[1 - \frac{V_{X1}}{T_1 - T_2}\{T_1 \exp(-\frac{t}{T_1}) - T_2 \exp(-\frac{t}{T_2})\}]$$

$$V_Y^{(t)} = \frac{V_{Y1} - V_{Y2}}{T_1 - T_2}\{T_1 \exp(-\frac{t}{T_1}) - T_2 \exp(-\frac{t}{T_2})\} + V_{Y2}$$

Therefore, the coordinates of the tool path at time t are calculated from the following equations:

$$X(t) = \int_0^t V_X(t)dt - X_0$$

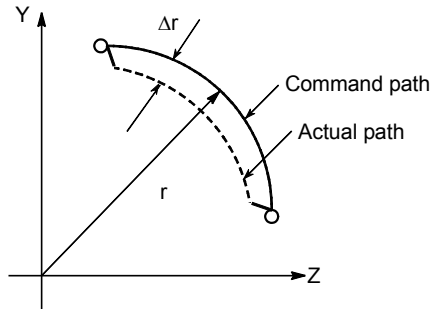
$$= \frac{V_{X2} - V_{X1}}{T_1 - T_2}\{T_1^2 \exp(-\frac{t}{T_1}) - T_2^2 \exp(-\frac{t}{T_2})\} - V_{X2}(T_1 + T_2 - t)$$

$$Y(t) = \int_0^t V_Y(t)dt - Y_0$$

$$= \frac{V_{Y2} - V_{Y1}}{T_1 - T_2}\{T_1^2 \exp(-\frac{t}{T_1}) - T_2^2 \exp(-\frac{t}{T_2})\} - V_{Y2}(T_1 + T_2 - t)$$

E.4 RADIUS DIRECTION ERROR AT CIRCLE CUTTING

When a servo motor is used, the positioning system causes an error between input commands and output results. Since the tool advances along the specified segment, an error is not produced in linear interpolation. In circular interpolation, however, radial errors may be produced, sepecially for circular cutting at high speeds.
This error can be obtained as follows:



$$\Delta r = \frac{1}{2}(T_1^2 + T_2^2(1-\alpha^2))\frac{V^2}{r} \quad \text{..... (1)}$$

Δr: Maximum radius error (mm)
v : Feedrate (mm/sec)
r : Circle radius (mm)
T_1: Exponential acceleration/deceleration time constantat cutting (sec) (T=0)
T_2: Time constant of positoning system (sec) (Inverse of positon loop gain)
α : Feed forward coefficient (%)

In the case of bell-shaped acceleration/deceleration and linear acceleration/deceleration after cutting feed interpolation, an approximation of this radius error can be obtained with the following expression:

Linear acceleration/deceleration after cutting feed interpolation

$$\Delta r = (\frac{1}{24}T_1^2 + \frac{1}{2}T_2^2(1-\alpha^2))\frac{V^2}{r}$$

Bell-shaped acceleration/deceleration after cutting feed interpolation

$$\Delta r = (\frac{1}{48}T_1^2 + \frac{1}{2}T_2^2(1-\alpha^2))\frac{V^2}{r}$$

Thus, the radius error in the case of bell-shaped acceleration/deceleration and linear acceleration/deceleration after interpolation is smaller than in case of exponential acceleration/deceleration by a factor of 12, excluding any error caused by a servo loop time constant.

Since the machining radius r (mm) and allowable error Δr (mm) of the workpiece is given in actual machining, the allowable limit feedrate v (mm /sec) is determined by equation (1).
Since the acceleration/deceleration time constant at cutting which is set by this equipment varies with the machine tool, refer to the manual issued by the machine tool builder.

F SETTINGS AT POWER-ON, IN THE CLEAR STATE, OR IN THE RESET STATE

Either the clear state or reset state is entered during a reset is set by bit 6 (CLR) of parameter No. 3402 (0: reset state/1: clear state).
The symbols in the figure below have the following meanings.

○ : The setting remains unchanged or the operation continues.
× : The setting is deleted or the operation stops.

Item		Power-on	Clear state	Reset state
Settings	Offset	○	○	○
	Set data	○	○	○
	Parameter	○	○	○
Various data items	Program in memory	○	○	○
	Information of the block read in advance during automatic operation	×	×	×
	N (sequence number)	× (Note 7)	× (Notes 1, 2, 7)	○ (Note 1, 2)
	One-shot G code	×	×	×
	Modal G code	Initial value (Note 3) G20 and G21 are set to the state at the previous power-off.	Initial value (Note 3) G20 and G21 are not changed (Note 2).	○ (Note 2 and 6)
	F	0	0	○ (Note 2)
	S, T, M	×	○ (Note 2)	○ (Note 2)
	K (count specification)	×	×	×
Coordinate	Workpiece coordinates	0 (Note 4)	○	○
Operation during execution	Movement	×	×	×
	Dwell	×	×	×
	M, S, T code output	×	×	×
	Tool position compensation	×	See "Tool Position Compensation at a Reset".	See "Tool Position Compensation at a Reset".
	Tool length compensation	×	Depending on the setting of bit 6 (LVK) of parameter No. 5003	Depending on the setting of bit 6 (LVK) of parameter No. 5003
	Tool radius/tool nose radius compensation	×	×	×
	Storing a called subprogram	×	× (Note 5)	×
Output signals	CNC alarm signal AL	"0"(when no alarm cause is present)	"0" (when no alarm cause is present)	"0" (when no alarm cause is present)
	Reference position return completion signal ZPx	×	○ (or × for emergency stop)	○(or × for emergency stop)
	S, T, B codes	×	○	○
	M code	×	×	×
	M, S, T, B strobe signals	×	×	×
	Spindle rotation signal (S analog signal)	×	○	○
	CNC ready signal MA	"1"	○	○
	Servo ready signal SA	"1" (when a servo alarm does not occur)	"1" (when a servo alarm does not occur)	"1" (when a servo alarm does not occur)
	Cycle start signal STL	×	×	×
	Feed hold signal SPL	×	×	×

Tool position compensation at a reset
○ : Cancelled.
× : Not canceled.

Compensation method		Bit 3 (LVC) of parameter No. 5006 and bit 7 (TGC) of parameter No. 5003			
		LVC=0 TGC=0	LVC=1 TGC=0	LVC=0 TGC=1	LVC=1 TGC=1
Tool movement	Wear compensation	×	○ (Axis movement)	×	○ (Axis movement)
	Geometry compensation				
Coordinate shift	Wear compensation	×	○	×	○
	Geometry compensation	×	×	○	○

NOTES

1 When the beginning position is found, the main program number is displayed.
2 If a reset is made during execution of a block, the states of the modal G code and modal address (such as N, F, S, T, or M) specified in the block are not reflected. The modal information specified in the previous and preceding blocks is kept.
3 The initial value of the modal G code depends on the following parameters. For details, refer to the parameter manual (B-63950).
- Parameter G01 (No.3402#0)
- Parameter G18 (No.3402#1)
- Parameter G19 (No.3402#2)
- Parameter G91 (No.3402#3)
- Parameter G23 (No.3402#7)
- Parameters No. 3406 to No. 3409

4 For detection of an absolute position, the value is generated from machine coordinates and a workpiece origin offset.
5 When a reset is made during execution of a subprogram, a return to the main program is performed. Execution at a midpoint in the subprogram is impossible.
6 When one of the following two settings, which hold the modal G code in group 1 by a reset is set:
- Reset state (bit 6 of parameter No. 3402 is 0)
- Clear state (bit 1 of parameter No. 3402 is 1) and

the modal G code in group 1 is held at a time of a reset (bit 1 of parameter No. 3406 is 1),
if a reset is made during execution of one of the following canned cycles, which cause cycle operation, the mode of the modal G code in group 1 is changed to the G01 mode.
- G90 : Outer surface/inner surface turning cycle
- G92 : Threading cycle
- G94 : Edge turning cycle

7 When bit 0 (SEK) of parameter No. 11353 is set to 1, sequence number N can be held even at power-on or in the clear state.

G CHARACTER-TO-CODES CORRESPONDENCE TABLE

Appendix G, "CHARACTER-TO-CODES CORRESPONDENCE TABLE", consists of the following sections:

G.1 CHARACTER-TO-CODES CORRESPONDENCE TABLE 2314
G.2 FANUC DOUBLE-BYTE CHARACTER CODE TABLE....2315

G.1 CHARACTER-TO-CODES CORRESPONDENCE TABLE

Character	Code	Comment	Character	Code	Comment
A	065		6	054	
B	066		7	055	
C	067		8	056	
D	068		9	057	
E	069			032	Space
F	070		!	033	Exclamation mark
G	071		”	034	Quotation mark
H	072		#	035	Sharp
I	073		$	036	Dollar sign
J	074		%	037	Percent
K	075		&	038	Ampersand
L	076		’	039	Apostrophe
M	077		(	040	Left parenthesis
N	078		)	041	Right parenthesis
O	079		*	042	Asterisk
P	080		+	043	Plus sign
Q	081		,	044	Comma
R	082		-	045	Minus sign
S	083		.	046	Period
T	084		/	047	Slash
U	085		:	058	Colon
V	086		;	059	Semicolon
W	087		<	060	Left angle bracket
X	088		=	061	Equal sign
Y	089		>	062	Right angle bracket
Z	090		?	063	Question mark
0	048		@	064	At mark
1	049		[	091	Left square bracket
2	050		]	093	Right square bracket
3	051		^	094	
4	052		_	095	Underscore
5	053				

G.2 FANUC DOUBLE-BYTE CHARACTER CODE TABLE

	00	02	04	06	08	0A	0C	0E	10	12	14	16	18	1A	1C	1E
0200	ぁ	け	ち	ぱ	め	材	面	号	ぉ	す	ど	べ	ら	形	削	度
0220	あ	げ	ぢ	ひ	も	を	最	仕	え	じ	と	へ	よ	成	切	速
0240	ぃ	こ	っ	び	ゃ	ん	小	上	ぇ	し	で	ぷ	ょ	穴	工	向
0260	い	ご	つ	ぴ	や	種	内	込	う	ざ	て	ぶ	ゆ	棒	加	方
0280	ぅ	さ	づ	ふ	ゅ	類	大	点	お	ず	な	ぺ	り	質	倣	送
02A0	が	ぜ	ぬ	ぼ	れ	法	途	開	か	せ	に	ほ	る	寸	正	量
02C0	き	そ	ね	ぽ	ろ	外	中	始	ぎ	ぞ	の	ま	ゎ	径	荒	深
02E0	く	た	は	み	わ	長	具	主	ぐ	だ	ば	む	素	端	番	軸
0300	回	領	源	規	設	仮	無	対	転	域	投	除	定	想	視	相
0320	数	診	入	隅	一	副	器	座	位	断	間	取	覧	行	原	標
0340	置	操	分	単	表	挿	登	示	決	作	秒	補	部	消	録	名
0360	線	引	運	独	合	山	処	変	直	手	自	能	炭	去	再	歯
0380	時	機	負	終	金	高	理	呼	円	械	荷	了	鋼	準	描	推
03A0	反	残	実	記	超	備	画	馬	現	移	使	角	硬	完	過	力
03C0	在	動	用	溝	先	後	容	系	指	次	寿	刃	付	弧	編	選
03E0	令	早	命	幅	摩	助	集	達	値	電	新	広	耗	択	未	閉

	00	02	04	06	08	0A	0C	0E	10	12	14	16	18	1A	1C	1E
0400	禁	復	帰	書	個	桁	稼	由	両	半	逃	底	逆	下	空	四
0420	触	平	代	辺	格	子	周	心	本	群	停	止	巾	微	状	路
0440	範	囲	倍	率	注	側	特	殊	距	離	連	続	増	隔	件	初
0460	期	条	経	握	圧	扱	陰	隠	右	押	横	黄	億	屋	化	何
0480	絵	階	概	該	巻	換	気	起	軌	技	疑	供	共	境	強	教
04A0	掘	繰	係	傾	型	検	権	研	肩	見	験	元	弦	減	孔	巧
04C0	控	更	校	構	根	左	差	雑	参	散	産	算	治	耳	式	失
04E0	修	十	従	勝	商	少	尚	昇	植	色	食	伸	信	侵	振	浸
0500	真	暗	以	意	異	影	鋭	越	価	可	科	果	箇	課	各	拡
0520	核	学	掛	漢	簡	観	関	含	却	客	休	急	業	曲	均	筋
0540	継	計	軽	言	限	互	降	採	済	細	姿	思	写	射	斜	者
0560	車	借	縦	重	出	述	術	渉	照	省	章	証	象	身	進	人
0580	図	違	印	沿	遠	央	奥	往	応	会	解	改	割	活	願	基
05A0	奇	寄	岐	既	近	区	矩	駆	偶	旧	求	球	究	級	欠	結
05C0	口	語	誤	交	厚	項	刻	告	黒	財	策	糸	試	資	事	持
05E0	似	釈	弱	受	収	純	順	所	序	剰	場	常	飾	水	錐	据
0600	制	整	製	前	全	然	則	属	即	他	多	存	探	短	徴	門
0620	鎖	調	頂	鉄	添	頭	同	導	道	熱	年	濃	発	抜	伴	郭
0640	必	百	複	物	文	聞	併	忘	末	密	有	余	裏	立	略	↓
0660	青	席	石	積	赤	接	折	粗	創	双	捜	太	体	待	態	▽▽
0680	替	段	知	地	致	遅	追	通	伝	得	読	凸	突	鈍	敗	般
06A0	杯	背	配	品	不	布	並	頁	別	片	返	勉	保	明	滅	給
06C0	木	目	歪	揺	様	溶	要	抑	良	輪	和	話	節	説	絶	性
06E0	千	専	浅	旋	総	走	退	台	第	題	卓	室	柱	鋳	丁	■■
0700	低	訂	肉	日	白	薄	比	皮	被	非	美	普	伏	歩	包	門
0720	問	絡	列	万	利	訳	礼	乱	放	枚	約	練	油	劣	例	郭
0740	戻	冷	垂	緑	紫	許	測	精	効	→	↗	↑	↖	←	↙	↓
0760	↘	↻	↻	⌒	◯	■	板	予	〃	家	装	管	粉	等	▽	▽▽
0780	▽▽	▽▽	貫	安	α	β	程	抗	張	任	破	損	御	足	守	般
07A0	納	義	丸	汎	固	毎	当	的	詳	鳥	適	論	額	縁	温	給
07C0	界	混	監	締	護	己	称	樹	脂	料	落	確	認	報	排	性
07E0	生	績	判	搬	砥	θ	島	壁	◈	▨	◈	◈	🕮	／／	〕〔	■■

	00	02	04	06	08	0A	0C	0E	10	12	14	16	18	1A	1C	1E
0800	阿	哀	愛	挨	逢	葵	悪	宛	案	闇	鞍	伊	依	偉	委	威
0820	慰	易	為	維	緯	茨	胃	鵜	医	井	育	一	稲	員	因	飲
0840	院	羽	雨	渦	唄	芋	浦	噂	雲	営	映	栄	永	泳	洩	英
0860	衛	液	益	駅	園	姻	延	演	炎	煙	鉛	塩	汚	王	岡	沖
0880	憶	腋	牡	乙	恩	院	穏	佳	夏	暇	架	歌	河	火	花	華
08A0	菓	貨	我	牙	芽	咽	賀	介	壊	廻	快	怪	海	灰	皆	貝
08C0	劾	害	慨	街	垣	園	殻	覚	較	革	楽	笠	括	滑	株	刈
08E0	乾	寒	刊	勧	喚	宛	官	干	幹	感	慣	敢	歓	汗	環	甘
0900	看	喜	缶	肝	還	館	岸	韓	陥	閑	鑑	眼	岩	顔	企	危
0920	緩	希	幾	揮	机	季	稀	祈	毅	棄	旗	徽	輝	騎	擬	犠
0940	議	菊	詰	脚	丘	弓	救	宮	吸	及	久	泣	牛	居	巨	拒
0960	挙	虚	魚	亭	亨	挟	橋	叫	協	競	京	況	狭	胸	興	郷
0980	鏡	響	驚	凝	局	琴	銀	錦	勤	玉	極	九	句	苦	馳	屑
09A0	屈	熊	君	訓	軍	掲	敬	契	兄	刑	郡	景	茎	警	芸	迎
09C0	劇	激	隙	潔	血	券	剣	兼	健	倹	月	圏	堅	建	憲	拳
09E0	犬	献	絹	県	謙	古	庫	幻	険	鍵	軒	戸	故	湖	狐	誇
0A00	好	孝	幸	康	弘	拘	攻									
0A20	講	購	郊	鉱	香	剛	克									
0A40	婚	査	砂	債	妻	彩	才									
0A60	坂	阪	咲	昨	索	錯	桜									
0A80	撤	讃	賛	酸	伺	刺	史									
0AA0	詞	詩	字	寺	磁	辞	七									
0AC0	酒	首	授	需	秋	習	臭									
0AE0	瞬	盾	巡	暑	女	傷	唱									
0B00	昭	招	承	笑	焼	焦	紹	衝	賞	乗	障	情				
0B20	寝	唇	職	申	審	森	神	芯	親	針	辛	尽				
0B40	遂	粋	吹	澄	杉	裾	世	是	勢	政	征	晴				
0B60	誓	西	声	税	請	静	昔	析	籍	跡	責	舌				
0B80	扇	戦	川	洗	栓	泉	染	潜	船	鮮	銑	組				
0BA0	争	巣	掃	騒	窓	草	像	臓	蔵	造	贈	息				
0BC0	尊	揃	其	堕	村	詑	妥	耐	帯	滞	怠	貸				
0BE0	託	濯	拓	脱	濁	奪	棚	誰	嘆	淡	担	弾				

	00	02	04	06	08	0A	0C	0E	10	12	14	16	18	1A	1C	1E
0C00	暖	男	談	池	築	畜	竹	筑	秩	茶	昼	虫	駐	貯	帳	庁
0C20	彫	挑	朝	町	脹	腸	跳	沈	珍	賃	塁	痛	塚	爪	吊	釣
0C40	庭	廷	提	釘	泥	摘	滴	笛	典	天	展	店	貼	殿	田	吐
0C60	塗	徒	都	砥	努	土	怒	倒	冬	凍	刀	島	東	湯	灯	答
0C80	筒	統	到	藤	討	踏	透	働	堂	胴	銅	峠	徳	毒	届	曇
0CA0	謎	鍋	縄	南	軟	難	二	句	乳	尿	念	燃	粘	悩	脳	農
0CC0	把	波	派	廃	拝	肺	買	売	博	拍	泊	舶	麦	肌	畑	八
0CE0	罰	版	犯	班	繁	販	飯	盤	否	彼	悲	扉	批	疲	秘	肥
0D00	費	避	飛	尾	鼻	菱	筆	俵	氷	票	評	病	浜	貧	敏	夫
0D20	婦	富	怖	浮	父	符	腐	武	舞	封	風	服	福	腹	払	沸
0D40	噴	憤	奮	紛	丙	兵	幣	柄	米	壁	癖	偏	便	捕	募	墓
0D60	母	簿	宝	崩	捧	泡	胞	芳	訪	豊	飽	亡	傍	剖	妨	帽
0D80	忙	房	暴	望	紡	肪	膨	防	北	僕	撲	釦	没	翻	磨	魔
0DA0	幕	膜	迄	満	味	魅	脈	妙	民	務	夢	矛	迷	鳴	免	綿
0DC0	模	茂	毛	盲	網	黙	紋	冶	夜	野	矢	役	薬	躍	諭	輸
0DE0	優	友	遊	郵	融	誉	預	幼	揚	曜	洋	葉	陽	養	浴	翼
0E00	螺	来	頼	欄	陸	律	流	留	粒	旅	療	稜	林	臨	隣	涙
0E20	累	励	鈴	暦	歴	烈	裂	労	漏	老	六	脇	惑	詫	湾	腕
0E40	斡	椅	菱	宇	嘘	閲	宴	欧	懐	拐	涯	穫	閣	潟	渇	冠
0E60	患	汽	貴	鬼	偽	戯	欺	喫	窮	糾	拠	漁	恐	狂	脅	仰
0E80	緊	愚	遇	靴	啓	慶	憩	携	撃	傑	嫌	懸	厳	雇	娯	洪
0EA0	紅	耕	航	貢	挫	催	栽	崎	柵	拶	傘	志	施	旨	至	誌
0EC0	識	狩	趣	就	秀	衆	襲	蹴	充	渋	緒	署	諸	叙	掌	訟
0EE0	鐘	壌	織	紳	酔	瀬	誠	繊	漸	繕	塑	礎	阻	奏	族	惰
0F00	戴	諾	叩	旦	誕	恥	仲	宙	忠	抽	兆	懲	抵	敵	撤	党
0F20	盗	糖	陶	闘	督	馴	覇	媒	爆	縛	髪	閥	泌	匹	府	敷
0F40	仏	慕	縫	乏	霧	盟	勇	誘	踊	裸	雷	卵	里	隆	慮	虜
0F60	寮	塁	隷	霊	恋	浪	郎	功	坑	々	⇧	⇔	⊾	∟	⊥	◇
0F80	`	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o
	p	q	r	s	t	u	v	w	x	y	z	{	:	}	~	▨
0FA0	Б	Г	Д	Ж	З	И	Й	К	Л	М	П	У	Ф	Ц	Ч	Ш
	Щ	Ъ	Ы	Ь	Э	Ю	Я		?	⊞	◇	▣				
0FC0	À	Á	Ã	Â	Æ	Ç	È	É	Ê	Ë	Ì	Í	Î	Ï	Ò	Ó
	Ô	Õ	Œ	Ø	Ù	Ú	Û	Ÿ	ß	à	á	â	ã	ä	å	æ
0FE0	Å	Ä	Ö	Ü	Ñ	¿	ç	è	é	ê	ë	ì	í	î	ï	ñ
	ò	ó	ô	õ	ö	œ	ø	ù	ú	û	ü	ÿ	¡			

	00	02	04	06	08	0A	0C	0E	10	12	14	16	18	1A	1C	1E
1000	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
1020	Q	R	S	T	U	V	W	X	Y	Z	a	b	c	d	e	f
1040	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v
1060	w	x	y	z	〃				γ	ε	μ	π	φ	ω	Δ	Σ
1080	Ω								ガ	ギ	グ	ゲ	ゴ	ザ	ジ	ズ	ゼ	ゾ	ダ	ヂ	ヅ	デ	ド
10A0	バ	ビ	ブ	ベ	ボ	パ	ピ	プ	ペ	ポ	ヴ	ヵ	ヶ	🕮	ァ	ア	ィ	イ	ゥ	ウ	ェ	エ
10C0	ォ	オ	カ	ガ	キ	ギ	ク	グ	ケ	ゲ	コ	ゴ	サ	ザ	シ	ジ
10E0	ス	ズ	セ	ゼ	ソ	ゾ	タ	ダ	チ	ヂ	ッ	ツ	ヅ	テ	デ	ト
1100	ド	ナ	ニ	ヌ	ネ	ノ	ハ	バ	パ	ヒ	ビ	ピ	フ	ブ	プ	ヘ
1120	ベ	ペ	ホ	ボ	ポ	マ	ミ	ム	メ	モ	ャ	ヤ	ュ	ユ	ョ	ヨ
1140	ラ	リ	ル	レ	ロ	ヮ	ワ	ヲ	ン	ヴ	ヵ	ヶ	ㇰ	ㇱ	ㇲ	ㇳ
1160	—	、	。	,	·	゛	：	；	?	!	゛	°	´	、	／	＼
1180	―	‖	｜	…	‥	‘	’	仝	々	〃	〇	ー	―	-	～	∥
11A0	～	‖	｜	…	‥	‘	’	“	”	（	）	〔	〕	［	］	｛
11C0	｝	〈	〉	《	》	「	」	『	』	【	】	＋	－	±	×	÷
11E0	＝	≠	＜	＞	≦	≧	∞	∴	♂	♀	°	′	″	℃	¥	$
1200	¢	£	%	♯	&	*	@	§	☆	★	○	●	◎	◇	◆	□
1220	■	△	▲	▽	▼	※	〒	→	←	↑	↓	=	〒	◇	◆	□
1240	1/1	2/2	3/3	4/4	5/5	6/6	㍾	㍽	mm	cm	km	mm²	m²	km²	cm³	m³
1260	mg	kg	cc	dl	l	kl	ms	μs	ns	HP	ps	Hz	(株)	©		
1280	亜	芦	尉	壱	逸	芋	姻	韻	詠	疫	悦	謁	猿	殴	翁	虞
12A0	卸	嫁	禍	彦	蚊	餓	悔	塊	戒	嚇	岳	檄	喝	褐	轄	且
12C0	勘	堪	棺	款	憾	艦	頑	忌	紀	飢	棋	宜	儀	吉	虐	朽
12E0	凶	峡	恭	矯	暁	斤	桐	菌	謹	襟	吟	隈	勲	薫	恵	渓
1300	蛍	鶏	鯨	遣	賢	繭	顕	玄	孤	枯	鼓	呉	悟	碁	后	恒
1320	皇	慌	酵	拷	豪	獄	昆	恨	紺	魂	墾	懇	佐	唆	詐	宰
1340	斎	歳	搾	桟	蚕	惨	暫	氏	司	社	肢	嗣	飼	雌	賜	諮
1360	児	侍	滋	慈	霊	疾	執	漆	舎	赦	遮	邪	蛇	勺	酌	爵
1380	寂	朱	珠	儒	囚	州	宗	拾	愁	酬	醜	汁	銃	獣	叔	淑
13A0	粛	塾	俊	旬	准	殉	循	潤	遵	庶	如	徐	升	召	匠	肖
13C0	償	抄	昌	晶	松	沼	宵	症	祥	硝	粧	詔	彰	礁	丈	冗
13E0	畳	嬢	譲	醸	殖	嘱	辱	臣	娠	慎	薪	仁	迅	甚	尋	炊

	00	02	04	06	08	0A	0C	0E	10	12	14	16	18	1A	1C	1E
1400	婿	租	賊	畜	坪	渡	屯	漠	漂	陛	貿	岬	憂	涼	楼	
1420	逝	祖	憎	稚	漬	斗	篤	迫	姫	墳	冒	漫	雄	硫	廊	郁
1440	牲	禅	霜	痴	津	澱	匿	伯	罷	雰	某	慢	裕	竜	朗	浩
1460	斉	薦	藻	壇	鎮	徹	童	賠	碑	覆	坊	抹	猶	柳	露	槻
1480	姓	遷	燥	鍛	陳	哲	洞	陪	卑	侮	褒	又	悠	履	炉	靖
14A0	随	拙	曹	逮	弔	帝	唐	寧	披	譜	砲	埋	幽	痢	錬	靱
14C0	錘	惜	桑	泰	著	貞	痘	忍	妃	附	俸	妹	唯	吏	廉	鐵
14E0	穂	隻	荘	胎	衷	亭	棟	妊	蛮	賦	抱	麻	癒	濫	麗	緞
1500	帥	脆	措	孫	逐	呈	奴	豚	鉢	苗	塀	謀	眠	羊	猟	賄
1520	衰	夕	疎	駄	窒	弟	桃	尼	伐	猫	弊	朴	娘	庸	陵	國
1540	睡	斥	壮	舵	嫡	邸	悼	弐	帆	賓	遍	牧	銘	窯	僚	揺
1560	穂	隻	荘	胎	衷	亭	棟	妊	畔	頻	舗	墨	妄	擁	糧	條
1580	錘	惜	桑	泰	著	貞	痘	忍	煩	瓶	暮	堀	猛	謡	厘	櫻
15A0	随	拙	曹	逮	弔	帝	唐	寧	頒	扶	邦	奔	匁	翌	倫	澤
15C0	髄	窃	喪	瀧	眺	艇	塔	婆	藩	赴	奉	凡	厄	羅	零	濾
15E0	枢	摂	葬	沢	潮	遥	搭	俳	晩	膚	峰	盆	愉	酪	齢	碌
1600	崇	仙	僧	但	聴	偵	謄	輩	蛮	賦	抱	麻	癒	濫	麗	緞
1620	菅	践	遭	丹	勅	提	豆	梅	妃	附	俸	妹	唯	吏	廉	鐵
1640	畝	銭	槽	胆	朕	送	騰	培	披	譜	砲	埋	幽	痢	錬	靱
1660	姓	遷	燥	鍛	陳	哲	洞	陪	卑	侮	褒	又	悠	履	炉	靖
1680	斉	薦	藻	壇	鎮	徹	童	賠	碑	覆	坊	抹	猶	柳	露	槻
16A0	牲	禅	霜	痴	津	澱	匿	伯	罷	雰	某	慢	裕	竜	朗	浩
16C0	逝	祖	憎	稚	漬	斗	篤	迫	姫	墳	冒	漫	雄	硫	廊	郁
16E0	婿	租	賊	畜	坪	渡	屯	漠	漂	陛	貿	岬	憂	涼	楼	

H ALARM LIST

Appendix H, "ALARM LIST", consists of the following itemss:

(1) Alarms on program and operation (PS alarm).....2322
(2) Background edit alarms (BG alarm) 2322
(3) Communication alarms (SR alarm).....2322
(4) Parameter writing alarm (SW alarm) 2361
(5) Servo alarms (SV alarm).....2361
(6) Overtravel alarms (OT alarm) 2368
(7) Memory file alarms (IO alarm) 2369
(8) Alarms requiring power to be turned off (PW alarm) 2369
(9) Spindle alarms (SP alarm).....2371
(10) Alarm list (serial spindle).....2374
(11) Overheat alarms (OH alarm) 2383
(12) Other alarms (DS alarm) 2384
(13) Malfunction prevention function alarms (IE alarm).....2388

(1) Alarms on program and operation (PS alarm)
(2) Background edit alarms (BG alarm)
(3) Communication alarms (SR alarm)

Alarm numbers are common to all these alarm types.
Depending on the state, an alarm is displayed as in the following examples:
PS"alarm number" Example: PS0003
BG"alarm number" Example: BG0085
SR"alarm number" Example: SR0001

Number	Message	Description
0001	TH ERROR	A TH error was detected during reading from an input device. The read code that caused the TH error and how many statements it is from the block can be verified in the diagnostics screen.
0002	TV ERROR	An error was detected during the single–block TV error. The TV check can be suppressed by setting TVC parameter No. 0000#0 to “0”.
0003	TOO MANY DIGIT	Data entered with more digits than permitted in the NC instruction word. The number of permissible digits varies according to the function and the word.
0004	INVALID BREAK POINT OF WORDS	NC word(s) address + numerical value not in word format. This alarm is also generated when a custom macro does not contain a reserved word, or does not conform to the syntax.
0005	NO DATA AFTER ADDRESS	NC word(s) address + numerical value not in word format. This alarm is also generated when a custom macro does not contain a reserved word, or does not conform to the syntax.
0006	ILLEGAL USE OF MINUS SIGN	A minus sign (–) was specified at an NC instruction word or system variable where no minus signal may be specified.
0007	ILLEGAL USE OF DECIMAL POINT	A decimal point (.) was specified at an address where no decimal point may be specified, or two decimal points were specified.
0009	IMPROPER NC-ADDRESS	An illegal address was specified, or parameter 1020 is not set.
0010	IMPROPER G-CODE	An unusable G code is specified.
0011	FEED ZERO (COMMAND)	The cutting feedrate instructed by an F code has been set to 0. This alarm is also generated if the F code instructed for the S code is set extremely small in a rigid tapping instruction as the tool cannot cut at the programmed lead.
0014	CAN NOT COMMAND G95	A synchronous feed is specified without the option for threading / synchronous feed. Modify the program.
0015	TOO MANY SIMULTANEOUS AXES	A move command was specified for more axes than can be controlled by simultaneous axis control. Either add on the simultaneous axis control extension option, or divide the number of programmed move axes into two blocks.
0020	OVER TOLERANCE OF RADIUS	An arc was specified for which the difference in the radius at the start and end points exceeds the value set in parameter No. 3410. Check arc center codes I, J and K in the program. The tool path when parameter No. 3410 is set to a large value is spiral.

Number	Message	Description
0021	ILLEGAL PLANE SELECT	The plane selection instructions G17 to G19 are in error. Reprogram so that same 3 basic parallel axes are not specified simultaneously. This alarm is also generated when an axis that should not be specified for plane machining is specified, for example, for circular interpolation or involute interpolation. To enable programming of 3 or more axes, the helical interpolation option must be added to each of the relevant axes.
0022	R OR I,J,K COMMAND NOT FOUND	The command for circular interpolation lacks arc radius R or coordinate I, J, or K of the distance between the start point to the center of the arc.
0025	CIRCLE CUT IN RAPID (F0)	F0 (rapid traverse in inverse feed or feed specified by an F code with 1–digit number) was specified during circular interpolation (G02, G03) or involute interpolation (G02.2, G03.2).
0027	NO AXES COMMANDED IN G43/G44	No axis is specified in G43 and G44 blocks for the tool length offset type C. Offset is not canceled but another axis is offset for the tool length offset type C. Multiple axes were specified for the same block when the tool length compensation type is C.
0028	ILLEGAL PLANE SELECT	The plane selection instructions G17 to G19 are in error. Reprogram so that same 3 basic parallel axes are not specified simultaneously. This alarm is also generated when an axis that should not be specified for plane machining is specified, for example, for circular interpolation or involute interpolation. To enable programming of 3 or more axes, the helical interpolation option must be added to each of the relevant axes.
0029	ILLEGAL OFFSET VALUE	Illegal offset No.
0030	ILLEGAL OFFSET NUMBER	An illegal offset No. was specified. This alarm is also generated when the tool shape offset No. exceeds the maximum number of tool offset sets in the case of tool offset memory B.
0031	ILLEGAL P COMMAND IN G10	The relevant data input or option could not be found for the L No. of G10. No data setting address such as P or R was specified. An address command not concerned with data setting was specified. An address varies with the L No. The sign or decimal point of the specified address is in error, or the specified address is out of range.
0032	ILLEGAL OFFSET VALUE IN G10	In setting an offset amount by G10 or in writing an offset amount by system variables, the offset amount was excessive.
0033	NO INTERSECTION AT CUTTER COMPENSATION	The intersection cannot be obtained by the intersection calculation in tool radius/tool nose radius compensation. Modify the program.
0034	NO CIRC ALLOWED IN STUP/EXT BLK	In tool radius/tool nose radius compensation, a startup or cancellation is performed in the G02 or G03 mode. Modify the program.

Number	Message	Description
0035	CAN NOT COMMANDED G31	- G31 cannot be specified. This alarm is generated when a G code (such as for tool radius/tool nose radius compensation) of group 07 is not canceled. - A torque limit skip was not specified in a torque limit skip command (G31P98 or P99). Specify the torque limit skip in the PMC window or the like. Or, specify the torque limit override by address Q.
0037	CAN NOT CHANGE PLANE IN G41/G42	The compensation plane G17/G18/G19 was changed during cutter or tool-nose radius compensation. Modify the program.
0038	INTERFERENCE IN CIRCULAR BLOCK	Overcutting will occur in tool radius/tool nose radius compensation because the arc start point or end point coincides with the arc center. Modify the program.
0039	CHF/CNR NOT ALLOWED IN G41,G42	Chamfering or corner R was specified with a start-up, a cancel, or switching between G41 and G42 in G41 and G42 commands (tool radius/tool nose radius compensation). The program may cause overcutting to occur in chamfering or corner R. Modify the program.
0041	INTERFERENCE IN CUTTER COMPENSATION	In tool radius/tool nose radius compensation, excessive cutting may occur. Modify the program.
0042	G45/G48 NOT ALLOWED IN CRC	Tool offset (G45 to G48) is commanded in tool radius compensation or 3-dimensional cutter compensation. Modify the program.
0043	ILLEGAL T-CODE COMMAND	On a system with a DRILL-MATE ATC installed, M06 is not specified in a block that specifies a T code. Alternatively, a T code beyond the allowable range is specified.
0044	G27-G30 NOT ALLOWED IN FIXED CYC	One of G27 to G30 is commanded in canned cycle mode. Modify the program.
0045	ADDRESS Q NOT FOUND (G73/G83)	In a high-speed peck drilling cycle (G73) or peck drilling cycle (G83), the amount of each-time cutting is not specified by address Q, or Q0 is specified. Modify the program.
0046	ILLEGAL REFERENCE RETURN COMMAND	A command for a return to the second, third or fourth reference position is error. (The address P command is in error.) Although an option for a return to the third or fourth reference position was not set, 3 or 4 was specified in address P.
0047	ILLEGAL AXIS SELECT	Two or more parallel axes (in parallel with a basic axis) have been specified upon start-up of 3-dimensional tool compensation or three-dimensional coordinate conversion.
0048	BASIC 3 AXIS NOT FOUND	Start-up of 3-dimensional tool compensation or three-dimensional coordinate conversion has been attempted, but the three basic axes used when Xp, Yp, or Zp is omitted are not set in parameter No. 1022.
0049	ILLEGAL COMMAND(G68,G69)	When three-dimensional coordinate conversion (G68 or G69) was specified, the tool compensation was not canceled. Or, programs of three-dimensional coordinate conversion (G68, G69) and tool compensation (G43, G44 or G49) were not nested. Or, the three-dimensional coordinate conversion was specified during the tool length compensation and another tool length compensation was specified.
0050	CHF/CNR NOT ALLOWED IN THRD BLK	Chamfering or corner R is commanded in the thread cutting block. Modify the program.
0051	MISSING MOVE AFTER CNR/CHF	Improper movement or the move distance was specified in the block next to the chamfering or corner R block. Modify the program.

Number	Message	Description
0052	CODE IS NOT G01 AFTER CHF/CNR	The block next to the chamfering or corner R block is not G01 (or vertical line). Modify the program.
0053	TOO MANY ADDRESS COMMANDS	In the chamfering and corner R commands, two or more of I, J, K and R are specified.
0054	NO TAPER ALLOWED AFTER CHF/CNR	A block in which chamfering in the specified angle or the corner R was specified includes a taper command. Modify the program.
0055	MISSING MOVE VALUE IN CHF/CNR	In chamfering or corner R block, the move distance is less than chamfer or corner R amount. Modify the program.
0056	NO END POINT & ANGLE IN CHF/CNR	In direct dimension drawing programming, both an end point and an angle were specified in the block next to the block in which only an angle was specified (Aa). Modify the program.
0057	NO SOLUTION OF BLOCK END	Block end point is not calculated correctly in direct dimension drawing programming. Modify the program.
0058	END POINT NOT FOUND	Block end point is not found in direct dimension drawing programming. Modify the program.
0060	SEQUENCE NUMBER NOT FOUND	[External data input/output] The specified number could not be found for program number and sequence number searches. Although input/output of a pot number of tool data or offset input was requested, no tool number was input after power on. The tool data corresponding to the entered tool number could not be found. [External workpiece number search] The program corresponding to the specified workpiece number could not be found. [Program restart] In the program restart sequence number specification, the specified sequence number could not be found.
0061	P OR Q COMMAND IS NOT IN THE MULTIPLE REPETIVE CYCLES BLOCK	Address P or Q is not specified in multiple repetitive cycle (G70, G71, G72, or G73) command.
0062	THE CUTTING AMOUNT IS ILLEGAL IN THE ROUGH CUTTING CYCLE	A zero or a negative value was specified in a multiple repetitive canned rough-cutting cycle (G71 or G72) as the depth of cut.
0063	THE BLOCK OF A SPECIFIED SEQUENCE NUMBER IS NOT FOUND	The sequence number specified by addresses P and Q in multiple repetitive cycle (G70, G71, G72, or G73) command cannot be searched.
0064	THE FINISHING SHAPE IS NOT A MONOTONOUS CHANGE(FIRST AXES)	In a shape program for the multiple repetitive canned rough-cutting cycle (G71 or G72), the command for the first plane axis was not a monotonous increase or decrease.
0065	G00/G01 IS NOT IN THE FIRST BLOCK OF SHAPE PROGRAM	In the first block of the shape program specified by P of the multiple repetitive canned cycle (G70, G71, G72, or G73), G00 or G01 was not specified.
0066	UNAVAILABLE COMMAND IS IN THE MULTIPLE REPETIVE CYCLES BLOCK	An unavailable command was found in a multiple repetitive canned cycle (G70, G71, G72, or G73) command block.
0067	THE MULTIPLE REPETIVE CYCLES IS NOT IN THE PART PROGRAM STORAGE	A multiple repetitive canned cycle (G70, G71, G72, or G73) command is not registered in a tape memory area.
0069	LAST BLOCK OF SHAPE PROGRAM IS AN ILLEGAL COMMAND	In a shape program in the multiple repetitive canned cycle (G70, G71, G72, or G73), a command for the chamfering or corner R in the last block is terminated in the middle.
0070	NO PROGRAM SPACE IN MEMORY	The memory area is insufficient. Delete any unnecessary programs, then retry.

Number	Message	Description
0071	DATA NOT FOUND	- The address to be searched was not found. - The program with specified program number was not found in program number search. - In the program restart block number specification, the specified block number could not be found. Check the data.
0072	DATA NOT FOUND	The number of programs to be stored exceeded 63 (basic), 125 (option), 200 (option), 400 (option) or 1000 (option). Delete unnecessary programs and execute program registration again.
0073	PROGRAM NUMBER ALREADY IN USE	The commanded program number has already been used. Change the program number or delete unnecessary programs and execute program registration again.
0074	PROGRAM NUMBER ALREADY IN USE	The program number is other than 1 to 9999. Modify the program number.
0075	PROTECT	An attempt was made to register a program whose number was protected. In program matching, the password for the encoded program was not correct. An attempt was made to select a program being edited in the background as the main program. An attempt was made to call a program being edited in the background as a subprogram.
0076	PROGRAM NOT FOUND	The specified program is not found in the subprogram call, macro call or graphic copy. The M, G, T or S codes are called by a P instruction other than that in an M98, G65, G66, G66.1 or interrupt type custom macro, and a program is called by a No. 2 auxiliary function code. This alarm is also generated when a program is not found by these calls.
0077	TOO MANY SUB,MACRO NESTING	The total number of subprogram and macro calls exceeds the permissible range. Another subprogram call was executed during an external memory subprogram call.
0078	SEQUENCE NUMBER NOT FOUND	The specified sequence No. was not found during sequence number search. The sequence No. specified as the jump destination in GOTO—— and M99P—— was not found.
0079	PROGRAM NOT MATCH	The program in memory does not match the program stored on tape. Multiple programs cannot be matched continuously when parameter No. 2200#3 is set to “1”. Set parameter No. 2200#3 to “0” before executing a match.
0080	G37 MEASURING POSITION REACHED SIGNAL IS NOT PROPERLY INPUT	- For machining center series When the tool length measurement function (G37) is performed, a measuring position reached signal goes 1 in front of the area determined by the ε value specified in parameter No.6254. Alternatively, the signal does not go 1. - For lathe When the automatic tool compensation function (G36, G37) is used, a measuring position reached signals (XAE1, XAE2) does not go 1 within the range determined by the ε value specified in parameters No.6254 and No.6255.

Number	Message	Description
0081	G37 OFFSET NO. UNASSIGNED	- For machining center series The tool length measurement function (G37) is specified without specifying an H code. Correct the program. - For lathe The automatic tool compensation function (G36, G37) is specified without specifying an T code. Correct the program.
0082	G37 SPECIFIED WITH H CODE	- For machining center series The tool length measurement function (G37) is specified together with an H code in the same block. Correct the program. - For lathe The automatic tool compensation function (G36, G37) is specified together with an T code in the same block. Correct the program.
0083	G37 IMPROPER AXIS COMMAND	- For machining center series An error has been found in axis specification of the tool length measurement function (G37). Alternatively, a move command is specified as an incremental command. Correct the program. - For lathe An error has been found in axis specification of the automatic tool compensation function (G36, G37). Alternatively, a command is specified as an incremental command. Correct the program.
0085	OVERRUN ERROR	The next character was received from the I/O device connected to reader/punch interface 1 before it could read a previously received character.
0086	DR OFF	During I/O process by reader/punch interface 1, the data set ready input signal of the I/O device (DR) was OFF. Possible causes are an I/O device not turn on, a broken cable, and a defective printed circuit board.
0087	BUFFER OVERFLOW	During a read by reader/punch interface 1, although a read stop command was issued, more than 10 characters were input. The I/O device or printed circuit board was defective.
0090	REFERENCE RETURN INCOMPLETE	1. The reference position return cannot be performed normally because the reference position return start point is too close to the reference position or the speed is too slow. Separate the start point far enough from the reference position, or specify a sufficiently fast speed for reference position return. 2. An attempt was made to set the zero position for the absolute position detector by return to the reference position when it was impossible to set the zero point. Rotate the motor manually at least one turn, and set the zero position of the absolute position detector after turning the CNC and servo amplifier off and then on again.
0091	MANUAL REFERENCE POSITION RETURN IS NOT PERFORMED IN FEED HOLD	Manual return to the reference position cannot be performed when automatic operation is halted. Perform the manual return to the reference position when automatic operation is stopped or reset.
0092	ZERO RETURN CHECK (G27) ERROR	The axis specified in G27 has not returned to zero. Reprogram so that the axis returns to zero.

Number	Message	Description
0094	P TYPE NOT ALLOWED (COORD CHG)	P type cannot be specified when the program is restarted. (After the automatic operation was interrupted, the coordinate system setting operation was performed.) Perform the correct operation according to the User's manual.
0095	P TYPE NOT ALLOWED (EXT OFS CHG)	P type cannot be specified when the program is restarted. (After the automatic operation was interrupted, the external workpiece offset amount changed.) Perform the correct operation according to the User's manual.
0096	P TYPE NOT ALLOWED (WRK OFS CHG)	P type cannot be specified when the program is restarted. (After the automatic operation was interrupted, the workpiece offset amount changed.) Perform the correct operation according to the User's manual.
0097	P TYPE NOT ALLOWED (AUTO EXEC)	P type cannot be directed when the program is restarted. (After power ON, after emergency stop or alarms 0094 to 0097 reset, no automatic operation is performed.) Perform automatic operation.
0098	G28 FOUND IN SEQUENCE RETURN	A command of the program restart was specified without the reference position return operation after power ON or emergency stop, and G28 was found during search. Perform the reference position return.
0099	MDI EXEC NOT ALLOWED AFT. SEARCH	After completion of search in program restart, a move command is given with MDI.
0109	FORMAT ERROR IN G08	A value other than 0 or 1 was specified after P in the G08 code, or no value was specified.
0110	OVERFLOW :INTEGER	An integer went out of range during arithmetic calculations.
0111	OVERFLOW :FLOATING	A decimal point (floating point number format data) went out of range during arithmetic calculations.
0112	ZERO DIVIDE	An attempt was made to divide by zero in a custom macro.
0113	IMPROPER COMMAND	A function which cannot be used in custom macro is commanded. Modify the program.
0114	ILLEGAL EXPRESSION FORMAT	The format used in an expression in a custom macro statement is in error. The parameter tape format is in error.
0115	VARIABLE NO. OUT OF RANGE	A number that cannot be used for a local variable, common variable, or system variable in a custom macro is specified. In the EGB axis skip function (G31.8), a non-existent custom macro variable number is specified. Or, the number of custom macro variables used to store skip positions is not sufficient. Alternatively, the header data in high-speed cycle machining is improper. This alarm is issued in the following cases. 1) The header corresponding to the specified call machining cycle number is absent. 2) The value of cycle connection information falls outside the allowable range (0 to 999). 3) The number of data items in the header falls outside the allowable range (1 to 65535). 4) The storage start data variable number of executable data falls outside the allowable ranges (#20000 to #85535/#200000 to #986431/#2000000 to #3999999). 5) The storage end data variable number of executable data falls outside the allowable ranges (#85535/#986431/#3999999). 6) The storage start data variable number of executable data is the same as the variable number used by the header.
0116	WRITE PROTECTED VARIABLE	An attempt was made in a custom macro to use on the left side of an expression a variable that can only be used on the right side of an expression.

Number	Message	Description
0118	TOO MANY BRACKET NESTING	Too many brackets “[ ]” were nested in a custom macro. The nesting level including function brackets is 5.
0119	ARGUMENT VALUE OUT OF RANGE	The value of an argument in a custom macro function is out of range.
0122	TOO MANY MACRO NESTING	Too many macro calls were nested in a custom macro.
0123	ILLEGAL MODE FOR GOTO/WHILE/DO	A GOTO statement or WHILE–DO statement was found in the main program in the MDI or DNC mode.
0124	MISSING END STATEMENT	The END instruction corresponding to the DO instruction was missing in a custom macro.
0125	MACRO STATEMENT FORMAT ERROR	The format used in a macro statement in a custom macro is in error.
0126	ILLEGAL LOOP NUMBER	DO and END Nos. in a custom macro are in error, or exceed the permissible range (valid range: 1 to 3).
0127	DUPLICATE NC,MACRO STATEMENT	An NC statement and macro statement were specified in the same block.
0128	ILLEGAL MACRO SEQUENCE NUMBER	The specified sequence No. could not be found for sequence number search. The sequence No. specified as the jump destination in GOTO-- and M99P-- could not be found.
0129	USE 'G' AS ARGUMENT	G is used as an argument in a custom macro call. G can be specified as an argument only in an every-block call (G66.1).
0130	NC AND PMC AXIS ARE CONFLICTED	The NC command and the PMC axis control command were conflicted. Modify the program or ladder.
0136	SPOS AXIS - OTHER AXIS SAME TIME	The spindle positioning axis and another axis are specified in the same block.
0137	M-CODE & MOVE CMD IN SAME BLK.	The spindle positioning axis and another axis are specified in the same block.
0138	SUPERIMPOSED DATA OVERFLOW	The total distribution amount of the CNC and PMC is too large during superimposed control for PMC axis control.
0139	CANNOT CHANGE PMC CONTROL AXIS	The PMC axis was selected for the axis for which the PMC axis is being controlled.
0140	PROGRAM NUMBER ALREADY IN USE	In the background, an attempt was made to select or delete the program being selected in the foreground. Perform the correct operation for the background edition.
0141	CAN NOT COMMAND G51 IN 3-D OFFSET	G51 (Scaling ON) is commanded in the 3-dimensional tool compensation mode. Modify the program.
0142	ILLEGAL SCALE RATE	The scaling rate is 0 times or 10000 times or more. Modify the setting of the scaling rate. (G51P_ … or G51I_J_K_ … or parameter (No. 5411 or 5421))
0143	COMMAND DATA OVERFLOW	An overflow occurred in the storage length of the CNC internal data. This alarm is also generated when the result of internal calculation of scaling, coordinate rotation and cylindrical interpolation overflows the data storage. It also is generated during input of the manual intervention amount.
0144	ILLEGAL PLANE SELECTED	The coordinate rotation plane and arc or tool radius·tool nose radius compensation plane must be the same. Modify the program.
0145	ILLEGAL USE OF G12.1/G13.1	The axis No. of plane selection parameter No. 5460 (linear axis) and No. 5461(axis of rotation) in the polar coordinate interpolation mode is out of range (1 to number of controlled axes).

Number	Message	Description
0146	ILLEGAL USE OF G-CODE	The modal G code group contains an illegal G code in the polar coordinate interpolation mode or when a mode was canceled. Only the following G codes are allowed: G40, G50, G69.1 An illegal G code was specified while in the polar coordinate interpolation mode. The following C codes are not allowed: G27, G28, G30, G30.1, G31 to G31.4, G37 to G387.3, G52, G92, G53, G17 to G19, G81 to G89, G68 In the 01 group, G codes other than G01, G02, G03, G02.2 and G03.2 cannot be specified.
0148	SETTING ERROR	Automatic corner override deceleration rate is out of the settable range of judgement angle. Modify the parameters (No.1710 to No.1714).
0149	FORMAT ERROR IN G10L3	In registration (G10L3 to G11) of tool life management data, an address other than Q1, Q2, P1, and P2 or an unusable address was specified.
0150	ILLEGAL LIFE GROUP NUMBER	The tool group number exceeded the maximum allowable value. The tool group number (P after specification of G10 L3;) or the group number given by the tool life management T code in a machining program.
0151	GROUP NOT FOUND AT LIFE DATA	The tool group specified in a machining program is not set in tool life management data.
0152	OVER MAXIMUM TOOL NUMBER	The number of tools registered in one group exceeded the maximum allowable registration tool number.
0153	T-CODE NOT FOUND	In registration of tool life data, a block in which the T code needs to be specified does not include the T code. Alternatively, in tool exchange method D, M06 is specified solely. Modify the program.
0154	NOT USING TOOL IN LIFE GROUP	- For the tool management command H99 or D99 was specified when no tool management data number is assigned to the spindle position. Modify the program. - For the tool life management command The H99 command, D99 command, or the H/D code set by parameters No. 13265 and No. 13266 was specified when no tool belonging to a group is used.
0155	ILLEGAL T-CODE COMMAND	In the machining program, the T code that is present in the block containing M06 does not correspond to the group currently being used. Modify the program.
0156	P/L COMMAND NOT FOUND	The P and L commands are not specified in the beginning of a program for setting a tool group. Modify the program.
0157	TOO MANY TOOL GROUPS	In registration of tool life management data, the group setting command block counts of P (group number) and L (tool life) exceeded the maximum group count.
0158	TOOL LIFE VALUE OUT OF RANGE	The life value that is being set is too large. Change the setting.
0159	ILLEGAL TOOL LIFE DATA	Tool life management data is corrupted for some reason. Register the tool data in the tool group or the tool data in the group again by G10L3; or MDI input.

Number	Message	Description
0160	MISMATCH WAITING M-CODE	A waiting M-code is in error. <1> When different M codes are specified for path 1 and path 2 as waiting M codes without a P command. <2> When the waiting M codes are not identical even though the P commands are identical <3> When the waiting M codes are identical and the P commands are not identical (This occurs when a P command is specified with binary value.) <4> When the number lists in the P commands contain a different number even though the waiting M codes are identical (This occurs when a P command is specified by combining path numbers.) <5> When a waiting M code without a P command (2-path waiting) and a waiting M code with a P command (3-or-more-path waiting) were specified at the same time <6> When a waiting M code without a P command was specified for 3 or more paths.
0161	ILLEGAL P OF WAITING M-CODE	P in a waiting M-code is incorrect. <1> When address P is negative <2> When a P value inappropriate for the system configuration was specified <3> When a waiting M code without a P command (2-path waiting) was specified in the system having 3 or more paths.
0163	ILLEGAL COMMAND IN G68/G69	G68 and G69 are not independently commanded in balance cut. An illegal value is commanded in a balance cut combination (address P).
0169	ILLEGAL TOOL GEOMETRY DATA	Incorrect tool figure data in interference check. Set correct data, or select correct tool figure data.
0175	ILLEGAL G07.1 AXIS	An axis which cannot perform cylindrical interpolation was specified. More than one axis was specified in a G07.1 block. An attempt was made to cancel cylindrical interpolation for an axis that was not in the cylindrical interpolation mode. For the cylindrical interpolation axis, set not “0” but one of 5, 6 or 7 (parallel axis specification) to parameter No. 1022 to instruct the arc with axis of rotation (ROT parameter No. 1006#1 is set to “1” and parameter No. 1260 is set) ON.
0176	ILLEGAL G-CODE USE(G07.1 MODE)	A G code was specified that cannot be specified in the cylindrical interpolation mode. This alarm also is generated when an 01 group G code was in the G00 mode or code G00 was instructed. Cancel the cylindrical interpolation mode before instructing code G00.

Number	Message	Description
0178	ILLEGAL COMMAND G05	The settings of bits 4 to 6 of parameter No.7501 are invalid or G05 was specified in any of the following mode. - Hypothetical axis interpolation (G07) - Cylindrical interpolation (G07.1) - Polar coordinate interpolation (G12.1) - Polar coordinates command (G16) - Spindle speed fluctuation detection (G26) - Tool radius · tool nose radius compensation (G41/G42) - Normal direction control (G41.1/G42.1) - Scaling (G51) - Programmable mirror image (G51.1) - Coordinate system rotation (G68) - Canned cycle (G81 to G89) - Constant surface speed control (G96) - Macro interruption(M96)
0179	PARAM. (NO.7510) SETTING ERROR	The number of controlled axes set by the parameter No. 7510 exceeds the maximum number. Modify the parameter setting value. The distribution of high-speed cycle machining stopped.
0190	ILLEGAL AXIS SELECTED (G96)	An illegal value was specified in P in a G96 block or parameter No. 5844.
0194	SPINDLE COMMAND IN SYNCHRO-MODE	A Cs contour control mode, spindle positioning command, or rigid tapping mode was specified during the spindle synchronous control mode or simple spindle synchronous control mode.
0197	C-AXIS COMMANDED IN SPINDLE MODE	The program specified a movement along the Cs-axis when the Cs contour control switching signal was off.
0199	MACRO WORD UNDEFINED	Undefined macro word was used. Modify the custom macro.
0200	ILLEGAL S CODE COMMAND	In the rigid tap, an S value was out of range or was not specified. The parameter (Nos. 5241 to 5243) setting is an S value which can be specified for the rigid tap. Correct the parameters or modify the program.
0201	FEEDRATE NOT FOUND IN RIGID TAP	The command F code for a cutting feedrate is a zero. If the value of F command is much smaller than that of the S command, when a rigid tap command is specified, this alarm is generated. This is because cutting is not possible by the lead specified by the program.
0202	POSITION LSI OVERFLOW	In the rigid tap, spindle distribution value is too large. (System error)
0203	PROGRAM MISS AT RIGID TAPPING	In the rigid tap, position for a rigid M code (M29) or an S command is incorrect. Modify the program.
0204	ILLEGAL AXIS OPERATION	In the rigid tap, an axis movement is specified between the rigid M code (M29) block and G84 (or G74) block. Modify the program.
0205	RIGID MODE DI SIGNAL OFF	Although a rigid M code (M29) is specified in rigid tapping, the rigid mode DI signal (DGN G061.0) is not ON during execution of the G84 (or G74) block. Check the PMC ladder diagram to find the reason why the DI signal is not turned on.
0206	CAN NOT CHANGE PLANE (RIGID TAP)	Plane changeover was instructed in the rigid mode. Modify the program.
0207	RIGID DATA MISMATCH	The specified distance was too short or too long in rigid tapping.

Number	Message	Description
0210	CAN NOT COMMAND M198/M99	1 The execution of an M198 or M99 command was attempted during scheduled operation. Alternatively, the execution of an M198 command was attempted during DNC operation. Modify the program. 2 The execution of an M99 command was attempted by an interrupt macro during pocket machining in a multiple repetitive canned cycle.
0212	ILLEGAL PLANE SELECT	The direct drawing dimensions programming is commanded for the plane other than the Z-X plane. Correct the program.
0213	ILLEGAL COMMAND IN SYNCHRO-MODE	In feed axis synchronization control, the following errors occurred during the synchronous operation. 1) The program issued the move command to the slave axis. 2) The program issued the manual operation to the slave axis. 3) The program issued the automatic reference position return command without specifying the manual reference position return after the power was turned on.
0214	ILLEGAL COMMAND IN SYNCHRO-MODE	Coordinate system is set or tool compensation of the shift type is executed in the synchronous control. Correct the program.
0217	DUPLICATE G51.2(COMMANDS)	G51.2 is further commanded in the G51.2 mode. Modify the program.
0218	NOT FOUND P/Q COMMAND	P or Q is not commanded in the G51.2 block, or the command value is out of the range. Modify the program. For a polygon turning between spindles, more information as to why this alarm occurred is indicated in DGN No. 471.
0219	COMMAND G51.2/G50.2 INDEPENDENTLY	G51.2 and 50.2 were specified in the same block for other commands. Modify the program in another block.
0220	ILLEGAL COMMAND IN SYNCHR-MODE	In the synchronous operation, movement is commanded by the NC program or PMC axis control interface for the synchronous axis. Modify the program or check the PMC ladder.
0221	ILLEGAL COMMAND IN SYNCHR-MODE	Polygon machining synchronous operation and axis control or balance cutting are executed at a time. Modify the program.
0222	DNC OP. NOT ALLOWED IN BG-EDIT	Input and output are executed at a time in the background edition. Execute a correct operation.
0224	ZERO RETURN NOT FINISHED	A reference return has not been performed before the start of automatic operation. (Only when bit 0 (ZRNx) of parameter No. 1005 is 0) Perform a reference position return.
0230	R CODE NOT FOUND	Cut depth R is not specified in the block including G161. Alternatively, the value specified for R is negative. Modify the program.
0231	ILLEGAL FORMAT IN G10 L52	Errors occurred in the specified format at the programmable-parameter input.
0232	TOO MANY HELICAL AXIS COMMAND	Three or more axes were specified as helical axes in the helical interpolation mode. Five or more axes were specified as helical axes in the helical interpolation B mode.
0233	DEVICE BUSY	When an attempt was made to use a unit such as that connected via the RS-232-C interface, other users were using it.
0241	ILLEGAL FORMAT IN G02.2/G03.2	The end point of an involute curve on the currently selected plane, or the center coordinate instruction I, J or K of the corresponding basic circle, or basic circle radius R was not specified.

Number	Message	Description
0242	ILLEGAL COMMAND IN G02.2/G03.2	An illegal value was specified in the involute curve. The coordinate instruction I, J or K of the basic circle on the currently selected plane or the basic circle radius R is “0”, or the start and end points are not inside the basic circle.
0243	OVER TOLERANCE OF END POINT	The end point is not positioned on the involute curve that passes through the start point, and this error exceeds the permissible error limit (parameter No. 2510).
0245	T-CODE NOT ALLOWED IN THIS BLOCK	One of the G codes, G50, G10, and G04, which cannot be specified in the same block as a T code, was specified with a T code.
0247	THE MISTAKE IS FOUND IN THE OUTPUT CODE OF DATA.	When an encrypted program is output, EIA is set for the output code. Specify ISO.
0250	TOOL CHANGE ILLEGAL Z AXIS COMMAND	A Z-axis move command was performed in the same block for M06 command.
0251	TOOL CHANGE ILLEGAL T COMMAND	An unusable T code was specified in M06Txx.
0253	G05 CAN NOT BE COMMANDED	A binary operation was specified during advanced preview control mode.
0300	ILLEGAL COMMAND IN SCALING	An illegal G code was specified during scaling. Modify the program. For the T system, one of the following functions is specified during scaling, this alarm is generated. - finishing cycle (G70 or G72) - outer surface rough-cutting cycle (G71 or G73) - end side rough-cutting cycle (G72 or G74) - closed loop cutting cycle (G73 or G75) - end side cutting-off cycle (G74 or G76) - outer surface or inner surface cutting-off cycle (G75 or G77) - multiple repetitive threading cycle (G76 or G78) - face drill cycle (G83 or G83) - face tap cycle (G84 or G84) - face boring cycle (G85 or G85) - side drill cycle (G87 or G87) - side tap cycle (G88 or G88) - side boring cycle (G89 or G89) - outer surface turning cycle or inner surface boring cycle (G77 or G20) - threading cycle (G78 or G21) - end side turning cycle (G79 or G24) (Specify G codes for systems B and C in that order.)
0301	RESETTING OF REFERENCE RETURN IS INHIBITED	Although parameter No. 1012#0 (IDGx) was set to 1 to inhibit the reference position from being set again for a return to the reference position without a dog, an attempt was made to perform a manual return to the reference position.
0302	SETTING THE REFERENCE POSITION WITHOUT DOG IS NOT PERFORMED	The reference position could not be set for a return to the reference position without a dog. Possible causes are: - The axis was not moved in the direction of a return to the reference position for jog feeding. - The axis was moved in the direction opposite to the direction of a manual return to the reference position.

Number	Message	Description
0303	REFERENCE POSITION RETURN IS NOT PERFORMED	When the setting of a reference position at any position was possible in Cs contour control (parameter CRF (No. 3700#0) = 1), a G00 command was issued for the Cs contour axis without a return to the reference position after the serial spindle was switched to Cs contour control mode. Perform a reference position return with a G28 command before issuing a G00 command.
0304	G28 IS COMMANDED WITHOUT ZERO RETURN	Although a reference position was not set, an automatic return to the reference position (G28) was commanded.
0305	INTERMEDIATE POSITION IS NOT ASSIGNED	Although a G28 (automatic return to the reference position), G30 (return to the second, third, or fourth reference position), or G30/1 (return to the floating reference position) command was not issued after power-up, G29 (return from the reference position) was commanded.
0306	MISMATCH AXIS WITH CNR/CHF	The correspondence between the moving axis and the I, J, or K command is incorrect in a block in which chamfering is specified.
0307	CAN NOT START REFERENCE RETURN WITH MECHANICAL STOPPER SETTING	An attempt was made to set a butt-type reference position for an axis for which to use the function to set a reference position without a dog.
0308	G72.1 NESTING ERROR	G72.1 was specified again during G72.1 rotation copying.
0309	G72.2 NESTING ERROR	G72.2 was specified again during G72.2 parallel copying.
0310	FILE NOT FOUND	The specified file could not be found during a subprogram or macro call.
0311	CALLED BY FILE NAME FORMAT ERROR	An invalid format was specified to call a subprogram or macro using a file name.
0312	ILLEGAL COMMAND IN DIRECT DRAWING DIMENSIONS PROGRAMMING	Direct input of drawing dimensions was commanded in an invalid format. An attempt was made to specify an invalid G code during direct input of drawing dimensions. Two or more blocks not to be moved exist in consecutive commands that specify direct input of drawing dimensions. Although non-use of commas (,) (parameter No. 3405#4 = 1) was specified for direct input of drawing dimensions, a comma was specified.
0313	ILLEGAL LEAD COMMAND	The variable-lead threading increment specified in address K exceeds the specified maximum value in variable-lead threading. Or, a negative lead value was specified.
0314	ILLEGAL SETTING OF POLYGONAL AXIS	An axis was specified invalidly in polygon turning. For polygon turning: A tool rotation axis is not specified. (Parameter No. 7610) For polygon turning between spindles: Valid spindles are not specified. (Parameter Nos. 7640 to 7643) - A spindle other than the serial spindle. - A spindle is not connected.
0315	ILLEGAL NOSE ANGLE COMMAND IS IN THE THREAD CUTTING CYCLE	An invalid tool tip angle is specified in a multiple repetitive canned threading cycle (G76).
0316	ILLEGAL CUTTING AMOUNT IS IN THE THREAD CUTTING CYCLE	An minimum depth of cut higher than the thread height is specified in a multiple repetitive canned threading cycle (G76).
0317	ILLEGAL THREAD COMMAND IS IN THE THREAD CUTTING CYCLE	A zero or a negative value is specified in a multiple repetitive canned threading cycle (G76) as the thread height or the depth of cut.

Number	Message	Description
0318	ILLEGAL RELIEF AMOUNT IS IN THE DRILLING CYCLE	Although an escape directions is set in a multiple repetitive canned cutting-off cycle (G74 or G75), a negative value is specified for Δd.
0319	THE END POINT COMMAND IS ILLEGAL IN THE DRILLING CYCLE	Although the Δi or Δk travel distance is set to 0 in a multiple repetitive canned cutting-off cycle (G74 or G75), a value other than 0 us specified for a U or W.
0320	ILLEGAL MOVEMENT AMOUNT/CUTTING AMOUNT IS IN THE DRILLING CYCLE	A negative value is specified in a multiple repetitive canned cutting-off cycle (G74 or G75) as Δi or Δk (travel distance/the depth of cut).
0321	ILLEGAL REPEATED TIME IS IN THE PATTERN REPEATING CYCLE	A zero or a negative value is specified in a multiple repetitive canned closed loop cycle (G73) as a repeated time.
0322	FINISHING SHAPE WHICH OVER OF STARTING POINT	An invalid shape which is over the cycle starting point is specified in a shape program for a multiple repetitive canned rough-cutting cycle (G71 or G72).
0323	THE FIRST BLOCK OF SHAPE PROGRAM IS A COMMAND OF TYPE II	Type II is specified in the first block of the shape program specified by P in a multiple repetitive canned rough-cutting cycle (G71 or G72). Z (W) command is for G71. X (U) command is for G72.
0324	THE INTERRUPTION TYPE MACRO WAS DONE IN THE MULTIPLE REPETIVE CYCLES	An interruption type macro was issued during the multiple repetitive canned cycle (G70, G71, G72, or G73).
0325	UNAVAILABLE COMMAND IS IN SHAPE PROGRAM	An usable command was issued in a shape program for a multiple repetitive canned cycle (G70, G71, G72, or G73).
0326	LAST BLOCK OF SHAPE PROGRAM IS A DIRECT DRAWING DIMENSIONS	In a shape program in the multiple repetitive canned cycle (G70, G71, G72, or G73), a command for direct input of drawing dimensions in the last block is terminated in the middle.
0327	MODAL THAT MULTIPLE REPETIVE CYCLES CANNOT BE DONE	A multiple repetitive canned cycle (G70, G71, G72, or G73) was commanded in a modal state in which a multiple repetitive canned cycle could not be commanded.
0328	ILLEGAL WORK POSITION IS IN THE TOOL NOSE RADIUS COMPENSATION	The specification for the blank side for a tool nose radius compensation (G41 or G42) is incorrect in a multiple repetitive canned cycle (G71 or G72).
0329	THE FINISHING SHAPE IS NOT A MONOTONOUS CHANGE(SECOND AXES)	In a shape program for the multiple repetitive canned rough-cutting cycle (G71 or G72), the command of the second plane axis was not a monotonous increase or decrease.
0330	ILLEGAL AXIS COMMAND IS IN THE TURNING CANNED CYCLE	An axis other than the plane is specified n a canned cycle(G90, G92, or G94).
0331	ILLEGAL AXIS NUMBER IN AX[]	An illegal value is specified for an AX[] axis number.
0332	ILLEGAL AXIS ADDRESS IN AXNUM[]	An illegal value is specified for an AXNUM[] axis address.
0333	TOO MANY SPINDLE COMMANDS	Multiple spindle commands could be found in the same block in using an expansion spindle name. Only one spindle could be commanded in the same block.
0334	OFFSET IS OUT OF EFFECTIVE RANGE	An offset data which was out of the effective range was specified. (malfunction prevention function)
0335	PLURAL M CODE	Multiple M codes are commanded simultaneously in a block for a wait function with peripheral devices by an M code.
0336	TOOL COMPENSATION COMMANDED MORE TWO AXES	For a tool length compensation C, an attempt was made to command the offset to other axes without canceling the offset. Or, for a tool length compensation C, multiple axes are specified in G43 or G44 block.
0337	EXCESS MAXIMUM INCREMENTAL VALUE	The command value exceeded the maximum amount of incremental. (malfunction prevention function)

Number	Message	Description
0338	CHECK SUM ERROR	An incorrect value was detected in a check sum. (malfunction prevention function)
0340	ILLEGAL RESTART(NANO SMOOTHING)	With manual absolute turned on, an attempt was made to restart the operation in nano smoothing mode after performing the manual interaction.
0341	TOO MANY COMMAND BLOCK (NANO SMOOTHING)	There are more blocks than can be commanded consecutively in nano smoothing mode.
0342	CUSTOM MACRO INTERRUPT ENABLE IN NANO SMOOTHING	A custom macro interrupt was enabled in nano smoothing mode. Or, nano smoothing mode was commanded with a custom macro interrupt enabled.
0343	ILLEGAL COMMAND IN NANO SMOOTHING	G43, G44, or G49 was commanded during a nano smoothing.
0344	CANNOT CONTINUE NANO SMOOTHING	An illegal command or operation by which a nano smoothing could not be continued was performed.
0345	TOOL CHANGE ILLEGAL Z AXIS POS	A tool change position on the Z-axis is incorrect.
0346	TOOL CHANGE ILLEGAL TOOL NUM	A tool change position is not set.
0347	TOOL CHANGE ILLEGAL COMMAND IN SAME BLK.	Tool changing is commanded twice or more in the same block.
0348	TOOL CHANGE Z AXIS POS NOT ESTABLISHED	A tool change spindle on the Z-axis is not set.
0349	TOOL CHANGE SPINDLE NOT STOP	A tool change spindle stop is not stopped.
0350	PARAMETER OF THE INDEX OF THE SYNCHRONOUS CONTROL AXIS SET ERROR.	An illegal synchronization control axis number (parameter No. 8180) is set.
0351	BECAUSE THE AXIS IS MOVING, THE SYNC CONTROL IS CAN'T BE USED.	While the axis being subject to synchronization control was moving, an attempt was made to start or cancel the synchronization control by a synchronization control axis selection signal.
0352	SYNCHRONOUS CONTROL AXIS COMPOSITION ERROR.	This error occurred when: 1) An attempt was made to perform synchronization control for the axis during a synchronization, composition, or superposition. 2) An attempt was made to synchronize a further great-grandchild for a parent-child-grandchild relation. 3) An attempt was made to operate synchronization control although a parent-child-grandchild relation was not set.
0353	THE INSTRUCTION WAS DONE FOR THE AXIS WHICH WAS NOT ABLE TO MOVE.	This error occurred when: - For synchronization 1) A move command was issued to the axis for which parameter No. 8163#7NUMx is set to 1. 2) A move command was issued to the slave axis. - For composition 1) A move command was issued to the axis for which parameter No. 8163#7NUMx is set to 1. 2) A move command was issued to the axis for which parameter No. 8162#7MUMx is set to 1.
0354	THE G28 WAS INSTRUCTED IN WITH THE REF POS NOT FIXED IN SYNC MODE	This error occurred when G28 was specified to the master axis being parking during synchronization control, but an axis reference position is not set for the slave axis.
0355	PARAMETER OF THE INDEX OF THE COMPOSITE CONTROL AXIS SET ERROR.	An illegal composite control axis number (parameter No. 8183) is specified.

Number	Message	Description
0356	BECAUSE THE AXIS IS MOVING, THE COMP CONTROL IS CAN'T BE USED.	While the axis being subject to composite control was moving, an attempt was made to start or cancel the composite control by a composite control axis selection signal.
0357	COMPOSITE CONTROL AXIS COMPOSITION ERROR.	This error occurred when an attempt was made to perform composite control for the axis during a synchronization, composition, or superposition.
0359	THE G28 WAS INSTRUCTED IN WITH THE REF POS NOT FIXED IN COMP MODE	This error occurred when G28 was specified to the composite axis during composite control, but a reference position is not set to the other part of the composition.
0360	PARAMETER OF THE INDEX OF THE SUPERPOS CONTROL AXIS SET ERROR.	An illegal superposition control axis number (parameter No. 8186) is specified.
0361	BECAUSE THE AXIS IS MOVING, THE SUPERPOS CONTROL IS CAN'T BE USED.	While the axis being subject to superposition control was moving, an attempt was made to start or cancel the superposition control by a superposition control axis selection signal.
0362	SUPERPOSITION CONTROL AXIS COMPOSITION ERROR.	This error occurred when: 1) An attempt was made to perform superposition control for the axis during a synchronization, composition, or superposition. 2) An attempt was made to synchronize a further great-grandchild for a parent-child-grandchild relation.
0363	THE G28 WAS INSTRUCTED IN TO THE SUPERPOS CONTROL SLAVE AXIS.	This error occurred when G28 was specified to the superposition control slave axis during superposition control.
0364	THE G53 WAS INSTRUCTED IN TO THE SUPERPOS CONTROL SLAVE AXIS.	This error occurred when G53 was specified to the slave axis being moved during superposition control.
0365	TOO MANY MAXIMUM SV/SP AXIS NUMBER PER PATH	The maximum control axis number or maximum control spindle number which could be used within a path was exceeded. (For a loader path, this alarm is generated if the number of axis per path is set to 5 or greater.)
0366	IMPROPER G-CODE IN TURRET METHOD	When the turret change tools method was selected (parameter No. 5040#3 (TCT) = 0), G43, G43.1, G43.4, G43.5, or G43.7 was commanded.
0367	3-D CONV. WAS COMMANDED IN SYNC MODE AS THE PARAMETER PKUx(NO.8162#2) IS 0.	A three-dimensional coordinate conversion was commanded during synchronization control when the parameter PKUx (No.8162#2) was 0.
0368	OFFSET REMAIN AT OFFSET COMMAND	When the ATC change tools method was selected (parameter No. 5040#3 (TCT) = 1) during G43, G43.1, G43.4, or G43.5 mode, G43.7 was commanded. Or, G43, G43.1, G43.4, or G43.5 was commanded during G43.7 mode.
0369	G31 FORMAT ERROR	- No axis is specified or tow or more axes are specified in the torque limit switch instruction (G31P98/P99). - The specified torque Q value in the torque limit switch instruction is out of range. The torque Q range is 1 to 254. - The high-speed continuous skip option is not present.

Number	Message	Description
0370	G31P/G04Q ERROR	1) The specified address P value for G31 is out of range. The address P range is 1 to 4 in a multistage skip function. 2) The specified address Q value for G04 is out of range. The address Q range is 1 to 4 in a multistage skip function. 3) P1-4 for G31, or Q1-4 for G04 was commanded without a multistage skip function option. 4) <T series > The specified value of address P of G72 or G74 falls outside the range. Address P ranges from 1 to 4 in the multistage skip function. P1-4 was specified in G72 or G74 even though the multistage skip function option is not present.
0371	ILLEGAL FORMAT IN G10 OR L50	In a command format for a programmable parameter input, an attempt was made to change the parameter for an encryption (No. 3220), key (No. 3221), or protection range (No.3222 or No.3223) as a "the encryption function for the key and program." Modify the program.
0372	REFERENCE RETURN INCOMPLETE	An attempt was made to perform an automatic return to the reference position on the orthogonal axis before the completion of a return to the reference position on the angular axis. However, this attempt failed because a manual return to the reference position during angular axis control or an automatic return to the reference position after power-up was not commanded. First, return to the reference position on the angular axis, then return to the reference position on the orthogonal axis.
0373	ILLEGAL HIGH-SPEED SKIP SIGNAL	In the skip commands (G31, G31P1 to G31P4) and dwell commands (G04, G04Q1 to G04Q4), the same high-speed signal is selected in different paths.
0374	ILLEGAL REGISTRATION OF TOOL MANAGER(G10)	G10L75 or G10L76 data was registered during the following data registration: - From the PMC window. - From the FOCAS2. - By G10L75 or G10L76 in another system. Command G10L75 or G10L76 again after the above operation is completed.
0375	CAN NOT ANGULAR CONTROL(SYNC:MIX:OVL)	Angular axis control is disabled for this axis configuration. 1) When some related axes under angular axis control are not in synchronous control mode or when one angular axis is not paired with the other angular axis or one Cartesian axis is not paired with the other Cartesian axis in synchronous control 2) When some related axes under composite control are not in composite control mode or when one angular axis is not paired with the other angular axis or one Cartesian axis is not paired with the other Cartesian axis in composite control 3) When related axes under angular axis control is switched to superposition control mode1)
0376	SERIAL DCL: ILLEGAL PARAMETER	1. When Parameter No.1815#1 is set to “1”, parameter No.2002#3 is set to “0” 2. The absolute-position detection function is enabled. (Parameter No.1815#5 is set to “1”.)
0387	ILLEGAL RTM DI/DO VAR	There is no DI/DO variable that has a specified signal address (alphabet, number).

Number	Message	Description
0389	ILLEGAL RTM SIGNAL BIT	Bits other than bits 0 to 7 cannot be specified with a DI/DO signal.
0391	RTM BRANCH OVER	The number of branches supported with real time custom macros was exceeded.
0392	TOO MANY SENTENCE CONTROL	Many reserved words (ZONCE, ZEDGE, ZWHILE, ZDO, ZEND, G65, M99) for RTM control were used in a real time macro command.
0393	NO SENTENCE CONTROL	In a real time macro command, there is no data to be assigned.
0394	ILLEGAL SENTENCE CONTROL	The matching of reserved words (ZONCE, ZEDGE, ZWHILE, ZDO, ZEND, G65, M99) for RTM control is incorrect.
0395	ILLEGAL NC WORD CONTROL	Control code G65 or M99 for calling a subprogram or returning from a subprogram is not coded correctly.
0396	ILLEGAL RTM SENTENCE CONTROL	In other than a real time macro command, a reserved word (ZONCE, ZEDGE, ZWHILE, ZDO, or ZEND) for RTM control is used.
0397	RTM BUFFER OVER	There is no buffer available for real time macro commands. Too many blocks read in advance are buffered as triggers used by real time macro commands.
0398	'ID OVER IN BUFFER	In blocks read in advance, there are too many real time macro commands with the same ID.
0399	'ID EXECUTION IN SAME TIME	An attempt was made to execute real time macro commands with the same ID by using the same NC statement as a trigger.
0400	ONESHOT CMDOVER	Too many one-shot real time macro commands are specified.
0401	EXEC CMD NUM OVER IN SAME TIME	The number of real time macro commands that can be executed simultaneously was exceeded
0402	ILLEGAL TOKEN FOR RTM	A token, variable, or function that is not supported by the real time custom macro function was detected.
0403	ACCESS TO RTM PROTECT VAR	An attempt was made to access a protected variable.
0404	RTM ERROR	An error related to a real time macro command occurred.
0406	CODE AREA SHORTAGE	The storage size of the real time macro area is insufficient.
0407	DOULE SLASH IN RTM MODE	In the compile mode, an attempt was made to set the compile mode again.
0408	G90 IS NOT PERMITTED	The absolute command cannot be specified.
0409	ILLEGAL AXIS NO	An invalid axis number is specified.
0410	MIDDLE POINT IS NOT ZERO	An intermediate point other than 0 is specified with G28.
0411	SIMULTANEOUSLY AXES OVER	The maximum number of axes that can be controlled simultaneously was exceeded.
0412	ILLEGAL G CODE	An unusable G code was used.
0413	ILLEGAL ADDRESS	An unusable address was used.
0414	ILLEGAL PMC AXIS NO.	An invalid PMC axis number is specified.
0415	GROUP IS IN USE	The group to which the specified axis belongs is already in used.
0416	UNABLE TO USE THE AXIS	The specified axis cannot be used.
0417	AXIS IS UNABLE TO MOVE	The specified axis is placed in the inoperative state.
0418	ILLEGAL FEED SETTING	An incorrect feedrate is set.
0419	ILLEGAL DISTANCE SETTING	A travel distance beyond the specifiable range is specified.
0420	CONSTANT NUMBER P	A subprogram is specified not by using a constant.
0421	ILLEGAL ARGUMENT G54	With G65, an invalid argument, L, is used.
0422	ILLEGAL ARGUMENT G54	With G65, an invalid argument is used.
0423	NO PMC AXIS CONTROL OPTION	The option for PMC axis control is missing.
0424	MULTIPLE AXES IN ONE GROUP	Multiple axes are using one group.
0425	ONE AXIS USE MULTIPLE GROU	One axis is using multiple groups.

Number	Message	Description
0429	ILLEGAL COMMAND IN G10.6	When retract was started in a threading block, a retract command had been issued for the long axis direction of threading.
0430	TOOL LIFE PAIRS ZERO	Tool life management group number parameter No.6813 is 0.
0431	ILLEGAL T/R DATA OF TOOL LIFE	The arbitrary group number (T) or remaining amount setting (R) is invalid.
0432	UNAVAILABLE POSTURE IN TPC	A command specifying an impossible tool posture was issued in tool posture control. Review the machine configuration and command.
0436	ILLEGAL PARAMETER IN WSC	An incorrect parameter was specified in compensation of workpiece placement error. - The basic three axes are not specified in parameter No.1022.
0437	ILLEGAL COMMAND IN WSC	An invalid command related to compensation of workpiece placement error was specified. - An illegal G code was specified in the workpiece placement error compensation mode. - There is an error in the modal setting used when the compensation of workpiece placement error is started. - G54.4 was not specified solely. - There is not the P command in the block including the G54.4 command. Alternatively, the value following P is out of the range. - Compensation of workpiece placement error was specified redundantly.
0438	ILLEGAL PARAMETER IN TOOL DIRC CMP	When compensation of workpiece placement error is performed in a 5-axis cutting machine (compensation in the tool direction (bit 0 (RCM) of parameter No. 11200 is 0)), an illegal parameter is set. - Acc./Dec. before interpolation is disabled. Set parameter No. 1660. - Acc./Dec. before rapid traverse interpolation is disabled. Set bit 1 of parameter No. 1401, bit 5 of parameter No. 1950, and parameter No. 1671. - The parameters (No.19680 to No.19714) for configuring the machine are incorrect. - The axis set by parameters No. 19681 and No. 19686 is not a rotation axis. - The basic three axes are not set in parameter No. 1022. - Bit 6 (TOS) of parameter No. 5006 is set to 0 when tool length compensation is performed during compensation of workpiece placement error. Set this parameter to 1.
0439	ILLEGAL COMMAND IN TOOL DIRC CMP	When compensation of workpiece placement error was performed in a 5-axis cutting machine (compensation in the tool direction (bit 0 (RCM) of parameter No. 11200 is 0)), an illegal command was issued. - An unspecifiable G code was specified. - There is an error in the modal setting used during startup. - An axis not related to 5-axis machining was specified. - The absolute coordinates of a rotation axis could not be obtained in the startup block of compensation of workpiece placement error or tool center point control.
0445	ILLEGAL AXIS OPERATION	The positioning command was issued in the speed control mode. Check the SV speed control mode signal.

Number	Message	Description
0446	ILLEGAL COMMAND IN G96.1/G96.2/G96.3/G96.4	G96.1, G96.2, G96.3, and G96.4 are specified in the block that includes other commands. Modify the program.
0447	ILLEGAL SETTING DATA	The live tool axis is incorrectly set. Check the parameter for the live tool control function by a servo motor.
0455	ILLEGAL COMMAND IN GRINDING	In grinding canned cycles: 1) <M series> The signs of the I, J, and K commands do not match. 2) <M series/T series > The amount of travel of the grinding axis is not specified.
0456	ILLEGAL PARAMETER IN GRINDING	Parameters related to grinding canned cycles are incorrectly set. Probable causes are given below. 1) <M series/T series> The axis number of the grinding axis is incorrectly set (parameters No. 5176 to No. 5179). 2) <M series> The axis number of the dressing axis is incorrectly set (parameters No. 5180 to No. 5183). 3) <M series/T series> The axis numbers of the cut axis, grinding axis, and dressing axis (only for the M series) overlap.
0460	ILLEGAL TORCH AXIS NUMBER	The axis number set in parameter No. 5490 (torch control axis) exceeds the number of control axes.
0461	ILLEGAL SETTING OF ROTATE AXIS FOR TORCH	The parameter setting (bit 0 of parameter No. 1006 = 1) of the rotation axis is not applied to the torch turning axis.
1014	ILLEGAL FORMAT OF PROGRAM NO.	Address O or N is not followed by a number.
1016	EOB NOT FOUND	EOB (End of Block) code is missing at the end of a program input in the MDI mode.
1018	M99 IN MAIN PROGRAM	A M99 was commanded during main program when the parameter AMM (No. 7712#4) = 1 was set.
1059	COMMAND IN BUFFERING MODE	The manual intervention compensation request signal MIGET became “1” when a advanced block was found during automatic operation. To input the manual intervention compensation during automatic operation, a sequence for manipulating the manual intervention compensation request signal MIGET is required in an M code instruction without buffering.
1077	PROGRAM IN USE	An attempt was made in the foreground to execute a program being edited in the background. The currently edited program cannot be executed, so end editing and restart program execution.
1079	PROGRAM FILE NOT FOUND	The program of the specified file No. is not registered in an external device. (external device subprogram call)
1080	DUPLICATE DEVICE SUB PROGRAM CALL	Another external device subprogram call was made from a subprogram after the subprogram called by the external device subprogram call.
1081	EXT DEVICE SUB PROGRAM CALL MODE ERROR	The external device subprogram call is not possible in this mode.
1091	DUPLICATE SUB-CALL WORD	More than one subprogram call instruction was specified in the same block.
1092	DUPLICATE MACRO-CALL WORD	More than one macro call instruction was specified in the same block.
1093	DUPLICATE NC-WORD & M99	An address other than O, N, P or L was specified in the same block as M99 during the macro modal call state.

Number	Message	Description
1095	TOO MANY TYPE-2 ARGUMENT	More than ten sets of I, J and K arguments were specified in the type–II arguments (A, B, C, I, J, K, I, J, K, ...) for custom macros.
1096	ILLEGAL VARIABLE NAME	An illegal variable name was specified. A code that cannot be specified as a variable name was specified. [#_OFSxx] does not match the tool offset memory option configuration.
1097	TOO LONG VARIABLE NAME	The specified variable name is too long.
1098	NO VARIABLE NAME	The specified variable name cannot be used as it is not registered.
1099	ILLLEGAL SUFFIX [ ]	A suffix was not specified to a variable name that required a suffix enclosed by [ ]. A suffix was specified to a variable name that did not require a suffix enclosed by [ ]. The value enclosed by the specified [ ] was out of range.
1100	CANCEL WITHOUT MODAL CALL	Call mode cancel (G67) was specified even though macro continuous–state call mode (G66) was not in effect.
1101	ILLEGAL CNC STATEMENT IRT.	An interrupt was made in a state where a custom macro interrupt containing a move instruction could not be executed.
1115	READ PROTECTED VARIABLE	An attempt was made in a custom macro to use on the right side of an expression a variable that can only be used on the left side of an expression.
1120	ILLEGAL ARGUMENT FORMAT	The specified argument in the argument function (ATAN, POW) is in error.
1124	MISSING DO STATEMENT	The DO instruction corresponding to the END instruction was missing in a custom macro.
1125	ILLEGAL EXPRESSION FORMAT	The description of the expression in a custom macro statement contains an error. A parameter program format error. The screen displayed to enter periodic maintenance data or item selection menu (machine) data does not match the data type.
1128	SEQUENCE NUMBER OUT OF RANGE	The jump destination sequence No. in a custom macro statement GOTO instruction was out of range (valid range: 1 to 99999999).
1131	MISSING OPEN BRACKET	The number of left brackets ([) is less than the number of right brackets (]) in a custom macro statement.
1132	MISSING CLOSE BRACKET	The number of right brackets (]) is less than the number of left brackets ([) in a custom macro statement.
1133	MISSING '='	An equal sign (=) is missing in the arithmetic calculation instruction in a custom macro statement.
1134	MISSING ','	A delimiter (,) is missing in a custom macro statement.
1137	IF STATEMENT FORMAT ERROR	The format used in the IF statement in a custom macro is in error.
1138	WHILE STATEMENT FORMAT ERROR	The format used in the WHILE statement in a custom macro is in error.
1139	SETVN STATEMENT FORMAT ERROR	The format used in the SETVN statement in a custom macro is in error.
1141	ILLEGAL CHARACTER IN VAR. NAME	The SETVN statement in a custom macro contacts a character that cannot be used in a variable name.
1142	TOO LONG V-NAME (SETVN)	The variable name used in a SETVN statement in a custom macro exceeds 8 characters.
1143	BPRNT/DPRNT STATEMENT FORMAT ERROR	The format used in the BPRINT statement or DPRINT statement is in error.

Number	Message	Description
1144	G10 FORMAT ERROR	The G10 L No. contains no relevant data input or corresponding option. Data setting address P or R is not specified. An address not relating to the data setting is specified. Which address to specify varies according to the L No. The sign, decimal point or range of the specified address are in error.
1145	G10.1 TIME OUT	The response to a G10.1 instruction was not received from the PMC within the specified time limit.
1146	G10.1 FORMAT ERROR	The G10.1 instruction format is in error.
1152	G31.9/G31.8 FORMAT ERROR	The format of the G31.9 or G31.8 block is erroneous in the following cases: - The axis was not specified in the G31.9 or G31.8 block. - Multiple axes were specified in the G31.9 or G31.8 block. - The P code was specified in the G31.9 or G31.8 block.
1153	CANNOT USE G31.9	G31.9 cannot be specified in this modal state. This alarm is also generated when G31.9 is specified when a group 07 G code (e.g. tool radius compensation) is not canceled.
1160	COMMAND DATA OVERFLOW	An overflow occurred in the position data within the CNC. This alarm is also generated if the target position of a command exceeds the maximum stroke as a result of calculation such as coordinate conversion, offset, or introduction of a manual intervention amount.
1180	ALL PARALLEL AXES IN PARKING	All of the axis specified for automatic operation are parked.
1196	ILLEGAL DRILLING AXIS SELECTED	An illegal axis was specified for drilling in a canned cycle for drilling. If the zero point of the drilling axis is not specified or parallel axes are specified in a block containing a G code in a canned cycle, simultaneously specify the parallel axes for the drilling axis.
1200	PULSCODER INVALID ZERO RETURN	The grid position could not be calculated during grid reference position return using the grid system as the one–revolution signal was not received before leaving the deceleration dog. This alarm is also generated when the tool does not reach a feedrate that exceeds the servo error amount preset to parameter No. 1841 before the deceleration limit switch is left (deceleration signal *DEC returns to “1”).
1202	NO F COMMAND AT G93	F codes in the inverse time specification mode (G93) are not handled as modal, and must be specified in individual blocks.
1223	ILLEGAL SPINDLE SELECT	1) An attempt was made to execute an instruction that uses the spindle although the spindle to be controlled has not been set correctly. 2) Interpolation type rigid tapping was specified in a path in which the Cs contour control function is not enabled.
1282	ILLEGAL COMMAND IN 3-D OFFSET	An illegal G code was specified in the 3-dimensional tool compensation mode.
1283	ILLEGAL IJK IN 3-D OFFSET	When bit 0 (ONI) of parameter No. 6029 is set to 1, I, J, and K commands are specified without the decimal point in three–dimensional tool compensation mode.
1298	ILLEGAL INCH/METRIC CONVERSION	An error occurred during inch/metric switching.

Number	Message	Description
1300	ILLEGAL ADDRESS	The axis No. address was specified even though the parameter is not an axis–type while loading parameters or pitch error compensation data from a tape or by entry of the G10 parameter. Axis No. cannot be specified in pitch error compensation data.
1301	MISSING ADDRESS	The axis No. was not specified even though the parameter is an axis–type while loading parameters or pitch error compensation data from a tape or by entry of the G10 parameter. Or, data No. address N, or setting data address P or R are not specified.
1302	ILLEGAL DATA NUMBER	A non–existent data No. was found while loading parameters or pitch error compensation data from a tape or by entry of the G10 parameter. An invalid address R value is specified in a pattern program for each machining purpose on the high–speed high–precision setting screen. This alarm is also generated when illegal word values are found.
1303	ILLEGAL AXIS NUMBER	An axis No. address exceeding the maximum number of controlled axes was found while loading parameters from a tape or by entry of the G10 parameter.
1304	TOO MANY DIGIT	Data with too many digits was found while loading parameters or pitch error compensation data from a tape.
1305	DATA OUT OF RANGE	Out–of–range data was found while loading parameters or pitch error compensation data from a tape. The values of the data setting addresses corresponding to L Nos. during data input by G10 was out of range. This alarm is also generated when NC programming words contain out–of–range values.
1306	MISSING AXIS NUMBER	A parameter which requires an axis to be specified was found without an axis No. (address A) while loading parameters from a tape.
1307	ILLEGAL USE OF MINUS SIGN	Data with an illegal sign was found while loading parameters or pitch error compensation data from a tape, or by entry of the G10 parameter. A sign was specified to an address that does not support the use of signs.
1308	MISSING DATA	An address not followed by a numeric value was found while loading parameters or pitch error compensation data from a tape.
1329	ILLEGAL MACHINE GROUP NUMBER	An machine group No. address exceeding the maximum number of controlled machine groups was found while loading parameters from a tape or by entry of the G10 parameter.
1330	ILLEGAL SPINDLE NUMBER	An spindle No. address exceeding the maximum number of controlled spindles was found while loading parameters from a tape or by entry of the G10 parameter.
1331	ILLEGAL PATH NUMBER	An path No. address exceeding the maximum number of controlled path was found while loading parameters from a tape or by entry of the G10 parameter.
1332	DATA WRITE LOCK ERROR	Could not load data while loading parameters, pitch error compensation data and work coordinate data from tape.
1333	DATA WRITE ERROR	Could not write data while loading data from tape.
1360	PARAMETER OUT OF RANGE (TLAC)	Illegal parameter setting. (Set value is out of range.)
1361	PARAMTER SETTING ERROR 1 (TLAC)	Illegal parameter setting. (axis of rotation setting)

Number	Message	Description
1362	PARAMETER SETTING ERROR 2 (TLAC)	Illegal parameter setting (tool axis setting)
1370	PARAMETER SETTING ERROR (DM3H-1)	Out–of–range data was set during setting of the three–dimensional handle feed parameter.
1371	PARAMETER SETTING ERROR (DM3H-2)	An illegal axis of rotation was set during setting of the three–dimensional handle feed parameter.
1372	PARAMETAR SETTING ERROR (DM3H-3)	An illegal master axis was set during setting of the three–dimensional handle feed parameter.
1373	PARAMETER SETTING ERROR (DM3H-4)	An illegal parallel axis or twin table was set during setting of the three–dimensional handle feed parameter.
1470	G40.1 –G42.1 PARAMETER MISS	A parameter setting related to normal direction control is illegal. The axis number of a normal direction controlled axis is set in parameter No. 5480, but that axis number is in the range of the number of controlled axes. The axis set as a normal direction controlled axis is not set as a rotation axis (ROTx, bit 0 of parameter No. 1006) = 1 and No.1022=0). Set the feedrate at which to insert rotation about a normal direction controlled axis in parameter No. 5481, in the range of 1 to 15000 mm/min.
1508	DUPLICATE M-CODE (INDEX TABLE REVERSING)	A function to which the same code as this M code is set exists. (index table indexing)
1509	DUPLICATE M-CODE (SPOS AXIS ORIENTATION)	A function to which the same code as this M code is set exists. (spindle positioning, orientation)
1510	DUPLICATE M-CODE (SPOS AXIS POSITIONING)	A function to which the same code as this M code is set exists. (spindle positioning, positioning)
1511	DUPLICATE M-CODE (SPOS AXIS RELEASE)	A function to which the same code as this M code is set exists. (spindle positioning, mode cancel)
1531	ILLEGAL USE OF DECIMAL POINT (F-CODE)	When the feedrate instruction contains valid data below the decimal point, the alarm is set and the F code contains valid data below the decimal point.
1532	ILLEGAL USE OF DECIMAL POINT (E-CODE)	When the feedrate instruction contains valid data below the decimal point, the alarm is set and the E code contains valid data below the decimal point.
1533	ADDRESS F UNDERFLOW (G95)	The feedrate for the hole drilling axis calculated from the F and S codes is too slow in the feed per single rotation mode (G95).
1534	ADDRESS F OVERFLOW (G95)	The feedrate for the hole drilling axis calculated from the F and S codes is too fast in the feed per single rotation mode (G95).
1535	ADDRESS E UNDERFLOW (G95)	The feedrate for the hole drilling axis calculated from the E and S codes is too slow in the feed per single rotation mode (G95).
1536	ADDRESS E OVERFLOW (G95)	The feedrate for the hole drilling axis calculated from the E and S codes is too fast in the feed per single rotation mode (G95).
1537	ADDRESS F UNDERFLOW (OVERRIDE)	The speed obtained by applying override to the F instruction is too slow.
1538	ADDRESS F OVERFLOW (OVERRIDE)	The speed obtained by applying override to the F instruction is too fast.
1539	ADDRESS E UNDERFLOW (OVERRIDE)	The speed obtained by applying override to the E instruction is too slow.
1540	ADDRESS E OVERFLOW (OVERRIDE)	The speed obtained by applying override to the E instruction is too fast.
1541	S-CODE ZERO	“0” has been instructed as the S code.
1542	FEED ZERO (E-CODE)	“0” has been instructed as the feedrate (E code).
1543	ILLEGAL GEAR SETTING	The gear ratio between the spindle and position coder, or the set position coder number of pulses is illegal in the spindle positioning function.

Number	Message	Description
1544	S-CODE OVER MAX	The S command exceeds the maximum spindle rotation number.
1548	ILLGAL AXIS MODE	The spindle positioning axis/Cs contour control axis was specified during switching of the controlled axis mode.
1561	ILLEGAL INDEXING ANGLE	The specified angle of rotation is not an integer multiple of the minimum indexing angle.
1564	INDEX TABLE AXIS – OTHER AXIS SAME TIME	The index table indexing axis and another axis have been specified in the same block.
1567	INDEX TABLE AXIS DUPLICATE AXIS COMMAND	Index table indexing was specified during axis movement or on an axis for which the index table indexing sequence was not completed.
1580	ENCODE ALARM (PSWD&KEY)	When an attempt was made to read a program, the specified password did not match the password on the tape and the password on tape was not equal to 0. When an attempt was made to punch an encrypted tape, the password was not in the range 0 to 99999999. The password parameter is No. 2210.
1581	ENCODE ALARM (PARAMETER)	When an attempt was made to punch an encrypted tape, the punch code parameter was set to EIA. Set parameter ISO (No. 0000#1) to “0”. An incorrect instruction was specified for program encryption or protection. This alarm is generated if an attempt is made to perform program editing, deletion, or range-specified punch-out in the protected range in the lock state. Or, a program outside the protected range is specified in rage specification punch-out in the unlock state. The protected range is defined from the program No. preset by parameter No. 3222 up to the program No. preset to parameter No. 3223. When both parameters are set to “0”, the protected range becomes O9000 to O9999.
1590	TH ERROR	A TH error was detected during reading from an input device. The read code that caused the TH error and how many statements it is from the block can be verified in the diagnostics screen.
1591	TV ERROR	An error was detected during the single–block TV error. The TV check can be suppressed by setting TVC parameter No. 0000#0 to “0”.
1592	END OF RECORD	The EOR (End of Record) code is specified in the middle of a block. This alarm is also generated when the percentage at the end of the NC program is read. For the program restart function, this alarm is generated if a specified block is not found.
1593	EGB PARAMETER SETTING ERROR	Error in setting a parameter related to the EGB (1) The setting of SYN, bit 0 of parameter No. 2011, is not correct. (2) The slave axis specified with G81 is not set as a rotation axis. (ROT, bit 0 of parameter No. 1006) (3) Number of pulses per rotation (Parameter (No. 7772 or No. 7773) or (No. 7782 or 7783) is not set.) (4) For a hobbing-machine-compatible command, parameter No. 7710 is not specified.

Number	Message	Description
1594	EGB FORMAT ERROR	Error in the format of the block of an EGB command (1) T (number of teeth) is not specified in the G81 block. (2) In the G81 block, the data specified for one of T, L, P, and Q is out of its valid range. (3) n the G81 block, only one of P and Q is specified. (4) In the G81.5 block, there is no command for the master or slave axis. (5) In the G81.5 block, data out of the specified range is specified for the master or slave axis.
1595	ILL-COMMAND IN EGB MODE	During synchronization with the EGB, a command that must not be issued is issued. (1) Slave axis command using G27, G28, G29, G30,G30.1, G33, G53, etc. (2) Inch/metric conversion command using G20, G21, etc.
1596	EGB OVERFLOW	An overflow occurred in the calculation of the synchronization coefficient.
1597	EGB AUTO PHASE FORMAT ERROR	Format error in the G80 or G81 block in EGB automatic phase synchronization (1) R is outside the permissible range.
1598	EGB AUTO PHASE PARAMETER SETTING ERROR	Error in the setting of a parameter related to EGB automatic phase synchronization (1) The acceleration/deceleration parameter is not correct. (2) The automatic phase synchronization parameter is not correct.
1618	ILLEGAL P-DATA(WHEEL WEAR COMPENSATION)	There is an error in P-data in compensation selection of grinding wheel wear compensation. Alternatively, the P command is not present.
1619	ILLEGAL AXIS(WHEEL WEAR COMPENSATION)	The compensation axis was switched in the grinding wheel wear compensation mode or compensation vector hold mode. Alternatively, parameters No. 5071 and No. 5072, which determine the axis to be subjected to grinding wheel wear compensation, are incorrectly set.
1805	ILLEGAL COMMAND	[I/O Device] An attempt was made to specify an illegal command during I/O processing on an I/O device. [G30 Zero Return] The P address Nos. for instructing No. 2 to No. 4 zero return are each out of the range 2 to 4. [Single Rotation Dwell] The specified spindle rotation is “0” when single rotation dwell is specified. [3-dimensional tool compensation] A G code that cannot be specified was specified in the 3-dimensional tool compensation mode. Scaling instruction G51, skip cutting G31 and automatic tool length measurement G37 were specified.
1806	DEVICE TYPE MISS MATCH	An operation not possible on the I/O device that is currently selected in the setting was specified. This alarm is also generated when file rewind is instructed even though the I/O device is not a FANUC Cassette.
1807	PARAMETER SETTING ERROR	An I/O interface option that has not yet been added on was specified. The external I/O device and baud rate, stop bit and protocol selection settings are erroneous.

Number	Message	Description
1808	DEVICE DOUBLE OPENED	An attempt was made to open a device that is being accessed.
1809	ILLEGAL COMMAND IN G41/G42	Specified direction tool length compensation parameters are incorrect. A move instruction for a axis of rotation was specified in the specified direction tool length compensation mode.
1820	ILLEGAL DI SIGNAL STATE	1. An each axis workpiece coordinate system preset signal was turned “1” in the state in which all axes on the path including the axis on which to perform preset with the each axis workpiece coordinate system were not stopped or in which a command was in execution. 2. When an M code for performing preset with an each axis workpiece coordinate system preset signal was specified, the each axis workpiece coordinate system preset signal was not turned “1”. 3. The auxiliary function lock is enabled. 4. When bit 6 (PGS) of parameter No. 3001 was set to 0 (M, S, T, and B codes are not output in the high speed program check mode), an M code for turning “1” an each axis workpiece coordinate system preset signal in the high speed program check mode was specified.
1823	FRAMING ERROR(1)	The stop bit of the character received from the I/O device connected to reader/punch interface 1 was not detected.
1830	DR OFF(2)	The data set ready input signal DR of the I/O device connected to reader/punch interface 2 turned OFF.
1832	OVERRUN ERROR(2)	The next character was received from the I/O device connected to reader/punch interface 2 before it could read a previously received character.
1833	FRAMING ERROR(2)	The stop bit of the character received from the I/O device connected to reader/punch interface 2 was not detected.
1834	BUFFER OVERFLOW(2)	The NC received more than 10 characters of data from the I/O device connected to reader/punch interface 2 even though the NC sent a stop code (DC3) during data reception.
1889	ILLEGAL COMMAND IN G54.3	An illegal command was issued in G54.3 block. (1) An attempt was made to command G54.3 in a mode in which it cannot be accepted. (2) The command was not issued in a single block.
1898	ILLEGAL PARAMETER IN G54.2	An illegal parameter (Nos. 6068 to 6076) was specified for fixture offset.
1912	V-DEVICE DRIVER ERROR (OPEN)	An error occurred during device driver control.
1960	ACCESS ERROR (MEMORY CARD)	Illegal memory card accessing This alarm is also generated during reading when reading is executed up to the end of the file without detection of the EOR code.
1961	NOT READY (MEMORY CARD)	The memory card is not ready.
1962	CARD FULL (MEMORY CARD)	The memory card has run out of space.
1963	CARD PROTECTED (MEMORY CARD)	The memory card is write–protected.
1964	NOT MOUNTED (MEMORY CARD)	The memory card could not be mounted.
1965	DIRECTORY FULL (MEMORY CARD)	The file could not be generated in the root directory for the memory card.
1966	FILE NOT FOUND (MEMORY CARD)	The specified file could not be found on the memory card.
1967	FILE PROTECTED (MEMORY CARD)	The memory card is write–protected.

Number	Message	Description
1968	ILLEGAL FILE NAME (MEMORY CARD)	Illegal memory card file name
1969	ILLEGAL FORMAT (MEMORY CARD)	Check the file name.
1970	ILLEGAL CARD (MEMORY CARD)	This memory card cannot be handled.
1971	ERASE ERROR (MEMORY CARD)	An error occurred during memory card erase.
1972	BATTERY LOW (MEMORY CARD)	The memory card battery is low.
1973	FILE ALREADY EXIST	A file having the same name already exists on the memory card.
1990	SPL:ILLEGAL AXIS COMMAND	The axis specified by the smooth interpolation (G5.1Q2) is illegal.
1993	SPL:CAN'T MAKE VECTOR	The end point and the 2 previous point are the same in generation of the 3–dimensional tool compensation vector by the end point for smooth interpolation.
1995	ILLEGAL PARAMETER IN G41.2/G42.2	The parameter settings (parameter Nos. 6080 to 6089) for determining the relationship between the axis of rotation and the rotation plane are incorrect.
1999	ILLEGAL PARAMETER IN G41.3	The parameter settings (parameter Nos. 6080 to 6089) for determining the relationship between the axis of rotation and the rotation plane are incorrect.
2002	NO KNOT COMMAND (NURBS)	Knot has not been specified, or a block not related to NURBS interpolation was specified in the NURBS interpolation mode.
2003	ILLEGAL AXIS COMMAND (NURBS)	An axis not specified as a control point was specified in the No. 1 block.
2004	ILLEGAL KNOT	There is an insufficient number of knot individual blocks.
2005	ILLEGAL CANCEL (NURBS)	The NURBS interpolation mode was turned OFF even though NURBS interpolation was not completed.
2006	ILLEGAL MODE (NURBS)	A mode that cannot be paired with the NURBS interpolation mode was specified.
2007	ILLEGAL MULTI-KNOT	Nested knots for each level can be specified for the start and end points.
2032	EMBEDDED ETHERNET/DATA SERVER ERROR	An error was returned in the built-in Ethernet/data server function. For details, see the error message screen of the built-in Ethernet or data server.
2051	#200-#499ILLEGAL P-CODE MACRO COMMON INPUT(NO OPTION)	An attempt was made to enter a custom macro common variable not existing in the system.
2052	#500-#549P-CODE MACRO COMMON SELECT(CANNOT USE SETVN)	The variable name cannot be entered. The SETVN command cannot be used with the P-CODE macro common variables #500 to #549.
2053	THE NUMBER OF #30000 IS UNMATCH	An attempt was made to enter a P-CODE-only variable not existing in the system.
2054	THE NUMBER OF #40000 IS UNMATCH	An attempt was made to enter an extended P-CODE-only variable not existing in the system.
2060	ILLEGAL PARAMETER IN G43.4/G43.5	The parameter for the pivot tool length compensation is incorrect.

Number	Message	Description
2061	ILLEGAL COMMAND IN G43.4/G43.5	An illegal command was specified in tool center point control. - A rotation axis command was specified in tool center point control (type 2) mode. - With a table rotary type or mixed-type machine, a I, J, or K command was specified in the tool center point control (type 2) command (G43.5) block. - A command that does not move the tool center point (only a rotation axis is moved) was specified for the workpiece in the G02 mode. - G43.4 or G43.5 was specified in the tool center point control mode. - When the workpiece coordinate system is set as the programming coordinate system (bit 5 (WKP) of parameter No. 19696 is 1), G02 or G03 was specified while the rotation axis was not perpendicular to the plane.
4010	ILLEGAL REAL VALUE OF OBUF :	The real value for a output buffer is in error.
5006	TOO MANY WORD IN ONE BLOCK	The number of words in a block exceeds the maximum. The maximum is 26 words. However, this figure varies according to NC options. Divide the instruction word into two blocks.
5007	TOO LARGE DISTANCE	Due to compensation, point of intersection calculation, interpolation or similar reasons, a movement distance that exceeds the maximum permissible distance was specified. Check the programmed coordinates or compensation amounts.
5009	PARAMETER ZERO (DRY RUN)	The dry run feedrate parameter No. 1410 or maximum cutting feedrate parameter No. 1422 for each axis has been set to 0.
5010	END OF RECORD	The EOR (End of Record) code is specified in the middle of a block. This alarm is also generated when the percentage at the end of the NC program is read.
5011	PARAMETER ZERO (CUT MAX)	The maximum cutting feedrate parameter No. 1430 has been set to 0.
5014	TRACE DATA NOT FOUND	A transfer could not be made because of no trace data.
5015	NO ROTATION AXIS	No rotation axis was found in a handle feed in the tool axis direction or in the tool axis right angle direction.
5016	ILLEGAL COMBINATION OF M CODES	M codes which belonged to the same group were specified in a block. Alternatively, an M code which must be specified without other M codes in the block was specified in a block with other M codes.
5018	POLYGON SPINDLE SPEED ERROR	In G51.2 mode, the speed of the spindle or polygon synchronous axis either exceeds the clamp value or is too small. The specified rotation speed ratio thus cannot be maintained. For polygon turning between spindles: More information as to why this alarm occurred is indicated in DGN No. 471.
5020	PARAMETER OF RESTART ERROR	An invalid value is set in parameter No. 7310, which specifies the axis order in which the tool is moved along axes to the machining restart position in dry run. A value ranging from 1 to the number of controlled axes may be set in this parameter.
5043	TOO MANY G68 NESTING	Three-dimensional coordinate conversion has been specified three or more times. To perform another coordinate conversion, perform cancellation, then specify the coordinate conversion.

Number	Message	Description
5044	G68 FORMAT ERROR	Errors for three-dimensional coordinate conversion command are: (1) No I, J, or K command was issued in three-dimensional coordinate conversion command block. (without coordinate rotation option) (2) All of I, J, or K command were 0 in three-dimensional coordinate conversion command block. (3) No rotation angle R was not commanded in three-dimensional coordinate conversion command block.
5046	ILLEGAL PARAMETER (S-COMP)	The setting of a parameter related to straightness compensation contains an error. Possible causes include: - A non-existent axis number is set in a moving or compensation axis parameter. - More than 128 pitch error compensation points are set between the furthest points in the negative and position regions. - The straightness compensation point numbers do not have correct magnitude relationships. - No straightness compensation point is found between the furthest pitch error compensation point in the negative region and that in the positive region. - The compensation per compensation point is either too large or too small.
5050	ILL-COMMAND IN G81.1 MODE	During chopping, a move command has been issued for the chopping axis.
5058	G35/G36 FORMAT ERROR	A command for switching the major axis has been specified for circular threading. Alternatively, a command for setting the length of the major axis to 0 has been specified for circular threading.
5060	ILLEGAL PARAMETER IN G02.3/G03.3	The axis parameter setting to perform an exponential interpolation is in error. Parameter No. 5641: A liner axis number for performing an exponential interpolation Parameter No. 5642: A rotation axis number for performing an exponential interpolation The settable value is 1 to the number of control axes, but it must not be duplicated.
5061	ILLEGAL FORMAT IN G02.3/G03.3	The exponential interpolation command (G02.3/G03.3) has a format error. The command range for address I or J is -89.0 to -1.0 or +1.0 to +89.0. No I or J is specified or out-of -range value is specified. No address R, or 0 is specified.
5062	ILLEGAL COMMAND IN G02.3/G03.3	The value specified in an exponential interpolation command (G02.3/03.3) is illegal. A value that does not allow exponential interpolation is specified. (For example, the value for In is 0 or negative.)
5064	DIFFERRENT AXIS UNIT	Circular interpolation has been specified on a plane consisting of axes having different increment systems.
5065	DIFFERRENT AXIS UNIT(PMC AXIS)	Axes having different increment systems have been specified in the same DI/DO group for PMC axis control. Modify the setting of parameter No. 8010.

Number	Message	Description
5066	RESTART ILLEGAL SEQUENCE NUMBER	A sequence number from 7000 to 7999 was read during the search for the next number in a restart program for the back or restart function.
5068	FORMAT ERROR IN G31P90	No travel axis was specified. Two or more travel axes were specified.
5073	NO DECIMAL POINT	No decimal point has been specified for an address requiring a decimal point.
5074	ADDRESS DUPLICATION ERROR	The same address has been specified two or more times in a single block. Alternatively, two or more G codes in the same group have been specified in a single block.
5085	SMOOTH IPL ERROR 1	A block for specifying smooth interpolation contains a syntax error.
5110	IMPROPER G-CODE (AICC MODE)	An unspecifiable G code was specified in the AI contour control mode.
5115	ILLEGAL ORDER (NURBS)	There is an error in the specification of the rank.
5116	ILLEGAL KNOT VALUE (NURBS)	Monotone increasing of knots is not observed.
5117	ILLEGAL 1ST CONTROL POINT (NURBS)	The first control point is incorrect. Or, it does not provide a continuity from the previous block.
5118	ILLEGAL RESTART (NURBS)	After manual intervention with manual absolute mode set to on, NURBS interpolation was restarted.
5122	ILLEGAL COMMAND IN SPIRAL	A spiral interpolation or conical interpolation command has an error. Specifically, this error is caused by one of the following: 1) L = 0 is specified. 2) Q = 0 is specified. 3) R/, R/, C is specified. 4) Zero is specified as height increment. 5) Zero is specified as height difference. 6) Three or more axes are specified as the height axes. 7) A height increment is specified when there are two height axes. 8) Q is specified when radius difference = 0. 9) Q < 0 is specified when radius difference > 0. 10) Q > 0 is specified when radius difference < 0. 11) A height increment is specified when no height axis is specified.
5123	OVER TOLERANCE OF END POINT IN SPIRAL	The difference between a specified end point and the calculated end point exceeds the allowable range (parameter 3471).
5124	CAN NOT COMMAND SPIRAL	A spiral interpolation or conical interpolation was specified in any of the following modes: 1) Scaling 2) Polar coordinate interpolation 3) In tool radius·tool nose radius compensation mode, the center is set as the end point.
5130	NC AND SUPERIMPOSE AXIS CONFLICT	In the PMC superposition axis control, the NC command and The PMC axis control command were conflicted. Modify the program and the ladder.
5131	NC COMMAND IS NOT COMPATIBLE	The PMC axis control and three-dimensional coordinate conversion or a polar coordinate interpolation were specified simultaneously.
5132	CANNOT CHANGE SUPERIMPOSED AXIS	The superposition axis was selected for the axis for which the PMC superposition axis is being controlled.

Number	Message	Description
5195	DIRECTION CAN NOT BE JUDGED	Measurement is invalid in the tool compensation measurement value direct input B function. [For 1-contact input] 1. The recorded pulse direction is not constant. - The machine is at a stop in the offset write mode. - The servo power is off. - Pulse directions are diverse. 2. The tool is moving along the two axes (X-axis and Y-axis). [For the movement direction discrimination specification] 1. The recorded pulse direction is not constant. - The machine is at a stop in the offset write mode. - The servo power is off. - Pulse directions are diverse. 2. The tool is moving along the two axes (X-axis and Z-axis). 3. The direction indicated by the tool compensation write signal does not match the movement direction of the axis.
5196	ILLEGAL AXIS OPERATION	During HPCC or during the execution of a 5-axis-related function, an unavailable function was used.
5220	REFERENCE POINT ADJUSTMENT MODE	In case of distance coded linear scale I/F, the reference point auto setting parameter (No.1819#2) is set to "1". Move the machine to reference position by manual operation and execute manual reference return.
5242	ILLEGAL AXIS NUMBER	A master axis number or a slave axis number was not set correctly when the flexible synchronization control mode was turned from off to on during automatic operation.
5243	DATA OUTRANGE	A gear ratio was not set correctly when the flexible synchronization control mode was turned from off to on during automatic operation.
5244	TOO MANY DI ON	- During automatic operation, the mode signal was not turned on or off if the flexible synchronization control mode was turned from off to on or or from on to off, after an M code was executed. - An attempt was made to turn on or off the flexible synchronization control mode when the axis was not stopped.

Number	Message	Description
5245	OTHERAXIS ARE COMMANDED	- For a flexible synchronization control group for which a PMC axis was a master axis, an attempt was made to turn on the synchronous mode during time other than automatic operation. - An attempt was made to turn on a synchronization group for which an PMC axis was a master axis when there existed a flexible synchronization control group for which a non-PMC, normal axis was a master axis. - The master and slave axes as synchronous axes overlap the EGB axis. - The master and slave axes as synchronous axes overlap the chopping axis. - The master and slave axes as synchronous axes overlap the axis related to angular axis control. - The master and slave axes as synchronous axes overlap the axis related to composite control. - The master and slave axes as synchronous axes overlap the axis related to superposition control. - The slave axis as a synchronous axis overlaps the axis related to synchronization control. - The reference position return mode is turned on (was turned on).
5255	G12.4/G13.4 FORMAT ERROR	The specified P, I, and K are incorrect or I is less than K.
5257	G41/G42 NOT ALLOWED IN MDI MODE	Tool radius/tool nose radius compensation was specified in MDI mode. (Depending on the setting of the parameter MCR (No. 5008#4))
5303	TOUCH PANEL ERROR	The touch panel is not connected correctly, or the touch panel cannot be initialized when the power is turned on. Correct the cause then turn on the power again.
5305	ILLEGAL SPINDLE NUMBER	In a spindle select function by address P for a multiple spindle control, 1) Address P is not specified. 2) Parameter No.3781 is not specified to the spindle to be selected. 3) An illegal G code which cannot be commanded with an S_P_; command is specified. 4) A multi spindle cannot be used because the parameter EMS (No. 3702#1) is 1. 5) The spindle amplifier number of each spindle is not set in parameter No. 3717. 6) A prohibited command for a spindle was issued (parameter No. 11090). 7) An invalid value is set in parameter No. 11090.
5312	ILLEGAL COMMAND IN G10 L75/76/77	One of formats in G10L75, G10L76, or G10L77 to G11 commands is in error, or the command value is out of data range. Modify the program.
5316	TOOL TYPE NUMBER NOT FOUND	A tool with the specified tool-type number could not be found. Modify the program or register the tool.
5317	ALL TOOL LIFE IS OVER	The lives of all tools with the specified tool-type number have expired. Replace the tool.
5320	DIA./RAD. MODE CAN'T BE SWITCHED .	In any of the following states, diameter/radius specification was switched: 1) When a buffered program is being executed 2) When a movement is being made on the axis

Number	Message	Description
5329	M98 AND NC COMMAND IN SAME BLOCK	A subprogram call which is not a single block was commanded during canned cycle mode.
5339	ILLEGAL FORMAT COMMAND IS EXECUTED IN SYNC/MIX/OVL CONTROL.	1. The value of P, Q, or L specified by G51.4/G50.4/G51.5/G50.5/G51.6/G50.6 is invalid. 2. A duplicate value is specified by parameter No. 12600.
5346	RETURN TO REFERENCE POINT	The coordinate establishment of the Cs contour control axis is not made. Perform a manual reference position return. 1. When Cs coordinate establishment is made for the Cs-axis for which the Cs-axis reference position status signal CSPENx is 0 2. When positional information is not sent from the spindle amplifier 3. When the servo off state is entered during the start of Cs-axis coordinate establishment 4. When the Cs-axis is subjected to synchronous control or superposition control 5. When the emergency stop state is entered during coordinate establishment 6. When an attempt is made to release composite control for the Cs axis being subjected to coordinate establishment 7. When an attempt is made to start synchronous, composite, or superposition control for the Cs axis being subjected to coordinate establishment.
5360	TOOL INTERFERENCE CHECK ERROR	This alarm is issued when interference with another tool is caused by a data modification based on G10 data input or file reading or when an attempt is made to modify the tool figure data of a tool registered in the cartridge.
5361	ILLEGAL MAGAZINE DATA	Tools stored in the cartridge are interfering with each other. Reregister the tools in the cartridge, or modify the tool management data or tool figure data. If this alarm is issued, no tool interference check is made when tools are registered in the cartridge management table. Moreover, empty pot search operation does not operate normally. If this alarm is issued, the power must be turned off before operation is continued.
5362	CONVERT INCH/MM AT REF-POS	An inch/metric conversion was performed at a position other than the reference position. Perform an inch/metric conversion after returning to the reference position.

Number	Message	Description
5364	ILLEGAL COMMAND IN PROGRAM CHECK	(1) An unspecifiable G code was specified in the high-speed program check mode. (2) The angular axis control option or customer's board option is enabled. (3) One of the following operations was performed. - Chopping in the high-speed program check mode - Starting the high-speed program check mode during chopping - High-speed cycle machining in the high-speed program check mode - Reference position return of an axis for which the reference position is not established, in the high-speed program check mode (4) Switching of PMC axis selection signal EAX*<G0136> was performed. (5) G10 was specified for bit 3 (PGR) of parameter No. 3454 in the high-speed program check mode. (6) G10 was specified for bit 6 (PGS) of parameter No. 3001 in the high-speed program check mode.
5365	NOT CHANGE OF PROGRAM CHECK MODE	(1) Switching of high-speed program check input signal PGCK<Gn290.5> was performed during execution of the program.
5381	INVALID COMMAND IN FSC MODE	An attempt was made to issue the following commands: - When the reference position for the master axis under flexible synchronization control has not been established, G28 command for the master axis. - G27/G28/G29/G30/G30.1/G53 command for a slave axis.
5391	CAN NOT USE G92	Workpiece coordinate system setting G92 (or G50 for the lathe system G-code system A) cannot be specified. (1) After tool length compensation was changed by tool length compensation shift type, G92 was specified when no absolute command is present. (2) G92 was specified in the block in which G49 is present.
5406	G41.3/G40 FORMAT ERROR	(1) The G41.3 or G40 block contains a move command. (2) The G41.3 block contains a G or M code that suppresses buffering.
5407	ILLEGAL COMMAND IN G41.3	(1) In the G41.3 mode, a G code of group 01 other than G00 and G01 is specified. (2) In the G41.3 mode, an offset command (a G code of group 07) is specified. (3) The block next to G41.3 (startup) specifies no movement.
5408	G41.3 ILLEGAL START_UP	(1) In a mode of group 01 other than G00 and G01, G41.3 (startup) is specified. (2) The included angle between the tool vector and move vector is 0 or 180 degrees at the time of startup.
5420	ILLEGAL PARAMETER IN G43.4/G43.5	A parameter related to tool center point control is illegal. - Acceleration/deceleration before interpolation is disabled. Set parameter No. 1660. - Rapid traverse acceleration/deceleration before interpolation is disabled. Set bit 1 of parameter No. 1401, bit 5 of parameter No. 19501, parameter No. 1671, and parameter No. 1672.

Number	Message	Description
5421	ILLEGAL COMMAND IN G43.4/G43.5	An illegal command was specified in tool center point control. - A rotation axis command was specified in tool center point control (type 2) mode. - With a table rotary type or mixed-type machine, a I,J,K command was specified in the tool center point control (type 2) command (G43.5) block. - A command that does not move the tool center point (only a rotation axis is moved) was specified for the workpiece in the G02 mode. - When the workpiece coordinate system is set as the programming coordinate system (bit 5 (WKP) of parameter No. 19696 is 1), G02 or G03 was specified while the rotation axis was not perpendicular to the plane. - A G code not specifiable during the tool center point control mode was specified. - The modal code used to specify tool center point control is incorrect. - During the tool center point control mode, an axis not related to tool center point control was specified. - When bit 0 (RCM) of parameter No. 11200 is set to 0 to disable tool direction compensation, tool center point control is specified during the workpiece setting error compensation/tilted working plane command mode. - When tool posture control is enabled under tool center point control (type 2), a command is specified to set a tool posture near a singular point. (This alarm may be suppressed with bit 3 (NPC) of parameter No. 19696.) Check the machine configuration and specification. - When tool posture control is enabled under tool center point control (type 1), a rotary axis angular displacement that disables tool posture control is specified. Check the machine configuration and specification. - During tool center point control (type 2) or tool posture control, NURBS interpolation is specified. Check the specification.
5422	EXCESS VELOCITY IN G43.4/G43.5	An attempt was made to make a movement at an axis feedrate exceeding the maximum cutting feedrate by tool center point control.
5424	ILLEGAL TOOL DIRECTION	The rotation axis position for specifying the tool axis direction is not ±90° × n (n = 0, 1, 2,・・・).
5425	ILLEGAL OFFSET VALUE	The offset number is incorrect.
5430	ILLEGAL COMMAND IN 3-D CIR	In a modal state in which three-dimensional circular interpolation cannot be specified, a three-dimensional circular interpolation (G02.4/G03.4) is specified. Alternatively, in three-dimensional circular interpolation mode, a code that cannot be specified is specified.
5432	G02.4/G03.4 FORMAT ERROR	A three-dimensional circular interpolation command (G02.4/G03.4) is incorrect.
5433	MANUAL INTERVENTION IN G02.4/G03.4 (ABS ON)	In three-dimensional circular interpolation mode (G02.4/G03.4), manual intervention was made when the manual absolute switch was on.
5435	PARAMETER OUT OF RANGE (TLAC)	Illegal parameter setting. (Set value is out of range.) Check parameters No. 19655, No. 19656, No. 19657, and No. 1022.

Number	Message	Description
5436	ILLEGAL PARAMETER SETTING OF ROTARY AXIS(TLAC)	Illegal parameter setting. (axis of rotation setting)
5437	ILLEGAL PARAMETER SETTING OF MASTER ROTARY AXIS(TLAC)	Illegal parameter setting. (master axis of rotation setting)
5445	CAN NOT COMMAND MOTION IN G39	Corner circular interpolation (G39) of tool radius/tool nose radius compensation is not specified alone but is specified with a move command.
5446	NO AVOIDANCE AT G41/G42	Because there is no interference avoidance vector, the interference check avoidance function of tool radius/tool nose radius compensation does not work.
5447	DANGEROUS AVOIDANCE AT G41/G42	The interference check avoidance function of tool radius/tool nose radius compensation operation will lead to danger.
5448	INTERFERENCE TO AVD. AT G41/G42	In the interference check avoidance function of tool radius/tool nose radius compensation, a further interference occurs for an already created interference avoidance vector.
5456	TOO MANY G68.2 NESTING	Tilted working plane command G68.2 was specified more than once. To perform another coordinate conversion, perform cancellation, then specify the coordinate conversion.
5457	G68.2 FORMAT ERROR	A G68.2 format error occurred.
5458	ILLEGAL USE OF G53.1	- G53.1 was specified preceding G68.2. - G53.1 needs to be specified solely. - There is no angle solution for the rotation axis that controls the tool direction in the +Z-axis direction of the feature coordinate system. - In the setting by which compensation in the tool direction is not performed (when bit 0 (RCM) of parameter No. 11200 is 0), G53.1 was specified in the workpiece placement error compensation mode.
5459	MACHINE PARAMETER INCORRECT	- The parameter No. 19665 to No. 19667, No. 19680 to No. 19744 for configuring the machine are incorrect. - The axis specified with parameter No. 19681 or No. 19686 is not a rotary axis. - In parameter No. 1022, the basic three axes are not set. - A rotary axis end point found by the NC with tool center point control type 2, 3-dimensional cutter compensation type 2, or the tilted working plane command is not within the range set by parameter No. 19741 to No. 19744. - No rotary axis end point was found with tool center point control type 2 or 3-dimensional cutter compensation type 2. Check the machine configuration and specification. - On a machine whose rotary axis is a virtual axis, tool center point control type 2 or 3-dimensional cutter compensation type 2 was specified. - When programming is performed in the workpiece coordinate system, tool center point control type 2 or 3-dimensional cutter compensation type 2 is specified.

Number	Message	Description
5460	ILLEGAL USE OF 3-DIMENSIONAL CUTTER COMPENSATION	- In the 3-dimensional cutter compensation mode (except the tool side offset function for a tool rotation type machine), a move command other than G00/G01 is specified. - With a table rotation type machine, when bit 1 (PTD) of parameter No. 19746 is set to 1, a plane selection is made with an axis other than the basic three axes at the start of 3-dimensional cutter compensation. - When bit 1 (SPG) of parameter No. 19607 is set to 1, there is a discrepancy between the machine type set in parameter No. 19680 and a G code specifying 3-dimensional cutter compensation (G41.2, G42.2, G41.4, G42.4, G41.5, or G42.5). - With a machine that is not of the tool rotation type, G41.3 is specified. - When bit 5 (WKP) of parameter No. 19696 is set to 0, and bit 4 (TBP) of parameter No. 19746 is set to 0, 3-dimensional cutter compensation and tool center point control are used at the same time. - A rotation axis command is specified in the 3-dimensional cutter compensation mode (type 2). - With a table rotation type or mixed type machine, IJK is specified in a block that specifies 3-dimensional cutter compensation (type 2) (G41.6/G42.6). - An illegal G code is specified in the 3-dimensional cutter compensation mode. - When 3-dimensional cutter compensation is specified, the modal state is illegal. - When the table coordinate system is set as the programming coordinate system, table rotate and 3-dimensional cutter compensation are specified after the start of tool center point control. - There is a difference in type1/type 2 specification between 3-dimensional cutter compensation and tool center point control. - When 3-dimensional cutter compensation and tool center point control are used at the same time, one of these functions that is specified earlier than the other is canceled earlier.
5461	ILLEGAL USE OF G41.2/G42.2/G41.5/G42.5	A move command other than G00 or G01 was performed during 3-dimensional cutter compensation in a mixed-type machine.
5462	ILLEGAL COMMAND (G68.2/G69)	(1) The modal setting used when G68.2 or G69 is specified is incorrect. (2) An unspecifiable G code was specified in the G68.2 mode. (3) The offset vector of tool radius/tool nose radius compensation is not canceled when G68.2 or G69 is specified.
5463	ILLEGAL PARAMETER IN 3-DIMENSIONAL CUTTER COMPENSATION	A parameter related to 3-dimensional cutter compensation is illegal. - Acceleration/deceleration before interpolation is disabled. Set parameter No. 1660. - Rapid traverse acceleration/deceleration before interpolation is disabled. Set bit 1 (LRP) of parameter No. 1401, bit 5 (FRP) of parameter No. 19501, and parameter Nos. No.1671 and 1672.

(4) Parameter writing alarm (SW alarm)

Number	Message	Description
SW0100	PARAMETER ENABLE SWITCH ON	The parameter setting is enabled (PWE, one bit of parameter No. 8000 is set to “1”). To set the parameter, turn this parameter ON. Otherwise, set to OFF.

(5) Servo alarms (SV alarm)

Number	Message	Description
SV0001	SYNC ALIGNMENT ERROR	In feed axis synchronization control, the amount of compensation for synchronization exceeded the parameter (No. 8325) setting value. This alarm occurs only for a slave axis.
SV0002	SYNC EXCESS ERROR ALARM 2	In feed axis synchronization control, the amount of synchronization error exceeded the parameter (No. 8332) setting value. When the synchronization is not completed after power-up, the determination is made by the parameter value (No. 8332) multiplied by the parameter (No. 8330) multiplier. This alarm occurs only for a slave axis only.
SV0003	SYNCHRONOUS/COMPOSITE/SUPERIMPOSED CONTROL MODE CAN'T BE CONTINUED	Since as axis in synchronization, composition, or superposition mode caused a servo alarm, the mode could not be continued, If one of the axes in a mode causes a servo alarm, all axes relating to the axis enter the servo-off state. This alarm is generated to enable the cause of the servo-off state to be checked.
SV0004	EXCESS ERROR (G31)	The amount of positional deviation during torque limit skip command operation exceeded the limit value of the parameter No.6287.
SV0005	SYNC EXCESS ERROR (MCN)	In feed axis synchronization control, for synchronization, the difference value of the machine coordinate between a master and slave axes exceeded the parameter (No. 8314) setting value. This alarm occurs for a master or slave axis.
SV0006	ILLEGAL TANDEM AXIS	For the slave axis under tandem control, absolute position detection is set (parameter bit 5 (APC) of parameter No. 1815 = 1).
SV0007	SV ALM ANOTHER PATH(MULTI AMP.)	When a multiaxis amplifier was used in a multipath system across paths, a servo alarm occurred on an axis belonging to another path. When a system with two or more paths and multiple servo axes between paths are controlled by a multiaxis amplifier, if a servo alarm occurs on an axis belonging to another path of the same amplifier, the MCC of the amplifier goes down and SV0401 V-READY OFF occurs on an axis belonging to the local path in the same amplifier. Since SV0401 is caused by a servo alarm occurred on an axis in another path, SV0007 is caused together to clearly indicate the fact. The axis belonging to another path in the same amplifier resolves the cause of the servo alarm.

Number	Message	Description
SV0301	APC ALARM: COMMUNICATION ERROR	Since the absolute-position detector caused a communication error, the correct machine position could not be obtained. (data transfer error) The absolute-position detector, cable, or servo interface module is thought to be defective.
SV0302	APC ALARM: OVER TIME ERROR	Since the absolute-position detector caused an overtime error, the correct machine position could not be obtained. (data transfer error) The absolute-position detector, cable, or servo interface module is thought to be defective.
SV0303	APC ALARM: FRAMING ERROR	Since the absolute-position detector caused a framing error, the correct machine position could not be obtained. (data transfer error) The absolute-position detector, cable, or servo interface module is thought to be defective.
SV0304	APC ALARM: PARITY ERROR	Since the absolute-position detector caused a parity error, the correct machine position could not be obtained. (data transfer error) The absolute-position detector, cable, or servo interface module is thought to be defective.
SV0305	APC ALARM: PULSE ERROR	Since the absolute-position detector caused a pulse error, the correct machine position could not be obtained. The absolute-position detector, or cable is thought to be defective.
SV0306	APC ALARM: OVER FLOW ERROR	Since the amount of positional deviation overflowed, the correct machine position could not be obtained. Check to see the parameter No. 2084 or No. 2085.
SV0307	APC ALARM: MOVEMENT EXCESS ERROR	Since the machine moved excessively, the correct machine position could not be obtained.
SV0360	ABNORMAL CHECKSUM(INT)	The checksum alarm occurred on the built–in Pulsecoder.
SV0361	ABNORMAL PHASE DATA(INT)	The phase data abnormal alarm occurred on the built–in Pulsecoder.
SV0362	ABNORMAL REV. DATA(INT)	The speed count abnormal alarm occurred on the built-in Pulsecoder.
SV0363	ABNORMAL CLOCK(INT)	The clock alarm occurred on the built–in Pulsecoder.
SV0364	SOFT PHASE ALARM(INT)	A digital servo soft detected an abnormality on the built in Pulsecoder.
SV0365	BROKEN LED(INT)	The digital servo software detected abnormal data on the built–in Pulsecoder.
SV0366	PULSE MISS(INT)	A pulse error occurred on the built–in Pulsecoder.
SV0367	COUNT MISS(INT)	A count error occurred on the built–in Pulsecoder.
SV0368	SERIAL DATA ERROR(INT)	The communications data could not be received from the built–in Pulsecoder.
SV0369	DATA TRANS. ERROR(INT)	A CRC error or stop bit error occurred in the communications data from the built–in Pulsecoder.
SV0380	BROKEN LED(EXT)	Separate detector error
SV0381	ABNORMAL PHASE (EXT)	An abnormal alarm in the position data occurred on the separate detector.
SV0382	COUNT MISS(EXT)	A count error occurred on the separate detector.
SV0383	PULSE MISS(EXT)	A pulse error occurred on the separate detector.
SV0384	SOFT PHASE ALARM(EXT)	The digital servo software detected abnormal data on the separate detector.

Number	Message	Description
SV0385	SERIAL DATA ERROR(EXT)	The communications data could not be received from the separate detector.
SV0386	DATA TRANS. ERROR(EXT)	A CRC error or stop bit error occurred in the communications data from the standalone detector.
SV0387	ABNORMAL ENCODER(EXT)	An abnormality occurred on a separate detector. For more information, contact the scale manufacturer.
SV0401	IMPROPER V_READY OFF	Although the ready signal (PRDY) of the position control was ON, the ready signal (VRDY) of the velocity control was OFF.
SV0403	CARD/SOFT MISMATCH	The combination of the axis control card and the servo software is incorrect. Probable causes are given below. - The correct axis control card is not attached. - The correct servo software is not installed in flash memory.
SV0404	IMPROPER V_READY ON	Although the ready signal (PRDY) of the position control was OFF, the ready signal (VRDY) of the velocity control was ON.
SV0407	EXCESS ERROR	The difference value of the amount of positional deviation for the synchronization axis exceeded the setting value. (during synchronization control only)
SV0409	DETECT ABNORMAL TORQUE	An abnormal load was detected on the servo motor, or during Cs axis or spindle positioning. The alarm can be canceled by RESET.
SV0410	EXCESS ERROR (STOP)	The amount of positional deviation during stopping exceeded the parameter (No. 1829) setting value.
SV0411	EXCESS ERROR (MOVING)	The amount of positional deviation during traveling became excessive than the parameter setting value.
SV0413	LSI OVERFLOW	The counter for the amount of positional deviation overflowed
SV0415	MOTION VALUE OVERFLOW	The velocity exceeding the travel velocity limit was commanded.
SV0417	ILL DGTL SERVO PARAMETER	A digital serve parameter setting is incorrect. When bit 4 of diagnosis information No. 203 is 1, an illegal parameter was detected by the servo software. Identify the cause with reference to diagnosis information No. 352. When bit 4 of diagnosis information No. 203 is 0, the CNC software detected an illegal parameter. Probable causes are given below (see diagnosis information No. 280). 1) The value specified in parameter No. 2020 as the motor model falls outside the specified range. 2) The motor rotation direction in parameter No. 2022 is not set to a correct value (111 or -111). 3) The speed feedback pulse count per motor rotation in parameter No. 2023 is set to a negative or other incorrect value. 4) The position feedback pulse count per motor rotation in parameter No. 2024 is set to a negative or other incorrect value.
SV0420	SYNC TORQUE EXCESS	In feed axis synchronization control, for synchronization, the difference value of torque between a master and slave axes exceeded the parameter (No. 2031) setting value. This alarm occurs for a master axis.
SV0421	EXCESS ERROR(SEMI-FULL)	The difference between the feedback from the semi and full sides exceeded the setting of parameter No.1729.

Number	Message	Description
SV0422	EXCESS VELOCITY IN TORQUE	In torque control, the commanded permissible velocity was exceeded.
SV0423	EXCESS ERROR IN TORQUE	In torque control, the total permissible move value specified as a parameter was exceeded.
SV0430	SV MOTOR OVERHEAT	The servo motor has overheated.
SV0431	CNV. OVERLOAD	PSM : Overheat β series SVU : Overheat
SV0432	CNV. LOW VOLT CONTROL	PSM : The control power supply voltage has dropped. PSMR : The control power supply voltage has dropped. β series SVU : The control power supply voltage has dropped.
SV0433	CNV. LOW VOLT DC LINK	PSM : Low DC link voltage PSMR : Low DC link voltage α series SVU : Low DC link voltage β series SVU : Low DC link voltage
SV0434	INV. LOW VOLT CONTROL	SVM : Low control power voltage
SV0435	INV. LOW VOLT DC LINK	SVM : Low DC link voltage
SV0436	SOFTTHERMAL(OVC)	The digital servo software detected a software thermal (OVC).
SV0437	CNV. OVERCURRENT POWER	PSM : Overcurrent on input circuit section.
SV0438	INV. ABNORMAL CURRENT	SVM : Motor overcurrent α series SVU : Motor overcurrent β series SVU : Motor overcurrent
SV0439	CNV. OVER VOLT DC LINK	PSM : The DC link voltage is too high. PSMR : The DC link voltage is too high. β series SVU : The DC link voltage is too high.
SV0440	CNV. EX DECELERATION POW.	PSMR : Excessive generative discharge α series SVU : Excessive generative discharge, or abnormal error in generative power circuit
SV0441	ABNORMAL CURRENT OFFSET	The digital servo software detected an abnormality in the motor current detection circuit.
SV0442	CNV. CHARGE FAILURE	PSM : The spare charge circuit for the DC link is abnormal. PSMR : The spare charge circuit for the DC link is abnormal.
SV0443	CNV. COOLING FAN FAILURE	PSM : Internal cooling fan failure. PSMR : Internal cooling fan failure. β series SVU : Internal cooling fan failure.
SV0444	INV. COOLING FAN FAILURE	SVM : Internal cooling fan failure.
SV0445	SOFT DISCONNECT ALARM	The digital servo software detected a disconnected Pulsecoder.
SV0446	HARD DISCONNECT ALARM	The hardware detected a disconnected built–in Pulsecoder.
SV0447	HARD DISCONNECT(EXT)	The hardware detected a disconnected separate detector.
SV0448	UNMATCHED FEEDBACK ALARM	The sign of the feedback signal from the standalone detector is opposite to that from the feedback signal from the built–on Pulsecoder.
SV0449	INV. IPM ALARM	SVM : The IPM (Intelligent Power Module) detected an alarm. α series SVU : The IPM (Intelligent Power Module) detected an alarm.
SV0453	SPC SOFT DISCONNECT ALARM	Software disconnection alarm of the α Pulsecoder. Turn off the power to the CNC, then remove and insert the Pulsecoder cable. If this alarm is issued again, replace the Pulsecoder.

Number	Message	Description
SV0454	ILLEGAL ROTOR POS DETECT	The magnetic pole detection function terminated abnormally. The magnetic pole could not be detected because the motor did not run.
SV0456	ILLEGAL CURRENT LOOP	An attempt was made to set the current loop that could not be set. The amplifier pulse module in use does not comply with HIGH SPEED HRV. Or, requirements to control are not satisfied in the system.
SV0458	CURRENT LOOP ERROR	The specified current loop differs from the actual current loop.
SV0459	HI HRV SETTING ERROR	For two axes whose servo axis numbers (parameter No. 1023) are consecutively even and odd numbers, HIGH SPEED HRV control is possible for one axis and impossible for the other.
SV0460	FSSB DISCONNECT	The FSSB connection was discontinued. Probable causes are: 1. The FSSB connection cable was disconnected or broken. 2. The amplifier was turned off . 3. In the amplifier, the low-voltage alarm occurred.
SV0462	SEND CNC DATA FAILED	The correct data could not be received on a slave side because of the FSSB communication error.
SV0463	SEND SLAVE DATA FAILED	The correct data could not be received in the servo software because of the FSSB communication error.
SV0465	READ ID DATA FAILED	A read of the ID information for the amplifier has failed at power-on.
SV0466	MOTOR/AMP. COMBINATION	The maximum current of an amplifier is different to that of a motor. Probable causes are: 1. The connection command for an amplifier is incorrect. 2. The parameter (No.2165) setting is incorrect
SV0468	HI HRV SETTING ERROR(AMP)	An attempt was made to set up HIGH SPEED HRV control for use when the controlled axis of an amplifier for which HIGH SPEED HRV control could not be used.
SV0474	EXCESS ERROR(STOP:SV)	The servo detected that the positional deviation during a stop exceeded the setting (parameters No. 1839 and No. 1842) in the n-axis.
SV0475	EXCESS ERROR(MOVE:SV)	The servo detected that the positional deviation during a travel exceeded the setting (parameters No. 1838 and No. 1841) in the n-axis.
SV0476	ILLEGAL SPEED CMD.(SV)	The servo detected that the speed command exceeded the safety speed (parameters No. 13821 to No. 13824 (during position control) or parameters No. 13826 to No. 13829 (during speed control)) during safety monitoring (when safety monitoring request signal *VLDVx is 0) in the n-axis. Keep the safety speed.
SV0477	ILLEGAL MACHINE POS.(SV)	The servo detected that the machine position fell outside the safety area (setting by parameters No. 13831 to No. 13838) during safety monitoring (when safety monitoring request signal *VLDVx is 0) in the n-axis. Keep the safety area. A machine position check is performed only on the axis for which the reference position has been established. The axis for which the reference position is not established is not subjected to a machine position check.

Number	Message	Description
SV0478	ILLEGAL AXIS DATA(SV)	The servo detected that an error occurred during transfer of axis data in the n-axis. When an alarm occurred because the configuration of the servo amplifier was changed, set the axis number for the servo amplifier (set bit 4 of parameter No. 2212 of the corresponding axis to 1 and then 0 and turn off the power of the entire system). When using a multiaxis amplifier, this operation may not clear the alarm. In this case, repeat this operation for the axes on which the alarm persists. If an alarm occurs due to a cause other than the above, replace the servo amplifier.
SV0481	SAFETY PARAM ERROR(SV)	The servo detected that a safety parameter error occurred in the n-axis.
SV0484	SAFETY FUNCTION ERROR(SV)	A safety function error related to the servo was detected in the n-axis. - The servo or CNC detected that the safety function was not executed in the servo. - The result of a servo safety function check did not match the result of a CNC safety function check. - An error occurred during a test of the CPU of the servo. - An error occurred during a check of RAM of the servo.
SV0488	SELF TEST OVER TIME	An MCC interruption test was not complete within the set period of time (parameter No. 1946). Check the MCC contact.
SV0489	SAFETY PARAM ERROR(CNC)	The CNC detected that a safety parameter error occurred in the n-axis.
SV0490	SAFETY FUNCTION ERROR(CNC)	A CNC safety function error occurred in the n-axis. - The servo detected that the safety function was not executed in the CNC. - The result of a servo safety function check did not match the result of a CNC safety function check.
SV0494	ILLEGAL SPEED CMD.(CNC)	The CNC detected that the speed command exceeded the safety speed (parameters No. 13821 to No. 13824 (during position control) or parameters No. 13826 to No. 13829 (during speed control)) during safety monitoring (when safety monitoring request signal *VLDVx is 0) in the n-axis. Keep the safety speed.
SV0495	ILLEGAL MACHINE POS.(CNC)	The CNC detected that the machine position fell outside the safety area (set range by parameters No. 13831 to No. 13838) during safety monitoring (when safety monitoring request signal *VLDVx is 0) in the n-axis. Keep the safety area. A machine position check is performed only on the axis for which the reference position has been established.

Number	Message	Description
SV0496	ILLEGAL AXIS DATA(CNC)	The CNC detected that an error occurred during transfer to axis data. When an alarm occurred because the configuration of the servo amplifier was changed, set the axis number for the servo amplifier (set bit 4 of parameter No. 2212 of the corresponding axis to 1 and 0 again and turn off the power of the entire system). When using a multiaxis amplifier, this operation may not clear the alarm. In this case, repeat this operation for the axes on which the alarm persists. If an alarm occurs due to a cause other than the above, replace the servo amplifier.
SV0498	AXIS NUMBER NOT SET(CNC)	The CNC detected that the axis number of the n-axis was not set for the servo amplifier. The axis number is set automatically, so turn off the power of the entire system.
SV0600	INV. DC LINK OVER CURRENT	SVM : DC link overcurrent. β SVU : DC link overcurrent.
SV0601	INV. RADIATOR FAN FAILURE	SVM : Radiator cooling fan failure. β SVU : Radiator cooling fan failure.
SV0602	INV. OVERHEAT	SVM : The servo motor has overheated.
SV0603	INV. IPM ALARM(OH)	SVM : The IPM (Intelligent Power Module) detected an overheat alarm. β SVU : The IPM (Intelligent Power Module) detected an overheat alarm.
SV0604	AMP. COMMUNICATION ERROR	The communication between SVM and PSM is in error.
SV0605	CNV. EX. DISCHARGE POW.	PSMR : The motor regenerative power is too much.
SV0606	CNV. RADIATOR FAN FAILURE	PSM : External radiator cooling fan failure. PSMR : External radiator cooling fan failure.
SV0607	CNV. SINGLE PHASE FAILURE	PSM : The input power supply has a missing phase. PSMR : The input power supply has a missing phase.
SV0646	ABNORMAL ANALOG SIGNAL(EXT)	An error occurred in the analog 1Vp-p output of the separate detector. The separate detector, cable, or separate detector interface unit may be failed.
SV1025	V_READY ON (INITIALIZING)	The ready signal (VRDY) of the velocity control which should be OFF is ON while the servo control is ON.
SV1026	ILLEGAL AXIS ARRANGE	The parameter for servo axis arrange is not set correctly. A negative value, duplicate value, or greater value than the number of control axes was set to the parameter No. 1023 "The servo axis number of each axis."
SV1055	ILLEGAL TANDEM AXIS	In tandem control, the setting of the parameter No. 1023 is incorrect.
SV1056	ILLEGAL TANDEM PAIR	In tandem control, the setting of the parameter TDM (No.1817#6) is incorrect.
SV1067	FSSB:CONFIGURATION ERROR(SOFT)	An FSSB configuration error occurred (detected by software). The connected amplifier type is incompatible with the FSSB setting value.
SV1068	DUAL CHECK SAFETY ALARM	An alarm that turns off the MCC of the entire system occurred in the Dual Check Safety function.
SV1069	EXCESS ERROR(SERVO OFF:CNC)	The CNC detected that the positional deviation during servo-off exceeded the set value (parameter No. 1840) in the n-axis.
SV1070	EXCESS ERROR(SERVO OFF:SV)	The servo detected that the positional deviation during servo-off exceeded the set value (parameter No. 1840) in the n-axis.

Number	Message	Description
SV1071	EXCESS ERROR(MOVE:CNC)	The CNC detected that the positional deviation during a travel exceeded the set value (parameters No. 1838 and No. 1841) in the n-axis.
SV1072	EXCESS ERROR(STOP:CNC)	The CNC detected that the positional deviation during a stop exceeded the set value (parameters No. 1839 and No. 1842) in the n-axis.
SV1100	S-COMP. VALUE OVERFLOW	The amount of compensation for the straightness exceeded a maximum value of 32767.
SV5134	FSSB:OPEN READY TIME OUT	In the initialization, the FSSB could not be in an open ready sate. The axis card is thought to be defective.
SV5136	FSSB:NUMBER OF AMP. IS INSUFFICIENT	The number of amplifier identified by the FSSB is insufficient than the number of control axes. Or, the setting of the number of axes or the amplifier connection is in error.
SV5137	FSSB:CONFIGURATION ERROR	An FSSB configuration error occurred. The connecting amplifier type is incompatible with the FSSB setting value.
SV5139	FSSB:ERROR	Servo initialization has not completed successfully. It is probable that an optical cable failed or a connection between the amplifier and another module failed.
SV5197	FSSB:OPEN TIME OUT	The initialization of the FSSB was completed, but it could not be opened. Or, the connection between the CNC and the amplifier in is incorrect.
SV5197	FSSB:OPEN TIME OUT	The FSSB could not be opened although the CNC permitted the opening of the FSSB. Check the connection between the CNC and the amplifier.
SV5311	FSSB:ILLEGAL CONNECTION	1. This alarm is issued if axes, whose servo axis numbers (parameter No. 1023) are even and odd numbers, are allocated to the amplifiers connected to the FSSBs of different paths. 2. This alarm is issued if an attempt is made to set up for use of the pulse modules connected to the FSSBs of different paths. And the system did not satisfy the requirements for performing HIGH SPEED HRV control.

(6) Overtravel alarms (OT alarm)

Number	Message	Description
OT0500	+ OVERTRAVEL (SOFT 1)	Exceeded the positive side stored stroke check 1.
OT0501	- OVERTRAVEL (SOFT 1)	Exceeded the negative side stored stroke check 1.
OT0502	+ OVERTRAVEL (SOFT 2)	Exceeded the positive side stored stroke check 2. Or, in the chuck tail stock barrier, an entry to the inhibited area was made during movement in the positive direction.
OT0503	- OVERTRAVEL (SOFT 2)	Exceeded the negative side stored stroke check 2. Or, in the chuck tail stock barrier, an entry to the inhibited area was made during movement in the negative direction.
OT0504	+ OVERTRAVEL (SOFT 3)	Exceeded the positive side stored stroke check 3.
OT0505	- OVERTRAVEL (SOFT 3)	Exceeded the - side stored stroke check 3.
OT0506	+ OVERTRAVEL (HARD)	The stroke limit switch in the positive direction was triggered. This alarm is generated when the machine reaches the stroke end. When this alarm is not generated, feed of all axes is stopped during automatic operation. During manual operation, only the feed of the axis on which the alarm occurred is stopped.

Number	Message	Description
OT0507	- OVERTRAVEL (HARD)	The stroke limit switch in the negative direction was triggered. This alarm is generated when the machine reaches the stroke end. When this alarm is not generated, feed of all axes is stopped during automatic operation. During manual operation, only the feed of the axis on which the alarm occurred is stopped.
OT0508	INTERFERENCE:+	A tool moving in the positive direction along the n axis has fouled another tool post.
OT0509	INTERFERENCE:-	A tool moving in the negative direction along the n axis has fouled another tool post.
OT0510	+ OVERTRAVEL (PRE-CHECK)	The tool exceeded the limit in the negative direction during the stroke check before movement.
OT0511	- OVERTRAVEL (PRE-CHECK)	The tool exceeded the limit in the positive direction during the stroke check before movement.
OT0514	(n) INTERFERENCE:+	An interference occurred on the plus side of the n-axis in a rotation area interference check.
OT0515	(n) INTERFERENCE:-	An interference occurred on the minus side of the n-axis in a rotation area interference check.
OT1710	ILLEGAL ACC. PARAMETER (OPTIMUM TORQUE ACC/DEC)	The permissible acceleration parameter for the optimum torque acceleration/deceleration is in error. A possible cause is either of the following: (1) The ratio of a negative acceleration to a positive acceleration is not more than the limit value. (2) The time to reduce to a velocity of 0 exceeded the maximum time.

(7) Memory file alarms (IO alarm)

Number	Message	Description
IO1001	FILE ACCESS ERROR	The resident–type file system could not be accessed as an error occurred in the resident–type file system.
IO1002	FILE SYSTEM ERROR	The file could not be accessed as an error occurred in the CNC file system.
IO1030	CHECK SUM ERROR	The checksum of the CNC part program storage memory is incorrect.
IO1032	MEMORY ACCESS OVER RANGE	Accessing of data occurred outside the CNC part program storage memory range.
IO1104	OVER MAXIMUM TOOL LIFE PAIRS	The maximum number of tool life management pairs is exceeded. Modify the setting of the maximum number of tool life management pairs in parameter No. 6813.

(8) Alarms requiring power to be turned off (PW alarm)

Number	Message	Description
PW0000	POWER MUST BE OFF	A parameter was set for which the power must be turned OFF then ON again.
PW0001	X-ADDRESS(*DEC) IS NOT ASSIGNED.	The X address of the PMC could not be assigned correctly. This alarm may occur in the following case: - During the setting of parameter No. 3013, the X address could not be assigned correctly for the deceleration dog (*DEC) for a return to the reference position.

Number	Message	Description
PW0002	PMC address is not correct(AXIS).	The address to assign the axis signal is incorrect. This alarm may occur in the following case: - The parameter No.3021 setting is incorrect.
PW0003	PMC address is not correct(SPINDLE).	The address to assign the spindle signal is incorrect. This alarm may occur in the following case: - The parameter No.3022 setting is incorrect.
PW0004	SETTING THE LOADER SYSTEM PATH IS NOT CORRECT.	The loader system could not be assigned correctly. The parameter No. 984 setting is incorrect. - The number of loader systems and the number of systems specified to the loader system in the parameter No. 984#0(LCP) does not match. - The parameter No. 984#0 of the system 1 is set to 1.
PW0006	POWER MUST BE OFF (ILL-EXEC-CHK)	The malfunction prevention function detected an alarm to require the power off.
PW0007	X-ADDRESS(SKIP) IS NOT ASSIGNED	The X address of PMC could not be assigned correctly. Possible causes are: - During the set of parameter No. 3012, the skip signal of the X address was not assigned correctly. - During the set of parameter No. 3019, the address other than the skip signal of the X address was not assigned correctly.
PW0008	CPU SELF TEST ERROR(DCS PMC)	On the DCS PMC side: - An error was detected by the CPU self diagnosis function. - An error was detected by the RAM check function.
PW0009	CPU SELF TEST ERROR(PMC)	On the PMC side: - An error was detected by the CPU self diagnosis function. - An error was detected by the RAM check function.
PW0010	SAFE I/O CROSS CHECK ERROR(DCS PMC)	On the DCS PMC side, an error was detected by the I/O cross check function in system-defined safety-related DI/DO.
PW0011	SAFE I/O CROSS CHECK ERROR(PMC)	On the PMC side, an error was detected by the I/O cross check function in system-defined safety-related DI/DO.
PW0012	USER I/O CROSS CHECK ERROR(DCS PMC)	On the DCS PMC side, an error was detected by the I/O cross check function in user-defined safety-related DI/DO.
PW0013	USER I/O CROSS CHECK ERROR(PMC)	On the PMC side, an error was detected by the I/O cross check function in user-defined safety-related DI/DO.
PW0014	CPU TEST ALARM (CNC)	An error occurred in a test of the CPU of the CNC.
PW0015	SAFETY PARAM ERROR	The CNC detected that an error occurred in a safety parameter for other than servo axes or spindle axes.
PW0016	RAM CHECK ERROR	An error was detected in a RAM check of the CNC.
PW0017	INEXECUTION OF SAFETY FUNCTIONS	The safety function was not executed normally in the CNC.
PW0018	CRC CHECK ERROR	An error was detected in a CRC check of the CNC.

Number	Message	Description
PW1102	ILLEGAL PARAMETER (I-COMP.)	The parameter for setting slope compensation is incorrect. This alarm occurs in the following cases: - When the number of pitch error compensation points on the axis on which slope compensation is executed exceeds 128 between the most negative side and most positive side - When the size relationship between the slope compensation point Nos. is incorrect - When the slope compensation point is not located between the most negative side and most positive side of pitch error compensation - When the compensation per compensation point is too small or too great.
PW1103	ILLEGAL PARAMETER (S-COMP.128)	The parameter for setting 128 straightness compensation points or the parameter compensation data is incorrect,
PW5046	ILLEGAL PARAMETER (S-COMP.)	The parameter for setting straightness compensation is incorrect.
PW5390	R-ADDRESS SETTING IS ILLEGAL	The start address or the set range of the PMC R addresses set by parameters No.13541 and No.13542 is invalid.

(9) Spindle alarms (SP alarm)

Number	Message	Description
SP0740	RIGID TAP ALARM : EXCESS ERROR	The positional deviation of the stopped spindle has exceeded the set value during rigid tapping.
SP0741	RIGID TAP ALARM : EXCESS ERROR	The positional deviation of the moving spindle has exceeded the set value during rigid tapping.
SP0742	RIGID TAP ALARM : LSI OVERFLOW	An LSI overflow has occurred for the spindle during rigid tapping.
SP0752	SPINDLE MODE CHANGE ERROR	This alarm is generated if the system does not properly terminate a mode change. The modes include the Cs contour control, spindle positioning, rigid tapping, and spindle control modes. The alarm is activated if the spindle control unit does not respond correctly to the mode change command issued by the NC.
SP0754	ABNORMAL TORQUE	An abnormal load was detected in a spindle motor. The alarm can be canceled by RESET.
SP0755	SAFETY FUNCTION ERROR	The CNC CPU detected that the safely function of the n-th spindle was not executed. Alternatively, the result of a CNC safety function check did not match the result of a spindle safety function check.
SP0756	ILLEGAL AXIS DATA	The CNC CPU detected that the connection state and the hardware setting of the spindle amplifier were incompatible on the n-th spindle. If an alarm occurs because of the configuration change of the spindle amplifier , set the spindle amplifier correctly.
SP0757	SAFETY SPEED OVER	The CNC CPU detected that during safety monitoring (when safety monitoring request signal *VLDPs is 0), the spindle motor speed was greater than the safety speed (parameter No. 4372, 4438, 4440, or 4442) on the n-th spindle. Operate within the safety speed.
SP1202	SPINDLE SELECT ERROR	In a multi spindle control, the spindle number other than the valid spindle number was selected by a position coder select signal. An attempt was made to select the spindle number of the system having no valid spindle.

Number	Message	Description
SP1210	TOOL CHANGE SP MOTION OVERFLOW	The amount of distribution to a spindle is too much. (specific to the FANUC ROBODRILL)
SP1211	TOOL CHANGE SP ORTN EXCESS ERROR	During a tool change, a too much orientation error was detected for the spindle. (specific to the FANUC ROBODRILL)
SP1212	TOOL CHANGE SP MOVE EXCESS ERROR	During a tool change, a too much moving error was detected for the spindle. (specific to the FANUC ROBODRILL)
SP1213	TOOL CHANGE SP STOP EXCESS ERROR	During a tool change, a too much stop error was detected for the spindle. (specific to the FANUC ROBODRILL)
SP1214	TOOL CHANGE SP ILLEGAL SEQUENCE	During changing tools, an abnormal spindle sequence was detected. (specific to the FANUC ROBODRILL)
SP1220	NO SPINDLE AMP.	Either the cable connected to a serial spindle amplifier is broken, or the serial spindle amplifier is not connected.
SP1221	ILLEGAL MOTOR NUMBER	The spindle No. and the motor No. are incorrectly matched.
SP1224	ILLEGAL SPINDLE-POSITION CODER GEAR RATIO	The spindle–position coder gear ratio was incorrect.
SP1225	CRC ERROR (SERIAL SPINDLE)	A CRC error (communications error) occurred in communications between the CNC and the serial spindle amplifier.
SP1226	FRAMING ERROR (SERIAL SPINDLE)	A framing error occurred in communications between the CNC and the serial spindle amplifier.
SP1227	RECEIVING ERROR (SERIAL SPINDLE)	A receive error occurred in communications between the CNC and the serial spindle amplifier.
SP1228	COMMUNICATION ERROR (SERIAL SPINDLE)	A communications error occurred between the CNC and the serial spindle amplifier.
SP1229	COMMUNICATION ERROR SERIAL SPINDLE AMP.	A communications error occurred between serial spindle amplifiers (motor Nos. 1 and 2, or motor Nos. 3–4).
SP1231	SPINDLE EXCESS ERROR (MOVING)	The position deviation during spindle rotation was greater than the value set in parameters.
SP1232	SPINDLE EXCESS ERROR (STOP)	The position deviation during spindle stop was greater than the value set in parameters.
SP1233	POSITION CODER OVERFLOW	The error counter/speed instruction value of the position coder overflowed.
SP1234	GRID SHIFT OVERFLOW	Grid shift overflowed.
SP1240	DISCONNECT POSITION CODER	The analog spindle position coder is broken.
SP1241	D/A CONVERTER ERROR	The D/A converter for controlling analog spindles is erroneous.
SP1243	ILLEGAL SPINDLE PARAMETER SETTING(GAIN)	The setting for the spindle position gain is incorrect.
SP1244	MOTION VALUE OVERFLOW	The amount of distribution to a spindle is too much
SP1245	COMMUNICATION DATA ERROR	A communication data error was detected on the CNC.
SP1246	COMMUNICATION DATA ERROR	A communication data error was detected on the CNC.
SP1247	COMMUNICATION DATA ERROR	A communication data error was detected on the CNC.
SP1700	SAFETY PARAM ERROR	The CNC detected that a safety parameter error occurred in the n-th spindle.
SP1969	SPINDLE CONTROL ERROR	An error occurred in the spindle control software.
SP1970	SPINDLE CONTROL ERROR	Initialization of spindle control ended in error.
SP1971	SPINDLE CONTROL ERROR	An error occurred in the spindle control software.
SP1972	SPINDLE CONTROL ERROR	An error occurred in the spindle control software.
SP1974	ANALOG SPINDLE CONTROL ERROR	An error occurred in the spindle control software.

Number	Message	Description
SP1975	ANALOG SPINDLE CONTROL ERROR	An position coder error was detected on the analog spindle.
SP1976	SERIAL SPINDLE COMMUNICATION ERROR	The amplifier No. could not be set to the serial spindle amplifier.
SP1977	SERIAL SPINDLE COMMUNICATION ERROR	An error occurred in the spindle control software.
SP1978	SERIAL SPINDLE COMMUNICATION ERROR	A time–out was detected during communications with the serial spindle amplifier.
SP1979	SERIAL SPINDLE COMMUNICATION ERROR	The communications sequence was no longer correct during communications with the serial spindle amplifier.
SP1980	SERIAL SPINDLE AMP. ERROR	Defective SIC–LSI on serial spindle amplifier
SP1981	SERIAL SPINDLE AMP. ERROR	An error occurred during reading of the data from SIC–LSI on the analog spindle amplifier side.
SP1982	SERIAL SPINDLE AMP. ERROR	An error occurred during reading of the data from SIC–LSI on the serial spindle amplifier side.
SP1983	SERIAL SPINDLE AMP. ERROR	Could not clear on the spindle amplifier side.
SP1984	SERIAL SPINDLE AMP. ERROR	An error occurred during re–initialization of the spindle amplifier.
SP1985	SERIAL SPINDLE CONTROL ERROR	Failed to automatically set parameters
SP1986	SERIAL SPINDLE CONTROL ERROR	An error occurred in the spindle control software.
SP1987	SERIAL SPINDLE CONTROL ERROR	Defective SIC–LSI on the CNC
SP1988	SPINDLE CONTROL ERROR	An error occurred in the spindle control software.
SP1989	SPINDLE CONTROL ERROR	An error occurred in the spindle control software.
SP1996	ILLEGAL SPINDLE PARAMETER SETTING	The spindle was assigned incorrectly. Check to see the following parameter. (No.3716 or 3717)
SP1998	SPINDLE CONTROL ERROR	An error occurred in the spindle control software.
SP1999	SPINDLE CONTROL ERROR	An error occurred in the spindle control software.

(10) Alarm list (serial spindle)

When a serial spindle alarm occurs, the following number is displayed on the CNC.

NOTE
*1 Note that the meanings of the SPM indications differ depending on which LED, the red or yellow LED, is on. When the red LED is on, the SPM indicates a 2-digit alarm number. When the yellow LED is on, the SPM indicates an error number that designates a sequence problem (for example, when a rotation command is entered with the emergency stop state not released).
See "Error Codes (Serial Spindle)."
*2 For serial spindle alarms with a number not listed, refer to the following documents depending on the spindle motor to which a connection is actually made.
- FANUC AC SPINDLE MOTOR αi series Maintenance Manual (B-65285JA)
- Technical Report etc.

Number	Message	SPM indication (*1)	Faulty location and remedy	Description
SP9001	SSPA:01 MOTOR OVERHEAT	01	1 Check and correct the peripheral temperature and load status. 2 If the cooling fan stops, replace it.	The thermostat embedded in the motor winding operated. The internal temperature of the motor exceeds the specified level. The motor is used in excess of the continuous rating, or the cooling component is abnormal.
SP9002	SSPA:02 EX DEVIATION SPEED	02	1 Check and correct the cutting conditions to decrease the load. 2 Correct parameter No. 4082.	The motor speed cannot follow a specified speed. An excessive motor load torque is detected. The acceleration/deceleration time in parameter No. 4082 is insufficient.
SP9003	SSPA:03 DC-LINK FUSE IS BROKEN	03	1 Replace the SPM unit. 2 Check the motor insulation status. 3 Replace the interface cable.	The PSM becomes ready (00 is indicated), but the DC link voltage is too low in the SPM. The fuse in the DC link section in the SPM is blown. (The power device is damaged or the motor is ground-fault.) The JX1A/JX1B connection cable is abnormal.
SP9004	SSPA:04 POWER SUPPLY ERROR	04	Check the state of the input power supply to the PSM.	The PSM found a missing power supply phase. (PSM alarm 5)
SP9006	THERMAL SENSOR DISCONNECT	06	1 Check and correct the parameter. 2 Replace the feedback cable.	The temperature sensor of the motor is disconnected.

Number	Message	SPM indication (*1)	Faulty location and remedy	Description
SP9007	SSPA:07 OVER SPEED	07	Check for a sequence error. (For example, check whether spindle synchronization was specified when the spindle could not be turned.)	The motor speed has exceeded 115% of its rated speed. When the spindle axis was in position control mode, positional deviations were accumulated excessively (SFR and SRV were turned off during spindle synchronization.)
SP9009	SSPA:09 OVERHEAT MAIN CIRCUIT	09	1 Improve the heat sink cooling status. 2 If the heat sink cooling fan stops, replace the SPM unit.	Abnormal temperature rise of the power transistor radiator
SP9010	SSPA:10 LOW VOLT INPUT POWER	09	1 Replace the cables. 2 Replace the spindle amplifier control printed circuit board.	The control power voltage of the spindle amplifier is reduced.
SP9011	SSPA:11 OVERVOLT POWER CIRCUIT	11	1 Check the selected PSM. 2 Check the input power voltage and change in power during motor deceleration. If the voltage exceeds 253 VAC (for the 200-V system) or 530 VAC (for the 400-V system), improve the power supply impedance.	Overvoltage of the DC link section of the PSM was detected. (PSM alarm indication: 7) PSM selection error. (The maximum output specification of the PSM is exceeded.)
SP9012	SSPA:12 OVERCURRENT POWER CIRCUIT	12	1 Check the motor insulation status. 2 Check the spindle parameters. 3 Replace the SPM unit.	The motor output current is abnormally high. A motor-specific parameter does not match the motor model. Poor motor insulation
SP9013	SSPA:13 CPU DATA MEMORY FAULT	13	Replace the SPM control printed circuit board.	Abnormality in an SPM control circuit component is detected. (RAM within the SPM is abnormal.)
SP9015	SSPA:15 SPINDLE SWITCHING FAULT	15	1 Check and correct the ladder sequence. 2 Replace the switching MC.	The switch sequence in spindle switch/output switch operation is abnormal. The switching MC contact status check signal and command do not match.
SP9016	SSPA:16 RAM ERROR	16	Replace the SPM control printed circuit board.	Abnormality in an SPM control circuit component is detected. (RAM for external data is abnormal.)
SP9018	SSPA:18 SUMCHECK ERROR PROGRAM ROM	18	Replace the SPM control printed circuit board.	Abnormality in an SPM control circuit component is detected. (Program ROM data is abnormal.)
SP9019	SSPA:19 EXCESS OFFSET CURRENT U	19	Replace the SPM unit.	Abnormality in an SPM component is detected. (The initial value for the U phase current detection circuit is abnormal.)

Number	Message	SPM indication (*1)	Faulty location and remedy	Description
SP9020	SSPA:20 EXCESS OFFSET CURRENT V	20	Replace the SPM unit.	Abnormality in an SPM component is detected. (The initial value of the V phase current detection circuit is abnormal.)
SP9021	POS SENSOR POLARITY ERROR	21	Check and correct the parameters. (No. 4000#0, 4001#4)	The polarity parameter setting of the position sensor is wrong.
SP9024	SSPA:24 SERIAL TRANSFER ERROR	24	1 Place the CNC-to-spindle cable away from the power cable. 2 Replace the cable.	The CNC power is turned off (normal power-off or broken cable). An error is detected in communication data transferred to the CNC.
SP9027	SSPA:27 DISCONNECT POSITION CODER	27	1 Replace the cable. 2 Re-adjust the BZ sensor signal.	1 The spindle position coder (connector JY4) signal is abnormal. 2 The signal amplitude (connector JY2) of the MZ or BZ sensor is abnormal. (Unconnected cable, adjustment error, etc.)
SP9029	SSPA:29 OVERLOAD	29	Check and correct the load status.	Excessive load has been applied continuously for a certain period of time. (This alarm is issued also when the motor shaft has been locked in the excitation state.)
SP9030	SSPA:30 OVERCURRENT INPUT CIRCUIT	30	Check and correct the power supply voltage.	Overcurrent is detected in PSM main circuit input. (PSM alarm indication: 1) Unbalanced power supply. PSM selection error (The maximum PSM output specification is exceeded.)
SP9031	SSPA:31 MOTOR LOCK OR DISCONNECT DETECTOR	31	1 Check and correct the load status. 2 Replace the motor sensor cable (JY2 or JY5).	The motor cannot rotate at a specified speed. (A level not exceeding the SST level for the rotation command has existed continuously.) Abnormality in the speed detection signal.
SP9032	SSPA:32 SIC-LSI RAM FAULT	32	Replace the SPM control printed circuit board.	Abnormality in an SPM control circuit component is detected. (The LSI device for serial transfer is abnormal.)
SP9033	SSPA:33 SHORTAGE POWER CHARGE	33	1 Check and correct the power supply voltage. 2 Replace the PSM unit.	Charging of direct current power supply voltage in the power circuit section is insufficient when the magnetic contractor in the amplifier is turned on (such as open phase and defective charging resistor).

Number	Message	SPM indication (*1)	Faulty location and remedy	Description
SP9034	SSPA:34 ILLEGAL PARAMETER	34	Correct a parameter value according to the manual. If the parameter number is unknown, connect the spindle check board, and check the indicated parameter.	Parameter data exceeding the allowable limit is set.
SP9036	SSPA:36 OVERFLOW ERROR COUNTER	36	Check whether the position gain value is too large, and correct the value.	An error counter overflow occurred.
SP9037	SSPA:37 ILLEGAL SETTING VELOCITY DETECTOR	37	Correct the value according to the parameter manual.	The setting of the parameter for the number of pulses in the speed detector is incorrect.
SP9041	SSPA:41 ILLEGAL 1REV SIGN OF POSITION CODER	41	1 Check and correct the parameter. 2 Replace the cable. 3 Re-adjust the BZ sensor signal. Re-adjust the BZ sensor signal.	1 The 1-rotation signal of the spindle position coder (connector JY4) is abnormal. 2 The 1-rotation signal (connector JY2) of the MZ or BZ sensor is abnormal. 3 Parameter setting error
SP9042	SSPA:42 NO 1REV SIGN OF POSITION CODER	42	1 Replace the cable. 2 Re-adjust the BZ sensor signal.	1 The 1-rotation signal of the spindle position coder (connector JY4) is disconnected. 2 The 1-rotation signal (connector JY2) of the MZ or BZ sensor is disconnected.
SP9043	SSPA:43 DISCONNECT POSITION CODER DEF. SPEED	43	Replace the cable.	The differential speed position coder signal (connector JY8) in SPM type 3 is abnormal.
SP9047	SSPA:47 ILLEGAL SIGNAL OF POSITION CODER	47	1 Replace the cable. 2 Re-adjust the BZ sensor signal. 3 Correct the cable layout (vicinity of the power line).	1 The A/B phase signal of the spindle position coder (connector JY4) is abnormal. 2 The A/B phase signal (connector JY2) of the MZ or BZ sensor is abnormal. The relationship between the A/B phase and 1-rotation signal is incorrect (Pulse interval mismatch).
SP9049	SSPA:49 DEF. SPEED IS OVER VALUE	49	Check whether the calculated differential speed value exceeds the maximum motor speed.	In differential speed mode, the speed of the other spindle converted to the speed of the local spindle has exceeded the allowable limit (the differential speed is calculated by multiplying the speed of the other spindle by the gear ratio).
SP9050	SSPA:50 SYNCRONOUS VALUE IS OVER SPEED	50	Check whether the calculated value exceeds the maximum motor speed.	In spindle synchronization, the speed command calculation value exceeded the allowable limit (the motor speed is calculated by multiplying the specified spindle speed by the gear ratio).

Number	Message	SPM indication (*1)	Faulty location and remedy	Description
SP9051	SSPA:51 LOW VOLT POWER CIRCUIT	51	1 Check and correct the power supply voltage. 2 Replace the MC.	Input voltage drop was detected. (PSM alarm indication: 4) (Momentary power failure or poor MC contact)
SP9052	SSPA:52 ITP FAULT 1	52	1 Replace the SPM control printed circuit board. 2 Replace the spindle interface printed circuit board in the CNC.	NC interface abnormality was detected (the ITP signal stopped).
SP9053	SSPA:53 ITP FAULT 2	53	1 Replace the SPM control printed circuit board. 2 Replace the spindle interface printed circuit board in the CNC.	NC interface abnormality was detected (the ITP signal stopped).
SP9054	SSPA:54 OVERCURRENT	54	Review the load state.	An overload current was detected.
SP9055	SSPA:55 ILLEGAL POWER LINE	55	1 Replace the magnetic contactor. 2 Check and correct the sequence.	The power line state signal of the magnetic contactor for selecting a spindle or output is abnormal.
SP9056	COOLING FAN FAILURE	56	Replace the SPM unit.	The cooling fan in the SPM control circuit stopped.
SP9057	CONV. EX. DECELERATION POW.	57	1 Decrease the acceleration/deceleration duty. 2 Check the cooling condition (peripheral temperature). 3 If the cooling fan stops, replace the resistor. 4 If the resistance is abnormal, replace the resistor.	An overload was detected in the regenerative resistance. (PSMR alarm indication: 8) Thermostat operation or short-time overload was detected. The regenerative resistor was disconnected, or an abnormal resistance was detected.
SP9058	CNV. OVERLOAD	58	1 Check the PSM cooling status. 2 Replace the PSM unit.	The temperature of the radiator of the PSM has increased abnormally. (PSM alarm indication: 3)
SP9059	CNV. COOLING FAN FAILURE	59	Replace the PSM unit.	The cooling fan in the PSM stopped. (PSM alarm indication: 2)
SP9061	SERIAL SPINDLE ALARM	61	Check parameter settings.	The error between the semi-closed and full-closed sides when the dual position feedback function is used is too large.
SP9065	SERIAL SPINDLE ALARM	65	1 Check parameter settings. 2 Check sensor connections and signals. 3 Check power line connections.	The move distance is too long when the magnetic pole is confirmed (synchronization spindle)
SP9066	COM. ERROR BETWEEN SP AMPS	66	1 Replace the cable. 2 Check and correct the connection.	An error was found in communication between amplifiers.

Number	Message	SPM indication (*1)	Faulty location and remedy	Description
SP9069	SAFETY SPEED OVER	69	1 Check the specified speed. 2 Check parameter settings. 3 Check the sequence.	In the state in which safety speed monitoring was enabled, the system detected that the motor speed exceeded the safety speed or detected an error during a free-run stop.
SP9070	ILLEGAL AXIS DATA	70	1 Check connections (JA7A of the second spindle requires a dedicated connector). 2 Replace the SPM control printed-circuit board.	An error was detected in an axis number check.
SP9071	SAFETY PARAMETER ERROR	71	Replace the SPM control printed-circuit board.	An error was detected in an axis parameter check.
SP9072	MISMATCH RESULT OF MOTOR SPEED CHECK	72	1 Replace the SPM control printed-circuit board. 2 Replace the spindle interface printed circuit board in the CNC.	A mismatch was detected between the safety speed check results of the SPM and those of the CNC.
SP9073	MOTOR SENSOR DISCONNECTED	73	1 Replace the feedback cable. 2 Check the shield processing. 3 Check and correct the connection. 4 Adjust the sensor.	The motor sensor feedback signal is not present.
SP9074	CPU TEST ERROR	74	Replace the SPM control printed-circuit board.	An error was detected in a CPU test.
SP9075	CRC ERROR	75	Replace the spindle amplifier control printed circuit board.	An error occurred in a spindle amplifier CRC test.
SP9076	INEXECUTION OF SAFETY FUNCTIONS	76	Replace the SPM control printed-circuit board.	The SPM detected that safety functions were not executed.
SP9077	MISMATCH RESULT OF AXIS NUMBER CHECK	77	1 Replace the SPM control printed-circuit board. 2 Replace the spindle interface printed circuit board in the CNC.	A mismatch was detected between the axis number check results of the SPM and those of the CNC.
SP9078	MISMATCH RESULT OF SAFETY PARAMETER CHECK	78	1 Replace the SPM control printed-circuit board. 2 Replace the spindle interface printed circuit board in the CNC.	A mismatch was detected between the safety parameter check results of the SPM and those of CNC.
SP9079	INITIAL TEST ERROR	79	Replace the spindle amplifier control printed circuit board.	An error was detected in a spindle amplifier initial test.
SP9080	ALARM AT THE OTHER SP AMP.	80	Remove the cause of the alarm of the remote SPM.	During inter-SPM communication, an alarm was generated on the remote SPM.
SP9081	1-ROT MOTOR SENSOR ERROR	81	1 Check and correct the parameter. 2 Replace the feedback cable. 3 Adjust the sensor.	The one-rotation signal of the motor sensor cannot be correctly detected.
SP9082	NO 1-ROT MOTOR SENSOR	82	1 Replace the feedback cable. 2 Adjust the sensor.	The one-rotation signal of the motor sensor is not generated.

Number	Message	SPM indication (*1)	Faulty location and remedy	Description
SP9083	MOTOR SENSOR SIGNAL ERROR	83	1 Replace the feedback cable. 2 Adjust the sensor.	An irregularity was detected in a motor sensor feedback signal.
SP9084	SPNDL SENSOR DISCONNECTED	84	1 Replace the feedback cable. 2 Check the shield processing. 3 Check and correct the connection. 4 Check and correct the parameter. 5 Adjust the sensor.	The spindle sensor feedback signal is not present.
SP9085	1-ROT SPNDL SENSOR ERROR	85	1 Check and correct the parameter. 2 Replace the feedback cable. 3 Adjust the sensor.	The one-rotation signal of the spindle sensor cannot be correctly detected.
SP9086	NO 1-ROT SPNDL SENSOR	86	1 Replace the feedback cable. 2 Adjust the sensor.	The one-rotation signal of the spindle sensor is not generated.
SP9087	SPNDL SENSOR SIGNAL ERROR	87	1 Replace the feedback cable. 2 Adjust the sensor.	An irregularity was detected in a spindle sensor feedback signal.
SP9088	COOLING RADI FAN FAILURE	88	Replace the SPM external cooling fan.	The external cooling fan stopped.
SP9089	COOLING RADI FAN FAILURE	89	1 Check the connection between the SPM and the submodule SM (SSM). 2 Replace the submodule SM(SSM). 3 Replace the SPM control printed-circuit board.	Submodule SM (SSM) error (synchronous spindle)
SP9110	AMP COMMUNICATION ERROR	b0	1 Replace the communication cable between amplifier and module. 2 Replace the SPM or PSM control printed circuit board.	Communication error between amplifier and module
SP9111	CONV. LOW VOLT CONTROL	b1	Replace the PSM control printed circuit board.	Low converter control power supply voltage (PSM indication = 6)
SP9112	CONV. EX. DISCHARGE POW.	b2	1 Check the regenerative resistance. 2 Check the motor selection. 3 Replace the PSM	Excessive converter regenerative power (PSM indication = 8)
SP9113	CONV. COOLING FAN FAILURE	b3	Replace the cooling fan.	Stopped cooling fan of the converter radiator (PSM indication = A)
SP9120	COMMUNICATION DATA ERROR	C0	1 Replace the communication cable between CNC and SPM. 2 Replace the SPM control printed circuit board. 3 Replace the CNC side spindle interface printed circuit board.	Communication data alarm

Number	Message	SPM indication (*1)	Faulty location and remedy	Description
SP9121	COMMUNICATION DATA ERROR	C1	1 Replace the communication cable between CNC and SPM. 2 Replace the SPM control printed circuit board. 3 Replace the CNC side spindle interface printed circuit board.	Communication data alarm
SP9122	COMMUNICATION DATA ERROR	C2	1 Replace the communication cable between CNC and SPM. 2 Replace the SPM control printed circuit board. 3 Replace the CNC side spindle interface printed circuit board.	Communication data alarm
SP9123	SERIAL SPINDLE ALARM	C3	Replace the submodule SW(SSW).	Submodule SW (SSW) error (spindle switching)
SP9135	SAFETY SPEED ZERO ERROR(SP)	D5	Perform operation within the safety speed zero range.	The motor position exceeded the safety speed zero monitoring width.
SP9136	MISMATCH RESULT OF SAFETY SPEED ZERO CHECK(SP)	D6	Replace the spindle amplifier control printed circuit board.	The spindle amplifier speed zero determination result did not match the CNC speed zero determination result.

Error codes (serial spindle)

NOTE
*1 Note that the meanings of the SPM indications differ depending on which LED, the red or yellow LED, is on. When the yellow LED is on, an error code is indicated with a 2-digit number. An error code is indicated in the CNC diagnosis, No.712. When the red LED is on, the SPM indicates the number of an alarm generated in the serial spindle.
→ See “(10) Serial spindle alarms (SP alarm).”

SPM indication (*1)	Faulty location and remedy	Description
01	Although neither *ESP (emergency stop signal; there are two types of signals including the input signal and PSM contact signal) nor MRDY (machine ready signal) is input, SFR (forward rotation signal)/SRF (reverse rotation signal)/ORCM (orientation command) is input.	Check the *ESP and MRDY sequence. For MRDY, pay attention to the parameter setting regarding the use of the MRDY signal (parameter No. 4001#0).
03	The parameter settings are such that a position sensor is not used (position control not performed) (bits 3, 2, 1, 0 of parameter No. 4002 = 0, 0, 0, 0), but a Cs contour control command is input. In this case, the motor is not excited.	Check the parameter settings.
04	The parameter settings are such that a position sensor is not used (position control not performed) (bits 3, 2, 1, 0 of parameter No.4002 = 0, 0, 0,), but a servo mode (rigid tapping, spindle positioning, etc.) or spindle synchronization command is input. In this case, the motor is not excited.	Check the parameter settings.
05	The orientation function option parameter is not specified, but ORCM (orientation command) is input.	Check the orientation function parameter settings.
06	The output switching control function option parameter is not specified, but low-speed characteristic winding is selected (RCH = 1).	Check the output switching control function parameter settings and the power line state check signal (RCH).
07	A Cs contour control command is input, but SFR (clockwise rotation command)/SRV (counterclockwise rotation command) is not input.	Check the sequence.
08	A servo mode (rigid tapping, spindle positioning, etc.) control command is input, but SFR (clockwise rotation command)/SRV (counterclockwise rotation command) is not input.	Check the sequence.
09	A spindle synchronization command is input, but SFR (clockwise rotation command)/SRV (counterclockwise rotation command) is not input.	Check the sequence.
10	A Cs contour control command is input, but another mode (servo mode, spindle synchronization, or orientation) is specified.	Do not switch to another mode during a Cs contour control command. Before moving to another mode, cancel the Cs contour control command.

SPM indication (*1)	Faulty location and remedy	Description
11	A servo mode (rigid tapping, spindle positioning, etc.) command is input, but another mode (Cs contour control, spindle synchronization, or orientation) is specified.	Do not switch to another mode during a servo mode command. Before moving to another mode, cancel the servo mode command.
12	A spindle synchronization command is input, but another mode (Cs contour control, servo mode, or orientation) is specified.	Do not switch to another mode during a spindle synchronization command. Before moving to another mode, cancel the spindle synchronization command.
14	Both SFR (clockwise rotation command) and SRV (counterclockwise rotation command) are input at the same time.	Issue either of them.
17	The speed detector parameter settings (bits 2, 1, and 0 of parameter No. 4011) are not valid. □There is no corresponding speed detector.	Check the parameter settings.
18	The parameter settings are such that a position sensor is not used (position control not performed (bits 3, 2, 1, and 0 of parameter No. 4002), but position coder system orientation is issued.	Check the parameter settings and the input signal.
24	If index is performed continuously in position coder system orientation, an incremental operation is performed first (INCMD = 1), then an absolute position command (INCMD = 0) is input.	Check INCMD (incremental command). If an absolute position command is to follow, be sure to perform absolute position command orientation first.
29	The parameter settings are such that the shortest-time orientation function is used (bit 6 of parameter No. 4018 = 0, Nos. 4320 to 4323 ≠0).	In the α*i* series spindle amplifier, the shortest-time orientation function cannot be used. Use normal-system orientation.
31	The hardware configuration is such that the spindle FAD function cannot be used. In this case, the motor is not activated.	Check the CNC model.
33	The hardware configuration is such that the spindle EGB function cannot be used. In this case, the motor is not activated.	Check the CNC model.
34	Both the spindle FAD function and the spindle EGB function are enabled. In this case, the motor is not activated.	The two functions cannot be used at the same time. Enable either function only.
34	The submodule SM (SSM) is faulty or the connection between SPM and SSM is in error.	Submodule SM (SSM) error (synchronous spindle)

NOTE
*2 PSM contact signal
Between ESP1 and ESP2 on the PSM
Contact open: Emergency stop
Contact closed: Normal operation

(11) Overheat alarms (OH alarm)

Number	Message	Description
OH0700	LOCKER OVERHEAT	CNC cabinet overheat
OH0701	FAN MOTOR STOP	PCB cooling fan motor abnormality

Number	Message	Description
OH0704	OVERHEAT	Spindle overheat due to detection of changes in the spindle speed - When the cutting load is large, offload the cutting conditions. - Check if the cutting tool became dull. - Check if the spindle amplifier malfunctions.

(12) Other alarms (DS alarm)

Number	Message	Description
DS0001	SYNC EXCESS ERROR (POS DEV)	In feed axis synchronization control, the difference in the amount of positional deviation between the master and slave axes exceeded the parameter (No. 8323) setting value. This alarm occurs only for the slave axis.
DS0002	SYNC EXCESS ERROR ALARM 1	In feed axis synchronization control, the difference in the amount of synchronization between the master and slave axes exceeded the parameter (No. 8331) setting value. This alarm occurs only for the slave axis.
DS0003	SYNCHRONIZE ADJUST MODE	The system is in the synchronize adjust mode.
DS0004	EXCESS MAXIMUM FEEDRATE	The malfunction prevention function detected the command in which a value exceeding the maximum speed was specified.
DS0005	EXCESS MAXIMUM ACCELERATION	The malfunction prevention function detected the command in which a value exceeding the maximum acceleration was specified.
DS0006	ILLEGAL EXECUTION SEQUENCE	The malfunction prevention function detected an illegal execution sequence.
DS0007	ILLEGAL EXECUTION SEQUENCE	The malfunction prevention function detected an illegal execution sequence.
DS0008	ILLEGAL EXECUTION SEQUENCE	The malfunction prevention function detected an illegal execution sequence.
DS0009	ILLEGAL EXECUTION SEQUENCE	The malfunction prevention function detected an illegal execution sequence.
DS0010	ILLEGAL REFERENCE AREA	The malfunction prevention function detected an invalid reference area.
DS0011	ILLEGAL REFERENCE AREA	The malfunction prevention function detected an invalid reference area.
DS0012	ILLEGAL REFERENCE AREA	The malfunction prevention function detected an invalid reference area.
DS0013	ILLEGAL REFERENCE AREA	The malfunction prevention function detected an invalid reference area.
DS0014	TOOL CHANGE DETECT MACHINE LOCK	A machine lock is turned on for the Z axis for which the tool is being changed.
DS0015	TOOL CHANGE DETECT MIRROR IMAGE	A mirror image is turned on for the Z axis for which the tool is being changed.
DS0016	SERIAL DCL:FOLLOW-UP ERROR	(1) The settings of parameters No.1883 and No.1884 fall outside the range. (2) The current position at establishment of the origin subtracted by the distance between the reference positions (detection unit) exceeded ±2147483647. Change the current position or reference position to prevent this situation.

Number	Message	Description
DS0017	SERIAL DCL:REF-POS ESTABLISH ERR	The travel amount at the FL speed at establishment of the origin exceeded the setting of parameter No. 14010.
DS0018	SERIAL DCL:MISMATCH(SSYNC CTRL)	Of the master and slave axes for feed axis synchronous control, one axis is a linear scale with the origin and the other is not a linear scale with the origin. In such a configuration, the feed axis synchronous control selection signal (SYNC<Gn138> or SYNCJ <Gn140>) needs to be set to 0 to establish the origin.
DS0020	REFERENCE RETURN INCOMPLETE	An attempt was made to perform an automatic return to the reference position on the perpendicular axis before the completion of a return to the reference position on the angular axis. However, this attempt failed because a manual return to the reference position during angular axis control or an automatic return to the reference position after power-up was not commanded. First, return to the reference position on the angular axis, then return to the reference position on the perpendicular axis.
DS0022	DUAL CHECK SAFETY IS NOT WORKED	Bit 6 (DCE) of parameter No. 1902 has set the Dual Check Safety function to be disabled.
DS0023	ILLEGAL PARAMETER (I-COMP VAL)	The setting of the inclination compensation parameter is incorrect. The compensation per compensation point is too large or too small.
DS0024	UINT SIGNAL WAS ILLEGALLY INPUT	An interruption custom macro was started during movement to the machining restart position at the dry run speed.
DS0025	G60 CANNOT BE EXECUTED	The state of a mirror image is different between the time when look-ahead of a block for unidirectional positioning was performed and the time when execution of the block was started, so unidirectional positioning cannot be performed. Modify the program.
DS0026	MISMATCH OF ANGULAR AXIS(D.C.S)	On angular axis control, one of the angular/perpendicular axes is the scale with ref-pos, and the other of them is not the scale with ref-pos. Such system is not admired.
DS0027	MISMATCH OF SYNCHRONOUS AXIS(D.C.S)	Master/slave axes of axis synchronous control, one of them is the linear scale with distance-coded reference marks, and the other of them is not the linear scale with distance-coded reference marks. Please establish reference position with the input signal SYNCn<G138>, SYNCJn<G140> or parameter setting to 0.
DS0029	UNAVAILABLE ROT AXIS MOVE IN TPC	In tool posture control, the rotation axis fell outside the set operation range (parameters No. 19741 to No. 19744). Review the machine configuration and command.
DS0030	TOOL DIRC CMP IMPOSSIBLE	When tool center point control is performed in the angular surface machining command mode or workpiece placement error compensation is performed in a 5-axis cutting machine, if settings are made to make compensation in the tool direction (bit 0 (RCM) of parameter No. 11200 is 1), the position of the rotation axis could not be obtained.

Number	Message	Description
DS0059	SPECIFIED NUMBER NOT FOUND	[External data I/O] The No. specified for a program No. or sequence No. search could not be found. There was an I/O request issued for a pot No. or offset (tool data), but either no tool numbers have been input since power ON or there is no data for the entered tool No. [External workpiece No. search] The program corresponding to the specified workpiece No. could not be found.
DS0131	TOO MANY MESSAGE	An attempt was made to display an external operator message or external alarm message, but five or more displays were required simultaneously.
DS0132	MESSAGE NUMBER NOT FOUND	An attempt to cancel an external operator message or external alarm message failed because the specified message number was not found.
DS0133	TOO LARGE NUMBER	A value other than 0 to 4095 was specified as the external operator message or the external alarm message number.
DS0300	APC ALARM: NEED REF RETURN	A setting to zero position for the absolute position detector (association with reference position and the counter value of the absolute position detector) is required. Perform the return to the reference position. This alarm may occur with other alarms simultaneously. In this case, other alarms must be handled first.
DS0306	APC ALARM: BATTERY VOLTAGE 0	The battery voltage of the absolute position detector has dropped to a level at which data can no longer be held. Or, the power was supplied to the Pulsecoder for the first time. The battery or cable is thought to be defective. Replace the battery with the machine turned on.
DS0307	APC ALARM: BATTERY LOW 1	The battery voltage of the absolute position detector has dropped to a level at which a replacement is required. Replace the battery with the machine turned on.
DS0308	APC ALARM: BATTERY LOW 2	The battery voltage of the absolute position detector dropped to a level at which a replacement was required in the past. (including during power off) Replace the battery with the machine turned on.
DS0309	APC ALARM: REF RETURN IMPOSSIBLE	An attempt was made to set the zero point for the absolute position detector by MDI operation when it was impossible to set the zero point. Rotate the motor manually at least one turn, and set the zero position of the absolute position detector after turning the CNC and servo amplifier off and then on again.
DS0310	NOT ON RETURN POINT	The return position recorded during retraction is not reached during recovery. The position may be displaced during recovery due to a machine lock or mirror image. Perform the operation again after making a reset.
DS0405	ZERO RETURN END NOT ON REF	The axis specified in automatic zero return was not at the correct zero point when positioning was completed. Perform zero return from a point whose distance from the zero return start position to the zero point is 2 or more revolutions of the motor. Other probable causes are: - The positional deviation after triggering the deceleration dog is less than 128. - Insufficient voltage or malfunctioning Pulsecoder.

Number	Message	Description
DS1120	UNASSIGNED ADDRESS (HIGH)	The upper 4 bits (EIA4 to EIA7) of an external data I/O interface address signal are set to an undefined address (high bits).
DS1121	UNASSIGNED ADDRESS (LOW)	The lower 4 bits (EIA0 to EIA3) of an external data I/O interface address signal are set to an undefined address (low bits).
DS1124	OUTPUT REQUEST ERROR	OUTPUT REQUEST ERROR An output request was issued during external data output, or an output request was issued for an address that has no output data.
DS1128	DI.EIDLL OUT OF RANGE	The numerical value input by external data input signals ED0 to ED31 has exceeded the permissible range.
DS1130	SEARCH REQUEST NOT ACCEPTED	No requests can be accepted for a program No. or a sequence No. search as the system is not in the memory mode or the reset state.
DS1131	EXT-DATA ERROR (OTHER)	[External Data I/O] An attempt was made to input tool data for tool offset by a tool No. during loading by the G10 code.
DS1150	A/D CONVERT ALARM	A/D converter malfunction
DS1184	PARAMETER ERROR IN TORQUE	An invalid parameter was set for torque control. The torque constant parameter is set to “0”.
DS1185	OVER MAXIMUM FEED	The maximum cutting feedrate or rapid traverse feedrate was exceeded in G54.3.
DS1448	ILLEGAL PARAMETER (D.C.S.)	The setting value of parameter for reference marks is satisfied the following any conditions. - The absolute-position detection function is enabled. - Either parameter 1821 (mark-1 interval) or parameter 1882 (mark-2 interval) is set to 0. - Parameters 1821 and 1882 have identical settings. - The difference between the settings made for parameters 1821 and 1882 is greater than or equal to twice either setting. - The setting value of parameters 1883 and 1884 are over the valid data range.
DS1449	REFERENCE MARK ARE DIFFERENT FROM PARAMETER	In case of distance coded linear scale I/F, the actual interval of reference marks is different from parameter (No.1821,1882) setting value.
DS1450	ZERO RETURN NOT FINISHED	1st reference position return (CDxX7 to CDxX0: 17h (Hex)) was specified when the manual reference position return was not executed with the reference position return function enabled (parameter ZRN (No. 1005#0) set to “0”).
DS1451	IMPROPER PMC AXIS COMMAND	The PMC axes cannot be controlled in this state.
DS1512	EXCESS VELOCITY	The feedrate of the linear axis during polar coordinate interpolation exceeded the maximum cutting feedrate.
DS1514	ILLEGAL MOTION IN G12.1 MODE	In a hypothetical axis direction compensation during the polar coordinate interpolation mode, an attempt is made to travel to the area in which the travel cannot be made.
DS1553	EXCESS VELOCITY IN G43.4/G43.5	The axis rate was attempt to exceed the maximum cutting feedrate and travel by the pivot tool length compensation.

Number	Message	Description
DS1710	ILLEGAL ACC. PARAMETER (OPTIMUM TORQUE ACC/DEC)	There are errors in the parameters of permissible acceleration for Optimum Torque Acceleration/Deceleration. One of the following is the cause. 1) The ratio of the acceleration for deceleration to the acceleration for the acceleration is lower than the limited value. 2) The time to decelerate to 0 is larger than the maximum.
DS1931	MACHINE PARAMETER INCORRECT	One of parameters Nos. 19665 to 19667 and Nos.19680 to 19744 used to configure the machine contains an error.
DS1932	DI.THML SIGNAL ON	One of the parameters used to configure the machine is rewritten while the tool direction thermal displacement compensation function is enabled.
DS1933	NEED REF RETURN(SYNC:MIX:OVL)	The relation between a machine coordinate of an axis in synchronization, composition, or superposition control, and the absolute, or relative coordinate was displaced. Perform the manual return to the reference position.

(13) Malfunction prevention function alarms (IE alarm)

Number	Message	Description
IE0001	+ OVERTRAVEL (SOFT 1)	The malfunction prevention function detected that stored stroke check 1 on the positive side was exceeded.
IE0002	- OVERTRAVEL (SOFT 1)	The malfunction prevention function detected that stored stroke check 1 on the negative side was exceeded.
IE0003	+ OVERTRAVEL (SOFT 2)	The malfunction prevention function detected that stored stroke check 2 on the positive side was exceeded.
IE0004	- OVERTRAVEL (SOFT 2)	The malfunction prevention function detected that stored stroke check 2 on the negative side was exceeded.
IE0005	+ OVERTRAVEL (SOFT 3)	The malfunction prevention function detected that stored stroke check 3 on the positive side was exceeded.
IE0006	- OVERTRAVEL (SOFT 3)	The malfunction prevention function detected that stored stroke check 3 on the negative side was exceeded.
IE0007	EXCESS MAXIMUM REV. DATA	The malfunction prevention function detected the command in which a value exceeding the maximum speed was specified.
IE0008	ILLEGAL ACC/DEC	The malfunction prevention function detected the acceleration/deceleration error.
IE0009	ILLEGAL MCN COODINATE	The malfunction prevention function detected the displacement of a machine coordinate in the check point.

PC TOOL FOR MEMORY CARD PROGRAM OPERATION/EDITING

Appendix I, "PC TOOL FOR MEMORY CARD PROGRAM OPERATION/EDITING", consists of the following sections:

I.1 PC TOOL FOR MEMORY CARD PROGRAM OPERATION/EDITING.....2390
I.2 NAMING RULES.....2401
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I.1 PC TOOL FOR MEMORY CARD PROGRAM OPERATION/EDITING

Overview

By using this PC tool, you can make the memory card program file ("FANUCPRG.BIN") which is needed for the function "Memory Card Program Operation/Editing".
The maximum size of the memory card program file is 2048 Mbytes (2 Gbytes). The "Memory Card Program Operation/Editing" function needs the memory card which has the memory card program file on the FAT formatted memory card.
This PC tool can be operated on a PC in the marketplace and with following OS :

- Windows(R)NT4.0 Workstation (SP5 or later)
- Windows(R)2000 Professional

And acceptable specification is as followings :

- Memory : 32Mbytes or more
- Hard disk : 10Mbytes or more free space and additional space for the memory card program file

I.1.1 Usage Notes

Before using this PC tool, please make sure there is no [temp] folder on the same place of this PC tool.
[temp] folder is created and used by this PC tool as work folder.
If [temp] folder is created, please do not access this folder.
[temp] folder and the files located in this folder will be deleted by this PC tool.

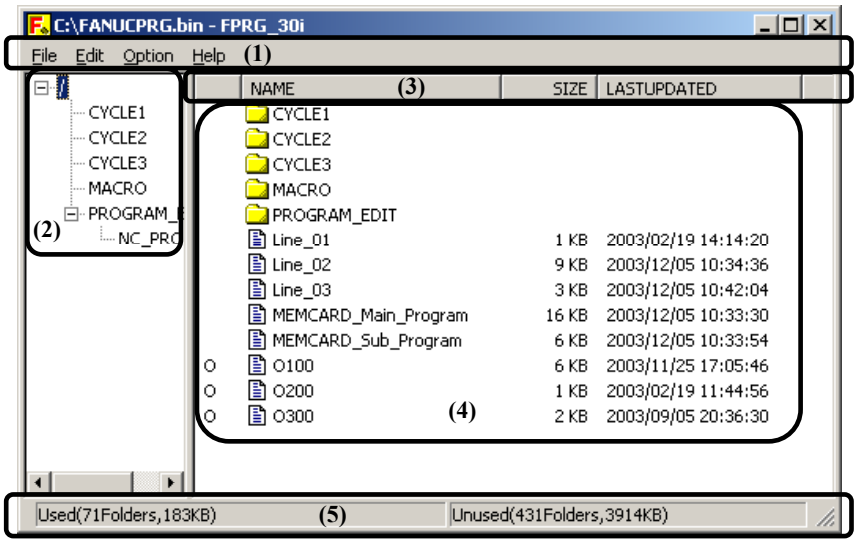
I.1.2 List of Functions of PC Tool

- Browsing the folders of the memory card program file
- Adding a program to the memory card program file by dropping the program into this PC tool from Explorer etc. (hereinafter referred to as "drop in")
- Extracting a program in the memory card program file as a text file on Windows file system by dropping out from this PC tool into Explorer etc. (hereinafter referred to as "drop out")
- Renaming a program in the memory card program file
- Deleting a program in the memory card program file
- Creating a new folder into the memory card program file
- Renaming a folder in the memory card program file
- Deleting a folder in the memory card program file
- Display of free space on the memory card program file
- Sorting list view of the memory card program file

I.1.3 Explanation Of Operations

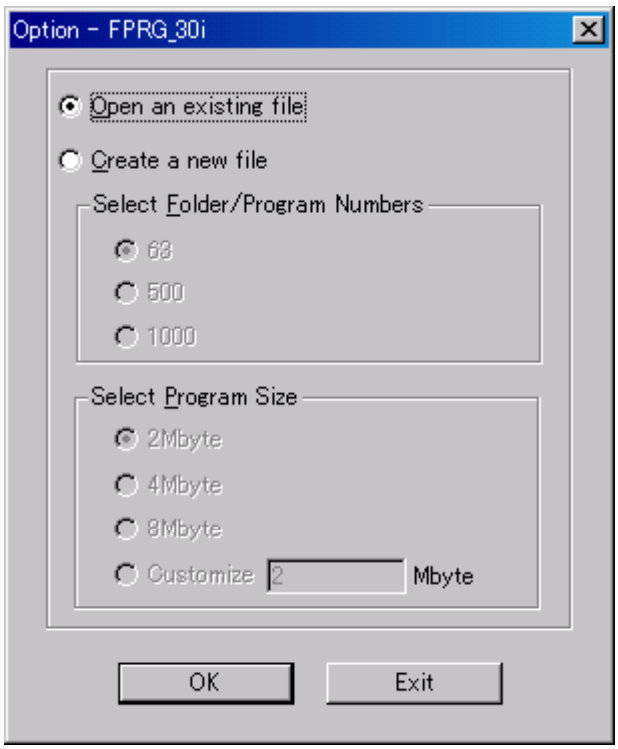
- Outline of screen

1) Menu bar : The menu of this PC tool is displayed.
2) Tree view : Browsing the folders of the memory card program file.
3) Column : Attributes of each file or folder in the memory card program file.
4) List view : Contents of selected folder are displayed.
5) Status bar : Used and Unused space in the memory card program file are displayed.

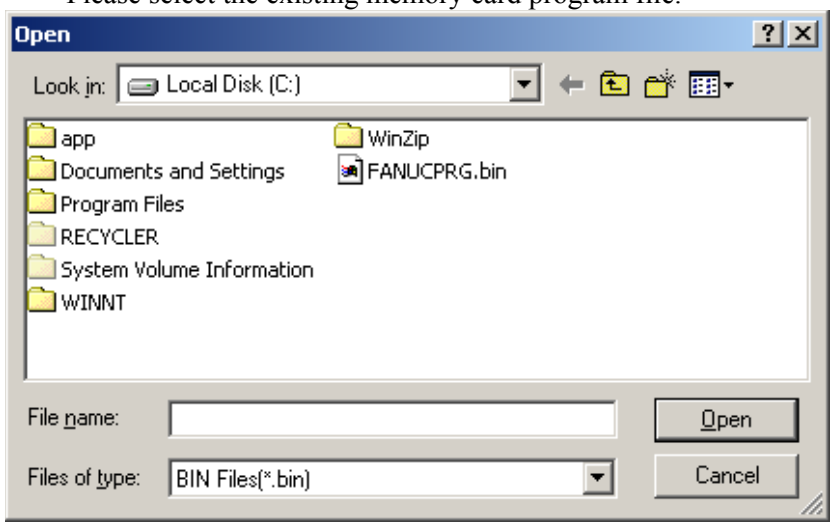


- Initial Option dialogue window

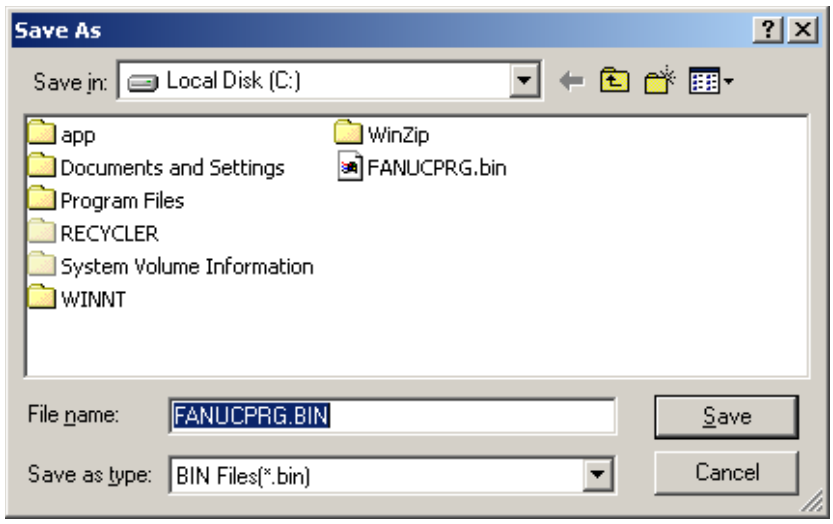
When this PC tool starts up, the Option dialogue window is displayed. Please select "Open an existing file" or "Create a new file".



- **When "Open an existing file" is selected**
  After OK button pushed, "Open" dialogue window is displayed.
  Please select the existing memory card program file.



- **When "Create a new file" is selected**
  After OK button pushed, "Save As" dialogue window is displayed.
  Please create a new memory card program file on the selected folder.



When the new the memory card program file is created, the following items need to be selected:

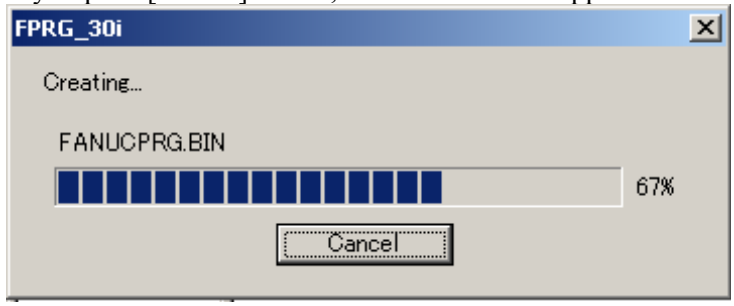
- Folder/Program Numbers
- Program Size

"Folder/Program Numbers" can be selected among 63 / 500 / 1000. The default value is 63.

"Program Size" can be selected among 2Mbyte, 4Mbyte, 8Mbyte, and Customize. The default value is 2Mbyte.

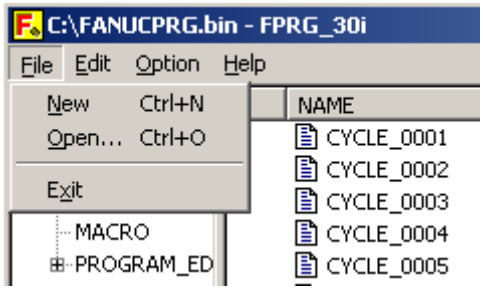
NOTE
1 When "Customize" size is selected, it is available to cover the range from 2Mbyte to 2048Mbyte.
2 Though the maximum size is 2048Mbyte, a slight loss exists by system use.
3 The number of "Folders" in the status bar includes number of program file and folder.

During creating of the memory card program file, the progress bar is being displayed. This progress bar is also displayed during executing of Drop-in and Drop-out.
If you push [Cancel] button, the execution is stopped.



- Menu

File menu

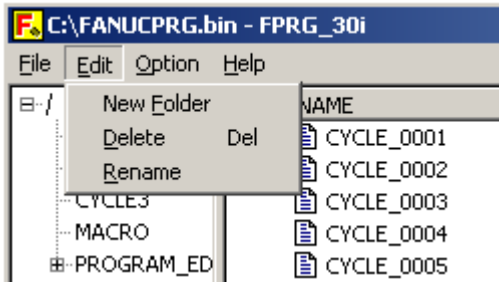


[New]
Create a new memory card program file.

[Open...]
Open the existing memory card program file.

[Exit]
Terminate this PC tool.

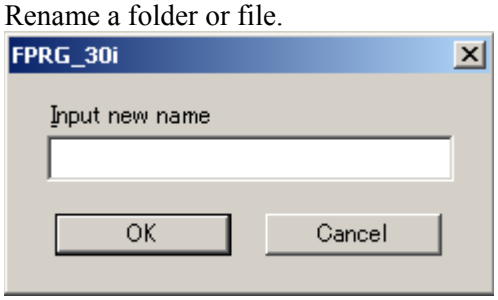
Edit menu



[New Folder]
Create new folder. It is available during Tree view selected.
Up to seven hierarchical levels starting from the user root folder (/USER) are permitted.
/USER/PATH1/Aaa/Bbb/Ccc/Ddd/O123
1 2 3 4 5 6 7(not folder)

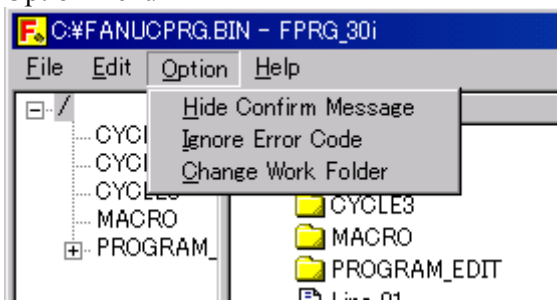
[Delete]
Delete program files or folders.
If you delete a folder, all folders and program files in the folder will be deleted.

[Rename]
Rename a folder or file.



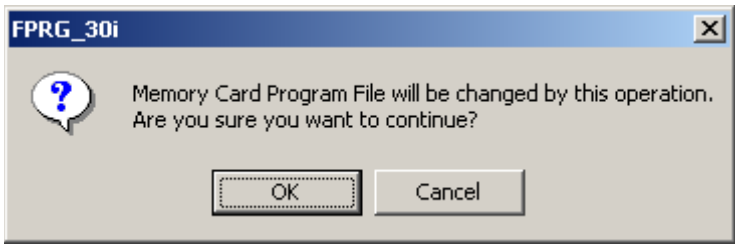
NOTE
For naming folder and program file, characters which can be used are limited. Please refer to "Naming rules".

Option menu



[Hide Confirm Message]

When the following operations are executed, the following Confirm Message is popped up before the memory card program file is modified.



1. Delete folder or program file
2. Rename folder or program file
3. Drop-in program file
4. Add folder

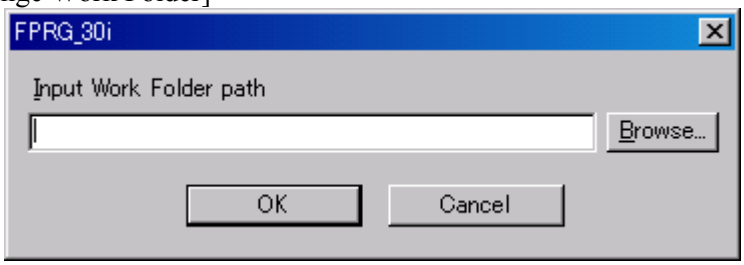
If the [OK] button is pushed, the operation advances.
If the [Cancel] button is pushed, the operation is abandoned.

When the [Hide Confirm Message] in menu bar is checked, the Confirm Message is not displayed and the operation advances at once. The default setting is the Confirm Message displayed.

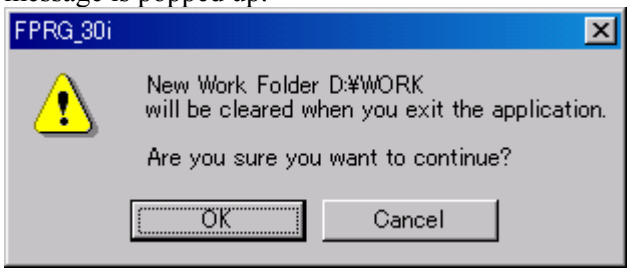
[Ignore Error Code]

When the [Ignore Error Code] in menu bar is checked, program file which includes unusable characters can be dropped in. The unusable characters in the program file will be ignored and will not be written into the memory card program file.

[Change Work Folder]

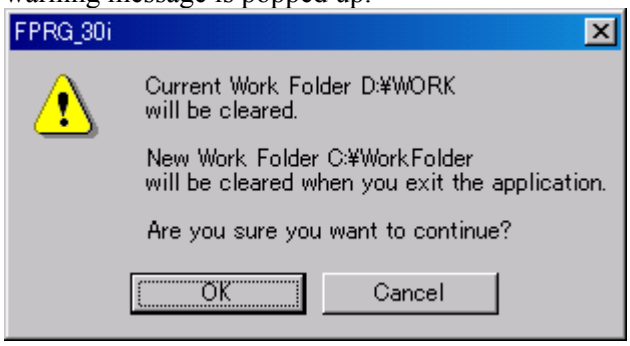


Work folder is used for temporarily keeping the dropped out files. If work folder has no enough free space, Drop-out will not be executed. To avoid this, you can check this option and change the work folder to an area where there is enough free space. With default setting, work folder [temp] will be created on the same place of this PC tool "FPRG_30i.exe".
If you changed the default work folder, the following warning message is popped up.



If the [OK] button is pushed, the operation advances.
If the [Cancel] button is pushed, the operation is abandoned.

If the work folder has been changed before, the following warning message is popped up.

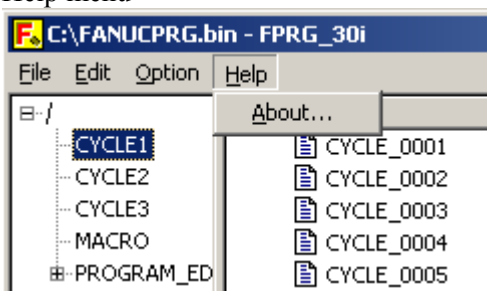


If the [OK] button is pushed, the operation advances.
If the [Cancel] button is pushed, the operation is abandoned.

⚠ WARNING

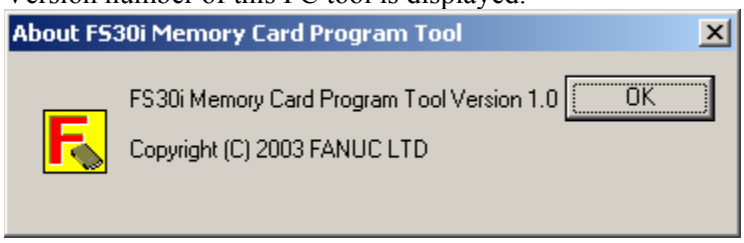
1 At this PC tool terminated, all files in the work folder are deleted.
2 During this PC tool executing, do not access the files in the work folder. If the access is done, there is no assurance of normal operation.

Help menu



[About...]

Version number of this PC tool is displayed.



- Mouse Operation

[Drop-in and Drop-out]

- Drop-in from Explorer
  NC program can be added by dropping files including the NC files into the List view window of this PC tool from Explorer

  NC program name and update time is the same as the Dropped-in files.

  If "Oxxxxxxxx" or "<xxxx>" exists at the top of files, These "Oxxxxxxxx" and "<xxxx>" become NC program name. If not exist, each file name becomes NC program name.

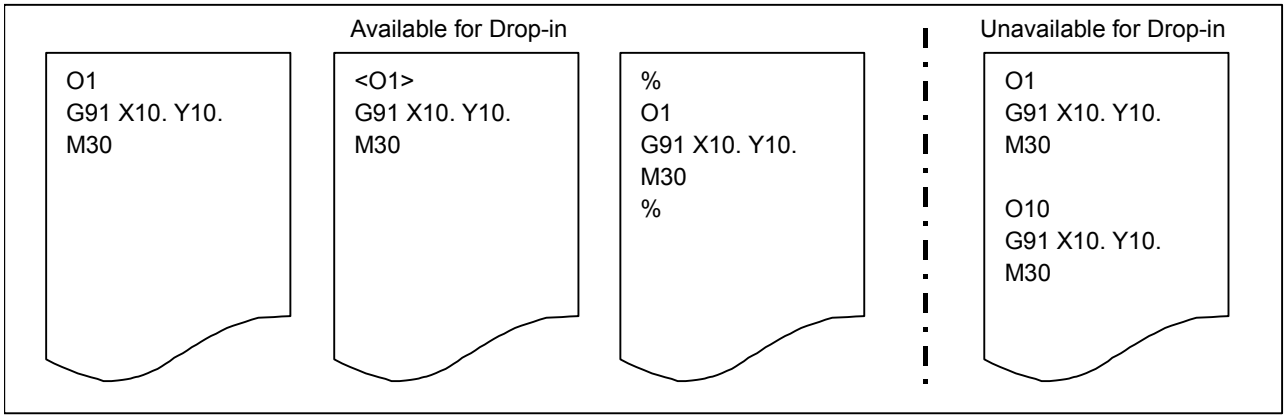
Examples			
Name of external file	**Top of external file**	**Name of internal file**	**Program number**
O1234	N10G00	O1234	1234
O123N10G00	N10G00	O123N10G00	Not an O number program
test.txt	O1234N10G00	O1234	1234
test.txt	<O1234>	O1234	1234
test.txt	<O1234N10>	O1234N10	Not an O number program
O1234	<O1234N10>	O1234N10	Not an O number program
O001234	N10G00	O1234	1234
O001234N10G00	N10G00	O001234N10G00	Not an O number program
test.txt	O001234	O1234	1234
test.txt	<O001234>	O1234	1234
test.txt	<O001234N10G00>	O001234N10G00	Not an O number program

NOTE
1 For naming program file, please refer to the following chapter "Naming rules of Program file".
2 For usable characters in Program file, please refer to the following chapter "Rules of characters in Program file".
3 The updating time of program file is available from 1997 to 2037.

This PC tool can check contents of dropped-in program file according to "Rules of characters in Program file". However, this PC tool does not check grammar of NC program.

The program file can only have one NC program. Therefore, the NC program output from CNC cannot be dropped in directly.

Example of Program



NOTE
1 If the same named program file exists, another same named program file cannot be dropped in.
2 If free space on the memory card program file is lacking, new program file cannot be dropped in.
3 If the NC program name is not compliant with "Naming rules of Program file", the program file can not be dropped in.

If the File name acts as Program number, "O" is displayed on the first row of list view.

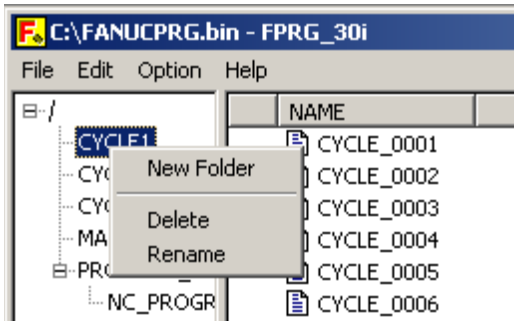
- Drop-out from List view window
  Drop-out from list view of this PC tool to Explorer is available.

NOTE
Do not drop out to Work folder. If dropped out to Work folder, this PC tool cannot continue to function normally.

- Pop-up menu

Pop-up menu is displayed by clicking the right mouse button.

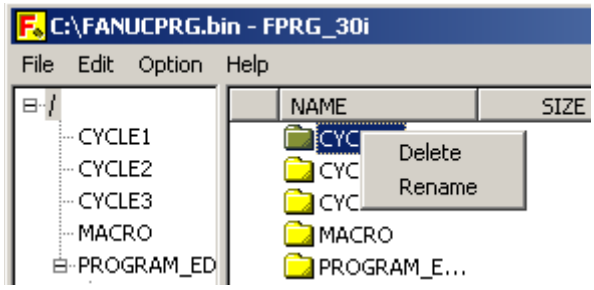
- Focus on Tree view



  Clicking "New Folder", a new folder is created on selected folder.
  Clicking "Delete", the selected folder is deleted.
  Clicking "Rename", the selected folder is renamed.

If clicking on root folder, "Delete" and "Rename" are not activated.

- Focus on List view



- Clicking "Delete", the selected folder or program file is deleted.
- Clicking "Rename", the selected folder or program file is renamed.

- Display of free space on the memory card program file ("FANUCPRG.BIN")

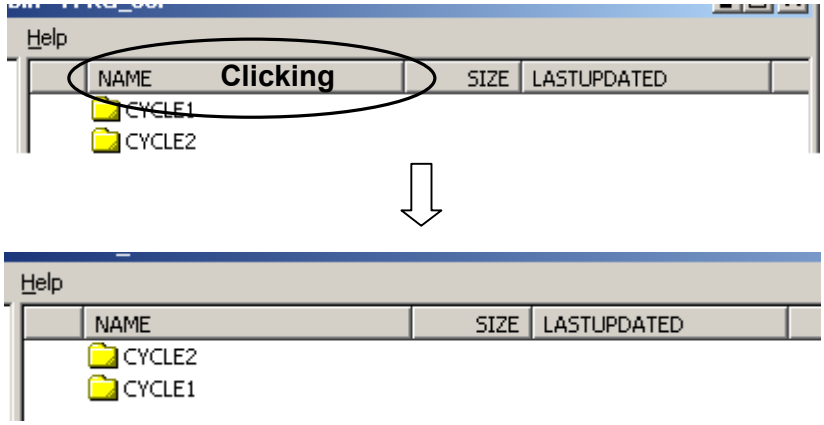
Number of used folder, size of used space, number of unused folder, and size of free space are displayed on status bar in the lower portion of the screen.

When a new memory card program file is created, two reserved folder are also created. Therefore, the number of used folder is two. However it does not show that number of unused folder is reduced.

The display of status bar is renewed at creating or deleting a folder, dropping-in from Explorer, and deleting a program file.

- Sorting list view of the memory card program file

When a column is being clicked, the list view of the memory card program file is being sorted by the column key in ascending or descending order.
The initial display is sorted by NAME key in ascending order.



I.2 NAMING RULES

Overview

Naming rules of folder and program file are described as follows.

I.2.1 Naming Rules of Program File

Here are Naming rules of Program file:

- Program file name can have a maximum of 32 characters.
- Program file name can have following characters.
  Alphabet(Upper and lower case letter), numeric character,
  "-"(minus), "+"(plus), "_"(under bar), "."(dot)
  "." and ".." can not be used since these are reserved for system use.

- The File name acts as Program number

When the File name is "O"+ 1-99999999, the file name acts as Program number.

Example)

"O123"	Program number 123
"O1"	Program number 1
"O3000"	Program number 3000
"O99999999"	Program number 99999999
"O0123"	Program number 123

The file name does not act as Program number, and can be dropped in:

"ABC"	Top of character is not upper case letter of "O".
"o123"	Top of character is not upper case letter of "O".
"O0123XY"	Characters following the "O" contains alphabet.

The file name does not act as Program number, and cannot be dropped in:

"O123456789"	Numeric characters exceed 8 digits.

NOTE
1 Program file name cannot be repeated in a Folder.
2 If program file name starts with "O" and the next eight characters are all numeric characters, the "0" (zero) after the "O" will be deleted.

I.2.2 Naming Rules Of Folder

Here are Naming rules of Folder:

- Folder name can have a maximum of 32 characters.
- Folder name can have following characters.
  Alphabet(Upper and lower case letter), numeric character,
  "-"(minus), "+"(plus), "_"(under bar), "." (dot)
  "." and ".." can not be used since these are reserved for system use.

NOTE
Folder name cannot be repeated in a Folder.

I.3 RULES OF CHARACTERS IN PROGRAM FILE

Overview

Words in parentheses "( )" in Program file are treated as comments.
The mark of comment start "(" is named "Control-out".
The mark of comment end ")" is named "Control-in".
"Control-out" and "Control-in" must make a pair. The order is 1st - "Control-out" and 2nd -"Control-in". And nested parentheses is not available.

NOTE

1 When a program file is dropped in, space code (0x20 SPC) , tab code (0x09 HT) , carriage return code(0x0d CR) and percent code(0x25 %) is deleted. If "%" is found in Control-in, characters between "%" and next "LF" (0x0a) is deleted.
2 The front of program number ":" is changed to "O"(O as in Oscar) while the program file dropped in.
3 The program file can only have one NC program.

I.3.1 Usable Characters in Program File

- Usable characters in Control-in

List of ANSI(ASCII) codes of usable characters(hexadecimal form)

Code	Character	Code	Character	Code	Character	Code	Character
0a	LF	3f	?	58	X	74	t
23	#	40	@	59	Y	75	u
26	&	41	A	5a	Z	76	v
28	(	42	B	5b	[	77	w
29	)	43	C	5d	]	78	x
2a	*	44	D	5f	_	79	y
2b	+	45	E	61	a	7a	z
2c	,	46	F	62	b		
2d	-	47	G	63	c		
2e	.	48	H	64	d		
2f	/	49	I	65	e		
30	0	4a	J	66	f		
31	1	4b	K	67	g		
32	2	4c	L	68	h		
33	3	4d	M	69	i		
34	4	4e	N	6a	j		
35	5	4f	O	6b	k		
36	6	50	P	6c	l		
37	7	51	Q	6d	m		
38	8	52	R	6e	n		
39	9	53	S	6f	o		
3a	:	54	T	70	p		
3c	<	55	U	71	q		
3d	=	56	V	72	r		
3e	>	57	W	73	s		

NOTE
In the Control-in, "O", ":", and "<" can not be used at top of the line except for the 1st line.

- Usable characters in Control-out(characters in parentheses)

List of ANSI(ASCII) codes of usable characters(hexadecimal form)

Code	Character	Code	Character	Code	Character	Code	Character
0a	LF	3c	<	55	U	71	q
20	SPC	3d	=	56	V	72	r
22	“	3e	>	57	W	73	s
23	#	3f	?	58	X	74	t
24	$	40	@	59	Y	75	u
26	&	41	A	5a	Z	76	v
27	‘	42	B	5b	[	77	w
2a	*	43	C	5d	]	78	x
2b	+	44	D	5f	_	79	y
2c	,	45	E	61	a	7a	z
2d	-	46	F	62	b		
2e	.	47	G	63	c		
2f	/	48	H	64	d		
30	0	49	I	65	e		
31	1	4a	J	66	f		
32	2	4b	K	67	g		
33	3	4c	L	68	h		
34	4	4d	M	69	i		
35	5	4e	N	6a	j		
36	6	4f	O	6b	k		
37	7	50	P	6c	l		
38	8	51	Q	6d	m		
39	9	52	R	6e	n		
3a	:	53	S	6f	o		
3b	;	54	T	70	p		

I.4 ERROR MESSAGE AND NOTE

Error may occur when using this application, hereafter explains the error messages and gives relative instructions.

I.4.1 List of Error Message

When an error occurred, the error message box is displayed as follows.

Message	Remarks
Failed to open the file you specified.	If failed to open once again, the file may be broken.
Failed to read or write to the specified file.	
There is insufficient disk space.	There is no enough free space to create a new memory card program file or to put drop out files to work folder. In the latter case, please refer to chapter "Menu" [Change Work Folder]
File name is not correct.	Please refer to the chapter "Naming rules of Program file".
Input name is already exists.	Please input another name.
File name is already exists.	Please input another name.
Input name is not correct.	Please refer to the chapter "Naming rules".
Please input an integer between 2 and 2048	The size of the memory card program file is available from 2Mbyte to 2048Mbyte.
An illegal character is included in the specified file.	Please refer to the chapter "Rules of characters in Program file".
Last update time of the specified file is unsupported.	The updating time of program file is available from 1997 to 2037.
The memory card program file you specified cannot be identified.	The specified file format is not the memory card program file
There is insufficient free folder.	Please delete unnecessary program files or folders.
There is insufficient free program space.	Please delete unnecessary program files.
Folder deeper than 7 cannot be created.	
Root folder cannot be deleted.	
Root folder cannot be renamed.	
Failed to create work folder.	Please check whether the [temp] work folder is able to create or not. In default, the [temp] work folder is created on the same place of this PC tool "FPRG_30i.exe".
Process has been cancelled.	
The specified work folder is not found.	The Drop-out cannot be executed. Terminate this PC tool. And check the setting of Work Folder on the Option dialogue window.
Only one instance of this application can be executed.	Double activation of this PC tool cannot be executed.

I.4.2 Note

- Folder and Program Numbers

Folder and Program Numbers can be selected among 63 / 500 / 1000.
For selecting the Numbers of 500 or 1000, the option "Registered programs expan. On the memory card" is needed in CNC side.

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